Modular Koszul Duality.
Volume IV: Arithmetic, Interfaces, and Trace Formulas

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Abstract

Arithmetic realization compares the operations that produce an invariant before evaluating it. We study coefficient algebras with specified modules, local conditions, units and functionals, together with the comparison maps that preserve these data. Explicit Gaussian, cubic and matrix examples distinguish a scalar equality from a comparison of products, higher operations or intermediate states.

The algebraic constructions include relative coefficient envelopes, composition of interfaces and their extension–restriction corings. Descent is proved under explicit completeness and detection hypotheses. For a strict cospan of differential graded algebras over a commutative ring, the two endpoint products admit an integral all-arity isomorphism of bar coalgebras. Its module and bimodule forms retain every ordered product of local primitives. The comparison extends to restricted inverse limits of finite coefficient bars; an unrestricted product over tensor lengths can fail continuity.

The arithmetic examples concern Galois cochains and local conditions, Hall and Hecke operations, coefficient duality, categorical traces and Chern–Simons comparisons. Their geometric and automorphic identifications require separately specified maps. The analytic endpoint is the complete signed Weil functional, with its polar, archimedean and prime contributions. Positive forms on auxiliary algebras do not identify this functional without a compatible evaluation theorem. The native arithmetic comparisons and positivity on the full Weil test domain remain distinct construction problems.

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Frame

What a trace forgets

A trace assigns a number to an operation. To recover the operation from that number, one must know what the trace has forgotten. Four elementary calculations will guide the arithmetic constructions in this volume. They retain, respectively, a primitive of a relation, a higher product, an intermediate module state, and a geometric condition on an arithmetic operator. Each calculation identifies data that a comparison must retain.

Gaussian reduction and multiplication

Fix a real number $\hbar > 0$. The normalized Gaussian functional on polynomials is

$$\mathbb{E}_\hbar[p] = \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} p(x) e^{-x^2/(2\hbar)}\, dx.$$

Integration by parts has no boundary term and gives

$$\mathbb{E}_\hbar[xp - \hbar p'] = 0. \tag{1}$$

This relation defines a two-term cochain complex

$$C^{-1} = \mathbb{R}[x]\varepsilon, \qquad C^0 = \mathbb{R}[x], \qquad d(p\varepsilon) = xp - \hbar p'.$$

There are no other degrees. Its differential is injective: if p has degree n and nonzero leading coefficient, then $xp - \hbar p'$ has degree $n+1$ with the same leading coefficient. The relations

$$[x] = 0, \qquad [x^{n+1}] = \hbar n [x^{n-1}], \qquad n \geq 1,$$

show that $H^0(C)$ is spanned by $[1]$. Equation (1) and $\mathbb{E}_\hbar[1] = 1$ show that $[1] \neq 0$, so

$$H^{-1}(C) = 0, \qquad H^0(C) \cong \mathbb{R}.$$

Under this identification,

$$\mathbb{E}_\hbar[x^{2r+1}] = 0, \qquad \mathbb{E}_\hbar[x^{2r}] = \hbar^r \frac{(2r)!}{2^r r!}.$$

The recurrence proves the formula without evaluating another integral.

Polynomial multiplication does not descend to this quotient. Indeed,

$$d\varepsilon = x, \qquad d(x\varepsilon) = x^2 - \hbar,$$

so $[x] = 0$ but $[x^2] = \hbar[1] \neq 0$. Equivalently, $\mathbb{E}_\hbar[x]^2 = 0$ and $\mathbb{E}_\hbar[x^2] = \hbar$. The differential is not a derivation for the graded product on $\mathbb{R}[x] \otimes \Lambda(\varepsilon)$, where $|\varepsilon| = -1$ and $\varepsilon^2 = 0$: the Leibniz rule applied to $x\varepsilon$ would omit the term $-\hbar$. Thus a valid reduction of complexes need not be a reduction of algebras.

The specified primitives $p\varepsilon$ record which relations have been imposed and how products encounter them.

Even when integration produces an effective quadratic action, that action does not determine the normalization. Let $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$, and let

$$Q = \begin{pmatrix} A & B \\ B^t & C \end{pmatrix}$$

be real symmetric and positive definite. Here C is invertible. Completing the square gives

$$\int_{\mathbb{R}^m} \exp\left[-\frac{x^t A x + 2x^t B y + y^t C y}{2\hbar}\right] dy \\ = (2\pi\hbar)^{m/2} (\det C)^{-1/2} \exp\left[-\frac{x^t (A - BC^{-1}B^t) x}{2\hbar}\right].$$

To verify both factors, substitute $y + C^{-1}B^t x$ for y and diagonalize the positive matrix C. Each resulting one-dimensional integral contributes $\sqrt{2\pi\hbar/c_i}$, where c_i is an eigenvalue of C. Already with $B = 0$, varying C leaves the effective action $x^t A x/2$ unchanged and changes its amplitude. A comparison of prepared states must therefore preserve both the quadratic response and the determinant factor.

A cubic relation in a strict differential graded algebra

Let k be a field of characteristic zero and $A = k[x]/(x^3)$, with augmentation $x \mapsto 0$. A free resolution of the augmentation module k, in nonpositive cohomological degrees, is

$$\cdots \xrightarrow{x} A \xrightarrow{x^2} A \xrightarrow{x} A \longrightarrow k \longrightarrow 0.$$

Write $P^{-n} = Ae_n$ for $n \geq 0$, and

$$d_P e_n = \begin{cases} x e_{n-1}, & n \text{ odd}, \\ x^2 e_{n-1}, & n > 0 \text{ even}. \end{cases}$$

Exactness follows from $\ker(x : A \to A) = (x^2)$ and $\ker(x^2 : A \to A) = (x)$.

The graded endomorphisms of P form an associative differential graded algebra E, with composition as product and differential

$$D(T) = d_P T - (-1)^{|T|} T d_P.$$

Define maps U, H of degree one and V of degree two by

	n odd	$n > 0$ even
Ue_n	e_{n-1}	$-xe_{n-1}$
He_n	0	$-e_{n-1}$

$$Ve_n = e_{n-2} \quad (n \geq 2).$$

Each map is zero when the indicated output degree is absent. The two parities give the identities

$$DU = DV = 0, \qquad U^2 = -xV = DH, \qquad UH + HU = -V. \tag{2}$$

For example, on e_2, both U^2 and DH give $-xe_0$, while $(UH + HU)e_2 = -e_0$. The same computations repeat every two degrees.

Applying $\mathrm{Hom}_A(-,k)$ to P gives one copy of k in each nonnegative degree and zero differential. Hence $\mathrm{Ext}^n_A(k,k) \cong k$ for every $n \geq 0$. The augmentation $P \to k$ identifies $H^*(E)$ with these groups. Here one uses the elementary property that a bounded-above complex of projective modules maps every acyclic complex to an acyclic Hom complex: a cocycle is made null-homotopic successively from the highest nonzero degree down, lifting at each step through the surjection onto cycles. This applies to the cone of $P \to k$.

Write $u = [U]$ and $v = [V]$. The maps V^r and UV^r map to nonzero elements of $\mathrm{Hom}_A(P^{-2r}, k)$ and $\mathrm{Hom}_A(P^{-(2r+1)}, k)$, respectively. Also $UV = VU$, since their coefficients have the same parity. Together with $U^2 = DH$, this proves

$$H^*(E) = \Lambda_k(u) \otimes_k k[v], \qquad |u| = 1, \quad |v| = 2.$$

There is more information in E than in this binary product. For a degree-one cocycle U with $U^2 = DH$, the cohomology class of $UH + HU$ is its triple Massey product, with this sign convention. It is closed because

$$D(UH + HU) = -U(DH) + (DH)U = 0.$$

Changing either primitive changes its class by an element of $uH^1(E) + H^1(E)u$. This indeterminacy is zero here, because $H^1(E) = ku$ and $u^2 = 0$. Thus

$$\langle u, u, u \rangle = \{-v\} \neq \{0\}. \tag{3}$$

In the graded algebra $H^*(E)$ with zero differential, the same triple product is $\{0\}$, by choosing both primitives zero. Maps of differential graded algebras preserve the displayed construction and its indeterminacy. A quasi-isomorphism therefore preserves this single-valued product under its cohomology identification. Consequently E cannot be joined to $H^*(E)$ with zero differential by a chain of quasi-isomorphisms. Its multiplication was strictly associative from the start. Strict associativity has retained the cubic information in a differential and its primitives; it has not removed it.

An intermediate state in an ordered product

Let e_1, e_2 be the standard basis of k^2, and let E_{ij} send e_j to e_i and the other basis vector to zero. Put $a = E_{12}$, $b = E_{21}$. Then

$$aba = a, \qquad aab = 0. \tag{4}$$

The first word carries e_2 to e_1, then to e_2, then to e_1, reading actions from right to left. The reordered word kills every vector. The order of the intermediate states is part of the operation.

The normalized trace $\tau(M) = \mathrm{Tr}(M)/2$ gives $\tau(ab) = \tau(ba) = 1/2$, and both words in (4) have trace zero. A later input distinguishes them:

$$\tau(abab) = \tfrac{1}{2}, \qquad \tau(aabb) = 0.$$

Even replacing a two-factor product by the scalar operator with the same trace changes subsequent composition. The linear projection $p(M) = \tau(M)I$ satisfies

$$p(ab)a = \tfrac{1}{2}a \neq aba.$$

Thus tracing a collision before composing it with the remaining inputs can discard a coefficient that those inputs still use.

POSITIVE COUNTING DATA WITHOUT THE REQUIRED ABSOLUTE VALUES

Consider the integral matrix

$$M = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \qquad N_n = 5^n + 1 - \operatorname{Tr}(M^n), \qquad n \geq 1.$$

For a positive integer d, let $\mu(d)$ be the Möbius function: it is zero if a prime square divides d, and is $(-1)^r$ if d is the product of r distinct primes. Define

$$a_d = \frac{1}{d} \sum_{e \mid d} \mu(d/e) N_e. \tag{5}$$

Every a_d is a positive integer. For integrality, regard M as the adjacency matrix of a finite directed multigraph. The number $\operatorname{Tr}(M^n)$ counts closed walks of length n with a marked starting point. Grouping them by their least period gives

$$\operatorname{Tr}(M^n) = \sum_{d \mid n} d\, b_d(M),$$

where $b_d(M)$ is the number of primitive cyclic walks of length d. The same statement holds for the one-vertex matrices (5) and (1). Möbius inversion therefore expresses a_d as the difference of three integers.

To prove positivity, write $\lambda = (5 + \sqrt{5})/2$ and $\lambda' = (5 - \sqrt{5})/2$. Both are positive, their sum is 5, and $\operatorname{Tr}(M^n) = \lambda^n + (\lambda')^n$. Consequently $0 < N_n \leq 5^n + 1$ and $N_n \geq 5^n - 2\lambda^n$. Every proper divisor of d is at most $d/2$, so

$$\sum_{\substack{e \mid d \\ e < d}} N_e \leq 2 \sum_{e=1}^{\lfloor d/2 \rfloor} 5^e < 5 \cdot 5^{d/2}.$$

Since $\lambda/5 < 3/4$, equation (5) gives, for $d \geq 4$,

$$d a_d > 5^d \left(1 - 2(3/4)^d - 5^{1-d/2}\right) > 0.$$

The bracket increases with d, and its value at $d = 4$ is $107/640$. Direct calculation gives

$$a_1 = 1, \quad a_2 = 5, \quad a_3 = 25, \quad a_4 = 110, \quad a_5 = 500.$$

Take a_d formal closed points of degree d. Their zeta series, defined by a formal Euler product, is

$$\begin{aligned} Z(t) &= \prod_{d \geq 1} (1 - t^d)^{-a_d} = \exp\left(\sum_{n \geq 1} \frac{N_n}{n} t^n \right) \\ &= \frac{1 - 5t + 5t^2}{(1 - t)(1 - 5t)}. \end{aligned}$$

Each coefficient uses only finitely many factors. The middle identity follows from $N_n = \sum_{d \mid n} d a_d$. The last follows by diagonalizing M, or by the formal identity $-\log\det(I - tM) = \sum_{n \geq 1} \operatorname{Tr}(M^n) t^n / n$.

These counting data admit positive forms of the usual Hall type. At a point of degree d, take finite-dimensional nilpotent $\mathbb{F}_{5^d}[u]$-modules. An object L is a finite direct sum of such local modules over

the formal points. Each automorphism group is finite. On the complex vector space with basis their isomorphism classes, the form

$$\left\langle \sum_L z_L [L], \sum_L w_L [L] \right\rangle = \sum_L \frac{\overline{z_L} w_L}{|\operatorname{Aut}(L)|}$$

is positive definite on finite sums. Extensions and subobjects are finite counting problems at each point; no algebraic curve has been used to supply them.

There is also a reciprocal polynomial and an alternating form:

$$P(t) = 1 - 5t + 5t^2 = 5t^2 P(1/(5t)), \qquad M^t J M = 5J, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Nevertheless the reciprocal roots λ, λ' of P do not have absolute value $\sqrt{5}$. Positive extension-counting forms and these two identities thus do not imply the weight-one absolute-value condition. They have not supplied a geometric Frobenius realization with that property.

The arithmetic comparisons that follow must retain the data exposed by these calculations. A relation has a specified primitive; a product can have a nonzero higher operation; a collision can leave an intermediate state; and a positive auxiliary form need not be the arithmetic functional to be realized. The comparison maps, their coefficient domains, and the operations they preserve are therefore part of the mathematical object studied in this volume.

Preface

Arithmetic operations produce quantities that can also appear as characters, partition functions or spectral traces. The subject of this volume is the comparison before evaluation: the coefficient algebras, modules, relative conditions and geometric operations that could explain such an equality. Its source and target must be constructed, and the map between them must preserve the structures consumed by later operations. Equality of the final numbers is then a consequence with a specified mathematical domain.

The prologue gives four reasons for retaining this information. Gaussian integration preserves a relation complex while changing both products and normalization. The cubic algebra has a strict differential graded model with a nonzero triple product. A matrix coefficient lost by an early trace changes a subsequent four-input evaluation. Positive extension-counting forms and a reciprocal polynomial can coexist with the wrong absolute values for a geometric weight-one operator. These are calculations on explicit carriers; each identifies data that the corresponding scalar has omitted.

Chapter 1 defines comparisons preserving an algebra, a module, a selected vector and a functional. It separates objectwise existence, uniqueness and coherent choice. The coefficient ring and its topology enter the definition: a finite torsion ring admits no unital map to a characteristic-zero field, and taking an inverse limit before rationalization is a different construction.

The algebraic core then retains intermediate coefficient states through relative interfaces. Chapter 8 constructs their composition under the stated central-action and derived tensor hypotheses. Chapter 9 identifies the coring introduced by restriction of coefficients and proves descent under explicit completeness and detection hypotheses. An empty coefficient and a local unit have different carriers; an adjunction counit need not identify a restricted composite with its original interface.

Relative arithmetic classes also retain primitives. For a restriction map $f : A \to B$, a relative cocycle (a, y) satisfies $da = 0$ and $f(a) = dy$. Multiplying two such classes can use one or the other local primitive. Chapter 11 constructs the two products and their integral comparison in every arity, including the induced module and bimodule operations. No division is needed. A restricted inverse limit over finite coefficient rings preserves the comparison; an unrestricted product over word lengths need not do so. Its interpretation as a derived tensor has its own flatness hypothesis.

The comparison questions from the first three volumes retain their specific invariants. For an automorphic product with seed constant coefficient $c_N(0)$, the weight identity

$$w_N = \frac{c_N(0)}{2}$$

is one possible numerical comparison. Interpreting this weight as a Borcherds–Kac–Moody invariant, written κ_{BKM}, requires the actual denominator construction and its normalization. Reading the same seed coefficient twice is not a second construction. Likewise the affine expression

$$\kappa = \frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee},$$

for a simple Lie algebra $\mathfrak{g}$, level k and dual Coxeter number $h^\vee$, is an invariant to compare in its stated normalization. It is not, by itself, a screening-cohomology comparison.

For the Virasoro shadow model, the proposed comparisons involve the rational functions

$$c_S = -\frac{218}{45}, \qquad |\omega|^2(c) = \frac{c^2(5c+22)}{4(45c+218)}, \qquad S_5(c) = -\frac{48}{c^2(5c+22)}.$$

The pole and denominators specify excluded parameter values for these formulas. If an actual function H satisfies $H^2 = t^4 Q_c$, then, on a domain where the quotients are defined,

$$U = (\log H)' - \frac{2}{t} = \frac{Q_c'}{2Q_c}, \qquad U' + 2U^2 = \frac{Q_c''}{2Q_c}.$$

The second identity follows by differentiating the first. Identifying these functions with a shadow tower, a Borel invariant or an operator-product coefficient requires the respective construction. The four quantities (K, K^κ, K_c, K_g), the conductor, complementarity, central-charge and ghost-number invariants, likewise have different definitions. Their comparisons, the determinant at $p = 691$, and the eight-form Gritsenko–Cléry family enter with the source hypotheses stated for each object.

For the finite-type, freely generated finite-jet Khan–Zeng Poisson vertex models, the shifted-cotangent Batalin–Vilkovisky construction compares the classical master equation with the Hamiltonian condition. The quantum master equation, reflected weights, boundary vertex algebra and higher operadic lift require additional analytic and geometric constructions. The spectral-decomposition retract, Fulton–MacPherson boundary terms, reflection conventions and the lift map belong to those hypotheses; the classical calculation does not supply them. The holomorphic–topological comparison also concerns the seven presentations of the two-coloured chiral–topological Swiss-cheese datum $\mathsf{SC}^{\mathrm{ch,top}}$. The operadic, Koszul-dual, factorization, BV/BRST and convolution descriptions require their actual comparison maps. The Drinfeld-centre and derived-geometric descriptions further require the specified enrichment and representability hypotheses.

On the Calabi–Yau side, a specified two-stage construction has the form

$$\Phi_d = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}}.$$

The first map produces the declared factorization object; the second uses the chosen cycle Σ_{d-1}, curve C and its specialization data. Comparisons with a K3 Hall–Drinfeld object $\mathbf{H}_{\Delta_5}$, Joyce's vertex algebra and Kontsevich–Pandharipande–Soibelman orientation data, the Maulik–Nekrasov–Okounkov–Pandharipande invariants, or a framed Lagrangian theory each require their own maps. The proposed K3 representation targets are the full rank-24 Mukai module, a Heisenberg–Mukai quotient, and an intermediate quantum vertex W-algebra. Identifying any of these targets requires the evaluation map, its preserved form and the specified $\mathfrak{so}(4,20)$ action. An S^3-framing likewise belongs to the Lagrangian data. A proved specialization does not identify all the other constructions with it.

The analytic arithmetic endpoint is the full signed Weil functional, with both polar evaluations, the archimedean term and every prime power. A positive state on an auxiliary algebra gives a realization of that functional only after a product- and involution-preserving comparison and its evaluation identity have been proved on the full test domain. Chapter 119 separates the stated arithmetic comparisons from the remaining Weil positivity condition; its list is not an exhaustion theorem for every possible realization problem.

The topology of a proposed spectral recipient can already exclude a comparison. The graph norm of multiplication by t on $L^2(\mathbb{R}, dt)$ is

$$\|f\|^2_{\mathrm{gr}} = \int_{\mathbb{R}} (1+t^2)|f(t)|^2\, dt.$$

Every continuous functional is integration against a measurable function q with $\int |q(t)|^2/(1+t^2)\,dt < \infty$. Such a function is locally integrable. If its distribution is supported on a locally finite set, it must vanish, since that set has Lebesgue measure zero. Put $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, where ζ is the Riemann zeta function. In particular the graph dual cannot carry nonzero point or finite-jet residues on

$$\Gamma_{\xi}^{\mathrm{crit}} = \{\gamma \in \mathbb{R} : \xi(\tfrac{1}{2} + i\gamma) = 0\}.$$

A spectral-flow construction using these residues needs a stronger specified domain, such as a suitable rigged, Paley–Wiener, de Branges or reproducing-kernel space, or a boundary theory with continuous trace and jet maps. It must also retain multiplicities, the full Hadamard divisor where needed, and compatibility with the post-Tate quotient. The choice of recipient is therefore part of the arithmetic comparison itself.

Part I

The Realization Formalism

Chapter 1

Marked realization problems

An arithmetic construction can determine an algebra, a module, and a trace without determining a geometric realization of them. A comparison must specify which of these data it preserves. It must also specify the maps used when the inputs vary. These requirements are already distinct for two-dimensional modules over the dual numbers.

1.1 A character does not determine a module

Let k be a field of characteristic zero, and put $A = k[\epsilon]/(\epsilon^2)$. On the vector space k^2, with basis e_1, e_2, consider the representations

$$\rho_0(a + b\epsilon) = aI, \qquad \rho_1(a + b\epsilon) = aI + bN, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Both are unital algebra maps, since $N^2 = 0$. Their characters, the functions obtained by taking the matrix trace, agree on every element of A:

$$\operatorname{Tr} \rho_0(a + b\epsilon) = \operatorname{Tr} \rho_1(a + b\epsilon) = 2a. \tag{1.1}$$

They are nevertheless nonisomorphic as A-modules. An intertwining isomorphism g would satisfy $g\rho_0(\epsilon) = \rho_1(\epsilon)g$, hence $Ng = 0$, contradicting invertibility. This remains impossible if one permits an automorphism of A: such an automorphism sends ϵ to $c\epsilon$ for some $c \in k^\times$, so its nonzero nilpotent action remains nonzero.

The vector e_2 gives a further useful mark. It generates the whole module under ρ_1, since $Ne_2 = e_1$; under ρ_0 it generates only ke_2. The common character has forgotten an extension and a cyclic-vector condition. Even equality of all these scalar evaluations does not provide the module comparison.

1.2 The data carried by a comparison

Fix a commutative ring R. Complexes are cohomological and all tensor products in this section are over R. A *marked algebra with module and functional* is a tuple

$$\mathcal{A} = (A, M, v, \tau),$$

where A is a unital associative differential graded algebra, M is a left differential graded A-module, $v : R \to M$ is a chain map, and $\tau : A \to R$ is a chain map of degree zero. The chosen vector $v(1)$ is therefore closed. If the functional is required to be cyclic, this means

$$\tau(ab) = (-1)^{|a||b|}\tau(ba)$$

for homogeneous inputs. Cyclicity is an extra hypothesis; it is not part of the meaning of an arbitrary functional.

A comparison $(f, g) : \mathcal{A} \to \mathcal{A}'$ consists of a unital differential graded algebra map $f : A \to A'$ and a chain map $g : M \to M'$ such that

$$g(am) = f(a)g(m), \qquad g \circ v = v', \qquad \tau' \circ f = \tau. \tag{1.2}$$

An isomorphism requires both maps to be invertible. A comparison need not be an isomorphism: its image and the operations lost in its kernel can be part of the question.

The two distinguished maps in this tuple have different types. The algebra unit is a map $R \to A$. The selected vector is a map $R \to M$. Unitality of the action says $1_A m = m$; it does not choose v. In geometric constructions an empty coefficient and a local vacuum likewise need their own maps on their respective carriers.

Additional operations enlarge the tuple and the equations (1.2). For example, an endomorphism $F_A : A \to A$ is preserved when $f F_A = F_{A'} f$; an operator F_M on the module requires $g F_M = F_{M'} g$. An involution requires its own compatibility. If a multiplication is associative only through specified higher homotopies, the comparison contains components compatible with those homotopies. Keeping only its first component is then a different comparison problem.

PROPOSITION 1.2.1. The displayed marked objects and their comparisons form a category. Every comparison preserves an ordered module operation and its final evaluation whenever the latter is defined by the retained maps. In particular,

$$g(a_1 \cdots a_r v(1)) = f(a_1) \cdots f(a_r) v'(1), \qquad \tau'(f(a_1) \cdots f(a_r)) = \tau(a_1 \cdots a_r).$$

Proof. Compose the algebra maps and the module maps separately. Substitution in each equation of (1.2) verifies it for the composite; identity maps satisfy the same equations. Ordinary composition of chain maps gives associativity. The first displayed identity follows by induction using the module equation. The second uses multiplicativity of f and then $\tau' f = \tau$. No factors have been interchanged, so no new Koszul sign occurs. □

The converse is false if only scalar identities are retained. Equation (1.1) gives characters agreeing on every input while the modules are not isomorphic. The ordered matrix calculation in the prologue shows why even a scalar identity at one arity may fail to survive insertion into a larger operation.

1.3 Coefficient change and geometric carriers

A change of coefficient ring begins with an actual unital map $R \to S$ of commutative rings. Ordinary extension gives

$$A_S = S \otimes_R A, \qquad M_S = S \otimes_R M, \qquad v_S = 1 \otimes v, \qquad \tau_S = 1 \otimes \tau,$$

using $S \otimes_R R \cong S$. Its algebra product and action are the induced ones. Tensoring the equations (1.2) gives a comparison over S. This algebraic construction does not require flatness. Extension depending only on quasi-isomorphism classes uses derived tensor products, with compatible algebra and module resolutions when needed.

For a prime ℓ and $r \geq 1$, there is no unital ring map $\mathbb{Z}/\ell^r \to K$ to a field K of characteristic zero: the relation $\ell^r 1 = 0$ would have to hold in K. More generally, if an abelian group V is annihilated by ℓ^r, then

$$V \otimes_{\mathbb{Z}} K = 0,$$

because $v \otimes c = \ell^r v \otimes (c/\ell^r) = 0$. Thus direct rationalization of finite-torsion cochains loses their coefficients.

The inverse-limit construction is different. With the usual reduction maps,

$$\mathbb{Z}_\ell = \varprojlim_r \mathbb{Z}/\ell^r, \qquad \mathbb{Q}_\ell = \mathbb{Z}_\ell[1/\ell].$$

Every term of the inverse system tensors to zero over $\mathbb{Z}$ with $\mathbb{Q}$, whereas $\mathbb{Z}_\ell \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}_\ell \neq 0$: an integer prime to ℓ is already a unit in $\mathbb{Z}_\ell$. Tensoring with $\mathbb{Q}$ has therefore not commuted with this inverse limit. There is also no continuous unital field map $\mathbb{Q}_\ell \to \mathbb{C}$ with their usual topologies. Such a map fixes the rational numbers, but the sequence ℓ^n tends to zero in $\mathbb{Q}_\ell$ and has unbounded complex absolute value. This argument concerns continuity, not the existence of an abstract field embedding.

Geometry introduces another change of carrier. If an arithmetic input is defined over R and the position curve X is defined over a field k, the coefficient functor and its interaction with the operations must be specified. Even after a scalar map $R \to k$ has been chosen, a coefficient algebra alone gives no sheaves on the Cartesian powers X^I, for finite ordered sets I, no restrictions to collision strata, and no maps for inserting one configuration into another.

For a chosen family of sheaves A_I, write $\boldsymbol{Q}_n(A_{I_1}, \dots, A_{I_n}; A_J)$ for its declared complex of allowed operations. The definition of $\boldsymbol{Q}_n$ must specify supports, pole or growth bounds, tensor products and completions, as appropriate to the example. A geometric comparison is then a system of maps on these actual complexes preserving their differentials, substitutions and specified units. A primitive in a larger Hom complex need not belong to $\boldsymbol{Q}_n$. Consequently an obstruction being exact after forgetting support or continuity does not provide a comparison in the chosen category.

1.4 Existence, uniqueness, and coherent choice

Suppose that a category of realizations and a category of retained data have been constructed, together with a forgetful functor. A groupoid is a category in which every arrow is invertible. To ask for realizations up to isomorphism, retain all objects and only invertible arrows. Denote the resulting groupoids by $\mathscr{R}$ and $\mathscr{D}$, and the functor between them by $U : \mathscr{R} \to \mathscr{D}$. For $d \in \mathscr{D}$, its *realization groupoid* has objects

$$\operatorname{Ob} \operatorname{Real}_U(d) = \{(r, \alpha) : r \in \mathscr{R},\ \alpha : U(r) \xrightarrow{\sim} d\}.$$

An arrow from (r, α) to (r', α') is an arrow $h : r \to r'$ satisfying $\alpha' U(h) = \alpha$. This description is the homotopy fibre of the map of classifying spaces of the two groupoids.

Existence means that this groupoid has an object. Uniqueness up to isomorphism means that it is connected. Uniqueness with a unique compatible isomorphism means that it is equivalent to the one-object groupoid with only its identity arrow; its classifying space is then contractible. Automorphisms of a realization can prevent this last condition even when every realization is isomorphic to it.

In a setting with higher morphisms the analogous definition is

$$\operatorname{Real}_U(d) = \operatorname{hofib}_d(U^{\simeq} : \mathscr{R}^{\simeq} \to \mathscr{D}^{\simeq}),$$

where the superscript denotes the space retaining objects and all equivalences. That formula presupposes the categories and U; it does not construct them. A comparison required to be noninvertible belongs to the relevant category of arrows, with its prescribed source and target, rather than to the displayed space of identifications.

A functor is fully faithful when its maps on all sets of arrows are bijections. It is essentially surjective when every target object is isomorphic to an image object.

PROPOSITION 1.4.1. For a functor between groupoids, every realization groupoid is equivalent to a point if and only if the functor is fully faithful and essentially surjective.

Proof. Essential surjectivity supplies an object (r, α) in each fibre. Full faithfulness supplies the unique arrow between any two such objects, namely the lift of $(\alpha')^{-1}\alpha$. This makes each fibre equivalent to a point.

Conversely, nonemptiness of the fibres gives essential surjectivity. Given $\beta : U(r) \to U(r')$, consider the two objects (r, β) and $(r', 1)$ in the fibre over $U(r')$. Its equivalence to a point supplies exactly one arrow between them. By its defining equation, this is exactly one lift of β. Thus U is fully faithful. □

Objectwise existence alone does not give coherent choices. For an explicit example, let $C_n = \mathbb{Z}/n\mathbb{Z}$, written additively, and let BC_n be the groupoid with one object and automorphism group C_n. Reduction modulo two gives

$$U : BC_4 \longrightarrow BC_2.$$

Its realization groupoid has two objects, corresponding to the two identifications in C_2. Reduction $C_4 \to C_2$ is surjective, so they are isomorphic. The automorphism group of either is the kernel $\{0, 2\} \cong C_2$. Thus the fibre is equivalent to BC_2; in particular it is nonempty.

A functorial section, even one whose composite with U is only naturally isomorphic to the identity, would give a group homomorphism $s : C_2 \to C_4$ splitting reduction. The natural isomorphism changes the composite by conjugation, which is trivial in C_2. But the image of the nonzero element would have to have order dividing two, so it is 0 or 2; both reduce to zero. No such section exists. The missing compatibility is on arrows, although the objectwise existence problem has been solved.

1.5 EVALUATION AFTER THE COMPARISON

The functional equation in (1.2) is a commutative triangle on a common domain. If the two functionals are defined independently, its proof identifies their evaluations. If instead τ is defined to be $\tau' f$, the same triangle holds by definition and gives a transported functional. It has not identified an independently prescribed arithmetic one. Both constructions are legitimate, with different conclusions.

For positivity the algebraic structure being preserved is decisive. Let A, B be complex algebras with conjugate-linear involutions $*$ satisfying $(ab)^* = b^*a^*$, and let $f : A \to B$ preserve the product and involution. Suppose that τ_B is linear on a subspace containing every $f(a)^*f(a)$, and $\tau_B(f(a)^*f(a)) \geq 0$ for all $a \in A$. If $\tau_A(a^*a) = \tau_B(f(a^*a))$ on the same inputs, then

$$\tau_A(a^*a) = \tau_B(f(a)^*f(a)) \geq 0.$$

This implication uses the actual product and involution identities. For an unbounded weight, the displayed expressions must lie in its domain; no value at the unit is inferred. A proposed positive trace realization of the Weil functional must supply this map and this evaluation identity on its full test domain.

The construction of each side is part of the comparison problem. Two arguments may use the same homological algebra without being circular; disjoint lists of references can still conceal a target defined from the desired answer. Independence here requires the arithmetic input and the comparison object to be specified without defining one by the invariant that the comparison is meant to explain. The remaining task is to construct the operation-preserving maps and their coherent behavior under change of inputs.

Chapter 2

Words, collisions, and the two units

Consider three ordered positions (x, y, x), with $x \neq y$. Neither cut 1|23 nor cut 12|3 separates the positions on its two sides. The obstruction is the equality of the first and last positions. A construction based on consecutive cuts must therefore retain a nonconsecutive collision. At four positions, the resulting support has two components, and their coefficients meet in a nonsplit extension. These examples determine the geometric data needed before a bar construction can be compared with fields.

We first construct the tensor product associated with ordered cuts. Its unit is supported at the empty configuration. The local vacuum is a different object, present at every number of positions. We then compute the collision supports and their coefficient complexes, and compare a fixed word with the space of all local orders. Finally, we separate these position spaces from the discrete set of arithmetic places.

2.1 The tensor product of ordered cuts

Let k be a field of characteristic zero and X a nonempty smooth separated curve of finite type over k. Write $\mathcal{D}(Y)$ for the unbounded derived category of right algebraic differential modules on a smooth scheme Y, with its stable enhancement; $\mathcal{D}(k) = \mathcal{D}(\operatorname{Spec} k)$ is the derived category of k-vector spaces. All inverse images, direct images and tensor products below are derived. We use the coherent differential-module formalism of external products, extraordinary inverse images $f^!$, direct images f_*, and localization for an open immersion. Thus base-change isomorphisms include their compatibility with composition. This is the formalism used in chiral Koszul duality.[1] Our cohomological convention is $H^q(M[s]) = H^{q+s}(M)$.

Put $[n] = \{1, \dots, n\}$. A monotone surjection $\alpha : [n] \twoheadrightarrow [m]$ defines a diagonal map

$$d_\alpha : X^m \longrightarrow X^n, \qquad (y_1, \dots, y_m) \longmapsto (y_{\alpha(1)}, \dots, y_{\alpha(n)}).$$

A *word coefficient system* consists of $F_0 \in \mathcal{D}(k)$, objects $F_n \in \mathcal{D}(X^n)$ for $n > 0$, and coherent equivalences $d_\alpha^! F_n \simeq F_m$. Coherent means that these equivalences and all their composition homotopies are part of the datum. Its category is

$$\mathcal{W}_X = \mathcal{D}(k) \times \varprojlim_{\substack{[n]\neq\varnothing \\ \text{monotone surjections}}} \mathcal{D}(X^n),$$

where a surjection acts by $d_\alpha^!$. The empty configuration is a separate factor.

For nonnegative p, q with $p + q = n$, let $U_{p,q} \subset X^n$ be the open locus $x_i \neq x_j$ whenever $i \leq p < j$, and write $j_{p,q} : U_{p,q} \hookrightarrow X^n$. If $p = 0$ or $q = 0$, this is the identity open immersion. Define the *cut product* by

$$(F \star_{\mathrm{cut}} G)_n = \bigoplus_{p+q=n} j_{p,q,*} j_{p,q}^!(F_p \boxtimes G_q). \tag{2.1}$$

[1] J. Francis and D. Gaitsgory, *Chiral Koszul duality*, Selecta Mathematica **18** (2012), 27–87, §1.4; arXiv:1103.5803.

Here $X^0 = \operatorname{Spec} k$, so for $n > 0$ the empty terms are the ordinary coefficient tensors $F_0 \otimes_k G_n$ and $F_n \otimes_k G_0$.

PROPOSITION 2.1.1 *(Cut product).* Formula (2.1) defines an associative monoidal structure on $\mathcal{W}_X$. Its unit is

$$K_0 = k, \qquad K_n = 0 \quad (n > 0).$$

For $r \geq 1$ systems $F^{(1)}, \dots, F^{(r)}$, its iterated product in arity n is the sum over consecutive ordered decompositions of $[n]$ into r blocks, localized where positions in different blocks are distinct.

Proof. Let $j : U \hookrightarrow Y$ be an open immersion with closed complement $i : Z \hookrightarrow Y$. Localization gives $i_* i^! M \to M \to j_* j^! M$, and $i^! j_* = 0$. For $N \in \mathcal{D}(U)$ and $f : Y' \to Y$, transitivity of extraordinary inverse images therefore shows that $f^! j_* N$ has zero extraordinary restriction to $f^{-1}Z$. Its restriction to $f^{-1}U$ is the pullback of N. Localization and the full faithfulness of extension from an open set identify it with $j'_* f_U^! N$, where $j' : f^{-1}U \hookrightarrow Y'$. These identifications are coherent under composition.

Apply this to $f = d_\alpha$ and $U = U_{p,q}$. If the cut separates two labels in the same fibre of α, its inverse image is empty. Otherwise it is the inverse image of a unique cut of $[m]$, and the coefficient equivalences for F and G give the corresponding summand in arity m. Thus the displayed sum is a word coefficient system.

In an iterated product, successive localizations require exactly that positions in distinct final blocks be disjoint. Each parenthesization therefore gives the same sum, with the same coefficient and open immersion in every summand. The coherent base-change maps identify these sums; their further compositions agree on the same final open set. This proves every associativity coherence. If one input is K, only its empty block survives, and the remaining summand is the other input. These identifications give the unit coherences. □

Call F *positive* if $F_0 = 0$. For positive inputs, all blocks in the proposition are nonempty. There are $\binom{n-1}{r-1}$ such decompositions when $1 \leq r \leq n$, and none when $r > n$: choose their $r - 1$ cuts among the $n - 1$ gaps. This finiteness concerns the number of geometric blocks. It places no bound on an auxiliary bar length in a coefficient algebra at the empty configuration.

2.2 The empty unit and the local vacuum

There is also a pointwise tensor product of differential modules, using the diagonal $\Delta_{X^n} : X^n \hookrightarrow X^n \times X^n$:

$$(F \otimes^! G)_n = \Delta_{X^n}^! (F_n \boxtimes G_n).$$

Its unit is the dualizing system

$$J_0 = k, \qquad J_n = p_n^! k = \omega_{X^n}[n] \quad (n > 0), \tag{2.2}$$

where $p_n : X^n \to \operatorname{Spec} k$. Transitivity gives $d_\alpha^! J_n = J_m$. The shift in (2.2) is part of the right differential-module convention; J_n is not the unshifted canonical bundle.

A counital coassociative cut coalgebra is an object with a coproduct $C \to C \star_{\mathrm{cut}} C$ and a counit $C \to K$, satisfying their coherent identities. A coaugmentation is a coalgebra map $K \to C$. We call it connected when $C_0 = k$ and the empty components of counit and coaugmentation are identities.

Proposition 2.2.1 *(Vacuum coalgebra).* The system J is a connected cut coalgebra. Its component at a cut is the localization map

$$J_n \longrightarrow j_{p,q,*}J_{U_{p,q}} = j_{p,q,*}j_{p,q}^{!}(J_p \boxtimes J_q).$$

The maps $K \to J \to K$ are identity at the empty configuration and zero in positive arity.

Proof. The adjunction for $j_{p,q}$ gives the displayed map. Two successive coproducts localize on the locus where three consecutive blocks are pairwise disjoint. Both orders of localization give its adjunction map. The same argument works for every number of blocks, proving the coherent coalgebra identities. The cuts with an empty block are identities. They prove the counit and coaugmentation identities. □

A *local vacuum marking* of a connected cut coalgebra is a specified coalgebra map $u : J \to C$. The coaugmentation $K \to C$ determines only its empty value. Even if a nonzero unary vacuum is specified, extending it to such a marking can fail.

Example 2.2.2 *(A unary vacuum that does not extend).* Let $\Delta_n : X \hookrightarrow X^n$ be the small diagonal and put

$$P_0 = k, \qquad P_n = \Delta_{n,*}J_1 \quad (n > 0).$$

Closed base change and Kashiwara equivalence give $d_\alpha^{!}P_n = P_m$. Every positive cut has empty intersection with the small diagonal. Give P zero coproduct at these cuts and identity coproduct at each cut with an empty block. This is a connected cut coalgebra: an iterated coproduct with two or more nonempty output blocks is zero in every parenthesization.

There is no coalgebra map $u : J \to P$ with $u_1 = \mathrm{id}_{J_1}$. Indeed, restrict its binary coproduct identity to $U = X^2 \setminus \Delta_2$. The path through P_2 is zero, whereas the path through $J_1 \boxtimes J_1$ is the identity of J_U. This object is nonzero, since a smooth nonempty curve has two distinct geometric points. The contradiction concerns the specified unary identity. The map $J \to K \to P$, whose positive components vanish, still exists.

2.3 Collisions that block every cut

Write $\Delta_{ij} \subset X^n$ for the equality $x_i = x_j$. For $n \geq 2$, define the reduced closed subscheme

$$Z_n = \left(\bigcap_{p=1}^{n-1} \bigcup_{i \leq p < j} \Delta_{ij} \right)_{\mathrm{red}}. \tag{2.3}$$

Thus a tuple lies in Z_n exactly when every consecutive cut is crossed by an equality. Put $Z_1 = X$. The first cases are

$$Z_2 = \Delta_{12}, \qquad Z_3 = \Delta_{13}, \qquad Z_4 = \Delta_{14} \cup (\Delta_{13} \cap \Delta_{24}). \tag{2.4}$$

For the last equality, a tuple with $x_1 \neq x_4$ must still cross its first and last gaps. If it does not have both $x_1 = x_3$ and $x_2 = x_4$, these two requirements force $x_1 = x_2$ and $x_3 = x_4$. The middle cut then forces all four coordinates equal, contrary to $x_1 \neq x_4$. The converse inclusions follow by checking the three gaps.

There is an equally explicit description in every arity. A matching on $[n]$ is a set of disjoint pairs (a, b), with $a < b$. Associate to it the closed subscheme $\Delta_M = \bigcap_{(a,b)\in M} \Delta_{ab}$. The pair (a, b) crosses the gaps $a, a+1, \ldots, b-1$. Call M spanning if its pairs cross every gap, and irredundant if deleting any pair destroys this property.

THEOREM 2.3.1 *(Components of the cut support).* Assume X is geometrically integral and $n \geq 2$. The irreducible components of Z_n are the Δ_M for irredundant spanning matchings M. A matching with r pairs has codimension r, and there are

$$\binom{n-2}{2r-2}, \qquad 1 \leq r \leq \lfloor n/2 \rfloor,$$

such components. For $n \geq 3$ their total number is 2^{n-3}.

Proof. The equalities of a geometric tuple partition $[n]$ into blocks. In a block with at least two labels, replace all its equalities by the equality of its smallest and largest labels. This retains exactly the gaps crossed by that block. The resulting pairs are disjoint. If the tuple lies in Z_n, they form a spanning matching; remove redundant pairs to obtain M. Consequently Z_n is the union of the Δ_M in the statement.

Order the pairs by their left endpoints, $a_1 < \cdots < a_r$. An interval contained in another is redundant, so their right endpoints also increase. Spanning gives $a_1 = 1$, $b_r = n$, and $a_{i+1} < b_i$: equality of these endpoints is excluded because this is a matching. A pair has a gap not crossed by its neighbours exactly when $b_{i-1} < a_{i+1}$, for $1 < i < r$. Thus, for $r > 1$, the endpoints occur in precisely the order

$$1 = a_1 < a_2 < b_1 < a_3 < b_2 < \cdots < a_r < b_{r-1} < b_r = n.$$

Conversely this order makes the matching spanning and gives every pair a private gap. Choosing its $2r-2$ interior endpoints therefore gives the binomial coefficient above. For $r = 1$, the unique pair is $(1, n)$.

Each Δ_M is isomorphic to X^{n-r}, hence is irreducible of codimension r. Containment between two such diagonals would make the partition into pairs for one refine that for the other. Since a pair has no nontrivial proper refinement except deletion, irredundancy excludes a strict containment. They are therefore exactly the components. Summing the even binomial coefficients of order $n-2$ gives 2^{n-3} when $n \geq 3$. □

2.4 COEFFICIENT COMPLEXES ON THE COLLISION SUPPORT

A support records where a coefficient can live. It does not record the coefficient or its maps along intersections. We make those maps explicit for $X = \mathbb{A}^1_k$. Let $R = k[z_1, \ldots, z_n]$. For this calculation we use complexes of R-modules. Their localizations have natural left differential-module actions. Tensoring with $\omega_{X^n}[n]$ gives the corresponding supported right-module coefficients of J_n; the scalar complexes below carry no such shift. For $f \in R$, put

$$L_f = [\, R \longrightarrow R[f^{-1}] \,],$$

in cohomological degrees 0, 1, with the localization differential. The augmentation $\epsilon_f : L_f \to R$ is identity in degree zero and zero in degree one. This flat complex represents $R\Gamma_{V(f)}R$, the derived functor of sections supported on $V(f)$. For a general coefficient complex Q, $L_f \otimes_R Q$ represents $R\Gamma_{V(f)}Q$.

Define

$$g_p = \prod_{i \leq p < j} (z_i - z_j), \qquad C_n = \bigotimes_{p=1}^{n-1} L_{g_p}.$$

This is a bounded complex of flat modules, with 2^{n-1} localization summands. Iterated local cohomology gives $C_n \simeq R\Gamma_{Z_n}R$. Local cohomology depends on the reduced support. The ideal generated by the g_p need not be radical: at $n = 3$, with $x = z_1 - z_2$ and $y = z_2 - z_3$, it is

$$(x(x+y),\, y(x+y)) = (x+y)(x,\, y),$$

whose radical is $(x + y)$.

The component description gives another model that displays the attachments. Choose a total order on the finite set $\mathcal{M}_n$ of irredundant spanning matchings. For $M \in \mathcal{M}_n$, let

$$K_M = \bigotimes_{(a,b)\in M} L_{z_a - z_b},$$

with pairs ordered by their left endpoints. For a nonempty subset $A = \{M_0 < \cdots < M_p\}$, put $K_A = K_{M_0} \otimes_R \cdots \otimes_R K_{M_p}$. Repeated equations remain repeated tensor factors in this formula.

Proposition 2.4.1 *(Component model).* The complex

$$S_n = \bigoplus_{\varnothing \neq A \subseteq \mathcal{M}_n} K_A[|A| - 1] \tag{2.5}$$

represents $R\Gamma_{Z_n} R$. On $K_A[p]$, its internal differential is $(-1)^p d_{K_A}$; its other differential is $\sum_{j=0}^{p} (-1)^j \partial_j$, where ∂_j applies the augmentation of K_{M_j} and deletes that factor. The latter differential is zero when $p = 0$. The augmentations give a natural span of quasi-isomorphisms

$$C_n \longleftarrow C_n \otimes_R S_n \longrightarrow S_n. \tag{2.6}$$

The same assertions hold after tensoring with any complex Q.

Proof. The tensor of the L_f's computes local cohomology supported on the intersection of their zero sets. Thus K_A represents $R\Gamma_{\cap_{M\in A}\Delta_M} R$. The augmentation of a factor induces the inclusion of sections supported on this intersection into sections supported on the intersection with that factor omitted. These are the maps ∂_j.

Two deletions commute before their alternating signs, so the horizontal differential squares to zero. It commutes with each unshifted internal differential; the factor $(-1)^p$ makes the two total differentials anticommute. For two closed sets, the resulting complex is the local-cohomology Mayer–Vietoris triangle

$$R\Gamma_{Z\cap Z'} R \longrightarrow R\Gamma_Z R \oplus R\Gamma_{Z'} R \longrightarrow R\Gamma_{Z\cup Z'} R.$$

Iterate this triangle over the finite component set. At each step the new intersections are exactly the summands containing the new component, with the displayed deletion maps. This proves (2.5).

Local cohomology supported on Z_n is idempotent. Indeed, its localization triangle becomes the identity triangle on an object whose restriction to the complement of Z_n vanishes. Apply this to C_n and S_n. The two counits are exactly the arrows of (2.6), so both are quasi-isomorphisms. Every term used is flat and the complexes are bounded. Tensoring the construction with an arbitrary complex therefore computes its derived tensor and preserves these quasi-isomorphisms. □

At the generic point of a component with r pairs, invert the equations that exclude the other components and choose independent normal coordinates $u_1, \ldots, u_r$ for the pairs. Its local cohomology is concentrated in degree r, represented by negative powers of all the u_i. In particular it contains $1/(u_1 \cdots u_r)$; the iterated residue of $du_1 \cdots du_r/(u_1 \cdots u_r)$, in that order, is one. This follows directly by tensoring the r two-term localization complexes, or by their Laurent monomial basis.

The idempotence just used is a property of local cohomology. It must be distinguished from derived intersection. For example, over $k[u]$, the free resolution $[\, k[u] \xrightarrow{u} k[u] \,]$ of k, in degrees -1, 0, gives

$$k \otimes^{\mathbf{L}}_{k[u]} k \simeq k \oplus k[1], \qquad L_u \otimes_{k[u]} L_u \simeq L_u.$$

The first complex retains a normal Tor class. The second retains all principal parts along the same closed support.

2.5 THE FOUR-POINT ATTACHMENT AND NORMAL DERIVATIVES

At four positions choose $a = z_1 - z_4, b = z_1 - z_3, c = z_2 - z_4$, and the spectator coordinate z_4. Suppress the spectator polynomial factor and put $R = k[a, b, c]$. Formula (2.4) becomes

$$Z_4 = V(a) \cup V(b, c) = V(ab, ac).$$

Its coefficient complex is

$$R \xrightarrow{f \mapsto (f,f)} R[(ab)^{-1}] \oplus R[(ac)^{-1}] \xrightarrow{(f,g) \mapsto g - f} R[(abc)^{-1}]. \tag{2.7}$$

Its terms have cohomological degrees 0, 1, 2.

PROPOSITION 2.5.1 *(Nonsplit attachment).* Let

$$M = R[b^{-1}, c^{-1}] / \big(R[b^{-1}] + R[c^{-1}]\big).$$

The only nonzero cohomology of (2.7) is

$$H^1_{Z_4}(R) = R[a^{-1}]/R, \qquad H^2_{Z_4}(R) = M[a^{-1}].$$

The attachment of $V(b, c)$ to the union is the nonsplit exact sequence of R-modules

$$0 \longrightarrow M \longrightarrow M[a^{-1}] \longrightarrow H^3_{(a,b,c)}(R) \longrightarrow 0.$$

Proof. Inside $R[(abc)^{-1}]$, the two middle localizations intersect in $R[a^{-1}]$. This computes H^1. Their quotient in the last term has basis $a^i b^{-j} c^{-l}$, with $i \in \mathbb{Z}$ and $j, l > 0$, which gives $M[a^{-1}]$. There is no degree-zero kernel because localization of the polynomial ring is injective.

The module M is free over $k[a]$, with basis $b^{-j} c^{-l}$ for $j, l > 0$. Its quotient after inverting a has basis with all three exponents negative. This is the top local cohomology at (a, b, c), proving exactness. Every element of this quotient is killed by a power of a, whereas multiplication by a is invertible on $M[a^{-1}]$. Every R-linear map from the quotient back to the middle term is consequently zero. The quotient contains the nonzero class $1/(abc)$, so the surjection has no section. □

The intersection $V(a, b, c)$ cannot therefore be replaced by an independent direct summand while preserving the coefficient action. Even at one normal coordinate, a scalar residue loses this action. To see it in the right differential-module convention, set

$$P = k[u, u^{-1}]\, du / k[u]\, du, \qquad e_r = [u^{-r-1} du] \quad (r \geq 0).$$

The right Weyl action is

$$e_0 u = 0, \qquad e_r u = e_{r-1}\ (r > 0), \qquad e_r \partial_u = (r + 1) e_{r+1}.$$

Indeed, right differentiation of a density $f\, du$ is $-f'\, du$. The formulas satisfy $\partial_u u - u \partial_u = 1$ and give $e_0 \partial_u^r = r! e_r$. All normal derivatives are present. The scalar residue sends e_0 to one and e_r to zero for $r > 0$, but its kernel is not stable under u: $\operatorname{Res}(e_1 u) = 1$. It is therefore insufficient as a replacement for this coefficient module.

2.6 The outer residue

The nonconsecutive support already changes a three-position calculation. Put $x = z_1 - z_2$, $y = z_2 - z_3$, and consider

$$f(x, y) = \frac{1}{x(x + y)}.$$

Let L be the coefficient of $x^{-1}y^{-1}$ in the expansion for $|x| < |y|$, and R the same iterated coefficient for $|y| < |x|$. These inequalities specify formal Laurent expansions; no analytic limiting operation is required. For the outer collision use $u = x + y = z_1 - z_3$. Let N be the coefficient of $u^{-1}y^{-1}$ in the expansion for $|u| < |y|$. The volume form is unchanged: $dx \wedge dy = du \wedge dy$.

Proposition 2.6.1. With these coordinates and orientations,

$$L = 1, \qquad R = 0, \qquad N = -1, \qquad L - R + N = 0.$$

Proof. The three expansions are respectively

$$\sum_{r \geq 0} (-1)^r x^{r-1} y^{-r-1}, \qquad \sum_{r \geq 0} (-1)^r y^r x^{-r-2}, \qquad -\sum_{r \geq 0} u^{r-1} y^{-r-1}.$$

Their indicated coefficients are 1, 0, −1. □

This residue identity does not by itself construct an associativity homotopy. In a specified operation complex Q, closed degree-zero operations L, R, N satisfying $L - R + N = 0$ admit $h \in Q^{-1}$ with $dh = L - R = -N$ exactly when $[N] = 0$ in $H^0(Q)$. For $Q = k$ in degree zero, the three scalars just computed satisfy the identity and admit no such h. A field comparison must specify the operation complex, its supported coefficients and the primitive there.

2.7 All local orders and the word comparison

A fixed word has consecutive cuts. To retain every local ordering at a collision, let I be a nonempty finite set and take a copy X^I_σ of X^I for each total order σ on I. For two orders define

$$V_{\sigma,\rho} = X^I \setminus \bigcup_{\substack{i,j \in I \\ \sigma \text{ and } \rho \text{ order } i,j \text{ differently}}} \Delta_{ij}.$$

Glue the two copies by the identity on this open subset, and denote the resulting scheme by $\mathscr{R}_I$. The chart coordinates induce $q_I : \mathscr{R}_I \to X^I$. For $X = \mathbb{A}^1_{\mathbb{C}}$, this is the configuration space used for raviolo vertex algebras.[2]

Proposition 2.7.1 *(Ordered configuration atlas).* These identifications satisfy the gluing cocycle. The map $q_I : \mathscr{R}_I \to X^I$ is a local isomorphism, and $\mathscr{R}_I$ is smooth and quasi-compact with quasi-compact intersections of affine opens. Over a geometric tuple whose equality blocks are B, its fibre has

$$\prod_B |B|!$$

points. In particular the fibre at a threefold collision has six points.

[2] L. Alfonsi, H. Kim and C. A. S. Young, *Raviolo vertex algebras, cochains and conformal blocks*, §4.1; arXiv:2401.11917.

Proof. On a triple overlap, if two orders disagree on a pair, at least one of their comparisons with the third order disagrees on that pair. Thus $V_{\sigma,\rho} \cap V_{\rho,\tau} \subseteq V_{\sigma,\tau}$. All transition maps are identities, which proves the cocycle. The charts give smoothness and the local-isomorphism assertion. There are finitely many charts; their intersections are complements of finitely many diagonals in the noetherian scheme X^I, so the asserted quasi-compactness follows.

Over a fixed tuple, two charts are identified exactly when they give the same order within each equality block. Every choice of orders on these blocks extends to a total order on I. The choices on different blocks are independent, proving the fibre formula. □

A surjection $\alpha : I \twoheadrightarrow J$ with an order on each fibre defines $\delta_\alpha : \mathscr{R}_J \to \mathscr{R}_I$. On a chart ordered by τ on J, expand each label to its ordered fibre and use the usual diagonal map of coordinates. The inverse image of an overlap is precisely the overlap for the two orders on J, so these chart maps glue. Composing such surjections concatenates the ordered fibres and gives the same diagonal map. These collision maps need not be closed immersions. For the ordered fibres $(1,3)|(2)$, their image in the chart $1 < 2 < 3$ is $\Delta_{13} \setminus \Delta_{123}$, which is not closed in that chart.

Define $\mathcal{C}_X$ to be the category of coherent systems $F_I \in \mathcal{D}(\mathscr{R}_I)$ with $\delta_\alpha^! F_I \simeq F_J$, together with a separate empty coefficient. Differential modules on this possibly nonseparated scheme are defined by descent from its charts. For an ordered decomposition $I = I_1 \sqcup I_2$, let U_{I_1,I_2} be the locus of disjoint positions in different blocks. On that locus, orders between the blocks are identified by the gluing, so the atlas is the corresponding open part of $\mathscr{R}_{I_1} \times \mathscr{R}_{I_2}$. Define $F \star G$ by summing the localized external products over all such ordered decompositions, including empty blocks. The proof of the cut-product proposition applies: a decomposition splitting a collision fibre has empty pullback; every surviving decomposition descends to the quotient. All iterates localize on the same disjoint-block locus. This gives a symmetric monoidal product, whose symmetry exchanges the two blocks.

Restriction to the standard chart $1 < \cdots < n$ defines $\mathrm{res} : \mathcal{C}_X \to \mathcal{W}_X$.

PROPOSITION 2.7.2 *(Projection to consecutive cuts).* There are natural maps

$$\iota_{F,G} : \mathrm{res}\, F \star_{\mathrm{cut}} \mathrm{res}\, G \longrightarrow \mathrm{res}(F \star G),$$
$$\pi_{F,G} : \mathrm{res}(F \star G) \longrightarrow \mathrm{res}\, F \star_{\mathrm{cut}} \mathrm{res}\, G, \qquad \pi_{F,G}\iota_{F,G} = \mathrm{id}.$$

The maps ι make restriction lax monoidal, and the maps π make it colax monoidal. They transport algebra and coalgebra structures, respectively. Neither comparison is an equivalence in general.

Proof. On a standard chart the atlas product is the finite direct sum over all ordered decompositions of $[n]$. Include the summands whose first block is a prefix, or project onto those summands. Under a monotone collision, a surviving prefix is exactly the inverse image of a quotient prefix. The maps therefore respect the word coefficient equivalences.

Every iterated inclusion or projection selects the same final consecutive decomposition with the same localized coefficient. This proves their monoidal coherences, including those for empty blocks, and projection after inclusion is identity. For two positive systems equal to the dualizing system in positive arity, the separated two-point fibre of the atlas product is $J_U \oplus J_U$, indexed by $1|2$ and $2|1$. The cut product has only the $1|2$ summand. Thus π has a nonzero kernel there, and ι has a nonzero cokernel. □

With n positive inputs at n distinct positions, the same calculation gives $n!$ atlas summands and one cut summand. A bar–cobar theorem for the cut product therefore does not identify the corresponding construction on the full atlas. Any such comparison must retain its actual maps on the omitted orders and their collision coefficients.

2.8 Arithmetic places and position geometry

An arithmetic place is a label for a local coefficient object. It need not be a variable position on the curve carrying a field. This distinction already appears in the simplest functors of points. For every scheme S, there is exactly one map $S \to \operatorname{Spec} \mathbb{Z}$. Consequently the functor of nonempty finite subsets of $\operatorname{Spec} \mathbb{Z}(S)$ is terminal. The closed points $\operatorname{Spec} \mathbb{F}_p \to \operatorname{Spec} \mathbb{Z}$ use different test schemes; they are not several elements of this set for one fixed S.

Let $\mathbb{P}$ be the set of rational primes and $V = \mathbb{P} \sqcup \{\infty\}$ the abstract set of places. Its nonempty finite subsets and its nonempty finite multisets are the discrete sets

$$\operatorname{Fin}_+(V), \qquad \operatorname{Mult}_+(V) = \coprod_{n \geq 1} V^n/\mathfrak{S}_n.$$

The coproduct in the second formula records multiplicity; it does not require transition maps between different symmetric powers. For a commutative ring A, let $\mathcal{D}(A)$ here denote the derived category of A-modules. Given complexes $H_v \in \mathcal{D}(A)$, finite tensor products give

$$F(S) = \bigotimes_{v \in S, A}^{\mathbf{L}} H_v, \qquad F(S \sqcup T) \simeq F(S) \otimes_A^{\mathbf{L}} F(T) \quad (S \cap T = \varnothing). \tag{2.8}$$

We include the empty set by $F(\varnothing) = A$. The associators and symmetries of derived tensor give all coherences, with the usual Koszul signs on complexes. This supplies factorization for disjoint labels. Maps at coincident positions, normal derivatives and residue operations require a specified position geometry and further coefficient maps.

To assemble infinitely many places, units must also be specified. More generally, let $\mathcal{E}$ be a cocomplete symmetric monoidal ∞-category with unit $\mathbf{1}$, and let $H_v \in \mathcal{E}$. Choose a cofinite subset $V^{\mathrm{sph}} \subseteq V$ and maps $u_v : \mathbf{1} \to H_v$ for $v \in V^{\mathrm{sph}}$. Let $\operatorname{Fin}^{\mathrm{sph}}(V)$ be the poset of finite subsets containing $V \setminus V^{\mathrm{sph}}$, ordered by inclusion. For $S \subseteq T$, tensor with the maps u_v at the new places to obtain

$$\iota_{S,T} : \bigotimes_{v \in S} H_v \longrightarrow \bigotimes_{v \in T} H_v.$$

The colimit

$$\bigotimes_{v \in V}{}' (H_v, u_v) = \varinjlim_{S \in \operatorname{Fin}^{\mathrm{sph}}(V)} \bigotimes_{v \in S} H_v$$

is called the *Tate restricted tensor product* for these specified reference maps. It is a colimit of a covariant diagram.

Indeed, $\iota_{T,U}\iota_{S,T} = \iota_{S,U}$ coherently: both maps insert the same u_v's in the same labelled factors, and tensor coherence identifies their parenthesizations. The index poset is nonempty and filtered, since the exceptional set is finite and the union of two allowed finite sets is allowed. This constructs the diagram and its colimit. To obtain an algebra, coalgebra or monoidal category by this construction, its reference maps must preserve that structure and its colimit must be taken in a coefficient category that supports the required operations.

PROPOSITION 2.8.1 *(Three finite-place coefficient models).* Let p be a rational prime. Define the completed tensor below to mean the ordinary p-adic completion. Then

$$\varprojlim_r (\mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{Z}_p)/p^r \simeq \mathbb{Z}_p, \tag{2.9}$$

$$\mathbb{F}_p \otimes^{\mathbf{L}}_{\mathbb{Z}} \mathbb{F}_p \simeq \mathbb{F}_p \oplus \mathbb{F}_p[1] \quad \text{as a complex,} \tag{2.10}$$

$$\mathbb{Z}_p[[t_1, t_2]]/(t_1 - t_2) \simeq \mathbb{Z}_p[[t]].$$

The kernel in the last line is the nonzero principal ideal $(t_1 - t_2)$.

Proof. Modulo p^r, the first tensor product is $(\mathbb{Z}/p^r) \otimes_{\mathbb{Z}} \mathbb{Z}_p = \mathbb{Z}/p^r$; multiplication induces this isomorphism compatibly for all r. Taking the inverse limit proves (2.9). The free resolution $[\, \mathbb{Z} \xrightarrow{p} \mathbb{Z} \,]$ of $\mathbb{F}_p$, in degrees -1, 0, becomes a complex with zero differential after tensoring with $\mathbb{F}_p$. This proves (2.10). Finally the formal change of variables $u = t_1 - t_2$, $t = t_2$ identifies the last ring before quotienting with $\mathbb{Z}_p[[u, t]]$. Evaluation at $u = 0$ has kernel (u), proving the last assertion. □

The completed base tensor, the derived closed-point intersection and the formal position disc thus retain different data. The first identity does not eliminate the normal class in the second or the position variable in the third. A finite-place field construction can use a chosen formal disc; crystalline or prismatic structures are further possible choices when their Frobenius data are required. None is forced solely by the discrete place index or by the first tensor identity. The arithmetic comparison must specify which position space and which coefficient operations its maps preserve.

Chapter 3

Bar–cobar and coefficient topology

A nonempty ordered configuration of n positions has $n-1$ gaps. A word of positive geometric factors therefore has length at most n. This bound makes the cut bar and cobar finite at each configuration. It does not bound the length of a resolution inside one coefficient module. Such a resolution can be infinite even at the empty configuration, where there are no geometric gaps.

We first prove the duality controlled by positive cuts. We then construct the coefficient bar with its signs and augmentation, and determine when it computes a derived tensor. The final sections compare three ways of allowing infinite words. A series may terminate at every finite precision while having unbounded length. An unrestricted product permits still more series but can destroy the continuity of an otherwise valid comparison.

3.1 Positive words and finite duality

Use the word category $\mathcal{W}_X$, cut product $\star_{\mathrm{cut}}$, and empty unit K of Chapter 2. Write $\star$ for this product throughout the present section. For $p \geq 0$, put

$$\mathcal{W}_{\geq p} = \{F \in \mathcal{W}_X : F_n = 0 \text{ for } n < p\}.$$

Thus $\mathcal{W}_{\geq 1}$ consists of the positive systems.

Lemma 3.1.1 *(Finite positive words).* The cut product sends $\mathcal{W}_{\geq p} \times \mathcal{W}_{\geq q}$ to $\mathcal{W}_{\geq p+q}$. In particular, $F_1 \star \cdots \star F_r \star M$ vanishes in arity $n < r$ when all F_i are positive, whether or not M is positive. If $V_r \in \mathcal{W}_{\geq r}$, then

$$\bigoplus_{r\geq 0} V_r \xrightarrow{\sim} \prod_{r\geq 0} V_r, \qquad \left(\bigoplus_{r\geq 0} V_r\right)_n = \bigoplus_{r=0}^{n} (V_r)_n.$$

Proof. A nonzero summand in a cut product assigns at least p positions to its first factor and q to its second. The mixed assertion uses one position for each positive factor and none necessarily for M.

The finite sums $V_n = \bigoplus_{r\leq n}(V_r)_n$ form a word system: for a monotone surjection $[n] \twoheadrightarrow [m]$, exact extraordinary pullback takes this sum to $\bigoplus_{r\leq n}(V_r)_m$, whose terms with $r > m$ vanish. Since these pullbacks preserve colimits, V is the coproduct. A family of maps $T \to V_r$ gives, in each arity, a map to the finite product defining V_n. These maps commute with collisions after every projection, and hence commute with them as maps to V. The same argument applies to coherent homotopies. It proves the product universal property. No assertion about pullback of an arbitrary infinite product is used. □

An augmented associative algebra in $\mathcal{W}_X$ is an associative algebra object A with an algebra map $A \to K$. It is *connected* if this map is an equivalence at the empty configuration. Its augmentation ideal $a = \mathrm{fib}(A \to K)$ is then positive, and the unit splits $A \simeq K \oplus a$. For a connected coaugmented coalgebra C, put $c = \mathrm{cofib}(K \to C)$. All algebra and coalgebra objects in this section retain their higher associativity or coassociativity homotopies.

The *cut bar* $B_{\mathrm{cut}}A$ is the geometric realization of the two-sided simplicial bar of K, A, K. Its faces multiply or augment, and its degeneracies insert the unit. The *cut cobar* $\Omega_{\mathrm{cut}}C$ is the totalization of the dual cosimplicial construction, whose cofaces comultiply or coaugment and whose codegeneracies apply the counit. For positive reduced parts their normalized terms are

$$\begin{aligned} B_{\mathrm{cut}}A &= K \oplus \bigoplus_{r\geq 1} (a[1])^{\star r}, \\ \Omega_{\mathrm{cut}}C &= K \oplus \bigoplus_{r\geq 1} (c[-1])^{\star r}. \end{aligned} \tag{3.1}$$

These expressions include the normalized differentials and higher attaching maps. They do not replace a homotopy-coherent algebra by its graded object with zero differential.

THEOREM 3.1.2 *(Finite cut duality).* Cut bar and cut cobar induce inverse equivalences between connected augmented associative algebras and connected coaugmented coassociative coalgebras in $(\mathcal{W}_X, \star, K)$. Each normalized construction in arity n has at most n positive factors. No global bound on word length or on coefficient cohomological degrees is imposed.

Proof. The differential-module categories are presentable, and the extraordinary pullbacks defining $\mathcal{W}_X$ are accessible and preserve colimits. Their limit is presentable. The associative bar–cobar adjunction therefore supplies coherent maps

$$u_A : A \longrightarrow \Omega_{\mathrm{cut}}B_{\mathrm{cut}}A, \qquad v_C : B_{\mathrm{cut}}\Omega_{\mathrm{cut}}C \longrightarrow C.$$

Here bar is left adjoint to totalized cobar. This adjunction uses the full mapping spaces of algebra and coalgebra objects.[1]

Splitting off the unit removes the degenerate simplicial terms; the normalized term in degree r is $a^{\star r}[r]$. The normalized cosimplicial term is $c^{\star r}[-r]$. By Lemma 3.1.1, the latter totalization tower stabilizes at step n in arity n. Its finite limits commute with collision pullback. The resulting compatible family is the totalization in $\mathcal{W}_X$: maps into a word system are the limit of its component mapping spaces, and that limit commutes with the cosimplicial limit. This proves (3.1) on the stated carrier.

Filter A increasingly by $F_0A = K$, $F_rA = A$ for $r \geq 1$. In its associated graded, a has weight one and zero multiplication. The induced filtration on $\Omega_{\mathrm{cut}}B_{\mathrm{cut}}A$ counts original a-factors. At two factors, the single gap is either inside a bar block or between two cobar factors. Splitting the block joins these two copies of the coefficient by an identity map, with its suspension sign. For r factors there are $r-1$ such gaps. The coefficient differential and block splitting preserve r; multiplication of original factors lowers it. The associated graded is thus the cube that changes an inside gap to an outside gap, with an identity map on the common coefficient at each edge. The suspension signs give opposite signs around each square. It is the tensor product of $r-1$ two-term identity complexes, tensored with the shifted coefficient $a^{\star r}$. For $r > 1$ it is contractible. The weights zero and one are K and a, and the associated graded map of u_A is the identity on these terms.

For C, use the decreasing filtration $F^0C = C$, $F^1C = c$, $F^2C = 0$, where the splitting by the counit regards c as a subobject. The associated graded reduced coalgebra is primitive. In $B_{\mathrm{cut}}\Omega_{\mathrm{cut}}C$, the corresponding filtration counts original c-factors. Cobar splitting increases this count, while bar multiplication preserves it. On the associated graded the same gap cube runs from outside bar cuts to inside product

[1] J. Lurie, *Higher Algebra*, Theorem 5.2.2.17 and Remark 5.2.2.19, pp. 827–828, author's text. Warning 5.2.2.16 explains why no preservation of both arbitrary realizations and totalizations by the tensor product is assumed.

gaps. Its weights greater than one are contractible; v_C is the identity on the surviving weights zero and one.

After evaluation in arity n, both filtrations are finite: terms with more than n original positive factors vanish. Their cofiber sequences prove that u_A and v_C are equivalences there. Component evaluation is jointly conservative, so they are equivalences of word systems. The construction in filtered objects retains coherent maps. Higher products lower the first filtration and higher coproducts raise the second; the same associated graded calculation applies. The adjunction gives the triangle identities and hence the asserted equivalence. □

Corollary 3.1.3. For the vacuum coalgebra J of Proposition 2.2.1, connected cut coalgebras equipped with a coalgebra map $J \to C$ correspond to connected augmented cut algebras equipped with an algebra map $\Omega_{\mathrm{cut}} J \to A$.

Proof. An equivalence of categories induces an equivalence on the categories of objects under any fixed object. Apply Theorem 3.1.2 to the object J. □

Thus the local vacuum marking survives as an algebra map with source $\Omega_{\mathrm{cut}} J$. It is different from the augmentation $A \to K$. The theorem concerns the cut tensor on words. A comparison with the full ordered atlas, or with prescribed field operations, must preserve the additional geometry and maps identified in Chapter 2.

3.2 The coefficient bar and its twisting maps

Let R be a commutative ring and A a unital differential graded R-algebra with a specified augmentation $\epsilon : A \to R$. Write $\bar{A} = \ker \epsilon$, so $A = R \oplus \bar{A}$ as a complex. All tensor products in this section are ordinary tensors over R. The cohomological differential has degree one. Put $sV = V[1]$ and $tV = V[-1]$; the maps s and t have degrees minus one and one.

The *coefficient bar* is

$$B_R A = T^c_R(s\bar{A}) = R \oplus \bigoplus_{r \geq 1} (s\bar{A})^{\otimes_R r}.$$

The superscript c denotes the tensor coalgebra: its coproduct deconcatenates each word at every position, including both ends. The empty word gives the coaugmentation and counit. Its reduced coproduct $\bar{\Delta}$ omits the end cuts. A coaugmented coalgebra is *conilpotent* if every element of its reduced part is killed by a sufficiently long iterated reduced coproduct. This property is called *conilpotence*. Finite words make $T^c_R(s\bar{A})$ conilpotent.

Define its differential as the coderivation whose components with one output factor are

$$b_1(sa) = -s(d_A a), \qquad b_2(sa, sb) = (-1)^{|a|} s(ab). \tag{3.2}$$

Inserting either degree-one component after $w_1, \dots, w_i$ has sign $(-1)^{|w_1|+\cdots+|w_i|}$. On degree-zero algebra inputs, the multiplication differential is therefore

$$b[a_1|\cdots|a_r] = \sum_{i=1}^{r-1} (-1)^{i-1} [a_1|\cdots|a_i a_{i+1}|\cdots|a_r].$$

The notation $[a_1|\cdots|a_r]$ abbreviates the tensor of the suspended entries; its degree is $\sum_i |a_i| - r$.

For a coaugmented counital differential graded coalgebra C, write $\bar{C} = \ker(C \to R)$. Its direct-sum coefficient cobar is the tensor algebra

$$\Omega_R C = T_R(t\bar{C}), \qquad d(tc) = -t(d_C c) + \sum_{\bar{\Delta} c = \sum c' \otimes c''} (-1)^{|c'|+1} (tc')(tc''). \tag{3.3}$$

The displayed map extends by the graded Leibniz rule. The coalgebra coproduct takes values in the ordinary algebraic tensor, so it gives a finite sum on each element.

PROPOSITION 3.2.1. The differentials in (3.2) and (3.3) square to zero. For conilpotent C, there are natural bijections

$$\mathrm{Hom}_{\mathrm{dgAlg}_R}(\Omega_R C, A) \cong \mathrm{Tw}_R(C, A) \cong \mathrm{Hom}_{\mathrm{dgCoalg}_R}(C, B_R A),$$

where algebra maps preserve augmentations, coalgebra maps preserve coaugmentations and counits, and $\mathrm{Tw}_R(C, A)$ consists of degree-one maps $\tau : \bar{C} \to \bar{A}$ satisfying

$$d_A \tau + \tau d_C + \mu_A(\tau \otimes \tau)\bar{\Delta} = 0. \tag{3.4}$$

The tensor of the two degree-one maps contributes $(-1)^{|c'|}$ on $c' \otimes c''$.

Proof. For the bar square, disjoint insertions cancel by their degree-one signs. The overlapping multiplication terms on three inputs give $(-1)^{|b|} s((ab)c - a(bc))$. Mixed terms vanish by the Leibniz identity for d_A, and the internal square is zero. These are all possibilities. The corresponding cobar calculation uses that $\bar{\Delta}$ is a chain map and is coassociative: the two orders of splitting one factor have opposite signs. Coderivation and derivation compatibility hold by construction.

A map from the tensor algebra $\Omega_R C$ is determined by $tc \mapsto \tau(c)$. Substitution in (3.3) shows that its chain-map condition is exactly (3.4). A coalgebra map to the tensor coalgebra is determined by its length-one component $s\tau$; its length-r component must be

$$(s\tau)^{\otimes r} \bar{\Delta}^{(r-1)}.$$

Conilpotence makes the sum of these components finite on each input. Coassociativity proves the coalgebra identity. The length-one part of its chain-map equation is (3.4), with (3.2); the coderivation identity then proves the equation in every length. These constructions are inverse and natural. □

This is a strict adjunction $\Omega_R \dashv B_R$ for algebraic tensor constructions and conilpotent coalgebras. It is distinct from the general enhanced adjunction between geometric bar and totalized cobar used in the finite cut theorem. On positive word systems the finiteness theorem makes both comparisons equivalences.

PROPOSITION 3.2.2 *(An algebra resolution).* For every augmented differential graded R-algebra A, the algebra map $\varepsilon_A : \Omega_R B_R A \to A$ that sends $t(sa)$ to a, and a cobar generator represented by a longer bar word to zero, is a quasi-isomorphism.

Proof. The desuspension $B_R A \to A$ is a twisting map, so the preceding proposition gives ε_A. Filter by the number r of original $\bar{A}$-entries. The target filtration is $F_0 A = R$, $F_r A = A$ for $r \geq 1$. The associated graded source is the gap cube of the finite cut proof, now with coefficient $\bar{A}^{\otimes_R r}$. Each gap contributes a two-term identity complex. Contracting the first gap contracts their tensor product for $r > 1$, over any ring R, without a flatness hypothesis. In weights zero and one, ε_A is the identity. Induction through the finite filtration stages proves a quasi-isomorphism at each stage. Every element belongs to some stage, and filtered unions of R-modules are exact. Passing to that union proves the result. No convergence of an infinite product is asserted. □

3.3 Which derived tensor the bar computes

A complex V of R-modules is *K-flat* if $N \otimes_R V$ is acyclic for every acyclic complex N. This condition concerns the whole complex; for an unbounded complex, degreewise flatness alone does not suffice.

Proposition 3.3.1. If $\bar{A}$ is K-flat over R, then

$$B_R A \simeq R \otimes^{\mathbf{L}}_A R.$$

The two modules R use the specified augmentation of A.

Proof. Let $\tau : B_R A \to A$ desuspend a word of length one and vanish elsewhere. On $L = A \otimes_R B_R A$, put

$$d_L(a \otimes c) = d_A a \otimes c + (-1)^{|a|} a \otimes bc - (-1)^{|a|} \sum a\tau(c_{(1)}) \otimes c_{(2)}.$$

The twisting identity proves $d_L^2 = 0$. Let $\epsilon_L : L \to R$ augment the length-zero term and vanish on positive lengths. With $\bar{a} = a - \epsilon(a)1$, define

$$h\big(a \otimes [a_1|\cdots|a_r]\big) = -1 \otimes [\bar{a}|a_1|\cdots|a_r].$$

The internal terms cancel in $d_L h + h d_L$, as do contractions away from the first slot. The first-slot term gives $\bar{a} \otimes c$; if $r > 0$, the terms that merge the inserted slot with the next one leave $\epsilon(a) \otimes c$. Thus $d_L h + h d_L = 1 - \iota\epsilon_L$, where ι includes the empty word. This proves $L \simeq R$.

The increasing length filtration has quotients $A \otimes_R (s\bar{A})^{\otimes_R r}$. Tensoring one with an acyclic right A-module N gives $N \otimes_R (s\bar{A})^{\otimes_R r}$, which is acyclic by K-flatness. The filtration is split on graded modules; induction and exact filtered unions show that L is a K-flat left A-resolution. Tensoring it with the right augmentation module kills the first-slot term and leaves $B_R A$. □

For a nonunital differential graded algebra P, the tensor bar on sP means the reduced bar of the unitization $R \oplus P$, augmented to R. It need not be the reduced bar of a unital algebra structure already present on P. Indeed, if P has a closed unit e, inserting se at the beginning of a nonempty word contracts this tensor bar onto R: $b(se \otimes c) + se \otimes bc = c$. A reduced bar of P itself instead requires a chosen augmentation and its kernel. This distinction determines the meaning of the endpoint bars in Chapter 11.

Example 3.3.2 *(An infinite bar at the empty configuration).* Take $A = k \oplus ka$, with $|a| = 1$, zero differential and $a^2 = 0$. Then $v = sa$ has degree zero and

$$B_k A = \bigoplus_{r \geq 0} k v^{\otimes r}, \qquad b = 0.$$

Put this algebra at the empty component of $\mathcal{W}_X$ and zero at every positive arity. Its cut bar is exactly this coefficient bar at the empty component. The augmentation is not an equivalence there, so the algebra is outside the connected class of Theorem 3.1.2. Arbitrarily long coefficient words survive in the same cohomological degree.

3.4 Length products and continuous Taylor maps

For R-modules V_n, the direct sum $\bigoplus_n V_n$ contains finite sums of words, whereas $\prod_n V_n$ contains an arbitrary word in every length. Give each V_n the discrete topology and the product its product topology. A linear map of direct sums has components $F_{mn} : V_n \to W_m$.

PROPOSITION 3.4.1 *(Continuity in length).* A linear map $F : \bigoplus_n V_n \to \bigoplus_m W_m$ extends to a continuous map of the products of discrete modules if and only if, for every m, only finitely many F_{mn} are nonzero. Such an extension is unique.

More generally, for a prime ℓ, let $R = \mathbb{Z}_\ell$, give each V_n its ℓ-adic topology, and assume every W_m is separated and complete for that topology. An R-linear F extends continuously to the products if and only if, for each m, j,

$$F_{mn}(V_n) \subseteq \ell^j W_m \quad \text{for all but finitely many } n.$$

Proof. Continuity of the m-th coordinate at zero implies that its kernel contains a basic product neighborhood. Outside finitely many input coordinates that neighborhood contains each whole module, so the corresponding F_{mn} vanish. Conversely, finite row sums define a continuous map coordinate by coordinate.

For the adic statement, apply the same argument to the m-th output modulo $\ell^j W_m$. The asserted condition makes $\sum_n F_{mn}(x_n)$ converge in W_m, uniformly modulo each ℓ^j; the finite remaining terms are continuous because they are R-linear. This constructs the extension. The direct sum is dense in the product, and the target is separated, proving uniqueness in both cases. □

A degree-zero coaugmentation-preserving map between tensor coalgebras is specified by its Taylor components $f_r : V^{\otimes r} \to W$. Its value on a word is the sum over consecutive subdivisions, applying one f_r to each block. Every finite word has finitely many subdivisions, but an output of length one can receive inputs of every length. Proposition 3.4.1 detects precisely this possible failure of continuity.

For example, let $V = \mathbb{Z}v$ in degree zero and set $f_r(v^{\otimes r}) = v$ for every $r \geq 1$. The resulting coalgebra map is defined on finite words. The sums $z_N = \sum_{r=1}^{N} v^{\otimes r}$ converge in the product of discrete length spaces, but the length-one output is Nv, which does not converge. In the relative identity example of Proposition 11.7.2, these are the Taylor components of the actual endpoint comparison.

A product also changes the recipient of deconcatenation. For $\widehat{T}(V) = \prod_{r\geq 0} V^{\otimes r}$, the natural recipient is

$$\prod_{p,q\geq 0} (V^{\otimes p} \otimes V^{\otimes q}),$$

with (p, q)-component given by cutting length $p + q$. It need not be $\widehat{T}(V) \otimes \widehat{T}(V)$. Over a characteristic-zero field, take $V = kv$ and the element $z = \sum_{r\geq 0} r!v^{\otimes r}$. Its coproduct coefficient matrix is $((p + q)!)_{p,q\geq 0}$. The leading d-square minor has determinant

$$\det\big((p + q)!\big)_{0\leq p,q<d} = \prod_{i=0}^{d-1} (i!)^2 \neq 0.$$

To see this, factor $p!$ from row p; the q-th column is then a monic polynomial of degree q in p, so its determinant is the Vandermonde determinant. Every element of an algebraic tensor is a finite sum of separated tensors and gives a finite-rank coefficient matrix. Thus Δz cannot belong to that algebraic tensor.

3.5 FINITE PRECISION AND DERIVED INVERSE LIMITS

Put $R_j = \mathbb{Z}/\ell^j$, and let A_j be a compatible tower of augmented differential graded R_j-algebras. Write $B_j = B_{R_j} A_j$. Define

$$\widehat{B} = \varprojlim_j B_j, \qquad \widehat{B} \mathbin{\widehat{\otimes}} \widehat{B} := \varprojlim_j (B_j \otimes_{R_j} B_j). \tag{3.5}$$

These are specified inverse limits, with discrete finite-precision terms. No identification with another completed tensor is implicit. The recipient of an iterated coproduct with r outputs is $\varprojlim_j B_j^{\otimes_{R_j} r}$. An element has only finitely many nonzero lengths at each precision, although that finite set can grow with j.

Proposition 3.5.1. A compatible family of differential graded coalgebra maps $F_j : B_j \to B'_j$ induces a continuous map $\widehat{F} : \widehat{B} \to \widehat{B}'$ preserving the coproduct with the recipients in (3.5). If every F_j has a compatible strict inverse, then $\widehat{F}$ is a continuous isomorphism.

Proof. Define the map by its projections. The chain and coproduct identities hold after every projection and therefore in the inverse limit. The output at precision j depends only on the input at that precision, proving continuity. The compatible inverses define its continuous inverse in the same way. □

For $A_j = R_j \oplus R_j a$ with the square-zero structure of Example 3.3.2, the element

$$z = \sum_{r \geq 1} \ell^r v^{\otimes r}$$

belongs to $\widehat{B}$. It is not elementwise conilpotent: for any number of cuts, a sufficiently long word has a nonzero coefficient in the corresponding tensor recipient. The Taylor map with $f_r(v^{\otimes r}) = v$ is nevertheless continuous on this restricted inverse limit. Its length-one value on z is $\ell(1-\ell)^{-1}v$. At precision j this is a finite sum, so no analytic summation is part of its definition.

The preceding proposition is about strict maps of complexes. A claim about their cohomology requires a different argument. For a countable tower of complexes M_j, with transition maps $q_j : M_{j+1} \to M_j$, define

$$R\varprojlim_j M_j = \operatorname{Cone}\left(\prod_j M_j \xrightarrow{1-T} \prod_j M_j\right)[-1], \qquad (Tx)_j = q_j x_{j+1}.$$

This is the derived inverse limit of the tower of module complexes.

Proposition 3.5.2. If every q_j is surjective in each cohomological degree, then the ordinary inverse limit represents this derived inverse limit. For each q there is an exact sequence

$$0 \longrightarrow \varprojlim{}^1{}_j H^{q-1}(M_j) \longrightarrow H^q(\varprojlim_j M_j) \longrightarrow \varprojlim_j H^q(M_j) \longrightarrow 0,$$

where $\varprojlim^1$ is the cokernel of $1-T$ on the product of the indicated cohomology groups.

Proof. Surjectivity permits the recursive solution of $x_j - q_j x_{j+1} = y_j$, starting with any x_1. Thus $1-T$ is degreewise surjective and its kernel is the ordinary inverse limit. The resulting short exact sequence of complexes identifies this kernel with the shifted cone. Products of modules are exact, so their cohomology is the product of the cohomologies. The long exact sequence of the cone gives the displayed sequence. □

In particular, commuting cohomology with the limit requires the stated $\varprojlim^1$ term to vanish. Scalar extension need not commute with the limit either:

$$\mathbb{Q}_\ell \otimes_{\mathbb{Z}_\ell} \varprojlim_j \mathbb{Z}_\ell/\ell^j = \mathbb{Q}_\ell, \qquad \varprojlim_j \left(\mathbb{Q}_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell/\ell^j\right) = 0.$$

The first identity uses ℓ-adic completeness; in the second, inverting ℓ kills each finite quotient.

3.6 Analytic bounds and the other bar constructions

There is a useful analytic counterpart to the continuity criterion. For real or complex Banach spaces V, W, use the completed projective tensor powers $V^{\widehat{\otimes}_\pi r}$. For $\rho > 0$, let

$$T^+_\rho(V) = \left\{ (x_r)_{r\geq 0} : \sum_{r\geq 0} \rho^r \|x_r\|_\pi < \infty \right\}.$$

The displayed weighted sum is its norm, and the length-zero summand is the scalar field. Suppose continuous Taylor maps $f_r : V^{\widehat{\otimes}_\pi r} \to W$ satisfy $\|f_r\| \leq M L^{r-1}$ for constants $M, L > 0$.

PROPOSITION 3.6.1. The subdivision map determined by these Taylor maps, acting as the identity in length zero, extends to a bounded linear map $T^+_\rho(V) \to T^+_\sigma(W)$ whenever $L + \sigma M \leq \rho$. Its norm is at most one, and on the positive-length part it is at most $\sigma M / \rho$.

Proof. There are $\binom{r-1}{m-1}$ consecutive subdivisions of a length-r word into m nonempty blocks. Each corresponding operator has norm at most $M^m L^{r-m}$. Hence its entire weighted output has norm at most

$$\sum_{m=1}^{r} \binom{r-1}{m-1} (\sigma M)^m L^{r-m} \|x_r\|_\pi = \sigma M (L + \sigma M)^{r-1} \|x_r\|_\pi \leq \frac{\sigma M}{\rho} \rho^r \|x_r\|_\pi.$$

Sum over r, retaining the scalar term separately. Finite words are dense, so the bounded extension is unique. □

This estimate concerns the Taylor map. Continuity of the differential and coproduct in the chosen recipients must also be checked. For example, give the completed two-index tensor the norm $\sum_{p,q\geq 0} \sigma^{p+q} \|z_{pq}\|_\pi$. Deconcatenation from radius ρ to this recipient, for $\sigma < \rho$, is bounded by $\sup_{r\geq 0} (r+1)(\sigma/\rho)^r$. At equal radii that bound need not be finite. Algebraic identities on finite words extend when all maps in those identities have continuous extensions to a common domain and a separated target.

A *transverse bar* uses another specified monoidal category $(\mathcal{E}, \otimes_{\mathrm{tr}}, \mathbf{1}_{\mathrm{tr}})$ and an augmented associative algebra T for that tensor. If geometric realizations exist and the tensor preserves them, its two-sided bar is

$$B_{\mathrm{tr}} T = \mathbf{1}_{\mathrm{tr}} \otimes^{\mathbf{L}}_{T} \mathbf{1}_{\mathrm{tr}}.$$

Its augmentation, coefficient category and convergence refer to this tensor. The positive-cut bound applies to it only if an additional compatible positive filtration is supplied.

Finally, for a covariant diagram F of complexes of k-vector spaces on a small category $\mathcal{I}$, its simplicial replacement has terms

$$B_n(\mathcal{I}, F) = \bigoplus_{i_0 \to \cdots \to i_n} F(i_0).$$

Its direct-sum total complex has $\mathrm{Tot}^q = \bigoplus_{n\geq 0} B_n^{q+n}$ and differential $(-1)^n d_{\mathrm{int}} + \sum_i (-1)^i d_i$, where the d_i are the simplicial faces. It computes the derived colimit. In Chapter 103, the ordered incidence construction is of this kind. It needs an actual coefficient diagram on its indexing category. Identifying that derived colimit with the homology of a geometric or physical field theory is a further comparison.

Chapter 4

Recognition and higher comparison

Two products can agree on cohomology while their primitives produce different triple operations. The cubic example in the prologue makes this distinction inside a strict differential graded algebra: $U^2 = DH$, but $UH + HU = -V$ represents a nonzero cohomology class. Strict associativity has encoded the higher operation; it has not removed it. A comparison must transport the primitive as well as the binary product.

Collision geometry imposes a further requirement. When the first and third positions meet, the second input remains between them. Moving an endpoint past that input uses an exchange map. Two binary products can agree while these exchanges permit different normal poles. We construct this example below, then give the complexes and equations that decide whether a specified partial comparison extends. The obstruction belongs to the operations allowed by that comparison, with its boundary values fixed.

4.1 Simultaneous operations and their recipients

Let k have characteristic zero and let X be the smooth curve of Chapter 2. For a finite nonempty set I, write $j_I : U_I \hookrightarrow X^I$ for the open immersion of the locus where every pair of positions is distinct, and $\Delta_I : X \hookrightarrow X^I$ for the small diagonal. For complexes of right differential modules M_i, N on X, define the complex of *simultaneous chiral operations* by

$$Q_X(M_i; i \in I; N) = \mathrm{RHom}_{\mathcal{D}(X^I)}\left(j_{I,*}j_I^! \boxtimes_{i\in I} M_i, \Delta_{I,*}N\right). \tag{4.1}$$

The external tensors and geometric functors are derived. The output retains the whole differential module on the diagonal, including its normal derivatives. Degree-zero cycles represent cochain operations in a chosen differential graded model; a degree-minus-one cochain is a possible homotopy between such operations. Passing to a scalar residue is a further map. For ordinary modules, the degree-zero construction is the chiral operation space of Beilinson and Drinfeld.[1]

To compose, let $\pi : I \twoheadrightarrow A$ have nonempty fibres I_a, so that $I = \coprod_{a\in A} I_a$, and let $d_\pi : X^A \to X^I$ identify positions in each block. Let $j_\pi : U_\pi \hookrightarrow X^I$ be the open immersion where positions in different blocks are distinct. Given operations

$$g_a \in Q_X(M_i; i \in I_a; L_a), \qquad f \in Q_X(L_a; a \in A; N),$$

[1] A. Beilinson and V. Drinfeld, *Chiral Algebras*, §§3.1.1–3.1.2, pp. 157–158. The composition below is the derived form of their displayed composition of chiral operations.

their composite is induced by

$$\begin{aligned} j_{I,*}j_I^!\boxtimes_I M_i &\simeq j_{\pi,*}j_\pi^!\boxtimes_A\left(j_{I_a,*}j_{I_a}^!\boxtimes_{I_a} M_i\right)\\ &\xrightarrow{\boxtimes g_a} j_{\pi,*}j_\pi^!\boxtimes_A \Delta_{I_a,*}L_a\\ &\simeq d_{\pi,*}j_{A,*}j_A^!\boxtimes_A L_a \xrightarrow{d_{\pi,*}f} \Delta_{I,*}N. \end{aligned}$$

The middle equivalence is open–closed base change. Every parenthesization of a nested partition uses the same disjoint open set and the same composite diagonal map. Coherent base change therefore makes these compositions associative, with all their homotopies. The unary identity is the identity of a differential module. Permuting labels transports the same diagram. Thus the operation complexes form a coherent coloured operad: the colours are the chosen differential modules, and an operation has its indicated list of input colours and output colour. Forgetting permutations leaves its nonsymmetric operad on ordered inputs. For equations between cochains, choose a differential graded operad model compatible with these geometric functors and their base-change maps.

The word carrier has a different endomorphism operad. For word systems $F_\lambda \in \mathcal{W}_X$, its complexes are

$$\mathcal{P}_F(\lambda_1,\dots,\lambda_r;\nu) = \mathrm{RHom}_{\mathcal{W}_X}(F_{\lambda_1}\star\cdots\star F_{\lambda_r}, F_\nu), \qquad r\geq 0, \tag{4.2}$$

where $\star = \star_{\mathrm{cut}}$ and the empty tensor is K. Substitution is $f\circ(g_1\star\cdots\star g_r)$, with the coherent cut associator. On homogeneous blocks x_i, the tensor of maps has sign $(-1)^{\sum_{i<j}|g_j||x_i|}$. The cut construction of Chapter 2 proves its operad coherences: both composites retain the same final consecutive decomposition and localized coefficient. Its arity-zero maps have source K, not the local vacuum J.

A field comparison has to relate independently specified operations to (4.2). For a chosen family whose operations are represented by a differential graded operad $\mathcal{Q}^{\mathrm{adm}}$, it includes maps

$$\rho_r : \mathcal{Q}^{\mathrm{adm}}(\lambda_1,\dots,\lambda_r;\nu) \longrightarrow \mathcal{P}_F(\lambda_1,\dots,\lambda_r;\nu) \tag{4.3}$$

compatible with differentials and the required substitutions. Here *admissible* means a specified complex of operations, with its coefficient domains, pole conditions and topology. In a meromorphic differential-module family it can be a subcomplex of (4.1); closure under the required substitutions must be proved. A fixed pole bound need not have this property, since compositions can increase pole order. Monodromic or analytic fields require their own coefficient recipient before such a map is defined.

Collision restrictions, exchanges, translation and local-vacuum insertion give additional diagrams for ρ. When their commutativity is only up to homotopy, the homotopies and their higher compatibilities are part of the comparison. The construction of either operad above does not produce (4.3). Nor does the singleton F_1 determine its target, which uses the whole word system.

Example 4.1.1 *(An undetected positive component).* For a differential module M, let $I(M)_n = \Delta_{n,*}M$ for $n>0$, with zero empty component. Closed base change gives its monotone collision equivalences. Put $I = I(J_X)$, where $J_X = p_X^!k$, and $H = I\star I$. Then $H_1 = 0$, while H_2 is nonzero on the separated two-point locus. The free augmented cut algebras

$$\mathbb{T}(I)\longrightarrow\mathbb{T}(I\oplus H), \qquad \mathbb{T}(E) = K\oplus\bigoplus_{r\geq 1}E^{\star r},$$

agree at the singleton and differ at two positions by the extra summand H_2. Each sum is finite in every geometric arity. Consequently singleton evaluation is not conservative even on connected augmented cut algebras.

4.2 The endpoint exchange at a ternary collision

Let $B = k[\epsilon]/(\epsilon^2)$, concentrated in cohomological degree zero. Give it an auxiliary parity by making 1 even and ϵ odd. For $\sigma = \pm 1$, define

$$R_\sigma(a \otimes b) = \sigma^{p(a)p(b)} b \otimes a$$

on the two homogeneous basis vectors, and extend linearly. The parity here specifies the exchange; it is not a cohomological Koszul sign. Both exchanges have the same unit crossings, are involutive, and satisfy the braid relation. For three basis vectors the two braid words have the same sign $\sigma^{p(a)p(b)+p(a)p(c)+p(b)p(c)}$. They also satisfy the two multiplication compatibility identities

$$R_\sigma(1 \otimes \mu) = (\mu \otimes 1)(1 \otimes R_\sigma)(R_\sigma \otimes 1),$$
$$R_\sigma(\mu \otimes 1) = (1 \otimes \mu)(R_\sigma \otimes 1)(1 \otimes R_\sigma).$$

When the multiplied entries are not both ϵ, their parities add; when both are ϵ, both sides vanish. This proves the identities on a basis. The binary multiplication μ is the same for both signs, and $\mu R_\sigma = \mu$.

To collide the first and third input while retaining the middle input, define

$$q_\sigma = (\mu \otimes 1)(1 \otimes R_\sigma) : B^{\otimes 3} \longrightarrow B \otimes B.$$

The first output is the product of the endpoints; the second is the unchanged middle state after exchange. For $\xi = \epsilon \otimes \epsilon \otimes 1 - 1 \otimes \epsilon \otimes \epsilon$, one obtains

$$q_+(\xi) = 0, \qquad q_-(\xi) = 2\epsilon \otimes \epsilon. \tag{4.4}$$

Thus an operation that vanishes for one exchange need not vanish for the other, although the binary products agree.

This discrepancy gives actual differential modules with different allowed poles. On the affine-line three-point chart put

$$Y = \operatorname{Spec} k[z_1, z_2, z_3, (z_1 - z_2)^{-1}, (z_2 - z_3)^{-1}], \qquad u = z_1 - z_3, \qquad T = B^{\otimes 3}.$$

Inside $T \otimes_k \omega_Y[u^{-1}]$, define right differential modules

$$\mathcal{M}_\sigma = T \otimes_k \omega_Y + (\ker q_\sigma) \otimes_k \omega_Y[u^{-1}]. \tag{4.5}$$

The kernels are constant coefficient spaces, so multiplication by regular functions and differentiation preserve these modules. In a coordinate density, the right derivative differentiates the coefficient with a minus sign. Every finite normal derivative of an allowed pole remains allowed.

Proposition 4.2.1 *(Binary agreement and a ternary failure).* Both modules in (4.5) restrict to the same meromorphic module off $u = 0$. The identity of that module extends to no map $\mathcal{M}_+ \to \mathcal{M}_-$, and to no map in the opposite direction. Their common binary coefficient extension, obtained from $\ker \mu$, does not decide this ternary extension problem.

Proof. Localizing at u makes the first summand in (4.5) the whole meromorphic module. The target is a submodule of that localization and has no u-torsion. A map restricting to the identity must therefore be the identity on every section: the difference vanishes after localization and hence vanishes already.

But $u^{-1}\xi\, dz_1 dz_2 dz_3$ belongs to $\mathcal{M}_+$ and not to $\mathcal{M}_-$, by (4.4). A nonzero constant coefficient of u^{-1} cannot become regular at $u = 0$. For the reverse direction, use $\eta = \epsilon \otimes \epsilon \otimes 1 + 1 \otimes \epsilon \otimes \epsilon$: $q_-(\eta) = 0$ and $q_+(\eta) = 2\epsilon \otimes \epsilon$. At two inputs the construction uses the common map $\mu : B^{\otimes 2} \to B$, so its kernel extension is identical. □

The proposition fixes the identity on the marked meromorphic coefficient space. It makes no assertion about isomorphisms that change those markings. It also constructs binary and ternary coefficient extensions, not a complete field system in every arity. An all-arity realization must additionally supply the collision maps and verify every substitution among them.

4.3 The complex controlling a higher comparison

Let R be a commutative ring. We use the suspended convention of Chapter 3. An augmented A_∞-algebra is specified here by a complex $V = s\bar{A}$ and a degree-one coderivation

$$b_A : T^c_R(V) \longrightarrow T^c_R(V), \qquad b_A^2 = 0, \qquad b_A(1) = 0.$$

Its Taylor components are maps $b_{A,r} : V^{\otimes r} \to V$ of degree one, with $b_{A,1}$ the differential on V. The augmented state algebra is the unitization $R \oplus V[-1]$: its new unit acts by the identity under binary multiplication, and every higher operation vanishes when one of its inputs is that unit. All displayed inputs belong to the augmentation ideal. For a differential graded algebra, only b_1, b_2 occur, with $b_1(sa) = -s(da)$ and $b_2(sa, sb) = (-1)^{|a|}s(ab)$.

For a second such algebra put $W = s\bar{B}$. A normalized A_∞-map is a coaugmentation-preserving degree-zero coalgebra map $F : T^c_R(V) \to T^c_R(W)$ satisfying $b_B F = F b_A$. Its components $f_r : V^{\otimes r} \to W$ determine it by consecutive subdivisions. Define the actual cochain complex

$$E_r = \mathrm{Hom}_R(V^{\otimes r}, W), \qquad \partial f = b_{B,1} f - (-1)^{|f|} f\, b^{(r)}_{A,1},$$

where $b^{(r)}_{A,1}$ is the tensor differential. For derived comparisons use specified cofibrant coefficient models, or replace this Hom by its derived model. The formulas below concern the chosen chain models, with no unmentioned resolution change.

Suppose $f_1, \dots, f_{n-1}$ already satisfy the comparison equations below length n. Define

$$\begin{aligned} O_n = &\sum_{\substack{q \geq 2,\ i_1+\cdots+i_q = n \\ i_j \geq 1}} b_{B,q}(f_{i_1} \otimes \cdots \otimes f_{i_q}) \\ &- \sum_{\substack{u+v+w=n \\ v \geq 2}} f_{u+1+w}(1^{\otimes u} \otimes b_{A,v} \otimes 1^{\otimes w}) \in E^1_n. \end{aligned} \tag{4.6}$$

All components on the right have index less than n, except the already specified algebra operations. The signs are tensor composition signs: insertion of $b_{A,v}$ after a homogeneous prefix has the sign given by its total degree. No division by two occurs.

THEOREM 4.3.1 *(One higher component).* The cochain O_n is closed. A component f_n extending the comparison through length n exists precisely when $[O_n] = 0$ in $H^1(E_n)$. It must satisfy

$$\partial f_n = -O_n.$$

When a solution exists, all solutions form a torsor for $Z^0(E_n)$. Modulo changes $f_n \mapsto f_n + \partial h$, with $h \in E_n^{-1}$ and all lower components fixed, they form a torsor for $H^0(E_n)$.

Proof. Extend the chosen components to a coalgebra map by setting the remaining Taylor components to zero. Put $D = b_B F - F b_A$. The square-zero equations imply

$$b_B D + D b_A = 0.$$

The map D is a coderivation over F. Since its length-one output vanishes on inputs shorter than n, the only part of this identity with n inputs and one output is $b_{B,1} O_n + O_n b_{A,1}^{(n)} = 0$. This is $\partial O_n = 0$. In the equation with one output and n inputs, the new f_n occurs only in the unary terms; all other terms are (4.6). This gives the displayed equation. Subtracting two solutions gives a degree-zero cocycle, and quotienting by degree-zero boundaries gives the final assertion. □

For strict differential graded inputs, the first equation is

$$\partial f_2 = f_1 b_{A,2} - b_{B,2}(f_1 \otimes f_1).$$

Thus f_2 is the multiplicativity homotopy in suspended coordinates. At length three, f_2 enters on both sides of the next equation. Choosing its cohomology class independently of that equation can obstruct the extension.

Example 4.3.2 *(A nonzero first ternary obstruction).* Let $R = k$, and take $V = kx \oplus ky$ with $|x| = -1$, $|y| = -2$. On $T^c(V)$, let the first coderivation be zero and let the second have the sole component

$$b_{B,3}(x, x, x) = y,$$

vanishing on inputs containing y. Its square is zero: nested insertions vanish, while disjoint degree-one insertions cancel. With $f_1 = 1_V$, both unary and binary comparison equations hold, for any f_2. But $O_3(x, x, x) = y$, and $\partial = 0$ on E_3. Hence no higher comparison with this linear term exists. The corresponding unsuspended states have degrees zero and minus one.

There is a version on the actual word carrier. Take the positive system $I = I(J_X)$ of Example 4.1.1, put $H = I^{\star 3}$, and use reduced states $G = I \oplus H[1]$. On the suspended states $G[1] = I[1] \oplus H[2]$, define b_3 to be the degree-one shifted identity

$$(I[1])^{\star 3} \simeq H[3] \longrightarrow H[2],$$

and zero if an input belongs to $H[2]$. Together with the internal differential it squares to zero by the same cancellation. Compare it to the structure with $b_3 = 0$ and the same unary differential and zero binary product. The first ternary defect projects to the identity of H. It is not a boundary, because H_3 is nonzero on the separated three-point locus. Thus the identity of the underlying word system and the agreement of binary operations do not extend to an A_∞-map. This example uses positive systems and lies within the geometric finiteness regime; coefficient completion is not its obstruction.

4.4 Fixed boundary values and allowed homotopies

Geometric conditions restrict the possible f_n. A restriction to an open set, a prescribed normal coefficient or a vacuum value can be represented, on fixed chain models, by a cochain map $r_n : E_n \to T_n$. Let $t_n \in T_n^0$ be the prescribed value of the new component. Compatibility with its chain equation requires

$$\partial t_n = -r_n O_n.$$

Several restrictions can be placed in the same target diagram; its maps and compatibility homotopies must be included when forming the derived restriction map.

First suppose r_n is degreewise surjective and require the strict boundary value $r_n f_n = t_n$. Put $K_n = \ker r_n$, and choose any degree-zero lift $\widetilde{f_n}$ of t_n.

THEOREM 4.4.1 *(The relative obstruction).* The cochain

$$o_n = O_n + \partial \widetilde{f_n}$$

is a degree-one cocycle in K_n. Its class in $H^1(K_n)$ is independent of the lift. The comparison extends with the prescribed boundary value if and only if this class vanishes. When it does, solutions modulo homotopies in K_n^{-1}, with all lower data fixed, form a torsor for $H^0(K_n)$.

Proof. The boundary equation gives $r_n o_n = 0$, and $\partial O_n = 0$ gives $\partial o_n = 0$. Changing the lift by $a \in K_n^0$ changes o_n by ∂a. A correction $a \in K_n^0$ makes $f_n = \widetilde{f_n} + a$ a solution precisely when $\partial a = -o_n$. Subtraction of two solutions and then quotienting by relative boundaries gives the asserted torsor. □

Without surjectivity, a strict lift of t_n is a separate first requirement. If the boundary is instead allowed to agree up to a specified homotopy, the correct complex is the *homotopy fibre*

$$\mathrm{Fib}(r_n)^q = E_n^q \oplus T_n^{q-1}, \qquad d(e,h) = (\partial e, r_n e - \partial h).$$

The pair $(O_n, -t_n)$ is closed. Its class vanishes precisely when there are $f_n \in E_n^0$, $h_n \in T_n^{-1}$ with

$$\partial f_n = -O_n, \qquad r_n f_n = t_n - \partial h_n.$$

Indeed, take the negative of the E_n-component of a primitive of $(O_n, -t_n)$, and retain its $T_n[-1]$ component. This proves the equivalence directly, including the sign of the boundary homotopy. Replacing a strict boundary condition by this homotopy condition changes the comparison problem unless that replacement was allowed.

Example 4.4.2 *(An inadmissible primitive).* Let E have basis h in degree zero and c in degree one, with $\partial h = c$. Let $T = kh$ in degree zero and let $r : E \to T$ be the quotient by kc. The obstruction $O = c$ is exact in E. With prescribed boundary $t = 0$, however, $K = kc$ has $H^1(K) = kc$, so no allowed primitive exists. The unique ambient solution $-h$ has the wrong boundary value. This is the distinction between finding a primitive and finding one in the permitted support or vacuum sector.

COROLLARY 4.4.3 *(Extension through all lengths).* Suppose the restriction complexes are specified at every length, each r_n is degreewise surjective, and $H^1(K_n) = 0$ for $n \geq 2$. An admissible unary comparison extends to a normalized A_∞-map if compatible boundary values can be supplied successively for its partial lifts. If contracting homotopies $h_n : K_n \to K_n[-1]$, satisfying $\partial h_n + h_n \partial = 1$, are supplied, one can take

$$f_n = \widetilde{f_n} - h_n o_n.$$

Over a field, a quasi-isomorphism f_1 gives a quasi-isomorphism of the resulting direct-sum bar complexes.

Proof. Apply Theorem 4.4.1 recursively. For the displayed choice, closedness of o_n gives $\partial h_n o_n = o_n$. At each finite word length the coalgebra formula uses only finitely many components, so the resulting map is defined on the direct-sum bars and satisfies the chain identity there.

Filter these bars by word length. On the associated graded, the comparison is $f_1^{\otimes r}$ in length r. Over a field tensor powers preserve quasi-isomorphisms. Finite-stage induction and exact filtered unions prove the final assertion. □

The hypotheses include compatibility of the prescribed boundary values with the partial solution chosen at the previous stage. Vanishing cohomology does not supply incompatible boundary data. Nor does it by itself imply uniqueness in a full mapping space: $H^0(K_n)$ describes this stage's choices, and negative-degree cohomology can describe homotopies between those choices. For completed coefficient bars, the same construction must also satisfy the continuity conditions of Chapter 3.

4.5 Operations, deformations, and a recognition problem

There is a parallel obstruction for constructing the operations themselves. Fix b_1 on a suspended coefficient complex V and choose $b_2, \dots, b_{n-1}$ satisfying $b^2 = 0$ below length n. Then

$$C_n = \sum_{\substack{i+j=n+1 \\ i,j\geq 2}} \sum_{u=0}^{i-1} b_i(1^{\otimes u} \otimes b_j \otimes 1^{\otimes(i-1-u)}) \in \operatorname{Hom}_R(V^{\otimes n}, V)^2.$$

Its differential is zero. To see this without dividing by any integer, use the identity $b\,b^2 = b^2 b$ for the partial coderivation. Its first nonzero component is $b_1 C_n - C_n b_1^{(n)} = 0$. A new operation b_n must satisfy $\partial b_n = -C_n$, so the obstruction is in degree two and the choices, modulo boundaries, are in degree one. This differs by one degree from the comparison problem, where f_n has degree zero.

In characteristic zero these equations also have a Lie-algebra form. Let

$$\mathfrak{g} = \prod_{r\geq 2} \operatorname{Hom}_k(V^{\otimes r}, V), \qquad d_{\mathfrak{g}} = [b_1, -].$$

Identify a collection with its coderivation of $T^c(V)$. The bracket is the graded commutator of coderivations, projected back to one output. Give arity r weight $r-1$, and let $F^p\mathfrak{g}$ consist of collections supported in weights at least p. Then $[F^p, F^q] \subset F^{p+q}$, and the product is complete and separated for this filtration. Each fixed-arity bracket coefficient is a finite sum.

A degree-one $\alpha \in \mathfrak{g}$ gives square-zero $b_1 + \alpha$ precisely when

$$d_{\mathfrak{g}}\alpha + \tfrac{1}{2}[\alpha, \alpha] = 0.$$

If α solves this equation modulo F^p, its curvature in F^p/F^{p+1} is a degree-two cocycle. The identity $d_{\mathfrak{g}}C + [\alpha, C] = 0$ follows by expanding the curvature, and its leading term proves closedness. A correction $\beta \in F^p\mathfrak{g}^1$ changes that class by $d_{\mathfrak{g}}\beta$. This gives the same extension criterion as the integral coderivation equation. For deformations around a complete nonzero solution, the controlling differential is instead $d_\alpha = d_{\mathfrak{g}} + [\alpha, -]$; retaining that twist is essential. These formulas apply to an admissible operation subcomplex only if it is closed under the differential and the required insertions.

To compare an entire coloured operation system, one can specify a differential graded operad resolution $\mathcal{R}$ with successive free generators e, each having differential in the previously constructed part. If a partial map $\rho : \mathcal{R} \to \mathcal{P}_F$ is fixed, extending it over a generator of degree d asks for

$$x \in \mathcal{P}_F^d, \qquad dx = \rho(de).$$

The right side is closed because $d^2 e = 0$. Its class in $H^{d+1}(\mathcal{P}_F)$, or in the relative complex with the prescribed boundary values, is exactly the obstruction. Freeness then extends the map to composites; the chain identity on generators implies it on every composite by the Leibniz rule. This is an explicit obstruction calculation once the generators, their differentials and their operation recipients are supplied. A name for an operad or a Hochschild group supplies none of those data.

Question 4.5.1 *(Recognition of an arithmetic family).* For an independently specified arithmetic coefficient family, construct its simultaneous operation complexes and their geometric recipients, then the maps (4.3) on their full coefficient domains. Compare every differential, supported restriction, normal derivative, endpoint exchange and local-vacuum insertion used by the family, with compatible nested substitutions. Retain arithmetic operators and a state or trace when these marks have been independently defined. Determine the image of the resulting comparison, or prove an equivalence on a stated class of objects and full mapping spaces.

The preceding constructions locate this question's successive equations and give an extension theorem when their relative cohomology vanishes. The ternary pole example shows why the binary field product cannot replace the simultaneous operation system. The cubic example shows why a strict presentation cannot replace the comparison of primitives. Both distinctions persist when a scalar character or trace already agrees.

Chapter 5

Regular coefficient models

An associative coefficient algebra already produces fields: translate the first input and multiply it by the second. Their products have no poles. Nevertheless, their realization on collision strata must retain the order of the inputs. For the matrix units

$$a = c = E_{12}, \qquad b = E_{21}, \qquad ac = 0, \qquad abc = E_{12},$$

combining the outer coefficients before allowing the middle one to act changes a nonzero operation into zero. The obstruction concerns a partial collision, where the first and third positions agree and the second remains separate.

We construct coefficients on every such stratum by retaining each consecutive run of a position in the original word. Their principal-parts maps form a complex, and their collision and cut maps are coherent. A specified presentation permits recovery of the original differential algebra. We then construct a different coefficient system using an explicit exchange. It compresses the runs while retaining their evaluated products. The compression has a computable kernel; it need not be an equivalence.

5.1 Regular fields and the information in a word

Fix a field k of characteristic zero and $X = \mathbb{A}^1_k$ with coordinate z. A *differential coefficient algebra* is a unital differential graded associative k-algebra $(B, d, \mu, 1)$, with $1 \neq 0$, together with a degree-zero derivation T commuting with d. No finite-dimensionality or boundedness condition is imposed on B. Tensor products in this chapter are algebraic over k, unless another coefficient category is specified.

Put

$$Y_B(a,u)b = (e^{uT}a)b = \sum_{r\geq 0} \frac{u^r}{r!}(T^r a)b \quad \text{in } B[[u]]. \tag{5.1}$$

This is a formal power series, even when T is not locally nilpotent. Its coefficient in every fixed degree is defined. Writing $Y_B(a,u)b = \sum_{n\in\mathbb{Z}} a_{(n)}b\, u^{-n-1}$, we obtain

$$a_{(-r-1)}b = \frac{(T^r a)b}{r!} \quad (r \geq 0), \qquad a_{(n)}b = 0 \quad (n \geq 0).$$

In particular, $a_{(-1)}b = ab$.

Proposition 5.1.1. The fields (5.1) preserve the differential and satisfy the vacuum and translation identities

$$Y_B(1,u)b = b, \qquad Y_B(a,u)1 = e^{uT}a, \qquad Y_B(Ta,u)b = \partial_u Y_B(a,u)b,$$

$$T(Y_B(a,u)b) = Y_B(Ta,u)b + Y_B(a,u)Tb.$$

Their simultaneous product, rooted at the last position, is

$$m_n(b_1, \dots, b_n; z) = \prod_{i=1}^{n} e^{(z_i - z_n)T} b_i \tag{5.2}$$

in the original order. Consecutive substitutions give this same power series.

Proof. The unit and derivation identities give the displayed formulas coefficient by coefficient. For example,

$$\begin{aligned} Y_B(Y_B(a,u)b,v)c &= e^{vT}\big((e^{uT}a)b\big)c \\ &= (e^{(u+v)T}a)(e^{vT}b)c = Y_B(a,u+v)Y_B(b,v)c. \end{aligned}$$

There are no negative powers to obstruct the substitution. At several consecutive blocks, the same derivation identity moves each block to the common root. Associativity removes the parentheses. The graded Leibniz rule and $dT = Td$ prove compatibility with the differential. □

The order in (5.2) remains part of the operation. For constant fields, changing it replaces ab by ba. A rule relating those expressions requires additional maps.

To measure the information between occurrences, leave r inputs open and define

$$\Gamma_{B,r} : B^{\otimes(r+1)} \longrightarrow \mathrm{Hom}_k(B^{\otimes r}, B), \qquad \Gamma_{B,r}(a_0, \dots, a_r)(x_1, \dots, x_r) = a_0 x_1 a_1 \cdots x_r a_r.$$

PROPOSITION 5.1.2 *(Matrix coefficients with open inputs).* For $B = \mathrm{Mat}_d(k)$, the map $\Gamma_{B,r}$ is an isomorphism. A linear encoding of its source that retains every evaluation therefore has dimension at least d^{2r+2}. For $r = 1$ and $d > 1$, one copy of B cannot supply that encoding.

Proof. On matrix units, the value is

$$\Gamma_{B,r}(E_{i_0 j_0}, \dots, E_{i_r j_r})(E_{p_1 q_1}, \dots, E_{p_r q_r}) = \left(\prod_{\ell=1}^{r} \delta_{j_{\ell-1}, p_\ell} \delta_{q_\ell, i_\ell} \right) E_{i_0 j_r}.$$

Every elementary linear map between the indicated tensor bases occurs exactly once. This proves the isomorphism. If $\Gamma_{B,r}$ factors as $B^{\otimes(r+1)} \to L \to \mathrm{Hom}_k(B^{\otimes r}, B)$, its first map is injective, which proves the dimension bound. □

There is a stronger constraint when the encoding is required to be natural in the coefficient algebra.

PROPOSITION 5.1.3 *(Naturality retains the open word).* Suppose a quotient

$$q_B : B^{\otimes(r+1)} \twoheadrightarrow Q_B$$

is given for every differential coefficient algebra, is natural under injective unital maps commuting with d, T, and permits $\Gamma_{B,r}$ to factor through it. Then every q_B is an isomorphism. For finite-dimensional ordinary algebras, it suffices to require naturality under injections into other finite-dimensional algebras.

Proof. Adjoin noncommuting symbols $X_1, \dots, X_r$ with $dX_i = TX_i = 0$. The free B-algebra on these symbols is the tensor algebra over B of $B \otimes_k k\{X_1, \dots, X_r\} \otimes_k B$. Its summand with symbol word $X_1 \cdots X_r$ is a copy of $B^{\otimes(r+1)}$. Thus

$$a_0 \otimes \cdots \otimes a_r \longmapsto a_0 X_1 a_1 \cdots X_r a_r$$

is injective. If $w \in \ker q_B$, naturality sends w into the kernel of the corresponding quotient for the free algebra. Evaluating its open inputs at $X_1, \dots, X_r$ therefore gives zero. Injectivity implies $w = 0$. For finite-dimensional B, quotient the free algebra by symbol words of length greater than r. This algebra is finite-dimensional and retains both the injection of B and the separating summand. □

At one fixed algebra the kernel can be nonzero. For $B = k[\epsilon]/(\epsilon^2)$, $\epsilon \otimes 1 - 1 \otimes \epsilon$ evaluates to zero on every element of B. Under $\epsilon \mapsto E_{12}$, its evaluation on E_{21} becomes $E_{11} - E_{22}$. Enlarging the allowed intermediate inputs reveals information that the original evaluations lost.

5.2 Run coefficients and their residue maps

Use the word category $\mathcal{W}_X$ and its cut product from Chapter 2. For a partition π of an ordered set I, read its equality blocks along the word and remove only adjacent repetitions. The resulting list is the list of *runs*. Write $\ell(\pi)$ for its length and $\beta_\pi(r)$ for the block occupied by run r. At three positions,

π	run list	coefficient
$12\|3$	$(12, 3)$	$B^{\otimes 2}$
$1\|23$	$(1, 23)$	$B^{\otimes 2}$
$13\|2$	$(13, 2, 13)$	$B^{\otimes 3}$.

The equality $z_1 = z_3$ does not make those two occurrences adjacent in the original word.

A coarsening $\pi \le \rho$ relabels every run by its ρ-block. Multiply the factors in each newly consecutive run, retaining their order. This defines

$$p_{\pi\rho} : B^{\otimes \ell(\pi)} \longrightarrow B^{\otimes \ell(\rho)}. \tag{5.3}$$

For example, $p_{1|2|3,13|2}$ is the identity and $p_{13|2,123}(a, b, c) = abc$.

Lemma 5.2.1. The maps (5.3) are chain maps and obey

$$p_{\rho\sigma}\, p_{\pi\rho} = p_{\pi\sigma}.$$

For each block C of π, put $T_C = \sum_{\beta_\pi(r)=C} T^{(r)}$. Along a merger, $p_{\pi\rho}$ intertwines the sum of the input block translations with the output block translation.

Proof. A final run is an interval of the original word. An intermediate coarsening merely parenthesizes the ordered product on that interval. Associativity gives the composition identity. The two derivation identities give the assertions for d and T. No coefficient factors are permuted in this argument. □

Let X^π have one coordinate per equality block, ordered by first occurrence. Let

$$i_\pi : X^\pi \hookrightarrow X^I, \qquad j_\pi : \mathrm{Conf}_\pi(X) \hookrightarrow X^\pi$$

be the diagonal embedding and the open subset where different blocks occupy distinct positions. On X^{π}, set

$$M_{\pi}(B) = \omega_{X^{\pi}} \otimes_{k} B^{\otimes \ell(\pi)}.$$

The right differential-module structure is

$$(f\,\omega_{\pi} \otimes \mathbf{b})\partial_{z_C} = -\partial_{z_C} f\,\omega_{\pi} \otimes \mathbf{b} - f\,\omega_{\pi} \otimes T_C \mathbf{b}, \qquad \omega_{\pi} = dz_{C_1} \wedge \cdots \wedge dz_{C_{|\pi|}}. \tag{5.4}$$

Define the graded object

$$C^{\mathrm{run}}_{I}(B) = \bigoplus_{\pi \in \Pi(I)} i_{\pi,*} j_{\pi,*} j_{\pi}^{*} M_{\pi}(B)[|\pi|], \qquad C^{\mathrm{run}}_{\varnothing}(B) = k. \tag{5.5}$$

The sum is finite for each I. A coefficient of internal degree q in its r-block term has total degree $q - r$.

We specify the boundary, including its normal module. For a normal coordinate u, let e_j, $j \geq 0$, denote the class of u^{-j-1} times its normal density. In right-module notation,

$$e_0 u = 0, \qquad e_j u = e_{j-1}\ (j > 0), \qquad e_j \partial_u = (j+1) e_{j+1}.$$

Writing $\delta = e_0$, equivalently $e_j = \delta\, \partial_u^{j} / j!$. The map pp_u takes the negative part of a Laurent expansion to this full normal module. It retains all negative powers.

For an elementary merger of blocks $i < j$, take $u = z_{C_i} - z_{C_j}$, retain z_{C_j} as the merged coordinate, and put the merged block in position i. The face on unshifted coefficients is

$$D_{\pi\rho}(f \otimes \mathbf{b}) = (-1)^{j-2}\, \mathrm{pp}_u\big(f\, p_{\pi\rho} e^{u T_{C_i}} \mathbf{b}\big). \tag{5.6}$$

All other differences are expanded normally before taking principal parts. The sign is fixed by $\omega_{\pi} = (-1)^{j-2} du \wedge \omega_{\rho}$. Normal transfer extends this formula to the indicated diagonal direct image. A fixed localized input has a finite pole order, so only finitely many terms of the exponential contribute to its principal part. The output therefore has algebraic coefficients even for unrestricted T.

On the r-block summand the internal differential is $(-1)^{r} d_{B^{\otimes \ell(\pi)}}$. The total differential is this map plus the sum of the elementary faces. It has degree one. To prove that it squares to zero, the normal contacts must first be compared in a common target.

The three normal channels

Put $s = z_1 - z_3$, $t = z_2 - z_3$ and $r = s - t$. For any vector space P, take

$$\mathcal{G}(P) = P[[s,t]][(st(s-t))^{-1}].$$

Power series are formed before localization, so an element has a finite pole bound but can have infinitely many independent state coefficients. The full contact module $E_{s,t}(P)$ consists of finite sums of $p\, s^{-a} t^{-b}$, $a, b \geq 1$, with their normal densities and right differential actions.

The change from contact coordinates (r,t) to (s,t) is

$$J(r^{-a} t^{-b}) = \sum_{j=0}^{b-1} \binom{a+j-1}{j} s^{-a-j} t^{-b+j}.$$

Indeed, J sends the simple contact to $s^{-1}t^{-1}$, the derivative in r to that in s, and the old derivative in t to $\partial_s + \partial_t$. Applying those derivatives gives the displayed formula. Define

$$\begin{aligned} L(F) &= J\,\mathrm{pp}_t\,\mathrm{pp}_r\,F(r+t,t) && (r \ll t),\\ R(F) &= \mathrm{pp}_s\,\mathrm{pp}_t\,F(s,t) && (t \ll s),\\ N(F) &= \mathrm{pp}_t\,\mathrm{pp}_s\,F(s,t) && (s \ll t). \end{aligned}$$

The first principal part in each expression is the rightmost one. These are the channels 12, 23 and 13, expressed in one contact module.

Lemma 5.2.2. For every $F \in \mathcal{G}(P)$,

$$L(F) - R(F) + N(F) = 0.$$

Proof. Write $F = G/(s^a t^b (s-t)^c)$. A monomial surviving two principal parts has total degree at most -2. Consequently numerator terms of degree greater than $a+b+c-2$ contribute to none of the three maps. Only a finite polynomial part of G matters.

Over $k[t,t^{-1}]$, the polynomials s and $s-t$ are coprime. Partial fractions reduce to polynomials in s, terms $a_j(t)s^{-j}$, and terms $b_j(t)(s-t)^{-j}$, with Laurent coefficients in t. A polynomial in s gives zero. For $a_j(t)s^{-j}$, the expansion at $r=0$ is regular, so $L=0$ and $R=N=s^{-j}\,\mathrm{pp}_t(a_j)$.

For $b_j(t)(s-t)^{-j}$, the expansion in $s \ll t$ is regular in s, so $N=0$. In the other chamber,

$$(s-t)^{-j} = \sum_{n\geq 0} \binom{j+n-1}{n} s^{-j-n}t^n.$$

A term $t^{-\beta}$ of b_j, with $\beta \geq 1$, therefore gives $\sum_{n<\beta}\binom{j+n-1}{n}s^{-j-n}t^{n-\beta}$ in R. This is exactly $J(r^{-j}t^{-\beta})$, its contribution to L. The identity follows. □

For $F = 1/(s(s-t))$, the three values are $s^{-1}t^{-1}$, 0, $-s^{-1}t^{-1}$. Deleting the outer channel leaves a nonzero defect. For $F = 1/((s-t)^2t^2)$, they are

$$L(F) = R(F) = s^{-2}t^{-2} + 2s^{-3}t^{-1}, \qquad N(F) = 0.$$

The second term comes from the normal-coordinate transfer; keeping only the scalar residue would lose it.

Theorem 5.2.3 *(The regular residue complex).* The total differential on (5.5) is differential-linear and squares to zero. Unital differential algebra maps commuting with T induce chain maps by their tensor powers on the runs.

Proof. The derivative of $e^{uT_{C_i}}$ is $T_{C_i}e^{uT_{C_i}}$. Together with (5.4) and Lemma 5.2.1, this identifies the input and output tangential connections in every face. In the normal direction, $\mathrm{pp}_u(-\partial_u f)$ is right multiplication by ∂_u; differentiation of the Taylor factor gives precisely the input translation term. Multiplication by u agrees with the lowering action on the e_j. Thus the face extends to a differential-module map.

The internal square is zero. Each unshifted face is a chain map, and the two adjacent block numbers have opposite internal signs. The mixed terms therefore cancel. Two successive mergers either join disjoint pairs or join three blocks. For a final partition σ, translate each original run from its original

block coordinate to the retained coordinate of its σ-block. Denote this formal Taylor operator by $E_{\pi/\sigma}$. The derivation identity gives, for every intermediate ρ,

$$p_{\rho\sigma}E_{\rho/\sigma}\; p_{\pi\rho}E_{\pi/\rho} = p_{\pi\sigma}E_{\pi/\sigma}.$$

Intermediate products only parenthesize the same factors, and their translation displacements add. On a fixed localized input, each compared contact uses finitely many Taylor coefficients, so this formal identity determines the actual normal distributions.

The common coefficient can now be taken through every route. For disjoint pairs the two normal orientations have opposite signs. For three blocks the oriented sum is the identity of Lemma 5.2.2, with any remaining coefficients as spectators. These two cases exhaust the square of the boundary. Maps of differential algebras preserve every product, Taylor coefficient and internal differential in these formulas, proving the final assertion. □

5.3 COLLISION COHERENCE AND FRAMED RECOGNITION

The complexes just constructed live on the actual position charts. To obtain an object of $\mathcal{W}_X$, their exceptional collision comparisons must also be supplied.

THEOREM 5.3.1. The complexes $C_I^{\mathrm{run}}(B)$ have canonical coherent equivalences

$$d_\alpha^! C_I^{\mathrm{run}}(B) \simeq C_J^{\mathrm{run}}(B)$$

for every monotone surjection $\alpha : I \twoheadrightarrow J$. They form a connected coaugmented cut coalgebra. The construction is functorial in differential coefficient algebras and preserves quasi-isomorphisms.

Proof. A partition term survives $d_\alpha^!$ only when each α-fibre lies in one equality block. Otherwise that collision lands in a diagonal removed by j_π, and the exceptional pullback vanishes by $i^! j_* = 0$. A surviving partition is uniquely $\pi = \alpha^{-1}(\bar\pi)$. Since the fibres of α are consecutive, deleting their repetitions identifies the run lists of π and $\bar\pi$. The closed diagonal i_π factors through d_α; the closed-embedding identity $d_\alpha^! d_{\alpha,*} \simeq \mathrm{id}$ and open base change identify the surviving coefficient with its $\bar\pi$-term. Its density and shift are still $\omega_{X^{\bar\pi}}$ and $[|\bar\pi|]$. Formula (5.6) identifies the surviving differentials.

Under two collisions, the same partitions survive as under their composite, and the same adjacent repetitions are deleted. The transitivity maps for closed direct image and open base change therefore give all collision coherences.

For a consecutive cut $I = A|D$, restrict to the open where every position in A differs from every position in D. A partition with a block crossing the cut has empty support there. Every surviving partition is $\pi_A \sqcup \pi_D$, and its runs are the concatenation of the two run lists. Thus there is an equivalence

$$C_I^{\mathrm{run}}(B)|_{U_{A,D}} \simeq \big(C_A^{\mathrm{run}}(B) \boxtimes C_D^{\mathrm{run}}(B)\big)|_{U_{A,D}}.$$

Use the canonical tensor-shift isomorphisms when identifying the differentials. Explicitly, the map from $M[a] \otimes N[b]$ to $(M \otimes N)[a+b]$ multiplies $x \otimes y$, with x of unshifted degree $|x|$, by $(-1)^{b|x|}$. The block densities occur in their concatenated order.

Localization followed by this equivalence, summed over cuts, defines the coproduct. For three or more output blocks every parenthesization restricts to the same disjoint-block open and concatenates the same run coefficients. This proves coassociativity and its compatibility with collisions. Empty cuts are identity maps on the nonempty coefficient and use $C_\varnothing = k$, giving the counit and coaugmentation. More than $|I|$ positive output blocks are impossible.

Finally filter by the number of equality blocks, using the terms with at most r blocks for the r-th stage. The differential preserves this finite filtration. On its associated graded, a quasi-isomorphism of B's acts by finite tensor powers over a field, followed by derived coefficient functors. It is a quasi-isomorphism on every term, and hence on the finite total object. □

The four-point partial strata display the coherence before the general proof is needed again:

π	run pattern	coarsening from $1\vert2\vert3\vert4$
$12\vert34$	$AABB$	$a\otimes b\otimes c\otimes d\mapsto ab\otimes cd$
$13\vert24$	$ABAB$	$a\otimes b\otimes c\otimes d\mapsto a\otimes b\otimes c\otimes d$
$14\vert23$	$ABBA$	$a\otimes b\otimes c\otimes d\mapsto a\otimes bc\otimes d.$

Merging either pair first gives the displayed map. The final collision gives $abcd$ in every case. The corresponding normal residue routes still carry their orientation signs.

The coefficient unit $k\to B$ supplies a coalgebra map $C^{\mathrm{run}}(k)\to C^{\mathrm{run}}(B)$. On each partition it inserts the coefficient unit in every run. This source retains nonempty coefficients: already its singleton is $\omega_X[1]$. It is distinct from the isolated empty unit K. The singleton unit and every tested regular mode are therefore retained without assigning the empty coefficient the value B.

Theorem 3.1.2 now gives an associative geometric algebra

$$A^{\mathrm{run}}(B)=\Omega_{\mathrm{cut}}C^{\mathrm{run}}(B),\qquad B_{\mathrm{cut}}A^{\mathrm{run}}(B)\simeq C^{\mathrm{run}}(B).$$

Its multiplication is cobar concatenation. The field multiplication of B is recovered instead from the framed two-point boundary

$$r_B(f;a,b)=\mathrm{pp}_u\big(f(e^{uT}a)b\big),\qquad a_{(n)}b=[e_0]\,r_B(u^n;a,b).$$

The test u^{-1} recovers ab; all negative-index modes are retained in the same way. This recovery uses the full normal module, not evaluation of a distribution at $u=0$.

Definition 5.3.2. A *split presentation with run coefficients* consists of the following data for a complex E with a commuting degree-zero operator T: the literal direct sums (5.5) with E in every run; the connections (5.4); a nonzero closed degree-zero singleton vector 1 with $T1=0$; the saturated-partition collision identifications of Theorem 5.3.1, including their transitivity; and the cut maps given by concatenation. It also has a differential extending the shifted internal differential and a binary contact r. The contact is *regular* when $[e_0]r(u^n;a,b)=0$ for every $n\geq 0$. A morphism is the literal tensor power, on each run coefficient, of one singleton chain map commuting with T and 1, compatible with all the displayed maps. Morphisms are not identified up to chain homotopy.

The definition supplies the carrier and its comparisons. It does not assume a multiplication on E. Recover the candidate product by $\mu(a,b)=[e_0]r(u^{-1};a,b)$.

Theorem 5.3.3 *(Recognition with a specified presentation).* A split presentation with regular binary contact is isomorphic to $C^{\mathrm{run}}(B)$ as a framed presentation exactly when the following equations hold: μ is associative with unit 1; d and T are derivations of μ; every elementary face is (5.6) for this product; and the total differential has no further components. The inverse construction is $B=(E,d,T,\mu,1)$. These constructions give an equivalence of the stated strict categories.

Proof. Every condition holds for the constructed realization. Conversely the first two conditions make E a differential coefficient algebra. Its fields are $(e^{uT}a)b$. The face equations identify every normal contact, and the condition on the remaining components identifies the full differential. The prescribed splitting identifies the coefficient modules, their shifts and connections. The prescribed saturation and cut maps identify the exceptional comparisons, their coherences and the coproduct. This gives an isomorphism of all the framed data.

A morphism of presentations recovers its singleton map. Compatibility with $r(u^{-1}; a, b)$ makes this map multiplicative, while the other singleton requirements preserve d, T, 1. Conversely such an algebra map preserves every construction. The two assignments are inverse on morphisms as well as objects. □

Prescribing the collision comparisons is necessary. In the scalar presentation, multiply the comparison for $\alpha : I \twoheadrightarrow J$ by $\lambda^{|I|-|J|}$, where $\lambda \in k^\times$. The exponents add under composition and under cut decomposition, so all their coherences remain valid. The coefficients and differentials are unchanged. For $\lambda \neq 1$, no unital singleton map $k \to k$ induces a framed isomorphism with the original presentation.

Even the unframed word carrier has coefficients invisible in arities one and two. Choose distinct $a, b \in k$. For a rational point $p \in X^n$, let δ_p be its differential-module direct image. The system

$$F_0 = 0, \qquad F_n = \bigoplus_{\substack{r_1,r_2,r_3 \geq 1 \\ r_1+r_2+r_3=n}} \delta_{a^{r_1}b^{r_2}a^{r_3}}$$

has canonical monotone-collision comparisons. Indeed, the exceptional pullback of a summand vanishes unless the point is constant on each collision fibre. Every surviving point is the unique expansion of a corresponding point of the shorter word, and closed direct image gives the transitive identifications. Thus $F_1 = F_2 = 0$ but $F_3 = \delta_{(a,b,a)} \neq 0$. There are $\binom{n-1}{2}$ summands for $n \geq 3$. Binary data alone are not conservative on this carrier.

Strict differentials retain higher products

The recognition theorem is a statement about literal presentations. For

$$B = k1 \oplus (ku \xrightarrow{d} kv), \qquad |u| = -1, \quad |v| = 0, \quad du = v, \quad dv = 0, \quad T = 0,$$

give the second summand square-zero multiplication. The unital maps that multiply u, v by a scalar λ are distinct strict algebra maps. The augmentation $B \to k$ is a quasi-isomorphism and has the same composite with every such map. Consequently these maps all become the identity after localization. Strict faithfulness does not imply faithfulness of strict maps into a derived category. This example does not decide the faithfulness of the functor from the already localized source.

The cubic algebra in the prologue supplies a different test. Its strict endomorphism algebra E, with $T = 0$, is an allowed coefficient algebra. Hence $C^{\mathrm{run}}(E)$ is defined with no finite state truncation. The two-point contact recovers composition in E, and its coefficient differential retains the identities

$$DU = DV = 0, \qquad U^2 = DH, \qquad UH + HU = -V.$$

In particular the single-valued product $\langle [U], [U], [U] \rangle = \{-[V]\}$ from (3) remains encoded by the product and its primitive. It is nonzero. Replacing E by its graded cohomology algebra with zero differential would replace that product by zero.

There is therefore no zigzag of framed quasi-isomorphisms between these two presentations: taking the singleton and its recovered product would give a zigzag of quasi-isomorphisms of differential graded

algebras, contradicting the computed Massey product. A higher presentation can express the same information in a ternary operation. A strict presentation expresses it in the differential and the chosen primitives. Neither description justifies discarding it.

5.4 An exchange that permits contextual compression

Retaining every run is one construction. A different one keeps one coefficient for each occupied position and allows the intervening coefficients to enter a merger. Its basic map is

$$c_B(a \otimes b) = ab \otimes 1 + 1 \otimes ab - a \otimes b.$$

This is the $R_{1,1,1}$ member of the Dăscălescu–Nichita operators.[1] We shall need its multiplication identities as well as its braid relation.

Lemma 5.4.1. For every unital differential graded algebra B, c_B is a degree-zero chain map, natural in unital algebra maps, and

$$c_B^2 = 1, \qquad \mu c_B = \mu, \qquad c_B(a \otimes 1) = 1 \otimes a, \qquad c_B(1 \otimes a) = a \otimes 1,$$
$$(c_B)_{12}(c_B)_{23}(c_B)_{12} = (c_B)_{23}(c_B)_{12}(c_B)_{23},$$
$$c_B(1 \otimes \mu) = (\mu \otimes 1)(1 \otimes c_B)(c_B \otimes 1),$$
$$c_B(\mu \otimes 1) = (1 \otimes \mu)(c_B \otimes 1)(1 \otimes c_B).$$

Proof. Multiplication and insertion of the closed unit are degree-zero chain maps. They give chain compatibility and naturality. The first line follows by substitution. Both braid words on $a \otimes b \otimes c$ reduce to

$$abc \otimes 1 \otimes 1 + 1 \otimes ab \otimes c + a \otimes bc \otimes 1 + 1 \otimes 1 \otimes abc$$
$$- ab \otimes c \otimes 1 - 1 \otimes a \otimes bc - a \otimes b \otimes c.$$

The last two identities reduce, respectively, to $abc \otimes 1 + 1 \otimes abc - a \otimes bc$ and $abc \otimes 1 + 1 \otimes abc - ab \otimes c$. Only associativity and the unit have been used. □

Adjacent c_B's therefore give an action of the symmetric group on each $B^{\otimes n}$. Multiplication is invariant under these specified crossings. This assertion does not replace c_B by the usual graded flip.

For a coarsening $\pi \leq \rho$, start with one B-factor per π-block. Stably group the blocks belonging to the same ρ-block by adjacent c_B's, then multiply each group in order. Denote the resulting map by

$$q_{\pi\rho} : B^{\otimes|\pi|} \longrightarrow B^{\otimes|\rho|}.$$

Sorting the runs of each π-block together and multiplying them similarly gives $\kappa_\pi : B^{\otimes \ell(\pi)} \to B^{\otimes|\pi|}$.

Proposition 5.4.2. These maps are independent of the adjacent-transposition expressions used for the prescribed grouping and obey

$$q_{\rho\sigma} q_{\pi\rho} = q_{\pi\sigma}, \qquad \kappa_\rho p_{\pi\rho} = q_{\pi\rho} \kappa_\pi.$$

They are compatible with reordering the blocks by the specified symmetric-group action.

[1] G. Massuyeau and F. F. Nichita, *Yang–Baxter operators arising from algebra structures and the Alexander polynomial of knots*, Theorem 2.3, p. 2; arXiv:math/0404010.

Proof. The braid, involution and distant-commutation relations identify all words for a fixed permutation. Crossings inside a group disappear after multiplication because $\mu c_B = \mu$. The two multiplication identities in Lemma 5.4.1 move a merger through a crossing.

Apply these identities to two successive grouping diagrams. Move crossings before the mergers and combine successive multiplications by associativity. The result groups the original inputs directly into their final blocks. This proves the first identity. Applying the same argument to the original run list proves the second. Starting with a reordered list instead gives the stated order compatibility. □

At the first nonconsecutive collision this gives

$$\kappa_{13|2}(a,b,c) = abc \otimes 1 + a \otimes bc - ab \otimes c, \qquad \mu\kappa_{13|2}(a,b,c) = abc. \tag{5.7}$$

For $a = c = E_{12}$, $b = E_{21}$, the result is

$$E_{12} \otimes 1 + E_{12} \otimes E_{22} - E_{11} \otimes E_{12}.$$

Its product is E_{12}. Each term has used the middle input, so this map does not factor through the blind outer multiplication $a \otimes c \mapsto ac$. The units in the formula are coefficient units at retained positions; they do not remove a position.

For two interlaced pairs,

$$q_{13|24}(a,b,c,d) = abc \otimes d + a \otimes bcd - ab \otimes cd.$$

Its product is $abcd$. Taking $a = c = E_{12}$ and $b = d = E_{21}$ gives

$$2E_{12} \otimes E_{21} - E_{11} \otimes E_{11}, \qquad \mu q_{13|24} = E_{11},$$

whereas $ac \otimes bd = 0$. For the nested pair pattern,

$$q_{14|23}(a,b,c,d) = abcd \otimes 1 + a \otimes bcd - abc \otimes d.$$

Both orders of the two mergers give these formulas, by the proved grouping identities.

The exchange contains more than its product

For $B = k[x]/(x^2)$, both c_B and the usual flip τ have the same composite with multiplication. In the basis $1 \otimes 1, 1 \otimes x, x \otimes 1, x \otimes x$, c_B fixes the first vector, exchanges the middle two and negates the last. Hence

$$\operatorname{Tr} c_B = 0, \qquad \operatorname{Tr} \tau = 2.$$

The two exchanges are not linearly conjugate.

One can measure the difference from the ordinary flip for every finite-dimensional ordinary algebra. Put $n = \dim B$, $a_n = n(n-1)/2$, $q = \dim[B,B]$, and

$$s(b) = b \otimes 1 + 1 \otimes b, \qquad R_B = \tau c_B = s\mu - \tau.$$

Here $[B,B]$ is the linear span of all commutators. Write $J_m(\lambda)$ for a Jordan block of size m with eigenvalue λ.

Proposition 5.4.3. The operator $R_B^2 - 1$ is square-zero and has rank q. The Jordan form of R_B is

$$qJ_2(1) \oplus (n + a_n - 2q)J_1(1) \oplus a_nJ_1(-1).$$

Consequently

$$\operatorname{rank}(R_B - 1) = a_n + q, \qquad \operatorname{Tr} R_B = n, \qquad R_B^2 = 1 \iff B \text{ is commutative.}$$

Proof. Since $\tau s = s$ and $\mu s = 2\,\mathrm{id}$,

$$R_B^2 - 1 = s(\mu - \mu\tau).$$

The map s is injective and $(\mu - \mu\tau)s = 0$. This proves the square-zero and rank assertions.

On $\operatorname{Sym}^2 B$, the map $s\mu/2$ is a projection onto sB. The operator R_B is 1 on this image and -1 on its kernel, which has dimension a_n. On $\bigwedge^2 B$, it is $1 + s\mu$, with $s\mu$ taking values in sB and having rank q. Thus R_B restricts to identity plus a square-zero operator of rank q on $sB \oplus \bigwedge^2 B$, a space of dimension $n + a_n$. This gives the displayed blocks and all their consequences. □

For $B = \operatorname{Mat}_d(k)$, commutators span the trace-zero matrices, so $q = d^2 - 1$. At $d = 2$, the multiplicities of $J_2(1), J_1(1), J_1(-1)$ are 3, 4, 6. These are invariants of the specified coefficient exchange. Equality after multiplication forgets them.

5.5 Descent and the kernel of compression

First take $T = 0$. On each word chart of the ordered atlas $\mathscr{R}_I$ of Section 2.7, replace the run coefficient in (5.5) by $B^{\otimes|\pi|}$. In its faces replace $p_{\pi\rho}$ by $q_{\pi\rho}$. Call the resulting chart complex $C_I^{\mathrm{ctx}}(B)$. The same three-channel identity proves that its differential squares to zero: every route to a final partition has the single coefficient map $q_{\pi\sigma}$.

On an overlap of two charts, a surviving equality block has the same internal order in both charts. Only its order relative to other, separated blocks can change. Use the action generated by c_B to change that block order, together with the corresponding density and tensor-shift identifications. These are specified transition maps.

Theorem 5.5.1. For $T = 0$, these transitions glue the chart complexes to a coherent exceptional collision system on $\mathscr{R}_I$. It is a connected coaugmented coalgebra for the atlas convolution $\star$. The maps κ_π define a natural map of cut coalgebras

$$C^{\mathrm{run}}(B) \longrightarrow \operatorname{res} C^{\mathrm{ctx}}(B), \tag{5.8}$$

where restriction uses the colax comparison of Proposition 2.7.2. The map is an isomorphism in arities one and two.

Proof. The symmetric-group relations prove the transition cocycle on every triple overlap. The grouping identities make transitions commute with all surviving faces. A target stratum absent from an overlap has zero restricted face. Hence these are chain isomorphisms on the full overlaps.

For an ordered-fibre collision, choose an expanded chart where its fibres are consecutive. The saturated partitions are precisely the pullbacks of partitions of the shorter word. Their densities, shifts and block coefficients agree. The argument in Theorem 5.3.1 therefore gives the exceptional comparison there. Compatibility with transitions makes it a comparison on the actual atlas collision map, and saturation under composite collisions proves transitivity.

On an arbitrary cross-disjoint decomposition $I = A \sqcup D$, the surviving partitions are $\pi_A \sqcup \pi_D$. Regroup their coefficient blocks using c_B. This is invertible, and localization into the resulting external product defines the coproduct. All iterates use the same final disjoint open and the same regrouping. The braid and multiplication identities give coassociativity, collision compatibility and the empty-unit identities.

The identity $\kappa_\rho p_{\pi\rho} = q_{\pi\rho}\kappa_\pi$ proves that (5.8) is a chain map and preserves collisions. On a consecutive cut, sorting the runs splits into the sorting on its two sides; this proves compatibility with the cut coproduct. At one or two positions every run list has one occurrence of each equality block, so each κ_π is the identity. □

This construction supplies a chosen exchange. It does not extend a different prescribed overlap map by leaving it unchanged.

PROPOSITION 5.5.2 *(A split compression kernel).* The coefficient map $\kappa = \kappa_{13|2}$ is a split surjection for every unital differential graded algebra. A section and the resulting projection onto its kernel are

$$s_0(x \otimes y) = x \otimes y \otimes 1, \qquad P = 1 - s_0\kappa.$$

Explicitly,

$$\begin{aligned} P(a,b,c) = a \otimes b \otimes c - abc \otimes 1 \otimes 1 - a \otimes bc \otimes 1 \\ + ab \otimes c \otimes 1. \end{aligned}$$

For an ordinary algebra of finite dimension d, $\dim \ker \kappa = d^3 - d^2$.

Proof. Substituting $c = 1$ in (5.7) gives $\kappa s_0 = \mathrm{id}$. Thus $P^2 = P$, $\kappa P = 0$, and P is identity on $\ker \kappa$. All maps commute with the differential. The dimension follows from the split surjection. □

On the open equality stratum $S = \{z_1 = z_3 \neq z_2\}$, all other partition terms vanish under exceptional restriction. Both remaining terms have shift $[2]$. Consequently

$$i_S^! \operatorname{Cone}\big(C_3^{\mathrm{run}}(B) \to (\operatorname{res} C^{\mathrm{ctx}}(B))_3\big) \simeq \omega_S \otimes \ker \kappa[3].$$

The extra shift is the shift of a kernel in the cone of a split surjection. For $d > 1$, this proves that compression is not an equivalence.

PROPOSITION 5.5.3 *(An obstruction to unchanged descent).* Let B be an ordinary finite-dimensional algebra with $T = 0$ and $\dim B = d > 1$. The run complexes on the separate word charts cannot descend to $\mathscr{R}_I$ with their chart objects unchanged.

Proof. The charts 123 and 132 overlap wherever $z_2 \neq z_3$, and both contain S. The run list of $13|2$ is ABA in the first chart and AB in the second. Their exceptional restrictions to S are

$$\omega_S[2] \otimes B^{\otimes 3}, \qquad \omega_S[2] \otimes B^{\otimes 2}.$$

They have ranks d^3 and d^2 in the same degree. An overlap equivalence would induce a quasi-isomorphism between them, which is impossible. □

For a differential graded algebra the comparison must use cohomology, not the dimensions of the underlying complexes. The same argument applies when $H^*(B)$ has finite total dimension greater than one. In contrast, the square-zero contractible extension of k considered above has run complexes quasi-isomorphic to the scalar ones. Its underlying state dimensions give no obstruction of this kind.

5.6 Translation frames and the algebraic boundary

The run construction allowed every derivation T. For contextual descent, translation also enters the overlap maps. Suppose first that T is locally nilpotent, meaning that for each $b \in B$ some power $T^N b$ is zero. Then

$$E_z = e^{zT} : O_X \otimes B \longrightarrow O_X \otimes B$$

is an algebraic change of frame, with inverse E_{-z}, and

$$\partial_z E_z = E_z(\partial_z + T).$$

This is an identity of operators on sections. It changes the translation connection to the constant connection.

Conversely, an algebraic horizontal frame $E(z)$ with $E(0) = 1$ is necessarily this one. Writing $E(z) = \sum_n z^n E_n$ on each input, horizontality gives $(n+1)E_{n+1} = E_n T$, hence $E_n = T^n/n!$. Polynomiality on every vector is exactly local nilpotence. Thus the hypothesis is precise for this method of constructing an algebraic frame.

On a partition put

$$E_{\pi,\mathrm{run}} = \bigotimes_{r=1}^{\ell(\pi)} e^{z_{\beta_\pi(r)} T}, \qquad E_{\pi,\mathrm{block}} = \bigotimes_{C \in \pi} e^{z_C T}.$$

Conjugate every contextual face and overlap map by its source and target frames, and set

$$\kappa_\pi^T = E_{\pi,\mathrm{block}}^{-1} \kappa_\pi E_{\pi,\mathrm{run}}. \tag{5.9}$$

Theorem 5.6.1. For locally nilpotent T, the contextual chart complexes with these conjugated maps descend to the ordered atlas and form a connected $\star$-coalgebra. Formula (5.9) gives a map of cut coalgebras from $C^{\mathrm{run}}(B)$ to its restriction. It retains every regular field coefficient.

Proof. All frames are algebraic and invertible on sections and commute with d. At a merged position,

$$e^{zT} \mu = \mu(e^{zT} \otimes e^{zT}).$$

Conjugating a face therefore gives exactly its normal Taylor transport. Intermediate frames cancel in every composition identity, including the transition cocycle, the grouping equations and the collision and coproduct comparisons. Thus Theorem 5.5.1 applies in the translation frames. At two positions the resulting contact is $\mathrm{pp}_u(f(e^{uT}a)b)$, so its Laurent tests recover all the original modes. □

For two positions t, 0, written in the exchanged order 0, t, the transition is

$$\begin{aligned} c_T(t)(a \otimes b) = {} & (e^{tT}a)b \otimes 1 + 1 \otimes a(e^{-tT}b) \\ & - (e^{tT}a) \otimes (e^{-tT}b). \end{aligned}$$

The reverse transition uses $-t$. Conjugating the constant relations proves

$$c_T(-t)c_T(t) = 1, \qquad c_{12}(v)c_{23}(u+v)c_{12}(u) = c_{23}(u)c_{12}(u+v)c_{23}(v).$$

These identities hold on all states; they are the cancellation of intermediate Taylor frames around the two placed diagrams.

At the partial collision with coordinates $(t, 0, t)$, formula (5.9) becomes

$$\begin{aligned}\kappa^{T}_{13|2}(a, b, c; t) = a(e^{-tT}b)c \otimes 1 + a \otimes b(e^{tT}c)\\ - a(e^{-tT}b) \otimes e^{tT}c.\end{aligned} \tag{5.10}$$

The binary evaluation $x \otimes y \mapsto (e^{tT}x)\,y$ gives $(e^{tT}a)b(e^{tT}c)$: the second and third terms cancel, and the first gives that ordered product. Local nilpotence makes the displayed coefficient algebraic.

Example 5.6.2 *(A nonconstant polynomial exchange).* Take $B = k\langle x, y\rangle$, $Tx = 0$, $Ty = x$. Translation lowers the number of y's in a word, so it is locally nilpotent. Here $Y_B(y, t)\,y = y^2 + txy$, and

$$\begin{aligned}c_T(t)(y \otimes y) = y^2 \otimes 1 + 1 \otimes y^2 - y \otimes y\\ + t(xy \otimes 1 - 1 \otimes yx - x \otimes y + y \otimes x) + t^2 x \otimes x.\end{aligned}$$

This is a polynomial transition on the original algebraic overlap, with the proved braid and reverse-transition identities. Its position dependence is given by the stated Taylor frames; no nontrivial full-turn monodromy follows from it.

For a derivation that is not locally nilpotent, the same formula can instead be only formal. For example, on $B = \mathrm{Mat}_2(k)$ take $T = \mathrm{ad}\,\mathrm{diag}(1, 0)$. Then

$$c_T(t)(E_{12} \otimes E_{21}) = e^{t}E_{11} \otimes 1 + e^{t}1 \otimes E_{11} - e^{2t}E_{12} \otimes E_{21}.$$

Its formal coefficients satisfy the same identities. They are not algebraic functions on $\mathbb{A}^2 \setminus \Delta$. The run construction still exists, because each of its normal faces uses only a finite Taylor jet.

A compression excluded by unrestricted translation

An *algebraic two-state expression* for an output at positions $t, 0$, rooted at zero, is a finite sum with coefficients in $k((t))$ of

$$E^{\mathrm{L}}_t(x, y) = (e^{tT}x)\,y, \qquad E^{\mathrm{R}}_t(x, y) = x(e^{tT}y),$$

where the pairs $x, y \in B$ are fixed states. A coefficientwise series in $(B \otimes B)[[t]]$ is a different permitted input.

THEOREM 5.6.3. Let

$$B = k\langle a_i, b_i, c_i \mid i \geq 0\rangle, \qquad Ta_i = a_{i+1}, \quad Tb_i = b_{i+1}, \quad Tc_i = c_{i+1}.$$

The ordered output

$$F(t) = (e^{tT}a_0)b_0(e^{tT}c_0)$$

has no algebraic two-state expression.

Proof. Project to the span of words $a_i b_j c_k$ and identify that span with $k[p, q, r]$ by $a_i b_j c_k \mapsto p^i q^j r^k$. The output becomes $e^{t(p+r)}$. In a binary expression, the translated state is a contiguous prefix or suffix of the output word. Its projection is therefore a finite sum of terms

$$f(t)P(p, q, r)e^{tL}, \qquad L \in \{0, p, p + q, p + q + r, r, q + r\},$$

with polynomial P and Laurent f. This includes empty factors and arbitrary descendants in the two fixed states. Other generator patterns contribute zero.

Set $D = \partial_p + 2\partial_q + 4\partial_r$. The six exponential eigenvalues are st, $s \in S = \{0, 1, 3, 4, 6, 7\}$. If all the finitely many polynomials P have degree at most d, the commuting operator

$$\prod_{s \in S} (D - st)^{d+1}$$

kills the proposed expression. Applied to $e^{t(p+r)}$, it instead gives

$$(80t^6)^{d+1} e^{t(p+r)} \neq 0,$$

since D has eigenvalue $5t$ there. These calculations hold in $k((t))[[p, q, r]]$. They contradict the proposed finite expression. □

The obstruction fixes the binary coefficient class in advance. It does not exclude every compression for a non-nilpotent translation. For a commutative coefficient algebra, multiplying the runs of each block already gives horizontal block coefficients, using the ordinary flip. The derivation and commutativity identities make these maps compatible with all coarsenings. For instance, $B = k[x]$, $Tx = x$, has no polynomial horizontal frame, yet

$$(e^{tT} a)b(e^{tT} c) = (e^{tT}(ac))b$$

is a two-state expression.

For general B, allowing the formal expression (5.10) changes the coefficient tensor and its completion. Descent then requires actual continuous collision functors and overlap maps on that coefficient category, together with a comparison preserving all the field modes, vacuum data and cut maps. The algebraic obstruction above leaves that completed comparison open. The regular run construction itself needs none of those additional choices.

The next construction introduces a specified singular local field factor. Its poles alter the normal principal parts. The ordered coefficient words, their run maps and the distinctions between the empty unit, the local vacuum and a prescribed state remain necessary.

Chapter 6

Singular backgrounds and vacuum data

The simplest singular current already separates a vacuum from a scalar augmentation. At nonzero level κ, its first product satisfies

$$J_{(1)}J = \kappa \mathbf{1}.$$

Every first product in the scalar vertex algebra is zero. A unital field map from this current algebra to the scalars is therefore impossible. Nevertheless, the current has a vacuum, a vacuum expectation and a geometric realization with scalar empty coefficient. These are different structures.

We first construct that realization from all collision strata. We then allow infinitely many current species. Completing each energy space retains the local field operations, including quadratic fields with Hilbert–Schmidt coefficients. A change of coordinate can require self-contractions that are discontinuous in this completion. This leads to two further coefficient domains, one controlling projective tensor norms and one retaining contraction values as independent central coordinates.

Gaussian preparations give actual normalized states in a specified Hilbert sector. Their existence and their transport require their own estimates. We also determine when a formal state parameter may be identified with an insertion coordinate. The resulting completed products retain convergent mode identities on lower-bounded modules.

6.1 A current with a vacuum and no scalar augmentation

Let k be a field of characteristic zero and fix $\kappa \in k^{\times}$. The state space of one current is

$$H_\kappa = k[x_1, x_2, \dots], \qquad \mathbf{1} = 1.$$

Every state is a polynomial in finitely many variables. Give x_m energy m. For $m > 0$, define operators

$$a_{-m} f = x_m f, \qquad a_m f = \kappa m \frac{\partial f}{\partial x_m}, \qquad a_0 = 0.$$

They obey

$$[a_m, a_n] = \kappa m \delta_{m+n,0}, \qquad a_m \mathbf{1} = 0 \quad (m \geq 0). \tag{6.1}$$

The translation operator is

$$T = \sum_{m \geq 1} m x_{m+1} \frac{\partial}{\partial x_m}.$$

This sum is finite on each state. It satisfies $T\mathbf{1} = 0$ and $[T, a_m] = -m a_{m-1}$.

Put

$$J(z) = \sum_{m \in \mathbb{Z}} a_m z^{-m-1}, \qquad J^{[m]}(z) = \frac{\partial_z^{m-1} J(z)}{(m-1)!} \quad (m \geq 1).$$

Normal ordering, denoted by colons, moves negative modes to the left of positive modes. Define the field of a monomial by

$$Y(x_{m_1}\cdots x_{m_r},z) =: J^{[m_1]}(z)\cdots J^{[m_r]}(z) :, \qquad Y(\mathbf{1},z)=\mathrm{id}.$$

Linearity defines Y on H_κ. In the notation $Y(a,z)b=\sum_n a_{(n)}b\,z^{-n-1}$, the current state is $J=x_1$.

PROPOSITION 6.1.1. These formulas make H_κ a local vertex algebra with vacuum $\mathbf{1}$ and translation T. For states of energies E, F, the locality exponent $E+F$ suffices. The elementary contraction is

$$J^{[m]}(z)J^{[n]}(w)\sim\frac{\kappa(-1)^{m-1}(m+n-1)!}{(m-1)!(n-1)!(z-w)^{m+n}}. \tag{6.2}$$

Here $\sim$ records only the negative powers of $z-w$.

Proof. Split J into its negative- and positive-mode parts. Equation (6.1) gives

$$[J_+(z),J_-(w)]=\kappa\sum_{m\ge1}mz^{-m-1}w^{m-1}=\iota_{z,w}\frac{\kappa}{(z-w)^2},$$

where $\iota_{z,w}$ means expansion in powers of w/z. Commuting the positive modes in one normally ordered monomial past the negative modes in another gives a sum over cross pairings. A pairing uses a factor at most once. The unpaired factors remain normally ordered. Differentiating the displayed contraction gives (6.2).

This expansion is the Wick formula. A pair of energies m, n contributes a pole of order $m+n$. Thus states of energies E, F have total pole order at most $E+F$. The two orders of the fields expand the same rational contractions in the opposite chambers. Multiplication by $(z-w)^{E+F}$ removes their difference.

On the vacuum, the negative part of J is $\sum_{r\ge0}x_{r+1}z^r=e^{zT}x_1$. Differentiation and the product rule give $Y(a,z)\mathbf{1}=e^{zT}a$ for every monomial. The commutator relation with T gives $[T,Y(a,z)]=\partial_zY(a,z)=Y(Ta,z)$.

For three monomials, every nested Wick expansion enumerates the same pairings, with the same uncontracted factors. After clearing the finite pairwise pole bound, their coefficient identities agree. The chamber expansions of this identity are the vertex Jacobi identity. All its coefficients on a fixed state use finitely many annihilators. Thus lower truncation, the vacuum, translation and locality identities hold on the stated polynomial space.

These are also the explicit formulas of the Heisenberg construction.[1] □

For example,

$$Y(J,u)J=\kappa u^{-2}\mathbf{1}+(e^{uT}J)J, \qquad J_{(1)}J=\kappa\mathbf{1}, \qquad J_{(0)}J=0.$$

The product on the right in the regular term is polynomial multiplication of creation states. It should not be substituted for every field product.

PROPOSITION 6.1.2. There is no unital vertex-algebra map $H_\kappa\to k$, where k has its constant fields and zero translation. The polynomial constant-term functional $\epsilon_0:H_\kappa\to k$ still defines a normalized vacuum expectation.

[1] E. Frenkel, *Vertex Algebras and Algebraic Curves*, §§2.1–2.4, arXiv:math/0007054.

Proof. A field map would preserve the first product and the vacuum. It would send $J_{(1)}J$ to κ, but the first product of any two scalar states is zero. This contradicts $\kappa \neq 0$. The functional ϵ_0 is defined on every polynomial and has $\epsilon_0(\mathbf{1}) = 1$. Applying it to products of fields gives the Wick vacuum coefficients. No preservation of every field product is asserted for this functional. □

The construction extends to a finite-dimensional space of species with a specified symmetric pairing. For orthogonal level-one currents, contractions of different species vanish. Their vacuum tensor product is the tensor product of their vertex algebras. The creation-state energy has Hilbert series

$$\prod_{m\geq 1}(1-q^m)^{-r}$$

for r species. The coefficient of each power of q is finite. An arithmetic energy weighting or a chosen charge is additional data on this space.

A finite-place coefficient field and its diagonal

The same construction over $\mathbb{Q}_p$ has the nonzero formal pole $\kappa(z-w)^{-2}$. Its insertion variable is a coordinate on an auxiliary curve over $\mathbb{Q}_p$. It is independent of the p-adic topology on its coefficient field. An analytic interpretation requires specified norms and convergence domains.

Two completions of a tensor product illustrate why this distinction matters. Set $R = \mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{Z}_p$, with multiplication map $\mu : R \to \mathbb{Z}_p$. Define its p-adic completion by $\widehat{R}_p = \varprojlim_r R/p^r R$.

Proposition 6.1.3. Multiplication induces $\widehat{R}_p \simeq \mathbb{Z}_p$. The ordinary multiplication map $R \to \mathbb{Z}_p$ is not injective. After inverting p, its diagonal ideal has nonzero conormal module.

Proof. For every $r \geq 1$, tensoring with $\mathbb{Z}/p^r$ gives

$$R/p^r R \cong (\mathbb{Z}_p/p^r) \otimes_{\mathbb{Z}} (\mathbb{Z}_p/p^r) \cong \mathbb{Z}/p^r.$$

These multiplication isomorphisms commute with reduction, proving the completion formula. No interchange of an ordinary tensor with an inverse limit is used.

Choose $a \in \mathbb{Z}_p \setminus \mathbb{Q}$. In $\mathbb{Q}_p \otimes_{\mathbb{Q}} \mathbb{Q}_p$, the element $a \otimes 1 - 1 \otimes a$ is nonzero. Indeed a $\mathbb{Q}$-linear functional $\ell : \mathbb{Q}_p \to \mathbb{Q}$ with $\ell(1) = 0, \ell(a) = 1$ detects it. Its antecedent in R is therefore nonzero, while multiplication kills it. The reduction formula also shows that it vanishes in every $R/p^r R$.

For the conormal assertion, choose a transcendental over $\mathbb{Q}$. Such elements exist because $\mathbb{Z}_p$ is uncountable and algebraic numbers are countable. Include a in a transcendence basis. The derivation sending a to 1 and all other basis elements to zero extends across the separable algebraic extension to $\mathbb{Q}_p$. Hence $\Omega_{\mathbb{Q}_p/\mathbb{Q}} \neq 0$.

If I is the kernel of multiplication on $\mathbb{Q}_p \otimes_{\mathbb{Q}} \mathbb{Q}_p$, the defining universal derivation gives

$$I/I^2 \cong \Omega_{\mathbb{Q}_p/\mathbb{Q}}.$$

It sends the class of $1 \otimes a - a \otimes 1$ to da. This proves nontriviality of the first infinitesimal neighborhood of that diagonal. □

Thus completion in p and completion in the diagonal ideal concern different directions. On the auxiliary line $\mathbb{A}^1_{\mathbb{Q}_p}$, the diagonal instead has normal parameter $u = z - w$. Its formal neighborhood has coefficient ring $\mathbb{Q}_p[z][[u]]$, and its principal parts retain every $u^{-r-1}du$. The formal current pole uses this parameter. The p-adic completion formula alone cannot eliminate these operations.

PROPOSITION 6.1.4. Any family of characteristic-zero fields (k_v) has an abstract common field extension over $\mathbb{Q}$. The formal Heisenberg algebra over $\mathbb{Q}$ also base changes to each k_v.

Proof. Form the filtered union

$$R_\infty = \varinjlim_{S \text{ finite}} \bigotimes_{v \in S, \mathbb{Q}} k_v,$$

where the maps insert units. Each finite tensor is a nonzero $\mathbb{Q}$-vector space. The transition maps are injective: a linear functional on the new field taking 1 to 1 gives a retraction. Hence R_∞ is a nonzero commutative ring.

Choose a maximal ideal. The quotient field receives a unital map from every k_v. Each such map is injective, since its source is a field. The explicit mode formulas and their identities commute with scalar extension, proving the final assertion. □

This field construction makes no topological or canonical choice. For example, an abstract embedding $\mathbb{Q}_p \to \mathbb{C}$ cannot be continuous: $p^r \to 0$ in $\mathbb{Q}_p$, while its images are the unbounded real numbers p^r. Likewise the adelic projection $\mathbb{A}_\mathbb{Q} \to \mathbb{Q}_p$ has the opposite direction from scalar extension of a $\mathbb{Q}_p$-algebra to the adeles. A placewise module construction must specify its actual rings, maps and restricted topology. Hecke actions and automorphic comparison maps are further structures on those coefficients.

6.2 The residue coalgebra of the singular background

Let $X = \mathbb{A}^1_k$. We use the word carrier $\mathcal{W}_X$, its cut tensor, and the partition embeddings i_π, j_π of Chapter 5. Order the blocks of π by first occurrence.

For $r = |\pi|$, put

$$M_\pi(H_\kappa) = \omega_{X^\pi} \otimes_k H_\kappa^{\otimes r}.$$

Its right derivative in the coordinate of block C is minus the scalar derivative minus T on its H_κ-factor. All external coefficient tensors in this construction are algebraic over k.

Define

$$(C_\kappa)_I = \bigoplus_{\pi \in \Pi(I)} i_{\pi,*} j_{\pi,*} j_\pi^* M_\pi(H_\kappa)[|\pi|], \qquad (C_\kappa)_\varnothing = k. \tag{6.3}$$

The state complex has zero differential. Its r-block term lies in cohomological degree $-r$.

For a merger of blocks $i < j$, take $u = z_{C_i} - z_{C_j}$, retain z_{C_j}, and place the merged block in position i. Replace its two states by $Y(v_i, u)v_j$, leaving the other block states in their original relative order. The face is

$$D_{\pi\rho}(f \otimes \mathbf{v}) = (-1)^{j-2} \operatorname{pp}_u \big(f \otimes v_1 \otimes \cdots \otimes Y(v_i, u)v_j \otimes \cdots \widehat{v_j} \cdots \big). \tag{6.4}$$

The principal part uses the full right normal module $e_m = u^{-m-1}du = \delta \partial_u^m / m!$. Other coordinate differences are expanded normally before taking the principal part. Normal transfer extends the map to the diagonal direct image. For each input, lower truncation and the finite pole order of f make its output a finite normal sum.

THEOREM 6.2.1. The sum of the faces (6.4) is a differential. The objects (6.3) have coherent exceptional comparisons for monotone collisions and form a connected coaugmented cut coalgebra C_κ.

Proof. Translation covariance of the field identifies the input and output tangential connections. In the normal direction the identity $e_m \partial_u = (m+1)e_{m+1}$, together with $Y(Ta, u)b = \partial_u Y(a, u)b$, gives differential linearity of every face.

Two consecutive faces either merge disjoint pairs or merge three blocks. In the first case the coefficient operations commute, and the two normal orientations have opposite signs. In the second case expand all three inputs by the Wick formula before taking any principal parts. There is one finite collection of contraction patterns. Each pattern is a simultaneous coefficient with a finite product of pairwise denominators and a formal regular numerator.

Locality permits its expansion in each of the three collision orders. The oriented contacts therefore obey $L - R + N = 0$, by Lemma 5.2.2, with the uncontracted states as coefficients. The normal-coordinate transfer in that lemma is part of the comparison. These two cases exhaust the square of the differential.

Let $\alpha : I \twoheadrightarrow J$ be monotone. Exceptional pullback kills a partition term unless each α-fiber lies in one of its blocks. Every other term meets a diagonal removed by j_π. Every surviving partition is $\pi = \alpha^{-1}(\bar{\pi})$. Its block-state tensor, density, shift and residue faces agree with the $\bar{\pi}$-term. Closed direct image and open base change give the exceptional equivalence.

Saturation for two collisions is saturation for their composite, so the same identifications give all the required coherences.

On the cross-disjoint open of a consecutive cut $I = A|D$, a surviving partition is uniquely $\pi_A \sqcup \pi_D$. Its states and densities split into the two consecutive blocks. Localization and this equivalence define the cut coproduct.

For any further cuts all composites use the same disjoint open, partition decomposition and ordered tensor factors. Thus the coproduct is coassociative and commutes with collisions. Empty cuts use the specified coefficient k. There are at most $|I|$ positive output blocks, which proves conilpotence in each arity. □

At three positions put $H_X = \omega_X \otimes_k H_\kappa$, and let each L denote the localized two-block coefficient with two H_κ-factors. Writing j for the three-point configuration open, the terms are

$$\begin{aligned}(C_\kappa)^{-3} &= j_* j^*(H_X^{\boxtimes 3}),\\ (C_\kappa)^{-2} &= i_{12,*} L_{12|3} \oplus i_{23,*} L_{1|23} \oplus i_{13,*} L_{13|2},\\ (C_\kappa)^{-1} &= i_{123,*} H_X.\end{aligned}$$

The third summand is present even though the cut tensor uses consecutive cuts. For independent currents J, P, the outer contraction in $J(z_1)P(z_2)J(z_3)$ is

$$\frac{\kappa P(z_2)}{(z_1 - z_3)^2}.$$

It survives on 13|2. Both adjacent contractions vanish. The scalar vacuum expectation of this term is zero. Thus that expectation does not retain the supported operation.

Insert $\mathbf{1}$ in every block of the scalar residue coalgebra $C^{\mathrm{run}}(k)$ from the preceding chapter. Since $Y(\mathbf{1}, u)\mathbf{1} = \mathbf{1}$, these maps commute with every face, collision and cut. They give a coalgebra map

$$C^{\mathrm{run}}(k) \longrightarrow C_\kappa. \tag{6.5}$$

Its singleton is $\omega_X[1] \to \omega_X \otimes H_\kappa[1]$, induced by the local vacuum. The source is not the isolated empty unit K.

Definition 6.2.2. The *Heisenberg geometric background* on the word carrier is the augmented associative algebra

$$A_\kappa = \Omega_{\mathrm{cut}} C_\kappa.$$

The vacuum background is $A_{\mathrm{vac}} = \Omega_{\mathrm{cut}} C^{\mathrm{run}}(k)$.

COROLLARY 6.2.3. There is an associative map $A_{\mathrm{vac}} \to A_\kappa$, and

$$B_{\mathrm{cut}} A_\kappa \simeq C_\kappa, \qquad (A_\kappa)_\varnothing = k.$$

The framed two-point boundary of this bar recovers every mode of H_κ.

Proof. Apply cut cobar to (6.5). The connected coalgebra satisfies the positive-cut hypotheses of Theorem 3.1.2. That theorem gives the bar comparison. Every positive cobar word vanishes at the empty input, so only its unit remains there.

In the framed two-point boundary,

$$r(f; a, b) = \mathrm{pp}_u(fY(a, u)b), \qquad [e_0] r(u^n; a, b) = a_{(n)} b.$$

All integer Laurent tests are allowed. This proves the mode recovery, including the central first product. □

The augmentation $A_\kappa \to K$ is a geometric augmentation of a cut cobar. It is not a unital vertex-algebra map $H_\kappa \to k$, which Proposition 6.1.2 excludes. Likewise cobar multiplication is concatenation. The field products are recovered from the framed contacts. The three-channel boundary identity does not identify those contacts with strictly associative composition of two consecutive residues.

For a unital differential coefficient algebra B, one may instead retain both the local field factor and the runs of B. The partition coefficient is

$$\omega_{X^\pi} \otimes H_\kappa^{\otimes|\pi|} \otimes B^{\otimes \ell(\pi)}.$$

Its merger combines Y_{H_κ} with the ordered run map and Taylor factor of (5.6). The same three-channel proof applies: every route has the same ordered B-coefficient after its formal translations. This gives $C^{\mathrm{run}}(H_\kappa, B)$ and its cut cobar, with empty coefficient k.

The relative envelope of Chapter 8 is $A_\kappa \otimes_k B$, with empty coefficient B. These are distinct constructions. An equivalence between them would already imply $B \simeq k$ after evaluation at the empty word. For an ordinary algebra of dimension greater than one this is impossible. A proposed comparison must therefore specify its relative base and its effect on the field contacts. The notation of scalar extension alone supplies neither.

6.3 THE DERIVED COEFFICIENT OF THE LOCAL VACUUM

There is a second geometric vacuum construction. The dualizing coalgebra J of Proposition 2.2.1 has $J_n = \omega_{X^n}[n]$ in positive arity. Its cut coproducts are localization maps. Write

$$Q_n = (\Omega_{\mathrm{cut}} J)_n, \qquad n > 0.$$

This definition uses J itself. Identifying it with the scalar residue background would require a comparison with $C^{\mathrm{run}}(k)$ preserving the coalgebra and collision maps.

We compute Q_n, including its support extensions. Let X now be any smooth separated geometrically connected curve over k, and put $Y_n = X^n$. Use the closed support Z_n of (2.3). Let U_t be the open where every position to the left of cut t differs from every position to its right. For $S \subseteq \{1, \dots, n-1\}$, put $U_S = \bigcap_{t \in S} U_t$, with inclusion j_S, and $U_\varnothing = Y_n$.

Define the localization cube

$$\mathsf{C}_n^r = \bigoplus_{|S|=r} j_{S,*}\omega_{U_S}, \qquad d_{S,S\cup\{t\}} = (-1)^{|\{s\in S : s<t\}|}\mathrm{loc}.$$

All derivatives normal to a diagonal are retained. Each localization permits arbitrary finite pole order. No completion in arity or pole order is used.

Proposition 6.3.1. For $n \geq 2$, there are natural comparisons

$$\mathsf{C}_n \simeq R\Gamma_{Z_n}\omega_{Y_n}, \qquad Q_n \simeq \mathsf{C}_n[n-1].$$

They retain the cube and its augmentation. With ordered density wedges and cobar desuspensions, the second comparison is the vertexwise sign $(-1)^{n|S|}$.

Proof. The opens U_t cover $Y_n \setminus Z_n$. The cube including its empty vertex computes the fiber of restriction to this union. Every intersection is locally a principal divisor complement, so its localization is exact. This proves the first comparison.

A set S gives $|S|+1$ consecutive cobar blocks. Their density shifts sum to $[n]$, while desuspending every block contributes $[-|S|-1]$. Thus its coefficient lies in degree $|S|+1-n$, as in $\mathsf{C}_n[n-1]$. Fix the suspension identification

$$(tc_1)\otimes\cdots\otimes(tc_b) \longmapsto (-1)^{\sum_{a=1}^{b}(b-a)|c_a|}(c_1\otimes\cdots\otimes c_b)[-b], \qquad t=[-1].$$

Splitting a block at the new cut t has sign $(-1)^{1+a}$, where $a = |\{s \in S : s < t\}|$. The shifted cube has sign $(-1)^{n-1+a}$. The identity

$$(-1)^{n(|S|+1)}(-1)^{1+a} = (-1)^{n-1+a}(-1)^{n|S|}$$

proves the claimed chain isomorphism. Both maps use the same localization at every face, including the empty vertex. □

An order with divisor intersections

For partitions p, q of $[n]$, write $p \leq q$ when every p-block lies in a q-block. Their join is $p \vee q$, and $c(p) = n - |p|$. Then $\Delta_p \cap \Delta_q = \Delta_{p\vee q}$. Call a partition indecomposable when its blocks cross every consecutive cut.

By Theorem 2.3.1, the minimal indecomposable partitions are the crossing chains

$$1 = i_1 < i_2 < j_1 < i_3 < j_2 < \cdots < i_c < j_{c-1} < j_c = n,$$

whose nonsingleton blocks are (i_a, j_a). The one-pair chain is $(1, n)$. Let $\mathcal{A}_n$ be their set, ordered lexicographically by

$$(-j_1, i_1, -j_2, i_2, \dots, -j_c, i_c). \tag{6.6}$$

Write $N \prec M$ for this order. No such list properly extends another: the last right endpoint is already n.

For an indecomposable p, replace each nonsingleton block by its hull $(\min B, \max B)$. Start with the hull containing 1. At a current right endpoint $r < n$, choose the eligible hull with largest right endpoint, where eligibility means

$$\min B < r < \max B.$$

Continue to n, and leave unused positions singleton. Denote this refinement by $g(p)$.

LEMMA 6.3.2. The partition $g(p)$ is the least crossing chain refining p. For $N \prec M$, let e be the first pair of N that differs from M, and put $p = M \vee e$. Then

$$g(p) \prec M, \qquad M \vee g(p) = p \leq M \vee N, \qquad c(p) = c(M) + 1.$$

Proof. The cut at r is crossed by a block with $\min B \leq r < \max B$. Equality on the left is impossible: r already belongs to the previously chosen block, whose maximum is r. Thus an eligible hull exists. Distinct blocks have distinct maxima. The right endpoint strictly increases, so the process terminates at n.

Write its selected pairs as (i_a, j_a). For $a \geq 3$, one has $i_a > j_{a-2}$. Equality would share an endpoint. A smaller value would make this hull eligible one step earlier, with a larger right endpoint than the greedy choice. Together with $i_a < j_{a-1} < j_a$, these inequalities give the crossing-chain order.

Compare the output with any chain refining p. At the first differing pair, the greedy right endpoint is at least the competing one. For the first pair this follows from the hull of the block containing 1. For a later pair, the competing pair lies in an eligible hull. If the right endpoints agree, they belong to the same block, and the greedy left endpoint is its minimum. This proves lexicographic minimality.

Now let the first differing pair be $e = (i, j)$, and let the corresponding pair of M be (i_a, j_a). Since $N \prec M$, either $j > j_a$, or $j = j_a$ and $i < i_a$. Suppose $g(M \vee e) = M$. In the first position, the new block containing 1 has maximum larger than j_1, contradicting this supposition.

At a later position put $r = j_{a-1}$. The block containing e has $\min B \leq i < r < j \leq \max B$. It is eligible.

A larger right endpoint contradicts greedy maximality. If its maximum equals j_a, uniqueness of that maximum forces this block to be chosen, with left endpoint at most $i < i_a$. This is again a contradiction. Hence $g(p) \prec M$.

Two distinct minimal indecomposable partitions are incomparable. Therefore

$$M < M \vee g(p) \leq p.$$

Adding one pair raises partition rank by at most one. The strict inequality forces $M \vee g(p) = p$ and $c(p) = c(M) + 1$. Since e is a pair of N, one also has $p \leq M \vee N$. □

For $M \in \mathcal{A}_n$, define

$$\mathcal{D}_M = \{\sigma : \sigma \text{ covers } M \text{ and } g(\sigma) \prec M\}.$$

A cover merges exactly two M-blocks. The exchange lemma proves the attachment equation

$$D_M := \Delta_M \cap \bigcup_{N \prec M} \Delta_N = \bigcup_{\sigma \in \mathcal{D}_M} \Delta_\sigma. \tag{6.7}$$

Indeed every earlier intersection lies in a cover supplied by the lemma. Conversely Δ_σ lies in both Δ_M and the earlier $\Delta_{g(\sigma)}$.

Each equality of two distinct block coordinates is a Cartier divisor on the smooth product Δ_M. Consequently D_M is empty or is the support of an effective Cartier divisor. On the affine line its equation is

$$h_M = \prod_{\substack{B<B' \\ g(M/(B\sim B'))\prec M}} (y_B - y_{B'}), \qquad D_M = V(h_M).$$

Blocks are ordered by their least label. The empty product is 1.

The support maps and exact width

Suppose $i : M \hookrightarrow Y$ is a closed embedding of smooth varieties of codimension c, and $A \subset Y$ is closed with $M \not\subset A$. If $D = A \cap M$ is a divisor support or is empty, there is a natural triangle

$$R\Gamma_A \omega_Y \longrightarrow R\Gamma_{A\cup M} \omega_Y \longrightarrow i_+ \omega_M(*D)[-c] \longrightarrow R\Gamma_A \omega_Y[1]. \tag{6.8}$$

Here i_+ is closed direct image with all finite normal derivatives.

To construct it, use the support square

$$\begin{array}{ccc} R\Gamma_D \omega_Y & \longrightarrow & R\Gamma_M \omega_Y \\ \downarrow & & \downarrow \\ R\Gamma_A \omega_Y & \longrightarrow & R\Gamma_{A\cup M} \omega_Y. \end{array}$$

Mayer–Vietoris makes this a homotopy pushout. Smooth purity identifies the upper row with

$$i_+ R\Gamma_D^M \omega_M[-c] \longrightarrow i_+ \omega_M[-c].$$

Its cofiber is $i_+ \omega_M(*D)[-c]$. Equality of the horizontal cofibers gives (6.8), including its connecting map.

The normalization can be checked directly. For ordered normal parameters $u_1, \dots, u_c$, the two-term localization complexes have cohomology only in degree c, with

$$\left[\frac{du_1 \wedge \cdots \wedge du_c \wedge \omega_M}{u_1^{a_1+1} \cdots u_c^{a_c+1}}\right] \longmapsto \frac{\omega_M \otimes \partial_{u_1}^{a_1} \cdots \partial_{u_c}^{a_c}}{a_1! \cdots a_c!}.$$

Right differentiation of densities gives the positive factorials. Normal-coordinate changes transform the density and residue determinant together, so these formulas glue.

The width of a nonzero bounded complex is the difference between its extreme nonzero cohomological degrees.

Theorem 6.3.3. For every $n \geq 2$,

$$\begin{aligned} H^j(Q_n) &= 0 \quad \text{unless } 2 - n \leq j \leq \lfloor n/2 \rfloor + 1 - n, \\ H^{c+1-n}(Q_n) &\neq 0 \quad (1 \leq c \leq \lfloor n/2 \rfloor). \end{aligned}$$

Thus $\operatorname{width}(Q_n) = \lfloor n/2 \rfloor - 1$. The successive support maps in the order (6.6) have cofibers

$$i_{M,+} \omega_{\Delta_M}(*D_M)[-c(M)]$$

before the common shift $[n-1]$.

Proof. Attach the components in their specified order. Each has codimension at most $m = \lfloor n/2 \rfloor$. Equation (6.7) and triangle (6.8) construct the asserted tower. Its factors have cohomology only in degrees between 1 and m. The long exact sequences propagate this bound. Proposition 6.3.1 then gives the vanishing range after the shift.

For each $1 \le c \le m$, the component count in Theorem 2.3.1 supplies a chain M of length c. Choose a geometric point with equality partition exactly M. No other minimal indecomposable partition occurs there. Locally Z_n is therefore this smooth codimension-c diagonal. Its class

$$\left[\frac{\omega_{Y_n}}{u_1 \cdots u_c}\right]$$

is nonzero in the normal Čech quotient.

Faithful extension to a geometric field detects nonzero sheaves over k. Thus every asserted degree occurs. □

The upper sheaf bound also holds for any quasicoherent coefficient over any field. At a codimension-c component, its support has c local equations. Its intersection with the preceding union has those equations and one lifted divisor equation. Their Čech complexes vanish above c and $c + 1$, respectively. The support Mayer–Vietoris sequence therefore propagates vanishing above $\lfloor n/2 \rfloor$. This argument uses no characteristic-zero purity.

Shifts preserve width. Consequently no arity-dependent shifts can put all Q_n into a fixed finite range of cohomological degrees. For $n \ge 4$, even a shifted heart object cannot represent Q_n.

The four-position extension

Take $X = \mathbb{A}^1_k$ and coordinates

$$a = z_1 - z_4, \quad b = z_1 - z_3, \quad c = z_2 - z_4, \quad s = z_4.$$

Then $Z_4 = V(a) \cup V(b, c)$. The ordered attachments first add 14, then 13|24 along its divisor $a = 0$. Put $E_x = k[x, x^{-1}]/k[x]$.

Suppressing the common density and $k[s]$-factor, triangle (6.8) gives

$$H^1_{Z_4} = E_a \otimes k[b, c], \qquad H^2_{Z_4} = k[a, a^{-1}] \otimes E_b \otimes E_c.$$

All other local cohomology vanishes. In particular there is an exact sequence

$$0 \longrightarrow k[a] \otimes E_b \otimes E_c \longrightarrow H^2_{Z_4} \longrightarrow E_a \otimes E_b \otimes E_c \longrightarrow 0.$$

This sequence does not split even as a sequence of ordinary modules. The quotient has a nonzero element killed by a, whereas multiplication by a is invertible on the middle module. A module section cannot preserve that relation. After the shift $[3]$, the cohomology of Q_4 lies in degrees $-2, -1$.

The tower retains its support and connecting maps. Its ordered factors cannot replace the coefficient by a direct sum. The ordering of components is auxiliary. Comparisons of these particular towers under cuts and monotone collisions must additionally preserve their extension maps and recover the operations of the unfiltered coefficient.

6.4 Infinitely many species at fixed energy

For the analytic constructions set $k = \mathbb{C}$ and normalize the level to one. Let H be a real separable Hilbert space. Its complexification $H_{\mathbb{C}}$ has both the Hermitian norm and the complex bilinear extension $(\ ,\ )$ of the real inner product. The bilinear pairing supplies contractions. The Hermitian norm supplies estimates. For $m \geq 1$, let H_m be its copy of energy m, with squared norm $m\|v\|^2$.

Define the Hilbert energy sector and the algebraic energy sum by

$$\mathcal{V}_E = \bigoplus_{\sum m r_m = E} \widehat{\bigotimes}_{m\geq 1} \widehat{\mathrm{Sym}}^{r_m} H_m, \qquad \mathcal{V} = \bigoplus_{E\geq 0} \mathcal{V}_E. \tag{6.9}$$

The symmetric tensors have the Fock normalization: a product of r identical unit creators has squared norm $r!$. Each displayed sum at fixed E has finitely many occupancy patterns. Only the sum over all energies remains algebraic. Denote its Hilbert completion by $\widehat{\mathcal{V}}$.

The creation and annihilation modes satisfy

$$[a_m(v), a_n(w)] = m\delta_{m+n,0}(v,w), \qquad a_0(v) = 0, \qquad a_m(v)\mathbf{1} = 0 \quad (m > 0).$$

They are complex linear in the species vector. For real vectors the paired creation and annihilation operators are adjoint. On finite species tensors set

$$J_v(z) = \sum_m a_m(v) z^{-m-1}, \qquad Y\left(\prod_{j=1}^{p} a_{-m_j}(v_j)\mathbf{1}, z\right) =: \prod_{j=1}^{p} \frac{\partial_z^{m_j-1} J_{v_j}(z)}{(m_j-1)!} : .$$

We next extend each coefficient of this formula.

Lemma 6.4.1. Contract r specified legs of a Hilbert tensor $a \in H_{\mathbb{C}}^{\widehat{\otimes} p}$ against r specified legs of $b \in H_{\mathbb{C}}^{\widehat{\otimes} q}$. The resulting tensor has norm at most $\|a\|\|b\|$. Consequently each mode extends to a bounded bilinear map

$$\mathcal{V}_E \times \mathcal{V}_F \longrightarrow \mathcal{V}_{E+F-n-1}, \qquad (a,b) \longmapsto a_{(n)}b. \tag{6.10}$$

A negative-energy sector is understood to be zero.

Proof. In a real orthonormal basis group the contracted indices together. A coefficient of the contraction has the form $\sum_\gamma a_{\alpha\gamma} b_{\gamma\beta}$. Cauchy–Schwarz gives

$$\sum_{\alpha,\beta} \left| \sum_\gamma a_{\alpha\gamma} b_{\gamma\beta} \right|^2 \leq \left(\sum_{\alpha,\gamma} |a_{\alpha\gamma}|^2 \right) \left(\sum_{\gamma,\beta} |b_{\gamma\beta}|^2 \right).$$

The same inequality applies to the complex bilinear contraction. Symmetrization is an orthogonal projection.

At fixed E, F, n, the Wick formula has finitely many occupancy, pairing and Taylor patterns. Their output energy is $E + F - n - 1$, so no unbounded choice of remaining creation modes occurs. The mode weights and Fock factorials contribute only a finite constant for these fixed indices. Summing the bounds proves the stated extension. □

To linearize a bounded bilinear coefficient, we use the completed *projective* tensor product of its two input sectors. No bounded linear map on their Hilbert tensor product is inferred from bilinear boundedness.

THEOREM 6.4.2. The extended coefficient maps make $\mathcal{V}$ a local vertex algebra. Its vacuum is $\mathbf{1}$, and its translation satisfies

$$T\mathbf{1} = 0, \qquad [T, a_{-m}(v)] = m a_{-m-1}(v).$$

For energies E, F, the locality exponent $E + F$ is independent of a species cutoff.

Proof. Output energy proves lower truncation in (6.10). Translation changes only finitely many occupancy patterns on $\mathcal{V}_E$, and is bounded into $\mathcal{V}_{E+1}$. Its identities and the vacuum identities therefore extend from finite species.

In a Wick contraction the pole order of a pair of derivative currents is the sum of their energies. Thus $E + F$ bounds the total pole order before taking any species limit. Clear these poles and apply locality to a third fixed-energy state. Each coefficient is a finite sum of bounded compositions of mode maps. It vanishes on finite species and hence on their dense completion. The mode Jacobi identities extend in the same way: the energy bound makes each of their fixed-index sums finite, uniformly in the approximants. □

There is a precise restriction in this argument. Contraction between two tensors is bounded. Contraction of two legs of one arbitrary Hilbert tensor need not be. For an orthogonal projection P_N of rank N, its tensor norm is $\sqrt{N}$, while its trace is N. Vacuum normal ordering has removed such internal contractions from the fixed-coordinate definition. They will return under general coordinate changes.

Quadratic fields and all their poles

For a real symmetric Hilbert–Schmidt operator A on H, define

$$q_A = \sum_{i,j} (e_i, Ae_j)\, a_{-1}(e_i) a_{-1}(e_j)\mathbf{1}, \qquad W_A(z) = Y(q_A, z).$$

The Hilbert–Schmidt norm is $\|A\|_2^2 = \sum_{i,j} |(e_i, Ae_j)|^2$.

PROPOSITION 6.4.3. The state $q_A \in \mathcal{V}_2$ is independent of the real orthonormal basis and satisfies

$$\|q_A\|^2 = 2\|A\|_2^2.$$

Conversely, for bounded symmetric A, convergence of $q_{P_N A P_N}$ for finite-rank orthogonal $P_N \to 1$ strongly implies that A is Hilbert–Schmidt.

Proof. The two pairings of two creation factors give the norm formula in finite rank. The intrinsic identification of symmetric Hilbert–Schmidt operators with symmetric Hilbert tensors then extends it and proves basis independence. For the converse the norm formula makes the compressions Cauchy in Hilbert–Schmidt norm. Their limit agrees with A in every matrix element, by strong convergence of P_N. □

For symmetric A, B, set

$$S_{AB} = \tfrac{1}{2}(AB + BA), \qquad D_{AB}(w) = \sum_{i,j} (e_i, ABe_j) : \partial J_{e_i}(w) J_{e_j}(w) : .$$

The product AB is trace class, hence Hilbert–Schmidt, so the latter is the field of an energy-three state.

Theorem 6.4.4. The full singular quadratic product is

$$W_A(z)W_B(w) \sim \frac{2\operatorname{Tr}(AB)}{(z-w)^4} + \frac{4W_{S_{AB}}(w)}{(z-w)^2} + \frac{4D_{AB}(w)}{z-w}. \tag{6.11}$$

Its coefficients converge in their Hilbert energy sectors under Hilbert–Schmidt approximation.

Proof. The two double contractions give the fourth-order pole. The four single contractions give

$$\frac{4}{(z-w)^2}\sum_{i,j}(e_i, ABe_j) : J_{e_i}(z)J_{e_j}(w) : .$$

Its constant Taylor term is $4W_{S_{AB}}$, since the equal-position creation factors are symmetric. Its linear term is $4D_{AB}$. All remaining Taylor terms are regular. The inequality $\|AB\|_1 \le \|A\|_2\|B\|_2$ controls both the trace and the coefficient tensors, and passes the finite-species calculation to the limit. □

The simple pole retains the order of the two operators. Indeed

$$D_{AB} + D_{BA} = \partial W_{S_{AB}}, \qquad 4D_{AB} = 2\partial W_{S_{AB}} + 2(D_{AB} - D_{BA}).$$

For $A = \operatorname{diag}(1, 0)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, the last term is

$$2(: \partial J_1 J_2 : - : \partial J_2 J_1 :).$$

Its two state components have modes $(-2, -1)$, are orthogonal and each have squared norm two. The total squared norm is sixteen. The Jordan product alone would miss this nonzero field.

For orthogonal rank-one projections P_1, P_2, the outer contraction in $W_{P_1}(z_1)W_{P_2}(z_2)W_{P_1}(z_3)$ is

$$\frac{2W_{P_2}(z_2)}{(z_1 - z_3)^4}.$$

All contractions meeting the middle species vanish. The normal test $(z_1 - z_3)^3$ therefore gives the nonzero residue $2W_{P_2}$, with squared state norm eight. Its scalar vacuum expectation is zero. This is the quadratic version of the supported operation lost by scalar evaluation.

On the specified affine coordinate, the right module $\mathcal{M} = \omega_X \otimes \mathcal{V}$ has derivative $-\partial_z - T$. Its local chiral operation is

$$j_* j^*(\mathcal{M} \boxtimes \mathcal{M}) \longrightarrow \Delta_! \mathcal{M}, \qquad f(u, z)(a \boxtimes b) \longmapsto \operatorname{pp}_u(f(u, z)Y(a, u)b).$$

Here the external tensor is algebraic when $\mathcal{V}$ is regarded as a vector space. Fixed-energy continuous coefficients linearize on projective external tensors.

Lower truncation makes every displayed normal sum finite. Translation proves differential linearity. Skew symmetry and Jacobi prove the chiral identities in the full normal module. This construction uses its stated auxiliary curve. It does not identify that curve with the support of an arithmetic random field.

The center and the energy grading

The ordinary vertex center consists of states whose field has no singular product with any state. Skew symmetry gives the equivalent condition that every nonnegative mode of every state annihilates the central state.

PROPOSITION 6.4.5. The ordinary vertex center of $\mathcal{V}$ is $\mathbb{C}\mathbf{1}$.

Proof. A central state is killed by every positive current mode $a_m(h)$. At fixed occupancy, these modes contract each occupied leg against h. The nondegenerate Hilbert pairing makes a tensor with all such contractions zero equal to zero. Different occupancy patterns remain distinguishable after the contracted mode is fixed. All positive-energy components therefore vanish. The vacuum is central. □

PROPOSITION 6.4.6. For a finite-rank orthogonal projection P, $L_P = \frac{1}{2}W_P$ satisfies the Virasoro operator product with central charge rank P. If H is infinite dimensional, no state of $\mathcal{V}$ is a conformal vector implementing its energy grading.

Proof. Since $P^2 = P$, (6.11) becomes

$$L_P(z)L_P(w) \sim \frac{\operatorname{rank}(P)/2}{(z-w)^4} + \frac{2L_P(w)}{(z-w)^2} + \frac{\partial L_P(w)}{z-w}.$$

Its grading concerns the projected species. The state has squared norm $\operatorname{rank}(P)/2$.

A conformal vector ω implementing the given energy must have energy two, because $L_0\omega = 2\omega$. Every vector there is $q_A + a_{-2}(v)\mathbf{1}$, for a complex symmetric Hilbert–Schmidt tensor A. Its mode (1) acts on the energy-one species space by $2A$. The improvement field ∂J_v contributes $-a_0(v) = 0$ to that mode. The required identity action would force $2A = 1_H$, which is not Hilbert–Schmidt in infinite dimension. □

Thus vacuum and translation survive without an energy-grading conformal state. The scalar ordinary center just computed also leaves the derived center and its geometric actions as separate objects of comparison.

A prime-labeled current cover

For each prime p, take two orthogonal level-one currents $a^{(p)}, b^{(p)}$. Their rank-two vertex algebra $\mathcal{H}^{(p)}$ has conformal state

$$\omega_p = \frac{1}{2}\Big((a^{(p)}_{-1})^2 + (b^{(p)}_{-1})^2\Big)\mathbf{1}_p.$$

The preceding calculation gives central charge two and integer oscillator grading $L_0^{(p)}$. For finite sets of primes S, the maps

$$\bigotimes_{p\in S} \mathcal{H}^{(p)} \longrightarrow \bigotimes_{p\in S'} \mathcal{H}^{(p)}, \qquad S \subset S',$$

insert the vacuum in each new factor. They preserve all field products and translation. Their filtered union defines the algebraic vacuum-restricted product

$$\mathcal{H}_{\mathbb{P}} = \bigotimes_{p}{}' \mathcal{H}^{(p)}.$$

Every state uses finitely many primes. For a finite set S, its conformal state is $\sum_{p\in S}\omega_p$, of central charge $2|S|$. The infinite sum is not a state of $\mathcal{H}_{\mathbb{P}}$.

There is also the logarithmically weighted operator

$$H_{\mathrm{ar}} = \sum_{p} (\log p) L_{\mathrm{o}}^{(p)}$$

on this algebraic product. Each state makes this sum finite. It assigns energy $m \log p$ to a mode-m creator at p. Its weights differ from the oscillator grading. Rescaling ω_p does not preserve its Virasoro normalization.

With species space $H = \ell^2(\mathbb{P}; \mathbb{R}^2)$, the algebraic product is dense in each Hilbert energy sector constructed above. An extension of H_{ar} to a closed Hilbert operator requires its own domain. A supported map from this current cover to independently defined arithmetic coefficients is further data.

6.5 Coordinate changes and coefficient norms

A coordinate change replaces the kernel subtracted in a normally ordered field. For a quadratic field this replacement contracts its two coefficient indices. The resulting trace determines whether the change acts on the chosen completion. Assume throughout this and the following sections that H is infinite dimensional.

Let f be a locally invertible holomorphic germ. The difference of the two current kernels is

$$H_f(z,u) = \frac{f'(z)f'(u)}{(f(z)-f(u))^2} - \frac{1}{(z-u)^2}.$$

It is regular along the diagonal. The Taylor expansion of $f(u+s)$ through order three gives

$$H_f(z,z) = \tfrac{1}{6}\{f,z\}, \qquad \{f,z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2}\left(\frac{f''(z)}{f'(z)}\right)^2.$$

Inserting the intermediate kernel under a second coordinate change gives the exact cocycle

$$H_{f\circ g}(z,u) = H_g(z,u) + g'(z)g'(u)H_f(g(z),g(u)). \tag{6.12}$$

Consequently a finite-rank quadratic coefficient has the transformation law

$$W_A^{(z)}(z) = f'(z)^2 W_A^{(w)}(f(z)) + \tfrac{1}{6}\operatorname{Tr}(A)\{f,z\}\mathbf{1}, \qquad w = f(z). \tag{6.13}$$

The sign follows by subtracting the old vacuum kernel from the pullback of the new one.

For a germ fixing zero, put

$$B_f(m,n) = [z^{m-1}u^{n-1}]H_f(z,u), \qquad m,n \geq 1.$$

The coordinate map on a finite-species state sums over its internal pairings. A paired pair of legs in modes m,n contributes its species pairing times $B_f(m,n)$. The unpaired legs undergo the finite triangular mode substitution defined below.

Theorem 6.5.1. If some $B_f(m,n) \neq 0$, this coordinate map on finite species is not closable in the Hilbert energy spaces of (6.9). Precisely the fractional-linear germs fixing zero act continuously on every bounded-energy space with the specified current and vacuum.

Proof. For an orthonormal sequence (e_i), consider

$$u_N^{mn} = \frac{1}{N}\sum_{i=1}^{N} a_{-m}(e_i)a_{-n}(e_i)\mathbf{1}.$$

Its squared norm is $2m^2/N$ if $m = n$ and mn/N otherwise. Thus $u_N^{mn} \to 0$. Its uncontracted coordinate image also tends to zero, because only finitely many bounded mode substitutions occur. Its contracted image is the fixed scalar $B_f(m,n)\mathbf{1}$. The closure of the graph therefore contains a nonzero vector over zero.

All coefficients $B_f(m,n)$ vanish exactly when H_f vanishes as a germ. In that case $\{f,z\} = 0$. Put $v = f''/f'$. Its equation is $v' = v^2/2$. Integration gives a fractional-linear function, including the affine case.

Conversely direct substitution of $f(z) = (az+b)/(cz+d)$, $ad - bc \neq 0$, gives $H_f = 0$. Its coordinate map and inverse then consist only of bounded mode substitutions on every finite set of energy sectors. □

The germ $f(z) = z + z^4$ has $B_f(1,1) = 0$ but $B_f(1,2) = 2$. Testing only the Schwarzian at the marked point therefore misses an obstruction at higher energy. Restricting to traceless finite-rank quadratics also fails to give a field algebra: for $A = \operatorname{diag}(1,-1)$, the quadratic product contains A^2, whose trace is two.

Projective tensors and all coordinate jets

For a finite tensor define its projective norm by

$$\|u\|_\pi = \inf_{u=\sum_\alpha v_{\alpha,1}\otimes\cdots\otimes v_{\alpha,p}} \sum_\alpha \prod_j \|v_{\alpha,j}\|.$$

Write $\widehat{\otimes}_\pi$ for the completion and $\widehat{\operatorname{Sym}}_\pi^{r}$ for the image of its continuous symmetrizing projection. Use the same Fock factorial weights as before and set

$$\mathcal{V}_E^\pi = \bigoplus_{\sum m r_m = E} \widehat{\bigotimes}_{\pi,m\geq 1} \widehat{\operatorname{Sym}}_\pi^{r_m} H_m, \qquad \mathcal{V}^\pi = \bigoplus_E \mathcal{V}_E^\pi.$$

Use the sum norm at each energy and the locally convex direct-sum topology over energies.

PROPOSITION 6.5.2. Every internal and cross contraction is bounded on the corresponding projective tensor sectors. The canonical map $\mathcal{V}_E^\pi \to \mathcal{V}_E$ is injective with dense image. For real symmetric quadratic coefficients,

$$\|q_A\|_\pi = \sqrt{2}\|A\|_1,$$

so this sector consists of symmetric trace-class operators. The field operations, vacuum and translation make $\mathcal{V}^\pi$ a local vertex algebra, and its map to $\mathcal{V}$ is a vertex homomorphism.

Proof. On a simple tensor, contraction multiplies the remaining factors by a pairing bounded by the two removed norms. Taking the infimum over decompositions proves the bound. Iterate it for a matching.

Finite species projections converge to the identity in projective norm, first on simple tensors and then by density and their uniform bound. If a projective tensor maps to zero in the Hilbert tensor product, each finite species projection is zero. Its projective limit is therefore zero. Finite species tensors also prove density of the image.

A real symmetric spectral decomposition gives the upper trace-norm bound. Pairing with the bounded sign operator gives the reverse bound. The Fock factor is $\sqrt{2}$. Each field coefficient at fixed energies has finitely many bounded Wick patterns. The finite-species identities extend by density, with the uniform locality exponent already proved. The two completions agree on their common finite-species subspace, so their inclusion preserves the coefficients. □

The unpaired mode substitution for a germ $f(0) = 0, f'(0) \neq 0$, is

$$T_f(m, j) = [z^{m-1}]f'(z)f(z)^{j-1} = \frac{m}{j}[z^m]f(z)^j, \qquad 1 \le j \le m.$$

The pair coefficient also has the expression

$$B_f(m, n) = mn[z^m u^n] \log \frac{f(z) - f(u)}{z - u}.$$

Indeed two derivatives of the logarithm give H_f. Its constant is immaterial.

Write $x_m(v) = a_{-m}(v)\mathbf{1}$. Define $R_f\mathbf{1} = \mathbf{1}$.

For a creation monomial sum over all partial matchings of its legs. An unpaired $x_m(v)$ contributes $\sum_{j\le m} T_f(m, j)x_j(v)$. A pair $x_m(v), x_n(w)$ contributes $(v, w)B_f(m, n)$. Multiply within each matching and sum. Energy at most E uses only the $(E + 1)$-jet of f.

Theorem 6.5.3. These formulas define bounded maps on every bounded-energy part of $\mathcal{V}^\pi$, with

$$R_f R_g = R_{f\circ g}, \qquad R_f^{-1} = R_{f^{-1}}.$$

For the recentered germ $f_z(u) = f(z + u) - f(z)$, they obey

$$R_f Y(a, z) R_f^{-1} = Y(R_{f_z} a, f(z)), \tag{6.14}$$

$$\partial_z R_{f_z} + f'(z) T R_{f_z} = R_{f_z} T. \tag{6.15}$$

The identities are coefficientwise in the coordinate variable.

Proof. Every matching is bounded by Proposition 6.5.2. Unpaired jets compose by the chain rule. In an iterated matching a pair is either formed at the first change or formed after that change. The kernel cocycle (6.12) adds these two contributions to the one-step pair. Partitioning matchings in this way proves composition at every tensor degree. The identity and inverse germs give the remaining group identities.

In finite species, the current transforms as a one-form. Wick expansion of a composite field adds exactly the difference of its normal-ordering kernels. This is the same matching rule, so it proves (6.14). Continuity at fixed coefficients extends it to projective tensors.

Applying it to the vacuum gives $R_f e^{zT} = e^{f(z)T} R_{f_z}$. Differentiation proves (6.15). These formal identities require no analytic exponential on the algebraic energy sum. □

For $n \ge -1$, the infinitesimal operators on finite species are

$$L_n = \sum_{m>\max(n,0),i} m x_{m-n,i}\partial_{x_{m,i}} + \frac{1}{2}\sum_{\substack{r+s=n\\ r,s\ge 1},i} rs\,\partial_{x_{r,i}}\partial_{x_{s,i}}.$$

Here $x_{m,i} = x_m(e_i)$. The sums on each bounded-energy sector are continuous by the contraction bounds. Direct commutation gives $[L_m, L_n] = (m-n)L_{m+n}$, with zero bracket when $m = n = -1$.

The only possible central cases have $m + n = 0$ and $m \in \{-1, 0, 1\}$, whose central coefficient vanishes. Thus this is the indicated part of the Witt action.

In particular

$$L_{-1} = T, \qquad L_0|_{\mathcal{V}_E^\pi} = E, \qquad L_2 q_A = \operatorname{Tr}(A)\mathbf{1}.$$

The operators L_{-1}, L_0, L_1 have no internal contractions and also act on the Hilbert sectors. Their existence does not require a state whose field implements the full action.

Descent to a curve

On a smooth complex analytic curve, use holomorphic sections which locally take values in a finite sum of the Banach sectors $\mathcal{V}_E^\pi$. For $w = f(z)$, glue their components by

$$s_w(f(z)) = R_{f_z} s_z(z).$$

The finite jet matrices are holomorphic in the root and locally bounded at each energy. Their composition law proves the overlap cocycle. Equation (6.15) glues $\nabla = d + T\,dz$. This connection raises the local energy bound by one and is flat on the curve.

Use completed projective external tensors, finite pole orders and finite normal derivatives for the local chiral operation. Its chart formula is the principal part of $h(u,z)Y(a,u)b$. Tensoring the left module with ω_X gives the right differential module. A normal class is still $\partial\,\partial_u^r/r!$. Covariance transports the expansion, density and all normal jets together. Translation and Jacobi therefore glue the analytic topological chiral operation.

There is a separate algebraic version on a smooth algebraic curve over $\mathbb{C}$. Regard $\mathcal{V}^\pi$ as a vector space and use algebraic external tensors. On a state of energy at most E, every word in positive L_n's with total index greater than E vanishes. The finitely many remaining words and the energy projections span a finite-dimensional stable subspace. Its coordinate action is algebraic and factors through a finite jet group.

Associate these subspaces to the formal-coordinate torsor and take their filtered union. The same identities descend its connection and finite principal-parts operation. The two versions use different external tensors.

One may instead keep the Hilbert sectors and choose a projective atlas, whose transitions are fractional linear. Their internal pair coefficients vanish, so the same descent works with the bounded Hilbert coefficient maps and projective external tensors. The projective atlas is part of this choice. On $\mathbb{P}^1$, inversion has the recentered germ

$$\frac{1}{p+u} - \frac{1}{p} = -\frac{u}{p(p+u)}, \qquad p \neq 0,$$

which is fractional linear. An affine uniformizing atlas of an elliptic curve gives another example. On a fixed such atlas, $\mathcal{V}^\pi \to \mathcal{V}$ is dense, injective and compatible with the operations, but is not a topological equivalence in infinite species.

Finally let P be a projective connection, with components $P_z = f'(z)^2 P_w(f(z)) + \{f, z\}$. For symmetric trace-class A, (6.13) shows that

$$\left(W_A^{(z)}(z) - \frac{\operatorname{Tr}(A)}{6} P_z(z)\mathbf{1}\right)(dz)^2$$

is coordinate independent. Changing P changes this field by the corresponding scalar quadratic differential. For a positive non-trace-class A, this subtraction has no value in the stated coefficient domain. It requires a trace, an operation stronger than the cross contractions which defined W_A.

6.6 A completion retaining the contractions

The sequence $q_{P_N/N}$ had Hilbert limit zero and trace one. To retain both limits, complete the finite-species states in the norm of their state tensor together with all their internal contractions. The new limit will be a central state, distinct from zero and from the vacuum.

Introduce symmetric variables

$$c_{mn} = c_{nm}, \qquad \deg c_{mn} = m + n, \qquad \Gamma = \mathbb{C}[c_{mn} : 1 \le m \le n].$$

Each homogeneous part Γ_d is finite dimensional. Define

$$\mathcal{W}_E = \bigoplus_{d=0}^{E} \mathcal{V}_{E-d} \otimes \Gamma_d, \qquad \mathcal{W} = \bigoplus_{E \ge 0} \mathcal{W}_E.$$

Give central monomials an orthonormal basis and use the finite Hilbert sum at each energy. Any choice of norms on these finite-dimensional central factors gives the same topology.

On finite-species oscillator polynomials put

$$\Delta_c = \frac{1}{2} \sum_{m,n \ge 1} c_{mn} \sum_i \partial_{x_m(e_i)} \partial_{x_n(e_i)}, \qquad \sigma = e^{\Delta_c}, \qquad j(a) = \sigma a.$$

The exponential terminates on each polynomial. Every derivative pair contracts two species legs by their intrinsic pairing, so the definition is independent of the orthonormal basis. The norm $\|j(a)\|_{\mathcal{W}_E}$ is the norm of all its contraction coefficients. Its constant coefficient in c is a, so it is indeed a norm.

The completed field operations

Define the central translation by

$$D_\Gamma c_{mn} = m c_{m+1,n} + n c_{m,n+1}.$$

The translation of $\mathcal{W}$ is $\mathsf{T} = T_x + D_\Gamma$, where $T_x x_m(v) = m x_{m+1}(v)$. For two modes set

$$P_m(z; v) = \sum_{r \ge 0} \binom{m + r - 1}{r} x_{m+r}(v) z^r,$$

$$\kappa_{mn}(z) = \frac{(-1)^{m-1}(m + n - 1)!}{(m - 1)!(n - 1)!} z^{-m-n},$$

$$G_{mn}(z) = \sum_{r \ge 0} \binom{m + r - 1}{r} c_{m+r,n} z^r.$$

The scalar κ_{mn} is the level-one contraction already computed. The new term G_{mn} retains its regular contraction data.

Define $Y_{\mathcal{W}}(a, z)b$ by matchings between the oscillator legs of its two polynomial inputs. A pair of modes m, n and species v, w contributes

$$(v, w)\big(\kappa_{mn}(z) + G_{mn}(z)\big). \tag{6.16}$$

An unpaired left leg contributes $P_m(z;v)$. An unpaired right leg stays at the root.

Translate the left central coefficient by e^{zD_Γ} and multiply it by the right one. The product of these terms, summed over partial matchings, is the field. Only contractions between separate Hilbert tensors occur in this rule.

PROPOSITION 6.6.1. The coefficient maps make $\mathcal{W}$ a local vertex algebra with vacuum 1 and translation T. Its fixed-energy mode maps are bounded bilinear, with locality exponent $E+F$ on input energies E, F. Both the finite-species map j and the projection

$$\pi : \mathcal{W} \longrightarrow \mathcal{V}, \qquad c_{mn} \longmapsto 0,$$

are vertex homomorphisms on their stated domains.

Proof. On finite species start with the tensor product of the Heisenberg vertex algebra and the commutative differential algebra (Γ, D_Γ). Its field on central coefficients is $(e^{zD_\Gamma}\gamma)\gamma'$. Conjugate its field operation by σ, whose polynomial inverse is $e^{-\Delta_c}$. The commutator $[T_x, \partial_{x_m}] = -(m-1)\partial_{x_{m-1}}$ shows, after reindexing, that $[T_x, \Delta_c] = -[D_\Gamma, \Delta_c]$. Thus σ commutes with T.

In the conjugated product, the internal contractions of σ^{-1} on each input cancel those of σ on that same input inside the output. The remaining pairs connect the two inputs. A translated left leg gives the new term G_{mn}, and the old singular pair gives κ_{mn}. This proves precisely (6.16).

At fixed energies and output coefficient, there are finitely many pairings, Taylor terms and central monomials. The cross contraction bound proves their continuity. The singular order is at most $E+F$, independently of species rank. The finite-species identities therefore extend by density as in Theorem 6.4.2. Conjugation proves the assertion about j. Evaluation at $c=0$ proves the assertion about π. □

THEOREM 6.6.2. The energywise completion of the finite-species states in the norm $a \mapsto \|j(a)\|_{\mathcal{W}_E}$ is exactly $\mathcal{W}$. The projection π is surjective and has the bounded linear section $v \mapsto v \otimes 1$, which is not a vertex homomorphism. The map j extends to a continuous, dense, proper vertex embedding $\mathcal{V}^\pi \hookrightarrow \mathcal{W}$.

Proof. For the states u_N^{mn} used in the coordinate obstruction,

$$j(u_N^{mn}) = u_N^{mn} + c_{mn}, \qquad \|u_N^{mn}\|^2 = \begin{cases} 2m^2/N, & m = n, \\ mn/N, & m \neq n. \end{cases}$$

Their limits put every c_{mn} in the closure of the image of j. That closure is stable under the continuous field coefficients. On central states the (-1)-product is polynomial multiplication, so it contains all central monomials.

For an oscillator polynomial a, the difference $j(a) - a$ is a sum of central monomials times polynomials with fewer oscillator legs. Induction on the number of legs therefore puts a, and then every ac^ν, in the closure. Such polynomials are dense in every finite Hilbert sum $\mathcal{W}_E$. This proves the completion assertion.

The linear section is not a vertex map. For a unit species, the completed current (-1)-product is $x_1^2 + c_{11}$. The section sends its old product to x_1^2, omitting the regular contraction.

Every internal contraction is bounded on projective tensors, so j extends continuously to $\mathcal{V}^\pi$. Its composite with π is the injective projective-to-Hilbert map. It is therefore injective. It has dense image because it contains the finite-species image of j. Its composite with π is injective, so the image contains no nonzero c_{mn}. The image is therefore proper. □

In the quadratic mode-one sector, this embedding is the dense graph

$$\mathcal{S}_1^{\mathrm{sym}}(H) \longrightarrow \mathcal{S}_2^{\mathrm{sym}}(H) \oplus \mathbb{C}c_{11}, \qquad A \longmapsto (A, \operatorname{Tr} A).$$

Here $\mathcal{S}_p$ denotes the Schatten class with summable p-th powers of singular values. The completion has an independent second coordinate. It has not defined a trace on every old Hilbert–Schmidt operator.

The central algebra and ordinary actions

THEOREM 6.6.3. The ordinary vertex center of $\mathcal{W}$ is Γ. With its translation it is the free commutative differential algebra on one generator of each positive even energy:

$$\Gamma = \mathbb{C}[D_\Gamma^r \chi_k : k \geq 1,\ r \geq 0], \qquad \chi_k = c_{kk}, \qquad \deg \chi_k = 2k. \tag{6.17}$$

Here a differential algebra means an algebra with a derivation. D_Γ is not a square-zero differential.

Proof. The regular term G_{mn} does not change the positive current modes. They remain $a_n(v) = n\partial_{x_n(v)}$, $n > 0$. Their common kernel on Hilbert oscillator tensors is the vacuum. Since the central monomial basis at each energy is finite, the same argument coefficient by coefficient puts every central state in Γ. Conversely its elements are central in the transported finite-species algebra. Continuity extends centrality to all of $\mathcal{W}$.

Identify the linear span of the c_{mn} with $\mathbb{C}[u, v]^{\mathfrak{S}_2}$ by

$$c_{mn} \longmapsto \frac{u^{m-1}v^{n-1} + u^{n-1}v^{m-1}}{2(m-1)!(n-1)!}.$$

Under this map D_Γ is multiplication by $u + v$. The symmetric polynomial ring $\mathbb{C}[u + v, uv]$ is free over $\mathbb{C}[u + v]$ on the powers of uv. Up to nonzero factorials, these are the images of c_{kk}. Taking the symmetric algebra proves both generation and algebraic independence in (6.17). □

For instance,

$$c_{12} = \tfrac{1}{2} D_\Gamma c_{11}, \qquad c_{13} = \tfrac{1}{4}(D_\Gamma^2 c_{11} - 2c_{22}).$$

The additional even-energy generators are needed to recover all two-mode contractions.

Let $B = \bigoplus_{r \geq 0} B_r$ be a commutative energy-graded algebra with Banach sectors, bounded multiplication and a bounded degree-one derivation D_B. Give it the commutative vertex operation $Y_B(b, z)b' = (e^{zD_B} b)b'$. In $B \otimes \mathcal{W}$, complete each external sector projectively and retain the algebraic energy sum. An ordinary action is an energy-preserving unital vertex homomorphism $\rho : B \otimes \mathcal{W} \to \mathcal{W}$, bounded on each sector and satisfying $\rho(1 \otimes w) = w$.

PROPOSITION 6.6.4. These actions are naturally in bijection with bounded graded algebra maps $\phi : B \to \Gamma$ commuting with the two derivations. The corresponding action is

$$\rho(b \otimes w) = \phi(b)_{(-1)} w.$$

Proof. The two tensor factors have no singular cross products. Since ρ fixes $\mathcal{W}$, its value on $b \otimes 1$ is central and hence lies in Γ. Preservation of (-1)-products and translation gives the stated algebra map. Conversely central multiplication gives the displayed bounded vertex homomorphism. The identity $b \otimes w = (b \otimes 1)_{(-1)}(1 \otimes w)$ proves uniqueness. Both constructions commute with maps of B. □

This universal property fixes ordinary continuous vertex actions and their coefficient topology. A derived center also requires its action spaces and higher comparisons. A factorizing transverse actor additionally requires the specified coactions and collision maps. Neither object is defined by this ordinary-action property alone.

Coordinates on the completed space

Use the coordinate coefficients $T_f(m, j)$ and $B_f(m, n)$ already defined. On the polynomial coordinates of $\mathcal{W}$, set

$$R_f x_m(v) = \sum_{j \le m} T_f(m, j) x_j(v),$$
$$R_f c_{mn} = \sum_{j \le m,\, k \le n} T_f(m, j) T_f(n, k) c_{jk} + B_f(m, n). \tag{6.18}$$

Extend by ordinary polynomial substitution. At every energy bound, this uses only finite-dimensional central maps and finitely many bounded Hilbert mode maps.

THEOREM 6.6.5. These substitutions extend continuously to $\mathcal{W}$, preserve bounded energy, and satisfy

$$R_f j = j R_f^{\mathrm{fin}}, \qquad R_f R_g = R_{f \circ g}, \qquad R_f^{-1} = R_{f^{-1}}.$$

Here R_f^{fin} is the finite-species matching formula. They also satisfy

$$R_f Y_{\mathcal{W}}(a, z) R_f^{-1} = Y_{\mathcal{W}}(R_{f_z} a, f(z)),$$
$$\partial_z R_{f_z} + f'(z) \mathsf{T} R_{f_z} = R_{f_z} \mathsf{T}.$$

Thus locally bounded-energy holomorphic sections glue on a smooth complex curve, with connection $d + \mathsf{T}\, dz$ and their continuous field coefficients.

Proof. Expand $j(a)$ into its self-pair matchings. For each pair, (6.18) chooses either the transported central coefficient or the scalar change in the vacuum kernel. Summing these choices is exactly the finite-species coordinate matching followed by j. This proves the first identity. The kernel and unpaired-jet cocycles prove composition, and the inverse germ proves inversion.

The field identity holds on the dense finite-species image by its established covariance. All coefficients are bounded at fixed energies, so it holds on $\mathcal{W}$. Applying it to the vacuum and differentiating gives the connection identity. Holomorphic dependence of the finite jets, their composition law and that identity glue the claimed sections and flat connection. □

For the generating series $C(z, u) = \sum_{m,n \ge 1} c_{mn} z^{m-1} u^{n-1}$, the central law reads

$$R_f C(z, u) = f'(z) f'(u) C(f(z), f(u)) + H_f(z, u). \tag{6.19}$$

In particular

$$R_f c_{11} = f'(0)^2 c_{11} + \tfrac{1}{6}\{f, 0\}, \qquad L_2 c_{11} = 1.$$

The infinitesimal operators for $n \ge -1$ are derivations of these polynomial coordinates:

$$L_n x_m(v) = \mathbf{1}_{m>n}\, m x_{m-n}(v),$$
$$L_n c_{pq} = \mathbf{1}_{p>n}\, p c_{p-n,q} + \mathbf{1}_{q>n}\, q c_{p,q-n} + pq\, \mathbf{1}_{p+q=n}.$$

Excluded terms are omitted. For $n = -1$ this is T. Differentiating the coordinate law gives the same positive-half Witt brackets.

Proposition 6.6.6. The old Fock form has a unique continuous extension along j, namely

$$\langle a, b\rangle_{0} = \langle \pi a, \pi b\rangle_{\mathrm{Fock}}.$$

Its radical is $\ker \pi$, which is not preserved by all coordinate changes. No conformal vector in $\mathcal{W}$ implements the displayed operator L_{2}.

Proof. The form agrees with the old one on $j(\mathcal{V}_{\mathrm{fin}})$. Density proves uniqueness on bounded-energy sectors. Its radical is the vertex ideal generated by the positive-degree central variables. But $L_{2}c_{\mathrm{II}} = 1$ sends one such variable to the vacuum. Every singular mode of every field annihilates a central state. A conformal vector implementing L_{2} would therefore give $L_{2}c_{\mathrm{II}} = 0$, a contradiction. □

In particular no positive-definite Hilbert form on this completion extends that specified vacuum form. A separately chosen state with different values on Γ is a different question.

A chosen bidifferential and its quotients

Choose a symmetric local bidifferential

$$\mathcal{B}(s,t) = \big((s-t)^{-2} + b(s,t)\big)\,ds\,dt,$$

with b regular near the diagonal. Its divided jets at the moving root p are

$$b_{mn}(p) = \frac{\partial_{s}^{m-1}\partial_{t}^{n-1}b(s,t)}{(m-1)!(n-1)!}\bigg|_{s=t=p}.$$

Take formal sections $\mathcal{W}_{E}[[p]]$, or holomorphic sections locally in finitely many energy sectors. Scalar functions carry the commutative vertex operation with translation ∂_{p}.

Proposition 6.6.7. The relations $c_{mn} = b_{mn}(p)$ generate a vertex ideal stable under $\mathsf{T} + \partial_{p}$. The quotient has the coefficient space of locally bounded-energy $\mathcal{V}$-valued sections, with the regular contractions G_{mn} evaluated at these jets. The quotients glue wherever $\mathcal{B}$ glues.

Proof. Differentiating both variables at the root gives

$$\partial_{p} b_{mn} = m b_{m+1,n} + n b_{m,n+1}.$$

Translation therefore takes each relation to a sum of such relations. Their coefficients are central, so the ideal is a vertex ideal. Evaluation in the finite central polynomial at each energy gives the quotient. Its kernel is this ideal: subtract the evaluated value of a monomial one factor at a time. The transformation law of the regular part of $\mathcal{B}$ is (6.19), which proves compatibility on overlaps. □

On $\mathbb{P}^{1}$, the bidifferential $ds\,dt/(s-t)^{2}$ has $b = 0$ in projective coordinates and recovers the Hilbert construction with that atlas. A choice on another curve is additional geometric data.

Two choices b, b' need not have a continuous current-preserving comparison on the unchanged Hilbert sectors. Such a comparison must send a finite-rank quadratic state by

$$q_{A} \longmapsto q_{A} + \mathrm{Tr}(A)(b'_{\mathrm{II}} - b_{\mathrm{II}}),$$

as follows by comparing the current (-1)-products. At a root where the jet difference is nonzero, the sequence $q_{P_{N}/N}$ proves nonclosability. Higher mode pairs test the other jet differences.

Both quotient families exist. Their specified comparison requires a stronger tensor domain. On $\mathcal{V}^{\pi}$, all the needed contractions are bounded and the comparison does extend.

6.7 GAUSSIAN PREPARATIONS AND THEIR OBSERVABLES

A vacuum specifies the field algebra. A preparation instead specifies a normalized vector and the observables on which its expectations exist. We construct such vectors in the first oscillator of $\widehat{\mathcal{V}}$, and then compute their scalar field matrix elements.

The first oscillator and Gaussian multiplication

Choose independent standard real Gaussians (ξ_i) in a real orthonormal basis (e_i) of H. The L^2 sum $\xi(h) = \sum_i (h, e_i)\xi_i$ has covariance $\mathbb{E}[\xi(h)\xi(k)] = (h, k)$. Let $\mathcal{F}_\xi$ be the completed sigma field they generate. For a symmetric tensor a of order p, define its multiple Gaussian integral $I_p(a)$ by polarization and the generating identity

$$e^{t\xi(h) - t^2\|h\|^2/2} = \sum_{p \geq 0} \frac{t^p}{p!} I_p(h^{\otimes p}).$$

Multiplying two such exponentials and taking their expectation gives

$$\mathbb{E}[I_p(a) I_q(b)] = \delta_{pq}\, p! \langle a, b \rangle. \tag{6.20}$$

This identity extends I_p to the symmetric Hilbert tensor power. Complex linear extension gives the corresponding complex L^2 space.

Write $a \widetilde{\otimes}_r b$ for contraction of r legs followed by symmetrization. Comparing coefficients in the same exponential identity gives

$$I_p(a) I_q(b) = \sum_{r=0}^{\min(p,q)} r! \binom{p}{r} \binom{q}{r} I_{p+q-2r}(a \widetilde{\otimes}_r b). \tag{6.21}$$

The coefficient chooses r legs on each side and a bijection between them. For finite tensors this proves the formula directly.

Under Hilbert tensor approximation, its right side converges in L^2, by Lemma 6.4.1. Its left side converges in L^1, by Cauchy–Schwarz and (6.20). The limits agree. Iterating the formula shows that finite sums of these completed chaoses form an algebra and belong to every finite L^p space.

Let $\mathcal{V}^{(1)} \subset \mathcal{V}$ consist of states using only mode-one creators, and let Π_1 be the projection onto this subspace. The map

$$\Phi\big(a_{-1}(h_1) \cdots a_{-1}(h_p)\mathbf{1}\big) = I_p\big(\mathrm{Sym}(h_1 \otimes \cdots \otimes h_p)\big)$$

preserves its Fock norm. Its product comparison is

$$\Phi\big(\Pi_1 Y(a, z) b|_{z=1}\big) = \Phi(a)\Phi(b), \qquad a, b \in \mathcal{V}^{(1)}.$$

Indeed each cross contraction contributes z^{-2}. The projection keeps only the constant mode-one term of every uncontracted transported creator. The result is the finite Laurent polynomial whose coefficients occur in (6.21). Thus projection precedes evaluation at $z = 1$.

For a unit current state j, this reads

$$\Pi_1 Y(j, z) j = j^2 + z^{-2}\mathbf{1}, \qquad \Phi(j^2) = \xi^2 - 1.$$

The contracted term restores ordinary Gaussian multiplication. The resulting product on $\mathcal{V}^{(1)}$ is associative, since Φ transports it to that multiplication. It differs from the vertex (-1)-product. In particular, Π_1 is not a vertex homomorphism.

Proposition 6.7.1. The map Φ extends to a unitary

$$U : \mathcal{F}_1 \xrightarrow{\sim} L^2(\Omega, \mathcal{F}_\xi, \mu),$$

where $\mathcal{F}_1$ is the Hilbert completion of $\mathcal{V}^{(1)}$. On finite-particle sectors, $a_{-1}(h) + a_1(h)$ becomes multiplication by $\xi(h)$. For a real symmetric Hilbert–Schmidt operator A, define

$$\mathsf{Q}_A = U^{-1} M_{I_2(A)} U, \qquad \mathrm{Dom}(\mathsf{Q}_A) = \{\psi : I_2(A) U\psi \in L^2\}.$$

This is a closed self-adjoint operator. On each finite-particle sector it equals

$$\sum_{i,j} A_{ij}\big(a_{-1}(e_i)a_{-1}(e_j) + 2a_{-1}(e_i)a_1(e_j) + a_1(e_i)a_1(e_j)\big).$$

Proof. The chaos isometry proves isometry. Hermite polynomials are dense for finitely many Gaussian coordinates. Conditional expectations onto increasing finite coordinate sigma fields converge in $L^2(\mathcal{F}_\xi)$, proving surjectivity. The one- and two-tensor product formulas give the two operator identities in finite rank.

On the p-particle sector, pair creation and pair annihilation have norm bounds

$$\sqrt{(p+1)(p+2)}\|A\|_2, \qquad \sqrt{p(p-1)}\|A\|_2.$$

The middle term has norm at most $2p\|A\|$. Hence the finite-rank expressions converge on every such sector. Multiplication by the real measurable function $I_2(A)$, on its displayed maximal domain, is self-adjoint. This specifies the closed operator and its domain. □

The sigma field in this unitary is essential. An additional independent centered sign is orthogonal to every Gaussian polynomial. It does not belong to the range of U in an enlarged probability space.

Connected contractions of quadratic fields

Put $Q_A = I_2(A)$. For variables with all moments, their joint cumulant is

$$\mathrm{cum}(Z_1, \dots, Z_n) = \sum_{\pi} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} \mathbb{E} \prod_{i\in B} Z_i,$$

where π ranges over partitions of $\{1, \dots, n\}$. The same partition expression defines a connected vacuum coefficient of fields.

Proposition 6.7.2. For symmetric Hilbert–Schmidt $A_1, \dots, A_n$, $n \geq 2$, one has

$$\mathrm{cum}(Q_{A_1}, \dots, Q_{A_n}) = 2^{n-1} \sum_{\sigma \in S_{\{2,\dots,n\}}} \mathrm{Tr}(A_1 A_{\sigma(2)} \cdots A_{\sigma(n)}).$$

Introduce symmetric independent variables $k_{ij} = k_{ji}$. For $i_1 = 1$, $i_j = \sigma(j)$ and $i_{n+1} = i_1$, the polynomial

$$2^{n-1} \sum_{\sigma \in S_{\{2,\dots,n\}}} \mathrm{Tr}(A_{i_1} \cdots A_{i_n}) \prod_{j=1}^{n} k_{i_j i_{j+1}} \tag{6.22}$$

gives this cumulant at $k_{ij} = 1$. At $k_{ij} = (z_i - z_j)^{-2}$, it gives the connected vacuum coefficient of the quadratic fields $W_{A_i}(z_i)$.

Proof. The partition sum cancels disconnected Wick graphs. Each remaining vertex has degree two, so the connected graph is a cycle. For $n \geq 3$, an unoriented cyclic order has 2^n choices of labeled legs. Its two orientations give the coefficient 2^{n-1} in the permutation sum. Reversal preserves the trace because the matrices are symmetric. For $n = 2$, there are two pairings and a double edge.

The species sum around each cycle is the displayed trace. Two Hilbert–Schmidt factors have trace-class product. The other factors are bounded. Thus each trace is continuous under Hilbert–Schmidt approximation. The Gaussian moments and the field coefficients converge by their contraction bounds. The two choices of edge kernel give the two evaluations of the same polynomial. □

For three fields, the formula is

$$\langle W_A(z_1)W_B(z_2)W_C(z_3)\rangle_0 = \frac{8\,\mathrm{Tr}(ABC)}{z_{12}^2 z_{23}^2 z_{31}^2}, \qquad z_{ij} = z_i - z_j.$$

Higher arity retains the order of noncommuting coefficients inside each trace. The Gaussian specialization is an evaluation of independent edge variables. It does not require complex positions with every pairwise contraction equal to one.

Normalized squeezed vectors

Let K be a real symmetric Hilbert–Schmidt operator with $\|K\| < 1$. Define the creation series

$$S_K = \exp\left(\frac{1}{2}\sum_{i,j} K_{ij}\, a_{-1}(e_i)a_{-1}(e_j)\right)\mathbf{1}, \qquad \Omega_K = \det(1-K^2)^{1/4} S_K. \tag{6.23}$$

The determinant is the Fredholm determinant: K^2 is trace class and $1 - K^2$ is strictly positive. The exponential is first interpreted in finite rank.

For real self-adjoint Hilbert–Schmidt A with $1 + A \geq \epsilon 1 > 0$, put

$$D_A = \det{}_2(1+A)^{1/2} e^{-I_2(A)/2}, \qquad \det{}_2(1+A) = \prod_j (1+a_j)e^{-a_j}.$$

Here a_j are its real eigenvalues. The logarithms of the product factors have a quadratic zero at zero, so they are summable.

THEOREM 6.7.3. The vectors in (6.23) exist in $\mathcal{F}_1$, independently of the finite-rank approximations, and

$$\|S_K\|^2 = \det(1-K^2)^{-1/2}, \qquad \|\Omega_K\| = 1.$$

The function D_A is a positive density of integral one. With

$$K = -A(2+A)^{-1},$$

one has $U\Omega_K = \sqrt{D_A}$. For two such preparations, with kernels K, L,

$$\langle \Omega_K, \Omega_L\rangle = \frac{\det(1-K^2)^{1/4}\det(1-L^2)^{1/4}}{\det(1-KL)^{1/2}} = \mathbb{E}\sqrt{D_A D_B}.$$

The square root is positive and continued from $K = L = 0$. No commutativity assumption is imposed.

Proof. For one oscillator the squared norm is

$$\sum_{m\ge 0}\frac{(2m)!}{4^m(m!)^2}k^{2m}=(1-k^2)^{-1/2}.$$

Diagonalization proves the finite-rank norm formula. The Hermite generating identity also gives

$$US_K=\det(1+K)^{-1/2}\exp\Big(\frac{1}{2}\xi^tK(1+K)^{-1}\xi\Big).$$

For the overlap of two such functions, the Gaussian precision is

$$P=1-K(1+K)^{-1}-L(1+L)^{-1}.$$

Each of the two subtracted self-adjoint operators has upper bound strictly less than $1/2$. Thus $P>0$, and direct multiplication gives

$$(1+K)P(1+L)=1-KL.$$

Gaussian integration proves $\langle S_K,S_L\rangle=\det(1-KL)^{-1/2}$. This argument fixes both the order and the branch.

Finite-rank compressions converge in Hilbert–Schmidt norm. Their squares and products converge in trace norm, so the displayed determinants converge. The overlap of two approximations to the same kernel makes their squeezed vectors Cauchy. It also proves independence of the approximation. Their fixed-particle components are the continuous tensor polynomials in K from (6.23).

For finite-rank A_N, the density is

$$D_{A_N}=\det(1+A_N)^{1/2}e^{-\xi^tA_N\xi/2},\qquad \mathbb{E}D_{A_N}=1.$$

Choose $p>1$ sufficiently close to one that $1+pA$ remains uniformly positive. Compressions retain common spectral bounds, and

$$\mathbb{E}D_{A_N}^p=\det{}_2(1+A_N)^{p/2}\det{}_2(1+pA_N)^{-1/2}.$$

The logarithm is uniformly bounded by a constant times $\|A_N\|_2^2$. Indeed the scalar logarithmic ratio has a quadratic zero at zero on the common bounded spectral interval.

The chaos isometry gives $I_2(A_N)\to I_2(A)$ in L^2. Regularized determinant convergence follows from

$$\log(1+A)-A=-\int_0^1 tA^2(1+tA)^{-1}\,dt.$$

Squares converge in trace norm, and the resolvents are uniformly bounded.

The densities therefore converge in probability. The L^p bound makes them uniformly integrable, so they converge in L^1. Their limit is D_A and has integral one. Their positive square roots converge in L^2.

On the bounded spectrum of A, separated from -1, the function $-a/(2+a)$ has absolute value strictly below one. It defines a Hilbert–Schmidt strict contraction. For $k=-a/(2+a)$, the one-coordinate identity is

$$(1-k^2)^{1/4}(1+k)^{-1/2}e^{k\xi^2/(2(1+k))}=(1+a)^{1/4}e^{-a\xi^2/4}.$$

The map $A\mapsto -A(2+A)^{-1}$ is continuous in Hilbert–Schmidt norm on this positive domain, by the resolvent identity. Passing to the two limits proves the square-root formula. The overlap identity follows by unitarity. □

The prepared covariance of the quadratures is $(1 + A)^{-1}$. More generally, for a bounded positive operator C, put $X(f) = \xi(C^{1/2}f)$. Its covariance in this preparation is

$$C^{1/2}(1 + A)^{-1}C^{1/2}.$$

Finite Gaussian characteristic functions give this formula before taking limits. The L^1 density limit preserves those bounded characteristic functions. The L^p estimates and Hölder's inequality also preserve every fixed polynomial expectation.

PROPOSITION 6.7.4. Suppose $A \geq 0$ is compact but not Hilbert–Schmidt. For $K = -A(2 + A)^{-1}$, the unnormalized squeezed norm product diverges. The normalized finite spectral preparations converge weakly to zero and have no norm limit in $\mathcal{F}_1$.

Proof. The eigenvalues $a_j \to 0$ give $|k_j| = a_j/(2 + a_j)$, comparable to a_j. Thus $\sum_j k_j^2 = \infty$, and

$$-\frac{1}{2}\sum_j \log(1 - k_j^2) = \infty.$$

The normalized vacuum overlap of a tail, $\prod_j (1 - k_j^2)^{1/4}$, tends to zero.

The overlap with any vector on finitely many eigenmodes contains this tail factor. Such vectors are dense, and all preparations have norm one. Hence their weak limit is zero. A norm limit would have norm one and the same weak limit, a contradiction. □

Prepared field matrix elements

The vector Ω_K usually has infinitely many energies. We use it to define scalar matrix elements, without defining $Y(\Omega_K, z)$ on all of $\widehat{\mathcal{V}}$.

Let N count particles in the first oscillator. If $s > 0$ and $e^{2s}\|K\| < 1$, then

$$\|e^{s\mathsf{N}}\Omega_K\|^2 = \det(1 - K^2)^{1/2} \det(1 - e^{4s}K^2)^{-1/2} < \infty.$$

This follows from $e^{s\mathsf{N}}S_K = S_{e^{2s}K}$. The overlap proof also gives convergence of the finite-rank preparations in this weighted norm. It controls every particle moment.

Define

$$N_K = K^2(1 - K^2)^{-1}, \qquad M_K = K(1 - K^2)^{-1}.$$

The first operator is trace class. The second is Hilbert–Schmidt. Writing $\langle O\rangle_K = \langle \Omega_K, O\Omega_K\rangle$, the first-mode contractions are

$$\langle a_{-1}(v)a_1(u)\rangle_K = (v, N_K u), \qquad \langle a_1(v)a_1(u)\rangle_K = (v, M_K u).$$

To see the full Wick rule, put $R = (1 - K^2)^{-1/2}$ and

$$b(v) = a_1(Rv) - a_{-1}(KRv).$$

These operators satisfy the canonical commutation relations and annihilate Ω_K. The identity $a_1(v) = b(Rv) + b(KRv)^*$ reduces its finite-rank contractions to vacuum contractions. The weighted bound passes each fixed polynomial mode identity to the limit. Higher oscillator modes remain in their vacuum.

Theorem 6.7.5. For real species vectors v, u, the scalar current matrix element is

$$\begin{aligned}\langle J_v(z)J_u(w)\rangle_K = \frac{(v,u)}{(z-w)^2} + (v, N_K u)(z^{-2}+w^{-2}) \\ + (v, M_K u)(1 + z^{-2}w^{-2}).\end{aligned} \tag{6.24}$$

For finitely many finite-energy states, finite oscillator and species cutoffs of their scalar matrix elements converge uniformly on compact subsets of

$$|z_1| > \cdots > |z_n| > 0.$$

The limits continue meromorphically. Their possible poles lie on pairwise diagonals and at the prepared punctures $z_i = 0$.

Proof. The vacuum current contraction is the convergent series $\sum_{m>0} m z^{-m-1} w^{m-1} = (z-w)^{-2}$. The two creation–annihilation orderings in mode one add the two N_K terms. The two anomalous contractions add the two M_K terms. This proves (6.24). The geometric series and every fixed derivative converge uniformly on compact nested annuli.

Each finite-energy state has finitely many occupancy types. Wick expansion gives finitely many contraction patterns for those types. Within one insertion, vacuum normal ordering removes the identity self-contraction. The remaining self kernels are derivatives of the prepared-minus-vacuum kernel. They are Hilbert–Schmidt in species. Cross kernels are bounded species operators.

Absorb each bounded cross operator into one tensor leg. Regard every self kernel as a separate two-leg Hilbert tensor. The resulting contraction graph has no self edge. Merge its vertices successively, contracting all edges between the two groups at each step. Lemma 6.4.1 bounds the result by the tensor norms.

The bounds are uniform on compact nested annuli. This proves the cutoff convergence. The finitely many positional factors are derivatives of the displayed rational kernel, so they have precisely the stated possible poles. □

For a quadratic field, the self-contraction is

$$\langle W_A(z)\rangle_K = 2z^{-2}\operatorname{Tr}(AN_K) + (1+z^{-4})\operatorname{Tr}(AM_K).$$

Both products have traces. Let $G_K(z,w)$ be the species operator of the current kernel in (6.24). The connected quadratic n-point coefficient is

$$2^{n-1}\sum_{\sigma\in S_{\{2,\dots,n\}}} \operatorname{Tr}\left(\prod_{j=1}^{n} A_{i_j} G_K(z_{i_j}, z_{i_{j+1}})\right),$$

with the cyclic indices from (6.22). The product retains its indicated order. Every trace contains two Hilbert–Schmidt A_i factors, which justify the same cycle proof.

For quadratures, the covariance instead becomes

$$1 + 2N_K + 2M_K = (1+K)(1-K)^{-1} = (1+A)^{-1}.$$

Thus the Gaussian and field comparisons concern specified observables. The puncture terms also show that the prepared state need not be translation invariant.

6.8 TRANSPORT OF GAUSSIAN PREPARATIONS

Coordinate covariance on finite-energy states does not itself transport an infinite-energy preparation. The cubic coordinate flow isolates the additional convergence problem:

$$f_t(z) = \frac{z}{\sqrt{1 - 2tz^2}}.$$

On finite-energy states its coordinate action is e^{tL_2}. On mode-one polynomials,

$$L_2 = \frac{1}{2}\sum_i \partial_{x_i}^2.$$

The sum is finite on each polynomial.

For a symmetric strict contraction K, write

$$K_t = K(1 - tK)^{-1}.$$

Every transport formula below uses a common real interval about zero on which its resolvents exist and all displayed response kernels remain strict contractions. Scalar branches start at one when $t = 0$.

THEOREM 6.8.1. If K is real symmetric and trace class, finite-species cubic transport has the Fock limit

$$\lim_{N\to\infty} e^{tL_{2,N}} S_{P_N K P_N} = \det(1 - tK)^{-1/2} S_{K_t}. \tag{6.25}$$

Here $P_N \to 1$ strongly are finite-rank orthogonal projections, and $L_{2,N}$ contracts only species in $P_N H$. If $K \geq 0$ is Hilbert–Schmidt but not trace class, the vacuum coefficients of these outputs diverge for sufficiently small $t > 0$.

Proof. In finite rank, insert $F_t \exp(x^t K_t x/2)$ in the heat equation. Comparison of its quadratic and constant terms gives

$$\dot{K}_t = K_t^2, \qquad \frac{d}{dt}\log F_t = \frac{1}{2}\operatorname{Tr} K_t.$$

The initial conditions yield the displayed resolvent and ordinary determinant. Their Gaussian series converges in Fock norm on the stipulated interval.

Trace-class compressions converge in trace norm. The resolvent identity gives convergence of their responses in that norm. Determinant continuity and the squeezed overlap formula prove (6.25).

For positive K and small positive t,

$$\log\det(1 - tP_N K P_N)^{-1/2} \geq \frac{t}{2}\operatorname{Tr}(P_N K P_N).$$

The traces tend to infinity if K is not trace class. Indeed strong convergence first captures any fixed finite spectral subspace. Those subspaces have unbounded traces. The determinant is the vacuum coefficient, so the proposed Hilbert limit cannot exist. □

For negative kernels the same normalization can have a zero limit instead of diverging. If $K \leq 0$ is Hilbert–Schmidt but not trace class, then for small $t > 0$,

$$e^{tL_{2,N}} S_{P_N K P_N} \longrightarrow 0.$$

To prove this, put $K_N = P_N K P_N$ and $q = \|K\|$. The response kernels converge in Hilbert–Schmidt norm and have a common strict-contraction bound. Their squeezed norms are consequently bounded. Meanwhile

$$\log\det(1 - tK_N)^{-1/2} \le -\frac{t}{2(1+tq)}\operatorname{Tr}|K_N| \longrightarrow -\infty,$$

using $\log(1+x) \ge x/(1+x)$. For small $t < 0$, the reverse sign gives diverging vacuum coefficients.

Normalized inputs Ω_{K_N} have the same zero positive-time limit, since their normalizing constants have a positive limit. Normalizing each output separately instead gives $\Omega_{(K_N)_t} \to \Omega_{K_t}$. This last operation changes the scalar normalization. It does not make the zero limit invertible.

These limits concern the specified Gaussian families. They do not construct a closable generator on full Hilbert Fock space. The finite-energy sequence in Theorem 6.5.1 still obstructs that extension.

A regularized scalar and its composition defect

For Hilbert–Schmidt K, define

$$\widehat{F}_t(K) = \det{}_2(1 - tK)^{-1/2}, \qquad \log\widehat{F}_t(K) = \frac{1}{2}\sum_{r\ge 2}\frac{t^r}{r}\operatorname{Tr}(K^r).$$

The series converges when $\|tK\| < 1$. It omits the possibly undefined term $t\operatorname{Tr}(K)/2$. That omission changes composition.

Proposition 6.8.2. On the common domain for $s, t, s+t$,

$$\widehat{F}_t(K)\widehat{F}_s(K_t) = \widehat{F}_{t+s}(K)\exp\Big[-\frac{st}{2}\operatorname{Tr}\big(K^2(1-tK)^{-1}\big)\Big]. \tag{6.26}$$

The multiplier depends on K.

Proof. The resolvent identities give

$$(1 - sK_t)(1 - tK) = 1 - (s+t)K, \qquad K_t - K = tK^2(1-tK)^{-1}.$$

In finite rank, ordinary determinants multiply. Their regularizing exponentials leave $-s\operatorname{Tr}(K_t - K)/2$ in the logarithm. This is the displayed exponent.

Hilbert–Schmidt approximation gives trace-norm convergence of the squared kernels and uniformly bounded resolvents. The logarithmic series converge locally uniformly, so the same identity holds in the stated domain. At $K = 0$ the multiplier is one. For nonzero positive rank-one K and positive s, t, it is not one. □

Consequently this multiplier cannot be removed on every squeezed line by a normalization depending only on the coordinate parameters. A map assigned to each such line also does not define a linear map on Fock space.

A chosen reference preparation gives a coherent comparison of lines along its orbit. Fix K_0, put $K_t = K_0(1 - tK_0)^{-1}$, and set

$$g(t) = \widehat{F}_t(K_0), \qquad \mathcal{R}(s,t) = g(t)/g(s).$$

The difference $K_s - K_0$ is trace class, and (6.26) gives

$$\mathcal{R}(s,t) = \widehat{F}_{t-s}(K_s)\exp\Big[\frac{t-s}{2}\operatorname{Tr}(K_s - K_0)\Big].$$

By its ratio definition, $\mathcal{R}(t,u)\mathcal{R}(s,t) = \mathcal{R}(s,u)$. This construction retains the reference preparation and its trace correction.

For a general coordinate germ, only two coefficients act on mode-one polynomials:

$$\lambda = f'(0), \qquad b = B_f(1,1) = \{f,0\}/6.$$

Assume these numbers are real and $\lambda \neq 0$. Let D_λ multiply a creation monomial of length p by λ^p. Each paired leg contributes b, and each unpaired leg contributes λ. Thus $R_f = D_\lambda e^{bL_2}$ on those polynomials.

For trace-class K, the corresponding cutoff limit is

$$R_f S_K = \det(1-bK)^{-1/2} S_{\lambda^2 K(1-bK)^{-1}}.$$

It has the same common-domain qualifications as the cubic formula. When $b = 0$, the formula holds for Hilbert–Schmidt K, provided $\|\lambda^2 K\| < 1$.

Kernel composition gives

$$b_{f\circ g} = b_g + \lambda_g^2 b_f, \qquad \lambda_{f\circ g} = \lambda_f \lambda_g.$$

These data determine the first-oscillator transport. Higher oscillator modes still detect the other coefficients $B_f(m,n)$.

Retaining a trace with the preparation

The contraction completion permits a more general transport datum. Retain modes $1,\dots,d$, with $d < \infty$, and put

$$y_m(v) = x_m(v)/\sqrt{m}, \qquad \mathcal{H} = \mathbb{R}^d \otimes H, \qquad C_{mn} = c_{mn}/\sqrt{mn}.$$

The central matrix C acts identically on species. For trace-class block operators on $\mathcal{H}$, define the partial trace

$$\operatorname{tr}_H(A)_{mn} = \operatorname{Tr}_H(A_{mn}).$$

It is bounded from the trace class to the finite matrix space. No such trace is assigned to an arbitrary Hilbert–Schmidt block operator.

Fix a real symmetric Hilbert–Schmidt strict contraction K on $\mathcal{H}$. Choose independently a real symmetric matrix τ of size d. If K is trace class, its canonical choice is $\tau = \operatorname{tr}_H(K)$.

LEMMA 6.8.3. Every pair (K,τ) has finite-rank symmetric approximations K_N with

$$\|K_N - K\|_2 \longrightarrow 0, \qquad \operatorname{tr}_H(K_N) = \tau.$$

The approximations are uniformly strict contractions for sufficiently large N. If $K \geq 0$ is not trace class, approximations with bounded prescribed traces cannot all be positive.

Proof. Compress K to the first N species in each mode, obtaining K_N°. Put $\delta_N = \tau - \operatorname{tr}_H(K_N^\circ)$. Choose an orthogonal reservoir of M_N further species and set

$$K_N = K_N^\circ + \delta_N \otimes P_{\mathrm{res}}/M_N, \qquad M_N > N^2\|\delta_N\|_F^2 + N + 1.$$

The added term has partial trace δ_N and Hilbert–Schmidt norm $\|\delta_N\|_F/\sqrt{M_N} < 1/N$. Hence $K_N \to K$ in Hilbert–Schmidt norm, and therefore in operator norm. The strict-contraction bound follows.

For positive approximations and any fixed finite-rank projection Q,

$$\operatorname{Tr}(QKQ) = \lim_N \operatorname{Tr}(QK_NQ) \le \liminf_N \operatorname{Tr} K_N.$$

The supremum of the left side is infinite for a positive non-trace-class K. Thus bounded prescribed traces require signed reservoir corrections in that case. □

In the product of energy sectors define

$$\begin{aligned}\mathscr{S}_{K,\tau}(y,C) = \exp\left[\frac{1}{2}\operatorname{tr}_d(C\tau) + \frac{1}{2}\sum_{r\ge 2}\frac{\operatorname{Tr}((CK)^r)}{r}\right] \\ \cdot \exp\left[\frac{1}{2}y^tK(1-CK)^{-1}y\right].\end{aligned} \tag{6.27}$$

The inverse is expanded in the positive-energy central variables. The traces for $r \ge 2$ exist because each contains two Hilbert–Schmidt factors.

Proposition 6.8.4. The formula defines an element of $\prod_E \mathcal{W}_E$. For every sequence from Lemma 6.8.3, it is the coefficientwise limit of jS_{K_N}. Under the sectorwise extension of π, it gives the Hilbert squeezed vector S_K.

Proof. In finite rank the Gaussian heat equation gives

$$e^{\Delta_c}e^{y^tKy/2} = \det(1-CK)^{-1/2}e^{y^tK(1-CK)^{-1}y/2}.$$

Separating the determinant's linear term gives (6.27). Closed cycles of length at least two have trace-class products. Open paths give Hilbert–Schmidt kernels, whose products give the required Hilbert tensors.

At fixed energy only finitely many paths, cycles and oscillator degrees occur. Their coefficients converge in trace norm or Hilbert tensor norm. The one-edge cycles converge by the prescribed partial trace τ. This proves the limit and its independence of the approximating sequence. At $C = 0$ the formula is S_K. Theorem 6.7.3 applies to the one-particle space $\mathcal{H}$. □

Response, trace and scalar under composition

A finite-mode Gaussian transformation consists of $T \in \mathrm{GL}_d(\mathbb{R})$ and $B \in \mathrm{Sym}_d(\mathbb{R})$. On polynomials it substitutes $y \mapsto Ty$ after contracting pairs with B. On the completed polynomial coordinates this is

$$y \longmapsto Ty, \qquad C \longmapsto TCT^t + B.$$

A coordinate germ gives such a pair by

$$T_{ij} = \sqrt{j/i}\,T_f(i,j), \qquad B_{ij} = B_f(i,j)/\sqrt{ij}.$$

An arbitrary matrix pair need not arise from a curve coordinate.

Write $\widetilde{T} = T \otimes 1_H$ and $\widetilde{B} = B \otimes 1_H$. Define

$$K^B = K(1-\widetilde{B}K)^{-1},$$

$$K' = \widetilde{T}^tK^B\widetilde{T}, \tag{6.28}$$

$$\tau' = T^t\tau T + \operatorname{tr}_H\big(\widetilde{T}^t(K^B-K)\widetilde{T}\big), \tag{6.29}$$

$$F_{T,B}(K,\tau) = e^{\operatorname{tr}_d(B\tau)/2}\det{}_2(1-\widetilde{B}K)^{-1/2}. \tag{6.30}$$

The resolvent identity

$$K^B = (1 - K\widetilde{B})^{-1}K$$

shows that the response is symmetric. The partial trace in (6.29) is legitimate:

$$K^B - K = K(1 - \widetilde{B}K)^{-1}\widetilde{B}K \in \mathcal{S}_1(\mathcal{H}). \tag{6.31}$$

Its two outside kernels are Hilbert–Schmidt.

Use a common domain where $\|\widetilde{B}K\| < 1$ and every response kernel is a strict contraction. These conditions must hold for each intermediate and composite transformation. The logarithmic determinant series beginning at degree two fixes its branch at the identity.

THEOREM 6.8.5. On this domain, the finite-rank transformations have the Gaussian cutoff limit

$$\mathscr{S}_{K,\tau} \longmapsto F_{T,B}(K,\tau)\mathscr{S}_{K',\tau'}.$$

If (T_2, B_2) is applied first, followed by (T_1, B_1), their composite data are

$$T = T_2T_1, \qquad B = B_2 + T_2B_1T_2^t. \tag{6.32}$$

The one-step and two-step responses and trace matrices agree, and

$$F_{T_2,B_2}(K,\tau)F_{T_1,B_1}(K_2,\tau_2) = F_{T,B}(K,\tau). \tag{6.33}$$

These identities hold under every finite regrouping of admitted transformations. No commutativity assumption is needed.

Proof. Finite-dimensional Gaussian heat integration followed by linear substitution gives (6.28) and the scalar $\det(1 - \widetilde{B}K)^{-1/2}$. Separating its linear trace term gives (6.30) when $\tau = \operatorname{tr}_H K$. Taking the partial trace of the response gives (6.29).

Apply the approximations of Lemma 6.8.3. Resolvents converge in operator norm, responses in Hilbert–Schmidt norm, and (6.31) in trace norm. For $r \geq 2$, the logarithmic series obeys

$$|\operatorname{Tr}((\widetilde{B}K)^r)| \leq \|\widetilde{B}K\|_2^2\|\widetilde{B}K\|^{r-2}.$$

It converges locally uniformly. Hence the scalar and trace matrix converge.

The Gaussian lift proposition gives the sectorwise limit. The overlap formula gives its Hilbert limit after applying π.

The substitutions on y, C give (6.32). The resolvent identity gives the same composition for K'. For the trace update, subtract the purely linear transform of K from the final response. Insert the intermediate response between them and take the partial trace. The two trace-class differences telescope, and finite mode matrices pass outside the partial trace. This proves the trace law.

It remains to check the scalar, including the sign of its regularized determinant. Put

$$A = \widetilde{B}_2K, \qquad D = \widetilde{T}_2\widetilde{B}_1\widetilde{T}_2^t, \qquad Q = DK.$$

By similarity, the second determinant has argument $1 - Q(1 - A)^{-1}$. The exact factorization is

$$(1 - Q(1 - A)^{-1})(1 - A) = 1 - A - Q.$$

For Hilbert–Schmidt U, V, the regularized product identity is

$$\det{}_2((1-U)(1-V)) = \det{}_2(1-U)\det{}_2(1-V)e^{-\operatorname{Tr}(UV)}.$$

In finite rank this follows by separating the linear trace from each ordinary determinant. Approximation in Hilbert–Schmidt norm extends it, since UV converges in trace norm.

This is also the $p = 2$ product formula for regularized Fredholm determinants.[2]

The two inverse square roots therefore leave the multiplier

$$\exp\Big[-\frac{1}{2}\operatorname{Tr}\big(Q(1-A)^{-1}A\big)\Big].$$

The trace update contributes its inverse:

$$\begin{aligned}\frac{1}{2}\operatorname{tr}_d(B_1\tau_2) - \frac{1}{2}\operatorname{tr}_d(T_2B_1T_2^t\tau) &= \frac{1}{2}\operatorname{Tr}\big(D(K^{B_2}-K)\big)\\ &= \frac{1}{2}\operatorname{Tr}\big(Q(1-A)^{-1}A\big).\end{aligned}$$

All traced products are trace class. The exponents cancel, proving (6.33). Associativity of the matrix pair law then proves every finite regrouping on its common domain. □

For $d = 1$, $T = 1$ and $B = t$, the construction becomes

$$K_t = K(1-tK)^{-1}, \qquad \tau_t = \tau + \operatorname{Tr}(K_t - K), \qquad F_t = e^{t\tau/2}\det{}_2(1-tK)^{-1/2}.$$

Differentiating the convergent series gives

$$\dot K_t = K_t^2, \qquad \dot\tau_t = \operatorname{Tr}(K_t^2), \qquad \frac{d}{dt}\log F_t = \tau_t/2.$$

The trace-class difference replaces an undefined $\operatorname{Tr} K$. Forgetting τ recovers the defect in (6.26).

Two trace data $\tau, \widehat\tau$ for the same K project to the same Hilbert vector S_K. Their transformed Hilbert limits instead differ by

$$\exp\Big[\frac{1}{2}\operatorname{tr}_d(B(\tau-\widehat\tau))\Big].$$

If this scalar is not one, no continuous operator on the old Hilbert space can realize both cutoff limits. Signed reservoirs do not alter the divergence of the original positive-cutoff sequence for a positive non-trace-class kernel. They specify a different trace datum.

Even the response endpoints do not determine the scalar. On one occupied oscillator take $0 < \kappa < 1$, $K = \tau = \kappa$, $T = a > 0$ near one, and $B = (1-a^2)/\kappa$. Then

$$K' = \kappa, \qquad F = a^{-1}.$$

The coordinate germ $f(z) = az/\sqrt{1-2Bz^2}$ has these first-mode data. The coordinate arrow and its normalization must therefore be retained even when the response returns to its initial kernel.

These formulas define a Gaussian domain in $\prod_E \mathcal{W}_E$. They do not extend every coordinate change to that unrestricted product. Coordinate contractions can lower energy without bound. The Gaussian resolvent is what supplies convergence on this domain.

[2] T. Britz, A. Carey, F. Gesztesy, R. Nichols, F. Sukochev and D. Zanin. *The product formula for regularized Fredholm determinants*, equations (1.3)–(1.5), arXiv:2007.12834.

Unitary implementation is a separate condition

Let a bounded invertible real symplectic map act on a one-particle Hilbert space. Relative to its chosen complex structure, write α for its linear block and β for its antilinear block. Shale's criterion says that its action on the canonical commutation relations has a unitary Fock implementer exactly when β is Hilbert–Schmidt.[3] A formal coordinate germ need not define such a bounded symplectic Hilbert map.

If the mode action is identical in every species, then $\beta = \beta_{\mathrm{mode}} \otimes 1_H$. For a unit mode vector v with $\beta_{\mathrm{mode}} v \neq 0$,

$$\sum_i \|\beta(v \otimes e_i)\|^2 = \sum_i \|\beta_{\mathrm{mode}} v\|^2 = \infty.$$

Thus this action has no unitary implementer. When $\beta = 0$ and α is unitary, ordinary second quantization does provide one.

For the uniform one-mode squeeze $\alpha = 5\,1_H/4$, $\beta = 3\,1_H/4$, the finite-species kernel is $K_N = 3P_N/5$, and

$$\|\beta_N\|_2^2 = 9N/16, \qquad \langle \mathbf{1}, \Omega_{K_N} \rangle = (16/25)^{N/4} \longrightarrow 0.$$

This distinguishes unitary implementation from convergence of selected preparations and from covariance of finite-energy fields.

A continuous prepared functional on a bounded-energy projective sector can be transported by precomposition with R_f. Preservation of positivity additionally requires its adjoint and positive cone to be preserved. This covector construction alone supplies neither a global state nor a physical sewing map.

6.9 State families and completed field operations

The preceding completions control species tensors or specified Gaussian families. A formal parameter raises a different question. When may that parameter be identified with an insertion coordinate? An arbitrary formal series does not suffice.

Return to a characteristic-zero field k with discrete coefficients. For the current $J \in H_\kappa$, introduce a parameter ξ, independent of the insertion coordinate u. The state family

$$A(\xi) = \sum_{r \geq 0} \frac{(-1)^r \xi^r}{(r+1)!} T^r J$$

exists in $H_\kappa[[\xi]]$. Its coefficientwise field on J has vacuum polar part

$$\kappa \sum_{r \geq 0} \xi^r u^{-r-2} \mathbf{1}.$$

The partial state sums converge ξ-adically. After the substitution $\xi = u$, their $u^{-2}\mathbf{1}$ coefficient is $N\kappa$ for a sum of N terms. It does not stabilize. Thus this substitution is not continuous on all formal state families with discrete scalars.

The exact domain for diagonal evaluation

Let V be a vertex algebra over k. The same discussion permits a differential graded vertex algebra, with all series taken in each cohomological degree. Give $V((u))$ the topology with neighborhoods $u^q V[[u]]$, $q \in \mathbb{Z}$. For $a, b \in V$, put

$$v_b(a) = \mathrm{ord}_u\, Y(a,u)b, \qquad \mathrm{ord}_u\, 0 = +\infty.$$

[3] L. Bruneau and J. Dereziński, *Bogoliubov Hamiltonians and one-parameter groups of Bogoliubov transformations*. Definition 3.1, Assumption 3.A and Theorem 3.2, arXiv:math-ph/0511069.

Define

$$\mathscr{A}_Y = \left\{ \sum_{r\ge 0} \xi^r a_r : r + \nu_b(a_r) \longrightarrow +\infty \text{ for every } b \in V \right\}. \tag{6.34}$$

The parameter ξ has cohomological degree zero. For a finite set $B \subset V$ and $q \in \mathbb{Z}$, let $U_{B,q}$ require

$$r + \nu_b(a_r) \ge q \quad \text{for every } r \ge 0,\ b \in B.$$

These subspaces define a linear topology on $\mathscr{A}_Y$. Equip $\mathrm{Hom}_k(V, V((u)))$ with pointwise Laurent convergence.

Proposition 6.9.1. Polynomial families are dense in $\mathscr{A}_Y$. Their diagonal evaluation extends uniquely to a continuous map

$$\mathrm{ev}_Y : \mathscr{A}_Y \longrightarrow \mathrm{Hom}_k(V, V((u))), \qquad \mathrm{ev}_Y(A)(u)b = \sum_{r\ge 0} u^r Y(a_r, u)b.$$

The limit condition in (6.34) is necessary and sufficient for convergence on every input. The domain is preserved by coefficient translation T, by ∂_ξ, and by the differential in the graded case. Evaluation commutes with that differential and satisfies

$$\mathrm{ev}_Y((T + \partial_\xi)A) = \partial_u \,\mathrm{ev}_Y(A), \qquad [T, \mathrm{ev}_Y(A)] = \mathrm{ev}_Y(TA).$$

Proof. The limit condition makes the series converge in the complete Laurent space on each fixed input. Conversely, successive differences of convergent partial sums must tend to zero, giving the same condition. The image of $U_{B,q}$ sends every $b \in B$ into $u^q V[[u]]$. This proves continuity.

Truncation tails tend to zero in every $U_{B,q}$, so polynomials are dense. The target is Hausdorff, proving uniqueness. The source is Hausdorff as well: the vacuum identity gives $\nu_{\mathbf{1}}(a) = 0$ for every nonzero a.

Translation covariance gives $\nu_b(Ta) \ge \nu_b(a) - 1$. Reindexing gives the same loss of at most one for ∂_ξ. For homogeneous a, the differential identity gives

$$\nu_b(da) \ge \min\{\nu_b(a), \nu_{db}(a)\}.$$

These estimates preserve the domain and justify termwise differentiation.

The product rule on $u^r Y(a_r, u)$ gives the first translation identity. Summing $[T, Y(a_r, u)] = Y(Ta_r, u)$ gives the second. The graded Hom differential gives the differential assertion. □

There are admissible infinite families with unbounded derivative order. For example,

$$\widetilde{A}(\xi) = (1 + \xi) \sum_{r\ge 0} \frac{(-1)^r \xi^{2r}}{(r+1)!} T^r J$$

belongs to $\mathscr{A}_Y$, since $\nu_b(T^r J) \ge \nu_b(J) - r$. Its vacuum coefficient on J after evaluation is

$$\kappa u^{-2}(1 + u) \sum_{r\ge 0} u^r = \kappa u^{-2} \frac{1 + u}{1 - u}.$$

This is expanded at $u = 0$. The parameter degree now outruns the derivative pole order.

A weight completion closed under every mode

Suppose V also has a nonnegative integer weight grading with

$$V=\bigoplus_{h\ge 0}V_h,\quad \mathbf{1}\in V_0,\quad T(V_h)\subset V_{h+1},\quad d(V_h)\subset V_h,\quad (V_h)_{(n)}V_{h'}\subset V_{h+h'-n-1}.$$

Negative weight spaces are zero. Weight is independent of cohomological degree, and its spaces need not be finite dimensional. Let $w(a)$ be the largest weight of a nonzero component of a, with $w(0)=-\infty$.

Put

$$\mathscr{A}_{\mathrm{wt}}(V)=\left\{\sum_{r\ge 0}\xi^r a_r : r-w(a_r)\longrightarrow+\infty\right\}. \tag{6.35}$$

Define

$$\gamma(A)=\inf_{r\ge 0}\{r-w(a_r)\},\qquad \mathcal{F}^q=\{A:\gamma(A)\ge q\},\qquad \gamma(0)=+\infty.$$

The $\mathcal{F}^q$ are the neighborhoods of zero. Each sum is still taken in a fixed cohomological degree.

On polynomials $W=V\otimes k[\xi]$, use the tensor-product fields

$$Y_W(\xi^r a,z)(\xi^s b)=(\xi+z)^r\xi^s Y_V(a,z)b. \tag{6.36}$$

The second factor has its commutative differential-algebra vertex structure. Translation on W is $D=T+\partial_\xi$. In an iterated field, the shifted parameter becomes $\xi+z_2+z_0$. This equals $\xi+z_1$ under $z_1=z_2+z_0$. Thus the tensor product retains the vertex identities.

THEOREM 6.9.2. The space $\mathscr{A}_{\mathrm{wt}}(V)$ is the Hausdorff completion of W for the degree $\deg(\xi^r a)=r-h$, $a\in V_h$. Every mode extends to the continuous bilinear product

$$A_{[n]}B=\sum_{r,s\ge 0}\sum_{j=0}^{r}\binom{r}{j}\xi^{r+s-j}(a_r)_{(n+j)}b_s. \tag{6.37}$$

Each parameter coefficient is a finite sum, and

$$\gamma(A_{[n]}B)\ge\gamma(A)+\gamma(B)+n+1,\qquad \gamma(A_{[n]}B)-n\longrightarrow+\infty\quad(n\to+\infty). \tag{6.38}$$

The vacuum, differential and translation extend with their usual identities. For homogeneous A,B,C and integers p,q,n, the three series in the identity

$$\begin{aligned}\sum_{j\ge 0}\binom{p}{j}(A_{[n+j]}B)_{[p+q-j]}C=&\sum_{j\ge 0}(-1)^j\binom{n}{j}A_{[p+n-j]}(B_{[q+j]}C)\\&-(-1)^{n+|A||B|}\sum_{j\ge 0}(-1)^j\binom{n}{j}B_{[q+n-j]}(A_{[p+j]}C)\end{aligned} \tag{6.39}$$

converge and satisfy the displayed equality. Every fixed coefficient of ξ receives only finitely many summands from each series.

Proof. Decompose the polynomial space as

$$W=\bigoplus_{\delta\in\mathbb{Z}}W_\delta,\qquad W_\delta=\bigoplus_{\substack{r,h\ge 0\\ r-h=\delta}}\xi^r V_h.$$

Its completion consists of sums bounded below in δ, with finitely many pairs (r, h) below every degree cutoff. This is equivalent to the limit condition in (6.35). For fixed r, every weight satisfies $r - h \le r$, so its coefficient remains a finite sum in V. This proves completeness, Hausdorffness and density of the polynomial subspace.

Expanding (6.36) gives the mode formula on polynomials. A nonzero summand at parameter exponent $\ell = r + s - j$ has weight at most $w(a_r) + w(b_s) - n - j - 1$. Hence

$$\ell - w((a_r)_{(n+j)}b_s) \ge (r - w(a_r)) + (s - w(b_s)) + n + 1.$$

Nonnegative output weight also gives

$$(r - w(a_r)) + (s - w(b_s)) \le \ell - n - 1.$$

The two sequences on the left tend to infinity and are bounded below. Only finitely many pairs (r, s) can contribute to a fixed ℓ. Then j is determined.

As $\ell \to \infty$, any contributing pair satisfies $r + s \ge \ell$. Its sum of surpluses therefore tends to infinity. This proves that the product belongs to the completion, with the first bound in (6.38). The bound also proves continuity.

Approximate A, B by polynomials A_0, B_0 with errors of arbitrarily large γ. For all sufficiently large n, $(A_0)_{[n]}B_0 = 0$. Apply the first bound to

$$(A - A_0)_{[n]}B + (A_0)_{[n]}(B - B_0).$$

After subtracting n, its lower bound can be made arbitrarily large, uniformly in n. This proves the second limit.

The differential preserves the filtration. D lowers it by at most one. Thus their polynomial identities extend continuously.

Each summand in the three iterated products of (6.39) has filtration lower bound

$$\gamma(A) + \gamma(B) + \gamma(C) + p + q + n + 2.$$

This bound is independent of j. Replacing any input by a high-filtration error raises it uniformly. Approximate all three inputs by polynomials with bounded-below filtrations. Their three series are finite, by ordinary mode truncation. Beyond the last polynomial term, every completed summand therefore lies in any prescribed $\mathcal{F}^N$, once the approximations are sufficiently close.

The series converge and inherit the polynomial Borcherds identity.

Finally, an element of $\mathcal{F}^{L+1}$ has zero ξ^L coefficient, since that coefficient would have negative weight. The preceding convergence proof places all sufficiently late summands there. This proves the asserted finiteness of each parameter coefficient. □

Modules and the lower weight bound

Let $M = \bigoplus_{\ell \ge b_0} M_\ell$ be a weight-graded V-module, where $b_0 \in \mathbb{Z}$, the differential preserves weight, and

$$(V_h)_{(m)}M_\ell \subset M_{h+\ell-m-1}.$$

Equip $\mathrm{Hom}_k(M, M((u)))$ with pointwise Laurent convergence.

PROPOSITION 6.9.3. Polynomial evaluation extends continuously to

$$\widehat{Y}_M : \mathscr{A}_{\mathrm{wt}}(V) \longrightarrow \mathrm{Hom}_k(M, M((u))), \qquad \widehat{Y}_M(A,u)b = \sum_{r\geq 0} u^r Y_M(a_r,u)b.$$

Its mode on $b \in M_\ell$ is the finite sum

$$A^M_{[m]}b = \sum_{r\geq 0}(a_r)_{(m+r)}b, \tag{6.40}$$

and

$$\mathrm{ord}_u\, \widehat{Y}_M(A,u)b \geq \gamma(A) + b_0 - \ell.$$

The module identities hold with the completed products (6.37). Their sums are locally finite on each fixed module vector. The action is natural in strict weight-preserving module maps, commutes with the differential, and intertwines D with ∂_u.

Proof. The r-th summand of (6.40) has weight at most

$$\ell - m - 1 - (r - w(a_r)).$$

It vanishes when this is less than b_0. The surplus limit proves finiteness and the common Laurent lower bound. That same bound proves continuity. On polynomials this is the tensor-product module where $f(\xi)$ acts by $f(u)$.

In a module Borcherds identity, each composite summand sends a weight-ℓ input to weights at most

$$\ell - p - q - n - 2 - \gamma(A) - \gamma(B),$$

independently of the summation index. A sufficiently high-filtration error in either input annihilates every summand. Replace both inputs by polynomials at that precision. The polynomial identity is finite on the given vector and agrees term by term with the completed one. This proves local finiteness and the identity. The differential, naturality and translation statements follow from the polynomial identities by the same estimate. □

The regular module gives $\mathscr{A}_{\mathrm{wt}}(V) \subset \mathscr{A}_Y$. It also contains the derivative-controlled families with

$$a_r \in \sum_{j=0}^{d_r} T^j E, \qquad r - d_r \longrightarrow +\infty,$$

where E is a fixed finite-dimensional state space. If c is its maximum weight, then $w(a_r) \leq c + d_r$. The stated inclusion follows directly.

The lower module bound is substantial. Consider the weak Heisenberg module

$$M = k[y, x_2, x_3, \ldots], \qquad w(y) = -1, \quad w(x_m) = m,$$

with operators

$$J_1 = y, \qquad J_{-1} = -\kappa\partial_y, \qquad J_0 = 0, \qquad J_m = m\kappa\partial_{x_m}, \quad J_{-m} = x_m \quad (m \geq 2).$$

The identities $[y, -\kappa\partial_y] = \kappa$ and $[m\kappa\partial_{x_m}, x_m] = m\kappa$ give the current commutators. Each polynomial is annihilated by sufficiently large positive modes. The normally ordered derivative fields therefore define a weak H_κ-module.

At each output power only finitely many negative-mode indices can contribute, after the finitely many possible positive modes have acted. Wick expansion proves the module field identities on this polynomial space.

Put $a_r = J^r_{-1}\mathbf{1}$ in the state algebra and

$$A(\xi) = \sum_{r\ge 0} \xi^{2r} a_r, \qquad A_N(\xi) = \sum_{r=0}^{N} \xi^{2r} a_r.$$

The state weight is $w(a_r) = r$. Thus $A_N \to A$ in the weight completion, with $\gamma(A - A_N) = N + 1$. On the constant polynomial $1 \in M$,

$$[u^0] \sum_{r=0}^{N} u^{2r} Y_M(a_r, u)1 = \sum_{r=0}^{N} y^r.$$

Indeed $Y_M(a_r, u) =: J(u)^r :$. After normal ordering, all positive modes except $J_1 = y$ annihilate 1. The coefficient of u^{-2r} comes only from r copies of yu^{-2}. Every negative-mode factor has a nonnegative power of u and gives a larger exponent. This proves the displayed formula.

Its successive constant coefficients differ by $y^{N+1} \neq 0$. The evaluations are not Cauchy modulo $uM[[u]]$. Consequently no continuous extension of polynomial evaluation exists on this module with the discrete Laurent target. The constant polynomial $1 \in M$ is not a highest-weight vacuum: $J_1 1 = y$. The obstruction concerns this module and target, while the lower-bounded action remains defined.

Common-scale fields and the geometric boundary

For n insertions put $z_i = t x_i$ with $x_1 = 1$, and let

$$R_n = k[x_2, \dots, x_n][\, x_i^{-1}, (x_i - x_j)^{-1} : 2 \le i \le n,\ 1 \le i < j \le n \,].$$

The first inverses keep the insertions away from zero. The second keep their positions distinct.

For homogeneous states of weights h_i and an input $b \in M_\ell$, the simultaneous rational product has its t^d coefficient in

$$M_{\ell + \sum_i h_i + d} \otimes R_n.$$

To obtain it, clear the finitely many pairwise locality poles and use the module field identities to compare orders. Place each insertion in turn at the rightmost position. Lower truncation on b then gives a lower bound for that variable's exponent. The pole-clearing factors are polynomials, so a common lower bound survives in every variable.

The resulting regular coefficients have fixed total degree and finitely many monomials at each degree. Dividing by the cleared factors gives the same rational coefficient in every chamber. The weight bound makes its t-order at least $b_0 - \ell - \sum_i h_i$.

For families $A_i = \sum_{r\ge 0} \xi^r a_{i,r}$, their term of parameter degrees r_i therefore has order at least

$$b_0 - \ell + \sum_i (r_i - w(a_{i,r_i})).$$

Only finitely many tuples lie below any cutoff. Hence these simultaneous fields converge in

$$(M \otimes R_n)((t))$$

for all $A_i \in \mathscr{A}_{\mathrm{wt}}(V)$. This gives a common-scale evaluation on every lower-bounded module.

The completed products need not have finite mode truncation. In H_κ, take

$$A(\xi) = \sum_{r\geq 0} \frac{\xi^{2r}}{r!} T^r J.$$

Its coefficient of ξ^{2r} has weight $r+1$, so it belongs to the weight completion. For $n \geq 1$,

$$A_{[n]}J = \kappa \sum_{r\geq n-1} (-1)^r (r+1) \binom{2r}{r+1-n} \xi^{r+n-1}\mathbf{1}.$$

This follows from $(T^r J)_{(r+1)}J = (-1)^r (r+1)!\kappa\mathbf{1}$ and the completed mode formula. Its first term is $(-1)^{n-1} n\kappa\xi^{2n-2}\mathbf{1}$. Thus every displayed mode is nonzero, while

$$\gamma(A_{[n]}J) - n = n - 2 \longrightarrow +\infty.$$

The mode identities converge despite the absence of finite singular-mode truncation.

The same example exhibits a displaced collision in its common-scale field. Its vacuum coefficient with $J(z_2)$ is

$$\sum_{r\geq 0} \frac{z_1^{2r}}{r!} \partial_{z_1}^r \frac{\kappa}{(z_1 - z_2)^2} = \frac{\kappa}{(z_1 + z_1^2 - z_2)^2}.$$

For $z_1 = t$, $z_2 = tx_2$, the right side is expanded as $\kappa t^{-2}(1 - x_2 + t)^{-2}$ in $R_2((t))$. Its displaced pole explains why closed completed modes do not automatically define residue maps on the original collision carrier.

Question 6.9.4. For a specified completion and its module domain, construct collision extensions and normal residue maps compatible with the completed field products and differential-module coefficients. For independently defined arithmetic coefficients, construct a supported comparison preserving all collision channels, both vacuum maps and the prepared functional. Which geometric data determine compatible bidifferentials and trace matrices, and which comparisons preserve the scalar under the prescribed sewing of states?

The algebraic Heisenberg background already has its cut bar comparison and framed field recovery. Its analytic completions retain the operations proved on their specified domains above. An arithmetic use must additionally construct the coefficient and state comparison in the question. Positivity of an independently defined Weil form, a derived factorizing center and Hilbert unitary implementation remain separate requirements.

Chapter 7

Comparison problems for the first three volumes

7.1 Independent comparison

The constructions of the first three volumes give particular comparison problems of the kind defined in Chapter 1. Their coefficient categories, modules, operations and evaluation maps must be compared before the resulting scalar identities can be interpreted geometrically.

Fixing an invariant functor specifies what an evaluation retains. A comparison on the original category can imply an equality of such invariants. Recovering a comparison from that equality requires further hypotheses on the functor; the character example of Section 1.1 shows the failure when it forgets extensions.

7.2 The realization pair

Definition 7.2.1 *(Comparison of invariants).* Let $\mathscr{C}$ be a category and $I : \mathscr{C} \to \mathrm{Set}$ a functor. A *comparison of invariants* is a pair (d, e), where d consists of objects A, B and a map $f : A \to B$, and e consists of specified elements $a \in I(A)$, $b \in I(B)$ with $I(f)(a) = b$. When a statement T in a family X asks for this identification, write $\mathrm{Real}(X, T) = (d, e)$ for a chosen pair. The map f is part of the data. Neither its existence nor its invertibility follows from the numerical values of a, b.

Proposition 7.2.2 *(Coherent comparison of a diagram).* Let J be a small category, let $U : \mathscr{R} \to \mathscr{D}$ be a functor between categories of realizations and retained data, and let $d : J \to \mathscr{D}$ be a diagram. For each object j, choose $r_j \in \mathscr{R}$ and an isomorphism $\alpha_j : U(r_j) \to d(j)$. Suppose that every arrow $a : j \to k$ has a chosen lift $r_a : r_j \to r_k$, with

$$\alpha_k U(r_a) = d(a)\alpha_j, \qquad r_{1_j} = 1_{r_j}, \qquad r_{ba} = r_b r_a.$$

Then these data define a functor $r : J \to \mathscr{R}$ and a natural isomorphism $Ur \cong d$. Conversely every such functor and natural isomorphism supplies the displayed data. If U is fully faithful, the chosen objects and isomorphisms determine all the lifts and their identities uniquely.

Proof. The unit and composition identities are exactly the functor axioms; the first equation is naturality of α. Conversely these axioms hold for every functor and natural isomorphism. If U is fully faithful, the map $\alpha_k^{-1} d(a) \alpha_j$ has exactly one lift to $\mathscr{R}$. Faithfulness then proves the unit and composition identities by applying U to both sides. Objectwise existence without these hypotheses is insufficient, as the nonsplit $BC_4 \to BC_2$ example in Section 1.4 shows. □

Definition 7.2.3 *(A restricted family of comparison problems).* Fix a category $\mathscr{I}$ of specified parameterized constructions and their specialization maps. Let $\mathscr{I}_{\mathrm{nd}}$ be its full subcategory obtained by omitting the critical affine level $k = -h^\vee$, the Yangian parameter value $\Upsilon = 0$, the chosen admissible logarithmic sectors, the super-Yangian coproduct comparison, and the five Calabi–Yau comparison constructions

$R_2, \ldots, R_6$ described below. The meaning of the parameters and sectors is that of the respective construction. This is a restriction of the comparison problem; no existence or regularity theorem is included in the notation.

The omitted cases need their own coefficient categories and maps. Their omission gives no automatic comparison on the remaining objects. In particular, a failure of one normalization at a critical parameter does not prove failure of every realization at that parameter.

7.3 SEVEN STRUCTURAL COMPARISON PROBLEMS

The seven families considered here concern Koszul reflection, the coderived–contraderived correspondence, derived centres, higher-genus obstructions, Hochschild cohomology, celestial boundary algebras, and reconstruction from higher invariants. Each comparison must use the hypotheses and the mathematical recipient of the particular source statement. A common name for the families gives no common existence theorem.

Open Problem 7.3.1 *(Structural realization comparisons).* For each specified object in one of these seven families, construct an independent realization of the retained operations and its comparison map. Determine its coefficient domain, its preserved structures, and its compatibility with specialization. Where only scalar invariants can presently be compared, determine which additional operations the comparison would have to retain.

For Koszul reflection, a filtered twisting-morphism construction and a factorization-homology construction provide possible starting objects. A comparison on an ordered Orlik–Solomon model would have to identify their differentials and tensor conventions, and then prove the claimed compatibility with gluing. Neither the existence of the two constructions nor equality of their graded dimensions supplies this map.

For the coderived–contraderived correspondence, specify the curved coalgebra, the two categories of modules, and the totalizations in the adjunction. An abstract adjoint functor theorem can construct an adjoint under its hypotheses; an equivalence additionally requires the unit and counit to be equivalences on the stated categories. The chain-level bicomplex and the categorical adjunction must be compared with these conditions intact.

For a derived centre, a mapping-stack or shifted-symplectic construction can serve as a comparison only after its stack, tangent complexes and shift have been defined on the chosen objects. The required map must identify the centre's operations and the specified duality involution. A symplectic form on a mapping stack does not itself construct that identification.

For a higher-genus obstruction expressed in terms of a Hodge class, Riemann–Roch and curvature formulas can compute characteristic classes of specified bundles on the universal curve. A comparison must identify the actual obstruction class with that characteristic class, including any term contributed by gluing different channels. The Hodge-bundle calculation alone supplies neither identification.

For Hochschild cohomology, the Heisenberg, Virasoro and affine examples have distinct candidate comparison complexes: Lie-algebra cochains, BRST cochains and relative cohomology complexes. Each needs its own map to the bar model. A proposed concentration statement must specify the completed or bounded complex to which it applies and retain any quotient or Koszul hypotheses. A finite-family calculation gives no concentration theorem for every chiral algebra.

For a celestial boundary algebra, an operator-product calculation in a holomorphic gauge theory gives local coefficients on its specified background. To compare it with an independently defined boundary

algebra, construct the boundary fields, their allowed singular products and the reduction map, and prove their compatibility with composition. A matching $w_{1+\infty}$ coefficient is an invariant of such a comparison, not its construction.

For reconstruction from a sequence of higher invariants, the comparison must show that the invariant functor distinguishes the permitted objects and morphisms. A Calabi–Yau categorification or an identification of numerical characters can give part of the data. One must still prove convergence, recover the retained operations, and control automorphisms. The realization groupoids of Section 1.4 separate existence from these uniqueness requirements.

7.4 Objectwise comparisons and their scope

Remark 7.4.1 *(Objectwise comparison criterion).* For a fixed statement, a comparison of evaluations exists when the actual constructions and calculation in Definition 7.2.1 are supplied. Restricting the parameter domain or changing the references does not supply them. Even after they have been supplied for each object, a functorial comparison still requires the lifted arrows and composition identities of Proposition 7.2.2.

Remark 7.4.2 *(Realization boundary).* A comparison of evaluations is weaker than a marked realization of the algebra, modules, higher operations and state. The two-dimensional modules in Section 1.1 have identical characters and no module isomorphism. Thus even a complete list of the chosen scalar comparisons would not establish all the structural comparisons above.

7.5 Boundary strata

The following are additional comparison problems for the omitted sectors. They are not an exhaustion of all possible failures of existence, uniqueness or continuity.

Degenerate admissible logarithmic sector

The Creutzig–Kanade–Linshaw admissible logarithmic sector for admissible representations of affine $\mathfrak{sl}_2$ at $k + 2 = p/q$ with $\gcd(p, q) = 1$, $q \geq 2$, lies outside the chosen restricted family. A comparison in this sector must retain the logarithmic representation structure. An argument using only a generic semisimple model does not supply that comparison.

Super-Yangian chain-level chiral coproduct

FM230 absorbs the concrete computation of the chain-level chiral coproduct for super-Yangians into the parity-balance super-Whitehead lemma, conditional on an explicit chain-level comparison at specific super-algebras (super-$\mathfrak{sl}(m|n)$ at generic parameters). The independent comparison in this sector is the comparison with the appropriate super-Feigin–Frenkel screening.

Vol III CY-C conditional bridges

Vol III gives six constructions to the quantum vertex chiral group $G(K3 \times E)$: R_1 the CY-to-chiral framed assignment $\Phi_3^{(\Sigma_2, C)}$; R_2 the Borcherds multiplicative lift; R_3 the Mukai-lattice VOA; R_4 the Kummer construction; R_5 the $\mathcal{N} = (2, 2)$ sigma-model half-twist; R_6 the BLLPR Schur-limit construction. Only R_1 is a Φ_3 application; CY-C is the assertion that R_2–R_6 identify as (Σ_2, C)-specialisations of one canonical Stage-1 datum. R_1 carries a proof of convergence at $d = 2$ via the framed Stage-1 output. The other five constructions carry *conditional* forms tied to convergence hypotheses:

(i) R_2 (Borcherds multiplicative lift): conditional on convergence of the Igusa cusp form product formula at genus 2 against the R_1 Stage-1 normalisation.

(ii) R_3 (Mukai-lattice VOA): conditional on identification of the Mukai-lattice OPE with the $\Phi_3^{(\Sigma_2,C)}$ chiral output on the $K3 \times E$ slice.

(iii) R_4 (Kummer construction): conditional on identification of the Kummer-orbifold chiral data with the smooth $K3$ chiral specialisation of the common Stage-1 datum.

(iv) R_5 (sigma-model half-twist): conditional on convergence of the $\mathcal{N} = (2,2)$ half-twisted vertex algebra to the same (Σ_2, C)-specialised Stage-1 output.

(v) R_6 (BLLPR Schur limit): conditional on identification of the Schur-limit construction at the (Σ_2, C) datum with the common Stage-1 specialisation.

For $R_2, \dots, R_6$ the realization datum is conditional precisely on the corresponding convergence isomorphism. Once that isomorphism is proved, the same datum carries *realized*.

7.6 Boundary Problems

The following problems retain these particular omitted comparisons. Their resolution requires the indicated maps, not only a choice of reference or a numerical identity.

Open Problem 7.6.1 (Admissible logarithmic realization). Construct a Vol IV realization pair for every statement in Vols I-III whose statement depends on admissible logarithmic representations of affine Lie algebras at $k + h^\vee \in \mathbb{Q}_{>0}$ with rational level. A functorial comparison must also supply compatible maps on specializations, as in Proposition 7.2.2. The comparison must preserve the Creutzig–Kanade–Linshaw logarithmic structure. A semi-infinite cohomology construction is a possible comparison recipient only after its coefficient category, extension data and map from the chosen chiral construction have been supplied.

Open Problem 7.6.2 (Super-Yangian chain-level chiral coproduct). Construct a Vol IV realization pair for every statement in Vol I concerning the chain-level chiral coproduct of super-Yangians $Y(\mathfrak{sl}(m|n))^{\mathrm{ch}}$, outside the domain of FM230's absorption into parity-balance super-Whitehead. A concrete super-Feigin–Frenkel screening construction for $Y(\mathfrak{sl}(2|1))^{\mathrm{ch}}$ at generic parameters suffices as the seed.

Open Problem 7.6.3 (Five of six CY-C bridges). Prove each of the five conditional CY-C constructions R_2–R_6 of Vol III (Borcherds multiplicative lift, Mukai-lattice VOA, Kummer construction, $\mathcal{N} = (2,2)$ sigma-model half-twist, BLLPR Schur-limit construction) by proving the specific convergence isomorphism attaching each construction to the common Stage-1 (Σ_2, C)-specialisation of Φ_3. Each discharge is a standalone result: proof of the Igusa cusp form product convergence against the R_1 normalisation; identification of the Mukai-lattice OPE with the framed Stage-1 chiral output; identification of the Kummer-orbifold chiral data with the smooth $K3$ chiral specialisation; and the analogous equalities for the sigma-model and Schur-limit constructions.

7.7 Structural Consequences Inherited from Vols I–III

The constructions below give further comparison questions in the first three volumes. A comparison of their scalar values must be distinguished from the additional maps required to identify their operators, categories and state spaces.

The four-conductor split and the Koszul-self-dual trinity

Off the Koszul-self-dual locus the single Koszul conductor $K(\mathcal{A}) = -c_{\mathrm{ghost}}(\mathrm{BRST}(\mathcal{A}))$ of Vol I equation (G1) separates into four distinct conductors: K (BRST ghost charge), $K^\kappa := \kappa + \kappa^!$ (Theorem-C complementarity ceiling, with values (0, 13, 250/3, 25/3, 8) on (G, L, C, M, B), the M-row value 25/3 conditional), K_c (central-charge trace), and K_g (ghost-number anomaly). The trinity $K = K_c = K_g$ holds only on the Koszul-self-dual subcategory admitting both BRST resolution and a Borcherds-lift input on a Calabi–Yau target.

Independent comparison. The four conductors are tested against four independent sources: K against the BRST ghost resolution, K^κ against the archetype rows of the landscape census (Kac–Moody trace-form, Virasoro Zamolodchikov norm, $\beta\gamma$ contact coefficient, $\mathcal{W}_N$ Drinfeld–Sokolov reduction chain, $K3$ Mukai pairing via Bruinier 2002 Heegner reciprocity), K_c against Kac–Wakimoto 1988 central-charge tables, and K_g against ghost-partition primary Euler-character computations. Disjointness of the four paths is the realization witness of the four-conductor *separation*; coincidence of three of them on Koszul-self-dual witnesses the trinity.

Universal chain-homotopy normalisation

The Loday–Vallette contracting homotopy on the bar–cobar Quillen pair, specialised to a chirally Koszul $\mathcal{A}$, rescales by the complementarity ceiling: $h_{\mathcal{A}} = h_{\mathrm{LV}}/(\kappa + \kappa^!)$. Per archetype: h_{G} trivial (Heisenberg, trace-form $\kappa + \kappa^! = 0$); h_{L} Sugawara twist $1/(2(k + h^\vee))$, diverging at critical level; h_{C} free-field OPE rescaling; h_{M} Zamolodchikov $c(5c + 22)/10$, singular at $c = 0$, $c = -22/5$ Yang–Lee, $c = -218/45$ secondary Borel.

Independent comparison. The value $h_{\mathcal{A}}$ is obtained from the Loday–Vallette universal homotopy (Loday–Vallette 2012 Theorem 11.3.2) divided by the Theorem-C ceiling, and is compared against per-archetype primary-source computations (Sugawara twist literature; free-field OPE contractions; Zamolodchikov *Infinite additional symmetries in two-dimensional conformal quantum field theory* 1985 for Virasoro norm).

Shadow-tower exact closed forms

The Borel-radius function on the Virasoro shadow tower is $|\omega|^2(c) = c^2(5c + 22)/[4(45c + 218)]$, large-$c$ expansion $|\omega|(c) = c/6 - 1/27 + O(c^{-1})$, with Stokes pole $c_S = -218/45$ (zero of $180c + 872$). The unweighted shadow coefficient is the Vol I closed form

$$S_6(\mathrm{Vir}_c) \;=\; \frac{4(240c + 1031)}{c^3(5c + 22)^2}$$

is forced by the master relation $2S_2S_6 + 2S_3S_5 + S_4^2 = 0$ reading off the $[t^6]$ coefficient of H^2; the Riccati-metric weighted companion is $\widehat{S}_6^{\mathrm{wt}}(\mathrm{Vir}_c) = 80(45c + 193)/[3c^3(5c + 22)^2]$. The master equation $H^2 = t^4 Q_c$ is quadratic-algebraic; the Riccati structure lives on $U = (\log H)' - 2/t$, satisfying $U' + 2U^2 = Q_c''/(2Q_c)$.

Independent comparison. The value $|\omega|^2(c)$ is obtained by iterated Borel summation of the shadow tower on the Riccati-in-U substitution and is checked at $c \in \{1, -22/5, 0, -218/45\}$ against the Borel radius computed from the Zamolodchikov norm formula $c(5c + 22)/10$ and the degeneration of the quintic resolvent at $c = -22/5$. The Riccati substitution and the Zamolodchikov norm table are disjoint primary sources.

Shadow-Galois Chenevier determinant at $p = 691$

Shadow tower level $r = 11$, Kubota–Leopoldt irregular prime $p = 691$:

$$\det \rho_{S_{11}(V_1(\mathfrak{g})),\,691} \;=\; \chi_{\mathrm{cyc}}^{11} \cdot \varepsilon_{B_{12}} \pmod{691},$$

via Kubota–Leopoldt p-adic L-function plus Herbrand–Ribet.
Independent comparison. The determinant $\det \rho_{S_{11}}$ at $p = 691$ is computed from the shadow-tower motivic realisation and checked against $B_{12} = -691/2730$ and the Herbrand–Ribet character table from Washington *Introduction to Cyclotomic Fields* 1997 (primary source disjoint from the shadow-tower derivation).

Drinfeld double global obstruction class

For $g \geq 2$ the universal Drinfeld double $D(\mathcal{A}, \mathcal{A}^!)$ carries an explicit cohomological obstruction

$$\mathrm{obs}_{\mathrm{double}}^{(1)} \;\in\; H^2\big(\mathfrak{grt}^{\mathrm{ell}},\, \mathfrak{sp}(\mathcal{A}) \otimes \mathfrak{sp}(\mathcal{A}^!)\big), \qquad \omega_{\mathrm{ell}} = \wp(\tau)\langle \cdot, \cdot \rangle_{\mathfrak{sp}},$$

vanishing if and only if $r_{\max}(\mathcal{A}) = 2$ (class G only); classes L, C, M are $g \geq 1$ obstructed.
Independent comparison. $\mathrm{obs}_{\mathrm{double}}^{(1)}$ is computed on each archetype via the Enriquez $\mathrm{GRT}_1^{\mathrm{ell}}(\mathbb{Q})$ construction (Enriquez 2008 *Elliptic associators*) and compared with vanishing at class G (Heisenberg) and nonzero obstruction at classes L, C, M via a disjoint computation on the elliptic 2-cocycle $\wp(\tau)\langle \cdot, \cdot \rangle_{\mathfrak{sp}}$.

Eight-form Gritsenko–Cléry spread and cover assignment

Weights $w(N) \in \{5, 2, 1, 1, 1/2, 1, 1/4, 0\}$, Fourier zero-coefficients $c_N(0) \in \{10, 4, 2, 2, 1, 2, 1/2, 0\}$, universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ for $N \in \{1, 2, 3, 4, 6\}$ integer weight and the half/quarter-integer continuations $N \in \{5, 7\}$; cover assignment $\mathrm{Sp}_4(\mathbb{Z})/\mathrm{Mp}_4(\mathbb{Z})/\widetilde{\mathrm{Mp}}_4$ by weight parity via Weil representation plus Stone–von Neumann at signature $(2, 3)$.
Independent comparison. $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is evaluated at each row using **distinct primary sources**: Gritsenko 1999 *Reflective modular forms in algebraic geometry* for $N = 1$ (Δ_5 weight); Gritsenko–Nikulin 1998 *Automorphic forms and Lorentzian–Kac–Moody algebras* for $N \in \{2, 3, 4, 6\}$; Gritsenko 1994 $\Gamma_0(5)^+$ for $N = 5$ Borcea–Voisin; Scheithauer 2015 Duke for $N = 7$ quarter-integer; Mongardi–Tari–Wandel 2020 for $N = 8$ terminal fibre. The eight sources are pairwise distinct, each serving exactly one row. The identity $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ is not a theorem at any N; the independent comparison rejects any proof which uses this additive split.

Three-factor universal trace identity with ζ-regularisation

On the Koszul-self-dual subcategory admitting BRST resolution and Calabi–Yau target supporting a Borcherds product,

$$\mathrm{tr}_{\mathrm{ghost}}^{\zeta}(Q_{\mathrm{BRST}}^2) \;=\; \mathrm{tr}_{\mathrm{Pentagon}} \;=\; \omega_{\mathrm{Borcherds}} \;=\; c_N(0)/2.$$

ζ-regularisation via Dabholkar–Harvey BPS counting (IIA $K3 \times T^2$, $\mathcal{N} = 4$, $D = 4$ BPS supermultiplicity). At $N = 1$ four paths converge on $c_1(0)/2 = 5$; at $N = 2$ three paths converge on $c_2(0)/2 = 2$.
Independent comparison. Four disjoint calculations give the four paths at $N = 1$: Vol I BRST ghost anomaly ($K(\Phi_3(K3 \times E))$); Vol III Gritsenko Δ_5 weight; Vol II Pentagon face of the $\mathrm{SC}^{\mathrm{ch,top}}$ boundary (Heptagon face 4 / Dimofte slab trace); Dabholkar–Harvey 2007 *BPS states and their supermultiplicity*. Their sources are pairwise disjoint and their values all equal $c_1(0)/2 = 5$, the common Borcherds–Kac–Moody central-charge value at $N = 1$.

Two-stage factorisation of the Calabi–Yau-to-chiral functor

$$\Phi_d \;=\; \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}},$$

with Stage 1 outputting a canonical E_d-holomorphic factorisation algebra pinned by Kontsevich–Tamarkin E_d-formality (category-level closed at $d \le 3$, open at $d \ge 4$) plus Costello–Gwilliam–Li locality; Stage 2 specialising via factorisation homology $\int_{\Sigma_{d-1}}$ over a $(d-1)$-cycle, restriction to a reference curve C. PTVV-forced native operadic level: $d = 1 \Rightarrow n = \infty$, $d = 2 \Rightarrow n = 2$, $d \ge 3 \Rightarrow n = 1$.
Independent comparison. The Stage-1 calculation builds the E_d-FA from the Kontsevich–Tamarkin side (primary source: Kontsevich 1999 *Operads and motives in deformation quantization*; Tamarkin 2003 *Action of the Grothendieck–Teichmüller group on the operad of little disks*); the Stage-2 calculation builds the E_1-chiral shadow from the factorisation-homology side (primary source: Ayala–Francis–Tanaka 2017 *Factorization homology of stratified spaces*; Costello–Gwilliam *Factorization algebras in quantum field theory* 2021). Pairwise disjointness is checked, the PTVV shift $(2-d)$ is confirmed via Pantev–Toën–Vaquié–Vezzosi 2013 primary-source evaluation on each d, and cross-verifies the Dunn–Lurie native-level dispatch from Lurie HA.5.1.2.2.

K3 Hall–Drinfeld chiral-bialgebra equivalence

$$\mathrm{Sp}^{\mathrm{ch}}_{K3,E}(\Phi_3^{\mathrm{FA}}(\mathrm{Perf}(K3 \times E))) \;\simeq\; \mathbf{H}_{\Delta_5}$$

is an equivalence at the chiral-bialgebra level, stronger than graded-dimension coincidence. The six canonical constructions R_1–R_6 of Vol III's six-construction taxonomy (R_1 framed Stage-1 $\Phi_3^{(\Sigma_2,C)}$, R_2 Borcherds multiplicative lift, R_3 Mukai-lattice VOA, R_4 Kummer construction, R_5 $\mathcal{N}=(2,2)$ sigma-model half-twist, R_6 BLLPR Schur limit) form a common limit cone on the chiral bialgebra $\mathbf{H}_{\Delta_5}$, with R_1 the proved generator and R_2–R_6 carrying convergence hypotheses (Chapter 7.5).
Independent comparison. The six constructions give six disjoint calculations of the chiral-bialgebra structure on $\Phi_3(D^b(K3 \times E))$, and the comparison is natural isomorphism of the six outputs. The Fake-Monster rank obstruction excludes any attempt to realise $\mathrm{II}_{25,1}$-Leech-lattice chiral data on a compact CY_3: the bound $h^{1,1}(X) \le h^{1,1}(\text{fibre}) + 2h^{0,2}(\text{fibre}) \le 22 < 24$ is a cohomological obstruction.

CoHA-to-$W_{1+\infty}$ triangle with two λ-parameters

$\mathrm{CoHA}(\mathbb{C}^3) = Y^+(\hat{\mathfrak{gl}}_1)$ primitive (Schiffmann–Vasserot 2013); Drinfeld double $Y(\hat{\mathfrak{gl}}_1)$ affine Yangian of $\mathfrak{gl}_1$ in Drinfeld currents (Tsymbaliuk 2017); $W_{1+\infty}$ via ev_λ evaluation (Gaiotto–Rapčák 2017). Two λ-parameters are distinguished at all times: $\lambda_{\mathrm{Tr}} = (\varepsilon_1+\varepsilon_2)/\varepsilon_3$ (truncation, fixed to -1 under the CY_3 slice) versus λ_W (Gaiotto–Rapčák central-charge, free). $\mathrm{CoHA}(\mathbb{C}^3)$ and $W_{1+\infty}$ are not equal as algebras.
Independent comparison. Three disjoint character calculations occur: CoHA Euler characteristic (Kontsevich–Soibelman 2008); Yangian PBW character (Drinfeld 1987); $W_{1+\infty}$ vacuum module character (Frenkel–Kac–Radul–Wang 1995 *$W_{1+\infty}$ and $W(\mathfrak{gl}_N)$ with central charge N*). The λ-parameter discipline is part of the statement: λ_{Tr} and λ_W are distinct parameters.

Three-tier topologisation discipline

Tier I cohomological and Tier II chain-level on quasi-isomorphic models are unconditional. Tier III strict chain-level on the original complex is closed for classes G, L, C and open for class M (the F3 frontier). Sugawara explicit at $V_1(\mathfrak{sl}_2)$: $T_{\mathrm{Sug}} = (1/6)(:J^0J^0: + :J^+J^-: + :J^-J^+:)$, $G_{\mathrm{Sug}} = (1/3)\sum_a :\bar{c}_a J^a:$, $[Q_{\mathrm{tot}}, G_{\mathrm{Sug}}] = T_{\mathrm{Sug}}$ on BRST cohomology.
Independent comparison. Three comparisons per class, one per tier: Tier I (Feigin–Frenkel 2011 cohomological E_3 on $H^*(\mathrm{Obs}^{\mathrm{BV}}_{\mathrm{hCS}})$); Tier II (Fresse 2017 Theorem 12.3.A homotopy transfer); Tier III (explicit

chain-level antighost primitive per archetype). The three tiers have pairwise disjoint source sets; the class-M Tier III form is conditional on F3 resolution.

Khan–Zeng finite-jet BV/PVA sector

For finite-type freely generated nonlinear finite-jet Poisson vertex algebras (PVAs), Vol II constructs the shifted-cotangent BV action on the half-space and identifies the classical master equation with the PVA Hamiltonian condition. This is a constructed classical statement. The all-loop quantum master equation, reflected weight identity, boundary vertex algebra, and E_3-lift are conditional on the analytic strong deformation retract (SDR) datum: finite-jet quotient compatibility, image-choice Stokes cancellation, field-parity normalisation, and self-image renormalisation.
Independent comparison. The classical comparison builds the shifted-cotangent BV action from the PVA Hamiltonian density and checks CME equality against the lambda-bracket Hamiltonian condition. The quantum comparison is absent until it supplies the analytic SDR on every jet-weight quotient, proves compatibility in the inverse limit, and verifies the reflected weight identity independently of the classical CME derivation. Thus there are two forms: constructed classical realization for CME/PVA, and conditional quantum realization for QME/boundary-VA/E_3.

Joyce–KPS compact Calabi–Yau threefold sector

For a generic smooth projective Calabi–Yau threefold X, Vol III uses Joyce's vertex algebra V_X on the derived moduli stack and the Kontsevich–Pandharipande–Soibelman orientation to place MNOP in the E_2-centre of the E_3 factorisation algebra on the constructed module surfaces. The S^3-framing statement for generic compact Calabi–Yau Lagrangians is chain-level on the Costello–Paquette defect-chiral model. A comparison with the global Hall–Borcherds or Drinfeld envelope is an additional bridge, not a source of the local construction.
Independent comparison. The compact-CY3 comparison checks the Joyce V_X construction and KPS orientation data against the MNOP centre automorphism. A separate chain-level comparison checks the S^3-framing through the defect-chiral model. A third comparison may test Hall–Borcherds/Drinfeld compatibility, but that bridge is not a proof source for the Joyce–KPS construction itself.

Post-Tate Hadamard support and the M_t-graph dual

The arithmetic spectral-flow branch separates the critical-line ordinate support

$$\Gamma_\xi^{\mathrm{crit}} = \{\gamma \in \mathbb{R} : \xi(1/2 + i\gamma) = 0\}$$

from the full Hadamard divisor. Multiplicities are forgotten in this support. The ordinary M_t-graph topology (the Hilbert-space topology induced by the graph norm of the multiplication operator M_t on $L^2(\mathbb{R}, dt)$, i.e. the $(1 + t^2)\,dt$ inner product) does not admit a non-zero continuous graph-dual functional supported on $\Gamma_\xi^{\mathrm{crit}}$, even after finite-jet residue data are allowed. The obstruction is functional analytic: the graph dual is represented through the resolvent chart by L^2-data, while a locally finite critical-line support has measure zero.
Independent comparison. The ordinary M_t-graph dual after the post-Tate quotient gives the L^2-support obstruction. A surviving construction must produce a continuous Hadamard trace-and-jet map $\Phi_\xi \to \ell^2(\Gamma_\xi^{\mathrm{crit}}, w)$ via a rigged, Paley–Wiener, de Branges, reproducing-kernel, nonlocal L^2, or BV–BFV boundary topology, prove post-Tate compatibility and adjoint-domain compatibility, and verify that the closed condition arises from the analytic structure and is not imposed by fiat.

Boundary frontiers F1–F10

The ten boundary frontiers are the complement of the present independent domain: (F1) logarithmic $\mathcal{W}(p)$ triplet tempering; (F2) periodic CDG for non-simply-laced admissible levels; (F3) original-complex Tier III for class M; (F4) super-complementarity canonical pairing; (F5) irregular-singular KZB at $\overline{\mathcal{M}}_{g,n}$ boundaries; (F6) Theorem B at nodal boundary without Koszul hypothesis; (F7) MC4 completion / H3 failure regime; (F8) Stage-1 chain level at $d \geq 4$; (F9) category-level E_d-formality at $d \geq 4$; (F10) bracket-level BPS Lie algebra on $K3 \times E$.

When any of F1–F10 is proved in Vol I–III, its realization pair $\mathrm{Real}(X, T) = (d, e)$ is the corresponding independent comparison in Vol IV. Until then it remains a boundary stratum.

Chapter 8

Relative coefficient envelopes

A coefficient algebra records which intermediate states may be composed. Passing to a scalar trace can erase that information. To retain it in a geometric realization, we shall attach the coefficient algebra to a fixed geometric algebra and transport bimodules together with their compositions.

The first calculation already determines the construction. Let A and E be ordinary unital associative algebras over a field k, let P be a right E-module, and let Q be a left E-module. Then

$$(A \otimes_k P) \otimes_{A \otimes_k E} (A \otimes_k Q) \longrightarrow A \otimes_k (P \otimes_E Q), \qquad (a \otimes p) \otimes (a' \otimes q) \longmapsto aa' \otimes (p \otimes q)$$

is an isomorphism. Its inverse sends $a \otimes (p \otimes q)$ to $(a \otimes p) \otimes (1_A \otimes q)$. Balancing moves the second A-factor into the first without interchanging them. Thus A need not be commutative.

For complexes, derived tensors retain the higher relations among the coefficients. For geometric objects, the same calculation uses the given geometric multiplication. We will construct the comparison on the two-sided bars that define those derived tensors. This specifies its signs, its unit, and its compatibility with any finite string of interfaces.

8.1 Coefficients in a monoidal category

Fix a commutative ring k. Write $D(k)$ for the stable ∞-category of cochain complexes of k-modules with quasi-isomorphisms inverted. Differentials have degree 1; the shift satisfies $M[n]^i = M^{i+n}$.

Let $(C, \star, K)$ be a stable presentable monoidal ∞-category. Assume that $\star$ preserves colimits separately in both variables and that $D(k)$ acts centrally on C. Here centrality means that a scalar complex can pass through a geometric factor, with the usual coherent Koszul interchange. It supplies no interchange of two geometric factors. We write $X \otimes_k V$ for the action of $V \in D(k)$ on $X \in C$, and

$$\underline{V} = K \otimes_k V$$

for the corresponding scalar object of C. All tensor products in this chapter are derived unless an ordinary representative is specified. Element formulas refer to flat dg representatives of the structural maps.

Fix a unital associative algebra object A in C. For a unital dg k-algebra B, define

$$R_B = A \otimes_k B.$$

We call R_B the *relative coefficient envelope* with background A. The product uses central interchange followed by the two multiplications. On homogeneous representatives it is

$$(a \otimes b)(a' \otimes b') = (-1)^{|b||a'|} aa' \otimes bb'.$$

The algebra unit is the composite $K \to A \to A \otimes_k B$ obtained from the units of A and B. The object K remains the monoidal unit.

For another unital dg k-algebra E, a dg B–E bimodule P gives an R_B–R_E bimodule

$$J_{B,E}(P) = A \otimes_k P$$

with differential and actions

$$\begin{aligned} d(a \otimes p) &= d_A a \otimes p + (-1)^{|a|} a \otimes d_P p, \\ (a \otimes b)(x \otimes p) &= (-1)^{|b||x|} ax \otimes bp, \qquad (8.1) \\ (x \otimes p)(a' \otimes e) &= (-1)^{|p||a'|} xa' \otimes pe. \qquad (8.2) \end{aligned}$$

We call such a bimodule an *interface* from its left algebra to its right algebra. Two interfaces compose by balanced tensor over their common algebra.

The signs in (8.1)–(8.2) are forced by the positions of the factors. For example, in a right action the coefficient p passes through a', so an odd p and an odd a' contribute a minus sign. A degree-r coefficient bimodule map $f : P \to P'$ is transported by

$$J(f)(a \otimes p) = (-1)^{r|a|} a \otimes f(p). \qquad (8.3)$$

Associativity, the commuting left and right coefficient actions, and the Koszul rule verify the bimodule and dg-functor identities.

8.2 The comparison of the two bars

Let B, E, D be unital dg k-algebras, and let ${}_BP_E$ and ${}_EQ_D$ be dg bimodules. The coefficient derived tensor $P \otimes^{\mathbb{L}}_E Q$ is the realization of the simplicial two-sided bar

$$\mathcal{B}_n(P, E, Q) = P \otimes_k E^{\otimes_k n} \otimes_k Q. \qquad (8.4)$$

For $n \geq 1$, the faces are the right action on P, the $n-1$ adjacent products in E, and the left action on Q, in that order. Degeneracies insert 1_E. In dg representatives we use the direct-sum total

$$|\mathcal{B}_\bullet| = \bigoplus_{n \geq 0} \mathcal{B}_n[n], \qquad d_{\mathrm{tot}}\big|_{\mathcal{B}_n[n]} = (-1)^n d_{\mathrm{int}} + \sum_{i=0}^{n} (-1)^i d_i, \qquad (8.5)$$

where the face differential out of column 0 is zero. The module structure on each shifted column is the shifted module structure. In particular, a left action of degree r acquires the sign $(-1)^{nr}$ in column n.

Independently, the geometric balanced tensor

$$J_{B,E}(P) \star^{\mathbb{L}}_{R_E} J_{E,D}(Q)$$

is the realization of

$$\mathcal{L}_n = (A \otimes_k P) \star (A \otimes_k E)^{\star n} \star (A \otimes_k Q).$$

Its factors are in their displayed order. Define $\phi_n : \mathcal{L}_n \to A \otimes_k \mathcal{B}_n(P, E, Q)$ by

$$\begin{aligned} &(a_0 \otimes p)|(a_1 \otimes e_1)|\cdots|(a_n \otimes e_n)|(a_{n+1} \otimes q) \\ &\qquad \longmapsto (-1)^{\epsilon} (a_0 a_1 \cdots a_{n+1}) \otimes (p|e_1|\cdots|e_n|q), \\ &\epsilon = |p| \sum_{i=1}^{n+1} |a_i| + \sum_{j=1}^{n} |e_j| \sum_{i=j+1}^{n+1} |a_i|. \end{aligned} \qquad (8.6)$$

Each summand in ϵ records one scalar factor passing through a later geometric factor. The A-factors are multiplied in their original order.

Lemma 8.2.1. The maps ϕ_n form a simplicial dg bimodule map. After realization they give a natural comparison

$$\Phi_{P,Q} : J_{B,E}(P) \star^{\mathbb{L}}_{R_E} J_{E,D}(Q) \longrightarrow J_{B,D}(P \otimes^{\mathbb{L}}_{E} Q).$$

Proof. Consider an interior face, which combines the pairs (a_j, e_j) and (a_{j+1}, e_{j+1}). Multiplication before the shuffle contributes $|e_j||a_{j+1}|$. After multiplication these two entries have degrees $|a_j| + |a_{j+1}|$ and $|e_j| + |e_{j+1}|$. The difference between the two crossing counts is exactly $|e_j||a_{j+1}|$. Hence the face commutes with ϕ_n. At the first and last faces the same count gives the right and left module-action signs. Degeneracies insert closed units of degree zero.

Central interchange and multiplication in A are chain maps. They therefore commute with the internal differential. The endpoint actions also commute with the map: both routes move the same scalar factors through the same geometric factors and then apply the same ordered product.

One totalization sign remains. The canonical chain map

$$|A \otimes_k \mathcal{B}_\bullet| \longrightarrow A \otimes_k |\mathcal{B}_\bullet|$$

multiplies a term with geometric degree $|a|$ in column n by $(-1)^{n|a|}$. For the differential of A, changing $|a|$ by one contributes $(-1)^n$, as required by (8.5). For a face, changing n to $n-1$ contributes $(-1)^{|a|}$, the tensor-differential sign on the target. The internal coefficient differential is unchanged. Composing this identification with $|\phi_\bullet|$ gives the asserted dg bimodule map. □

Theorem 8.2.2 *(Relative coefficient composition).* Under the hypotheses of Section 8.1, the comparison $\Phi_{P,Q}$ is an equivalence. For every finite string of coefficient bimodules, the comparisons satisfy the associativity and unit coherences. The identity coefficient bimodule B is sent to the regular interface R_B.

Proof. Resolve Q by the augmented free left-E bar

$$F_n = E \otimes_k E^{\otimes_k n} \otimes_k Q \longrightarrow Q.$$

After forgetting its left E-action, insertion of 1_E at the left gives an extra degeneracy. The augmented simplicial identities contract its realization, so $|F_\bullet| \simeq Q$.

Tensoring with A gives a resolution of $J_{E,D}(Q)$. Its term in degree n is the free left-R_E module

$$R_E \star \underline{E^{\otimes_k n} \otimes_k Q}.$$

Balancing it with $J_{B,E}(P)$ leaves

$$A \otimes_k P \otimes_k E^{\otimes_k n} \otimes_k Q.$$

The resulting faces are exactly the faces of (8.4). Their realization is $J_{B,D}(P \otimes^{\mathbb{L}}_{E} Q)$.

To identify the particular comparison, map the full free R_E-bar resolution of $J(Q)$ to $A \otimes_k F_\bullet$ by multiplying its A-entries in order. The crossing signs are those in (8.6). The preceding lemma verifies the faces, degeneracies, differentials, and outer actions. Both augmentations are equivalences; hence the realized comparison between the resolutions is an equivalence. Derived balancing preserves this equivalence. The resulting map is $\Phi_{P,Q}$.

For three interfaces, use their common bisimplicial bar. Both parenthesizations move each scalar coefficient through every later geometric entry and then multiply the geometric entries in order. Already in bar degree zero the exponents are

$$|p||a'| + (|p| + |q|)|a''| \quad \text{and} \quad |q||a''| + |p|(|a'| + |a''|),$$

which agree. In arbitrary bar degrees the two exponents count the same pairs of entries.

The same construction on the common multisimplicial bar of any finite string gives one comparison before a parenthesization is chosen. Every parenthesization induces that comparison, so all associativity diagrams commute. For units, $J_{B,B}(B) = R_B$ with its regular actions, and the two unit comparisons are the augmentations contracted by insertion of 1_B. These are the same degeneracies in every multibar. Realization therefore supplies the unit and associativity coherences together. □

The proof uses the geometric product of A, scalar interchange, and colimits. Consequently, any additional operation that commutes with these maps commutes with $\Phi_{P,Q}$. In a collision model this requires the specified collision comparisons for A; the algebraic formula alone does not construct them.

8.3 Recovery at the empty configuration

Suppose that C has a colimit-preserving strong monoidal functor

$$e_\circ : C \longrightarrow D(k)$$

compatible with scalar action, and fix an algebra equivalence $e_\circ(A) \simeq k$. In a word-convolution category, $e_\circ$ is evaluation at the empty configuration.

PROPOSITION 8.3.1. There are natural identifications

$$e_\circ(R_B) \simeq B, \qquad e_\circ(J_{B,E}(P)) \simeq P.$$

Under these identifications, $e_\circ(\Phi_{P,Q})$ is the identity comparison on $P \otimes^{\mathbb{L}}_E Q$. Empty evaluation also recovers every transported coefficient map and every diagram of such maps.

Proof. Compatibility with scalar action gives $e_\circ(A \otimes_k P) \simeq e_\circ(A) \otimes_k P \simeq P$. Since $e_\circ$ is strong monoidal and preserves colimits, it carries the geometric bar to the coefficient bar. The product of the A-entries in (8.6) becomes multiplication of copies of the scalar unit. The comparison is therefore the identity under the canonical unit identifications. The same argument applied to (8.3) recovers coefficient maps and their compositions. □

In particular, a homotopy between two transported coefficient maps gives a homotopy between the original maps by applying $e_\circ$. Recovery does not assert that every geometric bimodule map is induced by a coefficient map. The geometric category can contain additional maps.

The empty algebra of R_B is B. This retained coefficient is the reason that arbitrary coefficient interfaces compose with their original units. Replacing it by k changes the comparison problem.

8.4 Endomorphisms and the orientation of composition

A useful coefficient system is supplied by endomorphisms of one algebra. Let B be an ordinary unital k-algebra, and let $\alpha : B \to B$ be a unital endomorphism. Define

$$Q_\alpha = {}_\alpha B_1, \qquad b \cdot x \cdot c = \alpha(b)xc.$$

For endomorphisms α, β, an element $t \in B$ satisfying

$$t\alpha(b) = \beta(b)t \qquad (b \in B)$$

defines a bimodule map $Q_\alpha \to Q_\beta, x \mapsto tx$. Thus actual intertwiners, including their relations, become maps of interfaces.

Proposition 8.4.1. There is a natural bimodule isomorphism

$$Q_\alpha \otimes_B Q_\beta \xrightarrow{\sim} Q_{\beta\circ\alpha}, \qquad x \otimes y \longmapsto \beta(x)\,y.$$

It also computes the derived tensor. Consequently,

$$J(Q_\alpha) \star^{\mathbb{L}}_{R_B} J(Q_\beta) \simeq J(Q_{\beta\circ\alpha})$$

with the coherences of Theorem 8.2.2.

Proof. The balancing relation is $xb \otimes y = x \otimes \beta(b)\,y$. Its two images are both $\beta(x)\beta(b)\,y$. The left action of b on the image is $\beta(\alpha(b))$, and the right action is unchanged. The inverse is $y \mapsto 1_B \otimes y$. The right B-module underlying Q_α is free of rank one, so its ordinary tensor computes the derived tensor. The last assertion is the relative composition theorem. □

Composition is therefore $\beta \circ \alpha$ in the displayed order. A tensor category of endomorphisms using $\alpha \circ \beta$ must have its tensor order reversed before this functor is called monoidal. This convention also determines how associators and duality maps are transported.

Every specified identity among intertwiner diagrams survives under J, including pentagon and evaluation identities when they occur in the input. Neither a braiding nor a tensor product of independent representations is supplied by balanced interface composition.

8.5 Scalar change

Let $k \to k'$ be a map of commutative rings. Assume that a colimit-preserving monoidal scalar-change functor from C to a corresponding category C' is given and is compatible with the central scalar actions. Denote it by $X \mapsto X_{k'}$. Then

$$(R_B)_{k'} \simeq A_{k'} \otimes_{k'} (B \otimes^{\mathbb{L}}_k k'), \qquad J(P)_{k'} \simeq A_{k'} \otimes_{k'} (P \otimes^{\mathbb{L}}_k k').$$

These identifications preserve $\Phi_{P,Q}$. Indeed, scalar change preserves the two bar realizations and their common crossing map. Its coherence with a second scalar change follows on the same diagrams.

For arithmetic coefficient towers, this supplies each finite-precision comparison once the geometric scalar-change functors have been constructed. Continuity of an inverse limit is a further statement: it requires the chosen completed tensor and the corresponding limit comparison. The finite bar formulas do not interchange an arbitrary limit with realization.

Chapter 9

Connected descent and seam corings

Over a field k, relative coefficient envelopes retain the coefficient algebra at the empty configuration. A connected envelope instead has empty coefficient k. This change is already visible when two relative regular interfaces are restricted to a connected envelope and composed: their empty coefficient is $B \otimes_k B$, and comparison with composition over the relative envelope applies multiplication

$$B \otimes_k B \longrightarrow B.$$

For $B = \mathrm{Mat}_2(k)$, the source has dimension sixteen and the target dimension four. The twelve-dimensional kernel survives although every B-module is projective.

The object that records this difference is an extension–restriction coring. It occurs at every junction of restricted interfaces. We first construct it and compute its counit defect. A separate matrix calculation then shows that the failure cannot be removed by choosing a different strong unital lift to the same connected envelopes. Finally, under explicit hypotheses on the arity filtration, we reconstruct connected modules from the whole coring coaction. Reconstruction uses all its iterates; multiplication alone does not retain the required data.

9.1 The coring of an algebra map

Let $(C, \star, K)$ be a stable presentable monoidal ∞-category whose product preserves colimits separately in each variable. Fix a unital associative algebra map

$$f : G \longrightarrow R$$

in C. Module categories and tensor products are derived. We write $X \star_G Y$ for the realization of the two-sided bar of a right G-module X and a left G-module Y. Thus

$$X \star_G Y = \big|[n] \longmapsto X \star G^{\star n} \star Y\big|.$$

No interchange of two geometric factors is part of this definition.

Define the R–R bimodule

$$T_f = R \star_G R.$$

Its outer actions multiply the two outside R-factors. Multiplication gives a bimodule map

$$\epsilon : T_f \longrightarrow R.$$

To define a coproduct, insert f into the middle of $R \star_G G \star_G R$:

$$\Delta : T_f \longrightarrow R \star_G R \star_G R \simeq T_f \star_R T_f. \tag{9.1}$$

This map is called the *comultiplication* of the coring. On an ordinary flat representative, this is the familiar formula

$$r \otimes s \longmapsto (r \otimes 1) \otimes_R (1 \otimes s).$$

Equation (9.1), rather than an underived tensor formula, specifies the map in general.

PROPOSITION 9.1.1. The maps Δ and ϵ make T_f a counital coassociative coring in R-bimodules, with all higher coherences. For the extension–restriction adjunction

$$F = R \star_G - : \mathrm{Mod}_G(C) \rightleftarrows \mathrm{Mod}_R(C) : U,$$

the comonad FU is $T_f \star_R -$.

Proof. Two successive insertions of f give the same ordered tensor $R \star_G R \star_G R \star_G R$. Applying ϵ on either side of Δ removes one inserted unit and gives the identity. For any number of insertions, the comparisons are induced by the same multisimplicial bar. Its unit and associativity maps give the coherences after realization.

For an R-module N, associativity of the balanced tensor gives

$$FU(N) = R \star_G U(N) \simeq (R \star_G R) \star_R N.$$

The adjunction counit is the action of R, hence is induced by ϵ. If $\eta : \mathrm{Id} \to UF$ is the adjunction unit, the comultiplication $F\eta U$ inserts f and is induced by Δ. These identify the comonad structure as well as its functor. □

A coring coaction on N is a map $N \to T_f \star_R N$ with its counit, coassociativity, and higher compatibilities. We use this coherent notion throughout. The coring need not have a bimodule coaugmentation $R \to T_f$. For example, at a scalar extension $k \to B$, the map $b \mapsto b \otimes 1$ is left B-linear but generally fails right B-linearity.

9.2 COMPOSITION AFTER RESTRICTION

Let X be a right R-module and Y a left R-module. We also write U for restriction of right modules. Restriction gives a natural comparison

$$\chi_{X,Y} : U(X) \star_G U(Y) \longrightarrow X \star_R Y.$$

It applies f to every intermediate G-factor in the two-sided bar. Let $D_f = \mathrm{cofib}(\epsilon : T_f \to R)$.

THEOREM 9.2.1 *(Composition with its seam coring).* There are natural equivalences

$$\begin{aligned} U(X) \star_G U(Y) &\simeq X \star_R T_f \star_R Y, \\ \mathrm{cofib}(\chi_{X,Y}) &\simeq X \star_R D_f \star_R Y. \end{aligned} \tag{9.2}$$

Under the first equivalence, $\chi_{X,Y}$ applies the counit at the middle factor. For a finite string of restricted bimodules, one copy of the corresponding coring occurs at every intermediate algebra. These identifications are compatible with regrouping the string.

Proof. Insert the regular R-modules next to X and Y:

$$X \star_R (R \star_G R) \star_R Y \simeq (X \star_R R) \star_G (R \star_R Y) \simeq U(X) \star_G U(Y).$$

The equivalences follow by realizing the common multibar, in which all factors remain in their displayed order. Multiplication of the middle two R-factors is exactly the bar comparison defining $\chi_{X,Y}$. Derived tensoring preserves cofibres, giving (9.2). The same multibar proves the assertion for every finite string and all its parenthesizations. □

Apply this to the relative envelopes of Chapter 8. Suppose actual unital comparison maps

$$\theta_B : G_B \longrightarrow R_B = A \otimes_k B$$

have been supplied for the coefficient algebras under consideration. For a B–E bimodule P, restrict $J_{B,E}(P) = A \otimes_k P$ along both endpoint maps and denote the result by $I^{\theta}_{B,E}(P)$. For a composable E–D bimodule Q,

$$I^{\theta}(P) \star_{G_E} I^{\theta}(Q) \simeq \mathrm{Res}\big(J(P) \star_{R_E} T_{\theta_E} \star_{R_E} J(Q)\big).$$

The comparison to $I^{\theta}(P \otimes^{\mathbb{L}}_E Q)$ applies the counit and then Theorem 8.2.2. Its unit comparison is $\theta_B : G_B \to R_B$. Consequently restriction gives a lax composition law. For a selected pair P, Q, this law is an equivalence exactly when the right-hand side of (9.2) vanishes for $X = J(P), Y = J(Q)$. A nonzero D_{θ_E} does not by itself decide every such pair.

9.3 The empty coefficient and noncommutative differentials

For this section, let k be a field. Assume a colimit-preserving strong monoidal functor $e_{\circ} : C \to \mathcal{D}(k)$ is given, and that

$$e_{\circ}(G) = k, \qquad e_{\circ}(R) = B, \qquad e_{\circ}(f) = (k \longrightarrow B),$$

where B is an ordinary nonzero unital k-algebra. Since $e_{\circ}$ preserves the balanced bar,

$$e_{\circ}(T_f) = B \otimes_k B, \qquad e_{\circ}(\epsilon) = \mu_B.$$

Here $\mu_B(b \otimes c) = bc$.

Define the bimodule of noncommutative one-forms by

$$\Omega^{1}_{\mathrm{nc}}(B/k) = \ker\big(\mu_B : B \otimes_k B \to B\big),$$

with outer B-actions, and put

$$d_{\mathrm{nc}} b = 1 \otimes b - b \otimes 1.$$

Commutativity of B does not require the two actions on this bimodule to coincide; it does not identify it with the Kähler differential module.

Proposition 9.3.1. There is an equivalence

$$e_{\circ}(D_f) \simeq \Omega^{1}_{\mathrm{nc}}(B/k)[1].$$

The kernel is zero exactly when the unit $k \to B$ is an isomorphism. If $\dim_k B = r < \infty$, then

$$\dim_k \Omega^{1}_{\mathrm{nc}}(B/k) = r^2 - r.$$

In particular this dimension is $d^4 - d^2$ for $B = \mathrm{Mat}_d(k)$.

Proof. Multiplication has the k-linear section $b \mapsto 1 \otimes b$. Its cofibre is therefore its kernel shifted by one, and its rank is $\dim_k B$. This proves the equivalence and the finite-dimensional formula.

Choose a k-linear functional $\lambda : B \to k$ with $\lambda(1) = 1$. If the kernel vanishes, then $1 \otimes b - b \otimes 1 = 0$ for every b. Applying $\lambda \otimes \mathrm{id}$ gives $b = \lambda(b)1$. Thus $B = k1$. The converse follows from multiplication in k. □

The universal property explains which information this kernel carries. For a derivation $D : B \to M$ into a B-bimodule, define

$$h_D\left(\sum_i a_i \otimes b_i\right) = \sum_i a_i D(b_i) \quad \text{when } \sum_i a_i b_i = 0.$$

The derivation identity makes h_D right B-linear; left linearity is immediate. Moreover $h_D(d_{\mathrm{nc}} b) = D(b)$, and

$$\sum_i a_i \, d_{\mathrm{nc}} b_i = \sum_i a_i \otimes b_i$$

on the kernel. These formulas prove existence and uniqueness of the factorization $D = h_D d_{\mathrm{nc}}$. For a nonscalar b, the tensor $d_{\mathrm{nc}} b$ is nonzero: a functional vanishing on 1 and taking b to 1, applied to its first factor, gives -1.

Thus $B \neq k$ prevents the global counit $T_f \to R$ from being an equivalence. The converse requires information about the positive configurations. In a family for which $G_k = R_k$ and $\theta_k = \mathrm{id}$, it holds: the counit is then $R_k \star_{R_k} R_k \simeq R_k$. Empty evaluation alone proves no such converse for an arbitrary map $G \to R$.

9.4 A matrix obstruction to strong connected transport

The preceding defect concerns the specified restriction along f. We now test whether some other assignment could preserve all coefficient bimodule compositions while keeping the connected envelopes.

Let k have characteristic zero, let $X = \mathbb{A}^1_k$, and write $\mathcal{D}(X)$ for the derived category of right differential modules on X, enhanced as a stable ∞-category. Consider

$$\mathcal{T}_X = \mathcal{D}(k) \times \mathcal{D}(X)$$

with symmetric monoidal product

$$(U, M) \odot (V, N) = (U \otimes_k V,\ U \otimes_k N \oplus M \otimes_k V).$$

The two components have arities zero and one. Two arity-one factors have zero product in this category. The symmetry only moves scalar complexes past differential modules, with Koszul signs. It is not a statement that an intrinsic tensor product of two differential modules vanishes.

Suppose a colimit-preserving strong monoidal truncation $\tau_{\leq 1} : \mathcal{C} \to \mathcal{T}_X$ is given. It sends balanced bars to balanced bars and therefore preserves inverse bimodule interfaces.

For a unital algebra S in $\mathcal{T}_X$, let $\mathrm{CH}(S; L)$ be its Hochschild chain object with coefficients in an S-bimodule L. It is the realization of the cyclic bar with term $L \odot S^{\odot n}$ in degree n. The last face uses the symmetry to act at the left endpoint.

LEMMA 9.4.1. For $S = (k, M)$, with its scalar unit and actions, the arity-one component is

$$\mathrm{CH}(S; S)_1 \simeq M \oplus M[1].$$

Proof. The unit splits, with reduced object $\overline{S} = (0, M)$. The normalized bar terms are $S \odot \overline{S}^{\odot n}[n]$. At arity one only $n = 0, 1$ survive. They are M and $M[1]$, and their boundary is $1m - m1 = 0$. The split unit-degenerate subcomplex is contractible. Thus the normalization computes the full realization and retains the internal differential on both copies of M. □

LEMMA 9.4.2. For bimodules ${}_S P_T$ and ${}_T Q_S$ in $\mathcal{T}_X$, there is a natural equivalence

$$\mathrm{CH}(S; P \odot_T Q) \simeq \mathrm{CH}(T; Q \odot_S P).$$

Consequently inverse bimodule interfaces preserve the Hochschild chain object.

Proof. Resolve the balanced tensor and the Hochschild chains independently. The resulting bisimplicial terms on the two sides are

$$P \odot T^{\odot q} \odot Q \odot S^{\odot p}, \qquad Q \odot S^{\odot p} \odot P \odot T^{\odot q}.$$

Rotate the two displayed blocks. On homogeneous tensors the sign is

$$(-1)^{(|p_0| + \sum_j |t_j|)(|q_0| + \sum_i |s_i|)},$$

where $p_0 \in P$ and $q_0 \in Q$. Interior faces multiply the same adjacent factors. The two endpoint faces in the T-direction use the right action on P and the left action on Q; after rotation one becomes the cyclic endpoint face. The S-direction uses the other two actions in the same way. The Koszul symmetry supplies exactly their endpoint signs, and inserted units have degree zero. Thus the diagrams agree after their two indices are exchanged.

For the totalization convention (8.5), exchanging degrees p, q adds $(-1)^{pq}$. Indeed, their two bar differentials have signs $b_p + (-1)^p b_q$ before the exchange and $b_q + (-1)^q b_p$ afterwards. The internal differential has sign $(-1)^{p+q}$ on both sides. This verifies the chain comparison. Iterated realizations commute. If $P \odot_T Q \simeq S$ and $Q \odot_S P \simeq T$, substitution proves the last assertion. □

Let ω_X denote the right differential module of top forms, placed in cohomological degree zero.

THEOREM 9.4.3. Suppose connected envelopes G_B, for ordinary unital k-algebras B, satisfy

$$\tau_{\le 1}(G_B) = (k, \omega_X \otimes_k B)$$

with the indicated unit and scalar actions. There is no assignment on all coefficient bimodules, their maps, and equivalences that sends the regular bimodule B to G_B and every derived coefficient composition to an equivalence of composites over these envelopes. The obstruction already occurs for $\mathrm{Mat}_d(k)$ and k, with $d \ge 2$.

Proof. Set $B = \mathrm{Mat}_d(k)$, $e = E_{11}$, $P = Be$, and $Q = eB$. Matrix multiplication gives

$$P \otimes_k Q \simeq B, \qquad Q \otimes_B P \simeq k.$$

For the first map the basis tensors $E_{i1} \otimes E_{1j}$ map to E_{ij}. For the second, $E_{1i} \otimes E_{j1} = \delta_{ij}(e \otimes e)$; the inverse sends 1 to $e \otimes e$. All modules are projective on the balanced sides, so these also compute the derived tensors. Associativity of matrix multiplication gives their two Morita triangle identities.

The proposed assignment would yield inverse bimodule interfaces between G_B and G_k. Applying $\tau_{\leq 1}$ and Lemma 9.4.2 would identify their Hochschild chain objects. By Lemma 9.4.1, the arity-one components would be

$$(\omega_X \otimes_k B) \oplus (\omega_X \otimes_k B)[1] \simeq \omega_X \oplus \omega_X[1].$$

Their degree-zero cohomology differential modules have underlying O_X-ranks d^2 and 1, which is impossible. □

This obstruction applies to every assignment with the stated composition and unit requirements. It does not require its empty component to equal the original bimodule. It leaves the relative construction, lax restriction, and restricted classes of coefficient interfaces available.

9.5 Reconstruction along an arity filtration

Reconstruction asks whether the extended module $F(M)$, equipped with its canonical coring coaction, determines M. The answer below permits unbounded cohomological degrees. Its convergence comes from arity.

For this section k is any field. Assume that objects of C have nonnegative arities and that truncation to arities at most N gives exact strong monoidal functors

$$q_N : C \longrightarrow C_{\leq N}.$$

Assume $C_{\leq 0} = \mathcal{D}(k)$, with K supported at arity zero and $K_0 = k$. Assume these functors preserve limits and colimits, jointly detect equivalences, and have the same properties on the module and bimodule categories in use. Products add lower bounds on arity: a product of r positive-arity objects vanishes in arities less than r. These are hypotheses on the enhanced carrier. They must be verified before applying the theorem to a geometric coefficient system.

Let $G \to K$ be an algebra augmentation inducing $G_0 = k$ at arity zero. Regard K as a G-bimodule through this augmentation. Assume also that a G-bimodule map to K is null whenever its arity-zero component is null as a k-bimodule map. This holds if extension by zero at arity zero is right adjoint to arity-zero evaluation on G-bimodules.

Let $f : G \to R$ be a unital algebra map whose arity-zero unit $k \to R_0$ splits in $\mathcal{D}(k)$. Put

$$I = \mathrm{fib}(G \to R), \qquad u : I \to G, \qquad J = \mathrm{fib}(G \to K).$$

Here J has positive arity. In this section every power of I or J is balanced over G, in its written order.

LEMMA 9.5.1. The map u factors through J, and

$$q_N\big(u^{\star_G(N+1)} : I^{\star_G(N+1)} \longrightarrow G\big) = 0. \tag{9.3}$$

Proof. The splitting of $k \to R_0$ makes the fibre map $I_0 \to k$ null. The assumption on maps to K therefore makes the composite $I \to G \to K$ null as a bimodule map. The fibre sequence defining J gives the factorization.

Each term of the multibar for $J^{\star_G r}$ contains r positive-arity J-factors separated by nonnegative-arity G-factors. It vanishes below arity r. Truncation preserves its realization, so $q_N(J^{\star_G(N+1)}) = 0$. Tensoring the factorization of u with itself proves (9.3). □

The nilpotence concerns a map. Its source need not vanish. For example, for $k \to \mathrm{Mat}_2(k)$ at arity zero,

$$I_{\mathrm{o}} \simeq (\mathrm{Mat}_2(k)/k)[-1], \qquad \dim_k H^r(I_{\mathrm{o}}^{\otimes r}) = 3^r.$$

Every one of these tensor powers is nonzero, while the fibre map to k is null.

For a left G-module M, and $r \geq 1$, form the punctured ordered cube

$$Q_r(M) = \lim_{\varnothing \neq S \subseteq \{1,\dots,r\}} E_1(S) \star_G \cdots \star_G E_r(S) \star_G M, \qquad E_j(S) = \begin{cases} R, & j \in S, \\ G, & j \notin S. \end{cases} \tag{9.4}$$

The arrows insert f. Restriction to subsets of $\{1, \dots, r\}$ gives $Q_{r+1}(M) \to Q_r(M)$.

Theorem 9.5.2 *(Ordered reconstruction).* Under the stated arity and splitting hypotheses, there are natural fibre sequences

$$I^{\star_G r} \star_G M \longrightarrow M \longrightarrow Q_r(M),$$

and a natural equivalence

$$M \simeq \lim_r Q_r(M).$$

Consequently $F = R \star_G -$ detects equivalences.

Proof. Adjoin the empty subset to the cube. Its initial vertex is M. Taking a fibre in each of its r directions replaces that direction's $G \to R$ by I. Exactness of balanced tensor therefore identifies the total fibre with $I^{\star_G r} \star_G M$. This proves the fibre sequence.

A transition on these fibres applies u to one terminal I-factor. After q_N, any $N + 1$ successive transitions are null by (9.3). The inverse limit of this fibre tower is zero. Indeed, mapping any test object into it gives a tower of spectra whose homotopy groups have uniformly zero transition composites. Both the inverse limit and the first derived inverse limit of each such group tower vanish. The Milnor exact sequence makes the limit mapping spectrum contractible. Preservation of limits by q_N and joint detection now give the reconstruction equivalence.

If $FM = 0$, choose the rightmost R-factor in any nonempty vertex of (9.4). All factors to its right are regular G-modules, so that vertex contains $R \star_G M = 0$. Every $Q_r(M)$ is zero, and reconstruction gives $M = 0$. Apply this to the fibre of a map to prove detection of equivalences. □

There is no uniform nilpotence exponent in this argument: the bound $N + 1$ is applied at fixed arity before taking the arity limit. Nor does it bound coefficient resolution length. That length can remain infinite even at arity zero.

9.6 Effective coring descent

Conservativity of extension of scalars detects equivalences, but it does not alone imply that every coherent coring coaction is effective. The remaining condition is preservation of the totalizations that become split after extension.

A split augmented cosimplicial object has an extra codegeneracy contracting its augmentation. A tower is called *pro-constant* if its formal inverse system is equivalent to a constant object in the pro-category. For towers, the mapping spaces there are

$$\mathrm{Map}_{\mathrm{pro}}(\{X_i\}, \{Y_j\}) = \lim_j \operatorname*{colim}_i \mathrm{Map}(X_i, Y_j).$$

Two facts will be used. Constant pro-objects in a stable idempotent-complete category are closed under finite cofibres, shifts, and retracts. Also, the partial-totalization tower of a split augmented cosimplicial object is pro-constant at its augmentation object. The latter follows because the split augmentation remains a limit diagram under every functor; applying the constant embedding into the pro-category gives the assertion for the tower of finite totalizations.[1]

LEMMA 9.6.1. At fixed arity at most N, let $Z^\bullet$ be a cosimplicial G-module such that $FZ^\bullet$ extends to a split augmented cosimplicial R-module. The partial-totalization tower of $Z^\bullet$ is pro-constant. Every exact functor preserves its totalization.

Proof. Work in the truncated module category and suppress q_N. Call a G–G bimodule P admissible if

$$\{\mathrm{Tot}_n(P \star_G Z^\bullet)\}_n$$

is pro-constant. Since finite totalizations are finite stable limits, admissible bimodules are closed under finite cofibres, shifts, and retracts. The split augmentation makes R admissible.

If P is admissible and Q is any bimodule, the exact functor $Q \star_G -$ preserves each finite totalization. Its extension to pro-objects carries a constant tower to a constant tower. Hence $Q \star_G P$ is admissible.

Put $d = N + 1$. Factor the null map $u^{\star_G d}$ through

$$I^{\star_G d} \longrightarrow I^{\star_G (d-1)} \longrightarrow \cdots \longrightarrow I \longrightarrow G,$$

applying u on the right. The successive cofibres are $I^{\star_G j} \star_G R$, $0 \le j < d$. They are admissible. The cofibre of the composite is a finite extension of them, and G is a retract of this cofibre because the composite is null. Thus G is admissible.

A pro-constant tower has inverse limit equal to its constant value. An exact functor preserves finite totalizations and carries that formal constant tower to the corresponding constant tower of its image. It therefore preserves this totalization, without needing to preserve arbitrary products. □

For an F-split cosimplicial object in the full category, apply the lemma after each q_N. Since q_N preserves limits and balanced tensors, it identifies the image of the comparison

$$F\,\mathrm{Tot}\, Z^\bullet \longrightarrow \mathrm{Tot}\, FZ^\bullet \tag{9.5}$$

with the equivalence in the truncated category. Joint detection proves (9.5) is an equivalence in the full category.

Write $K_f = FU = T_f \star_R -$. A coherent T_f-comodule means a coalgebra for this comonad; denote its category by $\mathrm{Coalg}_{K_f}(\mathrm{Mod}_R(C))$.

THEOREM 9.6.2 *(Effective descent).* Under the hypotheses of Section 9.5, extension of scalars with its canonical coaction gives an equivalence

$$\mathrm{Mod}_G(C) \simeq \mathrm{Coalg}_{K_f}\big(\mathrm{Mod}_R(C)\big), \qquad M \longmapsto (FM, F\eta_M).$$

For a coherent comodule (N, γ), its inverse is

$$\mathcal{R}_f(N,\gamma) = \mathrm{Tot}_{[n]\in\Delta}\, UK_f^n N. \tag{9.6}$$

No cohomological boundedness assumption is imposed.

[1] See A. Mathew, *The Galois group of a stable homotopy theory*, Example 3.11 and the proof of Proposition 3.22, arXiv:1404.2156.

Proof. The adjunction supplies the comonad counit $\epsilon : K_f \to \mathrm{Id}$ and comultiplication $\delta = F\eta U$. In the cosimplicial object of (9.6), the cofaces out of degree n are

$$d^0 = \eta_{UK_f^n N}, \qquad d^i = UK_f^{i-1}\delta_{K_f^{n-i}N}\ (1 \le i \le n), \qquad d^{n+1} = UK_f^n\gamma.$$

The codegeneracies apply the counit in each possible position. Adjunction and coaction coherences give the entire cosimplicial diagram.

Mapping a G-module into its totalization and applying the adjunction at each vertex identifies the resulting mapping space with maps of coherent K_f-coalgebras from $(FM, F\eta_M)$ to (N, γ). Thus $\mathcal{R}_f$ is right adjoint to the comparison functor.

After F, the defining diagram has the split augmentation

$$N \xrightarrow{\gamma} K_f N \rightrightarrows K_f^2 N \longrightarrow \cdots,$$

whose extra codegeneracy is the counit at the left end. The limit comparison already proved gives an equivalence

$$F\,\mathrm{Tot}\,UK_f^\bullet N \simeq \mathrm{Tot}\,K_f^{\bullet+1}N \simeq N.$$

It respects coactions because it is induced by the full cosimplicial diagram. This is the counit of the comparison adjunction. It is therefore an equivalence.

The triangle identity shows that the comparison functor sends its unit to an equivalence. The underlying functor of the comparison is F, which detects equivalences by Theorem 9.5.2. Hence the unit is also an equivalence, proving the theorem. □

The totalization in (9.6) is a limit. Let N^n denote cosimplicial normalization in degree n. In dg representatives the totalization uses the product

$$\prod_{n\ge 0} \mathrm{N}^n(UK_f^\bullet N)[-n].$$

This product belongs to the descent construction; it is not a replacement of the direct-sum realizations used to define balanced tensor products. The counit of the equivalence is the map from this whole totalization. It is generally different from the individual map $K_f N \to N$.

Over a field, a nonacyclic unital dg algebra R_0 has a split unit as a complex. Indeed its unit cannot be a boundary: a primitive for 1, multiplied by any cocycle, would make every cocycle a boundary. Splitting complexes over the field then supplies a chain map $R_0 \to k$ taking 1 to 1. Thus, on a carrier satisfying the arity hypotheses and the hypothesis on maps to K, every such $G \to R$ has effective coring descent. Over a general arity-zero algebra S, the corresponding argument requires a splitting of $S \to R_0$ in $\mathcal{D}(S\text{-}S)$. A splitting of underlying scalar complexes does not supply that bimodule splitting.

Chapter 10

Arithmetic coefficients and local conditions

A global cohomology class can satisfy a local condition because its restriction differs from an allowed local class by a coboundary. The primitive of that coboundary belongs to the relative cocycle. Retaining it gives an explicit complex, and compatible local multiplications give the two products of Chapter 11. Forgetting it before taking products loses part of the arithmetic input.

Coefficients must also be retained before taking limits. Every finite ℓ-power torsion group vanishes after tensoring with $\mathbb{Q}_\ell$, whereas their inverse limit can have positive $\mathbb{Z}_\ell$-rank. We therefore construct the finite coefficient complexes first, specify their transition maps, and only then complete. Deformation functors impose a further requirement: all local and global objects must be defined on the same coefficient category. Their tangent complexes give local conditions on cochains. Their geometric representability is a separate question.

10.1 The arithmetic curve and its coefficients

Fix a number field K, its ring of integers $\mathcal{O}_K$, and a rational prime ℓ. A finite place v is a nonzero prime ideal of $\mathcal{O}_K$. Its completion K_v has valuation ring $\mathcal{O}_v$, maximal ideal $\mathfrak{m}_v$, and finite residue field k_v of characteristic p_v. The arithmetic residue prime p_v and the coefficient prime ℓ serve different purposes. They may coincide, but no such equality is imposed here.

The scheme $\operatorname{Spec}\mathcal{O}_K$ has dimension one. Its closed points are the finite places. Its generic point has field K. Archimedean completions are additional data, not additional closed points of this scheme.

The adele ring $\mathbb{A}_K$ is the restricted product of the completions, with respect to $\mathcal{O}_v$ at finite places. An adele is a tuple of local elements. Identifying such a tuple with a coordinate for chiral insertions would require a geometric construction and its transition maps.

Let S be a finite set of places containing the archimedean places and the places over ℓ. Write S_f for its finite part. Let $G_{K,S}$ be the Galois group of the maximal extension of K unramified outside S, and let G_{K_v} be the absolute Galois group of K_v.

Choose embeddings of algebraic closures defining continuous homomorphisms

$$j_v : G_{K_v} \longrightarrow G_{K,S} \qquad (v \in S_f).$$

These homomorphisms need not be injective. Strict restriction maps use the chosen j_v. Conjugate choices require cochain homotopies when the resulting complexes are compared.

Put $R_r = \mathbb{Z}/\ell^r\mathbb{Z}$, for $r \geq 1$. Complexes are cohomological, with differential of degree one and $M[1]^n = M^{n+1}$. Tensor products are over the displayed coefficient ring and use the Koszul sign rule. An ordinary tensor product denotes the algebraic tensor product. A derived or completed tensor product will be specified explicitly.

10.2 CONTINUOUS COCHAINS WITH THEIR MULTIPLICATION

Let G be profinite, R a commutative ring with trivial G-action, and M a discrete continuous $R[G]$-module. The normalized continuous cochains $C^n(G, M)$ are the continuous functions $G^n \to M$ vanishing when any argument is the identity. Set $C^0(G, M) = M$ and $C^n(G, M) = 0$ for $n < 0$.

Their differential is

$$\begin{aligned}(dc)(g_1, \dots, g_{n+1}) = {} & g_1 c(g_2, \dots, g_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i c(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ & + (-1)^{n+1} c(g_1, \dots, g_n). \end{aligned} \tag{10.1}$$

In degree zero this says $(dm)(g) = gm - m$. Continuous group cohomology is the cohomology of this complex.[1]

Suppose now that M is an associative unital R-algebra and that G acts by R-algebra automorphisms. For cochains c, c' of degrees n, m, define the ordered cup product by

$$(c \smile c')(g_1, \dots, g_{n+m}) = c(g_1, \dots, g_n)(g_1 \cdots g_n) c'(g_{n+1}, \dots, g_{n+m}). \tag{10.2}$$

No commutativity of M is needed. When the action is trivial, both action factors in (10.1) and (10.2) disappear.

PROPOSITION 10.2.1. The differential squares to zero. Under the algebra hypothesis, $C^\bullet(G, M)$ is a unital associative differential graded R-algebra. Restriction along continuous group homomorphisms, with the induced coefficient action, preserves its differential, product, and unit.

Proof. Write $d = \sum_i (-1)^i d_i$ for the faces in (10.1). For $i < j$, the identity $d_j d_i = d_i d_{j-1}$ pairs two terms of d^2 with opposite signs. The identity for the first two faces uses $g(hm) = (gh)m$. The adjacent inner faces use associativity in G. Every term is paired, so $d^2 = 0$. If an argument is the identity, the two neighboring faces cancel and all remaining terms vanish by normalization.

In the differential of a cup product, split the faces at the boundary of its two argument blocks. The faces in the first block give $(dc) \smile c'$, and those in the second give $(-1)^n c \smile dc'$. The extra faces at the boundary cancel. The action on the second factor in (10.2) makes this cancellation valid also for a nontrivial coefficient action. This proves Leibniz's rule.

For three cochains the two bracketings evaluate the same three blocks. The second value is acted on by the first block, and the third by the first two blocks. The action is by algebra automorphisms, so these expressions agree. The element 1_M in degree zero is a closed two-sided unit. Substitution of a group homomorphism in each argument respects every displayed formula. □

PROPOSITION 10.2.2. For trivial coefficients R_r, reduction gives degreewise surjections and isomorphisms

$$C^n(G, R_{r+1}) \otimes_{R_{r+1}} R_r \cong C^n(G, R_r).$$

These maps preserve the differential, cup product, and restrictions. Each module $C^n(G, R_r)$ is flat over R_r.

[1] J. S. Milne, *Class Field Theory*, version 4.03 (2020), Chapter II, §4, pp. 86–89, gives the continuous-cochain and finite-quotient conventions.

Proof. A continuous function from a compact space to a discrete space has finite image and is constant on a finite clopen partition. Lift each value from R_r to R_{r+1}, choosing the zero lift for the zero value. This preserves normalization.

A function reducing to zero has values in $\ell^r R_{r+1}$. Choosing preimages of its finitely many values shows that the kernel is $\ell^r C^n(G, R_{r+1})$. This proves the tensor assertion. All operations use only coefficient addition and multiplication.

Fix a finite clopen partition of G^n. Normalized functions constant on its cells form a finite free R_r-module. A basis consists of the characteristic functions of cells disjoint from all tuples with an identity argument. Refinement gives a filtered system whose union is all normalized continuous cochains. Filtered colimits of flat modules are flat. □

The finite models are compatible with the profinite topology. For trivial coefficients, every continuous cochain factors through $(G/U)^n$ for some open normal subgroup $U \subset G$. Indeed, its finite clopen fibers admit a finite cover by products of cosets of open subgroups. A common open normal subgroup refining those subgroups makes the function constant on the required product cosets. Normalization then holds on G/U, because a coset of the identity has the same value as an actual identity argument.

Consequently

$$C^\bullet(G, R_r) = \varinjlim_U C^\bullet(G/U, R_r),$$

with inflation as transition maps. The differential and cup product are compatible with this expression.

This flatness assertion concerns modules in individual degrees. It does not prove that an unbounded complex is K-flat, meaning that tensoring it with any acyclic complex remains acyclic. The resolution hypotheses of Proposition 3.3.1 are still needed when the coefficient bar is used to compute a derived tensor product.

10.3 Local conditions and retained primitives

Fix a discrete continuous $G_{K,S}$-module M over R. A cochain local condition at v consists of a complex L_v and a chain map

$$g_v : L_v \longrightarrow C^\bullet(G_{K_v}, M).$$

Set $A = C^\bullet(G_{K,S}, M)$, $B = \prod_{v \in S_f} C^\bullet(G_{K_v}, M)$, and $L = \prod_{v \in S_f} L_v$. Let $f = (j_v^*)_v$ and $g = (g_v)_v$. The products are finite, so they are also direct sums of complexes. Define the Selmer complex for these cochain data by

$$\begin{aligned} C^n_{\mathcal{L}} &= A^n \oplus L^n \oplus B^{n-1}, \\ D(a, c, y) &= (da, dc, f(a) - g(c) - dy). \end{aligned} \tag{10.3}$$

The symbol $\mathcal{L}$ records the chosen complexes and maps. A subspace of local cohomology alone does not specify these data.

Proposition 10.3.1. The complex $C^\bullet_{\mathcal{L}}$ is the homotopy fiber of $f - g : A \oplus L \to B$. In particular it has the exact sequence

$$\cdots \longrightarrow H^{n-1}(B) \longrightarrow H^n(C^\bullet_{\mathcal{L}}) \longrightarrow H^n(A) \oplus H^n(L) \xrightarrow{f_* - g_*} H^n(B) \longrightarrow \cdots .$$

A cocycle is a pair of global and allowed local cocycles together with a specified primitive of their difference after restriction.

Proof. The third component of D^2 is $f(da) - g(dc) - d(f(a) - g(c) - dy) = 0$. Projection onto $A \oplus L$ has kernel $B[-1]$, with differential $-d_B$. The resulting short exact sequence is the mapping-fiber sequence for $f - g$, and its connecting map is $f_* - g_*$. The cocycle equations are exactly those stated. □

Three choices exhibit the range of this definition. The zero condition $L_v = 0$ asks for a primitive of the local restriction. The full condition $L_v = C^\bullet(G_{K_v}, M)$, with g_v the identity, imposes no restriction. If every condition is full, the map $a \mapsto (a, f(a), 0)$ is a quasi-isomorphism $A \to C^\bullet_{\mathcal{L}}$: its complementary fiber is the contractible fiber of 1_B.

For a finite place, write I_v for inertia in G_{K_v}. The unramified cochain condition is

$$L_v = C^\bullet(G_{K_v}/I_v, M^{I_v}) \xrightarrow{\text{inflation}} C^\bullet(G_{K_v}, M). \tag{10.4}$$

The quotient acts on the inertia invariants, and inflation followed by their inclusion defines the map. If M is a coefficient algebra with Galois action by automorphisms, then M^{I_v} is a unital subalgebra and inflation preserves the cup product.

General local conditions are complexes. When A, L, B are strict dg algebras and f, g are algebra maps, there are additional products. For $x = (a, c, y)$ of degree n, they are

$$\mu_L(x, x') = (aa', cc', (-1)^n f(a)y' + yg(c')),$$
$$\mu_R(x, x') = (aa', cc', yf(a') + (-1)^n g(c)y').$$

Proposition 11.1.1 proves their dg identities. Chapter 11 constructs the comparison in every tensor length, including the module and bimodule maps. These conclusions apply to a local condition only after its algebra structure and the multiplicativity of g_v have been given.

If the cospan is unital, $u = (1_A, 1_L, 0)$ is the native unit of both products. For the zero local condition, the displayed triple provides no such common unit.

In either case the external unitization

$$R \oplus C^\bullet_{\mathcal{L}}, \qquad (r, x)(r', x') = (rr', rx' + r'x + xx'),$$

has augmentation $(r, x) \mapsto r$ and new unit $(1, 0)$. When u exists, $(0, u)$ is an idempotent in its augmentation ideal. It is not the new unit. The coefficient bar uses this external augmentation. Local vacua, scalar augmentations, and empty geometric cuts retain their distinct definitions.

A comparison of cochain models may respect restriction only up to homotopy. For the zero local condition, suppose algebra maps $\alpha : A \to A'$, $\beta : B \to B'$ and a degree-minus-one map $H : A \to B'$ satisfy $f'\alpha - \beta f = dH + Hd$. Then

$$(a, y) \longmapsto (\alpha(a), \beta(y) + H(a))$$

is a chain map of relative complexes. Its multiplicative defect is computed in Proposition 36.6.1. A nonzero defect requires higher comparison maps before one obtains a bar-coalgebra map. A chain homotopy alone is not that construction.

10.4 Completion before rationalization

Give $\mathbb{Z}_\ell = \varprojlim_r R_r$ its inverse-limit topology. A map $G^n \to \mathbb{Z}_\ell$ is continuous exactly when each of its reductions is continuous. Hence normalized cochains satisfy

$$C^\bullet_{\mathrm{cts}}(G, \mathbb{Z}_\ell) = \varprojlim_r C^\bullet(G, R_r).$$

The differential, cup product, and restrictions pass to this limit. For a general family of local conditions $L_{v,r}$, transition maps and coefficient base-change properties must be supplied as part of that family.

Proposition 10.4.1. Let E_r be a countable tower of complexes with degreewise surjective transition maps. There is an exact sequence

$$0 \longrightarrow \varprojlim{}^{1}{}_r H^{n-1}(E_r) \longrightarrow H^n(\varprojlim_r E_r) \longrightarrow \varprojlim_r H^n(E_r) \longrightarrow 0.$$

Here $\varprojlim^1$ is the cokernel of $1 - \text{shift}$ on the product of the indicated cohomology groups.

Proof. The map $1 - \text{shift} : \prod_r E_r \to \prod_r E_r$ is surjective in every degree. To solve $x_r - t_r(x_{r+1}) = y_r$, choose x_1 and lift $x_r - y_r$ recursively along the surjection $t_r : E_{r+1} \to E_r$. Its kernel is the inverse limit. Thus

$$0 \longrightarrow \varprojlim_r E_r \longrightarrow \prod_r E_r \xrightarrow{1-\text{shift}} \prod_r E_r \longrightarrow 0$$

is an exact sequence of complexes. Products of modules are exact, so their cohomology is the product of the cohomology groups. The resulting long exact sequence gives the assertion. □

The hypothesis holds for the coefficient cochains and their relative complexes with zero or full local conditions. It must be checked for other conditions. Even under this hypothesis, the possible $\varprojlim^1$ term prevents replacing the cohomology of the limit by the limit of the cohomology groups without further argument.

For a compatible tower of relative dg algebras P_r over R_r, let $\mathcal{B}_r$ be the coefficient bar of $R_r \oplus P_r$. It is the direct sum of its finite tensor words, with the signs of (3.2). Its coefficient completion and coproduct target are

$$\widehat{\mathcal{B}} = \varprojlim_r \mathcal{B}_r, \qquad \widehat{\mathcal{B}}\,\widehat{\otimes}\,\widehat{\mathcal{B}} := \varprojlim_r (\mathcal{B}_r \otimes_{R_r} \mathcal{B}_r). \tag{10.5}$$

Every element has finite word support at each coefficient precision. This is smaller than an unrestricted product over word lengths. The differential, coproduct, and endpoint comparisons pass to (10.5) by their compatible finite-stage formulas, as proved in Chapters 3 and 11.

Each finite-stage bar is conilpotent: sufficiently many reduced cuts kill any finite sum of words. The limit need not be conilpotent. Take a free rank-one module with basis a, where $|a| = 1$. The degree-zero element $\sum_{n\geq 1} \ell^n (sa)^{\otimes n}$ has finite support modulo every ℓ^r. Every iterated reduced coproduct has a nonzero term. Its existence depends on coefficient completion, not on a bound on geometric configurations.

Example 10.4.2. Each $R_r \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ is zero, since ℓ^r annihilates its first factor and is invertible on the second. But

$$(\varprojlim_r R_r) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell = \mathbb{Q}_\ell.$$

The constant degree-zero cochains already exhibit this difference. There is no unital ring map $R_r \to \mathbb{Q}_\ell$: the relation $\ell^r 1 = 0$ would persist in the target. Thus a characteristic-zero realization cannot be constructed by rationalizing these finite coefficient algebras separately.

10.5 RESTRICTION, TRANSFER, AND TENSOR SUPPORT

Restriction and coefficient reduction are multiplicative in the models above. Additive transfer has a different behavior. If $H \subset G$ has finite index n, corestriction on degree-zero cohomology with trivial $\mathbb{Z}_\ell$-coefficients is multiplication by n. For finite groups this is the coset sum. For profinite groups the same formula uses the finite coset space.

PROPOSITION 10.5.1. If $n > 1$, corestriction with trivial $\mathbb{Z}_\ell$-coefficients is not multiplicative. There is no A_∞ morphism of these cochain algebras whose induced cohomology map is corestriction.

Proof. On the class of the degree-zero unit, multiplicativity would give $n = \mathrm{cor}(1 \cdot 1) = \mathrm{cor}(1)\,\mathrm{cor}(1) = n^2$. The ring $\mathbb{Z}_\ell$ has characteristic zero, so this fails for $n > 1$. The arity-two identity for an A_∞ morphism makes its first component multiplicative on cohomology, yielding the same contradiction. For index ℓ, reduction modulo ℓ^2 already detects the failure: $\ell \neq \ell^2 = 0$. □

Multiplicative norm operations, when constructed, have different axioms and cannot be substituted for this additive transfer without specifying their coefficient and algebra structures.

Changing the coefficient prime creates another separation. An abelian group is ℓ-primary torsion if every element is killed by some power of ℓ. For different primes, the derived tensor of two such groups vanishes.

PROPOSITION 10.5.2. If M_ℓ and M_p are respectively ℓ-primary and p-primary torsion groups, with $\ell \neq p$, then $M_\ell \otimes^{\mathbf{L}}_{\mathbb{Z}} M_p = 0$. For equal primes the corresponding assertion is false.

Proof. The resolution $0 \to \mathbb{Z} \xrightarrow{\ell^a} \mathbb{Z} \to \mathbb{Z}/\ell^a \to 0$, tensored with M_p, is acyclic: multiplication by ℓ^a is invertible on every p-primary group. Every finitely generated subgroup of M_ℓ is a finite sum of cyclic ℓ-power groups. Filtered unions and their exact tensor and Tor functors extend the calculation to M_ℓ. There is no higher Tor over $\mathbb{Z}$. For equal primes,

$$\mathbb{F}_\ell \otimes^{\mathbf{L}}_{\mathbb{Z}} \mathbb{F}_\ell \simeq \mathbb{F}_\ell \oplus \mathbb{F}_\ell[1],$$

since the differential in the same two-term resolution becomes zero. The shift puts its Tor group in cohomological degree -1. □

This calculation concerns tensor products over $\mathbb{Z}$. It neither supplies comparison maps between local geometric categories nor makes two factors with the same coefficient field disjoint.

10.6 DEFORMATION FUNCTORS ON ONE COEFFICIENT CATEGORY

Local conditions on representations supply cochain conditions by linearization. To form their global pullback, first fix a finite field k of characteristic ℓ, its Witt ring $W(k)$, and a continuous residual representation

$$\bar{\rho} : G_{K,S} \longrightarrow \mathrm{GL}_d(k).$$

Use a single category $\mathrm{Art}_{W(k)}$ of augmented simplicial Artinian local $W(k)$-algebras with residue field k. Deformation functors take values in spaces and invert weak equivalences of these coefficient algebras. A local theory over the Witt ring of a different residue characteristic is not automatically a functor on this category.

For a profinite group G with the specified residual representation, write $\mathrm{Def}_G(A)$ for its derived continuous deformation space. Concretely, it is the homotopy fiber at $\bar\rho$ of

$$\mathrm{Map}_{\mathrm{cts}}(BG, B\mathrm{GL}_d(A)) \longrightarrow \mathrm{Map}_{\mathrm{cts}}(BG, B\mathrm{GL}_d(k)),$$

using the continuous derived mapping-space model.

Here BG is the classifying space of G, and the fiber retains an identification of the reduction with $\bar\rho$. For ordinary Artinian A, objects are lifts of $\bar\rho$. Arrows are conjugations by matrices reducing to the identity. These arrows retain infinitesimal automorphisms.

Keeping a basis and forbidding those conjugations gives the framed deformation problem instead.[2]

Let $\mathrm{ad}\bar\rho = M_d(k)$, with action $g \cdot X = \bar\rho(g) X \bar\rho(g)^{-1}$. The tangent complex of a deformation functor records its linear response to square-zero coefficient extensions. With the cohomological convention here, its degrees $-1, 0, 1$ describe infinitesimal automorphisms, deformations, and obstructions.

Proposition 10.6.1. The tangent complex of Def_G at $\bar\rho$ is

$$T\mathrm{Def}_G \simeq C^\bullet(G, \mathrm{ad}\bar\rho)[1].$$

In particular the unframed first-order deformation space is $H^1(G, \mathrm{ad}\bar\rho)$, its infinitesimal automorphisms are $H^0(G, \mathrm{ad}\bar\rho)$, and the framed first-order space is $Z^1(G, \mathrm{ad}\bar\rho)$.

Proof. Over $k[\epsilon]/\epsilon^2$, write a lift as $\rho_\epsilon(g) = (1 + \epsilon c(g))\bar\rho(g)$. The representation equation becomes

$$c(gh) = c(g) + g \cdot c(h),$$

the cocycle equation in (10.1). Conjugation by $1 + \epsilon X$ changes c by $X - g \cdot X = -dX$. Quotienting gives H^1, while the stabilizer is $\ker d = H^0$. Before quotienting, the lifts form Z^1.

The same calculation applies to a simplicial square-zero module N. The kernel of $\mathrm{GL}_d(k \oplus N) \to \mathrm{GL}_d(k)$ is the additive simplicial module $M_d(N)$, since products of two elements of N vanish. Conjugation induces the adjoint action.

The continuous cosimplicial mapping model has normalized cochains $C^\bullet(G, \mathrm{ad}\bar\rho \otimes_k N)$. Passing to the classifying space shifts this linear complex by one. This identifies the linear response on square-zero modules, hence the stated tangent complex. Automorphisms remain in degree -1. □

Fix a determinant lift $\delta : G_{K,S} \to W(k)^\times$. There are two distinct meanings of fixed determinant. One may require the determinant character to have the value δ, while allowing all strict GL_d-conjugations. Alternatively, one may retain an identification of the determinant representation with δ and require arrows to preserve that identification. The second problem is the homotopy fiber of the determinant map of deformation functors. Denote it by $\mathrm{Def}^{\mathrm{or}}_{G,\delta}$.

Proposition 10.6.2. For the determinant-identified problem,

$$T\mathrm{Def}^{\mathrm{or}}_{G,\delta} \simeq C^\bullet(G, \mathrm{ad}^0\bar\rho)[1], \qquad \mathrm{ad}^0\bar\rho = \ker(\mathrm{tr} : M_d(k) \to k).$$

This holds also when ℓ divides d.

[2] S. Galatius and A. Venkatesh, *Derived Galois deformation rings*, arXiv:1608.07236, §§4–5 and 7, construct the derived continuous deformation functors and their tangent complexes. Their §9 imposes local conditions by homotopy pullbacks.

Proof. For a square-zero matrix, expansion of the determinant gives $\det(1 + \epsilon X) = 1 + \epsilon \operatorname{tr} X$. The tangent map of determinant is therefore the trace map

$$C^\bullet(G, \operatorname{ad}\bar{\rho})[1] \longrightarrow C^\bullet(G, k)[1].$$

Trace is equivariant and surjective. It is surjective on normalized cochains in every degree: lift a scalar value a to $\operatorname{diag}(a, 0, \dots, 0)$. This choice preserves continuity and normalization, though it need not be equivariant as a splitting. The kernel complex is $C^\bullet(G, \operatorname{ad}^0\bar{\rho})$, so the homotopy fiber is quasi-isomorphic to that kernel. No division by d occurs. □

For the determinant-value problem modulo all strict conjugations, the first-order quotient is instead

$$Z^1(G, \operatorname{ad}^0\bar{\rho}) \,/\, d\big(M_d(k)\big),$$

and its infinitesimal automorphisms are $H^0(G, \operatorname{ad}\bar{\rho})$. For a trivial two-dimensional representation, these automorphisms have dimension four. Preserving a determinant identification leaves the trace-zero automorphisms, of dimension three, also in characteristic two. The notation for the deformation problem must therefore specify which arrows are permitted.

10.7 The Selmer pullback and its tangent

Use the determinant-identified convention throughout this section. Write $\mathcal{D} = \operatorname{Def}^{\mathrm{or}}_{G_{K,S},\vartheta}$ and $\mathcal{D}_v = \operatorname{Def}^{\mathrm{or}}_{G_{K_v},\vartheta|_v}$. Choose pointed local deformation functors $\mathcal{D}_v^{\mathcal{L}}$ and actual morphisms $\mathcal{D}_v^{\mathcal{L}} \to \mathcal{D}_v$, all on $\operatorname{Art}_{W(k)}$. Define

$$\mathcal{D}_{\mathcal{L}} = \mathcal{D} \times^h_{\prod_{v\in S_f} \mathcal{D}_v} \prod_{v\in S_f} \mathcal{D}_v^{\mathcal{L}}. \tag{10.6}$$

The symbol $\times^h$ denotes a homotopy pullback of functors to spaces. It imposes compatibility together with its homotopy, rather than equality only on isomorphism classes.

THEOREM 10.7.1. Suppose each chosen local functor has tangent complex $L_v[1]$, and its morphism to $\mathcal{D}_v$ induces a chain map $g_v : L_v \to C^\bullet(G_{K_v}, \operatorname{ad}^0\bar{\rho})$. Then

$$T\mathcal{D}_{\mathcal{L}} \simeq C^\bullet_{\mathcal{L}}(G_{K,S}, \operatorname{ad}^0\bar{\rho})[1],$$

where the complex on the right is (10.3) with these actual maps. Geometric representability and Noetherianity are additional conditions on this formal deformation functor.

Proof. Evaluate (10.6) on every square-zero coefficient extension and take the fiber at its base point. Homotopy limits commute with homotopy limits, so this operation preserves the displayed pullback. The associated tangent spectra, and thus their complexes, give the fiber of

$$C^\bullet(G_{K,S}, \operatorname{ad}^0\bar{\rho})[1] \oplus \prod_v L_v[1] \longrightarrow \prod_v C^\bullet(G_{K_v}, \operatorname{ad}^0\bar{\rho})[1].$$

The arrow is restriction minus the chosen local map. Its mapping fiber is exactly (10.3), shifted by one. □

If each g_v is an inclusion of complexes, the quotient complex $C^\bullet(G_{K_v}, \operatorname{ad}^0\bar{\rho})/L_v$ gives an equivalent description of the same fiber. Without such inclusions the quotient notation has no supplied meaning. The mapping cone remains valid. The unramified deformation condition gives a concrete example when

both the residual local representation and $\delta|_v$ are unramified: inflate the deformation functor of G_{K_v}/I_v. Its tangent map is the inflation map of (10.4).

The tangent coefficients do not usually carry an associative product. In rank two the trace-zero matrices E_{12} and E_{21} have product E_{11}, whose trace is one. Thus $\mathrm{ad}^0\bar{\rho}$ is not an associative subalgebra of $M_2(k)$. A Selmer tangent complex therefore does not acquire the associative bar construction merely from its deformation-theoretic name. A separate coefficient algebra, and a comparison respecting its operations, are needed.

Representability, Noetherianity, perfectness, and cotangent amplitude are further conditions on the deformation problem. Obstructions in $H^2_{\mathcal{L}}$, which appear in tangent degree one, do not by themselves rule out quasi-smoothness. For a derived scheme, quasi-smoothness means that its cotangent complex is perfect of amplitude $[-1, 0]$. This permits tangent degree one.

Example 10.7.2. Let $A = k[x]/(x^2)$. The element x^2 is a non-zero-divisor in $k[x]$, so Spec A is a local complete intersection and is quasi-smooth. Its cotangent complex is

$$\mathbb{L}_{A/k} \simeq [A \xrightarrow{2x} A],$$

in degrees $-1, 0$. At $x = 0$ the differential vanishes, and the tangent has $H^1 = k$.

This example works also in characteristic two. Hence vanishing of the obstruction space is not equivalent to quasi-smoothness. Stack automorphisms require their own amplitude convention in addition to this scheme calculation.

10.8 Real places and the range of cohomology

Only finite places entered the relative product above. A theory of compact support over a number field must also specify its complexes at archimedean places, including any modified Tate-cohomology convention. At a complex place the absolute Galois group is trivial. At a real place it is C_2, the group of order two. Coefficients of characteristic two make the latter distinction unavoidable.

Proposition 10.8.1. For the trivial C_2-module $k = \mathbb{F}_2$, $H^n(C_2, k) = k$ for every $n \geq 0$.

Proof. If σ generates C_2, set $\eta = \sigma - 1$. Then $k[C_2] = k[\eta]/(\eta^2)$. A free resolution of the trivial module is

$$\cdots \xrightarrow{\eta} k[C_2] \xrightarrow{\eta} k[C_2] \xrightarrow{\eta} k[C_2] \longrightarrow k \longrightarrow 0.$$

The kernel and image of multiplication by η are both $k\eta$, so the resolution is exact. Applying $\mathrm{Hom}_{k[C_2]}(-, k)$ makes every differential zero. □

Consequently the unmodified real local cochains cannot be assigned a cohomological bound of two at the coefficient prime two. Neither a finite-place calculation nor a formal Selmer pullback supplies the missing archimedean convention. A comparison with de Rham or chiral objects at infinity must also specify its scalar field and geometric carrier.

10.9 Tame symbols and their recipients

Residues provide arithmetic operations with explicitly different targets at different places. For a field F, the Milnor group $K_2^M(F)$ is the abelian group generated by symbols $\{a, b\}$, for $a, b \in F^\times$. Its defining relations are bilinearity in the multiplicative arguments and the Steinberg relation $\{a, 1 - a\} = 0$ for $a \neq 0, 1$.

Let $v : F^\times \to \mathbb{Z}$ be a discrete valuation with residue field κ_v. Write $a = \pi^m u$ and $b = \pi^n w$, where π is a uniformizer and u, w are units. Set

$$\partial_v\{a, b\} = (-1)^{mn}\overline{w}^m\overline{u}^{-n} \in \kappa_v^\times. \tag{10.7}$$

This convention gives $\partial_v\{\pi, u\} = \bar{u}$.

PROPOSITION 10.9.1. Formula (10.7) is independent of the uniformizer and defines a homomorphism $K_2^M(F) \to \kappa_v^\times$.

Proof. Replacing π by πt replaces the unit parts by ut^{-m} and wt^{-n}. The extra powers of $\bar{t}$ cancel. Addition of valuations and multiplication of unit parts prove bilinearity. It remains to evaluate $\{a, 1 - a\}$.

If $v(a) > 0$, then $1 - a$ is a unit of residue one, so the symbol is one. If $v(a) = 0$, both arguments are units unless $v(1 - a) > 0$. In that case $\bar{a} = 1$, with the same result. If $m = v(a) < 0$, then $v(1 - a) = m$ and the ratio of their unit parts has residue -1. The symbol becomes $(-1)^{m^2+m} = 1$. Thus every Steinberg relation is respected. □

For $F = \mathbb{Q}$, the finite symbols land in $\bigoplus_p \mathbb{F}_p^\times$, where all but finitely many components are one. There is also a real sign symbol

$$h_\infty\{a, b\} = \begin{cases} -1, & a < 0 \text{ and } b < 0, \\ 1, & \text{otherwise.} \end{cases}$$

Signs make it bilinear, and a and $1 - a$ cannot both be negative, so it too respects the Steinberg relation.

THEOREM 10.9.2 *(Tate's rational-field calculation).* The combined homomorphism is an isomorphism

$$(h_\infty, (\partial_p)_p) : K_2^M(\mathbb{Q}) \xrightarrow{\sim} \{\pm 1\} \oplus \bigoplus_p \mathbb{F}_p^\times.$$

This is the classical computation of $K_2(\mathbb{Q})$, using the identification of field K_2 with its Milnor presentation.[3] The finite symbols alone omit the order-two class $\{-1, -1\}$: each finite valuation of its arguments is zero, while its real symbol is -1. The real factor is therefore necessary even before any comparison with geometric residues is attempted.

A product formula needs a common recipient. On the rational function field $F(t)$, the normed tame symbols satisfy

$$\prod_{x \in |\mathbb{P}^1_F|} N_{\kappa(x)/F}\big(\partial_x \xi\big) = 1 \quad (\xi \in K_2^M(F(t))). \tag{10.8}$$

Here every factor belongs to $F^\times$, the point at infinity is included, and only finitely many factors differ from one. This is the tame-symbol reciprocity theorem for the projective line.[4]

For example, if $\xi = \{t, u\}$, with $u \in F^\times$, the symbols at zero and infinity are u and u^{-1}. All others are one.

The finite residue fields of a number field do not come with these norm maps to a single ground field. Multiplying elements of $\mathbb{F}_p^\times$ for varying p without a specified recipient is not a reciprocity law. Furthermore, Spec O_K has no codimension-two points. Its dimension-one residue diagram cannot be relabelled as the two-step flag diagram of an arithmetic surface. The latter requires an actual surface, its flags, and the corresponding successive residue and norm maps.

[3] C. Weibel, *The K-book*, Chapter III, §6, especially Theorem 6.5.1 and the real symbol on pp. 231–234.

[4] Weibel, *The K-book*, Chapter III, Example 6.5.3, p. 236.

One can choose the incidence poset of finite subsets of finite places, with inclusions as arrows. It records which places occur together. It is distinct from the scheme configuration functor of pairwise disjoint sections of a curve. In particular it supplies neither collision differential forms nor their operation homotopies. Comparing the Steinberg and geometric Arnold relations requires a map between their complexes. Its compatibility with the relevant operations must also be proved.

10.10 Iwasawa coefficients and coalgebra length

For a procyclic group $\Gamma \cong \mathbb{Z}_\ell$, fix a topological generator γ. Its completed group ring is

$$\Lambda = \mathbb{Z}_\ell[[\Gamma]] := \varprojlim_{r,n} R_r[\Gamma/\ell^n\Gamma].$$

The choice of γ identifies this ring with a power-series ring. If Γ comes from a cyclotomic extension, this means its chosen procyclic factor. The full cyclotomic group can also have a finite factor. In $\mathbb{Z}_\ell^\times$, the standard procyclic subgroup is $1 + \ell\mathbb{Z}_\ell$ for odd ℓ, and $1 + 4\mathbb{Z}_2$ for $\ell = 2$.

Proposition 10.10.1. The map $\gamma \mapsto 1 + T$ induces a topological ring isomorphism $\Lambda \cong \mathbb{Z}_\ell[[T]]$.

Proof. The finite quotients are

$$R_r[T]\,/\,\big((1+T)^{\ell^n} - 1\big).$$

The ideals $(\ell^r, (1+T)^{\ell^n} - 1)$ are cofinal with the powers of (ℓ, T). To see one inclusion, fix a power $(\ell, T)^N$, take $r \geq N$, and increase n. For each $1 \leq j < N$, the identity

$$\binom{\ell^n}{j} = \frac{\ell^n}{j}\binom{\ell^n - 1}{j - 1}$$

gives valuation at least $n - v_\ell(j)$. Thus every coefficient of degree below N lies in the required power of ℓ.

Conversely, modulo the displayed ideal, T^{ℓ^n} belongs to $\ell R_r[T]$, because every intermediate binomial coefficient is divisible by ℓ. Hence $T^{r\ell^n} = 0$, and also $\ell^r = 0$. A sufficiently large power of (ℓ, T) therefore vanishes in each finite quotient. The two completions agree, and completion of $\mathbb{Z}[T]$ at (ℓ, T) is $\mathbb{Z}_\ell[[T]]$. □

An augmented dg Λ-algebra has the coefficient bar of Chapter 3. Any additional completion must again name its tensor target. A trace class in Hochschild homology does not, by itself, provide that algebra or its augmentation.

For a finitely generated torsion Λ-module M, define

$$\mu(M) = \operatorname{length}_{\Lambda_{(\ell)}} M_{(\ell)}.$$

The localization $\Lambda_{(\ell)}$ is a discrete valuation ring with uniformizer ℓ. This invariant concerns a module. Conilpotence concerns iterated reduced coproducts of a coaugmented coalgebra. Even for elementary ordinary Λ-coalgebras, the two conditions are independent.

Proposition 10.10.2. There is a conilpotent coaugmented Λ-coalgebra whose coaugmentation quotient has $\mu = 1$. There is also a nonconilpotent one whose quotient has $\mu = 0$.

Proof. For the first example take $C = \Lambda \oplus M$, with $M = \Lambda/(\ell)$, counit projection, and every element of M primitive: $\Delta(m) = 1 \otimes m + m \otimes 1$. Its reduced coproduct is zero, while $M_{(\ell)}$ has length one.

For the second take $C = \Lambda \oplus Mx$, with $M = \Lambda/(T)$, and set

$$\Delta(x) = 1 \otimes x + x \otimes 1 + x \otimes x, \qquad \epsilon(x) = 0.$$

The relation $Tx = 0$ is respected by this formula. The element $1 + x$ is group-like, which proves coassociativity. The counit identities follow directly. Every iterated reduced coproduct of x is a nonzero tensor power, because $M^{\otimes_\Lambda n} = M$. But T is invertible in $\Lambda_{(\ell)}$, so $M_{(\ell)} = 0$ and $\mu(M) = 0$. □

These are ordinary coalgebras over Λ. Their quotient modules are not required to be flat. They rule out using the module condition $\mu = 0$ as a definition of conilpotence in this category. A duality theorem with Iwasawa coefficients must specify its algebra and coalgebra classes, tensor, and completion. It must then identify the unit and counit maps it claims to invert.

Chapter II

Relative products and all-arity gluing

A relative cocycle consists of a cocycle and a primitive of its restriction. Multiplication therefore has a choice: use the primitive of the first input, or that of the second. The resulting products agree on cohomology, but the chain comparison contains ordered products of arbitrarily many primitives. These higher terms also change module actions. They must be retained when relative classes are composed.

We construct the comparison over an arbitrary commutative ring. Its formulas use finite sums on each tensor word and require no division. This gives an integral comparison of relative cochains, their modules, and their bimodules. Passing to an inverse limit of finite coefficient rings preserves the comparison. Passing instead to an unrestricted product over tensor lengths can destroy its continuity.

II.1 The two endpoint products

Let R be a commutative ring. All complexes are cohomological, their differentials have degree one, and tensor products use the Koszul rule. Fix a strict cospan of associative differential graded R-algebras

$$A \xrightarrow{f} B \xleftarrow{g} C.$$

The algebras may be nonunital. Define the relative complex

$$P^n = A^n \oplus C^n \oplus B^{n-1}, \qquad D(a, c, y) = (d_A a, d_C c, f(a) - g(c) - d_B y). \tag{II.1}$$

A cocycle (a, c, y) thus includes a specified primitive y of $f(a) - g(c)$. The identity $D^2 = 0$ follows from the chain-map identities for f and g. In the derived category of complexes, P is the fibre of $f - g : A \oplus C \to B$. This statement specifies its underlying complex; its multiplicative structure is the additional construction below.

For homogeneous $x = (a, c, y) \in P^p$ and $x' = (a', c', y')$, put

$$\begin{aligned} \mu_L(x, x') &= (aa', cc', (-1)^p f(a)\, y' + y\, g(c')), \\ \mu_R(x, x') &= (aa', cc', y f(a') + (-1)^p g(c)\, y'). \end{aligned} \tag{II.2}$$

The symbols L and R name these displayed choices throughout the chapter. In the relative case $C = 0$, their third components reduce to $(-1)^p f(a)\, y'$ and $y f(a')$.

Proposition II.1.1. Both (P, D, μ_L) and (P, D, μ_R) are differential graded associative algebras. If A, B, C and both maps are unital, their common unit is $e = (1_A, 1_C, 0)$. The bilinear map of degree minus one

$$K(x, x') = (0, 0, (-1)^{p+1} y\, y')$$

satisfies

$$DK + K(D \otimes 1 + 1 \otimes D) = \mu_L - \mu_R. \tag{II.3}$$

Consequently the two products induce the same multiplication on $H^*(P)$.

Proof. Write $|x'| = q$. The third component of either bracketing of the left product on x, x', x'' is

$$(-1)^{p+q} f(a) f(a') y'' + (-1)^{p} f(a) y' g(c'') + y g(c') g(c'').$$

For the right product the common third component is

$$y f(a') f(a'') + (-1)^{p} g(c) y' f(a'') + (-1)^{p+q} g(c) g(c') y''.$$

The other components are associative because A and C are. In the left Leibniz identity the two mixed terms $f(a)g(c')$ cancel; the remaining terms give the differential of the third component in (II.2). In the right identity the cancelling terms are $g(c)f(a')$. This proves compatibility with D. Substitution proves both unit identities.

The third components of the three terms on the left of (II.3) are

$$\begin{aligned} &(-1)^{p}(d_B y)\, y' - y(d_B y'), \\ &(-1)^{p}\big(f(a) - g(c) - d_B y\big)\, y', \\ &- y\big(f(a') - g(c') - d_B y'\big). \end{aligned}$$

Their sum is the difference of the two products. On cocycles this difference is the boundary $DK(x, x')$. □

The left product also has a finite interval model. Let I_R have basis e_0, e_1 in degree zero and h in degree one, with

$$d e_0 = -h, \quad d e_1 = h, \quad d h = 0, \qquad e_i e_j = \delta_{ij} e_i, \qquad e_0 h = h = h e_1,$$

and $e_1 h = h e_0 = h^2 = 0$. Its unit is $e_0 + e_1$. The evaluations $\mathrm{ev}_i : I_R \to R$ send e_i to one and the other basis elements to zero. The graded algebra is the path algebra of one arrow from 0 to 1, placed in degree one. Its differential is the graded commutator with h, so Leibniz and $d^2 = 0$ follow from $h^2 = 0$. The constant inclusion $R \to I_R$ is a quasi-isomorphism: after quotienting by $R(e_0 + e_1)$, the differential identifies two free rank-one modules.

PROPOSITION II.1.2. The map

$$(a, c, y) \longmapsto \big(a, c, f(a) e_0 + g(c) e_1 + (-1)^{p} y h\big), \qquad |(a, c, y)| = p,$$

is a strict isomorphism from (P, D, μ_L) onto the dg algebra

$$\big\{(a, c, \beta) \in A \times C \times (B \otimes_R I_R) : \mathrm{ev}_0\, \beta = f(a), \quad \mathrm{ev}_1\, \beta = g(c)\big\}.$$

Here evaluations act on the interval factor, and $b e_i$ and $b h$ denote tensors in $B \otimes_R I_R$.

Proof. Every triple in the target has the indicated unique form. Its edge coefficient after differentiation is $(-1)^{p+1}(f(a) - g(c) - d_B y)$. In a product of degrees p, q, the edge coefficient is

$$(-1)^{q} f(a) y' + (-1)^{p+q} y g(c').$$

This is $(-1)^{p+q}$ times the third component of μ_L. □

11.2 The integral bar comparison

Let $s : P \to P[1]$ have degree minus one, and write $W = sP$. Thus $|sx| = |x| - 1$; the third coordinate of $s(a, c, y)$ identifies with $y \in B$ in this degree. The tensor coalgebra is

$$T^c_R(W) = \bigoplus_{n \geq 0} W^{\otimes_R n}.$$

Its coproduct Δ cuts a word at every position, including the two ends. Its coaugmentation $\eta : R \to T^c_R(W)$ includes the empty word, and its counit ϵ projects onto that word. On the kernel of ϵ, removing the two end cuts gives the reduced coproduct. The property that a sufficiently long iteration kills each element of this kernel is called *conilpotence*; the tensor coalgebra has this property.

Define $\mathcal{B}_L$ and $\mathcal{B}_R$ to be this coalgebra with degree-one coderivations determined by

$$b_1(sx) = -sDx, \qquad b_{2,L/R}(sx, sx') = (-1)^{|x|} s\mu_{L/R}(x, x'). \tag{11.4}$$

Insertion of either component after $w_1, \ldots, w_j$ has sign $(-1)^{|w_1|+\cdots+|w_j|}$. All other components vanish, and the differential of the empty word is zero. The dg algebra identities imply $b^2_L = b^2_R = 0$.

These are the bars of the underlying nonunital algebras. Equivalently, they are the reduced bars of their unitizations $U_{L/R} = R \oplus P$, with multiplication

$$(r, p)(r', p') = (rr', rp' + r'p + \mu_{L/R}(p, p'))$$

and augmentation given by projection onto R. No augmentation of P is chosen. No flatness or boundedness assumption is needed for these coalgebras.

Theorem 11.2.1 *(Integral endpoint comparison).* There are inverse dg coalgebra isomorphisms

$$F : \mathcal{B}_L \longrightarrow \mathcal{B}_R, \qquad G : \mathcal{B}_R \longrightarrow \mathcal{B}_L.$$

Their components with output in W are

$$\begin{aligned} F_1 &= G_1 = 1_W, \\ F_n(w_1, \ldots, w_n) &= s(0, 0, y_1 \cdots y_n), \qquad n \geq 2, \\ G_n &= (-1)^{n-1} F_n, \qquad n \geq 2, \end{aligned} \tag{11.5}$$

where $w_i = s(a_i, c_i, y_i)$. Both maps preserve the coaugmentation, counit and increasing filtration by tensor length. They commute with strict maps of cospans and ordinary scalar extension. If the cospan is unital, every higher component vanishes when an input is se. In particular, the two endpoint algebras are strictly unital A_∞-isomorphic in that case.

Proof. On an input of length N, define F by summing over all partitions into consecutive nonempty blocks. Apply F_i to a block of length i, and tensor the block values in their original order. Set $F(1) = 1$. Cutting an output between blocks is equivalent to cutting the input there and partitioning its two pieces. This proves the coalgebra identity. Every sum is finite, and no output is longer than the input.

For the chain identity, set $|w_i| = q_i$ and $Q_j = q_1 + \cdots + q_j$, with $Q_0 = 0$. Write $b^{\otimes}_1$ for the tensor differential on an input word. Since

$$\big(b_1 s(a, c, y)\big)_B = d_B y - f(a) + g(c),$$

the differential of the ordered product cancels its internal $d_B y_i$ terms against $F_n b_1^{\otimes}$. For $n \geq 2$, the third component of $b_1 F_n - F_n b_1^{\otimes}$ is

$$\sum_{j=1}^{n} (-1)^{Q_{j-1}} y_1 \cdots y_{j-1} \big(f(a_j) - g(c_j) \big) y_{j+1} \cdots y_n. \tag{II.6}$$

At arity two,

$$\begin{aligned} (b_{2,L})_B &= f(a_1) y_2 - (-1)^{q_1} y_1 g(c_2), \\ (b_{2,R})_B &= -(-1)^{q_1} y_1 f(a_2) + g(c_1) y_2. \end{aligned}$$

Their difference is exactly (II.6). At arity $n \geq 3$, an interior split in $b_{2,R}(F_i, F_{n-i})$ has two purely local outputs and is zero. The two end splits give

$$g(c_1) y_2 \cdots y_n - (-1)^{Q_{n-1}} y_1 \cdots y_{n-1} f(a_n).$$

Adding these to (II.6) leaves

$$\begin{aligned} &\sum_{j=1}^{n-1} (-1)^{Q_{j-1}} y_1 \cdots y_{j-1} f(a_j) y_{j+1} \cdots y_n \\ &\quad - \sum_{j=2}^{n} (-1)^{Q_{j-1}} y_1 \cdots y_{j-1} g(c_j) y_{j+1} \cdots y_n. \end{aligned}$$

This is the third component of the source binary contribution $F_{n-1} b_{2,L}$, including its insertion signs. The other components vanish in these arities. Thus $b_R F = F b_L$ after projection to W. The two sides are coderivations relative to F, so this projection determines their equality. Explicitly, induction on input length makes their remaining difference primitive; a primitive of the tensor coalgebra with zero projection to W is zero.

Construct G by the same partition formula. On length $n \geq 2$, the component in W of FG is the local product $s(0, 0, y_1 \cdots y_n)$ times

$$\sum_{k=1}^{n} \binom{n-1}{k-1} (-1)^{n-k} = 0.$$

For GF the sign is $(-1)^{k-1}$, and the sum again vanishes. The linear components are identities. The uniqueness of coalgebra maps with prescribed components now gives $FG = GF = 1$; hence G is also a chain map.

A strict map of cospans takes an ordered product of the y_i to the ordered product of their images. Ordinary scalar extension preserves the same formulas and finite sums. Finally the local coordinate of e is zero. This proves the unit assertion. □

The comparison contains more than the binary homotopy (II.3). For example, its arity-three component is $F_3(w_1, w_2, w_3) = s(0, 0, y_1 y_2 y_3)$. Let $\tau_{L/R} : \mathcal{B}_{L/R} \to P$ be the degree-one map that desuspends the length-one part and kills every other length. Then $\tau_R F$ evaluates every higher product in (II.5). Those terms carry the change of endpoint into representations.

11.3 Transporting all module operations

For a graded R-module M, the tensor product $M \otimes_R T^c_R(W)$ is a right comodule by deconcatenation on the word factor. Denote its coaction by ρ. A right A_∞ module over (P, μ_R) is a degree-one map Q_M satisfying

$$\rho Q_M = (Q_M \otimes 1 + 1 \otimes b_R)\rho, \qquad Q_M^2 = 0.$$

It is determined by its components $\beta_n : M \otimes_R W^{\otimes_R n} \to M$ of degree one, including the differential β_0. On a word, Q_M sums the operations β_j on the module entry and the first j factors, together with $(-1)^{|m|} m \otimes b_R$ on the word. All these sums are finite. The graded maps in the module category are the graded comodule maps $T : M \otimes \mathcal{B}_R \to N \otimes \mathcal{B}_R$, with differential $Q_N T - (-1)^{|T|} T Q_M$.

In these unsuspended-module conventions a strict dg right action is encoded by

$$\beta_0(m) = d_M m, \qquad \beta_1(m; sx) = (-1)^{|m|+1} mx, \qquad \beta_n = 0 \quad (n > 1). \tag{11.7}$$

The minus sign is fixed by (11.4): on $m \otimes sx \otimes sx'$, the two action terms in Q_M^2 have opposite coefficients $(-1)^{|x|}$ and $(-1)^{|x|+1}$. On one bar factor the identity is the dg module Leibniz rule.

Strict unitality means $\beta_1(m; se) = (-1)^{|m|+1} m$ and vanishing of β_n, for $n > 1$, when a word input is se. A normalized map has the analogous vanishing in every positive word arity.

Theorem 11.3.1. Pullback along F gives an isomorphism of the dg categories of right A_∞ modules over (P, μ_R) and (P, μ_L), with inverse pullback along G. It leaves the underlying complex unchanged. Its positive-arity operations are

$$\beta^F_n(m; w_1, \dots, w_n) = \sum_{\substack{i_1 + \cdots + i_k = n \\ i_j > 0}} \beta_k\big(m; F_{i_1}(w_1, \dots), \dots, F_{i_k}(\dots, w_n)\big). \tag{11.8}$$

For unital cospans it preserves strict units and normalized module maps.

Proof. Let $C^M_F = 1_M \otimes F$ and $C^M_G = 1_M \otimes G$. Define

$$Q^F_M = C^M_G Q_M C^M_F.$$

The identity $b_R F = F b_L$ transports the comodule identity to one over b_L, and $(Q^F_M)^2 = C^M_G Q^2_M C^M_F = 0$. Projection to the length-zero module factor satisfies $\mathrm{pr}_M\, C^M_G = \mathrm{pr}_M$. Its composite with Q^F_M is therefore exactly (11.8). No extra signs occur because every F_i has degree zero.

A homogeneous comodule map $T : M \otimes \mathcal{B}_R \to N \otimes \mathcal{B}_R$ is sent to $C^N_G T C^M_F$. Conjugation preserves composition and the differential on maps $Q_N T - (-1)^{|T|} T Q_M$. Conjugation by the inverse maps reverses both constructions. This proves the assertion for the entire dg category.

A unit input in a surviving partition must be a singleton, since every higher F_i vanishes on it. The unit identities for the outer operation then give exactly these properties after pullback. The same argument applies to G. □

A strict action can acquire operations in every arity: $\beta^F_n(m; w_1, \dots, w_n) = \beta_1(m; F_n(w_1, \dots, w_n))$. The identity linear term of F therefore does not permit the module comparison to be replaced by the identity on its underlying complex.

II.4 RELATIVE BIMODULES

An interface has actions on both sides. To compare them, fix two cospans

$$A_0 \xrightarrow{f_A} A_1 \xleftarrow{g_A} A_2, \qquad E_0 \xrightarrow{f_E} E_1 \xleftarrow{g_E} E_2.$$

Let M_i be a dg A_i–E_i bimodule, and choose chain maps $\phi : M_0 \to M_1$, $\psi : M_2 \to M_1$ with

$$\phi(ame) = f_A(a)\phi(m)f_E(e), \qquad \psi(ame) = g_A(a)\psi(m)g_E(e)$$

on the respective domains. Write P_A, P_E for their relative algebra complexes, and put

$$P_M^n = M_0^n \oplus M_2^n \oplus M_1^{n-1}, \qquad D_M(m_0, m_2, v) = (dm_0, dm_2, \phi(m_0) - \psi(m_2) - dv).$$

For $x = (a_0, a_2, y) \in P_A^p$, $m = (m_0, m_2, v) \in P_M^q$, and $u = (u_0, u_2, z) \in P_E$, the endpoint actions are

$$\begin{aligned}
x \cdot_L m &= (a_0 m_0, a_2 m_2, (-1)^p f_A(a_0)v + y\psi(m_2)),\\
x \cdot_R m &= (a_0 m_0, a_2 m_2, y\phi(m_0) + (-1)^p g_A(a_2)v),\\
m \cdot_L u &= (m_0 u_0, m_2 u_2, (-1)^q \phi(m_0)z + v g_E(u_2)),\\
m \cdot_R u &= (m_0 u_0, m_2 u_2, v f_E(u_0) + (-1)^q \psi(m_2)z).
\end{aligned}$$

For each choice of endpoint these make P_M a dg bimodule for the matching products on P_A and P_E. The proof below includes the compatibility of the two actions.

The corresponding bar bicomodule is

$$\mathcal{B}^M_{L/R} = T^c_R(sP_A) \otimes_R sP_M \otimes_R T^c_R(sP_E).$$

Its left and right coactions cut before or after the middle entry. Its differential inserts the unary differentials, the algebra products, and the two actions, with the suspended conventions of (II.4) on every entry, including sP_M. In particular this middle entry is suspended, unlike the module entry in (II.7).

THEOREM II.4.1. There is a dg bicomodule isomorphism $\Phi : \mathcal{B}^M_L \to \mathcal{B}^M_R$ over the coalgebra maps F_A, F_E of Theorem II.2.1. Its component with no algebra inputs is the identity on sP_M. For $r + t \geq 1$, its component with r left and t right inputs is

$$\Phi_{r,t}(sx_1, \dots, sx_r; sm; su_1, \dots, su_t) = s(0, 0, y_1 \cdots y_r\, v\, z_1 \cdots z_t). \tag{II.9}$$

Its inverse has higher components $\Psi_{r,t} = (-1)^{r+t}\Phi_{r,t}$ and identity linear term. The maps commute with strict diagram maps and ordinary scalar extension. They preserve strict units in the algebra inputs when the cospans and bimodules are unital.

Proof. Form the triangular dg algebras

$$T_i = \begin{pmatrix} A_i & M_i \\ 0 & E_i \end{pmatrix}, \qquad \begin{pmatrix} a & m \\ 0 & u \end{pmatrix}\begin{pmatrix} a' & m' \\ 0 & u' \end{pmatrix} = \begin{pmatrix} aa' & am' + mu' \\ 0 & uu' \end{pmatrix}.$$

Their entrywise differential has the dg algebra Leibniz property because the two module actions do. The maps ϕ, ψ, together with the algebra maps, give a strict cospan $T_0 \to T_1 \leftarrow T_2$. Its relative complex

has diagonal components P_A, P_E and off-diagonal component P_M. The two endpoint products give exactly the four displayed actions. Their dg bimodule identities are thus the corresponding entries of the dg algebra identities.

In the tensor bar of this triangular relative algebra, take words with upper-left entries, then one upper-right entry, then lower-right entries. They form $\mathcal{B}^M_{L/R}$, and its differential preserves this submodule. Applying the partition formula for the triangular algebra preserves it too. The unique block containing the upper-right entry has local component $y_1 \cdots y_r v z_1 \cdots z_t$, without any interchange of factors. This proves (11.9). The inverse sign is $(-1)^{(r+t+1)-1}$.

The chain and inverse identities restrict from those of the triangular bars. Deconcatenation gives

$$\Delta_L \Phi = (F_A \otimes \Phi)\Delta_L, \qquad \Delta_R \Phi = (\Phi \otimes F_E)\Delta_R.$$

Finally, strict diagram maps preserve every ordered local product, scalar extension preserves the finite formulas, and a gluing unit has local component zero. These give the remaining assertions. □

An algebra local condition for arithmetic cochains must provide a dg algebra and a multiplicative map into the local cochains. A linear local condition may instead enter the bimodule construction once its two actions and the compatible maps are supplied. Its cone as a complex does not determine either action.

11.5 The homological meaning of the bar

The relative complex and its tensor bar have different meanings. The complex P computes the relative cohomology of the cospan. The bar computes a derived tensor of the augmentation module of $U = R \oplus P$ under the K-flatness hypothesis of Proposition 3.3.1.

Proposition 11.5.1. If P is K-flat over R, then for either endpoint product its bar represents $R \otimes^{\mathbf{L}}_U R$, where R is a left or right U-module through the augmentation.

Proof. For either endpoint product, the augmentation ideal of $U = R \oplus P$ is P, and the differential and bar signs are those of (3.2). Apply Proposition 3.3.1 to this augmented algebra. □

Degreewise flatness does not replace K-flatness for an unbounded complex. Without that hypothesis the explicit bar and the endpoint isomorphism still exist; the proposition does not identify that bar with a derived tensor.

The unital gluing case gives a useful distinction even without flatness. If P has a closed unit e, its underlying nonunital bar contracts onto the empty word. On a nonempty word c, set $h(c) = se \otimes c$, and set $h(1) = 0$. Since $b_1(se) = 0$ and $b_2(se, sx) = sx$, direct substitution gives

$$bh + hb = 1 - \eta\epsilon.$$

Thus $H^*(\mathcal{B}) = R$, although $H^*(P)$ can be much larger. Algebraically the same distinction appears in $U \cong R \times P$, via $(r, p) \mapsto (r, re + p)$. To form a reduced bar of an augmented unital algebra P itself, one instead uses its specified augmentation ideal. No such choice is implicit in the endpoint construction.

II.6 HOMOTOPY-COMMUTING COEFFICIENT MAPS

Strict maps of cospans preserve the preceding constructions. A comparison commuting only up to cochain homotopy has an additional product defect, visible already in the relative case $C = 0$. Write $P(f) = A \oplus B[-1]$ for this specialization of (II.1). Alongside $f : A \to B$, choose a dg algebra map $f' : A' \to B'$. Let $\alpha : A \to A'$ and $\gamma : B \to B'$ be dg algebra maps, and let $H : A \to B'$ have degree minus one with

$$f'\alpha - \gamma f = dH + Hd.$$

Define $T : P(f) \to P(f')$ by $T(a, y) = (\alpha(a), \gamma(y) + H(a))$.

PROPOSITION II.6.1. The map T is a cochain map. For the left endpoint products its multiplicativity defect has zero first component and local component

$$\begin{aligned}(-1)^p &(f'\alpha - \gamma f)(a)\gamma(y') \\ &+ (-1)^p f'\alpha(a)H(a') - H(aa'), \qquad |(a, y)| = p.\end{aligned}$$

It induces a strict tensorwise map of the relative bars if and only if this expression vanishes on every pair of inputs.

Proof. The local component of $D'T - TD$ is $(f'\alpha - \gamma f)(a) - dH(a) - H(da) = 0$. Expanding $\mu'_L(Tx, Tx') - T\mu_L(x, x')$ gives the displayed formula. Projection of the strict bar-map identity onto one output factor on two inputs is precisely multiplicativity; on one input it is the chain-map identity. These two identities suffice in every length because the bar differentials have only unary and binary components. □

If the defect is nonzero, higher components must be constructed in the permitted operation spaces. Closedness of an obstruction in a larger Hom complex does not provide a primitive with prescribed support, pole bounds, units or continuity. The endpoint comparison above supplies such components for its explicit algebraic problem; it does not supply them for an arbitrary homotopy-commuting square.

II.7 FINITE PRECISION AND CONTINUOUS GLUING

Fix a prime ℓ, put $R_j = \mathbb{Z}/\ell^j$, and take compatible strict cospans over the rings R_j. Write $W_j = sP_j$ and $\mathcal{B}_{L/R,j} = T^c_{R_j}(W_j)$. Define their restricted inverse limits by

$$\widehat{\mathcal{B}}_{L/R} = \varprojlim_j \bigoplus_{n \geq 0} W_j^{\otimes_{R_j} n}.$$

Each finite-precision module is discrete; the limits have the inverse-limit topology. An element has only finitely many nonzero word lengths at any fixed precision, although the number of such lengths may grow with j. The completed tensor recipient is defined here to be

$$\widehat{\mathcal{B}} \,\widehat{\otimes}\, \widehat{\mathcal{B}} := \varprojlim_j (\mathcal{B}_j \otimes_{R_j} \mathcal{B}_j). \tag{II.10}$$

This definition makes no identification with a different completed tensor product.

For a compatible tower of modules or relative bimodules, use

$$\widehat{C}(M) = \varprojlim_j (M_j \otimes_{R_j} \mathcal{B}_j), \qquad \widehat{\mathcal{B}}^M = \varprojlim_j \mathcal{B}_j^M,$$

with the same topology. Their coaction recipients are, respectively, $\varprojlim_j (C(M_j) \otimes_{R_j} \mathcal{B}_j)$ and the two limits of $\mathcal{B}_{A,j} \otimes_{R_j} \mathcal{B}_j^M$ and $\mathcal{B}_j^M \otimes_{R_j} \mathcal{B}_{E,j}$. The subscripts A, E distinguish the left and right coefficient bars.

Theorem 11.7.1. The maps F_j, G_j induce inverse continuous maps $\widehat{F} : \widehat{\mathcal{B}}_L \to \widehat{\mathcal{B}}_R$ and $\widehat{G}$. They preserve the differential and coproduct, with recipient (11.10). Compatible module and bimodule bars admit the corresponding continuous comparisons, with coaction recipients the inverse limits of their finite-precision tensor recipients.

Proof. At precision j, each element is a finite sum of words, and each word has finitely many consecutive partitions. The maps are therefore defined there and satisfy all chain, inverse and coaction identities. Strict compatibility with coefficient reduction gives the maps on inverse limits. Every identity holds after each projection, hence in the limit. Each output at precision j depends only on the input at the same precision, proving continuity. The same argument applies to $1_M \otimes F_j$, to module conjugation and to the triangular bicomodule maps. □

No interchange of cohomology with inverse limit occurs in this proof. Such an interchange would need its own hypotheses on derived inverse limits. Ordinary conilpotence can also fail in the completed object. For example, let $P = \mathbb{Z}_\ell a$, with $|a| = 1$, zero differential and zero product. The element

$$z = \sum_{n \geq 1} \ell^n (sa)^{\otimes n}$$

exists in the restricted bar. Every iterated reduced coproduct has a nonzero component, obtained by cutting a sufficiently long word. Each finite coefficient bar is conilpotent, while z is not.

Proposition 11.7.2. The map F need not extend continuously to the unrestricted product $\prod_{n \geq 0} W^{\otimes_R n}$ with discrete word coordinates.

Proof. Take the relative identity map $f : \mathbb{Z} \to \mathbb{Z}$, with $C = 0$, and let $v = s(0, 1)$. It has degree zero and satisfies $F_n(v^{\otimes n}) = v$ for every $n \geq 1$. The partial sums $z_N = \sum_{n=1}^{N} v^{\otimes n}$ converge in the product of word spaces: each coordinate eventually stabilizes. The length-one coordinate of $F(z_N)$ is Nv, which does not converge in the discrete module $\mathbb{Z}v$. Continuity of an extension would contradict this sequence. □

In the restricted ℓ-adic bar the weighted series $\sum_{n \geq 1} \ell^n v^{\otimes n}$ is defined. Its length-one image is $\ell(1 - \ell)^{-1} v$. The denominator is an ℓ-adic unit; it arises by summing the convergent coefficients, not in the definition of any Taylor component. The comparison is integral at every finite precision. An analytic realization must separately bound the higher components in its chosen seminorms and compare its completed tensors with the recipient in which the coproduct is defined.

Chapter 12

Hall correspondences and primitive kernels

A Hall product counts extensions. Its associativity comes from flags: the two bracketings forget different steps of the same filtration. Recovering a geometric operation from the count requires more data. Automorphisms determine the integration weight, and a vanishing scalar sum need not come from an acyclic complex.

The first primitive already exhibits both issues. Its coefficients on a nonsplit extension and a split extension cancel under Green's coproduct. Placing those coefficients on separate components leaves nonzero cohomology. A Fourier kernel on the extension line instead joins the two strata by a Gysin morphism. We construct that kernel, extend it to every cut of a flag, and determine what is lost when the flag is forgotten.

Quiver cohomology gives another extension product. Its Euler factors, cohomological grading and vanishing cycles must be retained before comparing it with arithmetic fields. An explicit shuffle relation will also distinguish a valid product from a proposed ordinary cooperation. The arithmetic comparison remains a problem about these operations and their maps, beyond equality of numerical invariants.

12.1 Extension counts and the tensor crossing

Let $\mathcal{A}$ be an essentially small abelian category with finite sets $\mathrm{Hom}(M,N)$ and $\mathrm{Ext}^1(M,N)$. Write $a_M = |\operatorname{Aut} M|$. Define $F^L_{M,N}$ to be the number of subobjects $N' \subset L$ satisfying $N' \simeq N$ and $L/N' \simeq M$. Thus the first index names the quotient and the second names the subobject. Let $H(\mathcal{A})$ be the free vector space over a field k on isomorphism classes u_M.

Proposition 12.1.1. The formula

$$u_M u_N = \sum_{[L]} F^L_{M,N} u_L$$

defines an associative multiplication with unit u_0. Let $K_0(\mathcal{A})$ be the abelian group generated by object classes, subject to $[L] = [M] + [N]$ for each short exact sequence. For any bicharacter $b : K_0(\mathcal{A}) \times K_0(\mathcal{A}) \to k^\times$, the product $u_M *_b u_N = b([M],[N])u_M u_N$ is also associative. Here a bicharacter is multiplicative in each additive argument.

Proof. The finite extension set leaves finitely many possible middle objects. The finite Hom set bounds the number of embeddings into each one. The coefficient of u_L in either triple product counts flags

$$0 \subset L_1 \subset L_2 \subset L$$

with successive quotients P, N, M. The two bracketings forget L_1 or L_2, so their coefficients agree. The twist factors agree because

$$b(M,N)b(M+N,P) = b(N,P)b(M,N+P).$$

The zero object has precisely the unit subobject and quotient. □

For the coproduct, assume that $\mathcal{A}$ is linear over $\mathbb{F}_q$, hereditary, and has finite-dimensional Hom and Ext spaces. Hereditary means that $\operatorname{Ext}^i = 0$ for $i > 1$. Assume also that every object has finitely many subobjects, as for finite-dimensional modules over a finite field. Then

$$\langle M, N\rangle = \dim_{\mathbb{F}_q} \operatorname{Hom}(M, N) - \dim_{\mathbb{F}_q} \operatorname{Ext}^1(M, N)$$

is additive in short exact sequences. Put $(M, N) = \langle M, N\rangle + \langle N, M\rangle$ and $v = \sqrt{q}$. Work over $\mathbb{Q}(v)$ with the twisted product $u_M * u_N = v^{\langle M,N\rangle} u_M u_N$.

THEOREM 12.1.2 *(Green's compatibility).* The coproduct and counit

$$\Delta u_L = \sum_{[M],[N]} v^{\langle M,N\rangle} \frac{a_M a_N}{a_L} F^L_{M,N}\, u_M \otimes u_N, \qquad \varepsilon(u_L) = \delta_{L,0}, \tag{12.1}$$

make this Hall algebra a braided bialgebra. For homogeneous classes α, β, the crossing is

$$c(x_\alpha \otimes y_\beta) = v^{(\alpha,\beta)}\, y_\beta \otimes x_\alpha.$$

In particular, multiplicativity of Δ uses

$$(x_\alpha \otimes y_\beta)(x_\gamma \otimes y_\delta) = v^{(\beta,\gamma)} (x_\alpha * x_\gamma) \otimes (y_\beta * y_\delta). \tag{12.2}$$

All sums in these formulas are finite.

Proof. The automorphism factors in an iterated coproduct cancel, leaving the same three-step flags in either order. This proves coassociativity.

For multiplicativity, Green's formula counts pairs $N, N' \subset L$ through their four subquotients

$$L/(N + N'), \quad N/(N \cap N'), \quad N'/(N \cap N'), \quad N \cap N'.$$

If the first and last are A, D, the groupoid of extension-square fillings contributes

$$\frac{|\operatorname{Ext}^1(A, D)|}{|\operatorname{Hom}(A, D)|} = q^{-\langle A,D\rangle}.$$

Heredity removes the degree-two obstruction to a filling.

Substituting these weights into (12.1) leaves the crossing (12.2), by Euler additivity. This is Green's extension-counting theorem with its automorphism normalization. See Ringel, *Green's theorem on Hall algebras*, Part I, §§1 and 8. The zero-object terms prove both counit identities. □

For finite vector spaces let $s = u_{\mathbb{F}_q}$. Counting lines gives

$$s * s = v(q + 1) u_{\mathbb{F}_q^2}, \qquad [s \otimes s]\Delta u_{\mathbb{F}_q^2} = v^{-1}.$$

Here $[s \otimes s]$ denotes the coefficient of that basis vector. Thus $[s \otimes s]\Delta(s * s) = q + 1$. Expanding $\Delta(s)^2$ gives $1 + q$, since the crossed term has weight $v^2 = q$. An ordinary flip would give 2. Even this smallest example fixes the tensor multiplication used by the theorem.

A hereditary category can have infinitely many subobjects, as for coherent sheaves on a curve. Its coproduct requires a specified tensor completion and locally finite products. The finite extension-square identity persists. It does not by itself define multiplication on arbitrary coefficientwise infinite sums.

The hereditary hypothesis also has content. For $R = \mathbb{F}_q[\epsilon]/(\epsilon^2)$ and $S = R/(\epsilon)$, the resolution with every differential multiplication by ϵ gives $\operatorname{Ext}^n_R(S, S) = \mathbb{F}_q$ for every $n \geq 0$. The alternating sums of Ext dimensions oscillate. The Hall product still exists, but this infinite Ext sequence does not define a finite Euler twist.

12.2 The local primitive and its Green norm

Fix a prime power Q and the category of finite-dimensional nilpotent $\mathbb{F}_Q[t]$-modules. A partition $\lambda = (\lambda_1, \dots, \lambda_l)$ names the module $M_\lambda = \bigoplus_i \mathbb{F}_Q[t]/(t^{\lambda_i})$. Write $u_\lambda = u_{M_\lambda}$ and $|\lambda| = \sum_i \lambda_i$. The length $\ell(\lambda)$ is its number of Jordan blocks. The Euler form is zero, so Green's multiplication and coproduct use the ordinary tensor product here.

Work over $\mathbb{C}$ with the positive Hermitian Green form

$$(u_\lambda, u_\mu)_G = \frac{\delta_{\lambda\mu}}{a_{M_\lambda}},$$

conjugate-linear in the first argument. The coproduct is adjoint to multiplication. For $r \geq 0$, put $e_r = Q^{r(r-1)/2} u_{(1^r)}$, with $e_0 = 1$, and set $E(z) = \sum_{r\geq 0} e_r z^r$. Subspace counting gives

$$\Delta u_{(1^r)} = \sum_{i+j=r} Q^{-ij} u_{(1^i)} \otimes u_{(1^j)}, \qquad \Delta E(z) = E(z) \otimes E(z).$$

These are formal power series, with finite-dimensional pieces in each module dimension.

Theorem 12.2.1 *(Classical Hall primitives).* The algebra is polynomial on the e_r, $r > 0$. Define p_r by

$$\log E(z) = \sum_{r\geq 1} \frac{(-1)^{r-1}}{r} p_r z^r. \tag{12.3}$$

Then

$$p_r = \sum_{|\lambda|=r} \left(\prod_{j=1}^{\ell(\lambda)-1} (1 - Q^j) \right) u_\lambda. \tag{12.4}$$

Each p_r is primitive, and the homogeneous primitives of dimension r form its one-dimensional span. Their Green norms are

$$(p_r, p_s)_G = \delta_{rs} \frac{r}{Q^r - 1}. \tag{12.5}$$

Proof. The polynomial presentation is the classical Hall theorem. Its triangular construction uses products of semisimple classes whose dimensions are the successive kernel increments of t. The leading term has the kernel filtration of M_λ. Every other term has larger kernel dimensions. In each fixed dimension the resulting triangular matrix is invertible. Commutativity follows by dualizing exact sequences, since every nilpotent Jordan module is isomorphic to its linear dual.

The coefficient formula (12.4) is the classical Newton identity in this basis. In the present normalization, it is Lemma 3.5 of Shimoji–Yanagida, *A study of symmetric functions via derived Hall algebra*, arXiv:1812.06033. Its coefficients are fixed by the vertical-strip Hall numbers, rather than by rescaling a primitive after construction. Taking the logarithm of the group-like identity for E proves primitivity. The Newton relations identify the polynomial generators e_r and p_r triangularly.

In a characteristic-zero polynomial algebra with primitive generators, a primitive polynomial f satisfies $f(X + Y) = f(X) + f(Y)$. Differentiating in each Y variable at zero makes every partial derivative of f constant. Hence f is linear in the p_r. The dimension grading gives the asserted homogeneous span.

Finally, the order of the general linear group gives

$$(E(u), E(v))_G = \sum_{r\geq 0} \frac{Q^{r(r-1)/2} (\bar{u}v)^r}{\prod_{j=1}^r (Q^j - 1)} = \prod_{j\geq 1} (1 - Q^{-j} \bar{u}v)^{-1}.$$

The product identity follows coefficientwise from $(1-t)P(Qt) = P(t)$ and $P(0) = 1$. Primitivity and adjunction make distinct primitive dimensions orthogonal and give factorial pairings for powers. Taking logarithms of this identity and using (12.3) yields (12.5). □

In dimension two, writing $S = \mathbb{F}_Q[t]/(t)$ and $J_2 = \mathbb{F}_Q[t]/(t^2)$ gives

$$u_S^2 = u_{J_2} + (Q+1)u_{S\oplus S}, \qquad p_2 = u_{J_2} - (Q-1)u_{S\oplus S}.$$

The middle Green coefficients of u_{J_2} and $u_{S\oplus S}$ are $(Q-1)/Q$ and $1/Q$. They cancel for p_2. The cancellation uses automorphism orders and a signed linear combination of classes. It does not yet specify a complex or a homotopy on an extension space.

12.3 Flag convolution and a nonacyclic primitive class

Let $\mathcal{M}_n$ be a finite skeleton of the groupoid of n-dimensional nilpotent $\mathbb{F}_Q[t]$-modules. For a characteristic-zero field k, consider

$$\mathscr{D}_n = D^b \operatorname{Fun}(\mathcal{M}_n, \operatorname{Vect}_k^{\mathrm{fd}}) \simeq \prod_{|\lambda|=n} D^b \operatorname{Rep}_k^{\mathrm{fd}}(\operatorname{Aut} M_\lambda).$$

A functor on a groupoid assigns a vector space to each object and an invertible map to each isomorphism. Its restriction to one component is a representation of that object's automorphism group. Let $\mathcal{E}_{m,n}$ be the groupoid of short exact sequences with quotient dimension m and subobject dimension n. Write p for its endpoint map and q for its middle-object map.

Pullback followed by left Kan extension defines

$$\mathscr{F} \diamond \mathscr{G} = q_! p^*(\mathscr{F} \boxtimes \mathscr{G}).$$

Left Kan extension sums over the relevant finite groupoid fiber, taking coinvariants under its stabilizers. Characteristic zero makes these finite-group coinvariants exact. The two iterated convolution functors are push–pull along the same three-step flag groupoid. Base change gives associativity. Full flags give its higher coherences.

Let δ_M be the trivial representation on the component of M, extended by zero. At L, the fiber of $\delta_M \diamond \delta_N$ is the permutation representation on subobjects counted by $F^L_{M,N}$. Taking dimensions recovers the Hall product. For the reverse functor $p_! q^*$, coinvariants of the trivial representation of a finite group have dimension one. Groupoid cardinality instead assigns the reciprocal group order.

PROPOSITION 12.3.1. The zero-differential complex

$$\mathscr{P}_2 = \delta_{J_2} \oplus \delta_{S\oplus S}^{\oplus(Q-1)}[1]$$

has Euler class p_2. After applying $p_! q^*$ and forgetting endpoint actions, its (S,S) component has

$$H^0 = k^{Q-1}, \qquad H^{-1} = k^{Q-1}.$$

It is not acyclic.

Proof. There are Q extension classes of S by S. Each has stabilizer $\operatorname{Hom}(S,S)$ of order Q. The $Q-1$ nonzero classes have middle module J_2. The remaining class is split. Ordinary Kan extension therefore gives dimensions $Q-1$ and 1. It preserves the zero differential, so the shifted split summands give exactly the two displayed cohomology groups.

Green's scalar coefficients instead cancel as $(Q-1)/Q - (Q-1)/Q = 0$. □

The two source summands lie on different connected components of $\mathcal{M}_2$. This category has no morphism between them that could provide a cancelling differential. A geometric extension space can connect the split point to nonsplit extensions. The next construction retains that connection and its integration weight at the level of complexes.

12.4 A Fourier kernel on the extension space

Fix $\mathbb{F}_q$, a prime $\ell \nmid q$, and $K = \overline{\mathbb{Q}}_\ell$. We use bounded constructible complexes with Weil structure. Thus Frobenius acts on their stalk cohomology, and its alternating trace is defined. Our Tate convention gives $K(1)$ trace q^{-1}. A cohomological shift $[1]$ changes the trace sign. The notation $Rf_!$ denotes the derived direct image with proper supports, and Rf_* the ordinary derived direct image.

Choose a nontrivial additive character $\psi : \mathbb{F}_q \to K^\times$. Its Artin–Schreier sheaf $\mathcal{L}_\psi$ is the rank-one summand of the finite additive cover selected by ψ. For a d-dimensional vector space V, $d > 0$, let $p_V : V \times (V^\vee \setminus 0) \to V$ be projection. The pairing defines

$$\mathcal{P}_V = R p_{V!} \mathcal{L}_{\psi\langle v,\xi\rangle}[1]. \tag{12.6}$$

The shift is $[1]$ in every dimension. We use unshifted Fourier integration before inserting this shift.

Theorem 12.4.1. Let $i : \{0\} \hookrightarrow V$ and $j : V \setminus 0 \hookrightarrow V$. There is a triangle

$$i_* K(-d)[-2d] \xrightarrow{\text{Gys}} K_V \longrightarrow \mathcal{P}_V \longrightarrow . \tag{12.7}$$

The first arrow is the Gysin map normalized by compact trace, and $\mathcal{P}_V \simeq R j_* K_{V\setminus 0}$. Its nonzero stalk cohomology is

$$H^0(\mathcal{P}_V)_v = K \quad (v \in V), \qquad H^{2d-1}(\mathcal{P}_V)_0 = K(-d).$$

Its trace is 1 away from zero and $1 - q^d$ at zero, while $R\Gamma_c(V, \mathcal{P}_V) = 0$.

Proof. Fourier integration over all of $V^\vee$ gives $i_* K(-d)[-2d]$. Indeed, a nonzero v supplies a nonconstant linear coordinate whose Artin–Schreier sheaf has zero compact cohomology. At $v = 0$, the coefficient is constant on $\mathbb{A}^d$. Localization at the zero covector gives (12.7), with the first arrow initially defined by this integral.

Integrate in the other order. Each $\xi \neq 0$ again supplies a nonconstant Artin–Schreier coordinate, so the entire compact cohomology vanishes. The first arrow consequently induces the trace isomorphism on compact cohomology.

Purity identifies it with the normalized Gysin generator. The localization triangle $i_* i^! K_V \to K_V \to R j_* j^* K_V$ identifies its cone. At zero, the two terms have degrees $2d$ and 0. Their cone has the two stated degrees. The odd degree $2d - 1$ contributes $-q^d$ to the trace. □

The vanishing used in this proof is a vanishing of complexes. For the Artin–Schreier cover of $\mathbb{A}^1$, its deck translations act trivially on the sole compact cohomology of the covering affine line. Every nontrivial character summand therefore vanishes. Splitting off a nonconstant coordinate proves the higher-dimensional case. This also fixes the normalization independently of a character sum. Compare Laumon's Fourier formalism, §§1.1.3 and 1.2.1.

For the extension line of S by S, the operator is

$$\begin{pmatrix} 0 & v \\ 0 & 0 \end{pmatrix}, \qquad v \in \mathbb{A}^1.$$

Its middle module is J_2 when $v \neq 0$ and $S \oplus S$ when $v = 0$. Thus the trace of $\mathcal{P}_{\mathbb{A}^1}$ gives the coefficients of p_2, and integration over the extension line is zero as a complex. The Gysin arrow supplies the information absent from the disconnected complex of Proposition 12.3.1.

The construction works for a vector bundle $E \to B$ by using $E \times_B E^\vee$ and removing the zero section of the dual. It commutes with base change. Bundle automorphisms act contragrediently on the dual and preserve the pairing. The kernel therefore carries equivariance for the endpoint gauge action. This does not yet make it an object on the stack of middle modules.

PROPOSITION 12.4.2. For the split vector group $U = \mathbb{G}_a^h$,

$$R\Gamma_c(BU, K) = K(h)[2h].$$

Its trace is q^{-h}. On a product of an extension space with BU, proper-support integration includes this tensor factor.

Proof. The smooth atlas $\operatorname{Spec} \mathbb{F}_q \to BU$ has Čech nerve U^n. Each term has ordinary cohomology K. The normalized complex of these constants has only K in degree zero. The smooth stack has dimension $-h$ and dualizing complex $K(-h)[-2h]$. Verdier duality therefore gives the displayed compact cohomology. The product assertion follows from proper-support Künneth. □

For extensions of S by S, automorphisms fixing the endpoints form $\mathbb{G}_a$. The nonsplit and split trace contributions are $(q-1)/q$ and $(1-q)/q$. The attaching map makes their compact pushforward zero before and after tensoring with $K(1)[2]$. This geometric factor recovers the reciprocal stabilizer weight that ordinary finite-group coinvariants do not supply.

12.5 EVERY CUT OF A FLAG

For a vector space E, write the unshifted Fourier transform as

$$\mathsf{F}_E(\mathcal{K}) = R\operatorname{pr}_{E^\vee!}(\operatorname{pr}_E^* \mathcal{K} \otimes \mathcal{L}_{\psi\langle x,\xi\rangle}).$$

Integrating twice gives pullback by -1 followed by $(-\dim E)[-2\dim E]$. The rank-normalized transform is $\mathsf{F}_E[\dim E]$.

Let $0 = F_0 \subset F_1 \subset \cdots \subset F_s = V$ be a flag with positive graded dimensions n_i. Define

$$\mathfrak{g} = \operatorname{End}(V), \quad \mathfrak{p} = \{x : xF_i \subset F_i\}, \quad U = \{x : xF_i \subset F_{i-1}\}, \quad \mathfrak{l} = \bigoplus_i \operatorname{End}(F_i/F_{i-1}).$$

Let $i : \mathfrak{p} \hookrightarrow \mathfrak{g}$ and $q_0 : \mathfrak{p} \to \mathfrak{l}$ be the inclusion and quotient. Put $r = \dim U = \sum_{i<j} n_i n_j$. Trace pairings identify the full matrix spaces with their duals.

PROPOSITION 12.5.1. For every constructible complex $\mathcal{K}$ on $\mathfrak{l}$,

$$\mathsf{F}_{\mathfrak{g}}(i_* q_0^* \mathcal{K}) \simeq i_* q_0^* \mathsf{F}_{\mathfrak{l}}(\mathcal{K})(-r)[-2r].$$

For rank-normalized transforms, only the Tate twist remains. The comparison commutes with base change of flags, proper forgetting of flags, and finite flag refinements.

Proof. Choose an adapted frame. Integrating the strictly upper blocks forces the corresponding lower blocks of the dual matrix to vanish. This says $U^\perp = \mathfrak{p}$ for the trace pairing. The affine integral on this

locus contributes $K(-r)[-2r]$. The remaining phase is the trace pairing on $\mathfrak{l}$. Since $\dim \mathfrak{g} = \dim \mathfrak{l} + 2r$, rank normalization removes exactly that shift.

The pairings and subbundles are intrinsic, so the calculation glues over the flag scheme. Its incidence projection to $\mathfrak{g}$ is proper. Proper-support base change commutes the projection with Fourier integration. Every refinement integrates each strictly upper block once, with the same phase. Fubini and the coherent composition of the six operations identify all parenthesizations. □

Full matrix support matters here. The Fourier transform of a skyscraper at zero is constant on the dual matrix space. It need not retain nilpotent support.

For $1 \le p < s$, set

$$U_p = \{x : x(V) \subset F_p,\ x(F_p) = 0\} \subset U, \qquad H_p = U_p^{\perp} \subset U^\vee.$$

In an adapted splitting, U_p consists of blocks $i \le p < j$.

Define

$$\Omega = U^\vee \setminus \bigcup_{p=1}^{s-1} H_p, \qquad \mathcal{P}_F = \mathsf{F}_{U^\vee}(j_! K_\Omega)[s-1],$$

where $j : \Omega \hookrightarrow U^\vee$. A covector belongs to Ω exactly when its restriction to each whole cross-cut subspace is nonzero. For $s = 1$, set $\mathcal{P}_F = K$ on $U = 0$. This is an operation unit. A vacuum at a retained position is additional data.

Theorem 12.5.2. Every cut quotient $\rho_p : U \to U/U_p$ satisfies $R\rho_{p!}\mathcal{P}_F = 0$. The kernel and these comparisons are equivariant for the flag-preserving group. Successive quotient integrations have compatible Fubini comparisons.

Proof. Integrate first along U_p. A covector nonzero on U_p gives zero compact cohomology. A covector zero on U_p belongs to H_p, where $j_! K_\Omega$ restricts to zero. Proper-support base change proves the vanishing on the quotient. The flag group preserves each U_p and its annihilator, hence preserves the kernel. Iterated integration integrates over the same sum of cut subspaces, so its comparisons agree. □

The full flag is visible even in its scalar trace. For $J \subset \{1, \dots, s-1\}$, put $S_J = \sum_{p \in J} U_p$, $r_J = \dim S_J$ and $D = \dim U$. Inclusion–exclusion and character orthogonality give

$$k_F(x) = (-1)^{s-1} \sum_J (-1)^{|J|} q^{D - r_J} 1_{S_J}(x). \tag{12.8}$$

For three one-dimensional pieces, with coordinates $(a, b, c) = (x_{12}, x_{13}, x_{23})$, this becomes

$$k_3(x) = q^3 1_{x=0} - q 1_{c=0} - q 1_{a=0} + 1.$$

A nonzero outer entry b, with $a = c = 0$, gives $1 - 2q$. A single nonzero adjacent entry gives $1 - q$. Both operators have Jordan type $(2, 1)$, so the trace remembers the flag beyond the middle module.

For four one-dimensional pieces let $\delta_{ij} = 1_{x_{ij}=0}$. Formula (12.8) gives

$$\begin{aligned} k_4 = 1 &- q(\delta_{12} + \delta_{23} + \delta_{34}) + q^2 \delta_{12}\delta_{34} \\ &+ q^3(\delta_{23}\delta_{24}\delta_{34} + \delta_{12}\delta_{13}\delta_{23}) - q^6 1_{x=0}. \end{aligned}$$

Meeting every cut is weaker than connecting the graph of labels. The two edges 13 and 24 meet every consecutive cut but form a disconnected graph.

Comparison with the scalar Hall primitive

For a complete flag in dimension n, define

$$p_n(x) = \prod_{j=1}^{\ell(\lambda(x))-1} (1 - q^j), \qquad x \in U,$$

where $\lambda(x)$ is the Jordan partition. These are the coefficients in (12.4). For functions on the dual space, use $\mathsf{F}f(x) = \sum_\xi f(\xi)\psi(\langle x, \xi\rangle)$. Its inverse has opposite phase and factor q^{-D}.

PROPOSITION 12.5.3. For each proper cut and every $x \in U$,

$$\sum_{u \in U_p(\mathbb{F}_q)} p_n(x + u) = 0.$$

Hence $\mathsf{F}^{-1}p_n$ is supported on Ω. For $n = 3$, let $i_{13} : L_{13} \hookrightarrow U$ be the line $a = c = 0$, and write $\mathcal{P} = \mathcal{P}_{\mathbb{A}^1}$. Then

$$p_3 = \operatorname{tr}(\mathcal{P}_F \oplus i_{13*}\mathcal{P}(-1)[-2]). \tag{12.9}$$

Both summands have zero proper-support image under either cut. In dual coordinates (α, β, γ),

$$p_3 = \mathsf{F}(1_\Omega - q^{-1}1_{\{\beta \neq 0\}}).$$

Proof. Fix the diagonal blocks A, C at a cut. Changing a splitting changes its cross block B by $AZ - ZC$. The cokernel is $\operatorname{Ext}^1_{\mathbb{F}_q[t]}(C, A)$, so every extension class has the same number of cross-block representatives. The Green coefficient is the sum of the primitive's middle-object coefficients over these classes, divided by $|\operatorname{Hom}_{\mathbb{F}_q[t]}(C, A)|$. Primitivity makes it zero. Fourier transformation identifies this vanishing with zero restriction to H_p.

For three blocks, $ac \neq 0$ gives type (3), a nonzero operator with $ac = 0$ gives type (2, 1), and zero gives type (1, 1, 1). Their primitive coefficients are 1, $1 - q$ and $(1 - q)(1 - q^2)$. Subtracting from k_3 gives

$$k_3 - p_3 = -q1_{L_{13}} + q^2 1_{\{0\}} = -q \operatorname{tr}(i_{13*}\mathcal{P}).$$

The twist (-1) multiplies trace by q and $[-2]$ preserves its sign. The invariant line lies in both cut subspaces. Its cut pushforward has fiber $R\Gamma_c(\mathbb{A}^1, \mathcal{P}) = 0$. Finally, summing first over α, γ gives

$$\mathsf{F}1_{\{\beta \neq 0\}} = q^2 1_{L_{13}}(q1_{b=0} - 1),$$

which proves the last formula. □

The direct sum in (12.9) realizes the scalar primitive and its cut vanishing. It supplies no induction-compatible comparison of Hall products. The trace identity also supplies no distinguished map whose cone is the outer-line summand.

12.6 The attaching maps and their contractions

The flag kernel also has a construction over the integral flag scheme. Use constructible rational Beilinson motives, with their oriented six operations and purity for regular closed immersions. These operations and realizations are the setting of Cisinski–Déglise, *Triangulated categories of mixed motives*, Theorems 14.4.1 and 15.2.1 and Chapter 17. No integral Artin–Schreier sheaf is being assumed.

Over the flag scheme, U_p and S_J are vector subbundles of U: their adapted-frame descriptions have constant rank and glue. Let $i_J : S_J \hookrightarrow U$ and $h_J = D - r_J$. Define

$$B_J = i_{J*}\mathbf{1}_{S_J}(-h_J)[-2h_J].$$

Purity identifies B_J with $i_{J*}i_J^!\mathbf{1}_U$.

For $p \notin J$, the counit of the nested closed immersion gives a face $B_J \to B_{J\cup\{p\}}$. If its codimension is c, this is the Gysin map $\mathbf{1}_{S_J}(-c)[-2c] \to \mathbf{1}_{S_{J\cup\{p\}}}$, with closed direct image understood. Composition of the counits gives a coherent commuting cube.

Place B_J in external degree $|J|$, with total graded term $B_J[-|J|]$. The face in direction p has sign

$$\epsilon(J, p) = (-1)^{\#\{j\in J : j<p\}}.$$

The total differential is $d_{\mathrm{Tot}} = (-1)^{|J|}d_{\mathrm{int}} + d_{\mathrm{cube}}$. Use a compatible complex model of the finite cube and put $\mathcal{M}_F = \mathrm{Tot}(B_\bullet)[s-1]$. The Gysin morphisms remain part of this totalization.

Proposition 12.6.1. The ℓ-adic realization of $\mathcal{M}_F$ on a finite-field fiber is $\mathcal{P}_F$. The complex realization is the same total Gysin cube, retaining the Tate markings. On the fiber over a fixed complex flag, use the right differential-module convention $\mathrm{DR}(\omega_S) = \mathbb{C}_S[\dim S]$. Its vertex is

$$(i_J)_+\omega_{S_J}[-r_J - 2h_J], \qquad \text{with Tate marking } (-h_J). \tag{12.10}$$

Proof. On $U^\vee$, the tensor product of localization fibers $[K_{U^\vee} \to K_{H_p}]$ is $j_!K_\Omega$. Its J-vertex is K_{H_J} in external degree $|J|$, where $H_J = \bigcap_{p\in J} H_p = S_J^\perp$. Fourier integration of this vertex gives B_J: the integral vanishes off S_J and contributes $(-h_J)[-2h_J]$ on it. Each restriction face becomes the Gysin face normalized by compact trace. Thus Fourier integration identifies the entire cubes, including their total differentials and final shift.

Realization preserves the six operations and fundamental classes. Applying DR to (12.10) gives $i_{J*}\mathbb{C}_{S_J}[-2h_J]$, as required. For the absolute differential module over a smooth complex base B of flags, the shift is instead $[-\dim B - r_J - 2h_J]$, because $\dim S_J = \dim B + r_J$. For a normal coordinate t, the sequence

$$0 \longrightarrow \omega \longrightarrow \omega(*t) \longrightarrow i_+\omega_{t=0} \longrightarrow 0$$

represents the codimension-one Gysin map after these shifts. Tensoring its normal-coordinate models gives higher codimension. □

The supported differential module retains every finite normal derivative. If δ is its normal generator, then

$$(\delta\,\partial_t^a)t = a\,\delta\,\partial_t^{a-1}.$$

These are finite sums of derivatives, with no product completion. The comparison identifies the constructed cube in two realizations. It does not identify their whole coefficient categories.

Theorem 12.6.2. After quotient integration in a nonempty set P of cut directions, the total Gysin cube admits homotopies h_p, $p \in P$, with

$$dh_p + h_p d = 1, \qquad h_p^2 = 0, \qquad h_p h_q + h_q h_p = 0 \quad (p \neq q).$$

They are compatible with the trace-normalized comparisons for further cut integrations.

Proof. Set $U_P = \sum_{p\in P} U_p$ and let $\rho_P : U \to U/U_P$. The pushed vertex is supported on $\overline{S}_J = \rho_P(S_J)$. Its twist and shift are determined by

$$h_J + \dim(S_J \cap U_P) = D - \dim \overline{S}_J.$$

Both this integer and the support depend only on $J \setminus P$.

For $p \in P$, affine trace identifies the pushed p-face with the identity. The remaining faces are Gysin maps of the image subbundles, with common factor $(-\dim U_P)[-2\dim U_P]$. The pushed cube is therefore equivalent to the remaining cube tensored with $[K \xrightarrow{1} K]$ in every direction of P.

Write e_J for the external label of a vertex in this tensor model. Before the final shift define

$$h_p(e_J) = \begin{cases} \epsilon(J \setminus \{p\}, p)e_{J\setminus\{p\}}, & p \in J, \\ 0, & p \notin J. \end{cases}$$

Its coefficient map is the inverse of the identity comparison.

The p-face contributes the identity to $dh_p + h_p d$. Every other face cancels with the reversed order, and the internal differential cancels because label removal changes $|J|$ by one. Removing the same label twice gives zero. Reversing two distinct removals changes the sign. This proves all three identities.

The final shift multiplies both d and h_p by $(-1)^{s-1}$. Composing affine traces gives the trace for the composite quotient. The common orientation therefore makes these homotopies compatible with further integrations. □

The omitted outer covector locus

For three one-dimensional pieces, the two binary groupings use the dual opens

$$O_L = \{\alpha \neq 0,\ (\beta,\gamma) \neq (0,0)\}, \qquad O_R = \{\gamma \neq 0,\ (\alpha,\beta) \neq (0,0)\}.$$

Their union O misses the nonempty locus $Z = \{\alpha = \gamma = 0,\ \beta \neq 0\}$ in Ω. Extending by zero to $U^\vee$, the two triangles are

$$K_{O_L\cap O_R,!} \longrightarrow K_{O_L,!} \oplus K_{O_R,!} \longrightarrow K_{O,!} \longrightarrow,$$
$$K_{O,!} \longrightarrow K_{\Omega,!} \longrightarrow K_{Z,!} \longrightarrow .$$

Fourier transform with the common shift $[2]$ preserves them. The last term has trace $q1_{b=0} - 1$, so it is nonzero. These are extension coordinates. A position-space 13|2 operation requires a separate map preserving the corresponding supports.

12.7 Descent after forgetting the flag

Assume the finite field has characteristic different from two. Let $\mathcal{N}_2 \subset \mathfrak{sl}_2$ be the nilpotent cone, with punctured inclusion j_N. The extension line maps into it by $f(v) = \left(\begin{smallmatrix} 0 & v \\ 0 & 0 \end{smallmatrix}\right)$.

THEOREM 12.7.1. There is no constructible complex $\mathcal{F}$ on $\mathcal{N}_2$ with $j_N^*\mathcal{F} \simeq K$ and $f^*\mathcal{F} \simeq \mathcal{P}_{\mathbb{A}^1}$, whose restriction to $\mathbb{G}_m$ agrees with that constant marking.

Proof. The cone is $\mathbb{A}^2/\{\pm 1\}$, and the quotient is finite étale away from zero. Characteristic-zero coefficients make invariants exact. The scalar involution preserves the orientation class, so purity gives stalk cohomology K in degree zero and $K(-2)$ in degree three for $R\,j_{N*}K$ at zero. In particular, its degree-one stalk vanishes.

An $\mathcal{F}$ as stated would give, by adjunction and base change,

$$\mathcal{P}_{\mathbb{A}^1} \longrightarrow f^* R j_{N*} K \longrightarrow R j_* K_{\mathbb{G}_m} \simeq \mathcal{P}_{\mathbb{A}^1}.$$

The composite is the identity on $\mathbb{G}_m$. Since $R j_*$ is fully faithful, it is the identity everywhere. Its degree-one stalk would factor the identity of $K(-1)$ through zero, a contradiction. □

There is a further obstruction without an open marking. Work over $\overline{\mathbb{F}}_q$ and let $G = \mathrm{GL}_n$. For a parabolic P with Levi quotient L, use

$$\mathcal{N}_L \xleftarrow{p} \mathcal{N}_P \xrightarrow{i} \mathcal{N}_n, \qquad \mathcal{N}_P = \mathcal{N}_n \cap \mathrm{Lie}\, P.$$

Parabolic restriction $\mathsf{R}_L^G = R p_! i^*$ first forgets equivariance from G to P, then identifies P- and L-equivariance on $\mathcal{N}_L$. Use its perverse-exact normalization, and fix a maximal torus $T \subset B \subset G$.

Proposition 12.7.2. If $\mathcal{F} \in D^b_G(\mathcal{N}_n, K)$ and $\mathsf{R}_T^G \mathcal{F} = 0$, then $\mathcal{F} = 0$. For $n \geq 2$, no nonzero object of this category has all proper parabolic restrictions zero.

Proof. Parabolic restriction is perverse-exact and has parabolic induction as right adjoint. See Achar–Mautner, arXiv:1207.7044, Remark 2.1 and Proposition 3.1. Every simple G-equivariant perverse sheaf is a summand of the Springer sheaf $\mathsf{I}_T^G K$. Indeed, nilpotent centralizers in GL_n are connected, so these simples are indexed by partitions. The ordinary Springer injection is a bijection because irreducible characteristic-zero representations of S_n have the same indexing.

For a simple $\mathcal{L}$, its nonzero inclusion into $\mathsf{I}_T^G K$ gives a nonzero map $\mathsf{R}_T^G \mathcal{L} \to K$ by adjunction. Exactness and finite length show that restriction kills no nonzero perverse sheaf. It commutes with perverse cohomology, so boundedness proves the assertion for $\mathcal{F}$. Transitivity factors restriction to T through successive proper cuts. □

These statements concern the ordinary nilpotent recipient and its specified restriction functors. The marked Fourier construction retains an extension or cotangent variable. Its cut integrals are different operations, so its positive vanishing theorem survives.

The failure is also visible after a proper forgetting map. Let $\widetilde{\mathcal{N}}_2$ consist of pairs (N, L) with $N(V) \subset L$ and $N(L) = 0$. Its projection $\pi : \widetilde{\mathcal{N}}_2 \to \mathcal{N}_2$ forgets L. The extension line bundle over $\mathbb{P}^1$ carries the relative kernel $\mathcal{P}$.

Proposition 12.7.3. The compact cohomology of $f^* R\pi_* \mathcal{P}$ is nonzero. Its Frobenius trace is $q(1-q)$, although integration over the original extension fiber is zero.

Proof. A regular nilpotent has one stable line with nonzero extension coordinate, giving trace 1. At zero, all $q+1$ lines are stable and each contributes $1-q$. Summing on the marked line therefore gives $(q-1) + (q+1)(1-q) = q(1-q) \neq 0$. The nonzero trace detects nonzero cohomology. Including the unipotent factor $K(1)[2]$ divides the trace by q and leaves it nonzero. □

For linear grouping itself, put $\widetilde{\mathcal{P}}_V = \mathcal{P}_V[-1]$. The inclusion of the product of punctured duals into the punctured total dual gives

$$\widetilde{\mathcal{P}}_V \boxtimes \widetilde{\mathcal{P}}_W \longrightarrow \widetilde{\mathcal{P}}_{V \oplus W}.$$

All parenthesizations include the same locus where every original covector block is nonzero, so they compose coherently. The two omitted punctured axes give the triangle

$$(\mathcal{P}_V \boxtimes \mathcal{P}_W)[-1] \longrightarrow \mathcal{P}_{V \oplus W} \longrightarrow (\mathcal{P}_V \boxtimes K_W) \oplus (K_V \boxtimes \mathcal{P}_W) \longrightarrow .$$

This follows by applying Fourier integration to their localization triangle, retaining the single shift in (12.6).

A linear map $g : V \to W$ likewise has the comparison

$$\mathsf{F}_W(Rg_!\mathcal{G}) \simeq (g^\vee)^*\mathsf{F}_V(\mathcal{G}).$$

Both sides integrate the phase $\langle g(v), \eta\rangle$. Rank normalization adds $[\dim W - \dim V]$ on the right. These linear comparisons do not construct the nonlinear Hall incidences or a vanishing-cycle comparison.

12.8 Cohomology of the extension correspondence

An extension count records the size of a fiber. Equivariant cohomology also records its characteristic classes. Even the charge-one space already distinguishes the two constructions: a one-dimensional vector space has automorphism group $\mathbb{C}^\times$, and $H^*(B\mathbb{C}^\times;\mathbb{Q}) = \mathbb{Q}[x]$, with $|x| = 2$. There is one class in every nonnegative even degree.

Let Q be a finite quiver, with vertex set I and a_{ij} arrows from i to j. For a dimension vector $d \in \mathbb{N}^I$, put

$$X_d = \bigoplus_{i,j} \mathrm{Hom}(\mathbb{C}^{d_i}, \mathbb{C}^{d_j})^{\oplus a_{ij}}, \qquad G_d = \prod_i GL_{d_i}, \qquad \mathfrak{M}_d = [X_d/G_d].$$

Fix standard subspaces of dimension d_i in $\mathbb{C}^{d_i+e_i}$. Let $X_{d,e} \subset X_{d+e}$ be the representations preserving them, and let $P_{d,e} \subset G_{d+e}$ preserve the subspaces. The two maps

$$\mathfrak{M}_d \times \mathfrak{M}_e \xleftarrow{p} [X_{d,e}/P_{d,e}] \xrightarrow{q} \mathfrak{M}_{d+e}$$

remember the subrepresentation and quotient, or forget the subspaces. The map q is proper: its fibers are closed subvarieties of products of Grassmannians. The map p retains the extension coordinates and the unipotent automorphisms of the sequence.

Write

$$\chi(d,e) = \sum_i d_i e_i - \sum_{i,j} a_{ij} d_i e_j.$$

The cohomological Hall product acts on $\bigoplus_d H^*(\mathfrak{M}_d;\mathbb{Q})$. It combines pullback along p, the normal Euler class of $X_{d,e}$, and proper flag pushforward. Its cohomological degree is $-2\chi(d,e)$. Since X_d contracts equivariantly to zero,

$$H^*(\mathfrak{M}_d;\mathbb{Q}) = \mathbb{Q}[x_{i,r} : i \in I,\ 1 \le r \le d_i]^{\prod_i S_{d_i}}.$$

Here $x_{i,r}$ are Chern roots of degree two.

THEOREM 12.8.1 *(Cohomological Hall shuffle product).* For f of charge d and g of charge e, the product is

$$f * g = \sum_{\text{vertexwise shuffles}} f(x)g(y) \frac{\displaystyle\prod_{i,j}\prod_{r=1}^{d_i}\prod_{s=1}^{e_j}(y_{j,s} - x_{i,r})^{a_{ij}}}{\displaystyle\prod_i \prod_{r=1}^{d_i}\prod_{s=1}^{e_i}(y_{i,s} - x_{i,r})}. \tag{12.11}$$

The sum is a symmetric polynomial. It defines an associative unital product. The unit is $1 \in H^*(\mathfrak{M}_0) = \mathbb{Q}$.

Proof. The normal directions to the invariant-subspace locus are maps from the subspace at i to the quotient at j, one for each arrow $i \to j$. Their equivariant Euler class is the numerator. The tangent directions of the flag variety give the denominator.

The fixed points of its maximal torus are precisely the choices of d_i roots at each vertex. Equivariant localization of the proper pushforward therefore gives (12.11). The pushforward is defined before localization, so its sum is polynomial. This is the extension-correspondence calculation of Kontsevich–Soibelman, arXiv:1006.2706, §§2.2–2.4.

For three factors, either parenthesization chooses the same ordered three-block partition at every vertex. Every pair of blocks contributes the same Euler quotient exactly once. The terms, including their degrees, agree. An empty block contributes no Euler factors and has one shuffle. This proves associativity and the unit laws. □

Suppose now that $a_{ij} = a_{ji}$. The charge shift

$$A_d = H^*(\mathfrak{M}_d; \mathbb{Q})[-\chi(d,d)]$$

makes multiplication degree zero. Thus a polynomial of degree r has cohomological degree $2r + \chi(d,d)$. Exchanging its two blocks in (12.11) gives $f * g = (-1)^{\chi(d,e)} g * f$. We first use the symmetric tensor convention whose crossing on charges d, e is $(-1)^{\chi(d,e)}$.

There is also an ordinary super convention. Put

$$\tau(d,e) = \chi(d,e) + \chi(d,d)\chi(e,e) \pmod 2$$

and choose a bilinear form ψ modulo two with $\psi(d,e) + \psi(e,d) = \tau(d,e)$. Such a form exists because τ is alternating. The product $f \star g = (-1)^{\psi(d,e)} f * g$ is supercommutative for the usual cohomological parity. Its associativity follows from bilinearity of ψ. The choice changes signs, not the graded spaces.

Example 12.8.2. For one vertex with no arrows the kernel in (12.11) is $1/(y-x)$. Let $e_r = x^r$ in charge one. Then

$$e_r * e_s = \frac{x^r y^s - x^s y^r}{y - x}, \qquad |e_r| = 2r + 1.$$

The algebra is the exterior algebra on all e_r, $r \geq 0$. Indeed, the product of $e_{r_1}, \dots, e_{r_n}$ is, up to sign, their alternant divided by the Vandermonde determinant. For strictly increasing nonnegative exponents these quotients give the Schur basis of the symmetric polynomials. Repeated exponents give zero. This proves both spanning and linear independence.

For one vertex with one loop the kernel is 1 and $|e_r| = 2r$. Products are symmetrized monomials. Their multiplicity factorials are invertible over $\mathbb{Q}$, so the algebra is the polynomial algebra $\mathbb{Q}[e_0, e_1, \dots]$. Neither example has only one generator in charge one.

12.9 Primitive fields for a symmetric quiver

The dual of the Hall product is a coproduct. In the symmetric, zero-potential case it belongs to a vertex algebra with a precise primitive description. All duals in this section are graded duals:

$$V = \bigoplus_{d,k} (A_d^{-k})^\vee.$$

Each summand is finite dimensional. The full linear dual, with its unrestricted product over degrees, is a different recipient.

Use vertex algebras with lower-truncated fields $Y(a,z)=\sum_n a_{(n)}z^{-n-1}$, vacuum 1, and translation D. They satisfy $Y(1,z)=\mathrm{id}$, $Y(Da,z)=[D,Y(a,z)]=\partial_z Y(a,z)$ and $Y(a,z)1=e^{zD}a$, together with mutual locality. For a pair of states, locality of order $N\geq 0$ is equivalent to $a_{(n)}b=0$ for $n\geq N$. The symmetric tensor convention is the charge crossing fixed above.

Let e_i denote the ith standard dimension vector. Start with states v_i of charge e_i, parity $\chi(e_i,e_i) \bmod 2$ and cohomological degree $-\chi(e_i,e_i)$. Impose locality orders $N_{ij}=\max\{0,-\chi(e_i,e_j)\}$. The principal free vertex algebra P_Q is the universal vertex algebra with these generating states and relations.

The singular products $a_{(n)}b$, $n\geq 0$, and translation form a *vertex Lie algebra*. Its axioms are the sesquilinearity, skew-symmetry and Jacobi identities of the singular part of a field. Its universal enveloping vertex algebra $U^{\mathrm{ch}}(P)$ is generated by P with these prescribed singular products and translation. The coefficient-mode construction appears in the proof.

THEOREM 12.9.1 *(Dotsenko–Mozgovoy).* For a finite symmetric quiver with zero potential and rational coefficients, V is canonically a connected cocommutative vertex bialgebra and $P_Q\simeq V$. Its coproduct is dual to the Hall product. If

$$P=\{a\in V:\Delta a=a\otimes 1+1\otimes a\},$$

then P is a vertex Lie algebra and $U^{\mathrm{ch}}(P)\simeq V$. Moreover, D has cohomological degree -2, and

$$P\simeq W\otimes\mathbb{Q}[D],\qquad W=P/DP,\qquad \dim W_d<\infty.$$

Every nonzero W_d^k has $k\equiv\chi(d,d) \pmod 2$.

After choosing homogeneous lifts of W, graded duality gives

$$\mathcal{A}\simeq \mathrm{Sym}_{\chi}(W^{\vee}\otimes\mathbb{Q}[u]),\qquad |u|=2, \tag{12.12}$$

as algebras. With the product $\star$, this is a free supercommutative algebra in the ordinary parity convention. The choice of lifts makes this last presentation noncanonical.

Proof. We recall the maps and the separate finiteness argument in Dotsenko–Mozgovoy, arXiv:2108.10338, Proposition 5.3 and Theorems 5.6–5.7. For a sequence $i_1,\dots,i_m$ of vertices with total charge d, set $N(i,j)=-\chi(e_i,e_j)$. The map from $P_Q^{\vee}$ to symmetric polynomials is

$$\xi\longmapsto \prod_{r<s}(z_r-z_s)^{N(i_r,i_s)}\big\langle \xi, Y(v_{i_1},z_1)\cdots Y(v_{i_m},z_m)1\big\rangle.$$

Its value is symmetric under simultaneous permutations of vertices and variables. The vacuum removes negative powers of the last variable. Symmetry removes them from every variable. The graded functional has finite degree support, so the expression is polynomial.

When $N(i,i)=-1$, the state is odd. Self-locality of order zero already implies self-locality of order -1.

The map is injective because coefficients of the products of generating fields span P_Q. In the lattice realization, the creation part of these products is an exponential whose coefficients are monomials in power sums. These generate the symmetric polynomial ring over $\mathbb{Q}$, proving surjectivity.

Declare the v_i primitive. Expand the coproduct by assigning each factor of the ordered field product to one of two blocks. The cross factor is exactly the Euler quotient in (12.11). Thus the comparison preserves the coproduct, not only graded dimensions.

For primitive a, b of charges d, e, compatibility of the fields and coproduct gives

$$\Delta(Y(a,z)b) = Y(a,z)b \otimes 1 + 1 \otimes Y(a,z)b \\ + e^{zD}a \otimes b + (-1)^{\chi(d,e)} b \otimes e^{zD}a.$$

The two mixed terms are regular. Hence every singular coefficient $a_{(n)}b$, $n \geq 0$, is primitive. Translation preserves primitives as well. These operations give the vertex Lie algebra P.

Positive charge makes the coalgebra connected: more than $\sum_i d_i$ positive factors annihilate its iterated reduced coproduct in charge d. The connected vertex Milnor–Moore theorem then identifies $U^{\mathrm{ch}}(P)$ with V.

For clarity, the coefficient Lie algebra of P is

$$\mathfrak{g}_P = P \otimes \mathbb{Q}[t, t^{-1}]/(D + \partial_t), \qquad [a[m], b[n]] = \sum_{j \geq 0} \binom{m}{j} (a_{(j)}b)[m + n - j].$$

Its negative part has $a[-r] = (D^{r-1}a)[-1]/(r-1)!$ for $r \geq 1$. PBW symmetrization identifies the coalgebra of the vacuum induced module with $\mathrm{Sym}_{\chi}(\mathfrak{g}_P^-)$. In particular, this coalgebra statement concerns every negative mode.

The mode presentation also has an operator T, defined on generators by $T(v_i[n]) = -v_i[n+1]$, with $[D, T] = |d|\,\mathrm{id}$ in positive charge d. Its weights are bounded below, with D raising and T lowering weight. After dividing T by $|d|$, the following elementary argument proves that D acts freely. If $Da = 0$, the finite span of $a, Ta, T^2a, \ldots$ is stable under D, T. The trace of $[D, T] = 1$ would equal its dimension and also vanish, so $a = 0$.

Choose a graded complement to DP. Induction on weight gives spanning by its D-iterates. Cancellation of the lowest power of D proves their independence.

This argument alone does not prove $\dim W_d < \infty$. The framed CoHA character identity in the cited theorem implies that $(1 + q + \cdots + q^{w \cdot d - 1})\,\mathrm{ch}(W_d)$ is a Laurent polynomial, for a framing w with $w \cdot d > 0$. The series $\mathrm{ch}(W_d)$ is bounded below and has nonnegative coefficients. The product can have finite support only if the series does. The lattice weights give the asserted parity. Finally, degreewise duality of the PBW coalgebra, with the chosen lifts of W, proves (12.12). □

The algebra conclusion is also Efimov's freeness theorem, arXiv:1103.2736, Theorem 1.1. The vertex construction explains its translation strings. The quotient P/DP has a Lie bracket induced by the zeroth product. It must not be identified with a cohomological BPS Lie algebra merely because their graded characters give the same invariants.

12.10 A potential and the perverse filtration

A potential changes the coefficients of the correspondence. Let $W \in \mathbb{C}Q/[\mathbb{C}Q, \mathbb{C}Q]$ be a finite linear combination of cyclic paths. It defines the regular function $f_d = \mathrm{Tr}(W)$ on $\mathfrak{M}_d$. The Jacobi algebra is the path algebra modulo the cyclic derivatives $\partial_a W$ for arrows a. Its representation moduli form the critical locus of f_d. Critical cohomological Hall theory uses vanishing cycles of f_d, retaining their monodromy.

Choose numbers ζ_i with positive imaginary parts. For $d \neq 0$, set

$$Z_\zeta(d) = \sum_i \zeta_i d_i, \qquad \mu_\zeta(d) = -\frac{\mathrm{Re}\, Z_\zeta(d)}{\mathrm{Im}\, Z_\zeta(d)}.$$

A representation is ζ-semistable when every nonzero proper subrepresentation has slope at most that of the full representation. Fix a slope μ. Write Λ^{ζ}_{μ} for zero together with the dimension vectors of that slope, and $\mathfrak{M}_d^{\zeta\text{-ss}}$ for the stack of semistable representations.

Require ζ to be *μ-generic*: $\chi(d,e) = \chi(e,d)$ for all $d, e \in \Lambda^{\zeta}_{\mu}$. Fix a Serre subcategory $\mathcal{S}$, closed under subobjects, quotients and extensions. Require its semistable objects to form a locally closed substack $\mathfrak{M}_d^{\zeta\text{-ss},\mathcal{S}}$. These choices enter both the Hall algebra and its primitive term.

We use rational monodromic mixed Hodge modules. This coefficient category retains the Hodge and weight filtrations together with vanishing-cycle monodromy. Put $\mathbb{L} = H_c(\mathbb{A}^1;\mathbb{Q})$ and use its monodromic square root $\mathbb{L}^{1/2}$, of cohomological degree one.

Let ϕ_f^{mon} be the perverse-normalized monodromic vanishing-cycle functor. Assume that every critical point of f_d lies over zero. Without this assumption, replace it by the sum of vanishing cycles over all critical values.

Define

$$\mathcal{A}^{\zeta,\mathcal{S}}_{W,\mu} = \bigoplus_{d\in\Lambda^{\zeta}_{\mu}} H_c\left(\mathfrak{M}_d^{\zeta\text{-ss},\mathcal{S}}, \phi^{\mathrm{mon}}_{f_d}\mathbb{Q}_{\mathfrak{M}_d^{\zeta\text{-ss}}} \otimes \mathbb{L}^{-\dim \mathfrak{M}_d^{\zeta\text{-ss}}/2}\right)^{\vee}.$$

The functor and dimension shift are taken on the ambient smooth semistable stack before restriction to $\mathcal{S}$. Stack cohomology uses equivariant finite-dimensional approximations. The dual is the degreewise dual of compactly supported cohomology. The extension correspondence and Thom–Sebastiani isomorphism define the critical Hall multiplication on this object.

Let $M_d^{\zeta\text{-ss}}$ be the coarse moduli space. Its normalized intersection complex IC_d is Verdier self-dual. On a smooth component it is $\mathbb{L}^{-\dim M_d/2}\mathbb{Q}_{M_d}$. Define the BPS sheaf to be $\phi^{\mathrm{mon}}_{f_d}\mathrm{IC}_d$ if the stable locus is nonempty, and zero otherwise.

Denote by $\mathcal{B}^{\zeta,\mathcal{S}}_{W,\mu}$ the direct sum of the duals of its compactly supported cohomology restricted to the coarse loci selected by $\mathcal{S}$. The remaining universal factor is

$$H(B\mathbb{C}^{\times})_{\mathrm{vir}} = \mathbb{L}^{1/2} \otimes H^*(B\mathbb{C}^{\times};\mathbb{Q}).$$

Its half-Tate shift is part of the normalization.

THEOREM 12.10.1 *(Davison–Meinhardt).* With these hypotheses there is a PBW isomorphism of charge-graded, cohomologically graded monodromic mixed Hodge objects

$$\mathrm{Sym}_{\tau}\left(H(B\mathbb{C}^{\times})_{\mathrm{vir}} \otimes \mathcal{B}^{\zeta,\mathcal{S}}_{W,\mu}\right) \xrightarrow{\sim} \mathcal{A}^{\zeta,\mathcal{S}}_{W,\mu}.$$

Here the symmetry is the Koszul sign multiplied by $(-1)^{\tau(d,e)}$, with τ as above on Λ^{ζ}_{μ}. The perverse associated graded of the Hall algebra is free commutative for this symmetry. After a bilinear sign modification $\psi + \psi^{\mathrm{T}} = \tau$ of multiplication, that associated graded is free supercommutative for the ordinary Koszul symmetry. The displayed PBW map concerns underlying graded objects. It need not be an algebra isomorphism for the original Hall multiplication.

This is the absolute form of Theorems A and C of Davison–Meinhardt, arXiv:1601.02479. The perverse filtration comes from the map from the semistable stack to its coarse moduli space. It is not the filtration by total dimension alone. The PBW map uses the BPS embedding and symmetrized Hall multiplication. The original algebra can therefore have nonzero commutators whose leading perverse symbols commute.

The recipient allows infinitely many cohomological degrees. For each fixed charge and weight, only finitely many cohomological degrees occur, and the weights are bounded below at fixed charge. Symmetric powers are formed by sums in this recipient. They are not unrestricted products of all cohomological degrees. The zero-potential algebra theorem (12.12) retains its stronger conclusion in its own finite symmetric-quiver setting.

12.11 An ordinary coproduct obstruction on a shuffle algebra

The preceding vertex coproduct is dual to a symmetric Hall product. A deformed shuffle multiplication poses a different compatibility problem. Its charge-one elements can already prevent an ordinary counital coproduct from being multiplicative.

Work over

$$R = \mathbb{Q}[h_1, h_2], \qquad h_3 = -h_1 - h_2, \qquad a = -(h_1^2 + h_1h_2 + h_2^2), \qquad b = -h_1h_2(h_1 + h_2).$$

Put $N(v) = \prod_i (v + h_i) = v^3 + av + b$. For $j = 0, 1$, let

$$\mathcal{A}^{(j)} = \bigoplus_{n \geq 0} R[x_1, \dots, x_n]^{S_n}, \qquad K_j(v) = v^{2j-1} N(v).$$

These are algebraic direct sums. Charge is n. All cochain degrees and parities are zero. There is no completion or inversion of parameters. The product is

$$(F *_j G)(X) = \sum_{I \sqcup J = [d+e],\ |I| = d} F(X_I) G(X_J) \prod_{i \in I, k \in J} K_j(x_k - x_i).$$

There is no factorial denominator. For $j = 0$, opposite residues cancel along each true diagonal, so the sum is polynomial. For $j = 1$ every summand is polynomial. Ordered three-block partitions prove associativity in either case. The unit is $1_0 \in A_0 = R$, distinct from the constant polynomial in charge one.

The binary image remembers the coefficient ring. Write $c = (x + y)/2$, $v = y - x$, $t = v^2$ and $B = R[c, t]$. Then $R[x, y] = B \oplus vB$, and the multiplication map is

$$\mu_j(A + vC) = 2t^j \big((t + a)A + bC\big). \tag{12.13}$$

Consequently its image is $t^j(t + a, b)$. The complex $[B^2 \xrightarrow{\mu_j} B]$ in degrees $-1, 0$ has

$$H^{-1} = B\,(b, -(t + a)), \qquad H^0 = B/t^j(t + a, b).$$

To compute the kernel, reduce $(t + a)A + bC = 0$ modulo $t + a$. The quotient $R[c]$ is a domain, so C is divisible by $t + a$. Cancellation gives the displayed generator. Multiplication by t^j adds no kernel.

The two cokernels have different parameter annihilators:

$$\operatorname{Ann}_R B/(t + a, b) = (b), \qquad \operatorname{Ann}_R B/t(t + a, b) = 0.$$

For the first, set $t = -a$. For the second, evaluation at $t = 0$ maps the quotient onto the faithful R-module $R[c]$. Thus the two binary lattices cannot be interchanged over R by an operation-preserving charge-graded equivalence.

THEOREM 12.11.1. Neither $\mathcal{A}^{(j)}$ admits an R-linear, charge-preserving, unital algebra map

$$\Delta : \mathcal{A}^{(j)} \longrightarrow \mathcal{A}^{(j)} \otimes_R^{\mathbb{L}} \mathcal{A}^{(j)}$$

with ordinary Koszul multiplication and two-sided counit given by projection to charge zero. The same holds for the subalgebra generated by charge one. No coassociativity hypothesis is needed.

Proof. Set $e_m = x^m \in A_1$. The difference of the two kernel orders gives

$$[f, g]_{*_j} = 2bt^j \frac{f(x)g(y) - g(x)f(y)}{y - x}.$$

In particular,

$$[e_0, e_3] = 2bt^j(x^2 + xy + y^2), \qquad [e_1, e_2] = 2bt^j xy,$$
$$[e_0, e_1] = 2bt^j, \qquad e_0 *_j e_0 = 2t^j(t + a).$$

They satisfy the exact relation

$$[e_0, e_3] - 3[e_1, e_2] + a[e_0, e_1] = b(e_0 *_j e_0). \tag{12.14}$$

Charge preservation and both counits force $\Delta e_m = e_m \otimes 1_0 + 1_0 \otimes e_m$. In the ordinary tensor product, mixed terms in the commutator of two even primitive elements cancel. The $(1, 1)$ component of the coproduct of the left side of (12.14) is therefore zero. The right side gives $2b\, e_0 \otimes e_0$, which is nonzero in $R[x, y]$.

Every ambient charge component is free over R, so its derived tensor is ordinary. For the charge-one generated subalgebra the same detecting component is still the free tensor $R[x, y]$. Taking cohomology preserves the contradiction. At $(h_1, h_2, h_3) = (1, 1, -2)$ its coefficient is $2b = -4$. The proof neither divides by b nor asserts persistence at $b = 0$. □

The tensor in this theorem is essential. A rational replacement can carry a compatible cooperation. For ordered blocks X, Y, set

$$Q^{(j)}_{d,e}(X, Y) = \prod_{x \in X, y \in Y} K_j(y - x), \qquad \Delta^{\mathrm{rat}}_{d,e} F = \frac{F(X, Y)}{Q^{(j)}_{d,e}(X, Y)}, \qquad r(v) = \frac{v^3 + av - b}{N(v)}.$$

Use polynomial rings localized at all spectral differences and their translates $v \pm h_i$, then take the blockwise symmetric invariants. No parameter is inverted by itself, but closed spectral loci have been removed. Take F of bidegree (d, e) and G of bidegree (d', e'). Their target product is the sum of

$$F(X_1, Y_1)G(X_2, Y_2) \prod_{x \in X_1, x' \in X_2} K_j(x' - x)$$
$$\cdot \prod_{y \in Y_1, y' \in Y_2} K_j(y' - y) \prod_{y \in Y_1, x' \in X_2} r(y - x')$$

over the two prescribed ordered splittings of X and Y. The last factor changes the interchange of the middle two blocks.

PROPOSITION 12.11.2. This target multiplication is associative and unital. The maps Δ^{rat} are multiplicative, coassociative in the corresponding three-block localizations, and counital at charge zero.

Proof. In a shuffle term, division by the total cross kernel cancels the factors between X_1 and Y_2. The remaining middle ratio is $K_j(x' - y)/K_j(y - x') = r(y - x')$, for $y \in Y_1$, $x' \in X_2$. This is exactly the displayed target product.

For three factors, either parenthesization contains the same internal kernels and the same earlier-right, later-left exchange factors. This proves associativity. Iterating cooperation divides by $Q_{12}Q_{13}Q_{23}$ in both orders. Empty products give the unit and both counits. □

For (12.14), the new mixed term satisfies

$$(1 - r(v))v(v^2 + a) = b(1 + r(v)),$$

as follows directly from $1 - r(v) = 2b/N(v)$. It supplies the term absent from the ordinary tensor calculation.

Nevertheless,

$$\Delta^{\mathrm{rat}}_{1,1}(1_2) = \begin{cases} v/N(v), & j = 0, \\ 1/(vN(v)), & j = 1, \end{cases}$$

is not polynomial. Localizing an equation that vanishes on a shifted collision annihilates that closed restriction. Extending this rational cooperation to supported coefficients remains a separate problem. Such an extension must retain the integral images (12.13) and derived collision complexes. An ordinary coefficient bialgebra and a strong coalgebra for cut convolution have different tensors. The theorem excludes the latter only when its marked comparison induces the former.

12.12 The arithmetic Hall comparison

The constructions above specify several recipients of Hall data. The scalar Hall algebra retains automorphism-weighted extension counts. The Fourier kernel retains an attaching morphism and compatible primitive contractions. Cohomological Hall theory retains characteristic classes. A potential additionally retains vanishing cycles and their monodromy. A comparison must name which of these operations it preserves.

Question 12.12.1. Construct an independently defined arithmetic Hall category and an operation-preserving functor to the intended geometric or field category. Its extension correspondences must induce the specified Hall multiplication and primitive cut maps. It must retain automorphism and Tate normalizations, place labels, substitutions and nonconsecutive collision supports. The empty unit and marked local vacuum require their own compatible maps. All comparisons must respect composition.

For a critical CoHA target, construct the arithmetic quiver, potential, stability data and normalized coefficient comparison. For a field target, construct the actual fields, translation and supported products on the proposed states.

The flag kernels solve the primitive cut problem with the flag retained. Their descent failures locate the additional datum needed by an unmarked recipient. The symmetric-quiver theorem constructs primitive fields on its graded dual. Neither statement constructs an arithmetic Hall functor or an equivalence with an independently specified arithmetic category.

The formal Euler expansion also keeps a precise scope. On $\operatorname{Re} s > 1$,

$$\log \zeta(s) = \sum_p \sum_{k \geq 1} \frac{p^{-ks}}{k}, \qquad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p)\, p^{-ks}.$$

The coefficients of the first series are rational. Logarithms appear after differentiation. Neither series specifies a finite cyclic word in a path algebra. A potential requires such a cyclic class, or a specified completion and a trace map on it. A class in cyclic homology or an exponential Hodge recipient would require its own construction and comparison with the Euler series. The scalar identity supplies no canonical lift of this kind.

Chapter 13

Hecke, Adams, and Witt operations

An operation on mode indices does not determine its arithmetic meaning. Divisor substitutions, index shifts and Hecke operators act on different structures, even when their first values agree. Their composition laws separate them. Integral lattices, distinguished states and completed domains give further distinctions.

We begin with three explicit formulas. The divisor substitutions have a noncommutative algebra of boundary corrections. A second divisor formula gives a genuine linear Hecke action. The index shifts preserve the integral symmetric-function lattice and satisfy additional Frobenius congruences. Big Witt coordinates retain this integral information when ghost values lose it. Finally, the radial transform and weighted operator domains specify which comparisons survive Hilbert completion.

13.1 Three operations on mode indices

Let R be a commutative unital ring and S a finite set of species. Define the free module and its symmetric algebra by

$$M_R = \bigoplus_{s\in S,\ k\geq 1} R x_{s,k}, \qquad P_R = \mathrm{Sym}_R(M_R).$$

These objects have cohomological degree zero. Each generator has particle degree one and weight k. The unit $1 \in P_R$ is the polynomial vacuum. All polynomials have finite support.

The species index and the index of an operation are independent.

For $n \geq 1$, define two unital R-algebra endomorphisms by

$$D_n x_{s,k} = \sum_{d|(n,k)} d\, x_{s,nk/d}, \qquad \psi_n x_{s,k} = x_{s,nk}. \tag{13.1}$$

The first is the *divisor substitution*. The second is the *index shift*. The sums are finite and use integer coefficients. Thus both definitions commute with scalar extension.

For an arbitrary species set, take the union over its finite subsets. For $S = \varnothing$, both maps are the identity on R.

A third formula defines an R-linear endomorphism of M_R:

$$\mathsf{H}_n x_{s,k} = \sum_{d|(n,k)} d\, x_{s,nk/d^2}. \tag{13.2}$$

The index is integral because d divides both n and k. The squared denominator changes the composition law. We suppress the species when examining one summand.

Example 13.1.1. For a prime p, all three maps send x_1 to x_p. Their next values are

$$D_p x_p = x_{p^2} + p x_p, \qquad \psi_p x_p = x_{p^2}, \qquad \mathsf{H}_p x_p = x_{p^2} + p x_1.$$

Consequently

$$(D_p^2 - D_{p^2})x_1 = px_p, \qquad (D_p^2 - D_{p^2} - pI)x_1 = p(x_p - x_1).$$

Both differences are nonzero over $\mathbb{Z}$. The index shifts satisfy $\psi_m\psi_n = \psi_{mn}$ on every generator, hence on P_R. The linear Hecke relation will instead include the additional term pI.

Extending a linear generator map $A : M_R \to M_R$ to $\mathrm{Sym}(A) : P_R \to P_R$ preserves composition. It does not preserve addition as a function of A. Already

$$\mathrm{Sym}(2A)(x^2) = 4(Ax)^2, \qquad 2\,\mathrm{Sym}(A)(x^2) = 2(Ax)^2.$$

Moreover, every such extension fixes the vacuum. Over $\mathbb{Z}$, the putative polynomial Hecke relation

$$\mathrm{Sym}(\mathsf{H}_p)^2 = \mathrm{Sym}(\mathsf{H}_{p^2}) + pI$$

already fails on 1, where its two sides give 1 and $1 + p$. Linear operator relations and multiplicative substitutions must therefore retain their different carriers.

13.2 Composition and the boundary of a prime chain

Fix a prime p and $u \geq 1$ with $(u, p) = 1$. Write $e_r = x_{up^r}$ for $r \geq 0$, and let A_a be the restriction of D_{p^a} to this free submodule. Then $A_0 = I$ and

$$A_a e_r = \sum_{i=0}^{\min(a,r)} p^i e_{r+a-i}.$$

The lower endpoint $r = 0$ truncates the sum. Farther along the chain, the same operators act by polynomials in the forward shift. Their noncommutativity is confined to the truncated region.

THEOREM 13.2.1 *(Prime-chain composition).* For $a, b, r \geq 0$, put $[v]_+ = \max(v, 0)$ and

$$N_{a,b}(r, t) = \mathbf{1}_{t \leq r+b}\big[\min(b, r, t) - \max(0, t - a) + 1\big]_+.$$

Over every commutative coefficient ring,

$$A_a A_b e_r = \sum_{t=0}^{a+b} p^t N_{a,b}(r, t) e_{r+a+b-t}.$$

For $1 \leq a < b$, the boundary commutator is

$$[A_a, A_b]e_1 = -p^{a+1}e_b. \tag{13.3}$$

Thus, over a nonzero characteristic-zero ring, A_a and A_b commute exactly when $a = b$ or $ab = 0$.

Proof. A term of $A_a A_b e_r$ chooses integers

$$0 \leq i \leq \min(b, r), \qquad 0 \leq j \leq \min(a, r + b - i).$$

Its coefficient is p^{i+j} and its output index is $r + a + b - i - j$. Fixing $t = i + j$ leaves the conditions

$$t \leq r + b, \qquad \max(0, t - a) \leq i \leq \min(b, r, t).$$

Their number is $N_{a,b}(r,t)$, proving the first formula.

For the commutator, first use $A_b e_1 = e_{b+1} + p e_b$. Applying A_a gives successive coefficients $1, 2p, \dots, 2p^a, p^{a+1}$. In the reverse order, $A_a e_1 = e_{a+1} + p e_a$. Since $b \geq a+1$, its final coefficient is $2p^{a+1}$. All other coefficients agree. Their difference proves (13.3).

The nonzero integer p^{a+1} remains nonzero in a characteristic-zero ring. The commuting cases are immediate, and exchanging a, b covers the other order. □

For example, the entire first commutator is

$$[A_1, A_2]e_r = \begin{cases} -p^2 e_2, & r = 1, \\ 0, & r \neq 1. \end{cases}$$

For $r \geq 2$, both operators have their untruncated polynomial form. The two remaining basis vectors give the displayed boundary term.

Corollary 13.2.2 *(Independent prime coordinates).* Set $a_p = v_p(m)$, $b_p = v_p(n)$ and $r_p = v_p(k)$. Then

$$D_m D_n x_k = \sum_{(t_p)} \left(\prod_{p \mid mn} p^{t_p} N_{a_p,b_p}(r_p, t_p) \right) x_{mnk/\prod_{p\mid mn} p^{t_p}}, \qquad 0 \leq t_p \leq a_p + b_p.$$

Terms with zero coefficient are omitted. In particular, $D_m D_n = D_{mn}$ when $(m,n) = 1$. For a nonempty species set over a nonzero characteristic-zero ring,

$$\begin{aligned} D_m D_n = D_{mn} \quad &\Longleftrightarrow \quad (m,n) = 1, \\ [D_m, D_n] = 0 \quad &\Longleftrightarrow \quad \text{for every } p,\ a_p b_p = 0 \text{ or } a_p = b_p. \end{aligned}$$

Over $\mathbb{F}_\ell$, omit the condition at $p = \ell$ in both criteria.

Proof. Multiplication or division by a power of p leaves every other prime valuation unchanged. The divisor choices therefore factor over primes, giving the formula and the coprime statement. Equality on generators extends multiplicatively to P_R.

If $a_p, b_p > 0$, then

$$A_{a_p} A_{b_p} e_0 = \sum_{j=0}^{\min(a_p,b_p)} p^j e_{a_p+b_p-j}.$$

Its $j = 1$ term is absent from $A_{a_p+b_p} e_0$. Selecting the highest terms at the other primes isolates its nonzero coefficient. This proves the first necessity.

For the second, use x_p at a prime with unequal positive exponents. At other primes, both orders agree on their initial vector. Their highest terms and (13.3) again isolate a nonzero power of p.

In characteristic ℓ, the terms with positive powers of ℓ vanish, so $D_{\ell^a} = \psi_{\ell^a}$ on $P_{\mathbb{F}_\ell}$. At every other prime, the same powers of p remain nonzero. This proves precisely the stated exception. □

The index shifts interact with the divisor substitutions by

$$\psi_m D_n x_k = \sum_{d \mid (n,k)} d\, x_{mnk/d}, \qquad D_n \psi_m x_k = \sum_{d \mid (n,mk)} d\, x_{mnk/d}.$$

In characteristic zero they commute exactly when $(m, n) = 1$. Coprimality equates the divisor conditions. If a common prime exists, $k = 1$ supplies an extra term with a nonzero integer coefficient.

There is also a monoid action retaining the prime divisor maps. Define $\Theta_n = \prod_p D_p^{v_p(n)}$. Different prime maps commute, so $\Theta_m \Theta_n = \Theta_{mn}$.

For $a \geq 1$, its prime-chain values are

$$A_1^a e_r = \sum_{j=0}^{a} \binom{a}{j} p^j e_{r+a-j} \quad (r \geq 1), \qquad A_1^a e_0 = \sum_{j=0}^{a-1} \binom{a-1}{j} p^j e_{a-j}.$$

The first formula is the binomial theorem on the invariant tail. The second follows after $e_0 \mapsto e_1$. This action changes the higher operations. Its prime maps retain the Frobenius obstruction computed below.

13.3 The algebra of linear divisor operators

The prime-chain algebra admits a finite test for each proposed relation. Each test controls a finite boundary region and the entire infinite tail.

Let $U e_r = e_{r+1}$, $B e_0 = 0$ and $B e_r = e_{r-1}$ for $r \geq 1$. Then $BU = I$ and $U^i B^i$ projects onto the tail $r \geq i$. Consequently

$$A_a = U^a \sum_{i=0}^{a} p^i B^i, \qquad A_{a+1} = U A_a + p^{a+1} U^{a+1} B^{a+1}.$$

These identities place the generated algebra inside the algebra of the shift and its partial inverse. They need not generate that entire ambient algebra.

Put $q_a(z) = \sum_{i=0}^{a} p^i z^{a-i}$ for $a \geq 1$. For a word $w = Z_{a_l} \cdots Z_{a_1}$, define

$$d(w) = \sum_j a_j, \qquad q_w = \prod_j q_{a_j},$$

and

$$B_w(r, s) = \sum_{\substack{0 \leq i_j \leq a_j,\ \sum_j (a_j - i_j) = s \\ i_j \leq r + \sum_{h<j} (a_h - i_h) \text{ for each } j}} p^{\sum_j i_j}.$$

The rightmost factor acts first. For the empty word, set $d(w) = 0$, $q_w = 1$ and $B_w(r, s) = \mathbf{1}_{s=0}$.

THEOREM 13.3.1 *(Tail and boundary presentation).* Let $F = \sum_w c_w w$ be a finite element of $R\langle Z_1, Z_2, \ldots \rangle$, and set $d = \max_w d(w)$. For $F = 0$, take $d = 0$. Then $F(A_1, A_2, \ldots) = 0$ if and only if

$$\sum_w c_w q_w(z) = 0 \quad \text{in } R[z],$$
$$\sum_w c_w B_w(r, s) = 0 \quad (0 \leq r < d,\ 0 \leq s \leq d).$$

These conditions define the kernel ideal $I_{p,R}$, giving

$$R[A_1, A_2, \ldots] \simeq R\langle Z_1, Z_2, \ldots \rangle / I_{p,R}.$$

Every finite expression has a finite coefficient test. No finite generating set for the ideal is asserted.

Proof. Expanding the successive divisor choices gives $w(A)e_r = \sum_s B_w(r,s)e_{r+s}$. For $r \geq d$, every intermediate index is at least r. All choices $0 \leq i_j \leq a_j$ are then allowed. Their coefficients are exactly those of $q_w(z)$. The polynomial identity therefore tests all vectors with $r \geq d$.

The remaining equations test every earlier basis vector. Together they are precisely the kernel of evaluation $Z_a \mapsto A_a$. The kernel is a two-sided ideal, and the quotient is its image. □

Equivalently, every evaluated expression is uniquely $q(U) + E$. Here E is a finite matrix, supported in columns $r < d$ and rows $r \leq r + s \leq r + d$. Uniqueness follows by testing arbitrarily large e_r. The tail-symbol map onto $R[z]$ is surjective, since $q_1 = z + p$. Its kernel consists of the finite matrices which occur in the generated algebra. The commutator $[A_1, A_2]$ gives a nonzero example over $\mathbb{Z}$.

For finitely many primes, use a basis indexed by their valuation vectors. Word coefficients are products of the corresponding prime-chain coefficients. In each coordinate, separate the finitely many boundary values from the unrestricted tail. This gives finitely many mixed boundary tests and polynomial identities for every finite expression. Every expression in the full prime family involves only finitely many primes.

13.4 The linear Hecke law and its determinant

The partial inverse enters differently in (13.2). On M_R, define

$$U_p x_k = x_{pk}, \qquad B_p x_k = \begin{cases} x_{k/p}, & p \mid k, \\ 0, & p \nmid k. \end{cases}$$

Then $B_p U_p = I$, while $U_p B_p$ projects onto the p-divisible indices. At distinct primes all four pairs commute. The linear operators satisfy

$$\mathsf{H}_{p^r} = \sum_{j=0}^{r} p^j U_p^{r-j} B_p^j.$$

Multiplying by $U_p + pB_p$ gives

$$\mathsf{H}_p \mathsf{H}_{p^r} = \mathsf{H}_{p^{r+1}} + p\mathsf{H}_{p^{r-1}} \qquad (r \geq 1).$$

Indeed, the final backward-shift term supplies $p^{r+1}B_p^{r+1}$. Every other backward-shift term cancels a forward shift using $B_p U_p = I$, leaving $p\mathsf{H}_{p^{r-1}}$.

Proposition 13.4.1 *(Universal determinant identity).* For independent indeterminates T, d, define

$$H_0 = 1, \qquad H_1 = T, \qquad H_{r+1} = TH_r - dH_{r-1} \quad (r \geq 1).$$

Then in $\mathbb{Z}[T, d]$,

$$H_r H_s = \sum_{j=0}^{\min(r,s)} d^j H_{r+s-2j}.$$

The identity also holds after substituting any commuting pair of endomorphisms of a module over a commutative ring.

Proof. Substitute $T = u + v$ and $d = uv$. The recurrence gives $H_r = \sum_{i=0}^{r} u^i v^{r-i}$. For $r \leq s$, the coefficient of $u^i v^{r+s-i}$ in the product is

$$1 + \min(r, i, r + s - i) \qquad (0 \leq i \leq r + s).$$

The right side has one contribution for each $0 \le j \le \min(r, i, r+s-i)$, so the coefficients agree.

The substitution is injective: in lexicographic order with $u > v$, $(u+v)^a (uv)^b$ has leading monomial $u^{a+b}v^b$. These leading monomials are distinct for distinct pairs (a, b). The identity therefore holds before specialization, including $r = s$. Evaluation at commuting endomorphisms preserves it. □

COROLLARY 13.4.2. Over every commutative coefficient ring,

$$\mathsf{H}_m \mathsf{H}_n = \sum_{d \mid (m,n)} d\, \mathsf{H}_{mn/d^2}.$$

Proof. At each prime, specialize the proposition to (H_p, pI). The divisor formula factors H_n over its prime powers. Multiplication of the local identities proves the global formula. □

This is a representation of the stated linear operator algebra. The algebra generated by all U_p, B_p also retains these relations in its regular representation. Neither construction identifies its multiplication with the vertex products of a Heisenberg algebra.

13.5 SATAKE PARAMETERS AND THE RADIAL STATE

The determinant in the universal identity has a geometric normalization. Fix $K_p = \mathrm{GL}_2(\mathbb{Z}_p)$ and give it Haar volume one. The complex spherical Hecke algebra is

$$\mathcal{H}_p = C_c(K_p \backslash \mathrm{GL}_2(\mathbb{Q}_p) / K_p, \mathbb{C}),$$

The product is convolution. The two characteristic functions used below are

$$T_p = \mathbf{1}_{K_p \,\mathrm{diag}(p,1) K_p}, \qquad \mathsf{z}_p = \mathbf{1}_{K_p \,\mathrm{diag}(p,p) K_p}.$$

THEOREM 13.5.1 *(Normalized spherical Satake isomorphism).* The normalized Satake isomorphism identifies

$$\mathcal{H}_p \simeq \mathbb{C}[X^{\pm 1}, Y^{\pm 1}]^{S_2}, \qquad T_p \longmapsto \sqrt{p}(X+Y), \qquad \mathsf{z}_p \longmapsto XY.$$

At a character, put $\alpha = \sqrt{p}X$ and $\beta = \sqrt{p}Y$. Then $\alpha\beta = p\mathsf{z}_p$, using the scalar central character on the right. Trivial central character gives $\mathsf{z}_p = 1$. Temperedness is the additional condition $|X| = |Y| = 1$.

This is the convention of Gross, *On the Satake isomorphism*, Proposition 3.6, equations (3.13)–(3.14), and Section 8. In particular, trivial central character does not imply temperedness. The higher elements $H_{p^r} \in \mathcal{H}_p$ are specified by the complete symmetric Satake polynomials:

$$\sum_{r \ge 0} H_{p^r} t^r = (1 - T_p t + p\mathsf{z}_p t^2)^{-1}.$$

This follows by multiplying $(1 - \sqrt{p}Xt)$ and $(1 - \sqrt{p}Yt)$ in the Satake algebra and using injectivity. These higher elements need not be individual double-coset characteristic functions.

For a fixed pair $\alpha\beta = p$, the diagonal matrix $S_p = \mathrm{diag}(\alpha, \beta)$ gives

$$\mathrm{Tr}(\mathrm{Sym}^r S_p) = \sum_{j=0}^{r} \alpha^{r-j}\beta^j = H_r(\alpha + \beta, p).$$

This follows from the monomial eigenbasis of the symmetric power. The matrix, its scalar traces and the operators H_n are three separate representations of the parameter formulas. In a basis a, b, the line $\mathbb{C}(a+b)$ is S_p-invariant exactly when $\alpha = \beta$. Its image is $\alpha a + \beta b$, which proves the criterion.

If the pair is tempered, write

$$\{\alpha, \beta\} = \{\sqrt{p}\, e^{i\theta_p}, \sqrt{p}\, e^{-i\theta_p}\}.$$

The angle is defined modulo 2π and sign.

For any family of such angles, the product

$$L_\theta(s) = \prod_p \frac{1}{1 - 2\cos\theta_p\, p^{-s} + p^{-2s}}$$

converges absolutely and locally uniformly for $\operatorname{Re} s > 1$. Indeed, its logarithm is the absolutely convergent sum

$$\sum_p \sum_{m \geq 1} \frac{2\cos(m\theta_p)}{m} p^{-ms}.$$

On every closed smaller half-plane, its absolute values are bounded by a constant times $\sum_p p^{-\operatorname{Re} s}$.

This constructs the local-parameter Euler product on that domain. An automorphic identification or continuation beyond it requires additional arithmetic data. Replacing unitary parameters by α, β shifts the Dirichlet variable by one half.

The integral cyclic module of a tree

Fix a rooted $(p+1)$-regular tree $(\mathcal{T}_p, o)$. Use counting measure on its vertices and let S_r be the sphere of radius r about o. Its size is

$$s_0 = 1, \qquad s_r = (p+1)p^{r-1} \quad (r \geq 1).$$

The radial Hilbert space $\mathcal{R}_p$ consists of square-summable functions constant on each sphere. Its orthonormal basis is $e_r = s_r^{-1/2} \mathbf{1}_{S_r}$.

Adjacency is $(Tf)(v) = \sum_{w \sim v} f(w)$. Cauchy–Schwarz bounds its norm on the vertex Hilbert space by $p+1$, and symmetry makes it self-adjoint. Counting edges gives

$$\begin{aligned} Te_0 &= \sqrt{p+1}\, e_1, \\ Te_1 &= \sqrt{p+1}\, e_0 + \sqrt{p}\, e_2, \\ Te_r &= \sqrt{p}(e_{r-1} + e_{r+1}) \quad (r \geq 2). \end{aligned}$$

Distance defines a filtration, not a grading preserved by T.

Define monic integer polynomials by

$$R_0 = 1, \quad R_1 = x, \quad R_2 = x^2 - (p+1), \qquad R_{r+1} = xR_r - pR_{r-1} \quad (r \geq 2).$$

The recurrence gives $R_r(T)e_0 = \mathbf{1}_{S_r}$. Monicity then proves an integral cyclic-module isomorphism

$$\mathbb{Z}[x] \xrightarrow{\sim} \bigoplus_{r \geq 0} \mathbb{Z}\mathbf{1}_{S_r}, \qquad f \longmapsto f(T)e_0. \tag{13.4}$$

No prime is inverted in this statement. The normalized Hilbert basis instead uses real square roots.

THEOREM 13.5.2 *(Radial transform with its state).* Put $x(\theta) = 2\sqrt{p}\cos\theta$ for $0 \le \theta \le \pi$, and define

$$d\nu_p(\theta) = \frac{2p(p+1)\sin^2\theta}{\pi(p^2+1-2p\cos 2\theta)}\,d\theta.$$

The measure has mass one. The assignment

$$\mathcal{U}_p e_r = \frac{R_r(x(\theta))}{\sqrt{s_r}}$$

extends to a unitary $\mathcal{R}_p \xrightarrow{\sim} L^2([0,\pi],\nu_p)$. It sends e_0 to 1 and intertwines T with multiplication by x. In particular, $\|T|_{\mathcal{R}_p}\| = 2\sqrt{p}$, and

$$\langle e_0, f(T)e_0\rangle = \int_0^\pi f(2\sqrt{p}\cos\theta)\,d\nu_p(\theta)$$

for every bounded Borel function on the spectral interval.

Proof. The tridiagonal recurrence makes e_0 cyclic. After deleting its row and column, the remaining Jacobi operator has every off-diagonal coefficient equal to $\sqrt{p}$. Its first diagonal resolvent entry satisfies

$$g(z) = \frac{1}{z - pg(z)} = \frac{z - \sqrt{z^2-4p}}{2p}.$$

The branch is analytic off $[-2\sqrt{p}, 2\sqrt{p}]$ and asymptotic to z. It is selected by $g(z) \sim z^{-1}$.

Taking the Schur complement at the original root gives

$$m(z) = \langle e_0, (z-T)^{-1}e_0\rangle = \frac{1}{z-(p+1)g(z)}.$$

For $|x| < 2\sqrt{p}$, Stieltjes inversion yields

$$-\frac{1}{\pi}\operatorname{Im} m(x+i0) = \frac{(p+1)\sqrt{4p-x^2}}{2\pi((p+1)^2-x^2)}.$$

A pole outside the interval would require $g^2 = 1$ and $z = \pm(p+1)$. The chosen branch instead has $g(\pm(p+1)) = \pm p^{-1}$. At either endpoint the denominator is nonzero because $p > 1$. Thus m has continuous finite boundary values on the interval and analytic continuation across its complement.

The inversion formula gives only the displayed absolutely continuous measure. Its mass is one by $m(z) \sim z^{-1}$. Changing variables to $x = 2\sqrt{p}\cos\theta$ gives ν_p.

The spectral theorem and (13.4) imply

$$\int R_r(x)R_s(x)\,d\nu_p = \langle \mathbf{1}_{S_r}, \mathbf{1}_{S_s}\rangle = s_r\delta_{rs}.$$

The normalized polynomials are therefore orthonormal. They span all polynomials by monicity. Polynomial approximation on the compact interval proves their density in L^2, hence the asserted unitary.

The recurrence gives intertwining on finite sums, then everywhere by boundedness. The spectral support is the entire closed interval, so the operator norm is $2\sqrt{p}$. Bounded Borel functional calculus proves the state identity. □

For the Bruhat–Tits tree of $\mathrm{PGL}_2(\mathbb{Q}_p)$, vertices are homothety classes of lattices in $\mathbb{Q}_p^2$. Adjacency is represented by $pL \subset L' \subset L$ with $[L : L'] = p$. Lines in L/pL give $p+1$ neighbors. The maximal compact subgroup has volume one, so this adjacency is the Hecke operator of $\mathrm{diag}(p, 1)$. Pullback to GL_2 makes z_p act as the identity. The radial transform consequently preserves both the spherical Hecke action and its root state.

Proposition 13.5.3 *(Distance operators).* Define $\mathcal{D}_r$ by

$$(\mathcal{D}_r f)(v) = \sum_{d(v,w)=r} f(w).$$

Then $\mathcal{D}_r = R_r(T)$ and

$$H_k(T, p) = \sum_{j=0}^{\lfloor k/2 \rfloor} \mathcal{D}_{k-2j}.$$

Their root expectations satisfy

$$\langle e_0, H_k(T, p) e_0 \rangle = \begin{cases} 1, & k \text{ even}, \\ 0, & k \text{ odd}. \end{cases}$$

On $\mathcal{R}_p$, the generating series converges in operator norm for $|t| < p^{-1/2}$, and

$$\left\langle e_0, (I - tT + pt^2 I)^{-1} e_0 \right\rangle = \frac{1}{1 - t^2}.$$

Proof. Every row and column of $\mathcal{D}_r$ has s_r entries, so it is bounded. Counting the first step gives

$$T\mathcal{D}_1 = \mathcal{D}_2 + (p+1)I, \qquad T\mathcal{D}_r = \mathcal{D}_{r+1} + p\mathcal{D}_{r-1} \quad (r \geq 2).$$

Hence $\mathcal{D}_r = R_r(T)$. With $H_{-1} = H_{-2} = 0$, the recurrences give $R_r = H_r - H_{r-2}$. Telescoping proves the sum formula. Only $\mathcal{D}_0 e_0$ has a nonzero inner product with e_0, giving the expectations.

The spectral values are

$$H_k(2\sqrt{p}\cos\theta, p) = p^{k/2} \sum_{j=0}^{k} e^{i(k-2j)\theta}.$$

They include the coinciding-parameter cases $\theta = 0, \pi$. Thus $\|H_k(T, p)\| \leq (k+1)p^{k/2}$ on $\mathcal{R}_p$. This proves norm convergence in the stated disk. Multiplying the series by $I - tT + pt^2 I$ proves the inverse formula, and its state can be evaluated term by term. □

At $p = 2$, the first normalized radial polynomials are

$$1, \qquad x/\sqrt{3}, \qquad (x^2 - 3)/\sqrt{6},$$

whereas $H_2(x, 2) = x^2 - 2$. There are three two-step returns to the root. Of the four-step returns, nine return after two steps and six first reach distance two. Therefore

$$\int x^2 \, d\nu_2 = 3, \qquad \int x^4 \, d\nu_2 = 15, \qquad \int H_2 \, d\nu_2 = 1, \qquad \int R_2 \, d\nu_2 = 0.$$

The distinction persists before taking a spectral integral.

There is also a distinction between two cyclic Hecke representations. Give the free complex mode space the Hermitian form $\langle x_k, x_l \rangle = k\delta_{kl}$ and complete it. Then $U_p^* = pB_p$, and $\mathsf{H}_p = U_p + pB_p$ is bounded and self-adjoint. At its unit vector x_1,

$$\langle x_1, \mathsf{H}_p^2 x_1 \rangle = p, \qquad \langle e_0, T^2 e_0 \rangle = p + 1.$$

The first value follows from $\mathsf{H}_p^2 x_1 = x_{p^2} + p x_1$. Thus the common Hecke recursion does not give an intertwiner preserving these two specified states.

Finite sets of primes

For a finite prime set F, tensor the radial maps to obtain

$$\mathcal{U}_F : \widehat{\bigotimes}_{p \in F} \mathcal{R}_p \xrightarrow{\sim} L^2\left([0, \pi]^F, \prod_{p \in F} \nu_p \right).$$

It sends $\bigotimes_p e_{r_p}$ to the product of the normalized radial polynomials. It preserves each local Hecke operator and sends the tensor root state to integration against the product measure. The integral cyclic lattices are the finite tensor products of the sphere-sum lattices.

Adjoining a prime uses $v \mapsto v \otimes e_0$ on the tree side and $f \mapsto f \otimes 1$ on the spectral side. These isometries commute with $\mathcal{U}_F$ and preserve the states. Their algebraic union therefore has an isometric comparison, which extends to its Hilbert completion. Each finite-support Hecke operation acts on its own natural factors. No bound uniform over all primes is required or asserted.

13.6 The integral symmetric-function lattice

The index shifts already form a monoid action on P_R. Their integral Frobenius property depends on the lattice. For one species over $\mathbb{Z}$,

$$\psi_p x_1 - x_1^p = x_p - x_1^p \notin p P_{\mathbb{Z}}.$$

The two monomials remain distinct after reduction modulo p. We now construct the larger lattice on which the same rational operations satisfy the required congruences.

Inside $P_{\mathbb{Q}}$, define $h_0 = 1$ and

$$H(t) = \sum_{r \geq 0} h_r t^r = \exp\left(\sum_{k \geq 1} \frac{x_k t^k}{k} \right), \qquad \Lambda = \mathbb{Z}[h_1, h_2, \ldots].$$

The coefficients are defined formally, one degree at a time. Differentiation gives Newton's recursion

$$r h_r = \sum_{k=1}^{r} x_k h_{r-k}.$$

It expresses x_r as an integral polynomial in $h_1, \ldots, h_r$. Conversely, $h_r = x_r/r$ plus a polynomial in earlier x_k over $\mathbb{Q}$.

Thus the h_r are algebraically independent and

$$P_{\mathbb{Z}} \subsetneq \Lambda, \qquad P_{\mathbb{Q}} = \Lambda \otimes \mathbb{Q}.$$

For instance, $h_2 = (x_1^2 + x_2)/2$ does not belong to $P_{\mathbb{Z}}$. We call Λ the *Hall lattice*. For several species, take its finite tensor products.

To identify its standard operations, specialize $x_k = z_1^k + \cdots + z_N^k$. Then $H(t) = \prod_{i=1}^{N}(1 - z_i t)^{-1}$, so h_r is the complete symmetric polynomial of degree r. These symmetric polynomials stabilize when zero variables are adjoined. Every fixed weight is detected once N is sufficiently large. Substitution $z_i \mapsto z_i^n$ therefore preserves Λ and induces $\psi_n x_k = x_{nk}$.

In characteristic p, the same substitution is the Frobenius map on integer symmetric polynomials modulo p. Hence

$$\psi_p(f) \equiv f^p \pmod{p\Lambda} \qquad (f \in \Lambda). \tag{13.5}$$

Special lambda operations

The full integral structure includes operations beyond their rational power sums. We specify the convention explicitly. For two finite alphabets $X = (z_i)$ and $Y = (w_j)$, let $e_n(X)$ denote the elementary symmetric polynomial. There are universal integer polynomials P_n and $Q_{m,n}$ determined by

$$e_n(\{z_i w_j\}_{i,j}) = P_n(e_1(X), \dots, e_n(X); e_1(Y), \dots, e_n(Y)),$$

and

$$e_m(\{z_{i_1} \cdots z_{i_n} : i_1 < \cdots < i_n\}) = Q_{m,n}(e_1(X), \dots, e_{mn}(X)).$$

The fundamental theorem of symmetric polynomials gives their integral coefficients and stability as the alphabets grow.

Definition 13.6.1. A *special λ-ring* is a commutative unital ring A with operations $\lambda^n : A \to A$ for $n \geq 0$. Writing $\lambda_t(a) = \sum_{n \geq 0} \lambda^n(a) t^n$, their identities are

$$\lambda^0(a) = 1, \qquad \lambda^1(a) = a, \qquad \lambda_t(a + b) = \lambda_t(a)\lambda_t(b), \qquad \lambda_t(1) = 1 + t,$$

together with the product and composition identities given by P_n and $Q_{m,n}$ above. Its *Adams operations* are the integral coefficient operations in

$$-t\frac{(\lambda_{-t}(a))'}{\lambda_{-t}(a)} = \sum_{n \geq 1} \psi_n(a) t^n.$$

Here the prime means formal differentiation in t.

The alphabet identities show that each Adams operation is a unital ring endomorphism and $\psi_m \psi_n = \psi_{mn}$. They also give the Frobenius congruences. These are identities of universal integer polynomials, so they remain valid after evaluation in every special λ-ring. The ordinary symmetric functions give such a structure on Λ. Their Adams operations are precisely the index shifts above.

Theorem 13.6.2 *(Torsion-free Adams criterion).* Let the additive group of A be torsion-free. A family of unital ring endomorphisms ψ_n determines a unique special λ-structure if

$$\psi_1 = I, \qquad \psi_m \psi_n = \psi_{mn}, \qquad \psi_p(a) \equiv a^p \pmod{pA} \quad \text{for every prime } p.$$

This is Wilkerson's criterion, as stated in Borger, *The basic geometry of Witt vectors, I*, Section 1.17. Over a $\mathbb{Q}$-algebra its reconstruction takes the form

$$\lambda_t(a) = \exp\left(\sum_{n \geq 1} \frac{(-1)^{n-1} \psi_n(a) t^n}{n}\right).$$

Over $\mathbb{Z}$, the shifts therefore give the special structure on Λ, but fail the necessary congruence on $P_{\mathbb{Z}}$. This failure concerns the specified shifts. For example, the different maps $x_k \mapsto x_k^n$ give commuting Frobenius lifts on $P_{\mathbb{Z}}$. They define the structure in which each variable is a line element, meaning $\lambda_t(x_k) = 1 + x_k t$.

Example 13.6.3 *(The torsion hypothesis).* On $\mathbb{F}_p$, let every ψ_n be the identity. These ring maps commute and satisfy all the displayed Frobenius congruences. At p, this uses $a^p = a$. At other primes q, the ideal $q\mathbb{F}_p$ is the whole ring.

They cannot come from a special λ-structure. Such a structure would give

$$1 = \lambda_t(0) = \lambda_t(p \cdot 1) = \lambda_t(1)^p = (1+t)^p = 1 + t^p,$$

which is false in $\mathbb{F}_p[[t]]$. Thus commuting Frobenius lifts alone do not replace the full integral operations on a ring with additive torsion.

PROPOSITION 13.6.4 *(Divisor substitutions on the Hall lattice).* Every D_n preserves Λ, and

$$D_n H(t) = \prod_{d|n} \psi_n H(t^d). \tag{13.6}$$

Nevertheless D_p is not a Frobenius lift on this lattice:

$$D_p h_p = \psi_p h_p + \psi_p h_1 \equiv h_p^p + h_1^p \not\equiv h_p^p \pmod{p\Lambda}.$$

Proof. In $P_{\mathbb{Q}}[[t]]$, set $k = dj$ in the logarithm to obtain

$$\sum_{k\geq 1} \frac{D_n x_k}{k} t^k = \sum_{d|n} \sum_{j\geq 1} \frac{x_{nj}}{j} t^{dj}.$$

Exponentiation gives (13.6). Every factor has integral Hall coefficients, proving preservation of Λ. For $n = p$, only the two stated contributions have degree p. Apply (13.5) to them. The term h_1^p is nonzero in the polynomial ring $\Lambda/p\Lambda$. □

In particular, the equality $D_{p^a} = \psi_{p^a}$ on $P_{\mathbb{F}_p}$ does not imply equality after reducing Λ. The inclusion $P_{\mathbb{Z}} \subset \Lambda$ is not an identification of their integral coordinates or reductions.

There is a useful comparison which does preserve the algebraic images. Define a triangular substitution on P_R by

$$C_n x_{s,k} = \sum_{d|(n,k)} d\, x_{s,k/d}.$$

Its leading term is $x_{s,k}$, and every other mode index is smaller. Induction therefore constructs an integral polynomial inverse on every generator, over every coefficient ring.

The same logarithmic calculation gives

$$C_n H(t) = \prod_{d|n} H(t^d).$$

Its coefficient of t^r is h_r plus an integral polynomial in $h_1, \ldots, h_{r-1}$. Thus C_n is also an automorphism of the Hall lattice.

Direct evaluation on the generators now proves

$$D_n = \psi_n C_n, \qquad D_n(P_R) = \psi_n(P_R), \qquad D_n(\Lambda) = \psi_n(\Lambda). \tag{13.7}$$

Equal images do not identify the two actions or their composition laws. Nor does the triangular inverse have to survive a Hilbert completion.

13.7 Big Witt coordinates and the cofree construction

The previous exponential formulas live first over $\mathbb{Q}$. Big Witt coordinates give integral formulas before any scalar extension. For a commutative ring R, put $W(R) = \prod_{r\geq 1} R$ as a set. Associate to $a = (a_r)$ the series

$$A_a(t) = \prod_{r\geq 1}(1 - a_r t^r)^{-1} \in 1 + tR[[t]]. \tag{13.8}$$

Each coefficient involves finitely many factors. Its coefficient of t^r is a_r plus a polynomial in earlier coordinates. Induction therefore gives a natural bijection $W(R) \simeq 1 + tR[[t]]$ over every ring.

The *ghost components* are

$$w_n(a) = \sum_{r|n} r\, a_r^{n/r}, \qquad t\frac{A_a'(t)}{A_a(t)} = \sum_{n\geq 1} w_n(a) t^n.$$

The derivative identity follows by differentiating the factors. It involves no division by an integer in R.

Proposition 13.7.1. If the additive group of R is torsion-free, the ghost map is injective. Over $\mathbb{F}_p$, the nonzero vector with $a_p = 1$ and all other coordinates zero has every ghost equal to zero. Its series is $(1 - t^p)^{-1} \neq 1$.

Proof. If two vectors agree below n, equality of their nth ghosts gives $n(a_n - b_n) = 0$. Torsion-freeness gives equality at n, and induction proves injectivity. For the stated vector over $\mathbb{F}_p$, the only possible ghost contribution is p when $p \mid n$. It vanishes, although the coefficient of t^p in the series is one. □

We construct the ring operations on the series themselves. Write $E_r(a) = (1 - at^r)^{-1}$. Witt addition is ordinary multiplication of series: $A +_W B = AB$. Its zero is the series 1, and its negative is A^{-1}.

Witt multiplication, denoted $\odot$, is determined by

$$E_r(a) \odot E_s(b) = \left(1 - a^{s/g} b^{r/g} t^l\right)^{-g}, \qquad g = (r,s), \quad l = \operatorname{lcm}(r,s), \tag{13.9}$$

and distributivity over Witt addition. For two arbitrary coordinate products, multiply these factors over all pairs (r,s). Since $l \geq r, s$, each coefficient uses finitely many pairs. The construction is therefore a universal integral polynomial operation.

Theorem 13.7.2 *(The big Witt ring).* These operations make $W(R)$ a natural commutative unital ring. Its multiplicative unit is $E_1(1)$. Every $w_n : W(R) \to R$ is a ring homomorphism.

Proof. Logarithmic differentiation makes each ghost additive under series multiplication. For one factor,

$$w_n(E_r(a)) = \begin{cases} r a^{n/r}, & r \mid n, \\ 0, & r \nmid n. \end{cases}$$

The ghost of (13.9) is $g l\, a^{n/r} b^{n/s}$ when $l \mid n$, and zero otherwise. Since $gl = rs$, this is the product of the two ghosts. Distributivity then proves multiplicativity for arbitrary series.

It remains to verify the ring identities without assuming injectivity on the given coefficient ring. Fix a coefficient degree and regard the finitely many input coordinates as independent variables over $\mathbb{Z}$. This polynomial ring is torsion-free, so its ghosts are injective. Commutativity, associativity, distributivity and the unit law hold on ghosts, hence on these universal coordinate polynomials. Specializing their integer coefficients proves the identities over every R. □

The Teichmüller element is $[a] = E_1(a)$. It satisfies $[a] \odot [b] = [ab]$, but need not preserve addition in R. For $m \geq 1$, define Verschiebung by $V_m A(t) = A(t^m)$. It is additive for $+_W$. Define Frobenius by its values on the coordinate factors:

$$F_m E_r(a) = \left(1 - a^{m/g} t^{r/g}\right)^{-g}, \qquad g = (m, r), \tag{13.10}$$

and extend by products of series. A coefficient through degree N uses only $r \leq mN$, so this also defines an integral operation coefficientwise.

PROPOSITION 13.7.3 *(Frobenius and Verschiebung).* Each F_m is a natural unital ring endomorphism of $W(R)$, and

$$w_n(F_m a) = w_{mn}(a), \qquad w_n(V_m a) = \begin{cases} m\, w_{n/m}(a), & m \mid n, \\ 0, & m \nmid n. \end{cases}$$

Their identities over every coefficient ring include

$$F_m F_n = F_{mn}, \qquad V_m V_n = V_{mn}, \qquad F_m V_m = mI,$$
$$F_m V_n = g V_{n/g} F_{m/g}, \qquad g = (m, n).$$

The integer multiple means repeated Witt addition. In particular, $F_m[a] = [a^m]$.

Proof. For a single factor, (13.10) has nth ghost $r a^{mn/r}$ precisely when $r \mid mn$. This proves the Frobenius formula. Substituting t^m in the logarithmic derivative proves the Verschiebung formula. The ring-map property and all displayed identities follow on ghosts. Each operation was already constructed by universal integer polynomials. The torsion-free universal-ring argument from the theorem therefore proves the identities before specialization to arbitrary R. □

Plethysm and the full comonad

The ring $\Lambda = \mathbb{Z}[h_1, h_2, \ldots]$ represents the series functor:

$$W(R) \simeq \mathrm{Hom}_{\mathrm{Ring}}(\Lambda, R), \qquad f \longmapsto \sum_{n \geq 0} f(h_n) t^n.$$

The values $f(h_n)$ are series coefficients, not the Witt coordinates a_n. They are related by the triangular integral bijection (13.8). The power sum $x_n \in \Lambda$ evaluates to w_n.

We need the integral composition operation on Λ. For $h \in \Lambda$, write its specialization to a finite alphabet as an integer sum of monomials, $h(X) = \sum_M c_M M$. Define

$$\sum_{n \geq 0} (h_n \circ h)(X) t^n = \prod_M (1 - Mt)^{-c_M}.$$

Negative multiplicities are allowed, using the inverse formal series. The coefficients are integral symmetric polynomials and stabilize when zero variables are added.

Extend $g \mapsto g \circ h$ as a ring homomorphism from Λ. This operation is *plethysm*. Its unit is $h_1 = x_1$.

Plethysm is associative. One way to check this precisely is to embed Λ in $\mathbb{Q}[x_1, x_2, \ldots]$. The defining formula gives $x_n \circ h = \psi_n(h)$, where $\psi_n x_k = x_{nk}$ and rational coefficients are fixed. These substitutions commute and satisfy $\psi_n(g \circ h) = (\psi_n g) \circ h$. Evaluating a polynomial in the x_n therefore proves $(f \circ g) \circ h = f \circ (g \circ h)$. Since the operation was constructed integrally, the equality lies in Λ.

Theorem 13.7.4 *(Cofree special lambda rings).* The functor W is right adjoint to the forgetful functor from special λ-rings to commutative rings. Its counit and comultiplication are

$$\varepsilon_R(f) = f(h_1), \qquad (\Delta_R f)(g)(h) = f(h \circ g). \tag{13.11}$$

Thus $\Delta_R : W(R) \to W(W(R))$. It satisfies both counit identities and $\Delta_{W(R)}\Delta_R = W(\Delta_R)\Delta_R$. Its outer ghosts are $w_n(\Delta_R f) = F_n f$ as elements of $W(R)$.

Proof. For $f \in W(R)$ and $g \in \Lambda$, define the operation $g(f) \in W(R)$ by $g(f)(h) = f(h \circ g)$. For fixed g, the right side is a ring homomorphism in h, by the definition of plethysm. The universal symmetric-polynomial identities for sums and products show that $g \mapsto g(f)$ is a ring homomorphism into $W(R)$. These identities are exactly the series addition and multiplication constructed above. Associativity of plethysm supplies composition of the operations. Taking the elementary symmetric operations therefore makes $W(R)$ a special λ-ring.

Let A be a special λ-ring and $\phi : A \to R$ a ring map. For $a \in A$, evaluation of symmetric operations gives the map

$$\widehat{\phi}(a)(g) = \phi(g(a)) \qquad (g \in \Lambda).$$

The sum and product identities make $\widehat{\phi} : A \to W(R)$ a ring map. The composition identities make it a λ-ring map. Its counit is ϕ, since $h_1(a) = a$.

Conversely, the counit determines a λ-ring map into $W(R)$. Applying it after the operation g recovers evaluation at g. This proves the natural adjunction.

Applying this construction to the identity on the underlying ring of $W(R)$ gives the stated Δ_R. The equalities $h_1 \circ g = g$ and $h \circ h_1 = h$ prove the two counit identities separately. Evaluating the two coassociativity maps on three elements of Λ reduces them to the two associations of the same plethysm. Finally, $h \circ x_n = \psi_n(h)$. Evaluation of the outer power sum x_n in (13.11) therefore gives precisely the coefficient action F_n. □

Equivalently, a special λ-structure on a ring A is a ring map $\eta : A \to W(A)$ satisfying

$$\varepsilon_A \eta = I, \qquad \Delta_A \eta = W(\eta)\eta.$$

Indeed, its coordinates define the complete symmetric operations. The fact that η is a ring map gives their sum and product identities. The second displayed equality gives composition, and the first fixes the degree-one operation. Inverting the complete symmetric series at $-t$ recovers the elementary operations of Definition 13.6.1. The converse follows by evaluation of those operations.

The two universal constructions have different variances:

$$\mathrm{Hom}_{\lambda\text{-Ring}}(\Lambda, A) \simeq A, \qquad \mathrm{Hom}_{\lambda\text{-Ring}}(A, W(R)) \simeq \mathrm{Hom}_{\mathrm{Ring}}(A, R).$$

The first says that Λ is free on one distinguished element. The second is the cofree property of W. On rings with additive torsion, the full map η and its composition identity retain data which its ghost operations can lose. Borger's descent interpretation uses this integral structure and its forgetful functor. It does not create an arithmetic comparison from a rational index-shift formula.

13.8 PRODUCTS, GRADINGS, AND ALGEBRAIC COMPLETIONS

The polynomial ring has a primitive Hopf structure over R:

$$\Delta x_{s,k} = x_{s,k} \otimes 1 + 1 \otimes x_{s,k}, \qquad \epsilon(x_{s,k}) = 0, \qquad \mathcal{S}(x_{s,k}) = -x_{s,k}.$$

Here $\mathcal{S}$ denotes the antipode. Each of D_n, ψ_n and C_n preserves this structure. Their generator images are primitive, and the identities of algebra maps can be checked on generators. They also preserve the unit, augmentation and particle degree.

On the Hall lattice, the same coproduct is

$$\Delta h_r = \sum_{i+j=r} h_i \otimes h_j.$$

Preservation follows from the rational power-sum identity and the torsion-free integral inclusion. This is the ordinary symmetric-function Hopf structure. An arithmetic extension correspondence would require its own comparison.

PROPOSITION 13.8.1 *(Weights and completions).* For a nonempty species set and any nonzero coefficient ring, D_n and ψ_n are injective on P_R. For $n > 1$, neither is surjective. The index shift sends weight w to weight nw. The divisor substitution sends weight w into weights between w and nw. Both act on the finite-weight quotients and continuously on

$$\widehat{P}_R^{\mathrm{wt}} = \prod_{w \geq 0} P_{R,w}.$$

They also extend to the coefficient-adic completion $\varprojlim_j P_{\mathbb{Z}}/\ell^j P_{\mathbb{Z}}$ for every prime ℓ.

Proof. Let f have highest nonzero weight w. The highest-weight term of $D_n f$ is ψ_n applied to the weight-w component, and it has weight nw. Index substitution is injective on the monomial basis, so this component cannot vanish over a nonzero coefficient ring. This proves injectivity of both maps. For $n > 1$, the same highest-weight argument excludes a preimage of x_1.

Every term of $D_n x_k$ has weight between k and nk. Multiplication proves the weight bounds on monomials. Thus the ideal of weights greater than W is preserved. Its quotient has finite rank for a finite species set. An output through weight W depends only on inputs through weight W, which proves continuity on the weight completion.

Finally, the integral maps preserve every ideal $\ell^j P_{\mathbb{Z}}$. Their compatible quotient maps give the coefficient-adic extensions. □

A finite subalgebra generated by modes $k \leq N$ is not invariant when $n > 1$: the highest term of $D_n x_N$ is x_{nN}. The finite-weight quotient is a different construction. The coefficient-adic completion is also different from the weight product. In monomial coordinates, its coefficients lie in $\mathbb{Z}_\ell$, with only finitely many nonzero coefficients modulo each ℓ^j. The unrestricted weight product imposes no such decay.

Other symmetric-function operations give different preservation laws. For one species over $\mathbb{Q}$, the internal coproduct is the algebra map $\delta x_k = x_k \otimes x_k$. It is preserved by ψ_n. For D_n with $n > 1$, the image of x_n contains both x_{n^2} and $n x_n$. The tensor square of that image has cross terms absent from $\delta(D_n x_n)$, so D_n does not preserve δ.

For a partition λ with multiplicities m_k, write $x_\lambda = \prod_k x_k^{m_k}$ and $z_\lambda = \prod_k k^{m_k} m_k!$. The internal product is the bilinear operation

$$x_\lambda * x_\mu = \delta_{\lambda\mu} z_\lambda x_\lambda.$$

Neither family preserves it for $n > 1$. Indeed, $x_1 * x_1 = x_1$, whereas $x_n * x_n = n x_n$. These nonvanishing obstructions use characteristic zero. They do not transfer unchanged to a torsion coefficient ring.

13.9 Oscillator intertwiners and vertex products

For one species, define polynomial operators

$$a_{-k} f = x_k f, \qquad a_k f = k \partial_{x_k} f \quad (k \geq 1), \qquad a_0 = 0.$$

They obey $[a_k, a_{-l}] = k \delta_{kl}$, with central element acting as one. Different species commute. The Hall pairing is $\langle x_\lambda, x_\mu \rangle = \delta_{\lambda\mu} z_\lambda$. Over $\mathbb{C}$, its Hermitian version is positive and makes a_k the adjoint of a_{-k} on polynomials.

The substitutions intertwine creation by x_k with creation by its image. Their annihilation intertwiners require a separate calculation. For the index shifts, differentiation gives

$$\psi_n a_{-k} = a_{-nk} \psi_n, \qquad a_{nk} \psi_n = n \psi_n a_k, \qquad a_j \psi_n = 0 \quad (n \nmid j). \tag{13.12}$$

Sending both positive and negative modes to their index multiples would multiply the commutator by n. Over $\mathbb{Q}$, the assignments

$$a_{-k} \longmapsto a_{-nk}, \qquad a_k \longmapsto a_{nk}/n$$

instead define a central-charge-one Heisenberg embedding. It is intertwined by ψ_n, but does not preserve the Hall adjoint.

Proposition 13.9.1 *(An integral annihilation lift).* Fix $n \geq 1$. Define integers $c_{k,i}$ in increasing order of i by

$$c_{k,i} = 0 \quad (n \nmid i), \qquad c_{k,nj} = k \delta_{kj} - \sum_{\substack{d \mid (n,j) \\ d > 1}} d \, c_{k,nj/d}.$$

The polynomial derivations

$$\mathcal{A}_k^{(n)} = \sum_{i \geq 1} c_{k,i} \partial_{x_i}$$

are well-defined and commute. They satisfy

$$\mathcal{A}_k^{(n)} (D_n x_j) = k \delta_{kj}, \qquad \mathcal{A}_k^{(n)} D_n = D_n a_k.$$

Together with multiplication by $D_n x_j$, they give a central-charge-one oscillator representation.

Proof. Every term on the right of the recursion has smaller index. It therefore defines all coefficients integrally. Each polynomial uses finitely many variables, so only finitely many derivatives contribute to its image. Constant-coefficient derivations commute.

Substituting (13.1) into the first identity gives exactly the recursion. The chain rule proves the second identity on all polynomials. Commuting a derivation past multiplication by $D_n x_j$ gives its constant derivative $k \delta_{kj}$, proving the oscillator relations. □

Over $\mathbb{Q}$, the lift can be written $\mathcal{A}_k^{(n)} = \sum_i (c_{k,i}/i) a_i$. Integrality concerns the polynomial derivation, not these coefficients in the fixed annihilation modes. The lift is generally not adjoint-preserving. Both unscaled substitutions send x_1 to x_n, whose squared norm is n. Creation by x_n and its Hall adjoint therefore have commutator n, rather than one. This obstructs the specified adjoint-preserving assignment when $n > 1$.

PROPOSITION 13.9.2 *(The vertex obstruction).* Over $\mathbb{Q}$ or $\mathbb{C}$, neither D_n nor ψ_n is an endomorphism of the usual Heisenberg vertex algebra for $n > 1$. Both remain polynomial algebra and primitive Hopf endomorphisms.

Proof. The standard translation operator satisfies $\partial x_k = k x_{k+1}$. Every vertex homomorphism preserves it, since $\partial v = v_{(-2)}1$. Here

$$\partial \psi_n x_1 = n x_{n+1}, \qquad \psi_n \partial x_1 = x_{2n},$$

and

$$\partial D_n x_1 = n x_{n+1}, \qquad D_n \partial x_1 = x_{2n} + 2\mathbf{1}_{2|n} x_n.$$

The values differ for every $n > 1$. This argument does not require preservation of a chosen conformal vector.

The same failure is visible in the singular products. Writing $a(z) = Y(x_1, z)$, one has

$$Y(x_n, z) = \frac{\partial_z^{n-1} a(z)}{(n-1)!}, \qquad Y(x_n, z) Y(x_n, w) \sim \frac{(-1)^{n-1}(2n-1)!}{((n-1)!)^2 (z-w)^{2n}}.$$

Differentiate $a(z)a(w) \sim (z-w)^{-2}$ to obtain this coefficient. For $n > 1$, the higher pole cannot be the image of the original central second-order pole under a vacuum-preserving vertex map. The polynomial and Hopf assertions were proved above. □

For a finite species set, the index shifts nevertheless give a vacuum-preserving action of $\mathbb{N}^\times$ on the underlying vector space. This means endomorphisms with $\psi_1 = I$ and $\psi_m \psi_n = \psi_{mn}$. It does not assert an action by vertex endomorphisms. The divisor family fails even that monoid law in characteristic zero.

13.10 HILBERT CLOSURES AND A WEIGHTED THRESHOLD

Fix a nonempty finite species set and complete the complex polynomial space in the Hermitian Hall form. It is the symmetric Hilbert Fock space $\mathcal{F} = \bigoplus_{j \geq 0} \mathcal{F}^{(j)}$, with squared monomial norms $\prod_{s,k} k^{m_{s,k}} m_{s,k}!$. Its one-particle space has $\|x_{s,k}\|^2 = k$. The map $\psi_n/\sqrt{n}$ is an isometry there.

Each divisor summand $x_k \mapsto d\mathbf{1}_{d|k} x_{nk/d}$ has norm $\sqrt{nd}$. Its nonzero basis images are orthogonal with that constant norm ratio. Consequently, for $d_n = D_n|_{M_\mathbb{C}}$,

$$\|d_n\| \leq \sum_{d|n} \sqrt{nd}.$$

Both maps are bounded on one particle and on every fixed particle sector after taking symmetric tensor powers.

PROPOSITION 13.10.1 *(Closed full-Fock operators).* For $A = d_n$ or $A = \psi_n|_{M_\mathbb{C}}$, the polynomial operator $\mathrm{Sym}(A)$ is closable. Its closure is $\Gamma(A) = \bigoplus_{j \geq 0} \mathrm{Sym}^j(A)$, with domain

$$\mathrm{Dom}\,\Gamma(A) = \left\{ (v_j) \in \mathcal{F} : \sum_{j \geq 0} \|\mathrm{Sym}^j(A) v_j\|^2 < \infty \right\}.$$

For $n > 1$, both closures are unbounded. For the index shift, the domain condition is exactly $\sum_j n^j \|v_j\|^2 < \infty$.

Proof. A direct sum of bounded sector maps on the displayed domain is closed. Finite particle truncation approximates a vector and its image. Within each retained sector, finite-mode polynomials approximate both, since the sector map is bounded. Thus polynomials are a core and the direct sum is precisely their closure.

For the index shift, each sector map multiplies norms by $n^{j/2}$. Both substitutions send x_1^j to x_n^j, giving

$$\frac{\|D_n x_1^j\|}{\|x_1^j\|} = \frac{\|\psi_n x_1^j\|}{\|x_1^j\|} = n^{j/2}.$$

This proves unboundedness for $n > 1$ and the stated domain formula. □

The normalized substitution $\widetilde{\psi}_n x_{s,k} = n^{-1/2} x_{s,nk}$ gives a different positive result. It is an isometry on every particle sector and their Hilbert direct sum. It is also a complex polynomial Hopf map and a monoid action. Sending both a_{-k} and a_k to their index multiples divided by $\sqrt{n}$ preserves the Heisenberg relations and Hall adjoints. Equation (13.12), with this normalization, shows that $\widetilde{\psi}_n$ intertwines the resulting embedding. The normalization changes the integral structure and still fails ordinary translation for $n > 1$.

The kernel created by completion

The triangular automorphism C_n in (13.7) has a bounded one-particle extension. For $B_d x_k = \mathbf{1}_{d|k} x_{k/d}$, one has

$$c_n = C_n|_M = \sum_{d|n} dB_d, \qquad \|c_n\| \le \sum_{d|n} \sqrt{d}, \qquad c_n = \prod_{p|n} c_{p^{v_p(n)}}.$$

The norm estimate uses $\|B_d\| = d^{-1/2}$. The factorization follows from $B_d B_e = B_{de}$.

Theorem 13.10.2 *(One-particle kernel and range).* The operator c_n is a bounded surjection on the completed one-particle space. The divisor operator has closed range

$$\operatorname{Ran} d_n = \overline{\operatorname{span}}\{x_{nk} : k \ge 1\} = \operatorname{Ran}(\psi_n|_M), \qquad \ker d_n = \ker c_n.$$

For $n > 1$, this kernel is infinite-dimensional on the full one-species space. On a prime chain $e_r = x_{up^r}$ and for $a \ge 1$,

$$c_{p^a} = \sum_{j=0}^{a} p^j B^j, \qquad \ker c_{p^a} = \operatorname{span}_{\mathbb{C}} \left\{ \sum_{r \ge 0} (\zeta/p)^r e_r : \zeta^{a+1} = 1,\ \zeta \ne 1 \right\}.$$

The prime-chain kernel has dimension a. For $n = 1$, the operators are the identity and the kernels vanish.

Proof. On the chain, $\|e_r\|^2 = up^r$ and $\|B\| = p^{-1/2}$. For each stated root, the geometric vector has squared norm $u \sum_{r \ge 0} p^{-r} = u/(1 - p^{-1})$. It is a B-eigenvector with eigenvalue ζ/p, so c_{p^a} kills it. Conversely, a kernel vector $\sum_r b_r e_r$ satisfies

$$\sum_{j=0}^{a} p^j b_{r+j} = 0 \qquad (r \ge 0).$$

Its first a coefficients determine all the others. The distinct roots ζ/p give independent solutions by the Vandermonde determinant, exhausting this solution space.

Let $Ue_r = e_{r+1}$, so $BU = I$ and $\|U\| = \sqrt{p}$. For $\lambda = \zeta/p$, the operator

$$R_\lambda = U(I - \lambda U)^{-1}$$

is a bounded right inverse of $B - \lambda$. Indeed, $(B - \lambda)U = I - \lambda U$ and $\|\lambda U\| = p^{-1/2} < 1$. The inverse is a norm-convergent Neumann series.

Factor

$$c_{p^a} = p^a \prod_{\substack{\zeta^{a+1}=1 \\ \zeta \neq 1}} (B - \zeta/p).$$

Multiplying the right inverses in reverse order, with factor p^{-a}, gives a bounded right inverse of c_{p^a}. Its norm bound is independent of u. It therefore extends over the orthogonal sum of all prime chains. The finite product over prime factors proves surjectivity of c_n.

The factorization $d_n = \psi_n c_n$ and the scaled isometry $\psi_n|_M$ prove the range and kernel assertions. For $n > 1$, choose one nontrivial prime-power factor. Its kernel vectors on infinitely many disjoint prime chains are independent. The other factors commute with it, so all these vectors lie in $\ker c_n$. □

The polynomial inverse has a different domain. For example,

$$C_p^{-1} e_r = \sum_{j=0}^{r} (-p)^j e_{r-j}, \qquad \frac{\|C_p^{-1} e_r\|^2}{\|e_r\|^2} = \sum_{j=0}^{r} p^j.$$

Cancellation in $(I + pB)\sum_{j=0}^{r}(-pB)^j e_r = e_r$ proves the first formula. Orthogonality proves the second.

More generally, the polynomial inverse of C_n is not closable on this Hilbert space when $n > 1$. Choose $0 \neq v \in \ker c_n$ and finite truncations $v_N \to v$. Then $y_N = c_n v_N \to 0$, while $C_n^{-1} y_N = v_N \to v \neq 0$. The graph closure contains $(0, v)$ and cannot be an operator graph. The same argument applies to the algebraic inverse of d_n on its polynomial image.

These one-particle kernels also lie in $\ker \Gamma(d_n)$. By contrast, $\Gamma(\psi_n|_M)$ is injective, since each particle sector is a nonzero scalar multiple of an isometry. The polynomial maps remain injective throughout. No extension of polynomial multiplication to all Hilbert vectors is asserted.

The exact weighted boundary

Retain a prime chain $e_r = x_{up^r}$ and its finite linear span D. For $s \geq 0$, define

$$\mathcal{H}_s = \left\{ \sum_{r \geq 0} b_r e_r : \sum_{r \geq 0} (up^r)^{1+2s} |b_r|^2 < \infty \right\}.$$

If $Ee_r = up^r e_r$, this is $\mathrm{Dom}(E^s)$ in $\mathcal{H}_0$, with norm $\|E^s v\|_0$. Since $E \geq 1$, this norm is equivalent to its graph norm. Write $\mathcal{H}_{s,\geq a}$ for the closed tail spanned by e_r, $r \geq a$. Its codimension is a.

THEOREM 13.10.3 *(Weighted inverses).* Fix $a \geq 1$. The same polynomial maps c_{p^a} and d_{p^a} extend boundedly to every $\mathcal{H}_s$, with the following properties.

(i) For $0 \le s < 1/2$, c_{p^a} is onto $\mathcal{H}_s$, and d_{p^a} is onto $\mathcal{H}_{s,\ge a}$. Their common kernel has basis

$$g_\zeta = \sum_{r\ge 0} (\zeta/p)^r e_r, \qquad \zeta^{a+1} = 1, \quad \zeta \ne 1.$$

Their polynomial inverses are not closable.

(ii) For $s = 1/2$, both maps are injective. Their ranges are dense, proper and nonclosed in $\mathcal{H}_s$ and $\mathcal{H}_{s,\ge a}$, respectively. Their inverses on these actual ranges are closed and unbounded. They are the graph closures of the polynomial inverses.

(iii) For $s > 1/2$, c_{p^a} is boundedly invertible on $\mathcal{H}_s$, and d_{p^a} is a bounded isomorphism onto $\mathcal{H}_{s,\ge a}$. These inverses extend the polynomial inverses.

For $a = 0$, both maps are the identity for every s.

Proof. Set $b = p^{s+1/2}$, $q = p^{1/2-s}$ and $Q_a(z) = 1 + z + \cdots + z^a$. The vectors $f_r = e_r/(u^{s+1/2}b^r)$ form an orthonormal basis. Let $Vf_r = f_{r+1}$ and $L = V^*$. The raw shifts become $U = bV$ and $pB = qL$, so

$$c_{p^a} = Q_a(qL), \qquad d_{p^a} = b^a V^a Q_a(qL). \tag{13.13}$$

This proves boundedness with bounds $\sum_{j=0}^a q^j$ and $b^a \sum_{j=0}^a q^j$.

The isometry V^a identifies the space with its tail. It therefore suffices to analyze $Q_a(qL)$.

For $q > 1$, its kernel recurrence has distinct characteristic roots ζ/q, with $\zeta^{a+1} = 1$ and $\zeta \ne 1$. All resulting geometric sequences are square-summable. In raw coordinates, they are the stated g_ζ, with

$$\|g_\zeta\|_s^2 = \frac{u^{1+2s}}{1 - p^{2s-1}}.$$

The recurrence and Vandermonde argument prove that they form a basis.

Each $L - \zeta/q$ has bounded right inverse $V(I - (\zeta/q)V)^{-1}$, by its Neumann series. Factoring $Q_a(qL)$ and composing these right inverses in reverse order gives a bounded right inverse G for c_{p^a}. Then

$$d_{p^a}(b^{-a} G L^a) = V^a L^a,$$

the projection onto the tail. This proves both surjectivity assertions. Finite truncations of a nonzero kernel vector prove nonclosability of the polynomial inverses, as above.

At $q = 1$, kernel sequences are combinations of the periodic sequences ζ^r. A nonzero periodic sequence is not square-summable, so the kernel vanishes. The constant coefficient of Q_a is one. Its formal reciprocal acts on D, since L is locally nilpotent there. Thus $c_{p^a}(D) = D$, and the range is dense.

Fix one root $\zeta \ne 1$ and, for $N > a$, set

$$v_N = N^{-1/2} \sum_{r=0}^{N-1} \zeta^r f_r.$$

Its norm is one. The first $N - a$ coefficients of $Q_a(L)v_N$ vanish. At most a coefficients remain, each bounded by $(a+1)/\sqrt{N}$. Therefore

$$\|Q_a(L)v_N\| \le \frac{\sqrt{a}(a+1)}{\sqrt{N}} \longrightarrow 0.$$

An injective bounded operator with closed range is bounded below, by the bounded inverse theorem on its range. The sequence excludes this possibility. The dense range is consequently proper and nonclosed. Equation (13.13) gives the same conclusion inside the tail for d_{p^a}.

The inverse of an injective bounded operator on its range has closed graph: interchange the coordinates of the original closed graph. Here these inverses are unbounded by the preceding sequence.

For any $x \in \mathcal{H}_s$, choose $x_N \in D$ with $x_N \to x$. Boundedness gives $c_{p^a} x_N \to c_{p^a} x$. Hence $(c_{p^a} x_N, x_N) \to (c_{p^a} x, x)$ in the inverse graph. The same argument applies to d_{p^a}. This proves that the polynomial inverses have precisely these graph closures.

Finally, for $q < 1$, the identity $(1 - z)Q_a(z) = 1 - z^{a+1}$ gives

$$c_{p^a}^{-1} = (I - qL)(I - (qL)^{a+1})^{-1}, \qquad \|c_{p^a}^{-1}\| \le \frac{1+q}{1-q^{a+1}}.$$

The inverse is a norm-convergent Neumann series. On the tail, the inverse of d_{p^a} is $c_{p^a}^{-1} b^{-a} L^a$. Both agree with the polynomial inverses on finite vectors. □

All these bounded maps retain the common finite domain D as a core. Their polynomial formulas do not change with s. The completed kernels and inverses change because the allowed coefficient decay changes. In particular, excluding g_ζ from $\mathcal{H}_s$ at $s \ge 1/2$ does not remove its kernel class in $\mathcal{H}_0$.

13.11 Operations and states in the arithmetic comparison

The positive radial state belongs to its specified Hecke representation. A phase average on oscillator states gives a different operator, even when it uses the same measure. To see this directly, take two species with polynomial space $\mathbb{C}[a_k, b_k : k \ge 1]$ and the Hermitian Hall norm. The monomial $\prod_k a_k^{r_k} b_k^{s_k}$ has charge $m = \sum_k (r_k - s_k)$.

Let V_θ multiply its charge space by $e^{im\theta}$. These are unitary operators on the Hilbert completion. The strong average

$$A_p = \int_0^\pi V_\theta \, d\nu_p(\theta)$$

is a contraction. Its eigenvalue in charge m is $c_m = \int_0^\pi e^{im\theta} \, d\nu_p$.

Symmetry about $\pi/2$ gives

$$c_1 = i \int_0^\pi \sin\theta \, d\nu_p \in i\mathbb{R}_{>0}.$$

Thus this phase average cannot be unitarily intertwined with the self-adjoint adjacency operator while retaining the specified operators. The obstruction concerns this comparison, not every arithmetic realization.

The energy convention also matters. For species labeled by primes, integer mode weight k and arithmetic energy $k \log p$ define different diagonal operators. The integral monomial lattice survives both gradings, although the real energy operator need not preserve it.

The full oscillator partition function is $\prod_{k \ge 1} \zeta(k\beta)$ for $\beta > 1$. Keeping only mode one instead gives $\zeta(\beta)$. That subspace is not preserved by the index shifts or divisor substitutions. The operator domains and direct occupation sums establishing these statements are given in Section 28.7.

For comparison, on $\ell^2(\mathbb{N}_{>0})$ let $\mu_n \delta_N = \delta_{nN}$ and $H_{\mathrm{BC}} \delta_N = \log N \, \delta_N$. These commuting isometries satisfy

$$[H_{\mathrm{BC}}, \mu_n] = \log n \, \mu_n$$

on finite-support vectors. They move δ_1 to δ_n. The index substitutions instead fix their polynomial vacuum, and their arithmetic energy satisfies $H\psi_n = n\psi_n H$. A vacuum- and energy-preserving identification cannot equate these two specified actions for $n > 1$.

Question 13.11.1 *(Marked arithmetic realization).* Construct an arithmetic trace carrier independently of these local models, with its coefficient theory, integral form, selected trace class and functional. Construct its operations and an intertwiner with a specified local or restricted tensor-product representation. The map must preserve the distinguished vector, the functional, the determinant normalization and every chosen operation. For a Hilbert comparison, specify the domains, adjoints and completed tensor product. For a vertex comparison, retain translation and all singular products.

For example, the unramified comparison of Conjecture 27.9.1 requires the actual maps Φ, Q, Q_p, normalized local vectors and compatible linear functionals. An arbitrary family of Satake pairs does not construct those arithmetic objects or maps. Neither the root-state identity nor the Witt ghost formulas identify a full Weil functional.

There is also an immediate operation obstruction to some proposed maps. In characteristic zero, an injective intertwiner cannot identify the divisor family with commuting Adams operations or a commuting semigroup of isometries. Equation (13.3) would map a nonzero vector to zero. A quotient, a restricted domain or a different operation system remains a separate possibility with its own kernel and comparison calculation.

Chapter 14

Hall fields on curves

A torsion sheaf on a curve splits into its parts at closed points. Hall multiplication and the Green form split with it. This elementary decomposition produces operators whose exchange coefficients count points over every finite extension of the ground field. The coefficient depends on the normalization of both operators. Keeping the Hall creators and their Green adjoints gives a mass series. Changing their normalization gives the curve zeta function.

We follow these operators into three recipients. Polynomial Hall states carry local Heisenberg fields, and finite power covers retain their residue maps. Collective counting modes instead lead to fields with finitely many multiplicative collision divisors after reciprocal normalization. The unchanged Green exchange has a meromorphic difference equation and a positive coherent state. Derived intersections of the collision divisors explain why their scalar equations and supported differential modules require different comparisons.

Prime-indexed Macdonald spaces separate the choice of local parameters from a geometric Hall realization. A final nilpotent-coefficient example constructs the scalar kernel and its finite-jet comparison, then locates the missing critical Hall map.

14.1 Closed points and Green adjunction

Let C be a smooth projective geometrically connected curve over $\mathbb{F}_q$. For a closed point x, write $d_x = [k(x) : \mathbb{F}_q]$ and $Q_x = q^{d_x}$. A choice of parameter and coefficient field identifies the completed local ring with $\mathbb{F}_{Q_x}[[t]]$. Its finite-length modules are the finite nilpotent $\mathbb{F}_{Q_x}[t]$-modules. The resulting Hall algebra is independent of this presentation. The presentation names its Jordan partitions.

Use complex coefficients. For a prime power Q, let H_Q have basis u_M indexed by nilpotent modules and put $a_M = |\operatorname{Aut} M|$. The product and positive Hermitian form are

$$u_M u_N = \sum_L F^L_{M,N} u_L, \qquad \langle u_M, u_N \rangle_G = \frac{\delta_{MN}}{a_M},$$

where $F^L_{M,N}$ counts submodules of type N with quotient M. The form is conjugate-linear in its first argument. The torsion Euler form is zero. Thus the Green coproduct is adjoint to this product without an Euler twist.

The local primitive theorem, with its normalization fixed in Theorem 12.2.1, gives

$$\begin{aligned} H_Q &= \mathbb{C}[p_1, p_2, \ldots], \qquad \Delta p_r = p_r \otimes 1 + 1 \otimes p_r, \\ p_r &= \sum_{|\lambda|=r} \prod_{j=1}^{\ell(\lambda)-1} (1 - Q^j) u_\lambda, \qquad \langle p_r, p_s \rangle_G = \delta_{rs} \frac{r}{Q^r - 1}. \end{aligned} \tag{14.1}$$

In particular the coefficient of the indecomposable module of length r in p_r is one.

The elementary generating series gives another way to see the normalization. Put

$$e_r = Q^{r(r-1)/2} u_{1^r}, \qquad E(t) = \sum_{r \geq 0} e_r t^r, \qquad e_0 = 1.$$

Subspace counting gives $\Delta u_{1^r} = \sum_{i+j=r} Q^{-ij} u_{1^i} \otimes u_{1^j}$, so E is group-like.

Products of semisimple classes are triangular in the Jordan basis. The successive kernel dimensions specify a filtration lowered by t. Equality with these dimensions forces the kernel filtration itself. Its coefficient is one. Other terms have larger kernel dimensions. This proves the polynomial presentation in each finite Hall degree.

The logarithm then gives the primitive generators through

$$\log E(t) = \sum_{r \geq 1} \frac{(-1)^{r-1}}{r} p_r t^r.$$

Since $|\mathrm{GL}_r(\mathbb{F}_Q)| = Q^{r(r-1)/2} \prod_{j=1}^{r} (Q^j - 1)$, their norms can also be read from

$$\langle E(u), E(v) \rangle_G = \sum_{r \geq 0} \frac{Q^{r(r-1)/2} (\bar{u} v)^r}{\prod_{j=1}^{r} (Q^j - 1)} = \prod_{j \geq 1} (1 - Q^{-j} \bar{u} v)^{-1}.$$

The last equality follows from $(1 - t) P(Qt) = P(t)$ and $P(0) = 1$.

Primitivity and Hopf adjunction give factorial norms for powers of a primitive. Taking logarithms gives (14.1).

PROPOSITION 14.1.1. On the polynomial domain, multiplication M_r by p_r has Green adjoint

$$D_r = \frac{r}{Q^r - 1} \partial_{p_r}, \qquad [D_r, M_s] = \delta_{rs} \frac{r}{Q^r - 1} I.$$

The inverse pairing, completed by nonnegative Hall weight, is

$$\sum_{\lambda} u_\lambda \otimes a_\lambda u_\lambda = \exp\left(\sum_{r \geq 1} \frac{Q^r - 1}{r} p_r \otimes p_r \right). \tag{14.2}$$

Proof. The adjoint of multiplication by a primitive is a derivation. Its value on p_s is the scalar $\delta_{rs} r/(Q^r - 1)$, which determines the derivation on H_Q. Iterating adjunction makes the monomials orthogonal, with

$$\left\| \prod_r p_r^{m_r} \right\|_G^2 = \prod_r m_r! \left(\frac{r}{Q^r - 1} \right)^{m_r}.$$

Sum each monomial tensored with its dual to obtain the exponential. The Hall basis gives the left side of (14.2). Each weight space is finite dimensional, so both expressions define the same completed tensor. □

The torsion Hall space of C is the algebraic tensor product

$$\mathcal{F}_C = \bigotimes_{x \in |C|}{}' H_{q^{d_x}}, \qquad \deg p_{x,r} = r d_x.$$

The prime means that a tensor is the vacuum 1 at all but finitely many points. Hom and extension groups between different supports vanish, so products and Green forms factor.

There are finitely many closed points of bounded degree. Hence each weight space is finite dimensional.

Define, for $n > 0$,

$$N_n = \sum_{d_x \mid n} d_x = \#C(\mathbb{F}_{q^n}),$$
$$A_{-n} = \sum_{d_x \mid n} d_x M_{x,n/d_x}, \qquad B_n = \sum_{d_x \mid n} d_x D_{x,n/d_x}.$$

All sums are finite. A point of degree d has exactly d geometric points over $\mathbb{F}_{q^n}$ when $d \mid n$, and none otherwise. Operators at different points commute. Therefore

$$[B_n, A_{-m}] = \delta_{nm} \frac{nN_n}{q^n - 1} I, \qquad B_n = A_{-n}^{*G}.$$

Two positive modes commute, as do two negative modes.

14.2 Three normalizations of the exchange

For the unchanged creators and Green adjoints set

$$\Gamma_-(w) = \exp\left(\sum_{n>0} \frac{A_{-n} w^n}{n}\right), \qquad \Gamma_+^G(z) = \exp\left(\sum_{n>0} \frac{B_n z^{-n}}{n}\right).$$

These are coefficientwise series in the independent formal variables w, z^{-1}. Each coefficient is an operator on $\mathcal{F}_C$. Creation may equivalently be evaluated in the completion by weight. Annihilation is locally finite on every polynomial state.

Theorem 14.2.1. Put

$$K_G(t) = \exp\left(\sum_{n>0} \frac{N_n t^n}{n(q^n - 1)}\right), \qquad Z_C(t) = \exp\left(\sum_{n>0} \frac{N_n t^n}{n}\right).$$

Then

$$\Gamma_+^G(z)\Gamma_-(w) = K_G(w/z)\Gamma_-(w)\Gamma_+^G(z), \tag{14.3}$$
$$K_G(qt) = Z_C(t) K_G(t),$$
$$Z_C(t) = \prod_x (1 - t^{d_x})^{-1}, \qquad K_G(t) = \prod_{j \geq 1} Z_C(q^{-j} t).$$

The last product uses convergent logarithmic coefficients at the fixed numerical value $q > 1$.

Proof. The commutator of the two exponents is the central series $\sum_{n>0} N_n (w/z)^n / [n(q^n - 1)]$. Formal conjugation of the exponentials proves (14.3). Subtracting the logarithms at qt and t proves the difference equation. Writing $n = r d_x$ in $\log Z_C$ gives $\sum_x \sum_{r>0} t^{r d_x}/r$, the logarithm of the Euler product. Finally $1/(q^n - 1) = \sum_{j \geq 1} q^{-jn}$ proves the product for K_G in each logarithmic coefficient. □

There are two ways to obtain Z_C itself as an exchange factor. First keep the creators and replace B_n by $J_n = (q^n - 1) B_n$. The corresponding annihilation vertex is

$$\Gamma_+^\zeta(z) = \Gamma_+^G(z/q)\Gamma_+^G(z)^{-1} = \exp\left(\sum_{n>0} \frac{J_n z^{-n}}{n}\right). \tag{14.4}$$

It has exchange factor $Z_C(w/z)$ with $\Gamma_-(w)$. Its modes are no longer the Green adjoints of the fixed creators.

Second use the positive real square root and put

$$\widehat{A}_{-n} = \sqrt{q^n - 1}\, A_{-n}, \qquad \widehat{B}_n = \sqrt{q^n - 1}\, B_n.$$

They remain Green adjoints and satisfy $[\widehat{B}_n, \widehat{A}_{-m}] = \delta_{nm} n N_n I$. Their half-vertices again exchange by $Z_C(w/z)$, with both operators changed. Locally this is the adjoint pair

$$a_{-r} = \sqrt{Q^r - 1}\, M_r, \qquad a_r = \frac{r}{\sqrt{Q^r - 1}}\, \partial_{p_r}.$$

All three constructions have a common polynomial domain. No everywhere-defined operator on the Green Hilbert completion has been asserted. The dilation $z \mapsto z/q$ in (14.4) is a spectral change of variable. Absolute Frobenius is the different map $z \mapsto z^q$.

An independent oscillator polynomial space $\mathbb{C}[u_{x,r} : x \in |C|, r > 0]$ is identified with $\mathcal{F}_C$ by $u_{x,r} \mapsto p_{x,r}$. This preserves weight, multiplication and vacuum. It sends multiplication by $u_{x,r}$ to $M_{x,r}$ and $r\partial_{u_{x,r}}$ to $(q^{rd_x} - 1)D_{x,r}$. Thus it intertwines the one-sided normalized modes and all formal coefficients of their half-vertices. Specified residue operations require their own comparison below.

Example 14.2.2. For $C = \mathbb{P}^1$, $N_n = q^n + 1$ and $Z_C(t) = ((1 - t)(1 - qt))^{-1}$.

For the smooth projective cubic

$$E/\mathbb{F}_2 : \quad y^2 + y = x^3 + x + 1,$$

the first four point counts are $1, 5, 13, 25$. The equation $y^2 + y = b$ has two solutions exactly when $\mathrm{Tr}_{\mathbb{F}_{2^n}/\mathbb{F}_2}(b) = 0$. For $n = 1, 2, 3, 4$ there are respectively $0, 2, 6, 12$ such values of x, and the projective curve has one point at infinity. Consequently

$$Z_E(t) = 1 + t + 3t^2 + 7t^3 + 15t^4 + O(t^5).$$

These counts determine the exchange through degree four. The full formula $Z_E(t) = (1 - 2t + 2t^2)/((1 - t)(1 - 2t))$ also uses the classical rationality and degree-two numerator theorem for an elliptic curve.

An elementary modification

The full Hall algebra of coherent sheaves contains line bundles as well as torsion sheaves. Fix a rational point x and a line bundle L. Extensions of $\mathcal{O}_x$ by L have middle sheaf $L \oplus \mathcal{O}_x$ or $L(x)$: the extension space is one dimensional and all nonzero classes give $L(x)$.

In the split sheaf the relevant copies of L are the graphs of the q maps $L \to \mathcal{O}_x$. In $L(x)$ there is the unique subobject $L(x)(-x) = L$. Conversely $\mathrm{Ext}^1(L, \mathcal{O}_x) = 0$, and the split middle sheaf has its unique torsion subobject. Hence

$$u_{\mathcal{O}_x} u_L = q\, u_{L \oplus \mathcal{O}_x} + u_{L(x)}, \qquad u_L u_{\mathcal{O}_x} = u_{L \oplus \mathcal{O}_x}.$$

With $v = \sqrt{q}$, twist multiplication by $u_A * u_B = v^{\chi(A,B)} u_A u_B$. Here $\chi(\mathcal{O}_x, L) = -1$ and $\chi(L, \mathcal{O}_x) = 1$, so

$$[u_{\mathcal{O}_x}, u_L]_* = v^{-1} u_{L(x)}.$$

On $\mathbb{P}^1$, summing over its $q+1$ rational points gives

$$\left[\sum_x u_{O_x}, u_{O(n)}\right]_* = (v+v^{-1})u_{O(n+1)}.$$

This operation retains the elementary modification. A presentation of a quantum loop algebra requires its other generators and relations. An arithmetic trace comparison additionally requires its category, state and functional.

14.3 Local fields on every polynomial Hall state

Return to one point, with residue cardinality Q. Set $h_{-r} = M_r$, $h_r = (Q^r-1)D_r = r\partial_{p_r}$ for $r>0$, and $h_0 = 0$. Thus

$$[h_r, h_s] = r\delta_{r+s,0}I, \qquad h_r^{*G} = (Q^r-1)h_{-r} \quad (r>0).$$

Define a derivation T by $T(p_r) = r\,p_{r+1}$ and define

$$H(z) = \sum_{r\in\mathbb{Z}} h_r z^{-r-1}.$$

For a polynomial monomial set

$$Y(p_{r_1}\cdots p_{r_l}, z) =: \prod_{j=1}^{l} \frac{\partial_z^{r_j-1}H(z)}{(r_j-1)!} :, \qquad Y(1,z) = I. \tag{14.5}$$

Normal ordering places creation modes to the left. This defines the field on every state. Positive modes act locally finitely on a polynomial input, and the creation series has a lower bound on its powers of z.

Proposition 14.3.1. The fields (14.5), with vacuum 1 and translation T, give the free Heisenberg vertex algebra on H_Q. Every finite ordered product has a common normally ordered Wick expression, with rational contractions and finite poles along pairwise diagonals. Every nesting expands that expression and retains its nonconsecutive contractions.

Proof. Commuting an annihilator past a creator gives the scalar $r\delta_{rs}$. Summing gives $(z-w)^{-2}$ in the expansion $|w|<|z|$. For fixed polynomial states, commute each annihilator through later creators. The terms are indexed by partial matchings of the elementary factors. Matched factors contribute the appropriate derivatives of $(z-w)^{-2}$ and unmatched factors remain normally ordered. There are finitely many matchings and derivative orders, so one rational denominator clears every term.

Regrouping partitions the same matchings according to the blocks joined by their edges. Both nested expansions are therefore expansions of the same rational expression. Empty matchings give the vacuum identities.

Directly, $[T, h_m] = -mh_{m-1}$, which proves translation covariance. The two expansions of each contraction differ by a derivative of the formal delta distribution. A finite power of $z-w$ kills the commutator. These identities prove locality and associativity for all the fields in (14.5). □

Hall multiplication is represented by creation modes: multiplication by p_r is h_{-r}. It need not agree with the vertex (-1)-product on composite states. For example,

$$(p_1^2)_{(-1)}p_1 = p_1^3 + 2p_3, \qquad (p_1^2)p_1 = p_1^3.$$

The additional term is the constant coefficient of $2z^{-2}\sum_{r>0} p_r z^{r-1}$ in $:H(z)^2:\,p_1$.

Finite power covers

At a closed point of degree d, pull back the current density along $s = z^d$ on $\mathbb{G}_m$:

$$J_x(z) = dz^{d-1} H_x(z^d), \qquad J_n^x = \begin{cases} d\, h^x_{n/d}, & d \mid n, \\ 0, & d \nmid n. \end{cases}$$

The map is finite étale over $\mathbb{C}$ on the torus. Its extension across zero is ramified, so the domain matters. The common contraction is

$$\frac{d^2 z^{d-1} w^{d-1}}{(z^d - w^d)^2} = \sum_{\zeta^d = 1} \frac{\zeta}{(z - \zeta w)^2}. \tag{14.6}$$

Differentiate in w the identity $\sum_{\zeta^d=1} (z - \zeta w)^{-1} = dz^{d-1}/(z^d - w^d)$ to obtain this formula. Different point colors commute.

The coefficient of z^{rd-1} in $J_x(z)1$ is $d p_{x,r}$. Products of such creation coefficients recover all finite-support Hall states. Normal products and derivatives have the pulled-back Wick expansions. Derivative orders a, b give pole order at most $a + b + 2$ at each divisor $z = \zeta w$. This is a bound for each specified input. The states have no common finite derivative bound.

Fields, quasi-modules and residue maps

For a complex vector space W write $\mathcal{E}(W) = \mathrm{Hom}(W, W((z)))$. A subset is *quasi-local* if, for each pair of its fields, a nonzero polynomial in their two coordinates annihilates the commutator. It is *quasi-compatible* if each finite ordered product becomes a map to $W[[z_1, \dots, z_m]][z_1^{-1}, \dots, z_m^{-1}]$ after multiplication by a finite product of nonzero two-variable polynomials. Formula (14.6) proves quasi-locality of the point-labeled fields on $\mathcal{F}_C$. Their finite Wick expressions prove quasi-compatibility.

Li's reconstruction theorem gives a vertex algebra *of fields* from a quasi-local subset of $\mathcal{E}(W)$, including its closure under any chosen subgroup of dilations. Evaluation gives a faithful quasi-module on W.[1]

Concretely, if $p(z_1, z)$ clears the product, then

$$Y_{\mathcal{E}}(a(z), u)b(z) = \iota_{z,u}\big(p(z+u, z)^{-1}\big)\, p(z_1, z) a(z_1) b(z)\big|_{z_1 = z+u}. \tag{14.7}$$

The symbol $\iota_{z,u}$ means expansion in $\mathbb{C}((z))((u))$. Clearing first makes the substitution defined. The theorem makes its coefficients independent of the clearing polynomial and proves closure and the vertex identities. Substitution at $z_1 = \alpha z + u$ gives the dilation operations. The original Hall space is the quasi-module, and has not been identified with this vertex algebra of fields.

For finitely many point colors, the pole group is generated by finitely many roots of unity. The whole point-labeled construction is their filtered union: each operation uses finitely many denominators and colors.

There are also actual supported residue maps on the power-cover sector. Start with the Heisenberg fields of Proposition 14.3.1. At the collision of two positions, with $u = s_i - s_j$, the map on states and a multiplier f is

$$r_{ij}(f; \dots, a, \dots, b, \dots) = \mathrm{pp}_u\big(f Y(a, u) b\big).$$

[1] H. Li, *A new construction of vertex algebras and quasi modules for vertex algebras*. arXiv:math/0404110, Theorems 4.9 and 5.4. The closure includes the identity field. Faithfulness here follows from its inclusion in $\mathcal{E}(W)$.

Here pp_u retains the negative powers of u. The other states stay at their positions. Regular functions act on principal parts by their finite normal Taylor expansions. For example, for $t \neq 0$ and $r, \ell \geq 1$,

$$(u-t)^{-\ell}[u^{-r}] = (-1)^{\ell} \sum_{m=0}^{r-1} \binom{\ell+m-1}{m} t^{-\ell-m}[u^{m-r}].$$

This retains the spectator poles and every finite normal derivative.

Translation covariance makes the maps horizontal for the state connection $\partial_s + T$. Residues of disjoint pairs commute. For three positions, the three iterated principal parts come from the same rational Wick expression. The principal-parts relation between its three collision channels is their residue identity. Thus one has the full residue diagram, including its attachments, before taking a cover. These statements concern the polynomial Heisenberg states with their finite pole bounds.

Pull back this diagram along

$$(\mathbb{G}_m)^n \longrightarrow (\mathbb{G}_m)^n, \qquad (z_1, \dots, z_n) \longmapsto (z_1^d, \dots, z_n^d).$$

Finite étale base change preserves its right differential modules, residue maps and their relations. The diagonal $s_i = s_j$ pulls back to the divisors $z_i = \zeta z_j$, with their actual intersections and pulled-back attachments. Pullback of the current density is exactly (14.6). The cut tensor here excludes $z_i^d = z_j^d$ across a cut. It would change if only $z_i = z_j$ were excluded.

For finitely many colors, use the independent local factors and their labeled elementary insertions. Their finite Wick matchings give the corresponding coefficient diagrams. Composite fields at a retained root use (14.7). This supplies an operation comparison for this specified cover sector. It leaves a field presentation of an arbitrary geometric cut object as a separate question.

14.4 Collective counting fields

Sum the point-labeled currents coefficientwise:

$$J(z) = \sum_x J_x(z) = \sum_{n \in \mathbb{Z}} J_n z^{-n-1}, \qquad J_{-n} = A_{-n}, \quad J_n = (q^n - 1)B_n \quad (n > 0).$$

Set $J_0 = 0$. A fixed mode involves only points whose degrees divide its nonzero index, so it acts on the polynomial Hall space. On a fixed input only finitely many positive modes act nontrivially. The commutators are

$$[J_n, J_{-m}] = \delta_{nm} n N_n I \qquad (n, m > 0).$$

Proposition 14.4.1. A linear operator T on this mode representation can satisfy $[\mathsf{T}, J_n] = -nJ_{n-1}$ for all integers n only if $N_{n+1} = N_n$ for every $n \geq 1$.

Proof. Apply $[\mathsf{T}, -]$ to $[J_{n+1}, J_{-n}] = 0$ and use Jacobi. The resulting equality is

$$0 = -(n+1)[J_n, J_{-n}] + n[J_{n+1}, J_{-n-1}] = n(n+1)(N_{n+1} - N_n)I.$$

The coefficient field has characteristic zero. □

For $\mathbb{P}^1$, the difference $N_{n+1} - N_n = q^n(q-1)$ is nonzero. Thus the unchanged collective current has no ordinary vertex translation. The degree derivation $L_0(p_{x,r}) = r d_x p_{x,r}$ does satisfy $[L_0, J(z)] = (z\partial_z + 1)J(z)$. It implements dilation.

Reciprocal normalization

Write a rational counting function with constant term one as

$$Z(t) = \prod_{\lambda \in \Lambda} (1 - \lambda t)^{-\epsilon_\lambda}, \qquad \epsilon_\lambda \in \mathbb{Z},$$

with finitely many nonzero parameters. Its current contraction is

$$C(z, w) = \frac{1}{zw} (t\partial_t)^2 \log Z(t) = \sum_{\lambda \in \Lambda} \frac{\epsilon_\lambda \lambda}{(z - \lambda w)^2}, \qquad t = w/z. \tag{14.8}$$

This follows by twice differentiating each $-\log(1 - \lambda t)$.

For the curve, the classical rationality theorem gives

$$Z_C(t) = \frac{\prod_{i=1}^{2g} (1 - \alpha_i t)}{(1 - t)(1 - qt)}.$$

The numerator parameters α_i occur in the current contraction as well as $1, q$. They must therefore occur among its allowed collision gains. Poles of the current contraction are not determined just by poles of Z_C.

The full Laurent field needs both orders of a contraction. The expression (14.8) need not be symmetric under z, w interchange, so quasi-locality of each J_x alone does not establish quasi-locality of the infinite sum. Rationality and reciprocal eigenvalue pairing give a different collective field.

Choose $\sqrt{q} > 0$ and define

$$\widetilde{J}_{-n} = A_{-n}, \qquad \widetilde{J}_n = q^{-n/2}(q^n - 1) B_n, \qquad c_n = q^{-n/2} N_n.$$

These formulas use $n > 0$. Set $\widetilde{J}_0 = 0$.

The signed multiset

$$\widetilde{\Lambda} = \{q^{-1/2}, q^{1/2};\ \alpha_1/q^{1/2}, \dots, \alpha_{2g}/q^{1/2}\}$$

has signs $+, +, -, \dots, -$. The pairing $\alpha \leftrightarrow q/\alpha$ makes this multiset invariant under inversion. This uses reciprocal duality. No assertion about the absolute values of the α_i is needed.

THEOREM 14.4.2. The field $\widetilde{J}(z) = \sum_n \widetilde{J}_n z^{-n-1}$ has the symmetric rational contraction

$$\widetilde{C}(z, w) = \sum_{\lambda \in \widetilde{\Lambda}} \frac{\epsilon_\lambda \lambda}{(z - \lambda w)^2}.$$

Its commutator is annihilated by $\prod_{\lambda \in \widetilde{\Lambda}} (z - \lambda w)^2$. Its dilates generate a quasi-local system, and hence a vertex algebra of fields with faithful quasi-module $\mathcal{F}_C$.

Proof. Inversion invariance and

$$\frac{\lambda}{(w - \lambda z)^2} = \frac{\lambda^{-1}}{(z - \lambda^{-1} w)^2}$$

give symmetry. The coefficient of $z^{-n-1} w^{n-1}$ in the forward expansion is $n c_n$. Inversion invariance gives the same constants in the reverse expansion.

Their difference is

$$\sum_\lambda \epsilon_\lambda \partial_w \big(z^{-1} \delta(\lambda w / z) \big), \qquad \delta(t) = \sum_{m \in \mathbb{Z}} t^m.$$

Each summand is killed by $(z - \lambda w)^2$. Every finite set of dilates has a finite set of these factors. The quasi-local reconstruction theorem therefore applies. □

Translation in this vertex algebra acts on its fields through (14.7). The Hall space is a quasi-module, and its new positive modes also differ from J_n. Thus the theorem imposes no impossible translation on the unchanged representation in Proposition 14.4.1. The collective modes retain the counting sequence. Individual closed-point labels remain in the separate fields J_x.

The point-labeled and aggregate rational fields together are quasi-compatible. For a fixed point of degree d, their mixed contraction uses only $n \in d\mathbb{Z}$. Its coefficients are finite combinations of n times geometric progressions. It therefore sums to a rational function. Aggregate contractions use (14.8) and its rescalings.

Wick expansion clears a finite common denominator in every finite ordered product. Li's nonlocal reconstruction then gives a nonlocal vertex algebra of fields and a faithful Hall quasi-module.[2]

Pairwise quasi-locality of this combined set has not been assumed. The native Green current has an infinite-product kernel. It uses the analytic recipient below.

A finite-dimensional trace comparison

For a finite-dimensional graded vector space $H = \bigoplus_i H^i$ and a grading-preserving endomorphism F, graded symmetry gives

$$\sum_{n\geq 0} \operatorname{str}(F; \operatorname{Sym}^n H)t^n = \prod_i \det(1 - tF|H^i)^{(-1)^{i+1}}.$$

Indeed, after extending scalars and triangularizing, an even eigenvector contributes $(1 - \alpha t)^{-1}$. An odd eigenvector generates an exterior algebra and contributes $1 - \alpha t$. Multiply the factors and descend the coefficients. Jordan off-diagonal entries do not change the traces.

The two even eigenvalues $1, q$, in degrees zero and two, reproduce the series of $\mathbb{P}^1$. A more discriminating example is $y^2 = x^3 - x$ over $\mathbb{F}_5$, which has seven affine points and one point at infinity. The matrices

$$F = \begin{pmatrix} 0 & -5 \\ 1 & -2 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & -1 \\ -1 & 5 \end{pmatrix}$$

satisfy $F^t H F = 5H$. The leading principal minors of H are $1, 4$, so H is positive definite. This particular operator has eigenvalues of modulus $\sqrt{5}$ and determinant polynomial $1 + 2t + 5t^2$.

Identifying it with a geometric Frobenius representation still requires the cohomological realization. The point count and positive matrix alone do not construct it.

14.5 Gain equations and their derived intersections

Let k have characteristic zero and let $G \subset k^\times$ be a subgroup containing the chosen collision gains. Work on $X = (\mathbb{G}_m)^n$, with coordinate ring $S = k[z_1^{\pm 1}, \dots, z_n^{\pm 1}]$. For a finite oriented graph Γ, including repeated edges, an edge $e = (i, j, \lambda_e)$ names the equation $f_e = z_i - \lambda_e z_j$. A component is *balanced* if the oriented product of gains around every cycle is one. Reversing an edge replaces its gain by its inverse.

The scalar derived intersection is the Koszul complex

$$K_\Gamma = (S \otimes \Lambda(\xi_e : e \in E(\Gamma)), d), \qquad |\xi_e| = -1, \qquad d\xi_e = f_e.$$

For pole coefficients one instead inverts a finite collection of nonzero f_e. The union of these localization rings records functions with finitely many allowed poles. It does not assert that an infinite arrangement is a finite-type scheme.

[2] H. Li, *Nonlocal vertex algebras generated by formal vertex operators*. arXiv:math/0502244v2, Theorem 2.20. Quasi-compatibility is Definition 2.10.

THEOREM 14.5.1. Suppose Γ has n vertices, e edges and c components. If all components are balanced, a spanning forest gives

$$K_\Gamma \simeq k[y_1^{\pm 1}, \dots, y_c^{\pm 1}] \otimes \Lambda(\eta_1, \dots, \eta_{e-n+c}), \qquad |\eta_j| = -1, \quad d\eta_j = 0.$$

If a component is unbalanced, K_Γ is contractible.

Proof. Removing leaves of a spanning forest eliminates one coordinate at each step by a nonzero scalar multiple of its neighbor. Its $n - c$ equations form a regular sequence, with quotient the torus on the root coordinates. In a balanced component, every remaining equation is a constant linear combination of forest equations. Subtract this combination from its exterior generator. The resulting invertible triangular change splits off one closed generator for each extra edge.

An unbalanced cycle instead leaves $(1 - \lambda)\, y$ with $\lambda \neq 1$ and y invertible. Thus its equations generate the unit ideal. If $\sum_e g_e f_e = 1$, then exterior multiplication by $\beta = \sum_e g_e \xi_e$ satisfies $d(\beta \wedge -) + \beta \wedge d = 1$. This contracts the complete complex. □

For the balanced triangle,

$$f_1 = z_1 - \lambda z_2, \quad f_2 = z_2 - \mu z_3, \quad f_3 = z_1 - \lambda\mu z_3 = f_1 + \lambda f_2,$$
$$\eta = \xi_3 - \xi_1 - \lambda \xi_2, \qquad d\eta = 0.$$

Quotient by the forest equations and send ξ_1, ξ_2 to zero. The image of η is a nonzero exterior generator, so it cannot have been a boundary. If the third gain is instead $\nu \neq \lambda\mu$, the explicit primitive of the unit is

$$\beta = \frac{\xi_3 - \xi_1 - \lambda \xi_2}{(\lambda\mu - \nu) z_3}, \qquad d\beta = 1.$$

The forest basis computes the cohomology. Equation operations belong to the unreduced complex. A ring map $S \to S'$ and a matrix $U = (u_{te})$ satisfying $f_e \mapsto \sum_t u_{te} f'_t$ give the cochain map $\xi_e \mapsto \sum_t u_{te} \xi'_t$. Composition is matrix multiplication with coefficient substitution.

For a clustering $\alpha : I \twoheadrightarrow J$ and scales $\gamma_i \in G$, the substitution $z_i = \gamma_i y_{\alpha(i)}$ gives

$$f_{ij,\lambda} \longmapsto \gamma_i \big(y_{\alpha(i)} - \lambda \gamma_j \gamma_i^{-1} y_{\alpha(j)} \big).$$

The exterior generator has the same factor γ_i. An internal zero equation retains its exterior generator. An internal unit equation contracts. An equation that becomes zero cannot remain inverted, so a collision requires its restriction complex.

The divided exponential chart

In this spectral coefficient family, regard q as a formal variable. For gains q^a, with integers a, introduce a formal parameter $\hbar$ and use $q = e^{\hbar}$, $z_i = e^{\hbar x_i}$. In $k[x_1, \dots, x_n][[\hbar]]$ the divided equation is

$$\frac{e^{\hbar x_i} - e^{\hbar(x_j + a)}}{\hbar} = (x_i - x_j - a) U_{ij,a}(\hbar), \qquad U_{ij,a}(0) = 1.$$

The quotient denotes a power series, with no inversion of $\hbar$.

Rescaling the exterior generator by the unit $U_{ij,a}^{-1}$ gives the affine equation. A forest quotient is $k[y_1, \dots, y_c][[\hbar]]$, which is flat over $k[[\hbar]]$. Balanced cycles retain the excess exterior factors. An

unbalanced cycle telescopes to a nonzero integer, so its complex contracts. After inverting $\hbar$, this is the pulled-back multiplicative arrangement up to units.

The special fiber is the shifted arrangement $x_i - x_j = a$. Undivided equations would all reduce to zero. This constructs a family of spectral coefficient arrangements. Extending Green operators across $\hbar = 0$ further requires a lattice and a normalization, since $q^n - 1$ becomes $\hbar$ times a unit. No family of arithmetic curves is supplied by the coordinate substitution.

Scalar operations on the pole ring

Fix colors α and rational kernels $K_{\alpha\beta}(z, w)$ with finitely many allowed gain poles. For functions on two ordered lists of colors define

$$(f \diamond g)(z_1, \dots, z_{m+n}) = f(z_1, \dots, z_m)\, g(z_{m+1}, \dots, z_{m+n}) \prod_{i \le m < j} K_{\alpha_i \alpha_j}(z_i, z_j).$$

This is an associative product, with the empty list as unit: either bracketing of three blocks includes each cross-block factor exactly once. Summing over order-preserving shuffles gives an associative product on symmetric colored functions.

For graded colors, include the Koszul permutation sign. Both bracketings have the same three-block shuffles and signs.

Inverse exchanges may require gains from zeros as well as poles of the original kernels. The scalar construction then uses the enlarged group. A Hall comparison must also supply the maps of coefficients, orientations and any numerator conditions required by its recipient. The scalar product proves none of these from an equality of kernels.

14.6 Supported coefficients with excess

For one regular equation f, the scalar Koszul complex and the localization complex have different purposes. The latter is $L_f = [S \to S[f^{-1}]]$ in degrees 0, 1. Its map is horizontal for the usual left differential action. Localization gives

$$L_f \otimes^{\mathbb{L}}_S L_f \simeq L_f :$$

tensor the triangle $L_f \to S \to S[f^{-1}]$ with L_f and use the acyclicity of $L_f \otimes_S S[f^{-1}]$. In contrast, $S/(f) \otimes^{\mathbb{L}}_S S/(f)$ has a nonzero degree-minus-one Tor class.

For $f = z_i - \lambda z_j$, $i \neq j$, the Koszul complex $[S \xrightarrow{f} S]$ in degrees -1, 0 cannot admit a differential-module enhancement with the same object of $D(S)$. Such an enhancement would give its cohomology $S/(f)$ the stated S-action and a left differential action. But f acts by zero, whereas $[\partial_{z_i}, f] = 1$ would force their commutator to be the identity on this nonzero quotient.

We construct a supported recipient that also retains the excess. For a balanced graph let $V = k^{E(\Gamma)}$, with basis ξ_e, and view the equations as linear forms on $A_0 = k^n$. Put

$$F : V \longrightarrow A_0^*, \quad F(\xi_e) = f_e, \qquad W = \operatorname{im} F, \quad E = \ker F, \quad T_\Gamma = W^\perp.$$

Here E denotes the space of equation relations. If $r = \dim W$ and $c = \dim T_\Gamma$, then $r = n - c$ and $Z = T_\Gamma \cap X \cong (\mathbb{G}_m)^c$.

Let $L = \det T_\Gamma^*$ be its constant tangent density line. The density-valued scalar source is $\mathcal{K}_\Gamma = L \otimes K_\Gamma$, with

$$H^{-p}(\mathcal{K}_\Gamma) = \omega_Z \otimes_k \Lambda^p E, \qquad \omega_Z = \mathcal{O}_Z \otimes_k L.$$

These cohomology identifications are independent of a forest: wedge closed vectors of E with a degree-zero class. A forest splitting proves that this map is an isomorphism.

All complexes are cohomological. $C[m]^j = C^{j+m}$ with differential $(-1)^m d_C$.

Normal derivatives and the ordinary module

For $i : Z \hookrightarrow X$ define the right differential module

$$\mathcal{N} = i_+\omega_Z = i_*\big(\omega_Z \otimes_{\mathcal{D}_Z} \mathcal{D}_{Z\to X}\big), \qquad \mathcal{D}_{Z\to X} = \mathcal{O}_Z \otimes_{i^{-1}\mathcal{O}_X} i^{-1}\mathcal{D}_X.$$

It lies in cohomological degree zero.

Choose complementary linear coordinates $(u_1, \dots, u_r, y_1, \dots, y_c)$ with $Z = (u = 0)$, and set $\tau = dy_1 \wedge \cdots \wedge dy_c$. On the actual open chart its sections have the normal form

$$\mathcal{N} = \bigoplus_{\alpha\in\mathbb{N}^r} \mathcal{O}_Z\, \delta_{\tau,\alpha}, \qquad \delta_{\tau,\alpha} = \left[\frac{du_1 \wedge \cdots \wedge du_r \wedge \tau}{\prod_j u_j^{\alpha_j+1}}\right]. \tag{14.9}$$

The brackets denote local cohomology in degree r, placed in degree zero in $\mathcal{N}$. The residue of $du_1 \wedge \cdots \wedge du_r/(u_1 \cdots u_r)$ is one. Equivalently, $\delta_{\tau,\alpha} = \delta_{\tau,0}\partial_u^\alpha/\alpha!$. Thus

$$\begin{aligned} \delta_{\tau,\alpha} u_j &= \delta_{\tau,\alpha-e_j}, \qquad & \delta_{\tau,\alpha}\partial_{u_j} &= (\alpha_j+1)\delta_{\tau,\alpha+e_j}, \\ (b\delta_{\tau,\alpha})\partial_{y_j} &= -\partial_{y_j}(b)\delta_{\tau,\alpha}. \end{aligned} \tag{14.10}$$

The first expression is zero if $\alpha_j = 0$.

These formulas follow from $(a\Omega)\partial = -\partial(a)\Omega$ for a constant top form Ω. They satisfy the right Weyl relation $(m\partial_{u_i})u_j - (mu_j)\partial_{u_i} = \delta_{ij}m$. Every section has finite normal order. All finite orders occur. No infinite normal-derivative tail is introduced.

A constant change $u' = Au$ preserves the normalized order-zero residue and transforms derivatives by $\partial_{u'_i} = \sum_j (A^{-1})_{ji}\partial_{u_j}$. Both numerator density and denominator must transform. A tangent change $y' = By$ multiplies the density by $\det B$. These rules follow by testing the fractions against normal polynomials: $\delta_{\tau,\alpha}$ extracts the coefficient of u^α with its tangent density.

If $I = (f_e)$, the actual support complex is

$$C_I(\omega_X) = \bigotimes_{e\in E(\Gamma)} [S \longrightarrow S[f_e^{-1}]] \otimes_S \omega_X.$$

It has only $H^r = \mathcal{N}$. For forest equations this is the tensor product of the one-variable principal-parts calculation.

A redundant $f \in I$ adds no cohomology. Localization of the supported finite jets at f is zero. Thus tensoring with $[S \to S[f^{-1}]]$ changes the complex by an acyclic cone. For a zero equation, the localization is zero and the extra factor is just S in degree zero. Therefore $C_I(\omega_X)[r] \simeq \mathcal{N}$, including repeated equations.

There is a scalar cochain map

$$\varepsilon : \mathcal{K}_\Gamma \longrightarrow \mathrm{Res}_S \mathcal{N}, \qquad \tau a \longmapsto a|_Z \delta_{\tau,0},$$

which is zero on positive exterior length. Normal annihilation makes it S-linear and a cochain map. It includes ω_Z as the normal socle, the part killed by I.

Every degree-zero morphism from $\mathcal{K}_\Gamma$ to this ordinary target kills negative cohomology. More generally, $\mathcal{N}[m]$ has cohomology only in degree $-m$. It cannot receive both a nonzero vacuum class in degree zero and a nonzero excess class in degree -1. Replacing complexes by resolutions, changing density by a unit, or retaining more normal derivatives leaves this degree argument unchanged.

A derived supported complex

Let $D = \Gamma(X, \mathcal{D}_X)$. Define

$$\mathcal{P}_\Gamma = L \otimes_k \Lambda(V \oplus T_\Gamma) \otimes_k D, \qquad |V| = |T_\Gamma| = -1. \tag{14.11}$$

For $v \in V$ and $t \in T_\Gamma$ put $\mathsf{a}(v) = F(v)$ and $\mathsf{a}(t) = \partial_t$. On an ordered wedge the differential is

$$d(\tau \otimes a_1 \wedge \cdots \wedge a_m \otimes Q) = \sum_{j=1}^{m} (-1)^{j-1} \tau \otimes a_1 \wedge \cdots \widehat{a}_j \cdots \wedge a_m \otimes \mathsf{a}(a_j) Q.$$

Multiplication by $\mathsf{a}(a_j)$ is on the left of Q. The differential is right D-linear.

The operators commute because $[\partial_t, F(v)] = F(v)(t) = 0$, so $d^2 = 0$. The tangent exterior factors impose the Spencer relations.

There is an actual S-linear cochain map

$$j_\Gamma : \mathcal{K}_\Gamma \longrightarrow \operatorname{Res}_S \mathcal{P}_\Gamma, \qquad \tau a\, v_1 \wedge \cdots \wedge v_p \longmapsto \tau \otimes (v_1 \wedge \cdots \wedge v_p) \otimes a. \tag{14.12}$$

Functions commute with the $F(v_i)$, which proves its cochain equation. The scalar source is not being given a differential action by this map.

Theorem 14.6.1. There are canonical right D-module isomorphisms

$$H^{-p}(\mathcal{P}_\Gamma) \cong \Lambda^p E \otimes_k \mathcal{N}.$$

On cohomology, j_Γ includes $\omega_Z \otimes \Lambda^p E$ at normal order zero. This inclusion is injective and its right D-span is the whole group. It retains the vacuum, every excess class and, by differentiation, every finite normal descendant.

Proof. Choose V_0 complementary to E and a basis of T_Γ. The closed factor ΛE splits off. The remaining right Koszul complex uses the commuting operators $u_1, \dots, u_r, \partial_{y_1}, \dots, \partial_{y_c}$.

Filter D by differential order, assigning weight zero to normal exterior generators and one to tangent exterior generators. Its symbol complex is the commutative Koszul complex of

$$u_1, \dots, u_r, p_{y_1}, \dots, p_{y_c} \quad \text{in } S[p_1, \dots, p_n].$$

This is a regular sequence. To solve a negative-degree cycle, lift a boundary for its leading symbol and subtract it. The filtration degree decreases and is bounded below in every cochain degree, so the process terminates.

The degree-zero quotient is

$$L \otimes_k D \Big/ \left(\sum_i u_i D + \sum_j \partial_{y_j} D \right).$$

Normal ordering identifies it with the transfer module $i_+\omega_Z$: the tangent density relations remove tangent derivatives and the normal derivatives give (14.9). Wedging with closed elements of E gives a

canonical map $\Lambda^p E \otimes H^0(\mathcal{P}_\Gamma) \to H^{-p}(\mathcal{P}_\Gamma)$. The splitting proves that it is an isomorphism, removing the choice of complement from the conclusion. Formula (14.12) sends the source to the normal socle. The jet basis proves injectivity, and right normal differentiation generates the entire target cohomology. □

Projection in exterior degree zero defines a right D-linear map $\pi : \mathcal{P}_\Gamma \to \mathcal{N}$ by $\pi(\tau \otimes Q) = \delta_{\tau,0} Q$. Its cochain equation is exactly the normal and tangent annihilation of $\delta_{\tau,0}$. Thus $\pi j_\Gamma = \varepsilon$. If $E \neq 0$, π is not a quasi-isomorphism: it kills the negative excess.

Universal differential induction is a further useful recipient,

$$\mathcal{U}_\Gamma = \mathcal{K}_\Gamma \otimes_S D = L \otimes \Lambda V \otimes D.$$

Normal ordering makes D free as a left S-module. Induction therefore preserves the scalar cohomology and is functorial for every S-linear equation map, including matrices with variable coefficients and unchanged density. Inclusion of tangent exterior degree zero gives $\mathcal{U}_\Gamma \to \mathcal{P}_\Gamma$. The latter complex imposes the tangent relations while retaining the excess.

In one normal coordinate, let $A = k[u]$ and $E_u = A[u^{-1}]/A$. Left-to-right conversion gives $(u^{-n} \otimes du)\partial_u = n u^{-n-1} \otimes du$. Thus $\delta_m = u^{-m-1} \otimes du$ agrees with (14.10), and $[D \xrightarrow{u\cdot} D]$ in degrees -1, 0 is its free right D-resolution. This compares the supported normal coefficient resolutions without changing their derivative actions.

Supported dual functionals

The lost excess still has a supported dual functional. On the linear normal chart, with $B = \mathcal{O}_Z$, one has

$$H^p \operatorname{RHom}_S(\mathcal{K}_\Gamma, \mathcal{N}) \cong B \otimes_k \Lambda^p E^*, \qquad 0 \le p \le \dim E. \tag{14.13}$$

All other degrees vanish.

To compute it, the bounded free source allows ordinary Hom. Its degree-p differential is $-(-1)^p \phi d$. Rescaling the dual basis in degree p by $(-1)^{p(p+1)/2}$ gives the normal cochain Koszul differential.

For one normal variable this is $[\mathcal{N} \xrightarrow{u} \mathcal{N}]$ in degrees 0, 1. Multiplication by u is surjective, with kernel $B\delta_0$. The right inverse $R\delta_m = \delta_{m+1}$ satisfies $uR = 1$ and $Ru = 1 - \iota p$, where p extracts δ_0.

These homotopies are B-linear. The socle inclusion is S-linear and a quasi-isomorphism. Tensoring the normal contractions, with their cochain signs, leaves the socle and ΛE^*. The density line cancels, proving (14.13).

For the triangle the functional

$$\ell(dy\,\xi_3) = \delta_{dy,0}, \qquad \ell(dy\,\xi_1) = \ell(dy\,\xi_2) = 0$$

is closed and evaluates nontrivially on $dy\,\eta$. A Hom boundary evaluates to zero on that closed class. It therefore defines a nonzero map $\mathcal{K}_\Gamma \to \mathcal{N}[1]$, which kills the degree-zero vacuum. Derivative order and cohomological degree remain different gradings.

Keeping differential-module maps adds the tangent de Rham complex. On $Z = (\mathbb{G}_m)^c$ the bounded free complex (14.11) gives

$$\operatorname{RHom}_D(\mathcal{P}_\Gamma, \mathcal{N}) \simeq (\Omega^\bullet(Z), d_{\mathrm{dR}}) \otimes_k \Lambda(E^*), \qquad |E^*| = 1.$$

The same normal contractions commute with tangent differentiation. The tangent Spencer dual has differential $-d_{\mathrm{dR}}$ after the preceding degree normalization. Multiplying tangent degree a by $(-1)^a$ gives d_{dR}. The resulting degree-m cohomology is $\bigoplus_{a+p=m} H^a_{\mathrm{dR}}(Z) \otimes \Lambda^p E^*$.

Indeed $d(y^j) = j\, y^j\, d\log y$ contracts every nonzero Laurent weight. Weight zero leaves 1, $d\log y$, and tensoring gives $H^\bullet_{\mathrm{dR}}(Z) = \Lambda^\bullet\langle d\log y_1, \dots, d\log y_c\rangle$.

Equation operations and tangent connections

Suppose $F : V \to A_0^*$ and $F' : V' \to A_0^*$ have the same image, and let $U : V \to V'$ be constant with $F = F'U$. The assignments

$$v \longmapsto Uv, \qquad t \longmapsto t, \qquad Q \longmapsto Q, \qquad \tau \longmapsto \tau$$

define a right D-linear map $\mathcal{P}_F \to \mathcal{P}_{F'}$. The normal differentials agree by $F = F'U$, and constant coefficients commute with tangent differentiation. Thus the map is a cochain map and the square with the two maps j commutes strictly. It acts on cohomology as $\Lambda^p(U|_E) \otimes I_{\mathcal{N}}$, so it intertwines the excess action and composes by matrix multiplication. It may kill an excess class when $U|_E$ does.

An invertible linear coordinate map gives the same comparison on open domains that it actually identifies. It pulls back the density, sends tangent vectors by the inverse coordinate map, and transforms every normal derivative by the chain rule. These operations compose strictly. Diagonal scalings and coordinate permutations preserve the torus. An arbitrary invertible linear map need not preserve it.

For example, in the triangle take $\lambda = 2$, $\mu = 4$ and $u = z_1 - 2z_2$, $v = z_2 - 4z_3$, $y = z_3$. Under

$$z_1 = 4w_1, \qquad z_2 = \tfrac{1}{2}w_2, \qquad z_3 = w_3,$$

put $u' = w_1 - \frac{1}{4}w_2$, $v' = w_2 - 8w_3$, $y' = w_3$. The equations give $u = 4u'$, $v = \frac{1}{2}v'$ and $\eta \mapsto 4\eta'$, where $\eta' = \xi_3' - \xi_1' - \frac{1}{4}\xi_2'$.

The density-normalized jets transform by

$$\delta_{dy,(m,n)} \longmapsto 4^{-m}2^{n}\delta_{dy',(m,n)}.$$

Indeed the normal Jacobian is 2 and the denominator factor is $4^{m+1}(1/2)^{n+1}$. The vacuum is unchanged. Omitting the Jacobian would already multiply it by $1/2$.

The tangent connection restricts variable equation operations. On the same triangle let an equation automorphism fix ξ_1, ξ_2 and send

$$\xi_3 \longmapsto (1 - y)\xi_1 + \lambda(1 - y)\xi_2 + y\xi_3.$$

It fixes the equations and sends η to $y\eta$.

In supported cohomology, however, $m = \eta\delta_{dy,0}$ satisfies $m\partial_y = 0$ whereas $(ym)\partial_y = -m \neq 0$. No right D-linear cochain map on the fixed $\mathcal{P}_\Gamma$ can induce this action. Homotopies do not change the induced cohomology map.

Raw induction $\mathcal{U}_\Gamma$ remains functorial for this scalar equation map. Removing its free tangent derivatives while retaining the map requires additional connection data.

Equation symmetries can give a second obstruction, even when determinants are unchanged. Repeat the third triangle edge and put $\eta_i = \xi_{i+2} - \xi_1 - \lambda\xi_2$, $i = 1, 2$. Fix the forest generators and act by $\eta_1 \mapsto t\eta_1$, $\eta_2 \mapsto t^{-1}\eta_2$. This automorphism fixes the equations, ambient coordinates, normal quotient and tangent density, and has determinant one.

If a target assigns it the identity at $t = 2$, a natural homogeneous comparison kills both excess classes. Their images would have to equal twice and half themselves. Equation, tangent and normal determinant twists leave this argument unchanged. The invariant top wedge is not killed by this argument.

For the ordinary module on the split torus, $\mathrm{End}_D(\mathcal{N}) = k$. An endomorphism sends its normal socle generator to $b\delta_{\tau,0}$. The normal relations force the socle, and the tangent relations force every derivative of b to vanish. Right normal differentiation then determines the whole map.

Consequently a coherent action of the full $\mathrm{SL}(E)$ for $\dim E = 2$ on any single shift $\mathcal{N}[m]$ is a character to $k^\times$, and is trivial. To see this last assertion directly, conjugation by $\mathrm{diag}(2, 1/2)$ makes every upper or lower elementary unipotent a commutator. These unipotents generate $\mathrm{SL}_2(k)$.

Thus this naturality obstruction applies to the ordinary target with its full equation action. A derived target carrying ΛE has the nontrivial excess action instead.

With three copies of the third edge, a cyclic permutation is even. A natural comparison with identity target action kills the nonzero difference $\xi_3 - \xi_4 = (1 - \sigma)\eta_1$.

A nontransverse clustering

Again take the gains 2, 4, 8 and restrict to

$$\iota_Y : Y = (\mathbb{G}_m)^2 \longrightarrow X, \qquad (x, y) \longmapsto (2x, x, y).$$

This is $u = 0$, and $g = x - 4y$ is the surviving normal coordinate. The three equations become $(0, g, 2g)$. Writing $\zeta = \xi_1$ and $\eta = \xi_3 - \xi_1 - 2\xi_2$, one gets

$$\iota_Y^* K_\Gamma = K(g) \otimes \Lambda(\zeta, \eta).$$

The excess η is inherited. ζ is created by this nontransverse restriction.

Let $\mathcal{N}_X$ have normal coordinates (u, v) and let $\mathcal{N}_Y$ have normal coordinate $g = v$, both with density dy along the common stratum. For the derived pullback of underlying modules,

$$\mathcal{O}_Y \otimes^{\mathbb{L}}_{\mathcal{O}_X} \mathrm{Res}\, \mathcal{N}_X \simeq \mathcal{N}_Y[1]. \tag{14.14}$$

This convention is an $\mathcal{O}$-module pullback, rather than a dimension-normalized differential inverse image. Resolve $\mathcal{O}_Y$ by $[\mathcal{O}_X \xrightarrow{u} \mathcal{O}_X]$. After tensoring, multiplication by u is surjective on $\mathcal{N}_X$ and has kernel $\bigoplus_{n\geq 0} k[y^{\pm 1}]\delta_{dy,(0,n)}$. The residual v, ∂_v, y, ∂_y actions are exactly those of $\mathcal{N}_Y$, proving (14.14) and retaining every finite v-descendant.

Under the derived pullback of ε, the new class $dy\,\zeta$ maps to minus the supported vacuum in degree -1. Indeed, if $d\theta = u$ is the resolution generator, then $dy(\xi_1 - \theta)$ is a closed lift and maps to $-\theta\delta_{dy,0}$. The inherited $dy\,\eta$ already maps to zero before restriction, so still maps to zero. Thus inherited excess and a new Tor class have different images. No evaluation of a distribution on its singular support has been used.

Changing the third gain to 16 gives $d(\xi_3 - \xi_1 - 2\xi_2) = -8z_3$. The primitive $-(\xi_3 - \xi_1 - 2\xi_2)/(8z_3)$ of the unit remains a primitive after this clustering, with $z_3 = y$. Both scalar complexes contract and both supports are empty.

Contact differentials and the geometric comparison

A meromorphic coefficient map also needs its contact differential. Let $C, \mathcal{L}, J$ be cochain complexes, let $W : C \to \mathcal{L}$ be a cochain map, and suppose a supported extension has graded form $\mathcal{L} \oplus J$ and differential

$$d(a, s) = (d_{\mathcal{L}} a, d_J s + Ra), \qquad d_J R + R d_{\mathcal{L}} = 0.$$

Its pullback along W has the actual differential

$$E_W = C \oplus J, \qquad d(c, s) = (d_C c, d_J s + RWc).$$

The map $(c, s) \mapsto (Wc, s)$ is a cochain map because $d_J RW + RW d_C = 0$. A map to the localized component alone omits this term.

With two independent normals, the localization complex has both faces and their corner:

$$S \longrightarrow S[f^{-1}] \oplus S[g^{-1}] \longrightarrow S[(fg)^{-1}], \qquad d_0(a) = (a, a), \quad d_1(b, c) = c - b.$$

The sign makes $d_1 d_0 = 0$ and determines the corresponding connecting signs in the supported extension.

The maps ε, j_Γ and the supported dual functional now have different, explicit meanings. An independently specified geometric coefficient recipient must still receive a map from $\mathcal{P}_\Gamma$ preserving the required excess representation, tangent connection and vacuum density. For noninvertible root substitutions, it must give the transfer or supported correspondence and its action on the changing excess. Compatibility is required with successive substitutions, regular/localized/supported triangles and field attachments.

The variable equation map and (14.14) are concrete tests. The ordinary target cannot satisfy the excess requirement. The constructed derived coefficient target meets it for the constant equation and invertible coordinate operations above.

14.7 The analytic Green difference equation

Let $q > 1$ and let $Z(t)$ be rational with $Z(0) = 1$. The product

$$K(t) = \prod_{j \geq 1} Z(q^{-j} t)$$

defines a meromorphic function on $\mathbb{C}$, regular at zero with $K(0) = 1$. On a compact set, its tail factors are $1 + O(q^{-j})$ uniformly. Sufficiently far along the tail they have neither zeros nor poles. The product and its derivatives therefore converge normally there. The finite initial product is rational.

Reindexing gives $K(qt) = Z(t)K(t)$. The normalized holomorphic solution near zero is unique: the quotient of two such solutions has $H(qt) = H(t)$, and each positive-degree Taylor coefficient is killed by $q^n - 1$. For torsion Hall data this is the analytic function with formal expansion K_G.

Let $D_Z \subset \mathbb{C}^\times$ be the finite set of zeros and poles of Z, and put $U = \mathbb{C}^\times \setminus q^{\mathbb{Z}} D_Z$. Define

$$Z_n(t) = \begin{cases} \prod_{j=0}^{n-1} Z(q^j t), & n > 0, \\ 1, & n = 0, \\ \prod_{j=n}^{-1} Z(q^j t)^{-1}, & n < 0. \end{cases}$$

Then $Z_{m+n}(t) = Z_m(q^n t) Z_n(t)$. These factors also define the rank-one rational difference module with relation $\sigma e = Z(t)e$, where $\sigma f(t) = f(qt)\sigma$. Evaluation $e \mapsto K$ is a solution in meromorphic functions.

Proposition 14.7.1. The action $(t, v) \mapsto (q^n t, Z_n(t)v)$ descends the trivial line on U to a holomorphic line on $U/q^{\mathbb{Z}}$. The connection $d - d \log K$ descends with it, and K is a nowhere-zero horizontal section. For $Z(t) = ((1 - t)(1 - qt))^{-1}$ this analytic trivialization cannot be obtained from a rational function.

Proof. The action of $q^{\mathbb{Z}}$ on $\mathbb{C}^\times$ is free and properly discontinuous: logarithmic radius changes by $\log q$. Its zero/pole orbits are locally finite. Every Z_n is a holomorphic unit on U, so the cocycle gives transition isomorphisms. In logarithmic coordinates the quotient is $\mathbb{C}/(2\pi i \mathbb{Z} + \log q\, \mathbb{Z})$ with finitely many points removed.

The difference equation gives $q^*(d \log K) = d \log K + d \log Z$, the connection transformation. The product has no zeros or poles on U, and the same equation makes K the stated section.

If a rational R satisfied $R(qt)/R(t) = Z(t)$, write $r_b = \operatorname{ord}_b R$. The divisor of its ratio has order $r_{qb} - r_b$ at b. Its sum on each nonzero q-orbit is zero because the divisor of R has finite support. For the displayed Z, the orbit $q^{\mathbb{Z}}$ instead has two simple poles, at $1, q^{-1}$, and no zeros. Its sum is -2, a contradiction. □

Thus this line with connection is analytically trivial on the punctured elliptic quotient. The rational difference equation has not produced nontrivial monodromy there. Nor is the quotient by $q^{\mathbb{Z}}$ a quotient by every Frobenius ratio: a larger gain group need not act properly discontinuously.

Between polynomial Hall states, any finite product of the native Green half-vertices has a matrix element equal to a finite Laurent polynomial times factors $K(w/z)$. Normal ordering gives one factor for each annihilation–creation pair. The remaining matrix element is finite because its two test states have finite weight. The product construction continues these expressions meromorphically away from their locally finite pole divisors. Both bracketings give the same expression, proving associativity on these matrix elements. This domain statement asserts no boundedness of an unsmeared half-vertex on the entire Green Hilbert space.

14.8 HALL MASS AND POSITIVE STATES

Put $h_m = \sum_{|\lambda|=m} u_\lambda$, $h_0 = 1$, and $\mathcal{H}_Q(t) = \sum_{m\ge 0} h_m t^m$. The group-like Hall series satisfies

$$\mathcal{H}_Q(t) = \exp\left(\sum_{r\ge 1} \frac{p_r t^r}{r}\right). \tag{14.15}$$

Indeed the coefficient of $u_M \otimes u_N$ in $\Delta \sum_L u_L$ is

$$\sum_L \frac{a_M a_N}{a_L} F^L_{M,N} = \frac{|\operatorname{Ext}^1(M,N)|}{|\operatorname{Hom}(M,N)|} = 1.$$

The first equality is orbit–stabilizer for extension classes, and the last uses the zero torsion Euler form. Hence $\mathcal{H}_Q$ is group-like and its logarithm is primitive. In degree r it is a multiple of p_r.

Its indecomposable coefficient is $\sum_{j=1}^{r} (-1)^{j-1}\binom{r-1}{j-1}/j = 1/r$: each composition of r gives the unique corresponding flag in the indecomposable module. The normalization of p_r proves (14.15).

In the nonnegative weight completion set

$$v_s = \prod_x \mathcal{H}_{q^{d_x}}(s^{d_x}) = \sum_{[M]} s^{\deg M} u_M, \qquad \deg M = \sum_x d_x \operatorname{length} M_x.$$

The sum ranges over finite-support torsion objects. Their Green squared norm is a_M^{-1}.

THEOREM 14.8.1. Suppose $N_n \le C_0 q^n$. For $0 \le r < 1$,

$$K_G(r) = \sum_{[M]} \frac{r^{\deg M}}{a_M} = \exp\left(\sum_{n\ge 1} \frac{N_n r^n}{n(q^n-1)}\right) < \infty.$$

Thus v_s belongs to the Green Hilbert completion for $|s|^2 = r < 1$, with squared norm $K_G(r)$. For $0 < r < 1$ the weights $\mu_r(M) = r^{\deg M}/(a_M K_G(r))$ form a probability measure. For $0 < r < q^{-1}$,

$$\frac{\mu_{qr}(M)}{\mu_r(M)} = \frac{q^{\deg M}}{Z_C(r)}, \qquad \mathbb{E}_{\mu_r}[q^{\deg M}] = Z_C(r). \tag{14.16}$$

Proof. The factorial primitive norms give

$$\|\mathcal{H}_Q(s)\|_G^2 = \exp\left(\sum_{r\ge 1} \frac{|s|^{2r}}{r(Q^r-1)}\right).$$

Multiply over the closed points and write $n = rd_x$. The resulting logarithm is the one in the theorem. The growth assumption bounds its summands by a constant times r^n/n, so it converges for $r < 1$. Orthogonality in the Hall basis gives the mass sum. At real nonnegative r, partial products and partial sums increase to their common finite limit.

For $r < q^{-1}$, both series converge. Subtracting their logarithms gives

$$\frac{K_G(qr)}{K_G(r)} = Z_C(r).$$

Substitution into the two densities proves their likelihood ratio. Summing proves the expectation. □

A finite map $C \to \mathbb{P}^1$ gives the required growth bound by bounding every geometric fiber by its degree. This argument uses no spectral purity. At $r = 0$ the measure is concentrated on the zero object.

Local independence and finite support

Write $K_x(r) = \sum_\lambda r^{d_x|\lambda|}/a_\lambda(q^{d_x})$. The local measures $r^{d_x|\lambda|}/(a_\lambda K_x(r))$ are independent and their product gives μ_r. It introduces no infinite-support objects almost surely, since

$$\sum_x \Pr(M_x \ne 0) = \sum_x (1 - K_x(r)^{-1}) \le \sum_x \log K_x(r) = \log K_G(r) < \infty.$$

The first Borel–Cantelli lemma leaves only finitely many nonzero local factors almost surely. Independence is present in the construction, though this last implication does not require it.

The unit vector $\Psi_s = K_G(|s|^2)^{-1/2} v_s$ defines

$$\omega_s(P) = \langle \Psi_s, P\Psi_s\rangle_G$$

on finite polynomials in finitely many creators and Green annihilators. Every selected mode has a finite coherent parameter, so every such moment exists. The mean is

$$D_{x,m} v_s = \frac{s^{md_x}}{q^{md_x}-1} v_s.$$

Normal ordering gives all correlations from this mean and the unchanged Green commutators. In particular $\omega_s(P^{*G}P) \ge 0$. Infinite smearings require their own test functions and convergence domain.

The degree Hamiltonian and the ordinary trace

Let $L_0 u_M = (\deg M) u_M$ with its closed self-adjoint domain, specified by square summability after multiplication by $\deg M$. Its nonnegative integer eigenvalues have the finite Hall weight multiplicities. Equation (14.16) becomes the quadratic-form expectation

$$\|q^{L_0/2}\Psi_s\|_G^2 = Z_C(|s|^2), \qquad q|s|^2 < 1.$$

This requires the form domain of q^{L_0}. Membership in the smaller operator domain has not been assumed. The degree spectrum has been specified independently of zeta zeros.

The ordinary Gibbs trace instead counts orthonormal states once:

$$\operatorname{Tr}_{\mathcal{F}_C}(e^{-\beta L_\circ}) = \prod_{x,r\geq 1}(1-e^{-\beta r d_x})^{-1} = \prod_{r\geq 1} Z_C(e^{-\beta r}), \qquad \beta > \log q.$$

Its logarithm is $\sum_{n\geq 1} N_n/[n(e^{\beta n}-1)]$, which converges under the growth assumption. The Gibbs density is positive and trace class. In contrast, $K_G(r) = \prod_{j\geq 1} Z_C(q^{-j}r)$ is the automorphism-weighted mass and the norm of a coherent vector. The Green norm of the unnormalized basis does not insert automorphism weights into the ordinary trace. Neither positive functional has been identified with the complete signed Weil functional.

A stochastic extension product

For fixed torsion objects M, N, choose an extension class uniformly in $\operatorname{Ext}^1(M,N)$. Its middle term has law

$$P(L \mid M,N) = \frac{|\operatorname{Ext}^1(M,N)_L|}{|\operatorname{Ext}^1(M,N)|} = \frac{a_M a_N}{a_L} F^L_{M,N}.$$

The weights sum to one because the torsion Euler form is zero. In the rescaled basis $v_M = a_M u_M$, Hall multiplication reads $v_M v_N = \sum_L P(L \mid M,N) v_L$. Consequently

$$\sum_L P(L \mid M,N)P(K \mid L,P) = \sum_L P(L \mid N,P)P(K \mid M,L).$$

Both sides are $a_M a_N a_P / a_K$ times the same three-step flag count. Higher regroupings are the same higher flag counts, so this gives all finite stochastic associativity identities. An enhanced primitive object still requires its cochain or sheaf-level comparison.

14.9 Prime-indexed Macdonald and Hall constructions

The decomposition by closed points suggests a comparison with symmetric functions at each prime. This construction has two independent choices: the set of tensor factors and the parameters inside each factor. A single pair of parameters cannot equal several distinct pairs simultaneously. This elementary fact does not prohibit a family of Macdonald spaces or a map of their underlying algebras. The relevant obstruction appears only after specifying which gradings and operations the map must preserve.

Pairings and the degree obstruction

Let $K = \mathbb{Q}(q_p, t_p : p \text{ prime})$ be the field of rational functions in algebraically independent variables q_p, t_p. Each rational function involves finitely many variables. Put $\Lambda_{(p)} = K[p_1^{(p)}, p_2^{(p)}, \dots]$, with $\deg p_r^{(p)} = r$. For a partition λ, let $m_r(\lambda)$ be the number of parts equal to r, and set

$$p_\lambda^{(p)} = \prod_r (p_r^{(p)})^{m_r(\lambda)}, \qquad z_\lambda = \prod_r r^{m_r(\lambda)} m_r(\lambda)!.$$

Definition 14.9.1 *(Macdonald pairing).* The symmetric bilinear form on $\Lambda_{(p)}$ is specified by

$$\langle p_\lambda^{(p)}, p_\mu^{(p)} \rangle_p = \delta_{\lambda\mu} z_\lambda \prod_{j=1}^{\ell(\lambda)} \frac{1-q_p^{\lambda_j}}{1-t_p^{\lambda_j}}.$$

This is the normalization of Macdonald, *Symmetric Functions and Hall Polynomials*, VI, equation (1.5).

Definition 14.9.2. Define

$$\Lambda^{\mathrm{prim}} = \bigotimes_{p}{}' \Lambda_{(p)} = K[p_r^{(p)} : p \text{ prime},\ r \geq 1].$$

The prime denotes the algebraic tensor product with unit 1 in all but finitely many factors. Give it the tensor product pairing and total degree $\deg p_r^{(p)} = r$.

The form is nondegenerate. Every polynomial lies in a finite tensor product, where its power-sum monomials have nonzero diagonal norms. For real parameters $0 < q_p, t_p < 1$, the same diagonal formula defines a positive Hermitian form on the complex polynomial space, with conjugation in the first argument. Positivity holds for every finite linear combination. It requires no infinite tensor product completion.

Proposition 14.9.3. Let $\Lambda = K[p_1, p_2, \ldots]$, graded by $\deg p_r = r$. A tensor product of two or more $\Lambda_{(p)}$ admits no injective degree-preserving K-linear map to Λ. After forgetting degree, Λ^{prim} and Λ are isomorphic as K-algebras.

Proof. The degree-one space of Λ has dimension one. Each retained prime contributes an independent degree-one generator to the source, proving the obstruction. For the second assertion, choose a bijection b from the pairs (p, r) to the positive integers. The assignments $p_r^{(p)} \mapsto p_{b(p,r)}$ and their inverse extend uniquely to mutually inverse polynomial algebra maps. They need not preserve degree, the pairing or any marked operator. □

Thus $q_p = p^{-s}$ and $t_p = p^{-s-1}$, for real $s > 0$, give a well-defined family of positive pairings. They do not specify one constant pair (q, t) independent of p. An arithmetic comparison must say whether it preserves these pairings and the prime labels. Parameter variation alone is not a nonexistence theorem.

Adjoints, weights and Macdonald bases

Write $c_{p,r} = (1 - q_p^r)/(1 - t_p^r)$. Let $M_{p,r}$ denote multiplication by $p_r^{(p)}$. Its adjoint is

$$D_{p,r} = r c_{p,r} \partial_{p_r^{(p)}}, \qquad [D_{p,r}, M_{\ell,s}] = \delta_{p\ell}\delta_{rs} r c_{p,r} I.$$

Indeed, the squared norm of $(p_r^{(p)})^m$ is $m!(r c_{p,r})^m$. Comparing consecutive norms proves adjunction. This also identifies exactly which coefficient changes when the pairing changes.

Proposition 14.9.4. At $t_p = q_p$, one has $c_{p,r} = 1$. Along $t_p = q_p^2$,

$$c_{p,r} = \frac{1}{1 + q_p^r} \longrightarrow 1 \quad \text{as } q_p \longrightarrow 0.$$

Both give the ordinary Heisenberg squared norm r for $p_r^{(p)}$ in the stated specialization or limit. For any positive numbers w_p, replacing $c_{p,r}$ by w_p^r defines another positive form, with

$$\|p_\lambda^{(p)}\|^2 = z_\lambda w_p^{|\lambda|}.$$

Proof. The two formulas for $c_{p,r}$ follow by cancellation. Every power-sum monomial then has the claimed factorial norm. All its factors are positive in the weighted form. □

The choice $w_p = \log p$ yields $\|p_r^{(p)}\|^2 = r(\log p)^r$. It is extra metric data. Over the real numbers, the map $p_r^{(p)} \mapsto w_p^{r/2} p_r^{(p)}$ is an algebra isometry from this weighted space to the ordinary Hall inner-product space. It need not commute with the usual translation $T\,p_r^{(p)} = r\,p_{r+1}^{(p)}$: its two orders of application give factors $w_p^{r/2}$ and $w_p^{(r+1)/2}$. A conformal comparison must also transport translation and the chosen conformal vector.

Example 14.9.5. In the ordinary Heisenberg pairing, $p_r^{(p)}/\sqrt{r}$ has norm one. This is a statement about a Fock vector. For adjacency on the $(p+1)$-regular tree, the root spectral measure instead has moments

$$\int x^2\,d\nu_p = p+1, \qquad \int x^4\,d\nu_p = (p+1)(2p+1).$$

There are $p+1$ two-step returns. A four-step return either returns at step two, giving $(p+1)^2$ paths, or first reaches distance two, giving $(p+1)p$ paths. Odd moments vanish because the tree is bipartite. These state moments differ from the norms of individual power-sum vectors. Their Hecke recurrence is the one in Proposition 13.5.3.

The tensor product basis

Let m_λ be the monomial symmetric function. In dominance order, $\mu \le \lambda$ means $\sum_{i\le j}\mu_i \le \sum_{i\le j}\lambda_i$ for every j, with equal total size. Macdonald's theorem, VI (4.7), gives the unique orthogonal basis $P_\lambda(x;q,t)$ over $\mathbb{Q}(q,t)$ which is monic triangular in m_λ for this order. At specialized parameters the coefficients must remain defined.

Definition 14.9.6. For a family of partitions with finite support, put

$$P^{\mathrm{arith}}_{(\lambda_p)} = \bigotimes_p{}' P_{\lambda_p}(x^{(p)};q_p,t_p), \qquad P_\varnothing = 1.$$

These tensors form an orthogonal K-basis. Both expansion and linear independence reduce to a finite set of prime factors. For two primes, the degree-two basis consists of

$$P_{(2)}\otimes 1,\quad P_{(1,1)}\otimes 1,\quad P_{(1)}\otimes P_{(1)},$$
$$1\otimes P_{(2)},\quad 1\otimes P_{(1,1)}.$$

A single symmetric-function space has degree-two dimension two, with basis p_2, p_1^2. Symmetric polynomials in two variables also have dimension two in that degree. The three ordinary monomials x_1^2, x_1x_2, x_2^2 belong to the full polynomial space, whose symmetric subspace has basis $x_1^2+x_2^2$, x_1x_2.

DAHA and its geometric comparisons

The double affine Hecke algebra, abbreviated DAHA, acts on a fixed number of Laurent variables. This number is independent of the set of primes indexing a tensor product. The distinction can be seen directly in its polynomial representation.

Theorem 14.9.7 *(Polynomial representation).* Over $K_0 = \mathbb{Q}(q^{1/2}, t^{1/2})$, the DAHA of type GL_n has a faithful representation on $K_0[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$. In the convention of Schiffmann–Vasserot, arXiv:0905.2555, section 1.4, its operators are

$$\begin{aligned} X_i &= x_i, \\ T_i &= t^{1/2} s_i + \frac{t^{-1/2} - t^{1/2}}{x_i/x_{i+1} - 1}(s_i - 1), \\ Y_i &= T_i \cdots T_{n-1} \omega T_1^{-1} \cdots T_{i-1}^{-1}. \end{aligned}$$

Here s_i exchanges x_i, x_{i+1}, $\partial_i f(x) = f(x_1, \dots, qx_i, \dots, x_n)$ and $\omega = s_{n-1} \cdots s_1 \partial_1$. The spherical subalgebra is $e\mathbb{H}_n e$, where e is the normalized finite Hecke symmetrizer.

This is the standard faithful polynomial representation theorem. The displayed divided difference preserves Laurent polynomials, since $(s_i - 1)f$ is divisible by $x_i - x_{i+1}$. On symmetric polynomials the first spherical operator becomes

$$\sum_{i=1}^{n} \left(\prod_{j \neq i} \frac{t^{-1/2} x_i - t^{1/2} x_j}{x_i - x_j} \right) \partial_i .$$

Its eigenvectors are $P_\lambda(x; q, t^{-1})$, in this convention, with eigenvalues $t^{-(n-1)/2} \sum_i q^{\lambda_i} t^{i-1}$. The inverse in the second Macdonald parameter is part of the normalization.

Remark 14.9.8. The Weyl group acts inside each alphabet. Adams operations can also be defined factor by factor by $\psi_m(p_r^{(p)}) = p_{mr}^{(p)}$, fixing the coefficient field. Operators on distinct tensor factors commute. These definitions pose no incompatibility between prime labels and DAHA rank. An intertwiner between the two operator families remains additional data, as does a geometric interpretation of the factorwise Adams operations.

Two geometric results specify which Hall algebra enters this comparison. Let $T = (\mathbb{C}^\times)^2$ act on the affine plane, and let $L = \bigoplus_{n \geq 0} K_T(\mathrm{Hilb}^n \mathbb{A}^2)$ with equivariant parameters localized to their fraction field. Let H be the convolution algebra generated by the tautological line-bundle powers on adjacent nested Hilbert schemes and the Adams operations of the tautological bundles.

Theorem 14.9.9 *(Schiffmann–Vasserot).* In arXiv:0905.2555, Theorem 3.1 identifies H with the central specialization $\mathcal{E}_c$ of the elliptic Hall algebra at $c = (1, q^{1/2} t^{1/2})$. Its representation on L is the standard symmetric-function representation.

The stable centrally extended degenerate spherical DAHA SH^c of arXiv:1202.2756 has an integral form at $\kappa = 1$. For its specialization SH_1^c, Theorem F.10 gives

$$\mathrm{SH}_1^c \cong U(W_{1+\infty}) \otimes_{\mathbb{C}} \mathbb{C}[c_1, c_2, \dots].$$

Here $W_{1+\infty}$ is the universal central extension of the Lie algebra of regular differential operators on the circle.

The second construction uses the degeneration $q, t \longrightarrow 1$ with $t = q^{-\kappa}$, together with its integral form and central parameters. Its result is an enveloping algebra. A bare substitution $q = 1$ in a rational shuffle formula does not specify this degeneration. Neither geometric theorem constructs the arithmetic Hochschild trace class $\mathcal{V}^{\mathrm{prim}}$.

A rational shuffle construction

One can construct a prime-indexed shuffle algebra without making the geometric identification. In the field K above, put

$$\omega_p(x, y) = \frac{(1 - q_p x/y)(1 - t_p^{-1} x/y)}{(1 - x/y)(1 - q_p t_p^{-1} x/y)}.$$

For $n \geq 0$, let $\mathscr{S}_{p,n} = K(x_1, \ldots, x_n)^{S_n}$, with $\mathscr{S}_{p,0} = K$. For symmetric rational functions $f \in \mathscr{S}_{p,r}$ and $g \in \mathscr{S}_{p,s}$, define

$$(f * g)(x_1, \ldots, x_{r+s}) = \sum_{\substack{I \sqcup J = \{1,\ldots,r+s\} \\ |I| = r}} f(x_I) g(x_J) \prod_{i \in I,\, j \in J} \omega_p(x_i, x_j).$$

This is the sum over subsets, with no factorial normalization.

Definition 14.9.10. Set $\mathscr{S}_p = \bigoplus_{n \geq 0} \mathscr{S}_{p,n}$ and $\mathscr{S}^{\mathrm{prim}} = \bigotimes'_p \mathscr{S}_p$, with unit 1 in degree zero. Separately, a specified family of spherical elliptic Hall algebras $\mathrm{SH}(q_p, t_p)$ defines

$$\mathbb{H}^{\mathrm{arith}} = \bigotimes_p{}' \mathrm{SH}(q_p, t_p).$$

All factors are extended to the common coefficient field.

PROPOSITION 14.9.11. The displayed shuffle product is associative and unital. In $\mathscr{S}^{\mathrm{prim}}$, factors at distinct primes commute, and their cross-prime shuffle coefficient is one. The same commutation holds in $\mathbb{H}^{\mathrm{arith}}$ by its tensor product definition.

Proof. Both orders of a triple shuffle sum over ordered decompositions $I \sqcup J \sqcup L$. Their summand is $f(x_I) g(x_J) h(x_L)$ times the kernels for $I \times J$, $I \times L$ and $J \times L$. Each pair occurs once in both expressions. The scalar kernels commute, proving associativity. The empty block proves the unit law. Multiplication in distinct tensor factors introduces no kernel. □

This proves a property of the specified tensor product. It does not forbid other arithmetic constructions with cross-prime operations. Identifying a shuffle subalgebra with a positive elliptic Hall half requires its presentation and wheel conditions. Recovering the full algebra also requires the negative and Cartan parts and their relations. Associativity of the rational kernel alone proves none of these identifications.

PROPOSITION 14.9.12. The algebraic prime tensor space carries the nondegenerate Macdonald pairing, the tensor Macdonald basis and the factorwise Adams operations above. Its independent parameters are compatible with a prime-indexed family of shuffle or spherical Hall algebras. For the rational shuffle construction, there are no mixed-prime coefficients beyond tensor multiplication.

Proof. Each tensor and each operation uses finitely many factors. Nondegeneracy and the basis assertion reduce to those factors. The Adams substitutions preserve polynomial multiplication and satisfy $\psi_m \psi_n = \psi_{mn}$. The final assertion is Proposition 14.9.11. □

An arithmetic realization must still construct a map to the Hochschild trace carrier and state which operators, pairing, grading and trace functional it preserves. The polynomial space and its prime labels supply a possible domain for such a map. They do not identify its target.

14.10 A geometric Hall kernel on finite differential jets

The curve construction uses a finite field cardinality q. The following example uses a separate nilpotent parameter ϵ. It compares the equivariant cohomological Hall algebra of the three-loop quiver with *zero potential* to an explicitly constructed field algebra on differential jets. The critical algebra for $\mathrm{Tr}(Z[X,Y])$ has different coefficients and requires a further oriented map.

Let $k = \mathbb{Q}(a,b)$, $h = a + b$, and $R_N = k[\epsilon]/(\epsilon^N)$, $N \geq 1$. Give the three arrows weights $\epsilon a, \epsilon b, -\epsilon h$. This is a base change of the two-dimensional equivariant parameter ring. It inverts neither ϵ nor $h\epsilon$. Put

$$\begin{aligned} \ell &= \epsilon h, & P(s) &= s^2 - \ell^2, \\ K_{\mathrm{o}}(s) &= \frac{(s+\epsilon a)(s+\epsilon b)(s-\ell)}{s}, & E_{\mathrm{o}}(s) &= K_{\mathrm{o}}(s)^{-1}, \\ E_{\mathrm{pl}}(s) &= \frac{s(s+\ell)}{(s+\epsilon a)(s+\epsilon b)}, & r(s) &= \frac{E_{\mathrm{pl}}(s)}{E_{\mathrm{pl}}(-s)}. \end{aligned}$$

Then $E_{\mathrm{o}} = E_{\mathrm{pl}}/P$ and $E_{\mathrm{o}}(s)/E_{\mathrm{o}}(-s) = r(s)$. Expand in ϵ first, with the indicated spectral variable inverted.

The charge-d coefficient algebra is $A_d = R_N[x_1, \dots, x_d]^{\mathfrak{S}_d}$. The representation space before base change is $\mathrm{End}(\mathbb{C}^d)^3$. Its equivariant cohomology has this polynomial description.

The invariant-subspace correspondence gives

$$(F * G)(X) = \sum_{\substack{I \sqcup J = [d+e] \\ |I| = d}} F(X_I)\, G(X_J) \prod_{i \in I,\, j \in J} K_{\mathrm{o}}(x_j - x_i). \tag{14.17}$$

There is no factorial divisor in this proper flag pushforward.

At a coordinate flag, the three moving arrow Euler factors are $y - x + \epsilon a$, $y - x + \epsilon b$, $y - x - \ell$. The flag tangent factor is $y - x$. Their ratio is K_{o}. This is the shuffle calculation of Theorem 12.8.1 with these weights.

Proposition 14.10.1. The product (14.17) is polynomial and associative. Its inverse-Euler spectral coefficient map is

$$\Delta^{\mathrm{o}}_{d,e}(s)F = \prod_{i,j} E_{\mathrm{o}}(s + y_j - x_i)\, F(x - s, y) \in (A_d \otimes_{R_N} A_e)((s^{-1})). \tag{14.18}$$

It is rationally coassociative. In charge split $(1,1)$,

$$1 * 1 = 2\big((y-x)^2 + \epsilon^2(ab - h^2)\big), \qquad \Delta^{\mathrm{o}}_{1,1}(s)1 = E_{\mathrm{o}}(s + y - x).$$

Proof. Proper pushforward is defined before equivariant localization, so the shuffle sum is polynomial. Directly, residues on each possible root diagonal cancel between the two flags exchanged there. Either bracketing of three factors sums over the same ordered three-block partitions with one K_{o} per ordered cross-block pair. This proves associativity with the displayed normalization.

The inverse Euler factors and translation of the first block give (14.18). For a triple split, both composites have one factor for each pair, at parameters $s, t, s+t$, and evaluate the input at $(x - s - t, y - t, z)$. They agree before Laurent expansion. Finally $K_{\mathrm{o}}(v) + K_{\mathrm{o}}(-v)$ gives the displayed product in charge two. □

With roots and ϵ of degree two, a root polynomial of degree m in charge d has shifted Hall degree $2m - 2d^2$. The product is homogeneous and all charges have even parity. This grading is distinct from the cochain grading, which places the differential Hopf states below in degree zero.

There are two reasons to change the polynomial carrier before constructing these fields. Before nilpotent base change, take the three arrow weights to be 1, 1, -2. Then

$$r(s) = \frac{(s+2)(s-1)^2}{(s-2)(s+1)^2}, \qquad \operatorname{Res}_{s=2} r(s)\, ds = \frac{4}{9}.$$

Its expansion has an infinite nonzero tail. A coefficientwise injective marking preserves that tail. Thus the unchanged exchange cannot become a finite Laurent polynomial on the same pair of states. Completing only the charge grading leaves this charge-two obstruction unchanged, and a modification supported only at $s = 0$ cannot cancel the pole at $s = 2$.

Also, if $T^M a = 0$ and $Y(Ta, z)b = \partial_z Y(a, z)b$, then $\partial_z^M Y(a, z)b = 0$. Over a $\mathbb{Q}$-algebra the falling factorial of a negative integer is a unit. A lower-truncated Laurent series with this property is therefore a polynomial of degree less than M. The root derivative $-\sum_i \partial_{x_i}$ is locally nilpotent on A_d. Finite jets reverse this behavior: the transpose derivative raises jet order.

A difference primitive and its Hopf form

We first realize the normalized factor E_{pl}. There is a unique formal series $F(z) \in (\epsilon/z)k[[\epsilon/z]]$ satisfying

$$F(z - \ell) - F(z) = \log E_{\mathrm{pl}}(-z).$$

To prove it, write $t = \epsilon/z$, $F = \sum_{n\geq 1} f_n t^n$, and $g_j = (a^j + b^j - h^j)/j$. The right side is $\sum_{j\geq 2} g_j t^j$. Comparison of coefficients gives the recurrence

$$f_{j-1} = \frac{g_j - \sum_{n=1}^{j-2} f_n \binom{j-1}{j-n} h^{j-n}}{(j-1)h} \qquad (j \geq 2).$$

Its denominator is a unit in k, and no division by ϵ occurs. This construction uses h^{-1} and has not been specialized to $h = 0$. The identity $E_{\mathrm{pl}}(z - \ell) = E_{\mathrm{pl}}(-z)$ shows that $-F(-z)$ solves the same equation. Uniqueness makes F odd.

With $\varpi(z) = \exp F(z)$, this gives

$$\varpi(z)\varpi(-z) = 1, \qquad \frac{\varpi(z-\ell)}{\varpi(z)} = E_{\mathrm{pl}}(-z), \qquad \frac{\varpi(z)^2}{\varpi(z-\ell)\varpi(z+\ell)} = r(z). \tag{14.19}$$

For example, $F(z) = -(ab/h)\epsilon z^{-1} + (a^2b^2/(6h))\epsilon^3 z^{-3} + O(\epsilon^5)$. Reduce all these series modulo ϵ^N.

Take the commutative, cocommutative Hopf algebra

$$H_N = R_N[g, g^{-1}, b_0, b_1, \ldots],$$

where g is group-like, the b_i are primitive, $\varepsilon(g) = 1, \varepsilon(b_i) = 0, Sg = g^{-1}$ and $Sb_i = -b_i$. The derivation $T_H g = g b_0, T_H b_i = b_{i+1}$ is also a coderivation and satisfies $\varepsilon T_H = 0$. Every algebraic coproduct is finite.

For integers m, n and finite lists of formal variables u_i, v_j, define a bilinear form by differentiation at zero in those variables:

$$\rho_z^H\left(g^m e^{\sum_i u_i b_i}, g^n e^{\sum_j v_j b_j}\right) = \exp B_z^H,$$
$$B_z^H = mnF + n\sum_i u_i F^{(i+1)} + m\sum_j (-1)^{j+1} v_j F^{(j+1)} + \sum_{i,j} (-1)^{j+1} u_i v_j F^{(i+j+2)}.$$

Here $F^{(i)} = \partial_z^i F$. Differentiation, rather than an unnormalized coefficient, specifies the factorial convention on monomials. Every coefficient of B_z^H is divisible by ϵ, so the form has values in $R_N[z, z^{-1}]$, including on negative powers of g.

The identities needed below can be stated explicitly. For a Hopf algebra with translation T, write $a(z) = e^{zT} a$, and sum over coproduct subscripts. They are

$$\begin{aligned} \rho_z(1, a) &= \varepsilon(a) = \rho_z(a, 1), \\ \rho_z(ab, c) &= \rho_z(a, c_1)\rho_z(b, c_2), \\ \rho_z(a, bc) &= \rho_z(a_1, c)\rho_z(a_2, b), \\ \rho_z(a_1, b_1)a_2(z)b_2 &= b_1 a_1(z)\rho_z(a_2, b_2), \\ \rho_z(Ta, b) &= \partial_z \rho_z(a, b) = -\rho_z(a, Tb), \\ \rho_z(a_1, b_1)\rho_{-z}(b_2, a_2) &= \varepsilon(a)\varepsilon(b). \end{aligned} \tag{14.20}$$

The form ρ^H satisfies all six. The formal exponential inputs are group-like, and B_z^H is additive in each input list. This proves the two multiplicativities after differentiation. A zero list gives the unit rule.

Commutativity and cocommutativity give the fourth identity. The first-input translation rule on (g, g) is

$$\rho_z^H(T_H g, g) = \varpi F' = \varpi'.$$

The second-input derivative has the opposite sign. On the b_i, it follows by differentiating the displayed entries. Derivation and coderivation extend it to products and inverses.

Finally $F^{(i)}(-z) = (-1)^{i+1} F^{(i)}(z)$ makes the two exponents cancel when inputs are exchanged and z is replaced by $-z$. This proves the last, convolution identity.

Restoring the even Euler factor

The factor P^{-1} in E_0 changes the contraction while canceling from the exchange. To retain it, use a second commutative, cocommutative Hopf algebra

$$B_N = R_N[U, U^{-1}, V, V^{-1}, c_{\alpha,i} : \alpha = 1, 2,\ i \geq 0].$$

Let $g_1 = U$, $g_2 = V$ be group-like and the $c_{\alpha,i}$ primitive. Set $T_B g_\alpha = g_\alpha c_{\alpha,0}$, $T_B c_{\alpha,i} = c_{\alpha,i+1}$, with the corresponding group-like and primitive antipodes and counits. Write $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $L(z) = P'(z)/P(z)$.

The form is specified by

$$\begin{aligned}\rho^B_z(g_\alpha, g_\beta) &= P(z)^{J_{\alpha\beta}},\\ \rho^B_z(c_{\alpha,i}, g_\beta) &= J_{\alpha\beta}L^{(i)}(z),\\ \rho^B_z(g_\alpha, c_{\beta,j}) &= (-1)^{j+1}J_{\alpha\beta}L^{(j)}(z),\\ \rho^B_z(c_{\alpha,i}, c_{\beta,j}) &= (-1)^{j+1}J_{\alpha\beta}L^{(i+j+1)}(z).\end{aligned}$$

Precisely, its value on $g^m e^{\sum p_{\alpha,i}c_{\alpha,i}}$ and $g^n e^{\sum q_{\beta,j}c_{\beta,j}}$ is $P^{m^t Jn}\exp B^B_z$. Here $m, n \in \mathbb{Z}^2$, $g^m = U^{m_1}V^{m_2}$, and the auxiliary variable lists are finite. Set

$$B^B_z = \sum_{\alpha,\beta} J_{\alpha\beta}\Big(n_\beta \sum_i p_{\alpha,i}L^{(i)} + m_\alpha \sum_j (-1)^{j+1}q_{\beta,j}L^{(j)} + \sum_{i,j}(-1)^{j+1}p_{\alpha,i}q_{\beta,j}L^{(i+j+1)}\Big).$$

Again differentiate in the auxiliary variables at zero. For each coefficient only finitely many exponential terms contribute. Integral powers of P and derivatives of L are finite Laurent polynomials modulo ϵ^N.

This form also satisfies (14.20). Additivity of B^B and bilinearity of $m^t Jn$ prove multiplicativity. Units and commutation follow as before. Translation starts from $(P^J)' = JP^J L$.

For the last identity, P is even, J is alternating, and $L^{(i)}(-z) = (-1)^{i+1}L^{(i)}(z)$. The powers of P and all exponential terms cancel under reversal. This proof does not require L to be divisible by ϵ.

The shear

$$\sigma U = UV, \quad \sigma V = V, \quad \sigma c_{1,i} = c_{1,i} + c_{2,i}, \quad \sigma c_{2,i} = c_{2,i}$$

is a Hopf automorphism commuting with T_B and preserving the form. Its matrix $A = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ satisfies $A^t JA = J$. Its inverse sends U to UV^{-1} and $c_{1,i}$ to $c_{1,i} - c_{2,i}$, fixing V and the $c_{2,i}$.

On $\widetilde{H}_N = H_N \otimes_{R_N} B_N$ take the product form, total translation T, and

$$\vartheta = e^{\ell T_H} \otimes \sigma.$$

The exponential is finite because $\ell^N = 0$. It is a Hopf automorphism preserving the form: simultaneous translation differentiates a scalar value by $\partial_z - \partial_z = 0$. Such an automorphism is the *level* used in this algebraic construction. It is additional data on the specified Hopf carrier.

Put $G = gU$, $G(v) = e^{vT}G$ and $J_m = T^m G$. The subalgebra $W^+_N = R_N[J_0, J_1, \ldots]$ is freely polynomial. After $U = V = 1$, $c_{\alpha,i} = 0$, the J_m become $T^m_H g$. Their ratios by g are triangular polynomials in $b_0, \ldots, b_{m-1}$ with coefficient one on b_{m-1}. Localization at g proves algebraic independence.

Give each J_m charge one and denote charge d by $W^+_{N,d}$. The ambient charge assigns g degree one and $U, V, b_i, c_{\alpha,i}$ degree zero. It restricts to this grading and is preserved by ϑ.

The field and its charge bounds

For any level θ preserving the form, define

$$\delta_\theta(a) = a_2 \otimes a_3 S^{-1}(\theta a_1) = a_{(0)} \otimes a_{(1)}.$$

The field and exchange on the algebraic Hopf carrier are

$$\begin{aligned}Y_\theta(a, z)b &= a_1(z)b_{(0)}\rho_{-z}(b_{(1)}, a_2),\\ \mathcal{S}_\theta(z)(a \otimes b) &= a_{(0)} \otimes b_{(0)}\rho_z(a_{(1)}, b_{(1)}).\end{aligned}$$

All sums are finite.

The unit rule gives $Y_\vartheta(1, z) = I$ and $Y_\vartheta(a, z)1 = e^{zT}a$. The translation rule and total coderivation rule for δ_ϑ give covariance. For this commutative Hopf algebra, we check the remaining field identities on group-like generating series. Auxiliary differentiation then gives them on all states.

Theorem 14.10.2. The fields at level ϑ preserve W_N^+, add creation charges, and satisfy

$$Y_\vartheta(G, z)G = E_0(-z)G(z)G, \qquad \mathcal{S}_\vartheta(z)(G \otimes G) = r(z)G \otimes G,$$

$$Y_\vartheta\Big(\prod_i G(u_i), z\Big)\prod_j G(v_j) = \prod_i G(z + u_i)\prod_j G(v_j)\prod_{i,j} E_0(-z - u_i + v_j). \tag{14.21}$$

The exchange of the two blocks multiplies by $\prod_{i,j} r(z + u_i - v_j)$. Every finite product has a common rational expansion with the field associativity and exchange identities. For charges d, e and total input jet order M, a pole bound is $2de + N - 1 + M$.

Proof. The coaction of G is $G \otimes C$, where

$$C = G\vartheta(G)^{-1} = \frac{g}{e^{\ell T_H} g}\, V^{-1}.$$

Equations (14.19) and the auxiliary form give $\rho_{-z}(C, G) = E_{\mathrm{pl}}(-z)P(z)^{-1} = E_0(-z)$ and $\rho_z(C, C) = r(z)$. The auxiliary contribution to the latter is $\rho_z^B(V^{-1}, V^{-1}) = 1$. This proves the generator formulas. Multiplicativity and translation give the block formulas. Auxiliary derivatives express every coefficient in the polynomial algebra on the J_m.

For three or more blocks, either nesting includes the same contractions between each pair of elementary generating states, with the same translated regular product. This proves weak associativity as equality of expansions of a common rational germ.

Interchanging adjacent blocks uses $r(s)E_0(-s) = E_0(s)$. Reversal uses $r(s)r(-s) = 1$. Three exchanges multiply the same scalar factors, giving the Yang–Baxter identity, and splitting a block factors the same product, giving compatibility with a field insertion.

Differentiating proves these identities on every polynomial jet input. Clearing the finitely many pairwise denominators gives the corresponding exchange locality, with a bound depending on the two inputs.

Each contraction has baseline inverse degree two. For de pairs the baseline is $2de$. Its further inverse degree equals its ϵ degree and is at most $N - 1$. Input differentiation adds at most M. This proves the stated lower Laurent bound and all required truncations. □

Modulo ϵ^4,

$$E_0(-z) = z^{-2} + (h^2 - ab)\epsilon^2 z^{-4} - abh\epsilon^3 z^{-5}, \qquad r(z) = 1 + 2abh\epsilon^3 z^{-3}.$$

The even factor changes the zeroth-order contraction from 1 to z^{-2}, while leaving the cubic exchange unchanged.

Transposing the Hall coefficient map

Let $A_d[m]$ denote homogeneous root degree m, and put $A_d^\circ = \bigoplus_m \mathrm{Hom}_{R_N}(A_d[m], R_N)$. This is the continuous dual of $\prod_m A_d[m]$ with its degree topology. Pair it with $W_{N,d}^+$ by

$$\langle J_{m_1} \cdots J_{m_d}, F\rangle = \partial_{x_1}^{m_1} \cdots \partial_{x_d}^{m_d} F\big|_{x=0}.$$

Monomial symmetric functions and unordered derivative lists pair diagonally with nonzero factorials. Hence this pairing is perfect in every degree and gives $\iota_d : \mathcal{A}_d^\circ \cong W_{N,d}^+$.

THEOREM 14.10.3. For finite-jet inputs of charges d, e,

$$Y_{\mathcal{S}}(\iota_d\alpha, z)\iota_e\beta = \iota_{d+e}\big((\Delta_{d,e}^\circ(-z))^\circ(\alpha\otimes\beta)\big).$$

The right side is defined coefficientwise on polynomial tests and lies in $W_{N,d+e}^+((z))$. Translation is $T = -D^\circ$ for $D = -\sum_i \partial_{x_i}$. The exchange is the transpose of the inverse Hall multiplier at parameter $-z$.

Proof. Pair (14.21) with F. The result is

$$\prod_{i,j} E_0(-z-u_i+v_j)F(z+u_1,\dots,z+u_d,v_1,\dots,v_e).$$

It is exactly (14.18) at $s=-z$, evaluated at $(x,y)=(u,v)$. Every finite auxiliary derivative proves the asserted equality.

For each Laurent coefficient only a finite derivative order of the polynomial test is needed, so its coefficient belongs to the restricted dual. The preceding pole bound gives a common lower bound on the fixed input. Transposing $-\partial_x$ sends the mth derivative functional to minus the $(m+1)$st one. Finally P is even, and $r(-s)^{-1} = r(s)$ gives the stated exchange variance. □

With the opposite Hall grading, $\deg(J_{m_1}\cdots J_{m_d}) = 2d^2 - 2\sum_i m_i$. Translation has degree -2, and $\deg z = 2$ makes the transpose homogeneous. Ordinary multiplication in the ambient Hopf algebra has a different charge-dependent grading. Its cochain degree remains zero.

Charge completion gives $\prod_d W_{N,d}^+$ with outputs in $\prod_d W_{N,d}^+((z))$. Each fixed total charge has finitely many splittings, but there is no uniform pole bound over all charges. The coefficient map (14.18) has not been extended to every formal root series: translation by s does not preserve the root-adic topology.

The level map on the ambient Hopf carrier is different: $(\theta_2\otimes 1)\Delta$ sends G to $\theta_2(G)\otimes G$. Both output factors have creation charge one. A Hall split has output charges adding to its input charge. Moreover $\mathcal{S}(G) = (e^{\ell T_H}g)UV$ need not lie in W_N^+.

Thus the transpose theorem identifies Hall charge cooperation with a field product on finite jets. It does not identify it with a level-adding coproduct. The corresponding field tensor, if that further comparison is sought, must retain its spectral twist and this change of charge.

14.11 Moving centers and Cartan operations

Assume $N \geq 4$ and extend the coefficients to

$$R = \mathbb{C}(a,b)[a_{\mathrm{fr}}, \beta, \beta^{-1}, y_1, y_2, \kappa, \epsilon]/(\epsilon^N), \qquad \lambda = a_{\mathrm{fr}} + \epsilon b.$$

The affine parameter κ and labels y_1, y_2 are retained independent coefficients. They are not the root coordinates of the Hall correspondence.

On polynomial Hall coefficients define the connection

$$\nabla_d = \beta\left(\partial_{a_{\mathrm{fr}}} + \sum_{i=1}^{d}\partial_{x_i}\right).$$

On a tensor product over R this is the tensor connection over the single coefficient derivation $\beta\partial_{a_{\mathrm{fr}}}$. Every kernel depends only on root differences, so the simultaneous root derivative kills it. Translation of the first block by s also commutes with this derivative. Thus both the Hall product and (14.18) are horizontal.

The coordinates $\xi_i = x_i - \lambda$ are horizontal. Completion along this moving section therefore has a continuous dual with horizontal basis

$$v_{\mathbf{m},\lambda}(F) = \partial_{x_1}^{m_1} \cdots \partial_{x_d}^{m_d} F\big|_{x_i=\lambda}.$$

Indeed the chain rule gives

$$\beta\partial_{a_{\mathrm{fr}}}\langle v_{\mathbf{m},\lambda}, F\rangle = \langle v_{\mathbf{m},\lambda}, \nabla_d F\rangle.$$

Sending this basis to $\prod_i J_{m_i}$ gives the moving-center version of the transpose theorem. Its proof evaluates F at $(\lambda + z + u, \lambda + v)$, with the same scalar kernels. The first insertion has the horizontal character $F \mapsto F(\lambda)$. On $F = x$ both sides of the horizontality identity are β.

Evaluation at a nonnilpotent λ is not a finite linear combination of derivatives at zero. These are jets at a moving section, with no implicit $e^{\lambda T}$ on the fixed algebraic state space. The state derivative T raises jet order. The coefficient connection fixes the displayed basis.

The Cartan extension of the Hall coproduct

Let $\mathsf{H} = R[\mathsf{p}_0, \mathsf{p}_1, \ldots]$ have primitive generators acting on A_d by multiplication by $\sum_i x_i^n$. These act as derivations of the Hall product, because the power sums add on concatenated root lists. Form the smash product $\mathsf{B} = (\bigoplus_d A_d)\#\mathsf{H}$ with relations $[\mathsf{p}_n, F] = (\sum_i x_i^n)F$. For a spectral variable p define

$$\Phi(p) = \exp\left(\sum_{n\ge 0} \frac{(-1)^n}{n!}\mathsf{p}_n \partial_p^n \log r(p)\right). \tag{14.22}$$

Every coefficient in p^{-1} uses finitely many p_n. On a finite root list L, substitution $\mathsf{p}_n = \sum_{l\in L} l^n$ gives $\Phi_L(p) = \prod_{l\in L} r(p-l)$. This is a root-addition current. A framed highest-weight current is further data.

Proposition 14.11.1. In the localized spectral tensor target, the formulas

$$\Delta^{\mathrm{ext}}_{d,e}(s)F = \Delta^{\circ}_{d,e}(s)F \prod_{j=1}^{e} \Phi_L(y_j + s),$$

$$\Delta^{\mathrm{ext}}(s)\mathsf{p}_n = \sum_{i=0}^{n} \binom{n}{i}(-s)^{n-i}\mathsf{p}_i \otimes 1 + 1 \otimes \mathsf{p}_n$$

give a unital algebra coproduct on B. The coefficients of each Φ_L lie in the left output factor, to the right of its Hall coefficient in the displayed product. The coproduct is rationally coassociative and horizontal when $\nabla\mathsf{p}_n = \beta n \mathsf{p}_{n-1}$ and $\nabla\mathsf{p}_0 = 0$.

Proof. For multiplicativity, split the first input into root blocks A, C and the second into B, D, with A, B in the left output. Cancellation of cross Euler factors leaves

$$\prod_{b\in B, c\in C} \frac{E_0(s + c - b)}{E_0(-s + b - c)} = \prod_{b\in B, c\in C} r(s + c - b).$$

When the separate extended coproducts are multiplied, the smash relation moves the first Cartan factor past B and replaces its list L by $L \sqcup B$. The factor changes by precisely this product. All other factors coincide. The even factor P cancels from the ratio.

The binomial formula preserves the Cartan relations with shifted roots.

Evaluation on all finite root lists is jointly faithful. At length at least the largest positive power-sum index, those power sums are algebraically independent. Varying the length tests the polynomial in p_0. This remains valid coefficientwise modulo ϵ^N. The root calculation therefore proves the algebra identity. For a triple split, both sides have Cartan factor $\Phi_L(Y+s)\Phi_L(Z+s+t)\Phi_M(Z+t)$ and the same Euler factors, proving coassociativity.

Finally differentiating (14.22) by $\nabla \mathsf{p}_n = \beta n \mathsf{p}_{n-1}$ gives $-\beta\partial_p\Phi$. It cancels the derivative $\beta\partial_{y_j}\Phi_L(y_j+s)$. The raw Euler part is horizontal, and the binomial formula commutes with the same connection. This proves horizontality on all generators. □

The Cartan operation also has a direct state realization. For $C = G\mathcal{S}(G)^{-1}$ set

$$\mathcal{U}(p)x = x_{(0)}\rho_p(C, x_{(1)}).$$

It is an invertible algebra operator with

$$\mathcal{U}(p)G(v) = r(p-v)G(v), \qquad \mathcal{U}(p)1 = 1.$$

Multiplicativity follows from the group-like C and the multiplicative coaction. C^{-1} gives its inverse. It preserves W_N^+ coefficientwise.

Pairing its products and derivatives with polynomial tests shows that multiplication by $\prod_i r(p-x_i)$ has finite-jet transpose $\mathcal{U}(p-\lambda)$ at the moving center. In particular $(\beta\partial_{a_{\mathrm{fr}}} + \beta\partial_p)\mathcal{U}(p-\lambda) = 0$.

In the Laurent domain with p large compared with z, the creation field satisfies

$$\mathcal{U}(p)Y_{\mathcal{S}}(G,z)\mathcal{U}(p)^{-1} = r(p-z)Y_{\mathcal{S}}(G,z). \tag{14.23}$$

Apply both sides to a product of $G(v_j)$. The field adds the regular factor $G(z)$ and its contractions. The Cartan operator multiplies this new factor by $r(p-z)$ and cancels its factors on the input. Differentiation proves the relation on all finite jets. At $p - z = \beta u$, the cubic coefficient is $2abh\beta^{-3}u^{-3}$.

14.12 Framing markings and the supported current

The next calculation retains a moment complex with both odd relation directions. Before localizing in a, b, use

$$R^{\circ} = \mathbb{C}[a, b, a_{\mathrm{fr}}, \beta, \beta^{-1}, y_1, y_2, \kappa, \epsilon]/(\epsilon^N), \qquad N \geq 4, \quad \lambda = a_{\mathrm{fr}} + \epsilon b, \quad c = ab(a+b).$$

Its generic base change is the ring R above.

The two odd directions occur already in a rank-two framed moment chart. Take

$$X = \begin{pmatrix} x_1 & B_0 \\ 0 & x_2 \end{pmatrix}, \quad Y = \begin{pmatrix} v_1 & D_0 \\ 0 & v_2 \end{pmatrix}, \quad i = \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}, \quad j = (1, 0).$$

The four entries of $[X,Y] + ij$ are i_1, $(x_1-x_2)D_0 - (v_1-v_2)B_0$, i_2, 0. Eliminating the two regular Koszul pairs for i_1, i_2 leaves odd generators η, χ with

$$d\eta = (x_1-x_2)D_0 - (v_1-v_2)B_0, \qquad d\chi = 0.$$

The elimination is explicit: for $d\zeta = i$, the contraction sends i^m to ζi^{m-1} for $m \geq 1$. Tensor the two contractions with their Koszul signs.

On the open $B_0 \neq 0$, the change of basis $\mathrm{diag}(1, \gamma)$ preserves j and the flag. It scales B_0, D_0 by γ^{-1}, so uniquely fixes $B_0 = 1$. Derived restriction to $v_1 = v_2 = 0$ then leaves $d\eta = uD_0$, $u = x_1 - x_2$, and still $d\chi = 0$. The zero-equation generator survives the restriction.

This specifies a moment coefficient chart. Orientations for a Hall pushforward have not been supplied by the matrix calculation.

In position coordinates z, w, t, put $u = z - w$, $v = t - w$, and define

$$\mathcal{T} = \bigoplus_{p,r\geq 1} R^0[w]u^{-p}v^{-r}, \qquad \mathcal{M} = \mathcal{T}[D_0] \otimes \Lambda(\eta, \chi), \quad d\eta = uD_0, \quad d\chi = 0.$$

Here η, χ have degree -1, D_0 degree zero. Principal-part multiplication discards nonnegative powers: $u\,u^{-1} = v\,v^{-1} = 0$ in their respective factors. There is only a direct sum of normal orders.

For $\mathcal{T}_1 = \ker(u : \mathcal{T} \to \mathcal{T})$,

$$\begin{aligned} H^0(\mathcal{M}) &= \mathcal{T}, \\ H^{-1}(\mathcal{M}) &= \eta\mathcal{T}_1[D_0] \oplus \chi\mathcal{T}, \\ H^{-2}(\mathcal{M}) &= \eta\chi\mathcal{T}_1[D_0]. \end{aligned} \tag{14.24}$$

Indeed u is surjective on $\mathcal{T}$, so the image of uD_0 is $D_0\mathcal{T}[D_0]$ and its kernel is $\mathcal{T}_1[D_0]$. Tensor with the closed χ factor. The degree-zero and χ summands use $D_0 = 0$.

The required cubic contact class is

$$\gamma_N = 2c\epsilon^3\beta^{-3}u^{-3}v^{-1} \in H^0(\mathcal{M}). \tag{14.25}$$

It is nonzero: c is a non-zero-divisor in the polynomial coefficient ring and $\epsilon^3 \neq 0$ at $N \geq 4$.

The three normal coefficients of a marking

A scalar marking is an action of $R^0[\ell_{\mathrm{fr}}]$ on $\mathcal{M}$ in which ℓ_{fr} multiplies by $s(u, w) = \sum_{j\geq 0} a_j(w)u^j$, with $a_j \in R^0[w]$. Every principal part has finite u-pole order, so this action is defined without completing $\mathcal{T}$. It commutes with d. It specifies a coefficient action, not the Hopf level constructed in the preceding section.

Proposition 14.12.1. The relation $(s - \lambda)\gamma_N = 0$ holds exactly when

$$a_0 - \lambda, \quad a_1, \quad a_2 \in \epsilon^{N-3}R^0[w].$$

At all compatible precisions, the coefficientwise criterion is

$$s = \lambda + u^3h(u, w).$$

Proof. Multiplication by u^j kills u^{-3} for $j \geq 3$, and

$$(s - \lambda)\gamma_N = 2c\epsilon^3\beta^{-3}\big((a_0 - \lambda)u^{-3} + a_1u^{-2} + a_2u^{-1}\big)v^{-1}.$$

The three principal parts are independent. The annihilator of $c\epsilon^3$ is (ϵ^{N-3}), proving the criterion. At all compatible precisions, each fixed ϵ^m coefficient of these three series vanishes by taking $N > m + 3$. The converse is immediate. □

In particular an unramified normal marking, for which a_1 is a unit, cannot preserve the contact. Even $s = \lambda + \beta u$, with the correct diagonal character, leaves the defect $2c\epsilon^3\beta^{-2}u^{-2}v^{-1}$. Modulo ϵ^4 at $a = b = \beta = 1$ its coefficient is 4. The ramified marking $s = \lambda + \beta u^3$ does preserve the contact and gives the nonzero cochain map $1 \mapsto \gamma_N$ from $R^\circ[0]$ with character λ. It has different translation behavior.

Let $\nabla = \partial_w$ at fixed u, v, fixing the coefficient ring and the moment generators. It commutes with d and differentiates simultaneous translation of z, w, t. Call a marking translation-covariant with slope β when $[\nabla, m_s] = \beta I$.

THEOREM 14.12.2. Such a marking has $s = \beta w + b(u)$ with $b(u) \in R^\circ[[u]]$. For every $\lambda' \in R^\circ$, multiplication by $s - \lambda'$ is injective on every $H^i(\mathcal{M})$. Consequently a cochain map from a complex whose cohomology has character λ induces zero on cohomology if it intertwines the marking, even up to homotopy.

Proof. The commutator condition is $\partial_w s = \beta$. Coefficientwise integration in characteristic zero gives the displayed form.

On $\mathcal{T}$, take the largest u-pole of a nonzero element. Its leading coefficient is multiplied by $\beta w + b(0) - \lambda'$, a non-zero-divisor because its leading coefficient in w is the unit β. It remains nonzero.

On $\mathcal{T}_1[D_0]$, u acts by zero and the same polynomial proves injectivity. Formula (14.24) treats the closed odd factors as well.

For an intertwining map Φ, compatibility on cohomology gives $(s - \lambda)H(\Phi)(x) = H(\Phi)((\ell_{\mathrm{fr}} - \lambda)x) = 0$. Injectivity forces $H(\Phi)(x) = 0$. A compatibility homotopy gives this same equality. □

Conjugating the marking by a cochain isomorphism or adding $dH + Hd$ leaves the injectivity conclusion unchanged on cohomology. The ramified marking above has $[\nabla, m_s] = 0$, so it is outside this theorem's covariance hypothesis.

The moving framing fiber

Set

$$f = \beta(w + u) - \epsilon y_1 - \lambda, \qquad Q = \mathcal{M}/f\mathcal{M}.$$

Multiplication by f is injective on the terms and cohomology of $\mathcal{M}$: its highest w coefficient is multiplied by the unit β. Thus the ordinary quotient represents the derived framing fiber and $H^*(Q) = H^*(\mathcal{M})/fH^*(\mathcal{M})$. It eliminates w by $w = (\lambda + \epsilon y_1)/\beta - u$. In degree zero the remaining basis is $u^{-p}v^{-r}$ over R°, and $\tilde{\gamma}_N \neq 0$.

The operator $\nabla = \partial_w$ does not descend: $\nabla f = \beta$ would force $Q = 0$ if the ideal were stable. Nor can an R°-linear operator on Q have commutator βI with its marking, which is the scalar λ.

The connection

$$\nabla_{\mathrm{fr}} = \partial_w + \beta\partial_{a_{\mathrm{fr}}} \tag{14.26}$$

does descend. It annihilates f, commutes with the moment differential, and satisfies $[\nabla_{\mathrm{fr}}, m_\lambda] = \beta I$. Its coefficient rule is $\nabla_{\mathrm{fr}}(rx) = \beta(\partial_{a_{\mathrm{fr}}} r)x + r\nabla_{\mathrm{fr}}x$. This changes both the recipient and the kind of connection. It does not contradict the obstruction on the unchanged complex.

The Cauchy comparison and both odd directions

Work first over $A = R^\circ[z, w] = R^\circ[u, w]$. For the current coordinate t define

$$P_z = A[t, (t - z)^{-1}]/A[t], \qquad P_w = A[t, (t - w)^{-1}]/A[t],$$

$$P_{zw} = A[t, ((t - z)(t - w))^{-1}]/A[t], \quad \mathcal{K}_{zw} = [P_z \oplus P_w \xrightarrow{(a,b)\mapsto a-b} P_{zw}].$$

The two terms have degrees -1, 0 and use proper rational representatives at $t = \infty$. There is a relative differential-module quasi-isomorphism $\Theta : \mathcal{K}_{zw} \to \mathcal{T}[0]$, zero in degree -1. In degree zero, expand at $v = 0$, take its v-principal part, and then take its u-principal part. Explicitly,

$$\Theta\left(\frac{1}{v^a (v-u)^b}\right) = (-1)^b \sum_{j=0}^{a-1} \binom{b+j-1}{j} u^{-b-j} v^{j-a}. \tag{14.27}$$

Here is a direct proof on the stated quotient. Monic division by $(t-z)(t-w)$ makes P_{zw} free over A. P_z, P_w have their usual inverse-power bases. After inverting u, partial fractions separate the two poles.

The integral summands have zero intersection. Coefficientwise in ϵ, a function belonging to both has poles only at z and only at w. It therefore has no pole and, being proper, is zero.

Expansion gives (14.27) and kills both single-pole summands. At fixed $a + b$, its matrix is triangular with diagonal entries $(-1)^b$, so it surjects onto every negative u, v monomial.

If its value vanishes, the w-principal part has no u-pole and lies in P_w. Subtracting it leaves a member of P_z. This proves the kernel assertion.

Principal parts commute with derivatives, since differentiating a regular power creates no pole. The physical derivatives are $\partial_z = \partial_u$ and $\partial_w^{\mathrm{phys}} = \partial_w - \partial_u - \partial_v$, which proves relative differential linearity.

The derived diagonal restriction of $\mathcal{K}_{zw}$ has P_w in degree -1 and zero in degree zero: its differential becomes $(a, b) \mapsto a - b$ on two copies of P_w. In contrast, the ordinary restriction $\mathcal{T}/u\mathcal{T}$ is zero because u is surjective on $\mathcal{T}$.

Let $K_f = A[D_0] \otimes \Lambda(\eta, \chi)$ with $d\eta = uD_0$, $d\chi = 0$. It is bounded with free A-terms, so $1 \otimes \Theta : K_f \otimes_A \mathcal{K}_{zw} \to \mathcal{M}$ is a quasi-isomorphism, fixing both odd generators. The tensor differential $d(\xi \otimes a) = d\xi \otimes a + (-1)^{|\xi|} \xi \otimes da$ and A-linearity prove its cochain equation. It retains all the cohomology in (14.24). It does not provide separate commuting position connections on K_f: with D_0 fixed, $[\partial_u, d]\eta = D_0$.

Now pass to the generic coefficient ring R and the current G already constructed. Define

$$G_r = \frac{(\beta T)^{r-1} G}{(r-1)!}, \qquad \mathcal{J}_G = \bigoplus_{r \geq 1} R G_r, \qquad \mathcal{Q}_G = \left(\bigoplus_{p \geq 1} R u^{-p}\right) \otimes_R \mathcal{J}_G[D_0] \otimes \Lambda(\eta, \chi).$$

Let $vG_1 = 0$, $vG_r = G_{r-1}$ for $r > 1$, and let w act by $(\lambda + \epsilon y_1)/\beta - u$. Keep $d\eta = uD_0$, $d\chi = 0$. The coefficient connection is $\beta \partial_{a_{\mathrm{ft}}}$, fixing this displayed basis.

Theorem 14.12.3. There is an isomorphism of coefficient complexes with connection

$$\Psi : \mathcal{Q} \otimes_{R^0} R \longrightarrow \mathcal{Q}_G,$$
$$u^{-p} v^{-r} D_0^m \eta^e \chi^{e'} \longmapsto u^{-p} G_r D_0^m \eta^e \chi^{e'}.$$

It retains both odd moment directions and satisfies $\Psi(-\partial_v) = \beta T \Psi$. The composition with $1 \otimes \Theta$, followed by derived base change to $f = 0$, is a quasi-isomorphism for the single common-translation connection (14.26). It sends the cubic contact to

$$\Psi(\tilde{\gamma}_N) = 2abh\epsilon^3 \beta^{-3} u^{-3} G_1.$$

Proof. Elimination of w gives a unique finite expression in the displayed basis. The G_r are independent, so Ψ is an isomorphism of modules. It preserves u, D_0, η, χ and the specified v, w actions, hence

the differential. After eliminating w, ∇_{fr} differentiates only the coefficients of this basis, which proves horizontality. The derivative identity follows from $-\partial_v v^{-r} = r v^{-r-1}$ and $\beta T G_r = r G_{r+1}$.

Simultaneous translation is $\partial_z + \partial_w^{\mathrm{phys}} + \partial_t = \partial_w$ at fixed u, v. It fixes $u D_0$. The Cauchy map commutes with it and with the coefficient derivative. f is horizontal. Derived base change preserves its quasi-isomorphism. Injectivity of f on terms and cohomology identifies its derived target with the ordinary quotient $\boldsymbol{Q}$.

For the contact, expand the actual rational section $(r(\beta(z-t)) - 1)(t-w)^{-1}$. Its cubic coefficient is $2abh\epsilon^3\beta^{-3}/((z-t)^3(t-w))$. Formula (14.27), with $z - t = u - v$, gives the positive class $2abh\epsilon^3\beta^{-3}u^{-3}v^{-1}$. Applying Υ gives the stated value. For v^{-r}, the same expansion retains the first r Taylor coefficients, precisely the corresponding Leibniz derivatives of (14.23). □

Under the formal marking $x_i = \beta z_i - \epsilon y_i$, the exact Cartan argument is $\beta(z_1 - z_2) - \epsilon(y_1 - y_2)$. Its correction first occurs at order ϵ^4, so the cubic normalization above is unchanged. The common connection becomes $\sum_i \partial_{z_i} + \beta\partial_{a_{\mathrm{fr}}}$, and on the supported chart it is (14.26). The separate derivative βT differentiates the relative output current. These formulas do not identify plane eigenvalues in the moment chart with tautological Chern roots of a Hall correspondence.

The surviving classes include $\eta u^{-1} G_1$ and $\eta\chi u^{-1} G_1$ in degrees $-1, -2$. Hence the comparison cannot factor through a degree-zero polynomial projection. It also supplies only the common connection, with the separate-position obstruction still given by $[\partial_u, d]\eta = D_0$.

14.13 The remaining geometric comparison

The scalar charge-zero coaugmentation is 1. A local vacuum is a separate state marking. On the test module $R\nu \otimes_R W_N^+$, take fields $1 \otimes Y_S$ and distinguished state $\nu \otimes 1$. Its first creation state is $\nu \otimes G_1$, on which the first tautological coefficient acts by $F(\lambda)$. Then the identity field preserves $\nu \otimes 1$.

This specifies the field test module and its framing character. It does not identify it with a geometric framed module.

Problem 14.13.1. Construct an oriented critical Hall/Hecke coefficient family whose two-point map lands in $K_f \otimes_A \mathcal{K}_{zw}$ and whose derived framing fiber maps through Theorem 14.12.3. Require the Cartan and creation operations to give (14.23), with cubic contact (14.25). Preserve the framing character $F(\lambda)$, both odd moment directions, the common coefficient connection, the scalar coaugmentation and the local vacuum marking. Construct the comparison over R, and its descent to R° before localization in a, b.

On the same coefficient family, construct the nonconsecutive 13|2 comparison and the required small-diagonal homotopies. A comparison with a level-adding coproduct must additionally specify its different charge map and its twisted field tensor target.

The zero-potential Euler calculation and the finite-jet transpose are positive constructions on their named carriers. The supported moving fiber also has a constructed comparison, including its two odd directions and one common connection. The oriented map from actual critical Hall coefficients is still missing. A physical identification must distinguish the nilpotent coefficient parameter ϵ from a Moyal parameter or a loop parameter. It must also distinguish the exchange contact from a gauge anomaly. Equal scalar Euler factors alone give none of these further maps.

Chapter 15

Marked finite quantum dilogarithms

Consider two simple objects $\mathbf{1}$, ρ with

$$\rho \otimes \rho \simeq \mathbf{1} \oplus \rho.$$

An additive strong tensor functor to ordinary finite-dimensional vector spaces would give a positive integer D satisfying $D^2 = 1 + D$. No such integer exists. Nevertheless, the rule has a realization by finite bimodules, whose tensor product is balanced over an algebra with two idempotents. Its associativity maps retain information that the dimension equation forgets.

A real quadratic unit produces a family of such examples. The input is a finite function obtained from a meromorphic quantum dilogarithm. Its values become coefficients of explicit fusion maps. A normalized network recovers those values from the category. We construct the category over the field of its normalized arithmetic values, then construct its finite bimodule realization. Along the way, a rank-one correction separates a finite operator identity from the categorical pentagon.

15.1 The arithmetic function

Write $\mathrm{e}(z) = \exp(2\pi i z)$. For $|q| \neq 1$, define the q-Pochhammer symbol by

$$(x;q)_\infty = \begin{cases} \displaystyle\prod_{j \geq 0} (1 - xq^j), & |q| < 1, \\ \displaystyle\prod_{j \geq 1} (1 - xq^{-j})^{-1}, & |q| > 1. \end{cases}$$

The first product converges locally uniformly in x. The second defines a meromorphic function of x. For $\gamma = \left(\begin{smallmatrix} a_\gamma & b_\gamma \\ c_\gamma & d_\gamma \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$ and integers m, n, put

$$\Phi_{\gamma,m,n}(z;\tau) = \frac{(q^m \mathrm{e}(z);q)_\infty}{(\widetilde{q}^{\,n} \mathrm{e}(z/(c_\gamma\tau + d_\gamma));\widetilde{q})_\infty}, \qquad q = \mathrm{e}(\tau), \quad \widetilde{q} = \mathrm{e}(\gamma\tau). \tag{15.1}$$

Initially $\tau \notin \mathbb{R}$, and the quotient is interpreted meromorphically in z.

The modular quantum dilogarithm has a meromorphic continuation to

$$(z,\tau) \in \mathbb{C}^2, \qquad c_\gamma\tau + d_\gamma \notin \mathbb{R}_{\leq 0}.$$

We use the continuation of Radchenko–Wheeler, equation (1) and Section 2.2.[1]

[1] D. Radchenko and C. Wheeler, *Real quadratic fields and finite quantum dilogarithms I*. Version 2. arXiv:2609.21892v2. The product conventions are in Appendix A.1. The arithmetic identities are in Theorems 1–2 and Section 4. The fusion reconstruction is Theorem 12 and Appendix B.

For $c_\gamma > 0$, fix the Dedekind eta multiplier

$$\mu_\gamma = \frac{\eta(\gamma w)}{\eta(w)\sqrt{c_\gamma w + d_\gamma}}, \qquad w \in \mathbb{H},$$

using the holomorphic square root with positive value on the positive real axis. Here $\eta(w) = \mathrm{e}(w/24)\prod_{r\geq 1}(1-\mathrm{e}(rw))$. The quotient defining μ_γ is independent of w and is a twenty-fourth root of unity.

A quadratic lattice and its finite group

Fix a real quadratic field $K \subset \mathbb{R}$. Write x' for the other real embedding of $x \in K$. Let $I \subset K$ be a free abelian subgroup of rank two. Choose a totally positive unit $\varepsilon > 1$ satisfying $\varepsilon I = I$. Its norm is one.

Define

$$G_I = (\varepsilon - 1)^{-1} I/I, \qquad N = |G_I| = \mathrm{Tr}_{K/\mathbb{Q}}(\varepsilon) - 2.$$

Indeed, multiplication by $\varepsilon - 1$ identifies this group with $I/(\varepsilon - 1)I$. The absolute determinant is $|\mathrm{Nm}(\varepsilon - 1)| = \mathrm{Tr}(\varepsilon) - 2 > 0$.

An oriented basis (β_1, β_2) of I satisfies $\beta_1\beta_2' - \beta_1'\beta_2 > 0$. Choose one with β_2 totally positive, and write

$$\varepsilon\beta_1 = a_\gamma\beta_1 + b_\gamma\beta_2, \qquad \varepsilon\beta_2 = c_\gamma\beta_1 + d_\gamma\beta_2.$$

The resulting matrix γ lies in $SL_2(\mathbb{Z})$. For $\tau = \beta_1/\beta_2$, one has

$$\gamma\tau = \tau, \qquad c_\gamma\tau + d_\gamma = \varepsilon, \qquad c_\gamma = \frac{\varepsilon - \varepsilon^{-1}}{\tau - \tau'} > 0.$$

Thus τ is the attractive real fixed point, inside the domain of the continued function.

For $x \in G_I$, choose a representative in K. There are unique integers u, v with

$$(\varepsilon^{-1} - 1)x = u\beta_1 + v\beta_2.$$

For a nonzero class define

$$E_I(x) = \mu_\gamma \Phi_{\gamma,u,0}(x/\beta_2; \tau), \qquad E_I(0) = \sqrt{\varepsilon} > 0. \tag{15.2}$$

The continuation in (15.1) precedes this evaluation. At the real fixed point the two separate products need not converge. Their formally matching arguments therefore cannot be cancelled.

The representative and oriented-basis independence of (15.2) follows from the modular cocycle and its index-shift identities. We use the precise lattice formulation and change-of-basis theorem of the reciprocity paper.[2] In particular, the prescribed meromorphic function selects the arithmetic values. The scalar identities below do not alone specify that selection.

[2] *Stark units for real quadratic fields and reciprocity laws*, 4 October 2026, Definition 1, Theorems 1–2, Proposition 1 and Appendix B, `https://www.danrad.net/papers/stark-reciprocity.pdf`.

The Gaussian and Fourier normalization

Let $\omega_I : I \times I \to \mathbb{Z}$ be the alternating form with $\omega_I(\beta_1, \beta_2) = 1$, extended $\mathbb{Q}$-bilinearly to K. For $t \in I$, set

$$\vartheta_I(t) = \begin{cases} 1, & t \in 2I, \\ -1, & t \notin 2I. \end{cases}$$

The Gaussian and its polarization are

$$Q_I(x) = \vartheta_I((\varepsilon - 1)x)\mathrm{e}\big(\tfrac{1}{2}\omega_I(x, \varepsilon x)\big), \tag{15.3}$$

$$\chi_I(x, y) = \frac{Q_I(x + y)}{Q_I(x)Q_I(y)} = \mathrm{e}\big(\omega_I(x - \varepsilon x, y)\big). \tag{15.4}$$

Both are well defined on G_I.

To see the descent of Q_I, use

$$\vartheta_I(t + s)\vartheta_I(t)\vartheta_I(s) = (-1)^{\omega_I(t,s)}, \qquad t, s \in I,$$

and $\omega_I(\varepsilon x, \varepsilon y) = \omega_I(x, y)$. Replacing x by $x + t$ changes the sign and exponential by inverse factors. The same calculation gives (15.4). One has $Q_I(0) = 1$, $Q_I(-x) = Q_I(x)$, and all values of Q_I are roots of unity.

The bicharacter χ_I is nondegenerate. Indeed,

$$\omega_I(x - \varepsilon x, y) = \omega_I(\varepsilon x, (\varepsilon - 1)\, y).$$

If its exponential is one for every $y \in (\varepsilon - 1)^{-1} I$, unimodularity of ω_I gives $\varepsilon x \in I$. Hence $x \in I$.

For a function $f : G_I \to \mathbb{C}$, use the Fourier transform

$$\widehat{f}(y) = \frac{1}{\sqrt{N}} \sum_{x \in G_I} f(x)\chi_I(x, -y),$$

with the positive square root of N. Orthogonality of the characters gives $\widehat{\widehat{f}}(x) = f(-x)$.

Theorem 15.1.1 *(Arithmetic input).* Let $E = E_I$, $Q = Q_I$, and $\chi = \chi_I$. For a twenty-fourth root of unity λ, the arithmetic values satisfy

$$\begin{aligned} E(g)E(-g) &= Q(g)^{-1} \qquad (g \neq 0), \\ \widehat{E}(g) &= \lambda Q(g)E(g), \\ \chi(x, y)E(x)E(y) &= -\widehat{E}(0) + \frac{1}{\sqrt{N}} \sum_t E(y - t)E(t)Q(t)E(x - t). \end{aligned}$$

At zero the separate normalization is

$$E(0)^2 = \sqrt{N}\, E(0) + 1. \tag{15.5}$$

Every $E(g)$ is an algebraic unit.

Let m be the exponent of G_I, and let $\mathcal{O} = (I : I) = \mathbb{Z} + c\mathcal{O}_K$. The normalized values $e_{I,g} = E_I(g)/E_I(0)$ belong to $\mathcal{O}_L^\times$, where $L = H^+_{\mathcal{O},m}$ is the narrow ray class field modulo $m\mathcal{O}$. For an integral invertible $\mathcal{O}$-ideal $\mathfrak{a}$ prime to cm,

$$\mathrm{Art}(\mathfrak{a})(e_{I,g}) = e_{\mathfrak{a}^{-1}I,g}. \tag{15.6}$$

The right-hand label is the class of the same representative. The Artin symbol uses arithmetic Frobenius.

The scalar identities and algebraicity are the Radchenko–Wheeler theorems. The unit and reciprocity assertions are Theorems 1–2 of the reciprocity paper. These are imported arithmetic results. The identity (15.5) also follows directly from $\varepsilon + \varepsilon^{-1} = N + 2$. It remains part of the input when $N \le 4$, where a bare cubic equation can have additional solutions.

15.2 The arithmetic coefficient field

Suppress I in the notation. Put $u = \sqrt{\varepsilon} > 0$ and $s = \sqrt{N} > 0$. Then $s = u - u^{-1}$. Define

$$b_g = \frac{E(g)}{s}, \qquad b_0 = \frac{\varepsilon}{\varepsilon - 1} = \frac{\varepsilon - 1}{N}, \qquad b_g = b_0 e_g. \tag{15.7}$$

Consequently the coefficient field is

$$F_I = K(b_g : g \in G_I) = K(e_g : g \in G_I) \subseteq L.$$

Neither u nor s needs to belong to F_I. Every coefficient used in the algebraic construction below does.

Set

$$d = -b_0^{-1} = \varepsilon^{-1} - 1, \qquad \beta = \sum_g b_g, \qquad a = \frac{\beta}{N b_0}, \qquad c_1 = \frac{b_0}{\beta}. \tag{15.8}$$

Fourier evaluation at zero gives $\beta = \lambda u \neq 0$. Thus $a = \lambda/s$, $c_1 = \lambda^{-1}/s$, and $a c_1 = 1/N$. The notation c_1 distinguishes this coefficient from the conductor c.

Reflection recovers the Gaussian over the same field:

$$Q(0) = 1, \qquad Q(g) = \frac{1}{N b_g b_{-g}} \quad (g \neq 0).$$

Its polarization therefore has values in F_I. The arithmetic identities become

$$\begin{gathered} b_0^2 - b_0 = 1/N, \qquad d^2 = N(d+1), \\ \sum_x b_x \chi(x, -y) = N a Q(y) b_y, \\ \sum_t b_{y-t} b_t Q(t) b_{x-t} - \chi(x, y) b_x b_y = \beta/N. \end{gathered} \tag{15.9}$$

These follow by substituting $E(g) = s b_g$. In particular, the constant in (15.9) retains the Fourier normalization.

Example 15.2.1. Take $I = \mathbb{Z}[\varepsilon]$ with $\varepsilon = (3 + \sqrt{5})/2$. Then $N = 1$, G_I is trivial, and

$$b_0 = \frac{1 + \sqrt{5}}{2}, \qquad d = \frac{1 - \sqrt{5}}{2}, \qquad a = c_1 = 1.$$

The fusion rule will be $\rho^2 \simeq \mathbf{1} \oplus \rho$. The roots b_0, d will give its positive dimension and the specified duality closure, respectively.

15.3 The Leavitt presentation

Let $\mathcal{B}_I$ be the unital F_I-algebra generated by

$$X_g,\ X'_g,\ T_g,\ T'_g, \qquad g \in G_I,$$

with relations

$$X'_g X_h = T'_g T_h = \delta_{g,h} \mathrm{I}, \qquad X'_g T_h = T'_g X_h = 0,$$
$$\sum_g (X_g X'_g + T_g T'_g) = \mathrm{I}.$$

Primes denote algebraic dual generators, without any complex-conjugate or adjoint operation. This is the Leavitt algebra on $2N$ generator pairs.

For $g \in G_I$, define

$$\alpha_g(X_h) = X_{g+h}, \qquad \alpha_g(X'_h) = X'_{g+h},$$
$$\alpha_g(T_h) = \chi(g,h)^{-1} T_h, \qquad \alpha_g(T'_h) = \chi(g,h) T'_h.$$

These formulas preserve the relations and give $\alpha_g \alpha_h = \alpha_{g+h}$. Also define

$$U_g = \sum_h \chi(g,h)^{-1} X_h X'_h + \sum_h T_{h-g} T'_h. \tag{15.10}$$

Direct multiplication gives

$$U_0 = \mathrm{I}, \qquad U_g U_h = U_{g+h}, \qquad \alpha_g(U_h) = \chi(g,h) U_h.$$

Thus each U_g is invertible, with inverse U_{-g}.

The noninvertible generator of the category is an endomorphism ρ of $\mathcal{B}_I$. Its four families of images are

$$\rho(X_g) = U_g \left(d^{-1} \sum_h X_h + \sum_h Q(h) T_h T_{-h} \right) U_{-g}, \tag{15.11}$$

$$\rho(X'_g) = d^{-1} U_g \left(\sum_h X'_h + \sum_h Q(h)^{-1} T'_{-h} T'_h \right) U_{-g}, \tag{15.12}$$

$$\begin{aligned} \rho(T_g) = {} & \frac{a}{d} \sum_{h,k} \chi(h-g,k) X_h T'_k + \frac{c_1}{Q(g)} \sum_{h,k} \chi(g+h,k)^{-1} T_k X_h X'_h \\ & + \sum_{h,k} Q(h) b_{h+g} \chi(g,k)^{-1} T_{h+k} T_{-h} T'_k, \end{aligned} \tag{15.13}$$

$$\begin{aligned} \rho(T'_g) = {} & c_1 \sum_{h,k} \chi(h-g,k)^{-1} T_k X'_h + a Q(g) \sum_{h,k} \chi(g+h,k) X_h X'_h T'_k \\ & + \sum_{h,k} \frac{Q(g+h)}{Q(h)} b_{-h-g} \chi(g,k) T_k T'_{-h} T'_{h+k}. \end{aligned} \tag{15.14}$$

All sums run over G_I. All four formulas are over F_I.

The tensor category and its duality

For unital endomorphisms σ, τ of $\mathcal{B}_I$, put

$$\mathrm{Hom}(\sigma,\tau) = \{x \in \mathcal{B}_I : x\sigma(z) = \tau(z)x \text{ for every } z \in \mathcal{B}_I\}.$$

Composition is multiplication, and tensor product of objects is composition of endomorphisms. For $x : \sigma \to \sigma'$ and $y : \tau \to \tau'$, set

$$x \otimes y = x\sigma(y) = \sigma'(y)x : \sigma\tau \longrightarrow \sigma'\tau'. \tag{15.15}$$

This gives a strict tensor category. Let $\mathcal{C}_I$ be the additive idempotent completion of its subcategory generated by the α_g and ρ.

A fusion category is a rigid semisimple linear tensor category with finitely many simple objects, finite-dimensional Hom spaces and a simple tensor unit. It is split over F_I when the endomorphism field of every simple object is F_I.

THEOREM 15.3.1. Equations (15.11)–(15.14) define a unital endomorphism. The category $\mathcal{C}_I$ is a split rigid fusion category over F_I. Its simple objects are the pairwise distinct α_g, $g \in G_I$, and ρ, with rules

$$\alpha_g\alpha_h = \alpha_{g+h}, \qquad \alpha_g\rho \simeq \rho \simeq \rho\alpha_g, \qquad \rho^2 \simeq \bigoplus_g \alpha_g \oplus \rho^{\oplus N}. \tag{15.16}$$

For $\rho^\vee = \rho$, one may choose coevaluation X_0 and evaluation dX_0'.

Proof. Start with the complex Leavitt reconstruction of Radchenko–Wheeler, Section 5.4, Theorem 12 and Appendix B. In its notation make the change

$$X_g = \sqrt{d}\, S_g, \qquad X_g' = S_g'/\sqrt{d}.$$

The Leavitt relations are unchanged. Substitution gives exactly (15.11)–(15.14). The two mixed coefficients in $\rho(T_g)$ become $a/d, c_1$. Those in $\rho(T_g')$ become c_1, a. The quadratic terms in $\rho(X_g)$ lose their square root, and both terms in $\rho(X_g')$ acquire d^{-1}.

The reconstruction theorem proves the defining relations and

$$\alpha_g\rho = \rho, \qquad \rho\alpha_g = \mathrm{Ad}(U_g)\rho, \qquad \rho^2(z) = \sum_g X_g\alpha_g(z)X_g' + \sum_h T_h\rho(z)T_h'. \tag{15.17}$$

Faithful extension of scalars from F_I to $\mathbb{C}$ descends these equalities.

The same extension descends the Hom calculation. The intertwiner equations need only be imposed on the $4N$ algebra generators. Each Hom space is consequently the kernel of one F_I-linear map

$$\mathcal{B}_I \longrightarrow \mathcal{B}_I^{4N}.$$

A field extension preserves this kernel, even though $\mathcal{B}_I$ is infinite-dimensional. The complex calculation therefore gives $\mathrm{End}(\alpha_g) = \mathrm{End}(\rho) = F_I$ and zero Hom spaces between distinct displayed simples.

The last formula in (15.17) splits ρ^2, with inclusions X_g, T_h and projections X_g', T_h'. The Leavitt relations give their orthogonality and completeness. Every tensor word therefore splits into the listed simples. Finite sums and idempotent summands give a split semisimple category with finite-dimensional Hom spaces.

The dual of α_g is α_{-g}, with both maps 1. For ρ, the two snake equations are

$$dX_0'\rho(X_0) = 1, \qquad \rho(dX_0')X_0 = 1.$$

For the first, the X_0 term contributes $d\, d^{-1} = 1$, and every T-term vanishes. The second follows by the dual contraction in (15.12). This proves rigidity over the stated field. □

For the fusion rules above, the Frobenius–Perron dimension is the positive dimension function assigning dimension one to each invertible simple object.

Proposition 15.3.2. The Frobenius–Perron dimension of ρ is $\delta = \varepsilon - 1 = Nb_0$, while the specified duality closure of 1_ρ is $d = \varepsilon^{-1} - 1 < 0$. Moreover,

$$\varepsilon = 1 + Nb_0 = \frac{b_0}{b_0 - 1}, \qquad K = \mathbb{Q}(b_0).$$

There is no additive strong tensor functor from $\mathcal{C}_I$ to ordinary finite-dimensional vector spaces over any field.

Proof. The positive dimension satisfies $\delta^2 = N(1 + \delta)$. The identity $(\varepsilon - 1)^2 = N\varepsilon$ selects $\delta = \varepsilon - 1$. The other root is d. The chosen cup and cap have composite $dX_0'X_0 = d$. Formula (15.7) recovers the period and its quadratic field.

Under the proposed tensor functor, every invertible α_g would have dimension one. The image of ρ would have a nonzero integer dimension D, since its square contains the tensor unit. Then $D^2 = N(D + 1)$. It follows that $D + 1$ divides D^2, hence divides 1, which is impossible for $D > 0$. No faithfulness assumption is needed. □

An integral coefficient algebra

Put $\mathcal{S} = \mathcal{O}_L[1/N]$. The normalized units e_g and $b_0(b_0 - 1) = 1/N$ show that every b_g is a unit of $\mathcal{S}$. Also $\beta = \lambda u$ is an algebraic unit. Since $\beta \in L$, it belongs to $\mathcal{O}_L^\times$. Thus d, a, c_1, $Q(g)$ and all their required inverses lie in $\mathcal{S}$.

Proposition 15.3.3. The Leavitt presentation and all four formulas for ρ are defined over $\mathcal{S}$. The underlying Leavitt algebra is free as an $\mathcal{S}$-module. Its extension to L agrees with the extension of the F_I-presentation to L.

Proof. For the freeness assertion, combine the X and T generators into $s_1, \dots, s_{2N}$, with dual generators $t_1, \dots, t_{2N}$. Use the reduction rules

$$t_i s_j \longmapsto \delta_{ij}, \qquad s_{2N} t_{2N} \longmapsto 1 - \sum_{i<2N} s_i t_i.$$

Choose a length and lexicographic order with s_{2N} larger than the other positive generators. Every reduction decreases this order. The irreducible words are 1, pure positive or negative words, and positive words followed by negative words without the forbidden middle pair $s_{2N}t_{2N}$.

The only nontrivial overlap types are $t_i s_{2N} t_{2N}$ and $s_{2N} t_{2N} s_j$. Reducing either end of the first gives $\delta_{i,2N} t_{2N}$. Reducing either end of the second gives $\delta_{2N,j} s_{2N}$. Disjoint reductions commute. Hence the irreducible words give a basis over any coefficient ring.

All displayed coefficients lie in $\mathcal{S}$. The free normal form embeds this algebra into its scalar extension over L. The endomorphism identities already hold there, so they hold over $\mathcal{S}$. □

This integral model permits specialization away from N. It makes no semisimplicity assertion about the resulting category over a residue field. Primes dividing N require an additional integral construction.

15.4 FUSION BASES AND THE ARITHMETIC NETWORK

The fusion rule alone does not specify associativity coefficients. Here the Leavitt intertwiners give complete labeled splitting maps. For simple objects i, j, k, let $v_{ij}^{k,\mu} : k \to i \otimes j$ and $\bar{v}_{ij}^{k,\mu} : i \otimes j \to k$ denote dual inclusions and projections. Choose them as follows:

$i \otimes j,\ k$	$v_{ij}^{k,\mu}$	$\bar{v}_{ij}^{k,\mu}$
$\alpha_g\alpha_h,\ \alpha_{g+h}$	1	1
$\alpha_g\rho,\ \rho$	1	1
$\rho\alpha_g,\ \rho$	U_g	U_{-g}
$\rho^2,\ \alpha_g$	X_g	X'_g
$\rho^2,\ \rho$ in channel h	T_h	T'_h

They satisfy

$$\bar{v}_{ij}^{k,\mu} v_{ij}^{l,\nu} = \delta_{kl}\delta_{\mu\nu} 1_k, \qquad \sum_{k,\mu} v_{ij}^{k,\mu} \bar{v}_{ij}^{k,\mu} = 1_{i\otimes j}.$$

For four simple objects i, j, k, l, form bases of $\mathrm{Hom}(l, i \otimes j \otimes k)$:

$$L_{p,\mu,\nu} = (v_{ij}^{p,\mu} \otimes 1_k) v_{pk}^{l,\nu}, \qquad R_{q,\alpha,\beta} = (1_i \otimes v_{jk}^{q,\alpha}) v_{iq}^{l,\beta}.$$

Their dual projections are denoted by $\bar{L}, \bar{R}$. The full recoupling matrix is

$$F_{ijk}^{l}[(q,\alpha,\beta),(p,\mu,\nu)] = \bar{R}_{q,\alpha,\beta} L_{p,\mu,\nu}, \qquad (F^{-1})[(p,\mu,\nu),(q,\alpha,\beta)] = \bar{L}_{p,\mu,\nu} R_{q,\alpha,\beta}. \quad (15.18)$$

Each entry is a scalar in $\mathrm{End}(l) = F_I$. Formula (15.15) and Leavitt reduction compute every entry from the stated generators.

The completeness identities give $F^{-1}F = FF^{-1} = 1$. For four tensor factors, every composite of successive changes of basis equals the direct change between its first and last fusion trees. The two routes around the pentagon therefore agree. The unit choices give the triangle equations in the same way. A strict endomorphism presentation thus produces nontrivial recoupling matrices on its chosen simple-object skeleton.

Recovering the normalized values

Consider the endomorphism of ρ

$$\mathcal{N}_g = T'_{\mathrm{o}}(T'_{\mathrm{o}} \otimes 1_\rho)(1_\rho \otimes T_g)T_{\mathrm{o}} = T'_{\mathrm{o}}T'_{\mathrm{o}}\rho(T_g)T_{\mathrm{o}}. \quad (15.19)$$

The second expression is in the strict presentation. In a general presentation the first expression includes its specified associativity maps.

PROPOSITION 15.4.1. The network satisfies

$$\mathcal{N}_g = b_g 1_\rho, \qquad \mathcal{N}_g\mathcal{N}_{\mathrm{o}}^{-1} = e_g 1_\rho.$$

The scalars are intrinsic to the category with labeled invertible simples. They are preserved by linear tensor equivalences respecting those labels.

Proof. Substitute (15.13). Every mixed X-term vanishes. The right contraction forces $k = 0$, and the next forces $h = 0$. The remaining coefficient is b_g.

To recover the marking, choose isomorphisms $a_g : \alpha_g \otimes \rho \to \rho$. The equations

$$(a_g \otimes 1_\rho)(1_{\alpha_g} \otimes T_0) = T_0 a_g \quad \text{for every } g$$

define a line in $\mathrm{Hom}(\rho, \rho^2)$. In the explicit basis the action is $T_h \mapsto \chi(g,h)^{-1} T_h$. Nondegeneracy of χ makes the common invariant space exactly $F_I T_0$.

Choose $u_g : \rho \to \rho \otimes \alpha_g$. Normalize it so that

$$(1_\rho \otimes a_g)(u_g \otimes 1_\rho)$$

acts identically on $\mathrm{Hom}(\mathbf{1}, \rho^2)$. This fixes the endomorphism U_g.

Set $T_g = U_{-g} T_0$, and choose the dual projections T'_g. Only a common rescaling of the T_g remains. Its two factors cancel the two inverse factors in (15.19). The construction is therefore preserved by the stated equivalences. □

The unit summand normalizes U_g, but T_0 itself belongs to $\mathrm{Hom}(\rho, \rho^2)$. These are different channels. The specified duality closure is

$$dX'_0(1_\rho \otimes \mathcal{N}_g)X_0 = d\, b_g = -e_g. \tag{15.20}$$

Thus the closure recovers a normalized arithmetic unit, with its negative sign.

This closure cannot be replaced by a normalized cyclic functional on the ambient Leavitt algebra. Indeed,

$$\sum_{i=1}^{2N} [t_i, s_i] = (2N - 1)1.$$

In characteristic zero, the unit has zero image in

$$HH_0(\mathcal{B}_I) = \mathcal{B}_I / [\mathcal{B}_I, \mathcal{B}_I].$$

Every linear functional vanishing on commutators therefore sends 1 to zero. The marked categorical closure in (15.20) has a different domain.

Relabelling invertible simples relabels b_g. A semilinear tensor equivalence applies its field embedding to each network coefficient. Recovering the unnormalized $E(g)$ also requires the chosen $\sqrt{N}$. The normalized coefficients and their ratios are already defined over F_I.

15.5 The finite Heisenberg operators

The multiplicity space

$$H_I = \mathrm{Hom}(\rho, \rho^2) = \bigoplus_{h \in G_I} F_I T_h$$

carries two group actions:

$$p^g(T_h) = U_g T_h = T_{h-g}, \qquad q^g(T_h) = \alpha_{-g}(T_h) = \chi(g,h) T_h.$$

Their commutation rule is

$$p^g q^h = \chi(g,h) q^h p^g.$$

Because $Q(g+h) = Q(g)Q(h)\chi(g,h)$, the formula

$$(p+q)^g = Q(g)q^g p^g$$

also gives a group action. Here $p+q$ denotes this action, rather than the sum of two operators.

For any group action r and function $f : G_I \to F_I$, define

$$f(r) = \frac{1}{N} \sum_{x,y \in G_I} f(y)\chi(x,-y)r^x. \tag{15.21}$$

Character orthogonality gives $f(q)T_h = f(h)T_h$. Let δ_0 denote the indicator of zero, and put

$$P_0 = \delta_0(q), \qquad P_{\mathrm{av}} = \delta_0(p), \qquad D = P_0 P_{\mathrm{av}}.$$

The first two are projections, respectively onto $F_I T_0$ and the invariant vectors for translation. Their product acts by

$$D\left(\sum_h v_h T_h\right) = \frac{1}{N}\left(\sum_h v_h\right) T_0, \qquad D^2 = \frac{D}{N}, \qquad \mathrm{Tr}(D) = \frac{1}{N}.$$

For $N > 1$, the rank-one operator D is not an idempotent.

Although E need not be F_I-valued, its Fourier transform is:

$$\eta(g) := \widehat{E}(g) = NaQ(g)b_g \in F_I.$$

THEOREM 15.5.1. On H_I, one has

$$\eta(p)\eta(q) - \eta(q)\eta(p+q)\eta(p) = -N\beta D. \tag{15.22}$$

The tensor-unit projection of ρ^2 acts as zero on H_I, whereas $D \neq 0$.

Proof. Fourier inversion and (15.21) give

$$\eta(p) = \sum_x b_x p^{-x}, \qquad \eta(q) = \sum_y b_y q^{-y}, \qquad \eta(p+q) = \sum_t b_t Q(t) q^{-t} p^{-t}.$$

Order every term as $q^{-y}p^{-x}$. The coefficient of that term in the difference is

$$\chi(x,y)b_x b_y - \sum_t b_{y-t} b_t Q(t) b_{x-t} = -\frac{\beta}{N},$$

by (15.9). On the other hand,

$$D = \frac{1}{N^2} \sum_{x,y} q^{-y} p^{-x}.$$

This proves the factor N in (15.22).

The tensor-unit projection is $X_0 X_0'$. For every $v : \rho \to \rho^2$, the composite $X_0' v$ lies in $\mathrm{Hom}(\rho, \mathbf{1}) = 0$. Its action on H_I vanishes. By contrast, $D(T_0) = T_0/N \neq 0$. □

Example 15.5.2. Let $I = \mathbb{Z}[\sqrt{3}]$, $\varepsilon = 2 + \sqrt{3}$, and orient I by $(\sqrt{3}, 1)$. The nonzero class of $G_I \simeq \mathbb{Z}/2$ is represented by $x = (\sqrt{3} - 1)/2$. Since

$$(\varepsilon - 1)x = 1, \qquad \omega_I(x, \varepsilon x) = \frac{1}{2},$$

formula (15.3) gives $Q(x) = -i$.

Reflection gives $2b_1^2 = i$. Among its two signs, the Fourier condition $|\beta| = \sqrt{\varepsilon}$ selects

$$b_0 = \frac{1 + \sqrt{3}}{2}, \qquad b_1 = \frac{1 + i}{2}, \qquad \beta = 1 + \frac{\sqrt{3} + i}{2}.$$

In the basis T_0, T_1,

$$\eta(p) = \begin{pmatrix} b_0 & b_1 \\ b_1 & b_0 \end{pmatrix}, \qquad \eta(q) = \begin{pmatrix} \beta & 0 \\ 0 & b_0 - b_1 \end{pmatrix}.$$

Put $Q_q = \operatorname{diag}(1, -i)$. Then $\eta(p + q) = Q_q^{-1}\eta(p)Q_q$, and multiplication yields

$$\eta(p)\eta(q) - \eta(q)\eta(p + q)\eta(p) = -\beta \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = -2\beta D.$$

In particular, both the chosen multiplicity space and the normalization of the correction are visible in two dimensions.

The identity (15.22) is not an associativity failure of C_I. Its recoupling matrices satisfy the categorical pentagon. Moreover, the nonzero correction cannot be the image of a boundary under a chain map to $\operatorname{End}(H_I[0])$. That target has no term in degree -1. A comparison with supported chiral operations must retain a larger complex and specify its reduction to this multiplicity space.

15.6 Finite bimodule realization

We now construct the finite realization promised at the beginning. Let k be a field of characteristic zero, and let $\mathcal{C}$ be a split fusion category over k. Choose simple representatives $(S_i)_{i \in \mathcal{I}}$, with $S_0 = \mathbf{1}$, and set

$$A = \bigoplus_{i \in \mathcal{I}} k e_i, \qquad \omega(X)_{ij} = \operatorname{Hom}_{\mathcal{C}}(S_i, X \otimes S_j).$$

Declare $e_i \omega(X) e_j = \omega(X)_{ij}$. For $f : X \to X'$, the map $\omega(f)$ is postcomposition with $f \otimes 1$. Each $\omega(X)$ is a finite-dimensional A-bimodule.

Write

$$a_{X,Y,Z} : (X \otimes Y) \otimes Z \longrightarrow X \otimes (Y \otimes Z)$$

for the associator of $\mathcal{C}$. For $u : S_i \to X \otimes S_j$ and $v : S_j \to Y \otimes S_k$, define

$$\Phi_{X,Y}(u \otimes_A v) = a^{-1}_{X,Y,S_k}(1_X \otimes v)u. \tag{15.23}$$

The output belongs to $\operatorname{Hom}(S_i, (X \otimes Y) \otimes S_k)$. The balanced tensor retains precisely the matching middle labels.

Theorem 15.6.1. The maps $\Phi_{X,Y}$ are isomorphisms. Together with $\omega(\mathbf{1}) \simeq A$, they make

$$\omega : \mathcal{C} \longrightarrow \operatorname{Bimod}_A^{\mathrm{fd}}$$

a faithful exact strong monoidal functor. It need not be full.

Proof. For each $Y \otimes S_k$, choose splitting maps

$$v_{j\alpha} : S_j \longrightarrow Y \otimes S_k, \qquad w_{j\alpha} : Y \otimes S_k \longrightarrow S_j$$

with

$$w_{j\alpha} v_{l\beta} = \delta_{jl}\delta_{\alpha\beta} 1_{S_j}, \qquad \sum_{j,\alpha} v_{j\alpha} w_{j\alpha} = 1.$$

The inverse to (15.23) is

$$h \longmapsto \sum_{j,\alpha} (1_X \otimes w_{j\alpha}) a_{X,Y,S_k} h \otimes_A v_{j\alpha}.$$

Composing in one order uses the sum of the splitting idempotents. Composing in the other uses their pairwise orthogonality. A change of splitting basis acts by inverse matrices on the two factors, so the displayed sum is independent of that choice.

For three objects, both iterated tensorators compose the same three maps from simple objects. The associator pentagon equates the two parenthesizations. The same argument handles every finite tensor string, and the unitors give $\omega(\mathbf{1}) \simeq A$. The column $j = 0$ is $\bigoplus_i \mathrm{Hom}(S_i, X)$. It detects every morphism out of X, proving faithfulness. Split semisimplicity gives exactness.

Finally, $\mathrm{End}_{A-A}(A) = A$, whereas $\mathrm{End}_C(\mathbf{1}) = k$. For more than one simple label, the induced map on these endomorphism spaces is not surjective. □

An associator detected by its endpoint

Take the pointed category with labels $\mathbb{Z}/2$ and scalar associator

$$c(g,h,l) = (-1)^{ghl}.$$

Its pentagon is the mod-two three-cocycle equation.

The bimodule $\omega(g)$ has one-dimensional entries at $i = g + j$. On the component with right endpoint r, the tensorator is multiplication by $c(g,h,r)^{-1}$. Its coherence equation is

$$c(g,h,l)c(g,h,l+r)^{-1}c(g+h,l,r)^{-1} = c(h,l,r)^{-1}c(g,h+l,r)^{-1}.$$

At $g = h = l = 1$, forgetting the endpoint factors would leave the false equality $-1 = 1$. The strict ambient category of bimodules thus retains the nontrivial source associator through its tensorator.

The finite arithmetic detector

Apply the construction to C_I over $k = F_I$, with labels $\mathcal{I} = G_I \sqcup \{\rho\}$. Let W_X be the matrix with entries $(W_X)_{ij} = \dim_k \omega(X)_{ij}$. The fusion rule gives

$$W_{\alpha_g} = P_g \oplus [1], \qquad W_\rho = \begin{pmatrix} 0_{N\times N} & \mathbf{1} \\ \mathbf{1}^t & N \end{pmatrix},$$

where P_g is the translation permutation on G_I. Matrix multiplication records the balanced tensor product:

$$W_\rho^2 = \sum_g W_{\alpha_g} + NW_\rho, \qquad \dim_k \omega(\rho) = 3N.$$

For $N = 2$, this is

$$W_\rho = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Its entries sum to 6, while the entries of W_ρ^2 sum to 18. The ordinary vector-space tensor square would have dimension 36.

Transport every map in (15.19) through ω, using Φ for each tensor product. The resulting endomorphism M_g of $\omega(\rho)$ satisfies

$$M_g = b_g \mathrm{I}_{\omega(\rho)}, \qquad b_g = \frac{\mathrm{Tr}_k(M_g)}{3N}, \qquad E(g) = \frac{\sqrt{N}}{3N}\,\mathrm{Tr}_k(M_g).$$

The first recovery takes place over F_1. The second uses the chosen radical. Ratios recover e_g without it.

This matrix trace works because the specified composite ends in the simple object ρ and is scalar there. It does not identify ordinary matrix trace with the duality closure (15.20) on arbitrary diagrams. The finite realization also keeps its chosen categorical intertwiners, rather than every bimodule map.

Finally, it is not a finite-dimensional unital representation of $\mathcal{B}_I$. Such a representation on V would give $V \simeq V^{\oplus 2N}$ from the Leavitt generators. Equivalently, matrix trace applied to $\sum_i [t_i, s_i] = (2N-1)1$ gives $0 = (2N-1)\dim_k V$, impossible in characteristic zero.

The arithmetic values have therefore been retained by actual fusion maps, a normalized network and a finite balanced tensor. A further field realization must transport these maps and their duality, rather than only the scalar detector. Its supported operation complex must also explain how the correction $-N\beta D$ arises after reduction.

Chapter 16

Arithmetic fusion as singular interfaces

The marked network of Chapter 15 has value b_g, and its specified duality closes it to $-e_g$. Insert the same junctions into two Heisenberg currents. Their contraction produces the double pole

$$-\frac{e_g}{(z_1 - z_3)^2}.$$

The intervening insertions remain part of the ordered coefficient product. Thus the singular amplitude retains a marked fusion composition that its scalar residue alone would not determine.

We construct this realization on all oscillator states and every collision stratum. It has infinite-dimensional coefficient spaces. A second realization uses the finite bimodules of the preceding chapter, together with the categorical symmetry that singles out their allowed maps. Tensoring them with the E_8 lattice theory gives finite conformal grades and convergent relative sewing.

For the order-two arithmetic category, two further realizations have a single conformal vacuum in their ambient vertex algebra. Changing the quadratic embedding gives the E_6 module category at affine $\mathfrak{sl}_2$ level ten. The original embedding is realized by the Virasoro theory of central charge $-39/10$. We compare their actual junctions, full descendant fields and collision coefficients. The coefficient algebra, the finite bimodule, the conformal state space and the conformal-block bundle will remain separate objects throughout.

16.1 The junction and its coefficient field

Use the arithmetic data of Section 15.1 and write $F = F_I$, $B = \mathcal{B}_I$, $C = C_I$ and $G = G_I$. Thus F has characteristic zero and

$$G = (\varepsilon - 1)^{-1} I/I, \qquad N = |G| = \varepsilon + \varepsilon^{-1} - 2.$$

The generators X_g, X'_g, T_g, T'_g, their endomorphisms α_g, ρ, and the elements U_g are those of Section 15.3. Primes denote algebraic generators, not adjoints. In the notation used below,

$$e_g = E_g/E_0, \quad b_g = E_g/\sqrt{N}, \quad b_0 = \frac{\varepsilon - 1}{N}, \quad d = -b_0^{-1} = \varepsilon^{-1} - 1. \tag{16.1}$$

The positive roots obey $\sqrt{N} = \sqrt{\varepsilon} - \varepsilon^{-1/2}$. Put $\beta = \sum_g b_g$, $a = \beta/(N b_0)$ and $c = b_0/\beta$. Here c is the coefficient denoted c_1 in (15.8). The Gaussian Q and bicharacter χ retain the conventions of (15.4).

Tensor categories and their functors are linear over their stated fields, except where semilinearity is specified. Complexes are cohomologically $\mathbb{Z}$-graded, with the Koszul sign for tensor products of homogeneous maps.

The endomorphism presentation gives

$$\alpha_g \rho = \rho, \qquad \rho \alpha_g = \mathrm{Ad}(U_g)\rho, \qquad \alpha_g(U_h) = \chi(g,h) U_h. \tag{16.2}$$

Here $\mathrm{Ad}(u)(z) = uzu^{-1}$. The first two identities are proved in Theorem 15.3.1. The last follows by applying α_g to (15.10). The specified cup and cap are X_0 and dX_0'.

The fusion ring records the quadratic period. For $r = [\rho]$ and $S = \sum_g [\alpha_g]$, the relations $r^2 = Nr + S$ and $rS = Nr$ give

$$(1 + r)(1 - r + S) = 1.$$

The positive dimension character takes $1 + r$ to ε, while the specified duality dimension takes it to ε^{-1}. The inverse is a virtual class, not an inverse object for $1 \oplus \rho$.

The marked network of Proposition 15.4.1 is

$$\mathcal{N}_g = T_0' T_0' \rho(T_g) T_0. \tag{16.3}$$

Its value and its duality closure are

$$\mathcal{N}_g = b_g 1, \qquad dX_0' \rho(\mathcal{N}_g) X_0 = db_g = -e_g. \tag{16.4}$$

The calculation following (15.20) shows why this closure must retain the marked object: in the ambient algebra,

$$\sum_g \big([X_g', X_g] + [T_g', T_g] \big) = (2N - 1)1,$$

so the unit has zero Hochschild class.

For $\varepsilon = 2 + \sqrt{3}$, the simples are 1, g, ρ, with $g^2 = 1$, $g\rho = \rho g = \rho$ and $\rho^2 = 1 \oplus g \oplus \rho^{\oplus 2}$. Their coefficients are

$$b_0 = \frac{1 + \sqrt{3}}{2}, \qquad b_1 = \frac{1 + i}{2}, \qquad d = 1 - \sqrt{3}.$$

The second marked network closes to the negative of

$$e_1 = \frac{(\sqrt{3} - 1)(1 + i)}{2}, \qquad e_1^4 + 2e_1^3 + 2e_1^2 - 2e_1 + 1 = 0. \tag{16.5}$$

The singular construction will recover this value as the coefficient of a double pole.

16.2 Operator coefficients at a nonconsecutive collision

A coefficient multiplying a field may be noncommutative. This changes the operation at a collision that passes an intervening insertion. For example, take $A = C = E_{12}$ and $D = E_{21}$ in $\mathrm{Mat}_2(F)$. Then $AC = 0$ while $ADC = E_{12}$. The algebra B contains such matrix units: choose two of its positive generators L_1, L_2, and put $E_{ij} = L_i L_j'$. This gives a matrix corner with identity $E_{11} + E_{22}$.

The exchange of Section 5.4 retains this context. It is the linear map

$$c_B : B \otimes B \longrightarrow B \otimes B, \qquad c_B(x, y) = xy \otimes 1 + 1 \otimes xy - x \otimes y.$$

Let m denote multiplication in B. Direct expansion gives

$$\begin{aligned} c_B^2 = 1, \quad & mc_B = m, \quad c_B(1, x) = (x, 1), \quad c_B(x, 1) = (1, x), \\ & (c_B)_{12}(c_B)_{23}(c_B)_{12} = (c_B)_{23}(c_B)_{12}(c_B)_{23}, \\ & c_B(1 \otimes m) = (m \otimes 1)(1 \otimes c_B)(c_B \otimes 1), \\ & c_B(m \otimes 1) = (1 \otimes m)(c_B \otimes 1)(1 \otimes c_B). \end{aligned} \tag{16.6}$$

For example, the two braid words on $x \otimes y \otimes z$ both give

$$\begin{aligned} xyz \otimes 1 \otimes 1 - xy \otimes z \otimes 1 + 1 \otimes 1 \otimes xyz + 1 \otimes xy \otimes z \\ + x \otimes yz \otimes 1 - 1 \otimes x \otimes yz - x \otimes y \otimes z. \end{aligned}$$

No dimension hypothesis occurs in these identities.

Let f be a map between finite ordered sets. Define q_f by three steps. Sort the inputs into the fibers of f with adjacent c_B's. Multiply inside each fiber, and insert 1_B in an empty fiber. The relations (16.6) imply

$$q_g q_f = q_{gf}, \qquad q_{\mathrm{id}} = 1. \tag{16.7}$$

To prove this, use the braid and involution relations to change a sequence of adjacent crossings. Crossings inside one fiber disappear by $mc_B = m$. The last two relations move a merger through a crossing. Every composite therefore reduces to the sorting diagram for the composite map, followed by the multiplications in its fibers. Associativity and the unit equations remove the remaining choices. These moves also prove naturality in every unital algebra homomorphism $B \to B'$.

At three inputs, with the remaining blocks ordered as $2|13$, this gives

$$q_{2|13}(x, y, z) = xy \otimes z + 1 \otimes xyz - x \otimes yz, \qquad mq_{2|13}(x, y, z) = xyz.$$

At four inputs,

$$q_{13|24}(x, y, z, w) = xyz \otimes w + x \otimes yzw - xy \otimes zw.$$

The intermediate factors have not been replaced by the product of the colliding entries. This is the coefficient information needed for the geometric residue identities.

16.3 A singular construction in every arity

Use the level-one Heisenberg vertex algebra V of Section 6.1, now over F. Its state space is $F[x_1, x_2, \ldots]$, its vacuum is 1, and its current modes are

$$J_{-m} = x_m, \qquad J_m = m\partial_{x_m} \quad (m > 0), \qquad J_0 = 0.$$

Write $J = x_1$ for the current state. The translation operator is $T_V = \sum_{m>0} m x_{m+1} \partial_{x_m}$. The field of x_m is $\partial_z^{m-1} J(z)/(m-1)!$. The field of a monomial is the normal ordered product of these fields. The commutator $[J_m, J_n] = m\delta_{m+n,0}$ gives

$$J(z)J(w) \sim \frac{1}{(z-w)^2}.$$

The double pole is the contraction $\sum_{m>0} m z^{-m-1} w^{m-1} = (z-w)^{-2}$ in the region $|z| > |w|$. Thus the pole and its normalization are fixed by the oscillator commutator. The arithmetic coefficients will multiply this contraction. All states are polynomials, and all products have an input-dependent finite pole order. No completion of the coefficient algebra is imposed.

Take $X = \mathbb{A}^1_F$ and $H = \omega_X \otimes_F V$. We use right differential modules, with

$$(f(z)\,dz \otimes v)\partial_z = -f'(z)\,dz \otimes v - f(z)\,dz \otimes T_V v.$$

For $t = z_1 - z_2$, the local-current face is the full principal-part map

$$r_V(f; v, w) = \mathrm{pp}_{t=0}\big(f Y_V(v, t) w\big).$$

Normal derivatives are retained: t^{-q-1} corresponds to $\partial\,\partial_t^{q}/q!$, where $\partial t = 0$. Thus multiplication by t lowers the principal-part order and right multiplication by ∂_t raises it with coefficient $q+1$.

We use the local-current Chevalley–Cousin construction of Beilinson–Drinfeld, §3.4.11. Its input is the chiral differential module of V, with the above full faces. The two-step face sum vanishes: disjoint collisions cancel with their determinant signs, and a three-block collision is the chiral Jacobi identity. For the Heisenberg algebra these identities also follow from Wick expansion and the central commutator of the modes. This local-current input is distinct from the ordered coefficient multiplication in B.

For n labeled positions, the ordered collision scheme R_n has one chart X^n for each ordering. Identify two charts away from the diagonals on which their orderings disagree.

A partition π has one coordinate for each block. Write $i_\pi : X^\pi \to X^n$ for its diagonal and j_π for the distinct-block open subset of X^π. Define

$$Q_\pi(V) = (i_\pi)_*(j_\pi)_* j_\pi^*(\boxtimes_{A\in\pi} H).$$

Order the blocks by their last entries and retain the later root at a merger. Orient its normal coordinate by $z_i - z_j$ for $i < j$. With the ordered density line the face that removes block i has sign $(-1)^{i-1}$. These conventions fix the signed current face $d^V_{\pi,\eta}$ for every elementary coarsening $\pi \lessdot \eta$.

The associative construction uses consecutive cuts in a fixed word, meaning a finite ordered list of insertion positions. To specify its carrier, let $\mathrm{OrdSurj}_+$ have the nonempty finite ordered sets as objects and monotone surjections as arrows. Put

$$\mathcal{W}_X = D(F) \times \varprojlim_{\mathrm{OrdSurj}_+} D\,\mathrm{Mod}(X^n),$$

The limit is taken in stable ∞-categories. We use unbounded right differential modules with quasi-coherent underlying modules. For $\alpha : [n] \twoheadrightarrow [m]$, the transition is extraordinary inverse image along the diagonal $d_\alpha(x)_i = x_{\alpha(i)}$. An object includes the transition identifications and their coherences. The factor $D(F)$ records the empty configuration independently.

For $0 < p < n$, let $U_p \subset X^n$ be the open set where the first p positions are distinct from the remaining positions. Write $j_p : U_p \hookrightarrow X^n$. The cut product is

$$(M \star N)_n = \bigoplus_{p=0}^{n} (j_p)_*\,(M_p \boxtimes N_{n-p})\big|_{U_p}.$$

An empty block means scalar tensoring on all of X^n. The unit is F at the empty configuration and zero elsewhere. A strong cut coalgebra has a counit and coassociative cut maps whose restrictions to each U_p are isomorphisms. For a connected coaugmented such coalgebra C, its cut cobar is the tensor algebra $T_\star(C_+[-1])$ on its positive part. Writing tc for desuspension, its differential is the derivation with

$$d(tc) = -t(d_C c) + \sum_{\overline{\Delta} c = c'\otimes c''} (-1)^{|c'|+1}(tc')(tc'').$$

Here $\overline{\Delta}$ is the sum of the nonempty cut maps.

THEOREM 16.3.1. Let B be a nonzero unital associative F-algebra concentrated in degree zero. The chartwise formula

$$C_n(V,B) = \bigoplus_{\pi\vdash[n]} Q_\pi(V)[|\pi|] \otimes_F B^{\otimes|\pi|}, \qquad d = \sum_{\pi\lessdot\eta} d^V_{\pi,\eta}\otimes q_{\pi,\eta} \tag{16.8}$$

defines a complex on every ordered chart and glues on R_n. The complexes on the standard word charts form a connected strong cut coalgebra in $\mathcal{W}_X$, with $C_0(V, B) = F$. This coalgebra is natural in unital algebra maps. Its cut cobar

$$\mathcal{A}_B = \Omega_{\mathrm{cut}} C(V, B)$$

is an associative algebra in the ordered cut-convolution category. Its binary field system is

$$Y(v \otimes b, z)(w \otimes c) = Y_V(v, z)w \otimes bc, \qquad T = T_V \otimes 1. \tag{16.9}$$

Proof. For fixed π and ν separated by two mergers, (16.7) makes every coefficient composite equal to $q_{\pi,\nu}$. The corresponding component of d^2 is therefore

$$\left(\sum_{\pi \lessdot \eta \lessdot \nu} d^V_{\eta,\nu} d^V_{\pi,\eta} \right) \otimes q_{\pi,\nu} = 0.$$

On a chart overlap use the usual current permutation and determinant identification, and use c_B to reorder coefficient blocks. The braid relation gives the overlap cocycle, and (16.7) gives compatibility with the differential. A partition containing an internal pair whose orders disagree has empty support on that overlap.

Under an ordered collision $\alpha : [n] \twoheadrightarrow [m]$, a partition term survives exceptional pullback only when it is saturated for α. Otherwise its distinct-block open factor is pulled to a collision, and $i^! j_* = 0$ makes that term vanish. The surviving terms are exactly $\pi = \alpha^{-1}(\eta)$, with the same coefficient factors and faces as $C_m(V, B)$. Closed-diagonal base change identifies their differential modules. For composite collisions the same saturated partitions survive, so the comparisons compose.

On the open subset where the first p positions are distinct from the last $n - p$, every partition crossing that cut disappears. A remaining partition is uniquely a partition of each block. Its coefficient complex is the external product of the two block complexes. These are the strong cut comparisons, and their associativity follows from splitting into three blocks.

The cobar is the tensor algebra on the shifted positive coefficient, with differential given by d and these cut maps. At a fixed configuration its positive tensor length is bounded by the arity. No infinite sum in that length is needed.

All arguments take place in the category of differential modules allowing arbitrary coefficient vector spaces. Tensoring a differential module by a vector space means the corresponding direct sum. Localization and closed-diagonal direct image commute with these sums. Thus neither an algebraic dual of B nor a finite-rank Riemann–Hilbert assertion is used.

For a finitely generated algebra, filter by word length and put the total word-length filtration on $B^{\otimes s}$. The maps c_B and q_f preserve it. Each finite-arity coefficient diagram is the filtered colimit of its finite-dimensional coefficient diagrams. This applies to the Leavitt algebra, even though its finite word-length spaces are not subalgebras. Naturality and the binary assertion follow directly from the formulas. □

The positive-cut contraction of Section 3.1 retains every nonconsecutive stratum. Here its normal derivatives and coefficient maps can be written explicitly. The cobar retracts onto a complex supported on

$$Z_n = \bigcap_{p=1}^{n-1} \bigcup_{i \le p < j} \Delta_{ij}; \qquad Z_3 = \Delta_{13}, \quad Z_4 = \Delta_{14} \cup (\Delta_{13} \cap \Delta_{24}).$$

The differential modules on these strata retain all normal derivatives. For a partition π, let $J(\pi)$ be the cuts crossed by none of its blocks. A cobar term is indexed by a subset $Q \subseteq J(\pi)$, with coefficient

$$\big(Q_\pi(V) \otimes B^{\otimes|\pi|}\big)[|\pi| - |Q| - 1].$$

If $r_{\le c}$ is the number of blocks to the left of a cut c, adding $c \notin Q$ has sign $(-1)^{\#\{q\in Q: q<c\}+1-r_{\le c}}$. Merging root-indexed blocks $i < j$ in the ℓth consecutive segment has sign $(-1)^{\ell-1+i}$ and coefficient map $r^V_{ij} \otimes q_{ij}$, where r^V_{ij} is the unsigned current face. Write v and δ for these two parts of the differential.

For $J(\pi) \neq \varnothing$, put $c_0 = \min J(\pi)$. Define h_0 to remove c_0 with sign $(-1)^{1-r_{\le c_0}}$ when $c_0 \in Q$, and to be zero otherwise. It is also zero when $J(\pi) = \varnothing$.

Let i include the terms with no admissible cut, and let p_0 project onto them. Adding and removing c_0 gives $vh_0 + h_0 v = 1 - ip_0$. All other cut terms cancel in pairs. Merging blocks cannot create an admissible cut, so these terms form a subcomplex for δ. The perturbed projection and homotopy are

$$p = p_0 \sum_{q=0}^{n-1} (-\delta h_0)^q, \qquad H = h_0 \sum_{q=0}^{n-1} (-\delta h_0)^q.$$

Each residue lowers the number of blocks, which proves finiteness. With these maps one has $dH + Hd = 1 - ip$. After suspension, put $\bar H = -sHs^{-1}$, $\bar i = sis^{-1}$ and $\bar p = sps^{-1}$. If b_2 is the suspended cobar product, its transferred operations are specified in every arity by

$$F_1 = \bar i, \quad \Lambda_n = \sum_{a+b=n} b_2(F_a \otimes F_b), \quad F_n = -\bar H \Lambda_n, \quad B_n = \bar p \Lambda_n \quad (n \ge 2).$$

These are finite homological perturbation formulas. The coefficient maps commute with the contraction, so algebra homomorphisms preserve the full transferred system.

For the arithmetic algebra, (16.9) realizes (16.3) with a genuine pole. Let ϵ_V extract the constant Fock coefficient. It is a linear functional, not a character of the vertex algebra. In the expansion region $|z_1| > |z_3|$,

$$\begin{aligned}(\epsilon_V \otimes 1) Y(J \otimes T'_0, z_1) Y(1 \otimes T'_0, z_2) Y(J \otimes \rho(T_g), z_3) Y(1 \otimes T_0, z_4)(1 \otimes 1) \\ = \frac{b_g}{(z_1 - z_3)^2}\, 1_B.\end{aligned}$$

Closing the coefficient line multiplies this by d, giving $-e_g/(z_1 - z_3)^2$. The rational function determines the other expansion regions. Arithmetic enters through the marked endomorphisms and junctions. The Heisenberg level remains one.

16.4 Relative interfaces and the connected comparison

The field realization retains compositions of coefficients. For its tensor structure, recall the bimodules of Section 8.4 with the following conventions. Put $R = B^{\mathrm{op}}$. For an endomorphism σ of B, let P_σ have underlying vector space B and actions

$$a^{\mathrm{op}} \cdot b \cdot c^{\mathrm{op}} = \sigma(c) b a.$$

An intertwiner $x : \sigma \to \tau$ acts by $b \mapsto xb$. Every left R-linear map has this form, and right R-linearity is exactly the intertwining equation. Hence this construction is fully faithful. The tensor comparison is

$$P_\sigma \otimes_R P_\tau \longrightarrow P_{\sigma\tau}, \qquad b \otimes c \longmapsto \sigma(c) b. \tag{16.10}$$

It is balanced, has inverse $z \mapsto z \otimes 1$, and is associative because $\sigma(\tau(e)c)b = \sigma\tau(e)\sigma(c)b$. It takes tensor products of intertwiners to $x\sigma(y)$. The unit is $P_{\mathrm{id}} = R$. Every P_τ is free as a left R-module, so the same formula computes the derived balanced product.

Let $A = \mathcal{A}_F$ be the Heisenberg cut cobar, let $\star$ denote ordered cut convolution, and put

$$\mathfrak{R} = A \otimes_F R, \qquad \mathfrak{P}_\sigma = A \otimes_F P_\sigma.$$

Coefficient vector spaces act centrally on differential modules. The A-factors are always multiplied in their original order.

Theorem 16.4.1. There is a faithful strong tensor functor

$$C \longrightarrow \mathrm{Bimod}_{\mathfrak{R}}(\mathcal{W}_X), \qquad \sigma \longmapsto \mathfrak{P}_\sigma,$$

where $\mathcal{W}_X$ is the derived ordered cut-convolution category and the target tensor product is $\star^{\mathbb{L}}_{\mathfrak{R}}$. It preserves the specified duality and all fusion-network evaluations.

Proof. For coefficient bimodules of this form, the two-sided bar gives

$$(A \otimes P) \star^{\mathbb{L}}_{A \otimes R} (A \otimes Q) \simeq A \otimes (P \otimes^{\mathbb{L}}_R Q). \tag{16.11}$$

The comparison multiplies the A-factors in simplicial degree n in their original order and leaves the coefficient bar $P \otimes R^{\otimes n} \otimes Q$. One can compute using the free left coefficient resolution of Q. After extension by A its terms are free left $A \otimes R$-modules. The augmented resolution contracts by its extra unit degeneracy. This proves (16.11).

Equivalently, for P_σ and P_τ it is the internal version of the explicit free-module isomorphism (16.10).

All parenthesizations for three or more factors realize the same multisimplicial bar. Multiplying the geometric factors in order gives the same map in each case, proving the associativity and unit coherences.

Combine this comparison with (16.10) and extend to finite sums and retracts. Evaluation at the empty configuration recovers P_σ and its coefficient maps, proving faithfulness. The images of the cup and cap obey the same snake equations. Applying the functor to (16.4) proves the assertion about network values. □

Here $\mathfrak{R}_\emptyset = R$. This empty algebra is part of the tensor comparison. There is also a connected construction with empty coefficient F, and the relation between them is an actual algebra map. Apply Theorem 16.3.1 to R, and set $G_R = \mathcal{A}_R$. If $m_\pi : R^{\otimes|\pi|} \to R$ multiplies in block order, then

$$m_\eta q_{\pi,\eta} = m_\pi.$$

Let $\tau : C(V, F)_+ \to A$ be the universal cobar twisting map. On the π-term define $t_R = \tau \otimes m_\pi$. The last identity makes it compatible with residues, and on a cut the total product is the product of the two block products. Thus t_R obeys the twisting equation and induces

$$\theta_R : G_R \longrightarrow \mathfrak{R}. \tag{16.12}$$

On a cobar monomial it retains the geometric monomial and multiplies all coefficient factors in their original order. This construction takes place in the word category $\mathcal{W}_X$ and is natural in R. Its empty component is the unit map $F \to R$.

Intertwiners have a linear connected realization as well. For an R-bimodule P, form the square-zero algebra $R \ltimes P$. Give R weight zero and P weight one. Multiplication and the exchange preserve total weight, so $\mathcal{A}_{R\ltimes P}$ is weight graded. Its weight zero part is G_R. Its weight one part, denoted $L_R(P)$, is a G_R-bimodule.

Each coefficient in weight one contains exactly one P-factor. Consequently $P \mapsto L_R(P)$ is additive and linear on bimodule maps. Its one-point component is $H \otimes P$, so it is faithful on degree-zero coefficient maps.

Taking weight one in (16.12) for $R \ltimes P$ gives the comparison

$$L_R(P) \longrightarrow \mathrm{Res}_{\theta_R}(A \otimes P). \tag{16.13}$$

It is the identity at one point. At the empty configuration it is $0 \to P$.

This extraction of weight one is necessary. For example, with $M_\sigma = {}_\sigma B_B$, an intertwiner x gives an algebra map

$$B \ltimes M_\sigma \longrightarrow B \ltimes M_\tau, \qquad (b, m) \longmapsto (b, xm).$$

The algebra map has the identity on its weight zero part, so it is not linear as a function of x. Its weight one map is linear. Such maps realize the intertwiners, but do not identify their transported tensor product with $\bigstar^{\mathbb{L}}_{G_R}$.

16.5 Analytic states and relative sewing

Fix an embedding $F \hookrightarrow \mathbb{C}$ and extend scalars along it. The oscillator construction has an analytic realization on all states. This also gives an analytic meaning to the relative interfaces.

For an R-bimodule P put

$$\mathsf{H}(P) = V \otimes P, \qquad \mathsf{H}(P)_n = V_n \otimes P, \qquad \widehat{\mathsf{H}(P)} = \prod_{n\geq 0} \mathsf{H}(P)_n.$$

Here $\deg x_m = m$. States belong to the direct sum. Outputs of fields belong to the product. The restricted dual is $\bigoplus_{n\geq 0} \mathsf{H}(P)_n^*$, with the full algebraic dual in each degree.

The conformal vector is $\frac{1}{2}x_1^2 \otimes 1$ in $\mathsf{A} = \mathsf{H}(R)$. Its central charge is one.

An *oscillator interface* is $\mathsf{H}(P)$ with its Heisenberg action and commuting left and right R-actions. Morphisms commute with these three actions. If $f : P \otimes_R Q \to S$ is a bimodule map, define

$$\mathcal{I}_f(v \otimes p, z)(w \otimes q) = Y_V(v, z)w \otimes f(p \otimes_R q). \tag{16.14}$$

A balanced intertwining operator means an intertwining operator for the Heisenberg vertex algebra, with these left, right, and balancing identities on its coefficient factors. This definition includes every oscillator descendant.

THEOREM 16.5.1. The oscillator interfaces have the following properties.

(i) Their morphisms are exactly $1_V \otimes f$, for bimodule maps f. Every balanced intertwining operator is uniquely of the form (16.14). Consequently their relative fusion is represented by

$$\mathsf{H}(P) \boxtimes_{\mathsf{A}} \mathsf{H}(Q) = \mathsf{H}(P \otimes_R Q).$$

(ii) Products and iterates of their fields converge absolutely after pairing with any restricted-dual state, in the usual nested-disc regions. They are expansions of rational functions. Every genus-zero sewing of these interfaces respects their tensor associativity and units.

(iii) The functor $\sigma \mapsto \mathsf{H}(P_\sigma)$ is fully faithful and strong tensor. It preserves the specified cups, caps, and marked arithmetic junctions on all states. Its tensor-product field is

$$\mathcal{I}_{\sigma,\tau}(v \otimes b, z)(w \otimes c) = Y_V(v, z)w \otimes \sigma(c)b.$$

(iv) Taking full normal principal parts gives the relative chiral interfaces of Theorem 16.4.1. The comparison commutes with every collision, consecutive cut, and cobar differential.

Proof. The vectors killed by all positive Heisenberg modes in $\mathsf{H}(P)$ are exactly $1 \otimes P$. A map commuting with the modes is determined there, because the negative modes generate every polynomial state. The coefficient actions give the bimodule condition. This proves the morphism assertion.

For an intertwining operator $\mathcal{I}$, translation gives $\partial_z \mathcal{I}(1 \otimes p, z) = 0$. Its value on $1 \otimes q$ has weight zero, hence is $1 \otimes f(p, q)$. The Heisenberg commutator and iterate identities determine its values on all states by induction on the number of negative modes. They give exactly (16.14). Balancing says $f(pr, q) = f(p, rq)$, and the outer actions say that the induced map on $P \otimes_R Q$ is a bimodule map. This proves the universal property in (i).

Wick expansion of finitely many polynomial states has finitely many contraction patterns. Each contraction is a derivative of $(z_i - z_j)^{-2}$. Pairing with a restricted-dual state leaves a finite sum of rational functions. Their Taylor expansions converge absolutely when each inner disc is contained in its outer disc and avoids the remaining insertions. The finitely many coefficient products lie in a finite-dimensional subspace of the algebraic coefficient space, so no norm or completion of R is needed. Changing a tree changes only the expansion region of the same function and the bracketing of the same balanced coefficient product.

This proves (ii).

Apply (i) to (16.10). Its associativity, unit, and intertwiner equations prove (iii), first on endomorphisms and then on finite sums and retracts. The cups and caps are the corresponding coefficient maps tensored with 1_V. Their snake identities therefore hold on every descendant.

Finally, the state identification sends $x_{m_1} \cdots x_{m_l} \otimes p$ to $J_{-m_1} \cdots J_{-m_l} 1 \otimes p$. It intertwines translation and all fields. It therefore identifies the right differential-module actions and every full principal part. The coarsenings multiply the same ordered coefficients on both sides. The cut maps select the same partitions. These identities identify the whole Cousin diagram and hence its cobar, proving (iv). □

The sum over oscillator descendants in a sewing is explicit. For a partition $\lambda = (1^{m_1} 2^{m_2} \cdots)$ put $x_\lambda = \prod_j x_j^{m_j}$ and $z_\lambda = \prod_j j^{m_j} m_j!$. Normalize the Heisenberg pairing by $\langle 1, 1\rangle = 1$ and $\langle J_{-m} v, w\rangle = \langle v, J_m w\rangle$. Identify the opposite-coordinate dual Fock space through this pairing.

Its sewing tensor is

$$\mathcal{P}(t) = \sum_\lambda \frac{t^{|\lambda|}}{z_\lambda} x_\lambda \otimes x_\lambda, \qquad |\lambda| = \sum_j j m_j.$$

Coefficient factors are contracted over R, by the relative evaluation. They are not summed over a vector-space basis of R. The Wick argument in the proof establishes the convergence of the resulting matrix coefficients on a sewing annulus. Omitting the positive oscillator degrees would change, for example, the expansion

$$\frac{1}{(z - w)^2} = \sum_{n \geq 1} n\, z^{-n-1} w^{n-1}.$$

Thus every arithmetic category in the construction has a full analytic ordered realization, with nonconstant fields and all descendants. It is a relative theory with infinite-dimensional weight-zero coefficients.

This qualification cannot be removed by truncating the same coefficient action. A finite-dimensional unital B-module W would satisfy $W \simeq W^{\oplus 2N}$ by the Leavitt generators, hence $\dim W = 2N \dim W$ and $W = 0$. A realization with finite-dimensional homogeneous state spaces must use a different coefficient carrier.

16.6 Finite coefficients and categorical symmetry

Semisimplicity provides a finite carrier. Its extra structure is essential: the allowed maps must respect the right action of the fusion category. Without this condition a finite matrix realization introduces new junctions.

Continue over $\mathbb{C}$. Choose representatives $(S_i)_{i\in I}$ of the simple objects of C, with $S_0 = 1$, and put

$$R_0 = \bigoplus_{i\in I} \mathbb{C}e_i, \qquad e_i \mathsf{P}_X e_j = \mathrm{Hom}_C(S_i, X \otimes S_j).$$

The e_i are orthogonal idempotents with sum one. A map $x : X \to Y$ acts on P_X by $f \mapsto (x \otimes 1)f$. With $a_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$, the tensor comparison is

$$\mu_{X,Y} : \mathsf{P}_X \otimes_{R_0} \mathsf{P}_Y \longrightarrow \mathsf{P}_{X\otimes Y}, \qquad f \otimes g \longmapsto a^{-1}_{X,Y,S_k}(1_X \otimes g)f, \tag{16.15}$$

where $f : S_i \to X \otimes S_j$ and $g : S_j \to Y \otimes S_k$.

To describe the retained symmetry, associate to a finite R_0-bimodule P the linear functor

$$T_P(S_j) = \bigoplus_i S_i \otimes_{\mathbb{C}} e_i P e_j.$$

Here tensoring an object with a vector space means the corresponding finite direct sum. Extend T_P to sums and maps by linearity.

A right C-module structure on T_P consists of natural isomorphisms

$$\lambda_{M,Z} : T_P(M \otimes Z) \longrightarrow T_P(M) \otimes Z$$

compatible with the unit and associator. Explicitly, the two maps from $T_P((M \otimes Z) \otimes W)$ to $(T_P(M) \otimes Z) \otimes W$ agree:

$$(\lambda_{M,Z} \otimes 1_W)\lambda_{M\otimes Z,W} = a^{-1}_{T_P(M),Z,W}\lambda_{M,Z\otimes W}T_P(a_{M,Z,W}).$$

A morphism must commute with every $\lambda_{M,Z}$. These are finite conditions: simple M, Z, W and bases of their junction spaces determine all the displayed maps and equations by additivity.

Let $\mathcal{S}_C$ be this category. Its tensor product is composition of module functors, expressed on coefficients by $P \otimes_{R_0} Q$. For P_X, the functor is $L_X(M) = X \otimes M$, and its module structure is $\lambda^X_{M,Z} = a^{-1}_{X,M,Z}$.

THEOREM 16.6.1. The maps (16.15) give a tensor equivalence

$$C \xrightarrow{\sim} \mathcal{S}_C, \qquad X \longmapsto (\mathsf{P}_X, \lambda^X).$$

In particular the symmetry-compatible junctions between these finite bimodules are exactly the original junctions of C. The equivalence preserves the specified duality and spherical evaluations.

Proof. Choose inclusions $g_{j,\alpha} : S_j \to Y \otimes S_k$ and projections $\bar{g}_{j,\alpha}$ with $\sum_{j,\alpha} g_{j,\alpha}\bar{g}_{j,\alpha} = 1$. The inverse of (16.15) sends $h : S_i \to (X \otimes Y) \otimes S_k$ to

$$\sum_{j,\alpha} (1_X \otimes \bar{g}_{j,\alpha}) a_{X,Y,S_k} h \otimes g_{j,\alpha}.$$

The two composite identities follow from this resolution of the identity. They also prove independence of its choice. The associativity coherence is the pentagon for a, and the unit coefficient is $\mathsf{P}_1 = R_0$.

Every linear functor of the finite semisimple category is recovered from its matrix of spaces $\mathrm{Hom}(S_i, T(S_j))$. The same resolution proves that composition of functors is the balanced product of their matrices. Thus the definition of $\mathcal{S}_C$ retains exactly the module functors and their module-natural maps.

For a module functor T, its structure map at $(1, M)$ identifies $T(M)$ naturally with $T(1) \otimes M$. The module-functor equation makes this identification compatible with the right action. For a module-natural map $\eta : L_X \to L_Y$, the same equation gives $\eta_M = \eta_1 \otimes 1_M$, with the unit constraints understood. Evaluation at 1 is therefore an inverse equivalence, on both objects and maps. Composition identifies $L_X L_Y$ with $L_{X\otimes Y}$ through a^{-1}. This proves the tensor assertion.

Transport the given evaluations, coevaluations, and pivotal maps through this equivalence. Their defining equations are preserved, as is every closed composition. This gives the asserted duality. □

For the near-group rules the size of the construction is explicit. Order the simples by G followed by ρ. The matrix of dimensions of P_ρ is

$$D_\rho = \begin{pmatrix} 0_{N\times N} & \mathbf{1} \\ \mathbf{1}^{\mathrm{t}} & N \end{pmatrix}.$$

For $g \in G$, D_g is the permutation $h \mapsto g + h$ on the G-block and is the identity on the ρ-entry. Hence $\dim R_0 = N + 1$ and $\dim \mathsf{P}_\rho = 3N$.

If $\mathbf{d} = (1, \dots, 1, d)^{\mathrm{t}}$, then

$$D_g\mathbf{d} = \mathbf{d}, \qquad D_\rho\mathbf{d} = d\mathbf{d}, \qquad d^2 = N + Nd.$$

The last identity also explains the relative dimension. Closing a coefficient line over the j-th boundary weights its i-th summand by d_i/d_j. The total is d_X. These weights are the regular-module trace $\mathrm{tr}_{S_i}(1) = d_i$ of the given spherical category. They need not be positive.

The symmetry condition cannot be suppressed. Ordinary bimodule maps $R_0 \to R_0$ form $\mathbb{C}^{\mathcal{I}}$. A module-natural endomorphism of the identity has scalars c_i satisfying $c_i = c_0$ by evaluating its compatibility at $(1, S_i)$. Its space is consequently $\mathbb{C}$, as required by the tensor unit of C.

The reconstructed weak Hopf action

The categorical symmetry has an equivalent description by a finite weak Hopf algebra. It uses the left-insertion tensorator (16.15). The choice matters for its associator coefficients.

Let P denote the vector-space-valued functor underlying the preceding construction and put

$$H_C = \mathrm{Nat}(\mathsf{P}, \mathsf{P}) = \bigoplus_{s\in\mathcal{I}} \mathrm{End}_{\mathbb{C}}(\mathsf{P}_{S_s}).$$

Multiplication is composition. These matrices need not be R_0-bilinear. An element h acts on every P_X, with its action on sums and retracts determined by naturality.

Let $\pi_{X,Y}$ be the balanced quotient and let

$$j_{X,Y}(p \otimes_{R_0} q) = \sum_i pe_i \otimes e_i q$$

be its section. The coproduct is uniquely specified by its action on all pairs of simple representations:

$$(\Delta h)_{X,Y} = j_{X,Y}\mu_{X,Y}^{-1} h_{X\otimes Y}\mu_{X,Y}\pi_{X,Y}. \tag{16.16}$$

In particular $\Delta(1) = E$, where $E(p \otimes q) = \sum_i pe_i \otimes e_i q$. With $\epsilon_0(e_i) = 1$, its counit is

$$\epsilon(h) = \epsilon_0(h_1(1_{R_0})), \qquad \epsilon(1) = |\mathcal{I}|,$$

where $\mathcal{I}$ denotes the chosen set of simple labels.

These formulas define a weak Hopf algebra. Here "weak" means that the coproduct preserves multiplication but its unit is the idempotent E, and the counit obeys the corresponding weak counit identities. Its representation tensor product is $E(U \otimes V)$, with the action given by Δ. Its unit representation is R_0. This is the dual form of Hayashi's finite semisimple reconstruction, Theorems 3.1 and 4.1.

The identities are visible in the formulas. Multiplicativity uses $\pi j = 1$. Coassociativity uses the associativity of μ and the equality of the two projections onto matching triples of boundary idempotents. The weak unit and counit equations reduce to $e_i e_j = \delta_{ij} e_i$ and $\epsilon_0(e_i) = 1$.

The antipode is the transpose action on the dual coefficients. More explicitly, let

$$b_X(q, p) = \epsilon_0 \mathsf{P}(\mathrm{ev}_X)\mu_{X^*,X}\pi_{X^*,X}(q \otimes p).$$

This pairing between P_{X^*} and P_X is nondegenerate. Define $S(h)$ by $b_X(q, S(h)_X p) = b_X(h_{X^*}q, p)$. The duality snake equations give the antipode equations, with the source and target counital projections in place of the scalar unit.

Every finite H_C-module is a sum of the displayed simple modules P_{S_s}. Its intertwiners are precisely the maps coming from C, and (16.16) identifies its tensor product with (16.15). Thus $C \simeq \mathrm{Rep}(H_C)$ as a tensor category. Equivalently, one can impose this weak Hopf action instead of the module-functor symmetry maps.

For the near-group rules,

$$\dim H_C = N(N+1)^2 + 9N^2.$$

At $N = 2$ this is 54. On $\mathsf{P}_\rho \otimes \mathsf{P}_\rho$, the idempotent E has rank 18 in dimension 36. Its role cannot be replaced by the identity.

The associator in a pointed example

Let G_0 be a finite abelian group and let $\Omega : G_0^3 \to \mathbb{C}^\times$ be a normalized three-cocycle. In $\mathrm{Vec}_{G_0}^{\Omega}$, the simple label g has coefficient basis $v_{g,j}$, with left endpoint $g + j$ and right endpoint j. The tensorator is

$$\mu(v_{g,h+j} \otimes_{R_0} v_{h,j}) = \Omega(g, h, j)^{-1} v_{g+h,j}.$$

Thus a nontrivial associator survives even though every simple fusion multiplicity is one.

Write E^{g}_{ab} for the matrix sending $v_{g,b}$ to $v_{g,a}$ and killing the other vectors and simple blocks. The reconstruction gives

$$\begin{aligned} E^{g}_{ab}E^{h}_{cd} &= \delta_{g,h}\delta_{b,c}E^{g}_{ad}, \\ \Delta(E^{s}_{ab}) &= \sum_{g+h=s} \frac{\Omega(g,h,a)}{\Omega(g,h,b)} E^{g}_{h+a,h+b} \otimes E^{h}_{ab}, \\ \epsilon(E^{g}_{ab}) &= \delta_{g,0}, \\ S(E^{s}_{ab}) &= \frac{\Omega(s,-s,b+s)}{\Omega(s,-s,a+s)} E^{-s}_{b+s,a+s}. \end{aligned} \tag{16.17}$$

The counit is nonzero on every matrix unit in the unit block, including the off-diagonal ones.

Indeed, composing μ, the matrix unit and μ^{-1} contributes first $\Omega(g,h,b)^{-1}$, then $\Omega(g,h,a)$. Matching the endpoints gives the displayed matrix indices. Every matrix unit in the unit block sends 1_{R_0} to a vector of coordinate sum one. The dual pairing pairs $v_{-g,j+g}$ with $v_{g,j}$, with factor proportional to $\Omega(-g,g,j)^{-1}$. Its transpose gives the antipode. The evaluation normalization cancels in the ratio.

The identity

$$\Omega(h,k,j)\Omega(g,h+k,j)\Omega(g,h,k) = \Omega(g+h,k,j)\Omega(g,h,k+j)$$

is the equality of the iterated tensorators. It proves coassociativity directly. The matching-endpoint projections and duality snakes give the remaining weak Hopf identities.

For $G_0 = \mathbb{Z}/n$, choose representatives $0,\dots,n-1$ and a primitive root ζ_n. The carry cocycle is

$$\Omega(g,h,j) = \zeta_n^{g\lfloor (h+j)/n \rfloor}.$$

The cocycle equation expresses equality of the total carry under two parenthesizations. At $n=3$, the $(g,h)=(1,2)$ summand of $\Delta(E^{0}_{01})$ and one antipode value are

$$\zeta_3^{-1}E^{1}_{20}\otimes E^{2}_{01}, \qquad S(E^{1}_{02}) = \zeta_3^{-1}E^{2}_{01}.$$

These values distinguish the ratio in (16.17) from its inverse. Even when $\Omega = 1, \Delta(1)$ remains the matching-endpoint projection.

16.7 Finite-dimensional conformal grades for the general family

We now combine this finite coefficient carrier with an explicit conformal theory. Let $\Lambda = E_8$ be the positive definite lattice

$$\Lambda = \{x \in \mathbb{Z}^8 \cup (\mathbb{Z}+\tfrac{1}{2})^8 : \textstyle\sum_i x_i \in 2\mathbb{Z}\}.$$

Its even unimodular form gives the holomorphic lattice vertex algebra $W = V_\Lambda$ of central charge eight, as in Dong–Mason.

Its full space of states is

$$W = \bigoplus_{\alpha\in\Lambda} \operatorname{Sym}\left(\bigoplus_{n>0} \mathbb{C}^8 t^{-n}\right) e^{\alpha}.$$

A monomial $h^{1}_{-n_1}\cdots h^{r}_{-n_r}e^{\alpha}$ has weight $\sum n_i + (\alpha,\alpha)/2$. Positive definiteness makes each weight space finite dimensional.

For completeness, choose a normalized lattice cocycle ϵ with $\epsilon(\alpha,\beta)/\epsilon(\beta,\alpha) = (-1)^{(\alpha,\beta)}$. The currents obey $[h_m, k_n] = m(h,k)\delta_{m+n,0}$, and the lattice fields are

$$Y(e^{\alpha},z) = \exp\Big(\sum_{n>0}\frac{\alpha_{-n}}{n}z^{n}\Big)\exp\Big(-\sum_{n>0}\frac{\alpha_{n}}{n}z^{-n}\Big)e_{\alpha}z^{\alpha_0},$$
$$e_{\alpha}e^{\beta} = \epsilon(\alpha,\beta)e^{\alpha+\beta}.$$

Normal products with differentiated currents define the fields of all other states. The translation is L_{-1} and the conformal vector is $\frac{1}{2}\sum_{a=1}^{8}(h^a_{-1})^2 1$ for an orthonormal basis (h^a).

For $(P,\lambda)\in\mathcal{S}_C$ define $\mathsf{V}(P) = W\otimes P$. Retain the R_0 actions and the finite module-functor maps λ. A field is required to respect them. For a compatible junction $f : P\otimes_{R_0} Q\to S$ it is

$$\mathcal{J}_f(u\otimes p,z)(v\otimes q) = Y_W(u,z)v\otimes f(p\otimes_{R_0} q). \tag{16.18}$$

The algebra of boundary vacua is $W\otimes R_0$. Its weight-zero space is R_0. Completions are products over conformal weights, and states themselves remain finite sums.

Equivalently, H_C acts on the coefficient factors and the fields obey the concrete covariance equation

$$h\,\mathcal{J}_f(u\otimes p,z)(v\otimes q) = \sum_{(h)}\mathcal{J}_f(u\otimes h_{(1)}p,z)(v\otimes h_{(2)}q),\qquad \Delta h = \sum_{(h)} h_{(1)}\otimes h_{(2)}.$$

The weak coproduct projects to the matching boundary sectors before the field is applied.

The sewing domain

A sewing datum starts with finitely many compact pointed Riemann surfaces, possibly disconnected. At each paired point choose coordinate discs of radii r_j, r'_j, with all discs pairwise disjoint. Glue their coordinates by $u_j v_j = t_j$, where

$$0 < |t_j| < r_j r'_j.$$

Each initial connected component must have a marked point that remains unsewn. Fix the local coordinates, contragredient pairings and the initial conformal block. Fractional powers of t_j use a chosen branch of $\log t_j$.

For a C_2-cofinite vertex algebra and finitely generated modules, Gui's theorem gives absolute, locally uniform convergence of the full propagator sums on this domain. Here C_2-cofiniteness means that the quotient by the span of $u_{(-2)}v$ is finite dimensional. The theorem allows simultaneous sewings that create handles. It concerns the specified conformal block. It does not choose a globally normalized scalar partition function.

If an initial component has no remaining insertion, first insert a vacuum. Apply propagation of vacua, then remove the auxiliary mark through the propagation isomorphism. The hypothesis is not removed by simply omitting the point. All finite simultaneous-sewing assertions below use this domain.

THEOREM 16.7.1. Every arithmetic fusion category considered here has a fully faithful analytic ordered realization $X\mapsto \mathsf{V}(\mathsf{P}_X,\lambda^X)$ with finite-dimensional homogeneous state spaces. Its symmetry-compatible relative fusion is

$$\mathsf{V}(\mathsf{P}_X)\boxtimes_{W\otimes R_0}\mathsf{V}(\mathsf{P}_Y) = \mathsf{V}(\mathsf{P}_{X\otimes Y}),$$

with tensorator (16.15) on coefficients. Its junctions, duality, and all fusion-tree sewings recover C. All descendant sewings on the domain of Section 16.7 converge absolutely and locally uniformly, including the permitted sewings that create handles. Taking full collision principal parts gives relative interfaces in the ordered word category, with compatible cut-cobar comparisons in every arity.

Proof. The lattice algebra is simple and its vacuum module has scalar endomorphisms. A W-module map between finite sums of this module is therefore $1_W \otimes f$. The space of vacuum-to-vacuum intertwining operators is one dimensional, generated by Y_W. Consequently every balanced field between these finite sums has the form (16.18). Its balancing and symmetry identities say precisely that f is a morphism in $\mathcal{S}_C$ out of $P \otimes_{R_0} Q$. Theorem 16.6.1 now proves full faithfulness, the tensor assertion, and preservation of the specified duality.

An iterated product is a finite sum of W conformal blocks multiplied by compositions of coefficient maps. The vertex-algebra associativity identifies their analytic continuations. In fusion-tree bases the coefficient comparison is the original associator of C. The module-functor pentagon makes this identity compatible with every retained symmetry map and every further cut.

Positive definiteness makes the lattice algebra regular by Dong–Li–Mason, Theorem 3.16. Abe–Buhl–Dong, Theorem 4.5, then gives rationality and C_2-cofiniteness. All the state spaces in the construction are finite sums of its vacuum module. Its propagator is the sum of the identity tensors of W_n, multiplied by t^n. The coefficient contractions use the transported relative evaluations. They are finite dimensional, so each sewn expression is a finite linear combination of ordinary W sewings.

Gui's Theorem 13.1 applies to each summand with all the hypotheses of Section 16.7. In particular, each initial connected component retains an unsewn marked point. It permits simultaneous sewings and disconnected initial surfaces, including sewings that create handles. Finite linear combinations preserve absolute and locally uniform convergence. The sewing identities on coefficients and states consequently remain valid after these sums.

Finally apply the local-current Cousin construction to the ordinary lattice chiral algebra and denote its cut cobar by A_W. The proof of Theorem 16.3.1 with scalar coefficients uses only translation, finite pole orders, and chiral Jacobi, all of which hold for W. Set

$$\mathfrak{R}_W = A_W \otimes R_0, \qquad \mathfrak{P}_X = A_W \otimes \mathsf{P}_X.$$

The relative bar comparison (16.11) gives

$$\mathfrak{P}_X \star^{\mathbb{L}}_{\mathfrak{R}_W} \mathfrak{P}_Y \simeq A_W \otimes (\mathsf{P}_X \otimes_{R_0} \mathsf{P}_Y) \xrightarrow{1 \otimes \mu_{X,Y}} \mathfrak{P}_{X \otimes Y}.$$

There are no higher coefficient Tor groups because R_0 is semisimple.

The normal principal part of (16.18) is the lattice-current face tensored with its coefficient junction. At a nonconsecutive collision the full ordered coefficient tensor is retained as a spectator. Thus no interchange of the two interface labels is needed. Every geometric face commutes with the finite symmetry maps, and the relative bar comparison commutes with every cut. These identities prove the geometric assertion on the word carrier. □

For example, take a unit-norm current in W and close the arithmetic junction of (16.5) on its coefficient line. The resulting two-current amplitude is $-e_1(z-w)^{-2}$. All its higher oscillator and lattice descendants are present. At $N = 2$, the three sector multiplicities are 3, 3, 6. Since $\dim W_1 = 248$, their first positive grades have dimensions 744, 744, 1488.

This is a theory with specified categorical symmetry and several boundary vacua. The underlying ordinary lattice-module category does not encode C by itself. The finite symmetry maps are part of the field theory, and their compatible junctions encode precisely C. In particular this construction makes no assertion that C is the category of local modules of a vertex algebra.

16.8 THE OBSTRUCTION AT THE EMPTY CONFIGURATION

The defect of restriction along (16.12) is already visible at the empty configuration. Set

$$\mathcal{T}_R = \mathfrak{R} \star^{\mathbb{L}}_{G_R} \mathfrak{R}, \qquad \epsilon_R : \mathcal{T}_R \to \mathfrak{R}.$$

Associativity of relative tensor products gives, for two $\mathfrak{R}$-bimodules P, Q,

$$\operatorname{Res} P \star^{\mathbb{L}}_{G_R} \operatorname{Res} Q \simeq \operatorname{Res}\Big(P \star^{\mathbb{L}}_{\mathfrak{R}} \mathcal{T}_R \star^{\mathbb{L}}_{\mathfrak{R}} Q \Big).$$

The comparison with $\operatorname{Res}(P \star^{\mathbb{L}}_{\mathfrak{R}} Q)$ uses ϵ_R. At the empty configuration it is multiplication $R \otimes_F R \to R$. For $r \notin F$ 1 its kernel contains the nonzero element $1 \otimes r - r \otimes 1$. A linear functional taking 1 to zero and r to one proves nonvanishing. The restricted relative unit also has empty coefficient R, whereas the G_R-bimodule unit has coefficient F.

Changing the particular connected bimodules cannot remove this problem.

THEOREM 16.8.1. Let D be an algebra in $\mathcal{W}_X$ with $D_\emptyset \simeq F$ as a unital algebra. Let $\mathcal{D}$ be a rigid category with the fusion rules

$$\alpha_g \alpha_h = \alpha_{g+h}, \qquad \alpha_g \rho = \rho \alpha_g = \rho, \qquad \rho^2 = \bigoplus_{g \in G} \alpha_g \oplus \rho^{\oplus N}, \quad |G| = N.$$

For $N > 0$, there is no additive strong unital tensor functor $\mathcal{D} \to \mathrm{Bimod}_D(\mathcal{W}_X)$ for the derived balanced cut tensor product.

Proof. Empty evaluation is strong monoidal for that product. Indeed the empty component of its two-sided bar is

$$P_\emptyset \otimes_F D_\emptyset^{\otimes n} \otimes_F Q_\emptyset,$$

whose realization is $P_\emptyset \otimes^{\mathbb{L}}_F Q_\emptyset$. It takes the unit to F. A proposed functor would therefore give a strong tensor functor to the ordinary $\mathbb{Z}$-graded derived category $D(F)$.

Images of rigid objects are dualizable. A coevaluation over a field is a finite sum of homogeneous tensors. The snake identity expresses every cohomology class in the span of their components. Thus these images are perfect complexes.

An invertible object of $D(F)$ is a shift of a one-dimensional vector space. Since each α_g has finite tensor order, its shift is zero and its Euler characteristic is one.

If a is the integer Euler characteristic of the image of ρ, additivity and multiplicativity give $a^2 = N + Na$. This is impossible because

$$(N+1)^2 < N^2 + 4N < (N+2)^2 \qquad (N \geq 1).$$

Thus the dimension polynomial has no integer root. No faithfulness hypothesis was needed. □

The theorem concerns the stated balanced tensor product. Fusion of rooted probes defined by collision or nearby-cycle operations is a different tensor product and is not excluded. Likewise, the coefficient exchange c_B is not a braiding of C. It need not even be linear for the left regular B-action: the difference between $c_B(ab, c)$ and $a\, c_B(b, c)$ is $1 \otimes abc - a \otimes bc$. These operations occur on different carriers.

16.9 Finite recoupling and the E_6 connection

The multiplicity space $M = \mathrm{Hom}(\rho, \rho^2) = \bigoplus_h FT_h$ carries

$$p^g T_h = T_{h-g}, \qquad q^g T_h = \chi(g,h) T_h.$$

Let P_0 project to FT_0, and let P_{av} project to the constant vector by averaging. The rank-one operator $D = P_0 P_{\mathrm{av}}$ obeys

$$D\left(\sum_h v_h T_h\right) = N^{-1} \sum_h v_h T_0, \qquad D^2 = N^{-1} D.$$

For $N > 1$ it is not idempotent. Nor is it the tensor-unit projection in ρ^2: that projection is $X_0 X_0'$, which kills every T_h.

The corrected identity of Radchenko–Wheeler, Section 4.2, acts on M. Its right-hand side is $-N\beta D$, as in (15.22).

The geometric associativity homotopy has a different type. If $dh = L - R$ in a differential graded operation complex, every chain map from that complex to $\mathrm{End}_F(M[0])$ sends $L - R$ to zero. It therefore cannot send this boundary to the nonzero corrected finite-pentagon difference. A comparison through a larger channel system would have to specify the intermediate operations and the compression. The two corrections cannot be identified by a chain map to this degree-zero space.

Allowing negative degrees does not remove this obstruction if the comparison must preserve the full finite operator algebra.

Proposition 16.9.1. Let E be a unital differential graded algebra and $\iota : \mathrm{End}_F(M) \to Z^0(E)$ a unital algebra map. If $\iota(D)$ is exact, then E is acyclic.

Proof. Write E_{ij} for the matrix units in the basis $(T_g)_{g \in G}$. The displayed formula for D gives

$$E_{i0} D E_{0i} = N^{-1} E_{ii}, \qquad N \sum_i E_{i0} D E_{0i} = 1.$$

If $dh = \iota(D)$, the degree -1 element $H = N \sum_i \iota(E_{i0}) h \iota(E_{0i})$ satisfies $dH = 1$, since the matrix units have closed images. Left multiplication by H contracts E: $d(Hx) + H\,dx = x$. Hence every cohomology group vanishes. □

The matrix units are already generated by the finite Heisenberg pair. Indeed, Fourier averaging of the q^g gives P_0, and translates $p^{-i} P_0 p^j$ give E_{ij}. Thus a comparison retaining the Heisenberg operators and their products cannot make the nonzero correction exact in a recipient with nonzero cohomology. A geometric comparison must retain the larger channel system and specify how its compression fails to preserve either differentials or compositions.

Compression and the supported operation complex

Compression can produce a nonzero difference from an identity on a larger space. Suppose $\mathsf{AB} = \mathsf{CDE}$ there. Let $P^2 = P$ and $Q = 1 - P$. Inserting $P + Q$ at every intermediate position gives

$$\begin{aligned}(P\mathsf{A}P)(P\mathsf{B}P) - (P\mathsf{C}P)(P\mathsf{D}P)(P\mathsf{E}P) \\ = P\mathsf{C}Q\mathsf{D}P\mathsf{E}P + P\mathsf{C}P\mathsf{D}Q\mathsf{E}P \\ + P\mathsf{C}Q\mathsf{D}Q\mathsf{E}P - P\mathsf{A}Q\mathsf{B}P.\end{aligned}$$

All four terms record discarded intermediate channels. To obtain $-N\beta D$, one must construct the larger supported maps and the specified projection P, then identify their displayed sum with that operator.

The supported construction retains more than the total diagonal. At three positions it includes

$$[\,\mathcal{V}_{2|13} \xrightarrow{-r_{2,(13)}} \mathcal{V}_{123}\,]$$

in degrees -1, 0, with outer-collision homotopy r_{13}. At four positions it also retains $13|24$. Normal derivatives, the positions of spectator states, the empty cut unit and the local vacuum are part of the comparison.

There is an obstruction calculation for constructing these operations themselves. Let $\mathcal{E}$ be a differential graded nonsymmetric operad of admissible supported operations. Require closure under insertions and impose the translation, differential-module, cut and vacuum conditions in its definition. Use suspended operations, let b_1 be the differential, and write $\partial = [b_1, -]$.

Suppose degree-one operations $b_2, \dots, b_{n-1}$ satisfy all identities below arity n. Define

$$\mathfrak{o}_n = \left(\sum_{i=2}^{n-1} b_i\right)^2_{\text{arity } n} \in \mathcal{E}(n)^2.$$

Then $\partial\mathfrak{o}_n = 0$, and b_n exists exactly when

$$[\mathfrak{o}_n] = 0 \quad \text{in } H^2(\mathcal{E}(n)).$$

It must satisfy $\partial b_n = -\mathfrak{o}_n$. With the lower operations fixed, solutions modulo normalized degree-zero homotopy form an $H^1(\mathcal{E}(n))$-torsor.

Indeed, for $B = b_1 + \cdots + b_{n-1}$, associativity of composition gives $[B, B^2] = 0$. All lower-arity components of B^2 vanish, leaving $[b_1, \mathfrak{o}_n] = 0$ in arity n. Adding b_n changes this component by ∂b_n. The difference of two solutions is a degree-one cocycle, and a normalized homotopy changes it by a boundary.

This is the obstruction to a new operation. The higher-map problem of Theorem 4.3.1 instead has obstruction in degree one and choices in degree zero. Neither cohomology is determined by the rigidity of the finite fusion category or by a calculation on position-free states.

The full recoupling space is larger and is explicitly available. For the quadratic example of (16.5), $\mathrm{Hom}(\rho, \rho^3)$ has dimension six. Its left and right tree bases are

$$L_g = X_g, \quad L_{h,k} = T_h T_k, \qquad R_g = \rho(X_g)U_g, \quad R_{h,k} = \rho(T_h)T_k.$$

Their projections are $L'_g = X'_g$, $L'_{h,k} = T'_k T'_h$, $R'_g = U_{-g}\rho(X'_g)$ and $R'_{h,k} = T'_k\rho(T'_h)$. The change-of-basis matrix is $\mathsf{F}_{ij} = R'_i L_j$. Its inverse has entries $L'_i R_j$. In the order 0, 1, (0, 0), (0, 1), (1, 0), (1, 1), put $r = (1+\sqrt{3})/2$, $t = (1+i)/2$, $\bar{t} = (1-i)/2$ and $c = (1-i)(\sqrt{3}+i)/4$. Direct substitution gives

$$\mathsf{F} = \begin{pmatrix} -r & -r & -r & 0 & 0 & -ir \\ -r & r & 0 & -ir & -r & 0 \\ c & c & r & 0 & 0 & t \\ c & -c & 0 & t & r & 0 \\ c & c & \bar{t} & 0 & 0 & ir \\ -c & c & 0 & -ir & -\bar{t} & 0 \end{pmatrix}. \tag{16.19}$$

Put $a_F = (2+\sqrt{3}+i)/4$. In the same order, the inverse is

$$\mathsf{F}^{-1} = \begin{pmatrix} -r & -r & -a_F & -a_F & -a_F & a_F \\ -r & r & -a_F & a_F & -a_F & -a_F \\ 1 & 0 & r & 0 & t & 0 \\ 0 & -i & 0 & \bar{t} & 0 & ir \\ 0 & 1 & 0 & r & 0 & -t \\ -i & 0 & \bar{t} & 0 & -ir & 0 \end{pmatrix}.$$

Multiplying the two displayed matrices gives the identity. Equivalently, this follows by inserting the resolutions $\sum_i L_i L_i' = \sum_i R_i R_i' = 1$ between their entries.

For every number of inputs the analogous tree bases lie in one intertwiner space. Their change-of-basis matrices obey the pentagon because both products compare the same two bases. The relative chiral functor of Theorem 16.4.1 transports these actual maps and their coherence.

There is an independent singular realization at affine $\mathfrak{sl}_2$ level ten. Write V_j for its fusion label of highest weight j. The E_6 algebra object $V_0 \oplus V_6$ has a module tensor subcategory of type $(\mathbb{Z}/2, 2)$, as in Thornton, Example 1.2.

Its free-module images are

$$V_0 \mapsto 1, \quad V_2 \mapsto \rho, \quad V_4 \mapsto g \oplus \rho, \quad V_6 \mapsto 1 \oplus \rho, \quad V_8 \mapsto \rho, \quad V_{10} \mapsto g.$$

These follow from free-module adjunction and the even $SU(2)_{10}$ fusion rules. Its positive dimension is $1+\sqrt{3}$. It is the other embedding of the dimension parameter in (16.1).

The primary three-input blocks leading to ρ have ranks 3, 2, 1, for total highest weights 2, 4, 6. They come from $V_2^{\otimes 3} = V_0 \oplus 3V_2 \oplus 2V_4 \oplus V_6$ before the free-module functor.

Fix the spin-one basis v_0, v_1, v_2 with $Hv_j = (2-2j)v_j$, $Ev_1 = 2v_0$, $Ev_2 = v_1$, $Fv_0 = v_1$ and $Fv_1 = 2v_2$. The remaining raising and lowering actions vanish. Put $w_{abc} = v_a \otimes v_b \otimes v_c$. Use the ordered bases

$$\begin{aligned} 2: \quad & w_{002} - \tfrac{1}{2} w_{011} + w_{020}, \\ & 4w_{002} - w_{011} - w_{101} + w_{110}, \\ & w_{002} - \tfrac{1}{2} w_{101} + w_{200}; \\ 4: \quad & -w_{001} + w_{010}, \ -w_{001} + w_{100}; \\ 6: \quad & w_{000}. \end{aligned}$$

On these covariant highest-weight spaces the Knizhnik–Zamolodchikov (KZ) connection is

$$\nabla = d - \frac{1}{12}\left(\frac{\Omega_{12}}{z} + \frac{\Omega_{23}}{z-1}\right) dz, \qquad \Omega = \tfrac{1}{2} H \otimes H + E \otimes F + F \otimes E.$$

For highest weight two one obtains the matrices

$$\Omega_{12} = \begin{pmatrix} -2 & 4 & 0 \\ 1 & 0 & 1 \\ 0 & 4 & -2 \end{pmatrix}, \qquad \Omega_{23} = \begin{pmatrix} -4 & -4 & 0 \\ 0 & -2 & -1 \\ 0 & 0 & 2 \end{pmatrix}.$$

For highest weight four they are $\left(\begin{smallmatrix} 0 & 2 \\ 2 & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} -2 & -2 \\ 0 & 2 \end{smallmatrix}\right)$. For highest weight six both equal (2).

These matrices are obtained by restricting the ordinary spin-one Casimirs to the kernel of the total raising operator. The exponents at zero are $-1/3, -1/6, 1/6$, then $-1/6, 1/6$, then $1/6$. The system of conformal-block functionals uses the dual connection.

This differential system has nontrivial local monodromy. Its comparison with arithmetic fusion requires the module junctions as well as the ambient affine channels. We construct those junctions first.

16.10 The normalized E_6 tensor comparison

Put $s = \sqrt{3} > 0$, $q = e^{\pi i/12}$, and $[n] = \sin(n\pi/12)/\sin(\pi/12)$. Write $[n]! = \prod_{j=1}^{n}[j]$ and $[0]! = 1$. Let $\mathcal{D}_q$ be the semisimplified category of tilting $U_q(\mathfrak{sl}_2)$-modules, with simple objects $V_0, \dots, V_{10}$. The admissible channels of $V_a \otimes V_b$ are

$$|a-b|,\ |a-b|+2, \dots, \min(a+b, 20-a-b). \tag{16.20}$$

All recouplings below have even external labels.

We specify the trivalent normalization explicitly. For an admissible triple put

$$D(a,b,c) = \frac{[(-a+b+c)/2]![(a-b+c)/2]![(a+b-c)/2]!}{[(a+b+c)/2+1]!}.$$

For a, b, c, d and intermediate labels e, f, put

$$(A_1, A_2, A_3, A_4) = \tfrac{1}{2}(a+b+e,\ a+d+f,\ c+b+f,\ c+d+e),$$
$$(B_1, B_2, B_3) = \tfrac{1}{2}(a+b+c+d,\ a+c+e+f,\ b+d+e+f).$$

The rational Racah normalization is

$$(F_d^{abc})_{fe} = (-1)^{(a+b+c+d)/2}[f+1]D(b,c,f)D(a,f,d) \times \sum_{z=\max A_i}^{\min(B_1,B_2,B_3,10)} \frac{(-1)^z[z+1]!}{\prod_i [z-A_i]! \prod_j [B_j - z]!}. \tag{16.21}$$

An inadmissible entry is zero. Columns are left-tree inclusions and rows are right-tree projections. Equivalently, the column of right-tree projections is F_d^{abc} times the column of left-tree projections. The counterclockwise exchange in channel c is

$$R_c^{ab} = (-1)^{(a+b-c)/2} q^{(c(c+2)-a(a+2)-b(b+2))/4}.$$

Let $A_E = V_0 \oplus V_6$. Its unit is the inclusion of V_0. In the normalization (16.21), define its multiplication by the unit constraints and

$$m_{66}^{0} = -\frac{1}{3} + \frac{s}{6}, \qquad m_{66}^{6} = 1. \tag{16.22}$$

Every other component is zero. This is the connected commutative separable E_6 algebra of Kirillov–Ostrik, §6.

To see the normalization, write $m_{66}^{0} = \alpha$. The channel $e = 2$ of the associativity equation on $V_6^{\otimes 3} \to V_6$ gives

$$\alpha = -\frac{(F_6^{666})_{6,2}}{(F_6^{666})_{0,2}} = -\frac{1}{3} + \frac{s}{6}.$$

Substitution in the remaining channels gives associativity. The two nonzero channels of m_{66} have $R_0^{66} = R_6^{66} = 1$. The uniqueness of the E_6 algebra then identifies this multiplication with the conformal-extension algebra.

Write $\mathcal{D}_q^+$ for the full even-label subcategory and $\mathcal{M}_E = (\mathcal{D}_q^+)_{A_E}$ for its right A_E-modules. Their tensor product is over A_E. The left action on a right module is obtained using the displayed braiding. The free-module functor is $F_E(V) = V \otimes A_E$. Its three simple images are

$$M_1 = F_E(V_0), \qquad M_g = F_E(V_{10}), \qquad M_\rho = F_E(V_2).$$

Adjunction gives their endomorphism spaces and the free-module decompositions displayed above. Every module is a summand of a free module because A_E is separable. These decompositions therefore exhaust the simple objects. They give $M_\rho \otimes_{A_E} M_\rho = M_1 \oplus M_g \oplus 2M_\rho$.

The free-module junctions have a concrete basis. For $a, b, c \in \{0, 10, 2\}$, use adjunction to write

$$\mathrm{Hom}_{A_E}(F_E(V_a) \otimes_{A_E} F_E(V_b), F_E(V_c)) = \mathrm{Hom}_{\mathcal{D}_q}(V_a \otimes V_b, V_c \otimes A_E).$$

Its basis element $p_{ab}^{c;e,t}$ projects to V_e and includes it in $V_c \otimes V_t$, where $t \in \{0, 6\}$ and e occurs in both tensor products. Order these elements by t and then e. In particular the two $\rho\rho \to \rho$ junctions have $(e, t) = (2, 0), (4, 6)$.

To compose such maps, move the first A_E factor past the next input by R and multiply the two A_E factors by (16.22). This prescription and (16.21) determine every module recoupling.

Let $\sigma : F \to \mathbb{C}$ send $\sqrt{3}$ to $-\sqrt{3}$ and fix i. For the order-two example $F = \mathbb{Q}(\sqrt{3}, i)$. Write $C^\sigma = C \otimes_{F,\sigma} \mathbb{C}$. The arithmetic projection bases are those supplied by $1, U_g, X'_g, T'_h$ in the Leavitt presentation. Unit junctions have their unit normalization on both sides.

We give the comparison on the six remaining binary spaces. Put

$$a_* = \frac{(1-i)(-s+i)}{2}, \qquad S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

and

$$P = \begin{pmatrix} 1 & \dfrac{(1-i)(3+s(2-i))}{2} \\ -\dfrac{(1+i)(s+1)}{2} & \dfrac{(1-i)^2(3+si)}{2} \end{pmatrix}.$$

Define changes of projection basis N^{a}, N^{q} by

$(a, b; c)$	$N^{\mathrm{a}}_{ab;c}$	$N^{\mathrm{q}}_{ab;c}$
$(g, g; 1)$	1	$-\frac{1}{2} + \frac{s}{3}$
$(g, \rho; \rho)$	1	$-i$
$(\rho, g; \rho)$	1	1
$(\rho, \rho; 1)$	a_*	$1 - \frac{s}{2} + \frac{i}{2}$
$(\rho, \rho; g)$	a_*	$-3 - 3i(2+s)$
$(\rho, \rho; \rho)$	S	P

and set

$$K_{ab;c} = (N^{\mathrm{a}}_{ab;c})^{-1} N^{\mathrm{q}}_{ab;c}, \qquad p^{\mathrm{a}}_{ab;c} \longmapsto K_{ab;c}\, p^{\mathrm{q}}_{ab;c}. \tag{16.23}$$

The projection bases in this equation are columns. The inclusions transform by the inverse matrices, preserving their pairing.

THEOREM 16.10.1. The junction maps (16.23) extend to a tensor equivalence

$$\mathcal{E} : C^{\sigma} \xrightarrow{\sim} \mathcal{M}_{E}, \qquad (1, g, \rho) \longmapsto (M_{1}, M_{g}, M_{\rho}).$$

They preserve the spherical duality and every marked network value.

Proof. Fix a triple (a, b, c) and output d. Applying K at both vertices changes the projection bases by matrices L_K and R_K for the two trees. The required equation is

$$F^{\mathrm{a}} L_K = R_K F^{\mathrm{q}}. \tag{16.24}$$

The quantum computation takes place in

$$\mathrm{Hom}_{\mathcal{D}_q}((V_a \otimes V_b) \otimes V_c, V_d \otimes A_E).$$

Choose its ambient basis by the output label $v \in \{0, 6\}$, the total channel j, and the first-pair channel x. For a left module tree, write its two junction labels as (x, t) and (y, u), with intermediate simple V_e. The coefficient vanishes unless the ambient first-pair channel is the first junction channel x. Its remaining coefficient is

$$\left(\sum_{f} (F_j^{ect})^{-1}_{yf} \, R_f^{tc} \, (F_j^{etc})_{fx} \right) (F_j^{dut})_{vy} \, m^{v}_{ut}.$$

For a right module tree with junction labels (f, t), (y, u), the coefficient is

$$(F_j^{abc})_{fx} \, (F_j^{aet})^{-1}_{yf} \, (F_j^{dut})_{vy} \, m^{v}_{ut}.$$

Inadmissible channels contribute zero. These are respectively the crossing of the first algebra factor and the two associativity changes before multiplying the algebra factors. Writing their columns as L and R gives $F^{\mathrm{q}} = (L^{-1}R)^{\mathrm{t}}$.

Apply the two normalizations N^{a} and N^{q} to the Leavitt composites and to the preceding free-module prescription. Both give the following matrices. Every unit block and every block not listed is the identity. Zero-dimensional blocks are omitted:

$$F_{\rho}^{g\rho g} = F_{g}^{\rho g \rho} = -1, \qquad F_{\rho}^{g\rho\rho} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$F_{\rho}^{\rho g \rho} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad F_{\rho}^{\rho\rho g} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

$$F_{1}^{\rho\rho\rho} = u \begin{pmatrix} 1 & b \\ 1 & -b \end{pmatrix}, \qquad F_{g}^{\rho\rho\rho} = u \begin{pmatrix} b & 1 \\ -b & 1 \end{pmatrix},$$

$$b = -i, \qquad u = \frac{(1 - i)(1 - si)}{4}.$$

For the remaining block put

$$\varphi = \frac{s - 1}{2}, \quad w = \frac{(s - 1)(1 + is)}{8}, \quad y = \frac{1 + i(s - 2)}{4}, \quad z = \frac{-s - i}{4}.$$

In the channel order used in (16.19), that block is

$$F_{\rho}^{\rho\rho\rho}=\begin{pmatrix} \varphi & \varphi & -wb & w & w & -wb \\ \varphi & -\varphi & -wb & w & -w & wb \\ 1 & 1 & -yb & z & y & -zb \\ 1 & 1 & -zb & y & z & -yb \\ 1 & -1 & -yb & z & -y & zb \\ -1 & 1 & zb & -y & z & -yb \end{pmatrix}.$$

For example the arithmetic block is obtained from (16.19) by $\sqrt{3}\mapsto -s$, followed by the two-vertex changes induced by a_* and S. The quantum block is obtained by the free-module composition with $m^{\circ}_{66}=-1/3+s/6$. The displayed scalar, two-dimensional, and six-dimensional blocks exhaust the possible triples. They prove (16.24).

All the matrices K are invertible. Equation (16.24), together with the identity on unit junctions, is the associativity and unit equation for a tensor functor between the two semisimple categories. It therefore defines $\mathcal{E}$ on sums and matrices of simple objects. The three simple images are distinct and exhaustive, so the functor is an equivalence.

There is no nontrivial tensor automorphism of the identity in either category. Indeed its scalar on ρ satisfies both $\lambda_{\rho}^{2}=1$ and $\lambda_{\rho}^{2}=\lambda_{\rho}$, and $\lambda_{g}\lambda_{\rho}=\lambda_{\rho}$. Thus all scalars are one.

The spherical structure is consequently unique. Its dimension on ρ is $\sigma(d)=1+s$, equal to $\dim_{\mathcal{D}_q}(\mathcal{M}_{\rho})/\dim_{\mathcal{D}_q}(\mathcal{A}_E)$. Transporting duality along the tensor equivalence preserves this structure. In particular it preserves cups, caps, and their closed composites. □

The embedding σ is essential. Before changing the embedding, the global dimension is $2+(1-s)^2=6-2s$. That of $\mathcal{M}_E$ is $6+2s$. These are invariants of the fusion categories, so there is no complex-linear tensor equivalence at the original embedding. Replacing i by $-i$ as well also changes the category: the trace of $F_{1}^{\rho\rho\rho}$ changes from $(1-is)/2$ to $(1+is)/2$. It cannot be removed by a binary-junction change of basis.

16.11 Affine descendants and analytic sewing

Let $H_j=L_{\widehat{\mathfrak{sl}}_2}(10,j)$, $0\le j\le 10$. Its lowest-weight space is the ordinary $\mathfrak{sl}_2$-module of highest weight j, and its conformal weights are

$$h_j+n,\qquad h_j=\frac{j(j+2)}{48},\quad n\ge 0.$$

The states are finite sums of negative current modes applied to that space, modulo the integrable null relation

$$E_{-1}^{11-j}v_j=0 \tag{16.25}$$

and the submodule it generates. All homogeneous spaces are finite dimensional. Outputs lie in their product completion. The translation is L_{-1}, and the Sugawara central charge is $5/2$.

Huang–Lepowsky, Theorems 3.7–3.13, give these modules their vertex tensor category $\mathcal{D}_{10}$.

Fix the ribbon equivalence

$$\Phi:\mathcal{D}_q\xrightarrow{\sim}\mathcal{D}_{10},\qquad \Phi(V_j)=H_j,$$

with counterclockwise braiding and vacuum unit normalization. We use Finkelberg's equivalence with the correction in his erratum. The corrected equivalence uses rigidity and the prescribed opposite tensor-category convention before fixing the counterclockwise ribbon structure. Its numerical range includes $\kappa = k + h^\vee \geq h^\vee + 3$. For $\mathfrak{sl}_2$ at $k = 10$, one has $h^\vee = 2$ and $\kappa = 12$, so the bound holds.

The rational trivalent basis (16.21) is transported through this equivalence. This specifies the normalization of every affine intertwining operator used below. A basis obtained by separately setting all three-point highest-weight coefficients to one need not have that normalization.

Write $\Phi_2 : \Phi(X) \boxtimes \Phi(Y) \to \Phi(X \otimes Y)$ for its tensor comparison. The algebra $\widehat{A}_E = \Phi(A_E)$ is the conformal extension $L_{\widehat{\mathfrak{sp}}_4}(1, 0)$. Its underlying H_0-module is $H_0 \oplus H_6$ (Kirillov–Ostrik, Example 5.3 and Section 6).

For an A_E-module M, let $U(M)$ denote its underlying object of $\mathcal{D}_q$ and set $\widehat{M} = \Phi(U(M))$. In particular,

$$\begin{aligned} \widehat{M_1} &= H_0 \oplus H_6, \\ \widehat{M_g} &= H_4 \oplus H_{10}, \\ \widehat{M_\rho} &= H_2 \oplus H_4 \oplus H_6 \oplus H_8. \end{aligned} \tag{16.26}$$

These are full affine state spaces. The last is an ordered, nonlocal $\widehat{A}_E$-sector. It is not an ordinary local module of the extension.

We next define its fields and its relative sewing, including the balancing map. Choose the special Frobenius coproduct of A_E with $m\Delta = 1_{A_E}$, and put $e = \Delta\eta$. It is uniquely determined by m and by the counit $(3 + s)$ times projection to V_0. Explicitly, put $D_E = 3 + s$ and $\alpha = -1/3 + s/6$. In the inclusion bases dual to the projection bases its nonzero components are

$$\Delta_0^{00} = \Delta_6^{06} = \Delta_6^{60} = D_E^{-1}, \qquad \Delta_0^{66} = \frac{D_E - 1}{D_E \alpha}, \qquad \Delta_6^{66} = \frac{D_E - 2}{D_E}.$$

For right modules M, N let r_M, r_N be their actions and $\ell_N = r_N c_{A_E,N}$. The balancing idempotent is

$$P_{M,N} = (r_M \otimes \ell_N)(1_M \otimes e \otimes 1_N). \tag{16.27}$$

Its image is $M \otimes_{A_E} N$. Denote the quotient onto this image by $q_{M,N}$.

For a junction $p : M \otimes_{A_E} N \to P$, define

$$\mathcal{Y}_p = \mathcal{Y}_{\Phi(U(p) q_{M,N})\,\Phi_2}. \tag{16.28}$$

Here $\mathcal{Y}_f$ is the intertwining operator corresponding to the H_0-module map $f : \widehat{M} \boxtimes \widehat{N} \to \widehat{P}$. Thus (16.28) is a definition on all descendants, through the universal property of the vertex tensor product. For an arithmetic projection $p^{\mathrm{a}}_{ab;c,\gamma}$ its operator is

$$\mathcal{Y}^{\mathrm{a}}_{ab;c,\gamma} = \sum_\beta (K_{ab;c})_{\gamma\beta} \mathcal{Y}^{\mathrm{q}}_{ab;c,\beta}. \tag{16.29}$$

The arithmetic sector corresponding to a has the state space $\widehat{M_a}$ of (16.26).

THEOREM 16.11.1. Equations (16.26)–(16.29) give an analytic ordered realization of $\mathcal{C}^\sigma$. The comparison is the identity on the full state spaces and is K on binary junctions. It preserves translation, conformal

grading, relative duality, collision expansions, and sewing in every arity. All products and iterates, paired with restricted-dual states, converge absolutely on their nested-disc domains. Their continuations have the level-ten conformal-block monodromy. Sewing on the domain of Section 16.7 converges absolutely and locally uniformly, including the allowed sewings that create handles.

Proof. The Frobenius and separability equations make (16.27) idempotent and identify its image with the balanced tensor product. They also identify the two iterated balancing quotients of $M \otimes N \otimes P$. Applying the strong tensor functor Φ gives the corresponding identity of affine module maps. The vertex-tensor associativity isomorphism identifies those maps with the products and iterates of (16.28). Its defining identity holds for every state, by Huang–Lepowsky, Theorems 3.7–3.8.

Apply (16.24) to that identity. The changes on its two binary vertices are exactly L_K and R_K, so (16.29) identifies the two complete operator products. Any two binary trees are joined by associativity moves. The pentagon identifies the resulting maps between trees, and the unit constraints handle vacuum insertions. This proves the comparison in every arity. All maps are affine module maps and preserve L_{-1}, L_0, and the null submodules.

For clarity, the sewing along a module H_j uses the tensor

$$\mathcal{P}_j(t) = \sum_{n \geq 0} t^{h_j+n} \sum_{\alpha} u_{j,n,\alpha} \otimes u^{\vee}_{j,n,\alpha}. \tag{16.30}$$

The second basis is dual to the first in the contragredient module, with the specified opposite local coordinate. A branch of $\log t$ fixes the fractional powers. Direct sums, the balancing idempotents, and the relative evaluations specify the analogous sewing for the three sectors.

No descendant degree is omitted. The identity tensor in each finite-dimensional grade is independent of its basis. Thus the junction change K and its dual inverse cancel at an internal edge. The spherical comparison of Theorem 16.10.1 handles cups and caps.

The integrable affine vertex algebra H_0 is rational and C_2-cofinite. Its finite sums $\widehat{M}_a$ are finitely generated modules. Genus-zero absolute convergence is part of the cited vertex tensor theorem.

Sew local coordinates $uv = t$ on disjoint coordinate discs of radii r and r'. The full sums (16.30) converge absolutely and locally uniformly by Gui's Theorem 13.1. Use $0 < |t| < rr'$ and all the hypotheses of Section 16.7. It allows simultaneous sewings and disconnected initial surfaces. The balancing and junction maps form finite sums of such conformal blocks, so the same convergence holds here. Equality before sewing consequently remains equality after each further sewing. □

This construction also has ordinary boundary fields. For boundary conditions $M, N \in \{M_1, M_g, M_\rho\}$ their state space is

$$\mathsf{H}_{MN} = \bigoplus_{j=0}^{10} \operatorname{Hom}_{A_E}(V_j \otimes M, N) \otimes H_j.$$

Composition of the module maps gives their operator-product coefficients, and the corresponding affine intertwining operators give their position dependence.

If $P = M_1 \oplus M_g \oplus M_\rho$, these spaces are the matrix entries of the internal endomorphism algebra of P in $\mathcal{D}_{10}$. Its associativity gives a grading-restricted conformal open-string vertex algebra by Huang–Kong, Theorem 4.3. The three boundary identity states account for its three-dimensional weight-zero space. Its conformal transformations and disc sewings are those of Huang–Kong, Theorem 5.4.

16.12 Collision coefficients of the affine realization

The fractional powers of the affine fields belong to their coefficient connection. They must be canceled before taking a normal principal part. For labels i, j, k, put $a_{ij}^{k} = h_k - h_i - h_j$. On this binary channel write

$$\mathcal{Y}_{ij}^{k}(u) = u^{a_{ij}^{k}} \Theta_{ij}^{k}(u).$$

On homogeneous input states, Θ_{ij}^{k} has integral powers and a finite lower bound.

The coefficient symbol $\xi_{ij}^{k} = u^{-a_{ij}^{k}}$ satisfies $\partial_u \xi_{ij}^{k} = -a_{ij}^{k} u^{-1} \xi_{ij}^{k}$. The collision map is

$$r_{ij}(f \xi_{ij}^{k}; v, w) = \mathrm{pp}_{u=0}\big(f \Theta_{ij}^{k}(v,u) w\big). \tag{16.31}$$

The derivative of ξ_{ij}^{k} and the translation identity of the field make this a map of right differential modules. In particular the term $a_{ij}^{k} u^{-1} \Theta_{ij}^{k}$ in the field derivative is retained.

For more points, the symbols form the inverse conformal-block coefficient system. It is the finite-rank integrable subbundle of the Knizhnik–Zamolodchikov connection. One algebraic description is useful. Let $U_j \subset H_j$ be the lowest-weight space and fix output l. The coefficient at positions $z_1, \dots, z_r$ is the subspace of $\mathrm{Hom}_{\mathfrak{sl}_2}(U_{j_1} \otimes \cdots \otimes U_{j_r}, U_l)$ whose invariant functional τ obeys

$$\tau\left(\left(\sum_{i} z_i E^{(i)}\right)^{n-l} b \otimes v_l^{\vee}\right) = 0 \quad \text{for every lowest-space tensor } b. \tag{16.32}$$

The vector $v_l^{\vee}$ is highest in the module at infinity.

This is the null relation (16.25) moved from infinity by the current Ward identity. The integrable null relations at the other points give the same kernel. It is therefore an algebraic bundle with its induced flat connection, by Beauville, Sections 3–4.

Let $\mathcal{K}_{\mathbf{j}}^{l}$ denote this coefficient bundle, with the inverse-field convention just specified. On a stratum with blocks π, the coefficient with states is

$$(i_\pi)_*(j_\pi)_*\Big(\bigoplus_{\mathbf{j}} \mathcal{K}_{\mathbf{j}}^{l} \otimes \boxtimes_{B \in \pi}(\omega_X \otimes H_{j_B})\Big)[|\pi|]. \tag{16.33}$$

Sewing frames decompose this bundle near a collision over the allowed intermediate labels. The faces in these frames are (16.31), with the determinant signs used in (16.8).

Algebraic coefficients and radial sewing frames

The bundle in (16.32) is algebraic. Its defining matrix is polynomial in the positions, and its kernel has the constant rank given by the integrable fusion multiplicity. A matrix of constant rank on the reduced configuration variety has locally free kernel, image and cokernel. Thus the kernel is locally a direct summand of the constant lowest-state tensor bundle.

It is invariant under common translation and nonzero dilation of the positions. Under translation by c, the operator $\sum_i z_i E^{(i)}$ becomes $\sum_i z_i E^{(i)} + c \sum_i E^{(i)}$. The summands commute. Invariance moves each occurrence of the total raising operator to the highest vector at infinity, where it vanishes. Dilation multiplies the null equation by a nonzero scalar.

The Casimir connection preserves this algebraic kernel. Locally, analytic horizontal conformal blocks span every fiber. Writing an algebraic section in such a frame shows that its covariant derivative lies in

the same kernel at every point. It is therefore an algebraic one-form with values in the subbundle. The inherited connection has regular singularities along diagonals.

Consider a cut with left and right roots y', y'', and put $d = y' - y'' \neq 0$. Contract the two groups by

$$y_B = y' + \lambda v_B \quad (B \text{ on the left}), \qquad y_B = y'' + \lambda v_B \quad (B \text{ on the right}).$$

The root coordinates have $v_B = 0$. The v_B record the internal shapes and λ their common scale. Let A_c be the ring of rational functions of these shapes and roots, with poles allowed only on internal diagonals and $d = 0$.

For total output l, the fused coefficient is

$$\mathcal{K}_c^l = \bigoplus_{a,b=0}^{10} \mathcal{K}_{ab}^l \otimes \mathcal{K}_{\mathbf{j}'}^a \otimes \mathcal{K}_{\mathbf{j}''}^b.$$

Only the integrable channels occur. Compose the integrable germs of the two groups with the two-root germ. The result gives a radial sewing map

$$J_c : \mathcal{K}_c^l \otimes A_c[[\lambda]] \longrightarrow \widehat{\mathcal{K}}_{\mathbf{j}'\mathbf{j}''}^l. \tag{16.34}$$

Its constant term is the composition of the three lowest-state maps. Ward recursion on the integrable modules defines every higher coefficient, including at resonant levels.

Proposition 16.12.1. The map (16.34) is an isomorphism over $A_c[[\lambda]]$, intertwining the fused and full connections. Frames for nested collisions and cuts compose. For fixed shape and roots it converges when

$$|\lambda| \max_B |v_B| < |d|/4.$$

It continues to a holomorphic isomorphism over the region where every left position has larger real part than every right position.

Proof. Insertion of the three germs gives rational Taylor coefficients, with precisely the allowed poles. Ward identities hold at every original point. Uniqueness of the integrable germ determines the series coefficient by coefficient.

The constant term is injective by semisimple composition of lowest-state maps. Locally choose a retraction L_0 of this constant term. Then $L_0 J_c = 1 + O(\lambda)$ is invertible, and $(L_0 J_c)^{-1} L_0$ is a left inverse. The image lies in the kernel of the polynomial null-equation matrix.

Its rank equals the rank of that kernel, since

$$\sum_{a,b} N_{ab}^l N_{\mathbf{j}'}^a N_{\mathbf{j}''}^b = N_{\mathbf{j}'\mathbf{j}''}^l$$

by associativity of integrable fusion. The quotient of the ambient free module by the split image is projective. The null kernel modulo the image is a rank-zero submodule of this torsion-free quotient, so it vanishes. This proves the formal isomorphism.

Two nested frames give the same inserted Ward germ. Their coefficients are finite sums in each state degree, so associativity of insertion identifies the series. Translation of the germ identifies the connections.

Contracting horizontality with $\lambda\partial_\lambda$ gives a regular-singular linear equation

$$\lambda \frac{dJ_c}{d\lambda} = J_c\big(C + R(\lambda)\big) - C^{\circ} J_c.$$

Here C is the sum of internal Casimirs and C° its fused action. Each cross term of R has the form

$$t_{ij}\frac{\lambda(v_i - v_j)}{d + \lambda(v_i - v_j)}, \qquad t_{ij} = \frac{\Omega_{ij}}{12}.$$

It is holomorphic for $|\lambda| \max |v_B| < |d|/2$. Above the finitely many differences of eigenvalues of C and C°, the recursion has an inverse bounded by a constant times the inverse degree. The usual power-series majorant therefore converges on every smaller disc.

The finitely many resonant degrees do not affect convergence: their coefficients have already been fixed by the germs.

Scale contraction keeps the two groups separated in real part. Horizontal continuation from the convergent neighborhood extends J_c through that region. Flatness and uniqueness identify the continuations on intersecting neighborhoods. □

Resonance is a reason to use the germs. For example, at level four the three spin-one inputs with spin-one output have pair exponents $-2/3, -1/3, 1/3$. The degree-one recursion between the first and third channels is singular. Its right side vanishes, since one Lie-algebra generator changes the intermediate spin by at most one. The differential equation alone leaves a free coefficient. The specified integrable germ selects it.

The coupled product on full cut opens

For this construction a state system records a color in $\{0, \dots, 10\}$ and a state factor on every block of every partition. Its coefficient diagram attaches $\mathcal{K}^{l}_{\mathbf{j}}$ to the entire ordered color word. It retains the state factors, densities and block-number shift of (16.33). The maps are generated by maps on states and by collision residues with arbitrary spectator blocks.

Concatenate the two color words and their state factors. Attaching the coefficient bundle of this whole word defines the coupled cut product:

$$\mathcal{K}\mathscr{S}' \star_{\mathcal{K}} \mathcal{K}\mathscr{S}'' = \mathcal{K}(\mathscr{S}' \boxtimes \mathscr{S}''). \tag{16.35}$$

The state tensor is external, but the coefficient bundle includes all cross-Casimir terms. The unit is $\mathbb{C}$ in color zero at the empty word.

A residue of one factor extends by the residue in the enlarged configuration, with the other states as spectators. The radial comparison determines this extension. Rational expansion is injective, state modules are free, and coefficient bundles lie inside constant bundles. The nested-frame identity proves compatibility with compositions of the generating maps.

Thus $\star_{\mathcal{K}}$ is a graded monoidal product. The Koszul rule makes its product differential square to zero. Both parenthesizations carry the same whole-word bundle, so the associator on these diagrams is the identity.

For a consecutive cut c, project onto the partitions having no block that crosses c. A discarded block remains crossing under every coarsening. Hence this is a chain map. The discarded summands have diagonal support outside the cut open U_c. On U_c, every retained term on both sides has the same bundle $\mathcal{K}^{l}_{\mathbf{j}'\mathbf{j}''}$. Consequently the cut map is an isomorphism on the full algebraic open, before radial completion.

Repeated cut projections commute, and the empty word supplies the counit. The resulting strong cut coalgebra has an associative cobar for $\star_{\mathcal{K}}$. Its positive-cut contraction uses the signs and finite sums already given after Theorem 16.3.1. On suspended singleton inputs the transferred operation is

$$B_n(sx_1, \dots, sx_n) = s\, r_{1n}(x_1, \dots, x_n), \qquad n \geq 2;$$

in particular the ternary operation retains the nonconsecutive residue r_{13}.

The radial frame identifies this cut map with the fusion-channel cut after completion. Horizontal frames of two iterated cuts differ by the integrable channel associator. Thus an identity associator on whole-word coefficient diagrams and a nontrivial channel associator describe the same maps in different presentations.

Algebraic reorderings and their monodromy

Let σ, τ be two orders of the insertion labels. For a partition with r blocks, let w carry its root order for σ to its root order for τ. Write c_w for the ordinary permutation of the lowest-state tensor factors. The transition is

$$f \otimes e \otimes x_1 \otimes \cdots \otimes x_r \longmapsto \operatorname{sgn}(w)\, f \otimes e\, c_w^{-1} \otimes x_{w^{-1}(1)} \otimes \cdots \otimes x_{w^{-1}(r)}. \tag{16.36}$$

The sign is the determinant sign of the shifted block factors.

The Casimir connection and infinity null equation are equivariant for these algebraic reorderings. On the chart overlap, a partition containing two labels ordered differently has empty support. For every surviving collision the two colliding roots keep their relative order. Its coordinate, integrable Ward germ and full principal part therefore agree under (16.36). If the deleted root moves from place i to place p, omitting it changes the permutation sign by $(-1)^{i+p}$. This converts the face sign $(-1)^{i-1}$ to $(-1)^{p-1}$.

Hence the order maps commute with the differential. They satisfy the cocycle identity by permutation coherence. For an ordered collision, only saturated partitions survive extraordinary pullback, and their root orders agree. Thus the transition also commutes with every collision comparison.

On a cut-admissible partition, reordering either side is $w_A \times w_B$, with sign $\operatorname{sgn}(w_A)\operatorname{sgn}(w_B)$. The order maps therefore glue the coupled products and their cut isomorphisms. Interchanging the two groups gives a symmetry whose sign on r_A, r_B blocks is $(-1)^{r_A r_B}$. The cut coalgebra is cocommutative for this reordering symmetry. This symmetry does not remove the monodromy of its coefficient bundle.

A half turn describes the order map in continued horizontal frames. At a pair of roots, continuation of u^a through a counterclockwise semicircle contributes $e^{\pi i a}$. The normalized integral-power germ changes by translation and ordinary reordering. The exponent is retained by the coefficient connection. Clockwise continuation gives $e^{-\pi i a}$. Their ratio is the full turn $e^{2\pi i a}$.

These matrices do not generally define transitions on coefficients at the same point for the fixed product (16.35). The distinction already appears for three defining $\mathfrak{sl}_2$ colors.

Proposition 16.12.2. At an integral level $k \geq 2$, the block bundle with three defining inputs and defining output is irreducible of rank two. Consider a flat coefficient transition tensored with state reordering. Compatibility with the fixed coupled product forces one scalar on both binary channels relative to ordinary reordering. The half turn has different scalars and cannot be that transition.

Proof. Put $h = (k+2)^{-1}$. Both singlet and triplet channels are integrable, so the rank is two. On the slice $(0, x, 1)$, the residues at $0, 1, \infty$ have eigenvalues

$$\{-3h/2, h/2\}, \qquad \{-3h/2, h/2\}, \qquad \{0, 2h\}.$$

All exponent differences are nonintegral.

An invariant line would extend to a line bundle whose degree is minus a sum of residue eigenvalues. This sum is one of $-3h, -h, h, 3h$, none an integer when $k + 2 \geq 4$. Thus no invariant line exists.

A flat endomorphism of this irreducible complex local system is scalar. Compatibility with the cut 12|3 extends the binary transition over the third defining input. The two channels therefore have the same scalar relative to reordering. The half turn instead has eigenvalues

$$-e^{-3\pi ih/2}, \qquad e^{\pi ih/2}.$$

After division by the reordering signs, their ratio is $e^{-2\pi ih} \neq 1$. This contradicts the necessary equality. □

At level one only the singlet channel remains. The same rank-two argument applies at irrational h to the full defining-color KZ system: the four possible residue sums are again nonintegral. The proposition fixes the coefficient transition and the coupled product. It makes no assertion about a different tensor product with separately constructed comparison maps.

The rank-two example also distinguishes local meridians from full monodromy. At level two put $\zeta = e^{\pi i/4}$. The two local full turns can be represented as

$$A = \zeta Z, \qquad B = \zeta X, \qquad Z^2 = X^2 = 1, \quad ZX = -XZ.$$

Each has order eight and $ABA^{-1}B^{-1} = -1$. The group they generate on the slice has order sixteen:

$$\{\zeta^a Z^b X^c : a \in \mathbb{Z}/8,\ b, c \in \mathbb{Z}/2,\ a \equiv b + c \pmod 2\}.$$

The full twist of the three moving points contributes ζ^{-31}, so the full pure-braid image has order thirty-two. Projectively it is the Klein four-group.

Indeed, the exponent differences on the three-punctured sphere are $1/2$. Its projective local monodromies are distinct involutions with involutive product. Irreducibility forces their matrix lifts to anticommute.

Adjoining eighth roots of all differences kills the individual local meridians. That abelian cover still contains their commutator loop, whose action is -1. The cover corresponding to the kernel of the full monodromy trivializes the bundle. On it the continued horizontal frames are single valued and the reorderings are constant half-turn matrices. Killing local eigenvalues alone would miss this remaining sign.

PROPOSITION 16.12.3. The full ambient affine coefficient system gives a strong cut coalgebra with the terms (16.33) and the coupled cut product. It glues on the ordered collision schemes. Mark it with the A_E-module junctions on its sewing charts. The arithmetic junction comparison (16.29) identifies these marked residue diagrams, their cut maps, and their cobars. It retains every normal derivative and the vacuum.

Proof. An affine Ward identity moves a negative current mode at one insertion to regular expansions at the others. Induction on total degree therefore determines every germ from its lowest-space value. The singular-vector equation (16.32) is exactly the condition that this germ descend to the integrable quotients. At a collision, inserting a binary integrable germ into an external germ gives its factorization over the allowed intermediate labels. The factorization is an isomorphism by the integrable conformal-block sewing and factorization theorems of Tsuchiya–Ueno–Yamada, Theorems 6.2.1 and 6.2.5. It uses the full tensor (16.30).

For two disjoint collisions the two principal-part maps commute before their determinant signs are inserted, so their signed composites cancel. For three blocks the three routes are the three expansions

of one Ward germ. The two-variable Cousin boundary identity therefore gives their signed sum zero. These statements prove that the sum of the elementary faces squares to zero. They include a collision of the first and third blocks with the second retained as a spectator.

For a consecutive cut, discard exactly the partitions having a block on both sides. Such a block remains crossing under every coarsening, so the projection is a chain map. Its discarded terms have diagonal support outside the cut open. On every retained term the coupled product has the coefficient bundle of the same whole word. The cut map is consequently an isomorphism there. Repeated projections commute and the empty word is the counit.

On two ordered charts, reorder the lowest-space tensor factors and precompose their coefficient functionals by the inverse permutation. The Casimir connection is equivariant for these algebraic reorderings. If a colliding pair has the same relative order in both charts, its Ward germ and principal part are carried to each other. If its order is reversed, its diagonal is absent from the overlap. Thus the reorderings give a cocycle of complexes.

In the continued horizontal frames of Section 16.12, these algebraic maps are represented by half-turn matrices. They are not pointwise coefficient transitions for the fixed coupled product.

Finally (16.29) is an equality of full intertwining operators. Its coefficient symbols transform by the dual inverse of K, preserving their contraction with fields. It therefore commutes with (16.31) and with every spectator extension. The cut maps are the same projections on both sides. The resulting identification of coalgebras identifies their cobars. □

The ambient coefficient bundle is essential here. A projection onto a chosen fusion channel in a sewing frame need not extend to a single-valued flat projection on the whole configuration space. The construction retains that bundle and marks the module junctions on the sewing charts. It does not replace the bundle by constant channel spaces or assert a global splitting into them.

The Heisenberg, lattice, and affine constructions are distinct conformal theories, with central charges 1, 8, and 5/2. Their full state spaces are not identified. They realize the same specified junctions on different conformal carriers. The affine construction also supplies nontrivial conformal-block monodromy at the embedding σ.

16.13 The original embedding in a Virasoro theory

The original arithmetic embedding has $d = 1 - \sqrt{3}$. It therefore requires a different conformal carrier from the positive affine theory. Such a carrier is provided by the Virasoro minimal model

$$W_- = \mathrm{Vir}(12, 5), \qquad c_- = 1 - \frac{6(12-5)^2}{12 \cdot 5} = -\frac{39}{10}.$$

Let U_j be its simple module with Kac label $(j+1, 1)$, $0 \le j \le 10$. Its lowest conformal weight is

$$h_j^- = \frac{(5(j+1) - 12)^2 - 49}{240} = \frac{5j(j+2)}{48} - \frac{j}{2}.$$

Its states are the negative Virasoro modes on a lowest vector, modulo the two singular-vector submodules at levels $j+1$ and $4(11-j)$. This specifies all descendants, including their null relations. The category is rational and C_2-cofinite.

Huang's Theorems 2.1 and 3.8–3.10 give its intertwining operators the convergent vertex tensor structure. Creutzig–Huang–Yang, Section 7, give the related admissible tensor-category formulation.

Put $q_- = e^{5\pi i/12}$. The cyclotomic automorphism $\gamma : q \mapsto q_-$ satisfies $\gamma(\sqrt{3}) = -\sqrt{3}$ and $\gamma(i) = i$. Write $\mathcal{D}^+_{q_-}$ for the even quantum-group category. There is a ribbon equivalence

$$\Psi : \mathcal{D}^+_{q_-} \xrightarrow{\sim} \langle U_0, U_2, U_4, U_6, U_8, U_{10}\rangle, \qquad V_j \longmapsto U_j. \tag{16.37}$$

Here and below the rational Racah bases are transported through Ψ to normalize the Virasoro intertwining operators.

We explain why the parameter and the ribbon structure in this comparison are fixed. The fusion rules of $U_0, \dots, U_{10}$ are (16.20). The object U_1 generates them and $U_1^2 = U_0 \oplus U_2$. Its counterclockwise exchanges on these two channels, computed from their leading primary powers, are

$$e^{\pi i(h^-_0 - 2h^-_1)} = e^{3\pi i/8} = -q_-^{-3/2}, \qquad e^{\pi i(h^-_2 - 2h^-_1)} = e^{5\pi i/24} = q_-^{1/2}.$$

Categories with SL_2 Verlinde rules are reconstructed from the quantum-group categories, up to a cocycle on the parity grading. This is the classification in Etingof–Gelaki–Nikshych–Ostrik, Section 8.27.5.

In the Temperley–Lieb reconstruction, these two eigenvalues fix the braid generator. The successive Jones–Wenzl projections use

$$[n]_- = \frac{\sin(5n\pi/12)}{\sin(5\pi/12)}, \qquad [n]_- \neq 0 \ (1 \le n \le 11), \quad [12]_- = 0.$$

Their path matrix units give the A_{11} fusion paths. The next projection vanishes. Thus the comparison is full on the generated category. A possible parity cocycle restricts to the constant one on the even subcategory. There the twists agree exactly, since $e^{2\pi i h^-_j} = q_-^{j(j+2)/2}$ for even j. This proves (16.37), including its braiding and twists.

In particular it fixes the orientation of analytic continuation, which the fusion rules alone would not determine.

Apply γ to the algebra and junction matrices of Section 16.10. In $\mathcal{D}^+_{q_-}$ the algebra is again $V_0 \oplus V_6$, now with

$$m^0_{66} = -\frac{1}{3} - \frac{\sqrt{3}}{6}, \qquad m^6_{66} = 1, \qquad \dim A_- = 3 - \sqrt{3}.$$

Its associativity, commutativity, separability, and all junction comparison equations are polynomial identities over the cyclotomic field, so they are preserved by γ. The comparison from the original arithmetic category has binary matrices

$$K^-_{ab;c} = \gamma(K_{ab;c}). \tag{16.38}$$

Indeed γ restricts to σ on F, and applying it to C^σ restores the original embedding.

Transport A_- and its three modules through Ψ. Their full Virasoro state spaces are

$$\begin{aligned} \mathsf{H}^-_1 &= U_0 \oplus U_6, \\ \mathsf{H}^-_g &= U_4 \oplus U_{10}, \\ \mathsf{H}^-_\rho &= U_2 \oplus U_4 \oplus U_6 \oplus U_8. \end{aligned} \tag{16.39}$$

Let q be the balancing quotient defined by the conjugate of (16.27). For a module junction p, its full field is the intertwining operator of the module map $\Psi(pq)\Psi_2$. The arithmetic field is K^- times this column of fields.

Theorem 16.13.1. Equations (16.38) and (16.39) give an analytic ordered realization of the order-two arithmetic category at its original complex embedding. It has central charge $-39/10$ and relative dimensions $1, 1, 1-\sqrt{3}$. The comparison preserves every junction, all descendants, translation, relative duality, and collision and sewing operations in every arity. Its sewings on the domain of Section 16.7 converge absolutely and locally uniformly, including finite simultaneous sewings that create handles.

Proof. The ribbon equivalence (16.37) identifies the commutative separable algebra and its balanced module tensor product. The conjugate of (16.24) gives all its binary and ternary comparison maps. The proof of Theorem 16.11.1 now applies with the Virasoro vertex tensor structure in place of the affine one. It is an equality of intertwining operators on entire modules, so it includes their singular-vector quotients and every descendant.

The module dimensions can be checked without a choice of junction basis. The categorical dimension of U_j is

$$(-1)^j \frac{\sin(5(j+1)\pi/12)}{\sin(5\pi/12)}.$$

Divide the sums for the three modules in (16.39) by $3-\sqrt{3}$. They give $1, 1, 1-\sqrt{3}$. The unique spherical structure on the near-group category is therefore the original arithmetic one. In particular the closed marked junction has value $-e_1$.

The sewing tensor on U_j is

$$\sum_{n\geq 0} t^{h_j^- + n} \sum_{\alpha} v_{j,n,\alpha} \otimes v^{\vee}_{j,n,\alpha}.$$

Every grade is finite dimensional. The possible negative lowest weights contribute only fixed powers of t on a chosen logarithmic branch. Rationality and C_2-cofiniteness allow Gui, Theorem 13.1 to be applied with the hypotheses of Section 16.7. This gives the asserted convergence on that domain.

On a binary collision divide the field by $u^{h_k^- - h_i^- - h_j^-}$ and pair it with the inverse coefficient symbol before taking its full principal part. The derivative of that symbol retains the conformal exponent, as in (16.31). The full intertwiner identity commutes with this operation and with every spectator insertion. The Virasoro null-vector equations supply its conformal-block connection. Their sewing identities give the same Cousin boundary and cut identities. Thus the comparison also preserves the collision diagrams. □

For even $j = 0, 2, 4, 6, 8, 10$, the lowest weights are respectively $0, -1/6, 1/2, 2, 13/3, 15/2$. These weights and the central charge belong to the new Virasoro construction. The Galois automorphism acts on the finite junction data. The independently defined Virasoro vertex tensor theory supplies the full analytic lift. These sewings use algebraic contragredients. No positive Hilbert metric is assumed.

16.14 The connected realization beyond order two

The relative E_8 construction works for every arithmetic category considered here. Its vacuum algebra has $N+1$ idempotents. The affine and Virasoro constructions give complete order-two realizations on independently defined conformal state spaces. These results leave a different first example.

Question 16.14.1. For $K = \mathbb{Q}(\sqrt{21})$ and $\varepsilon = (5+\sqrt{21})/2$, one has $N = 3$. Construct a connected rational nonlocal conformal realization of C_I, with its specified module fusion, marked junctions, duality and convergent sewing on full descendant spaces. Which rooted, collision or nearby-cycle tensor realizes its fusion?

Theorem 16.8.1 excludes the stated strong unital realization by the scalar-empty balanced cut tensor. It does not exclude the alternative tensor asked for here. Likewise, the supported recoupling comparison must retain its larger channel system and specify the compression that produces $-N\beta D$. Its first missing map is an operation-preserving supported comparison with that declared compression.

Arithmetic field automorphisms act on the finite junction coefficients. Transport of a conformal theory requires, in addition, its actual tensorator, state spaces and analytic sewing. This distinction is already visible in the different affine and Virasoro conformal weights.

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Chapter 17

Artin transport, norms and specialization

The arithmetic fusion category has two kinds of transport. An Artin automorphism moves its marked junctions from one lattice to another. A finite lattice covering instead multiplies arithmetic values over periodic orbits. We construct both maps and compare their tensor properties.

The distinction already appears for $\varepsilon = 2 + \sqrt{3}$ and $I = \mathbb{Z}[\sqrt{3}]$. The category at this period has simples 1, g, ρ, with

$$g^2 = 1, \qquad g\rho = \rho g = \rho, \qquad \rho^2 = 1 + g + 2\rho.$$

Changing to period ε^2 gives a positive map of fusion rings that sends the new noninvertible simple to $1 + g + 4\rho$. The unit occurs once in this image. Tensorator coherence would force a commutation relation on that one-dimensional space which two nonzero scalars cannot satisfy. The ring map survives. Its linear tensor lift does not.

We first transport the full marked category under reciprocity. Periodic points then organize multiplicative norms and their parameter algebras. The unit multiplicity gives obstructions to tensor lifts of every even period ratio. A further cocycle calculation excludes period three for the lattice above. Finally, norms of lines and indexed tensors provide positive multiplicative constructions, with their own coefficient domains.

17.1 Reciprocity on the marked family

Use the arithmetic data of Chapter 15. Thus K is a real quadratic field, $I \subset K$ is an oriented rank-two lattice, and $\varepsilon > 1$ is a totally positive norm-one unit preserving I. Write

$$G_I = (\varepsilon - 1)^{-1} I/I, \qquad N = |G_I| = \varepsilon + \varepsilon^{-1} - 2.$$

Let m be the exponent of G_I, and write $\mathcal{O} = (I : I) = \mathbb{Z} + c_0 \mathcal{O}_K$ for its multiplier order, with conductor c_0.

The normalized arithmetic values are $e_{I,g} = E_{I,\varepsilon}(g)/E_{I,\varepsilon}(0)$. They lie in the narrow ray field $L = H^+_{\mathcal{O},m}$. We use arithmetic Artin symbols: at an unramified prime their reduction acts by the residue-field norm power.

The reciprocity input is Theorem 2 of *Stark units for real quadratic fields and reciprocity laws*. It is stated in (15.6). The coefficients $b_{I,g} = b_0 e_{I,g}$, with $b_0 = (\varepsilon - 1)/N$, define the Leavitt algebra B_I and the marked category C_I. The algebra isomorphism below transports all four generator types X_g, X'_g, T_g, T'_g, including their algebraic duals.

Now let $\mathfrak{a}$ be an integral invertible $\mathcal{O}$-ideal prime to $c_0 m$, and put $J = \mathfrak{a}^{-1} I$. The natural label map is

$$\phi_{\mathfrak{a}} : G_I \longrightarrow G_J, \qquad x + I \longmapsto x + J.$$

Lemma 17.1.1. The map $\phi_{\mathfrak{a}}$ is an isomorphism, and J has multiplier order $\mathcal{O}$.

Proof. Coprimality gives $z \in \mathfrak{a}$ with $z \equiv 1 \pmod{mO}$. The integer $h = \mathrm{Nm}(z) = z\bar{z}$ belongs to $\mathfrak{a}$ and is prime to m. Therefore $h(J/I) = 0$. The kernel of $\phi_{\mathfrak{a}}$ is killed by both h and m, so is zero. Both groups have order N, proving surjectivity. If $tJ \subseteq J$, multiplication by the invertible ideal $\mathfrak{a}$ gives $tI \subseteq I$, and the converse follows by multiplication by $\mathfrak{a}^{-1}$. □

Extend coefficients to L, and write $\sigma_{\mathfrak{a}} = \mathrm{Art}(\mathfrak{a})|_L$. Since $b_0 \in K$, the arithmetic law gives

$$\sigma_{\mathfrak{a}}(b_{I,g}) = b_{J,\phi_{\mathfrak{a}}g}.$$

All reconstruction coefficients are rational expressions in the b_g and N. Thus Q, χ, a, c_1, β transform by the same label map, while d is fixed.

THEOREM 17.1.2. There is a $\sigma_{\mathfrak{a}}$-semilinear strong tensor equivalence

$$\Phi_{\mathfrak{a},I} : C_{I,L} \longrightarrow C_{\mathfrak{a}^{-1}I,L}$$

preserving ρ, sending α_g to $\alpha_{\phi_{\mathfrak{a}}g}$, and preserving the specified duality maps and normalized networks. For two such ideals,

$$\Phi_{\mathfrak{b},\mathfrak{a}^{-1}I}\Phi_{\mathfrak{a},I} = \Phi_{\mathfrak{a}\mathfrak{b},I}$$

on the explicit models.

Proof. Apply $\sigma_{\mathfrak{a}}$ to scalars and $\phi_{\mathfrak{a}}$ to every generator label. This defines a semilinear algebra isomorphism $\Theta_{\mathfrak{a}} : B_{I,L} \to B_{J,L}$. The coefficient formulas show $\Theta_{\mathfrak{a}}\rho_I = \rho_J\Theta_{\mathfrak{a}}$ and the corresponding equality for α_g. Thus intertwiners map to intertwiners, and (15.15) is preserved. Finite direct sums and idempotent splittings are preserved as well.

The zero label and d are fixed, proving duality compatibility. The normalization of Proposition 15.4.1 is preserved. Finally, Artin symbols multiply and both composed label maps send a representative to its class in $(\mathfrak{a}\mathfrak{b})^{-1}I$. The two semilinear maps therefore agree on scalars and generators. □

This is transport on the lattice-indexed family. Identifying its targets with one fixed category would require lattice identifications. This construction lifts the arithmetic law (15.6) to the marked category.

The scalar normalization matters for the unnormalized values. Choose an extension of $\sigma_{\mathfrak{a}}$ to $L(\sqrt{\varepsilon})$. For this extension, reciprocity reads

$$\sigma_{\mathfrak{a}}\big(E_{I,\varepsilon}(x)\big) = \frac{\sigma_{\mathfrak{a}}(\sqrt{\varepsilon})}{\sqrt{\varepsilon}}\, E_{\mathfrak{a}^{-1}I,\varepsilon}(x).$$

The possible sign disappears for $e_{I,x}$. It cannot be dropped from a statement about the raw values.

Transport of singular interfaces and finite symmetry

Let A be the level-one Heisenberg cut cobar over K used in Chapter 16. Put $R_I = B_I^{\mathrm{op}}$. For an endomorphism η of B_I, let P_η have underlying space B_I and actions

$$a^{\mathrm{op}}\, b\, c^{\mathrm{op}} = \eta(c)ba.$$

The relative realization is

$$\mathfrak{R}_I = A_L \otimes_L R_{I,L}, \qquad \mathfrak{P}_\eta = A_L \otimes_L P_{\eta,L}.$$

The map $\Theta_{\mathfrak{a}}$ transports these coefficients. The system A is defined over K, so its coordinates, translation, residues, local vacuum and cut maps are fixed.

Let P, Q be composable coefficient bimodules over $R = R_{I,L}$. In bar degree n, the comparison

$$(A_L \otimes P) \bigstar^{\mathbb{L}}_{A_L \otimes R} (A_L \otimes Q) \simeq A_L \otimes (P \otimes^{\mathbb{L}}_R Q)$$

multiplies the $n+2$ geometric factors in order. It leaves the coefficient bar $P \otimes R^{\otimes n} \otimes Q$. Thus it commutes with $\Theta_{\mathfrak{a}}$ in every degree. The common multibar proves the same assertion for every associativity and unit constraint.

Naturality also transports the connected comparison (16.12) and its weight-one interface (16.13). All normal derivatives and spectator states are retained.

The finite receiver has an equally explicit transport. Take split finite semisimple rigid categories over characteristic-zero fields k, k', with simple units. Let $F : \mathcal{C} \to \mathcal{C}'$ be a specified $\sigma : k \to k'$-semilinear tensor equivalence. Choose its identifications of simple objects, with permutation π of their labels. They give semilinear maps

$$U_i : P_{S_i} \longrightarrow P'_{S'_{\pi(i)}}$$

compatible with the tensorators, units and duality pairings of Section 16.6. The reconstructed algebras are $H_{\mathcal{C}} = \bigoplus_i \operatorname{End}_k(P_{S_i})$ and $H_{\mathcal{C}'}$.

Proposition 17.1.3. The formula

$$\Theta_\sigma(h)_{\pi(i)} = U_i h_i U_i^{-1}$$

defines a semilinear weak Hopf isomorphism. These maps compose with the marked equivalences.

Proof. Conjugation preserves multiplication. The compatibility of U with the balanced tensorator and matching-endpoint projection preserves the coproduct (16.16). Compatibility with the unit and its coordinate-sum functional preserves the counit. Compatibility with the duality pairings preserves their transposes, hence the antipode. Composing the maps U gives the stated composition law. □

In matrices after scalar twisting, this formula is $U_i\, \sigma(h_i)\, U_i^{-1}$, where U_i now denotes the linear matrix of the twisted map. The Artin equivalences supply the required identifications: they preserve X_{o}, dX'_{o} and every marked network. The assertion therefore includes the weak unit, categorical junctions and arithmetic markings.

17.2 Coherence on a fixed presentation

Transport of a marked algebra has two distinct forms. Base change moves the whole presentation to its conjugate. Transport on one fixed presentation first requires identifications with those conjugates, and their associativity has an obstruction. We compute this obstruction before discussing multiplicative norms, whose coherence uses different data.

Let k be a field and let B be an associative unital k-algebra with specified generators, relations and marked intertwiners. Include all roots and normalization constants in k. An automorphism σ of k carries these data to their coefficientwise conjugates. Polynomial relations and compositions are preserved, so this construction gives coherent semilinear transport on the family of conjugate presentations. Identifying its value with a prescribed arithmetic relabelling requires maps of the marked presentations themselves.

The central associativity class

Regard a group Γ acting on k as a discrete group. Suppose we are given σ-semilinear algebra automorphisms ϕ_σ of B and units $u_{\sigma,\tau} \in B^\times$ satisfying

$$\phi_\sigma \phi_\tau = \mathrm{Ad}(u_{\sigma,\tau})\phi_{\sigma\tau}, \qquad \phi_1 = 1, \qquad u_{1,\sigma} = u_{\sigma,1} = 1. \tag{17.1}$$

The last two unit equalities are part of the normalization. Set

$$c_{\sigma,\tau,\nu} = (u_{\sigma,\tau}u_{\sigma\tau,\nu})^{-1}\phi_\sigma(u_{\tau,\nu})u_{\sigma,\tau\nu}. \tag{17.2}$$

The restrictions of the ϕ_σ to $Z(B)$ define an honest Γ action, because inner automorphisms are trivial on the center. Write $\sigma(z) = \phi_\sigma(z)$ for $z \in Z(B)$.

PROPOSITION 17.2.1 *(Associativity of a fixed presentation).* The elements (17.2) form a normalized 3-cocycle with values in this Γ-module $Z(B)^\times$. Replacing $u_{\sigma,\tau}$ by $z_{\sigma,\tau}u_{\sigma,\tau}$ for a normalized central 2-cochain changes c by

$$(\delta z)_{\sigma,\tau,\nu} = \frac{\sigma(z_{\tau,\nu})z_{\sigma,\tau\nu}}{z_{\sigma,\tau}z_{\sigma\tau,\nu}}. \tag{17.3}$$

For the supplied automorphisms and inner comparisons, coherent implementers with $c = 1$ exist exactly when $[c] = 0$ in $H^3(\Gamma, Z(B)^\times)$.

Proof. Both $u_{\sigma,\tau}u_{\sigma\tau,\nu}$ and $\phi_\sigma(u_{\tau,\nu})u_{\sigma,\tau\nu}$ implement the same comparison between $\phi_\sigma\phi_\tau\phi_\nu$ and $\phi_{\sigma\tau\nu}$. Their ratio is therefore central.

Compare the five parenthesizations of four automorphisms using these implementers. The central triple ratios commute with every remaining factor. The two paths around the pentagon give

$$\sigma(c_{\tau,\nu,\kappa})c_{\sigma,\tau\nu,\kappa}c_{\sigma,\tau,\nu} = c_{\sigma\tau,\nu,\kappa}c_{\sigma,\tau,\nu\kappa}.$$

This is the multiplicative cocycle equation.

The unit equalities in (17.1) make c equal to 1 when any argument is the identity. Substitution of zu into (17.2) gives (17.3). Finally, two units implementing the same inner automorphism differ by a central unit. These cochains exhaust the choices for fixed ϕ. Their coboundaries therefore give exactly the stated vanishing criterion. □

When $c = 1$, the direct sum

$$B \rtimes_{\phi,u} \Gamma = \bigoplus_{\sigma\in\Gamma} BU_\sigma,$$
$$(bU_\sigma)(dU_\tau) = b\phi_\sigma(d)u_{\sigma,\tau}U_{\sigma\tau}$$

is a unital associative algebra over k^Γ. Indeed expansion of a triple product uses $\phi_\sigma\phi_\tau = \mathrm{Ad}(u_{\sigma,\tau})\phi_{\sigma\tau}$ to reduce its two parenthesizations to the two products of implementers in (17.2). For general c, the right-associated product is the left-associated one multiplied in its coefficient by $c_{\sigma,\tau,\nu}$. This also makes the defect of associativity explicit.

Normalization of ϕ_1 alone is insufficient. For example, take $B = \mathbb{Q}$, $\Gamma = \mathbb{Z}/3$ written additively, all $\phi = 1$, and choose $u_{0,1} = 2$, $u_{0,2} = 3$, with all other $u = 1$. The inner relation still holds, but $c_{0,1,1} = 3/2$. The units at an identity argument are central and can be normalized first. The displayed c becomes normalized only after this correction.

The strict maps on an arithmetic family in Theorem 17.1.2 and the fixed-presentation datum (17.1) have different domains. A family action does not by itself supply the identifications and inner implementers needed to define this class. Conversely, the class does not obstruct coefficientwise transport on the whole conjugate family.

17.3 Periodic points and normalized norms

The lattice operations have a geometric organization independent of the chiral background. Form the real two-torus

$$\mathbb{T}_I = (K \otimes_{\mathbb{Q}} \mathbb{R})/I.$$

Multiplication by ε induces a hyperbolic automorphism u_I with eigenvalues $\varepsilon, \varepsilon^{-1}$. For every integer $r \geq 1$,

$$\operatorname{Fix}(u_I^r) = (\varepsilon^r - 1)^{-1} I/I, \qquad N_r = |\operatorname{Fix}(u_I^r)| = \operatorname{Tr}(\varepsilon^r) - 2.$$

Indeed, a lift is fixed precisely when $(\varepsilon^r - 1)x \in I$. The real-linear map $\varepsilon^r - 1$ is invertible, and its determinant gives the stated cardinality.

The formal periodic-point zeta function is

$$\exp\left(\sum_{r\geq 1} \frac{N_r}{r} t^r\right) = \frac{(1-t)^2}{1 - \operatorname{Tr}(\varepsilon)t + t^2}.$$

This follows by expanding the logarithms of $(1-\varepsilon t)$, $(1-\varepsilon^{-1}t)$ and $(1-t)^2$.

An inclusion $I \subset J$ of lattices preserved by ε gives an equivariant finite covering $f : \mathbb{T}_I \to \mathbb{T}_J$. A fiber over a point fixed by u_J^r is permuted by u_I^r. Its elements need not all have period r. The orbit lengths determine the norm.

More generally, let u denote compatible automorphisms on spaces with finite equivariant coverings. A multiplicative periodic datum with values in a field M consists of $w(x, r) \in M^\times$, for $r \geq 1$ and $u^r x = x$, satisfying

$$w(ux, r) = w(x, r), \qquad w(x, rs) = w(x, r)^s \quad (s \geq 1). \tag{17.4}$$

For an equivariant finite covering f, define

$$(f_* w)(z, r) = \prod_{O \in u^r \backslash f^{-1}(z)} w(y_O, r|O|), \tag{17.5}$$

where y_O is any representative of the orbit. The first identity in (17.4) makes the choice irrelevant.

Proposition 17.3.1. The datum $f_* w$ satisfies (17.4). For composable equivariant finite coverings, $(g \circ f)_* w = g_*(f_* w)$. If the covers have constant degree and preserve a distinguished fixed point 0, the normalized operations

$$N_f^\circ(w)(z, r) = \frac{(f_* w)(z, r)}{(f_* w)(0, r)}$$

also satisfy $N_g^\circ N_f^\circ = N_{g\circ f}^\circ$.

Proof. Applying u permutes the orbits in a fiber and preserves their weights. To verify the power law, an orbit of length s under u^r splits into $\gcd(s, t)$ orbits under u^{rt}. Each new weight is the old weight raised to $t/\gcd(s, t)$. Their product is its t-th power.

For transitivity, fix a point of period dividing r in the last space. An orbit in the middle fiber has length s. Above one of its representatives, an orbit under u^{rs} has some length t. It gives one orbit of length st in the composite fiber, and every composite orbit is obtained uniquely this way. Both products therefore have the same factors $w(x, rst)$.

Put $v = f_* w$. Before applying g_*, normalization divides the factor on an orbit of length s by $v(0, rs) = v(0, r)^s$. The total denominator is $v(0, r)^{\deg(g)}$, independent of the target point. It cancels when the result is normalized at zero. Ordinary transitivity then proves normalized transitivity. □

For the arithmetic system take $M = \overline{\mathbb{Q}}$ and

$$w_I(x, r) = \frac{E_{I,\varepsilon^r}(x)}{E_{I,\varepsilon^r}(0)}.$$

The period identities give (17.4). Proposition 3 of *Stark units for real quadratic fields and reciprocity laws* gives the product over the orbits in a covering fiber. Dividing that identity by its value at zero removes its root-of-unity multiplier and proves

$$N_f^0(w_I) = w_J.$$

The product of the orbits' zero-value normalization factors is $\varepsilon^{r \deg(f)/2}$, independent of their decomposition. Thus replacing raw values by w_I gives the same normalized law.

If $H \subset G_{I,\varepsilon}$ and $J/I = H$, the law reduces, for $x \in G_{I,\varepsilon}$, to

$$e_{J,\varepsilon}(\pi x) = \prod_{h \in H} \frac{e_{I,\varepsilon}(x + h)}{e_{I,\varepsilon}(h)}.$$

The denominator is the zero-point normalization. The image of G_I/H need not fill G_J. The other points are handled by the longer orbits in (17.5).

One can encode each weight as a one-dimensional M-space with its scalar automorphism. The products then become tensor products of lines, and normalization tensors with the dual reference line. This realizes the norm law on lines with automorphisms. It does not construct tensor functors between the fusion categories.

For example, take $J = (\varepsilon - 1)^{-1} I$. Then $G_I = J/I$, so its image in G_J consists only of zero. But $|G_J| = N$. If $N > 1$, every other fixed point of $\mathbb{T}_J$ has a fiber with no fixed lift. The longer orbits in (17.5) are necessary.

The arithmetic identity uses quotient coverings of lattices. It is not invariant under every equivariant map of tori. For the order-two example of (16.5), translation by the nonzero fixed point exchanges the values $1, e_1$. Its normalized norm sends e_1 to $e_1^{-1} \neq e_1$.

The algebra of normalized periodic parameters

These nonlinear operations become algebra maps on parameter spaces. Define the Laurent algebra

$$\mathcal{P}_I = K[z_{x,r}^{\pm 1} : r \geq 1, x \in \mathrm{Fix}(u^r)] \big/ (z_{ux,r} - z_{x,r}, z_{x,rs} - z_{x,r}^s, z_{0,r} - 1),$$

where the power relation is imposed only when $u^r x = x$. The pullback for f is the finite monomial formula

$$f^*(z_{z,r}) = \frac{\prod_{O \subset f^{-1}(z)} z_{y_O, r|O|}}{\prod_{O \subset f^{-1}(0)} z_{y_O, r|O|}}.$$

The orbit argument proves that it respects the relations and that these pullbacks compose. Applying Theorem 16.3.1 over K gives all-arity chiral algebra maps $\mathcal{A}_{\mathcal{P}_J} \to \mathcal{A}_{\mathcal{P}_I}$. Arithmetic evaluation to $\overline{K}$ commutes with them. For any finite collection of periods, Artin transport is compatible under the common coefficient extension and its stated coprimality conditions. A single finite ray class field is not asserted for all periods at once.

17.4 Changing the period

Write $G_r = (\varepsilon^r - 1)^{-1} I / I$. The quotient map $I/(\varepsilon^r - 1)I \to I/(\varepsilon - 1)I$ gives a canonical surjection

$$\pi_r : G_r \longrightarrow G_1, \qquad x \longmapsto (1 + \varepsilon + \cdots + \varepsilon^{r-1})x.$$

This is distinct from the inclusion $G_1 \subset G_r$ used in the arithmetic power identity.

Let R_{G_1} be the integral fusion ring with basis α_g, ρ and rules (15.16). Put

$$A_G = \sum_{g \in G_1} \alpha_g, \qquad W_r = (1 + \rho)^r - 1.$$

The element W_r is positive in the fusion basis, since it is a sum of positive tensor powers with nonnegative integer coefficients.

Theorem 17.4.1. There is a positive unital ring homomorphism

$$\Psi_r : R_{G_r} \longrightarrow R_{G_1}, \qquad \alpha_h \longmapsto \alpha_{\pi_r(h)}, \qquad \rho_r \longmapsto W_r.$$

These homomorphisms compose under successive changes of period.

Proof. In $R_{G_1} \otimes \mathbb{Q}$, the element $e = A_G/N$ satisfies $e^2 = e$, $e\rho = \rho$, and $(1 - e)\rho = 0$. In the component with unit e, set $z = e + \rho$. The fusion rule gives

$$z^2 - (N + 2)z + e = 0, \qquad z + z^{-1} = (N + 2)e.$$

The second equality uses the inverse $(N + 2)e - z$. The recurrence for $z^r + z^{-r}$, together with $N + 2 = \mathrm{Tr}(\varepsilon)$, yields $z^r + z^{-r} = \mathrm{Tr}(\varepsilon^r)e$.

Since $1 + \rho = (1 - e) + z$ has orthogonal components, $W_r = z^r - e$. Consequently

$$W_r^2 = N_r W_r + (N_r/N) A_G. \tag{17.6}$$

The integer N_r/N is the size of $\ker \pi_r$. Thus

$$\sum_{h \in G_r} \alpha_{\pi_r(h)} = (N_r/N) A_G,$$

and (17.6) is exactly the image of the source fusion relation. The relations involving invertibles follow from the group map and $\alpha_g W_r = W_r$.

The rational calculation proves an integral identity because the fusion basis is free over $\mathbb{Z}$.

For successive periods r, s, the label multipliers multiply to $1 + \varepsilon + \cdots + \varepsilon^{rs-1}$. On the other generator the composite is $(1 + W_r)^s - 1 = (1 + \rho)^{rs} - 1$. This proves composition on every basis element. □

Both roots of the dimension equation are respected:

$$W_r(\varepsilon - 1) = \varepsilon^r - 1, \qquad W_r(\varepsilon^{-1} - 1) = \varepsilon^{-r} - 1.$$

For $K = \mathbb{Q}(\sqrt{5})$, $I = \mathcal{O}_K$, and $\varepsilon = (3 + \sqrt{5})/2$, one has $N = 1$, $N_2 = 5$, and $G_2 \simeq \mathbb{Z}/5$. The map sends every invertible simple at period two to the unit and sends ρ_2 to $1 + 3\rho$. Indeed,

$$(1 + 3\rho)^2 = 10 + 15\rho = 5 + 5(1 + 3\rho).$$

A tensor lift requires maps on multiplicity spaces. Choose the fusion bases of (15.18) and let $\mathscr{F}$ be the additive functor specified by Υ_r. Its tensorator would consist of invertible maps

$$J_{X,Y} : \mathscr{F}(X) \otimes \mathscr{F}(Y) \longrightarrow \mathscr{F}(X \otimes Y)$$

satisfying

$$\begin{aligned} &\mathscr{F}(a_{X,Y,Z}) J_{X\otimes Y,Z}(J_{X,Y} \otimes 1) \\ &\qquad = J_{X,Y\otimes Z}(1 \otimes J_{Y,Z}) a_{\mathscr{F}(X),\mathscr{F}(Y),\mathscr{F}(Z)}. \end{aligned} \tag{17.7}$$

There are also the two unit equations. The finite fusion bases make these polynomial equations with invertibility conditions. The next obstruction uses the tensorator only on invertible objects and the unit channel.

17.5 Quotient maps and their multiplicities

Let $\mathcal{C}_N$ and $\mathcal{C}_M$ have the near-group rules above, with groups G, H of orders N, M, and let $\pi : G \twoheadrightarrow H$ have kernel of order ℓ, so $N = \ell M$. Suppose an additive strong tensor functor F sends α_g to $\alpha_{\pi(g)}$. Write τ for the noninvertible simple in the target and $S = \sum_{h\in H}[\alpha_h]$. Since $\alpha_g \rho = \rho$, the class of $F\rho$ is invariant under all translations of H. Thus

$$[F\rho] = aS + r[\tau], \qquad a, r \in \mathbb{Z}_{\geq 0}.$$

The two coefficients contain all possible multiplicities. No additional simple objects exist in the specified target.

THEOREM 17.5.1 *(Exact fusion-ring obstruction).* A positive unital ring homomorphism with the indicated map on invertible simples exists if and only if there are integers $a \geq 0$, $r > 0$ satisfying

$$r = \ell - 2a, \qquad (M + 4)a(\ell - a) = \ell(\ell - 1). \tag{17.8}$$

Consequently the existence of an additive strong tensor functor with that invertible-sector map requires (17.8). The equations do not alone construct its tensorator.

Proof. In the target $S^2 = MS$, $S\tau = M\tau$, and $\tau^2 = S + M\tau$. Hence

$$(aS + r\tau)^2 = (Ma^2 + r^2)S + (2arM + r^2 M)\tau.$$

The image of the source relation is

$$\ell S + N(aS + r\tau).$$

First $r \neq 0$. Otherwise the positive number aM would solve $d^2 = Nd + N$, whose positive root $(N + \sqrt{N^2 + 4N})/2$ is irrational: $N^2 + 4N$ lies strictly between $(N + 1)^2$ and $(N + 2)^2$ for $N \geq 1$.

Comparing the τ coefficients and dividing by rM gives $r = \ell - 2a$. Substitution in the S coefficient gives the second equation.

Conversely these computations verify every source relation, including the group actions, so the assignment defines the stated ring homomorphism. It is nonnegative and sends the unit to the unit. □

For a proper quotient, $a = 0$ is impossible. For $\ell = 2, 3, 4$, the equations and $r > 0$ have no solutions: at $\ell = 2$ there is no positive $a < \ell/2$. At $\ell = 3$ the only candidate is $a = 1$, which would force $M = -1$. At $\ell = 4$ it forces $M = 0$. In particular, there is no such tensor functor from the $N = 4$ near-group category to the $M = 2$ category along a quotient of their invertible sectors. This obstructs the additive tensor functor itself.

There are nontrivial passing ring tests. For $a = 1$, the equations give

$$M = \ell - 4, \qquad N = \ell(\ell - 4), \qquad r = \ell - 2 \qquad (\ell \geq 5).$$

Thus $N = 12$, $M = 2$, $\ell = 6$ admits the positive ring assignment $\rho \mapsto S + 4\tau$. This passing ring test does not assert existence of every numerical near-group category, and no tensor functor is inferred.

For two given categories, the remaining problem is a finite system of tensorator equations on their multiplicity spaces. Those equations use both associators. The finite receiver of Section 15.6 turns these into matrices, but it does not construct a solution.

For the fixed target $M = 2$, the criterion has a useful arithmetic form:

$$3r^2 = \ell(\ell + 2), \qquad (\ell + 1)^2 - 3r^2 = 1, \qquad a = (\ell - r)/2.$$

In particular, define integers x_j, r_j by $x_j + r_j\sqrt{3} = (2 + \sqrt{3})^j$ and put $\ell_j = x_j - 1$. Then

$$(N, M, \ell, a, r) = (2\ell_j, 2, \ell_j, (\ell_j - r_j)/2, r_j)$$

is a passing based-ring datum for every $j \geq 1$. The first proper cases are $(12, 2, 6, 1, 4)$ and $(50, 2, 25, 5, 15)$.

The recurrence $x_{j+1} = 2x_j + 3r_j$, $r_{j+1} = x_j + 2r_j$ preserves $x_j^2 - 3r_j^2 = 1$. Reduction modulo two gives $\ell_j \equiv r_j$, and $\ell_j \geq r_j$ for $j \geq 1$. These facts prove integrality and nonnegativity of a. These maps specialize the positive period maps of Theorem 17.4.1 to the quadratic unit $2 + \sqrt{3}$. They retain the Pell constraint on the multiplicities.

An additional quick obstruction follows from Frobenius–Perron dimensions. A tensor functor preserves this positive dimension character. Alternatively, this follows here directly by applying the target positive dimension to the source relations. Write $d_N = (N + \sqrt{N^2 + 4N})/2$ and define d_M similarly. They satisfy

$$d_N = aM + rd_M.$$

Unequal quadratic fields $\mathbb{Q}(\sqrt{N^2 + 4N})$ and $\mathbb{Q}(\sqrt{M^2 + 4M})$ prohibit such a map. Equality of these fields is necessary, not sufficient. Here the dimension condition follows directly from the two fusion relations, without a separate categorical reconstruction.

The unit channel also constrains a lift when its multiplicity exceeds one. We use the arithmetic endomorphism presentation of (16.2). Its source group is denoted by G, its noninvertible object by ρ, and its nondegenerate bicharacter by χ. The intertwiners satisfy $U_g U_h = U_{g+h}$ by (15.10). All functors below are linear over a common field F of characteristic zero. For a subgroup $H_0 \subset G$, write

$$H_0^{\perp} = \{g \in G : \chi(g, h) = 1 \text{ for every } h \in H_0\}.$$

PROPOSITION 17.5.2 *(The full unit multiplicity).* Let F be a strong tensor functor and let $H_0 \subset G$ be a subgroup whose invertible objects have image isomorphic to the unit. Suppose

$$a = \dim_F \mathrm{Hom}(1, \mathsf{F}\rho)$$

is finite and positive. Then

$$\chi(g,h)^a = 1 \quad (g,h \in H_0), \qquad aH_0 \subset H_0^{\perp}. \tag{17.9}$$

If H_0 has at most d generators, then

$$|H_0| \mid [G : H_0]a^d. \tag{17.10}$$

Proof. Strictify the target, choose the unit identifications and put $M = \mathsf{F}\rho$. Write $J_{X,Y} : \mathsf{F}X \otimes \mathsf{F}Y \to \mathsf{F}(XY)$ for the tensorator. For $g, h \in H_0$, define invertible maps

$$A_g = J_{\alpha_g,\rho}, \qquad B_h = J^{-1}_{\rho,\alpha_h}\mathsf{F}(U_h).$$

Coherence on $(\alpha_g, \rho, \alpha_h)$ gives

$$J_{\alpha_g,\rho\alpha_h} = J_{\rho,\alpha_h} A_g J^{-1}_{\rho,\alpha_h}.$$

Naturality for $U_h : \rho \to \rho\alpha_h$, together with $\alpha_g(U_h) = \chi(g,h)U_h$, then gives

$$A_g B_h = \chi(g,h) B_h A_g.$$

These maps act by postcomposition on the entire space $\mathrm{Hom}(1, M)$. Taking determinants gives (17.9).

Write $H_0[a]$ for the kernel of multiplication by a on H_0. Its order divides a^d, by the cyclic decomposition of H_0. Moreover aH_0 is a subgroup of $H_0^{\perp}$, whose order is $[G : H_0]$ by nondegeneracy. The identity $|H_0| = |H_0[a]|\,|aH_0|$ proves (17.10). □

In particular, $a = 1$ forces $H_0 \subset H_0^{\perp}$. For period doubling,

$$N_2 = N(N+4), \qquad |\ker \pi_2| = N + 4, \qquad W_2 = S + (N+2)\rho.$$

The unit multiplicity is one, so a tensor lift would require $(N+4)^2 \le N(N+4)$. Their difference is $4(N+4) > 0$. This proves the obstruction in the opening example. For the Fibonacci target, two nonzero scalars would have to satisfy a nontrivial fifth-root commutation relation.

For period maps the kernel has at most two generators. The resulting divisibility condition excludes every even ratio of periods.

THEOREM 17.5.3 *(Even ratios of periods).* Let I be a full rank-two lattice in a real quadratic field and let $\varepsilon > 1$ be a norm-one unit with $\varepsilon I = I$. Put

$$G_r = (\varepsilon^r - 1)^{-1} I/I, \qquad N_r = \varepsilon^r + \varepsilon^{-r} - 2, \qquad N = N_1.$$

Let $\pi_r : G_r \to G_1$ be multiplication by $(\varepsilon^r - 1)/(\varepsilon - 1)$. Assume that the arithmetic near-group categories for the periods in question are given with the nondegenerate bicharacters and endomorphism presentations above. If $r \ge 2$ is even, the positive period ring map of Theorem 17.4.1,

$$\alpha_g \longmapsto \alpha_{\pi_r(g)}, \qquad \rho_r \longmapsto W_r = (1+\rho)^r - 1,$$

has no linear strong tensor lift. The same conclusion holds for $G_{rs} \to G_s$ whenever the ratio r is even, and after any characteristic-zero extension of the coefficient field.

Proof. Put $t = N + 2$, $V_j = \varepsilon^j + \varepsilon^{-j}$, and define $u_{-1} = 0$, $u_0 = 1$ and $u_j = t u_{j-1} - u_{j-2}$ for $j \geq 1$. In the target fusion ring write $S = \sum_{g \in G_1} \alpha_g$. The relations $S^2 = NS$, $S\rho = N\rho$ and $\rho^2 = S + N\rho$ give

$$W_r = a_r S + u_{r-1}\rho, \qquad N a_r = u_{r-1} - u_{r-2} - 1. \tag{17.11}$$

For example, multiplication of W_r by $1 + \rho$, followed by addition of ρ, gives the formula for W_{r+1}. Thus all these multiplicities are integers. Evaluating in the two roots ε, ε^{-1} also gives

$$N a_r = \frac{(\varepsilon^r - 1)(\varepsilon^{r-1} - 1)}{\varepsilon^{r-1}(\varepsilon + 1)} > 0 \qquad (r \geq 2).$$

The kernel H_0 of π_r has order N_r/N and index N in G_r. It is a subgroup of $(\mathbb{Q}/\mathbb{Z})^2$, so it has at most two generators. A lift would have unit multiplicity a_r.

Proposition 17.5.2 would therefore make

$$\frac{N^2 a_r^2}{N_r} = \frac{V_{r-1} - 2}{N + 4} \tag{17.12}$$

an integer. The equality follows by inserting the preceding formula and $N_r = (\varepsilon^r - 1)^2/\varepsilon^r$.

But $V_0 = 2$, $V_1 = t$ and $V_j = tV_{j-1} - V_{j-2}$. Modulo $N + 4$, this gives $V_j \equiv 2(-1)^j$. For even r the numerator on the right of (17.12) is congruent to -4. Since $N + 4 \geq 5$, it is not divisible by $N + 4$.

Replacing ε by ε^s gives the assertion at base period s. The unit multiplicity and the root-of-unity commutator relations are unchanged by extension of coefficients. □

For an odd ratio $r = 2k + 1$, the same quotient is

$$\frac{V_{2k} - 2}{N + 4} = N u_{k-1}^2.$$

Thus every odd ratio passes this determinant test. In the Pell family above, $j = 2m$ gives

$$\ell_j = 6r_m^2, \qquad a_j = r_m(3r_m - x_m), \qquad \frac{2a_j^2}{\ell_j} = 4r_m^2 - 2x_m r_m + \frac{1}{3},$$

whereas $j = 2m + 1$ gives $\ell_j = (x_m + 3r_m)^2$, $a_j = r_m(x_m + 3r_m)$ and $2a_j^2/\ell_j = 2r_m^2$. For the standard Pell lattice, the first odd ratio has a further obstruction: coherence relates the two projective kernel actions.

Lemma 17.5.4 *(Reciprocal kernel cocycles).* In Proposition 17.5.2, suppose the target unit has endomorphism field F. Write $J_{X,Y} : \mathsf{F}X \otimes \mathsf{F}Y \to \mathsf{F}(XY)$ for the tensorator. Choose $\mathsf{F}\alpha_g = 1$ for $g \in H_0$ and write $c(g, h) \in F^\times$ for the tensorator on (α_g, α_h). Put

$$T_h = J_{\rho,\alpha_h}, \qquad A_g = J_{\alpha_g,\rho}, \qquad B_h = T_h^{-1}\mathsf{F}(U_h).$$

Then c is a scalar 2-cocycle and

$$\begin{aligned} A_g A_h &= c(g, h) A_{g+h}, \\ B_g B_h &= c(g, h)^{-1} B_{g+h}, \\ A_g B_h &= \chi(g, h) B_h A_g. \end{aligned}$$

These identities hold on $\mathrm{Hom}(1, \mathsf{F}\rho)$.

Proof. Coherence on three invertible objects gives the cocycle identity. Coherence on $(\alpha_g, \alpha_h, \rho)$ gives the first operator identity, since $\alpha_h \rho = \rho$. For the second, the source has

$$(U_g \otimes 1_{\alpha_h}) U_h = U_{g+h}.$$

Naturality and coherence give, respectively,

$$\mathsf{F}(U_g \otimes 1) T_h = J_{\rho\alpha_g, \alpha_h} \mathsf{F}(U_g), \qquad J_{\rho\alpha_g, \alpha_h} T_g = c(g,h) T_{g+h}.$$

Consequently $\mathsf{F}(U_{g+h}) = c(g,h) T_{g+h} B_g B_h$. Cancelling T_{g+h} proves the second identity. The third is the relation used in Proposition 17.5.2. Postcomposition restricts all three to the unit channel. □

PROPOSITION 17.5.5 *(The third period of the standard Pell lattice).* For $I = \mathbb{Z}[\sqrt{3}]$ and $\varepsilon = 2 + \sqrt{3}$, the positive period map $G_3 \to G_1$ has no linear strong tensor lift between the specified arithmetic categories. The conclusion holds after any characteristic-zero extension of coefficients.

Proof. Extend the field to an algebraic closure. Here $|G_3| = 50$ and $1 + \varepsilon + \varepsilon^2 = 5\varepsilon$. Thus the kernel is

$$H_0 = (\tfrac{1}{5} I)/I \simeq (\mathbb{Z}/5)^2,$$

with basis $\varepsilon/5, 1/5$. Formula (17.11) gives $W_3 = 5S + 15\rho$, so a lift would have a five-dimensional unit channel.

On this kernel the arithmetic Gaussian is given in Radchenko–Wheeler, Section 4.4, p. 20:

$$Q\big((k\varepsilon + l)/5\big) = \zeta^{k^2 - kl + l^2}, \qquad \zeta = e^{2\pi i/5}.$$

Hence $\chi(g,h) = \zeta^{g^{\mathrm{t}} K h}$ with $K = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ over $\mathbb{F}_5$.

The commutator of a scalar cocycle on H_0 is an alternating bicharacter. It therefore has the form $c(g,h)c(h,g)^{-1} = \zeta^{v \det(g,h)}$ for some $v \in \mathbb{F}_5$. By Lemma 17.5.4, the operators $A_g B_h$ give a projective representation of $H_0 \oplus H_0$ with alternating commutator matrix

$$\Omega_v = \begin{pmatrix} vJ & K \\ -K & -vJ \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Its Pfaffian is $-v^2 - 3$, which is nonzero in $\mathbb{F}_5$: $2 = -3$ is not a square. The commutator form is nondegenerate.

Choose a symplectic basis for this four-dimensional form. The corresponding operators form two independent Weyl pairs of order five. Rescale their scalar fifth powers to one. The two commuting clock operators then diagonalize, and the two shift operators carry each joint eigenspace through twenty-five distinct weights of equal dimension. Every nonzero representation consequently has dimension divisible by twenty-five. This contradicts the five-dimensional unit channel. □

Question 17.5.6. For the arithmetic categories attached to $\varepsilon = 2 + \sqrt{3}$, which odd periods $j \geq 3$ admit a linear strong tensor lift of the positive period map? Its noninvertible image is $a_j S + r_j \tau$, where $a_j = (\ell_j - r_j)/2$ and the integers ℓ_j, r_j are defined above. Construct invertible maps on the finite multiplicity bimodules satisfying the unit and associator equations, or exhibit an obstruction. For $I = \mathbb{Z}[\sqrt{3}]$, Proposition 17.5.5 makes $j = 5$ the first remaining case.

17.6 Multiplicative norms on coefficients

The preceding obstructions concern additive strong tensor functors between the specified fusion categories. Two multiplicative constructions remain available. The first acts on invertible modules. The second tensors an indexed family of algebras and bimodules.

Norms of invertible modules

Let R be a commutative ring and let E/R be finite étale of constant rank d. For an invertible E-module L define the invertible R-module

$$N_{E/R}(L) = \det_R(L) \otimes_R \det_R(E)^{-1}. \tag{17.13}$$

The source and target are the Picard groupoids of invertible modules and their isomorphisms. Their monoidal operation is tensor product.

Theorem 17.6.1 *(Tensor and tower coherence of the norm).* The construction (17.13) is strong symmetric monoidal:

$$N(L \otimes_E M) \simeq N(L) \otimes_R N(M), \qquad N(E) \simeq R.$$

For finite etale towers $R \to E \to F$, it has coherent comparisons $N_{F/R} \simeq N_{E/R} N_{F/E}$. All these maps commute with arbitrary base change on this finite etale domain.

Proof. After an etale splitting, $E = R^d$ and $L = (L_1, \dots, L_d)$. Formula (17.13) becomes the tensor product of the lines L_i. A permutation of sheets has the same determinant sign in $\det_R(L)$ and $\det_R(E)$, so the signs cancel. Thus the local tensor comparison is invariant under every sheet permutation.

For a tower, split all sheets at once. Either norm is the tensor product over the same final sheets, grouped first by their fibers over E or directly over R. Associativity and symmetry of tensor products identify these expressions and every longer grouping. They are invariant under changes of splitting and hence descend by faithful flat descent. The same sheetwise description proves base change and all tensor and tower coherences. □

On automorphisms of the unit line, this is $u \mapsto \det_R(m_u)$, the multiplicative norm of units. The determinant polynomial on all elements is not additive. For $E = R \times R$, it is $(a, b) \mapsto ab$. The two orthogonal idempotents have norm zero while their sum has norm one. An additive functor on all modules cannot induce this map on endomorphisms of the unit. A transfer on a Hall or fusion category therefore needs its own construction or a specified multiplicative subcategory.

For comparison, a finite scalar norm has an exact normalized tower identity. If $\pi : G \twoheadrightarrow H$ is a surjection of finite groups and u has values in a commutative group of units, set

$$N_\pi(u)(h) = \prod_{\pi(g)=h} u(g), \qquad \mathcal{N}_\pi(u)(h) = N_\pi(u)(h)/N_\pi(u)(1_H).$$

For $G \xrightarrow{\pi} H \xrightarrow{\rho} K$ one has

$$\mathcal{N}_\rho(\mathcal{N}_\pi(u)) = \mathcal{N}_{\rho\pi}(u). \tag{17.14}$$

Every fiber of ρ has cardinality $|\ker \rho|$. The intermediate denominator therefore occurs to the same power in every fiber product. Final normalization cancels it, proving (17.14). The group identity here is the reference point for normalization.

The indexed tensor norm on coefficients

An indexed tensor norm has a different target and is nonadditive. Fix a field k and a finite set S. For dg algebras B_s, E_s, D_s over k and composable dg bimodules ${}_{B_s}P_{sE_s}$ and ${}_{E_s}Q_{sD_s}$, define

$$N_S(B_s) = \bigotimes_{s\in S} B_s, \qquad N_S(P_s) = \bigotimes_{s\in S} P_s.$$

Fix an order temporarily to write formulas. The symmetric dg tensor comparison makes different orders canonically equivalent, with their Koszul signs. The left and right actions on the tensor of bimodules are the ordered tensor actions.

THEOREM 17.6.2 *(Composition of indexed norms).* There is a coherent derived comparison

$$\left(\bigotimes_s P_s\right) \otimes^{\mathbb{L}}_{\bigotimes_s E_s} \left(\bigotimes_s Q_s\right) \simeq \bigotimes_s (P_s \otimes^{\mathbb{L}}_{E_s} Q_s).$$

It preserves regular bimodules, arbitrary finite towers of indexing sets, and permutations of the sheets. It does not assert additivity in the repeated input P of $P \mapsto P^{\otimes S}$.

Proof. Use free bar resolutions or K-flat representatives. On homogeneous tensors the comparison is

$$(\otimes_s p_s) \otimes (\otimes_s q_s) \longmapsto (-1)^{\epsilon} \otimes_s (p_s \otimes q_s), \qquad \epsilon = \sum_{s<t} |q_s||p_t|.$$

Balancing on sheet s becomes precisely balancing over E_s after this permutation. The tensor differential commutes with a Koszul permutation. The same permutation applied to the common multibar compares the derived tensors. Its realizations agree because finite tensors over a field preserve the required colimits and because geometric realizations over multiple simplicial indices commute.

For three interfaces, the two routes make the same final permutation of all factors. Thus their signs agree. Regular bimodules give the unit comparison. Grouping a finite set into fibers and then grouping those fibers leaves the final tensor and its permutation unchanged, proving tower coherence. □

Over a characteristic-zero field, this construction descends along a finite étale algebra. Choose a splitting cover with sheet set S. The symmetric tensor action of sheet permutations gives descent data for the algebras and bimodules.

For invertible modules in degree zero, it recovers the Picard norm, with normalization $\det(f_*L) \otimes \det(f_*\mathcal{O})^{-1}$ for the finite étale map f. On arbitrary objects it is polynomial rather than additive. For example,

$$(P \oplus Q)^{\otimes S} \cong \bigoplus_{U\subseteq S} \left(\bigotimes_{s\in U} P\right) \otimes \left(\bigotimes_{s\notin U} Q\right),$$

with factors kept in sheet order. The mixed summands exhibit the failure of additivity.

Let $\mathscr{A}$ be a fixed singular background over k, as in Chapter 16. To obtain interfaces, apply the relative realization after taking this coefficient norm:

$$N_S(B_s) \longmapsto \mathscr{A} \otimes N_S(B_s), \qquad N_S(P_s) \longmapsto \mathscr{A} \otimes N_S(P_s).$$

Theorem 17.6.2 and Theorem 8.2.2 give its relative interface composition. The construction retains one copy of $\mathscr{A}$. An identification of $\mathscr{A}^{\otimes S}$ with $\mathscr{A}$ is not among its maps.

Sheet permutations act on the coefficient tensor. They supply no exchange of positive geometric word factors. The target is the relative envelope with the indexed coefficient algebra. Its comparison with another period's arithmetic fusion category is an additional construction.

17.7 Integral transport and residue fields

The integral presentation of Proposition 15.3.3 also transports. For a fixed period, put $\mathcal{S} = O_L[1/N]$. The field automorphism $\sigma_{\mathfrak{a}}$ preserves this ring, and all four generator formulas are defined over it.

Proposition 17.7.1. The map $\Theta_{\mathfrak{a}}$ restricts to a semilinear isomorphism of the integral Leavitt presentations over $\mathcal{S}$. It preserves their endomorphisms, displayed splittings, cups, caps and marked network equations. For a prime ideal $\mathfrak{q} \subset \mathcal{S}$, it induces an isomorphism between the reductions at $\mathfrak{q}$ and $\sigma_{\mathfrak{a}}(\mathfrak{q})$, over the induced residue-field isomorphism.

Proof. The map permutes the generator labels and conjugates their coefficients. Each normalized coefficient is a unit in $\mathcal{S}$, and its conjugate is the corresponding coefficient for $\mathfrak{a}^{-1}I$. The inverse label map and inverse field automorphism give the inverse integral presentation map.

All the stated equations are identities between finite expressions over $\mathcal{S}$. Their validity follows either from the defining relations or from the free normal form of Proposition 15.3.3. Reducing these expressions preserves every identity. The semilinear map carries the ideal $\mathfrak{q}B_I$ onto $\sigma_{\mathfrak{a}}(\mathfrak{q})B_{\mathfrak{a}^{-1}I}$, so it descends to the claimed isomorphism. □

This statement concerns the presentations and their displayed maps. Intertwiner spaces are kernels of linear equations. Their dimensions can increase after reduction, so semisimplicity and the full list of simple objects require a separate argument. The localization excludes primes dividing N. An integral model at those primes requires an additional construction.

A finite collection of periods can be treated over a common number field containing their coefficients and every chosen normalization. Inverting the product of their N_r gives a common ring for the displayed formulas. The Artin ideals must satisfy the coprimality conditions for each period. No single finite ray field is asserted for all periods simultaneously.

17.8 The remaining tensor comparison

The constructions have distinct domains. Reciprocity gives semilinear equivalences on a marked family. Orbit norms give compatible maps of parameter algebras. Picard norms and indexed tensor norms give multiplicative maps on their specified coefficient categories. The positive period ring map retains both dimension characters, while its tensorator must solve (17.7).

For the standard lattice $\mathbb{Z}[\sqrt{3}]$, every even period ratio and period three are excluded by the preceding proofs. At period five the proposed image is

$$W_5 = 76S + 209\rho, \qquad N_5 = 722, \qquad |\ker \pi_5| = 361.$$

Its determinant test passes: $N^2 76^2/722 = 32$. The reciprocal kernel relations and the full associator equation remain to be solved on these multiplicity spaces. For other lattices, the restricted arithmetic Gaussian must also be determined before applying the period-three argument.

Question 17.8.1. Construct a multiplicative correspondence between the marked arithmetic fusion systems whose scalar evaluation is the normalized orbit norm. Specify its coefficient category, supported field operations, unit and comparison with the indexed norm. Does it preserve the arithmetic junctions and their full collision diagrams?

The parameter-algebra maps already realize the scalar norm in every arity. The question asks for its comparison with the marked fusion objects and their fields.

17.9 References

- D. Radchenko and C. Wheeler. *Real quadratic fields and finite quantum dilogarithms. I*. arXiv:2609.21892v2, especially Section 4.4 and the fusion construction in Section 5.
- *Stark units for real quadratic fields and reciprocity laws*. 4 October 2026, the reciprocity and distribution paper, Theorem 2, Propositions 2–3 and Appendix A.

Chapter 18

Intrinsic geometric and automorphic comparison

A local Hecke polynomial determines two eigenvalues. It does not determine an automorphic representation, the extensions between representations, or the category on which the Hecke operators act. For example, the functor

$$D(k[t]) \longrightarrow D(k[t]) \times D(k[t]), \qquad M \longmapsto (M, 0),$$

is fully faithful and respects a faithful scalar action on the target. Its image misses every nonzero object in the second factor. This elementary example separates the scalar comparison from categorical reconstruction.

The arithmetic constructions of the preceding chapters provide operators, coefficient algebras, marked interfaces and their composition maps. An intrinsic realization must identify these data with independently defined geometric or automorphic objects. This chapter constructs the formal and local comparisons that enter that problem. It retains the deformation groupoids, central actions, extension classes and tensor transitions needed to pass between them.

The coefficients matter throughout. Smooth representations with characteristic-zero coefficients away from the residue characteristic have Bernstein blocks and Weil–Deligne parameters. Smooth p-power-torsion representations have a different spectral recipient, built from étale (φ, Γ)-modules. Global determinant deformations use one fixed coefficient ring at every place. Each comparison below specifies the regime in which its maps are defined.

18.1 Marked objects and categorical recognition

Let $\mathscr{D}$ be an ∞-category of arithmetic data. An object may contain a coefficient algebra, selected modules, Hecke or Galois actions, intertwiners, a distinguished state and a scalar evaluation on a specified domain. Morphisms preserve these markings. Let $\mathscr{R}$ be a category of proposed geometric realizations and let $U : \mathscr{R} \to \mathscr{D}$ retain the data. Write $\mathscr{D}^{\simeq}$ for its maximal ∞-groupoid, and similarly for $\mathscr{R}$.

Definition 18.1.1. For $d \in \mathscr{D}$, the space of marked realizations is

$$\mathrm{Real}_U(d) = \mathrm{hofib}_d\big(\mathscr{R}^{\simeq} \xrightarrow{U^{\simeq}} \mathscr{D}^{\simeq}\big).$$

A point consists of $r \in \mathscr{R}$ and an equivalence $U(r) \simeq d$, with their higher identifications.

Nonemptiness expresses existence. Contractibility expresses uniqueness with coherent identifications. A functorial realization requires a functor $S : \mathscr{D} \to \mathscr{R}$ and an equivalence $US \simeq \mathrm{id}_{\mathscr{D}}$. Objectwise choices do not define S on noninvertible morphisms. When the desired comparison is a map, its realizations instead retain that map in the appropriate arrow category.

This definition applies to several different arithmetic problems. The relative singular models retain a chosen geometric background. The marked fusion models also retain their associator, duality and distinguished arithmetic junction. A global automorphic realization must in addition preserve its ramified and

archimedean actions, coefficient topology and evaluation functional. A scalar identity obtained after evaluation supplies one consequence of such a map.

Coefficients and retained operations

A coefficient change must be a map in the stated coefficient category. For example,

$$(\mathbb{Z}/\ell^r) \otimes_{\mathbb{Z}} \mathbb{Q} = 0, \qquad \left(\varprojlim_r \mathbb{Z}/\ell^r\right) \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}_\ell.$$

Thus rationalization before completion destroys this coefficient tower. Likewise, over $\mathbb{F}_p$, the big Witt vector with only $a_p = 1$ is nonzero although all its ghosts $\sum_{d|n} d\, a_d^{n/d}$ vanish. The integral operations require their original coordinates.

Even an equivalence of underlying categories need not preserve a chosen central action. Give $D(k)$ the two $k[t]$-actions $t = 0$ and $t = 1$. Its identity equivalence cannot intertwine them: the proposed equality on the object k would be $0 = 1$. The action and its specified compatibility are therefore part of the realization datum.

Comparison on generators

THEOREM 18.1.2 *(Comparison on compact generators).* Let $\mathcal{C}, \mathcal{D}$ be presentable stable ∞-categories, and let $F : \mathcal{C} \to \mathcal{D}$ be exact and colimit-preserving. Let $\{P_i\}_{i \in I}$ be a set of compact generators of $\mathcal{C}$. Here generation means that their smallest full stable subcategory closed under all small colimits is $\mathcal{C}$. Assume each $F(P_i)$ is compact and that the maps of mapping spectra

$$\mathrm{Map}_{\mathcal{C}}(P_i, P_j) \longrightarrow \mathrm{Map}_{\mathcal{D}}(F(P_i), F(P_j))$$

are equivalences for all i, j. Then F is fully faithful. It is an equivalence if the objects $F(P_i)$ also generate $\mathcal{D}$.

Proof. Fix i. Compactness and exactness imply that both mapping-spectrum functors in the comparison, with second variable X, preserve coproducts and finite colimits. They therefore preserve all colimits between stable categories. The objects X where the comparison is an equivalence form a localizing subcategory of $\mathcal{C}$. It contains every P_j, so it is all of $\mathcal{C}$.

Fix X and vary the first argument. Mapping spectra convert colimits in this argument into limits. Hence the objects Y for which

$$\mathrm{Map}_{\mathcal{C}}(Y, X) \longrightarrow \mathrm{Map}_{\mathcal{D}}(F(Y), F(X))$$

is an equivalence form a localizing subcategory containing all P_i. This proves full faithfulness. Its essential image is closed under colimits and finite cones, since diagrams in that image lift through a fully faithful functor. If the image contains generators of $\mathcal{D}$, it is all of $\mathcal{D}$. □

For an E-linear differential graded enhancement, the hypothesis is a quasi-isomorphism on each derived Hom complex induced by the chosen functor. An equality of dimensions or a match in degree one is weaker. Use the augmentation modules $k = k[t]/(t)$ and $k = (k[\epsilon]/(\epsilon^2))/(\epsilon)$. Then k has $\mathrm{Ext}^1(k,k) = k$ over both $k[t]$ and $k[\epsilon]/(\epsilon^2)$. Over $k[t]$, its length-one free resolution makes all higher Ext groups zero. Over the dual numbers, the periodic resolution with differential multiplication by ϵ gives $\mathrm{Ext}^n(k,k) = k$ for every $n \geq 0$.

The local categorical Langlands question requires the actual source subcategory, spectral support condition, comparison functor, and objects P_i. Frobenius parameters and invariant vectors supply inputs for this question. A categorical equivalence follows only after the derived Hom and generation hypotheses have been established on those categories.

18.2 Determinants and local parameter spaces

A representation gives characteristic polynomials after every scalar extension. A determinant polynomial law retains these compatible polynomials even when no free representation has been chosen. Its definition works in characteristic two, where a trace alone does not contain every characteristic coefficient.

Definition 18.2.1 *(Chenevier determinant and associated trace).* Let R be a commutative ring, Γ a group and $d \geq 1$. An R-valued determinant of dimension d is a multiplicative, homogeneous polynomial law of degree d from $R[\Gamma]$ to R. Thus it specifies maps $D_A : A[\Gamma] \to A$ for every commutative R-algebra A, natural in A, with $D_A(1) = 1$, $D_A(xy) = D_A(x)D_A(y)$ and $D_A(ax) = a^d D_A(x)$. Write

$$D_{R[t]}(t - x) = \sum_{j=0}^{d} (-1)^j \Lambda_j(x) t^{d-j}.$$

The associated trace is the R-linear map $T = \Lambda_1$. The notation (T, D) retains this dependence. The characteristic polynomial gives the Cayley–Hamilton relations in the quotient by their coefficients after polynomial base change. These relations need not hold in $R[\Gamma]$ itself. For a profinite Γ and a topological coefficient ring, continuity means that all characteristic-polynomial coefficient functions on Γ are continuous.

Proposition 18.2.2 *(determinantal deformation rings).* Let k be a finite field of characteristic p_0, and put $W = W(k)$. Fix a continuous residual determinant $\bar{D}$ of dimension d of a profinite group Γ. On finite local W-algebras A with residue field k, let $\mathcal{D}_\Gamma(A)$ be the set of continuous determinants lifting $\bar{D}$. This covariant functor is prorepresented by a profinite local W-algebra R_Γ^{Ch}:

$$\mathcal{D}_\Gamma(A) = \mathrm{Hom}_{W,\mathrm{cont}}(R_\Gamma^{\mathrm{Ch}}, A).$$

Assume that every open subgroup has only finitely many continuous homomorphisms to $\mathbb{Z}/p_0\mathbb{Z}$.

This is Mazur's finiteness condition, and it makes the ring complete Noetherian. A closed deformation condition represented by a quotient is imposed by that quotient. The representing formal space is $\mathrm{Spf}\, R_\Gamma^{\mathrm{Ch}}$. For $\Gamma = G_{\mathbb{Q},S}$ with finite S, and for local Galois groups in characteristic zero, the finiteness condition holds. If the residual representation is absolutely irreducible and defined over k, determinant deformations agree with its representation deformations up to strict equivalence.

Attribution. Chenevier, arXiv:0809.0415, §1.1 and §1.10 define polynomial laws and their characteristic coefficients. Proposition 3.3, p. 42, proves prorepresentability. Example 3.6 and Proposition 3.7, p. 43, give the finiteness assertions. Example 3.4 and Theorem 2.22 give the absolutely irreducible comparison. Bellaïche–Chenevier, *Families of Galois representations and Selmer groups*, Chapter 1, provides the related deformation-theoretic setting. A chosen local condition must be shown to define the quotient used in any particular application. □

The Frobenius parameter stack

Let E be a field of characteristic zero and let $d \in E^\times$. For an E-algebra A, consider a rank-two projective A-module V with an automorphism F satisfying $\wedge^2 F = d\ \mathrm{id}_{\wedge^2 V}$. Isomorphisms intertwine F. No trivialization of $\wedge^2 V$ is part of the data. This defines the quotient stack

$$\mathcal{U}_d = [Y_d / GL_2], \qquad Y_d = \operatorname{Spec} E[x_{11}, x_{12}, x_{21}, x_{22}]/(x_{11}x_{22} - x_{12}x_{21} - d),$$

where GL_2 acts by conjugation. It is the algebraic parameter stack for an unramified rank-two Weil representation with chosen generator acting by F.

Put $t = \operatorname{tr} F$. The regular semisimple open substack $\mathcal{U}_d^{\mathrm{rs}}$ is defined by $t^2 - 4d \neq 0$. The ordered eigenvalue scheme and diagonal torus are

$$T_d = \operatorname{Spec} E[a, a^{-1}, (a^2 - d)^{-1}], \qquad T = \mathbb{G}_m^2.$$

The involution w sends a to d/a and exchanges the factors of T.

PROPOSITION 18.2.3 *(Eigenvalues and eigenline automorphisms).* There is an equivalence

$$\mathcal{U}_d^{\mathrm{rs}} \simeq [T_d / (T \rtimes \langle w \rangle)],$$

where T acts trivially on T_d. The scheme of unordered eigenvalues is

$$U_d = \operatorname{Spec} E[t, (t^2 - 4d)^{-1}].$$

It has dimension one. Over an algebraically closed field, each geometric object of $\mathcal{U}_d^{\mathrm{rs}}$ has automorphism group $\mathbb{G}_m^2$.

Proof. After a finite étale base change, the polynomial $X^2 - tX + d$ has distinct roots a, b, with $b = d/a$. The endomorphisms

$$e_a = \frac{F - b}{a - b}, \qquad e_b = \frac{F - a}{b - a}$$

are complementary idempotents. Their images are rank-one projective modules, and F acts on them by a and b. Conversely, two line bundles with these scalar actions give (V, F).

Changing their frames gives the T-action. Exchanging their order gives the involution. Descent gives the stated quotient stack.

The map $a \mapsto t = a + d/a$ identifies the unordered roots. Its coordinate extension satisfies $a^2 - ta + d = 0$. Inverting $t^2 - 4d$ makes this a finite étale double cover with the stated involution. The invariant ring is $E[t, (t^2 - 4d)^{-1}]$. Each automorphism preserves the two eigenspaces and scales them independently. □

The atlas Y_d has dimension three, since the determinant equation has nonzero differential on invertible matrices. Thus $\mathcal{U}_d$ has dimension $3 - 4 = -1$ as an algebraic stack. Its eigenvalue scheme has a different dimension and omits stabilizers. If a trivialization of $\wedge^2 V$ is required, the quotient uses SL_2 instead. Its regular semisimple stabilizer is $\mathbb{G}_m$. These are distinct moduli conventions.

An unramified deformation ring

Let O be a complete discrete valuation ring of mixed characteristic, with finite residue field k of characteristic p. Fix a nonarchimedean local field L. Write G_L for its absolute Galois group and I_L for inertia, so $G_L/I_L \simeq \widehat{\mathbb{Z}}$. Fix $\overline{F} \in GL_2(k)$ and $d \in O^\times$ lifting $\det \overline{F}$.

A test algebra is a local Artinian O-algebra A with residue field k. An unramified framed deformation sends arithmetic Frobenius to $F_A \in GL_2(A)$, with $F_A \bmod \mathfrak{m}_A = \overline{F}$ and $\det F_A = d$. Here arithmetic Frobenius acts on a residue field by its cardinality power. Inertia acts trivially.

Proposition 18.2.4 *(Framed deformations and distinct residual roots).* Choose a matrix lift $F_0 \in M_2(O)$ of $\overline{F}$. The framed deformation functor is represented by

$$R_d^{\square} = O[[X_{11}, X_{12}, X_{21}, X_{22}]]/(\det(F_0 + X) - d).$$

This ring is formally smooth of relative dimension three. Suppose additionally that $\overline{F} = \operatorname{diag}(\bar{a}, \bar{b})$, with $\bar{a} \neq \bar{b}$.

Choose $a_0 \in O^\times$ lifting $\bar{a}$ and put $b_0 = d/a_0$. Then

$$R_d^{\square} \simeq O[[x, y, z]], \qquad F = \begin{pmatrix} a_0 + x & y \\ z & (d + yz)/(a_0 + x) \end{pmatrix}.$$

Strict conjugacy classes of these deformations are represented by $O[[u]]$. The corresponding ordered eigenvalues are $a = a_0 + u$ and $b = d/a$.

Proof. Every test algebra is finite. A matrix in its finite group $GL_2(A)$ defines a unique continuous homomorphism from $\widehat{\mathbb{Z}}$. Thus specifying F_A is equivalent to specifying the unramified representation. The matrix entries and determinant equation give the first ring.

At least one cofactor of $\overline{F}$ is nonzero. The derivative in the corresponding matrix coordinate is a unit. Solving for that coordinate proves formal smoothness and leaves three parameters. In the displayed diagonal case, $a_0 + x$ is a unit and the determinant equation solves directly for the lower-right entry.

For each F_A, Hensel lifting gives the unique characteristic root a reducing to $\bar{a}$. The other root is $b = d/a$. Since $a - b$ is a unit, the projectors above split A^2 into two free rank-one eigenspaces. Their bases can be chosen to reduce to the standard residual bases.

The resulting change of basis is the identity modulo $\mathfrak{m}_A$, so it is a strict conjugation. Two diagonal lifts are strictly conjugate exactly when their ordered roots agree. The choice of $a - a_0 \in \mathfrak{m}_A$ therefore represents the set of strict conjugacy classes. □

The last ring represents classes, not the deformation groupoid. For a diagonal lift, strict automorphisms contain $(1 + \mathfrak{m}_A)^2$. A quotient stack retains these automorphisms. The construction concerns unramified lifts only. A crystalline lifting condition or a larger Emerton–Gee component requires its own map and coefficient conventions.

18.3 Ordinary quotients and characteristic-zero types

Fix $p \geq 5$. Let $E/\mathbb{Q}_p$ be finite, with integer ring O, uniformizer ϖ, and finite residue field k. Put $G = G_{\mathbb{Q}_p}$ and let $I \subset G$ be inertia. Fix a continuous representation $\bar{\rho} : G \to \mathrm{GL}_2(k)$, a basis of k^2, and a continuous character $\delta : G \to O^\times$ lifting $\det \bar{\rho}$.

Give an Artinian local O-algebra A with residue field k the discrete topology. A framed lift is a continuous homomorphism $\rho_A : G \to \mathrm{GL}_2(A)$ reducing to the specified matrix representation $\bar{\rho}$.

Framed lifts are not divided by conjugation. The framed deformation ring $R^{\square}$ is a complete Noetherian local O-algebra with universal representation $\rho^{\square}$.

Its existence is the framed representability statement in (Kisin, *The structure of potentially semi-stable deformation rings*, §1.2). Set

$$R = R^{\square,\delta} = R^{\square}/\overline{(\det \rho^{\square}(g) - \delta(g) : g \in G)}.$$

The bar denotes closure in the maximal-ideal topology. This quotient represents framed lifts of determinant δ. Write its universal matrices as

$$\rho_R(g) = \begin{pmatrix} a_g & b_g \\ c_g & d_g \end{pmatrix}.$$

Assume a chosen residual quotient $\bar{q} : k^2 \twoheadrightarrow k(\bar{\eta})$ is unramified. Here $k(\bar{\eta})$ is the one-dimensional representation with character $\bar{\eta}$, and unramified means $\bar{\eta}|_I = 1$. Choose the basis so that $\bar{q} = (0, 1)$. Then

$$\bar{\rho}(g) = \begin{pmatrix} \bar{\alpha}(g) & * \\ 0 & \bar{\eta}(g) \end{pmatrix}, \qquad \bar{\alpha} = \bar{\delta}\bar{\eta}^{-1}.$$

An ordinary lift relative to $\bar{q}$ is a framed lift together with a free rank-one quotient lifting $\bar{q}$, on which inertia acts trivially. Quotients are identified up to an equivariant target isomorphism. Each such quotient has a unique normalized row $q = (t, 1)$, where $t \in \mathfrak{m}_A$. Its character is allowed to vary with A.

PROPOSITION 18.3.1 *(explicit ordinary deformation equations).* Suppose $\bar{\alpha} \neq \bar{\eta}$. Choose $g_0 \in G$ with $\bar{\alpha}(g_0) \neq \bar{\eta}(g_0)$. There is a unique $t_0 \in \mathfrak{m}_R$ satisfying

$$b_{g_0} t_0^2 + (d_{g_0} - a_{g_0}) t_0 - c_{g_0} = 0. \tag{18.1}$$

Let $J_{\mathrm{ord}} \subset R$ be the closed ideal generated by

$$\begin{aligned} b_g t_0^2 + (d_g - a_g) t_0 - c_g \quad & (g \in G), \\ t_0 b_h + d_h - 1 \quad & (h \in I). \end{aligned}$$

Then $R^{\mathrm{ord}} = R/J_{\mathrm{ord}}$ represents ordinary lifts relative to $\bar{q}$. The quotient line is uniquely determined by the framed representation. Thus forgetting the quotient gives a closed immersion $\operatorname{Spf} R^{\mathrm{ord}} \hookrightarrow \operatorname{Spf} R$.

Proof. The polynomial in (18.1) vanishes at zero modulo $\mathfrak{m}_R$. Its derivative there is $\bar{\eta}(g_0) - \bar{\alpha}(g_0) \neq 0$. Completeness and Hensel's lemma give the unique root t_0. For completeness, successive correction modulo powers of $\mathfrak{m}_R$ solves a linear equation with this invertible coefficient.

These corrections converge in R. If two roots lie in $\mathfrak{m}_R$, their difference times $b_{g_0}(t + t') + d_{g_0} - a_{g_0}$ is zero. The second factor is a unit, so the roots agree.

For a row $q = (t, 1)$, stability of its kernel under $\rho_A(g)$ is equivalent to

$$q\rho_A(g) = \eta_A(g)q, \qquad \eta_A(g) = tb_g + d_g,$$

where the matrix entries are specialized to A. The first component is exactly $b_g t^2 + (d_g - a_g)t - c_g = 0$. Since $\eta_A(g)$ reduces to $\bar{\eta}(g) \neq 0$, it is a unit. Surjectivity of q and multiplication of the matrices show that η_A is a character. It is continuous because ρ_A is continuous. The equations indexed by I make it unramified.

Every ordinary quotient satisfies the equation for g_0. The uniqueness argument over the Artinian local ring A forces t to be the image of t_0. The remaining equations are precisely the images of J_{ord}. Conversely, a map $R^{\mathrm{ord}} \to A$ gives the quotient $(t_0, 1)$ with these properties.

Both constructions commute with local O-algebra maps of Artinian coefficient rings. The construction also applies to complete Noetherian local coefficient rings by passage to their Artinian quotients. The normalized rows are compatible by uniqueness. □

The kernel of this quotient is freely generated by $(1, -t_0)$. Consequently every ordinary lift has an exact sequence

$$0 \longrightarrow A(\delta\eta_A^{-1}) \longrightarrow A^2_{\rho_A} \longrightarrow A(\eta_A) \longrightarrow 0, \qquad \eta_A|_I = 1. \tag{18.2}$$

Indeed the action on the kernel is multiplication by $a_g - b_g t_0$, and the determinant is the product of the two characters. The sequence need not split. To prescribe an unramified lift $\eta : G \to O^\times$, add the equations $t_0 b_g + d_g = \eta(g)$ for $g \in G$.

Remark 18.3.2 *(coincident residual characters).* Without $\bar\alpha \neq \bar\eta$, the quotient must be retained as part of the deformation problem. That problem is still represented by

$$R[[t]]/\overline{\big(b_g t^2 + (d_g - a_g)t - c_g \ (g \in G), \quad t b_h + d_h - 1 \ (h \in I)\big)}.$$

Here t maps into the maximal ideal of each coefficient ring. The same row calculation proves the assertion.

For the trivial residual representation and trivial determinant, take the constant trivial lift to $k[u]/(u^2)$. The rows $(au, 1)$ for $a \in k$ give distinct ordinary quotient lines lifting $(0, 1)$. Thus the forgetful map need not be injective on deformation functors when the residual characters coincide.

Hodge types on characteristic-zero lifts

The ordinary equations make sense over Artinian coefficient rings. Hodge and inertial types require characteristic-zero period modules. For a continuous representation V on a finite-dimensional vector space over a finite extension E'/E, put

$$D_{\mathrm{dR}}(V) = (B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V)^G.$$

Here B_{dR} is Fontaine's filtered de Rham period field.

The representation is de Rham when $\dim_{E'} D_{\mathrm{dR}}(V) = \dim_{E'} V$. We use filtration degrees as Hodge–Tate weights, so the cyclotomic character ε has weight -1. A type $\mathbf{h} = \{h_1, h_2\}$ prescribes the dimensions of $\mathrm{gr}^j D_{\mathrm{dR}}(V)$. Equivalently, Kisin's cocharacter is the conjugacy class of $z \mapsto \mathrm{diag}(z^{-h_1}, z^{-h_2})$.

Let $B_{\mathrm{cris}} \subset B_{\mathrm{st}}$ be Fontaine's crystalline and semistable period rings. For a finite extension $K'/\mathbb{Q}_p$ with maximal unramified subfield K'_0, set

$$D_{\mathrm{st},K'}(V) = (B_{\mathrm{st}} \otimes_{\mathbb{Q}_p} V)^{G_{K'}}, \qquad D_{\mathrm{cris},K'}(V) = (B_{\mathrm{cris}} \otimes_{\mathbb{Q}_p} V)^{G_{K'}}.$$

The representation is potentially semistable, respectively potentially crystalline, when for some K' the corresponding module is free of rank $\dim_{E'} V$ over $K'_0 \otimes_{\mathbb{Q}_p} E'$. Its potentially semistable period module is

$$D_{\mathrm{pst}}(V) = \varinjlim_{K'} D_{\mathrm{st},K'}(V).$$

Inertia acts linearly on this module over $\overline{\mathbb{Q}}_p^{\mathrm{nr}} \otimes_{\mathbb{Q}_p} E'$.

An inertial type is a continuous representation $\tau : I \to \mathrm{GL}_2(E)$ with open kernel which extends to the Weil group. The representation V has this type when its inertia representation on $D_{\mathrm{pst}}(V)$ is isomorphic to τ after scalar extension. Here $\overline{\mathbb{Q}}_p^{\mathrm{nr}}$ is the maximal unramified extension of $\mathbb{Q}_p$ inside its chosen algebraic closure. This is the period-module action, not the inertia action on V.

Choose E large enough to define τ and the Hodge type. Kisin constructs quotients $R_{\mathrm{pst}}^{\square,\mathbf{h},\tau}$ and $R_{\mathrm{pcris}}^{\square,\mathbf{h},\tau}$ of $R^{\square}$. Their characteristic-zero quotients impose the potentially semistable and potentially crystalline type conditions on finite local characteristic-zero coefficient algebras. In particular, their field-valued points specify lifts of these types. These are Kisin's Theorems 1.2.1–1.2.2, pp. 6–7.

For $\star \in \{\mathrm{pst}, \mathrm{pcris}\}$ define the actual fixed-determinant intersection ring and its generic fiber by

$$S_\star = R_\star^{\square,\mathbf{h},\tau} \otimes_{R^{\square}} R, \qquad X_\star = \operatorname{Spec} S_\star[1/p].$$

Both factors are quotient rings, so this tensor product is an ordinary quotient of $R^{\square}$. It is already complete and Noetherian. A continuous map $S_\star \to \mathcal{O}_{E'}$ gives a framed lattice of determinant δ whose rational representation has the indicated type.

Conversely, choose a frame of such a lattice reducing to the given residual frame. Kisin's factorization criterion applies over E'. The defining ideal then vanishes in the torsion-free ring $\mathcal{O}_{E'}$, giving the required integral map. Here coefficient extensions carry a fixed residue-field embedding, and all integral maps are continuous local $\mathcal{O}$-algebra maps.

The scheme $X_\star$ also retains the infinitesimal structure of the intersection. No flatness or reducedness of $S_\star$ is needed for this construction. Incompatible determinant and type conditions can give an empty generic fiber.

The schematic closure of the generic fiber in $\operatorname{Spec} S_\star$ is defined by the p-power torsion ideal. Its coordinate ring is $S_\star/S_\star[p^\infty]$, where $S_\star[p^\infty] = \{x : p^n x = 0 \text{ for some } n\}$. This quotient is p-torsion-free, hence flat over the discrete valuation ring $\mathcal{O}$. It has no effect after inverting p, but can change the special fiber. The integral intersection and its generic fiber must therefore remain distinct.

Under the distinguished quotient hypothesis, define

$$X_\star^{\mathrm{ord}} = \operatorname{Spec}\big((S_\star/J_{\mathrm{ord}}S_\star)[1/p]\big), \qquad X_\star^{\mathrm{rest}} = X_\star \setminus X_\star^{\mathrm{ord}}.$$

The first is a closed subscheme and the second its open complement. This gives a two-piece decomposition within the specified lifting space. The ordinary condition is relative to the chosen residual quotient $\bar{q}$.

COROLLARY 18.3.3 *(ordinary lifts of a prescribed type).* A continuous point $S_\star \to \mathcal{O}_{E'}$ lies in the ordinary piece precisely when its framed lattice has an unramified quotient lifting $\bar{q}$. On this piece its rational representation has an exact sequence

$$0 \longrightarrow E'(\delta\eta^{-1}) \longrightarrow V \longrightarrow E'(\eta) \longrightarrow 0, \qquad \eta|_I = 1,$$

and has the prescribed Hodge and inertial types.

Proof. Factoring through R^{ord} is equivalent to the ordinary quotient property by the proposition. Apply (18.2) and invert p. The quotient $R_\star^{\square,\mathbf{h},\tau}$ imposes the type. If the equations vanish only after inverting p, they already vanish in the torsion-free ring $\mathcal{O}_{E'}$. □

For the potentially Barsotti–Tate problem in the Tate-module convention, take $\star = \mathrm{pcris}$ and $\mathbf{h} = \{-1, 0\}$, with a specified inertial type. The dual convention has weights $\{0, 1\}$. Other Hodge types require other quotient rings. These spaces do not give an exhaustive decomposition of all residual points.

They describe lifts above a fixed residual representation. For example, with $a = p - 1$, the representations $1 \oplus \varepsilon$ and $\varepsilon^a \oplus \varepsilon^{1-a}$ have the same reduction and determinant. Their Hodge–Tate multisets are $\{0, -1\}$ and $\{1 - p, p - 2\}$, respectively.

18.4 A local categorical realization

The formal lifting spaces above have natural maps into a spectral stack at the coefficient prime. The determinant identification belongs to this stack. It is more structure than the determinant value used for the Frobenius quotient stack.

To compare these deformation spaces with categorical local Langlands, fix a continuous central character $\zeta : \mathbb{Q}_p^\times \to O^\times$. Use local reciprocity sending a uniformizer to geometric Frobenius, and require the already fixed determinant to satisfy $\delta = \zeta \varepsilon^{-1}$ as a Galois character. These are the conventions of (Dotto–Emerton–Gee, arXiv:2603.26887, §1.3, p. 8).

Let $\mathcal{A}_\zeta$ be smooth $\mathrm{GL}_2(\mathbb{Q}_p)$ representations on p-power torsion O-modules with central character ζ. Let $\mathcal{X}_\delta$ classify rank-two projective étale (φ, Γ)-modules with a determinant identification. This identification is an isomorphism with the rank-one module attached to δ. Morphisms preserve this isomorphism.

This is the fiber-product definition in (Dotto–Emerton–Gee, arXiv:2603.26887, Definition C.1.1, p. 265). The Dotto–Emerton–Gee functor is

$$F : D(\mathcal{A}_\zeta) \longrightarrow \mathrm{Ind}\, D^b_{\mathrm{coh}}(\mathcal{X}_\delta).$$

It is exact, colimit-preserving and fully faithful (Dotto–Emerton–Gee, arXiv:2603.26887, Theorem 1.1.1 and its following paragraph, p. 3).

The functor has an actual kernel. Write $O[[G]]_\zeta$ for the completed group algebra with central character ζ, where $G = GL_2(\mathbb{Q}_p)$. Dotto–Emerton–Gee construct a pro-coherent right $O[[G]]_\zeta$-module $\mathcal{L}_\infty$ on $\mathcal{X}_\delta$, and on the bounded finitely presented source their functor is

$$F(\pi) = \mathcal{L}_\infty \otimes^{\mathbb{L}}_{O[[G]]_\zeta} \pi.$$

The right action has a specified variance.

The initial Colmez construction has left central character $\zeta \varepsilon^{-2}$. Twisting by $(\zeta^{-1}\varepsilon) \circ \det$ changes this to ζ^{-1}. Precomposition with $g \mapsto g^{-1}$ then gives the right ζ-action.

On a smooth formal chart $\mathrm{Spf}\, A$, the kernel comes from the compatible systems attached to the étale modules over A/I^n. Their coefficient base change and descent construct the global pro-coherent object. Bounded coherence of the displayed tensor and full faithfulness are separate theorems of that construction. Their locators are Definitions 5.1.31–5.1.33, Proposition 5.3.23 and Theorem 5.5.1. A symbol for the tensor product alone would prove neither assertion.

Let $\bar\theta$ be the residual pseudorepresentation of $\bar\rho$. Complete $\mathcal{X}_\delta$ along the fiber over $\bar\theta$. This classifies Galois representations deforming that pseudorepresentation, with the specified determinant identification (Dotto–Emerton–Gee, §1.1.6, p. 4). The universal representation defines a morphism

$$\mathrm{Spf}\, R \longrightarrow \widehat{\mathcal{X}}_{\delta,\bar\theta} \longrightarrow \mathcal{X}_\delta$$

as follows.

Reduce ρ_R modulo each power of its maximal ideal. Over each finite coefficient ring, apply the equivalence between continuous Galois representations and étale (φ, Γ)-modules. Retain the determinant

identification sending the wedge of the two frame vectors to 1 in the character module of δ. The construction preserves tensor products and base change. See Dotto–Emerton–Gee, Appendix C.2, especially (C.2.1)–(C.2.2), p. 268, and Remark C.2.17, p. 272.

The compatible objects define the displayed formal morphism. Their residual pseudorepresentation is $\bar{\theta}$, so the map factors through the indicated completion. A change of basis preserving the determinant identification gives an isomorphism in that stack. Scalar automorphisms ± 1 remain.

The ordinary and type quotients give morphisms into this diagram. For the type quotients these are the maps from $\mathrm{Spf}\, S_\star$. The ordinary intersection uses $\mathrm{Spf}(S_\star/J_{\mathrm{ord}}S_\star)$. These are morphisms of formal moduli problems, whereas $X_\star$ denotes the separately defined scheme after inverting p. They do not identify the source formal schemes with the stack, which retains automorphisms.

An ordinary or potentially Barsotti–Tate comparison must specify full stable source and target subcategories. Essential surjectivity first requires a proof that F restricts between them. The deformation maps alone do not specify these categories or their generators. The functor on the entire target is not an equivalence (Dotto–Emerton–Gee, arXiv:2603.26887, Remark 1.1.2, p. 3). This statement does not decide equivalence on a specified smaller target.

PROPOSITION 18.4.1 *(generation criterion for a restricted functor).* Let $\mathcal{C}, \mathcal{D}$ be presentable stable ∞-categories. Suppose an exact colimit-preserving fully faithful functor $F_0 : \mathcal{C} \to \mathcal{D}$ is supplied. If objects $P_i \in \mathcal{C}$ have images generating $\mathcal{D}$ under colimits, shifts and cofibers, then F_0 is an equivalence.

Proof. Full faithfulness makes the essential image a full subcategory equivalent to $\mathcal{C}$. Every diagram in that image can therefore be lifted to $\mathcal{C}$. Preservation of colimits makes its colimit in $\mathcal{D}$ another image object. Exactness gives closure under shifts and cofibers. The essential image contains the specified generators and hence all of $\mathcal{D}$. □

The local arithmetic construction needed for this criterion is the identification of $F(P_i)$ with actual spectral objects, together with a proof of target generation. If lifting conditions overlap, their union must generate all required extension objects. A direct-sum assembly additionally requires orthogonality. Neither modularity of a lift nor a ring of scalar actions establishes these categorical properties.

Line and quotient conventions. The ordinary lifting problem just constructed retains an unramified *quotient*. Its kernel has character $\delta\eta^{-1}$ and need not be unramified. An ordinary-line condition instead asks for an unramified subrepresentation.

Dualizing a quotient of V gives an unramified subrepresentation of $V^\vee$, with determinant δ^{-1}. It does not identify the two conditions on the same V. Any use of the line convention in the comparison below must specify this dualization or another proved flag comparison, including the central-character and Hodge-sign changes.

Smooth objects with the required central character

Suppose the local residual rank-two representation is unramified at p, and use the determinant convention $\det \bar{\rho} = \chi\bar{\varepsilon}^{-1}$. Under geometric reciprocity, $\bar{\varepsilon}(u) = \bar{u}$ for $u \in \mathbb{Z}_p^\times$ and $\bar{\varepsilon}(p) = 1$. Consequently $\chi(u) = \bar{u}$. An induction from two unramified characters has a different central character on units. The following induction gives actual objects in the required torsion category.

PROPOSITION 18.4.2 *(central-character-compatible smooth source objects).* Fix a finite extension of $\mathbb{Q}_p$ with integer ring O and residue field k. Choose a continuous lift $\zeta : \mathbb{Q}_p^\times \to O^\times$ of χ. Under geometric reciprocity, set $\delta = \zeta\varepsilon^{-1}$, where ε is the p-adic cyclotomic character. Let A be a finite local

Artinian O-algebra with residue field k. Write ζ_A for the induced character, and choose an unramified character $\eta_A : \mathbb{Q}_p^\times \to A^\times$.

Put $G = GL_2(\mathbb{Q}_p)$ and let B be its upper triangular subgroup. The unnormalized smooth induction

$$I_A = \mathrm{Ind}_B^G(\zeta_A \eta_A^{-1} \otimes \eta_A)$$

consists of locally constant $f : G \to A$ satisfying

$$f\left(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} g\right) = \zeta_A(a)\eta_A(a)^{-1}\eta_A(d) f(g), \qquad (g_0 f)(g) = f(g g_0).$$

It is a nonzero smooth p-power-torsion O-representation with central character ζ. It has no $GL_2(\mathbb{Z}_p)$-fixed vectors. In particular, no square-root modulus factor is used over A.

Proof. The finite coefficient characters have open kernels. Choose a principal congruence subgroup $K_m \subset GL_2(\mathbb{Z}_p)$ so that the inducing character θ_A of B is trivial on $B \cap K_m$. Define $f(bk) = \theta_A(b)$ on BK_m and zero outside. Two factorizations differ by $B \cap K_m$, so this is well-defined. The image of K_m in the Hausdorff space $B\backslash G$ is compact and open, hence closed as well.

The resulting zero extension is locally constant and $f(1) = 1$. Restrict a locally constant induced function to the compact group $GL_2(\mathbb{Z}_p)$. A finite clopen partition refines its local constancy neighborhoods. It is therefore fixed on the right by some sufficiently small open subgroup. This proves smoothness.

The ring A is killed by a power of p, so the same is true of I_A. A central scalar x acts by $\zeta_A(x)\eta_A(x)^{-1}\eta_A(x) = \zeta_A(x)$. Choose a unit u whose reduction is not 1. The element $\zeta_A(u) - 1$ is a unit of A. If a vector were fixed by $GL_2(\mathbb{Z}_p)$, it would be annihilated by this unit, hence vanish. □

A determinant twist cannot turn an unramified principal series into this residual central-character family when p is odd. On the cyclic group $\mathbb{F}_p^\times$, such a twist changes the central exponent by $2a$. The required exponent is 1. The congruence $2a = 1 \pmod{p-1}$ has no solution. This remains an obstruction after coefficient extension and, by reduction, for the finite rings A. The characteristic-zero unramified invariant-vector calculation has a different central-character carrier.

Let C_{ind} be the stable colimit closure of these induced objects in $D(\mathcal{A}_\zeta)$. It is a well-defined source category. To identify it with an ordinary comparison category one must identify its spectral image and generators, including the chosen ordinary flag. An invariant line and an invariant quotient are distinct conditions.

18.5 Quadratic induction and ordinary lifting spaces

A quadratic global induction can become reducible at a nonsplit local place. The local character determines this behavior. Ordinary lifting conditions require further data: a residual quotient line, a determinant, and a characteristic-zero lifting space. These data specify the local objects whose images under a categorical Langlands functor must be compared.

Restriction of a quadratic induction

Let $K/\mathbb{Q}$ be quadratic. Fix a place v of $\mathbb{Q}$ and compatible decomposition groups. Let L be a coefficient field, and let $\psi : G_K \to L^\times$ be continuous. Here G_F denotes the absolute Galois group of a field F, and L is equipped with its specified field topology. All representation spaces carry the corresponding finite-dimensional topology. Induction from an open subgroup of finite index means ordinary algebraic induction, with its induced continuous action.

If v splits in K, the double-coset decomposition gives

$$\operatorname{Ind}_{G_K}^{G_{\mathbb{Q}}} \psi\Big|_{G_{\mathbb{Q}_v}} \simeq \psi_w \oplus \psi_{w'},$$

where w, w' are the two places above v. The two local characters can coincide. If v is nonsplit, there is one place w above v, and the same decomposition gives

$$\operatorname{Ind}_{G_K}^{G_{\mathbb{Q}}} \psi\Big|_{G_{\mathbb{Q}_v}} \simeq \operatorname{Ind}_{G_{K_w}}^{G_{\mathbb{Q}_v}} \psi_w.$$

These formulas follow by separating functions according to their support on the double cosets $G_K \backslash G_{\mathbb{Q}} / G_{\mathbb{Q}_v}$. There are two cosets in the split case and one in the nonsplit case.

To determine the second representation, let H be an open normal subgroup of index two in a profinite group G. Choose $s \in G \setminus H$. For a continuous character $\chi : H \to L^\times$, put

$$\chi^s(h) = \chi(s^{-1}hs), \qquad c = \chi(s^2).$$

On the induced basis $e_1 = 1 \otimes 1$, $e_2 = s \otimes 1$, the action is

$$\rho(h) = \begin{pmatrix} \chi(h) & 0 \\ 0 & \chi^s(h) \end{pmatrix}, \qquad \rho(s) = \begin{pmatrix} 0 & c \\ 1 & 0 \end{pmatrix}. \tag{18.3}$$

PROPOSITION 18.5.1 *(local irreducibility criterion).* If $\chi \neq \chi^s$, then $\operatorname{Ind}_H^G \chi$ is absolutely irreducible. If $\chi = \chi^s$, its reducibility over L is equivalent to the existence of a root of $T^2 - c$ in L.

Suppose $\chi = \chi^s$ and $b^2 = c$ with $b \in L$. There is a continuous extension $\widetilde{\chi} : G \to L^\times$ with $\widetilde{\chi}|_H = \chi$ and $\widetilde{\chi}(s) = b$. If $\operatorname{char} L \neq 2$, then

$$\operatorname{Ind}_H^G \chi \simeq \widetilde{\chi} \oplus (\widetilde{\chi}\, \omega_{G/H}),$$

where $\omega_{G/H}$ is trivial on H and sends s to -1. In characteristic two the induced representation is a nonsplit self-extension of $\widetilde{\chi}$.

Proof. When the characters differ, choose $h \in H$ with $\chi(h) \neq \chi^s(h)$. Any invariant line over an algebraic closure must be one of the two eigenspaces of $\rho(h)$. The matrix $\rho(s)$ interchanges them, so neither is G-stable.

When the characters agree, H acts by scalars. A G-stable line therefore exists precisely when $\rho(s)$ has an eigenvalue in L. Its characteristic and minimal polynomials are both $T^2 - c$. The prescribed values of $\widetilde{\chi}$ respect conjugation by s and the relation $s^2 \in H$, hence define a character.

It is continuous because H is open and χ is continuous. In characteristic different from two the two roots $b, -b$ are distinct, which gives the displayed direct sum. In characteristic two the minimal polynomial is $(T - b)^2$. The matrix is not scalar, so its unique eigenline has no invariant complement. □

Thus over an algebraically closed field, local irreducibility in the nonsplit case is equivalent to $\psi_w \neq \psi_w^s$. Inertness and ramification alone do not decide this equality. Over a smaller coefficient field, an invariant character can instead give an irreducible representation which becomes reducible after scalar extension. Over coefficient rings the matrices in (18.3) still define induction. The field criterion applies after specialization and does not assert an integral direct-sum decomposition.

For a ramified quadratic extension of $\mathbb{Q}_p$ with $p \geq 5$, the ramification index is two, so the extension is tame. A characteristic-zero inducing character can still be nontrivial on wild inertia. For a continuous character valued in $\overline{\mathbb{F}}_p^{\times}$ with the discrete topology, its image is finite of order prime to p. Its restriction to wild inertia is therefore trivial. Since the quadratic extension is tame, wild inertia lies in the inducing subgroup and acts trivially on the induced residual representation.

Example 18.5.2 *(an inert place with reducible restriction).* Let L_0 be the splitting field of $X^3 - 2$. Eisenstein's criterion at 2 and the discriminant -108 give $\mathrm{Gal}(L_0/\mathbb{Q}) = S_3$. The quadratic subfield is $K = \mathbb{Q}(\sqrt{-3})$. Over $\overline{\mathbb{F}}_5$, induction of a nontrivial character of A_3 gives an absolutely irreducible representation of S_3. Complex conjugation is a transposition, so this representation is odd.

The prime 5 is unramified in L_0, and

$$X^3 - 2 = (X - 3)(X^2 + 3X + 4) \pmod 5.$$

The quadratic discriminant is 3, which is nonsquare modulo 5. Frobenius is consequently a transposition. Also $-3 = 2 \pmod 5$ is nonsquare, so 5 is inert in K. The local representation factors through the order-two decomposition group and splits into its $+1$ and -1 Frobenius eigenspaces.

Lemma 18.5.3 *(restriction of a dihedral residual representation).* For a quadratic induction, the split restriction is the sum of the two local characters, which can coincide. At a nonsplit place the restriction is induced from the local quadratic subgroup. Over an algebraically closed coefficient field, it is irreducible exactly when its two conjugate characters differ. Ramification of the quadratic extension alone specifies no Hodge type. At $p \geq 5$ that quadratic extension is tame, and a discrete mod-p character is trivial on wild inertia.

Proof. The double-coset decomposition, matrix computation and characteristic-two case in Proposition 18.5.1 prove these statements. The inert prime 5 example shows why inertness alone cannot imply irreducibility. A comparison with the Colmez Montreal functor, including its actual block and coefficient category, remains additional arithmetic data. It is not a consequence of the restriction formula. □

18.6 Residual parameters and deformation objects

The complex parameters of residual automorphic representations and the mod-p parameters of a deformation stack have different coefficient fields. Their comparison requires more than the same two-dimensional notation. Even over $\mathbb{C}$, the two SL_2 factors in a local Arthur parameter have different roles.

Definition 18.6.1 *(The non-tempered parameter).* At a finite place, use the domain $W_F \times SL_2(\mathbb{C})_{\mathrm{D}} \times SL_2(\mathbb{C})_{\mathrm{A}}$. The parameter takes values in $GL_2(\mathbb{C})$. Here W_F is the Weil group, the factor marked D is the Deligne factor, and the factor marked A is the additional Arthur factor. The trivial-character rank-two non-tempered parameter is

$$\psi_1(w, g_{\mathrm{D}}, g_{\mathrm{A}}) = g_{\mathrm{A}}. \tag{18.4}$$

The associated Langlands parameter is obtained by substituting $\mathrm{diag}(|w|^{1/2}, |w|^{-1/2})$ in the Arthur factor. This convention records its non-tempered exponents.

Definition 18.6.2 *(The monodromy parameter).* On the same domain, the standard rank-two Deligne parameter, regarded as an Arthur parameter trivial on its additional factor, is

$$\psi_0(w, g_{\mathrm{D}}, g_{\mathrm{A}}) = g_{\mathrm{D}}. \tag{18.5}$$

The distinction from (18.4) lies in which factor acts. Restricting both displays to an unnamed SL_2 would erase that distinction. A determinant twist changes the Weil part and does not exchange the two factors.

Remark 18.6.3 *(Local and global classification).* Arthur's parameter formalism separates the Deligne and Arthur factors. His endoscopic classification concerns orthogonal and symplectic groups, with GL_n classification as input. The residual classification for GL_n is the theorem of Moœglin–Waldspurger. For global data one can specify a formal cuspidal datum $\mu \boxtimes \nu_2$, where ν_2 is the standard two-dimensional representation.

This notation does not construct a global Langlands group or a map to a residual-characteristic stack. Packet assertions must specify twists, central character and local-global realization. In particular, (18.5) does not construct an automorphic representation Steinberg at every finite place.

The direct sum of the trivial and mod-p cyclotomic characters admits several independent cyclotomic extension directions. Their dimensions, and the automorphisms of their deformations, depend on the prime. These data determine what a comparison with a distinguished extension class must preserve. The constructions below give the continuous local deformation groupoids, explicit Kummer classes, and a formal family integrating those classes.

First-order deformations and their automorphisms

Fix a separable closure K^{sep} of $K = \mathbb{Q}_p$, and put $\Gamma = \mathrm{Gal}(K^{\mathrm{sep}}/K)$, with its profinite topology. Let $k/\mathbb{F}_p$ be a finite field with the discrete topology. Choose a primitive pth root ζ_p, with $\zeta_2 = -1$, and define

$$g(\zeta_p) = \zeta_p^{\omega(g)}, \qquad \omega : \Gamma \longrightarrow \mathbb{F}_p^{\times} \longrightarrow k^{\times}.$$

Write $k(r)$ for the one-dimensional k-representation with character ω^r.

In particular, $\omega = 1$ when $p = 2$. Fix the ordered basis e_1, e_2 of $V = k^2$, and set

$$\rho_0(g) = \begin{pmatrix} 1 & 0 \\ 0 & \omega(g) \end{pmatrix}, \qquad \mathfrak{a} = \mathrm{End}_k(V), \qquad \mathfrak{a}_0 = \ker(\mathrm{tr} : \mathfrak{a} \to k).$$

Write E_{ij} for the matrix sending e_j to e_i and the other basis vector to zero. The action on $\mathfrak{a}$ is $gX = \rho_0(g) X \rho_0(g)^{-1}$. All cochains and representations are continuous.

For a finite discrete Γ-module N, use the inhomogeneous cochains $C^n(\Gamma, N) = \mathrm{Map}_{\mathrm{cts}}(\Gamma^n, N)$, with $C^0(\Gamma, N) = N$. Their differential is

$$\begin{aligned}(df)(g_1, \dots, g_{n+1}) = {} & g_1 f(g_2, \dots, g_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ & + (-1)^{n+1} f(g_1, \dots, g_n).\end{aligned}$$

Thus $dX(g) = gX - X$ and $dc(g,h) = gc(h) - c(gh) + c(g)$. Write $Z^1(\Gamma, N) = \ker(d : C^1 \to C^2)$, $B^1(\Gamma, N) = dN$, and $H^i(\Gamma, N) = H^i(C^\bullet(\Gamma, N))$.

Set $A = k[\epsilon]/(\epsilon^2)$, with the discrete topology. A framed deformation is a continuous representation $\rho : \Gamma \to \mathrm{GL}_2(A)$ reducing to ρ_0 in the specified basis. Morphisms preserve this entire A-basis. A residual-marked unframed deformation is a free rank-two A-module with a continuous Γ-action and a specified identification of its reduction with V. Its morphisms induce the identity on V.

Proposition 18.6.4 *(Continuous first-order groupoids).* Framed deformations form the discrete groupoid on $Z^1(\Gamma, \mathfrak{a})$. Residual-marked unframed deformations form the action groupoid

$$[Z^1(\Gamma, \mathfrak{a})/\mathfrak{a}], \qquad X : c \longmapsto c - dX.$$

Its isomorphism classes form $H^1(\Gamma, \mathfrak{a})$. The automorphism group of each object is the additive group $H^0(\Gamma, \mathfrak{a})$.

Proof. Every framed lift has a unique expression

$$\rho_c(g) = (1 + \epsilon c(g))\rho_0(g).$$

Multiplication gives

$$\rho_c(g)\rho_c(h) = \big(1 + \epsilon(c(g) + gc(h))\big)\rho_0(gh).$$

The homomorphism condition is therefore $dc = 0$. A basis lifting e_1, e_2 identifies any residual-marked deformation with such a matrix representation.

Two such choices differ by $1 + \epsilon X$, and

$$(1 + \epsilon X)\rho_c(g)(1 - \epsilon X) = \rho_{c-dX}(g).$$

The quotient and automorphism statements follow from this equality. An automorphism fixes c exactly when $dX = 0$. □

If the residual marking is forgotten, the residual centralizer also acts. Write

$$C = \{B \in \mathrm{GL}_2(k) : B\rho_0(g) = \rho_0(g)B \text{ for all } g\}.$$

Every permissible change of basis is $(1 + \epsilon X)B$, and its action is

$$c \longmapsto BcB^{-1} - dX.$$

Thus the acting group is $\mathfrak{a} \rtimes C$.

For odd p, the character ω is nontrivial and C is the group of invertible diagonal matrices. For $p = 2$, it is $\mathrm{GL}_2(k)$. The stabilizer of c consists of the pairs satisfying $BcB^{-1} - dX = c$. At $c = 0$, this is $H^0(\Gamma, \mathfrak{a}) \rtimes C$.

A formal scheme, viewed as a groupoid-valued functor, has only identity automorphisms at its points. It cannot represent these unframed groupoids, which retain nonzero infinitesimal automorphisms. This assertion concerns the groupoids, rather than representability of their sets of isomorphism classes.

Determinant character and determinant identification

Fixing a determinant character restricts the objects of a deformation problem. Choosing an identification of determinant representations also restricts its morphisms. To separate these conditions, fix

$$\delta_\eta(g) = \omega(g)(1 + \epsilon\eta(g)), \qquad \eta \in Z^1(\Gamma, k).$$

Here the action on k is trivial, so η is a continuous additive homomorphism.

For a character $\delta : \Gamma \to A^\times$, write $A(\delta)$ for the free rank-one module with that action. The constant determinant lift has $\eta = 0$. The identity

$$\det \rho_c(g) = \omega(g)(1 + \epsilon \operatorname{tr} c(g))$$

holds in every characteristic.

PROPOSITION 18.6.5 *(The two fixed-determinant groupoids).* For the constant lift $\delta_0 = \omega$, the residual-marked deformation groupoid with fixed determinant character is

$$\mathcal{D}_{\mathrm{char}}(A) = [Z^1(\Gamma, \mathfrak{a}_0)/\mathfrak{a}].$$

Fix the residual orientation $e_1 \wedge e_2$. If objects include an identification $\lambda : \det(V_A) \xrightarrow{\sim} A(\omega)$ lifting that orientation, and arrows preserve λ, the groupoid is

$$\mathcal{D}_{\det}(A) = [Z^1(\Gamma, \mathfrak{a}_0)/\mathfrak{a}_0].$$

Both groupoids have isomorphism classes $H^1(\Gamma, \mathfrak{a}_0)$. Their automorphism groups are respectively $H^0(\Gamma, \mathfrak{a})$ and $H^0(\Gamma, \mathfrak{a}_0)$.

Proof. The fixed-character equation is $\operatorname{tr} c = 0$. Every strict conjugation preserves the determinant, and $\operatorname{tr}(dX) = 0$. This proves the first groupoid formula. For the second, choose a lifted basis with $\lambda(e_1 \wedge e_2) = 1$. Such a basis exists because

$$\det : 1 + \epsilon\mathfrak{a} \longrightarrow 1 + \epsilon k$$

is surjective: $\operatorname{diag}(1 + \epsilon t, 1)$ has determinant $1 + \epsilon t$. Changes between these bases satisfy $\operatorname{tr} X = 0$, which proves the second formula.

The trace admits the Γ-equivariant splitting

$$k \longrightarrow \mathfrak{a}, \qquad t \longmapsto tE_{11}.$$

Indeed, E_{11} is invariant and has trace one. Hence $\mathfrak{a} = \mathfrak{a}_0 \oplus kE_{11}$ and $d\mathfrak{a} = d\mathfrak{a}_0$, including when $p = 2$. The two orbit sets are therefore $H^1(\Gamma, \mathfrak{a}_0)$. The equations $dX = 0$, with the respective trace restrictions, give the stated automorphism groups. □

For a general η, the cocycle $c_\eta(g) = \eta(g)E_{11}$ has trace η. The object spaces are therefore the affine translates $c_\eta + Z^1(\Gamma, \mathfrak{a}_0)$, with the same gauge formulas. A determinant-compatible framing requires $\lambda(e_1 \wedge e_2) = 1$, and gives the discrete groupoid on $Z^1(\Gamma, \mathfrak{a}_0)$ when $\eta = 0$. If a full framing and an independent determinant identification are both retained, the discrete object set is instead $Z^1(\Gamma, \mathfrak{a}_0) \times k$: the second coordinate records $\lambda(e_1 \wedge e_2) = 1 + \epsilon t$. For a chosen determinant identification, forgetting the residual marking allows residual changes in $C \cap \mathrm{SL}_2(k)$.

The coefficient decomposition is

$$\mathfrak{a} = kE_{11} \oplus kE_{22} \oplus k(-1)E_{12} \oplus k(1)E_{21},$$
$$\mathfrak{a}_0 = k(E_{11} - E_{22}) \oplus k(-1)E_{12} \oplus k(1)E_{21}.$$

At $p = 2$, the diagonal trace-zero generator is I, and every summand has trivial action. At $p = 3$, the two off-diagonal characters coincide, but their matrix subspaces remain distinct.

The first-order tangent groupoid is presented by the two-term complex

$$[\mathfrak{a} \xrightarrow{-d} Z^1(\Gamma, \mathfrak{a})] \quad \text{in degrees } [-1, 0].$$

Its degree -1 cohomology records automorphisms and its degree 0 cohomology records deformation classes. This is the truncation through degree zero of $C^\bullet(\Gamma, \mathfrak{a})[1]$, with the cohomological shift convention $(C[1])^i = C^{i+1}$ and shifted differential $-d$. Thus $H^2(\Gamma, \mathfrak{a})$ occurs in degree one of the full shifted complex.

For a determinant identification, replace $\mathfrak{a}$ by $\mathfrak{a}_0$. For the fixed-character problem alone, the two-term complex is $[\mathfrak{a} \xrightarrow{-d} Z^1(\Gamma, \mathfrak{a}_0)]$. Identifying any full cochain complex with the tangent complex of a specified derived stack requires an equivalence of the underlying moduli problems.

Kummer classes at every prime

The lower-left summand has an explicit arithmetic description. For $a \in K^\times$, choose $\beta_a \in K^{\mathrm{sep}}$ with $\beta_a^p = a$, and define

$$\frac{g(\beta_a)}{\beta_a} = \zeta_p^{\kappa_a(g)}, \qquad \kappa_a(g) \in \mathbb{F}_p \subset k.$$

The action on roots of unity gives

$$\kappa_a(gh) = \kappa_a(g) + \omega(g)\kappa_a(h).$$

This cocycle is continuous, since β_a and ζ_p lie in a finite Galois extension. Replacing β_a by $\zeta_p^r \beta_a$ changes it by the coboundary $dr(g) = \omega(g)r - r$.

Proposition 18.6.6 *(Explicit cyclotomic cohomology).* The Kummer map gives an isomorphism

$$(\mathbb{Q}_p^\times / \mathbb{Q}_p^{\times p}) \otimes_{\mathbb{F}_p} k \xrightarrow{\sim} H^1(\Gamma, k(1)), \qquad [a] \otimes 1 \longmapsto [\kappa_a].$$

For odd p, a basis is $[\kappa_p]$, $[\kappa_{1+p}]$. For $p = 2$, a basis is $[\kappa_2]$, $[\kappa_{-1}]$, $[\kappa_5]$. The dimensions are respectively two and three.

Proof. Multiplication of roots makes the Kummer map additive modulo pth powers. If $\kappa_a = dr$, replacing β_a by $\zeta_p^{-r}\beta_a$ makes it invariant. Hence a is a pth power in K. Conversely, Hilbert 90 implies that every continuous μ_p-valued cocycle is $g(\beta)/\beta$ for some $\beta \in (K^{\mathrm{sep}})^\times$. Then $\beta^p \in K^\times$. This proves the isomorphism over $\mathbb{F}_p$.

The required Hilbert 90 statement reduces to a finite Galois extension L/K through which the cocycle factors. If z_σ is a multiplicative cocycle for $\mathrm{Gal}(L/K)$, choose $a \in L$ so that

$$b = \sum_\sigma z_\sigma \sigma(a) \neq 0.$$

Such an a exists by linear independence of distinct field automorphisms. The cocycle identity gives $\tau(b) = z_\tau^{-1} b$.

Consequently $z_\tau = \tau(b^{-1})/b^{-1}$, as required. This is the proof of Milne, *Class Field Theory*, Proposition II.1.22, p. 67. A basis of the finite extension $k/\mathbb{F}_p$ decomposes each continuous cochain into finitely many coordinate cochains. Tensoring with k therefore commutes with cohomology.

For odd p, the multiplicative decomposition is

$$\mathbb{Q}_p^\times = p^{\mathbb{Z}} \times \mu_{p-1} \times (1 + p\mathbb{Z}_p).$$

The valuation supplies the first factor. Hensel lifting of the distinct nonzero residues supplies the roots of unity in the second factor. The pth-power map on μ_{p-1} is bijective.

The logarithm and exponential give inverse homomorphisms

$$\log : 1 + p\mathbb{Z}_p \xrightarrow{\sim} p\mathbb{Z}_p.$$

For $x, y \in p\mathbb{Z}_p$, the estimates

$$v_p(x^n/n) \geq n - v_p(n), \qquad v_p(y^n/n!) \geq n - v_p(n!)$$

prove convergence of the logarithm and exponential, using $v_p(n!) \leq n/(p-1)$. Their formal inverse and product identities pass to these convergent series. Logarithm carries pth powers to $p^2\mathbb{Z}_p$, and $\log(1+p) \equiv p \pmod{p^2}$. Thus valuation and the principal-unit quotient give the basis $[p]$, $[1+p]$.

At $p = 2$, each odd unit has a unique sign making it congruent to one modulo four. Hence

$$\mathbb{Q}_2^\times = 2^{\mathbb{Z}} \times \{1, -1\} \times (1 + 4\mathbb{Z}_2).$$

The preceding convergence estimates, with input valuation at least two, give $\log(1 + 4\mathbb{Z}_2) = 4\mathbb{Z}_2$. Squaring becomes multiplication by two, and $\log(5) \equiv 4 \pmod{8}$. The valuation, sign, and principal-unit factors give the independent classes $[2]$, $[-1]$, $[5]$. □

In particular, $p = 3$ gives the basis $[\kappa_3]$, $[\kappa_4]$. The choice of ζ_p identifies μ_p with $\mathbb{F}_p(1)$. Changing it rescales all numerical cocycles by one nonzero scalar. The intrinsic classes take values in $\mu_p \otimes_{\mathbb{F}_p} k$.

The unit classes form an arithmetic subspace, and valuation gives an exact sequence

$$0 \longrightarrow (\mathbb{Z}_p^\times/(\mathbb{Z}_p^\times)^p) \otimes_{\mathbb{F}_p} k \longrightarrow H^1(\Gamma, k(1)) \xrightarrow{v_p} k \longrightarrow 0.$$

The unit subspace has dimension one for odd p and two for $p = 2$. The specified uniformizer p supplies the splitting $1 \mapsto [\kappa_p]$. The quotient itself does not specify a splitting.

There is also an unramified condition. If $I \subset \Gamma$ is inertia, define its subspace as

$$H^1_{\mathrm{ur}}(\Gamma, k(1)) = \operatorname{im}\bigl(H^1(\Gamma/I, k(1)^I) \to H^1(\Gamma, k(1))\bigr).$$

For odd p, this is zero: $\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$ is totally ramified of degree $p-1$, since $((1+T)^p - 1)/T$ is Eisenstein at p. Thus $k(1)^I = 0$.

At $p = 2$, the unramified subspace is the line $k[\kappa_5]$. Indeed, $(1 + \sqrt{5})/2$ satisfies $x^2 - x - 1 = 0$, whose reduction is separable and irreducible over $\mathbb{F}_2$. It therefore generates the unramified quadratic extension. Since $\Gamma/I \cong \widehat{\mathbb{Z}}$, its continuous additive characters into k form a one-dimensional k-space.

An explicit formal family

Each cyclotomic cocycle defines both an extension of residual representations and a tangent direction at their split sum. For $b \in Z^1(\Gamma, k(1))$, set

$$r_b(g) = \begin{pmatrix} 1 & 0 \\ b(g) & \omega(g) \end{pmatrix}.$$

The line ke_2 is invariant, and the quotient is trivial. Thus r_b gives an exact sequence

$$0 \longrightarrow k(1) \longrightarrow E_b \longrightarrow k \longrightarrow 0.$$

Conversely, a vector-space splitting of any such extension gives this matrix form. Changing the splitting changes b by a coboundary. Consequently

$$\operatorname{Ext}^1_{\mathrm{Rep}_k^{\mathrm{cts}}(\Gamma)}(k, k(1)) \cong H^1(\Gamma, k(1)),$$

where the category consists of finite-dimensional discrete continuous k-representations. The order of the two endpoint characters is part of this identification.

The coefficient inclusion $u \mapsto uE_{21}$ and projection onto the lower-left entry commute with the cochain differential. They induce an injection with a specified left inverse

$$j : H^1(\Gamma, k(1)) \hookrightarrow H^1(\Gamma, \mathfrak{a}_0).$$

The class $j([b])$ is represented by $\rho_{\epsilon b}(g) = \left(\begin{smallmatrix} 1 & 0 \\ \epsilon b(g) & \omega(g) \end{smallmatrix}\right)$.

Theorem 18.6.7 *(Formal cyclotomic family).* Set $d = 2$ and $(a_1, a_2) = (p, 1 + p)$ for odd p. For $p = 2$, set $d = 3$ and $(a_1, a_2, a_3) = (2, -1, 5)$. Choose the Kummer cocycles above and set

$$R_{\mathrm{gal}} = k[[t_1, \ldots, t_d]], \qquad u(g) = \sum_{i=1}^{d} t_i \kappa_{a_i}(g).$$

With the maximal-ideal topology on R_{gal}, the matrices

$$\rho_{R_{\mathrm{gal}}}(g) = \begin{pmatrix} 1 & 0 \\ u(g) & \omega(g) \end{pmatrix}$$

define a continuous representation reducing to ρ_0, with determinant ω and the standard determinant identification. Its tangent map is injective, with image $j(H^1(\Gamma, k(1)))$.

Proof. The cocycle equation for u is exactly the lower-left entry of $\rho_{R_{\mathrm{gal}}}(gh) = \rho_{R_{\mathrm{gal}}}(g)\rho_{R_{\mathrm{gal}}}(h)$. The finitely many cocycles factor through a common finite Galois quotient, so the representation is continuous for the maximal-ideal topology. Its determinant and reduction follow from the diagonal entries.

Give each Artinian local k-algebra B with residue field k the discrete topology. Use the same residual-marked deformation definition on free rank-two B-modules. A continuous local homomorphism $R_{\mathrm{gal}} \to B$ specializes these matrices to a deformation. This is compatible with homomorphisms between such algebras and therefore defines a map from the functor of points of Spf R_{gal} to the determinant-identified deformation groupoid.

A tangent homomorphism sends t_i to ϵx_i. Its class is $j(\sum_i x_i[\kappa_{a_i}])$. The Kummer basis and the left inverse of j prove injectivity. □

Changing β_{a_i} to $\zeta_p^{r_i}\beta_{a_i}$ adds dr_i to the corresponding cocycle. Set $s = \sum_i t_i r_i$. Conjugation by $1 - sE_{21}$ carries the old formal family to the new one, since its lower-left entry changes by ds. This calculation holds over R_{gal}, without a first-order truncation. The formal family therefore changes by an explicit isomorphism under these root choices.

This family fixes both diagonal characters and a line and quotient. It is not asserted to represent the full deformation problem. At $p = 2$, the underlying residual representation is trivial, and its unmarked form has no preferred decomposition into two lines. The inclusion through E_{21} uses the stated frame or flag. For odd p, the two distinct character subspaces are intrinsic, although their scalar coordinates still depend on the chosen bases.

Dimensions and comparison maps

The Kummer calculation determines one direct summand without local duality. The full dimension table also uses finite-module local duality and the local Euler characteristic theorem. These are Milne,

Arithmetic Duality Theorems, Corollary I.2.3, p. 28, and Theorem I.2.8, p. 31. For a finite-dimensional continuous $k[\Gamma]$-module N, they imply

$$h^2(N) = h^0(N^\vee(1)), \qquad h^0(N) - h^1(N) + h^2(N) = -\dim_k N,$$

where $h^i(N) = \dim_k H^i(\Gamma, N)$ and $N^\vee = \mathrm{Hom}_k(N, k)$.

The field trace pairing identifies the underlying finite-module dual with the additive module of $N^\vee(1)$. The Euler formula follows by substituting $|N| = p^{[k:\mathbb{F}_p]\dim_k N}$ into $\chi(\Gamma, N) = (\mathbb{Z}_p : |N|\mathbb{Z}_p)^{-1}$. Both theorems apply at $p = 2$, since the base field has characteristic zero. For $N = k(r)$, they give

$$h^1(k(r)) = 1 + \mathbf{1}_{\omega^r=1} + \mathbf{1}_{\omega^{1-r}=1}.$$

The matrix decomposition and $\dim B^1(\Gamma, N) = \dim N - h^0(N)$ now give

	$p \geq 5$	$p = 3$	$p = 2$
$h^1(k)$	2	2	3
$h^1(k(1))$	2	2	3
$h^1(k(-1))$	1	2	3
$h^0(\mathfrak{a})$	2	2	4
$h^0(\mathfrak{a}_0)$	1	1	3
$h^1(\mathfrak{a})$	7	8	12
$h^1(\mathfrak{a}_0)$	5	6	9
$\dim Z^1(\Gamma, \mathfrak{a})$	9	10	12
$\dim Z^1(\Gamma, \mathfrak{a}_0)$	7	8	9
$h^2(\mathfrak{a})$	1	2	4
$h^2(\mathfrak{a}_0)$	1	2	3

These dimensions concern the unrestricted local continuous problem. A crystalline, finite-flat, ordinary, or global Selmer condition requires its own definition before imposing a further subspace.

The explicit formal family supplies a local Galois target for a comparison. A specified nonzero Kummer class selects a one-parameter subfamily. The unit condition selects a line for odd p and a plane at $p = 2$. To identify an analytic residue line with one of these lines, construct a map carrying the residue to the specified nonzero cohomology class.

A coherent extension requires further categorical data. Suppose an exact k-linear functor from $\mathrm{Rep}_k^{\mathrm{cts}}(\Gamma)$ to an abelian category has identified images of k and $k(1)$. It sends E_b to an extension with those endpoints. An exact functor between specified stable categories instead sends its associated triangle to a triangle. Either construction gives a map on extension classes.

An isomorphism requires a calculation of that map, including its kernel and image. A tangent vector alone does not define two coherent objects or a functor between their categories. On $\mathrm{Spec}\, k[t]$, for instance, the tangent space at zero is k, whereas $\mathrm{Ext}^1(k[t], k[t]) = 0$. For the residue module k_0, the resolution

$$0 \longrightarrow k[t] \xrightarrow{t} k[t] \longrightarrow k_0 \longrightarrow 0$$

instead gives $\mathrm{Ext}^1(k_0, k_0) = k$. Thus the endpoints of a coherent extension are indispensable.

A comparison from a complex representation category also requires arithmetic coefficient data. There is no unital field map $k \to \mathbb{C}$, since it would carry $p \cdot 1_k = 0$ to $p \cdot 1_{\mathbb{C}} \neq 0$. There is no unital map in

the reverse direction either, since p is invertible in $\mathbb{C}$. Reduction requires an integral coefficient ring and an integral object or stable lattice. A residue-field map to k must then induce the required map on extension classes. A global Galois comparison additionally requires a ramification set, local conditions, and the restriction maps into the local cohomology computed here.

Finally, scalar twisting gives

$$(1 \oplus \omega) \otimes \omega^{-1} = \omega^{-1} \oplus 1.$$

This is $1 \oplus 1$ only at $p = 2$. At $p = 3$, exchanging summands identifies it with the original representation. For any scalar character χ, conjugation gives $\mathrm{Ad}(\rho_0 \otimes \chi) = \mathrm{Ad}(\rho_0)$, while the determinant changes by χ^2. Hence a scalar twist need not preserve a fixed-determinant fiber, and no extra determinant twist enters the adjoint cochain formulas.

The formal stack and its framed presentation

The local deformation groupoids above retain the residual representation and its automorphisms. At the split cyclotomic point, the residual-marked determinant-identified groupoid has

$$\pi_0 \mathcal{D}_{\det}(k[\epsilon]/(\epsilon^2)) = H^1_{\mathrm{cts}}(\Gamma, \mathrm{ad}^0(1 \oplus \omega)).$$

Here π_0 means the set of isomorphism classes, with its vector-space structure from the cocycle description. Fixing only the determinant character gives the same isomorphism classes at this split point, but retains the larger automorphism group $H^0(\Gamma, \mathfrak{a})$. This distinction follows from the equivariant trace splitting through E_{11}, including in characteristic two.

For the Emerton–Gee moduli problem, a framed lifting ring supplies a versal presentation of the formal deformation groupoid. Emerton–Gee, Proposition 3.6.3, p. 76, identifies its self-fiber product using the change-of-basis group completed along the residual centralizer. The pointed formal problem instead uses completion along the identity. It does not replace the unframed groupoid by a formal scheme with no automorphisms. A derived enhancement, wild-inertia conditions and an identification with its full tangent complex require the corresponding moduli comparison. The computed local complex retains this requirement.

Tangent complexes and spectral extension classes

A tangent direction describes deformation of one point. An extension between two sheaves requires two objects and a category of morphisms. To express the proposed comparison, suppose a derived stack $\mathcal{X}$ over k has been specified, with a cotangent complex $\mathbb{L}_{\mathcal{X}}$ admitting the dual used below. At a point x, set $T_x = R\,\mathrm{Hom}_k(x^*\mathbb{L}_{\mathcal{X}}, k)$. Its degree-zero cohomology is the first-order tangent space, while degree minus one records infinitesimal automorphisms. Write $\mathbb{T}^1_x = H^0(T_x)$ for this convention.

Suppose in addition that a spectral category $\mathcal{A}_{\mathrm{sp}}$ has been constructed, with objects $\mathcal{O}_{\mathrm{St}}$ and $\mathcal{O}_{\mathbf{1}}$ and specified support. These names alone do not define sheaves. For example, an ind-coherent category with nilpotent singular support requires a derived stack for which its singularity space and the specified nilpotent subset are defined. The desired comparison is

$$L_{\mathrm{cyc}} \longrightarrow \mathrm{Ext}^1_{\mathcal{A}_{\mathrm{sp}}}(\mathcal{O}_{\mathrm{St}}, \mathcal{O}_{\mathbf{1}}), \qquad L_{\mathrm{cyc}} \subseteq \mathbb{T}^1_{x_{\mathrm{cyc}}}.$$

Here the subspace L_{cyc}, its dimension and the displayed map are constructions still required. A cotangent calculation does not identify this Ext group between two different sheaves.

Remark 18.6.8 *(The reverse spectral comparison).* For a point x_{triv} representing $1 \oplus 1$, the opposite orientation asks for a specified subspace $L_{\mathrm{triv}} \subseteq \mathbb{T}^{1}_{x_{\mathrm{triv}}}$ and a map

$$L_{\mathrm{triv}} \longrightarrow \mathrm{Ext}^{1}_{\mathcal{A}_{\mathrm{sp}}}(\mathcal{O}_{\mathbf{1}}, \mathcal{O}_{\mathrm{St}}).$$

The two endpoints and their support must be constructed in this category. The explicit scalar-twist calculation above does not identify this problem with the cyclotomic problem at odd primes. At $p = 2$, equality of the residual representations still does not choose the two sheaves or the map. An Arthur parameter over $\mathbb{C}$ supplies neither spectral comparison by itself.

18.7 Blocks, extensions and geometric support

A local categorical comparison must specify its block algebra and its spectral support. The ordinary center, the full endomorphism algebra of a generator and the deformation ring are different objects. Their distinctions persist even when every simple object has the expected parameter.

Characteristic-zero Bernstein decomposition

THEOREM 18.7.1 *(Bernstein–Deligne).* Let G be a connected reductive p-adic group and fix an algebraically closed coefficient field k of characteristic zero. Let $\mathrm{Mod}_G^{\mathrm{sm}}$ denote its smooth representations over k. This category decomposes as a product:

$$\mathrm{Mod}_G^{\mathrm{sm}} = \prod_{\mathfrak{s} \in \mathfrak{B}(G)} \mathrm{Block}_{\mathfrak{s}},$$

Here $\mathfrak{B}(G)$ is the set of inertial equivalence classes of supercuspidal supports. The decomposition is realized by a central idempotent decomposition of the Bernstein center $Z(\mathrm{Mod}_G^{\mathrm{sm}}) = \prod_{\mathfrak{s}} Z_{\mathfrak{s}}$ into a product of integral domains.

Attribution. Bernstein–Deligne (1984), in the proceedings *Representations of reductive groups over a local field*. For the block decomposition and torsor of unramified twists modulo self-twists, see also Helm, arXiv:1201.1874, Theorems 3.4–3.5, pp. 7–8. □

PROPOSITION 18.7.2 *(central-character fibers of Bernstein blocks).* Let k be algebraically closed of characteristic zero, and let $\chi : Z(G) \to k^{\times}$ be a smooth central character. For a Bernstein block $\mathcal{B}_{\mathfrak{s}}$, put

$$\mathcal{B}_{\mathfrak{s}}^{\chi} = \{M \in \mathcal{B}_{\mathfrak{s}} : z|_M = \chi(z)\,\mathrm{id}_M \text{ for every } z \in Z(G)\}.$$

Then

$$\mathrm{Rep}_k^{\mathrm{sm},\chi}(G) \simeq \prod_{\mathfrak{s}} \mathcal{B}_{\mathfrak{s}}^{\chi},$$

with zero factors omitted.

Let $\delta_z \in Z_{\mathfrak{s}}$ denote the central operation supplied by $z \in Z(G)$, and let

$$I_{\mathfrak{s},\chi} = (\delta_z - \chi(z) : z \in Z(G)) \subseteq Z_{\mathfrak{s}}.$$

The ring $Z_{\mathfrak{s}}/I_{\mathfrak{s},\chi}$ acts on $\mathcal{B}_{\mathfrak{s}}^{\chi}$. This is a fiber condition inside the block.

Proof. Each Bernstein projector commutes with the action of $Z(G)$. Thus every component of an object with central character χ again has that central character. Conversely, a family of block objects reconstructs a smooth representation by its direct sum, and the sum satisfies the same scalar equations. This proves the categorical decomposition.

The generators of $I_{\mathfrak{s},\chi}$ act as zero on the indicated subcategory, so the central action factors through the quotient. No claim that taking the categorical center commutes with this quotient is needed. □

For $G = GL_2(F)$, fix a uniformizer ϖ and normalized parabolic induction. In the unramified principal-series block write $a = \eta_1(\varpi)$, $b = \eta_2(\varpi)$ for its two unramified inducing characters, so $t = a + b$ and $d = ab$. The modulus character in normalized induction is trivial on scalar matrices. Consequently the central operation of $u \in O_F^\times$ is 1, and that of ϖ is d, in the ring $k[t, d, d^{-1}]$.

These equalities hold as central operations: their values on all supercuspidal supports are respectively 1 and ab, and the block ring is reduced. If χ is unramified, put $c = \chi(\varpi) \in k^\times$. Then $I_{\mathfrak{s},\chi} = (d - c)$ and the fiber has the quotient scalar action $k[t, d, d^{-1}]/(d - c)$. It is nonzero, for instance on any irreducible constituent of the normalized induction of the unramified characters with values 1, c at ϖ.

If χ is ramified, choose $u \in O_F^\times$ with $\chi(u) \neq 1$. The relation $1 - \chi(u)$ is an invertible scalar, so the quotient and the fixed-character fiber category are zero. Thus fixing central character neither selects a whole Bernstein block nor gives a nonzero fiber in every block.

The exact Bernstein projectors extend degreewise to ordinary complexes of smooth characteristic-zero representations. They preserve quasi-isomorphisms because they are exact, and therefore act on the derived category. This elementary extension does not identify that category with a separately defined geometric Hecke category.

For $GL_2(F)$, take algebraically closed characteristic-zero ℓ-adic coefficients with $\ell \neq p$. Use compatible reciprocity and induction normalizations. The ordinary Bernstein center then identifies with scalar functions on the continuous Weil parameter stack. This includes residue primes two and three.

The four inertial cases retain different parameter rings: irreducible inertia, two fixed characters, an interchanged pair, and scalar inertia. They cannot be recovered from a Frobenius matrix with its inertia action discarded. The ring construction and its full coefficient hypotheses are in Proposition 116.2.2.

That scalar image need not exhaust the center of the spectral category. Scalar GL_2 acts trivially on the parameter coordinates, but equivariant coherent objects retain central weights. After Ind-completion, projections to these weight factors give central operations. A determinant twist changes weight by two.

The weight-zero projector separates this operation from scalar multiplication. See Proposition 116.2.5. Choosing one weight removes this particular obstruction, without proving a categorical equivalence.

The coefficient-prime block algebra

Use the coefficient and reciprocity conventions of Paškūnas, *The image of Colmez's Montreal functor*. Here $p \geq 5$, $G = GL_2(\mathbb{Q}_p)$, and O is a finite extension's integer ring with residue field k. Fix a full central character ζ. Let $\mathcal{A}_\zeta^{\mathrm{lfin}}$ be the smooth O-torsion representations which are locally of finite length: each vector generates a finite-length $O[G]$-module.

A block $\mathcal{B}$ is a connected component of the extension relation on irreducible objects, allowing extensions in both directions. Let $J_\mathcal{B}$ be the injective envelope of the sum of its irreducibles. Its Pontryagin dual $P_\mathcal{B} = J_\mathcal{B}^\vee$ is a projective generator of the corresponding compact category. Put

$$E_\mathcal{B} = \mathrm{End}(P_\mathcal{B}) = \mathrm{End}_G(J_\mathcal{B})^{\mathrm{op}}.$$

Endomorphisms use ordinary composition. This is a pseudocompact ring, with topology given by its finite-length quotients. The functor $\tau \mapsto \mathrm{Hom}(P_\mathcal{B}, \tau^\vee)$ is an anti-equivalence from the block to

compact right $E_{\mathcal{B}}$-modules. The right action is precomposition. These compact modules and the locally finite torsion category are different from the finite-length admissible Banach category.

After a coefficient extension splitting the irreducibles, the four block forms in this theorem are as follows. The first is a supersingular singleton. The second consists of

$$\mathrm{Ind}_B^G(\chi_1 \otimes \chi_2\omega^{-1}), \qquad \mathrm{Ind}_B^G(\chi_2 \otimes \chi_1\omega^{-1}), \qquad \chi_1\chi_2^{-1} \notin \{1, \omega, \omega^{-1}\}.$$

The third is the singleton $\mathrm{Ind}_B^G(\chi \otimes \chi\omega^{-1})$.

The fourth consists of

$$\eta \circ \det, \qquad \mathrm{Sp} \otimes (\eta \circ \det), \qquad \mathrm{Ind}_B^G(\omega \otimes \omega^{-1}) \otimes (\eta \circ \det).$$

Here ω is the residual cyclotomic character under the paper's class-field convention. The associated semisimple Galois parameters are, respectively, absolutely irreducible, $\chi_1 \oplus \chi_2$, $\chi \oplus \chi$, and $\eta \oplus \eta\omega$. In particular, the last is not scalar when $p \geq 5$.

THEOREM 18.7.3 *(Paškūnas, Theorems 1.5–1.7).* For these blocks the center of $E_{\mathcal{B}}$ is the fixed-determinant pseudodeformation ring $R^{\mathrm{ps},\zeta\varepsilon}_{\mathrm{tr}\,\bar\rho}$. The algebra $E_{\mathcal{B}}$ is finite over its center and, on the stated complement of the reducible locus, is a matrix algebra.

For a supersingular singleton, the full algebra is $R^{\zeta\varepsilon}_{\bar\rho}$. For the generic reducible block, let ρ_1, ρ_2 be the two nonsplit extensions of its distinct characters in opposite directions. Their universal fixed-determinant deformations satisfy

$$E_{\mathcal{B}} \simeq \mathrm{End}_{G_{\mathbb{Q}_p}}(\rho_1^{\mathrm{un}} \oplus \rho_2^{\mathrm{un}}).$$

The corresponding dualized and twisted Montreal images of $J_{\mathcal{B}}$ are the indicated universal representation or sum of the two universal representations.

The compact-module convention is that of the paper's §2. Pontryagin duality reverses the endomorphism ring. Keeping this opposite is necessary when comparing the noncommutative block algebras.

The determinant is $\zeta\varepsilon$ in this theorem. The DEG convention above uses $\zeta\varepsilon^{-1}$ and its specified kernel twist. A comparison must retain the dualizations, twists and reciprocity conventions which relate the two functors.

The same paper's Theorem 1.4 gives a different positive equivalence. Fix an absolutely irreducible admissible unitary non-ordinary Banach representation Π. Take the finite-length block with every irreducible subquotient isomorphic to Π. It is equivalent to finite-length modules over the fixed-determinant deformation ring of its associated characteristic-zero Galois representation. It does not identify every torsion block with one commutative deformation ring.

For a finite sum $P = P_1 \oplus P_2$ of nonzero projective covers of distinct simples, $\mathrm{End}(P)$ contains the orthogonal projection idempotents. It cannot be a commutative local ring. A dimension or regularity calculation for its center therefore does not give that calculation for its full endomorphism algebra.

Generic localization and first-order classes

PROPOSITION 18.7.4 *(extension vanishing over the fraction field).* Let R be a commutative Noetherian domain, let $K = \mathrm{Frac}(R)$, and let M, N be finite R-modules. Then

$$\mathrm{Ext}^1_R(M, N) \otimes_R K \simeq \mathrm{Ext}^1_K(M \otimes_R K, N \otimes_R K) = 0.$$

In particular, the statement applies to a regular local domain. It asserts vanishing at the generic point, not at every regular point.

Proof. Noetherianity gives a free resolution of M with finitely generated terms: take a finite set of generators for each successive kernel. Localization is exact, and homomorphisms from each finite free term commute with localization. Localizing the resulting Hom complex therefore computes the displayed extension group over K. Every short exact sequence of vector spaces over K splits. □

Corollary 18.7.5 *(arithmetic transport on a specified generic fiber).* Suppose an arithmetic block category is exactly equivalent to finite modules over a commutative Noetherian domain R. Assume its specified generic-fiber category corresponds to finite $K = \mathrm{Frac}(R)$-modules. Then generic-fiber extensions vanish. If a spectral locus is also identified with this localization by compatible functors, the same vanishing holds in its corresponding category.

Proof. An exact equivalence identifies extension classes. Apply Proposition 18.7.4 and the stated localization comparison. Neither regularity alone nor flatness alone constructs the arithmetic or spectral equivalence. □

Local self-extensions

Proposition 18.7.6 *(Local self-extensions and conditional arithmetic transport).* For a commutative Noetherian local ring $(A, \mathfrak{m}, k)$, there is a natural isomorphism

$$\mathrm{Ext}^1_A(k,k) \cong \mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k).$$

Take distinct simples S, S' in an arithmetic ordinary block, and put $A = A_\rho$. Assume a specified k-linear map identifies $\mathrm{Ext}^1_{\mathrm{Block}_\rho}(S, S')$ with a subspace D_ρ of the displayed dual cotangent space. Then its dimension is $\dim_k D_\rho$. If in addition $\dim_k \mathfrak{m}_\rho/\mathfrak{m}_\rho^2 \le 3$ is established, this gives the bound $\dim_k D_\rho \le 3$. Breuil–Mezard multiplicities alone supply neither this class comparison nor the bound on the embedding dimension.

Proof. Apply $\mathrm{Hom}_A(-,k)$ to $0 \to \mathfrak{m} \to A \to k \to 0$. The restriction $\mathrm{Hom}_A(A,k) \to \mathrm{Hom}_A(\mathfrak{m},k)$ is zero, since $\mathfrak{m}k = 0$, and $\mathrm{Ext}^1_A(A,k) = 0$. Thus the connecting map identifies $\mathrm{Hom}_A(\mathfrak{m},k)$ with $\mathrm{Ext}^1_A(k,k)$. Every such homomorphism kills $\mathfrak{m}^2$, giving the displayed dual cotangent space. Concretely, a functional gives the extension obtained by pushing out $0 \to \mathfrak{m} \to A \to k \to 0$ along $\mathfrak{m} \to k$.

The arithmetic conclusion follows from the separately assumed class comparison. There is no subtraction by a number of defining relations: for $A = k[\epsilon]/(\epsilon^2)$, the embedding dimension is one and $\mathrm{Ext}^1_A(k,k) = k$. This space is annihilated by $\mathfrak{m}$, and is not a nonzero free $A/\mathfrak{m}^2$-module. Every simple module over a commutative local ring is its residue field. Comparison with distinct arithmetic simples therefore requires the specified additional map. □

The reduced total-quotient version also has a precise domain. For a reduced Noetherian ring R, its total quotient ring is a finite product of fields. Localization of finite-module Ext therefore gives vanishing of every positive extension degree over that semisimple ring. Reducedness cannot be replaced by the hypersurface condition. For

$$R = k[[t,u,\epsilon]]/(\epsilon^2),$$

the generic coefficient algebra is $K[\epsilon]/(\epsilon^2)$, with $K = \mathrm{Frac}(k[[t,u]])$. Its simple module has a periodic free resolution with differential ϵ, so $\mathrm{Ext}^1(K,K) = K$.

For a complete local algebra over a discrete valuation ring with uniformizer ϖ, the relative first-order tangent is

$$\operatorname{Hom}_k\bigl(\mathfrak{m}/(\mathfrak{m}^2+\varpi R),k\bigr).$$

The absolute self-extension space uses $\mathfrak{m}/\mathfrak{m}^2$. For $R=O[[x]]$, these dimensions are one and two, respectively. They coincide for a coefficient problem over k, not for every mixed-characteristic deformation problem.

A comparison with an extension between two arithmetic simples must specify both endpoints, the functor or class map, and its normalization. A commutative local ring has only its residue field as a simple module. It does not label several arithmetic simple pairs by itself. Matching one extension dimension gives no higher-product comparison. The polynomial and dual-number examples in Section 18.1 agree in degree one and differ in every higher degree.

Support and compact generation

The localizing subcategory generated by $k=k[t]/(t)$ in $D(k[t])$ disappears after inverting t. The object $k[t]$ survives. Thus a family of image objects can have correct pointwise descriptions and still fail to generate its intended target.

For a quasi-smooth derived scheme X, singular support lies in its scheme of singularities, built from $H^1(T_X)$ over X^{cl}. Its action is the degree-two Hochschild action. It is not a locus inside X defined merely by naming a degree-one Galois class. On derived Artin stacks, the automorphism directions must also be retained when specifying cotangent amplitude and the smooth-local definition of support.

Definition 18.7.7 *(Residual gerbe).* Let x be a point of a locally Noetherian algebraic stack X. Its *residual gerbe* $\mathcal{G}_x$ is the reduced locally Noetherian substack whose underlying space consists of x alone. The inclusion is a monomorphism. This characterizes the gerbe up to its unique equivalence inside X. Existence and uniqueness are The Stacks Project, Tags 06MT and 0H22.

The stabilizers of rank-two parameters give concrete tests. An irreducible representation has scalar centralizer. A distinct split pair has a two-dimensional torus. A scalar representation has GL_2 centralizer.

For a nonsplit self-extension with nilpotent matrix N, the centralizer is $\{aI+bN : a\neq 0\}$. It includes scalar automorphisms and a unipotent subgroup. A rational representative neutralizes its residual gerbe. An arbitrary gerbe need not have such a chosen representative. Determinant twists shift scalar weights by two and do not generate all weights from one of them.

A finite-stratification argument can prove compact generation when the support-compatible recollement functors are supplied. In particular, an extension functor preserves compact objects if its right adjoint preserves filtered colimits. Apply this fact at each stratum, then use the recollement triangles to generate the ambient category. Per-stratum compactness alone does not prove the needed compactness after extension.

At residue primes two and three, a comparison requires the actual block algebra and an enhanced functor into the specified spectral support category. Full derived endomorphisms and target generation must then be compared. One simple object does not imply semisimplicity. A type, an irreducible representation, an inertial class, a deformation point and an irreducible geometric component are different objects. None is substituted for another in this criterion.

A nilpotent local model

Proposition 18.7.8 *(the nilpotent cone in rank two).* Let K be an algebraically closed field of characteristic zero. The nilpotent cone in $\mathfrak{sl}_2(K)$ has dimension two. Its nonzero locus is the regular nilpotent orbit, of codimension one in $\mathfrak{sl}_2(K)$. The zero orbit has dimension zero.

Proof. Write

$$X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}.$$

Then $X^2 = (a^2 + bc)\mathrm{I}$, so nilpotency is the equation $a^2 + bc = 0$ in $\mathbb{A}^3_K$. On $b \neq 0$ it is parametrized by a, b, with $c = -a^2/b$. The quadric is irreducible: a factorization into linear forms would give a quadratic form of rank at most two, whereas $a^2 + bc$ has rank three.

Thus the cone has dimension two. Every nonzero nilpotent matrix has one Jordan block and is conjugate to $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. A conjugating matrix can be rescaled to determinant one because K is algebraically closed. The nonzero locus is therefore one $SL_2(K)$-orbit of dimension two. The remaining orbit is $\{0\}$. □

Remark 18.7.9 *(the arithmetic local model).* This calculation does not determine the dimension of an arithmetic Selmer space. Such a conclusion requires a constructed local-model morphism and its relative dimension. A nonempty transition space cannot be a subspace of both a closed locus and its open complement, since their intersection is empty. In singular-support theory the support condition lies in the scheme of singularities of a quasi-smooth derived space.

It is not, by definition, a locus where the underlying representations have unipotent monodromy. The relevant constructions are Arinkin–Gaitsgory, arXiv:1201.6343, §§2.3, 4.1 and 8.2. §11.1 defines the global nilpotent cone for the geometric local-system stack. An arithmetic analogue requires its own space, singular-support subset and functors.

Filtration length and geometric codimension

Example 18.7.10 *(a Nygaard filtration on a point).* On the prism $(\mathbb{Z}_p, (p))$, take the Frobenius lift $\varphi = \mathrm{id}$. The Nygaard submodules defined by $\varphi(x) \in p^i\mathbb{Z}_p$ are

$$N^{\geq i}\mathbb{Z}_p = p^i\mathbb{Z}_p \quad (i \geq 0), \qquad \mathrm{gr}^i_N\,\mathbb{Z}_p = \mathbb{F}_p.$$

Every nonnegative graded degree is nonzero, although $\operatorname{Spec}\mathbb{F}_p$ has dimension zero. Thus a geometric codimension bound does not by itself truncate the Nygaard filtration in degree two.

For a filtered complex C with decreasing filtration F, the usual cohomological spectral sequence begins with

$$E_1^{a,b} = H^{a+b}(\mathrm{gr}^a_F\, C). \tag{18.6}$$

Convergence of Equation (18.6) to $H^{a+b}(C)$ requires appropriate completeness, exhaustiveness and convergence hypotheses on the filtered complex. A comparison with a cellular filtration additionally requires a filtered chain map and a proof of its effect on graded pieces. The dimension calculation above supplies neither a map nor a spectral-sequence degeneration statement.

Lifting cycles and automorphic hypotheses

Fixed-type cycle theorems supply positive geometric information without identifying categorical generators. In Emerton–Gee's cocharacter notation, the minimal regular type is BT = (1, 0), corresponding to filtration degrees $\{-1, 0\}$ above. For $GL_2(K)$, with $K/\mathbb{Q}_p$ finite and $p > 2$, Theorem 8.6.2 gives the geometric Breuil–Mézard formula for potentially Barsotti–Tate types and arbitrary inertial type.

Write $\mathcal{X}_2^{\mathbf{k}}$ for the component indexed by the Serre weight $\mathbf{k}$. Call $\mathbf{k}$ Steinberg when $k_{\sigma,1} - k_{\sigma,2} = p - 1$ at every embedding. In that case put $\widetilde{k}_{\sigma,1} = \widetilde{k}_{\sigma,2} = k_{\sigma,2}$. The cycles in that formula are

$$\mathcal{Z}_{\mathbf{k}} = \begin{cases} [\mathcal{X}_2^{\mathbf{k}}], & \mathbf{k} \text{ not Steinberg,} \\ [\mathcal{X}_2^{\mathbf{k}}] + [\mathcal{X}_2^{\widetilde{\mathbf{k}}}], & \mathbf{k} \text{ Steinberg.} \end{cases}$$

The special-fiber cycle for a type is the sum of these cycles with the Jordan–Hölder multiplicities of its reduced type. The crystalline and semistable versions use their respective types. The additional component in the Steinberg case is part of the theorem.

The general geometric formula of their Conjecture 8.2.2 has a different scope. Their Theorem 8.1.4 supplies finite cycle support for each fixed regular type. It does not assert smooth divisors or transverse intersections. Nor does a cycle determine a projective generator and all its morphisms. The range $2 \leq k \leq p - 1$ gives dual weights $(0, k - 1)$, including $(0, p - 2)$ at its endpoint. It does not reduce the full supersingular lifting problem to two weights.

There are also positive modularity theorems with specific global hypotheses. For example, Taylor's Theorem A in *On icosahedral Artin representations II* applies to an irreducible continuous $\rho : G_{\mathbb{Q}} \to GL_2(\mathbb{C})$ with $\det \rho(c) = -1$. If its projective image is A_5, impose two further conditions. Its projective inertia image at three has odd order. Its projective decomposition image at five is unramified of order two. Then ρ is modular and its Artin L-function is entire.

This theorem does not give a local compact generator for every residual A_5 block. Potential automorphy, ordinary modularity and local-global compatibility theorems retain their residual, regularity, cyclotomic-irreducibility and local hypotheses. Potential automorphy alone supplies no tower of cyclic extensions for descent. Cyclic descent requires the actual invariance and descent data at each step. The finite-field PSL_2 and PGL_2 image cases and overlaps of lifting loci also remain in a proposed coverage argument.

A categorical comparison can use these results after constructing the relevant image objects and supports. For any collection of overlapping lifting conditions, take the localizing closure of the actual image objects, including their extensions. A direct-sum decomposition additionally requires orthogonality. The generator criterion then applies on this specified category. It does not infer coverage from a list of arithmetic names.

18.8 Geometric Hecke categories and spectral actions

Several geometric constructions realize a full representation category or a distinguished subcategory. Their coefficient theories and geometric spaces are part of their statements. The equal-characteristic affine flag, the Weil–Deligne parameter stack and the p-adic determinant-identified stack give different recipients for an arithmetic comparison.

Bundles and elementary modifications

Every finite projective $\mathbb{Z}$-module is free. A finite projective module is a direct summand of a finite free module. A submodule of a free module over the principal ideal domain $\mathbb{Z}$ is free. Thus the groupoid of

rank-two bundles on Spec $\mathbb{Z}$ has one isomorphism class, with automorphism group $GL_2(\mathbb{Z})$. This is a statement about bundles over that base. It does not identify their moduli functor on every test scheme with a constant classifying stack.

Elementary modifications still occur. The inclusion $p\mathbb{Z} \oplus \mathbb{Z} \subset \mathbb{Z}^2$ has quotient $\mathbb{F}_p$ and becomes an isomorphism away from p. The existence of these lattice correspondences is compatible with freeness of both endpoint bundles. Their convolution, its sheaf realization and its scalar function are separate constructions.

The affine flag and Springer correspondences

Let G be a split reductive group over an algebraically closed field k, and fix $\ell \neq \operatorname{char} k$. Use constructible $\overline{\mathbb{Q}}_\ell$-complexes on the equal-characteristic affine flag spaces of $G(k((t)))$. Let I be an Iwahori subgroup, I° its pro-unipotent radical, and $T = I/I^\circ$. Write D_{II} for the I-equivariant derived category on $G(k((t)))/I$. Write $D_{I^\circ I}$ for the I°-equivariant category on this same space. Finally, $D^{\mathrm{un}}_{I^\circ I^\circ}$ is the category on $G(k((t)))/I^\circ$ with I°-equivariance on the left and unipotent T-monodromy on the right.

Let $\check{G}$ be the Langlands dual group over $\overline{\mathbb{Q}}_\ell$, with Lie algebra $\check{\mathfrak{g}}$. Its Grothendieck and Springer resolutions are

$$\widetilde{\mathfrak{g}} = \{(\mathfrak{b}, x) : x \in \mathfrak{b}\}, \qquad \widetilde{\mathcal{N}} = \{(\mathfrak{b}, x) : x \in \operatorname{rad}(\mathfrak{b})\}.$$

Set

$$\mathrm{St} = \widetilde{\mathfrak{g}} \times_{\check{\mathfrak{g}}} \widetilde{\mathfrak{g}}, \qquad \mathrm{St}' = \widetilde{\mathfrak{g}} \times_{\check{\mathfrak{g}}} \widetilde{\mathcal{N}}.$$

For a map to $\check{\mathfrak{g}}$, the subscript $\mathcal{N}$ on coherent sheaves denotes set-theoretic nilpotent support.

Theorem 18.8.1 *(Bezrukavnikov).* There are equivalences

$$D^{\mathrm{un}}_{I^\circ I^\circ} \simeq D^b \mathrm{Coh}^{\check{G}}_{\mathcal{N}}(\mathrm{St}),$$
$$D_{I^\circ I} \simeq D^b \mathrm{Coh}^{\check{G}}(\mathrm{St}'),$$
$$D_{II} \simeq \mathrm{DGCoh}^{\check{G}}\big(\widetilde{\mathcal{N}} \times^{\mathbb{L}}_{\check{\mathfrak{g}}} \widetilde{\mathcal{N}}\big).$$

The first and third preserve convolution. The middle equivalence preserves both convolution actions. With the corresponding Iwahori–Whittaker conditions, one also has the module-category equivalences

$$D^I_{\mathrm{IW}} \simeq D^b \mathrm{Coh}^{\check{G}}(\widetilde{\mathcal{N}}), \qquad D^{I^\circ,\mathrm{un}}_{\mathrm{IW}} \simeq D^b \mathrm{Coh}^{\check{G}}_{\widetilde{\mathcal{N}}}(\widetilde{\mathfrak{g}}).$$

They preserve the respective actions above.

These are Theorems 1–2 of *On two geometric realizations of an affine Hecke algebra*. The Whittaker character is nondegenerate on the finite simple-root quotients and trivial on the affine-root quotient, with the left and right conventions of that paper. The monodromic convolution category in the first line need not contain a unit. The Iwahori category in the third line is unital. The derived fiber product in that line retains its higher Tor. The two other displayed Steinberg products have their ordinary scheme descriptions under the stated resolution hypotheses.

In particular, the Springer-resolution category is a module category, and the Iwahori category has a Steinberg carrier. The Iwahori derived category is not the bounded derived category of its perverse heart in general. Passing to functions over a finite field also requires Frobenius structures and the sheaf–function map. These are substantive parts of any comparison with an arithmetic Hecke category.

Coherent Springer objects

Let H be reductive over $L = \overline{\mathbb{Q}}_\ell$ or $\mathbb{C}$. Let $\widehat{\mathcal{N}}$ be the formal completion of its Lie algebra along the nilpotent cone, and let $\mu : \widetilde{\mathcal{N}} \to \widehat{\mathcal{N}}$ be the Springer map followed by completion. Use H_{gr} for H together with the dilation action, and let $\mathcal{L}$ denote derived loops. The coherent Springer object is

$$\mathcal{S}_H = (\mathcal{L}\mu)_* \mathcal{O}_{\mathcal{L}(\widetilde{\mathcal{N}}/H_{\mathrm{gr}})} \quad \text{on } \mathcal{L}(\widehat{\mathcal{N}}/H_{\mathrm{gr}}).$$

THEOREM 18.8.2 *(Ben-Zvi–Chen–Helm–Nadler).* The full derived endomorphism algebra of $\mathcal{S}_H$ is the affine Hecke algebra $\mathcal{H}_H$, concentrated in degree zero. Tensoring with $\mathcal{S}_H$ defines a fully faithful functor

$$D(\mathcal{H}_H) \longrightarrow \mathrm{IndCoh}\big(\mathcal{L}(\widehat{\mathcal{N}}/H_{\mathrm{gr}})\big)$$

whose image is generated by $\mathcal{S}_H$. It is compatible with parabolic induction. The anti-spherical module maps to the projection of the dualizing object onto this Springer subcategory.

For $q \in L^\times$ not a root of unity, let

$$\mathcal{L}^u_{q,H} = [\{(g, N) : N \in \mathcal{N},\ gNg^{-1} = qN\}/H]$$

be the unipotent parameter stack and let $\mathcal{S}_{q,H}$ be the corresponding coherent specialization. There is a fully faithful functor

$$\mathrm{Perf}(\mathcal{H}_{q,H}) \longrightarrow \mathrm{Coh}(\mathcal{L}^u_{q,H})$$

defined by $\mathcal{S}_{q,H}$. It takes the anti-spherical module to $\mathcal{O}_{\mathcal{L}^u_{q,H}}$ and preserves the parabolic induction correspondences.

The assertions are Theorems 1.7 and 1.9 of *Coherent Springer theory and categorical Deligne–Langlands*. At a non-root of unity the affine Hecke algebra has finite cohomological dimension, so its perfect and bounded coherent derived categories agree in this statement. The formal derived loop carrier before specialization and the classical reduced q-parameter stack are different objects. In particular, the unspecialized anti-spherical image is the stated projection of a dualizing object.

For GL_n, the same work gives a larger positive comparison. Let F be a nonarchimedean local field and $\mathcal{L}_{F,M}$ the stack of continuous Weil–Deligne parameters for a Levi subgroup M of GL_n, over $\mathbb{C}$. It retains formal deformation spaces of parameters. Theorem 1.12 gives full embeddings

$$D(\mathrm{Rep}^{\mathrm{sm}}_{\mathbb{C}}(M(F))) \longrightarrow \mathrm{IndCoh}(\mathcal{L}_{F,M}).$$

They send irreducible cuspidal representations to the parameter skyscraper sheaves prescribed by local Langlands, and commute with parabolic induction through its pull–push correspondence. These properties characterize the supplied embeddings. No equivalence with their whole ind-coherent targets is asserted.

The spectral action on the Fargues–Fontaine curve

Let F have residue cardinality q, let ℓ differ from its residue characteristic, and let L be a field over $\mathbb{Q}_\ell(\sqrt{q})$. For a reductive group G/F, write $\widehat{G}$ for its dual. The parameter stack $[Z^1(W_F, \widehat{G})_L/\widehat{G}]$ uses continuous Weil cocycles and the specified Weil action on $\widehat{G}$. It is not generally quasi-compact.

Fargues–Scholze, Corollary X.1.3, constructs an action

$$\mathrm{Perf}([Z^1(W_F, \widehat{G})_L/\widehat{G}]) \curvearrowright D_{\mathrm{lis}}(\mathrm{Bun}_G, L)^\omega.$$

Here Bun_G is the bundle stack on the Fargues–Fontaine curve, and ω denotes compact objects of the indicated category of lisse sheaves. The action is compactly supported: on each object it factors through parameters trivial on some sufficiently small open subgroup of wild inertia. Its restrictions along the representation functors recover the Hecke actions, functorially in every finite set of legs.

For integral coefficients, their Theorem X.0.1 also imposes $\ell \nmid |\pi_1(\widehat{G})_{\mathrm{tor}}|$. The rational action is a theorem. The equivalence formulated in their Conjecture X.1.4 is a further assertion about a Whittaker-generated functor and its right adjoint. Neither statement changes D_{lis} into coherent sheaves on Bun_G, or identifies the parameter stack with the coefficient-prime stack $\mathcal{X}_\partial$.

Factorization and Whittaker equivalences

The critical-level comparison has a stronger factorization form. Use a reductive group on a smooth characteristic-zero curve. With the twisting conventions of *Proof of the geometric Langlands conjecture II*, Theorem 6.1.4 gives

$$\mathrm{KL}(G)_{\mathrm{crit}} \simeq \mathrm{IndCoh}^*(\mathrm{Op}_{\check{G}}^{\text{mon-free}})$$

as factorization categories. Here KL is the spherical critical Kac–Moody category.

A monodromy-free oper consists of a local system extending to the disk, together with its oper structure on the punctured disk. It is not an unrestricted monodromic local system. Corollary 6.1.9 proves that this functor is strictly unital. The regular-oper structure object maps to the critical vacuum.

At nondegenerate levels, Campbell–Dhillon–Raskin construct the spherical and affine-flag DG equivalences

$$\mathrm{Whit}_\kappa^{\mathrm{sph}} \simeq \widehat{\check{\mathfrak{g}}}_{\check{\kappa}}\text{-mod}^{L^+\check{G}}, \qquad \mathrm{Whit}_\kappa^{\mathrm{aff}} \simeq \widehat{\check{\mathfrak{g}}}_{\check{\kappa}}\text{-mod}^{\check{I}}.$$

The left categories are twisted Whittaker D-modules on the affine Grassmannian and flag variety. The right categories are Harish–Chandra DG module categories for the displayed affine Lie algebra and subgroup. The dual level uses the critical shifts of their §2.7.

Their good-level hypothesis has a concrete form. On each simple factor an irrational level is good. For a rational level $\kappa^\circ = (-h^\vee + a/b)\kappa_{\mathrm{bas}}$, with $(a,b) = 1$, the denominator b must be coprime to

type	A_n	B_n	C_n	D_n	E_6	E_7	E_8	F_4	G_2
$n(\mathfrak{g})$	1	2	2	2	6	6	30	6	6.

This is Proposition 3.4.5. These DG equivalences in that paper set aside the factorization enhancement. They are not a theorem about arbitrary quantum parameters on an arithmetic base.

Gaitsgory–Lysenko's small local equivalence has a third precise recipient. Choose their geometric metaplectic datum on a smooth curve. Let q be its restricted Weyl-invariant root-of-unity quadratic form, and H the associated dual datum. Fix the required trivialization at the marked point.

Retain the nondegeneracy and sheaf-theory hypotheses of that datum. Nondegeneracy means that $q(\alpha)$ has order different from one for every coroot α of G, as in Definition 2.3.4. Their graded Hecke category is formed using the metaplectic Satake action of $\mathrm{Rep}(H)$ and restriction to its Cartan T_H. Theorem 19.2.5 identifies it with the renormalized category of factorization modules:

$$\mathrm{Hecke}^\bullet(\mathrm{Whit}_{q,x}(G)) \simeq \left(\Omega_q^{\mathrm{small}}\text{-FactMod}\right)^{\mathrm{ren}}.$$

The renormalization retains the specified Artinian structure and compact objects. Replacing it by the unrenormalized category changes the statement. The allowed sheaf theories include the stated D-module, constructible and ℓ-adic settings. The factorization-category strengthening is distinguished in their §0.4.5 from the categorical theorem they formulate.

These are geometric positive comparisons with specified units, levels and enhancements. A value p^{-s} in a Hecke spectrum does not identify their quantum parameter with an arithmetic one. An arithmetic use requires the coefficient and operation-preserving map, including its collision and unit data.

18.9 Normalized intertwining for unramified principal series

An Eisenstein comparison must distinguish the normalization of an intertwining operator from its passage to a category of representations. For unramified principal series over a nonarchimedean field, the operator can be computed on every locally constant vector. The computation also determines its poles and its degenerations.

Let F be a finite extension of $\mathbb{Q}_p$. Write O for its ring of integers, ϖ for a uniformizer, and $q = |O/\varpi O|$. Normalize $|\varpi| = q^{-1}$ and the additive Haar measure by $\mathrm{vol}(O) = 1$. Fix an additive character $\psi : F \to \mathbb{C}^\times$ that is trivial on O and nontrivial on $\varpi^{-1}O$.

The operator below does not depend on ψ. This choice fixes the additive convention for subsequent local-factor comparisons. All representations and function spaces have coefficients in $\mathbb{C}$.

Put $G = \mathrm{GL}_2(F)$, $K = \mathrm{GL}_2(O)$, and let B be the upper triangular subgroup. For unramified smooth characters $\chi_1, \chi_2 : F^\times \to \mathbb{C}^\times$, let $I(\chi_1, \chi_2)$ consist of locally constant functions $f : G \to \mathbb{C}$ satisfying

$$f\left(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} g\right) = |a/d|^{1/2} \chi_1(a)\chi_2(d) f(g).$$

The action is right translation, $(\pi(g)f)(h) = f(hg)$. Thus $I(\chi_1, \chi_2)$ is normalized smooth induction.

Set

$$z = \frac{\chi_1(\varpi)}{\chi_2(\varpi)}, \qquad w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad n(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

For $|z| < 1$, define

$$(M_z f)(g) = \int_F f(wn(x)g)\, dx. \tag{18.7}$$

The target is $I(\chi_2, \chi_1)$. The subscript z specifies the ratio parameter. The source and target characters remain part of the operator's data.

The compact picture and ball averages

Restriction to K identifies every $I(\chi_1, \chi_2)$ with

$$V = C^\infty(X; \mathbb{C}), \qquad X = (B \cap K)\backslash K = \mathbb{P}^1(F).$$

Here C^∞ means locally constant. The identification of X sends a matrix to its bottom row, up to scalar. It follows from $G = BK$ and the triviality of unramified characters on $O^\times$.

A primitive row has at least one unit coordinate. For primitive representatives $u, v \in O^2$, define

$$\rho([u], [v]) = |\det(u, v)|.$$

This function is independent of the primitive representatives and is invariant under the right action of K. Let ν be the K-invariant probability measure on X.

For $n \geq 1$, reduction modulo ϖ^n gives a finite set

$$X_n = \mathbb{P}^1(O/\varpi^n O), \qquad |X_n| = (q+1)q^{n-1}.$$

Let $V_n \subset V$ be the functions that factor through X_n. Put $V_0 = \mathbb{C}\mathbf{1}$ and $H = L^2(X, \nu)$, with inner product linear in the first variable. The orthogonal projection $E_n : H \to V_n$ averages over each reduction fiber.

The projection E_0 is global averaging. The fibers are the balls $\rho(x, y) \le q^{-n}$, each of measure

$$b_n = \frac{1}{(q+1)q^{n-1}}.$$

Every locally constant function factors through some X_n. Consequently $V = \bigcup_{n\ge 0} V_n$ is dense in H.

Define mutually orthogonal finite-dimensional spaces

$$H_0 = V_0, \qquad H_n = V_n \cap V_{n-1}^{\perp} \quad (n \ge 1).$$

Then

$$V = \bigoplus_{n\ge 0}^{\mathrm{alg}} H_n, \qquad H = \widehat{\bigoplus}_{n\ge 0} H_n.$$

In particular, $\dim H_1 = q$ and $\dim H_n = (q+1)(q-1)q^{n-2}$ for $n \ge 2$.

Lemma 18.9.1 *(Integral kernel and meromorphic continuation).* For $|z| < 1$, the integral in (18.7) converges absolutely. It defines a smooth G-intertwiner. In the compact picture it is

$$M_z = (1 + q^{-1})E_0 + (z - q^{-1}) \sum_{n\ge 1} z^{n-1} E_n. \tag{18.8}$$

The series converges in operator norm on H.

On the orthogonal summands it acts by

$$M_z|_{H_0} = c(z), \qquad c(z) = \frac{1 - q^{-1}z}{1 - z}, \qquad M_z|_{H_n} = \frac{(z - q^{-1})z^{n-1}}{1 - z} \quad (n \ge 1).$$

These formulas give a meromorphic continuation on every V_N. On every nonzero V_N, its only pole in $z \in \mathbb{C}^\times$ is the simple pole at $z = 1$. Here meromorphic continuation means continuation on each finite level, not a family of bounded operators on H for all parameters.

Proof. Let $x_0 = [(0, 1)]$. For $|x| \le 1$, the matrix $wn(x)$ belongs to K. For $|x| > 1$, the factorization

$$wn(x) = \begin{pmatrix} -x^{-1} & 1 \\ 0 & x \end{pmatrix} \begin{pmatrix} 1 & 0 \\ x^{-1} & 1 \end{pmatrix}$$

shows that the inducing factor on $|x| = q^n$ is $q^{-n}z^n$. The shell has additive measure $(1 - q^{-1})q^n$. Since compact-picture functions are bounded, the tail is dominated by $a \sum_{n\ge 1} |z|^n$ for a finite constant a. The same argument applies to each right translate of f.

The integral commutes with right translations. Changing x to $x + y$ proves invariance under left multiplication by $n(y)$. For $a, d \in F^\times$, use

$$wn(x) \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} = \begin{pmatrix} d & 0 \\ 0 & a \end{pmatrix} wn(xd/a).$$

The substitution $u = xd/a$ contributes $|a/d|$.

The resulting factor is $|a/d|^{1/2}\chi_2(a)\chi_1(d)$, which is the target induction law. Right local constancy is preserved by the integral. Thus M_z is the stated smooth intertwiner.

The chart $[(1, x)]$, $x \in O$, has measure $q(q+1)^{-1}dx$. The chart $[(y, 1)]$, $y \in \varpi O$, has the same density. For example, reduction modulo ϖ^N divides either chart into sets of additive measure q^{-N} and ν-measure $((q+1)q^{N-1})^{-1}$. This proves the density assertion on locally constant functions.

In the second chart, the substitution $y = x^{-1}$ has Jacobian $dx = |y|^{-2}dy$. On $|y| = q^{-n}$, the resulting density in the raw integral is $q^n z^n\, dy$. Thus its kernel, relative to ν, has value $1 + q^{-1}$ when $\rho(x_0, y) = 1$, and value $(1+q^{-1})q^n z^n$ when $\rho(x_0, y) = q^{-n}$. The diagonal has measure zero. Right K-equivariance gives the same radial kernel about every point.

For $\rho(x, y) = q^{-m}$, telescope the kernel as

$$(1+q^{-1})\left[1 + \sum_{n=1}^{m}\left(q^n z^n - q^{n-1}z^{n-1}\right)\right].$$

Integration of the nth ball indicator is $b_n E_n$. Since $(1+q^{-1})b_n = q^{-n}$, its coefficient is $(z - q^{-1})z^{n-1}$. Absolute convergence justifies the sum and proves (18.8). The projection norms are one, so the series also converges in operator norm.

On H_0, every E_n acts as the identity. On H_m, $m \geq 1$, the projection E_n is zero for $n < m$ and is the identity for $n \geq m$. Summing the geometric series gives the displayed eigenvalues. Each V_N is a finite sum of these spaces.

The rational formulas therefore give its meromorphic continuation. At $z = 1$, every numerator has nonzero value $1 - q^{-1}$. The pole is simple on every nonzero summand. □

Normalization, composition, and domains

The spherical vector is $\mathbf{1} \in H_0$. The scalar in the preceding lemma equals the unramified local ratio

$$c(z) = \frac{L(0, \chi_1\chi_2^{-1})}{L(1, \chi_1\chi_2^{-1})}, \qquad L(u, \chi_1\chi_2^{-1}) = (1 - zq^{-u})^{-1}.$$

Normalizing the spherical vector therefore requires division by $c(z)$. Define the meromorphic family

$$R_z = c(z)^{-1}M_z.$$

THEOREM 18.9.2 *(The normalized unramified intertwiner).* The compact-picture eigenvalues of R_z are

$$r_0(z) = 1, \qquad r_n(z) = z^{n-1}\frac{z - q^{-1}}{1 - q^{-1}z} \quad (n \geq 1). \tag{18.9}$$

The operator is holomorphic on each V_N away from $z = q$. It has a simple pole on every H_n with $n \geq 1$ at $z = q$. The value at $z = 1$ is id_V.

The value at $z = q^{-1}$ is E_0. For $z \notin \{q, q^{-1}\}$, the smooth maps satisfy

$$R_{z^{-1}}R_z = \mathrm{id}_V, \qquad R_z R_{z^{-1}} = \mathrm{id}_V\,.$$

Here the second operator uses the swapped inducing characters.

If χ_1 and χ_2 are unitary, R_z extends uniquely to a unitary intertwiner between their induced Hilbert representations on H. Its Hilbert domain is all of H. In particular, the equal-character point $z = 1$ is regular.

Proof. Division of the preceding eigenvalues by $c(z)$ gives (18.9). This calculation cancels the apparent singularity at $z = 1$. At $z = q$, the denominator has a simple zero and the numerator is nonzero on every H_n, $n \geq 1$. Substitution gives the stated values at 1 and q^{-1}.

For $n \geq 1$, direct cancellation gives

$$r_n(z)r_n(z^{-1}) = \frac{(z - q^{-1})(1 - q^{-1}z)}{(1 - q^{-1}z)(z - q^{-1})} = 1.$$

The identity also holds on H_0. Every smooth vector has finitely many components, so both compositions are defined on all of V at the stated parameters. Intertwining holds first on the convergence domain of the raw integral.

On each smooth vector and each fixed group element, the compact-picture formulas are rational in the two nonzero unramified character values. For a fixed $g \in G$, right translation sends V_N into some V_M independently of the unramified character values. Indeed, an open subgroup of $K \cap gK_Ng^{-1}$ fixes the translated vectors, where $K_N = 1 + \varpi^N M_2(\mathcal{O})$ for $N \geq 1$. On the resulting finite partition of K, the Iwasawa factors have fixed diagonal valuations.

Their character values are Laurent monomials. The integral intertwining identity therefore continues meromorphically. This also proves intertwining at every regular specialization.

If the inducing characters are unitary, $|z| = 1$. Then $|z - q^{-1}| = |1 - q^{-1}z|$, so $|r_n(z)| = 1$ for every n. The orthogonal decomposition of H gives a unique unitary extension. Its inverse has parameter z^{-1}.

For completeness, the induced action preserves the stated Hilbert norm. In the open Bruhat chart put $F_f(x) = f(wn(x))$. For unitary inducing characters, the chart measures above give

$$\|f|_K\|_H^2 = \frac{q}{q+1}\int_F |F_f(x)|^2\,dx.$$

Right translation by $n(y)$ gives $F_f(x + y)$.

Right translation by $\mathrm{diag}(a, d)$ gives

$$|d/a|^{1/2}\chi_1(d)\chi_2(a)F_f(xd/a).$$

Right translation by w, away from $x = 0$, gives

$$|x|^{-1}\chi_1(-x^{-1})\chi_2(x)F_f(x^{-1}).$$

Translations, dilations, and inversion preserve the displayed norm with these factors and their respective Jacobians. These elements generate G.

Each smooth vector has an open stabilizer. Density and unitarity therefore imply strong continuity of the extended action. The smooth equivariance of R_z extends by density to these induced unitary representations. □

For nonunitary parameters, the same compact-picture formulas also specify an operator on a fixed Hilbert space. This assertion does not equip the induced representation with an invariant positive inner product.

Proposition 18.9.3 *(Closed compact-picture realization).* For $z \neq q$, write $h = \sum_{n\geq 0} h_n$ with $h_n \in H_n$. The closure of $R_z|_V$ on H is the normal operator with domain

$$\mathrm{Dom}(\overline{R}_z) = \left\{h \in H : \sum_{n\geq 0} |r_n(z)|^2\|h_n\|^2 < \infty\right\}, \qquad \overline{R}_z h = \sum_{n\geq 0} r_n(z)h_n.$$

The space V is a core. This operator is bounded exactly when $|z| \le 1$. The composition assertions of the preceding theorem hold on V. On the unit circle they hold on all of H.

Proof. The orthogonal sum of scalar operators with the displayed maximal domain is closed and normal. Its adjoint has coefficients $\overline{r_n(z)}$ and the same domain. Truncating the expansion after $n = N$ converges in its graph norm. Thus this maximal operator is the closure of its restriction to V.

Boundedness is equivalent to $\sup_n |r_n(z)| < \infty$. Formula (18.9) proves boundedness for $|z| \le 1$. For $|z| > 1$ and $z \ne q$, the nonzero factor $(z - q^{-1})/(1 - q^{-1}z)$ is fixed and $|z|^{n-1}$ is unbounded. □

Reducibility and the unitary axis

The poles of a normalization and the reducibility of the induced representation are different loci. They can both be determined in the present unramified family.

PROPOSITION 18.9.4 *(Reducibility locus).* The smooth representation $I(\chi_1, \chi_2)$ is reducible exactly when

$$\chi_1\chi_2^{-1} = |\cdot| \quad \text{or} \quad \chi_1\chi_2^{-1} = |\cdot|^{-1}.$$

Equivalently, $z \in \{q^{-1}, q\}$. For $z = q^{-1}$, the regular intertwiner R_z has image H_0 in $I(\chi_2, \chi_1)$ and kernel in $I(\chi_1, \chi_2)$ equal to $\bigoplus_{n\ge1}^{\mathrm{alg}} H_n$. For $z = q$, the swapped regular intertwiner $R_{z^{-1}}$ has image H_0 in $I(\chi_1, \chi_2)$. The stated kernel at $z = q^{-1}$ and image at $z = q$ are nonzero proper invariant subspaces of $I(\chi_1, \chi_2)$.

Proof. The kernel and image assertions follow from (18.9) and the regular intertwining identity. They prove reducibility at the two displayed ratios. Away from these ratios, apply Bernstein–Zelevinsky's irreducibility criterion to the two characters of $\mathrm{GL}_1(F)$. This is *Induced representations of reductive p-adic groups I*, Theorem 4.2, p. 454. The induction conventions are in §1.8(a), p. 444, and §§2.2–2.3, p. 446.

For the upper triangular parabolic, their modulus on the unipotent radical is $|a/d|$, so their square-root factor is the one used here. Every character of $\mathrm{GL}_1(F)$ is irreducible and cuspidal. The criterion excludes exactly the relations $\chi_j = |\cdot|\chi_i$ for distinct i, j. Thus it proves irreducibility on the stated complement. Only this direction of their criterion is used. □

For the symmetric family $\chi_1 = |\cdot|^{s/2}$, $\chi_2 = |\cdot|^{-s/2}$, one has $z = q^{-s}$. The raw integral converges for $\Re(s) > 0$. Its poles occur at

$$s \in \frac{2\pi i}{\log q}\mathbb{Z}.$$

On every smooth finite level, the normalized operator is regular on $\Re(s) \ge 0$.

Its poles occur at

$$s \in -1 + \frac{2\pi i}{\log q}\mathbb{Z},$$

and reducibility occurs at

$$s \in \left(1 + \frac{2\pi i}{\log q}\mathbb{Z}\right) \cup \left(-1 + \frac{2\pi i}{\log q}\mathbb{Z}\right).$$

Neither reducibility locus meets the unitary axis $\Re(s) = 0$. These statements concern local unramified induction. Poles of a global Eisenstein series require a separate global construction.

Example 18.9.5 *(The first nonconstant vectors).* Let $q = 2$ and $z = -1$. The three reduction fibers of $X \to \mathbb{P}^1(\mathbb{F}_2)$ have equal measure. On V_1, the raw integral and normalized operator are

$$M_{-1} = \frac{1}{4}\begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}, \qquad R_{-1} = \frac{1}{3}\begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}.$$

The spherical scalar is $c(-1) = 3/4$.

The normalized matrix fixes constants and negates their orthogonal complement. Its square is the identity. At $z = 1/2$, the normalized matrix is the rank-one averaging projection. At $z = 1$, its continued value is the identity matrix.

The automorphic comparison

A categorical Eisenstein comparison requires categories and functors in addition to these operators. One must specify the coefficient field, the local parameter spaces, and the global automorphic quotient with its central character and level. The global category must carry its Hecke actions and its Eisenstein subcategory. A comparison functor must identify the relevant derived mapping objects and determine its essential image. The shared global induction parameter must also be imposed on local families.

The construction above supplies the full smooth intertwiner and its Hilbert extension only for unramified nonarchimedean induction. Ramified characters and the archimedean place require their own operators, normalizations, and analytic domains. A global restricted product requires compatible distinguished vectors and an explicit global family. The categorical comparison further requires functors implementing these maps and their Weyl coherences. No equivalence of automorphic categories follows from the compact-picture Hilbert isomorphism alone.

The place-indexed operator notation

Definition 18.9.6 *(the raw place-indexed operator).* For a finite local field F_v and unramified characters $\chi = (\chi_1, \chi_2)$, use normalized smooth induction

$$I_v(s, \chi) = \mathrm{Ind}_{B(F_v)}^{GL_2(F_v)}(\chi_1|\cdot|^{s/2} \otimes \chi_2|\cdot|^{-s/2}).$$

Put $\alpha_1 = \chi_1|\cdot|^{s/2}$, $\alpha_2 = \chi_2|\cdot|^{-s/2}$ and $z = \alpha_1(\varpi)/\alpha_2(\varpi)$. With $\chi^w = (\chi_2, \chi_1)$, the notation is

$$M_v(s, \chi) = M_z : I_v(s, \chi) = I(\alpha_1, \alpha_2) \longrightarrow I(\alpha_2, \alpha_1) = I_v(-s, \chi^w).$$

The integral is defined for $|z| < 1$ and continued on each smooth finite level by Lemma 18.9.1.

Definition 18.9.7 *(the normalized place-indexed operator).* On the same source and target, define

$$N_v(s, \chi) = R_z = c(z)^{-1} M_z, \qquad c(z) = \frac{1 - q^{-1}z}{1 - z}.$$

Cancellation and continuation at $z = 1$ are part of Theorem 18.9.2. The expression is not a pointwise product of undefined values there. Its poles, domains and unitary specializations are those proved in that theorem.

18.10 Algebraic parameters for unramified induction

The normalized unramified intertwiner varies with the inducing characters. A category of coefficient complexes records this variation and its Weyl symmetry. Its relation to smooth induction is a functor defined by tensoring with universal smooth induction. The resulting category concerns local algebraic parameters. An automorphic comparison requires further constructions.

Retain the nonarchimedean field F, residue cardinality q, group $G = \mathrm{GL}_2(F)$, and compact subgroup K of the unramified calculation. In particular, $|\varpi| = q^{-1}$ and $\mathrm{vol}(\mathcal{O}) = 1$. Let a, b denote the values at ϖ of the two unramified characters. Their ratio is $z = a/b$. To make both normalized intertwining directions regular and invertible, put

$$A = \mathbb{C}[a^{\pm 1}, b^{\pm 1}, (a - qb)^{-1}, (b - qa)^{-1}], \qquad T = \operatorname{Spec} A.$$

Thus T omits precisely the two divisors $z = q, q^{-1}$. The Weyl group $W = \{1, w\}$ acts through the ring involution $\sigma(a) = b$, $\sigma(b) = a$. The involution exchanges the two inverted divisors. At a complex point (a_0, b_0), define $\chi_{a_0}(x) = a_0^{v_F(x)}$, where $v_F(\varpi) = 1$. The corresponding representation is the normalized induction $I(\chi_{a_0}, \chi_{b_0})$.

The ratio alone determines the normalized operator in the compact picture. It does not determine the representation. Replacing (a_0, b_0) by (ta_0, tb_0) twists the representation by $\chi_t \circ \det$. For example, the central element $\varpi\,\mathrm{id}$ acts by $a_0 b_0$ before the twist and by $t^2 a_0 b_0$ afterwards. Both characters must therefore remain in the parameter space.

Coefficient complexes and their unit

For a ring D, write $\mathcal{D}(D)$ for the stable ∞-category of unbounded cochain complexes of left D-modules, with quasi-isomorphisms inverted. The differential has degree 1. We use $P[1]^n = P^{n+1}$ with differential $-d_P$. For commutative D, tensor products are derived over D. Their symmetry has sign $(-1)^{ij}$ on degrees i, j. Let $\mathrm{Perf}(D)$ be a small skeleton of the full subcategory generated by $D[0]$ under finite sums, shifts, cones, and retracts.

PROPOSITION 18.10.1 *(The coefficient category).* The category $\mathrm{Perf}(A)$ is stable, idempotent complete, and symmetric monoidal under $\otimes^{\mathbf{L}}_A$. There is a canonical equivalence

$$\mathrm{Ind}(\mathrm{Perf}(A)) \simeq \mathcal{D}(A).$$

The compact objects of $\mathcal{D}(A)$ are exactly $\mathrm{Perf}(A)$. The monoidal unit is $A[0]$. The functor

$$e : \mathcal{D}(\mathbb{C}) \longrightarrow \mathcal{D}(A), \qquad U \longmapsto A \otimes_{\mathbb{C}} U$$

is a colimit-preserving symmetric monoidal pointing and preserves compact objects.

Proof. The category of derived modules is stable and presentable. For a commutative ring it identifies symmetric monoidally with modules over the associated discrete ring spectrum. This is Lurie, *Higher Algebra* (2017), Theorem 7.1.2.13, pp. 1212–1213. Its shifts of A detect zero objects, since $\mathrm{Hom}_{\mathcal{D}(A)}(A[-n], P) = H^n(P)$.

These shifts are compact because cohomology commutes with filtered colimits of modules. The module compact-generation theorem identifies their finite stable retract closure with all compact objects and gives the Ind equivalence. See Lurie, *Higher Algebra* (2017), Definition 7.2.4.1 and Proposition 7.2.4.2, pp. 1260–1261. For ordinary rings these are bounded complexes of finitely generated projective modules (The Stacks Project, Tag 07LQ, Lemma 15.80.1 and Proposition 15.80.3).

Tensoring two such complexes gives another such complex. The associativity, unit, and symmetry are those of derived tensor product. Extension of scalars preserves colimits and takes a finite complex of finite-dimensional vector spaces to a finite free complex. Its tensor and unit identities prove the assertion about e. □

A point of T gives the specialization functor $\mathbb{C} \otimes^{\mathbf{L}}_{A,(a_0,b_0)} (-)$. The point module $A/(a - a_0, b - b_0)$ is also perfect. Its length-two Koszul resolution follows because $a - a_0, b - b_0$ form a regular sequence. Indeed, the quotient by the first element is a localization of $\mathbb{C}[b^{\pm 1}]$, and the second element is nonzero there.

The inverted elements remain nonzero at the chosen point. This point module differs from the monoidal unit A. For two distinct points, its tensor product with the other point module is zero, as follows by localizing at their disjoint supports. The unit cannot have this property.

The universal smooth representation

Let $X = (B \cap K)\backslash K$ and $V = C^\infty(X; \mathbb{C})$, where B is the upper triangular subgroup and functions are locally constant. The compact-picture calculation gives $V = \bigoplus_{n \geq 0} H_n$ and $V_N = \bigoplus_{n=0}^{N} H_n$, with $H_0 = \mathbb{C}\mathbf{1}$ and finite-dimensional H_n. Put $E = A \otimes_{\mathbb{C}} V$ and $E_N = A \otimes V_N$. All these modules are in cochain degree zero.

Proposition 18.10.2 *(Induction over the parameter ring).* There is an A-linear smooth action π_A of G on E. Its specialization at (a_0, b_0) is $I(\chi_{a_0}, \chi_{b_0})$. The normalized intertwiner defines an A-linear isomorphism

$$R : E \longrightarrow E^{\mathrm{sw}}, \qquad R|_{A \otimes H_0} = 1, \qquad R|_{A \otimes H_n} = z^{n-1} \frac{z - q^{-1}}{1 - q^{-1} z} \quad (n \geq 1),$$

where E^{sw} has the inducing parameters b, a. Its inverse has the displayed coefficients with z replaced by z^{-1}. The coefficient module E is free and is not compact in $\mathcal{D}(A)$. Every E_N is compact and K-stable.

Proof. Use the universal unramified characters $\chi_a(x) = a^{v_F(x)}$ and $\chi_b(x) = b^{v_F(x)}$ in the induction law. The factor $|x/y|^{1/2}$ is a fixed positive real scalar. For $g \in G$, extend a function from K to $BK = G$ by this law, then translate it on the right by g. Independence of the Iwasawa factorization follows because the characters and modulus are trivial on $B \cap K$. For each fixed g, N, compactness and local constancy give a finite partition on which all diagonal valuations are constant.

The resulting coefficients are Laurent monomials in a, b. Thus $\pi_A(g) E_N \subset E_M$ for some finite M. Right translation satisfies the group law before restriction to K, and consequently defines an action on E. Each vector has an open stabilizer in K. Specializing the induction law proves the fiber assertion.

The stated coefficients belong to A and are units in A. The normalized intertwiner calculation proves the intertwining identity and $r_n(z^{-1}) r_n(z) = 1$. These identities apply over A: for fixed g, N they are finite matrix identities in rational functions of a, b. They hold at every regular complex specialization, hence hold in A. This proves the asserted isomorphism and inverse.

Each E_N is finite free. The decomposition of V makes E free of infinite rank. If E were compact, its identity would factor through some term of $E = \operatorname{colim}_N E_N$. That would express an infinite-rank free module as a retract of a finite-rank module, which is impossible after tensoring with the fraction field of A. The compact-picture K-action preserves each V_N. The full G-action need not preserve these finite levels. □

Let $\mathcal{D}_{\mathrm{sm}}(G, A)$ denote the derived ∞-category of smooth A-linear G-representations. Here smooth means that every vector has an open stabilizer, and G acts trivially on A. Tensoring with the flat module E gives an actual functor

$$\mathcal{F} : \mathcal{D}(A) \longrightarrow \mathcal{D}_{\mathrm{sm}}(G, A), \qquad P \longmapsto E \otimes_A P.$$

It is well-defined on quasi-isomorphisms since E is free over A.

Its specialization is $I(\chi_{a_\circ}, \chi_{b_\circ}) \otimes_{\mathbb{C}} (\mathbb{C} \otimes^{\mathbf{L}}_A P)$. The spherical map $A \to E, c \mapsto c\mathbf{1}$, is K-equivariant. It is generally not G-equivariant when A has trivial G-action. For example, the central element $\mathfrak{Z}$ id acts on E by ab. Thus neither the spherical vector nor the representation E replaces the coefficient category's monoidal unit.

Weyl descent

The involution σ acts on coefficient complexes by restriction of scalars. On the underlying complex the new scalar action is $c \cdot m = \sigma(c)m$. Applying this operation twice is the identity. It preserves derived tensor products and the unit, with their usual canonical identifications. This is a specified coherent action of W on $\mathcal{D}(A)$. Its descent category is the homotopy limit $\mathcal{D}(A)^{hW}$.

PROPOSITION 18.10.3 *(An explicit descent category).* Let $D = A \rtimes W = A \oplus Au$, with multiplication given by $u^2 = 1$ and $uc = \sigma(c)u$ for $c \in A$. There is an equivalence

$$\mathcal{D}(A)^{hW} \simeq \mathcal{D}(D).$$

This category is stable and presentable. An object is compact if and only if its underlying A-complex is perfect. Its symmetric monoidal product is derived tensor over A, with diagonal Weyl action. Its compact monoidal unit is A with $u(c) = \sigma(c)$.

Proof. A strict D-module is an A-module with a semilinear involution θ: $\theta(cm) = \sigma(c)\theta(m)$ and $\theta^2 = 1$. The same description applies to complexes with a chain involution. To identify the derived category with coherent descent, consider the evaluation functor from the homotopy limit to $\mathcal{D}(A)$. It is conservative and has the free-equivariant left adjoint $P \mapsto P \oplus \sigma^* P$ with its permutation action.

Colimits are computed on underlying complexes since σ^* preserves them. In particular, evaluation preserves geometric realizations of split simplicial objects. The monadicity theorem therefore applies (Lurie, *Higher Algebra* (2017), Theorem 4.7.3.5, p. 685). Its monad is $P \mapsto P \oplus \sigma^* P$, with multiplication from W.

This is also the monad of restriction from D-modules to A-modules, whose left adjoint is $D \otimes^{\mathbf{L}}_A (-)$. Restriction is conservative and preserves geometric realizations. The two monadic descriptions prove the equivalence, including all higher coherences.

As a left or right A-module, D is free of rank two. Thus every perfect D-complex becomes perfect over A. Conversely, a D-complex M is a retract of $D \otimes_A M$. The action map is a retraction of the D-linear chain map

$$m \longmapsto \tfrac{1}{2}(1 \otimes m + u \otimes \theta(m)).$$

The relations $uc = \sigma(c)u$ and $\theta^2 = 1$ verify linearity directly. If the underlying A-complex is perfect, extension of scalars makes $D \otimes^{\mathbf{L}}_A M$ perfect over D. The retract is therefore perfect. The compact-object characterization in the coefficient proposition applies to D as well.

The diagonal involution on $M \otimes^{\mathbf{L}}_A N$ is $\theta_M \otimes \theta_N$. Semilinearity makes it well-defined, and the usual tensor constraints are equivariant. The module A with involution σ is the unit and is perfect over A. This proves its compactness after descent. □

The involution is not free on T. Every point (a_0, a_0) is fixed and belongs to T. The modules A with involutions σ and $-\sigma$ specialize there to the trivial and sign representations of W. These are not isomorphic: an intertwining scalar would satisfy $c = -c$. Consequently descent retains stabilizer representations. It cannot be replaced by the category of modules on the coarse orbit space through ordinary pullback, which has trivial stabilizer action on every fiber.

The representation E has its own compatible descent datum. Let $S : E \to E$ apply σ to coefficients and fix V. Define $\Theta = SR$, using the common compact-picture module. Then

$$\Theta(cf) = \sigma(c)\Theta(f), \qquad \Theta^2 = 1, \qquad \Theta\pi_A(g) = \pi_A(g)\Theta.$$

Indeed, $S\pi_A(g) = \pi_A^{\mathrm{sw}}(g)S$ and $R\pi_A(g) = \pi_A^{\mathrm{sw}}(g)R$.

Conjugating R by S replaces z by z^{-1}, so its inverse is obtained. These identities prove all three assertions. Moreover, $\Theta(c\mathbf{1}) = \sigma(c)\mathbf{1}$. For an equivariant coefficient complex (P, θ), the formula $(E \otimes_A P, \Theta \otimes \theta)$ therefore defines equivariant smooth induction. This is a construction of a functor, without an assertion of full faithfulness or essential surjectivity.

A common complex parameter

For the symmetric inducing characters $|\cdot|^{s/2}, |\cdot|^{-s/2}$, the two parameters are $a = q^{-s/2}$ and $b = q^{s/2}$. Here $q^s = \exp(s \log q)$, with the real logarithm of q. The ratio $z = q^{-s}$ has period $2\pi i/\log q$. The ordered pair (a, b) has period $4\pi i/\log q$. A shift by the smaller period changes both character values by -1. This explains why the ratio does not encode the whole family.

Let $\Omega = \{s \in \mathbb{C} : |\operatorname{Re} s| < 1\}$ and $B_\Omega = \mathcal{O}(\Omega)$, treated as an ordinary commutative ring. There is a ring map

$$\phi_q : A \longrightarrow B_\Omega, \qquad a \longmapsto e^{-s \log q/2}, \qquad b \longmapsto e^{s \log q/2}.$$

The inverted factors have no zeros in Ω because $|q^{-s}| = q^{-\operatorname{Re} s}$ lies strictly between q^{-1} and q. Their reciprocals are holomorphic, which proves that the map is defined.

The involution $\tau(h)(s) = h(-s)$ satisfies $\phi_q\sigma = \tau\phi_q$. Derived extension of scalars gives a compact-preserving functor

$$B_\Omega \otimes^{\mathbf{L}}_A (-) : \mathcal{D}(A) \longrightarrow \mathcal{D}(B_\Omega)$$

and an equivariant version on descent categories. The latter sends (P, θ) to the complex with involution $h \otimes p \mapsto \tau(h) \otimes \theta(p)$. All finite-level matrices of E and R pull back to holomorphic matrices.

Evaluating at s gives exactly the smooth representation and normalized operator at $z = q^{-s}$. On $s \in i\mathbb{R}$, the previously constructed Hilbert extension is unitary. No topology on B_Ω or Hilbert completion is part of this scalar-extension functor.

For a finite set of finite places S, allow independent local character parameters and put $A_S = \bigotimes_{v \in S, \mathbb{C}} A_v$. The common symmetric family gives a single map

$$\phi_S : A_S \longrightarrow B_\Omega, \qquad \bigotimes_v f_v \longmapsto \prod_v \phi_{q_v}(f_v).$$

It imposes the same variable s at every place. For $S \subset S'$, scalar extension along $A_S \to A_{S'}$ supplies compatible coefficient functors, and $\phi_{S'}$ restricts to ϕ_S.

Thus these maps also define a map from the filtered union of the rings A_S. They give actual base-change functors for independent local parameters. The compatible Weyl action is the simultaneous swap at all places. An individual local swap generally does not preserve the common family.

For example, with $q_2 = 2$ and $q_3 = 3$, the independent unitary ratios $(z_2, z_3) = (1, -1)$ do not come from any common s. The first equation forces $s = 2\pi i n/\log 2$ for an integer n. The second would give $2n \log 3 = (2m + 1) \log 2$ for an integer m, after changing signs of the indices if necessary. Exponentiation contradicts unique factorization of rational numbers. Thus tensoring independent unitary axes does not impose the global parameter.

After common base change, put $E_{v,\Omega} = B_\Omega \otimes_{A_v} E_v$. Write $R_{v,\Omega}$ for the scalar extension of R_v. The local semilinear involution $\Theta_v = S_v R_v$ induces

$$\Theta_{v,\Omega}(h \otimes f) = \tau(h) \otimes \Theta_v(f).$$

This formula is balanced because $\phi_{q_v} \sigma_v = \tau \phi_{q_v}$.

It squares to the identity, commutes with G_v, and fixes $\mathbf{1}_v$. For finite S, put $E_{S,\Omega} = \bigotimes_{v \in S, B_\Omega} E_{v,\Omega}$. The B_Ω-linear operator $\bigotimes_{v \in S} R_{v,\Omega}$ has target the family with every inducing pair swapped. The semilinear involution on $E_{S,\Omega}$ is instead defined by

$$\Theta_S\Big(\bigotimes_{v \in S} f_v\Big) = \bigotimes_{v \in S} \Theta_{v,\Omega}(f_v).$$

For $S = \varnothing$, set $E_{S,\Omega} = B_\Omega$ and $\Theta_S = \tau$. For nonempty S, moving a scalar between two tensor factors changes both resulting expressions by its image under τ. Thus the displayed map is balanced and τ-semilinear. Applying it twice gives the identity on pure tensors, hence on the whole module.

It commutes with the action of $\prod_{v \in S} G_v$. Insertion of $\mathbf{1}_v$ defines compatible maps as S grows. These maps intertwine Θ_S and $\Theta_{S'}$ because each $\Theta_{v,\Omega}$ fixes the inserted vector. This also holds for the insertion from the empty set, by semilinearity.

The algebraic filtered colimit is the restricted tensor product of these smooth families over finite places, relative to $\mathbf{1}_v$. It therefore inherits a semilinear involution from the maps Θ_S. It carries the restricted product group relative to K_v: outside finitely many places an element belongs to K_v and fixes $\mathbf{1}_v$. Every tensor and every group action is computed at some finite stage. No infinite product of operator norms is needed for this algebraic construction.

The automorphic comparison problem

The common family above fixes the two global inducing characters to be trivial before the powers of the absolute value are inserted. For general inducing characters, their local components must arise from specified Hecke characters, and the Weyl action must also swap that character pair. Independent local characters do not by themselves supply these global data.

An analytic coefficient category still requires a choice of sheaf theory, topology, growth conditions, and derived completion on a complex parameter domain. The ring category $\mathcal{D}(B_\Omega)$ specifies none of those choices. An automorphic target further requires the adelic quotient, central character, level, archimedean factor, and an Eisenstein functor with analytic domains. To compare that target with the parameter category, one must construct the functor on complexes and compare its derived mapping objects and essential image.

Compactness of coefficient complexes does not prove admissibility or finite generation over local Hecke algebras after induction. The local smooth family, its coherent Weyl descent, and common-parameter base change provide the input maps for this comparison. They do not identify a category of automorphic representations.

Remark 18.10.4 *(global normalizations and the pairing comparison).* The local operator $N_v(s,\chi) = R_z$ divides the raw operator by its spherical scalar $c(z)$. A global product requires Hecke characters, compatible spherical insertions, the archimedean and ramified factors, and an analytic continuation on a specified global family. For the trivial level-one Eisenstein datum, put

$$\Lambda(u) = \pi^{-u/2}\Gamma(u/2)\zeta(u), \qquad \xi(u) = \tfrac{1}{2}u(u-1)\Lambda(u).$$

The classical scattering factor is $\Lambda(2u-1)/\Lambda(2u)$, with symmetric induction variable $s = 2u-1$.

The entire-completion ratio instead satisfies

$$\frac{\xi(2u-1)}{\xi(2u)} = \frac{u-1}{u}\frac{\Lambda(2u-1)}{\Lambda(2u)}.$$

The two phases are different. For a general inducing character, the global epsilon product retains its character and additive convention. It is not universally 1.

An operator, a scalar phase and a Plancherel density are different data. The Koszul–Petersson transport requires its actual pairing map and measure identity. If an even measure on the full axis has density $\frac{1}{2}w(t)dt$, pushforward under $t \mapsto |t|$ has density $w(t)dt$ off zero. This folding identity does not determine w.

Definition 18.10.5 *(local degeneration and the regular unitary locus).* For the symmetric unramified family at a finite place, define

$$\Sigma_v^{\mathrm{exc}} = \left(1 + \frac{2\pi i}{\log q_v}\mathbb{Z}\right) \cup \left(-1 + \frac{2\pi i}{\log q_v}\mathbb{Z}\right).$$

These are reducibility parameters. The normalized operator has poles only on the second set and is a projection at the first set. The non-exceptional unitary locus is all of $i\mathbb{R}$. The point 0 is regular and 1 is not on the unitary axis.

Remark 18.10.6 *(local degeneration and global residues).* A residue of a global Eisenstein series is not a pole of every normalized local operator. Its parameter, global quotient and central character must be specified. The local kernel/image sequences at $z = q^{\pm 1}$ are given by Proposition 18.9.4. Comparison of their extension classes with arithmetic deformation data remains a separate coefficient-compatible construction.

18.11 The regular unramified family

The normalized unramified intertwiner loses invertibility at two character ratios. Its cone records the resulting kernels and quotients, including those created by specialization. To define this cone across both ratios, fix the scalar used to clear the denominator.

Let F be a finite extension of $\mathbb{Q}_p$ with ring of integers O, uniformizer ϖ, and residue cardinality q. Set $|\varpi| = q^{-1}$ and normalize additive Haar measure by $\mathrm{vol}(O) = 1$. Put $G = \mathrm{GL}_2(F)$, $K = \mathrm{GL}_2(O)$, and let B be the upper triangular subgroup. The coefficient ring and ratio are

$$R = \mathbb{C}[a^{\pm 1}, b^{\pm 1}], \qquad z = a/b.$$

Every coefficient ring in this local-family construction is an ordinary commutative $\mathbb{C}$-algebra. It has no analytic topology. The universal unramified characters are $\chi_a(x) = a^{v_F(x)}$ and $\chi_b(x) = b^{v_F(x)}$, with $v_F(\varpi) = 1$.

Let $E = I_R(\chi_a, \chi_b)$ be the smooth normalized induction. Its elements are locally constant functions $f : G \to R$ satisfying

$$f\left(\begin{pmatrix} u & v \\ 0 & t \end{pmatrix} g\right) = |u/t|^{1/2} a^{v_F(u)} b^{v_F(t)} f(g).$$

The square root is positive real, and G acts by right translation. Restriction to K identifies its coefficient module with $R \otimes_{\mathbb{C}} V$, where

$$X = (B \cap K)\backslash K = \mathbb{P}^1(F), \qquad V = C^\infty(X; \mathbb{C}).$$

Here C^∞ means locally constant. Indeed, $G = BK$, and the inducing factor is trivial on $B \cap K$. For a fixed $g \in G$, compactness partitions K into finitely many pieces with constant diagonal valuations in an Iwasawa decomposition of kg. Right translation therefore has Laurent polynomial coefficients on every finite space of functions.

This constructs the action over R and proves its compatibility with arbitrary coefficient extension. Every vector is fixed by an open subgroup of K. Define $E^{\mathrm{sw}} = I_R(\chi_b, \chi_a)$ in the same way. These are different G-actions on the same coefficient module.

Let ν be the K-invariant probability measure on X. For $N \geq 1$, let V_N be the functions pulled back from $X_N = \mathbb{P}^1(\mathcal{O}/\varpi^N\mathcal{O})$, and put $V_0 = \mathbb{C}\mathbf{1}$. Averaging on reduction fibers defines projections $P_N : V \to V_N$. Let P_0 be global averaging, and define

$$H_0 = V_0, \qquad H_n = \ker(P_{n-1}|_{V_n}) \quad (n \geq 1).$$

Thus $V = \bigoplus_{n\geq 0} H_n$ and $V_N = \bigoplus_{n=0}^{N} H_n$. Write $H_+ = \bigoplus_{n\geq 1} H_n$. The dimensions are $\dim H_0 = 1$, $\dim H_1 = q$, and $\dim H_n = (q+1)(q-1)q^{n-2}$ for $n \geq 2$. These are K-stable spaces. They need not be G-stable.

The unramified integral calculation fixes the normalization. With $w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $n(x) = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$, it gives

$$M_z f(g) = \int_F f(wn(x)g)\, dx, \qquad c(z) = \frac{1 - q^{-1}z}{1 - z}, \qquad R_z = c(z)^{-1} M_z.$$

The integral is initially defined at complex specializations with $|z| < 1$. Its rational continuation, on each V_N, satisfies

$$R_z|_{H_0} = 1, \qquad R_z|_{H_n} = z^{n-1}\frac{z - q^{-1}}{1 - q^{-1}z} \quad (n \geq 1). \tag{18.10}$$

For clarity, the compact-picture calculation underlying this formula is

$$M_z = (1 + q^{-1})P_0 + (z - q^{-1})\sum_{n\geq 1} z^{n-1} P_n.$$

A reduction fiber at depth n has measure $((q+1)q^{n-1})^{-1}$. For primitive rows $u, v \in \mathcal{O}^2$, set $\rho([u],[v]) = |\det(u,v)|$. The radial integral kernel has successive values $(1 + q^{-1})q^n z^n$ at distance q^{-n}. Subtracting successive values gives the displayed projection coefficients.

On H_m, $m \geq 1$, only the terms $n \geq m$ remain. Their geometric sum is $(z - q^{-1})z^{m-1}/(1 - z)$. On constants the sum is $c(z)$, which proves (18.10). The integral commutes with right translation and exchanges the inducing characters by the substitution $x \mapsto xd/u$ for a diagonal matrix $\mathrm{diag}(u, d)$. Its intertwining identity consequently continues as a rational identity on finite spaces of vectors.

Set

$$\alpha = 1 - q^{-1}z, \qquad \beta = z - q^{-1}, \qquad h = (1 - q^{-1}z)(1 - q^{-1}z^{-1}) = z^{-1}\alpha\beta.$$

The chosen regular map is

$$D = \alpha R_z = (1 - z)M_z : E \longrightarrow E^{\mathrm{sw}}, \qquad D|_{R\otimes H_0} = \alpha, \quad D|_{R\otimes H_n} = z^{n-1}\beta. \tag{18.11}$$

The equalities with meromorphic operators mean equality of rational families. The right-hand coefficients define D over all of R, including $z = 1$. Multiplication by an additional nonunit would change the cone. Thus the scalar in (18.11) is part of the construction.

Example 18.11.1. Let $q = 2$ and $N = 1$. The three residue directions have equal measure. In their characteristic-function basis, put $J = (1)_{i,j=1}^{3}$. The projection onto constants is $P = J/3$, so

$$D_1 = \alpha P + \beta(1 - P) = \frac{1}{2}\begin{pmatrix} z & 1 - z & 1 - z \\ 1 - z & z & 1 - z \\ 1 - z & 1 - z & z \end{pmatrix}, \qquad \det D_1 = (1 - z/2)(z - 1/2)^2.$$

At $z = 1/2$ its value is $J/4$, with kernel the two-dimensional sum-zero space. At $z = 2$ its value is $\frac{3}{2}(1 - P)$, with kernel the constants. At $z = 1$ its value is $\frac{1}{2}\,\mathrm{id}$. The matrix with swapped inducing characters is $D_1(z^{-1})$, and

$$D_1(z^{-1})D_1(z) = (1 - z/2)(1 - z^{-1}/2)\,\mathrm{id}\,.$$

Indeed, both matrices preserve constants and their sum-zero complement. Multiplying the two eigenvalues on either space gives the displayed scalar. The two distinct kernels determine the two support summands of the general cone.

Proposition 18.11.2. The map D is an R-linear smooth G-intertwiner on all of E. Let $D^{\mathrm{sw}} : E^{\mathrm{sw}} \to E$ be the same regular construction with (a, b) replaced by (b, a). Then

$$D^{\mathrm{sw}}D = h\,\mathrm{id}_E, \qquad DD^{\mathrm{sw}} = h\,\mathrm{id}_{E^{\mathrm{sw}}}\,. \tag{18.12}$$

On H_0 the coefficient of D^{sw} is β/z. On H_n, $n \geq 1$, it is $z^{-n}\alpha$. After extension to $R[h^{-1}]$, the normalized operator is $R_z = \alpha^{-1}D$ and its inverse is $(1 - q^{-1}z^{-1})^{-1}D^{\mathrm{sw}}$.

Proof. Every vector has only finitely many H_n components, so (18.11) defines a map on all smooth vectors. For fixed g, N, both sides of the intertwining identity lie in some $R \otimes V_M$. Their coefficients are Laurent polynomials. The rational intertwining identity established above proves their equality in the fraction field of R, hence in R.

Replacing z by z^{-1} gives the stated opposite coefficients. Their products are h on every summand, which proves (18.12). Inverting h inverts both α and β, and the normalized composition follows. □

The ordered pair cannot be replaced by its ratio in the representation data. The central element $\varpi\,\mathrm{id}$ acts by ab. Replacing (a, b) by (ta, tb) leaves z fixed and tensors the representation with $\chi_t \circ \det$. The swap in Proposition 18.11.2 always exchanges the two characters.

18.12 The cone and its coefficient extensions

Use cochain complexes, with differentials of degree 1 and $P[1]^j = P^{j+1}$ with differential $-d_P$. The cone of the map between degree-zero representations is

$$C = \mathrm{Cone}(D) = [\, E \xrightarrow{D} E^{\mathrm{sw}} \,], \qquad C^{-1} = E, \quad C^0 = E^{\mathrm{sw}}.$$

Let $\mathcal{D}_{\mathrm{sm}}(G, R)$ denote the derived category of smooth R-linear representations of G. Its objects are complexes, with quasi-isomorphisms inverted.

The action on coefficients is trivial. The displayed complex is an object of this category. Its terms are free as R-modules.

THEOREM 18.12.1. The map D is injective over R. Writing $Q_{\mathrm{loc}} = \operatorname{coker} D$, the augmentation $C \to Q_{\mathrm{loc}}[0]$ is a G-equivariant quasi-isomorphism. As an (R, K)-module,

$$Q_{\mathrm{loc}} \cong (R/(\alpha)) \otimes H_0 \;\oplus\; (R/(\beta)) \otimes H_+. \tag{18.13}$$

Its support is the disjoint union of the two divisors $z = q$ and $z = q^{-1}$. It is annihilated by h. The two support summands in (18.13) are G-stable.

For every commutative R-algebra S, the complex

$$C_S = [\, S \otimes_R E \xrightarrow{1 \otimes D} S \otimes_R E^{\mathrm{sw}} \,]$$

computes the derived coefficient extension $S \otimes^{\mathbf{L}}_R C$. As S-modules with K-action, its cohomology is

$$H^{-1}(C_S) = (\mathrm{ann}_S\, \alpha) \otimes_{\mathbb{C}} H_0 \oplus (\mathrm{ann}_S\, \beta) \otimes_{\mathbb{C}} H_+,$$
$$H^0(C_S) = (S/\alpha S) \otimes_{\mathbb{C}} H_0 \oplus (S/\beta S) \otimes_{\mathbb{C}} H_+.$$

There is no cohomology in other degrees. Here α, β, z denote their images in S.

Proof. The ring R is a domain. Every eigenvalue in (18.11) is nonzero, so multiplication is injective on each free summand. The direct sum remains injective. Since z is a unit, the image on $R \otimes H_n$ is $\beta(R \otimes H_n)$ for $n \geq 1$. This proves (18.13) and the quasi-isomorphism.

The identity

$$\alpha + q^{-1}\beta = 1 - q^{-2} \tag{18.14}$$

shows that the two defining ideals are comaximal. Both quotient summands are nonzero, so their supports are exactly the stated divisors. Scalar multiplication by $h = z^{-1}\alpha\beta$ kills them. The Chinese remainder idempotents of $R/(\alpha\beta)$ commute with G. Their images therefore give the G-stable support summands. This argument makes no claim that individual H_n are G-stable.

A bounded complex of free coefficient modules computes derived extension of scalars. Tensoring (18.11) with S gives the displayed complex. The factors z^{n-1} remain units. Computing kernels and cokernels of scalar multiplication on each direct summand gives the formulas. □

For a coefficient map $S \to T$, the maps $s \otimes f \mapsto \bar{s} \otimes f$ define a G-equivariant chain map $C_S \to C_T$. They compose strictly, and $T \otimes^{\mathbf{L}}_S C_S \simeq C_T$. These are the specialization maps of this family. In particular, evaluation at a point $x = (a_0, b_0) \in (\mathbb{C}^\times)^2$ gives

$$C_x = [\, I(\chi_{a_0}, \chi_{b_0}) \xrightarrow{D_x} I(\chi_{b_0}, \chi_{a_0}) \,].$$

This statement includes nonflat coefficient maps. The global equality $\ker D = 0$ does not imply $\ker D_x = 0$. The new kernel is $\mathrm{Tor}^R_1(\mathbb{C}_x, Q_{\mathrm{loc}})$. Distinct closed points have no specified specialization map between their fibers. The maps above arise from actual coefficient homomorphisms.

18.13 The representations on the exceptional divisors

The support decomposition permits a full G-equivariant description. Let $J = C^\infty(B\backslash G;\mathbb{C})$ be unnormalized induction of the trivial character, with right translation. Its constant functions form the trivial representation. Define the Steinberg representation here by the quotient

$$\mathrm{St} = J/\mathbb{C}\mathbf{1}.$$

Only this definition is needed below.

Its compact-picture coefficient space is V/H_0. For any invertible coefficient t, write $\chi_t(x) = t^{v_F(x)}$. Define free R-modules with smooth G-action by

$$L_\alpha = R \otimes (\chi_{q^{1/2}b} \circ \det), \qquad L_\beta = (R \otimes_{\mathbb{C}} \mathrm{St}) \otimes (\chi_{q^{1/2}a} \circ \det).$$

The first notation means a rank-one free module on which g acts by $(q^{1/2}b)^{v_F(\det g)}$. The second combines the indicated scalar with the usual action on St.

Theorem 18.13.1. There is an isomorphism of smooth R-linear G-representations

$$Q_{\mathrm{loc}} \cong L_\alpha/\alpha L_\alpha \oplus L_\beta/\beta L_\beta. \tag{18.15}$$

Consequently, in $\mathcal{D}_{\mathrm{sm}}(G,R)$,

$$C \simeq [\, L_\alpha \xrightarrow{\alpha} L_\alpha \,] \oplus [\, L_\beta \xrightarrow{\beta} L_\beta \,], \tag{18.16}$$

with both source terms in degree -1. The equivalence is specified by the augmentations of both sides to (18.15). It remains an equivalence after every derived coefficient extension.

Proof. On $\beta = 0$, the ordered parameters satisfy $b = qa$. The target induction has ordered pair $(b,a) = (qa,a)$. Put $\lambda = q^{1/2}a$. Its normalized induction factor is

$$|u/t|^{1/2}(qa)^{v_F(u)}a^{v_F(t)} = \lambda^{v_F(ut)}.$$

Thus $E^{\mathrm{sw}}|_{\beta=0}$ is J tensored with $\chi_\lambda \circ \det$. The image of D there is precisely the constants, since $\alpha|_{\beta=0} = 1 - q^{-2}$ is nonzero. Taking the quotient gives $L_\beta/\beta L_\beta$. This computation holds over the divisor ring $R/(\beta)$.

On $\alpha = 0$, the ordered pair is $(a,b) = (qb,b)$. The source $E|_{\alpha=0}$ is J tensored with $\chi_{q^{1/2}b} \circ \det$, by the same induction-law calculation. The target has ratio q^{-1}. The regular normalized intertwiner from this target to the source is P_0, by (18.10).

Its image is the constant line in the source, and its kernel is H_+. The map D from the source to the target has image H_+, since all its positive-level coefficients are units on this divisor. Therefore P_0 identifies the cokernel with $L_\alpha/\alpha L_\alpha$. Together with the G-stable support decomposition, this proves (18.15).

Multiplication by α on L_α and by β on L_β is injective. The two displayed complexes are free coefficient resolutions of the corresponding summands of Q_{loc}. Their augmentations and $C \to Q_{\mathrm{loc}}$ give the claimed equivalence. All terms of the two resolutions and of C are coefficient-flat. Hence these resolutions also compute every derived coefficient extension. □

PROPOSITION 18.13.2. At a complex point $x = (a_0, b_0)$, put $z_0 = a_0/b_0$. If $z_0 \notin \{q, q^{-1}\}$, then C_x is acyclic. At the two exceptional ratios, define

$$W_x = \begin{cases} \chi_{q^{1/2}b_0} \circ \det, & z_0 = q, \\ \mathrm{St} \otimes (\chi_{q^{1/2}a_0} \circ \det), & z_0 = q^{-1}. \end{cases}$$

Then $H^{-1}(C_x) \cong W_x \cong H^0(C_x)$ as smooth G-representations.

In the full category $\mathcal{D}_{\mathrm{sm}}(G, \mathbb{C})$,

$$C_x \simeq W_x[1] \oplus W_x.$$

In the compact picture, the kernel and cokernel spaces are H_0 at $z_0 = q$ and H_+ at $z_0 = q^{-1}$. The cokernel means the quotient of the target by the complementary image.

Proof. Outside the two divisors every coefficient of D_x is nonzero, so the map is invertible on V. At $z_0 = q$, the H_0 coefficient is zero and all positive-level coefficients are nonzero. At $z_0 = q^{-1}$ the reverse holds. This computes kernels and cokernels as K-spaces.

Applying coefficient extension to (18.16) identifies the full derived G-objects. On the vanishing divisor the relevant Koszul differential becomes zero. The other differential becomes an invertible scalar, so its complex is contractible. The surviving representation is exactly W_x. □

Remark 18.13.3 *(The central character).* The action on E, E^{sw}, C, and Q_{loc} satisfies

$$\pi(t\,\mathrm{id}) = (ab)^{v_F(t)}\,\mathrm{id} \qquad (t \in F^\times).$$

One may require this identity strictly on every object and complex. It defines a smaller abelian category of smooth representations and its derived category. The cone C, its augmentation to Q_{loc}, and its coefficient specializations belong to that category, with the coefficient character specialized when necessary.

The lifts L_α and L_β have central characters $(qb^2)^{v_F(t)}$ and $(qa^2)^{v_F(t)}$, respectively. These equal $(ab)^{v_F(t)}$ only after imposing their respective divisor equations. Thus (18.16) is an equivalence in the full category stated in Theorem 18.13.1. The universal lifts therefore do not prove a splitting inside the strict universal-central-character category.

At a fixed exceptional complex point, however, the same actual complex C_x does split in the strictly specialized category. The next construction supplies a projective resolution and the two quasi-isomorphisms that establish this assertion. Universal marked-family compatibility remains a separate question.

18.14 STRICT CENTRAL CHARACTER AT A COMPLEX POINT

At an exceptional unramified character ratio, the regular intertwining complex has two nonzero cohomology groups. A splitting must preserve the central character on every representation and every morphism. The lattice tree supplies a projective resolution inside that category and removes the degree-two obstruction.

Use the same local field and operator as above. More explicitly, let F be a finite extension of $\mathbb{Q}_p$. Write O for its ring of integers, ϖ for a uniformizer, and q for its residue cardinality. Normalize $v_F(\varpi) = 1$ and $|\varpi| = q^{-1}$.

Set $G = \mathrm{GL}_2(F)$, with center $Z = F^\times\,\mathrm{id}$. In this fixed-point splitting construction, representations are complex vector spaces with smooth group action: each vector has an open stabilizer. All tensor products are algebraic over $\mathbb{C}$.

Fix $a_0, b_0 \in \mathbb{C}^\times$ with $z_0 = a_0/b_0 \in \{q, q^{-1}\}$. Define $\chi_t(u) = t^{v_F(u)}$ for $t \in \mathbb{C}^\times$. Let $E_x = I(\chi_{a_0}, \chi_{b_0})$ be normalized induction from the upper triangular subgroup B. Thus its locally constant functions satisfy

$$f\left(\begin{pmatrix} u & v \\ 0 & t \end{pmatrix} g\right) = |u/t|^{1/2} \chi_{a_0}(u)\chi_{b_0}(t) f(g), \qquad (\pi(g)f)(h) = f(hg),$$

with the positive square root.

Write $E_x^{\mathrm{sw}} = I(\chi_{b_0}, \chi_{a_0})$. Both representations have central character

$$\zeta(t\,\mathrm{id}) = (a_0 b_0)^{v_F(t)}.$$

Let $\mathcal{A}_\zeta$ be the abelian category of smooth complex G-representations on which every t id acts exactly by this scalar. Its morphisms are G-equivariant linear maps.

Kernels and cokernels in smooth representations preserve the stated identity. Write $\mathcal{D}^b(\mathcal{A}_\zeta)$ for its bounded derived category. Use cochain complexes with differential of degree 1 and $P[1]^j = P^{j+1}$ with differential $-d_P$.

The regular intertwiner is the specialization of $D_z = (1 - q^{-1}z)R_z = (1 - z)M_z$, with additive measure $\mathrm{vol}(O) = 1$ in the raw intertwining integral M_z. The source and target are the ordered inducing pairs and their swaps. The established compact-picture formula is

$$D_z|_{H_0} = 1 - q^{-1}z, \qquad D_z|_{H_n} = z^{n-1}(z - q^{-1}) \quad (n \geq 1).$$

Here H_0 is the constant line on $\mathbb{P}^1(F)$.

For $n \geq 1$, H_n consists of functions on $\mathbb{P}^1(O/\varpi^n O)$ with zero average on every fiber of reduction to level $n - 1$. At level zero, reduction means global averaging. These spaces form the algebraic direct sum of all locally constant functions. The regular map $D_x : E_x \to E_x^{\mathrm{sw}}$ is a smooth G-intertwiner.

The complex under consideration is

$$C_x = [\, E_x \xrightarrow{D_x} E_x^{\mathrm{sw}} \,], \qquad C_x^{-1} = E_x, \quad C_x^0 = E_x^{\mathrm{sw}}. \tag{18.17}$$

Its kernel and cokernel are both isomorphic to

$$W_x = \begin{cases} \chi_{q^{1/2}b_0} \circ \det, & z_0 = q, \\ \mathrm{St} \otimes (\chi_{q^{1/2}a_0} \circ \det), & z_0 = q^{-1}, \end{cases} \tag{18.18}$$

where $\mathrm{St} = C^\infty(B\backslash G; \mathbb{C})/\mathbb{C}\mathbf{1}$ uses unnormalized induction of the trivial character. The operator and these kernel and quotient identifications are the input to the splitting problem. The following construction applies to every two-term complex in $\mathcal{A}_\zeta$.

Removing the fixed central character

Set

$$\lambda = \begin{cases} q^{1/2}b_0, & z_0 = q, \\ q^{1/2}a_0, & z_0 = q^{-1}. \end{cases} \qquad \eta = \chi_\lambda \circ \det.$$

In either case $\lambda^2 = a_0 b_0$. The character η is smooth, and its restriction to Z equals ζ. Let $H = G/Z = \mathrm{PGL}_2(F)$, with the quotient topology.

LEMMA 18.14.1. Twisting by η^{-1} and descending through $G \to H$ gives an exact equivalence

$$\mathcal{A}_\zeta \longrightarrow \mathrm{Rep}_{\mathrm{sm},\mathbb{C}}(H).$$

Its inverse inflates an H-representation and twists by η. Under this equivalence, W_x becomes the trivial representation when $z_0 = q$ and the Steinberg quotient when $z_0 = q^{-1}$.

Proof. For $V \in \mathcal{A}_\zeta$, every central element acts trivially on $V \otimes \eta^{-1}$. Its action therefore factors uniquely through H. If a vector has open stabilizer U in G, the image of UZ is open in H and fixes that vector. Thus the descended action is smooth.

Conversely, inflation of a smooth H-action is smooth, and the twist gives central character ζ. The two constructions are inverse on objects and morphisms. Their underlying vector-space functors are exact. Formula (18.18) gives the final assertion. □

It remains to construct projective resolutions for smooth representations of H. The quotient by the full center is essential: the stabilizers of lattice classes in H are compact.

The lattice tree

An O-lattice in F^2 is the O-span of an F-basis. Vertices are homothety classes $[L]$ of such lattices. Two classes are adjacent if they have representatives satisfying

$$\varpi L \subset M \subset L, \qquad \dim_{O/\varpi O}(M/\varpi L) = 1.$$

This is a symmetric relation on classes. Indeed, the reverse inclusion can be represented by $\varpi M \subset \varpi L \subset M$. The group H acts on this graph by its action on lattice classes.

LEMMA 18.14.2. This graph is a tree. Each vertex has $q + 1$ neighbors.

Proof. The neighbors of $[L]$ are exactly the lines in $L/\varpi L$, so there are $q + 1$. Fix the root $[L_0]$, where $L_0 = Oe_1 \oplus Oe_2$. Each class has a unique representative $L \subset L_0$ with $L \not\subset \varpi L_0$. Multiplication by a unit does not change a lattice, which proves uniqueness. Choose a primitive vector $u \in L$ and complete it to a basis u, v of L_0.

The image of L in L_0/Ou is an ideal $\varpi^n O$. Subtracting an Ou component from a lift of its generator gives

$$L = Ou \oplus O\varpi^n v, \qquad n \geq 0.$$

The integer n is the valuation of the determinant of this normalized lattice relative to L_0. It is independent of the chosen basis and will be its height. Height zero gives the root.

For $n > 0$, the line generated by $\varpi^n v$ in $L/\varpi L$ gives the neighboring lattice $O\varpi u \oplus O\varpi^n v$. After normalization, its height is $n - 1$. Every other line is generated by $u + c\varpi^n v$ for one residue $c \in O/\varpi O$. Its neighboring lattice is

$$O(u + c\varpi^n v) \oplus O\varpi^{n+1} v,$$

which is normalized and has height $n + 1$.

Every neighbor of the root has height one. Thus every positive-height vertex has exactly one neighbor of lower height. Repeated lower-height steps connect it to the root. A cycle would have a maximal-height vertex with two neighbors of lower height, which is impossible. □

An element of H can exchange the endpoints of an edge. Insert a midpoint in each edge to obtain the barycentric subdivision T. Orient each new edge from its lattice vertex to its midpoint. The H-action preserves these two vertex types and therefore preserves the chosen orientations.

Let $v_0 = [L_0]$ and $v_1 = [\mathcal{O}e_1 \oplus \mathcal{O}\varpi e_2]$. Let m_0 be the midpoint between them. The stabilizer K of v_0 is the image of $\mathrm{GL}_2(\mathcal{O})$ in H. It is compact and open.

The stabilizer I of the incidence (v_0, m_0) fixes both v_0 and v_1. In K, it is the inverse image of the stabilizer of a line modulo ϖ, hence compact and open. The class of

$$s = \begin{pmatrix} 0 & 1 \\ \varpi & 0 \end{pmatrix}$$

exchanges v_0 and v_1 and satisfies $s^2 = 1$ in H. The midpoint stabilizer is $J = I \sqcup sI$ and is compact and open.

The action is transitive on lattice vertices, on midpoints, and on the oriented edges of T. The first assertion follows by sending one lattice basis to another. Transitivity on incident pairs follows because K acts transitively on lines modulo ϖ.

Example 18.14.3 *(The first ball when $q = 2$).* The first ball about v_0 is a star with three leaves v_1, v_2, v_3. Write m_i for the midpoint of $v_0 v_i$. The subdivision has seven vertices and six oriented edges $e_i = (v_0, m_i)$ and $f_i = (v_i, m_i)$, for $1 \le i \le 3$. Its boundary map and augmentation are

$$\partial e_i = [m_i] - [v_0], \qquad \partial f_i = [m_i] - [v_i], \qquad \epsilon[v] = 1.$$

Define a linear map h from vertices to edge chains by

$$h[v_0] = 0, \qquad h[m_i] = e_i, \qquad h[v_i] = e_i - f_i.$$

Then

$$\partial h = \mathrm{id} - [v_0]\epsilon, \qquad h\partial = \mathrm{id}\,.$$

Both identities follow by applying the displayed formulas to each basis vector. Thus ∂ has rank six and its image is $\ker \epsilon$. This finite ball is not H-stable. The same finite-path calculation proves exactness for arbitrary finite chains in the entire tree.

Projective cellular chains

For a compact open subgroup $U \subset H$, let $\mathrm{Rep}_{\mathrm{sm},\mathbb{C}}(U)$ denote its smooth complex representations. An object is projective if every morphism from it to a quotient lifts through the quotient map. The following property depends on the coefficient field $\mathbb{C}$.

Lemma 18.14.4. Every smooth complex representation of U is projective in $\mathrm{Rep}_{\mathrm{sm},\mathbb{C}}(U)$.

Proof. The orbit of any vector under U is finite because its stabilizer is open and U is compact. The orbit spans a finite-dimensional U-submodule. A basis of this submodule has a common open stabilizer. Its normal core is an open normal subgroup, so the action factors through a finite quotient of U.

For an invariant subspace, average a linear projection over the finite quotient to obtain an equivariant projection. Its kernel is an invariant complement. Induction on dimension therefore decomposes this submodule into irreducibles. Hence every smooth U-module is a sum of finite-dimensional irreducible submodules.

Choose a maximal direct sum of such submodules. If it does not contain the whole module, a finite-dimensional submodule contributes a simple summand outside it. That simple summand has zero intersection with the maximal direct sum, a contradiction. Thus the whole module is a direct sum of finite-dimensional irreducibles.

Each such irreducible τ is projective. To verify the lifting property explicitly, let $Y \twoheadrightarrow Z$ be an epimorphism and let $f : \tau \to Z$ be U-equivariant. Choose lifts in Y of the images of a basis of τ and extend them to a linear map $\ell : \tau \to Y$. The U-orbits of these finitely many lifts span a finite-dimensional submodule Y_0.

Choose an open normal subgroup $N \subset U$ acting trivially on both Y_0 and τ. Then

$$\widetilde{f}(v) = \frac{1}{|U/N|} \sum_{u \in U/N} u\, \ell(u^{-1}v)$$

is well-defined, U-equivariant, and lifts f. Only finite-dimensional smooth data are averaged. Finally, morphisms out of a direct sum can be lifted on its summands and then assembled. A direct sum of projectives is therefore projective. □

For $M \in \mathrm{Rep}_{\mathrm{sm},\mathbb{C}}(U)$, define compact induction $\mathrm{c\text{-}Ind}_U^H M$ by finite sums of symbols $[g, m]$, linear in m, with relations

$$[gu, m] = [g, um] \qquad (g \in H,\ u \in U).$$

The group acts by $h[g, m] = [hg, m]$. This is the algebraic finite-support model on H/U, valid because U is open. It is smooth: the stabilizer of $[g, m]$ contains $gU_m g^{-1}$, where U_m is an open stabilizer of m in U. A finite sum has the intersection of finitely many such open stabilizers.

The map determined by $[g, m] \mapsto gf(m)$ gives the adjunction

$$\mathrm{Hom}_H(\mathrm{c\text{-}Ind}_U^H M, V) \cong \mathrm{Hom}_U(M, V|_U).$$

Its inverse restricts a morphism to the symbols $[1, m]$. Restriction is exact, so compact induction takes projectives to projectives. Lemma 18.14.4 applies to every compact open cell stabilizer of T.

For a discrete set Y, write $C_c(Y; \mathbb{C})$ for its finitely supported functions, with basis vectors denoted $[y]$. Let T_0 and T_1 be the vertices and the chosen oriented edges of T. For a smooth H-representation V, define

$$P_1(V) = C_c(T_1; \mathbb{C}) \otimes V, \qquad P_0(V) = C_c(T_0; \mathbb{C}) \otimes V,$$

with diagonal action $g([c] \otimes v) = [gc] \otimes gv$. For an oriented edge $e = (u, m)$, define

$$\partial([e] \otimes v) = ([m] - [u]) \otimes v, \qquad \epsilon([u] \otimes v) = v.$$

Both maps commute with H, and $\epsilon\partial = 0$.

THEOREM 18.14.5. For every smooth complex H-representation V, the sequence

$$0 \longrightarrow P_1(V) \xrightarrow{\partial} P_0(V) \xrightarrow{\epsilon} V \longrightarrow 0 \tag{18.19}$$

is exact and its first two terms are projective smooth H-representations. Consequently,

$$\mathrm{Ext}^j_{\mathrm{Rep}_{\mathrm{sm},\mathbb{C}}(H)}(V, U) = 0 \qquad (j \geq 2)$$

for all smooth V, U.

Proof. Fix a root vertex o of T. Let p_v be the finite edge chain along the unique path from o to v. Its coefficient is -1 wherever the path opposes the chosen edge orientation. Then $\partial p_v = [v] - [o]$.

The map $[v] \otimes w \mapsto p_v \otimes w$ gives $\partial h = \mathrm{id} - ([o] \otimes -)\epsilon$. Thus every chain of augmentation zero is a boundary. The augmentation is surjective because $w = \epsilon([o] \otimes w)$. A nonzero finite-support edge cycle would have a leaf in the finite subtree spanned by its support.

At that leaf, its boundary equals the nonzero coefficient of its unique incident supported edge, up to sign. This is impossible, so ∂ is injective. These arguments also apply to vector coefficients and use only finite sums. The root-path contraction proves exactness and is not claimed equivariant.

For a chosen cell c with stabilizer U, the map

$$\text{c-}\operatorname{Ind}_U^H(V|_U) \longrightarrow C_c(Hc;\mathbb{C}) \otimes V, \qquad [g,v] \longmapsto [gc] \otimes gv$$

is an equivariant isomorphism. The relation $[gu, v] = [g, uv]$ verifies that it is well-defined. Its inverse sends $[gc] \otimes w$ to $[g, g^{-1}w]$ and is independent of the representative g. The three cell orbits therefore give

$$P_1(V) \cong \text{c-}\operatorname{Ind}_I^H(V|_I), \qquad P_0(V) \cong \text{c-}\operatorname{Ind}_K^H(V|_K) \oplus \text{c-}\operatorname{Ind}_J^H(V|_J).$$

They are smooth projectives by Lemma 18.14.4 and the compact-induction adjunction. Applying $\operatorname{Hom}(-, U)$ to this projective resolution computes Ext and leaves no terms in degrees $j \geq 2$. □

The constant-coefficient resolution above needs no finite generation or admissibility hypothesis. Its projective terms can have infinite multiplicity. Schneider–Stuhler develop cellular resolutions with coefficient systems of compact-subgroup invariants in *Representation theory and sheaves on the Bruhat–Tits building*. See Section II.2, pp. 121–122, and Theorem II.3.1, p. 123. Here the rank-one tree and constant coefficients give the stated resolution directly.

The obstruction and the splitting maps

Lemma 18.14.1 transports the resolution and projectivity into $\mathcal{A}_\zeta$. For $V \in \mathcal{A}_\zeta$, apply (18.19) to its descended twist $V \otimes \eta^{-1}$, then inflate and tensor every term and map with η. This gives a projective resolution

$$0 \longrightarrow P_1^\zeta(V) \xrightarrow{d} P_0^\zeta(V) \xrightarrow{\epsilon} V \longrightarrow 0$$

inside $\mathcal{A}_\zeta$. In particular, $\operatorname{Ext}^j_{\mathcal{A}_\zeta}(V, U) = 0$ for all $j \geq 2$. Every central element acts by ζ on both projective terms.

Theorem 18.14.6. Let $A \xrightarrow{D} B$ be a morphism in $\mathcal{A}_\zeta$. Place A in degree -1 and B in degree 0, and set $U = \ker D, V = \operatorname{coker} D$. There is an isomorphism

$$[\, A \xrightarrow{D} B \,] \simeq U[1] \oplus V \qquad \text{in } \mathcal{D}^b(\mathcal{A}_\zeta).$$

It can be represented by two explicit quasi-isomorphisms through the cellular projective resolution of V.

Proof. Write $P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} V$ for the resolution just constructed. Let $p : B \twoheadrightarrow V$ and $i : U \hookrightarrow A$ be the quotient and kernel maps. Since P_0 is projective, choose a morphism $f_0 : P_0 \to B$ with $pf_0 = \epsilon$. The identity $pf_0 d = \epsilon d = 0$ shows that $f_0 d$ takes values in $\operatorname{im} D$.

Projectivity of P_1 lifts that map through the epimorphism $A \twoheadrightarrow \operatorname{im} D$. Thus there is $f_1 : P_1 \to A$ satisfying

$$Df_1 = f_0 d. \tag{18.20}$$

All objects, quotient maps, and lifts lie in $\mathcal{A}_\zeta$.

Let $P = [\,P_1 \xrightarrow{d} P_0\,]$, in degrees -1, 0. Equation (18.20) defines a chain map

$$\Phi : U[1] \oplus P \longrightarrow [\,A \xrightarrow{D} B\,], \qquad \Phi^{-1}(u, v) = i(u) + f_1(v), \quad \Phi^0 = f_0.$$

The superscripts denote cochain degrees. Since d is injective, $H^{-1}(U[1] \oplus P) = U$ and $H^0(U[1] \oplus P) = V$. The map Φ induces the identity on these groups, using $pf_0 = \epsilon$.

It is therefore a quasi-isomorphism. The augmentation of P gives another quasi-isomorphism

$$\Psi : U[1] \oplus P \longrightarrow U[1] \oplus V, \qquad \Psi^{-1}(u, v) = u, \quad \Psi^0 = \epsilon.$$

The roof

$$[\,A \xrightarrow{D} B\,] \xleftarrow{\Phi} U[1] \oplus P \xrightarrow{\Psi} U[1] \oplus V$$

proves the asserted derived isomorphism. The compact averaging construction in Lemma 18.14.4, followed by compact induction, constructs the required lifts after choices of irreducible summands and linear lifts. No averaging over the noncompact group is used. □

The associated obstruction is the class of the exact two-extension

$$0 \longrightarrow U \xrightarrow{i} A \xrightarrow{D} B \xrightarrow{p} V \longrightarrow 0$$

in $\mathrm{Ext}^2_{\mathcal{A}_\zeta}(V, U)$. Equivalently, it is the connecting morphism $V \to U[2]$ in the cohomological truncation triangle. Theorem 18.14.5 makes this group zero, and the roof above realizes the splitting by actual maps. The construction proves existence and depends on the choices of lifts. No canonical splitting or compatibility between different parameters is asserted.

PROPOSITION 18.14.7 *(The exceptional fibers).* For the exceptional pair fixed in (18.17), there is an isomorphism

$$C_x \simeq W_x[1] \oplus W_x \qquad \text{in } \mathcal{D}^b(\mathcal{A}_\zeta).$$

Its degree-two obstruction vanishes in the strictly fixed-central-character category.

Proof. The kernel and cokernel identifications (18.18) are morphisms inside $\mathcal{A}_\zeta$. Apply Theorem 18.14.6 to the actual map D_x. The resulting roof uses the projective resolution twisted by the stated η, so every term has central character ζ. □

This is a pointwise statement over $\mathbb{C}$. The compact projectivity proof uses division by finite group orders and does not establish a theorem over arbitrary coefficient rings. The character η is fixed at the chosen complex point. Universal-central-character splitting over the ordered-character ring requires a separate construction. So do canonical Weyl-compatible pointwise splittings and a comparison with global Eisenstein categories.

18.15 NORMAL EQUATIONS AND SPECIALIZATION MAPS

The opposite operator provides a direct comparison of these two cohomology groups. Put $T = R/(h)$. Equation (18.12) gives a well-defined map

$$\overline{D^{\mathrm{sw}}} : \mathrm{coker}(D_T) \longrightarrow \ker(D_T), \qquad [y] \longmapsto D_T^{\mathrm{sw}} y.$$

It is an isomorphism of smooth T-linear G-representations.

To prove this, decompose T by the two comaximal ideals. On $\alpha = 0$, its surviving coefficient is $\beta/z = 1 - q^{-2}$ on H_0. On $\beta = 0$, its surviving coefficients are $z^{-n}\alpha$ on H_n, $n \geq 1$. All are units in the corresponding divisor ring. The same isomorphism holds after further coefficient extension from T.

There is also a comparison determined by a chosen equation of one divisor. For $t = \alpha$ or $t = \beta$, let $\overline{R} = R/(t)$. For $x \in \ker D_{\overline{R}}$, choose a lift $\widetilde{x} \in E$. The vector $D\widetilde{x}$ is divisible by t in the free module E^{sw}.

Define

$$\delta_t : \ker D_{\overline{R}} \longrightarrow \operatorname{coker} D_{\overline{R}}, \qquad x \longmapsto [\, D\widetilde{x}/t \,].$$

Changing the lift by tu changes the quotient by Du, so the map is well-defined. Equivariance follows by applying $D\pi(g) = \pi^{\mathrm{sw}}(g)D$ to the chosen lift. This is the connecting map for the coefficient sequence $0 \to R/(t) \xrightarrow{t} R/(t^2) \to R/(t) \to 0$ applied to C. It also identifies the kernel calculated from the free resolution of Q_{loc} with its t-torsion using the resolution $[R \xrightarrow{t} R]$ of $R/(t)$.

The map δ_α has coefficient 1 on H_0. The map δ_β has coefficient z^{n-1} on H_n, $n \geq 1$, evaluated on $\beta = 0$. Thus both are isomorphisms. They are compatible with subsequent extension along the respective divisor.

They satisfy

$$\delta_t \overline{D^{\mathrm{sw}}} = (h/t)|_{t=0}\ \mathrm{id}, \qquad (h/\alpha)|_{\alpha=0} = 1 - q^{-2}, \quad (h/\beta)|_{\beta=0} = q(1 - q^{-2}).$$

Replacing the local equation t by ut, with u a unit, multiplies δ_t by u^{-1} on the divisor. The normal equation is therefore part of this specialization comparison.

18.16 Finite levels and matrix factorizations

The finite spaces V_N retain the integral's compact-level refinement. Set $E_N = R \otimes V_N$ and $E_N^{\mathrm{sw}} = R \otimes V_N$, with their respective induced compact actions. Restriction gives $D_N : E_N \to E_N^{\mathrm{sw}}$ and $D_N^{\mathrm{sw}} : E_N^{\mathrm{sw}} \to E_N$. Inclusions $V_N \hookrightarrow V_{N+1}$ and averaging maps $P_N : V_{N+1} \to V_N$ commute with both operators because they preserve each H_n.

They therefore induce maps of the cones and of their coefficient extensions. The inclusions give

$$C = \operatorname*{colim}_N C_N, \qquad C_N = [E_N \xrightarrow{D_N} E_N^{\mathrm{sw}}].$$

Filtered colimits of modules are exact, so kernels, cokernels, and the displayed coefficient extensions commute with this colimit.

For $N \geq 1$, let $K_N = 1 + \varpi^N M_2(\mathcal{O})$, and set $K_0 = K$. Then $E^{K_N} = E_N$ and $(E^{\mathrm{sw}})^{K_N} = E_N^{\mathrm{sw}}$. Compact averaging is defined over every $\mathbb{C}$-algebra and is exact on smooth representations. Indeed, any finite set of vectors is fixed by an open subgroup, and averaging reduces to a finite sum with invertible denominator.

It follows that $C_N = C^{K_N}$ and $\operatorname{coker} D_N = Q_{\mathrm{loc}}^{K_N}$. Full G-translation can move E_N into a larger E_M. The finite complexes carry K-actions, whereas their colimit carries the full G-action.

A finite-free matrix factorization of $h \in R$ consists of finite-free modules M_0, M_1 and maps $d_0 : M_0 \to M_1$, $d_1 : M_1 \to M_0$ satisfying $d_1 d_0 = h\,\mathrm{id}$ and $d_0 d_1 = h\,\mathrm{id}$. Thus (D_N, D_N^{sw}) is a finite-free K-equivariant matrix factorization of h. Its rank is $\dim V_N$ in each parity. The full pair (D, D^{sw}) has infinite-rank free terms.

PROPOSITION 18.16.1. Each finite-level matrix factorization (D_N, D_N^{sw}) is contractible through K-equivariant maps. Its cokernel is a projective $R/(h)$-module. The ordinary two-term cone C_N is nevertheless nonzero in the derived category for every $N \geq 0$.

Proof. Put $c = 1 - q^{-2}$. Write $d_0 = D_N$, $d_1 = D_N^{\mathrm{sw}}$. A contraction consists of maps $s_0 : M_0 \to M_1$ and $s_1 : M_1 \to M_0$ with $ds + sd = \mathrm{id}$ on the two-parity module. On H_0 take

$$s_1 = c^{-1}, \qquad s_0 = q^{-1} z / c.$$

Then $s_1 d_0 + d_1 s_0 = (\alpha + q^{-1}\beta)/c = 1$. On H_n, $n \geq 1$, take

$$s_1 = q^{-1} z^{-(n-1)} / c, \qquad s_0 = z^n / c.$$

Here $s_1 d_0 + d_1 s_0 = (q^{-1}\beta + \alpha)/c = 1$ as well. All coefficients commute, so the identity also holds on the other parity. These are Laurent coefficients and preserve the K-stable decomposition.

The ring $R/(h)$ is $R/(\alpha) \times R/(\beta)$ by (18.14). The cokernel is free of rank one on the first component and rank $\dim V_N - 1$ on the second. It is therefore projective over their product. As an R-module it is nonzero, since the first component is nonzero. This cokernel is $H^0(C_N)$, which proves the last assertion. □

The contraction above is asserted only for the coefficient modules with K-action. Its diagonal formulas do not establish a contraction commuting with the full G-action. A matrix factorization uses both opposite maps and satisfies $d^2 = h$. The ordinary cone uses only D and has the nonzero cohomology (18.13). These are distinct homological constructions.

The regular intertwiner therefore defines a smooth equivariant degeneration complex on the entire ordered-character torus. Its cohomology consists of a character family on one divisor and a Steinberg quotient family on the other. Its derived fibers and their specialization comparisons are given above. A category comparison for global Eisenstein representations still requires a global target and a functor whose derived mapping objects and essential image are determined. The local degeneration complex supplies no such equivalence by itself.

18.17 Local Yoneda classes and finite exterior products

A local nonsplit principal series defines a degree-one extension between its Steinberg quotient and its constant subrepresentation. At two distinct finite places, the exterior product of these extensions has degree two. An infinite restricted tensor introduces a second condition: its distinguished local vectors must be fixed by the prescribed compact subgroups at almost every place. These two requirements distinguish a finite exterior product of Steinberg representations from an adelic extension of principal series.

The local sequence and its invariant vectors

Fix the coefficient field $\mathbb{C}$. For a finite prime p, put $G_p = GL_2(\mathbb{Q}_p)$, $K_p = GL_2(\mathbb{Z}_p)$, and let B_p be the upper triangular Borel subgroup. Give these groups their usual p-adic topologies. Write $X_p = B_p \backslash G_p \simeq \mathbb{P}^1(\mathbb{Q}_p)$, with its compact quotient topology and right G_p-action. The identification sends a matrix to the line spanned by its bottom row.

Let

$$E_p = C^{\mathrm{lc}}(X_p, \mathbb{C}), \qquad (\rho_p(g)f)(x) = f(xg).$$

Here C^{lc} means locally constant functions, without an analytic completion. This is unnormalized smooth induction of the trivial character of B_p. With normalized induction and $|p|_p = p^{-1}$, its ordered inducing characters are $|\cdot|_p^{-1/2}$ and $|\cdot|_p^{1/2}$. Thus, if symmetric induction is written with characters $(|\cdot|_p^{s/2}, |\cdot|_p^{-s/2})$, this sequence occurs at $s = -1$.

The constants give the exact sequence

$$0 \longrightarrow \mathbb{C} \xrightarrow{i_p} E_p \longrightarrow \mathrm{St}_p \longrightarrow 0, \qquad \mathrm{St}_p = E_p/\mathbb{C}\mathbf{1}_p. \tag{18.21}$$

This is the usual rank-one Steinberg quotient. See Casselman, *Introduction to the Theory of Admissible Representations of p-Adic Reductive Groups*, Section 8.1, p. 70. Every term has trivial central character.

We use $\mathrm{Rep}_{\mathbb{C}}^{\mathrm{sm}}(G_p)$ for the abelian category of all smooth complex representations. Its full subcategory on which the center acts trivially is identified with $\mathrm{Rep}_{\mathbb{C}}^{\mathrm{sm}}(PGL_2(\mathbb{Q}_p))$. Extension groups below specify which of these categories is used.

Lemma 18.17.1. The sequence (18.21) is nonsplit. Moreover,

$$\mathrm{Hom}_{G_p}(E_p, \mathbb{C}) = 0, \qquad \mathrm{Hom}_{G_p}(\mathrm{St}_p, \mathbb{C}) = 0, \qquad \mathrm{St}_p^{K_p} = 0.$$

The same statements hold after passage to $PGL_2(\mathbb{Q}_p)$.

Proof. Consider a G_p-invariant linear functional $\ell : E_p \to \mathbb{C}$. Projective p-adic balls are translates of one another under G_p. Affine transformations relate the balls in $\mathbb{Q}_p$. Inversion relates them to the balls containing infinity. A ball is a disjoint union of p smaller balls.

Consequently the common value c of ℓ on ball indicators satisfies $c = pc$, so $c = 0$. Locally constant functions on X_p are finite linear combinations of these indicators, since X_p is compact and the balls form a clopen basis. Hence $\ell = 0$. A splitting of (18.21) would provide such a functional with $\ell(\mathbf{1}_p) = 1$. A morphism $\mathrm{St}_p \to \mathbb{C}$ would pull back to a morphism $E_p \to \mathbb{C}$, proving the first two assertions.

Every line in $\mathbb{Q}_p^2$ has a primitive integral representative that extends to an integral basis. Thus K_p acts transitively on X_p and $E_p^{K_p} = \mathbb{C}\mathbf{1}_p$. Invariants under a compact open subgroup are exact for smooth complex representations. Indeed, normalized Haar averaging of a smooth vector is a finite sum, and averaging a lift of an invariant vector gives an invariant lift.

Applying invariants to (18.21) gives $\mathrm{St}_p^{K_p} = 0$. This averaging argument is also Casselman, Proposition 2.1.7, p. 20. All actions factor through $PGL_2(\mathbb{Q}_p)$, and the same proof applies there. □

Write α_p for the Yoneda class of (18.21). It is a specified nonzero element of $\mathrm{Ext}^1(\mathrm{St}_p, \mathbb{C})$. No residue operator is needed to define it, and no dimension assertion about that group is required below.

Two finite places and the exterior degree

Let $p \neq q$ be finite primes. External tensor products over $\mathbb{C}$ are written $\boxtimes$ and carry the separate actions of the two groups. Tensoring the two local sequences gives the exact sequence

$$\begin{aligned} 0 \longrightarrow \mathbb{C} \boxtimes \mathbb{C} \xrightarrow{d_2} (E_p \boxtimes \mathbb{C}) \oplus (\mathbb{C} \boxtimes E_q) \xrightarrow{d_1} E_p \boxtimes E_q \\ \longrightarrow \mathrm{St}_p \boxtimes \mathrm{St}_q \longrightarrow 0, \end{aligned} \tag{18.22}$$

where

$$d_2 = (i_p \boxtimes 1, -1 \boxtimes i_q), \qquad d_1(x, y) = (1 \boxtimes i_q)x + (i_p \boxtimes 1)y.$$

Unlike the cochain indexing of the local cone, the following totals use homological degrees and differentials of degree -1.

These signs are the tensor differential for homological complexes, $\partial(x \otimes y) = \partial x \otimes y + (-1)^{\deg x} x \otimes \partial y$. In particular $d_1 d_2 = 0$. The two complexes $[\mathbb{C} \to E_p]$ and $[\mathbb{C} \to E_q]$, in homological degrees $1, 0$, have homology only in degree zero, with values St_p and St_q. Tensoring over a field preserves this assertion. It proves the exactness of (18.22).

The sequence (18.22) is a Yoneda two-extension, not a short exact sequence with these endpoints. Its class is the exterior Yoneda product of α_p and α_q, with the displayed tensor sign convention. The following argument proves that this class is nonzero without replacing either local representation by a finite-dimensional model.

PROPOSITION 18.17.2. Let S be a nonempty finite set of finite primes, $n = |S|$, and $\mathrm{St}_S = \boxtimes_{p \in S} \mathrm{St}_p$. Put $G_S = \prod_{p \in S} G_p$. In $\mathrm{Rep}^{\mathrm{sm}}_{\mathbb{C}}(G_S)$ one has

$$\mathrm{Ext}^r_{G_S}(\mathrm{St}_S, \mathbb{C}) = 0 \quad (0 \le r < n).$$

The exterior product of the classes α_p is nonzero in $\mathrm{Ext}^n_{G_S}(\mathrm{St}_S, \mathbb{C})$. It is represented by the total tensor of the n sequences (18.21). Both assertions also hold in the category where each local center acts trivially, namely $\mathrm{Rep}^{\mathrm{sm}}_{\mathbb{C}}(\prod_{p \in S} PGL_2(\mathbb{Q}_p))$.

Proof. For either of the groups G_p and $PGL_2(\mathbb{Q}_p)$, compact induction $\text{c-Ind}^G_K \mathbb{C}$ from a compact open subgroup is projective. A G-map from this compact induction to V is determined by the image of the indicator of K. That image can be any element of V^K. Thus its Hom functor is $V \mapsto V^K$, which is exact by averaging.

Every smooth vector is fixed by a compact open subgroup. Taking one such compact induction for each vector gives enough projectives. Resolutions can therefore use sums of these compact inductions.

Choose such a resolution $P_{p,\bullet} \to \mathrm{St}_p$, with homological indices at least zero. Products of compact open subgroups are compact open, and

$$\left(\text{c-Ind}^G_K \mathbb{C}\right) \boxtimes \left(\text{c-Ind}^H_L \mathbb{C}\right) \simeq \text{c-Ind}^{G \times H}_{K \times L} \mathbb{C}.$$

Both sides are finite-support functions on the product of the relevant discrete coset spaces. Hence the total external tensor of the chosen resolutions is a projective resolution of St_S.

Let $C_{p,\bullet} = (P_{p,\bullet})_{G_p}$ be its coinvariant complex (or use $PGL_2(\mathbb{Q}_p)$ in the fixed-central-character category). For any representation V, a map $V \to \mathbb{C}$ factors uniquely through V_G. It follows that

$$\mathrm{Ext}^r_G(\mathrm{St}_p, \mathbb{C}) \simeq \mathrm{Hom}_{\mathbb{C}}(H_r(C_{p,\bullet}), \mathbb{C}). \tag{18.23}$$

Here the algebraic dual is exact over the field $\mathbb{C}$.

No topological or continuous dual is used. Coinvariants of an external tensor for a product group are the tensor of the coinvariants. Consequently the same argument identifies the product-group Ext with the dual homology of $\bigotimes_{p \in S} C_{p,\bullet}$.

Lemma 18.17.1 and (18.23) imply $H_0(C_{p,\bullet}) = 0$. The Künneth formula for bounded-below complexes of vector spaces gives

$$H_r\left(\bigotimes_{p \in S} C_{p,\bullet}\right) \simeq \bigoplus_{\substack{(r_p) \in \mathbb{Z}^S_{\ge 0} \\ \sum_p r_p = r}} \bigotimes_{p \in S} H_{r_p}(C_{p,\bullet}).$$

For completeness, one can split each vector-space complex as its homology with zero differential plus a contractible complex. Tensoring these splittings proves the formula. For $r < n$, every summand has some $r_p = 0$ and is zero. This proves the vanishing assertion.

Each nonzero class α_p is a nonzero functional on $H_1(C_{p,\bullet})$. Choose a homology element on which it is nonzero. The product functional is nonzero on the tensor of these elements, which lies in degree n. It represents the exterior Yoneda class.

Lift the projective resolution's augmentation to E_p. Its degree-one boundary then factors through $\mathbb{C}$, giving a chain map to $[\mathbb{C} \to E_p]$. The resulting degree-one cocycle represents α_p. Tensoring these chain maps gives exactly the total extension and the product cocycle, with the tensor signs already specified. Thus the exterior class is nonzero.

This proof uses genuinely compact open subgroups even for GL_2. It does not treat compact-mod-center subgroups as compact, and it makes no assertion about vanishing above degree n in the unrestricted GL_2 category. □

18.18 The restricted tensor condition

Let

$$G_f = GL_2(\mathbb{A}_f) = \prod_p{}'(G_p, K_p).$$

Its restricted-product topology has neighborhoods of the identity that contain $\prod_{p\notin S} K_p$ for some finite S. A usual algebraic restricted tensor representation uses nonzero distinguished vectors $e_p \in V_p^{K_p}$ for all but finitely many p. Its vectors are finite linear combinations of tensors whose factors equal e_p almost everywhere. For $g = (g_p) \in G_f$, both $g_p \in K_p$ and the distinguished factor condition hold almost everywhere, making the coordinate action well-defined.

Proposition 18.18.1. The local representations St_p do not admit a nonzero restricted tensor representation with the standard hyperspecial pointings at almost every finite prime. More generally, suppose a smooth G_f-representation V has factorizations $V \simeq \mathrm{St}_p \otimes W_p$ for every finite p. If G_p acts only on the first factor, then $V = 0$.

Proof. The required pointing space is zero at every prime by Lemma 18.17.1. For the second assertion, a vector in a smooth G_f-representation is fixed by an open subgroup. That subgroup contains K_p for every p outside a finite set. But the stipulated factorization implies

$$V^{K_p} = \mathrm{St}_p^{K_p} \otimes W_p = 0.$$

This equality follows from averaging on the first tensor factor, or from writing any tensor as a finite sum with linearly independent second factors. Thus the vector is zero. □

Choosing Iwahori-fixed vectors at infinitely many places does not supply these hyperspecial pointings. The restricted product relative to those Iwahori subgroups is a different subgroup and topology. No passage to an unspecified completed tensor product is part of this construction.

For a finite set S, a genuine smooth adelic representation is

$$\mathrm{St}_S^{(f)} = \left(\boxtimes_{p\in S} \mathrm{St}_p\right) \boxtimes \left(\bigotimes_{p\notin S}{}' \mathbb{C}\right),$$

where each omitted place uses the trivial representation and vector 1. It is inflation from G_S under the continuous projection $G_f \to G_S$. The extension of Proposition 18.17.2 inflates with it and remains nonzero: restriction along the inclusion $G_S \hookrightarrow G_f$ is exact and recovers the original Yoneda class. This representation has Steinberg factors precisely at the specified finite set. It is not an all-place Steinberg representation.

18.19 An adelic principal-series extension

The local principal series E_p do have compatible distinguished vectors, namely the constant functions $\mathbf{1}_p$. These vectors are fixed by all of G_p. For finite S, write $E_S = \boxtimes_{p\in S} E_p$, inflated to G_f with trivial factors outside S. If $S \subset T$, define

$$E_S \longrightarrow E_T, \qquad x \longmapsto x \otimes \bigotimes_{p\in T\setminus S} \mathbf{1}_p.$$

The superscript res below means restricted tensor product. It does not name an automorphic residual spectrum. These maps are injective and G_f-equivariant. The canonical associativity and symmetry identifications make their composites agree for $S \subset T \subset U$.

Put $E^{\mathrm{res}} = \operatorname{colim}_S E_S$. Each vector lies in some E_S and is smooth at the finitely many primes of S. The full groups outside S fix it. Thus the colimit representation is smooth. Its central character is trivial.

PROPOSITION 18.19.1. Let $X = \prod_p \mathbb{P}^1(\mathbb{Q}_p)$ with its compact product topology. There is a G_f-equivariant isomorphism

$$E^{\mathrm{res}} \simeq C^{\mathrm{lc}}(X, \mathbb{C})$$

for the coordinatewise right action. The exact sequence

$$0 \longrightarrow \mathbb{C} \longrightarrow E^{\mathrm{res}} \longrightarrow Q_{\mathrm{ad}} \longrightarrow 0, \qquad Q_{\mathrm{ad}} = E^{\mathrm{res}}/\mathbb{C}, \tag{18.24}$$

is nonsplit in $\mathrm{Rep}^{\mathrm{sm}}_{\mathbb{C}}(G_f)$ and in its trivial-central-character category. Both E^{res} and Q_{ad} are admissible: their invariants under every compact open subgroup are finite-dimensional.

Proof. For finite S, products of functions identify E_S with $C^{\mathrm{lc}}(\prod_{p\in S} X_p, \mathbb{C})$. Surjectivity follows by covering clopen subsets of the compact finite product by finitely many clopen rectangles and refining this cover to a finite partition. Evaluations separate every finite-dimensional space of functions. For each space involved, choose a basis and finitely many evaluation points giving an injective matrix. The tensor of these matrices is injective, proving injectivity.

Every locally constant function on X has a finite cover by basic open cylinders on which it is constant. Such a finite cover exists by compactness. The union of the finite coordinate sets defining the cylinders is finite. Two points equal in these coordinates belong to the same cylinders and therefore have equal function values.

The function thus depends on only finitely many coordinates. This proves surjectivity from the colimit, and injectivity follows from the finite-stage statement. The insertion maps become pullback along coordinate projections. Each $g \in G_f$ acts by the coordinate homeomorphisms $x_p \mapsto x_p g_p$, so the isomorphism is equivariant. Smoothness also follows from the finite-coordinate description established above.

A compact open subgroup contains $U_S \times \prod_{p\notin S} K_p$ for some finite S. Here U_S is a product of local compact open subgroups at the places of S. Invariance under the tail K_p forces a finite-coordinate

function to be independent of those tail coordinates, because K_p is transitive on X_p. Its invariants therefore lie in $E_S^{U_S}$.

Each local compact open subgroup has open orbits on the compact space X_p, hence finitely many orbits. It follows that $E_S^{U_S}$ is finite-dimensional. This proves admissibility of E^{res}. Compact-open invariants are exact by the same finite averaging argument, so its quotient Q_{ad} is also admissible.

A splitting of (18.24) would give a G_f-invariant retraction $E^{\mathrm{res}} \to \mathbb{C}$. Its restriction to the embedded $E_{\{p\}}$ would be a G_p-invariant retraction of the constants, contradicting Lemma 18.17.1. The same argument applies when the full center acts trivially. □

The quotient Q_{ad} retains mixed finite-place terms. To describe them, set $F_{-1}E^{\mathrm{res}} = 0$ and, for $m \geq 0$, let $F_m E^{\mathrm{res}}$ be the sum of the images of E_T over finite sets T with $|T| \leq m$. This is an increasing, exhaustive filtration by G_f-subrepresentations, with $F_0 E^{\mathrm{res}} = \mathbb{C}$.

Proposition 18.19.2. For every $m \geq 0$ there is a canonical isomorphism

$$\mathrm{gr}_m^F E^{\mathrm{res}} \simeq \bigoplus_{\substack{T \text{ finite} \\ |T|=m}} \mathrm{St}_T^{(f)},$$

where the empty-set factor is $\mathbb{C}$. The induced exhaustive filtration $Q_{\mathrm{ad},m} = F_m E^{\mathrm{res}}/\mathbb{C}$ of Q_{ad} starts with $Q_{\mathrm{ad},0} = 0$ and has these same associated graded pieces for $m \geq 1$.

Proof. At a finite stage S, use the two-step filtration $\mathbb{C}\mathbf{1}_p \subset E_p$ in every tensor factor. Take a class in $\boxtimes_{p\in T} \mathrm{St}_p$, where $T \subset S$ has size m. Lift its factors to E_p and insert constants elsewhere. This defines a class in the mth associated graded.

Changing a lift adds a tensor from the preceding filtration term, so this map is canonical and equivariant. It is an isomorphism: choose arbitrary vector-space complements to the constants in each E_p and expand the finite tensor product. The summands with precisely m complement factors give the asserted graded piece. The complements prove bijectivity but do not define the map.

The canonical maps are compatible with the constant insertions as S grows. Taking the filtered colimit gives the displayed direct sum, since each vector involves only finitely many places. Quotienting by the constant subrepresentation gives the statement for Q_{ad}. □

In particular there is a nonsplit degree-one extension

$$0 \longrightarrow \mathbb{C} \longrightarrow F_1 E^{\mathrm{res}} \longrightarrow \bigoplus_p \mathrm{St}_{\{p\}}^{(f)} \longrightarrow 0.$$

Pulling it back to any single summand gives (18.21), inflated to G_f. This degree-one construction has a direct sum as its quotient. The full quotient Q_{ad} additionally contains every finite mixed tensor through its filtration. Neither quotient is the all-place tensor $\otimes'_p \mathrm{St}_p$.

Indeed, the local quotient maps $E_p \to \mathrm{St}_p$ send the distinguished constant vectors to zero. Using those zero vectors as insertions makes every finite tensor map zero after a new prime is added. The resulting colimit is zero. Thus the quotient maps cannot turn (18.24) into a nonzero all-place Steinberg extension.

18.20 ARCHIMEDEAN AND AUTOMORPHIC SCOPE

All preceding Ext groups belong to explicitly stated categories of smooth finite-adelic or finite-product representations. Tensoring with the trivial representation of $GL_2(\mathbb{R})$ gives a specified archimedean factor and an action of $GL_2(\mathbb{A})$, without adding another extension class. A nontrivial archimedean extension requires its own smooth topological representation category or Harish–Chandra module category, its central character, and its tensor conventions. An additional degree-one exterior factor contributes an additional degree when its exterior product is defined. It does not erase the finite-place degrees above.

A representation of $GL_2(\mathbb{A})$ is not thereby a realization on $GL_2(\mathbb{Q})\backslash GL_2(\mathbb{A})$. Such a realization requires a specified automorphic function space or enhanced category, central quotient and support conditions, and a comparison functor. The symbols $\mathrm{Ext}^1_{[GL_2]}$ alone do not specify that category. The constructions here give genuine local and adelic group representations and their Yoneda extensions. They neither identify a residue with an extension class nor supply an arithmetic or automorphic comparison.

18.21 GALOIS TOPOLOGY AND THE GEOMETRIC RECIPIENT

For a finite field $\mathbb{F}_q$, its absolute Galois group is procyclic, generated topologically by arithmetic Frobenius. For a nonarchimedean local field F, the corresponding exact sequence is

$$1 \longrightarrow I_F \longrightarrow G_F \longrightarrow G_{\mathbb{F}_q} \longrightarrow 1.$$

Its kernel is inertia. The tame quotient of inertia has its prime-to-residue-characteristic Tate twist, whereas wild inertia is a pro-p group. Neither is this Frobenius quotient.

Precomposition with a quotient $G \to Q$ embeds representations of Q among those representations of G on which the kernel acts trivially. It does not descend an arbitrary representation. Similarly, a representation of a product of groups has commuting component images, with continuity for the chosen topologies. Independently chosen local representations need not define such a product representation.

Unramified twists make the distinction visible without wild ramification. Two different unramified characters of W_F have the same trivial inertia restriction and different Frobenius eigenvalues. Thus inertia alone does not determine the full parameter. On a general test scheme, a mapping stack also retains torsors and their descent. A conjugacy set on field-valued points does not define that stack.

Finite cyclic and ordinary loops

Let k be algebraically closed of characteristic zero. For the finite cyclic group C_n,

$$\mathrm{Map}(BC_n, B\mathbb{G}_m) \simeq \coprod_{\zeta \in \mu_n} B\mathbb{G}_m.$$

Indeed, a character of C_n is determined by an nth root of unity. Its automorphism group is $\mathbb{G}_m$. This is distinct from ordinary derived loops

$$\mathcal{L}(B\mathbb{G}_m) = \mathrm{Map}(B\mathbb{Z}, B\mathbb{G}_m) \simeq \mathbb{G}_m \times B\mathbb{G}_m.$$

It is also distinct from an infinitesimal neighborhood of the identity diagonal.

If $n \mid m$, the quotient $C_m \to C_n$ induces

$$\mathrm{Map}(BC_n, X) \longrightarrow \mathrm{Map}(BC_m, X)$$

by precomposition. The arrows reverse those of the groups. A continuous homomorphism from $\widehat{\mathbb{Z}}$ to a discrete group has finite image, since a compact discrete subgroup is finite. A topology on the coefficient group can change that condition. The ambient prestack and topology must therefore be fixed before a profinite loop comparison.

Selmer deformation functors

Fix one finite residual field k, one coefficient ring $W(k)$, a residual representation, determinant convention and a finite ramification set. Let $\mathcal{F}_{\mathrm{glob}}$ and $\mathcal{F}_v$ be the continuous deformation functors on the same category of simplicial Artin coefficient rings. Give every local condition an actual map $\mathcal{L}_v \to \mathcal{F}_v$. Then the Selmer functor is

$$\mathcal{F}_{\mathrm{Sel}} = \mathcal{F}_{\mathrm{glob}} \times^{h}_{\prod_v \mathcal{F}_v} \prod_v \mathcal{L}_v.$$

Its construction and tangent fiber are proved in Theorem 10.7.1. A cochain quotient notation is available when compatible subcomplexes have actually been chosen. Local stacks over unrelated residue-characteristic bases cannot be inserted into this product without coefficient comparisons.

Framing retains cocycles. A determinant identification gives the fiber of the trace map on the full shifted cochain complexes. Fixed determinant value modulo arbitrary strict GL_2 conjugacy has the different gauge convention described above. Automorphisms and obstructions remain in their respective degrees. At coefficient characteristic two, real decomposition groups also require their specified archimedean or modified Tate condition.

Formal pullback notation proves neither representability nor a cotangent-amplitude bound. For example, the dual numbers over a characteristic-zero field have a two-term cotangent complex and a nonzero obstruction tangent at their closed point. Such an obstruction is allowed in a quasi-smooth derived scheme. Likewise, smoothness of X does not make its derived loop space classical: for $X = \mathbb{A}^1_k$,

$$\mathcal{O}(\mathcal{L}X) \simeq k[t] \otimes \Lambda(dt), \qquad |dt| = -1.$$

Cotangent comparisons on finite products use the pullbacks of each factor's cotangent complex over the common base. Changes of charts use the transitivity triangle. An inverse-limit formula or an ind-coherent refinement requires its own continuity theorem.

18.22 Restriction to two places

Unramified matrices describe a local deformation problem. Global restriction imposes further relations between places and can have several lifts over the same local pair. The following example retains absolute irreducibility of the residual representation and fixes its determinant.

Fix distinct finite primes p, q and a coefficient prime $p_0 \neq p, q$. Let k be a finite field of characteristic p_0, and put $W = W(k)$. Choose a finite set S of places of $\mathbb{Q}$ containing p, q, p_0, ∞. Write $G_{\mathbb{Q},S}$ for the Galois group of the maximal extension unramified outside S.

Fix a continuous representation $\bar{\rho} : G_{\mathbb{Q},S} \to GL_2(k)$ which is absolutely irreducible, and let $\bar{D}$ be its determinant polynomial law. Here a determinant of dimension two is a multiplicative homogeneous polynomial law of degree two on the group algebra, as in Definition 18.2.1. Its characteristic coefficients, rather than a trace alone, define the object in all residual characteristics.

Choose decomposition-group maps $G_{\mathbb{Q}_p}, G_{\mathbb{Q}_q} \to G_{\mathbb{Q},S}$. Choose closed local and global deformation conditions preserved by restriction. These may include a fixed determinant character lifting $\det \bar{\rho}$. Let A be a finite local W-algebra with residue field k. Write $\mathcal{D}_{\mathrm{glob}}(A)$ and $\mathcal{D}_v(A)$ for the corresponding continuous determinant lifts, where $v = p, q$. The local residual restrictions need not be absolutely irreducible.

Chenevier's representability theorem gives complete Noetherian local rings R^{Ch}, R^{Ch}_p, R^{Ch}_q for these functors. The precise hypotheses are those of Proposition 18.2.2. Its source is Chenevier, Proposition 3.3 and Example 3.4, p. 42, and Example 3.6 and Proposition 3.7, p. 43. Each closed condition must be

represented by the quotient used here. These are formal deformation rings over one coefficient ring W. An identification with functions on a local or global spectral stack is additional geometric data.

A pair of indistinguishable local restrictions

PROPOSITION 18.22.1 *(nonuniqueness with absolutely irreducible residual representation).* There are distinct continuous rank-two determinant deformations over $\mathbb{Z}/8\mathbb{Z}$ with the same absolutely irreducible reduction and fixed determinant character. Their restrictions at 3 and 5 coincide.

Proof. Let K be the splitting field of $X^3 - X - 1$ over $\mathbb{Q}$. The polynomial has no root modulo 2, hence is irreducible over $\mathbb{Q}$. Its discriminant is -23, so its Galois group is S_3. An integral standard representation of this group is generated by

$$A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Direct multiplication gives $A^3 = B^2 = 1$ and $BAB = A^{-1}$. Modulo 2, the two eigenlines of A over $\mathbb{F}_4$ are interchanged by B. Thus this residual representation is absolutely irreducible. Inflate the integral representation to $\rho : G_{\mathbb{Q}} \to GL_2(\mathbb{Z}_2)$.

Let χ be the quadratic character of $L = \mathbb{Q}(\sqrt{241})$. For $v = 3, 5$, the polynomial $X^2 - 241$ has the simple root 1 modulo v. A root u_r modulo v^r has a unique lift modulo v^{r+1} in its residue class. The correction $u_r + cv^r$ solves an equation with invertible coefficient $2u_r$ modulo v. Taking the limit gives a root in $\mathbb{Z}_v$.

Hence $\chi|_{G_{\mathbb{Q}_v}} = 1$ for both places. Set $\rho' = \chi\rho$. Then

$$\bar{\rho}' = \bar{\rho}, \qquad \det \rho' = \chi^2 \det \rho = \det \rho.$$

The restrictions of ρ and ρ' at 3 and 5 agree. Both are unramified outside $S = \{2, 3, 5, 23, 241, \infty\}$.

The unique quadratic subfield of K is $\mathbb{Q}(\sqrt{-23})$, which differs from L. Consequently $K \cap L = \mathbb{Q}$, and the composite extension has Galois group $S_3 \times C_2$. Choose g whose image is $(1, -1)$. Then

$$\det(T - \rho(g)) = T^2 - 2T + 1, \qquad \det(T - \rho'(g)) = T^2 + 2T + 1.$$

The traces are 2 and 6 modulo 8. Thus the two determinant laws modulo 8 lift the same residual law and are distinct. Their local restrictions and fixed determinant character agree, as asserted. □

The example lies in a single finite-ramification deformation problem. A more restrictive arithmetic condition could exclude one lift. Uniqueness on such a condition would require its own proof.

18.23 RESTRICTION, CLOSED IMAGE AND FIBERS

The local product functor is represented by

$$B_{p,q} = R_p^{\mathrm{Ch}} \widehat{\otimes}_W R_q^{\mathrm{Ch}} = \varprojlim_r (R_p^{\mathrm{Ch}}/\mathfrak{m}_p^r) \otimes_W (R_q^{\mathrm{Ch}}/\mathfrak{m}_q^r).$$

The completion is for the sum of the maximal ideals. Restriction is a natural transformation $\mathcal{D}_{\mathrm{glob}} \to \mathcal{D}_p \times \mathcal{D}_q$, so its map on representing rings has the opposite direction.

Proposition 18.23.1 *(the formal restriction diagram).* Restriction induces a continuous local W-algebra homomorphism

$$r^{\sharp}_{p,q} : B_{p,q} \longrightarrow R^{\mathrm{Ch}}, \qquad x \otimes y \longmapsto r^{\sharp}_{p}(x) r^{\sharp}_{q}(y). \tag{18.25}$$

For the map in Equation (18.25), put $J_{p,q} = \ker r^{\sharp}_{p,q}$ and $C_{p,q} = B_{p,q}/J_{p,q}$. Then

$$B_{p,q} \twoheadrightarrow C_{p,q} \hookrightarrow R^{\mathrm{Ch}}, \qquad \operatorname{Spf} R^{\mathrm{Ch}} \longrightarrow \operatorname{Spf} C_{p,q} \hookrightarrow \operatorname{Spf} B_{p,q}.$$

For a continuous local map $b : B_{p,q} \to A$, the fiber over the prescribed local pair is represented over A by $R^{\mathrm{Ch}} \widehat{\otimes}_{B_{p,q},b} A$.

Proof. Restricting the universal determinant gives maps $r^{\sharp}_{v} : R^{\mathrm{Ch}}_{v} \to R^{\mathrm{Ch}}$. Their images commute, and their W-actions agree. The completed tensor-product property gives $r^{\sharp}_{p,q}$ with the displayed formula. The kernel is closed. The quotient is complete and injects into R^{Ch}.

Since the rings are profinite, its image is closed. The formal-spectrum diagram reverses these maps. Let A' be a finite local A-algebra. A map from the completed base change to A' is a map $R^{\mathrm{Ch}} \to A'$ agreeing with the prescribed $B_{p,q}$-action through A. This proves the fiber assertion. □

Definition 18.23.2 *(the closed image of two-prime restriction).* The formal scheme $\operatorname{Spf} C_{p,q}$ is the closed image of restriction inside the local deformation product. A proposed two-prime Selmer algebra $R^{\mathrm{Ch}}_{p,q}$ requires a specified $B_{p,q}$-algebra structure and comparison $R^{\mathrm{Ch}}_{p,q} \to C_{p,q}$ before the two can be identified. The quotient $C_{p,q}$ itself imposes the kernel relations only.

Example 18.23.3 *(a flat closed-image map with a nonliftable point).* Let $B = k[[x]]$ and $R = k[[t]]$, with $x \mapsto t^2$. Every series in t is uniquely $f(t^2) + t\,g(t^2)$. Thus R is free of rank two over B, and the map is injective. Its kernel quotient is B itself.

Nevertheless, the local point $B \to k[\epsilon]/(\epsilon^2)$, $x \mapsto \epsilon$, has no lift to R. A local image of t is $a\epsilon$, whose square is zero. Flatness does not turn the closed image into the image functor on these Artinian points.

Example 18.23.4 *(an ordinary fiber with nontrivial derived structure).* For $B = k[[x]] \to R = k$, $x \mapsto 0$, the fiber over the residue point has ordinary coordinate ring k. The free resolution $[B \xrightarrow{x} B]$, in cohomological degrees -1, 0, gives

$$k \otimes^{\mathbb{L}}_{B} k \simeq k \oplus k[1], \qquad H^{-1}(k \otimes^{\mathbb{L}}_{B} k) = k.$$

Thus the derived fiber retains information lost by the ordinary fiber. The same calculation proves that this map is not flat.

Remark 18.23.5 *(the information retained by restriction).* The kernel quotient records equations satisfied by local pairs. The fibers record their global lifts. Proposition 18.22.1 shows that those lifts need not be unique, so $r^{\sharp}_{p,q}$ need not be surjective. In a derived deformation theory the fiber uses derived completed tensor product. None of these constructions identifies a Nygaard filtration or supplies a comparison with prismatic sheaves.

18.24 The pairwise pullback and its central actions

Definition 18.24.1 *(completed external product and pullback).* Let M_p, M_q be finite modules over the local deformation rings. Their completed external product is the finite $B_{p,q}$-module

$$M_{p,q} = \varprojlim_r (M_p/\mathfrak{m}_p^r M_p) \otimes_W (M_q/\mathfrak{m}_q^r M_q).$$

The module obtained by ordinary restriction pullback is

$$M_{p,q}^{\mathrm{res}} = R^{\mathrm{Ch}} \widehat{\otimes}_{B_{p,q}} M_{p,q} = \varprojlim_r \frac{R^{\mathrm{Ch}} \otimes_{B_{p,q}} M_{p,q}}{\mathfrak{m}_{\mathrm{glob}}^r (R^{\mathrm{Ch}} \otimes_{B_{p,q}} M_{p,q})}. \tag{18.26}$$

The algebraic base change is finite over R^{Ch} and therefore already complete. Equation (18.26) also specifies its topology.

PROPOSITION 18.24.2 *(the action on the formal pullback).* Let Z_p, Z_q be unital W-algebras acting R_p^{Ch}-linearly and R_q^{Ch}-linearly on M_p, M_q, respectively. Each acting endomorphism is continuous for the maximal-ideal topology. Then $M_{p,q}^{\mathrm{res}}$ has a canonical action of $Z_p \otimes_W Z_q$, given by

$$(z_p \otimes z_q)(a \otimes (m_p \otimes m_q)) = a \otimes (z_p m_p \otimes z_q m_q).$$

This action commutes with the R^{Ch}-action.

Proof. Finite presentations of M_p, M_q show that $M_{p,q}$ is finite over $B_{p,q}$. Base change gives a finite module over R^{Ch}. Local linearity makes the displayed action compatible with the balancing relations and with every quotient in the inverse limits. It therefore defines an action on the completed tensors.

The two local actions affect separate factors, so they commute. Multiplication on the first factor commutes with both. No topology on the abstract algebras Z_p, Z_q is needed. □

The associated coherent sheaf on $\mathrm{Spf}\, R^{\mathrm{Ch}}$ is the ordinary pullback of the completed external product. The derived construction replaces the external product and pullback by their derived completed versions. Its boundedness requires additional Tor conditions. The ordinary pullback is exact only on a domain where the required higher Tor groups vanish.

PROPOSITION 18.24.3 *(transport of ordinary central actions).* For a W-linear category $\mathcal{C}$, let $Z(\mathcal{C}) = \mathrm{End}(\mathrm{id}_{\mathcal{C}})$ be its ordinary center. A W-linear equivalence $F : \mathcal{C} \to \mathcal{D}$ induces an isomorphism

$$Z(\mathcal{C}) \xrightarrow{\sim} Z(\mathcal{D}). \tag{18.27}$$

For a fully faithful functor, this statement applies to its essential image. It does not identify the center of a larger target category.

Proof. Choose a quasi-inverse G and the unit and counit isomorphisms. A natural endomorphism z of $\mathrm{id}_{\mathcal{C}}$ defines an endomorphism of $\mathrm{id}_{\mathcal{D}}$ by conjugating FzG with the counit. Naturality gives a central action, and composition and addition are preserved. This defines the map in Equation (18.27). The construction with G is its inverse. Restricting a fully faithful functor to its essential image makes it an equivalence. □

Remark 18.24.4 *(local coefficients and Bernstein blocks).* For smooth representations of $GL_2(\mathbb{Q}_v)$ in characteristic zero with coefficient prime different from v, the Bernstein decomposition has the form

$$\mathrm{Rep}(GL_2(\mathbb{Q}_v)) = \prod_{\mathfrak{s}} \mathrm{Rep}_{\mathfrak{s}}(GL_2(\mathbb{Q}_v)). \tag{18.28}$$

For the blocks in Equation (18.28), the ordinary scalar comparison and its precise domain are proved in Proposition 116.2.2. Its image in the spectral categorical center is the scalar subalgebra of Proposition 116.2.5. Using a Bernstein center as Z_v in the formal pullback requires an action over the chosen coefficient ring and on the actual module M_v. A categorical comparison additionally needs a functor with specified essential image. A local (φ, Γ)-module at residue prime v is not identified here with a deformation over $W(k)$ of coefficient characteristic $p_0 \neq v$.

18.25 Equivariant isomorphisms on formal charts

The module $M^{\mathrm{res}}_{p,q}$ is constructed from local data. Let N be an independently specified global module. The pairwise comparison asks for an isomorphism from the former to the latter which intertwines their central actions. Equal determinant laws alone do not solve this problem.

Example 18.25.1 *(extension data invisible to determinant laws).* Let $R = k[[t]]$ and $M = N = R^2$. Give them actions of $R[z]$ by

$$z|_M = 0, \qquad z|_N = J = \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix}.$$

For every polynomial f, one has $f(J) = f(0)1 + f'(0)J$, so both actions have characteristic polynomial $(X - f(0))^2$ for $f(z)$. The determinant polynomial laws agree after every coefficient extension.

The actions also coincide modulo t. An invertible intertwiner U would satisfy $JU = 0$, contradicting $J \neq 0$. These are reducible algebra representations. They exhibit the loss of extension data for coherent objects. They do not contradict reconstruction of a representation from an absolutely irreducible residual determinant.

Proposition 18.25.2 *(the scheme of equivariant isomorphisms).* Let $(R, \mathfrak{m}, k)$ be a complete Noetherian local ring. Let M, N be free R-modules of rank n with actions of an R-algebra generated by $a_1, \dots, a_s$. Choose bases and let $P_i, Q_i \in \mathrm{Mat}_n(R)$ denote the action matrices on M, N. The functor of equivariant isomorphisms is represented by

$$\mathrm{Isom}(M, N) = \mathrm{Spec}\, \frac{R[u_{ab}, d]}{(d \det(U) - 1,\ (Q_i U - U P_i)_{ab} : i, a, b)}. \tag{18.29}$$

Here $U = (u_{ab})$, with $1 \leq a, b \leq n$ and $1 \leq i \leq s$.

Proof. Over any commutative R-algebra A, a matrix U is invertible exactly when its determinant is a unit. The variable d specifies the unique inverse of that determinant. Intertwining the generators is equivalent to intertwining the whole algebra. Thus maps from the coordinate ring in Equation (18.29) to A are exactly the equivariant isomorphisms $M \otimes_R A \to N \otimes_R A$, naturally in A. □

Proposition 18.25.3 *(successive lifting obstructions).* In the preceding situation, put $I_r = \mathfrak{m}^r/\mathfrak{m}^{r+1}$ for $r \geq 1$. Let U_r be an invertible intertwiner modulo $\mathfrak{m}^r$. Choose a lift $\widetilde{U}$ modulo $\mathfrak{m}^{r+1}$ and set

$$E_i = Q_i \widetilde{U} - \widetilde{U} P_i \in \mathrm{Mat}_n(I_r).$$

Write bars for reduction modulo $\mathfrak{m}$, and define

$$L_r : \mathrm{Mat}_n(I_r) \longrightarrow \mathrm{Mat}_n(I_r)^s, \qquad V \longmapsto (\overline{Q}_i V - V \overline{P}_i)_i.$$

The class $[(E_i)_i] \in \operatorname{coker} L_r$ is independent of the lift. It vanishes precisely when this U_r extends to an invertible intertwiner modulo $\mathfrak{m}^{r+1}$. A compatible sequence of such lifts determines an equivariant isomorphism over R.

Proof. Changing $\widetilde{U}$ by $V \in \mathrm{Mat}_n(I_r)$ changes the defects by $L_r(V)$, since $\mathfrak{m} I_r = 0$. Thus the class is independent of the lift and vanishes exactly when a correction solves every intertwining equation. The determinant of a corrected lift remains invertible modulo the maximal ideal, hence is a unit. Completeness gives the matrix limit of a compatible sequence. The intertwining equations pass to the limit, and its determinant is a unit because its residue is. Conversely an equivariant isomorphism over R gives such a compatible sequence by reduction. □

The obstruction depends on the specified partial lift U_r. Vanishing for a compatible sequence, rather than separate choices at unrelated orders, constructs the limit. For general coherent complexes the comparison uses a derived equivariant Hom complex. The finite free criterion does not assert that the intended arithmetic sheaves satisfy its hypotheses.

18.26 Global actions and geometric comparison

PROPOSITION 18.26.1 *(the action generated by local centers).* Let E be a commutative ring and Z_v commutative unital E-algebras. For finite sets S of places put $A_S = \bigotimes_{v \in S, E} Z_v$, with transition maps inserting units. Set

$$A = \varinjlim_{S \text{ finite}} A_S. \tag{18.30}$$

Pairwise commuting unital actions $\alpha_v : Z_v \to \mathrm{End}_E(M)$ on an E-module induce a unique action

$$\alpha : A \to \mathrm{End}_E(M), \qquad \alpha(\otimes_{v \in S} z_v) = \prod_{v \in S} \alpha_v(z_v).$$

Its image is $A/\ker \alpha$. Natural actions on an E-linear category instead give a map to its ordinary center.

Proof. The product of the commuting actions is multiplicative and E-multilinear on each finite product. The tensor-product property gives an action of A_S. Inserting a unit changes no endomorphism, so these maps are compatible. The colimit in Equation (18.30) therefore gives α. The quotient assertion is the first isomorphism theorem. Products of natural endomorphisms are natural, which proves the categorical assertion. □

PROPOSITION 18.26.2 *(faithfulness and generation for a global scalar center).* Suppose the preceding actions define $\alpha : A \to Z(C)$ on an E-linear category. Then

$$A/\ker \alpha \xrightarrow{\sim} \operatorname{im} \alpha \subseteq Z(C). \tag{18.31}$$

This image is the entire ordinary center exactly when every central endomorphism is generated by the finite local actions. The map from A itself is an isomorphism exactly when it is also faithful. No identification with R^{Ch} is implicit in this statement.

Proof. The quotient isomorphism in Equation (18.31) follows from the preceding proposition. Surjectivity means precisely that every central endomorphism belongs to this image. Injectivity means $\ker \alpha = 0$. Together these are exactly the isomorphism conditions. □

Remark 18.26.3 *(first-order fixed-determinant deformations).* Let $V = k^2$ with action $\bar\rho$, and let $\operatorname{ad}\bar\rho = \operatorname{End}_k(V)$ with conjugation action. A lift over $k[\epsilon]/(\epsilon^2)$ has the form $(1+\epsilon c(g))\bar\rho(g)$. Multiplicativity is the continuous cocycle equation for c. The fixed-determinant condition is $\operatorname{tr} c(g) = 0$.

Strict change of basis changes c by the coboundary of an element of $\operatorname{ad}\bar\rho$. Thus the tangent space before additional local conditions is

$$\frac{Z^1_{\mathrm{cts}}(G_{\mathbb{Q},S}, \operatorname{ad}^0\bar\rho)}{d\ \operatorname{ad}\bar\rho}.$$

The denominator consists of traceless cocycles. When 2 is invertible, scalar subtraction identifies it with $d\ \operatorname{ad}^0\bar\rho$, giving the usual $H^1_{\mathrm{cts}}(G_{\mathbb{Q},S}, \operatorname{ad}^0\bar\rho)$.

At characteristic two this simplification needs justification. Additional deformation conditions restrict this tangent problem. A tangent-space comparison does not identify the complete rings.

18.27 Frobenius coefficients and global reconstruction

Unramified characteristic polynomials provide scalar functions on a Galois deformation space. The full collection determines the complete local ring with its deformation topology. A finite generating subset exists, but its choice depends on the deformation problem.

Let K be a number field, let k be a finite field of characteristic p_0, and put $W = W(k)$. Fix a finite set S containing the infinite places and all places above p_0. Write $G = G_{K,S}$ for the Galois group of the maximal extension of K unramified outside S. Fix a continuous absolutely irreducible representation $\bar\rho : G \to \mathrm{GL}_2(k)$ and a continuous character $\delta : G \to W^\times$ lifting $\det\bar\rho$.

For a finite local W-algebra A with residue field k, a deformation is a continuous lift $\rho : G \to \mathrm{GL}_2(A)$ with determinant δ_A. Two lifts are strictly equivalent if they are conjugate by an element of $1 + M_2(\mathfrak{m}_A)$. Let $R = R^{\delta}_{\bar\rho,S}$ be the complete Noetherian local W-algebra representing these deformation classes. Equivalently, R represents continuous rank-two determinant laws lifting $\det\bar\rho$ with determinant character δ.

The representability and comparison hypotheses are those of Proposition 18.2.2. In particular, Mazur's finiteness condition gives Noetherianity. Let ρ^u be a universal lift, let $T = \operatorname{tr}\rho^u$, and let $\mathfrak{m}$ be the maximal ideal of R. All topologies on R below are $\mathfrak{m}$-adic.

For a finite place $v \notin S$, use arithmetic Frobenius Frob_v, acting on residue fields by the $\#k(v)$-power map. Its conjugacy class in G is well defined. Put

$$t_v = T(\mathrm{Frob}_v), \qquad d_v = \delta(\mathrm{Frob}_v), \qquad P_v(X) = X^2 - t_vX + d_v.$$

These conventions involve no choice of a square root of $\#k(v)$.

The distinction between a representation and its determinant law is already visible over $\mathbb{F}_2[\epsilon]/(\epsilon^2)$. For the cyclic group $C_2 = \langle s\rangle$, the representations

$$\rho_0(s) = I_2, \qquad \rho_1(s) = (1+\epsilon)I_2$$

have the same trace and determinant on every element. Their determinant laws agree, since

$$\det\rho_1(a+bs) = (a+b(1+\epsilon))^2 = (a+b)^2 = \det\rho_0(a+bs)$$

after every commutative coefficient extension. They are not conjugate, since both images of s are scalar and they differ. Thus equality of determinant laws need not determine a representation when the residual representation is reducible.

A framing retains a chosen basis, so framed deformations are lifts without quotienting by strict conjugacy. Over $A = \mathbb{F}_2[\epsilon]/(\epsilon^2)$, consider the constant lift $\sigma_0 : \mathrm{GL}_2(\mathbb{F}_2) \to \mathrm{GL}_2(A)$ of the natural representation. Its reduction is absolutely irreducible: the matrices

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

have different unique invariant lines over $\overline{\mathbb{F}}_2$. Put $P = I_2 + \epsilon E_{12}$ and $\sigma_1 = P\sigma_0 P^{-1}$.

Since $\epsilon^2 = 0$ and the characteristic is 2,

$$PUP^{-1} = U, \qquad PVP^{-1} = V + \epsilon(E_{12}V - VE_{12}) = V + \epsilon I_2.$$

Both lifts have determinant 1. Conjugation preserves all traces and determinant laws, but the displayed images of V differ. These are the same unframed deformation and distinct framed deformations.

LEMMA 18.27.1 *(trace determination with absolutely irreducible reduction).* Let A be a commutative local ring with residue field k. Suppose $\rho_1, \rho_2 : G \to \mathrm{GL}_2(A)$ reduce to the same absolutely irreducible representation over k. If $\operatorname{tr}\rho_1(g) = \operatorname{tr}\rho_2(g)$ for every $g \in G$, then ρ_1 and ρ_2 are strictly equivalent. The assertion includes residue characteristic 2.

Proof. The k-linear span of the residual group matrices is $M_2(k)$. Here is a proof in dimension two. Extend scalars to an algebraic closure $\bar{k}$ and let E be the algebra spanned by the group image. Absolute irreducibility says that E acts irreducibly.

It contains a nonscalar matrix, and subtracting an eigenvalue produces a nonzero rank-one matrix $u\varphi$. Irreducibility gives $Eu = \bar{k}^2$ and $\varphi E = (\bar{k}^2)^*$ over the algebraic closure. Indeed, the annihilator of φE is invariant and cannot be the whole space. Thus the matrices $a(u\varphi)b$ span all endomorphisms. Descent of dimension gives the asserted span over k.

Choose $g_1, \dots, g_4 \in G$ whose residual matrices form a basis. For $a = 1, 2$, the matrices $A_{a,i} = \rho_a(g_i)$ form an A-basis of $M_2(A)$, since their coordinate determinant is a unit. The trace pairing is perfect over every commutative ring:

$$\operatorname{tr}(E_{ij}E_{rs}) = \delta_{jr}\delta_{is}.$$

Hence the common Gram matrix $Q_{ij} = \operatorname{tr}\rho_a(g_i g_j)$ is invertible.

The coordinates $c_i(g)$ of $\rho_a(g)$ in this basis satisfy

$$\sum_i c_i(g) Q_{ij} = \operatorname{tr}\rho_a(g g_j).$$

They are therefore independent of a. The A-linear map $\Phi(A_{1,i}) = A_{2,i}$ sends $\rho_1(g)$ to $\rho_2(g)$ for every g. Since products of basis matrices are group matrices, Φ is a unital algebra isomorphism. Its reduction is the identity.

Put $F_{ij} = \Phi(E_{ij})$, $v = F_{11}e_1$, and $w = F_{21}v$. The matrix $P = (v, w)$ is congruent to I_2 modulo $\mathfrak{m}_A$, so it is invertible. The matrix-unit relations give $F_{ij}P = PE_{ij}$. Consequently $\Phi(X) = PXP^{-1}$, which proves strict equivalence. No division by 2 occurs. □

For example, over $\mathbb{F}_2$ the matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad UV$$

form a basis. Their trace Gram matrix is

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix},$$

whose determinant is 1, although $\operatorname{tr}(I) = 0$.

Lemma 18.27.2 *(unramified coefficients determine continuous determinants).* Consider two continuous rank-two determinant laws over a complete separated local coefficient ring. They agree if and only if their characteristic polynomials agree at every $v \notin S$. For deformations of $\bar{\rho}$ with fixed determinant δ, equality of Frobenius traces suffices. Their representation classes then agree.

Proof. Chebotarev's theorem implies that the union of the conjugacy classes of Frob_v, for $v \notin S$, is dense in G. Indeed, every conjugacy class in every finite quotient of G occurs at infinitely many such places. The inverse image of any nonempty open subset of a finite quotient therefore meets this union. Trace and determinant are continuous conjugacy-invariant functions. Equality on Frobenius classes consequently implies equality on G.

The passage from group coefficients to polynomial laws is integral. For a rank-two determinant law D with trace T_D and determinant character d, its value after any commutative coefficient extension $A \to C$ is

$$D_C\left(\sum_{i=1}^{r} a_i g_i\right) = \sum_i a_i^2 d(g_i) + \sum_{i<j} a_i a_j d(g_j) T_D(g_i g_j^{-1}).$$

To derive the mixed coefficient, multiply $Ug_i + Vg_j$ on the right by g_j^{-1} and use multiplicativity. The characteristic polynomial of $g_i g_j^{-1}$ gives

$$D(Ug_i + Vg_j) = d(g_i)U^2 + d(g_j)T_D(g_i g_j^{-1})UV + d(g_j)V^2.$$

A homogeneous quadratic polynomial has only diagonal and mixed coefficients, so this determines the displayed formula for every finite sum. Thus equality of (T_D, d) implies equality of the laws. The previous lemma proves the assertion about representation classes. □

Theorem 18.27.3 *(global scalar reconstruction from Frobenius coefficients).* Define the polynomial parameter algebra

$$H^S = W[\mathsf{T}_v, \mathsf{D}_v, \mathsf{D}_v^{-1} : v \notin S]$$

and its fixed-determinant quotient

$$H^{S,\delta} = H^S/(\mathsf{D}_v - d_v : v \notin S) \simeq W[\mathsf{T}_v : v \notin S].$$

The map

$$\eta : H^{S,\delta} \longrightarrow R, \qquad \mathsf{T}_v \longmapsto t_v$$

has $\mathfrak{m}$-adically dense image.

Choose $a_v \in W$ lifting $\operatorname{tr} \bar{\rho}(\mathrm{Frob}_v)$, put $\mathfrak{n} = (p_0, \mathsf{T}_v - a_v : v \notin S)$, and define

$$\widehat{H}^{S,\delta} = \varprojlim_n H^{S,\delta}/\mathfrak{n}^n.$$

The induced continuous homomorphism $\widehat{\eta} : \widehat{H}^{S,\delta} \to R$ is surjective. It gives the topological quotient

$$R \simeq \widehat{H}^{S,\delta}/\ker \widehat{\eta}.$$

Moreover, finitely many places $v_1, \dots, v_d \notin S$ give a continuous surjection

$$W[[X_1, \dots, X_d]] \longrightarrow R, \qquad X_i \longmapsto t_{v_i} - a_{v_i}.$$

The same density conclusion holds after any quotient imposing a closed deformation condition.

Proof. Let B be the closure of $W[t_v : v \notin S]$ in R. It is a closed subring. Chebotarev and continuity show that $T(g) \in B$ for every $g \in G$. The ring B is profinite, with finite quotients $B_n = B/(B \cap \mathfrak{m}^n)$.

It is local with residue field k. Indeed, if $b \in B$ has nonzero residue, choose a unit $u \in W$ with the same residue. Writing $b = u(1 - x)$ with $x \in B \cap \mathfrak{m}$, the convergent series $u^{-1} \sum_{j \geq 0} x^j$ belongs to B and inverts b.

The quadratic formula in the previous proof defines the universal determinant law over B, because all its coefficients belong to B. Its multiplicativity follows as a polynomial identity: the identities hold after the injection $B[\mathbf{X}] \hookrightarrow R[\mathbf{X}]$, and hence already hold over B. This also proves the identities after arbitrary coefficient extensions, without a flatness assumption. Its coefficient functions are continuous for the subspace topology on B, and its determinant character is δ.

The representing property gives compatible maps $f_n : R \to B_n$. Their inverse limit is a continuous map $f : R \to B$. If $i : B \hookrightarrow R$ is the inclusion, then $i \circ f$ induces the universal determinant over each $R/\mathfrak{m}^n$. Uniqueness in the representing property gives $i \circ f = \mathrm{id}_R$. Therefore $B = R$. This argument uses finite local quotients of B and does not assume that B is Noetherian before proving the equality.

Density implies that the classes of $t_v - a_v$ span $\mathfrak{m}/(\mathfrak{m}^2 + p_0 R)$ over k. In detail, approximate any element of $\mathfrak{m}$ modulo $\mathfrak{m}^2 + p_0 R$ by a polynomial in the t_v. Substituting $t_v = a_v + (t_v - a_v)$ leaves only its linear part in this quotient. Noetherianity makes this vector space finite dimensional, so choose finitely many places whose classes form a basis.

Nakayama's lemma gives $\mathfrak{m} = (p_0, t_{v_1} - a_{v_1}, \dots, t_{v_d} - a_{v_d})$. Completeness then gives the displayed power-series surjection: approximate an element successively modulo $\mathfrak{m}^n$, lifting its degree-n correction to $(p_0, X_1, \dots, X_d)^n$. The resulting corrections converge in the power-series ring.

The polynomial map η sends $\mathfrak{n}^n$ into $\mathfrak{m}^n$, so it extends to $\widehat{\eta}$. The preceding finite power-series map factors through $\widehat{H}^{S,\delta}$, proving surjectivity. If F_n is the kernel of the projection $\widehat{H}^{S,\delta} \to H^{S,\delta}/\mathfrak{n}^n$, then $\widehat{\eta}(F_n) = \mathfrak{m}^n$. The inclusion from left to right follows by construction.

The reverse inclusion follows from the finite power-series surjection on the nth powers of the maximal ideals. Thus the quotient topology is exactly the $\mathfrak{m}$-adic topology. Density passes to every quotient of R. □

Remark 18.27.4 *(topology and global relations).* If $A_0 = \eta(H^{S,\delta})$, its completion for the induced filtration $A_0 \cap \mathfrak{m}^n$ is canonically R:

$$\varprojlim_n A_0/(A_0 \cap \mathfrak{m}^n) \simeq R.$$

The kernel of $\widehat{\eta}$ records the global deformation relations. The theorem does not identify it with the closure of the kernel of η. Such an identification requires a separate comparison of completion and quotient. The finite set of generating places depends on the deformation problem. An arbitrarily prescribed pair of places need not generate its ring.

Proposition 18.27.5 *(obstruction to a product of deformation components).* The fixed-residual ring R has no decomposition as a product of two nonzero unital rings. More generally it cannot be a unital restricted product containing the characteristic idempotents of two nonzero factors.

Proof. If $e^2 = e$ in a local ring, one of e and $1 - e$ is a unit. The equation $e(1 - e) = 0$ then gives $e = 1$ or $e = 0$. A product of two nonzero rings contains the nontrivial idempotent $(1, 0)$. The same obstruction applies whenever a restricted product contains these characteristic idempotents. □

The algebra $H^{S,\partial}$ consists of formal characteristic-polynomial parameters. An automorphic action requires a separate module or category and a homomorphism from these parameters to its endomorphisms. An identification of its completed action image with R additionally requires equality of the two kernels. The scalar reconstruction gives no such equality. It also supplies no local type decomposition, flatness assertion, prismatic filtration, untilt descent, or identification with a higher center.

Example 18.27.6 *(a split generic fiber of a local ring).* Put $R = \mathbb{Z}_2[[x]]/(x(x-2))$. It is local with residue field $\mathbb{F}_2$. The relation $x^m = 2^{m-1}x$ for $m \geq 1$ reduces every series uniquely to $a + bx$ with $a, b \in \mathbb{Z}_2$. The reductions converge 2-adically.

Uniqueness follows by evaluating at $x = 0$ and $x = 2$: $a = 0$ and $2b = 0$ imply $a = b = 0$. After inverting 2, these evaluations give

$$R[1/2] \xrightarrow{\sim} \mathbb{Q}_2 \times \mathbb{Q}_2, \qquad a + bx \longmapsto (a, a + 2b). \tag{18.32}$$

The inverse in Equation (18.32) sends (u, v) to $u + (v - u)x/2$. Integrally, the image consists of pairs in $\mathbb{Z}_2^2$ with $v - u \in 2\mathbb{Z}_2$, so $(1, 0)$ is not in the image. The generic-fiber splitting therefore does not split R.

This distinction is necessary before relating a Bernstein component to a formal deformation space. Such a comparison must specify whether it concerns the integral ring, its generic fiber, a quotient, or an action image. A correspondence of closed points establishes none of those ring identifications by itself.

Remark 18.27.7 *(coefficient variants).* The same proof applies over a complete discrete valuation ring $\mathcal{O}$ finite over W. Fix its finite residue field, residual representation and determinant character. Replace p_0 by a uniformizer of $\mathcal{O}$, and assume the corresponding complete Noetherian representing ring exists. For a deformation problem over k itself, omit the uniformizer from the residual ideal. These are statements about the indicated coefficient problems. A comparison between their representing rings requires its own base-change map and proof.

The density argument uses only the representing determinant functor. Absolute irreducibility is used to identify that functor with representation deformations and to obtain strict conjugacy from traces.

The polynomial-law identities are Chenevier, arXiv:0809.0415v2, Example 1.8 and Lemma 1.9, pp. 10–11. Definition–Proposition 2.18 and Theorem 2.22, pp. 32–34, give the Cayley–Hamilton and absolutely irreducible comparisons. Proposition 3.3 through Proposition 3.7, pp. 42–43, supply the deformation-ring assertions. The number-field density theorem is stated in Stevenhagen–Lenstra, *Chebotarev and his density theorem*, Appendix, p. 35. The finite set of generating places in the theorem is not a prescribed pair: the two-prime nonuniqueness example of Proposition 18.22.1 retains its different quantifiers.

The remaining global automorphic comparison

PROPOSITION 18.27.8 *(a specified action quotient and its comparison).* Use the coefficient ring and deformation problem of Theorem 18.27.3. Suppose an automorphic module or category has a complete separated commutative W-algebra A_{act} of scalar actions. Assume there is a continuous surjection

$$\theta : \widehat{H}^{S,\partial} \longrightarrow A_{\mathrm{act}}.$$

Give A_{act} the quotient topology for this map.

If $\ker\theta = \ker\widehat{\eta}$, there is a unique topological W-algebra isomorphism $A_{\mathrm{act}} \simeq R$ intertwining the maps from $\widehat{H}^{S,\partial}$. To identify this algebra with the full ordinary center of the global category, additionally require that its central action be faithful and exhaust that center. An identification of higher centers additionally requires a specified equivalence of their derived algebras, not only this degree-zero isomorphism.

Proof. Both algebras are topological quotients of the same algebra by the same ideal. The identity of the source descends to the claimed isomorphism, with inverse induced in the same way. It is a homeomorphism because both topologies are quotient topologies. Surjectivity of the source map proves uniqueness.

A faithful central action identifies an algebra with its image. Equality with the full center requires that image to be the whole center. The argument has made no assertion about higher cohomology or operations. □

The Frobenius theorem constructs $\widehat{\eta}$ and its scalar quotient. It does not construct θ or compare these kernels. The unframed Galois deformation problem is fixed at one residual representation. An automorphic spherical locus, local Bernstein types and their coefficient changes must be specified independently. Almost-everywhere sphericity does not provide the missing ring maps.

18.28 COHERENT GLUING AND ITS HYPOTHESES

A restriction diagram reconstructs objects only when it retains their overlap and extension maps. Closed and open restrictions already show the issue on the affine line. A formal neighborhood provides a positive coherent gluing theorem.

Closed and open restriction on the affine line

Let k be a field and put $A = k[t]$, $C = k[t, t^{-1}]$. Let $X = \operatorname{Spec} A$, $Z = V(t)$, and $U = D(t)$. Write $\mathrm{mod}_{\mathrm{fg}}(R)$ for the category of finitely generated left modules over a ring R. The intersection $Z \times_X U$ is empty, since its coordinate ring is $(A/(t))[t^{-1}] = 0$. Consequently a fiber product of categories over the category of this intersection is a product.

PROPOSITION 18.28.1. The restriction functor

$$\mathrm{mod}_{\mathrm{fg}}(A) \longrightarrow \mathrm{mod}_{\mathrm{fg}}(k) \times \mathrm{mod}_{\mathrm{fg}}(C), \qquad M \longmapsto (M/tM, M[t^{-1}]),$$

is not full. The corresponding derived restriction functor on perfect complexes is not full either.

Proof. The object A maps to (k, C). Its endomorphism homomorphism is

$$A \longrightarrow k \times C, \qquad f \longmapsto (f(0), f).$$

The endomorphism $(1, 0)$ does not belong to its image. The same calculation applies to degree-zero derived endomorphisms. □

The extension retained by localization

Let $D(A)$ be the stable k-linear ∞-category obtained by inverting quasi-isomorphisms of unbounded cochain complexes of A-modules. Cohomological shifts satisfy $H^n(M[1]) = H^{n+1}(M)$. Let $D_t(A)$ be its full subcategory of complexes whose cohomology modules are elementwise killed by powers of t. Extension of scalars $L = C \otimes^{\mathbf{L}}_A -$ is left adjoint to restriction of scalars $j_* : D(C) \to D(A)$.

Because C is a localization, it is flat and $C \otimes_A C = C$. Thus $Lj_* = \mathrm{id}$ and j_* is fully faithful. Define

$$\Gamma_t M = \mathrm{fib}(M \longrightarrow j_* LM).$$

Localization is exact, so $L\Gamma_t M = 0$.

Exactness of localization on modules identifies $\ker L$ with $D_t(A)$. If $K \in D_t(A)$, adjunction gives $\mathrm{Map}_{D(A)}(K, j_*N) = 0$ for every $N \in D(C)$. Applying mapping spaces to the defining fiber sequence therefore shows that Γ_t is right adjoint to $D_t(A) \hookrightarrow D(A)$.

Define a stable ∞-category $\mathcal{G}$ whose objects are

$$(K, N, \delta), \qquad K \in D_t(A), \quad N \in D(C), \quad \delta : j_*N \longrightarrow K[1].$$

Precisely, $\mathcal{G}$ is the pullback of $\mathrm{Fun}(\Delta^1, D(A)) \to D(A) \times D(A)$ along $(N, K) \mapsto (j_*N, K[1])$. Morphisms include the homotopies in the resulting commutative squares, rather than merely an equality in a homotopy category.

Proposition 18.28.2. There is an equivalence of stable ∞-categories

$$D(A) \simeq \mathcal{G}.$$

It sends M to $(\Gamma_t M, LM, \delta_M)$, where δ_M is the connecting morphism of the localization triangle. Its inverse sends (K, N, δ) to $\mathrm{fib}(\delta)$.

Proof. A triple defines the fiber sequence

$$K \longrightarrow \mathrm{fib}(\delta) \longrightarrow j_*N \xrightarrow{\delta} K[1].$$

Applying L identifies $L\,\mathrm{fib}(\delta)$ with N. Under this identification, the middle map is the unit of $L \dashv j_*$: by adjunction it corresponds to the identity of N. Its fiber is therefore $\Gamma_t\,\mathrm{fib}(\delta) = K$.

These identifications recover the connecting morphism δ. Conversely, the fiber of the connecting map in a localization triangle is its middle term. Fiber constructions are functorial in a stable ∞-category, so these identifications give mutually inverse functors, including their mapping spaces. □

For $M = A$, the complex $\Gamma_t A$ is represented by $[A \to C]$ in degrees 0, 1. Hence $\Gamma_t A \simeq (C/A)[-1]$, and the connecting map is

$$C \longrightarrow C/A.$$

Its fiber is A in degree zero. This nonzero quotient map is the extension datum that reconstructs A.

The two objects C and $\Gamma_t A$ are not perfect over A: their nonzero cohomology modules are not finitely generated. Thus this localization construction takes place in $D(A)$. It does not present perfect complexes as a product of perfect complexes on the two strata. Nor can the supported category be replaced by $D(k)$ under closed pushforward. Indeed, the resolution $0 \to A \xrightarrow{t} A \to k \to 0$ gives $\mathrm{Ext}^1_A(k, k) = k$, whereas $\mathrm{Ext}^1_k(k, k) = 0$.

Coherent gluing across a formal neighborhood

To retain coherent modules, replace the closed point by its formal neighborhood. Put

$$B = k[[t]], \qquad D = k((t)), \qquad C = k[t, t^{-1}].$$

The maps $A \to B$, $A \to C$, $B \to D$, and $C \to D$ are the inclusions. The overlap $\operatorname{Spec} D$ is the punctured formal disk.

It is not the empty intersection $Z \cap U$. The ring B is the coordinate ring of the t-adic formal disk. Here its coherent category means finitely generated B-modules, with their t-adic topology. Tensor products displayed below are ordinary algebraic tensor products unless explicitly derived.

Let $\mathcal{P}$ be the category of triples

$$(P, Q, \eta), \qquad P \in \operatorname{mod}_{\mathrm{fg}}(B), \quad Q \in \operatorname{mod}_{\mathrm{fg}}(C), \quad \eta : D \otimes_B P \xrightarrow{\sim} D \otimes_C Q.$$

Morphisms are compatible pairs of module homomorphisms.

THEOREM 18.28.3. The functor

$$\operatorname{mod}_{\mathrm{fg}}(A) \longrightarrow \mathcal{P}, \qquad M \longmapsto (B \otimes_A M, C \otimes_A M, \mathrm{can})$$

is an equivalence. Its inverse sends a triple to

$$\{(p, q) \in P \oplus Q : \eta(1 \otimes p) = 1 \otimes q\}.$$

Proof. Every Laurent series is the sum of a power series and a Laurent polynomial. Their intersection is $k[t]$. Consequently

$$0 \longrightarrow A \longrightarrow B \oplus C \xrightarrow{(b,c) \mapsto b - c} D \longrightarrow 0$$

is exact. The modules B, C, D are flat over A.

For C, D this follows from localization and torsion-freeness over the principal ideal domain A. A torsion-free module over a principal ideal domain is the filtered union of its finite free submodules. It is therefore flat, which also applies to B. Since D is flat, tensoring the displayed sequence with any A-module M remains exact.

In particular M is the indicated fiber product of its two scalar extensions. A compatible pair of homomorphisms on these extensions sends this fiber product into the corresponding fiber product for the target. The resulting A-linear homomorphism recovers the pair, since the elements $1 \otimes m$ generate both scalar extensions. This proves full faithfulness.

For essential surjectivity, use the structure theorem for finitely generated modules over the principal ideal domains B and C. A torsion summand $B/(t^m)$ descends from $A/(t^m)$, whose scalar extension to C vanishes. A torsion summand of Q has the form $C/(f)$, where $f \in k[t]$ and $f(0) \neq 0$. It descends from $A/(f)$, whose extension to B vanishes because f is a unit in B. Both kinds of torsion disappear over D. After splitting off these summands, P and Q are free of the same rank r.

Choose $Q = C^r$ and view P as a B-lattice $\Lambda \subset D^r$ through η. There is an integer $n \geq 0$ such that

$$t^n B^r \subset \Lambda \subset t^{-n} B^r.$$

Set $M = \Lambda \cap C^r$. Intersecting the bounds with C^r gives

$$t^n A^r \subset M \subset t^{-n} A^r.$$

Thus M is a finite free A-module of rank r, and $C \otimes_A M = C^r$. For $\lambda \in \Lambda$, truncate its Laurent expansion below degree n. The truncation q is a Laurent polynomial, and $\lambda - q \in t^n B^r \subset \Lambda$. Hence $q \in M$ and $\Lambda = M + t^n B^r$.

The map $B \otimes_A M \to \Lambda$ is therefore surjective. It is injective: an A-basis of M becomes a basis of C^r, so its determinant remains nonzero over the field D. This proves $B \otimes_A M = \Lambda$ with the specified overlap map. Adding the torsion summands proves essential surjectivity. Full faithfulness identifies the stated fiber-product construction with a quasi-inverse on all triples. □

For a k-linear category $\mathcal{C}$, write $Z(\mathcal{C}) = \mathrm{End}(\mathrm{id}_{\mathcal{C}})$ for its ordinary center, with multiplication given by composition.

Corollary 18.28.4. Under the preceding equivalence,

$$Z(\mathrm{mod}_{\mathrm{fg}}(A)) \simeq B \times_D C \simeq Z(\mathrm{mod}_{\mathrm{fg}}(B)) \times_{Z(\mathrm{mod}_{\mathrm{fg}}(D))} Z(\mathrm{mod}_{\mathrm{fg}}(C)).$$

The comparison maps on centers are the indicated inclusions of rings.

Proof. For every commutative ring R, a natural endomorphism z of $\mathrm{mod}_{\mathrm{fg}}(R)$ is multiplication by $a = z_R(1) \in R$. Indeed, for each $m \in M$, apply naturality to $R \to M$, $1 \mapsto m$. This yields $z_M(m) = am$. Thus the center is R, compatibly with scalar extension. The exact ring sequence above gives $A = B \times_D C$. □

Effective equivariant descent

Let X be a specified global parameter space in a fixed coefficient theory, and let $U_\bullet \to X$ be an augmented simplicial diagram. Fix the coherent or ind-coherent support subcategories being used. Suppose restriction is an equivalence

$$\mathcal{C}(X) \xrightarrow{\sim} \mathrm{Tot}\, \mathcal{C}(U_\bullet).$$

An object of the totalization includes the objects on the charts, their isomorphisms on overlaps and every higher compatibility.

Applying the inverse equivalence constructs its global object. Full faithfulness identifies the mapping spaces and gives uniqueness as a descent object. Coherence follows from this equivalence. It does not follow from an arbitrary filtered colimit of coherent objects in an ind-category.

For a derived pairwise construction there are two separate Tor conditions if ordinary tensors are to compute it. One condition makes the external product of the two local objects ordinary. The other makes its pullback to the global base ordinary. Perfectness of the local objects and flatness of the pullback do not imply the first condition. For $W = k[[t]]$, both local modules k are perfect over W, but

$$k \otimes^{\mathbb{L}}_W k \simeq k \oplus k[1].$$

An identity pullback is flat and retains this negative-degree term.

Suppose, in addition, a compatible family of algebra actions $A_S \to Z^\circ(\mathcal{C})$ is given for every finite set S of places. The ordinary center here consists of natural endomorphisms of the identity on the homotopy category. Their colimit $A = \mathrm{colim}_S A_S$ acts, because every element and relation comes from a finite stage. An equivalence which intertwines each action intertwines the assembled action by that same property.

For derived actions, the corresponding input consists of maps of the specified derived algebras and coherent comparisons. The identity

$$\mathrm{Map}(\operatorname*{colim}_S A_S, Z) \simeq \lim_S \mathrm{Map}(A_S, Z)$$

assembles these comparisons, including their homotopies. Equal degree-zero actions alone do not supply them. All finite sets enter this indexing system. The poset of pairs alone is not filtered.

Question 18.28.5. For the continuous Selmer deformation problem over one coefficient base, construct the geometric charts, overlap correspondences and effective equivariant descent above. Compare the local spectral objects with their arithmetic coefficient objects, preserving the full local parameters, the specified support and the global actions. If a prismatic refinement is required, construct its filtered objects and cross-prime maps on this same recipient.

The ordinary formal pullback modules and their central actions have already been constructed. This question asks for their geometric identification and descent. An actual Nygaard filtration on a local prism does not, by itself, provide cross-prime maps.

18.29 Centers, categorical traces and character classes

Fix a commutative coefficient ring k. Let C be a stable presentable k-linear monoidal category whose tensor preserves colimits in each variable. Its categorical center and trace are

$$Z_{\mathrm{cat}}(C) = \mathrm{Fun}^{\mathrm{L}}_{C\otimes C^{\mathrm{op}}}(C, C), \qquad \mathrm{Tr}_{\mathrm{cat}}(C) = C \otimes_{C\otimes C^{\mathrm{op}}} C.$$

Here tensors are in k-linear presentable categories, and the functors preserve colimits.

Both expressions are categories. The center is braided and the trace has a circle action. For an associative k-algebra A, the scalar Hochschild complexes instead have the forms

$$C^{\bullet}(A, A) = \mathrm{RHom}_{A\otimes^{\mathbb{L}} A^{\mathrm{op}}}(A, A), \qquad C_{\bullet}(A) = A \otimes^{\mathbb{L}}_{A\otimes^{\mathbb{L}} A^{\mathrm{op}}} A.$$

Their degree-zero center and trace are still different objects. An identity trace of a chosen perfect module is a class in HH_0, not the whole Hochschild complex.

For a perfect derived stack X, Ben-Zvi–Francis–Nadler, Theorem 1.7, prove

$$Z_{\mathrm{cat}}(\mathrm{QCoh}(X)) \simeq \mathrm{QCoh}(\mathcal{L}X) \simeq \mathrm{Tr}_{\mathrm{cat}}(\mathrm{QCoh}(X)).$$

Here perfect means that X has affine diagonal and $\mathrm{QCoh}(X) = \mathrm{Ind}\,\mathrm{Perf}(X)$. For $X = \mathrm{Spec}\,A$, scalar Hochschild chains of $\mathrm{Perf}(A)$ identify with functions on the derived self-intersection $X \times^{\mathbb{L}}_{X\times X} X$. This affine algebra calculation and the categorical theorem occur at different levels.

A commutative algebra can still have nontrivial Hochschild information: $HH_1(k[t]) = k[t]\,dt$. Nor does an E_1-algebra automatically have E_2 operations. These observations rule out deductions of arithmetic trace identities from spherical commutativity alone.

The finite Hecke character theorem

Let G be a complex reductive group, B a Borel subgroup, and N its unipotent radical. Put

$$\mathcal{H}_G = D(B\backslash G/B).$$

Its unipotently monodromic version $\widetilde{\mathcal{H}}_G$ is the full colimit-generated subcategory of $D(N\backslash G/N)$ generated by pullbacks from $\mathcal{H}_G$. Both use convolution.

The horocycle correspondence

$$G/\mathrm{Ad}\,G \xleftarrow{p} G/\mathrm{Ad}\,B \xrightarrow{\delta} B\backslash G/B$$

defines the Harish–Chandra transform $p_* \partial^!$. Let Ch_G be the full subcategory of $D(G/\mathrm{Ad}\, G)$ generated under colimits by its image. Ben-Zvi–Nadler, Theorem 1.8 of *The character theory of a complex group*, prove

$$Z_{\mathrm{cat}}(\mathcal{H}_G) \simeq \mathrm{Ch}_G \simeq \mathrm{Tr}_{\mathrm{cat}}(\mathcal{H}_G),$$

and the same for $\widetilde{\mathcal{H}}_G$. These Hecke categories are semi-rigid and have the specified pivotal and Calabi–Yau structures. Semi-rigid means compactly generated with dualizable compact generators. It does not require a compact unit.

A dualizable Hecke module has a character object in Ch_G. This is a positive categorical character construction. It is not an identification of $HH(D(G/\mathrm{Ad}\, G))$ with functions on an unspecified flat locus. An arithmetic identity trace requires its own source category, chosen object and comparison to the intended arithmetic recipient.

Continuity and arithmetic trace comparisons

There is an elementary continuity model on a small differential graded presentation. In an orthogonal union of small K-flat dg categories, every cyclic composable word stays in one block. The Hochschild complex is therefore the direct sum of the block complexes, compatibly with all faces, degeneracies and the Connes operator. Perfect completion retains finite block support. This proves the corresponding circle-equivariant trace formula on that carrier.

An infinite coordinatewise sum of nonzero unital algebras has no unit, and a categorical product is not that algebraic sum. A multiplicative trace comparison consequently needs its actual monoidal enhancement. For a filtered dg diagram, a Hochschild continuity statement likewise requires its compact or Morita presentation and the specified maps on cyclic bars. Neither a cofiltered limit of geometric spaces nor its functions is exchanged with an arithmetic completed trace by notation.

Question 18.29.1. Construct the arithmetic representation or Hecke category, its chosen object and identity trace, and an operation-preserving map to the specified geometric trace recipient. Compare the coefficient theory, completion, singular support and circle action. If a scalar evaluation is prescribed, prove that the map intertwines it on its stated domain.

18.30 Restricted tensors and local actions

A finite collection of local categories has a tensor product over a common coefficient category. Passing to all places requires maps which insert the omitted factors. These maps must preserve the operations being assembled.

Write $\mathcal{V} = \mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ for the symmetric monoidal ∞-category of stable presentable categories and colimit-preserving functors. Its tensor represents functors preserving colimits separately in each input. Its unit is Sp, with sphere object $\mathbb{S}$. The same construction can be made in a specified linear version, whose tensor unit is the corresponding module category.

Fix a small coherent one-colored ∞-operad $\mathcal{O}$, for example E_n or E_∞. Let V be a set of places, and let $E \subset V$ be finite. For each v, give C_v an $\mathcal{O}$-algebra structure in $\mathcal{V}$. For $v \notin E$, supply a morphism

$$\eta_v : \mathrm{Sp} \longrightarrow C_v \quad \text{in } \mathrm{Alg}_{\mathcal{O}}(\mathcal{V}), \qquad u_v = \eta_v(\mathbb{S}).$$

The map includes every operation and its coherences. In the unital E_n cases it sends the unit to the unit, so $u_v \simeq \mathbb{1}_{C_v}$.

Let J_E be the filtered poset of finite subsets $S \subset V$ containing E. Define

$$C_S = \bigotimes_{v \in S} C_v, \qquad \iota_{S,S'} = \mathrm{id} \otimes \bigotimes_{v \in S' \setminus S} \eta_v.$$

For a chain $S \subset S' \subset S''$, both composites insert the same maps at the same places. Tensor coherence supplies their identification and all higher compatibilities. Thus these are a diagram of O-algebras.

THEOREM 18.30.1. The underlying restricted tensor

$$C'_V = \operatorname*{colim}_{S \in J_E} C_S \quad \text{in } \mathcal{V}$$

is the underlying object of the colimit in $\mathrm{Alg}_O(\mathcal{V})$. Every finite-stage map is an O-algebra map. Fixing the full diagram and its underlying colimit cocone, the space of lifts to this structured colimit cocone is contractible.

A canonical single-place map $C_v \to C'_V$ is supplied directly when $E \subset \{v\}$. In general, such a map additionally requires pointings $\mathrm{Sp} \to C_w$ for $w \in E \setminus \{v\}$. The theorem makes no full-faithfulness assertion about these maps.

Proof. Tensor product in $\mathcal{V}$ preserves colimits separately. For a filtered diagram and each finite arity, its diagonal indexing functor is cofinal. Hence tensor powers commute with its colimit. The forgetful functor from O-algebras therefore creates filtered colimits, by the operadic colimit theorem of *Higher Algebra*, §3.2.3. Apply it to the displayed diagram. The structure maps of its colimit are morphisms of O-algebras.

An initial object in a category of cocones has a contractible space of choices. A comparison between two structured colimit cocones has the underlying equivalence determined by their fixed colimit cocones. The forgetful functor detects equivalences. Thus the space asserted in the theorem concerns the entire coherent cocone, and is contractible.

If $E \subset \{v\}$, the singleton is an object of J_E. Otherwise the extra maps give

$$C_v \longrightarrow \bigotimes_{w \in E \cup \{v\}} C_w \longrightarrow C'_V.$$

The first arrow inserts the additionally supplied pointings. Its compatibility with larger finite sets follows from tensor coherence. It depends on these choices. □

The need for a unital map is already visible at one place. In $C = D(k) \times D(k)$, with componentwise tensor, the object $u = (k, 0)$ is a compact central idempotent algebra under the unit (k, k). The insertion $D(k) \to C$ with value u does not preserve that unit. One can instead use the monoidal subcategory with unit u, or a separately specified lax or nonunital theory. These change the recipient of the construction.

Nor is a unit insertion generally fully faithful. Extension of scalars $D(k) \to D(k[t])$ sends the unit to the unit, but its map on unit endomorphisms is $k \to k[t]$. Unitality and full faithfulness are different conditions.

Factorization, coefficient change and iteration

For a partition $V = V_1 \sqcup V_2$, finite subsets separate as pairs of finite subsets. Cofinality and separate colimit preservation give

$$C'_V \simeq C'_{V_1} \otimes C'_{V_2},$$

using $E \cap V_i$ in each factor. The same proof handles every finite partition, with its associativity and permutation coherences: all maps are induced by the corresponding finite tensor diagrams.

Suppose colimit-preserving $\mathcal{O}$-algebra maps $F_v : \mathcal{C}_v \to \mathcal{D}_v$ come with coherent identifications $F_v\eta_v \simeq \eta'_v$. Their finite tensor products commute with every insertion. Taking their colimit gives the induced map of restricted tensors. An enlargement of the exceptional finite set restricts to a cofinal subdiagram and leaves the colimit unchanged.

If $B : \mathcal{V} \to \mathcal{W}$ is a symmetric monoidal, colimit-preserving functor between the chosen coefficient categories, then

$$B(\mathcal{C}'_V) \simeq \operatorname*{colim}_{S \in J_E} \bigotimes_{v \in S} B(\mathcal{C}_v).$$

The comparison retains $B(\eta_v)$, including their nullary compatibilities. This is the coefficient-change map. The same argument commutes two filtered systems of finite subsets. It gives iterated restricted tensors whenever the stated pointings define both diagrams and the paired indexing systems are cofinal.

Dunn additivity identifies an E_{m+n}-structure with a compatible E_m-structure in E_n-algebras. When this full structure is supplied at each finite stage and on the insertion maps, the preceding colimit transports it. An E_m-structure by itself supplies no E_{m+n}-structure. A matrix algebra in degree zero is a simple test: an E_2-algebra has commutative degree-zero homology, whereas $\mathrm{Mat}_2(k)$ is noncommutative.

Module categories and compact objects

For E_∞-ring spectra R_v over a common E_∞-ring A, extension of scalars gives the pointings of Mod_A-linear module categories. The universal property of the module-category tensor gives

$$\bigotimes_{v \in S, \mathrm{Mod}_A} \mathrm{Mod}_{R_v} \simeq \mathrm{Mod}_{\otimes_{v \in S, A} R_v}.$$

Here the tensor on rings is the derived spectral tensor over A. The finite comparisons are natural under unit insertion. Module categories preserve this filtered algebra colimit, so

$$\bigotimes_{v \in V}' \mathrm{Mod}_{R_v} \simeq \mathrm{Mod}_{\operatorname{colim}_S \otimes_{v \in S, A} R_v}.$$

This formula does not replace a tensor of rings by their product, or an arithmetic completion by an uncompleted tensor.

For compact generation, suppose every $\mathcal{C}_S$ is compactly generated and each insertion preserves compact objects. Write $\mathcal{C}_S^c$ for its small idempotent-complete stable category of compacts. The colimit is then

$$\mathcal{C}'_V \simeq \mathrm{Ind}\left(\operatorname*{colim}_{S \in J_E} \mathcal{C}_S^c\right),$$

where the inner colimit is taken in small idempotent-complete stable categories. Indeed, Ind-completion identifies that category with compactly generated presentable stable categories and compact-preserving colimit functors.

The universal property on both sides gives the displayed equivalence. A selected spherical generator only produces the sector it generates. Taking $\mathrm{Ind}(\mathcal{C}^c)$ does not remove a continuous spectrum from an already compactly generated category.

Actions at separate places

Let $T_v : C_v \to C_v$ be an exact colimit-preserving endofunctor at one place. The subposet of finite sets containing v is cofinal. On that diagram T_v acts on the v-factor and the identity acts on the other factors. It commutes with every insertion, which occurs away from v. Thus it defines an exact colimit-preserving endofunctor on C'_V.

Finitely many such actions extend on the cofinal diagram containing all their places. The tensor interchange gives their coherent commutation. No monoidal structure on the individual endofunctors is needed. A construction on a diagram which omits an acting place, or a specified simultaneous infinite action, additionally needs compatibility with the relevant pointings. A Hecke correspondence is therefore usable through its actual endofunctor. It need not be a unital monoidal endofunctor.

Finite labeled disks

The finite-set tensor diagram gives a discrete factorization construction: disjoint sets of labels tensor independently. A geometric factorization algebra additionally requires a space and categories on its opens or configurations. It also requires maps at their collisions.

There is also a distinction within a finite disk construction. Putting a fixed algebra $A_S = \bigotimes_{v \in S} A_v$ on each of r disjoint disks gives $A_S^{\otimes r}$. Putting A_v on one disk carrying label v gives one factor per label. Their products need not agree.

For example, for $A_S = k[x]$, two copies give $k[x_1, x_2]$, whereas a single labeled factor gives $k[x]$. A varying-label factorization diagram must retain this assignment and the maps for every labeled embedding. The finite-set colimit theorem alone supplies no arithmetic collision comparison.

18.31 TENSOR SECTORS AND EXTENSION DATA

An orthogonal decomposition into local sectors gives a direct-sum decomposition of their tensor product. Even with two sectors at each of two places, tensor distributivity gives four terms:

$$(\mathcal{A}_0 \oplus \mathcal{A}_1) \otimes (\mathcal{B}_0 \oplus \mathcal{B}_1) \simeq (\mathcal{A}_0 \otimes \mathcal{B}_0) \oplus (\mathcal{A}_0 \otimes \mathcal{B}_1) \oplus (\mathcal{A}_1 \otimes \mathcal{B}_0) \oplus (\mathcal{A}_1 \otimes \mathcal{B}_1).$$

The middle two terms record different sector choices at the two places. A relation between those choices requires additional structure.

Orthogonal decompositions

Let $\mathcal{V}$ be a symmetric monoidal ∞-category with small colimits. Assume its tensor product preserves colimits separately in each variable. Write $\mathbf{1}$ for its tensor unit and $\coprod$ for its categorical coproduct. The following argument applies to the ∞-category of presentable stable ∞-categories and colimit-preserving functors.

Its tensor represents functors preserving colimits separately in each variable. See Lurie, *Higher Algebra* (2017), Remarks 4.8.1.18 and 4.8.1.24, and Proposition 4.8.2.18. For finite families of stable categories, this coproduct is their orthogonal direct sum, denoted $\oplus$.

Indeed, in a finite product $\prod_i C_i$, every object is the finite direct sum of its components. A colimit-preserving functor out of this product is therefore determined by its restrictions to the factors. Conversely, functors $F_i : C_i \to \mathcal{D}$ extend by $(x_i)_i \mapsto \bigoplus_i F_i(x_i)$. This proves the coproduct universal property. Distinct factors have zero mapping spectra.

Proposition 18.31.1 *(Finite sector decomposition).* Let S be a finite set. For each $v \in S$, fix a finite set I_v and a specified equivalence $C_v \simeq \coprod_{r \in I_v} C_{v,r}$ in $\mathcal{V}$. There is an equivalence, induced by the given sector inclusions,

$$\bigotimes_{v \in S} C_v \simeq \coprod_{\alpha \in \prod_{v \in S} I_v} \bigotimes_{v \in S} C_{v,\alpha(v)}.$$

For $S = \varnothing$, both sides are $\mathbf{1}$.

Proof. For one factor the assertion is the specified decomposition. Tensoring that decomposition with another factor preserves its coproduct. Applying the same fact to the new factor gives the Cartesian product of the two index sets. Induction proves the formula. The maps are induced by the sector inclusions, so the equivalence respects maps that preserve the specified decompositions. □

Write $D(k)$ for the stable ∞-category of unbounded cochain complexes of k-vector spaces, with quasi-isomorphisms inverted. For example, in complex-linear presentable stable categories, put $\mathcal{A}_i = \mathcal{B}_j = D(\mathbb{C})$ and tensor over $D(\mathbb{C})$. This category is the tensor unit, so the result is $D(\mathbb{C})^{\oplus 4}$. Keeping only the two equal-index factors gives $D(\mathbb{C})^{\oplus 2}$. These categories are not equivalent.

Their compact objects have Grothendieck groups $\mathbb{Z}^4$ and $\mathbb{Z}^2$, respectively. Here the Grothendieck group imposes the relation $[y] = [x] + [z]$ for each exact triangle $x \to y \to z$. In $\mathrm{Perf}(\mathbb{C})$, splitting a bounded complex into its cohomology identifies this group with $\mathbb{Z}$, by Euler characteristic. Finite products give the stated ranks. An equivalence preserves compact objects and exact triangles, hence would preserve these groups.

The restricted tensor construction also depends on the sector of each insertion. Let V be a small set. For each $v \in V$, fix a finite nonempty set I_v, an element $\mathrm{o}_v \in I_v$, and a specified decomposition of C_v as above. Fix a morphism $e_v : \mathbf{1} \to C_{v,\mathrm{o}_v}$ in $\mathcal{V}$. For finite $S \subset V$, set

$$C_S = \bigotimes_{v \in S} C_v.$$

For $S \subset T$, tensor with the morphisms e_v, followed by sector inclusion, for $v \in T \setminus S$. The monoidal coherence maps give a diagram on the poset of finite subsets of V. Define its restricted product by

$$C^{\mathrm{res}} = \operatorname*{colim}_{S \subset V,\, S \text{ finite}} C_S.$$

This is an algebraic colimit in $\mathcal{V}$. No analytic or topological completion is implicit.

Let I^{fin} consist of functions α with $\alpha(v) \in I_v$ whose support $\mathrm{supp}(\alpha) = \{v : \alpha(v) \neq \mathrm{o}_v\}$ is finite. For such α, define

$$C^{\mathrm{res}}_\alpha = \operatorname*{colim}_{\substack{S \subset V \text{ finite} \\ S \supset \mathrm{supp}(\alpha)}} \bigotimes_{v \in S} C_{v,\alpha(v)},$$

with the same distinguished insertions.

Proposition 18.31.2 *(Restricted sector decomposition).* Under these hypotheses, the sector inclusions induce an equivalence

$$C^{\mathrm{res}} \simeq \coprod_{\alpha \in I^{\mathrm{fin}}} C^{\mathrm{res}}_\alpha.$$

Proof. For each α, extend its displayed diagram to all finite S by assigning the initial object when S does not contain its support. Call the resulting diagram E_α. At a fixed finite S, only functions supported in S contribute noninitial terms. The finite sector decomposition therefore gives $C_S \simeq \coprod_\alpha E_\alpha(S)$.

The insertion lies in sector 0_v, so the transition maps preserve this identification of diagrams. Colimits commute with coproducts. Thus

$$\operatorname{colim}_S C_S \simeq \coprod_\alpha \operatorname{colim}_S E_\alpha(S).$$

The finite sets containing a fixed support form a cofinal subposet: $S \mapsto S \cup \operatorname{supp}(\alpha)$ supplies a finite upper bound compatible with inclusions. Each inner colimit is consequently C_α^{res}. □

A general insertion need not lie in one sector. If it has several components, its transition maps need not preserve the finite-support indexing above. The finite sector formula still holds. The restricted colimit must then retain the actual transition maps between those finite decompositions.

A nonsplit extension

Orthogonal direct sums exclude extensions between distinct factors. The quiver $1 \to 2$ gives a concrete extension that must be retained when orthogonality is absent. Let k be a field, with no restriction on its characteristic. A representation is a linear map $u : A \to B$ of k-vector spaces. A morphism is a pair (f, g) satisfying $gu = u'f$. Kernels and cokernels are computed at the two vertices.

Consider

$$S_1 = (k \to 0), \qquad S_2 = (0 \to k), \qquad P = (k \xrightarrow{1} k).$$

There is an exact sequence of representations

$$0 \longrightarrow S_2 \longrightarrow P \longrightarrow S_1 \longrightarrow 0.$$

It does not split: a morphism $S_1 \to P$ has source component $f : k \to k$ satisfying $f = 0$. It cannot be a section.

Moreover, $\operatorname{Hom}(P, (A \to B)) \cong A$ and $\operatorname{Hom}(S_2, (A \to B)) \cong B$. Both functors are exact, so P and S_2 are projective. The sequence is therefore a projective resolution of S_1. Applying $\operatorname{Hom}(-, S_2)$ gives

$$\operatorname{Hom}(P, S_2) = 0, \qquad \operatorname{Hom}(S_2, S_2) = k, \qquad \operatorname{Ext}^1(S_1, S_2) = k.$$

The displayed sequence represents the image of $1 \in k$ under the connecting isomorphism. Projectivity of S_2 also gives $\operatorname{Ext}^n(S_2, S_1) = 0$ for every $n \geq 0$. Thus one direction is orthogonal, while the opposite direction carries an extension.

Equivalently, these are left modules over

$$T = \left\{ \begin{pmatrix} a & 0 \\ m & b \end{pmatrix} : a, b, m \in k \right\}, \qquad \begin{pmatrix} a & 0 \\ m & b \end{pmatrix} \begin{pmatrix} a' & 0 \\ m' & b' \end{pmatrix} = \begin{pmatrix} aa' & 0 \\ ma' + bm' & bb' \end{pmatrix}.$$

The two diagonal idempotents recover A and B. The lower-left matrix unit acts by $u : A \to B$. Associativity gives exactly the representation relations. Replacing this algebra by $k \times k$ removes its lower-left entry and the extension above.

Stable gluing

To retain extensions in a stable category, the connecting morphism must be part of each object. Let $\mathcal{A}$ and $\mathcal{B}$ be stable ∞-categories and let $F : \mathcal{A} \to \mathcal{B}$ be exact. All mapping objects below are spectra, written $\operatorname{map}_{\mathcal{A}}(-, -)$ and similarly for other categories. The shift $[1]$ denotes suspension. For cochain complexes it has $K[1]^n = K^{n+1}$ and differential $-d_K$.

Define $\mathcal{G}(F)$ to be the pullback

$$\mathcal{G}(F) = (\mathcal{A} \times \mathcal{B}) \underset{\mathcal{B}\times\mathcal{B}}{\times} \mathrm{Fun}(\Delta^1, \mathcal{B}),$$

where the first map is $(a, b) \mapsto (Fa, b)$ and the second records the source and target of an arrow. Thus an object is (a, b, u) with $u : Fa \to b$. A morphism consists of $f : a \to a'$, $g : b \to b'$, and a specified homotopy from $u'Ff$ to gu. The pullback retains all higher coherences.

Proposition 18.31.3 *(Mapping spectra and extensions).* The category $\mathcal{G}(F)$ is stable, with finite limits and colimits computed in the two components. For $x = (a, b, u)$ and $x' = (a', b', u')$,

$$\mathrm{map}_{\mathcal{G}(F)}(x, x') \simeq \mathrm{fib}\Big(\mathrm{map}_{\mathcal{A}}(a, a') \oplus \mathrm{map}_{\mathcal{B}}(b, b') \xrightarrow{d} \mathrm{map}_{\mathcal{B}}(Fa, b')\Big). \tag{18.33}$$

Here $d(f, g) = u'Ff - gu$.

The exact fully faithful functors

$$i(b) = (0, b, 0), \qquad j(a) = (a, 0, 0)$$

satisfy

$$\mathrm{map}(i(b), j(a)) = 0, \qquad \mathrm{map}(j(a), i(b)) \simeq \Omega\,\mathrm{map}_{\mathcal{B}}(Fa, b) \simeq \mathrm{map}_{\mathcal{B}}(Fa[1], b).$$

Every object has a natural exact triangle

$$i(b) \longrightarrow (a, b, u) \longrightarrow j(a) \xrightarrow{\delta_u} i(b)[1].$$

Under the induced equivalence $\mathrm{map}(j(a), i(b)[1]) \simeq \mathrm{map}_{\mathcal{B}}(Fa, b)$, the connecting morphism δ_u corresponds to u. The choice of sign in this last equivalence is fixed by this exact triangle.

Proof. Exactness of F allows componentwise finite limits and colimits to supply the required arrows. The zero components give a zero object. Pushout squares are pullback squares componentwise, proving stability. A map between two arrows is a homotopy-commutative square.

Its mapping space is the homotopy equalizer of the two compositions $u'Ff$ and gu. The same statement after every suspension gives the homotopy equalizer of mapping spectra. In spectra this is the fiber of their difference, which proves (18.33).

For two objects in the image of i, the formula reduces to $\mathrm{map}_{\mathcal{B}}$. For two in the image of j, it reduces to $\mathrm{map}_{\mathcal{A}}$. The two cross formulas follow by substituting the zero components. In particular, the fiber of $0 \to E$ is ΩE, not E.

The cofiber of $i(b) \to (a, b, u)$ is $(a, 0, 0)$ componentwise. Its connecting morphism includes the homotopy encoded by u. The suspension equivalence identifies that homotopy with u and fixes the stated sign convention. □

The triangle splits precisely when δ_u is nullhomotopic, equivalently when u is nullhomotopic. Indeed, a nullhomotopy identifies its fiber with $i(b) \oplus j(a)$. Conversely, a splitting makes the connecting morphism zero in the homotopy category. When $F = 0$, $\mathcal{G}(F) \simeq \mathcal{A} \times \mathcal{B}$.

Taking $\mathcal{A} = \mathcal{B} = D(k)$ and $F = \mathrm{id}$ gives the arrow category of complexes. Here $h\mathcal{G}(F)$ denotes the homotopy category. For $a = b = k$ in degree zero, the second cross formula gives

$$\mathrm{Hom}_{h\mathcal{G}(F)}(j(k), i(k)[1]) \cong k.$$

The object $(k, k, 1)$ gives its nonzero connecting class. This recovers the extension mechanism in the quiver calculation, including its degree.

Reconstruction from a semiorthogonal filtration

The same description applies to a category whose extension structure is already specified. Let C be stable and let $i : \mathcal{B} \to C$ and $j : \mathcal{A} \to C$ be exact fully faithful functors. Assume

$$\mathrm{map}_C(i(b), j(a)) = 0 \quad \text{for all } a \in \mathcal{A},\ b \in \mathcal{B}.$$

Assume every $x \in C$ admits an exact triangle

$$i(b) \longrightarrow x \longrightarrow j(a) \longrightarrow i(b)[1].$$

These hypotheses state the orientation of the semiorthogonal filtration. They are stronger than generation by the two subcategories alone.

Let $\mathcal{E}$ be the full pullback of the arrow category of C whose arrows have source $j(a)$ and target $i(b)[1]$. Its objects are triples (a, b, δ) with $\delta : j(a) \to i(b)[1]$. Then

$$\mathrm{map}_{\mathcal{E}}((a,b,\delta),(a',b',\delta')) \simeq \mathrm{fib}\big(\mathrm{map}_{\mathcal{A}}(a,a') \oplus \mathrm{map}_{\mathcal{B}}(b,b') \longrightarrow \mathrm{map}_C(j(a), i(b')[1])\big),$$

where the map sends (f, g) to $\delta' j(f) - i(g)[1]\delta$.

PROPOSITION 18.31.4 *(Semiorthogonal reconstruction).* Under these hypotheses, the functor

$$\mathcal{E} \longrightarrow C, \qquad (a,b,\delta) \longmapsto \mathrm{fib}(\delta)$$

is an equivalence. Suppose, in addition, that i has an exact left adjoint $L : C \to \mathcal{B}$. Put $F = (Lj)[-1]$. Then $C \simeq \mathcal{G}(F)$, with the given embeddings i, j.

Proof. A connecting arrow has a fiber triangle $i(b) \to \mathrm{fib}(\delta) \to j(a)$. The filtration hypothesis gives essential surjectivity. To prove full faithfulness, consider two such triangles with middle objects x, x'. Applying $\mathrm{map}_C(-, j(a'))$ to the first triangle gives an equivalence

$$\mathrm{map}_C(j(a), j(a')) \xrightarrow{\sim} \mathrm{map}_C(x, j(a')),$$

because the mapping spectrum from $i(b)$ to $j(a')$ is zero. Consequently a map $h : x \to x'$ determines, with contractible choices, a map $j(f) : j(a) \to j(a')$ and a homotopy between $j(f)(x \to j(a))$ and $x \xrightarrow{h} x' \to j(a')$.

Taking fibers gives a map $i(g) : i(b) \to i(b')$. Taking cofibers supplies the compatible square of connecting arrows. Conversely, a square of connecting arrows gives h by taking fibers. The fiber and cofiber universal properties make these constructions inverse on mapping spaces, including all their higher homotopies. Thus the functor is fully faithful.

For the additional assertion, adjunction and exactness give

$$\mathrm{map}_C(j(a), i(b)[1]) \simeq \mathrm{map}_{\mathcal{B}}(Lj(a), b[1]) \simeq \mathrm{map}_{\mathcal{B}}(F(a), b).$$

This natural equivalence identifies objects, squares, and their higher coherences in $\mathcal{E}$ with those in $\mathcal{G}(F)$. □

In particular, a recollement can supply such a filtration without supplying an orthogonal direct sum. Use the convention $i_* : \mathcal{B} \to C$ for the closed embedding and $j_! : \mathcal{A} \to C$ for the open extension

functor. The recollement adjunctions include $i^* \dashv i_* \dashv i^!$ and $j_! \dashv j^* \dashv j_*$. Its defining vanishing $j^* i_* = 0$ gives $\mathrm{map}(j_! a, i_* b) = 0$, and its exact triangle is

$$j_! j^* x \longrightarrow x \longrightarrow i_* i^* x \longrightarrow j_! j^* x[1].$$

The first assertion of the proposition applies with the roles of the closed and open categories interchanged. It retains the connecting arrow $i_* i^* x \to j_! j^* x[1]$. It does not require a left adjoint to $j_!$ or assert one exists.

The semiorthogonal description becomes an orthogonal direct sum if the other cross mapping spectrum also vanishes. Then every connecting spectrum is zero, and $\mathcal{E} \simeq \mathcal{A} \times \mathcal{B}$. Without this vanishing, tensor assembly must retain the gluing functors or connecting spectra and their composition data. The two sector names alone do not determine the category.

For automorphic categories, applying these propositions requires actual local decompositions or semiorthogonal filtrations and specified insertion functors. An identification with a global automorphic category also requires a comparison functor preserving the Hecke actions, parameter constraints, and extension data. Neither tensor distributivity nor semiorthogonal reconstruction constructs that comparison functor.

18.32 Adelic factorization and automorphic realization

The restricted tensor construction has an algebraic representation theorem as a model. For a restricted product of local groups, choose nonzero reference vectors in the almost-everywhere spherical lines of given local representations. The algebraic restricted tensor is the colimit of finite tensors under insertion of these vectors.

Flath's factorization theorem says that every admissible irreducible adelic representation has this form. For a number field, the archimedean factors in its algebraic form are $(\mathfrak{g}, K)$-modules. Cuspidality is not a hypothesis of this factorization theorem. Changing the nonzero vectors in the almost-everywhere spherical lines changes the concrete insertions but not the isomorphism class of the restricted tensor. It does not choose canonical vectors.

The unitary Hilbert restricted tensor is a different completion. Its reference vectors are unit vectors and its inner product is defined first on the algebraic finite tensors. An infinite tensor without these reference choices has not been specified by listing its local factors.

Neither construction places every independent tuple of local representations in automorphic functions. For GL_1, the idele character must be trivial on the diagonal number field's multiplicative group. For GL_n, automorphic realization likewise imposes its global relations, growth and spectral conditions. The local tuples and their tensor are positive constructions before that additional comparison.

The real place

Let $\mathfrak{g}_{\mathbb{C}}$ be the complexified Lie algebra of the real group and let K be a maximal compact subgroup. A Harish–Chandra module has compatible $\mathfrak{g}_{\mathbb{C}}$- and K-actions. Admissibility asks for finite multiplicity of each irreducible K-type. It does not ask for only finitely many types.

The category of admissible modules is not closed under arbitrary direct sums. An infinite sum of trivial representations has an infinite-dimensional trivial K-isotype. Consequently it is not the asserted presentable category merely by being admissible. One algebraic recipient is the Ind-completion of a chosen small bounded derived category of finite-length Harish–Chandra modules. Its analytic realization and comparison with Hilbert representations require their own functors.

A spherical principal series is one object in a family, not the tensor unit of this category. On a cofinal finite-place diagram which always includes infinity, the smallest stage is the specified archimedean category. In a linear tensor theory the genuinely empty tensor is the coefficient module category.

The real Weil group is the nonsplit extension generated by $\mathbb{C}^\times$ and j, with

$$j^2 = -1, \qquad jzj^{-1} = \bar{z}.$$

For a lift zj of the nontrivial Galois element, $(zj)^2 = -|z|^2 \neq 1$, proving that this extension does not split. Using $\mathbb{C}^\times \rtimes C_2$ would change its representation theory.

Strong approximation for SL_2, with its specified omitted places and noncompactness hypotheses, can simplify a group quotient. For GL_2 the determinant still gives the idele-class quotient. Neither group-theoretic statement identifies all independent local representation data with an automorphic category.

Comparisons of enhanced categories

Let $(\mathcal{C}_S)$ and $(\mathcal{D}_S)$ be supplied diagrams of stable presentable categories on finite sets of places. Suppose exact colimit-preserving equivalences $F_S : \mathcal{C}_S \simeq \mathcal{D}_S$ intertwine every insertion, with their higher coherences. Taking the colimit gives

$$\operatorname*{colim}_S \mathcal{C}_S \simeq \operatorname*{colim}_S \mathcal{D}_S.$$

Indeed, their inverse diagrams give an inverse colimit functor, and the coherent units and counits pass to these colimits. This is a comparison of the supplied diagrams. Identifying either colimit with an independently specified geometric or automorphic category is additional data.

For a global functor $F : \mathcal{C}_{\mathrm{arith}} \to \mathcal{D}_{\mathrm{aut}}$, Theorem 18.1.2 gives a sufficient test. Compare the full mapping spectra of compact generators and preserve their compactness. Prove that their images generate the intended target. Natural compatibility with finite-place Hecke, ramified, Whittaker and archimedean operations must also be supplied. A match of central characters does not provide it.

The cuspidal, residual and continuous summands in a Hilbert spectral decomposition have a precise analytic meaning. Hilbert orthogonality does not imply vanishing of cross mapping spectra in an enhanced category. The nonsplit principal-series and adelic extensions above exhibit the data a categorical gluing must retain. Such a gluing uses its actual connecting functor, all mixed finite-place sectors and coherent transitions.

Question 18.32.1. Construct an enhanced global automorphic category and a geometric recipient for the fixed Selmer problem, together with the comparison functor. Identify the local and global generators, their full derived endomorphisms and the essential image. Prove compatibility with finite and archimedean actions, Whittaker realization, coefficient completion and the specified trace or pairing.

The local full embeddings, determinant-ring reconstruction, formal patching and restricted tensor theorems supply parts of this construction. Their hypotheses do not state a necessity theorem for every possible arithmetic realization. A different geometric recipient or a restricted natural domain remains possible when its comparison maps are constructed.

18.33 Formal operators and arithmetic recognition

Formal spectral variables make sense over a common coefficient ring. Their existence is separate from identifying the resulting algebra with arithmetic operators. In particular, Spec $\mathbb{Z}$ is not discrete: its generic

point is dense. The lack of a chosen comparison between different local fields does not prevent the formal expression $u - v$ over $\mathbb{Z}$.

An integral RTT construction

Let R be a commutative ring and set

$$T_{ij}(u) = \delta_{ij} + \sum_{r \geq 1} t_{ij}^{(r)} u^{-r}.$$

Define an R-algebra by the coefficient relations

$$(u - v)[T_{ij}(u), T_{kl}(v)] = T_{kj}(u)T_{il}(v) - T_{kj}(v)T_{il}(u).$$

The denominator has been cleared, so these are integral relations among the formal coefficients. Equivalently, they are RTT with $\mathcal{R}(u - v) = 1 - P/(u - v)$, where P exchanges the two auxiliary factors, in the corresponding rational formal notation.

Proposition 18.33.1. These relations admit the coproduct and counit

$$\Delta T_{ij}(u) = \sum_{a} T_{ia}(u) \otimes T_{aj}(u), \qquad \epsilon(T_{ij}(u)) = \delta_{ij}.$$

The formal matrix inverse defines the antipode. There is an evaluation homomorphism to $U_R(\mathfrak{gl}_n)$ given by $T_{ij}(u) \mapsto \delta_{ij} + E_{ij}u^{-1}$.

Proof. Suppose T and S satisfy RTT and their coefficients commute with one another. Move the R-matrix through T_1T_2, and then through S_1S_2. This proves RTT for their matrix product. The coefficients in the two tensor factors commute, so the intermediate factors can be reordered as required.

This proves that the coproduct preserves the relations. Associativity of matrix multiplication gives coassociativity. The identity matrix gives the counit.

The constant term of T is the identity. Its inverse is therefore defined coefficientwise, with each coefficient a finite sum of products of the generators. Inverting RTT gives the opposite-algebra relation required of an antipode, and matrix inversion gives both antipode identities. Finally, substitute the displayed evaluation. The relation reduces to

$$[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{il}E_{kj},$$

which holds in the enveloping algebra over R. □

This is a formal algebraic construction. It neither identifies primes with spectral coordinates nor produces an arithmetic coproduct on an independently specified Hecke category. A proposed shuffle obstruction must likewise specify its kernel and the identity which fails. A name for a zeta-valued kernel is not an obstruction proof.

Bethe algebras in specified representations

A Bethe algebra is a commutative subalgebra of the Yangian. Its image on a chosen representation is a quotient of that subalgebra. It is not thereby a quotient of the entire Yangian.

The discrete Wronski theorem gives a precise positive model. Fix distinct nonzero $q_1, \dots, q_N \in \mathbb{C}$ and nonnegative integers λ_i with $\sum_i \lambda_i = n$. Let

$$f_i(u) = q_i^u(u^{\lambda_i} + p_{i,1}u^{\lambda_i - 1} + \cdots + p_{i,\lambda_i}).$$

The $p_{i,j}$ are coordinates on an affine n-space.

For $\mathrm{Wr}(g_1, \dots, g_N) = \det(g_i(u - j + 1))_{i,j}$, write

$$\mathrm{Wr}(f_1(u-1), \dots, f_N(u-1)) = \prod_i q_i^{u-1} \prod_{i<j} (q_j^{-1} - q_i^{-1}) \left(u^n + \sum_{s=1}^{n} (-1)^s \Sigma_s u^{n-s} \right).$$

The map given by the Σ_s is the discrete Wronski map. Let $O^q_{\lambda,\mathbf{a}}$ be the coordinate ring of its scheme-theoretic fiber at $\mathbf{a} = (a_1, \dots, a_n)$.

Choose $b_1, \dots, b_n \in \mathbb{C}$ with $b_i \neq b_j + 1$ for $i > j$, and put $a_s = e_s(b_1, \dots, b_n)$. For the tensor of vector evaluation modules

$$V(\mathbf{b}) = \mathbb{C}^N(b_1) \otimes \cdots \otimes \mathbb{C}^N(b_n),$$

Mukhin–Tarasov–Varchenko, Theorem 5.2 and Corollary 5.4, identify $V(\mathbf{b})_\lambda$ with the regular module of $O^q_{\lambda,\mathbf{a}}$. This identifies its Bethe action for the diagonal twist q.

The resulting Bethe image is maximal commutative and Frobenius. Their Theorem 6.3 and Corollary 6.4 give the polynomial Wronski version for the untwisted singular weight space. These specified fibers and evaluation modules are the recipients of the theorem.

In contrast, a local spherical Hecke algebra for GL_n is a Laurent symmetric Satake algebra with its chosen normalization. Its determinant generator is invertible. Already for GL_1, valuation translation acts on a bilateral lattice, whereas a unilateral creation operator has no inverse at the vacuum. Arbitrary characters of a tensor of local Satake algebras specify local parameter tuples. Their global automorphic and Galois realizations still require the maps of Question 18.32.1. An oper or critical-center comparison must also specify its actual Lie algebra, representation and spectral curve.

18.34 Scalar invariants from chiral geometry

A numerical invariant can be computed before its proposed geometric realization is constructed. The examples below retain the scalar calculation and identify the additional object it would have to describe.

Example 18.34.1 *(A rational function and its square root).* Define

$$W(c) = \frac{c^2(5c + 22)}{4(45c + 218)}.$$

It has a simple pole at $c = -218/45$, a double zero at $c = 0$, and a simple zero at $c = -22/5$. These statements follow from the numerator and denominator, which have no common root. For $c > 0$, let $\Omega(c)$ be its positive square root.

Then

$$\Omega(c) = \frac{c}{6} - \frac{1}{27} + O(c^{-1}) \qquad (c \longrightarrow +\infty).$$

Indeed,

$$W(c) = \frac{c^2}{36} \left(1 - \frac{4}{9c} + O(c^{-2}) \right),$$

and the positive square-root expansion gives the result. This computation specifies a rational function and a real branch. Identifying either with a Borel invariant requires its defining transform and an equality on the common domain.

There is a separate differential identity. Let $Q(t)$ and $H(t)$ be twice differentiable functions on a domain where t, Q, H are nonzero, and suppose $H^2 = t^4 Q$. Then

$$U = \frac{H'}{H} - \frac{2}{t} = \frac{Q'}{2Q}, \qquad U' + 2U^2 = \frac{Q''}{2Q}.$$

The first equality follows by logarithmic differentiation.

Differentiating $Q'/(2Q)$ proves the second, including the cancellation of $(Q')^2/(2Q^2)$. The identity is valid for any such Q, H. It therefore does not identify a particular chiral operation by itself.

A weighting also changes the scalar coefficients. The two functions

$$S_6(c) = \frac{4(240c + 1031)}{c^3(5c + 22)^2}, \qquad \widehat{S}_6(c) = \frac{80(45c + 193)}{3c^3(5c + 22)^2}$$

satisfy

$$\widehat{S}_6(c) - S_6(c) = \frac{4(180c + 767)}{3c^3(5c + 22)^2}.$$

They cannot be interchanged without a comparison of their weighting conventions. If a formal series $H(t) = \sum_{r \geq 2} S_r t^r$ satisfies $[t^6]H^2 = 0$, its coefficients instead obey the literal equation

$$2S_2S_4 + S_3^2 = 0.$$

The equation $2S_2S_6 + 2S_3S_5 + S_4^2 = 0$ is the coefficient equation in degree eight. Which degree vanishes must be determined from the specified right-hand side of H^2.

Example 18.34.2 *(The Bernoulli numerator* 691*).* Define $B_n \in \mathbb{Q}$ by

$$\frac{x}{e^x - 1} = \sum_{n \geq 0} B_n \frac{x^n}{n!}.$$

Multiplication by $e^x - 1$ gives $B_0 = 1$ and

$$\sum_{j=0}^{m} \binom{m+1}{j} B_j = 0 \qquad (m \geq 1).$$

Successive substitution yields $B_1 = -1/2$, vanishing of the odd values from B_3 through B_{11}, and

n	2	4	6	8	10	12
B_n	1/6	−1/30	1/42	−1/30	5/66	−691/2730.

This is an exact rational calculation.

It specifies neither a two-dimensional Galois representation nor its determinant law. A proposed representation attached to a shadow coefficient must first be constructed, with its coefficient ring and continuous action. Its characteristic polynomials can then be compared by the determinant methods of this chapter.

An automorphic product has a similarly limited numerical test: a seed coefficient $c(0)$ gives the proposed weight $c(0)/2$. An identification with a chiral central invariant also requires the product, its transformation law and the comparison map. A form of half-integral or quarter-integral weight requires its actual multiplier system or covering group. The numerical weight alone chooses none of these structures.

18.35 PHYSICAL REALIZATIONS AND SCALAR TESTS

Gauge-theoretic constructions give further possible recipients for arithmetic data. The Kapustin–Witten construction starts with a smooth geometric curve and a twisted four-dimensional gauge theory. The Costello–Witten–Yamazaki construction uses a topological surface and a complex curve with a chosen meromorphic one-form, together with a partial connection and its boundary conditions. Replacing that curve by an arithmetic compactification requires new geometric and physical data. The absence of those data does not prove nonexistence of every arithmetic bulk theory.

The protected chiral construction of Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees assigns a chiral algebra to the protected sector of a four-dimensional $\mathcal{N} = 2$ superconformal theory. Its normalization has $c_{2d} = -12c_{4d}$ and $k_{2d} = -k_{4d}/2$. A precise finite-current obstruction requires a finite-dimensional weight-one space. Under that hypothesis, there cannot be an infinite linearly independent family of weight-one currents, by dimension alone. This argument does not establish that hypothesis for every possible superconformal theory.

The oscillator commutator and its logarithmic energy are independent normalizations. For example,

$$[a_{p,m}, a_{q,n}] = m\delta_{pq}\delta_{m+n,0}, \qquad E = \sum_p (\log p) N_p$$

retain level one at each prime while weighting the occupation energy by $\log p$. They do not force Chern–Simons levels $1/\log p$. Nor does finite rank of a root lattice bound the number of states or generators in all of its infinite-dimensional representations.

A stronger concrete automorphic model comes from the odd Maaß eigenfunctions on the Dirichlet half-modular domain. The Eisenstein continuum of the full modular quotient does not satisfy that odd boundary condition. For a fixed Hecke eigenstate, assume its normalized parameters are $e^{i\theta_p}, e^{-i\theta_p}$. Then on two charged oscillators at each prime,

$$\operatorname{Tr}\left(e^{i\theta_p(N_{p,1}-N_{p,2})} p^{-s(N_{p,1}+N_{p,2})}\right) = \frac{1}{(1-e^{i\theta_p}p^{-s})(1-e^{-i\theta_p}p^{-s})}.$$

The local identity is an ordinary geometric-series calculation. Under these unit-modulus parameters, the product is absolutely convergent for $\operatorname{Re} s > 1$. Real phases retain the Ramanujan hypothesis. General Satake parameters require the corresponding convergence bound.

In the Hartnoll–Yang model the dilatation wavefunction contains the completed L-function as a factor. Its zeros are wavefunction zeros. They are not, on that account, eigenvalues or a density of states of the closed dilatation operator. A fixed-prime spectral average, a fixed-state prime-distribution statement and a large-prime continuum limit have separate hypotheses. In particular, the formal multiplicity $(p-1)/2$ is not an integer at $p = 2$.

The evaluation map remains part of the comparison

For a finite set of primes, an exterior generator of energy $\log p$ contributes $1 - p^{-s}$ to a supertrace. The natural dual Ext generator instead has energy $-\log p$ and contributes $1 - p^{s}$. Assigning it positive energy is an additional grading. The restricted exterior limits have their different absolute-convergence half-planes, $\operatorname{Re} s > 1$ and $\operatorname{Re} s < -1$. They do not identify unrestricted Ext over an infinite polynomial algebra with that restricted direct limit.

Likewise, a local Whittaker pairing has its full measure and adjoint-L normalization. A common nonzero complex rescaling preserves an equality of sesquilinear pairings, but does not establish positivity. A regularized global product is not an ordinary convergent Euler product at one. The arithmetic comparison must supply the actual maps and their evaluation domains.

Satake estimates also have a precise scope. For a unitary normalized GL_2 parameter with extremal exponent θ_p, the extremal symmetric-m exponent is $m\theta_p$. The general rank-four and rank-five estimates give $5/34$ through the symmetric cube and $3/26$ through the symmetric fourth. Its refined estimate gives $7/64$. Conditional cuspidal automorphy of the Maaß symmetric ninth would give $11/202$ by the general rank-ten bound. Automorphy for an unbounded sequence of these symmetric powers would force $\theta_p = 0$.

These are implications under the indicated automorphy hypotheses. Newton–Thorne's symmetric-power theorem concerns non-CM regular algebraic forms. A holomorphic companion does not multiply a Maaß exponent by its own symmetric-power degree. The identity

$$\Lambda^2 \operatorname{Sym}^3(V) \simeq \left(\operatorname{Sym}^4(V) \otimes \det V\right) \oplus (\det V)^3$$

retains both summands.

It does not supply missing cuspidality or temperedness. In Ramakrishnan's tensor-product criterion, nondihedral twist-related pairs are also noncuspidal. A bound on local Satake exponents still requires an actual comparison estimate before it gives a summable adelic error.

The arithmetic realization problem therefore retains the object, its operators and their full comparison, followed by the scalar evaluation. The finite coefficient, local representation and formal gluing constructions above remain positive at those levels. The automorphic trace, physical bulk and analytic positivity comparisons require their separately stated maps.

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Chapter 19

Marked duality and coefficient polarity

Arithmetic duality must transport an operation before it can explain a functional equation. Its first datum is a pairing between specified objects. The pairing determines the direction of a transpose and the tensor product in which evaluation is defined. A topology then decides which transposes remain continuous. These choices already distinguish several dualities on the polynomial oscillator representation.

We begin with finite coefficients, where evaluation is explicit. We then pass to compact modules, marked derived modules and coalgebras. The completed oscillator shows why a continuous coaction can exist without a continuous evaluation on the proposed tensor. Full collision coefficients admit a different duality, whose field realization can be constructed on a marked charged-current domain. The arithmetic question is to compare such operations with theta and Mellin reflection while retaining the vacuum and both polar terms.

19.1 The pairing fixes the dual

Let R be a commutative ring. For a finite projective R-module P, put $P^\vee = \mathrm{Hom}_R(P, R)$. Evaluation is $P^\vee \otimes_R P \to R, f \otimes p \mapsto f(p)$.

A map $a : P \to Q$ has transpose $a^\vee : Q^\vee \to P^\vee, f \mapsto f \circ a$. Thus

$$\langle a(p), f\rangle = \langle p, a^\vee(f)\rangle, \qquad (ba)^\vee = a^\vee b^\vee. \tag{19.1}$$

These formulas determine variance. They do not assert that a given bilinear form identifies P with its dual.

Proposition 19.1.1 *(Finite projective polarity).* Evaluation gives natural isomorphisms

$$P \xrightarrow{\sim} P^{\vee\vee}, \qquad Q^\vee \otimes_R P^\vee \xrightarrow{\sim} (P \otimes_R Q)^\vee.$$

The second isomorphism uses evaluation in the order of the displayed factors. For a ring map $R \to R'$, the natural map

$$P^\vee \otimes_R R' \longrightarrow \mathrm{Hom}_{R'}(P \otimes_R R', R')$$

is an isomorphism. These statements extend to perfect complexes, with derived tensors and the cohomological Koszul signs.

Proof. For $P = R^n$ the evaluation matrix in a basis and its dual is the identity. The tensor assertion and coefficient comparison are the corresponding identities for finite matrices. A finite projective module is a direct summand of a finite free module, so the identities restrict to it.

Apply the same argument to bounded complexes of finite projectives. For a degree-j functional the dual differential is $d(f) = (-1)^{j+1} f\, d_P$. This makes evaluation a chain map. Finite cones and direct summands preserve all three comparisons. □

If C is a finite projective coalgebra, its coproduct transposes to an associative multiplication on $C^\vee$. The coalgebra counit transposes to the algebra unit. Conversely, an algebra multiplication transposes to a coproduct on its finite projective dual. Each identity is obtained by applying the same duality to its defining diagram. This is coefficient transposition. The cobar algebra of C instead uses tensor words in its shifted coaugmentation quotient and a differential determined by the reduced coproduct.

An integral divided-power example

Consider $R[x]$ with x primitive: $\Delta x = x \otimes 1 + 1 \otimes x$. Give x weight one, and write f_n for the coefficient functional on x^n. The *restricted graded dual* is $\bigoplus_{n \geq 0} R f_n$, the direct sum of the dual weight spaces.

Transposing the coproduct gives

$$f_a f_b = \binom{a+b}{a} f_{a+b}, \qquad f_1^n = n! f_n, \qquad f_0 = 1.$$

Indeed the coefficient of $x^a \otimes x^b$ in $\Delta(x^{a+b})$ is the binomial coefficient. The identity

$$\binom{a+b}{a}\binom{a+b+c}{a+b} = \frac{(a+b+c)!}{a!b!c!} = \binom{b+c}{b}\binom{a+b+c}{a}$$

proves associativity over the integers, and hence over every R. It involves no division in R.

Over a field of characteristic zero the substitution $f_n \mapsto y^n/n!$ identifies this algebra with a polynomial algebra. Over $\mathbb{F}_p$, however, $f_1^p = 0$ while $f_p \neq 0$. The integral dual therefore retains divided powers. For several primitive variables the formula is the product of the corresponding binomial coefficients. The energy grading of the oscillator makes each weight finite, so the same calculation applies weight by weight.

The full linear dual of $\bigoplus_n R x^n$ is instead $\prod_n R f_n$. Its multiplication still has finite sums in each output weight. Whether its evaluation extends to a tensor completion is a separate question, considered in Section 19.6.

19.2 Compact coefficients and their discrete duals

For a locally compact Hausdorff abelian group G, its Pontryagin dual is $G^* = \mathrm{Hom}_{\mathrm{cts}}(G, \mathbb{R}/\mathbb{Z})$, with the compact-open topology. The classical duality theorem says that evaluation $G \to G^{**}$ is a topological isomorphism. It exchanges compact and discrete groups. See Roeder, *Category theory applied to Pontryagin duality*, Theorems 2, 4 and 5. The specialization needed for Iwasawa coefficients has an elementary proof through finite groups.

Fix a prime p and put $D_p = \mathbb{Q}_p/\mathbb{Z}_p$, with the discrete topology. A *profinite $\mathbb{Z}_p$-module* is an inverse limit of finite p-primary modules with its inverse-limit topology. For such an M, and for a discrete p-primary torsion module N, define

$$M^\vee = \mathrm{Hom}_{\mathrm{cts}}(M, D_p), \qquad N^\vee = \mathrm{Hom}_{\mathbb{Z}_p}(N, D_p).$$

The first dual is discrete. Give the second the inverse-limit topology obtained by restricting characters to finite submodules of N. Every finitely generated submodule of N is finite.

THEOREM 19.2.1 *(Compact and discrete coefficient duality).* These duals are exact contravariant inverse equivalences between profinite $\mathbb{Z}_p$-modules and discrete p-primary torsion modules. Their double-dual comparisons are evaluation maps.

Proof. For $\mathbb{Z}/p^n$, a character is determined by the image of 1, an element of $D_p[p^n]$. The pairing of these two cyclic groups is perfect. Finite direct sums prove the assertion for every finite p-group.

A continuous character on M has an open kernel, so it factors through a finite quotient. Consequently

$$M^\vee = \varinjlim_{U} (M/U)^\vee, \qquad N^\vee = \varprojlim_{N_0 \subset N} N_0^\vee,$$

where U runs through open submodules and N_0 through finite submodules. Taking a second dual and using the finite evaluation identities recovers M with its topology and N with its discrete topology. The comparisons are natural.

To prove exactness, let $L \subset M$ be a closed submodule and let $\chi : L \to D_p$ be continuous. Choose an open submodule U of M such that $L \cap U \subset \ker \chi$. The character of $L/(L \cap U) \subset M/U$ extends to M/U, since D_p is divisible and hence injective as a $\mathbb{Z}$-module. Pulling back gives a continuous extension to M. The kernel of restriction is exactly the dual of M/L. Double duality gives exactness on the discrete side as well. □

The same D_p describes circle characters of a profinite p-group. Choose an arc around zero in $\mathbb{R}/\mathbb{Z}$ containing no nonzero subgroup. The inverse image of this arc contains an open subgroup U.

Its image must be zero. Every character therefore factors through a finite p-group and has p-primary torsion image. Here D_p is identified with the p-power torsion subgroup of the circle. It is given its discrete topology in the preceding duality.

The dual tensor and the Iwasawa algebra

For profinite modules define

$$M \mathbin{\widehat{\otimes}}_{\mathbb{Z}_p} N = \varprojlim_{U,V} (M/U) \otimes_{\mathbb{Z}_p} (N/V).$$

For discrete p-primary modules use the transported product

$$A \odot B = (A^\vee \mathbin{\widehat{\otimes}}_{\mathbb{Z}_p} B^\vee)^\vee. \tag{19.2}$$

Its unit is D_p, since $D_p^\vee = \mathbb{Z}_p$. Associativity and symmetry are transported through the natural double-dual maps. Thus a compact unital algebra Λ has a dual coalgebra $C = \Lambda^\vee$ for $\odot$, with $\Delta : C \to C \odot C$ and counit $C \to D_p$.

Dualizing an action $M \mathbin{\widehat{\otimes}} \Lambda \to M$ gives a coaction $M^\vee \to M^\vee \odot C$. Its associativity diagram becomes the coassociativity diagram. Dualizing again recovers the action. Hence continuous compact right Λ-modules and discrete right C-comodules for $\odot$ are exactly anti-equivalent. Exactness extends this comparison to bounded derived categories, with the dual differential used above.

The transported tensor is essential. Ordinary tensor gives $D_p \otimes_{\mathbb{Z}_p} D_p = 0$: if $p^n a = 0$, write $b = p^n c$ and compute $a \otimes b = p^n a \otimes c = 0$. Moreover $\operatorname{Hom}_{\mathbb{Z}_p}(D_p, \mathbb{Z}_p) = 0$. Thus D_p cannot be a nonzero ordinary counital $\mathbb{Z}_p$-coalgebra. Its counit for $\odot$ takes values in the correct unit D_p.

Let G be a profinite group and put

$$\Lambda(G) = \mathbb{Z}_p[[G]] = \varprojlim_{U,n} (\mathbb{Z}/p^n)[G/U],$$

where U is open and normal. The finite coefficient pairing identifies the dual of $(\mathbb{Z}/p^n)[G/U]$ with the functions $G/U \to D_p[p^n]$. Therefore

$$\Lambda(G)^\vee = C_{\mathrm{lc}}(G, D_p).$$

Every locally constant function on the compact group has finite image and factors through some finite quotient. Its values are killed by a common power of p. This proves surjectivity of the displayed identification as well as injectivity.

For left translation on $\Lambda(G)$, choose the contragredient left action on its dual. It is $(g \cdot f)(h) = f(g^{-1}h)$. This follows by evaluating a character on the group-ring basis element $[h]$. The full local abelian Galois group can be used for G, but it is not the chosen procyclic group whose Iwasawa algebra is $\mathbb{Z}_p[[T]]$.

There is also a natural identity

$$(M/pM)^\vee = M^\vee[p].$$

Indeed a character vanishes on pM exactly when multiplication by p kills it. This is an identity of ordinary modules. Derived reduction retains additional Tor groups.

Matlis duality and the coefficient object

For a complete Noetherian local ring $(R, \mathfrak{m}, k)$, let E_R be an injective hull of k. Thus E_R is injective and contains k as an essential submodule: every nonzero submodule meets k. Define the *Matlis dual* by $D_R(M) = \mathrm{Hom}_R(M, E_R)$. Its coefficient object is E_R, rather than R.

THEOREM 19.2.2 *(Matlis duality).* The functor D_R is an exact anti-equivalence between finite R-modules and Artinian R-modules. Evaluation identifies either kind of module with its double dual.

Proof. The injectivity of E_R makes D_R exact. The structural facts are that E_R is Artinian, that $\mathrm{End}_R(E_R) = R$, and that every Artinian M has a finite copresentation $0 \to M \to E_R^a \to E_R^b$. They are proved in the Stacks Project, Tags 08Z6–08Z8. A finite module has a presentation $R^b \to R^a \to M \to 0$.

Dualizing twice and using $D_R(R) = E_R$, $D_R(E_R) = R$ proves biduality by kernels and cokernels. The same argument applied to the copresentation proves biduality on the Artinian side. Naturality of evaluation proves that the two functors are inverse. See also Tag 08Z9. □

Completeness enters the identity for $\mathrm{End}_R(E_R)$. Without it that endomorphism ring is $\widehat{R}$. Indeed $E_R = \bigcup_n E_R[\mathfrak{m}^n]$, and endomorphisms restrict to the finite-length duality on each quotient. Their inverse limit is $\varprojlim_n R/\mathfrak{m}^n$. For $R = \mathbb{Z}_p$, the injective hull is D_p, so Matlis duality agrees with the preceding pairing on finite $\mathbb{Z}_p$-modules. Its stated domain is smaller: $\bigoplus_{n\geq 0} \mathbb{F}_p$ is discrete p-primary torsion but is not Artinian.

Replacing E_R by R can lose the entire module. For example, $\mathrm{Hom}_{\mathbb{Z}_p}(\mathbb{F}_p, \mathbb{Z}_p) = 0$, whereas $D_{\mathbb{Z}_p}(\mathbb{F}_p) \cong \mathbb{F}_p$. A claim of biduality must therefore specify its dualizing object and its module class.

19.3 MARKED MODULES AND BALANCED INTERFACES

The dual of a module can retain a chosen boundary object rather than a scalar coefficient ring. The domain of biduality is then determined by the objects generated from that boundary.

Let A be a dg algebra and P a right A-module. Choose a derived model for

$$E_P = \mathrm{RHom}_A(P, P),$$

with multiplication given by composition. Then P is an (E_P, A)-bimodule. Write $\mathrm{thick}_A(P)$ for the smallest full stable subcategory of $D(A)$ containing P and closed under direct summands. Its objects

are obtained by finitely many shifts, cones and summands. A left E_P-module is *perfect* when it lies in $\mathrm{thick}_{E_P}(E_P)$.

Proposition 19.3.1 *(Duality generated by a marked module).* The functors

$$F_P = \mathrm{RHom}_A(-, P), \qquad G_P = \mathrm{RHom}_{E_P}(-, P)$$

give inverse anti-equivalences $\mathrm{thick}_A(P)^{\mathrm{op}} \simeq \mathrm{Perf}(E_P)$. The first Hom has a left E_P-action by post-composition. The second has a right A-action through P.

Proof. Both functors are exact and preserve finite direct summands. They send P to the regular module E_P and E_P to P. The evaluation maps from an object to its double Hom are equivalences on these generators. The class where either evaluation is an equivalence is closed under shifts, cones and summands. Hence both evaluations are equivalences on the stated categories. The evaluation identities give the inverse functors on mapping complexes and their compositions. □

Now let S be a Noetherian ring of finite global dimension and let A be an ordinary augmented S-algebra, finite over S. Assume that its augmentation ideal I is nilpotent. A complex of right A-modules is *coefficient-perfect* when its restriction to S is perfect. Put $E = \mathrm{RHom}_A(S, S)$.

Theorem 19.3.2 *(The coefficient-perfect domain).* Coefficient-perfect A-modules form $\mathrm{thick}_A(S)$. The functor $F_A(M) = \mathrm{RHom}_A(M, S)$ therefore gives an anti-equivalence from this category to perfect left E-modules. It sends S to E and A to the augmentation module S of E.

Proof. The object S is coefficient-perfect, and this property is preserved by finite cones and summands. This proves one inclusion. Conversely, a coefficient-perfect complex has bounded, finite S-module cohomology. A finite Postnikov filtration reduces the claim to each cohomology module M. Filter M by the powers of I.

Since I is nilpotent, there are finitely many quotients, each a finite S-module with A-action through the augmentation. Every such quotient has a finite projective S-resolution, so belongs to $\mathrm{thick}_A(S)$. Finite extensions prove the other inclusion. Apply Proposition 19.3.1 with $P = S$. The boundary values follow from the two displayed derived Hom complexes. □

Let $C = \mathrm{thick}_A(S)$. Its pro-category retains an inverse system as an object, including its transition maps. The extension of the preceding duality is

$$\mathrm{Pro}(C)^{\mathrm{op}} \simeq \mathrm{Ind}(\mathrm{Perf}(E)) \simeq D(E), \qquad \{M_i\} \longmapsto \varinjlim_i \mathrm{RHom}_A(M_i, S).$$

The direct limit is taken over the opposite of the cofiltered index category.

To see the first equivalence, apply the small anti-equivalence to the formula for maps of pro-objects, $\varprojlim_j \varinjlim_i \mathrm{Map}(M_i, N_j)$. The resulting formula is precisely that for the corresponding ind-objects, with the arguments reversed. The second equivalence uses the regular module E as a compact generator of $D(E)$. Taking the inverse limit of $\{M_i\}$ inside $D(A)$ before dualizing is a different operation.

Two tests for the domain

For $A = k[\epsilon]/(\epsilon^2)$, with k a field, the augmentation has the free resolution

$$\cdots \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \longrightarrow k.$$

The kernel and image of each multiplication map are both $k\epsilon$, which proves exactness. Applying $\mathrm{Hom}_A(-, k)$ gives zero differentials and one copy of k in every nonnegative degree.

The periodic extension class generates their Yoneda products: splicing its first n steps gives the nonzero class in degree n. Consequently

$$E \simeq k\langle t\rangle, \qquad |t| = 1, \qquad t^2 \neq 0.$$

To obtain the dg model, lift the degree-one class to the endomorphism complex of the resolution. Its components are alternating scalar identities, so it anticommutes with the differential.

Its powers represent the nonzero extension classes just computed. The resulting map from the free associative dg algebra $k\langle t\rangle$ is a quasi-isomorphism. This algebra is not required to be graded commutative. The nonzero Ext groups in arbitrarily high degrees show that k is coefficient-perfect and is not perfect over A.

The same small duality sends $\mathrm{Perf}(A)$ to $\mathrm{thick}_E(k)$. Its ind-extension generates $\mathrm{Loc}_E(k)$, the full stable subcategory closed under colimits and generated by k. Inverting t annihilates k, and hence this whole localizing subcategory, but does not annihilate E. Thus the algebra-perfect and coefficient-perfect extensions differ. In this example k is perfect over E, as the two-cell resolution given by multiplication by t shows.

Nilpotence of the augmentation ideal also matters. Take $A = k[x]$ and $E = \Lambda_k(\eta)$, with $|\eta| = 1$ and zero differential. The augmentation $E \to k$ has a semifree resolution with generators e_n, $n \geq 0$, of degree zero and

$$de_0 = 0, \qquad de_n = e_{n-1}\eta \quad (n \geq 1).$$

Every element is a finite sum of generators with coefficients in E.

All positive-degree cycles are boundaries, and its degree-zero cohomology is generated by e_0. Applying $\mathrm{Hom}_E(-, k)$ gives $\prod_{n\geq 0} k$ in degree zero. The endomorphism $e_n \mapsto e_{n-1}$, with $e_{-1} = 0$, identifies composition with multiplication of formal power series. Hence

$$\mathrm{RHom}_A(A, k) = k, \qquad \mathrm{RHom}_E(k, k) = k[[x]].$$

Each formal series acts by a finite sum on every e_n. The second dual has completed $k[x]$. It has not returned the polynomial algebra. Retaining the tower $\{A/(x^n)\}_n$ records the missing topology.

The identity interface

Choose right modules P_A, P_B, P_C over dg algebras A, B, C, and let $E_A = \mathrm{RHom}_A(P_A, P_A)$, with analogous notation for B, C. An (A, B)-bimodule K acts on right modules by $M \mapsto M \otimes^{\mathbb{L}}_A K$. Assume

$$P_A \otimes^{\mathbb{L}}_A K \in \mathrm{thick}_B(P_B), \qquad P_B \otimes^{\mathbb{L}}_B H \in \mathrm{thick}_C(P_C)$$

for a second bimodule ${}_BH_C$. Define the transformed interfaces

$${}_{E_B}L_{K\,E_A} = \mathrm{RHom}_B(P_A \otimes^{\mathbb{L}}_A K, P_B), \qquad {}_{E_C}L_{H\,E_B} = \mathrm{RHom}_C(P_B \otimes^{\mathbb{L}}_B H, P_C).$$

Their left actions are postcomposition and their right actions are precomposition. These choices fix the order of the product.

Proposition 19.3.3 *(Composition of marked interfaces).* Composition induces a natural equivalence

$$L_H \otimes^{\mathbb{L}}_{E_B} L_K \xrightarrow{\sim} L_{K \otimes^{\mathbb{L}}_B H}, \qquad g \otimes f \longmapsto g \circ (f \otimes 1_H).$$

It respects iterated composition. The identity A-bimodule is sent to the regular E_A-bimodule.

Proof. The formula is balanced because acting on g by precomposition has the same effect as acting on f by postcomposition. Put $Q = P_A \otimes^{\mathbb{L}}_A K$. Both sides are exact contravariant functors of $Q \in \operatorname{thick}_B(P_B)$. At $Q = P_B$, the map is the unit isomorphism for tensor over E_B.

It is therefore an equivalence on the thick closure. For three interfaces, both parenthesizations compose the same three maps. The derived associativity constraints give the same comparison. For $K = A$, the definition gives $L_A = \mathrm{RHom}_A(P_A, P_A) = E_A$ with its two regular actions. □

The augmentation boundary and the identity interface have different roles. The first becomes a regular dual module in Theorem 19.3.2. The second becomes the unit bimodule for balanced composition. Replacing that bimodule by its augmentation would change the composition law. A field realization must preserve the chosen boundary action as well as these units.

19.4 Comodules, contramodules and the direction of a map

A coalgebra has a second correspondence that is covariant in its module argument. Let C be a counital coalgebra over a field k. A left comodule has a coaction $M \to C \otimes_k M$.

A *left contramodule* has instead a map

$$\pi : \mathrm{Hom}_k(C, P) \longrightarrow P.$$

It satisfies the unit identity induced by the counit of C and the associativity identity obtained from $\mathrm{Hom}_k(C, \mathrm{Hom}_k(C, P))$. The two maps in that identity are composition with π and precomposition with the coproduct, using tensor–Hom adjunction. For an infinite coalgebra this retains operations on all linear maps from C, not only finite linear combinations.

For a right comodule N and a left contramodule P, their *contratensor product* $N \odot_C P$ is the cokernel of

$$N \otimes_k \mathrm{Hom}_k(C, P) \rightrightarrows N \otimes_k P.$$

One map is $n \otimes f \mapsto n \otimes \pi(f)$. The other is $n \otimes f \mapsto \sum n_{(0)} \otimes f(n_{(1)})$, where $\delta n = \sum n_{(0)} \otimes n_{(1)}$. The sums here are finite on each comodule element. The subscript C distinguishes this quotient from the transported tensor $\odot$ of (19.2).

The regular coalgebra is a bicomodule. It defines functors

$$\Phi_C(P) = C \odot_C P, \qquad \Psi_C(M) = \mathrm{Hom}_C(C, M), \qquad \Phi_C \dashv \Psi_C. \tag{19.3}$$

The Hom consists of colinear maps.

The left coaction on $\Phi_C(P)$ comes from the first factor of the coproduct. The right coaction on C induces the contramodule operation on $\Psi_C(M)$. In ungraded notation it sends $F : C \to \mathrm{Hom}_C(C, M)$ to $c \mapsto \sum F(c_{(2)})(c_{(1)})$. Tensor–Hom adjunction identifies the two balancing conditions and proves the adjunction in (19.3).

For a dg coalgebra, use the same constructions in complexes with Koszul signs. In the homotopy category of dg comodules, take the thick subcategory generated by total complexes of exact triples and closed under direct sums. The quotient is the *coderived category*.

The objects in this subcategory are called coacyclic. The *contraderived category* uses exact triples of contramodules and closure under products. Its corresponding objects are contraacyclic. These are the acyclic classes relevant to the unbounded correspondence.

THEOREM 19.4.1 *(The field-coefficient correspondence).* For a dg coalgebra over a field, the derived functors of (19.3) give inverse covariant equivalences

$$\mathbb{L}\Phi_C : D^{\mathrm{ctr}}(C\text{-contra}) \rightleftarrows D^{\mathrm{co}}(C\text{-comod}) : \mathbb{R}\Psi_C.$$

Proof. The coderived category is modeled by dg comodules that are injective after forgetting the differential. The contraderived category is modeled by contramodules that are projective after forgetting the differential. These resolution statements are Positselski, *Two kinds of derived categories*, Theorem 4.4. They concern the graded comodule and contramodule categories.

For a graded vector space V, the functors send the free contramodule $\mathrm{Hom}_k(C, V)$ to the cofree comodule $C \otimes_k V$, and conversely. The comparisons follow from evaluation, coproduct and counit. Graded projectives are summands of free contramodules, and graded injectives are summands of cofree comodules. Thus the comparisons extend to these two classes, commute with their differentials and give inverse functors on the two homotopy models. This is the correspondence theorem of that source, Section 5.2. □

For $C = k$, the functor Ψ_C is the identity on complexes. It is not vector-space dualization. Even for a finite coalgebra, colinear Hom cannot be replaced by all linear maps.

Let $C = k1 \oplus kc$, with c primitive. A colinear endomorphism has the form

$$f(1) = a1, \qquad f(c) = b1 + ac.$$

Indeed the coaction identity first forces the displayed value on 1 and then the coefficient of c. Thus $\dim_k \mathrm{Hom}_C(C, C) = 2$, whereas $\dim_k \mathrm{Hom}_k(C, C) = 4$.

The injective condition also retains the coaction. The inclusion $k1 \subset C$ has no comodule retraction. If $r(1) = 1$, the colinearity equation on c would read $1 \otimes r(c) = 1 \otimes r(c) + c \otimes 1$. Thus the trivial comodule k is not injective, although its underlying vector space is injective. Underlying coefficient injectivity alone cannot justify the resolutions in the theorem.

Bar and cobar supply additional algebra comparisons

Let A be an augmented dg algebra over k, and let BA be its normalized bar coalgebra. In this subsection use left A-modules. Write $\tau : BA \to A$ for the canonical degree-one twisting cochain.

Its identity is

$$d_A\tau + \tau d_{BA} + \mu_A(\tau \otimes \tau)\Delta = 0,$$

where the tensor of maps includes its Koszul sign. For a module M, the twisted tensor $BA \otimes_\tau M$ has the ordinary tensor differential plus the term applying the coproduct, τ and the module action. The identity above makes its differential square to zero. Twisted Hom is defined by the adjoint formula.

The resulting equivalences are

$$D(A\text{-mod}) \simeq D^{\mathrm{co}}(BA\text{-comod}), \qquad D(A\text{-mod}) \simeq D^{\mathrm{ctr}}(BA\text{-contra}). \tag{19.4}$$

They are Positselski's Theorem 6.3. One composite is the reduced bar resolution of M. Its augmentation is a quasi-isomorphism.

For the other composite, filtration by the number of bar entries reduces the unit comparison to the cofree coalgebra resolution. The corresponding Hom argument uses products and the contraderived acyclic class. Section 6.4 gives the analogous statement for a conilpotent dg coalgebra C and $A = \Omega C$. The normalized algebra bars and their signs were constructed in Chapter 3.

The two equivalences in (19.4) intertwine the comodule–contramodule correspondence. On a twisted cofree object this is the explicit identity

$$\Psi_C(C \otimes_\tau M) \cong \mathrm{Hom}_\tau(C, M).$$

Thus coefficient transposition, bar–cobar comparison and comodule–contramodule correspondence have specified connecting maps. They remain different functors. Replacing the field by $\mathbb{Z}_p[[T]]$ requires an exact structure and resolution hypotheses over that ring.

Changing the coalgebra

Let $f : C \to D$ be a coalgebra map. Corestriction sends a left C-comodule to a left D-comodule by

$$M \xrightarrow{\partial_M} C \otimes M \xrightarrow{f \otimes \mathrm{I}} D \otimes M.$$

It has the same underlying vector space. A C-contramodule likewise becomes a D-contramodule by

$$\mathrm{Hom}_k(D, P) \xrightarrow{\mathrm{Hom}(f,P)} \mathrm{Hom}_k(C, P) \xrightarrow{\pi} P.$$

Both functors go from C to D. Applying the word "opposite" to f does not produce a linear map $D \to C$.

The reverse comodule functor uses cotensor. Give C its (C, D)-bicomodule structure through f. For a left D-comodule N, set

$$f^! N = C \,\square_D\, N = \ker\big(C \otimes N \rightrightarrows C \otimes D \otimes N\big).$$

The maps use the right D-coaction on C and the left D-coaction on N.

It is right adjoint to corestriction. For example, the adjoint of $u : M \to N$ is $m \mapsto \sum m_{(-1)} \otimes u(m_{(0)})$. The inverse adjunction map applies the counit of C. These formulas prove the adjunction identities.

On contramodules the reverse functor is $f_! Q = \mathrm{Cohom}_D(C, Q)$. For a left D-comodule L, define $\mathrm{Cohom}_D(L, Q)$ as the cokernel of two maps

$$\mathrm{Hom}_k(D \otimes L, Q) \rightrightarrows \mathrm{Hom}_k(L, Q).$$

One precomposes with the coaction of L. The other uses

$$\mathrm{Hom}_k(D \otimes L, Q) \cong \mathrm{Hom}_k(L, \mathrm{Hom}_k(D, Q))$$

and then the contraaction of Q. For $L = C$, the right C-coaction induces the left C-contramodule structure on the quotient. This is left adjoint to the contramodule corestriction.

Proposition 19.4.2 *(Naturality with derived adjoints).* For dg coalgebras over a field there is a natural comparison

$$\mathbb{R}\Psi_C \, \mathbb{R}f^! \simeq \mathbb{L}f_! \, \mathbb{R}\Psi_D : D^{\mathrm{co}}(D\text{-comod}) \longrightarrow D^{\mathrm{ctr}}(C\text{-contra}).$$

Here the derived functors use the graded injective and projective models of Theorem 19.4.1.

Proof. For a graded-injective D-comodule N, the comparison is induced by

$$\mathrm{Hom}_C(C, C \,\square_D\, N) \cong \mathrm{Hom}_D(C, N) \cong \mathrm{Cohom}_D(C, \mathrm{Hom}_D(D, N)).$$

The first isomorphism is the cotensor adjunction. For a cofree $N = D \otimes V$, the second is the evaluation identification with $\mathrm{Hom}_k(C, V)$.

It extends to injective summands. The cotensor functor preserves graded injectives, and its cohomomorphism counterpart preserves graded projectives. Thus the isomorphisms commute with differentials and pass to the two homotopy models. This is the naturality comparison of Positselski, Section 5.4. □

There is a related comparison when algebra maps and coalgebra maps preserve a specified acyclic twisting cochain. If $\tau : C \to A$, $\sigma : D \to B$, $g : A \to B$ and $f : C \to D$ satisfy $\sigma f = g\tau$, then extension over g corresponds, under the twisted tensor equivalences, to corestriction over f. Restriction over g corresponds to derived cotensor. The chain identities follow by substituting $\sigma f = g\tau$ in the twisted differentials. See Positselski, Section 6.9. The equation is an additional compatibility of actual maps.

Example 19.4.3 *(A linear splitting does not reverse a coaction).* Let D have basis $1, c, d$ and coproduct

$$\Delta c = 1 \otimes c + c \otimes 1, \qquad \Delta d = 1 \otimes d + c \otimes c + d \otimes 1.$$

The counit kills c, d. Coassociativity follows by expanding these formulas, or by dualizing multiplication in $k[x]/(x^3)$. The subspace $C = k1 \oplus kc$ is a subcoalgebra.

The linear retraction $r : D \to C$ fixing $1, c$ and killing d preserves the counit, but

$$\Delta_C r(d) = 0, \qquad (r \otimes r)\Delta_D d = c \otimes c \neq 0.$$

It is not a coalgebra map. Applying $1 \otimes r$ to the right regular coaction of D gives a map $\delta : D \to D \otimes C$ whose coassociativity defect on d is

$$(\delta \otimes 1)\delta(d) - (1 \otimes \Delta_C)\delta(d) = 1 \otimes c \otimes c.$$

Thus even a counital splitting of underlying modules does not give the reverse corestriction functor.

For a finite projective coalgebra map, the algebra transpose $f^\vee : D^\vee \to C^\vee$ does exist. Ordinary restriction of modules along it goes from $C^\vee$-modules to $D^\vee$-modules. Extension in the other direction changes the underlying module. The evaluation identity (19.1) remains valid without implying any of the incorrect reverse functors.

19.5 Completion and detection of a comparison

Integral duality requires control of the entire coefficient tower. Let R be a commutative Noetherian ring and $I \subset R$ an ideal. The derived I-adic completion of $M \in D(R)$ is

$$\mathbb{L}\Lambda_I M = \mathrm{Rlim}_n(M \otimes^{\mathbb{L}}_R R/I^n). \tag{19.5}$$

A complex is *derived I-complete* when its natural map to (19.5) is an equivalence. For Noetherian R this is the reflection into the complexes satisfying $\mathrm{RHom}_R(R[1/f], M) = 0$ for every $f \in I$. The equivalence of these definitions is the Noetherian derived-completion theorem, Stacks Project, Tag 0922.

Write $K_n = M \otimes^{\mathbb{L}}_R R/I^n$. The inverse-limit term is visible in the exact sequence

$$0 \longrightarrow \varprojlim_n {}^1H^{j-1}(K_n) \longrightarrow H^j(\mathbb{L}\Lambda_I M) \longrightarrow \varprojlim_n H^j(K_n) \longrightarrow 0.$$

Indeed the derived limit is the fiber of $1 - s : \prod_n K_n \to \prod_n K_n$, where s applies the transition to the next coordinate. Products of modules are exact. The long exact cohomology sequence therefore identifies the two outer terms with the cokernel and kernel of $1 - s$.

The first derived limit vanishes for a surjective cohomology tower, since $(1 - s)x = y$ can then be solved recursively. It also vanishes under the Mittag–Leffler condition: for each fixed index the images of all sufficiently late transition maps agree. This is weaker than surjectivity of each transition.

The reduction tower of $M = [\mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_p]$, in degrees zero and one, gives an example:

$$H^0(M \otimes^{\mathbb{L}}_{\mathbb{Z}_p} \mathbb{Z}/p^n) = p^{n-1}\mathbb{Z}_p/p^n \cong \mathbb{F}_p.$$

Every transition on this group is zero. Its eventual image is zero, so it is Mittag–Leffler but is not a surjective tower. Each term of the complex is flat over $\mathbb{Z}_p$.

Ordinary quotients can lose all the cohomology in the answer. For $D_p = \mathbb{Q}_p/\mathbb{Z}_p$, one has $D_p/p^n D_p = 0$, but the two-term resolution of $\mathbb{Z}/p^n$ gives

$$D_p \otimes^{\mathbb{L}}_{\mathbb{Z}_p} \mathbb{Z}/p^n \simeq D_p[p^n][1], \qquad \mathbb{L}\Lambda_{(p)} D_p \simeq \mathbb{Z}_p[1].$$

The transition on the kernels is multiplication by p. Writing a kernel element as a/p^n identifies it with the reduction $\mathbb{Z}/p^{n+1} \to \mathbb{Z}/p^n$. This tower is surjective, with inverse limit $\mathbb{Z}_p$. Our shift convention places the displayed cohomology in degree -1.

For finite cohomology modules there is a useful simplification. If every $H^j(M)$ is finite over R, then

$$H^j(\mathbb{L}\Lambda_I M) \cong \varprojlim_n H^j(M)/I^n H^j(M).$$

This is Stacks Project, Tag 0A06.

The Noetherian Artin–Rees theorem makes completion exact on finite modules. Finite free resolutions and truncations give the assertion for complexes. In particular finite cohomology over $\mathbb{Z}_p[[T]]$ is already p-adically complete. This does not justify replacing the derived tensor in (19.5) by every chosen termwise quotient.

The unit and counit must be the maps tested

Proposition 19.5.1 *(Conservativity of reduction).* Let $u : M \to N$ be a map of derived I-complete complexes. If $u \otimes^{\mathbb{L}}_R R/I$ is an equivalence, then u is an equivalence.

Proof. The cone K is derived complete, since derived completion is exact. The hypothesis says $K \otimes^{\mathbb{L}}_R R/I = 0$. It follows that $K \otimes^{\mathbb{L}}_R V = 0$ for every R/I-module V, by associativity of derived tensor. The finite filtration of R/I^n by powers of I has such successive quotients. Induction therefore gives $K \otimes^{\mathbb{L}}_R R/I^n = 0$ for every n. Taking the derived limit and using completeness gives $K = 0$. □

Suppose now that an integral adjunction $F \dashv G$ is defined between stable coefficient categories. Suppose both categories have exact reduction functors, conservative on the complete objects under consideration. Assume F, G preserve these objects and that their reductions form an adjunction $\bar{F} \dashv \bar{G}$.

The comparison must identify the reductions of the actual unit and counit with the unit and counit of $\bar{F} \dashv \bar{G}$.

Under these hypotheses, if the reduced adjunction is an equivalence, then the original adjunction is an equivalence on that complete domain. Its unit and counit have zero reduced cones, so conservativity makes both maps equivalences. For coderived or contraderived recipients, this argument requires an internal detection statement for those cones. An underlying ordinary derived complex suffices only when the forgetful comparison detects them.

A compatible tower gives another positive construction. Let $C_n \simeq \mathcal{D}_n$ be equivalences of categories, equipped with coherent comparisons to their transition functors. They form an equivalence of diagrams, so induce

$$\varprojlim_n C_n \simeq \varprojlim_n \mathcal{D}_n.$$

The limits retain their descent data and mapping spaces. Equivalences chosen independently at each level do not yet give the required equivalence of diagrams.

Limits of reduction

Conservativity detects equivalences of supplied maps. It is not faithfulness: multiplication by p on $\mathbb{Z}_p$ is nonzero and reduces to zero. Nor does reduction make a lift unique. For $R = \mathbb{Z}_p[[T]]$, the modules

$$M_0 = R/(T), \qquad M_1 = R/(T - p)$$

have the same reduction modulo p. They are not isomorphic as R-modules, since their annihilator ideals differ. Even a fixed lift has distinct automorphisms 1 and $1 + p$ with the same reduction.

There is a second distinction between scalar and power-series coefficients. The module $R/(T)$ is free over $\mathbb{Z}_p$, but the free R-resolution $[R \xrightarrow{T} R]$ gives

$$\operatorname{Tor}_1^R(R/(T), R/(T)) = R/(T).$$

It is not flat over R. Its Iwasawa invariant μ is zero, because T becomes invertible at the height-one prime (p). Thus $\mu = 0$ supplies neither flatness nor the required base-change comparison.

The coefficient ring $\Lambda = \mathbb{Z}_p[[T]]$ is regular local of dimension two. The parameters p, T form a regular sequence, and their Koszul complex resolves the residue field in length two. A minimal free resolution of a finite module has zero differentials after reduction to the residue field. Its terms then compute Tor. Computing Tor instead with the Koszul resolution shows that every term above degree two is zero, by Nakayama's lemma.

Finite Λ-modules therefore have projective dimension at most two. In particular this holds for Λ/J for every ideal J, since Λ is Noetherian. Dimension shifting and Baer's injectivity criterion then bound the injective dimension of every Λ-module by two. These bounds can support resolution arguments, but the exact comodule and contramodule structures must still be specified. In particular the module condition $\mu = 0$ and coalgebra conilpotence are independent, as the two explicit coalgebras in Proposition 10.10.2 show.

For an abelian number field, form the Iwasawa module from the p-primary class groups of its cyclotomic tower. Ferrero–Washington proves that its μ invariant vanishes. Transferring this invariant to another arithmetic object requires a comparison of the specified coefficient modules, for example a pseudo-isomorphism. Neither the oscillator basis nor the coproduct alone provides that comparison.

19.6 Completed coordinate coactions and Fock fields

The transpose of a coaction can have infinitely many output coordinates even when each input state is a polynomial. This does not prevent a continuous coordinate coaction. It does force a choice of recipient.

Let k be a discrete field of characteristic zero and let H be a Hopf algebra with bijective antipode s. Suppose a right comodule is an increasing union of finite subcomodules, $M = \bigcup_N M_N$. Define

$$D_M = \varprojlim_N M_N^*, \qquad \mathscr{T}_r(M) = \varprojlim_N (M_N^* \otimes_k H^{\otimes r}).$$

The finite-stage spaces are discrete. An element of $\mathscr{T}_r(M)$ is equivalently a linear map $M \to H^{\otimes r}$, with the topology of restriction to finite stages. In particular the target of a completed coaction is $\mathscr{T}_1(M)$. It need not be $D_M \otimes_k H$.

Use the following tensor convention on right comodules:

$$\delta(m \otimes n) = m_{(0)} \otimes n_{(0)} \otimes n_{(1)} m_{(1)}. \tag{19.6}$$

The order of the two Hopf-algebra coefficients is part of the convention. Associativity of H makes this a monoidal tensor.

Proposition 19.6.1 *(The continuous coordinate coaction).* There is a natural continuous map

$$\widehat{\delta}_M : D_M \longrightarrow \mathscr{T}_1(M), \qquad (\widehat{\delta}_M f)(m) = \sum f(m_{(0)}) \mathsf{s}^{-1}(m_{(1)}). \tag{19.7}$$

It is counital and coassociative in the corresponding coordinate recipients. It gives a natural tensor comparison

$$D_{M \otimes N} \cong D_N \widehat{\otimes} D_M := \varprojlim_{a,b} (N_b^* \otimes M_a^*),$$

where both sides retain their inverse systems of comodules.

Proof. In a basis e_j of M_N, write $\delta e_j = \sum_i e_i \otimes u_{ij}$. The coaction identity is $\Delta u_{ij} = \sum_l u_{il} \otimes u_{lj}$. The proposed dual coaction sends e^i to $\sum_j e^j \otimes \mathsf{s}^{-1}(u_{ij})$. Since the antipode reverses coproduct,

$$\Delta \mathsf{s}^{-1}(u_{ij}) = \sum_l \mathsf{s}^{-1}(u_{lj}) \otimes \mathsf{s}^{-1}(u_{il}).$$

This is precisely coassociativity of the dual formula. The counit identity follows from the identity coefficient matrix. Restriction to a smaller subcomodule commutes with the formula, so it defines the inverse-limit map. Each finite projection uses only $f|_{M_N}$, which proves continuity.

At a finite pair of stages, $g \otimes f$ corresponds to the functional $m \otimes n \mapsto f(m)g(n)$. Its Hopf coefficient is compatible because

$$\mathsf{s}^{-1}(n_{(1)} m_{(1)}) = \mathsf{s}^{-1}(m_{(1)}) \mathsf{s}^{-1}(n_{(1)}).$$

This is the coefficient of $g \otimes f$ under (19.6). Passing through the restriction maps gives the tensor comparison. A map of the exhaustive filtered unions takes each finite-dimensional stage into a finite stage, so its transpose is continuous and all comparisons are natural. □

The reversal is visible in the four-dimensional Hopf algebra generated by g, x, with

$$g^2 = 1, \quad x^2 = 0, \quad gx = -xg, \qquad \Delta g = g \otimes g, \quad \Delta x = 1 \otimes x + x \otimes g.$$

Its inverse antipode has $\mathsf{s}^{-1}(x) = -gx$. Consequently $\mathsf{s}^{-1}(xg) = -x$, while $\mathsf{s}^{-1}(x)\mathsf{s}^{-1}(g) = x$. Keeping the original tensor order changes this coefficient by $2x$.

The full binomial coaction

Take $F_K = k[x_1, x_2, \ldots]$, with $K \neq 0$, and give x_n energy n. Let $H = k[b_2, b_3, \ldots]$ with every b_j primitive, and set

$$\delta_K(x_n) = x_n \otimes 1 - nK\, 1 \otimes b_{n+1}.$$

Extending as an algebra map gives a right coaction. The subspace of energy at most N is finite-dimensional and stable under it.

Let f_α extract the coefficient of $x^\alpha = \prod_j x_j^{\alpha_j}$, where α has finite support. Applying (19.7) to the binomial expansion gives

$$\widehat{\delta}_K(f_\alpha) = \sum_{\beta \geq \alpha} \binom{\beta}{\alpha} f_\beta \otimes \prod_{j \geq 1} (jK b_{j+1})^{\beta_j - \alpha_j}, \qquad \binom{\beta}{\alpha} = \prod_j \binom{\beta_j}{\alpha_j}.$$

Each finite-energy projection has finitely many terms. For the vacuum functional, $(\widehat{\delta}_K f_0)(x_1^n) = (Kb_2)^n$. The arbitrarily high powers show why the algebraic restricted-dual tensor is too small for this formula.

The construction also works over a formal coefficient tower. Suppose that H_d and $M_{N,d}$ are compatible Hopf algebras and finite free subcomodules over $k[\hbar]/(\hbar^d)$, with bijective antipodes. The same finite matrix proof gives a coaction with target

$$\varprojlim_{N,d} \left(M_{N,d}^\vee \otimes_{k[\hbar]/(\hbar^d)} H_d \right).$$

For example, a translation derivation satisfying $Tb_n = nb_{n+1}$ gives primitive tails

$$\beta_{K,n} = \frac{b_n - e^{K\hbar T} b_n}{\hbar} = -\sum_{r \geq 1} \frac{K^r \hbar^{r-1}}{r!} T^r b_n.$$

This quotient is defined by the series and does not invert $\hbar$. On the polynomial Hopf algebra with primitive b_n, these tails define $\delta_K(x_n) = x_n \otimes 1 + 1 \otimes \beta_{K,n}$. At each precision only finitely many terms of each tail remain. The limit recovers a continuous coordinate coaction on $\prod_\alpha k[[\hbar]] f_\alpha$.

Evaluation imposes another topology

Let $M = \bigoplus_{n \geq 0} M[n]$, with each $M[n]$ finite, and let $D^{\mathrm{res}} = \bigoplus_n M[n]^*$. Consider the matrix completion

$$\mathscr{E}(M) = \prod_{p,q \geq 0} M[p] \otimes M[q]^*, \tag{19.8}$$

with discrete finite blocks. The finite sums of the identity tensors on $M[0], \ldots, M[N]$ converge to an element of $\mathscr{E}(M)$ with identity diagonal blocks and zero off-diagonal blocks. This is the formal diagonal coevaluation.

PROPOSITION 19.6.2 *(Failure of continuous contraction).* If infinitely many $M[n]$ are nonzero, ordinary evaluation on $M \otimes D^{\mathrm{res}}$ has no continuous extension to (19.8).

Proof. Choose distinct $n_j \to \infty$, vectors $e_j \in M[n_j]$ and functionals f_j with $f_j(e_j) = 1$. The tensors $e_j \otimes f_j$ converge to zero in the product topology, since every fixed finite collection of blocks is eventually zero. Their evaluations are all 1, which does not converge to zero in k. This contradicts continuity. □

The argument persists over $k[[\hbar]]$ with the product of the $\hbar$-adic block topologies: the constant scalar 1 still does not approach zero. Exchanging the two tensor orders is a homeomorphism of the underlying matrix products, so it does not repair continuity. No assertion about that flip as a comodule map is needed.

Weighting the coevaluation by q^n gives a different map. Its zigzag on $M[n]$ is q^n id rather than id. Such a weighted sewing operator can be useful, but it is not the coevaluation of ordinary duality. The coordinate coaction above and the diagonal coevaluation therefore do not make this matrix completion into a rigid tensor category.

The inverse coordinate and the central level

On F_K use the Heisenberg modes $J_{-n} = x_n$, $J_n = nK\partial_{x_n}$ for $n > 0$, and $J_0 = 0$. Then $[J_m, J_n] = mK\delta_{m+n,0}$. For an operator A, its transpose is $(A^t f)(v) = f(Av)$. At the same coordinate the evaluation-compatible Lie action is $-J_m^t$. It has level $-K$. It is not lower truncated on the restricted dual: the positive modes send f_0 to $-nKf_{e_n}$ for every $n > 0$.

In the inverse coordinate $w = z^{-1}$, the weight-one contragredient instead has

$$J^\infty(w) = -w^{-2}J(w^{-1})^t, \qquad J_m^\infty = -J_{-m}^t. \tag{19.9}$$

This is the primary-current specialization of Huang, *Introduction to vertex operator algebras, III*, equation (1.26). Its properties here can be checked directly.

Proposition 19.6.3 *(Restricted contragredient Fock module).* The current (19.9) is lower truncated on $F_K^{\mathrm{res}} = \bigoplus_n F_K[n]^*$. It has level K and is isomorphic to F_K by

$$x^\alpha \longmapsto \left(\prod_j (-jK)^{\alpha_j}\alpha_j!\right) f_\alpha. \tag{19.10}$$

On the full product $\prod_n F_K[n]^*$ the same current need not be lower truncated.

Proof. For $n > 0$ its modes are

$$J_n^\infty f_\alpha = -f_{\alpha-e_n}, \qquad J_{-n}^\infty f_\alpha = -(\alpha_n + 1)nKf_{\alpha+e_n}.$$

The first expression is zero when $\alpha_n = 0$. Positive modes lower dual energy, so sufficiently large ones kill every vector in the direct sum. These formulas also prove (19.10), whose coefficients are nonzero because k has characteristic zero and $K \neq 0$. Transposition reverses commutators, giving

$$[J_m^\infty, J_n^\infty] = -[J_{-m}, J_{-n}]^t = mK\delta_{m+n,0}.$$

Normal ordering and differentiation then identify all the Heisenberg module fields.

For the full product, take the functional f with $f(x_n) = 1$ for every $n > 0$ and zero on the other monomials. Its vacuum component is $(J^\infty(w)f)(1) = -\sum_{n\geq 1} w^{-n-1}$. It has no lower Laurent bound, proving the last assertion. □

The inverse coordinate restores truncation on the restricted dual and restores the original level. For a tensor that adds levels and has scalar unit of level zero, it is therefore not an inverse-level dual when $K \neq 0$. The field-module isomorphism does not assert compatibility with the completed coordinate coaction. Finding a category in which the desired evaluation, coaction and field maps coexist remains a further construction. The two explicit failures above concern their stated recipients.

19.7 FULL COLLISION COEFFICIENTS

A singular field needs a coefficient category that retains all its translates. Suppose $Y(a,z)b$ has a nonzero pole over a field of characteristic zero. A relation $p(T)a = 0$ would give $p(\partial_z)Y(a,z)b = 0$. If the lowest Laurent term is vz^{-q} and p has leading term ct^d, the coefficient of z^{-q-d} is

$$c(-q)(-q-1)\cdots(-q-d+1)v \neq 0.$$

No lower derivative contributes there. Thus the translates $T^n a$ are linearly independent.

On $X = \mathbb{A}^1_k$, the right differential-module action is

$$(f\,dz \otimes a)\partial_z = -f'\,dz \otimes a - f\,dz \otimes Ta. \tag{19.11}$$

The translation orbit therefore gives a free rank-one $\mathcal{D}_X$-submodule of $\omega_X \otimes V$. It is perfect over $\mathcal{D}_X$, although it is not holonomic: its characteristic variety is the full cotangent bundle. Holonomicity, perfectness and dualizability for a collision tensor are different conditions.

At finite formal precision a nonzero pole can have a torsion coefficient. Over $k[\hbar]/(\hbar^2)$ the state $\hbar J$ can have a nonzero pole and is still killed by $\hbar$. For a scalar-extended family, freeness of the translation orbit over the parameter ring follows if the pole remains nonzero modulo $\hbar$. A least-order coefficient of any proposed relation then contradicts the argument over k.

The retained perfect diagram

Let U be a smooth affine chart of dimension d, and set $\mathcal{D} = \Gamma(U, \mathcal{D}_U)$. A perfect right $\mathcal{D}$-complex is a summand of a finite complex of finite free right modules. A coordinate volume form converts left modules to right modules by $f^t = f$, $\partial_i^t = -\partial_i$ and $(ab)^t = b^t a^t$. Define

$$P^\dagger = \big(\mathrm{RHom}_{\mathcal{D}}(P, \mathcal{D})\big)^t[d].$$

The canonical line gives the same conversion without a chosen volume form. Finite projective evaluation proves $(P^\dagger)^\dagger \simeq P$. The two shifts cancel because duality reverses shifts. The same finite calculation commutes with restriction to an open chart and with external products. Dimensions add in the latter comparison.

Every derived $\mathcal{D}$-module has a filtered presentation $M \simeq \varinjlim_\alpha P_\alpha$ by perfect objects. The presentation is in the derived category, and need not be a union of submodules. Define its full coefficient dual and the inverse construction by

$$M^\sharp = \{P_\alpha^\dagger\}_\alpha, \qquad \mathbb{S}\{Q_\beta\} = \varinjlim_\beta Q_\beta^\dagger. \tag{19.12}$$

The braces retain the pro-object. The fact used here is that the derived module category is the ind-completion of its perfect objects, since the regular module is a compact generator.

PROPOSITION 19.7.1 *(Full coefficient polarity).* The functors in (19.12) give a presentation-independent anti-equivalence

$$D(\mathcal{D}\text{-right})^{\mathrm{op}} \simeq \mathrm{Pro}(\mathrm{Perf}_{\mathrm{r}}(\mathcal{D})).$$

Proof. For $M = \varinjlim_\alpha P_\alpha$ and $N = \varinjlim_\beta Q_\beta$, the mapping formula is

$$\begin{aligned}\mathrm{Map}_{\mathrm{Pro}}(N^\sharp, M^\sharp) &\simeq \varprojlim_\alpha \varinjlim_\beta \mathrm{Map}(Q_\beta^\dagger, P_\alpha^\dagger)\\ &\simeq \varprojlim_\alpha \varinjlim_\beta \mathrm{Map}(P_\alpha, Q_\beta) \simeq \mathrm{Map}(M, N).\end{aligned}$$

The last equivalence uses compactness of P_α. The formula respects composition. Applying it to any pro-perfect diagram identifies that diagram with the dual of its image under $\mathbb{S}$. It proves both inverse comparisons and independence of the presentation. □

The pro-object contains more information than its ordinary inverse limit. For the Weyl algebra $D = k\langle u, \partial\rangle/([\partial, u] - 1)$, the right-module localization has presentation

$$D[u^{-1}] = \varinjlim(D \xrightarrow{L_u} D \xrightarrow{L_u} \cdots), \qquad a \text{ at stage } n \longmapsto u^{-n}a.$$

Its coefficient dual is the tower of $D[1]$ with transitions L_u. No transition to a fixed stage is zero, so this pro-object is nonzero. Its underived inverse limit is zero: each coordinate of a compatible sequence lies in $\bigcap_n u^n D = 0$, as follows from polynomial coefficients in Weyl normal form. No vanishing of a derived inverse limit is asserted.

The vacuum has another finite resolution, $[D \xrightarrow{L_\partial} D] \to \omega_X$. It gives $\omega_X^\dagger \simeq \omega_X$, whereas $D^\dagger = D[1]$. Thus the full dual retains both the vacuum and the free translation orbit with their respective shifts.

Transporting the collision tensor

Use the ordered collision schemes $\mathscr{R}_I$ and their exceptional collision pullbacks. A coefficient system assigns a right differential complex to each scheme, together with coherent comparisons along these pullbacks. Let C denote this category. Define $C^\sharp$ by applying Proposition 19.7.1 on affine charts and gluing. Opposite categories preserve limits, so the affine comparisons give $C^{\mathrm{op}} \simeq C^\sharp$ on the descent categories.

For any geometric functor F, its transported functor is $F^\sharp = \sharp F \mathbb{S}$. In particular collision pullback becomes $\delta^\dagger = \sharp \delta^! \mathbb{S}$, and open direct image becomes $j_{!,\mathrm{pro}} = \sharp j_* \mathbb{S}$. The latter is a functor to pro-perfect coefficients. It is not an assertion that ordinary $j_!$ exists on every large differential-module coefficient in this form.

For an ordered decomposition $I = G \sqcup H$, let $U_{G|H}$ be the open where the two groups of points are disjoint, and let $j_{G|H}$ be its inclusion. Internal collisions within a group remain allowed. The atlas convolution $\star$ sums the corresponding open direct images over all ordered decompositions. Its transported tensor is

$$(E \star_\sharp F)_I = \bigoplus_{I=G\sqcup H} j_{G|H,!,\mathrm{pro}}\,(E_G \boxtimes^\sharp F_H)\Big|_{U_{G|H}}. \tag{19.13}$$

Here the external product uses the two-index perfect diagrams. The sum is finite. Equivalently, $E \star_\sharp F = (\mathbb{S}E \star \mathbb{S}F)^\sharp$. Associativity and every collision comparison are transported through the same coefficient duality.

The tensor unit K has coefficient k at empty support and zero at nonempty support. Thus $K^\sharp = K$. The word-cut tensor, whose decompositions are consecutive prefixes and suffixes, is a separate construction. In the atlas tensor, $13 \mid 2$ and $2 \mid 13$ are present. Transposition retains these components and all normal derivatives rather than replacing a full collision by an ordered word.

For finite lists there is consequently an equivalence of operation spaces

$$\mathrm{Map}_{C}\left(\bigstar_{i} X_i, \bigstar_{j} Y_j\right) \simeq \mathrm{Map}_{C^\sharp}\left(\bigstar_{j}^{\sharp} Y_j^{\sharp}, \bigstar_{i}^{\sharp} X_i^{\sharp}\right).$$

It follows directly from the mapping equivalence and (19.13). It reverses composition and retains identities, homotopies and their coherences.

Associative algebras in the two polarities

An augmented coefficient algebra is called connected here when its empty coefficient is k. Its augmentation ideal then has zero empty coefficient. On a configuration I, a tensor word in that ideal has at most $|I|$ nonempty factors. This support bound is the finiteness condition for normalized bar and cobar, independently of the dimension of any individual coefficient.

THEOREM 19.7.2 *(Associative coefficient duality).* On the connected augmented domains for the full atlas tensor, normalized bar and full coefficient duality give an anti-equivalence

$$\mathrm{Alg}^{\mathrm{aug,conn}}_{E_1}(C, \star)^{\mathrm{op}} \simeq \mathrm{Alg}^{\mathrm{aug,conn}}_{E_1}(C^\sharp, \star_\sharp).$$

Its functor is $\mathcal{K}(A) = (B_\star A)^\sharp$, and its inverse is $E \mapsto \mathbb{S}B_{\star_\sharp}E$.

Proof. The bar coproduct transposes to the multiplication of $\mathcal{K}(A)$ through (19.13). The bar counit becomes its unit, and its coaugmentation becomes its augmentation. For a connected coalgebra C, dualizing the normalized bar of $C^\sharp$ gives $\mathbb{S}B_{\star_\sharp}(C^\sharp) \simeq \Omega_\star C$. Suspensions reverse, multiplication transposes to the coproduct, and outer deconcatenation becomes concatenation.

The bar–cobar comparison is finite on every I. Filter a composite by its original positive-support entries. For more than one entry, each gap is either an outer cut or an inner product gap. On the associated graded these two choices form a two-term identity complex.

Contract the first gap, with the tensor sign. The single-entry comparison is the identity, and the finite filtration proves the unit and counit equivalences. This is the finite-support bar–cobar argument of Chapter 3 applied to all ordered decompositions. Its triangles, together with coefficient biduality, give the two inverse functors. □

For a formal family the same proof applies at every $R_m = k[\hbar]/(\hbar^m)$ using relative differential operators $\mathcal{D}_U \otimes_k R_m$. Finite perfect duality does not require R_m to be regular. Keep the coefficient-reduction functors and their transported pro-diagrams. No exchange of a pro-limit, an adic limit and a Laurent expansion is used.

This theorem constructs a second associative coefficient algebra with an inverse comparison. The Lie–commutative chiral correspondence concerns other algebra objects and does not supply this associative comparison. A realization of its output by fields must additionally identify the field products, exchange, tensor and local vacuum. The next construction supplies such a realization with explicit markings.

19.8 A field realization with a bar contact marking

The field data must be strong enough to recover the input algebra from its dual. For charged currents the relevant data are a quadratic current core, its translation and jet derivations, and the contact operations retained in the bar. We work over $k = \mathbb{C}$ at the fixed precision $R_2 = k[\hbar]/(\hbar^2)$. The parameter has degree, charge and weight zero. Position on the affine line is denoted by z and field separation by u.

Charged jets and their fields

Let E be a bounded finite-dimensional complex in nonpositive cohomological degrees. Give $e_n = e \otimes t^n$ charge one and weight $n+1$. On $\mathrm{Sym}_k(E \otimes k[t])$ define even derivations

$$T e_n = (n+1)e_{n+1}, \qquad \partial_i e_n = e_{n-i+1} \quad (i \geq 1), \qquad e_j = 0 \ (j < 0). \tag{19.14}$$

They kill the unit, commute with the differential and satisfy $[\partial_i, \partial_j] = 0$ and $[\partial_i, T] = (i-1)\partial_{i-1}$, with $\partial_0 = 0$. The derivation ∂_1 measures charge.

A *charged jet algebra* is a quotient of this dg algebra by an ideal homogeneous in charge and weight and stable under the displayed derivations. Require its charge-zero part to be k. Each weight piece is then a bounded finite-dimensional complex. The quotient presentation is not an extra marking.

The charge-one generators are recovered intrinsically. On A_1 put $D = \partial_2$. Then $[D, T] = 1$ and D is locally nilpotent. The formulas

$$\pi = \sum_{j \geq 0} \frac{(-1)^j}{j!} T^j D^j, \qquad a = \sum_{n \geq 0} \frac{T^n}{n!} \pi(D^n a)$$

give $A_1 = E_A \otimes k[t]$, where $E_A = \ker(D|_{A_1})$.

Both sums are finite on every input. Commuting D through T^j cancels adjacent terms and proves $D\pi = 0$. The binomial identity proves the second formula. Successive powers of D prove its uniqueness. Thus E_A and the kernel of $\mathrm{Sym}(E_A \otimes k[t]) \to A$ are recovered from A and its jet actions.

After scalar extension to R_2, define

$$\begin{aligned} C_1(u) &= \sum_{p,q \geq 0} (-1)^{p+1} \binom{p+q}{p} u^{-p-q-1} \partial_{p+1} \otimes \partial_{q+1}, \\ E(u) &= 1 - \hbar C_1(u), \qquad Y_A(a,u)b = \mu_A(e^{uT} \otimes 1)E(u)(a \otimes b), \\ \mathcal{S}_A(u) &= 1 + 2\hbar C_1(u). \end{aligned} \tag{19.15}$$

On fixed algebraic inputs the contraction is finite. Taylor translation contributes nonnegative powers, so the field output still has a finite lower Laurent bound. The symbol $\mathcal{S}_A$ denotes unpermuted exchange.

An input morphism is an R_2-linear unital dg algebra map preserving charge and commuting with T and every ∂_i. In particular it preserves multiplication, $f\mu_A = \mu_{A'}(f \otimes f)$. These equations imply that it preserves the fields and exchange in (19.15). The weak equivalences are state quasi-isomorphisms.

The first interacting one-seed quotient is

$$A_0 = k[e_0, e_1, \ldots]/I^3 = k \oplus V \oplus Q, \qquad I = (e_0, e_1, \ldots), \quad Q = \mathrm{Sym}^2 V. \tag{19.16}$$

Its differential is zero. The field of e_0 satisfies

$$Y_{A_0}(e_0, u)e_0 = (1 + \hbar/u) \sum_{n \geq 0} e_n e_0 u^n.$$

Its $\hbar u^{-1}$ coefficient is $e_0^2 \neq 0$. The quotient by I^2 would kill this interaction.

The marked current core

Put $\mathfrak{g} = \mathfrak{sl}_2(k)$, with $[h, e] = 2e$, $[h, f] = -2f$ and $[e, f] = h$. An independent current core consists of a discrete nonpositive complex L with bounded finite-dimensional nonnegative weight pieces and

completed generators $W = (L[1])^\vee$. Here the dual of a discrete space is its full degreewise dual with product topology. The dual of a completed space is its continuous degreewise dual. Thus $W^\vee[-1] = L$.

The finite-weight generator space is $W_{\mathrm{pol}} = \bigoplus_q (L_q[1])^*$. Its polynomial core is $\mathrm{Sym}(W_{\mathrm{pol}})$, and its completion is the product of the weight pieces in each cohomological degree. Require a continuous square-zero derivation whose values on generators have linear and quadratic parts. If $|v_i| = a_i$, $d_L v_i = \sum_k D_i^k v_k$ and $[v_i, v_j] = \sum_k f_{ij}^k v_k$, our sign convention is

$$dc^k = \sum_i (-1)^{a_i} D_i^k c^i - \frac{1}{2} \sum_{i,j} (-1)^{a_i(1-a_j)} f_{ij}^k c^i c^j.$$

The linear, quadratic and cubic components of $d^2 = 0$ give the differential, derivation and Jacobi identities on L.

Mark a scalar-current map dual to a dg Lie map $j : \mathfrak{g} \to L$, and require translation to kill $j(\mathfrak{g})$. For $x \cdot v = [j(x), v]$, require

$$\mathfrak{g} \otimes \mathrm{Hom}_{\mathfrak{g}}(\mathfrak{g}, L) \xrightarrow{\sim} L. \tag{19.17}$$

Charge, translation and all jet actions are retained. The recovered multiplicity space must satisfy the charged jet conditions above. This is a condition on the supplied core and its operations.

For a charged jet algebra A, the forward core is constructed from

$$L_A = \mathfrak{g} \otimes A, \quad [x \otimes a, y \otimes b] = [x, y] \otimes ab, \quad d_{L_A} = 1 \otimes d_A, \quad j_A(x) = x \otimes 1.$$

Its translation and jets are $1 \otimes T$ and $1 \otimes \partial_i$. The corresponding derivations on the dual generators are their negative transposes, denoted T_B, D_i.

Let $C^{(n)}$ be charge n of the polynomial core, and set

$$\begin{aligned} \Lambda_n &= k[x_1, \dots, x_n]^{S_n}, & d_n &= \prod_{i<j} (x_i - x_j)^2, \\ B^{\mathrm{mer}} &= \bigoplus_{n \geq 0} C^{(n)} \otimes_{\Lambda_n} \Lambda_n[d_n^{-1}]. \end{aligned} \tag{19.18}$$

Here $d_0 = d_1 = 1$. The variables x_i are constituent dual-jet variables, not positions on the affine line. Their power sums act by $p_m \mapsto -D_{m+1}$, including $p_0 = n$. This symmetric-ring action is determined by the marked jets.

Multilinear polarization explains the action and the localization. At charge n, distribute the n labeled constituents among current generators. Their jet indices give monomials in $x_1, \dots, x_n$.

Take symmetric invariants with cohomological Koszul signs. Multiplication uses the normalized projector $(n!)^{-1} \sum_{\sigma \in S_n} \sigma$. The pairing of x^j with et^j is one, and repeated even constituents contribute their factorials. There is no additional binomial factor.

For a free jet algebra this identifies the charge piece with $(K_n \otimes k[x_1, \dots, x_n])^{S_n}$, where K_n records the finite current and constituent data. For a jet quotient its dual core is a submodule of this free polarized module. Multiplication by d_n is therefore injective. The current differential changes only how the same constituents are grouped. It leaves their variables fixed. It is Λ_n-linear and extends to (19.18).

For two groups of charges p, q, the joint recipient also inverts every cross difference $x_a - y_b$. For field parameters it inverts their translated differences as well. At the fixed precision only finitely many denominator factors occur in any element. Put $w(u) = 1 - \hbar/u$ and define

$$\begin{aligned} \Omega_{p,q}(v) &= \prod_{a,b} w(v + x_a - y_b), & \mu_w &= \mu_0 \Omega_{p,q}(0), \\ Y_B(b, v)c &= \mu_w(e^{vT_B} b \otimes c). \end{aligned} \tag{19.19}$$

The map μ_0 is ordinary current multiplication extended to this joint localization. The native vacuum is the empty symmetric monomial.

A current morphism is an R_2-linear unital dg algebra map of the polynomial cores, linear on the marked generator spaces. It preserves the scalar-current map and charge, commutes with T_B and every D_i, and extends continuously to the completed cores. Its extension to the localizations is then unique. These conditions preserve the product, fields, exchange and vacuum in (19.19). The polynomial core is retained as a marking. It need not be a subalgebra for the deformed product μ_w.

The objects on both sides are scalar extensions from k to R_2. Their morphisms need not be extended from k. After extension, generator duals are R_2-linear with the same topologies. The Frobenius pairing $(a, b) \mapsto [\hbar](ab)$ identifies these dual calculations with exact k-linear duality.

PROPOSITION 19.8.1 *(The two field structures).* The input and current fields just defined satisfy the differential, vacuum, translation, associativity and exchange identities on their stated recipients. Their exchanges are unitary and satisfy Yang–Baxter and the field hexagon.

Proof. The commuting jet derivations give $[T \otimes 1, C_1(u)] = -\partial_u C_1(u)$ and $[T \otimes 1 + 1 \otimes T, C_1(u)] = 0$. Each derivation kills the vacuum. The product rule splits a contraction against a product into its two contractions. Thus either association of three fields uses the same three pair factors, after translation of their separations. The reversed ratio of the two input factors is $\mathcal{S}_A$.

On the current side, normalized symmetrization absorbs each intermediate projector. Both associations have the factor $\Omega_{p,q}(0)\Omega_{p,r}(0)\Omega_{q,r}(0)$. The differential leaves constituent variables fixed, and total translation kills differences. Hence both are derivations of the deformed product.

Empty groups give the two vacuum identities. The exchange is $\Omega_{q,p}(-v)^{21}/\Omega_{p,q}(v)$. Reversing twice gives its inverse. Commuting pair factors give Yang–Baxter, and partitioning their pair sets gives the hexagon. Stability under the jet derivations carries the input identities through the quotient ideals. □

Formal equivalences and the contact coalgebra

The completed core has a specified notion of weak equivalence. Let S be a finite-dimensional nonpositive augmented commutative dg k-algebra with nilpotent augmentation ideal $\mathfrak{m}_S$. Put

$$\Omega_n = k[t_0, \dots, t_n, dt_0, \dots, dt_n]/(\textstyle\sum t_i - 1, \sum dt_i), \qquad d(t_i) = dt_i,$$

where $|t_i| = 0$ and $|dt_i| = 1$. For the completed untwisted core $\widehat{C}_B$ with positive symmetric-degree ideal I_B, consider the simplicial set $X_B(S)$. Its n-simplices are continuous unital R_2-linear dg algebra maps satisfying

$$\begin{aligned} \phi &: \widehat{C}_B \longrightarrow R_2 \otimes S \otimes \Omega_n, \\ \phi(I_B) &\subset R_2 \otimes \mathfrak{m}_S \otimes \Omega_n. \end{aligned} \tag{19.20}$$

In particular $\phi(ab) = \phi(a)\phi(b)$. The target is discrete. These test maps need not preserve the generator splitting or the jet markings. A current morphism $B \to B'$ is a *formal core equivalence* if precomposition gives a weak equivalence $X_{B'}(S) \to X_B(S)$ for every S. This defines the weak equivalences on the current side.

The generator images identify (19.20) with the Maurer–Cartan simplicial set of $(\mathfrak{g} \otimes A_{R_2}) \otimes \mathfrak{m}_S$: the chain-map equation is $d\alpha + \frac{1}{2}[\alpha, \alpha] = 0$. Nilpotence makes all products finite. Continuity gives finite weight support at each simplex. The nilpotent Maurer–Cartan invariance theorem sends a state quasi-isomorphism to an equivalence of these tests. See Getzler, *Lie theory for nilpotent L_∞-algebras*, Theorem 4.8.

Conversely, the test $S = k[\varepsilon]/(\varepsilon^2)$ has, at the zero point,

$$\pi_n X_B(S) = H^{1-n}(\mathfrak{g} \otimes A_{R_2})\varepsilon \quad (n \geq 1), \qquad \pi_0 X_B(S) = *.$$

The bracket is zero in this test, and the polynomial de Rham complex computes the shifted linear complex. Every state degree occurs because the states are nonpositive. Thus a formal core equivalence detects the recovered state quasi-isomorphism. This uses the marked core and does not replace it by ordinary cohomology of the unmarked meromorphic algebra.

Now form the full contact coalgebra C_A of the input fields. Its singleton is $(\omega_X \otimes A_{R_2})[1]$ with right action (19.11). For every ordered partition retain its open and closed direct images and all normal derivatives. A merger uses the principal part of the field, together with exchange to bring nonconsecutive blocks together.

In particular it retains

$$r_A(u^m; a, b) = \mathrm{pp}(u^m Y_A(a, u)b) \qquad (m \in \mathbb{Z}).$$

Here pp takes the strictly negative Laurent part. Fields have expansion $Y_A(a, u)b = \sum_n (a_{(n)}b)u^{-n-1}$, so

$$a_{(n)}b = [u^{-1}]r_A(u^n; a, b).$$

Negative multipliers recover the regular modes as well. The coefficient differential and all its coherences come from the full contact construction, not from the extraction of a single residue.

Define $U_*(A) = \Omega_\star C_A$. If tc is the degree-one desuspension of a reduced coalgebra element, its cobar differential is

$$d(tc) = -t(d_C c) + \sum_{\overline{\Delta}c = c' \otimes c''} (-1)^{|c'|+1}(tc')(tc''). \tag{19.21}$$

Every ordered-subset coproduct is included. The empty word is the unit of $U_*(A)$.

PROPOSITION 19.8.2 *(The bar contact marking).* The identity at empty support and the reduced map

$$\eta_A(c) = \sum_{r \geq 1} (s\tau)^{\star r} \overline{\Delta}^{(r-1)} c, \qquad \tau(c) = tc, \quad |s| = -1,$$

give a natural equivalence $C_A \to B_\star U_*(A)$ preserving the entire contact diagram.

Proof. Use bar signs $b_1(sa) = -s(da)$ and $b_2(sa, sb) = (-1)^{|a|}s(ab)$. In (19.21), a split has sign $(-1)^{|c'|+1}$. It cancels the bar multiplication joining the corresponding adjacent factors. Internal differentials have the suspension sign in b_1. Coassociativity identifies the iterated cuts, proving the chain and coalgebra identities.

Use the finite gap filtration of Theorem 19.7.2. A single original entry gives the identity. A longer word has a contractible first gap. On I the filtration ends at $|I|$, proving equivalence. Every map is a composite of the full coefficient maps, so it retains collisions, spectators and normal derivatives. Naturality gives compatibility with input morphisms. □

Write $\mathcal{N}_*^{\mathrm{bc}}$ for the input category with this bar contact marking, and $\mathcal{N}_\sharp^{\mathrm{cj}}$ for the marked current category just defined. The bar marking realizes the field operations inside $B_\star U_* A$. It is not an identification with bare multiplication on a singleton of the cobar algebra.

For $J = C_k$, the state unit and augmentation give $v_A : J \to C_A$ and $a_A : C_A \to J$, with $a_A v_A = 1$. The bar pointing is $\eta_A v_A : J \to B_\star U_* A$. It is distinct from the empty-support tensor unit K.

Native multiplication and every collision

The current product has a tensor comparison determined by its own pair multipliers. In the ambient current category, the tensor of the cores of A and A' is the core of $A \times A'$. Its localization includes the cross differences between the two groups.

Let $F_{B,C} = \Omega_{B,C}(0)$ and let τ be the Koszul permutation. The deformed crossing and product on this joint tensor are

$$\sigma_w = \tau(F^{21}_{C,B})^{-1} F_{B,C}, \qquad \mu_{B\otimes_w C} = (\mu_B \otimes \mu_C)(1 \otimes \sigma_w \otimes 1).$$

Removing the native cross multiplier has tensor comparison $U_2 = F^{-1}_{B,C}$. For three groups, both tensor comparisons contain the same three commuting cross factors.

This proves their associativity. An empty group gives the unit. The charge-zero-k slice is not closed under this tensor, since $(A \times A')_0 = k \times k$. All tensor assertions here use the ambient category.

Let C°_A denote the contact coalgebra for the regular fields $\mu_A(e^{uT} \otimes 1)$, and write $C^w_A = C_A$ for the deformed one. On an ordered configuration multiply all pair operators $E(u)$ from (19.15). Call this operator E_A.

Translation covariance makes it horizontal, so it extends to closed direct images and all normal derivatives. The product rule at a merger gives, before taking principal parts,

$$\mu_{ab} e^{uT_a} E_{ac}(z_b + u - z_c) E_{bc}(z_b - z_c) = E_{(ab)c}(z_b - z_c)\mu_{ab} e^{uT_a}.$$

The internal factor $E_{ab}(u)$ is precisely the input contraction. This proves compatibility with adjacent mergers and their Laurent multipliers.

For a nonconsecutive merger, exchange supplies the passage through the intervening block. For 13, put $u = z_1 - z_3$, $t = z_2 - z_3$ and $X_{12}(u - t) = E_{12}(u - t)E_{21}(t - u)^{-1}$. The corresponding equality is

$$r^\circ_{13} E_{12}(u - t)E_{13}(u)E_{23}(t) = E_{2,(13)}(t)\,\mathrm{pp}_{u=0}\,\mu_{13} e^{uT_1} E_{13}(u) X_{12}(u - t).$$

It remains valid after every Laurent multiplier. Commuting the pair factors proves the higher collision comparisons. Thus $E_A : C^w_A \to C^\circ_A$ is an isomorphism of full coefficient complexes.

For a cut $G \sqcup H$, let $W_{G,H}$ be the product of the inverse input contractions between the two groups. Cancellation of the internal pair factors gives

$$(E_G \boxtimes E_H)\Delta^w_{G,H} E_A^{-1} = W_{G,H}\Delta^\circ_{G,H}. \tag{19.22}$$

Every ordered-subset cut occurs. For three groups, both composites contain $W_{G,H}W_{G,J}W_{H,J}$. These are identities of the full diagrams, including spectators and normal derivatives.

Recover the state algebra A_B from a current core and put $G(B) = (C^\circ_{A_B})^\sharp$. This constructs the ordinary contact system before coefficient duality is applied. Colored partitions identify the contact system of $A \times A'$ with the two-color coefficient tensor.

Dualizing this comparison and the native map U_2 gives J°. Define its deformed tensor comparison by $J^w = J^\circ(W_{G,H})^\sharp$ on each cut component. The pair factor is forced by expanding the native multiplier $w(v + x_a - y_b)$ in the constituent jets. Its first-order expansion is the negative transpose of $C_1(v)$, so it is the same $W_{G,H}$ as in (19.22).

Consequently actual native multiplication has the comparison

$$G(\mu_w)J^w = (\Delta^\circ_{G,H})^\sharp (W_{G,H})^\sharp. \tag{19.23}$$

Indeed removing the cross multiplier turns native multiplication into ordinary core multiplication. The latter transposes the diagonal $A \to A \times A$, whose colored-partition map is the cut coproduct. Restoring the native multiplier gives the right side of (19.23). Equation (19.22) now shows that $E_A^\sharp$ identifies this realized algebra with $(C_A^w)^\sharp$. Denote the resulting operation-preserving realization by $U_\sharp$.

THEOREM 19.8.3 *(Bar-marked charged-current duality).* At precision R_2 there is an anti-equivalence

$$D : (\mathcal{N}_*^{\mathrm{bc}})^{\mathrm{op}} \xrightarrow{\sim} \mathcal{N}_\sharp^{\mathrm{cj}}, \qquad D(A) = B_A^{\mathrm{mer}}.$$

Its inverse recovers the R_2-algebra

$$R(B) = \mathrm{Hom}_{\mathfrak{g}_{R_2}}(\mathfrak{g}_{R_2}, W_B^\vee[-1]).$$

There is a natural coefficient equivalence

$$\mathcal{K}(U_* A) = (B_\star \Omega_\star C_A)^\sharp \xrightarrow{\eta_A^\sharp} C_A^\sharp \xrightarrow{\sim} U_\sharp D(A). \tag{19.24}$$

It preserves the full operations and both native vacuum identities. It takes the bar pointing to its outgoing transpose to $J^\sharp$. The comparison remains an anti-equivalence after the declared weak equivalences are inverted, including their mapping spaces.

Proof. The adjoint representation of $\mathfrak{sl}_2$ has

$$\mathrm{End}_{\mathfrak{g}}(\mathfrak{g}) = k, \qquad \mathrm{Hom}_{\mathfrak{g}}(\mathfrak{g} \otimes \mathfrak{g}, \mathfrak{g}) = k[\,,\,].$$

Its exterior square is the adjoint representation. Its symmetric square is the sum of the five-dimensional irreducible and the invariant line.

Thus (19.17) forces the recovered bracket to be $[x, y] \otimes m(a, b)$. Graded antisymmetry makes m graded commutative. Jacobi on $e \otimes a, e \otimes b, f \otimes c$ gives $a(bc) = (-1)^{|a||b|} b(ac)$, and hence

$$(ab)c = (-1)^{(|a|+|b|)|c|} c(ab) = (-1)^{|b||c|} a(cb) = a(bc).$$

The scalar current supplies the unit.

Equivariance makes the differential, translation and each marked jet derivation equal to $1 \otimes d_A, 1 \otimes T$ and $1 \otimes \partial_t$. The Lie derivation identities become the product derivation identities. A transposed core morphism is $1 \otimes f$. Its bracket equation is exactly multiplicativity of f. Its other equations give the stated input morphism conditions. This recovers objects and maps.

The inverse comparisons are $a \mapsto (x \mapsto x \otimes a)$ and inverse dual evaluation of $\mathfrak{g} \otimes R(B) \to L_B$. They extend uniquely through the completed core and its localization. They commute with the symmetric-ring actions and pair multipliers, so preserve the actual native fields and products. Their triangles hold on generators and therefore on the complete presentations.

The formal-test calculation identifies the two classes of weak equivalences. The exact R_2-dual is justified by its Frobenius pairing. Thus the inverse functors descend to the localized categories. The universal property of localization retains the inverse comparisons and all mapping spaces.

Finally (19.23) and the bar marking give (19.24). The three-group factor identity proves coherence with the tensor comparisons. The ambient field unit, obtained from the zero state algebra, is R_2 and realizes to K. Its unit map supplies the native vacuum.

The scalar-current object B_k instead realizes to $J^\sharp$. The map to that object is $v_A^\sharp$, and $v_A^\sharp \eta_A^\sharp = (\eta_A v_A)^\sharp$. The augmentation supplies its section. This proves the pointing and both vacuum assertions. □

The theorem uses the bar contact marking, all jet actions and the split quadratic current core. An arbitrary enhancement with the same unmarked coefficient algebra is a larger recognition problem. Even inside the stated categories, coefficient framings and coefficient arrows contain information that an unframed state isomorphism forgets.

19.9 The cubic enhancement space and its arrows

An enhancement of a coefficient algebra P is a native object A together with an equivalence $U(A) \simeq P$. Work in the categories localized at their declared weak equivalences. The space of enhancements is the homotopy fiber, over P, of the map between the maximal subgroupoids of the native and coefficient categories.

Take A_0 from (19.16), and put $C_0 = C_{A_0}$, $P_0 = \Omega_\star C_0$ and $E_0 = C_0^\sharp$. We write coefficient maps in coalgebra coordinates on C_0. Bar–cobar identifies them with maps of P_0. Transposition identifies them, with composition reversed, with maps of E_0. A scalar-pointed map commutes with both maps induced by $k \to A_0 \to k$.

All coefficient endomorphisms

Let $\mathfrak{m} = V \oplus Q$. For $v \in A_0[z]$ and $p \in k[z]$, define

$$J_n(v) = \frac{(\partial_z + T)^n}{n!} v, \qquad S_p(e_n) = J_n(pe_0) = \sum_{j=0}^{n} \frac{p^{(j)}(z)}{j!} e_{n-j}.$$

Extend S_p multiplicatively, fixing $k[z]$. Horizontality of a singleton map $F(z)$ means $[\partial_z + T, F] = 0$. Its field equation is

$$F(y)Y(a,u)b = Y(F(y+u)a,u)F(y)b. \tag{19.25}$$

The shifted position of the first input is essential.

For this full contact system, a coefficient map is determined by its singleton. The residue-lattice description proves this. On n positions the contact complex is a module in degree $-n$. It embeds in the open direct image from the fully distinct locus.

The module is the intersection of the pair residue kernels. Cuts determine a map on that locus. The embedding then determines its extension. Maps between equally shifted modules have no negative Ext groups, so their mapping spaces are discrete. Limits over the collision diagram retain this discreteness.

For completeness, concentration of the contact complex follows by induction on the number of positions. Off the small diagonal it splits into two smaller groups. The remaining cokernel is a principal-part residue on the small diagonal. Insert the vacuum in one colliding input to lift each principal part.

Its other residues vanish by the residue-kernel condition. This proves surjectivity and concentration. The finite complex of flat coefficients remains exact under coefficient extension. This also justifies its use over R_2.

Naturality of every Laurent-multiplied binary boundary gives (19.25). Both cut orders retain unmultiplied exchange. Conversely a horizontal singleton satisfying these field and exchange equations is placed at each block. The field equation intertwines adjacent residues, exchange intertwines nonconsecutive passages, and horizontality retains normal derivatives. It therefore extends to the full coefficient map.

Theorem 19.9.1 *(Coefficient endomorphisms of the cubic system).* Every scalar-pointed coefficient endomorphism is uniquely $F_{p,v}$, with $p \in k[z]$ and $v \in \mathfrak{m}[z]$, where

$$F_{p,v}(1) = 1, \qquad F_{p,v}(e_n) = J_n(pe_0) + \hbar J_n(v), \tag{19.26}$$

extended multiplicatively. Composition is

$$(p, v) \circ (q, w) = (pq, qv + S_p(w)). \tag{19.27}$$

Proof. Reduce F modulo $\hbar$. The constant field coefficient makes F_0 an augmented algebra map. Let $f = F_0(e_0)$. If its largest jet index is $N > 0$, then $\partial_{N+1} f = e_0 \partial f / \partial e_N \neq 0$. In charge at most two, multiplication by e_0 does not kill this nonzero derivative.

The exchange equation on $e_0 \otimes e_0$ is

$$C_1(u)(f(y+u) \otimes f(y)) = -u^{-1} f(y+u) \otimes f(y).$$

The left side has nonzero coefficient

$$[u^{-2N-1}] = (-1)^{N+1} \binom{2N}{N} \partial_{N+1} f(y) \otimes \partial_{N+1} f(y).$$

The two tensor factors are free over $k[y]$. The right side has at most a simple pole, a contradiction. Hence $N = 0$.

Write $f = pe_0 + be_0^2$. The simple-pole equation is $\partial_1 f \otimes \partial_1 f = f \otimes f$. Its bidegree $(2, 2)$ coefficient gives $4b^2 = b^2$, hence $b = 0$. Horizontality then forces $F_0 = S_p$. The displayed formula for S_p shows that it commutes with every jet derivation and satisfies the field equation.

Write $F = S_p + \hbar G$. A first-order correction and a quantum contraction together have order $\hbar^2$, so the remaining equation is classical. It says $G(1) = 0$ and $G(ab) = G(a)S_p(b) + S_p(a)G(b)$.

Horizontality determines $G(e_n) = J_n(v)$ from $v = G(e_0) \in \mathfrak{m}[z]$. Such a derivation preserves I^3, since each generator image has positive charge. The singleton criterion proves existence and uniqueness of the full map. Evaluating a composite on e_0 gives (19.27). □

For example $S_z(e_n) = ze_n + e_{n-1}$, where $e_{-1} = 0$. The term e_0 in $S_z(e_1)$ is forced by horizontality. This map commutes with all jets but is not a constant native state map.

If only the empty unit and counit are fixed, there is also a map Z vanishing on every nonempty coefficient. Every other map is locally unital and has (19.26), with $v \in A_0[z]$ permitted when $p = 0$. Indeed the field equation makes $F(1)$ an idempotent, hence zero or one. A zero value forces the map to vanish on nonempty support.

For value one and $p \neq 0$, a scalar part v_{sc} would give $3\hbar p^2 v_{sc} e_0^2$ in the cubic relation, so it must vanish. For $p = 0$, products of two positive generator images already vanish, and impose no such condition. Composition still has the displayed formula, and Z is absorbing. In particular $F_{0,v} F_{0,w} = F_{0,w_{sc}}$. The scalar-pointed local projection $F_{0,0}$ preserves the local vacuum and differs from Z.

The entire enhancement fiber

An invertible coefficient map has $p = a \in k^\times$. In the coordinates $F_{a,av}$ its group is

$$\Gamma = \mathfrak{m}[z] \rtimes k^\times, \qquad (v, a)(w, b) = (v + \vartheta_a w, ab), \quad \vartheta_a(w_V + w_Q) = w_V + a w_Q. \tag{19.28}$$

The native endomorphisms are exactly $s_\lambda(e_n) = \lambda e_n$, $\lambda \in R_2$. Charge and $\ker(\delta_2|_V) = ke_0$ determine the seed image, and translation and multiplication determine the map. The native automorphism image is therefore $H = \{(be_0, a) : b \in k,\ a \in k^\times\}$.

Theorem 19.9.2 *(Cubic enhancement fibers).* Within the two stated native categories, every enhancement of P_0, respectively E_0, is equivalent to A_0, respectively $D(A_0)$, with a coefficient framing. Its enhancement space is

$$\mathrm{Enh}_*(P_0) \simeq \mathrm{Enh}_\sharp(E_0) \simeq (\Gamma/H)_{\mathrm{disc}} \simeq (\mathfrak{m}[z]/ke_0)_{\mathrm{disc}}. \tag{19.29}$$

The quotient Γ/H is a right-coset set.

Proof. Bar–cobar turns an enhancement into $C_A \simeq C_0$. Singleton cohomology and faithful scalar extension by $k[z]$ and R_2 force $H^i(A) = 0$ for $i \neq 0$. The map $A \to H^0(A)$ is a charged jet quasi-isomorphism: its boundary ideal is stable under all markings. We may therefore work with a degree-zero algebra.

The residue-lattice criterion gives an invertible horizontal polynomial singleton. Modulo $\hbar$, its constant field coefficient is an algebra isomorphism over $k[z]$. Evaluation at $z = 0$ gives $A \cong A_0$ as ordinary algebras.

If $d = \dim E_A$, its map on augmentation indecomposables identifies $\mathcal{D}_X^d$ with $\mathcal{D}_X$. These are the free Weyl modules generated by the seeds. For the Bernstein filtration, $\dim_k F_m\mathcal{D}_X = (m+1)(m+2)/2$. An isomorphism and its inverse have bounded filtration displacement, so comparison of the leading quadratic growth gives $d = 1$.

The ordinary algebra isomorphism transports $\mathfrak{m}^3 = 0$ and the isomorphism $\mathrm{Sym}^2(\mathfrak{m}/\mathfrak{m}^2) \to \mathfrak{m}^2$. Generation in charge one therefore leaves no quadratic relation and kills every cubic product. The one-dimensional seed gives precisely the ideal I^3. Thus A is a charged jet copy of A_0.

Maps between degree-zero objects in this localized nonpositive dg category are their ordinary discrete sets. Indeed H^0 is left adjoint to the inclusion of degree-zero objects. It preserves weak equivalences and has identity counit there. Coefficient maps are discrete by the lattice criterion.

The enhancement fiber is consequently the right action groupoid $\Gamma//H$. This action is free, so the groupoid is equivalent to the discrete coset set. Right multiplication by (be_0, c) sends (v, a) to $(v + be_0, ac)$, giving the last identification in (19.29). The native duality proves the same assertion on the opposite side. □

The set in (19.29) is not a quotient group. For $a \neq 1$,

$$(e_0^2, 1)(0, a)(e_0^2, 1)^{-1} = ((1-a)e_0^2, a) \notin H.$$

The transitive action groupoid of Γ on its coset set is BH. It differs from the enhancement fiber, whose action by H on Γ is free.

Relative arrows and a higher-precision defect

For a realization $U : \mathcal{N} \to \mathcal{A}$ and a coefficient chain $a_0 \to \cdots \to a_n$, take the homotopy fiber of

$$\mathrm{Fun}([n], \mathcal{N})^{\simeq} \longrightarrow \mathrm{Fun}([n], \mathcal{A})^{\simeq}$$

over that chain. The superscript means the maximal subgroupoid. This *relative chain space* retains the enhancements at all vertices and their compatible native arrows. The case $n = 0$ is the enhancement space above.

PROPOSITION 19.9.3 *(Lifting coefficient arrows).* For fixed frames g_s, $g_t \in \Gamma$, a coefficient map F has a native lift exactly when

$$F = g_t s_\lambda g_s^{-1} \qquad \text{for some } \lambda \in R_2.$$

The lifting space is empty or a point. Allowing both frames to vary, $F_{p,v}$ lifts somewhere precisely when $p \in k^\times$, or $p = 0$ and $v \in k e_0$. The map Z never lifts.

Proof. Theorem 19.9.2 supplies all objects, and the maps s_λ are all their native endomorphisms. Distinct λ have distinct singleton values, which proves the fixed-frame assertion. If λ has nonzero reduction it is a unit, and changing one frame produces every coefficient automorphism.

For $\lambda = \hbar b$, let a_s, a_t be the classical scalings of the two frames. The composition law gives

$$g_t s_{\hbar b} g_s^{-1} = s_{\hbar b a_t/a_s}.$$

Every first-order correction in a frame acquires a second factor of $\hbar$ and vanishes. Only the constant e_0 direction can therefore lift in this case. Nonconstant p and the map Z meet neither possibility. □

For a chain (F_i), all lifts are consequently described by $F_i = g_i s_{\lambda_i} g_{i-1}^{-1}$, modulo changes at each vertex

$$g_i \mapsto g_i h_i^{-1}, \qquad s_{\lambda_i} \mapsto h_i s_{\lambda_i} h_{i-1}^{-1}, \qquad h_i \in H.$$

The action on the frames is free. This gives a discrete relative chain space.

Faces compose the adjacent scalars and degeneracies insert 1. The anti-equivalences and (19.24) identify it with the relative chain space over $\mathcal{K}(a_n) \to \cdots \to \mathcal{K}(a_0)$. Indeed a frame g becomes $(g^{-1})^\sharp$, and

$$(g_s^{-1})^\sharp s_\lambda^\sharp g_t^\sharp = (g_t s_\lambda g_s^{-1})^\sharp.$$

This identity retains empty lifting spaces as well as the nonempty ones.

For instance S_z has no lift under any reframing. Neither does F_{0,e_1}, whose singleton satisfies

$$F_{0,e_1}(e_n) = \hbar(n+1)e_{n+1}, \qquad [\partial_2, F_{0,e_1}](e_0) = \hbar e_0.$$

The latter preserves the coefficient field diagram but not the marked native jet action. Its relative space is empty on both sides of the duality.

The fixed first-order precision is also substantive. Define a horizontal derivation on A_0 by

$$D(e_n) = \sum_{i+j=n} e_i e_j.$$

The formula satisfies $[D, T] = 0$, as follows by differentiating the two factors and collecting their coefficients. The map $1 + \hbar D$ is a coefficient automorphism over R_2.

PROPOSITION 19.9.4 *(A frame obstructed at the next precision).* Keep the exchange kernel $w(u) = 1 - \hbar/u$. The frame $1 + \hbar D$ has no extension over $R_3 = k[\hbar]/(\hbar^3)$ preserving exchange, even with an arbitrary position-dependent second-order correction.

Proof. Write a proposed extension as $1 + \hbar D + \hbar^2 K$. Exchange has expansion $1 + 2\hbar C_1 + \hbar^2 \mathcal{S}_2$. Its order-two intertwining defect is forced to be $2[C_1, D \otimes 1 + 1 \otimes D]$. Both the new correction K and $\mathcal{S}_2$ commute with the identity at this order.

On $e_0 \otimes e_0$ the defect is

$$-\frac{2}{u}(e_0^2 \otimes e_0 + e_0 \otimes e_0^2) \neq 0.$$

The terms have separate bidegrees $(2,1)$ and $(1,2)$. Multiplying the two tensor factors would kill both by $I^3 = 0$, but exchange acts before that multiplication. No second-order correction can remove the displayed defect. □

This is an obstruction to extending the specified frame with the fixed exchange kernel. The native duality and the exhaustive fiber calculation remain assertions at R_2. A compatible tower requires its own objects, surviving frames and comparison maps at every precision. More general native enhancements also require a separate recognition theorem or an obstruction on their declared domain.

19.10 Fields can forget a jet marking

The constituent identities in the charged-current presentation contain more information than the commutation relations of its jet derivations. We now vary those identities while retaining the entire rational field. This gives a larger marked category in which the standard recognition statement fails, together with a duality on an explicit family inside that category.

Continue over $R_2 = \mathbb{C}[\hbar]/(\hbar^2)$. Use the cubic algebra $A = R_2 \oplus V_{R_2} \oplus \operatorname{Sym}^2 V_{R_2}$ of (19.16), with $V = \bigoplus_{n\geq 0} \mathbb{C}e_n$. The differential is zero, e_n has charge one and weight $n+1$, and $\hbar$ has weight zero.

Keep T and the standard δ_i of (19.14), and set $N = \delta_1$. On charge one,

$$\delta_{m+1} = \delta_2^m.$$

This identity concerns compositions on V. It does not identify δ_{m+1} with a power of a derivation on the whole algebra.

Definition 19.10.1 *(Filtered jet marking).* A *filtered jet marking* on a nonpositive commutative dg state algebra consists of degree-zero chain derivations T, J_i satisfying

$$J_1 = N, \qquad [J_i, J_j] = 0, \qquad [J_i, T] = (i-1)J_{i-1}, \qquad J_0 = 0.$$

They annihilate the unit and preserve charge. Each J_i is bounded for the weight filtration.

Only finitely many act nontrivially on each state. The rational fields and exchange are (19.15) with δ_i replaced by J_i. Morphisms are constant R_2-linear dg algebra maps preserving charge, translation, every J_i, the fields and exchange. Weak equivalences are the state quasi-isomorphisms among these maps.

In particular $[d, J_i] = 0$ is required. Exact operator weight $1 - i$ and the constituent identities on charge one are additional conditions. The following family satisfies neither additional condition when its parameter is nonzero.

Proposition 19.10.2 *(Equal fields with distinct markings).* For $\kappa \in \mathbb{C}$, put

$$J_1^\kappa = N, \qquad J_2^\kappa = \delta_2, \qquad J_i^\kappa = \delta_i + \hbar\kappa \binom{i-1}{2} \delta_{i-2} \quad (i \geq 3).$$

These are filtered jet markings on A. Write A^κ for the resulting marked algebra. Their rational fields, exchange and full collision coefficients are identical. Distinct parameters remain inequivalent after inverting marked state quasi-isomorphisms. For $\kappa \neq 0$ they are not equivalent there to a standard charged-jet quotient.

Proof. The operators are linear combinations of commuting derivations. Translation covariance follows from

$$\binom{i-1}{2}(i-3) = (i-1)\binom{i-2}{2}$$

for $i \geq 4$. The added term at $i = 3$ is $\hbar\kappa N$, which commutes with T. On e_n both terms vanish for $i > n+3$. The derivation rule proves finite support on every state. Their two weight shifts are $1-i$ and $3-i$.

The contraction in (19.15) has an external factor $\hbar$. Each change of a marked operator has another factor $\hbar$. Thus $\hbar C_1^{J^\kappa} = \hbar C_1^{\delta}$ over the common base R_2. The fields agree before principal parts are taken. Every iterate, normal derivative, spectator and collision comparison therefore agrees.

On the charge-one complex define

$$\Xi_2 = J_3 - J_2^2.$$

It acts on cohomology because the markings are chain maps. In this family

$$\Xi_2^\kappa = \hbar\kappa\, 1_{V_{R_2}}.$$

An invertible intertwiner between parameters κ, λ would imply $\hbar(\kappa-\lambda)1_{V_{R_2}} = 0$. Freeness forces $\kappa = \lambda$. The same reasoning applies to every zigzag, since its cohomology maps are invertible intertwiners. A standard charged-jet quotient has $\Xi_2 = 0$, which gives the last assertion. □

The smallest calculation is $J_2 e_0 = 0$ and $J_3^1 e_0 = \hbar e_0$. The interaction is also nonzero:

$$[\hbar u^{-1}]Y(e_0, u)e_0 = e_0^2.$$

The quotient by the square of the augmentation ideal would retain a marking defect but kill this product. Nor can the infinite jet orbit be replaced by a nonzero finite-dimensional charge-one space in characteristic zero: $[J_2, T] = 1$ would contradict the trace of a commutator.

The chain condition detects the distinction

If the marking need not commute with the differential, a contractible extension can erase the preceding distinction. Let $\widetilde{A}$ be the graded commutative algebra on species e_n, v_n, u_n, truncated at total charge three. Give e_n, v_n degree zero and u_n degree -1, with $du_n = v_n$ and $de_n = dv_n = 0$. All species have weight $n+1$. Use standard shifts and translation on each.

On their linear span set $Me_n = 0$, $Mv_n = e_n + v_n$, $Mu_n = u_n$. Set $\widetilde{J_1} = N$ and $\widetilde{J_2} = \partial_2$. For $i \geq 3$ define on generators

$$\widetilde{J_i}x_n = x_{n-i+1} + \hbar\binom{i-1}{2}Mx_{n-i+3},$$

then extend as derivations. The shifts commute with M, so these derivations commute and obey the same translation identity. Their fields are still the standard dg fields: a correction to a marking acquires a second $\hbar$ in the contraction.

There are marked dg algebra maps

$$A^0 \xrightarrow{i_0} \widetilde{A} \xleftarrow{i_1} A^1, \qquad i_0(e_n) = e_n, \quad i_1(e_n) = e_n + v_n.$$

To verify their quasi-isomorphism property, let p kill u_n, v_n and fix e_n. Let K be the degree--1 derivation $K(v_n) = u_n$, $K(e_n) = K(u_n) = 0$. Then $dK + Kd$ counts the factors from the two contractible species.

On a monomial containing $r > 0$ such factors put $H = K/r$, and put $H = 0$ on pure e-monomials. Here r is one or two. Consequently

$$dH + Hd = 1 - i_0 p.$$

This proves that i_0, p are quasi-isomorphisms. Since $pi_1 = 1$, so is i_1.

The marking nevertheless fails the chain condition:

$$[d, \widetilde{J_3}]u_0 = \hbar v_0 - \hbar(e_0 + v_0) = -\hbar e_0.$$

The cohomologous representatives e_0 and $e_0 + v_0$ have images differing by a nonzero cohomology class. Thus $\widetilde{J_3}$ does not induce a cohomology operator. This span is excluded precisely by the chain condition in Definition 19.10.1.

All markings with fixed first jets

Theorem 19.10.3 *(Polynomial classification on the cubic carrier).* Fix A, T, N and $J_2 = \delta_2$. Every filtered jet marking reducing to δ_i modulo $\hbar$ has the unique form

$$\begin{aligned} J^c_{m+1} &= \delta_{m+1} + \hbar \sum_{r=2}^{m} \binom{m}{r} c_r \delta_{m-r+1}, \\ c(t) &= \sum_{r \geq 2} c_r \frac{t^r}{r!} \in t^2\mathbb{C}[t]. \end{aligned} \tag{19.30}$$

All these markings have the same fields. Distinct polynomials give distinct marked equivalence classes, including under chain-compatible nonpositive dg zigzags. Exact operator weight $-m$ for J_{m+1} forces $c = 0$.

Proof. Write $D = \delta_2$. Commutation with D puts $J_{m+1}e_0$ in $\ker(D|_V) = R_2 e_0$, say $a_m e_0$. The first two markings give $a_0 = 1$, $a_1 = 0$. Reduction gives $a_m = \hbar c_m$ for $m \geq 2$. Finite support on e_0 makes (c_m) finitely supported.

Translation covariance is the recursion

$$(n+1)J_{m+1}e_{n+1} = TJ_{m+1}e_n + mJ_m e_n.$$

Starting at e_0, induction gives

$$J_{m+1}e_n = \sum_{q=0}^{\min(m,n)} \binom{m}{q} a_{m-q} e_{n-q}.$$

The induction uses $(n-q+1)\binom{m}{q} + m\binom{m-1}{q-1} = (n+1)\binom{m}{q}$. Substitute the a_m to obtain (19.30). The derivation property determines the operator on all products.

Conversely the displayed operators commute. The identity $\binom{m}{r}(m-r) = m\binom{m-1}{r}$ proves translation covariance. If $c_r = 0$ for $r > R$, then $J_{m+1}e_n = 0$ for $m > n + R$. The same external-$\hbar$ calculation as before proves equality of fields.

A marked equivalence induces an invertible cohomology map commuting with D, T. It acts on the D-kernel by a unit $a \in R_2^\times$. Intertwining $J_{m+1}e_0$ gives $\hbar(c_m - c'_m)a = 0$, hence $c_m = c'_m$. Finally an operator of weight $-m$ sends e_0 to weight $1 - m$. There is no such charge-one state for $m \geq 2$, forcing every c_m to vanish. □

This classifies markings on the fixed cubic carrier, with the specified first jets and reduction. It does not classify all algebras realizing its coefficient diagram. Write A^c for the algebra with polynomial marking c.

Compensated moments and the current partner

The current partner retains the marked operators without treating them as constituent power sums. Use the polynomial Chevalley core of $\mathfrak{sl}_2 \otimes A$ from Section 19.8. Its negative transposes are

$$T_B = -(1 \otimes T)^\vee, \qquad D_i^c = -(1 \otimes J_i^c)^\vee.$$

On total charge n, let $p_m = \sum_{a=1}^n x_a^m$, with $p_0 = n$. Constituent polarization gives

$$P_m^c = -D_{m+1}^c = p_m + \hbar \sum_{r=2}^{m} \binom{m}{r} c_r\, p_{m-r},$$

$$\widehat{P}_m = P_m^c - \hbar \sum_{r=2}^{m} \binom{m}{r} c_r\, P_{m-r}^c = p_m.$$

The second equality uses $\hbar^2 = 0$. It supplies the ordinary action of $\Lambda_n = \mathbb{C}[x_1, \dots, x_n]^{S_n}$.

Define B^c by localizing the charge-n core at $d_n = \prod_{a<b}(x_a - x_b)^2$ using this compensated action, with $d_0 = d_1 = 1$. Use the same cross differences and their declared translates for multiple inputs. Define the product and fields by the pair multiplier

$$\Omega_{p,q}(v) = \prod_{a=1}^{p} \prod_{b=1}^{q} \left(1 - \frac{\hbar}{v + x_a - y_b}\right), \qquad \mu_w = \mu_0 \Omega_{p,q}(0).$$

They are independent of c. Each D_i^c is a finite linear combination of standard current derivations, so it extends through these localizations and remains a derivation of μ_w. The three-input multiplier is again $\Omega_{p,q}\Omega_{p,r}\Omega_{q,r}$. Thus the differential, associativity, exchange and vacuum identities hold on the same recipients.

A *compensated current presentation* retains the split core, scalar current, charge, translation, D_i^c, compensated spectral action and rational recipients. Its maps preserve these structures and are linear on core generators. It has the formal-core weak equivalences defined in Section 19.8. Standardization replaces D_{m+1}^c by $\widehat{D}_{m+1} = -\widehat{P}_m$. It leaves the underlying map, core, differential, fields and recipients fixed. Every specified morphism preserves $\widehat{P}_m$, so standardization is a functor to the standard current category.

Apply $U_\sharp$ to this standardized presentation, using the algebra structure induced by its native product and tensor comparison. This defines $U_\sharp$ on the displayed compensated category. Standardization is not claimed to be an equivalence of marked fields or to make this category closed under tensor products.

Let C_0 be their common full coefficient coalgebra and $P_0 = \Omega_\star C_0$. The bar marking and the native product comparison give a natural equivalence

$$(B_\star P_0)^\sharp \xrightarrow{\gamma^\sharp} C_0^\sharp \xrightarrow{\theta_c} U_\sharp(B^c). \tag{19.31}$$

Here θ_c is the inverse of the standard current comparison after standardization. It preserves all collisions, cuts, normal derivatives, spectators and local pointings. The bar pointing remains the composite $J \to C_0 \to B_\star P_0$, separate from the empty-support unit.

Raw moments would give the wrong localization. In the one-parameter family and charge one,

$$P_2^\kappa - (P_1^\kappa)^2 = \hbar\kappa\, 1, \qquad D_3^\kappa + (D_2^\kappa)^2 = -\hbar\kappa\, 1.$$

Since $p_2 = p_1^2$ in $\Lambda_1 = \mathbb{C}[x]$, the raw assignment $p_m \mapsto P_m^\kappa$ does not define an action when $\kappa \neq 0$. Compensation repairs this equation while retaining the marked operator invariant.

Proposition 19.10.4 *(All arrows in the displayed family).* The native maps $A^c \to A^d$ are exactly

$$s_\alpha(1) = 1, \quad s_\alpha(e_n) = \alpha e_n, \quad s_\alpha(e_i e_j) = \alpha^2 e_i e_j,$$
$$\hbar\alpha(c_r - d_r) = 0 \quad (r \geq 2),$$

with $\alpha \in R_2$. Current pullback gives an anti-equivalence between the displayed state and compensated current categories. It commutes with (19.31).

Proof. Charge and D preservation put the image of e_0 in $R_2 e_0$. Translation determines every e_n, and multiplication determines the quadratic states. Intertwining J_{r+1} on e_0 gives the condition on α. Formula (19.30) makes it sufficient on all generators, and product defects acquire further powers of α.

For the reverse construction, scalar-current equivariance makes a transposed linear core map $1 \otimes f$. The quadratic Chevalley equation says $f(ab) = f(a)f(b)$. The scalar-current condition gives the unit. Translation and the marked derivations recover the remaining input equations. This recognition calculation does not use the higher constituent identities.

Pullback by s_α scales charge n by α^n. It commutes with the compensated spectral action and extends through the same localizations. If $c \neq d$, the condition forces $\alpha \in \hbar\mathbb{C}$.

Charge-one images then have a factor $\hbar$, and all images of charge at least two vanish. The residual marked differences therefore vanish on the image, without inverting a zero divisor. The native unit and counit are inverse evaluation on the adjoint-isotypic core, as in Theorem 19.8.3. Their triangles and the standardized comparison prove the assertions. □

For equal parameters every α occurs. For unequal parameters exactly $\alpha \in \hbar\mathbb{C}$ occurs. Two arrows between unequal parameters compose to s_0, the scalar projection followed by its section. This map preserves the native unit. Isomorphisms require equal parameters and $\alpha \in R_2^\times$.

The coefficient frame group Γ and subgroup H of (19.28) are unchanged. Thus the framed subspace supported on these displayed classes is

$$\coprod_{c(t) \in t^2\mathbb{C}[t]} (\Gamma/H)_{\mathrm{disc}}.$$

For frames g_s, g_t, a coefficient arrow F has a native lift between parameters c, d exactly when

$$F = g_t s_\alpha g_s^{-1}, \qquad \hbar\alpha(c_r - d_r) = 0 \quad (r \geq 2).$$

Its relative space is then a point and is otherwise empty. The proof of Proposition 19.9.3 applies in each component: chain compatibility makes cohomology preserve all markings, and the frame action is free. Duality sends g to $(g^{-1})^\sharp$ and reverses arrows. Faces compose the scalars and degeneracies insert their identities.

These cosets are sets over the fixed ring R_2, not a quotient group or an exhaustion of arbitrary unmarked native fields. Recovering a current core and its constituent moments from intrinsic field axioms remains a different recognition problem.

19.11 Curvature is a separate comparison

Coefficient transposition and a contragredient module have specified outputs. Neither identifies two curved field algebras. That comparison must include changes of state. A translation class of curvature can remain invariant under all such changes.

Let K be a Noetherian domain containing a characteristic-zero field. A curved vertex superalgebra here has a degree-one derivation d and a degree-two state w with

$$d^2 = w_{(0)}, \qquad dw \in TB.$$

Complete its PBW state space coefficientwise in an independent degree-zero parameter: $\widehat{B} = B_K[[\varepsilon]]$.

Each coefficient is a finite sum of states. Field identities hold at every finite ε-precision. This completion contains $e^{\varepsilon T}a$. It does not contain an arbitrary derivative series in the coefficient of ε^0.

For $\xi \in \widehat{B}^1$, the changed differential and curvature are

$$d^\xi = d + \xi_{(0)}, \qquad w^\xi = w + d\xi + \tfrac{1}{2}\xi_{(0)}\xi.$$

A comparison is a continuous unital graded field isomorphism preserving T and all modes. It intertwines the changed differentials and carries one changed curvature to the other modulo actual translation. There is no weight-preservation requirement.

These changes are well defined at the chosen completion. Operations are finite Cauchy sums at each precision, and

$$T\widehat{B} = (TB_K)[[\varepsilon]], \qquad \widehat{B}/T\widehat{B} = (B_K/TB_K)[[\varepsilon]].$$

Indeed one may choose a translation primitive separately in each coefficient. The zero-mode identities give $(d^\xi)^2 = (w^\xi)_{(0)}$. In the Lie superalgebra $\widehat{B}/T\widehat{B}$, odd Jacobi then gives $d^\xi w^\xi = 0$. Thus its lift is an actual translation.

A detector on a commutative quotient

Consider

$$C_K = \Lambda_K(\theta, T\theta, T^2\theta, \ldots), \qquad |\theta| = 1, \qquad \partial\theta = \theta T\theta.$$

Its fields are $Y(a,u)b = (e^{uT}a)b$. All nonnegative modes vanish. The odd Leibniz rule gives $\partial^2\theta = 0$, so $\partial^2 = 0$. Represent a degree-two monomial $(T^i\theta)(T^j\theta)$, $i < j$, by the antisymmetric polynomial $X^iY^j - X^jY^i$. Translation becomes multiplication by $X + Y$.

Evaluation at $Y = -X$ has kernel exactly $(X + Y)$ times the antisymmetric polynomials. Division by the monic polynomial $X + Y$ preserves antisymmetry of the quotient. Consequently

$$\ell_3([p]) = -\tfrac{1}{2}[X^3]\,p(X,-X)$$

is well defined on C_K^2/TC_K^2. Its values include

$$\ell_3([\theta T^3\theta]) = 1, \qquad \ell_3([T\theta T^2\theta]) = -1, \qquad \ell_3([\theta T\theta]) = 0.$$

The first two are compatible with $T(\theta T^2\theta) = T\theta T^2\theta + \theta T^3\theta$. Extend the detector coefficientwise to the ε-completion.

LEMMA 19.11.1 *(Automorphisms of the quotient).* Every continuous graded $K[[\varepsilon]]$-linear differential field automorphism of $C_K[[\varepsilon]]$ is $\theta \mapsto e^{sT}\theta$ for $s \in \varepsilon K[[\varepsilon]]$. It induces the identity modulo actual translation.

Proof. The image of θ is $P(T)\theta$, where $P \in K[T][[\varepsilon]]$ is a unit. Commutation with ∂ gives

$$P(X)P(Y)(Y - X) = P(X + Y)(Y - X).$$

Cancel $Y - X$ in the domain. Invertibility gives $P(0) = 1$. Differentiation at $Y = 0$ yields $P'(X) = sP(X)$, so $P(X) = e^{sX}$.

Its coefficient at ε^0 must be polynomial in X. Characteristic zero and the domain condition force s to have zero constant coefficient. Conversely such an exponential and its inverse are defined in the specified completion. For any completed state a,

$$(e^{sT} - 1)a = T\left(\sum_{n\geq 1} \frac{s^n}{n!} T^{n-1} a\right).$$

The displayed primitive converges ε-adically. □

Theorem 19.11.2 *(An invariant of changed curvature).* Suppose a family of curved field algebras has closed graded field ideals I preserved by every graded field isomorphism between its members. Assume $dI \subset I$ and identify their completed quotients with $(C_K[[\varepsilon]], \delta)$. If

$$\pi(w) = \frac{c}{24}\theta T^3 \theta,$$

then $24\ell_3([\pi(w^{\xi})]) = c$ for every change of state. Every comparison above preserves c.

Proof. Write $\pi(\xi) = a(T, \varepsilon)\theta$. Zero modes vanish in the quotient, so the changed differential remains δ there. The curvature class is

$$[\pi(w^{\xi})] = \frac{c}{24}[\theta T^3 \theta] + a(0, \varepsilon)[\theta T \theta].$$

Every positive power of T in $\delta\pi(\xi)$ is an actual translation. The detector kills the second summand. A comparison descends to a differential automorphism of the quotient. The lemma makes its action on translation classes the identity, proving invariance. □

This obstruction is conditional on the intrinsic quotient and the specified completion. A reflection $c \mapsto c'$ can be implemented by such a comparison only when $c = c'$. Enlarging the algebra or changing the recipient may change the problem. An uncurving of a larger algebra therefore requires an additional comparison with the original one.

19.12 The arithmetic evaluation

An arithmetic application needs a map from its categorical pairing to an analytic pairing. We first specify the analytic reflection and its normalization. The remaining question can then be stated as a comparison of actual maps.

Half densities and Mellin reflection

On $\mathbb{R}_{>0}$ use Haar measure dt/t. Define

$$\mathcal{H} = L^2(\mathbb{R}_{>0}, t^{1/2}\, dt/t), \qquad (\mathcal{J}f)(t) = t^{-1/2} f(t^{-1}).$$

The map

$$W : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dx), \qquad (Wf)(x) = e^{x/4} f(e^x)$$

is unitary by the substitution $t = e^x$.

The factor $e^{x/4}$ is the square root of the density relating the two measures. Direct calculation gives $W\mathcal{J}W^{-1}h(x) = h(-x)$. Thus

$$\mathcal{J}^2 = 1, \qquad \mathcal{J}^\dagger = \mathcal{J}.$$

The adjoint here belongs to the displayed Hilbert pairing.

For $f \in C_c^\infty(\mathbb{R}_{>0})$ define

$$(\mathcal{M}f)(s) = \int_0^\infty f(t)t^{s/2}\frac{dt}{t}.$$

This is entire. Substitution $t = u^{-1}$ proves

$$\mathcal{M}(\mathcal{J}f)(s) = (\mathcal{M}f)(1-s). \tag{19.32}$$

Complex conjugation satisfies $\mathcal{M}(\overline{f})(s) = \overline{\mathcal{M}f(\overline{s})}$. More explicitly,

$$\mathcal{M}f(1/2 + i\tau) = \int_{\mathbb{R}} (Wf)(x)e^{i\tau x/2}\,dx.$$

These formulas fix both the reflection and its adjoint. Using unweighted inversion on the same Mellin convention would give $s \mapsto -s$ instead.

The Gaussian and both boundary poles

Fix the Fourier convention $\widehat{f}(y) = \int_{\mathbb{R}} f(x)e^{-2\pi ixy}\,dx$. For $f_t(x) = e^{-\pi tx^2}$,

$$\widehat{f_t}(y) = t^{-1/2}e^{-\pi y^2/t}.$$

For completeness, integration by parts at $t = 1$ gives $\widehat{f_1}'(y) = -2\pi y\widehat{f_1}(y)$. The Gaussian integral is one, as follows by squaring it and using polar coordinates. The differential equation gives the formula at $t = 1$, and scaling proves it for every $t > 0$.

Periodize f_t over $\mathbb{Z}$. The series and all derivatives converge uniformly on compact sets. Its Fourier coefficient at m is $\widehat{f_t}(m)$. Both Fourier series converge absolutely, so evaluation at zero gives the theta identity

$$\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t} = t^{-1/2}\theta(t^{-1}).$$

This is a lattice theta series. The oscillator character $\prod_{n\geq 1}(1-q^n)^{-1}$ is a different series: its coefficient at q is one, whereas that of $\sum_{n\in\mathbb{Z}} q^{n^2}$ is two.

PROPOSITION 19.12.1 *(Completed Mellin evaluation).* For $\operatorname{Re} s > 1$,

$$\Lambda_\zeta(s) := \pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2}\int_0^\infty (\theta(t)-1)t^{s/2}\frac{dt}{t}.$$

Its meromorphic continuation is

$$\begin{aligned}\Lambda_\zeta(s) = &-\frac{1}{s} - \frac{1}{1-s}\\ &+ \frac{1}{2}\int_1^\infty (\theta(t)-1)\left(t^{s/2} + t^{(1-s)/2}\right)\frac{dt}{t}.\end{aligned}$$

It has residues -1 and 1 at zero and one, respectively, and $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$.

Proof. Termwise integration is absolutely convergent for $\operatorname{Re} s > 1$. Substitution $u = \pi n^2 t$ in each positive-integer term gives $\pi^{-s/2}\Gamma(s/2)n^{-s}$. The two signs of n cancel the factor $1/2$.

Split the integral at one. On $(0,1)$ use

$$\theta(t) - 1 = t^{-1/2}(\theta(t^{-1}) - 1) + t^{-1/2} - 1.$$

Inverting t in the first summand gives the integral with exponent $(1-s)/2$ over $(1,\infty)$. The last two terms integrate to $1/(s-1) - 1/s$. The remaining integral is entire: the theta tail decays exponentially, uniformly against every power and every power of $\log t$ on compact s-sets. The formula proves the residues and reflection. □

The theta tail is outside $\mathcal{H}$: near zero it has leading term $t^{-1/2}$. Its Mellin evaluation therefore uses the displayed subtraction and continuation. It cannot be obtained by applying an everywhere-defined Hilbert pairing to that state.

On boundary monomials, $\mathcal{J}1 = t^{-1/2}$ and $\mathcal{J}(t^{-1/2}) = 1$. Their integrals over $(0,1)$ produce the two polar terms. An arithmetic comparison must retain these boundary contributions as well as its native vacuum. No identification of that vacuum with either boundary monomial has been assumed.

The entire normalization $\xi(s) = s(s-1)\Lambda_\zeta(s)/2$ removes the poles and is still reflection invariant. Multiplication by this polynomial is not a construction of the missing arithmetic evaluation. Likewise Euler inversion to $1/\zeta(s)$ changes the scalar function without constructing the involution in (19.32).

Detecting a central defect

Let V be a complex vector space, $J : V \to V$ an involution, and $E : V \to \mathscr{M}(\mathbb{C})$ a linear map to meromorphic functions. Fix $v, w \in V$ and scalars $\kappa, \kappa^!$. The following criterion keeps the categorical construction separate from its consequence.

Proposition 19.12.2 *(Detection of a specified central defect).* Suppose

$$v - Jv = (\kappa + \kappa^!)w, \qquad E(v)(s) = \Lambda_\zeta(s), \qquad E(Jv)(s) = \Lambda_\zeta(1-s),$$

and $E(w)$ is not identically zero. Then $\kappa + \kappa^! = 0$ if and only if $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$. The same equivalence holds for the entire normalization ξ.

Proof. Apply E to the defect equation:

$$\Lambda_\zeta(s) - \Lambda_\zeta(1-s) = (\kappa + \kappa^!)E(w)(s).$$

A scalar multiple of a nonzero meromorphic function vanishes exactly when the scalar does. The polynomial $s(s-1)/2$ is nonzero and reflection invariant, which proves the assertion for ξ. □

For the forward implication, equivariance $E(Jx)(s) = E(x)(1-s)$ and the defect equation suffice. For the converse, detection of the stated defect is necessary. For example, on $\mathbb{C}^2$ let $J(x,y) = (y,x)$ and $E(x,y) = x + y$. The evaluations of $v = (1,0)$ and Jv agree, but $v - Jv = (1,-1) \neq 0$ lies in $\ker E$. Full injectivity of E is unnecessary. Nonvanishing on the specified defect line is the relevant hypothesis.

The scalar reflection has been proved independently above. Consequently this proposition tests a proposed arithmetic construction: if its defect and evaluation identities hold, its level sum must vanish. An equivalence of categories without those identities gives no value for that sum.

Local assembly and the remaining comparison

Suppose presentable stable R-linear categories $C_v, \mathcal{D}_v$ have chosen objects c_v, d_v at each place. For a finite set F take their tensor products over R. Use insertion of the chosen objects as transition functors. Compatible local equivalences $\Psi_v : C_v \simeq \mathcal{D}_v$ with coherent identifications $\Psi_v(c_v) \simeq d_v$ give an equivalence

$$\varinjlim_{F} \bigotimes_{v \in F, R} C_v \simeq \varinjlim_{F} \bigotimes_{v \in F, R} \mathcal{D}_v.$$

Indeed they give an equivalence at every finite tensor stage and commute with its transition maps. Colimits preserve this equivalence of diagrams.

For arithmetic categories one must construct the common coefficient action, local functors and compatible pointings. A unital map $\mathbb{F}_p \to \mathbb{C}$ cannot supply a coefficient comparison, since it would send $p \cdot 1 = 0$ to $p \cdot 1 \neq 0$. A common integral base with two derived fibers is a different, possible construction. Identifying the categorical colimit with an analytic restricted tensor product also requires its own map.

Question 19.12.3 *(Completed arithmetic duality and evaluation).* Construct the archimedean coefficient complex and the pairing of a functorial completed arithmetic Koszul duality. Construct an operation-preserving comparison to theta and Mellin reflection with its specified half-density normalization and adjoint. The comparison must transport both polar terms and the native vacuum, as well as the products, collision operations and completions on its declared domain.

The first missing map is from this arithmetic coefficient complex and its duality pairing to the analytic recipient with the evaluation above. To obtain a level criterion, it must also supply the defect identity and detect its defect line. The coefficient anti-equivalence, the marked native examples and the scalar theta theorem each provide positive structures. Their comparison remains the mathematical construction asked for in the question.

REFERENCES FOR THE DUALITY THEOREMS

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Chapter 20

Centers, actions, and representation tensors

A center records operations that commute with a specified action. Its definition therefore depends on the objects allowed to act and on the meaning of commutation. An algebra homomorphism need not transport even an ordinary center. An equivalence of module categories can transport a derived center, provided it retains the coefficient action. These two facts determine the first comparison we need.

For collision coefficients there is another choice. Convolution combines inputs on disjoint configurations, whereas diagonal tensor combines coefficients at the same configuration. Their units differ. The center of a convolution algebra and the universal diagonal actor thus solve different problems. Factorization and a specified local vacuum impose further conditions on an actor.

We first construct scalar Hochschild centers and their Morita comparisons. Normal differential modules then give explicit centers whose products, brackets, and coefficient dependence can be computed. We next compare the word and collision constructions and examine diagonal actions. The finite fusion example retains both its braided center and the distinguished algebra that recovers its boundary category. These data specify what the arithmetic comparison must preserve.

20.1 Maps and centers

Let k be a field and let A be a unital associative k-algebra. Its *ordinary center* is

$$Z_{\mathrm{alg}}(A) = \{a \in A : ab = ba \text{ for every } b \in A\}.$$

It is a commutative algebra. A homomorphism $f : A \to B$ sends it into the centralizer of $f(A)$ in B. That centralizer can be larger than the center of B.

Example 20.1.1 *(A central element with noncentral image).* Take

$$A = k \times k, \qquad B = M_2(k), \qquad f(a,b) = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}.$$

The element $(1,0)$ belongs to $Z_{\mathrm{alg}}(A)$. Its image is E_{11}, and

$$E_{11}E_{12} - E_{12}E_{11} = E_{12} \neq 0.$$

Thus f does not restrict to a map $Z_{\mathrm{alg}}(A) \to Z_{\mathrm{alg}}(B)$. The centralizer of the diagonal subalgebra is the diagonal subalgebra itself, whereas $Z_{\mathrm{alg}}(B) = kI_2$.

The derived construction retains this distinction. Put $A^e = A \otimes_k A^{\mathrm{op}}$, so that left A^e-modules are A-bimodules. The *scalar Hochschild center* is

$$Z_k(A) = R\,\mathrm{Hom}_{A^e}(A, A).$$

For a differential graded algebra the tensor and Hom in this formula are derived. We use cohomological grading, with $M[r]^i = M^{i+r}$.

For an ordinary algebra, a Hochschild p-cochain is a k-linear map $A^{\otimes p} \to A$. Its differential is

$$\begin{aligned}(bf)(a_1, \dots, a_{p+1}) = {} & a_1 f(a_2, \dots, a_{p+1}) \\ & + \sum_{i=1}^{p} (-1)^i f(a_1, \dots, a_i a_{i+1}, \dots, a_{p+1}) \\ & + (-1)^{p+1} f(a_1, \dots, a_p) a_{p+1}.\end{aligned} \tag{20.1}$$

In degree zero this is $b(a)(x) = xa - ax$. Hence $H^0 Z_k(A) = Z_{\mathrm{alg}}(A)$.

PROPOSITION 20.1.2 *(The first two Hochschild groups).* For an ordinary unital algebra,

$$H^0 Z_k(A) = Z_{\mathrm{alg}}(A), \qquad H^1 Z_k(A) = \mathrm{Der}_k(A)/\mathrm{Inn}(A).$$

Here a derivation satisfies $D(ab) = aD(b) + D(a)b$, and an inner derivation has the form $b \mapsto ab - ba$.

Proof. The degree-zero calculation was just made. For a one-cochain D,

$$(bD)(a, b) = aD(b) - D(ab) + D(a)b.$$

Its vanishing is the derivation identity. The image from degree zero consists exactly of the inner derivations, with an immaterial change of sign in their parameter. □

A map $f : A \to B$ gives a mixed complex $C^\bullet(A, B)$, in which A acts on B through f. It receives two natural cochain maps:

$$C^\bullet(A, A) \longrightarrow C^\bullet(A, B) \longleftarrow C^\bullet(B, B). \tag{20.2}$$

The first applies f to the output. The second restricts all inputs along f. Both preserve cup products. They need not give a map between the two endpoint centers.

The mixed complex also explains a limitation of insertion operations. A cochain $A^{\otimes p} \to B$ outputs an element of B. Another such cochain expects an element of A in each input. Inserting the first into the second is therefore undefined without additional data. The cup product uses multiplication in B and has no such typing problem.

20.2 The full cochain structure

The derived center remembers more than its cohomology ring. The bar resolution makes its additional operations explicit. For an ordinary unital algebra its normalized cochains are those that vanish whenever an input is 1. Normalization does not remove degree-zero cochains.

The cup product of cochains of arities p and q is

$$(f \smile g)(a_1, \dots, a_{p+q}) = f(a_1, \dots, a_p)\, g(a_{p+1}, \dots, a_{p+q}). \tag{20.3}$$

For an ungraded algebra, define

$$f \circ g = \sum_{i=0}^{p-1} (-1)^{i(q-1)} f(1^{\otimes i} \otimes g \otimes 1^{\otimes (p-1-i)}).$$

The *Gerstenhaber bracket* is

$$[f, g]_{\mathrm{G}} = f \circ g - (-1)^{(p-1)(q-1)} g \circ f. \tag{20.4}$$

It has cohomological degree -1. On one-cochains it is the commutator of endomorphisms.

Higher *braces* $f\{g_1, \dots, g_r\}$ insert the g_i into disjoint slots of f, in the indicated order. The sum runs over all allowed positions. For a differential graded algebra, suspend the inputs and use the Koszul sign of each insertion. This convention includes the internal differential and avoids assigning signs by arity alone.

An E_n-algebra is an algebra over chains on the little n-disks operad, or its equivalent derived operad.

Theorem 20.2.1 *(The cochain center).* The Hochschild center has an E_2-algebra structure. Its product and degree--1 bracket on cohomology are (20.3) and (20.4). In particular, the cup product is graded commutative on cohomology, and the bracket is a graded derivation of that product.

This is the cochain form of the Deligne theorem. McClure–Smith identify the unit, cup, and brace operad with little two-cubes chains. Their result is Theorem 1.1 of *A solution of Deligne's Hochschild cohomology conjecture*. Their construction is integral. In the derived setting, apply the construction to a suitable cofibrant algebra model.

The E_2 structure is not the assertion that a selected cochain cup product is strictly commutative.

The obstruction to a higher disk structure

Let $A = k[x, y, z]$ in characteristic zero. The derivation $D = \partial_x$ is a Hochschild one-cocycle, and x is a degree-zero cocycle. Direct insertion gives

$$[D, x]_{\mathrm{G}} = 1. \tag{20.5}$$

There are no inner derivations in this commutative algebra, and 1 is nonzero in its center.

Proposition 20.2.2 *(A restriction obstruction).* The specified Hochschild E_2 structure on $Z_k(k[x, y, z])$ cannot be obtained by restriction of an E_3 structure along the standard inclusion of little disks.

Proof. The space of two little two-disks has the homotopy type of S^1. Its degree-one homology class gives the Gerstenhaber bracket. The corresponding space for three-disks has the homotopy type of S^2, whose degree-one homology is zero. Restriction of an E_3 structure would therefore make that bracket zero. Equation (20.5) contradicts this consequence. □

A volume form can identify polyvectors with differential forms and furnish additional operators. It does not change the degree of this nonzero bracket. The proposition concerns the specified E_2 structure and the specified restriction map. It does not exclude every differently structured object with the same underlying graded vector space.

Completion in cochain length

For an augmented differential graded algebra, write $\bar{A}$ for the augmentation ideal and $BA = \bigoplus_{q \geq 0} (\bar{A}[1])^{\otimes q}$ for its bar coalgebra. Its differential combines the internal differential with multiplication, using suspended signs. The complete Hochschild cochain complex is

$$\mathrm{Hom}(BA, A) = \prod_{q \geq 0} \mathrm{Hom}((\bar{A}[1])^{\otimes q}, A).$$

The product is forced by Hom from a direct sum. Replacing it by a direct sum can remove cochains.

Let $\tau : BA \to A$ be the universal twisting cochain. Convolution uses deconcatenation in BA and multiplication in A. The Hochschild differential is

$$d_{\mathrm{Hoch}} = \partial + [\tau, -].$$

The equation $\partial\tau + \tau * \tau = 0$ implies $d^2_{\mathrm{Hoch}} = 0$. The two exterior terms in (20.1) are part of this twist.

At each fixed input length, convolution has finitely many cuts. The complete product therefore defines these operations without a convergence assumption on the sequence of cochain lengths. A completion in state weight, pole order, or coefficient parameters is a separate choice.

20.3 Coefficient-linear Morita comparison

We now specify the structure that transports the center. Let $(C, \star, K)$ be a presentable stable k-linear symmetric monoidal category. Assume that $\star$ preserves colimits separately. An algebra and its module actions are unital, and relative tensors are derived.

For algebras A, B in C, let ${}_B\mathrm{Bimod}_A(C)$ denote the category of objects with a left B-action and a commuting right A-action. Write $\mathrm{Mod}_A(C)$ for left A-modules. A B-A bimodule P defines

$$P \star_A - : \mathrm{Mod}_A(C) \longrightarrow \mathrm{Mod}_B(C).$$

It preserves colimits and is C-linear: tensoring by a coefficient object commutes coherently with this functor.

THEOREM 20.3.1 *(Coefficient-linear Eilenberg–Watts comparison).* The assignment $P \mapsto P \star_A -$ gives an equivalence

$${}_B\mathrm{Bimod}_A(C) \simeq \mathrm{Fun}^L_C\big(\mathrm{Mod}_A(C), \mathrm{Mod}_B(C)\big). \tag{20.6}$$

For a C-linear colimit-preserving functor F, the inverse sends F to $F(A)$ with its induced right A-action. Composition corresponds to relative tensor.

Proof. The free left A-module on $X \in C$ is $A \star X$. Every module M is the geometric realization of its free simplicial bar resolution. Its terms have the form $A \star A^{\star q} \star M$, with faces supplied by multiplication and the action.

Put $P = F(A)$. Right multiplication on the regular module A is an A-module action. Applying F gives the right A-action on P. The left B-action is part of its value in $\mathrm{Mod}_B(C)$. Coefficient linearity identifies $F(A \star X)$ with $P \star X$.

Apply F to the free simplicial resolution. Preservation of colimits identifies its realization with $F(M)$. The resulting simplicial object is the two-sided bar construction computing $P \star_A M$. This gives a natural equivalence $P \star_A M \simeq F(M)$.

A natural transformation between two such functors is determined on the free modules by its value at A and coefficient linearity. Compatibility with right multiplication says precisely that this value is a bimodule map. The free resolutions extend it to all modules, including its higher homotopies.

Finally, the two iterated bar realizations for a composite can be realized in either order. Their common bisimplicial object gives $(Q \star_B P) \star_A M \simeq Q \star_B (P \star_A M)$. Its associativity is inherited from the same multibar construction. □

The identity functor corresponds to the regular A-bimodule. Its endomorphisms have composition and the coherent multiplication induced by composition of functors. Their interaction is the E_2 structure on the center. Thus the coefficient action is part of the comparison, even when the underlying module categories are equivalent as abstract categories.

Definition 20.3.2. The *internal convolution center* of A, denoted $Z^{\mathrm{int}}_{\star}(A)$, represents the functor

$$W \longmapsto \mathrm{Map}_{A\text{-}A}(W \star A, A). \tag{20.7}$$

The bimodule structure on $W \star A$ uses the symmetry to move A past W. Representation means a natural equivalence with $\mathrm{Map}_{\mathcal{C}}(W, Z^{\mathrm{int}}_{\star}(A))$ for every $W \in \mathcal{C}$.

The functor $W \mapsto W \star A$ preserves colimits. Its right adjoint therefore supplies the representing object. The center theorem gives this object its E_2 structure. In Lurie's *Higher Algebra*, Corollary 5.3.1.15 gives existence, and Theorem 5.3.1.30 identifies its underlying object through module maps. For associative algebras these modules are bimodules.

The coefficient action gives the scalar center

$$Z_k(A) = R\,\mathrm{Hom}_{\mathcal{C}}(K, Z^{\mathrm{int}}_{\star}(A)).$$

The internal center is an object of $\mathcal{C}$. The scalar center is a complex over k. Evaluating at K does not identify these two recipients.

Theorem 20.3.3 *(Morita invariance with coefficients).* Suppose that P is a B-A bimodule and Q an A-B bimodule, with coherent equivalences

$$P \star_A Q \simeq B, \qquad Q \star_B P \simeq A.$$

Then conjugation gives equivalences of internal and scalar E_2 centers:

$$Z^{\mathrm{int}}_{\star}(A) \simeq Z^{\mathrm{int}}_{\star}(B), \qquad Z_k(A) \simeq Z_k(B).$$

The equivalences depend on the supplied Morita data and commute with their composition.

Proof. The functor $M \mapsto P \star_A M \star_A Q$ identifies the categories of A- and B-bimodules. It takes A to B. For every coefficient object W, symmetry and coefficient linearity identify its value on $W \star A$ with $W \star B$. It therefore identifies the functors represented in (20.7). The internal equivalence follows by Yoneda.

Conjugation also identifies composition of coefficient-linear endofunctors. Under (20.6), both centers are endomorphisms of the corresponding identity functor. Thus the equivalence preserves the coherent operations giving the E_2 structure. Evaluation at K gives the scalar assertion.

For a composite Morita equivalence, both constructions are the same iterated relative tensor. The multibar associativity in the preceding theorem identifies them, with its full coherence. □

For differential graded algebras this also has a cochain description. Keller, *Derived invariance of higher structures on the Hochschild complex*, Section 4.6, constructs the comparison in the homotopy category of brace algebras. Its hypotheses include the required fully faithful derived tensor functors. An arbitrary homomorphism such as Example 20.1.1 does not satisfy this conclusion.

Matrices retain the scalar center

Let $B = M_n(A)$. The column module $P = A^n$ is a B-A bimodule. Its right-A dual is the row A-B bimodule

$$Q = \mathrm{Hom}_{\mathrm{Mod}\text{-}A}(P, A).$$

Matrix multiplication gives $P \otimes_A Q \simeq B$ and $Q \otimes_B P \simeq A$. Hence $Z_k(M_n(A)) \simeq Z_k(A)$ as E_2 algebras.

An explicit cochain comparison uses matrix paths. For a p-cochain f on A, define

$$\big(\widetilde{f}(X_1, \dots, X_p)\big)_{i_0 i_p} = \sum_{i_1, \dots, i_{p-1}} f\big((X_1)_{i_0 i_1}, \dots, (X_p)_{i_{p-1} i_p}\big).$$

For $p = 0$ put $\widetilde{a} = a I_n$. Matrix multiplication is concatenation of these paths.

PROPOSITION 20.3.4. The matrix-path map preserves the Hochschild differential, cup product, and all braces. It induces the Morita equivalence on scalar centers.

Proof. Each inner term of the Hochschild differential contracts two neighboring edges. Summing their common index is matrix multiplication. The first and last terms attach an edge at the corresponding endpoint. These descriptions prove compatibility with the differential.

The cup product splits a path into two consecutive pieces and multiplies their values. A brace inserts shorter paths into selected slots of a longer one. The index sums are the same before and after either operation. Matrix units have degree zero, so the suspended signs are unchanged.

The corner $E_{11} B E_{11} = A$ and the two bimodules above implement the inverse derived equivalence. The matrix-path formula is its map on the bar cochains. One can also compare ordinary cochains with cochains relative to $M_n(k)$ using the separability element

$$\sum_{i=1}^{n} E_{i1} \otimes E_{1i}.$$

Its product is I_n, and it commutes with the two actions of $M_n(k)$. Inserting it contracts the complement of the relative bar resolution. This identifies the displayed map with the Morita quasi-isomorphism. □

No factor $1/n$ occurs. The comparison is valid even when the coefficient characteristic divides n. It concerns the scalar Hochschild center of the matrix algebra. If a geometric algebra is required to have scalar empty coefficient, replacing that coefficient by $M_n(k)$ changes the geometric class.

20.4 RELATIVE SCALAR EXTENSION

Let C now carry a central action of $D(k)$, compatible with its tensor. Let A_0 be an algebra in C. For a differential graded k-algebra B, set

$$A_B = A_0 \otimes_k B.$$

For a B'-B bimodule P, set $J(P) = A_0 \otimes_k P$ with its induced $A_{B'}$-A_B actions.

Assume the coefficient action has the relative-tensor comparison

$$(A_0 \otimes Q) \star_{A_{B'}} (A_0 \otimes P) \simeq A_0 \otimes (Q \otimes^{\mathbf{L}}_{B'} P).$$

Its coherence is part of the central coefficient action. It can be checked on the common two-sided bar construction.

Suppose also that empty evaluation $e_\varnothing : C \to D(k)$ is strong monoidal, preserves the realizations in question, and satisfies $e_\varnothing(A_0) = k$. It then sends $J(P)$ back to P, including its two coefficient actions.

Theorem 20.4.1 *(A relative center retract).* Under these hypotheses there are maps of scalar E_2 algebras

$$Z_k(B) \xrightarrow{J_B} Z_C(A_B) \xrightarrow{e_\varnothing} Z_k(B), \qquad e_\varnothing J_B = \mathrm{id}.$$

Here $Z_C(A_B) = R\,\mathrm{Hom}_{A_B\text{-}A_B}(A_B, A_B)$ is formed in C. The statement makes $Z_k(B)$ a retract. It does not assert that J_B is essentially surjective on geometric bimodules.

Proof. The relative-tensor comparison makes J a monoidal functor on the endomorphism category of the object B in the bimodule bicategory. It sends the regular B-bimodule to the regular A_B-bimodule. It therefore induces a map on endomorphisms of these units, preserving their E_2 operations.

Empty evaluation gives the reverse monoidal functor on these objects. The identity $e_\varnothing J(P) = P$ is natural and compatible with composition. Its value on unit endomorphisms proves the retract identity.

On bar cochains the forward map multiplies the A_0 inputs and applies the coefficient cochain to the B inputs. Centrality moves the coefficients past the A_0 factors. The resulting Koszul signs are exactly those in the tensor product and suspended insertion conventions. The same formula commutes with every cut, cup, and brace. □

In particular, a nonzero coefficient-center class remains nonzero after extension. Its empty evaluation recovers the original class. A class supported on nonempty configurations may lie outside the image of J_B.

This construction is useful for arithmetic coefficient algebras: it places their actual Hochschild operations in a geometric recipient. It does not construct an identification with every internal center, every diagonal actor, or a chosen physical bulk algebra. Each such identification has its own mapping property.

20.5 The center of a homomorphism diagram

The two arrows in (20.2) have a common target. Their homotopy fiber can therefore be formed as a complex. To obtain its higher operations, we realize the diagram by one algebra.

Let $f : A \to B$ be a unital homomorphism of ordinary k-algebras. Regard B as a B-A bimodule by right multiplication through f. Define the *triangular algebra*

$$T_f = \begin{pmatrix} A & 0 \\ B & B \end{pmatrix}, \qquad \begin{pmatrix} a & 0 \\ m & b \end{pmatrix}\begin{pmatrix} a' & 0 \\ m' & b' \end{pmatrix} = \begin{pmatrix} aa' & 0 \\ mf(a') + bm' & bb' \end{pmatrix}.$$

Its unit is $\mathrm{diag}(1, 1)$. The two diagonal idempotents are denoted e_A, e_B.

Put $u \mapsto f_* u$ for the first map of (20.2), and $v \mapsto f^* v$ for the second. Define a cochain complex $\mathcal{F}_f$ by

$$\mathcal{F}_f^n = C^n(A, A) \oplus C^n(B, B) \oplus C^{n-1}(A, B),$$

with differential

$$d(u, v, h) = (bu, bv, f_* u - f^* v - bh). \tag{20.8}$$

The sign on bh is the shift sign. Since both comparison maps commute with b, one has $d^2 = 0$.

Theorem 20.5.1 *(A triangular realization).* There is a quasi-isomorphism of complexes

$$Z_k(T_f) \simeq \mathcal{F}_f.$$

Consequently the homotopy fiber has an E_2 structure through the center of T_f. This assertion does not assign the ordinary brace formulas to mixed cochains.

Proof. The subalgebra $S = ke_A \oplus ke_B$ is separable over k. Its separability element is $e_A \otimes e_A + e_B \otimes e_B$. Inserting this element gives the usual contraction from the ordinary bar resolution to the relative bar resolution over S. Thus relative Hochschild cochains compute $Z_k(T_f)$.

An S-balanced tensor is a composable sequence of matrix entries. Such a sequence either stays at one diagonal vertex or crosses once through the B-A entry. It cannot cross twice, because the opposite off-diagonal entry is zero. The two constant-vertex parts are $C^\bullet(A, A)$ and $C^\bullet(B, B)$.

The part with one crossing has terms

$$\mathrm{Hom}_k(B^{\otimes q} \otimes B \otimes A^{\otimes p}, B), \qquad p, q \geq 0.$$

It occupies total Hochschild degree $p + q + 1$. Its differential is the two-sided bar differential for the bimodule B, together with the two endpoint actions. Without those endpoint terms, it computes $R\,\mathrm{Hom}_{B\text{-}A}(B, B)$ shifted by one.

Resolve the right A-action on B by the bar resolution, retaining its free left B-action. Evaluation on $1 \in B$ then identifies this derived Hom with $C^\bullet(A, B)$. The extra left-B bar direction contracts by its unit degeneracy. The contraction respects the right-A differential, so the same comparison applies to the total complex.

The remaining endpoint maps apply an A-cochain through f and restrict a B-cochain along f. Their orientations are opposite. With the shifted mixed term, the total differential is exactly (20.8). This proves the quasi-isomorphism. Transport the E_2 structure from $Z_k(T_f)$ along this equivalence. No mixed insertion with mismatched source and target is required. □

The degree-zero part can be read directly from the algebra.

PROPOSITION 20.5.2. There is an isomorphism

$$Z_{\mathrm{alg}}(T_f) = \{(a, b) \in Z_{\mathrm{alg}}(A) \times Z_{\mathrm{alg}}(B) : f(a) = b\}.$$

For the diagonal homomorphism of Example 20.1.1, this center is k.

Proof. Commutation with e_A and e_B forces the off-diagonal entry of a central element to vanish. Commutation with the two diagonal subalgebras makes its remaining entries central. Commutation with an off-diagonal element $m \in B$ is $mf(a) = bm$ for every m. Taking $m = 1$ gives $f(a) = b$, and centrality of b makes this condition sufficient.

For the stated example, $b = \lambda I_2$. The equality $\mathrm{diag}(a_1, a_2) = b$ forces $a_1 = a_2 = \lambda$. □

The triangular center thus retains the compatibility of the two actions. Merely taking the product of the endpoint centers would lose that condition. Conversely, it would be incorrect to call every ordinary fiber of cochain complexes a strict brace algebra. The triangular model supplies the operations that make the comparison meaningful.

20.6 THE REGULAR AND PUNCTURED NORMAL LINES

The simplest collision coordinate already gives a center with a nontrivial bracket. Let k have characteristic zero, and put

$$D = k\langle u, \partial\rangle/(\partial u - u\partial - 1), \qquad R = k[u], \qquad L = k[u, u^{-1}].$$

Both R and L are left D-modules with their derivative connections. Write $C_2 = \mathrm{thick}_D(R \oplus L)$ for the full differential graded category obtained by shifts, finite cones, and direct summands. Its scalar center

is the derived endomorphism complex of its diagonal bimodule. We will compute the center together with its cup product and Gerstenhaber bracket.

For $v \in D$, let r_v denote right multiplication by v. This is a map of free left D-modules. The two resolutions

$$P_0 = [D \xrightarrow{r_\partial} D] \longrightarrow R, \qquad P_1 = [D \xrightarrow{r_{u\partial+1}} D] \longrightarrow L \tag{20.9}$$

have terms in degrees -1, 0 and augmentations 1, u^{-1}. The term 1 in the second differential is essential: $(u\partial + 1)u^{-1} = 0$, whereas $u\partial(u^{-1}) = -u^{-1}$.

Lemma 20.6.1. The resolutions in (20.9) are exact. They give

$R\operatorname{Hom}_D(-,-)$	R	L
R	k	$k \oplus k[-1]$
L	0	$k \oplus k[-1]$.

Proof. The Weyl algebra is a domain, so the two differentials are injective. The first cokernel is the polynomial module. In the second cokernel the Euler relation reduces mixed monomials to powers of u and powers of ∂. Their images are

$$u^{m-1} \quad (m \geq 0), \qquad (-1)^n n! u^{-n-1} \quad (n \geq 1).$$

These form a Laurent basis, proving the second assertion about cokernels.

Applying Hom to the resolutions gives the operators ∂ and $u\partial + 1$ on R and L. Polynomial integration makes $\partial : R \to R$ surjective with kernel k. On L its cokernel is the residue line represented by u^{-1}. Finally, $(u\partial + 1)u^m = (m+1)u^m$. This operator is invertible on R, and on L it has the single exceptional mode $m = -1$ in both kernel and cokernel. □

The four complexes alone do not specify their composition. To retain composition, take the graded path algebra T on two objects. Give it an arrow $a : 0 \to 1$ of degree zero and a loop $\theta : 1 \to 1$ of degree one. Set

$$\theta^2 = 0, \qquad b = \theta a, \qquad T^0 = ke_0 \oplus ke_1 \oplus ka, \qquad T^1 = k\theta \oplus kb.$$

The differential is zero.

Products are written in composition order. Apart from identities, the sole nonzero product of basis elements is $\theta a = b$.

Proposition 20.6.2. There is a strict quasi-isomorphism

$$T \longrightarrow \operatorname{End}_D(P_0 \oplus P_1).$$

Consequently C_2 is equivalent to $\operatorname{Perf}(T)$, using right T-modules.

Proof. Send e_i to the two projections. The arrow a is represented by a map $\alpha : P_0 \to P_1$ with $\alpha^{-1} = 1$ and $\alpha^0 = r_u$. Its chain equation is $\partial u = u\partial + 1$. Represent θ by the degree-one identity $\Theta : P_1^{-1} \to P_1^0$ and b by $\Theta\alpha$.

These maps are closed and satisfy the multiplication relations strictly. After augmentation, α is the inclusion $R \to L$, while Θ and $\Theta\alpha$ represent the two residue classes. Thus the map induces isomorphisms in all four Hom corners. Hom from $P_0 \oplus P_1$ and its tensor adjoint identify the generators and their Hom complexes. The identification extends to their thick closures. □

The completed center and its bracket

Use the normalized two-sided bar relative to the separable algebra $S = ke_0 \oplus ke_1$. Its middle term is the direct sum of finite composable words in $(T/S)[1]$. Taking Hom therefore gives a product over arities. We denote the resulting complete cochain algebra by $C(T)$.

Here are explicit sign conventions for the calculation. An n-cochain of internal degree r has total degree $n + r$. Its differential is $D = (-1)^r b_H$, where

$$\begin{aligned}(b_H f)(v_1, \dots, v_{n+1}) = {} & (-1)^{r|v_1|} v_1 f(v_2, \dots, v_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(v_1, \dots, v_i v_{i+1}, \dots, v_{n+1}) \\ & + (-1)^{n+1} f(v_1, \dots, v_n) v_{n+1}.\end{aligned}$$

For an m-cochain g of internal degree s, the cup product is

$$(f \smile g)(v_1, \dots, v_{n+m}) = (-1)^{s(\sum_{i=1}^{n} |v_i| - n)} f(v_1, \dots, v_n) g(v_{n+1}, \dots, v_{n+m}).$$

These formulas express the suspended-bar operations on unsuspended inputs.

THEOREM 20.6.3. The scalar center of C_2 has a continuous associative differential graded model

$$k[[x]] \otimes \Lambda(\eta), \qquad |x| = 0, \quad |\eta| = 1, \qquad d = 0.$$

Its Gerstenhaber bracket is determined by

$$[\eta, x] = -x, \qquad [x, x] = [\eta, \eta] = 0.$$

Proof. Give θ, b weight one and e_0, e_1, a weight zero. For a cochain let d be input weight minus output weight. Every nonempty composable word has one of the forms

$$\theta^{\otimes n}, \qquad \theta^{\otimes (n-1)} \otimes a, \qquad \theta^{\otimes (n-1)} \otimes b.$$

The output has the same endpoints.

For each $d \geq 1$, the cochain block in total degrees 0, 1, 2 is

$$k^2 \xrightarrow{M} k^3 \xrightarrow{N} k, \qquad M = \begin{pmatrix} 0 & 0 \\ -1 & 1 \\ -1 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix}. \tag{20.10}$$

The degree-zero basis sends $\theta^{\otimes d}$ to e_1 and $\theta^{\otimes (d-1)} \otimes b$ to a. The degree-one basis has outputs θ, a, b on the corresponding words of input weights $d + 1, d, d + 1$. The degree-two basis sends $\theta^{\otimes (d+1)} \otimes a$ to b. The multiplication $\theta a = b$ gives the interior faces, and the endpoint faces give the other entries of (20.10).

For $d = 0$ the degree-zero basis is e_0, e_1 and the first matrix is $-M$. Weight -1 consists of an identity map $k \to k$ in degrees one and two. Its source is the nullary θ and its target sends a to b. There are no further blocks.

For $d \geq 0$ choose representatives and projections

$$\begin{aligned} i_0(c) &= (c, c), & p_0(p, q) &= p, \\ i_1(c) &= (c, 0, c), & p_1(r, s, t) &= t - s. \end{aligned}$$

A homotopy on the complement is

$$h_1(r,s,t) = \begin{cases} (0,s), & d>0, \\ (0,-s), & d=0, \end{cases} \qquad h_2(z) = (z,0,0).$$

The matrices give $Dh + hD = 1 - ip$ and $h^2 = hi = ph = 0$. Use the inverse identity in weight -1. The maps preserve weight, so their product contracts the complete cochain complex.

To identify the product, define derivations X, Y by

	e_0	e_1	θ	a	b
X	0	0	e_1	0	a
Y	0	0	θ	0	b.

Their internal degrees are $-1, 0$ and their total degrees are $0, 1$. They are closed, $X \smile Y = Y \smile X$, and $Y \smile Y = 0$. The cochains $X^{\smile d}$ and $Y \smile X^{\smile d}$ give the representatives just chosen. Thus $x \mapsto X$, $\eta \mapsto Y$ is a continuous multiplicative quasi-isomorphism. On each fixed word, only finitely many terms of a formal series contribute.

The unary insertion bracket is the graded commutator. Since $YX - XY = -X$, it gives $[\eta, x] = -x$. The biderivation identity determines the remaining brackets. □

In this description η acts as $-x\partial_x$. For formal series f, g,

$$[\eta f, g] = -xfg', \qquad [\eta f, \eta g] = \eta x(gf' - fg').$$

The associative model and this cohomology bracket do not assert formality of the full E_2 structure. The braces still live on the complete bar cochains. Replacing their product by a direct sum would give $k[x]$ in place of $k[[x]]$ and would omit legitimate maps from the bar.

Restriction to the full subcategory generated by L has target $\Lambda(\theta)[[x]]$. It is

$$x \longmapsto x, \qquad \eta \longmapsto \theta x.$$

Restriction to R sends x, η to zero and preserves constants. The regular object therefore forces the odd coefficient at the punctured object to vanish at the formal origin.

The same conclusion follows from the relative bar triangle

$$C(T) \longrightarrow k \oplus C(\Lambda(\theta)) \longrightarrow R\operatorname{Hom}_{\Lambda(\theta)}(\Lambda(\theta)a, \Lambda(\theta)a).$$

A composable word crosses from 0 to 1 at most once. Its crossing terms form the shifted module bar. The module $\Lambda(\theta)a$ is free, so this bar contracts to $\Lambda(\theta)$. On the diagonal models, the last map is $(c, f) \mapsto c - f(0)$. Its kernel is $k \oplus x\Lambda(\theta)[[x]]$, which recovers $\eta = \theta x$.

Removing nullary cochains

Deformation problems can impose conditions that exclude the primitives used by the full center. Fix the two objects, hence the inclusion $S \subset T$, and retain only positive-arity cochains. These describe uncurved S-linear deformations with zero nullary comparison term. Write their complex as $C_{\geq 1,S}(T)$.

PROPOSITION 20.6.4. The only nonzero groups of $C_{\geq 1,S}(T)$ are

$$H^0 = xk[[x]], \qquad H^1 = k[[x]]\eta \oplus k\sigma, \qquad H^2 = kq.$$

Here $\sigma(a) = a$, $\sigma(b) = b$, and $q(a) = b$, with other generator values zero. For the normalized positive-arity complex relative to k, they are instead

$$\begin{aligned} H^0 &= xk[[x]], \\ H^1 &= k[[x]]\eta \oplus k\sigma \oplus kDa, \\ H^2 &= kD\theta \oplus kDb. \end{aligned}$$

The symbol D in the last expressions is the Hochschild differential of a nullary element in the full complex.

Proof. The positive-weight blocks do not change. Removing the two nullary identities leaves their former boundary σ in degree one and removes the constant degree-zero class. Removing the nullary θ leaves q in degree two.

For the k-relative complex, use the exact sequence

$$0 \to C_{\geq 1,k}(T) \to C_k(T) \to T \to 0.$$

The map on degree-zero cohomology is $f(x) \mapsto f(0)1_T$, and the classes $\eta k[[x]]$ have zero nullary component. The long exact sequence contributes $T^0/k1_T$ in degree one and T^1 in degree two. Represent the former by e_0, a. The connecting maps give the displayed classes. They are distinct: $(Db)(e_0) = b$, while $(D\theta)(e_0) = 0$ and $(D\theta)(a) = b$. □

Let $A = k[[s]]$ and give $T \otimes_k A$ the differential

$$d_s a = sb, \qquad d_s\theta = d_s b = d_s e_i = 0. \tag{20.11}$$

This is a differential graded algebra, since $d_s = s[\theta, -]$ and $\theta^2 = 0$. Its underlying graded module is free, but

$$H^0(T_s) = Ae_0 \oplus Ae_1, \qquad H^1(T_s) = A\theta \oplus (A/(s))b.$$

Even modulo s^2, there is no uncurved A_∞ map $T_s \to T \otimes A$ with $F_0 = 0$ and $F_1 = 1 \bmod s$. Its unary equation would give $F_1 d_s = 0$, whereas its value on a is $sb \neq 0 \bmod s^2$. Higher components do not enter this equation. The identity $D\theta = q$ in the full center does not remove the obstruction when the nullary term is forbidden.

Changing the connection residue

The deformation has a geometric realization. Let $D_A = A\langle u, \partial\rangle/(\partial u - u\partial - 1)$, take $R_A = A[u]$ with its derivative connection, and put

$$L_s = A[u, u^{-1}], \qquad \partial f = f' - \frac{s}{u} f.$$

These are polynomial and Laurent polynomial modules over A. There is no completion in u.

Theorem 20.6.5. The category generated by $R_A \oplus L_s$ is equivalent to $\mathrm{Perf}(T_s)$. For $K = k((s))$,

$$\mathrm{Perf}(T_s \otimes_A K) \simeq \mathrm{Perf}(K) \oplus \mathrm{Perf}(\Lambda_K(\theta)).$$

The family T_s is not A-linearly Morita equivalent to the constant family $T \otimes_k A$.

Proof. Replace P_1 by

$$P_{1,s} = [D_A \xrightarrow{r_{u\partial+1+s}} D_A], \qquad P_{1,s} \longrightarrow L_s, \quad 1 \longmapsto u^{-1}.$$

Its cokernel calculation uses

$$\partial^n u^{-1} = (-1)^n \prod_{j=1}^{n} (j+s) u^{-n-1}.$$

All $j + s$ are units of A, so these vectors and the nonnegative powers give a Laurent basis.

Keep the maps $\alpha, \Theta, \Theta\alpha$ used above. Now

$$d_{\mathrm{Hom}}\alpha = r_{u\partial+1+s} - r_{\partial u} = s\Theta\alpha.$$

Thus they define a strict map from T_s to the endomorphism algebra of the two resolutions.

In the corner from R_A to L_s, all Laurent modes except $m = 0$ contract under $u^m \mapsto (m - s)u^{m-1}$. The remaining complex is $[A \xrightarrow{s} A]$, with the Hom sign chosen as in (20.11). The reverse corner is acyclic because $m + 1 + s$ is a unit for $m \geq 0$. The two diagonal corners are A and $A \oplus A[-1]$. The map is therefore a quasi-isomorphism, and the generator argument proves the categorical equivalence.

Over K, the ideal spanned by a, b is contractible. Quotienting gives the algebra $K \times \Lambda_K(\theta)$. Its degree-zero Hochschild center is $K \times K[[x]]$. The constant family, computed directly over K, has center $K[[x]]$.

The former ring has a nontrivial idempotent and the latter has none. Morita equivalence would preserve these rings. An A-linear equivalence would remain one over K, so it is also excluded. □

The ordinary inclusion $R_A \to L_s$ fails to respect the connections by $-s/u$. The equation $d_s a = sb$ records that failure. The proof computed the two generic centers directly. It did not interchange completed Hochschild cochains with extension of scalars.

20.7 Independent directions and a triple collision

Independent normal coordinates allow the preceding calculation to tensor. Let D_d be the Weyl algebra of $R_d = k[u_1, \ldots, u_d]$ and set

$$G_d = \bigoplus_{I \subseteq \{1,\ldots,d\}} R_d\left[\left(\prod_{i \in I} u_i\right)^{-1}\right].$$

Every summand has its derivative connection, and the empty product is one.

Theorem 20.7.1. The scalar center of $\mathrm{thick}_{D_d}(G_d)$, with product-total Hochschild cochains, has the associative differential graded model

$$k[[x_1, \ldots, x_d]] \otimes \Lambda(\eta_1, \ldots, \eta_d), \qquad d = 0,$$

where $|x_i| = 0$, $|\eta_i| = 1$. The cohomology bracket is

$$[\eta_i, x_j] = -\delta_{ij} x_j, \qquad [\eta_i, \eta_j] = [x_i, x_j] = 0.$$

Proof. The generator is the external tensor product of the d modules $k[u_i] \oplus k[u_i, u_i^{-1}]$. Tensor the bounded free resolutions and their strict endomorphism models. They identify the generated category with $\mathrm{Perf}(T^{\otimes d})$.

Resolve the diagonal bimodule by the tensor product of the relative bars. At each multiweight the cochain spaces are finite dimensional and bounded. Their contractions tensor: for two factors the homotopy is $h_1 \otimes 1 + t_1 p_1 \otimes h_2$, with the graded tensor sign. Taking the product over multiweights gives precisely the displayed formal series.

The multiplicative representatives tensor as well. The shuffle comparison with the bar of $T^{\otimes d}$ identifies insertions up to coherent homotopy. The bracket is the sum of the factor brackets, with zero brackets between different factors. □

This theorem applies to two independent pair collisions, for example $12|34$ with constant tangential coefficients. At a triple collision the equations u, v, and $u - v$ are dependent. There is an additional shape coordinate.

The separated shape

Write three distinct points as

$$z_3 = z, \qquad z_1 = z + u, \qquad z_2 = z + ut.$$

Their configuration space is $\mathbb{A}^1_z \times \mathbb{G}_{m,u} \times V$, where

$$V = \operatorname{Spec} k[t, t^{-1}, (t-1)^{-1}].$$

The coordinate u measures scale, and t retains the two finite collision channels.

The algebraic de Rham algebra of V has a strict multiplicative model

$$B = k \oplus k\epsilon_0 \oplus k\epsilon_1, \qquad |\epsilon_i| = 1, \qquad \epsilon_i \epsilon_j = 0,$$

via $\epsilon_0 \mapsto dt/t$ and $\epsilon_1 \mapsto dt/(t-1)$. Partial fractions integrate all polynomial terms and all poles of order at least two. The two simple residues remain independent. This proves the quasi-isomorphism. The scale factor contributes $\Lambda(du/u)$, giving de Rham dimensions $1, 3, 2$.

PROPOSITION 20.7.2. Let $F = k\langle\langle x, y \rangle\rangle$ be the completed free associative algebra. The complete normalized Hochschild cochains of B identify with

$$F \xrightarrow{d} F \oplus F, \qquad df = ([x, f], [y, f]),$$

in degrees zero and one. The degree-one ideal has square zero and componentwise left and right F-actions. Consequently

$$H^0 C(B) = k, \qquad H^1 C(B) = (F \oplus F)/\{([x, f], [y, f]) : f \in F\}.$$

There is no other cohomology. The component of H^1 of output word length m has dimension

$$2 \quad (m = 0), \qquad 4 \quad (m = 1), \qquad 3 \cdot 2^{m-1} \quad (m \geq 2).$$

Proof. The two suspended augmentation generators have degree zero. Scalar outputs give the series F in total degree zero, and the two odd outputs give $F \oplus F$ in degree one. All interior bar faces vanish. The endpoint faces are the two commutators, while the cup product concatenates words and multiplies outputs.

A homogeneous word polynomial commuting with x is a polynomial in x. Comparing xf with fx first excludes words ending in y. Moving each final x to the front then excludes every occurrence of y. Commutation with y leaves only constants. Applying this in each length gives $Z(F) = k$.

At output length $m \geq 2$, the differential is injective on scalar words of length $m-1$, hence has rank 2^{m-1}. Its rank is zero for $m = 1$, and it has no source for $m = 0$. Subtract these ranks from $2 \cdot 2^m$. All maps respect the finite length components, so the image is the product of its images in these components. The computation therefore respects the completion. □

This is the center of the separated shape corner. Identifying the full triple coefficient category requires its remaining objects and collision maps.

A class invisible to every pair

The difference also appears in ordinary global de Rham cohomology. For charges $q_i \in k$, consider the connection

$$\nabla_n = d + c \sum_{i<j} q_i q_j \, d\log(z_i - z_j)$$

on the distinct n-point configuration. After separating a scale coordinate u, its radial residue is $\lambda_n = c \sum_{i<j} q_i q_j$. The differential on the Laurent mode u^m is multiplication by $m + \lambda_n$, followed by du/u. If $\lambda_n \notin \mathbb{Z}$, division by these scalars contracts the radial complex.

Proposition 20.7.3. For $c = 1/3$ and $q_1 = q_2 = q_3 = 1$, all two-point global algebraic de Rham complexes are acyclic. The three-point complex has exactly

$$H^1 = k\left[u^{-1}\frac{dt}{t}\right], \qquad H^2 = k\left[u^{-1}\frac{du}{u} \wedge \frac{dt}{t}\right].$$

Proof. The pair residue is $1/3$ and the triple residue is one. In the coordinates above, the connection is

$$d + \frac{du}{u} + \frac{1}{3}\left(\frac{dt}{t} + \frac{dt}{t-1}\right).$$

Let $D_{\mathrm{sh}} f = f' + (1/(3t) + 1/(3(t-1)))f$.

A pole of order $j \geq 2$ at $p = 0, 1$ with leading coefficient a is removed by subtracting

$$D_{\mathrm{sh}}\left(\frac{a(t-p)^{1-j}}{1/3 - (j-1)}\right).$$

This introduces at most a simple pole at the other point. Remove a polynomial leading term at^j with $D_{\mathrm{sh}}(at^{j+1}/(j+1+2/3))$.

The remainder is $A_0/t + A_1/(t-1)$. Subtracting $D_{\mathrm{sh}}(3A_1)$ leaves $(A_0 - A_1)/t$. For example,

$$\frac{1}{t^2(t-1)^2} = D_{\mathrm{sh}}\left(\frac{3-6t^2}{2t(t-1)}\right) + \frac{2}{t}.$$

There is no nonzero rational horizontal function: its order at zero would have to be $-1/3$. Nor is $1/t$ in the image. A primitive with a pole would produce a pole of higher order with nonzero coefficient. It must therefore be polynomial. A positive degree would produce a polynomial part, and a constant gives equal coefficients at the two simple poles. Thus the shape cohomology is exactly $k\,dt/t$ in degree one.

The radial contraction removes all modes except u^{-1} and $u^{-1}du/u$. Polynomial integration contracts the translation coordinate. Tensoring the three contractions gives the stated result. □

The exceptional integral residues can all be included in the calculation. Let $a_{12}, a_{13}, a_{23} \in k$ and $\lambda = a_{12} + a_{13} + a_{23}$. All functions and forms in the following statement have finite Laurent support.

PROPOSITION 20.7.4. For the connection $d + \sum_{i<j} a_{ij}\, d\log(z_i - z_j)$, the global de Rham dimensions are

	H^0	H^1	H^2
$\lambda \notin \mathbb{Z}$	0	0	0
$\lambda \in \mathbb{Z},\ (a_{12}, a_{13}, a_{23}) \notin \mathbb{Z}^3$	0	1	1
$(a_{12}, a_{13}, a_{23}) \in \mathbb{Z}^3$	1	3	2.

All other groups vanish.

Proof. The scale differential is $m + \lambda$ on u^m. When λ is integral it retains one class in degrees zero and one. The shape operator is

$$Df = \left(f' + \left(\frac{a}{t} + \frac{b}{t-1}\right)f\right)dt, \qquad a = a_{23}, \quad b = a_{12}.$$

Choose an integer $M \geq 1$ such that $a - j, b - j$, and $j + a + b$ are nonzero for every integer $j > M$. Let V_M consist of functions with poles of order at most M at each puncture and polynomial degree at most M. Let W_M consist of one-forms with pole orders at most $M + 1$ and polynomial degree at most $M - 1$. Then

$$D(V_M) \subset W_M, \qquad \dim V_M = 3M + 1, \qquad \dim W_M = 3M + 2.$$

The leading coefficients just listed remove every pole or polynomial term outside this finite complex. Conversely, a function outside V_M has a leading term whose derivative lies outside W_M with nonzero coefficient. Thus the finite inclusion induces an isomorphism on cohomology.

A nonzero horizontal rational function has orders $-a, -b$ at the two punctures, so it exists exactly when both residues are integral. In that case it is a scalar multiple of $t^{-a}(t-1)^{-b}$. The dimension difference gives shape dimensions $(1, 2)$ in the integral case and $(0, 1)$ otherwise. Tensor with the two scale classes. For integral λ, integrality of a_{12}, a_{23} is equivalent to integrality of all three residues. □

Pairwise global cohomology therefore does not recover the complete coefficient diagram. The example does not negate factorization at the coefficient level. Its conclusion concerns the loss caused by taking global cohomology separately on pairs.

The relative residue class

Localization also gives a relative complex

$$\mathcal{R} = \operatorname{fib}(R\operatorname{Hom}_D(R, R) \longrightarrow R\operatorname{Hom}_D(R, L)).$$

It records a map to the regular coefficient together with a nullhomotopy after localization. The constant terms cancel, while the residue line survives one degree higher:

$$\mathcal{R} \simeq k[-2].$$

In $\mathrm{Cone}(k \to \Omega^\bullet(L))[-1]$ the generator is $(0, du/u)$. It cannot be a boundary because no Laurent polynomial has derivative u^{-1}.

Thus a relative map can be nonzero even when forgetting its specified nullhomotopy kills the underlying map. An application to pointed actions must supply its actual input maps into $\mathcal{R}$ and their normalizations. The one-dimensional cohomology does not choose them.

20.8 Centers over a parameter ring

A finite bar model makes coefficient change especially explicit. It also separates the cup product from the insertion bracket. Let

$$R = k[h_1, \dots, h_r], \qquad I = (h_1, \dots, h_r), \qquad S = R/I,$$

where $r \geq 1$ and k has characteristic zero. Let W have basis $u_1, \dots, u_r$, all in cohomological degree one. Write L_i for exterior multiplication by u_i, and form the complex

$$M = R \otimes_k \Lambda W, \qquad d_M = \sum_i h_i L_i.$$

The anticommutation of the L_i gives $d_M^2 = 0$. The complex M is an arrow complex. Its differential is not a derivation of the auxiliary exterior multiplication. Its cohomology is S in degree r.

Take the triangular differential graded algebra

$$T_R = R\varepsilon_0 \oplus R\varepsilon_1 \oplus M, \qquad \varepsilon_1 M \varepsilon_0 = M, \qquad M^2 = 0.$$

We compute its relative scalar center $Z_R(T_R) = R\,\mathrm{Hom}_{T_R \otimes_R T_R^{\mathrm{op}}}(T_R, T_R)$. All cochains here are R-linear.

A finite model with both operations

Put $E = \mathrm{End}_R(M)$, with differential $\delta f = d_M f - (-1)^{|f|} f d_M$. The normalized bar relative to $S_0 = R\varepsilon_0 \oplus R\varepsilon_1$ has just two terms,

$$T_R \otimes_{S_0} T_R, \qquad T_R \otimes_{S_0} M[1] \otimes_{S_0} T_R.$$

There are no longer terms because $M \otimes_{S_0} M = 0$. Reversing the sign of its one-arrow generator gives the finite free cochain model

$$\begin{aligned} Z^n &= (R^2)^n \oplus E^{n-1}, \\ d(a, b; f) &= (0, 0; (b - a)\,\mathrm{id} - \delta f). \end{aligned} \tag{20.12}$$

The endpoint summands occur only in degree zero. The model has total rank $2 + 4^r$ over R.

Use the source-first opposite-cup convention. The cup product and unit are

$$(a, b; f)(a', b'; f') = (aa', bb'; fb' + af'), \qquad 1 = (1, 1; 0). \tag{20.13}$$

The differential identity on endpoints is $bb' - aa' = (b - a)b' + a(b' - a')$. Together with R-linearity of δ, it proves the differential Leibniz rule. Associativity follows from the commuting endpoint scalar actions.

Let $\sigma f = (0, 0; f)$, of degree $|f| + 1$. The insertion convention is

$$[\sigma f, \sigma g] = \sigma\big(f g - (-1)^{|f||g|} g f\big).$$

Endpoint scalars have zero bracket with one-arrow cochains.

These formulas follow by inserting an arrow operation into another arrow operation. They give a bracket of degree -1. Its differential and Jacobi identities reduce to the graded commutator identities in E. In particular, composition in E remains visible even though two pure one-arrow cochains have zero cup product.

Integral cohomology and contraction brackets

Let ι_i be contraction by the dual of u_i. The operators satisfy

$$L_i L_j + L_j L_i = 0, \qquad \iota_i \iota_j + \iota_j \iota_i = 0, \qquad \iota_i L_j + L_j \iota_i = \delta_{ij} \,\mathrm{id}\,.$$

The ordered operators $L_A \iota_B$, for subsets $A, B \subseteq \{1, \dots, r\}$, form an R-basis of E. For one generator, 1, L, ι, $L\iota$ span all two-by-two matrices. Graded tensor products give the general assertion.

Moreover,

$$\delta L_i = 0, \qquad \delta \iota_i = h_i \,\mathrm{id}\,.$$

For fixed L_A, the contraction factors therefore form a Koszul resolution. Hence

$$H^q(E) = S \otimes_k \Lambda^q W \quad (0 \le q \le r),$$

with no other cohomology.

THEOREM 20.8.1. The relative Hochschild groups are

$$\begin{aligned} \mathrm{HH}^0_R(T_R) &= R \times_S R, \\ \mathrm{HH}^1_R(T_R) &= 0, \\ \mathrm{HH}^{q+1}_R(T_R) &= S \otimes_k \Lambda^q W \quad (1 \le q \le r). \end{aligned}$$

They vanish in all other degrees. The positive-degree ideal has square-zero cup product, and a degree-zero pair acts through its common residue in S.

Writing $t_i = [(0, h_i; \iota_i)]$, the degree-zero ring has presentation

$$R[t_1, \dots, t_r]/(h_i t_j - h_j t_i,\ t_i t_j - h_i t_j : 1 \le i, j \le r).$$

Proof. The fiber complex (20.12) gives the map $R^2 \to H^0(E) = S$, $(a, b) \mapsto b - a$. It is surjective with kernel $R \times_S R$. Its remaining cohomology is the positive cohomology of E shifted by one. The cup formula proves the product statements.

Each representative t_i is closed and maps to $(0, h_i)$. Every element of the fiber product is a diagonal scalar plus $(0, c)$ with $c \in I$. Thus these elements generate. The quadratic relations reduce a polynomial to $a + \sum_i b_i t_i$. It maps to zero exactly when $a = 0$ and $\sum_i b_i h_i = 0$. The Koszul syzygies of the regular sequence generate these relations, giving the presentation. □

The exterior powers in the theorem describe modules of classes. Their exterior product is not the cup product in the center. The proof works for any regular sequence in a commutative $\mathbb{Q}$-algebra, with the corresponding free exterior module.

For $A = \{a_1 < \cdots < a_q\}$, let $\alpha_A = [\sigma L_{a_1} \cdots L_{a_q}]$. It has degree $q + 1$. The contraction relations give, for $q \geq 2$,

$$[t_i, \alpha_A] = \begin{cases} (-1)^{j-1} \alpha_{A \setminus \{i\}}, & i = a_j, \\ 0, & i \notin A. \end{cases}$$

For $q = 1$ the result is zero in cohomology: the commutator is a scalar identity, which is the boundary of an endpoint difference. The L_i supercommute, as do the ι_i, so brackets between two positive classes or two degree-zero classes vanish. The bracket with R also vanishes.

For two parameters, put $\omega = \alpha_{\{1,2\}}$. Then

$$[t_1, \omega] = \alpha_{\{2\}}, \qquad [t_2, \omega] = -\alpha_{\{1\}}.$$

The positive classes have zero mutual cup products, and the t_i annihilate them under cup product. The bracket still detects the contraction of an arrow operation.

Derived specialization

The finite model gives an operation-preserving base-change theorem. For every commutative derived R-algebra B,

$$Z_R(T_R) \otimes^{\mathbb{L}}_R B \simeq Z_B(T_R \otimes^{\mathbb{L}}_R B). \tag{20.14}$$

Indeed, both terms of the bimodule bar are induced from bounded finite free R-complexes. Tensor–Hom base change holds on each term and commutes with the bar differential, evaluation, and insertions. If one h_i is inverted, $h_i^{-1}\iota_i$ contracts M. The center then reduces to the two scalar endpoint sectors.

At the common zero of the parameters, put $V = \Lambda W$ and $T_0 = T_R \otimes_R k$. The arrow differential becomes zero.

Theorem 20.8.2. The entire singular-fiber center has cohomology

$$\mathrm{HH}^*_k(T_0) = k \ltimes \big(\mathrm{End}_k(V)/k\,\mathrm{id}\big)[-1].$$

The ideal has square-zero cup product and the shifted graded-commutator bracket. The unit has zero bracket. Its dimensions are

$$\dim_k \mathrm{HH}^n_k(T_0) = \begin{cases} 1 + \binom{2r}{r-1}, & n = 0, \\ \binom{2r}{r} - 1, & n = 1, \\ \binom{2r}{r+n-1}, & \text{otherwise}, \end{cases}$$

where out-of-range binomial coefficients are zero. The total dimension is 4^r.

Proof. In (20.12), the remaining differential sends the endpoint difference to id, now in center degree one. The diagonal scalar is the unit. All other classes are one-arrow cochains modulo this identity. The formulas for cup products and brackets descend to the stated operations.

For internal degree p,

$$\dim_k \mathrm{End}^p(\Lambda W) = \sum_j \binom{r}{j}\binom{r}{j+p} = \binom{2r}{r+p}.$$

Shift by one, remove the scalar identity in degree one, and add the unit in degree zero. This gives the formula. □

For $r = 2$, the dimensions in degrees $-1, 0, 1, 2, 3$ are $1, 5, 5, 4, 1$. Ordinary specialization of the integral degree-zero ring has dimension only three, with basis $1, t_1, t_2$. Thus (20.14) requires derived specialization before cohomology.

There is also no R-linear deformation retraction of the displayed finite complex onto its integral cohomology with zero differential. The nonzero group $\mathrm{HH}^2_R(T_R)$ is annihilated by I, while Z^2 is free. Every R-linear map from that group into Z^2 is zero, since h_1 acts injectively on Z^2. Such a map cannot have a left inverse. This excludes the specified contraction, without deciding all possible formality zigzags.

A nonzero bracket obstruction

Take $r = 2$ and the basis $v_0 = 1, v_1 = u_1, v_2 = u_2, v_3 = u_1u_2$, of degrees $0, 1, 1, 2$. Define a degree-one endomorphism by

$$f(v_0) = v_1, \qquad f(v_1) = v_3, \qquad f(v_2) = f(v_3) = 0.$$

The arrow differential ϵf defines a first-order deformation over $k[\epsilon]/(\epsilon^2)$. Its square vanishes modulo ϵ^2, and no two arrows compose.

PROPOSITION 20.8.3. This first-order deformation has no extension over $k[\epsilon]/(\epsilon^3)$ on the same graded module, even with uncurved A_∞ corrections having the same first-order Hochschild class. The obstruction is

$$\frac{1}{2}[\sigma f, \sigma f] = \sigma f^2 \neq 0 \quad \text{in } \mathrm{HH}^3_k(T_0).$$

Its cup square is zero.

Proof. The endomorphism square is nonzero: $f^2(v_0) = v_3$. Adding $\epsilon^2 g$ to the arrow differential leaves its square $\epsilon^2 f^2$ modulo ϵ^3.

Allowing higher-arity operations gives the second-order equation

$$d_{\mathrm{Hoch}} g + \tfrac{1}{2}[\sigma f, \sigma f] = 0.$$

The class of σf^2 is nonzero in Theorem 20.8.2: its internal degree is two, so it cannot be a scalar identity. The obstruction class depends only on the first-order cohomology class. No such correction exists. The cup square vanishes by (20.13). □

The original Koszul directions do integrate: $(xL_1 + yL_2)^2 = 0$ over $k[x, y]$. The insertion bracket distinguishes these directions from f. Under the scalar extension and empty-coefficient retraction constructed earlier, the nonzero obstruction remains nonzero in the scalar center of the extended algebra. Applying the retraction recovers it, including its bracket. Interpreting that image as a diagonal action still requires a map between the two action problems.

20.9 DUALITY CHANGES THE VARIANCE OF A CENTER

Return to the ordered collision category $(C, \star, K)$. Its convolution combines coefficients on disjoint subsets, and K is supported at the empty configuration. Let A be an augmented associative algebra with $A_\varnothing = k$, and let $C = BA$. The full coefficient duality of Chapter 19 is an anti-equivalence

$$(-)^\sharp : C^{\mathrm{op}} \simeq C^\sharp, \qquad (V \star W)^\sharp \simeq V^\sharp \star_\sharp W^\sharp.$$

It retains the entire pro-perfect diagram, including its transition maps. Write S for the inverse anti-equivalence and $E = C^\sharp$ for the transposed bar algebra.

A central operation is a bimodule map $V \star A \to A$. The coefficient-linear bar comparison carries it to $V \star C \to C$ in bicomodules. Transposition gives

$$E \longrightarrow V^\sharp \star_\sharp E.$$

Thus the coefficient has moved from the source to the target. This is the variance of a central coaction.

Theorem 20.9.1. Let $Z^{\mathrm{int}}(A)$ be the internal center and put $Q_A = Z^{\mathrm{int}}(A)^\sharp$. For every $W \in C^\sharp$,

$$R\,\mathrm{Hom}_{C^\sharp}(Q_A, W) \simeq R\,\mathrm{Hom}_{E\text{-}E}(E, W \star_\sharp E).$$

The object Q_A has the counital coalgebra structure dual to composition in the center. Its universal map is a coherent coaction in E-bimodules,

$$\gamma_A : E \longrightarrow Q_A \star_\sharp E.$$

Proof. The three natural equivalences are

$$\begin{aligned} R\,\mathrm{Hom}(Q_A, W) &\simeq R\,\mathrm{Hom}_C(\mathsf{S}W, Z^{\mathrm{int}}(A)) \\ &\simeq R\,\mathrm{Hom}^{C\text{-}C}((\mathsf{S}W) \star C, C) \\ &\simeq R\,\mathrm{Hom}_{E\text{-}E}(E, W \star_\sharp E). \end{aligned}$$

The middle step is the coefficient-linear bar comparison, and the last transposes the actual bicomodule maps. The identity of Q_A gives γ_A. Naturality says that the map associated with $q : Q_A \to W$ is $(q \star_\sharp \mathrm{I}_E)\gamma_A$.

Composition makes the evaluation of the internal center an associative unital action. Transpose its identities and coherent homotopies. They become the coassociativity and counit identities of γ_A. Every comparison respects the original collision diagram. □

An arbitrary map $q : Q_A \to W$ gives a bimodule map with coefficient W. Suppose W is counital. The displayed map is a counital coaction precisely when q is a coalgebra map. Explicitly, this requires

$$\epsilon_W q = \epsilon_{Q_A}, \qquad \Delta_W q = (q \star_\sharp q)\Delta_{Q_A},$$

together with their higher coherences. Apply the natural equivalence of the theorem to K, $W \star_\sharp W$, and all higher tensor powers to obtain this equivalence of coherent structures. The zero bimodule map, for example, fails the counit identity when $E \neq 0$.

An internal center of E would represent the different functor $W \mapsto R\,\mathrm{Hom}_{E\text{-}E}(W \star_\sharp E, E)$. The existence of Q_A does not settle that representability question.

Scalar centers and reflection

Scalarization tests only the tensor unit. For bounded holonomic coefficients, Verdier duality gives the algebra $\mathsf{K}_* A = \mathbb{D}BA$ for the transported product $\star_!$. There is an equivalence

$$Z^{\mathrm{sc}}(A) \simeq Z^{\mathrm{sc}}(\mathsf{K}_* A)^\sigma$$

of E_2-algebras, where σ reflects one multiplication direction.

To see the reflection, first use the coefficient-linear equivalence between bimodules over A and bicomodules over BA. The regular bimodule becomes the regular bicomodule. The derived cotensor of two bicomodules is computed by normalized terms $P \star \overline{C}^{\star q} \star Q$. At a configuration I, positivity of $\overline{C}$ leaves only $q \leq |I|$.

Verdier duality therefore transposes a finite limit into the finite realization that computes the balanced tensor product over $\mathbb{D}C$. It preserves the units and iterated interface compositions with their coherences. It reverses the vertical composition of maps. The two compositions that give the E_2 structure consequently have one direction reflected.

On cohomology this preserves the cup product and reverses the Gerstenhaber bracket. Multiplication by $(-1)^n$ in degree n identifies the resulting graded Gerstenhaber algebras abstractly. That observation alone supplies no chain-level E_2 comparison without the reflection. Internal polarity, scalar reflection, and opposite algebra formation remain separately typed constructions.

A point-supported test

Fix a point x of the curve. Put the same differential-module point direct image on each positive ordered chart, at the tuple $(x, \dots, x)$, with its Kashiwara collision identifications. Such systems, together with an arbitrary empty coefficient, form a category $D(k) \times D(k)$. Write an object as (U_0, U_1). Its tensor is

$$(U_0, U_1) \star (V_0, V_1) = \big(U_0 \otimes V_0,\ (U_0 \otimes V_1) \oplus (U_1 \otimes V_0)\big).$$

Two positive factors have zero product: the disjoint open misses their common support. The remaining collision identifications have no extra mapping data, since positive ordered sets and monotone surjections have the singleton as terminal object.

For a vector space M, take

$$A_M = (k, M[-1]), \qquad E_M = (BA_M)^\sharp = (k, M^\sharp).$$

The bar has only the empty and singleton terms. Here $M^\sharp$ is the pro-object of duals of finite-dimensional subspaces of M.

LEMMA 20.9.2. For $W(V) = (V^\sharp, 0)$, there is a natural equivalence

$$R\operatorname{Hom}_{E_M\text{-}E_M}(W(V) \star_\sharp E_M, E_M) \simeq V \oplus \operatorname{Hom}_k(M, V \otimes M)[-1].$$

For vector spaces V, M, these summands lie in degrees zero and one with zero differential.

Proof. Let $N = (0, M^\sharp)$. Since $N \star_\sharp N = 0$, the bimodule bar has two free terms. Its positive part is the split exact sequence

$$M^\sharp \xrightarrow{n \mapsto (n,-n)} M^\sharp \oplus M^\sharp \xrightarrow{(a,b) \mapsto a+b} M^\sharp.$$

The first free term has resolution degree -1.

Tensoring by $W(V)$ and applying free bimodule adjunction gives the two coefficient mapping objects V and $\operatorname{Hom}_k(M, V \otimes M)$. The left and right scalar actions agree, so the Hom differential is zero. The resolution degree places the second summand in degree one. □

For $M = k$, all relevant internal objects exist within these point categories, but they have different coefficients:

$Z^{\mathrm{int}}(A_k)$	$(k \oplus k[-1], k[-1])$
Q_{A_k}	$(k \oplus k[1], k[1])$
$Z^{\mathrm{int}}(E_k)$	$(k \oplus k[-1], k)$.

The positive test has no additional bar term, while the empty test acquires the shifted endomorphism term. Duality reverses its shift. Maps from the unit to the center and maps from the coHom to the unit still give the same scalar complex $k \oplus k[-1]$.

Proposition 20.9.3. For $M = \bigoplus_{r \geq 0} k e_r$, the algebra E_M has no internal center representing all pro-perfect coefficient tests. Its universal central coaction still exists.

Proof. A representing object would turn the restricted tests into $R\operatorname{Hom}_k(T, V)$, where T is its transposed empty coefficient. This functor preserves products in V. So would its degree-one cohomology.

The product comparison in the lemma would then be surjective:

$$\operatorname{Hom}_k\left(M, \left(\prod_{r \geq 0} k\right) \otimes M\right) \longrightarrow \prod_{r \geq 0} \operatorname{End}_k(M).$$

It misses the family $f_r(e_0) = e_r$, $f_r(e_j) = 0$ for $j > 0$. The value at e_0 on the left is a finite tensor sum. All its coordinates lie in one finite-dimensional subspace of M, whereas the e_r do not. This contradicts product preservation. The coaction follows independently from Theorem 20.9.1. □

20.10 A holonomic representing kernel

Finite-dimensional duality suggests a positive criterion: $\operatorname{Hom}(M, V \otimes N) \simeq \operatorname{Hom}(M \otimes N^*, V)$. The output N is dualized, while the test V remains arbitrary. For holonomic differential modules, direct image and Verdier duality give the corresponding relative kernel.

Work over $\mathbb{C}$ on smooth schemes of finite type. Let $\mathsf{H}(Y) = D^b_{\mathrm{hol}}(\mathcal{D}_Y)$. Use $P \otimes^! Q = \Delta^!_Y(P \boxtimes Q)$, with its density and shift conventions. The two required identities are

$$R\operatorname{Hom}_{\mathcal{D}_Y}(M, L) \simeq R\Gamma_{\mathrm{dR}}(Y, \mathbb{D}_Y M \otimes^! L),$$
$$p_*(P \otimes^! p^! L) \simeq p_* P \otimes^! L.$$

The first allows bounded coherent M and arbitrary L. The second is the projection formula for quasi-compact morphisms.

These are the Verdier pairing and projection formula of Drinfeld–Gaitsgory, *On some finiteness questions for algebraic stacks*, Sections 5.2.6 and 5.3.1. We use preservation of bounded holonomicity under inverse image, direct image, and duality. On affine space the first two are Miličić, *Lectures on Algebraic Theory of D-Modules*, Chapter I, Theorems 14.6 and 14.7.

Lemma 20.10.1. For $p : U \to Y$, $q : U \to Z$, and $M \in \mathsf{H}(U)$, $N \in \mathsf{H}(Z)$, put

$$P_{p,q}(M, N) = \mathbb{D}_Y\, p_*(\mathbb{D}_U M \otimes^! q^! N).$$

This object is bounded holonomic and, for every $V \in \mathrm{DMod}(Y)$,

$$R\operatorname{Hom}_{\mathcal{D}_U}(M, p^! V \otimes^! q^! N) \simeq R\operatorname{Hom}_{\mathcal{D}_Y}(P_{p,q}(M, N), V).$$

Proof. External product, inverse image, and duality preserve bounded holonomicity. So does algebraic direct image. The tensor product is the inverse image of an external product along the diagonal. This proves the first assertion.

For the mapping equivalence, apply the Verdier pairing, then the projection formula:

$$R\Gamma_{\mathrm{dR}}(U, \mathbb{D}_U M \otimes^! p^! V \otimes^! q^! N) \\ \simeq R\Gamma_{\mathrm{dR}}(Y, V \otimes^! p_*(\mathbb{D}_U M \otimes^! q^! N)).$$

Apply the pairing again to the bounded holonomic kernel. At no step was V required to be holonomic. Naturality gives the coherent composition comparisons. □

In particular, $f_! M = \mathbb{D}_Z f_* \mathbb{D}_Y M$ is left adjoint to $f^!$ on these source objects, against arbitrary targets. Properness is not required for this assertion.

Retaining the collision diagram

Call a coefficient system *configurationwise holonomic* if its empty coefficient is perfect and each nonempty coefficient is bounded holonomic. The bounds may depend on the configuration. Let $\mathcal{C}_{\mathrm{h}}$ be the ordered collision category formed from $\mathrm{Ind}\,\mathsf{H}(\mathscr{R}_I)$, with a $D(k)$ empty coefficient.

Bounded holonomic modules are compact in $\mathrm{DMod}(\mathscr{R}_I)$: they are bounded coherent, and the compactness statement is Drinfeld–Gaitsgory, Section 5.1.17. Thus the inclusion $b_I : \mathrm{Ind}\,\mathsf{H}(\mathscr{R}_I) \to \mathrm{DMod}(\mathscr{R}_I)$ is fully faithful, preserves colimits, and has a right adjoint r_I. For every collision morphism f,

$$r_Y f^! \simeq f^! r_Z.$$

Indeed, testing both sides on bounded holonomic H gives

$$R\,\mathrm{Hom}(H, f^! L) \simeq R\,\mathrm{Hom}(f_! H, L) \simeq R\,\mathrm{Hom}(f_! H, r_Z L).$$

These tests generate the ind-category. The equivalences are adjunction mates of the coherent composition maps.

On ind-holonomic modules, $f^!$ preserves colimits and is also a right adjoint to the ind-extension of $f_!$, so it preserves limits. The coherent limit category $\mathcal{C}_{\mathrm{h}}$ is therefore presentable with componentwise limits and colimits. Its inclusion b has right adjoint r, and evaluation e_I has a left adjoint $e_{I,!}$.

PROPOSITION 20.10.2. For configurationwise holonomic M, N, there is an object $\mathsf{T}(M, N) \in \mathcal{C}_{\mathrm{h}}$ such that, for every $V \in \mathcal{C}$,

$$R\,\mathrm{Hom}_{\mathcal{C}}(b\mathsf{T}(M, N), V) \simeq R\,\mathrm{Hom}_{\mathcal{C}}(M, V \star N).$$

Proof. For a split $I = A \sqcup B$, let j be its disjoint open immersion and p, q the projections from that open to $\mathscr{R}_A, \mathscr{R}_B$. The split summand of the mapping object on the right is

$$R\,\mathrm{Hom}_{\mathcal{D}_{\mathscr{R}_A}}(P_{p,q}(j^! M_I, N_B), V_A).$$

This follows from open adjunction and the preceding lemma. The kernel is bounded holonomic, so these maps depend only on $r_A V_A$.

Define the finite sum

$$T_I = \bigoplus_{I = A \sqcup B} e_{A,!} P_{p,q}(j^! M_I, N_B).$$

Its mapping functor gives the mapping object at I. The collision maps induce, by enriched Yoneda, the contravariant coherent diagram of the T_I. Take its colimit, including the separate empty term, to obtain $\mathsf{T}(M, N)$.

Mapping out of the colimit gives the required coherent limit over configurations. The adjunction $b \dashv r$ extends the equivalence from ind-holonomic tests to all V. The construction retains the collision homotopies. □

The center of the transposed algebra

Let A be configurationwise holonomic, augmented, with $A_{\varnothing} = k$. For $s \geq 0$ form $T_s = \mathsf{T}(A^{\star s}, A)$, where $A^{\star 0} = K$. Finite convolution powers remain configurationwise holonomic. The two-sided bar gives a simplicial object $T_\bullet$, with faces and degeneracies induced by multiplication and unit. Put $T_A = b|T_\bullet|$.

Theorem 20.10.3. For $E = (BA)^\sharp$, the object $T_A^\sharp$ represents the internal center on every pro-perfect probe:

$$R\operatorname{Hom}_{C^\sharp}(W, T_A^\sharp) \simeq R\operatorname{Hom}_{E\text{-}E}(W \star_\sharp E, E).$$

The equivalence includes evaluation, the internal E_2 operations, and the ordered collision coherences.

Proof. The augmented two-sided bar contracts by its unit. Free bimodule adjunction and the proposition give

$$\begin{aligned} R\operatorname{Hom}_{A\text{-}A}(A, V \star A) &\simeq \operatorname*{Tot}_s R\operatorname{Hom}(A^{\star s}, V \star A) \\ &\simeq \operatorname*{Tot}_s R\operatorname{Hom}(bT_s, V) \\ &\simeq R\operatorname{Hom}(T_A, V). \end{aligned}$$

This keeps the entire realization. It assumes neither a uniform arity bound nor bounded holonomicity of T_A itself.

Set $V = \mathsf{S}W$. Coefficient-linear bar comparison and transposition identify the first line with $R\operatorname{Hom}_{E\text{-}E}(W \star_\sharp E, E)$. The last line becomes $R\operatorname{Hom}_{C^\sharp}(W, T_A^\sharp)$. These identifications are natural for all probes.

The identity of T_A corresponds to a universal map $\rho : A \to T_A \star A$. Its iteration gives the coproduct of T_A, and the identity of A gives the counit. There is a second composition from $\rho \star_A^{\mathbb{L}} \rho$, using

$$(T_A \star A) \star_A^{\mathbb{L}} (T_A \star A) \simeq (T_A \star T_A) \star A.$$

Convolution preserves the bar realization of the balanced product. The two compositions share a unit and satisfy coherent interchange. They give the E_2-coalgebra operations on T_A, which transpose to the center operations on $T_A^\sharp$. The same adjunctions retain collision coherences. □

The universal central coaction and this internal center now coexist. They solve the two mapping problems with opposite variance. Neither construction identifies the actor of an independently specified bulk theory.

20.11 Free generators and their support

A two-term bimodule resolution computes the center of a free algebra without suppressing the support of its generators. First work in the word category $(\mathcal{W}_X, \otimes_{\mathrm{cut}}, K)$, where the tensor uses consecutive cuts. For an object M, put

$$A = T_{\mathrm{cut}}(M) = \bigoplus_{n \geq 0} M^{\otimes_{\mathrm{cut}} n}, \qquad M^{\otimes_{\mathrm{cut}} 0} = K.$$

This is an algebraic direct sum. Multiplication concatenates the factors, with no permutation.

PROPOSITION 20.11.1. There is a cofiber sequence of bimodules

$$A \otimes_{\mathrm{cut}} M \otimes_{\mathrm{cut}} A \xrightarrow{\partial} A \otimes_{\mathrm{cut}} A \xrightarrow{\mu} A,$$

where $\partial(a \otimes m \otimes b) = am \otimes b - a \otimes mb$. Consequently the scalar center is

$$Z_{\mathrm{word}}(A) \simeq \mathrm{fib}\big(R\,\mathrm{Hom}_{\mathcal{W}_X}(K, A) \longrightarrow R\,\mathrm{Hom}_{\mathcal{W}_X}(M, A)\big).$$

On a degree-zero coefficient $c : K \to A$, the arrow is $m \mapsto mc - cm$.

Proof. Fix tensor length n. The middle term has one copy of the word coefficient for each cut $c_0, \dots, c_n$. The first term has one copy for each marked factor $e_0, \dots, e_{n-1}$. The maps are

$$\partial e_i = c_{i+1} - c_i, \qquad \mu c_i = 1.$$

This is the augmented interval complex. The section $1 \mapsto c_0$ has homotopy $h(c_i) = e_0 + \cdots + e_{i-1}$. Tensoring with the word coefficient preserves the contraction, including its internal differential. Summing over n proves the cofiber sequence. Apply the derived free bimodule adjunction to obtain the fiber formula. □

For a differential module N, let iN be its positive diagonal insertion, with coefficient $\Delta_{n*}N$ on a word of length n. Kashiwara equivalence identifies all these coefficients with N. The collision index has the singleton as terminal object, so

$$R\,\mathrm{Hom}_{\mathcal{W}_X}(iN, G) \simeq R\,\mathrm{Hom}_{\mathcal{D}_X}(N, G_1).$$

For $A = T_{\mathrm{cut}}(iN)$, $A_\varnothing = k$ and $A_1 = N$. The scalar commutator is zero. Thus

$$Z_{\mathrm{word}}(T_{\mathrm{cut}}(iN)) \simeq k \oplus R\,\mathrm{Hom}_{\mathcal{D}_X}(N, N)[-1].$$

Any internal self-extension of N remains in this formula, shifted by one center degree.

Example 20.11.2 *(Two diagonal generators).* On $X = \mathbb{A}^1_k$, take $N = \omega_X \otimes_k k^2$ with trivial connection. As a right differential module, $\omega_X = D/\partial_z D$. Its free resolution has differential $P \mapsto \partial_z P$. Applying Hom into ω_X gives $k[z] \xrightarrow{-\partial_z} k[z]$. Polynomial integration therefore gives

$$H^0 Z_{\mathrm{word}}(A) = k, \qquad H^1 Z_{\mathrm{word}}(A) = M_2(k),$$

and no further groups.

A matrix acts by the derivation that applies it in each tensor position. The bracket is the matrix commutator. No nonzero matrix is an inner boundary, because the only degree-zero coefficients are scalars. Positive cup products vanish, since $H^2 = 0$.

On $\mathbb{G}_m$, the derivative on $k[z, z^{-1}]$ has the additional cokernel generated by z^{-1}. The same two generators then give $H^2 = M_2(k)$ as well, from the de Rham residue dz/z.

Placing the two generators at the empty word gives a different result. Let $M = K \otimes_k (kx \oplus ky)$. Then $A = K \otimes_k F$ for the ordinary free algebra $F = k\langle x, y\rangle$, and its center complex is

$$F \longrightarrow F \oplus F, \qquad c \longmapsto (xc - cx, yc - cy).$$

It has $H^0 = k$ and $H^1 = \mathrm{Der}(F)/\mathrm{Inn}(F)$, with no higher groups. The degree-one space is infinite dimensional: the derivations $x \mapsto y^m$, $y \mapsto 0$ are independent modulo inner derivations. Abelianization kills commutators and sends their first components to the independent polynomials y^m.

The difference comes from support. A diagonal generator can only take values compatible with the diagonal mapping adjunction. A generator at the empty word can take values in the entire ordinary free algebra.

Perfect differential modules

The same interval resolution applies to the full collision convolution. Let M be a perfect right $\mathcal{D}_X$-complex on $X = \mathbb{A}^1_{\mathbb{C}}$. Let G_M be its positive diagonal family, with one small-diagonal direct image of M on each ordered chart. Define

$$\mathfrak{A}_M = T_\star(G_M), \qquad \mathfrak{C}_M = K \oplus G_M[1], \qquad \mathfrak{E}_M = \mathfrak{C}_M^\sharp.$$

The reduced coproduct of $\mathfrak{C}_M$ is zero. Its cobar is $\mathfrak{A}_M$, and its map to $B\mathfrak{A}_M$ sends $g[1]$ to the singleton bar word. Bar–cobar comparison identifies these coalgebras. At a fixed configuration the word filtration is finite.

Write $[G, N]_\star$ for internal Hom and $B_M = R\,\mathrm{Hom}_{\mathcal{D}_X}(M, M)$ for the scalar endomorphism complex. The interval resolution, tested on every coefficient, gives

$$Z^{\mathrm{int}}(\mathfrak{A}_M) \simeq \mathrm{fib}\Big(\mathfrak{A}_M \xrightarrow{\mathrm{ad}} [G_M, \mathfrak{A}_M]_\star\Big).$$

Here ad is adjoint to the graded commutator. Transposing this fiber gives the coHom

$$Q_{\mathfrak{A}_M} \simeq \mathrm{cofib}\Big([G_M, \mathfrak{A}_M]^\sharp_\star \longrightarrow \mathfrak{A}_M^\sharp\Big),$$

with its universal central coaction.

Lemma 20.11.3. For every complex V over $\mathbb{C}$,

$$R\,\mathrm{Hom}_{\mathfrak{A}_M\text{-}\mathfrak{A}_M}(\mathfrak{A}_M, (VK) \star \mathfrak{A}_M) \simeq V \oplus (V \otimes B_M)[-1].$$

Proof. Apply the free bimodule resolution to the indicated target. The resolution-degree-zero term is V. The two endpoint actions agree, so the map to the other term is zero.

Only the length-one part of $\mathfrak{A}_M$ contributes to maps from G_M. Every longer word is extended by j_* from an open missing the small diagonal. Open adjunction makes its mapping complex from G_M zero. The empty word also receives no map from this positive object. These vanishings respect every collision map.

Kashiwara equivalence and the contractible index of ordered surjections identify the remaining maps with $R\,\mathrm{Hom}_{\mathcal{D}_X}(M, V \otimes M)$. On the affine line, a perfect M is a retract of a finite complex of finite free modules. Thus V can be moved through this finite Hom complex, giving $V \otimes B_M$. The bar degree shifts it by one. □

Theorem 20.11.4. If $\mathfrak{E}_M$ has an internal center on all pro-perfect probes, then B_M is perfect over $\mathbb{C}$. In particular, $\mathfrak{E}_{\mathcal{D}_X}$ has no such center, although each configuration coefficient is the dual of a perfect differential-module complex.

Proof. Coefficient-linear bar comparison and transposition turn the lemma into the tests of a proposed center on $W(V) = (VK)^\sharp$. Representability would make $V \mapsto V \oplus (V \otimes B_M)[-1]$ preserve products. Its second summand would also preserve them.

The complex B_M is bounded. If $H^q(B_M)$ contained independent vectors $b_0, b_1, \ldots$, the family (b_i) would not lie in the image of

$$\left(\prod_{i\ge 0} \mathbb{C}\right) \otimes H^q(B_M) \longrightarrow \prod_{i\ge 0} H^q(B_M).$$

Every tensor on the left uses one finite-dimensional span for all coordinates. Products of vector spaces are exact, so this is also the product obstruction on cohomology. Every $H^q(B_M)$ must therefore be finite dimensional.

For $M = \mathcal{D}_X$, the complex B_M is the Weyl algebra $\mathbb{C}\langle z, \partial\rangle/(\partial z - z\partial - 1)$ in degree zero. The sequence ∂^i violates the common finite differential-order bound.

The coefficient perfectness is a different statement. On a smooth chart the direct image of the free module along a small diagonal has a finite Koszul resolution in its normal coordinates. The normal coordinate operators form a regular sequence by Weyl normal ordering. Finite sums, shifts, cones, retracts, and the finite-cell dual preserve perfectness. This proves the coefficient assertion. □

For scalar probes alone the condition is also sufficient. If B is a bounded complex over k, the functor $V \mapsto V \oplus (V \otimes B)[-1]$ is corepresented by $k \oplus B^\vee[1]$ exactly when B is perfect. Finite duality proves sufficiency, and the same product argument proves necessity. Extending this scalar representation to all collision probes still requires compatible geometric kernels.

The scalar center of $\mathfrak{A}_{\mathcal{D}_X}$ remains $k \oplus B_M[-1]$. A right-linear endomorphism a extends to a derivation D_a of the free convolution algebra. Its bracket is

$$[D_a, D_b] = D_{ab-ba}, \qquad [D_\partial, D_z] = D_1 \neq 0.$$

Here D_1 multiplies a word by its length. It is not an inner boundary, since degree-zero coefficients are scalars.

Coefficient transposition sends z to z and ∂ to $-\partial$. Hence the transposed commutator is -1, displaying the reflected bracket sign on a nonzero class.

These free-algebra examples have no prescribed field realization or normalized local vacuum. To apply the same test to a field algebra, its actual relations must enter the endomorphism and representability calculation.

20.12 Heisenberg states and the word center

The Heisenberg residue algebra has the entire vacuum module at one position. It is therefore different from a free word algebra on two diagonal current generators. The vacuum, products of currents, and all translations must remain in the coefficient module.

Let k have characteristic zero. Take $B = k[\kappa]$, or a field over k with a specified level κ. Let $g = (g_{ij})$ be a nondegenerate symmetric form on a rank-two current space. The even currents satisfy

$$[a^i_\lambda a^j] = \kappa g_{ij} \lambda \mathbf{1}.$$

Their state module and translation are

$$V_\kappa = B[x^i_m : i = 1, 2,\ m \ge 1], \qquad x^i_m = a^i_{-m}\mathbf{1}, \qquad T = \sum_{i,m\ge 1} m x^i_{m+1} \partial_{x^i_m}.$$

Every state is a polynomial in finitely many variables. There is no completion in state weight. The right differential module $\mathscr{V}_\kappa$ has action

$$(f\,dz \otimes v)\partial_z = -f'\,dz \otimes v - f\,dz \otimes Tv.$$

All maps are relative to B.

Let $C(V_\kappa)$ be the word residue coalgebra and $A_\kappa = \Omega_{\mathrm{cut}} C(V_\kappa)$. The equation $xy - yx = \kappa \mathbf{1}_K$ cannot present this algebra on positive generators at nonzero level. Applying the augmentation would give $0 = \kappa$. The Heisenberg bracket instead takes values in the positive local vacuum line inside $\mathscr{V}_\kappa$. That line is distinct from the empty-word unit.

The full degree-one kernel

Put $\nabla = \partial_z + T$. For a pair of polynomials $P = (P_1, P_2) \in B[z, x_m^i]^2$, define

$$\mathcal{K}_{ij}(P) = \sum_{s=1}^{2} \sum_{m \geq 1} \frac{1}{(m-1)!} \Big(g_{is} \lambda^m \partial_{x_m^s} P_j - g_{js} (-\lambda - \nabla)^m \partial_{x_m^s} P_i \Big). \tag{20.15}$$

The operator ∇ in the second term acts on its coefficient to the right. All sums are finite. A *chiral derivation* is a differential-module endomorphism δ such that

$$\delta r_{12} = r_{12}(\delta \boxtimes 1 + 1 \boxtimes \delta)$$

on the entire localized two-input coefficient, for every multiplier.

Theorem 20.12.1. The full scalar word center satisfies

$$\begin{aligned} H^q Z_{\mathrm{word},B}(A_\kappa) &= 0 \quad (q < 0), \\ H^0 Z_{\mathrm{word},B}(A_\kappa) &= B, \\ H^1 Z_{\mathrm{word},B}(A_\kappa) &\cong \{P \in B[z, x_m^i]^2 : \kappa \mathcal{K}_{ij}(P) = 0 \text{ for all } i, j\}. \end{aligned}$$

The degree-one class has $\delta_z a^i = P_i(z)$ and acts by the sum of δ over the state blocks of every partition. There is no quotient by non-scalar inner derivations.

Proof. Use the field retract in the normalized two-sided bar. A map from a partition with r blocks to one with t blocks vanishes unless the source partition refines the target. Otherwise some coordinate difference annihilates the source and is inverted on the target. Open adjunction gives zero.

For q consecutive positive factors, the source shift is $r - q$, the target shift is $t - 1$, and Hochschild length adds q. Thus a coefficient Ext class of degree e has total degree $e + r - t + 1$. This is at least one. For $q \geq 2$, a cut source can map to an indecomposable target only after a merger, so the degree is at least two.

The uniform lower bounds survive the complete products and coherent collision limits. The length-zero term is B, with zero scalar commutator. This proves the first two claims.

In degree one only ordinary unary maps on equal partitions remain. Their one-block restriction is one map $\delta : \mathscr{V}_\kappa \to \mathscr{V}_\kappa$. The binary equation says precisely that it is a chiral derivation. The simple-pole two-vacuum input also gives $\delta \mathbf{1} = 0$.

Subtract its blockwise sum. Induct on the number of blocks. Every merger of an indecomposable partition remains indecomposable, so commutation with residues makes every merger of the difference

zero. The Heisenberg field map is faithful on polynomial states, with arbitrary spectators and Laurent multipliers. Consequently the difference is zero.

For completeness, this faithfulness can be checked before quantization. In the associated graded by current number, the field is $(e^{uT}a)b$. On any finite collection of jets, sufficiently high translated jets give an invertible coefficient minor: the binomial coefficient matrix has nonzero Vandermonde leading determinant. Hence the two-field map is injective.

Wick contractions lower current number by two, so the same filtered argument proves injectivity at every level. Over $k[\kappa]$, torsion-freeness reduces to the fraction field. Multiplying by every power of the separation tests every field coefficient through its residue.

It remains to characterize δ. The differential-module and field identities are

$$\begin{aligned} \partial_z \delta_z + [T, \delta_z] &= 0, \\ \delta_z Y(v,u)w &= Y(\delta_{z+u}v, u)w + Y(v,u)\delta_z w. \end{aligned}$$

The positive current modes are $a^i_m = m\kappa \sum_s g_{is}\partial_{x^s_m}$, while $a^i_0 = 0$. Applying δ to the current bracket gives exactly $\kappa\mathcal{K}_{ij}(P) = 0$. Skew symmetry on a moving coefficient replaces T by ∇, explaining the second term of (20.15).

Conversely, put

$$\begin{aligned} Q^i_n(P;z) &= \sum_{\ell \ge 0} \frac{(\partial_z^\ell P_i(z))_{(n+\ell)}}{\ell!}, \\ \delta_P \mathbf{1} &= 0, \\ \delta_P(x^i_m v) &= x^i_m \delta_P(v) + Q^i_{-m}(P;z)v. \end{aligned} \tag{20.16}$$

Over $B[\epsilon]/(\epsilon^2)$, the kernel equation says that $Y(a^i,u) + \epsilon Y(P_i(z+u),u)$ satisfies the Heisenberg commutators. Its nonnegative modes kill the vacuum. The induced vacuum module therefore makes the recursion independent of the order of negative modes. Commuting through normally ordered differentiated fields proves the derivation identity on every state, including all contractions.

Finally

$$[T, Q^i_n] = -nQ^i_{n-1} - \partial_z Q^i_n$$

shows that $\partial_z\delta_P + [T, \delta_P]$ commutes with the currents and kills the vacuum, hence vanishes. The blockwise sum commutes with cuts, residues, and the first-cut contraction, so it is a cocycle for all transferred operations. The degree bounds leave no additional degree-one classes or inner identifications. □

Constant linear images $P_i = \sum_j M_{ij}x^j_1$ give the subspace

$$\kappa(Mg + gM^{\mathrm{t}}) = 0.$$

For the hyperbolic form this has dimension one at a nonzero field-valued level, and dimension four at level zero. It is only a subspace of the full kernel. For every level, $P_i = c_i(z)$ gives

$$\delta_P x^i_m = \frac{c_i^{(m-1)}(z)}{(m-1)!},$$

and $P_i = x_2^i$ gives translation. At level zero every pair of differential polynomials is allowed, with

$$\delta_P = \sum_{i,m\geq 1} \frac{\nabla^{m-1} P_i}{(m-1)!} \partial_{x_m^i}.$$

At nonzero level, reconstructing nonlinear current images requires the quantum terms in (20.16). For one current, the image $P = x_1 x_2$ is allowed and

$$\delta_P(x_1^2) = 2x_1^2 x_2 + \kappa x_4.$$

A polynomial derivation of the state multiplication alone would miss the last term.

A central state also need not extend to a center endomorphism of the word cobar. The regular right module is free on K, so its degree-zero right-module endomorphisms are $\mathrm{Hom}(K, A) = B$. If such an endomorphism restricts to $c_{(-1)}$ on states, evaluation on the vacuum forces $c = b\mathbf{1}$. At level zero every state is vertex-central, but only these scalar states extend in this manner. Applying a scalar separately to each block is a different assignment. The scalar 2 would act by 4 on two blocks, whereas 2 id acts by 2.

The level variation

In the strict word cobar, the state modules and concatenation are independent of κ. Only the residue differential D_κ varies. Define the B-linear family of cochains $L_\kappa = \partial_\kappa D_\kappa$. The derivative here defines a family of maps. A cochain does not differentiate its scalar coefficients. Let N count current factors in every state block.

Differentiating $D_\kappa^2 = 0$ and the differential derivation identity shows that L_κ is a degree-two Hochschild cocycle. A Wick term with c contractions contains κ^c and lowers current number by $2c$. Hence

$$bN = 2\kappa L_\kappa.$$

At a nonzero field-valued level the class is zero.

At level zero, its value on two holomorphic currents has a nonzero double-pole vacuum coefficient. Every undeformed field is regular there, so the binary boundary of a state endomorphism has zero such coefficient. The degree bounds in the preceding proof exclude additional derived primitives. Thus the class is nonzero at zero level. Over $k[\kappa]$, reduction modulo κ detects every coefficient not divisible by κ. Its annihilator is exactly (κ).

Varying the symmetric pairing similarly gives $(B/(\kappa))^3$ in degree two for rank two. Higher poles give further classes, as follows.

20.13 Degree two on words and collision coefficients

Let $\mathcal{V}_\pi$ denote the localized partition coefficient with its normal differential operators and residue maps $r_{ij} : \mathcal{V}_\pi \to \mathcal{V}_{\pi/(i\sim j)}$. We first give a presentation of the entire word degree-two group. It permits one component for every finite word.

Let $\mathfrak{G}_B$ consist of differential-module maps $\psi_\pi : \mathcal{V}_\pi \to \mathcal{V}_\pi$ compatible with monotone collisions, zero at the empty word, and additive at every admissible cut:

$$\psi_{\pi'|\pi''} = \psi_{\pi'} \boxtimes 1 + 1 \boxtimes \psi_{\pi''}.$$

Let $\mathfrak{R}_B$ consist of maps $h_{\pi;ij} : \mathcal{V}_\pi \to \mathcal{V}_{\pi/(i\sim j)}$ with the residue cut-coderivation and collision-base-change identities, and the following linearized residue equations:

$$\begin{aligned} 0 &= h_{kl}r_{ij} + r_{kl}h_{ij} - h_{ij}r_{kl} - r_{ij}h_{kl}, \\ 0 &= h_{(ij),l}r_{ij} + r_{(ij),l}h_{ij} - h_{i,(jl)}r_{jl} - r_{i,(jl)}h_{jl} \\ &\quad + h_{j,(il)}r_{il} + r_{j,(il)}h_{il}. \end{aligned} \tag{20.17}$$

The first equation is for disjoint pairs and the second for $i < j < l$. Each subscript after a merger denotes the surviving root. No skew-symmetry or fixed-vacuum condition is added to h. Define

$$(\partial\psi)_{\pi;ij} = r_{ij}\psi_\pi - \psi_{\pi/(i\sim j)}r_{ij}.$$

THEOREM 20.13.1. For the full Heisenberg word residue algebra,

$$H^2 Z_{\mathrm{word},B}(A_\kappa) \cong \mathfrak{R}_B/\partial\mathfrak{G}_B.$$

Every class is represented on the unchanged partition modules by replacing r_{ij} with $r_{ij} + \epsilon h_{\pi;ij}$. Two such representatives give the same class exactly when they differ by $\partial\psi$.

Proof. The two displayed equations are the first-order terms in the disjoint-merger and three-block residue identities. The cut and collision identities make the new differential a compatible coderivation. Thus each h defines a residue coalgebra over the dual numbers and a Hochschild class of its cobar. Conjugating coefficients by $1 - \epsilon\psi$ gives the stated boundary.

To account for derived classes, identify the bar of the field retract with the cofree cut coalgebra. Its degree-two coderivations have only two possible components: ordinary maps for a single merger and Ext^1 between equal partitions. Represent the latter by flat module extensions E_π over the dual numbers. The cocycle equation says precisely that residues lift between these extensions and that the lifted differential squares to zero.

There are no extra attaching maps between pieces separated by two or more mergers: their internal degrees would be negative. Negative Ext vanishes, and the opposite support direction vanishes by open adjunction. The same argument removes higher attaching homotopies. Degree-zero coproducts allow only equal-partition cut maps, which remain isomorphisms over the dual numbers.

The one-position extension is trivialized by its vacuum attachment. Explicitly, if its connection perturbation is S, the lifted vacuum identity gives $(\partial_z + T)F_0 = S\mathbf{1}$. This is the condition that kills its class in $\mathrm{coker}(\partial_z + T)$. A splitting of the free complement then gives the module trivialization. Collision compatibility trivializes every one-block coefficient.

For a larger indecomposable partition, split off position one from its block. The remaining indecomposable intervals have smaller word size, so induction trivializes their extensions. The lifted residue that merges the new singleton back, restricted to its horizontal vacuum, gives a splitting of the target extension. The simple-pole vacuum residue is surjective on principal parts, so this splitting is an isomorphism modulo ϵ and on both steps of the ϵ-filtration. It is therefore an isomorphism.

Choose the trivializations first on block patterns with no adjacent positions in the same block. Every other pattern contracts to one of these by collapsing maximal constant intervals. Kashiwara equivalence transports the choice, and the terminal contraction removes additional descent choices. Extend across cuts by tensor products. All extensions are now trivial, leaving just the single-merger changes h.

Finally, a degree-one primitive can have only ordinary equal-partition components. Its coderivation and collision conditions are exactly those defining $\mathfrak{G}_B$. Its boundary is $\partial\psi$. The constructions are finite at each word size, and their product retains the full quotient. □

All finite central-polynomial variations

Let $\boldsymbol{Q}_B$ be the space of polynomial bilinear forms

$$Q(\lambda) = \sum_{d \geq 0} C_d \lambda^d, \qquad Q(\lambda) = -Q(-\lambda)^{\mathrm{t}}.$$

Thus $C_d^{\mathrm{t}} = (-1)^{d+1} C_d$: even powers have alternating coefficients and odd powers have symmetric coefficients. Only finite polynomials are allowed. Over $B[\epsilon]/(\epsilon^2)$, consider

$$[a_\lambda^i a^j]_\epsilon = (\kappa g_{ij} \lambda + \epsilon Q_{ij}(\lambda))\mathbf{1}.$$

The state module, vacuum, and translation are fixed. No requirement of conformal weight preservation is imposed.

This is a flat field deformation on every polynomial state. To construct it explicitly, write $(p)_d = p(p-1)\cdots(p-d+1)$ and set

$$J_p^i = x_{-p}^i \cdot \quad (p < 0),$$

$$J_p^i = \kappa p \sum_j g_{ij} \partial_{x_p^j} + \epsilon \sum_{d=0}^{p} (p)_d \sum_j (C_d)_{ij} \partial_{x_{p-d+1}^j} \qquad (p \geq 0).$$

The first term is zero for $p = 0$. The commutator is

$$[J_p^i, J_q^j] = \kappa p g_{ij} \delta_{p+q,0} + \epsilon \sum_{d \geq 0} (p)_d (C_d)_{ij} \delta_{p+q,d-1}.$$

For the reversed order, use $(d-1-q)_d = (-1)^d (q)_d$. When both indices are nonnegative and $p + q = d - 1$, the factor $(p)_d$ is zero. This checks all mode pairs.

The commutator is central and has finite pole order. Moreover, $[T, J_p^i] = -p J_{p-1}^i$ and $J^i(u)\mathbf{1} = e^{uT} a^i$. The first identity uses $(p)_d (p-d) = p(p-1)_d$.

Assign to a polynomial state the normally ordered product of its differentiated current fields. Moving nonnegative modes through negative modes gives a finite Wick expansion. The commutators are central, so regrouping three collections of modes gives the same expansion. This proves the field identities on all states, with the prescribed vacuum and translation.

Differentiate all residues to obtain a word cocycle Γ_Q^{w}. For collision coefficients, put $L = \iota_{\mathrm{ord}} \mathscr{V}_\kappa$. Its residue bracket is a Lie bracket for $\star$, and the collision cobar is the envelope $U_\star L$. Differentiating its bracket gives the Chevalley cocycle

$$\Gamma_Q^{\mathrm{c}} : \Lambda_\star^2 L \longrightarrow L \longrightarrow U_\star L.$$

Its differential is the linearized Jacobi identity.

Write $Q = C_0 + Q_+$ with $Q_+(0) = 0$. Set

$$M(t) = -\frac{Q_+(-t)}{2t} g^{-1}, \qquad P(z) = M(T)a + z C_0 g^{-1} a,$$

and define the polynomial state endomorphism N_Q by

$$N_Q \mathbf{1} = 0, \qquad N_Q x_m^i = \frac{(\partial_z + T)^{m-1} P_i(z)}{(m-1)!}.$$

Extend by the polynomial derivation rule and then by sums over blocks. Sesquilinearity gives

$$\kappa\lambda\big(M(-\lambda)g + gM(\lambda)^{\mathrm{t}}\big) = \kappa Q_{+}(\lambda).$$

For the constant alternating term, the two moving points give

$$\frac{\kappa}{u^{2}}\big((z+u)C_{\mathrm{o}} + zC_{\mathrm{o}}^{\mathrm{t}}\big) = \frac{\kappa C_{\mathrm{o}}}{u}.$$

The all-state Wick construction then proves

$$bN_{Q} = \kappa\Gamma_{Q}^{\mathrm{w}}, \qquad d_{\mathrm{CE}}N_{Q} = \kappa\Gamma_{Q}^{\mathrm{c}}. \tag{20.18}$$

At level zero, evaluating on holomorphic currents gives the independent principal parts

$$\sum_{d} d!(C_{d})_{ij}[u^{-d-1}]\mathbf{1}.$$

All undeformed fields are regular, so every incoming ordinary binary boundary vanishes on this input.

The degree bounds exclude derived alternatives. In the collision Chevalley filtration, a total-degree-one primitive is an ordinary map $L \to U_{\star}L$. Support forces its values into L. The same word bound was used above. Thus the kernel of $Q \mapsto [\Gamma_{Q}]$ is exactly $\kappa\boldsymbol{Q}_{B}$ on either carrier.

For $Q = \lambda^{3}H$ with H symmetric, the primitive is $P = \frac{1}{2}Hg^{-1}T^{2}a$. The mode variation is $p(p-1)(p-2)H$ and its value at $p = 3$ is $6H$, the coefficient of the fourth-order pole. The falling factorial is required by translation.

Products of the full derivation kernel

Let $\mathfrak{d}_{B}$ be the entire kernel in Theorem 20.12.1. Let Z_{w} be the word scalar center and Z_{c} the scalar center of $U_{\star}L$. The collision Chevalley resolution identifies its degree-one group with the same chiral derivations. Support puts a map $L \to U_{\star}L$ in the primitive part. Its cocycle equation is the chiral derivation identity. The empty term is B, so it introduces no inner quotient.

THEOREM 20.13.2. For $\alpha = \mathrm{w}, \mathrm{c}$ there are injections

$$\Lambda_{B}^{2}\mathfrak{d}_{B} \oplus \boldsymbol{Q}_{B}/\kappa\boldsymbol{Q}_{B} \longrightarrow H^{2}Z_{\alpha},$$

given by derivation cup products and the classes Γ_{Q}^{α}. Their common parameter module identifies these submodules canonically. On the exterior summands, the identification preserves the degree-one Gerstenhaber action.

Proof. External products of independent differential-module endomorphisms at two distinct positions are independent. Over a field, finitely many evaluations on states and output coordinate functionals detect any finite independent family. Apply them separately in the two variables. Over $k[\kappa]$, the derivation module and its tensor square are torsion-free, since this ring is a principal ideal domain. The fraction-field argument then proves the assertion. Antisymmetrization embeds the exterior square.

For words, the two-singleton component of a cup is $r_{12}(\delta \boxtimes \eta)$. Its sum with the exchanged component is

$$r_{12}(\delta \boxtimes \eta - \eta \boxtimes \delta).$$

This detector vanishes on boundaries with zero unary component. Faithfulness of the field map with all Laurent multipliers, followed by the external-product injection, detects every finite wedge sum. Central-polynomial variations have zero detector because their local brackets remain skew-symmetric.

For collision coefficients, the Chevalley cup is

$$(\delta \smile \eta)(x, y) = \delta x\, \eta y - \delta y\, \eta x.$$

Its PBW-degree-two symbol on separated positions is $\delta \boxtimes \eta - \eta \boxtimes \delta$. A degree-two boundary with zero unary component comes from an ordinary map $L \to U_\star L$. Support forces this map into L, so its boundary has zero separated PBW-degree-two symbol. The cocycle Γ^{c}_Q also takes values in L.

The symbol therefore detects the wedge coefficient of any relation. The pole detector and (20.18) then detect its central coefficient.

The constructions retain the current space, pairing, vacuum, coordinate, and state module and commute with changes of current basis. Finally,

$$[\theta, \delta \smile \eta] = [\theta, \delta] \smile \eta + \delta \smile [\theta, \eta]$$

on both carriers. This is the natural action on $\Lambda^2 \mathfrak{d}_B$. □

At nonzero field-valued level the central-polynomial summand vanishes. At level zero it is the entire Q_k. In particular, cubic central polynomials lie outside the span of pairing variations and all derivation cups. The theorem does not classify the remaining word residue quotient or extend the comparison to it. Nor does it claim that arbitrary nonlinear derivations preserve the central-polynomial summand.

20.14 The collision center at every configuration

The internal collision center admits a stronger support theorem. Work over a characteristic-zero field on $X = \mathbb{A}^1_k$. Let V be an ordinary unital vertex algebra, $\mathscr{V}$ its right differential module, and

$$L = \iota_{\mathrm{ord}} \mathscr{V}, \qquad U = U_\star L, \qquad C = \mathrm{CE}_\star(L), \qquad Z = Z^\star(U).$$

The envelope and its coefficients lie in the heart. Its center uses the complete Chevalley cochains, equivalently the twisted bar Hom.

Theorem 20.14.1. For every such V, $Z_I = 0$ at every nonempty configuration I. The empty coefficient gives an equivalence of algebras

$$Z_\varnothing K \xrightarrow{\sim} Z.$$

Proof. For right differential modules $T_1, \dots, T_n$, use the positive spectator object

$$W_n = \iota_{\mathrm{ord}} T_1 \star \cdots \star \iota_{\mathrm{ord}} T_n.$$

Distinguish its first factor. In each partition summand, let B be that factor's block and a its first coordinate.

Insert a new vacuum at z_q, with simple pole u^{-1}, $u = z_q - a$. Retain the output block containing B. In the normal direction this uses the quotient

$$k[u, u^{-1}] \longrightarrow \Delta_u = k[u, u^{-1}]/k[u].$$

Writing $e_j = \partial_u^j[u^{-1}]$, the Euler operator $u\partial_u + 1$ has eigenvalue $-j$ on e_j. Its degree-zero normal projection selects e_0.

Apply this projection before taking the remaining tangential derived Hom. It therefore defines an extraction operator R_p for arbitrary spectator modules, with no compactness assumption on them.

The Chevalley differential has three kinds of terms. Vacuum action on the output block containing B gives the normal quotient above, whose projection is the identity. Old actions and brackets commute with the normal projection and contribute $-dR_p$. The bracket of the new vacuum with an old generator has zero residue. The old separation t from B is invertible, so $(s+t)^{-1}$ is regular in the new collision parameter s. Thus

$$R_{p+1}d + dR_p = 1.$$

A target partition that splits a source block receives no map, by the support argument used above. A collision across two spectator blocks annihilates the source summand, since it is an open direct image and $\delta^! j_* = 0$ there. Chart changes preserve the internal order of B. These observations make the extraction compatible with the full collision diagram.

They also apply before the coefficient resolutions are totalized, so the identity holds in derived Hom. The Chevalley filtration has $p, q \geq 0$ and only finitely many terms in each total degree. Its contracted rows imply $R\operatorname{Hom}(W_n, Z) = 0$.

We now detect the coefficients inductively. Let F be positive and vanish at all configurations of size less than n. Stratify each ordered collision scheme by the number of distinct coordinates. On a stratum with $r < n$ coordinates, collision compatibility identifies the exceptional restriction with that of $F_{[r]}$, which is zero. Hence local cohomology on the locus of at most $n-1$ coordinates vanishes.

On the n-coordinate stratum, the labeled partition index has the collapse to $[n]$ as terminal object. Its coherent mapping limit is therefore the identity decomposition at $[n]$. If O_n is the distinct configuration open, this gives

$$R\operatorname{Hom}(W_n, F) \simeq R\operatorname{Hom}_{\mathcal{D}_{O_n}}((T_1 \boxtimes \cdots \boxtimes T_n)|_{O_n}, j_n^! F_{[n]}).$$

All overlap homotopies remain in this equivalence.

Apply the formula to the positive part of Z, assuming the lower coefficients already vanish, and take $T_i = \mathcal{D}_X$. The source on O_n is $\mathcal{D}_{O_n}$. The open O_n is affine, so the vanishing of this derived Hom forces $j_n^! Z_{[n]} = 0$. The lower-coordinate local cohomology also vanishes. The localization triangle gives $Z_{[n]} = 0$. Induction proves the claim, and the isolated empty factor gives the algebra equivalence. □

The simple pole is necessary in the contraction. Replacing it by u^{-2} gives $[u^{-2}] = -e_1$, whose projection to e_0 is zero. The identity term in the contraction then disappears. The theorem also uses the envelope of an ordinary unital vertex algebra in the heart. It makes no claim for arbitrary convolution algebras outside that domain.

For any $W \in C$, the theorem gives

$$R\operatorname{Hom}_C(W, Z) \simeq R\operatorname{Hom}_k(W_\varnothing, Z_\varnothing).$$

Thus the scalar center contains all the surviving information, while its positive evaluation is zero. If an algebra $\mathcal{B}$ has $\mathcal{B}_\varnothing = k$, every algebra map $Z \to \mathcal{B}$ factors through its unit and an augmentation $Z_\varnothing \to k$. For the diagonal actor cobar introduced below, the augmentation matching the convolution evaluation therefore gives a coherent scalar comparison. The normalized homotopy complex has an empty-supported source and positive target, so it is zero. The corresponding homotopy is unique up to a contractible choice.

Global modes in degree one

The support theorem does not make the scalar center small. The Chevalley filtration gives

$$H^i Z_\varnothing = 0 \ (i < 0), \qquad H^0 Z_\varnothing = k, \qquad H^1 Z_\varnothing = \mathrm{Der}(L).$$

Indeed, in total degree one the term $\mathrm{Ext}^1(K, U)$ is zero. An ordinary map $L \to U$ has primitive values by support, and its Chevalley differential is the derivation equation. The incoming scalar commutator is zero.

In coordinates a derivation is an endomorphism h_z of $k[z] \otimes V$ commuting with $\partial_z + T$. Put $\psi = h_0$. Then $h_z = e^{-z \operatorname{ad}_T}\psi$, polynomial on each state, and the residue condition is equivalent to

$$[\psi, Y(a,x)] = Y(e^{-x \operatorname{ad}_T}\psi(a), x).$$

All Laurent multipliers are required to recover this field identity from residues.

For any state a and $n \geq 0$, its global mode is

$$h_z^{a,n} = \sum_{m=0}^{n} \binom{n}{m} z^m a_{(n-m)} = \mathrm{Res}_y (y+z)^n Y(a,y).$$

The translation identity $[T, a_{(j)}] = -j a_{(j-1)}$ and the mode commutator formula prove the criterion. Thus every global mode is a degree-one class.

For the rest of this section take the rank-one Heisenberg algebra with $[J_\lambda J] = \kappa\lambda$, and put $u^{(m)} = m! x_{m+1}$. Then $V = k[u, u', u'', \ldots]$, $T u^{(m)} = u^{(m+1)}$, and

$$J_n = \kappa n! \frac{\partial}{\partial u^{(n-1)}} \quad (n \geq 1).$$

At level zero a derivation is determined freely by its image of x_1, so $H^1 Z_\varnothing \cong k[z] \otimes V_0$. At nonzero level there is a complete global-mode description.

Theorem 20.14.2. Suppose $\kappa \neq 0$. On $V[t]$ put $\nabla = \partial_t + T$. The map

$$\gamma : V[t] \longrightarrow \mathrm{Der}(L), \qquad \sum_n t^n a_n \longmapsto \sum_n h^{a_n, n},$$

is surjective with kernel $\nabla V[t] + k x_1$. Consequently

$$H^1 Z_\varnothing \cong V[t] / (\nabla V[t] + k x_1).$$

Proof. For a polynomial v define its linearization and variational derivative by

$$\ell_v = \sum_{m \geq 0} \frac{\partial v}{\partial u^{(m)}} \nabla^m, \qquad \frac{\delta v}{\delta u} = \sum_{m \geq 0} (-\nabla)^m \frac{\partial v}{\partial u^{(m)}}.$$

The formal adjoint of $\sum p_m \nabla^m$ is $\sum (-\nabla)^m \circ p_m$. Integration by parts gives $\varphi P(\chi) - P^*(\varphi)\chi \in \nabla V[t]$. Inspection of the highest jet gives $\ker \nabla = k$ and $\ell_{\nabla v} = \nabla \ell_v$.

The variational kernel is exactly $\nabla V[t]$. To prove the reverse inclusion, decompose v by polynomial current degree. For a homogeneous piece v_r with $r > 0$, Euler's identity and integration by parts give $r v_r \equiv u \, \delta v_r / \delta u$ modulo $\nabla V[t]$. The current-degree-zero part lies in $k[t] = \partial_t k[t]$. Also, if ℓ_f is self-adjoint, then

$$f = \frac{\delta}{\delta u}\left(\int_0^1 u \, f(\lambda u) \, d\lambda \right),$$

where $f(\lambda)$ scales every current jet by λ. Differentiating $\lambda f(\lambda)$ under the polynomial integral proves this identity.

For a derivation h, write $h_z x_1 = \sum_j z^j v_j$ and $v = \sum_j t^j v_j$. The field criterion gives $[\psi, J(x)] = \sum_j x^j Y(v_j, x)$. This commutator determines h: it determines its action on every negative-mode monomial and h kills the vacuum.

Since $[J(y), J(w)]$ is scalar, its commutator with ψ vanishes. Expand the resulting identity in derivatives of the formal delta function. The coefficient identity is

$$\sum_{i \geq 1} \frac{\lambda^i}{i!} J_i v = \sum_{i \geq 1} \frac{(-\lambda - \nabla)^i}{i!} J_i v.$$

The coefficients are unique by residue extraction. Using the displayed formula for J_i, this says that $\ell_v \nabla$ is self-adjoint.

Apply both sides to 1. It follows that $\partial v / \partial u = c \in k$, so $v = cu + \nabla f$. Self-adjointness becomes

$$2c\nabla = \nabla(\ell_f^* - \ell_f)\nabla.$$

If the middle operator were nonzero, the right side would have order at least two, because leading coefficients multiply in the domain $V[t]$. Hence $c = 0$ and $\ell_f = \ell_f^*$. The polynomial integral above gives $f = \partial a / \partial u$.

Skew symmetry and the global-mode formula give the current image associated with $\gamma(a)$ as $\kappa \nabla(\partial a / \partial u)$. Thus $h = \gamma(\kappa^{-1} a)$. Finally $\gamma(a) = 0$ exactly when $\partial a / \partial u \in k$. The variational kernel identifies these a with $\nabla V[t] + ku$. □

The energy operator is also visible here. For the Heisenberg conformal vector $\omega = J_{-1}^2 \mathbf{1}/(2\kappa)$, one has $\omega_{(0)} = T$ and $\omega_{(1)} = E$. Its first global mode is $E + zT$. Scalar polynomial current images give infinitely many further classes. Thus degree one is infinite dimensional at every level.

The part of highest PBW degree

The PBW splitting is $U = \bigoplus_{r \geq 0} \operatorname{Sym}_\star^r L$. Let $Z_\varnothing^{(r)}$ denote cochains whose values lie in the rth summand. The internal differential acts through the inputs, and the twisting term is the adjoint action of L on the outputs. Both preserve PBW degree. Therefore

$$Z_\varnothing = \prod_{r \geq 0} Z_\varnothing^{(r)}, \qquad Z_\varnothing^{(0)} = k.$$

Support forces a cochain with r output blocks to have at least r input blocks. Hence

$$H^i Z_\varnothing^{(r)} = 0 \quad (i < r).$$

In degree r, the entire group is the space of closed ordinary maps

$$\Lambda_\star^r L \longrightarrow \operatorname{Sym}_\star^r L.$$

There is no boundary to divide by in this part.

Let $k[O_r] = k[z_1, \dots, z_r, (z_l - z_m)^{-1}]$. Such maps are equivalent to $k[O_r]$-linear endomorphisms P of $k[O_r] \otimes V^{\otimes r}$ that commute with $\nabla_l = \partial_{z_l} + T^{(l)}$ and alternate under simultaneous permutation

of positions and factors. The Chevalley cocycle condition is the derivation identity in each factor. In the first factor it is

$$\begin{aligned} P(Y(a,x)b\otimes w) &= Y^{(1)}(a,x)P(b\otimes w)\\ &\quad + Y^{(1)}(P_{z_1+x}(a\otimes w),x)b. \end{aligned} \tag{20.19}$$

Here the subscript $z_1 + x$ denotes Taylor expansion, and $Y^{(1)}$ acts on the first state factor.

The detection lemma in the support theorem identifies the maps on O_r. In the Chevalley differential, choose the component in which two specified inputs merge. Its three terms are exactly (20.19). All components arise this way. This proves the criterion with all collision maps retained.

Proposition 20.14.3. For every $r \geq 1$, cup product gives an injection

$$\Lambda^r_k \operatorname{Der}(L) \hookrightarrow H^r Z_\varnothing.$$

Its highest-PBW-degree component is the alternating external product of the derivations.

Proof. The alternating product satisfies (20.19) and commutes with the connections. It is therefore closed.

Evaluate at distinct points $c_1, \dots, c_r$. A derivation is determined by its value at one point, since $h_{c+x} = e^{-x\,\mathrm{ad}_T} h_c$. Tensor products of independent state endomorphisms remain independent, as in the preceding cup calculation. Thus the alternating external product is injective. The highest-PBW-degree part has no incoming boundaries.

The cup uses the unshuffle coproduct. Under the suspension convention $\widetilde{P}(s y_1 \cdots s y_r) = (-1)^{r(r-1)/2} P(y_1, \dots, y_r)$, its symbol is precisely this alternating product. This identifies the cup class asserted. □

For rank-one Heisenberg, every positive degree is consequently infinite dimensional. At level zero the highest-PBW-degree part is all alternating values $F \in k[O_r] \otimes V_0^{\otimes r}$ on $x_1^{\otimes r}$, extended as differential derivations in each factor. At nonzero level the analogous description has a variational condition.

Mode sequences and rational classes

A *mode sequence* is a sequence $(A_m)_{m\geq 0}$ of endomorphisms of V, eventually zero on each state, satisfying

$$[T, A_m] = -m A_{m-1}, \qquad [A_m, Y(b,x)] = \sum_{j\geq 0} \binom{m}{j} x^{m-j} Y(A_j b, x),$$

where $A_{-1} = 0$. For sequences $A^1, \dots, A^r$ and $f \in k[O_r]$, define

$$P^f_{A^1,\dots,A^r} = \sum_{m_1,\dots,m_r\geq 0} \partial_{z_1}^{(m_1)} \cdots \partial_{z_r}^{(m_r)} f\; A^1_{m_1} \otimes \cdots \otimes A^r_{m_r}. \tag{20.20}$$

The sum is finite on each tensor of states. The first sequence identity cancels the two terms in $[\nabla_l, P]$. The second, together with Taylor's formula, gives (20.19). Alternating this operator therefore gives a closed map of highest PBW degree.

At nonzero level, the current sequence and $f = (z_1 - z_2)^{-1}$ give an already alternating operator with

$$P^f_{J,J}(x_1 \otimes x_1) = -2\kappa^2 (z_1 - z_2)^{-3} \mathbf{1} \otimes \mathbf{1}.$$

Every finite sum of derivation cups has polynomial value on this input. Thus this rational class lies outside their span.

There is one additional type of sequence:

$$\varphi_m = -\frac{J_{m+1}}{m+1}.$$

It satisfies the sequence identities because $J_0 = 0$. For the same f,

$$P^f_{\varphi,\varphi}(x_1 \otimes x_1) = \frac{\kappa^2}{z_1 - z_2}\mathbf{1} \otimes \mathbf{1}.$$

This class cannot be a sum of operators using only state mode sequences. Their vacuum coefficient on this input lies in $\partial_{z_1}\partial_{z_2}k[O_2]$, by skew symmetry and $J_0 = 0$. Writing $w = z_1 - z_2$, no ∂_w-derivative of a Laurent polynomial has a w^{-1} term.

PROPOSITION 20.14.4. At nonzero rank-one Heisenberg level, every mode sequence is

$$A_m = a_{(m)} + c\varphi_m$$

for a state a and $c \in k$. The scalar c is unique and a is unique modulo the vacuum line.

Proof. Each A_m gives a global derivation by the sequence identities. A translation-commuting derivation has constant h_z. The global-mode classification and its variational kernel then express its value as $a_{(0)} + cJ_1$: the variational derivative of a representing polynomial has the form $f_0 + ct$.

Grade states by energy, $\deg x_n = n$. Take a homogeneous sequence whose A_m has degree $\delta - m$. If it vanishes before $m = p$, then A_p commutes with translation and is $b_{(0)} + cJ_1$. For $p \geq 1$, the states

$$b' = \frac{(-T)^p b}{p!}, \qquad e = \frac{(-T)^{p-1}x_1}{p!}$$

have mode sequences vanishing before p and taking values $b_{(0)}, J_1$ at p. Subtract them. For $p = 0$, subtract the modes of b and the appropriate multiple of φ.

Energy forces this procedure to terminate once $p \geq \delta + 2$. Only the component $\delta = -1$ can retain φ.

There are only finitely many nonzero homogeneous sequences. Their values on x_1 have finitely many energy components and eventually vanish in m. These values determine all the endomorphisms. For uniqueness, energy -1 separates the φ term. If all nonnegative modes of a vanish, skew symmetry on x_1 and the largest nonzero $J_n a$ force every $J_n a$ to vanish. Then a is scalar. □

THEOREM 20.14.5. At nonzero rank-one Heisenberg level, every closed map $\Lambda^r_\star L \to \mathrm{Sym}^r_\star L$ is a finite sum of alternations of (20.20), using state modes or φ in each factor.

Equivalently, evaluation on $x_1^{\otimes r}$ identifies $H^r Z^{(r)}_\varnothing$ with alternating $F \in k[O_r] \otimes V^{\otimes r}$ such that $\ell^{(1)}_F \nabla_1$ is self-adjoint. The superscript means linearization in the first state factor, with all other factors as coefficients.

Proof. The slot identity first makes P vanish when an input is the vacuum. Commuting a negative current mode through it then reduces its value to its value on x_1 in that factor. Induction over the state energies shows that F determines P. The formal-delta commutator calculation used for global modes, now over the coefficient ring of the other factors, proves self-adjointness of $\ell^{(l)}_F \nabla_l$ for every l.

The variational argument works over any differential domain (B, ∂). Its kernel statements become

$$\ker(\partial + T) = B^{\partial}, \qquad \ker(\delta/\delta u) = (\partial + T)(B \otimes V) + B.$$

If $\ell_v(\partial + T)$ is self-adjoint, then $v = (\partial + T)f + b$ with $b \in B$ and ℓ_f self-adjoint. The proof is the same integration by parts and leading-order calculation as above. The current-degree-zero term now remains in B.

Apply this to a homogeneous current-multidegree component of F. In every factor with positive current degree, F lies in the image of ∇_l. These images intersect as their product: if $F = \nabla_1 \cdots \nabla_{s-1} g$ and $\delta_s F = 0$, injectivity of the other ∇_l on positive current degree gives $\delta_s g = 0$. Thus g also lies in the image of ∇_s.

Write $F = (\prod \nabla_l)\, g$ over those factors. Their injectivity and commutation with the other adjoints imply that $\ell_g^{(l)}$ is self-adjoint. Apply the polynomial variational integral in each factor to write $g = (\prod \delta_l)h$. It preserves the self-adjointness conditions in the other factors.

Expand h as a finite sum of pure state tensors with coefficients in $k[O_r]$. State-mode sequences evaluate on the current as $\kappa \nabla_l \delta_l$. φ evaluates as $-\kappa$ in factors of current degree zero. After dividing these nonzero scalar factors, (20.20) constructs a closed operator with value F. Alternation and division by $r!$ give the prescribed symmetry. Since F determines the operator, this proves both descriptions. □

20.15 Degree-two rigidity and the degree-three remainder

The highest-PBW-degree calculation does not by itself determine the lower-PBW-degree groups. For rank-one Heisenberg coefficients, the remaining degree-two group vanishes at nonzero level. In degree three, derived extension terms can be removed at every level, leaving two ordinary cohomology groups to compute.

Removing the derived term in degree two

At every level the state differential module splits as

$$\mathscr{V} = \omega_X \oplus \mathscr{F}, \qquad \mathscr{F} = W \otimes_k \mathcal{D}_X.$$

To obtain the splitting, take the vacuum projection along positive energy. Translation is injective on the positive-energy part. A graded complement W to its image gives $V_+ = \bigoplus_{n \geq 0} T^n W$, which is the free differential-module summand. Consequently

$$\operatorname{Ext}^1_C(L, L) \cong \operatorname{coker}(\partial_z + T : k[z] \otimes V \to k[z] \otimes V), \qquad \operatorname{Ext}^q_C(L, L) = 0 \quad (q \geq 2).$$

The free summand has no higher Ext, and the vacuum summand has its two-term de Rham resolution.

The first Chevalley differential is injective on this Ext^1. Represent a class by $[v] \circ p$, with p the vacuum projection. Restrict its differential to two vacuum inputs. The three terms become three signed copies of $\Delta_*[v] \circ \nu$, where

$$0 \to \omega_{X^2} \to j_* j^* \omega_{X^2} \xrightarrow{\nu} \Delta_* \omega_X \to 0.$$

Their coefficient is odd, hence nonzero in characteristic zero.

Composition with ν is injective on the relevant Ext group. The regular term has no map to $\Delta_* \mathscr{V}$. A normal derivative annihilates its vacuum generator and acts injectively on the diagonal target. Kashiwara equivalence then detects $[v]$.

It follows that $H^2 Z^{(1)}_{\varnothing}$ is the second cohomology of the ordinary complex

$$B^p = \operatorname{Hom}_C(\Lambda^p_\star L, L), \qquad d = d_\ell.$$

The possible derived term dies on the second page, and no later differential enters total degree two.

From all words to chiral operations

Let C^p_{ch} be the space of alternating differential-module operations

$$j_* j^*(\mathscr{V}^{\boxtimes p}) \longrightarrow \Delta_* \mathscr{V},$$

with the Chevalley differential of the chiral bracket. Here $p \geq 1$ and every localized multiplier is retained. A binary operation is *vacuum normalized* when it vanishes if either input lies in the local vacuum line.

An ordinary cocycle in B^2 has a two-block operation ρ_n for each reduced word of length n. Its linearized Jacobi equation is (20.17), with the lengths induced by the corresponding three-label word. The vacuum attachment

$$j_* j^*(\omega_X \boxtimes \mathscr{V}) \longrightarrow \Delta_* \mathscr{V}$$

has regular kernel. Every map from the regular kernel to the diagonal target is zero. Thus restriction of ρ_n to either vacuum axis is uniquely the attachment followed by an endomorphism λ_n or Υ_n.

The two families $(ij)^k l$ and $(ji)^k jl$ have six merger lengths

$$(2k, 2, 2; 2, 2k, 2k+1), \qquad (2k+1, 2, 2; 2, 2k+1, 2k+2).$$

Insert the vacuum in each of their three blocks, using a simple pole on just one attachment. The linearized Jacobi equation first makes $\lambda_n = \Upsilon_n = \ell$ one scalar, since the only horizontal state is the vacuum. Subtract $\ell\beta = d_\beta(\ell\,\mathrm{id})$. The remaining operations vanish on both vacuum axes. The same two families then give $\rho_n = \rho_2$ for every $n \geq 2$.

This common operation is alternating. With the vacuum in the third block, write the three iterated principal-part routes as L, R, N, in their residue orientations. They satisfy

$$L - R + N = 0.$$

The cocycle identity compares $L(G) - R(G) + N(H)$, where G, H are the two exchanged field series. Subtract the three-channel identity. All Laurent multipliers then give $N(H - G) = 0$, which detects every series coefficient. Thus $H = G$, the chiral skew-symmetry identity. The word 123 gives its chiral cocycle equation.

Restriction to the word 12 also detects boundaries. Therefore

$$H^2 C^\bullet_{\mathrm{ch}} \simeq H^2 B^\bullet. \tag{20.21}$$

This reduction uses the horizontal-state hypothesis. For the commutative vertex algebra $k[y]$ with $T = \partial_y$, the section $y - z$ is horizontal, so that hypothesis fails.

Vanishing at nonzero level

THEOREM 20.15.1. At nonzero rank-one Heisenberg level,

$$H^2 C^\bullet_{\mathrm{ch}} = 0, \qquad H^2 Z^{(1)}_{\varnothing} = 0.$$

Hence $H^2 Z_{\varnothing} = H^2 Z^{(2)}_{\varnothing}$, and the complete degree-two collision group is the space of closed maps described in Theorem 20.14.5 for $r = 2$.

Proof. Insert a global mode $\alpha \in V[t]$ as a first input and take its residue. This defines $\iota_\alpha : C^p_{\mathrm{ch}} \to C^{p-1}_{\mathrm{ch}}$ for $p \geq 2$. Expanding the Chevalley differential with this input gives the Cartan identity

$$\mathcal{L}_\alpha = d\iota_\alpha + \iota_\alpha d.$$

The output action and the actions on each old input give $\mathcal{L}_\alpha$. The remaining terms are the differential of the insertion. Residue commutes with the old operations by base change.

Give a separation and a position weight -1, and state x_n energy n. The mode $t\omega$ acts on a homogeneous binary cochain of degree e by multiplication by e. Every nonzero-degree cocycle component is therefore a boundary, with primitive $e^{-1}\iota_{t\omega}A_e$. The sum of these primitives is finite on each input and coefficient: the residue uses only two coefficients of the field series.

Use the word reduction to start with a vacuum-normalized degree-zero cocycle. The mode ω acts as minus the position derivative. On endomorphisms of the free complement $\mathscr{F}$, integration in z solves $\mathcal{L}_\omega B = \iota_\omega A$ with $B\mathbf{1} = 0$. Then $A - dB$ is translation invariant and remains vacuum normalized.

The corresponding first-order fields $Y_\epsilon = Y + \epsilon\Phi_A$ satisfy the Borcherds identities: these are the coefficients of the residue Jacobi identity with all Laurent multipliers. Energy and skew symmetry force the current bracket to be $[J^\epsilon_m, J^\epsilon_n] = (\kappa + \epsilon\eta)m\delta_{m+n,0}$. Higher singular terms have negative state energy, and the simple-pole coefficient is its own negative.

Rescale the current modes by $1 - \epsilon\eta/(2\kappa)$. They now satisfy the original Heisenberg relations and kill the vacuum in nonnegative degree. The induced Fock module gives an isomorphism $\theta = 1 + \epsilon\psi$ that fixes the vacuum and commutes with translation.

Conjugate Y_ϵ by θ. Its current modes are the original ones. The Borcherds identity reconstructs every negative-mode state field from these currents, so the conjugated fields are exactly Y. The coefficient of ϵ says $A = d(-\psi)$.

This proves $H^2C^\bullet_{\mathrm{ch}} = 0$. Equation (20.21) and the removal of the derived term give the second vanishing. The PBW degree bound leaves only the highest-degree part. □

At level zero the central-polynomial classes already show that the lower-PBW-degree group need not vanish.

Formal brackets on the fixed coefficient module

Let $R = k[[t_1, \dots, t_d]]$ for finitely many degree-zero parameters, with maximal ideal $\mathfrak{m}$. Complete only in these parameters. The polynomial state module and all word sizes remain as above. For an endomorphism $h \in B^1$, fix the sign convention

$$(d_\beta h)(x, y) = \beta(hx, y) + \beta(x, hy) - h\beta(x, y).$$

Theorem 20.15.2. At $\kappa \neq 0$, every formal Lie bracket

$$\beta_R = \beta + \sum_{|\alpha| \geq 1} \mathbf{t}^\alpha b_\alpha, \qquad b_\alpha \in B^2,$$

on the fixed $L[[\mathbf{t}]]$ has a continuous automorphism $\Phi = 1 \bmod \mathfrak{m}$ such that

$$\Phi\beta_R = \beta(\Phi \star \Phi).$$

If every correction is vacuum normalized, Φ fixes the local vacuum line.

Proof. Suppose the bracket has already been made equal to β modulo $\mathfrak{m}^n$. Its order-n error b_n is a d_β-cocycle: two correction terms have order at least $2n \geq n+1$. The space $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is finite dimensional, so the vanishing of $H^2B^\bullet$ gives h_n with $d_\beta h_n = b_n$. For $g_n = 1 + h_n$,

$$g_n(\beta + b_n)(g_n^{-1} \star g_n^{-1}) = \beta + b_n - d_\beta h_n = \beta \pmod{\mathfrak{m}^{n+1}}.$$

The sign follows from $g_n^{-1} = 1 - h_n$ to this order.

The products $g_N \cdots g_1$ stabilize modulo every power of $\mathfrak{m}$. Their limit is Φ, with inverse given by the compatible reverse products. Each parameter coefficient uses finitely many compositions of maps in C, so it preserves all collisions.

For a normalized error, every primitive annihilates the vacuum. Indeed, $h\mathbf{1} = c\mathbf{1}$ by horizontality. Evaluating $d_\beta h$ on the simple-pole two-vacuum input gives $c + c - c = c$. Normalization forces $c = 0$. Conjugation by $1 + h_n$ therefore preserves both vacuum attachments, by their naturality. Induction gives the last assertion. □

Tensor powers of $\Phi[1]$ give an isomorphism of the completed Chevalley coalgebras, and applying Φ in every cobar factor gives an isomorphism of their cobars. The unshuffle coproduct and all cut maps commute with these tensor powers. At a fixed configuration, only finitely many nonempty factors occur, so the maps commute with parameter completion. In the normalized case they are under the vacuum coalgebra and its cobar.

This is rigidity of strict Lie brackets on the fixed differential-module family. Deformations of the underlying module, or arbitrary higher algebra operations, are separate questions.

The remaining degree-three groups

Keep any level $\kappa \in k$. The vacuum splitting gives $L = L_0 \oplus L_F$, with $L_0 = \iota_{\mathrm{ord}}\omega_X$ and $L_F = \iota_{\mathrm{ord}}\mathscr{F}$. On the first page for PBW degree r, write

$$E_{(r)}^{a,b,q} = \mathrm{Ext}_C^q\Big(\Lambda_\star^a L_0 \star \Lambda_\star^b L_F, \mathrm{Sym}_\star^r L\Big).$$

The differential has parts adding a vacuum input, adding a free input, and replacing a vacuum input by two free inputs.

THEOREM 20.15.3. Every derived term of total degree three vanishes on the second Chevalley page. Thus

$$\begin{aligned} H^3 Z_\varnothing \cong\ & H^3 \mathrm{Hom}_C(\Lambda_\star^\bullet L, L) \\ & \oplus H^3 \mathrm{Hom}_C(\Lambda_\star^\bullet L, \mathrm{Sym}_\star^2 L) \\ & \oplus H^3 Z_\varnothing^{(3)}. \end{aligned}$$

The first two summands use ordinary maps and the Chevalley differential. The third consists of closed maps $\Lambda_\star^3 L \to \mathrm{Sym}_\star^3 L$.

Proof. First, two free inputs have no positive Ext into PBW degree one or two. For degree one, the normal source is the localization $\mathcal{D}_{X^2}[u^{-1}]$, the colimit of free modules under left multiplication by u. Its Hom into $\Delta_*\mathscr{V}$ is the derived inverse limit of right multiplication by u. On $\mathscr{V} \otimes k[\partial_u]$ this is $m\partial_u^n \mapsto n\, m\partial_u^{n-1}$, which is surjective.

The countable tower has no derived limit in positive degrees. Products over the free basis are exact. For PBW degree two, restriction to O_2 has free source on an affine scheme, giving the same vanishing.

To treat vacuum inputs, use the natural exact sequence

$$0 \to \omega_X \boxtimes N \to j_* j^*(\omega_X \boxtimes N) \xrightarrow{\nu_N} \Delta_* N \to 0.$$

Restriction in Ext has kernel the image of $\mathrm{Ext}^1_X(N, \mathscr{V})$ under composition with ν_N. For free N, restriction is therefore injective.

Replace every occurrence of a label a in a reduced two-label word w by two vacuum labels $y_0 y_1$. Localize only along $z_{y_0} = z_{y_1}$. Of the Chevalley terms adding a vacuum, exactly three survive. Naturality of the attachment identifies each with a signed copy of the restricted class followed by ν. Their sum is an odd multiple.

The same regular-to-diagonal Hom vanishing makes attachment injective on Ext. Thus a class with one vacuum and one free input in the kernel must vanish.

For two vacuum inputs and PBW degree one, the restricted class is likewise zero. It therefore has the form $\Delta_* \eta_n \circ \beta_n$ at a reduced word of length n, with $\eta_n \in \mathrm{Ext}^1_X(\omega_X, \mathscr{V})$. Let K_L, K_R, K_N be the three vacuum merger routes. They obey $K_L - K_R + K_N = 0$. The first two are independent: on $1/(s(s-t))$ their values are $s^{-1}t^{-1}$, 0, and on $1/(st)$ they are 0, $s^{-1}t^{-1}$.

This independence persists with the Ext coefficients. Separate the two normal coordinates from the tangential one. The normal localization is holonomic and has a bounded resolution by finite projectives. The tangential vacuum has its Spencer resolution. The Künneth map therefore embeds normal degree zero tensored with tangential degree one into total degree one.

The cocycle equation forces the three sums of the relevant η_n to agree. The two word families displayed above give $\eta_{2k+1} = \eta_{2k}$ and $\eta_{2k+2} = \eta_{2k+1}$. All η_n are one class, and its family is the boundary of that class in arity one.

For PBW degree two, restriction from the two-block open-image target loses no class. Apply the same three surviving terms to the component in which the two vacua merge. They force the restriction to vanish and hence the class itself to vanish.

Now take a cocycle in $E_1^{2,1}$. The two-free-input part is zero. In PBW degree one, subtract the boundary just constructed from its two-vacuum part. The one-vacuum part then vanishes by injectivity. In PBW degree two, both vacuum parts are already detected injectively. Thus $E_2^{2,1} = 0$.

The terms $E_1^{1,2}$ and $E_1^{0,3}$ also vanish. The only possible later source into $E^{3,0}$ is the degree-two derived term already removed. For PBW degree three only the highest-degree ordinary maps occur. First-quadrant connectivity excludes extra limit terms. □

The proof uses the vacuum splitting with free complement, not the numerical value of the level. At nonzero level the final summand has the explicit description of Theorem 20.14.5. The two ordinary degree-three groups remain to be computed. Their reduction to ordinary cochains is not a vanishing theorem.

20.16 The ordinary transverse endomorphism monoid

The diagonal action problem uses another tensor product. Write

$$(F \otimes^! G)_I = \Delta_I^!(F_I \boxtimes G_I)$$

on the ordered collision category. Its unit is the vacuum coefficient system J, whose empty coefficient is k. The convolution unit K is instead supported only at the empty configuration.

Both products come from the same external product at different relative positions. Convolution extends from the disjoint open. Diagonal tensor restricts to the ordered branch where each input position is stacked with its copy. For one position, the two ordered diagonal branches give the localization triangle

$$\delta_{12,*}(M \otimes^! N) \oplus \delta_{21,*}(M \otimes^! N) \longrightarrow q^!(M \boxtimes N) \longrightarrow j_* j^! q^!(M \boxtimes N).$$

The connecting maps are their residues. Thus diagonal tensor cannot be replaced by disjoint convolution.

Let Coalg denote the enhanced coaugmented convolution coalgebras in the bar–cobar domain, with fixed empty coefficient k. The forward interchange

$$\zeta : (F \star G) \otimes^! (H \star L) \longrightarrow (F \otimes^! H) \star (G \otimes^! L)$$

keeps common ordered decompositions. It tensors coproducts and makes $(\mathrm{Coalg}, \otimes^!, J)$ symmetric monoidal.

There is also a reverse binary map χ including the common-decomposition summands. It is compatible with collision pullback and all positive-arity associativity diagrams. Indeed, a final decomposition occurs precisely when the decompositions in every row agree, independently of parenthesization. On that summand the maps are the same external tensor identifications. Hence

$$\zeta\chi = 1, \qquad \chi\zeta = \text{projection to common decompositions.}$$

For a cut open j, consider $j_*M \otimes^! j_*N \to j_*(M \otimes^! N)$. The source has zero exceptional restriction to the complement and restricts to $M \otimes^! N$ on the open. The comparison is therefore an equivalence. This supplies the comparison on each summand. The binary reverse map alone gives no inverse to the full interchange or to its unit comparisons.

Definition 20.16.1. A *transverse monoid* is an associative unital algebra object F of $(\mathrm{Coalg}, \otimes^!, J)$. It has coalgebra maps

$$m : F \otimes^! F \to F, \qquad u : J \to F$$

with coherent associativity and unit constraints. A transverse action on C is a left module structure $F \otimes^! C \to C$ in this category.

The coalgebra condition on m is

$$\Delta_F m = (m \star m)\zeta(\Delta_F \otimes^! \Delta_F),$$

including its coherent homotopies. At one position, after converting right modules to left modules with the prescribed density shift, this is an ordinary horizontal associative multiplication over $\mathcal{O}_X$. It is not a product at distinct positions.

The closed category and its universal action

THEOREM 20.16.2. The category Coalg is presentable, and $- \otimes^! C$ preserves colimits. It has a right adjoint $\underline{\mathrm{Hom}}_{\mathrm{Coalg}}(C, -)$. The object

$$\mathcal{E}(C) = \underline{\mathrm{Hom}}_{\mathrm{Coalg}}(C, C)$$

is a transverse monoid with evaluation action. For every transverse monoid F, evaluation induces an equivalence between monoid maps $F \to \mathcal{E}(C)$ and transverse actions of F on C.

Proof. Bar–cobar identifies Coalg with augmented algebras having fixed empty coefficient. These are nonunital algebras on the positive ideal in a presentable category whose convolution preserves colimits separately. Their category is presentable.

Colimits of coalgebras are computed on reduced parts: in the opposite category this is the fact that limits of algebras are computed on underlying objects. The diagonal tensor has reduced part $c \otimes^! d$, and extraordinary inverse image and external product preserve colimits. Thus $- \otimes^! C$ preserves colimits and has the stated right adjoint.

Composition of the mapping object gives an associative unital endomorphism object. The unit corresponds to id_C. Under the adjunction, the associative and unital diagrams for a map into this object are exactly the module diagrams for its adjoint evaluation. Applying the same adjunction to all iterated products identifies their coherent homotopies. This proves the equivalence of action spaces. □

At one position,

$$\mathcal{E}(C)_{\{1\}} \simeq \underline{\mathrm{Hom}}_X(C_{\{1\}}, C_{\{1\}}).$$

To check this, a bar word with two positive factors vanishes at one position. The single remaining term is the pointwise mapping object, and the bar and cobar shifts cancel. Diagonal insertion and Kashiwara adjunction identify its evaluation with internal Hom on X.

Its untwisted cochains

For positive coefficient objects c, a, let $\underline{\mathrm{Hom}}_!(c, a)$ represent maps $V \otimes^! c \to a$. Take c to be the reduced part of a coalgebra C and a the augmentation ideal of an algebra A. The mapping object then has a convolution cup product. Its adjoint is the composite

$$\begin{aligned}(V \star W) \otimes^! c &\longrightarrow (V \star W) \otimes^! (c \star c) \\ &\xrightarrow{\zeta} (V \otimes^! c) \star (W \otimes^! c) \longrightarrow a \star a \xrightarrow{\mu_A} a,\end{aligned}$$

using the reduced coproduct and the two evaluations.

Proposition 20.16.3. There is a natural algebra equivalence

$$\Omega \underline{\mathrm{Hom}}_{\mathrm{Coalg}}(C, BA) \simeq K \oplus \underline{\mathrm{Hom}}_!(c, a).$$

On strict representatives, the differential on the right is the ordinary mapping-object differential and the product is the cup just defined.

Proof. For positive V, the bar of the free convolution algebra $T_\star V$ is $K \oplus V[1]$ with primitive positive part. Tensoring it diagonally with C still gives zero reduced coproduct. The interchange retains only common decompositions. The primitive factor has no decomposition into two positive parts. Its cobar is therefore $T_\star(V \otimes^! c)$.

The free adjunctions identify maps from V into the augmentation ideal on the left with maps $V \otimes^! c \to a$. This gives the underlying mapping object for every test V. The cup is associative because the coproduct, interchange, and product are coherent.

Let π be the bar twisting cochain of this cup algebra. Compose it with evaluation to get $\tau = \mathrm{ev}(\pi \otimes^! 1)$. The defining cup identity turns the Maurer–Cartan equation of π into the twisting equation of τ. Thus evaluation defines the coalgebra map to BA. Its adjoint is the identity on the augmentation mapping object, so it is the required algebra equivalence. □

Taking $A = \Omega C$ gives a concrete model for $\Omega\mathcal{E}(C)$. Its unit point $J \to \mathcal{E}(C)$ corresponds to the identity of C. The twisting cochain is the element expressing this point. It is not added to the differential of the pointwise cochain algebra. The convolution Hochschild center instead has its twisted differential $\partial + [\tau, -]$.

Constant symmetries and exchange

For a coassociative coalgebra H over k, put $H^J_\varnothing = k$ and $H^J_I = J_I \otimes_k H$ for nonempty I. For a decomposition into two nonempty parts, its coproduct is that of J tensored with Δ_H. For an empty part, it is the identity. This construction is symmetric monoidal. Thus a bialgebra H gives a transverse monoid H^J, with unit induced by 1_H.

An action of H^J is a family of horizontal actions $H \otimes C_I \to C_I$ compatible with collisions and

$$\Delta_{A,B}(h \cdot c) = \sum (h_{(1)} \otimes h_{(2)}) \cdot \Delta_{A,B}(c).$$

The universal property gives its map to $\mathcal{E}(C)$. Over the disjoint open, factorization of H^J would make Δ_H an equivalence. For an ordinary bialgebra this forces $H = k$: the two counit maps would be the same inverse, so $h \otimes 1 = 1 \otimes h$ and $h = \epsilon(h)1$.

If C is factorizing, its two ordered coproduct components define an exchange R_C. The coalgebra action identity gives

$$R_C \Delta_H(h) = \Delta_H^{\mathrm{op}}(h) R_C.$$

For a factorizing transverse actor F, the analogous identity is

$$R_C(\alpha_1 \boxtimes \alpha_1) = (\alpha_1 \boxtimes \alpha_1)(R_F \otimes^! R_C)$$

on the disjoint two-point locus. Both follow by commuting the action with the two coproduct components and inverting one factorization map. The first identity is a constraint on a given bialgebra action. It does not reconstruct a Hopf algebra from the exchange.

A dual for diagonal tensor

Transport diagonal tensor through bar–cobar and write the resulting product on algebras as $\diamond$, with unit $\mathbb{U} = \Omega J$. Define

$$A^\dagger = [A, \mathbb{U}]_\diamond.$$

Then

$$\mathrm{Map}(Y, A^\dagger) \simeq \mathrm{Map}(Y \diamond A, \mathbb{U}) \simeq \mathrm{Map}_{\mathrm{Coalg}}(BY \otimes^! BA, J).$$

The last expression is a pairing of bar coalgebras with values in the vacuum coalgebra. Its augmentation ideal is $\underline{\mathrm{Hom}}_!(\overline{BA}, \overline{\mathbb{U}})$. At one position,

$$(A^\dagger)_{\{1\}} \simeq \underline{\mathrm{Hom}}_X(a_{\{1\}}[1], \omega_X).$$

Evaluation and symmetry give a natural map $A \to A^{\dagger\dagger}$, and $\mathbb{U}^\dagger \simeq \mathbb{U}$. The bidual map requires its own hypotheses to be an equivalence. This internal diagonal dual retains positive configurations, whereas internal convolution Hom into K sees only the empty coefficient.

20.17 Factorization and the pointed action problem

Let $\mathrm{Coalg}_{\mathrm{fact}}$ be the full subcategory whose ordered coproduct maps are equivalences on every disjoint open. It is closed under diagonal tensor and contains J. A *local center* of a factorizing coalgebra C is a final transverse monoid acting on C within this subcategory. It is final in the category of actors together with their actions, not merely among the underlying coalgebras.

For a pointed factorizing coalgebra $u : J \to F$, define

$$\Phi_C(F, u) = \mathrm{Map}_{C/}(F \otimes^! C, C).$$

The source lies under C by $u \otimes^! 1_C$, and the target by its identity. Thus a point is a coalgebra map whose restriction along the specified vacuum is identified with the identity of C. Representability of this pointed functor supplies a local center and its monoid structure. This is the centralizer construction of Lurie's *Higher Algebra*, Proposition 5.3.1.8.

If L_C is a local center, its action gives $\beta : L_C \to \mathcal{E}(C)$. For every factorizing transverse monoid F,

$$\mathrm{Map}_{\mathrm{Alg}}(F, L_C) \xrightarrow{\sim} \mathrm{Map}_{\mathrm{Alg}}(F, \mathcal{E}(C)).$$

Indeed, the fiber over any action map is the space of maps to the final action, hence contractible. The ordinary endomorphism monoid $\mathcal{E}(C)$ is itself a local center exactly when it factorizes.

Obstructions to a factorization right adjoint

Factorizing coalgebras are closed under sifted colimits. For each ordered decomposition, the coproduct comparison commutes with colimits, and the diagonal of a sifted indexing category is cofinal in every finite power. Thus the factorization equivalences survive. Arbitrary coproducts do not have this property.

The reduced part of $J \sqcup J$ is the sum of two copies of that of J. At two distinct positions its factorization map has two summands in the source and four in the target. Its image consists of the two diagonal summands. The two cross terms remain in the cofiber. Therefore $J \sqcup J$ is not factorizing.

It follows that the full inclusion $\mathrm{Coalg}_{\mathrm{fact}} \subset \mathrm{Coalg}$ has no right adjoint. If it had one, the colimit of any diagram of objects in its image would again lie in that image. To see this, factor the colimit cocone through the counit. The universal property gives a section of the counit, and full faithfulness makes the reverse composite the identity. The coproduct just computed contradicts this consequence.

There is an independent obstruction already at the singleton algebra.

Proposition 20.17.1. On a connected nonempty curve, a factorizing transverse monoid cannot have singleton algebra $M_n(k) \otimes J_{\{1\}}$ for $n \geq 2$. Consequently, if B is a constant commutative algebra with $2 \leq \dim_k B < \infty$, then $\mathcal{E}(C_B)$ does not factorize.

Proof. For a constant singleton algebra R, the two-position localization boundary is the normal residue tensored with a linear map $\tau : R \otimes R \to R$. We check that diagonal tensor tensors these coefficient maps, including derived terms.

In a normal coordinate u, put $P = \mathcal{O}[u^{-1}]$ and $D = P/\mathcal{O}$. After the density and cohomological shifts, a coefficient map $\tau : V \to R$ has complex $[V \otimes P \to R \otimes D]$ in degrees zero and one. Split the linear map into the elementary arrows $k \xrightarrow{1} k$, $k \to 0$, and $0 \to k$. Their complexes are respectively $\mathcal{O}$, P, Q, where $Q = [\mathcal{O} \to P] \simeq D[-1]$.

Localization gives

$$P \otimes^{\mathbb{L}}_{\mathcal{O}} P = P, \qquad P \otimes^{\mathbb{L}}_{\mathcal{O}} Q = 0, \qquad Q \otimes^{\mathbb{L}}_{\mathcal{O}} Q \simeq Q.$$

The last identity follows by tensoring $Q \to O \to P$ with Q. These identities give the tensor of arbitrary coefficient arrows.

Naturality for the transverse product and its unit therefore makes τ a unital algebra homomorphism $R \otimes R \to R$. For $R = M_n(k)$, its source is $M_{n^2}(k)$, which is simple. A unital map must be injective, but $n^4 > n^2$. This is impossible.

For C_B, the singleton of the ordinary endomorphism monoid is $\mathrm{End}_k(B)$, with composition and identity. The first assertion applies. □

Linearity alone would not suffice for this obstruction. When n is invertible, the map $r \otimes s \mapsto n^{-1}\operatorname{tr}(s)r$ is unital and linear. It kills $1 \otimes E_{12}$ and $1 \otimes E_{21}$, but sends their product to $n^{-1}I$. It fails precisely the required multiplicativity.

The inclusion of factorizing transverse monoids also has no right adjoint. Take $B = k \times k$ and the coproduct of two copies of C_B as transverse monoids. Its singleton algebra is $B * B$, and its two-position open algebra is $(B \otimes B) * (B \otimes B)$. These free products also compute the derived coproducts: the augmentations split, and the alternating tensor description of the free product is over the field k.

Let x, y name the two inclusions and put $e = (1, 0)$. The factorization map to $(B * B) \otimes (B * B)$ kills

$$[x(e \otimes 1),\, y(1 \otimes e)],$$

because the images belong to different tensor factors.

The source element is nonzero. Represent its two idempotents by

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Use a character on each unused tensor factor. The commutator maps to $PQ - QP = E_{12} \neq 0$. Thus the coproduct fails factorization, and the same counit argument excludes a right adjoint.

The positive result in the heart

Let $\mathcal{A}$ be an $\mathcal{O}_X$-flat chiral algebra. Its chiral center $Z_{\mathrm{ch}}(\mathcal{A})$ consists of sections z for which the chiral bracket vanishes on every regular tensor $z \boxtimes a$. This definition tests regular inputs. It does not require all meromorphic residues to vanish. Flatness makes the center a commutative chiral subalgebra. After changing to left modules, $Z_{\mathrm{ch}}(\mathcal{A})^{\ell}$ is a commutative differential algebra and acts on $\mathcal{A}$.

For a factorization algebra $\mathcal{B}$, let $C_{\mathcal{B}}$ be its shifted right-module coalgebra on the ordered collision schemes. Its vacuum is $J \to C_{\mathcal{B}}$. On a chart with $|I|$ positions, the module lies in degree $-|I|$. This is the factorization heart. Diagonal tensor of such objects lies in degrees at most $-|I|$, since in left-module notation it is derived tensor over functions.

Mapping spaces from those tensor powers into a heart object are discrete. They are determined by top cohomology on each chart. The maps agree on the disjoint open, and their difference would have diagonal-supported image. Factorization modules have no such sections, so the maps glue uniquely. Coalgebra maps are therefore maps of factorization structures, and being under J is exactly preserving units.

THEOREM 20.17.2. For every factorization algebra $\mathcal{B}$, the space

$$\mathrm{Map}_{C_{\mathcal{A}}/}(C_{\mathcal{B}} \otimes^{!} C_{\mathcal{A}}, C_{\mathcal{A}})$$

is discrete. Its points are the chiral algebra maps from the chiral algebra of $\mathcal{B}$ to $Z_{\mathrm{ch}}(\mathcal{A})$.

For a commutative unital differential algebra N, transverse actions of C_N on $C_{\mathcal{A}}$ are exactly the differential-algebra maps $N \to Z_{\mathrm{ch}}(\mathcal{A})^{\ell}$. The center coalgebra with its central action is final among these actors.

Proof. The factorization dictionary just described makes the mapping space discrete. The equivalence between factorization and chiral algebras, and the tensor universal property, are Beilinson–Drinfeld, *Chiral Algebras*, Sections 3.4.9 and 3.4.15. A map out of the tensor product corresponds to two commuting chiral maps. The condition under $C_{\mathcal{A}}$ fixes the second map as the identity. The first therefore takes values in the chiral center.

For commutative N, the two composites expressing associativity have the same restrictions to the three factors $N, N, \mathcal{A}$. Uniqueness in the tensor universal property makes them equal. Thus every central map is an action, with singleton formula $b \otimes a \mapsto \varphi(b) \cdot a$. Commutative chiral maps are commutative differential-algebra maps, by Sections 3.3.3 and 3.3.9 of the same reference.

An equivariant map of two such actors is a map $g : N \to N'$ with $\varphi' g = \varphi$. For $N' = Z_{\mathrm{ch}}(\mathcal{A})^{\ell}$ and $\varphi' = \mathrm{id}$, there is exactly one, namely $g = \varphi$. □

In the heart, a factorization coalgebra admits a transverse monoid structure exactly when its factorization algebra is commutative. The structure is then unique. Indeed, let e be a transverse unit. The sections $e \boxtimes \mathbf{1}$ and $\mathbf{1} \boxtimes e$ restrict to e on the diagonal. Their product away from the diagonal is $\mathbf{1} \boxtimes \mathbf{1}$. No diagonal-supported section can alter this equality.

Restriction gives $m(e, e) = \mathbf{1}$, while the unit law gives $m(e, e) = e$. Thus $e = \mathbf{1}$. The product is a factorization map restricting to the identity on each unit axis. This is equivalent to commutativity and determines the product uniquely (Beilinson–Drinfeld, Section 3.4.20).

The map from this center coalgebra to $\mathcal{E}(C_{\mathcal{A}})$ is injective on singleton cohomology. Evaluate its action on the vacuum: $z \cdot \mathbf{1} = z$. The composite is the inclusion of the chiral center into $\mathcal{A}$.

Every actor bounded above by the heart

The same central object represents more than the ordinary heart actors. Say that F is *bounded above by the factorization heart* if, on every chart, $H^j(F_I) = 0$ for $j > -|I|$. No uniform lower bound is required.

Theorem 20.17.3. For every pointed factorizing F with this bound, central action gives an equivalence of discrete spaces

$$\mathrm{Map}_{J/}(F, C_{Z_{\mathrm{ch}}(\mathcal{A})}) \xrightarrow{\sim} \Phi_{C_{\mathcal{A}}}(F, u).$$

Its inverse is restriction to the vacuum of $C_{\mathcal{A}}$. Consequently the center coalgebra is final among all factorizing transverse monoids with this bound that act on $C_{\mathcal{A}}$.

Proof. The degree bound makes all mapping spaces into the heart target discrete and reduces maps to top cohomology. In left-module notation, an action gives $a_I : M_I \otimes A_I \to A_I$. Let e be the image of the pointed unit and set $\zeta(f) = a_1(f \otimes \mathbf{1})$. The pointed condition says $a_1(e \otimes a) = a$.

The two-position localization boundary of F gives a chiral operation μ_F on M_1. Tensor the localization triangle with $C_{\mathcal{A},2}$ and use naturality of its connecting map under the action. Apply the resulting identity to $(f \boxtimes e)\varphi$ and to the two sections $\mathbf{1} \boxtimes a$, $a \boxtimes \mathbf{1}$ of A_2. Their diagonal restrictions agree, so

$$\mu_{\mathcal{A}}((\zeta(f) \boxtimes a)\varphi) = \mu_{\mathcal{A}}((a_1(f \otimes a) \boxtimes \mathbf{1})\varphi).$$

For $\varphi = 1$, the right side is zero, making $\zeta(f)$ central. For $\varphi = (z_1 - z_2)^{-1}$, the unit residue gives $a_1(f \otimes a) = \zeta(f) \cdot a$. Both sides are horizontal maps, so the result is coordinate independent.

Restriction to the vacuum gives a coalgebra map $g : F \to C_{\mathcal{A}}$. On distinct positions it is the external product of the central singleton maps. Inside the localized external state module, the center factorization module is

$$Z(\mathcal{A})^{\ell}_{X^I} = A_I \cap j_{I,*} j_I^*(Z(\mathcal{A})^{\ell \boxtimes I}).$$

This follows by restricting the first Chevalley–Cousin differential to the central subalgebra (Beilinson–Drinfeld, Sections 3.4.11–3.4.12). Thus g factors through the center coalgebra. Its collision and coproduct identities follow from those in $C_{\mathcal{A}}$ by the injective inclusion.

The reconstructed central action agrees with the original on the distinct locus. Their difference would have diagonal-supported image in A_I, which is zero. Evaluation on the vacuum recovers g, giving the inverse.

For a monoid F, apply the same equivalence to $F \otimes^! F$. Associativity of the action makes g preserve multiplication. The pointing makes it preserve the unit. All iterated mapping spaces remain discrete, so the coherences are unique. This proves finality. □

If a local center on all factorizing actors exists, it receives this ordinary center. Their mapping functors agree on actors with the stated bound. Above the heart, the reduction to top cohomology fails. Already $\pi_1 \operatorname{Map}(k[-1], k) = k$ although $H^0(k[-1]) = 0$. The following Heisenberg actions retain precisely such derived information.

20.18 Derived actions of the Heisenberg boundary

Fix a field k of characteristic zero. The degree-one operator in the following calculation comes from varying a charge. It is invisible in the ordinary center, although it acts on a resolution of the vacuum.

Let $\mathfrak{h}_\kappa$ have generators J_n, $n \in \mathbb{Z}$, and a central element $\mathbf{c}$, with

$$[J_m, J_n] = \kappa m \delta_{m+n,0} \mathbf{c}.$$

A *smooth module* has $\mathbf{c} = 1$, and each vector is annihilated by all sufficiently large J_n. Write Sm_κ for their abelian category. These are precisely the modules for the Heisenberg vertex algebra V_κ: normal products reconstruct the fields from the smooth current action. On the vacuum,

$$V_\kappa = k[x_1, x_2, \dots], \qquad J_{-n} = x_n, \quad J_n = \kappa n \partial_{x_n} \quad (n \geq 1), \qquad J_0 = 0.$$

Smoothness does not require local nilpotence of each positive mode.

PROPOSITION 20.18.1 *(Continuous vacuum cochains).* For $M \in \mathsf{Sm}_\kappa$, put

$$C^\bullet_{\mathrm{sm}}(M) = M \otimes \Lambda(\xi_0, \xi_1, \dots), \qquad |\xi_n| = 1, \qquad d(m \otimes \omega) = \sum_{n \geq 0} J_n m \otimes \xi_n \wedge \omega.$$

The exterior algebra consists of finite sums of monomials. Then

$$\operatorname{Ext}^i_{\mathsf{Sm}_\kappa}(V_\kappa, M) \simeq H^i C^\bullet_{\mathrm{sm}}(M).$$

Proof. The sum defining d is finite on every element. The nonnegative modes commute, so $d^2 = 0$. Let $\mathfrak{b}$ be their span and let $\mathfrak{a}_p$ be the span of J_n for $n \geq p$. The displayed complex is the filtered union of

$$C^\bullet(\mathfrak{b}/\mathfrak{a}_p, M^{\mathfrak{a}_p}).$$

Its degree-zero cohomology is $M^{\mathfrak{b}}$, which is $\mathrm{Hom}(V_\kappa, M)$ by the induced presentation of the vacuum.

It remains to check acyclicity on injectives in the smooth category. For $S_p = k[J_0, \dots, J_{p-1}]$, induction from $\mathfrak{b}$, with central charge one, is exact and takes S_p-modules to smooth modules. It is left adjoint to $M \mapsto M^{\mathfrak{a}_p}$. Thus an injective smooth module Y has $Y^{\mathfrak{a}_p}$ injective over S_p.

The finite Koszul complex has no positive cohomology on this module. Filtered unions are exact, so $C^\bullet_{\mathrm{sm}}(Y)$ has no positive cohomology. The exact cochain functors and dimension shifting now identify their cohomology with Ext. □

Theorem 20.18.2 *(Charge and vacuum extensions).* At nonzero level,

$$\mathrm{Ext}^*_{\mathsf{Sm}_\kappa}(V_\kappa, V_\kappa) = \Lambda(\xi_0)$$

as a graded algebra. Its nonzero degree-one class is the extension $V_\kappa[\varepsilon]/(\varepsilon^2)$ with $J_0 = \varepsilon$. For the Fock module F_μ with nonzero charge $J_0 = \mu$, all groups $\mathrm{Ext}^i(V_\kappa, F_\mu)$ vanish. At zero level the vacuum cochain differential vanishes, giving

$$\mathrm{Ext}^*_{\mathsf{Sm}_0}(V_0, V_0) \simeq V_0 \otimes \Lambda(\xi_0, \xi_1, \dots)$$

as graded vector spaces.

Proof. For $\kappa \neq 0$, the vacuum differential is $\sum_{n\geq 1} \kappa n \xi_n \partial_{x_n}$. The degree-minus-one derivation

$$\sigma(\xi_n) = \frac{x_n}{\kappa n} \quad (n \geq 1), \qquad \sigma(\xi_0) = \sigma(x_n) = 0$$

satisfies $d\sigma + \sigma d = N$, where N counts the x_n and the ξ_n with $n \geq 1$. Dividing by N contracts each positive-degree summand. The remaining complex is $\Lambda(\xi_0)$ with zero differential.

In the extension displayed in the statement, J_0 is nonzero. It would vanish on a split extension of two vacuum modules. The extension therefore spans the one-dimensional Ext group. Its square is zero because Ext in degree two vanishes.

For F_μ, the extra differential is $\mu \xi_0 \wedge$. $\mu^{-1}\partial_{\xi_0}$ contracts the complex. At $\kappa = 0$ every nonnegative mode kills the vacuum, proving the final assertion. □

The degree-one class differentiates the family F_μ at $\mu = 0$. Requiring $J_0 = 0$, or requiring it to act semisimply, excludes this extension. The ambient module category is therefore part of the answer. The smooth module

$$k[x_1, x_1^{-1}, x_2, x_3, \dots]$$

also shows why smoothness cannot be replaced silently by local nilpotence: $J_1 = \kappa \partial_{x_1}$ is not locally nilpotent when $\kappa \neq 0$.

The vacuum with a moving root

Let $X = \mathbb{A}^1_k$ with coordinate y. A *module with a moving root* is a left D_X-module with an $\mathcal{O}_X$-linear smooth mode action satisfying

$$[\nabla_{\partial_y}, J_n] = -nJ_{n-1}.$$

This equation makes the current horizontal when its argument is measured from the moving root. The vacuum is $\mathcal{V} = \mathcal{O}_X \otimes V_\kappa$, with connection $\partial_y + T$. On its cochains define

$$T x_n = n x_{n+1}, \qquad T\xi_n = (n+1)\xi_{n+1}.$$

This derivation commutes with d.

THEOREM 20.18.3 *(Internal endomorphisms of the moving vacuum).* The differential D_X-module

$$\mathcal{Z}^\ell = \mathcal{O}_X \otimes C^\bullet_{\mathrm{sm}}(V_\kappa), \qquad \nabla_{\partial_y} = \partial_y + T$$

represents the derived functor

$$R\,\mathrm{Hom}_{\mathrm{mov}}(N \otimes_{\mathcal{O}_X} \mathcal{V}, \mathcal{V}) \simeq R\,\mathrm{Hom}_{D_X}(N, \mathcal{Z}^\ell)$$

for every left D_X-module N. For $\kappa \neq 0$, translation is null-homotopic:

$$K(\xi_n) = \kappa^{-1} x_{n+1}, \qquad K(x_n) = 0, \qquad dK + Kd = T.$$

Consequently the cohomology is $\mathcal{O}_X \otimes \Lambda(\xi_0)$ with trivial connection.

Proof. The functor $N \mapsto N \otimes \mathcal{V}$ is exact. Its right adjoint takes nonnegative-mode invariants, with the induced connection. Thus the desired representing object is the derived invariant complex.

To compute it in the moving category, let $A_\kappa = U(\mathfrak{h}_\kappa)/(\mathbf{c} - 1)$ and adjoin y, D with relations

$$[D, y] = 1, \qquad [y, J_n] = 0, \qquad [D, J_n] = -nJ_{n-1}.$$

The last derivation preserves the Heisenberg relations: its value on the central commutator relation is $-\kappa m(m + n - 1)\delta_{m+n-1,0} = 0$. The Ore normal form is free as a right A_κ-module, with basis $y^a D^r$. Induction is therefore exact.

It also preserves smoothness. Indeed,

$$J_n D^r = \sum_{j=0}^{r} \binom{r}{j} n(n-1)\cdots(n-j+1) D^{r-j} J_{n-j}.$$

A vector killed by all modes of index at least p induces a vector $y^a D^r \otimes m$ killed by all modes of index at least $p + r$. Hence the forgetful functor preserves injectives.

Proposition 20.18.1 computes their derived invariants by the displayed cochains. The computation respects the connection. Applying an injective resolution proves the adjunction.

Finally, both $dK + Kd$ and T are derivations. On generators,

$$(dK + Kd)x_m = mx_{m+1}, \qquad (dK + Kd)\xi_m = (m+1)\xi_{m+1}.$$

Thus T acts trivially on cohomology, as asserted. □

This theorem gives an internal Hom in the moving module category. To obtain a transverse factorizing action, commutation must hold on a boundary resolution and across all configurations. We now construct that additional structure.

A quartet resolution

Assume $\kappa \neq 0$. Let $\Lambda[c]$ be the commutative vertex superalgebra on an odd degree-one field and its translates. Set

$$\mathcal{Z}_\kappa = V_\kappa \otimes \Lambda[c], \qquad \xi_n = \frac{T^n c}{n!}, \qquad d = (:cJ:)_{(0)}, \qquad dJ = \kappa Tc, \quad dc = 0.$$

A zero mode is a derivation of all vertex operations. These formulas therefore define a differential vertex algebra whose complex is $C^\bullet_{\mathrm{sm}}(V_\kappa)$. Its bracket $[J_\lambda J] = \kappa\lambda$ is retained.

Let E have even fields t, p of degree zero and odd fields η, ι of degrees 1, −1. Its nonzero generating brackets are

$$[p_\lambda t] = 1, \qquad [\iota_\lambda \eta] = 1,$$

together with those forced by graded skewsymmetry. Give this Weyl–Clifford vertex algebra the differential

$$Q = (:\eta p:)_{(0)}, \qquad Qt = \eta, \quad Q\iota = p, \quad Q\eta = Qp = 0.$$

All state spaces here are algebraic polynomial or exterior spaces. Each state involves finitely many jets.

Theorem 20.18.4 *(Commuting currents on a resolution).* In $W = V_\kappa \otimes E$, with differential Q, the maps

	J	c
$i : V_\kappa \longrightarrow W$	$J + p$	
$j : \mathcal{Z}_\kappa \longrightarrow W$	$J - \kappa T t$	$-\eta$

preserve the differential vertex operations and have commuting images. The map i is a quasi-isomorphism. If

$$W^{\mathfrak{b}} = \{w : (J + p)_n w = 0 \text{ for all } n \geq 0\},$$

then $j : \mathcal{Z}_\kappa \to W^{\mathfrak{b}}$ is also a quasi-isomorphism. The resulting derived endomorphism comparison takes $[c]$ to $[\xi_0]$ and preserves the Yoneda product.

Proof. Write $U = J - \kappa T t$. The decisive brackets are

$$[(J + p)_\lambda (J + p)] = [U_\lambda U] = \kappa\lambda, \qquad [(J + p)_\lambda U] = \kappa\lambda - \kappa\lambda = 0.$$

Here $[p_\lambda T t] = \lambda$. Also $Q(J + p) = 0$ and $QU = -\kappa T \eta$, proving compatibility with the differentials.

Use state coordinates x_n, π_n, τ_m, $\bar\eta_m$, $\bar\iota_n$ with $n \geq 1$, $m \geq 0$. They are respectively the creation modes of J, p, t, η, ι. The last two are odd. The differential on W is

$$Q = \sum_{m \geq 0} \bar\eta_m \partial_{\tau_m} + \sum_{n \geq 1} \pi_n \partial_{\bar\iota_n}.$$

The operator $\sum_m \tau_m \partial_{\bar\eta_m} + \sum_n \bar\iota_n \partial_{\pi_n}$ contracts the nonconstant quartet monomials after division by their total number of generators. Thus $H(E) = k$. Since $p = Q\iota$, the map i induces the identity on $H(W) = V_\kappa$.

The nonnegative modes of $J + p$ are

$$(J + p)_0 = \partial_{\tau_0}, \qquad (J + p)_n = \kappa n \partial_{x_n} + \partial_{\tau_n}.$$

In the coordinates $v_n = x_n - \kappa n \tau_n$ they become the derivatives ∂_{τ_m}. Their common kernel is

$$k[v_n, \pi_n : n \geq 1] \otimes \Lambda(\bar\eta_m : m \geq 0, \bar\iota_n : n \geq 1).$$

Its differential has $Qv_n = -\kappa n \bar\eta_n$ and $Q\bar\iota_n = \pi_n$.

The image of j consists of the v_n, $\bar\eta_m$ factors. The remaining pairs are contractible. This is a decomposition of complexes, without an assertion that the two vertex subalgebras commute.

The continuous invariant complex in the τ coordinates is a polynomial de Rham complex. Hence

$$\mathcal{Z}_\kappa \longrightarrow W^{\mathfrak{b}} \longrightarrow C^\bullet_{\mathrm{sm}}(W) \longleftarrow C^\bullet_{\mathrm{sm}}(V_\kappa)$$

is a zigzag of quasi-isomorphisms. To justify the totalizations, give each coordinate and each cochain generator weight equal to its mode index. At fixed weight r, the quartet degree lies between $-r$ and $r+1$, and the cochain degree lies between 0 and $r+1$. Both filtrations are finite.

Finally, in the total complex,

$$(Q + d_{\mathrm{CE}})\tau_0 = \bar{\eta}_0 + \xi_0.$$

The class of $j(c) = -\bar{\eta}_0$ is therefore $[\xi_0]$. Both cohomology algebras are exterior on this class, with the same unit, which proves the product assertion. □

The correction to the current is necessary. On the original cochain complex, $[d, J_{-1}]\mathbf{1} = \kappa\xi_1$. The field p makes the boundary current closed, and $-\kappa T t$ cancels its bracket with the cochain current.

There is also a binary product in the homotopy category. In $\mathcal{Z}_\kappa \otimes E$, with differential $d + Q$, the two commuting maps are

	J	c
b_1	$J + p$	c
b_2	$J - \kappa T t$	$c - \eta$.

Their product, followed by the inverse of the quartet quasi-isomorphism, is unital in the homotopy category. The two unit homotopies are explicit over $k[s, ds]$:

	J	c
$b_1(s)$	$J + s p + ds\, \iota$	c
$b_2(s)$	$J - s\kappa T t$	$c - s\eta - ds\, t$.

The signs follow from the tensor differential: $d(ds\, \iota) = -ds\, p$ and $d(ds\, t) = -ds\, \eta$. On cohomology this is the exterior product.

A binary operation and its unit homotopies do not yet give an associative object with all coherences. The construction below supplies them and specifies its order: its multiplication is the opposite of the binary operation just displayed.

A factorizing ghost action

For a differential vertex algebra B, write $C_B = C^{\mathrm{CE}}_\star(L_B)$ for its Chevalley collision coalgebra. Let $R = \Lambda[c]$, $F = C_R$, and $C = C_{V_\kappa}$. The multiplication of the commutative differential algebra R makes F a transverse monoid. Its vacuum is the pointing $J \to F$.

We first justify the tensor comparison used in the action. For a finite configuration I, put

$$P_I(t) = \prod_{i \in I} (t - z_i), \qquad H_I = \varinjlim_m \big(\mathcal{O}_{X^I}[t]/P_I(t)^m\big)^\vee.$$

The factorization lattice of R is $\Lambda_{\mathcal{O}_{X^I}}(H_I)$, with generators in degree one. The finite jet modules are free because P_I^m is monic. Exterior powers and filtered colimits give flat graded components. With zero differential this is a K-flat complex.

Chinese remainders give factorization on disjoint opens. On collision strata, the positive multiples of the changed root multiplicities are cofinal, giving the collision maps. The finite Cousin resolution identifies the lattice with C_R in the normalization used here. The same construction gives, naturally,

$$C_{R^{\otimes a}} \otimes^! C_B \simeq C_{R^{\otimes a} \otimes B}.$$

It respects multiplication, units, and polynomial homotopies.

The finite Cousin direction has at most $|I|$ terms, so its passage to the internal differential is valid. External tensor over k, localization, and finite symmetric-group coinvariants preserve quasi-isomorphisms. In particular, the maps C_{i_b} below are equivalences.

Let $W_b = V_\kappa \otimes E^{\otimes b}$ and $i_b(J) = K_b = J + \sum_{j=1}^{b} p_j$. On the cube of polynomial forms

$$\Omega_b = k[s_1, \dots, s_{b-1}] \otimes \Lambda(ds_1, \dots, ds_{b-1})$$

define rows recursively by

$$M_1 = e_1, \qquad M_i = s_{i-1}e_i + (1 - s_{i-1})M_{i-1}.$$

Every row has sum one. Put forms to the left of fields and set

$$\gamma_i = -\sum_j \big(M_{ij}\eta_j + (dM_{ij})t_j\big).$$

Proposition 20.18.5 *(The coherent ghost action).* The assignments $c_i \mapsto \gamma_i$ and $J \mapsto K_b$ give maps

$$R^{\otimes b} \otimes V_\kappa \longrightarrow W_b \otimes \Omega_b$$

whose faces define a unital transverse action of F on C. Its action map is the roof

$$C_R \otimes^! C_{V_\kappa} \longrightarrow C_{W_1} \xleftarrow{\sim} C_{V_\kappa}, \qquad c \mapsto -\eta_1, \quad J \mapsto J + p_1.$$

Proof. For the total differential $D = Q + d$, $D(M\eta + dM\, t) = 0$: the two $dM\,\eta$ terms cancel. The γ_i commute, and

$$[K_b \lambda \gamma_i] = -\sum_j dM_{ij} = 0.$$

Also $DK_b = 0$ and $[K_b \lambda K_b] = \kappa\lambda$. Thus the assignments preserve all generating operations.

At $s_i = 0$ the adjacent rows coincide. This face multiplies the corresponding ghost factors. The target comparison replaces one quartet by

$$p \mapsto p_i + p_{i+1}, \quad \iota \mapsto \iota_i + \iota_{i+1}, \quad t \mapsto t_i, \quad \eta \mapsto \eta_i.$$

It preserves brackets and differentials, lies under the boundary, and is a quasi-isomorphism.

At $s_i = 1$ the recurrence restarts at e_{i+1}. This face composes the actions of the two consecutive blocks. In a common resolution their boundary current is K_b.

These face identities agree on intersections. Repeated merges add p, ι and retain the first t, η, independently of parentheses. Repeated splits use the same separate quartets, and disjoint merges and splits commute. Thus the cubes give all associative coherences.

More explicitly, the cubical resolution of a nonunital monoid has terms $[0,1]^{b-1} \odot F^{\otimes^! b}$. Zero faces multiply adjacent inputs. Multiplication concatenates strings with intervening parameter one. Shrinking every parameter to zero contracts its augmentation to F. Our zero and one faces give a multiplicative map from this resolution.

Polynomial forms define actual mapping-space cubes as follows. Pull them back along every simplex of a cube. Each polynomial simplex algebra has cohomology k by its Euler contraction. The resulting diagrams $C_{W_b \otimes \Omega_n}$ receive objectwise equivalences from the constant boundary diagram. Invert this

natural transformation in the functor category. Naturality in faces, products, and target comparisons preserves all the identities already proved.

When every actor input is the vacuum, the roof is the identity under C, with its coherences. The quasi-unit theorem in *Higher Algebra*, Theorem 5.4.3.5, supplies the unital refinement with this pointed restriction. □

PROPOSITION 20.18.6 *(An action outside the ordinary center).* The ghost action does not factor through the diagonal tensor unit J. In particular, the ordinary center of V_κ does not represent the pointed action functor on all derived factorizing actors.

Proof. Put $\mathfrak{g} = L_{V_\kappa}$ and $\mathfrak{h} = L_{R\otimes V_\kappa}$. The one-ghost summand is $N[-1]$, where N has left state module

$$\bigoplus_{m\geq 0} k e_m \otimes V_\kappa, \qquad T(e_m \otimes v) = (m+1)e_{m+1}\otimes v + e_m \otimes Tv.$$

For a pointed map, bar–cobar gives a map $U_\star\mathfrak{h} \to U_\star\mathfrak{g}$ under $U_\star\mathfrak{g}$. Restrict to $\mathfrak{h}$, project by PBW symmetrization to $\mathfrak{g}$, and restrict to $N[-1]$. The adjoint action preserves symmetric degree, so this defines a homotopy invariant

$$\chi : \pi_0\Phi_C(F,u) \longrightarrow \operatorname{Ext}^1_{\mathfrak{g}\text{-Mod}}(N,\mathfrak{g}).$$

Negating the ghost negates χ.

For our roof, the extension is represented by

$$\nu(e_m\otimes v) = -\frac{1}{m!} : (T^m\eta)i(v) : .$$

A middle module is

$$\{(n,w)\in N\oplus W^0 : Qw = \nu(n)\}/\{(0,Qv) : v\in W^{-1}\}.$$

With kernel inclusion $v\mapsto [(0,-i(v))]$, it represents the indicated connecting class for the usual cochain cone $d(b,a) = (d_B b + f(a), -d_A a)$. Changing this kernel marking changes the sign, without changing whether it splits.

Lift $e_0\otimes \mathbf{1}$ by $-t$. Then

$$Q(-t) = -\eta, \qquad (J+p)_0(-t) = -\mathbf{1}.$$

The same zero mode vanishes on the quotient and on the kernel. Changing the lift cannot remove the displayed vacuum class. The vacuum is not a Q-boundary. Hence the extension is nonsplit.

Main-diagonal differential modules inherit their usual heart, and convolution with $\mathfrak{g}$ is exact there. This heart extension therefore gives a nonzero derived Ext class.

The automorphism negating the boundary current conjugates the action to the action with $c\mapsto -c$. It is realized on the resolution by negating J, t, p, η, ι. Thus conjugation negates the nonzero invariant χ. An action factoring through J is unchanged by boundary conjugation. In characteristic zero these conclusions are incompatible.

Finally, the positive modes on V_κ are the nonzero multiples of ∂_{x_n}, so its ordinary center is $k\mathbf{1}$ and its center coalgebra is J. This proves the asserted failure of representability. □

Tensor products and parameter algebras

The preceding tensor comparison extends to free fields over k. Suppose B has finitely many homogeneous generators, a PBW basis of their jets, central polynomial generating brackets, and a differential linear in the jets. Its factorization lattice has a PBW filtration whose graded pieces are symmetric powers of the free principal-parts modules. Each fixed PBW degree is a bounded complex of free modules. Extensions and filtered unions preserve K-flatness, so every lattice B_I is K-flat.

For two such algebras, the generating Lie algebras have a common constant and otherwise commute. Their vacuum modules give

$$(A \otimes_k B)_I = A_I \otimes_{\mathcal{O}_{X^I}} B_I.$$

The identities respect collisions, disjoint factorization, units, and symmetry. Realization contracts one diagonal tensor unit, giving a symmetric monoidal functor $B \mapsto C_B$. Polynomial homotopies of the form $B \otimes_k \Omega_b$ are also preserved.

The coefficient field in this assertion matters. A family over a differential parameter algebra need not be flat for derived tensor, even when its underlying graded algebra is free.

Proposition 20.18.7 *(A parameter tensor with extra cohomology).* Let $\Omega = k[s, ds]$, with $ds = ds$, and define

$$A_+ = \Omega[a_n : n \geq 0], \qquad A_- = \Omega[b_n : n \geq 0], \qquad Da_n = ds\, a_n, \quad Db_n = -ds\, b_n.$$

All generating vertex brackets vanish, and translation shifts each subscript by one and kills Ω. The augmentations $A_\pm \to \Omega$ are quasi-isomorphisms. Nevertheless,

$$H^0(A_+ \otimes_\Omega A_-) = B_0, \qquad H^{i\neq 0}(A_+ \otimes_\Omega A_-) = 0,$$

where B_0 is the charge-zero part of $k[a_i, b_j]$ for $\mathrm{ch}(a_i) = 1$, $\mathrm{ch}(b_j) = -1$. It has the presentation

$$B_0 = \frac{k[z_{ij} : i, j \geq 0]}{(z_{ij}z_{kl} - z_{il}z_{kj})}, \qquad z_{ij} = a_i b_j, \qquad Tz_{ij} = z_{i+1,j} + z_{i,j+1}.$$

For r row indices and s column indices, its polynomial degree-q dimension is

$$\binom{r+q-1}{q}\binom{s+q-1}{q}.$$

The derived-to-ordinary comparison on cohomology is $k \to B_0$. Neither A_+ nor A_- is K-flat over Ω.

Proof. On a monomial of charge c, the coefficient complex is

$$k[s] \xrightarrow{\partial_s + c} k[s]\, ds.$$

For $c \neq 0$ its inverse on a polynomial f is the finite sum

$$(\partial_s + c)^{-1} f = \sum_{n\geq 0} \frac{(-1)^n}{c^{n+1}} \partial_s^n f.$$

For $c = 0$, differentiation is surjective with kernel k. In each $A_\pm$, only the vacuum has charge zero. In the tensor, every balanced monomial does. This proves the cohomology calculation and, more precisely, the quasi-isomorphism $\Omega \otimes_k B_0 \to A_+ \otimes_\Omega A_-$.

Every balanced monomial is a product of $a_i b_j$. The minor relations exchange the paired column indices of two factors. They put each monomial into the pairing of its sorted row and column multisets. Distinct pairs of multisets have distinct images in $k[a_i, b_j]$, proving that these relations generate the kernel. This argument applies to infinitely many variables because each polynomial uses finitely many. Counting the two multisets gives the dimension formula, and the Leibniz rule gives translation.

The inclusions $\Omega \to A_{\pm}$ compute their derived tensor, whose cohomology is k. The comparison is the inclusion of constants into B_0. Tensoring either augmentation with the other algebra kills the nonzero class $a_0 b_0$. It therefore fails to preserve that quasi-isomorphism, proving both failures of K-flatness. □

The coherent cochain monoid

In the boundary resolution W_b, define

$$\mathcal{Z}^{(b)} = W_b^{\mathfrak{b}}, \qquad K_b = J + \sum_{j=1}^{b} p_j.$$

For an order-preserving surjection $\phi : \{1, \dots, b'\} \to \{1, \dots, b\}$, let $\phi_*(j) = \min \phi^{-1}(j)$. The merge map fixes J and sends

$$p_j \mapsto \sum_{\phi(l)=j} p_l, \quad \iota_j \mapsto \sum_{\phi(l)=j} \iota_l, \quad t_j \mapsto t_{\phi_*(j)}, \quad \eta_j \mapsto \eta_{\phi_*(j)}.$$

It maps W_b to $W_{b'}$ and preserves the boundary current. The shift $r_a : W_b \to W_{a+b}$ sends J to K_a and quartet j to quartet $a + j$. Define

$$\mu_{a,b} : \mathcal{Z}^{(a)} \otimes W_b \longrightarrow W_{a+b}, \qquad z \otimes w \longmapsto {:}\, z\, r_a(w) \,{:}.$$

LEMMA 20.18.8 *(Products through boundary resolutions).* The merge maps are quasi-isomorphisms on both W_b and $\mathcal{Z}^{(b)}$. The latter is freely generated by

$$J - \kappa T t_1, \quad t_j - t_1 \ (j \geq 2), \quad p_j, \eta_j, \iota_j.$$

The products $\mu_{a,b}$ preserve differentials and vertex operations, are strictly associative, and commute with merges. Their restrictions map $\mathcal{Z}^{(a)} \otimes \mathcal{Z}^{(b)}$ to $\mathcal{Z}^{(a+b)}$. The b cochain inputs and one boundary input have images

$$J^{(i)} \mapsto U_i = J + \sum_{l<i} p_l - \kappa T t_i, \qquad c^{(i)} \mapsto -\eta_i, \qquad J \mapsto K_b.$$

The currents $U_1, \dots, U_b, K_b$ commute pairwise, and each has level κ.

Proof. A merged sum of p_l pairs with the retained $t_{\phi_*(j)}$ through exactly one term. The same holds for ι, η. This checks brackets. The formulas also commute with Q.

Every quartet index occurs once, so K_b maps to $K_{b'}$. The least-element rule is compatible with composition. Since all i_b are quasi-isomorphisms, so are the merges.

The nonnegative boundary modes are $\sum_j \partial_{\tau_0^{(j)}}$ and $\kappa n \partial_{x_n} + \sum_j \partial_{\tau_n^{(j)}}$. Use coordinates $x_n - \kappa n \tau_n^{(1)}$ and $\tau_n^{(j)} - \tau_n^{(1)}$ for $j \geq 2$. The modes become derivatives in $\tau_n^{(1)}$, so their common kernel has the stated free generators. Continuous polynomial de Rham cochains in these variables compare each commutant

to the derived invariants of W_b. The finite weight filtrations used in Theorem 20.18.4 apply for every finite b. Naturality and the equivalences i_b then show that merges are equivalences on commutants.

The image of r_a commutes with $\mathcal{Z}^{(a)}$: its current is K_a, and its other fields use later quartets. Thus normal multiplication defines a vertex algebra map. The identity $r_a r_b = r_{a+b}$ proves associativity. Merges preserve these shifts and inclusions, giving naturality.

Finally, shift the two images of j successively. This gives the displayed U_i and ghosts. For $i < j$, the $\kappa\lambda$ contribution from their common J cancels the pairing of $-\kappa T t_i$ with p_i. The same cancellation proves $[U_{i\,\lambda} K_b] = 0$. Their self-brackets retain $\kappa\lambda$. □

Theorem 20.18.9 *(A factorizing cochain monoid and its action).* The coalgebra $C_{\mathcal{Z}_\kappa}$ has a unital transverse monoid structure and a transverse action on C_{V_κ}. The b-fold product is represented by

$$\mathcal{Z}_\kappa^{\otimes b} \longrightarrow \mathcal{Z}^{(b)} \xleftarrow{\sim} \mathcal{Z}^{(1)} \xleftarrow[j]{\sim} \mathcal{Z}_\kappa.$$

The action is the quartet roof of Theorem 20.18.4. It restricts along $C_R \to C_{\mathcal{Z}_\kappa}$ to the pointed ghost action, and does not factor through J.

Proof. The free-field tensor comparison transports the preceding strict products and action to factorizing coalgebras. Let $\mathcal{P}$ be the category of nonempty finite ordered sets and order-preserving surjections. The two towers are functors on $\mathcal{P}^{\mathrm{op}}$, with lax multiplication given by concatenation. They take every arrow to an equivalence.

Their coherent descent can be seen in the operad for a nonunital algebra and a left module. The symmetric monoidal envelope of the pair of towers has fibers that are finite products of $\mathcal{P}^{\mathrm{op}}$. Each is contractible, since the singleton is final in $\mathcal{P}$. Localizing at the fiber arrows therefore gives the envelope of the algebra-module operad. The strict tower data descend through this localization. This proves associativity with all coherences, and gives the product roofs in the statement.

The vacuum is a quasi-unit. Right multiplication is the inclusion into the two-quartet term. Left multiplication is the shift r_1. On the image of j, the inclusion agrees with the merge. Two polynomial homotopies join the merge to the shift:

	J	c
	$J - \kappa T((1-s)t_1 + st_2)$	$-(1-s)\eta_1 - s\eta_2 - ds(t_2 - t_1)$
	$J + sp_1 + ds\,\iota_1 - \kappa T t_2$	$-\eta_2.$

They preserve brackets, differential, and commutation with K_2.

The vacuum action on the module is also an equivalence, since $r_1 i_1 = i_2$. The quasi-unit theorems (*Higher Algebra*, Theorem 5.4.3.5 and Proposition 5.4.3.14) refine both structures to unital ones, uniquely up to contractible choice.

Transport along j and i_1 gives the stated objects. Let $m : W_1 \to W_2$ be the merge map. The identity

$$\mu_{1,1}(j(z) \otimes i_1(v)) =: j(z) i_2(v) := m\big(: j(z) i_1(v) :\big)$$

identifies the action roof. Its restriction has $c \mapsto -\eta_1$ and $J \mapsto J + p_1$. Proposition 20.18.6 excludes factorization through J. □

The action at one position is the full evaluation of the moving-vacuum internal Hom. Indeed, its adjoint is $z \mapsto j(z)$ into the resolved invariants. In their continuous cochains, compare

$$f_0 : (J, c) \mapsto (J + p, c), \qquad f_1 : (J, c) \mapsto (J - \kappa T t, -\eta).$$

They are joined through $(J, c) \mapsto (J, c)$ by

$$(J + s p + ds\, \iota, c), \qquad (J - s\kappa T t, (1 - s)c - s\eta - ds\, t).$$

Here the total cochain differential has $Dt = \eta + c$.

Integrating the ds coefficient gives a chain homotopy commuting with translation. Thus the adjoint is the identity of $\mathcal{Z}^{\ell}$, and evaluation induces

$$R\operatorname{Hom}_{D_X}(N, \mathcal{Z}^{\ell}) \simeq R\operatorname{Hom}_{\mathrm{mov}}(N \otimes_{O_X} \mathcal{V}, \mathcal{V})$$

for every complex N. This universal property at one position does not establish finality of the factorizing actor on all configurations.

The order of the coherent product can also be checked on a common resolution. Put $\mathcal{T} = C^{\bullet}_{\mathrm{sm}}(W_2) = W_2 \otimes \Lambda[c]$. Its differential satisfies

$$DJ = \kappa T c, \quad Dt_l = \eta_l + c, \quad D\iota_l = p_l, \quad D p_l = D\eta_l = Dc = 0.$$

The tower roof and the exchanged binary roof map into $\mathcal{T}$ through quasi-isomorphisms. Their unit maps are joined by the two homotopies used above.

A family of two-input maps is given by

$$J^{(i)} \longmapsto J + \sum_l \big(\beta_l^{(i)} p_l + \kappa \epsilon_l^{(i)} T t_l + ds\, \partial_s \beta_l^{(i)} \iota_l\big),$$
$$c^{(i)} \longmapsto (1 + \epsilon_1^{(i)} + \epsilon_2^{(i)})c + \sum_l \big(\epsilon_l^{(i)} \eta_l + ds\, \partial_s \epsilon_l^{(i)} t_l\big).$$

It preserves brackets precisely when

$$\beta^{(i)} \cdot \epsilon^{(i)} = 0, \qquad \beta^{(1)} \cdot \epsilon^{(2)} + \beta^{(2)} \cdot \epsilon^{(1)} = -1,$$

and every cross pairing is constant in s.

The tower starts at

$$(\beta^{(1)}, \epsilon^{(1)}, \beta^{(2)}, \epsilon^{(2)}) = ((0, 0), (-1, 0), (1, 0), (0, -1)).$$

Change successively

$$\epsilon^{(2)} = (0, s - 1), \quad \beta^{(2)} = (1, 2s), \quad \epsilon^{(1)} = (-1 + 2s, -s), \quad \beta^{(1)} = (s, s),$$

keeping the other three vectors at their current values. Each segment satisfies the bracket conditions. The endpoint is the exchanged binary map. Thus the tower product equals the opposite binary product in the localization at quasi-isomorphisms. Its induced cohomology product remains the exterior product.

A coherent loop that prevents finality

The cochain monoid has the expected moving-vacuum evaluation. A local center must also represent the homotopies of actions on separated configurations. An actor in degree two detects a homotopy that the cochain monoid misses.

Let

$$N = k[t_0, t_1, \ldots], \qquad |t_n| = 2, \qquad T t_n = t_{n+1}, \qquad d = 0,$$

with zero vertex brackets. Its augmentation ϵ kills the t_n. The coalgebra $F_2 = C_N$ is a commutative transverse monoid. Its lattice on X^I is $\mathrm{Sym}_{\mathcal{O}_{X^I}}(H_I[-2])$. It is K-flat by the same finite-jet argument as for the ghost, and gives natural equivalences

$$F_2^{\otimes^! p} \otimes^! C \simeq C_{N^{\otimes p} \otimes V_\kappa}.$$

Let α_0 be the augmentation action and $g_0 : F_2 \to J \to C_{\mathcal{Z}_\kappa}$ its factorization through the unit.

Put $\mathfrak{h}_N = L_{N \otimes V_\kappa}$ and $\mathfrak{g} = L_{V_\kappa}$. The derived deformation complex at α_0 is the relative convolution complex

$$\mathcal{T}_{\mathrm{rel}} = \mathrm{Hom}^{\mathrm{der}}\big(\overline{C^{\mathrm{CE}}_\star(\mathfrak{h}_N)}, U_\star\mathfrak{g}\big)_{\mathrm{rel}}, \qquad d_{\tau_0} = d + [\tau_0, -].$$

Here *relative* means vanishing on $C^{\mathrm{CE}}_\star(\mathfrak{g})$, and all coefficient Homs remain derived. Completion is by products over the total polynomial degree G in the t_n. The base twisting cochain τ_0 is the augmentation on singleton inputs.

This filtration is complete, starts at $G = 1$, and is added by convolution. Thus the complex is a pronilpotent differential graded Lie algebra. Derived bar–cobar identifies

$$\pi_1\Phi_C(F_2, u) = \big(H^0\mathcal{T}_{\mathrm{rel}}, \mathrm{BCH}\big), \qquad \pi_n\Phi_C(F_2, u) = H^{1-n}\mathcal{T}_{\mathrm{rel}} \quad (n \geq 2)$$

at the augmentation. A closed degree-zero cochain h gives the loop $\tau_0 + ds\, h$, with class $\exp[h]$.

Because τ_0 is Lie-valued, PBW output degree is preserved. In output degree a, a component has a partition into a blocks, determined on their distinct locus. Its differential acts inside each block. This assertion uses support and localization, rather than a factorization of Hom for infinite-dimensional coefficient modules.

Lemma 20.18.10 *(The degree bound).* In total t-degree G and output degree a, the complex is concentrated in degrees at least $a - 2G$. For $G = 1$, $a = 2$, closed degree-zero cochains have ordinary binary coefficients, and no degree-zero boundaries occur.

Proof. With n inputs and coefficient extension degree e, suspension changes the input degree from $2G$ to $2G - n$. The cochain degree is therefore $e + n - 2G$. The underlying coefficient modules admit derived Hom in degrees $e \geq 0$. Support forces $n \geq a$: a source on at most n blocks restricts to zero on an a-point locus when $n < a$. Derived adjunction gives the same vanishing.

At each finite configuration the number of blocks is finite. Products and the configuration limit preserve the bound. For $G = 1$, $a = 2$, degree zero forces $n = 2$, $e = 0$, and degree minus one does not occur. □

Define the augmentation derivation $\ell : N \to k$ by $\ell(t_n) = \delta_{n0}$, vanishing on other polynomial degrees. It obeys

$$\ell(Ta) = 0, \qquad \ell(ab) = \ell(a)\epsilon(b) + \epsilon(a)\ell(b).$$

The corresponding singleton cochain is

$$\kappa_1\big(s(t_n \otimes v)\big) = \delta_{n0} v.$$

It has degree -1 and is closed.

Indeed, $t_0 \mapsto u$, $t_{n>0} \mapsto 0$ defines the first-order vertex map with $|u| = 2$, $u^2 = 0$. The coefficient of u in its Maurer–Cartan equation is $d_{\tau_0}\kappa_1 = 0$. The degree bound excludes boundaries below it, and its value on $s(t_0 \otimes \mathbf{1})$ is nonzero.

On the vacuum, $J_1 = \kappa \partial_{x_1}$ commutes with T and is a derivation of every vertex operation. It kills $\mathbf{1}$ and sends J to $\kappa\mathbf{1}$. Define the two-output cochain

$$h = \kappa_1 \boxtimes J_1 = -\big(\kappa_1 * (J_1\tau_0)\big)^{(2)}.$$

On two separated singleton blocks its value is

$$h\big(s(t_n \otimes v), s(\mathbf{1} \otimes w)\big) = \delta_{n0}\, v \boxtimes J_1 w.$$

The negative convolution sign cancels the sign from passing the degree-one second cochain across the degree-one first input. Both factors are closed, so h is closed of degree zero.

LEMMA 20.18.11 *(The separated vacuum coefficient).* There is a degree-minus-one chain map $\mathsf{P} : \mathcal{T}_{\mathrm{rel}}^{(a)} \to \mathcal{T}_{\mathrm{rel}}^{(a-1)}$ with

$$d_{\tau_0}\mathsf{P} + \mathsf{P}d_{\tau_0} = 0, \qquad \mathsf{P}(h) = \kappa\kappa_1.$$

Its value on the degree-zero, $G = 1$, two-output part of every class induced by a pointed map to $C_{\mathcal{Z}_\kappa}$ is zero.

Proof. Insert a separated singleton at z carrying $\mathbf{1} \otimes J$, use test function one there, and apply the vacuum coefficient to its output. Expand the resulting rational function at $z = \infty$ and take its constant term. Place the suspended current first, including its permutation sign. On a degree-r cochain multiply this operation by $(-1)^{r+1}$. This defines P.

The input J is a free translation generator, since it has weight one and TV_+ has no weight-one state. Hence insertion is defined on derived coefficient resolutions. Extraction commutes with the coefficient differential and with derivatives and residues in the remaining variables. A bracket inside the singleton block has no second input. The two products with $\tau_0(J) = J$ cancel on the separated locus. The remaining differential is the one on the other blocks.

Insertion has degree -1. The displayed sign gives the asserted anticommutation. These operations preserve finite normal order and G, so extend to the completion.

For h, the inserted current gives $J_1 J = \kappa\mathbf{1}$, constant in z. The permutation sign cancels the sign in P, leaving $\kappa\kappa_1$.

For the image assertion, use the quartet commutant model of the cochain action. Its transferred twisting cochain is Lie-valued and supported on small diagonals. Its linear term is the weight-preserving contraction of

$$m(\zeta, v) = \zeta_{(-1)} i(v).$$

Here is a transfer retaining these properties. Take the quartet contraction (i, π, H) in coordinates that keep $i(V_\kappa)$ fixed. Then $QH + HQ = 1 - i\pi$, $Hi = \pi H = H^2 = 0$, and $H\mathbf{1} = 0$.

On n suspended factors let K_n be the signed sum of the shifted H, divided by the number of factors of positive quartet degree. Set it to zero when that number is zero. It satisfies

$$d_Q K_n + K_n d_Q = 1 - \mathrm{Sym}^n(i\pi).$$

With $P_1 = s\pi s^{-1}$, define recursively

$$P_n = \left(\frac{1}{2} \sum_{n_1+n_2=n} \beta_V (P_{n_1} \otimes P_{n_2})\Delta_{n_1,n_2} - P_{n-1}\delta_W\right) K_n.$$

Here β_V is the suspended bracket, ∂_W the Chevalley bracket differential, and Δ the unshuffle coproduct.

The expression in parentheses is a cycle and vanishes on $\mathrm{Sym}^n(i)$. The contraction identity therefore proves the coalgebra-map equation inductively. It gives $P\,\mathrm{CE}(i) = 1$. Each component is a finite sum of bracket and contraction composites, with values in the Lie object.

The higher components vanish on a vacuum input. Indeed, the vacuum has quartet degree zero, and its bracket with the output equals the sum of its brackets with the inputs. This is the partial-fraction residue identity around those inputs. The two remaining terms in the recursion cancel, with the suspended vacuum sign. Thus $\sigma = \vartheta_V P\,\mathrm{CE}(m)$ is the required normalized twisting cochain, where ϑ_V is the canonical Lie-valued twisting cochain.

Free localized normal resolutions have surjective normal derivative. Their inverse limits and products preserve this comparison. Its derivatives retain the normalization.

Differentiate composition at the augmentation. At a separated input carrying $\mathbf{1} \otimes J$, either the base point supplies J, or the variable block supplies $\zeta_{(-1)}(J + p)$ before contraction. Both have positive conformal weight. Their vacuum coefficient vanishes, since the resolution has nonnegative weights and the contraction preserves weight.

This is also a derived vanishing: the map factors through the subcomplex of positive weights before applying coefficient resolutions. By the degree bound, its closed degree-zero two-output representative is ordinary. Its vacuum coefficient is already zero before taking the constant term at infinity. This proves the assertion. □

Theorem 20.18.12 *(Failure of finality of the cochain actor).* The mapping space from the augmentation action (F_2, α_0) to the cochain action $(C_{\mathcal{Z}_\kappa}, \alpha_{\mathcal{Z}})$ has at least two connected components. Consequently this cochain actor is not a local center on the full category of factorizing coalgebras.

Proof. The class $[h]$ is missed by the map on pointed-action fundamental groups. Its probe is the nonzero class $\kappa[\kappa_1]$, whereas every image has zero probe. A source loop is represented by $ds\,K'$ with K' closed of degree zero. Only the derivative of composition contributes, since $(ds)^2 = 0$. The exponential identification for the complete Lie algebra therefore shows that $\exp[h]$ is not an image loop.

We must also retain action coherences. For p actor factors put

$$\epsilon_p(a_1, \dots, a_p) = \prod_i \epsilon(a_i), \qquad \ell_p(a_1, \dots, a_p) = \sum_i \ell(a_i) \prod_{j \neq i} \epsilon(a_j).$$

Let κ_p be the resulting singleton cochain and $\tau_0^{(p)}$ the augmentation twisting cochain. Then

$$h_p = -\big(\kappa_p * (J_1 \tau_0^{(p)})\big)^{(2)}$$

is closed. Over a square-zero parameter δ of degree one, $\tau_0^{(p)} + \delta h_p$ is a twisting cochain.

Multiplication faces use the derivation rule for ℓ. Unit faces use $\ell(1) = 0$. Composition uses

$$\ell_{p+q} = \ell_p \otimes \epsilon_q + \epsilon_p \otimes \ell_q.$$

Terms with two copies of δ vanish. On every parenthesized word, the linear term is therefore the same sum over the factor on which ℓ acts.

The free associative bar resolution has constant compatible fillers on all simplices and its usual extra unit degeneracy. It gives all action coherences. Extending $\delta \mapsto ds$ produces a loop of transverse actions based at α_0, whose underlying pointed loop is $\exp[h]$.

The mapping space in the statement is the homotopy fiber over α_0 of

$$\mathrm{Map}_{\mathrm{Alg}}(F_2, C_{\mathcal{Z}_\kappa}) \longrightarrow \mathrm{Act}(F_2, C).$$

It contains g_0. The loop just constructed is not the image of a loop on the left, since forgetting multiplication would contradict the pointed obstruction. The homotopy-fiber sequence therefore gives at least two components. A final action would make this fiber contractible. □

This obstruction concerns the specified cochain actor. It does not exclude every possible local center, or claim that individual actions cannot lift. The missing homotopy changes the external-product identifications at a separated boundary input. Fixing those identifications coherently defines a further representability problem. So does constructing a larger actor that retains their deformations.

20.19 The marked center of the affine E_6 category

A finite tensor category supplies a different central action. Its sectors may change the boundary object. The affine E_6 realization makes that action explicit, including the algebra that recovers the boundary and the fields that implement its junctions.

Work over $\mathbb{C}$ at the unitary embedding of Theorem 16.11.1. Put

$$V = L_{\widehat{\mathfrak{sl}}_2}(10, 0), \qquad \mathcal{D} = \mathrm{Rep}(V).$$

Its simple modules H_j, $0 \le j \le 10$, have lowest weights $h_j = j(j+2)/48$. Retain the connected commutative separable algebra $A = H_0 \oplus H_6$ with its specified multiplication.

Let $\mathcal{M} = \mathcal{D}_A$ be the category of right A-modules. Our balanced tensor gives a right module N the left action

$$\ell_N = r_N c_{A,N}.$$

This convention is part of the center computation.

The even component C has simples 1, g, ρ and

$$g^2 = 1, \qquad g\rho = \rho g = \rho, \qquad \rho^2 = 1 + g + 2\rho.$$

With $d = 1 + \sqrt{3}$, their dimensions are $1, 1, d$, and $\dim C = 6 + 2\sqrt{3}$. The local modules form $\mathcal{E} = \mathrm{Rep}(W)$ for $W = L_{\widehat{\mathfrak{so}}_5}(1, 0)$:

$$W_0 = H_0 \oplus H_6, \quad W_v = H_4 \oplus H_{10}, \quad W_s = H_3 \oplus H_7.$$

Their dimensions are $1, 1, \sqrt{2}$ and their lowest weights are $0, 1/2, 5/16$.

Let $S = L_{\widehat{\mathfrak{so}}_{11}}(1, 0)$. Its three simples S_0, S_v, S_s have the same dimensions and weights $0, 1/2, 11/16$. The affine realization identifies $\mathrm{Rep}(S)$ with $\mathcal{E}^{\mathrm{rev}}$, where reversal means $c^{\mathrm{rev}}_{X,Y} = c^{-1}_{Y,X}$. This is the reversed Ising category in Bischoff's center realization, Corollary 4.7 and Section 5.

Set $T = V \otimes S$ and $\mathcal{T} = \mathcal{D} \boxtimes \mathrm{Rep}(S)$. The invertible object $J_* = H_{10}S_v$ has order two and integral weight three. Choose the unitary commutative algebra $L = 1 \oplus J_*$ with its vacuum unit. Its vertex extension is

$$U = H_0S_0 \oplus H_{10}S_v, \qquad c_U = 8.$$

Here H_jS_a abbreviates the external tensor module $H_j \otimes S_a$. The category tensor is the specified vertex tensor product, not the algebraic tensor of states. The tensor-product and extension theorems apply to these strongly rational affine algebras.

An object of the *Drinfeld center* $Z(C)$ is an object M with a natural half-braiding $\gamma_N : M \otimes_A N \to N \otimes_A M$ satisfying the hexagon. Its forgetful functor retains the object while forgetting this family.

Theorem 20.19.1 *(Ten central sectors).* There is a braided equivalence

$$\Psi : \mathrm{Rep}(U)^{\mathrm{rev}} \xrightarrow{\sim} Z(\mathcal{C}).$$

The central forgetful functor F is obtained before descent from

$$\widetilde{F}(X,Y) = (X \otimes A) \otimes_A Y, \qquad X \in \mathcal{D}, \quad Y \in \mathcal{E}.$$

Its half-braiding with a right A-module N is induced by

$$X \otimes Y \otimes N \xrightarrow{1 \otimes c_{Y,N}} X \otimes N \otimes Y \xrightarrow{c_{N,X}^{-1} \otimes 1} N \otimes X \otimes Y.$$

Associators are inserted between these displayed tensors. Restrict to $\mathcal{C}$ and descend by the trivial character of $\widetilde{F}(L) \simeq \mathbb{C}[\mathbb{Z}/2] \otimes 1$.

The ten simples and their forgetful images are

	restriction to T	F	dim	h mod 1
x_0	$H_0S_0 \oplus H_{10}S_v$	1	1	0
x_2	$H_2S_0 \oplus H_8S_v$	ρ	d	$1/6$
x_4	$H_4S_0 \oplus H_6S_v$	$g + \rho$	$d+1$	$1/2$
x_6	$H_6S_0 \oplus H_4S_v$	$1 + \rho$	$d+1$	0
x_8	$H_8S_0 \oplus H_2S_v$	ρ	d	$2/3$
x_{10}	$H_{10}S_0 \oplus H_0S_v$	g	1	$1/2$
y_1	$(H_1 \oplus H_9)S_s$	ρ	d	$3/4$
y_3	$(H_3 \oplus H_7)S_s$	$1 + g + \rho$	$d+2$	0
z_+	H_5S_s	ρ	d	$5/12$
z_-	H_5S_s	ρ	d	$5/12$.

Sums in this table are direct sums. The twist in $Z(\mathcal{C})$ is $e^{-2\pi i h}$. The universal categorical central action is $(Z, M) \mapsto F(Z) \otimes_A M$ with this half-braiding.

Proof. Apply the center theorem for a connected étale algebra to $\mathcal{D}^{\mathrm{rev}}$. With our module tensor convention, the theorem of Davydov–Müger–Nikshych–Ostrik gives

$$\mathcal{D}^{\mathrm{rev}} \boxtimes \mathcal{E} \simeq Z(\mathcal{M}).$$

Its two crossings give the displayed formula. Locality of Y, expressed by $r_Y c_{A,Y} c_{Y,A} = r_Y$, preserves the balancing relations. The braiding hexagons then give the half-braiding hexagon.

The unit checks the orientation. For $Y = N = A$ the inverse crossing cancels the crossing in the left unit constraint. A positive crossing would leave the double braiding $c_{A,X}c_{X,A}$. On the H_6 channel of $H_2 \boxtimes H_6$ its scalar is

$$e^{2\pi i(h_6 - h_2 - h_6)} = e^{-\pi i/3} \neq 1.$$

It would therefore fail the unit constraint.

The central object $H_{10}S_v$ forgets to $g \otimes_A g = 1$. Its central structure is the nontrivial parity character. The fusion rules force g, ρ to have neutral degree in any grading, while $H_1 \otimes A$ generates the odd part.

Thus the universal grading of $\mathcal{M}$ is $\mathbb{Z}/2$. The center theorem for graded fusion categories (Gelaki–Naidu–Nikshych, Corollary 3.7) identifies the neutral de-equivariantized center with $Z(C)$.

Concretely, an L-action becomes an involution t after applying $\widetilde{F}$. The idempotent $(1+t)/2$ selects its vacuum character. The half-braiding of J_* is trivial on C, so it descends to this image. The vertex extension theorem identifies $\mathcal{T}_L^0$ with $\mathrm{Rep}(U)$ as a braided tensor category. Reversing the ambient braiding reverses the local-module braiding, proving the claimed orientation.

The monodromy of J_* on $H_j S_a$ is $(-1)^j$ for $a = 0, v$, and $(-1)^{j+1}$ for $a = s$. There are seventeen local simples before extension. Eight orbits have size two. The fixed object $H_5 S_s$ has two equivariant structures, producing z_+, z_-. The multiplication of L fixes $t^2 = 1$. Changing the sign of its normalized fixed-point intertwiner exchanges these two labels.

To compute F, let $\widetilde{I}$ be right adjoint to $\widetilde{F}$. Rigidity and induction give

$$\mathrm{Hom}_{\mathcal{M}}(\widetilde{F}(X,Y),1) \simeq \mathrm{Hom}_{\mathcal{D}}(X, \mathrm{Res}\, Y^*).$$

Consequently

$$\widetilde{I}(1) = (H_0 \oplus H_6)S_0 \oplus (H_4 \oplus H_{10})S_v \oplus (H_3 \oplus H_7)S_s.$$

The images of H_0S_0, $H_{10}S_0$, H_2S_0 are $1, g, \rho$. By the projection formula, the multiplicity of one of these images in $\widetilde{F}(Q)$ is the coefficient of its corresponding T-module in $\widetilde{I}(1) \boxtimes Q$. Use

$$H_a \boxtimes H_b = \bigoplus_{\substack{|a-b| \le j \le \min(a+b,20-a-b) \\ j \equiv a+b \ (2)}} H_j$$

and $S_v^2 = S_0$, $S_v S_s = S_s$, $S_s^2 = S_0 + S_v$. This gives the forgetful column. At the fixed point the image is 2ρ. Each resolved summand has dimension d and therefore image ρ.

Dimensions and weights follow by induction. The sum of their squared dimensions is $(6+2\sqrt{3})^2$. Finally, a central tensor functor is exactly a tensor functor with the specified half-braidings. Its factorization through $Z(C)$ gives the universal categorical action. □

This action has actual affine fields. For $p : F(Z) \otimes_A M \to N$, take the intertwining operator associated with

$$\mathrm{Res}\, F(Z) \boxtimes \mathrm{Res}\, M \longrightarrow \mathrm{Res}(F(Z) \otimes_A M) \xrightarrow{\mathrm{Res}\, p} \mathrm{Res}\, N.$$

The first arrow is the balanced quotient defined by the separability idempotent. Vertex tensor associativity transports its identities to every descendant coefficient. The X constituent crosses clockwise and the local Y constituent counterclockwise.

The parameter states are those of $\mathrm{Res}\, F(Z)$. A field with the states of the U-module Z has another domain.

The distinguished algebra and the boundary sectors

Let I be right adjoint to F and put $B = I(1)$. Its multiplication is part of the central data: if $\epsilon : F(B) \to 1$ is the counit, then $\mu_B : B \boxtimes B \to B$ is adjoint to

$$F(B \boxtimes B) \simeq F(B) \otimes_A F(B) \xrightarrow{\epsilon \otimes \epsilon} 1.$$

The unit is adjoint to 1_1. A direct-sum decomposition alone would not specify this algebra.

Proposition 20.19.2 *(The marked algebra).* With the preceding multiplication,

$$B = x_0 \oplus x_6 \oplus y_3, \qquad F(B) = 31 \oplus g \oplus 2\rho.$$

It is connected, commutative, separable, of dimension $6 + 2\sqrt{3}$, and has trivial twist. Using the same positive balancing convention on right B-modules,

$$\mathrm{Rep}(U)_B \simeq \mathcal{C}, \qquad \mathrm{Rep}(U)_B^0 \simeq \mathrm{Vec}\,.$$

Its vertex extension is the holomorphic algebra $L_{\widehat{\mathfrak{e}_8}}(1,0)$.

Proof. Adjunction identifies the multiplicity of Z in $I(1)$ with the multiplicity of 1 in $F(Z)$. The table gives the displayed sum. The adjoint of the central forgetful functor carries its canonical connected commutative separable algebra, with the multiplication just defined. The canonical-algebra theorem recovers $\mathcal{C}$ as its module category.

Commutativity for a braiding also implies commutativity for its reverse. The canonical module tensor uses $c_{N,B}^{-1}$ in the central category. Since that category is $\mathrm{Rep}(U)^{\mathrm{rev}}$, this is the positive crossing $c_{B,N}$ of U-modules. The identity $\dim(B)^2 = \mathrm{Dim}\,\mathrm{Rep}(U)$ makes the local module category trivial. All three summands of B have integral weight.

The vertex extension theorem constructs a simple CFT-type algebra $\mathbb{V}_B$ with multiplication μ_B. Its restriction to T is $\widetilde{I}(1)$. It has a one-dimensional weight-zero space. The other two U-summands begin in weight one.

Apply Creutzig–Kanade–McRae's gluing theorem, Theorem 5.12 and Corollary 5.13, with S as the first factor. Its distinct simple modules pair with the V-modules $\mathrm{Res}\,W_a$, and

$$\dim \mathrm{Hom}_V(V, \mathrm{Res}\,W_a) = \delta_{a0}.$$

It proves strong rationality of this extension.

Every irreducible extension module is grading-restricted. Its restriction to the regular algebra T has finite multiplicities, since each conformal-weight space is finite. The local-module equivalence therefore places it in $\mathrm{Rep}(U)_B^0 = \mathrm{Vec}$. Thus the vacuum is the only irreducible module.

The central charge is eight, so Dong–Mason's classification, Theorem 1, identifies it with $L_{\widehat{\mathfrak{e}_8}}(1,0)$. This constructs the embedding from μ_B. It requires no uniqueness assertion about another embedding into the same holomorphic algebra. □

The other simple right B-modules are determined by the same adjunction:

$$\begin{aligned} I(g) &= x_{10} \oplus x_4 \oplus y_3, \\ I(\rho) &= x_2 \oplus x_4 \oplus x_6 \oplus x_8 \oplus y_1 \oplus y_3 \oplus z_+ \oplus z_-. \end{aligned}$$

Their complete restrictions are

$$\begin{aligned} I(1)|_T &= (H_0 \oplus H_6)S_0 + (H_4 \oplus H_{10})S_v + (H_3 \oplus H_7)S_s, \\ I(g)|_T &= (H_4 \oplus H_{10})S_0 + (H_0 \oplus H_6)S_v + (H_3 \oplus H_7)S_s, \\ I(\rho)|_T &= (H_2 \oplus H_4 \oplus H_6 \oplus H_8)(S_0 \oplus S_v) \\ &\qquad + (H_1 \oplus H_3 \oplus 2H_5 \oplus H_7 \oplus H_9)S_s. \end{aligned}$$

These are full module identities, including descendants. Their S_0-multiplicity spaces are the original three affine sectors. Only $I(1)$ is local. The other two are ordered boundary sectors, rather than ordinary modules of the holomorphic extension.

The extension also realizes fields with full U-module inputs. For $K_M = I(M)$, the projection formula is

$$\kappa_{Z,M} : Z \boxtimes K_M \xrightarrow{\sim} I(F(Z) \otimes_A M).$$

It is adjoint to $1 \otimes \epsilon_M$. Rigidity and adjunction identify, for every Q,

$$\begin{aligned} \mathrm{Hom}(Q, Z \boxtimes I(M)) &\simeq \mathrm{Hom}_C(F(Z)^* \otimes_A F(Q), M) \\ &\simeq \mathrm{Hom}(Q, I(F(Z) \otimes_A M)). \end{aligned}$$

Hence $\kappa_{Z,M}$ is an isomorphism. The counit construction makes it B-linear and compatible with iterated tensor products.

For a junction $p : F(Z) \otimes_A M \to N$, the field is

$$\mathcal{J}_p = \mathcal{Y}_{I(p)\kappa_{Z,M}} : Z \otimes K_M \longrightarrow \overline{K_N}\{u\}.$$

The bar completes the output by conformal weight. The braces allow the prescribed fractional powers. The vertex tensor correspondence gives all descendant coefficients, associativity, and the central crossing. At $Z = U$ this is the ordinary module field Y_{K_M}. It can have singular products.

The mirror states cannot be removed by an operation-preserving vacuum projection. Already on $V \otimes S$, let $P = 1_V \otimes \epsilon_S$ project to the mirror vacuum. For normalized mirror currents a, b,

$$P(a_{(1)}b) = 1_V, \qquad P(a)_{(1)}P(b) = 0.$$

The extension has weight-one dimension $3 + 55 + 7 + 5 \cdot 11 + 4 \cdot 32 = 248$. The mirror currents alone contribute 55 states. The multiplicity-space identification does not delete their vertex operations.

The pointed field action and its obstruction

For a vertex algebra Q, define

$$Z_{\mathrm{vert}}(Q) = \{q : q_{(n)}v = 0 \text{ for every } v \in Q,\ n \geq 0\}.$$

A pointed action of Q on W means a unital vertex map $Q \otimes W \to W$ restricting to the identity on $1_Q \otimes W$. It need not preserve a conformal vector.

PROPOSITION 20.19.3 *(The current double-pole obstruction).* Pointed vertex maps $Q \otimes W \to W$ are in bijection with vertex maps $Q \to \mathbb{C}1$. In particular, there is no pointed map $U \otimes W \to W$. The derived pointed mapping space

$$\mathrm{Map}_{C_W/}(C_U \otimes^! C_W, C_W)$$

is empty.

Proof. If w is central, skewsymmetry makes $Y(v,z)w$ regular. Taking $v = \omega_W$ gives $L_0 w = 0$. Since W is of CFT type, $Z_{\mathrm{vert}}(W) = \mathbb{C}1$.

The image of $q \otimes 1$ under a pointed map must commute with $1 \otimes W$ in every nonnegative product. It therefore lies in this center. Conversely, a vertex map $\varphi : Q \to \mathbb{C}1$ defines the pointed action $q \otimes w \mapsto \varphi(q)_{(-1)}w$. These constructions are inverse.

Take standard currents e, f in $L_{\widehat{\mathfrak{sl}}_2}(10, 0) \subset U$, with $(e, f) = 1$. They satisfy

$$e_{(1)}f = 10\, 1_U.$$

All nonnegative products of scalar vacuum states vanish. A unital map $U \to \mathbb{C}1$ would send this identity to $10 = 0$, which is impossible.

The affine vertex algebras give flat chiral algebras. The heart theorem identifies their pointed factorization mapping space with the discrete space of chiral maps from U to the chiral center of W. The same current relation excludes such a map locally. Thus no higher coherence supplies a missing point. Replacing the entire pointed diagram by equivalent objects preserves this empty mapping space. □

A regular action of a commutative vertex algebra on all three boundary sectors, compatible with their ordered junctions, is consequently scalar. Its restriction to W is a map of commutative differential algebras $\varphi : R^{\ell} \to \mathcal{O}_X$. Inserting the vacuum in the junction $W \otimes_A M \to M$ forces the action on every M to be $r \otimes m \mapsto \varphi(r)m$. If it commutes with translations, φ comes from a vertex map $R \to \mathbb{C}$.

The categorical central action remains valid. It takes a central sector to a tensor functor on boundary objects, often changing the object. Its universal property does not give a pointed field map on the fixed vacuum W. The ordinary transverse endomorphism monoid of C_W also exists by the earlier construction. The extra factorization and pointing conditions determine the obstruction just proved.

Derived current classes

Scalar regular actions do not exhaust derived central cohomology. Let $A_W = \Omega_{\mathrm{cut}} C(W)$ be the word cobar of the ordered residue coalgebra. Its scalar word center uses product totalization in word size and Hochschild length, with algebraic input state spaces.

Proposition 20.19.4 *(Polynomial current derivations).* The scalar word center satisfies

$$H^{i<0} Z_{\mathrm{word}}(A_W) = 0, \qquad H^0 Z_{\mathrm{word}}(A_W) = \mathbb{C}.$$

There is a natural injection of Lie algebras

$$\mathfrak{so}_5[z] \hookrightarrow H^1 Z_{\mathrm{word}}(A_W), \qquad x \otimes f \longmapsto \delta_{x,f}(z) = \sum_{n \geq 0} \frac{f^{(n)}(z)}{n!} x_{(n)}.$$

On a current state a,

$$\delta_{x,f}(a) = f(z)[x, a] + f'(z)(x, a)\mathbf{1}.$$

The invariant form gives long roots squared length two. Constant conformal-weight-preserving vertex derivations form the ten-dimensional subspace of constant currents.

Proof. The negative and degree-zero assertions are the general word-center calculation above. The translation relation gives

$$\partial_z \delta_{x,f} + [T, \delta_{x,f}] = 0.$$

The current commutator formula and Taylor expansion give

$$[\delta_{x,f}(z), Y(a, u)] = Y(\delta_{x,f}(z + u)a, u).$$

These are precisely the chiral derivation equations. Summing the operator over state blocks commutes with all mergers, cuts, and normal derivatives. The all-word derivation identification therefore gives the class, with no inner-derivation quotient.

The affine relation at level one gives its value on a. If a polynomial combination vanishes, its weight-one part commutes with every element of $\mathfrak{so}_5$. The Lie center of $\mathfrak{so}_5[z]$ is zero, proving injectivity. For nonnegative modes the central term of their commutator vanishes. The derivative Leibniz rule yields

$$[\partial_{x,f}, \partial_{y,g}] = \partial_{[x,y],fg}.$$

A constant weight-preserving derivation restricts to a derivation of the simple Lie algebra W_1. It is inner there and extends uniquely, because the currents generate W. Thus it is $a_{(0)}$ for a unique $a \in \mathfrak{so}_5$. □

The same construction with $\mathfrak{sl}_2[z]$ acts on every affine boundary summand H_j. For a junction intertwiner $\mathcal{Y}_p$, its Jacobi identity reads

$$\partial^N_{x,f}(z)\mathcal{Y}_p(m,u) - \mathcal{Y}_p(m,u)\partial^M_{x,f}(z) = \mathcal{Y}_p(\partial^{F(Z)}_{x,f}(z+u)m, u).$$

The balancing maps are affine module maps, so these identities respect their composites and central crossing. On the level-ten currents the scalar term is $10f'(z)(x,a)\mathbf{1}$. It retains the level in the field operations. It does not prove that these classes survive a different endomorphism complex for the full marked transverse center.

The remaining comparison must retain the root, full collision atlas, and distinguished algebra $I(1)$ with its multiplication. It also requires a passage to rational boundary sectors. The center equivalence supplies the categorical action and its intertwiners. The double-pole calculation excludes replacing this passage by a pointed action of C_U on the original vacuum.

20.20 Representation tensors from specified fields

A field exchange becomes a representation intertwiner only after the tensor action has been specified. Central contractions give a model in which both the positive construction and the possible mismatch can be computed on all modes.

Let $P = k[x_0, x_1, \dots]$, with $Tx_i = x_{i+1}$ and $\mathrm{wt}(x_i) = i + 1$. Introduce a formal parameter $\hbar$ and a central state κ with $T\kappa = d\kappa = 0$. For a Laurent kernel $K(z)$, prescribe

$$Y_K(x_0, z)x_0 = (e^{zT}x_0)x_0 + \hbar\kappa K(z).$$

The simple pole $K = z^{-1}$ is nonlocal. The symmetric double pole $K = z^{-2}$ gives the usual central local current bracket.

We retain a resolution of this polynomial algebra. Let $A_{\mathrm{ext}} = \Lambda(\theta_i : i \geq 0)$, where $|\theta_i| = 1$ and $\mathrm{wt}(\theta_i) = i + 1$. The weight-restricted dual

$$E = (B_{\mathrm{alg}} A_{\mathrm{ext}})^\circ$$

is freely associative on y_I for nonempty increasing finite subsets I of $\mathbb{N}$. Its differential is

$$|y_I| = 1 - |I|, \qquad dy_I = \sum_{I = A \sqcup B} (-1)^{|A|+1} \epsilon(A,B)\, y_A y_B.$$

The sum uses ordered nonempty parts, and $\epsilon(A,B)$ is their exterior permutation sign. Thus $x_i = y_{\{i\}}$ and $dy_{ij} = x_i x_j - x_j x_i$.

Lemma 20.20.1 *(The finite-weight resolution).* Sending singleton generators to x_i and higher generators to zero gives a quasi-isomorphism $\pi : E \to P$. It uses a direct sum of finite weight duals.

Proof. Let $\mathrm{Sym}^c(\bigoplus_i k t_i)$ have primitive degree-zero generators of weights $i+1$. The map to the bar is

$$j(t_{i_1}\cdots t_{i_r}) = \sum_{\sigma\in S_r} [s\theta_{i_{\sigma(1)}}|\cdots|s\theta_{i_{\sigma(r)}}].$$

Adjacent interior products cancel in pairs, since the θ_i are odd. Deconcatenation matches the unshuffle coproduct, including repeated-index multiplicities.

The free A_{ext}-module complex $A_{\mathrm{ext}}\otimes k[t_i]$, with $dt_i = \theta_i$, resolves k. Its contraction on positive total generator degree N is $N^{-1}\sum_i t_i\partial_{\theta_i}$. The normalized bar is another semifree resolution. The map $1\otimes j$ intertwines their differentials, including the endpoint term $\sum_i \theta_i\partial_{t_i}$, and lifts the identity on k. Lifting through the free filtrations gives a homotopy inverse, weight by weight.

Every weight is finite-dimensional. Dualize there with $\langle x^\alpha, t^\beta\rangle = \alpha!\delta_{\alpha\beta}$, which converts the unshuffle coproduct into polynomial multiplication. The dual is π. Taking the direct sum of the weight comparisons proves the assertion. □

Extend translation to E by raising each index of y_I in turn, with a repeated index giving zero and exterior reordering carrying its sign. Let $\partial_m(y_I) = 1$ for $I = \{m\}$ and zero otherwise. These graded derivations satisfy

$$[d,T] = [d,\partial_m] = 0, \qquad [T,\partial_m] = -\partial_{m-1}, \qquad [\partial_m,\partial_n] = 0, \qquad \partial_{-1} = 0.$$

On the bar, the transposed operators lower an exterior index and insert $s\theta_m$ in every gap. Those operations commute with its differential, proving the identities. The map π intertwines them.

Put

$$K_{mn}(z) = \partial_z^m(-\partial_z)^n K(z), \qquad C_K(z) = \hbar\kappa \sum_{m,n\ge 0} K_{mn}(z)\partial_m\otimes\partial_n.$$

With the ordered coefficient multiplication μ_E, define

$$Y_K(a,z)b = \mu_E(e^{zT}\otimes 1)e^{C_K(z)}(a\otimes b).$$

Every contraction lowers polynomial degree in both inputs, so the exponential is finite on polynomial states. The same formula with μ_P defines the polynomial fields.

All parenthesized ordered products expand the common expression

$$\mu_r\left(\bigotimes_i e^{z_i T}\right)\exp\left(\sum_{i<j} C_K^{ij}(z_i - z_j)\right).$$

The pair contractions commute, and their derivation rule gives the product and iterate identities. Thus both carriers are differential nonlocal vertex algebras, and π preserves every field mode. The completion is $(E\otimes k[\kappa])[[\hbar]]$: each parameter coefficient has finite polynomial and energy support. There is no unrestricted energy product.

For inputs of energies a, b, a term with q contractions and a kernel of pole order p has pole order at most $a + b + (p-2)q$. Modulo $\hbar^N$, also $q \le \min(a, b, N-1)$. This proves the required Laurent bounds at each precision.

The regular action is faithful, since $Y_E(a,z)1 = e^{zT}a$ has constant term a. Its operator-field multiplication uses regularized substitution $w_1 = w + u$ after multiplying by a sufficient power of $w_1 - w$. The common Wick expression proves

$$\mathcal{Y}(Y_E(a,w),u)Y_E(b,w) = Y_E(Y_E(a,u)b,w).$$

This retains the full substitution, not only a principal part.

The universal level and additive module tensor

The coproduct is over $k[[\hbar]]$, with κ a state:

$$\Delta\kappa = \kappa\otimes 1 + 1\otimes\kappa, \qquad \Delta y_I = y_I\otimes 1 + 1\otimes y_I.$$

Its target has independent level states κ_L, κ_R. The counit kills them and all y_I.

PROPOSITION 20.20.2 *(The universal vertex bialgebra).* These formulas make $(E\otimes k[\kappa])[[\hbar]]$ a counital differential nonlocal vertex bialgebra. The map π is a coalgebra map and an all-mode field quasi-isomorphism. The induced module tensor adds levels.

Proof. Each dy_I is a sum of graded commutators of primitive generators, so Δ is a chain map. Coassociativity and the coefficient counit hold on generators. For each m,

$$(\partial_m\otimes 1)\Delta = \Delta\partial_m = (1\otimes\partial_m)\Delta.$$

Each outer derivative separately gives this identity. Their sum would double it.

The target contractions therefore contribute $\hbar(\kappa_L+\kappa_R)K_{mn}$, exactly the image of the source contraction. Exponentiation proves

$$\Delta Y_K = (Y_K\otimes Y_K)(1\otimes\tau\otimes 1)(\Delta\otimes\Delta).$$

Every contracted term contains κ, so the counit kills it. The uncontracted term is its ordinary augmentation. Equivariance of π proves the remaining claims. □

A nonzero fixed-level ideal $(\kappa - c)$ is not a coideal: its image is $\kappa_L + \kappa_R - c$, while setting both target levels to c gives c. This does not prevent maps from level $c_1 + c_2$ to the tensor of levels c_1, c_2.

An admissible module here is topologically free, $M = M_0[[\hbar]]$, with continuous fields lower truncated on each input modulo every $\hbar^N$. For two such modules define $M\widehat{\boxtimes}N = (M_0\otimes_k N_0)[[\hbar]]$ and

$$Y_{M\widehat{\boxtimes}N}(a,z)(m\otimes n) = \sum_{(a)}(-1)^{|a_{(2)}||m|}Y_M(a_{(1)},z)m\otimes Y_N(a_{(2)},z)n.$$

At each precision only finitely many coproduct terms occur, and two lower-bounded Laurent series have finite coefficient convolution. The bialgebra identity proves module associativity. Coassociativity proves the pentagon.

A derived version uses K-flat models and the specified completed derived tensor. This is an additive-level module tensor, with its stated carrier.

Normal-ordering transport

Let $H(z) \in k[[z]]$ be even and put

$$H_{mn} = (-1)^n H^{(m+n)}(0), \qquad D_H = \frac{1}{2}\sum_{m,n} H_{mn}\partial_m\partial_n, \qquad W_H = e^{\hbar\kappa D_H}.$$

The matrix is symmetric and $H_{m+1,n} + H_{m,n+1} = 0$.

Proposition 20.20.3 *(Fields, tensors, and formal states).* The map W_H is invertible, preserves the vacuum, differential, and translation, and satisfies

$$\begin{aligned} W_H Y_K(a,z)b &= Y_{K+H}(W_H a, z) W_H b, \\ \Delta W_H &= (W_H \otimes W_H)\Delta, \qquad \pi W_H = W_H \pi, \\ W_{H_2} W_{H_1} &= W_{H_1+H_2}, \qquad W_H^{-1} = W_{-H}. \end{aligned}$$

Define a coefficient-valued functional independently by zero odd moments and

$$\lambda_H(x_{i_1}\cdots x_{i_{2r}}) = (-\hbar\kappa)^r \sum_{\text{pairings}} \prod_{\{a,b\}} H_{i_a i_b}, \qquad \lambda_H(1) = 1.$$

Then $\lambda_H = \lambda_0 W_H^{-1}$, where λ_0 sets currents to zero while retaining κ.

Proof. All contractions commute with d. The recurrence for H_{mn} makes $[T, D_H] = 0$. The second-order Leibniz rule gives

$$D_H \mu_E = \mu_E\left(D_H \otimes 1 + 1 \otimes D_H + \sum_{m,n} H_{mn}\partial_m \otimes \partial_n\right).$$

Exponentiate and move the cross term past e^{zT}. Its Taylor series is $K_{mn}(z) + \partial_z^m(-\partial_z)^n H(z)$, proving the field identity.

For the coproduct, each outer D_H equals ΔD_H after Δ. The source level splits as $\kappa_L + \kappa_R$, giving precisely the two target normal-ordering operators. Their commutation proves the composition and inverse formulas.

Finally, the constant term of $D_H^r/r!$ counts complete pairings. The factor $1/2$ removes their two orientations and $r!$ removes their order. This proves the stated moment formula, including repeated indices. □

The formal state has a separate bar realization. Let I_m insert $s\theta_m$ into every bar gap. Adjacent exterior products cancel, so I_m commutes with the bar differential. Insertion operators commute. The vector

$$\nu_H = \exp\left(-\frac{\hbar\kappa}{2}\sum_{m,n} H_{mn} I_m I_n\right)1$$

lies in the product over output weights, with a finite formula at each weight and parameter order.

Its pairing against weight-restricted dual observables is a continuous chain functional. The identity $I_m j(f) = j(t_m f)$ and the factorial pairing give $\langle -, \nu_H\rangle = \lambda_H \pi$. Thus this comparison follows from two specified constructions of the state.

These functionals are formal and coefficient-valued. They retain κ and detect contractions, unlike the bialgebra counit. For example,

$$\lambda_H(x_0^2) = -\hbar\kappa H(0), \qquad \lambda_H(Y_{K+H}(x_0, z)x_0) = \hbar\kappa K(z).$$

The second identity cancels the new regular contraction.

On a curve chart, a regular symmetric kernel $H(z, w)$ has diagonal jets $H_{mn}(s)$ satisfying $\partial_s H_{mn} = H_{m+1,n} + H_{m,n+1}$. Hence $[\partial_s + T, D_H(s)] = 0$. For supplied kernels whose differences are $H_{\alpha\beta}$, the identity

$$W_{H_{\beta\gamma}} W_{H_{\alpha\beta}} = W_{H_{\alpha\gamma}}$$

gives horizontal gluing of fields and comparisons. It assumes the kernels and their coordinate identifications. A projective connection alone does not supply a global bidifferential or state. A regular symmetric change leaves the exchange-antisymmetric part of the kernel unchanged.

Changing the tensor action

At unit level and $K(z) = z^{-1}$, the polynomial state-exchange operator is

$$\mathcal{S}(z) = \exp\left(-2\hbar \sum_{m,n} K_{mn}(z)\, \partial_m \otimes \partial_n\right).$$

The additive current acts by $J_{-1}^{\mathrm{tot}} = M_{x_0} \otimes 1 + 1 \otimes M_{x_0}$. They do not commute:

$$[\mathcal{S}(z), J_{-1}^{\mathrm{tot}}](x_0 \otimes 1) = -\frac{2\hbar}{z}\, 1 \otimes 1.$$

The ordinary flip does intertwine the cocommutative additive action. Thus field exchange and the displayed module tensor have different intertwiners.

A change of action gives a positive comparison. Decorate each admissible module by commuting even chain operators D_m lowering an exhaustive nonnegative energy filtration by $m + 1$. Each algebraic state has bounded energy. Morphisms preserve these operators and the field action.

For a finite set of spectral slots I, set

$$R_I = k[z_i : i \in I][(z_i - z_j)^{-1} : i \neq j], \qquad M_I = (R_I \otimes_k \bigotimes_i M_{i,0})[[\hbar]].$$

The field variable w is independent of the z_i. For an order σ define

$$C_{ij} = \hbar \sum_{m,n} K_{mn}(z_i - z_j) D_m^{(i)} D_n^{(j)}, \qquad F_\sigma = \exp\left(\sum_{i \prec_\sigma j} C_{ij}\right), \qquad \rho_\sigma(a, w) = F_\sigma \rho_0(a, w) F_\sigma^{-1}.$$

The coefficient $\hbar$ is specified tensor data. It is not forced by an arbitrary coproduct.

A term with q contractions consumes at least $\sum(m + n + 2)$ units of energy and has pole order $\sum(m + n + 1)$. Its pole order is at most $\sum_i a_i - q$ on inputs of energies a_i. Also $q < N$ modulo $\hbar^N$. These bounds make $F_\sigma^{\pm 1}$ continuous on the stated carrier and preserve each lower Laurent bound in w.

THEOREM 20.20.4 *(Coherent tensor transport on separated slots).* For ordered blocks B, D, the maps $F_{B,D} = \exp(\sum_{i \in B, j \in D} C_{ij})$ give a monoidal equivalence between the original decorated tensor actions and ρ_σ. If $P_{\tau\sigma}$ is the graded permutation of carriers, then

$$c_{\tau\sigma} = F_\tau P_{\tau\sigma} F_\sigma^{-1}$$

intertwines every field mode and satisfies all block hexagons and braid identities. On two slots it is

$$c^{F}_{M,N}(z) = \tau \exp\left(-2\hbar \sum_{m,n} K_{mn}(z) D_m \otimes D_n\right).$$

Proof. All pair contractions commute, including those sharing a slot. For three blocks, both parenthesizations give

$$\exp(C_{B,D} + C_{B,H} + C_{D,H}).$$

Every higher parenthesization counts each pair once. This proves the pentagon. Empty pair sets give the unit.

The original permutation intertwines ρ_0. Conjugating this identity proves the new intertwining relation on every mode. Consecutive order changes telescope, while splitting blocks partitions the same pair set. These facts prove the braid and hexagon identities. Finally $K_{nm}(-z) = -K_{mn}(z)$ gives the two-slot formula. The energy and parameter bounds justify every exponential. □

This transported category is monoidally equivalent to the original symmetric category. Its nonidentity formula does not by itself construct an inequivalent representation category.

For two regular polynomial factors put $M = M_{x_0^{(1)} + x_0^{(2)}}$ and $C = C_{12}$. Then

$$[C, M] = \hbar A(z), \qquad [C, A(z)] = 0,$$

where $A(z) = \sum_m K_{m0}(z) D_m^{(1)} + \sum_n K_{0n}(z) D_n^{(2)}$. The transported creation modes are

$$J^{\pm}_{-1} = M \pm \hbar A(z), \qquad e^{-2C} J^{+}_{-1} = J^{-}_{-1} e^{-2C}.$$

On polynomials in the two lowest generators this reads $x + y \pm \hbar z^{-1}(\partial_x + \partial_y)$.

For the full current,

$$J^{\pm}(w) = J^{0}(w) \pm \hbar \sum_{n \geq 0} \big(K_{n0}(z - w) D_n^{(1)} + K_{0n}(z + w) D_n^{(2)}\big),$$

expanded in nonnegative powers of w. The first commutator gives the correction and the second vanishes. The fields of the higher y_I are unchanged: the decorations kill those generators and their translates. The negative-degree resolution cells are retained.

Equivariance of π gives a quasi-isomorphism $\Pi_I = \pi^{\widehat{\otimes} I}$ intertwining all transported fields, tensorators, and exchanges. The finite-weight contraction extends coefficientwise to the parameter completion.

There are compatible formal states as well. Define λ_σ by complete matchings between distinct slots, assigning $-\hbar K_{mn}(z_i - z_j)$ to a paired occurrence $x_m^{(i)}, x_n^{(j)}$ with $i \prec_\sigma j$. The bar-side definition uses

$$\nu_\sigma = \exp\left(-\hbar \sum_{i \prec_\sigma j} \sum_{m,n} K_{mn}(z_i - z_j) I_m^{(i)} I_n^{(j)}\right) 1^{\otimes I}.$$

The pairing-count argument proves

$$\lambda_\sigma = \lambda_0^{\otimes I} F_\sigma^{-1}, \qquad \lambda_\sigma^{E} = \lambda_\sigma \Pi_I, \qquad \lambda_\tau c_{\tau\sigma} = \lambda_\sigma.$$

For four lowest currents,

$$\lambda_{1234}(x_0^{\otimes 4}) = \hbar^2 \left(\frac{1}{z_{12} z_{34}} + \frac{1}{z_{13} z_{24}} + \frac{1}{z_{14} z_{23}}\right).$$

At $(1,3,7,13)$ it is $29\hbar^2/240$. These are normalized formal functionals on the separated tensor spaces, with no positivity assertion.

On the actual order-chart overlap, in one fixed tensor order,

$$G_{\tau\sigma} = F_\tau F_\sigma^{-1}.$$

Only pairs whose orders differ occur in its exponent. Their differences are precisely the ones inverted on that overlap. On regular E, P coefficients the connection is $\nabla_i = \partial_{z_i} + T_i$. The kernel recurrence gives $[T_i, C_{ij}] = -\partial_{z_i} C_{ij}$, so $G_{\tau\sigma}$ is horizontal. Cancellation gives its cocycle identity.

These maps glue differential modules, but they do not supply the unchanged tensor-power collision equivalence. Modulo $\hbar$, diagonal restriction of the free two-point module leaves $P \otimes P$. Its energy-one piece has dimension two, whereas that of P has dimension one. The normal Koszul complex adds no differential that changes these ranks. Thus an energy-preserving collision equivalence with singleton P is impossible on this carrier. Additional collision relations or coefficients are needed.

An actual spectral symmetry Hopf algebra

Let $\mathfrak{t}$ have generators T, p_m, $m \geq 0$, with

$$[p_m, p_n] = 0, \qquad [T, p_m] = -p_{m-1}, \qquad p_{-1} = 0.$$

Give $\mathscr{H} = U(\mathfrak{t})[[\hbar]]$ the primitive coproduct. On modules with T raising energy and p_m lowering it by $m+1$, define

$$\mathcal{R}(z) = \exp\left(-2\hbar \sum_{m,n} K_{mn}(z)\, p_m \otimes p_n\right).$$

This is an operator in the completion defined by these modules. It need not be a finite tensor in $\mathscr{H}^{\otimes 2}$.

The exponent commutes with each Δp_m. Its two commutators with T are opposite z-derivatives, so it also commutes with ΔT. Therefore $\tau\mathcal{R}(z)$ is a module intertwiner. Applying the primitive coproduct splits its exponent, proving both hexagons. The pair exponents commute, giving Yang–Baxter at arguments u, $u+v$, v and

$$\mathcal{R}^{21}(-z)\mathcal{R}(z) = 1.$$

For E, P, take $p_m = \partial_m$. The field has shifted equivariance

$$hY_E(a,z)b = \sum Y_E\big((e^{-z\,\mathrm{ad}\,T} h_{(1)})a, z\big) h_{(2)} b.$$

This follows by moving $h_{(1)}$ through e^{zT}. The shifted coproduct commutes with the contractions. It identifies the symmetry acting in this model. The earlier creation-mode defect prevents identifying that symmetry with an arbitrary vertex-module tensor action.

Noncommuting contractions and the singlet order

Permutation operators give a precise limit of the commuting-contraction argument. On $(k^2)^{\otimes 3}$, scalar normalization of $e^{\hbar P/u}$ gives $1 + f(u)P$ with $f(u) = \tanh(\hbar/u)$. Direct multiplication yields

$$\begin{aligned}
&(1 + aP_{12})(1 + bP_{13})(1 + cP_{23}) \\
&\quad - (1 + cP_{23})(1 + bP_{13})(1 + aP_{12}) \\
&= (ac - b(a+c))(P_{12}P_{23} - P_{23}P_{12}).
\end{aligned}$$

For $a = f(u)$, $b = f(u+v)$, $c = f(v)$, the first scalar defect is

$$-\frac{\hbar^4(u^2+uv+v^2)}{3u^2v^2(u+v)^2} + O(\hbar^6).$$

The commutator matrix has rank four. At $u = v = 1$ its order-four coefficient is $-1/4$. A scalar prefactor multiplies both Yang–Baxter sides by the same factor and cannot remove this defect.

By contrast, $f(u) = \hbar/u$ satisfies $ac = b(a+c)$. Hence

$$R_{\mathrm{rat}}(u) = \frac{u\,1 + \hbar P}{u+\hbar}$$

satisfies Yang–Baxter and $R^{21}_{\mathrm{rat}}(-u)R_{\mathrm{rat}}(u) = 1$. The integral RTT Hopf algebra and the specified Bethe-algebra representation of Section 18.33 retain this positive algebraic domain. Neither existence nor nonexistence of an arithmetic realization follows from the absence of a subtraction operation between local fields.

Likewise, the quadratic Hecke relation $(\check R - q)(\check R + q^{-1}) = 0$ gives

$$\check R - \check R^{-1} = (q - q^{-1})1.$$

For $C(z) = z\check R - z^{-1}\check R^{-1}$ it implies

$$C(z^{-1})C(z) = (q^2 + q^{-2} - z^2 - z^{-2})1.$$

This calculation remains valid at repeated roots. Yang–Baxter additionally requires a Hecke braid representation. The involution $\mathrm{diag}(1,1,1,-1)$ at $q = 1$ satisfies the quadratic relation and fails the adjacent braid relation.

For the singlet, let

$$j(v) = (e_0 \otimes e_1 - e_1 \otimes e_0) \otimes v, \qquad \pi = \tfrac{1}{2} j^{\mathrm{t}}, \qquad R(u) = 1 + \hbar P/u.$$

Then

$$R_{23}(u+\hbar)R_{13}(u)\,j = \frac{u+2\hbar}{u+\hbar}\,j, \qquad \pi R_{13}(u)R_{23}(u+\hbar) = \frac{u+2\hbar}{u+\hbar}\,\pi.$$

Indeed, for $A = P_{13}$, $B = P_{23}$, $(A+B)\,j = j$ and $BAj = -Aj$. With $a = \hbar/u$, $b = \hbar/(u+\hbar)$, the unwanted coefficient is $a - b - ab = 0$. In the reverse order it is $a - b + ab \neq 0$.

The normalization preserving this cup and cap therefore obeys

$$f(u)f(u+\hbar) = \frac{u+\hbar}{u+2\hbar}.$$

It has a unique solution with constant term one at infinity: each coefficient is twice the new unknown plus earlier terms. Writing $t = \hbar/u$ gives

$$F(t) = 1 - \tfrac{1}{2}t + \tfrac{5}{8}t^2 - \tfrac{11}{16}t^3 + \cdots, \qquad F(t)F(-t) = \frac{1}{1-t^2}.$$

For the last identity, put $G(t) = F(t)F(-t)$. The shift equation implies $G(t)G(t/(1+t)) = (1+t)/((1-t)(1+2t))$. Both G and $(1-t^2)^{-1}$ solve it with constant term one. The same triangular uniqueness proves equality. Thus the normalized exchange is formally unitary.

For fixed $\hbar \neq 0$, there is no rational solution. If d_n is its order at $-2\hbar + n\hbar$, the equation requires $d_n + d_{n+1}$ to be -1 at $n = 0, 1$ at $n = 1$, and zero elsewhere. Finite support forces its largest nonzero index into $\{0,1\}$ and its smallest into $\{1,2\}$. Only d_1 could remain, but it would have to equal both -1 and 1. Meromorphic normalizations require separate analytic choices. The algebraic cup calculation also leaves the comparison with a chosen Yangian evaluation-module duality to its specified coproduct and parameter conventions.

20.21 HALF-BRAIDINGS AND THEIR PROBES

The objects against which a half-braiding is tested are part of its definition. For geometric words, the allowed maps of the labeled base matter as well.

Let X be smooth and separated of finite type, with arbitrarily many distinct closed k-points. Use word objects with monotone collision equivalences, the prefix/suffix cut tensor, and its isolated empty unit. Require comparison maps over the fixed labeled base. For a point x, let P_x have zero empty coefficient and positive coefficient $(x, \dots, x)_* k$.

PROPOSITION 20.21.1 *(The absolute point probes).* If $F \star P_x \simeq P_x \star F$ for every closed point x, then every positive coefficient of F vanishes.

Proof. Choose the least positive n with $F_n \neq 0$. In arity $n+1$, restrict to $(X \setminus \{x\})^n \times \{x\}$. On the left, the cut after n contributes $F_n|_{(X\setminus\{x\})^n}$. All smaller positive cuts vanish by minimality. The empty term has constant support and also vanishes.

On the right the remaining positive cut requires the first coordinate to be x, so it vanishes too. Thus F_n restricts to zero on each $(X \setminus \{x\})^n$.

Choose $n+1$ distinct points. Every n-tuple omits at least one, so these opens form a finite cover of X^n. Descent gives $F_n = 0$, a contradiction. □

This result uses the cut tensor, all point probes, and the fixed labeled base. It does not apply to the symmetric convolution using all ordered decompositions.

Restricting to one point gives a positive alternative. Objects supported at the constant tuple x correspond to pairs of complexes (V_0, V_1), because the positive collision diagram has terminal singleton. Their tensor is

$$(V_0, V_1) \star (W_0, W_1) = (V_0 \otimes W_0,\ V_0 \otimes W_1 \oplus V_1 \otimes W_0).$$

Two positive factors give zero on the cut-disjoint open. Every nonzero iterated term therefore has at most one positive factor. The Koszul flip on these terms supplies a symmetry and all its coherences. Nonzero positive objects have half-braidings in this relative probe category.

Charge categories and excess central objects

Even a braided category can have central objects outside its canonical central image. For finite-dimensional $\mathbb{Z}$-graded vector spaces, choose any bicharacter braiding. On the unit, the scalars T^n on charge n define a half-braiding for every $T \in \mathbb{C}^\times$. Its hexagon is $T^{n+m} = T^n T^m$. The canonical central image with underlying unit has trivial half-braiding, so the examples with $T \neq 1$ are additional central objects. A fixed-level Heisenberg realization would require its own fusion tensor and monoidal comparison.

There is a finite modular example with all coherences visible. Let $\ell > 0$ be even and $G = \mathbb{Z}/\ell\mathbb{Z}$. For representatives $0 \le a, b < \ell$, define

$$c(a,b) = \left\lfloor \frac{a+b}{\ell} \right\rfloor, \qquad [a+b] = a + b - \ell c(a,b).$$

On finite-dimensional G-graded vector spaces set

$$\omega(a,b,c) = (-1)^{ac(b,c)}, \qquad R(a,b) = e^{\pi i ab/\ell}, \qquad \theta(a) = e^{\pi i a^2/\ell}.$$

The first scalar is the associator from left to right parenthesization, the second multiplies the interchange, and the third is the twist.

Proposition 20.21.2 *(The finite cyclic center).* These data define a modular ribbon category $\mathcal{C}_\ell$. Its simple objects have dimension one. The center has ℓ^2 simples, indexed by $(a, j) \in G^2$, with canonical central image $(a, 0)$. The braided functor

$$\mathcal{C}_\ell \boxtimes \mathcal{C}_\ell^{\mathrm{rev}} \longrightarrow \mathcal{Z}(\mathcal{C}_\ell)$$

is an equivalence and sends (a, b) to $([a + b], -b)$.

Proof. The carry identity

$$c(b, c) + c([b + c], d) = c(c, d) + c(b, [c + d])$$

expresses the quotient and remainder of $b + c + d$. The pentagon exponent discrepancy is $\ell c(a, b)c(c, d)$, which is even. The two hexagons are

$$R(a, [b + c]) = \frac{\omega(b, a, c)}{\omega(a, b, c)\omega(b, c, a)} R(a, b)R(a, c),$$
$$R([a + b], c) = \frac{\omega(a, b, c)\omega(c, a, b)}{\omega(a, c, b)} R(a, c)R(b, c).$$

Symmetry of the carry cancels two factors in each ratio. The remaining sign is exactly the phase from reducing the sum modulo ℓ. Without the associator, the first hexagon already fails at $\ell = 2, a = b = c = 1$.

Balancing follows from

$$\theta([a + b]) = R(a, b)R(b, a)\theta(a)\theta(b).$$

The exponent discrepancy is $\pi i[-2c(a, b)(a + b) + \ell c(a, b)^2]$. Choose dual $[-a]$, evaluation one, and coevaluation $\omega(a, [-a], a)^{-1}$. The carry identities prove both triangles, and $\theta([-a]) = \theta(a)$ gives dual compatibility. The ribbon dimension is

$$\theta(a)R(a, [-a])\omega(a, [-a], a)^{-1} = 1.$$

Double braiding is the Fourier matrix $e^{2\pi i ab/\ell}$, with inverse $\ell^{-1}e^{-2\pi i ab/\ell}$, proving nondegeneracy.

For a central object of degree a, divide its half-braiding with degree b by $R(a, b)$. The hexagon makes the remaining operators a representation of G. It splits into characters $e^{2\pi i jb/\ell}$, giving the stated simple labels. The canonical image has $j = 0$.

Using the original crossing on the first tensor factor and the inverse crossing on the second gives $([a + b], -b)$. The associator factors are those just checked. The mixed double braiding cancels, so this is braided. It is bijective on simples and their scalar Hom spaces, hence is an equivalence of semisimple categories. □

The even-level condition has an integral normalization. With minimum electric charge one and $F/(2\pi)$ representing c_1, the action $\ell(4\pi)^{-1}\int A \wedge dA$ changes under a four-dimensional extension by

$$\exp\left(2\pi i\,\frac{\ell}{2}\int (F/(2\pi))^2\right).$$

Even ℓ makes this one on every oriented extension. The hyperplane line bundle on $\mathbb{CP}^2$ proves necessity. In this bosonic convention the class is $(\ell/2)c_1^2 \in H^4(BU(1); \mathbb{Z})$. The cell structure of $\mathbb{CP}^\infty$ gives $H^*(BU(1); \mathbb{Z}) = \mathbb{Z}[c_1]$ and $H^3 = 0$.

For the marked boundary current with $J(z)J(w) \sim \ell(z - w)^{-2}$, the stress tensor $(2\ell)^{-1} : JJ :$ has double-contraction coefficient $2\ell^2/(4\ell^2) = 1/2$, hence Virasoro central charge one. The current commutator, integral charge lattice, and quadratic form still retain ℓ. An unmarked rescaling does not preserve these data.

20.22 ARITHMETIC CENTERS AND GEOMETRIC COMPARISONS

The arithmetic constructions require the same distinctions. An ordinary center acts by natural endomorphisms. A scalar Hochschild center is a complex with its higher operations. A categorical center is itself a category. An identity trace is a class in a trace recipient. A comparison must specify which of these objects it carries.

For a stable presentable k-linear monoidal category C, with tensor preserving colimits separately, put

$$Z_{\mathrm{cat}}(C) = \mathrm{Fun}^{\mathrm{L}}_{C\otimes C^{\mathrm{op}}}(C, C),$$
$$\mathrm{Tr}_{\mathrm{cat}}(C) = C \otimes_{C\otimes C^{\mathrm{op}}} C.$$

The tensor and functors are in presentable k-linear categories. The first has a braided tensor structure. The second carries a circle action. For an ordinary algebra A, the scalar complexes $R\,\mathrm{Hom}_{A^e}(A, A)$ and $A \otimes^{\mathbf{L}}_{A^e} A$ occur one categorical level lower. They are not identified by their shared name Hochschild.

The precise positive geometric comparisons of Section 18.29 apply here. For a perfect derived stack X,

$$Z_{\mathrm{cat}}(\mathrm{QCoh}(X)) \simeq \mathrm{QCoh}(\mathcal{L}X) \simeq \mathrm{Tr}_{\mathrm{cat}}(\mathrm{QCoh}(X)).$$

For the complex reductive Hecke category $D(B\backslash G/B)$, the center and trace identify with the character category generated by the Harish–Chandra transform. The same holds for its specified unipotently monodromic version. These are the theorems of Ben-Zvi–Francis–Nadler and Ben-Zvi–Nadler on their respective geometric carriers. They do not identify a category with a ring of functions on an unspecified arithmetic parameter space.

Even a smooth affine line has derived loops:

$$O(\mathcal{L}\mathbb{A}^1) = k[x] \otimes^{\mathbf{L}}_{k[x_1,x_2]} k[x] \simeq k[x, \eta], \qquad |\eta| = -1, \quad d = 0.$$

The diagonal Koszul resolution gives this formula. Thus smoothness of X does not make its derived loop space classical.

For finite cyclic loops of $B\mathbb{G}_m$, the complex points instead have μ_n components, each with stabilizer $\mathbb{C}^\times$. A quotient of finite cyclic groups reverses the induced mapping-stack arrow. These constructions do not form formal neighborhoods of the identity by notation alone. An arithmetic loop comparison must specify its source topology and retain the actual Weil and inertia data.

Scalar actions and the full categorical center

Assume characteristic zero, with coefficient prime different from the residue prime. The stated GL_2 Bernstein theorem identifies the ordinary Bernstein center with functions on its continuous Weil parameter stack. Its normalization retains inertia and Frobenius. The fixed-character and coefficient-prime comparisons are different theorems.

For the latter, the ring

$$E_{\mathcal{B}} = \mathrm{End}(P_{\mathcal{B}}) = \mathrm{End}_G(J_{\mathcal{B}})^{\mathrm{op}}$$

of Section 18.7.3 is pseudocompact and finite over its pseudodeformation center. In the generic reducible case its full algebra retains the two opposite extension representations. The center does not replace that algebra.

A scalar action on a spectral category need not exhaust its center. For example, objects of $\mathrm{QCoh}(B\mathbb{G}_m)$ decompose into integer weights. Each sequence $(a_n)_{n\in\mathbb{Z}}$ acts naturally by a_n on weight n. The weight-zero projector is a central operation that is not multiplication by a scalar function on $B\mathbb{G}_m$, whose global

functions are k. In the scalar stabilizer of a GL_2 parameter, a determinant twist changes weight by two. It cannot identify every weight sector or eliminate these projectors.

Similarly, a matrix-algebra description on a generic locus gives a Morita equivalence there, with its center. It does not extend across a singular locus without a compatible functor and its inverse. A regular ring alone does not make extensions vanish at every point: for $A = k[t]$,

$$\operatorname{Ext}^1_A(k,k) = k.$$

Localization to $k(t)$ kills this supported module. The generic and supported categories retain different information.

Gluing central actions

Formal patching supplies a positive ordinary-center calculation with actual overlap maps. For

$$A = k[t], \quad B = k[[t]], \quad C = k[t,t^{-1}], \quad D = k((t)),$$

the equivalence of finitely generated module categories in Theorem 18.28.3 uses triples (P, Q, η) with an isomorphism over D. The ordinary center is

$$Z(\operatorname{mod}_{\mathrm{fg}} A) \simeq B \times_D C = A.$$

Indeed, a natural endomorphism on finitely generated R-modules is multiplication by its value on $1 \in R$. The identity $B \cap C = A$ inside D then gives the formula.

The punctured formal disk is an actual overlap. It is not the empty intersection of a closed point with its complementary open.

Higher centers need more than this scalar equalizer. An equivalence of categories with effective descent transports central operations together with their coherences. An isomorphism of rings of parameters does not supply the categorical descent equivalence. For a derived external product followed by pullback, ordinary formulas also need two Tor-independence conditions, one for each step. Already

$$k \otimes^{\mathbf{L}}_{k[[t]]} k \simeq k \oplus k[1]$$

has an extra term although both factors are perfect. An identity pullback preserves it.

For quasi-smooth characteristic-zero charts, singular support is a cone in

$$\operatorname{Sing}(X) = \operatorname{Spec}_{X^{\mathrm{cl}}} \operatorname{Sym} H^1(T_X).$$

Its linear functions act in Hochschild degree two. It is not an unspecified closed locus in X itself. An arithmetic category with this support therefore needs the derived charts, their coefficient theory, and the compatible cone. The finite-stratum generator argument also needs extension functors preserving compactness and the actual gluing morphisms. The localization and recollement constructions of Chapter 18 retain those extensions.

The restricted tensor theorem of Section 18.30 gives another positive assembly. Its finite-stage insertion maps preserve the specified operadic structures, including units. The filtered colimit transports that structure. Local endofunctors act on cofinal diagrams containing their places and commute through tensor interchange. Neither scalar commutativity nor these separate-place actions creates collision maps or raises an E_1 structure to E_2.

The operad of little intervals has an integral associative comparison. Each ordering component is a nonempty convex region of endpoints. Its chains therefore augment quasi-isomorphically to $R[S_n]$ in every arity. Substitution respects orderings.

Rectification of an algebra along this comparison, when available, does not make the algebra formal. The cubic example of (3) retains its nonzero triple Massey product in a strict associative model. For centers, the chain-level E_2 operations, coefficient base, and comparison maps must likewise be retained. A rational operadic comparison alone supplies no integral replacement or arithmetic geometric action.

20.23 The marked arithmetic comparison

Let C_I be the arithmetic fusion category of Chapter 15, over its specified field F_I. Write

$$U_I : Z(C_I) \longrightarrow C_I, \qquad U_I \dashv R_I, \qquad B_I = R_I(\mathbf{1}).$$

The distinguished algebra B_I is defined by adjunction: its multiplication is adjoint to

$$U_I(B_I \otimes B_I) \simeq U_I(B_I) \otimes U_I(B_I) \xrightarrow{\epsilon\otimes\epsilon} \mathbf{1},$$

and its unit is adjoint to $\imath_{\mathbf{1}}$. Its module category reconstructs the boundary category. Consequently the comparison object is the pair $(Z(C_I), B_I)$ with this multiplication and its forgetful action.

Suppose an independently defined boundary or bulk construction provides a braided category $\mathcal{B}_I$ and an algebra $\mathfrak{b}_I$ in it. A marked comparison consists of a braided equivalence

$$\Xi_I : Z(C_I) \xrightarrow{\sim} \mathcal{B}_I$$

and an algebra isomorphism $\xi_I : \Xi_I(B_I) \xrightarrow{\sim} \mathfrak{b}_I$. Using the tensor comparison of Ξ_I, it must satisfy

$$\xi_I \circ \Xi_I(\mu_{B_I}) = \mu_{\mathfrak{b}_I} \circ (\xi_I \otimes \xi_I), \qquad \xi_I \circ \Xi_I(\eta_{B_I}) = \eta_{\mathfrak{b}_I}.$$

The induced module-category comparison must intertwine the central action on the specified boundary objects. An unmarked braided equivalence does not provide these additional identifications.

The arithmetic marking also has semilinear transport. For an ideal $\mathfrak{a}$ in the domain of (15.6), put $\sigma = \mathrm{Art}(\mathfrak{a})$ and $I' = \mathfrak{a}^{-1}I$. The normalized network coefficients satisfy $\sigma(e_{I,g}) = e_{I',g}$, with the stated label transport.

A tensor transport lifting this law must be σ-semilinear on morphisms and preserve the marked junctions and duality. Denote such a transport by $\mathsf{A}_{\mathfrak{a}} : C_I \to C_{I'}$. It induces a center transport and, by the adjunction, a marked algebra isomorphism

$$Z(\mathsf{A}_{\mathfrak{a}})(B_I) \simeq B_{I'}.$$

Its multiplication compatibility follows by transporting the counit definition above.

The boundary or bulk construction must supply its own semilinear transport $\mathsf{T}_{\mathfrak{a}} : \mathcal{B}_I \to \mathcal{B}_{I'}$. The required square is a coherent braided isomorphism

$$\mathsf{T}_{\mathfrak{a}}\Xi_I \simeq \Xi_{I'} Z(\mathsf{A}_{\mathfrak{a}}),$$

compatible with the algebra maps ξ_I, $\xi_{I'}$, the units, and composition of Artin transports. If these tensor functors have not been constructed, the scalar reciprocity identity is only their necessary coefficient condition.

Question 20.23.1 *(The marked center comparison).* Construct the independently specified boundary or bulk center, its distinguished algebra and central action, and the equivalences Ξ_I, ξ_I with their semilinear transport compatibilities. On a field realization, retain the root and full collision diagrams. Construct the separated products and action, and identify the passage to the rational boundary sectors.

The affine E_6 theorem solves the tensor-category center problem in that realization and identifies its distinguished algebra and mirror extension. The pointed obstruction shows why the corresponding U-state algebra cannot simply act on the original vacuum in the required pointed manner. The derived Heisenberg example independently shows that even the correct one-position endomorphism object may miss action homotopies on separated configurations. Both results constrain the comparison without replacing its original domain.

A solution must therefore provide the stated maps, or an invariant incompatible with every permitted marked comparison. Equality of fusion rules, characters, scalar networks, or ordinary centers does not decide it. The trace comparison, the sewing of higher-genus amplitudes, and physical bulk reconstruction require their own operation-preserving maps as well.

References

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The fusion-center construction uses Davydov–Müger–Nikshych–Ostrik, *The Witt group of non-degenerate braided fusion categories*, arXiv:1009.2117, Sections 3–4. Gelaki–Naidu–Nikshych, *Centers of graded fusion categories*, arXiv:0905.3117, Corollary 3.7. And Bischoff, *A remark on CFT realization of quantum doubles of subfactors. Case index less than four*, arXiv:1506.02606, Corollary 4.7 and Section 5.

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Chapter 21

Enhanced traces and sewing

A matrix trace annihilates commutators. The trace of a category can retain an entire complex and a circle action. An arithmetic comparison must specify which of these objects it uses, which class it evaluates, and how the evaluation reaches a scalar partition function. Sewing requires compatible gluing maps as well.

Two elementary examples distinguish these data. In a matrix algebra, the identity need not generate degree-zero Hochschild homology. In a Frobenius algebra, changing the functional can change the sphere value without changing multiplication. Both phenomena matter when a local vacuum or a distinguished arithmetic class is part of the comparison.

We begin with ordinary algebras over a field. Tensor products in the first constructions are algebraic. The later categorical, completed, and Hilbert constructions will have their own coefficients and domains. The centers and actions of Chapter 20 act on specified objects. Passing to their traces is a further operation.

21.1 A trace and the class it evaluates

Let k be a field and A a nonzero unital associative k-algebra. Write $[A, A]$ for the k-linear span of commutators $ab - ba$. Define

$$HH_0(A) = A/[A, A].$$

This quotient is a vector space. In general $[A, A]$ is not an ideal, so the quotient has no multiplication induced by A.

A *cyclic functional* on A is a k-linear map $\tau : A \to k$ satisfying $\tau(ab) = \tau(ba)$ for every a, b. It is equivalent to a linear functional on $HH_0(A)$. The class $[1_A]$ and the value $\tau(1_A)$ are separate data. A normalized cyclic functional has $\tau(1_A) = 1$.

Proposition 21.1.1 *(Matrix trace and the regular object).* For every integer $d \geq 1$, matrix trace induces an isomorphism

$$HH_0(M_d(A)) \longrightarrow HH_0(A), \qquad [X] \longmapsto \left[\sum_i X_{ii}\right].$$

Its inverse sends $[a]$ to $[aE_{11}]$. Under this isomorphism,

$$[1_{M_d(A)}] \longmapsto d[1_A].$$

Here E_{ij} is the matrix with entry one in position (i, j) and zero elsewhere.

Proof. The trace of a matrix commutator is a sum of commutators in A:

$$\operatorname{Tr}(XY - YX) = \sum_{i,j} [X_{ij}, Y_{ji}].$$

Thus trace descends to the quotient.

Conversely, $[a, b]E_{11} = [aE_{11}, bE_{11}]$, so the proposed inverse is well defined. For $i \neq j$,

$$aE_{ij} = [aE_{ij}, E_{jj}],$$

and for $i \neq 1$,

$$aE_{ii} - aE_{11} = [aE_{i1}, E_{1i}].$$

Every matrix class therefore equals the class of its trace placed in the $(1,1)$ entry. The two maps are inverse. The formula for the identity follows by adding its d diagonal entries. □

For $A = k$ of characteristic two and $d = 2$, the space $HH_0(M_2(k))$ is one dimensional, but its identity class is zero. Consequently no cyclic functional on this algebra has value one on the identity. Over a field in which d is invertible, the normalized matrix functional $d^{-1}\operatorname{Tr}$ does have that value. The division by d is part of its normalization.

The matrix comparison has an object-theoretic interpretation. The usual equivalence between perfect $M_d(A)$-modules and perfect A-modules carries the regular matrix module to $A^{\oplus d}$. Its identity class accordingly becomes $d[1_A]$. An equivalence of categories does not require these two regular objects to correspond. An additional object marking can impose that requirement.

21.2 Chains, cyclic rotation, and completion

Put $\overline{A} = A/k1$. The normalized Hochschild chains are

$$C_n(A) = A \otimes_k \overline{A}^{\otimes n} \quad (n \geq 0), \qquad C_n(A) = 0 \quad (n < 0).$$

Write an elementary tensor as $a_0[a_1|\cdots|a_n]$. An entry equal to 1 inside the brackets is zero. The differential lowers degree and is

$$\begin{aligned} b\big(a_0[a_1|\cdots|a_n]\big) = {} & a_0a_1[a_2|\cdots|a_n] \\ & + \sum_{i=1}^{n-1}(-1)^i a_0[a_1|\cdots|a_ia_{i+1}|\cdots|a_n] \\ & + (-1)^n a_n a_0[a_1|\cdots|a_{n-1}]. \end{aligned}$$

The formula in positive degree descends from the unnormalized chains $A^{\otimes(n+1)}$ by quotienting the tensors with a unit in a bracket position. Set $b = 0$ on degree zero.

Definition 21.2.1 *(The regular Hochschild trace).* The *Hochschild chain complex with regular coefficients* is $(C_\bullet(A), b)$. Its homology is $HH_\bullet(A)$. The coefficient bimodule is A itself, with left and right multiplication. Replacing that bimodule changes the trace complex.

Associativity proves $b^2 = 0$. In the unnormalized complex, composing two multiplication faces in either order gives the same result after renumbering the later face. Their alternating signs are opposite. This includes the face multiplying the last entry into the first. In degree zero the resulting homology is exactly $A/[A, A]$.

The circle operation is already visible on these chains. Define Connes' operator $B : C_n(A) \to C_{n+1}(A)$ by

$$B\big(a_0[a_1|\cdots|a_n]\big) = \sum_{i=0}^{n}(-1)^{ni}1[a_i|\cdots|a_n|a_0|\cdots|a_{i-1}]. \tag{21.1}$$

For $i = 0$, the bracket is $[a_0|a_1|\cdots|a_n]$. Thus

$$B(a) = 1[a], \qquad B(a_0[a_1]) = 1[a_0|a_1] - 1[a_1|a_0].$$

Proposition 21.2.2 *(The mixed complex).* The operators satisfy

$$b^2 = 0, \qquad B^2 = 0, \qquad bB + Bb = 0.$$

In particular the pair of differentials (b, B) contains more information than $HH_\bullet(A)$ as a graded vector space.

Proof. The multiplication-face cancellation proves the first identity. Every term of B has leading coefficient one. Applying B again places that coefficient inside the bracket, so every resulting term is degenerate. This proves $B^2 = 0$.

For the remaining identity, expand the boundary of each cyclic rotation in (21.1). Its two outer faces cancel the corresponding outer faces from the adjacent rotation. Each interior multiplication then occurs once in bB and once in Bb. Moving the multiplication past the cyclic starting point changes its face index by one, with the rotation sign $(-1)^{ni}$ accounting for the wrap. The signs of these two terms are opposite. All terms cancel, proving $bB = -Bb$. □

More generally, a *mixed complex* over k is a homologically graded vector space C with two operators. They have degrees $|b| = -1$, $|B| = 1$ and satisfy these three identities. This definition also applies to the cyclic complexes of differential graded categories. Their internal differentials and permutations carry the Koszul signs of their graded morphisms.

For a mixed complex, let u have degree -2. Define its negative cyclic complex by

$$CC^-_m(C) = \prod_{j\geq 0} C_{m+2j}u^j, \qquad d = b + uB.$$

The mixed identities give $d^2 = 0$. The product in this formula is part of the definition.

Its periodic cyclic complex is $CC^{\mathrm{per}}(C) = CC^-(C)[u^{-1}]$, with the same differential. Inverting u is a filtered colimit after forming the displayed product. These two limiting operations must be respected by any comparison involving these complexes.

Example 21.2.3 *(A product created by negative cyclic completion).* For $i \geq 0$, let $C^{(i)}$ be one copy of k in homological degree $2i$, with $b = B = 0$. Then

$$CC^-_0\left(\bigoplus_{i\geq 0} C^{(i)}\right) = \prod_{i\geq 0} k, \qquad \left(\bigoplus_{i\geq 0} CC^-(C^{(i)})\right)_0 = \bigoplus_{i\geq 0} k.$$

The natural map is the inclusion of finite-support sequences and is not surjective. All differentials vanish, so the discrepancy also occurs in homology. An equivariant direct-sum decomposition of Hochschild chains therefore does not, by itself, give a direct-sum decomposition after negative cyclic completion.

A cyclic functional $\tau : A \to k$ is a map of mixed complexes from $C_\bullet(A)$ to k in degree zero, with zero B on the target. The chain condition is $\tau b(a_0[a_1]) = 0$. All other degrees give zero, and the compatibility with B follows from the degrees. This functional is an evaluation of the mixed complex, rather than the complex itself.

Categorical trace and Morita comparison

For a dg algebra A over a commutative ring k, the derived regular trace is $A \otimes^{\mathrm{L}}_{A^e} A$. Choose a dg model with K-flat morphism complexes over k: tensoring with each complex must preserve quasi-isomorphisms. A cofibrant replacement provides such a model for the bar tensors.

This is the Hochschild trace of the identity of its perfect-module category. The complex, with its circle action, is distinct from a single class associated to an object. For example a perfect object has its Hochschild Chern character, giving the additive map $K_0(\mathrm{Perf}\ A) \to HH_0(A)$. For the regular object A, this class is $[1_A]$.

The derived mixed complex is invariant under dg Morita equivalence. More precisely, Keller's Theorem 5.2 in *On differential graded categories* makes the mixed-complex construction a functor on the Morita category and identifies $C(A)$ with $C(\mathrm{Perf}_{\mathrm{dg}}\ A)$. This theorem concerns the enhanced dg categories and their derived equivalences. An equivalence of lists of simple objects does not meet its hypotheses.

For matrix algebras the comparison has an explicit cyclic-chain formula. On unnormalized degree-n chains, set

$$\mathrm{tr}_{\mathrm{cyc}}(X_0 \otimes \cdots \otimes X_n) = \sum_{i_0,\ldots,i_n} (X_0)_{i_0 i_1} \otimes (X_1)_{i_1 i_2} \otimes \cdots \otimes (X_n)_{i_n i_0}.$$

Matrix multiplication proves compatibility with every face, including the cyclic last face. Inserting an identity matrix collapses an adjacent index and gives the scalar-unit degeneracy. Rotating the indices proves compatibility with cyclic rotation. Thus it descends to a map of mixed complexes. In degree zero it is precisely Proposition 21.1.1.

The corner inclusion used there gives an inverse on HH_0. It is not a unital cyclic inverse on normalized chains. It sends 1_A to E_{11}, whereas the matrix-algebra unit is $\sum_i E_{ii}$. The enhanced Morita comparison accounts for the full units and their homotopies. It preserves the derived trace while transporting each chosen object marking to the object actually corresponding under the equivalence.

21.3 A functional that supplies sewing

To contract two boundary entries, a functional needs a nondegenerate pairing. This condition can be imposed and calculated in finite dimension before asking for an arithmetic or analytic realization.

Definition 21.3.1 *(Commutative Frobenius algebra).* A *commutative Frobenius algebra* over k is a finite-dimensional commutative unital k-algebra A with a linear functional $\varepsilon : A \to k$ such that

$$(a, b)_\varepsilon = \varepsilon(ab)$$

is nondegenerate. Choose dual bases e_i, f_i for this pairing and put

$$c = \sum_i e_i \otimes f_i, \qquad E = \sum_i e_i f_i.$$

The tensor c is the inverse pairing, and E is the *handle element*. Both are independent of the chosen dual bases.

The multiplication m joins two inputs. The pairing contracts two entries, and c supplies the reverse contraction. Define

$$\Delta(a) = (a \otimes 1)c.$$

This is the linear map adjoint to multiplication for the Frobenius pairing. In particular its normalization is determined by ε, not by multiplication alone.

PROPOSITION 21.3.2 *(Finite sewing identities).* The maps m, Δ, the algebra unit and ε satisfy the unit and counit identities, associativity and coassociativity, and

$$\Delta(ab) = (a \otimes 1)\Delta(b) = (1 \otimes b)\Delta(a).$$

Furthermore,

$$m\Delta(a) = aE.$$

The closed contraction formed from one unit, g successive handles and one counit is

$$Z_g = \varepsilon(E^g) \qquad (g \geq 0).$$

Proof. The inverse-pairing identity is $\sum_i \varepsilon(af_i)e_i = a$. It proves the counit identity and shows that the adjoint of m is the displayed Δ. Taking adjoints of associativity proves coassociativity, since the tensor-product pairing is nondegenerate.

The inverse tensor obeys $(a \otimes 1)c = (1 \otimes a)c$. To verify this equality, pair both tensors with $x \otimes y$. Each value is $\varepsilon(axy)$. Nondegeneracy proves the equality, and then proves both Frobenius identities for $\Delta(ab)$. Multiplying the two factors of $\Delta(a)$ gives aE. Iterating this handle map on 1 and applying ε gives the formula for Z_g. □

Example 21.3.3 *(Semisimple weights).* Let $A = k^r$, with orthogonal primitive idempotents e_i. Choose nonzero $w_i \in k$ and set $\varepsilon(e_i) = w_i$. The dual basis is $w_i^{-1}e_i$, so

$$\Delta(e_i) = w_i^{-1}e_i \otimes e_i, \qquad E = \sum_i w_i^{-1}e_i, \qquad Z_g = \sum_i w_i^{1-g}.$$

In particular $Z_0 = \sum_i w_i$ and $Z_1 = r$ in k. The algebra does not determine the weights. The counit and the inverse pairing retain them through every contraction.

Example 21.3.4 *(A nilpotent algebra with a nondegenerate pairing).* Let $A = k[\epsilon]/(\epsilon^2)$ and choose $\varepsilon(1) = a$, $\varepsilon(\epsilon) = 1$. Its Gram matrix in the basis $(1, \epsilon)$ is

$$\begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix},$$

whose determinant is -1 in every characteristic. The dual basis is $(\epsilon, 1 - a\epsilon)$. Consequently

$$E = 2\epsilon, \qquad Z_0 = a, \qquad Z_1 = 2, \qquad Z_g = 0 \quad (g \geq 2).$$

These are algebraic contractions. Over $\mathbb{R}$, the displayed Gram matrix is indefinite. Nondegeneracy supplies sewing, but does not assert positivity of a Hilbert-space pairing.

The finite identities specify what must be transported in a larger comparison: the functional, inverse pairing, unit, and gluing maps. An equality of closed scalar values cannot recover all these maps. For infinite-dimensional or completed objects, defining the inverse pairing and proving convergence of its contractions are additional parts of the construction.

21.4 Spherical traces and marked closures

A spherical category supplies a trace by its duality maps. It need not supply a cyclic functional on an algebra used to present its objects. The arithmetic fusion example makes this distinction explicit.

Let C be a split spherical fusion category over a characteristic-zero field k. Thus C is a rigid semisimple k-linear tensor category with finitely many simple objects, finite-dimensional morphism spaces, and simple unit. Splitness means $\mathrm{End}(S) = k$ for each simple S. Its chosen pivotal duality identifies left and right categorical traces. Write S_i for its simple representatives and

$$d_i = \mathrm{tr}_{S_i}(1_{S_i})$$

for their spherical dimensions. For an object $X \simeq \bigoplus_i S_i^{\oplus m_i}$, an endomorphism has multiplicity matrices $f_i \in M_{m_i}(k)$.

Proposition 21.4.1 *(Trace on multiplicity spaces).* The categorical trace is

$$\mathrm{tr}_X(f) = \sum_i d_i \,\mathrm{Tr}_k(f_i).$$

In particular it is a cyclic functional on $\mathrm{End}(X)$. The coefficients d_i are determined by the chosen duality.

Proof. Choose inclusions $u_{i\alpha} : S_i \to X$ and projections $v_{i\alpha} : X \to S_i$ with

$$v_{i\alpha} u_{j\beta} = \delta_{ij}\delta_{\alpha\beta} 1_{S_i}, \qquad \sum_{i,\alpha} u_{i\alpha} v_{i\alpha} = 1_X.$$

Expand f using these splittings. Cyclicity of the spherical trace gives $\mathrm{tr}_X(u_{i\alpha}v_{i\beta}) = \mathrm{tr}_{S_i}(v_{i\beta}u_{i\alpha}) = \delta_{\alpha\beta} d_i$. The diagonal matrix entries therefore give the displayed sum. Ordinary matrix trace is cyclic, proving the last assertion. □

For the order-two arithmetic category of Chapter 15, the fusion rules are

$$g^2 = 1, \qquad g\rho = \rho g = \rho, \qquad \rho^2 = 1 + g + 2\rho.$$

The constructed spherical structures have $d_1 = d_g = 1$ and

$$d_\rho = 1 - \sqrt{3} \quad \text{or} \quad d_\rho = 1 + \sqrt{3}$$

at the two indicated embeddings of the coefficient field. Indeed multiplicativity and additivity of dimension require $d_\rho^2 = 2 + 2d_\rho$. Their global dimensions and traces on the sum of the simples are respectively

$$\sum_i d_i^2 = 6 \mp 2\sqrt{3}, \qquad \mathrm{tr}_{1\oplus g\oplus\rho}(1) = 3 \mp \sqrt{3}.$$

These are different contractions. The unitary realization in Chapter 20 uses $1 + \sqrt{3}$. The original arithmetic embedding retains $1 - \sqrt{3}$ and the nonunitary realization of Chapter 16.

One cannot change the sign of a single dimension while keeping that category and its pivotal structure. More precisely, any tensor automorphism of the identity functor acts on the three simples by scalars $\eta_1 = 1, \eta_g, \eta_\rho$. The unit and ρ summands of ρ^2 require

$$\eta_\rho^2 = 1, \qquad \eta_\rho^2 = \eta_\rho.$$

Hence $\eta_\rho = 1$, and the g summand gives $\eta_g = 1$. Two pivotal structures differ by such an automorphism, so the existing pivotal structure is unique. Passing to the other field embedding changes the coefficient data of the category.

The algebra $\mathrm{End}(\bigoplus_i S_i)$ is k^r. When every d_i is nonzero, its spherical trace makes this commutative algebra a Frobenius algebra with weights $w_i = d_i$. Example 21.3.3 then gives its finite contractions. This construction records a trace on the sum of the simples. Recovering the fusion tensor, its associator or its geometric sewing requires those maps as additional data.

The scalar unit of a Leavitt algebra

Let L_r be the unital algebra with generators $s_1, \dots, s_r, t_1, \dots, t_r$ and relations

$$t_i s_j = \delta_{ij} 1, \qquad \sum_i s_i t_i = 1.$$

The t_i are algebraic dual generators. No complex conjugation is part of this notation.

Proposition 21.4.2 *(A vanishing unit class).* If $r - 1$ is invertible in k, then $[1] = 0$ in $HH_0(L_r)$. Consequently every cyclic functional on L_r vanishes at 1.

Proof. The defining relations give the literal commutator identity

$$\sum_{i=1}^{r} [t_i, s_i] = r1 - \sum_i s_i t_i = (r-1)1.$$

Divide by $r - 1$ and pass to $L_r/[L_r, L_r]$. □

This argument concerns one class. It does not assert that $HH_0(L_r)$ is zero. It also has the stated coefficient restriction. If $r = 1$, the algebra is $k[s, s^{-1}]$, and the identity class is nonzero.

For the arithmetic algebra $\mathcal{B}_I$ of Section 15.3, one has $r = 2N$. Its characteristic-zero coefficient field makes $2N - 1$ invertible. Nevertheless its marked categorical closure recovers the arithmetic values. Retain the object ρ, its duality maps X_0, X_0', and the scalar network $\mathcal{N}_g = b_g 1_\rho$. With $d = -1/b_0$ and $e_g = b_g/b_0$, the closure is

$$\mathrm{cl}_\rho(a) = dX_0' \rho(a) X_0, \qquad -\mathrm{cl}_\rho(\mathcal{N}_g) = e_g.$$

The second identity follows from $X_0' X_0 = 1$ and the fact that ρ fixes scalars. This functional has domain $\mathrm{End}(\rho)$ and uses its specified duality. Its value is compatible with the vanishing ambient unit class because it is a different trace construction.

The finite balanced-tensor realization of Theorem 15.6.1 retains this network as well. Its matrix detector is

$$M_g = b_g 1_{\omega(\rho)}, \qquad b_g = \frac{\mathrm{Tr}_k(M_g)}{3N}.$$

This formula uses the scalar endomorphism and the particular dimension $\dim_k \omega(\rho) = 3N$. It is not a replacement for categorical trace on arbitrary diagrams. The functor takes values in bimodules over k^{N+1}, with their balanced tensor. It is not a finite-dimensional unital representation of $\mathcal{B}_I$.

21.5 Extensions and regular Hochschild traces

The trace of a nonsemisimple category requires its actual complex. Extensions between simple modules give useful information, but they need not survive regular Hochschild trace. Two small algebras show both behaviors.

Let A be a finite-dimensional algebra over a field k. There are finitely many simple right A-modules up to isomorphism: each occurs among the composition factors of the regular module. If M, N are finite-dimensional, then $\operatorname{Ext}^n_A(M,N)$ is finite-dimensional for each n. Indeed successive surjections from finite free modules give a projective resolution with finite-dimensional terms. Applying $\operatorname{Hom}_A(-,N)$ proves degreewise finiteness. There is no resulting bound on the total graded dimension.

The dual numbers

Put $A = k[\epsilon]/(\epsilon^2)$ and $S = A/(\epsilon)$. Multiplication by ϵ has kernel and image $k\epsilon$, so

$$\cdots \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \longrightarrow S \longrightarrow 0$$

is an exact free resolution. Applying $\operatorname{Hom}_A(-,S)$ makes every differential zero. Thus

$$\operatorname{Ext}^n_A(S,S) = k \qquad (n \geq 0).$$

The algebra has radical square zero, but its simple module has infinite projective dimension. In characteristic two it is also $k[C_2]$, under $\epsilon \mapsto g - 1$.

For the regular trace, use the enveloping algebra $A^e = A \otimes_k A^{\mathrm{op}}$ and the coefficient bimodule A. The bar resolution identifies

$$HH_n(A) = \operatorname{Tor}_n^{A^e}(A,A).$$

Here both occurrences of A retain their regular actions.

PROPOSITION 21.5.1 *(The complete regular trace of the dual numbers).* For every field k,

$$HH_0(A) = A, \qquad HH_{2r+1}(A) = A/(2\epsilon A) \quad (r \geq 0), \qquad HH_{2r}(A) = \operatorname{ann}_A(2\epsilon) \quad (r \geq 1).$$

If $\operatorname{char} k \neq 2$, each positive-degree group has dimension one. If $\operatorname{char} k = 2$, then $HH_n(A) = A$ in every nonnegative degree.

Proof. Write $A^e = k[x,y]/(x^2,y^2)$, with x acting on the left and y on the right. Set $u = x - y$ and $v = x + y$. There is a free bimodule complex

$$\cdots \xrightarrow{u} A^e \xrightarrow{v} A^e \xrightarrow{u} A^e \xrightarrow{\mu} A \longrightarrow 0, \qquad \mu(x) = \mu(y) = \epsilon.$$

The differential out of odd degree is u and that out of positive even degree is v. Their composites vanish.

For $z = \alpha + \beta x + \gamma y + \eta xy$,

$$uz = \alpha(x-y) + (\gamma - \beta)xy, \qquad vz = \alpha(x+y) + (\beta + \gamma)xy.$$

Consequently

$$\ker u = \langle x+y, xy\rangle = \operatorname{im} v, \qquad \ker v = \langle x-y, xy\rangle = \operatorname{im} u.$$

Also $\ker \mu = \langle x-y, xy\rangle$. These statements remain valid in characteristic two, and prove exactness.

Tensoring with the regular A sends u to zero and v to 2ϵ. The resulting alternating differential gives the asserted quotients and annihilators. If two is invertible, the annihilator is ϵA. If two is zero, every differential vanishes.

The normalized Hochschild formula gives the same calculation without a change of resolution. Since $\overline{A} = k\epsilon$, interior products in a normalized chain vanish, and the two outer faces give

$$b_n\big(c[\epsilon|\cdots|\epsilon]\big) = (1 + (-1)^n)c\epsilon[\epsilon|\cdots|\epsilon].$$

The bracket on the right has $n-1$ entries. In particular the cyclic last face is responsible for the factor two in even degree. □

In characteristic two, the simple-module Ext groups have dimension one while the regular Hochschild groups have dimension two. The coefficient module and the traced resolution are indispensable parts of either statement.

A directed algebra and cyclic endpoint matching

Let Q be a finite quiver, J its arrow ideal, and $B = kQ/J^2$. Concatenate paths from left to right and use right modules. For a vertex i, put $P_i = e_iB$ and $S_i = e_iB/e_iJ$. The radical of P_i is one copy of S_j for each arrow $i \to j$. Iterating projective covers therefore gives

$$\dim_k \operatorname{Ext}^n_B(S_i, S_j) = \#\{\text{oriented length-}n\text{ paths from } i \text{ to } j\}.$$

To justify the last step, every differential in this minimal resolution lands in a radical. A map to a simple module kills that radical, so the Hom complex has zero differential. All groups of degree at least two vanish precisely when the quiver has no composable pair of arrows.

Take

$$Q = (0 \xrightarrow{a} 1 \xrightarrow{b} 2), \qquad B = kQ/(ab).$$

Then $P_0 = \langle e_0, a\rangle$, $P_1 = \langle e_1, b\rangle$, and $P_2 = \langle e_2\rangle$. The sequence

$$0 \longrightarrow P_2 \xrightarrow{e_2 \mapsto b} P_1 \xrightarrow{e_1 \mapsto a} P_0 \longrightarrow S_0 \longrightarrow 0$$

is exact: the middle kernel is kb because $ab = 0$. It gives

$$\operatorname{Ext}^1_B(S_0, S_1) = k, \qquad \operatorname{Ext}^1_B(S_1, S_2) = k, \qquad \operatorname{Ext}^2_B(S_0, S_2) = k.$$

The product of the two adjacent degree-one classes is nonzero. The exact sequence $0 \to S_1 \to P_0 \to S_0 \to 0$ identifies its Yoneda product with the dimension-shifting isomorphism $\operatorname{Ext}^1_B(S_1, S_2) \simeq \operatorname{Ext}^2_B(S_0, S_2)$. Thus the algebra relation $ab = 0$ does not make this product zero.

For comparison, the path algebra kQ before imposing $ab = 0$ has $\operatorname{rad} P_0 \simeq P_1$ and $\operatorname{rad} P_1 \simeq P_2$. Its simple modules have projective resolutions of length at most one. It is a different algebra, and its P_0 has Loewy length three.

Proposition 21.5.2 *(Regular trace of the directed model).* For $B = kQ/(ab)$ as above,

$$HH_0(B) = ke_0 \oplus ke_1 \oplus ke_2, \qquad HH_n(B) = 0 \quad (n \geq 1).$$

Proof. Set $S = ke_0 \oplus ke_1 \oplus ke_2$ and $J = ka \oplus kb$. The relative normalized bar terms $P_n = B \otimes_S J^{\otimes_S n} \otimes_S B$ are

$$\begin{aligned} P_0 &= \bigoplus_{i=0}^{2} Be_i \otimes_k e_i B, \\ P_1 &= (Be_0 \otimes_k e_1 B) \oplus (Be_1 \otimes_k e_2 B), \\ P_2 &= Be_0 \otimes_k e_2 B, \end{aligned}$$

and $P_n = 0$ for $n \geq 3$. Each summand is a projective bimodule, being an idempotent summand of $B \otimes_k B$. The differentials are

$$\begin{aligned} d_1(l \otimes j \otimes r) &= lj \otimes r - l \otimes jr, \\ d_2(l \otimes a \otimes b \otimes r) &= la \otimes b \otimes r + l \otimes a \otimes br. \end{aligned}$$

The interior term of d_2 vanishes because $ab = 0$.

For completeness, the dimensions of these terms are 8, 4, 1. A basis of P_1 consists of

$$\begin{aligned} q_1 &= e_0 \otimes a \otimes e_1, & q_2 &= e_0 \otimes a \otimes b, \\ q_3 &= e_1 \otimes b \otimes e_2, & q_4 &= a \otimes b \otimes e_2. \end{aligned}$$

Their images are

$$a \otimes e_1 - e_0 \otimes a, \quad a \otimes b, \quad b \otimes e_2 - e_1 \otimes b, \quad -a \otimes b.$$

The first three are independent. The kernel of d_1 is $k(q_2 + q_4)$, and d_2 sends its generator to $q_2 + q_4$.

Multiplication $P_0 \to B$ is surjective with a three-dimensional kernel. This proves exactness in every characteristic.

Regular trace sends each summand to

$$B \otimes_{B^e} (Be_i \otimes_k e_j B) \simeq e_j B e_i.$$

The forward map is $c \otimes (l \otimes r) \mapsto rcl$. The inverse is $t \mapsto t \otimes (e_i \otimes e_j)$. The balancing relations prove that these maps are inverse.

All off-diagonal terms here vanish, because a directed path has no return path. The diagonal terms are ke_i. The traced complex is consequently k^3 in degree zero and zero in every positive degree. □

The relation contributes a nonzero degree-two bimodule-resolution term. Taking the regular trace kills that term by its endpoints. An arithmetic use of simple-module extensions must therefore construct the map into its trace complex and determine the image of the Yoneda products.

21.6 Tensoring local trace complexes

Local degree bounds do not simply add their positive parts. Fix a commutative ring R. We use homological shifts, so a module $M[1]$ is concentrated in degree one. Suppose two complexes have specified splittings

$$C_2 \simeq B_2 \oplus W_2[1], \qquad C_3 \simeq B_3 \oplus W_3[1],$$

Here B_2, W_2, B_3, W_3 are R-modules in degree zero. The subscripts label two local factors.

Proposition 21.6.1 *(All four tensor terms).* There is an equivalence

$$\begin{aligned} C_2 \otimes^{\mathrm{L}}_R C_3 \simeq\ & B_2 \otimes^{\mathrm{L}}_R B_3 \\ & \oplus (W_2 \otimes^{\mathrm{L}}_R B_3)[1] \\ & \oplus (B_2 \otimes^{\mathrm{L}}_R W_3)[1] \\ & \oplus (W_2 \otimes^{\mathrm{L}}_R W_3)[2]. \end{aligned}$$

If all remaining local factors assemble into a degree-zero complex D, their contribution is the derived tensor of D with every displayed term.

Proof. Choose projective resolutions of the modules. Distribute the tensor product over the two finite direct sums. The tensor differential is $d(x \otimes y) = dx \otimes y + (-1)^{|x|} x \otimes dy$, and the two shifts in the last term add to two. Associativity of derived tensor gives the assertion about D. □

Over a field, take $B_2 = B_3 = k$, $W_2 = k$, and $W_3 = k^2$. The result is

$$k \oplus k^3[1] \oplus k^2[2].$$

If x spans W_2 and y, z form a basis of W_3, then $x \otimes y$ and $x \otimes z$ are independent degree-two cycles. Every differential is zero, so they are not boundaries. Anticommutativity does not cancel products in distinct factors. Taking unit reference vectors at almost all primes still permits non-unit entries at both selected primes.

Integral coefficients can give a different result, for a specific algebraic reason.

Proposition 21.6.2 *(Distinct-primary vanishing).* If M_2 is a 2-primary torsion abelian group and M_3 is a 3-primary torsion abelian group, then

$$M_2 \otimes^{\mathrm{L}}_{\mathbb{Z}} M_3 = 0.$$

If B_2, B_3 are flat abelian groups, the tensor of $B_2 \oplus M_2[1]$ and $B_3 \oplus M_3[1]$ is concentrated in degrees zero and one.

Proof. The free resolution of $\mathbb{Z}/2^a$ is $[\mathbb{Z} \xrightarrow{2^a} \mathbb{Z}]$. After tensoring with $\mathbb{Z}/3^b$, its differential is invertible. Its tensor product and first Tor therefore vanish. Finite primary groups are sums of such cyclic groups. Every primary torsion abelian group is the filtered union of its finite subgroups, and tensor and Tor commute with these filtered colimits.

There is no higher Tor over $\mathbb{Z}$: the kernel of a surjection from a free abelian group is free. This proves the first assertion. Flatness removes higher Tor from the other three terms of Proposition 21.6.1. □

For $B_2 = B_3 = \mathbb{Z}$, $M_2 = \mathbb{Z}/2$, and $M_3 = (\mathbb{Z}/3)^2$, the result is

$$\mathbb{Z} \oplus \big(\mathbb{Z}/2 \oplus (\mathbb{Z}/3)^2\big)[1].$$

This calculation does not replace the common-field example. They have different coefficients. An arithmetic comparison needs its integral models, change-of-coefficient maps, and a proof that the chosen completion preserves the asserted tensor.

Question 21.6.3 *(Arithmetic trace assembly).* Construct the local arithmetic categories and regular trace complexes, the maps from their extension data, and the completed restricted tensor that assembles them globally. Determine whether the mixed term survives, vanishes by a coefficient argument, or is removed by a specified quotient. In the last case, construct the quotient map and prove its compatibility with the trace operations. A degree-one truncation and the full trace have different comparison problems.

21.7 ORTHOGONAL BLOCKS AND CYCLIC DECOMPOSITION

Let k now be a commutative ring, and let $(\mathcal{A}_i)_{i\in I}$ be small k-linear differential graded categories with K-flat morphism complexes. Here K-flat means that tensoring with the complex preserves quasi-isomorphisms. This hypothesis lets ordinary tensor products compute the derived tensors used below.

Form the orthogonal union $\mathcal{A}$: objects belong to one of the $\mathcal{A}_i$, and the morphism complex between different blocks is zero. Let $\mathrm{Perf}(\mathcal{A})$ be the category obtained from its representable modules by finite sums, shifts, cones and retracts. Write $C_\bullet(\mathcal{B})$ for the derived Hochschild mixed complex of a dg category $\mathcal{B}$.

PROPOSITION 21.7.1 *(Cyclic block decomposition).* There is an S^1-equivariant equivalence

$$C_\bullet(\mathrm{Perf}(\mathcal{A})) \simeq \bigoplus_{i\in I} C_\bullet(\mathrm{Perf}(\mathcal{A}_i)).$$

Every perfect module has finite block support.

Proof. In cyclic degree n, the bar construction is the sum of

$$\mathcal{A}(X_n, X_0) \otimes_k \mathcal{A}(X_{n-1}, X_n) \otimes_k \cdots \otimes_k \mathcal{A}(X_0, X_1)$$

over its cyclic words of objects. A word using two blocks has a zero morphism factor, including when the change of block occurs at the closing pair. Every surviving word therefore lies in one block. Composition faces, unit degeneracies, internal differentials and cyclic rotation preserve it.

Thus the cyclic object itself decomposes as the direct sum of the block cyclic objects. The K-flat assumption computes the derived tensors, and realization commutes with direct sums. The cyclic structure, hence the circle action, is preserved. Passing to perfect modules is the dg Morita invariance of this derived trace.

A representable is supported in one block. Each finite sum, shift, cone or retract preserves finite support, so every perfect object has it. Conversely, a finite-support family of perfect block modules is perfect over $\mathcal{A}$. This identifies the compact model used in the formula. □

For a compactly generated presentable category, this statement applies when its compact objects have the specified model. A product description of a presentable category alone does not identify the relevant trace or its compact support. Moreover the proposition is an equivalence of cyclic complexes. A multiplicative trace comparison needs a compatible monoidal enhancement and preservation of its operations.

Even coordinatewise multiplication on the right has a unit only under an additional finiteness condition. If infinitely many blocks are nonzero unital algebras, every element of their direct sum omits some block. It cannot act as identity on that block. The infinite coordinatewise direct sum is then nonunital. Finally, negative cyclic completion need not preserve this sum, as Example 21.2.3 shows.

For arithmetic block comparisons, the coefficient ring, dg models and indexing maps must be given before applying this decomposition. The perfect-stack loop comparisons of Chapter 18 also retain their own geometric hypotheses. The topological circle, a finite cyclic classifying stack, and a profinite mapping object define different constructions. Passing between them requires comparison maps and continuity theorems on the chosen carriers.

Filtered continuity on a specified trace carrier

The same finite-word argument applies to a filtered union of dg categories. Every cyclic word, morphism tensor and relation appears at a common finite stage. For K-flat models over a common ring, tensoring and filtered colimits commute. Thus the cyclic complex of that union is the filtered colimit of its cyclic complexes, before taking homology or cyclic completion.

There is also a categorical version with a different output. Let $\mathcal{V}$ be a presentable symmetric monoidal category whose tensor preserves colimits separately. For an associative algebra object A of $\mathcal{V}$, define

$$\mathrm{Tr}_{\mathcal{V}}(A) = \left| [n] \longmapsto A^{\otimes(n+1)} \right|.$$

The faces multiply adjacent entries, including the cyclic closing pair, and the degeneracies insert the unit. The bars denote geometric realization of the cyclic object.

For $\mathcal{V} = D(k)$ this is a derived Hochschild complex. For $\mathcal{V} = \mathrm{Pr}^{\mathrm{L}}_{k,\mathrm{st}}$ it is a category, as in Section 18.29.

Proposition 21.7.2 *(Filtered continuity of cyclic realization).* Let $(A_i)_{i\in I}$ be a filtered diagram of associative algebra objects in $\mathcal{V}$, and let A be its colimit. Then there is a natural circle-equivariant equivalence

$$\operatorname*{colim}_i \mathrm{Tr}_{\mathcal{V}}(A_i) \simeq \mathrm{Tr}_{\mathcal{V}}(A).$$

If the diagram has compatible E_2 structures, this equivalence preserves the induced E_1 multiplication on the trace.

Proof. Filtered colimits of associative algebras are computed on their underlying objects. Indeed all operations have finite arity, and the diagonal $I \to I^m$ is cofinal for every finite m. Separate preservation of colimits therefore gives

$$A^{\otimes(n+1)} \simeq \operatorname*{colim}_i A_i^{\otimes(n+1)}.$$

These equivalences preserve multiplication, the unit and permutations. They consequently identify every cyclic face, degeneracy and rotation.

Geometric realization is a colimit and commutes with the filtered colimit. Realizing the degreewise equivalence proves the assertion with its circle action. For E_2 inputs, use their description as associative algebras in associative algebras. The cyclic construction in one direction retains the multiplication in the other, with its interchange coherences. The same colimit argument preserves these operations and proves the last assertion. □

To apply this result to an arithmetic formal presentation, one must identify its category with the stated colimit of its finite-stage categories. The transition functors must preserve the convolution structure being used. A closed immersion of parameter spaces does not by itself establish this categorical equivalence or its support conditions.

After that input, the trace-continuity step is positive. If $T_i \simeq Q_i$ are compatible finite-stage trace comparisons and $\mathrm{colim}_i Q_i \simeq Q$ is independently proved, their colimits give $T \simeq Q$. The loop-space computation and passage of a singular-support condition to Q remain separate hypotheses. Functions on an inverse limit, a pro-étale mapping object, and negative cyclic completion cannot be exchanged with these colimits without their own arguments.

An arithmetic determinant trace is different again. The trace coefficient of a universal Chenevier determinant is a function on a deformation ring. Identifying it with the invariant trace of a universal (φ, Γ)-module requires a comparison of those coefficient objects. Compatibility with a specified Nygaard

filtration is an additional equation on that comparison. Neither scalar Hochschild continuity nor the categorical trace theorem constructs it.

21.8 ARITHMETIC SHIFTS, PHASES, AND OCCUPATION TRACES

The Bost–Connes algebra gives a concrete arithmetic representation. Its occupation trace, algebraic Hochschild cochains, and smooth Hochschild cochains can all be studied. Relating any of them to an independently defined arithmetic trace category is a further comparison.

Definition 21.8.1 *(The Bost–Connes algebra).* Let A_{BC} be the complex unital C^*-algebra generated by isometries μ_n, for $n \geq 1$, and unitaries $e(\gamma)$, for $\gamma \in \mathbb{Q}/\mathbb{Z}$, with relations

$$\begin{aligned} \mu_1 &= 1, & \mu_m\mu_n &= \mu_{mn}, & \mu_n^*\mu_n &= 1, \\ e(0) &= 1, & e(\gamma)e(\eta) &= e(\gamma+\eta), & e(\gamma)^* &= e(-\gamma), \\ \mu_m^*\mu_n &= \mu_n\mu_m^* & &\text{if } (m,n)=1, & & \\ \mu_n e(\gamma)\mu_n^* &= \frac{1}{n}\sum_{n\delta=\gamma} e(\delta). & & & & \end{aligned}$$

In particular $\mu_n\mu_n^* = e_n = n^{-1}\sum_{n\delta=0} e(\delta)$. The time evolution is

$$\sigma_t(\mu_n) = n^{it}\mu_n, \qquad \sigma_t(e(\gamma)) = e(\gamma).$$

On $\mathcal{L} = \ell^2(\mathbb{N}^\times)$, with orthonormal basis δ_m, define

$$\mu_n\delta_m = \delta_{nm}, \qquad e(\gamma)\delta_m = \exp(2\pi i m\gamma)\delta_m.$$

We use the same notation for the abstract generators and these operators once faithfulness is proved.

PROPOSITION 21.8.2 *(The faithful occupation representation).* These operators define a faithful representation of A_{BC} on $\mathcal{L}$. The range projection $\mu_n\mu_n^*$ selects the integers divisible by n. Its spectrum is $\{0,1\}$ for $n > 1$.

Proof. The adjoint shift sends δ_m to $\delta_{m/n}$ if $n \mid m$, and to zero otherwise. This gives the isometry, semigroup and coprime-adjoint relations. The transfer relation is the finite character identity

$$\frac{1}{n}\sum_{n\delta=\gamma}\exp(2\pi i m\delta) = \begin{cases} \exp(2\pi i(m/n)\gamma), & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

It also gives the range projection. For $n > 1$, its range contains δ_n and its kernel contains δ_1, so both spectral values occur.

To prove faithfulness, let $K = \prod_p U(1)$, with the product topology. For $z \in K$, write $z(n) = \prod_p z_p^{v_p(n)}$. The gauge action α_z fixes the phases and sends μ_n to $z(n)\mu_n$. It preserves the defining relations.

The finite normal sums

$$A_{\mathrm{alg}} = \mathrm{span}_{\mathbb{C}}\{\mu_r e(\gamma)\mu_s^* : r,s \geq 1,\ \gamma \in \mathbb{Q}/\mathbb{Z}\}$$

form a dense star subalgebra. Indeed the transfer identities give $e(\gamma)\mu_n = \mu_n e(n\gamma)$ and its adjoint. If $s = ds'$, $u = du'$, and $(s', u') = 1$, multiplication becomes

$$(\mu_r e(\gamma)\mu_s^*)(\mu_u e(\eta)\mu_v^*) = \mu_{ru'} e(u'\gamma + s'\eta)\mu_{vs'}^*.$$

Adjoint exchanges r, s and negates γ. The gauge orbit of each normal term is continuous. Density and isometry prove continuity on every element of A_{BC}.

A normal term has gauge character $z(r/s)$. Haar averaging therefore kills it unless $r = s$, in which case the transfer relation puts it in the phase algebra D. Thus $E(a) = \int_K \alpha_z(a)\, dz$ is a conditional expectation onto D. The representation of $C^*(\mathbb{Q}/\mathbb{Z}) = C(\widehat{\mathbb{Z}})$ on $\mathcal{L}$ evaluates at the positive integers. Those integers meet every residue class and are dense in $\widehat{\mathbb{Z}}$. Their evaluations are faithful, so $D = C(\widehat{\mathbb{Z}})$ and the representation is faithful on D.

The expectation itself is faithful. For $a \geq 0$ and a state ω, the function $z \mapsto \omega(\alpha_z(a))$ is continuous and nonnegative. If $E(a) = 0$, its integral vanishes. Full support of Haar measure makes its value at the identity zero. States separate positive elements, so $a = 0$.

Finally, $V_z \delta_m = z(m)\delta_m$ implements α_z in the occupation representation π. If $\pi(a) = 0$, then $\pi(E(a^*a)) = 0$. Faithfulness on D, followed by faithfulness of E, gives $a^*a = 0$ and $a = 0$. □

Normalized creation and higher modes

For each prime p, let $\mathscr{F}_p = \ell^2(\mathbb{N}_0)$, with number states $|r\rangle_p$. Form the Hilbert restricted tensor product with vacuum reference vectors $|0\rangle_p$. On finite occupation vectors, define

$$a_p|r\rangle_p = \sqrt{r}\,|r-1\rangle_p, \qquad a_p^\dagger|r\rangle_p = \sqrt{r+1}\,|r+1\rangle_p, \qquad N_p = a_p^\dagger a_p.$$

Unique prime factorization gives a unitary

$$U : \mathcal{L} \longrightarrow \widehat{\bigotimes_p}{}' \mathscr{F}_p, \qquad U\delta_m = \bigotimes_p |v_p(m)\rangle_p.$$

The arithmetic prime shift is

$$U\mu_p U^{-1} = a_p^\dagger (N_p + 1)^{-1/2}.$$

The diagonal factor cancels the creation coefficient, leaving an isometry. This identity begins on finite occupation vectors and extends boundedly. The unnormalized creation operator is unbounded.

For $z \in \mathbb{C}$, the coherent vector

$$\exp(-|z|^2/2) \sum_{k\geq 0} \frac{z^k}{\sqrt{k!}} U\delta_{p^k}$$

has norm one. It lies in the arithmetic algebra orbit of the vacuum: apply the norm-convergent series $\exp(-|z|^2/2)\sum_{k\geq 0} z^k (U\mu_p U^{-1})^k/\sqrt{k!}$. The ratio test proves convergence of its scalar coefficient norms.

For a finite prime set F, consider the bosonic current Fock space

$$\mathscr{F}_F^{\mathrm{cur}} = \Gamma_s\big(\ell^2(F \times \mathbb{N}_{\geq 1})\big).$$

Here Γ_s is the Hilbert direct sum of symmetric powers. Mode j corresponds to the normalized Heisenberg creation operator $a_{-j}^{(p)}/\sqrt{j}$. Splitting the one-particle space into modes $j = 1$ and $j \geq 2$ gives

$$\mathscr{F}_F^{\mathrm{cur}} \simeq L_F \widehat{\otimes} \mathscr{F}_F^{\geq 2}, \qquad L_F = \widehat{\bigotimes}_{p\in F} \mathscr{F}_p.$$

The orthonormal number-state bases prove this unitary directly, and it preserves the grading assigning degree j to a mode-j creation.

Let $\mathbb{N}_F$ be the integers supported on F. The phases and shifts μ_n, $n \in \mathbb{N}_F$, act on $L_F \simeq \ell^2(\mathbb{N}_F)$ and extend by the identity on $\mathscr{F}_F^{\geq 2}$. Every closed subspace $L_F \widehat{\otimes} K_0$, with $K_0 \subset \mathscr{F}_F^{\geq 2}$, is invariant under these operators and their adjoints. If F is nonempty, the higher-mode vacuum is properly contained in larger such stable sectors.

This restricted representation need not be faithful on all phases. If $q \notin F$, the nonzero projection

$$e_q = \frac{1}{q} \sum_{j=0}^{q-1} e(j/q)$$

acts as zero on L_F, since no integer in $\mathbb{N}_F$ is divisible by q. Also the higher-mode factor by itself is not the full vacuum Heisenberg vertex algebra: it omits every mode-one creation.

The Gibbs trace and its domain

Define the occupation Hamiltonian by

$$H\delta_m = (\log m)\delta_m, \qquad \operatorname{Dom} H = \left\{ \xi : \sum_m (\log m)^2 |\xi_m|^2 < \infty \right\}.$$

This is a self-adjoint diagonal operator with finite-support core. For a real $\beta > 1$,

$$\operatorname{Tr}(e^{-\beta H}) = \sum_{m \geq 1} m^{-\beta} = \prod_p (1 - p^{-\beta})^{-1} = \zeta(\beta). \tag{21.2}$$

The series is positive and convergent. Unique factorization proves the Euler product by increasing finite prime sets. At $\beta = 1$ the trace is the divergent harmonic series.

With the KMS convention $\varphi(ab) = \varphi(b\sigma_{i\beta}(a))$ for analytic elements, the Gibbs state is

$$\varphi_\beta(a) = \zeta(\beta)^{-1} \operatorname{Tr}(ae^{-\beta H}).$$

Indeed $e^{-\beta H} a = \sigma_{i\beta}(a) e^{-\beta H}$ on analytic normal terms, and trace cyclicity proves the identity. The full Bost–Connes classification gives a unique β-KMS state for $0 < \beta \leq 1$. For $\beta > 1$ its extremal states are indexed by embeddings of the cyclotomic field into $\mathbb{C}$. These are Bost–Connes, Theorems 5 and 25, with the critical case in Section 7. At the critical temperature, existence of a KMS state does not give a trace-class density in the occupation representation.

The spherical $GL_2(\mathbb{Q}_p)$ measure and the Jordan-quiver groupoid mass have their own constructions in Chapters 13 and 12. Relating those objects to the Bost–Connes system requires an operator or categorical comparison preserving their states and normalizations. Their prime indexing supplies no such map.

21.9 Gauge classes in Hochschild cohomology

We now use cochains with regular coefficients on A_{alg}. These compute a cohomology algebra, whereas the earlier cyclic chains compute a trace complex. All cochains in this first calculation are algebraic $\mathbb{C}$-multilinear maps.

Let

$$W = \operatorname{Hom}_{\mathbb{Z}}(\mathbb{Q}_{>0}^{\times}, (\mathbb{C}, +)) = \prod_p \mathbb{C}.$$

A family $\lambda = (\lambda_p)_p$ evaluates on a positive rational by $\lambda(g) = \sum_p v_p(g)\lambda_p$. Every such sum has finite support, even if the family does not.

Proposition 21.9.1 *(Algebraic gauge derivations).* The formula

$$D_\lambda(\mu_r e(\gamma)\mu_s^*) = \lambda(r/s)\mu_r e(\gamma)\mu_s^*$$

defines a derivation. The map $\lambda \mapsto [D_\lambda]$ injects W into $HH^1_{\mathrm{alg}}(A_{\mathrm{alg}}, A_{\mathrm{alg}})$. Every nonzero D_λ is unbounded in the inherited C^*-norm and has no bounded spatial implementer on $\mathcal{L}$. Moreover $HH^0_{\mathrm{alg}}(A_{\mathrm{alg}}, A_{\mathrm{alg}}) = \mathbb{C}$.

Proof. Gauge averaging against characters decomposes the normal algebra as

$$A_{\mathrm{alg}} = \bigoplus_{g \in \mathbb{Q}^\times_{>0}} (A_{\mathrm{alg}})_g, \qquad (A_{\mathrm{alg}})_g (A_{\mathrm{alg}})_h \subseteq (A_{\mathrm{alg}})_{gh}.$$

This uses homogeneous components, and does not assert independence of normal terms of the same degree. Multiplying the component of degree g by $\lambda(g)$ is consequently well defined. Since $\lambda(gh) = \lambda(g) + \lambda(h)$, it is a derivation.

One-cocycles are derivations and one-coboundaries are inner derivations. If $D_\lambda = [b, -]$ for a bounded operator b, the norm-one isometries μ_p^k give

$$k|\lambda_p| = \|D_\lambda(\mu_p^k)\| \le 2\|b\| \qquad (k \ge 1).$$

Thus $\lambda_p = 0$ for every prime. This also proves injectivity on classes by taking differences, and proves unboundedness for every nonzero family.

An operator commuting with all phases is diagonal: any two distinct integer indices are separated by some character $\exp(2\pi i m\gamma)$. Write it as $T\delta_m = t_m \delta_m$. Commutation with every shift requires $t_{nm} = t_m$, hence $t_n = t_1$. The represented commutant is scalar. Faithfulness gives the asserted center and degree-zero cohomology. □

The derivation preserves adjoint exactly when every λ_p is purely imaginary. This follows by comparing its values on μ_p and μ_p^*. The product $\prod_p \mathbb{C}$ parametrizes algebraic derivations. It is not a convergent sum of bounded operators.

Finite cycles detecting every exterior degree

There is a character $\varepsilon : A_{\mathrm{BC}} \to \mathbb{C}$ defined by

$$\varepsilon(\mu_n) = \varepsilon(\mu_n^*) = \varepsilon(e(\gamma)) = 1.$$

The presentation verifies every relation, including the transfer average of n ones. This is a norm-one unital star homomorphism. It is different from the vacuum vector state, which takes μ_p to zero.

Let $\mathbb{C}_\varepsilon$ be the bimodule with left and right actions through ε. Composition of a regular-coefficient cochain with ε is a cochain map. For clarity, the two differentials are

$$\begin{aligned}(bF)(a_1, \dots, a_{n+1}) &= a_1 F(a_2, \dots, a_{n+1}) \\ &\quad + \sum_{i=1}^{n} (-1)^i F(a_1, \dots, a_i a_{i+1}, \dots, a_{n+1}) \\ &\quad + (-1)^{n+1} F(a_1, \dots, a_n) a_{n+1},\end{aligned}$$

and

$$\begin{aligned} b\big(c[a_1|\cdots|a_n]\big) = {} & c\varepsilon(a_1)[a_2|\cdots|a_n] \\ & + \sum_{i=1}^{n-1}(-1)^i c[a_1|\cdots|a_i a_{i+1}|\cdots|a_n] \\ & + (-1)^n \varepsilon(a_n) c[a_1|\cdots|a_{n-1}]. \end{aligned}$$

Evaluation makes these formulas dual after changing cochain coefficients by ε.

For distinct primes $P = (p_1, \dots, p_n)$, write $s_j = \mu_{p_j}$ and set

$$c_P = \sum_{\pi \in S_n} \operatorname{sgn}(\pi) 1[s_{\pi(1)}|\cdots|s_{\pi(n)}].$$

This is a cycle with coefficients in $\mathbb{C}_\varepsilon$. For an interior face, exchange the adjacent multiplied factors: they commute and their permutation signs are opposite. For the outer faces, pair a word with its cyclic rotation. Rotation has sign $(-1)^{n-1}$ and the last face has sign $(-1)^n$, so those terms cancel as well. For $n = 1$ the boundary is directly zero.

Therefore

$$L_P(F) = \sum_{\pi \in S_n} \operatorname{sgn}(\pi) \varepsilon\big(F(s_{\pi(1)}, \dots, s_{\pi(n)})\big)$$

vanishes on every regular-coefficient coboundary.

THEOREM 21.9.2 *(An exterior algebra of gauge classes).* Cup product defines an injective graded-algebra map

$$\Lambda^*_{\mathbb{C}} W \hookrightarrow HH^*_{\mathrm{alg}}(A_{\mathrm{alg}}, A_{\mathrm{alg}}), \qquad \lambda \longmapsto [D_\lambda].$$

The exterior algebra consists of finite sums of finite wedges. It has no completion in this statement.

Proof. The cup of n derivations is the cochain

$$F_{\lambda_1,\dots,\lambda_n}(a_1, \dots, a_n) = D_{\lambda_1}(a_1) \cdots D_{\lambda_n}(a_n).$$

The cochain differential obeys the graded product rule, as follows by cancellation of its interior multiplication terms. Hence this cup is a cocycle. For two derivations D, E, expansion of the Leibniz rules gives

$$b(DE)(a, b) = -D(a)E(b) - E(a)D(b).$$

Their classes anticommute, and their squares vanish over $\mathbb{C}$. This constructs the exterior-algebra map.

The finite cycle evaluation is exactly

$$L_P(F_{\lambda_1,\dots,\lambda_n}) = \det\big(\lambda_i(p_j)\big)_{1 \le i,j \le n}.$$

Every exterior element belongs to $\Lambda^* U$ for some finite-dimensional subspace $U \subset W$. Prime-coordinate functionals separate points of U, so they span U^*. Otherwise their span would have a nonzero annihilator whose every prime coordinate vanished. A finite selection already spans U^*, and its exterior products detect every nonzero element of $\Lambda^n U$.

The corresponding L_P proves injectivity in degree n. The character detects degree zero. □

For two distinct primes, the cycle is $1[\mu_p|\mu_q] - 1[\mu_q|\mu_p]$. Its boundary is zero term by term after commuting the shifts, and $L_{(p,q)}(D_p \smile D_q) = 1$, where D_p is the coordinate derivation. For three primes the six-term alternating cycle gives $L_{(p,q,r)}(D_p \smile D_q \smile D_r) = 1$. Thus these lower bounds are nonzero in every degree. They are not a classification of all Hochschild classes.

21.10 Smooth algebras and the time generator

The topological domain changes the gauge calculation. For a finite prime set F, let B_F consist of the elements whose orbit under the finite torus $\mathbb{T}^F \subset K$ is norm-smooth. Angular derivatives are denoted δ_p, so $\delta_p(\mu_q) = i\delta_{pq}\mu_q$ and $D_p = \delta_p/i$. For a nonnegative multiindex α supported in F, put

$$q_\alpha(a) = \|\delta^\alpha a\|, \qquad q_0(a) = \|a\|.$$

Define

$$B_{\mathrm{g}} = \bigcap_{F \text{ finite}} B_F$$

with all these finite-support seminorms. This uses smoothness on ordinary finite-dimensional tori.

Proposition 21.10.1 *(Smooth gauge cohomology).* The spaces B_F and B_{g} are Fréchet star algebras, norm-dense in A_{BC}. For jointly continuous multilinear cochains with regular coefficients, there are injections

$$\Lambda^*\left(\bigoplus_{p\in F} \mathbb{C}D_p\right) \hookrightarrow HH^*_{\mathrm{cont}}(B_F, B_F), \qquad \Lambda^*\left(\bigoplus_{p} \mathbb{C}D_p\right) \hookrightarrow HH^*_{\mathrm{cont}}(B_{\mathrm{g}}, B_{\mathrm{g}}).$$

Proof. For B_F, a Cauchy sequence in every seminorm gives orbit maps that are Cauchy in every uniform derivative norm. Smooth Banach-valued functions on a compact torus are complete. The fundamental theorem of calculus identifies the uniform limit of first derivatives with the derivative of the uniform limit. Iteration handles every order. The limit orbit is $\alpha_z(a)$ for the norm limit a. Thus B_F is complete.

There are countably many finite-support multiindices. A Cauchy sequence for B_{g} converges in every B_F, and all limits agree with its norm limit. This proves completeness and metrizability. The iterated product rule

$$\delta^\alpha(ab) = \sum_{\beta\le\alpha} \binom{\alpha}{\beta} (\delta^\beta a)(\delta^{\alpha-\beta} b)$$

proves continuous multiplication. Real angular derivatives commute with adjoint. Normal sums lie in every smooth algebra and are norm-dense.

Each permitted D_p is continuous, with its seminorm bounded by the seminorm having one additional p derivative. For B_{g} this follows by differentiating on $F \cup \{p\}$ before restricting to F. Finite cup cochains are consequently continuous. The character is continuous because $|\varepsilon(a)| \le q_0(a)$. The finite cycles and determinant evaluations of Theorem 21.9.2 therefore detect these continuous cohomology classes as well. □

Proposition 21.10.2 *(The continuity criterion).* The algebraic D_λ extends to a continuous linear map $B_{\mathrm{g}} \to A_{\mathrm{BC}}$ exactly when λ has finite support. It then extends as a continuous derivation $B_{\mathrm{g}} \to B_{\mathrm{g}}$. On B_F the criterion is support contained in F.

Proof. Continuity would give

$$\|D_\lambda(a)\| \le C \max_{\alpha\in E} q_\alpha(a)$$

for some finite set E of multiindices, including zero. Let F_0 contain their supports. For $q \notin F_0$, every positive derivative on the right vanishes on μ_q^k, while its norm is one. Hence $k|\lambda_q| \le C$ for all k, forcing $\lambda_q = 0$. On B_F the same argument excludes every $q \notin F$. Conversely a finite linear combination of the permitted continuous derivatives has the required extension. □

This criterion concerns this particular family of derivations. It does not classify every continuous derivation of these algebras.

It also separates gauge smoothness from physical-time smoothness. Enumerate distinct primes p_j and form

$$a = \sum_{j \geq 1} 2^{-j} \mu_{p_j}^{2^j}.$$

The series converges in norm. Only finitely many terms move under any fixed finite prime torus, so $a \in B_{\mathrm{g}}$.

If its time derivative existed in norm, applying it to δ_1 would give coordinate $i \log p_j$ at each distinct basis vector $\delta_{p_j^{2^j}}$. Those coordinates are not square summable. Thus a is not even norm-differentiable under σ.

Closed differentiation in physical time

Let δ be the norm generator of σ:

$$\operatorname{Dom} \delta = \left\{ a : \lim_{t \to 0} \frac{\sigma_t(a) - a}{t} \text{ exists in norm} \right\}.$$

Every normal term is entire analytic, and

$$\delta(\mu_r e(\gamma) \mu_s^*) = i \log(r/s) \mu_r e(\gamma) \mu_s^*.$$

The domain is dense and proper. For a direct one-prime example of properness, use $\sum_{j \geq 1} 2^{-j} \mu_p^{2^j}$. The derivative on δ_1 would have the same nonzero coordinate $i \log p$ at infinitely many states.

The generator is closed. For a in its domain, differentiating the orbit gives the Bochner integral identity

$$\sigma_t(a) - a = \int_0^t \sigma_s(\delta a)\, ds.$$

If $a_n \to a$ and $\delta a_n \to b$ in norm, isometry permits passage to the limit. Division by t and norm continuity then give $\delta a = b$.

The norm-smooth orbit algebra is

$$A_\sigma^\infty = \bigcap_{j \geq 0} \operatorname{Dom} \delta^j, \qquad p_k(a) = \sum_{j=0}^{k} \|\delta^j a\|.$$

Repeated differentiation identifies this intersection with smoothness of the orbit. Closedness of δ, applied successively to a Cauchy sequence of derivatives, proves completeness. The iterated Leibniz rule proves continuous multiplication, and adjoint is continuous. Thus it is a dense Fréchet star algebra. The estimate $p_k(\delta a) \leq p_{k+1}(a)$ makes δ continuous on it.

The class of δ in $HH^1_{\mathrm{cont}}(A_\sigma^\infty, A_\sigma^\infty)$ is nonzero. An inner implementer would be a bounded element of A_{BC}, contrary to the prime-power estimate in Proposition 21.9.1. Here the algebraic family is $\lambda_p = i \log p$. Its infinite support also explains why it does not extend continuously from the gauge-smooth topology.

In the occupation representation it has an unbounded spatial implementer. Normal terms preserve $\operatorname{Dom} H$, since their nonzero action changes the logarithmic weight by the constant $\log(r/s)$. On that domain,

$$[H, \mu_r e(\gamma) \mu_s^*] = \log(r/s) \mu_r e(\gamma) \mu_s^*, \qquad \delta = i[H, -].$$

Truncation in the weighted norm proves the finite-support core assertion for H and justifies passage from basis vectors to its maximal domain. The scalar commutant computed above and the bicommutant theorem give $A_{\mathrm{BC}}'' = B(\mathcal{L})$ in this representation. A GNS representation from a specified KMS state is another carrier and need not have this weak closure.

Logarithmic dilation and an actual circle quotient

Logarithmic coordinates give a useful comparison of continuous actions. The map

$$Uf(t) = f(e^t) : L^2(\mathbb{R}_{>0}, dr/r) \xrightarrow{\sim} L^2(\mathbb{R}, dt)$$

is unitary. For $D_a f(r) = f(e^{-a} r)$, substitution gives $U D_a U^{-1} F(t) = F(t - a)$.

Fix $L > 0$. The quotient group $\mathbb{R}_{>0}/e^{L\mathbb{Z}}$ instead becomes $\mathbb{R}/L\mathbb{Z}$. With measure dt on $[0, L)$, its orthonormal Fourier basis is

$$L^{-1/2} e^{2\pi i n t/L}, \qquad n \in \mathbb{Z}.$$

Inverse translation has eigenvalue $e^{-2\pi i n a/L}$ on this basis. The logarithm gives the quotient isomorphism, and Fourier orthogonality proves the normalization.

This circle Hilbert space is not a nonzero periodic subspace of $L^2(\mathbb{R}, dt)$. The squared norm of a periodic function is the sum of its identical integrals over disjoint periods. It is finite only when the function is zero almost everywhere.

A period L for Bost–Connes time would require $L \log n \in 2\pi\mathbb{Z}$ for every surviving shift. It cannot hold for both $n = 2$ and $n = 3$. Otherwise $\log 2/\log 3$ would be rational, implying $2^a = 3^b$ for positive integers a, b, contrary to unique factorization. A circle KMS comparison therefore needs its actual quotient or representation, periodic dynamics, state and intertwiner. The logarithmic coordinate alone does not construct these data.

Question 21.10.3 *(The arithmetic Hochschild comparison).* Specify an arithmetic category or algebra, its coefficient ring, trace bimodule, grading, topology and completion. Construct its comparison with the Bost–Connes normal algebra or one of the stated smooth algebras. Determine which gauge classes survive and whether the arithmetic theory has an integral degree-one carrier generated by logarithmic prime data. An integral lattice, a degree bound, or a map to a Witt cotangent complex requires its own construction.

For a Hilbert realization, fix a pre-Hilbert space L_{alg} and an arithmetic trace complex $(C_\bullet, b)$. Give a cycle map $J : L_{\mathrm{alg}} \to Z_d(C_\bullet)$ and a positive semidefinite form q on $H_d(C_\bullet)$. If

$$q([Jv], [Jw]) = \langle v, w \rangle,$$

then $v \mapsto [Jv]$ induces an isometry from the completion of L_{alg} into the completion of $H_d(C_\bullet)/\ker q$. Density is needed for surjectivity onto a proposed sector. The action on homology and its intertwining with J are additional data. The occupation representation by itself does not supply them.

21.11 Explicit positive local pairings

A bilinear evaluation becomes a Hermitian form only after specifying an antilinear map in one slot. We use inner products linear in the first variable throughout this section. The following models calculate the map, its normalization, and its domain.

The Segal form and an archimedean line

Let $\mathscr{F}_{\mathrm{alg}}$ be the Heisenberg vacuum module generated by α_{-n}, $n \geq 1$, with

$$[\alpha_m, \alpha_n] = m\delta_{m+n,0}, \qquad \alpha_n\Omega = 0 \quad (n > 0).$$

Normalize $H_\infty(\Omega, \Omega) = 1$ and impose $\alpha_{-n}^\dagger = \alpha_n$. For a finite occupation array $\mathbf{r} = (r_1, r_2, \ldots)$, put $v_{\mathbf{r}} = \prod_n \alpha_{-n}^{r_n}\Omega$. Commuting each annihilation operator to the right gives

$$H_\infty(v_{\mathbf{r}}, v_{\mathbf{s}}) = \delta_{\mathbf{r},\mathbf{s}} \prod_n n^{r_n} r_n!.$$

The factorial counts repeated contractions of the same mode. The factor n comes from its commutator. Equivalently, for two monomials of lengths k, l,

$$H_\infty(\alpha_{-n_1}\cdots\alpha_{-n_k}\Omega, \alpha_{-m_1}\cdots\alpha_{-m_l}\Omega) = \delta_{kl} \sum_{\pi\in S_k} \prod_{i=1}^{k} n_i \delta_{n_i, m_{\pi(i)}}.$$

This defines a positive form and its Segal Hilbert completion.

Let J_∞ conjugate coefficients in the monomial basis. The Gram matrix is real, so J_∞ is an isometric antilinear involution. The bilinear form

$$B_\infty(v, w) = H_\infty(v, J_\infty w)$$

is bounded by $\|v\|\|w\|$, extends to the completion, and satisfies $B_\infty(v, J_\infty w) = H_\infty(v, w)$. The complex-linear identity cannot replace J_∞: on the vacuum, $B_\infty(\Omega, i\Omega) = i$ while $H_\infty(\Omega, i\Omega) = -i$.

There is also an explicit archimedean comparison of lines. On $\mathbb{R}^\times$ use measure $da/|a|$ and set

$$W_0(a) = |a|^{1/2} K_0(2\pi|a|), \qquad K_0(x) = \int_0^\infty e^{-x\cosh u}\, du \quad (x > 0).$$

This is the parameter-zero spherical Whittaker function in the chosen normalization. Its squared norm is $\pi/4$. To see the constant, Tonelli gives

$$\int_0^\infty K_0(x)^2\, dx = \int_0^\infty \int_0^\infty \frac{du\, dv}{\cosh u + \cosh v}.$$

Put $a = (u + v)/2$ and $b = (u - v)/2$. The Jacobian is two, the region is $a \geq 0$, $|b| \leq a$, and the denominator is $2\cosh a \cosh b$. With $F(a) = \int_0^a \operatorname{sech} b\, db$, the integral is

$$2\int_0^\infty F'(a)F(a)\, da = F(\infty)^2 = \frac{\pi^2}{4}.$$

Scaling $x = 2\pi|a|$ and including both signs gives $\|W_0\|^2 = \pi/4$.

Hence

$$\mathbb{C}\Omega \longrightarrow \mathbb{C}W_0, \qquad z\Omega \longmapsto \frac{2z}{\sqrt{\pi}} W_0$$

is an isometry. Extending it to a full Whittaker or arithmetic representation requires an intertwiner for the specified actions. A line isometry does not determine one.

Weighted occupations and two isometries

Fix a prime p, and define

$$H_{\mathrm{occ},p} = \ell^2(\mathbb{N}_0, p^{-j}), \qquad H_{\mathrm{occ}}(v,w) = \sum_{j\ge 0} p^{-j} v_j \overline{w_j}.$$

Its coordinate vector δ_j has squared norm p^{-j}. The raw shift $R\delta_j = \delta_{j+1}$ has norm $p^{-1/2}$, and $\sqrt{p}\,R$ is an isometry. Coordinate conjugation J_{occ} gives the continuous bilinear form

$$B_{\mathrm{occ}}(v,w) = \sum_{j\ge 0} p^{-j} v_j w_j, \qquad B_{\mathrm{occ}}(v, J_{\mathrm{occ}} w) = H_{\mathrm{occ}}(v,w).$$

Cauchy–Schwarz proves absolute convergence and continuity.

Let E_{ij} be the matrix units on $\ell^2(\mathbb{N}_0)$. Choose a multiplicative Haar measure on $\mathbb{Q}_p^\times$ and put $m_p = \mathrm{vol}(\mathbb{Z}_p^\times) > 0$. There are isometries

$$D\delta_j = p^{-j/2} E_{jj}, \qquad T\delta_j = p^{-j/2} m_p^{-1/2} \mathbf{1}_{p^j \mathbb{Z}_p^\times}. \tag{21.3}$$

Their ranges are the diagonal Hilbert–Schmidt subspace and the closed radial-function subspace supported on nonnegative valuations, respectively.

Indeed matrix units are Hilbert–Schmidt orthonormal. The disjoint shells $p^j \mathbb{Z}_p^\times$ each have multiplicative measure m_p. Both displayed families therefore have Gram matrix $p^{-j}\delta_{jk}$. Finite spans are dense in the stated ranges, proving the isometric extensions.

Their basis vectors are fixed by adjunction or function conjugation. Thus

$$H_{\mathrm{occ}}(v,w) = \mathrm{Tr}(Dv(Dw)^*) = \int_{\mathbb{Q}_p^\times} Tv(a)\overline{Tw(a)}\, d^\times a.$$

Also $B_{\mathrm{occ}}(v,w) = \mathrm{Tr}(DvDw)$.

The diagonal Hilbert–Schmidt range is not invariant under left multiplication by the shift. For example, $VE_{jj} = E_{j+1,j}$. Consequently this isometry does not identify the whole prime GNS representation. Nor can its dual basis be realized as $\mu^*_{p^j}\delta_1$ in the occupation representation: those vectors vanish for $j > 0$.

The spherical norm depends on the representation

Fix a normalized unramified tempered representation π_p of $GL_2(\mathbb{Q}_p)$, with Satake parameters $|\alpha| = |\beta| = 1$ and an additive character of conductor $\mathbb{Z}_p$. Its spherical Whittaker function normalized by $W(1) = 1$ has shell values

$$W(\mathrm{diag}(p^j, 1)) = p^{-j/2} h_j(\alpha,\beta), \qquad h_j(\alpha,\beta) = \sum_{r=0}^{j} \alpha^{j-r}\beta^r \quad (j \ge 0),$$

and vanishes at negative valuations. We use this standard spherical Whittaker formula with the stated normalization.

Proposition 21.11.1 *(The full spherical norm).* For $\mathrm{Re}\, s > 0$,

$$Z_p(s) = \int_{\mathbb{Q}_p^\times} |W(a)|^2 |a|^{s-1}\, d^\times a = m_p \frac{L_p(s, \pi_p \times \widetilde{\pi}_p)}{\zeta_p(2s)}.$$

In particular,

$$C_p = \|W\|^2 = m_p(1 + p^{-1}) L_p(1, \mathrm{Ad}\,\pi_p) > 0.$$

Proof. Shell integration gives $Z_p(s) = m_p \sum_{j\ge 0} |h_j|^2 p^{-js}$. The bound $|h_j| \le j+1$ proves absolute convergence on the indicated half-plane. Put $r = \alpha/\beta$. For $|x| < 1$, geometric summation gives

$$\sum_{j\ge 0} |h_j|^2 x^j = \frac{1+x}{(1-x)(1-rx)(1-r^{-1}x)}.$$

For $r \ne 1$, expand $|h_j|^2 = (2 - r^{j+1} - r^{-j-1})/|1-r|^2$ and sum its three geometric series. For $r = 1$, use $\sum_{j\ge 0}(j+1)^2 x^j = (1+x)/(1-x)^3$.

The Rankin–Selberg denominator is $(1-x)^2(1-rx)(1-r^{-1}x)$. Multiplication by $1-x^2$ proves the formula with $x = p^{-s}$. Since

$$L_p(s, \pi_p \times \widetilde{\pi}_p) = \zeta_p(s) L_p(s, \operatorname{Ad} \pi_p),$$

evaluation at one gives C_p. It is a positive norm, and its denominator factors are nonzero because $p^{-1} < 1$. □

For $d^\times a = da/|a|$ with additive unit volume, $m_p = 1 - p^{-1}$ and $C_p = (1-p^{-2}) L_p(1, \operatorname{Ad}\pi_p)$. In contrast, the compact subgroup volume for the invariant matrix measure $|\det g|^{-2} \prod dg_{ij}$ is

$$\operatorname{vol}(GL_2(\mathbb{Z}_p)) = \frac{(p^2-1)(p^2-p)}{p^4} = (1-p^{-1})(1-p^{-2}).$$

Reduction modulo p proves this formula by counting the two independent columns. It is not the Whittaker norm, which depends on α/β. Inserting an additional $|a|^{-1}$ into the norm integral changes the question to $Z_p(0)$. For $\alpha = \beta = 1$, the resulting shell series is $m_p \sum_{j\ge 0}(j+1)^2$ and diverges.

The radial isometry of (21.3) identifies

$$W = T\left(\sqrt{m_p} \sum_{j\ge 0} h_j(\alpha,\beta)\, \delta_j\right).$$

This vector converges in $H_{\mathrm{occ},p}$ and has squared norm C_p. It is generally different from $T\delta_0$. Unit normalization uses $W/\sqrt{C_p}$. Dividing a pairing by the adjoint factor alone gives spherical squared norm $m_p(1+p^{-1})$, rather than one.

21.12 The prime Gibbs state and its modular conjugation

The full prime cyclic space can also be constructed explicitly. Let $K_0 = \ell^2(\mathbb{N}_0)$, $V e_j = e_{j+1}$, and $H_p e_j = j \log p\, e_j$, with maximal domain $\{\xi : \sum j^2 |\xi_j|^2 < \infty\}$. For $\beta > 0$, put

$$q = p^{-\beta}, \qquad r_j = (1-q)q^j, \qquad \rho e_j = r_j e_j.$$

The positive injective operator ρ has trace one. Its state on the Toeplitz algebra $\mathcal{T} = C^*(V)$ is

$$\varphi_{p,\beta}(a) = \operatorname{Tr}(\rho a), \qquad \varphi_{p,\beta}(V^m V^{*n}) = \delta_{mn} q^m.$$

The physical time is $\alpha_t(a) = e^{itH_p} a e^{-itH_p}$, with $\alpha_t(V) = p^{it} V$.

This is the unique β-KMS state on $(\mathcal{T}, \alpha)$. On analytic monomials, $\rho a = \alpha_{i\beta}(a)\rho$, so the KMS identity follows by trace cyclicity. Conversely that identity with second entry one kills each monomial of nonzero degree. With first entry V^m and second entry V^{*m}, it gives the value q^m. Density of the analytic monomials proves uniqueness.

Proposition 21.12.1 *(The Hilbert–Schmidt GNS model).* The GNS space and representation are

$$\mathcal{H}_{p,\beta} = \mathrm{HS}(K_0), \qquad \pi(a)X = aX, \qquad \Omega = \rho^{1/2},$$

with inner product $\langle X, Y\rangle = \mathrm{Tr}(XY^*)$. The GNS map is $[a] \mapsto a\rho^{1/2}$. The vector Ω is cyclic and separating for left $B(K_0) = \pi(\mathcal{T})''$.

Proof. The map is isometric because its inner product is $\mathrm{Tr}(\rho b^* a) = \varphi_{p,\beta}(b^* a)$. The matrix units $E_{ij} = V^i(1 - VV^*)V^{*j}$ belong to $\mathcal{T}$, and $E_{ij}\rho^{1/2} = \sqrt{r_j}E_{ij}$. They span a dense subspace of $\mathrm{HS}(K_0)$. Finite matrix corners strongly approximate bounded operators, proving the weak-closure assertion. If $a\Omega = 0$, every column of a vanishes because $r_j > 0$ for all j. Thus Ω is separating. □

Define the antilinear operator $S_0(a\Omega) = a^*\Omega$ on $\mathcal{T}\Omega$.

Proposition 21.12.2 *(Closed modular operators and their domains).* Set $JX = X^*$. For $X = \sum x_{ij}E_{ij}$, the positive modular operator satisfies

$$\Delta E_{ij} = q^{i-j}E_{ij}, \qquad \mathrm{Dom}\,\Delta^a = \left\{X : \sum_{i,j} q^{2a(i-j)}|x_{ij}|^2 < \infty\right\} \quad (a \in \mathbb{R}).$$

One has $\overline{S_0} = J\Delta^{1/2}$ and

$$\Delta^{it}X = e^{-i\beta t H_p} X e^{i\beta t H_p}.$$

The modular action on left multiplication is physical time $\alpha_{-\beta t}$. Both Δ and Δ^{-1} are unbounded.

Proof. On matrix units, $S_0 E_{ij} = q^{(i-j)/2}E_{ji}$ antilinearly. Truncation of a convergent weighted double sum makes finite matrix sums graph-dense in the displayed domain for $a = 1/2$. For bounded a, the vector $a\Omega$ belongs to that domain and the formula gives $a^*\Omega$. It follows that the closure is precisely $J\Delta^{1/2}$. Diagonal spectral calculus proves all powers and the unitary group. Its eigenvalues range through $q^{\mathbb{Z}}$, proving both unboundedness assertions. □

The logarithm has eigenvalue $(i - j)\log q$ on E_{ij} and maximal domain $\{X : \sum |(i - j)\log q|^2|x_{ij}|^2 < \infty\}$. The conjugation acts on prime-shift vectors by

$$J(V^k\Omega) = p^{\beta k/2}V^{*k}\Omega, \qquad J(V^{*k}\Omega) = p^{-\beta k/2}V^k\Omega.$$

Expand $V^k\Omega = \sum_j \sqrt{r_j}E_{j+k,j}$ and use $r_{j+k} = q^k r_j$ to verify the factors. In particular $\|V^k\Omega\| = 1$ and $\|V^{*k}\Omega\| = q^{k/2}$. The algebra star operation alone is not this antiunitary on the cyclic space.

For left and right multiplication, $JL_aJ = R_{a^*}$. The continuous bilinear form $B(X, Z) = \mathrm{Tr}(XZ)$ satisfies

$$B(X, JY) = \mathrm{Tr}(XY^*) = \langle X, Y\rangle.$$

The product of two Hilbert–Schmidt operators is trace class and $|\mathrm{Tr}(XZ)| \le \|X\|_2\|Z\|_2$, which proves the asserted continuity.

Proposition 21.12.3 *(Comparison with a specified prime cyclic space).* Let $(\mathcal{A}, \alpha, \psi_\beta)$ be a dynamical C^*-algebra with a β-KMS state and an isometry μ_p satisfying $\alpha_t(\mu_p) = p^{it}\mu_p$. The map $\iota_p : \mathcal{T} \to \mathcal{A}$ sending V to μ_p induces a unitary

$$U_p : \mathrm{HS}(K_0) \longrightarrow \overline{\pi_{\psi_\beta}(C^*(\mu_p))\Omega_{\psi_\beta}}, \qquad a\rho^{1/2} \longmapsto \pi_{\psi_\beta}(\iota_p(a))\Omega_{\psi_\beta}.$$

It preserves the prime algebra action, cyclic vector, and closed modular operators of this restricted cyclic representation, including when $\beta = 1$.

Proof. The pulled-back state is a KMS state on $\mathcal{T}$, so uniqueness identifies it with $\varphi_{p,\beta}$. The displayed map is therefore isometric on $\mathcal{T}\rho^{1/2}$ and has dense image in the specified cyclic subspace. It extends to a unitary. It intertwines multiplication and $a\Omega \mapsto a^*\Omega$ on the core, hence their closures and polar decompositions. □

The target is the prime-generated cyclic subspace, not the whole global GNS space. Its identification says nothing yet about additional phase operators or an arithmetic automorphic placement. For finitely many primes, tensoring the densities and matrix formulas gives their finite Gibbs and GNS products for every $\beta > 0$. The global occupation heat trace still diverges at $\beta = 1$, as (21.2) shows.

21.13 Restricted tensor pairings and arithmetic placement

Choose complex Hilbert spaces (H_v, e_v) indexed by the places, with unit reference vectors. Suppose each has a conjugation J_v, with $J_v e_v = e_v$, and a bilinear form $B_v(x, J_v y) = H_v(x, y)$. The algebraic restricted tensor consists of elementary tensors equal to e_v outside a finite set.

PROPOSITION 21.13.1 *(Assembly of normalized forms).* There are forms

$$H(\otimes'_v x_v, \otimes'_v y_v) = \prod_v H_v(x_v, y_v), \qquad B(\otimes'_v x_v, \otimes'_v y_v) = \prod_v B_v(x_v, y_v),$$

and a conjugation $J = \otimes'_v J_v$. They extend to the Hilbert restricted tensor product, where $H(x, y) = B(x, Jy)$. Its reference vector has squared norm one.

Proof. At each finite set of places, use the Hilbert tensor form. Enlarging the set tensors with unit reference vectors, so its insertion maps are isometries. The directed union is a pre-Hilbert space. The local conjugations give compatible antilinear isometries, whose square is the identity. They extend to a conjugation on its completion.

Define $B(x, y) = H(x, Jy)$ there. It is bounded and agrees with the displayed finite products. All omitted reference factors equal one. □

Raw local norms cannot be multiplied without addressing convergence. For the invariant matrix measure on $GL_n(\mathbb{Q}_p)$,

$$\operatorname{vol}(GL_n(\mathbb{Z}_p)) = \prod_{j=1}^{n} (1 - p^{-j}).$$

Counting independent columns modulo p proves this identity. The product of its $j = 1$ factors tends to zero. The factors with $j \geq 2$ multiply to $\prod_{j=2}^{n} \zeta(j)^{-1}$.

A Tamagawa measure inserts its convergence factors, global residue normalization, and rational-character quotient. It is not the raw product of these volumes. Nor does the volume of an automorphic quotient determine the Petersson norm of a specified cusp form.

For example, with the Tamagawa measure disintegrated along $|\det|$ using dt/t, the volume of

$$GL_2(\mathbb{Q})\backslash GL_2(\mathbb{A})^1, \qquad GL_2(\mathbb{A})^1 = \ker |\det|_{\mathbb{A}},$$

is one. This is the classical Tamagawa number in its rational-character normalization, as in Lai, Section 1.7 and Theorem 7.1.

In that formula the split diagonal torus has Tamagawa number one. Restricting its character (a, b) to the scalar torus and the derived diagonal torus gives $(a + b, a - b)$, a sublattice of index two. The central characters trivial on their intersection μ_2 are also a sublattice of index two. The two index factors cancel, giving $\tau(GL_2) = \tau(\mathbb{G}_m^2) = 1$.

This global volume calculation does not normalize the vector W or an individual automorphic form.

The local spherical factors are likewise representation dependent. If a specified global adjoint L-function continues holomorphically to one with $c_\pi = L(1, \operatorname{Ad}\pi) \neq 0$, then $c_\pi^{-1}B$ is a well-defined rescaled pairing. Rescaling both sides of a pairing identity by that same scalar preserves the identity. It does not prove ordinary convergence of the local product at one.

For tempered unramified parameters and $\sigma > 1$, the adjoint logarithmic series has the bound

$$\sum_p \sum_{j \geq 1} \frac{3p^{-j\sigma}}{j} \leq \frac{3}{1 - 2^{-\sigma}} \sum_p p^{-\sigma} < \infty.$$

This gives an initial half-plane of absolute convergence. Continuation and nonvanishing at one are additional analytic inputs for the representation in question. Finitely many ramified factors require their own normalization and pole conditions.

Hypothesis 21.13.2 *(Arithmetic pairing comparison).* Fix an arithmetic state space V, a dual coefficient space $V^!$, a bilinear evaluation $B^{\mathrm{Ar}} : V \otimes V^! \to \mathbb{C}$, and an antilinear map $\sigma : V \to V^!$. Specify an injective placement ι of a common dense domain into an automorphic Hilbert space with fixed central character and fixed Haar normalization. At each place require maps on both pairing carriers, preserving the stated actions. They must carry the evaluation to a chosen local model and σ to its conjugation.

Require the maps to preserve the reference vectors at almost all places and to identify the placed local Petersson form with the model Hermitian form. Specify the resulting adelic factorization with a common global scalar. Finally require continuous extension on the stated completion, or a specified regularized extension compatible with those maps. Every included ramified, residual, or continuous sector requires its own local maps and normalization.

Proposition 21.13.3 *(The conditional Koszul–Petersson identity).* Under Hypothesis 21.13.2, on its specified domain,

$$\langle \iota v, \iota w \rangle_{\mathrm{Pet}} = B^{\mathrm{Ar}}(v, \sigma w).$$

If the placement is into an actual L^2 space, this transported Hermitian form is positive definite.

Proof. For elementary restricted tensors, the local model identity $H_v(x, y) = B_v(x, J_v y)$ gives equality at each place. The reference normalization and common global scalar make their finite products equal. The stipulated extension carries the equality to its stated domain. If $v \neq 0$, injectivity gives $\|\iota v\|^2 > 0$, proving positivity. □

The holomorphic sector supplies concrete local examples. Let $f = \sum_{n \geq 1} a_n q^n$ be a normalized holomorphic cuspidal newform of weight $k \geq 2$, level N and nebentypus χ. At $p \nmid N$, write

$$X^2 - a_p X + \chi(p) p^{k-1} = (X - \alpha_p^{\mathrm{alg}})(X - \beta_p^{\mathrm{alg}}).$$

Deligne's Theorem 8.2 in *La conjecture de Weil I* gives $|\alpha_p^{\mathrm{alg}}| = |\beta_p^{\mathrm{alg}}| = p^{(k-1)/2}$. The unitary parameters $\alpha_p = p^{-(k-1)/2}\alpha_p^{\mathrm{alg}}$ and $\beta_p = p^{-(k-1)/2}\beta_p^{\mathrm{alg}}$ therefore satisfy the hypotheses of the spherical norm calculation.

For a cuspidal weight-one eigenform, Deligne–Serre's Theorem 4.1 constructs the associated continuous complex Galois representation. Its finite-image Frobenius eigenvalues are roots of unity, giving the same unitary conclusion at unramified primes. This is their Corollary 4.2. The archimedean weight-one representation is a limit of discrete series. It is not a discrete-series representation of weight at least two.

These theorems identify the unramified local parameters. They do not supply the arithmetic placement, ramified pairing maps or adelic extension in Hypothesis 21.13.2. Once those data are given on a selected holomorphic sector, the preceding proposition proves its pairing identity. The same implication applies to a selected Maass sector with its own local models and maps.

The positivity implication itself needs no temperedness assumption. Temperedness enters the separate local spherical model above. Using that model for a Maass sector requires the relevant local bounds as well as the arithmetic maps. A local exponent estimate is neither such a map nor an adelic summability statement. For example $\sum_p p^{-25/64}$ diverges, so this power cannot be an absolutely summable prime error bound.

The hypothesis preserves the original comparison problem even on a normalized cuspidal core. An adjunction alone does not identify its counit target with $\mathbb{C}$, its dual carrier with Bost–Connes operators, or its conjugation with an automorphic one. If a placement U intertwines the model conjugation with function conjugation on a dense domain, continuity extends that equality to the completion. In a fixed central-character space, function conjugation naturally lands in the conjugate-character space, which must be included in the specified target.

Positive forms also do not by themselves make an action unitary. One must prove the adjoint identities for its operators on invariant domains and establish any closures used by the representation. The bounded local isometries above provide the pairings and lines they state. The arithmetic intertwiners remain to be constructed.

PROPOSITION 21.13.4 *(Two dense realizations of one pairing).* Let E be a complex vector space with a positive semidefinite Hermitian form q. Suppose linear maps $I : E \to H_1$ and $J : E \to H_2$ have dense images in Hilbert spaces and satisfy

$$q(v, w) = \langle Iv, Iw\rangle_1 = \langle Jv, Jw\rangle_2.$$

Then $Iv \mapsto Jv$ extends uniquely to a unitary $H_1 \to H_2$.

Proof. Both kernels are $\{v : q(v, v) = 0\}$. The proposed map is consequently well defined and isometric on the dense image of I. Extend by continuity. An isometry from a complete space has closed range, and this range contains the dense image of J. It is therefore all of H_2. Density proves uniqueness. □

For a Petersson–chiral comparison, one must construct E, B, σ, I, J with $q(v, w) = B(v, \sigma w)$ on the stated automorphic and chiral carriers. An Arnold or Knizhnik–Zamolodchikov connection further requires bundles over an identified base and a map preserving their connections. A Hilbert-space isometry does not by itself compare those connections.

Meromorphic comparison

Suppose two scalar pairings of specified meromorphic families of vectors extend to meromorphic functions on a connected parameter domain. If the actual comparison identifies them on a nonempty open subset, they agree everywhere meromorphically. Locally clear their poles and apply the identity theorem. Their Laurent coefficients, pole orders and residues then agree.

This implication requires both families and the comparison. The local spherical function $Z_p(s) = m_p L_p(s, \pi_p \times \widetilde{\pi}_p)/\zeta_p(2s)$ uses a rank-four Rankin–Selberg factor. It is not the rank-two standard factor $(1-\alpha p^{-s})^{-1}(1-\beta p^{-s})^{-1}$. It does not imply a universal assertion that every arithmetic pairing has only two Tate boundary poles. A meromorphic line bundle alone has no pole divisor: that divisor belongs to a specified meromorphic section in its local trivializations.

21.14 Instanton sums, Fock traces, and their comparison

An equivariant instanton integral and a trace on symmetric powers weight partitions differently. Their distinction is visible before taking an arithmetic product.

Let $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots)$ be a partition, with $\lambda_i = 0$ beyond its last row. For a box $s = (i, j)$, define

$$A_\lambda(s) = \lambda_i - j, \qquad L_\lambda(s) = \lambda'_j - i.$$

For a box outside the diagram these integers can be negative. In particular the leg of a box in one partition, measured against a different partition, is not necessarily nonnegative.

For a rank r framing, let $\boldsymbol{\lambda} = (\lambda^{(1)}, \ldots, \lambda^{(r)})$, $|\boldsymbol{\lambda}| = \sum_\alpha |\lambda^{(\alpha)}|$, and choose equivariant parameters $a_1, \ldots, a_r, \epsilon_1, \epsilon_2$. Write $\epsilon_+ = \epsilon_1 + \epsilon_2$ and

$$E_{\alpha\beta}(s) = a_\alpha - a_\beta - \epsilon_1 L_{\lambda^{(\beta)}}(s) + \epsilon_2 (A_{\lambda^{(\alpha)}}(s) + 1).$$

The pure vector-multiplet fixed-point sum is

$$Z_{\mathrm{inst}} = \sum_{\boldsymbol{\lambda}} q^{|\boldsymbol{\lambda}|} \prod_{\alpha,\beta=1}^{r} \prod_{s \in \lambda^{(\alpha)}} \frac{1}{E_{\alpha\beta}(s)(\epsilon_+ - E_{\alpha\beta}(s))}. \tag{21.4}$$

This is a formal series in the instanton-counting parameter q, with rational equivariant coefficients. Specialization requires its denominators to be defined. In the pure-gauge scale convention of Nekrasov–Okounkov, $q = \Lambda^{2r}$. Keeping q explicit avoids confusing that convention with a change in rank. Matter multiplets contribute additional fixed-point factors.

The localization construction uses framed sheaves on $\mathbb{C}^2$, or its projective compactification with a framing at infinity. This is the equivariant instanton sum of Nekrasov and Okounkov. For the signed arm and leg convention, see Alday, Gaiotto and Tachikawa, Appendix B.1.

Proposition 21.14.1 *(The fixed-parameter large-equivariant limit).* For $r = 1$, the one-box coefficient is $1/(\epsilon_1\epsilon_2)$. If $\epsilon_1 \neq 0$ and q are fixed while $\epsilon_2 \to \infty$, the series (21.4) tends coefficientwise in q to 1.

Proof. For the single box, $A = L = 0$ and $E = \epsilon_2$, $\epsilon_+ - E = \epsilon_1$. For a general box in a single partition,

$$E = -\epsilon_1 L + \epsilon_2 (A + 1), \qquad \epsilon_+ - E = \epsilon_1 (L + 1) - \epsilon_2 A.$$

Here $A, L \geq 0$. The first factor has nonzero leading coefficient in ϵ_2. If $A > 0$, the second also grows linearly. If $A = 0$, it is the nonzero constant $\epsilon_1(L+1)$.

Every nonempty partition therefore contributes a coefficient tending to zero. There are only finitely many partitions of each fixed size, so their sum tends to zero as well. □

Thus retaining a term $1 + q$ requires a different scaling or normalization. Such a scaling must be specified and tested on all charges. Even an actual two-term local series would not be the bosonic Euler factor $(1-q)^{-1}$.

The hook-weighted specialization

On the slice $\epsilon_1 = -\epsilon_2 = \hbar \neq 0$, the rank-one denominator of each box is $-\hbar^2 h_\lambda(s)^2$, where $h_\lambda(s) = A_\lambda(s) + L_\lambda(s) + 1$. After setting $x = q/(-\hbar^2)$, the normalized sum is

$$Z_{\mathrm{hook}}(x) = \sum_\lambda \frac{x^{|\lambda|}}{\prod_{s\in\lambda} h_\lambda(s)^2}.$$

The hook-length formula for the irreducible S_n-representation indexed by λ says $f^\lambda = n!/\prod_s h_\lambda(s)$. Decomposing the regular representation of S_n gives $\sum_{\lambda\vdash n}(f^\lambda)^2 = n!$. Consequently

$$\sum_{\lambda\vdash n}\prod_{s\in\lambda} h_\lambda(s)^{-2} = \frac{1}{n!}, \qquad Z_{\mathrm{hook}}(x) = e^x.$$

This is the Plancherel specialization in Nekrasov–Okounkov, Section 3.2. For $n = 2$, the two diagrams each contribute $1/4$, so the coefficient is $1/2$, already different from the coefficient one of $(1-x)^{-1}$.

Substituting $x = p^{-s}$ and multiplying over primes gives $\exp(\sum_p p^{-s})$ when $\operatorname{Re} s > 1$. It retains the first prime-power term of the logarithm of the Euler product. The higher prime powers are different coefficients.

The all-charge Fock trace

Let V be finite-dimensional over $\mathbb{C}$, and let $T \in \operatorname{End}(V)$. Then

$$\sum_{n\geq 0} q^n \operatorname{Tr}(\operatorname{Sym}^n T) = \det(1-qT)^{-1}$$

as a formal series. Triangularize T. On monomials in a triangular basis, the diagonal entries are products of its eigenvalues. Summing their nonnegative multiplicities gives the product of geometric series, proving the identity without a semisimplicity assumption. For the Hilbert direct sum of symmetric powers, $|q|\|T\| < 1$ is a sufficient trace-class condition.

For one oscillator with $T = 1$, every charge occurs once, so its trace is $(1-q)^{-1}$. At $q = p^{-s}$ this is the primon local factor. For a specified unramified Frobenius operator F_p on a rank-r space, the corresponding local trace is $\det(1-p^{-s}F_p)^{-1}$. This rank is the dimension of that coefficient space. It is not inferred from the number of primes in an indexing set.

For $\operatorname{Re} s > 1$, choose the logarithm of $\zeta(s)$ that tends to zero as $\operatorname{Re} s \to +\infty$. Absolute convergence gives

$$\log\zeta(s) = \sum_p \sum_{k\geq 1} \frac{p^{-ks}}{k}. \tag{21.5}$$

The bound $\sum_{p,k} p^{-k\sigma}/k \leq (1-2^{-\sigma})^{-1}\sum_p p^{-\sigma}$ justifies rearrangement. The identical argument applies to the specified unramified factors of a character in a half-plane of absolute convergence.

One may define a local scalar $\mathcal{W}_p(s) = -\log(1-p^{-s})$ by this formula. Identifying it with a Nekrasov–Shatashvili superpotential requires an equivariant family and the actual scaled limit defining that superpotential. The equality (21.5) alone does not construct such a family.

The arithmetic instanton comparison

An arithmetic curve and the framed complex surface used in localization are different geometric inputs. A proposed arithmetic self-product must specify its base category, structure sheaf, moduli problem,

equivariant action, fixed loci and integration theory. Listing prime pairs does not construct those data. Likewise a countable family of local oscillators does not determine a gauge group or its Coulomb branch.

The real parameter in a metrized arithmetic line can be recorded by its Arakelov degree. A unitary character supplies separate phase data. This gives meaningful scalar parameters, but no identification with the Coulomb differences $a_\alpha - a_\beta$ in (21.4) until the comparison maps are supplied. An arithmetic intersection pairing must first specify its arithmetic surface, metrized bundles, and cycle classes.

Question 21.14.2 *(From an arithmetic moduli problem to Euler factors).* Construct an arithmetic moduli problem with an equivariant or motivic integration theory and a comparison to the framed instanton construction. Specify the coefficient ring, potential or vanishing-cycle data when used, and every parameter scaling. Determine whether its all-charge trace is the prescribed Frobenius determinant and prove that comparison on the actual objects. The fixed-parameter limit and hook-weighted specialization above do not supply it.

21.15 Geometric transfer operators and Liouville scaling

The geometric R-matrix construction contains commuting transfer operators, but it retains more than their scalar traces. Suppose $R_{F,W}(u)$ is a compatible family of invertible R-matrices satisfying the Yang–Baxter equation. For a finite-dimensional auxiliary space F and an endomorphism g_F, define

$$T_F(g,u) = \operatorname{Tr}_F\big((g_F \otimes 1)R_{F,W}(u)\big).$$

Assume the family g commutes with the R-matrices on each pair of auxiliary spaces. The Yang–Baxter equation conjugates the two ordered products by the auxiliary R-matrix. Taking the trace over both auxiliary factors therefore gives

$$[T_{F_1}(g,u_1), T_{F_2}(g,u_2)] = 0.$$

For infinite-dimensional auxiliary spaces, the trace needs its specified grading, completion, or convergence condition.

Maulik–Okounkov construct such operators from stable envelopes of Nakajima varieties. Their Theorem 1.3.1 identifies the cohomological degree-two part of the Baxter algebra with modified quantum multiplication by tautological degree-two classes. The modification includes the theta-characteristic sign on curve classes. In the divisor formula, the nonclassical part has the form

$$-\hbar \sum_{\theta\cdot\alpha>0} (\lambda,\alpha)\frac{q^\alpha}{1-q^\alpha} e_\alpha e_{-\alpha},$$

with dual root-space bases and the scalar term fixed by the unit condition. This is a geometric theorem on its equivariant cohomology carrier.

For the Jordan quiver, with one vertex and one loop, the variety $\mathcal{M}(r,n)$ parametrizes framed rank-r torsion-free sheaves on $\mathbb{P}^2$ with second Chern class n. Its Yangian belongs to the affine $\widehat{\mathfrak{gl}}_1$ construction. The integer r is the framing rank and labels the representation. It does not replace the quiver by affine $\mathfrak{gl}_r$. At $r = 1$, the Hilbert-scheme cohomology has its Heisenberg Fock realization. Stable envelopes relate the rank-r cohomology to r such bosonic factors on the equivariant domain.

A comparison with arithmetic Hecke operators must specify the geometric representation, arithmetic representation, parameter map and intertwiner. Commutativity of a Baxter algebra does not erase the R-matrix or its tensor structure. Identifying a scalar instanton partition sum with a holonomy trace requires another comparison. These are additional equalities to construct.

The semiclassical Liouville equation

For Liouville theory with a reference metric $\widetilde{g}$, use

$$S_L(\phi) = \frac{1}{4\pi}\int \left(|d\phi|^2_{\widetilde{g}} + Q\widetilde{R}\phi + 4\pi\mu e^{2b\phi}\right) d\,\mathrm{vol}_{\widetilde{g}}, \qquad Q = b + b^{-1}.$$

The stress tensor and primary weights are

$$T(z) = -(\partial\phi)^2 + Q\partial^2\phi, \qquad c = 1 + 6Q^2, \qquad \Delta_\alpha = \alpha(Q-\alpha)$$

for $V_\alpha = e^{2\alpha\phi}$. These formulas retain the background charge.

Set $\phi_c = 2b\phi$ and hold $\lambda = \pi\mu b^2$ fixed as $b \to 0$. Then the leading rescaled action is

$$b^2 S_L = \frac{1}{16\pi}\int \left(|d\phi_c|^2 + 2\widetilde{R}\phi_c + 16\lambda e^{\phi_c}\right) d\,\mathrm{vol}_{\widetilde{g}} + O(b^2),$$

with the corresponding boundary terms when the surface is represented by a punctured or unbounded coordinate chart. In a flat coordinate its equation of motion is

$$\partial\bar\partial\phi_c = 2\lambda e^{\phi_c}.$$

The exponential interaction survives at fixed nonzero λ. Thus this semiclassical limit is not a free-boson equation. Sending b to zero also sends c to infinity. It does not by itself construct a contraction of Virasoro conformal blocks to free-boson correlators.

The distinction is visible in primary weights. For a heavy scaling $\alpha = \eta/b$,

$$\Delta_\alpha = \frac{\eta(1-\eta)}{b^2} + \eta.$$

For a light scaling $\alpha = b\sigma$,

$$\Delta_\alpha = \sigma + b^2\sigma(1-\sigma).$$

An unscaled nonzero α has a different asymptotic behavior. A proposed limiting operator must specify which scaling it uses.

The original Alday–Gaiotto–Tachikawa correspondence relates specified four-dimensional gauge theories and their matter to Liouville blocks, with the required $U(1)$ factors and parameter conventions. The relation $b^2 = \epsilon_1/\epsilon_2$ is part of such conventions. Applying it does not turn every rank-one instanton sum into an oscillator trace. An arithmetic version would require its own theory, comparison functor, and controlled parameter limit. The positive geometric constructions and the failed fixed-parameter limit leave that question open.

21.16 TATE INTEGRALS AND MEROMORPHIC LINES

Tate theory supplies local integrals, Euler products, and a global functional equation. These are precise scalar constructions. We first retain them on their actual domains, then describe what geometric sewing would require in addition.

Normalize multiplicative Haar measure on $\mathbb{Q}_p^\times$ by $\mathrm{vol}(\mathbb{Z}_p^\times) = 1$. For an unramified character χ_p with $\chi_p(p) = \alpha_p$, shell integration gives

$$Z_p(\mathbf{1}_{\mathbb{Z}_p}, \chi_p, s) := \int_{\mathbb{Q}_p^\times} \mathbf{1}_{\mathbb{Z}_p}(x)\chi_p(x)|x|^s\, d^\times x = \sum_{j\ge 0} \alpha_p^j p^{-js} = \frac{1}{1-\alpha_p p^{-s}}$$

when $|\alpha_p p^{-s}| < 1$. Using $\mathbf{1}_{\mathbb{Z}_p^\times}$ instead gives one. Thus changing the local test function can remove the unramified Euler factor.

At the real place, with $dx/|x|$ and $\Phi_\infty(x) = e^{-\pi x^2}$,

$$Z_\infty(\Phi_\infty, 1, s) = 2\int_0^\infty e^{-\pi x^2} x^{s-1}\, dx = \pi^{-s/2}\Gamma(s/2) \qquad (\operatorname{Re} s > 0).$$

The substitution $u = \pi x^2$ proves the last equality. For the trivial global character, their product is

$$\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s) \qquad (\operatorname{Re} s > 1).$$

Proposition 21.16.1 *(Finite removal and reinsertion).* For a finite prime set S, put

$$P_S(s) = \prod_{p\in S}(1 - p^{-s}), \qquad \Phi_S(s) = P_S(s)\Lambda_\zeta(s).$$

Using the unit-shell test function at the places in S gives Φ_S in the half-plane of absolute convergence. Reinserting their standard unramified tests multiplies by $P_S(s)^{-1}$. The identities extend meromorphically.

Proof. The local shell computation replaces precisely the factors indexed by S. Absolute convergence allows the finite factors to be separated from the remaining product. The meromorphic continuation of both sides preserves the identity. □

For $S = \{2, 3\}$, replacing the two factors by specified local parameters gives

$$\Lambda_\zeta(s)\frac{(1 - 2^{-s})(1 - 3^{-s})}{(1 - \alpha_2 2^{-s})(1 - \alpha_3 3^{-s})}.$$

It equals $\Lambda_\zeta(s)$ when both parameters are one. Otherwise it is the stated partial replacement. A full character L-function needs its Euler factors at every place, its conductor, and the archimedean parity factor. Arbitrary choices at two primes do not specify a global character.

The functional equation and its normalization

For completeness, the scalar zeta symmetry follows from the Gaussian Poisson identity

$$\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t} = t^{-1/2}\theta(t^{-1}).$$

For $\operatorname{Re} s > 1$, termwise integration gives

$$\Lambda_\zeta(s) = \frac{1}{2}\int_0^\infty (\theta(t) - 1)t^{s/2}\frac{dt}{t}.$$

Split at one and apply the theta identity to the first integral. The result is

$$\Lambda_\zeta(s) = \frac{1}{s(s-1)} + \frac{1}{2}\int_1^\infty (\theta(t) - 1)\left(t^{s/2} + t^{(1-s)/2}\right)\frac{dt}{t}. \tag{21.6}$$

The tail decays exponentially, so the integral is entire in s, locally uniformly with all derivatives. This proves meromorphic continuation, simple poles at zero and one with residues -1 and 1, and $\Lambda_\zeta(s) = \Lambda_\zeta(1 - s)$.

The entire normalization is $\xi(s) = \frac{1}{2}s(s-1)\Lambda_\zeta(s)$.

For a primitive Dirichlet character χ of conductor f, let $\chi(-1) = (-1)^a$, $a \in \{0, 1\}$. The completed function and root number are

$$\Lambda(s,\chi) = \left(\frac{f}{\pi}\right)^{(s+a)/2} \Gamma\Big(\frac{s+a}{2}\Big) L(s,\chi), \qquad \varepsilon(\chi) = \frac{\tau(\chi)}{i^a\sqrt{f}},$$

where $\tau(\chi) = \sum_{x \bmod f} \chi(x) e^{2\pi i x/f}$. Tate's global functional equation, with these conventions, is

$$\Lambda(s,\chi) = \varepsilon(\chi)\Lambda(1-s,\overline{\chi}).$$

Its adelic proof uses Fourier transform and Poisson summation, together with the local character and measure data. An imprimitive character requires the corresponding removed local factors. It cannot be inserted unchanged in the primitive formula.

These identities construct meromorphic sections of scalar lines. For instance, let $\mathcal{M}$ be the field of meromorphic functions of s, and attach the line $\mathcal{M}e_S$ to each finite set S. The map deleting a prime $p \in S$ is multiplication by $(1-p^{-s})^{-1}$. Maps for different primes commute, so this is a functor from the finite-deletion category to meromorphic lines. The sections $\Phi_S e_S$ are compatible with deletion.

The unmarked involution is $s \mapsto 1-s$. On a marked section it has the explicit factor

$$\Phi_S(s) = \frac{P_S(s)}{P_S(1-s)}\Phi_S(1-s).$$

This is an actual comparison of the specified lines and sections. Its source is the finite-deletion category. It is not yet a category of arithmetic surfaces with boundary sewings.

A scattering phase and a positive measure

The two completed zeta normalizations also give different scattering ratios. Put

$$m_\Lambda(s) = \frac{\Lambda_\zeta(2s-1)}{\Lambda_\zeta(2s)}, \qquad m_\xi(s) = \frac{\xi(2s-1)}{\xi(2s)}.$$

For the standard weight-zero Eisenstein series on $SL_2(\mathbb{Z})\backslash\mathbb{H}$, the constant term is $y^s + m_\Lambda(s)\,y^{1-s}$. The ratio using the entire function instead satisfies

$$m_\xi(s) = \frac{s-1}{s} m_\Lambda(s).$$

This follows by substituting $\xi(w) = \frac{1}{2}w(w-1)\Lambda_\zeta(w)$ and cancelling. Thus the two scalar conventions cannot be interchanged in a claimed normalization of an intertwining operator.

Both ratios satisfy $m(s)m(1-s) = 1$ meromorphically. On $s = 1/2 + it$, their modulus is one wherever they are finite and nonzero. Indeed the functional equation and real conjugation give

$$\xi(2s-1) = \xi(1-2it) = \overline{\xi(2s)},$$

and the same calculation applies to Λ_ζ. The rational factor $(s-1)/s$ also has unit modulus on this line. At the removable central value, $m_\Lambda(1/2) = -1$ and $m_\xi(1/2) = 1$. The residues $-1, 1$ in (21.6) prove the first value, and $\xi(0) = \xi(1) = 1/2$ proves the second.

Write $\Gamma_{\mathbb{R}}(w) = \pi^{-w/2}\Gamma(w/2)$. There is a precise Euler product for the first ratio:

$$m_{\Lambda}(s) = \frac{\Gamma_{\mathbb{R}}(2s-1)}{\Gamma_{\mathbb{R}}(2s)} \prod_{p} \frac{1-p^{-2s}}{1-p^{1-2s}} \qquad (\operatorname{Re} s > 1).$$

Both zeta products converge absolutely on this half-plane, so taking their quotient proves the formula.

Meromorphic continuation defines its other values. It does not assert absolute Euler-product convergence on $\operatorname{Re} s = 1/2$. Ramified representations require their actual local L- and epsilon-factors, character and measure data. They are not obtained by appending arbitrary phases to this unramified zeta ratio.

To see what phase removal proves, work on a domain of the unitary axis where $\xi(2s) \neq 0$ and put

$$d(s) = |\xi(2s)|^{-2}, \qquad P(s) = 2d(s), \qquad K(s) = m_{\xi}(s)^{-1}d(s).$$

These are specified scalar functions. They satisfy

$$d(s) \leq |P(s) - K(s)| \leq 3d(s), \qquad m_{\xi}(s)K(s) = d(s) = P(s)/2.$$

The triangle and reverse-triangle inequalities prove the bounds, since $|m_{\xi}| = 1$ and $d > 0$.

Multiplication by the phase removes that phase. It leaves the factor two in the proposed positive coefficient. Realizing either scalar as an actual continuous-spectrum pairing remains another question.

A measured comparison must preserve more than its diagonal density. For example, suppose P, K are positive measurable functions on a measured space (X, ν). If $|c|^2 P = K$ almost everywhere, then

$$f \longmapsto cf : L^2(X, K\,d\nu) \xrightarrow{\sim} L^2(X, P\,d\nu)$$

is unitary. Integrating $|cf|^2 P = |f|^2 K$ proves isometry. Division by the nonzero c gives its inverse. For the phase-valued K above, positivity is not available before the form and its normalization have been specified.

A proposed arithmetic extension therefore needs a bivariate Hermitian pairing, its domain, the measured spectral decomposition, and an intertwiner realizing the positive form. Multiplication by $m((s_1 + s_2)/2)$ alone need not preserve the Hermitian symmetry in the two parameters. A norm identity on individual formal spectral vectors does not provide the mixed pairings or convergence of wave packets.

On a specified decomposition $V = V_{\mathrm{disc}} \oplus V_{\mathrm{cont}}$, one may define the correction to be the identity on the first summand. If both pairings make the summands orthogonal, comparisons on the two summands combine to a comparison on V. Expanding the pairing of $v_d + v_c$ and $w_d + w_c$ proves this assertion: both mixed terms vanish. Without that orthogonality, the mixed comparisons must also be given.

A selected residual line is a particularly simple case. If the placement carries its arithmetic vector v_0 to a unit vector and $B^{\mathrm{Ar}}(v_0, \sigma v_0) = 1$, then both forms on av_0, bv_0 equal $a\overline{b}$. This calculation does not identify every residual automorphic summand with one vacuum. Nor does discreteness of a zeta zero set place its trace in this automorphic discrete summand. That placement requires the actual spectral comparison.

Finally, a scalar functional equation does not transport obstruction classes. A specified linear map $F : H \to H_{\mathrm{Ar}}$ does carry a relation $c + c^{!} = 0$ to $F(c) + F(c^{!}) = 0$. Applying this fact to duality classes requires the obstruction complexes and a comparison defined for the corrected pairing. The identity $m(s)m(1-s) = 1$ supplies no such map.

Discriminants and residues

Let K be a number field of signature (r_1, r_2) and discriminant d_K. In the standard Minkowski embedding $K \hookrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$, use Lebesgue measure $dx\,dy$ on each complex coordinate. Then

$$\operatorname{covol}(\mathcal{O}_K) = 2^{-r_2}\sqrt{|d_K|}. \tag{21.7}$$

To prove the factor, choose an integral basis and form its complex embedding matrix. Its determinant has absolute value $\sqrt{|d_K|}$ by the discriminant identity. Replacing each conjugate pair by real and imaginary parts divides the absolute determinant by two. There are r_2 such pairs, giving (21.7).

Thus

$$\log \operatorname{covol}(\mathcal{O}_K) = \frac{1}{2}\log|d_K| - r_2 \log 2.$$

In particular $\frac{1}{2}\log|d_K|$ is neither an integer genus nor the half-logarithm of this covolume.

Metrics and measure conventions matter for arithmetic Euler characteristics as well. One can define the real discriminant expression

$$D_K = \log|d_K| - r_1 \log(\pi/4) - r_2 \log(\pi/2).$$

It is a specified normalization of discriminant data. Identifying it with a metrized determinant of cohomology or an arithmetic Euler characteristic requires the actual complex, metrics and comparison formula. The expression alone supplies none of them.

The analytic class-number formula gives the positive residue

$$\operatorname*{Res}_{s=1} \zeta_K(s) = \frac{2^{r_1}(2\pi)^{r_2} h_K R_K}{w_K \sqrt{|d_K|}}.$$

Here h_K is the ideal class number, R_K the regulator, and w_K the number of roots of unity. The standard rank-zero regulator convention is $R_K = 1$. This is a scalar residue. For example, for $K = \mathbb{Q}(i)$ it is $\pi/4$, using $h_K = R_K = 1$, $w_K = 4$, and $|d_K| = 4$. It is therefore not in general an integer dimension of conformal blocks.

21.17 The maps required for sewing

The finite Frobenius construction supplied an inverse pairing and explicit gluing maps. For a family of geometric or arithmetic objects, sewing requires their analogues, compatible under repeated gluing.

One possible coefficient for an anomaly is a determinant line. For a bounded complex $C^\bullet$ of finite-dimensional spaces, define

$$\det C^\bullet = \bigotimes_i (\det C^i)^{(-1)^i}.$$

Odd exponents mean dual lines. Cancellation of the boundary and coboundary subspaces gives its canonical identification with $\bigotimes_i (\det H^i(C))^{(-1)^i}$. Exact sequences or exact triangles give corresponding determinant-line isomorphisms with their graded signs.

A trivialization, compatible with those isomorphisms, is additional data. It cannot be replaced just by an orientation word or by a scalar functional equation.

Intermediate labels and coends

The intermediate objects in sewing belong to a specified category. For a small k-linear category C, a right module $F : C^{\mathrm{op}} \to \mathrm{Vect}_k$ and a left module $G : C \to \mathrm{Vect}_k$ have a coend

$$F \otimes_C G = \int^{a \in C} F(a) \otimes_k G(a).$$

Concretely it is the quotient of $\bigoplus_a F(a) \otimes_k G(a)$ by the relations $F(f)(x) \otimes y = x \otimes G(f)(y)$ for $f : a \to b$, $x \in F(b)$ and $y \in G(a)$. In a dg setting, the derived coend is computed by its bar construction with all strings of composable morphisms.

If C is split finite semisimple, the expression reduces to a finite sum over its simples with the specified endomorphism-field balancing. Without semisimplicity, replacing the derived coend by a sum over simple isomorphism classes can lose extensions and higher Tor. A measured direct integral instead requires a measurable family, a measure, Hilbert completions and a convergence theorem. It is a different construction.

For local Weil–Deligne data, one must fix the local field, coefficient field, rank, topology, Frobenius convention, monodromy, and allowable inertia types. A list of all such isomorphism classes does not specify a coend or a measure. Likewise local Langlands does not, by itself, construct a sewing functor on arithmetic surfaces or identify its objects with WZW labels.

The finite and archimedean actions also have different types. A change of coefficient embedding can transport Galois labels when its semilinear maps are given. A Hecke correspondence acts on its specified automorphic or sheaf-theoretic carrier. The quotient $G(\mathbb{A})/G(K)$ is not a group in general, so it cannot serve as a symmetry group without a separate action construction. Inertia and wild inertia remain part of ramified local data.

Question 21.17.1 *(Arithmetic sewing).* Construct the source category of marked arithmetic objects and its gluing operations. Give the intermediate coefficient category, the two variances of the boundary modules, and the derived coend or measured integral that supplies their gluing. Construct its map to the coefficient object of the glued arithmetic space. Require compatible units, dualities, determinant lines and repeated-sewing isomorphisms.

Compare the resulting scalar evaluations with the Tate sections on their common domains, including the conductor, archimedean and local normalization factors. Both the separating sewing and the self-gluing that creates a handle are part of this question.

The meromorphic-line functor above provides finite Euler-factor compatibility and functional-equation symmetry. Those results leave the stated geometric maps to be constructed. Conversely, a geometric sewing theorem can exist on a different specified coefficient category without yet giving the arithmetic comparison.

A finite analytic sewing construction

There is a positive analytic example already constructed in Theorem 16.7.1. It uses the holomorphic lattice algebra $W = V_{E_8}$ and the finite marked coefficient objects (P, λ) of the arithmetic fusion category. For a compatible junction $f : P \otimes_{R_\circ} Q \to S$, the field is

$$\mathcal{J}_f(u \otimes p, z)(v \otimes q) = Y_W(u, z)v \otimes f(p \otimes_{R_\circ} q).$$

The coefficient tensor is balanced over the boundary algebra $R_\circ$. Its duality maps and associators retain the original finite fusion data.

The sewing domain is explicit. Start with finitely many compact pointed Riemann surfaces, possibly disconnected. Choose disjoint coordinate discs of radii r_j, r'_j at each paired marking and glue by $u_j v_j = t_j$, with $0 < |t_j| < r_j r'_j$. Each initial connected component retains an unsewn marked point. Fix the initial conformal block, contragredient pairings, coordinates and branches for fractional powers.

On this domain the descendant propagator sums converge absolutely and locally uniformly, including simultaneous sewings that create handles. The reason is that each coefficient contraction is finite-dimensional: the sewn expression is a finite linear combination of ordinary W conformal block sewings. The lattice algebra is rational and C_2-cofinite, and its modules here are finite sums of the vacuum module. Gui's sewing theorem applies with the stated retained-marking hypothesis. Finite sums preserve its convergence and sewing identities.

If an initial component has no unsewn mark, propagation of vacua first inserts one. Its propagation isomorphism removes it afterward. The hypothesis is not removed by silently omitting the point. This theorem supplies actual analytic sewing for the constructed marked fusion realization. It does not choose a globally normalized partition scalar or the arithmetic coend in Question 21.17.1.

21.18 Genus, rank, and coefficient data

The genus of a smooth proper curve controls the rank of its first cohomology. A field degree, a framing rank, and a Siegel parameter are different inputs, even when their numerical values coincide.

For a smooth projective curve C/K of genus g, the Jacobian has dimension g and its first cohomology has rank $2g$. At a complex embedding its Hodge dimensions are (g, g), and $\deg \omega_C = 2g - 2$. Arithmetic intersection and determinant formulas additionally use a regular arithmetic model, metrics, singular fiber contributions and analytic torsion. The topological formula $2 - 2g$ is not by itself an arithmetic Euler characteristic or a Koszul level identity.

Several possible coefficient constructions should therefore be kept distinct.

(i) A smooth proper curve gives its Jacobian's weight-one cohomology. An arithmetic surface model permits metrized intersections and a determinant of cohomology, subject to their actual hypotheses.

(ii) A totally imaginary field K of degree $2g$ gives a Minkowski lattice in $\mathbb{C}^g$ after selecting one embedding from each conjugate pair. Its quotient is a complex torus. An abelian variety requires a positive integral Riemann form. In a CM construction one specifies a CM type and polarization. This torus is not the weight-zero motive whose L-function is ζ_K.

(iii) A rank-$2g$ ordinary crystalline representation has its coefficient field, Hodge–Tate multiset, ordinary filtration and semilinear Frobenius. Its moduli dimension depends on the imposed type and deformation conditions. Frobenius is not automatically an operator with square -1.

(iv) A theta construction on a principally polarized abelian g-fold uses a chosen level and theta line. Its level-n section space has dimension n^g, whereas its first cohomology has rank $2g$. Comparing these carriers requires an actual map. The Steinberg relation $\{a, 1 - a\} = 0$ in $K_2^M(K)$, for $a \neq 0, 1$, is a field relation and does not supply that map.

(v) A surface factorization construction needs both local operations and their interchange coherences. Two separately named E_1 structures do not give an E_2 structure until the hypotheses of the iterated-algebra construction hold. Double-residue and completed duality comparisons must preserve the specified coefficients and signs.

(vi) A Siegel or other Shimura construction uses its actual automorphic representation and L-group representation. For GSp_{2g}, the standard representation of the dual GSpin_{2g+1} has dimension $2g+1$, while the spin representation has dimension 2^g. Neither is automatically the rank-$2g$ Jacobian representation.

These alternatives name different objects. They permit comparison problems. They do not define six presentations of an already identified arithmetic object. In particular an almost-complex structure belongs to a real tangent bundle of an analytic space, whereas a crystalline Frobenius belongs to its specified coefficient realization. An arithmetic scheme does not acquire the former from the latter by notation.

Weights and the zero-rank case

For a weight-m realization in motivic normalization, the expected functional-equation involution is $s \mapsto m+1-s$. Its existence and normalization must be established for the chosen L-function. The Dedekind zeta function is weight zero and has $s \mapsto 1-s$, independently of $[K:\mathbb{Q}]$. The Jacobian H^1 function is weight one and, when its completed functional equation is available, uses $s \mapsto 2-s$. A spin or standard automorphic function has the shifts and conductor of its own chosen normalization.

Gamma factors likewise come from the Hodge or archimedean representation. The number g alone does not select them. For example, with $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ and $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$, the Dedekind completion is

$$\Lambda_K(s) = |d_K|^{s/2}\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\zeta_K(s).$$

A curve's weight-one completion also requires its conductor and all local factors, including bad reduction.

At rank zero the determinant $\det(1 - tF \mid 0)$ is one and $\mathrm{Sym}(0) = k$. Its all-charge trace is one. For $\mathbb{P}^1$, the Jacobian is trivial and $H^1 = 0$. Its H^1 Euler factor is also one.

The full zeta function of $\mathbb{P}^1_{\mathbb{F}_q}$ is instead

$$Z(\mathbb{P}^1_{\mathbb{F}_q}, t) = \frac{1}{(1-t)(1-qt)},$$

because $\#\mathbb{P}^1(\mathbb{F}_{q^n}) = q^n + 1$. It includes degrees zero and two. Over a number field the corresponding finite Euler product is $\zeta_K(s)\zeta_K(s-1)$ in its initial half-plane. None of these is the H^1 factor.

A number field cannot have degree zero. Thus setting $2g = 0$ in a totally imaginary degree-$2g$ family does not produce $\mathbb{Q}$. A Riemann Tate sector or a rank-two Wilson label can be added as separate data. The empty tensor contributes only its scalar unit.

For positive rank, specify the Frobenius carrier as well. If $V_\ell J$ is the Tate module of an abelian variety, its dual $H^1_{\text{ét}}(J, \mathbb{Q}_\ell)$ has the dual action. Arithmetic and geometric Frobenius are inverse. On the cohomological geometric convention for weight one, polarization pairs eigenvalues with product q. On the Tate-module geometric convention the product is q^{-1}. These distinctions persist through the determinant trace and any Wilson comparison.

21.19 Naturality and stabilization

A genus-wise identity of scalar functions does not define maps between coefficient objects of different rank. We state the chain-level construction that does give such comparisons when its maps are known.

Let $\mathcal{D}$ be a specified category of arithmetic coefficient systems. An object includes its local complexes, reference vectors, local actions, completed or algebraic tensors, and all merger and residue maps being used. Morphisms preserve these data. If conformal vectors or pairings are part of the structure, their

preservation is an additional equation. Composition of such morphisms remains compatible because its defining squares compose. No compatibility of a completion is inferred solely from the presence of almost-everywhere reference vectors.

Suppose two stage constructions have differentials $D_S^{(a)} = D_{\mathrm{o},S}^{(a)} + \tau_S^{(a)}$, $a = 1, 2$, with

$$(D_{\mathrm{o},S}^{(a)})^2 = 0, \qquad D_{\mathrm{o},S}^{(a)}\tau_S^{(a)} + \tau_S^{(a)}D_{\mathrm{o},S}^{(a)} + (\tau_S^{(a)})^2 = 0.$$

The second identity is the precise condition for the total differential to square to zero. Its validity for two complexes does not yet give a map between them.

PROPOSITION 21.19.1 *(Comparison from natural stage maps).* Over a common coefficient ring, suppose quasi-isomorphisms $\Xi_{\mathsf{A},S} : C_S^{(1)}(\mathsf{A}) \to C_S^{(2)}(\mathsf{A})$ are given. Require naturality in A, compatibility with the stage transitions, and

$$\Xi D_{\mathrm{o},S}^{(1)} = D_{\mathrm{o},S}^{(2)}\Xi, \qquad \Xi\tau_S^{(1)} = \tau_S^{(2)}\Xi.$$

Then their algebraic filtered colimits are naturally quasi-isomorphic. Any additional completed comparison requires preservation of this quasi-isomorphism by the chosen completion.

Proof. Adding the two equations gives the chain-map identity. Transition compatibility defines the colimit map. Filtered colimits of modules are exact, so homology commutes with these colimits and the map remains a quasi-isomorphism. The naturality squares descend from those at each stage. The final assertion specifies the extra limiting operation and its needed property. □

For ordered, arithmetic-chiral, Deninger-type and bulk-boundary constructions, the complexes and stage maps must each be given on their independent carriers. Defining one assignment to be another produces an identity natural transformation, but not an independent geometric comparison. An independently defined bulk theory also needs its boundary functor and the comparison of its operations. Frobenius and duality pairings are transported only when the specified maps preserve them.

Changing the rank

For categories $\mathcal{D}_g$, supply an actual stabilization functor $S_g : \mathcal{D}_g \to \mathcal{D}_{g+1}$. Let $F_{\star,g}$ be specified chain functors and V_2 a fixed graded coefficient space with zero differential. Suppose there are natural quasi-isomorphisms

$$q_{\star,g} : F_{\star,g}(\mathsf{A}) \otimes V_2 \longrightarrow F_{\star,g+1}(S_g\mathsf{A}).$$

Over a field, algebraic tensoring with V_2 gives

$$H(F_{\star,g} \otimes V_2) \simeq H(F_{\star,g}) \otimes V_2.$$

One proof splits each complex as its cohomology and a contractible complex. Tensor its contracting homotopy with 1_{V_2}. For other coefficients or completed tensors, the corresponding Künneth statement is an additional hypothesis.

If the maps $q_{\star,g}$ commute with the comparison maps η up to specified chain homotopies, then their cohomology isomorphisms commute with η. In particular,

$$\eta_{g+1}q_{\mathrm{ch},g} \simeq q_{\partial,g}(\eta_g \otimes 1)$$

becomes equality on homology. The same argument transports specified Frobenius actions, retractions and pairings, provided their compatibility is included. This gives uniformity for every g where all those data exist.

An oscillator splitting alone does not construct the maps $q_{\star,g}$ for a differential whose differential includes maps coupling residues at different places. Even a trivial action does not imply flatness: for $\Lambda = \mathbb{Z}_p[[T]]$ and $M = \Lambda/(T)$, the resolution

$$0 \longrightarrow \Lambda \xrightarrow{T} \Lambda \longrightarrow M \longrightarrow 0$$

gives $\operatorname{Tor}_1^{\Lambda}(M, M) = M$. The p-primary Iwasawa μ-invariant of this module is zero, but M is not Λ-flat. In several variables the corresponding completion and duality hypotheses must still be proved.

A block-diagonal group map $GL_{2g} \times GL_2 \hookrightarrow GL_{2g+2}$ also does not give a tensor decomposition of the ambient current algebra. The latter has off-diagonal matrix currents absent from the two block Lie algebras. A retraction of coefficient complexes requires a chosen splitting and a chain map. It is not an automatic retraction of vertex algebras.

Three different diagonal constructions

For a rank-r Heisenberg vertex algebra V_r at level one, the tensor $V_r \otimes V_r$ has rank $2r$ and standard central charge $2r$. Writing the two current families as $J_i^{(1)}$ and $J_i^{(2)}$, their unscaled sums obey

$$(J_i^{(1)} + J_i^{(2)})(z)(J_j^{(1)} + J_j^{(2)})(w) \sim \frac{2\delta_{ij}}{(z-w)^2}.$$

They therefore do not give a level-one diagonal map. The normalized sums $(J_i^{(1)} + J_i^{(2)})/\sqrt{2}$ do give a Heisenberg embedding, but its image conformal vector is the Sugawara vector of only the diagonal currents. It is different from the full tensor conformal vector: the complementary anti-diagonal currents contribute central charge r.

The coordinatewise product $V_r \times V_r$ is a different vertex algebra. Its vacuum is $(1, 1)$ and its conformal vector is (ω, ω). The map $v \mapsto (v, v)$ preserves both and all products. Its central term is $(r/2)(1, 1)$, so the central charge remains r. Its weight-zero space is two-dimensional, unlike that of the rank-$2r$ Heisenberg tensor product. Neither construction is the free product merely by its notation.

Nor is the reduced bar an additive construction on arbitrary coefficient algebras. Let $A = k \oplus kx$ and $B = k \oplus ky$ have square-zero augmentation ideals. The square-zero algebra $C = k \oplus (kx \oplus ky)$ has zero reduced-bar differential. Its bar length-two term has basis

$$[x|x], \quad [x|y], \quad [y|x], \quad [y|y].$$

The direct sum of the positive reduced bars of A and B has only the first and last terms in that length. The mixed terms survive in homology. Thus an additive diagonal claim for bar constructions needs the actual carrier and comparison. It cannot follow from linearity of one residue operator.

Evaluating a duality defect

The same requirement applies to a proposed all-genus level identity. Suppose an independently defined defect $\mathfrak{d}_g$ has a specified scalar evaluation $\mathcal{E}_g$ such that

$$\mathcal{E}_g(\mathfrak{d}_g)(s) = \Lambda_g(s) - \varepsilon_g \Lambda_g(w_g - s).$$

If the functional equation holds, the evaluated defect vanishes. Concluding $\mathfrak{d}_g = 0$ requires that $\mathcal{E}_g$ detect zero on the actual defect space. This criterion is immediate from injectivity, and preserves the need to construct both the evaluation and detection. Euler factors, weights, and a scalar functional equation alone do not provide them.

21.20 The level-dependent Hecke trace

A Hecke trace on a finite space of cusp forms is another concrete scalar. Its relation to a categorical arithmetic trace requires a map carrying the chosen endomorphism to that Hecke operator. The classical trace itself retains the level in both its elliptic and cusp terms.

Let $N \geq 1$, $k \geq 2$, and $n \geq 1$. Write

$$\Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) : a \equiv d \equiv 1,\ c \equiv 0 \pmod{N} \right\}.$$

Let $S_k(\Gamma_1(N))$ be its holomorphic cusp forms and T_n the usual Hecke operator. One precise normalization is obtained by decomposing under diamond characters χ and using on $S_k(\Gamma_0(N), \chi)$

$$f|T_n = \sum_{\sigma \in \Gamma_0(N)\backslash \Delta_n} (\det \sigma)^{k-1} \chi(a_\sigma)\, f|_k \sigma,$$

where

$$\Delta_n = \{\sigma \in M_2(\mathbb{Z}) : \det \sigma = n,\ N \mid c_\sigma,\ (a_\sigma, N) = 1\},$$

$$f|_k \begin{pmatrix} a & b \\ c & d \end{pmatrix}(z) = (cz + d)^{-k} f\left(\frac{az + b}{cz + d}\right).$$

This convention includes indices not coprime to N.

Define the polynomials $P_j(t, n)$ by

$$\frac{1}{1 - tx + nx^2} = \sum_{j \geq 0} P_j(t, n) x^j.$$

Thus $P_0 = 1$, $P_1 = t$, and $P_j = tP_{j-1} - nP_{j-2}$. If $\alpha + \beta = t$ and $\alpha\beta = n$, then $P_{k-2}(t, n) = (\alpha^{k-1} - \beta^{k-1})/(\alpha - \beta)$ when the roots are distinct. The polynomial defines the repeated-root value.

Let φ be Euler's totient, and put

$$I(N) = N \prod_{p \mid N} (1 + p^{-1}), \qquad \sigma_{1,N}(n) = \sum_{\substack{d \mid n \\ (d,N)=1}} \frac{n}{d}.$$

For $u \mid N$ and $u^2 \mid t^2 - 4n$, define

$$B_N(u, t, n) = \begin{cases} I(N)/I(N/u), & Nu \mid t - n - 1, \\ 0, & \text{otherwise}, \end{cases} \qquad D_N(u, t, n) = \sum_{d \mid u} \mu(d) B_N(u/d, t, n).$$

Here μ is the Möbius function. For integers a, d, set

$$\Psi_N(a, d) = \sum_{\substack{rs = N \\ r \mid a-1,\ s \mid d-1}} \varphi((r, s))\varphi(N/(r, s)).$$

For $D > 0$, let $H(D)$ be the Hurwitz class number of positive definite integral binary quadratic forms of discriminant $-D$, including imprimitive forms. A class has weight the reciprocal of its stabilizer order in $PSL_2(\mathbb{Z})$. Thus $H(3) = 1/3$, $H(4) = 1/2$. Set $H(0) = -1/12$, and set $H(D) = 0$ for a nonintegral argument or an inadmissible discriminant.

Theorem 21.20.1 *(Eichler–Selberg at level $\Gamma_1(N)$).* With the preceding conventions,

$$\begin{aligned}\operatorname{Tr}(T_n | S_k(\Gamma_1(N))) = {} & \delta_{k,2}\sigma_{1,N}(n) \\ & - \frac{\varphi(N)}{2} \sum_{\substack{t\in\mathbb{Z} \\ t^2\le 4n}} P_{k-2}(t,n) \sum_{\substack{u\mid N \\ u^2\mid t^2-4n}} H\left(\frac{4n-t^2}{u^2}\right) D_N(u,t,n) \\ & - \frac{1}{4} \sum_{\substack{a,d>0 \\ ad=n}} \min(a,d)^{k-1}\left(\Psi_N(a,d) + (-1)^k \Psi_N(-a,-d)\right).\end{aligned}$$

This holds for all $n \ge 1$, including bad-level indices.

The formula is Popa's Theorem 5 in *On the trace formula for Hecke operators on congruence subgroups, II*. Its elliptic level factor is D_N. The terms $t^2 = 4n$ include the scalar contribution through $H(0)$. The last line and the weight-two term retain the cusp and Eisenstein corrections.

For $N = 1$, $D_1 = 1$, $\Psi_1 = 1$, and the even-weight formula reduces to

$$\delta_{k,2}\sigma_1(n) - \frac{1}{2} \sum_{t^2\le 4n} P_{k-2}(t,n) H(4n-t^2) - \frac{1}{2} \sum_{ad=n} \min(a,d)^{k-1}.$$

At $n = 1$, $k = 2$, the class-number sum is $2H(0) + 2H(3) + H(4) = 1$. The three contributions are 1, $-1/2$, $-1/2$, giving $\dim S_2(SL_2(\mathbb{Z})) = 0$. At $k = 12$, the class-number sum is -3, and the trace is $3/2 - 1/2 = 1$. These finite examples check the normalization. They do not replace the level-dependent formula.

A universal arithmetic trace family needs a coefficient space over its actual weight and level base. Its specializations to cusp-form spaces must commute with every T_n. Matching the traces does not construct the eigenobjects, their extensions, or their categorical endomorphisms.

21.21 Trace functionals and their analytic domains

To compare an ordered bar character with an analytic trace, first turn its homology into a scalar on a specified domain. For example, suppose H_i are Hilbert realizations and $T_i(s)$ are holomorphic trace-class families on a half-plane Ω, with $\sum_i \|T_i(s)\|_1$ locally uniformly convergent. Then

$$Z(s) = \sum_i (-1)^i \operatorname{Tr} T_i(s)$$

is holomorphic there. Continuity of trace and locally uniform convergence prove the assertion. A regularized character needs its own prescription and normalization. Homology itself is not this scalar.

Suppose a comparison identifies the local factors and proves $Z(s) = \xi(s)$ after the archimedean and polar normalizations. If both sides have meromorphic continuations, the identity theorem preserves that equality. The argument principle then computes the same zero distribution. It does not add an independent Hilbert spectral realization.

A common compact test class

Take an even $f \in C_c^\infty(\mathbb{R})$ and define

$$h(z) = \int_{\mathbb{R}} f(u) e^{izu}\, du, \qquad H(s) = h(-i(s - \tfrac{1}{2})).$$

The function h is entire and rapidly decreasing in each horizontal strip. With $\psi_\Gamma = \Gamma'/\Gamma$, the compact Weil explicit formula is

$$\begin{aligned}\sum_\rho H(\rho) &= h(i/2) + h(-i/2) \\ &\quad + \frac{1}{2\pi} \int_{\mathbb{R}} h(t) \big(\operatorname{Re} \psi_\Gamma(1/4 + it/2) - \log \pi\big)\, dt \\ &\quad - \sum_p \sum_{j \geq 1} (\log p) p^{-j/2} \big(f(j \log p) + f(-j \log p)\big).\end{aligned}$$

Zeros are counted with multiplicity, without assuming they are on the critical line. The prime sum is finite because f is compactly supported. Rapid decay in strips and the classical $O(T \log(T + 2))$ zero count give absolute convergence of the zero sum.

The gamma integral converges by its logarithmic growth.

For a positively oriented rectangle Γ_T avoiding zeros on its boundary, the argument principle gives

$$\frac{1}{2\pi i} \int_{\Gamma_T} H(s) \frac{\xi'(s)}{\xi(s)}\, ds = \sum_{\rho \text{ inside } \Gamma_T} H(\rho).$$

The entire function ξ is nonzero at zero and one. Its logarithmic derivative has no polar residues there. If one instead uses Λ_ζ, its poles at zero and one contribute with negative multiplicity. The choice determines the boundary terms.

When an independent arithmetic construction has $Z = \xi$ on the proved common domain, its argument-principle functional is the one just displayed. A comparison with a distributional scaling trace also requires equality of that trace on this same test class, including the gamma, prime and polar terms. No expansion of $-\zeta'/\zeta$ on the critical line is used: its Dirichlet series starts in $\operatorname{Re} s > 1$.

A finite trace difference with a negative value

Even complete unitary scattering does not force a positive trace distribution. On $L^2(\mathbb{R}, d\lambda) \oplus \mathbb{C}$, take

$$A_0 = M_\lambda \oplus 0, \qquad A_1 = M_\lambda \oplus a, \qquad a > 0,$$

with common maximal domain $\operatorname{Dom} M_\lambda \oplus \mathbb{C}$. Their difference is positive and rank one.

Both absolutely continuous subspaces are the first summand. The wave operators are the identity there, so scattering is complete and its scattering operator is one.

Nevertheless,

$$\operatorname{Tr}(h(A_1) - h(A_0)) = h(a) - h(0) = \int_0^a h'(\lambda)\, d\lambda.$$

The spectral shift is $\mathbf{1}_{(0,a)}$. The corresponding trace distribution is $\delta_a - \delta_0$.

This distribution is negative on an explicit Fourier square. Choose $\phi \in C_c^\infty(\mathbb{R})$ with $\int \phi = 1$, and put $f(u) = \phi(u) + \phi(u - \pi/a)$. Then

$$h_f(z) = (1 + e^{i\pi z/a}) h_\phi(z), \qquad |h_f(a)|^2 - |h_f(0)|^2 = -4.$$

The scattering phase is still $e^{-2\pi i \mathbf{1}_{(0,a)}} = 1$ almost everywhere. Thus its unit modulus does not determine the sign of the trace functional.

A restriction to a smaller test space needs its own comparison. If a Hermitian form is known on $\ker m$ for a moment functional m, and $m(\psi) = 1$, write $f = f_0 + m(f)\psi$. Then

$$q(f,f) = q(f_0,f_0) + 2\,\mathrm{Re}\big(\overline{m(f)}\, q(f_0,\psi)\big) + |m(f)|^2 q(\psi,\psi).$$

The complementary value and mixed pairing are needed to extend a positivity statement. They cannot be discarded by a mean-zero reduction. The complete arithmetic Weil form and its spectral comparison retain these terms.

21.22 The enhanced arithmetic comparisons

The constructions above provide several actual traces: a mixed complex, a marked categorical closure, finite Frobenius contractions, Gibbs traces, and classical Hecke traces. Each has its own coefficients, unit, completion and evaluation. Their arithmetic comparison must retain those data.

The enhanced Hall question asks for a functor realizing the correspondence operations and their collision coherences on an independently defined arithmetic carrier. The marked-center question asks for the map preserving the chosen center, object and action. Nonlinear norm transport requires its maps on parameters and their multiplicative distribution laws. These are different from constructing all-genus sewing.

An integral Frobenius trace comparison further requires the commuting square between the chiral operations, their arithmetic sheaf realizations, the Frobenius actions and the trace evaluations. An automorphy comparison must identify the actual chiral functions with independently arithmetic ones, including their Euler factors, analytic continuation, functional equation and bounds. Neither follows from an equal scalar character.

These maps are the remaining mathematical questions. The finite counterexamples specify what a comparison must preserve. The positive constructions specify data available for such a comparison. The gluing maps, their inverses where an equivalence is claimed, and their compatibility under sewing still have to be constructed on the stated arithmetic domains.

References

For the dg mixed complex and its Morita invariance, see Keller, *On differential graded categories*, Section 5.3, Theorem 5.2. The categorical perfect-stack and Hecke character theorems have their separate sources and hypotheses in Section 18.29.

See Bost–Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*. The phase transition is in Theorems 5 and 25 and Section 7. The spherical norm uses the unramified Whittaker formula of Casselman–Shalika, *The unramified principal series of p-adic groups II: The Whittaker function*. For the global volume normalization, see Lai, *Tamagawa number of reductive algebraic groups*, Section 1.7 and Theorem 7.1.

The holomorphic parameter bounds are Deligne, *La conjecture de Weil I*, Theorem 8.2, and Deligne–Serre, *Formes modulaires de poids 1*, Theorem 4.1 and Corollary 4.2. The Tate integrals and functional equation use Tate, *Fourier analysis in number fields and Hecke's zeta-functions*, in Cassels–Fröhlich, *Algebraic Number Theory*. The compact explicit formula uses the normalization of Theorem 78.1.2.

The fixed-point sum and hook specialization are Nekrasov–Okounkov, *Seiberg–Witten theory and random partitions*, arXiv:hep-th/0306238, Sections 3.1–3.2. The matter and rank-two gauge/Liouville

conventions are Alday–Gaiotto–Tachikawa, *Liouville correlation functions from four-dimensional gauge theories*, arXiv:0906.3219, including Appendix B.1. The geometric transfer operators and divisor theorem are Maulik–Okounkov, *Quantum groups and quantum cohomology*, arXiv:1211.1287, Theorem 1.3.1. For the Liouville action and heavy/light limits, see Harlow–Maltz–Witten, *Analytic continuation of Liouville theory*, arXiv:1108.4417, Section 2.

The analytic sewing input is Gui, *Convergence of sewing conformal blocks*, arXiv:2011.07450, Theorem 13.1, with its component-marking hypothesis. The finite fusion and conformal realizations used here are constructed in Chapters 15 and 16. The level-dependent Hecke formula is Popa, *On the trace formula for Hecke operators on congruence subgroups, II*, arXiv:1706.02691, Theorem 5. Its conventions include indices divisible by the level.

Chapter 22

Arithmetic Chern–Simons and boundary realization

A gauge theory assigns an amplitude to a field and sums over fields when regions are glued. These are two different operations. For the group $C_2 = \{1, -1\}$, the sign representation has Wilson values 1 and -1 on the two circle holonomies. With trivial action their normalized average is zero. Specifying one holonomy instead gives one of the two values. An arithmetic Wilson comparison must determine which operation produces the Frobenius trace.

Finite gauge groups make the distinction exact. Automorphisms determine the integration measure, a three-cocycle determines the action, and transgression determines the boundary line. Over a number field, arithmetic duality supplies a different integration map in cohomological degree three. It gives a scalar Chern–Simons invariant and torsors over local representation spaces. In the abelian case it also gives a linking pairing whose Gaussian sum computes Wilson correlations. These are constructions on specified arithmetic objects.

We develop these finite theories before passing to congruence towers. The transition maps expose the lifting measures required by a profinite limit. A free holomorphic–topological model then supplies an explicit physical boundary comparison, with its current algebra, coefficient complex and Gaussian cap. The final question is to construct arithmetic bulk and boundary objects for which these operations agree. Neither an Euler product nor a boundary character supplies those maps.

The prerequisites are group cohomology, finite groupoids and the coefficient and trace distinctions of Chapters 10 and 21. Continuous cochains are used for profinite groups. Tensor products are algebraic unless a completion is specified.

22.1 Finite gauge fields and their integration

A *finite groupoid* is a groupoid with finitely many isomorphism classes and finite automorphism groups. Its integral of an isomorphism-invariant complex function is

$$\int_{\mathcal{X}} f = \sum_{[x]\in\pi_0\mathcal{X}} \frac{f(x)}{|\operatorname{Aut}_{\mathcal{X}}(x)|}, \qquad |\mathcal{X}| = \int_{\mathcal{X}} 1. \tag{22.1}$$

The symbol BG denotes the one-object groupoid with automorphism group G. Thus $|BG| = 1/|G|$. For a finite G-set X, the action groupoid $X//G$ has objects X and arrows the action of G. Orbit–stabilizer gives

$$\int_{X//G} f = \frac{1}{|G|} \sum_{x\in X} f(x).$$

For a connected space with fundamental group Γ, finite gauge fields with group G form $\mathcal{F}_G = \operatorname{Hom}(\Gamma, G)//G$, whenever this groupoid is finite. An object is a homomorphism σ. Its automorphism group is the centralizer of $\sigma(\Gamma)$. In the arithmetic case Γ is a specified profinite fundamental group

and the homomorphisms are continuous. Ramification conditions are part of that specification. They cannot be recovered from the finite group alone.

Integration over homotopy fibers

Let $p : \mathcal{X} \to \mathcal{Y}$ be a functor of finite groupoids. Its *homotopy fiber* at y has objects (x, ϕ), where $\phi : p(x) \xrightarrow{\sim} y$. An arrow $a : (x, \phi) \to (x', \phi')$ satisfies $\phi' p(a) = \phi$. Define

$$(p_! f)(y) = \int_{\mathcal{X}_y} f(x), \qquad (p^* g)(x) = g(p(x)).$$

Keeping ϕ is necessary for the measure.

THEOREM 22.1.1 *(Finite integration).* For functors of finite groupoids the following identities hold:

$$\int_{\mathcal{Y}} p_! f = \int_{\mathcal{X}} f, \qquad (qp)_! = q_! p_!, \qquad p_!(f\ p^* g) = (p_! f)\, g.$$

In a homotopy Cartesian square

$$\begin{array}{ccc} \mathcal{X}' & \xrightarrow{q'} & \mathcal{X} \\ {\scriptstyle p'}\downarrow & & \downarrow{\scriptstyle p} \\ \mathcal{Y}' & \xrightarrow[q]{} & \mathcal{Y}, \end{array}$$

one has $q^* p_! = p'_! q'^*$.

Proof. Fix an isomorphism class $[x]$ and a representative y of $[p(x)]$. The set $\mathrm{Iso}(p(x), y)$ has $|\operatorname{Aut} y|$ elements. The contribution of the component over $[x]$ to $\mathcal{X}_y$ is its action groupoid under $\operatorname{Aut} x$, and hence has mass $|\operatorname{Aut} y|/|\operatorname{Aut} x|$. Integration over $[y]$ cancels the numerator and gives (22.1).

For a composite, an object in an iterated fiber consists of x, y, an isomorphism $p(x) \to y$, and an isomorphism $q(y) \to z$. Composing the latter with the image of the former identifies the iterated fiber with the fiber of qp, with its automorphisms. Applying the same count proves composition. The projection formula follows because $g(p(x)) = g(y)$ in the fiber. Finally, the fiber of p' at y' is equivalent to $\mathcal{X}_{q(y')}$. This equivalence preserves the integrand and proves base change. □

For example, $\mathrm{pt} \times^h_{BG} \mathrm{pt}$ is the discrete set G. The two trivial objects must be supplied with an isomorphism, and that is an element of G. Its mass is $|G|$. The pullback of the two sets of isomorphism classes is one point and has mass one. It gives the wrong gluing factor as soon as G is nontrivial.

Boundary lines and adjoints

A Hermitian line L on a groupoid $\mathcal{Y}$ assigns a Hermitian one-dimensional space to each object and unitary transport to each arrow. The transport is functorial. Its invariant sections form a Hilbert space

$$H(\mathcal{Y}, L) = \Gamma(\mathcal{Y}, L), \qquad \langle s, t \rangle = \sum_{[y]} \frac{\langle s(y), t(y) \rangle_{L_y}}{|\operatorname{Aut} y|}. \tag{22.2}$$

The pairing is linear in its first variable. If an automorphism acts nontrivially on L_y, every invariant section vanishes at y. The corresponding component contributes no state.

Pullback of sections along p has an adjoint. For $s \in H(X, p^*L)$, it is

$$(p_!s)(y) = \sum_{[(x,\phi)]\in\pi_0 X_y} \frac{L(\phi)s(x)}{|\operatorname{Aut}(x,\phi)|}. \tag{22.3}$$

The transport makes each summand lie in L_y. Fubini and unitarity give $\langle p_!s, t\rangle = \langle s, p^*t\rangle$. Thus orientation reversal, when it conjugates the action and reverses the field span, reverses the corresponding operator by its Hilbert adjoint.

This integral is distinct from categorical pushforward of a representation. For $p : BG \to \mathrm{pt}$, the invariants of the trivial complex representation are $\mathbb{C}$, of dimension one. The integral of the constant function is $1/|G|$. Both operations exist. Identifying one with a trace of the other requires a specified trace construction and normalization.

Circle and surface counts

For the circle, $\mathcal{F}_G(S^1) = G//G$, with conjugation action, so its mass is one. The space of class functions on this groupoid has dimension the number of conjugacy classes. These two numbers already distinguish a partition mass from a state-space dimension.

For a closed oriented genus-g surface, the untwisted mass is

$$|\mathcal{F}_G(\Sigma_g)| = \frac{1}{|G|}\#\left\{(a_i,b_i)_{i=1}^g : \prod_i [a_i,b_i] = 1\right\} = |G|^{2g-2}\sum_{\rho\in\widehat{G}} (\dim\rho)^{2-2g}. \tag{22.4}$$

Here $\widehat{G}$ is the set of irreducible complex representations.

To verify the formula, put $K = \sum_{a,b\in G}[a,b] \in \mathbb{C}[G]$. For an irreducible representation of dimension d, averaging $\rho(a)\rho(b)\rho(a)^{-1}$ over a gives $|G|\chi_\rho(b)I/d$. Character orthogonality then gives $\rho(K) = |G|^2 I/d^2$. The coefficient of the identity in K^g is $|G|^{-1}$ times its regular trace. Since ρ occurs d times in the regular representation, division by the additional $|G|$ gives (22.4).

For S_3, whose irreducible dimensions are 1, 1, 2, the torus and genus-two masses are respectively 3 and 81. In three dimensions a different count occurs:

$$|\mathcal{F}_G(T^3)| = \frac{\#\{(a,b,c) : a,b,c \text{ commute pairwise}\}}{|G|} = \sum_{[a]} \#\widehat{C_G(a)}.$$

For fixed a, the number of commuting pairs in its centralizer is $|C_G(a)|$ times the number of its conjugacy classes. Summing over the class of a proves the equality. For S_3 the three centralizers are S_3, C_2, C_3, and the answer is $3 + 2 + 3 = 8$.

22.2 Three-cocycles and finite topological gluing

Let G be finite. A normalized multiplicative three-cocycle $\omega : G^3 \to U(1)$ is one when any argument is the identity and satisfies

$$\omega(h,k,l)\omega(g,hk,l)\omega(g,h,k) = \omega(gh,k,l)\omega(g,h,kl). \tag{22.5}$$

For an oriented triangulated closed three-manifold, assign $g_{ij} \in G$ to oriented edges, with $g_{ji} = g_{ij}^{-1}$ and $g_{ij}g_{jk} = g_{ik}$ on each face. The action phase is

$$a(g) = \prod_{[0123]} \omega(g_{01}, g_{12}, g_{23})^{\epsilon_{0123}}, \qquad Z_\omega(M) = |G|^{-N_0}\sum_{\text{flat } g} a(g),$$

where N_0 is the number of vertices and ϵ_{0123} records orientation. The factor $|G|^{-N_0}$ integrates the action of vertex gauge transformations. It remains necessary when the action has stabilizers.

The two-to-three Pachner identity is precisely (22.5). In a one-to-four move, an extension of a fixed flat boundary field has $|G|$ choices for the new vertex label. The cocycle identity makes each extended phase equal to the original phase, and the new vertex factor cancels these choices. Normalization handles degenerate simplices. This gives the finite Dijkgraaf–Witten state sum. Boundary amplitudes require the line determined by the same cocycle.

The degree of transgression

Suppose a space K and its field space admit an evaluation map $\mathrm{ev} : K \times \mathcal{F}_G(K) \to BG$. For a chain $c \in C_d(K)$, define the transgression of an additive cochain u by $(\tau_c u)(a) = \mathrm{ev}^* u(c \times a)$. The cross-product boundary formula gives

$$\delta \tau_c u = (-1)^d \big(\tau_c \delta u - \tau_{\partial c} u\big). \tag{22.6}$$

Indeed, evaluating δu on $c \times a$ gives $u(\partial c \times a) + (-1)^d u(c \times \partial a)$. Solving for the second term proves the identity.

For a degree-three cocycle, integration over a closed three-manifold gives a degree-zero phase. Over a closed surface it gives a degree-one cocycle, hence a flat line on the field groupoid. Over a circle it gives a degree-two cocycle, hence twisting data for representations of that field groupoid. At a point the degree-three class supplies an associator.

The degrees are

dimension integrated	degree remaining	recipient
3	0	action phase
2	1	line on fields
1	2	twisted representation category
0	3	monoidal associator.

In particular, the boundary of a tubular neighborhood of a line in a three-manifold is a surface. Its transgression is a line on surface fields. The twisted category belongs one categorical degree lower, at a circle, with the indicated transgression.

For a three-chain with boundary, (22.6) says that the boundary of its action cochain is the boundary-line cocycle. Its exponential is therefore a section of that line, rather than an invariant scalar on unrestricted boundary fields. Changing the orientation conjugates the line and the phase. These constructions are the finite topological ones of Freed–Quinn, Theorems 1.7 and 2.13. They use oriented manifolds and the actual integration of their chains. An arithmetic analogue must supply its own duality and boundary integration.

THEOREM 22.2.1 *(Finite gluing on specified field data).* Suppose regions have finite field groupoids, boundary Hermitian lines, and invariant action sections. Assume gluing regions gives the homotopy pullback of their field groupoids, and that the action sections contract under the boundary-line pairing.

Assign to a boundary its Hilbert space (22.2). For a region from Σ_0 to Σ_1, let r_i be restriction of fields and let a be the section of $r_0^* L_0^* \otimes r_1^* L_1$. Then

$$T_M s = (r_1)_! \big(a\, r_0^* s\big) \tag{22.7}$$

composes under gluing.

A cylinder whose field restriction is the identity and whose section is the identity gives the identity operator. Disjoint unions give tensor products. Reversing a region and conjugating its action gives T_M^*.

Proof. In the composite of two expressions (22.7), base change replaces restriction followed by integration with integration over the homotopy pullback. The two action sections multiply and contract there to the glued section. Fubini combines the two integrations. The cylinder and disjoint-union assertions follow from the identity functor and multiplication of automorphism orders. Finally, (22.3) is adjoint to pullback, and the adjoint of multiplication by a line section uses its Hermitian conjugate. This reverses the span and proves the last assertion. □

Descent of fields supplies the homotopy pullback in this theorem. The action sections and their contraction are additional hypotheses. An equality of field groupoids without those sections does not determine the glued amplitude.

22.3 The arithmetic action and its boundary torsor

Let F be a totally imaginary number field containing μ_n, put $X = \operatorname{Spec} O_F$, and fix $\zeta : \mathbb{Z}/n \xrightarrow{\sim} \mu_n$. Arithmetic duality gives an invariant map

$$\operatorname{inv}_\zeta : H^3_{\text{ét}}(X, \mathbb{Z}/n) \xrightarrow{\sim} \tfrac{1}{n}\mathbb{Z}/\mathbb{Z}.$$

The choice of ζ identifies the constant coefficients with the roots-of-unity coefficients in this map.

Write $\pi = \pi_1^{\text{ét}}(X)$. For a finite group G and $c \in Z^3(G, \mathbb{Z}/n)$, define

$$\operatorname{CS}_c(\sigma) = \operatorname{inv}_\zeta\big(j^3\sigma^*[c]\big), \qquad j^3 : H^3_{\text{cont}}(\pi, \mathbb{Z}/n) \longrightarrow H^3_{\text{ét}}(X, \mathbb{Z}/n). \tag{22.8}$$

This is Kim's finite arithmetic action. The displayed comparison j^3 is used as a map. No equivalence between the entire group cohomology and étale cohomology is needed.

The value in (22.8) is a scalar in $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$. Before evaluation, the input is a degree-three class. The coefficient sheaf $\mathbb{G}_m$ appearing in arithmetic duality is not the finite gauge group G. Nor does cohomological degree three make the one-dimensional scheme X a real three-manifold.

Conjugation acts trivially on the pulled-back cohomology class, so the action descends to the finite field groupoid $\operatorname{Hom}_{\text{cont}}(\pi, G)//G$. Its finite partition value, with the measure already fixed, is

$$Z_{G,c}(X) = \sum_{[\sigma]} \frac{\exp(2\pi i \operatorname{CS}_c(\sigma))}{|C_G(\sigma(\pi))|}.$$

Finiteness here is the arithmetic finiteness of the specified representation set, as follows from bounded-degree extension finiteness with the stated ramification conditions. A different arithmetic region needs its own such assertion.

An explicit gauge homotopy

The boundary construction needs conjugation on cochains, including its composition. Use the inhomogeneous differential with trivial G-action on $\mathbb{Z}/n$. For a two-cochain b,

$$(db)(g,h,k) = b(h,k) - b(gh,k) + b(g,hk) - b(g,h).$$

Put $\mathrm{Ad}_a\, g = a g a^{-1}$. For a normalized three-cocycle c, define

$$h_a(g,h) = c(a^{-1}, \mathrm{Ad}_a\, g, \mathrm{Ad}_a\, h) - c(g, a^{-1}, \mathrm{Ad}_a\, h) + c(g, h, a^{-1}),$$
$$H_{a,b}(g) = c(b^{-1}, a^{-1}, \mathrm{Ad}_{ab}\, g) - c(b^{-1}, \mathrm{Ad}_b\, g, a^{-1}) + c(g, b^{-1}, a^{-1}). \tag{22.9}$$

The cochain identities are

$$dh_a = c \circ \mathrm{Ad}_a - c, \qquad h_{ab} - h_a \circ \mathrm{Ad}_b - h_b = dH_{a,b}. \tag{22.10}$$

They follow by expanding the differential and cancelling the three-cocycle identities on the successive simplices of a prism. The alternating signs in (22.9) are essential. These are also the explicit conjugation homotopies in Chung–Kim–Kim–Park–Yoo, Appendix B, in the inhomogeneous convention above.

If $db = \sigma^* c$, conjugation by a sends the primitive to $b + \sigma^* h_a$, a primitive for $(\mathrm{Ad}_a\, \sigma)^* c$. The second identity in (22.10) makes these maps compose after quotienting primitives by two-coboundaries. Thus the construction retains the gauge action on the torsor itself.

Local torsors and the global section

Let S be a nonempty finite set of finite places containing every place dividing n. Put

$$U = \operatorname{Spec} \mathcal{O}_F[1/S], \qquad \pi_S = \pi_1^{\text{ét}}(U).$$

Choose restriction maps from $\pi_v = \mathrm{Gal}(\overline{F}_v / F_v)$ to π_S. For a local field $\sigma_v : \pi_v \to G$, consider

$$P_v(\sigma_v) = \{ b_v \in C^2(\pi_v, \mathbb{Z}/n) : db_v = \sigma_v^* c \} / B^2. \tag{22.11}$$

Local cohomological dimension gives $H^3(\pi_v, \mathbb{Z}/n) = 0$, so this set is nonempty. The difference of two primitives is a two-cocycle. Consequently $P_v(\sigma_v)$ is a torsor for $H^2(\pi_v, \mathbb{Z}/n)$, identified by local duality and ζ with $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$.

Take the product of these torsors and push it out along the sum of the local invariant maps. The result is a $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$-torsor $P_S(\sigma_S)$. Here the pushout of a torsor means its contracted product along the specified homomorphism of acting groups. Exponentiation along $t \mapsto e^{2\pi i t}$ makes the associated Hermitian complex line. Formula (22.9) defines its maps under local conjugations.

For a global $\sigma : \pi_S \to G$, the same cohomological dimension theorem gives a primitive b_+ with $db_+ = \sigma^* c$. Its local restrictions determine an element

$$\mathrm{CS}_{U,c}(\sigma) = [\mathrm{res}_v\, b_+]_{v \in S} \quad \text{in } P_S(\mathrm{res}\, \sigma). \tag{22.12}$$

Changing b_+ by a global two-cocycle does not change this element: global reciprocity says that the sum of its local invariants is zero. Changing it by a coboundary already has no effect in (22.11). The gauge homotopies make (22.12) an invariant section over the restriction functor from global to local fields.

The arithmetic input is vanishing in degree three, local invariants in degree two, and reciprocity for restrictions of global degree-two classes. It supplies the torsor and its section. Gluing arbitrary arithmetic bordisms would add the corresponding regions, field descent and compatible action-section identifications of Theorem 22.2.1.

The decomposition formula

Suppose now that σ is unramified on X. At each $v \in S$, its local restriction factors through $\pi_v/I_v \cong \widehat{\mathbb{Z}}$. This group has cohomological dimension one for the finite coefficients in use. Thus its degree-three cocycle has a primitive $b_{-,v}$ unique modulo two-coboundaries. Inflation gives a canonical class of unramified local primitives in (22.11).

Theorem 22.3.1 *(Arithmetic decomposition).* With the coefficients and hypotheses above,

$$\mathrm{CS}_c(\sigma) = \sum_{v \in S} \mathrm{inv}_v[b_{-,v} - \mathrm{res}_v\, b_+].$$

Each bracket is a two-cocycle class. The right side is independent of the chosen global primitive and of the representatives of the local primitive classes.

Proof. Use the compact-support mapping fiber, with differential

$$C_c^j = C^j(\pi_S, \mathbb{Z}/n) \oplus \prod_{v \in S} C^{j-1}(\pi_v, \mathbb{Z}/n), \qquad d(a,b) = (da, \mathrm{res}\, a - db).$$

The pair $(\sigma^* c, b_-)$ represents the unramified class on X. Subtracting $d(b_+, 0)$ gives $(0, b_- - \mathrm{res}\, b_+)$. Arithmetic duality identifies its invariant with the sum of the displayed local invariants. Reciprocity removes a change in b_+, and local coboundaries remove a change in b_-. This is the mapping-fiber proof of Chung–Kim–Kim–Park–Yoo, Theorem 1.1 and §4. □

Both primitive classes belong to the same boundary torsor. Their difference is a scalar. Reversing the difference reverses the action, so the compact-support differential fixes a necessary sign.

Compact p-adic coefficients

There is also a torsor construction with compact coefficients. For an odd prime p, write $\Gamma = \mathbb{Z}_p^\times = \Delta \times \Gamma_1$, where Δ has order $p-1$ and $\Gamma_1 = 1 + p\mathbb{Z}_p$. Let A be a p-adic Lie group with a homomorphism $t : A \to \Gamma$, and let χ be the cyclotomic character. Fix an actual continuous automorphism σ of Γ and restrict to representations ρ satisfying $t\rho = \sigma\chi$. Gauge transformations are taken in $\ker t$.

Let $\mathbb{Z}_p[[\Gamma]]_\rho$ have the left action through $t\rho$. A compatible choice of roots of unity gives a basis ζ of $\mathbb{Z}_p(1)$. There is an equivariant isomorphism

$$\mathbb{Z}_p[[\Gamma]]_\rho \longrightarrow \Lambda = \mathbb{Z}_p[[\Gamma]] \widehat{\otimes} \mathbb{Z}_p(1), \qquad [\gamma] \longmapsto [\sigma^{-1}\gamma] \otimes (\sigma^{-1}\gamma)\zeta. \tag{22.13}$$

The first factor on the right has left action through χ, and the second has the Tate action. To check equivariance, put $u = \chi(g)$. The image of $[\sigma(u)\gamma]$ is $[u\sigma^{-1}\gamma] \otimes u(\sigma^{-1}\gamma)\zeta$, which is exactly the action of g on the displayed image.

For a continuous class $c \in H^3(A, \mathbb{Z}_p[[\Gamma]])$, use (22.13) on the pulled-back cocycle. Under vanishing of degree-three continuous cohomology with these compact coefficients, its local primitives form $H^2(\pi_v, \Lambda)$-torsors. The augmentation $\Lambda \to \mathbb{Z}_p(1)$, followed by the local invariant $H^2(\pi_v, \mathbb{Z}_p(1)) \cong \mathbb{Z}_p$, pushes their product to a $\mathbb{Z}_p$-torsor. A global primitive gives a section by the same reciprocity argument. Gauge maps use the conjugation prism with this coefficient action.

A sufficient setting uses finite quotient modules that compute the continuous compact-coefficient cochains by an inverse sequence. Assume finite degree-two cohomology and vanishing degree-three cohomology. The finite degree-two systems satisfy the Mittag–Leffler condition. The derived inverse-limit exact sequence then gives degree-three vanishing. The augmentations, local duality and reciprocity

must be compatible with this sequence. These hypotheses describe the coefficient passage used in the p-adic construction of Arithmetic Chern–Simons II, §3.

The automorphism condition in (22.13) matters. A p-adic unit exponent acts invertibly on Γ_1, but does not automatically define an automorphism of the full Γ. At $p = 5$, the integer exponent 2 is a 5-adic unit and sends -1 to 1.

For an integer exponent m, the full power map is an automorphism precisely when $\gcd(m, p(p-1)) = 1$. One may instead specify a power automorphism on Γ_1 and an independent automorphism on Δ. The resulting torsor construction is positive on this domain. It supplies no complex Hilbert measure without additional scalar and pairing data.

22.4 Abelian linking and Wilson correlations

The abelian theory has a computable arithmetic correlation, beyond the definition of a finite sum. Assume now $\mu_{n^2} \subset F$ and retain the trivialization ζ. Put

$$V = H^1(X, \mathbb{Z}/n), \qquad V^\vee = \operatorname{Hom}(V, \tfrac{1}{n}\mathbb{Z}/\mathbb{Z}).$$

Artin–Verdier duality identifies $V^\vee$ with $\operatorname{Ext}^2_X(\mathbb{Z}/n, \mathbb{G}_m)$ and with $\operatorname{Cl}(F)/n$. Let δ be the Bockstein from $0 \to \mu_n \to \mu_{n^2} \to \mu_n \to 0$. Define $D : V \to V^\vee$ by

$$(Dy)(x) = B(x, y) := \operatorname{inv}_\zeta\big(x \smile \delta(\zeta_* y)\big), \tag{22.14}$$

where the notation for the invariant includes the indicated coefficient pairing.

Proposition 22.4.1 *(Arithmetic linking pairing).* The form B is symmetric. Its radical is $K = \ker D$. For $\xi, \eta \in \operatorname{im} D$, the expression

$$\operatorname{ht}_n(\xi, \eta) = B(x, y), \qquad Dx = \xi, \quad Dy = \eta,$$

is well defined and symmetric. A homologically n-trivial ideal means an ideal whose class modulo n lies in $\operatorname{im} D$. It need not have zero class modulo n.

Proof. The compatible trivialization modulo n^2 identifies the constant and roots-of-unity Bocksteins. Their cup-product identity in degree one is

$$\delta_2(x \smile \zeta_* y) = \delta' x \smile \zeta_* y - x \smile \delta(\zeta_* y).$$

In this arithmetic setting $\delta_2 = 0$. Indeed the n-adic duality group $H^3(X, \mathbb{Z}_n(1)) \cong \mathbb{Z}_n$ is torsion free, so reduction in degree two from $\mathbb{Z}_n(1)$ to μ_n is surjective. It factors through μ_{n^2}, which makes the Bockstein zero. Graded commutativity between degrees two and one now gives $B(x, y) = B(y, x)$. This is the argument of Chung–Kim–Kim–Pappas–Park–Yoo, Lemmas 2.1–2.2.

Perfect duality shows that $B(x, y) = 0$ for all x exactly when $Dy = 0$. Symmetry gives the same radical in the other variable. Changing either lift by an element of K therefore leaves the height unchanged. □

Let $G = \mathbb{Z}/n$, and let $u \in H^1(G, \mathbb{Z}/n)$ be the identity character. For the three-class $c = u \smile \delta' u$, naturality of the Bockstein identifies the arithmetic action with $\mathrm{CS}_c(x) = B(x, x)$. An ideal class $\xi \in V^\vee$ supplies the Wilson character

$$W_\xi(x) = \exp(2\pi i \langle x, \xi \rangle).$$

Under class field theory this evaluates the unramified character x on the Artin class of the ideal. Its correlation in the quadratic action is therefore an actual arithmetic finite sum.

Theorem 22.4.2 *(Finite Gaussian correlation).* Let p be odd and let V be an a-dimensional $\mathbb{F}_p$-vector space. Let $B : V \times V \to \mathbb{F}_p$ be symmetric with radical K of dimension b. Write $\psi(t) = e^{2\pi i t/p}$, $r = a - b$, and $G_p = \sum_{t \in \mathbb{F}_p} \psi(t^2)$. For $\lambda \in V^*$, put

$$S(\lambda) = \sum_{x \in V} \psi\big(B(x,x) + \lambda(x)\big).$$

If $\lambda|_K \neq 0$, then $S(\lambda) = 0$. Otherwise choose z with $\lambda(x) = B(x,z)$. If $\overline{B}$ is the induced nondegenerate form on V/K, then

$$S(\lambda) = p^b \left(\frac{\det \overline{B}}{p}\right) G_p^r \, \psi\big(-B(z,z)/4\big). \tag{22.15}$$

The determinant is taken in a basis of V/K. Its Legendre symbol is independent of that basis. The empty determinant and G_p^0 are one.

Proof. On a coset of K, the quadratic term is constant. The sum of the additive character $\psi\lambda$ over K is zero unless $\lambda|_K = 0$. In the latter case, nondegeneracy on V/K supplies z, unique modulo K.

Since 2 is invertible,

$$B(x,x) + B(x,z) = B(x + z/2, x + z/2) - B(z,z)/4.$$

Translation leaves the sum unchanged.

Diagonalize $\overline{B}$ over $\mathbb{F}_p$. Each diagonal coefficient $d \neq 0$ contributes $\sum_t \psi(dt^2) = (d/p) G_p$: counting square roots of an element gives $1 + (u/p)$, and the substitution $u \mapsto du$ proves this identity. The radical contributes p^b. Multiplying proves (22.15). □

The classical Gauss evaluation is $G_p = \epsilon_p \sqrt{p}$, with $\epsilon_p = 1$ for $p \equiv 1 \pmod 4$ and $\epsilon_p = i$ for $p \equiv 3 \pmod 4$, for the displayed additive character and the positive real square root. Thus the magnitude in (22.15) is $p^{(a+b)/2}$. The formula with this constant is Theorem 1.1 of the abelian arithmetic Chern–Simons paper.

Apply the theorem to (22.14), after identifying $\frac{1}{p}\mathbb{Z}/\mathbb{Z}$ with $\mathbb{F}_p$. If each $\xi_j = Dz_j$, then $z = \sum_j z_j$ represents $\lambda = \sum_j \xi_j$, and

$$\frac{S(\sum_j \xi_j)}{S(0)} = \psi\left(-\frac{1}{4} \sum_{i,j} \mathrm{ht}_p(\xi_i, \xi_j)\right).$$

Here $S(0) \neq 0$. In the action groupoid every abelian field has automorphism group $\mathbb{Z}/p$, so the absolute amplitude is $S(\lambda)/p$. This factor cancels in the normalized correlation. The factor $1/4$, the radical selection rule and the distinction between absolute and normalized amplitudes are all forced by the calculation.

At $p = 2$, this completion of the square is unavailable. A quadratic refinement requires separate treatment.

The linking form also has a class-field description. A $\mathbb{Z}/n$-torsor on X determines the invertible O_F-module on which the group acts through ζ. Its ideal class is the class-invariant map $\Psi : V \to \mathrm{Cl}(F)$. With the Artin convention of the duality pairing, the map D is Ψ followed by reduction modulo n. Thus the pairing is the evaluation of one unramified character on the class invariant of the other.

The proof in the same paper, Theorem 4.1, identifies the dual Bockstein with $\mathrm{Cl}(F)[n] \to \mathrm{Cl}(F)/n$ by Yoneda products. This gives the same arithmetic action, with its coefficient trivialization retained.

Local boundary conditions for linking

There is a relative version on U. For coefficient objects G on U, F_S on its specified local boundary, and a map $f : F_S \to i^*G$, its cochain complex is

$$C^j(U, G \times_S F_S) = C^j(U, G) \oplus C^j(\partial U, F_S) \oplus C^{j-1}(\partial U, i^*G),$$

with

$$d(c, b, a) = (dc, db, da + (-1)^j (f b - \operatorname{res} c)).$$

For degrees i, j, one cup product is

$$(c, b, a) \smile (c', b', a') = \big(c \smile c', b \smile b', (-1)^j a \smile f(b') + \operatorname{res}(c) \smile a'\big).$$

Expanding the differential verifies its Leibniz identity. Replacing the third component by $(-1)^j a \smile \operatorname{res}(c') + f(b) \smile a'$ changes the product of cocycles by $d(0, 0, (-1)^i a \smile a')$. Thus the two choices induce the same product in cohomology.

Take $G = \mathbb{Z}/n$, $F_S = \mu_{n^2}$, and let f be reduction followed by ζ^{-1}, with all places dividing n in S. The resulting degree-one classes specify characters with chosen local lifts to μ_{n^2}. Their connecting map lands in $H^2_c(U, \mathbb{Z}/n)$. The relative duality pairing and this connecting map define the height as above. The degree-two connecting map vanishes in this setting, and the Bockstein product identity proves symmetry. This is the relative construction in §3 of the same paper.

It also gives a local computation without confusing a knot longitude with inertia. If an ideal class $[I]$ is the connecting image of a character x, its height against $J = \prod_{v \notin S} \mathfrak{p}_v^{e_v}$ is

$$\operatorname{ht}_{S,n}(I, J) = \sum_{v \notin S} e_v \, x(\operatorname{Art}_v(\varpi_v)).$$

The Artin maps here use the normalization paired with the chosen duality invariant. If x is represented by a Kummer class, each summand is the corresponding local norm-residue symbol with that same normalization. The connecting image and reciprocity map must be fixed before replacing this formula by a Hilbert-symbol convention.

22.5 Wilson values and Frobenius

Let $\sigma : \Gamma \to G$ be a gauge field and let $\tau : G \to \mathrm{GL}(V_\tau)$ be a representation over a specified field. These maps have different sources. For a specified conjugacy class of loops γ in Γ, the Wilson function is

$$W_{\tau,\gamma}(\sigma) = \operatorname{tr} \tau(\sigma(\gamma)).$$

It is invariant under gauge conjugation.

For a complex gauge representation and an invariant phase a, its normalized expectation is

$$\langle W_{\tau,\gamma} \rangle_a = \frac{\displaystyle\int_{\mathcal{F}_G} a W_{\tau,\gamma}}{\displaystyle\int_{\mathcal{F}_G} a}, \qquad \int_{\mathcal{F}_G} a \neq 0. \tag{22.16}$$

The denominator condition is part of the definition. A complex phase can make it vanish: two discrete fields with phases 1 and -1 already give zero.

At trivial action, (22.16) is a groupoid average. It equals a selected field value when the observable is constant on the support of the weighted sum. Conditioning on one field component also gives its value. It does not concentrate on a chosen field merely because the action is zero. On $\mathcal{F}_{C_2}(S^1) = C_2//C_2$, each holonomy has weight $1/2$, so the sign character has expectation zero and square expectation one.

For fixed fields, the tensor identity is $W_{\tau\otimes\nu,\gamma} = W_{\tau,\gamma}W_{\nu,\gamma}$. Its expectation is the correlation of the two observables, not generally the product of their expectations. The sign character example proves this distinction in dimension one.

The unramified quotient

For a nonarchimedean local field F_v with residue field $\mathbb{F}_q$, write

$$1 \longrightarrow I_v \longrightarrow G_{F_v} \longrightarrow \widehat{\mathbb{Z}} \longrightarrow 1.$$

Arithmetic Frobenius is the element of the quotient acting as $x \mapsto x^q$ on residue-field extensions. Geometric Frobenius is its inverse. A lift to G_{F_v} is well defined modulo inertia. Its image in an unramified representation is therefore well defined up to conjugacy.

The tame inertia quotient is $I_v/P_v \cong \prod_{\ell\neq p} \mathbb{Z}_\ell(1)$, where P_v is wild inertia and p is the residue characteristic. It is not the unramified quotient $\widehat{\mathbb{Z}}$. In the knot analogy, inertia gives a meridian and a Frobenius lift gives a longitude. On an unramified rank-d representation every inertia element has trace d, while the Frobenius trace can be different.

Nor does $p \nmid N$ make a representation into $\mathrm{GL}_d(\mathbb{Z}/N)$ unramified. The quadratic extension $\mathbb{Q}_3(\sqrt{3})/\mathbb{Q}_3$ is totally ramified. Its quadratic character has values $\{\pm 1\} \subset \mathbb{F}_5^\times$, although $3 \nmid 5$. Unramifiedness is a condition on the kernel of the representation.

Let E_w/F_v have residue degree f. Its arithmetic Frobenius maps to the f-th power of Frobenius on the unramified quotient of F_v. The same power rule holds in geometric convention. It gives the corresponding trace-of-power identity on an unramified carrier. A map between arbitrary formal neighborhoods at different rational primes has no such interpretation. Indeed, when $p \neq q$, there is no unital $\mathbb{Z}$-algebra map $\mathbb{Z}_p \to \mathbb{Z}_q$: the integer q is a unit in the source and not in the target.

The closed immersion $\operatorname{Spec}\mathbb{F}_p \hookrightarrow \operatorname{Spec}\mathbb{Z}$ has scheme codimension one. The knot analogy is a comparison of cohomology, fundamental groups and local duality. It does not change that codimension or furnish a topological tubular neighborhood with all its gauge-theoretic operations.

Coefficient fields and congruences

The tautological representation of $\mathrm{GL}_d(\mathbb{Z}/N)$ has trace in $\mathbb{Z}/N$. For $N > 1$, there is no unital ring map from this ring to $\mathbb{C}$. Thus it is not itself a complex Wilson label. A complex representation of the same finite group has its own character, which must be compared to the modular matrix trace by additional data.

A continuous ℓ-adic representation on a finite-dimensional E-vector space, with $E/\mathbb{Q}_\ell$ finite, admits a stable $\mathcal{O}_E$-lattice when its image is compact. Fix such a lattice L and a uniformizer ϖ. The traces on $L/\varpi^r L$ are compatible and satisfy

$$\operatorname{tr}\rho(g) = \varprojlim_r \operatorname{tr}\big(\rho(g) \mid L/\varpi^r L\big) \quad \text{in } \mathcal{O}_E.$$

This is one ℓ-adic congruence tower. It does not produce reductions modulo every integer or complex characters of every congruence group.

If a trace lies in a number field, one can specify its ℓ-adic and complex embeddings. The topology still matters. Under any field embedding fixing $\mathbb{Q}$, the sequence ℓ^r, which tends to zero ℓ-adically, maps to the same real integers and diverges in $\mathbb{C}$. An abstract scalar embedding therefore cannot turn ℓ-adic convergence into complex convergence.

An arithmetic Wilson identity now has a precise form. Choose a gauge representation τ_N, a Frobenius longitude and a scalar realization. Then compare its normalized expectation with the trace of the arithmetic representation. A profinite version additionally needs the actual transition maps and convergence in the chosen scalar topology. Equality of ranks supplies none of these identities.

Power traces and Euler factors

Let F be an endomorphism of a d-dimensional vector space over a characteristic-zero field. In formal power series,

$$\det(1 - TF)^{-1} = \exp\left(\sum_{m\geq 1} \frac{\operatorname{tr}(F^m)}{m} T^m\right). \tag{22.17}$$

After scalar extension, put F in triangular form. Multiplying $(1 - T\alpha_i)^{-1}$ and expanding its logarithm gives the right side. Equivalently, differentiating logarithmically and fixing the constant term proves the identity over the original field.

Thus all power traces determine the Euler polynomial and the semisimple conjugacy class over an algebraic closure. A single trace does not. The finite-order matrices $\operatorname{diag}(1, -1)$ and $\operatorname{diag}(i, -i)$ both have trace zero, but their polynomials are $1 - T^2$ and $1 + T^2$. Even all power traces do not detect a Jordan extension: I_2 and a nontrivial unipotent Jordan block have trace two in every positive power.

Characteristic zero is also necessary for this reconstruction from unrestricted power traces. Over $\mathbb{F}_3$, the matrices I_3 and $-I_3$ have all power traces zero, while their Euler polynomials are $1 - T^3$ and $1 + T^3$. Division by m in (22.17) cannot be performed in that coefficient ring. Integral Witt coordinates retain additional data beyond their ghost traces.

For an unramified ℓ-adic representation at v, $\ell \neq p$, the chosen geometric Frobenius gives

$$L_v(s, \rho) = \det(1 - q^{-s}\rho(\operatorname{Frob}_v^{\mathrm{geom}}))^{-1}.$$

A Wilson formula for every power would determine this factor after the coefficient comparison. A Wilson formula for the first trace alone would not.

For a Weil–Deligne representation (r, N) at a ramified place, the local factor instead uses the Frobenius-stable space $(\ker N)^{I_v}$:

$$L_v(s, r, N) = \det(1 - q^{-s} r(\operatorname{Frob}_v^{\mathrm{geom}}) \mid (\ker N)^{I_v})^{-1}.$$

The conductor and epsilon factor are further local data. In particular an epsilon factor also depends on the additive character and measure convention. It is not inserted into this determinant as an unspecified correction to the first trace.

If a realization is pure of motivic weight m, its complex Frobenius eigenvalues have absolute value $q^{m/2}$. Unitary normalization multiplies them by $q^{-m/2}$. This normalization is separate from a functional-equation center of a global L-function and from an energy grading on a Fock space. For example $\mathbb{Q}_\ell(-1)$ has motivic weight two in the cohomological geometric convention. A formal scaling parameter labeled one does not change it.

An elliptic curve at five

The curve $E : y^2 = x^3 - x$ over $\mathbb{F}_5$ has affine fiber sizes 1, 1, 2, 2, 1 over $x = 0, 1, 2, 3, 4$. The point at infinity gives $\#E(\mathbb{F}_5) = 8$, so $a_5 = 5 + 1 - 8 = -2$. For $\ell \neq 5$, arithmetic Frobenius on $V_\ell E = T_\ell E \otimes \mathbb{Q}_\ell$ has trace -2 and determinant 5. Its inverse has trace $-2/5$ and determinant $1/5$.

The cohomological carrier

$$H^1_{\text{ét}}(E_{\overline{\mathbb{F}}_5}, \mathbb{Q}_\ell) = \operatorname{Hom}(V_\ell E, \mathbb{Q}_\ell)$$

therefore has geometric Frobenius trace -2 and determinant 5. Its local Euler polynomial is $1 + 2T + 5T^2$. Inertia on the good-reduction representation is trivial and has trace two, so the meridian and longitude give different observables in this example.

The Tate-module polarization takes values in $\mathbb{Q}_\ell(1)$. Dualizing gives a cohomological polarization with paired geometric eigenvalue product 5.

For a genus-g Jacobian the corresponding cohomological rank is $2g$. In a Siegel setting the standard and spin representations instead have ranks $2g + 1$ and 2^g. Their Wilson labels and Euler factors must use the actual representation.

The substitution $g = 0$ gives a rank-zero Jacobian carrier, whose trace is zero and whose symmetric-algebra character is one. A nontrivial rank-two arithmetic gauge sector requires a separate representation.

For any finite-dimensional carrier V, the graded trace on $\operatorname{Sym}(V)$ is the inverse determinant in (22.17). A Heisenberg Fock space with creation modes at every positive energy has the different character $\prod_{r \geq 1} \det(1 - q^r F)^{-1}$, when the trace is defined. The oscillator grading must therefore be specified before either expression is called a boundary character. The arithmetic equality between such a character and an Euler factor remains a map and trace comparison on the chosen realization.

22.6 Congruence towers and Hilbert limits

For a fixed rank d, reduction gives surjections $G_N = \mathrm{GL}_d(\mathbb{Z}/N) \to G_M$ when $M \mid N$. Their group limit is $\mathrm{GL}_d(\widehat{\mathbb{Z}})$. Passing from groups to fields, actions and state spaces requires more maps. In particular, a field at level M can fail to lift to level N.

For a prime-power step, put $G_r = \mathrm{GL}_d(\mathbb{Z}/\ell^r)$, $r \geq 1$. Then

$$|G_r| = \ell^{d^2(r-1)} \prod_{j=0}^{d-1} (\ell^d - \ell^j), \qquad \ker(G_{r+1} \to G_r) = 1 + \ell^r M_d(\mathbb{F}_\ell).$$

The kernel has the additive multiplication law of $M_d(\mathbb{F}_\ell)$, since $2r \geq r + 1$. It is abelian but generally not central: conjugation by the permutation matrix exchanging two basis vectors sends E_{12} to E_{21}. The cardinality formula follows by counting ordered independent columns modulo ℓ and then lifting each matrix entry. An invertible matrix modulo ℓ remains invertible over $\mathbb{Z}/\ell^r$.

The lifting measure

Consider an extension of finite groups $1 \to K \to E \xrightarrow{p} G \to 1$, and a finitely presented group Γ, or a profinite group with the specified finite representation sets. For $\rho : \Gamma \to G$, let $\mathcal{L}_\rho$ be the homotopy fiber of $\operatorname{Hom}(\Gamma, E)//E \to \operatorname{Hom}(\Gamma, G)//G$. The isomorphism to ρ is part of this fiber.

Proposition 22.6.1 *(Lifting mass).* Suppose K is abelian. If ρ has a lift, conjugation through a lift defines a Γ-module K_ρ, and

$$|\mathcal{L}_\rho| = \frac{|H^1(\Gamma, K_\rho)|}{|H^0(\Gamma, K_\rho)|}.$$

If there is no lift, the mass is zero. For compatible action phases $a_E = p^* a_G$, finite integration gives

$$Z_E = \sum_{[\rho]} \frac{a_G(\rho)|\mathcal{L}_\rho|}{|C_G(\rho(\Gamma))|}. \tag{22.18}$$

Proof. Choose a lift $\widetilde{\rho}$. Any other lift has the form $z(g)\widetilde{\rho}(g)$, and the homomorphism law is precisely the one-cocycle equation for z with values in K_ρ. The homotopy fiber is equivalent to these lifts modulo K-conjugation. Lift an isomorphism to the target field into E to make it the identity.

Conjugation changes z by a one-coboundary. Its stabilizer is $H^0(\Gamma, K_\rho)$. This proves the mass formula.

The empty case is immediate. Equation (22.18) is Theorem 22.1.1 with the phase constant on each fiber. □

For a central kernel $K = (\mathbb{F}_\ell)^e$ with trivial action, the mass is $|K|^{b_1 - b_0}$, where the b_i are the dimensions of $H^i(\Gamma, \mathbb{F}_\ell)$. For the point it is $1/|K|$, not one.

Centrality also does not remove the obstruction to lifting. Take $E = C_4$, $G = C_2$, and $\Gamma = C_2$. The trivial homomorphism to G has two lifts, each with two automorphisms in the fiber, so its mass is one. The identity homomorphism has no lift: an odd element of C_4 cannot be the image of an element of order two. Its mass is zero. No constant fiber factor works for both fields.

If all masses equal a specified $m_{N,M}$, then $Z_N = m_{N,M} Z_M$. For the chosen normalization $\widetilde{Z}_N = |G_N|^{-\chi} Z_N$, one obtains

$$\widetilde{Z}_N = |K|^{-\chi} m_{N,M} \widetilde{Z}_M.$$

The exponent χ must come from a specified cohomology theory. Compatibility requires the displayed factor to be one. The equation uses no division by a possibly zero partition value, and a scalar equality alone does not define transition maps on state spaces.

The phases also require compatible levels. Pullback of a fixed cocycle from a quotient gives one such datum on all finer quotients. If one instead requires $c_N = p_{N,M}^* c_M$ for every $M \mid N$, including $M = 1$, every normalized three-class is zero, since the bottom group is trivial. Nontrivial continuous finite-coefficient classes are naturally represented on a sufficiently fine finite quotient and pulled back above it. More general transition laws require their own cochains and coherences. A complex scalar called a level is not by itself such a family of finite-group three-classes.

A positive profinite Hilbert construction

There is a direct limit of finite Hilbert spaces when the measure and transition maps are known. Let $G_{r+1} \twoheadrightarrow G_r$ be a sequence of finite groups, with kernels K_r, and use uniform probability measure on each group. Set

$$H_r = L^2(G_r, |G_r|^{-1}\,\text{counting measure}), \qquad (i_r f)(g) = f(p_r g).$$

Then i_r is an isometry. Its adjoint is conditional averaging,

$$(P_r f)(h) = \frac{1}{|K_r|} \sum_{p_r g = h} f(g), \qquad P_r i_r = 1.$$

Both assertions follow by partitioning G_{r+1} into its equally sized fibers.

Theorem 22.6.2 *(Haar realization).* For $G = \varprojlim_r G_r$, normalized Haar measure μ has the given uniform measures as its finite pushforwards. Pullback identifies the Hilbert completion of $\operatorname{colim}_r (H_r, i_r)$ with $L^2(G, \mu)$. The finite conjugation-invariant subspaces have dense union in the conjugation-invariant subspace of $L^2(G, \mu)$.

Proof. The compatible uniform measures assign mass $1/|G_r|$ to each cylinder over a point of G_r. They define the normalized invariant measure on the compact profinite group. Every finite pullback is isometric by this cylinder formula. Locally constant cylinder functions separate points and are dense in continuous functions, hence in $L^2(G, \mu)$. This proves the completion statement.

The conditional expectation onto level r commutes with conjugation, because its fibers and Haar measure do. These orthogonal projections converge strongly to the identity on the dense union just described. Applying them to an invariant function proves density in the invariant subspace. □

This construction uses probability on the group elements. It does not replace the groupoid measure on arithmetic representations or the action phase in (22.18). To apply it to arithmetic states one must construct equally precise measures, isometries and adjoints there.

The bounded part of an inverse system

An inverse diagram always defines a pro-object in its specified category. A topological inverse limit need not have a Hilbert norm. For example, the coordinate projections $\mathbb{C}^{r+1} \to \mathbb{C}^r$ have inverse limit $\mathbb{C}^{\mathbb{N}}$ with its product topology. Every basic neighborhood constrains finitely many coordinates and contains a whole line in an unconstrained coordinate. It cannot be a bounded neighborhood for any norm inducing that topology. Passing to the associated condensed vector spaces does not supply a missing Hilbert topology.

The orthogonal projections above do have a distinguished Hilbert part. The following statement also applies to finite-dimensional state spaces once isometric transitions have been constructed.

Proposition 22.6.3 *(Bounded compatible vectors).* Let H_r be Hilbert spaces with isometric maps i_r, let $P_r = i_r^*$, and let H be the Hilbert completion of their increasing union. Then

$$H \xrightarrow{\sim} \left\{ (x_r) : P_r x_{r+1} = x_r, \sup_r \|x_r\| < \infty \right\}, \qquad \|x\|^2 = \lim_r \|x_r\|^2,$$

is an isometric isomorphism, using the projections from H on the left and the displayed norm on the right.

Proof. The projections onto the increasing closed subspaces H_r converge strongly to the identity of H. They give a compatible sequence with the asserted norm.

Conversely, view a compatible sequence in the increasing union. For $s \geq r$, orthogonality gives

$$\|x_s - x_r\|^2 = \|x_s\|^2 - \|x_r\|^2.$$

The squared norms increase and are bounded, so this sequence is Cauchy. Its limit has projections x_r, and its norm is the stated limit. This is inverse to the first map. □

For the coordinate system, the bounded part is ℓ^2, a proper subspace of the product limit. Specifying this norm condition is a substantive part of the construction. A Hilbert limit does not ensure trace continuity. The identity on $\mathbb{C}^r$ has trace r, while the identity on ℓ^2 is not trace class. Coherent finite operators alone do not define an infinite partition trace.

22.7 TWISTED TRACES AND A FINITE MODULAR EXAMPLE

The degree-two transgression on circle fields has a categorical recipient. Let $\mathcal{X}$ be a finite groupoid and let τ be a normalized complex multiplicative two-cocycle on its composable arrows. Its twisted groupoid algebra has basis u_a, one element for each arrow, and multiplication

$$u_a u_b = \tau(a,b)u_{ab} \quad \text{when } a,b \text{ are composable}, \qquad u_a u_b = 0 \quad \text{otherwise}.$$

The cocycle identity gives associativity, and the identity arrows give its unit. A finite skeleton may be used if the groupoid has more than finitely many objects.

The *inertia groupoid* $I\mathcal{X}$ has objects (x,g), with $g \in \mathrm{Aut}(x)$, and arrows $h : (x,g) \to (y, hgh^{-1})$. Conjugation in the twisted algebra gives a line on this groupoid. On a one-object component its transport is

$$u_g \longmapsto \frac{\tau(h,g)\tau(hg,h^{-1})}{\tau(h,h^{-1})} u_{hgh^{-1}}. \tag{22.19}$$

The same formula applies to composable arrows. Denote this line by L_τ. Its degree-zero homology means the direct sum of its fibers modulo these transport relations.

PROPOSITION 22.7.1 *(Finite twisted Hochschild trace).* Over $\mathbb{C}$, the twisted groupoid algebra satisfies

$$\mathrm{HH}_0(\mathbb{C}_\tau[\mathcal{X}]) \cong H_0(I\mathcal{X}, L_\tau), \qquad \mathrm{HH}_j(\mathbb{C}_\tau[\mathcal{X}]) = 0 \quad (j>0).$$

The same degree-zero trace is obtained from its category of perfect modules.

Proof. Degree-zero Hochschild homology is the quotient of the algebra by commutators. Commuting with the identity idempotents kills every non-loop arrow. For a loop, $[u_h, u_g u_h^{-1}]$ identifies its class with the conjugate loop in (22.19). Conversely the cyclic relation between any two composable arrows is one of these transported loop relations. This gives the first isomorphism.

Each connected component is Morita equivalent to a twisted algebra of a finite automorphism group H. It is semisimple: average a linear projection onto a submodule over conjugation by the invertible operators u_h. Projective scalar factors cancel under conjugation, so the average is module-linear. Equivalently, the element

$$\frac{1}{|H|}\sum_{h\in H} u_h \otimes u_h^{-1}$$

splits multiplication as a bimodule map. The algebra is therefore projective over its enveloping algebra, and positive Hochschild homology vanishes. Morita invariance proves the remaining assertions. □

The characteristic assumption is essential. Over $\mathbb{F}_p$, the untwisted algebra of C_p is $\mathbb{F}_p[\epsilon]/(\epsilon^p)$, whose module of differentials, and hence its first Hochschild homology, is nonzero. A derived trace cannot then be replaced by its degree-zero inertia quotient.

For the finite topological theory, $\mathcal{X}$ can be the groupoid of circle fields, with τ the transgression of its three-cocycle. Its category of twisted representations and its Hochschild trace have the types just described. A trace comparison with an arithmetic Hecke category still requires that arithmetic category, its coefficient realization and the actual map on cyclic complexes. Equal dimensions would not define such a map.

The double of C_2

The simplest nontrivial modular example can be given without an arithmetic hypothesis. Let $A = C_2$, written additively, and let $\widehat{A}$ be its two complex characters. Consider finite-dimensional vector spaces graded by $A \times \widehat{A}$. Simple objects are (a, χ), tensor product adds a and multiplies characters, and the associator is the identity. On simple objects set

$$c_{(a,\chi),(b,\psi)} = \chi(b), \qquad \theta_{(a,\chi)} = \chi(a).$$

The bicharacter law gives both hexagon identities. The balancing identity is

$$(\chi\psi)(a+b) = \chi(a)\psi(b)\chi(b)\psi(a),$$

so these data give a ribbon category with all simple dimensions one.

If a simple object has trivial monodromy with every simple, varying b forces $\chi = 1$ and varying ψ forces $a = 0$. Thus the braiding is nondegenerate.

In the order $(0,1)$, $(0,\mathrm{sgn})$, $(1,1)$, $(1,\mathrm{sgn})$, the normalized matrices are

$$S = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad T = \mathrm{diag}(1,1,1,-1). \tag{22.20}$$

Character orthogonality gives $S^2 = 1$, and direct multiplication gives $(ST)^3 = 1$.

These matrices can also be obtained from the untwisted finite gauge states on a torus. Its four fields are $(a,b) \in A^2$, each with automorphism group A. Normalize their delta functions by the measure (22.2) and Fourier transform in the second holonomy. The resulting basis is indexed by (a,χ).

Exchanging the two holonomies acts by $S_{(a,\chi),(b,\psi)} = \chi(b)\psi(a)/2$, while a Dehn twist acts by $\chi(a)$. This constructs the displayed comparison of finite gauge and modular data. Its dimension is four, agreeing with $|\mathcal{F}_{C_2}(T^3)| = 8/2 = 4$.

A finite semisimple complex category has finite-dimensional HH_0, with one basis vector for each simple object. It cannot be isomorphic as a vector space to a target containing a nonzero polynomial Fock space with infinitely many independent creation states. This excludes that literal finite-category trace lift. It leaves open an infinite category, an appropriate completion, a selected finite trace sector, or a compatible family of finite categories. Each choice changes the recipient and requires its own trace map.

22.8 Ordered incidence and local current algebras

An order on places gives an algebraic bar construction before it gives an arithmetic field theory. Fix a finite ordered set $S = \{s_1 < \cdots < s_r\}$. An interval partition of S is determined by a subset of the $r-1$ possible cuts. Let $\mathcal{I}_S$ have these partitions as objects and one arrow from a partition to each coarsening obtained by deleting cuts. The one-block partition t_S is terminal.

Let $D_S : \mathcal{I}_S \to \mathrm{Ch}(k)$ be any covariant diagram of chain complexes. Its simplicial replacement has

$$B_n(D_S) = \bigoplus_{P_0 \to \cdots \to P_n} D_S(P_0).$$

The first face applies the first diagram map. The other faces omit an intermediate object or the last object. Degeneracies insert identities. Write $\partial = \sum_i (-1)^i d_i$. With internal homological degree q, the direct-sum totalization has degree $n+q$ and differential $\partial + (-1)^n d_{\mathrm{int}}$.

PROPOSITION 22.8.1 *(Contraction of the incidence bar).* The augmentation induced by $P \longrightarrow t_S$ is a quasi-isomorphism

$$\mathrm{Tot}_{\oplus}\, B_\bullet(D_S) \longrightarrow D_S(t_S).$$

It holds for unbounded internal complexes. If the indexing category is the disjoint union of the $\mathcal{I}_S$, its homotopy colimit is $\bigoplus_S D_S(t_S)$.

Proof. Include $D_S(t_S)$ as the terminal summand in degree zero. On a simplex of degree n, append its unique arrow to t_S and multiply by $(-1)^{n+1}$. Denote this map by h_n. The faces cancel in pairs, except the last face, and in degree zero the terminal augmentation. Consequently $\partial h + h\partial = 1 - i\epsilon$.

The internal differential commutes with appending a terminal arrow. Its totalization signs in adjacent simplicial degrees are opposite, so the same identity holds for the total differential. Every vector uses finitely many simplices in the direct sum. No convergence or lower bound on internal degree is used. The contraction preserves S, proving the final assertion. □

For example, if all coefficient complexes are k and the available places are p, q, the nonempty components $\{p\}, \{q\}, \{p, q\}$ give k^3. The restricted tensor product of the two copies of k, with unit reference vectors, is k. The incidence category has no arrows inserting a new place. Such arrows must be supplied before its colimit can express restricted tensor assembly.

The simplicial replacement also records partitions of one ordered set, not all collision diagrams on a curve. Its order is the chosen combinatorial order on places. A comparison with physical time order or with factorization on collision strata requires maps that preserve those additional operations.

An actual tensor construction of vertex algebras

There is a positive local construction once current algebras have been supplied. For every place p, choose a finite-dimensional complex Lie algebra $\mathfrak{g}_p$, an invariant symmetric bilinear form and a level k_p. The affine brackets are

$$[x_m, y_n] = [x, y]_{m+n} + m\delta_{m+n,0} k_p(x, y).$$

The induced vacuum module $V^{k_p}(\mathfrak{g}_p)$ is a vertex algebra at every level. For a finite set S, take its tensor product over $p \in S$. Inserting the chosen vacua gives maps for inclusions of finite sets, and hence a restricted tensor vertex algebra

$$V_{\mathrm{res}} = \mathrm{colim}_{S\ \mathrm{finite}} \bigotimes_{p\in S} V^{k_p}(\mathfrak{g}_p). \tag{22.21}$$

Every vector and every pair of inputs belong to one finite tensor stage. Their vertex operation, translation and vacuum identities are defined there, are preserved by enlargement, and therefore define the colimit. Locality for a pair is the finite tensor locality bound at that stage. This proves the construction without forming an infinite product of fields on one input.

A conformal vector needs more. For a simple Lie algebra with the standard normalization, Sugawara requires $k_p + h_p^\vee \neq 0$. A relation between two shifted levels, $\kappa_p + \kappa_p^! = 0$, does not imply either is nonzero. Moreover the sum of conformal vectors over infinitely many nontrivial places need not belong to the algebraic restricted tensor product. A conformal completion must be specified if that sum is intended.

Satake eigenvalues do not themselves supply these Lie algebras, levels, vacuum modules or conformal vectors. They can give a finite-dimensional Frobenius carrier and its symmetric-algebra trace. Identifying that carrier with a current-algebra sector, and identifying (22.21) with an arithmetic Hochschild trace realization, are separate construction problems.

22.9 An explicit physical boundary model

A physical boundary comparison should identify cochains and operations before taking a character. The free abelian holomorphic–topological theory gives such an example. Its base is the actual half-space $M = [0,\infty)_t \times \mathbb{C}_z$, with incoming orientation $-dt \wedge dx \wedge dy$, where $z = x + iy$.

Partial forms use dt, $d\bar z$, with

$$D = dt\,\partial_t + d\bar z\,\partial_{\bar z}, \qquad \partial = \partial_z, \qquad \tau_+(a) = \frac{1}{2\pi i}\int_M^{dt\,dx\,dy} dz \wedge a.$$

The subscript records the auxiliary positive orientation used in the integral. Take superfields Φ, η of total degrees zero and one.

The incoming action and boundary condition are

$$S_\kappa = -\tau_+\Big(\eta D\Phi + \frac{\kappa}{2}\eta\partial\eta\Big), \qquad i^*\eta = 0.$$

We fix $\kappa \in \mathbb{C}^\times$ and $\hbar = 1$ for the Wilson-jet comparison below. The action is quadratic. No interacting arithmetic field theory is being inferred from it.

Regular observables and the ending line

The tangent fields are $E = \Omega^\bullet_{\mathrm{HT}} e_\Phi \oplus \Omega^\bullet_{\mathrm{HT}} e_\eta$, with $|e_\Phi| = 0$, $|e_\eta| = -1$, and

$$\begin{aligned} Q(\alpha e_\Phi) &= D\alpha e_\Phi, \\ Q(\beta e_\eta) &= D\beta e_\eta + (-1)^{\deg\beta+1}\kappa\partial\beta e_\Phi. \end{aligned}$$

Compactly supported fields may meet $t = 0$, with the relative condition on every tangential e_η component. The incoming pairing is

$$B_-(\alpha e_\Phi, \beta e_\eta) = -\tau_+(\alpha\beta), \qquad B_-(\beta e_\eta, \alpha e_\Phi) = (-1)^{\deg\alpha}\tau_+(\beta\alpha),$$

and vanishes on equal summands. On the suspension, $|s\xi| = |\xi| - 1$, put $d(s\xi) = -sQ\xi$ and $b_-(s\xi, s\upsilon) = (-1)^{|\xi|+1}B_-(\xi,\upsilon)$.

For $U = [0,\infty)\times V$, let $\mathcal{O}(U) = \bigoplus_{m\ge 0}\mathrm{Sym}^m_b(E_c(U)[1])$. At each field number, complete the smooth joint kernels in the projective topology at fixed compact support. Then take the limit over compact supports. The sum over field numbers is algebraic.

Let Δ_{b_-} contract unordered pairs once, with Koszul signs. The quantum differential is $d + \Delta_{b_-}$. Smoothness makes its diagonal contractions defined. A point-supported composite is outside this space.

Write a test field as $\gamma(t) + dt\,\beta(t)$, and let a superscript (0) mean Dolbeault degree zero. For $n \ge 0$, the ray and endpoint derivations are

$$\begin{aligned} \ell_n(s\xi) &= \frac{1}{n!}\int_0^\infty \partial_z^n \beta_\eta^{(0)}(t,0)\,dt, && |\ell_n| = 1, \\ r_n(s\xi) &= \frac{1}{n!}\int_0^\infty \partial_z^n \beta_\Phi^{(0)}(t,0)\,dt, && |r_n| = 0, \\ e_n(s\xi) &= \frac{1}{n!}\partial_z^n \gamma_\Phi^{(0)}(0,0), && |e_n| = 1. \end{aligned}$$

Extend them as constant derivations on kernels.

Integration of the normal derivative gives

$$[d, \ell_n] = [d, e_n] = 0, \qquad [d, r_n] = \kappa(n+1)\ell_{n+1} - e_n. \tag{22.22}$$

The first two identities use the relative boundary condition. The minus sign in the last uses $\int_0^\infty \partial_t \gamma(t)\, dt = -\gamma(0)$. All constant derivations graded-commute with one another and with Δ_{b_-}.

For a finite $F \subset \mathbb{Z}_{>0}$, introduce the commutative differential graded coefficient algebra

$$Y_F = \mathbb{C}[B_0, B_n : n \in F] \otimes \Lambda(c_{n-1} : n \in F), \quad Q_Y B_n = \kappa n c_{n-1}, \quad Q_Y B_0 = Q_Y c_j = 0. \tag{22.23}$$

The B_n have degree zero and the c_j degree one. Coefficients in operator expressions act by graded left multiplication. The module has the regular right Y_F-action. All tensors with these coefficients are algebraic.

PROPOSITION 22.9.1 *(The endpoint correction).* Put

$$\begin{aligned} D_0 &= d + \Delta_{b_-} + Q_Y + B_0 \ell_0, \\ L_F &= \sum_{n \in F} \frac{B_n}{\kappa n} r_{n-1}, \\ D_F &= D_0 + \sum_{n \in F} \left(B_n \ell_n + c_{n-1} r_{n-1} - \frac{B_n}{\kappa n} e_{n-1} \right). \end{aligned}$$

Then $D_F = e^{-L_F} D_0 e^{L_F}$, so $D_F^2 = 0$. The exponential is finite on each input and fixes the unit and the right coefficient action. Omitting the endpoint term gives the nonzero square $\sum_{n \in F} c_{n-1} e_{n-1}$.

Proof. The coefficient differential of $B_n \ell_n$ gives $\kappa n c_{n-1} \ell_n$. The commutator of d with $c_{n-1} r_{n-1}$ gives its negative plus $c_{n-1} e_{n-1}$, by (22.22). The endpoint correction cancels the latter term.

More precisely, $[D_0, L_F] = D_F - D_0$, and its commutator with L_F is zero. The conjugation formula follows. Each r_n lowers field number and kills the unit, so its exponential terminates and is right-linear. For $F = \{1\}$, a suspended e_Φ test function equal to one near the endpoint detects the omitted-term defect as c_0. □

All jets with corrected transition maps

For $F \subset F'$, coefficient inclusion alone need not commute with the source differential. The corrected map is

$$j_{F,F'} = e^{-(L_{F'} - L_F)} \iota_{F,F'}.$$

The L_F commute, so these maps are injective cochain maps and satisfy $j_{F',F''} j_{F,F'} = j_{F,F''}$.

For $Y_\infty = \operatorname{colim}_F Y_F$, there is an explicit isomorphism

$$\operatorname{colim}_F (\mathcal{O}(U) \otimes Y_F, D_F) \xrightarrow{\sim} (\mathcal{O}(U) \otimes Y_\infty, D_0), \qquad [u]_F \longmapsto e^{L_F} u. \tag{22.24}$$

The inverse takes v to $[e^{-L_F} v]_F$ at any finite stage containing its coefficients. Enlarging the stage leaves this class unchanged. Each map is continuous at fixed field number and support, and preserves products on tangentially disjoint opens.

This is not the literal all-jets differential on an unchanged smooth observable. On a small disk, choose a compact test whose ray derivatives all equal one. Its raw source term contains $\sum_{n>0} c_{n-1}$, which is not in the polynomial coefficient algebra. Formula (22.24) instead compares the representatives $u - \sum_{n \in F} B_n/(\kappa n)$ at different cutoffs. Their differentials agree under the corrected maps.

The boundary period map

Normal contraction reduces D_0 to the boundary complex

$$\mathcal{A}(V) = \Big(\mathrm{Sym}_b(\Omega_c^{0,\bullet}(V)[1]) \otimes Y_\infty, d_1 + \Delta_\mu + B_0\epsilon_0 + Q_Y\Big), \tag{22.25}$$

where $d_1(s\alpha) = -s\bar\partial\alpha$, $\epsilon_0(s\phi) = \phi(0)$ on functions, and ϵ_0 vanishes on top forms. The boundary pairing is $\mu(s\phi, s\psi) = -\kappa\tau_{\mathbb{C}}(\phi\partial\psi)$, for a function ϕ and a $(0,1)$-form ψ, with $\tau_{\mathbb{C}}(a) = (2\pi i)^{-1}\int dz \wedge a$. The reversed order is given by graded symmetry.

Here the normal contraction can be constructed with compact support. Choose a smooth compact profile $\rho(t)$ of integral one and set $F_\rho(t) = \int_0^t \rho(r)\,dr$. On the complementary e_η boundary component the linear maps are

$$\begin{aligned} P_N(\gamma + dt\,\beta) &= \int_0^\infty \beta_\eta(r)\,dr, \\ I_N g &= dt\,\rho g + (F_\rho - 1)Q_{\mathrm{rel}} g, \\ h_N(\gamma + dt\,\beta) &= -\int_t^\infty \beta(r)\,dr + (1 - F_\rho(t))\int_0^\infty \beta_\eta(r)\,dr. \end{aligned}$$

In the last term the result lies in the e_η summand, and $Q_{\mathrm{rel}} = Q - D$. The fundamental theorem of calculus gives $P_N I_N = 1$ and $Qh_N + h_N Q = 1 - I_N P_N$. The correction in h_N makes its relative boundary value zero and keeps its normal support compact.

For a boundary form a of Dolbeault degree j, set $i(sa) = sI_N((-1)^j a e_\eta)$. On a test field with Dolbeault degree j, put $p(s\xi) = s((-1)^j P_N\xi)$, and set $h(s\xi) = -sh_N\xi$. Then $pi = 1$, $dh + hd = 1 - ip$, and $hi = ph = h^2 = 0$.

Let H_2 be the graded symmetrization of $h \otimes 1 + ip \otimes h$, and put $P_2 = -b_- H_2$. Its Hom differential is $b_- - p^*\mu$. Thus $p_0 = \mathrm{Sym}(p)e^{\partial_{P_2}}$ is the quantum normal projection to (22.25).

The exponential contracts pairs and is finite on each field-polynomial input. Conjugating the linear contraction gives a homotopy inverse. On the kernel summand, the symmetric-algebra homotopy is obtained by dividing the derivation induced by h by the positive number of kernel factors. These formulas give a continuous contraction on the stated smooth joint kernels and commute with disjoint products.

For a disk D_R centered at zero, define the exterior germ space

$$\mathscr{P}(D_R) = \mathrm{colim}_{r<R}\left\{\sum_{n\geq 1} a_n w^{-n} : \sum_n |a_n| r^{-n} < \infty\right\}.$$

Each stage has the displayed weighted ℓ^1 norm. The linking maps for increasing radii are compact.

For a compact top form a, its Cauchy transform is

$$C(a)(w) = \tau_z \frac{a(z)}{w - z},$$

expanded outside its support. Set $C = 0$ on shifted functions.

The compact Dolbeault complex maps to this germ space by a continuous quasi-isomorphism. Explicitly, a germ has representative $s(\bar\partial\chi)f$ on a slightly larger annulus, where χ is one near the inner disk. If the exterior germ of a top form is zero, Cauchy convolution is its compact primitive. Cauchy estimates on two separated radii make both maps continuous. Applying this contraction in each kernel variable and averaging over permutations proves the statement for every completed symmetric power.

Let $k_1(z) = 1/z$ as a locally integrable distribution and $k_2 = -\partial_z k_1$. Put

$$P(sa, sb) = \kappa\tau_z\tau_w\big(k_2(z-w)a(z)b(w)\big), \qquad \gamma(sa) = \tau_z(a(z)/z)$$

on top forms and zero on other degrees. The distributional Cauchy identity gives $d_{\mathrm{Hom}}P = \mu$ and $d_{\mathrm{Hom}}\gamma = \epsilon_0$. Consequently $N_{B_0} = e^{\partial_P + B_0\partial_\gamma}$ conjugates the boundary differential to $d_1 + Q_Y$. Only finitely many contractions act on each input, and the distributions are paired with smooth kernels in independent variables.

The resulting period map on the corrected limit is

$$\Pi[u]_F = \mathrm{Sym}(C)N_{B_0}p_0e^{L_F}u. \tag{22.26}$$

Its target is the algebraic sum of completed symmetric powers of $\mathscr{P}(D_R)$, tensored algebraically with Y_∞.

THEOREM 22.9.2 *(Rotation-finite boundary comparison).* Rotate z and its form factors geometrically, with an additional weight one on e_Φ. Give B_n and c_{n-1} weight n, and B_0 weight zero. Let $\mathcal{J}_{\mathrm{rot}}(D_R)$ be the algebraic sum of the circle eigenspaces of the corrected source. Then (22.26) restricts to a continuous right Y_∞-linear quasi-isomorphism

$$\Pi_{\mathrm{rot}} : \mathcal{J}_{\mathrm{rot}}(D_R) \longrightarrow (F_{\mathrm{pol}} \otimes Y_\infty, Q_Y), \qquad F_{\mathrm{pol}} = \mathbb{C}[x_1, x_2, \ldots].$$

It sends 1 to 1, is compatible with centered disk inclusions, and carries separated contour currents to

$$J_{-n} = x_n, \qquad J_n = \kappa n\partial_{x_n} \quad (n > 0), \qquad J_0 = B_0. \tag{22.27}$$

For every algebraic differential graded left Y_∞-module L, ordinary tensoring over Y_∞ computes the derived tensor product and preserves this quasi-isomorphism.

Proof. The corrected transition isomorphism, normal contraction, Gaussian conjugation and compact Dolbeault contraction prove the comparison on the full germ target. They commute with rotations. A monomial $w_1^{-n_1}\cdots w_m^{-n_m}$ has weight $-\sum n_i$. For a fixed sum, only finitely many symmetric monomials occur, and m is bounded by that sum. Thus the finite-circle-spectrum part of the germ target is exactly F_{pol}. Each input has finitely many coefficient monomials, so adjoining Y_∞ preserves this conclusion.

To restrict the quasi-isomorphism, take a cocycle of finite circle spectrum in its acyclic mapping cone. Choose a primitive in the full cone and apply the finite Fourier projector onto the cocycle's weights. The rotated compact support stays in one larger compact disk. The integral therefore exists in a single smooth-kernel or germ stage and commutes with the differential. It gives a primitive of finite spectrum.

A current smeared on an outer circle has period

$$\sum_{n>0} x_n z^{n-1} + B_0 z^{-1} + \kappa \sum_{n>0} n z^{-n-1}\partial_{x_n}.$$

The uncontracted insertion gives the first sum, its charge contraction gives the middle term, and $P(sa_z, sa) = -\kappa\partial_z C(a)(z)$ gives the last. Contour coefficients yield (22.27). Moving the circle without crossing the inner support gives a cochain homotopy by Stokes' formula.

Finally filter the source by field number. Its differential does not raise that number. Every associated graded piece is a complex of vector spaces tensored with the regular coefficient algebra, and is homotopically flat over that algebra. The exhaustive, bounded-below filtration and exact filtered unions give homotopical flatness of the source. The target is free as a graded coefficient module. Tensoring the quasi-isomorphism with any algebraic differential graded module therefore preserves it, including unbounded modules. □

The Gaussian cap

On the boundary write $\Phi_\Sigma = \phi + d\bar z\,\phi_b$ and $\eta_\Sigma = c + d\bar z\,a_b$. Here $c(z)$ is the boundary ghost. The c_j in (22.23) are line coefficients. Put $x_n = \partial_z^{n-1}\phi(0)/(n-1)!$. The incoming boundary charge, including the endpoint, is

$$S^F_{\mathrm{in}} = -\frac{1}{\pi}\int c(\partial_{\bar z}\phi + \kappa\partial_z a_b)\,dx\,dy + \mu_F(0),$$
$$\mu_F(0) = B_0 c(0) + \sum_{n\in F}\left(\frac{B_n\partial_z^n c(0)}{n!} + c_{n-1}x_n\right).$$

The coefficient differential Q_Y is external to the boundary Poisson bracket, since the line coefficients are Casimirs. Its contribution $\pi\kappa n c_{n-1}\partial_z^n g(0)/n!$ to the square on $\phi_b(g)$ cancels the contribution $-\pi\kappa c_{n-1}\partial_z^n g(0)/(n-1)!$ from the line-ghost term. The remaining square is zero on every field generator.

Let $\mathcal{P}_{\mathrm{sm}}(V)$ be the polynomial functionals of ϕ, ϕ_b with compact smooth kernels. For a finite jet set F and $N \ge \max(F\cup\{1\})$, set

$$W_{F,N} = Y_F[[x_1,\dots,x_N]], \qquad \mathcal{H}^F_{\mathrm{in}} = \operatorname{colim}_N\big(\mathcal{P}_{\mathrm{sm}}(V)\otimes_{\mathbb{C}} W_{F,N}\big).$$

Completion is by jet weight, $\operatorname{wt} x_n = n$, before taking the union in N. Each weight coefficient is polynomial in the line coefficients. The tensor with smooth functionals is algebraic.

In the incoming polarization, momenta act as $\widehat{a_b} = \pi\delta/\delta\phi$ and $\widehat{c} = \pi\delta/\delta\phi_b$, with derivatives on the right. Quantizing the charge defines Ω^F_{in} on this space. A mixed contraction differentiates one of the finitely many jet variables and lowers weight by at most N. Its other factor takes finitely many derivatives of a smooth kernel at the junction. Smooth-field contractions restrict smooth joint kernels to diagonals. The operator is therefore defined and continuous at each fixed N.

The square of $Q_Y + \Omega^F_{\mathrm{in}}$ is zero. Each pair of charge terms shares at most one position–momentum contraction, so its commutator quantizes the classical bracket. The two-bulk-term contribution pairs a symmetric mixed-derivative kernel with two antisymmetric ghosts and vanishes. The endpoint bracket cancels $Q_Y\mu_F(0)$ by the displayed factorial identity.

On the pure junction subspace $W_F = \operatorname{colim}_N W_{F,N}$, every momentum term contains $\delta/\delta\phi_b$ and vanishes. The resulting differential is

$$D_{\mathrm{cap}} = Q_Y + \sum_{n\in F} x_n c_{n-1}.$$

Its multiplication term is the ghost part of the endpoint charge. Omitting it would violate the relative master equation of the coupled boundary and line system.

Consider a cap e^f with f polynomial in finitely many x_n, normalized by $f|_{x=0} = 0$. It is closed precisely when

$$f = -\sum_{n\in F}\frac{B_n x_n}{\kappa n} + h, \qquad h \in \mathbb{C}[B_0, x_1, x_2, \dots].$$

Indeed $D_{\mathrm{cap}}e^f = e^f(Q_Y f + \sum x_n c_{n-1})$, so the displayed equation integrates the coefficient derivatives.

Under rotation, x_n has weight $-n$. The first sum has weight zero and every nonconstant monomial of h has negative weight. Thus the unique rotation-invariant normalized cap in this class is

$$G_F = \exp\left(-\sum_{n\in F}\frac{B_n x_n}{\kappa n}\right).$$

The constant state is not closed when $F \neq \varnothing$. Gluing the linear exponent of G_F by the BV pairing produces exactly $-\sum(B_n/(\kappa n))e_{n-1}$, the endpoint correction of Proposition 22.9.1. Changing its sign leaves square $2\sum c_{n-1}e_{n-1}$.

For all jets, use the degreewise creation-weight completion $\prod_{N\geq 0}(F_N \otimes Y_\infty^d)$, where $F_N \subset F_{\mathrm{pol}}$ has creation weight N. In it set

$$G = \exp\left(-\sum_{n>0} \frac{x_n B_n}{\kappa n}\right), \qquad \mathsf{P}_\kappa = G(F_{\mathrm{pol}} \otimes Y_\infty).$$

Every weight coefficient is finite. On this specified image, multiplication by G is an invertible cochain map from Q_Y to $D_G = Q_Y + \sum_{n>0} x_n c_{n-1}$. Hence $G\Pi_{\mathrm{rot}}$ sends the supported unit to the cap and carries the currents to

$$\widehat{J}_{-n} = x_n, \qquad \widehat{J}_n = \kappa n \partial_{x_n} + B_n, \qquad \widehat{J}_0 = B_0.$$

Coefficientwise differentiation proves $D_G(Gf) = GQ_Y f$ and $\widehat{J}_n(Gf) = G\kappa n \partial_{x_n} f$. The chosen image P_κ, the full completion and the raw polynomial complex are different carriers.

The same formulas detect a possible Ward error. For $m > 0$, the raw operator $O_m = -(\widehat{J}_m + B_m)$ has

$$[D_G, O_m] = -\kappa m c_{m-1}, \qquad O_m G = -B_m G.$$

Adding the coefficient term B_m gives the cochain map $-\widehat{J}_m$. A residue identity for a raw insertion therefore cannot replace the correction that makes the operation a cochain map.

The order of state completion also matters. If all jet variables are completed before adjoining smooth functionals, take a smooth test equal to $(1-z)^{-1}$ near the endpoint. Applying the two-momentum part of the boundary operator to $\phi_b(g)\sum_{n=1}^{M} x_n$ has scalar term $\pi\kappa M(M+1)/2$. It does not stabilize even in weight zero. The finite-jet completion used to construct the cap avoids this undefined operation.

Rigidity in the regular polynomial class

The cap also explains which local choices survive rotation. Let $\mathfrak{J}$ be the complex of constant derivations obtained by taking finitely many normal, holomorphic and antiholomorphic derivatives of test fields at the endpoint. Its differential is $[d, -]$. The relative boundary condition makes its cohomology

$$H^1(\mathfrak{J}) = \bigoplus_{n\geq 0} \mathbb{C}[e_n], \qquad H^j(\mathfrak{J}) = 0 \quad (j \neq 1).$$

To see this, take the restricted dual of formal field jets, graded by normal and antiholomorphic order and rotation weight. Each piece is finite. The relative e_η normal complex $t\mathbb{C}[[t]] \xrightarrow{\partial_t} \mathbb{C}[[t]]dt$ is acyclic. The unrestricted e_Φ normal complex has constant cohomology, and its formal Dolbeault complex has cohomology $\mathbb{C}[[z]]$. Dualizing gives precisely the holomorphic evaluation classes $[e_n]$.

Consider junction corrections of degree one in $(Y_F \otimes \mathfrak{J}) \oplus Y_F$, the second summand acting by multiplication. Write $X_F^\circ = -\sum(B_n/(\kappa n))e_{n-1}$. The square-zero corrections are exactly

$$X = X_F^\circ + \partial Y + \sum_{n\geq 0} h_n e_n - Q_Y(q), \tag{22.28}$$

where $\partial = Q_Y + [d, -]$, Y is a degree-zero coefficient-valued constant derivation, the sum is finite, $h_n \in \mathbb{C}[B_0]$, and $q \in \mathbb{C}[B_0, B_n : n \in F]$. Indeed the square is $\partial(X - X_F^\circ)$. The polynomial de Rham contraction gives $H(Y_F) = \mathbb{C}[B_0]$, and the preceding jet calculation identifies all its degree-one classes. This proves (22.28).

The term ∂Y is removed by the finite conjugation e^{Y}, which fixes the unit and the regular right action. The classes $h_n e_n$ have weight $-(n+1)$, and a nonzero $Q_Y(q)$ has positive coefficient weight. Rotation invariance therefore forces both to vanish and permits a weight-zero Y. In this regular polynomial class, a rotation-invariant junction is isomorphic to the one used in Theorem 22.9.2. The conclusion concerns this local source and its admissible corrections. It is not a generation theorem for every boundary module.

A coefficient twist can change the polynomial ground. With correction $X_F^{\circ} - Q_Y(q)$ and $q \notin \mathbb{C}[B_0]$, the period differential becomes $Q_Y - Q_Y(q)$. A polynomial ground killed by all positive currents would satisfy $\partial_{B_n} f = (\partial_{B_n} q) f$.

For a variable with $\partial_{B_n} q \neq 0$, the right side has total polynomial degree at least that of nonzero f, while the left side has smaller degree. Thus no normalized polynomial ground exists.

Completing in the positive B_n permits the ground $e^{q - q|_{B_{>0}=0}}$. This repairs the comparison on a different coefficient space and changes the selected vector.

22.10 Sewing states and evaluating the formal comparison

For an algebraic differential graded line module L, write $\Phi_\kappa(L) = F_{\mathrm{pol}} \otimes L$ with its differential and the currents (22.27). Two endpoints carry two copies of the coefficient algebra. On $U = M \otimes N'$, their zero modes commute. Put $N = B_0' B_0'' / \kappa$. The state family for boundary fusion has carrier $F_{\mathrm{pol}} \otimes U$ and connection

$$\nabla_{\mathrm{st}} = d + \rho(N)\, d \log z,$$

where ρ denotes the coefficient action and $z \neq 0$ is the separation.

For scalar charges a, b, its normalized current operation on the first ground state is

$$K_{a,b}(z) f(x) |b\rangle = \exp\!\left(\frac{a}{\kappa} \sum_{n>0} \frac{x_n z^n}{n} \right) f(x_n - a z^{-n}) |a+b\rangle. \tag{22.29}$$

The output is completed in creation weight. Each weight coefficient on a polynomial input is a Laurent polynomial in z. Commuting $\kappa m \partial_{x_m}$ and x_m through the two factors verifies $[J_m, K_{a,b}(z)] = a z^m K_{a,b}(z)$, where the commutator uses the input and output current actions.

Negative-current descendants of the first input are determined by the normal-ordered insertion rule

$$I(J_{-n} u, z) = \frac{\partial_z^{n-1} J^-(z)}{(n-1)!} I(u,z) + I(u,z) \frac{\partial_z^{n-1} J^+(z)}{(n-1)!}.$$

Here J^- contains negative modes and J^+ nonnegative modes. Truncation on polynomial inputs and completion of output weight make each expression defined.

Removing $K_{a,b}$ from an operation satisfying these identities leaves a map commuting with all nonzero oscillators. It is determined by the image of the ground state. More generally, for a coefficient module with locally nilpotent positive modes and positive-mode truncation on each vector, the oscillator decomposition is $F_{\mathrm{pol}} \otimes \Omega_+$, where $\Omega_+ = \bigcap_{m>0} \ker J_m$. Thus normalized Ward operations are represented by the state family above.

On the charge-a Fock space, translation is

$$L_{-1}^{(a)} = \frac{a}{\kappa} x_1 + \sum_{n>0} n x_{n+1} \partial_{x_n}.$$

Substituting (22.29) into translation covariance makes its ground coefficient satisfy $\partial_z h = (N/z) h$. Its connection is therefore $d - \rho(N) d \log z$. Formally at $z = 1$, its normalized horizontal coefficient is $\exp(N \log(1 + (z-1)))$.

For scalar charges it becomes $z^{ab/\kappa}$ on a logarithmic cover. A counterclockwise half-loop transports states by $e^{-\pi i ab/\kappa}$ and operations by $e^{\pi i ab/\kappa}$. The ordinary tensor product of two level-κ Heisenberg representations has level 2κ under the diagonal current. The fusion just specified remains at level κ and adds the charges.

The jet twist

The classical Gauss constraint alone does not impose the holomorphic Ward identity. On a disk of radius R with two smaller disks of radii r', r'' removed, the partial connection $a = d\bar{t}$ is closed. With zero sources its mode-one contour defect is $\kappa(R^2 - r'^2 - r''^2)$. This follows from $\oint (t - c)d\bar{t} = -2\pi i r^2$ around a counterclockwise circle. At $\kappa = 2$, $R = 2$, $r' = r'' = 1/4$, the defect is $31/4$. The quantum comparison above uses the constructed current operations, not that constraint alone.

The line-side family has an additional jet twist. In the two coefficient algebras set

$$s(z) = \frac{1}{\kappa} \sum_{p+m \geq 1} (-1)^{p+1} \binom{p+m}{p} \frac{B'_p B''_m}{(p+m) z^{p+m}}, \qquad r = Q_Y s, \qquad h = \frac{N}{z} + \partial_z s.$$

Both indices are nonnegative. Coefficients outside a chosen finite jet set are zero when that restriction is imposed. For all jets the sum is a power series of positive order in $q = z^{-1}$.

Each coefficient is a finite sum. Its generating scalar is $\kappa^{-1} \log(1 + (u - v)/z)$, with $u^p v^m$ replaced by $B'_p B''_m$. In particular $Q_Y s = r$ and $\partial_z s = h - N/z$, with all signs fixed.

THEOREM 22.10.1 *(Formal line-to-state comparison).* Replace the coefficient complex U degreewise by $U^d((q))$, the Laurent series with a finite lower bound and arbitrary coefficient vectors. On $F_{\mathrm{pol}} \otimes U((q))$, the map

$$E(q) = e^{-\rho(s(z))}$$

is an isomorphism from the differential d_U and connection $\partial_z + \rho(N)/z$ to $d_U + \rho(r)$ and $\partial_z + \rho(h)$. It commutes with the currents and their rooted residue operations, and preserves the identity action of the vertex vacuum. It need not fix a chosen cyclic vector.

Proof. The differential graded module identity gives $[d_U, \rho(s)] = \rho(r)$. The even operator $\rho(s)$ commutes with $\rho(r)$ and $\rho(h)$, because the coefficient algebra is graded commutative. Thus

$$e^{\rho(s)}(d_U + \rho(r))e^{-\rho(s)} = d_U, \qquad e^{\rho(s)}(\partial_z + \rho(h))e^{-\rho(s)} = \partial_z + \rho(N)/z.$$

There are no further commutators. Positive q-order makes each coefficient of the exponential and its inverse finite. This proves the formulas on the stated Laurent completion for every algebraic module.

The map acts only on coefficients and commutes with all oscillator modes and the total zero mode. It therefore commutes with normal-ordered current operations and their residues. Applied to a ground vector u, it gives $e^{-\rho(s)}u$, which proves the final distinction. □

For finitely many endpoints, the pair twists commute. On a common formal separation domain, the exponential of their sum is independent of parenthesization. After removing it, the state connection is

$$d + \sum_{i<j} \frac{B_{\mathrm{o}}^{(i)} B_{\mathrm{o}}^{(j)}}{\kappa} d \log(z_i - z_j).$$

This proves formal refinement coherence on that domain. It does not evaluate an arbitrary all-jets series at numerical separations.

An exact evaluation criterion

Fix finitely many jets and write $\rho(s) = S(q) = \sum_{k=1}^{K} A_k q^k$, with commuting degree-zero operators A_k. In this paragraph, an exponential is defined when its partial sums on each vector converge within a finite-dimensional subspace. The operators are *jointly locally finite* when $\mathbb{C}[A_1, \dots, A_K]u$ is finite dimensional for every u.

Proposition 22.10.2 *(Algebraic evaluation).* At a fixed nonzero separation, $e^{-S(q)}$ is defined on algebraic states exactly when $S(q)$ is locally finite. The formal map preserves $U \otimes \mathbb{C}((q))$ exactly when the A_k are jointly locally finite. This is equivalent to evaluation being defined at every nonzero separation. Evaluation at K distinct nonzero values of q already suffices. On this domain a removable singularity of the comparison at $z = 0$ occurs exactly when every A_k vanishes, in which case the comparison is the identity.

Proof. Successive differences of partial sums on u recover $S(q)^n u/n!$. Their finite-dimensional convergence is therefore equivalent to local finiteness, and the inverse is the matrix exponential on each finite orbit. These orbits may be enlarged by those of $d_U u$, $\rho(r)u$, $\rho(r)d_U u$. The relations $[d_U, S] = \rho(r)$, $[S, \rho(r)] = 0$ and $\rho(r)^2 = 0$ then justify the cochain identity on a finite-dimensional stable space.

A vector-valued series lies in the algebraic scalar extension precisely when its coefficients span a finite-dimensional space. A finite joint orbit clearly suffices.

For necessity, identify the cyclic coefficient module of a nonzero u with the quotient

$$B = \mathbb{C}[t_1, \dots, t_K]/\mathrm{Ann}(u).$$

If the coefficients of $\exp(-\sum q^k t_k)$ span a finite space, evaluation at characters of B puts all scalar functions $\exp(-\sum q^k \chi(t_k))$ in a finite-dimensional space.

Exponentials of distinct polynomials with zero constant term are linearly independent. Choose a ray avoiding the finitely many leading real-part coincidences. Divide a dependence by its fastest growing term and let the radius tend to infinity. Proceed inductively.

Thus B has finitely many characters. The Nullstellensatz makes each polynomial $\prod_\chi (t_k - \chi(t_k))$ nilpotent in B. A power is a monic relation for t_k, so reducing all exponents shows that B is finite dimensional.

If $S(q_j)$ are locally finite at K distinct nonzero q_j, their commuting joint orbit is finite on each vector, by their monic relations. The matrix (q_j^k) is invertible, so each A_k is a linear combination of these operators. This proves joint local finiteness. Finally on each finite joint orbit $e^{-S(q)}$ is entire in q. Removability at $z = 0$ makes its entries bounded at infinity. Liouville's theorem makes it the constant identity, and the formal logarithm gives $S = 0$. □

The coefficientwise completion is strictly larger in a basic example. Let $M = \mathbb{C}[b] \otimes \Lambda(c)$, $d_M = \kappa c \partial_b$, with $B_0 = 0$, $B_1 = b$, $c_0 = c$, and all higher coefficients zero. Take a second module of scalar charge one. Then $s = b/(\kappa z)$, $r = c/z$, and

$$e^{-s}1 = \sum_{n \geq 0} \frac{(-1)^n b^n q^n}{n! \kappa^n}$$

lies in $M((q))$ but not in $M \otimes \mathbb{C}((q))$.

At fixed $z \neq 0$, the untwisted degree-zero cohomology is $\mathbb{C}$. The twisted equation $\kappa f'(b) + f(b)/z = 0$ has no nonzero polynomial solution, by its highest coefficient. The two algebraic complexes are not quasi-isomorphic. The formal theorem remains valid on its larger coefficient space.

Coincidence and nested regions

Universal coefficients also admit a distributional interpretation with fixed normalization. For $k \geq 1$, define

$$T_k = \frac{(-1)^{k-1}}{(k-1)!}\partial_z^{k-1}(1/z).$$

It extends z^{-k} and satisfies $\bar{\partial}T_k = \pi(-1)^{k-1}\partial_z^{k-1}\delta_0/(k-1)!$. At finite jet set, complete the coefficient algebra in the positive-jet ideal. Each coefficient of e^{-s} is a finite linear combination of powers z^{-k}, so this formula extends it coefficientwise and fixes its contact derivatives. It does not multiply distributions with the same singular support or evaluate the whole coefficient series on an arbitrary module.

For a smooth radial approximate identity ρ_r of mass one, these fixed coefficients satisfy $T_k * \rho_r \to T_k$ with error $O(r^2)$ on each compact smooth test. Taylor expansion of the test function has no first-moment contribution by radial symmetry. A finite-order distribution bound controls the second remainder. This gives a precise local smoothing statement with the same contact normalization.

Composition of Laurent expansions has its own domain. For example,

$$\frac{1}{(z_1 - z_2)^2} = \sum_{n\geq 0}(n+1)\frac{z_2^n}{z_1^{n+2}} \qquad (|z_2| < |z_1|).$$

The series converges with all derivatives on compact subsets of this nested region.

No distributional convergence holds on an open set crossing the boundary. In the ratio variable, take a radial test centered at a point of modulus greater than one. Holomorphicity makes its integral the value at the center, which does not tend to zero. The existence of the distribution $T_2(z_1 - z_2)$ does not enlarge the convergence domain of this particular composition series. Wick products inherit the corresponding nested domains, while their formal refinement identities use the common coefficient completion specified above.

Finally, the comparison map and the binary operation have different coincidence behavior. All positive coefficients may act by zero, making $E = 1$, while the current operation still has a pole. For scalar charges with $ab/\kappa \notin \mathbb{Z}$, the state connection has monodromy $e^{-2\pi i ab/\kappa}$, and $z^{ab/\kappa}K_{a,b}$ is defined on branches. It has no single-valued distributional extension across a disk about zero agreeing with those branches. These are conditions on the operation and its coefficient local system, not a failure of the formal coefficient isomorphism.

22.11 ARITHMETIC BULK AND BOUNDARY COMPARISONS

The finite arithmetic torsor, the topological state sum and the free physical boundary complex have now been constructed on their stated domains. An arithmetic bulk theory relating them needs a common geometric and categorical recipient. Write $\mathscr{X}$ for a specified arithmetic site and $j : \mathscr{B}_\infty \to \mathscr{X}$ for a proposed archimedean boundary morphism. An orientation consists of a dualizing coefficient object and a trace in its specified degree. In a suitable coefficient theory, it is a map $R\Gamma_c(\mathscr{X}, \omega_{\mathscr{X}}) \to k[-3]$. It must be compatible with the restriction and relative duality used by the action.

At the boundary, distinguish a linear category $\mathcal{B}$, its Hochschild complex, a marked trace class, and a chosen state realization E_∞. For example, the intended primary arithmetic datum can be a marked class in the Hochschild trace of a specified Iwahori–Hecke category. Extracting E_∞ from that trace requires its degree, coefficient theory and completion. The trace class is not a vertex algebra or a category of its own modules. A boundary comparison must give maps of these actual objects preserving the marked class and the operations under consideration.

Three arithmetic identities then have different content. The boundary identity compares the designated trace or state realizations. The Wilson identity compares normalized field integrals with Frobenius traces in the chosen coefficients. The partition identity compares a defined scalar trace with a completed L-function.

For the Riemann normalization the latter function is

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Its polar-removal factor cannot be omitted when the comparison is made with ξ rather than $\pi^{-s/2}\Gamma(s/2)\zeta(s)$. An identity as a function of s needs an analytic family of observables or traces and a common domain. Renaming a fixed finite-group level s does not produce such a family.

Finite towers

A tower construction starts with actual finite field functors and coherent transition maps on their states and actions. Proposition 22.6.1 computes the measure required by a reduction. A coherent inverse diagram defines a pro-object in the chosen functor category. If a represented limit exists and its composition maps preserve the finite gluing identities, it defines the corresponding limiting assignment.

A Hilbert realization can be constructed under the isometric and boundedness hypotheses of Theorem 22.6.2 and Proposition 22.6.3. Arithmetic field spaces need their own measures and maps satisfying those hypotheses. The boundary, Wilson and partition comparisons are further identities on that limit.

Even the untwisted closed arithmetic example distinguishes the finite sum from a zeta partition formula. The étale fundamental group of Spec $\mathbb{Z}$ is trivial: a nontrivial finite extension unramified at all finite primes would contradict the discriminant bound. Its finite G-field groupoid is BG, so its untwisted mass is $1/|G|$. Additional ramification, boundary or analytic data are needed for a nonconstant arithmetic partition family.

Modular categories

A finite semisimple ribbon category with nondegenerate braiding gives the algebraic input to a Reshetikhin–Turaev theory, together with its anomaly and tangential structure. The C_2 example above provides explicit such data. To apply Z_{RT} to arithmetic objects, construct an arithmetic bordism category. Supply also a symmetric monoidal functor C into the appropriate topological bordism category. Their composition $Z_{\mathrm{RT}}C$ is then an arithmetic assignment.

This composition preserves the monoidal and gluing identities by functoriality. Its boundary trace and Wilson labels agree with the intended arithmetic ones only under maps establishing those equalities. An arithmetic decomposition of a closed object could identify its amplitude with an operator trace. A separate character formula could then identify that trace with $\xi(s)$. A modular category alone supplies neither the arithmetic bordism functor nor the analytic character identity. If the target contains infinitely many Fock states, the finite trace restriction following (22.20) must also be addressed.

Factorization from fields

There is a useful positive construction before imposing factorization descent. Suppose $U \mapsto \mathcal{F}(U)$ is a contravariant field-groupoid functor on opens, finite on the regions in use, and takes finite disjoint unions to products. Define $A(U) = \mathbb{C}^{\pi_0\mathcal{F}(U)}$. For disjoint $U_i \subset V$, set

$$\mu_{U_i,V}(f_1 \otimes \cdots \otimes f_r)(b) = \prod_i f_i(b|_{U_i}). \tag{22.30}$$

Restriction of fields and multiplication of functions prove associativity of these maps, their symmetry and the empty-input unit. Thus they define a prefactorization algebra of finite field functions. Weighted integration is an additional functional on it. Compatibility of that functional with gluing uses the action and the homotopy-fiber measure.

Descent of the field groupoids does not ensure factorization descent of these function spaces. A circle gives a counterexample. Take finite C_2-bundles and the cover by all complements of one point. It is a Weiss cover: every finite set of points lies in one member. Each nonempty finite intersection is a finite union of intervals and has one isomorphism class of bundles, although it retains automorphisms.

Every function space in its Čech diagram is therefore $\mathbb{C}$, with identity transition maps. The cover nerve has every finite simplex, so its homotopy colimit is $\mathbb{C}$.

On the circle itself there are two holonomies, and the function space is $\mathbb{C}^2$. The descent map has image the constants and misses the sign character.

Consequently the arithmetic factorization problem can begin with the maps (22.30), but must construct or replace their descent in the intended recipient. Derived or categorical coefficients can retain data lost by functions on isomorphism classes. Their factorization maps, support and continuity still need a proof. This issue already occurs before an archimedean Hilbert extension.

Fully dualizable objects

The framed cobordism hypothesis starts from a symmetric monoidal $(\infty, 3)$-category and a fully dualizable object. It uses the object's dual, evaluation, coevaluation and the required adjoints through the morphism levels, with their coherences. For the Morita three-category of finite tensor categories, Douglas–Schommer-Pries–Snyder prove that separable tensor categories are fully dualizable. In characteristic zero this includes finite semisimple tensor categories. Their theorem gives a framed three-dimensional theory. An oriented extension uses the appropriate homotopy fixed-point structure.

A categorical arithmetic recipient with trace and restricted-tensor conditions must itself be constructed: composition must exist and preserve those conditions. A monoidal comparison with the ordinary Morita category must transport the actual duality and adjunction data. A list of candidate objects and bimodules does not prove these facts. Once they hold, the framed theory exists in that target. An arithmetic bordism comparison, the boundary trace map and the Frobenius labels remain separate inputs. Agreement with a proposed RT theory requires comparison of their extended data in a common category.

A six-functor trace

Suppose an arithmetic space $X^\diamond$ has coefficient categories $C(-)$. Let its structure morphism $f : X^\diamond \to \mathrm{pt}$ admit the required functors. For $B \in C(\mathrm{pt})$, the expression $f_! f^! B$ then has a definite type. If it is dualizable, its identity has trace

$$\mathrm{Tr}(1_{f_! f^! B}) \in \mathrm{End}_{C(\mathrm{pt})}(1).$$

A scalar value uses a specified map from this endomorphism ring to the scalar field. A regularized trace, if needed in place of dualizability, requires its own domain and normalization.

This produces a trace expression on the supplied geometry. The global arithmetic diamond and its archimedean boundary are additional inputs. So are the realization of the Hecke trace as B and the Euler-product trace identity. Local geometric Langlands and local diamond geometry provide arithmetic objects that a comparison may use. They do not by themselves define all of this global data.

Comparing Hilbert realizations

A common positive pairing does give a strong comparison once the state spaces have been constructed. Let E_∞ be a specified complex trace realization, with a positive semidefinite Hermitian form q_∞. Write its nullspace as N_q.

Proposition 22.11.1 *(Boundary Hilbert transport).* Suppose $I_B : E_\infty \to H_B$ and $I_P : E_\infty \to H_P$ have dense ranges and satisfy

$$\langle I_B v, I_B w\rangle = q_\infty(v, w) = \langle I_P v, I_P w\rangle.$$

There is a unique unitary $T : H_B \to H_P$ with $T I_B = I_P$. If closed operators A_B, A_P have cores $I_B\mathcal{D}$, $I_P\mathcal{D}$, respectively, and $T A_B I_B v = A_P I_P v$ for $v \in \mathcal{D}$, then

$$T \operatorname{Dom} A_B = \operatorname{Dom} A_P, \qquad T A_B = A_P T.$$

For normal operators this also intertwines their bounded spectral functions.

Proof. Cauchy–Schwarz makes N_q orthogonal to all of E_∞. Both realization kernels equal N_q, so $I_B v \mapsto I_P v$ is a well-defined isometry on dense subspaces. It extends uniquely onto H_P. The graph of each closed operator is the closure of its graph on the specified core. Applying $T \oplus T$ gives the domain identity and the full intertwining relation. The normal functional calculus then proves the last assertion. □

Boundary operations must preserve N_q to descend even algebraically. Their closed generators need the domains in this proposition. A further identification with logarithmic scaling on $L^2(\mathbb{R}_{>0}, dx/x)$ requires another unitary with its domain identity. Polynomial Fock energy has eigenvectors, whereas the full-line logarithmic translation generator has none. Those representations cannot be identified by a unitary preserving these generators.

Positive Hilbert descent is also distinct from the complete Weil pairing. Let $\mathcal{T}$ be the chosen Weil test algebra with involution $*$. Its complete signed functional W includes the polar, archimedean and prime terms.

If a map $J : \mathcal{T} \to E_\infty$ satisfies

$$q_\infty(Jf, Jg) = W(f * g^*) \qquad (f, g \in \mathcal{T}), \tag{22.31}$$

then $W(f * f^*) \geq 0$. This follows by setting $g = f$. The substantive assertion is (22.31) on the full required test domain, with its completion and normalization. A positive form on an auxiliary finite gauge or oscillator space does not supply it. Conversely, positivity of the full Weil form does not construct the arithmetic bordism functor.

22.12 Other physical realizations and the remaining maps

Arithmetic Chern–Simons is one possible physical formulation. It does not exclude others. The class-$\mathcal{S}$ construction compactifies the six-dimensional $(2, 0)$ theory on a complex curve, with its twist and defect data, to obtain four-dimensional $\mathcal{N} = 2$ theories.

For an unpunctured complex curve its Hitchin base uses $\bigoplus_i H^0(C, K_C^{d_i})$. Punctures change the allowed poles. Replacing K_C by a metrized arithmetic dualizing line does not specify a Higgs field, Hitchin equations or a six-dimensional compactification. Krull dimension cannot be added to a real manifold dimension to prove that every arithmetic version is impossible. The question is to construct the geometry, fields and comparison that would make such a version defined.

Likewise, a local Hecke character is a label, not the one-dimensional defect carrying it. Local class field theory relates its appropriate finite or Weil realization to local arithmetic data. Turning it into a boundary module requires a functor to the specified defect category, including its ramified action. The local character space can itself have positive-dimensional families. Choosing a point in it does not assign a dimension to the physical defect.

The protected chiral construction of Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees starts with an actual four-dimensional superconformal theory and its protected cohomology. Its algebra and Schur trace retain those fields and symmetries. An arithmetic Euler trace has a simpler independent construction: on the finite occupation states with energies $\sum_p n_p \log p$, the trace of e^{-sH} is $\prod_p (1 - p^{-s})^{-1}$ for $\operatorname{Re} s > 1$. This does not identify the space with protected cohomology. Frobenius is an endomorphism, not automatically an integer grading. A symbol t^{Frob} requires a defined functional calculus or a specified graded realization.

The arithmetic comparison therefore remains an object and map problem. The required data are the arithmetic bulk geometry, field action, coherent measures and gluing. One must construct the boundary trace realization, its pairing and the Frobenius Wilson comparison. The partition trace also requires its analytic domain.

The finite torsors, abelian linking correlations, Haar realization and free physical period map give positive constructions in four specified settings. Their extension to one arithmetic theory requires the maps just identified. The full Weil comparison adds a further identity on its complete test algebra.

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Chapter 23

The Weil algebra and its domain

The explicit formula equates arithmetic and spectral evaluations of a test function. The arithmetic evaluation contains every prime power, a gamma integral and two polar values. The spectral evaluation sums over the zeros of the completed zeta function. Applying the formula to convolution products produces a Hermitian form. Its positivity on the full test algebra is equivalent to the Riemann hypothesis.

The domain is part of this assertion. Compact support makes each prime sum finite. Exponential decay permits Gaussian tests and controls the infinite prime sum. The ordinary Schwartz topology controls the gamma integral but does not control the critically weighted prime measure. We shall construct the relevant extensions and the maps between their domains.

Two examples explain why these choices matter. The polar form has rank two and both signs. The finite-prime form also has a two-dimensional subspace with both signs. Their contributions enter the complete formula with specified signs. A positive realization must reproduce that complete form.

Fourier inversion, elementary distribution theory and complex analysis are the prerequisites. We use Lebesgue measure on the additive real line. Hermitian forms are linear in their first argument. The convolution algebras below have no unit unless a unitization is explicitly taken.

23.1 Compact tests and Mellin reflection

For $R > 0$, let $\mathscr{A}_R$ be the space of complex smooth functions supported in $[-R, R]$, with seminorms

$$\|f\|_{R,j} = \sup_{x\in\mathbb{R}} |f^{(j)}(x)|, \qquad j \geq 0.$$

The compact test space is

$$\mathscr{A}_c = C_c^\infty(\mathbb{R}) = \operatorname*{colim}_{R\to\infty} \mathscr{A}_R$$

with its usual strict locally convex inductive-limit topology. A convergent sequence lies in one $\mathscr{A}_R$ and converges there in every derivative seminorm.

Definition 23.1.1 *(The compact convolution algebra).* For $f, g \in \mathscr{A}_c$, set

$$(f * g)(x) = \int_{\mathbb{R}} f(y)g(x-y)\,dy, \qquad f^*(x) = \overline{f(-x)}. \tag{23.1}$$

The Fourier transform and its real-axis inverse are

$$h_f(z) = \int_{\mathbb{R}} f(x)e^{izx}\,dx, \qquad f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} h_f(t)e^{-itx}\,dt.$$

For an entire function h, write $h^\sharp(z) = \overline{h(\bar{z})}$.

PROPOSITION 23.1.2 *(Products, support and involution).* Convolution and $f \mapsto f^*$ make $\mathscr{A}_c$ a commutative topological complex $*$-algebra. They satisfy

$$\operatorname{supp}(f * g) \subset \operatorname{supp} f + \operatorname{supp} g, \qquad h_{f*g} = h_f h_g, \qquad h_{f^*} = h_f^\sharp.$$

For $f \in \mathscr{A}_R$, h_f is entire and obeys

$$|h_f(z)| \le C_{R,N}(f) e^{R|\operatorname{Im} z|}(1 + |z|)^{-N} \qquad (N \ge 0),$$

where $C_{R,N}$ is a continuous seminorm. In particular, h_f and all its derivatives decrease rapidly on every closed horizontal strip.

Proof. The support assertion follows directly from the integral. Fubini proves associativity, commutativity and the product identity. Reflection and conjugation give the involution identity. Differentiation may be performed under convolution on a fixed compact support. The resulting estimates prove continuity on the compact test space.

Compact support also permits differentiation of h_f at every complex argument. Repeated integration by parts gives

$$(iz)^N h_f(z) = (-1)^N \int f^{(N)}(x) e^{izx}\, dx.$$

There are no boundary terms. Combine this bound for $|z| \ge 1$ with the integral bound for $|z| \le 1$. Applying the same argument to $x^j f(x)$ proves the assertion about derivatives. □

We give the Fourier image $h(\mathscr{A}_c)$ the transported topology. An entire function in this image cannot vanish on a real interval unless it is zero. Exact restriction to a bounded frequency interval therefore does not preserve this Fourier image.

The map $\epsilon(f) = \int f$ is multiplicative. It does not provide a convolution unit. Indeed, a unit $e \in \mathscr{A}_c$ would satisfy $h_e(t) h_f(t) = h_f(t)$ for every f. At each real t, some compact test has nonzero transform, so $h_e(t) = 1$. This contradicts rapid decay.

The Dirac distribution at zero is the unit in a larger distribution algebra. Smooth functions $b_\delta(x) = \delta^{-1} b(x/\delta)$, with $\int b = 1$, instead form an approximate identity on $\mathscr{A}_c$.

The multiplicative coordinate $u = e^x$ introduces a half-density. For $f \in \mathscr{A}_c$, define

$$\phi_f(u) = u^{-1/2} f(\log u), \qquad \Phi_f(s) = \int_0^\infty \phi_f(u) u^s \, \frac{du}{u} = h_f\big(-i(s - \tfrac{1}{2})\big).$$

Thus ϕ_f is compactly supported on $\mathbb{R}_{>0}$. Multiplicative convolution uses the measure du/u.

PROPOSITION 23.1.3 *(The two reciprocal maps).* The assignment $f \mapsto \phi_f$ preserves convolution. Its induced antilinear involution and its linear reciprocal map are

$$\phi^*(u) = u^{-1} \overline{\phi(u^{-1})}, \qquad \phi^\iota(u) = u^{-1} \phi(u^{-1}).$$

Their Mellin transforms are

$$\Phi_{\phi^*}(s) = \overline{\Phi_\phi(1 - \bar{s})}, \qquad \Phi_{\phi^\iota}(s) = \Phi_\phi(1 - s).$$

In particular,

$$\Phi_{f * g^*}(s) = \Phi_f(s) \overline{\Phi_g(1 - \bar{s})}.$$

Proof. In multiplicative convolution substitute $u = e^x$, $v = e^y$. The two half-density factors multiply to $e^{-x/2}$, leaving $(f * g)(x)$. Substitution $u \mapsto u^{-1}$ in the Mellin integral gives both reciprocal identities. The last formula follows from Fourier multiplication. □

Mellin reflection is linear. A Hermitian pairing also needs complex conjugation. Equivalently, a form q linear in its first argument is a complex-linear functional on $\mathscr{A}_c \otimes_{\mathbb{C}} \overline{\mathscr{A}_c}$, where the bar denotes the conjugate vector space. Replacing this factor by $\mathscr{A}_c$ changes the scalar rule: a bilinear diagonal changes sign under $f \mapsto if$, whereas a Hermitian diagonal does not.

23.2 Prime powers and exponential weights

Let $\Lambda(n)$ be the von Mangoldt function: $\Lambda(p^k) = \log p$ for primes p and integers $k \geq 1$, and $\Lambda(n) = 0$ otherwise. The logarithmic prime measures are

$$\mu_\sigma = \sum_{n \geq 2} \Lambda(n) n^{-\sigma} \delta_{\log n}, \qquad \sigma \in \mathbb{R}.$$

Every compact interval meets finitely many atoms. Each μ_σ therefore acts continuously on $\mathscr{A}_c$.

Lemma 23.2.1 *(Elementary prime mass).* Put $\Psi(X) = \sum_{n \leq X} \Lambda(n)$. For integers $m \geq 1$,

$$2m \log 2 - \log(2m + 1) \leq \Psi(2m).$$

Moreover, $\Psi(X) \leq X \log X$ for $X \geq 2$. Consequently, $\Psi(X) \geq cX$ for all sufficiently large X, with some $c > 0$.

Proof. The largest coefficient in $(1 + 1)^{2m}$ is $\binom{2m}{m}$, so $\binom{2m}{m} \geq 4^m/(2m + 1)$. Its p-adic valuation is

$$v_p\binom{2m}{m} = \sum_{k \geq 1} \left(\left\lfloor \frac{2m}{p^k} \right\rfloor - 2 \left\lfloor \frac{m}{p^k} \right\rfloor \right).$$

Each summand is zero or one and vanishes for $p^k > 2m$. Taking logarithms gives $\log \binom{2m}{m} \leq \Psi(2m)$. The upper bound follows from $\Lambda(n) \leq \log n$. Monotonicity, with $m = \lfloor X/2 \rfloor$, gives the final lower bound. □

Theorem 23.2.2 *(The exact temperedness threshold).* The positive measure μ_σ is tempered exactly when $\sigma \geq 1$. It has finite total mass exactly when $\sigma > 1$. In particular, the critical measure $\mu_{1/2}$ is not tempered.

Proof. For $\sigma \geq 1$ and a Schwartz function v,

$$|\mu_\sigma(v)| \leq \sup_x (1 + |x|)^3 |v(x)| \sum_{n \geq 2} \frac{\log n}{n(1 + \log n)^3}.$$

The series converges by the integral test. For $\sigma > 1$, the stronger bound $\sum (\log n) n^{-\sigma} < \infty$ proves finite mass. At $\sigma = 1$, partial summation gives

$$\sum_{n \leq X} \frac{\Lambda(n)}{n} = \frac{\Psi(X)}{X} + \int_1^X \frac{\Psi(u)}{u^2}\, du.$$

The lower bound of the lemma makes this quantity diverge.

A positive tempered measure has polynomially bounded mass on $[-R, R]$. To see this, test against $v(x/R)$, where $v \geq 0$ is a fixed compact smooth function equal to one on $[-1, 1]$. Each of its finitely many required Schwartz seminorms grows polynomially in R.

If $0 \leq \sigma < 1$, then

$$\mu_\sigma([0, R]) \geq e^{-\sigma R}\Psi(e^R) \geq ce^{(1-\sigma)R}$$

for large R. If $\sigma < 0$, the lower bound ce^R already suffices. These estimates contradict polynomial growth. They also exclude finite mass. □

The Euler product gives a transform on its actual convergence domain. For $\operatorname{Re} s > 1$,

$$\zeta(s) = \sum_{n\geq 1} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \qquad -\frac{\zeta'(s)}{\zeta(s)} = \sum_{n\geq 2} \Lambda(n) n^{-s}. \tag{23.2}$$

Absolute convergence proves the product and permits logarithmic differentiation on compact subsets of this half-plane. Thus $-\zeta'/\zeta$ is the Laplace transform of μ_0 there. With Fourier convention $\int e^{-itx}\, d\mu(x)$, the transform of μ_σ, $\sigma > 1$, is $-\zeta'(\sigma + it)/\zeta(\sigma + it)$.

The seminorm estimate in the theorem and dominated convergence give $\mu_{1+\varepsilon} \to \mu_1$ weakly in tempered distributions. Fourier continuity gives the corresponding boundary transform. Meromorphic continuation of $-\zeta'/\zeta$ does not extend the original Laplace integral beyond its convergence domain.

23.3 THETA CONTINUATION AND THE ZERO DIVISOR

We normalize the completed zeta function before forming its logarithmic derivative. Let

$$\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}, \qquad r(t) = \frac{\theta(t) - 1}{2} \qquad (t > 0).$$

The Gaussian transform gives the theta identity

$$\theta(t) = t^{-1/2}\theta(t^{-1}). \tag{23.3}$$

One proof periodizes $e^{-\pi t x^2}$. Its m-th Fourier coefficient is $t^{-1/2}e^{-\pi m^2/t}$. Both periodic sums converge uniformly, and uniqueness of Fourier coefficients gives the identity at $x = 0$.

The Gaussian transform itself follows by differentiating its integral with respect to frequency and using its value at zero.

For $\operatorname{Re} s > 1$, termwise integration yields

$$\Lambda_\zeta(s) := \pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty r(t) t^{s/2}\, \frac{dt}{t}.$$

Split the integral at one and use (23.3) in the lower part. The result is

$$\Lambda_\zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty r(t)\left(t^{s/2} + t^{(1-s)/2}\right) \frac{dt}{t}.$$

The integral is entire because $r(t)$ decreases exponentially. It follows that Λ_ζ is meromorphic, with residues -1 at zero and 1 at one.

Definition 23.3.1 *(The normalized entire function).* Set

$$\xi(s) = \frac{1}{2}s(s-1)\Lambda_\zeta(s) = \frac{1}{2} + \frac{s(s-1)}{2}\int_1^\infty r(t)\left(t^{s/2} + t^{(1-s)/2}\right)\frac{dt}{t}. \tag{23.4}$$

Its zeros are counted with their analytic multiplicities. For a zero ρ, put $z_\rho = -i(\rho - \frac{1}{2})$.

The integral proves

$$\xi(0) = \xi(1) = \frac{1}{2}, \qquad \xi(1-s) = \xi(s), \qquad \xi(\bar{s}) = \overline{\xi(s)}.$$

The Euler product has no zero for $\operatorname{Re} s > 1$. The functional equation therefore places all zeros of ξ in $0 \le \operatorname{Re} s \le 1$. The associated parameters satisfy $|\operatorname{Im} z_\rho| \le 1/2$.

There is no real zero of ξ. For $0 < s < 1$, pair consecutive terms of the alternating series:

$$\eta(s) = \sum_{m\ge 1}\left((2m-1)^{-s} - (2m)^{-s}\right) > 0.$$

Dirichlet convergence and analytic continuation give $\eta(s) = (1-2^{1-s})\zeta(s)$. Hence $\zeta(s) \ne 0$ on that interval. The endpoints and the remaining real line follow from the displayed values, the Euler product and reflection. Consequently no z_ρ lies on the imaginary axis.

Lemma 23.3.2 *(Growth and summability).* For $R \ge 2$,

$$\log \max_{|s|\le R} |\xi(s)| \le CR\log(R+2).$$

The number of zeros in $|s| \le R$, with multiplicity, is $O(R\log(R+2))$. In particular, $\sum_\rho |\rho|^{-2} < \infty$, and the corresponding inverse-square sum over z_ρ converges. There are infinitely many zeros.

Proof. On $t \ge 1$, one has $r(t) \le Ce^{-\pi t}$. For $|s| \le R$, the two powers in (23.4) are bounded by $2t^{(R+1)/2}$. The elementary maximum of $t^A e^{-\pi t/2}$ is $(2A/(\pi e))^A$. This bounds the integral by $\exp(CR\log(R+2))$, including its quadratic prefactor.

Jensen's formula at radius $2R$, with $\xi(0) = 1/2$, bounds the number of zeros in radius R: each contributes at least $\log 2$. Summing over dyadic annuli proves inverse-square summability. Translation by $1/2$ gives the assertion for z_ρ.

If the zero set were finite, Hadamard factorization for an entire function of order at most one would give $\xi(s) = e^{as+b}P(s)$ for a polynomial P. Along positive real s, Stirling's formula and $\zeta(s) \to 1$ instead give $\log|\xi(s)| = \frac{1}{2}s\log s + O(s)$. This contradicts that representation. □

Proposition 23.3.3 *(The genus-one normalization).* Let γ_E be Euler's constant and put

$$B = -1 - \frac{\gamma_E}{2} + \frac{1}{2}\log(4\pi).$$

Then

$$\xi(s) = \frac{1}{2}e^{Bs}\prod_\rho (1-s/\rho)e^{s/\rho}, \qquad \frac{\xi'(s)}{\xi(s)} = B + \sum_\rho\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right). \tag{23.5}$$

The factors and paired summands converge locally uniformly away from their zeros or poles.

Proof. The preceding growth estimate permits Hadamard factorization of genus one. Inverse-square summability gives local normal convergence. At zero the constant factor is $1/2$, and the logarithmic derivative of every elementary factor vanishes. The linear exponential coefficient is therefore $\xi'(0)/\xi(0)$.

Use $\Gamma(1+s/2) = (s/2)\Gamma(s/2)$ to write

$$\xi(s) = (s-1)\Gamma(1+s/2)\pi^{-s/2}\zeta(s).$$

The values $\zeta(0) = -1/2$, $\zeta'(0) = -\frac{1}{2}\log(2\pi)$ and $\Gamma'(1) = -\gamma_E$ give the stated B. For the two zeta values, expand the equivalent functional equation

$$\zeta(s) = 2^s\pi^{s-1}\sin(\pi s/2)\Gamma(1-s)\zeta(1-s).$$

The product $\Gamma(1-s)\zeta(1-s)$ is $-s^{-1} + O(s)$. The remaining factor is $\frac{s}{2}(1 + s\log(2\pi) + O(s^2))$. This proves both values.

Local differentiation of the normally convergent canonical product proves the logarithmic derivative formula. □

The summands in (23.5) are paired. The formula makes no separate absolute-convergence assertion for $\sum_\rho 1/\rho$. This convention will also determine the normalization of spectral determinants.

The argument term in zero counting

Let $N(T)$ count the zeros ρ with $0 < \operatorname{Im}\rho \le T$, including multiplicity. For $T > 0$ that is not a zero ordinate, define

$$\vartheta(T) = \operatorname{Im}\log\Gamma\left(\tfrac{1}{4} + \tfrac{iT}{2}\right) - \frac{T}{2}\log\pi, \qquad S(T) = \frac{1}{\pi}\Delta_{2\to 2+iT\to 1/2+iT}\arg\zeta.$$

The gamma logarithm is the analytic branch on the right half-plane that is real on the positive real axis. The zeta argument starts at zero at $s = 2$.

PROPOSITION 23.3.4 *(The zero-count normalization).* Away from zero ordinates,

$$N(T) = 1 + \frac{\vartheta(T)}{\pi} + S(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8} + S(T) + O(T^{-1}).$$

At a zero ordinate, the formula for the stated convention $N(T)$ uses the right limit of S. The classical bound $S(T) = O(\log T)$ gives

$$N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + O(\log T).$$

Proof. Use the rectangle with real parts -1, 2 and imaginary parts 0, T. There are no zeros on its boundary. The real bottom edge contributes no argument change, since ξ is real and nonzero there. Reflection $s \mapsto 1 - \bar{s}$ and conjugation of ξ show that the remaining argument change is twice the change along

$$C_T: \quad 2 \longrightarrow 2 + iT \longrightarrow \tfrac{1}{2} + iT.$$

The argument principle therefore gives $N(T) = \pi^{-1}\Delta_{C_T}\arg\xi$.

At the endpoint, the continuous arguments of s and $s-1$ sum to π. The factorization $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ now gives the first equality. In particular, the horizontal segment contributes to $S(T)$.

Stirling's expansion at $z = 1/4 + iT/2$ gives

$$\vartheta(T) = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} - \frac{\pi}{8} + \frac{1}{48T} + O(T^{-3}).$$

For the constant, the term $(z - \frac{1}{2})\log z - z$ contributes $-\frac{1}{4}\arg z \to -\pi/8$. Thus $7/8 = 1 - 1/8$. Taking right limits proves the convention at zero heights.

The argument bound is the classical theorem. A precise primary estimate is Hasanalizade–Shen–Wong, *Counting zeros of the Riemann zeta function*, Theorem 1.3 and Corollary 1.4, arXiv:2107.06506. Their Corollary 1.2 also supplies the displayed estimate for $N(T)$. □

This count controls ordinary convergent sums. It does not regularize divergent positive-power moments. For example, put

$$M_n(T) = \sum_{0<\operatorname{Im}\rho\le T} \left(\tfrac{1}{4} + (\operatorname{Im}\rho)^2\right)^n.$$

For every integer $n \ge 0$, partial summation gives

$$M_n(T) = \frac{T^{2n+1}}{2\pi(2n+1)}\log\frac{T}{2\pi} - \frac{T^{2n+1}}{2\pi(2n+1)^2} + O_n(T^{2n}\log T).$$

Indeed, integrate $(1/4 + t^2)^n$ against $dN(t)$. The error $O(\log(2+t))$ in the count contributes $O_n(T^{2n}\log T)$. In the main integral, replacing $(1/4 + t^2)^n$ by t^{2n} gives a smaller error. The integral of $t^{2n}\log(t/(2\pi))/(2\pi)$ is the displayed main term.

Every $M_n(T)$ diverges. A finite arithmetic moment functional therefore requires an additional, compatible regularization.

23.4 The complete arithmetic functional

Write $\psi_\Gamma = \Gamma'/\Gamma$ and define the real even functions

$$a(t) = \operatorname{Re}\psi_\Gamma\left(\tfrac{1}{4} + \tfrac{it}{2}\right) - \log\pi, \qquad k(t) = -a(t). \tag{23.6}$$

They are smooth on the real line and have at most logarithmic growth.

For $f \in \mathscr{A}_c$, set

$$\begin{aligned} P(f) &= h_f(i/2) + h_f(-i/2),\\ W_\infty(f) &= \frac{1}{2\pi}\int_{\mathbb{R}} h_f(t)k(t)\,dt,\\ W_{\mathrm{fin}}(f) &= \sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\big(f(\log n) + f(-\log n)\big),\\ W(f) &= P(f) - W_\infty(f) - W_{\mathrm{fin}}(f). \end{aligned} \tag{23.7}$$

The prime sum is finite. The gamma integral is absolutely convergent. These formulas define continuous complex-linear functionals on the compact test space. The sign of W_∞ is fixed by its density k.

Theorem 23.4.1 *(The complete explicit formula).* For every $f \in \mathscr{A}_c$,

$$W(f) = \sum_\rho h_f(z_\rho). \tag{23.8}$$

The sum over zeros is absolutely convergent, with multiplicity. For $f, g \in \mathscr{A}_c$, the form

$$q_W(f,g) = W(f * g^*) = \sum_\rho h_f(z_\rho)\overline{h_g(\bar{z}_\rho)} \tag{23.9}$$

is Hermitian and linear in its first argument.

Proof. Put $H(s) = \Phi_f(s)$, and choose $1 < c < 3/2$. Let

$$G(s) = \frac{\Lambda_\zeta'(s)}{\Lambda_\zeta(s)} = -\frac{1}{2}\log\pi + \frac{1}{2}\psi_\Gamma(s/2) + \frac{\zeta'(s)}{\zeta(s)}.$$

Inside the strip $1 - c \le \operatorname{Re} s \le c$, its residues are the zero multiplicities and -1 at each of $0, 1$. The functional equation gives $G(1-s) = -G(s)$.

We justify the contour limit. For all sufficiently large integers j, choose $T_j \in [j, j+1]$ at distance at least j^{-2} from every zero ordinate. The excluded intervals have total length $O(\log j/j) < 1$, by Lemma 23.3.2.

On the horizontal segments at heights $\pm T_j$, (23.5) gives a polynomial bound for ξ'/ξ. The zeros with $|\rho| \le 2T_j$ are bounded using the chosen distance. For the others,

$$\left|\frac{1}{s-\rho} + \frac{1}{\rho}\right| \le C\frac{|s|}{|\rho|^2}.$$

The rational correction $G = \xi'/\xi - 1/s - 1/(s-1)$ has the same bound. Rapid vertical-strip decay of H makes both horizontal integrals tend to zero.

Apply the residue theorem to $G(s)H(s)$ on that rectangle. Reflection of the left vertical side gives

$$\sum_\rho H(\rho) - H(0) - H(1) = \frac{1}{2\pi i}\int_{\operatorname{Re} s = c} G(s)\big(H(s) + H(1-s)\big)\,ds.$$

Absolute convergence of the zero sum follows from strip decay and the zero count.

On the right line, the Dirichlet series in (23.2) is absolutely convergent. Termwise integration is permitted. Fourier inversion gives

$$\frac{1}{2\pi i}\int_{\operatorname{Re} s = c} n^{-s}H(s)\,ds = n^{-1/2}f(\log n).$$

The reflected test gives $n^{-1/2}f(-\log n)$. The negative sign in ζ'/ζ therefore produces $-W_{\mathrm{fin}}(f)$.

Move the gamma part of the integral to $\operatorname{Re} s = 1/2$. Its logarithmic derivative has no pole between these lines. The two reflected integrals combine to

$$\frac{1}{2\pi}\int_{\mathbb{R}} h_f(t)\big(\operatorname{Re}\psi_\Gamma(\tfrac{1}{4} + \tfrac{it}{2}) - \log\pi\big)\,dt = -W_\infty(f).$$

Finally, $H(0) = h_f(i/2)$ and $H(1) = h_f(-i/2)$. This proves (23.8).

Fourier multiplication proves the paired formula. The zero multiset is invariant under $\rho \mapsto 1 - \bar{\rho}$, which sends z_ρ to $\bar{z}_\rho$. Reindexing the absolutely convergent sum proves Hermitian symmetry. □

In (23.9), conjugation acts at $\bar{z}_\rho$. Replacing the summand by $|h_f(z_\rho)|^2$ would change the form when z_ρ is nonreal. Under the Riemann hypothesis every z_ρ is real, and that specialization is valid.

23.5 Exponential extensions and their topology

Gaussian tests require a domain larger than $\mathscr{A}_c$. The growth threshold in Theorem 23.2.2 indicates the needed exponential margin.

Definition 23.5.1 *(Exponential test spaces).* For $a_0 > 1/2$, let

$$\mathscr{A}_{a_0} = \left\{ f \in C^\infty(\mathbb{R}) : p_{a_0,j}(f) := \int_{\mathbb{R}} e^{a_0|x|} |f^{(j)}(x)|\, dx < \infty \text{ for every } j \ge 0 \right\}.$$

Give this space the topology of its displayed seminorms. Put

$$\mathscr{A}_{\exp} = \bigcup_{a_0 > 1/2} \mathscr{A}_{a_0}, \qquad \mathscr{A}_\infty = \bigcap_{a_0 > 1/2} \mathscr{A}_{a_0}.$$

The union has the final locally convex topology of the steps $\mathscr{A}_{1/2+1/n}$. The intersection has all the exponential seminorms. Convolution and the involution are as in (23.1).

Theorem 23.5.2 *(Continuity on each exponential step).* Each $\mathscr{A}_{a_0}$ is a complete Fréchet convolution $*$-algebra, and $\mathscr{A}_c$ is dense in it. The transform h_f is holomorphic for $|\operatorname{Im} z| < a_0$. All its derivatives decrease rapidly on smaller closed strips.

Every term of (23.7) extends continuously to this step. There is a constant C_{a_0} such that

$$|W(f)| \le C_{a_0}\big(p_{a_0,0}(f) + p_{a_0,1}(f) + p_{a_0,2}(f)\big). \tag{23.10}$$

The explicit formula and its Hermitian pairing remain valid, with absolutely convergent prime and zero sums.

Proof. The inequality $e^{a_0|x+y|} \le e^{a_0|x|} e^{a_0|y|}$ gives

$$p_{a_0,j}(f * g) \le p_{a_0,j}(f)\, p_{a_0,0}(g).$$

The involution preserves each seminorm. A Cauchy sequence has weighted L^1 limits for every derivative. Distributional differentiation identifies successive limits. Local one-dimensional Sobolev regularity then gives a smooth representative with those derivatives. This proves completeness.

Choose $\chi \in C_c^\infty(\mathbb{R})$ equal to one near zero. Leibniz's rule shows $\chi(x/R)f(x) \to f(x)$ in every seminorm. The errors are integrable derivative tails multiplied by bounded cutoff derivatives and nonpositive powers of R. This proves density.

On a strip $|\operatorname{Im} z| \le b < a_0$, the spare weight $e^{(a_0-b)|x|}$ controls every polynomial in x. Differentiation in z and integration by parts in x give the asserted holomorphy and rapid decay. The two polar evaluations are bounded by $p_{a_0,0}(f)$. On the real line,

$$|h_f(t)| \le \min\{p_{a_0,0}(f),\ |t|^{-2} p_{a_0,2}(f)\}.$$

Logarithmic growth of k therefore controls the gamma integral.

The function $e^{a_0|x|} f(x)$ has an integrable weak derivative. Integrating from infinity on either half-line gives

$$\sup_x e^{a_0|x|} |f(x)| \le a_0\, p_{a_0,0}(f) + p_{a_0,1}(f).$$

Consequently the absolute prime sum is at most

$$2\bigl(a_0 p_{a_0,0}(f) + p_{a_0,1}(f)\bigr) \sum_{n\ge 2} \frac{\log n}{n^{a_0+1/2}}.$$

This series converges because $a_0 > 1/2$, proving (23.10).

Strip decay and the zero count similarly control the zero sum. Pass to the limit through the compact cutoffs to obtain the explicit formula. Convolution continuity proves the paired assertion. □

The extensions agree on overlapping steps, so they define W on $\mathscr{A}_{\exp}$. All assertions about passage of a quadratic inequality below use a specified step or the compact test topology. No completeness claim for the union is needed.

The intersection $\mathscr{A}_\infty$ is the space of smooth functions with rapid exponential decay. Its topology is equivalently given by

$$\sup_x e^{m|x|}(1+|x|)^N |f^{(j)}(x)|, \qquad m, N, j \in \mathbb{N}.$$

Indeed, one extra exponential weight bounds a polynomial weight and an integral. Conversely, the weighted L^1 bounds for two consecutive derivatives bound the weighted supremum as in the proof above. Compact cutoffs converge in all these seminorms.

The gamma functional alone extends continuously to Schwartz space. The prime functional does not have such an extension: its positive measure contains $\mu_{1/2}$ and its reflection. Thus Schwartz approximation alone does not justify passing each arithmetic term to a limit.

PROPOSITION 23.5.3 *(The regular convolution norm).* Convolution by $f \in \mathscr{A}_{\exp}$ acts boundedly on $L^2(\mathbb{R})$, with operator norm $\|h_f\|_\infty$. The completion for this norm is $C_0(\mathbb{R})$ under Fourier transform.

Proof. Plancherel turns convolution into multiplication by h_f. Since $f \in L^1$, its transform is continuous and vanishes at infinity. To prove density, start with $H \in C_c^\infty(\mathbb{R})$ in the frequency variable. Its inverse Fourier transform v is Schwartz. Compact cutoffs v_R converge to v in L^1, so $\|h_{v_R} - H\|_\infty \le \|v_R - v\|_1 \to 0$. Finally, C_c^∞ is uniformly dense in C_0. □

23.6 THE POLAR AND FINITE-PRIME FORMS

For $f \in \mathscr{A}_{\exp}$, write

$$\ell_-(f) = \int f(x) e^{-x/2}\,dx, \qquad \ell_+(f) = \int f(x) e^{x/2}\,dx.$$

Both are continuous on every exponential step.

PROPOSITION 23.6.1 *(The full polar matrix).* One has

$$P(f * g^*) = \ell_-(f)\overline{\ell_+(g)} + \ell_+(f)\overline{\ell_-(g)}.$$

On compact tests, the map $(\ell_-, \ell_+) : \mathscr{A}_c \to \mathbb{C}^2$ is surjective. The polar form has rank two, with one positive and one negative direction. Its kernel is the common kernel of the two moments. On the complex even tests it is the positive form $2\ell_+(f)\overline{\ell_+(g)}$.

Proof. Evaluate $h_f h_g^\sharp$ at $i/2$ and $-i/2$. A linear relation between the two moment functionals would make $c_- e^{-x/2} + c_+ e^{x/2}$ vanish identically. Its value and derivative at zero force $c_- = c_+ = 0$. The map therefore has two-dimensional range.

The matrix on that range is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Its eigenvalues are 1, -1. Evenness gives $\ell_- = \ell_+$, which proves the last assertion. □

A right inverse $S : \mathbb{C}^2 \to \mathscr{A}_c$ gives the continuous projection $1 - S(\ell_-, \ell_+)$ onto the polar kernel. A single vector with norm square $|\ell_+(f)|^2$ does not realize the full polar form.

The prime term has a different finite-support realization. For $R > 0$, let $j_R : L^2([-R, R]) \to L^2(\mathbb{R})$ extend by zero. Write $T_a v(x) = v(x - a)$, and define

$$B_R = \sum_{2 \le n \le e^{2R}} \frac{\Lambda(n)}{\sqrt{n}} j_R^* (T_{\log n} + T_{-\log n}) j_R.$$

Proposition 23.6.2 *(The finite-prime operator).* The operator B_R is bounded and self-adjoint, with $\|B_R\| \le 2 \sum_{n \le e^{2R}} \Lambda(n) n^{-1/2}$. For smooth f, g supported in $(-R, R)$,

$$W_{\mathrm{fin}}(f * g^*) = \langle B_R f, g \rangle = \langle |B_R|^{1/2} f, \operatorname{sgn}(B_R) |B_R|^{1/2} g \rangle.$$

These pairings agree under enlargement of the support interval.

Proof. Translations are unitary and $T_a^* = T_{-a}$, proving the operator claims and the norm bound. A change of variables gives

$$(f * g^*)(a) = \int f(x) \overline{g(x - a)}\, dx = \langle T_{-a} j_R f, j_R g \rangle.$$

This vanishes for $|a| \ge 2R$. Summing at both signs of each prime-power logarithm proves the first identity and compatibility. The spectral theorem for B_R, with $\operatorname{sgn}(0) = 0$, proves the second identity. □

Example 23.6.3 *(Two translated bumps).* Put $L = \log 2$. Choose a nonzero real smooth b supported in $(-\varepsilon, \varepsilon)$, where $2\varepsilon < \min\{\log 2, \log(3/2)\}$. On the span of b, $T_L b$, the finite-prime form has matrix

$$\frac{\log 2}{\sqrt{2}} \|b\|_2^2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Indeed, $b * b^*$ is supported in $[-2\varepsilon, 2\varepsilon]$ and has value $\|b\|_2^2$ at zero. Only the two shifts $\pm \log 2$ meet the off-diagonal correlations.

Hence

$$W_{\mathrm{fin}}\big((b \pm T_L b) * (b \pm T_L b)^*\big) = \pm \sqrt{2} \log 2\, \|b\|_2^2.$$

This excludes a positive Hilbert realization of the finite-prime form on all compact tests. In the full form that term is subtracted. Its negative value is therefore not a negative full Weil square.

Prime powers remain essential even in such small examples. Two sufficiently narrow bumps separated by $\log 4$ have an off-diagonal contribution $\Lambda(4)/\sqrt{4} = (\log 2)/2$. Retaining only primes would erase that contribution.

23.7 The gamma kernel and its auxiliary form

The signed gamma density k has both signs. Its absolute value cannot be absorbed into an unspecified positive measure without changing the arithmetic functional. To see its shape explicitly, set $c_n = n + 1/4$. The digamma series gives

$$a(t) = a(0) + \sum_{n\ge 0} \frac{t^2/4}{c_n(c_n^2 + t^2/4)}, \qquad a(0) = -\gamma_E - \frac{\pi}{2} - 3\log 2 - \log \pi. \tag{23.11}$$

Here and below $a = -k$, as in (23.6). See DLMF 5.7.6 and 5.4.19.

PROPOSITION 23.7.1 *(The gamma signs).* The function a is even and strictly increasing on $(0, \infty)$. It has its minimum at zero and satisfies

$$a(t) = \log \frac{|t|}{2\pi} + O(t^{-2}) \qquad (|t| \to \infty).$$

Thus $k(0) > 0$, and $k(t) \to -\infty$. The function $|\Gamma(1/4 + it/2)|^2$ is not a constant multiple of $k(t)$.

Proof. Subtract the digamma series at zero. This gives (23.11), whose summands are nonnegative and strictly increasing in $|t|$. Their derivatives converge uniformly on compact intervals:

$$a'(t) = \sum_{n\ge 0} \frac{c_n t/2}{(c_n^2 + t^2/4)^2} > 0 \qquad (t > 0).$$

The value at zero follows from the reflection and duplication identities for ψ_Γ. Stirling's expansion $\psi_\Gamma(z) = \log z - 1/(2z) + O(|z|^{-2})$, on the vertical line $\operatorname{Re} z = 1/4$, gives the asymptotic. Finally, the gamma absolute square is strictly positive everywhere, whereas k changes sign. □

The phase and the absolute square also have different derivatives:

$$\frac{d}{dt} \arg \Gamma(\tfrac{1}{4} + \tfrac{it}{2}) = \frac{1}{2} \operatorname{Re} \psi_\Gamma(\tfrac{1}{4} + \tfrac{it}{2}), \qquad \frac{d}{dt} \log |\Gamma(\tfrac{1}{4} + \tfrac{it}{2})|^2 = -\operatorname{Im} \psi_\Gamma(\tfrac{1}{4} + \tfrac{it}{2}).$$

These follow by taking the imaginary and real parts of $(i/2)\psi_\Gamma$. A local continuous phase is sufficient. For $\Lambda_\infty(s) = \pi^{-s/2}\Gamma(s/2)$, the arithmetic density is

$$k(t) = -2 \operatorname{Re} \frac{\Lambda_\infty'}{\Lambda_\infty}(\tfrac{1}{2} + it).$$

This specifies the operation and normalization that a local archimedean comparison must preserve.

PROPOSITION 23.7.2 *(The kernel away from the origin).* On compact tests supported away from zero, the distribution $P - W_\infty$ is integration against

$$K(x) = 2\cosh(x/2) - \frac{e^{-|x|/2}}{1 - e^{-2|x|}}, \qquad x \ne 0. \tag{23.12}$$

In particular, if $0 < b < c < \log 2$ and $f \ne 0$ is even, nonnegative and supported in $(-c, -b) \cup (b, c)$, then

$$W_{\mathrm{fin}}(f) = 0, \qquad W_\infty(f) = 2\int_b^c \frac{e^{-x/2}}{1 - e^{-2x}} f(x)\, dx > 0.$$

Proof. The polar functional is integration against $2\cosh(x/2)$. Use the integral identity

$$\psi_\Gamma(z) + \gamma_E = \int_0^\infty \frac{e^{-v} - e^{-zv}}{1 - e^{-v}}\, dv, \qquad \operatorname{Re} z > 0,$$

as in DLMF 5.9.16. For a test vanishing near zero, constant Fourier terms contribute zero. Fourier inversion of the cosine term, followed by $v = 2|x|$, gives the second term of (23.12) with its negative sign. Truncating the v-integral first justifies the calculation.

The support condition removes its singular endpoint, and exponential decay controls infinity. The final assertion follows because no prime-power logarithm lies in the stated support. □

A second gamma density

Define

$$b(t) = \operatorname{Re}\psi_\Gamma\left(\tfrac{1}{4} + \tfrac{it}{2}\right) + \log\pi = 2\log\pi - k(t),$$

and the auxiliary form

$$G_+(f,g) = \frac{1}{2\pi}\int_{\mathbb{R}} h_f(t)\overline{h_g(t)}b(t)\, dt.$$

It is continuous on Schwartz space. Plancherel gives

$$W_\infty(f * g^*) = 2\log\pi\,\langle f, g\rangle_{L^2} - G_+(f,g).$$

The function b is strictly increasing in $|t|$, with

$$b(0) = -\gamma_E - \frac{\pi}{2} - 3\log 2 + \log\pi < 0, \qquad b(t) = \log(|t|\pi/2) + O(t^{-2}).$$

It therefore has a unique positive zero, denoted t_b.

The interval $0.84 < t_b < 0.86$ follows from a positive-tail estimate. Put

$$S_N(t) = b(0) + \sum_{n=0}^{N} \frac{t^2/4}{c_n(c_n^2 + t^2/4)}.$$

Then

$$0 \le b(t) - S_N(t) \le \frac{t^2}{8(N+1/4)^2}.$$

At $N = 20$, the rational bounds

$$0.577 < \gamma_E < 0.578, \qquad 3.1415 < \pi < 3.1416,$$
$$0.6931 < \log 2 < 0.6932, \qquad 1.1447 < \log\pi < 1.1448$$

give

$$-0.024 < S_{20}(0.84) < -0.020, \qquad 0.017 < S_{20}(0.86) < 0.020.$$

Both tails are less than 0.00023. The signs and monotonicity prove the interval.

For a compact test, define the two nonnegative energies

$$A_b(f) = \int_{|t|<t_b} |h_f(t)|^2(-b(t))\, dt, \qquad B_b(f) = \int_{|t|>t_b} |h_f(t)|^2 b(t)\, dt.$$

Their difference gives $G_+(f,f) = (B_b(f) - A_b(f))/(2\pi)$. The sufficient high-frequency condition is

$$2t_b|b(0)| \sup_{|t|\le t_b} |h_f(t)|^2 \le B_b(f). \tag{23.13}$$

It bounds $A_b(f)$ by $B_b(f)$, and hence implies $G_+(f,f) \ge 0$.

PROPOSITION 23.7.3 *(Compact high-frequency tests).* The class defined by (23.13) contains nonzero real even compact tests. It is a proper subset of $\mathscr{A}_c$.

Proof. Choose a real even compact smooth χ with $\int \chi \neq 0$. For $M > 0$, let $f_M(x) = \chi(x)\cos(Mx)$. Then

$$h_{f_M}(t) = \tfrac{1}{2}\big(h_\chi(t+M) + h_\chi(t-M)\big).$$

Schwartz decay makes the left side of (23.13) tend to zero faster than every inverse power of M.

Near $t = M$, the translate $h_\chi(t-M)$ has modulus bounded below on a fixed interval, while $h_\chi(t+M) \to 0$ uniformly there. Since $b(t) \sim \log M$ on that interval, $B_b(f_M) \geq c \log M$ for large M. This proves the first assertion.

For the second, set $v_L(x) = L^{-1/2}\chi(x/L)$. A change of variables gives

$$G_+(v_L, v_L) = \frac{1}{2\pi}\int_{\mathbb{R}} |h_\chi(t)|^2 b(t/L)\,dt \longrightarrow b(0)\|\chi\|_2^2 < 0.$$

Logarithmic growth and Schwartz decay justify dominated convergence. Thus v_L fails (23.13) for all sufficiently large L. □

The relation to full Weil positivity retains the arithmetic terms. Put $R(f) = P(f * f^*) - W_{\mathrm{fin}}(f * f^*)$. Then

$$q_W(f,f) = R(f) - 2\log\pi\,\|f\|_2^2 + \frac{B_b(f) - A_b(f)}{2\pi}.$$

Thus $q_W(f,f) \geq 0$ is equivalent to

$$\frac{A_b(f)}{2\pi} + 2\log\pi\,\|f\|_2^2 \leq \frac{B_b(f)}{2\pi} + R(f).$$

Auxiliary positivity controls only the difference of the two energies.

A nonnegative majorant $v \geq -b$ on $[-t_b, t_b]$ gives the useful bound $A_b(f) \leq \int_{-t_b}^{t_b} |h_f|^2 v$. For support in $[-R, R]$, Cauchy–Schwarz gives

$$\sup_t |h_f(t)|^2 \leq 2R\|f\|_2^2, \qquad A_b(f) \leq 4t_b R|b(0)|\|f\|_2^2.$$

These are sufficient estimates for the corresponding terms. They place no bound on $R(f)$.

23.8 GAUSSIAN NORMALIZATION AND CROSS TERMS

All derivatives of a Gaussian have rapid exponential decay. Theorem 23.5.2 therefore applies to the entire Gaussian family and to its convolution products. For $\lambda > 0$, set

$$g_\lambda(x) = e^{-\lambda x^2/2}.$$

Completing the square gives

$$h_{g_\lambda}(t) = \sqrt{\frac{2\pi}{\lambda}}\,e^{-t^2/(2\lambda)}, \qquad (g_\lambda * g_\mu)(x) = \sqrt{\frac{2\pi}{\lambda+\mu}}\,e^{-\lambda\mu x^2/[2(\lambda+\mu)]}.$$

These identities include all normalization factors.

Define the normalized signed gamma integral

$$H(\lambda) = \frac{1}{\sqrt{\pi\lambda}} \int_{\mathbb{R}} e^{-t^2/\lambda} k(t)\, dt.$$

If $\nu = 2\lambda\mu/(\lambda+\mu)$, then

$$W_\infty(g_\lambda * g_\mu) = \sqrt{\frac{2\pi}{\lambda+\mu}}\, H(\nu). \tag{23.14}$$

This follows by inserting the Fourier product in the integral with coefficient $1/(2\pi)$.

Theorem 23.8.1 *(A convergent Gaussian series).* Let $\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-u^2}\, du$. For every $\lambda > 0$,

$$H(\lambda) = k(0) - \sum_{n\ge 0} \left(\frac{1}{c_n} - \frac{2\sqrt{\pi}}{\sqrt{\lambda}} e^{4c_n^2/\lambda} \operatorname{erfc}(2c_n/\sqrt{\lambda}) \right). \tag{23.15}$$

Each summand is nonnegative and at most $\lambda/(8c_n^3)$. The series converges uniformly on compact λ-intervals.

Proof. The identity (23.11) expresses $k(0) - k(t)$ as a sum of nonnegative functions. Tonelli permits integration of this sum against the Gaussian. For $d > 0$,

$$\int_{\mathbb{R}} \frac{e^{-t^2/\lambda}}{t^2+d^2}\, dt = \sqrt{\pi} \int_0^\infty \frac{e^{-d^2 u}}{\sqrt{u+1/\lambda}}\, du = \frac{\pi}{d} e^{d^2/\lambda} \operatorname{erfc}(d/\sqrt{\lambda}).$$

The first equality uses $(t^2+d^2)^{-1} = \int_0^\infty e^{-u(t^2+d^2)}\, du$. The substitution $v = d\sqrt{u+1/\lambda}$ gives the second. Set $d = 2c_n$ to obtain (23.15).

Replacing $c_n^2 + t^2/4$ by c_n^2 bounds its summand by

$$\frac{1}{\sqrt{\pi\lambda}} \int e^{-t^2/\lambda} \frac{t^2}{4c_n^3}\, dt = \frac{\lambda}{8c_n^3}.$$

This proves uniform convergence. □

Proposition 23.8.2 *(The signed Gaussian threshold).* The function H is continuous and strictly decreasing on $(0,\infty)$. Its limits at zero and infinity are $k(0) > 0$ and $-\infty$, respectively. It has a unique zero $\lambda_c > 0$. Consequently,

$$W_\infty(g_\lambda * g_\lambda) \ge 0 \quad \Longleftrightarrow \quad \lambda \le \lambda_c.$$

For $a_j \ge 0$ and $0 < \lambda_j \le \lambda_c$,

$$W_\infty\left(\left(\sum_j a_j g_{\lambda_j} \right) * \left(\sum_j a_j g_{\lambda_j} \right)^* \right) \ge 0.$$

Proof. After $t = \sqrt{\lambda}\, u$, $H(\lambda)$ is the average of $k(\sqrt{\lambda}\, u)$ against $\pi^{-1/2} e^{-u^2}\, du$. Strict radial decrease of k gives strict decrease of H. Logarithmic growth supplies an integrable majorant on compact parameter intervals and near zero. Monotone convergence applied to $k(0) - k(\sqrt{\lambda}\, u)$ gives divergence to infinity as $\lambda \to \infty$. This proves the threshold assertion.

If both $\lambda, \mu \le \lambda_c$, their harmonic mean $2\lambda\mu/(\lambda+\mu)$ is at most λ_c. Every cross pairing in (23.14) is therefore nonnegative. Expand the finite sum with coefficients $a_i a_j \ge 0$.

□

The polar and prime values on a Gaussian square are

$$P(g_\lambda * g_\lambda) = \frac{4\pi}{\lambda} e^{1/(4\lambda)},$$
$$W_{\mathrm{fin}}(g_\lambda * g_\lambda) = 2\sqrt{\frac{\pi}{\lambda}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-\lambda(\log n)^2/4}.$$

The sum converges for every positive λ. The full form subtracts both the prime and signed gamma terms from the polar term. Positivity of the gamma cone alone gives no sign for that complete expression.

The heat normalization

For $t > 0$, let

$$p_t(x) = \frac{1}{\sqrt{4\pi t}} e^{-x^2/(4t)}, \qquad h_{p_t}(z) = e^{-tz^2}.$$

The explicit formula becomes

$$\begin{aligned}\sum_\rho e^{-tz_\rho^2} = 2e^{t/4} - \frac{\log \pi}{\sqrt{4\pi t}} + \frac{1}{2\pi}\int_{\mathbb{R}} e^{-tu^2} \operatorname{Re} \psi_{\Gamma}\left(\tfrac{1}{4} + \tfrac{iu}{2}\right) du \\ - \frac{1}{\sqrt{\pi t}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-(\log n)^2/(4t)}.\end{aligned} \tag{23.16}$$

Each expression is ordinarily convergent.

On a compact interval of positive t, every parameter derivative has an integrable or summable majorant. For the prime sum, a Gaussian in $\log n$ dominates every inverse power of n. For the zero sum, $|e^{-tz_\rho^2}| \leq e^{t/4} e^{-t(\operatorname{Im}\rho)^2}$. The zero count proves its summability.

The difference between the two sides of (23.16) is zero by the explicit formula. Its vanishing is distinct from nonnegativity of q_W. The convolution law $p_s * p_t = p_{s+t}$ evaluates cross terms, but does not determine their sign.

Changing the Fourier scale

For the convention $\widehat{f}_{2\pi}(u) = \int f(x) e^{-2\pi i u x}\, dx$, define

$$(\mathcal{R}f)(x) = \frac{1}{2\pi} f\left(-\frac{x}{2\pi}\right).$$

A change of variables gives

$$h_{\mathcal{R}f} = \widehat{f}_{2\pi}, \qquad \mathcal{R}(f * g) = \mathcal{R}f * \mathcal{R}g, \qquad \mathcal{R}(f^*) = (\mathcal{R}f)^*.$$

If f has integrable derivatives weighted by $e^{b_0|x|}$ with $b_0 > \pi$, then $\mathcal{R}f \in \mathscr{A}_{b_0/(2\pi)}$. Thus the complete arithmetic functional in this convention is

$$W_{2\pi}(f) = W(\mathcal{R}f).$$

Its prime term is

$$-\frac{1}{2\pi} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} \left(f\left(\frac{\log n}{2\pi}\right) + f\left(-\frac{\log n}{2\pi}\right)\right),$$

and its gamma integral is $(2\pi)^{-1}\int \widehat{f_{2\pi}}(u)a(u)\,du$. Both the sampling points and the measure factor change.

For $f_\ell(x)=e^{-\pi x^2/\ell}$, one has $\widehat{f}_{\ell,2\pi}(u)=\sqrt{\ell}e^{-\pi\ell u^2}$. Multiplying (23.16) at $t=\pi\ell$ by $\sqrt{\ell}$ gives

$$\sqrt{\ell}\sum_\rho e^{-\pi\ell z_\rho^2}=2\sqrt{\ell}e^{\pi\ell/4}-\frac{\log\pi}{2\pi}+\frac{\sqrt{\ell}}{2\pi}\int e^{-\pi\ell u^2}\operatorname{Re}\psi_\Gamma\!\left(\tfrac14+\tfrac{iu}{2}\right)du$$
$$-\frac{1}{\pi}\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}e^{-(\log n)^2/(4\pi\ell)}.$$

This also fixes the constants in the two-scale Gaussian identity $f_\ell * f_m=\sqrt{\ell m/(\ell+m)}\,f_{\ell+m}$.

The auxiliary Gaussian cone

The auxiliary density b has its own Gaussian average. Put

$$I(L)=\int_{\mathbb{R}} e^{-2\pi u^2/L}b(u)\,du,\qquad m_L=\sqrt{L/2}.$$

Then

$$\frac{I(L)}{m_L}=2\log\pi-H\!\left(\frac{L}{2\pi}\right).\tag{23.17}$$

The normalized integral is strictly increasing and has a unique zero $L_b>0$. Its equation is $H(L_b/(2\pi))=2\log\pi$. This is a different level from $H(\lambda_c)=0$.

The individual integrals

$$J_n(L)=\int e^{-2\pi u^2/L}\frac{c_n}{c_n^2+u^2/4}\,du=2\pi e^{8\pi c_n^2/L}\operatorname{erfc}\!\left(2c_n\sqrt{2\pi/L}\right)$$

give the absolutely convergent paired series

$$I(L)=m_L(-\gamma_E+\log\pi)+\sum_{n\ge0}\left(\frac{m_L}{n+1}-J_n(L)\right).$$

Subtracting the value of the digamma series at zero improves its tail:

$$\frac{I(L)}{m_L}=b(0)+\sum_{n\ge0}\left(\frac{1}{c_n}-\frac{J_n(L)}{m_L}\right),\qquad 0\le\frac{1}{c_n}-\frac{J_n(L)}{m_L}\le\frac{L}{16\pi c_n^3}.$$

The tail starting at $n=N\ge1$ is at most $L/[32\pi(N-3/4)^2]$. These formulas follow from Theorem 23.8.1 with $\lambda=L/(2\pi)$. The difference $1/(n+1)-1/c_n$ is absolutely summable, which justifies regrouping.

The same real Gaussian integral also gives, for $\alpha,d>0$,

$$\int_{\mathbb{R}}\frac{e^{-\alpha u^2}}{d-iu}\,du=\pi e^{\alpha d^2}\operatorname{erfc}(d\sqrt{\alpha}).$$

Indeed, $(d-iu)^{-1}=(d+iu)/(d^2+u^2)$, and the imaginary part integrates to zero.

The small- and large-parameter limits are

$$\frac{I(L)}{m_L}=b(0)+\frac{L}{8\pi}b''(0)+O(L^2)\quad(L\downarrow0),\qquad \frac{I(L)}{m_L}=\frac12\log L+O(1)\quad(L\to\infty).$$

For the first, scale $u = \sqrt{L/(2\pi)}v$ and use the second moment $1/2$ of $\pi^{-1/2}e^{-v^2}\,dv$. Taylor expansion near zero and the Gaussian tail justify the remainder. For the second, use $b(u) = \log|u| + O(1)$ outside a bounded interval. After scaling, $\log|v|$ is Gaussian-integrable. The shrinking central interval contributes $O(L^{-1/2}\log L)$ after subtraction of $\frac{1}{2}\log L$.

To state the auxiliary cone in the 2π convention, define

$$G_{2\pi}(f,g) = \frac{1}{2\pi}\int \widehat{f}_{2\pi}(u)\overline{\widehat{g}_{2\pi}(u)}b(u)\,du, \qquad F_L(x) = e^{-\pi Lx^2}.$$

For $L, M > 0$, direct multiplication of their Fourier transforms gives

$$G_{2\pi}(F_L, F_M) = \frac{I(2LM/(L+M))}{2\pi\sqrt{LM}}. \tag{23.18}$$

PROPOSITION 23.8.3 *(The exact auxiliary cone).* The diagonal $G_{2\pi}(F_L, F_L)$ is nonnegative exactly for $L \geq L_b$, and vanishes exactly at $L = L_b$. For a finite sum $f = \sum_j a_j F_{L_j}$ with $a_j \geq 0$ and $L_j \geq L_b$, one has $G_{2\pi}(f,f) \geq 0$. A nonzero such sum has zero value precisely when every active parameter equals L_b. Nonnegativity also holds on the Schwartz closure of this cone.

Proof. The diagonal formula is $I(L)/(2\pi L)$. Its sign follows from (23.17). For $L, M \geq L_b$, their harmonic mean is at least L_b, so every cross term in (23.18) is nonnegative. Expand the finite sum.

An active parameter strictly above L_b has a strictly positive diagonal term. If all active parameters equal L_b, every term vanishes. Schwartz continuity proves the closure assertion. □

For example, take $L_b/2 < L < L_b$ and $M > L_bL/(2L - L_b)$. Then $M > L_b$ and $2LM/(L + M) > L_b$, but $I(L) < 0$. The value of $G_{2\pi}(F_L + \varepsilon F_M, F_L + \varepsilon F_M)$ remains negative for sufficiently small $\varepsilon > 0$. This lies outside the stated cone.

Neither Gaussian cone is a complex vector space. Its nonzero elements are positive on all of $\mathbb{R}$, and hence are not compactly supported. The span of centered Gaussians lies in the closed even Schwartz subspace, so it is not dense in the full Schwartz space. These facts do not prevent their use in $\mathscr{A}_\infty$. They specify which density and positivity claims are available.

For clarity, the two Fourier normalizations obey

$$\frac{1}{2\pi}\int \widehat{f}_{2\pi}(u)\overline{\widehat{g}_{2\pi}(u)}k(u)\,du = \frac{\log\pi}{\pi}\langle f, g\rangle_{L^2} - G_{2\pi}(f,g),$$
$$W_\infty(f * g^*) = \int \widehat{f}_{2\pi}(u)\overline{\widehat{g}_{2\pi}(u)}k(2\pi u)\,du.$$

The second identity follows from $h_f(2\pi u) = \widehat{f}_{2\pi}(-u)$. A change of convention rescales the kernel argument as well as the measure.

23.9 THE COMPLETE POSITIVITY CRITERION

The signs of the individual arithmetic terms do not decide the sign of their sum. The spectral formula does decide it once every zero lies on the critical line. The converse requires a test detecting a nonreal ordinate while controlling every other zero.

Let Σ be the set of distinct ordinates $z_\rho = -i(\rho - 1/2)$, and let m_z be their multiplicities. The functional equation and conjugation make Σ invariant under $z \mapsto -z$ and $z \mapsto \bar{z}$, preserving multiplicity. It lies in $|\operatorname{Im} z| \le 1/2$ and is locally finite. Equation (23.9) becomes

$$q_W(f,g) = \sum_{z\in\Sigma} m_z h_f(z)\overline{h_g(\bar{z})}. \tag{23.19}$$

This absolutely convergent formula holds on every exponential test step.

An individual nonreal pair has an indefinite two-dimensional form. Polynomial interpolation and a Gaussian control the other coordinates.

Theorem 23.9.1 *(A negative test from an off-line zero).* Suppose $z_0 = \gamma_0 + i\eta_0 \in \Sigma$, with $\eta_0 \neq 0$. Then a polynomial times a modulated Gaussian belongs to $\mathscr{A}_\infty$ and has negative Weil diagonal. There is also a compact smooth test with negative Weil diagonal.

Proof. Choose $R > 1$, and let Z_R consist of the distinct points of Σ with $|\operatorname{Re} z - \gamma_0| \le R$. It is finite and conjugation invariant. The polynomial

$$P_R(z) = \prod_{w\in Z_R\setminus\{z_0\}} \frac{z-w}{z_0-w} - \prod_{w\in Z_R\setminus\{\bar{z}_0\}} \frac{z-w}{\bar{z}_0-w}$$

takes the values $1, -1$ at $z_0, \bar{z}_0$, respectively, and vanishes on the other points of Z_R. For $\tau \ge 1$, put

$$H_\tau(z) = P_R(z)e^{-\tau((z-\gamma_0)^2+\eta_0^2)},$$
$$f_\tau(x) = e^{-\tau\eta_0^2} P_R(i\partial_x)\frac{e^{-x^2/(4\tau)-i\gamma_0 x}}{\sqrt{4\pi\tau}}.$$

Fourier differentiation gives $h_{f_\tau} = H_\tau$.

Every derivative of f_τ has Gaussian decay. Its two selected transform values remain $1, -1$, so their contribution to the diagonal is $-2m_{z_0}$.

For $z = \gamma + i\eta \notin Z_R$, define

$$d(z) = (\gamma-\gamma_0)^2 - \eta^2 + \eta_0^2, \qquad c = R^2 - \tfrac{1}{4} + \eta_0^2 > 0.$$

The strip bound gives $d(z) \ge c$. The absolute value of the remaining contribution is at most

$$e^{-2(\tau-1)c} \sum_{z\notin Z_R} m_z |P_R(z)P_R(\bar{z})| e^{-2d(z)}. \tag{23.20}$$

The sum is finite.

Indeed, the polynomial has polynomial growth on the fixed strip, the zero count is $O(T\log(2+T))$, and the Gaussian controls the resulting sum over unit intervals. Choose τ large enough that (23.20) is less than m_{z_0}. Then $q_W(f_\tau, f_\tau) < -m_{z_0}$.

Fix this test and multiply it by compact smooth cutoffs tending to one. The products converge in each $\mathscr{A}_{a_0}$. The continuity in Theorem 23.5.2 preserves strict negativity for all sufficiently large cutoffs. □

Theorem 23.9.2 *(Weil positivity on its full domain).* The following assertions are equivalent:

(i) Every zero of ξ has real part $1/2$.

(ii) $q_W(f,f) \ge 0$ for every $f \in \mathscr{A}_c$.

(iii) $q_W(f,f) \geq 0$ for every $f \in \mathscr{A}_{\exp}$.

(iv) For some fixed $a_0 > 1/2$, a subset dense in $\mathscr{A}_{a_0}$ has nonnegative Weil diagonal.

The second assertion is the complete arithmetic inequality

$$W_\infty(f * f^*) \leq P(f * f^*) - W_{\mathrm{fin}}(f * f^*) \qquad (f \in \mathscr{A}_c).$$

Proof. If all ordinates are real, (23.19) is a sum of absolute squares on every exponential step. This implies the third assertion and hence the second and fourth. Continuity extends the fourth assertion to its whole step, which contains all compact smooth tests. Theorem 23.9.1 shows that the second assertion excludes every nonreal ordinate. Finally, the arithmetic definition of W gives the displayed inequality. □

PROPOSITION 23.9.3 *(Hilbert realization of a positive restriction).* Let $D \subset \mathscr{A}_{\exp}$ be a complex vector subspace on which q_W is positive semidefinite. Its nullspace $N_D = \{f \in D : q_W(f,f) = 0\}$ is a linear subspace. The completion of D/N_D is a Hilbert space H_D, and the canonical linear map $I_D : D \to H_D$ satisfies

$$\langle I_D f, I_D g \rangle = q_W(f,g).$$

Conversely, any such map to a positive Hilbert space implies positivity on D. In particular, existence of this realization on the full compact test algebra is equivalent to the Riemann hypothesis.

Proof. Positivity of $q_W(f + tg, f + tg)$ for complex t gives Cauchy–Schwarz. A null vector therefore pairs to zero with every vector. This proves that the quotient form is well-defined and positive definite. Its metric completion has the asserted property. The converse follows by taking the diagonal, and the last assertion uses Theorem 23.9.2. □

This quotient construction starts from the Weil form. It supplies no independent arithmetic reason for its positivity. An arithmetic Hilbert space, constructed by a different method, must still be connected to it by the displayed pairing identity on the stated domain.

The compact domain also distinguishes this criterion from rational tests used in Li's criterion. For an integer $n \geq 1$,

$$1 - (1 - 1/s)^n = \frac{n}{s} + O(s^{-2}) \qquad (|s| \to \infty).$$

After $s = 1/2 + iz$, this is neither entire nor rapidly decreasing on horizontal strips. It is not h_f for a compact smooth test. A formula involving these rational tests needs its own limiting prescription. It is not obtained by substituting them into the compact explicit formula without a convergence argument.

23.10 MOMENT EQUATIONS AND THE TOPOLOGY OF APPROXIMATION

A proposed arithmetic comparison may first be defined on tests satisfying moment equations. Their closed kernel can often be approximated exactly, but this does not make it dense in the full domain. We give the projection and determine its closure in two different topologies.

For a compact test put

$$M_f(s) = \int_{\mathbb{R}} f(x) e^{sx}\, dx = h_f(-is).$$

On the real even compact tests, define

$$\ell(f) = \left(\int f(x)\, dx,\ \int f(x) \cosh x\, dx,\ \int f(x) x \sinh x\, dx \right), \qquad K = \ker \ell.$$

Thus K imposes $M_f(0) = M_f(1) = M_f'(1) = 0$. Evenness already gives $M_f'(0) = 0$. These are the double-zero equations at 0, 1 for an even transform.

All three functionals are continuous on $\mathscr{A}_\infty$. They are also continuous on each $\mathscr{A}_{a_0}$ with $a_0 > 1$.

Proposition 23.10.1 *(A continuous moment projection).* There are real even compact tests η_0, η_1, η_2 with $\ell_i(\eta_j) = \delta_{ij}$. Hence

$$\Pi f = f - \sum_{j=0}^{2} \ell_j(f)\eta_j$$

is a continuous projection onto the moment kernel. That kernel is closed of codimension three, both on compact tests and on each of the exponential spaces just specified. For any $A > 0$, the η_j may all be chosen with support in $[-A, A]$.

Proof. Write $w(x) = (1, \cosh x, x \sinh x)$. The matrix with columns $w(0), w(1), w(2)$ has determinant

$$2\sinh(1)(\cosh(1) - 1)^2 > 0.$$

Even mollified point masses at 0, ± 1, ± 2, with total mass one at each pair, give a moment matrix converging to this matrix. For small mollifier support, its inverse supplies the η_j. Continuity of the three moments then proves the projection and codimension assertions.

To prescribe the support, choose a smooth even $\chi \geq 0$, supported in $[-A, A]$ and positive on $(-A, A)$. The Gram matrix

$$G_{ij} = \int \chi(x) w_i(x) w_j(x)\, dx$$

is positive definite. A nonzero linear combination of the three analytic functions cannot vanish on an interval. Analytic continuation and their distinct constant, exponential and linear-exponential growth show their independence. Set $\eta_j = \chi \sum_i (G^{-1})_{ij} w_i$. These tests have the required moments and support. □

If $f_j \to f \in K$ in the compact-test topology, then $\Pi f_j \to f$. In an exponential space, even compact cutoffs followed by Π prove that compact members of the kernel are dense in that kernel. A nonnegative bump has nonzero mean, so the kernel is not dense in the ambient compact-test space. A moment correction converges to zero only when the limiting function already has the required moments.

Proposition 23.10.2 *(The finite moment quotient).* On the complex even compact convolution algebra the map

$$f \longmapsto \big(M_f(0), M_f(1) + M_f'(1)\varepsilon\big), \qquad \varepsilon^2 = 0,$$

is a continuous surjective algebra homomorphism onto $\mathbb{C} \times \mathbb{C}[\varepsilon]/(\varepsilon^2)$. Its kernel is the complexification of K, and its involution is coefficient conjugation with $\varepsilon^* = \varepsilon$. A complex-linear functional on this quotient is positive on squares exactly when it has the form

$$\lambda(a, b + c\varepsilon) = \alpha a + \beta b, \qquad \alpha, \beta \geq 0.$$

Proof. Fubini gives $M_{f*g} = M_f M_g$. Evaluation at zero and the product rule at one give the algebra law. The moment right inverse proves surjectivity. Evenness makes the involution conjugation at these real evaluation points.

Write $\lambda(a, b + c\varepsilon) = \alpha a + \beta b + \gamma c$. Positivity on the unital quotient makes λ Hermitian, by expanding squares of $1 + tx$ and $1 + itx$ for real t. Thus its coefficients are real. Its value at the square of $(a, b + c\varepsilon)$ is

$$\alpha|a|^2 + \beta|b|^2 + 2\gamma \operatorname{Re}(b\bar{c}).$$

Taking $b = 1$ and varying c forces $\gamma = 0$. The two remaining coefficients must be nonnegative, which also suffices. □

This quotient describes the imposed moments. It does not assert that W factors through the quotient. The polar evaluations occur at $\pm 1/2$, not at $0, 1$. For a nonzero nonnegative even compact test v, the function

$$f = D^2(D^2 - 1)^2 v, \qquad D = d/dx,$$

lies in K, whereas

$$M_f(1/2) = \frac{9}{64} M_v(1/2) > 0, \qquad P(f * f^*) = 2M_f(1/2)^2 > 0.$$

Integration by parts gives $M_f(s) = s^2(s^2 - 1)^2 M_v(s)$, proving every assertion.

THEOREM 23.10.3 *(The weaker Schwartz closure).* The closure of K in the real even Schwartz space is

$$\left\{ f \in \mathcal{S}(\mathbb{R}) : f \text{ is real and even, } \int f = 0 \right\}.$$

In the relative Schwartz topology on compact tests, intersect this set with $\mathscr{A}_c$. The two exponential moment equations need not survive a limit in this weaker topology.

Proof. The mean is Schwartz-continuous, so it vanishes on every such limit. Choose a nonnegative even compact test β with $\int \beta = 1$, and put

$$A = \int \beta(x) \cosh x \, dx > 0, \qquad B = \int \beta(x) x \sinh x \, dx.$$

For $R > 0$, set

$$\beta_R(x) = \tfrac{1}{2}[\beta(x - R) + \beta(x + R)], \qquad v_R = \beta_R - \beta.$$

Translation gives

$$\ell(v_R) = \big(0, A(\cosh R - 1), B(\cosh R - 1) + AR \sinh R\big).$$

The determinant of the last two coordinates of v_R, v_{2R} is

$$2A^2 R \sinh R (\cosh R - 1)^2 > 0.$$

Thus a fixed pair of these moments can be prescribed by $c_R v_R + d_R v_{2R}$, where $c_R = O(e^{-R})$ and $d_R = O(e^{-2R})$. Every Schwartz seminorm of v_R grows at most polynomially in R. The correction therefore tends to zero in the Schwartz topology.

For a compact real even mean-zero f, prescribe the negatives of its last two moments. Adding the correction gives elements of K converging to f. General real even mean-zero Schwartz functions are approximated by even compact cutoffs with their means subtracted times β. For the nth approximation, choose R so large that the correcting term has its first n Schwartz seminorms below $1/n$. This diagonal choice proves the asserted closure. □

The complete Weil form is continuous in the compact and exponential topologies used above. The critical prime measure is not tempered by Theorem 23.2.2. Consequently, the weaker Schwartz closure does not itself pass the complete Weil inequality to its limits.

Proposition 23.10.4 *(The missing cross inequalities).* Let a complex vector space decompose as $D = K_0 \oplus E$, and let q be a Hermitian form, linear in its first argument. It is positive semidefinite on D exactly when

$$q(u,u) \ge 0, \qquad q(e,e) \ge 0, \qquad |q(u,e)|^2 \le q(u,u)q(e,e) \quad (u \in K_0,\ e \in E).$$

Proof. Necessity is Cauchy–Schwarz. Conversely, the three conditions give

$$q(u+e,u+e) \ge \bigl(\sqrt{q(u,u)} - \sqrt{q(e,e)}\bigr)^2 \ge 0.$$

Every vector has this form. □

The complex moment kernel has a three-dimensional complement among even tests. Its positivity leaves both that complement and all cross terms to determine. Odd tests remain an additional part of the full domain. The exact polar kernel has a two-dimensional complement on the full space. Finite codimension alone supplies none of these inequalities.

Similarly, if maps $I_\alpha : D_\alpha \to H_\alpha$ to positive Hilbert spaces satisfy $q_W(f,g) = \langle I_\alpha f, I_\alpha g\rangle$ on their respective complex domains, their union has nonnegative diagonal. Its closure in a fixed exponential step has the same property. Their linear span requires the cross terms between different domains. The Hermitian matrix $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ is positive on each coordinate axis and negative on $(1,-1)$. Thus separate positive diagonal identities do not suffice for the span.

23.11 The maximal positive convolution ideal

Convolution stability gives a stronger classification than positivity on a subspace. A positive ideal must vanish at every nonreal ordinate. To prove this assertion, we first construct tests with exactly prescribed values on a finite part of the zero set.

Lemma 23.11.1 *(A paired product and its Fourier inverse).* The product

$$\mathcal{P}(z) = \prod_{\substack{\omega \in \Sigma \\ \operatorname{Re}\omega > 0}} (1 - z^2/\omega^2)^{m_\omega}$$

converges locally uniformly to an entire function with precisely the zeros Σ and their multiplicities. It is real on the real axis and satisfies

$$|\mathcal{P}(z)| \le \exp\bigl(C(1+|z|)\log(2+|z|)\bigr).$$

For a finite $E \subset \Sigma$, division by $\prod_{\omega\in E}(z-\omega)^{m_\omega}$ gives an entire function R_E, nonzero at each point of E, with the same type of bound. For $t > 0$ and any polynomial p, $e^{-tz^2} R_E(z)\,p(z)$ is the Fourier transform of a member of $\mathscr{A}_\infty$.

Proof. There are no ordinates on the imaginary axis. Pairing ω and $-\omega$ therefore accounts for all zeros. Inverse-square summability proves normal convergence of the quadratic factors. Conjugation permutes these factors and proves reality on the real axis.

Let $n(u)$ count the factors by modulus with multiplicity, and choose $u_0 > 0$ below the modulus of every zero. For $R \geq 2$,

$$\log \max_{|z| \leq R} |\mathcal{P}(z)| \leq \int_{u_0}^{\infty} \log(1 + R^2/u^2)\, dn(u)$$
$$= 2R^2 \int_{u_0}^{\infty} \frac{n(u)}{u(u^2 + R^2)}\, du.$$

Split at R and use $n(u) = O(u \log(2 + u))$. Both integrals are $O(R \log R)$.

Removing the full order of each selected zero leaves a nonzero value there. Outside a fixed disk the dividing polynomial has a polynomial lower bound. Inside that disk the entire quotient is bounded.

Put $H(z) = e^{-tz^2} R_E(z)\, p(z)$. On each horizontal strip, the negative quadratic in $\operatorname{Re} z$ dominates the other factors. Cauchy's estimates on a slightly larger strip give rapid decay of all derivatives. Define

$$f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} H(r) e^{-irx}\, dr.$$

For $x > 0$ shift to $\operatorname{Im} z = -b$, and for $x < 0$ to $\operatorname{Im} z = b$. The vertical integrals tend to zero.

The same shift applied to $z^j H(z)$ gives $|f^{(j)}(x)| \leq C_{b,j} e^{-b|x|}$. Since b is arbitrary, $f \in \mathscr{A}_\infty$. Fourier inversion and holomorphic continuation give $h_f = H$. □

THEOREM 23.11.2 *(Exact finite-value isolation).* For a finite $E \subset \Sigma$ and prescribed complex numbers v_ω, there is $f \in \mathscr{A}_\infty$ such that $h_f(\omega) = v_\omega$ on E and $h_f = 0$ on $\Sigma \setminus E$.

Proof. For any $t > 0$, take the inverse Fourier transform of

$$e^{-tz^2} R_E(z) \sum_{\omega \in E} \frac{v_\omega e^{t\omega^2}}{R_E(\omega)} \prod_{\substack{\eta \in E \\ \eta \neq \omega}} \frac{z - \eta}{\omega - \eta}.$$

All denominators are nonzero. The polynomial interpolates the required values after division by the nonvanishing prefactor, and R_E retains every other zero. Lemma 23.11.1 supplies the required inverse transform. □

Full multiplicities are removed in R_E, although the prescription specifies only values. Removing one linear factor at a multiple zero would leave the prefactor zero there. Exact isolation here uses rapid exponential decay. Compact cutoffs preserve a strict inequality, but need not preserve these exact spectral values.

Write $\Sigma_0 = \Sigma \cap \mathbb{R}$ and $\Sigma_{\mathrm{nr}} = \Sigma \setminus \mathbb{R}$. A positive convolution $*$-ideal means a complex algebraic ideal $J \subset \mathscr{A}_{\exp}$, stable under $*$, with $q_W(f, f) \geq 0$ for every $f \in J$. Define

$$\mathcal{I}_{\mathrm{nr}} = \{f \in \mathscr{A}_{\exp} : h_f(z) = 0 \text{ for every } z \in \Sigma_{\mathrm{nr}}\}.$$

THEOREM 23.11.3 *(The largest positive ideal).* The ideal $\mathcal{I}_{\mathrm{nr}}$ is the largest positive convolution $*$-ideal of $\mathscr{A}_{\exp}$. It is closed in the final test topology and in every exponential step. On this ideal,

$$W(f) = \sum_{\gamma \in \Sigma_0} m_\gamma h_f(\gamma), \qquad q_W(f, g) = \sum_{\gamma \in \Sigma_0} m_\gamma h_f(\gamma) \overline{h_g(\gamma)}. \tag{23.21}$$

If a $*$-ideal contains f with $h_f(z_0) \neq 0$ at a nonreal ordinate, it contains r with $q_W(r, r) = -2m_{z_0}$.

Proof. Each evaluation at a point of Σ is continuous on every step, hence on the final topology. Fourier multiplication and conjugation show that the intersection of the indicated kernels is a closed $*$-ideal. The explicit formula gives (23.21) and its positivity.

For maximality put $u = h_f(z_0) \neq 0$. Exact isolation supplies $e_{z_0}, e_{\bar{z}_0}$ whose transforms on Σ are the respective coordinate vectors. In the given ideal form

$$r = u^{-1} f * e_{z_0} - \bar{u}^{-1} f^* * e_{\bar{z}_0}.$$

Its only nonzero transform values are 1 at z_0 and -1 at $\bar{z}_0$, since $h_{f^*}(\bar{z}_0) = \bar{u}$. Their equal multiplicities give the asserted negative value. Every positive ideal must therefore lie in $\mathcal{I}_{\mathrm{nr}}$. □

This theorem classifies positive ideals. A positive subspace need not contain the two convolution products used in its proof, so it need not lie in $\mathcal{I}_{\mathrm{nr}}$. The theorem neither asserts nor excludes the existence of a nonreal zero.

The regular completions lose the distinction between this ideal and the whole test algebra. To see this, form the off-real subproduct

$$B_{\mathrm{nr}}(z) = \prod_{\substack{\omega \in \Sigma \\ \operatorname{Re}\omega > 0,\ \operatorname{Im}\omega > 0}} \left[(1 - z^2/\omega^2)(1 - z^2/\bar{\omega}^2) \right]^{m_\omega}.$$

The empty product equals one. The preceding product estimate applies.

Its zeros are precisely the nonreal ordinates with their multiplicities, and for real r,

$$B_{\mathrm{nr}}(r) = \prod_{\substack{\operatorname{Re}\omega > 0 \\ \operatorname{Im}\omega > 0}} |1 - r^2/\omega^2|^{2m_\omega} > 0.$$

For fixed $t > 0$, let $b_t \in \mathscr{A}_\infty$ have transform $\beta_t(z) = e^{-tz^2} B_{\mathrm{nr}}(z)$, and put $\mathcal{D}_t = b_t * \mathscr{A}_{\exp}$. This is a principal ideal in an algebra without a unit.

Theorem 23.11.4 *(Density in the regular completions).* The ideal $\mathcal{D}_t$ is a positive convolution $*$-ideal contained in $\mathcal{I}_{\mathrm{nr}}$. Its Fourier image is dense in $\mathcal{S}(\mathbb{R})$. Consequently,

$$\overline{\mathcal{D}_t}^{L^1} = L^1(\mathbb{R}), \qquad \overline{\mathcal{D}_t}^{L^2} = L^2(\mathbb{R}), \qquad \overline{h(\mathcal{D}_t)}^{\|\cdot\|_\infty} = C_0(\mathbb{R}).$$

The same three closures hold for $\mathcal{I}_{\mathrm{nr}}$. If a nonreal ordinate exists, $\mathcal{I}_{\mathrm{nr}}$ is nevertheless proper and closed in the test topology.

Proof. The function β_t is real on the real axis, so $b_t^* = b_t$. Commutativity makes $\mathcal{D}_t$ a $*$-ideal. Its Fourier factor vanishes on every nonreal ordinate, proving inclusion and positivity.

Take $H \in C_c^\infty(\mathbb{R})$ in the Fourier variable, and put $u = H/\beta_t$. Strict positivity of β_t on the real axis makes u smooth and compactly supported. Define

$$\varphi_\epsilon(r) = (4\pi\epsilon)^{-1/2} e^{-r^2/(4\epsilon)}, \qquad u_\epsilon = u * \varphi_\epsilon, \qquad H_\epsilon = \beta_t u_\epsilon.$$

Convolution here is in the Fourier variable. Its inverse transform is $\check{u}_\epsilon(x) = e^{-\epsilon x^2} \check{u}(x)$, which belongs to $\mathscr{A}_\infty$. Thus $H_\epsilon \in h(\mathcal{D}_t)$.

Gaussian convolution converges to the identity in the Schwartz topology. This follows from translation continuity near zero and the polynomially weighted Gaussian tail outside a fixed neighborhood.

Multiplication by the fixed Schwartz function β_t is continuous, so $H_\epsilon \to H$ in that topology. Smooth cutoffs make compact smooth functions dense in Schwartz space. Fourier inversion and its continuous dense inclusions into L^1, L^2 and C_0 give the three closures.

If z_0 is nonreal, the continuous evaluation $h_f(z_0)$ vanishes on $\mathcal{I}_{\mathrm{nr}}$ but not on a Gaussian. This proves properness in the test topology. □

Neither ideal has an internal unit. Such a unit would act identically on a uniformly dense Fourier subalgebra of $C_0(\mathbb{R})$. Its transform would then equal one on all of $\mathbb{R}$. Nor is test-topology density of $\mathcal{D}_t$ in $\mathcal{I}_{\mathrm{nr}}$ asserted. The generator vanishes to full analytic multiplicity, while the latter ideal requires only value vanishing.

These ideals depend on the zero set. Their unconditional construction does not supply an arithmetic positive domain independently of that set. More decisively, density in any one exponential test step would imply full positivity by Theorem 23.9.2. The regular density theorem therefore proves a different assertion.

23.12 The independent arithmetic comparison

The analytic target is now fixed: the convolution algebra, its topology, its involution, and the complete signed functional. An arithmetic source may approach this target through a trace, a cohomological pairing, an automorphic norm or a boundary construction. Each approach must specify its map on the same tests.

Question 23.12.1 *(A full arithmetic realization).* Construct an arithmetic complex vector space V, a Hermitian pairing B, and a complex-linear map $I : \mathscr{A}_c \to V$, such that

$$B(If, Ig) = P(f * g^*) - W_\infty(f * g^*) - W_{\mathrm{fin}}(f * g^*) \qquad (f, g \in \mathscr{A}_c). \tag{23.22}$$

Determine whether B is positive semidefinite on $I(\mathscr{A}_c)$ by the arithmetic construction. For an exponential extension, construct the continuous map and prove the same identity on a specified step.

An operator or weight realization needs an actual convolution representation on a common domain that preserves the involution. Its trace or weight must be finite on the tested products. A comparison with a bilinear categorical pairing also needs the conjugate coefficient factor that turns it into a Hermitian form. Local determinants, a scalar functional equation and matching characters do not specify these maps.

A geometric realization asks for a continuous $*$-algebra map into a proper positive trace ideal constructed from independent arithmetic operations. The comparison must preserve the involution, normal derivatives, collision maps and both vacuum data of its source. Arithmetic-weight and arity-radius continuity are additional requirements. The resulting counting weight or self-adjoint shift must be identified with a native arithmetic operation. Positivity of the scalar Weil form alone does not construct this stronger realization.

Partial domains retain their exact positive content. A norm identity on a moment kernel proves positivity on that kernel. A Gaussian cone gives its tested inequalities. A positive ideal has the classification above. To reach the full criterion, one must also prove the missing cross inequalities and density in a topology controlling every term of (23.22). Both polar evaluations, all prime powers and the signed gamma integral remain part of that equality.

Part II

The Arithmetic Realization Branch

Chapter 24

The Arithmetic Realization Branch

24.1 Identity

The arithmetic branch of Volume IV carries the extension of the chiral-bar-cobar programme from smooth complex curves $C/\mathbb{C}$ to the Arakelov-compactified arithmetic curve $\overline{C} = \overline{\mathrm{Spec}\,\mathbb{Z}}$. Its object of study is the Hochschild trace class of the identity on the global arithmetic Iwahori Hecke category,

$$\mathcal{V}^{\mathrm{prim}} \;:=\; \mathrm{HH}_\bullet\big(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\,\mathrm{glob}}\big),$$

studied through five conditional technical windows: (i) a classical Kim arithmetic Chern–Simons torsor/line (arXiv:1510.05818) whose possible boundary factorization shadow is a comparison problem for $\mathcal{V}^{\mathrm{prim}}$; (ii) an automorphic/spectral window controlled by a Selmer-derived global deformation stack with fixed residual representation, determinant, and local conditions; (iii) a Bethe-algebra heuristic in which primes are lattice sites and Hecke operators are transfer matrices, with the RTT/Yangian Hopf structure absent at the arithmetic level; (iv) a Heisenberg enhancement of the Bost–Connes quantum statistical system; and (v) a factorization algebra on the archimedean place $\mathbb{C}_\infty$ valued in $\widehat{\mathbb{Z}}$-modules with commuting Adams operations ψ_p. Each window is a comparison problem for $\mathcal{V}^{\mathrm{prim}}$; their mutual equivalence is the conditional content of the Hochschild-trace layer.

This branch states the five theorems A–D+H in their arithmetic avatar and isolates three precise open conjectures whose resolution closes the cross-Langlands loop. The arithmetic avatars A^{arith}–D^{arith}, H^{arith}, and $F1$–$F3$ form the realization diagram over $\overline{\mathrm{Spec}\,\mathbb{Z}}$. Agreement over both $C/\mathbb{C}$ and $\overline{\mathrm{Spec}\,\mathbb{Z}}$ is a structural assertion, not a metaphor.

24.2 Form Discipline

Every claim in the arithmetic branch carries one of five tags.

Theorem. A claim stated with a proof body. In the arithmetic branch this is reserved for statements whose proof transports from the complex case with no new input beyond replacement of $C/\mathbb{C}$ by the archimedean fibre of $\overline{C}$.

Conditional. A claim with a complete strategy of proof and all ingredients in the primary literature, whose full write-up requires a substantial treatment.

Conjecture. A claim plausible on the strength of multiple cross-consistency checks but without a complete proof strategy.

Heuristic. A pattern match between classical and arithmetic structures with no derivation. Heuristic claims are essential only as guidance, never as premises.

Obstruction. A failed candidate whose failure is a theorem: the obstruction is a named computation, missing structure, or non-vanishing class.

Only theorem, conditional theorem, conjecture, heuristic, and obstruction enter the stable language of the branch. A failed candidate remains in the text only after its failure has been made mathematical.

24.3 Local-to-Global Form

The branch has three layers: local obstruction theory, the Hochschild-trace construction of $\mathcal{V}^{\mathrm{prim}}$, and the three conjectural comparison maps completing the cross-Langlands loop.

Local obstruction theory

Chapter 25 isolates the finite Heisenberg cover $\mathcal{H}^{\mathrm{cov}}_{\mathbb{P}} = \otimes^{!}_{p}\, \mathcal{H}^{(p)}$ from the Hochschild trace class $\mathcal{V}^{\mathrm{prim}}$. The next five chapters compare this cover with independent source families (Drinfeld+Kapranov, Kazhdan+Beilinson, Gelfand+Borger, Connes+Soibelman, Costello+Gwilliam). Each comparison gives one obstruction and one surviving structural core:

(i) The arithmetic curve is Arakelov-compactified $\overline{\mathrm{Spec}\,\mathbb{Z}}$, not $\mathrm{Spec}\,\mathbb{Z}$.

(ii) Formal Heisenberg fields exist on an auxiliary line over $\mathbb{Q}_p$ (Chapter 26). The arithmetic finite-place coefficients require a separate comparison with those field operations.

(iii) The Steinberg relation $\{p, 1-p\} = 0$ holds in $K_2^M(\mathbb{Q})$. Comparing its residue complex with chiral collision operations requires maps preserving those operations (Chapter 33).

(iv) The Hecke-Adams lift is GL_1 Adams, not GL_2 Hecke; the GL_2 structure lives in the rank-2 Heisenberg via the Satake eigenvalue pair (α_p, β_p) with $\alpha_p\beta_p = p$.

(v) The correct local model is the Iwahori-equivariant affine Hecke category, not the spherical Mirković–Vilonen category; spherical collapses the OPE.

Hochschild trace synthesis

The central theorem of the branch places $\mathcal{V}^{\mathrm{prim}}$ as the Hochschild trace class of the identity endofunctor on the global affine Iwahori Hecke category (Chapter 38). Under this placement, ten classical positions receive arithmetic analogues: Segal sewing is occupied by Euler-product factorization (Chapter 32); the Verlinde scalar becomes the class-number residue; the mapping-class position is occupied by Galois labels and Hecke symmetries; the modular S-matrix position is occupied by the Γ-factor plus functional equation; the conformal-block position is occupied by Tate integral lines.

Chapter 43 isolates the essential mathematical content of the branch: the arithmetic G-row complementarity

$$\kappa^{\mathrm{Ar}}_{\mathsf{G}} + \left(\kappa^{!}\right)^{\mathrm{Ar}}_{\mathsf{G}} \;=\; 0 \quad\Longleftrightarrow\quad \xi(s) \;=\; \xi(1-s),$$

under the arithmetic realization hypotheses of Theorem 43.5.1. These include an actual central-defect identity and a Mellin evaluation detecting that defect. Poisson summation proves the scalar functional equation. Constructing those maps from arithmetic duality remains a separate problem.

Comparison conjectures

Chapter 42 isolates the three conjectures whose proof closes the cross-Langlands loop:

(i) Global arithmetic Arinkin–Gaitsgory equivalence, linking $\mathrm{D}([GL_2])^{\text{sph-fin}}$ to $\mathrm{IndCoh}_{\mathrm{Nilp}}$ on the Selmer-derived global spectral stack attached to a fixed $(\bar{\rho}, \det, \mathcal{L})$.

(ii) E_2-enhancement of the global Iwahori Hecke category via Scholze diamonds, supplying Dunn additivity for the trace.

(iii) Arithmetic Ben-Zvi–Nadler identification of $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\,\mathrm{glob}})$ with $\mathcal{V}^{\mathrm{prim}}$ as an E_1-algebra over the full $\mathrm{Weil}/(\varphi, \Gamma)$ target, with $B\widehat{\mathbb{Z}}$ retained only as an inertia shadow.

These three conjectures link Kim's classical 3D-CS input (arXiv:1510.05818), Bezrukavnikov–Bernstein's Hochschild presentation, and Beilinson's bar–cobar presentation. $\mathcal{V}^{\mathrm{prim}}$ is a Hochschild trace class. Realizing it by fields requires coefficient and operation comparisons. The complex Heisenberg model at infinity and formal finite-place models provide candidates (Chapter 26).

24.4 Realization over $\overline{\mathrm{Spec}\,\mathbb{Z}}$

The arithmetic branch puts the same bar–cobar spine, Koszul complementarity, and obstruction tower on the Arakelov curve $\overline{\mathrm{Spec}\,\mathbb{Z}}$. If the complex and arithmetic bases carry the same identities after source-disjoint comparison, the invariant belongs to one-dimensional factorization itself, not to the analytic model in which it was first seen.

The Beilinson principle is the governing rule: a statement enters the realized corpus only after an independent comparison has separated its proof source from its verification source.

24.5 Source Decomposition

Fifteen primary-source families meet the arithmetic trace construction at distinct structural joints: Drinfeld+Kapranov, Kazhdan+Beilinson, Gelfand+Borger, Connes+Soibelman, Costello+Gwilliam, Etingof, Nekrasov, Segal, Beilinson–solo, Witten, Polyakov, Gaiotto, Bezrukavnikov+Bernstein, Kontsevich+Soibelman, and Drinfeld–solo. Their common remnant is a single mathematical object viewed through five windows, together with the three comparison conjectures above.

24.6 Conventions

Base. $\overline{C} = \overline{\mathrm{Spec}\,\mathbb{Z}}$ denotes the Arakelov compactification: closed points are rational primes $p \in \mathbb{P}$ plus the archimedean place ∞. Local rings: $\mathbb{Z}_p$ at p (uniformizer p, residue $\mathbb{F}_p$, Frobenius $x \mapsto x^p$); $\mathbb{R}$ at ∞ (with complex conjugation on $\mathbb{C}$).

Iwasawa. $\Lambda = \mathbb{Z}_p[[\Gamma]]$, where $\Gamma \cong \mathbb{Z}_p$ is the Galois group of a chosen cyclotomic $\mathbb{Z}_p$-extension. The full cyclotomic Galois group may also have a finite factor.

Adeles. $\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p' \mathbb{Q}_p$ (restricted product); $\mathbb{A}_{\mathbb{Q}}^{\times}$ the idele group; $\mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}$ the idele class group.

Hecke. $\mathcal{H}_p^{\mathrm{sph}} = C_c(GL_n(\mathbb{Z}_p)\backslash GL_n(\mathbb{Q}_p)/GL_n(\mathbb{Z}_p))$; Satake isomorphism $\mathrm{Sat}_p : \mathcal{H}_p^{\mathrm{sph}} \xrightarrow{\sim} \mathbb{Z}[\alpha_1^{\pm}, \dots, \alpha_n^{\pm}]^{S_n}$.

Satake pair ($\mathbf{GL_2}$). $\alpha_p, \beta_p \in \mathbb{C}^{\times}$ with $\alpha_p\beta_p = p$; trace $a_p = \alpha_p + \beta_p = 2p^{1/2}\cos\theta_p$ on unitary sector.

Kesten–McKay. $\nu_p(\theta) = (2/\pi)\cdot p(1+p)\sin^2\theta/(1+p^2-2p\cos 2\theta)$, $\theta \in [0,\pi]$; this is the Plancherel density of unramified $GL_2(\mathbb{Q}_p)$.

Arakelov degree. $\widehat{\deg}\, O(s\cdot[\infty]) = s \in \widehat{\mathrm{CH}}^1(\overline{C}) \otimes \mathbb{R} \cong \mathbb{R}$.

Completed zeta. $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ with functional equation $\xi(s) = \xi(1-s)$.

All references to Vols I–III use the native cross-volume citation. The branch has no internal dependency cycle; chapters are readable in order.

24.7 Imports and Assembly

This branch imports local theorems from the arithmetic Langlands programme: local geometric Satake (Fargues–Scholze, arXiv:2102.13459), the Emerton–Gee stack (arXiv:2012.12403), and Caraiani–Scholze vanishing. It assembles them into the E_1-chiral-algebra presentation natural to the bar–cobar programme;

the novel content is the assembly. Caraiani–Scholze, Fargues–Scholze, and Emerton–Gee develop the arithmetic Langlands programme independently; this branch is a consequent consumer.

An arithmetic character interpretation of the Hochschild trace class $\mathcal{V}^{\mathrm{prim}}$ requires its actual comparison with the specified L-function. A vertex or factorization realization must also supply its fields and operation maps. The local models of Chapters 26 and 33 provide inputs to these comparisons.

The Deninger $\mathbb{F}_1$-geometry supplies motivation and is cited as a consequent reference. At the overlaps (the adelic norm form $\prod_v |x|_v = 1$, the Connes–Consani arithmetic site as a categorical base, tropical semirings for the structure sheaf) the arithmetic branch is a formal consequence of Deninger's comparison geometry.

24.8 Arithmetic-Modular Extension: All-Genus Axis

The three-layer arithmetic branch treats the $g = 0$ row: Riemann $\xi(s) = \xi(1-s)$ as the functional-equation witness of Heisenberg G-row complementarity $\kappa^{\mathrm{Ar}} + (\kappa^!)^{\mathrm{Ar}} = 0$. A six-source-family comparison extends the branch to arbitrary $g \geq 0$. Chapter 102 defines the *arithmetic genus-g datum* $\mathcal{D}_g$ admitting six convergent presentations: Arakelov fibration (Faltings, Gillet–Soulé, Bost–Künnemann); Minkowski torus over totally imaginary K of degree $2g$ (Deninger, Connes–Consani, Bost–Connes–Marcolli); ordinary crystalline Emerton–Gee stack $\mathcal{X}_{2g}^{\mathrm{crys,ord}}$ (Fargues–Scholze, Bhatt–Scholze, Caraiani–Scholze); Hodge–Arakelov point of arithmetic Teichmüller $\mathcal{T}_g^{\mathrm{arith}}$ with theta-structure (Mochizuki); chiral-bar-cobar interior $\mathcal{V}^{\mathrm{prim,surf},g} = \mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw,surf}}_{GL_{2g},\mathrm{glob}})$ with E_2-structure via Dunn additivity (Beilinson–Drinfeld, Francis–Gaitsgory, Ben-Zvi–Nadler); Kudla–Rapoport Shimura–Siegel datum with g-fold arithmetic intersection of special cycles (Kudla, Bruinier, Li–Liu).

The essential conditional of the extension (Chapter 102.4.1) states: the Koszul self-duality identity $\kappa^{\mathrm{Ar},g} + (\kappa^!)^{\mathrm{Ar},g} = 0$ is preserved at every $g \geq 0$ and is equivalent to the motivic functional equation of the corresponding row. The genus enters the functional equation through the archimedean Γ-factor $\Gamma_{\mathbb{C}}(s)^g$, the rank $2g$ of the Heisenberg Fock oscillators, and the motivic weight $w_{\mathcal{D}_g} \in \{1, 2, 2g+1\}$ setting the functional-equation center (Dedekind zeta, Hasse–Weil L-function of the Jacobian, Siegel spin L-function of a GSp_{2g} cusp form); the Siegel-spin entry $w = 2g+1$ inherits the spinor L-function normalisation convention for vector-valued GSp_{2g} cusp forms (Arthur 2013) from Chapter 102. The $g = 0$ specialisation recovers the zero-placement axis of Chapter 44. The all-genus F-RH corollary (Chapter 102.5.1) is the common form of the zero-placement clauses for the classical zeta function, Hasse–Weil L-functions of Jacobians, Dedekind zeta functions of totally imaginary fields, and Siegel-spin L-functions for GSp_{2g} automorphic representations; it is not a proof of these hypotheses without the corresponding spectral positivity input.

The almost-complex-structure candidate is obstructed in every source class at a category-theoretic joint: an arithmetic scheme has no real tangent bundle, and Newlander–Nirenberg integrability is a theorem on real two-manifolds. The surviving operator is Frobenius φ_g on primitive cohomology, namely archimedean Frobenius, prismatic φ, or the chiral Koszul operator J^{Koz}; see Chapter 102.1.2. The extension is arithmetic-modular in the sense that the six windows are modular presentations of a single object $\mathcal{D}_g$, each supplying an independent verification path satisfying the independence of sources criterion. The residual conditions of Chapter 102.6 are ten genus-g comparison morphisms; each is a finite theorem condition in the corresponding source family.

24.9 Arithmetic Gravity and the E_1 Ordered Bar

The proposed arithmetic Kac–Moody enhancement requires the current operations and level comparisons of Question 103.3.1. A conformal vector alone does not construct those currents. The scalar identity $\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0$ also does not exclude a critical level.

Such an enhancement would be an input to a conjectural quantum Chern–Simons theory $\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}}$ on $\overline{\mathrm{Spec}\,\mathbb{Z}}$. The arithmetic three-space analogy comes from Mazur's duality picture and Morishita's primes-as-knots comparison. Kim's classical torsor construction motivates the finite state sums and the further profinite TQFT question. The boundary comparison with $\mathcal{V}^{\mathrm{prim}}$, the zeta scalar character, and the Beilinson–Drinfeld/Costello–Gwilliam (BD/CG) chiral-homology comparison are conditional. What is closed in Chapter 103 is the ordered incidence-bar calculation on the auxiliary ordered place set $\mathbb{P} \cup \{\infty\}$ (Chapter 103.5.4).

Over the arithmetic base the 3D topological gauge theory is Chern-Simons rather than holomorphic-twist: $\overline{\mathrm{Spec}\,\mathbb{Z}}$ has étale cohomological dimension 3 without a differential-geometric holomorphic structure. The complex Vol II Part VI 3D HT QFT bulk becomes the 3D arithmetic Chern-Simons bulk; the complex-boundary $\mathcal{V}$ becomes the arithmetic-boundary $\mathcal{V}^{\mathrm{prim}}$. The zero-placement clause of F-RH (Chapter 44.1.1) becomes a spectral statement on the partition-function zero locus of the bulk TQFT (Chapter 103.7.1); the all-genus F-RH corollary (Chapter 102.5.1) lifts to the bulk partition function valued in $\Lambda_{\mathcal{D}_g}(k^{\mathrm{Ar},g})$ for every row of the arithmetic-modular family (Chapter 103.8.1).

The arithmetic-gravity closure adds a non-formal theorem condition of residual condition to the all-genus programme (Chapter 103.9): quantum ACS construction on the étale site; bulk-boundary correspondence; ordered-incidence bar calculation plus conditional BD/CG comparison; and the positive Hilbert–Pólya/VB input. Together with Chapter 102's all-genus residual list, the extension of the arithmetic branch from the $g = 0$ axis to the all-genus arithmetic-modular family plus arithmetic-gravity bulk takes a non-formal theorem condition in four to six mathematical blocks.

Chapter 25

The Arithmetic Curve and the Heisenberg Cover

25.1 The Arithmetic Curve

Write $\overline{C} := \overline{\operatorname{Spec} \mathbb{Z}}$ for the Arakelov compactification: closed points are primes $p \in \mathbb{P}$ plus the archimedean place ∞. At p the completed local ring is $\mathbb{Z}_p$, with uniformizer p, residue field $\mathbb{F}_p$, and Frobenius $\mathrm{Frob}_p : \bar{\mathbb{F}}_p \to \bar{\mathbb{F}}_p$, $x \mapsto x^p$. At ∞ the completion is $\mathbb{R}$ with complex conjugation on $\mathbb{C}$. The generic stalk is $\mathbb{Q}$.

A smooth complex curve has local coordinates on open neighborhoods, with transition maps on their overlaps. It need not have one global coordinate. An adele $(x_\infty; x_2, x_3, x_5, \ldots) \in \mathbb{A}_{\mathbb{Q}}$ is instead a tuple of local elements, integral at almost every finite place. It becomes an insertion coordinate only if a geometric construction supplies that interpretation and its transition maps. Chapter 10 constructs continuous arithmetic coefficient complexes without assuming such an identification.

25.2 Λ-Ring Input

Definition 25.2.1 *(after Grothendieck, Borger).* For a $\mathbb{Z}$-torsion-free commutative ring R, a Λ-*structure* is equivalently given by ring endomorphisms $\psi_n : R \to R$, $n \in \mathbb{N}^\times$, such that $\psi_1 = \mathrm{id}$, $\psi_m \psi_n = \psi_{mn}$, and $\psi_p(x) \equiv x^p \pmod{p}$ for every prime p.

Theorem 25.2.2 *(Borger, arXiv:0801.1691; arXiv:0906.3146).* Borger interprets the forgetful functor from Λ-spaces to ordinary spaces as base change from $\mathbb{F}_1$ to $\mathbb{Z}$. For flat affine objects the structure is specified by commuting Frobenius lifts. This categorical interpretation does not construct a new tensor product or a descent datum on an unsupplied arithmetic object (Borger0906.3146, §§1.1–1.2).

Remark 25.2.3. The universal Λ-ring on one generator is the ring $\Lambda = \mathbb{Z}[h_1, h_2, \ldots]$ of symmetric functions. On Newton power sums the Adams operation acts by $\psi_n(p_m) = p_{mn}$ (Macdonald, *Symmetric Functions and Hall Polynomials*, Ch. I §8).

Remark 25.2.4. The big Witt vector functor $W : \mathrm{CRing} \to \mathrm{Ring}_\Lambda$ is right adjoint to the forgetful functor. $W(R)$ is the cofree Λ-ring on R. The ghost map $W(R) \to R^{\mathbb{N}^\times}$ is injective when R is $\mathbb{Z}$-torsion-free; injectivity is not asserted for arbitrary torsion rings (Borger0801.1691, §1.14).

On $\operatorname{Spec} \mathbb{Z}$, local translations are replaced by a monoid $\mathbb{N}^\times$ of commuting Adams operations $\{\psi_p\}_{p \in \mathbb{P}}$, one per prime. An action on a proposed arithmetic chiral object is additional data. The polynomial index-shift action and the different divisor substitutions are constructed in Chapter 28; neither alone supplies an action on the arithmetic trace class.

25.3 The Finite Heisenberg Cover

Definition 25.3.1 *(finite Heisenberg cover).* For each prime p, let $\mathcal{H}^{(p)}$ be the rank-2 Heisenberg vertex algebra with generators $a^{(p)}$, $b^{(p)}$ and OPE

$$\begin{aligned} a^{(p)}(z)\,a^{(p)}(w) &\sim \frac{1}{(z-w)^2}, \\ b^{(p)}(z)\,b^{(p)}(w) &\sim \frac{1}{(z-w)^2}, \\ a^{(p)}(z)\,b^{(q)}(w) &\sim 0 \quad \forall p, q. \end{aligned}$$

Each orthogonal current has level one. Use the ordinary rank-two conformal vector, of central charge two,

$$\omega_p = \tfrac{1}{2}\big((a^{(p)}_{-1})^2 + (b^{(p)}_{-1})^2\big)|0\rangle_p.$$

Its integer oscillator weight is distinct from the energy $H_p = (\log p)L_0^{(p)}$. The latter is an operator, not a rescaled conformal vector. The algebraic restricted tensor product $\bigotimes'_p \mathcal{H}^{(p)}$ uses the vacuum at almost every prime. It is a proposed cover; a map from it to the arithmetic trace class is additional data. An infinite sum of its conformal vectors is not a state in this algebraic restricted tensor product.

PROPOSITION 25.3.2 *(full and selected finite-place Euler factors).* On the Hall Hilbert completion at one prime p, let Q_p act on a monomial by the number of a-oscillators minus the number of b-oscillators; write $N_p = Q_p$ when this occupation charge is used below. For real θ_p and $\operatorname{Re} s > 0$,

$$\operatorname{Tr}(e^{-sH_p}e^{i\theta_p Q_p}) = \prod_{k\geq 1} \frac{1}{(1-e^{i\theta_p}p^{-ks})(1-e^{-i\theta_p}p^{-ks})}.$$

Selecting only the two modes $a^{(p)}_{-1}$, $b^{(p)}_{-1}$ gives

$$\frac{1}{1-2\cos\theta_p\, p^{-s}+p^{-2s}}.$$

The product over all primes converges absolutely for $\operatorname{Re} s > 1$. The selected factor agrees with the trigonometric primon Euler factor; it is not the full two-species oscillator trace. The selected space is not preserved by the substitutions of Chapter 28 for $n > 1$.

Proof. If the occupations are r_k, t_k, their energy is $\sum_k k(r_k + t_k)\log p$ and their charge is $\sum_k (r_k - t_k)$. Summing each occupation independently gives the geometric factors. Absolute convergence follows from the same geometric bounds as in Section 28.7; two species only double the bound on the logarithm. Keeping only $k = 1$ gives the second formula. In the vacuum Heisenberg module the zero modes a_0, b_0 vanish; Q_p is the explicitly defined occupation operator, not their difference. The first generator maps to mode n under both substitutions, proving the final assertion. □

25.4 The Divisor-Substitution Obstruction to GL_2-Hecke

PROPOSITION 25.4.1 *(divisor substitutions and Hecke recursion).* On the polynomial Fock carrier over $\mathbb{Z}$, the formula

$$T_n(a^{(p)}_{-k}) = \sum_{d\mid n,\, d\mid k} d\, a^{(p)}_{-nk/d}$$

extends as a unital ring map. It satisfies coprime multiplicativity, but not $T_m T_n = T_{mn}$ for all indices, and not the recursion $T_{p^2} = T_p^2 - p\,\mathrm{id}$.

Proof. The divisor $d = 1$ always contributes. Consequently

$$T_p a_{-1}^{(p)} = a_{-p}^{(p)}, \quad T_p^2 a_{-1}^{(p)} = a_{-p^2}^{(p)} + p a_{-p}^{(p)}, \quad T_{p^2} a_{-1}^{(p)} = a_{-p^2}^{(p)}.$$

The Adams discrepancy is $p a_{-p}^{(p)}$; the displayed Hecke recursion has discrepancy $p(a_{-p}^{(p)} - a_{-1}^{(p)})$. Both are nonzero in this free integral carrier. The coprime statement and the complete composition law are Corollary 28.2.2. The index-shift operations $\psi_n(a_{-k}^{(p)}) = a_{-nk}^{(p)}$ instead form an Adams action. A comparison with an arithmetic Hecke representation must specify its operators and preserve their actual relations. □

25.5 Polynomial Averaging and the Arithmetic Comparison

Definition 25.5.1 *(finite occupation-phase averaging).* Let S be a finite set of primes, and let $P_{S,\mathbb{C}}$ be the polynomial span of the two-species oscillator states at those primes. The occupation charges $N_p = Q_p$ defined above act diagonally on this carrier. Define the complex-linear operator

$$A_S(v) = \int_{[0,\pi]^S} \exp\left(i \sum_{p \in S} \theta_p N_p \right) v \prod_{p \in S} \nu_p(\theta_p)\, d\theta_p, \qquad v \in P_{S,\mathbb{C}},$$

where

$$\nu_p(\theta) = \frac{2}{\pi} \frac{p(1+p)\sin^2\theta}{1 + p^2 - 2p\cos 2\theta}.$$

The integral is defined coefficient by coefficient: each polynomial has finite support in the occupation basis, and the displayed density is bounded on $[0, \pi]$ for every prime p. Thus A_S is an endomorphism of the polynomial vector space. No quotient or arithmetic factorization structure is used in its definition.

Warning 25.5.2 *(arithmetic averaging requires a comparison).* The proposed arithmetic map

$$\mathrm{av}^{\mathrm{arith}} : \mathcal{V}^{E_1,\mathrm{arith}} \longrightarrow \mathcal{V}^{\mathrm{Hecke}}$$

requires the construction of its arithmetic E_1-chiral carrier, its Hecke quotient, and the map between them. These are additional requirements in Chapter 30. The formula for A_S does not identify those carriers or construct that arithmetic map. Theorem 27.6.1 constructs the spherical transform on radial tree functions and preserves its root state. Section 27.8 proves that the one-prime occupation-phase average has a nonreal eigenvalue. Thus it cannot be unitarily intertwined with the self-adjoint tree adjacency operator. The operadic averaging from ordered to symmetric chiral cohomology in Vol I has its own carriers and structure maps; the shared integral notation supplies no comparison between these operations.

25.6 The Depth Analogy

Heuristic. The following four rows name possible arithmetic shadows of the Vol I depth filtration. They do not classify Λ-rings.

- **Depth 2:** trivial ψ_n beyond ψ_1; *archetype G analogue* (primon Heisenberg only);
- **Depth 3:** finitely generated image in $\prod_p \mathbb{Z}$; *archetype L analogue* (arithmetic Kac–Moody / Iwasawa-type Λ-module);
- **Depth 4:** higher-derived Λ-structure engaging Witt filtration; *archetype C analogue* (arithmetic $\beta\gamma$);

• **Depth ∞:** full Connes–Consani arithmetic site; *archetype M analogue* (arithmetic Virasoro).

Remark 25.6.1. The depth numerology $\{2, 3, 4, \infty\}$ on the Vol I side is forced by explicit shadow-tower computation (Vol I, shadow-tower chapter). On the arithmetic side the corresponding derivation via a Λ-ring analogue of Mok's shadow-tower rigidity remains open; the classification above is a heuristic pattern match.

25.7 Boundary Conditions for the Five Arithmetic Analogues

Conjecture 25.7.1 *(A^{arith}).* There is an adjunction $\Omega^{\mathrm{ch,arith}} \dashv B^{\mathrm{ch,arith}}$ between the ∞-category of Λ-ring-enhanced E_1-algebras on $\overline{C}$ and the ∞-category of conilpotent Λ-coalgebras. On $\mathcal{V}^{\mathrm{prim}}$ this adjunction is an equivalence on the subcategory of Λ-ring objects with prime-adic convergence at every finite place.

Question 25.7.2 *(B^{arith}).* Construct the arithmetic adjunction of Conjecture 25.7.1 and determine its domain of coalgebra recovery. For a coaugmented coalgebra C, the required unit has the type

$$C \longrightarrow B^{\mathrm{ch,arith}}\Omega^{\mathrm{ch,arith}}(C).$$

Specify the coefficient topology and weak equivalences under which this map is an equivalence. Conilpotence concerns iterated reduced coproducts. A module-theoretic $\mu = 0$ hypothesis, when defined, is separate, by Proposition 10.10.2.

Conjecture 25.7.3 *(C^{arith}).* Define the arithmetic κ of a Λ-chiral algebra as its Arakelov first Chern class (sum of finite-place p-adic orders plus archimedean logarithmic defect). On the Koszul self-dual arithmetic subcategory, $\kappa + \kappa^! \in \{0, \log 2, \log 3, \ldots\} \cup \{0, 1, 2, \ldots\}$ with the precise ceiling unknown.

Warning 25.7.4 *(Sharpened in Chapter 43).* The correct arithmetic G-row ceiling is 0, and its content is the Riemann functional equation $\xi(s) = \xi(1 - s)$.

Conjecture 25.7.5 *(D^{arith}).* The genus-g obstruction for the arithmetic Maurer–Cartan element $\Theta_A^{\mathrm{arith}}$ equals $\kappa \cdot h_g^{\mathrm{Ar}}$ where h_g^{Ar} is the Arakelov height of the level-g component of $\overline{M}_{g,n}$.

Conjecture 25.7.6 *(H^{arith}).* $\mathrm{HH}^\bullet(\mathcal{V}^{\mathrm{prim}})$ is concentrated in degrees $\{0, 1, 2\}$:

$$\begin{aligned} \mathrm{HH}^0 &= \mathbb{Z} \quad (\Lambda\text{-ring augmentation}), \\ \mathrm{HH}^1 &= \bigoplus_p \log p \cdot \mathbb{Z} \quad (\text{Bost–Connes KMS generators}), \\ \mathrm{HH}^2 &= \text{a } \Lambda\text{-torsor related to Witt cotangent.} \end{aligned}$$

25.8 Boundary Conditions

Four boundary conditions govern the construction above.

(i) **p-adic locality.** The OPE locality axiom $[a(z), b(w)] = 0$ off-diagonal requires an interpretation of the formal variable $z - w$ at finite places of $\overline{C}$: see Chapter 26.

(ii) **Adelic factorization.** Beilinson–Drinfeld factorization on a curve employs $C^n/\mathfrak{S}_n$. On $\overline{C}$ the two candidate analogues are $W_n(\mathbb{Z})/\mathfrak{S}_n$ and $\mathrm{Spec}(\Lambda^{\otimes n})^{\mathfrak{S}_n}$; their comparison is settled in Chapter 30.

(iii) **Real place.** The archimedean place carries an $\mathrm{SL}(2, \mathbb{R})$ conformal quantum mechanics (Hartnoll–Yang); the non-archimedean places carry $\mathrm{SL}(2, \mathbb{Q}_p)$. The adelic $\mathrm{SL}(2)$-automorphic picture gluing these is developed in Chapter 32.

(iv) **Four-archetype derivation.** The shadow-tower depth numerology on the arithmetic side would require a Λ-ring analogue of Mok's shadow-tower rigidity. No such rigidity theorem is used below.

Chapter 26

Finite-place fields and arithmetic comparison

A formal Heisenberg algebra exists over every characteristic-zero field, including $\mathbb{Q}_p$. Its insertion coordinate belongs to an auxiliary curve over that field. A comparison with an arithmetic space must specify the support, coefficient maps and topology that relate them.

There are three separate choices. A logarithmic energy assigns real weights to states. A coefficient construction determines the ring over which local fields are assembled. A geometric realization specifies the diagonal and its normal directions. We examine these choices before stating the arithmetic comparison problem.

26.1 The formal algebra over a finite-place field

A vertex algebra over k has a state-field map

$$Y : V \otimes_k V \longrightarrow V((z)),$$

with a vacuum, translation and the formal field identities. No absolute value occurs in those identities. The construction in Section 6.1 gives two orthogonal level-one currents over $\mathbb{Q}$, and scalar extension gives

$$\mathcal{H}_k = \mathrm{Sym}_k(k[t]a \oplus k[t]b), \qquad [a_m, a_n] = [b_m, b_n] = m\delta_{m+n,0}, \quad [a_m, b_n] = 0.$$

The monomial $t^{m-1}a$ denotes a mode-m creation state.

Proposition 26.1.1. For every characteristic-zero field k, the displayed Fock space is a rank-two formal Heisenberg vertex algebra, with

$$a(z)a(w) \sim (z-w)^{-2}, \qquad b(z)b(w) \sim (z-w)^{-2}, \qquad a(z)b(w) \sim 0.$$

In particular it gives a formal vertex algebra over $\mathbb{Q}_p$.

Proof. Define negative modes by multiplication and positive modes by the corresponding weighted derivatives. Their commutators give the three contractions. Normal ordering and the finite Wick expansion prove locality, translation and the vacuum identities, as in Proposition 6.1.1. All formulas commute with extension from $\mathbb{Q}$ to k. □

The variable z remains formal. A norm on V, a completion of its tensors and a domain for z are additional analytic data. Neither the existence nor failure of one analytic completion changes this algebraic construction.

26.2 CONFORMAL NORMALIZATION AND LOGARITHMIC ENERGY

The standard conformal state is

$$\omega = \tfrac{1}{2}(a_{-1}^2 + b_{-1}^2)\mathbf{1}.$$

Its field T has central charge two and integer oscillator weights. For $T' = \lambda T$, its second pole is $2\lambda^2 T$, whereas the Virasoro normalization requires $2T' = 2\lambda T$.

Observation 26.2.1. On the nontrivial Heisenberg algebra, a scalar multiple $\lambda\omega$ implements the original conformal structure only when $\lambda = 1$. The real weight $\log p$ in $H_p = (\log p)L_0$ is external energy data. It does not define a rescaled conformal state over $\mathbb{Q}_p$.

Indeed the second pole forces $\lambda^2 = \lambda$. The value zero cannot give the required translation operator. This leaves $\lambda = 1$. The fourth pole alone would give $2\lambda^2$, but it cannot determine a conformal structure without the other poles.

The p-adic logarithm provides different data. Its convergent series on principal units extends to $\mathbb{Q}_p^\times$ after one chooses its value on p. The usual normalization sets that value to zero. This convention is not an interpretation of the real number $\log p$, nor is it the only possible choice of a logarithm on the valuation factor.

Convention 26.2.2. Use the standard rank-two conformal state at each finite place. When the coefficient space is complex and prime-labeled, use $(\log p)L_0$ as a separately specified energy operator. A comparison with a p-adic coefficient system must supply its own coefficient maps and normalizations.

26.3 COMMON COEFFICIENTS AND RESTRICTED TENSORS

For finite sets of places, tensor products require a common base ring and actual maps from that base. Since every $\mathbb{Q}_p$ is a $\mathbb{Q}$-algebra, the algebraic tensors

$$R_S = \bigotimes_{p \in S, \mathbb{Q}} \mathbb{Q}_p$$

are defined. Their transition maps insert units. They are injective as maps of rational vector spaces.

Observation 26.3.1. There is an abstract common field extension of all the $\mathbb{Q}_p$. This fact does not provide canonical or continuous coefficient embeddings, an adelic topology, or a restricted tensor of representations.

To construct such a field, take a maximal-ideal quotient of $\varinjlim_S R_S$, as in Proposition 6.1.4. The transition injections make this a nonzero ring. Each local field injects into the quotient because its map is unital. This construction retains its algebraic scope.

Three coefficient constructions illustrate the further choices. First, choose embeddings into a common characteristic-zero field and extend the formal Heisenberg factors there. Even an abstract embedding $\mathbb{Q}_p \to \mathbb{C}$ is discontinuous, since it sends the sequence $p^r \to 0$ to an unbounded real sequence. Extending scalars in this way does not transport the local norm.

Second, work over an adelic coefficient ring with specified modules and lattices. The canonical map is the projection $\mathbb{A}_\mathbb{Q} \to \mathbb{Q}_p$. It does not supply a unital map in the reverse direction for scalar extension. The finite-place inclusion supported only at p sends the unit to a proper idempotent. Its module convention must therefore be stated explicitly.

Third, use the rational model itself. The assignment

$$k \longmapsto \mathcal{H}_\mathbb{Q} \otimes_\mathbb{Q} k$$

is already a functor on characteristic-zero fields. A field embedding acts on coefficients and preserves the field operations. A descent datum for another form, or an arithmetic Galois action with Frobenius compatibility, is additional input.

Warning 26.3.2. None of these coefficient constructions produces a Hecke action or identifies the oscillator algebra with an arithmetic trace class. Those comparisons require independently defined targets and maps preserving their operations.

26.4 Completion in the prime and completion at the diagonal

Put $R = \mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{Z}_p$. The p-adic completion of this ring and the formal completion at its multiplication ideal use different ideals. The first is computed by reduction modulo powers of p.

Proposition 26.4.1. Multiplication induces

$$\varprojlim_r R/p^r R \cong \mathbb{Z}_p.$$

The ordinary multiplication map $R \to \mathbb{Z}_p$ has a nonzero kernel. After inverting p, its diagonal conormal module is $\Omega_{\mathbb{Q}_p/\mathbb{Q}} \neq 0$.

Proof. At every finite precision,

$$R/p^r R \cong (\mathbb{Z}_p/p^r) \otimes_{\mathbb{Z}} (\mathbb{Z}_p/p^r) \cong \mathbb{Z}/p^r.$$

Taking this inverse limit proves the first assertion. For $a \in \mathbb{Z}_p \setminus \mathbb{Q}$, the element $a \otimes 1 - 1 \otimes a$ is nonzero before completion. A rational linear functional taking 1 to zero and a to one detects it after inverting p. Multiplication and every finite reduction kill it.

The conormal module of multiplication on $\mathbb{Q}_p \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is the module of Kähler differentials. A derivation along an element of a transcendence basis is nonzero. Characteristic zero extends that derivation across the remaining algebraic extension. Thus this conormal module is nonzero, as in Proposition 6.1.3. □

Flatness does not make an ordinary tensor commute with this inverse limit. The nonzero kernel just exhibited shows why that interchange would give the wrong ordinary ring.

Corollary 26.4.2. The p-adic completion calculation does not imply that every finite-place field operation is locally constant. On the auxiliary line $\mathbb{A}^1_{\mathbb{Q}_p}$, the rank-two currents have nonzero principal-parts operations at the diagonal.

Proof. That diagonal has parameter $u = z - w$, with formal normal ring $\mathbb{Q}_p[z][[u]]$. The current product has principal part $u^{-2}du$. This is nonzero in its full normal module. Completion of the coefficient ring in p does not remove this independent normal variable. □

26.5 Choosing a local coefficient model

A locally constant coefficient system is a possible model. It must be specified rather than deduced from the preceding completion formula. A commutative coefficient algebra with a Hecke endomorphism is also different from the formal Heisenberg algebra, whose current first product is nonzero.

Definition 26.5.1. A constant local coefficient datum consists of a commutative unital coefficient algebra B_p, a chosen filtration or topology, and a specified compatible endomorphism T_p. Its constant vertex operation is $Y(b,z)c = bc$, with zero translation. This is one possible local datum for a comparison with an independently defined arithmetic object.

An E_0-algebra by itself is pointed data, without a binary multiplication supplied by that notation. The commutative product in the definition is an explicit additional structure. Its endomorphism is not simultaneously a Hecke and Adams operation unless the corresponding relations are proved.

THEOREM 26.5.2 *(Complex auxiliary realization).* The scalar extension $\mathcal{H}_{\mathbb{C}}$ gives a Heisenberg field algebra on the auxiliary complex line, with its nonzero diagonal principal-parts operation.

Proof. The formal field construction and normal residue operation are those of Sections 6.1 and 6.2 over $\mathbb{C}$. Translation gives differential linearity, and the field identities give the residue identities. □

Identifying this auxiliary realization with a specified archimedean arithmetic object requires a comparison map. The coefficient extension alone does not define that object.

26.6 The adelic and automorphic comparison

A restricted tensor representation starts with representations (π_v, V_v) of the local groups over a common field. At almost every finite place it also specifies a vector fixed by a chosen compact open subgroup. The tensor transition maps insert these vectors. A restricted product of group elements then acts because only finitely many factors move them.

This constructs an abstract representation on its stated tensor carrier. Realizing it as automorphic functions requires further global data. In particular, local oscillator fields do not already specify the $GL_2(\mathbb{Q}_v)$-representations or the required global realization.

Question 26.6.1. Choose arithmetic local coefficient objects, local group actions, and unramified reference vectors over a specified common base. Construct their restricted tensor and an intertwiner to an independently defined automorphic recipient. Can this intertwiner be compared with the local field operations, including their collision maps, vacuum data and coefficient topology?

Remark 26.6.2. A compatible adelic system may be a useful geometric recipient for this question. Its structure must determine the relevant local and global maps. Neither the coefficient-field choices nor the completion of $\mathbb{Z}_p \otimes \mathbb{Z}_p$ force that recipient or prove that singularities occur only at the archimedean place.

26.7 Formal loop spaces and the arithmetic problem

Remark 26.7.1. Kapranov and Vasserot construct formal loop spaces for schemes of finite type over a field. Their global versions form factorization monoids. For smooth schemes their geometric construction recovers the chiral de Rham complex and its vertex structure.[1] The field-based construction supplies a comparison model. An arithmetic extension must define its own loop objects, coefficient categories and factorization maps.

Remark 26.7.2. Reciprocity and cohomology determinants suggest further comparison data. A determinant line requires a specified perfect complex, and a reciprocity statement requires its actual local

[1] M. Kapranov and E. Vasserot, *Vertex algebras and the formal loop space*, Introduction, pp. 1–2, arXiv:math/0107143.

symbols and common recipient. An analogy with Langlands correspondence or a field theory does not provide these constructions. The automorphic problem above remains one possible formulation.

26.8 The local and global maps still required

The formal finite-place Heisenberg algebra, its auxiliary diagonal operation, and the p-adically completed coefficient tensor are explicit constructions. The ordinary tensor and diagonal conormal module retain the information that the p-adic completion forgets. Logarithmic energy preserves its separate real normalization.

A geometric arithmetic realization must relate these specified carriers through coefficient and support maps. An automorphic realization must additionally supply its group actions and global intertwiner. The choice of a common field, a locally constant model, or an adelic tensor leaves those maps to be constructed.

Chapter 27

Divisor Substitutions and Satake Parameters

The divisor substitutions on polynomial Fock space, Adams ring maps, and a spherical Hecke representation are different operator data. We specify the Satake normalization, compute the divisor discrepancy, and distinguish a Satake matrix from its scalar symmetric-power traces.

27.1 The Satake Pair and Why It Matters

Theorem 27.1.1 *(normalized spherical Satake convention).* Let $K = GL_2(\mathbb{Z}_p)$, normalize Haar measure by $\mathrm{vol}(K) = 1$, and use complex coefficients. The normalized spherical Satake isomorphism identifies $\mathcal{H}_p = C_c(K\backslash GL_2(\mathbb{Q}_p)/K, \mathbb{C})$ with $\mathbb{C}[X^{\pm 1}, Y^{\pm 1}]^{S_2}$. The characteristic functions of $K\,\mathrm{diag}(p,1)K$ and $K\,\mathrm{diag}(p,p)K$, denoted T_p and D_p, have images $p^{1/2}(X+Y)$ and XY. For a character of this algebra set $\alpha_p = p^{1/2}X$, $\beta_p = p^{1/2}Y$ at that character. Then $\alpha_p\beta_p = pD_p$, where the right side means its scalar value. Trivial central character gives $D_p = 1$. The condition $|X| = |Y| = 1$ is the additional tempered condition, not a consequence of trivial central character alone. These conventions are Gross, *On the Satake isomorphism*, Proposition3.6, equations(3.13)–(3.14) and §8.

Proposition 27.1.2 *(symmetric-power generating function).* Define the elements $H_{p^k} \in \mathcal{H}_p$ by their Satake images $h_k(p^{1/2}X, p^{1/2}Y)$, the complete symmetric polynomials. Then $H_{p^0} = 1$, $H_p = T_p$, and

$$\sum_{k\geq 0} H_{p^k} t^k = (1 - T_p t + pD_p t^2)^{-1}.$$

Proof. In the Satake algebra the generating series is $(1 - p^{1/2}Xt)^{-1}(1 - p^{1/2}Yt)^{-1}$. Multiply the denominators and use injectivity of the Satake map. This fixes the determinant normalization and the choice of the higher Hecke elements rather than identifying them with a single undeclared double-coset convention. □

At a character with $D_p = 1$ this series is $\big((1-\alpha_p t)(1-\beta_p t)\big)^{-1}$. The two parameters may coincide. In the tempered case their unitary versions $p^{-1/2}\alpha_p$, $p^{-1/2}\beta_p$ give the local trigonometric Euler factor; changing between the two normalizations shifts the Dirichlet variable by one half. This complex Satake convention is separate from the integral polynomial carrier used for the divisor substitutions below.

27.2 Adams operations and Hecke representations

Proposition 27.2.1. The Adams operations of a special λ-ring (Definition 13.6.1) satisfy $\psi_m\psi_n = \psi_{mn}$. They give an action of $(\mathbb{N}_{\geq 1}, \times)$, the free commutative monoid on the primes. Conversely, commuting Frobenius lifts determine the special structure under the torsion-free hypothesis of Theorem 13.6.2. Example 13.6.3 shows why that hypothesis cannot be omitted. This action need not extend to the group completion: the polynomial index shifts are not surjective for $n > 1$.

The full local spherical GL_1 Hecke algebra over $\mathbb{C}$ is $\mathbb{C}[\mathbb{Z}] = \mathbb{C}[z, z^{-1}]$. Valuation identifies the double cosets of $\mathbb{Q}_p^\times/\mathbb{Z}_p^\times$ with $\mathbb{Z}$, and, with the unit subgroup of volume one, convolution adds valuations. An Adams action supplies only the multiplicative monoid action; a comparison with a specified positive Hecke subalgebra is separate data.

Observation 27.2.2 *(distinct operator data).* An Adams action is an action of the multiplicative monoid by ring endomorphisms. It does not by itself specify a GL_2 Hecke representation, its Satake parameters, or its determinant operation. A Λ-ring can contain symmetric functions in two parameters; the missing datum is their comparison with the specified arithmetic operators, not an impossibility of encoding an unordered pair.

27.3 Explicit Computation: The Recursion Fails

Let $\mathcal{H}^{(p)}_{\mathrm{rk1}}$ denote the rank-1 Heisenberg algebra at prime p, with mode operators $a_n^{(p)}$ ($n \in \mathbb{Z}$) satisfying $[a_m^{(p)}, a_n^{(p)}] = m\,\delta_{m+n,0}$, acting on the Fock space $\mathcal{F}^{(p)}$ generated from the vacuum $|0\rangle$ (annihilated by $a_n^{(p)}$ for $n \geq 0$). For each $k \geq 1$, the state $a_{-k}^{(p)}|0\rangle$ spans the one-particle subspace of weight k. The full weight-k subspace also contains the multi-particle partition states.

The claimed formula for the action of a putative GL_2 Hecke operator T_n on this Fock space is

$$T_n(a_{-k}^{(p)}|0\rangle) \;=\; \sum_{d|n} d \cdot a_{-nk/d}^{(p)}|0\rangle \cdot \mathbf{1}_{d|k},$$

extended by $T_n(|0\rangle) = |0\rangle$.

Computation 27.3.1. $T_p(a_{-1}^{(p)}|0\rangle)$. Divisors of $n = p$ are $d \in \{1, p\}$. Only $d = 1$ satisfies $d \mid k = 1$. So

$$T_p(a_{-1}^{(p)}|0\rangle) \;=\; a_{-p}^{(p)}|0\rangle.$$

Computation 27.3.2. $T_p^2(a_{-1}^{(p)}|0\rangle)$. Apply T_p to $a_{-p}^{(p)}|0\rangle$: divisors $d \in \{1, p\}$, both satisfy $d \mid p$, so

$$T_p(a_{-p}^{(p)}|0\rangle) \;=\; a_{-p^2}^{(p)}|0\rangle + p \cdot a_{-p}^{(p)}|0\rangle.$$

Therefore

$$T_p^2(a_{-1}^{(p)}|0\rangle) \;=\; a_{-p^2}^{(p)}|0\rangle + p \cdot a_{-p}^{(p)}|0\rangle.$$

Computation 27.3.3. $T_{p^2}(a_{-1}^{(p)}|0\rangle)$. Divisors $d \in \{1, p, p^2\}$, only $d = 1$ satisfies $d \mid 1$. So

$$T_{p^2}(a_{-1}^{(p)}|0\rangle) \;=\; a_{-p^2}^{(p)}|0\rangle.$$

Observation 27.3.4 *(recursion discrepancy).* Testing $T_{p^2} = T_p^2 - p \cdot \mathrm{id}$ on $a_{-1}^{(p)}|0\rangle$:

$$\begin{aligned} T_p^2 - p \cdot \mathrm{id}\big|_{a_{-1}^{(p)}|0\rangle} &= a_{-p^2}^{(p)}|0\rangle + p \cdot a_{-p}^{(p)}|0\rangle - p \cdot a_{-1}^{(p)}|0\rangle, \\ T_{p^2}\big|_{a_{-1}^{(p)}|0\rangle} &= a_{-p^2}^{(p)}|0\rangle. \end{aligned}$$

The discrepancy is $p(a_{-p}^{(p)} - a_{-1}^{(p)})|0\rangle \neq 0$.

Corollary 27.3.5 *(falsification).* The formula $T_n(a_{-k}^{(p)}) = \sum_{d|n} d \cdot a_{-nk/d}^{(p)} \mathbf{1}_{d|k}$ fails both universal Adams multiplicativity and the displayed GL_2 Hecke recursion. It satisfies coprime multiplicativity and the prime-valuation composition law of Corollary 28.2.2. The preceding computations prove the two failures already on the first generator.

27.4 Diagonalisation onto the Satake Pair

A two-dimensional Satake model gives a different construction. Choose nonzero complex numbers α_p, β_p with $\alpha_p\beta_p = p$. This is parameter data, not a consequence of the rank of an oscillator algebra. In the tempered parametrization one can write $\alpha_p = p^{1/2}e^{i\theta_p}$, $\beta_p = p^{1/2}e^{-i\theta_p}$ with real θ_p.

Definition 27.4.1 *(Satake operator).* Let $F_p^1 = \mathbb{C}a_{-1}^{(p)}|0\rangle \oplus \mathbb{C}b_{-1}^{(p)}|0\rangle$. Define $S_p = \mathrm{diag}(\alpha_p, \beta_p)$ in this basis. The equation $\alpha_p\beta_p = p$ is imposed on the parameters. It is not encoded by multiplying a Heisenberg conformal vector by $\log p$; logarithmic energy is the separate convention of Convention 28.7.1.

Theorem 27.4.2 *(trace of the Satake operator).* The operator S_p has determinant p and trace $\alpha_p + \beta_p$, equal to $2p^{1/2}\cos\theta_p$ in the tempered parametrization. The line spanned by $(a_{-1}^{(p)} + b_{-1}^{(p)})|0\rangle$ is invariant precisely when $\alpha_p = \beta_p$.

Proof. The determinant and trace are those of the displayed diagonal matrix. Its value on the sum of the two basis vectors is proportional to that sum exactly when its two diagonal entries agree. □

Corollary 27.4.3 *(symmetric-power trace recursion).* For $k \geq 0$, set $h_k = \mathrm{Tr}(\mathrm{Sym}^k S_p)$. Then $h_0 = 1$, $h_1 = \alpha_p + \beta_p$, and

$$h_k = \sum_{j=0}^{k} \alpha_p^{k-j}\beta_p^j, \qquad h_{k+1} = (\alpha_p + \beta_p)h_k - ph_{k-1} \quad (k \geq 1).$$

Thus the scalar operators $H_{p^k} = h_k\,\mathrm{id}$ on a fixed one-dimensional space satisfy the displayed local Hecke recursion. This is a scalar-parameter representation, distinct from S_p on F_p^1 and from the divisor substitutions T_{p^k}.

Proof. The monomial basis of $\mathrm{Sym}^k F_p^1$ has eigenvalues $\alpha_p^{k-j}\beta_p^j$. Summing gives h_k. Multiplying this sum by $\alpha_p + \beta_p$ and subtracting $\alpha_p\beta_p h_{k-1}$ leaves exactly the sum for h_{k+1}. □

27.5 The Angle θ_p as Unramified Langlands Parameter

The angle θ_p first appeared in Definition 27.4.1 as the argument of the Satake parameters $\alpha_p = p^{1/2}e^{i\theta_p}$, $\beta_p = p^{1/2}e^{-i\theta_p}$. The question is whether θ_p is a free internal parameter of the Heisenberg algebra or is fixed by external (Langlands) data. For a specified unramified automorphic representation it is determined by its Satake pair, when that pair has the tempered normalization above. An arbitrary choice of angles does not construct such a representation.

Proposition 27.5.1 *(tempered determinant normalization).* If $\alpha_p\beta_p = p$ and $|\alpha_p| = |\beta_p| = p^{1/2}$, then the unordered pair is $\{p^{1/2}e^{i\theta_p}, p^{1/2}e^{-i\theta_p}\}$, where θ_p is defined modulo 2π and sign.

Proof. The number $\alpha_p/p^{1/2}$ has modulus one, so it is $e^{i\theta_p}$. The determinant condition forces the second normalized number to be $e^{-i\theta_p}$. Interchanging the two entries changes the sign of θ_p. □

Convention 27.5.2. $\{\theta_p\}_p$ is chosen once and for all as a map $\theta\colon \mathbb{P} \to \mathbb{R}/(2\pi\mathbb{Z},\ \theta \sim -\theta)$ (the angle modulo $\theta \leftrightarrow -\theta$, since $2\cos\theta = 2\cos(-\theta)$). This is the data of an unramified local Langlands parameter at each finite place. The Hartnoll–Yang L-function

$$L(s) \;=\; \prod_p \frac{1}{1 - 2\cos\theta_p\, p^{-s} + p^{-2s}}$$

is a formal Euler product. It is the standard L-function of an automorphic representation only if that representation and the comparison with its normalized local parameters are supplied; arbitrary angles do not establish automorphy or analytic continuation.

27.6 A radial Hecke comparison

The Satake trace recursion also admits a representation on radial functions on a regular tree. Its adjacency operator has the density used in Definition 25.5.1. Matching that density does not identify the operators. The radial tree representation gives an explicit comparison with the spherical Hecke action and its vector state. This is the classical spherical transform, with the distance and symmetric-power normalizations kept separate. The density is the Kesten–McKay measure. McKay, *The expected eigenvalue distribution of a large regular graph*, Linear Algebra Appl. 40 (1981), 203–216, Theorem 1.1, gives its graph-theoretic normalization.

Fix a prime p and a rooted $(p+1)$-regular tree $(\mathcal{T}_p, o)$. Use counting measure on its vertices and complex coefficients. Let S_n be the vertices at distance n from o, and write

$$s_0 = 1, \qquad s_n = (p+1)p^{n-1} \quad (n \geq 1), \qquad e_n = s_n^{-1/2}\mathbf{1}_{S_n}.$$

The radial Hilbert space $\mathcal{R}_p$ consists of the square-summable functions constant on each S_n. The vectors e_n form an orthonormal basis. All these vector spaces have cohomological degree zero. Distance gives a filtration, not a grading preserved by adjacency.

Let T be adjacency, $(Tf)(v) = \sum_{w\sim v} f(w)$. It is a bounded self-adjoint operator on the vertex Hilbert space, with norm at most $p+1$ by Cauchy–Schwarz. Counting edges between consecutive spheres gives

$$\begin{aligned} Te_0 &= \sqrt{p+1}\, e_1, \\ Te_1 &= \sqrt{p+1}\, e_0 + \sqrt{p}\, e_2, \\ Te_n &= \sqrt{p}\,(e_{n-1} + e_{n+1}) \quad (n \geq 2). \end{aligned}$$

In particular, two steps return to the root in $p+1$ ways. Thus the polynomial which produces the distance-two sphere is $T^2 - (p+1)I$. The symmetric-power Hecke polynomial is instead $T^2 - pI$. Their difference is the identity operator.

Define monic integral polynomials by

$$R_0(x) = 1, \quad R_1(x) = x, \quad R_2(x) = x^2 - (p+1), \qquad R_{n+1}(x) = xR_n(x) - pR_{n-1}(x) \quad (n \geq 2).$$

The preceding recurrence proves $R_n(T)e_0 = \mathbf{1}_{S_n}$. These formulas already give an integral $\mathbb{Z}[x]$-module isomorphism

$$\mathbb{Z}[x] \longrightarrow \bigoplus_{n\geq 0} \mathbb{Z}\mathbf{1}_{S_n}, \qquad f \longmapsto f(T)e_0.$$

Here x acts on the right by T. Monicity makes the R_n an integral basis, so this assertion needs no inversion of p. The normalized Hilbert comparison uses real square roots and complex scalars.

THEOREM 27.6.1 *(radial spherical transform).* Put $x(\theta) = 2\sqrt{p}\cos\theta$ for $0 \leq \theta \leq \pi$, and set

$$d\nu_p(\theta) = \frac{2p(p+1)\sin^2\theta}{\pi(p^2+1-2p\cos 2\theta)}\, d\theta, \qquad q_n(\theta) = \frac{R_n(x(\theta))}{\sqrt{s_n}}.$$

The assignment $\mathcal{U}_p e_n = q_n$ extends uniquely to a unitary

$$\mathcal{U}_p : \mathcal{R}_p \longrightarrow L^2([0,\pi],\nu_p).$$

It sends e_0 to 1 and satisfies the commutative square

$$\begin{array}{ccc} \mathcal{R}_p & \xrightarrow{T} & \mathcal{R}_p \\ \mathcal{U}_p \downarrow & & \downarrow \mathcal{U}_p \\ L^2(\nu_p) & \xrightarrow{M_x} & L^2(\nu_p) \end{array}$$

where M_x is multiplication by $x(\theta)$. Consequently $\|T|_{\mathcal{R}_p}\| = 2\sqrt{p}$, and for every bounded Borel function f on $[-2\sqrt{p}, 2\sqrt{p}]$,

$$\langle e_0, f(T)e_0\rangle = \int_0^\pi f(2\sqrt{p}\cos\theta)\, d\nu_p(\theta).$$

Proof. The tridiagonal recurrence makes e_0 cyclic for T on $\mathcal{R}_p$. We compute its spectral measure to determine the Hilbert norm of the polynomial map. Remove the first row and column of that tridiagonal matrix. Its tail has every off-diagonal coefficient equal to $\sqrt{p}$. For $\operatorname{Im} z > 0$, the first diagonal entry $g(z)$ of its resolvent $(z - J)^{-1}$ satisfies the Schur-complement identity

$$g(z) = \frac{1}{z - pg(z)} = \frac{z - \sqrt{z^2 - 4p}}{2p}.$$

The square root is analytic off $[-2\sqrt{p}, 2\sqrt{p}]$ and is asymptotic to z at infinity. The asymptotic $g(z) \sim z^{-1}$ selects this root. The root resolvent therefore equals

$$m(z) = \langle e_0, (z - T)^{-1} e_0\rangle = \frac{1}{z - (p+1)g(z)}.$$

For $|x| < 2\sqrt{p}$, substitution and Stieltjes inversion give

$$-\frac{1}{\pi}\operatorname{Im} m(x + i0) = \frac{(p+1)\sqrt{4p - x^2}}{2\pi((p+1)^2 - x^2)}.$$

There are no poles off this interval. A zero of the denominator would force $g^2 = 1$ and $z = \pm(p+1)$, whereas the chosen branch has $g(\pm(p+1)) = \pm p^{-1}$. At both endpoints the denominator is nonzero because $p > 1$. Thus m has continuous finite boundary values on the interval, including its endpoints, and analytic continuation across its complement. The inversion formula gives only the displayed absolutely continuous measure. Its total mass is one, since $m(z) \sim z^{-1}$. Changing variables $x = 2\sqrt{p}\cos\theta$ gives ν_p.

The spectral theorem now gives

$$\int R_n(x) R_m(x)\, d\nu_p = \langle R_n(T)e_0, R_m(T)e_0\rangle = s_n \delta_{nm}.$$

Hence the q_n are orthonormal. Polynomials are dense in continuous functions on the compact x interval and thus in its L^2 space. The monic R_n span all polynomials, so the q_n are complete. This proves that $\mathcal{U}_p$ is unitary. The recurrence proves intertwining first on finite sums and then everywhere by boundedness. The support of the measure is the full closed interval. Multiplication by x therefore has norm $2\sqrt{p}$. Bounded Borel functional calculus proves the state identity. □

27.7 HECKE OPERATIONS AND THE DISTINGUISHED STATE

For the Bruhat–Tits tree of $\mathrm{PGL}_2(\mathbb{Q}_p)$, vertices are homothety classes of $\mathbb{Z}_p$-lattices in $\mathbb{Q}_p^2$. Two vertices are adjacent when representatives satisfy $pL \subset L' \subset L$ with $[L : L'] = p$. The lines in L/pL give exactly $p+1$ neighbors. With the maximal compact subgroup of Haar volume one, its adjacency operator is the characteristic-function Hecke operator for $\mathrm{diag}(p, 1)$. Pullback to GL_2 fixes the central operator $\mathrm{diag}(p, p)$ to be the identity. Thus the representation has trivial central character.

Define $H_0(x) = 1$, $H_1(x) = x$ and $H_{k+1}(x) = xH_k(x) - pH_{k-1}(x)$ for $k \geq 1$. In the notation of Proposition 27.1.2, $H_k(T)$ represents H_{p^k} with $D_p = 1$. These are the complete-symmetric Satake elements in this central character. Their generating series and spectral values are

$$\sum_{k\geq 0} H_k(x)t^k = (1 - xt + pt^2)^{-1}, \qquad H_k(x(\theta)) = p^{k/2}\sum_{j=0}^{k} e^{i(k-2j)\theta}.$$

The second formula follows by multiplying the two geometric series with parameters $\sqrt{p}\,e^{i\theta}$ and $\sqrt{p}\,e^{-i\theta}$. It remains valid at $\theta = 0, \pi$, where the parameters coincide. The transform therefore intertwines every $H_k(T)$ with the stated multiplication operator, with the same cyclic state.

PROPOSITION 27.7.1 *(distance and symmetric-power elements).* Let $(D_n f)(v) = \sum_{d(v,w)=n} f(w)$ on the vertex Hilbert space. Then $D_n = R_n(T)$ and

$$H_k(T) = \sum_{j=0}^{\lfloor k/2\rfloor} D_{k-2j}, \qquad \langle e_0, H_k(T)e_0\rangle = \begin{cases} 1 & k \text{ even}, \\ 0 & k \text{ odd}. \end{cases}$$

On $\mathcal{R}_p$, for $|t| < p^{-1/2}$, the series converges in operator norm and

$$\left\langle e_0, (I - tT + pt^2 I)^{-1} e_0 \right\rangle = \frac{1}{1 - t^2}.$$

Proof. The distance operators are bounded because every row and column has s_n entries. Counting the first step gives $TD_1 = D_2 + (p+1)I$ and $TD_n = D_{n+1} + pD_{n-1}$ for $n \geq 2$. Thus $D_n = R_n(T)$. With $H_{-1} = H_{-2} = 0$, the recurrences give $R_n = H_n - H_{n-2}$ for every $n \geq 0$. Telescoping proves the sum formula. Since $D_n e_0 = \mathbf{1}_{S_n}$, only D_0 contributes to the root state. The spectral formula bounds $\|H_k(T)|_{\mathcal{R}_p}\|$ by $(k+1)p^{k/2}$. This proves norm convergence on $\mathcal{R}_p$ in the stated disk and permits evaluation of its state term by term. Multiplication of the series by $I - tT + pt^2 I$ proves the inverse formula on that space. □

For example, at $p = 2$,

$$q_0 = 1, \qquad q_1 = x/\sqrt{3}, \qquad q_2 = (x^2 - 3)/\sqrt{6}, \qquad H_2 = x^2 - 2.$$

The tree counts give $\int x^2 d\nu_2 = 3$ and $\int x^4 d\nu_2 = 15$. Indeed a four-step return either returns after two steps, giving nine walks, or first reaches distance two, giving six walks. Consequently $\int H_2 d\nu_2 = 1$ but $\int R_2 d\nu_2 = 0$, and $\int R_2^2 d\nu_2 = 6$. These values distinguish the two Hecke normalizations already at the first nontrivial higher element.

27.8 Finite sets of primes and the arithmetic comparison

For a finite prime set F, form the completed Hilbert tensor products $\mathcal{R}_F = \widehat{\bigotimes}_{p\in F}\mathcal{R}_p$ and the product probability measure $\nu_F = \prod_{p\in F}\nu_p$. The tensor product of the maps $\mathcal{U}_p$ is the unitary

$$\mathcal{U}_F : \mathcal{R}_F \longrightarrow L^2([0,\pi]^F, \nu_F), \qquad \bigotimes_p e_{n_p} \longmapsto \prod_p q_{n_p}(\theta_p).$$

It intertwines each local Hecke operation, hence every element of $\bigotimes_{p\in F}\mathbb{C}[T_p]$, with multiplication by its Satake polynomial. The tensor root vector maps to 1, so its vector state becomes integration against ν_F. The algebraic integral lattices are the finite tensor products of the sphere-sum lattices above. Adjoining a prime uses $v \mapsto v \otimes e_0$ on the tree side and $f \mapsto f \otimes 1$ on the spectral side. These isometric maps commute with $\mathcal{U}_F$ and preserve the distinguished states. They give an operation-compatible finite-stage system.

To specify the occupation-phase comparison, let $P_p = \mathbb{C}[a_k, b_k \mid k \geq 1]$ be the polynomial oscillator space. For finitely supported nonnegative integers r_k, s_k, give its monomial $\prod_k a_k^{r_k} b_k^{s_k}$ squared norm $\prod_k k^{r_k+s_k} r_k! s_k!$, with distinct monomials orthogonal. Complete this positive inner product to a Hilbert space $\mathcal{F}_p$. The charge of this monomial is $m = \sum_k (r_k - s_k)$. Multiplication by $e^{im\theta}$ on each charge space defines a unitary V_θ on $\mathcal{F}_p$. This is the one-prime occupation action in Definition 25.5.1. The strong integral $A_p = \int_0^\pi V_\theta \, d\nu_p(\theta)$ is bounded with norm at most one. It acts on a charge-m monomial by $c_m = \int_0^\pi e^{im\theta}\, d\nu_p(\theta)$. For $m = 1$, symmetry about $\pi/2$ gives

$$c_1 = i \int_0^\pi \sin\theta \, d\nu_p(\theta) \in i\mathbb{R}_{>0}.$$

Thus this averaging operator has a nonreal eigenvalue. It cannot be unitarily intertwined with the self-adjoint adjacency operator or with multiplication by its real spectral coordinate. This obstruction concerns those specified operators. It does not exclude a different arithmetic comparison.

The radial transform supplies a complex Hilbert Hecke representation, an integral cyclic module, and a compatible finite-prime state system. A comparison with the proposed primitive arithmetic trace class $\mathcal{V}^{\mathrm{prim}}$ requires constructing its arithmetic trace carrier and a map preserving the specified Hecke operations and trace pairing. Tame residue and Gersten operations require additional maps on common coefficients. No full Weil functional or global spectral realization is identified by the positive root state.

27.9 The Chiral Framework over $\overline{\mathrm{Spec}\,\mathbb{Z}}$

The matrices S_p and their scalar trace recursions define the local linear constructions above. Global arithmetic realization requires local representations, distinguished vectors, a restricted tensor-product construction and an intertwiner with the proposed trace carrier. Beilinson–Drinfeld factorization on a smooth characteristic-zero curve does not by itself supply those data over $\overline{\mathrm{Spec}\,\mathbb{Z}}$.

Conjecture 27.9.1 *(unramified arithmetic trace realization).* There is an arithmetic dg category $\mathscr{C}_{\mathrm{arith}}$ over $\mathbb{C}$, defined independently of its proposed Hecke realization. Let $C_\bullet = CC_\bullet(\mathscr{C}_{\mathrm{arith}})$ be its Hochschild chain complex, with homological differential of degree -1. The construction specifies a degree d, a subspace $M_{\mathrm{arith}} \subset H_d(C_\bullet)$ containing the primitive trace class $v_{\mathrm{prim}} = \mathcal{V}^{\mathrm{prim}}$, and a normalized linear trace functional $\tau_{\mathrm{arith}} : M_{\mathrm{arith}} \to \mathbb{C}$ with $\tau_{\mathrm{arith}}(v_{\mathrm{prim}}) = 1$. It constructs commuting local spherical Hecke actions on this subspace, with the central element D_p acting as the identity.

For each prime p, it also specifies a smooth representation Π_p of $GL_2(\mathbb{Q}_p)$ with trivial central character, a Whittaker realization, and the spherical Hecke module $M_p = \Pi_p^{GL_2(\mathbb{Z}_p)}$. Its distinguished

vector u_p and linear functional $\ell_p : M_p \to \mathbb{C}$ satisfy $\ell_p(u_p) = 1$. The arithmetic or automorphic construction must determine these representations, vectors, and functionals. Set

$$M_{\mathrm{aut}} = \bigotimes_p{}' (M_p, u_p), \qquad u = \bigotimes_p u_p, \qquad \ell = \bigotimes_p \ell_p.$$

These are algebraic restricted tensor products over $\mathbb{C}$. The normalization makes ℓ compatible with adjoining u_p. Each local Hecke algebra acts on its own tensor factor, so the algebraic restricted tensor product of the local Hecke algebras acts on M_{aut}.

There is a complex-linear isomorphism

$$\Phi : M_{\mathrm{arith}} \longrightarrow M_{\mathrm{aut}}, \qquad \Phi(v_{\mathrm{prim}}) = u, \qquad \Phi(hm) = h\Phi(m), \qquad \tau_{\mathrm{arith}}(m) = \ell(\Phi(m)).$$

The identities hold for every $m \in M_{\mathrm{arith}}$ and every finite-support spherical Hecke element h. In particular, the distinguished Hecke functionals agree: $\tau_{\mathrm{arith}}(hv_{\mathrm{prim}}) = \ell(hu)$. The oscillator cover additionally admits a complex-linear map $Q : \otimes'_p \mathcal{H}^{(p)} \to M_{\mathrm{arith}}$ sending its vacuum to v_{prim}, with complex-linear local maps $Q_p : \mathcal{H}^{(p)} \to M_p$ such that $Q_p(|0\rangle_p) = u_p$ and $\Phi Q = \otimes_p Q_p$.

The class v_{prim} is an element of the arithmetic trace carrier, whereas M_{aut} is a representation space. The conjecture asks for their comparison through Φ, together with the oscillator maps Q, Q_p. It does not identify the class with that space. All these arithmetic constructions remain open. The linear functionals above are not assumed positive. A positive state, Hilbert completion, spectral decomposition, ramified extension, and archimedean or full Weil comparison each require their own construction and compatibility theorem.

27.10 The Corrected Local Model and Its Consequences

The global realization in Conjecture 27.9.1 remains conjectural. The unconditional local statements are the divisor composition law, the distinct Adams monoid action, the symmetric-power trace recursion for a specified Satake matrix, and the radial spherical transform of Theorem 27.6.1. The latter intertwines the local Hecke operations and preserves the root state. An arithmetic Hecke action requires a comparison preserving its carrier and determinant operation. Neither the oscillator rank nor an equality of scalar traces constructs that comparison.

Chapter 28

Divisor sums and Adams operations on polynomial Fock space

An arithmetic operation on a Fock space must be specified before it can be compared with Frobenius, a Hecke correspondence, or a vertex operation. Two integral substitutions are relevant here. One sums over divisors of a mode index; the other multiplies that index. They agree on the first mode, but their compositions differ. We compute the divisor substitutions, then construct the Adams action and identify the structures retained by each. The calculation concerns a polynomial oscillator representation. A map to an arithmetic Hochschild trace class remains additional data.

28.1 The two substitutions

Let R be a commutative unital ring and let S be a finite set of oscillator species. Put

$$M_R = \bigoplus_{s\in S,\ k\geq 1} R x_{s,k}, \qquad P_R = \mathrm{Sym}_R(M_R) = R[x_{s,k} : s \in S, k \geq 1].$$

All polynomials and all sums defining an image of a generator have finite support. The vacuum is 1. Particle degree assigns degree one to every $x_{s,k}$; weight assigns weight k. The species index can label an arithmetic place, but is independent of the integer indexing an operation. Suppress it when there is only one species. Define unital R-algebra maps

$$T_n(x_{s,k}) = \sum_{d\mid n,\ d\mid k} d\, x_{s,nk/d}, \qquad \psi_n(x_{s,k}) = x_{s,nk} \qquad (n \geq 1). \tag{28.1}$$

The divisibility condition $d \mid k$ restricts the sum; integrality of nk/d already follows from $d \mid n$. Write $t_n = T_n|_{M_R}$ for the underlying linear map. Thus $T_n = \mathrm{Sym}(t_n)$. These definitions commute with change of coefficient ring. They also extend to any set of species as an algebraic direct limit of finite sets. For an empty set of species all maps are the identity on R. Counterexamples below use a nonempty species set and characteristic zero, unless a coefficient ring is specified otherwise.

First-mode calculations

For species indexed by the primes 2 and 3, the first-mode identities are

Computation 28.1.1. $T_2 x_{2,1} = \psi_2 x_{2,1} = x_{2,2}$.

Computation 28.1.2. $T_2 x_{3,1} = \psi_2 x_{3,1} = x_{3,2}$.

Computation 28.1.3. Both substitutions send $x_{2,1}x_{3,1}$ to $x_{2,2}x_{3,2}$, by their polynomial product rule.

These first-mode equalities do not determine the higher compositions.

Example 28.1.4. For a prime p,

$$T_p x_1 = x_p, \quad T_p x_p = x_{p^2} + p x_p, \quad T_{p^2} x_1 = x_{p^2}.$$

Consequently $T_p^2 \neq T_{p^2}$. The usual weight-two Hecke recursion also fails: its discrepancy on x_1 is $p(x_p - x_1)$. Neither statement is a failure of the distinct maps ψ_n, which satisfy $\psi_m \psi_n = \psi_{mn}$ on every generator and therefore on P_R.

28.2 Composition on a prime chain

Fix a prime p and an integer $u \geq 1$ prime to p. The submodule with basis $e_r = x_{up^r}$, $r \geq 0$, is invariant under t_{p^a}. Denote its restriction by A_a, including $A_0 = 1$. Then

$$A_a e_r = \sum_{i=0}^{\min(a,r)} p^i e_{r+a-i}. \tag{28.2}$$

The indices never decrease. This boundary at $r = 0$ is responsible for the noncommutativity.

THEOREM 28.2.1 *(prime-valuation composition).* For $a, b, r \geq 0$, set $[v]_+ = \max(v, 0)$ for an integer v, and put

$$N_{a,b}(r,t) = \mathbf{1}_{t \leq r+b} \big[\min(b, r, t) - \max(0, t-a) + 1\big]_+.$$

Then, over every coefficient ring,

$$A_a A_b e_r = \sum_{t=0}^{a+b} p^t N_{a,b}(r,t) e_{r+a+b-t}. \tag{28.3}$$

Subtracting the same expression with a, b interchanged gives the commutator. Over $\mathbb{Z}$ or a characteristic-zero field, A_a and A_b commute precisely when $a = b$ or $ab = 0$.

Proof. Apply A_b first. A term is specified by $0 \leq i \leq \min(b, r)$, followed by $0 \leq j \leq \min(a, r + b - i)$. Its coefficient is p^{i+j} and its index is $r + a + b - i - j$. For fixed $t = i + j$, the second upper bound is equivalent to $t \leq r + b$, and the remaining bounds are

$$\max(0, t - a) \leq i \leq \min(b, r, t).$$

Counting these integers proves the formula without division. If $0 < a < b$, take $r = a$ and $t = 2a$. In the commutator the coefficient of e_b is $-p^{2a} \min(a, b - a)$, which is nonzero over the stated rings. The cases $a = b$ and $ab = 0$ commute by definition; interchanging a, b covers the remaining case. □

In particular,

$$[A_1, A_2] e_r = \begin{cases} -p^2 e_2, & r = 1, \\ 0, & r \neq 1. \end{cases} \tag{28.4}$$

For $r \geq 2$ this follows either from the formula or from the polynomials $U + p$ and $U^2 + pU + p^2$, where $U e_r = e_{r+1}$. Direct substitution checks $r = 0, 1$.

Corollary 28.2.2 *(different primes and arbitrary indices).* Families belonging to different primes commute. If $(m,n)=1$, then $T_mT_n=T_{mn}$. More generally, write $a_p=v_p(m)$, $b_p=v_p(n)$, and $r_p=v_p(k)$. On a generator,

$$T_mT_nx_k=\sum_{(t_p)}\left(\prod_{p\mid mn}p^{t_p}N_{a_p,b_p}(r_p,t_p)\right)x_{mnk/\prod_{p\mid mn}p^{t_p}},\qquad 0\le t_p\le a_p+b_p. \tag{28.5}$$

Terms with zero coefficient are omitted. All nonzero terms have integral indices. On a monomial the image is the product of these finite sums.

Proof. Multiplication or division of an index by a power of p does not change its valuation at another prime. Thus the choices of divisors at distinct primes are independent. The divisor formula factors t_n as the product of the $A_{v_p(n)}$ acting in these independent coordinates. Apply the theorem in each coordinate and extend the resulting equality on generators multiplicatively. For coprime m,n, one of a_p,b_p vanishes at each prime, so the product is exactly the formula for T_{mn}. □

These formulas include coefficient exceptions. In characteristic p, $T_{p^a}=\psi_{p^a}$, since every summand with $i>0$ vanishes. Over other rings a displayed commutator coefficient may vanish through torsion. Formula (28.3), rather than the characteristic-zero noncommutativity assertion, determines those cases.

More generally, for $1\le a<b$, direct expansion gives

$$[A_a,A_b]e_1=-p^{a+1}e_b. \tag{28.6}$$

Indeed $A_be_1=e_{b+1}+pe_b$, so applying A_a gives successive coefficients $1,2p,\dots,2p^a,p^{a+1}$. In the reverse order $A_ae_1=e_{a+1}+pe_a$; since $b\ge a+1$, the expansion ends with $2p^{a+1}$ instead of p^{a+1}.

Over any nonzero characteristic-zero ring, the global criteria are

$$T_mT_n=T_{mn}\iff(m,n)=1,$$

and

$$[T_m,T_n]=0\iff\text{for each }p,\quad a_pb_p=0\text{ or }a_p=b_p.$$

For the first necessity, on x_1 a common prime contributes $A_aA_be_0=\sum_{j=0}^{\min(a,b)}p^je_{a+b-j}$, whereas $A_{a+b}e_0=e_{a+b}$. Select its $j=1$ term and the coefficient-one highest terms at the other primes; it cannot vanish. For the second, if $0<a_p<b_p$, apply the commutator to x_p. At every other prime the two orders agree on e_0; selecting their highest terms and using (28.6) gives a nonzero coefficient. Sufficiency follows from the independent prime factors. Over $\mathbb{F}_\ell$, both criteria have the corresponding exception at $p=\ell$: that prime imposes no condition. At every other prime, the same nonzero power-of-p test proves necessity. These assertions concern the polynomial carrier; reducing the larger Hall lattice below is a different operation.

The interaction with the Adams shifts is equally explicit:

$$\psi_mT_nx_k=\sum_{d\mid n,\,d\mid k}dx_{mnk/d},\qquad T_n\psi_mx_k=\sum_{d\mid n,\,d\mid mk}dx_{mnk/d}.$$

In characteristic zero these commute exactly for coprime m,n. Sufficiency follows from equality of the divisor conditions, and $k=1$ detects an extra positive coefficient when the indices have a common prime.

One can instead retain the prime substitutions and change the higher operations: define $U_n = \prod_p T_p^{v_p(n)}$. Then $U_m U_n = U_{mn}$, because different prime substitutions commute. This is another monoid action, distinct from both T_n and ψ_n. Its prime-power formula is

$$T_p^a e_r = \sum_{j=0}^{a} \binom{a}{j} p^j e_{r+a-j} \quad (r \geq 1), \qquad T_p^a e_0 = \sum_{j=0}^{a-1} \binom{a-1}{j} p^j e_{a-j} \quad (a \geq 1).$$

The first is the binomial theorem for $U + p$ on the invariant tail $r \geq 1$; the second follows after the first step $e_0 \mapsto e_1$. Its prime maps still fail the Frobenius tests below, so this change repairs only the composition law.

28.3 AN EXPLICIT PRESENTATION OF THE PRIME-CHAIN ALGEBRA

A presentation should distinguish relations in linear operators on M_R from composition of ring maps on P_R. We give a finite coefficient test for every proposed linear relation. It describes the entire prime-chain operator algebra, including its boundary terms.

A useful ambient realization uses $Ue_r = e_{r+1}$ and $Be_0 = 0$, $Be_r = e_{r-1}$ for $r \geq 1$. They satisfy $BU = 1$, and $U^i B^i$ projects onto the tail $r \geq i$. Therefore

$$A_a = \sum_{i=0}^{a} p^i U^{a-i} U^i B^i = U^a \sum_{i=0}^{a} p^i B^i, \qquad A_{a+1} = U A_a + p^{a+1} U^{a+1} B^{a+1}.$$

These identities locate the generated algebra inside the integral shift-and-partial-inverse operator algebra. They do not identify the subalgebra generated by the A_a with that whole ambient algebra.

For $a \geq 1$, put $q_a(z) = \sum_{i=0}^{a} p^i z^{a-i}$. For a word $w = Z_{a_\ell} \cdots Z_{a_1}$, write $D_w = \sum_j a_j$ and $q_w = \prod_j q_{a_j}$; the rightmost letter acts first. For $r, s \geq 0$, define the integer

$$B_w(r,s) = \sum_{\substack{0 \leq i_j \leq a_j,\ \sum_j (a_j - i_j) = s \\ i_j \leq r + \sum_{h<j}(a_h - i_h) \text{ for every } j}} p^{\sum_j i_j}. \tag{28.7}$$

For the empty word set $D_w = 0$, $q_w = 1$, and $B_w(r,s) = \mathbf{1}_{s=0}$. Every sum is finite, and $B_w(r,s) = 0$ if $s > D_w$.

THEOREM 28.3.1 *(finite-test presentation).* Let $\mathcal{F}_R = R\langle Z_1, Z_2, \ldots \rangle$. For a finite expression $F = \sum_w c_w w$, put $D = \max_w D_w$, taking $D = 0$ for the zero expression. Define $I_{p,R}$ by the following explicit coefficient conditions:

$$\sum_w c_w q_w(z) = 0 \quad \text{in } R[z], \tag{28.8}$$

$$\sum_w c_w B_w(r,s) = 0 \quad (0 \leq r < D,\ 0 \leq s \leq D). \tag{28.9}$$

Then $I_{p,R}$ is a two-sided ideal and

$$R[A_1, A_2, \ldots] \cong \mathcal{F}_R / I_{p,R}.$$

Here the left side is the unital R-subalgebra of $\mathrm{End}_R(\bigoplus_{r \geq 0} Re_r)$ generated by the indicated linear maps. This is an explicit, generally infinite presentation, not an assertion of finite generation or a finite list of relations.

Proof. Repeated use of (28.2) gives $w(A)e_r = \sum_s B_w(r,s)e_{r+s}$. If $r \geq D$, every intermediate index is at least r, hence every choice $0 \leq i_j \leq a_j$ is allowed. In that range $B_w(r,s)$ is the coefficient of z^s in $q_w(z)$. Thus (28.8) says exactly that $F(A)e_r = 0$ for all $r \geq D$, and (28.9) says the same for the remaining basis vectors. These conditions are therefore exactly the kernel of evaluation $Z_a \mapsto A_a$. The kernel is a two-sided ideal and the first isomorphism theorem proves the assertion. This argument works over any commutative R; all tests are finite equalities in R. □

Equivalently, an evaluated expression has a unique form $q(U) + E$, where $q \in R[z]$ and E is a finite matrix supported in columns $r < D$ and rows $r \leq r + s \leq r + D$. Uniqueness follows by applying the difference to arbitrarily large e_r. The symbol map onto $R[z]$ is surjective because $q_1 = z + p$. Its kernel consists of the finite matrices that occur in the generated algebra. No assertion that all finite matrices occur is needed. For $R = \mathbb{Z}$, the commutator in (28.4) is a nonzero element of this kernel.

For finitely many primes, the same coefficient construction applies on the basis indexed by their valuation vectors. Operators at different primes commute; their word coefficients are products of (28.7). Tail and boundary tests in each coordinate give an explicit presentation there as well. Every word in the full family involves only finitely many primes.

The multiplicative extension to P_R preserves composition and is faithful on the monoid of the generator maps: equality is tested on M_R. It does not preserve addition of operators. For example, if L is a generator map, then $\mathrm{Sym}(2L)(x^2) = 4(Lx)^2$, whereas $2\,\mathrm{Sym}(L)(x^2) = 2(Lx)^2$. Thus a linear relation from this presentation is not automatically a linear relation between operators on the entire polynomial ring.

28.4 Integral Adams operations and the Hall lattice

Proposition 28.4.1 *(Adams action on the monomial carrier).* The maps ψ_n form a monoid action by ring endomorphisms on every P_R. Over $\mathbb{Z}$, however, this action is not a system of Frobenius lifts:

$$\psi_p(x_1) - x_1^p = x_p - x_1^p \notin pP_{\mathbb{Z}}.$$

The two distinct monomials remain distinct modulo p. This excludes this particular Adams action from defining a special λ-ring on the monomial lattice. It does not exclude other λ-structures on that ring. For example, $\phi_n(x_{s,k}) = x_{s,k}^n$, with the identity on $\mathbb{Z}$, gives commuting Frobenius lifts and the special λ-structure in which each variable is a line element.

Theorem 28.4.2 *(torsion-free Adams criterion).* The torsion-free criterion is the following: a commuting system of unital ring endomorphisms with $\psi_m\psi_n = \psi_{mn}$, $\psi_1 = 1$, and $\psi_p(f) \equiv f^p \pmod{pR}$ determines a unique special λ-structure on a $\mathbb{Z}$-torsion-free ring. This is Wilkerson's Proposition 1.2, as identified in Borger, *The basic geometry of Witt vectors, I*, §1.17. Borger's §§1.5–1.6 distinguish commuting operations from Frobenius lifts. No torsion-free converse is asserted for arbitrary torsion rings.

Observation 28.4.3 *(rational coefficients).* Over a $\mathbb{Q}$-algebra the congruences are vacuous, but the commuting action is still substantive data. Its operations are determined by

$$\lambda_t(f) = \exp\left(\sum_{n\geq 1} \frac{(-1)^{n-1}\psi_n(f)t^n}{n}\right).$$

The two integral lattices

Inside $P_{\mathbb{Q}}$ for one species, define

$$H(t) = \sum_{r\geq 0} h_r t^r = \exp\left(\sum_{k\geq 1} \frac{x_k t^k}{k}\right), \qquad L_{\mathbb{Z}} = \mathbb{Z}[h_1, h_2, \ldots].$$

Observation 28.4.4 *(integral and rational lattices).* The symmetric-function identification $x_k = p_k$ gives $P_{\mathbb{Z}} \subsetneq L_{\mathbb{Z}}$ and $P_{\mathbb{Q}} = L_{\mathbb{Z}} \otimes \mathbb{Q}$. For instance $h_2 = (x_1^2 + x_2)/2$ belongs to the second lattice but not the first. These are the Newton and Hall lattice conventions of Macdonald, I.2, equations (2.10)–(2.14), and I.4, equations (4.5)–(4.9). For several species take the tensor product of these integral lattices.

THEOREM 28.4.5 *(Adams operations on the Hall lattice).* The standard Adams action on symmetric functions substitutes $(z_1, z_2, \ldots) \mapsto (z_1^n, z_2^n, \ldots)$; hence it preserves $L_{\mathbb{Z}}$ and acts on power sums by $x_k \mapsto x_{nk}$. For a prime p, reducing symmetric polynomials in finitely many variables modulo p proves the Frobenius congruence; passage through stable symmetric degrees proves it on $L_{\mathbb{Z}}$.

PROPOSITION 28.4.6 *(integrality of the divisor substitutions).* Every T_n also preserves $L_{\mathbb{Z}}$. Its complete-symmetric series satisfies

$$T_n H(t) = \prod_{d\mid n} \psi_n H(t^d). \tag{28.10}$$

Nevertheless T_p is not a Frobenius lift on this lattice:

$$T_p h_p = \psi_p h_p + \psi_p h_1 \equiv h_p^p + h_1^p \not\equiv h_p^p \pmod{pL_{\mathbb{Z}}}.$$

Proof. In the logarithm of $T_n H(t)$, set $k = dj$ in the summand indexed by $d \mid n$. This gives

$$\sum_{k\geq 1} \frac{T_n(x_k)t^k}{k} = \sum_{d\mid n} \sum_{j\geq 1} \frac{x_{nj} t^{dj}}{j}.$$

Exponentiating proves (28.10) in $P_{\mathbb{Q}}[[t]]$. Each factor has coefficients in $L_{\mathbb{Z}}$, so comparison of coefficients proves integrality. For $n = p$, the coefficient of t^p in the two factors is exactly the displayed sum. The Frobenius congruence for ψ_p gives its reduction, and h_1^p is nonzero in the polynomial ring $L_{\mathbb{Z}}/pL_{\mathbb{Z}}$. □

Thus passing to the Hall lattice repairs the integral Frobenius condition for ψ_n, but does not make the family T_n an Adams action. Both lattices and their reductions remain available for T_n, with the distinct fibres specified rather than identified.

28.5 PRODUCTS, COPRODUCTS, AND GRADINGS

Equip P_R with its primitive Hopf structure: $\Delta x_{s,k} = x_{s,k} \otimes 1 + 1 \otimes x_{s,k}$, $\epsilon(x_{s,k}) = 0$, and antipode $x_{s,k} \mapsto -x_{s,k}$. Both families are Hopf endomorphisms: their generator images are primitive, and equality of algebra maps can be checked on those generators. They preserve the unit, augmentation, and particle degree. On $L_{\mathbb{Z}}$ the same coproduct has $\Delta h_r = \sum_{i+j=r} h_i \otimes h_j$; its preservation follows from the rational identity and the torsion-free integral inclusion. This is the ordinary symmetric-function coproduct, not a constructed arithmetic Hall correspondence.

Over a characteristic-zero field both maps are injective and, for $n > 1$, not surjective. Indeed, on a polynomial of highest nonzero weight w, the highest-weight part of T_n is ψ_n applied to that part and has weight nw. The index substitution is injective, so that part cannot cancel. This proves injectivity and shows that x_1 has no preimage for $n > 1$. The same argument in fact works over any nonzero coefficient ring on the free polynomial carrier.

The Adams map carries weight w to weight nw. The divisor map carries a weight-w polynomial into weights between w and nw. It is generally inhomogeneous; already $T_p x_p = x_{p^2} + p x_p$. For $n > 1$, neither map preserves a nontrivial finite cutoff of the mode generators. Both preserve the ideals of weights greater than W, so they induce maps on the finite-weight quotient. For finite S, these quotients have finite rank over R. They also extend continuously to the weight completion $\prod_{w \geq 0} P_{R,w}$, since an output of weight at most W depends only on inputs of weight at most W. Their integral matrices give continuous extensions to $\varprojlim_j P_{\mathbb{Z}}/p^j P_{\mathbb{Z}}$ as well. This last completion imposes finite support modulo each p^j, and is not the unrestricted product over weights.

Other symmetric-function structures have different answers. Over $\mathbb{Q}$, the internal coproduct $\delta x_k = x_k \otimes x_k$ is preserved by ψ_n. For T_n, $n > 1$, take $k = n$: its image contains the distinct terms x_{n^2} and $n x_n$, and $(T_n \otimes T_n)\delta x_n$ has cross terms absent from $\delta T_n x_n$. Neither family preserves the internal product normalized by $x_k * x_l = \mathbf{1}_{k=l} k x_k$, since its value on the images of x_1, x_1 is $n x_n$, not x_n, for $n > 1$. These assertions are over characteristic zero; no nonvanishing statement is transferred to a torsion fibre without checking its coefficient.

28.6 Oscillators and vertex operations

On the polynomial representation set $a_{-k} f = x_k f$, $a_k f = k \partial_{x_k} f$ for $k \geq 1$, and $a_0 = 0$. Then $[a_k, a_{-l}] = k \delta_{kl}$, with central element acting as one. Species are independent. These integral identities, and the Hall form $\langle x_\lambda, x_\mu \rangle = \delta_{\lambda\mu} z_\lambda$, where $z_\lambda = \prod_k k^{m_k} m_k!$, fix the normalization.

The substitutions intertwine creation with creation by their generator images. They do not intertwine the fixed annihilation operators. For the Adams maps the exact formulas are

$$\psi_n a_{-k} = a_{-nk} \psi_n, \qquad a_{nk} \psi_n = n \psi_n a_k, \qquad a_j \psi_n = 0 \quad (n \nmid j).$$

They follow by differentiation. If both positive and negative modes are sent to their index multiples, their commutator is multiplied by n; this is not a central-charge-one Heisenberg map. Over $\mathbb{Q}$, sending a_k to a_{nk}/n, a_{-k} to a_{-nk}, and the central element to itself does give a Heisenberg embedding, intertwined by ψ_n. It does not preserve the involution $a_k^* = a_{-k}$.

There is also a precise non-adjoint oscillator lift for a fixed T_n. It illustrates what extra choices of annihilation entail. Define integers $c_{k,i}$, in increasing order of i, by

$$c_{k,i} = 0 \ (n \nmid i), \qquad c_{k,nj} = k \delta_{kj} - \sum_{\substack{d \mid n,\, d \mid j \\ d > 1}} d \, c_{k,nj/d}.$$

Every term on the right has smaller index. The derivation $D_k^{(n)} = \sum_{i \geq 1} c_{k,i} \partial_{x_i}$ is defined on every polynomial, since only its finitely many variables contribute. By construction

$$D_k^{(n)}(T_n x_j) = k \delta_{kj}, \qquad [D_k^{(n)}, \text{multiplication by } T_n x_j] = k \delta_{kj}.$$

These derivations commute, as do the creation maps. The chain rule gives $D_k^{(n)} T_n = T_n a_k$. Thus they give a central-charge-one oscillator representation in polynomial endomorphisms, generally using infinite

locally finite sums of annihilation modes over $\mathbb{Q}$: $D_k^{(n)} = \sum_i (c_{k,i}/i) a_i$. Over $\mathbb{Z}$, integrality refers to the polynomial derivation, not to its coefficients in the a_i. This is not an identification with the fixed Heisenberg generators or their adjoints. For instance, both T_n and ψ_n send x_1 to x_n, whose norm squared is n; its creation operator and its Hall adjoint have commutator n, rather than one. Therefore the adjoint-preserving assignment fails for $n > 1$.

PROPOSITION 28.6.1 *(translation and vertex operations).* Over $\mathbb{Q}$ or $\mathbb{C}$, the usual Heisenberg vertex algebra has translation $Dx_k = kx_{k+1}$. A vertex-algebra homomorphism preserves translation, because $Dv = v_{(-2)}1$, irrespective of whether it preserves a chosen conformal vector. Here

$$D\psi_n x_1 = n x_{n+1}, \qquad \psi_n D x_1 = x_{2n},$$

and

$$DT_n x_1 = n x_{n+1}, \qquad T_n D x_1 = x_{2n} + 2\mathbf{1}_{2|n} x_n.$$

For $n > 1$ these disagree. Equivalently, writing $a(z) = Y(x_1, z)$, one has

$$Y(x_n, z) = \frac{\partial_z^{n-1} a(z)}{(n-1)!}, \qquad Y(x_n, z) Y(x_n, w) \sim \frac{(-1)^{n-1}(2n-1)!}{((n-1)!)^2 (z-w)^{2n}}.$$

This is the derivative of $a(z)a(w) \sim (z-w)^{-2}$, and it cannot be the image of the latter OPE for $n > 1$. Thus neither family consists of vertex-algebra endomorphisms. Ordinary polynomial multiplication and primitive coproduct remain preserved; they do not encode all vertex products. When $n = 1$, both maps are the identity and all these obstructions disappear.

PROPOSITION 28.6.2 *(underlying monoid action on a finite vertex carrier).* For a finite set of species the underlying vector space of the Heisenberg vertex algebra, with its vacuum and maps ψ_n, is a vacuum-preserving $\mathbb{N}^\times$-module. The action is not by vertex endomorphisms for $n > 1$. The divisor family T_n is not such a monoid action in characteristic zero.

Proof. The index substitutions preserve 1 and obey the monoid law on polynomial generators. The preceding translation calculation proves the vertex assertion; $T_p^2 x_1 \neq T_{p^2} x_1$ proves the last statement. □

28.7 HILBERT COMPLETION AND ARITHMETIC ENERGY

Over $\mathbb{C}$, use the positive Hermitian Hall form with $\|x_{s,k}\|^2 = k$, and complete to the symmetric Hilbert Fock space $\mathcal{F} = \bigoplus_{j \geq 0} \mathcal{F}^{(j)}$, with the above monomial factorial normalization. On the one-particle space, $\psi_n/\sqrt{n}$ is an isometry. Each summand $d\,\mathbf{1}_{d|k} x_{nk/d}$ of $t_n x_k$ has operator norm $\sqrt{nd}$: its nonzero basis images are orthogonal and have that constant norm ratio. Hence

$$\|t_n\| \leq \sum_{d|n} \sqrt{nd}.$$

This proves boundedness on one particle, and therefore on each fixed particle sector after taking its symmetric tensor power.

On the full Fock space both substitutions are unbounded for $n > 1$:

$$\frac{\|T_n x_1^j\|}{\|x_1^j\|} = \frac{\|\psi_n x_1^j\|}{\|x_1^j\|} = n^{j/2}.$$

Their polynomial-domain operators are closable. More precisely, for $A = t_n$ or $A = \psi_n|_{M_{\mathbb{C}}}$, the closure is the direct sum of the bounded sector maps $\mathrm{Sym}^j(A)$, with domain

$$\left\{(v_j) \in \mathcal{F} : \sum_{j\geq 0} \|\mathrm{Sym}^j(A)v_j\|^2 < \infty\right\}.$$

Indeed this direct sum is closed. Finite particle truncation, followed by finite-mode polynomial approximation in each bounded sector, approximates both a vector and its image, so the polynomial domain is a core. For ψ_n the domain condition is exactly $\sum_j n^j\|v_j\|^2 < \infty$. Thus neither map is a bounded Hilbert-space isometry or a bounded Bost–Connes isometry on this normalization.

The separately normalized substitution $\widetilde{\psi}_n(x_{s,k}) = n^{-1/2}x_{s,nk}$ is a Hilbert isometry on every particle sector and on their direct sum. It is a complex polynomial Hopf map and a monoid action, but it changes the integral structure and still fails the translation identity for $n > 1$. Its existence alone supplies no Bost–Connes representation or trace comparison.

Convention 28.7.1 *(integer weight and arithmetic energy).* For a finite species set consisting of arithmetic primes, use:

(i) Integer oscillator weight k on $x_{p,k}$, extended additively to monomials.

(ii) Arithmetic energy $k\log p$ on $x_{p,k}$, again extended additively. This real grading is distinct from the integer weight.

The corresponding energy operator on monomials is

$$H\left(\prod_{p,k} x_{p,k}^{m_{p,k}}\right) = \left(\sum_{p,k} km_{p,k}\log p\right)\prod_{p,k} x_{p,k}^{m_{p,k}}.$$

Observation 28.7.2 *(energy and integral states).* The associated eigenspaces are spanned by integral monomials. The operator H itself need not preserve their integral lattice, but the real energy grading does not destroy that lattice. It does not rescale the vertex-algebra conformal vector. On polynomials, $H\psi_n = n\psi_n H$; T_n generally mixes energies.

For one oscillator species per arithmetic prime and a finite set S, direct summation of the occupation basis gives, for real $\beta > 0$,

$$\mathrm{Tr}_{\mathcal{F}_S}(e^{-\beta H}) = \prod_{p\in S}\prod_{k\geq 1}(1 - p^{-k\beta})^{-1}.$$

The logarithm converges by geometric comparison for finite S. For all arithmetic primes it converges for $\beta > 1$: the logarithm is bounded by a constant times $\sum_p p^{-\beta}$, which converges. The resulting product is $\prod_{k\geq 1}\zeta(k\beta)$, not $\zeta(\beta)$. At $0 < \beta \leq 1$, already the $k = 1$ factors diverge. Multiple oscillator species change the exponents of the factors. An infinite set of equally weighted species instead has infinite multiplicity in weight one and no finite conformal character.

For the energy operator just defined, its self-adjoint domain is the set of occupation expansions $\sum_\nu c_\nu e_\nu$ with $\sum_\nu E(\nu)^2|c_\nu|^2 < \infty$, where e_ν are normalized occupation vectors. The finite-support span is a core by truncation. The trace statements concern this positive diagonal operator. For all primes the Gibbs operator is not trace class at $\beta \leq 0$ either, since its diagonal entries do not tend to zero. Keeping only modes $k = 1$ identifies occupation states with positive integers $N = \prod_p p^{m_p}$, with energy $\log N$, and gives $\zeta(\beta)$ for $\beta > 1$. That subspace is not preserved by T_n or ψ_n, which send $x_{p,1}$ to $x_{p,n}$.

On $\ell^2(\mathbb{N}_{>0})$, the different operators $\mu_n \delta_N = \delta_{nN}$ are commuting isometries and satisfy $[H_{\mathrm{BC}}, \mu_n] = \log(n)\mu_n$ on finite-support vectors, where $H_{\mathrm{BC}}\delta_N = \log N\, \delta_N$. This follows immediately by applying both sides to δ_N. They move δ_1 to δ_n, whereas both polynomial substitutions fix the vacuum. Thus even the normalized Adams action does not yield a vacuum- and energy-preserving identification with these multiplicative shifts: its energy relation is dilation, and its vacuum is fixed. Other representations or operations require separate comparison.

28.8 The Common Algebraic Image and the Hilbert Kernel

The divisor maps and the index shifts have different composition laws, but have the same algebraic images. The difference can be expressed by a triangular change of generators. That change ceases to be invertible on the Hilbert completion, so polynomial injectivity must be kept separate from injectivity of the completed operator.

PROPOSITION 28.8.1 *(triangular factorization).* On the polynomial carrier over a commutative ring R, define

$$C_n(x_{s,k}) = \sum_{d\mid n,\, d\mid k} d\, x_{s,k/d},$$

and extend as a unital R-algebra map. Then C_n is an integral triangular Hopf automorphism and

$$T_n = \psi_n C_n, \qquad T_n(P_R) = \psi_n(P_R). \tag{28.11}$$

For the Hall lattice $L_{\mathbb{Z}}$ of one species,

$$C_n H(t) = \prod_{d\mid n} H(t^d). \tag{28.12}$$

Thus C_n is also an automorphism of that integral lattice, and $T_n(L_{\mathbb{Z}}) = \psi_n(L_{\mathbb{Z}})$. For finitely many species the statement holds on their tensor product.

Proof. The term $d = 1$ is $x_{s,k}$; every other term has smaller mode index. Inductively solve for each $x_{s,k}$ in terms of the $C_n(x_{s,j})$, $j \le k$. This constructs an inverse with integer coefficients, finite on every generator. Hence it defines an inverse on the countable polynomial ring and after every coefficient change. The generator images are primitive, so C_n is a Hopf map and its inverse is a Hopf map. Applying ψ_n to those images gives the literal divisor formula, proving (28.11).

To prove (28.12), take the logarithm in $P_{\mathbb{Q}}[[t]]$, set $k = dj$ in each divisor summand, and obtain

$$\sum_{k\ge 1} \frac{C_n(x_k)t^k}{k} = \sum_{d\mid n}\sum_{j\ge 1} \frac{x_j t^{dj}}{j}.$$

Exponentiation proves the identity. Its coefficient of t^r is h_r plus an integral polynomial in $h_1, \dots, h_{r-1}$: only the factor with $d = 1$ can contribute h_r, and it does so with coefficient one. Induction therefore constructs an integral inverse on the Hall generators as well. The asserted image equalities follow from surjectivity of C_n on the respective carriers. □

This is a change of generators, not an identification of the two operator families. For example, $T_p^2 \neq T_{p^2}$, whereas $\psi_p^2 = \psi_{p^2}$. Equal images do not equate their actions.

Theorem 28.8.2 *(one-particle Hilbert kernel and range).* Over $\mathbb{C}$, complete the one-particle span with $\|x_k\|^2 = k$. The linear restriction $c_n = C_n|_M$ extends to a bounded surjection on this Hilbert space. The bounded divisor operator t_n has closed range

$$\operatorname{Ran}(t_n) = \overline{\operatorname{span}}\{x_{nk} : k \geq 1\} = \operatorname{Ran}(\psi_n|_M), \qquad \ker t_n = \ker c_n.$$

For $n > 1$ that kernel is nonzero and, on the full one-species one-particle space, infinite-dimensional.

More precisely, on a prime chain $e_r = x_{up^r}$, $(u, p) = 1$, write $Be_0 = 0$, $Be_r = e_{r-1}$ for $r \geq 1$. For $a \geq 1$,

$$c_{p^a} = \sum_{j=0}^{a} p^j B^j, \qquad \ker c_{p^a} = \operatorname{span}_{\mathbb{C}} \left\{ v_\omega = \sum_{r \geq 0} (\omega/p)^r e_r : \omega^{a+1} = 1,\ \omega \neq 1 \right\}.$$

Its dimension on that chain is a. For $a = 0$ the operator is the identity with zero kernel. For $n = 1$, all the global operators are identities.

Proof. Define $B_d x_k = x_{k/d}$ when $d \mid k$, and zero otherwise. Its nonzero images are orthogonal with norm ratio $d^{-1/2}$. Thus $c_n = \sum_{d|n} dB_d$ is bounded, with norm at most $\sum_{d|n} \sqrt{d}$. The identities $B_d B_e = B_{de}$ imply $c_n = \prod_{p|n} c_{p^{v_p(n)}}$.

On the stated prime chain, $\|e_r\|^2 = up^r$, so B has norm $p^{-1/2}$ and the forward shift $Ue_r = e_{r+1}$ has norm $p^{1/2}$, with $BU = 1$. Each displayed v_ω is nonzero and square-summable, since $\|v_\omega\|^2 = u \sum_{r \geq 0} p^{-r} = u/(1 - p^{-1})$. It satisfies $Bv_\omega = (\omega/p)v_\omega$, whence $c_{p^a} v_\omega = (\sum_{j=0}^{a} \omega^j) v_\omega = 0$. Conversely a vector $\sum_r b_r e_r$ in the kernel satisfies $\sum_{j=0}^{a} p^j b_{r+j} = 0$ for all $r \geq 0$. The first a coefficients determine the rest. The a distinct roots ω/p give independent solutions by the Vandermonde determinant, so these square-summable solutions exhaust the kernel.

Surjectivity also has an explicit bounded proof. For $\lambda = \omega/p$,

$$(B - \lambda)U = 1 - \lambda U, \qquad R_\lambda = U(1 - \lambda U)^{-1}$$

is a right inverse of $B - \lambda$, since $\|\lambda U\| = p^{-1/2} < 1$. The inverse is its norm-convergent Neumann series. Factoring $c_{p^a} = p^a \prod_{\omega \neq 1,\ \omega^{a+1}=1} (B - \omega/p)$ and multiplying these right inverses in reverse order, then multiplying by p^{-a}, gives a bounded right inverse for c_{p^a}. Its norm bound is independent of the prime-free index u, so the construction extends to the orthogonal sum of all prime chains. Products over the finitely many primes therefore make c_n surjective. Since $\psi_n|_M$ is $\sqrt{n}$ times an isometry, the factorization $t_n = \psi_n c_n$ proves the range and kernel assertions. For $n > 1$, a prime-power factor has the kernels just exhibited; all other factors commute with it. Choosing infinitely many mutually disjoint prime chains gives infinitely many independent vectors in $\ker c_n$. □

The inverse of C_n on polynomials is not a bounded inverse on this Hilbert space. For example,

$$C_p^{-1} e_r = \sum_{j=0}^{r} (-p)^j e_{r-j}, \qquad \frac{\|C_p^{-1} e_r\|^2}{\|e_r\|^2} = \sum_{j=0}^{r} p^j.$$

These identities follow by cancellation of successive terms in $(1 + pB) \sum_{j=0}^{r} (-pB)^j e_r = e_r$. More generally, the algebraic inverse on the finite one-particle span is not even closable when $n > 1$. Indeed, choose a nonzero $v \in \ker c_n$ and finite truncations $v_N \to v$. Boundedness gives $y_N = c_n v_N \to 0$, whereas the polynomial inverse satisfies $C_n^{-1} y_N = v_N \to v \neq 0$. Thus the closure of its graph contains

$(0, v)$, which cannot belong to the graph of a linear operator. The same argument applies to the algebraic inverse of t_n on its polynomial image.

For $n > 1$, the closed full-Fock divisor operator $\Gamma(t_n)$ described in Section 28.7 also has nonzero kernel: any of these one-particle vectors lies in its domain and is annihilated. The corresponding closed operator $\Gamma(\psi_n|_M)$ is injective, since each particle sector is a nonzero scalar multiple of an isometry. Neither assertion contradicts injectivity of the polynomial maps. Nor does it claim that multiplication extends to all Hilbert vectors.

A Weighted Hilbert Threshold

The kernel vectors above contain arbitrarily high modes. Requiring additional decay of their coefficients can exclude them. On a single prime chain the exact boundary is accessible because both operators are polynomials in a backward shift.

Fix a prime p, an integer $u \geq 1$ coprime to p, and an integer $a \geq 1$. Retain $e_r = x_{up^r}$ and $\|e_r\|_0^2 = up^r$, and write D for their finite linear span. Let E be the positive diagonal operator $Ee_r = up^r e_r$. This E is the mode-index operator on the chain; it is distinct from the logarithmic arithmetic energy used in the partition functions. For a real number $s \geq 0$, define

$$\mathcal{H}_s = \left\{ \sum_{r\geq 0} d_r e_r : \sum_{r\geq 0} (up^r)^{1+2s} |d_r|^2 < \infty \right\}, \qquad \|v\|_s = \|E^s v\|_0.$$

Thus $\mathcal{H}_s = \mathrm{Dom}(E^s)$ as a subspace of $\mathcal{H}_0$, endowed with the displayed norm. Since $E \geq 1$, this norm is equivalent to its graph norm. Define the closed tail $\mathcal{H}_{s,\geq a} = \overline{\mathrm{span}}\{e_r : r \geq a\}$, of codimension a in $\mathcal{H}_s$.

THEOREM 28.8.3 *(weighted kernel, range, and inverse).* The polynomial operators c_{p^a} and t_{p^a} extend boundedly to every $\mathcal{H}_s$. Their behavior on these completions is as follows.

(i) If $0 \leq s < \frac{1}{2}$, then c_{p^a} is onto $\mathcal{H}_s$, and t_{p^a} is onto $\mathcal{H}_{s,\geq a}$. Both have the same a-dimensional kernel, spanned by
$$g_\zeta = \sum_{r\geq 0} (\zeta/p)^r e_r, \qquad \zeta^{a+1} = 1, \quad \zeta \neq 1.$$
Their polynomial inverses, defined on D and on $D \cap \mathcal{H}_{s,\geq a}$, respectively, are not closable.

(ii) If $s = \frac{1}{2}$, both maps are injective. The range of c_{p^a} is dense, proper, and nonclosed in $\mathcal{H}_s$; the range of t_{p^a} is dense, proper, and nonclosed in $\mathcal{H}_{s,\geq a}$. Their inverses on their actual ranges are closed and unbounded. The polynomial inverses are closable, and their graph closures are precisely these inverses.

(iii) If $s > \frac{1}{2}$, then c_{p^a} is boundedly invertible on $\mathcal{H}_s$, and t_{p^a} is a bounded isomorphism from $\mathcal{H}_s$ onto $\mathcal{H}_{s,\geq a}$. These bounded inverses extend the polynomial inverses.

For $a = 0$, both maps are the identity for every $s \geq 0$.

Proof. Put $b = p^{s+1/2}$, $q = p^{1/2-s}$, and $Q_a(z) = 1 + z + \cdots + z^a$. The vectors

$$f_r = \frac{e_r}{u^{s+1/2} b^r}, \qquad r \geq 0,$$

form an orthonormal basis of $\mathcal{H}_s$. Let $Vf_r = f_{r+1}$ and $L = V^*$. The raw forward shift on the e_r is bV, and the raw backward shift satisfies $pB = qL$. Consequently

$$c_{p^a} = Q_a(qL), \qquad t_{p^a} = b^a V^a Q_a(qL).$$

This proves boundedness, with norm bounds $\sum_{j=0}^{a} q^j$ and $b^a \sum_{j=0}^{a} q^j$, respectively. The isometry V^a identifies $\mathcal{H}_s$ with its closed tail, so it suffices to analyze $Q_a(qL)$.

Suppose first that $q > 1$. For $v = \sum_{r\ge 0} \gamma_r f_r$, the kernel equation is $\sum_{j=0}^{a} q^j \gamma_{r+j} = 0$ for every $r \ge 0$. Its characteristic roots are the distinct numbers ζ/q, where $\zeta^{a+1} = 1$ and $\zeta \neq 1$. The first a coefficients determine all others, and the Vandermonde determinant shows that the corresponding geometric sequences form a basis of the solution space. All belong to ℓ^2, since $q > 1$. In raw coordinates these are the displayed g_ζ, with

$$\|g_\zeta\|_s^2 = \frac{u^{1+2s}}{1 - p^{2s-1}}.$$

For each root, $L - \zeta/q$ has the bounded right inverse $V(I - (\zeta/q)V)^{-1}$, where the inverse is the norm-convergent Neumann series. Since $Q_a(qL) = q^a \prod_\zeta (L - \zeta/q)$, a bounded right inverse G of c_{p^a} is q^{-a} times the product of these right inverses in reverse order. Then

$$t_{p^a}\big(b^{-a} G L^a\big) = V^a L^a,$$

which is the orthogonal projection onto the tail. This proves both surjectivity statements. Finite truncations of any nonzero g_ζ give the nonclosability argument above, now in $\mathcal{H}_s$, for both polynomial inverses.

At $q = 1$, the same recurrence has solutions that are linear combinations of the periodic sequences ζ^r. A nonzero periodic sequence is not square-summable, so the kernel is zero. The constant coefficient of Q_a is one. Its formal reciprocal therefore acts on D, because L is locally nilpotent there. Thus $c_{p^a}(D) = D$, and its range is dense. Likewise $t_{p^a}(D) = D \cap \mathcal{H}_{s,\ge a}$. To see that the range is not closed, fix a root $\zeta \neq 1$ and let

$$v_N = N^{-1/2} \sum_{r=0}^{N-1} \zeta^r f_r, \qquad N > a.$$

Its norm is one. The first $N - a$ coefficients of $Q_a(L) v_N$ vanish, and the remaining at most a coefficients have absolute value at most $(a+1)/\sqrt{N}$. Hence

$$\|Q_a(L) v_N\|_s \le \frac{\sqrt{a}\,(a+1)}{\sqrt{N}} \longrightarrow 0.$$

An injective bounded operator with closed range is bounded below: its inverse on that range is bounded by the bounded inverse theorem. The displayed sequence rules this out, and therefore the dense range is both proper and nonclosed. Multiplication by $b^a V^a$ gives the same assertion within the tail for t_{p^a}.

An injective bounded operator has a closed inverse on its range, since interchanging the two coordinates of its closed graph preserves closedness. These inverses are unbounded by the preceding sequence. Their polynomial restrictions have those full inverse graphs as their closures: for any $x \in \mathcal{H}_s$, choose $x_N \in D$ with $x_N \to x$. Boundedness gives $c_{p^a} x_N \to c_{p^a} x$, so $(c_{p^a} x_N, x_N) \to (c_{p^a} x, x)$; the same argument applies to t_{p^a}. This also proves closability of the restrictions.

Finally suppose $q < 1$. The polynomial identity $(1 - z) Q_a(z) = 1 - z^{a+1}$ gives

$$c_{p^a}^{-1} = (I - qL)\big(I - (qL)^{a+1}\big)^{-1}, \qquad \|c_{p^a}^{-1}\| \le \frac{1+q}{1-q^{a+1}}.$$

The inverse is defined by a norm-convergent Neumann series. On the tail the inverse of t_{p^a} is $c_{p^a}^{-1} b^{-a} L^a$. Both inverses agree on finite vectors with the locally finite polynomial inverses, as follows either from the same formulas or from uniqueness of the inverse. □

All these bounded maps preserve the common finite domain D, which is a core for them in each $\mathcal{H}_s$. The polynomial maps have not changed. Their completed kernels and the possible extensions of their inverses depend on the norm: for $s \geq \frac{1}{2}$ the vectors g_ζ no longer belong to the carrier. This change of domain neither removes the kernel on $\mathcal{H}_0$ nor identifies these weighted spaces with an arithmetic realization.

28.9 The arithmetic comparison problem

Remark 28.9.1 *(local arithmetic structures).* There are now two explicit local inputs: the integral divisor substitutions with the composition law (28.5), and the honest Adams action on $P_{\mathbb{Q}}$ with its Frobenius-compatible Hall lattice $L_{\mathbb{Z}}$. Neither is a vertex action. On the polynomial underlying vector space, ψ_n gives an $\mathbb{N}^\times$-module; this terminology records only the monoid action, not a separate notion of a Λ-module or a trace theory.

Definition 28.9.2 *(module with multiplicative-monoid action).* An $\mathbb{N}^\times$-module is a module with endomorphisms ψ_n satisfying $\psi_1 = 1$ and $\psi_m \psi_n = \psi_{mn}$. This specifies no ring product or Frobenius congruence.

Question 28.9.3 *(arithmetic realization).* An arithmetic realization must specify its native carrier, integral form, operations, and comparison maps. Transporting ψ_n to a Hochschild trace class requires an actual intertwining map and the operations on that class; a polynomial presentation of another object does not supply them. If the proposed operations are instead T_n, their same-prime noncommutativity must be respected. In characteristic zero they cannot be identified by an injective intertwiner with commuting Adams operations or a commuting semigroup of Bost–Connes isometries. The commutator (28.4) is already an obstruction. A quotient or restricted subspace would require a specified map and a new calculation of its kernel and preserved operations.

Remark 28.9.4 *(integral descent and operator comparison).* Borger's descent interpretation concerns Λ-structures and their forgetful functor, not an automatic tensor product with a field of one element. See *Lambda-rings and the field with one element*, §§1.1–1.2. A Bost–Connes comparison additionally requires a C^*-representation, adjoints, time evolution, and trace or state intertwiners. Its standard $\zeta(\beta)$ partition function cannot be inferred from the full oscillator partition function above. The arithmetic trace comparison and the choice of an integral realization therefore remain specific construction problems; the local polynomial calculations neither settle nor rule out all their possible solutions.

28.10 Curve zeta functions in Hall operator exchange

The polynomial oscillator calculations admit a geometric source over a finite field. The required local category is the category of nilpotent modules, with its Hall multiplication and Green pairing. A torsion sheaf on a smooth projective curve is the direct sum of its parts supported at closed points. This decomposition turns local Hall operators into global operators. Their exchange coefficient counts the points over every finite extension of the ground field.

Let X be a smooth projective geometrically connected curve over $\mathbb{F}_q$. For a closed point x, put $d_x = [\mathbb{F}_q(x) : \mathbb{F}_q]$ and $Q_x = q^{d_x}$. The completion of the local ring is $\mathbb{F}_{Q_x}[[t]]$, after a choice of parameter and coefficient field. Its finite-length modules are nilpotent modules over $\mathbb{F}_{Q_x}[t]$.

Local Hall operators and their normalization

For a prime power Q, let H_Q be the complex Hall algebra of these nilpotent modules. Its basis element u_M denotes the isomorphism class of M. The coefficient of u_C in $u_A u_B$ counts submodules $B' \subset C$ with $B' \simeq B$ and $C/B' \simeq A$. Give it the positive Hermitian form, conjugate-linear in the first slot,

$$\langle u_M, u_N \rangle = \frac{\delta_{MN}}{|\operatorname{Aut} M|}.$$

The Green coproduct is adjoint to multiplication for this form. In this category it is cocommutative and multiplicative. These are the classical Hall bialgebra identities, on this untwisted category with zero Euler form. The polynomial presentation below is Steinitz–Hall's theorem; see Schiffmann, *Lectures on Hall algebras*, Theorem 2.6 and equation (2.9).

Write u_{1^r} for the zero endomorphism of $\mathbb{F}_Q^r$ and set

$$e_r = Q^{r(r-1)/2} u_{1^r}, \qquad E(z) = \sum_{r \geq 0} e_r z^r, \qquad e_0 = 1.$$

Subspace counting gives $\Delta u_{1^r} = \sum_{i+j=r} Q^{-ij} u_{1^i} \otimes u_{1^j}$. Thus $\Delta E(z) = E(z) \otimes E(z)$. The elements e_r, $r > 0$, freely generate H_Q. One can see the triangular presentation directly: for a Jordan partition λ, use the successive dimensions of $\ker t^i / \ker t^{i-1}$. The product of the corresponding semisimple classes counts filtrations lowered by t. Its coefficient on u_λ is one, since equality forces the kernel filtration. Every other term has larger kernel dimensions. The resulting triangular matrix is invertible in each finite dimension.

Define p_r by the formal identity

$$\log E(z) = \sum_{r \geq 1} \frac{(-1)^{r-1}}{r} p_r z^r.$$

They are primitive polynomial generators. Their norms can be computed without a choice of an orthonormal basis. Since $|\operatorname{GL}_r(\mathbb{F}_Q)| = Q^{r(r-1)/2} \prod_{j=1}^{r} (Q^j - 1)$,

$$\langle E(u), E(v) \rangle = \sum_{r \geq 0} \frac{Q^{r(r-1)/2} (\bar{u} v)^r}{\prod_{j=1}^{r} (Q^j - 1)} = \prod_{j \geq 1} (1 - Q^{-j} \bar{u} v)^{-1}.$$

The last equality follows by comparing coefficients in $(1 - t) P(Qt) = P(t)$, with constant coefficient one. Adjunction and primitivity make distinct primitive degrees orthogonal and give factorial pairings for their powers. Taking logarithms therefore gives

$$\langle p_r, p_s \rangle = \delta_{rs} \frac{r}{Q^r - 1}.$$

Consequently the following operators are adjoint on the polynomial domain:

$$a_{-r} = \sqrt{Q^r - 1}\, M_{p_r}, \qquad a_r = \frac{r}{\sqrt{Q^r - 1}} \frac{\partial}{\partial p_r}, \qquad [a_r, a_{-s}] = r \delta_{rs} I.$$

Here M_{p_r} means multiplication by p_r, and the square root is positive. No assertion that these unbounded operators act everywhere on the Hilbert completion is needed.

The global exchange

The torsion Hall algebra of X is the algebraic tensor product $H_X^{\mathrm{tor}} = \bigotimes_{x\in|X|}^{\mathrm{alg}} H_{Q_x}$, with vacuum 1 in all but finitely many factors. Distinct supports have no extensions or homomorphisms between them. Their Hall products and pairings therefore factor. Give a length-r local module degree rd_x. There are finitely many closed points of bounded degree, so each homogeneous piece of this algebra is finite-dimensional. Let $a_{x,\pm r}$ be the preceding operators on the x factor, and define

$$A_{-n} = \sum_{d|n} d \sum_{d_x=d} a_{x,-n/d}, \qquad A_n = A_{-n}^{\dagger} \qquad (n > 0).$$

These are finite sums on the common polynomial domain. A closed point of degree d contributes d geometric points over $\mathbb{F}_{q^n}$ exactly when $d \mid n$. Thus, writing $N_n = \#X(\mathbb{F}_{q^n})$,

$$[A_n, A_{-m}] = \delta_{nm} \sum_{d|n} d^2 \frac{n}{d} \#\{x : d_x = d\} I = \delta_{nm} n N_n I.$$

All commutators between two positive or two negative modes vanish.

Put $s = w^{-1}$, with z, s independent formal variables, and set

$$\Gamma_-(z) = \exp\left(\sum_{n>0} \frac{A_{-n}}{n} z^n\right), \qquad \Gamma_+(w) = \exp\left(\sum_{n>0} \frac{A_n}{n} w^{-n}\right).$$

Each coefficient is a finite sum of polynomial-domain operators. Products are identities in $\mathrm{End}_{\mathbb{C}}(H_X^{\mathrm{tor}})[[z, s]]$. For the formal zeta function

$$Z_X(t) = \exp\left(\sum_{n>0} \frac{N_n}{n} t^n\right) = \prod_{x\in|X|} (1 - t^{d_x})^{-1},$$

the logarithm of the Euler product proves the second equality coefficientwise. The central commutator of the two exponents equals $\log Z_X(zs)$. Expanding the exponentials, or differentiating their formal conjugation identity, therefore proves

$$\Gamma_+(w)\Gamma_-(z) = Z_X(z/w)\Gamma_-(z)\Gamma_+(w).$$

This is an operator identity before taking a trace. Repeated reordering uses the product of the central factors for all exchanged pairs. The statement specifies a formal expansion at $z/w = 0$; an analytic operator identity would require additional domain and convergence estimates.

For $X = \mathbb{P}^1$, counting gives $N_n = q^n + 1$, hence $Z_X(t) = ((1 - t)(1 - qt))^{-1}$. For the smooth projective cubic $E/\mathbb{F}_2 : y^2 + y = x^3 + x + 1$, the first four point counts are 1, 5, 13, 25. The trace criterion for $y^2 + y = b$ gives two solutions when $\mathrm{Tr}_{\mathbb{F}_{2^n}/\mathbb{F}_2}(b) = 0$, and none otherwise. For $n = 1, 2, 3, 4$, it gives respectively 0, 2, 6, 12 admissible x values, with one additional point at infinity. Thus

$$Z_E(t) = 1 + t + 3t^2 + 7t^3 + 15t^4 + O(t^5).$$

These coefficients determine the corresponding operator exchange to degree four. The full rational formula $Z_E(t) = (1 - 2t + 2t^2)/((1 - t)(1 - 2t))$ uses the classical elliptic-curve zeta theorem in addition to these finite counts.

Torsion acting on a line bundle

The full Hall algebra of coherent sheaves also contains the classes of line bundles. Fix a rational point x and a line bundle L. There are two middle sheaves in an extension of $\mathcal{O}_x$ by L: the split sheaf $L \oplus \mathcal{O}_x$, and $L(x)$. Indeed $\operatorname{Ext}^1(\mathcal{O}_x, L)$ is one-dimensional, and all nonzero classes have the latter middle sheaf. In the split middle sheaf, the relevant copies of L are the graphs of the q maps $L \to \mathcal{O}_x$. In $L(x)$ the relevant subobject is uniquely $L(x)(-x) = L$. Conversely, an extension of L by $\mathcal{O}_x$ splits because $\operatorname{Ext}^1(L, \mathcal{O}_x) = 0$. Its unique torsion subobject is $\mathcal{O}_x$. Therefore

$$u_{\mathcal{O}_x} u_L = q\, u_{L \oplus \mathcal{O}_x} + u_{L(x)}, \qquad u_L u_{\mathcal{O}_x} = u_{L \oplus \mathcal{O}_x}.$$

Let $v = \sqrt{q}$ and define the twisted product $u_A * u_B = v^{\chi(A,B)} u_A u_B$, where $\chi(A, B) = \dim \operatorname{Hom}(A, B) - \dim \operatorname{Ext}^1(A, B)$. Here $\chi(\mathcal{O}_x, L) = -1$ and $\chi(L, \mathcal{O}_x) = 1$. The ordinary commutator for this twisted product consequently satisfies

$$[u_{\mathcal{O}_x}, u_L]_* = v^{-1} u_{L(x)}.$$

On $\mathbb{P}^1$, all rational-point modifications of $\mathcal{O}(n)$ are isomorphic to $\mathcal{O}(n+1)$. Summing over the $q+1$ rational points gives

$$\left[\sum_{x \in \mathbb{P}^1(\mathbb{F}_q)} u_{\mathcal{O}_x}, u_{\mathcal{O}(n)} \right]_* = (v + v^{-1}) u_{\mathcal{O}(n+1)}.$$

This mixed Hall operation retains the elementary modification of the line bundle. Identifying a full quantum loop algebra additionally requires its remaining generators and relations. Neither this mixed operation nor the torsion exchange constructs a comparison with a global arithmetic trace category or with the full Weil functional.

Chapter 29

Bost–Connes Operators and Hochschild Trace Classes

Multiplication by a prime raises its occupation number by one. This arithmetic shift is a normalized creation operator on a bosonic Fock space. Together with the phases indexed by $\mathbb{Q}/\mathbb{Z}$, it gives a faithful representation of the Bost–Connes C^*-algebra. The object $\mathcal{V}^{\mathrm{prim}}$ is a Hochschild trace class. Relating it to this representation requires a specified comparison map, pairing and completion.

29.1 The Bost–Connes C^*-Algebra

Definition 29.1.1 *(Bost–Connes presentation).* The Bost–Connes Q-lattice C^*-algebra $C^*_{\mathbb{Q}}$ has generators

$$\{\mu_n\}_{n\in\mathbb{N}^\times}, \quad \{\mu_n^*\}_{n\in\mathbb{N}^\times}, \quad \{e(\gamma)\}_{\gamma\in\mathbb{Q}/\mathbb{Z}},$$

with relations

$$\begin{aligned}
\mu_1 &= 1, \qquad e(0)=1, \qquad e(\gamma)^* = e(-\gamma),\\
\mu_m^*\mu_n &= \mu_n\mu_m^* \quad \text{if } \gcd(m,n)=1,\\
\mu_n\mu_m &= \mu_{nm},\\
\mu_n^*\mu_n &= 1,\\
\mu_n\mu_n^* &= e_n = \frac{1}{n}\sum_{\gamma:\, n\gamma=0} e(\gamma),\\
e(\gamma_1)e(\gamma_2) &= e(\gamma_1+\gamma_2),\\
\mu_n e(\gamma)\mu_n^* &= \rho_n(e(\gamma)) = \frac{1}{n}\sum_{\delta:\, n\delta=\gamma} e(\delta).
\end{aligned}$$

The $e(\gamma)$ generate the group C^*-algebra $C^*(\mathbb{Q}/\mathbb{Z})$; the μ_n furnish an endomorphism semigroup action; $\mu_n^*\mu_n = 1$, $\mu_n\mu_n^* = e_n$ makes μ_n a Toeplitz isometry. The crossed product $C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^\times$ yields $C^*_{\mathbb{Q}}$.

Theorem 29.1.2 *(KMS states and the partition function, Bost–Connes Prop. 3.3 and Thm 5.4).* The KMS_β state φ_β on $C^*_{\mathbb{Q}}$ is unique for $0 < \beta \le 1$. For $\beta > 1$ the KMS_β states form a simplex with extremal points indexed by the symmetry-broken arithmetic boundary. In the standard Gibbs representation, for $\beta > 1$, the partition function is

$$Z_{\mathrm{BC}}(\beta) = \operatorname{tr}(e^{-\beta H}) = \sum_{n\ge 1} n^{-\beta} = \zeta(\beta),$$

where $H = \log N$ generates $\sigma_t(\mu_n) = n^{it}\mu_n$. The ζ-partition function is explicitly motivated by Julia's primon gas (*Statistical theory of numbers*, Springer Proc. Phys. 47, 1990).

29.2 Prime occupations and oscillator operators

Definition 29.2.1 *(Prime occupation space).* For each prime p, let $\mathscr{F}_p = \ell^2(\mathbb{N}_0)$ over $\mathbb{C}$, with orthonormal states $|r\rangle_p$ and vacuum $|0\rangle_p$. Let L_{alg} be the complex span of tensors $|\mathbf{r}\rangle = \bigotimes_p |r_p\rangle_p$ with finitely many nonzero occupations. Declare this basis orthonormal and write L for its Hilbert completion. On L_{alg}, define

$$a_p|\mathbf{r}\rangle = \sqrt{r_p}\,|\mathbf{r}-\mathbf{e}_p\rangle, \qquad a_p^\dagger|\mathbf{r}\rangle = \sqrt{r_p+1}\,|\mathbf{r}+\mathbf{e}_p\rangle, \qquad N_p = a_p^\dagger a_p,$$

where $\mathbf{e}_p$ has occupation one at p and zero elsewhere. The first expression is zero when $r_p = 0$. On this common invariant dense domain,

$$[a_p, a_q^\dagger] = \delta_{pq}\mathbb{1}, \qquad \langle a_p v, w\rangle = \langle v, a_p^\dagger w\rangle.$$

In particular, $|r\rangle_p = (a_p^\dagger)^r|0\rangle_p/\sqrt{r!}$. In a vacuum Heisenberg module these are occupations of the mode $a_{-1}^{(p)}$, rather than charges of the zero mode $a_0^{(p)}$.

Proposition 29.2.2 *(Arithmetic shifts on the occupation space).* Let $v_p(m)$ denote the exponent of p in m. The map

$$U : \ell^2(\mathbb{N}^\times) \longrightarrow L, \qquad U\delta_m = \bigotimes_p |v_p(m)\rangle_p$$

is unitary. Define $\mu_n\delta_m = \delta_{nm}$ on $\ell^2(\mathbb{N}^\times)$. For each prime,

$$U\mu_p U^{-1} = a_p^\dagger (N_p+1)^{-1/2},$$

where the right side is first evaluated on L_{alg} and then extended boundedly. For every $n \geq 1$, $\mu_n^*\mu_n = \mathbb{1}$. The operator $\mu_n\mu_n^*$ is the orthogonal projection onto the closed span of δ_m with $n \mid m$. Its spectrum is $\{0,1\}$ for $n > 1$ and $\{1\}$ for $n = 1$.

Proof. Unique factorization gives a bijection of the two orthonormal bases, so U extends to a unitary. The diagonal factor $(N_p+1)^{-1/2}$ cancels the creation coefficient $\sqrt{r_p+1}$. The resulting operator raises r_p by one with coefficient one. It preserves the norm on L_{alg} and therefore has a unique isometric extension. Directly,

$$\mu_n^*\delta_m = \begin{cases} \delta_{m/n}, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

This proves both projection identities. For $n > 1$, δ_n belongs to the range and δ_1 belongs to its orthogonal complement. Both spectral values occur. □

29.3 Arithmetic phases and the comparison carrier

Proposition 29.3.1 *(Bost–Connes operators on L).* Fix the character $\chi(\gamma) = e^{2\pi i\gamma}$ of $\mathbb{Q}/\mathbb{Z}$. Together with the transported shifts, the operators

$$e(\gamma)U\delta_m = \chi(m\gamma)U\delta_m$$

give a faithful representation of $C_{\mathbb{Q}}^*$ on L.

Proof. The phase operators are unitary and multiply by addition of their indices. The finite character sum is

$$\frac{1}{n}\sum_{n\delta=\gamma}\chi(m\delta)=\begin{cases}\chi((m/n)\gamma), & n\mid m,\\ 0, & n\nmid m.\end{cases}$$

These are exactly the diagonal entries of $\mu_n e(\gamma)\mu_n^*$. The semigroup and coprime adjoint relations hold on the occupation basis. Thus the universal presentation defines a representation π.

For faithfulness, give $K=\mathrm{Hom}(\mathbb{Q}_{>0}^{\times},U(1))$ its compact product topology, with one circle for each prime. Its gauge action α fixes $e(\gamma)$ and sends μ_n to $z(n)\mu_n$. These formulas preserve every defining relation. The linear span of $\mu_r e(\gamma)\mu_s^*$ is dense in $C_{\mathbb{Q}}^*$. Indeed, the transfer relation implies $e(\gamma)\mu_n=\mu_n e(n\gamma)$ and its adjoint identity. Writing $r=dr'$, $s=ds'$ with $\gcd(r',s')=1$ gives $\mu_r^*\mu_s=\mu_{s'}\mu_{r'}^*$. These identities reduce products of normal terms to the same form. A normal term has gauge degree zero exactly when $r=s$. The transfer relation then puts it in the phase algebra. Thus Haar averaging $E(a)=\int_K \alpha_z(a)\,dz$ is a conditional expectation onto the phase algebra D.

The universal group algebra $C^*(\mathbb{Q}/\mathbb{Z})$ maps onto D. Its representation on $\ell^2(\mathbb{N}^{\times})$ evaluates $C(\widehat{\mathbb{Z}})$ at the positive integers. Every residue class modulo every positive integer contains a positive integer, so these evaluation points are dense. This representation is faithful, and hence $D=C^*(\mathbb{Q}/\mathbb{Z})=C(\widehat{\mathbb{Z}})$.

The expectation E is faithful. For $a\geq 0$ and any state ω, the function $z\mapsto\omega(\alpha_z(a))$ is continuous and nonnegative. If $E(a)=0$, its integral is zero and its value at the identity is zero. States separate positive elements, so $a=0$. Finally, $V_z\delta_m=z(m)\delta_m$ implements the gauge action in π before transport through U. If $\pi(a)=0$, gauge equivariance gives $\pi(E(a^*a))=0$. Faithfulness on D gives $E(a^*a)=0$, and faithful averaging gives $a=0$. □

Remark 29.3.2 *(The map to a Hochschild trace class).* The preceding construction takes place on L. A Hilbert realization from Hochschild homology requires a specified complex $(C_\bullet,b)$ over $\mathbb{C}$, a degree d, and a map $J:L_{\mathrm{alg}}\to Z_d(C_\bullet,b)$. It also requires a positive sesquilinear form q on $H_d(C_\bullet,b)$, a null space $\mathcal{N}_q$, and the completion

$$\mathcal{H}_q=\overline{H_d(C_\bullet,b)/\mathcal{N}_q}.$$

The identity $q([Jv],[Jw])=\langle v,w\rangle_L$ extends J isometrically to $L\to\mathcal{H}_q$. Surjectivity onto a proposed closed sector requires density of the image. Identifying that sector or a class in it with $\mathcal{V}^{\mathrm{prim}}$ is further data. An action on this homology and its intertwining with J must also be specified. The occupation-space construction supplies none of these Hochschild comparison data.

Example 29.3.3 *(A coherent state in the arithmetic representation).* For a prime p and $z\in\mathbb{C}$, the vector

$$e^{-|z|^2/2}\sum_{k\geq 0}\frac{z^k}{\sqrt{k!}}U\delta_{p^k}$$

has norm one and lies in L. It is the image of $U\delta_1$ under $e^{-|z|^2/2}\sum_{k\geq 0}z^k(U\mu_pU^{-1})^k/\sqrt{k!}$. This operator series converges in norm, since the ratio of successive coefficient magnitudes is $|z|/\sqrt{k+1}$. Thus the coherent state belongs to the arithmetic algebra orbit of the vacuum.

29.4 The tensor decomposition by current modes

Proposition 29.4.1 *(Mode-one and higher-mode factors).* For a finite set F of primes, let $\mathscr{F}_F^{\mathrm{cur}}=\Gamma_s(\ell^2(F\times\mathbb{N}_{\geq 1}))$ be the complex bosonic Fock space, where Γ_s is the Hilbert direct sum of the symmetric

tensor powers. This model has one oscillator per prime and mode. The mode index $j \geq 1$ corresponds to $a_{-j}^{(p)}/\sqrt{j}$. Splitting the one-particle space into $j = 1$ and $j \geq 2$ gives the canonical unitary

$$\mathscr{F}_F^{\mathrm{cur}} \cong L_F \widehat{\otimes} \mathscr{F}_F^{\geq 2}, \qquad L_F = \widehat{\bigotimes}_{p \in F} \mathscr{F}_p, \qquad \mathscr{F}_F^{\geq 2} = \Gamma_s(\ell^2(F \times \mathbb{N}_{\geq 2})).$$

On algebraic number states it also preserves the grading that assigns degree j to a mode-j creation operator. Let $\mathbb{N}_F$ consist of the positive integers supported on F. The formulas $\mu_n \delta_m = \delta_{nm}$ for $n, m \in \mathbb{N}_F$ and $e(\gamma)\delta_m = \chi(m\gamma)\delta_m$ act on $L_F \cong \ell^2(\mathbb{N}_F)$. They extend to the displayed tensor product by the identity on $\mathscr{F}_F^{\geq 2}$.

Proof. Each normalized current number state determines a unique pair of occupation arrays, for modes one and modes at least two. This orthonormal basis bijection gives the unitary and preserves the sum of the mode degrees. The subspace $\ell^2(\mathbb{N}_F) \subset \ell^2(\mathbb{N}^\times)$ is invariant under the phases and under μ_n, μ_n^* for $n \in \mathbb{N}_F$. Restricting these operators and tensoring with the identity preserves their relations. □

COROLLARY 29.4.2 *(Stable descendant sectors).* For every closed subspace $K \subset \mathscr{F}_F^{\geq 2}$, the subspace $L_F \widehat{\otimes} K$ is invariant under the extended arithmetic representation and its adjoints. If $F \neq \varnothing$, the higher-mode vacuum sector is therefore not a maximal stable subspace.

Proof. Each operator has the form $A \otimes 1$ and preserves $L_F \widehat{\otimes} K$ together with its adjoint. When $F \neq \varnothing$, the higher-mode space contains both the vacuum and a nonzero mode-two state. The vacuum sector is properly contained in the full stable tensor product. □

29.5 KESTEN–MCKAY = PLANCHEREL OF UNRAMIFIED $GL_2(\mathbb{Q}_p)$

THEOREM 29.5.1 *(Kesten 1959; McKay 1981; Lubotzky 1994 Thm 7.4.5).* The Kesten–McKay distribution on $[-2\sqrt{p}, 2\sqrt{p}]$ for $q = p$ prime,

$$d\nu_p(x) \;=\; \frac{(p+1)\sqrt{4p - x^2}}{2\pi((p+1)^2 - x^2)}\, dx,$$

is the spectral measure of the adjacency operator on the $(p+1)$-regular tree, equivalently the p-adic Bruhat–Tits tree of $\mathrm{PGL}_2(\mathbb{Q}_p)$, equivalently the Plancherel measure of the spherical-function algebra on the tree.

COROLLARY 29.5.2. The Kesten–McKay averaging is the Plancherel decomposition of the $GL_2(\mathbb{Q}_p)$ spherical Hecke algebra, not a spectral theorem for the GL_1 Bost–Connes semigroup algebra. Any map from the Julia–Bost–Connes partition function to ν_p must pass through a specified $GL_1 \to GL_2$ correspondence.

Remark 29.5.3 *(arithmetic averaging and Plancherel comparison).* The finite occupation-phase operator A_S of Definition 25.5.1 acts on a polynomial oscillator space. It does not construct the proposed arithmetic carrier or quotient for $\mathrm{av}^{\mathrm{arith}}$. An arithmetic Plancherel interpretation requires a specified representation, its spectral transform, and an intertwining map from the intended arithmetic object. The operadic averaging $\mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$ of Vol I instead acts on chiral cohomology. A common integral form does not identify these carriers or supply a map between them.

29.6 The Soibelman Cohomological Hall Algebra Reading

A rank-n representation of the Jordan quiver over $\mathbb{F}_q$ is an endomorphism of $\mathbb{F}_q^n$. Change of basis acts by conjugation. Its finite groupoid mass weights each isomorphism class by the reciprocal of its automorphism-group order.

Proposition 29.6.1 *(finite-field mass of Jordan-quiver representations).* For a prime power q and $n \geq 1$,

$$\sum_{[A] \subset M_n(\mathbb{F}_q)} \frac{1}{|\operatorname{Cent}_{GL_n(\mathbb{F}_q)}(A)|} = \frac{q^{n^2}}{|GL_n(\mathbb{F}_q)|} = \prod_{i=1}^{n} (1 - q^{-i})^{-1}.$$

The sum is over conjugacy classes of matrices. In rank one its value is $q/(q-1)$.

Proof. An orbit of A has $|GL_n(\mathbb{F}_q)|/|\operatorname{Cent}_{GL_n(\mathbb{F}_q)}(A)|$ elements. Summing orbit sizes gives $|M_n(\mathbb{F}_q)| = q^{n^2}$. Counting successive independent columns gives

$$|GL_n(\mathbb{F}_q)| = \prod_{j=0}^{n-1} (q^n - q^j) = q^{n^2} \prod_{i=1}^{n} (1 - q^{-i}).$$

Division proves the formula and its rank-one specialization. □

This scalar groupoid mass is not a real spectral measure. A comparison with the Kesten–McKay measure requires a specified operator, state and moment or spectral-transform map. A Behrend-weighted or vanishing-cycle refinement also requires its own normalization. Kontsevich–Soibelman, arXiv:1006.2706, §§2.4–2.5, constructs the cohomological Hall product and its shuffle expression. That construction does not supply the spectral comparison described here.

Proposition 29.6.2 *(three-way comparison obstruction).* The composed arrow

$$\text{Julia primon gas} \xrightarrow{\text{Plancherel}} \text{Bost-Connes Hecke-Plancherel} \xrightarrow{\text{CoHA-motivic}} \text{Soibelman DT on } \mathcal{Q}_{\text{Jordan}}$$

is not defined by the preceding theorems. The Bost–Connes algebra is a GL_1 semigroup system, the Kesten–McKay measure is a GL_2 spherical Plancherel measure, and the Jordan-quiver CoHA is a motivic Hall algebra. Equality of Euler factors can be asserted only after a functor between these three structures has been constructed.

29.7 Gauge Derivations and Smooth Domains

The occupation representation supplies bounded arithmetic shifts and phases. Its time derivative is unbounded, so a Hochschild calculation must specify an algebra on which that derivative is defined. We use the finite normal-term algebra to construct gauge classes and their cup products. Smoothness under finite prime-circle actions gives a continuous version; smoothness under the time evolution gives a different coefficient algebra. Their distinction is visible on an explicit norm-convergent series. None of these constructions identifies an arithmetic Hochschild trace carrier.

Put $A = C^*_{\mathbb{Q}}$ over $\mathbb{C}$, represented faithfully on $\mathcal{L} = \ell^2(\mathbb{N}^\times)$ as in Proposition 29.3.1. Write its orthonormal basis as e_m, so that $\mu_n e_m = e_{nm}$ and $e(\gamma)e_m = \exp(2\pi i m\gamma)e_m$. Define the complex vector space of finite sums

$$A_{\text{alg}} = \operatorname{span}_{\mathbb{C}}\{\mu_r e(\gamma)\mu_s^* : r, s \geq 1,\ \gamma \in \mathbb{Q}/\mathbb{Z}\}.$$

It is a dense unital $*$-subalgebra of A. Indeed, if $d = \gcd(s,u)$, $s = ds'$, and $u = du'$, the defining relations give

$$\begin{aligned}(\mu_r e(\gamma)\mu_s^*)(\mu_u e(\eta)\mu_v^*)\\ = \mu_{ru'} e(u'\gamma + s'\eta)\mu_{vs'}^*.\end{aligned}$$

Here $\mu_s^*\mu_u = \mu_{u'}\mu_{s'}^*$, $e(\gamma)\mu_{u'} = \mu_{u'}e(u'\gamma)$, and $\mu_{s'}^* e(\eta) = e(s'\eta)\mu_{s'}^*$. Adjoint exchanges r, s and negates γ. Thus the displayed span is closed under multiplication and adjoint, and it contains all generators.

THEOREM 29.7.1 *(gauge classes in algebraic Hochschild cohomology).* Use algebraic $\mathbb{C}$-multilinear Hochschild cochains of A_{alg}, with coefficients in its regular bimodule. For every $\lambda = (\lambda_p)_{p\ \mathrm{prime}} \in \prod_p \mathbb{C}$, the formula

$$D_\lambda(\mu_r e(\gamma)\mu_s^*) = \left(\sum_p (v_p(r) - v_p(s))\lambda_p\right)\mu_r e(\gamma)\mu_s^*$$

defines a derivation. There is an injective linear map

$$\prod_p \mathbb{C} \hookrightarrow HH^1_{\mathrm{alg}}(A_{\mathrm{alg}}, A_{\mathrm{alg}}), \qquad \lambda \longmapsto [D_\lambda].$$

No boundedness or summability condition on the family is required. Every nonzero D_λ is unbounded for the inherited C^*-norm and cannot be implemented by a bounded operator on $\mathcal{L}$. Moreover $HH^0_{\mathrm{alg}}(A_{\mathrm{alg}}, A_{\mathrm{alg}}) = \mathbb{C}$. This is a lower bound in degree one, not a complete Hochschild calculation.

Proof. Let $K = \prod_p U(1)$, and for $z \in K$ put $z(n) = \prod_p z_p^{v_p(n)}$. The assignments $\alpha_z(\mu_n) = z(n)\mu_n$ and $\alpha_z(e(\gamma)) = e(\gamma)$ preserve the relations and define an action by $*$-automorphisms. It is norm continuous in z on each finite normal sum, and hence on A by uniform approximation and isometry. A normal term has character $z(r/s)$. Haar integration against the inverse character projects onto its homogeneous component; distinct rational degrees therefore give

$$A_{\mathrm{alg}} = \bigoplus_{g \in \mathbb{Q}^\times_{>0}} (A_{\mathrm{alg}})_g, \qquad (A_{\mathrm{alg}})_g (A_{\mathrm{alg}})_h \subseteq (A_{\mathrm{alg}})_{gh}.$$

This does not assert independence of normal terms of the same degree. Unique factorization makes $\lambda(g) = \sum_p v_p(g)\lambda_p$ an additive character of the multiplicative rational group: $\lambda(gh) = \lambda(g) + \lambda(h)$. All these sums are finite. Define D_λ on a homogeneous element of degree g by multiplication by $\lambda(g)$. The direct-sum decomposition proves well-definedness, and the character identity proves the Leibniz rule. In particular it annihilates the phases and sends μ_n to $\lambda(n)\mu_n$ and μ_n^* to $-\lambda(n)\mu_n^*$.

Algebraic Hochschild one-cocycles are precisely derivations and one-coboundaries are inner derivations. If $D_\lambda = [b, -]$ for a bounded operator b, the isometry μ_p^k gives

$$k|\lambda_p| = \|D_\lambda(\mu_p^k)\| = \|[b, \mu_p^k]\| \le 2\|b\| \qquad (k \ge 1).$$

Hence every $\lambda_p = 0$. Applying this to differences proves injectivity on Hochschild classes. The same estimate on the norm-one sequence μ_p^k proves unboundedness whenever a prime value is nonzero, and excludes a bounded extension $A \to \mathcal{A}$.

An operator commuting with all phases has vanishing matrix entries between distinct e_m, e_n: some $\gamma \in \mathbb{Q}/\mathbb{Z}$ has different characters at m, n. It is therefore diagonal, say $Te_m = t_m e_m$. Commutation with μ_n gives $t_{nm} = t_m$; setting $m = 1$ makes all entries equal. Thus the represented commutant is scalar. Faithfulness implies $Z(A_{\mathrm{alg}}) = \mathbb{C}$, which is its Hochschild degree zero. □

The product $\prod_p \mathbb{C}$ parametrizes actual derivations, not convergent infinite operator sums: every input has finite homogeneous support. A derivation in this family preserves adjoint exactly when every λ_p is purely imaginary, as follows by comparing its values on μ_p and μ_p^*. The preceding algebraic result does not compute continuous Hochschild cohomology of the Banach algebra A.

Higher Gauge Cup Products

The degree-one classes also detect higher cohomology. To prove this, we use a scalar character to change coefficients and then evaluate cup products on explicit finite cycles. This coefficient change is part of the argument; it is not an identification with Hochschild homology with regular coefficients.

There is a character $\varepsilon : A \to \mathbb{C}$ defined by

$$\varepsilon(\mu_n) = \varepsilon(\mu_n^*) = \varepsilon(e(\gamma)) = 1.$$

These values respect the unit, involution, phase multiplication, semigroup, coprime-adjoint and isometry relations. The transfer relation becomes one equal to the average of n ones. The universal property therefore gives a unital $*$-homomorphism, of norm one. It is not the vacuum vector state: the latter takes μ_p to zero.

Let $\mathbb{C}_\varepsilon$ be the bimodule with $a \cdot c \cdot b = \varepsilon(a)c\varepsilon(b)$. Our algebraic cochains have differential

$$\begin{aligned}(bF)(a_1, \dots, a_{n+1}) = {} & a_1 F(a_2, \dots, a_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i F(a_1, \dots, a_i a_{i+1}, \dots, a_{n+1}) \\ & + (-1)^{n+1} F(a_1, \dots, a_n) a_{n+1}.\end{aligned}$$

Composition with ε is a cochain map from regular coefficients to $\mathbb{C}_\varepsilon$, by multiplicativity. For scalar-coefficient chains, written $c[a_1|\cdots|a_n]$, use

$$\begin{aligned}b(c[a_1|\cdots|a_n]) = {} & c\varepsilon(a_1)[a_2|\cdots|a_n] \\ & + \sum_{i=1}^{n-1} (-1)^i c[a_1|\cdots|a_i a_{i+1}|\cdots|a_n] \\ & + (-1)^n \varepsilon(a_n) c[a_1|\cdots|a_{n-1}].\end{aligned}$$

Evaluation makes these two differential formulas dual.

Lemma 29.7.2 *(finite detecting cycles).* For distinct primes $P = (p_1, \dots, p_n)$, put $s_j = \mu_{p_j}$. Then

$$c_P = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) 1[s_{\sigma(1)}|\cdots|s_{\sigma(n)}]$$

is a cycle with coefficients in $\mathbb{C}_\varepsilon$. Consequently the functional on regular-coefficient cochains

$$L_P(F) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \varepsilon\big(F(s_{\sigma(1)}, \dots, s_{\sigma(n)})\big)$$

annihilates Hochschild coboundaries.

Proof. For each interior face, pair a permutation with the one exchanging the adjacent multiplied factors. Their signs are opposite and the boundary terms agree, since the s_j commute. Pair the first face of $(j_1, \dots, j_n)$ with the last face of its cyclic rotation $(j_2, \dots, j_n, j_1)$. Rotation has sign $(-1)^{n-1}$, whereas the last boundary sign is $(-1)^n$; the terms cancel because $\varepsilon(s_j) = 1$. For $n = 1$, the boundary is $\varepsilon(s_1) - \varepsilon(s_1) = 0$. The cochain map and the dual differential formulas now show that evaluation on c_P vanishes on every regular-coefficient coboundary. □

THEOREM 29.7.3 *(an algebraic exterior-algebra injection).* Let $W = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Q}^{\times}_{>0}, (\mathbb{C}, +)) = \prod_p \mathbb{C}$. The gauge derivations give an injective graded-algebra map

$$\Lambda^*_{\mathbb{C}} W \hookrightarrow HH^*_{\mathrm{alg}}(A_{\mathrm{alg}}, A_{\mathrm{alg}}), \qquad \lambda \longmapsto [D_\lambda].$$

The exterior algebra here is algebraic: its elements are finite sums of finite wedges, without completion. In particular the Hochschild cohomology is nonzero in every nonnegative degree.

Proof. The cup of n derivations is

$$F_{\lambda_1,\dots,\lambda_n}(a_1, \dots, a_n) = D_{\lambda_1}(a_1) \cdots D_{\lambda_n}(a_n).$$

Substitution into the differential gives $b(F \smile G) = bF \smile G + (-1)^{\deg F} F \smile bG$: the interior terms cancel except the two indicated coboundaries. Each derivation is a cocycle, so their cup is a cocycle. Using $D_{\lambda_i}(s_j) = \lambda_i(p_j) s_j$ gives

$$L_P(F_{\lambda_1,\dots,\lambda_n}) = \det\big(\lambda_i(p_j)\big)_{1 \le i,j \le n}.$$

For two derivations D, E, their composition is a one-cochain. Expanding both Leibniz rules yields

$$b(DE)(a, b) = -D(a)E(b) - E(a)D(b).$$

Thus their cohomology classes anticommute, and each square vanishes in characteristic zero. This defines the exterior-algebra map.

An exterior element involves a finite-dimensional subspace $U \subset W$. Prime-coordinate functionals separate its points; their span is U^*, since otherwise their common annihilator would contain a nonzero element with all prime coordinates zero. A finite subset therefore spans U^*, and wedges of its coordinates span $(\Lambda^n U)^*$. Every nonzero degree-n exterior element pairs nontrivially with some wedge of distinct prime coordinates. The corresponding determinant functional L_P detects its image in Hochschild cohomology. Degree zero is detected by $\varepsilon(1) = 1$. □

Example 29.7.4 *(degrees two and three).* For distinct primes p, q, the chain $1[\mu_p|\mu_q] - 1[\mu_q|\mu_p]$ has boundary

$$1[\mu_q] - 1[\mu_p\mu_q] + 1[\mu_p] - 1[\mu_p] + 1[\mu_q\mu_p] - 1[\mu_q] = 0.$$

Here D_p denotes the family with value one at p and zero elsewhere. The evaluation $L_{(p,q)}(D_p \smile D_q) = 1$ proves this cup is not a coboundary. For three distinct primes, among the six terms defining $L_{(p,q,r)}(D_p \smile D_q \smile D_r)$, only the identity assignment is nonzero, and its value is one. These are explicit higher classes, not a classification of all Hochschild classes.

The time evolution is $\sigma_t(\mu_n) = n^{it}\mu_n$, fixing the phases. Its norm-generator domain consists of those $a \in A$ for which $\lim_{t\to 0}(\sigma_t(a) - a)/t$ exists in norm; call that generator δ, and write A^∞_σ for its norm-smooth orbit algebra. Its closedness and smooth topology are proved below.

Finite-Prime and All-Prime Gauge Smoothness

For a finite prime set F, let $\mathbb{T}^F \subset K$ act by setting the other coordinates to one. Define

$$B_F = \{a \in A : z \mapsto \alpha_z(a) \text{ is norm-smooth on } \mathbb{T}^F\}.$$

For angular differentiation in the p-coordinate write δ_p; thus $\delta_p(\mu_q) = i\delta_{pq}\mu_q$ and $D_p = \delta_p/i$. For nonnegative multiindices α supported in F, put $q_\alpha(a) = \|\delta^\alpha a\|$, including $q_0(a) = \|a\|$. Define also

$$B_{\mathrm{g}} = \bigcap_{F \text{ finite}} B_F$$

with all finite-support multiindex seminorms. This definition uses ordinary smoothness on each finite-dimensional torus; it makes no unspecified infinite-dimensional differentiability assumption.

Proposition 29.7.5 *(smooth gauge algebras and their cup classes).* The algebras B_F and B_{g} are Fréchet $*$-algebras, dense in A for its C^*-norm. The derivations D_p, for $p \in F$ on B_F and all primes on B_{g}, are continuous. With jointly continuous complex multilinear cochains and regular coefficients, there are injections

$$\Lambda^*_{\mathbb{C}} \operatorname{span}_{\mathbb{C}}\{D_p : p \in F\} \hookrightarrow HH^*_{\mathrm{cont}}(B_F, B_F),$$

$$\Lambda^*_{\mathbb{C}}\left(\bigoplus_p \mathbb{C}D_p\right) \hookrightarrow HH^*_{\mathrm{cont}}(B_{\mathrm{g}}, B_{\mathrm{g}}).$$

Proof. If a_j is Cauchy for every q_α on B_F, its orbit maps $f_j(z) = \alpha_z(a_j)$ are Cauchy in every uniform derivative norm, since $\partial^\alpha f_j(z) = \alpha_z(\partial^\alpha a_j)$ and the action is isometric. The space of smooth Banach-valued functions on the torus is complete: the fundamental theorem of calculus in coordinate charts shows that uniform limits of functions and their first derivatives retain those derivatives, and iteration gives the assertion at every order. The limit f has $f(z) = \alpha_z(a)$, where $a = \lim_j a_j$ in norm, by continuity of each automorphism. Thus $a \in B_F$ and all seminorms converge. The iterated product rule

$$\partial^\alpha(ab) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} (\partial^\beta a)(\partial^{\alpha-\beta} b)$$

proves continuous multiplication. Adjoint commutes with real angular derivatives and is continuous; mixed derivatives commute by smoothness.

There are countably many finite prime sets and finite-support multiindices. A sequence Cauchy in all seminorms of B_{g} converges in every B_F; the limits coincide with its norm limit in A. This proves completeness of B_{g}, and the same product rule proves its Fréchet algebra structure. Finite normal sums belong to every B_F, giving norm density. For p and finite F, differentiate the orbit for $\mathbb{T}^{F\cup\{p\}}$ and restrict to $\mathbb{T}^F$. This shows that δ_p preserves B_{g}, with $q_\alpha(\delta_p a) = q_{\alpha+e_p}(a)$; the assertion on B_F is identical for $p \in F$.

The finite cup cochains are continuous by these estimates and continuity of multiplication. The character is continuous because $|\varepsilon(a)| \le q_0(a)$. The finite cycles of Lemma 29.7.2 therefore annihilate every continuous coboundary as well. Their determinant evaluations prove the two injections. No completed chain-complex comparison is needed. □

Proposition 29.7.6 *(which algebraic gauge derivations extend continuously).* An algebraic gauge derivation D_λ extends to a continuous linear map $B_{\mathrm{g}} \to A$ exactly when λ has finite support. In that case it extends as a continuous derivation $B_{\mathrm{g}} \to B_{\mathrm{g}}$. For B_F, the analogous criterion is that λ be supported in F.

Proof. Continuity of an extension $T : B_{\mathrm{g}} \to A$ gives

$$\|T(a)\| \le C \max_{\alpha \in E} q_\alpha(a)$$

for a finite set E of multiindices, enlarged to include zero. Let F contain their supports. If a prime $q \notin F$, then $q_0(\mu_q^k) = 1$, whereas every positive derivative in E is zero. Thus $k|\lambda_q| \le C$ for every k, forcing $\lambda_q = 0$. Conversely a finite linear combination of the continuous D_p is continuous into B_{g}. The same argument on B_F proves the second criterion. These criteria concern this family of algebraic derivations; they do not classify all continuous derivations. □

Example 29.7.7 *(gauge smoothness does not imply time differentiability).* Enumerate distinct primes p_j, and put

$$a = \sum_{j \ge 1} 2^{-j} \mu_{p_j}^{2^j}.$$

The series converges in norm. For each finite F, only finitely many summands move under $\mathbb{T}^F$, while the remaining norm-convergent sum is fixed. Hence $a \in B_{\mathrm{g}}$. If its time derivative existed in norm, its value on e_1 would have coordinate $i \log p_j$ at each distinct state $e_{p_j^{2^j}}$. These coordinates are not square summable. Thus $a \notin \mathrm{Dom}(\delta)$, and B_{g} is not contained in A_σ^∞. The time family $\lambda_p = i \log p$ also fails the finite-support continuity criterion. Gauge-smooth and time-smooth Hochschild complexes are therefore different specified constructions.

PROPOSITION 29.7.8 *(the closed time generator and its proper domain).* For $\sigma_t(\mu_n) = n^{it}\mu_n$ and $\sigma_t(e(\gamma)) = e(\gamma)$, let

$$\mathrm{Dom}(\delta) = \left\{ a \in A : \lim_{t \to 0} \frac{\sigma_t(a) - a}{t} \text{ exists in } A \right\}.$$

The norm generator δ is closed and densely defined, its domain is proper, and its restriction to A_{alg} is $D_{(i \log p)_p}$.

Proof. The gauge action makes σ pointwise norm continuous. Every normal term is entire analytic, with derivative $i \log(r/s)\mu_r e(\gamma)\mu_s^*$; hence the domain contains the dense normal algebra. Fix a prime p. The series

$$a = \sum_{j \ge 1} 2^{-j} \mu_p^{2^j}$$

converges in norm because the summand norms sum to one. Applied to e_1, its orbit difference quotient has coordinate

$$2^{-j} \frac{\exp(it2^j \log p) - 1}{t} \longrightarrow i \log p$$

at $e_{p^{2^j}}$. If the quotient converged in operator norm, its value on e_1 would converge to an ℓ^2 vector having this same nonzero coordinate at infinitely many orthonormal states. This is impossible, so $a \notin \mathrm{Dom}(\delta)$.

For a domain element, differentiation along the norm-continuous orbit gives the Bochner integral identity

$$\sigma_t(a) - a = \int_0^t \sigma_s(\delta a)\, ds.$$

If $a_n \to a$ and $\delta a_n \to b$ in norm, isometry permits passage to the limit in this identity. Dividing by t and using norm continuity of $\sigma_s(b)$ gives $\delta a = b$. This proves closedness. □

Proposition 29.7.9 *(a smooth Hochschild class).* The algebra of norm-smooth elements is

$$A_\sigma^\infty = \bigcap_{j\ge 0} \mathrm{Dom}(\delta^j), \qquad p_k(a) = \sum_{j=0}^{k} \|\delta^j a\|.$$

With these seminorms it is a Fréchet algebra, dense in A. The derivation $\delta : A_\sigma^\infty \to A_\sigma^\infty$ is continuous and represents a nonzero class in continuous Hochschild degree one with coefficients in this same smooth algebra.

Proof. Repeated differentiation of the orbit identifies smoothness with membership in every generator domain. If a sequence is Cauchy for every p_k, each $\delta^j a_n$ has a norm limit. Closedness of δ, applied successively, identifies these limits as $\delta^j a$. Thus the seminorm topology is complete. The iterated Leibniz rule

$$\delta^j(ab) = \sum_{i=0}^{j} \binom{j}{i} (\delta^i a)(\delta^{j-i} b)$$

proves closure under multiplication and its continuity; the generator also commutes with adjoint. The normal algebra consists of smooth elements and is norm dense in A. Finally $p_k(\delta a) \le p_{k+1}(a)$ proves continuity. An inner implementer in this smooth algebra would be a bounded element of A, contrary to the power estimate in Theorem 29.7.1 with $\lambda_p = i \log p$. □

Proposition 29.7.10 *(the occupation Hamiltonian and represented closure).* On $\mathcal{L}$ let $He_m = (\log m) e_m$, with maximal domain

$$\mathrm{Dom}(H) = \left\{ \xi \in \mathcal{L} : \sum_m (\log m)^2 |\xi_m|^2 < \infty \right\}.$$

This diagonal operator is self-adjoint, with finite-support core. Normal terms preserve its domain, and there

$$[H, \mu_r e(\gamma) \mu_s^*] = \log(r/s)\, \mu_r e(\gamma) \mu_s^*.$$

The unitaries e^{itH} implement σ_t, and the represented von Neumann closure of A is $B(\mathcal{L})$.

Proof. For a real diagonal multiplier, testing the adjoint against basis vectors gives precisely the stated square-summability domain; thus the maximal operator equals its adjoint. Truncating a vector in both the norm and weighted norm proves the core assertion. A normal term vanishes unless $s \mid m$, and otherwise sends e_m to a phase times $e_{rm/s}$. The logarithmic weights change by the constant $\log(r/s)$, which proves domain preservation and the commutator identity. Equivalently, prove the identity on the core and pass to graph limits by closedness. Exponentiating the diagonal weights gives the asserted implementation. The scalar commutant computed in Theorem 29.7.1 and the bicommutant theorem give the represented closure $B(\mathcal{L})$. □

Thus $\delta = i[H, -]$ on A_{alg} has an unbounded spatial implementer. The prime-power norm estimate in Theorem 29.7.1 excludes every bounded implementer, so δ is outer on A_{alg} and A_σ^∞. This occupation closure is a type-I-infinity factor. A factor obtained from another state by a KMS–GNS construction is a different carrier. Its type identifies none of the algebraic, Banach or Fréchet Hochschild complexes above. The algebraic and all-prime gauge-smooth lower bounds above are nonzero in every degree. They do not classify the full cohomology on these carriers or compute the other Hochschild theories.

Question 29.7.11 *(the arithmetic Hochschild comparison).* Construct the arithmetic category or algebra whose Hochschild invariants are to realize $\mathcal{V}^{\mathrm{prim}}$, specifying its coefficient ring, bimodule or categorical trace, topology, grading and completion. Construct the comparison with the Bost–Connes normal algebra or a specified smooth algebra and the cycle pairing of Remark 29.3.2. Determine which gauge classes survive under that comparison and whether an integral degree-one carrier generated by logarithmic prime data exists. Any proposed degree-two relation to a Witt cotangent complex requires its actual map and coefficient comparison. A trace class alone defines no Hochschild cochain algebra, and the complex injection above establishes neither an integral lattice nor concentration of the arithmetic theory in degrees zero through two.

29.8 Vertex-Algebraic Invariants of the Heisenberg Covers

THEOREM 29.8.1 *(universal Heisenberg-cover invariants).* The finite Heisenberg cover $\mathcal{H}^{\otimes\pi(N)}$ of $\mathcal{V}_N^{\mathrm{prim}}$ is a rank-$\pi(N)$ Heisenberg vertex algebra. It lies in the Heisenberg archetype G of the Vol I landscape census, with cover central charge $\kappa = \pi(N)$, depth $r_{\max} = 2$, and chiral Hochschild cohomology concentrated in degrees $\{0, 1, 2\}$. These are invariants of the cover, not of the trace class.

COROLLARY 29.8.2 *(placement of the finite covers).* The finite Heisenberg covers of $\mathcal{V}^{\mathrm{prim}}$ lie in archetype G (Heisenberg / Gaussian), with $\kappa_N = \pi(N)$. The Hilbert comparison with the Hochschild trace class requires the map, pairing and completion of Remark 29.3.2.

29.9 The Fock representation and the trace comparison

COROLLARY 29.9.1 *(Finite current Fock representation).* The finite current Fock space has the Hilbert tensor decomposition of Proposition 29.4.1. Its arithmetic shifts and phases act on the mode-one factor and extend by the identity on the higher-mode factor.

Proof. Apply Proposition 29.4.1 to the given finite set of primes. □

Remark 29.9.2 *(Scope of the module structure).* The tensor decomposition is a statement about current Fock spaces. It induces no tensor decomposition of a Hochschild trace class without a compatible comparison. The higher-mode factor alone is not the full vacuum Heisenberg vertex algebra: it omits all mode-one creations. Nor does the finite-prime restriction assert faithfulness for all arithmetic phases. If $q \notin F$, the nonzero Bost–Connes projection $e_q = q^{-1}\sum_{j=0}^{q-1} e(j/q)$ acts by zero on L_F, since q divides no element of $\mathbb{N}_F$.

Remark 29.9.3 *(the* B*-row threshold).* A proposed arithmetic chiral algebra should encode Hecke eigenvalues in its stress tensor. Comparing its chiral Hochschild cohomology with Bost–Connes cohomology requires a specified algebraic or smooth coefficient algebra and an actual subquotient map. The gauge classes above do not construct this map. Placement in the Borcherds/BKM B-row further requires verification of the hypotheses of the Vol III classification theorem (Theorem C, BKM column), whose corresponding value is $K^\kappa = 8$. The proposed passage from the Heisenberg-cover G-row uses a Borcherds-product encoding of Hecke data in the stress tensor, with the target relation $\Phi_N = c_N(0)/2$. Prime indexing alone supplies neither that encoding nor the Hochschild comparison.

29.10 Structural Conclusions

(i) Bost–Connes $C_{\mathbb{Q}}^*$ carries $\zeta(\beta)$ as Gibbs partition function for $\beta > 1$ (Theorem 29.1.2).

(ii) The oscillator operators and bounded arithmetic shifts act on the same prime occupation space. The shift is normalized creation, as in Proposition 29.2.2.

(iii) The arithmetic phases and shifts give the faithful representation of Proposition 29.3.1. Its identification with a Hochschild trace class requires the map, pairing and completion of Remark 29.3.2.

(iv) The current Fock space splits into mode-one and higher-mode factors. The arithmetic representation extends over the latter, with the stable sectors of Corollary 29.4.2.

(v) The Kesten–McKay measure describes the spherical $GL_2(\mathbb{Q}_p)$ Hecke spectrum. Relating it to the GL_1 Bost–Connes system or the Jordan-quiver CoHA requires the comparison maps described in Corollary 29.5.2 and Proposition 29.6.2.

(vi) The normal-term algebra has the independent gauge classes of Theorem 29.7.1. The time derivative requires its specified algebraic or smooth domain; the algebraic and all-prime gauge-smooth cup products are nonzero in every degree, without a full Hochschild classification.

(vii) The finite Heisenberg covers lie in archetype G with $\kappa = \pi(N)$, $r_{\max} = 2$; the B-row threshold requires Borcherds-product encoding of Hecke data (Corollary 29.8.2).

The vertex algebra structure is universal and non-arithmetic because it belongs only to the finite Heisenberg covers. The Bost–Connes representation gives the arithmetic shifts, phases and Gibbs partition function. Its relation to the Hochschild trace class requires the comparison of Remark 29.3.2. A genuine VA-arithmetic object encoding Hecke eigenvalues in the stress tensor would be a new object, not $\mathcal{V}^{\mathrm{prim}}$.

Chapter 30

Adelic Factorization Axioms

An adelic factorization construction must specify both its local coefficients and the spaces on which positions collide. The set of places supplies an index for the coefficients. A holomorphic curve, a formal disc or a derived arithmetic intersection supplies different position data. The elementary distinctions of Section 2.8 are the starting point.

30.1 The Ran Space of $\overline{\operatorname{Spec}\mathbb{Z}}$

Observation 30.1.1 *(naive Ran is terminal).* The functor of nonempty finite subsets of $X(S)$, applied to $X = \operatorname{Spec}\mathbb{Z}$ and arbitrary test schemes S, is terminal. Indeed,

$$S \mapsto \operatorname{Spec}\mathbb{Z}(S) = \operatorname{Hom}_{\mathrm{Ring}}(\mathbb{Z}, \Gamma(S, \mathcal{O}_S)) = \{*\}$$

(singleton). Its only nonempty finite subset is itself. The maps $\operatorname{Spec}\mathbb{F}_p \to \operatorname{Spec}\mathbb{Z}$ are points for different test schemes. They do not give several sections over a fixed S.

Definition 30.1.2 *(Discrete configurations of places).* Let $V = \mathbb{P} \sqcup \{\infty\}$, where $\mathbb{P}$ is the set of rational primes. The discrete place configurations are $\mathrm{Fin}_+(V)$. When multiplicities are retained, use $\mathrm{Mult}_+(V) = \coprod_{n\geq 1} V^n/\mathfrak{S}_n$. These are combinatorial index sets. They specify neither a scheme structure on an Arakelov compactification nor a normal neighbourhood of a position diagonal. We write $\overline{C}$ for the arithmetic curve only when its additional geometric data have been specified.

Remark 30.1.3. An order on V, or the multiplicity of a place in a multiset, does not produce a normal coordinate. Such coordinates belong to a separately chosen local position space. The cut construction of Chapter 2 uses a smooth curve and its differential modules; its hypotheses are additional to the choice of V.

30.2 Factorization for Disjoint Place Labels

Observation 30.2.1. Choose a common commutative coefficient ring A and complexes $H_v \in \mathcal{D}(A)$. Set $F(S) = \bigotimes^{\mathbf{L}}_{v\in S, A} H_v$ for a finite set of places. For disjoint sets S, T, the tensor associator gives $F(S \sqcup T) \simeq F(S) \otimes^{\mathbf{L}}_A F(T)$. Its coherences follow from those of derived tensor, as in (2.8).

Corollary 30.2.2 *(Scope of discrete factorization).* The preceding construction gives a coherent factorization for disjoint finite subsets of V. It requires no maps at coincident positions. Identifying it with a geometric arithmetic factorization system requires the position spaces, their collision coefficients and comparison maps in addition to the disjoint tensor identities.

Proof. Every map used in the construction is an associator or symmetry of the finite coefficient tensor. Its source and target are indexed by disjoint sets. No diagonal coefficient occurs among these maps. □

30.3 The Diagonal and $\mathbb{Z}_p \widehat{\otimes} \mathbb{Z}_p$

Observation 30.3.1. For the ordinary p-adic completion of the indicated tensor,

$$\mathbb{Z}_p \widehat{\otimes}_{\mathbb{Z}} \mathbb{Z}_p \;\cong\; \mathbb{Z}_p,$$

where the left side means $\varprojlim_r (\mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{Z}_p)/p^r$. Multiplication is an isomorphism modulo p^r for every r, since both factors then give $\mathbb{Z}/p^r$. The inverse limit proves the assertion. This computation concerns the coefficient base, with no position variable.

Remark 30.3.2. A chosen formal position disc has self-product $\mathbb{Z}_p[[t_1, t_2]]$ and nonzero diagonal ideal $(t_1 - t_2)$. The derived closed-point intersection instead has the normal class in $\mathbb{F}_p \otimes^{\mathbf{L}}_{\mathbb{Z}} \mathbb{F}_p \simeq \mathbb{F}_p \oplus \mathbb{F}_p[1]$. These are distinct coefficient models, as proved in Proposition 2.8.1. Crystalline or prismatic data can be added when a construction requires their Frobenius structure; the completed base tensor does not force that choice.

THEOREM 30.3.3 *(Coefficient completion and position diagonals).* The diagonal of a complex position line has normal coordinate $z_1 - z_2$. A formal position disc over $\mathbb{Z}_p$ has normal coordinate $t_1 - t_2$. Both ideals are nonzero, although the ordinary completed base tensor of Observation 30.3.1 is $\mathbb{Z}_p$.

Proof. In both position rings, change variables to the second coordinate and the difference of the two coordinates. Restriction to the diagonal evaluates the difference at zero and has the displayed nonzero kernel. The base-tensor assertion is the observation above. □

30.4 Pointed Data at Each Finite Place

Remark 30.4.1 *(pointed local data and factorization).* An E_0-algebra in vector spaces is a pointed vector space. The vacuum therefore equips each polynomial oscillator module with such an underlying datum. This does not construct a factorization algebra on Spf $\mathbb{Z}_p$. Classification results for locally constant factorization on manifolds require their geometric hypotheses; a formal scheme is not made a zero-dimensional manifold by this notation.

Definition 30.4.2 *(minimal adelic factorization structure).* The proposed adelic factorization datum consists of:

- At ∞: genuine holomorphic factorization algebra on $\mathbb{C}_\infty$ in the CG sense (Vol. I Def. 3.1.4), with Heisenberg OPE $j(z)\,j(w) \sim k_\infty/(z-w)^2$, $k_\infty = 1$.
- At each finite p: locally constant prestack on Spf $\mathbb{Z}_p$ with constant stalk $\mathcal{H}^{(p)}$ and a specified Adams endomorphism ψ_p. Its underlying pointed module is an E_0-datum; its geometric structure maps are additional requirements.
- Gluing data: restricted tensor product over spherical vectors $|0\rangle_p$ and explicit local-to-global comparison maps. An automorphic representation is further data, not a consequence of the tensor notation.

30.5 The Holomorphic-Topological Structure

Remark 30.5.1 *(Costello HT factorization, arXiv:1303.2632; Costello 2020).* Costello's holomorphic-topological factorization algebras live on $\mathbb{C}^{n_H} \times \mathbb{R}^{n_T}$ with n_H holomorphic and n_T topological directions. The axioms blend holomorphic OPE (on the $\mathbb{C}^{n_H}$ factor) and locally constant (E_{n_T}-algebra structure on the $\mathbb{R}^{n_T}$ factor).

Conjecture 30.5.2 *(adelic HT structure). Conjectural statement.* On $\overline{C}$ the natural HT structure is

$$\underbrace{\mathbb{C}_\infty}_{\text{holomorphic}} \times \underbrace{\operatorname{Spf} \widehat{\mathbb{Z}}}_{\text{adelic / } E_0 \text{ at each place}},$$

with the profinite completion $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ totally disconnected. Locally constant factorization on Spf $\widehat{\mathbb{Z}}$ decomposes as a product of E_0-stalks. The non-trivial chiral content concentrates at ∞.

Definition 30.5.3 *(relative factorization window with adelic base).* A relative factorization window for $\mathcal{V}^{\mathrm{prim}}$, when supplied, is a holomorphic factorization algebra on $\mathbb{C}_\infty$ *over the base* Spec $\widehat{\mathbb{Z}}$, with the base acting by Adams operations $\{\psi_p\}_p$ and Hecke operators $\{T_p\}_p$:

$$\mathcal{F}_{\mathrm{CG}}(\mathcal{V}^{\mathrm{prim}}) \in \mathrm{FactAlg}_{\mathbb{C}_\infty}\big(\mathrm{Mod}_{\widehat{\mathbb{Z}},\psi}\big),$$

where the coefficient category and its scalar comparison at infinity must be specified. The symbol $\mathrm{Mod}_{\widehat{\mathbb{Z}},\psi}$ records modules with a named compatible commuting action; it does not identify divisor substitutions with Adams maps. The full divisor family fails that commutation requirement by Equation 28.6.

30.6 Five Axioms (F1)–(F5) for the Minimal Adelic Factorization

Definition 30.6.1 *(Costello–Gwilliam arithmetic factorization).* The following are requirements on an arithmetic factorization datum, not consequences of the polynomial oscillator representation:

(F1) Archimedean factorization. Holomorphic factorization algebra on $\mathbb{C}_\infty$ in the sense of CG Vol. I Def. 3.1.4, with OPE $j(z)\,j(w) \sim k_\infty/(z-w)^2$, $k_\infty = 1$.

(F2) Coefficient category. Values in $\mathrm{Mod}_{\widehat{\mathbb{Z}},\psi}$: $\widehat{\mathbb{Z}}$-modules with commuting Adams operations $\psi_p \colon \mathcal{M} \to \mathcal{M}$ satisfying $\psi_p\psi_q = \psi_q\psi_p$ and $\psi_p|_{\mathbb{Z}_p} = \mathrm{id}$.

(F3) Hecke-eigen decomposition. Global sections decompose as a restricted tensor product

$$\mathcal{V}^{\mathrm{prim}}(\mathbb{C}_\infty) = \bigotimes_p{}' \mathcal{H}^{(p)}$$

with specified reference vectors $|0\rangle_p$ and separately specified intertwiners for its Adams and Hecke operators.

(F4) Archimedean–finite compatibility. The archimedean Heisenberg level is $k_\infty = 1$, hence $v_p(k_\infty) = 0$ for every finite prime. The finite arithmetic datum is carried by the Satake eigenvalue a_p, not by a p-adic valuation of the archimedean level.

(F5) Regular p-adic product. At each finite place p, the local factor $\mathcal{H}^{(p)}$ carries regular (non-singular) commutative multiplication and Adams endomorphism ψ_p among its specified operations. The product $\mathcal{H}^{(p)} \otimes \mathcal{H}^{(p)} \to \mathcal{H}^{(p)}$ is the commutative multiplication in the Fock module: a commutative algebra structure, distinct from a BD factorization-algebra fusion.

Remark 30.6.2 *(scope of the five conditions).* Conditions (F1)–(F5) specify proposed local objects, coefficient actions and comparisons. The commutative product and index-shift action on the finite polynomial carrier satisfy the algebraic part of (F5). The geometric structure maps in (F1), the coefficient comparison in (F2), and the global tensor and operator identifications in (F3) require separate constructions. No arithmetic factorization existence theorem follows from this local polynomial calculation.

30.7 Classical Factorization at Finite Places: Absence of a BV Action Functional

Remark 30.7.1 *(BV quantization requires action functional).* CG Vol. I §4 constructs a factorization algebra of observables as a BV-quantization of a classical field theory $(\mathcal{E}, S)$. For the free boson on $C/\mathbb{C}$: $\mathcal{E} = \Gamma(C, \Omega^1_C)[-1]$, action $S[\phi] = \int_C \phi \wedge \bar{\partial}\phi$, the quantum factorization algebra is the Heisenberg factorization algebra (CG Vol. I Thm 4.6.1; as a translation-equivariant factorization algebra on $\mathbb{C}$, this recovers the Heisenberg vertex algebra, but the two structures are distinct: factorization algebra ≠ vertex algebra).

Observation 30.7.2 *(arithmetic differential forms and the free-field obstruction).* The sheaf of Kähler differentials $\Omega^1_{\overline{\mathbb{Z}}/\mathbb{Z}}$ on $\overline{\operatorname{Spec}\,\overline{\mathbb{Z}}}$ is zero in the ordinary algebraic sense; Arakelov geometry replaces it with a metrised line bundle at the archimedean place. A classical free-field action at a finite place p would require a section of $\Omega^1_{\overline{C}/\mathbb{F}_1}$ in a neighbourhood of p; the requisite $\mathbb{F}_1$-geometry remains unconstructed.

Remark 30.7.3. Borger's Frobenius-lift criterion has the torsion-free scope stated in Theorem 28.4.2. The monomial and Hall integral lattices carry different reductions. The shifts ψ_n preserve both, but are Frobenius lifts only on the latter. The divisor maps T_n also preserve both lattices and fail that condition even on the Hall lattice. These operations supply no action functional, BV bracket or geometric factorization map.

THEOREM 30.7.4 *(finite polynomial Adams datum).* For each finite arithmetic place p, let $P_{\mathbb{Z}_p} = \mathbb{Z}_p[x_k : k \geq 1]$. Its ordinary product, vacuum 1, and substitutions $\psi_n(x_k) = x_{nk}$ give a unital commutative algebra with an $\mathbb{N}^\times$-action. The prescribed ψ_p is not a Frobenius lift on this monomial lattice. An action functional or factorization algebra on $\operatorname{Spf}\,\mathbb{Z}_p$ is additional data.

Proof. The polynomial product and substitutions are defined over $\mathbb{Z}_p$, and $\psi_m\psi_n(x_k) = x_{mnk}$ proves the monoid law on generators. The two distinct monomials in $\psi_p(x_1) - x_1^p = x_p - x_1^p$ remain distinct modulo p. This proves the claimed failure of this particular Frobenius lift; it proves no impossibility of other integral structures or field theories. □

30.8 Factorization Boundary

(i) The finite-subset functor of $\operatorname{Spec}\,\mathbb{Z}(S)$ is terminal. Finite sets and multisets of places are separate discrete index sets (Observation 30.1.1, Definition 30.1.2).

(ii) Finite coefficient tensors give coherent factorization for disjoint place labels. Their geometric collision maps require additional data (Corollary 30.2.2).

(iii) The completed tensor product $\mathbb{Z}_p \mathbin{\widehat{\otimes}}_{\mathbb{Z}} \mathbb{Z}_p \cong \mathbb{Z}_p$ makes the completed base diagonal an isomorphism. Formal position discs and derived closed-point intersections retain normal data (Observation 30.3.1, Theorem 30.3.3).

(iv) At each finite place the vacuum gives a pointed module. A factorization structure on a formal scheme is additional geometry (Remark 30.4.1).

(v) The proposed relative datum includes a holomorphic factorization algebra on $\mathbb{C}_\infty$ with $\widehat{\mathbb{Z}}$-module coefficients and commuting Adams operations, as in (F1)–(F5) (Definition 30.5.3, Definition 30.6.1).

(vi) The monomial finite-place carrier has an Adams action whose prime shift fails the Frobenius congruence. BV quantization requires a separately supplied classical action functional (Theorem 30.7.4).

A Costello–Gwilliam window, if constructed with the stated data, is a holomorphic factorization algebra on $\mathbb{C}_\infty$ with an adelic local system decoration (axioms (F1)–(F5)); the full Hochschild trace class itself is not thereby made a vertex algebra.

Chapter 31

The Arithmetic Instanton Partition Function

Nekrasov's partition function has two equivariant parameters and a Coulomb modulus. The Hartnoll–Yang $L(s)$ sees only the degenerate ($\epsilon_2 \to \infty, b \to 0$) arithmetic limit; the full Ω-background is absent.

31.1 Nekrasov's Z_{inst} on $\mathbb{C}^2$

Theorem 31.1.1 *(Nekrasov, hep-th/0206161 Thm 2; Nekrasov–Okounkov hep-th/0306238 §3).* . For $U(N)$ gauge theory on $\mathbb{C}^2$ with equivariant torus action by (t_1, t_2) on $(z_1, z_2) \in \mathbb{C}^2$, $\epsilon_i = \log t_i$:

$$Z_{\text{inst}}^{U(N)}(\vec{a}, \epsilon_1, \epsilon_2; \Lambda) = \sum_{\vec{\lambda}} \Lambda^{N|\vec{\lambda}|} \prod_{\alpha,\beta=1}^{N} \prod_{s \in \lambda^{(\alpha)}} \frac{1}{E_{\alpha\beta}(s)(\epsilon_1 + \epsilon_2 - E_{\alpha\beta}(s))}$$

with $E_{\alpha\beta}(s) = a_\alpha - a_\beta - \epsilon_1 L_{\lambda^{(\beta)}}(s) + \epsilon_2(A_{\lambda^{(\alpha)}}(s) + 1)$, where $A_\lambda(s)$ (resp. $L_\lambda(s)$) denotes the arm length (number of boxes strictly to the right of s in λ) and leg length (number of boxes strictly below s in λ). This is the Nekrasov–Okounkov arm-on-α, leg-on-β convention.

31.2 The Arithmetic Curve as a One-Dimensional Object: No Instanton Sector

Observation 31.2.1. $\mathbb{C}^2$ is real 4-dimensional with a T^2 torus action. The Arakelov-compactified arithmetic curve $\overline{C} = \overline{\operatorname{Spec} \mathbb{Z}}$ has arithmetic Krull dimension 1; arithmetic Chow groups $\widehat{\mathrm{CH}}^*(\overline{C})$ live only in degrees 0, 1. There is no $\widehat{\mathrm{CH}}^2$; no instanton class.

Corollary 31.2.2. . A $U(N)$-gauge theory on the arithmetic 1-dimensional $\overline{C}$ is 2d Yang–Mills in character with no instanton sector. The primon partition function $L(s)$ is the degenerate Nekrasov limit on the self-product $\overline{C} \times_{\mathbb{F}_1} \overline{C}$, not a direct $Z_{\text{inst}}^{U(2)}$ on $\overline{C}$.

31.3 The Self-Product $\overline{C} \times_{\mathbb{F}_1} \overline{C}$ as Arithmetic Surface

Principle 31.3.1 *(Deligne).* Arakelov-theoretic constructions on Spec $\mathbb{Z}$ become two-dimensional on the self-product. Durov's arXiv:0704.2030 and Connes–Consani arXiv:1405.4527 and arXiv:1502.05580 supply the $\mathbb{F}_1$ base over which $\overline{C} \times_{\mathbb{F}_1} \overline{C}$ is a surface.

Definition 31.3.2 *(arithmetic surface setup).* On the self-product $\overline{C} \times_{\mathbb{F}_1} \overline{C}$:

- Torus action: $\mathbb{G}_m(\mathbb{F}_1)^2 = \{1\}^2$ literally, or $(\mathbb{Z}_{>0}, \times)^2$ in Connes–Consani semigroup form.
- Fixed points: pairs (p, q) of primes; or single primes p on the diagonal.
- Equivariant weights: $\epsilon_p = \log p$ per prime.

31.4 RANK ANALYSIS: $U(1)^\infty$ PER PRIME

Observation 31.4.1. The primon construction $\mathcal{V}^{\mathrm{prim}}$ carries Coulomb-branch moduli $\{a_p\}_p$ indexed by primes. The gauge rank is $|\mathbb{P}| = \aleph_0$, and $L(s)$ is the arithmetic Nekrasov partition function of a $U(1)^{\mathbb{P}}$ (abelian, rank-$\aleph_0$) theory with one Coulomb factor per prime.

Definition 31.4.2 *(abelian arithmetic Nekrasov per prime).*

$$Z^{U(1)}_{\mathrm{inst},p}(a_p, \epsilon_1^{(p)}, \epsilon_2^{(p)}; \Lambda_p) \;=\; \sum_{\lambda} \Lambda_p^{|\lambda|} \prod_{s\in\lambda} \frac{1}{E_{\mathrm{II}}(s)\big(\epsilon_1^{(p)} + \epsilon_2^{(p)} - E_{\mathrm{II}}(s)\big)},$$

the $N = 1$ specialisation of Chapter 31.1.1 with $E_{\mathrm{II}}(s) = -\epsilon_1^{(p)} L_\lambda(s) + \epsilon_2^{(p)}(A_\lambda(s) + 1)$. On the CY / Nekrasov–Okounkov slice $\epsilon_1 + \epsilon_2 = 0$, $\epsilon_1 = -\epsilon_2 = \hbar$:

$$Z^{U(1),\mathrm{CY}}_{\mathrm{inst}} \;=\; \sum_{\lambda} \frac{\Lambda^{|\lambda|}}{\prod_{s\in\lambda} h_\lambda(s)^2} \;=\; \exp(\Lambda),$$

after absorbing the $\epsilon_1\epsilon_2$-normalisation into Λ. The equality is the hook-length identity $\sum_{\lambda\vdash n} \prod_{s\in\lambda} h_\lambda(s)^{-2} = 1/n!$.

COMPUTATION 31.4.3. At $\Lambda_p = p^{-s}$:

$$Z^{U(1),\mathrm{CY}}_{\mathrm{inst},p}(s) \;=\; \exp(p^{-s}).$$

The product over primes gives

$$\prod_p Z^{U(1),\mathrm{CY}}_{\mathrm{inst},p}(s) \;=\; \exp\!\left(\sum_p p^{-s}\right).$$

This is the first plethystic logarithmic term of the Euler product, not the Euler factor itself. The factor $(1 - p^{-s})^{-1}$ belongs to the arithmetic Fock degeneration below.

31.5 THE $\epsilon_2 \to \infty$ DEGENERATION

Definition 31.5.1 *(degenerate per-prime limit).* Set per prime p:

$$\epsilon_1^{(p)} = \log p, \qquad \epsilon_2^{(p)} = +\infty.$$

In this limit $1/\epsilon_2^{(p)} \to 0$ and every Young diagram with $|\lambda| \geq 2$ contributes 0. The sum collapses to

$$Z^{U(1)}_{\mathrm{inst},p}\big|_{\mathrm{arith}} \;=\; 1 + p^{-s}.$$

THEOREM 31.5.2 *(primon partition as degenerate Nekrasov).* . The Hartnoll–Yang primon partition function $\zeta(s)$ is obtained from a per-prime theory only if the charge-k contribution survives with weight p^{-ks} for every $k \geq 0$:

$$\boxed{\zeta(s) \;=\; \prod_p \sum_{k\geq 0} p^{-ks} \;=\; \prod_p (1 - p^{-s})^{-1}.}$$

For a rank-N local system ρ, the analogous Euler factor is $\det(1 - p^{-s}\rho_p(\mathrm{Frob}_p))^{-1}$; that is a separate rank-N extension, not the primon rank statement. The two-term $\epsilon_2 \to \infty$ degeneration gives $\prod_p(1 + p^{-s})$, not the Euler product. The Euler factor therefore belongs to the all-charge Fock specialization, not to the vanishing of all Young diagrams of size at least 2.

31.6 Arakelov Coulomb Modulus

Definition 31.6.1 *(arithmetic Coulomb modulus).* The complex parameter $s = \sigma + i\tau \in \mathbb{C}$ (as in p^{-s}) decomposes via the Arakelov degree map: its real part $\sigma = \mathrm{Re}(s)$ is the arithmetic first Chern class

$$\sigma \;=\; \widehat{c}_1\big(\mathcal{O}(\sigma\cdot[\infty])\big) \;\in\; \widehat{\mathrm{CH}}^1(\overline{C})\otimes\mathbb{R} \;\cong\; \mathbb{R},$$

under the canonical identification of $\widehat{\mathrm{CH}}^1(\overline{C})\otimes\mathbb{R}$ with $\mathbb{R}$ by the archimedean degree map; the imaginary part $\tau = \mathrm{Im}(s)$ parameterises the unitary direction (Hecke-character / spectral angle). The Nekrasov Coulomb modulus $\vec{a}$ is replaced by this single arithmetic modulus s. The Hecke-character tuple $\{\theta_p\}$ is the arithmetic analogue of the non-trivial Coulomb splitting $a_\alpha - a_\beta$ for $U(N)$.

Remark 31.6.2 *(Gillet–Soulé Arakelov pairing).* On $\overline{C}\times\overline{C}$ the height pairing at $\infty\times\infty$ is $s\cdot s'$ up to finite contributions (Gillet–Soulé *Arithmetic intersection theory*, Publ. IHES 1992).

31.7 Nekrasov–Shatashvili Limit: $\log\zeta(s) = \sum_p \mathcal{W}_p^{\mathrm{NS}}(s)$

Theorem 31.7.1 *(arithmetic NS identity).* . The regulated per-prime NS superpotential $\mathcal{W}_p^{\mathrm{NS}}(s) = \log(1-p^{-s})^{-1} = \sum_{k\ge1} p^{-ks}/k$ sums over primes to

$$\boxed{\log\zeta(s) \;=\; \sum_p \mathcal{W}_p^{\mathrm{NS}}(s).}$$

The Dedekind zeta logarithm is the sum over primes of the arithmetic Nekrasov–Shatashvili superpotential, evaluated at arithmetic Coulomb modulus s.

31.8 AGT: Semiclassical Liouville Degenerates to Free Boson

Remark 31.8.1 *(AGT obstruction).* AGT (Alday–Gaiotto–Tachikawa arXiv:0906.3219): $Z^{U(2)}_{\mathrm{inst}}(\vec{a},\epsilon_1,\epsilon_2;\Lambda) = \mathcal{B}^{\mathrm{Liouville}}_{c,\Delta_i}(\vec{q})$ with $c = 1+6(b+b^{-1})^2$, $b^2 = \epsilon_1/\epsilon_2$. In the arithmetic limit $\epsilon_2\to\infty$, $b\to 0$: $c\to\infty$. The AGT correspondence at finite b requires Liouville theory at finite central charge; its $b\to 0$ limit yields the semiclassical Liouville theory, which degenerates further to a free boson (Theorem 31.8.2).

Theorem 31.8.2 *(semiclassical-Liouville → free boson).* (Teschner arXiv:hep-th/0104158; Harlow–Maltz–Witten arXiv:1108.4417). In the $b\to 0$ limit, Virasoro degenerates to Heisenberg: the Liouville primary $V_\alpha = e^{2\alpha\varphi}$ with $\alpha = \alpha_0 + \alpha_1 b$ becomes a free-boson vertex operator of charge α_0; Virasoro conformal blocks become free-boson correlators.

Corollary 31.8.3 *(arithmetic AGT at $b\to 0$).* . Per prime,

$$Z^{U(1)}_{\mathrm{inst},p}\big|_{b^{(p)}\to 0}(\Lambda_p) \;=\; \mathrm{Tr}_{\mathrm{Sym}(\mathbb{C}a_{-1})}(\Lambda_p^{N_p}) \;=\; (1-\Lambda_p)^{-1}.$$

At $\Lambda_p = p^{-s}$ this gives the primon Heisenberg trace $(1-p^{-s})^{-1}$ per prime.

Theorem 31.8.4 *(Hartnoll–Yang primon trace = $b\to 0$ AGT).* .

$$L(s) \;=\; \prod_p Z^{U(1)}_{\mathrm{inst},p}\big|_{\mathrm{AGT},\,b\to 0}(s) \;=\; \prod_p \langle \mathcal{H}^{(p)}_{k=1}\rangle_{\mathrm{partition}}.$$

The rank-1 Heisenberg per prime is the semiclassical-Liouville degeneration of Virasoro at prime p, and the AGT map reduces to a free-boson partition at each place.

31.9 MAULIK–OKOUNKOV AND THE BETHE-HECKE SHADOW

THEOREM 31.9.1 *(Maulik–Okounkov arXiv:1211.1287, Thm 1.5.1).* . The R-matrix on equivariant cohomology of the instanton moduli space $\mathcal{M}_{N,k}$ generates the Yangian $Y(\widehat{\mathfrak{gl}}_N)$. $Z^{U(N)}_{\text{inst}}$ is the trace of the holonomy of the MO R-matrix.

Definition 31.9.2 *(primon Bethe-Hecke shadow).* The *primon Bethe-Hecke shadow* is the commutative Bethe/Hecke algebra acting on the arithmetic instanton model. Fixed-point data are indexed by N-tuples of Young diagrams per prime; the arithmetic action surviving the $\epsilon_2 \to \infty$ degeneration is through Hecke transfer operators, not an RTT R-matrix or Yangian Hopf structure.

THEOREM 31.9.3 *(L-function as MO trace).* . $L(s, \rho)$ is the Bethe/Hecke transfer trace in the arithmetic instanton shadow. The Maulik–Okounkov Yangian $Y(\widehat{\mathfrak{gl}}_N)$ governs the complex-geometric instanton count; its arithmetic avatar, in the degenerate limit, is the commutative Hecke algebra acting at each prime.

31.10 THE TOPOLOGICAL-SECTOR READING (COMPATIBLE WITH NEKRASOV)

Remark 31.10.1 *(Kurokawa/Deninger).* The identity $L(s) = \sum_{n\geq 1} n^{-s}$ expresses $\zeta(s)$ as a sum over arithmetic topological classes with p-adic valuation $v_p(n)$; the term n^{-s} carries the arithmetic class indexed by the prime factorization of n. From the $\mathbb{F}_1$-zeta perspective of Kurokawa (Proc. Japan Acad. 2005) and Deninger (1994), $\zeta(s) = \det(s - \Theta)^{-1}_{\text{reg}}$ for a hypothetical cohomological operator Θ.

THEOREM 31.10.2 *(motivic-topological compatibility).* . The motivic instanton partition function (Nekrasov–Okounkov §4; Behrend motivic DT arXiv:math/0507523) on the arithmetic surface localises in the $\epsilon_2 \to \infty$ limit to the $\mathbb{F}_1$-cohomological partition sum. Young diagrams collapse to their size $|\lambda|$, which labels the instanton number c_2^{Ar}, and the c_2^{Ar}-sum becomes the v_p-sum:

$$Z^{U(N),\text{arith}}_{\text{inst},p}(s, \rho_p) \;=\; \det(1 - p^{-s}\rho_p(\text{Frob}_p))^{-1}.$$

31.11 THE ($\epsilon_2 \to \infty$, $b \to 0$) DEGENERATE LIMIT: STRUCTURE OF $L(s)$

(i) The arithmetic curve $\overline{C}$ has arithmetic Krull dimension 1; Nekrasov's $\mathbb{C}^2$ is real dimension 4. The self-product $\overline{C} \times_{\mathbb{F}_1} \overline{C}$ is the arithmetic surface carrying the instanton data (Chapter 31.3.1).

(ii) The gauge group is $U(1)^{\mathbb{P}}$, abelian, rank $\aleph_0$ (Chapter 31.4.1).

(iii) The Ω-background per prime is $(\epsilon_1^{(p)}, \epsilon_2^{(p)}) = (\log p, +\infty)$ (Chapter 31.5.1).

(iv) The primon partition function $L(s)$ is the ($\epsilon_2 \to \infty$, $b \to 0$) degenerate Nekrasov limit (Chapter 31.5.2).

(v) The Coulomb modulus is the arithmetic first Chern class $s = \widehat{\deg}\, O(\sigma)$ (Chapter 31.6.1).

(vi) $\log \zeta(s) = \sum_p \mathcal{W}_p^{\text{NS}}(s)$: the zeta logarithm is the sum of arithmetic Nekrasov–Shatashvili superpotentials (Chapter 31.7.1).

(vii) In the $b \to 0$ AGT limit, the Liouville conformal block at prime p degenerates to the free-boson partition function $(1 - p^{-s})^{-1}$ (Chapter 31.8.3).

(viii) The Maulik–Okounkov R-matrix reduces, in the arithmetic degeneration, to the Hecke transfer algebra; $L(s, \rho)$ is the Hecke transfer trace (Chapter 31.9.3).

$L(s)$ is the ($\epsilon_2 \to \infty$, $b \to 0$) degenerate limit of the Nekrasov partition function: the Euler-factor data of the Ω-background survives; the full instanton moduli geometry contracts to fixed-point contributions labeled by $\epsilon_1^{(p)} = \log p$ per prime.

Chapter 32

The Arithmetic Modular Functor

Segal's modular functor assigns conformal blocks to marked Riemann surfaces and uses sewing along circles. The arithmetic curve has no such sewing category. Tate theory supplies the object occupying the same formal position.

32.1 Axioms and Their Translation

Each Segal axiom for a Riemann surface $\Sigma_{g,n}$ – finiteness of conformal blocks, sewing along circles, modular invariance, and the partition-function trace – has an arithmetic analogue on the Arakelov curve $(\overline{\operatorname{Spec} \mathcal{O}_K}, S)$. The entries below are a dictionary of roles. They are not literal identifications of the two categories.

Convention 32.1.1 *(arithmetic Segal dictionary).*

Geometric Segal	Arithmetic Segal
Riemann surface $\Sigma_{g,n}$	Arakelov curve $(\overline{\operatorname{Spec} \mathcal{O}_K}, S)$
Orientation	Trivialization of $\det R\Gamma$ (Deligne)
Genus contribution	Arakelov discriminant volume $\frac{1}{2}\log\lvert d_K\rvert$
Marked point	Place (finite or archimedean)
Sewing circle	Punctured formal disk $\operatorname{Spec} K_{\mathfrak{p}}$
Intermediate rep	Weil–Deligne parameter (ρ, N)
Sewing law	Euler-product factorization of $L(s,\chi)$
Monodromy $U(1)$	Frobenius $\alpha_p = \chi_p(p)$
$\mathrm{MCG}(\Sigma)$	arithmetic symmetry group acting on labels and Hecke lines
Modular S-matrix	Γ-factor + functional equation
Verlinde scalar	Class-number residue at $s = 1$
Conformal block at $\{V_i\}$	Tate integral $\prod_v \int \lvert\cdot\rvert^s \Phi_v$
$Z(\Sigma) = \operatorname{tr} q^{L_0 - c/24}$	finite Euler product $\prod_{\mathfrak{p}} (1 - N\mathfrak{p}^{-s})^{-1}$, completed by the archimedean Tate factors

32.2 Arakelov Discriminant as Lattice Volume

Observation 32.2.1 *(the discriminant term is not a genus).* The geometric genus satisfies $g = \frac{1}{2}\dim_{\mathbb{R}} H^1(\Sigma;\mathbb{R})$; equivalently, the Jacobian $J(\Sigma) = H^0(\Sigma,\Omega^1)^*/H_1(\Sigma,\mathbb{Z})$ is a g-dimensional torus whose Weil height is $\frac{1}{2}\log\det(\operatorname{Im}\tau)$ for the period matrix τ. The arithmetic analogue is $\frac{1}{2}\log\lvert d_K\rvert = \frac{1}{2}\log(\operatorname{covol}(\mathcal{O}_K \subset K \otimes_{\mathbb{Q}} \mathbb{R}))$. Both are half-logarithms of lattice covolumes; the functional is the same, the lattices differ. The number $\frac{1}{2}\log\lvert d_K\rvert$ is not an integer genus and is not a Betti number.

Definition 32.2.2 *(arithmetic Euler characteristic).*

$$\chi^{\mathrm{arith}}(\overline{X}) \;:=\; -\widehat{\deg}\,\omega_{\overline{X}} + \sum_{v\mid\infty} \log \|\cdot\|_v .$$

On $\operatorname{Spec} O_K$ with signature (r_1, r_2), $\chi^{\mathrm{arith}} = \log|d_K| - r_1\log(\pi/4) - r_2\log(\pi/2)$ by Minkowski–Odlyzko. Additive on disjoint unions of Arakelov surfaces.

32.3 Weil–Deligne Sewing

THEOREM 32.3.1 *(sewing law).* .

Here $\mathcal{F}^{\mathrm{arith}}$ denotes the arithmetic modular functor defined in Definition 32.8.1 (an adelically factorized assignment of meromorphic lines to Arakelov surfaces with marked places); and $\mathcal{V}^{\mathrm{prim}}$ denotes the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 (a class in K-theory, not a vertex algebra). The ℓ-adic coefficient system at a non-archimedean prime $\mathfrak{p}$ is the étale cohomology local system attached to $\mathcal{V}^{\mathrm{prim}}$ via the comparison isomorphism $H^*_{\text{ét}}(-, \mathbb{Q}_\ell)$ for any prime $\ell \neq p$; this is the coefficient ring of the Weil–Deligne representation, not an enrichment of $\mathcal{V}^{\mathrm{prim}}$ with vertex-algebra structure.

$$\mathcal{F}^{\mathrm{arith}}(\overline{X}_1 \cup_p \overline{X}_2) \;=\; \bigoplus_{(\rho,N)} \mathcal{F}^{\mathrm{arith}}(\overline{X}_1; \rho, N) \otimes_{\mathrm{End}(\rho)} \mathcal{F}^{\mathrm{arith}}(\overline{X}_2; \rho, N)^*$$

where (ρ, N) runs over isomorphism classes of Weil–Deligne representations of $W_{\mathbb{Q}_p}$ whose coefficients are ℓ-adic ($\ell \neq p$), ρ continuous, N the monodromy operator.

Remark 32.3.2 *(labels are Langlands parameters).* Segal's intermediate representation labels are local Galois representations. At archimedean places, sewing is along Hodge structures (Weil-group representations of $W_{\mathbb{R}}$ or $W_{\mathbb{C}}$). The correspondence between WZW representation labels and automorphic representations under local Langlands is:

$$\{\text{irr. admissible } G(\mathbb{Q}_p)\text{-reps}\} \;\leftrightarrow\; \{\text{Weil–Deligne } L\text{-parameters}\}.$$

The arithmetic functor uses local Langlands parameters in the Segal sewing position.

32.4 Arithmetic Symmetries

Observation 32.4.1. The arithmetic symmetry group has two complementary actions. The Galois group $\mathrm{Gal}(\overline{K}/K)$ controls label permutations: it acts on Weil–Deligne parameters at marked places, the arithmetic counterpart of the deck transformations on the geometric surface. The adelic quotient $\mathrm{GL}_n(\mathbb{A}_K)/\mathrm{GL}_n(K)$ acts through Hecke correspondences on automorphic lines. This pair is the arithmetic replacement for the geometric mapping-class action; it is not a mapping class group of a surface.

THEOREM 32.4.2 *(arithmetic symmetry action).* . There is an arithmetic symmetry action in which $\mathrm{Gal}(\overline{K}/K)$ acts on local Langlands labels and $\mathrm{GL}_n(\mathbb{A}_K)$ acts by Hecke correspondences. At an unramified prime p, Frobenius generates the quotient $\mathrm{Gal}(\bar{k}_p/k_p) \cong \widehat{\mathbb{Z}}$. This is only the unramified quotient of local monodromy; the inertia and wild inertia are not Dehn twists and are not discarded in ramified sewing.

32.5 Verlinde Scalar and the Class Number Formula

Theorem 32.5.1 *(arithmetic Verlinde).* (Tate's thesis, class number formula). The arithmetic Verlinde scalar attached to the empty-marking sector is the residue of the Dedekind zeta at $s = 1$:

$$\boxed{\mathsf{Ver}^{\mathrm{arith}}(K) \;:=\; \mathrm{Res}_{s=1}\zeta_K(s) \;=\; \frac{2^{r_1}(2\pi)^{r_2}h_K R_K}{w_K\sqrt{|d_K|}}.}$$

This scalar is generally not an integer dimension of conformal blocks; it is the class-number residue in the Verlinde position of the arithmetic analogy.

32.6 Explicit Computation: Conformal Blocks via Tate Integrals

Computation 32.6.1 *(conformal block at* $(\mathbb{Z}, \{p, \infty\}, \mathrm{triv})$*).* $\mathcal{F}^{\mathrm{arith}}(\mathrm{Spec}\,\mathbb{Z}, \{p, \infty\}, V_{\mathrm{vac}} \otimes \lambda_\infty)$ is spanned by the Tate integral

$$\Phi_p(s) \;=\; \Lambda_\zeta(s) \cdot (1 - p^{-s}),$$

where $\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed Riemann zeta (Tate's thesis, §3.4). Removing the p-th Euler factor from $\zeta(s)$ gives $\zeta(s)(1 - p^{-s})$; multiplication by $\pi^{-s/2}\Gamma(s/2)$ completes to $\Lambda_\zeta(s)(1 - p^{-s})$. Dimension: 1.

Observation 32.6.2. The vacuum conformal block at prime p is the completed zeta function with the p-th Euler factor removed. The sewing axiom at p reinserts it.

Computation 32.6.3 *(sewing at* $\{2, 3\}$*).* The vacuum conformal block at marked places $\{2, 3, \infty\}$ is $\Lambda_\zeta(s)(1 - 2^{-s})(1 - 3^{-s})$ (the completed zeta with the Euler factors at 2, 3 removed; Chapter 32.6.2). Sewing at the two finite places with unramified Satake parameters $\alpha_p = \chi_p(p) \in \{\pm 1\}$ reinserts the twisted Euler factors:

$$\Phi(s) \;=\; \Lambda_\zeta(s)\,\frac{(1 - 2^{-s})(1 - 3^{-s})}{(1 - \alpha_2 \cdot 2^{-s})(1 - \alpha_3 \cdot 3^{-s})},$$

where $\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$. For trivial Satake $(\alpha_2, \alpha_3) = (1, 1)$ one recovers $\Phi(s) = \Lambda_\zeta(s)$; for a character with local data specified only at 2 and 3, this is only a two-prime partial twist. The full completed L-function $\Lambda(s, \chi)$ is obtained only after replacing the Euler factor at every finite prime by $(1 - \chi(p)\,p^{-s})^{-1}$, together with the correct archimedean factor and conductor.

32.7 Functional Equation as Sewing Axiom

Theorem 32.7.1 *(Tate's global functional equation).* .

Let $\Lambda(s, \chi)$ denote the completed Dirichlet L-function with its conductor and parity-dependent archimedean factor. Then

$$\Lambda(s, \chi) \;=\; \varepsilon(\chi) \cdot \Lambda(1 - s, \chi^{-1}),$$

where $\varepsilon(\chi)$ is the root number, proved by adelic Poisson summation (Tate's thesis, §4.4). The arithmetic analogue of Segal sewing is the functional equation.

Remark 32.7.2 *(finiteness reading).* The content of sewing is not dimension 1 per se; it is the $s \leftrightarrow 1 - s$ involution on the one-dimensional line whose invariants are the values of the completed L-function.

32.8 FIVE-AXIOM ARITHMETIC MODULAR FUNCTOR

Definition 32.8.1. An *arithmetic modular functor* is a functor

$$\mathcal{F}\colon \mathcal{C}^{\mathrm{Arak}} \to \mathrm{Hom}(\mathbb{C}_s, \mathrm{Line})$$

(from Arakelov surfaces to lines depending meromorphically on a parameter s) satisfying:

(A1) Finiteness. $\mathcal{F}(\overline{X}, S, \{\chi_{\mathfrak{p}}\})$ is a 1-dimensional meromorphic line, with pole divisor prescribed by the completed L-function represented by the line.

(A2) Factorization. For independent places, $\mathcal{F}(\overline{X}, S, \{\chi_{\mathfrak{p}}\}) = \bigotimes_{\mathfrak{p}\in S} \mathcal{F}_{\mathfrak{p}}(\chi_{\mathfrak{p}})$.

(A3) Weil–Deligne sewing. For $S = S_1 \sqcup S_2$ along a parameter (ρ, N) at an extra prime $\mathfrak{q}$,

$$\mathcal{F}(\overline{X}, S \cup \{\mathfrak{q}\}) \;=\; \sum_{(\rho,N)} \mathcal{F}(\overline{X}_1, S_1 \cup (\rho, N)) \cdot \mathcal{F}(\overline{X}_2, S_2 \cup (\rho, N))^{-1},$$

summed over unramified (ρ, N).

(A4) Modular invariance. the Galois action on local labels and the Hecke action of $GL_n(\mathbb{A}_K)$ act on the functor.

(A5) Functional equation.

$$\mathcal{F}(\overline{X}, S, \chi)(s) \;=\; \varepsilon(\chi) \cdot \mathcal{F}(\overline{X}, S, \chi^{-1})(1 - s).$$

This is the arithmetic modular invariance: $\mathrm{SL}_2(\mathbb{Z})$ becomes the $\mathbb{Z}/2$ flip $s \leftrightarrow 1 - s$.

THEOREM 32.8.2 *(Tate line in the arithmetic Segal position).* (Tate's thesis). Tate's adelic construction gives a meromorphic L-line satisfying (A1)–(A5) in the zeta and Dirichlet sectors. No uniqueness is asserted before the source category and the local test-line normalizations are fixed.

The geometric and arithmetic Segal functors are different functors. The conjecture says that Tate's L-functor satisfies the arithmetic axioms occupying the Segal positions.

Chapter 33

Arithmetic bar constructions and residue comparisons

Continuous Galois cochains give an integral arithmetic input to the coefficient bar. Their relative products and coefficient limits are constructed in Chapter 10. A comparison with chiral collision operations requires residue complexes on the geometric carrier. It also requires compatible operations and maps identifying them with the coefficient model. The elementary residue and bar constructions below specify what is available before making that comparison.

33.1 Incidence of finite places

Observation 33.1.1. The ordered n-tuples of distinct finite places of Spec $\mathbb{Z}$ form a countable set. Giving this set the discrete topology produces a combinatorial configuration model. It is not an identification with the scheme configuration functor, and its distinct-place tuples contain no collision locus. A topology describing collisions must be supplied separately.

Definition 33.1.2 *(incidence poset of arithmetic places).* Let $\mathcal{I}_K$ be the poset of finite subsets of the finite places of K, ordered by inclusion. Its order complex has a simplex for every finite strict chain of such subsets. The simplicial cochains of this order complex are a combinatorial coefficient complex. They do not by definition carry the differential forms or residues of a geometric configuration space.

A motivic fundamental group is another construction with a specified base and category of motives. Relating its operations, or those of a Grothendieck–Teichmüller Lie algebra, to this incidence complex requires an actual map. Neither a common set of prime labels nor a formal resemblance between relations provides that map.

33.2 Steinberg symbols and reciprocity

Observation 33.2.1 *(constant differences).* For two distinct rational primes p, q, the difference $p - q$ is a constant in $\mathbb{Q}^\times$. Its relative differential in $\Omega^1_{\mathbb{Q}/\mathbb{Q}}$ is zero. Thus it does not supply the nonzero logarithmic form associated with moving insertion coordinates on a complex curve.

Theorem 33.2.2 *(Steinberg relation and the rational tame symbols).* In Milnor K-theory, $\{a, 1 - a\} = 0$ for $a \neq 0, 1$. In particular $\{p, 1 - p\} = 0$ in $K_2^M(\mathbb{Q})$. Tate's theorem identifies

$$K_2^M(\mathbb{Q}) \cong \{\pm 1\} \oplus \bigoplus_p \mathbb{F}_p^\times$$

by the real sign symbol together with the finite tame symbols. The finite symbols alone have an order-two kernel.

Proof. The first assertion is a defining relation of K_2^M. The isomorphism is the external theorem stated in Theorem 10.9.2. The class $\{-1,-1\}$ has trivial finite tame symbols and real sign -1, and that theorem identifies it as the kernel generator. □

This theorem does not identify the Steinberg relation with an Arnold relation on configuration-space forms. Such an identification would need a map on the complexes carrying those relations and a comparison of their induced operations.

THEOREM 33.2.3 *(dimension-one residues and their range).* For each finite place v of K, the normalized tame symbol is a homomorphism

$$\partial_v : K_2^M(K) \longrightarrow k_v^\times, \qquad \partial_v\{\pi_v, u\} = \bar{u}.$$

There is no codimension-two term on $\operatorname{Spec} O_K$. On $\mathbb{P}^1_F$, the normed tame symbols have common recipient $F^\times$ and obey (10.8).

Proof. Proposition 10.9.1 constructs the local homomorphisms and checks their Steinberg relations. The ring O_K has dimension one, proving the second assertion. The last assertion is the tame-symbol reciprocity theorem for the projective line with its norm maps, cited in §10.9. □

For a general degree n, the dimension-one residue diagram has terms $K_n^M(K)$ and $\bigoplus_v K_{n-1}^M(k_v)$. No additional codimension-two points are produced by writing a further arrow. A surface construction instead starts from a regular arithmetic surface X, a curve $C \subset X$, and a point $x \in C$. Its successive symbols require these flags and appropriate recipients and norm maps. A sum or product of symbols in different fields is undefined until those maps are specified. Exactness and a motivic comparison are additional statements with their own coefficient hypotheses.

33.3 THE COEFFICIENT TENSOR COALGEBRA

Fix a procyclic group $\Gamma \cong \mathbb{Z}_\ell$, with chosen generator, and put $\Lambda = \mathbb{Z}_\ell[[\Gamma]]$. Proposition 10.10.1 identifies it with $\mathbb{Z}_\ell[[T]]$. If the full cyclotomic group has a finite factor, that factor must either be retained in the group ring or removed by a stated choice. It is not part of the displayed procyclic identification.

Definition 33.3.1 *(coefficient bar over Λ).* Let A be an augmented associative dg Λ-algebra, with augmentation ideal I. In cohomological grading put $sI = I[1]$. Its coefficient bar is

$$B_\Lambda A = \bigoplus_{n \geq 0} (sI)^{\otimes_\Lambda n},$$

with deconcatenation coproduct and differential $b = b_1 + b_2$, where

$$b_1(sa) = -s(da), \qquad b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

Extend these maps as coderivations with the Koszul tensor convention. The empty word is the coaugmentation. Completion, when needed, is specified as in (10.5).

The identities $d^2 = 0$, Leibniz's rule, and associativity imply $b^2 = 0$, by Proposition 3.2.1. This is an associative coefficient construction. The primitive arithmetic $\mathcal{V}^{\mathrm{prim}}$ is a Hochschild trace class and is not thereby an augmented algebra A. Its identification with such an input requires a separately constructed object, product, and augmentation.

A proposed additional residue differential must act on a common complex with the bar differential. If a residue complex is tensored with $B_\Lambda A$, the tensor is a complex. A coalgebra structure on it also requires a compatible diagonal on that residue complex. Names such as an Iwasawa differential or a tame-symbol weight do not specify degree-one endomorphisms of this tensor.

Question 33.3.2 *(a mixed arithmetic differential).* Construct a common trigraded coalgebra and coderivations d_{Iw}, d_{res}, d_{loc} of the three coordinate degrees, realizing respectively the intended Iwasawa, residue, and local bar operations. Do they satisfy

$$d_i^2 = 0, \qquad d_i d_j + d_j d_i = 0 \quad (i \neq j)?$$

With these tridegrees, the equations are equivalent to $(d_{\mathrm{Iw}} + d_{\mathrm{res}} + d_{\mathrm{loc}})^2 = 0$, since its six homogeneous components have distinct tridegrees. The first requirement is construction of the displayed maps on the same carrier. Nilpotence is the subsequent equation.

Question 33.3.3 *(archimedean comparison).* Choose an archimedean realization of the arithmetic coefficient object and a chiral algebra on a specified complex curve. Construct and test a map from the realized arithmetic bar to that chiral bar, including its operations, scalar extension, and unit.

The elementary equality $\Omega^j_{\mathbb{R}/\mathbb{R}} = 0$ for $j > 0$ supplies no such comparison: the point $\operatorname{Spec}\mathbb{R}$ is not the complex curve on which chiral collisions take place. Real Galois cochains also differ from a trivial complex at $\ell = 2$, by Proposition 10.8.1.

33.4 Graded Koszulness and arithmetic purity

Let $A = \bigoplus_{j \geq 0} A_j$ be a connected graded algebra over a field k, so $A_0 = k$, with augmentation onto degree zero. The internal weight j is distinct from resolution degree i. By definition, A is Koszul when $\operatorname{Tor}_i^A(k,k)_j = 0$ for $i \neq j$. The coefficient bar resolves this Tor group under the hypotheses of Proposition 3.3.1.

Theorem 33.4.1 *(a conditional weight criterion).* Suppose k has characteristic zero and each A_j is finite-dimensional. Assume the total groups $\operatorname{Tor}_i^A(k,k)$ are finite-dimensional, and a bar-compatible operator F preserves internal weights. Fix an embedding of its eigenvalue field into $\mathbb{C}$ and a real number $q > 1$. Assume that every eigenvalue on a nonzero internal-weight-j bar term has absolute value $q^{j/2}$. Then A is Koszul if and only if every nonzero $\operatorname{Tor}_i^A(k,k)$ is pure of weight i, meaning that its eigenvalues have absolute value $q^{i/2}$.

Proof. The differential preserves internal weight and commutes with F. Every eigenvalue on its weight-j cohomology therefore still has absolute value $q^{j/2}$, since eigenvalues on a subquotient come from the original operator. If the Tor groups are diagonal, their purity follows. Conversely, $q^{j/2} \neq q^{i/2}$ when $i \neq j$. Purity in weight i thus forces every off-diagonal summand to vanish, which is the Koszul condition. □

An arithmetic application requires the graded algebra, its realization, and the operator. It must also identify internal grading with the stated arithmetic weights. Frobenius weights, Hodge filtrations, and Sen weights are attached to different objects. An equivalence between their purity conditions is not supplied by this graded argument. The proposed criterion for a p-adic Simpson realization of $\mathcal{V}^{\mathrm{prim}}$ therefore remains a comparison problem, including existence of that realization.

Remark 33.4.2 *(comparison of realizations).* A de Rham–étale comparison theorem has a specified geometric input and coefficient range. It does not identify arbitrary mixed Hodge modules over $\mathbb{C}$ with Frobenius modules over finite fields. Any purity argument here must carry those realization hypotheses before invoking a comparison.

33.5 The primary Mellin character

The positive integers under multiplication form the free commutative monoid on the primes. Let V have a basis e_m indexed by $m \geq 1$, with energy $He_m = (\log m)e_m$. Its Mellin character is the absolutely convergent sum $\chi_V(s) = \sum_{m\geq 1} m^{-s}$ for $\operatorname{Re} s > 1$. This grading records multiplicative labels, not homological degree.

THEOREM 33.5.1 *(the multiplicative-label character).* For the displayed graded space,

$$\chi_V(s) = \prod_p (1 - p^{-s})^{-1} = \zeta(s), \qquad \operatorname{Re} s > 1.$$

If a specified primary zero-mode quotient of an arithmetic bar is isomorphic to V with these energies, it has the same Mellin character. The existence of that quotient and isomorphism is an additional hypothesis.

Proof. For finitely many primes, expansion of the geometric series and unique factorization give the sum over integers supported on those primes. Absolute convergence of $\sum_m m^{-s}$ for $\operatorname{Re} s > 1$ permits the increasing-prime limit. This proves the Euler product. A grading-preserving isomorphism preserves the summands and their energies, proving the last assertion. □

Question 33.5.2 *(the Bost–Connes comparison).* Construct an augmented arithmetic Heisenberg coefficient algebra and its Koszul dual. Is there a comparison with the Bost–Connes system that preserves the intended multiplication, module category, and time evolution? A Mellin character equal to $\zeta(s)$ is one scalar condition on such a comparison. It does not determine these objects or imply their equivalence.

33.6 Complementarity and the functional equation

Put $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, using the meromorphic continuation of the zeta function. The classical functional equation is $\xi(s) = \xi(1-s)$. To derive it from an arithmetic duality one needs an evaluation map that turns the duality into reflection of the Mellin parameter.

Question 33.6.1 *(arithmetic complementarity).* Construct arithmetic Heisenberg and dual coefficient objects with defined central parameters $\kappa_{\mathsf{G}}^{\mathrm{Ar}}$ and $(\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}}$, together with an evaluation to completed Mellin data. Can that evaluation identify

$$\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0 \quad \text{with} \quad \xi(s) = \xi(1-s)?$$

It must include the archimedean gamma factor and the polar terms, and it must identify the two functions rather than merely their uncompleted Dirichlet coefficients.

The numerical equation on central parameters has no Mellin variable. Without the requested map it proves neither meromorphic continuation nor a functional equation. An arithmetic comparison preserving the full evaluation would explain these analytic operations. An equality of characters in the half-plane of convergence leaves it open.

33.7 Coalgebra duality with Iwasawa coefficients

The coderived category of comodules and the contraderived category of contramodules use different acyclicity conventions and different infinite operations. A contramodule over a coalgebra C has a contraaction from $\mathrm{Hom}(C, P)$ to P, obeying contraassociativity and a counit identity. It is not a comodule with the arrow reversed. A co–contra comparison requires a specified coefficient category and the corresponding derived constructions.

Question 33.7.1 *(an arithmetic co–contra comparison).* For a specified dg coalgebra C over Λ, construct the comodules, contramodules, and derived categories appropriate to its topology. Under what projectivity, conilpotence, and completeness hypotheses does the comparison give an equivalence

$$D^{\mathrm{co}}(C\text{-comod}) \simeq D^{\mathrm{ctr}}(C\text{-contramod})?$$

Determine separately when the coalgebra unit $C \to B_\Lambda \Omega_\Lambda C$ and algebra counit $\Omega_\Lambda B_\Lambda A \to A$ are equivalences in the chosen categories.

The bar takes an augmented algebra as input and the cobar takes a coaugmented coalgebra. Thus the expression $\Omega_\Lambda B_\Lambda C$, with only a coalgebra C supplied, is not the coalgebra unit. The coefficient adjunction and its algebra resolution are constructed in Chapter 3. Their extension to the intended arithmetic categories remains separate. Theorem 46.1.5 gives a lifting criterion once the required adjunction is constructed.

Remark 33.7.2 *(the module μ-invariant).* For a finitely generated torsion Λ-module M, the Iwasawa invariant is $\mu(M) = \mathrm{length}_{\Lambda_{(\ell)}} M_{(\ell)}$. An arbitrary coalgebra has no such invariant without choosing a module to which the definition applies. Proposition 10.10.2 gives a conilpotent coalgebra with quotient $\mu = 1$, and a nonconilpotent one with quotient $\mu = 0$. Thus this module invariant cannot replace the coalgebra hypotheses in the displayed question.

33.8 A concrete arithmetic coefficient bar

Definition 33.8.1 *(relative Galois coefficient bar).* For the strict Galois cochain cospan with specified multiplicative local conditions over R_r, let P_r be the complex (10.3), with either endpoint product. Its arithmetic coefficient bar is

$$\mathcal{B}_r = B_{R_r}(R_r \oplus P_r).$$

The augmentation is the external scalar projection. Its coefficient completion is $\varprojlim_r \mathcal{B}_r$, with tensor target (10.5), when the local conditions and maps form a compatible coefficient tower.

Theorem 33.8.2 *(properties of the coefficient construction).* The finite-stage bars have square-zero differential and coassociative counital coproduct. They are conilpotent, and the two endpoint choices are isomorphic by the integral all-arity comparison of Chapter 11. For a compatible tower that comparison extends to the specified coefficient completion. An interpretation as a derived tensor retains the resolution hypotheses of Proposition 3.3.1.

Proof. The dg identities for the endpoint products are Proposition 11.1.1. The coefficient-bar identities then apply. Finite words admit only finitely many reduced cuts. The all-arity maps and their coefficient continuity are the constructions of Chapter 11. □

33.9 The arithmetic comparison to be constructed

The continuous-cochain bar and the local tame symbols are defined on their displayed carriers. The first has a multiplication, augmentation, and ordered bar coproduct. The second changes coefficient fields along valuations. A chiral arithmetic comparison must relate these operations, not just their prime labels.

The common residue carrier and mixed maps are asked for in Question 33.3.2. Archimedean realization requires Question 33.3.3. The arithmetic weight realization is the hypothesis needed for Theorem 33.4.1.

The Mellin model has its stated character. Its identification with a primary bar quotient, its Bost–Connes duality, and its completed functional equation comparison remain separate.

Likewise, the proposed numerical complementarity values 0, 8, 13, 250/3, 25/3 require actual arithmetic objects and computed central parameters before they classify any examples. In particular, a proposed K3–Mukai Borcherds realization of the value eight is its own construction. Neither those numerical candidates nor the Iwasawa module invariant settle the coefficient, geometric, and co–contra comparisons above.

Chapter 34

The Automorphic Vertex of the Langlands–TQFT Triangle

Kapustin–Witten realizes geometric Langlands as S-duality in $\mathcal{N} = 4$ gauge theory on $\Sigma \times C$. The arithmetic trace class has a different problem: whether the Hartnoll–Yang BKL–primon correspondence supplies a bulk theory at all.

$\mathcal{V}^{\mathrm{prim}}$ is a Hochschild trace class whose graded character is $\zeta(s)$. Three TQFT frameworks (Kapustin–Witten, Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees (BLLPRvR), and Chern–Simons/WZW) offer candidate TQFT-bulk structures for the Langlands triangle. Each fails at a specific structural obstruction, placing $\mathcal{V}^{\mathrm{prim}}$ on the automorphic vertex rather than the TQFT vertex.

34.1 The Kapustin–Witten Obstruction: Absence of Complex Structure

Theorem 34.1.1 *(Kapustin–Witten setup).* . Geometric Langlands for $C/\mathbb{C}$ arises as S-duality of 4D $\mathcal{N} = 4$ super Yang–Mills (SYM) on $\Sigma \times C$ with Σ topologically twisted and C holomorphically twisted. Categories $D(\mathrm{Bun}_G)$ and $\mathrm{QCoh}(\mathrm{Loc}_{^L G})$ appear as boundary conditions; S-duality $\tau \mapsto -1/\tau$ exchanges them.

Observation 34.1.2 *(arithmetic obstruction).* $\overline{C} = \overline{\mathrm{Spec}\,\mathbb{Z}}$ is an arithmetic 1-scheme with closed-point residue fields $\mathbb{F}_p$ at each prime p, plus the archimedean place. The holomorphic twist of $\mathcal{N} = 4$ super Yang–Mills requires a complex structure on the compactified direction; $\overline{\mathrm{Spec}\,\mathbb{Z}}$ carries no such structure, its closed fibres being spectra of finite fields rather than complex-analytic curves.

Corollary 34.1.3 *(Kapustin–Witten is undefined over $\mathbb{Z}$).* . The expression "$\mathcal{N} = 4$ SYM on $\Sigma \times \overline{\mathrm{Spec}\,\mathbb{Z}}$" names no defined theory in 2026's mathematics. The Kapustin–Witten holomorphic twist requires a complex structure on the compactified direction; the topological Σ-coupling and the S-duality $\tau \mapsto -1/\tau$ are consequences. The arithmetic compactification $\overline{\mathrm{Spec}\,\mathbb{Z}}$ admits no such structure.

34.2 The BLLPRvR Obstruction: Infinite Flavour Rank

Theorem 34.2.1 *(BLLPRvR, arXiv:1312.5344).* . The Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees (BLLPRvR) correspondence assigns a vertex algebra (VA) $\chi(\mathcal{T})$ to any 4D $\mathcal{N} = 2$ superconformal field theory (SCFT) $\mathcal{T}$, with central charges satisfying $c_{2d} = -12c_{4d}$ and $k_{2d} = -\frac{1}{2}k_{4d}$.

Observation 34.2.2. The finite Heisenberg covers of the primon construction are generated by the modes $\{a_{p,n}\}_{p\in S, n\in\mathbb{Z}}$ for a finite set of primes S. The formal cover over all primes would have flavour group $\prod_{p\in\mathbb{P}} U(1)_p$, hence flavour rank $\aleph_0$. The Hochschild trace class $\mathcal{V}^{\mathrm{prim}}$ itself is not a vertex algebra, and the scalar $r_\zeta = 2$ of Chapter 35 is a determinant-rank residue rather than a two-dimensional central charge. No 4D $\mathcal{N} = 2$ SCFT has flavour symmetry of rank $\aleph_0$: the flavour rank is bounded by the gauge rank plus hypermultiplet content, both finite for any theory with a Lagrangian description.

COROLLARY 34.2.3. . The VA $\chi(\mathcal{T})$ assigned by BLLPRvR to any 4D $\mathcal{N} = 2$ SCFT $\mathcal{T}$ has finite flavour rank. The all-prime Heisenberg cover has countable flavour rank, while $\mathcal{V}^{\mathrm{prim}}$ is only its Hochschild trace class. Hence no BLLPRvR vertex algebra of a 4D $\mathcal{N} = 2$ SCFT is $\mathcal{V}^{\mathrm{prim}}$. The class-$\mathcal{S}$ construction on an infinite-genus or infinite-puncture arithmetic curve would require $\overline{\mathrm{Spec}\,\mathbb{Z}}$ to carry the complex structure of a Riemann surface, obstructed by Chapter 34.1.2.

34.3 The Chern–Simons Obstruction: Irrational Levels

Observation 34.3.1 *(non-integer level).* Witten's Chern–Simons/WZW correspondence gives $CS_k(G)$ on $\Sigma \times \mathbb{R}$ with boundary vertex operator algebra (VOA) $V_k(\mathfrak{g})$ at integer level $k \in \mathbb{Z}$. A boundary theory whose finite Heisenberg covers reproduce all prime Euler factors would require gauge group $\prod_{p\in\mathbb{P}} U(1)_p$ with effective levels $k_p = 1/\log p$. These levels are irrational; the Chern–Simons path integral on non-trivial 3-manifolds requires $k \in \mathbb{Z}$ for well-definedness.

COROLLARY 34.3.2. . No finite-level abelian Chern–Simons theory has a boundary vertex algebra whose all-prime Hochschild trace is $\mathcal{V}^{\mathrm{prim}}$.

34.4 The Belinski–Khalatnikov–Lifshitz Billiard: Thermodynamic Coarse-Graining

Observation 34.4.1 *(Selberg scattering versus primon partition).* The Selberg trace formula on $\mathrm{SL}(2,\mathbb{Z})\backslash\mathbb{H}^2$ expresses the spectral density of the Laplacian as a sum over Maass cusp forms plus a continuous Eisenstein series contribution. The Eisenstein scattering phase is $\zeta(2it)/\zeta(2it+1)$ on the critical line, a scattering-matrix entry; the primon partition function is $\prod_p (1-p^{-s})^{-1} = \zeta(s)$, with pole at $s = 1$. The two occurrences of ζ are structurally distinct: scattering amplitude versus thermodynamic partition function.

THEOREM 34.4.2 *(Hartnoll–Yang: thermodynamic coarse-graining).* (Hartnoll–Yang arXiv:2502.02661, §3). The primon–Belinski–Khalatnikov–Lifshitz (BKL) correspondence is a thermodynamic coarse-graining: at high temperature, the Selberg continuous spectrum above eigenvalue $1/4$ is replaced by the primon oscillator count. The spectral match is asymptotic, not exact.

34.5 The Automorphic Vertex

THEOREM 34.5.1 *($\mathcal{V}^{\mathrm{prim}}$ on the automorphic vertex).* . $\mathcal{V}^{\mathrm{prim}}$ occupies the automorphic vertex through the GL_2 Iwahori Hecke category. The abelian GL_1 class-field-theory row is only its determinant shadow:

$$\underbrace{\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}})}_{\mathcal{V}^{\mathrm{prim}}} \longrightarrow \underbrace{\{\text{automorphic characters of } GL_1(\mathbb{A}_{\mathbb{Q}})\}}_{\det\text{-shadow}} \longleftrightarrow \underbrace{\{1\text{-dimensional Galois representations}\}}_{\text{Artin reciprocity}}.$$

The Euler product $\prod_p (1-p^{-s})^{-1}$ is the abelian determinant trace. It is not an identification of $\mathcal{V}^{\mathrm{prim}}$ with the category of GL_1 automorphic representations.

34.6 Asymptotic Symmetries at the BKL End-of-Time Boundary

PROPOSITION 34.6.1 *(Damour–Henneaux–Nicolai hep-th/0207267).* . BKL dynamics for 4D pure gravity on T^3 reduces to null billiards on the Weyl chamber of AE_3, a rank-3 hyperbolic Kac–Moody algebra; the fundamental domain is a modification of $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}^2$.

Observation 34.6.2 *(rank obstruction for AE_3).* A soft-mode algebra at the BKL end-of-time boundary decomposes into Weyl-group representations of $W(AE_3)$, where AE_3 has root lattice of rank 3. The primon Heisenberg algebra is generated by $\{a_{p,n}\}_{p\in\mathbb{P},n\in\mathbb{Z}}$, with generator index set of cardinality $\aleph_0$. The two index sets are incommensurable; $\mathcal{V}^{\text{prim}}$ is the bosonic Fock quantisation of the Eisenstein continuous sector of the gravitational superspace, not the full $W(AE_3)$ soft-mode algebra. The Hartnoll–Yang thermodynamic match (Chapter 34.4.2) is a coarse-graining of the Eisenstein density of states, not a spectral identification.

34.7 The Vol III Φ_d Functor: Arithmetic Domain Boundary

Observation 34.7.1. Vol III's functor $\Phi_d = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi^{\mathrm{FA}}_d$ takes a Calabi–Yau d-fold over $\mathbb{C}$ as input. $\overline{\mathrm{Spec}\,\mathbb{Z}}$ is an arithmetic 1-scheme over $\mathbb{Z}$; it admits no Calabi–Yau structure. The functor Φ_d is therefore defined on the $\mathbb{C}$-analytic locus of the programme, with $\overline{\mathrm{Spec}\,\mathbb{Z}}$ outside its domain.

Remark 34.7.2. When a Calabi–Yau 3-fold carries an $\mathrm{SL}_2(\mathbb{Z})$ action on its elliptic-fibration moduli (for example, K3×E), the modular-form output of Φ_3 carries arithmetic data. The Borcherds product Φ_N ($N \in \{1, 2, 3, 4, 6\}$) lives on $\mathrm{Sp}_4(\mathbb{Z})$-moduli space; it is an automorphic object, occupying the automorphic vertex together with $\mathcal{V}^{\text{prim}}$.

34.8 Automorphic Placement

Theorem 34.8.1 *(automorphic placement of $\mathcal{V}^{\text{prim}}$).* . Each candidate TQFT-bulk framework for $\mathcal{V}^{\text{prim}}$ (Kapustin–Witten, BLLPRvR, holomorphic Chern–Simons, the Costello–Gaiotto holomorphically-twisted theory, and Vol III Φ_d) is obstructed by a specific structural obstruction listed in §§34.1–34.7. $\mathcal{V}^{\text{prim}}$ occupies the automorphic vertex of the Langlands–TQFT triangle (Chapter 34.5.1).

Remark 34.8.2 *(the Deninger programme).* A rigorous construction of $\overline{\mathrm{Spec}\,\mathbb{Z}}$ as a "curve over $\mathbb{F}_1$" with a Frobenius-type $\mathbb{R}_{>0}$-action, Deninger's $\mathbb{F}_1$-geometric programme, would supply the missing TQFT-bulk vertex. The arithmetic site of Connes–Consani (arXiv:1405.4527) provides one candidate foundation.

34.9 The Automorphic Vertex: Structural Consolidation

Each obstruction in §§34.1–34.7 is structural: the complex-structure requirement of Kapustin–Witten (Chapter 34.1.3), the finite flavour-rank bound of BLLPRvR (Chapter 34.2.3), the integrality requirement of Chern–Simons levels (Chapter 34.3.2), the thermodynamic rather than spectral character of the Hartnoll–Yang match (Chapter 34.4.2), and the $\mathbb{C}$-analytic domain of Φ_d (Chapter 34.7.1). The automorphic vertex placement (Chapter 34.5.1) is the consequence: $\mathcal{V}^{\text{prim}}$ is identified by Artin reciprocity with the automorphic side of the Langlands triangle, with the TQFT-bulk vertex remaining Deninger's open programme.

Chapter 35

Local central charge and global residue

A local central charge is a coefficient in a stress-tensor operator product. A pole order of a global scalar determinant is a different invariant. Both can equal two in a rank-two Heisenberg example, but their equality does not construct a global Virasoro action.

35.1 Rescaling and the conformal vector

Observation 35.1.1 *(Rescaling and Virasoro normalization).* Let T be a nonzero stress tensor on the Heisenberg algebra. For $T' = \lambda T$, its singular product is

$$T'(z)T'(w) \sim \lambda^2\left(\frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}\right).$$

The standard conformal identities for the original translation operator force $\lambda = 1$.

The second pole requires $\lambda^2 = \lambda$. If $\lambda = 0$, the zero state cannot implement the nonzero translation operator. Thus the fourth-pole coefficient $\lambda^2 c$ cannot be interpreted as a new central charge while the other conformal identities are left unsatisfied.

Remark 35.1.2. At a prime p, take the standard rank-two Sugawara state. Its oscillator grading is $L_0^{(p)}$, while the chosen arithmetic energy is

$$H_p = (\log p) L_0^{(p)}.$$

The real number $\log p$ weights this operator. It does not rescale the conformal state.

Theorem 35.1.3 *(The local central charge).* Each rank-two level-one Heisenberg factor has central charge two. The operator H_p has energy $m \log p$ on a mode-m creator. It is not the zero mode of a scalar-rescaled stress tensor satisfying the same conformal identities.

Proof. For two orthogonal currents, the stress tensor is $T = \frac{1}{2}(:a^2: + :b^2:)$. Its two independent double contractions give coefficient one at the fourth pole, so $c/2 = 1$. The single contractions give the required second and first poles, as in Proposition 6.4.6. The energy formula follows from the oscillator grading. The rescaling observation proves the final assertion. □

35.2 The sum over primes

Write $\mathcal{H}_{\mathbb{P}} = \bigotimes'_p \mathcal{H}^{(p)}$ for the algebraic restricted tensor using the vacua. A vector has nonvacuum factors at only finitely many primes. This is the explicit current cover of Chapter 6. It is separate from any proposed arithmetic trace class.

Observation 35.2.1. The formal sum $\sum_p \omega_p$ is not a vector of $\mathcal{H}_{\mathbb{P}}$. In the Hilbert completion its finite partial sums have squared norm equal to the number of primes included, so they do not converge there either.

The norm assertion follows because each ω_p has squared norm one and different prime sectors are orthogonal. It rules out the indicated sum, with its indicated normalization. The stronger absence of a conformal state implementing the whole Hilbert oscillator grading is proved in Proposition 6.4.6.

COROLLARY 35.2.2. The sums of local modes $\sum_p L_m^{(p)}$, for $m \geq -1$, act on $\mathcal{H}_{\mathbb{P}}$. For $m \leq -2$, these same sums do not define operators on its vacuum. Consequently the prescribed sums do not give a full Virasoro representation on this carrier.

Proof. The modes with $m \geq -1$ annihilate each local vacuum. Only the finitely many occupied prime factors contribute. For $m \leq -2$, each local vacuum receives a nonzero creation state. The infinite sum is not in the algebraic restricted tensor and has divergent Hilbert norm. □

35.3 THE PRIME ZETA FUNCTION AND ANALYTIC CONTINUATION

The prime zeta function is

$$P(s) = \sum_p p^{-s}, \qquad \operatorname{Re} s > 1.$$

Taking logarithms of the Euler product and applying Möbius inversion gives

$$\log \zeta(s) = \sum_{r \geq 1} \frac{P(rs)}{r}, \qquad P(s) = \sum_{r \geq 1} \frac{\mu(r)}{r} \log \zeta(rs). \tag{35.1}$$

Here μ is the Möbius function. The sums first converge absolutely in the stated half-plane. Local branches of the logarithms give the continuation where they are defined.

THEOREM 35.3.1 *(Natural boundary of prime zeta).* The line $\operatorname{Re} s = 0$ is a natural boundary for the analytic continuation of P. In particular P has no holomorphic continuation to a neighborhood of the origin.[1]

The real singularities at $s = 1/r$ occur when r is squarefree. The coefficient $\mu(r)$ matters: (35.1) does not assert a branch point for every integer denominator or every putative zero contribution. The natural-boundary theorem concerns the surviving singularities.

COROLLARY 35.3.2. Ordinary analytic continuation does not assign $P(0)$ a finite value. Hence it does not define a global central charge by $c = 2P(0)$.

Proof. Such an evaluation would require a holomorphic germ at zero. The natural-boundary theorem excludes that germ. Other regularization prescriptions require their own definitions and do not construct the missing stress-tensor modes. □

Remark 35.3.3. The value $\zeta(0) = -1/2$ belongs to the zeta function of all positive integers. Replacing P by ζ changes the indexing set of the regularized sum. It does not evaluate the sum of the local prime charges.

[1] C.-E. Fröberg, *On the prime zeta function*, BIT **8** (1968), 187–202. See also P. L. Krapivsky and J. M. Luck, *Zeta-regularization and natural boundaries*, §2.2, equations (2.11)–(2.12), arXiv:2606.24536.

35.4 Pole order and logarithmic residue

A specified scalar determinant can have a finite pole order. The meromorphic function itself, its leading Laurent coefficient, and the residue of its logarithmic derivative must be distinguished.

Theorem 35.4.1. Suppose a scalar determinant is identified near $s = 1$ with $D(s) = \zeta(s)^2 h(s)$, where h is holomorphic and $h(1) \neq 0$. Then its pole order is two, and

$$-\operatorname{Res}_{s=1} \frac{D'(s)}{D(s)} = 2, \qquad \lim_{s \to 1} (s-1)^2 D(s) = h(1).$$

The separately defined scalar $r_\zeta = 2 \operatorname{Res}_{s=1} \zeta(s)$ also equals two. None of these scalar calculations defines a Virasoro action.

Proof. Write $\zeta(s) = (s-1)^{-1} u(s)$, where u is holomorphic and $u(1) = 1$. Then

$$D(s) = (s-1)^{-2} u(s)^2 h(s), \qquad \frac{D'}{D} = -\frac{2}{s-1} + 2\frac{u'}{u} + \frac{h'}{h}.$$

The last two terms are regular near one. This proves the pole, limit and logarithmic-residue formulas. The residue of ζ is one, giving the stated value of r_ζ. □

To apply the theorem to an arithmetic trace, one must construct its determinant and prove the local factorization with $h(1) \neq 0$. The equality $r_\zeta = 2$ alone proves neither premise.

35.5 The surviving Witt modes

Theorem 35.5.1. For $m, n \geq -1$ with $m + n \geq -1$, the prescribed sums satisfy

$$[L_m, L_n] = (m-n) L_{m+n}, \qquad L_m = \sum_p L_m^{(p)}.$$

The bracket is zero when $m = n = -1$. The negative sums with index at most -2 remain undefined on the restricted vacuum.

Proof. On any vector, the contributing prime set is finite. Different prime factors commute, so their Witt brackets add. A Virasoro central term would require $m + n = 0$. In this index range the possible values of m are -1, 0, 1, and $m(m^2 - 1) = 0$. Thus the central term vanishes on these modes. The vacuum calculation above excludes the remaining prescribed sums. □

Remark 35.5.2. For a finite prime set S, the full local sum has central charge $2|S|$. When $|m| \geq 2$, its central commutator coefficient grows with $|S|$. There is no finite limit for this unrenormalized coefficient. This is a statement about the specified sums, not an obstruction to every other representation.

35.6 Mellin reflection and a modular comparison

Set

$$\Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s), \qquad \xi(s) = \tfrac{1}{2} s(s-1) \Lambda(s).$$

The first function retains the poles at zero and one. The second removes them.

THEOREM 35.6.1. The completed zeta functions satisfy $\Lambda(s) = \Lambda(1-s)$ and $\xi(s) = \xi(1-s)$. If an independently specified scalar trace is identified with one of these functions, it inherits that reflection with the same normalization. Such an identification does not by itself construct a genus-one state space or its modular action.

Proof. Let $\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}$. Poisson summation for the Gaussian gives $\theta(t) = t^{-1/2}\theta(1/t)$. Its Mellin integral initially converges for $\operatorname{Re} s > 1$:

$$\Lambda(s) = \frac{1}{2}\int_0^\infty (\theta(t)-1)t^{s/2}\frac{dt}{t}.$$

Split at one and use the theta identity on $(0,1)$. The result is

$$\Lambda(s) = -\frac{1}{s} - \frac{1}{1-s} + \frac{1}{2}\int_1^\infty (\theta(t)-1)\big(t^{s/2} + t^{(1-s)/2}\big)\frac{dt}{t}.$$

The integral is entire by exponential decay and is invariant under $s \mapsto 1-s$. The two rational terms exchange. Multiplication by $s(s-1)/2$ proves the assertion for ξ. Transport through a specified scalar identification gives the conditional trace consequence. □

A vertex-algebra genus-one trace instead has the form $\operatorname{Tr} q^{L_0 - c/24}$, on a specified graded state space and convergence domain. Relating its modular substitution to Mellin reflection requires a map between those constructions. A shared functional equation does not supply that map.

35.7 ARITHMETIC DEGREE AND A POSSIBLE ANOMALY COMPARISON

For a finite divisor $D = \sum_p n_p[p]$ on $\operatorname{Spec}\mathbb{Z}$, define its arithmetic degree by $\widehat{\deg} D = \sum_p n_p \log p$. Archimedean metric contributions are separate data for a metrized line bundle.

PROPOSITION 35.7.1. The finite divisor $[p]$ has degree $\log p$. For $\operatorname{Re} s > 1$, these degree weights satisfy

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{r\ge 1} (\log p)\, p^{-rs}.$$

Proof. The first equality is the definition of arithmetic degree. Differentiate the absolutely convergent logarithmic Euler product $\log\zeta(s) = \sum_{p,r} p^{-rs}/r$. Normal convergence on smaller half-planes justifies the derivative. □

A conformal state ω_p is a vector, not a metrized line bundle. A Deligne pairing requires geometric line bundles.

For example, let $\pi : Y \to S$ be a smooth projective curve family of integral complex schemes, with line bundles L, M. Its Deligne pairing is the following determinant line.[2]

$$\langle L, M\rangle = \det R\pi_*(L\otimes M) \otimes (\det R\pi_* L)^{-1} \otimes (\det R\pi_* M)^{-1} \otimes \det R\pi_* \mathcal{O}_Y.$$

A metric refinement must additionally specify metrics and norms. The scalar $\log p$ does not identify this line with a quadratic expression in the conformal state.

[2] I. Biswas, G. Schumacher and L. Weng, *Deligne pairing and determinant bundle*. Theorem 1, arXiv:1106.0255.

Remark 35.7.2. In the convention $\langle T^{\mu}_{\mu} \rangle = -cR/(24\pi)$, the two-dimensional Weyl anomaly describes metric variation of a quantum field theory.[3] An arithmetic comparison would assign metrized determinant lines to the relevant field data and compare their variation with this functional. Arithmetic degree alone does not prove such a variation formula.

35.8 The comparison data

The local current factors have central charge two. Their restricted product has the prescribed Witt modes with indices at least -1, and its unchanged negative sums fail on the vacuum. A scalar determinant with the specified zeta factor has pole order two, with the logarithmic residue computed above.

The arithmetic questions concern maps relating these objects. A trace identification must preserve its Mellin normalization and polar factors. A conformal or anomaly comparison must additionally construct the state space, modes or metrized lines whose operations are being compared.

[3] For the Euclidean effective action, see T. Bautista, A. Benevides and A. Dabholkar, *Nonlocal quantum effective actions in Weyl-flat spacetimes*. Equations (2.26) and (3.2), arXiv:1711.00135.

Chapter 36

Integral Relative Galois Cochains and Their Bar Coalgebras

36.1 The relative arithmetic problem

A global Galois cochain can become a coboundary at every specified local field. The local primitives belong to the data of its relative class. For a restriction map of cochain algebras $f : A \to B$, a relative cocycle is therefore a pair

$$(a, y), \qquad da = 0, \qquad f(a) = dy.$$

Products of such pairs must retain a local primitive. If a has degree n, the expression $(-1)^n f(a)\, y'$ is a primitive for $f(aa')$. This observation determines the associative product used below.

Fix a number field K, a rational prime ℓ, and a finite set S of places containing the archimedean places and all places above ℓ. Write S_f for its finite places. Let $G_{K,S}$ be the Galois group of the maximal extension unramified outside S, and let G_{K_v} be the absolute Galois group of the completion at v. Fix the embeddings of algebraic closures that define homomorphisms

$$j_v : G_{K_v} \longrightarrow G_{K,S} \qquad (v \in S_f).$$

They need not be injective. All strict functoriality statements refer to these fixed homomorphisms. Conjugate choices require the corresponding cochain homotopies.

Put $R_r = \mathbb{Z}/\ell^r\mathbb{Z}$ for $r \geq 1$, with trivial Galois action and discrete topology. The continuous cochain model is classical.[1] The cochains below are continuous normalized inhomogeneous cochains. They compute continuous group cohomology by definition. Only finite local places enter the relative complex. An identification with a compact-support theory requires its own archimedean conventions.

Complexes are cohomologically graded. The differential has degree 1. For homogeneous maps and elements, the tensor convention is

$$(u \otimes v)(x \otimes y) = (-1)^{|v||x|} u(x) \otimes v(y).$$

We write $sM = M[1]$ and $|sa| = |a| - 1$. Tensor products are algebraic over the displayed coefficient ring until a completion is explicitly introduced.

[1] J. S. Milne, *Class Field Theory*, version 4.03 (2020), Chapter II, §4, pp. 86–89.

36.2 CONTINUOUS COCHAINS AND THE PATH FIBER

For a profinite group G and a discrete commutative ring R, let $C^n(G, R)$ consist of continuous functions $G^n \to R$ that vanish when any argument is the identity. Set $C^0(G, R) = R$ and $C^n = 0$ for $n < 0$. The trivial action gives

$$\begin{aligned}(dc)(g_1, \dots, g_{n+1}) = {} & c(g_2, \dots, g_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i c(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ & + (-1)^{n+1} c(g_1, \dots, g_n).\end{aligned}$$

The product of degrees n, m is

$$(cc')(g_1, \dots, g_{n+m}) = c(g_1, \dots, g_n) c'(g_{n+1}, \dots, g_{n+m}).$$

LEMMA 36.2.1. These formulas define a unital differential graded R-algebra. Precomposition with a continuous homomorphism of profinite groups is a morphism of such algebras.

Proof. The terms in d^2 cancel in pairs. Deleting or multiplying the ith entry and then the jth entry agrees with the operation in reverse order with the shifted index. The two alternating signs differ by -1. Associativity handles the pair of adjacent multiplications. If an argument is the identity, its two adjacent terms cancel, so normalization is preserved. In $d(cc')$, the terms before the cut at n are $(dc)c'$. The terms after that cut are $(-1)^n c(dc')$. The two extra terms at the cut cancel. This proves the Leibniz rule. Both bracketings of a triple product evaluate the same three consecutive argument blocks. The constant 1 in degree zero is the unit. Precomposition respects each formula term by term. □

Apply this lemma to

$$A_r = C^\bullet(G_{K,S}, R_r), \qquad B_r = \prod_{v \in S_f} C^\bullet(G_{K_v}, R_r), \qquad f_r = (j_v^*)_v : A_r \longrightarrow B_r.$$

The finite product in B_r uses componentwise multiplication.

THEOREM 36.2.2. Let $f : A \to B$ be a morphism of differential graded algebras over a commutative ring R. The formulas

$$P(f)^n = A^n \oplus B^{n-1}, \qquad D(a, y) = (da, f(a) - dy), \tag{36.1}$$

$$(a, y)(a', y') = (aa', (-1)^n f(a) y') \qquad (|(a, y)| = n) \tag{36.2}$$

define an associative nonunital differential graded algebra. Its underlying complex is the relative complex $\mathrm{Cone}(f)[-1]$ with the sign convention in (36.1).

Proof. The square of D has second component $f(da) - df(a) = 0$. For three inputs of degrees n, m, k, either bracketing has second component $(-1)^{n+m} f(a) f(a') y''$. The first components agree by associativity in A. The second component of $D((a, y)(a', y'))$ is

$$f(a)f(a') - (-1)^n f(da) y' - f(a) dy'.$$

The expression $D(a, y)(a', y') + (-1)^n (a, y) D(a', y')$ gives the same formula. The first components agree by the Leibniz rule in A. The cone statement follows from the displayed differential. □

The product is asymmetric on chains. An interval algebra explains both this asymmetry and the sign. Define I_R by

$$I_R^0 = Re_0 \oplus Re_1, \quad I_R^1 = Rh, \quad de_0 = -h, \quad de_1 = h, \quad dh = 0,$$

with products

$$e_i e_j = \delta_{ij} e_i, \qquad e_0 h = h = h e_1, \qquad e_1 h = h e_0 = h^2 = 0.$$

Its unit is $e_0 + e_1$. The evaluations ev_i send e_i to 1, the other idempotent to 0, and h to 0.

Proposition 36.2.3. The algebra I_R is a differential graded algebra. There is a strict isomorphism of nonunital differential graded algebras

$$P(f) \longrightarrow \{(a, \beta) \in A \times (B \otimes I_R) : (\mathrm{id} \otimes \mathrm{ev}_0)\beta = f(a),\ (\mathrm{id} \otimes \mathrm{ev}_1)\beta = 0\},$$

given in degree n by

$$(a, y) \longmapsto (a, f(a)e_0 + (-1)^n y h).$$

Here be_i denotes $b \otimes e_i$.

Proof. The relations identify the graded algebra with the path algebra of the single directed edge $0 \to 1$, where that edge has degree one. Thus multiplication is associative. Checking the four idempotent products and the products with h verifies the Leibniz rule. Every element on the right has the unique indicated form. The coefficient of h in its differential is $(-1)^{n+1} f(a) + (-1)^n dy$, which equals $(-1)^{n+1}(f(a) - dy)$. In a product of two such elements, only $e_0 h = h$ contributes an edge term. That coefficient is $(-1)^m f(a) y'$. This equals $(-1)^{n+m}$ times the second component in (36.2). □

The constant inclusion $R \to I_R$ is a quasi-isomorphism: its quotient has differential an isomorphism between two rank-one modules. Thus the interval description is an integral path-fiber construction. It requires no polynomial integration or division by integers.

The opposite choice of endpoint gives a second relative product. Its relation to the first can be computed without changing the underlying cochain complex.

Proposition 36.2.4. On $P(f)$ the formula

$$\mu_R((a, y), (a', y')) = (aa', yf(a'))$$

also defines an associative differential graded product. Let μ_L denote (36.2). The degree-minus-one bilinear map

$$K((a, y), (a', y')) = (0, (-1)^{n+1} y y'), \qquad |(a, y)| = n,$$

satisfies

$$DK + K(D \otimes 1 + 1 \otimes D) = \mu_L - \mu_R.$$

Consequently both products induce the same product on relative cohomology.

Proof. For μ_R, both triple products have second component $yf(a')f(a'')$. Its Leibniz rule follows by expanding $d(yf(a'))$ with $|y| = n - 1$. For the homotopy identity, the three second components on the left are

$$\begin{aligned} &(-1)^n dy\, y' - y\, dy', \\ &(-1)^n (f(a) - dy) y', \\ &-y(f(a') - dy'). \end{aligned}$$

Their sum is $(-1)^n f(a)\,y' - y f(a')$, as required. On cocycles the difference of products is a boundary. $\square$

The right product is the specialization with vanishing second local term of the gluing-cone product

$$(u,a,b)(u',a',b') = (uu',\,aa',\,b\,r(u') + (-1)^{|u|}s(a)b').$$

Here r and s are differential graded algebra maps to the same target, and the cone differential is $(du, da, r(u) - s(a) - db)$. Proposition 36.2.4 compares the induced relative cohomology products. Theorem 36.4.2 extends this binary comparison to an isomorphism of the full bar coalgebras.

COROLLARY 36.2.5. Put $P_r = P(f_r)$. Its cohomology is the relative global-to-finite-local continuous Galois cohomology, with the exact sequence

$$\cdots \to H^{n-1}(B_r) \to H^n(P_r) \to H^n(A_r) \xrightarrow{H^n(f_r)} H^n(B_r) \to \cdots .$$

For $S_f \neq \varnothing$, $H^0(P_r) = 0$, and

$$0 \longrightarrow R_r^{S_f}/R_r \longrightarrow H^1(P_r) \longrightarrow \ker(H^1(A_r) \to H^1(B_r)) \longrightarrow 0,$$

where $R_r \to R_r^{S_f}$ is the diagonal inclusion.

Proof. The short exact sequence $B_r[-1] \to P_r \to A_r$ gives the long exact sequence. Both global and local degree-zero cohomology consist of constants, and their restriction is the diagonal inclusion. $\square$

36.3 The bar coalgebra and its universal map

The relative product supplies the multiplication needed for a bar differential. The tensor bar and its twisting-cochain description are classical.[2] For any nonunital differential graded R-algebra P, its unitization is

$$U = R \oplus P, \qquad (r,p)(t,q) = (rt,\, rq + tp + pq), \qquad \epsilon(r,p) = r.$$

The differential is zero on R and equals D on P. In particular $P = \ker\epsilon$. Define the tensor coalgebra

$$\mathcal{B}(U) = \bigoplus_{j\geq 0} (sP)^{\otimes j}, \qquad \Delta[v_1|\cdots|v_j] = \sum_{i=0}^{j} [v_1|\cdots|v_i] \otimes [v_{i+1}|\cdots|v_j].$$

The empty word is 1, the counit projects onto its coefficient, and the coaugmentation includes it. The reduced coproduct $\bar\Delta$ omits its two end terms. Repeated reduced coproducts vanish on each word, which is the conilpotence convention used here.

THEOREM 36.3.1. There is a unique coderivation b on $\mathcal{B}(U)$ with $b(1) = 0$ whose projections to sP are

$$b_1(sa) = -s(Da), \qquad b_2(sa \otimes sb) = (-1)^{|a|}s(ab), \qquad b_j = 0 \quad (j > 2).$$

It has degree one and satisfies $b^2 = 0$ over every commutative ring. Consequently $\mathcal{B}(U)$ is a conilpotent differential graded coalgebra.

[2] J.-L. Loday and B. Vallette, *Algebraic Operads* (2012), §2.2. The proof below fixes the cohomological signs and works over the displayed commutative ring.

Proof. On a word, insert b_1 in every one-letter position and b_2 in every adjacent two-letter position. An insertion after $v_1, \dots, v_i$ has sign $(-1)^{|v_1|+\cdots+|v_i|}$. Cutting a word before or after the inserted block proves the coderivation identity. Conversely that identity and projection to one-letter words determine the formula.

In the square, two insertions with disjoint supports cancel: reversing their order crosses two degree-one operations. The overlapping insertions have supports of lengths one, two, or three. On one letter the result is $sD^2a = 0$. On two letters, the three terms are

$$(-1)^{|a|+1}sD(ab) + (-1)^{|a|}s((Da)b) + s(aDb) = 0.$$

On three letters, the two terms are

$$(-1)^{|b|}s((ab)c) - (-1)^{|b|}s(a(bc)) = 0.$$

They vanish by the Leibniz rule and associativity. Every term in the square is of one of these types, so the proof covers every word length. It uses cancellation of opposite signs, including in characteristic two, and never divides by two. □

A twisting cochain packages the compatibility between a differential and a multiplication. For a coaugmented conilpotent differential graded coalgebra C and an augmented differential graded algebra U, it is a degree-one map $t : C \to U$ with $t(1) = 0$, $\epsilon t = 0$, and

$$d_U t + t d_C + \mu(t \otimes t)\bar{\Delta} = 0. \tag{36.3}$$

The tensor sign in this formula is part of the stated convention.

Theorem 36.3.2. The map $\tau : \mathcal{B}(U) \to U$ sending sa to a and all other lengths to zero is a twisting cochain. Composition with τ gives a bijection between differential graded coalgebra maps $C \to \mathcal{B}(U)$ preserving coaugmentation and twisting cochains $C \to U$.

Proof. On length one, (36.3) reads $Da - Da = 0$. On length two, its last two terms are $(-1)^{|a|}ab$ and $(-1)^{|a|-1}ab$. Other lengths contribute zero. For a twisting cochain t, define

$$F_t(c) = \epsilon_C(c)1 + \sum_{j \geq 1} (st)^{\otimes j}\bar{\Delta}^{(j-1)}(\bar{c}), \qquad \bar{c} = c - \epsilon_C(c)1.$$

Conilpotence makes the sum finite for each c. Coassociativity gives $\Delta F_t = (F_t \otimes F_t)\Delta$, and its one-letter projection is st. Those properties also prove uniqueness by applying repeated reduced coproducts and then projecting every output to length one. The difference $bF_t - F_t d_C$ satisfies the coderivation identity relative to F_t. Its one-letter projection is zero precisely when (36.3) holds. Repeated reduced coproducts show that a relative coderivation with zero one-letter projection is zero. This proves that F_t commutes with differentials and proves both directions. □

36.4 Changing the endpoint in every arity

A binary homotopy compares two products, but a bar map must also compare their compositions on longer words. Ordered products of the local primitives supply all the required components. For the identity map $f : R \to R$, put $u = s(1, 0)$ and $v = s(0, 1)$. Their degrees are -1 and 0. The bar conventions above give

$$b_1 u = -v, \qquad b_1 v = 0, \qquad b_{2,L}(u, v) = v, \qquad b_{2,R}(v, u) = -v.$$

Both binary operations send (u, u) to u; their other values on these generators vanish. The comparison below sends

$$[v|v] \longmapsto [v|v] + [v], \qquad [v|v|v] \longmapsto [v|v|v] + 2[v|v] + [v].$$

Each term contracts consecutive blocks of local primitives. This ordered contraction makes sense without commutativity of the algebras.

Let R be any commutative ring. Consider a cospan of associative differential graded R-algebras

$$A \xrightarrow{f} B \xleftarrow{g} C.$$

Put

$$P^n = A^n \oplus C^n \oplus B^{n-1}, \qquad D(a, c, y) = (d_A a, d_C c, f(a) - g(c) - d_B y). \tag{36.4}$$

For $x = (a, c, y)$ of degree p define

$$\mu_L(x, x') = (aa', cc', (-1)^p f(a)\, y' + y\, g(c')), \tag{36.5}$$

$$\mu_R(x, x') = (aa', cc', y f(a') + (-1)^p g(c)\, y'). \tag{36.6}$$

The relative case is obtained by taking $C = 0$ in the category of nonunital algebras. If f, g are unital, $(1_A, 1_C, 0)$ is a unit for both gluing products.

LEMMA 36.4.1. The two formulas (36.5) and (36.6) define differential graded associative algebras on (36.4).

Proof. The square of D vanishes because f and g commute with differentials. Write the degrees of three inputs as p, q, r. The third component of either parenthesization of μ_L is

$$(-1)^{p+q} f(a) f(a')\, y'' + (-1)^p f(a)\, y'\, g(c'') + y\, g(c')\, g(c'').$$

For μ_R the corresponding common component is

$$y f(a') f(a'') + (-1)^p g(c)\, y' f(a'') + (-1)^{p+q} g(c)\, g(c')\, y''.$$

The first two components are associative by assumption.

For the Leibniz identity of μ_L, the third component of $D\mu_L(x, x')$ is

$$\begin{aligned} f(a) f(a') - g(c) g(c') - (-1)^p f(d_A a)\, y' - f(a) d_B y' \\ - d_B y\, g(c') + (-1)^p y\, g(d_C c'). \end{aligned}$$

Expanding $\mu_L(Dx, x') + (-1)^p \mu_L(x, Dx')$ gives this expression; the terms $f(a)\, g(c')$ cancel. For μ_R, the third component of $D\mu_R(x, x')$ is

$$\begin{aligned} f(a) f(a') - g(c) g(c') - d_B y\, f(a') + (-1)^p y f(d_A a') \\ - (-1)^p g(d_C c)\, y' - g(c) d_B y'. \end{aligned}$$

The same expansion on the Leibniz side gives these terms, with the two occurrences of $g(c) f(a')$ cancelling. The first two components obey the original Leibniz identities. This proves the assertion over R without division. □

The endpoint isomorphism

For either product, first regard P as a nonunital algebra and use its augmented unitization $U = R \oplus P$. This fixes the meaning of the normalized bar even when the gluing algebra already has a unit. The normalized bar is

$$\mathcal{B}_L = T^c_R(sP) = \bigoplus_{n\geq 0} (sP)^{\otimes_R n}, \qquad \mathcal{B}_R = T^c_R(sP),$$

where $s : P \to P[1]$ has degree -1. Its coproduct is deconcatenation; the summand R is the empty word, counit target, and coaugmentation source. Its coderivation is determined by

$$b_1(sx) = -sDx, \qquad b_{2,L/R}(sx, sx') = (-1)^{|x|} s\mu_{L/R}(x, x'). \tag{36.7}$$

Higher Taylor coefficients of these differentials vanish. Lemma 36.4.1 implies $b^2_L = b^2_R = 0$ by the one-, two-, and three-input checks.

We identify the suspended module with

$$W = sP = sA \oplus sC \oplus B, \qquad s(a, c, y) \longleftrightarrow (sa, sc, y).$$

Thus, if $w_i = s(a_i, c_i, y_i)$ has degree q_i, then $|y_i| = q_i$. Define degree-zero maps $F_n : W^{\otimes n} \to W$ by

$$F_1 = \mathrm{id}_W, \qquad F_n(w_1, \dots, w_n) = s(0, 0, y_1 y_2 \cdots y_n) \quad (n \geq 2). \tag{36.8}$$

The multiplication in the last expression is the ordered multiplication of B. In particular, it introduces no reordering sign. Equation (36.8) uses the suspended convention (36.7); signs in an unsuspended convention must be obtained by the tensor rule, not copied without change.

Theorem 36.4.2 *(Integral endpoint change).* For every cospan of differential graded associative algebras over R, the maps (36.8) are the corestrictions of a coaugmentation- and counit-preserving isomorphism

$$\mathcal{F} : (\mathcal{B}_L, b_L, \Delta_{\mathrm{dec}}) \xrightarrow{\cong} (\mathcal{B}_R, b_R, \Delta_{\mathrm{dec}})$$

of differential graded coalgebras. Its inverse $\mathcal{G}$ has corestrictions

$$G_1 = \mathrm{id}_W, \qquad G_n(w_1, \dots, w_n) = (-1)^{n-1} s(0, 0, y_1 \cdots y_n) \quad (n \geq 2).$$

Both maps preserve the increasing filtration by word length. They are defined over the integers and require no division. In particular they are valid in characteristics two and three and over $\mathbb{Z}/\ell^r$.

Proof. The full coalgebra maps. For a word of length $N > 0$, partition its inputs into consecutive nonempty blocks. Explicitly,

$$\mathcal{F}[w_1|\cdots|w_N] = \sum_{N=n_1+\cdots+n_k} [F_{n_1}(w_1, \dots, w_{n_1})|\cdots|F_{n_k}(w_{N-n_k+1}, \dots, w_N)]. \tag{36.9}$$

Here $k \geq 1$, all $n_i \geq 1$, and $\mathcal{F}(1) = 1$. There are 2^{N-1} terms before any vanish. Cutting the output between two blocks is equivalent to first cutting the input there and independently partitioning the two resulting words. This proves

$$\Delta_{\mathrm{dec}}\mathcal{F} = (\mathcal{F} \otimes \mathcal{F})\Delta_{\mathrm{dec}}.$$

It also proves that (36.9) is the coalgebra map with the indicated corestrictions. The same construction gives $\mathcal{G}$. Since only finitely many partitions occur, algebraic tensor products and direct sums suffice.

The chain equation in every arity. Let $\pi_1 : T^c(W) \to W$ be projection to length one. Both $b_R\mathcal{F}$ and $\mathcal{F}b_L$ are coderivations relative to $\mathcal{F}$. Equality of their corestrictions therefore implies equality of the full maps. In this tensor coalgebra the implication can also be proved by induction on input length: all reduced coproduct terms of a difference are determined by shorter input words, leaving only a primitive difference; a primitive element of $T^c(W)$ with zero length-one component is zero.

Set

$$Q_j = q_1 + \cdots + q_j, \qquad Q_0 = 0.$$

In the coordinates above, the unary differential is

$$b_1(sa, sc, y) = (-sd_A a, -sd_C c, d_B y - f(a) + g(c)).$$

Here b_1^{tens} inserts b_1 at position j with sign $(-1)^{Q_{j-1}}$. For $n \geq 2$, write $U_n = b_1F_n - F_n b_1^{\mathrm{tens}}$. Its first two components vanish, and its third component is

$$\sum_{j=1}^{n} (-1)^{Q_{j-1}} y_1 \cdots y_{j-1} \big(f(a_j) - g(c_j)\big) y_{j+1} \cdots y_n. \tag{36.10}$$

Indeed, the terms containing $d_B y_j$ cancel by the graded Leibniz rule in B. This cancellation is the source of the higher comparison.

For $n = 1$ the chain equation holds because F_1 is the identity and the unary differentials agree. For $n = 2$, the third components of the binary bar operations are

$$\begin{aligned} b_{2,L}(w_1, w_2)_B &= f(a_1)\, y_2 - (-1)^{q_1} y_1 g(c_2), \\ b_{2,R}(w_1, w_2)_B &= -(-1)^{q_1} y_1 f(a_2) + g(c_1)\, y_2. \end{aligned}$$

Their first two components agree. Hence, as maps $W^{\otimes 2} \to W$,

$$b_{2,L} - b_{2,R} = U_2.$$

Thus

$$b_1F_2 + b_{2,R} = F_2 b_1^{\mathrm{tens}} + b_{2,L}.$$

Now let $n \geq 3$. The third component of the target binary contribution $\sum_{i=1}^{n-1} b_{2,R}(F_i, F_{n-i})$ is

$$g(c_1)\, y_2 \cdots y_n - (-1)^{Q_{n-1}} y_1 \cdots y_{n-1} f(a_n). \tag{36.11}$$

All interior splits vanish: both outputs then have only a B component, whose binary product in the gluing cone is zero. The third component of the source binary contribution is

$$\begin{aligned} &\sum_{j=1}^{n-1} (-1)^{Q_{j-1}} y_1 \cdots y_{j-1} f(a_j)\, y_{j+1} \cdots y_n \\ &\qquad - \sum_{j=2}^{n} (-1)^{Q_{j-1}} y_1 \cdots y_{j-1} g(c_j)\, y_{j+1} \cdots y_n. \end{aligned} \tag{36.12}$$

To see the second sum, move the sign of the inserted degree-one b_2 past its preceding suspended inputs, and then use the second term of $b_{2,L}$. No input factors are permuted.

Adding (36.11) to (36.10) gives exactly (36.12). The f term with $j = n$ and the g term with $j = 1$ are cancelled by the two endpoints. For $n \geq 3$ all first and second components of the remaining terms vanish. This proves the corestriction equation in every arity, and hence

$$b_R \mathcal{F} = \mathcal{F} b_L.$$

The inverse. The length-one component of either composite is the identity. For length $n \geq 2$, every contribution to its corestriction is an ordered product $y_1 \cdots y_n$ in the third component. An input with a vanishing local component gives zero in all such contributions. For $\mathcal{F}\mathcal{G}$, a partition into k blocks contributes sign $(-1)^{n-k}$, and there are $\binom{n-1}{k-1}$ such partitions. The total coefficient is

$$\sum_{k=1}^{n} \binom{n-1}{k-1} (-1)^{n-k} = (1-1)^{n-1} = 0.$$

For $\mathcal{G}\mathcal{F}$, the outer coefficient instead gives $(-1)^{k-1}$ and the same binomial cancellation applies. Thus both composites have exactly the identity corestrictions. Formula (36.9) proves that the full composites are identities. Since $\mathcal{F}$ is a chain map and $\mathcal{G}$ is its inverse, $\mathcal{G}$ is a chain map as well. □

Corollary 36.4.3. The two relative algebras of Proposition 36.2.4 are A_∞-isomorphic with identity first component, and their normalized bar coalgebras are isomorphic over R. If the gluing maps are unital, all higher components vanish when any input is the native gluing unit $(1_A, 1_C, 0)$.

Proof. The relative case is the specialization $C = 0$. A coalgebra map satisfying the bar chain equation is exactly an A_∞ map in the convention (36.7). The explicit inverse makes it an isomorphism in that sense. The higher coefficients depend on every local component; the local component of the native unit is zero. □

The isomorphism $\mathcal{F}$ preserves differential, deconcatenation, coaugmentation, counit, and the increasing word filtration. It does *not* say that the identity of P is a strict multiplicative map. Nor does it preserve the two original universal twisting cochains by an identity: if τ_R extracts a single suspended letter, then $\tau_R \mathcal{F}$ has higher components $(0, 0, y_1 \cdots y_n)$. Those components are part of the comparison, not terms to discard.

Strict naturality

A strict map of cospans consists of maps $\alpha : A \to A'$, $\gamma : C \to C'$, and $\beta : B \to B'$ with

$$\beta f = f' \alpha, \qquad \beta g = g' \gamma.$$

It induces $(a, c, y) \mapsto (\alpha a, \gamma c, \beta y)$ on the gluing cones. Because $\beta(y_1 \cdots y_n) = \beta(y_1) \cdots \beta(y_n)$, it commutes with every F_n and G_n, and therefore with both full coalgebra maps. If H_L and H_R denote the induced tensorwise bar maps, then

$$\begin{array}{ccc} \mathcal{B}_L & \xrightarrow{\mathcal{F}} & \mathcal{B}_R \\ {\scriptstyle H_L}\downarrow & & \downarrow{\scriptstyle H_R} \\ \mathcal{B}'_L & \xrightarrow{\mathcal{F}'} & \mathcal{B}'_R \end{array}$$

commutes. For a ring map $R \to S$, the canonical identifications $S \otimes_R \mathcal{B}_{L/R} \cong \mathcal{B}_{L/R}(S \otimes_R P)$ identify $S \otimes_R \mathcal{F}$ and $S \otimes_R \mathcal{G}$ with the corresponding maps after scalar extension. Tensor powers and direct

sums commute with this extension, so no flatness assumption is needed. The same formulas commute with compatible coefficient reductions. An arbitrary square commuting only up to an unspecified chain homotopy is not covered by this argument.

36.5 The derived tensor interpretation and its hypothesis

The underlying relative complex computes the cohomology in Corollary 36.2.5. The bar coalgebra has a different homological interpretation, for which flatness of complexes matters. A complex M over R is called K-flat if $N \otimes_R M$ is acyclic for every acyclic complex N. Degreewise flatness alone is not substituted for this condition on an unbounded complex.

THEOREM 36.5.1. Let $U = R \oplus P$ as above, and assume that the underlying complex P is K-flat over R. Then $\mathcal{B}(U)$ represents $R \otimes^{\mathbf{L}}_U R$. Without this hypothesis, Theorems 36.3.1 and 36.3.2 still construct the coalgebra and its universal twisting cochain, but this derived tensor assertion is not made.

Proof. On $L = U \otimes_R \mathcal{B}(U)$ put

$$d_L(u \otimes c) = d_U u \otimes c + (-1)^{|u|} u \otimes bc - (-1)^{|u|} \sum u\tau(c_{(1)}) \otimes c_{(2)},$$

where $\Delta c = \sum c_{(1)} \otimes c_{(2)}$. The twisting equation implies $d_L^2 = 0$ by expansion. In word notation, the last sum removes only the first letter. The augmentation $L \to R$ is ϵ_U on length zero and zero elsewhere.

Write $\bar{u} = u - \epsilon_U(u)1$. An R-linear homotopy of degree -1 is

$$h(u \otimes [sa_1|\cdots|sa_j]) = -1 \otimes [s\bar{u}|sa_1|\cdots|sa_j].$$

An entry $s0$ gives zero. In $d_L h + h d_L$, the internal differential on $\bar{u}$ cancels, as do insertions within the remaining word. The first multiplication gives $\bar{u} \otimes c$. The two terms that join $\bar{u}$ to a_1 leave $\epsilon_U(u) \otimes c$ when $j > 0$. Thus $d_L h + h d_L = \mathrm{id} - \iota\epsilon_L$, including $j = 0$. This proves that $L \to R$ is a quasi-isomorphism.

Filter L by word length at most j. Its successive quotients are $U \otimes_R (sP)^{\otimes j}$ with their internal differentials. If N is an acyclic right U-module, their tensors with N are $N \otimes_R (sP)^{\otimes j}$, which are acyclic by K-flatness. The filtration sequences are split on graded modules. Induction and exactness of filtered unions show that $N \otimes_U L$ is acyclic. Hence L is K-flat as a left U-module. Tensoring this resolution with the right augmentation module R kills its first-letter term and leaves exactly $(\mathcal{B}(U), b)$. □

For the arithmetic algebra $U_r = R_r \oplus P_r$, define $\mathcal{B}_r = \mathcal{B}(U_r)$. Its construction and functoriality do not require a K-flat replacement. Theorem 36.5.1 applies if such a hypothesis is established for the chosen arithmetic model. No identification of its bar cohomology with relative Galois cohomology is asserted: the latter is the cohomology of P_r itself.

36.6 Coefficient maps and completion

A commuting square of differential graded algebra maps

$$\alpha : A \to A', \qquad \beta : B \to B', \qquad \beta f = f' \alpha$$

induces $(a, y) \mapsto (\alpha(a), \beta(y))$ on the relative algebras. The formulas for D, multiplication, and b prove functoriality directly. This includes coefficient maps and compatible diagrams of continuous group

homomorphisms. Changing a local embedding by conjugation does not give such a strictly commuting square without additional choices.

A comparison of resolutions may commute only up to cochain homotopy. The correction to the relative map then has a measurable product defect.

Proposition 36.6.1. Let $\alpha : A \to A'$ and $\beta : B \to B'$ be differential graded algebra maps, and let $H : A \to B'$ have degree -1 with $f'\alpha - \beta f = dH + Hd$. Then

$$F : P(f) \to P(f'), \qquad F(a, y) = (\alpha(a), \beta(y) + H(a))$$

is a cochain map. For $x = (a, y)$ of degree n and $x' = (a', y')$, the first component of $F(x)F(x') - F(xx')$ is zero. Its second component is

$$\begin{aligned}(-1)^n &(f'\alpha - \beta f)(a)\beta(y') \\ &+ (-1)^n f'\alpha(a)H(a') - H(aa').\end{aligned}$$

In particular F induces the strict bar map constructed above if and only if this expression vanishes for every pair of inputs.

Proof. The second component of $D'F - FD$ is $(f'\alpha - \beta f)(a) - dH(a) - H(da) = 0$. Expanding the two products in (36.2) gives the displayed defect. A cochain map induces the tensorwise strict bar map precisely when it preserves the binary multiplication: projection of the bar-map equation on two letters is that identity. On one letter it is the cochain-map equation, and all other Taylor coefficients vanish. □

Thus cochain transport of a local primitive can preserve relative classes without supplying the multiplicativity required by the bar construction. A sheaf-cochain comparison must also control this defect or supply coherent higher components.

Proposition 36.6.2. For the Galois cochains with finite coefficients, reduction gives degreewise surjections and canonical isomorphisms

$$P_{r+1} \otimes_{R_{r+1}} R_r \cong P_r, \qquad \mathcal{B}_{r+1} \otimes_{R_{r+1}} R_r \cong \mathcal{B}_r.$$

They preserve differentials, relative products, coproducts, and universal twisting cochains.

Proof. A continuous map from a compact space to a finite discrete ring is constant on the parts of a finite clopen partition. Lift each of its values to R_{r+1}. A normalized cochain has value zero on every degenerate tuple, and choosing the zero lift preserves normalization. If a cochain reduces to zero, divide its finitely many values by ℓ^r and choose lifts. Its kernel is therefore exactly $\ell^r C^n(G, R_{r+1})$. This proves the cochain assertion. Finite products and the two summands of P_r commute with reduction. Tensor powers and direct sums do also. Every defining formula uses addition and multiplication of coefficients, which proves all the compatibilities. Each cochain module is also flat over its finite coefficient ring. Indeed, normalized functions constant on a fixed finite clopen partition form a finite free module: omit the cells meeting the degenerate tuples. Refining partitions gives a filtered union equal to all normalized cochains. Filtered colimits of flat modules are flat, since tensoring and filtered colimits preserve exact sequences. Thus every degree of P_r and $\mathcal{B}_r$ is flat. This statement concerns modules in each degree, not K-flatness of the whole complex. □

Continuous $\mathbb{Z}_\ell$-valued cochains are the inverse limits of these finite cochains. Define the degreewise inverse limit

$$\widehat{\mathcal{B}} = \varprojlim_r \mathcal{B}_r.$$

Its tensor target is specified as

$$\widehat{\mathcal{B}} \,\widehat{\otimes}\, \widehat{\mathcal{B}} := \varprojlim_r (\mathcal{B}_r \otimes_{R_r} \mathcal{B}_r).$$

This is the completion used for the coproduct; it makes no implicit identification with another completed tensor product.

THEOREM 36.6.3. The compatible maps on $\mathcal{B}_r$ give a continuous square-zero differential and a coassociative counital coproduct on $\widehat{\mathcal{B}}$. The coderivation identity holds in the displayed completed tensor target. Each reduction is conilpotent. Ordinary conilpotence of $\widehat{\mathcal{B}}$ need not hold.

Proof. Every identity holds after every projection to r. Equality in an inverse limit is componentwise equality, so all the stated identities follow. Degreewise surjectivity of the transitions allows each element of $\mathcal{B}_r$ to be lifted successively to all higher indices. Moreover the kernel of the projection is $\ell^r\widehat{\mathcal{B}}$: in coefficients modulo ℓ^{r+k} divide a kernel element by ℓ^r and reduce modulo ℓ^k. These divided elements are well-defined because degreewise flatness identifies the annihilator of ℓ^r with $\ell^k\mathcal{B}_{r+k}$. They are compatible as k varies. Thus the projections are the coefficient reductions.

For failure of ordinary conilpotence, take $R = \mathbb{Z}_\ell$, a free rank-one $P = Ra$ in degree one with zero differential and zero product. In its completed bar, $z = \sum_{j\geq 1} \ell^j (sa)^{\otimes j}$ is defined because it has finite support modulo every ℓ^r. Every iterated reduced coproduct of z has a nonzero component of sufficient word length. Hence z is not conilpotent. □

The theorem concerns identities in the completed complex. An exchange of cohomology with inverse limit needs a separate vanishing statement for the corresponding $\varprojlim^1$ of cohomology groups. No such exchange enters the construction.

COROLLARY 36.6.4 *(Restricted adic completion).* For the relative Galois algebras P_r, let $\mathcal{B}_{r,L}$ and $\mathcal{B}_{r,R}$ use the two endpoint products. The maps of Theorem 36.4.2 induce a continuous isomorphism

$$\widehat{\mathcal{F}} : \varprojlim_r \mathcal{B}_{r,L} \xrightarrow{\cong} \varprojlim_r \mathcal{B}_{r,R},$$

whose inverse is $\varprojlim_r \mathcal{G}_r$. It preserves the completed differential and coproduct in

$$\varprojlim_r (\mathcal{B}_{r,R} \otimes_{R_r} \mathcal{B}_{r,R}).$$

Proof. The maps and inverses commute with each transition. Their two composite identities and their differential and coproduct identities hold at every finite coefficient stage. Equality in the inverse limit is componentwise equality. The maps are continuous for the inverse-limit topology because their composites with the projections factor through the corresponding finite stage. No exchange of cohomology with inverse limit is used. □

A map preserving the *increasing* filtration by word length need not be continuous for the *decreasing* filtration defining an unrestricted product of word lengths. The endpoint-change map supplies a minimal test.

Proposition 36.6.5. In general $\mathcal{F}$ has no continuous extension from $T_R^c(W)$ to

$$\prod_{n\geq 0} W^{\otimes_R n}$$

with discrete word coordinates, even for the identity map $f : \mathbb{Z} \to \mathbb{Z}$ concentrated in degree zero.

Proof. Take the relative map $f : R \to R$ to be the identity and let

$$v = s(0, 1) \in sP.$$

Its degree is zero and $F_n(v^{\otimes n}) = v$ for all $n \geq 1$. The partial sums

$$z_N = \sum_{n=1}^{N} v^{\otimes n}$$

converge in the word-length product topology: each word coordinate eventually stabilizes. But the length-one coordinate of $\mathcal{F}(z_N)$ is Nv, which is not eventually constant in the discrete module Rv. Hence the image sequence does not converge and no continuous extension agreeing with $\mathcal{F}$ can exist. □

By contrast, $z = \sum_{n\geq 1} \ell^n v^{\otimes n}$ belongs to the restricted adic completion: modulo every ℓ^r only finitely many word lengths occur. The length-one coordinate of its image is the well-defined ℓ-adic sum

$$\sum_{n\geq 1} \ell^n v = \frac{\ell}{1-\ell}\, v.$$

The denominator here is a unit in $\mathbb{Z}_\ell$. It is not a division used in the definition of the coalgebra map. This example distinguishes two completions on the very same generators.

36.7 Corestriction and arithmetic comparison

Coefficient reduction changes the ring while preserving the cochain products. Corestriction along a field extension has a different multiplicative behavior. For coefficients $\mathbb{Z}_\ell$, cochains mean the inverse limits of the finite coefficient cochains.

Proposition 36.7.1. Let $H \subset G$ be an open subgroup of index $n > 1$. With trivial $\mathbb{Z}_\ell$ coefficients, corestriction cannot be induced by a multiplicative cochain map on the ordinary cochain algebras. Nor can it be the first component of an A_∞ morphism inducing corestriction on cohomology. For index ℓ, the multiplicativity obstruction remains nonzero modulo ℓ^2.

Proof. Corestriction in degree zero sums over the cosets, so it sends $1 \in H^0(H, \mathbb{Z}_\ell)$ to $n \in H^0(G, \mathbb{Z}_\ell)$. Multiplicativity would imply $n = n^2$, which is false in $\mathbb{Z}_\ell$. An A_∞ morphism has a multiplicative map on cohomology, since its second identity makes its product defect a boundary. It therefore has the same obstruction. For $n = \ell$, $n^2 - n$ has ℓ-adic valuation one. The defects are 2 modulo 4 and 6 modulo 9 for $\ell = 2, 3$. □

Thus a coefficient inverse limit of bar coalgebras does not supply a multiplicative corestriction limit over a cyclotomic field tower. An arithmetic Hecke trace class $\mathcal{V}^{\mathrm{prim}}$ requires a separate comparison with these relative Galois algebras. Such a comparison must specify a chain algebra, augmentation, coefficient maps, and compatible local products. The algebraic bar constructed here has a square-zero differential and

endpoint-independent coalgebra type. Identifying it with an arithmetic chiral bar additionally requires geometric operations and their compatibility with residue maps and completion.

A finite arithmetic Chern–Simons action instead evaluates a Galois cohomology class against arithmetic duality data. Chapter 37 addresses that action and its proposed quantum boundary extension. The relative complex here records finite-local primitives; it supplies neither archimedean compact-support conventions nor an arithmetic fundamental-class pairing. A comparison with states, operators, or the full Weil functional requires further maps preserving the corresponding pairings and analytic domains.

Chapter 37

Arithmetic Chern–Simons Holography

Class-$\mathcal{S}$, Higgs-branch vertex algebras, and Costello–Gaiotto boundary categories require a complex curve or a quantized boundary theory. Kim arithmetic Chern–Simons supplies only the classical arithmetic action line. Finite state sums and the conditional boundary category appear in Chapter 103; the profinite comparison with $\mathcal{V}^{\mathrm{prim}}$ remains open.

The relative Galois complex of Chapter 36 retains a global cocycle together with chosen finite-local primitives. Its endpoint comparison is an integral bar-coalgebra isomorphism. An arithmetic action additionally requires a duality pairing and its archimedean conventions. Those data do not follow from the bar differential.

37.1 Class $\mathcal{S}$ Fails Arithmetically

Theorem 37.1.1 *(Gaiotto, arXiv:0904.2715).* . Class-$\mathcal{S}$ builds 4D $\mathcal{N} = 2$ theories $T[\Sigma, \mathfrak{g}]$ by compactifying the 6D $(2, 0)$ theory $\mathfrak{X}[\mathfrak{g}]$ on a punctured Riemann surface Σ.

Observation 37.1.2 *(three obstructions).* Arithmetic class-$\mathcal{S}$ on $\overline{\mathrm{Spec}\,\mathbb{Z}} \times \mathbb{R}^4$ fails:

(i) 6D $(2, 0)$ requires a smooth oriented 6-manifold; $\overline{\mathrm{Spec}\,\mathbb{Z}} \times \mathbb{R}^4$ is topologically 5-dimensional (arithmetic 1-dim plus 4-dim real). Connes' spectral dimension of $\overline{\mathrm{Spec}\,\mathbb{Z}}$ is 1, not 2 (Connes–Consani arXiv:1405.4527).

(ii) The self-dual 2-form B on an M5-brane has no Arakelov analogue (no star on 2-forms on a 1-d arithmetic curve, where $\Omega^2 = 0$).

(iii) Hitchin-moduli Higgs branches on $\overline{\mathrm{Spec}\,\mathbb{Z}}$ are not defined by the class-$\mathcal{S}$ formula: the Arakelov dualizing sheaf is a metrised arithmetic line, not the holomorphic canonical bundle of a Riemann surface.

Corollary 37.1.3. . Class-$\mathcal{S}$ as a generating principle for arithmetic vertex algebras is obstructed.

37.2 The Hitchin Base Has No Class-$\mathcal{S}$ Arithmetic Specialization

Theorem 37.2.1 *(arithmetic Hitchin obstruction).* . The class-$\mathcal{S}$ Coulomb branch

$$C_{\Sigma,\mathfrak{g}} = \bigoplus_i H^0(\Sigma, K_\Sigma^{d_i})$$

uses the holomorphic canonical bundle of a complex curve. For $\overline{\mathrm{Spec}\,\mathbb{Z}}$ the dualizing object is the Arakelov line with metric at infinity. Its sections belong to Arakelov intersection theory, not to a Hitchin base. Hence the class-$\mathcal{S}$ Coulomb and Higgs branches do not vanish; rather, the formula defining them has no canonical arithmetic value.

Remark 37.2.2 *(why the replacement is three-dimensional).* This obstruction is not a proof that an arithmetic 4-dimensional theory is topological. It says that the class-$\mathcal{S}$ descent from a 6-dimensional theory

along a complex curve has no arithmetic curve to descend along. Kim's arithmetic Chern–Simons theory is therefore not the topological remnant of that construction; it is a separate 3-dimensional arithmetic field theory.

37.3 KIM'S ARITHMETIC CHERN–SIMONS

Definition 37.3.1 *(Kim 2015, arXiv:1510.05818).* For a number field F and finite Galois module M, arithmetic Chern–Simons theory assigns to a Galois representation $\rho\colon \pi_1^{\text{ét}}(\overline{\operatorname{Spec} \mathcal{O}_F}) \to G$ the action

$$\mathrm{CS}_F(\rho) \;=\; \langle \rho^* c, [\overline{\operatorname{Spec} \mathcal{O}_F}] \rangle \;\in\; H^3(\overline{\operatorname{Spec} \mathcal{O}_F}, \mu_N)$$

built from a class $c \in H^3(BG, \mu_N)$. This is a genuine 3D arithmetic Chern–Simons action, not yet an Atiyah–Segal quantum TQFT. The topology is the Mazur–Morishita dictionary: primes are knots and $\overline{\operatorname{Spec} \mathbb{Z}}$ is the arithmetic 3-manifold (Morishita arXiv:0904.3399).

Remark 37.3.2. Chung–Kim–Kim–Pappas–Park–Yoo arXiv:1706.03336 extends Kim's construction to higher-dimensional arithmetic manifolds. Arithmetic Chern–Simons supplies the rigorous classical action functional at the arithmetic vertex that Witten's analysis (Chapter 34.8.1) identifies as the ingredient Deninger's programme requires.

37.4 $\mathcal{V}^{\mathrm{prim}}$ AND THE ACS BOUNDARY SHADOW

THEOREM 37.4.1 *(finite arithmetic CS boundary shadow).* . Let G be a finite gauge group and $c \in H^3(BG, \mu_N)$ a Kim arithmetic Chern–Simons level. Kim's theory supplies a classical arithmetic CS torsor/action line for this finite datum. The expression $\mathrm{ACS}[\overline{\operatorname{Spec} \mathbb{Z}}, \mathbb{G}_m, 1]$ is not an instance of Kim's finite-gauge construction; it is a separate abelian extension problem. A quantum boundary category $\partial_q \mathrm{ACS}$, once constructed, has conjectural archimedean Heisenberg factorization shadow

$$\chi\big(\partial_q \mathrm{ACS}\big)(s) \;=\; \prod_p (1 - p^{-s})^{-1},$$

and is conjecturally compared with the Hochschild-trace candidate $\mathcal{V}^{\mathrm{prim}}$ (a Hochschild trace class; Chapter 33.5.1). The quantum boundary category at finite places is conditional, as in Chapter 103.

Form. Kim's construction is classical: it assigns Chern–Simons actions and torsors to Galois representations. Witten's CS/WZW boundary theorem transports to $\overline{\operatorname{Spec} \mathbb{Z}}$ only after replacing the complex boundary curve and its OPE by the arithmetic integral transform of Chapter 103. That chapter defines the finite-level boundary category by transgression, conditional on the finite arithmetic cobordism input; the displayed Euler product is then a character-level expectation for the renormalised profinite boundary comparison with the Hochschild-trace class $\mathcal{V}^{\mathrm{prim}}$. □

COROLLARY 37.4.2 *(character = ζ).* .

$$\chi(\partial_q \mathrm{ACS})(s) \;=\; \prod_p (1 - p^{-s})^{-1} \;=\; \zeta(s),$$

conditional on constructing the quantum boundary category and identifying its Hochschild trace with the Hochschild-trace candidate.

37.5 Wilson Lines Are Labeled by Ramified Hecke Characters

Theorem 37.5.1 *(Wilson-line labels as local Hecke data).* . Wilson lines in ACS wrapped around the "knot" corresponding to a prime p (Mazur–Morishita arithmetic topology) are 1-dimensional defects in the arithmetic 3-fold, each *labeled by* a local Hecke character χ_p of $\mathbb{Q}_p^\times$ (ramified or unramified). The label is a closed point of the universal family of local characters, and conjecturally a module over the quantum boundary category specialized at p. Inserting a Wilson line with label χ_p produces a boundary defect whose local datum at p is the ramification datum of χ_p (conjectural; the categorical boundary module is part of the conditional boundary-category input of Chapter 103). Wilson-line labels are 0-dimensional data; the Wilson line itself is 1-dimensional.

Remark 37.5.2 *(defect/puncture dictionary).* Class-$\mathcal{S}$ punctures correspond to 6D codimension-2 defects in the complex setting; arithmetic ramification corresponds to Wilson lines in arithmetic Chern–Simons. The structural dictionary "punctures = defects" becomes "Wilson lines = ramification" in the arithmetic setting, with the 6D/4D framing replaced by the 3D arithmetic TQFT framing.

37.6 Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees as Adelic Character

Theorem 37.6.1 *(arithmetic Schur index).* . Let $\mathcal{H}^{\mathrm{arith}}$ denote the conjectural arithmetic Hilbert space on which Frobenius operators Frob_p act (one per prime p), and let the Schur grading at p be $t_p^{\mathrm{Frob}_p}$ (the formal Frob_p-graded trace variable). The arithmetic Schur index is the formal adelic graded trace

$$\mathcal{I}^{\mathrm{arith}}(\{t_p\}) \;=\; \mathrm{Tr}_{\mathcal{H}^{\mathrm{arith}}} \prod_p t_p^{\mathrm{Frob}_p}.$$

Under the Hecke substitution $t_p = p^{-s}$, $\mathcal{I}^{\mathrm{arith}}(s) = \prod_p (1 - p^{-s})^{-1} = \zeta(s)$. This is an Euler-character computation, not a construction of the Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees Schur-index category.

Remark 37.6.2. The Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees correspondence takes holographic form only after the finite-gauge ACS datum, boundary category, and trace functor have been constructed. The arithmetic vertex algebra generating problem, for which class-$\mathcal{S}$ is obstructed (Chapter 37.1.3), admits arithmetic CS holography as a conditional comparison, not as a theorem of the present finite-gauge formalism.

37.7 Arithmetic CS as the Generating Principle

(i) Class-$\mathcal{S}$ obstructed: three 4D-gauge obstructions (Chapter 37.1.3).

(ii) The class-$\mathcal{S}$ Hitchin base has no canonical arithmetic specialization on $\overline{\mathrm{Spec}\,\mathbb{Z}}$ (Chapter 37.2.1).

(iii) Kim's arithmetic Chern–Simons (arXiv:1510.05818) is the classical 3D action functional (Chapter 37.3.1).

(iv) Finite state sums are constructed in Chapter 103; the renormalised profinite boundary category is conjectural, as is its comparison with $\mathcal{V}^{\mathrm{prim}}$ and its character $\zeta(s)$ (Chapter 37.4.1).

(v) Wilson lines are 1-dimensional defects labeled by ramified Hecke characters (= expected modules of the quantum boundary category specialized at p, 0-dimensional data) (Chapter 37.5.1).

(vi) BLLPRvR adelic character: $\zeta(s)$ is the conjectural Schur index character of arithmetic CS (Chapter 37.6.1).

Kim arithmetic $\mathbb{G}_m$-Chern–Simons on $\overline{\operatorname{Spec}\,\mathbb{Z}}$ at level 1 supplies the classical arithmetic action. Chapter 103 constructs finite state sums and states the boundary category as a conditional profinite problem; the renormalised comparison with $\mathcal{V}^{\mathrm{prim}}$ remains open.

Chapter 38

The Hochschild Trace of the Iwahori Hecke Category

The primary object is $\mathcal{V}^{\mathrm{prim}}$, the Hochschild trace class of the identity on the global arithmetic Iwahori Hecke category of GL_2. This chapter places $\mathcal{V}^{\mathrm{prim}}$ on the arithmetic Langlands diamond via Bezrukavnikov's affine Hecke equivalences (arXiv:1209.0403), Mirković–Vilonen geometric Satake (arXiv:math/0401222), Ben-Zvi–Nadler character theory (arXiv:0904.1247), Arinkin–Gaitsgory spectral decomposition (arXiv:1201.6343), the Emerton–Gee (φ, Γ) stack (arXiv:1908.07185), and Fargues–Scholze geometrization (arXiv:2102.13459).

38.1 The Naive $\mathrm{Bun}_{GL_2}(\mathrm{Spec}\,\mathbb{Z})$ Has Only One Coarse Point

Theorem 38.1.1 *(Minkowski).* . $\pi_1^{\text{ét}}(\mathrm{Spec}\,\mathbb{Z}) = 1$: every étale cover of Spec $\mathbb{Z}$ is trivial. Class number of $\mathbb{Q}$ is 1.

Corollary 38.1.2. . The groupoid $\mathrm{Bun}_{GL_2}(\mathrm{Spec}\,\mathbb{Z})$ has one object up to isomorphism, represented by $\mathbb{Z}^2$, and stabilizer $GL_2(\mathbb{Z})$. Thus

$$\mathrm{Bun}_{GL_2}(\mathrm{Spec}\,\mathbb{Z}) \simeq BGL_2(\mathbb{Z})$$

as a discrete stack, not as a point. The lost datum in the coarse quotient is exactly the arithmetic automorphism group; it does not produce the geometric Hecke correspondence supplied by a curve.

38.2 The Spherical Hecke Category Collapses the OPE

The coarse automorphic stack over Spec $\mathbb{Z}$ has one component and stabilizer $GL_2(\mathbb{Z})$. It has arithmetic monodromy, but no geometric curve along which a Hecke correspondence can move. Thus one is forced to ask which local Hecke category at a prime p carries the noncommutative structure visible to a Hochschild trace. Geometric Satake identifies the spherical category but reveals its commutativity.

Theorem 38.2.1 *(Mirković–Vilonen).* . Geometric Satake gives $\mathrm{Perv}_{GL_2(O)}(\mathrm{Gr}_{GL_2}) \xrightarrow{\sim} \mathrm{Rep}(\widehat{GL_2}) = \mathrm{Rep}(GL_2)$.

Observation 38.2.2 *(spherical Hecke commutativity).* The local spherical Hecke algebra $\mathcal{H}_p = C_c(GL_2(\mathbb{Z}_p)\backslash GL_2(\mathbb{Q}_p)/GL_2(\mathbb{Z}_p))$ is commutative: this is the content of the Satake isomorphism $\mathcal{H}_p \xrightarrow{\sim} \mathbb{C}[GL_2^\vee]^{GL_2^\vee}$. Its derived enhancement $\mathcal{H}^{(p)}$, an E_1-algebra in chain complexes (hence E_2 as a graded object under Dunn additivity), and Perv(Gr), a monoidal abelian category under convolution, are related by the Mirković–Vilonen equivalence; but the commutativity of $\mathcal{H}_p$ means the resulting Hochschild homology carries no automorphic spectral content beyond the centre.

38.3 The Iwahori Hecke Category Carries Convolution Data

Before invoking Bezrukavnikov's equivalence we record the Bernstein presentation that gives the chapter its name. The affine Hecke algebra $H_q = H_q(\widetilde{W})$ of the extended affine Weyl group $\widetilde{W} = W_0 \ltimes X^*(T)$ (with W_0 the finite Weyl group and $X^*(T)$ the character lattice of the maximal torus) is generated by:

- lattice translations θ_λ for $\lambda \in X^*(T)$, forming the Bernstein subalgebra $\mathbb{C}[X^*(T)] \subset H_q$;
- finite Hecke generators T_s for each simple reflection $s \in W_0$, satisfying the braid and quadratic $(T_s - q)(T_s + 1) = 0$ relations.

The interaction between the two layers is the *Bernstein relation*: for a simple reflection $s = s_\alpha$ with simple coroot $\alpha^\vee$ and $\lambda \in X^*(T)$ orthogonal to $\alpha^\vee$,

$$T_s\, \theta_\lambda\, T_s^{-1} \;=\; \theta_{s(\lambda)},$$

while for general λ the conjugation acquires a correction term proportional to $(\theta_\lambda - \theta_{s(\lambda)})/(1 - \theta_{-\alpha})$ expressing the failure of θ_λ to be s-invariant (Lusztig 1989, *Affine Hecke algebras and their graded version*, J. AMS 2, eq. (8.4)). The Bezrukavnikov equivalence below sends this noncommutative algebra structure to a coherent-sheaf convolution on the Steinberg variety; it is the categorical lift of the Bernstein presentation, and the chapter's framing "Bezrukavnikov–Bernstein–Langlands" records the three layers.

The commutative spherical category is replaced by the Iwahori-equivariant category, whose noncommutative convolution datum is precisely what Bezrukavnikov's equivalences encode on the coherent side.

THEOREM 38.3.1 *(Bezrukavnikov arXiv:1209.0403).* . Let $\widetilde{\mathcal{N}}$ denote the Springer resolution $\widetilde{\mathcal{N}} = T^*(GL_2/B)$ of the nilpotent cone, and let $\mathrm{St}_{GL_2} = \widetilde{\mathcal{N}} \times_{\mathcal{N}} \widetilde{\mathcal{N}}$ denote the Steinberg variety (the derived fibre product over the nilpotent cone). Equivalences

$$\mathrm{D}^b\mathrm{Coh}^{GL_2 \times \mathbb{G}_m}(\widetilde{\mathcal{N}}) \xrightarrow{\sim} \mathrm{D}^b\mathrm{Coh}(\mathrm{St}_{GL_2}) \xrightarrow{\sim} \mathrm{D}^b_{\mathrm{mix}}(\mathrm{Perv}_{\mathrm{Iw}}(\mathrm{Fl}_{GL_2})).$$

The Iwahori-fixed direction carries the noncommutative convolution datum needed for the Hochschild trace: the affine flag variety admits an Iwahori stratification indexed by the extended affine Weyl group $\widetilde{W}_{GL_2} = X^*(T) \rtimes W_f \cong \mathbb{Z}^2 \rtimes S_2$, where $X^*(T) = \mathbb{Z}^2$ is the character lattice of the standard torus of GL_2.

Definition 38.3.2 *(global Iwahori Hecke category).*

$$\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}} \;:=\; \bigotimes_p{}' \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p}, \qquad \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p} := \mathrm{D}^b_{\mathrm{mix}}\big(\mathrm{Perv}_{\mathrm{Iw}}(\mathrm{Fl}_{GL_2,\overline{\mathbb{F}}_p}), \mathrm{Fr}_p\big).$$

Remark 38.3.3 *(geometric local convention).* The p-adic Iwahori Hecke algebra is recovered from this geometric category by the sheaf–function dictionary. The coherent-side equivalences of Bezrukavnikov are geometric statements before this sheaf–function specialization.

38.4 The Emerton–Gee Spectral Side

The automorphic side is the global Iwahori Hecke category. The spectral side requires a global (φ, Γ)-module stack; Emerton–Gee construct the local building block, and the global Selmer-derived stack assembles them with fixed arithmetic data.

Theorem 38.4.1 *(Emerton–Gee arXiv:1908.07185).* . There is a Noetherian formal algebraic stack $\mathcal{X}^{\mathrm{EG}}_{GL_2}$ over $\operatorname{Spec}\mathbb{Z}_p$ whose points parametrise (φ,Γ)-modules of rank 2. The global spectral object used below is not a product or colimit of these local stacks; it is the Selmer-derived global deformation stack with fixed residual representation, determinant, and local conditions.

Theorem 38.4.2 *(arithmetic Arinkin–Gaitsgory).* (conditional locally, conjectural globally). Let $\mathrm{D}^{\text{sph-fin}}([GL_2])$ denote the spherically-generated full subcategory of the automorphic ∞-category $\mathrm{D}([GL_2])$ (functions on the adelic quotient $GL_2(\mathbb{A})$), and let $\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}$ denote the Selmer-derived global deformation stack of a fixed residual representation $\bar\rho\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to GL_2$, with fixed determinant and local conditions $\mathcal{L}$. Then:

$$\mathrm{D}^{\text{sph-fin}}([GL_2]) \;\simeq\; \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}).$$

Fargues–Scholze arXiv:2102.13459 supplies local geometrization, spectral action, and Hecke functoriality; Emerton–Gee supplies the local (φ,Γ) stack. The displayed automorphic-to-$\mathrm{IndCoh}_{\mathrm{Nilp}}$ equivalence is conditional locally and conjectural under global adelic descent.

38.5 The Hochschild Trace Placement

The automorphic and spectral sides being assembled, the placement of $\mathcal{V}^{\mathrm{prim}}$ is the Hochschild homology of the global Iwahori Hecke category.

Theorem 38.5.1 *(crowning placement).* .

$$\boxed{\mathcal{V}^{\mathrm{prim}} \;=\; \mathrm{HH}_\bullet\big(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}\big)}$$

E_1-algebra from Dunn additivity ($E_1\otimes E_1 = E_2$) applied to the E_2-monoidal structure on each local factor; Ben-Zvi–Nadler arXiv:0904.1247 identifies $\mathrm{HH}_\bullet(\mathrm{D}(G/G)) = \mathcal{O}(\mathrm{LocSys}_G(S^1))^{\mathrm{flat}}$.

Remark 38.5.2 *(adelic inertia circle).* The local Weil group fits into

$$1 \longrightarrow I_{\mathbb{Q}_p} \longrightarrow W_{\mathbb{Q}_p} \longrightarrow \mathbb{Z} \longrightarrow 1,$$

where the quotient is generated by Frobenius. Thus $\widehat{\mathbb{Z}}$ is the completed Frobenius quotient, not the inertia group. Tame inertia has its own prime-to-p quotient and wild inertia is pro-p. The local system stack $\mathrm{LocSys}_{GL_2}(S^1_p)$, if used as an arithmetic circle shadow, records the Frobenius quotient rather than inertia itself. Ben-Zvi–Nadler arXiv:0904.1247 proves the complex-group identification $\mathrm{HH}_\bullet(\mathrm{D}(G/G)) \simeq \mathcal{O}(\mathrm{LocSys}_G(S^1))^{\mathrm{flat}}$; the present arithmetic statement is its (φ,Γ)-module counterpart.

38.6 Connection to Gaiotto's 3D Chern–Simons Presentation

Proposition 38.6.1 *(three-way comparison).* . The three presentations of $\mathcal{V}^{\mathrm{prim}}$ are conjecturally equivalent:

(i) (Kim arXiv:1510.05818 / Gaiotto) arithmetic Chern–Simons at the classical level produces a boundary contribution whose partition function specializes to $\mathcal{V}^{\mathrm{prim}}$ on the arithmetic Langlands spectral side;

(ii) (Bezrukavnikov–Bernstein) $\mathcal{V}^{\mathrm{prim}} = \mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}})$;

(iii) (Costello–Gwilliam) the factorization window of $\mathcal{V}^{\mathrm{prim}}$ lies in $\mathrm{FactAlg}(\mathrm{Mod}_{\widehat{\mathbb{Z}},\psi})$ over the adelic curve, with the Whittaker and factorization conditions of Chapter 20.

These are conditional comparison identifications, not proved equivalences. Ben-Zvi–Nadler's Chern–Simons/trace/Hecke triangle (arXiv:0904.1247) is the complex template; the arithmetic gluing requires global adelic descent (Conjecture F3 of Chapter 21).

38.7 Hecke Trace = Eichler–Selberg

The Hochschild trace of $\mathcal{V}^{\mathrm{prim}}$ specializes at each weight-level pair (k, N) to the trace of a Hecke operator on the space of cusp forms; this specialization is the Eichler–Selberg trace formula.

THEOREM 38.7.1 *(Eichler–Selberg trace formula, termwise form).* . For each $N \geq 1$, $k \geq 2$:

$$\mathrm{tr}\big(T_p \mid S_k(\Gamma_1(N))\big) \;=\; \mathrm{Ell}_{p,k,N} + \mathrm{Hyp}_{p,k,N} + \delta_{\mathrm{id}}(p) + \delta_{\mathrm{Eis}}(p,k,N),$$

where

$$\mathrm{Ell}_{p,k,N} = -\sum_{t^2<4p} P_k(t,p) H_w(t^2-4p),$$

where $P_k(t,p) = (\alpha^{k-1} - \beta^{k-1})/(\alpha-\beta)$, $\alpha\beta = p$, $\alpha+\beta = t$, H_w denotes the Hurwitz weighted class number of binary quadratic forms of discriminant t^2-4p; $\mathrm{Hyp}_{p,k,N}$ is the hyperbolic/parabolic contribution, $\delta_{\mathrm{id}}(p)$ is the identity contribution, and $\delta_{\mathrm{Eis}}(p,k,N)$ is the Eisenstein boundary contribution.

THEOREM 38.7.2 *(universal trace-class tabulation).* . $\mathcal{V}^{\mathrm{prim}}$ represents the universal family of Hecke eigensystems on the global automorphic category $\mathrm{D}([GL_2])^{\text{sph-fin}}$: it is the object over the weight-level base $\mathcal{W} \times \mathrm{Spec}\,\mathbb{Z}[1/N]$ (where $\mathcal{W}$ is the weight space parametrizing integers $k \geq 2$) whose closed-point fibres are individual Hecke eigensystems. Restricted to $S_k(\Gamma_1(N))$ for fixed (k, N), the fibre of the universal family recovers the Eichler–Selberg trace formula coefficient by coefficient; individual Hecke eigenforms are the closed points of this family.

38.8 The Arithmetic Langlands Diamond for $\mathcal{V}^{\mathrm{prim}}$

(i) $\mathrm{Bun}_{GL_2}(\mathrm{Spec}\,\mathbb{Z}) \simeq BGL_2(\mathbb{Z})$ has one coarse point and nontrivial stabilizer (Chapter 38.1.2); it is too small geometrically to carry the required Hecke correspondence, forcing the passage to the Iwahori-equivariant setting.

(ii) The spherical Hecke algebra is commutative (Satake isomorphism); the Iwahori-equivariant affine Hecke category $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p}$ carries the noncommutative convolution datum (Chapter 38.3.1).

(iii) The spectral side is the Selmer-derived global stack, built from local Emerton–Gee/(φ,Γ) stacks with fixed residual representation, determinant, and local conditions.

(iv) Arithmetic Arinkin–Gaitsgory: Fargues–Scholze and Emerton–Gee supply local inputs; the automorphic-to-$\mathrm{IndCoh}_{\mathrm{Nilp}}$ equivalence is conditional locally and conjectural globally (Chapter 38.4.2).

(v) Crowning placement: $\mathcal{V}^{\mathrm{prim}} = \mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}})$ (Chapter 38.5.1).

(vi) Three-way comparison with Kim–Gaiotto arithmetic Chern–Simons and Costello–Gwilliam factorization (Chapter 38.6.1).

(vii) Hecke-trace tabulation recovers the Eichler–Selberg formula (Chapter 38.7.2).

$\mathcal{V}^{\mathrm{prim}}$ is an arithmetic Hochschild character. It encodes on the automorphic side the universal trace of the identity on the global Iwahori Hecke category of GL_2; on the spectral side, through the conjectural arithmetic Langlands equivalence, the structure sheaf $\mathcal{O}$ of the global Galois-deformation stack. The rank-2 Heisenberg presentation per prime is the Whittaker model of this trace, with the two generators diagonalising the Satake parameters.

Chapter 39

Arithmetic E_n Formality and CoHA

39.1 Chains on Little Intervals and Rectification

The operad of little intervals records ordered associative composition. Its chains can be compared with the associative operad over any coefficient ring. This statement concerns the operad; formality of an individual algebra is a different question.

Theorem 39.1.1 *(Little-interval chains).* . For any commutative ring R, the map

$$C_*(E_1; R) \longrightarrow \mathrm{Ass}_R$$

that sends each configuration to its ordering of the input intervals is an arity wise quasi-isomorphism of operads. Here singular chains use the shuffle map for products, and $\mathrm{Ass}_R(n) = R[S_n]$ lies in degree zero.

Proof. At arity n, the little-interval space has one component for each ordering of the intervals. In one such component, their left and right endpoints satisfy the ordered linear inequalities for nonoverlapping intervals of positive length. This region is convex and nonempty. Straight-line contraction to a fixed configuration shows that its singular chains have homology R in degree zero and zero otherwise, over any R. Substitution of intervals substitutes their orderings, so the chain augmentation is operadic. □

When the chosen algebra model categories admit rectification along this operad map, it replaces an E_1-algebra by an equivalent associative differential graded algebra. It does not replace that algebra by its cohomology with zero differential.

Corollary 39.1.2 *(Higher operations can survive rectification).* . Torsion-freeness of $H_*(E_1; \mathbb{Z}) = \mathrm{Ass}_{\mathbb{Z}}$ does not imply that an associative differential graded algebra is formal or that its minimal model has vanishing higher products.

Proof. For $A = k[x]/(x^3)$, the strict endomorphism algebra of its augmentation resolution has $H^*E = \Lambda(u) \otimes k[v]$, but $\langle u, u, u\rangle = \{-v\} \neq \{0\}$, with zero indeterminacy; see (2) and (3). Its cohomology algebra with zero differential has zero triple product. Quasi-isomorphisms preserve this single-valued Massey product, so the strict algebra is not formal. The comparison equations of Theorem 4.3.1 retain precisely the higher components needed to transport such data. □

39.2 E_2-Formality Fails Integrally

Theorem 39.2.1 *(Salvatore–Wahl; Cohen LNM 533; Fresse–Willwacher arXiv:1503.08699).* . Algebras over $C_\bullet(E_2; \mathbb{F}_p)$ carry non-trivial Dyer–Lashof power operations not encoded by the Gerstenhaber homology operad; these operations obstruct integral formality. Over $\mathbb{Q}$, Tamarkin's theorem (arXiv:math/9803025) establishes $C_\bullet(E_2; \mathbb{Q}) \xrightarrow{\sim} H_\bullet(E_2; \mathbb{Q}) = \mathrm{Ger}_{\mathbb{Q}}$ (where $\mathrm{Ger}_{\mathbb{Q}}$ is the Gerstenhaber operad over $\mathbb{Q}$), a torsor for the Grothendieck–Teichmüller group $\mathrm{GRT}_1(\mathbb{Q})$.

COROLLARY 39.2.2 *(no integral replacement by the Gerstenhaber operad).* . An integral E_2-algebra structure is not forbidden. What fails is the replacement

$$C_\bullet(E_2;\mathbb{Z}) \simeq \mathrm{Ger}_{\mathbb{Z}}.$$

Consequently an E_2-chiral-centre enhancement of $\mathcal{V}^{\mathrm{prim}}$ over $\mathbb{Z}$, if present, must retain the chain-level E_2-operad and its power operations. After tensoring with $\mathbb{Q}$, Tamarkin formality identifies the rational structure with a $\mathrm{GRT}_1(\mathbb{Q})$-torsor.

Remark 39.2.3 *(Kaledin Japanese cocycle).* Kaledin isolates the arithmetic obstruction: for an associative $\mathbb{Z}$-algebra A, the reduction $A \otimes \mathbb{F}_p$ may carry a non-trivial Hochschild A_∞-deformation class, the "Japanese cocycle," in $\mathrm{HH}^2(A \otimes \mathbb{F}_p)/\mathrm{HH}^2(A) \otimes \mathbb{F}_p$. Such classes are obstructions to naive integral strictification. They do not, by themselves, prove nonexistence of an integral E_2-algebra.

39.3 A Specified Strict Associative Model

A strict model is an actual differential graded algebra with an actual comparison to the proposed arithmetic operation system. The quasi-isomorphism of coefficient operads supplies neither the arithmetic algebra nor its geometric collision maps.

THEOREM 39.3.1. . For a differential graded associative R-algebra A, its displayed A_∞-presentation has only its differential and binary multiplication. A quasi-isomorphic minimal presentation can have nonzero higher operations; they must be retained by the comparison maps.

Proof. The differential graded identities are the A_∞ equations for $m_1 = d_A$, $m_2 = \mu_A$ and $m_n = 0$ for $n \geq 3$. The cubic example in the preceding corollary proves the second assertion. Its nonzero triple Massey product cannot disappear under a quasi-isomorphism. □

COROLLARY 39.3.2 *(Integral associative presentations).* . A supplied associative differential graded $\mathbb{Z}$-algebra gives an integral A_∞-algebra by the preceding construction. Identifying it with an arithmetic field system additionally requires the maps of (4.3), including their unit, collision and higher-substitution conditions.

Proof. No division occurs in the differential graded or suspended bar identities. The second assertion specifies the additional structure of that comparison: the coefficient algebra by itself has no maps to the independently given geometric operations. □

39.4 The Chiral Bracket via Adelic Residue

Definition 39.4.1 *(archimedean λ-bracket on the cover).* Let $\overline{C}$ denote the Arakelov compactification of the arithmetic curve, obtained by adjoining the archimedean place $|\cdot|_\infty$ to $\operatorname{Spec}\mathbb{Z}$. The Beilinson–Drinfeld λ-bracket is defined only on the archimedean Heisenberg cover, where a holomorphic diagonal and a complex residue exist:

$$\{a,b\}_\infty \;=\; \mathrm{Res}_{z=w}\,\omega_{a,b}(z,w).$$

No finite prime contributes a vertex-algebra residue on $\mathcal{V}^{\mathrm{prim}}$. A finite-place bracket would require a separate prismatic construction and is not part of the Hochschild trace class.

THEOREM 39.4.2 *(finite-place residue obstruction).* . The archimedean bracket above does not descend to a Beilinson–Drinfeld bracket on $\mathcal{V}^{\mathrm{prim}}$ over $\overline{C}$. Finite places are totally disconnected in the factorization topology used here, and the global object is an E_1 Hochschild trace class rather than a vertex algebra.

39.5 The arithmetic Hall comparison

A cohomological Hall comparison starts with an enhanced arithmetic category, its extension correspondences and its coefficient theory. A Hochschild trace class does not specify these data. Chapter 12 constructs the finitary and geometric Hall operations and separates their arithmetic comparison problems.

Theorem 39.5.1 *(Kontsevich–Soibelman).* For a finite quiver over $\mathbb{C}$ with zero potential, equivariant cohomology of its representation stacks has the extension product of Theorem 12.8.1. For a potential, the critical product uses normalized monodromic vanishing cycles, as defined in Section 12.10. The dimension shifts, coefficient category and chosen semistable or Serre loci are part of that construction.

Theorem 39.5.2 *(Davison–Meinhardt).* Assume the hypotheses of Theorem 12.10.1, including its stability and coefficient data. The critical Hall algebra has a PBW isomorphism of underlying graded objects with symmetric powers of $H(B\mathbb{C}^{\times})_{\mathrm{vir}} \otimes \mathcal{B}$. Its perverse associated graded is free commutative in the specified symmetry, or free supercommutative after the stated sign modification. An algebra isomorphism with a free supercommutative algebra holds for the finite symmetric, zero-potential quiver of Theorem 12.9.1.

Question 39.5.3 *(Arithmetic CoHA comparison).* Construct an arithmetic quiver with potential, or another independently defined Hall category, and the enhanced arithmetic object to compare with its BPS object. Specify the coefficient theory, stability and completion, and construct a comparison preserving the extension operations and their primitive maps. If places label vertices and Hecke correspondences supply arrows, construct the actual incidence spaces and their compatible finite-place assembly.

The Bost–Connes system has its own semigroup and time evolution. These data do not specify a quiver representation stack or a cohomological Hall product. Even a single isolated vertex with zero potential has the exterior algebra on infinitely many odd charge-one generators. One loop gives infinitely many even generators over $\mathbb{Q}$ (Example 12.8.2). A polynomial generator for each prime therefore does not describe these CoHAs. An arithmetic BPS identification requires the functor in Question 12.12.1.

39.6 Euler series and cyclic classes

Observation 39.6.1. On $\operatorname{Re} s > 1$ the absolutely convergent formulas are

$$\log \zeta(s) = \sum_{p} \sum_{k \geq 1} \frac{p^{-ks}}{k}, \qquad -\frac{\zeta'(s)}{\zeta(s)} = \sum_{p} \sum_{k \geq 1} (\log p)\, p^{-ks}.$$

The logarithms occur in the logarithmic derivative. An algebraic quiver potential is instead a cyclic class in a path algebra. Passing from either series to a potential requires a specified coefficient ring, any necessary completion, and an actual map compatible with traces.

Kontsevich–Soibelman's exponential Hodge theory provides a recipient for potential-dependent Hall invariants. Applying it to an arithmetic Euler series first requires the potential and its comparison map. A scalar identity does not supply a cyclic-homology class or that map.

39.7 Arithmetic Formality Stratification

The little-interval chain operad has the integral associative comparison of Theorem 39.1.1. Individual algebras can nevertheless retain the nonzero Massey products of Corollary 39.1.2. A supplied strict integral

algebra gives the presentation of Corollary 39.3.2; neither a geometric field realization nor a residue bracket follows from the operad comparison. The E_2 question and the arithmetic Hall comparison require their own operation data. Question 39.5.3 asks for the arithmetic quiver, potential, coefficient theory and BPS comparison. These constructions are required before identifying a trace class with an enhanced arithmetic object.

Chapter 40

The Arithmetic Yangian Collapses to Its Bethe Subalgebra

The Yangian $Y(\mathfrak{gl}_n)$ is controlled by the RTT presentation, an R-matrix, and a spectral parameter. An arithmetic Yangian has no uniform spectral difference $u - v$ across primes. The surviving structure is the commutative Bethe algebra together with the automorphic modularity condition.

The Yangian $Y(\mathfrak{gl}_n)$ (Drinfeld 1985, 1988) is a Hopf algebra deformation of $U(\mathfrak{gl}_n[z])$ defined by the RTT relation $R(u-v)(T(u)\otimes 1)(1\otimes T(v)) = (1\otimes T(v))(T(u)\otimes 1)R(u-v)$ with Yang's rational R-matrix $R(u) = 1 + P/u$ (here P the permutation operator on $V\otimes V$). The question is whether this structure admits an arithmetic analogue over Spec $\mathbb{Z}$. Four independent obstructions show it does not. The correct arithmetic replacement is the commutative Bethe subalgebra, whose spectrum is the automorphic spectrum of GL_n.

40.1 The Four Obstructions to a Naive Arithmetic Yangian

Theorem 40.1.1 *(four structural obstructions to the RTT model).* . Yang's additive RTT construction with R-matrix $R(u) = 1 + P/u$ has no arithmetic analogue satisfying the four requirements below:

(i) **Discrete-continuous mismatch.** The spectral parameter u is continuous and Spec $\mathbb{Z}$ is discrete; primes cannot be spectral parameters.

(ii) **No cross-prime subtraction.** $u - v$ has no analogue across distinct primes ($\mathbb{Q}_p \not\hookrightarrow \mathbb{Q}_q$ for $p \neq q$).

(iii) **No meromorphic 1-form on** Spec $\mathbb{Z}$**.** Costello 4D Chern–Simons (arXiv:1709.09993) requires a meromorphic 1-form ω on the spectral-parameter direction; Spec $\mathbb{Z}$ has none.

(iv) **No ζ-valued shuffle kernel.** Arithmetic shuffle-kernel constructions with ζ-valued kernels fail the cocycle condition (Feigin–Odesskii).

Corollary 40.1.2 *(no RTT arithmetic Yangian as Hopf algebra).* . $Y^{\mathrm{arith}}(\mathfrak{gl}_n)$ does not exist as a noncommutative Hopf algebra deforming $U(\mathfrak{gl}_n[z])$ with the specified additive spectral parameter, arithmetic R-matrix, and RTT relation. Other categorical or non-RTT notions are not excluded by this obstruction.

40.2 The Bethe-Algebra Collapse

The four obstructions force the question: what commutative quotient of the Yangian does admit an arithmetic analogue? The answer is the Bethe subalgebra, which over $\mathbb{C}$ is already the coordinate ring of opers on a disk.

Theorem 40.2.1 *(Frenkel–Reshetikhin; Mukhin–Tarasov–Varchenko; Kamnitzer–Pham–Weekes).* (Frenkel–Reshetikhin, *Commun. Math. Phys.* 197 (1998) 1–31; Mukhin–Tarasov–Varchenko, *Selecta Math.* 17 (2011); Kamnitzer–Weekes, arXiv:1811.11902). For $\mathfrak{g}$ simply-laced, the Bethe algebra $\mathcal{B}(Y(\mathfrak{g})) \subset Y(\mathfrak{g})$ (the maximal commutative subalgebra generated by higher transfer matrices) is the coordinate ring of ${}^L\mathfrak{g}$-opers on a formal disk D:

$$\mathcal{B}(Y(\mathfrak{g})) \;\cong\; \mathcal{O}(\mathrm{Op}_{{}^L\mathfrak{g}}(D)).$$

Definition 40.2.2 *(arithmetic Bethe algebra).* The *arithmetic Bethe algebra* $\mathcal{B}^{\mathrm{arith}}(GL_n)$ acting on $\mathcal{V}^{\mathrm{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori Hecke category, Chapter 24.1) is the commutative algebra generated by unramified Hecke operators $T_{p,1}, \dots, T_{p,n}$ for all primes p, together with Casimirs of $\mathfrak{gl}_n$ on each local factor $\mathcal{F}_p$ (the rank-n Heisenberg Fock space at the prime p). Equivalently,

$$\mathcal{B}^{\mathrm{arith}}(GL_n) \;=\; \bigotimes_p \mathcal{H}_p^{\mathrm{sph}}(GL_n) \;\cong\; \bigotimes_p \mathbb{C}[\alpha_1^{(p)}, \dots, \alpha_n^{(p)}]^{S_n}.$$

Theorem 40.2.3 *(Bethe-algebra shadow of $\mathcal{V}^{\mathrm{prim}}$).* . The surviving arithmetic object is the commutative Hecke spectral algebra $\mathcal{B}^{\mathrm{arith}}(GL_n)$. Its simultaneous eigenvectors are Hecke eigenforms, and its eigenvalues are Satake parameters. This is the Bethe-algebra shadow of an integrable system; no Bethe equations, transfer matrix, or Bethe-root reality theorem is asserted.

40.3 Primes Are Lattice Sites, Not Spectral Parameters

Principle 40.3.1 *(site-spectral duality).* In the XXZ spin chain from $U_q(\widehat{\mathfrak{sl}}_2)$, sites are discrete (labelled by $\mathbb{Z}$ or by particle number N); spectral parameter u is continuous ($\in \mathbb{C}$). Hecke eigenvalues live in $\mathbb{C}$ (spectral); primes live in $\mathbb{Z}_{\geq 2}$ (site).

Corollary 40.3.2 *(arithmetic spectral parameter is s).* . The arithmetic spectral parameter is the Langlands $s \in \mathbb{C}$, not a prime, not a p-adic number. The spectral direction is the archimedean fibre $\mathbb{C}_\infty$; the site direction is the Arakelov-compactified arithmetic curve $\overline{\mathrm{Spec}\,\mathbb{Z}}$.

Heuristic 40.3.3 *(4D Chern–Simons analogy, conditional on F2).* . Costello–Witten–Yamazaki 4D CS (arXiv:1709.09993) is a theory on $\Sigma \times C$ with Σ a topological real 2-manifold and C a complex curve carrying a meromorphic 1-form; its factorization algebra lives on Σ. The direct identification $\Sigma = \mathrm{Spec}\,\mathbb{Z}$ is dimensionally inconsistent on two counts. (i) $\overline{\mathrm{Spec}\,\mathbb{Z}}$ has étale cohomological dimension 3 (Artin–Verdier duality), not 2, so $\mathrm{Spec}\,\mathbb{Z} \times \mathbb{C}$ has total real dimension 5, not 4. (ii) Each finite place of $\overline{\mathrm{Spec}\,\mathbb{Z}}$ carries only locally constant (E_0) factorization structure (Chapter 30.4.1), so a 2-dimensional E_1-chiral factorization algebra of 4D CS type cannot live on the finite places at all; the E_1-chiral OPE concentrates at the archimedean fibre $\mathbb{C}_\infty$.

Remark 40.3.4 *(rerouting the 4D CS analogy through F2).* The honest analogy is conditional on Conjecture F2 (Chapter 42.2.1, perfectoid E_2-enhancement via Scholze diamonds). If F2 holds globally, the diamond side of the global Iwahori Hecke supplies an étale 2-site $S^{\text{ét}}$ with the Frobenius-semilinear arithmetic-complex datum of Chapter 58.1.2, and the CWY identification becomes

$$C = \mathbb{C} \text{ (spectral, } s\text{-plane)}, \qquad \Sigma = S^{\text{ét}} \text{ (Scholze-diamond 2-site, not Spec } \mathbb{Z} \text{ itself)},$$

with the E_1-chiral OPE constructed at $\mathbb{C}_\infty$ and extended across $S^{\text{ét}}$ by the F2 E_2-monoidal structure (Dunn additivity $E_1 \otimes E_1 = E_2$). Without F2 the 4D CS analogy is pattern-match only. The parallel 3D analogy with Kim arithmetic Chern–Simons (arXiv:1510.05818) matches $\overline{\mathrm{Spec}\,\mathbb{Z}}$ dimensionally

but carries no spectral direction, and is the source of the five-presentation item (i) in the manifesto (Chapter 24.1).

40.4 Automorphic Closure as Bethe Heuristic

Theorem 40.4.1 *(automorphic closure as Bethe coherence). Heuristic.* The eigenvalues $\alpha_1^{(p)}, \dots, \alpha_n^{(p)}$ satisfy no nontrivial relations at a single prime; they are arbitrary elements of $(\mathbb{C}^\times)^n$ modulo S_n. Across primes, they satisfy only the automorphic condition that $\{\alpha_i^{(p)}\}_p$ arises from a single Galois representation $\rho\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to GL_n(\mathbb{C})$ with $\alpha_i^{(p)}$ the eigenvalues of $\rho(\mathrm{Frob}_p)$. The rigidification is Bethe/Hecke coherence: the commutative Bethe algebra $\mathcal{B}^{\mathrm{arith}}(GL_n)$ diagonalises on automorphic eigenspaces labelled by Galois representations.

Remark 40.4.2 *(Ramanujan–Petersson, modularity).* The comparison with complex RTT systems is mnemonic. Modularity rigidifies a family of local Frobenius parameters by forcing them to arise from one global Galois representation; the arithmetic rigidification is commutative Bethe/Hecke and Galois-representational, not quadratic RTT.

40.5 Explicit Computation: $\mathfrak{gl}_1$ and $\mathfrak{gl}_2$

$\mathfrak{gl}_1$ *(Dirichlet characters)*

Proposition 40.5.1 *(rank-1 Hecke as a Fock adjacency model). Heuristic.* The spherical Hecke algebra $\mathcal{H}_p^{\mathrm{sph}}(GL_1) = \mathbb{C}[T_p^{\pm}]$ is the group algebra of the valuation lattice. In the rank-1 Fock model, the normalised adjacency operator $U_p = p^{-1/2}T_p + p^{1/2}T_p^{-1}$ acts by

$$U_p \cdot (a_{-1}^{(p)})^k|0\rangle_p \;=\; p^{-1/2}(a_{-1}^{(p)})^{k+1}|0\rangle_p + p^{1/2}(a_{-1}^{(p)})^{k-1}|0\rangle_p.$$

Eigenvalues are $\alpha_p + \alpha_p^{-1}$ with $\alpha_p \in \mathbb{C}^\times$.

Corollary 40.5.2. $\mathcal{B}^{\mathrm{arith}}(GL_1) = \mathbb{C}[T_2^{\pm}, T_3^{\pm}, T_5^{\pm}, \dots]$, polynomial algebra in countably many commuting variables. Eigenvectors = Hecke eigenforms = primitive elements of $\mathcal{V}^{\mathrm{prim}}$ labelled by global characters $\chi\colon \mathbb{A}_{\mathbb{Q}}^\times/\mathbb{Q}^\times \to \mathbb{C}^\times$.

$\mathfrak{gl}_2$ *(Hecke eigenforms)*

Proposition 40.5.3. $\mathcal{H}_p^{\mathrm{sph}}(GL_2) = \mathbb{C}[T_{p,1}, T_{p,2}^{\pm}]$ with

- $T_{p,1}$ eigenvalue $= \alpha_1^{(p)} + \alpha_2^{(p)}$ = trace of Frobenius in a 2-dim Galois rep;
- $T_{p,2}$ eigenvalue $= \alpha_1^{(p)}\alpha_2^{(p)}$ = determinant = central character at p.

Action on rank-2 Fock $\bigotimes_p \mathcal{F}_p^{(2)}$ via raising/lowering in each Heisenberg factor (Bump §3).

40.6 Local r-Matrix per Prime

Proposition 40.6.1 *(local classical r-matrix).* At each prime p, the Heisenberg level is $k = 1$, so $r_p^{\mathrm{arith}}(z) = \Omega_p/z$ with $z \in \mathbb{C}$ and Ω_p = the Casimir of the local $\mathfrak{gl}_n$-factor. The local classical r-matrices are independent across distinct primes; the discrete structure of Spec $\mathbb{Z}$ prevents their assembly into a global r-matrix.

Remark 40.6.2 *(automorphic condition replaces global r-matrix).* What rigidifies the local factors into a global object is the automorphic condition on Galois representations: the spectrum of $\mathcal{B}^{\mathrm{arith}}(GL_n)$ is the set of $\{\alpha_i^{(p)}\}_p$ arising from a single $\rho\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to GL_n(\mathbb{C})$. This is Bethe-spectrum coherence.

40.7 Yangian → Hecke = Langlands as Arithmetic Bethe Ansatz

Theorem 40.7.1 *(Hecke spectral algebra as Bethe shadow).* The classical Jacquet–Langlands correspondence (SLN 114, 1970) supplies the following Hecke spectral data:

- Hecke eigenforms are simultaneous eigenvectors of $\mathcal{B}^{\mathrm{arith}}$;
- Hecke eigenvalues are Satake parameters;
- Ramanujan–Petersson is a bound on Satake parameters;
- automorphic spectral decomposition of $L^2(GL_n(\mathbb{A})/GL_n(\mathbb{Q}))$ gives the diagonalisation of $\mathcal{B}^{\mathrm{arith}}$ on $\mathcal{V}^{\mathrm{prim}}$.

The arithmetic Bethe algebra $\mathcal{B}^{\mathrm{arith}}(GL_n)$ acts on $\mathcal{V}^{\mathrm{prim}}$, and the global Langlands correspondence is the arithmetic rigidity condition on its spectrum.

The Yangian $Y(\mathfrak{gl}_n)$ quantises an affine Lie algebra; the Hecke algebra quantises the automorphic side of Langlands. These are parallel quantum integrable systems; the arithmetic Bethe algebra $\mathcal{B}^{\mathrm{arith}}(GL_n) = \otimes_p \mathcal{H}_p^{\mathrm{sph}}(GL_n)$ is the commutative quotient that survives over $\mathrm{Spec}\,\mathbb{Z}$, and it is exactly the Hecke algebra. The $\kappa + \kappa^! = 0$ complementarity reappears as the unitarity of the Bethe spectrum and the functional equation, closing the loop with Chapter 33.

Chapter 41

Synthesis: Six Conditional Windows

$\mathcal{V}^{\mathrm{prim}}$ is the Hochschild trace class of the identity endofunctor on the global arithmetic Iwahori Hecke category of GL_2. It admits six conditional windows, five arithmetic avatars of the Vol I theorems, and the structure of Definition 41.6.1. Three conjectures state the remaining cross-Langlands comparisons.

41.1 The Synthesis

Conjecture 41.1.1 *(six conditional windows).* $\mathcal{V}^{\mathrm{prim}}$ admits the following six conditional windows on the Hochschild-trace candidate:

(i) **(Langlands / automorphic)** $\mathcal{V}^{\mathrm{prim}}$ is expected, under F1 and adelic categorical descent, to correspond to the structure sheaf of the Selmer-derived global spectral stack attached to a fixed residual representation $\bar{\rho}$, determinant, and local conditions. Individual Hecke characters are closed-point shadows of this family, not the family itself.

(ii) **(TQFT / Chern–Simons)** Kim arithmetic Chern–Simons supplies a classical arithmetic CS torsor/action line. A genuine quantum boundary category is an open construction. Thus the ACS window is retained only as a classical TQFT/factorization shadow whose Wilson labels are representation-level ramified Hecke data.

(iii) **(Categorical / Hochschild trace)** $\mathcal{V}^{\mathrm{prim}} = \mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}})$, the trace class of the identity on the global affine Iwahori Hecke category.

(iv) **(Integrability / Bethe algebra)** $\mathcal{V}^{\mathrm{prim}}$ has a Bethe-algebra heuristic: primes behave as lattice sites, Hecke operators as transfer matrices, and Hecke eigenforms as Bethe eigenvectors. There is no RTT arithmetic Yangian Hopf algebra; the surviving algebra is the commutative Hecke/Bethe algebra $\otimes_p \mathcal{H}_p^{\mathrm{sph}}$.

(v) **(Statistical mechanics)**

$$\begin{aligned}\mathcal{V}_N^{\mathrm{prim}} \cong\ & (\text{Bost–Connes at rank } \pi(N)) \\ & \otimes (\text{universal Heisenberg descendant tower of rank } \pi(N))\end{aligned}$$

All arithmetic content lies in the Bost–Connes/Hecke factor; the Heisenberg factor is a finite-cutoff universal model rather than a global VOA.

(vi) **(Factorization / homotopical)** $\mathcal{F}_{\mathrm{CG}}(\mathcal{V}^{\mathrm{prim}}) \in \mathrm{FactAlg}_{\mathbb{C}_\infty}(\mathrm{Mod}_{\widehat{\mathbb{Z}},\psi})$ with (F1)–(F5) of Definition 30.6.1: holomorphic factorization at the archimedean place, locally constant with Adams operations at each finite place.

These are comparison windows, not established equivalent presentations. Ben-Zvi–Nadler's identification $\mathrm{HH}_\bullet(\mathrm{D}(G/G)) = \mathcal{O}(\mathrm{LocSys}_G(S^1))^{\mathrm{flat}}$ is the complex template for gluing (1), (2), (3); the arithmetic version is F3 and must use the full Weil/(φ,Γ) target, with $B\widehat{\mathbb{Z}}$ only as an inertia shadow.

41.2 The Essential Theorem

THEOREM 41.2.1 *(G-row arithmetic complementarity = Riemann functional equation).* Assume the vector space, involution, central-defect identity, and meromorphic evaluation of Theorem 43.5.1, with the arithmetic levels as its two parameters. Assume in particular that the evaluation detects the specified defect vector. Then

$$\boxed{\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0 \iff \xi(s) = \xi(1-s).}$$

The intended arithmetic realization is the G-row with $\kappa = 0$. Identifying its Koszul dual with a Bost–Connes object requires the algebra, coefficient, completion and pairing maps of Question 46.5.4. To interpret the displayed biconditional, those maps must also identify the arithmetic central classes and their complementarity with the completed scalar character. The factor $\Gamma(s/2)\pi^{-s/2}$ is supplied by the Gaussian Mellin calculation. Interpreting it as a derived-center shift requires a separate compatible comparison. Neither the faithful Bost–Connes operator representation nor the scalar functional equation supplies these arithmetic identifications.

Proof. Apply Theorem 43.5.1 to the supplied arithmetic comparison. Its detection hypothesis gives the converse. Its defect identity and evaluation give the forward implication. □

Remark 41.2.2 *(the proposed arithmetic Koszul comparison).* The Vol I relation between Koszul duality and derived-centre complementarity, with values $\kappa + \kappa^! \in \{0, 8, 13, 250/3, 25/3\}$, motivates an arithmetic G-row comparison $\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0$ with $\xi(s) = \xi(1-s)$. Such a specialization requires an arithmetic carrier and a duality preserving the relevant classes and pairings. It must also explain the proposed derived-centre interpretation of $\Gamma(s/2)\pi^{-s/2}$. These comparisons remain construction problems even though the scalar identities are classical.

41.3 Five Arithmetic Comparison Problems

Conjecture 41.3.1 *(arithmetic avatars).* The following are proposed arithmetic counterparts of the five chiral-bar-cobar theorems; their arithmetic carriers and comparison maps are part of the conjecture:

$\mathcal{A}^{\mathrm{arith}}$ Bar–cobar adjunction $\Omega^{\mathrm{ch,Ar}} \dashv B^{\mathrm{ch,Ar}}$ on Iwasawa-twisted Λ-coalgebras with $\mu = 0$ and arithmetic E_1-algebras with Deligne-pure bar cohomology.

B^{arith} Arithmetic Positselski: $\Omega^{\mathrm{Ar}} B^{\mathrm{Ar}}(C) \xrightarrow{\sim} C$ on conilpotent Λ-coalgebras with $\mu = 0$.

C^{arith} Derived-centre complementarity: $\kappa + \kappa^! \in \{0, 8, 13, 250/3, 25/3\}$. G-row $= 0$ is the Riemann functional equation (Theorem 41.2.1). B-row $= 8$ is the Vol III Borcherds lift.

D^{arith} Obstruction tower = Arakelov height filtration: $\mathrm{obs}_g^{\mathrm{Ar}} = \kappa \cdot h_g^{\mathrm{Ar}}$.

H^{arith} Construct an arithmetic algebra or category, its coefficients and Hochschild complex before formulating an arithmetic concentration theorem. The proposed integral targets $\mathrm{HH}^0 = \mathbb{Z}$ and $\mathrm{HH}^1 = \bigoplus_p \log p \cdot \mathbb{Z}$, and a proposed Witt-cotangent description of degree two, require the comparison maps of Question 29.7.11. For the specified complex normal-term Bost–Connes algebra, Theorem 29.7.1 instead gives an injection $\prod_p \mathbb{C} \hookrightarrow HH^1_{\mathrm{alg}}$. The exterior-algebra injection of Theorem 29.7.3 further gives nonzero algebraic classes in every degree. These results neither construct the proposed integral carrier nor identify an arithmetic trace map.

Vol I theorem	Arithmetic avatar	Form
A: bar–cobar	A^{arith} on Λ-coalgebras	Conjecture
B: Positselski	B^{arith} on $\mu = 0$-coalgebras	Conjecture
C: $\kappa + \kappa^!$ ceiling	G-row = 0	Conditional
C: $\kappa + \kappa^!$ ceiling	B-row = 8 (arithmetic K3)	Conjecture
D: obstruction tower	Arakelov height filtration	Conjecture
H: Hochschild invariants	Arithmetic carrier and comparison maps	Open problem

Table 41.1: Arithmetic avatars of the five Vol I theorems.

41.4 Segal Axioms as Tate's Thesis

Theorem 41.4.1 *(Tate line in arithmetic Segal notation).* (Tate's thesis for the analytic identities). Tate's thesis supplies the adelic integral line, Euler product, meromorphic continuation, functional equation, and residue for the zeta and Dirichlet sectors. Placing these data in Segal notation gives the following dictionary:

Segal sewing position	= Euler-product factorization of $L(s, \chi)$
Verlinde scalar	= class-number residue
Modular S-matrix position	= Γ-factor plus functional equation
Genus contribution	= Arakelov discriminant volume $\frac{1}{2}\log\lvert d_K\rvert$
Mapping-class position	= Galois labels plus Hecke symmetries
Conformal-block position	= Tate integral line
Sewing label	= Weil–Deligne parameter
Marked point	= place
Orientation	= Deligne trivialization of $\det R\Gamma$
Closed scalar $Z_K(s)$	= finite Euler product for $\mathrm{Re}(s) > 1$, completed by the archimedean Γ-factor

The theorem asserts the analytic Tate line. It does not assert the existence of a full arithmetic Segal functor.

41.5 Rank Residue

Theorem 41.5.1 *(Polyakov: the determinant rank is a residue).*

$$r_\zeta(\mathcal{V}^{\mathrm{prim}}) \;=\; 2 \cdot \mathrm{Res}_{s=1}\zeta(s) \;=\; 2.$$

This is not a trace, not a sum, and not a global Virasoro central charge. The functional equation $\xi(s) = \xi(1-s)$ replaces modular invariance; the Arakelov height pairing replaces the Weyl anomaly.

41.6 The Coherent Arithmetic Structure

The six-window placement (Conjecture 41.1.1), the arithmetic modular functor (Definition 32.8.1), the ACS classical boundary-shadow window (Theorem 37.4.1), the Hochschild trace placement (Theorem 38.5.1), and the rank residue (Theorem 41.5.1) are not six unrelated structures: each builds part of a common tuple $(S^{\text{ét}}, \mathcal{V}, \mathrm{Mod}(\mathcal{V}), W, \mathcal{F}, Z, r_\zeta, \kappa)$. Definition 41.6.1 collects these data on

the Arakelov curve $\overline{C}$. The dimensional comparison below relates the universal construction to its specializations.

Definition 41.6.1 *(arithmetic conformal field theory).* An *arithmetic conformal field theory* (aCFT) on the Arakelov curve $\overline{C} = \overline{\operatorname{Spec} O_K}$ with gauge datum (G, k) is a tuple $(S^{\text{ét}}, \mathcal{V}, \operatorname{Mod}(\mathcal{V}), W, \mathcal{F}, Z, c, \kappa)$ satisfying (B)–(K) below.

(B) Base. A stratified étale arithmetic site $S^{\text{ét}}$ of operadic arithmetic dimension 2, equipped with a stratified arithmetic complex datum $\mathfrak{J}^{\text{Ar}}$ in the sense of Remark 58.1.2. Conditional on F2 (Conjecture 42.2.1):

- at the archimedean place: $S^{\text{ét}}|_\infty = \mathbb{C}_\infty$ with its standard complex structure ($\mathfrak{J}^{\text{Ar}}_\infty = i$);
- at a finite prime p: $S^{\text{ét}}|_p = \operatorname{Spd} \mathbb{Q}_p$ (Scholze diamond) with $\mathfrak{J}^{\text{Ar}}_p$ the Frobenius-semilinear untilt-orientation, together with the Adams operation ψ_p and the prismatic/Nygaard comparison filtration when the finite-place diagonal is thickened.

The global $S^{\text{ét}}$ glues the local pieces through the Scholze-diamond side of the global Iwahori Hecke category. Thus $\mathfrak{J}^{\text{Ar}}$ is literal almost-complex geometry only at ∞; at finite primes it is the Frobenius-oriented arithmetic substitute that makes the pointed E_2 Iwahori Hecke datum functorial. It is not a tangent-bundle endomorphism at a finite prime and carries no equation $J^2 = -1$.

(V) Chiral algebra. A factorization algebra

$$\mathcal{V} \in \operatorname{FactAlg}_{S^{\text{ét}}}(\operatorname{Mod}_{\widehat{\mathbb{Z}},\psi})$$

of arithmetic dimension 2, satisfying (F1)–(F5) of Definition 30.6.1: genuine E_1-chiral OPE at the archimedean fibre $\mathbb{C}_\infty$, locally constant (E_0) with commuting Adams operations ψ_p at each finite place. $\mathcal{V}$ is the 2-dimensional universal family whose fibre at each closed point x of $S^{\text{ét}}$ is a module category $\operatorname{Mod}(\mathcal{V})_x$.

(M) Module category. For each closed point $x \in S^{\text{ét}}$, a category $\operatorname{Mod}(\mathcal{V})_x$ of $\mathcal{V}$-modules supported at x. Objects of $\operatorname{Mod}(\mathcal{V})_x$ are 0-dimensional data: individual modules, i.e., closed points of the universal family (V). The primary datum $(\mathbb{G}_m, 1)$ gives $\operatorname{Mod}(\mathcal{V}^{\text{prim}})_p$ = local Hecke characters of $\mathbb{Q}_p^\times$ (ramified allowed).

(W) Wilson-line defects. The bulk theory $\operatorname{ACS}[\overline{C}, G, k]$ on the arithmetic 3-fold $\overline{C}$ carries 1-dimensional defects $W_x(M)$ supported on closed-point loci $x \subset \overline{C}$ (prime-knots K_p and the archimedean locus K_∞), each labeled by a module $M \in \operatorname{Mod}(\mathcal{V})_x$. The defect is 1-dimensional; the label is 0-dimensional; neither is $\mathcal{V}$ itself, which is 2-dimensional and lives on $S^{\text{ét}}$ (not on $\overline{C}$).

(F) Modular functor. A Segal-style functor

$$\mathcal{F} \colon \mathcal{C}^{\text{Arak}} \to \operatorname{Hom}(\mathbb{C}_s, \operatorname{Line})$$

satisfying axioms (A1)–(A5) of Definition 32.8.1 (finiteness, factorization, Weil–Deligne sewing, modular invariance under $\operatorname{Gal}(\overline{K}/K) \ltimes GL_n(\mathbb{A}_K)/GL_n(K)$, functional equation). The relation to (V)–(M) is: the conformal block $\mathcal{F}(\overline{X}, S, \{\chi_{\mathfrak{p}}\})$ is the space of global $\mathcal{V}$-correlators for the module insertion $\{\chi_{\mathfrak{p}}\}_{\mathfrak{p} \in S}$, each $\chi_{\mathfrak{p}} \in \operatorname{Mod}(\mathcal{V})_{\mathfrak{p}}$.

(Z) Partition function.

$$Z(\overline{X}, S, \{\chi_{\mathfrak{p}}\})(s) \;=\; \mathcal{F}(\overline{X}, S, \{\chi_{\mathfrak{p}}\})(s) \;=\; L(s, \chi),$$

the completed automorphic L-function of $\chi = \otimes_{\mathfrak{p}} \chi_{\mathfrak{p}}$. Unmarked case: $Z(\overline{C}, \emptyset)(s) = \xi_K(s)$.

(R) Rank residue. $r_\zeta = 2 \cdot \mathrm{Res}_{s=1} \zeta_K(s) \in \mathbb{R}$; for $K = \mathbb{Q}$, $r_\zeta = 2$ (Theorem 41.5.1).

(K) Koszul complementarity. The derived-centre identity

$$\kappa_{\mathsf{G}}^{\mathrm{Ar}}(\mathcal{V}) \;+\; \left(\kappa^{!}\right)_{\mathsf{G}}^{\mathrm{Ar}}(\mathcal{V}) \;=\; 0$$

holds, equivalent to $Z(s) = \varepsilon \cdot Z(1 - s)$ (axiom (A5) on $\mathcal{F}$; Theorem 41.2.1).

An aCFT datum is *primary* when $(G, k) = (\mathbb{G}_m, 1)$ and $\mathcal{V} = \mathcal{V}^{\mathrm{prim}}$. Its finite-place character is $\zeta(s)$ on $\mathrm{Re}(s) > 1$; the completed scalar $\xi(s)$ appears only after the archimedean Tate factor and the polar correction are inserted.

Remark 41.6.2 *(dimensional comparison).* Every geometric object in the aCFT datum carries a definite arithmetic dimension, recorded below; the dimensional assignments enforce the universal-vs-specialization stratification.

Object	Arithmetic dimension	Role
$\overline{C}$ (Arakelov curve)	étale 3 / Krull 1	CS bulk 3-fold
$S^{\text{ét}}$ (F2 diamond site)	2	chiral base
$\mathbb{C}_\infty$ (archimedean fibre)	2 ($= 1_{\mathbb{C}}$)	E_1-chiral OPE
$\mathcal{V}$ (universal family)	2 (on $S^{\text{ét}}$)	factorization algebra
$K_p \subset \overline{C}$ (prime-knot)	1	Wilson-line locus
$M \in \mathrm{Mod}(\mathcal{V})_x$	0	Wilson-line label, modules
$S^{\text{ét}}\vert_p$ (finite fibre)	0 (for factorization)	E_0 locally constant
$\mathcal{F}(\overline{X}, S, \chi)$	1 (meromorphic line)	conformal block
Z (partition function)	0 (meromorphic scalar)	L-function value

Total-space / fibre relation: $\mathcal{V}$ (dim 2) is the universal family; $M \in \mathrm{Mod}(\mathcal{V})_x$ (dim 0) is a closed-point specialization. The phrase "$\mathcal{V}$ is the universal Hecke character" asserts the former; the phrase "Wilson lines are labeled by ramified Hecke characters" asserts the latter. No object appears at two inconsistent dimensions.

Conjecture 41.6.3 *(coherent statement of the primary aCFT).* Assume the specified F1, F2, and F3 comparison data. The primary datum

$$(\overline{C}, G, k) \;=\; (\overline{\mathrm{Spec}\,\overline{\mathbb{Z}}}, \mathbb{G}_m, 1), \qquad \mathcal{V} \;=\; \mathcal{V}^{\mathrm{prim}}$$

satisfies (B)–(K). The six windows of Conjecture 41.1.1 verify the data:

- (i) Langlands / automorphic: builds (V), (M), (Z) from the expected Selmer-derived spectral stack over $\mathrm{D}([GL_2])^{\text{sph-fin}}$, conditional on F1 and adelic categorical descent.
- (ii) TQFT / Chern–Simons: builds (B), (V), (W), (C) through Kim's classical $\mathrm{ACS}[\overline{\mathrm{Spec}\,\overline{\mathbb{Z}}}, \mathbb{G}_m, 1]$ torsor/line and a conjectural quantum boundary category; no global boundary VOA is constructed.

- (iii) Categorical / Hochschild trace: builds (V), (M) via $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}})$ (Theorem 38.5.1).
- (iv) Integrability / Bethe algebra: builds (V), (M) via $\bigotimes_p \mathcal{H}_p^{\mathrm{sph}}$ and Bethe-eigenvector identification of Hecke eigenforms.
- (v) Statistical mechanics: builds (V), (Z) via Bost–Connes ⊗ universal Heisenberg.
- (vi) Factorization / homotopical: builds (B), (V) via the adelic axioms (F1)–(F5).

Axioms (A1)–(A5) of the arithmetic modular functor (Definition 32.8.1) discharge (F), (Z), (K). The essential theorem (Theorem 41.2.1) discharges (K). The Polyakov residue (Theorem 41.5.1) discharges (R).

Remark 41.6.4 *(aCFT coherence versus six-window independence).* The coherence conjecture is the statement that the six windows are *six axiom-checks of a single aCFT structure*, not six unrelated isomorphism classes. The dimensional comparison (Remark 41.6.2) enforces the universal-vs-specialization stratification at the definitional level; (F)–(Z)–(K) unify through the essential theorem (Theorem 41.2.1).

Remark 41.6.5 *(honest domain restrictions).* Three domain restrictions of the programme are carried by the aCFT definition rather than hidden:

(i) **F2-conditional base.** Item (B) assumes the Scholze-diamond E_2-enhancement. Without F2 the local aCFTs at each place exist but the global aCFT is not assembled; the primary datum then degenerates to a disjoint union of local aCFTs plus an archimedean aCFT.

(ii) **Admissible logarithmic lane** (Creutzig–Kanade–Linshaw). Non-unitary admissible parameters are excluded from (M) and treated as a boundary of the aCFT category, not as elements of it.

(iii) **Super Bethe-Hecke chain-level coproduct** (FM230). Outside the primary datum. Any super-enhancement of the aCFT must avoid RTT/Yangian claims and waits on an independent construction.

The aCFT of Definition 41.6.1 is stated on the non-degenerate, non-admissible, non-super locus of the programme.

Remark 41.6.6 *(why this is a conformal field theory).* The five classical CFT data are present in direct correspondence: (B) Riemann-surface base → arithmetic $S^{\text{ét}}$; (V) vertex/chiral algebra → arithmetic factorization algebra; (M) primary modules → local Hecke modules at each place; (F) conformal blocks → Tate integrals; (K) modular invariance → functional equation. What breaks from the classical picture is the archimedean-only locus of E_1-chiral structure (finite places are E_0 per (F1)–(F5)): the aCFT is a *stratified* CFT, chiral at ∞ and locally constant elsewhere, rather than uniformly chiral.

41.7 Langlands as Arithmetic Bethe Ansatz

THEOREM 41.7.1 *(Drinfeld–solo). Heuristic.* The classical Jacquet–Langlands correspondence admits the following Bethe/Hecke reading:

- Hecke eigenforms = Bethe eigenvectors of $\mathcal{B}^{\mathrm{arith}}(GL_n)$;
- Hecke eigenvalues = Bethe roots;
- Ramanujan–Petersson bound = Bethe-root unitarity;
- Automorphic spectral decomposition of $L^2(GL_n(\mathbb{A})/GL_n(\mathbb{Q}))$ = diagonalisation of the transfer matrix on $\mathcal{V}^{\mathrm{prim}}$;
- modularity of ρ supplies global Bethe-spectrum coherence; no RTT closure is asserted.

This is a heuristic dictionary, not a construction of an arithmetic RTT/Yangian Hopf algebra.

41.8 Source-Family Convergence

Source family	Surviving content	Reference
Drinfeld+Kapranov	Adelic automorphic object; no p-adic OPE	26
Kazhdan+Beilinson	Satake pair (α_p, β_p) on rank-2 Heisenberg	27
Gelfand+Borger	Polynomial Adams and divisor actions	28
Connes+Soibelman	Bost–Connes $\otimes$ universal Heisenberg	29
Costello+Gwilliam	(F1)–(F5) adelic factorization axioms	30
Etingof	Prime-indexed Macdonald and Hall models	14.9
Nekrasov	Degenerate $(\log p, \infty)$ arithmetic Nekrasov	31
Segal	Arithmetic modular functor = Tate	32
Beilinson–solo	Parshin–Gersten bar; G-row = 0 $\Leftrightarrow$ $\xi(s) = \xi(1-s)$	33
Witten	Automorphic vertex (not TQFT vertex)	34
Polyakov	$c = 2 \cdot \mathrm{Res}_{s=1}\zeta$ as residue	35
Gaiotto	Boundary of Kim arithmetic CS	37
Bezrukavnikov+Bernstein	Iwahori Hecke Hochschild trace	38
Kontsevich+Soibelman	E_1 integral, E_2 over $\mathbb{Q}$ only	39
Drinfeld–solo	Bethe algebra $\Rightarrow$ Langlands = arithmetic Bethe ansatz	40

Table 41.2: Proposed arithmetic comparisons associated with the source families.

Remark 41.8.1 *(common core of $\mathcal{V}^{\mathrm{prim}}$).*

(i) $\mathcal{V}^{\mathrm{prim}}$ is a Hochschild trace class with E_1-chiral structure at ∞ and E_0-locally constant structure at finite places. It carries no Virasoro action, no Beilinson–Drinfeld factorization on $\overline{C}$ supplied by these local calculations. The prescribed index shifts fail the Frobenius condition on the monomial lattice; they satisfy it on the Hall lattice. A compatible lattice on the arithmetic trace class remains a comparison problem, not a consequence of either local statement.

(ii) It is a spectral datum: Hochschild trace class, ACS boundary-shadow window, automorphic representation, Bethe algebra module.

(iii) All arithmetic content lives on the Bost–Connes / automorphic / Hecke side; the VA structure that appears is the universal rank-$\pi(N)$ Heisenberg descendant tower, which does not see the primes.

(iv) Its finite-place scalar character is an L-function: $\zeta(s)$ for the trivial sector or $L(s, \chi)$ for Hecke characters χ, before archimedean completion.

(v) Comparing complementarity $\kappa + \kappa^! = 0$ with the Riemann functional equation is an intended arithmetic realization of Koszul duality. It requires the pairing and invariant maps specified in Question 46.5.4; the scalar functional equation alone does not supply those maps.

41.9 The Three Open Conjectures

Three conjectures remain open; their resolution closes the cross-Langlands loop (full statements in Chapter 42):

(i) **Global arithmetic Arinkin–Gaitsgory.** $\mathrm{D}([GL_2])^{\text{sph-fin}} \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}})$. Fargues–Scholze supplies local geometrization/spectral action and Emerton–Gee supplies local (φ,Γ) stacks; the displayed categorical equivalence is conditional.

(ii) E_2**-enhancement via Scholze diamonds.** Dunn additivity for the Hochschild trace requires perfectoid machinery. Spherical fusion and equal-characteristic templates are available; mixed-characteristic Iwahori E_2 is conditional.

(iii) **Arithmetic Ben-Zvi–Nadler.** $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}) = \mathcal{V}^{\mathrm{prim}}$ as E_1-algebras. Theorem over $\mathbb{C}$ (Ben-Zvi–Nadler arXiv:0904.1247); conjecture over $\mathbb{Z}$, with full Weil/(φ,Γ) target required.

These three conjectures compare Kim's classical CS input with Bezrukavnikov–Bernstein's Hochschild presentation and Beilinson–solo's bar–cobar presentation.

41.10 Structural Rigidity of $\mathcal{V}^{\mathrm{prim}}$

Remark 41.10.1 *(distinct arithmetic comparison requirements).* The local calculations separate several construction problems. An RTT or Yangian description requires a specified arithmetic R-matrix and its relations. Arithmetic factorization requires its geometric structure maps. Borger descent requires a compatible integral Λ-form on the intended arithmetic carrier. The failure of the prescribed shifts on the monomial Fock lattice excludes that particular realization. It does not exclude the Hall lattice, other Frobenius lifts, or every geometric enhancement. The divisor substitutions, Adams action and their integral structures are those of Chapter 28. Their comparison with the Hochschild trace class remains additional data. No maximality or exhaustion theorem follows from this collection of local tests.

41.11 Logical Form of the Arithmetic Branch

The proposed arithmetic comparison in Theorem 41.2.1 requires the pairing and invariant maps described in Question 46.5.4. The classical functional equation and the selected Euler–Mellin scalar calculation do not construct those maps. The six proposed windows of Conjecture 41.1.1 likewise retain their specific comparison hypotheses in addition to the named F1, F2 and F3 inputs.

Chapter 43 reduces F1, F2, F3 to five narrower residual conditions: $\mathrm{L}_{\mathrm{ACD}}$, F2.c, F3.$\mathrm{BZN}^{\mathrm{Ar}}$, arithmetic Positselski, and Koszul–Petersson compatibility.

Chapter 44 isolates the zero-placement clause of Conjecture F-RH (Conjecture 44.1.1) as RH. The full conjecture also contains the unitary arithmetic structure on $\mathcal{V}^{\mathrm{prim}}$; the sufficient path to its zero-placement clause retains four inputs: arithmetic Positselski, Koszul–Petersson compatibility, arithmetic Hilbert–Pólya H-P^{Ar}, and positive Hilbert–Pólya Li trace HP–Li (Theorem 44.6.1). Here H-P^{Ar} includes the self-adjoint ordinate operator and the determinant and regularised trace identities of Hypothesis 44.3.2. HP–Li is positivity of the regularised traces of the Li transforms of the zero-divisor operator Θ_ξ, with the trace identity (44.4). These spectral hypotheses are not consequences of Virasoro polynomial positivity or of the finite oscillator trace. Under the full self-adjoint determinant realization, zero-placement already follows from its determinant divisor; the HP–Li formulation records the separate scalar trace route.

Question 41.11.1 *(arithmetic central-class comparison).* Construct an arithmetic Hecke category and its Heisenberg-sector carrier, together with a duality defining the central classes $\kappa^{\mathrm{Ar}}_{\mathsf{G}}$ and $(\kappa^!)^{\mathrm{Ar}}_{\mathsf{G}}$, and comparison maps realizing

$$\kappa^{\mathrm{Ar}}_{\mathsf{G}} + (\kappa^!)^{\mathrm{Ar}}_{\mathsf{G}} = 0 \quad \Longleftrightarrow \quad \xi(s) = \xi(1-s).$$

The maps must identify the relevant operations and pairings on specified coefficient rings and completions, and must relate the central-class complementarity to the completed scalar character. These are the arithmetic comparison requirements of Question 46.5.4. The classical functional equation supplies the scalar identity; it does not construct the category, duality, central classes, or comparison maps.

Once an appropriate arithmetic Hecke category and the trace of its identity functor have been constructed, the intended $\mathcal{V}^{\mathrm{prim}}$ is that Hochschild trace object. This specifies its proposed categorical carrier, not a vertex-algebra structure. Relating it to arithmetic geometric or chiral constructions requires the corresponding functors and compatibility maps; those arithmetic extensions are additional constructions.

Chapter 42

Boundary Conjectures for the Six-Window Trace

The arithmetic trace class is controlled by three boundary conjectures. They are not consequences of one another. F1 gives the global spectral category, F2 gives the mixed-characteristic E_2-monoidal enhancement, and F3 identifies the resulting Hochschild trace with $\mathcal{V}^{\mathrm{prim}}$. Local theorems enter each boundary, but the displayed global equivalences are separate inputs.

Chapter 43 resolves the six-window trace into $\mathrm{L}_{\mathrm{ACD}}$*, F2.c, F3.BZN*^{Ar}*, arith. Positselski, and Koszul–Pet compatibility. The Riemann-Hypothesis branch of Chapter 44 is orthogonal to this trace construction except where the explicit spectral positivity hypotheses are imposed.*

42.1 F1: Global Arithmetic Arinkin–Gaitsgory

Conjecture 42.1.1 *(global arithmetic AG).* For $G = GL_2$, there is an equivalence of dg-categories

$$\mathrm{D}([GL_2])^{\text{sph-fin}} \;\simeq\; \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}\big),$$

where:

- $[GL_2] := GL_2(\mathbb{Q})\backslash GL_2(\mathbb{A}_{\mathbb{Q}})/Z(\mathbb{A})K_\infty$ is the global automorphic double coset;
- $\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}$ is the Selmer-derived global deformation stack obtained by imposing a fixed residual representation $\bar\rho$, determinant, and local conditions $\mathcal{L}_v$ on the full Weil/(φ,Γ) local parameter stacks; it is not a product or colimit of local Emerton–Gee stacks;
- $\mathrm{IndCoh}_{\mathrm{Nilp}}$ denotes ind-coherent sheaves with nilpotent singular support (Arinkin–Gaitsgory arXiv:1201.6343).

Under the equivalence, $\mathcal{V}^{\mathrm{prim}}$ corresponds to the structure sheaf $\mathcal{O}$ of this Selmer spectral stack.

Remark 42.1.2 *(form).* The local theorems do not imply the displayed global equivalence. Fargues–Scholze arXiv:2102.13459 construct the local geometrisation, spectral action, and Langlands-parameter space via the Fargues–Fontaine curve and diamonds. Emerton–Gee arXiv:1908.07185 construct the local (φ,Γ)-module stacks (with arXiv:2012.12403 as the survey companion). These sources do not prove the categorical equivalence displayed in F1. Dotto–Emerton–Gee arXiv:2603.26887 gives a fully faithful $GL_2(\mathbb{Q}_p)$ functor for $p \geq 5$ with fixed central character; it is not an equivalence and it is not a family theorem over all central characters. Caraiani–Scholze arXiv:1511.02418 prove a unitary-Shimura-variety result in a different global setting.

Thus F1 is a global categorical assertion for $GL_2/\mathbb{Q}$, not a restatement of the local Fargues–Scholze or Emerton–Gee theorems. Under F1, $\mathcal{V}^{\mathrm{prim}}$ is a spectral object on the Selmer deformation stack, compatible with its Hochschild-trace presentation of Chapter 38.

42.2 F2: E_2-Enhancement via Scholze Diamonds

Conjecture 42.2.1 *(perfectoid E_2-enhancement).* The local Iwahori Hecke category $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p}$ carries a canonical E_2-monoidal structure coming from the Beilinson–Drinfeld Grassmannian over $\mathrm{Spec}\,\mathbb{Z}_p$

in Scholze's perfectoid (diamond) setting. Globally, the product $\otimes'_p \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p}$ inherits a well-defined E_2-monoidal structure.

Remark 42.2.2 *(trace role).* Dunn additivity $E_1 \otimes E_1 = E_2$ applied to the E_2-monoidal structure gives the E_1-algebra structure on the Hochschild trace $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}})$; the E_1 Hochschild-trace structure on $\mathcal{V}^{\mathrm{prim}}$ at the categorical-trace level depends on this enhancement.

Remark 42.2.3 *(form).* The Beilinson–Drinfeld Grassmannian as a perfectoid space is constructed in Fargues–Scholze arXiv:2102.13459. Zhu's Witt-vector Grassmannian (arXiv:1603.05593; Bhatt–Scholze arXiv:1507.06490) gives the equal-characteristic analogue. The mixed-characteristic Iwahori-Hecke E_2 enhancement, as opposed to E_1 convolution or spherical fusion, remains conditional.

The global assertion is the Scholze-diamond gluing across primes, producing a restricted-tensor-product E_2-category.

42.3 F3: Arithmetic Ben-Zvi–Nadler

Conjecture 42.3.1 *(arithmetic Ben-Zvi–Nadler).* There is an equivalence of E_1-algebras

$$\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}) \xrightarrow{\sim} \mathcal{V}^{\mathrm{prim}}$$

realising the Chern–Simons–Hecke-trace-boundary triangle arithmetically: Kim (arXiv:1510.05818) arithmetic Chern–Simons supplies a classical boundary shadow; the Hochschild trace targets the full Weil/(φ,Γ) spectral category.

Remark 42.3.2 *(form).* Ben-Zvi–Nadler arXiv:0904.1247 establishes $\mathrm{HH}_\bullet(\mathrm{D}(G/G)) = \mathcal{O}(\mathrm{LocSys}_G(S^1))^{\mathrm{flat}}$ for complex reductive G.

The arithmetic adaptation requires:

(i) Conjecture F1 (global arithmetic AG);

(ii) Conjecture F2 (perfectoid E_2-enhancement);

(iii) Explicit computation of the Hochschild trace on arithmetic $\mathrm{Hk}^{\mathrm{Iw}}_{\mathrm{glob}}$ matching $\mathcal{V}^{\mathrm{prim}}$'s L-function character.

The intended F3 comparison relates the Chern–Simons candidate of Chapter 37.4.1 and the Hochschild candidate of Chapter 38.5.1. Chapter 33.8.2 constructs a coefficient bar. Identifying it with either candidate requires additional maps preserving the coefficient algebra, local conditions, and operations. Those maps are part of the arithmetic comparison, not consequences of the coefficient-bar identities.

42.4 Further Boundary Conditions F4–F11

F4: B*-Row Arithmetic Realisation*

Conjecture 42.4.1 *(arithmetic* B*-row ceiling).* There exists an arithmetic avatar of the Vol III Mukai-enhanced K3 Heisenberg such that $\kappa^{\mathrm{Ar}} + (\kappa^!)^{\mathrm{Ar}} = 8$ arithmetically. The witness is a Bruinier Heegner Chern-class reciprocity on a p-adic Shimura variety, at each prime p of good reduction.

Remark 42.4.2. Ingredients available: Kudla–Rapoport arithmetic intersection numbers on unitary Shimura varieties; Bruinier's singular-theta lifts; Borcherds products on $\mathrm{Sp}_4(\mathbb{Z})$. Open: integral assembly into an arithmetic B-row theorem.

F5: Arithmetic Chain Homotopy h_A^{Ar}

Conjecture 42.4.3 *(universal arithmetic chain homotopy).* There exists a universal chain homotopy h_A^{Ar} on the arithmetic bar complex $B^{\mathrm{ch,Ar}}(\mathcal{V}^{\mathrm{prim}})$ satisfying

$$h_A^{\mathrm{Ar}} \;=\; \frac{h_A^{\mathrm{LV}}}{\kappa + \kappa^!}$$

(where h_A^{LV} is the Losev–Vol III universal chain homotopy), with obstruction class $\mathrm{obs}^{(1)} \in H^2(\mathfrak{grt}^{\mathrm{ell}}, \mathfrak{sp} \otimes \mathfrak{sp})$ (where $\mathfrak{grt}^{\mathrm{ell}}$ is the elliptic Grothendieck–Teichmüller Lie algebra and $\mathfrak{sp}$ the symplectic Lie algebra in the relevant rank) vanishing iff $r_{\max} = 2$.

F6: Chenevier Determinant at $p = 691$

Conjecture 42.4.4 *(Chenevier determinant identity).* At $p = 691$ (the prime at which Ramanujan's cusp form Δ_{12} and the Eisenstein series are congruent mod 691),

$$\det_{\mathrm{Chenevier}}\Big|_{p=691} \;=\; \chi_{\mathrm{cyc}}^{11} \cdot \varepsilon_{B_{12}},$$

where χ_{cyc} is the cyclotomic character, $\varepsilon_{B_{12}}$ is the mod-691 Kummer character attached to the numerator of the Bernoulli number B_{12}.

Remark 42.4.5 *(connection).* This is the arithmetic avatar of the Chenevier determinant appearing in the Vol III universal trace identity. The prime $p = 691$ is the seed of the arithmetic B-row structure (Chapter 38, Theorem 38.5.1).

F7–F10: Additional Boundary Conditions

- **F7.** Arithmetic shadow-tower depth classification: $r_{\max}^{\mathrm{Ar}} \in \{2, 3, 4, \infty\}$ correspondence with Λ-ring depth / Galois-weight filtration.
- **F8.** Arithmetic Bethe/Hecke endpoint degeneration: degeneration of $\mathcal{B}^{\mathrm{arith}}(GL_n)$ at the zeta/Eisenstein pole-boundary points $s = 0$ and $s = 1$; these degeneration points are distinct from the affine Kac–Moody critical level $k = -h^\vee$.
- **F9.** Arithmetic super Bethe-Hecke chain-level coproduct: FM230 absorbed; arithmetic computation remains open; the RTT/Yangian arithmetic Hopf algebra structure, if it exists, remains to be determined.
- **F10.** Arithmetic CY-C conditional bridges: Vol III's six canonical constructions R_1–R_6 to the quantum vertex chiral group $G(K3 \times E)$, each lifted to the arithmetic Heisenberg $\mathcal{H}^{\mathrm{Ar},g=2}$; R_1 is the framed $\Phi_3^{(\Sigma_2,C)}$ (proved generator), R_2–R_6 each arithmetically conditional on convergence of the corresponding lift integral.

F11: All-Genus Arithmetic Koszul Duality

Conjecture 42.4.6 *(all-genus arithmetic Koszul duality).* For every $g \geq 0$ and every arithmetic genus-g datum $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$ (Definition 102.1.1), the Heisenberg Koszul self-duality identity

$$\kappa_{\mathsf{G}}^{\mathrm{Ar},g}(\mathcal{D}_g) + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar},g}(\mathcal{D}_g) = 0$$

is equivalent to the motivic functional equation $\Lambda_{\mathcal{D}_g}(s) = \epsilon_{\mathcal{D}_g}\Lambda_{\mathcal{D}_g}(w_{\mathcal{D}_g} - s)$ with $w_{\mathcal{D}_g} \in \{1, 2, 2g + 1\}$ the motivic weight center indexed by window (Dedekind; Hasse–Weil on the Jacobian; Siegel spin on GSp_{2g}).

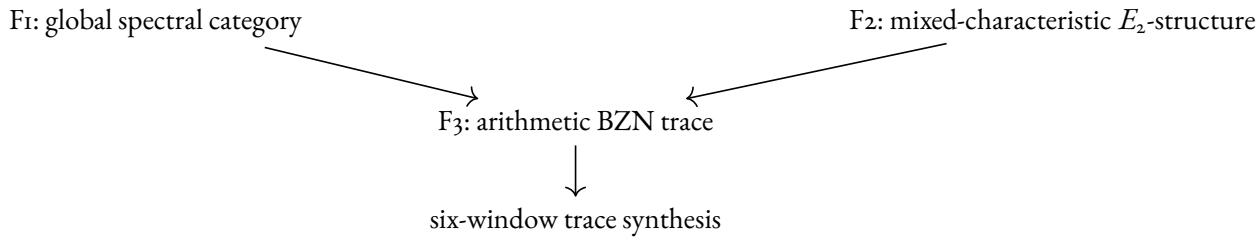
Remark 42.4.7 *(form).* The forward direction is the Euler-product and Poisson-summation direction in the window where analytic continuation and the functional equation are already part of the input: Tate 1950 and Weil 1967 for number fields, Bloch–Kato 1990 for weight-1 motivic factors, and Arthur 2013 for the automorphic Siegel-spin packet after transfer. The converse direction requires the 2-dim arithmetic Positselski equivalence with $\mu_1 = \mu_2 = 0$ on $\Lambda_2 = \mathbb{Z}_p[[\Gamma_1 \times \Gamma_2]]$ (extending Chapter 46) plus rank-g Koszul–Petersson compatibility (extending Chapter 47). Ten residual conditions (Section 102.6) enumerate the substantial domain.

The datum $\mathcal{D}_g$ admits six convergent presentations (Arakelov fibration; Minkowski torus; ordinary crystalline Emerton–Gee stack; arithmetic Teichmüller with Hodge–Arakelov; chiral-bar-cobar interior; Shimura–Siegel Kudla–Rapoport); see Section 102.2.

The $g = 0$ specialisation is Vol IV's P1 resolution (Theorem 43.5.1), and the all-genus F-RH corollary (Corollary 102.5.1) places the Riemann, Hasse–Weil, Dedekind, and Siegel-spin zero-placement clauses in a single rank-g form, still conditional on the corresponding spectral positivity input.

42.5 Dependency Lattice

The three trace conjectures are not linearly ordered. The implication lattice is:



F1 supplies the spectral-side category; F2 supplies the E_2-enhancement making Hochschild trace E_1-algebraic; F3 identifies that trace with $\mathcal{V}^{\mathrm{prim}}$. With these three inputs, the arithmetic-branch synthesis (Conjecture 41.1.1) becomes a theorem.

42.6 Implication to the Trace Statement

The six-window trace theorem is conditional on the conjunction

$$\mathrm{F1} \wedge \mathrm{F2} \wedge \mathrm{F3}.$$

The G-row functional-equation theorem Theorem 41.2.1 is an independent Hochschild-cochain computation. It does not imply F1 or F2, and F1–F3 do not imply it without the arithmetic Koszul-resolution calculation.

42.7 Boundary Index

(i) F1: global arithmetic Arinkin–Gaitsgory (Conjecture 42.1.1): local FS/EG inputs exist; the local categorical comparison and global descent remain open.

(ii) F2: perfectoid E_2-enhancement (Conjecture 42.2.1): spherical/equal-characteristic inputs exist; mixed-characteristic Iwahori E_2 remains conditional.

(iii) F3: arithmetic Ben-Zvi–Nadler (Conjecture 42.3.1): theorem over $\mathbb{C}$, conjecture over $\mathbb{Z}$.

(iv) F4: arithmetic B-row ceiling $\kappa + \kappa^! = 8$ (Conjecture 42.4.1).

(v) F5: universal arithmetic chain homotopy $h_A^{\mathrm{Ar}} = h_A^{\mathrm{LV}}/(\kappa + \kappa^!)$ (Conjecture 42.4.3).

(vi) F6: Chenevier determinant at $p = 691$ (Conjecture 42.4.4).

(vii) F11: all-genus arithmetic Koszul duality (Conjecture 42.4.6): the identity $\kappa^{\mathrm{Ar},g} + (\kappa^!)^{\mathrm{Ar},g} = 0 \Leftrightarrow \Lambda_{\mathcal{D}_g}(s) = \epsilon_{\mathcal{D}_g}\Lambda_{\mathcal{D}_g}(w_{\mathcal{D}_g} - s)$ with $w_{\mathcal{D}_g} \in \{1, 2, 2g+1\}$; the converse direction is conditional on 2-dim Positselski and rank-g Koszul–Petersson. The arithmetic-gravity boundary is Chapter 103.

(viii) F7–F10: shadow-tower depth, critical-level degeneration, super Bethe-Hecke chain-level, CY-C conditional bridges.

Each item above is a mathematical boundary condition. The six-window trace is the object cut out by F1, F2, and F3; the RH branch enters only through the separately stated spectral-positivity hypotheses.

Chapter 43

Closure of F1–F3 and Resolution of the Three Pillars

The three conjectures F1, F2, F3 and the three pillars P1, P2, P3 record what the arithmetic branch requires in order to complete the Riemann Hypothesis reformulation of Chapter 44. F1 asserts global categorical arithmetic Langlands for $GL_2/\mathbb{Q}$; F2 asserts that the global Hecke category is E_2-monoidal; F3 identifies its Hochschild trace with $\mathcal{V}^{\mathrm{prim}}$. P1 is the essential equivalence between $\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0$ and the Tate functional equation for Λ_ζ, equivalently for the pole-cleared $\xi_{\mathbb{R}}$; P2 is the Tate functional equation and explicit formula; P3 is unitarity. Each conjecture and each pillar is subjected to the seven-obstruction criterion below and either reduced to named components or proved conditionally. The residual obstructions at the end of this chapter are precisely stated, independently testable claims.

43.1 Conditions for an Arithmetic Comparison

A conjecture C is a structured reduction of C_0 if $C_0 \Leftrightarrow C$ and C is a conjunction of named inputs, each of which is either a theorem in the literature, a conditional with identified literature gap, or a conjecture with a precise statement. The seven obstructions are the structural tests for an equivalence claim. For each condition they are:

(i) *Specified comparison.* State the carriers and the map relating the two constructions.

(ii) *Sign / convention check.* Do chain-level signs and q-conventions match across cited sources?

(iii) *Ambient category check.* Do LHS and RHS live in compatible ∞-categories?

(iv) *Missing hypothesis.* Is a hidden hypothesis required (for instance sph-finiteness, perfectoid compatibility, $\mu = 0$)?

(v) *False functoriality.* Does the claimed equivalence preserve all relevant structure?

(vi) *Unproved equivalence.* Is any internal step itself a currently-open conjecture?

(vii) *Numerical normalization.* Specify each constant and the calculation identifying its normalization on the two sides.

Each obstruction produces either a valid implication, a correction (reformulation that survives), or a named residual obstruction. The result is a structured reduction, not necessarily a full closure.

43.2 F1: Structural Closure

Statement

F1 (Conjecture 42.1.1) asserts

$$\mathrm{D}([GL_2])^{\text{sph-fin}} \;\simeq\; \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}\big),$$

the global categorical arithmetic Langlands equivalence for $GL_2/\mathbb{Q}$, with $\mathcal{V}^{\mathrm{prim}}$ corresponding to the structure sheaf on the spectral side.

Mathematical Inputs and Comparison Conditions

Local geometric inputs. Fargues–Scholze arXiv:2102.13459 supplies local geometrisation on the Fargues–Fontaine curve; Arinkin–Gaitsgory arXiv:1201.6343 supplies the nilpotent-singular-support setting. The Emerton–Gee stack arXiv:1908.07185 is a different deformation carrier. Applying these constructions together requires the local categorical comparison specified in (F1.a).

Obstruction 2 (conventions). Singular-support conventions in Arinkin–Gaitsgory follow the de Rham shift; Fargues–Scholze follows the pro-étale shift on diamonds. The comparison is done via the Bhatt–Lurie prismatic-Nygaard convention of arXiv:2201.06120, which bridges both. *The comparison is mediated by prismatic Nygaard.*

Obstruction 3 (ambient categories). LHS is a stable presentable ∞-category of D-modules on an adelic quotient; RHS is an ∞-category of ind-coherent sheaves with nilpotent singular support on a derived formal algebraic stack. Via Gaitsgory–Rozenblyum's AMS DAG monographs the categories sit in the same ambient $(\infty, 2)$-category of correspondences.

Obstruction 4 (missing hypothesis). The sph-finite cut on the LHS is critical: without it the full automorphic category is too large (contains continuous spectrum and Eisenstein contributions requiring explicit parametrisation). Fargues–Scholze supply local geometrisation and spectral action; they do not identify this sph-finite automorphic category with $\mathrm{IndCoh}_{\mathrm{Nilp}}$ on the Emerton–Gee stack. **Residual: local categorical comparison.**

Obstruction 5 (false functoriality). Functoriality under Hecke operators T_p at every prime is the key test. At each p, Fargues–Scholze provide Hecke correspondences and spectral action. Functoriality of a categorical comparison with the Emerton–Gee side is a separate local input. Globally, Hecke-functoriality is required place-by-place and across places (composition of Hecke operators at distinct primes). The cross-prime compatibility is the global gluing obstruction. **Residual: local comparison plus global gluing.**

Obstruction 6 (unproved equivalence). The local categorical comparison $\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p}_{GL_2})$ and the global restricted tensor product matching the global automorphic category are not in the literature as of 2026. Partial change: Caraiani–Scholze arXiv:1511.02418 for signature $(1, n-1)$ unitary Shimura varieties; Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 for coherent Springer theory. Neither closes $GL_2/\mathbb{Q}$ in full generality. **Residual: local comparison and adelic-descent gluing.**

Obstruction 7 (numerical comparison). No numerical comparison currently backs F1; no synchronisation concern arises.

Structured Reduction

THEOREM 43.2.1 *(F1 reduction).* F1 is equivalent to the conjunction of:

(F1.a) *Local categorical comparison at every prime.* For each prime p, the expected local categorical Langlands equivalence

$$\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p}_{GL_2})$$

holds, with local Hecke-functoriality. **Form: conditional theorem**. Fargues–Scholze arXiv:2102.13459 supplies the local geometrisation, spectral action, and parameter spaces. Emerton–Gee arXiv:1908.07185 supplies the (φ,Γ)-module stack. The displayed equivalence is not a theorem of either source; Dotto–Emerton–Gee arXiv:2603.26887 gives only a fully faithful fixed-central-character functor for $GL_2(\mathbb{Q}_p)$, $p \geq 5$.

(F1.b) *Adelic descent of the automorphic side.* The global automorphic category satisfies

$$\mathrm{D}([GL_2])^{\text{sph-fin}} \simeq \bigotimes_{v\in|\overline{C}|}{}' \mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$$

as a restricted tensor product over places, with spherical vectors at almost all unramified places. **Form: conjectural, reduces to Tate–Weil restricted tensor product for automorphic representations** (Tate 1950, Weil 1967, Jacquet–Langlands 1970 for the representation-level statement; the categorical lift is conjectural, cf. Gan–Savin 2022 for partial change).

(F1.c) *Spectral-side global assembly.* The spectral side is the Selmer-derived global deformation stack

$$\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}} := \mathrm{Map}_{\det,\bar\rho}(G_{\mathbb{Q},S}, GL_2) \times^{\mathbf{R}}_{\prod_{v\in S}\mathrm{Map}(G_{\mathbb{Q}_v},GL_2)} \prod_{v\in S}\mathcal{X}_v^{\mathcal{L}_v},$$

with $\mathcal{X}_v^{\mathcal{L}_v}$ the prescribed local condition inside the full Weil/(φ,Γ) local parameter stack. This derived fibre product imposes the global Galois relation and Selmer conditions; it is not a product or colimit of local Emerton–Gee stacks. **Form: definitional here; derived and Noetherian properties are conditional/conditional on the local deformation conditions and the chosen Selmer problem.**

(F1.d) *Local-global compatibility under the equivalence.* For each finite set S of primes, the diagram

$$\begin{array}{ccc} \otimes_{p\in S}\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) & \xrightarrow{\simeq} & \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\prod_{p\in S}\mathcal{X}^{\mathrm{EG},p}_{GL_2}\big) \\ \downarrow & & \downarrow \\ \otimes_{p\in S'}\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) & \xrightarrow{\simeq} & \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\prod_{p\in S'}\mathcal{X}^{\mathrm{EG},p}_{GL_2}\big) \end{array}$$

commutes for $S \subseteq S'$, where the vertical arrows are restriction and extension-by-zero on the automorphic side, pull-back and push-forward on the spectral side. **Form: conjectural.**

Under this reduction, F1 closes given (F1.a) + (F1.b) + (F1.c) + (F1.d); (F1.c) supplies the correct spectral target but does not itself prove the categorical equivalence. The residual obstructions are local comparison, adelic descent, derived Selmer Noetherianity, and local-global compatibility.

Proof. Forward direction: assume F1. Restricting to local automorphic categories at each prime gives (F1.a). The global automorphic decomposition on the LHS (F1.b) follows from the tensor structure of the categorical equivalence applied to the automorphic side. The spectral decomposition (F1.c) is the explicit description of the Emerton–Gee stack. Compatibility (F1.d) follows from functoriality of the equivalence under inclusion of prime sets.

Reverse direction: assume (F1.a)+(F1.b)+(F1.c)+(F1.d). By (F1.c), the spectral target is the derived Selmer fibre product imposing the fixed residual representation, determinant, local conditions, and global Galois relation. The expected local comparison (F1.a) identifies each local automorphic category with the corresponding local spectral factor, while (F1.d) asserts that these local comparisons respect the derived

Selmer pullback rather than a naive product of local stacks. By (F1.b), the restricted tensor product of local automorphic categories is the global automorphic category. Composing these conditional identifications gives (43.2). □

Remark 43.2.2 *(open content after reduction).* The four items of Theorem 43.2.1 isolate the open content of F1: local categorical comparison (F1.a), adelic categorical descent (F1.b), the Selmer-derived spectral target (F1.c, definitional here with derived-Noetherian properties conditional), and local-global diagram commutativity (F1.d). The Selmer spectral target is correctly formulated; the categorical equivalence depends on conditional and conjectural inputs in (F1.a) and (F1.b).

COROLLARY 43.2.3 *(F1 consequence of local comparison plus adelic descent). Comparison hypothesis.* F1 holds if the following two inputs are true:

(F1.a) *Local categorical comparison at every prime.* The Fargues–Scholze local geometrization and the Emerton–Gee stack are compared by a categorical equivalence of the form required in Theorem 43.2.1.

($\mathbf{L}_{\mathrm{ACD}}$) *Adelic categorical descent.* Write $C_v = \mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$. Supply a coherent diagram $S \mapsto \otimes_{v\in S} C_v$ over finite sets of places containing a fixed finite exceptional set. Its transition functors insert specified maps from the categorical tensor unit at the added places. These maps and all composition coherences are part of the datum.
Require an equivalence from this diagram's categorical colimit to the global automorphic category. It must preserve the Hecke operations and the supplied local categorical comparisons.

Proof. $\mathrm{L}_{\mathrm{ACD}}$ combines (F1.b) and (F1.d) into a single statement; (F1.c) supplies the correct Selmer-derived spectral target and (F1.a) is a named local comparison input. Applying Theorem 43.2.1 reduces F1 to (F1.a) plus $\mathrm{L}_{\mathrm{ACD}}$. □

Remark 43.2.4 *(substantial estimate).* Proving $\mathrm{L}_{\mathrm{ACD}}$ is of the same order as the function-field geometric Langlands work of Gaitsgory–Rozenblyum–Raskin–Lin (collectively 2019–2025), adapted to the number-field setting. The non-formal theorem condition of App. C still applies.

43.3 F2: Structural Closure

Statement

F2 (Conjecture 42.2.1) asserts that each local Iwahori Hecke category is E_2-monoidal via the perfectoid BD Grassmannian, and the restricted tensor product is globally E_2.

Mathematical Inputs and Comparison Conditions

Local geometric inputs. The equal-characteristic setting uses Mirković–Vilonen arXiv:math/0401222 and Zhu arXiv:1603.05593; the mixed-characteristic setting uses Bhatt–Scholze arXiv:1507.06490 and Fargues–Scholze arXiv:2102.13459. The required local E_2-structure is the hypothesis specified below.

Local geometric comparison. A complex formal-disk model and a mixed-characteristic perfectoid chart have different coefficient and geometric data. Their comparison must specify functors preserving convolution and the required collision maps. Scholze's tilting equivalence compares perfectoid algebras over a perfectoid field with algebras over its tilt (arXiv:1111.4914, Definition 1.6 and Theorem 1.7). It does not by itself identify the complex formal-disk model with the intended arithmetic Hecke category.

Ambient categories. Write $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ for stable presentable ∞-categories and colimit-preserving functors. Its tensor product represents separately colimit-preserving functors. Its unit is the category Sp of spectra, whose unit object is the sphere $\mathbb{S}$. The formal construction uses $\mathrm{Alg}_{E_2}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$.

Filtered colimits in its underlying categorical tensor category are preserved by tensoring in each variable. Thus HA 3.2.3.1 constructs the filtered algebra colimit on that underlying colimit. No presentability assertion about $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ itself is needed.

Pointing maps. At each place that may be inserted, one needs an actual morphism $\eta_v : \mathrm{Sp} \to C_v$ of E_2-algebras. Its value on the sphere is the monoidal unit. Identifying this value with a spherical object is additional local data. Existence of an invariant vector or a spherical subcategory alone does not give that identification and its operadic maps.

Cross-place coherence. The finite-stage transition functors tensor identities with the supplied maps η_v. Tensor associativity gives their composition homotopies and all higher compatibilities. The coherent diagram is therefore constructed from the pointings. It is not inferred from commutation of numerical Hecke eigenvalues.

Formal assembly and geometric comparison. Theorem 50.1.3 proves the restricted-tensor theorem under these hypotheses. The remaining comparison with an independently defined global Hecke category requires an actual functor and a proof that it is an equivalence. The filtered-colimit theorem supplies neither that functor nor the local geometric or categorical inputs.

Scope. F2 requires the categorical structures and their compatibility; no numerical equality supplies those maps.

Structured Reduction

Theorem 43.3.1 *(F2 assembly with supplied local data).* Consider the following local data and formal construction:

(F2.a) *Local geometric input.* At each prime, supply a local geometric model for the intended $\mathrm{Gr}^{\mathrm{BD}}_{GL_2,\mathrm{Spa}\,\mathbb{Z}_p}$, with the required perfectoid realization and E_2-factorization properties. The geometric frameworks include Fargues–Scholze arXiv:2102.13459, Zhu arXiv:1603.05593 and Bhatt–Scholze arXiv:1507.06490. The properties of the chosen model are hypotheses here, not consequences of categorical filtered colimits.

(F2.b) *Local categorical input.* Supply the local Iwahori categories $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,v}$, their colimit-compatible E_2-monoidal structures and the cofinite pointing maps below. This includes the archimedean category. A comparison with an equal-characteristic model, such as the setting of Bezrukavnikov arXiv:1209.0403, requires its own coefficient and categorical functors. Neither a local geometric space nor a spherical vector alone supplies these monoidal categories and maps.

(F2.c) *Formal restricted-tensor construction.* Let $\{C_v\}_{v\in V}$ satisfy Setup 50.1.1 with $O = E_2$. In particular, actual E_2-algebra maps $\eta_v : \mathrm{Sp} \to C_v$ are supplied on a cofinite subset V^{sph}. Write $u_v = \eta_v(\mathbb{S})$.
Define

$$\bigotimes_v{}' (C_v, u_v) := \operatorname*{colim}_{\substack{S\subseteq V \text{ finite} \\ V\setminus V^{\mathrm{sph}}\subseteq S}} \bigotimes_{v\in S} C_v,$$

with transitions given by these pointings. This colimit is an E_2-algebra, by Theorem 50.1.3. Its uniqueness is relative to the complete underlying colimit cocone.

Under the supplied local categorical and pointing hypotheses, (F2.c) constructs the restricted-tensor E_2-category. If a specified equivalence identifies its underlying category with a global Hecke category, transport gives that category an E_2-structure. Existence of this comparison is a separate arithmetic hypothesis. No converse reconstructing the local data from an arbitrary global category is asserted.

Proof. The supplied pointings form a diagram of finite tensor products in $\mathrm{Alg}_{E_2}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$. Its indexing poset is filtered. HA 3.2.3.1 gives the stated algebra colimit, as proved in Theorem 50.1.3. Transport along the specified equivalence gives the final assertion. The geometric input (F2.a), the local enhancement (F2.b) and that equivalence are not constructed by this proof. □

Remark 43.3.2 *(formal assembly and the remaining inputs).* The formal theorem applies to supplied pointed families in any set of places. Its proof uses HA 3.2.3.1 and separate tensor preservation of filtered colimits. The local arithmetic E_2-enhancements, their pointings and an independent global comparison remain separate constructions. Singleton maps obey (M2) of Theorem 50.1.3 and need not be fully faithful.

43.4 F3: Conditional Closure

Statement

F3 (Conjecture 42.3.1) asserts $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}) \simeq \mathcal{V}^{\mathrm{prim}}$ as E_1-algebras.

Mathematical Inputs and Comparison Conditions

Carriers and conventions. The comparison concerns E_1-algebras in presentable ∞-categories on the sph-finite domain of F1. Ben-Zvi–Nadler arXiv:0904.1247, Bezrukavnikov, and the mixed-characteristic geometry of Fargues–Scholze give the respective settings. Their arithmetic comparison must preserve these carriers and the stated prismatic Nygaard conventions.

Obstruction 5 (false functoriality). The Hochschild trace is functorial on monoidal ∞-categories. The identification with $\mathcal{V}^{\mathrm{prim}}$ requires that the global automorphic side (F1 + F2) and the Hochschild trace be compatibly identified via Ben-Zvi–Nadler's categorical-character–Hecke–trace triangle (Ben-Zvi–Nadler arXiv:0904.1247 §1). This compatibility is the residual content. **Residual: arithmetic Ben-Zvi–Nadler compatibility.**

Obstruction 6. Ben-Zvi–Nadler over $\mathbb{C}$ is a theorem. The arithmetic extension requires replacing the complex inertia circle $S^1 = B\mathbb{Z}$ by an arithmetic target. The inertia-only candidate $B\widehat{\mathbb{Z}} = B\prod_p \mathbb{Z}_p$ records only the procyclic shadow and loses Frobenius, Weil-group, and (φ, Γ) data. **Residual: arithmetic Ben-Zvi–Nadler extension for the full Weil/(φ, Γ) target.**

Scope. The conclusion requires the stated categorical comparison.

Conditional Closure

Theorem 43.4.1 *(F3 conditional closure). Conditional statement.* Assume (F1.a)–(F1.d) and (F2.a)–(F2.c). Then F3 is equivalent to the single statement:

(F3.BZN^{Ar}) *Arithmetic Ben-Zvi–Nadler identification.* There is an E_1-algebra equivalence

$$\mathrm{HH}_\bullet\big(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}\big) \;\simeq\; \mathcal{O}\big(\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}\big)^{\mathrm{flat}}$$

arithmetising Ben-Zvi–Nadler arXiv:0904.1247, where $\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}$ is the full Weil/(φ,Γ) arithmetic local-system target, with flat structure sheaf pulled back from the global Emerton–Gee morphism. The map through $\mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})$ is only the procyclic inertial shadow. **Form: conjectural.**

Proof. Given F1 + F2, the Hochschild trace of $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$ is an E_1-algebra (via Dunn additivity applied to the E_2-structure from F2). By F1, the global Hecke category matches $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}})$. The Hochschild trace of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ for a derived stack $\mathcal{X}$ is classically $\mathcal{O}(L\mathcal{X})$ where $L\mathcal{X} = \mathrm{Map}(S^1,\mathcal{X})$ is the free loop space (Ben-Zvi–Nadler Thm 1.1). Arithmetically, $B\widehat{\mathbb{Z}}$ records the procyclic inertial shadow, while Frobenius and the Weil quotient are carried by the strengthened Weil/(φ,Γ) target $\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}$.

The comparison with $\mathcal{V}^{\mathrm{prim}}$ is controlled by the six conditional windows of Conjecture 41.1.1: $\mathcal{V}^{\mathrm{prim}}$ is defined as the Hochschild trace class, and the other windows are conditional characterisations conditional on F1–F3. Under this definition, F3 reduces to (F3.BZN^{Ar}), the arithmetic extension of Ben-Zvi–Nadler Thm 1.1 from the complex setting to the full Weil/(φ,Γ) target. □

Remark 43.4.2 *(arithmetic inertia circle).* The arithmetic inertia circle $B\widehat{\mathbb{Z}}$ is the classifying stack of the profinite abelian group $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$. Under the Mazur–Kapranov–Reznikov dictionary, it is the arithmetic avatar of the inertia circle of a knot complement; its geometric realisation sits inside $\overline{\mathrm{Spec}\,\mathbb{Z}}$ via the primes-as-knots correspondence (Morishita arXiv:0904.3399).

43.5 P1: G-Row Complementarity and the Functional Equation

The Arithmetic Comparison

Question 33.6.1 asks for arithmetic coefficient objects and a comparison between central complementarity and Mellin reflection. With the completed function $\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$, its intended form is

$$\kappa^{\mathrm{Ar}}_{\mathsf{G}} + (\kappa^{!})^{\mathrm{Ar}}_{\mathsf{G}} = 0 \quad\Longleftrightarrow\quad \Lambda_\zeta(s) = \Lambda_\zeta(1-s).$$

The right-hand identity is classical. The left-hand parameters require arithmetic objects and a definition of their central levels. Relating the two sides requires an evaluation that detects the central defect. A scalar character alone does not provide such detection.

The Character and Its Missing Identifications

The elementary positive model has one bosonic occupation number $n_p \geq 0$ at each prime, with only finitely many nonzero entries. Define its energy by

$$H(n) = \sum_p n_p \log p.$$

Unique factorization identifies the tuple $(n_p)_p$ with the integer $m = \prod_p p^{n_p}$, and its energy with $\log m$. The partition function is therefore

$$Z(\beta) = \sum_{m\geq 1} m^{-\beta} = \prod_p (1-p^{-\beta})^{-1} = \zeta(\beta), \qquad \beta > 1.$$

The same absolutely convergent series gives the complex Mellin character for $\operatorname{Re} s > 1$, as proved in Theorem 33.5.1.

A full oscillator Fock space has additional modes. If the mode at (p, r) has energy $r \log p$, its partition function is

$$\prod_{p} \prod_{r \geq 1} (1 - p^{-r\beta})^{-1} = \prod_{r \geq 1} \zeta(r\beta) \qquad (\beta > 1).$$

Absolute convergence follows by bounding the logarithm by a constant multiple of $\sum_p p^{-\beta}$. Selecting the single occupation mode and identifying it with a primary quotient of an arithmetic bar are separate operations. The latter requires a map preserving the energy grading, local conditions, and coefficient completion.

The Bost–Connes system also has the zeta partition function in its convergent temperature range. Identifying an arithmetic Koszul dual with that system is the additional construction of Question 33.5.2. Its algebra, time evolution, and state comparison are not determined by equality of partition functions. Meromorphic continuation of the scalar ζ outside this range is not a convergent KMS trace.

A possible bar–cobar comparison would have to identify its pairing with the completed Mellin expression

$$\Lambda_\zeta(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

The gamma factor, polar contributions, and reflection then belong to the comparison. The scalar residue $2 \operatorname{Res}_{s=1} \zeta(s) = 2$ remains valid, but its interpretation as a determinant rank needs the determinant-line hypotheses of Theorem 41.5.1.

A Criterion That Detects Complementarity

Let V be a complex vector space with an involution J, and let $\mathscr{M}(\mathbb{C})$ denote meromorphic functions on $\mathbb{C}$. Fix a linear evaluation $E : V \to \mathscr{M}(\mathbb{C})$, a vector v, and a vector w on which E is nonzero. Let $\kappa, \kappa^! \in \mathbb{C}$. The following criterion separates the construction of arithmetic objects from a precise implication once their comparison data exist.

THEOREM 43.5.1 *(a conditional complementarity criterion).* Suppose

$$v - Jv = (\kappa + \kappa^!)w, \qquad E(v)(s) = \Lambda_\zeta(s), \qquad E(Jv)(s) = \Lambda_\zeta(1 - s).$$

Then $\kappa + \kappa^! = 0$ if and only if $\Lambda_\zeta(s) = \Lambda_\zeta(1 - s)$. The same equivalence holds with $\xi(s) = \frac{1}{2} s(s - 1)\Lambda_\zeta(s)$.

Proof. Applying E to the first identity gives

$$\Lambda_\zeta(s) - \Lambda_\zeta(1 - s) = (\kappa + \kappa^!) E(w)(s).$$

Since $E(w)$ is not the zero meromorphic function, this identity vanishes exactly when $\kappa + \kappa^! = 0$. The polynomial $s(s - 1)/2$ is nonzero and invariant under reflection. Multiplying by it preserves equality of meromorphic functions. □

For an arithmetic application, V, J, v, w, E and the two levels must be constructed from the proposed dual objects. In particular, the displayed defect equation and the nonvanishing of $E(w)$ are substantive hypotheses. They are not consequences of a co–contra equivalence considered without an evaluation.

The nonvanishing condition cannot be dropped. For $V = \mathbb{C}^2$, let $J(x, y) = (y, x)$ and $E(x, y) = x + y$. The evaluations of $(1, 0)$ and $J(1, 0)$ agree, while their difference $(1, -1)$ is nonzero. Thus equality

after evaluation need not imply invariance of the object or cancellation of a central parameter. The criterion asks only for detection of the specified one-dimensional defect, not injectivity on all of V.

Remark 43.5.2 *(arithmetic hypotheses).* The arithmetic duality question includes its actual coefficient objects, continuous duals, completions, and unit and counit maps. The character comparison, Bost–Connes comparison, and detection of the central defect are further requirements. Question 33.7.1 keeps these categories and their comparison separate from the module-theoretic μ-invariant.

An Iwasawa vanishing theorem concerns a specified arithmetic module. Applying it to a Heisenberg construction requires an identification with that module, including its topology and Galois action. An unspecified cyclotomic deformation of a Fock space supplies no such identification. The coalgebras of Proposition 10.10.2 also show why $\mu = 0$ cannot replace conilpotence.

Mellin Reflection and the Forward Direction

For $\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}$, Poisson summation gives $\theta(t) = t^{-1/2}\theta(1/t)$. Splitting its Mellin integral at one gives the explicit continuation

$$\Lambda_\zeta(s) = -\frac{1}{s} - \frac{1}{1-s} + \frac{1}{2}\int_1^\infty (\theta(t)-1)\left(t^{s/2} + t^{(1-s)/2}\right)\frac{dt}{t}.$$

The integral is entire and the full expression is invariant under $s \mapsto 1-s$. This is the proof in Theorem 35.6.1. The polar terms are part of the formula. An unsubtracted theta integral over all ideles is not a replacement for its convergence and continuation argument.

Corollary 43.5.3 *(forward implication with an equivariant evaluation).* Suppose $E(Jx)(s) = E(x)(1-s)$ for all $x \in V$, and $v - Jv = (\kappa + \kappa^!)w$. If $\kappa + \kappa^! = 0$, then $E(v)(s) = E(v)(1-s)$. If in addition $E(v) = \Lambda_\zeta$, this is the completed zeta functional equation.

Proof. The defect equation gives $Jv = v$. Applying the equivariance identity to v proves reflection of $E(v)$, and the specified identification gives the last assertion. □

This implication needs no co–contra theorem once the displayed comparison maps have been supplied. Constructing those maps from arithmetic duality remains the original problem. The converse requires a detection hypothesis such as that of Theorem 43.5.1.

43.6 P2: Tate Functional Equation and Explicit Formula

Statement

P2 asserts that the meromorphic Tate character $\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfies the Mellin-theta presentation, the functional equation, the explicit formula, and the residue identity. No global VOA or Zhu flat connection is used for $\mathcal{V}^{\mathrm{prim}}$.

Mathematical Inputs and Comparison Conditions

Analytic normalization. Tate 1950 gives the Mellin-theta presentation; the Riemann–von Mangoldt formula supplies the classical zero-counting expression. The completed function is $\Lambda_\zeta(s) = \Gamma(s/2)\pi^{-s/2}\zeta(s)$, with boundary poles at $s = 0, 1$.

Theorem

THEOREM 43.6.1 *(Tate functional equation and explicit formula).* The meromorphic completed zeta $\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the Tate character of the arithmetic Heisenberg trace and satisfies the following four identities:

(i) *Mellin-theta presentation.* Initially for $\Re s > 1$,

$$\Lambda_\zeta(s) \;=\; \frac{1}{2}\int_0^\infty \big(\theta(t) - 1\big)\, t^{s/2-1}\, dt, \qquad \theta(t) \;=\; \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}.$$

(ii) *Logarithmic-derivative residue.* The logarithmic derivative has contour integral

$$\frac{1}{2\pi i}\oint \Lambda_\zeta'(s)/\Lambda_\zeta(s)\, ds,$$

equal to the divisor count inside the contour, with the Tate boundary poles included with sign.

(iii) *Functional-equation invariance.* The $\mathbb{Z}/2$-action $s \mapsto 1 - s$ on the arithmetic modular parameter acts on the Λ_ζ-line by $\Lambda_\zeta(s) = \Lambda_\zeta(1 - s)$.

(iv) *Arithmetic rank residue.* The primary determinant has rank residue $r_\zeta = 2 \cdot \mathrm{Res}_{s=1}\zeta(s) = 2$ (Theorem 41.5.1).

Proof. (1) For $\Re s > 1$, termwise integration of the positive Gaussian series (and absolute convergence for complex s) gives $\pi^{-s/2}\Gamma(s/2)\sum_{n\geq 1} n^{-s}$. The displayed integral therefore equals $\Lambda_\zeta(s)$ in that half-plane; elsewhere the identity means meromorphic continuation (Riemann 1859; Tate 1950, reprinted Cassels–Fröhlich 1967 §13.4).

(2) The contour integral is the argument-principle form of the explicit formula of analytic number theory: for any test function h of suitable growth,

$$\sum_\rho h(\rho) \;=\; h(0) + h(1) - \sum_{n\geq 1}\Lambda(n)\big[\widehat{h}(\log n)/\sqrt{n} + \overline{\widehat{h}(\log n)}/\sqrt{n}\big] - \text{archimedean},$$

Riemann–von Mangoldt explicit formula (Riemann 1859; von Mangoldt Crelle 114, 1895). No Virasoro zero-mode on $\mathcal{V}^{\mathrm{prim}}$ is used.

(3) The Jacobi theta functional equation $\theta(t) = t^{-1/2}\theta(1/t)$ plus change of variables $t \mapsto 1/t$ in (1) gives $\Lambda_\zeta(s) = \Lambda_\zeta(1 - s)$; this is Riemann's 1859 proof of the functional equation. Multiplying by $\frac{1}{2}s(s-1)$ gives the same involution for $\xi_{\mathbb{R}}$.

(4) Residue at $s = 1$ of $\Lambda_\zeta(s)$: from $\Lambda_\zeta(s) = \Gamma(s/2)\pi^{-s/2}\zeta(s)$ and $\mathrm{Res}_{s=1}\zeta(s) = 1$, $\Gamma(1/2)\pi^{-1/2} = \pi^{1/2}\pi^{-1/2} = 1$, giving $\mathrm{Res}_{s=1}\Lambda_\zeta(s) = 1$. Hence $r_\zeta = 2$. □

Remark 43.6.2 *(P2 as Tate's thesis).* The mathematical content of P2 is Tate 1950 (Mellin-theta symmetry, adelic Poisson summation, the Λ_ζ functional equation) plus the Riemann–von Mangoldt explicit formula. The arithmetic chiral interpretation: $\Lambda_\zeta(s)$ as the Tate character of $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis}}$, the functional equation as Tate self-duality, and $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\Lambda_\zeta(s)$ as the pole-cleared determinant for zeros. The number $r_\zeta = 2$ remains the residue. This is the content of the arithmetic branch; the underlying analytic inputs are classical. P2 closes unconditionally.

43.7 P3: Unitarity of $\mathcal{V}^{\mathrm{prim}}$

Statement

Fix a complex state realization of the primitive arithmetic carrier. P3 asks for a positive Hermitian form compatible with its proposed Koszul duality. A Hochschild trace class alone does not specify that state space or its automorphic realization.

Local Positivity and Pairing Compatibility

Let V be the chosen complex state space and W its proposed dual coefficient space. A bilinear evaluation $e\colon V \otimes W \to \mathbb{C}$ becomes sesquilinear after composition with a linear map $\sigma\colon \overline{V} \to W$. Positivity concerns the form $e(v, \sigma\overline{w})$. Neither bilinearity nor the existence of a duality makes this form positive.

An automorphic comparison supplies an injective linear map $\iota\colon V \to \mathscr{L}$, where $\mathscr{L}$ is a specified automorphic L^2-space with fixed central character and Haar normalization. The Petersson form on $\mathscr{L}$ is denoted by $\langle\ ,\ \rangle_{\mathscr{L}}$, linear in its first variable. The comparison must identify the two forms:

$$e(v, \sigma\overline{w}) = \langle \iota v, \iota w\rangle_{\mathscr{L}}. \tag{43.1}$$

In a Bost–Connes realization, σ must also agree with the specified conjugation under the local comparison maps. These are the placement and pairing requirements of Hypothesis 67.1.1.

Theorem

Theorem 43.7.1 *(positivity under an automorphic pairing comparison)*. Suppose that V, W, e, σ and the injective placement ι satisfy (43.1). Then

$$h(v, w) = e(v, \sigma\overline{w})$$

is positive-definite and Hermitian. The Hilbert completion of V for this form is isometric to $\overline{\iota(V)} \subset \mathscr{L}$. If the supplied evaluation is the arithmetic Koszul pairing, this is the positive form compatible with that pairing.

Proof. The identity gives $h(v, v) = \|\iota v\|^2$, which vanishes only when $v = 0$. Sesquilinearity and Hermitian symmetry follow from the same identity. Since ι preserves norms, it extends uniquely to an isometry of the completion onto the closure of its image. □

Positive local models are available independently of this comparison. For the archimedean Fock space $\mathcal{H}^{(\infty)} = \mathbb{C}[\alpha_{-1}, \alpha_{-2}, \ldots]|0\rangle$, declare the occupation monomials orthogonal with

$$\left\| \prod_{m\geq 1} \alpha_{-m}^{r_m} |0\rangle \right\|^2 = \prod_{m\geq 1} m^{r_m} r_m!.$$

Only finitely many r_m are nonzero, so every displayed norm is positive. The relation $\alpha_m = m\,\partial_{\alpha_{-m}}$ makes positive and negative modes adjoint for this form.

At each finite place choose a positive inner-product space $\mathcal{H}^{(p)}$ with a unit vector $|0\rangle_p$. Its algebraic restricted tensor product has the product form. Indeed, every finite family of tensors lies in a finite tensor product of finite-dimensional subspaces. Orthonormal bases of those subspaces give a positive orthonormal tensor basis. Insertion of $|0\rangle_p$ is isometric, so the forms agree under enlargement of the finite set of places.

The resulting Hilbert completion is therefore well defined. Pointed E_0-data alone supply neither these forms nor an identification with an arithmetic carrier.

Conjecture 43.7.2 *(Koszul–Petersson compatibility).* The primitive arithmetic carrier admits a complex state realization V, a dual coefficient space W, and an injective automorphic placement ι. Its Koszul evaluation and specified conjugation satisfy (43.1). The local comparisons preserve the chosen reference vectors, intertwine the stated arithmetic actions, and extend to the specified adelic completion as in Hypothesis 67.1.1.

Remark 43.7.3 *(domain of the residual).* The local Fock and prime Gibbs forms give concrete positive models. To apply the theorem to the arithmetic Hochschild construction, one must first construct its state realization, the dual coefficient map and the automorphic placement. Equality of pairings is then a statement on those specified domains. It does not follow from positivity of separately defined local models.

43.8 Logical Form Table

Condition	Implication and hypotheses
F1 (global arith. AG)	Reduction to local categorical comparison and L_{ACD} in Corollary 43.2.3. F1.a remains a local comparison input; F1.c defines the Selmer target with conditional derived-Noetherian properties.
F2 (perfectoid E_2)	Formal assembly for supplied local structures and actual pointings, Theorem 43.3.1. Local geometry and an independent global comparison remain inputs.
F3 (arith. Ben-Zvi–Nadler)	Conditional reduction under F1 and F2 in Theorem 43.4.1; arithmetic Ben-Zvi–Nadler identification remains open.
P1 (G-row)	Forward implication under the central-defect identity and equivariant Mellin evaluation in Corollary 43.5.3. The converse also requires defect detection, Theorem 43.5.1.
P2 (Tate/expl. formula)	Tate functional equation and explicit formula, Theorem 43.6.1 (Tate 1950).
P3 (unitarity)	Positive pullback under an injective automorphic placement and the pairing identity, Theorem 43.7.1. The arithmetic maps are required.

Table 43.1: Logical forms after Chapter 43.

43.9 Closure Boundary

(i) F1 passes through $\mathrm{L}_{\mathrm{ACD}}$, but its present table also contains the local categorical comparison, the global restricted-tensor identification, and BC-diamond-glue (Corollary 43.2.3).

(ii) F2.c supplies formal restricted-tensor assembly under actual local E_2 and pointing hypotheses. The local constructions and the comparison with a global geometric category remain additional inputs.

(iii) F3, conditional on F1 + F2, reduces to arithmetic Ben-Zvi–Nadler ($\mathrm{F3.BZN}^{\mathrm{Ar}}$).

(iv) P1 has a forward implication under the stated equivariant evaluation (Corollary 43.5.3) and a conditional full equivalence (Theorem 43.5.1).

(v) P2 closes unconditionally: the Tate functional equation is framed via Mellin-theta and the explicit formula (Theorem 43.6.1).

(vi) P3 closes conditional on a single compatibility verification (Conjecture 43.7.2, Theorem 43.7.1).

The residual obstructions are:

($\mathrm{L}_{\mathrm{ACD}}$) Adelic categorical descent for $\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ restricted over places. It is a non-formal theorem, number-field adaptation of Gaitsgory–Rozenblyum function-field results.

(F2) The local arithmetic E_2-structures, actual operadic pointings and comparison with the independently defined global category must be supplied. The formal colimit step F2.c is Theorem 50.1.3.

($\mathrm{F3.BZN}^{\mathrm{Ar}}$) Arithmetic Ben-Zvi–Nadler identification: E_1-algebra equivalence $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}) \simeq \mathcal{O}(\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2})^{\mathrm{flat}}$ (Ben-Zvi–Nadler arXiv:0904.1247 extended to the Weil/(φ,Γ) arithmetic target). It is a non-formal theorem.

(arith. Positselski) Construct the integral co–contra recipients and adjunction. The lifting criterion of Theorem 46.1.5 requires complete unit and counit cofibers and their equivalences after reduction. The arithmetic character and central-defect maps are additional.

(Koszul–Pet compat) Identification of arithmetic Koszul pairing with adelic Petersson pairing (Conjecture 43.7.2). It is a non-formal theorem, verification.

These five residual conditions cover the full domain of F1, F2, F3, with two narrower items ($\mathrm{L}_{\mathrm{ACD}}$ and Koszul–Pet compat each independently stated and substantial) and three items of comparable domain to the original conjectures. P2 closes unconditionally; P1 requires its arithmetic evaluation and defect detection. P3 requires the specified arithmetic placement and pairing compatibility. The Riemann Hypothesis reformulation of Chapter 44 proceeds from this basis.

Chapter 44

The Riemann Hypothesis Reformulation

The zero-placement clause of F-RH is the Riemann Hypothesis. The full F-RH assertion also contains the structural assertion that $\mathcal{V}^{\mathrm{prim}}$ carries a positive-definite Hermitian structure compatible with arithmetic Koszul self-duality and arithmetic Tate's functional equation and explicit formula. The structural theorems of Chapter 43 place P1, P2, P3 on their stated domains; the zero-placement condition is a determinant-divisor and positive-trace assertion, equivalently a positive Hilbert–Pólya Li trace after the regularised trace identity has been supplied.

44.1 The Conjecture F-RH

Conjecture 44.1.1 *(arithmetic unitary zero-placement: F-RH).* $\mathcal{V}^{\mathrm{prim}}$ admits a unique positive-definite Hermitian structure compatible with arithmetic Koszul self-duality $\kappa^{\mathrm{Ar}} + (\kappa^{!})^{\mathrm{Ar}} = 0$ and with Tate's completed-zeta character and explicit formula. Under this structure, the non-trivial zeros of the entire character $\xi_{\mathbb{R}}(s)$ of $\mathcal{V}^{\mathrm{prim}}$ lie on the self-dual locus $\mathrm{Re}(s) = 1/2$.

Remark 44.1.2 *(zero-placement and RH).* The Tate character supplied by P2 is the meromorphic completed zeta function

$$\Lambda_{\zeta}(s) := \pi^{-s/2}\Gamma(s/2)\zeta(s),$$

with simple poles at $s = 0, 1$. The entire Li–Hadamard object is

$$\xi_{\mathbb{R}}(s) := \frac{1}{2}s(s-1)\Lambda_{\zeta}(s).$$

Here ξ means $\xi_{\mathbb{R}}$ unless Λ_{ζ} is written explicitly. The zeros of $\xi_{\mathbb{R}}$ are exactly the non-trivial zeros of ζ; the gamma poles cancel the negative even trivial zeros of ζ, and the factor $s(s-1)$ removes the Tate poles. Thus the zero-placement assertion in F-RH is exactly RH. The existence and compatibility of the positive Hermitian structure are additional arithmetic structure, not a consequence of RH alone.

44.2 Three-Pillar Implication Structure

Theorem 44.2.1 *(zero-placement from three pillars plus Hilbert–Pólya–Li).* Suppose:

(P1) $\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^{!})_{\mathsf{G}}^{\mathrm{Ar}} = 0$ for the specified arithmetic realization, together with the defect identity and equivariant Mellin evaluation of Corollary 43.5.3.

(P2) $\xi_{\mathbb{R}}(s)$ satisfies Tate's functional equation and the Weil explicit formula, with rank residue $r_{\zeta} = 2$ (Theorem 43.6.1).

(P3) $\mathcal{V}^{\mathrm{prim}}$ carries a positive-definite Petersson inner product compatible with arithmetic Koszul self-duality (Theorem 43.7.1).

(H-P^{Ar}) The operator Θ_{ξ} of Hypothesis 44.3.2 has determinant divisor div $\xi_{\mathbb{R}}$ and the Weil–Connes trace distribution.

(HP–Li) The Li transforms of Θ_ξ have non-negative regularised trace.

Then the zero-placement clause of F-RH holds. Equivalently, RH holds.

Proof. P1 gives the functional equation $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s)$. P2 gives the explicit formula identifying the zero-side distribution with the prime and archimedean terms. P3 supplies the positive Hermitian domain. H-P^{Ar} places the zero divisor on the normal operator Θ_ξ, and HP–Li makes every Li trace non-negative. Li's theorem then gives RH, which is exactly the zero-placement clause. □

Remark 44.2.2 *(HP–Li as residual condition).* With P1, P2, P3 resolved to Theorem or Theorem-conditional, the zero-placement clause of F-RH reduces to the construction of the Hilbert–Pólya zero-divisor operator and to HP–Li positivity. This is the arithmetic form of the classical Weil–Li positivity problem, not a formal consequence of Virasoro representation theory.

44.3 Li-Positivity as Arithmetic Virasoro Unitarity

Li's Criterion

THEOREM 44.3.1 *(Li 1997, J. Number Theory 65).* RH is equivalent to $\lambda_n \geq 0$ for all $n \geq 1$, where

$$\lambda_n := \sum_\rho \left(1 - \left(1 - \frac{1}{\rho}\right)^n\right) = \frac{1}{(n-1)!}\frac{d^n}{ds^n}\left[s^{n-1}\log\xi(s)\right]\Big|_{s=1}$$

(sum over non-trivial zeros of ζ, counted with multiplicity).

Li's λ_n admits a direct Hilbert–Pólya transform interpretation. The transform lives on the zero-divisor operator Θ_ξ, not on the Virasoro/Fock energy operator.

Hilbert–Pólya Transform Computation

The Li criterion is a scalar positivity statement. To embed it in the arithmetic chiral-algebra formalism and exploit the Petersson inner product of P3, one requires an operator Θ_ξ on the arithmetic Heisenberg Hilbert completion whose regularised trace recovers the Li coefficients. The following hypothesis names this operator precisely.

Hypothesis 44.3.2 *(arithmetic Hilbert–Pólya, H-P^{Ar}).* There exists a Hilbert completion $\mathcal{H}_\xi$ of the primary arithmetic Heisenberg core, a dense invariant core $\mathcal{D}_\xi \subset \mathcal{H}_\xi$, and a self-adjoint unbounded operator

$$H_{\mathrm{HP}} : \mathcal{D}_\xi \subset \mathcal{H}_\xi \longrightarrow \mathcal{H}_\xi$$

in the Petersson/GNS inner product. Put

$$\Theta_\xi := \frac{1}{2}\,\mathrm{id} + iH_{\mathrm{HP}}.$$

The regularised determinant and trace distribution of Θ_ξ satisfy

$$\det_\infty(s\,\mathrm{id} - \Theta_\xi) = \xi_{\mathbb{R}}(s), \tag{44.1}$$

and, for every admissible Weil–Connes Paley–Wiener test function h,

$$\mathrm{Tr}_{\mathrm{reg}}\big(h(H_{\mathrm{HP}})\big) = \sum_\rho h(\gamma_\rho), \qquad \rho = \frac{1}{2} + i\gamma_\rho. \tag{44.2}$$

Here the notation $\rho = \frac{1}{2} + i\gamma_\rho$ is part of the hypothesis: self-adjointness of H_{HP} together with the determinant identity forces the zero divisor onto the critical line. The symbolic kernel $\sum_\rho e^{-t\rho}$ is only the distributional wave-kernel shorthand obtained from (44.2) by admissible test-function regularisation; it is not an ordinary heat trace of a non-negative self-adjoint operator.

Remark 44.3.3 *(form of H-P^{Ar}).* H-P^{Ar} is the Hilbert–Pólya conjecture stated inside the arithmetic chiral-algebra formalism. It is not derivable from P1, P2, P3 alone: P2 supplies the completed-zeta functional equation and explicit formula. Under its arithmetic placement and pairing hypotheses, P3 supplies a positive form on the specified state domain. A self-adjoint ordinate operator H_{HP} is an independent input. Its determinant identity (44.1) and distributional trace identity (44.2) are additional hypotheses.

Remark 44.3.4 *(primary-sector Witt–Virasoro reconciliation).* Chapter 35 constructs the positive-half Witt operators on the specified restricted tensor product of prime Fock spaces. The prescribed sums of negative Sugawara modes already fail on its vacuum. This rules out that summed action on that algebraic carrier. It does not classify all Virasoro realizations of an arithmetic state space.

The scalar $r_\zeta = 2$ is defined by $2\,\mathrm{Res}_{s=1}\zeta(s)$. Under the factorization hypothesis of Theorem 35.4.1, it also equals the primary determinant's pole order. Neither equality constructs a Virasoro central term. The Hilbert–Pólya operator H_{HP} is independently specified. Identifying it with a conformal zero-mode would require an intertwiner on the actual operator domains. The following conjecture asks for additional operators and a compatible state realization:

Conjecture 44.3.5 *(primary-Virasoro reconciliation).* There is a Hilbert state realization of $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}}$ with closed operators L^{PV}_m and comparison maps from the finite-place conformal structures. On a common invariant dense domain, the operators satisfy the Virasoro bracket

$$[L^{\mathrm{PV}}_m, L^{\mathrm{PV}}_n] \;=\; (m-n)L^{\mathrm{PV}}_{m+n} + \frac{c_{\mathrm{PV}}}{12}(m^3-m)\delta_{m+n,0}, \qquad c_{\mathrm{PV}} = 2.$$

The equality $c_{\mathrm{PV}} = r_\zeta$ is an extra compatibility condition with the rank residue of Theorem 35.4.1. The residue calculation alone does not construct L^{PV}_m.

Remark 44.3.6 *(operator dictionary).* Four operators must remain distinct. L^{Vir}_0 is the Fock/Virasoro conformal-energy operator. N_{BC} is the Bost–Connes Hamiltonian, with eigenvalues $\log m$ on the zero-mode lattice and partition function $\zeta(\beta)$. H_{HP} is the hypothetical self-adjoint determinant-trace operator. $\Theta_\xi = 1/2 + iH_{\mathrm{HP}}$ is the normal zero-divisor operator whose determinant divisor is $\mathrm{div}\,\xi_{\mathbb{R}}$. Conflating these four objects turns Euler-product positivity into a false Hilbert–Pólya argument.

Remark 44.3.7 *(Tate-pole and trivial-zero normalisation).* $\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ is meromorphic, not entire: the negative even trivial zeros of ζ are cancelled by gamma poles, and Λ_ζ has poles at 0 and 1. The entire function $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\Lambda_\zeta(s)$ removes these Tate poles and has as zeros exactly the non-trivial zeros. Thus H-P^{Ar} is a statement about the primary determinant $\xi_{\mathbb{R}}$, while the meromorphic Tate character Λ_ζ belongs to the cohomological/superdeterminant surface.

Proposition 44.3.8 *(Li coefficient = Hilbert–Pólya transform trace). (conditional on Hypothesis 44.3.2).* Assume H-P^{Ar}. For $n \geq 1$, define the normal Li-transform operator

$$\mathcal{L}_n := \mathrm{id} - \left(\mathrm{id} - \Theta_\xi^{-1}\right)^n. \tag{44.3}$$

Then

$$\lambda_n = \mathrm{Tr}_{\mathrm{reg}}(\mathcal{L}_n). \tag{44.4}$$

This is a trace identity for the Hilbert–Pólya zero-divisor operator Θ_ξ, not a polynomial identity in L_o^{Vir} or N_{BC}.

Proof. The determinant identity $\det_\infty(s\mathrm{id} - \Theta_\xi) = \xi_{\mathbb{R}}(s)$ identifies the determinant divisor of Θ_ξ, in the regularised determinant sense, with the non-trivial zero divisor. Applying the holomorphic functional calculus to $f_n(z) = 1 - (1 - z^{-1})^n$ gives

$$\mathrm{Tr}_{\mathrm{reg}}(f_n(\Theta_\xi)) = \sum_\rho \left(1 - \left(1 - \frac{1}{\rho}\right)^n\right),$$

which is Li's coefficient by Theorem 44.3.1. □

COROLLARY 44.3.9 *(Li positivity from positive HP trace). (conditional on P2, P3, H-P^{Ar}).* Assume the regularised Hilbert–Pólya trace is positive on the Weil positive cone and that the Li transforms $\mathcal{L}_n$ of (44.3) belong to the closure of that positive cone after the standard Li/Weil approximation. Then $\lambda_n \geq 0$ for every $n \geq 1$. The converse is *not* an operator-positivity theorem.

Proof. By the positivity assumption, $\mathrm{Tr}_{\mathrm{reg}}(\mathcal{L}_n) \geq 0$. The trace identity (44.4) identifies this number with λ_n. The reverse implication would require a determinate positive trace representation of the full zero measure from the scalar Li sequence; that is the classical Li/Weil positivity problem, not a consequence of P1–P3. □

THEOREM 44.3.10 *(positive Hilbert–Pólya trace implies zero-placement). (conditional on P2, P3, H-P^{Ar}).* Assume P2, P3, and the determinant and regularised trace identities of H-P^{Ar}. If HP–Li positivity holds:

(HP–Li) *Positive Hilbert–Pólya trace.* $\mathrm{Tr}_{\mathrm{reg}}(\mathcal{L}_n) \geq 0$ for every $n \geq 1$, where $\mathcal{L}_n$ is (44.3),

then the zero-placement clause of F-RH holds. Equivalently, RH holds. If the self-adjointness assertion in H-P^{Ar} is also assumed, then the determinant identity already gives zero-placement; the scalar Li route uses only the determinant and trace identities.

Proof. HP–Li gives $\lambda_n \geq 0$ for all $n \geq 1$ by Proposition 44.3.8. Li positivity is equivalent to RH (Li 1997, Theorem 44.3.1); this is exactly the zero-placement clause of F-RH (Remark 44.1.2). □

Remark 44.3.11 *(one-directional implication and operator discipline).* The polynomial $(L_o^{\mathrm{Ar}})^n - (L_o^{\mathrm{Ar}} - \mathrm{id})^n$ is *not* Li's transform; L_o^{Ar} would need to serve simultaneously as Fock energy, Bost–Connes Hamiltonian, and determinant-trace operator, three incompatible roles (see Remark 44.3.6). The correct statement is the one-directional HP–Li positivity ⇒ zero-placement, where H-P^{Ar} supplies Θ_ξ. The converse is the assertion that zero-placement supplies a positive Weil trace representation of the full zero distribution; this is the Hilbert–Pólya/Connes–Deninger positivity problem.

Remark 44.3.12 *(content of F-RH as a reformulation).* F-RH separates RH from the extra arithmetic structures carried by $\mathcal{V}^{\mathrm{prim}}$.

(i) The spectral object is made precise: the self-adjoint determinant form of H-P^{Ar} is a Hilbert–Pólya realisation of RH, while HP–Li positivity is Li's scalar RH-equivalent positivity written through the zero-divisor operator after the trace identity is fixed.

(ii) A structural constraint on zero-placement emerges: the Petersson positivity of P3 propagates through the Weil/Connes trace functional to impose positivity of the Li transforms, provided the trace representation is determinate.

(iii) The reformulation connects RH mechanisms (Connes, Deninger, Meyer) to positive-trace and factorisation methods, with the arithmetic explicit formula replacing compact-CFT modularity.

(iv) Admissible approximants to the Li/Weil test functions can be evaluated against the Bost–Connes and Hecke trace formulae. Positivity is a distributional trace question, not finite Fock positivity of T_n.

44.4 Comparison to Existing RH Mechanisms

The group $\mathbb{Q}^\times$ acts on $\mathbb{A}_\mathbb{Q}$ by multiplication. Its orbit quotient $\mathbb{A}_\mathbb{Q}/\mathbb{Q}^\times$ is called the *adele class space*. This quotient is distinct from the idele class group $\mathbb{A}_\mathbb{Q}^\times/\mathbb{Q}^\times$, which uses only invertible adeles.

Mechanism	Input	Analytic theorem
Hilbert–Pólya (folklore)	Self-adjoint op. with determinant divisor div ξ	Operator not exhibited
Connes (adele class space)	$\mathrm{KMS}_{\beta=1}$ Bost–Connes positivity	Trace-formula positivity
Deninger ($H^\bullet_\Delta$)	Purity of Frobenius on arith. H^1	Cohomology not constructed
Meyer (Schwartz space)	Mellin self-adjoint operator plus missing zero-divisor quotient	Operator not exhibited
Arakelov height-zeta	Height-function positivity	Equivalence to RH not established
Weil positivity	Positivity of Weil inner product	Known equivalent, no proof
Li positivity	$\lambda_n \geq 0$ for all $n \geq 1$	Known equivalent, no proof
Riemann–von Mangoldt	Explicit formula approach	Approaches RH asymptotically
Montgomery–Dyson	GUE statistics of zeros	Statistical, not deterministic
Vol IV zero-placement	**H-P^{Ar} + positive trace**	**Self-adjoint zero-divisor operator not constructed**

Table 44.1: The zero-placement clause placed against existing RH mechanisms.

Remark 44.4.1 *(relations).* (i) HP–Li positivity is Li positivity written on the Hilbert–Pólya zero-divisor operator Θ_ξ. It implies zero-placement, hence RH, via Theorem 44.3.10. A Virasoro polynomial T_n is not a substitute for this transform.

(ii) The zero-placement clause is governed by Weil positivity: Comparison of the Weil form with the Petersson form requires a specified linear map from the complete Weil test carrier to the arithmetic Petersson domain, preserving the involution and both forms. A restricted tensor-product presentation does not construct that form-preserving map. F-RH adds the arithmetic Hermitian, Koszul, and completed-zeta trace structures around this clause.

(iii) F-RH embeds Hilbert–Pólya explicitly as H-P^{Ar} (Hypothesis 44.3.2): it asks for a self-adjoint operator A, the positive form from P3, the determinant divisor of $\Theta_\xi = 1/2 + iA$, the regularised non-trivial-zero trace distribution, and $\mathbb{Z}/2$-symmetry under the arithmetic S-modular involution from P1. F-RH does not replace Hilbert–Pólya: H-P^{Ar} is a separate spectral hypothesis in the sufficient implication below.

(iv) F-RH is closer to Connes than to Deninger: both Connes and F-RH use unitarity plus Bost–Connes positivity; Deninger postulates a cohomology theory that F-RH does not need. The difference from Connes is that F-RH carries the positivity through arithmetic holomorphic-topological (HT) observables and a Weil-positive trace rather than through the bare trace-formula construction.

44.5 Residual Conditions

After the structural reduction of Chapter 43 and the F-RH reformulation above, the RH-relevant part is controlled by the following conditions. The spectral conditions imply zero-placement; the categorical conditions construct the trace object on which that problem is stated.

(HP–Li) *Positive Hilbert–Pólya trace.* $\mathrm{Tr}_{\mathrm{reg}}(\mathcal{L}_n) \geq 0$ for every $n \geq 1$, where $\mathcal{L}_n$ is the Li transform of Θ_ξ (Theorem 44.3.10). *Sufficient for RH given P2, P3, H-P*^{Ar}.

(H-P^{Ar}) *Arithmetic Hilbert–Pólya.* There is a self-adjoint operator A such that $\Theta_\xi = 1/2 + iA$ has determinant divisor $\mathrm{div}\,\xi_{\mathbb{R}}$ and Weil–Connes trace distribution equal to the non-trivial-zero distribution (Hypothesis 44.3.2). *Required for the HP–Li $\Rightarrow$ zero-placement chain; not derivable from P1, P2, P3.*

(arith. Positselski) Construct the arithmetic co–contra adjunction and verify the lifting criterion of Theorem 46.1.5. Its interpretation as P1 also requires the central-defect, Mellin-evaluation and detection maps of Theorem 43.5.1.

(Koszul–Pet compat) Identification of arithmetic Koszul pairing with adelic Petersson pairing (Conjecture 43.7.2). *Required for P3.*

($\mathrm{L}_{\mathrm{ACD}}$) Adelic categorical descent for $\mathrm{D}^{\text{sph-fin}}$ (Corollary 43.2.3). *Required for F1, hence for six-window placement.*

($\mathrm{F3.BZN}^{\mathrm{Ar}}$) Arithmetic Ben-Zvi–Nadler identification (Theorem 43.4.1). *Required for F3, hence for the categorical trace presentation of $\mathcal{V}^{\mathrm{prim}}$.*

Items (arith. Positselski), (Koszul–Pet compat), H-P^{Ar}, and HP–Li positivity lie on the P1–P2–P3 branch of the zero-placement clause and together form a sufficient set for RH in the chiral-algebra formalism. Items ($\mathrm{L}_{\mathrm{ACD}}$) and ($\mathrm{F3.BZN}^{\mathrm{Ar}}$) are independent of RH proper and lie on the six-window presentation; they would state $\mathcal{V}^{\mathrm{prim}}$ as a theorem-grade object but do not imply RH.

44.6 A Sufficient Path to RH

THEOREM 44.6.1 *(sufficient implication).* RH follows from the following implication in the chiral-algebra formalism:

$$\mathrm{P1} \xrightarrow{\text{Mellin comparison}} \text{P1 functional equation}$$

$$\mathrm{P2} \xrightarrow{\text{Tate 1950}} \text{P2 unconditional}$$

$$\mathrm{P3} \xrightarrow{\text{Koszul–Pet compat}} \text{P3 restricted unitarity}$$

$$(\mathrm{P1}, \mathrm{P2}, \mathrm{P3}, \text{H-P}^{\mathrm{Ar}}) \xrightarrow{\text{HP–Li}} \text{zero-placement} \xrightarrow{\Leftrightarrow} \text{RH}$$

Here the first arrow requires the central-defect identity and Mellin evaluation of Corollary 43.5.3. The remaining arithmetic inputs are Koszul–Petersson compatibility, arithmetic Hilbert–Pólya (H-P^{Ar}), and positive Hilbert–Pólya Li trace (HP–Li), on the domains stated above.

Proof. P2 is the completed-zeta character/explicit-formula input (Theorem 43.6.1). P1 is the functional-equation branch under the specified arithmetic evaluation (Corollary 43.5.3). P3 is the positive-form input conditional on the restricted Koszul–Petersson compatibility (Theorem 43.7.1). Given these inputs plus H-P^{Ar}, Theorem 44.3.10 gives HP–Li ⇒ zero-placement, and zero-placement ⇔ RH via Li's criterion (Remark 44.1.2, Theorem 44.3.1). □

Remark 44.6.2 *(analytic content of the implication).* **Mellin comparison** An arithmetic dual pair, its central-defect identity, and an equivariant evaluation giving the completed zeta function must be constructed. A co–contra equivalence alone does not supply these data.

Koszul–Pet compat non-formal theorem; compatibility of the arithmetic Koszul and adelic Petersson pairings.

H-P^{Ar} This is the Hilbert–Pólya operator-construction problem in the arithmetic Fock setting; the domain theorem alone does not supply the zero-divisor determinant.

HP–Li This is the positive-trace form of the classical Hilbert–Pólya / Weil positivity problem. If tractable through Li coefficients, the $n = 2$ case ($\lambda_2 \geq 0$) has been computed numerically to $n \sim 10^5$ zeros of ζ (Coffey 2005; Voros 2003 for the first few). The assertion for all n is Li's RH-equivalent positivity theorem.

The arithmetic formalism does not shorten the Hilbert–Pólya obstacle, but it places the positivity in a precise chain-level trace setting where factorisation techniques may be applied without conflating them with the zero-divisor theorem.

44.7 Relation to the Arithmetic Branch Essential Theorem

Under the arithmetic comparison and defect-detection hypotheses of Theorems 41.2.1 and 43.5.1, one has

$$\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0 \iff \xi(s) = \xi(1-s).$$

This gives the functional equation (geometric self-duality of $\xi_{\mathbb{R}}$), not RH (spectral zero-placement). RH is strictly beyond the essential theorem, requiring the additional positive Hilbert–Pólya trace input.

F-RH asks whether the arithmetic self-duality comparison, P3 positivity, and P2 explicit formula yield the required zero placement. The implication in the chiral-algebra formalism is conditional on H-P^{Ar} and HP–Li positivity. Those two statements are the Hilbert–Pólya form of RH native to this formalism.

44.8 The F-RH Reduction

(i) The zero-placement clause of F-RH (Conjecture 44.1.1) is exactly RH (Remark 44.1.2). F-RH also asks for the positive Hermitian structure compatible with arithmetic Koszul self-duality, Tate's functional equation, and the explicit formula.

(ii) The zero-placement clause is implied by H-P^{Ar} and HP–Li positivity together with P1 + P2 + P3 (Theorem 44.2.1).

(iii) The zero-placement clause is implied by HP–Li positivity (positive Hilbert–Pólya trace; Theorem 44.3.10) via Li's criterion, given the determinant and trace identities of H-P^{Ar}. If the full

determinant-trace H-P^{Ar} hypothesis is assumed, zero-placement is already part of the determinant-trace realisation. The converse direction from zero-placement to HP–Li is the positive-trace realisation problem.

(iv) Li's coefficients λ_n are Hilbert–Pólya transform traces of Θ_ξ (Proposition 44.3.8).

(v) Li positivity and Weil positivity govern the zero-placement clause. F-RH joins this clause to the arithmetic Hermitian, Koszul, and completed-zeta trace structures.

(vi) The sufficient implication to RH uses four inputs: arith. Positselski, Koszul–Pet compat, H-P^{Ar}, and HP–Li positivity (Theorem 44.6.1). This is the *four-input conditional conjunction* stated in Chapter 60 and made precise as Theorem 60.3.1: each input carries its own analytic content (five Positselski sub-items, KMS_1/GNS identification beyond the sph-fin cuspidal core, zeta-zero determinant-trace matching, RH-equivalent positive trace), and the conjunction implies zero-placement.

(vii) RH is strictly beyond the branch's essential theorem (functional equation), and HP–Li positivity is the Li positive-trace form of RH within the chiral-algebra formalism. The operator dictionary of Remark 44.3.11 preserves only the forward implication HP–Li⇒zero-placement; the converse is the positive determinant-trace realisation problem (Hypothesis 44.3.2) and does not help the forward direction.

Chapter 45

Arithmetic Virasoro Positivity Bootstrap

With the determinant form of Chapter 44 in place, the Virasoro-polynomial sequence $T_n^{\mathrm{Vir}} = (L_0^{\mathrm{Vir}})^n - (L_0^{\mathrm{Vir}} - \mathrm{id})^n$ is not the Li operator. It proves finite-sector Fock positivity, but it does not construct a Hilbert–Pólya operator and it does not give RH. The RH-strength input is the positive Hilbert–Pólya trace of $\mathcal{L}_n = \mathrm{id} - (\mathrm{id} - \Theta_\xi^{-1})^n$ from Chapter 44.

45.1 Statement Recalled

The Operator Sequence

Definition 45.1.1 *(formal Virasoro polynomial sequence).* . Let $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}}$ denote the primary (non-descendant) subspace of the arithmetic Heisenberg Fock, equipped with the Petersson inner product $\langle \cdot, \cdot \rangle^{\mathrm{Pet}}$ of Chapter 43.7.1. For every integer $n \geq 1$ define

$$T_n^{\mathrm{Vir}} \;:=\; (L_0^{\mathrm{Vir}})^n \;-\; (L_0^{\mathrm{Vir}} - \mathrm{id})^n \;\in\; \mathrm{End}\big(\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}}\big).$$

Here L_0^{Vir} is the Fock/Virasoro conformal-energy operator. It is not the Hilbert–Pólya determinant-trace operator H_{HP}, the Bost–Connes Hamiltonian N_{BC}, or the zero-divisor operator $\Theta_\xi = 1/2 + iH_{\mathrm{HP}}$. Thus T_n^{Vir} is a legitimate Fock polynomial on any common Virasoro spectral core, but it is not Li's transform. In the sequel, later occurrences of T_n mean T_n^{Vir} unless the Hilbert–Pólya Li transform $\mathcal{L}_n$ is written explicitly.

Remark 45.1.2 *(why T_n^{Vir} is not the Li object).* Two conventions for Li's criterion circulate in the literature. The original Li 1997 J. Number Theory paper defines $\lambda_n = \sum_\rho (1 - (1 - 1/\rho)^n)$; Bombieri–Lagarias J. Number Theory 77 (1999) uses the equivalent closed form $\lambda_n = (n-1)!^{-1}(d/ds)^n [s^{n-1} \log \xi(s)]\big|_{s=1}$. The corresponding operator is $\mathcal{L}_n = \mathrm{id} - (\mathrm{id} - \Theta_\xi^{-1})^n$, with Θ_ξ the normal zero-divisor operator of Chapter 44.3.2. A polynomial $(L_0^{\mathrm{Vir}})^n - (L_0^{\mathrm{Vir}} - \mathrm{id})^n$ does not become Li's transform by regularisation. Consequently, every claim below involving T_n^{Vir} is a formal Fock/Virasoro claim unless it is explicitly restated in terms of $\mathcal{L}_n$ and the positive Weil–Connes trace.

Conditional Chain: VB $\Rightarrow$ Li Positivity $\Leftrightarrow$ RH*

Proposition 45.1.3 *(formal Virasoro positivity is not Li positivity).* . Positivity of the formal Virasoro polynomial T_n^{Vir} on finite Fock sectors does not imply $\lambda_n \geq 0$, and therefore does not imply RH. The one-directional RH implication in Vol IV is instead

$$\mathrm{HP\text{–}Li} \Longrightarrow \lambda_n \geq 0 \;\forall n \Longleftrightarrow \mathrm{RH},$$

where HP–Li is the positive trace statement of Chapter 44.3.10.

Proof. Li's criterion is Li 1997 J. Number Theory, reproduced in Bombieri–Lagarias J. Number Theory 77 (1999). The Li summand is $1 - (1 - 1/\rho)^n$. The spectral transform realising this summand is therefore

$z \mapsto 1 - (1 - z^{-1})^n$ applied to the zero operator $z = \Theta_\xi$. The polynomial $x^n - (x-1)^n$ applied to a Virasoro energy $x = L_0^{\mathrm{Vir}}$ is a different function of a different operator. Positivity of the latter is a Fock calculation; it has no trace identity with Li's coefficients unless the Hilbert–Pólya trace identity has already been replaced by the correct Θ_ξ-transform. The HP–Li implication follows from Chapter 44.3.10. □

Remark 45.1.4 *(what is genuinely equivalent).* The known RH-equivalent positivity languages are Li positivity, Weil positivity for the number-field explicit formula, and the positive zero-measure conjecture for Θ_ξ, $\mathrm{VB}^*_{\mathrm{core}}$. Under explicit hypotheses these languages give the same positivity condition. None of the RH-equivalent positivity statements is unconditional here.

Remark 45.1.5 *(the level at which RH lives).* RH lives on the zero-divisor operator $\Theta_\xi = 1/2 + iH_{\mathrm{HP}}$, not on the Virasoro energy operator. The Bost–Connes Hamiltonian and the Virasoro L_0 give Euler weights and Fock energies; neither carries $\operatorname{div} \xi_{\mathbb{R}}$. A real spectral measure appears only after H-P^{Ar} supplies H_{HP} and identifies $\det_\infty(s - \Theta_\xi)$ with $\xi_{\mathbb{R}}(s)$. VB*, in its RH-strength form, is therefore a positivity statement for the Weil–Connes trace of functions of Θ_ξ, together with determinacy of the corresponding zero measure.

45.2 Obstructions

Virasoro levels

For $n = 1$ one has $T_1^{\mathrm{Vir}} = \mathrm{id}$. For $n = 2$ one has

$$(L_0^{\mathrm{Vir}})^2 - (L_0^{\mathrm{Vir}} - \mathrm{id})^2 = 2L_0^{\mathrm{Vir}} - \mathrm{id}.$$

Thus $T_2^{\mathrm{Vir}} \succeq 0$ is the Fock threshold $L_0^{\mathrm{Vir}} \succeq \frac{1}{2}\mathrm{id}$. It is not a statement about zeta zeros. Li's coefficient at level 2 is the trace of $z \mapsto 1 - (1 - z^{-1})^2$ on Θ_ξ, and equality between these two functional calculi would itself be an additional Hilbert–Pólya theorem.

Moment problem

For a real sequence $(m_n)_{n \geq 0}$, seek a positive Borel measure ν on $\mathbb{R}$ such that

$$m_n = \int_{\mathbb{R}} x^n \, d\nu(x) \qquad (n \geq 0).$$

This is called the *Hamburger moment problem*. Requiring support in $[0, \infty)$ gives the Stieltjes moment problem.

The Hamburger moment problem belongs to the regularised spectral measure of Θ_ξ. There is no identity

$$\mathrm{Tr}_{\mathrm{reg}}(T_n^{\mathrm{Vir}}) = n\lambda_n.$$

The valid moment sequence is obtained from the zero-divisor operator and the Weil–Connes trace. Stieltjes' theorem supplies the algebraic test: a sequence is the moment sequence of a positive measure on $[0, \infty)$ precisely when all Hankel and shifted Hankel minors are non-negative.

Kac determinant

At $c = 2$, the Kac parametrisation gives

$$1 - \frac{6}{m(m+1)} = 2, \qquad m(m+1) = -6.$$

The null weights form the countable set $\{h_{r,s}(2) : r, s \geq 1\}$. Outside this set the contravariant form is non-degenerate. Non-degeneracy is weaker than positivity; the Petersson sign is not determined by the non-vanishing of the Kac determinant.

Fock combinatorics

On the free-boson number state $|\mathbf{n}\rangle$, L_0^{Vir} has eigenvalue

$$m(\mathbf{n}) = \sum_{i \geq 1} i n_i.$$

For $m(\mathbf{n}) \geq 1$,

$$T_n^{\mathrm{Vir}} |\mathbf{n}\rangle = \big(m(\mathbf{n})^n - (m(\mathbf{n}) - 1)^n\big)|\mathbf{n}\rangle$$

has non-negative eigenvalue. The vacuum has $m = 0$; for even n the eigenvalue is negative. Hence the Fock positivity statement is a non-vacuum statement.

Weil trace

Weil positivity is a positivity theorem for the explicit formula and is equivalent to RH over number fields. In the Vol IV notation its operator representative is the positive regularised trace of $\mathcal{L}_n(\Theta_\xi)$, not a trace of T_n^{Vir}. Koszul–Petersson compatibility can identify the Weil trace pairing with a Petersson pairing on admissible arithmetic states; it cannot identify the Virasoro energy spectrum with the zero divisor.

de Branges datum

For an entire function E, put $E^*(z) = \overline{E(\overline{z})}$. It is called *Hermite–Biehler* when $|E(z)| > |E^*(z)|$ for $\operatorname{Im} z > 0$. For such E, the de Branges kernel is

$$K(w, z) = \frac{E(w)\overline{E(z)} - E^*(w)\overline{E^*(z)}}{2\pi i(\overline{z} - w)}.$$

The real entire function $\Xi(z) = \xi_{\mathbb{R}}(1/2 + iz)$ satisfies $\Xi^* = \Xi$, so Ξ itself is not a Hermite–Biehler function. A de Branges reformulation of RH requires a genuine datum E whose divisor controls the divisor of Ξ. The Hermite–Biehler condition is then part of the RH-strength input, not a consequence of formal Virasoro positivity.

45.3 Level-by-Level Positivity

Level $n = 1$: Unconditional

Theorem 45.3.1 *($n = 1$ arithmetic Virasoro positivity).* . $T_1^{\mathrm{Vir}} = (L_0^{\mathrm{Vir}}) - (L_0^{\mathrm{Vir}} - \mathrm{id}) = \mathrm{id}$. Hence $T_1 \succeq 0$ unconditionally on $\mathcal{V}_{\mathrm{Heis,prim}}^{\mathrm{prim}}$.

Proof. Direct computation. The operator is the identity. □

Remark 45.3.2 *(numerical consistency).* The ordinary trace of the identity on $\mathcal{V}_{\mathrm{Heis,prim}}^{\mathrm{prim}}$ is not defined without a trace-class or regularized-trace convention. Bombieri–Lagarias §2 gives $\lambda_1 = \frac{1}{2}(2 + \gamma - \log(4\pi)) \approx 0.02310$. The positivity of this number is consistent but not itself a proof of $T_1 \succeq 0$; the proof is the identity-operator identity. The number λ_1 has been known to be positive since Li 1997.

Level $n = 2$: Formal Threshold

THEOREM 45.3.3 *($n = 2$ Virasoro threshold).* . $T_2^{\mathrm{Vir}} = 2L_0^{\mathrm{Vir}} - \mathrm{id}$. Hence $T_2^{\mathrm{Vir}} \succeq 0$ on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}}$ if and only if $L_0^{\mathrm{Vir}} \succeq \frac{1}{2}\mathrm{id}$ on the same space.

Proof. Direct expansion: $T_2^{\mathrm{Vir}} = (L_0^{\mathrm{Vir}})^2 - (L_0^{\mathrm{Vir}} - \mathrm{id})^2 = 2L_0^{\mathrm{Vir}} - \mathrm{id}$. The inequality $T_2^{\mathrm{Vir}} \succeq 0$ is therefore equivalent to $2L_0^{\mathrm{Vir}} \succeq \mathrm{id}$, i.e. to $L_0^{\mathrm{Vir}} \succeq \frac{1}{2}\mathrm{id}$. □

COROLLARY 45.3.4 *($n = 2$ is not a zero-placement theorem).* . The implication

$$T_2^{\mathrm{Vir}} \succeq 0 \quad \Longrightarrow \quad \mathrm{Re}(\rho) = 1/2 \text{ for every non-trivial zero } \rho$$

is false without an additional Hilbert–Pólya identification of the Virasoro energy with the zero-divisor operator. H-P^{Ar} supplies Θ_ξ, not an equality $L_0^{\mathrm{Vir}} = \Theta_\xi$.

Proof. The spectrum of L_0^{Vir} is the conformal-energy spectral measure of the Fock representation. The operator Θ_ξ is constrained by the determinant divisor $\mathrm{div}\,\xi_{\mathbb{R}}$. Equality of these data would identify a non-negative Fock energy measure with a complex divisor. No such equality follows from P1, P2, P3, or the definition of T_2^{Vir}. The valid RH implication uses the Li transform $\mathcal{L}_n$ of Θ_ξ, as in Chapter 44.3.8. □

Remark 45.3.5 *(the spectral barrier).* The $n = 2$ Virasoro calculation is a threshold calculation inside the Fock representation. The RH barrier is the construction of the zero-divisor operator Θ_ξ and of a positive trace on its Li transforms. Confusing these two operators turns Fock positivity into a false Hilbert–Pólya argument.

Levels $n = 3, 4$: Bombieri-Lagarias Upper Bound

THEOREM 45.3.6 *(Bombieri–Lagarias asymptotic bound).* (Bombieri–Lagarias J. Number Theory 77 (1999):274–287, Theorem 2). For every $n \geq 1$,

$$\lambda_n \;\leq\; \frac{n(c_0 + \log n)}{2}, \qquad c_0 := 2 + \gamma - \log(4\pi) \approx 0.04619.$$

The matching lower bound $\lambda_n \geq -\frac{1}{2}n \log n - O(n)$ holds unconditionally.

COROLLARY 45.3.7 *(finite Li data are scalar data).* . For any finite set $F \subset \mathbb{N}$, the inequalities $\lambda_n \geq 0$ for $n \in F$ are scalar inequalities for the Li transforms $\mathcal{L}_n = 1 - (1 - \Theta_\xi^{-1})^n$. They do not imply operator positivity of T_n^{Vir}, and they do not imply RH. Bombieri–Lagarias bounds and finite numerical evaluations are therefore finite scalar controls, not Virasoro positivity theorems.

Proof. Li's criterion requires $\lambda_n \geq 0$ for every $n \geq 1$. A finite set of inequalities is therefore not equivalent to RH. The identity $\lambda_n = \mathrm{Tr}_{\mathrm{reg}}(\mathcal{L}_n)$ is a scalar trace identity for Θ_ξ. A positive trace of one operator does not force positivity of another operator: a diagonal matrix with eigenvalues -1 and 2 has positive trace and is not positive. The Virasoro polynomial T_n^{Vir} is a different functional calculus from $\mathcal{L}_n$. □

Remark 45.3.8 *(the trace-positivity gap).* The number-theoretic trace is attached to $\mathcal{L}_n(\Theta_\xi)$, not to T_n^{Vir}. Li positivity for all n is equivalent to RH. Operator positivity of a Virasoro polynomial is a separate Fock statement, and a scalar trace cannot bridge the two without a positive, determinate Weil–Connes representation of the zero measure.

45.4 Moment Problem Reformulation

The Shifted Zero Measure

Definition 45.4.1 *(off-line defect measure).* . Let $\rho = 1/2 + i\gamma_\rho$ denote a non-trivial zero of ξ, written with $\gamma_\rho \in \mathbb{C}$ (real for every zero iff RH). Under the symmetrisation $\rho \leftrightarrow 1 - \overline{\rho}$, the pair set $Z = \{\rho, 1 - \overline{\rho}\}$ pushes to the off-line defect measure ν_{def} on $\mathbb{R}_{\geq 0}$ via

$$\int f \, d\nu_{\mathrm{def}} := \sum_\rho f(|\mathrm{Im}\, \gamma_\rho|^2)$$

for test functions f of suitable growth, Bost–Connes-regularised. The measure ν_{def} is supported at 0 exactly when RH holds. It is a defect detector, not Li's positive moment measure.

Remark 45.4.2. Under RH, $\gamma_\rho \in \mathbb{R}$ and ν_{def} is supported at 0. This defect measure is not the spectral measure of H_{HP} and not the Li moment measure; it records only the distance of zeros from the critical line.

Li-Stieltjes Moment Formulation

Proposition 45.4.3 *(the defect measure is not a Li–Stieltjes moment model).* . The off-line defect measure ν_{def} of Chapter 45.4.1 does not give a Stieltjes moment representation of Li's coefficients or of $\mu_n := \sum_\rho \rho^n$. A Li–Stieltjes formulation requires the Hilbert–Pólya zero-divisor operator Θ_ξ, the Li transform $\mathcal{L}_n$, and the positive Weil–Connes trace of Chapter 44.3.2.

Proof. ν_{def} records $|\mathrm{Im}\, \gamma_\rho|^2$, which vanishes exactly on the critical line. Li's summand is instead $1 - (1 - 1/\rho)^n$, a holomorphic function of the zero ρ. No change of variables from the defect alone recovers this holomorphic transform with multiplicity. The Hilbert–Pólya trace identity is Chapter 44.3.8, which applies the Li transform to Θ_ξ. □

Hamburger Moment Determinacy

Theorem 45.4.4 *(Hamburger reformulation for the formal T_n).* . The following equivalence is a formal operator-theoretic template for a genuine positive moment problem; it is not the live RH path unless the measure is replaced by the Hilbert–Pólya zero measure of H_{HP} and the trace is the positive Weil–Connes trace:

(i) Formal positivity: $T_n^{\mathrm{Vir}} \succeq 0$ for all $n \geq 1$ on a fixed self-adjoint spectral model.

(ii) The associated spectral measure is a positive measure supported on $[1/2, \infty)$.

(iii) For the shifted moments $\nu_k := \int_{[1/2,\infty)} (x - 1/2)^k \, d\mu(x)$, all Hankel minors $H_n := \det(\nu_{i+j})_{0 \leq i,j \leq n}$ and shifted Hankel minors $H_n^{(1)} := \det(\nu_{i+j+1})_{0 \leq i,j \leq n}$ are non-negative for every $n \geq 0$.

This theorem is not an independent proof of RH and does not connect T_n^{Vir} to Li's coefficients.

Proof. (1) ⇔ (2) is the spectral theorem for the chosen self-adjoint model. Since

$$T_2^{\mathrm{Vir}}(x) = x^2 - (x-1)^2 = 2x - 1,$$

positivity for all n forces $x \geq 1/2$. Conversely, for $x = 1/2 + y$ with $y \geq 0$, $(y + 1/2)^n - (y - 1/2)^n$ is non-negative. This is a formal statement about that model, not about zeta zeros.

(2) ⇔ (3): Stieltjes moment-problem theorem (Shohat–Tamarkin 1943, Ch. III; Akhiezer 1965, §2.1): a sequence $\{m_k\}_{k\geq 0}$ is the moment sequence of a positive measure on $[0,\infty)$ iff all Hankel and shifted Hankel minors are non-negative. Apply it to the translated variable $y = x - 1/2$.

No implication to RH is asserted. □

Remark 45.4.5 *(moment information).* Chapter 45.4.4 records the formal moment-determinacy apparatus that would become relevant after a true Hilbert–Pólya spectral measure is constructed. In particular:

- Testing a genuine positive spectral measure level-by-level becomes testing Hankel determinants H_n level-by-level, a finite numerical procedure at each n.
- Known numerical bounds for Li coefficients do not test these Hankel determinants; they test a different holomorphic transform of the zero divisor.
- Known structural estimates on Li coefficients and Voros regularised sums do not supply lower bounds for the shifted Stieltjes minors above.
- The gap between known and needed becomes precisely specified: *unconditional* lower bounds on $H_n^{(1)}$ for all n.

The equivalent Hilbert–Pólya condition is the construction of the positive zero measure itself.

45.5 Local Kac Determinant at $c = 2$

The Kac Formula

THEOREM 45.5.1 *(Kac determinant formula).* (Kac 1978; Feigin–Fuchs 1990). For the Virasoro Verma module $M(c,h)$ of central charge c and highest weight h, the determinant of the contravariant form at level n is

$$\det_n^{c,h} = \prod_{\substack{r,s\geq 1\\ rs\leq n}} (h - h_{r,s}(c))^{p(n-rs)}$$

where $p(k)$ is the partition function and

$$h_{r,s}(c) = \frac{((m+1)r - ms)^2 - 1}{4m(m+1)}, \qquad c = 1 - \frac{6}{m(m+1)}.$$

The Arithmetic $c = 2$ Regime

PROPOSITION 45.5.2 *(Kac at $c = 2$).* For the local rank-2 free-boson Virasoro module, $c = 2$. In the Kac parametrisation the parameter m satisfies $1 - \frac{6}{m(m+1)} = 2$, hence $m(m+1) = -6$, giving the formal solution $m = (-1 \pm i\sqrt{23})/2$. The Kac null weights become

$$h_{r,s}(c=2) = \frac{((m+1)r - ms)^2 - 1}{4m(m+1)} = -\frac{((m+1)r - ms)^2 - 1}{24}.$$

Thus the possible null weights are the countable Kac set $\{h_{r,s}(2) : r,s \geq 1\} \subset \mathbb{C}$. No rationalisation to $\{0,-1,-2,\ldots\}$ is implied.

Proof. Direct substitution $c = 2$ in the Kac formula gives the displayed set. The free-boson realisation proves unitarity of the Fock module itself; it does not identify the Kac zero locus with a real arithmetic sequence. □

Corollary 45.5.3 *(no generic null-vector obstruction).* For generic primary weight h outside the countable Kac zero locus $\{h_{r,s}(2) : r, s \geq 1\}$, $\det_n^{c=2,h} \neq 0$ at every level n. This proves generic non-degeneracy for the local Virasoro Verma module. It does not prove Petersson positivity for the arithmetic primary object.

Unitarity Gap at Special Weights

Remark 45.5.4 *(the gap between non-degeneracy and positivity).* Chapter 45.5.3 establishes non-degeneracy, not positivity. Two Hermitian forms can be non-degenerate (no null vectors) without being positive-definite (a form of signature $(+, -, +, -, \ldots)$ is non-degenerate). The Friedan–Qiu–Shenker 1986 $c < 1$ analysis converts non-degeneracy to positivity via the Kac determinant *sign pattern*: when $c < 1$ is not in the discrete unitary series, the sign of $\det_n^{c,h}$ oscillates, producing at least one negative eigenvalue and refuting unitarity. The $c = 2$ case is outside the discrete $c < 1$ minimal unitary series, but it lies in the standard $c \geq 1$ free-boson unitary regime. Hence the FQS sign-oscillation argument for non-minimal $c < 1$ modules does not apply.

Theorem 45.5.5 *($c = 2$ free-field unitarity).* (Segal 1981; Frenkel–Ben-Zvi 2004). The free-boson rank-2 Fock module at central charge $c = 2$ is a unitary representation of the Virasoro algebra, with Hermitian form induced from the free-boson vacuum pairing $\langle \alpha_{-n_1} \cdots \alpha_{-n_k}|0\rangle, \alpha_{-m_1} \cdots \alpha_{-m_l}|0\rangle\rangle = \delta_{k,l} \sum_{\sigma \in S_k} \prod_i n_i \delta_{n_i, m_{\sigma(i)}}$. This is positive-definite by inspection.

Corollary 45.5.6 *(archimedean positivity of T_n^{Vir}).* On the *archimedean* sector $\mathcal{H}^{(\infty)}$ of the primary arithmetic Heisenberg Fock (the rank-2 free-boson Fock of Theorem 43.7.1 Step 1), $T_n^{\mathrm{Vir}} \succeq 0$ for every $n \geq 1$ on every non-vacuum number state.

Proof. The free-boson vacuum pairing of Theorem 45.5.5 is positive-definite; L_0^{Vir} acts diagonally on number states with eigenvalue $m = \sum_i i n_i \geq 0$ (with $m = 0$ precisely on the vacuum). On non-vacuum ($m \geq 1$) number states, $T_n^{\mathrm{Vir}} = (L_0^{\mathrm{Vir}})^n - (L_0^{\mathrm{Vir}} - \mathrm{id})^n$ acts by the scalar $m^n - (m-1)^n \geq 0$ (the function $x \mapsto x^n$ is monotone on $[0, \infty)$). Positive-semidefiniteness follows. □

Remark 45.5.7 *(the missing sector).* Corollary 45.5.6 establishes the archimedean sector. The full primary arithmetic Heisenberg Fock is *not* just the archimedean sector: the Bost–Connes zero-mode lattice $L = \ell^2(\mathbb{N}^\times)$ (Definition 29.2.1) enters multiplicatively, and the full primary space is $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}} \cong L \otimes \mathcal{H}^{(\infty)}_{\mathrm{primary}}$. Positivity of T_n^{Vir} on the $\mathcal{H}^{(\infty)}_{\mathrm{primary}}$ factor is Corollary 45.5.6. Positivity of a Virasoro polynomial on the L-factor is a statement about the Bost–Connes energy spectrum $\{\log m : m \geq 1\}$. It is not a statement about the zero divisor of ζ. The zero divisor enters only through the Hilbert–Pólya operator Θ_ξ and the Weil–Connes trace.

45.6 Fock-Combinatorial Identity

Explicit Action on Number States

Lemma 45.6.1 *(diagonal action on number states).* On the Fock number state $|\mathbf{n}\rangle := \prod_{i \geq 1} (\alpha_{-i}^{\infty})^{n_i}|0\rangle \big/ \prod_i \sqrt{i^{n_i} n_i!}$ (normalised with respect to the free-boson Fock inner product of Theorem 45.5.5), L_0^{Vir} acts as

$$L_0^{\mathrm{Vir}}|\mathbf{n}\rangle \;=\; \Big(\sum_{i \geq 1} i\, n_i\Big)|\mathbf{n}\rangle \;=:\; m(\mathbf{n})|\mathbf{n}\rangle.$$

Proof. $L_0 = \sum_{i\geq 1} \alpha_{-i}\alpha_i$ (Sugawara stress tensor zero mode for the free boson; Frenkel–Ben-Zvi 2004 §3.5). Applied to $(\alpha_{-i})^{n_i}|0\rangle$: by the canonical commutator $[\alpha_i, \alpha_{-i}] = i$, $\alpha_i(\alpha_{-i})^{n_i}|0\rangle = n_i \cdot i \cdot (\alpha_{-i})^{n_i-1}|0\rangle$, hence $\alpha_{-i}\alpha_i(\alpha_{-i})^{n_i}|0\rangle = in_i(\alpha_{-i})^{n_i}|0\rangle$. Summing over i gives the diagonal eigenvalue $\sum_i in_i$. □

The Symmetric-Function Identity

THEOREM 45.6.2 *(combinatorial positivity of T_n on Fock).* On every non-vacuum number state $|\mathbf{n}\rangle$ with $m(\mathbf{n}) \geq 1$,

$$T_n^{\mathrm{Vir}}|\mathbf{n}\rangle = [m(\mathbf{n})^n - (m(\mathbf{n})-1)^n]\,|\mathbf{n}\rangle,$$

and $m(\mathbf{n})^n - (m(\mathbf{n})-1)^n \geq 0$ for every integer $m(\mathbf{n}) \geq 1$ and every $n \geq 1$.

Proof. By Lemma 45.6.1, L_0^{Vir} acts by the scalar $m = m(\mathbf{n})$. Hence $(L_0^{\mathrm{Vir}})^n$ acts by m^n and $(L_0^{\mathrm{Vir}} - \mathrm{id})^n$ by $(m-1)^n$. The difference is $m^n - (m-1)^n$.

Non-negativity: the function $x \mapsto x^n$ is monotone non-decreasing on $[0,\infty)$ for $n \geq 1$; hence for $m \geq 1$ the strict inequality $m^n > (m-1)^n$ holds. For $m = 1$, the difference is exactly 1. □

Remark 45.6.3 *(combinatorial expansion).* Binomial expansion:

$$m^n - (m-1)^n = \sum_{k=0}^{n-1} \binom{n}{k} m^k (-1)^{n-1-k} = nm^{n-1} - \binom{n}{2} m^{n-2} + \cdots + (-1)^{n-1}.$$

For $m = 1$: the identity $1 - 0 = 1$; hence $T_n^{\mathrm{Vir}}|\mathbf{n}\rangle = |\mathbf{n}\rangle$ on weight-1 states for every $n \geq 1$. This is the *singleton-primary*-state positivity.

COROLLARY 45.6.4 *(Fock-level restricted positivity).* On the subspace $\mathcal{F}^{m\geq 1} \subseteq \mathcal{H}^{(\infty)}$ spanned by non-vacuum number states, $T_n^{\mathrm{Vir}} \succeq 0$ for every $n \geq 1$.

Proof. Direct from Theorem 45.6.2: T_n^{Vir} acts diagonally with non-negative eigenvalues; positivity follows from the diagonal characterisation. □

Remark 45.6.5 *(zero-mode separation).* Corollary 45.6.4 establishes non-vacuum archimedean Fock positivity for T_n^{Vir}. The primary arithmetic Heisenberg Fock also contains the Bost–Connes zero-mode lattice $\ell^2(\mathbb{N}^\times)$ lattice where μ_n acts via $\mu_n \delta_m = \delta_{nm}$; see Proposition 29.3.1. On this lattice the Bost–Connes Hamiltonian has Dirichlet-series spectrum $\{\log n : n \in \mathbb{N}^\times\}$ at KMS temperature β. This spectrum gives the Euler factor $\zeta(\beta)$; it is not the divisor of $\xi_{\mathbb{R}}$. The RH-strength input is the separate trace bridge from this Euler regularisation to Θ_ξ.

45.7 REDUCTION TO THE IRREDUCIBLE CORE OBSTRUCTION

The Core Statement

THEOREM 45.7.1 *(conditional reduction to the zero-operator core). (conditional on P2, P3, H-P^{Ar}, the primary tensor decomposition, and a positive regularised trace).* Under these hypotheses, the RH-strength positivity statement reduces to the following zero-operator statement:

(VB$^*_{\mathrm{core}}$) *Zero-operator spectral positivity.* The operator $\Theta_\xi = 1/2 + iH_{\mathrm{HP}}$ satisfies $\det_\infty(s - \Theta_\xi) = \xi_{\mathbb{R}}(s)$, and the regularised Weil–Connes trace on the Li transforms $\mathcal{L}_n(\Theta_\xi)$ is positive for every $n \geq 1$.

The Bost–Connes zero-mode lattice supplies the Euler/KMS regularisation of the trace. It is not the spectrum of Θ_ξ.

Proof. H-P^{Ar} gives the determinant identity for Θ_ξ. By Proposition 44.3.8, the Li coefficient λ_n is the regularised trace of $\mathcal{L}_n(\Theta_\xi)$. Positivity of these traces for every n is Li positivity, hence RH by Li's theorem. Conversely, RH is the support condition for the spectral measure of Θ_ξ. The Bost–Connes spectrum $\{\log m\}$ enters only through the regularisation giving the Euler product; no step identifies it with the determinant divisor of Θ_ξ. □

The Named Obstruction

Corollary 45.7.2 *(the irreducible core of VB*). (conditional on the hypotheses of Chapter 45.7.1).* The irreducible core is the positive zero-divisor measure problem for Θ_ξ. Replacing Θ_ξ by the Bost–Connes Hamiltonian N_{BC} replaces the zero divisor by the Euler eigenvalues and therefore destroys the Hilbert–Pólya statement.

Remark 45.7.3 *(core equivalence).* $(\mathrm{VB}^*_{\mathrm{core}})$ is not a *weaker* statement than VB* under the Hilbert–Pólya and positive-trace hypotheses; it is the zero-operator form of the same condition.

(i) The zero divisor is carried by Θ_ξ.
(ii) The Euler product is carried by the Bost–Connes KMS trace.
(iii) The equality between the two is the Hilbert–Pólya trace bridge.
(iv) The Bost–Connes phase transition at $\beta = 1$ records the pole of ζ; it is not a statement about phase transitions on the critical line.

45.8 Comparison to Classical RH Equivalents

Formulation	Input data	Positivity shape
Riemann 1859	$\xi(s)$ entire of order 1	Zero placement on line
Li 1997	$\lambda_n, n \geq 1$	Scalar sequence ≥ 0
Weil 1952	Test functions $f \in \mathcal{S}$	Inner product positivity
Báez-Duarte 2003	Finite linear combinations of $\{\rho^s\}$	Density in L^2
Lagarias 1999	de Branges datum over Ξ	Hermite–Biehler divisor control
Bombieri–Lagarias 1999	$\{\lambda_n\}$ with bounds	Explicit lower bounds
Connes 1999	$\mathcal{S}_{\mathbb{A}}/\mathbb{Q}^*$	Trace-formula positivity
Vol IV Virasoro	$\{T_n^{\mathrm{Vir}}\}_{n\geq 1}$ in primary Fock	Fock polynomial positivity
Vol IV HP–Li core	$\boldsymbol{\Theta_\xi}$ with Weil–Connes trace	**zero-measure positivity**

Table 45.1: RH-equivalent positivity formulations across the literature.

Remark 45.8.1 *(placement of VB* against its neighbours).* The formal Virasoro polynomial T_n^{Vir} does not imply Li's inequality. The RH-strength statement is HP–Li: positivity of the regularised Weil–Connes trace of $\mathcal{L}_n(\Theta_\xi)$ for every n. Weil 1952 and the de Branges formulations place the same zero-placement barrier in classical languages. The Vol IV core is the construction and positivity of the Θ_ξ-measure, with the Bost–Connes zero-mode serving as Euler regularisation rather than as the zero spectrum.

Proposition 45.8.2 *(Weil comparison under P3). (conditional on Koszul–Petersson compatibility Chapter 43.7.2 and the H-P^{Ar} trace condition).* Weil 1952 explicit-formula positivity and HP–Li positivity express the same RH-equivalent barrier after identifying the Weil distribution with the regularised trace of the zero-divisor operator Θ_ξ. The Petersson form enters through the Koszul–Petersson comparison of the trace functional; it does not turn T_n^{Vir} into a Li transform.

Proof. Weil 1952 gives $\lambda_n = \langle f_n, f_n \rangle^{\mathrm{Weil}}$ for f_n an explicit test function on the adele class space. H-P^{Ar} gives $\lambda_n = \mathrm{Tr}_{\mathrm{reg}}(\mathcal{L}_n(\Theta_\xi))$. Koszul–Petersson compatibility identifies the trace pairing with the Petersson pairing on the arithmetic primary sector whenever the admissible Weil test lies in that sector. Thus Weil positivity and HP–Li positivity are the same positivity problem after the trace bridge is supplied. No equality with $\mathrm{Tr}_{\mathrm{reg}}(T_n^{\mathrm{Vir}})$ is used. □

45.9 Zero-Divisor Condition

The Precise Condition

The RH-strength positivity statement is equivalent to the following condition:

Definition 45.9.1 *(the Vol IV zero-divisor condition).* The *Vol IV zero-divisor condition* is the following statement:

> There is a normal zero-divisor operator $\Theta_\xi = 1/2 + iH_{\mathrm{HP}}$ whose determinant divisor is $\mathrm{div}\,\xi_{\mathbb{R}}$, and whose Weil–Connes trace is supported on $1/2 + i\mathbb{R}$. The Bost–Connes zero-mode gives the Euler/KMS regularisation of the trace; it does not carry $\mathrm{div}\,\xi_{\mathbb{R}}$. ($\mathcal{R}_{\mathrm{IV}}$)

THEOREM 45.9.2 *($\mathcal{R}_{\mathrm{IV}}$ implies the Riemann Hypothesis).* $\mathcal{R}_{\mathrm{IV}} \Rightarrow$ RH. Conversely, RH gives only the zero-placement part of $\mathcal{R}_{\mathrm{IV}}$; it does not construct the normal zero-divisor operator, the determinant divisor, the Weil–Connes trace, or the Bost–Connes regularisation.

Proof. By definition, $\mathcal{R}_{\mathrm{IV}}$ includes a Θ_ξ-determinant divisor and Weil–Connes trace concentrated on the critical line. Since $\det_\infty(s - \Theta_\xi) = \xi_{\mathbb{R}}(s)$, this is exactly the zero-placement assertion for the nontrivial zeros of ζ, hence RH by Chapter 44.1.2. The converse would require the extra operator-realization data named in the statement; zero placement alone does not supply it. □

Virasoro Boundary

Remark 45.9.3 *(structural change).* The arithmetic Virasoro reduction gives:

(i) Closed $n = 1$ unconditionally (Chapter 45.3.1).
(ii) Reduced $n = 2$ to the formal Virasoro threshold $L_0^{\mathrm{Vir}} \succeq (1/2)\mathrm{id}$, not to zero placement (Chapter 45.3.4).
(iii) Established Fock-level positivity for the archimedean sector at every n (Chapter 45.6.4).
(iv) Reformulated the zero-measure problem as a Hamburger moment determinacy problem for Θ_ξ (Chapter 45.4.4).
(v) Proved the Kac determinant at $c = 2$ is non-zero at generic weights (Chapter 45.5.2), establishing absence of null-vector unitarity obstruction.
(vi) Placed HP–Li in correspondence with Weil positivity (Chapter 45.8.2) and the de Branges Hermite-Biehler framework.
(vii) The RH-strength condition is a single zero-measure positivity statement for Θ_ξ ($\mathrm{VB}^*_{\mathrm{core}}$, Chapter 45.7.1).

VB* itself remains open, as it is equivalent to RH.

Remark 45.9.4 *(VB^*_{core} boundary).* The assertion ($\mathrm{VB}^*_{\mathrm{core}}$) contains the following independent analytic inputs:

(i) A second KMS-positivity input beyond Bost–Connes: a constraint on the zero-divisor measure forcing critical-line concentration.

(ii) An arithmetic-modular bootstrap input: a Kac-formula-style analysis of the arithmetic Virasoro algebra on the Bost–Connes C^*-algebra, extracting unitarity from null-vector absence.

(iii) An $n \to \infty$ asymptotic input: a moment-generating-function bootstrap strong enough to close the $n \to \infty$ limit of λ_n scaling.

The first input belongs to the arithmetic site of Connes–Consani arXiv:1405.4527. The second is the C^*-algebraic analogue of Friedan–Qiu–Shenker. The third is an asymptotic moment problem.

45.10 Arithmetic Zhu Finiteness Test

The tempting shortcut is Verlinde finite-dimensionality: prove that the primary Bost–Connes core is an arithmetic Zhu-finite object, obtain a finite-dimensional Hermitian modular functor, and read positivity from ordinary linear algebra. The shortcut fails on the full zeta object.

Definition 45.10.1 *(Euler-faithful arithmetic Zhu quotient).* An *Euler-faithful arithmetic Zhu quotient* of the primary Bost–Connes core is a quotient Q of the zero-mode lattice $L = \ell^2(\mathbb{N}^\times)$ satisfying:

(i) the Bost–Connes operators descend:

$$\mu_n \overline{\delta}_m = \overline{\delta}_{nm}, \qquad \mu_n^* \overline{\delta}_m = \begin{cases} \overline{\delta}_{m/n}, & n \mid m, \\ 0, & n \nmid m, \end{cases}$$

where $\overline{\delta}_m$ is the image of $\delta_m \in L$;

(ii) the KMS Hamiltonian $H = \log N$ descends with $H\overline{\delta}_m = (\log m)\overline{\delta}_m$ whenever $\overline{\delta}_m \neq 0$;

(iii) the ordinary KMS character of the quotient is the full Bost–Connes Euler product:

$$\mathrm{Tr}_Q(e^{-\beta H}) = \sum_{m \geq 1} m^{-\beta} = \zeta(\beta) \qquad (\mathrm{Re}\,\beta > 1).$$

It is *Zhu finite* if $\dim_{\mathbb{C}} Q < \infty$.

Theorem 45.10.2 *(no Euler-faithful Zhu finiteness).* No Euler-faithful arithmetic Zhu quotient of the primary Bost–Connes core is Zhu finite.

Proof. If Q were finite-dimensional, the trace $\mathrm{Tr}_Q(e^{-\beta H})$ would be a finite sum $\sum_{j=1}^{r} a_j e^{-\beta E_j}$ after diagonalising the finite-dimensional self-adjoint operator H. In particular it extends to an entire function of β. Euler-faithfulness requires this finite sum to equal $\zeta(\beta)$ on the half-plane $\mathrm{Re}\,\beta > 1$. By the identity theorem the meromorphic continuation of ζ would be entire, contradicting its simple pole at $\beta = 1$. Hence Q is infinite-dimensional.

Equivalently, the faithful Bost–Connes action of Proposition 29.3.1 sends δ_1 to δ_m by $\mu_m \delta_1 = \delta_m$. Keeping the full Euler product keeps one independent KMS eigenline at every $m \in \mathbb{N}^\times$. A finite Zhu quotient must kill or identify infinitely many of these lines, and then it no longer has character $\zeta(\beta)$. □

Corollary 45.10.3 *(Verlinde finite-dimensionality cannot close full RH).* Arithmetic Zhu/Verlinde finite-dimensionality cannot prove $\mathrm{VB}^*_{\mathrm{core}}$ for the full $\mathcal{V}^{\mathrm{prim}}$ while preserving the completed zeta partition function. The only Zhu-finite quotients compatible with the Bost–Connes action are finite-prime or finite-conductor truncations; their characters are finite Euler products or finite Dirichlet polynomials, not $\xi(s)$.

Proof. By Theorem 45.10.2, a finite-dimensional quotient cannot be Euler-faithful. A quotient supported on a finite set S of primes has character $\prod_{p\in S}(1-p^{-\beta})^{-1}$, possibly with an archimedean Tate factor. Such a finite Euler product is a finite-level arithmetic modular-functor object; it is not the zeta object whose zero-placement is RH. Therefore finite-dimensional Hermitian linear algebra can test finite windows, but it cannot supply the missing positivity for the full Θ_ξ-spectral measure. □

Remark 45.10.4 *(central charge does not rescue Zhu finiteness).* The residue value $r_\zeta = 2$ of Chapter 35.4.1 equals a pole order under the stated determinant factorization. Chapter 35 also shows that the prescribed sums of negative Sugawara modes fail on the restricted tensor vacuum. These statements concern a scalar invariant and a particular proposed action. Neither supplies a finite-dimensional C_2-quotient or Zhu algebra. Thus $r_\zeta = 2$ supplies no C_2-cofiniteness or Zhu-finiteness hypothesis.

Remark 45.10.5 *(surviving direction).* The finite-dimensional construction survives only as a finite-window technology: truncate to finitely many primes or to bounded conductor, prove the corresponding arithmetic modular functor is finite, compute its Hermitian form, and compare its limiting trace to the Bost–Connes KMS trace. Any limiting theorem strong enough to recover $\mathrm{VB}^*_{\mathrm{core}}$ must add a new positivity or compactness input beyond Zhu finite-dimensionality at finite level.

45.11 Zero-Measure Conjecture

Conjecture 45.11.1 *(VB^*_{core}: the Vol IV form of RH). Conjectural statement.* The zero-divisor operator $\Theta_\xi = 1/2 + iH_{\mathrm{HP}}$, with $\operatorname{div}\det_\infty(s-\Theta_\xi) = \operatorname{div}\xi_{\mathbb{R}}$, has determinant divisor and Weil–Connes trace supported on $1/2 + i\mathbb{R}$. Equivalently, all non-trivial zeros of ζ lie on the critical line.

Remark 45.11.2 *(the conjecture is RH).* Chapter 45.11.1 is, verbatim, the Riemann Hypothesis. The arithmetic chiral-algebra formulation asks for a positive, determinate Weil–Connes trace and determinant divisor on the zero-divisor operator Θ_ξ. The Bost–Connes algebra supplies the Euler regularisation and KMS weight, in the sense of Connes' adelic trace formula; it does not supply the determinant divisor by itself.

Remark 45.11.3 *(size of* $\mathrm{VB}^*_{\mathrm{core}}$*).* Any unconditional proof of Chapter 45.11.1 would be an unconditional proof of RH, a Millennium Prize problem. No finite estimate is meaningful. Partial reductions have the following mathematical sizes:

Bombieri-Lagarias-style sharpening at specific n. substantial local theorem.

Kac-determinant analysis at $c = 2$ to low level. finite-level determinant computation.

Moment-problem Hankel analysis. unbounded Hamburger determinacy and lower bounds on all shifted Hankel minors.

Arithmetic-site incorporation. Connes–Consani arithmetic-site positivity; no unconditional proof of RH follows.

45.12 Operator Boundary

(i) The formal Virasoro polynomial sequence $\{T_n^{\mathrm{Vir}}\}$ gives Fock positivity statements, not Li transforms (Chapter 45.1.3).

(ii) The RH-strength path uses $\mathcal{L}_n(\Theta_\xi) = 1-(1-\Theta_\xi^{-1})^n$, the positive Weil–Connes trace, and Li's criterion (Chapter 45.4.4, Chapter 45.8.2).

(iii) $n = 1$ is unconditional: $T_1^{\mathrm{Vir}} = \mathrm{id}$ (Chapter 45.3.1).

(iv) $n = 2$ reduces to $L_0^{\mathrm{Vir}} \succeq (1/2)\mathrm{id}$, a formal Fock threshold rather than a zero-placement theorem (Chapter 45.3.3, Chapter 45.3.4).

(v) Finite Li data at $n \geq 3$ are scalar trace data for Θ_ξ; they do not imply Virasoro operator positivity or RH.

(vi) Fock-combinatorial positivity holds in the archimedean sector: $T_n^{\mathrm{Vir}} \succeq 0$ on non-vacuum number states for every n (Chapter 45.6.2).

(vii) The local Kac determinant at $c = 2$ does not vanish at generic weights, establishing absence of null-vector obstruction (Chapter 45.5.2, Chapter 45.5.3).

(viii) The RH-strength core is a single statement about the Θ_ξ-spectral measure: ($\mathrm{VB}^*_{\mathrm{core}}$, Chapter 45.7.1).

(ix) Arithmetic Zhu/Verlinde finite-dimensionality holds at finite windows only: any Euler-faithful quotient preserving the Bost–Connes character $\zeta(\beta)$ is infinite-dimensional (Chapter 45.10.2, Chapter 45.10.3).

(x) ($\mathrm{VB}^*_{\mathrm{core}}$) is the Vol IV form of RH, equivalent to $\mathrm{Re}(\rho) = 1/2$ for all non-trivial zeros (Chapter 45.9.2).

(xi) The condition $\mathcal{R}_{\mathrm{IV}}$ is the existence of the zero-divisor operator with positive, determinate Weil–Connes trace; it is equivalent to RH.

The unconditional Virasoro content is the identity at $n = 1$, the formal threshold at $n = 2$, and non-vacuum Fock positivity. The Riemann Hypothesis enters only through the zero-divisor operator Θ_ξ, the Li transforms, and the positive Weil–Connes trace. The equivalent condition is stated in two operator-theoretic forms (Chapter 45.11.1, Chapter 45.9.1) and is, in full, the Riemann Hypothesis.

Chapter 46

Arithmetic Co–Contra Comparison

An arithmetic co–contra comparison must specify its coalgebra, continuous dual, completion, and derived categories over a common coefficient ring. The classical field-coefficient correspondence provides a model for this question. The module-theoretic μ-invariant does not characterize conilpotence, as shown in Proposition 10.10.2.

For the proposed arithmetic Heisenberg application, a categorical comparison must also preserve the relevant pairing and central classes. Theorem 43.5.1 isolates the further Mellin evaluation and defect-detection hypotheses that imply $\kappa + \kappa^{!} = 0 \Longleftrightarrow \xi(s) = \xi(1-s)$. The scalar functional equation and the oscillator energy calculation do not construct those arithmetic maps.

46.1 Statement and Motivation

Review: Classical Positselski

Let k be a field and C a dg coalgebra over k. A left C-comodule has a coaction $M \to C \otimes_k M$. A left C-contramodule has a contraaction $\pi : \mathrm{Hom}_k(C, P) \to P$. The latter satisfies the associativity identity induced by $\Delta : C \to C \otimes_k C$ and the unit identity induced by $\epsilon : C \to k$. All maps commute with differentials, and tensor–Hom adjunctions use the Koszul signs.

The *coacyclic* dg comodules form the smallest triangulated subcategory containing totalizations of exact triples and closed under arbitrary direct sums. The *contraacyclic* dg contramodules are defined using products instead. Their Verdier quotients are the coderived and contraderived categories. These quotients need not agree with the ordinary derived categories, even though the coefficient ring is a field.

For a right comodule N and a left contramodule P, define

$$N \odot_C P = \mathrm{coker}\big(N \otimes_k \mathrm{Hom}_k(C, P) \rightrightarrows N \otimes_k P\big).$$

The arrows use the contraaction and the right coaction followed by evaluation, respectively. On homogeneous elements the latter evaluation is $c \otimes f \mapsto (-1)^{|c||f|} f(c)$. This quotient is the *contratensor product*. Write Hom_C for colinear homogeneous maps, with their Hom differential.

Theorem 46.1.1 *(Positselski,* Two kinds of derived categories, *§5.2).* The derived functors of

$$\Psi_C(M) = \mathrm{Hom}_C(C, M), \qquad \Phi_C(P) = C \odot_C P$$

give mutually inverse equivalences

$$\mathrm{D}^{\mathrm{co}}(C\text{-comod}_k) \simeq \mathrm{D}^{\mathrm{ctr}}(C\text{-contra}_k).$$

The functor Φ_C is left adjoint to Ψ_C.

Proof. Positselski's resolution theorem, §4.4, represents the two quotients by dg objects that are injective or projective after forgetting the differential. Graded injective comodules are summands of cofree objects

$C \otimes_k V$. Graded projective contramodules are summands of free objects $\operatorname{Hom}_k(C, V)$. The two functors exchange these free and cofree objects, with natural inverse maps. They therefore exchange their summands and the corresponding dg objects. The resulting equivalence of these homotopy categories proves the assertion. □

This correspondence is covariant in M. For $C = k$, its right adjoint is $M \mapsto M$, not the contravariant vector-space dual. The coalgebra dual $C^\vee$ and the cobar algebra ΩC are also different constructions. A topological description using $C^\vee$ must specify its continuous duality and its contramodules.

The Λ-Coefficient Target

Fix a prime p, a procyclic group $\Gamma \cong \mathbb{Z}_p$, and a topological generator γ. Set

$$\Lambda = \mathbb{Z}_p[[\Gamma]] \cong \mathbb{Z}_p[[T]], \qquad T = \gamma - 1, \qquad \mathfrak{m} = (p, T).$$

For the cyclotomic application, Γ is the Galois group of a chosen cyclotomic $\mathbb{Z}_p$-extension. The full cyclotomic Galois group may also have a finite factor. The ring Λ is complete, Noetherian, local and regular of dimension two, with residue field $\mathbb{F}_p$.

A pseudo-isomorphism is a map with finite kernel and cokernel. Every finitely generated Λ-module M admits one of the form

$$M \longrightarrow \Lambda^r \oplus \bigoplus_i \Lambda/p^{a_i}\Lambda \oplus \bigoplus_j \Lambda/f_j^{b_j}\Lambda.$$

Here $r \geq 0$, $a_i, b_j \geq 1$, and each $f_j \in \mathbb{Z}_p[T]$ is irreducible and distinguished. A monic polynomial is distinguished when its other coefficients are divisible by p. The Iwasawa invariants are $\mu(M) = \sum_i a_i$ and $\lambda(M) = \sum_j b_j \deg f_j$. Proposition 46.3.1 recalls the structure theorem.

Definition 46.1.2 *(arithmetic dg Λ-coalgebra).* An *arithmetic dg Λ-coalgebra* is a coaugmented dg coalgebra $C = \Lambda\mathbf{1} \oplus \overline{C}$ over Λ, with $\overline{C} = \ker \epsilon$, satisfying:

(i) *Local finiteness over* Λ: the underlying graded Λ-module C is a filtered union of finitely generated Λ-submodules $C_{\leq n}$ with $C = \varinjlim_n C_{\leq n}$ and each $C_{\leq n}/C_{\leq n-1}$ finitely generated.

(ii) *Comultiplication*: a graded Λ-linear map $\Delta\colon C \to C \otimes_\Lambda C$ with counit $\epsilon\colon C \to \Lambda$, satisfying the standard coassociativity / counit axioms, and compatible with the differential d.

(iii) *Conilpotence*: for every $c \in \overline{C}$, some iterated reduced coproduct vanishes on c. The reduced coproduct is

$$\overline{\Delta}(c) = \Delta(c) - \mathbf{1} \otimes c - c \otimes \mathbf{1} \in \overline{C} \otimes_\Lambda \overline{C}.$$

The notation $\mu(C) = \mu(\overline{C})$ requires $\overline{C}$ to be a finitely generated torsion Λ-module. For an infinite graded or completed object, one must instead specify the coefficient modules whose invariants are being taken.

Definition 46.1.3 *($\mu = 0$ arithmetic Λ-coalgebra).* Suppose $M = \overline{C}$ is a finitely generated torsion Λ-module. Then C is $\mu = 0$ when

$$\mu(M) = \operatorname{length}_{\Lambda_{(p)}} M_{(p)} = 0.$$

Equivalently, its structure theorem has no $\Lambda/p^a\Lambda$ summand. This is a statement up to pseudo-isomorphism, not the absence of all p-torsion. For example, $M = \Lambda/(p, T)$ is nonzero and killed by p, but $M_{(p)} = 0$ because T becomes invertible.

Remark 46.1.4 *(the role of $\mu = 0$).* The invariant measures torsion at the height-one prime (p). It does not control all torsion or derived base change. The module $\Lambda/(p, T)$ has $\mu = 0$ and nonzero p-torsion. The module $\Lambda/(T)$ also has $\mu = 0$, but its first Tor with itself is $\Lambda/(T)$. Proposition 46.3.5 computes these effects. Completion and reduction must therefore be checked on the actual comparison maps.

A precise lifting criterion uses completion inside the recipient category. In a stable Λ-linear ∞-category with sequential limits, put

$$X/p^n = \operatorname{cofib}(p^n : X \to X).$$

Call X *derived p-complete* when the natural map $X \to \varprojlim_n X/p^n$ is an equivalence. For $X \in \mathrm{D}(\Lambda)$, the quotient is $X \otimes^{\mathbb{L}}_{\Lambda} \Lambda/p^n$. In a coderived recipient, its intrinsic cofiber and limit are required. An underlying ordinary derived complex may forget information.

Theorem 46.1.5 *(lifting an arithmetic comparison).* Let $\mathcal{C}$ and $\mathcal{D}$ be specified stable Λ-linear ∞-categories with sequential limits. Suppose an exact Λ-linear adjunction $\Phi : \mathcal{D} \rightleftarrows \mathcal{C} : \Psi$ has been constructed. Write its unit and counit as

$$\eta_P : P \longrightarrow \Psi\Phi P, \qquad \varepsilon_M : \Phi\Psi M \longrightarrow M.$$

Assume their cofibers are derived p-complete in their respective categories, and that η_P/p and ε_M/p are equivalences for every P, M. Then

$$\Psi : \mathcal{C} \xrightarrow{\sim} \mathcal{D}, \qquad \Phi \simeq \Psi^{-1}.$$

The proof is given in Section 46.4. For the arithmetic problem, one must construct $\mathcal{C}$ and $\mathcal{D}$ from the intended integral comodules and contramodules, with their exact structures and topology. One must then construct the adjunction and verify these hypotheses. Module-theoretic $\mu = 0$ does not supply them. Compatibility with the algebraic bar–cobar adjunction is a further comparison of functors.

Remark 46.1.6 *(data for the arithmetic comparison).* The categorical construction and its zeta interpretation require different data:

(i) *Continuous duals.* Specify the coalgebra, its topology, its dual recipient, and the exact comodule and contramodule categories. Identify any proposed topological-algebra description with these categories.

(ii) *Derived completion.* Prove completeness of the unit and counit cofibers in those recipients. Ordinary module completion need not detect coderived equivalences.

(iii) *Reduction of the adjunction.* Construct the comparison with an actual adjunction over Λ/p, preserving its unit and counit. Prove that the reduced maps are equivalences. This requires detection of equivalences, not faithfulness of reduction on morphisms.

(iv) *Character and pairing.* Identify the selected one-mode character with ζ and construct its arithmetic pairing. The full oscillator character is $\prod_{k\geq 1} \zeta(k\beta)$. Neither scalar character constructs the central-defect or Mellin maps.

(v) *Bost–Connes comparison.* Construct the cobar algebra in the chosen coefficient category and compare it with the specified Bost–Connes algebra, action and pairing. Question 33.5.2 retains this problem. The continuous dual is not the cobar algebra by definition.

The first three requirements concern the lifting criterion. The last two concern its arithmetic interpretation. They are not an exhaustion theorem for all possible arithmetic realizations.

Section 46.4 presents the proposed co–contra construction. Its six coefficient and completion comparisons are in Section 46.2. Section 46.3 fixes the required Λ-module notions.

The arithmetic Heisenberg application and its level comparison are in Section 46.5. Section 46.7 retains the non-abelian Iwahori and Whittaker comparison conditions.

46.2 Obstruction Analysis Table

The following comparisons specify the coefficient, completion and pairing conditions in the arithmetic statement.

Obstruction 1: Base-Change Lift

The field correspondence is not a theorem over $\mathbb{F}_p[[T]]$, which is a discrete valuation ring. There is no unital map from a field to $\mathbb{Z}_p[[T]]$: positive characteristic is incompatible with characteristic zero, and a characteristic-zero field would make p invertible. Thus extension of scalars from a field cannot construct the integral comparison directly.

Reduction $\Lambda \to \Lambda/p$ is nevertheless useful. On derived p-complete objects, it detects equivalences, by Proposition 46.4.5. This proves a lifting criterion once the functors and their reduced unit and counit are available. It does not construct these functors, and reduction is not faithful on morphisms.

Obstruction 2: Exact Structures and Derived Recipients

Over Λ, an arithmetic comparison must select the admissible exact sequences of comodules and contramodules. The coacyclic and contraacyclic subcategories are then formed using these sequences and the appropriate sums or products. An injective resolution over the coefficient ring need not be an injective resolution in the comodule category. The corresponding distinction holds for projectives.

Selmer complexes furnish arithmetic complexes with specified local conditions. To use them here requires functors into the chosen comodule or contramodule recipient, preserving those conditions. Their existence does not follow from the notation $\mathrm{D}(\Lambda)$, or from a vanishing μ-invariant. Definitions 46.3.6 and 46.3.7 state the exact-category construction used below.

Obstruction 3: μ-Invariant Vanishing for the Heisenberg Bar

For a defined arithmetic bar object, the first question is which finitely generated torsion Λ-module carries the proposed μ-invariant. A filtered or completed complex has no single such invariant until the relevant cohomology or graded coefficient modules are specified.

There is an elementary vanishing calculation on a fixed coefficient model. Let $\Lambda = \mathbb{Z}_p[[T]]$, and let M be a finite free $\mathbb{Z}_p$-module of rank d, with trivial Γ-action. Then

$$M \cong (\Lambda/(T))^d, \qquad M_{(p)} = 0, \qquad \mu(M) = 0.$$

Indeed, $T \notin (p)$, so localization at (p) makes the annihilator T invertible. This proves vanishing for this module without identifying it with a cyclotomic Iwasawa module.

Given an actual augmented coefficient algebra with augmentation ideal I, one can form finite bar-length pieces

$$B_{\le n} = \bigoplus_{j=0}^{n} (I[1])^{\otimes_\Lambda j},$$

with cohomological shift $I[1]^q = I^{q+1}$. For positive bar lengths, applying that calculation requires the selected coefficients to be finite free over $\mathbb{Z}_p$ with trivial action. The length-zero coefficient is Λ itself.

A residue complex can be tensored into this construction only after its Λ-module structure and total differential have been supplied. Finite tame symbols, by themselves, provide neither structure.

Ferrero–Washington gives a vanishing theorem for specified cyclotomic Iwasawa modules. Its use requires identifying the arithmetic coefficient module with one in that theorem's domain. The identification must preserve the action and topology. Neither the name Heisenberg nor commutativity of Γ supplies such an identification. Proposition 10.10.2 further separates module-theoretic vanishing from coalgebra conilpotence.

Obstruction 4: Weight Compatibility Under Bar–Cobar

Weight comparison. The proposed arithmetic pairing would relate bar coefficients of energy $\log p$ to independently defined dual coefficients of energy $-\log n$. To interpret energy cancellation as $\kappa^{\mathrm{Ar}} + (\kappa^{!})^{\mathrm{Ar}} = 0$, one must construct the central-defect identity and evaluation of Theorem 43.5.1. Compatibility with a Bost–Connes Hecke operator μ_n is an additional action map.

A cyclotomic character supplies a Galois action only after that action is specified on the coefficient object. Its compatibility with a co–contra functor is additional data. Oscillator energies $k \log p$ belong to a different grading. The ungraded correspondence does not identify the two.

Weight calculation. Convention 28.7.1(ii) assigns energy $k \log p$ to $x_{p,k} = \alpha_{-k}^{(p)}$. Its linear dual can be given the negative energy. Evaluation then has total energy zero, as constructed in Theorem 46.5.1. The numerical identity $\log(p^k) = k \log p$ is also compatible with assigning $-\log(p^k)$ to a proposed dual symbol. It does not identify that symbol with a Bost–Connes isometry, construct a twisting cochain, or identify oscillator energy with a central level.

The full one-species-per-prime oscillator trace is $\prod_{k\geq 1} \zeta(k\beta)$ for $\beta > 1$. The selected mode-one occupation space has trace $\zeta(\beta)$, but is not invariant under the divisor or index-shift operations for index greater than one (Section 28.7). A pairing or arithmetic comparison must specify which of these carriers it uses.

Obstruction 5: Archimedean Completion

The completed zeta function contains the real factor $\pi^{-s/2}\Gamma(s/2)$. Fourier self-duality of the Gaussian and Poisson summation provide its analytic input. Theorem 35.6.1 proves the scalar reflection with its polar terms.

A categorical interpretation needs an archimedean coalgebra or algebra, its modules and its pairing. It also needs a comparison to the Fourier model. The local Fock space does not specify these data by itself. Section 46.6 gives the Gaussian calculation and a conditional assembly of actual local equivalences over a common coefficient ring.

Obstruction 6: Noncommutative Coefficients

For a compact p-adic group G, the completed group algebra $\mathbb{Z}_p[[G]]$ is a possible noncommutative coefficient ring. Its use for an Iwahori Hecke category requires an actual action and a comparison with that category. These are not consequences of the name of the group.

A coalgebra over a noncommutative base must specify left and right module structures. The resulting coring has a coproduct into its tensor square over the base. Its conilpotence, continuous Hom, completion and exact structures must be stated in that setting. The commutative structure theorem below does not provide them. The noncommutative and Whittaker questions in Section 46.7 retain these additional constructions.

Comparison Table

Input	Content	Conclusion
1	Reduction and lifting	complete cofibers and actual maps
2	Derived recipients	specify exact structures
3	$\mu = 0$ for specified bar modules	requires the module comparison
4	Weight pairing	oscillator evaluation constructed
5	Archimedean comparison	Fourier model. Categorical map needed
6	Noncommutative coefficients	coring and action required

TABLE 46.1: Arithmetic Positselski comparisons. The abelian comparison retains the data of Remark 46.1.6 and the pairing and invariant maps of Question 46.5.4. Non-abelian Iwahori level requires further data (Section 46.7).

46.3 IWASAWA PRELIMINARIES

The coefficient calculations use three facts about finitely generated Λ-modules: the structure theorem, the resolution bounds, and the actual Tor groups governing reduction. These module calculations do not identify the derived comodule and contramodule recipients.

Λ-Modules and Structure Theorem

PROPOSITION 46.3.1 *(Λ-module structure theorem).* Let $\Lambda = \mathbb{Z}_p[[\Gamma]]$ and let M be a finitely generated Λ-module. Then there is a pseudo-isomorphism (morphism with finite kernel and cokernel)

$$M \sim \Lambda^r \oplus \bigoplus_{i=1}^{s} \Lambda/p^{a_i}\Lambda \oplus \bigoplus_{j=1}^{t} \Lambda/f_j^{b_j}\Lambda,$$

where $r, s, t \in \mathbb{Z}_{\geq 0}$, $a_i \in \mathbb{Z}_{\geq 1}$, and $f_j \in \mathbb{Z}_p[\gamma - 1]$ are distinguished irreducible polynomials (Weierstrass polynomials). The invariants r, $\mu(M) = \sum a_i$, $\lambda(M) = \sum b_j \deg f_j$ are well-defined.

Proof. Apply Serre, *Classes des corps cyclotomiques*, §6, Lemmas 5–6. Up to finite kernel and cokernel, a finitely generated module splits into a reflexive module and height-one primary factors. The reflexive module is free because Λ is regular local of dimension two. Localization at a height-one prime is a discrete valuation ring, so each primary factor has its elementary-divisor decomposition. Weierstrass preparation identifies the height-one primes as (p) and the ideals (f) for distinguished irreducible f. Their localized multiplicities give the stated invariants. □

COROLLARY 46.3.2 *($\mu = 0$ criterion).* For a finitely generated Λ-module M, the invariant $\mu(M)$ is the length of the torsion submodule of $M_{(p)}$. It vanishes exactly when $M_{(p)}$ is free over $\Lambda_{(p)}$. If M is torsion, this is equivalent to $M_{(p)} = 0$.

Proof. Finite modules vanish after localization at (p). The structure theorem therefore becomes an isomorphism there. The summand Λ/p^a contributes length a. A distinguished polynomial reduces modulo p to a nonzero power of T, so it does not lie in (p) and becomes a unit. Its summand vanishes. The remaining free summand has rank r, proving all assertions. □

Injective and Projective Resolutions Over Λ

Proposition 46.3.3 *(projective resolution bound).* Every finitely generated Λ-module M has a finite free resolution of length at most two.

Proof. The regular sequence (p, T) gives the Koszul resolution

$$0 \longrightarrow \Lambda \xrightarrow{\binom{-T}{p}} \Lambda^2 \xrightarrow{(p\ \ T)} \Lambda \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

It follows that $\operatorname{Tor}_i^{\Lambda}(\mathbb{F}_p, M) = 0$ for $i > 2$. Construct a minimal free resolution of M by successively choosing minimal generators of its kernels. Each free module is finite because Λ is Noetherian. Every differential reduces to zero modulo $\mathfrak{m}$, so its i-th free module modulo $\mathfrak{m}$ is the i-th Tor group. Nakayama's lemma makes those free modules zero for $i > 2$. □

Proposition 46.3.4 *(injective resolution bound).* Every Λ-module M has an injective resolution $0 \to M \to I^0 \to I^1 \to I^2 \to 0$. No finite-generation condition on the injective terms is asserted.

Proof. For every ideal J, the preceding proposition gives $\operatorname{Ext}^3_{\Lambda}(\Lambda/J, M) = 0$. Start an injective resolution and let Q be the cokernel of its first two injective terms, so that $0 \to M \to I^0 \to I^1 \to Q \to 0$ is exact. Dimension shifting gives $\operatorname{Ext}^1_{\Lambda}(\Lambda/J, Q) = 0$ for every J. Baer's criterion makes Q injective. □

These are resolutions of modules over Λ. They do not supply the required injectives or projectives in an arithmetic comodule or contramodule category. In particular, the residue field is not an injective Λ-module: the same Koszul complex gives $\operatorname{Ext}^1_{\Lambda}(\mathbb{F}_p, \mathbb{F}_p) \cong \mathbb{F}_p^2$.

Proposition 46.3.5 *(derived reduction and its torsion).* Put $B = \Lambda/p = \mathbb{F}_p[[T]]$. For every Λ-module M,

$$M \otimes^{\mathbb{L}}_{\Lambda} B \simeq [\, M \xrightarrow{p} M \,], \qquad \operatorname{Tor}_1^{\Lambda}(M, B) = M[p],$$

where the displayed complex lies in degrees -1, 0, and $M[p] = \ker(p : M \to M)$. Higher Tor groups vanish. Reduction of a projective resolution therefore resolves the ordinary module M/pM exactly when $M[p] = 0$. The corresponding statements for $\Lambda/(T)$ use the map T and its kernel $M[T]$.

Proof. Tensor the free resolution $0 \to \Lambda \xrightarrow{p} \Lambda \to B \to 0$ with M. Its homology gives the assertion. Multiplication by T is also injective on Λ, so the same argument applies to $\Lambda/(T)$. □

For $M = \Lambda/(p, T)$, one has $\mu(M) = 0$ and $M[p] = M \neq 0$. Thus the positive-degree Tor term survives. For $N = \Lambda/(T)$, one has $\mu(N) = 0$ and

$$\operatorname{Tor}_1^{\Lambda}(N, N) = N \neq 0.$$

This second module is free over $\mathbb{Z}_p$ but is not flat over Λ. Neither $\mu = 0$ nor $\mathbb{Z}_p$-freeness can replace the required Tor calculation.

Co-Derived and Contra-Derived Λ-Categories

Let C be a specified dg Λ-coalgebra. Fix exact structures on its dg comodules and dg contramodules, including the admissible exact triples. For example, one may use triples split as graded Λ-modules. This gives a relative exact structure. An abelian or continuous version requires its own hypotheses.

Assume the chosen comodule category has admissible arbitrary sums and the contramodule category has admissible arbitrary products. Write Hot for the homotopy category of these dg objects.

Definition 46.3.6 *(Λ-coderived recipient).* Let $\mathrm{Ac}^{\mathrm{co}}_{\Lambda}(C)$ be the smallest thick triangulated subcategory of $\mathrm{Hot}(C\text{-comod}_{\Lambda})$ containing the totalizations of admissible exact triples and closed under arbitrary sums. Define

$$\mathrm{D}^{\mathrm{co}}_{\Lambda}(C\text{-comod}) = \mathrm{Hot}(C\text{-comod}_{\Lambda})/\mathrm{Ac}^{\mathrm{co}}_{\Lambda}(C).$$

The chosen exact structure is part of this notation.

Definition 46.3.7 *(Λ-contraderived recipient).* Let $\mathrm{Ac}^{\mathrm{ctr}}_{\Lambda}(C)$ be the smallest thick triangulated subcategory of $\mathrm{Hot}(C\text{-contra}_{\Lambda})$ containing the totalizations of admissible exact triples and closed under arbitrary products. Define

$$\mathrm{D}^{\mathrm{ctr}}_{\Lambda}(C\text{-contra}) = \mathrm{Hot}(C\text{-contra}_{\Lambda})/\mathrm{Ac}^{\mathrm{ctr}}_{\Lambda}(C).$$

A description by continuous $C^{\vee}$-contramodules requires an identification with this specified recipient.

Remark 46.3.8 *(comparison with Selmer complexes).* A Selmer complex is an object of an ordinary derived category with specified local-condition maps. Chapter 10 constructs such complexes and their coefficient comparisons. A realization in the recipients above must preserve those maps and specify which acyclic objects are annihilated. The distinction between acyclic and coacyclic objects already occurs for coalgebras over a field. Coefficient linear algebra does not remove it.

Question 46.3.9 *(enhancements and compact objects).* For the intended arithmetic coalgebra and exact structures, construct the stable enhancements of these recipients. Determine the complete subcategories on which the co–contra functors exist and the compact objects, if the relevant coproducts exist.

Compactness cannot mean concentration in degree zero. Compact objects in a stable category are closed under shifts. For example, $\Lambda[1]$ is compact in $\mathrm{D}(\Lambda)$ and is not a module concentrated in degree zero. The module resolution bounds do not resolve Question 46.3.9.

Iwasawa μ-Invariant and Ferrero–Washington

THEOREM 46.3.10 *(Ferrero–Washington 1979, Ann. Math. 109).* Let K be an abelian number field, let K_{∞}/K be its cyclotomic $\mathbb{Z}_p$-extension, and let K_n be its finite layers. With norm transition maps, put

$$X_K = \varprojlim_{n} \mathrm{Cl}(K_n)_p.$$

Identify $\mathrm{Gal}(K_{\infty}/K)$ with the chosen Γ. Then X_K is a finitely generated torsion Λ-module and $\mu(X_K) = 0$.

Remark 46.3.11 *(applicability to Heisenberg bar).* An application to an arithmetic Heisenberg bar must first specify its finitely generated torsion coefficient module M and its continuous Γ-action. It must then identify the relevant localization with that of a module in Theorem 46.3.10. Commuting creation operators and a trivial action on an oscillator lattice supply no such identification. The full Heisenberg Lie algebra itself has a nonzero central commutator.

Section 46.2 proves vanishing for finite free $\mathbb{Z}_p$-modules with trivial Γ-action directly. The next statement gives a different, conditional arithmetic route. Neither route supplies a conilpotence criterion.

Corollary 46.3.12 *(transport of arithmetic μ-vanishing).* Let M be a finitely generated torsion Λ-module selected from a specified arithmetic bar construction. Suppose M admits a pseudo-isomorphism to $X_K^{\oplus d}$, where K is abelian and X_K is as in Theorem 46.3.10. Then $\mu(M) = 0$.

Proof. The finite kernel and cokernel have support at (p, T). They vanish after localization at (p), giving

$$M_{(p)} \cong (X_K)_{(p)}^{\oplus d} = 0.$$

The last equality is Theorem 46.3.10. Taking length proves the assertion. The pseudo-isomorphism is an input, not a consequence of the name of the bar construction. □

46.4 Constructing and Detecting the Comparison

The Functors and Their Comparison Maps

The algebraic functors can be written before choosing a completed recipient. Their descent to a derived category, compatibility with reduction, and equivalence are separate assertions. We first give the functors, then prove the completion criterion. The algebraic bar–cobar adjunction and compatible quotient towers provide two further comparisons.

Step 1: The Algebraic Functors

Use the algebraic tensor product over Λ in this subsection. Let C be a dg coalgebra and use its left and right regular coactions. The contramodule and contratensor definitions above extend by replacing k with Λ. A continuous version requires its own Hom and tensor objects.

Definition 46.4.1 *(the colinear Hom functor).* For a dg left comodule M, put

$$\Psi_C(M) = \mathrm{Hom}_C(C, M) \subset \mathrm{Hom}_\Lambda(C, M), \qquad \mathrm{Hom}_\Lambda(C, M)^n = \prod_i \mathrm{Hom}_\Lambda(C^i, M^{i+n}).$$

The differential is $d(f) = d_M f - (-1)^{|f|} f\, d_C$. The right coaction on C induces the left contraaction on this colinear Hom complex. On morphisms, Ψ_C acts by postcomposition.

Definition 46.4.2 *(the contratensor functor).* For a dg left contramodule P, put

$$\Phi_C(P) = C \odot_C P = \mathrm{coker}\big(C \otimes_\Lambda \mathrm{Hom}_\Lambda(C, P) \rightrightarrows C \otimes_\Lambda P\big).$$

The two arrows are the contraaction and the right coaction followed by signed evaluation. The differential and the left coaction descend from $C \otimes_\Lambda P$.

Tensor–Hom adjunction identifies maps from this cokernel to M with contramodule maps from P to $\mathrm{Hom}_C(C, M)$. Thus $\Phi_C \dashv \Psi_C$ on the algebraic dg categories. Completion is not part of this calculation.

Lemma 46.4.3 *(descent through chosen resolution models).* Suppose full triangulated resolution subcategories $\mathcal{J} \subset \mathrm{Hot}(C\text{-comod}_\Lambda)$ and $\mathcal{P} \subset \mathrm{Hot}(C\text{-contra}_\Lambda)$ map equivalently to the chosen coderived and contraderived recipients. If $\Psi_C(\mathcal{J}) \subset \mathcal{P}$ and $\Phi_C(\mathcal{P}) \subset \mathcal{J}$, the restricted adjunction transports to those recipients.

Proof. Compose each restricted functor with the two specified resolution equivalences and their inverses. The Hom adjunction transports along the same equivalences, as do its unit, counit and triangle identities. □

Over a field, Theorem 46.1.1 supplies these resolution models and their equivalence. Over Λ, the models and their preservation must be constructed. Injectivity over Λ alone does not prove the lemma's hypotheses.

The contratensor quotient matters already in dimension two. Over a field, let $C = k\mathbf{1} \oplus kc$, with $\Delta c = \mathbf{1} \otimes c + c \otimes \mathbf{1}$. The colinear endomorphisms of C have basis

$$f_0 = \mathrm{id}_C, \qquad f_1(\mathbf{1}) = 0, \quad f_1(c) = \mathbf{1}.$$

Evaluation $C \otimes \mathrm{Hom}_C(C, C) \to C$ is surjective and has kernel

$$k(\mathbf{1} \otimes f_1) \oplus k(c \otimes f_1 - \mathbf{1} \otimes f_0).$$

These are precisely the contratensor relations, so the quotient is C. Omitting the quotient leaves a four-dimensional space in place of a two-dimensional one.

Step 2: Coefficient Change

LEMMA 46.4.4 *(two coefficient comparisons).* Let $R \to S$ be a map of commutative rings.

(i) If C is a coaugmented conilpotent dg R-coalgebra, then the algebraic scalar extension $C \otimes_R S$ is a coaugmented conilpotent dg S-coalgebra.

(ii) If E is a perfect complex of R-modules, then for every $M \in \mathrm{D}(R)$, the natural map

$$\mathrm{RHom}_R(E, M) \otimes^{\mathbb{L}}_R S \longrightarrow \mathrm{RHom}_S(E \otimes^{\mathbb{L}}_R S, M \otimes^{\mathbb{L}}_R S)$$

is an equivalence.

Proof. The coaugmentation splitting commutes with algebraic scalar extension. Each element of its reduced part is a finite sum of tensors $c_i \otimes s_i$. A common bound on the vanishing of the iterated reduced coproducts of the finitely many c_i proves conilpotence. No flatness of S is needed for this algebraic assertion.

For the second assertion, represent E by a bounded complex of finite projective modules. Its Hom complex is $E^\vee \otimes_R M$, and $E^\vee \otimes_R S \cong (E \otimes_R S)^\vee$. These identities give the stated derived map and its inverse. □

The second assertion concerns coefficient Hom. The functor Ψ_C also imposes colinearity, and a continuous version may involve an inverse limit. Indeed, $\mathrm{Hom}_\Lambda(\varinjlim C_i, M) = \varprojlim \mathrm{Hom}_\Lambda(C_i, M)$. Commutation of this limit, its derived functors and coefficient change requires an additional proof.

For $B = \Lambda/p$, a proposed comparison has the form

$$\begin{array}{ccc} C & \xrightarrow{\Psi} & \mathcal{D} \\ {\scriptstyle R_C}\downarrow & & \downarrow{\scriptstyle R_{\mathcal{D}}} \\ \overline{C} & \xrightarrow[\overline{\Psi}]{} & \overline{\mathcal{D}}. \end{array}$$

Here the barred recipients and the reduction functors must be defined. A natural equivalence making this square commute is part of the comparison. The corresponding square for Φ must also identify the units and counits. An equivalence over B, when supplied, is not the field-coefficient theorem: B is not a field.

Step 3: Detection by Derived Completion

Proposition 46.4.5 *(complete objects and reduction).* In a stable Λ-linear ∞-category with sequential limits, a derived p-complete object X is zero if $X/p = 0$. Consequently, a map whose cofiber is derived p-complete is an equivalence if its reduction modulo p is.

Proof. The factorization $p^n = p\, p^{n-1}$ gives a cofiber sequence

$$X/p^{n-1} \longrightarrow X/p^n \longrightarrow X/p.$$

Induction gives $X/p^n = 0$ for every n. Derived completeness then gives $X \simeq \varprojlim_n X/p^n = 0$. Apply this result to the cofiber of the given map. □

This proves conservativity on the stated complete objects. Faithfulness on morphisms fails. Take $M = \Lambda/(T) \cong \mathbb{Z}_p$. It is derived p-complete and has $\mu(M) = 0$, yet

$$p\,\mathrm{id}_M \neq 0, \qquad (p\,\mathrm{id}_M) \otimes^{\mathbb{L}}_{\Lambda} B = 0.$$

The two-term free resolution with differential T verifies the second equality directly after reduction.

Lifts are not unique either. The modules $\Lambda/(T)$ and $\Lambda/(T-p)$ are $\mathbb{Z}_p$-free, derived p-complete, and have $\mu = 0$. Their reductions are both $B/(T)$. They are not isomorphic over Λ: T acts as zero on the first and as multiplication by p on the second. Even the first lift has the nonidentity automorphism $(1+p)\mathrm{id}$ reducing to the identity.

Corollary 46.4.6 *(lifting a specified adjunction).* Consider an exact adjunction as in Theorem 46.1.5. Assume its unit and counit cofibers are derived p-complete and its reduced unit and counit are equivalences. Then the adjunction is an equivalence.

Proof. Proposition 46.4.5 makes the unit and counit equivalences. The triangle identities identify the two adjoint functors as inverses. □

An arithmetic application must verify these conditions in its specified recipients. Applying ordinary reduction to underlying complexes does not suffice unless that comparison is itself conservative on the relevant cofibers.

Step 4: The Algebraic Bar–Cobar Adjunction

Let A be an augmented dg Λ-algebra and C a coaugmented conilpotent dg Λ-coalgebra. Write $\overline{A} = \ker \epsilon_A$ and $\overline{C} = \ker \epsilon_C$, and use cohomological shifts $V[1]^i = V^{i+1}$. A twisting cochain is a degree-one map $\tau : \overline{C} \to \overline{A}$ satisfying

$$d_A\tau + \tau d_C + \mu_A(\tau \otimes \tau)\overline{\Delta}_C = 0.$$

The tensor of homogeneous maps includes its Koszul sign.

Theorem 46.4.7 *(algebraic Λ-bar–cobar adjunction).* With algebraic tensors and direct sums, the constructions

$$B^{\Lambda}A = \bigoplus_{n\geq 0} (\overline{A}[1])^{\otimes_{\Lambda} n}, \qquad \Omega^{\Lambda}C = \bigoplus_{n\geq 0} (\overline{C}[-1])^{\otimes_{\Lambda} n}$$

carry their bar and cobar differentials and satisfy

$$\mathrm{Hom}_{\mathrm{dgAlg}^{\mathrm{aug}}_{\Lambda}}(\Omega^{\Lambda}C, A) \cong \mathrm{Tw}_{\Lambda}(C, A) \cong \mathrm{Hom}_{\mathrm{dgCoalg}^{\mathrm{conil}}_{\Lambda}}(C, B^{\Lambda}A).$$

Thus $\Omega^{\Lambda} \dashv B^{\Lambda}$ on these algebraic dg categories.

Proof. An algebra map from the tensor algebra $\Omega^{\Lambda}C$ is determined by its restriction to $\overline{C}[-1]$. It commutes with differentials exactly when the corresponding τ satisfies the displayed twisting equation.

A coalgebra map to the tensor coalgebra $B^{\Lambda}A$ is determined by projection to $\overline{A}[1]$. Its length-n component applies that projection to the n-fold reduced coproduct. Conilpotence makes the resulting sum finite on each element. The chain-map condition is the same twisting equation. These two constructions are inverse and natural. □

No completion or derived equivalence is asserted by this algebraic adjunction. A completed version must justify the convergence of its sums and the weak equivalences it preserves. It is also distinct from the module adjunction $\Phi_C \dashv \Psi_C$.

Step 5: Unit and Counit

THEOREM 46.4.8 *(a complete criterion for unit and counit).* For the algebraic adjunction, the natural maps have types

$$\eta_C : C \longrightarrow B^{\Lambda}\Omega^{\Lambda}C, \qquad \varepsilon_A : \Omega^{\Lambda}B^{\Lambda}A \longrightarrow A.$$

Suppose the cone of either map, viewed in $\mathrm{D}(\Lambda)$, is derived p-complete, and that its derived reduction modulo p is a quasi-isomorphism. Then that map is a quasi-isomorphism over Λ.

Proof. Apply Proposition 46.4.5 in $\mathrm{D}(\Lambda)$ to the specified cone. The maps exist by the algebraic adjunction. Their completeness and reduction hypotheses must be checked for the chosen A and C. □

The coalgebra unit uses iterated *reduced* coproducts. The full coproduct of a primitive element retains terms with $\mathbf{1}$ in every arity, so conilpotence cannot truncate that full coproduct. Nor does an equivalence in a chosen coderived localization assert an ordinary quasi-isomorphism without a comparison of the two notions.

Step 6: Compatible Quotient Towers

LEMMA 46.4.9 *(limits of compatible equivalences).* Let $(\mathcal{C}_n)$ and $(\mathcal{D}_n)$ be inverse diagrams of stable categories. A compatible equivalence of diagrams $\Psi_n : \mathcal{C}_n \simeq \mathcal{D}_n$ induces

$$\varprojlim_n \mathcal{C}_n \simeq \varprojlim_n \mathcal{D}_n.$$

Restriction along a coinitial indexing functor preserves these limits. In particular, cofinal systems of open ideals give the same limit for a fixed diagram on those ideals.

Proof. The limit functor preserves equivalences of diagrams. Its universal property also gives invariance under coinitial restriction. For two systems of open ideals I_n, J_m, the quotients by $I_n \cap J_m$ give a common refinement with maps to the original quotients. □

The lemma requires the diagram and its compatibility maps. Separate equivalences at individual quotients do not supply a compatible equivalence of diagrams. Identifying an arithmetic recipient with this inverse limit is another hypothesis. The lemma gives no uniqueness for arbitrary lifts of a residual object.

The Comparison Criterion

Proof of Theorem 46.1.5. Let $K_P = \operatorname{cofib}(\eta_P)$ in $\mathcal{D}$ and $L_M = \operatorname{cofib}(\varepsilon_M)$ in C. They are derived p-complete by hypothesis. Their reductions vanish because the reduced unit and counit are equivalences. Proposition 46.4.5 gives $K_P = L_M = 0$. Thus both natural transformations are equivalences, and the adjunction exhibits Φ and Ψ as inverse functors. □

Remark 46.4.10 *(the dual arithmetic module).* A Selmer μ-invariant requires a specified finitely generated torsion Iwasawa module. Comparing it with the invariant on another coefficient object requires an arithmetic duality map and control of its kernel and cokernel. A pseudo-isomorphism of the relevant modules would identify their height-one localizations and hence their μ-invariants. Formal vector-space duality alone gives no such arithmetic pseudo-isomorphism.

46.5 Application: Pairings and Arithmetic Levels

The Required Arithmetic Comparison

For Theorem 43.5.1 to apply to arithmetic duality, the following assignments must be constructed:

- the arithmetic level $\kappa_{\mathsf{G}}^{\mathrm{Ar}}$ as a central invariant of a specified bar object.
- the dual level $(\kappa^{!})_{\mathsf{G}}^{\mathrm{Ar}}$ as a central invariant of its specified dual object.
- an involution and Mellin evaluation carrying their common central defect to the difference of the two completed characters.

The converse also requires that evaluation to detect the defect. A cobar algebra and a continuous dual algebra have different constructions. A comparison with a Bost–Connes algebra must specify which is used. The local weight calculation below proves evaluation compatibility for its oscillator model. It supplies none of these arithmetic identifications.

The Explicit Weight Computation

Theorem 46.5.1 *(evaluation pairing and arithmetic energy).* For a finite set S of arithmetic primes let $P_S = \mathbb{Z}[x_{p,k} : p \in S, k \geq 1]$, with oscillator weight k and energy $k \log p$ on $x_{p,k}$. Each weight component $P_{S,w}$ is finite free. Its restricted dual

$$P_S^{\circ} = \bigoplus_{w \geq 0} \operatorname{Hom}_{\mathbb{Z}}(P_{S,w}, \mathbb{Z})$$

has a perfect evaluation pairing in each weight. Give the dual of a monomial energy equal to the negative of that monomial's energy. Then evaluation has total energy zero.

If an arithmetic bar–cobar pairing and comparison maps identify these particular weight components and their pairings with $P_{S,w}$ and its restricted dual, the same energy cancellation holds on their images. An identification of a dual generator with μ_{p^k}, and an identification of this energy cancellation with $\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^{!})_{\mathsf{G}}^{\mathrm{Ar}} = 0$, are separate comparison hypotheses.

Proof. There are finitely many monomials of a given oscillator weight because S is finite. Write x^{ν} for their monomial basis and η_{ν} for the dual basis. Evaluation is $\eta_{\mu}(x^{\nu}) = \delta_{\mu\nu}$, hence perfect in every weight. Its only nonzero basis pairings have energies $E(\nu)$ and $-E(\nu)$, whose sum is zero. Transport through the specified pairing-preserving maps proves the conditional assertion. No arithmetic map is constructed by this transport calculation.

The distinction from a polynomial generator algebra can already be seen in one variable: dualizing the primitive coproduct of $\mathbb{Z}[x]$ gives $\eta_a\eta_b = \binom{a+b}{a}\eta_{a+b}$, so $\eta_1^k = k!\eta_k$. This follows by evaluating on $\Delta x^{a+b} = \sum_j \binom{a+b}{j} x^j \otimes x^{a+b-j}$. It is not the relation $\mu_{p^k} = \mu_p^k$ with the same integral basis. A different normalization or cobar differential requires its own construction. □

Global Euler-Product Check

PROPOSITION 46.5.2 *(global weight sum).* The formal cancellation of opposite energy labels over prime-power indices is

$$\sum_{p\ \text{prime}} \sum_{n\geq 1} \left[n \log p \cdot p^{-ns} + (-n \log p) \cdot p^{-ns} \right] = 0,$$

convergent absolutely for $\mathrm{Re}(s) > 1$. After termwise pairing it is the zero function, hence extends as zero. No meromorphic continuation of the unpaired prime-indexed sum is asserted.

Proof. Term-by-term: at each prime p and each $n \geq 1$, the two weights $n \log p$ and $-n \log p$ sum to zero. For $\sigma = \mathrm{Re}\, s > 1$, the unpaired absolute sum is

$$\sum_p (\log p) \frac{p^{-\sigma}}{(1-p^{-\sigma})^2} \leq (1-2^{-\sigma})^{-2} \sum_{m\geq 2} (\log m) m^{-\sigma} < \infty.$$

Thus pairing terms is justified in that half-plane, and their sum is zero. This scalar identity supplies no arithmetic pairing or level map. The identity $-\zeta'(s)/\zeta(s) = \sum_{p,k\geq 1} (\log p)\, p^{-ks}$ is a prime-power identity, not an identity for the prime-indexed sum alone. □

Remark 46.5.3 *(pairing and level comparison).* The local evaluation theorem proves energy cancellation. To use it in Theorem 43.5.1, one must construct the arithmetic pairing, the central-defect identity, and the Mellin evaluation, including its detection of the defect. Opposite energy labels alone do not identify these central classes. The five conditions of Remark 46.1.6 must be realized with those compatibility maps; a named categorical equivalence alone does not identify its numerical invariants.

The Arithmetic Comparison Required for P1

Question 46.5.4 *(level cancellation and the functional equation).* Construct the arithmetic pairing, its integral and completed coefficient models, and the comparison of its central obstruction classes with the Tate functional equation. The intended conclusion remains

$$\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0 \quad \Longleftrightarrow \quad \xi(s) = \xi(1-s).$$

The scalar functional equation is a theorem. The oscillator energy calculation does not prove either identification of the arithmetic classes with its two sides.

Remark 46.5.5 *(dependencies of the arithmetic comparison).* The pairing and invariant comparisons are required whenever Remark 43.5.2 and Question 33.6.1 and Theorem 44.6.1 invoke the arithmetic Positselski input. The continuous-dual, completion, lifting, character and Bost–Connes requirements remain distinct. Neither the formal sum of opposite energies nor the scalar zeta identity removes these maps.

46.6 Fourier Pairings and Adelic Assembly

The Archimedean Fourier Model

The real local factor enters the scalar functional equation through the Gaussian. Fix the Fourier convention

$$\widehat{f}(y) = \int_{\mathbb{R}} f(x) e^{-2\pi i x y}\, dx.$$

This integral uses real analysis and a specified measure. A comparison with an algebraic pairing must preserve that normalization.

Theorem 46.6.1 *(Gaussian Fourier identity).* For $t > 0$ and $f_t(x) = e^{-\pi t x^2}$,

$$\widehat{f_t}(y) = t^{-1/2} e^{-\pi y^2/t}.$$

Proof. For $t = 1$, differentiate under the integral and integrate by parts. The Gaussian decay eliminates the boundary term and gives $\widehat{f_1}'(y) = -2\pi y \widehat{f_1}(y)$. The value at zero is one: the square of the positive Gaussian integral is $2\pi \int_0^\infty r e^{-\pi r^2}\, dr = 1$. Solving the differential equation gives the formula at $t = 1$. The substitution $u = \sqrt{t}\, x$ proves the general case. □

Poisson summation gives $\theta(t) = t^{-1/2}\theta(1/t)$ for $\theta(t) = \sum_{m\in\mathbb{Z}} e^{-\pi m^2 t}$. The theta series counts a lattice quadratic form. The oscillator Fock space $\mathbb{C}[\alpha_{-1}, \alpha_{-2}, \ldots]|0\rangle$, with weight n on α_{-n}, has character $\prod_{n\geq 1}(1 - q^n)^{-1}$. The two series already differ at the coefficient of q: $\sum_{m\in\mathbb{Z}} q^{m^2}$ has coefficient two, whereas the oscillator character has coefficient one.

To compare the Fourier identity with co–contra duality, specify an archimedean coalgebra and a pairing-preserving map. Theorem 46.1.1 applies to an actual dg coalgebra over $\mathbb{C}$. It does not turn a Fock module, a theta series or a Fourier transform into that coalgebra or its categorical pairing.

Adelic Assembly

Fix a commutative coefficient ring R. For each place v, let $\mathcal{C}_v$, $\mathcal{D}_v$ be presentable stable R-linear categories, with chosen objects c_v, d_v. Use the tensor product of such categories characterized by R-bilinear functors preserving colimits in each variable.

For a finite place set F, put

$$\mathcal{C}_F = \bigotimes_{v\in F, R} \mathcal{C}_v, \qquad \mathcal{D}_F = \bigotimes_{v\in F, R} \mathcal{D}_v.$$

The empty tensor is Mod_R. Insertion of c_v, respectively d_v, defines the transition functors when a new place is added. Define the corresponding restricted tensor categories as colimits over these finite place sets.

Theorem 46.6.2 *(assembly of pointed local equivalences).* Suppose R-linear equivalences $\Upsilon_v : \mathcal{C}_v \simeq \mathcal{D}_v$ and identifications $\Upsilon_v(c_v) \simeq d_v$ give a compatible equivalence of the finite tensor diagrams. Then

$$\varinjlim_F \mathcal{C}_F \xrightarrow{\sim} \varinjlim_F \mathcal{D}_F.$$

Proof. Tensoring finitely many equivalences gives an equivalence $\mathcal{C}_F \simeq \mathcal{D}_F$. The pointing identifications and their assumed coherence make these maps an equivalence of diagrams. Taking the colimit preserves that equivalence. □

For arithmetic local categories, the common coefficient action, tensor recipient, local equivalences and compatible pointings must be constructed. The symbols $\mathbb{Q}_p$ and $\mathbb{C}$ do not provide a common field by themselves. The chosen objects also need not be scalar augmentations or vacua of an independently specified field theory. Identifying this categorical colimit with an analytic adelic restricted tensor product is a further comparison.

Remark 46.6.3 *(the adelic pairing and the functional equation).* The theta–Mellin calculation of Theorem 35.6.1 retains its polar terms and proves the scalar functional equation. To interpret it through an adelic bar pairing, construct an evaluation identifying the duality involution with Mellin reflection. The forward implication also requires the central-defect identity. The converse requires that evaluation to detect the specified defect, as in Theorem 43.5.1.

An adelic co–contra equivalence is a separate categorical comparison. Identifying its pairing with Fourier self-duality requires the actual coefficient and pairing maps. The scalar theta calculation does not construct them.

46.7 Residual Conditions

The integral lifting criterion and the local assembly theorem apply only after their functors, recipients and comparison maps have been supplied. The following questions ask for further arithmetic structures. The commutative comparison remains subject to the preceding hypotheses.

Non-Abelian Iwahori Level

For a compact p-adic group G, write $A = \mathbb{Z}_p[[G]]$. An A-coring is an A-bimodule C with counit $C \to A$ and a coassociative coproduct $C \to C \otimes_A C$. A continuous version replaces the tensor by a specified completed tensor and requires continuous structure maps.

Question 46.7.1 *(noncommutative arithmetic co–contra comparison).* For an arithmetic coring over A, construct exact comodule and contramodule recipients and an adjunction between them. Determine a coefficient and completion domain on which its unit and counit are equivalences. For $G = GL_2(\mathbb{Z}_p)$, construct the required comparison with the intended Iwahori Hecke category and its actions. If a noncommutative μ-invariant is used, specify its module and prove the precise role of its vanishing.

Remark 46.7.2 *(module structure and categorical comparison).* A noncommutative module structure theorem has its own hypotheses on the group, ring and torsion class. It does not by itself compare comodule and contramodule resolutions. Such a comparison needs the actual tensor and Hom functors, with the correct module sides. Even over the commutative ring Λ, Proposition 46.3.5 shows why $\mu = 0$ cannot replace the necessary Tor conditions.

Whittaker Model Compatibility

A Whittaker condition requires a specified unipotent group N, a character ψ, and an action on the recipient. It also requires a choice of equivariant or derived coinvariant construction. The coefficient ring must support that character.

Question 46.7.3 *(Whittaker-compatible co–contra comparison).* Construct an arithmetic co–contra comparison equipped with compatible (N, ψ)-actions and the required Hecke actions. Show that it induces an equivalence between the resulting Whittaker recipients,

$$\mathcal{C}^{N,\psi} \simeq \mathcal{D}^{N,\psi},$$

and construct its comparison with the chosen Satake and Whittaker functors. A scalar character identity alone does not define these equivariant functors.

Extension to Higher Rank

Replacing $\Gamma \cong \mathbb{Z}_p$ by $\mathbb{Z}_p^d$ changes the coefficient ring to $\Lambda_d = \mathbb{Z}_p[[T_1, \dots, T_d]]$. The co–contra comparison then needs categories and resolution hypotheses over this ring. An arithmetic vanishing theorem also requires a specified module attached to the chosen tower. The cyclotomic theorem does not supply that module or an induction in d.

Remark 46.7.4 *(higher-rank extension).* For a finitely generated torsion Λ_d-module, the length of its localization at (p) gives a corresponding height-one invariant. The module $\Lambda_d/(p)$ has length one there, although Γ is abelian. Thus abelianity alone does not force this invariant to vanish. Determining its value for an arithmetic tower and comparing that module with a bar coefficient are separate problems.

46.8 Positselski Boundary

(i) Theorem 46.1.5 gives a lifting criterion for a specified adjunction. Its unit and counit cofibers must be derived p-complete, and their reductions must vanish. The integral arithmetic categories and functors remain additional constructions. The field correspondence does not establish them by scalar extension or faithful reduction.

(ii) The comparisons of Section 46.2 have separate hypotheses. In particular, the weight calculation needs arithmetic pairing and invariant maps even in the abelian case. The nonabelian Iwahori extension is a further problem (Section 46.7).

(iii) The hypothesis $\mu(C) = 0$ concerns the specified torsion coefficient module. Section 46.2 proves a local lattice calculation. Corollary 46.3.12 gives an arithmetic consequence under an actual module comparison. Neither statement identifies an unspecified Heisenberg bar with a cyclotomic class-group module or characterizes conilpotence.

(iv) Theorem 46.5.1 constructs the restricted integral dual and its energy-zero evaluation pairing on finite-species polynomial components. Identifying this pairing with arithmetic bar–cobar data and its energies with central obstruction classes is additional work.

(v) The intended P1 comparison remains Question 46.5.4. Its operator and invariant maps are not consequences of the scalar functional equation.

(vi) Adelic extension retains the local categorical hypotheses and their compatible pointings. Its interpretation through the functional equation additionally requires the evaluation and central-defect maps of Remark 46.6.3. Gaussian Fourier self-duality and the scalar theta calculation provide analytic models, not those arithmetic identifications.

(vii) Residual condition (Section 46.7): non-commutative arithmetic Positselski for non-abelian Iwahori level (Question 46.7.1), Whittaker compatibility (Question 46.7.3) and higher-rank Iwasawa coefficients (Remark 46.7.4).

The intended arithmetic comparison relates central obstruction classes to the classical functional equation. The local constructions above do not identify those classes with the explicitly computed scalar pairing. The continuous-dual, completion, lifting, character and Bost–Connes requirements must therefore be supplied as actual compatible maps on the intended carrier. The arithmetic Positselski input on the sufficient path of Theorem 44.6.1 includes that comparison; it is not discharged by the equality of opposite energy labels.

Chapter 47

Koszul–Petersson Pairing Compatibility

An arithmetic Koszul pairing can be compared with a Petersson form only after their domains have been related. The notation $\mathcal{V}^{\mathrm{prim}}$ denotes the Hochschild trace class of the identity on the proposed global arithmetic Iwahori–Hecke category of GL_2. A state realization of this class, its dual coefficient space, and its automorphic placement are additional constructions. Chapter 43 proves positivity conditional on those maps and the pairing identity.

The Segal and prime Hilbert models provide explicit positive forms. An identification with the arithmetic bar–cobar pairing requires the injective placement, local pairing maps and completed adelic extension of Hypothesis 67.1.1. Every arithmetic Koszul–Petersson comparison in this chapter retains these hypotheses. The prime cyclic GNS construction gives an actual local map; it does not supply the arithmetic isometry.

47.1 Statement Recalled

Setup

Write $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis}}$ for the chosen local tensor model

$$\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis}} \;=\; \mathcal{H}^{(\infty)} \;\otimes\; \bigotimes_{p}{}' \mathcal{H}^{(p)}, \tag{47.1}$$

with archimedean Heisenberg Fock space $\mathcal{H}^{(\infty)} = \mathbb{C}[\alpha_{-1}, \alpha_{-2}, \dots] \cdot |0\rangle_\infty$ and finite-place factors $\mathcal{H}^{(p)} = C_c(GL_2(\mathbb{Z}_p)\backslash GL_2(\mathbb{Q}_p)/GL_2(\mathbb{Z}_p))$ under convolution. Normalize Haar measure by $\mathrm{vol}(GL_2(\mathbb{Z}_p)) = 1$, and use $|0\rangle_p = \mathbf{1}_{GL_2(\mathbb{Z}_p)}$ as reference vector. The form $\int f(g)\overline{h(g)}\,dg$ is positive on this space and gives the reference vector norm one. The restricted tensor product uses these vectors, so its product form is positive by the finite-tensor argument of Chapter 43.

Any additional operators ψ_p must be specified on their actual local carriers. Pointed E_0-data in Definition 26.5.1 do not determine Hecke operations or a Bost–Connes representation. Equation (47.1) defines a tensor model. Its identification with a state realization of the arithmetic Hochschild class remains part of the comparison hypothesis.

An additional arithmetic dg algebra comparison would identify the Koszul-dual carrier with the Bost–Connes Q-lattice system

$$\big(\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis}}\big)^{!} \;=\; \mathbb{A}^{\mathrm{BC}}$$

as asked in Question 33.5.2. The Bost–Connes presentation alone does not provide that comparison.

Let τ^{BC} be the algebraic Bost–Connes anti-involution,

$$\tau^{\mathrm{BC}}(\mu_n) = \mu_n^*, \qquad \tau^{\mathrm{BC}}(e(\gamma)) = e(-\gamma),$$

the $*$-operation of the Bost–Connes C^*-algebra (Bost–Connes §1.2). On the prime cyclic GNS space at $\beta = 1$, for $m = p^k$, the modular lift satisfies

$$J\pi(\mu_m)\Omega = m^{1/2}\pi(\mu_m^*)\Omega.$$

This is Proposition 67.2.4 on its Hilbert–Schmidt carrier. In arithmetic pairing formulae, σ_p^{BC} denotes the separately specified antilinear map compatible with that model under the comparison hypothesis. An action on additional phase generators requires the chosen full state and its representation.

The Two Pairings

Under the specified arithmetic carrier and dual-coefficient comparisons, the Koszul evaluation takes the bilinear form

$$\langle \cdot, \cdot \rangle^{\mathrm{Koszul}} \colon \mathcal{V}_{\mathrm{Heis}}^{\mathrm{prim}} \otimes \mathbb{A}^{\mathrm{BC}} \longrightarrow \mathbb{C}$$

obtained by transporting the chosen bar–cobar evaluation. An adjunction $\Omega^{\mathrm{Ar}} \dashv B^{\mathrm{Ar}}$ alone does not identify its evaluation target with $\mathbb{C}$ or its dual coefficient space with $\mathbb{A}^{\mathrm{BC}}$. Those maps belong to the hypotheses. A self-pairing further requires a map from the conjugate state space to that dual coefficient space.

Let V denote the chosen state realization, with injective placement ι into $L^2([GL_2])$. The transported Petersson form is

$$\langle \cdot, \cdot \rangle^{\mathrm{Pet}} \colon V \otimes \overline{V} \longrightarrow \mathbb{C}, \quad \langle v, w \rangle^{\mathrm{Pet}} = \int_{[GL_2]} (\iota v)(\gamma)\overline{(\iota w)(\gamma)}\, d\gamma.$$

Here $d\gamma$ is the chosen Tamagawa measure on $Z(\mathbb{A})\, GL_2(\mathbb{Q})\backslash GL_2(\mathbb{A})/K_\infty$. An adelic Whittaker realization must be constructed and compared with the arithmetic carrier before it supplies such a placement.

The Identity

THEOREM 47.1.1 *(Koszul–Petersson compatibility, domain-restricted).* Assume the arithmetic placement, local pairing maps and completed adelic extension of Hypothesis 67.1.1 on the stated normalized core. *(domain-restricted to the normalized sph-fin cuspidal core; see Remark 47.1.2).* For all $v, w \in \mathcal{V}_{\text{sph-fin,cusp}}^{\mathrm{prim}}$,

$$\langle v, w \rangle^{\mathrm{Pet}} = \langle v, \sigma^{\mathrm{BC}}(w) \rangle^{\mathrm{Koszul}}.$$

Equivalently, the adelic Petersson pairing equals the Koszul pairing composed with $\mathrm{id} \otimes \sigma^{\mathrm{BC}}$ under the identification $(\mathcal{V}^{\mathrm{prim}})^! = \mathbb{A}^{\mathrm{BC}}$, on the normalized sph-fin cuspidal core.

Remark 47.1.2 *(domain restriction and open sub-items).* Theorem 47.1.1 is conditional on the arithmetic placement and pairing comparisons, including on the normalized sph-fin cuspidal core. The explicit constructions below give weighted occupation, prime Gibbs, spherical Whittaker and archimedean vacuum-line models. They do not identify the full arithmetic bar carrier with an automorphic representation. That identification, its compatibility with the local actions and conjugations, and its completed adelic extension are the remaining comparison maps in Hypothesis 67.1.1. Ramified factors and the Eisenstein continuous spectrum require their own carriers and comparison maps; the spherical calculation does not supply them.

47.2 Pairing Domains and Normalizations

The Koszul pairing is bilinear, while Petersson is conjugate-linear in its second variable. Composition with the specified antilinear map has the required sesquilinearity. The completed arithmetic carrier, its local actions and the Haar normalization must be preserved by the comparison maps. At almost every finite place they must preserve the selected reference vector, so their tensor products are defined on the restricted tensor domain.

The prime Gibbs state and its modular conjugation are constructed in Chapter 67. They are not an adelic Plancherel identification. Neither the existence of spherical vectors nor the equality of scalar Euler factors constructs the arithmetic pairing maps. The remaining comparison is Question 67.1.2.

47.3 Archimedean Identification

The Fock Inner Product

The archimedean factor $\mathcal{H}^{(\infty)} = \mathbb{C}[\alpha_{-1}, \alpha_{-2}, \ldots] \cdot |0\rangle_\infty$ is the Heisenberg vacuum Fock module. Its canonical Hermitian form is the *Segal inner product*: the unique positive-definite sesquilinear form $\langle \cdot\,, \cdot \rangle_{\mathrm{Segal}}$ on $\mathcal{H}^{(\infty)}$ satisfying

$$\alpha_{-n}^{\dagger} \;=\; \alpha_n, \qquad \langle |0\rangle_\infty, |0\rangle_\infty \rangle_{\mathrm{Segal}} \;=\; 1, \qquad \alpha_n |0\rangle_\infty \;=\; 0 \;\text{ for } n \geq 1,$$

together with the Heisenberg CCR $[\alpha_m, \alpha_n] = m\delta_{m+n,0}$ (Segal 1981 Commun. Math. Phys. 80; Frenkel–Ben-Zvi 2004 §3.5; Kac 1998 *Vertex Algebras for Beginners* §2.5 Prop. 2.2).

Proposition 47.3.1 *(Segal form on monomials).* For monomials $v_1 = \alpha_{-n_1} \cdots \alpha_{-n_k}|0\rangle_\infty$, $v_2 = \alpha_{-m_1} \cdots \alpha_{-m_l}|0\rangle_\infty$ with $n_1 \geq \cdots \geq n_k \geq 1$, $m_1 \geq \cdots \geq m_l \geq 1$,

$$\langle v_1, v_2 \rangle_{\mathrm{Segal}} \;=\; \delta_{k,l} \cdot \sum_{\sigma \in S_k} \prod_{i=1}^{k} n_i \, \delta_{n_i, m_{\sigma(i)}}.$$

Equivalently, Segal's form is the L^2-inner product of the Wick-normal-ordered fields.

Proof. Wick's theorem applied to a product of creation/annihilation operators equates $\langle v_1, v_2 \rangle_{\mathrm{Segal}} = \langle |0\rangle_\infty, \alpha_{n_1} \cdots \alpha_{n_k} \alpha_{-m_1} \cdots \alpha_{-m_l} |0\rangle_\infty \rangle$. Normal-ordering and using the CCR to collect the contractions gives exactly (47.3.1). See Kac 1998 §2.5 Prop. 2.2 or Frenkel–Ben-Zvi 2004 §3.5.4. □

Bilinear and Hermitian Fock Forms

On the algebraic Heisenberg Fock space, let J_∞ conjugate coefficients in the monomial basis. Define the bilinear form

$$B_\infty(v, w) = H_{\mathrm{Segal}}(v, J_\infty w).$$

Its Gram matrix on monomials is the real permutation sum of Proposition 47.3.1.

Proposition 47.3.2 *(The Fock conjugation identity).* The form extends continuously to the Segal Hilbert completion, and

$$B_\infty(v, J_\infty w) = H_{\mathrm{Segal}}(v, w). \tag{47.2}$$

Proof. The monomial Gram matrix is real, so coefficient conjugation is an isometric involution. Hence $|B_\infty(v, w)| \leq \|v\|\|w\|$, which gives the extension, and $J_\infty^2 = 1$ gives the identity. Identifying an arithmetic bar–cobar pairing with this form requires maps on its two actual carriers, preserving the pairing and its operator domains. □

COROLLARY 47.3.3 *(Archimedean conjugation type).* A bilinear arithmetic pairing identified with B_∞ becomes the Segal Hermitian form after the transported coefficient conjugation in its second slot. That conjugation is antilinear; it cannot be replaced by the complex-linear identity on all vectors.

Proof. Apply (47.2); replacing w by iw also directly distinguishes the two types. □

An Explicit Archimedean Whittaker Line

Fix the function $W_0(a) = |a|^{1/2} K_0(2\pi|a|)$ and the measure $da/|a|$ on $\mathbb{R}^\times$. This specifies a spherical Whittaker function in the spectral-parameter-zero model; a different spectral parameter has a different Bessel order. The Hermitian form is

$$H_{\mathrm{Whit},\infty}(W, W') = \int_{\mathbb{R}^\times} W(a)\overline{W'(a)}\frac{da}{|a|}.$$

LEMMA 47.3.4 *(The K_0 line norm).* For this precise coefficient normalization, $H_{\mathrm{Whit},\infty}(W_0, W_0) = \pi/4$.

Proof. The classical integral representation $K_0(x) = \int_0^\infty e^{-x\cosh u}\, du$, $x > 0$ (NIST DLMF (10.32.9)), and Tonelli give

$$\int_0^\infty K_0(x)^2\, dx = \int_0^\infty \int_0^\infty \frac{du\, dv}{\cosh u + \cosh v}.$$

Put $a = (u+v)/2$, $b = (u-v)/2$. The Jacobian is two, $a \geq 0$, $|b| \leq a$, and the denominator is $2\cosh a \cosh b$. If $F(a) = \int_0^a \operatorname{sech} b\, db$, the integral is $2\int_0^\infty F'(a)F(a)\, da = F(\infty)^2 = \pi^2/4$. Consequently $\|W_0\|^2 = 2\int_0^\infty K_0(2\pi a)^2\, da = \pi/4$. □

PROPOSITION 47.3.5 *(Normalized archimedean vacuum-line comparison).* Let Ω_∞ be the unit Segal vacuum. The map $U_\infty(a\Omega_\infty) = 2aW_0/\sqrt{\pi}$ is an isometry of complex lines, with

$$H_{\mathrm{Whit},\infty}(U_\infty v, U_\infty w) = H_{\mathrm{Segal}}(v, w) \qquad (v, w \in \mathbb{C}\Omega_\infty).$$

Proof. Both sides are $a\overline{b}$ for $v = a\Omega_\infty$, $w = b\Omega_\infty$, by the computed norm. An extension intertwining the full arithmetic or Heisenberg actions with a Whittaker representation is an additional construction. Irreducibility statements for different group actions do not produce such an intertwiner. □

47.4 FINITE-PLACE IDENTIFICATION

Bost–Connes Hilbert Pairing

For each prime p, define the weighted occupation model $H_{\mathrm{occ},p} = \ell^2(\mathbb{N}_0, p^{-j})$, with basis δ_j and Hermitian form

$$\langle v, w\rangle_p^{\mathrm{BC}} = \sum_{j\geq 0} p^{-j} v_j \overline{w_j}, \qquad \langle \delta_j, \delta_k\rangle_p^{\mathrm{BC}} = p^{-j}\delta_{jk}. \tag{47.3}$$

The raw shift $R\delta_j = \delta_{j+1}$ satisfies $\|Rv\| = p^{-1/2}\|v\|$; the normalized shift $p^{1/2}R$ is an isometry. This definition makes no identification with a spherical Hecke factor or its arithmetic action. Such an identification must specify the carrier and intertwining maps separately.

Lemma 47.4.1 *(the actual prime GNS carrier).* The prime shift Gibbs state at $\beta = 1$ has GNS carrier $\mathrm{HS}(\ell^2(\mathbb{N}_0))$, cyclic vector $\rho_p^{1/2}$, and left multiplication representation, where $\rho_p e_j = (1 - p^{-1})\, p^{-j} e_j$. For a specified global critical KMS state, this carrier is unitarily identified with its prime-generated cyclic subspace. The weighted occupation pairing (47.3) is not thereby identified with the whole GNS representation or an automorphic Plancherel space.

Proof. The matrix-unit construction and the state-preserving cyclic unitary are Lemma 67.2.2 and Theorem 67.2.5. A further identification of pairings requires the maps in Hypothesis 67.1.1. □

The Koszul Pairing at p

An arithmetic local bar–cobar pairing has its own two carriers:

$$B_p^{\mathrm{Ar}} : V_p \otimes V_p^! \longrightarrow \mathbb{C}.$$

Identifying either carrier with a Bost–Connes module or algebra is part of the arithmetic comparison, not a consequence of the following weighted Hilbert model.

Proposition 47.4.2 *(Bilinear form on the weighted occupation carrier).* On $H_{\mathrm{occ}} = \ell^2(\mathbb{N}_0, p^{-j})$, let J be coordinate conjugation and set

$$B_{\mathrm{occ}}(v, w) = \sum_{j \geq 0} p^{-j} v_j w_j, \qquad B_{\mathrm{occ}}(v, Jw) = H_{\mathrm{occ}}(v, w).$$

This is a continuous bilinear form. Under $D\delta_j = p^{-j/2} E_{jj}$, it equals $\mathrm{Tr}(DvDw)$.

Proof. Cauchy–Schwarz bounds the absolute series by $\|v\|\|w\|$. The diagonal trace is that same series, and inserting coordinate conjugation gives the Hermitian form. This constructs the model pairing. It does not identify the arithmetic bar–cobar dual with vectors $\mu_{p^j}^* \delta_1$: for $j > 0$, those vectors are zero in the occupation representation. Transport of B_p^{Ar} requires actual maps on both carriers and compatibility with their actions. □

Bost–Connes vs. Petersson at p

Fix an unramified tempered representation and the spherical Whittaker normalization $W(1) = 1$. The Kirillov Hermitian form is

$$H_{\mathrm{Whit}}(\phi, \phi') = \int_{\mathbb{Q}_p^\times} W_\phi(a) \overline{W_{\phi'}(a)}\, d^\times a.$$

Here $m_p = \mathrm{vol}(\mathbb{Z}_p^\times)$ fixes the measure. This local representation-theoretic form is transported to an arithmetic Petersson realization only by the specified placement maps.

Lemma 47.4.3 *(Spherical norm and the compact subgroup volume).* The spherical norm is

$$C_p = H_{\mathrm{Whit}}(W, W) = m_p (1 + p^{-1}) L_p(1, \mathrm{Ad}\, \pi_p).$$

It is finite, positive, and representation-dependent. In contrast, the compact subgroup volume for additive-matrix unit measure is $(1 - p^{-1})(1 - p^{-2})$.

Proof. Lemma 56.7.4 proves the norm formula by the convergent spherical shell series. The compact volume is computed by reduction modulo p: the invertible matrices form $(p^2-1)(p^2-p)$ of the p^4 residue classes. These are different calculations and different measures. An extra factor $|a|^{-1}$ in the integral would instead evaluate $Z_p(0)$; for $\alpha=\beta=1$, its shell series is $m_p\sum(j+1)^2$, which diverges. □

PROPOSITION 47.4.4 *(conditional finite-place pairing comparison).* Assume the local placement and pairing maps of Hypothesis 67.1.1, with common measure normalization. If they realize the two local forms on the prime Hilbert–Schmidt carrier by the bilinear trace form B_p and $JX=X^*$, then

$$\langle \iota_p v,\iota_p w\rangle_p^{\mathrm{Pet}} = B_p(c_p v, Jc_p w).$$

Proof. On Hilbert–Schmidt operators, $B_p(X,JY)=\mathrm{Tr}(XY^*)$, so the stipulated maps transport the same Hermitian form. The weighted occupation form has a concrete diagonal realization: for $m=p^j$, send its basis vector to $D_j=p^{-j/2}E_{jj}$. Then $\mathrm{Tr}(D_jD_k^*)=\delta_{jk}p^{-j}$ and $JD_j=D_j$. The radial-function isometry of Theorem 56.7.5 uses the chosen multiplicative Haar mass explicitly; the spherical line is normalized by its full positive norm C_p. This verifies the normalization on the specified model. The diagonal subspace is not invariant under left multiplication by V, and it is not the whole prime GNS representation. Its arithmetic or automorphic placement remains an assumption. □

COROLLARY 47.4.5 *(Local compatibility under actual pairing maps).* At each place, suppose maps on the two arithmetic pairing carriers identify B_p^{Ar} with the bilinear model and transport the specified conjugation to its model conjugation. Suppose also that the placed Petersson form is the corresponding Hermitian form. Then, on that common domain,

$$H_p^{\mathrm{Pet}}(v,w)=B_p^{\mathrm{Ar}}(v,\sigma_p w).$$

The archimedean placement is included among these data.

Proof. The model identity is $H(x,y)=B(x,Jy)$. Apply the supplied pairing maps and their conjugation compatibility. At finite primes the weighted diagonal calculation above supplies this identity on its stated model, without identifying the entire arithmetic carrier. □

47.5 ADELIC ASSEMBLY

Tate's Restricted Tensor Product

Definition 47.5.1 *(Normalized restricted tensor product of pairings).* Choose local Hermitian spaces (H_v,e_v) with unit reference vectors. On elementary tensors equal to these references outside a finite set, define

$$H(\otimes_v' x_v,\otimes_v' y_v)=\prod_v H_v(x_v,y_v).$$

Only finitely many factors differ from one. If raw local reference vectors have nonunit norms, first specify their normalization or provide a separate global continuation and scalar comparison. The compact subgroup volume is not the norm of every spherical vector.

LEMMA 47.5.2 *(Tamagawa regularisation of the adelic product).* The ordinary product of the spherical finite-place volumes for GL_n is not the Tamagawa volume before regularisation. For GL_n over $\mathbb{Q}$, $\mathrm{vol}(GL_n(\mathbb{Z}_p))=\prod_{k=1}^n(1-p^{-k})$, so the $k=1$ factor is the divergent Euler factor $\zeta(1)^{-1}$. The factors with $k\ge 2$ converge to $\prod_{k=2}^n\zeta(k)^{-1}$. The finite non-zero value $\tau(GL_2)=1$ is obtained only after the completed-zeta residue, the archimedean Haar factor, and the quotient by the rational character have been inserted.

Proof. Weil 1967 *Basic Number Theory* Ch. VII §3 computes $\mathrm{vol}([GL_n])$ for the Tamagawa normalisation. For GL_2, the Tamagawa number is $\tau(GL_2) = 1$ by Weil's theorem (Platonov–Rapinchuk 1994 Ch. 3 §1.3, Theorem 3.4 for split groups over $\mathbb{Q}$); the computation uses the completed zeta function and the different in an essential way, not merely a residue. The full Tamagawa formalism is classical: see Weil 1967 and the full modern account in Platonov–Rapinchuk *Algebraic Groups and Number Theory* (1994) Ch. 3. □

Assembly of Petersson and Koszul

Theorem 47.5.3 *(Restricted tensor assembly of normalized local forms).* Let (H_v, e_v) be Hilbert spaces with specified unit reference vectors, conjugations $J_v e_v = e_v$, and bilinear forms $B_v(x, J_v y) = H_v(x, y)$. Their algebraic restricted tensor product has forms, for elementary tensors,

$$H(\otimes'_v x_v, \otimes'_v y_v) = \prod_v H_v(x_v, y_v),$$
$$B(\otimes'_v x_v, \otimes'_v y_v) = \prod_v B_v(x_v, y_v).$$

They extend to the Hilbert completion, and $H(x, y) = B(x, Jy)$. Their arithmetic realization requires the actual local placements, compatible adelic maps and common global scalar of Hypothesis 67.1.1.

Proof. Only finitely many factors differ from the reference unit values, so these products are finite. At each finite stage the Hilbert tensor form is positive definite and the embeddings obtained by tensoring with reference vectors are isometries. Their directed union has a Hilbert completion. Tensor products of the conjugations are compatible isometries and extend to a conjugation J; define $B(x, y) = H(x, Jy)$, which is bounded and agrees with the finite products. This constructs the restricted tensor model. It does not identify an arithmetic completed pairing with that model or assign a value to an unnormalized infinite product at $s = 1$. □

Corollary 47.5.4 *(global compatibility via local compatibility).* Assume the local comparison maps and the completed adelic realization of Theorem 47.5.3, with the same specified global normalization for both forms. The global identity (47.1.1) follows from the local identities (47.4.5) by taking the restricted tensor product. Explicitly,

$$\langle v, w\rangle^{\mathrm{Pet}} = \prod_v \langle v_v, w_v\rangle_v^{\mathrm{Pet}} = \prod_v \langle v_v, \sigma_v^{\mathrm{BC}}(w_v)\rangle_v^{\mathrm{Koszul}} = \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}},$$

using (47.5.3), (47.4.5), and (47.5.3), with $\sigma^{\mathrm{BC}} = \otimes_v \sigma_v^{\mathrm{BC}}$ the adelic restricted tensor product involution.

47.6 The Bost–Connes Anti-Involution and its GNS Lift

Definition and Basic Properties

The Bost–Connes algebraic anti-involution τ^{BC} is the adelic extension of its local versions τ_p^{BC}. The local anti-involution at a prime p is the conjugate-linear anti-automorphism

$$\tau_p^{\mathrm{BC}} \colon \mathbb{A}_p^{\mathrm{BC}} \to \mathbb{A}_p^{\mathrm{BC}}, \quad \mu_m \mapsto \mu_m^*, \quad \mu_m^* \mapsto \mu_m, \quad e(\gamma) \mapsto e(-\gamma).$$

On the prime cyclic GNS space at $\beta = 1$, for $m = p^k$, the corresponding antiunitary is not τ_p^{BC} itself but

$$J_{\varphi_1, p}\pi(\mu_m)\Omega_{\varphi_1} = m^{1/2}\pi(\mu_m^*)\Omega_{\varphi_1}, \qquad J_{\varphi_1, p}\pi(\mu_m^*)\Omega_{\varphi_1} = m^{-1/2}\pi(\mu_m)\Omega_{\varphi_1}.$$

The factors are forced by $S_{\varphi_1} = J_{\varphi_1}\Delta_{\varphi_1}^{1/2}$. At the archimedean place, the required antilinear map is coefficient conjugation in the normalized Segal basis, transported through the specified arithmetic comparison. It is not the complex-linear identity: for the unit vacuum Ω, the Hermitian value $H(\Omega, i\Omega) = -i$ whereas the bilinear value $B(\Omega, i\Omega) = i$. The relation is $H(v, w) = B(v, J_\infty w)$.

LEMMA 47.6.1 *(properties of the local anti-involution and its lift).* The local Bost–Connes anti-involution satisfies:

- $\tau_p^{\mathrm{BC}} \circ \tau_p^{\mathrm{BC}} = \mathrm{id}$ (involutive; Bost–Connes §1.2).
- τ_p^{BC} is $\mathbb{C}$-antilinear (it is the $*$-conjugation of the C^*-structure).
- The GNS lift $J_{\varphi_1,p}$ preserves the Bost–Connes Hilbert pairing: $\langle J_{\varphi_1,p}v, J_{\varphi_1,p}w\rangle_p^{\mathrm{BC}} = \overline{\langle v, w\rangle_p^{\mathrm{BC}}}$.

Proof. The first two properties are immediate from the defining formulas (47.6): involutivity from $(\mu_m^*)^* = \mu_m$ and $e(-(-\gamma)) = e(\gamma)$; $\mathbb{C}$-antilinearity from the $*$-conjugation structure. The third is Tomita–Takesaki antiunitarity for the KMS_1 GNS representation, with the half-density factors displayed above (Bost–Connes §2.2 Thm 25; Connes–Marcolli arXiv:math/0404128 §1.8). □

Adelic Involution

Definition 47.6.2 *(Restricted tensor conjugation).* For local conjugations $J_v e_v = e_v$ on the specified normalized Hilbert models, put $J = \otimes_v' J_v$ on the algebraic restricted tensor product. It extends isometrically to its completion. At infinity use coefficient conjugation on the Segal carrier; at a prime Gibbs factor use its Hilbert–Schmidt adjoint conjugation. Transport to the arithmetic carrier requires its specified maps.

PROPOSITION 47.6.3 *(Transport of conjugation to an automorphic domain).* Suppose an isometric placement U of the completed model into an automorphic Hilbert space intertwines its local conjugations with pointwise complex conjugation on a dense restricted tensor domain. Then this intertwining extends to the completion:

$$U(Jv) = \overline{U(v)}.$$

Proof. Both sides define continuous antilinear maps. Their equality on the dense domain gives equality everywhere. The local arithmetic intertwiners are hypotheses here; inversion of Satake parameters does not construct them from the Bost–Connes algebra involution. □

47.7 SPHERICAL-VECTOR CALCULATION

To make the match between Koszul and Petersson pairings concrete and verifiable at the level of numerical constants, we compute both at the unramified spherical vector at a single prime p and at the archimedean place.

Spherical Petersson at p

LEMMA 47.7.1 *(Spherical Whittaker value).* For the spherical function normalized by $W(1) = 1$,

$$H_{\mathrm{Whit}}(W, W) = C_p = m_p(1 + p^{-1})L_p(1, \mathrm{Ad}\,\pi_p).$$

Proof. This is Lemma 47.4.3. □

The Normalized Model Line

Lemma 47.7.2 *(Unit reference vectors and bilinear pairing).* The occupation reference vector δ_0, the Gibbs vector $\Omega = \sqrt{\rho}$, and the normalized spherical vector $W/\sqrt{C_p}$ each have norm one. The specified line maps between them are isometries. On the occupation model,

$$B_{\mathrm{occ}}(\delta_0, J\delta_0) = 1.$$

An arithmetic Koszul vacuum has this value only when its comparison maps carry it to the specified model reference vector.

Proof. The occupation Gram matrix gives the first norm, $\operatorname{Tr}\rho = 1$ gives the second, and the preceding lemma gives the third. The displayed bilinear identity follows from the coordinate formula. □

Global Reference Normalization

Proposition 47.7.3 *(Reference value in the restricted tensor model).* For the normalized restricted tensor model of Theorem 47.5.3, the reference vector $e = \otimes'_v e_v$ satisfies

$$H(e,e) = B(e,Je) = 1.$$

If its arithmetic realization multiplies both forms by a specified scalar c, both reference values become c. The scalar and realization must be constructed for that arithmetic representation.

Proof. Every finite-stage reference vector has norm one and the embeddings are isometries. The identity therefore holds in the completion; common scalar multiplication preserves it. A cuspidal Petersson norm is not determined by the volume of the whole automorphic quotient. The Tamagawa number consequently does not fix this scalar. □

47.8 Analytic Continuation and Poles

Pole Structure of Both Pairings

For the specified unramified tempered spherical model, the local meromorphic function is

$$Z_p(s) = m_p \frac{L_p(s, \pi_p \times \widetilde{\pi}_p)}{\zeta_p(2s)}.$$

Its rank-four Rankin–Selberg factor is $\zeta_p(s)L_p(s, \operatorname{Ad}\pi_p)$; it is not the rank-two standard factor $(1-\alpha p^{-s})^{-1}(1-\beta p^{-s})^{-1}$. A meromorphic arithmetic bar–cobar pairing requires its own family of carriers, domains and compatible comparison maps.

Proposition 47.8.1 *(Pole matching under a meromorphic comparison).* Suppose two scalar pairings obtained from specified meromorphic families of vectors extend to meromorphic functions on a connected parameter domain. If the actual pairing comparison identifies them on a nonempty open subset, they agree meromorphically, including their pole orders and residues.

Proof. Their difference is meromorphic and vanishes on an open subset. The identity theorem, after clearing poles locally, makes it zero on the connected domain. Equal Laurent expansions give the pole and residue statement. Neither the local spherical formula nor a partition function constructs the assumed arithmetic family or its comparison. In particular no universal assertion that all such pairings have only the two Tate boundary poles follows. □

No Exotic Poles

Remark 47.8.2 *(The meromorphic arithmetic comparison).* Once the specified meromorphic pairing comparison is constructed, its common Laurent coefficients determine all residues. Without that comparison, neither the pole divisor nor its equality with a Petersson family is supplied by the local Hilbert model. The full parameter-dependent arithmetic pairing remains the construction required by Proposition 47.8.1.

47.9 The Main Theorem

THEOREM 47.9.1 *(Koszul–Petersson pairing compatibility, domain-restricted).* Assume the arithmetic placement, local pairing maps and completed adelic extension of Hypothesis 67.1.1 on the stated normalized core. *Conditional statement. (domain-restricted to the normalized sph-fin cuspidal core and using the Tamagawa regularisation of Lemma 47.5.2; see Remark 47.1.2).* The identity (47.1.1)

$$\langle v, w\rangle^{\mathrm{Pet}} \;=\; \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}}$$

holds for all $v, w \in \mathcal{V}^{\mathrm{prim}}_{\text{sph-fin,cusp}}$, the normalized sph-fin cuspidal core of $\mathcal{V}^{\mathrm{prim}}$. Equivalently, the arithmetic Koszul pairing of the sph-fin cuspidal core with $(\mathcal{V}^{\mathrm{prim}})^! = \mathbb{A}^{\mathrm{BC}}$ coincides, after Bost–Connes involution on the dual, with the adelic Petersson inner product on $L^2([GL_2])$ restricted to the image of the sph-fin cuspidal core. Full-sector compatibility and KMS_1/GNS identification beyond this core remain open (Remark 47.1.2).

Proof. For every place $v \le \infty$, Corollary 47.4.5 gives $\langle v_v, w_v\rangle_v^{\mathrm{Pet}} = \langle v_v, \sigma_v^{\mathrm{BC}}(w_v)\rangle_v^{\mathrm{Koszul}}$. Theorem 47.5.3 provides the restricted tensor product assembly of both sides. The adelic involution $\sigma^{\mathrm{BC}} = \bigotimes'_v \sigma_v^{\mathrm{BC}}$ is well-defined on the restricted tensor product (Definition 47.6.2). Taking the product over v of the local identities yields

$$\langle v, w\rangle^{\mathrm{Pet}} = \prod_v \langle v_v, w_v\rangle_v^{\mathrm{Pet}} = \prod_v \langle v_v, \sigma_v^{\mathrm{BC}}(w_v)\rangle_v^{\mathrm{Koszul}} = \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}},$$

as in Corollary 47.5.4.

The comparison is first defined on the stipulated normalized restricted-tensor core. Its extension to the completed arithmetic carrier is part of Hypothesis 67.1.1. The spherical-volume product has the Tamagawa residue normalization of Lemma 47.5.2; this does not assert ordinary absolute convergence at $s = 1$. Where a continued pairing is used, both its domain and compatibility with the comparison maps must be supplied. Thus the displayed equality follows from the specified local identities and their assumed adelic extension. □

Remark 47.9.2 *(local forms and their comparison maps).* The spherical calculation fixes the multiplicative Haar measure and computes the Whittaker norm. The weighted occupation and diagonal Hilbert–Schmidt models have explicit isometries; at infinity the normalized vacuum line has the computed K_0-Whittaker norm. These are comparisons of the stated models and lines. Neither Stone–von Neumann uniqueness nor a local functional equation constructs the arithmetic bar-to-automorphic map. Equality of the arithmetic pairings requires the placement, conjugation and adelic extension in Hypothesis 67.1.1.

47.10 Consequence: P3 on the Sph-Fin Cuspidal Core

COROLLARY 47.10.1 *(P3 unitarity on the sph-fin cuspidal core).* Assume the arithmetic placement, local pairing maps and completed adelic extension of Hypothesis 67.1.1 on the stated normalized core.

Conditional statement. (domain-restricted to the normalized sph-fin cuspidal core and to the Tamagawa-regularised pairing). On the normalized sph-fin cuspidal core $\mathcal{V}^{\mathrm{prim}}_{\text{sph-fin,cusp}}$, the primary arithmetic Heisenberg is a unitary arithmetic chiral algebra: it carries a positive-definite Hermitian form

$$\langle \cdot, \cdot \rangle^{\mathrm{Pet}} \colon \mathcal{V}^{\mathrm{prim}}_{\text{sph-fin,cusp}} \otimes \overline{\mathcal{V}^{\mathrm{prim}}_{\text{sph-fin,cusp}}} \to \mathbb{C}$$

given by the adelic Petersson–Tamagawa inner product, and this form is compatible with arithmetic Koszul self-duality via

$$\langle v, w \rangle^{\mathrm{Pet}} \;=\; \langle v, \sigma^{\mathrm{BC}}(w) \rangle^{\mathrm{Koszul}}.$$

Full-sector P3 is not a consequence of the sph-fin cuspidal identity: the prime KMS_1/GNS cyclic-subspace comparison is the operator-algebraic statement of Theorem 67.2.5, while the ramified Rankin–Selberg and Eisenstein terms are separate domain conditions (Remark 47.1.2).

Proof. Positive-definiteness of $\langle \cdot, \cdot \rangle^{\mathrm{Pet}}$ was established place by place in the proof of Theorem 43.7.1: at ∞ by Segal Fock positivity; at each prime p by Bost–Connes $\mathrm{KMS}_{\beta=1}$ positivity; adelic assembly by Tate 1950 restricted tensor product preserves positive-definiteness.

Compatibility with arithmetic Koszul self-duality is Theorem 47.9.1, which is established on the normalised sph-fin cuspidal core. Combining: $\mathcal{V}^{\mathrm{prim}}$ carries both a positive-definite Hermitian form (Petersson) and an arithmetic Koszul self-duality pairing (arithmetic bar–cobar), and the two are compatible via Bost–Connes involution. □

Remark 47.10.2 *(form restriction for P3).* Theorem 43.7.1 is conditional on Conjecture 43.7.2. Its conditional transport is Theorem 47.9.1, a domain-restricted and Tamagawa-regularised theorem on the normalized sph-fin cuspidal core. P3 retains the same arithmetic comparison hypotheses: on the sph-fin cuspidal core, $\mathcal{V}^{\mathrm{prim}}_{\text{sph-fin,cusp}}$ is unitary with Petersson inner product compatible with the Koszul pairing; full-sector P3 also requires the ramified Rankin–Selberg and Eisenstein terms recorded in the determinacy comparison residual Section 60.2.

Impact on the Sufficient RH Implication

Corollary 47.10.3 *(sufficient RH path, four-input conditional).* Under the normalization and comparison hypotheses of Theorem 47.9.1, the sufficient closure path to RH in the arithmetic formalism (Theorem 44.6.1) is the four-input conditional conjunction of Theorem 60.3.1:

(i) The arithmetic comparison includes the central-defect identity and detecting Mellin evaluation of Theorem 43.5.1. Its co–contra component retains the coefficient requirements of Remark 46.1.6.

(ii) Koszul–Petersson compatibility (Theorem 47.9.1): domain-restricted to the normalized sph-fin cuspidal core and conditional on the Tamagawa-regularised assembly; the full-sector extension has separate ramified Rankin–Selberg and Eisenstein components, while the KMS_1/GNS component is the prime cyclic-subspace statement of Theorem 67.2.5.

(iii) Arithmetic Hilbert–Pólya H-P^{Ar} (Hypothesis 44.3.2): the independent spectral-matching problem.

(iv) Arithmetic Virasoro positivity VB* main direction (Theorem 44.3.10): the classical positive-trace problem, equivalent to RH.

Proof. After the determinacy comparison the sufficient RH path carries four independent conditional inputs. The preceding theorem reduces the second to the sph-fin domain and the Tamagawa-regularised assembly; the other three remain on the four-input RH branch residual list (Section C.10). □

Remark 47.10.4 *(remaining arithmetic comparison problems).* The conditional pairing theorem leaves the following mathematical problems:

Arith. Positselski non-formal theorem (Iwasawa-theoretic Positselski).

VB* the Li positivity / arithmetic Virasoro bootstrap problem.

$\mathrm{L}_{\mathrm{ACD}}$ non-formal theorem (adelic categorical descent; not on RH path).

F2.c non-formal theorem (Tate ∞-categorical restricted tensor; not on RH path).

F3.BZN^{Ar} non-formal theorem (arithmetic Ben-Zvi–Nadler; not on RH path).

The pairing conclusion of Theorem 47.9.1 retains its arithmetic placement, local comparison, and adelic extension hypotheses.

47.11 Pairing Constructions and Arithmetic Comparison

The explicit constructions are the Segal bilinear/Hermitian identity, the normalized K_{o} vacuum-line isometry, the weighted occupation and diagonal Hilbert–Schmidt pairing, and the radial-function isometry of Theorem 56.7.5. For $W(1) = 1$, the finite spherical norm is $m_p(1 + p^{-1})L_p(1, \mathrm{Ad}\,\pi_p)$; unit normalization uses this full factor. It is not the compact subgroup volume.

The normalized restricted tensor construction is proved in Theorem 47.5.3. Its identification with the actual arithmetic completed pairing, its global scalar, and its action-compatible automorphic placement remain required data. The domain-restricted Koszul–Petersson theorem retains these hypotheses; restricting to a cuspidal sph-fin core does not construct them. Meromorphic pole matching additionally requires an actual parameter-dependent comparison on an open domain.

The prime KMS/GNS theorem identifies a prime-generated cyclic subspace, not the whole arithmetic or automorphic Hilbert space. Ramified, archimedean full-carrier, and continuous-spectrum comparisons remain part of the original pairing problem. The sufficient RH implication still retains all four inputs of Corollary 47.10.3.

Chapter 48

Adelic Categorical Descent

$\mathrm{L}_{\mathrm{ACD}}$ (Corollary 43.2.3) is the essential adelic gluing statement for conjecture F1 (global arithmetic Arinkin–Gaitsgory for $GL_2/\mathbb{Q}$), with a substantial closure problem (App. C). This chapter decomposes $\mathrm{L}_{\mathrm{ACD}}$ into four subconditions (L_1)–(L_4). The formal restricted tensor product (L_1) is constructed here. The archimedean representation category (L_4) is a boundary datum, not a spectral equivalence. (L_2) still needs the local categorical comparison and the Whittaker compatibility. (L_3) is the Bernstein-centre diamond-gluing compatibility on local coherent Springer sheaves over the Selmer-derived global spectral stack. The main result (Theorem 48.2.7) is a conditional reduction, not a proof of $\mathrm{L}_{\mathrm{ACD}}$.

48.1 Statement Recalled

The Condition

Corollary 43.2.3 states:

> $\mathrm{L}_{\mathrm{ACD}}$ **(adelic categorical descent).** The assignment
>
> $$v \longmapsto \mathrm{D}^{\text{sph-fin}}\big([GL_2(\mathbb{Q}_v)]\big)$$
>
> (for v ranging over places of $\mathbb{Q}$, finite and archimedean) forms a Tate-restricted tensor product whose colimit is the global automorphic category $\mathrm{D}([GL_2])^{\text{sph-fin}}$; this restricted tensor product is compatible with Hecke-functoriality and with the local Fargues–Scholze spectral action together with the still-open F1.a local categorical comparison, so that
>
> $$\bigotimes_v{}' \mathrm{D}^{\text{sph-fin}}\big([GL_2(\mathbb{Q}_v)]\big) \;\simeq\; \mathrm{D}\big([GL_2]\big)^{\text{sph-fin}}$$
>
> as Hecke-equivariant stable presentable ∞-categories.

The restricted tensor product $\bigotimes'_v$ follows Tate 1950 / Weil 1967 at the level of spaces of automorphic forms. The canonical ∞-categorical definition is the filtered-colimit construction of Definition 50.1.2. The display below is mnemonic:

$$\bigotimes_v{}' C_v \;:=\; \varinjlim_{S \text{ finite}} \bigotimes_{v \in S} C_v \otimes \bigotimes_{v \notin S} \mathbb{1}_{u_v}, \tag{48.1}$$

where $u_v \in C_v$ is the distinguished unit (the unit of the unramified spherical Hecke category) at almost every finite place.

Context and Significance

$\mathrm{L}_{\mathrm{ACD}}$ is the essential adelic gluing statement for F1 (global arithmetic Arinkin–Gaitsgory, Conjecture 42.1.1). The structural reduction of F1 in Theorem 43.2.1 splits F1 into four sub-items (F1.a)–(F1.d). Only (F1.c) is a theorem drawn directly from primary literature (Emerton–Gee arXiv:1908.07185). Fargues–Scholze arXiv:2102.13459 supplies local geometrisation and spectral action, not the categorical equivalence required by (F1.a). The remaining items (F1.a), (F1.b), and (F1.d) are equivalent to the local comparison plus $\mathrm{L}_{\mathrm{ACD}}$ conditions whose closure removes F1.

$\mathrm{L}_{\mathrm{ACD}}$ is a **conjectural, non-formal theorem**. This chapter does not close $\mathrm{L}_{\mathrm{ACD}}$; it reduces it to a narrower and precisely-stated compatibility on local coherent Springer sheaves (Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321) glued across primes on the Selmer-derived global spectral stack.

48.2 Modular Reduction of $\mathrm{L}_{\mathrm{ACD}}$

Six independent lines of obstruction on $\mathrm{L}_{\mathrm{ACD}}$ are given below. Each line produces a structural decomposition, a partial closure, or a named residual, following the schema of Section 43.1.

Modular Decomposition of $\mathrm{L}_{\mathrm{ACD}}$

$\mathrm{L}_{\mathrm{ACD}}$ is the conjunction of four ingredients: (i) well-definedness of the restricted tensor product of local ∞-categories with distinguished units; (ii) Hecke-functoriality across places; (iii) compatibility of the Bernstein centre action across places under restricted tensor; (iv) archimedean-finite assembly (extension across the archimedean place). A structural closure requires separating each ingredient. The decomposition admits four disjoint subconditions, (L_1)–(L_4), each a precise statement at the ∞-categorical level.

THEOREM 48.2.1 *(modular decomposition of* $\mathrm{L}_{\mathrm{ACD}}$*).* $\mathrm{L}_{\mathrm{ACD}}$ contains the following four subconditions:

- (L_1) *Well-definedness of the restricted tensor product.* The assignment $v \mapsto C_v$ with $C_v = \mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ and distinguished units u_v (spherical vectors) defines a well-formed restricted tensor product $\otimes'_v C_v$ in the ∞-category $\mathrm{Pr}^{\mathrm{L,st}}$ of stable presentable ∞-categories.
- (L_2) *Hecke-functoriality lift across places.* For each finite set S of places, assume the local categorical comparison functors $C_v \to \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},v}_{GL_2})$ for the finite places $v \in S$. Their tensor product is required to give an equivalence $\otimes_{v\in S} C_v \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\prod_{v\in S} \mathcal{X}^{\mathrm{EG},v}_{GL_2})$ compatible with Hecke operators at distinct primes.
- (L_3) *Bernstein-centre diamond gluing.* The Bernstein-centre action $\mathcal{Z}_v \curvearrowright C_v$ (the centre of the local categorical Hecke algebra) extends to a global action $\mathcal{Z}_{\mathrm{glob}} \curvearrowright \otimes'_v C_v$ making the tensor product restricted-tensor-equivariant under the Bernstein centre.
- (L_4) *Archimedean extension.* The archimedean factor $C_\infty = \mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{R})])$ is the ind-completed derived category of admissible $(\mathfrak{gl}_2, O(2))$-modules of finite K-type. It carries the trivial K-spherical object as distinguished unit, and the restricted tensor product $\otimes'_v C_v$ extends the finite-place product $\otimes'_{p<\infty} C_p$ by tensoring with C_∞.

$\mathrm{L}_{\mathrm{ACD}}$ holds if (L_1) through (L_4) hold together with the identification of the resulting object with the global automorphic category $\mathrm{D}([GL_2])^{\text{sph-fin}}$.

Proof. Forward direction: assume L_{ACD}. The restricted tensor product (48.1) is well-defined by definition, giving (L_1). Hecke-functoriality across places is part of the L_{ACD} hypothesis, giving (L_2). Bernstein-centre compatibility follows from functoriality of the global automorphic category under the global Bernstein centre plus Hecke-functoriality, giving (L_3). Archimedean extension is explicit in the definition of L_{ACD} (the index v ranges over all places), giving (L_4).

Converse direction: assume (L_1)–(L_4) and the global identification just stated. By (L_1) and (L_4), $\otimes'_v C_v$ is a well-defined stable presentable ∞-category. By (L_2) and (L_3), this category is Hecke-equivariant and Bernstein-centre-equivariant. The identification with the global automorphic category follows from the assumed local-global compatibility. No theorem of Bernstein, Raskin, or geometric Langlands supplies this number-field identification without that input. □

Remark 48.2.2 *(form of the descent subconditions).* The four descent subconditions have the following mathematical forms:

(L_1) Theorem, closed in this chapter (§48.3).

(L_2) Conditional; reduces to the local categorical comparison plus Whittaker and Bernstein-centre compatibility (§48.4).

(L_3) Conjectural; the residual content of L_{ACD} after the decomposition of Theorem 48.2.1 (§48.5).

(L_4) Boundary datum; Jacquet–Langlands and Vogan define the archimedean representation category, not an Emerton–Gee spectral equivalence (§48.6).

Curve Precedent and Transport Obstructions

For a smooth complete curve over an algebraically closed field of characteristic zero, the de Rham geometric Langlands theorem gives the model for the categorical descent pattern. The sheaf-function analogue over finite fields is a different theorem. Neither theorem transports directly to the number-field case. The obstructions are:

(i) No archimedean place in the curve setting. Geometric Langlands uses the closed points of a curve. Number-field L_{ACD} requires the archimedean extension (L_4), which has no curve analogue.

(ii) Emerton–Gee integrality. The number-field spectral side uses the Emerton–Gee stack arXiv:1908.07185, which encodes integrality of Galois representations; de Rham geometric Langlands uses the stack $\mathrm{LocSys}_G^{\mathrm{dR}}(X)$, which is smoother and has weaker integrality content. The Bernstein-centre compatibility (L_3) is strictly more subtle in the number-field setting because the Emerton–Gee stack is not smooth over $\mathrm{Spec}\,\mathbb{Z}$.

(iii) Perfectoid diamonds. Fargues–Scholze's local theorem uses the Fargues–Fontaine curve and perfectoid diamonds; geometric Langlands uses ind-schemes. Gluing across finite places in the number-field case requires comparing perfectoid diamonds at distinct primes, a non-issue in the de Rham curve setting.

Theorem 48.2.3 *(curve transport, qualified). Comparison hypothesis.* The Gaitsgory–Rozenblyum–Raskin–Lin / AGKRRV proof of de Rham geometric Langlands gives the model for subconditions (L_1) and (L_2) as an analogy, not as a transport theorem (Raskin, "Chiral principal series categories I", Adv. Math. 388 (2021); Raskin, "D-modules on infinite-dimensional varieties", available as `samraskin.net/dmod.pdf`). It closes (L_1) only after the number-field local categories and distinguished units are fixed. It does not close (L_2) without the local categorical comparison of (F1.a) and (BC-diamond-glue). It does *not* close (L_3) (Bernstein centre compatibility is strictly more subtle in the

number-field setting due to Emerton–Gee stack non-smoothness) or (L_4) (no archimedean analogue on the curve).

Proof. (L_1): Raskin 2021 JAMS constructs restricted tensor products of local principal-series categories over a curve. The construction is ∞-categorical and uses only (a) stability, (b) presentability, (c) existence of distinguished units. These properties can be imposed on the number-field local categories. This gives the formal construction of (L_1), not the global automorphic identification.

(L_2): AGKRRV's Hecke-functoriality argument combines local geometric Langlands equivalences with Bernstein–Zelevinsky factorization on a curve. For number fields, Fargues–Scholze arXiv:2102.13459 supplies the local diamond geometrisation and spectral action. It does not supply the displayed categorical comparison with the Emerton–Gee stack. The Bernstein–Zelevinsky pattern applies only after the local comparison and Whittaker-model compatibility are constructed. Domain: §48.4.

(L_3): Number-field obstruction. The Bernstein centre $\mathcal{Z}^{\mathrm{Ar}}$ of the global arithmetic Iwahori Hecke category acts on $\otimes'_v C_v$ via the local Bernstein centres $\mathcal{Z}_v$, but compatibility across primes involves comparing Bernstein parameters at distinct primes through the Emerton–Gee stack. The curve analogue uses the de Rham stack, which is smoother. The number-field compatibility is the residual.

(L_4): No curve analogue. □

Sphericity-Finiteness Preserved Under Restricted Tensor

The "sph-fin" cut in $\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ restricts to categories with finite-type spherical sub-structure, excluding the continuous Eisenstein spectrum and broad classes of non-tempered representations. The condition is preserved under the restricted tensor product.

LEMMA 48.2.4 *(sph-fin preservation under restricted tensor).* Let $\{C_v\}$ be a family of stable presentable ∞-categories with distinguished units u_v at almost every v, and assume each C_v is "sph-fin" in the sense that: (i) C_v has a compact subcategory C_v^ω whose compact objects are finite-type over the local Hecke algebra $\mathcal{H}_v$; (ii) u_v is compact in C_v^ω; (iii) the compact objects generate C_v. Then the restricted tensor product $\otimes'_v C_v$ is likewise sph-fin.

Proof. By Lurie HA 5.3 (higher Deligne centralizers) and HA 4.8 (tensor products of presentable ∞-categories), the tensor product $\otimes_{v\in S} C_v$ over a finite set S inherits compact generation from its finite factors: a family of compact generators is given by tensor products of local compact generators. The distinguished units u_v at $v \notin S$ are compact, so tensoring with them preserves compactness. Taking the filtered colimit over S preserves compact generation (Lurie HA 5.3.2.9 on filtered colimits of compactly generated categories). Hence $\otimes'_v C_v$ is compactly generated, with compact subcategory the filtered colimit of finite tensor products of local compacts. The sph-fin condition is preserved. □

Cross-Place Hecke Commutativity

Commutativity of T_p and T_q at the level of cohomology is automatic; categorical commutativity in the ∞-setting is a coherence datum following from the E_2-monoidal structure on each local Hecke category (assuming F2). Cross-place compatibility requires showing that the global automorphic category decomposes as a restricted tensor product whose factor at each prime is the local Hecke category.

PROPOSITION 48.2.5 *(cross-place Hecke commutativity).* In the restricted tensor product $\otimes'_v C_v$, the local Hecke operators $T_p : C_p \to C_p$ extended by identity on factors at $v \neq p$ (Tate extension) commute: for distinct primes $p \neq q$,

$$(T_p \otimes \mathrm{id}_{\neq p}) \circ (T_q \otimes \mathrm{id}_{\neq q}) \;=\; (T_q \otimes \mathrm{id}_{\neq q}) \circ (T_p \otimes \mathrm{id}_{\neq p})$$

as endofunctors of $\bigotimes'_v C_v$, up to canonical homotopy.

Proof. Each T_p acts on the C_p-factor and by identity on all other factors. The universal property of the tensor product in $\mathrm{Pr}^{\mathrm{L,st}}$ (Lurie HA 4.8.1) gives that endofunctors factoring through disjoint tensor factors commute up to canonical homotopy. Since T_p factors through C_p and T_q through C_q, their extensions to the restricted tensor product commute. □

Non-Transference from Caraiani–Scholze

Caraiani–Scholze arXiv:1511.02418 prove a unitary Shimura variety analogue of $\mathrm{L}_{\mathrm{ACD}}$ for signature $(1, n-1)$ unitary groups $G = U(1, n-1)$ over CM fields. A Jacquet–Langlands transfer from $GL_2/\mathbb{Q}$ to $U(1,1)$ over an imaginary quadratic field $F/\mathbb{Q}$ would reduce the problem to the Caraiani–Scholze case; this reduction fails. Caraiani–Scholze's proof uses three inputs specific to unitary Shimura varieties:

(i) Calegari–Geraghty patching arXiv:1207.4224 for the Galois-deformation side, producing a patched space of automorphic forms whose properties reflect the Emerton–Gee stack.

(ii) The Kottwitz conjecture on cohomology of Shimura varieties in its form for unitary groups (Caraiani–Scholze 2017 §4).

(iii) Ihara-avoidance hypothesis on the existence of adequate Taylor–Wiles primes, required for patching.

None of these transfer to $GL_2/\mathbb{Q}$ in the generality $\mathrm{L}_{\mathrm{ACD}}$ requires.

Observation 48.2.6 *(non-transference of Caraiani–Scholze).* $GL_2/\mathbb{Q}$ has a Shimura variety over $\mathbb{Q}$ in the form of the modular curve $Y(\Gamma(N))$, but the Shimura variety is 1-dimensional, whereas Caraiani–Scholze's Ihara-avoidance argument requires the Shimura variety to be higher-dimensional (specifically $\geq n-1$ where $n = \mathrm{rk}\, G$). For $GL_2/\mathbb{Q}$, the Ihara-avoidance input is essentially degenerate. Caraiani–Scholze's proof cannot be transferred.

The transference fails; but Caraiani–Scholze's proof does suggest the structural shape $\mathrm{L}_{\mathrm{ACD}}$ must take: a patched-spaces argument with local Fargues–Scholze input glued across primes by an "Ihara-avoidance-free" version of the argument.

Reduction to a Single Compatibility

(L_1) is formal in presentable categories. (L_4) supplies the archimedean representation-theoretic boundary datum. (L_2) reduces to the local categorical comparison plus Whittaker-model compatibility; the remaining central compatibility is (L_3): Bernstein-centre diamond gluing. $\mathrm{L}_{\mathrm{ACD}}$ therefore reduces to the single compatibility stated below.

Theorem 48.2.7 *(conditional reduction of* $\mathrm{L}_{\mathrm{ACD}}$*). Conditional statement.* Assuming the local categorical comparison at all finite primes and the archimedean boundary datum, $\mathrm{L}_{\mathrm{ACD}}$ reduces to the following compatibility:

(BC-diamond-glue) *Bernstein-centre diamond gluing.* For each pair of primes $p \neq q$, the local Bernstein centres $\mathcal{Z}_p$ and $\mathcal{Z}_q$ (the centres of the local categorical Hecke algebras on the perfectoid Fargues–Fontaine curves at p and q) admit a common global refinement $\mathcal{Z}_{\{p,q\}}$ such that:

(a) $\mathcal{Z}_{\{p,q\}}$ embeds into $\mathcal{Z}_p \otimes \mathcal{Z}_q$ under the external tensor action on $C_p \otimes C_q$;

(b) the embedding is compatible with restriction to smaller prime sets $S \subsetneq \{p, q\}$;

(c) the inductive limit $\mathcal{Z}_{\mathrm{glob}} := \varinjlim_S \mathcal{Z}_S$ over finite sets S of primes agrees with the global Bernstein centre acting on the global automorphic category.

Proof. Forward direction: assume $\mathrm{L}_{\mathrm{ACD}}$. The global automorphic category $\mathrm{D}([GL_2])^{\text{sph-fin}}$ carries a global Bernstein-centre action. The restricted tensor product structure of $\mathrm{L}_{\mathrm{ACD}}$ expresses this global action as a filtered colimit of finite products of local Bernstein centres. Compatibilities (a)–(c) follow.

Reverse direction: assume (BC-diamond-glue), the local categorical comparisons, and the archimedean boundary datum. Then by Theorem 48.2.1, it suffices to close (L_1), (L_2), (L_3), (L_4). (L_1) closes in §48.3. (L_2) is supplied by the assumed local comparisons, up to local-global compatibility of Whittaker models, which (BC-diamond-glue) supplies via compatibility (b). (L_3) closes from (BC-diamond-glue) directly. (L_4) is the assumed archimedean boundary datum. Assembling, $\mathrm{L}_{\mathrm{ACD}}$ holds. □

Remark 48.2.8 *(domain of (BC-diamond-glue)).* (BC-diamond-glue) is strictly narrower than $\mathrm{L}_{\mathrm{ACD}}$. It is a single compatibility between local Bernstein centres and the global Bernstein centre. Partial change: Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 ("Coherent Springer theory and the categorical Deligne–Langlands correspondence") constructs the local Bernstein centres as coherent sheaves on the local Emerton–Gee stacks. (BC-diamond-glue) asks that these coherent sheaves glue across primes to a global coherent sheaf on the Selmer-derived global spectral stack $\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}$, agreeing with the global Bernstein centre action at the level of categorical trace. This is a gluing and centre-comparison problem. The reduction in Theorem 48.2.7 also retains the local categorical comparison, archimedean boundary datum and global identification; it does not construct these maps.

48.3 Subcondition (L_1): Restricted Tensor in ∞-Categories

Definition

Definition 48.3.1 *(restricted tensor product in* $\mathrm{Pr}^{\mathrm{L,st}}$*).* The canonical definition is Definition 50.1.2; the following formula is its older mnemonic form in the ACD notation. Let V be a countable indexing set, $\{C_v\}_{v\in V}$ a family of stable presentable ∞-categories, and let $u_v \in C_v$ be a distinguished unit at almost every v (i.e. at all but finitely many v in a fixed cofinite subset $V^{\mathrm{unr}} \subseteq V$). The *Tate-restricted tensor product* is

$$\bigotimes_{v\in V}{}' C_v \;:=\; \varinjlim_{S \text{ finite},\, V^{\mathrm{unr}}\supseteq V\setminus S} \;\bigotimes_{v\in S} C_v \otimes \bigotimes_{v\in V^{\mathrm{unr}}\setminus S} \mathbb{1}_{u_v},$$

where $\mathbb{1}_{u_v}$ denotes the unit ∞-category generated by u_v (a full subcategory of C_v).

THEOREM 48.3.2 *((L_1) closed).* The restricted tensor product of Definition 48.3.1 is a well-defined stable presentable ∞-category. Specifically:

(i) The filtered colimit $\varinjlim_S$ exists in $\mathrm{Pr}^{\mathrm{L,st}}$.

(ii) The resulting object is stable, presentable, and compactly-generated if each C_v is compactly-generated (Raskin, "D-modules on infinite-dimensional varieties", `samraskin.net/dmod.pdf`; Lurie HA 5.3.2).

(iii) Sphericity-finiteness is preserved (Lemma 48.2.4).

(iv) The restricted tensor product is functorial in the family $\{C_v, u_v\}$ under morphisms that preserve distinguished units.

Proof. **(i) Existence of the filtered colimit.** The ∞-category $\mathrm{Pr}^{\mathrm{L,st}}$ of stable presentable ∞-categories is a (large) ∞-category admitting all small colimits (Lurie HA 4.8.1.15). Filtered colimits exist.

(ii) Presentability and compact generation. Lurie HA 5.3.2.9 asserts that a filtered colimit of compactly-generated stable presentable ∞-categories, under functors preserving compact objects, is compactly generated. The transition maps in the defining filtered colimit are of the form "tensor with $\mathbb{1}_{u_v}$" for newly-included places v, which are left-adjoint (and preserve compact objects because the unit u_v is assumed compact, as in Lemma 48.2.4). Hence the colimit is compactly generated.

(iii) Stability. Filtered colimits of stable ∞-categories in $\mathrm{Pr}^{\mathrm{L,st}}$ are stable (Lurie HA 1.1.4.4: stable ∞-categories are closed under filtered colimits in Pr^{L}).

(iv) Sphericity-finiteness preservation. Lemma 48.2.4.

(v) Functoriality. A morphism of families $\{(C_v, u_v)\} \to \{(C'_v, u'_v)\}$ is a family of left adjoints $F_v : C_v \to C'_v$ sending u_v to u'_v. The induced map $\otimes_{v\in S} F_v : \otimes_{v\in S} C_v \to \otimes_{v\in S} C'_v$ is compatible with unit-extension to larger $S' \supseteq S$ because F_v sends distinguished units to distinguished units. Taking the filtered colimit yields the induced map $\otimes'_v F_v : \otimes'_v C_v \to \otimes'_v C'_v$. The construction is independent of the choice of cofinite unramified subset V^{unr} (any two choices agree on a further cofinite subset, and the filtered colimit is insensitive to such shrinking). □

Remark 48.3.3 *(comparison with Raskin's construction).* Raskin "Chiral principal series categories I", Adv. Math. 388 (2021), 107856, and Raskin "D-modules on infinite-dimensional varieties" (`samraskin.net/dmod.pdf`) construct restricted tensor products of principal-series categories over a smooth curve. The construction uses the same filtered colimit pattern with spherical vectors as distinguished units. The number-field analogue is not a direct transport. The same filtered colimit defines the formal restricted tensor product once the local categories and distinguished units are fixed. The non-trivial content is the identification with the global automorphic category, which is (L_2) and (L_3).

Corollary 48.3.4 *((L_1) holds).* (L_1) is a theorem, closed by Theorem 48.3.2 plus Lemma 48.2.4.

48.4 Subcondition (L_2): Hecke-Functoriality Across Places

Statement and Reduction

Theorem 48.4.1 *((L_2) reduction). Comparison hypothesis.* For each finite set S of primes, the Fargues–Scholze local geometry, after the local categorical comparison (F1.a), supplies comparison functors

$$\Phi_v^{\mathrm{loc}} \;:\; C_v \longrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{EG},v}_{GL_2}\big), \qquad v \in S$$

which glue to an equivalence

$$\Phi_S^{\mathrm{loc}} \;:\; \bigotimes_{v\in S} C_v \;\xrightarrow{\sim}\; \mathrm{IndCoh}_{\mathrm{Nilp}}\Big(\prod_{v\in S} \mathcal{X}^{\mathrm{EG},v}_{GL_2}\Big) \tag{48.2}$$

compatible with Hecke operators T_v at distinct primes. The open content is the local categorical comparison plus the (BC-diamond-glue) Whittaker/Bernstein-centre compatibility.

Proof. **Step 1: Local geometry exists.** At each prime p, Fargues–Scholze arXiv:2102.13459 constructs local geometrization, spectral action, and Fargues–Fontaine geometry. It does not by itself identify C_p with $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p}_{GL_2})$. That identification is the local comparison input (F1.a). At the archimedean place, the analogue is Jacquet–Langlands SLN 114 (see §48.6).

Step 2: Hecke-functoriality at each place. Local Fargues–Scholze includes Hecke-functoriality at each prime: the local Hecke algebra $\mathcal{H}_p$ acts on both sides of Φ_p^{loc} once the local comparison is supplied, and Φ_p^{loc} must be $\mathcal{H}_p$-equivariant.

Step 3: Tensor product of equivalences. Lurie HA 4.8.1 states that the tensor product in $\mathrm{Pr}^{\mathrm{L,st}}$ is functorial: a family of equivalences $\Phi_v : C_v \to \mathcal{D}_v$ produces an equivalence $\bigotimes_{v\in S} \Phi_v : \bigotimes_{v\in S} C_v \to \bigotimes_{v\in S} \mathcal{D}_v$. Applied to Φ_v^{loc} (whose existence is (F1.a)), this gives the equivalence Φ_S^{loc}.

Step 4: Tensor compatibility with Hecke. The tensor product of $\mathcal{H}_v$-equivariant functors is $\mathcal{H}_S := \otimes_{v\in S}\mathcal{H}_v$-equivariant in the appropriate sense. Hecke-functoriality across places follows from cross-place commutativity (Proposition 48.2.5).

Step 5: Spectral side decomposition. The spectral side $\mathrm{IndCoh}_{\mathrm{Nilp}}(\prod_{v\in S} \mathcal{X}_{GL_2}^{\mathrm{EG},v})$ decomposes as a tensor product of local factors because $\prod_{v\in S} \mathcal{X}_{GL_2}^{\mathrm{EG},v}$ is a product of derived formal stacks (Gaitsgory–Rozenblyum's AMS DAG monographs give continuity of $\mathrm{IndCoh}_{\mathrm{Nilp}}$ under products of derived stacks). Hence the right-hand side of (48.2) agrees with $\bigotimes_{v\in S} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},v})$.

Step 6: Whittaker-model compatibility. The cross-place compatibility requires checking that the Whittaker model on the automorphic side, which factorizes as a tensor product over places, matches the corresponding factorization on the spectral side. This factorization is where (BC-diamond-glue) enters: the Whittaker-model factorization is a compatibility between local Bernstein centres, exactly what (BC-diamond-glue) asserts via compatibility (b) of Theorem 48.2.7.

Thus (L_2) is not closed here. It is reduced to the local categorical comparison (F1.a) plus (BC-diamond-glue). □

Remark 48.4.2 *(finite-prime (L_2) is conditional).* For the subset of conditions involving only finite primes (no archimedean place), the tensor-product formalism is unconditional once the local categorical comparisons are supplied. Fargues–Scholze arXiv:2102.13459, Lurie HA 4.8.1, and Gaitsgory–Rozenblyum supply the local geometry, tensor functoriality, and product-continuity of $\mathrm{IndCoh}_{\mathrm{Nilp}}$. They do not prove the local comparison with the Emerton–Gee stack, nor the Whittaker compatibility encoded by (BC-diamond-glue).

48.5 Subcondition (L_3): Bernstein Centre Compatibility

Statement and Residual Content

THEOREM 48.5.1 *((L_3) is (BC-diamond-glue)).* (L_3) is equivalent to the (BC-diamond-glue) compatibility of Theorem 48.2.7.

Proof. Forward: (L_3) asserts that the local Bernstein-centre actions assemble to a global action on $\otimes'_v C_v$ compatible with restricted tensor. The global Bernstein centre $\mathcal{Z}_{\mathrm{glob}}$ must embed into the colimit $\varinjlim_S \otimes_{v\in S}\mathcal{Z}_v$; this is precisely compatibility (c) of (BC-diamond-glue).

Reverse: given (BC-diamond-glue), conditions (a), (b), (c) supply exactly the data required to construct the global action $\mathcal{Z}_{\mathrm{glob}} \curvearrowright \otimes'_v C_v$ as a filtered colimit of finite-S actions, with the compatibility required by (L_3). □

Transport of a Coherent Springer Object

THEOREM 48.5.2 *(A supplied local transport).* Let $\mathcal{S}_p$ be a coherent Springer object on its specified parameter stack Y_p, with a specified action of the local centre Z_p. Suppose an exact functor $T_p : \mathrm{Coh}(Y_p) \to \mathrm{Coh}(\mathcal{X}_p^{\mathrm{EG}})$ is supplied over compatible coefficients and preserves the required support and centre action. Then $\mathcal{F}_p = T_p\mathcal{S}_p$ is a coherent object with the transported Z_p-action.

Proof. The functor preserves coherence by its declared target, and its action on endomorphisms preserves composition and identities. The specified intertwining identifies the resulting action with the required local one. For a derived action the same assertion uses the supplied homotopy-coherent algebra map. □

Remark 48.5.3 *(The local and global comparison maps).* The coherent Springer construction of Ben-Zvi–Chen–Helm–Nadler is the source object on its own parameter stack. The functor T_p, the coefficient comparison and the arithmetic support condition are additional data; the theorem does not identify the parameter stack with the Emerton–Gee stack. Across primes, a common carrier, compatible transitions and effective coherent descent must be constructed. Agreement with the global arithmetic centre requires an equivariant comparison, beyond an equivalence of underlying categories.

Conjecture 48.5.4 *(BC-diamond-glue, concrete form).* Construct the local transports of Theorem 48.5.2, a common Selmer-derived carrier and compatible transition maps with effective coherent descent, so that the objects $\{\mathcal{F}_v\}_v$ descend to a global coherent sheaf $\mathcal{F}_{\mathrm{glob}}$ whose action on $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}})$ agrees, under the conjectural global arithmetic AG equivalence, with the action of the global Bernstein centre on $\mathrm{D}([GL_2])^{\text{sph-fin}}$.

Remark 48.5.5 *(domain of the residual).* Conjecture 48.5.4 is the residual content of $\mathrm{L}_{\mathrm{ACD}}$ after the reduction of Theorem 48.2.7. Its domain:

- Gluing of coherent sheaves across a product of stacks: non-formal theorem, standard techniques.
- Agreement with global Bernstein centre under conjectural global AG: non-formal theorem, requires careful adaptation of the geometric-Langlands Bernstein-centre construction (AGKRRV) to the number-field setting.
- Total condition of (BC-diamond-glue): non-formal theorem.

The full $\mathrm{L}_{\mathrm{ACD}}$ comparison additionally requires the local categorical comparisons, archimedean boundary datum and global identification.

48.6 Subcondition (L_4): Archimedean Extension

Statement

Theorem 48.6.1 *((L_4) archimedean boundary datum).* The archimedean factor $C_\infty = \mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{R})])$ is modelled by the ind-completed derived category of admissible $(\mathfrak{gl}_2, O(2))$-modules of finite K-type. It carries the trivial K-spherical object as distinguished unit and extends the finite-place restricted tensor product by direct tensoring.

Proof. **Step 1: Well-definedness.** The derived category of admissible $(\mathfrak{g}, K_\infty)$-modules, where $\mathfrak{g} = \mathfrak{gl}_2(\mathbb{R})$ and $K_\infty = O(2) \subseteq GL_2(\mathbb{R})$, has the standard Harish-Chandra description (Jacquet–Langlands SLN 114 §1–3; Vogan 1981 "Representations of Real Reductive Groups"). Its ind-completion is stable and presentable. The sph-fin cut restricts to finite K_∞-type objects.

Step 2: Distinguished unit. The spherical unit at $v = \infty$ is the trivial K_∞-spherical object. Spherical principal series are not unique; they are objects of the category, not the distinguished unit.

Step 3: Tensor product extension. The restricted tensor product of Definition 48.3.1 extends naturally by including $v = \infty$ in the index set V of places: $V = \mathbb{P} \cup \{\infty\}$ with cofinite unramified subset $V^{\mathrm{unr}} = \mathbb{P} \cup \{\infty\} \setminus S_{\mathrm{ram}}$ for the given ramification set S_{ram}. The archimedean factor enters as a tensor factor in each S containing ∞ and by identity-via-distinguished-unit in S excluding ∞.

Step 4: Hecke action at infinity. The archimedean Hecke action is represented by the centre of the universal enveloping algebra $U(\mathfrak{gl}_2(\mathbb{R}))$, commuting with finite-place Hecke action by construction (both act via representation-theoretic restriction to the adelic double coset).

Step 5: Archimedean local factors. Jacquet–Langlands SLN 114 §1–3 classify the archimedean local representations of $GL_2(\mathbb{R})$ and $GL_2(\mathbb{C})$. The sph-fin subcategory is finite-type in each isotypic component; Vogan's classification gives a complete parametrization by Harish-Chandra parameters.

This proves the archimedean boundary datum. It does not construct a number-field spectral equivalence at $v = \infty$. □

Remark 48.6.2 *(archimedean spectral side).* The spectral-side analogue at $v = \infty$ is a proposed Weil-parameter stack for $W_{\mathbb{R}} = \mathbb{C}^\times \rtimes \mathbb{Z}/2$. Vogan and Langlands classify the representations. They do not give an Emerton–Gee stack at infinity, and no local categorical equivalence at infinity is asserted here.

48.7 Curve Analogy: Obstructions and Transfers

The Curve Theorem

THEOREM 48.7.1 *(de Rham geometric Langlands, AGKRRV).* For a smooth complete curve X over an algebraically closed field of characteristic zero and a reductive group G, there is an equivalence

$$\mathrm{D\text{-}mod}(\mathrm{Bun}_G(X)) \;\simeq\; \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_G^{\mathrm{dR}}(X))$$

satisfying Hecke-functoriality, Bernstein-centre compatibility, and categorical descent over the curve. The proof combines Gaitsgory–Rozenblyum's AMS DAG monographs, Raskin's chiral principal series and D-module foundations, and the AGKRRV mathematical treatments on geometric Langlands.

Obstructions to Transport

THEOREM 48.7.2 *(obstructions to number-field transport).* The de Rham geometric-Langlands proof of Theorem 48.7.1 does not transport directly to the number-field case. The obstructions:

(Obs.1) *No archimedean place.* Geometric Langlands on a curve has no real place. (L_4) has no curve analogue and requires separate treatment (Jacquet–Langlands archimedean).

(Obs.2) *Non-smooth Emerton–Gee stack.* Number-field $\mathcal{X}^{\mathrm{EG}}_{GL_2}$ is formally algebraic but not smooth over $\mathrm{Spec}\,\mathbb{Z}$; the de Rham stack $\mathrm{LocSys}_G^{\mathrm{dR}}(X)$ is smooth in the standard geometric-Langlands range under mild hypotheses. This affects the Bernstein-centre compatibility (L_3).

(Obs.3) *Perfectoid diamonds vs ind-schemes.* Geometric Langlands uses algebraic and ind-algebraic loop spaces. Number-field local geometrisation uses perfectoid diamonds (Scholze arXiv:1709.07343). The gluing formalism differs, and the Bhatt–Lurie prismatic comparison arXiv:2201.06120 is required to bridge the two.

(Obs.4) *Integrality of Galois deformations.* De Rham local systems are geometric; number-field Galois deformations are integral (Emerton–Gee stack). The Bernstein-centre action incorporates integrality in a way the de Rham curve case does not.

COROLLARY 48.7.3 *(what the curve analogy supplies).* Under the de Rham curve proof, the following patterns survive in the number-field setting:

- (L_1) (restricted tensor product well-definedness): **formal pattern**. Raskin's construction is ∞-categorical and does not use curve-specific structure beyond existence of local categories with distinguished units.

- (L_2) (Hecke-functoriality gluing over a finite set of primes): **conditional pattern**. It applies only after the local categorical comparison is supplied. Fargues–Scholze replaces the local de Rham geometric input by diamond geometry and spectral action; the cross-place compatibility argument still needs the local comparison and (BC-diamond-glue).

The following do *not* transport:

- (L_3) (Bernstein-centre compatibility): obstructed by (Obs.2), (Obs.4). Resolution requires (BC-diamond-glue), Conjecture 48.5.4.
- (L_4) (archimedean extension): obstructed by (Obs.1). Resolution requires independent Jacquet–Langlands archimedean construction (§48.6).

48.8 Final Reduction: L_{ACD} to a Single Lemma

Statement

Theorem 48.8.1 *(L_{ACD} residual reduction). Comparison hypothesis.* L_{ACD} holds after the local categorical comparison (F1.a) and the BC-diamond-gluing compatibility lemma (Conjecture 48.5.4) are supplied.

Proof. By Theorem 48.2.1, L_{ACD} contains (L_1), (L_2), (L_3), and (L_4), together with the global identification.

(L_1) is a theorem (Corollary 48.3.4).

(L_2) is conditional: it needs the local categorical comparison (F1.a) and (BC-diamond-glue) for the cross-place Whittaker compatibility (Theorem 48.4.1, Steps 1 and 6).

(L_3) is controlled by the Bernstein-centre gluing datum (Theorem 48.5.1).

(L_4) is the archimedean boundary datum (Theorem 48.6.1).

Hence L_{ACD} reduces, in this chapter, to the conjunction of F1.a, (BC-diamond-glue), and the global identification after the formal construction of (L_1) and the archimedean datum (L_4). □

Dependency on Established Theorems

Proposition 48.8.2 *(dependency table of L_{ACD}).* The conditional reduction of L_{ACD} uses the following established theorems:

(i) Fargues–Scholze arXiv:2102.13459 (local geometrization at each prime).

(ii) Emerton–Gee arXiv:1908.07185 (local Emerton–Gee stack construction; the global target is Selmer-derived here).

(iii) Gaitsgory–Rozenblyum, *A Study in Derived Algebraic Geometry*, AMS Mathematical Surveys and Monographs 221, Vols. I–II ($(\infty, 2)$-category of correspondences, $\mathrm{IndCoh}_{\mathrm{Nilp}}$ continuity).

(iv) Raskin, *Chiral principal series categories I*, Adv. Math. 388 (2021), 107856, and *D-modules on infinite-dimensional varieties* (restricted tensor product machinery for principal-series categories).

(v) Lurie HA 4.8 (tensor products of presentable ∞-categories), 5.3 (compact generation, higher Deligne).

(vi) Scholze arXiv:1709.07343 (étale cohomology of diamonds).

(vii) Bhatt–Scholze arXiv:1905.08229 (prismatic cohomology).

(viii) Bhatt–Lurie arXiv:2201.06120 (absolute prismatic, Nygaard-filtration convention bridge).

(ix) Zhu arXiv:1603.05593 (Witt-vector affine Grassmannian).

(x) Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 (coherent Springer theory; local Bernstein centres as coherent sheaves).

(xi) Jacquet–Langlands SLN 114 (archimedean local factors).

(xii) Vogan 1981 "Representations of Real Reductive Groups" (archimedean classification).

Together with the local categorical comparison (F1.a), (BC-diamond-glue), and the global identification, these established results constitute the primary literature input for the reduction.

Remark 48.8.3 *(geometric inputs to descent).* The twelve established theorems of Proposition 48.8.2 decompose into the following mathematical topics:

Fargues–Scholze list: (1), (6), (7). Perfectoid / diamond geometry, Fargues–Fontaine curve.

Gaitsgory–Rozenblyum list: (3), (4). DAG and $(\infty, 2)$-categorical correspondences, restricted tensor.

Lurie list: (5). Higher algebra foundations.

Emerton–Gee list: (2). Galois deformation stacks.

Bhatt–Lurie list: (8). Prismatic Nygaard bridge.

Zhu list: (9). Witt-vector Grassmannian.

Coherent Springer theory list: (10). Local Bernstein centres as coherent sheaves.

Jacquet–Langlands + Vogan list: (11), (12). Archimedean local representation theory.

The local categorical comparison, Bernstein-centre gluing, and global restricted-tensor identification remain separate mathematical inputs.

48.9 Consequence: F1 Closure Map

The Full Chain

THEOREM 48.9.1 *(F1 residual chain via* $\mathrm{L}_{\mathrm{ACD}}$*). Comparison hypothesis.* F1 (global arithmetic Arinkin–Gaitsgory, Conjecture 42.1.1) reduces to the local categorical comparison (F1.a) plus (BC-diamond-glue) (Conjecture 48.5.4).

Proof. By Corollary 43.2.3, F1 reduces to F1.a plus $\mathrm{L}_{\mathrm{ACD}}$. By Theorem 48.8.1, $\mathrm{L}_{\mathrm{ACD}}$ reduces to F1.a plus (BC-diamond-glue). The theorem identifies the residual pair {F1.a, (BC-diamond-glue)} as the obstruction visible in this reduction. □

Formal Consequences

COROLLARY 48.9.2 *(consequent form). Comparison hypothesis.* Given the local categorical comparison (F1.a), (BC-diamond-glue), F2, and the corrected arithmetic BZN comparison, the following implications hold under those hypotheses:

- F1 (global arithmetic Arinkin–Gaitsgory).
- $\mathrm{F3.BZN}^{\mathrm{Ar}}$ (arithmetic Ben-Zvi–Nadler, conditional on F1 + F2 plus the full-target/formal-loop comparison).
- The six-window placement theorem (Conjecture 41.1.1) remains conditional on the same residual chain.

Remark 48.9.3 *(comparison of the six windows).* A proof of Conjecture 41.1.1 requires F1 (hence (BC-diamond-glue)), F2 (hence the ∞-categorical Tate restricted tensor lemma F2.c), and F3 (hence $\mathrm{F3.BZN}^{\mathrm{Ar}}$). With (BC-diamond-glue) and the local comparison supplied, F1 becomes a conditional

theorem. With F2.c supplied, F2 becomes a conditional theorem. F3 then requires F3.BZN^{Ar}. The six-window placement remains conditional on these comparison inputs.

Effect on $\mathcal{V}^{\mathrm{prim}}$ Spectral Identification

Corollary 48.9.4 *($\mathcal{V}^{\mathrm{prim}}$ spectral shadow). Comparison hypothesis.* Given the local categorical comparison, (BC-diamond-glue), and the corrected arithmetic BZN full-target comparison, the spectral shadow of $\mathcal{V}^{\mathrm{prim}}$ is

$$\mathcal{V}^{\mathrm{prim}} \rightsquigarrow \mathcal{O}(L\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}).$$

The further passage to $\mathcal{O}(\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}})$ requires a constant-loop restriction or a flat-locus truncation. It is not a consequence of Hochschild trace alone.

Proof. $\mathcal{V}^{\mathrm{prim}} = \mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}})$ (Theorem 38.5.1). Under F1, which here means the local comparison together with (BC-diamond-glue), $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}} \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}})$. Ben-Zvi–Nadler arXiv:0904.1247 computes $\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})) = \mathcal{O}(L\mathcal{X}) = \mathcal{O}(\mathrm{Map}(S^1,\mathcal{X}))$ for derived stacks $\mathcal{X}$. The constant-loop map $\mathcal{X} \to L\mathcal{X}$ gives a restriction from the loop shadow to $\mathcal{O}(\mathcal{X})$. This restriction is additional data. The categorical trace supplies the loop object. □

48.10 Residual Obstructions

Obstruction Decomposition

After the reduction of Theorem 48.2.7, the residual inputs for $\mathrm{L}_{\mathrm{ACD}}$, and hence for F1, are:

- **Local categorical comparison:** the passage from the pointwise local spectral action to the local categorical equivalence used in L_2.
- **Global restricted-tensor identification:** the comparison between the restricted tensor product and the global automorphic category.
- **(BC-diamond-glue):** Conjecture 48.5.4. The central open residual, non-formal theorem. Domain: gluing of coherent Springer sheaves across a product of local Emerton–Gee stacks, plus agreement with the global Bernstein centre acting on the global automorphic category.

Prior Hypotheses

(BC-diamond-glue) is independent of the remaining open items from the arithmetic branch of Vol IV:

- F2.c (F2 residual, ∞-categorical Tate tensor): independent branch.
- Arithmetic Positselski: independent, on the P1 branch.
- Koszul–Petersson compatibility: independent, on the P3 branch.
- VB* (arithmetic Virasoro positivity): independent, on the RH branch.

(BC-diamond-glue) is central on the F1 branch only after the local comparison and the global identification are fixed.

Primary-Literature Precedents

- Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321: *local* Bernstein centres as coherent sheaves on local Emerton–Gee stacks. This is the state of the art; (BC-diamond-glue) is the global assembly of these local constructions.

- Gan–Savin: partial change on restricted tensor product of local automorphic categories, with specific compatibilities verified for certain Jacquet–Langlands transfers; the global Bernstein-centre compatibility for $GL_2/\mathbb{Q}$ remains open.
- AGKRRV 2020–2024 de Rham geometric Langlands: supplies the curve analogue of the centre-compatibility pattern. It does not prove the number-field (BC-diamond-glue) statement. The surviving pattern is documented in Corollary 48.7.3.

Remaining Descent Inputs

Condition	Mathematical form
$\mathrm{L}_{\mathrm{ACD}}$	Open descent theorem, conditional on the remaining comparisons
(L_1)	Theorem, Corollary 48.3.4
(L_2)	Conditional on a non-formal theorem and the local categorical comparison
(L_3) = (BC-diamond-glue)	Conjectural residual requiring a non-formal theorem
(L_4)	Archimedean boundary datum, without a spectral equivalence

TABLE 48.1: Descent inputs in the reduction of Theorem 48.2.7.

The Bernstein-centre and Whittaker comparisons, the local categorical comparison, and the global identification remain necessary for $\mathrm{L}_{\mathrm{ACD}}$. The stated reductions do not prove full adelic descent without these inputs.

48.11 DESCENT BOUNDARY

(i) $\mathrm{L}_{\mathrm{ACD}}$ decomposes into four disjoint subconditions (L_1)–(L_4) (Theorem 48.2.1).

(ii) (L_1) (restricted tensor well-definedness) closes unconditionally via Lurie HA 4.8 + 5.3, Raskin 2021–2022 (Theorem 48.3.2, Corollary 48.3.4).

(iii) (L_4) (archimedean extension) supplies the representation-theoretic boundary datum via Jacquet–Langlands SLN 114 + Vogan 1981 (Theorem 48.6.1).

(iv) (L_2) (Hecke-functoriality gluing) is conditional: after the local categorical comparisons are supplied it reduces to (BC-diamond-glue), via tensor product formalism + Gaitsgory–Rozenblyum continuity (Theorem 48.4.1).

(v) (L_3) is the Bernstein-centre part of (BC-diamond-glue) (Theorem 48.5.1).

(vi) $\mathrm{L}_{\mathrm{ACD}}$ reduces to local comparison, (BC-diamond-glue), the archimedean boundary datum, and the global identification (Theorem 48.8.1).

(vii) F1 (global arithmetic Arinkin–Gaitsgory) is conditional on the same residual chain (Theorem 48.9.1).

(viii) Given the residual chain, the six-window placement theorem remains conditional on the stated comparison maps. The Hochschild trace of $\mathcal{V}^{\mathrm{prim}}$ sees the loop stack; passage to the Selmer-derived structure sheaf is an extra constant-loop or flat-locus restriction (Corollary 48.9.4).

(ix) The remaining gluing and comparison hypotheses are listed in Table 48.1.

$\mathrm{L}_{\mathrm{ACD}}$ reduces from one large conjecture to a formal tensor construction, local comparison, Bernstein-centre gluing, an archimedean boundary datum, and a global identification. The structural content is a reduction, not an equivalence theorem for F1. The geometric inputs are grouped by their mathematical role (Remark 48.8.3).

The residual (BC-diamond-glue) is a single compatibility check between Ben-Zvi–Chen–Helm–Nadler's local coherent Springer sheaves and the global Bernstein centre on the global automorphic category. It is central on the F1 six-window branch of the arithmetic branch, but it is not the only hypothesis.

48.12 Forward Pointers

- Chapter 43 gives the parent reduction of F1 (Theorem 43.2.1) into sub-items (F1.a)–(F1.d). Theorem 48.2.1 refines that reduction at the adelic-descent layer.
- Chapter 41, Conjecture 41.1.1, requires the (BC-diamond-glue) and F2/F3 comparison hypotheses in Corollary 48.9.2.
- Chapter 44 contains the RH reformulation F-RH; (BC-diamond-glue) sits on the six-window-placement branch (F1–F3 structural branch in Appendix C), not the RH-closure branch (four-input RH branch). (BC-diamond-glue) controls the categorical-trace presentation of $\mathcal{V}^{\mathrm{prim}}$; the RH-closure branch (four-input RH branch) remains on a separate dependency chain.
- Appendix C, Section C.9, records $\mathrm{L}_{\mathrm{ACD}}$ on the F1 branch. After the reduction of Theorem 48.2.7, the Bernstein-centre part is (BC-diamond-glue), with visible requires a non-formal theorem.

Chapter 49

Arithmetic Ben-Zvi–Nadler Identification

The arithmetic Ben-Zvi–Nadler candidate has a sharp obstruction. The procyclic stack $B\widehat{\mathbb{Z}}$ retains only the inertia shadow of the finite-prime topology; it forgets Frobenius, Weil-group, and (φ, Γ) data. The corrected target requires the full Weil/(φ, Γ) parameter stack, with the $B\widehat{\mathbb{Z}}$ construction playing the role of a forgetful procyclic shadow.

49.1 Statement

Conjecture 49.1.1 *(F3.BZN^{Ar}, full target). Conjectural statement. (conditional beyond F1 and F2, i.e. beyond* $\mathrm{L}_{\mathrm{ACD}}$ *and (F2.c) of Theorem 43.2.1 and Theorem 43.3.1).* There should be a canonical equivalence of E_1-algebras in stable presentable ∞-categories, after replacing the inertia-only target by the full Weil/(φ, Γ) parameter stack,

$$\mathrm{HH}_\bullet\big(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}\big) \xrightarrow{\sim} \mathcal{O}\big(\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}\big)^{\mathrm{flat}},$$

where:

(i) $\mathrm{HH}_\bullet$ is the derived Hochschild trace of the E_2-monoidal ∞-category $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$, with the E_2-monoidal structure supplied by F2;

(ii) $\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}$ is the full derived moduli stack retaining Weil-group and (φ, Γ) data at finite primes; its restriction to $B\widehat{\mathbb{Z}} = B\prod_p \mathbb{Z}_p$ is only a procyclic shadow;

(iii) the flat locus $(\cdot)^{\mathrm{flat}}$ is picked out by the Fargues–Scholze local Langlands compatibility at each prime plus the almost-everywhere-unramified condition at the archimedean place;

(iv) the structure sheaf $\mathcal{O}$ on the right is expected to pull back along the global Emerton–Gee comparison to the full parameter stack, not along the inertia-only shadow of Proposition 49.4.6.

Corollary 49.1.2 *(F3 closure under F1 + F2). Conjectural statement.* Given F1 and F2, F3 (Conjecture 42.3.1) reduces to the corrected arithmetic BZN comparison, the formal-loop compatibility, the flat-locus comparison, and the full Weil/(φ, Γ) target comparison. It is not closed by the inertia-only model.

Proof. This is the honest residual formulation. F1 and F2 supply the global spectral and E_2 inputs. The remaining inputs are the identification of the full Weil/(φ, Γ) parameter stack with the inertia shadow and the profinite formal-loop continuity required by the arithmetic Ben-Zvi–Nadler comparison. □

49.2 Comparison Table

The attempted proof of Conjecture 49.1.1 passes through five independent conditions. The independent comparison identifies which close and which yield residuals.

MKR shadow and the inertia circle. The classifying stack $B\widehat{\mathbb{Z}}$ is the classifying ∞-stack of the profinite abelian group $\widehat{\mathbb{Z}}$; its pro-étale fundamental group is $\pi_1^{\text{ét}}(B\widehat{\mathbb{Z}}) = \widehat{\mathbb{Z}}$. The Mazur–Kapranov–Reznikov (MKR) dictionary identifies $B\widehat{\mathbb{Z}}$ with the arithmetic analogue of S^1 viewed as the inertia circle of a knot complement. Closes (Proposition 49.3.3).

Loop-space identification for the inertia shadow. For an n-geometric derived stack $\mathcal{X}$, the arithmetic free loop space $L^{\text{Ar}}\mathcal{X} := \text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ is a well-defined derived stack (Toën–Vezzosi DAG framework). For $\mathcal{X} = BGL_2$, one computes $L^{\text{Ar}}BGL_2 = \text{Hom}(\widehat{\mathbb{Z}}, GL_2)/GL_2$ as a derived quotient stack. Closes (Proposition 49.4.4).

Arithmetic transport of the Ben-Zvi–Nadler machine. Ben-Zvi–Nadler prove $\text{HH}_\bullet(\text{IndCoh}(\mathcal{X}/G)) = \mathcal{O}(L(\mathcal{X}/G))^{\text{flat}}$ over $\mathbb{C}$ with S^1 as the topological circle. The proof factors through two ingredients: (i) the Costello–Francis–Lurie identification of the Hochschild trace of a symmetric monoidal ∞-category with factorization homology over S^1; (ii) the flatness assertion cutting out ind-coherent sheaves inside functions on the loop space. Ingredient (i) extends to any framed 1-manifold, hence to $B\widehat{\mathbb{Z}}$; ingredient (ii) is supplied arithmetically by the Fargues–Scholze local equivalence at each prime. Conditional (Theorem 49.5.5).

Dunn additivity and the E_1-structure. The global Iwahori Hecke category is E_2-monoidal by F2. Dunn additivity (Lurie HA 5.1.2.2) supplies the canonical E_1-algebra structure on $\text{HH}_\bullet$ of any E_2-monoidal category, and the arithmetic Ben-Zvi–Nadler map respects this structure via the factorization-homology presentation. Closes under F2 (Proposition 49.6.1).

Full Weil/(φ, Γ) target and the $\mathcal{V}^{\text{prim}}$ pullback. The structure sheaf on the full Weil/(φ, Γ) flat target should pull back along the Emerton–Gee comparison to $\mathcal{V}^{\text{prim}}$ via the spectral-side identification of F1. The forgetful map from the Emerton–Gee stack to the inertia shadow $\text{LocSys}^{\text{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})$ records only the procyclic restriction; the full Frobenius and (φ, Γ) data are retained only in the complete target. Open (Theorem 49.8.2).

The independent criterion for each condition is recorded in the corresponding subsection below.

49.3 The Arithmetic Inertia Circle $B\widehat{\mathbb{Z}}$

Definition

Definition 49.3.1 *(classifying stack of profinite $\widehat{\mathbb{Z}}$).* Let $\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} = \prod_p \mathbb{Z}_p$, the profinite completion of $\mathbb{Z}$. Its classifying ∞-stack $B\widehat{\mathbb{Z}}$ is

$$B\widehat{\mathbb{Z}} \;:=\; * \,/\, \widehat{\mathbb{Z}},$$

the derived quotient of a point by the profinite group $\widehat{\mathbb{Z}}$, formed in the ∞-category of pro-étale derived stacks over $\mathbb{Z}$. Equivalently, $B\widehat{\mathbb{Z}} = \varprojlim_n B(\mathbb{Z}/n\mathbb{Z})$ as a pro-étale limit.

Remark 49.3.2 *(two realisations).* $B\widehat{\mathbb{Z}}$ admits two useful realisations.

(i) *Galois realisation.* $B\widehat{\mathbb{Z}} = B\text{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ for any finite field $\mathbb{F}_q$; the absolute Galois group of any finite field is topologically generated by Frobenius and has profinite completion $\widehat{\mathbb{Z}}$.

(ii) *Product realisation.* Under $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$,

$$B\widehat{\mathbb{Z}} \;=\; \prod_p^{\text{proét}} B\mathbb{Z}_p.$$

The pro-étale product is taken in the ∞-category of pro-étale derived stacks (Bhatt–Scholze arXiv:1309.1198); it is well-defined because each $B\mathbb{Z}_p$ is a classifying stack of a profinite group, whose pro-étale cohomology is finitary.

MKR Dictionary and the Arithmetic S^1

Proposition 49.3.3 *(MKR dictionary for $B\widehat{\mathbb{Z}}$). Heuristic.* $B\widehat{\mathbb{Z}}$ is the arithmetic analogue of S^1 under the Mazur–Kapranov–Reznikov (MKR) dictionary:

(i) *Fundamental group.* $\pi_1^{\text{proét}}(B\widehat{\mathbb{Z}}) = \widehat{\mathbb{Z}}$, the profinite completion of $\pi_1(S^1) = \mathbb{Z}$.

(ii) *Higher homotopy.* $\pi_n(B\widehat{\mathbb{Z}}) = 0$ for $n \neq 1$; so $B\widehat{\mathbb{Z}}$ is a pro-$\mathrm{K}(\widehat{\mathbb{Z}}, 1)$, arithmetically the same homotopy type as S^1 profinitely completed.

(iii) *Cup-product structure.* Under étale cohomology $H^*(B\widehat{\mathbb{Z}}; \mathbb{Z}_\ell) = \mathbb{Z}_\ell \oplus \mathbb{Z}_\ell[-1]$ for every ℓ; this matches $H^*(S^1; \mathbb{Z}) = \mathbb{Z} \oplus \mathbb{Z}[-1]$ after pro-ℓ completion.

(iv) *Knot-complement interpretation.* Under Morishita's arXiv:0904.3399 realisation of $\operatorname{Spec} \mathbb{Z}_p$ as a knot complement with inertia circle $B\mathbb{Z}_p$, the product $B\widehat{\mathbb{Z}} = \prod_p B\mathbb{Z}_p$ is the product of all local inertia circles, i.e. the global inertia circle of $\overline{\operatorname{Spec} \mathbb{Z}}$.

Proof. (1) $\widehat{\mathbb{Z}}$ is a profinite abelian group, so $B\widehat{\mathbb{Z}}$ is a $\mathrm{K}(\widehat{\mathbb{Z}}, 1)$-space in the pro-étale topology. Taking the pro-étale fundamental group of a classifying stack of a profinite group returns the group itself (Bhatt–Scholze arXiv:1309.1198 Lem. 7.4.3).

(2) A classifying stack of an abelian group G has $\pi_n = G$ if $n = 1$ and $\pi_n = 0$ otherwise; this is the general fact for classifying stacks of discrete groups, transported to the pro-étale setting via pro-discrete presentation of $\widehat{\mathbb{Z}}$.

(3) $H^*(B\widehat{\mathbb{Z}}; \mathbb{Z}_\ell) = H^*_{\text{cts}}(\widehat{\mathbb{Z}}; \mathbb{Z}_\ell) = \mathbb{Z}_\ell[\epsilon]/(\epsilon^2)$ with ϵ of degree 1 (continuous group cohomology of $\widehat{\mathbb{Z}}$ coefficient in $\mathbb{Z}_\ell$; classical). This matches $H^*(S^1; \mathbb{Z}_\ell) = \mathbb{Z}_\ell[\epsilon]/(\epsilon^2)$.

(4) Morishita's arXiv:0904.3399 identifies the local inertia subgroup of $W_{\mathbb{Q}_p}$ at a prime p with the fundamental group of a small circle linking the "knot" corresponding to p; its classifying stack is $B\mathbb{Z}_p$ (or $BO_p^\times$ in the ramified extension, whose connected component is $B\mathbb{Z}_p$). Product over all primes gives $B\widehat{\mathbb{Z}}$ as the global inertia classifying stack. □

Remark 49.3.4 *(the archimedean place).* The archimedean place ∞ of $\overline{\operatorname{Spec} \mathbb{Z}}$ carries local inertia equal to the fixed-at-∞ subgroup of $W_{\mathbb{R}}$, which is $W_{\mathbb{C}} = \mathbb{C}^\times$; its profinite completion is trivial. Hence $B\widehat{\mathbb{Z}}$ captures only the finite-place inertia, which suffices for the Hochschild trace computation at hand because $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$ is defined over the finite adeles.

The Procyclic Carrier and Its Arithmetic Interpretation

Procyclic geometry and arithmetic topology. $B\widehat{\mathbb{Z}}$ is defined in the Bhatt–Scholze pro-étale formalism (arXiv:1309.1198). The arithmetic-topology interpretation is discussed by Morishita arXiv:0904.3399, Mazur 2000 Selecta, and Kapranov–Reznikov 1996.

Obstruction 2 (conventions). Pro-étale fundamental group convention: Bhatt–Scholze; matches Morishita's profinite inertia convention.

Obstruction 3 (ambient categories). $B\widehat{\mathbb{Z}}$ is a pro-étale ∞-stack. Derived stacks in the Toën–Vezzosi DAG setting admit pro-étale enhancements (Toën DAG §2.3); so $B\widehat{\mathbb{Z}}$ sits in the same ambient category as the Emerton–Gee stack.

Obstruction 4 (missing hypothesis). None: the definition requires only that $\widehat{\mathbb{Z}}$ be a profinite group, which is classical.

Scope of the analogy. The procyclic carrier alone does not supply an arithmetic trace equivalence.

The MKR dictionary condition closes as a definition of the procyclic shadow; its arithmetic interpretation remains heuristic.

49.4 Arithmetic Free Loop Space

Definition

Definition 49.4.1 *(arithmetic free loop space).* Let $\mathcal{X}$ be an n-geometric derived pro-étale stack over $\mathbb{Z}$ (Toën–Vezzosi, DAG framework). The *arithmetic free loop space of* $\mathcal{X}$ is the internal mapping stack

$$L^{\mathrm{Ar}}\mathcal{X} \;:=\; \mathrm{Map}\big(B\widehat{\mathbb{Z}}, \mathcal{X}\big),$$

the internal Hom in the ∞-category of derived pro-étale stacks.

Remark 49.4.2 *(analogy with complex free loop space).* Over $\mathbb{C}$, the free loop space is $L\mathcal{X} = \mathrm{Map}(S^1, \mathcal{X})$; since $S^1 \simeq B\mathbb{Z}$, this is $\mathrm{Map}(B\mathbb{Z}, \mathcal{X})$. Arithmetically we replace $\mathbb{Z}$ by its profinite completion $\widehat{\mathbb{Z}}$, reflecting the fact that all arithmetic loops (= monodromy actions) are profinite (Grothendieck's étale fundamental group is always profinite).

Remark 49.4.3 *(existence as a derived stack).* That $L^{\mathrm{Ar}}\mathcal{X}$ is a well-defined derived pro-étale stack follows from Toën–Vezzosi DAG V Prop. 4.1 (internal mapping stacks in DAG), applied to the cocartesian closed structure on the ∞-topos of pro-étale derived stacks (Bhatt–Scholze arXiv:1309.1198 §7). Explicitly, the hom $S \mapsto \mathrm{Map}_{\mathrm{proét}}(S \times B\widehat{\mathbb{Z}}, \mathcal{X})$ satisfies pro-étale descent in S by the standard axiom.

Computation for $\mathcal{X} = BGL_2$

PROPOSITION 49.4.4 *($L^{\mathrm{Ar}}BGL_2$).* There is a canonical equivalence of derived pro-étale stacks

$$L^{\mathrm{Ar}}\,BGL_2 \;\simeq\; \mathrm{Hom}^{\mathrm{cts}}\big(\widehat{\mathbb{Z}}, GL_2\big) \;/\; GL_2,$$

where GL_2 acts on $\mathrm{Hom}^{\mathrm{cts}}(\widehat{\mathbb{Z}}, GL_2)$ by conjugation and $\mathrm{Hom}^{\mathrm{cts}}$ denotes continuous group homomorphisms.

Proof. By definition of BG for a group scheme G, a map $B\widehat{\mathbb{Z}} \to BGL_2$ of classifying stacks of a discrete and a group-scheme object is a continuous group homomorphism $\widehat{\mathbb{Z}} \to GL_2$ up to GL_2-conjugation (which acts on the target by inner automorphism, corresponding to gauge transformations of the loop-space description). In derived pro-étale geometry this upgrades to the derived quotient $\mathrm{Hom}^{\mathrm{cts}}(\widehat{\mathbb{Z}}, GL_2)/GL_2$, where the action is by conjugation.

Explicitly, $\mathrm{Map}(B\widehat{\mathbb{Z}}, BGL_2)$ is the ∞-groupoid whose objects are GL_2-torsors on $B\widehat{\mathbb{Z}}$. Such a torsor is equivalent to a representation $\widehat{\mathbb{Z}} \to GL_2$ of the fundamental group (= $\widehat{\mathbb{Z}}$ by Proposition 49.3.3 (1)) modulo GL_2-equivalence. □

COROLLARY 49.4.5 *(identification with arithmetic GL_2-local systems on $B\widehat{\mathbb{Z}}$).*

$$L^{\mathrm{Ar}}\,BGL_2 \;=\; \mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}\big(B\widehat{\mathbb{Z}}\big) \;=\; \Big\{(\rho_p)_p \in \prod_p \mathrm{Hom}(\mathbb{Z}_p, GL_2) \;/\; \mathrm{conj}\Big\},$$

an arithmetic analogue of the classical character variety.

Proof. By Proposition 49.4.4 and Remark 49.3.2 (2), $\mathrm{Hom}^{\mathrm{cts}}(\widehat{\mathbb{Z}}, GL_2) = \mathrm{Hom}^{\mathrm{cts}}(\prod_p \mathbb{Z}_p, GL_2)$. Since GL_2 is algebraic (hence its topology is discrete on the points-functor but ℓ-adic on ℓ-adic points), a continuous homomorphism from $\prod_p \mathbb{Z}_p$ factors uniquely through a finite product for any finite-image representation, and in general decomposes as $(\rho_p)_p$ with almost all ρ_p unramified (i.e. trivial on $\mathcal{O}_p^\times$ under the local reciprocity map); this is the adelic restricted-product structure on $\widehat{\mathbb{Z}}$. □

The Selmer Inertia-Shadow Morphism

Proposition 49.4.6 *(Selmer target to inertia shadow).* There is an expected forgetful morphism of pro-étale derived stacks

$$\iota^{\mathrm{Sel}} : \mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}} \longrightarrow \mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})$$

which, at each prime p, should be the composition of the Emerton–Gee stack's universal Galois representation (φ,Γ)-module $\to GL_2$-rep of the inertia subgroup of $W_{\mathbb{Q}_p}$, followed by restriction to $\mathbb{Z}_p \subset W_{\mathbb{Q}_p}$ (the wild-tame inertia quotient composed with reciprocity).

Proof. At each prime p, the Emerton–Gee stack $\mathcal{X}^{\mathrm{EG},p}_{GL_2}$ parametrises (φ,Γ)-modules of rank 2 over the period ring of $\mathbb{Q}_p$. Each such module determines, via local reciprocity, a representation $W_{\mathbb{Q}_p} \to GL_2$ of the Weil group; its restriction to the inertia subgroup $I_p \subset W_{\mathbb{Q}_p}$, composed with the natural map $I_p \twoheadrightarrow \mathbb{Z}_p$ (quotient by wild inertia), is a representation $\mathbb{Z}_p \to GL_2$, hence a point of $\mathrm{Hom}^{\mathrm{cts}}(\mathbb{Z}_p, GL_2)/GL_2$.

Globally, the assembly factors through the Selmer-derived stack: the local inertia-shadow maps must be compatible with one global representation, fixed determinant, and the chosen local Selmer conditions. It is not a product or filtered colimit of local maps. □

The Flat Locus

Definition 49.4.7 *(provisional flat locus).* The *provisional flat locus* $\mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})^{\mathrm{flat}} \subset \mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})$ is the procyclic shadow of the full Fargues–Scholze or Emerton–Gee flat parameter space; the full space retains Frobenius, Weil, and (φ,Γ) data beyond what the procyclic shadow records. The provisional locus is cut out by:

(i) *Fargues–Scholze local compatibility.* At each prime p, $\rho_p : \mathbb{Z}_p \to GL_2$ extends to a representation of $W_{\mathbb{Q}_p}$ coming from a continuous (φ,Γ)-module (Fargues–Scholze arXiv:2102.13459 Thm. I.9).

(ii) *Almost-everywhere unramified condition.* For all but finitely many primes p, ρ_p is unramified (i.e. trivial on the maximal pro-p subgroup of $\mathbb{Z}_p$).

Remark 49.4.8 *(flat in the arithmetic sense).* The adjective "flat" in this context does not refer to flatness of modules; it refers to the arithmetic analogue of *flat connections* in the complex Ben-Zvi–Nadler setting. Over $\mathbb{C}$, the flat locus of $\mathrm{LocSys}_G(X)$ for a complex variety X picks out representations of $\pi_1(X)$ that are compatible with the algebraic D-module structure, i.e. come from flat connections. Arithmetically, the analogue is: representations of the fundamental group (= $\widehat{\mathbb{Z}}$) that come from (φ,Γ)-module data, which is the correct analogue of a "flat connection in the arithmetic topology".

Proposition 49.4.9 *(image of the Selmer inertia-shadow morphism).* Conjectural statement. The full Selmer comparison morphism is expected to factor through the flat full-parameter locus. The forgetful inertial shadow ι^{Sel} of Proposition 49.4.6 factors through the flat locus

$$\iota^{\mathrm{Sel}} : \mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}} \longrightarrow \mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})^{\mathrm{flat}}$$

as a map of pro-étale derived stacks. Whether this inertia-only restriction is a closed immersion is an open question: on points, the Colmez–Fontaine theorem shows the full (φ, Γ) data carries strictly more information than the procyclic restriction.

Proof. Factorization through the flat locus: by construction, each (φ, Γ)-module gives rise to a $W_{\mathbb{Q}_p}$-representation (Fontaine's theorem, reviewed in Emerton–Gee arXiv:1908.07185 §1.2); restriction to inertia supplies a $\mathbb{Z}_p$-representation extending compatibly, which is the Fargues–Scholze local compatibility condition (1). The almost-everywhere-unramified condition (2) is imposed in the Selmer datum by requiring unramified local conditions outside the finite set S.

Closed immersion residual. Locally at each prime, the full (φ, Γ) or Weil parameter retains Frobenius data beyond the inertia restriction. The map $\iota^{\mathrm{EG},p}$ is surjective onto the locally-admissible $\mathbb{Z}_p$-representations (those coming from (φ, Γ)-modules) by Fargues–Scholze Thm. I.9.6, and is injective on $\mathbb{C}$-points by the standard fact that a (φ, Γ)-module is determined by its Weil–Deligne or (φ, Γ) data up to finite data (Colmez–Fontaine 2000). The inertia restriction forgets Frobenius and unramified data; the closed-immersion property accordingly belongs to the full Weil/(φ, Γ) target. □

Loop-Space Carriers and Comparison Maps

Geometric carriers. Toën's derived mapping stacks and Bhatt–Scholze's pro-étale geometry (arXiv:1309.1198) provide the mapping-space settings. Emerton–Gee arXiv:1908.07185 and Fargues–Scholze arXiv:2102.13459 supply arithmetic deformation and spectral-action data; identifying these with a loop-space model requires the comparison morphism.

Obstruction 2 (conventions). (φ, Γ)-module convention: Fontaine. Local reciprocity convention: Artin arithmetic normalisation sends a uniformizer to arithmetic Frobenius; geometric Frobenius is its inverse and acts by p^{-1} on the Tate line. These are the Emerton–Gee conventions; they match Fargues–Scholze after the standard twist.

Obstruction 3 (ambient categories). $L^{\mathrm{Ar}} BGL_2$ and $\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}$ both live in pro-étale derived stacks; the morphism ι^{Sel} is in the same ambient ∞-category as the target.

Obstruction 4 (missing hypothesis). The almost-everywhere-unramified condition is essential; without it the product $\prod_p \mathrm{Hom}(\mathbb{Z}_p, GL_2)/GL_2$ is not locally of finite type. This is an explicit hypothesis in Definition 49.4.7. *hypothesis explicit.*

Obstruction 5 (false functoriality). The Emerton–Gee morphism is functorial in the chain $GL_2 \to GL_2$-rep $\to \mathbb{Z}_p$-rep $\to \widehat{\mathbb{Z}}$-rep, compatible with conjugation and with the inductive limit over finite sets of primes.

Obstruction 6 (unproved equivalence). The closed-immersion property requires the full Weil/(φ, Γ) target: $B\widehat{\mathbb{Z}}$ records only the procyclic shadow, omitting Frobenius and unramified data retained by the Colmez–Fontaine datum. **Residual: full target comparison.**

The loop-space condition closes for the procyclic mapping-stack computation. The Emerton–Gee comparison to the full arithmetic target remains open.

49.5 The Hochschild-Trace Theorem (Arithmetic)

Classical Ben-Zvi–Nadler

Theorem 49.5.1 *(Ben-Zvi–Nadler 2009, Theorem 1.1).* For a complex reductive group G, there is an equivalence of E_1-algebras

$$\mathrm{HH}_\bullet\big(\mathrm{D}(G/G)\big) \;\simeq\; \mathcal{O}\big(\mathrm{LocSys}_G(S^1)\big)^{\mathrm{flat}},$$

where G/G is the adjoint quotient, $\mathrm{LocSys}_G(S^1) = G/G$ as a derived stack, and the flat locus consists of flat G-connections (= all connections on S^1, i.e. the whole space).

Remark 49.5.2 *(structure of the classical proof).* Ben-Zvi–Nadler's proof has two ingredients.

(i) *Costello–Francis–Lurie.* For a symmetric monoidal ∞-category $\mathcal{C}$, $\mathrm{HH}_\bullet(\mathcal{C}) = \int_{S^1} \mathcal{C}$, factorization homology of $\mathcal{C}$ over S^1 (Costello 2004; Francis arXiv:1206.5522; Lurie HA 5.5.3).

(ii) *Loop-space identification.* For $\mathcal{C} = \mathrm{D}(\mathcal{X})$ the ∞-category of quasi-coherent sheaves on a derived stack $\mathcal{X}$, factorization homology over S^1 equals functions on the free loop space $\mathcal{O}(L\mathcal{X})$, with a flatness cut supplied by the requirement that functions come from quasi-coherent sheaves rather than arbitrary pro-coherent data.

Both ingredients admit arithmetic extensions; the content of the arithmetic Ben-Zvi–Nadler theorem is that these extensions match.

Arithmetic Extension of Costello–Francis–Lurie

Theorem 49.5.3 *(arithmetic factorization homology).* Conjectural statement. For an E_2-monoidal stable presentable ∞-category $\mathcal{C}$ in the ∞-category of pro-étale derived stacks, there is a canonical equivalence

$$\mathrm{HH}_\bullet(\mathcal{C}) \;\simeq\; \int_{B\widehat{\mathbb{Z}}} \mathcal{C}$$

as E_1-algebras, where the right-hand side is factorization homology of $\mathcal{C}$ over the pro-étale classifying stack $B\widehat{\mathbb{Z}}$ equipped with its cyclotomic/profinite orientation datum.

Proof. Factorization homology $\int_M \mathcal{C}$ is defined (Francis arXiv:1206.5522) as the left Kan extension along the inclusion of framed disks into framed manifolds, applied to $\mathcal{C}$ viewed as an E_n-algebra. The framed 1-manifold case requires an E_1-algebra structure on $\mathcal{C}$, which is supplied by the E_2-monoidal structure via Dunn additivity.

The Costello–Francis–Lurie theorem (Lurie HA 5.5.3.11) computes $\mathrm{HH}_\bullet(\mathcal{C}) = \int_{S^1} \mathcal{C}$ for a symmetric monoidal stable ∞-category $\mathcal{C}$, with S^1 the standard topological circle. For the arithmetic extension, S^1 is replaced by $B\widehat{\mathbb{Z}}$, which is a cyclotomically oriented pro-étale classifying stack (Proposition 49.3.3: $B\widehat{\mathbb{Z}}$ has the same homotopy type as S^1 pro-ℓ-completed). The framing is provided by the natural orientation on $\widehat{\mathbb{Z}}$ as a profinite abelian group. Frobenius acts through the Weil quotient and is recorded separately by the full $\mathrm{Weil}/(\varphi, \Gamma)$ target; it is not identified with an inertia generator.

The computation

$$\int_{B\widehat{\mathbb{Z}}} \mathcal{C} \;=\; \varinjlim_{n} \int_{B(\mathbb{Z}/n\mathbb{Z})} \mathcal{C}$$

follows from the pro-étale limit presentation $B\widehat{\mathbb{Z}} = \varprojlim_n B(\mathbb{Z}/n\mathbb{Z})$ (Definition 49.3.1) combined with the fact that factorization homology converts pro-étale limits of framed manifolds into colimits of factorization homology (by the left-Kan-extension definition).

Each $\int_{B(\mathbb{Z}/n\mathbb{Z})} C$ is the $\mathbb{Z}/n\mathbb{Z}$-Hochschild trace of C (Lurie HA 5.5.3 applied to the finite cyclic group); equivalently, the $\mathbb{Z}/n\mathbb{Z}$-coinvariants of the cyclic object computing $\mathrm{HH}_\bullet(C)$. The limit over n under divisibility recovers $\mathrm{HH}_\bullet(C)$ itself, because the cyclic object's $\mathbb{Z}$-coinvariants are $\mathrm{HH}_\bullet$ (classical) and pro-completion over n does not alter the value on finite-type categories. □

Remark 49.5.4 *(The trace and completion comparison).* Factorization homology supplies a trace construction on its declared coefficient category; pro-étale completion changes the carrier and its limiting diagrams. The arithmetic comparison must specify the transition maps and prove that the trace commutes with the required completion. The factorization-homology and pro-étale frameworks alone do not supply that interchange.

Arithmetic Loop-Space Identification

THEOREM 49.5.5 *(arithmetic loop-space identification, core HH identity).* Conjectural statement. Let $\mathcal{X}$ be a derived pro-étale formal algebraic stack, Noetherian, with $C = \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ the ∞-category of ind-coherent sheaves with nilpotent singular support. Then

$$\mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})\big) \;\simeq\; \mathcal{O}\big(L^{\mathrm{Ar}}\mathcal{X}\big)^{\mathrm{flat}}$$

as E_1-algebras, where $L^{\mathrm{Ar}}\mathcal{X} = \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ and the flat locus on the right is cut out by the condition that functions extend to global sections of the pro-ind-coherent structure sheaf.

Proof. We combine Theorem 49.5.3 with an arithmetic version of Ben-Zvi–Nadler's ind-coherent-sheaves/free-loop-space identification.

Step 1: HH to factorization homology. $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ is an E_2-monoidal stable presentable ∞-category (by the convolution structure on ind-coherent sheaves on derived stacks; Gaitsgory–Rozenblyum Vol. II Ch. IV). By Theorem 49.5.3,

$$\mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})\big) \;=\; \int_{B\widehat{\mathbb{Z}}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}).$$

Step 2: Factorization homology to loop-space functions. Ben-Zvi–Francis–Nadler arXiv:0805.0157 (integral transforms and categorical trace) establish

$$\int_M \mathrm{IndCoh}(\mathcal{X}) \;=\; \mathcal{O}\big(\mathrm{Map}(M, \mathcal{X})\big)$$

for M a framed compact 1-manifold and $\mathcal{X}$ a derived stack in the complex setting. The proof goes through a *spatial-loop-space* computation: factorization homology of ind-coherent sheaves on $\mathcal{X}$ over M is functions on the mapping stack $\mathrm{Map}(M, \mathcal{X})$, via the pull-push along the correspondence $\mathcal{X} \leftarrow \mathrm{Map}(M, \mathcal{X}) \to \mathrm{Map}(M, \mathcal{X})$ using IndCoh-functoriality. This proof uses only derived-geometry inputs: Toën–Vezzosi's DAG, Gaitsgory–Rozenblyum's Vol. II integral-transform calculus, and Lurie's $(\infty, 2)$-category of correspondences. None of these are specific to the complex setting; all admit pro-étale extensions.

Applied to $M = B\widehat{\mathbb{Z}}$ with the arithmetic DAG framework, this yields

$$\int_{B\widehat{\mathbb{Z}}} \mathrm{IndCoh}(\mathcal{X}) \;=\; \mathcal{O}\big(\mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})\big) \;=\; \mathcal{O}\big(L^{\mathrm{Ar}}\mathcal{X}\big).$$

Step 3: Nilpotent singular support and flatness cut. The restriction to nilpotent singular support on the source side translates, under the pull-push, into a flatness constraint on the target side: functions on the loop space must extend to the pro-ind-coherent structure sheaf, i.e. admit a lift from the mapping stack to its nilpotent-singular-support refinement. Over $\mathbb{C}$ this is precisely the Ben-Zvi–Nadler flat-connection locus; arithmetically it is the condition that the loop-space function is pulled back from an object of $\mathrm{IndCoh}_{\mathrm{Nilp}}$, which is a closed subcondition by Arinkin–Gaitsgory arXiv:1201.6343 for singular support applied to pro-étale derived stacks.

Step 4: E_1-algebra structure compatibility. The E_1-algebra structure on $\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}))$ comes from Dunn additivity applied to the E_2-monoidal structure (Lurie HA 5.1.2.2). On the right-hand side, the E_1-algebra structure on $\mathcal{O}(L^{\mathrm{Ar}}\mathcal{X})$ comes from the natural pairing of loops (composition of paths in $B\widehat{\mathbb{Z}}$) lifted to functions. The equivalence (49.5.5) is compatible with these E_1-structures because the factorization-homology model encodes the E_1-structure via the 1-dimensional factorization structure on $B\widehat{\mathbb{Z}}$. □

Remark 49.5.6 *(prismatic-Nygaard convention bridge).* The bridge between the pro-étale topology on $B\widehat{\mathbb{Z}}$ and the de Rham-style loop-space functions on $\mathcal{X}^{\mathrm{EG}}$ uses the Bhatt–Lurie prismatic-Nygaard filtration arXiv:2201.06120, which aligns pro-étale and de Rham conventions at the level of the arithmetic cotangent complex. The role is identical to its role in (F1) Obstruction 2 (Section 43.2).

Source Comparison: Arithmetic Transport of Ben-Zvi–Nadler

Trace and geometric transport. Factorization homology (Francis, arXiv:1206.5522), the Hochschild trace (Lurie, *Higher Algebra*, Section 5.5.3), and integral transforms (Ben-Zvi–Francis–Nadler, arXiv:0805.0157) give their stated categorical constructions. Arithmetic transport additionally needs the coefficient, singular-support and completion maps to the chosen Emerton–Gee or full Weil target.

Obstruction 2 (conventions). Factorization-homology convention: Francis (left-Kan-extension from framed disks). Singular-support convention: Arinkin–Gaitsgory. Pro-étale convention: Bhatt–Scholze. Prismatic-Nygaard bridge: Bhatt–Lurie. Their compatibility is part of the specified arithmetic comparison.

Obstruction 3 (ambient categories). E_2-monoidal stable presentable ∞-categories in pro-étale derived geometry. Factorization homology, mapping stacks, and IndCoh all live in this ambient $(\infty, 2)$-category (Gaitsgory–Rozenblyum).

Obstruction 4 (missing hypothesis). Noetherianness and derived formal algebraic stack structure on $\mathcal{X}$ are required to apply Arinkin–Gaitsgory singular support. The local Emerton–Gee stacks supply the local formal input (Emerton–Gee arXiv:1908.07185 Thm B.0.1), but the corrected global target is Selmer-derived. Its Noetherian and singular-support properties are additional conditional hypotheses.

Obstruction 5 (false functoriality). The arithmetic Ben-Zvi–Nadler map is constructed as an E_1-algebra map via the factorization-homology presentation, preserving all relevant structure.

Obstruction 6 (unproved equivalence). Step 2 invokes Ben-Zvi–Francis–Nadler arXiv:0805.0157 in the pro-étale setting; the pro-étale extension is not explicitly written in the 2009 paper but is a formal

consequence of the setup, since all inputs (DAG, correspondences, integral transforms) are topology-agnostic. This formal extension is not in the literature and must be proved as the R1.formal.b continuity theorem. **Residual: profinite formal-loop continuity.**

The arithmetic transport condition is conditional. The pro-étale lift of Ben-Zvi–Francis–Nadler and the compatibility with formal Noetherian Emerton–Gee stacks are residual inputs.

49.6 Dunn Additivity and E_1-Structure

PROPOSITION 49.6.1 *(Dunn additivity for the global Hecke category). (conditional on F2).* Given F2 (the perfectoid E_2-enhancement; see Theorem 43.3.1), the global Iwahori Hecke category $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$ is E_2-monoidal, and its Hochschild trace $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}})$ carries a canonical E_1-algebra structure obtained by Dunn additivity from the E_2-structure.

Proof. Dunn additivity (Lurie HA 5.1.2.2): for an E_2-monoidal stable presentable ∞-category C, $\mathrm{HH}_\bullet(C)$ is an E_1-algebra in stable presentable ∞-categories, with the E_1-structure induced by the E_2-monoidal convolution on C. The E_2-monoidal structure on $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$ is supplied by F2: the perfectoid BD Grassmannian with its two-dimensional factorization structure induces an E_2-monoidal structure on each local Hecke category (Theorem 43.3.1 (F2.a), (F2.b)), and the ∞-categorical Tate restricted tensor product (F2.c) assembles these into a global E_2-monoidal category.

Applied to $C = \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$, Dunn additivity gives the E_1-algebra structure on $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}})$. □

Remark 49.6.2 *(domain of F2-conditionality).* The conditionality on F2 is narrow: it enters only through the E_2-structure on the global Hecke category, which is precisely what F2 supplies. Without F2, the Hochschild trace is still defined (as a stable ∞-category) but its natural E_1-algebra structure requires the E_2-enhancement to avoid an ad hoc choice.

49.7 Main Theorem

THEOREM 49.7.1 *(F3.BZN^{Ar}, conditional on F1 and F2).* Conjectural statement. *(conditional on F1, F2, and the full-target/formal-loop comparison).* There should be a canonical equivalence of E_1-algebras

$$\Psi^{\mathrm{BZN,Ar}} : \mathrm{HH}_\bullet\big(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}\big) \xrightarrow{\sim} \mathcal{O}\big(\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}\big)^{\mathrm{flat}}.$$

Proof. The following is a conditional strategy; each step names its hypothesis explicitly.

Step 1: F1 identifies the Hecke category with $\mathrm{IndCoh}_{\mathrm{Nilp}}$ **of the Selmer-derived stack.** If F1 is proved after the local comparison and adelic descent residuals, it gives

$$\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}} \simeq \mathrm{D}([GL_2])^{\text{sph-fin}} \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}\big).$$

Here the first equivalence is the standard identification of the global Iwahori Hecke category with the sph-finite automorphic category (Bezrukavnikov arXiv:1209.0403 for local, lifted globally via restricted tensor product); the second is F1 itself.

Step 2: F2 supplies the E_2**-structure.** If F2 is proved after the local mixed-characteristic E_2 input and (F2.c): $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$ is E_2-monoidal, with the equivalence of Step 1 respecting the E_2-structure (both sides are E_2-monoidal and the equivalence is E_2-monoidal by functoriality of F1).

Step 3: Apply the arithmetic HH identity. If Theorem 49.5.5 is proved for $\mathcal{X} = \mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}$ with the required formal-Noetherian hypotheses, it yields

$$\mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}})\big) \;\simeq\; \mathcal{O}\big(L^{\mathrm{Ar}}\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}\big)^{\mathrm{flat}}.$$

Step 4: Full Weil/(φ,Γ) target comparison. The required comparison involves a morphism from the procyclic loop space to the full Weil/(φ,Γ) target. $\mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}})$ is the procyclic shadow of the arithmetic loop problem; the target $\mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})$ records only the inertia restriction, while the full Weil group data lives in $\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}$. The required morphism is:

$$L^{\mathrm{Ar}}\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}} \longrightarrow \mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2},$$

whose pullback on flat functions must identify $\mathcal{O}(L^{\mathrm{Ar}}\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}})^{\mathrm{flat}}$ with $\mathcal{O}(\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2})^{\mathrm{flat}}$. This is the full-target/formal-loop comparison named in the statement.

If Steps 1–4 are all supplied, concatenation gives:

$$\begin{aligned}\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}) &\overset{\mathrm{F1}}{\simeq} \mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}))\\ &\overset{\text{arith BZN}}{\simeq} \mathcal{O}(L^{\mathrm{Ar}}\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}})^{\mathrm{flat}}\\ &\simeq \mathcal{O}(\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2})^{\mathrm{flat}}.\end{aligned}$$

Each equivalence preserves the E_1-algebra structure (the first by F1 functoriality, the second by Theorem 49.5.5, the third by the still-open full-target/formal-loop comparison). The composition would be $\Psi^{\mathrm{BZN,Ar}}$.

The proof therefore leaves exactly the residuals listed in the statement: F1, F2, formal-loop continuity, and full-target pullback. □

Remark 49.7.2 *(Procyclic definition and conditional comparison).* The procyclic shadow is defined independently of F2. Given the E_2-structure required by F2, Dunn additivity gives the stated operadic consequence. The arithmetic Ben-Zvi–Nadler machinery, specifically Theorem 49.5.3, Theorem 49.5.5, Proposition 49.4.6, Proposition 49.4.9, remains conjectural until the full Weil/(φ,Γ) target and formal-loop continuity are established.

49.8 Consequence: F3 Closure under F1 + F2

Theorem 49.8.1 *(F3 theorem under F1 + F2).* Conjectural statement. Assume F1 (equivalently, $\mathrm{L}_{\mathrm{ACD}}$) and F2 (equivalently, (F2.c)). Then F3 (arithmetic Ben-Zvi–Nadler, Conjecture 42.3.1) has as its remaining input the corrected BZN comparison against the full Weil/(φ,Γ) target.

Proof. F3 asserts the E_1-algebra equivalence $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}) \xrightarrow{\sim} \mathcal{V}^{\mathrm{prim}}$. The displayed conjecture Theorem 49.7.1 would give the Hochschild side as functions on the full Weil/(φ,Γ) flat target. F1 would then identify the Emerton–Gee structure sheaf with $\mathcal{V}^{\mathrm{prim}}$. The corrected BZN comparison and the full-target pullback are the remaining inputs; their supply closes F3. □

THEOREM 49.8.2 *(identification of structure sheaf pullback with $\mathcal{V}^{\mathrm{prim}}$). (conditional on F1 and the full-target comparison).* The pullback of the structure sheaf on the full Weil/(φ, Γ) flat target along the Emerton–Gee comparison is expected to equal $\mathcal{V}^{\mathrm{prim}}$:

$$\iota^{\mathrm{EG},*}\mathcal{O} \;=\; \mathcal{V}^{\mathrm{prim}}.$$

Proof. By F1 (Conjecture 42.1.1), the spectral-side reading identifies $\mathcal{V}^{\mathrm{prim}}$ with the structure sheaf of the Emerton–Gee stack. The full-target comparison requires that the map from Emerton–Gee to the Weil/(φ, Γ) flat target has the required pullback on structure sheaves; the inertia-only map to $\mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})$ records only the procyclic restriction, and the full Weil group data must enter through the complete target. □

Dunn Additivity and Full-Target Pullback

Dunn additivity and E_1-structure.

Operadic input. Lurie HA 5.1.2 supplies Dunn additivity. Its use here requires the stated E_2-monoidal structure; it does not construct the perfectoid geometric input of F2.

Obstruction 2 (conventions). E_n-operad convention: Lurie HA (little cubes). E_2-monoidal category convention matches Fresse's operadic conventions via the Dunn–Lurie theorem.

Obstruction 3 (ambient categories). E_2-monoidal stable presentable ∞-categories in pro-étale derived geometry. Lurie HA applies unchanged in this ambient category.

Obstruction 4 (missing hypothesis). Stability and presentability of $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$ are supplied by Bezrukavnikov's mixed derived category at each prime (stable, presentable) plus the ∞-categorical Tate restricted tensor product preserving these properties (part of (F2.c)).

Obstruction 5 (false functoriality). Dunn additivity is a natural transformation of $(\infty, 2)$-functors, functorial in C.

Scope. The operadic conclusion uses the stated categorical structures.

The Dunn additivity condition closes under F2.

Full Weil/(φ, Γ) target and $\mathcal{V}^{\mathrm{prim}}$ pullback.

Pullback input. Pullback along closed immersions: Emerton–Gee arXiv:1908.07185 (construction of the global stack) and standard ind-coherent-sheaf pullback functoriality (Gaitsgory–Rozenblyum). F1 supplies the spectral-side reading of $\mathcal{V}^{\mathrm{prim}}$.

Obstruction 2 (conventions). Structure-sheaf pullback convention matches the standard adjunction pullback ι^* left adjoint to pushforward ι_*; in the ind-coherent setting (Gaitsgory–Rozenblyum) this is a shift-compatible convention.

Obstruction 3 (ambient categories). Ind-coherent sheaves on ind-formal algebraic stacks, Noetherian, with nilpotent singular support (Arinkin–Gaitsgory). Emerton–Gee stack satisfies all three constraints.

Obstruction 4 (missing hypothesis). The identification of $\mathcal{V}^{\mathrm{prim}}$ with the Emerton–Gee structure sheaf requires F1 (spectral-side reading). *hypothesis explicit as F1.*

Obstruction 5 (false functoriality). The pullback $\iota^{\mathrm{EG},*}$ is the standard functoriality, which preserves structure sheaves (trivially: $\iota^* \mathcal{O}_Y = \mathcal{O}_X$ for any morphism $\iota\colon X \to Y$).

Obstruction 6 (unproved equivalence). The identification of $\mathcal{V}^{\mathrm{prim}}$ with the Emerton–Gee structure sheaf is part of F1's content (spectral-side reading of the arithmetic Langlands equivalence sends the trivial automorphic side object to the structure sheaf). Classical for local Fargues–Scholze; F1 globalises. *Conditional on F1.*

The full-target pullback condition closes under F1.

49.9 Trace Constructions and Arithmetic Comparison Data

The comparison path for Theorem 49.7.1.

Factorization Homology and Categorical Traces

The derivation of Theorem 49.7.1 above uses:

- Francis arXiv:1206.5522 (factorization homology);
- Lurie HA 5.5.3 (Hochschild trace);
- Ben-Zvi–Francis–Nadler arXiv:0805.0157 (integral transforms);
- Emerton–Gee arXiv:1908.07185;
- Fargues–Scholze arXiv:2102.13459 (local equivalence);
- Bhatt–Scholze arXiv:1309.1198 (pro-étale);
- Toën DAG framework.

Arithmetic Topology and Hodge-Theoretic Comparisons

Related arithmetic constructions include:

- Mazur 2000 Selecta, Morishita arXiv:0904.3399 (MKR dictionary);
- Kapranov 1996 (arithmetic topology);
- Antieau–Nikolaus (TR, THH, arithmetic traces);
- Teleman 2020 (HKR in arithmetic);
- Bhatt–Morrow–Scholze Inv. Math. 218 (prismatic Nygaard);
- Scholze 2018 Forum Math. Pi 6 (p-adic Hodge theory);
- Costello 2007 PAMQ (A-model Hochschild trace).

Distinct Carriers and Required Maps

Remark 49.9.1 *(comparison of trace carriers).* The arithmetic-topology dictionary of Mazur–Kapranov–Reznikov–Morishita provides a topological analogy. Topological Hochschild, cyclic and restriction homology use equivariant spectra, while categorical traces and loop-space functions lie in their specified

derived categories. The constructions of Antieau–Nikolaus and Nikolaus–Scholze, Teleman's cohomological calculations, and the prismatic Nygaard comparison of Bhatt–Morrow–Scholze concern these different carriers. Using them together requires maps preserving the relevant coefficients, actions, filtrations and completion. A single-prime Hodge comparison such as that discussed in Remark 49.5.6 does not construct the global arithmetic Ben-Zvi–Nadler equivalence for the full Weil/(φ, Γ) target.

Comparison Sketch

Remark 49.9.2 *(the comparison path).* A non-formal comparison theorem would proceed as follows.

(i) *Teleman–Antieau–Nikolaus computation of* $\mathrm{HH}_\bullet$ *of the global Iwahori Hecke category via THH*$_\bullet$ of the Emerton–Gee stack regarded as a prismatic-Nygaard-polarised derived scheme. This path goes through the trace-methods framework of Nikolaus–Scholze, producing an E_1-algebra in genuine S^1-equivariant spectra, whose underlying object matches the factorization-homology model of Theorem 49.5.3.

(ii) *MKR-dictionary match of the arithmetic loop space.* From the topological analogy, the arithmetic free loop space The full-target arithmetic loop object for $\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}$ corresponds to the loop space of the 3-manifold analogue of the Emerton–Gee stack; Morishita's dictionary (arXiv:0904.3399 Ch. 3) reads off the correct moduli of local-system pairs.

(iii) *Comparison of Bhatt–Morrow–Scholze prismatic-Nygaard trace with the* E_1*-algebra* $\mathcal{O}(\mathrm{LocSys}^{\mathrm{Ar}})^{\mathrm{flat}}$. Both sides give an E_1-algebra in S^1-equivariant spectra; the match is a prismatic-Nygaard compatibility, encoded as the Bhatt–Lurie filtration of arXiv:2201.06120.

These local descriptions concern the stated carriers. Comparing them with the arithmetic trace of Theorem 49.7.1 requires maps preserving the cyclic action, products and filtrations; the descriptions alone do not supply those maps.

49.10 Residual Conditions

The independent comparison leaves F3 with genuine residual conditions. Besides F1 and F2, the proof needs the corrected arithmetic BZN comparison against the full Weil/(φ, Γ) target, the profinite formal-loop continuity theorem, and the flat-locus pullback comparison. Thus F3.BZN^{Ar} remains a residual of Section 43.9.

Remark 49.10.1 *(residual F3 ingredients).* The open ingredients are:

- the local categorical comparison (F1.a) and $\mathrm{L}_{\mathrm{ACD}}$ (F1's adelic categorical descent);
- the mixed-characteristic local E_2 Hecke enhancement and (F2.c) (∞-categorical Tate restricted tensor product);
- profinite formal-loop continuity for $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$;
- the full Weil/(φ, Γ) target comparison and flat-locus pullback;
- arithmetic Positselski (from P1's Step 3);
- Koszul–Petersson compatibility (from P3).

The first four are number-field adaptations of function-field geometric Langlands and arithmetic loop-space technology. The last two are bar-cobar-structural conditions internal to the arithmetic Heisenberg.

Domain: Archimedean Place

Remark 49.10.2 *(archimedean contribution).* Theorem 49.7.1 treats only the finite-place inertia (Remark 49.3.4). The archimedean contribution to $\mathcal{V}^{\mathrm{prim}}$ is the complex Heisenberg Fock on the archimedean fibre (Theorem 43.7.1 archimedean-place discussion), whose Hochschild trace is classical Ben-Zvi–Nadler over $\mathbb{C}$. The arithmetic-to-classical reduction at the archimedean place is exactly the classical Ben-Zvi–Nadler theorem Theorem 49.5.1, which is a theorem of 2009. No new work is needed for the archimedean fibre.

Domain: Non-GL_2 Groups

Remark 49.10.3 *(extension to GL_n).* The machinery extends to $G = GL_n$ for any $n \geq 1$, since:

- The definition of $B\widehat{\mathbb{Z}}$ and arithmetic free loop space is group-independent.
- Local Emerton–Gee stacks are defined for all n in Emerton–Gee arXiv:1908.07185; the global spectral target must still be Selmer-derived.
- Fargues–Scholze local geometrization and spectral action are available for GL_n (arXiv:2102.13459 Thm. I.9); the categorical comparison remains a separate input.

The only GL_2-specific input is F1 ($\mathrm{L}_{\mathrm{ACD}}$ is currently formulated for $GL_2/\mathbb{Q}$), which is a number-theoretic arithmetic-Langlands conjecture. With F1 upgraded to GL_n, Theorem 49.7.1 extends directly. The rank-2 structure is not used essentially.

49.11 Ben-Zvi–Nadler Boundary

(i) The arithmetic inertia circle $B\widehat{\mathbb{Z}}$ is well-defined as a pro-étale derived stack, has pro-étale fundamental group $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, and is the arithmetic analogue of S^1 under the MKR dictionary (Proposition 49.3.3).

(ii) The arithmetic free loop space $L^{\mathrm{Ar}}\mathcal{X} = \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ is a derived pro-étale stack for any n-geometric derived pro-étale stack $\mathcal{X}$ (Definition 49.4.1, Remark 49.4.3). For $\mathcal{X} = BGL_2$, $L^{\mathrm{Ar}}BGL_2 = \mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})$ (Proposition 49.4.4, Corollary 49.4.5).

(iii) The Emerton–Gee stack has an expected comparison to the full Weil/(φ, Γ) flat target; its restriction to $\mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})$ is a forgetful procyclic shadow; whether the restriction is a closed immersion is an open question (Proposition 49.4.6, Proposition 49.4.9).

(iv) Arithmetic factorization homology is conjectured to give $\mathrm{HH}_\bullet(C) = \int_{B\widehat{\mathbb{Z}}} C$ for any E_2-monoidal stable presentable ∞-category C in pro-étale derived geometry (Theorem 49.5.3).

(v) The core arithmetic HH identity Theorem 49.5.5 gives $\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})) = \mathcal{O}(L^{\mathrm{Ar}}\mathcal{X})^{\mathrm{flat}}$ for any Noetherian derived formal algebraic pro-étale stack $\mathcal{X}$; this remains conditional for formal Noetherian pro-étale stacks and requires the R1.formal.b/R1.formal.c inputs.

(vi) Dunn additivity gives the E_1-algebra structure on the Hochschild trace (Proposition 49.6.1), conditional on F2.

(vii) The main comparison Theorem 49.7.1 is conjectural and must target $\mathrm{LocSys}^{W,\varphi^\Gamma}_{GL_2}$, not the inertia-only shadow.

(viii) F3 remains conditional after F1 and F2 (Theorem 49.8.1); the residual is the corrected BZN comparison, formal-loop compatibility, and full-target pullback.

(ix) The arithmetic-topology dictionary, topological Hochschild invariants, cohomological calculations and prismatic Nygaard comparisons involve distinct carriers. Their use in the main theorem requires the specified structure-preserving maps (Remark 49.9.1, Remark 49.9.2).

(x) The machinery extends to GL_n for arbitrary $n \geq 1$ with only F1 as the group-specific input (Remark 49.10.3).

The six-window presentation of $\mathcal{V}^{\mathrm{prim}}$ (Chapter 41) remains conditional. F1 and F2 reduce F3 to a sharper residual: construct the corrected arithmetic BZN equivalence over the full Weil/(φ, Γ) target and prove the profinite formal-loop and flat-locus comparisons.

Chapter 50

The ∞-Categorical Tate Restricted Tensor Product

The restricted-tensor assertion F2.c is a formal assembly theorem. Its inputs are stable presentable categories with specified operadic structures and the actual pointing maps of Setup 50.1.1(D2). Finite tensor products of these maps form an operadic diagram. HA 3.2.3.1 constructs its filtered colimit on the underlying categorical colimit. No arithmetic input is used in that construction.

Applying it to arithmetic Iwahori categories still requires the local E_2-structures and the specified pointing maps. Neither an invariant vector nor a central idempotent alone supplies those maps. The assembly theorem does not construct geometric collision comparisons or an identification with an independently defined global Hecke category.

50.1 Statement of (F2.c)

Setup and Definition

Fix a small one-coloured ∞-operad $\mathcal{O}^{\otimes}$ (the chief cases are $\mathcal{O} = E_n$ for $n \geq 1$). Fix an indexing set V of *places*: in the target application $V = |\overline{C}|$ is the set of places of $\overline{\mathrm{Spec}\,\mathbb{Z}}$, but the statement is place-agnostic.

Write $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ for the ∞-category of stable presentable ∞-categories and colimit-preserving functors. Its tensor product represents functors preserving colimits in each input. The tensor unit is Sp, the ∞-category of spectra, and $\mathbb{S} \in \mathrm{Sp}$ is the sphere spectrum. In particular, an empty tensor of categories means Sp.

Setup 50.1.1 *(F2.c data).*

(D1) For each $v \in V$, a stable presentable ∞-category $C_v \in \mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ equipped with an $\mathcal{O}$-monoidal structure compatible with colimits in each variable: $C_v^{\otimes} \to \mathcal{O}^{\otimes}$ exhibits C_v as an $\mathcal{O}$-algebra in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$.

(D2) A cofinite subset $V^{\mathrm{sph}} \subseteq V$ and, for each $v \in V^{\mathrm{sph}}$, a specified morphism

$$\eta_v : \mathrm{Sp} \longrightarrow C_v \quad \text{in} \quad \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}).$$

Write $u_v = \eta_v(\mathbb{S})$. The morphism includes compatibility with every operation, including nullary operations, and all their coherences. Throughout this chapter, pointed spherical data means these maps, not merely their values on $\mathbb{S}$.

In the unital E_n cases, $n \geq 1$, necessarily $u_v \simeq \mathbb{1}_{C_v}$. A non-unit central idempotent under $\mathbb{1}_{C_v}$ does not give such a map into C_v. A construction using it would require a different unital target or a different notion of morphism.

Write $\mathrm{Fin}(V)$ for the filtered poset of finite subsets of V ordered by inclusion, and $\mathrm{Fin}^{\mathrm{sph}}(V)$ for the cofinal subposet consisting of finite $S \subseteq V$ with $V \setminus S \subseteq V^{\mathrm{sph}}$.

Definition 50.1.2 *(Tate restricted tensor product).* Under Setup 50.1.1, the *Tate restricted tensor product* of $\{C_v\}_v$ with pointed spherical data $\{(\eta_v, u_v)\}_{v\in V^{\mathrm{sph}}}$ is the colimit

$$\bigotimes_{v\in V}{}'(C_v, u_v) \;:=\; \varinjlim_{S\in \mathrm{Fin}^{\mathrm{sph}}(V)} \Big(\bigotimes_{v\in S} C_v\Big) \tag{50.1}$$

in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$. The transition morphism for $S \subseteq S'$ uses the pointing at each $v \in S' \setminus S$:

$$\iota_{S\subseteq S'} \;:\; \bigotimes_{v\in S} C_v \xrightarrow{\mathrm{id}\otimes \bigotimes_{v\in S'\setminus S}\eta_v} \bigotimes_{v\in S'} C_v, \qquad X \mapsto X \otimes \Big(\bigotimes_{v\in S'\setminus S} u_v\Big). \tag{50.2}$$

The right-hand side uses the Lurie tensor product in Pr^{L} of HA 4.8.1. It restricts to a symmetric monoidal structure on $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ by HA 4.8.2.18.

THEOREM 50.1.3 *(F2.c: ∞-categorical Tate restricted $\mathcal{O}$-tensor).* Fix a small coherent ∞-operad $\mathcal{O}^{\otimes}$. This includes E_n for $n \geq 1$ and E_∞. Under Setup 50.1.1, the restricted tensor product (50.1) carries a canonical $\mathcal{O}$-monoidal structure such that:

(M1) For each $S \in \mathrm{Fin}^{\mathrm{sph}}(V)$, the canonical map

$$\iota_S \;:\; \bigotimes_{v\in S} C_v \;\longrightarrow\; \bigotimes_{v\in V}{}'(C_v, u_v)$$

is an $\mathcal{O}$-monoidal functor.

(M2) If $V \setminus \{v\} \subseteq V^{\mathrm{sph}}$, the canonical functor $C_v \to \otimes'_{w\in V}(C_w, u_w)$ is $\mathcal{O}$-monoidal. For a general v, the same construction requires specified $\mathcal{O}$-algebra maps $\mathrm{Sp} \to C_w$ at the finitely many remaining places $w \in V \setminus (V^{\mathrm{sph}} \cup \{v\})$.

(M3) Fix the full diagram of finite tensor products and specified pointing maps, and fix its underlying colimit cocone in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$. The space of lifts of this entire cocone, including all higher coherences, to a colimit cocone in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$ is contractible.

Applied with $\mathcal{O} = E_2$ and $C_v = \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,v}$ and the pointing maps of (D2), this proves the formal transport step (F2.c) in Theorem 43.3.1. Both the local Iwahori E_2-structures and any identification $|0\rangle_v \simeq \eta_v(\mathbb{S})$ remain required inputs.

Tensor Products and Factorization Homology

The tensor-product construction of Theorem 50.1.3 uses Lurie HA 4.8 and the integral E_n-operads discussed by Fresse. Dunn additivity (HA 5.1.2.4; Dunn 1988), Ayala–Francis factorization homology, and Ginot's account of factorization algebras provide related descriptions of the monoidal structures (Section 50.12). Applying these descriptions requires the same units, coefficient category and colimit conventions.

50.2 Obstructions and Their Resolutions

Six questions isolate the ingredients of Theorem 50.1.3: operadic insertion, filtered colimits, discrete factorization, additivity, cocone uniqueness and the cofinite pointing condition.

The Pointing Required for Unit Insertion

Obstruction 50.2.1 *(operadic insertion).* Does a distinguished object u_v, or a central idempotent under the unit, suffice to make insertion an $\mathcal{O}$-monoidal functor?

Correction 50.2.2 *(actual pointing maps).* The required datum is the morphism η_v in Setup 50.1.1(D2). Tensoring these morphisms with identities gives the insertion functors, including their nullary and higher compatibilities. For unital E_n, a strong E_n-monoidal functor preserves the unit, so its value on $\mathbb{S}$ cannot be a non-unit central idempotent. For a nonzero field k, the object $(k, 0)$ of $\mathcal{D}(k \times k)$ is compact, central and idempotent under the unit (k, k). Insertion of $(k, 0)$ from Sp is not unital monoidal. Lemma 50.4.1 uses the specified maps, not an implication from central idempotence.

$\mathcal{O}$-Algebra Structure under Filtered Colimits

Obstruction 50.2.3 *(filtered colimit of $\mathcal{O}$-algebras).* (50.1) is a filtered colimit in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$. Does the $\mathcal{O}$-algebra structure propagate through such a colimit?

Correction 50.2.4 *(colimit preservation).* Lurie HA 3.2.3.1, applied to the ∞-operad $\mathcal{O}^{\otimes}$ and the symmetric monoidal ∞-category $(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}, \otimes)$, which admits filtered colimits preserved by tensoring in each variable: the forgetful functor $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}) \to \mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ preserves filtered colimits. Hence the colimit $\varinjlim_S (\otimes_{v \in S} \mathcal{C}_v)$ taken in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$ coincides with the colimit in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$, and the $\mathcal{O}$-algebra structure is preserved. Detailed in Proposition 50.5.1.

Factorization for Disjoint Subsets of V

Obstruction 50.2.5 *(factorization axiom).* Does the finite tensor diagram underlying (50.1) respect disjoint unions of place labels, and what geometric data does this supply?

Correction 50.2.6 *(discreteness of V).* The associator identifies the tensor for $S \sqcup T$ with the tensor of the two separate values whenever $S \cap T = \varnothing$. Spherical-unit insertions respect these identifications on the inclusions for which they are defined. This is the discrete factorization of Corollary 30.2.2. It specifies no position diagonal or normal coefficient. A comparison with a geometric factorization system must supply those data separately.

Dunn Additivity and the Restricted Tensor Product

Obstruction 50.2.7 *(Dunn additivity compatibility).* For $\mathcal{O} = E_{n_1}$ and $\mathcal{O}' = E_{n_2}$ with $n_1 + n_2 = n$, Dunn additivity (HA 5.1.2.4) gives $E_{n_1} \otimes E_{n_2} \simeq E_{n_1+n_2}$, with the tensor product of ∞-operads. The restricted tensor product of E_{n_1}-categories composed with the restricted tensor product of E_{n_2}-categories should give the restricted tensor product of $E_{n_1+n_2}$-categories. Does Theorem 50.1.3 respect this composition?

Correction 50.2.8 *(operad-variable functoriality).* Dunn additivity concerns the operad E_n itself (HA 5.1.2.4). For an operad product $\mathcal{O} \otimes \mathcal{O}'$, the restricted-tensor construction agrees with its iterated form. Both supplied structures and their interchange coherences must be retained. The corresponding algebra equivalence identifies the finite tensor diagrams and their filtered colimits. Proposition 50.7.6 gives the comparison for the little-cubes operads.

Cofinality Invariance of the $\mathcal{O}$-Structure

Obstruction 50.2.9 *(uniqueness modulo cofinal filtration).* The proof of Theorem 50.1.3(M3) claims uniqueness of the $\mathcal{O}$-monoidal structure up to contractible choice. Does this uniqueness hold unconditionally, or only modulo the choice of cofinal filtration $\mathrm{Fin}^{\mathrm{sph}}(V)$?

Correction 50.2.10 *(cofinality invariance).* A cofinal replacement of the filtered indexing category does not change the colimit of its operadic diagram. The resulting equivalence is the one compatible with the complete colimit cocones. Colimit cocones are initial objects in the appropriate cocone category, so their space of choices is contractible. This uses the universal property of colimits, not an ∞-topos structure on Cat_∞. See Proposition 50.9.1.

Ramified Places without a Spherical Unit

Obstruction 50.2.11 *(ramified archimedean places).* Setup 50.1.1 asks for a distinguished spherical unit u_v at *almost all* places $v \in V$. At a finite number of ramified places (for $GL_2/\mathbb{Q}$ typically the archimedean place and a finite set of primes where the Iwahori level is nontrivial), no spherical unit need exist. Does Theorem 50.1.3 still apply?

Correction 50.2.12 *(cofinite spherical locus).* Every $S \in \mathrm{Fin}^{\mathrm{sph}}(V)$ contains the finite complement $V \setminus V^{\mathrm{sph}}$. Thus every newly added place has a specified pointing map, and the finite-stage diagram is defined without choosing pointing maps at the remaining places. This does not supply every singleton map: the precise condition is (M2). An archimedean invariant vector does not by itself verify (D2). An actual operadic pointing is still required before that place can be inserted.

50.3 ∞-Operadic Preliminaries

The proof of Theorem 50.1.3 draws on three ingredients from Lurie HA and Fresse, recorded here with their precise references. Numbering follows the 2017 version of Lurie's *Higher Algebra*.

Symmetric Monoidal Structure on $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$

THEOREM 50.3.1 *(HA 4.8.1.15, HA 4.8.2.18).* The ∞-category Pr^{L} of presentable ∞-categories and colimit-preserving functors carries a canonical symmetric monoidal structure. Its tensor product is characterized by the universal property

$$\mathrm{Fun}^{\mathrm{L}}(\mathcal{C} \otimes \mathcal{D}, \mathcal{E}) \;\simeq\; \mathrm{Fun}^{\mathrm{L,L}}(\mathcal{C} \times \mathcal{D}, \mathcal{E})$$

(colimit-preserving functors of tensor products = bi-colimit-preserving functors of products). The full subcategory $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ of stable presentable ∞-categories is closed under this tensor product and carries a canonical symmetric monoidal structure with unit Sp (the ∞-category of spectra).

COROLLARY 50.3.2 *(HA 4.8.1.24 and HA 4.8.2.18).* The ∞-category $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ admits small colimits, and the tensor product ⊗ preserves colimits separately in each variable. In particular, ⊗ preserves filtered colimits in each variable.

Algebras over an ∞-Operad in a Presentable Symmetric Monoidal ∞-Category

THEOREM 50.3.3 *(HA 3.2.3.1 and 3.2.3.5).* For a small coherent ∞-operad $\mathcal{O}^{\otimes}$ and a presentable symmetric monoidal ∞-category $\mathcal{E}^{\otimes}$ with $\otimes_{\mathcal{E}}$ preserving colimits separately in each variable:

(i) $\mathrm{Alg}_{\mathcal{O}}(\mathcal{E})$ is presentable;
(ii) the forgetful functor $\mathrm{Alg}_{\mathcal{O}}(\mathcal{E}) \to \mathcal{E}$ preserves sifted colimits (and in particular filtered colimits).

Stability of $\mathcal{E}$ does not imply stability of its algebra category. For example, for a nonzero field k, the initial object of $\mathrm{Alg}_{E_1}(\mathcal{D}(k))$ is k and its terminal object is the zero algebra. They are not equivalent, so this category is not pointed and cannot be stable.

For the restricted tensor product, we need only filtered colimits. Corollary 50.3.2 supplies these in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$, compatibly with tensor products. Proposition 50.3.7 then constructs the colimit of the algebra diagram on its underlying colimit. This application uses no presentability assertion about $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ itself.

E_n-Operad and Integral Structure

Theorem 50.3.4 *(Fresse 2017* Homotopy of Operads *I.4.1; Lurie HA 5.1.0.4).* For each $n \geq 0$, let $E_n^{\otimes}$ be the little n-cubes ∞-operad. Its k-ary operations are parametrized by the space $\mathrm{Conf}_k(\mathbb{R}^n)$ of configurations of k ordered little n-cubes in the unit n-cube. This ∞-operad is coherent in the sense of HA 3.3.1.1. The standard inclusion $E_n^{\otimes} \to E_{n+1}^{\otimes}$ induces restriction from E_{n+1}-algebras to E_n-algebras. The relation with iterated algebra structures is the Dunn additivity equivalence below, not a forgetful morphism $E_{n+1} \to E_n$.

Theorem 50.3.5 *(Dunn additivity; HA 5.1.2.4; Dunn 1988 JPAA 50).* For any $n_1, n_2 \geq 0$,

$$E_{n_1}^{\otimes} \otimes E_{n_2}^{\otimes} \;\simeq\; E_{n_1+n_2}^{\otimes}$$

as ∞-operads, where $\otimes$ is the tensor product of ∞-operads (HA 2.2.5.6). In particular, an $E_{n_1+n_2}$-algebra in $\mathcal{E}$ is the same datum as an E_{n_1}-algebra in E_{n_2}-algebras in $\mathcal{E}$.

Compatibility of a Specified Pointing

Theorem 50.3.6 *(operadic pointing compatibility).* Let $\eta : \mathrm{Sp} \to C$ be a morphism in $\mathrm{Alg}_O(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$. For every O-operation, its compatibility with η, together with all substitution coherences, is part of this morphism. For unital E_n, $n \geq 1$, its nullary compatibility gives $\eta(\mathbb{S}) \simeq \mathbb{1}_C$.

Proof. A morphism of operadic algebras commutes with the operation maps and their coherences. The nullary operation of the canonical unital E_n-algebra Sp is the identity $\mathrm{Sp} \to \mathrm{Sp}$. That of C is its unit functor. Compatibility identifies η with that unit functor and gives the stated equivalence on $\mathbb{S}$. □

This is the pointing compatibility used in Lemma 50.4.1. It asserts no mapping-space identity involving the identity functor as an object of C.

Filtered Colimits of ∞-Operadic Algebras

Proposition 50.3.7 *(HA 3.2.3.1).* Let $O^{\otimes}$ be a small one-coloured coherent ∞-operad, and let $\mathcal{E}^{\otimes}$ be a symmetric monoidal ∞-category admitting small filtered colimits. Suppose $\otimes_{\mathcal{E}}$ preserves these colimits in each variable. Then the forgetful functor

$$U_O \;:\; \mathrm{Alg}_O(\mathcal{E}) \;\longrightarrow\; \mathcal{E}$$

preserves and creates filtered colimits. Concretely: a filtered diagram $F : I \to \mathrm{Alg}_O(\mathcal{E})$ has colimit whose underlying object in $\mathcal{E}$ is $\varinjlim_i U_O F(i)$, with O-algebra structure induced levelwise.

This is the mechanism of colimit-preservation (§50.5).

50.4 Unital Inclusion Functors

Inserting a spherical unit at a new place gives an O-monoidal functor; the following lemma makes this precise.

LEMMA 50.4.1 *(unit insertion is $\mathcal{O}$-monoidal).* Let $\mathcal{O}^{\otimes}$ be a coherent ∞-operad, and let $C_1, \dots, C_k$ and $C_{k+1}, \dots, C_m$ be $\mathcal{O}$-algebras in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ for some $k \le m$. For $i > k$ choose a spherical unit $u_i \in C_i$ in the sense of Setup 50.1.1(D2). Then the functor

$$\iota_{\{1,\dots,k\}\subseteq\{1,\dots,m\}} \;:\; \bigotimes_{i=1}^{k} C_i \longrightarrow \bigotimes_{i=1}^{m} C_i$$

defined by inserting u_i at position i for $i > k$ is an $\mathcal{O}$-monoidal functor.

Proof. The symmetric monoidal tensor product on $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$ is induced by the ambient tensor product. The insertion is the morphism

$$\bigotimes_{i\le k} C_i \simeq \Big(\bigotimes_{i\le k} C_i\Big) \otimes \mathrm{Sp}^{\otimes(m-k)} \xrightarrow{\mathrm{id}\otimes\bigotimes_{i>k}\eta_i} \bigotimes_{i\le m} C_i.$$

Every η_i is a specified morphism of $\mathcal{O}$-algebras, so this is also such a morphism. In particular, writing A and B for the source and target, respectively, each operation $c \in \mathcal{O}(r)$ satisfies

$$\iota \circ \mu_c^{A} \simeq \mu_c^{B} \circ \iota^{\otimes r}.$$

For $r = 0$ this is unit compatibility. The higher compatibilities are inherited from the supplied morphisms and the symmetric monoidal structure, rather than chosen independently for each operation. Tensor associativity supplies the compatible composites for every chain of finite subsets. □

COROLLARY 50.4.2 *(restriction of ι).* Assume the hypotheses of Lemma 50.4.1. For every sub-triangle $\{1,\dots,k\} \subseteq \{1,\dots,k'\} \subseteq \{1,\dots,m\}$ the composite $\iota_{\{1,\dots,k\}\subseteq\{1,\dots,m\}}$ equals $\iota_{\{1,\dots,k'\}\subseteq\{1,\dots,m\}} \circ \iota_{\{1,\dots,k\}\subseteq\{1,\dots,k'\}}$ as $\mathcal{O}$-monoidal functors under the comparison supplied by tensor associativity and the fixed pointing maps. These comparisons are coherent for longer chains.

Proof. Both sides are tensor products of the same identities and supplied maps η_i at the same positions. The associativity and unit coherences of the symmetric monoidal algebra category identify these composites, compatibly with longer chains. This is the specified coherent comparison, not an assertion that arbitrary homotopies between the underlying functors form a contractible space. □

Remark 50.4.3 *(two cells of the full diagram system).* Lemma 50.4.1 together with Corollary 50.4.2 upgrades the system $\{\iota_{S\subseteq S'}\}_{S\subseteq S'}$ to a diagram

$$\Phi_{\{C_v,u_v\}} \;:\; \mathrm{Fin}^{\mathrm{sph}}(V) \longrightarrow \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$$

covariant in the filtered poset ordered by inclusion. Its map for $S \subseteq S'$ inserts the specified spherical units in $S' \setminus S$. This is the diagram on which Proposition 50.5.1 will take colimits.

50.5 COLIMIT PRESERVATION

PROPOSITION 50.5.1 *($\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ of $\mathcal{O}$-categories is cocomplete under filtered colimits).* Let $\mathcal{O}^{\otimes}$ be a small one-coloured coherent ∞-operad. For every small filtered ∞-category I and every functor $\Phi : I \to \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$, the colimit $\varinjlim_i \Phi(i)$ exists in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$, and its underlying presentable stable ∞-category

is $\varinjlim_i (U_{\mathcal{O}}\Phi(i))$ in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$. Every $\mathcal{O}$-operation on the colimit is computed levelwise: given $c \in \mathcal{O}(r)$, the operation

$$\mu_c^{(\mathrm{colim})} \;:\; \prod_{j=1}^{r}\Big(\varinjlim_i \Phi(i)\Big) \;\longrightarrow\; \varinjlim_i \Phi(i)$$

is the filtered colimit of the family $\{\mu_c^{(\Phi(i))}\}_i$.

Proof. Put $C_i = U_{\mathcal{O}}\Phi(i)$ and $C = \varinjlim_i C_i$ in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$. Corollary 50.3.2 gives this colimit and preservation by tensoring. Hence Proposition 50.3.7, with $\mathcal{E} = \mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$, gives an algebra colimit whose underlying object is C.

The operations can be seen directly. For $r > 0$, the diagonal $I \to I^r$ is cofinal because I is filtered. Therefore

$$C^{\otimes r} \simeq \varinjlim_{(i_1,\dots,i_r)\in I^r} C_{i_1}\otimes\cdots\otimes C_{i_r} \simeq \varinjlim_{i\in I} C_i^{\otimes r}.$$

For $c \in \mathcal{O}(r)$, take the colimit of $\mu_c^{(i)} : C_i^{\otimes r} \to C_i$. The resulting map is the displayed operation on C, viewed as a functor preserving colimits in each input. Nullary operations pass through the coherent cocone from Sp; the nerve of a filtered category is contractible. Operadic composition and its homotopies pass through these same colimit comparisons. Thus the construction retains the full algebra structure, as asserted by HA 3.2.3.1. □

Corollary 50.5.2 *(restricted tensor product as colimit in* $\mathrm{Alg}_{\mathcal{O}}$*).* The diagram $\Phi_{\{C_v,u_v\}} : \mathrm{Fin}^{\mathrm{sph}}(V) \to \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$ of Remark 50.4.3 admits a colimit in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$, whose underlying presentable stable ∞-category is $\otimes'_{v\in V}(C_v, u_v)$ of Definition 50.1.2.

Proof. $\mathrm{Fin}^{\mathrm{sph}}(V)$ is a filtered ∞-poset (union of finite subsets is again finite; the empty set, or any initial element of $\mathrm{Fin}^{\mathrm{sph}}(V)$, gives non-emptiness). Apply Proposition 50.5.1. □

50.6 Proof of the Main Theorem

Proof of Theorem 50.1.3. **Step 1 (construction of the diagram).** Given the data of Setup 50.1.1, construct the diagram

$$\Phi \;:\; \mathrm{Fin}^{\mathrm{sph}}(V) \;\longrightarrow\; \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}), \qquad \Phi(S) \;:=\; \bigotimes_{v\in S} C_v$$

with transition maps $\iota_{S\subseteq S'}$ of (50.2). By Lemma 50.4.1, each $\iota_{S\subseteq S'}$ is an $\mathcal{O}$-monoidal functor. For every chain of inclusions, each composite tensors with the same supplied maps η_v at the same positions. Coherence of the tensor product gives all composition homotopies and their compatibilities; the triangular identity is Corollary 50.4.2. Thus Φ is a functor from the filtered ∞-poset $\mathrm{Fin}^{\mathrm{sph}}(V)$ into $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$.

Step 2 (existence of the $\mathcal{O}$-structured colimit). By Corollary 50.5.2, the colimit $\varinjlim_S \Phi(S)$ exists in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$, and its underlying object in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ is $\otimes'_{v\in V}(C_v, u_v)$ by Definition 50.1.2. This supplies the $\mathcal{O}$-monoidal structure promised in Theorem 50.1.3.

Step 3 (property (M1): each ι_S is $\mathcal{O}$-monoidal). The canonical map $\iota_S : \Phi(S) \to \varinjlim_{S'} \Phi(S')$ is a morphism in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$ by the universal property of the colimit applied inside the ∞-category of $\mathcal{O}$-algebras. Hence ι_S is $\mathcal{O}$-monoidal. This is (M1).

Step 4 (property (M2): a single-place map). If $V \setminus \{v\} \subseteq V^{\mathrm{sph}}$, the singleton $\{v\}$ belongs to $\mathrm{Fin}^{\mathrm{sph}}(V)$, so Step 3 applies directly. Otherwise put $E = V \setminus V^{\mathrm{sph}}$ and choose the additional maps in (M2). They give an $\mathcal{O}$-algebra map

$$C_v \longrightarrow \bigotimes_{w \in E \cup \{v\}} C_w \xrightarrow{\iota_{E\cup\{v\}}} \bigotimes'_{w \in V} (C_w, u_w).$$

The first arrow tensors with the chosen maps at $E \setminus \{v\}$; the second is supplied by (M1). Passing to a larger admissible finite set inserts only the original spherical-unit maps, so tensor coherence and the colimit cocone identify the composites. The resulting functor depends on the extra choices. Full faithfulness is not part of (M2).

Step 5 (property (M3): uniqueness up to contractible space of choices). By Step 2, the structured colimit is initial among cocones under Φ in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$. Here compatibility with (ι_S) includes the whole coherent cocone. Let another such structure be given on the fixed underlying colimit. Initiality supplies a contractible space of maps to its cocone. After forgetting the algebra structure, each is the equivalence determined by the two colimit cocones. The forgetful functor detects equivalences, so the structured map is itself an equivalence. Consequently every compatible structured cocone is initial, and the space of them is contractible. This proves (M3).

Steps 1–5 together prove Theorem 50.1.3. □

Remark 50.6.1 *(categorical content).* The proof displays the ∞-categorical Tate restricted tensor product as a filtered colimit in the ∞-category $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$. This is the ∞-categorical analogue of the classical Tate restricted direct product of locally compact abelian groups:

$$\prod'_{v \in V} G_v \;:=\; \varinjlim_{S \text{ finite}} \prod_{v \in S} G_v \times \prod_{v \notin S} K_v$$

with $K_v \subseteq G_v$ compact-open, via filtered union over finite ramification sets. Here a transition uses the inclusions $K_v \hookrightarrow G_v$ at the newly unrestricted places; the categorical tensor instead inserts the specified objects u_v. The common filtered indexing does not identify these different transition maps or their colimits.

50.7 Discrete Factorization and Geometric Comparison

Finite tensor products carry coherent decomposition maps for disjoint sets of place labels. We construct these maps together with the allowed unit insertions. A comparison with geometric factorization must also specify the coefficients and maps at colliding positions.

Factorization on Finite Sets of Places

Setup 50.7.1 *(finite subsets and multisets of V).* For the set V, let $\mathrm{Fin}_+(V)$ be its nonempty finite subsets. The set of nonempty finite multisets is instead $\coprod_{n\geq 1} V^n/\mathfrak{S}_n$. Forgetting multiplicities maps the latter onto the former. The diagram below is indexed by finite subsets, including the empty set, with explicitly specified arrows; no transition between symmetric powers is needed.

Definition 50.7.2 *(Discrete factorization with specified unit insertions).* Let $\mathrm{Fin}^{u}(V)$ have all finite subsets as objects, with an arrow $S \to T$ when $S \subseteq T$ and $T \setminus S \subseteq V^{\mathrm{sph}}$. Thus only places with a specified spherical-unit map may be inserted. A discrete factorization datum in the category of $\mathcal{O}$-algebras in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ is a diagram

$$\mathcal{A} \;:\; \mathrm{Fin}^{u}(V) \;\longrightarrow\; \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$$

satisfying the factorization axiom: for every pair of disjoint finite subsets $S_1, S_2 \subset V$,

$$\mathcal{A}(S_1 \sqcup S_2) \;\simeq\; \mathcal{A}(S_1) \otimes \mathcal{A}(S_2)$$

as O-algebras, coherently associative and symmetric, and natural in the allowed inclusions of disjoint subsets.

Theorem 50.7.3 *(Finite tensors give discrete factorization).* Under Setup 50.1.1, the finite tensors define a discrete factorization datum in the sense of Definition 50.7.2. Its restriction to $\mathrm{Fin}^{\mathrm{sph}}(V)$ is the diagram $\Phi_{\{C_v, u_v\}}$ of Remark 50.4.3. Its value on $S = \{v_1, \dots, v_k\}$ is $\bigotimes_{i=1}^{k} C_{v_i}$, and on a disjoint union $S_1 \sqcup S_2$ is $\mathcal{A}(S_1) \otimes \mathcal{A}(S_2)$ by associativity of the tensor product.

Proof. For each allowed inclusion, insert the maps η_v at the new places. Composing two inclusions inserts the same maps at the same places, giving the diagram. Associativity of the finite tensor identifies the value on a disjoint union with the tensor of its two values. Tensor coherence proves all refinements and permutations of this identification. On $\mathrm{Fin}^{\mathrm{sph}}(V)$, these maps are precisely the transitions defining Φ. □

Corollary 50.7.4 *(Filtered assembly of the finite tensors).* Under the hypotheses of Theorem 50.1.3,

$$\bigotimes_{v \in V}{}' (C_v, u_v) \;=\; \varinjlim_{S \in \mathrm{Fin}^{\mathrm{sph}}(V)} \Phi(S).$$

The colimit assembles the discrete factorization datum of Theorem 50.7.3 along its spherical-unit insertions. Identifying it with sections or homology of a geometric Ran object requires a separate comparison.

Remark 50.7.5 *(Position geometry is additional).* The discrete construction proves the disjoint tensor identities and their unit-insertion coherences. It does not compute the coefficient on a position diagonal. The formal-disc and derived closed-point models of Proposition 2.8.1 retain different normal data at a finite place. A geometric arithmetic realization must select and compare its local models, including the archimedean one in Remark 30.6.2.

Compatibility with Dunn Additivity

The restricted tensor product commutes with Dunn additivity (Theorem 50.3.5) in the following precise sense.

Proposition 50.7.6 *(Dunn additivity commutes with restricted tensor product).* Fix $n_1, n_2 \geq 0$. Let $\{C_v\}_{v \in V}$ be a family of $E_{n_1+n_2}$-monoidal stable presentable ∞-categories with the pointing maps of Setup 50.1.1(D2) for $E_{n_1+n_2}$. Then the restricted tensor product $\bigotimes'_v (C_v, u_v)$ constructed via Theorem 50.1.3 with $O = E_{n_1+n_2}$ coincides, as an $E_{n_1+n_2}$-monoidal ∞-category, with the iterated construction:

(a) View each C_v as an E_{n_1}-algebra in E_{n_2}-monoidal stable presentable ∞-categories (via Theorem 50.3.5).

(b) Apply Theorem 50.1.3 with $O = E_{n_2}$ to the underlying family to get $\bigotimes'_v C_v$ as an E_{n_2}-category;

(c) View the result as an E_{n_1}-algebra in E_{n_2}-categories, by functoriality of the restricted tensor product.

Via Theorem 50.3.5, this is an $E_{n_1+n_2}$-monoidal structure. It agrees with the $E_{n_1+n_2}$-structure of Theorem 50.1.3 applied directly.

Proof. Both constructions yield filtered colimits in the same ∞-category. The key is that the tensor product of ∞-operads $E_{n_1} \otimes E_{n_2}$ (Theorem 50.3.5) encodes the two supplied structures together with their interchange coherences. The colimit over S preserves these operations and coherences (Proposition 50.5.1). The induced $E_{n_1+n_2}$-structure on the colimit equals the $E_{n_1+n_2}$-structure of the direct construction. Formally: the forgetful functor $\mathrm{Alg}_{E_{n_1+n_2}} \simeq \mathrm{Alg}_{E_{n_1} \otimes E_{n_2}} \to \mathrm{Alg}_{E_{n_2}}$ commutes with filtered colimits (Proposition 50.3.7), so both iterated colimit computations agree. □

COROLLARY 50.7.7 *(special case: $E_1 \otimes E_1 = E_2$).* Suppose each C_v is E_1-monoidal in E_1-monoidal presentable stable ∞-categories (equivalently, E_2-monoidal by Theorem 50.3.5). Suppose the supplied pointings are maps for that entire structure. Then the restricted tensor product is E_2-monoidal. It is computed either by direct E_2-application of Theorem 50.1.3 or by nested E_1-applications.

Remark 50.7.8 *(role for F2).* In the F2 application, suppose $C_v = \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,v}$ has the required E_2-structure and pointing maps. Dunn additivity gives a factorization as an E_1-algebra in E_1-categories. The compatibility of Proposition 50.7.6 confirms that the global E_2-structure on $\bigotimes'_v \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,v}$ is well-defined either via direct E_2-restricted-tensor or via nested E_1-restricted-tensors.

50.8 APPLICATION TO THE IWAHORI HECKE CATEGORY

The Conditional Local Data

Setup 50.8.1 *(Iwahori Hecke family).* Fix $V = \mathbb{P} \cup \{\infty\}$. Suppose a stable presentable model $C_v = \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,v}$, its colimit-compatible E_2-monoidal structure, and the actual pointing maps of Setup 50.1.1(D2) have been specified. Write $|0\rangle_v = \eta_v(\mathbb{S})$ where the pointing is supplied. These are inputs. A class of a double coset or a fixed vector does not by itself construct such a categorical pointing.

LEMMA 50.8.2 *(the archimedean unit identification required for insertion).* Suppose C_∞ has the specified stable presentable E_2-monoidal structure. Its unit functor $\eta_\infty : \mathrm{Sp} \to C_\infty$ is an E_2-algebra morphism. To use a chosen object $|0\rangle_\infty$ as its value on $\mathbb{S}$, a unit identification $|0\rangle_\infty \simeq \mathbb{1}_{C_\infty}$ is required. Existence of an $O(2)$-fixed vector alone does not supply this identification.

Proof. The unit functor sends a spectrum E to its canonical spectral action on $\mathbb{1}_{C_\infty}$. Applying an operation commutes with the spectral action in each input, by colimit preservation. The unit identities then identify its value with the smash product of the input spectra acting on $\mathbb{1}_{C_\infty}$. The operadic coherences give the same identifications under substitution, including the nullary operation. Thus the functor is E_2-monoidal, and its value on $\mathbb{S}$ is the unit. This identifies the extra datum needed for a chosen object to serve as that value. □

The Conditional Assembly Corollary

COROLLARY 50.8.3 *(restricted-tensor E_2 transfer).* Under Setup 50.8.1, define

$$\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}} := \bigotimes'_{v \in V} (C_v, |0\rangle_v).$$

This object has the E_2-monoidal structure and admissible finite-stage maps of Theorem 50.1.3. A singleton map is available under exactly its condition (M2), including any additional pointing maps required there. Uniqueness is relative to the complete specified colimit cocone. The displayed formula is a definition, not a comparison with a separately constructed global arithmetic Hecke category.

Proof. The hypotheses give the diagram in $\mathrm{Alg}_{E_2}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$. Apply Theorem 50.1.3, including its singleton and cocone qualifications. The proof supplies neither the local E_2-structures nor pointing maps whose existence has not been assumed. □

Remark 50.8.4 *(local operations and geometric comparison remain inputs).* Discrete place labels do not construct the local E_2-operations, nor do they identify operations at geometric collision strata. Dunn additivity compares supplied operadic structures. It does not supply missing local structures from the discreteness of the place set. These requirements remain separate from F2.c.

Remark 50.8.5 *(archimedean pointing).* The archimedean factor may be included in every admissible finite stage without a pointing. Inserting that factor, or forming a singleton map that must insert it, requires the actual E_2-algebra map into that factor. A spherical vector in a representation or in a Fock-space presentation is not a proof of this categorical datum.

50.9 Uniqueness of the E_n Structure

Proposition 50.9.1 *(uniqueness for a fixed cocone).* Fix the full diagram $\Phi : \mathrm{Fin}^{\mathrm{sph}}(V) \to \mathrm{Alg}_{O}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$ and an underlying colimit cocone with vertex $C = \operatorname{colim} U\Phi$. The space of lifts of this whole cocone to a colimit cocone of O-algebras is contractible. The lift of Theorem 50.1.3 is one such lift.

Proof. HA 3.2.3.1, in the form of Proposition 50.3.7, creates the filtered colimit and detects a colimit cocone on underlying objects. Thus every lift of the fixed underlying colimit cocone is a structured colimit cocone. Any two structured colimit cocones have a contractible space of equivalences respecting the complete cocone. Forgetting the structure gives the corresponding equivalences of underlying colimit cocones, whose space is also contractible. Taking the fiber over the fixed underlying cocone proves the assertion. Merely specifying individual arrows or commuting triangles would not specify that complete cocone. □

Corollary 50.9.2 *(compatibility of available singleton maps).* If $V \setminus \{v\} \subseteq V^{\mathrm{sph}}$, the singleton map $\iota_v : C_v \to C$ is O-monoidal. For a general v, the same assertion holds after the additional pointing maps required by (M2) have been specified. In either case ι_v intertwines the local and global operations, with the coherences of its specified operadic cocone. No full faithfulness or identification of C_v with its image is asserted.

Proof. In the first case $\{v\}$ is an admissible finite stage, so (M1) applies. In the second case insert the additionally supplied pointings to reach an admissible stage and apply (M1). For example, for two places with categories Sp and $\mathrm{Mod}_{H\mathbb{Q}}$ and their unit maps, the singleton map from Sp is rationalization. It is not fully faithful: for any prime p, the nonzero spectrum $H\mathbb{F}_p$ maps to zero. It is nevertheless symmetric monoidal. Thus recovery of a local category from its image cannot follow from uniqueness. □

50.10 Base Change and Functoriality

The restricted tensor product is functorial in the family $\{C_v\}$ and covariant under base change.

Proposition 50.10.1 *(functoriality in the pointed family).* Let two families over the same set V satisfy Setup 50.1.1, and let $f_v : C_v \to C'_v$ be O-algebra morphisms. At every place where both pointings

are supplied, require a specified equivalence $f_v\eta_v \simeq \eta'_v$ as morphisms of O-algebras. Then these pointed morphisms induce an O-algebra map

$$\bigotimes_v{}' f_v : \bigotimes_v{}'(C_v, u_v) \longrightarrow \bigotimes_v{}'(C'_v, u'_v).$$

This construction is ∞-functorial in the pointed family. An equivalence of the objects $f_v(u_v)$ and u'_v without compatibility with the specified operadic pointings is not the hypothesis for a general O.

Proof. Use the common cofinal index consisting of finite subsets containing $(V \setminus V^{\mathrm{sph}}) \cup (V \setminus V'^{\mathrm{sph}})$. The finite tensor products $\bigotimes_{v\in S} f_v$, together with the given pointed equivalences, form a natural transformation of the complete operadic diagrams. Taking its colimit gives the asserted map. Functoriality follows from the functoriality of tensor products and of the colimit functor on diagram categories. □

PROPOSITION 50.10.2 *(base change along a colimit-preserving symmetric monoidal functor).* Let $F^{\otimes} : (\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}, \otimes) \to (\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}, \otimes)$ be a symmetric monoidal colimit-preserving functor. For a family as in Setup 50.1.1, define its transported pointing and distinguished object by

$$\eta_v^F : \mathrm{Sp} \simeq F(\mathrm{Sp}) \xrightarrow{F(\eta_v)} F(C_v), \qquad u_v^F = \eta_v^F(\mathbb{S}).$$

Then

$$F\big(\bigotimes_v{}'(C_v, u_v)\big) \;\simeq\; \bigotimes_v{}'(FC_v, u_v^F)$$

as O-monoidal ∞-categories.

Proof. F preserves both filtered colimits (colimit-preservation) and tensor products (symmetric monoidality). Both operations are separately applied to the colimit diagram Φ; their commutativity gives the equivalence. Formally:

$$F\big(\varinjlim_S \bigotimes_{v\in S} C_v\big) \;\simeq\; \varinjlim_S F\big(\bigotimes_{v\in S} C_v\big) \;\simeq\; \varinjlim_S \bigotimes_{v\in S} FC_v$$

by symmetric monoidality of F, and the transition maps $F\iota_{S\subseteq S'} = \iota^{(F)}_{S\subseteq S'}$ by functoriality of F applied to the pointed spherical maps η_v. □

Remark 50.10.3 *(scope of categorical base change).* Proposition 50.10.2 applies to a specified functor on stable presentable categories and to its transported pointing maps. It supplies an arithmetic change of base only after those categorical data have been constructed. The bar–cobar base-change comparisons in Volume I concern their own coefficient systems and hypotheses. Relating the two constructions requires a map between their carriers that preserves the relevant operations. Formal commutation with tensor products and colimits alone does not identify them.

50.11 EXAMPLES

Example 1: Module Categories and the Spectral Tensor Product

Example 50.11.1 *(the tensor colimit of module categories).* Let R_v be discrete commutative rings, put $A_v = HR_v$, and give $C_v = \mathrm{Mod}_{A_v}$ its usual symmetric monoidal structure and unit pointing. Define

$$A_{\mathrm{tens}} = \operatorname*{colim}_{S\in \mathrm{Fin}(V)} \bigotimes_{v\in S,\,\mathbb{S}} A_v \quad \text{in} \quad \mathrm{CAlg}(\mathrm{Sp}),$$

with unit insertions as transition maps and empty tensor $\mathbb{S}$. Then

$$\bigotimes_{v\in V}{}'(\mathrm{Mod}_{A_v}, A_v) \simeq \mathrm{Mod}_{A_{\mathrm{tens}}}$$

as symmetric monoidal stable presentable categories. This is a spectral tensor-colimit formula. It is not an identification with a profinite completion or with a classical adelic restricted product.

Proof. The module-category tensor equivalence identifies each finite stage with modules over $\bigotimes_{v\in S,\mathbb{S}} A_v$, and identifies the transition functors with extension of scalars. The module-category functor preserves filtered colimits (HA 4.8.5.13), with tensor compatibility given by HA 4.8.5.16. These identifications preserve the symmetric monoidal structures and unit maps, so passage to the filtered colimit gives the formula.

For example, if $V = \mathbb{P}$ and $R_p = \mathbb{Z}_p$, every prime ℓ becomes invertible in A_{tens}: add any factor $H\mathbb{Z}_p$ with $p \neq \ell$. Thus the output is rational, whereas $H\widehat{\mathbb{Z}}$ is not. In particular this construction does not recover $\mathrm{Mod}_{H\widehat{\mathbb{Z}}}$. The classical finite adeles also use integral subrings at the unramified places. That restricted-product diagram is not the diagram above with $A_p = H\mathbb{Q}_p$. □

Example 2: Conditional Function-Field Application

Example 50.11.2 *(function-field assembly).* Let $V = |X|$ for a smooth projective curve over $\mathbb{F}_q$. A family of stable presentable E_2-monoidal local categories with actual pointing maps satisfying (D2) has the restricted E_2-tensor product of Theorem 50.1.3. To apply this to chosen Hecke categories, their local E_2-structures and those maps must first be supplied. A comparison with Beilinson–Drinfeld factorization requires its geometric coefficients, collision maps and higher coherences separately. Discretizing $|X|$ does not provide that comparison or its missing operations.

Remark 50.11.3 *(no automatic comparison between arithmetic bases).* The filtered-colimit theorem applies over either indexing set once its categorical inputs are supplied. It gives no function-field-to-arithmetic comparison functor. Such a comparison must specify the functor, its pointing compatibility, and any required geometric factorization data.

Example 3: Conditional Arithmetic Iwahori Assembly

Example 50.11.4 *(arithmetic Iwahori application).* Under Setup 50.8.1, Corollary 50.8.3 gives an E_2-structure on the categorical restricted tensor product. The notation $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$ in that corollary denotes this defined object. An independently defined global arithmetic Hecke category requires a separate comparison. So does a restricted tensor product of representations. Neither comparison follows from filtered colimits.

50.12 The Finite Labeled-Disk Description

Setup 50.12.1 *(finite labeled coefficients).* Take $\mathcal{O} = E_n$, $n \geq 1$, and the data of Setup 50.1.1. For a finite set S, write $M_S = \coprod_{v\in S} \mathbb{R}^n_v$, retaining each place label. Assign coefficient C_v to its own labeled component. The finite labeled-disk value is

$$F^{\mathrm{lab}}(M_S) := \bigotimes_{v\in S} C_v.$$

This is a family of coefficients, not the single constant coefficient $\bigotimes_{v\in S} C_v$ on every disk. For admissible inclusions insert the specified maps η_v. Only this finite labeled diagram and its filtered colimit are being defined. No presentability assertion about $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ itself is used.

THEOREM 50.12.2 *(comparison with the finite labeled-disk diagram).* The diagram F^{lab} on $\mathrm{Fin}^{\mathrm{sph}}(V)$, with its specified insertions, is the finite tensor diagram Φ used in Theorem 50.1.3. Hence

$$\operatorname*{colim}_{S \in \mathrm{Fin}^{\mathrm{sph}}(V)} F^{\mathrm{lab}}(M_S) \simeq \bigotimes_{v \in V}{}' (C_v, u_v)$$

as E_n-algebras in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$, compatibly with the complete colimit cocones.

Proof. At each finite stage the values are identical. On each inclusion the morphisms are the same tensor products of identities and supplied η_v. Their higher compatibilities come from the same symmetric monoidal structure and the same pointing maps. Thus this is the same operadic diagram, and taking its filtered colimit gives the assertion. □

For comparison, ordinary factorization homology on framed disks with a single coefficient algebra A, when defined, satisfies

$$\int_{\coprod_{v \in S} \mathbb{R}^n} A \simeq A^{\otimes |S|}.$$

The disk overcategory has a terminal object. Applying the symmetric monoidal disk algebra gives the displayed tensor power. See Ayala–Francis, *A factorization homology primer*, Definitions 3.5 and 3.7.

Putting $A = \otimes_{v \in S} C_v$ in this formula therefore does not give $\otimes_{v \in S} C_v$. Nor is an integral with a single coefficient $A_V = \otimes'_v C_v$ over all the disks a replacement for the labeled diagram. The usual factorization homology construction is a left Kan extension of a disk algebra. Disk normalization and disjoint-union compatibility alone do not characterize its values on general manifolds.

Remark 50.12.3 *(scope of the description).* The labeled-disk description uses exactly the same finite diagram and colimit argument as the restricted tensor theorem. It is not a second independent proof, an Ext comparison, or a construction of geometric collision operations. A comparison with a factorization algebra on a geometric space requires its coefficient system and comparison morphisms separately.

50.13 REMARKS ON OPERADIC COHERENCE

Hypotheses Used by the Filtered-Colimit Argument

The application of HA 3.2.3.1 uses a small operad and filtered colimits in the ambient symmetric monoidal category. Tensoring must preserve those colimits in each variable. It requires neither operadic coherence in the sense of HA 3.3.1.1 nor presentability of the ambient category. The coherence assumption retained in some statements above is an extra hypothesis, not a premise needed for this argument. The separate presentability result HA 3.2.3.5 is not being applied to $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ itself. The preservation of higher operation coherences follows from taking the colimit of the supplied operadic diagram, not from recognizing operations in its cohomology.

Alternative Pointing Maps

Remark 50.13.1 *(non-spherical insertion).* A replacement pointing must be an actual morphism $\widetilde{\eta}_v : \mathrm{Sp} \to C_v$ in $\mathrm{Alg}_O(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$, with $\widetilde{u}_v = \widetilde{\eta}_v(\mathbb{S})$. Specified equivalences between pointing maps, in that mapping space, induce equivalences of the diagrams and their colimits. Equivalence of the underlying central objects alone is not the general operadic compatibility required. In the unital E_n cases every such pointing sends $\mathbb{S}$ to the actual monoidal unit. Thus a non-unit central object is not an alternative pointing of this kind.

Iterated Restricted Tensor Products

Proposition 50.13.2 *(iterated restricted tensor products).* Let $V = V_1 \sqcup V_2$ be a partition, with spherical data restricting to each V_i. Then

$$\bigotimes_{v \in V}{}'(C_v, u_v) \;\simeq\; \Big(\bigotimes_{v \in V_1}{}'(C_v, u_v)\Big) \otimes \Big(\bigotimes_{v \in V_2}{}'(C_v, u_v)\Big)$$

as O-monoidal ∞-categories.

Proof. Intersection with V_1 and V_2 identifies the indexing poset with $\mathrm{Fin}^{\mathrm{sph}}(V_1) \times \mathrm{Fin}^{\mathrm{sph}}(V_2)$. At each pair of finite stages, tensor associativity identifies the two values and their pointing maps. Separate preservation of filtered colimits by tensor products, from Corollary 50.3.2, identifies the underlying colimits. The same comparison preserves every operation in the supplied diagrams, so it is an equivalence of O-algebras. □

50.14 Completeness of F2.c and Residual Local Input

Theorem 50.14.1 *(F2.c assembly for supplied local Iwahori data).* F2.c proves the restricted-tensor assembly step for local Iwahori categories satisfying Setup 50.8.1, including the specified E_2-algebra pointings. Its uniqueness is relative to the complete colimit cocone. It does not construct any of these local inputs.

Proof. Apply Corollary 50.8.3 with its full hypotheses, which in turn is the E_2 specialization of Theorem 50.1.3. □

Remark 50.14.2 *(scope of the F2 conclusion).* F2.c is the formal assembly theorem. An arithmetic application requires the local E_2-enhancement F2.b and all pointing data of (D2). These remain hypotheses wherever they have not been constructed. Identifying the defined tensor colimit with an independently specified global geometric object remains a separate comparison.

Corollary 50.14.3 *(F3 after formal F2.c assembly).* Assume the full local E_2 and pointing hypotheses. Then F2.c supplies the formal restricted-tensor construction in the reduction of $F3$. The local input F2.b and any unverified pointing or global comparison requirements remain. F2.c also leaves the other hypotheses of Theorem 43.4.1 unchanged.

Proof. Substitute Theorem 50.14.1 only for its formal assembly conclusion in the cited reduction. That theorem assumes the local pointed operadic diagram and makes no independent arithmetic comparison. □

50.15 Tate Tensor Boundary

The categorical restricted tensor product is the filtered colimit of the finite tensor diagram of supplied O-algebras and actual operadic pointing maps. HA 3.2.3.1 applies because the ambient categorical tensor product preserves filtered colimits separately. Presentability of $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ itself is not needed.

The whole colimit cocone has a contractible space of structured lifts. This does not assert full faithfulness of singleton maps, and a singleton map requires the precise pointing condition (M2). The construction is functorial in pointed families. It commutes with the stated symmetric monoidal base changes and iterated tensor products, and respects Dunn additivity for supplied structures.

The finite labeled-disk description is the same diagram, not an independent derivation. The module-category example computes a spectral tensor colimit, not a profinite or adelic restricted product. Neither example supplies geometric collision maps.

F2.c is therefore a proved formal assembly theorem under its full hypotheses. The local arithmetic E_2-structures, pointing maps and any independent global comparison remain required inputs. The arithmetic comparison in F3 is likewise not absorbed by this categorical construction.

Chapter 51

Arithmetic Comparisons and Residual Structure

The six comparison problems of Chapter 43 are $\mathrm{L}_{\mathrm{ACD}}$ (adelic categorical descent), F2.c (∞-categorical Tate restricted tensor product), $\mathrm{F3.BZN}^{\mathrm{Ar}}$ (arithmetic Ben-Zvi–Nadler identification), arithmetic Positselski (co-contra equivalence over Λ), Koszul–Petersson compatibility, and VB* (arithmetic Virasoro positivity). Each comparison has a specified domain and additional conditions for extension beyond it. In the positivity problem, three different objects must be distinguished: a scalar Li coefficient, a transform of a zero-divisor operator, and a polynomial in a Fock energy operator. A positive scalar trace does not identify these operators or make each of their quadratic forms nonnegative.

51.1 First Residual Decomposition

F2.c: ∞-Categorical Tate Restricted Tensor Product

The formal restricted-tensor construction is proved in Theorem 50.1.3. Its inputs include actual operadic maps $\eta_v : \mathrm{Sp} \to \mathcal{C}_v$ on a cofinite set of places. These maps form the transition functors between finite tensors. The colimit is taken over that diagram, without an infinite tensor of unspecified unit-comparison arrows.

Definition 51.2.1 states the pointing used here, and Theorem 51.2.3 records the formal E_2 consequence. The specialization in Theorem 73.3.1 has the same inputs. A non-unit central idempotent does not replace them. The remaining arithmetic problem is to construct the local structures, their pointings and any independently required global geometric comparison.

$\mathrm{L}_{\mathrm{ACD}}$*: Adelic Categorical Descent*

$\mathrm{L}_{\mathrm{ACD}}$ in Chapter 43 asserts that the adelic categorical descent from local automorphic categories to the global automorphic category holds; but four subconditions are conflated into the single statement: (i) dg-categorical proof that the restricted tensor product with distinguished units preserves the colimit structure; (ii) proof that local Hecke-functoriality together with cross-prime commutativity implies global Hecke-functoriality on the colimit; (iii) sph-finite preservation under the colimit; (iv) strong approximation for the simply connected derived group, together with the determinant idele-class quotient for $GL_2/\mathbb{Q}$. Subcondition (iv) is a theorem after this separation: Kneser supplies the simply connected strong-approximation input for SL_2, while Tate's idelic formalism supplies the determinant quotient. The conditions (i)–(iii) depend on F2.c and are conjectural. Thus $\mathrm{L}_{\mathrm{ACD}}$ splits into the four subconditions $\mathrm{L}_{\mathrm{ACD}}.1$–$\mathrm{L}_{\mathrm{ACD}}.4$; see Theorem 51.2.4.

F3.BZN^{Ar}: Arithmetic Ben-Zvi–Nadler Identification

$\mathrm{F3.BZN}^{\mathrm{Ar}}$ proposes that the arithmetic analogue of the Ben-Zvi–Nadler Hochschild-trace identification holds. Three independent obstructions separate the complex-analytic source theorem from its arithmetic extension. (R1) Ben-Zvi–Nadler's proof over $\mathbb{C}$ uses holomorphic factorization on $S^1 = B\mathbb{Z}$; the arithmetic analogue requires locally-constant-with-Adams factorization on $B\widehat{\mathbb{Z}}$, and the survival of the

Hochschild-trace pairing under this qualitative change is not established via Gaitsgory–Rozenblyum DAG Vol. II for formal-Noetherian targets. (R2) The arithmetic free loop space requires a fixed descent topology; the rigid-analytic topology on the Emerton–Gee stack (arXiv:1908.07185 §2) is the canonical choice. (R3) The Mazur–Kapranov–Reznikov dictionary (primes as knots, $\overline{\mathrm{Spec}\,\mathbb{Z}}$ as 3-manifold) is a heuristic motivation; the technical content runs through the Emerton–Gee rigid-analytic construction. The arithmetic statement uses R2's topology, treats R3 as motivation, and isolates R1's formal-Noetherian extension as the open subcondition; see Theorem 51.2.5.

Arithmetic Co–Contra Comparison

An arithmetic co–contra correspondence must specify a coalgebra over $\Lambda = \mathbb{Z}_p[[\Gamma]]$, its continuous dual, and the categories and resolutions to be compared. The abelian nature of Γ does not identify an oscillator module with a cyclotomic Iwasawa module. Nor does the vanishing of a module's μ-invariant imply conilpotence of a coalgebra. Question 51.2.7 retains the comparison with these coefficient requirements.

Koszul–Petersson Compatibility: Sub-Claim Structure

The Koszul–Petersson compatibility asserted in Chapter 43 is the identity $\langle\cdot,\cdot\rangle^{\mathrm{Pet}} = \langle\cdot,\cdot\rangle^{\mathrm{Koszul}} \circ (\mathrm{id} \otimes \text{BC-inv.})$ where BC-inv. denotes the Bost–Connes involution on the dual. The assertion conflates seven independent sub-claims: (K1) definition of the arithmetic Koszul pairing via adic transfer of Vol I's derived-centre pairing; (K2) Hecke-equivariant embedding $\mathcal{V}^{\mathrm{prim}} \hookrightarrow L^2([GL_2])^{\text{sph-fin}}$; (K3) specification of the Bost–Connes anti-involution as the KMS_1 Tomita–Takesaki modular conjugation after the modular half-density normalisation; (K4) archimedean normalisation; (K5) reconciliation of the Bost–Connes KMS_1 pairing with the local Whittaker/Kirillov pairing at each finite prime p, up to the declared spherical norm factor $m_p(1 + p^{-1})L_p(1, \mathrm{Ad})$; (K6) agreement of adelic Tamagawa measure with the restricted tensor product of local Haar measures; (K7) confirmation that the sph-finite cut is the natural locus for the identification. The explicit model calculations fix the conjugations and local norms. Their transport to the arithmetic carrier still requires (K2), the arithmetic identifications in (K3)–(K5), and the adelic comparison in (K6). A local functional equation does not construct these maps. The Heisenberg self-duality and Bost–Connes self-duality are each properties of their respective objects; the sub-claim structure exhibits that their pairing requires a separate comparison. See Theorem 51.2.9.

VB: Arithmetic Virasoro Positivity and Spectral Hypotheses*

Scalar positivity $\lambda_n \geq 0$ and operator positivity $T_n \succeq 0$ are different assertions. Operator positivity means that $\langle T_n v, v\rangle \geq 0$ for every vector in the specified quadratic-form domain. It cannot be recovered from the sign of one trace. This distinction already occurs for bounded self-adjoint matrices, independently of any issue of moment determinacy.

For example, on $\mathbb{C}^2$ with its usual Hermitian inner product, set $A = \mathrm{diag}(0, 2)$, let I be the identity, and put $S_n = A^n - (A - I)^n$ for $n \geq 1$. Direct diagonal multiplication gives

$$S_n = \mathrm{diag}\big((-1)^{n+1}, 2^n - 1\big), \qquad \mathrm{Tr}(S_n) = \begin{cases} 2^n - 2, & n \text{ even}, \\ 2^n, & n \text{ odd}. \end{cases}$$

Every displayed trace is strictly positive, whereas $S_2 = \mathrm{diag}(-1, 3)$ has quadratic form -1 on the first coordinate vector. Here A is bounded, nonnegative and self-adjoint, and the ordinary trace is positive and faithful. Its moment measure is $\delta_0 + \delta_2$, where δ_a denotes unit mass at a. This measure is determinate:

any positive measure with the same moments has integral of $x^2(x-2)^2$ equal to zero, hence is supported on $\{0, 2\}$; its zeroth and first moments then determine both masses. Thus even compact spectral support and moment determinacy do not turn the scalar inequalities into operator positivity. No identification of these traces with the arithmetic Li coefficients is being made.

For these particular polynomials there is an exact operator criterion. Let A be self-adjoint on a Hilbert space $\mathcal{H}$, with spectral projections E_A, and use the common spectral domain

$$\mathcal{D}_{\mathrm{sp}} = \bigcup_{R>0} E_A([-R, R])\mathcal{H}.$$

Every power of A preserves this dense domain. Spectral truncations converge in the graph norm of every polynomial in A, so it is a common core. Interpret $p_n(A)$, where $p_n(x) = x^n - (x-1)^n$, first on $\mathcal{D}_{\mathrm{sp}}$ and then by its closed spectral-calculus realization. Then

$$p_n(A) \succeq 0 \quad \text{for all } n \geq 1 \quad \Longleftrightarrow \quad A \succeq \tfrac{1}{2} I.$$

Necessity follows from $p_2(A) = 2A - I$. For sufficiency, odd powers are strictly increasing on $\mathbb{R}$, so $p_n(x) > 0$ for odd n. If n is even and $x \geq \frac{1}{2}$, then $|x| \geq |x-1|$, hence $p_n(x) \geq 0$. The spectral calculus gives the asserted positivity. For an unbounded operator, a statement made only on a smaller common domain also needs a specified form core and closure. The scalar traces supply neither that domain nor the spectral lower bound.

The arithmetic Hilbert–Pólya hypothesis H-P^{Ar} (Hypothesis 44.3.2) supplies a separate spectral object: $\Theta_\xi = \frac{1}{2} I + i H_{\mathrm{HP}}$, with a self-adjoint ordinate operator, specified domain, determinant divisor and regularised trace. It does not identify H_{HP} with a Virasoro zero-mode L_0^{Ar}. Its Li transforms are

$$\mathcal{L}_n = I - (I - \Theta_\xi^{-1})^n, \qquad \lambda_n = \mathrm{Tr}_{\mathrm{reg}}(\mathcal{L}_n),$$

under the determinant and trace hypotheses of Proposition 44.3.8. If the self-adjoint ordinate operator and the determinant-divisor identity are both assumed, zero-placement already follows from the spectrum of Θ_ξ. The scalar Li route uses the determinant and trace identities separately. The forward implication of Theorem 44.3.10 is therefore positive HP–Li trace $\Rightarrow$ Li positivity $\Rightarrow$ zero-placement. The operator and trace positivity requirements can be separated explicitly. Under the self-adjoint H-P^{Ar} hypothesis, put $U = I - \Theta_\xi^{-1}$. At a real spectral ordinate t, its scalar value is $(-1/2 + it)/(1/2 + it)$, of modulus one. Thus U is unitary, and

$$\mathrm{Re}\, \mathcal{L}_n := \frac{\mathcal{L}_n + \mathcal{L}_n^*}{2} = \frac{(I - U^n)^*(I - U^n)}{2} \succeq 0.$$

The unsymmetrized $\mathcal{L}_n$ need not be self-adjoint. To pass from this real-part positivity to $\lambda_n \geq 0$, the regularised trace must be defined, compatible with adjoints and nonnegative on the relevant real-part cone, with the required continuity or Li/Weil approximation. These are additional trace hypotheses in the positive-trace route of Corollary 44.3.9; regularisation alone supplies no positivity property.

To deduce this implication from VB*, one would also need a comparison from the primary Fock operators and their positive forms to these Li transforms and the regularised trace, preserving the relevant positivity cone. No such comparison is supplied by the moment argument. Conversely, zero-placement alone does not construct the primary Virasoro operators, their common invariant domain, self-adjoint extensions, or this comparison. On the standard Fock space the zero-energy vacuum prevents positivity of every Virasoro polynomial. The precise positive spectral sector and the remaining arithmetic comparison are described below.

51.2 SHARP SUBCONDITIONS

Pointed E_n-monoidal Category

Definition 51.2.1 *(pointed E_n-monoidal category).* A *pointed E_n-monoidal stable presentable ∞-category* for $n \geq 1$ consists of an E_n-algebra C in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ and a specified morphism

$$\eta : \mathrm{Sp} \longrightarrow C \quad \text{in} \quad \mathrm{Alg}_{E_n}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}).$$

Write $u = \eta(\mathbb{S})$, where $\mathbb{S}$ is the sphere. The notation (C, u, η) retains this relation. Compatibility with the nullary operation gives $u \simeq \mathbb{1}_C$. Thus η is a functor with all operadic compatibilities, not an arbitrary arrow $\mathbb{1}_C \to u$ inside C.

Remark 51.2.2 *(why pointed).* The classical restricted direct product specifies local subgroups and their inclusions. The categorical tensor construction instead specifies transition functors by tensoring with the maps η_v. Their nullary and higher compatibilities make these functors operadic morphisms. A chosen object or central idempotent alone does not do so. The maps into the colimit need not be fully faithful.

With these pointings, the formal F2.c assertion is the following theorem.

THEOREM 51.2.3 *(F2.c with specified pointings).* Let C_v be stable presentable E_2-monoidal categories whose tensor products preserve colimits separately. On a cofinite subset $V^{\mathrm{sph}} \subseteq V$, supply pointings as in Definition 51.2.1. Put $E = V \setminus V^{\mathrm{sph}}$. Then

$$\bigotimes_v{}' (C_v, u_v) := \operatorname*{colim}_{\substack{S \subseteq V \text{ finite} \\ E \subseteq S}} \bigotimes_{v \in S} C_v$$

is an E_2-algebra in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$.

Transitions insert the supplied maps η_v. The finite-stage maps are E_2-monoidal, with uniqueness relative to the complete underlying colimit cocone. Singleton maps require the condition or additional pointings in (M2) of Theorem 50.1.3.

Proof. Tensoring the actual E_2-algebra maps with identities gives the finite-stage diagram in $\mathrm{Alg}_{E_2}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$. Its indexing poset is filtered. The categorical tensor product preserves filtered colimits separately, so HA 3.2.3.1 constructs its colimit on the underlying categorical colimit. Theorem 50.1.3 gives the cocone and singleton assertions with precisely these hypotheses. □

$\mathrm{L}_{\mathrm{ACD}}$ *Split into Four Named Subconditions*

THEOREM 51.2.4 *($\mathrm{L}_{\mathrm{ACD}}$ split).* $\mathrm{L}_{\mathrm{ACD}}$ is equivalent to the conjunction of:

($\mathrm{L}_{\mathrm{ACD}}$.1) *Dg-categorical colimit commutation.* The restricted tensor product of dg-categories with pointed units commutes with the sph-finite truncation. **Form: conjectural**. Formal assembly is supplied by Theorem 51.2.3, but it does not prove compatibility with the specified sph-finite truncation.

($\mathrm{L}_{\mathrm{ACD}}$.2) *Cross-prime Hecke coherence.* For distinct primes $p \neq q$, the local Hecke operators T_p and T_q commute on the restricted tensor product of local automorphic categories, with associativity and unit coherences. **Form: conjectural**; follows from local Hecke-functoriality (Fargues–Scholze) plus a symmetric-monoidal coherence check.

($\mathrm{L}_{\mathrm{ACD}}$.3) *Sph-finite preservation.* The sph-finite subcategory $\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ is preserved under the restricted-tensor colimit: the colimit of sph-finite categories embeds in the sph-finite subcategory of the colimit. **Form: conjectural**; plausibly automatic given ($\mathrm{L}_{\mathrm{ACD}}$.1).

(L_{ACD}.4) *Strong approximation and determinant quotient.* The simply connected group $SL_2/\mathbb{Q}$ satisfies strong approximation away from a nonempty set of places (Kneser 1966; Platonov–Rapinchuk 1994). The passage from SL_2 to GL_2 keeps the determinant idele-class quotient $\mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}$, which is the Tate line rather than a strong-approximation consequence. **Form: theorem.**

Under this split, L_{ACD} follows from (L_{ACD}.1) + (L_{ACD}.2) + (L_{ACD}.3); (L_{ACD}.4) is theorem.

F3.BZN^{Ar} Split into Three Sub-Obstructions

Theorem 51.2.5 *(Conditional arithmetic trace comparison).* Fix the coefficient category, trace construction and completions. Assume the following comparison data, with compatible circle actions and the operations required in F3:

(R1) For the specified spectral category C_X, an actual equivalence from its chosen Hochschild trace complex to $\Gamma(LX, \mathcal{O}_{LX})$. Here the geometric loop object is the derived self-intersection of the diagonal. An arithmetic or profinite loop object requires its own comparison.

(R2) The specified global Selmer-derived target X, with its coefficient, formal and singular-support conditions, and the categorical equivalence transporting the arithmetic trace to that of C_X.

(R3) An actual equivalence from this loop-function complex to the full Weil/(φ, Γ) target, preserving the stated completion and operations. The inertia object $\mathrm{Map}(B\widehat{\mathbb{Z}}, X)$ alone does not supply this equivalence.

Then composition gives the F3 trace comparison on the specified carrier. These are sufficient data; no formal-stack trace theorem is inferred from a categorical framework alone.

Proof. Transport the trace along the categorical equivalence in (R2), then apply (R1) and (R3). Their compatibility preserves the specified circle action, completion and operations. □

Proposition 51.2.6 *(The affine loop calculation).* Let k be a field and A a smooth finitely generated commutative k-algebra. For the small k-linear category $\mathrm{Perf}(A)$ and ordinary derived loops of $X = \mathrm{Spec}\, A$,

$$\mathrm{HH}_k(\mathrm{Perf}(A)) \simeq A \otimes^{\mathbf{L}}_{A\otimes_k A} A \simeq \Gamma(X \times^{\mathbf{R}}_{X\times_k X} X, \mathcal{O}).$$

Proof. The perfect generator A identifies the categorical trace with the trace of the diagonal A-bimodule. Resolve that bimodule by the two-sided bar resolution over $A \otimes_k A$; tensoring with A gives the Hochschild chain complex and the first formula. The derived affine fibre-product rule gives the second. The cyclic bar construction supplies the circle action. Extending this calculation to a formal completion or a nilpotent-support ind-coherent category requires the actual continuity and support comparison in (R1); smoothness of an arbitrary stack is not such a comparison. □

Arithmetic Positselski with Abelian Domain Restriction

Question 51.2.7 *(an integral co–contra comparison).* Let C be a specified complete coalgebra over $\Lambda = \mathbb{Z}_p[[\Gamma]]$. Construct its co-derived category of continuous comodules and the contra-derived category of contramodules over its specified continuous dual. Determine hypotheses on the coefficient modules, resolutions, conilpotent filtration, and inverse limits that give an equivalence

$$\mathrm{D}^{\mathrm{co}}(C\text{-comod}_\Lambda) \simeq \mathrm{D}^{\mathrm{ctr}}(C^\vee_{\mathrm{cont}}\text{-contramod}_\Lambda).$$

If additional chiral operations are imposed, construct their compatibility with this ordinary coefficient comparison.

A field-coefficient equivalence cannot be transferred by an unspecified unital map $k \to \mathbb{Z}_p[[\Gamma]]$. For $\Gamma \simeq \mathbb{Z}_p$, the target is $\mathbb{Z}_p[[T]]$, which has characteristic zero and in which p is not invertible. A field of positive characteristic cannot map to it unitally. A field of characteristic zero contains $1/p$, whose image would invert p. Thus there is no such field map. Reduction to $\mathbb{F}_p$ and extension to $\mathbb{Q}_p$ are different operations, and an integral comparison must control both when it uses them.

Question 51.2.8 *(the arithmetic Heisenberg coefficient object).* Construct the coalgebra, coefficient module, topology, and Γ-action for the proposed arithmetic Heisenberg realization. If an Iwasawa vanishing theorem is used, identify the resulting module with one in that theorem's domain and transport its μ-invariant. Prove separately the hypotheses of Question 51.2.7.

Ferrero–Washington concerns the Iwasawa modules attached to abelian number fields. A trivial Γ-action on an oscillator lattice does not give the required identification. A co–contra equivalence still leaves the P1 evaluation and defect identity to construct. The converse in Theorem 43.5.1 also requires defect detection.

Koszul–Petersson Compatibility Split into Seven Sub-Claims

THEOREM 51.2.9 *(Sufficient pairing comparison data).* The following data give Koszul–Petersson compatibility on the specified cuspidal core:

(K1) A defined arithmetic bilinear Koszul pairing on the chosen completed carrier and a continuous antilinear involution J, with Hermitian form $H(v, w) = B(v, Jw)$.

(K2) An actual Hecke-equivariant arithmetic placement into the chosen automorphic Hilbert space, with a dense comparison core.

(K3) Compatibility with local conjugations. On the prime Gibbs cyclic carrier at $\beta = 1$, $J(\mu_{p^k}\Omega) = p^{k/2}\mu^*_{p^k}\Omega$. This Hilbert–Schmidt carrier is distinct from the weighted occupation model; their arithmetic placement is supplied data.

(K4) The archimedean comparison. For the declared K_0-Whittaker function, its norm squared is $\pi/4$, giving an isometry of the normalized vacuum line. The full Fock comparison and transport of Segal coefficient conjugation must also be supplied.

(K5) The finite-place comparison with the declared multiplicative Haar measure. For normalized unramified tempered parameters, the spherical Whittaker norm is

$$C_p = m_p(1 + p^{-1})L_p(1, \mathrm{Ad}), \qquad m_p = \mathrm{vol}(\mathbb{Z}_p^\times).$$

The normalized line has unit norm; the weighted occupation/radial and diagonal Hilbert–Schmidt isometries are those of Chapter 56. Identifying the actual arithmetic pairing with them is an additional comparison.

(K6) Compatible normalized reference vectors, completed restricted tensor products, and an adelic placement preserving pairings and the specified global measure. Any continued scalar normalization must be finite and nonzero and applied to both forms.

(K7) The chosen cuspidal domain and its completed embedding are specified. No equality on the Eisenstein continuous spectrum is inferred from these spherical or discrete calculations.

Under these data the transported Hermitian Koszul form equals the Petersson form on the comparison core and hence on its Hilbert completion wherever both forms are continuous.

Proof. The local model identities are proved in Chapters 47 and 56. Under the assumed arithmetic intertwiners they are identities of the arithmetic local forms. Their products agree on normalized finite tensors by (K6); linearity and continuity extend equality to the stated completion. The antilinear map in (K1) converts the bilinear pairing to the required sesquilinear form. This is a conditional transport theorem, not a construction of (K2) or (K6). □

VB: Vacuum Obstruction and Positive Spectral Sectors*

The spectral comparison separates the following inputs:

- H-P^{Ar}, with the operator, domain, determinant and trace data of Hypothesis 44.3.2; these are additional to P1, P2 and P3.
- Positive HP–Li trace, which implies Li positivity and zero-placement by Corollary 44.3.9 and Theorem 44.3.10, under their stated hypotheses.
- An arithmetic comparison from a specified positive energy sector to this trace condition, with the operations preserved on that sector stated explicitly.

The conformal-energy normalization is decisive for the operator inequalities. Write L_0^{Vir} for the standard Fock energy and $T_n^{\mathrm{Vir}} = p_n(L_0^{\mathrm{Vir}})$, with $p_n(x) = x^n - (x-1)^n$ as above. The vacuum is a primary vector; the word "primary" does not exclude its energy-zero contribution.

Proposition 51.2.10 *(vacuum obstruction and maximal positive spectral sector).* In the normalization of Lemma 45.6.1, the nonzero vacuum Ω satisfies $L_0^{\mathrm{Vir}}\Omega = 0$. Consequently

$$T_n^{\mathrm{Vir}}\Omega = (-1)^{n+1}\Omega, \qquad \langle T_{2k}^{\mathrm{Vir}}\Omega, \Omega\rangle = -\|\Omega\|^2 < 0 \quad (k \geq 1).$$

Thus positivity of every T_n^{Vir} fails on the vacuum-containing Fock space, independently of any hypothesis on Θ_ξ.

For a specified self-adjoint energy operator A on $\mathcal{H}$, with spectral projections E_A, put

$$\mathcal{H}_+ = E_A([1/2, \infty))\mathcal{H}, \qquad \mathcal{D}_+ = \bigcup_{R \geq 1/2} E_A([1/2, R])\mathcal{H}.$$

Then $\mathcal{H}_+$ is the largest closed reducing subspace on which every closed spectral-calculus operator $p_n(A)$ is nonnegative, and $\mathcal{D}_+$ is a common spectral core for their restrictions. Here reducing means that the orthogonal projection onto the subspace commutes with every spectral projection of A.

Proof. On the vacuum, substitution of the eigenvalue zero gives $p_n(0) = (-1)^{n+1}$, proving the obstruction. On $[1/2, \infty)$, the polynomial sign calculation above gives $p_n \geq 0$. Spectral calculus therefore proves positivity on $\mathcal{H}_+$, and bounded spectral truncations give the stated common core. Conversely, on any closed reducing subspace with this positivity property, $p_2(A) = 2A - I \succeq 0$. Its spectral support lies in $[1/2, \infty)$, so the subspace is contained in $\mathcal{H}_+$. This proves maximality without a zero-placement hypothesis. □

For the standard free-boson Fock space, the energies are the nonnegative integers and the zero-energy space is the vacuum line. Hence $\mathcal{H}_+ = \Omega^\perp$. The non-vacuum positivity of Corollary 45.6.4 therefore extends to the closed polynomial realizations on this subspace. This is a statement about the archimedean Fock factor; it does not identify the full arithmetic energy operator or its domain.

Reducing the energy operator is weaker than preserving a Virasoro representation. In the $c = 2$ vacuum representation, write L_m for the Virasoro modes, so $L_0 = L_0^{\mathrm{Vir}}$ and $L_m\Omega = 0$ for $m > 0$. The Virasoro relation gives

$$L_2 L_{-2}\Omega = [L_2, L_{-2}]\Omega = (4L_0 + c/2)\Omega = \Omega.$$

In particular $L_{-2}\Omega \neq 0$; it has energy two because $[L_0, L_{-2}] = 2L_{-2}$. Thus a non-vacuum vector has a Virasoro image in the vacuum line. The vacuum complement is not a Virasoro subrepresentation. A shift $L_0 \mapsto L_0 + aI$ with all other modes fixed is not an alternative normalization of the same representation: $[L_1, L_{-1}] = 2L_0$ forces $a = 0$.

Remark 51.2.11 *(arithmetic sector and trace comparison).* An arithmetic positivity construction must specify a Hilbert sector, its self-adjoint conformal-energy operator and common core, establish $A \succeq I/2$ there, and state which arithmetic and Virasoro operations preserve that sector. The spectral projection alone supplies a reducing sector for A, not those additional operations. A different arithmetic representation must be constructed if the intended operations require more than the non-vacuum Fock sector. To imply HP–Li positivity, such a construction must also provide a comparison with the Li transforms of Θ_ξ and the regularised trace that preserves the required positive forms and domains. These are separate construction requirements, not consequences of RH or of moment determinacy.

51.3 First Logical Core

The six residuals from Chapter 43 have the following logical form.

F1 reduces from the monolithic $\mathrm{L_{ACD}}$ to the conjunction $\mathrm{L_{ACD}}.1 + .2 + .3$; subcondition $\mathrm{L_{ACD}}.4$ is the product of strong approximation for $SL_2/\mathbb{Q}$ and the determinant idele-class quotient for $GL_2/\mathbb{Q}$; the first is a theorem of Kneser (1966) and Platonov–Rapinchuk (1994).

F2.c is Theorem 51.2.3, with actual operadic pointings. Its arithmetic use still requires the local E_2-structures, those pointing maps and the comparison with any independently defined global category.

F3 follows from the compatible R1, R2 and R3 comparison data of Theorem 51.2.5. The finite-type trace model proved here is the affine Perf(A) calculation of Proposition 51.2.6. The formal trace, global target and full Weil/(φ, Γ) equivalences remain actual comparison inputs.

P1 (Theorem 43.5.1) requires the arithmetic central-defect identity, Mellin evaluation, and defect detection. Question 51.2.8 asks separately for the coefficient object needed to apply an Iwasawa vanishing theorem.

P2 is theorem and unchanged.

P3 follows from the seven specified comparison data in Theorem 51.2.9. The local model norms and conjugations are explicit; the arithmetic placement and completed adelic pairing comparison remain to be constructed.

For zero-placement, the available implication is positive HP–Li trace $\Rightarrow$ Li positivity $\Rightarrow$ RH (Theorem 44.3.10), with the determinant and trace identities of H-P^{Ar}. A derivation from VB* requires the additional positive comparison described above.

The relevant comparison theories are Positselski (2011) and Ferrero–Washington (1979) for the arithmetic Positselski abelian domain; Flath (1979), Weil (1967), and Godement–Jacquet (1972) for the Koszul–Petersson sub-claims; and Gaitsgory–Rozenblyum DAG Vol. II for F3.BZN^{Ar}. The spectral comparison must also distinguish scalar traces from operator quadratic forms.

51.4 Second Residual Decomposition

The categorical comparisons require higher coherence data in addition to their scalar identities. The spectral comparison also requires a separation between a zero-divisor operator and a primary Virasoro representation, as in Chapter 35.

F2.c: E_2-Algebra Structure on the Unit Map

The pointing is a morphism $\eta : \mathrm{Sp} \to C$ in $\mathrm{Alg}_{E_2}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$. It is not an object of $\mathrm{Alg}_{E_2}(C)_{\mathbb{1}/}$. Tensoring these functors forms the finite-stage diagram, including nullary and higher compatibilities. Uniqueness of its structured colimit is relative to the full underlying colimit cocone.

Dunn additivity $E_2 \otimes E_2 \simeq E_4$ compares two supplied E_2-structures with their interchange coherences to an E_4-structure. A single E_2-structure does not supply that additional data. The inclusion $E_2 \to E_4$ induces restriction from E_4-algebras to E_2-algebras. Neither this restriction nor the filtered-colimit theorem constructs a higher structure on an arbitrary E_2-category.

$\mathrm{L}_{\mathrm{ACD}}$*: Archimedean Stratum and Hecke-at-Infinity*

The first-comparison split into $\mathrm{L}_{\mathrm{ACD}}$.1–.4 covers only finite places. At $v = \infty$, the spherical condition is K_∞-finiteness / admissibility in the sense of Harish-Chandra (1966), not Iwahori-fixedness; the empty-S base case collapses to $D(BK_\infty)$, not to the trivial category. The completed Tate character $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ forces any honest $\mathrm{L}_{\mathrm{ACD}}$ to include a parallel archimedean factor. The new subcondition is:

($\mathrm{L}_{\mathrm{ACD}}$.0) *Archimedean stratum.* At $v = \infty$, the local factor is $D((\mathfrak{g}, K_\infty)\text{-mod}^{\text{sph-fin}})$ with admissibility (Harish-Chandra) replacing Iwahori-fixedness; $\mathrm{L}_{\mathrm{ACD}}$.2 extends to include $U(\mathfrak{gl}_2)^{K_\infty}$-equivariance commuting with the finite-prime Hecke operators T_p; the empty-S base is $\mathrm{Rep}(K_\infty)$ with $\eta : \mathbb{1} \to \mathrm{triv}_{K_\infty}$. **Form: conjectural.**

F3.BZN^{Ar}*: Formal-Noetherian Extension, Non-Semisimple Correction, and Loop-Space Convention*

The first-comparison subconditions R1–R3 require four refinements.

(D1) The finite-type model proved in Proposition 51.2.6 uses $\mathrm{Perf}(A)$, a smooth affine k-scheme and ordinary derived loops. Denote this precise model by R1.fin-type. The arithmetic R1.formal requirement is an actual comparison for the chosen completed, support-restricted category on the Selmer-derived target. It must supply continuity under completion and the loop-object comparison; neither the phrase "formal Noetherian" nor smoothness of an unspecified stack supplies them.

(D2) R2 requires an ind-level cotangent-complex calculation on the global Selmer-derived target. The local smoothness input of Emerton–Gee §2 does not by itself supply the global comparison or its descent compatibility.

(D3) New subcondition R4: the Iwahori Hecke category $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p}$ at wild primes $p \in \{2, 3\}$ is not semisimple; the Hochschild trace picks up $\mathrm{Ext}^{\mathrm{I}}$-contributions invisible to the Ben-Zvi–Nadler semisimple template (Etingof–Ostrik 2004). **Form: conjectural.**

(D4) R3 requires a derived-internal-Hom convention for profinite classifying stacks and comparison maps among the cofiltered-limit, derived internal-Hom and direct ind-constructions after the specified formal-Noetherian completion. Their agreement is part of the required comparison.

Arithmetic Positselski: Five Supplementary Subconditions

The coefficient comparison of Question 51.2.7 must be supplemented by further operations in the arithmetic theory. (C1) requires factorization-coalgebra coherence at each place. (C2) requires $d^2 = 0$

for the three-piece arithmetic differential $d^{\mathrm{Ar}} = d_{\mathrm{Iw}} + d_{\mathrm{Gersten}} + d_{\mathrm{bar}}^{\mathrm{local}}$ (the candidate cancellation input is Parshin reciprocity). (C3) requires Euler-product convergence for the infinite-product structure, with an actual convergence theorem. (C4) requires a comparison at the archimedean place. (C5) requires naturality of the co-contra equivalence in morphisms of abelian Λ-coalgebras.

C5 is required for any functorial use of a constructed co–contra comparison. P1 additionally needs the explicit evaluation in Theorem 43.5.1. Naturality alone supplies neither its central-defect identity nor its detection condition.

Koszul–Petersson Compatibility: Five Higher-Structural Sub-Claims

The scalar-level decomposition K1–K7 of the first comparison (Theorem 51.2.9) conceals five higher-structural conditions: (K1a) the categorical-to-Hilbert descent functor $F = \mathrm{RHom}(\mathbb{1}, -)$ is not exhibited; (K3a) Galois-KMS$_1$ equivariance requires the Connes–Marcolli (2004) passage from the Bost–Connes Galois symmetry to the half-density normalised Tomita–Takesaki conjugation; (K5a) defect-pairing commutativity at ramified primes is a potential anomaly in the Kim arithmetic Chern–Simons defect setup: the open question is whether the Kim arithmetic-CS defect structure is anomaly-free at primes where Hecke characters ramify; (K6a) the actual adelic measure comparison must be compatible with any finite nonzero continued $L(1, \mathrm{Ad}\pi)$ normalization applied to both pairings; (K7a) dimensional-reduction coherence ($3d$ adelic $\to$ $2d$ archimedean $\to$ $1d$ prime-knot) is stated without proof. K5a asks for the ramified defect-pairing calculation and its compatibility with the specified arithmetic comparison maps.

VB: Zero-Divisor Operators and Primary Virasoro Representations*

H-P^{Ar} (Hypothesis 44.3.2) concerns the self-adjoint ordinate operator H_{HP} and the normal operator $\Theta_\xi = \frac{1}{2}I + iH_{\mathrm{HP}}$. A primary Virasoro representation is a separate construction. Chapter 35 identifies three obstructions on the full Hochschild trace class $\mathcal{V}^{\mathrm{prim}}$: (i) only the modes $m \geq -1$ act as global finite sums on the restricted tensor product; (ii) the central term $c_{\mathrm{full}} = \sum_p c^{(p)}$ diverges; (iii) the scalar $r_\zeta = 2$ is a determinant-rank residue, not a trace or a global central charge. Restricting to the primary subspace does not by itself construct closed Virasoro operators or identify them with the spectral operator. The relevant distinctions in Chapter 44 are:

- Remark 44.3.4 separates H_{HP} from conformal modes and keeps $\mathcal{V}^{\mathrm{prim}}$ as a Hochschild trace class.
- Conjecture 44.3.5 asks for closed operators L_m^{PV} on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}}$ satisfying the Virasoro relations with central charge 2. Its equality with $r_\zeta = 2$ is additional compatibility data.
- Remark 44.3.6 distinguishes the Fock energy L_0^{Vir}, Bost–Connes Hamiltonian N_{BC}, ordinate operator H_{HP}, and zero-divisor operator Θ_ξ. Self-adjointness of H_{HP} means that its spectral ordinates are real; $\Theta_\xi - \frac{1}{2}I = iH_{\mathrm{HP}}$ is skew-adjoint.
- Remark 44.3.7 uses $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. The gamma poles cancel the trivial zeros and $s(s-1)$ removes the Tate poles. This normalization is distinct from restricting a Fock representation to a primary subspace.

The primary-Virasoro construction and the Weyl-asymptotic comparison remain separate questions. The latter asks whether the Riemann–von Mangoldt density $N(T) \sim (T/2\pi)(\log(T/2\pi) - 1)$ arises in a specified Berry–Keating xp, Connes adelic, or Meyer–Schwartz operator model.

51.5 Resulting Logical Core

The subcondition structure is as follows.

For F2.c, the pointing in Definition 51.2.1 is a morphism $\mathrm{Sp} \to \mathcal{C}$ of E_2-algebras in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$. It includes nullary compatibility. The colimit structure has a contractible space of choices relative to the entire underlying colimit cocone, as in Theorem 51.2.3.

For $\mathrm{L}_{\mathrm{ACD}}$: new subcondition $\mathrm{L}_{\mathrm{ACD}}.0$ (archimedean stratum, as above); $\mathrm{L}_{\mathrm{ACD}}.2$ is extended to include $U(\mathfrak{gl}_2)^{K_\infty}$-equivariance.

For F3.$\mathrm{BZN}^{\mathrm{Ar}}$: R1 splits into R1.fin-type (the stated smooth affine $\mathrm{Perf}(A)$ calculation) and R1.formal (the required completed, support-compatible trace comparison); R4 is the non-semisimple Hecke condition at wild primes $p \in \{2, 3\}$.

For arithmetic co–contra comparison, C1–C5 specify further operations and naturality. The P1 application also requires the central-defect identity, Mellin evaluation, and detection condition.

For Koszul–Petersson: five higher-structural sub-claims K1a–K7a are required; K5a asks whether the ramified defect pairing in the Kim arithmetic Chern–Simons construction is anomaly-free.

For VB^*, the primary-Virasoro construction (Conjecture 44.3.5), its common operator domain, and its comparison with Θ_ξ remain additional data. The Weyl-asymptotic problem concerns the spectral realization rather than a Virasoro central-charge computation.

The P1 criterion requires its arithmetic evaluation and defect detection (Theorem 43.5.1). P2 retains its stated domain. P3 requires the arithmetic placement, pairings, completions, and adelic maps of K1–K7 and their higher conditions. The local model calculations do not supply these comparisons.

The F1–F3 comparisons include $\mathrm{L}_{\mathrm{ACD}}.0$, R4, and R1.formal. The positive HP–Li trace implication in Theorem 44.3.10 requires H-P^{Ar} from Hypothesis 44.3.2. The primary Virasoro comparison remains Conjecture 44.3.5.

The sufficient trace route to RH uses the determinant and trace identities of H-P^{Ar} and HP–Li positivity, on the specified arithmetic Hermitian domain. Arithmetic Positselski (C1–C5), Koszul–Petersson compatibility (K1–K7, K1a–K7a), primary-Virasoro reconstruction and Weyl asymptotics are distinct comparison problems; their labels do not identify their carriers. A route through operator positivity must first specify its sector and preserved operations, and then supply the positive trace comparison. Proposition 51.2.10 rules out the all-Fock assertion under the vacuum normalization and constructs the maximal reducing positivity sector of a given self-adjoint energy operator. Its comparison with arithmetic operations and the HP–Li trace is the remaining problem of Remark 51.2.11.

The finite trace calculation and the operator dictionary have different roles. The calculation excludes a general scalar-trace-to-operator positivity argument. The dictionary states which arithmetic operators and maps a proof must actually construct. Neither removes the primary-Virasoro or spectral-realization problems.

The remaining open subconditions are primary-Virasoro reconciliation, the non-semisimple Hecke condition R4, the defect-pairing anomaly K5a, and Weyl-asymptotic class.

Chapter 52

$\mathrm{L}_{\mathrm{ACD}}$ Subconditions: Archimedean Stratum, Colimit Commutation, Cross-Prime Hecke, Sph-Finite Preservation

The modular decomposition of $\mathrm{L}_{\mathrm{ACD}}$ *(Theorem 48.2.1, Theorem 51.2.4) produces five subconditions:* $\mathrm{L}_{\mathrm{ACD}}.0$ *(archimedean stratum admissibility, isolated by the archimedean-factor analysis of Section 51.4),* $\mathrm{L}_{\mathrm{ACD}}.1$ *(dg-categorical colimit commutation),* $\mathrm{L}_{\mathrm{ACD}}.2$ *(cross-prime Hecke coherence including* $U(\mathfrak{gl}_2)^{K_\infty}$*-equivariance),* $\mathrm{L}_{\mathrm{ACD}}.3$ *(sph-finite preservation), and* $\mathrm{L}_{\mathrm{ACD}}.4$ *(strong approximation for* $SL_2/\mathbb{Q}$ *together with the determinant idele-class quotient for* $GL_2/\mathbb{Q}$*). Kneser 1966 and Platonov–Rapinchuk 1994 supply the simply connected part; Tate's idele formalism supplies the determinant line. This chapter separates the formal subconditions from the remaining comparison data: Harish-Chandra 1966 admissibility, Jacquet–Langlands SLN 114 archimedean theory, Vogan 1981 classification, Wallach 1988/1992 real-reductive-groups spectrum theory, Bernstein–Zelevinsky 1976 induced representations, and Lurie HA §5.1, §4.8 and Fresse 2017* E_n *operadic machinery; the Fargues–Scholze list is not invoked for the formal steps. The* ∞*-categorical Bernstein lift and the Bernstein-centre diamond-gluing compatibility remain open inputs.*

52.1 Subconditions and Independent Sources

The Subcondition List

The modular decomposition (Theorem 48.2.1) splits $\mathrm{L}_{\mathrm{ACD}}$ into (L_1)–(L_4); Theorem 51.2.4 relabels these as $\mathrm{L}_{\mathrm{ACD}}.1$–$\mathrm{L}_{\mathrm{ACD}}.4$:

- $\mathrm{L}_{\mathrm{ACD}}.1$ (colimit commutation);
- $\mathrm{L}_{\mathrm{ACD}}.2$ (cross-prime Hecke coherence);
- $\mathrm{L}_{\mathrm{ACD}}.3$ (sph-finite preservation);
- $\mathrm{L}_{\mathrm{ACD}}.4$ (strong approximation for SL_2 plus determinant idele-class quotient for GL_2).

The archimedean-factor analysis of Section 51.4 identifies a fifth subcondition $\mathrm{L}_{\mathrm{ACD}}.0$ (archimedean stratum admissibility) and tightens $\mathrm{L}_{\mathrm{ACD}}.2$ to include $U(\mathfrak{gl}_2)^{K_\infty}$-equivariance.

The local list of subconditions considered here is:

$\mathrm{L}_{\mathrm{ACD}}.0$: archimedean stratum with Harish-Chandra admissibility replacing Iwahori-fixedness;

$\mathrm{L}_{\mathrm{ACD}}.1$: colimit commutes with sph-finite truncation in $\mathrm{Pr}^{\mathrm{L,st}}$;

$\mathrm{L}_{\mathrm{ACD}}.2$: cross-prime Hecke coherence including $U(\mathfrak{gl}_2)^{K_\infty}$-equivariance;

$\mathrm{L}_{\mathrm{ACD}}.3$: sph-finite preservation under restricted-tensor colimit;

$\mathrm{L}_{\mathrm{ACD}}.4$: strong approximation for $SL_2/\mathbb{Q}$, with the determinant quotient retained for $GL_2/\mathbb{Q}$.

Independent Sources

The formal parts of $L_{ACD}.0$–$L_{ACD}.3$ are treated here via an independent list; the Fargues–Scholze geometrization construction is not invoked for those formal steps.

Primary sources used:

Harish-Chandra 1966 "Discrete series for semisimple Lie groups II", Acta Math. 116. Admissible representations of reductive Lie groups; the category of $(\mathfrak{g}, K)$-modules.

Jacquet–Langlands 1970 *Automorphic Forms on $GL(2)$*, SLN 114. Archimedean local theory for $GL_2(\mathbb{R})$ and $GL_2(\mathbb{C})$.

Vogan 1981 *Representations of Real Reductive Lie Groups*, change in Math. 15. Langlands–Vogan classification at $\mathbb{R}$.

Wallach 1988/1992 *Real Reductive Groups I, II*, Pure & Applied Math., Academic Press. Admissible spectrum theory.

Bernstein–Zelevinsky 1976 "Induced representations of reductive $\mathfrak{p}$-adic groups I", Ann. Sci. ENS. Bernstein decomposition of p-adic Hecke categories.

Lurie HA *Higher Algebra*, 2017 draft. §5.1 (E_n monoidal coherence), §4.8 (colimits in Pr^{L}).

Fresse 2017 *Homotopy of Operads and Grothendieck–Teichmüller Groups, Vol. II*. Integral E_n operadic machinery.

Secondary sources: Beilinson–Bernstein 1993 (localization; D-modules on G/B, archimedean localization), Nadler 2009 (microlocal sheaves on real flag varieties), Borel 1976 (admissible representations of semi-simple groups over local fields), Kneser 1966 (strong approximation for simply-connected groups), Platonov–Rapinchuk 1994 *Algebraic Groups and Number Theory*, Moeglin–Waldspurger 2016 (twisted trace formula).

Raskin 2021 JAMS supplies the geometric precedent for (L_1). For (L_2), the local input is Fargues–Scholze, while the function-field comparison remains an analogy. Gaitsgory–Rozenblyum supplies the $\mathrm{IndCoh}_{\mathrm{Nilp}}$ continuity input.

The categorical colimit argument uses Lurie HA §5.1 and §4.8, with the stated pointed-E_n hypotheses. The archimedean admissibility input uses Harish-Chandra, Vogan, and Wallach. Bernstein–Zelevinsky and Lurie HA §5.1.2 supply the local Hecke and symmetric-monoidal constructions. For $L_{ACD}.0$, $L_{ACD}.1$, and $L_{ACD}.3$, the geometric source set consists of Raskin, Fargues–Scholze, and Gaitsgory–Rozenblyum. The categorical and representation-theoretic set consists of Lurie, Harish-Chandra, Vogan, Wallach, and Bernstein–Zelevinsky. These sources concern different parts of the construction. A comparison requires the actual maps and compatibilities of Chapter 1; the separation of references does not provide them. Raskin's geometric descent result does not establish archimedean admissibility, and Wallach's representation theory does not construct colimits in $\mathrm{Pr}^{\mathrm{L,st}}$. The ∞-categorical lift required for Bernstein–Zelevinsky compatibility remains a separate lift datum.

52.2 Overview of Subcondition Closures

The formal subconditions $L_{ACD}.0$–$L_{ACD}.3$ are analysed in the four sections that follow; the compatibility with the completed Tate character $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ functional equation occupies Section 52.8.

$L_{ACD}.0$: *archimedean stratum*

At the archimedean place, the local category $\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{R})])$ is not defined by Iwahori-fixed vectors (as at finite primes) but by $(\mathfrak{g}, K_\infty)$-admissibility: the empty-S base case of the restricted tensor product

is $\mathrm{Rep}(K_\infty)$, not Vec, and any construction of the restricted tensor product $\bigotimes'_{v\in|\overline{C}|} C_v$ must carry a parallel archimedean factor.

The resolution is Harish-Chandra admissibility. The archimedean local category is $\mathrm{D}((\mathfrak{g}, K_\infty)\text{-mod}^{\text{sph-fin}})$ with admissibility replacing Iwahori-fixedness. Jacquet–Langlands SLN 114 §1–3 and Vogan 1981 classify the admissible $(\mathfrak{g}, K_\infty)$-modules for $GL_2(\mathbb{R})$; the sph-fin cut selects the modules finite over the archimedean Hecke algebra (the convolution algebra of bi-K_∞-finite distributions). The detailed construction and proof appear in Section 52.3.

$\mathrm{L}_{\mathrm{ACD}}$.1*: colimit commutation*

The filtered colimit defining $\bigotimes'_v C_v$ and the sph-fin subcategory inclusion $C_v^{\text{sph-fin}} \hookrightarrow C_v$ are a priori unrelated operations; it is not immediate that they commute.

F2.c tightened (Theorem 50.1.3) supplies the universal property needed: the restricted tensor product of pointed E_n-monoidal categories carries a canonical E_n-monoidal structure, unique up to a contractible space of choices, compatible with the insertion functors ι_S. Sph-finite truncation is defined by compact generation (Lemma 48.2.4), and Lurie HA 5.3.2.9 gives that filtered colimits of compactly generated categories along compact-generator-preserving functors are compactly generated. Hence the colimit commutes with the sph-fin inclusion. The proof appears in Section 52.4.

$\mathrm{L}_{\mathrm{ACD}}$.2*: cross-prime Hecke coherence*

For distinct primes $p \neq q$, the local Hecke operators T_p and T_q act on disjoint tensor factors of the restricted tensor product. The question is whether they commute up to coherent associators and unit coherences, and whether this extends to $U(\mathfrak{gl}_2)^{K_\infty}$-equivariance at the archimedean place.

Lurie HA 5.1.2.4 (Dunn additivity) and HA 5.1.2.26 (centrality of the unit in an E_n-monoidal category) supply the coherence: endofunctors of a tensor product acting on disjoint tensor factors commute up to a contractible space of coherence data. Extension to the archimedean place is immediate from the disjointness of C_∞ and each C_p as tensor factors. The proof appears in Section 52.5.

$\mathrm{L}_{\mathrm{ACD}}$.3*: sph-finite preservation*

Sph-finite truncation composes with the colimit to give the sph-finite truncation of the colimit (not merely an approximation), because this is the content of $\mathrm{L}_{\mathrm{ACD}}$.1: if the colimit commutes with sph-fin, then the sph-fin subcategory of the colimit equals the colimit of sph-fin subcategories. Lemma 48.2.4 handles the compact generation step. The proof appears in Section 52.6.

Archimedean character and the completed ξ-function

The archimedean factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ is the archimedean local L-factor of the trivial representation; Tate 1950 §4.3 computes it as the Mellin transform of the Gaussian. Thus the trace comparison, if it is constructed, must send the archimedean unit to the Gaussian Tate vector. The restricted tensor product is compared with $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$, not with $\zeta(s)$ alone. The necessary Tate calculation appears in Section 52.8; the categorical trace comparison is a separate input.

52.3 $\mathrm{L}_{\mathrm{ACD}}$.0: Archimedean Stratum Closure

Statement and Construction

Definition 52.3.1 *(archimedean admissible $(\mathfrak{g}, K_\infty)$-module).* Let $\mathfrak{g} = \mathfrak{gl}_2(\mathbb{R})$ and $K_\infty = O(2) \subseteq GL_2(\mathbb{R})$. A $(\mathfrak{g}, K_\infty)$-*module* is a real vector space V with a $\mathfrak{g}$-module structure and a compatible

K_∞-action such that: (i) the K_∞-action is locally finite; (ii) the differentiated K_∞-action agrees with the restriction of the $\mathfrak{g}$-action to $\mathrm{Lie}(K_\infty) = \mathfrak{k}$. The module is *admissible* if also: (iii) each K_∞-isotype $V(\sigma)$ for $\sigma \in \widehat{K_\infty}$ is finite-dimensional (Harish-Chandra 1966 §3).

Definition 52.3.2 *(archimedean sph-finite category).* The archimedean sph-finite category is

$$\mathcal{C}_\infty \;:=\; \mathrm{D}^{\text{sph-fin}}\big((\mathfrak{g}, K_\infty)\text{-mod}^{\text{adm}}\big)$$

the stable presentable ∞-category of admissible $(\mathfrak{g}, K_\infty)$-modules of finite type over the archimedean Hecke algebra $\mathcal{H}_\infty = \mathrm{End}_{(\mathfrak{g},K_\infty)}(V_{K_\infty}^{\mathrm{fin}})$, where $V_{K_\infty}^{\mathrm{fin}}$ ranges over admissible $(\mathfrak{g}, K_\infty)$-modules.

Theorem 52.3.3 *(L_{ACD}.0 closed).* The archimedean category $\mathcal{C}_\infty$ of Definition 52.3.2 is:

(i) a well-defined stable presentable ∞-category;

(ii) equipped with a distinguished unit $u_\infty \in \mathcal{C}_\infty$ (the K_∞-spherical unramified principal series) whose unit-comparison map $\eta_\infty : \mathbb{1}_{\mathcal{C}_\infty} \to u_\infty$ is canonical;

(iii) classifiable by Langlands–Vogan parameters (Vogan 1981 Chapters 2–5): the isomorphism classes of irreducible admissible $(\mathfrak{g}, K_\infty)$-modules correspond bijectively to triples $(\lambda, \mu, \varepsilon)$ with $\lambda \in \mathbb{C}$ (weight), $\mu \in \mathbb{Z}_{\geq 0}$ (discrete-series parameter), and $\varepsilon \in \{\pm 1\}$ (central character sign);

(iv) the empty-S base case of the restricted tensor product is $\mathrm{Rep}(K_\infty)$, not Vec.

Proof. **Step 1: Stability and presentability.** The category of $(\mathfrak{g}, K_\infty)$-modules is abelian (Wallach 1988 §0.1). Passing to the derived category and restricting to admissible modules (finite per K_∞-isotype) preserves stability (the derived category of an abelian category is stable). Presentability follows from Wallach 1988 Theorem 0.A.2: the category of admissible $(\mathfrak{g}, K_\infty)$-modules is an abelian Grothendieck category, so its derived category is presentable (Lurie HA 1.3.5.21).

Step 2: Distinguished unit. The unramified principal series $\mathrm{Ind}_{B(\mathbb{R})}^{GL_2(\mathbb{R})}(\chi_1 \boxtimes \chi_2)$ with $\chi_1 = \chi_2 = |\cdot|^0$ the trivial character gives the irreducible spherical unramified principal series; this is the archimedean analogue of the spherical Iwahori-fixed vector at a finite prime. Harish-Chandra 1958 "Spherical functions on a semisimple Lie group" identifies this module as the unique irreducible admissible representation of $GL_2(\mathbb{R})$ with a non-zero K_∞-fixed vector at weight 0. The unit-comparison map $\eta_\infty : \mathbb{1} \to u_\infty$ is the inclusion of the unit admissible module (the trivial character of the archimedean Hecke algebra) into the spherical unramified principal series.

Step 3: Langlands–Vogan classification. Vogan 1981 Theorem 5.1.1 gives the Langlands classification for $GL_2(\mathbb{R})$: the irreducible admissible $(\mathfrak{g}, K_\infty)$-modules are parametrized by Langlands data, which for $GL_2(\mathbb{R})$ reduces to the triple $(\lambda, \mu, \varepsilon)$. The classes are:

- principal series (including the spherical unramified one), parametrized by $\lambda \in \mathbb{C}$ and central character sign ε;
- discrete series D_k for $k \in \mathbb{Z}_{\geq 2}$, with lowest weight k;
- limits of discrete series $D_1^{\pm}$;
- irreducible unitary finite-dimensional modules (only the trivial module here).

Each class is K_∞-finite in each isotype, hence admissible. Wallach 1988 Ch. 5 organises these into the admissible spectrum.

Step 4: Base case identification. The empty-S stage of the restricted tensor product is $\bigotimes_{v \in \emptyset} \mathcal{C}_v = \mathbb{1}_{\mathrm{Pr}^{L,\mathrm{st}}} = \mathrm{Sp}$. The transition map $\iota_{\emptyset \subseteq \{\infty\}}$ inserts the spherical unit u_∞. The image of Sp under tensoring with u_∞ is $\mathrm{Sp} \otimes u_\infty$, which inside $\mathcal{C}_\infty$ is the full subcategory generated by u_∞. Since u_∞ is a K_∞-finite

admissible module, this subcategory is $\mathrm{Rep}(K_\infty)$ as a subcategory of admissible $(\mathfrak{g}, K_\infty)$-modules. The empty-S base is therefore $\mathrm{Rep}(K_\infty)$, not Vec. □

Compatibility with Finite-Place Sph-Fin Structure

Proposition 52.3.4 *(uniformity of the sph-fin definition across places).* The sph-fin condition at an archimedean place (Definition 52.3.2) is the natural admissibility restriction, and coincides with the finite-prime sph-fin condition (compact-generated ∞-category with finite-type compacts) under the identification of K_∞-finite admissible modules with finite-type modules over the archimedean Hecke algebra.

Proof. Wallach 1988 §0.A associates to each admissible $(\mathfrak{g}, K_\infty)$-module a module over the archimedean Hecke algebra $\mathcal{H}_\infty$, which is the convolution algebra of bi-K_∞-finite distributions on $GL_2(\mathbb{R})$. This gives a faithful embedding of admissible $(\mathfrak{g}, K_\infty)$-modules into modules over $\mathcal{H}_\infty$. Finite-type $\mathcal{H}_\infty$-modules correspond to K_∞-finite admissible modules (Harish-Chandra 1966 §4). Hence the archimedean sph-fin $C_\infty^{\text{sph-fin}}$ corresponds to the full subcategory of compactly generated $\mathcal{H}_\infty$-modules, matching the finite-prime definition. □

Corollary 52.3.5 *(archimedean factor enters the restricted tensor).* The archimedean factor C_∞ enters the restricted tensor product $\bigotimes'_{v\in|\overline{C}|} C_v$ on equal footing with the finite-prime factors, with distinguished unit u_∞ as in Theorem 52.3.3. The indexing set is $V = |\overline{C}| = \mathbb{P} \cup \{\infty\}$, and the cofinite unramified subset V^{sph} includes both finite unramified primes and the archimedean place.

Remark 52.3.6 *(archimedean admissibility tightens the (L_4) extension requirement).* The subcondition (L_4) of Theorem 48.2.1 was stated as "the archimedean factor extends the finite-place product by direct tensoring." L_{ACD}.0 tightens this: the archimedean factor is Harish-Chandra admissible rather than Iwahori-fixed, and the empty-S base case is $\mathrm{Rep}(K_\infty)$ rather than Vec. The original (L_4) statement is preserved as a corollary; the tightening is substantive, since no Iwahori subgroup of $GL_2(\mathbb{R})$ gives the natural finiteness cut at $v = \infty$.

52.4 L_{ACD}.1: Dg-Categorical Colimit Commutation

Statement

Theorem 52.4.1 (L_{ACD}.1 *closed given F2.c). (conditional on F2.c tightened, Theorem 50.1.3; F2.c tightened is itself a theorem, see Remark 52.7.2).* Let $\{(C_v, u_v, \eta_v)\}_v$ be a family of pointed E_2-monoidal stable presentable ∞-categories (in the sense of Definition 51.2.1, with η_v an E_2-algebra map per the tightening of Theorem 50.1.3) with $u_v = \mathbb{1}_{C_v}$ at almost every v. Let $V^{\text{sph}} \subseteq V$ be a cofinite subset of places. Assume each C_v carries a sph-fin subcategory $C_v^{\text{sph-fin}} \subseteq C_v$ compact-generating, and u_v is compact for $v \in V^{\text{sph}}$. Then the restricted tensor product commutes with sph-fin truncation:

$$\Big(\bigotimes_v{}' C_v\Big)^{\text{sph-fin}} \;\simeq\; \bigotimes_v{}' C_v^{\text{sph-fin}} \tag{52.1}$$

as stable presentable ∞-categories.

Proof. **Step 1: Both sides are compactly generated.** Theorem 50.1.3(M2) and Lemma 48.2.4 together show that $\otimes'_v C_v$ is compactly generated, with compact subcategory $\varinjlim_S \otimes_{v\in S} C_v^\omega$. The right-hand side of (52.1) is the restricted tensor product of sph-fin categories, itself compactly generated by Lemma 48.2.4.

Step 2: Compact objects match. Lurie HA 5.3.2.9 gives that a filtered colimit of compactly generated ∞-categories along compact-generator-preserving functors is compactly generated, with compact subcategory the filtered colimit of compact subcategories. Applied to the defining filtered colimit of the restricted tensor product:

$$\Big(\bigotimes_v{}' C_v\Big)^{\omega} \;=\; \varinjlim_{S} \bigotimes_{v\in S} C_v^{\omega}.$$

Step 3: Sph-fin is the compact subcategory. By the sph-fin definition in Lemma 48.2.4, the sph-fin condition *is* the compact-generation structure enforced by the distinguished spherical units. Hence

$$\Big(\bigotimes_v{}' C_v\Big)^{\text{sph-fin}} \;=\; \mathrm{Ind}\Big(\big(\bigotimes_v{}' C_v\big)^{\omega}\Big) \;=\; \mathrm{Ind}\Big(\varinjlim_{S} \bigotimes_{v\in S} C_v^{\omega}\Big).$$

Step 4: Ind commutes with filtered colim. Lurie HTT 5.3.5.15 (Ind-completion preserves filtered colimits of small ∞-categories). The Ind of a filtered colimit of small ∞-categories is the filtered colimit of Ind-completions. Hence

$$\mathrm{Ind}\Big(\varinjlim_{S} \bigotimes_{v\in S} C_v^{\omega}\Big) \;=\; \varinjlim_{S} \mathrm{Ind}\Big(\bigotimes_{v\in S} C_v^{\omega}\Big) \;=\; \varinjlim_{S} \bigotimes_{v\in S} C_v^{\text{sph-fin}} \;=\; \bigotimes_v{}' C_v^{\text{sph-fin}}.$$

Step 5: E_2-monoidal structure matches. F2.c tightened (Theorem 50.1.3(M3)) gives that the E_2-monoidal structure on the restricted tensor product is unique up to a contractible space of choices, and the uniqueness is cofinality-invariant. The E_2-structure on the LHS of (52.1) is inherited from $\otimes'_v C_v$ via restriction to the sph-fin subcategory; the E_2-structure on the RHS is inherited from $\otimes'_v C_v^{\text{sph-fin}}$. Both agree by the universal property of the canonical E_2-monoidal structure.

Assembling Steps 1–5: both sides of (52.1) are compactly generated stable presentable ∞-categories with isomorphic compact subcategories and isomorphic E_2-monoidal structures. The isomorphism follows. □

Remark 52.4.2 *(role of F2.c tightening).* L_{ACD}.1 is a consequence of F2.c tightened plus Lurie HA 5.3.2.9 and HTT 5.3.5.15. F2.c tightened supplies the universal property of the colimit's E_2-monoidal structure; HA 5.3.2.9 supplies the sph-fin preservation under the filtered colimit; HTT 5.3.5.15 supplies the Ind-filtered-colimit exchange. The three ingredients together close L_{ACD}.1. The source list is entirely Lurie HA + HTT, disjoint from Fargues–Scholze and from Raskin 2021 JAMS.

Explicit Commutation Square

Corollary 52.4.3 *(commutation square).* The following square in $\mathrm{Pr}^{L,\mathrm{st}}$ commutes up to equivalence:

$$\begin{array}{ccc} \big\{C_v^{\text{sph-fin}}\big\}_v & \xrightarrow{\otimes'} & \otimes'_v C_v^{\text{sph-fin}} \\ \big\downarrow & & \big\downarrow{\scriptstyle \simeq} \\ \big\{C_v\big\}_v & \xrightarrow{\otimes'} & \big(\otimes'_v C_v\big)^{\text{sph-fin}} \end{array}$$

where the vertical arrows are sph-fin inclusion / sph-fin truncation and the horizontal arrows are the Tate restricted tensor product.

Proof. Immediate from Theorem 52.4.1 plus functoriality of the restricted tensor product under compact-generator-preserving functors. □

52.5 $L_{ACD}.2$: Cross-Prime Hecke Coherence

Statement

Theorem 52.5.1 *(formal tensor-factor Hecke commutativity).* For a family $\{(C_v, u_v, \eta_v)\}_v$ of pointed E_2-monoidal stable presentable ∞-categories as in Theorem 52.4.1, equipped with local Hecke operators $T_v \in \mathrm{End}_{E_2}(C_v)$ (i.e. E_2-algebra endomorphisms of each factor) and archimedean $U(\mathfrak{gl}_2)^{K_\infty}$-action on C_∞:

(i) For any two distinct places $v \neq w$, the extensions $\widetilde{T}_v, \widetilde{T}_w : \otimes'_v C_v \to \otimes'_v C_v$ of the local operators by identity on all other factors commute up to canonical homotopy:

$$\widetilde{T}_v \circ \widetilde{T}_w \;\simeq\; \widetilde{T}_w \circ \widetilde{T}_v.$$

(ii) Associativity and unit coherences hold: for any three distinct places v_1, v_2, v_3, the square associators and unitors satisfy the standard symmetric-monoidal MacLane pentagon / triangle coherences (Lurie HA 5.1.2.6).

(iii) The archimedean $U(\mathfrak{gl}_2)^{K_\infty}$-equivariance commutes with every finite-prime T_p.

Remark 52.5.2 *(domain: formal versus global Hecke coherence).* The theorem establishes commutativity of local endofunctors on the restricted tensor product and the archimedean equivariance. Global Hecke functoriality of the automorphic category and the local-global Whittaker–Bernstein compatibility are distinct properties; both remain part of (BC-diamond-glue) and are outside the domain of Theorem 52.5.1.

Proof. **Step 1: Tensor-factor disjointness.** Each Hecke operator T_v acts on the single factor C_v of the restricted tensor product. The extension by identity on other factors, $\widetilde{T}_v := T_v \otimes \mathrm{id}_{\neq v}$, is an endofunctor of $\otimes'_v C_v$ that factors through the tensor factor at v. For distinct places $v \neq w$, the endofunctors $\widetilde{T}_v$ and $\widetilde{T}_w$ factor through disjoint tensor factors.

Step 2: Disjoint tensor factors commute up to canonical homotopy. Lurie HA 4.8.1 states that endofunctors of a tensor product $\mathcal{C} \otimes \mathcal{D}$ that factor through disjoint tensor factors commute up to a canonical homotopy, because they correspond to bi-colimit-preserving functors $\mathcal{C} \times \mathcal{D} \to \mathcal{C} \otimes \mathcal{D}$ and the product in the source is symmetric. Applied to $\widetilde{T}_v$ and $\widetilde{T}_w$ acting on $C_v \otimes C_w \otimes \otimes'_{u \neq v,w} C_u$: commutativity holds up to canonical homotopy.

Step 3: Higher coherences. Lurie HA 5.1.2.6 asserts that E_2-monoidal stable presentable ∞-categories form a $(\infty, 3)$-category in which the coherence data for a symmetric-monoidal pattern is contractible. The three-fold composition of Hecke operators at distinct places v_1, v_2, v_3 admits a unique coherent associator by the universal property of the restricted tensor product. The pentagon and triangle coherences are automatic.

Step 4: Archimedean equivariance. The $U(\mathfrak{gl}_2)^{K_\infty}$-action on C_∞ is the central Harish-Chandra action on the archimedean $(\mathfrak{g}, K_\infty)$-category. It factors through C_∞ alone. For any finite prime p, the Hecke operator T_p factors through C_p. The two factor through disjoint tensor factors (C_∞ and C_p). By Step 2, they commute up to canonical homotopy.

Full coherence. The full coherence of the symmetric-monoidal pattern is encoded by the requirement that $\otimes'_v C_v$ is an E_2-algebra object in $\mathrm{Pr}^{L,\mathrm{st}}$ carrying E_2-algebra endomorphisms $\widetilde{T}_v$ that factor through disjoint tensor factors. Lurie HA 5.1.2.4 (Dunn additivity $E_2 \otimes E_2 \simeq E_4$) would suggest that the composite space of such endomorphisms is naturally E_4-structured; however, restriction to the subspace of endomorphisms factoring through individual tensor factors (i.e. local Hecke operators) reduces

the structure back to E_2 per factor. This is the precise sense in which cross-prime Hecke coherence holds. □

Bernstein–Zelevinsky Compatibility

Conjecture 52.5.3 *(∞-categorical Bernstein lift).* The Bernstein–Zelevinsky decomposition of the abelian category of smooth $GL_2(\mathbb{Q}_p)$-representations lifts to the stable presentable local Hecke category C_p, with the Bernstein centre acting by orthogonal idempotent projectors and with the spherical finite subcategory generated by compact objects finite over the spherical Bernstein centre.

Proposition 52.5.4 *(Bernstein–Zelevinsky local compatibility).* Conditional on Conjecture 52.5.3. At each finite prime p, the local Hecke category C_p decomposes along Bernstein–Zelevinsky components (Bernstein–Zelevinsky 1976 Ann. ENS §2): each Bernstein component corresponds to a cuspidal datum $[M, \sigma]$ of a Levi $M \subseteq GL_2(\mathbb{Q}_p)$ and a cuspidal supercuspidal representation σ of M. The Hecke operator T_p preserves each Bernstein component.

Proof. Bernstein–Zelevinsky 1976 proves the decomposition of the category of smooth representations of $GL_2(\mathbb{Q}_p)$ into Bernstein components. Each component is invariant under the centre of the Hecke algebra, which includes the spherical Hecke operator T_p acting on the spherical principal series component (the unramified component). The passage from this abelian decomposition to C_p is exactly Conjecture 52.5.3. Under that datum, the idempotent projectors commute with T_p, so T_p preserves every lifted Bernstein component. □

Remark 52.5.5 *(role of Bernstein–Zelevinsky).* Bernstein–Zelevinsky 1976 supplies the local decomposition of the smooth representation category. After Conjecture 52.5.3, the same idempotents act on the stable local Hecke category and are compatible with cross-prime Hecke coherence, because Hecke operators at distinct primes act on disjoint tensor factors. The global statement is therefore conditional on the lift datum; it is not a consequence of the abelian Bernstein–Zelevinsky theorem alone.

52.6 $L_{ACD}.3$: Sph-Finite Preservation

Statement

Theorem 52.6.1 *($L_{ACD}.3$ automatic given $L_{ACD}.1$).* The sph-fin subcategory $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ is preserved under the restricted-tensor colimit: the colimit of sph-fin categories embeds in the sph-fin subcategory of the colimit, and this embedding is an equivalence:

$$\varinjlim_{S} \bigotimes_{v \in S} C_v^{\text{sph-fin}} \xrightarrow{\sim} \Big(\bigotimes_{v}{}' C_v \Big)^{\text{sph-fin}}.$$

Proof. Immediate from Theorem 52.4.1: the right-hand side equals $\otimes'_v C_v^{\text{sph-fin}}$, which by definition of the restricted tensor product is the filtered colimit of finite tensor products of sph-fin factors, i.e. the left-hand side. □

Corollary 52.6.2 *(sph-fin preserved fully).* The forgetful functor $\mathrm{Pr}^{L,\text{st,sph-fin}} \to \mathrm{Pr}^{L,\text{st}}$ preserves the restricted tensor product of pointed E_2-monoidal categories satisfying Setup 50.1.1.

Proof. Consequence of Theorem 52.6.1. □

Uniformity of the Sph-Fin Subcategory

Proposition 52.6.3 *(uniformity across finite primes).* Conditional on Conjecture 52.5.3. The sph-fin subcategory $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])$ at a finite prime p is the full subcategory of ∞-categorically compact objects generated by the spherical vector $|0\rangle_p$, and the compact generators are finite over the local spherical Hecke algebra.

Proof. Conjecture 52.5.3 supplies the compact generation and the finiteness over the spherical Bernstein centre. The spherical vector generates the unramified component, and Proposition 52.5.4 identifies that component inside the local Hecke category. □

Proposition 52.6.4 *(uniformity at infinity).* The archimedean sph-fin subcategory $D^{\text{sph-fin}}((\mathfrak{g}, K_\infty)\text{-mod}^{\text{adm}})$ is the full subcategory of compact objects generated by the spherical unramified principal series u_∞, and the compact generators are K_∞-finite admissible modules (equivalently, finite-type over the archimedean Hecke algebra $\mathcal{H}_\infty$).

Proof. Proposition 52.3.4 supplies the identification of K_∞-finite admissible modules with finite-type $\mathcal{H}_\infty$-modules. Wallach 1988 Theorem 0.A.2 gives the Grothendieck-category structure on admissible modules, hence compact generation of the derived category. □

52.7 Consequence: Conditional Reduction of L_{ACD}

The Residual Lemma

Corollary 52.7.1 *(L_{ACD} conditional obstruction decomposition). Conditional statement.* Assume F2.c tightened (Theorem 50.1.3), the ∞-categorical Bernstein lift *Conjecture* 52.5.3, the local categorical comparison datum for the finite-prime categories, the archimedean boundary datum of *Theorem* 48.6.1, and the global identification of the restricted tensor product with the automorphic category. Then the remaining Bernstein-centre part of L_{ACD} is (BC-diamond-glue). The latter decomposes by Theorem 56.3.2 into (BCglue.local), requiring the actual coefficient, support and local categorical transport; (BCglue.glue), requiring compatible transitions and effective coherent descent on the common carrier; and (BC-diamond-glue.core), requiring equivariance of the arithmetic comparison for the central actions, including higher homotopies when derived centres are used. Without these inputs, (BC-diamond-glue) alone does not imply L_{ACD}.

Proof. By the initial arithmetic comparison split of L_{ACD} (Theorem 51.2.4) plus the residual splitting refinement (Section 51.4), one has a normalisation implication from the following data:

$$\begin{gathered} L_{ACD}.0, \dots, L_{ACD}.4, \quad \text{local comparison}, \\ \text{global identification}, \quad \text{(BC-diamond-glue)} \\ \Longrightarrow \quad L_{ACD}. \end{gathered}$$

The formal items have the following form:

- $L_{ACD}.0$: archimedean boundary datum (Theorem 52.3.3);
- $L_{ACD}.1$: theorem given F2.c tightened (Theorem 52.4.1);
- $L_{ACD}.2$: formal Hecke commutativity, with the finite-prime Bernstein part conditional on Conjecture 52.5.3 (Theorem 52.5.1);
- $L_{ACD}.3$: theorem (Theorem 52.6.1);

- $L_{ACD}.4$: theorem (Kneser 1966 and Platonov–Rapinchuk 1994 for SL_2; Tate for the determinant idele-class quotient).

Thus this chapter removes only the formal part of the obstruction. The categorical comparison data and (BC-diamond-glue) remain separate inputs. □

Remark 52.7.2 *(F2.c tightened is a theorem).* F2.c tightened (Theorem 50.1.3) is stated as a theorem in Chapter 50 (Lurie HA + Fresse 2017, disjoint from Fargues–Scholze). It supplies only the filtered-colimit and E_2-tensor formalism. It does not supply the ∞-categorical Bernstein lift, the local categorical comparison, the archimedean boundary spectral comparison, the global identification, or (BC-diamond-glue).

Domain of (BC-diamond-glue)

Remark 52.7.3 *((BC-diamond-glue) required theorem).* By Remark 48.2.8, (BC-diamond-glue) requires effective gluing of the specified coherent Springer sheaves on a common arithmetic carrier, together with identification of their centre action with the global Bernstein centre. The full L_{ACD} comparison also needs the local categorical comparison, the ∞-categorical Bernstein lift, the archimedean boundary datum, and the global identification.

Remark 52.7.4 *(cumulative reduction).* The original L_{ACD} contains the full adelic descent problem. The modular decomposition of Theorem 48.2.1 isolates a Bernstein-centre gluing problem, but it does not prove that this problem is the only obstruction. After the present chapter the visible residuals are the local categorical comparison, the ∞-categorical Bernstein lift, the archimedean boundary comparison, the global identification, and (BC-diamond-glue).

52.8 Compatibility with $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ Factorization

The Completed L-Factor

The completed Riemann ξ-function is

$$\xi(s) \;=\; \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) \;=\; \tfrac{1}{2}s(s-1)\cdot\Gamma_{\mathbb{R}}(s)\zeta(s)$$

with archimedean local L-factor $\Gamma_{\mathbb{R}}(s) := \pi^{-s/2}\Gamma(s/2)$ and

$$\zeta(s) \;=\; \prod_p \frac{1}{1-p^{-s}} \;=\; \prod_p L_p(s)$$

with finite-prime local L-factors $L_p(s) = (1-p^{-s})^{-1}$. The completed ξ satisfies the functional equation

$$\xi(s) \;=\; \xi(1-s).$$

Any honest L_{ACD} for $GL_2/\mathbb{Q}$ must produce a trace class whose Tate comparison sees $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$, not merely $\zeta(s)$. Multiplication by $\frac{1}{2}s(s-1)$ then gives the entire function $\xi(s)$.

Tate Character Identification

Proposition 52.8.1 *(archimedean character = $\Gamma_{\mathbb{R}}$).* The Tate local zeta integral of the archimedean Gaussian, after the standard factor 2 normalization, is $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$, the archimedean local L-factor of the trivial representation.

Proof. Tate 1950 §4.3 (Mellin transform of the Gaussian) computes

$$\int_0^\infty e^{-\pi x^2} x^s \frac{dx}{x} \;=\; \tfrac{1}{2}\pi^{-s/2}\Gamma(s/2) \;=\; \tfrac{1}{2}\Gamma_{\mathbb{R}}(s).$$

The archimedean spherical unramified principal series has Whittaker model whose K_∞-fixed vector is the Gaussian (up to normalisation; Jacquet–Langlands SLN 114 §1.5). The local L-factor at infinity is the Mellin transform of the Gaussian, hence $\Gamma_{\mathbb{R}}(s)$. □

Proposition 52.8.2 *(finite-prime Tate character = $L_p(s)$).* With multiplicative Haar measure normalized by $\mathrm{vol}(\mathbb{Z}_p^\times) = 1$, the Tate local zeta integral of the characteristic function of $\mathbb{Z}_p$ is $L_p(s) = (1 - p^{-s})^{-1}$, the finite-prime local L-factor of the trivial representation.

Proof. Tate 1950 §2 computes the local p-adic integral: the spherical vector $|0\rangle_p$ evaluates to $(1 - p^{-s})^{-1}$ under the standard local Tate integral. □

Theorem 52.8.3 *(Tate product gives $\Lambda_\zeta(s)$).* The product of the normalized Tate local characters over all places of $\mathbb{Q}$ is

$$\prod_v Z_v(s) \;=\; \Gamma_{\mathbb{R}}(s)\prod_p L_p(s) \;=\; \Gamma_{\mathbb{R}}(s)\zeta(s) \;=\; \frac{\xi(s)}{\frac{1}{2}s(s-1)},$$

as a meromorphic identity. Thus the Tate comparison target is $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$.

Proof. This is Tate's product formula for the standard adelic test function:

$$\prod_v Z_v(s) \;=\; \Gamma_{\mathbb{R}}(s)\prod_p L_p(s) \;=\; \Gamma_{\mathbb{R}}(s)\zeta(s).$$

The standard completed ξ-function is obtained from this meromorphic factor by the elementary multiplier $\frac{1}{2}s(s-1)$. □

Functional Equation Compatibility

Corollary 52.8.4 *(functional equation preserved).* The completed Tate product $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ satisfies the functional equation $\Gamma_{\mathbb{R}}(s)\zeta(s) = \Gamma_{\mathbb{R}}(1-s)\zeta(1-s)$, equivalently $\xi(s) = \xi(1-s)$.

Proof. Theorem 52.8.3 gives Λ_ζ. Tate 1950 identifies the Fourier transform of the standard adelic test function with itself and obtains $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$. Multiplication by $\frac{1}{2}s(s-1)$ gives the displayed equation for ξ. □

Remark 52.8.5 *(significance of the parallel archimedean factor).* Without the parallel archimedean factor, the restricted tensor product $\otimes'_{p<\infty} C_p$ could at most compare with $\zeta(s)$, not with $\Lambda_\zeta(s)$. The finite-prime product satisfies no functional equation of its own; only the completed Tate product $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ does. Hence $\mathrm{L}_{\mathrm{ACD}}.0$ (archimedean stratum) is a *mathematical necessity*, not a cosmetic addition: it is required before any restricted-tensor construction can be compared with the completed Tate character.

Remark 52.8.6 *(trace comparison is extra structure).* The Tate calculation above is not a computation of Hochschild trace in the local Hecke categories. A categorical realization must still construct a trace-to-Tate comparison map and prove that the chosen local units map to the standard local test functions.

52.9 Residual Conditions

Residual LACD Conditions

L_{ACD} contains the local categorical comparison, the ∞-categorical Bernstein lift, the archimedean boundary comparison, the global identification, and (BC-diamond-glue).

Subcondition	Form	Source
L_{ACD}.0 (archimedean stratum)	**Boundary datum**	Harish-Chandra 1966; Vogan 1981; Wallach 1988
L_{ACD}.1 (colimit commutation)	**Theorem** (F2.c tightened)	Lurie HA 5.3.2.9 + F2.c
L_{ACD}.2 (cross-prime Hecke)	**Formal theorem; conditional Bernstein part**	Lurie HA; ∞-categorical BZ lift
L_{ACD}.3 (sph-fin preservation)	**Theorem** (from .1)	Immediate
L_{ACD}.4 (derived strong approximation and determinant quotient)	**Theorem**	Kneser 1966; Platonov–Rapinchuk 1994; Tate 1950
F2.c tightened	**Theorem under pointed E_2 hypotheses** (see Rem. 52.7.2)	Lurie HA; Fresse 2017
∞-categorical BZ lift	**Conjectural**	Conjecture 52.5.3
local categorical comparison and global identification	**Open**	F1.a and L_{ACD} input
(BC-diamond-glue)	**Conjectural**	Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321; construction ε

Table 52.1: Form of L_{ACD} components. Formal pieces are separated from comparison data.

F1 Closure Chain

Theorem 52.9.1 *(F1 residual after local comparison and descent hypotheses). Comparison hypothesis.* Assume F2.c tightened (Theorem 50.1.3), the local categorical comparison (F1.a), the ∞-categorical Bernstein lift, the archimedean boundary comparison, and the global automorphic descent hypotheses. Then F1 (global arithmetic Arinkin–Gaitsgory for $GL_2/\mathbb{Q}$, Conjecture 42.1.1) reduces to (BC-diamond-glue) (Conjecture 48.5.4). Without those hypotheses, (BC-diamond-glue) alone does not close F1.

Proof. By Theorem 43.2.1, F1 reduces to (F1.a) + (F1.b) + (F1.c) + (F1.d). The item (F1.a) is an assumed local categorical comparison; (F1.c) is the Emerton–Gee stack input. Items (F1.b) and (F1.d) combine into L_{ACD} (Corollary 43.2.3). By Corollary 52.7.1, L_{ACD} reduces to (BC-diamond-glue) only after the comparison hypotheses listed in the statement are supplied. □

Remark 52.9.2 *(form of F1 after this chapter).* F1 does not reduce to (BC-diamond-glue) alone. The reduction still contains the local comparison, the ∞-categorical Bernstein lift, the archimedean boundary comparison, and the global descent input. Construction ε addresses only the Bernstein-centre gluing part.

Hypothesis	Role in L_{ACD} reduction	Source
F2.c tightened	Universal property for colimit E_2-structure	Lurie HA, pointed E_2 hypotheses
Lurie HA 5.3.2.9	Sph-fin preservation under filtered colimit	Lurie HA
Lurie HA 5.1.2.4	Dunn additivity for E_n-operads	Lurie HA
Lurie HA 5.1.2.26	Unit centrality in E_n-monoidal cat	Lurie HA
Lurie HTT 5.3.5.15	Ind-completion preserves filtered colimits	Lurie HTT
Harish-Chandra 1966	Admissibility of $(\mathfrak{g}, K_\infty)$-mod	Primary source
Vogan 1981	Langlands–Vogan classification at $\mathbb{R}$	Primary source
Wallach 1988/1992	Real reductive groups spectrum theory	Primary source
Jacquet–Langlands SLN 114	Archimedean local representation	Primary source
Bernstein–Zelevinsky 1976	Bernstein decomposition of p-adic Hecke	Primary source
∞-categorical BZ lift	Stable lift of Bernstein projectors	Conjecture 52.5.3; conjectural
Tate 1950	Adelic Mellin/theta and functional equation	ξ-compatibility
Kneser 1966	Strong approximation for SL_2	L_{ACD}.4 derived subgroup theorem
Tate 1950	Idele-class determinant quotient for GL_2	L_{ACD}.4 determinant line
(BC-diamond-glue)	Bernstein-centre gluing	Construction ε, not yet theorem

Table 52.2: Hypotheses used in this chapter. The non-theorem inputs also include the local comparison, the archimedean boundary comparison, and the global identification.

Hypotheses Inventory

52.10 Conditional Reduction of L_{ACD}

The modular decomposition (Theorem 48.2.1, Theorem 51.2.4, Section 51.4) identifies five subconditions L_{ACD}.0–L_{ACD}.4, with L_{ACD}.4 combines Kneser's strong approximation for SL_2 with Tate's determinant quotient for GL_2. The formal pieces treated here are:

- L_{ACD}.0 as archimedean boundary datum via Harish-Chandra admissibility, Vogan 1981 classification, and Wallach 1988/1992 real-reductive-groups spectrum theory (Theorem 52.3.3);
- L_{ACD}.1 via F2.c tightened, Lurie HA 5.3.2.9, and HTT 5.3.5.15 (Theorem 52.4.1);
- L_{ACD}.2 (formal tensor-factor commutativity) via Lurie HA 5.1.2.4 (Dunn additivity), with finite-prime Bernstein compatibility conditional on Conjecture 52.5.3 (Theorem 52.5.1);
- L_{ACD}.3 from L_{ACD}.1 (Theorem 52.6.1).

The residual of L_{ACD} is not only (BC-diamond-glue). It also contains the local categorical comparison, the ∞-categorical Bernstein lift, the archimedean boundary comparison, and the global identification (Corollary 52.7.1).

The parallel archimedean factor (L_{ACD}.0) is required: without it, the restricted tensor product can at most compare with $\zeta(s)$, not with $\Gamma_{\mathbb{R}}(s)\zeta(s)$, and the Tate functional equation is invisible (Remark 52.8.5).

F1 (global arithmetic Arinkin–Gaitsgory for $GL_2/\mathbb{Q}$) reduces to (BC-diamond-glue) only under the stated comparison and descent hypotheses (Theorem 52.9.1).

The source list for L_{ACD}.0, L_{ACD}.1, and L_{ACD}.3 is disjoint from the Fargues–Scholze list: Harish-Chandra, Jacquet–Langlands, Vogan, Wallach (archimedean representation theory), Lurie HA, Fresse

2017 (higher operadic machinery), and Bernstein–Zelevinsky 1976 (Bernstein decomposition). The ∞-categorical Bernstein lift is not proved from these sources; it is the separate conjectural datum Conjecture 52.5.3.

Chapter 53

BZN^{Ar}: Formal Noetherian Hochschild and Derived Structure on Emerton–Gee

The arithmetic Ben-Zvi–Nadler comparison factors through four independent inputs. The finite-type theorem of Gaitsgory–Rozenblyum applies to smooth finite-type derived stacks. A candidate global spectral object is a Selmer deformation functor on one coefficient category. A formal geometric representative must be constructed before the formal comparison can be applied. That comparison requires a categorical Hochschild-colimit hypothesis, an arithmetic loop-space hypothesis, and a flat-singular-support hypothesis. The inertia shadow does not contain the full Weil/(φ,Γ) datum, and the primes 2 and 3 carry non-semisimple Hecke summands.

53.1 Formal Boundary of the Comparison

Structural Inputs

Chapter 49 isolates the desired arithmetic Hochschild trace equivalence

$$\mathrm{HH}_\bullet\big(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}\big) \;\simeq\; \mathcal{O}\big(\mathrm{LocSys}^{W,\varphi^\Gamma}_{GL_2}\big)^{\mathrm{flat}}$$

as an equivalence of E_1-algebras in stable presentable ∞-categories. The finite-type BZN theorem supplies only one factor of the argument. The arithmetic comparison requires the following additional inputs:

R1.fin-type. The Gaitsgory–Rozenblyum theorem (AMS DAG Vol. II Part V Ch. III) that $\mathrm{HH}_\bullet(\mathrm{IndCoh}(\mathcal{X})) \simeq \mathcal{O}(L\mathcal{X})$ holds for smooth finite-type derived stacks $\mathcal{X}$. *Form: theorem* (Gaitsgory–Rozenblyum). The Selmer-derived stack $\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}$ is initially a formal deformation functor. If its chosen geometric representative is formal Noetherian rather than finite-type, R1.formal below is the required extension.

R1.formal. The extension of the Gaitsgory–Rozenblyum theorem to formal Noetherian derived stacks. For $\mathcal{X}$ a formal Noetherian derived algebraic stack and $\mathcal{C} = \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$,

$$\mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})\big) \;\simeq\; \mathcal{O}\big(L^{\mathrm{Ar}}\mathcal{X}\big)^{\mathrm{flat}}.$$

Form: conjectural; a genuinely open extension requiring new arguments beyond the finite-type case.

R2. Construct the global Selmer deformation functor on a fixed coefficient category, with an identified determinant and actual maps from the local-condition functors. Its tangent is then the corresponding Selmer complex. Comparison with local parameter stacks, geometric representability, formal Noetherianity, and nilpotent-support compatibility are further requirements on the chosen local conditions.

R3. Comparison from the inertia shadow $\mathrm{Map}(B\widehat{\mathbb{Z}},-)$ to the full Weil/(φ,Γ) local-system target. *Form: conjectural*; the inertia-only identification via $\mathrm{Map}(B\widehat{\mathbb{Z}},-)$ constitutes a procyclic shadow of the full Weil/(φ,Γ) target.

R4. Non-semisimple summand at wild primes $p = 2, 3$: $\mathrm{Ext}^{\mathrm{I}}$ contributions to $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p})$ invisible to semisimple templates. *Form: conjectural*; carried by the wild Hecke category.

R2 and R1.formal

The conditions R2 and R1.formal are:

(i) **R2** defines a global Selmer deformation functor by a homotopy pullback. Its local functors must use the same residual coefficient field and determinant convention as the global functor. Local parameter stacks can be used only after their comparison maps with these functors are constructed. Geometric representability and derived Noetherianity remain requirements for the proposed spectral stack. They are not consequences of a product-colimit presentation.

(ii) **R1.formal** follows from three hypotheses (R1.formal.a, b, c). The first is a Hochschild trace theorem after the categorical colimit presentation of $\mathrm{IndCoh}_{\mathrm{Nilp}}$ is supplied. The second and third are respectively the arithmetic loop-space comparison and the flat-singular-support comparison.

R1.fin-type is external. R3 is not definitional: it is the passage from the procyclic inertia shadow to the full $\mathrm{Weil}/(\varphi, \Gamma)$ target. R4 is the non-semisimple wild-prime summand.

Derived Enhancement of the Emerton–Gee Stack

Emerton–Gee arXiv:1908.07185 is written in the language of Noetherian formal algebraic stacks in the sense of the Stacks Project, with the smoothness and mass-formula structure at the level of ordinary (non-derived) schemes. Gaitsgory–Rozenblyum DAG requires a derived enhancement: the cotangent complex must be computed as a perfect complex (or almost-perfect, in the formal case) in the ∞-category of quasi-coherent sheaves, and the ind-colimit presentation must preserve this structure at every stage. The translation is not automatic. It asks for cotangent-complex functoriality under the chosen formal presentation, compatibility with formal thickening, and preservation under the Selmer fibre product. These are hypotheses on the derived enhancement, not consequences of the underived mass formula alone.

Finite-Type Hypotheses in the Gaitsgory–Rozenblyum Argument

The Gaitsgory–Rozenblyum proof of $\mathrm{HH}_\bullet(\mathrm{IndCoh}(\mathcal{X})) \simeq \mathcal{O}(L\mathcal{X})$ uses three ingredients that each rest on finite-type smoothness and require new arguments in the formal Noetherian setting:

(i) Atiyah duality for IndCoh: the self-duality $\mathrm{IndCoh}(\mathcal{X})^\vee \simeq \mathrm{IndCoh}(\mathcal{X})$ under the Atiyah pairing, which requires $\mathcal{X}$ smooth finite-type.

(ii) Smooth-base-change for IndCoh, which also uses finite-type.

(iii) Loop-space duality: the free loop space $L\mathcal{X}$ of a finite-type smooth stack is itself smooth (with dimension equal to $\dim \mathcal{X}$, by the standard calculation).

For formal Noetherian $\mathcal{X}$, each ingredient requires a formal analogue: Atiyah duality must be extended to the formal setting (conjectural in general), smooth base-change must be replaced by formal-smooth base-change (subject of ongoing work, e.g., Mok 2025), and the loop space $L\mathcal{X}$ is in general not smooth finite-type.

The formal comparison therefore has the form of a criterion. Given the categorical colimit presentation of $\mathrm{IndCoh}_{\mathrm{Nilp}}$, Hochschild trace commutes with it; the loop and flat-locus comparisons remain separate geometric hypotheses.

53.2 Logical Form of the Inputs

The input structure is as follows:

R2 tangent complex. The continuous deformation functor has tangent $C^\bullet(G, \mathrm{ad}\bar{\rho})[1]$, retaining automorphisms in degree -1. A chosen identification of its determinant replaces the coefficients by $\mathrm{ad}^\circ \bar{\rho}$, including in residual characteristic two. A comparison with a derived local parameter stack must identify that stack's formal deformation functor first. See Chapter 53.3.1.

R2 global Galois constraint. Restriction of a global representation gives a morphism to the product of full local deformation functors. The pointed local-condition functors map to that same product, over the same category $\mathrm{Art}_{W(k)}$. Their homotopy pullback is (69.1), and its tangent is the mapping fiber of restriction minus the local tangent maps. The trace-zero notation in (69.2) uses a determinant identification preserved by arrows.

R2 geometric enhancement. A geometric representative must have the finiteness, cotangent, and singular-support properties needed by the intended category of sheaves. A smooth atlas is a smooth map to a stack. Its existence alone does not make the stack smooth over the coefficient base. Pointwise tangent bounds also do not prove perfectness of a family of cotangent complexes. These are additional hypotheses in Chapter 53.3.5.

R1.formal three inputs. The Gaitsgory–Rozenblyum theorem, ported from finite-type to formal Noetherian, requires three separate arguments: (i) HH-colim-continuity (whether $\mathrm{HH}_\bullet$ commutes with the ind-colimit presentation of the formal stack); (ii) arithmetic loop-space formal computation (whether $L^{\mathrm{Ar}}\mathcal{X} \simeq \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ admits a workable formal description); (iii) flat-locus pullback (whether the nilpotent-singular-support-flat locus on the loop space matches the Emerton–Gee flat locus of Chapter 49.4.7). The first is a trace theorem once the categorical colimit exists; the other two are geometric hypotheses. (See Chapter 53.4.1.)

R1.formal sub-reductions. R1.formal.a is the Hochschild trace statement for a given filtered colimit in the Morita $(\infty, 2)$-category. R1.formal.b reduces to the existence of a cotangent-complex formal loop-space identification (a precise statement in Lurie DAG and Gaitsgory–Rozenblyum Vol. II). R1.formal.c reduces to the compatibility of Arinkin–Gaitsgory nilpotent singular support with the (φ, Γ)-module Fargues–Scholze flat locus. (See Chapter 53.4.2 and Chapter 53.4.9.)

Consistency with the initial arithmetic comparison. The arithmetic comparison of Chapter 49.5.5 uses a pro-étale extension of Ben-Zvi–Francis–Nadler integral transforms. In the present notation this is the R1.formal.b loop-space hypothesis together with the R1.formal.c flat-locus hypothesis.

53.3 R2: The Emerton–Gee Cotangent Complex

The Local Formal Deformation Problem

Proposition 53.3.1 *(local tangent and the geometric comparison).* Let k be a finite field of characteristic ℓ, and let $\bar{\rho} : G_{\mathbb{Q}_p} \to \mathrm{GL}_2(k)$ be continuous. The unframed derived continuous deformation functor on $\mathrm{Art}_{W(k)}$ has tangent

$$T\mathrm{Def}_{G_{\mathbb{Q}_p}} \simeq C^\bullet(G_{\mathbb{Q}_p}, \mathrm{ad}\bar{\rho})[1].$$

A derived enhancement of a local parameter stack has this tangent at $\bar{\rho}$ if its formal deformation functor is equivalent to this one. The claimed equivalence must preserve the coefficient category, residual identification, and allowed arrows.

Proof. Proposition 10.6.1 computes the linearized representation equation and its infinitesimal conjugations, and then its response to simplicial square-zero modules. An equivalence of formal deformation functors identifies those responses, hence their tangent complexes. It is the required extra hypothesis for the stack comparison. □

The framed first-order tangent is Z^1, whereas strict conjugacy classes give H^1. In the unframed stack, H^0 remains in tangent degree -1. It is not removed from the tangent complex by being an automorphism group. With a determinant identification, Proposition 10.6.2 gives $C^\bullet(G_{\mathbb{Q}_p}, \mathrm{ad}^0\bar{\rho})[1]$. Fixing only the determinant value with all strict conjugations is a different deformation problem.

Remark 53.3.2 *(pointwise bounds and perfectness).* If the continuous cohomology groups are finite-dimensional and vanish outside 0, 1, 2, the tangent at this field-valued point is perfect of amplitude contained in $[-1, 1]$. Indeed, over a field a bounded complex with finite-dimensional cohomology is equivalent to a bounded complex of finite-dimensional vector spaces. Its H^0 produces tangent degree -1, and its H^2 produces degree one. Neither term disappears by changing the word used for the moduli object. This pointwise statement does not prove perfectness over a formal coefficient family. It also gives no uniform bound for real places at coefficient characteristic two, by Proposition 10.8.1.

A Common Recipient for Cotangent Comparisons

PROPOSITION 53.3.3 *(the pointed cotangent pro-system).* Let $X = \varinjlim_i X_i$ be an objectwise filtered colimit of pointed functors on derived affine k-schemes, with compatible k-points x_i. Suppose X_i has cotangent complex L_i at x_i. For a square-zero coefficient module N, the space of lifts of the point to $k \oplus N$ is

$$\mathrm{Der}_x(X, N) \simeq \varinjlim_i \mathrm{Map}_{D(k)}(L_i, N).$$

The transition maps give a well-typed inverse system $L_j \to L_i$ in $D(k)$. This assertion does not identify a cotangent object in $\mathrm{IndCoh}(X)$ without additional representability and categorical comparison data.

Proof. By definition, the lift space is the homotopy fiber at x of $X(k \oplus N) \to X(k)$. Filtered colimits of spaces commute with finite homotopy limits, so this fiber is the filtered colimit of the corresponding fibers for X_i. Each is represented by L_i by the cotangent universal property. A transition $X_i \to X_j$ induces $\mathrm{Map}(L_i, N) \to \mathrm{Map}(L_j, N)$, hence $L_j \to L_i$, compatibly with composition. □

Closed immersions do not make the cotangent complexes of their source and target equal by pullback. For the origin $i : \mathrm{Spec}\, k \hookrightarrow \mathbb{A}^1_k$, $i^*\mathbb{L}_{\mathbb{A}^1_k/k} = k$ in degree zero, whereas $\mathbb{L}_{k/k} = 0$. The relative cotangent complex is $k[1]$, and the transitivity triangle is $k \to 0 \to k[1]$. Thus an ind-presentation must retain its actual cotangent maps. There is no automatic inverse-limit formula obtained by moving objects from different sheaf categories into $\mathrm{IndCoh}(X)$.

COROLLARY 53.3.4 *(cotangents of a derived product).* Let X, Y be derived stacks over a common base $\mathrm{Spec}\, R$ with cotangent complexes, and write π_X, π_Y for the projections from their derived product. Then

$$\mathbb{L}_{X \times^{\mathbf{R}}_R Y/R} \simeq \pi_X^*\mathbb{L}_{X/R} \oplus \pi_Y^*\mathbb{L}_{Y/R}.$$

Proof. Fix a map from T to the derived product and a square-zero extension $T[N]$. Its lifts are exactly pairs of lifts to X and Y over R. Their spaces of derivations are therefore a product. The direct sum on the right represents this product of mapping spaces for every test T and coefficient module N. The cotangent universal property gives the stated equivalence. □

The Selmer Tangent and Its Geometric Requirements

Theorem 53.3.5 *(Selmer deformation functor).* Fix the common coefficient category, determinant identification, residual representation, and pointed local functors of Definition 69.1.1. Their homotopy pullback $\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\partial,\mathcal{L}}$ exists as a functor. If the local tangent maps are $L_v \to C^\bullet(G_{\mathbb{Q}_v}, \mathrm{ad}^\circ\bar\rho)$, its tangent is

$$T\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\partial,\mathcal{L}} \simeq C^\bullet_{\mathcal{L}}(G_{\mathbb{Q},S}, \mathrm{ad}^\circ\bar\rho)[1].$$

To use this functor as the geometric spectral target, the following remain additional requirements:

(i) a geometric representative and comparisons with the chosen local parameter stacks on the same coefficient category.

(ii) the Noetherian, perfectness, and cotangent-amplitude properties needed for the intended sheaf theory.

(iii) a definition of the specified nilpotent singular support and its compatibility with the Selmer cotangent maps.

Proof. The pullback is computed in functors to spaces. Its tangent is the fiber of the global and local tangent maps, as proved in Theorem 10.7.1. The further geometric assertions are hypotheses, not conclusions of this functor calculation. In particular, nonzero $H^2_{\mathcal{L}}$ is compatible with quasi-smoothness of a derived scheme, as shown in Example 10.7.2. □

Corollary 53.3.6 *(the inertia shadow does not determine the full target).* . Given Chapter 53.3.5, R3 still requires the full-target comparison:

$$\mathrm{Map}\big(B\widehat{\mathbb{Z}}, \mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}\big) \longrightarrow \mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}\big(B\widehat{\mathbb{Z}}\big),$$

as a procyclic shadow, and then a comparison with $\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}$.

Proof. By Chapter 49.4.1 and Chapter 49.4.5, $\mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}}) = L^{\mathrm{Ar}}BGL_2 = \mathrm{Map}(B\widehat{\mathbb{Z}}, BGL_2)$. The map from $\mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}})$ is obtained only after choosing an inertia-shadow comparison morphism $\iota^{\mathrm{Sel}}\colon \mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}} \to \mathrm{LocSys}^{\mathrm{Ar}}_{GL_2}(B\widehat{\mathbb{Z}})$ (Chapter 49.4.6). That inertia-only morphism retains only the procyclic structure, discarding Frobenius and (φ,Γ) data; the full comparison $\mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}) \simeq \mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}$ requires a separate argument (R3). □

Source and Convention Comparison for R2

The deformation functor. The construction uses continuous derived deformation functors on $\mathrm{Art}_{W(k)}$, with specified residual point and determinant convention. The local and global maps must exist there before their homotopy pullback is formed. A description by a local parameter stack requires a comparison of formal functors, not only an agreement of first-order dimensions.

Degrees and sheaf categories. The tangent of the unframed deformation functor is a shifted cochain complex. Automorphisms remain in degree -1.

A geometric cotangent complex lives initially in quasi-coherent sheaves on its own base. An identification with an ind-coherent object uses the relevant comparison functor and its hypotheses. The pointwise inverse system in Chapter 53.3.3 has the common recipient $D(k)$. It makes no global categorical identification.

Products and formal presentations. The product formula in Chapter 53.3.4 follows from pairs of derivations over a common base. A closed-immersion presentation has relative cotangent terms and must retain their transition maps. Neither formula proves formal Noetherianity or the singular-support compatibility needed by the arithmetic Hochschild comparison.

Remark 53.3.7 *(derived geometric enhancement).* An equivalence between a chosen derived local parameter stack and its continuous deformation functor would identify their tangents by Chapter 53.3.1. It would still leave the global representability, finiteness, and support requirements of Chapter 53.3.5. First-order tangent data, or an ordinary moduli presentation, do not supply that equivalence.

53.4 R1.formal: Formal Hochschild Criterion

The Three Inputs

Theorem 53.4.1 *(formal BZN criterion).* . Let $\mathcal{X}$ be a derived formal algebraic stack in the Gaitsgory–Rozenblyum sense, Noetherian, presented as an ind-colimit $\mathcal{X} = \varinjlim_i \mathcal{X}_i$ of finite-type derived stacks under closed immersions (as in the Emerton–Gee structure theorem). Let $C(\mathcal{X}) = \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$. Assume the finite-type BZN comparison on each $\mathcal{X}_i$, and assume the three conditions below. Then

$$\mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})\big) \;\simeq\; \mathcal{O}\big(L^{\mathrm{Ar}}\mathcal{X}\big)^{\mathrm{flat}},$$

namely R1.formal, holds.

(R1.formal.a) *HH-colim continuity.* The Hochschild trace preserves the ind-colimit presentation of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$:

$$\mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})\big) \;\simeq\; \varinjlim_i \mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_i)\big),$$

under the transition maps induced by closed immersions. **Form: theorem after the categorical colimit presentation is given** (trace continuity; proved in Chapter 53.4.2).

(R1.formal.b) *Arithmetic loop-space formal computation.* For a formal Noetherian derived stack $\mathcal{X}$ as above, the free loop space admits a formal ind-colimit description

$$L^{\mathrm{Ar}}\mathcal{X} \;\simeq\; \varinjlim_i L^{\mathrm{Ar}}\mathcal{X}_i$$

compatible with the closed immersions; equivalently, $L^{\mathrm{Ar}} = \mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ commutes with the ind-colimit presentation. **Form: conjectural** (Lurie ∞-topos formalism in the pro-étale formal setting; see Chapter 53.4.4).

(R1.formal.c) *Flat-locus pullback compatibility.* The flat locus $L^{\mathrm{Ar}}\mathcal{X}^{\mathrm{flat}}$ (cut out by the Arinkin–Gaitsgory nilpotent-singular-support condition lifted to the loop space) is compatible with the ind-colimit structure:

$$\mathcal{O}\big(L^{\mathrm{Ar}}\mathcal{X}\big)^{\mathrm{flat}} \;\simeq\; \varinjlim_i \mathcal{O}\big(L^{\mathrm{Ar}}\mathcal{X}_i\big)^{\mathrm{flat}},$$

with the flat loci on finite-type pieces matching the (φ, Γ)-module Fargues–Scholze flat locus of Chapter 49.4.7. **Form: conjectural** (Arinkin–Gaitsgory singular support extended to this formal Selmer setting; see Chapter 53.4.7).

Proof. Assume (R1.formal.a), (R1.formal.b), and (R1.formal.c). By (R1.formal.a),

$$\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})) \;\simeq\; \varinjlim_i \mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_i)).$$

By the Gaitsgory–Rozenblyum theorem applied term-wise,

$$\varinjlim_i \mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_i)) \;\simeq\; \varinjlim_i \mathcal{O}(L\mathcal{X}_i)^{\mathrm{flat}}.$$

By (R1.formal.b), $\varinjlim_i L\mathcal{X}_i \simeq L^{\mathrm{Ar}}\mathcal{X}$ (noting $L = L^{\mathrm{Ar}}$ on finite-type pieces via the reduction $S^1 \to B\widehat{\mathbb{Z}}$ in the pro-étale setting; this is the content of Chapter 49.5.3 on each finite-type piece). By (R1.formal.c),

$$\varinjlim_i \mathcal{O}(L\mathcal{X}_i)^{\mathrm{flat}} \;\simeq\; \mathcal{O}(L^{\mathrm{Ar}}\mathcal{X})^{\mathrm{flat}}.$$

Composing gives R1.formal. □

R1.formal.a: HH Colimit Continuity

Theorem 53.4.2 *(Hochschild trace along a categorical colimit).* . Let $\mathcal{X} = \varinjlim_i \mathcal{X}_i$ be a formal presentation by derived stacks under closed immersions. Assume that the canonical functor

$$\varinjlim_i \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_i) \longrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$$

is an equivalence in the E_2-monoidal Morita $(\infty, 2)$-category of stable presentable categories. Then

$$\mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})\big) \;\simeq\; \varinjlim_i \mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_i)\big)$$

as E_1-algebras in stable presentable ∞-categories.

Proof. Hochschild homology is the trace of the identity endofunctor in the Morita $(\infty, 2)$-category of stable presentable categories. The trace functor is symmetric monoidal and preserves filtered colimits of dualizable objects and dualizable morphisms; this is the form of Lurie HA 5.5.3 used here, and it is also obtained from the cyclic bar description of Toën. Applying the trace to the assumed categorical colimit gives the displayed equivalence. The E_1-algebra structure is transported by Dunn additivity from the E_2-monoidal convolution structure. □

Remark 53.4.3 *(where the geometry enters).* The theorem is purely categorical. It does not prove that $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ is the filtered colimit of the categories on the finite pieces. That assertion is a geometric compatibility statement for formal completion, singular support, and the closed-immersion transition functors.

R1.formal.b: Arithmetic Loop-Space Formal Computation (Conditional)

Proposition 53.4.4 *(R1.formal.b domain).* . R1.formal.b reduces to:

(b.1) The internal mapping stack $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ commutes with filtered colimits in derived pro-étale stacks under closed immersions.

(b.2) For a formal Noetherian derived stack $\mathcal{X}$ with $\mathcal{X} = \varinjlim_i \mathcal{X}_i$, the natural map $\varinjlim_i \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}_i) \to \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ is an equivalence in the pro-étale topology.

Both (b.1) and (b.2) are Conjectural.

Proof of reduction. R1.formal.b as stated is $L^{\mathrm{Ar}}\mathcal{X} \simeq \varinjlim_i L^{\mathrm{Ar}}\mathcal{X}_i$. By Chapter 49.4.1, $L^{\mathrm{Ar}}\mathcal{X} = \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$. The ind-colimit compatibility of this mapping stack is precisely (b.1) (the general statement about the mapping functor) plus (b.2) (verification of the specific map for the ind-colimit presentation of $\mathcal{X}$). Conversely, given (b.1) and (b.2), R1.formal.b is immediate. □

Remark 53.4.5 *(Compactness and the arithmetic mapping object).* An object M is compact when its space-valued mapping functor $\mathrm{Map}(M, -)$ preserves filtered colimits. Preservation by an internal mapping object must instead be checked after evaluation on the test objects of its ambient category. Neither assertion follows from finite cardinality of a group.

For a prime p, the space $B(\mathbb{Z}/p)$ is not compact in spaces. Let g generate $\mathbb{Z}/p$. The periodic group resolution has alternating differentials $1 - g$ and $1 + g + \cdots + g^{p-1}$. Applying homomorphisms to the trivial module $\mathbb{F}_p$ makes both differentials zero, so its cohomology is nonzero in every nonnegative degree.

A compact space is a retract, up to homotopy, of a finite complex, and has bounded cohomology. Thus this classifying space is not compact.

The profinite cyclic group uses all finite cyclic quotients with divisibility transition maps. The p-power tower describes only $\mathbb{Z}_p$. A presentation by classifying objects requires the specified topology, and an inverse limit does not by itself preserve compactness. Statements (b.1) and (b.2) require the actual comparison maps for the proposed formal arithmetic targets. The finite-module Artin–Rees theorem does not supply them, as the limit exchange in (80.12) illustrates.

Remark 53.4.6 *(size of the loop-space hypothesis).* The hypothesis has three parts: compactness of $B\widehat{\mathbb{Z}}$ against the relevant formal targets, commutation of $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ with the chosen formal-Noetherian ind-colimits, and compatibility with IndCoh-functoriality. Each part is a statement in the pro-étale topology. Finite-level mapping behavior and passage to the profinite limit each require their own proof.

R1.formal.c: Flat-Locus Pullback Compatibility (Conditional)

Proposition 53.4.7 *(R1.formal.c domain).* . R1.formal.c reduces to:

(c.1) *Arinkin–Gaitsgory nilpotent singular support extends to derived formal algebraic stacks.* The conic subcategory $\mathrm{IndCoh}_{\mathrm{Nilp}} \subset \mathrm{IndCoh}$ is defined for derived formal algebraic stacks in the GR DAG Vol. II sense, preserving the Arinkin–Gaitsgory functoriality (arXiv:1201.6343).

(c.2) *Flat locus preserved under ind-colimit.* For an ind-colimit $\mathcal{X} = \varinjlim_i \mathcal{X}_i$, the flat locus on the loop space commutes with the ind-colimit: $L^{\mathrm{Ar}}\mathcal{X}^{\mathrm{flat}} \simeq \varinjlim_i L^{\mathrm{Ar}}\mathcal{X}_i^{\mathrm{flat}}$.

(c.3) *Flat locus matches FS locus.* On each finite-type piece $\mathcal{X}_i$, the Arinkin–Gaitsgory flat locus of the loop space coincides with the (φ, Γ)-module Fargues–Scholze flat locus of Chapter 49.4.7.

All three statements are hypotheses.

Proof of reduction. R1.formal.c as stated is $\mathcal{O}(L^{\mathrm{Ar}}\mathcal{X})^{\mathrm{flat}} \simeq \varinjlim_i \mathcal{O}(L^{\mathrm{Ar}}\mathcal{X}_i)^{\mathrm{flat}}$, with the flat loci on finite-type pieces matching Chapter 49.4.7. Unpacking: this requires the flat locus to be *defined* on each finite-type $\mathcal{X}_i$ (which is a statement about nilpotent singular support extending to derived formal algebraic stacks; this is (c.1)), to *commute with ind-colimit* (which is (c.2)), and to *match the FS flat locus* on each finite-type piece (which is (c.3)). Conversely, (c.1) + (c.2) + (c.3) imply R1.formal.c by pasting the three statements together. □

Remark 53.4.8 *(flat support is a separate geometry).* Arinkin–Gaitsgory singular support is defined by the cotangent geometry of a quasi-smooth stack. The Fargues–Scholze flat locus is defined by (φ, Γ)-module geometry. Equality of the two loci on the Emerton–Gee finite pieces is a comparison theorem, not a definition. Compatibility with the formal colimit is another theorem condition.

Sufficient Criterion for F3.BZN^{Ar}

Theorem 53.4.9 *(sufficient criterion for F3.BZN^{Ar}).* . Suppose the structural splitting Chapter 51.2.5 is imposed. The conjunction

- R1.fin-type (Gaitsgory–Rozenblyum: theorem on smooth finite-type).
- R1.formal.a (Hochschild trace along the given categorical colimit).
- R1.formal.b (arithmetic loop-space formal computation: conjectural).
- R1.formal.c (flat-locus pullback compatibility: conjectural).
- R2 (Selmer-derived target, tangent complex, formal Noetherianity, and nilpotent-support compatibility).
- R3 (arithmetic free loop space identification: full-target comparison still required, Chapter 53.3.6).
- R4 (non-semisimple summand at wild primes $p = 2, 3$: wild Hecke summands included).

implies F3.BZN^{Ar}.

Proof. Chapter 51.2.5 expresses F3.BZN^{Ar} as R1 + R2 + R3 + R4, where R1 further splits into R1.fin-type + R1.formal (Chapter 53.1). R1.fin-type is Gaitsgory–Rozenblyum; R1.formal splits into R1.formal.a (Chapter 53.4.2, theorem) plus R1.formal.b (Chapter 53.4.4, conjectural) plus R1.formal.c (Chapter 53.4.7, conjectural) by Chapter 53.4.1. R2 supplies the Selmer target and tangent complex and includes the required finiteness and singular-support hypotheses. R3 supplies the full Weil/(φ, Γ) comparison. R4 inserts the wild-prime summands. These inputs are exactly the factors in the structural splitting, hence their conjunction implies F3.BZN^{Ar}. □

Source and Convention Comparison for R1.formal

Source independence. The argument for R1.formal draws on four disjoint source lists: Lurie HA 5.5.3 / 4.8 / 7.4 (HH, cotangent, colimits); Gaitsgory–Rozenblyum DAG Vol. II Part V Ch. III (HH on smooth finite-type); Arinkin–Gaitsgory arXiv:1201.6343 (nilpotent singular support); Fargues–Scholze arXiv:2102.13459 (local Langlands).

HH conventions. Lurie HA places HH on symmetric monoidal stable presentable ∞-categories in spectra; Gaitsgory–Rozenblyum uses a derived-category trace on IndCoh; Toën 2014 uses a direct simplicial-object computation. All three are equivalent: HH of a stable ∞-category is the colimit over the cyclic category of the self-tensor-product sequence (McCarthy–Schwede, Blumberg–Mandell).

Ambient category. All objects live in derived pro-étale stacks with the $(\infty, 2)$-category of correspondences for IndCoh-valued functoriality.

Hypotheses. The main hypothesis on $\mathcal{X}$ is Noetherianness together with derived-formal-algebraic structure. For the Selmer target this includes the formal Noetherian and nilpotent-support compatibility conditions of Chapter 53.3.5.

Functoriality. HH-colim continuity is a functorial statement on the ∞-category of stable presentable ∞-categories; the criterion by three inputs is itself functorial in $\mathcal{X}$.

Source of the implication. The criterion Chapter 53.4.1 is one-way. The finite-type theorem, Hochschild trace continuity, loop-space compatibility, and flat-locus compatibility imply the formal BZN comparison. The converse is not used.

53.5 Conditional Form of F3.BZN^{Ar}

Corollary 53.5.1 *(conditional arithmetic BZN comparison).* . F3.BZN^{Ar} follows from Chapter 53.4.9 and the conjunction of:

- R1.formal.b (arithmetic loop-space formal computation, conjectural).
- R1.formal.c (flat-locus pullback compatibility, conjectural).
- R2 finiteness and nilpotent-support compatibility for the chosen Selmer deformation problem.
- R3 (full-target loop-space identification, including the Weil/(φ, Γ) comparison).
- R4 (the non-semisimple Hecke summands at $p = 2, 3$).

R1.fin-type is the finite-type theorem. R1.formal.a is the trace theorem for a given categorical colimit. The other entries are geometric hypotheses.

Proof. By Chapter 53.4.9: the conjunction R1.fin-type + R1.formal.a + R1.formal.b + R1.formal.c + R2 + R3 + R4 implies F3.BZN^{Ar}, where R2 includes the finiteness and support hypotheses of Chapter 53.3.5. The corollary is the same implication with the finite-type theorem and trace theorem separated from the geometric hypotheses. □

Remark 53.5.2 *(formal-loop and flat-support inputs).* The pro-étale extension used in Chapter 49.5.5 contains two distinct geometries. The first is the commutation of Map$(B\widehat{\mathbb{Z}}, -)$ with the formal Selmer presentation. The second is the equality between the Arinkin–Gaitsgory flat singular-support locus and the Fargues–Scholze (φ, Γ)-flat locus.

Remark 53.5.3 *(pro-étale extension).* Chapter 49 Remark 49.5.4 notes that Chapter 49.5.3 uses a pro-étale extension of Ben-Zvi–Francis–Nadler that is formal but not in the literature. In the present notation that extension is the R1.formal.b hypothesis, together with the R1.formal.c comparison of flat loci.

53.6 Boundary Conditions

Form of the Inputs

The seven inputs have different logical weights. R1.fin-type is the Gaitsgory–Rozenblyum finite-type theorem. R1.formal.a is a trace theorem after the categorical colimit presentation is known. R2 fixes the Selmer target and tangent complex, but its use in formal BZN requires formal Noetherianity and nilpotent-support compatibility for the chosen local conditions. R1.formal.b, R1.formal.c, R3, and R4 are geometric hypotheses.

Logical Consequence

The arithmetic BZN comparison is therefore not a consequence of a product formula for local stacks. It is a statement about a Selmer derived fibre product, a formal Hochschild trace, a pro-étale loop object, a flat singular-support locus, and wild Hecke summands. The finite-type theorem and the trace theorem do not eliminate the loop, flat-locus, full-target, or wild-prime hypotheses.

Open Problems

The reduction of Chapter 53.4.9 poses three precise open problems.

(i) **R1.formal.b.** Does $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ commute with ind-colimits under closed immersions in derived pro-étale stacks? The relevant tools are Lurie's ∞-topos formalism (HTT 6, HA 4.4) combined with Bhatt–Scholze pro-étale (§7). A positive answer proves R1.formal.b; a counterexample restricts Chapter 49.5.3 to the cofinal finite-type subclass.

(ii) **R1.formal.c.** Does the Arinkin–Gaitsgory nilpotent-singular-support flat locus coincide with the (φ, Γ)-module Fargues–Scholze flat locus on each finite-type piece of the Emerton–Gee ind-colimit? The comparison between Fargues–Scholze arXiv:2102.13459 and Arinkin–Gaitsgory arXiv:1201.6343 can be checked on unramified crystalline representations with fixed Hodge–Tate weight.

(iii) **R4.** What are the $\mathrm{Ext}^{\mathrm{I}}$-contributions to $\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p})$ at wild primes $p = 2, 3$, and do they shift the arithmetic BZN identification? The Etingof–Ostrik 2004 finite-tensor-category framework applies; a model case is $p = 2$ with tame restriction to the depth-1 local category.

53.7 Formal Noetherian Boundary

The global spectral target for $\mathrm{F3.BZN}^{\mathrm{Ar}}$ is the Selmer-derived stack $\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}$ with tangent complex $C^\bullet_{\mathcal{L}}(G_{\mathbb{Q},S}, \mathrm{ad}^{\circ}\rho)[1]$ (Chapter 53.3.5); formal Noetherianity, quasi-smoothness, and nilpotent-support compatibility for the chosen local deformation problem remain hypotheses. R3 is a full-target hypothesis under R2 (Chapter 53.3.6).

R1.formal follows from three conditions (Chapter 53.4.1). R1.formal.a is Hochschild trace continuity along the given categorical colimit (Chapter 53.4.2). R1.formal.b is the arithmetic loop-space formal computation, and R1.formal.c is flat-locus pullback compatibility (Chapter 53.4.4 and Chapter 53.4.7).

Thus $\mathrm{F3.BZN}^{\mathrm{Ar}}$ is conditional on R1.formal.b, R1.formal.c, the finiteness and support part of R2, R3, and R4 (Chapter 53.5.1). The pro-étale extension of Ben-Zvi–Francis–Nadler is not a formal theorem of the finite-type theory; in this notation it is the loop-space hypothesis together with the flat-locus comparison.

Chapter 54

$\mathrm{BZN}^{\mathrm{Ar}}$: Loop-Space Convention and Wild-Prime Summands

The arithmetic trace comparison has two separate requirements. First, the internal mapping stack $\underline{\mathrm{Map}}(B\widehat{\mathbb{Z}}, \mathcal{X})$ must be compared with finite-quotient and ind-completed loop models under explicit $\widetilde{\text{formal}}$-convergence hypotheses. Second, local arithmetic categories at the primes two and three must be constructed with their coefficient rings, extension groups and trace maps.

The finite-algebra calculations below show why the second requirement cannot be replaced by a Loewy-length argument. A radical-square-zero algebra can have nonzero Ext in arbitrarily high degrees. Even when two local trace complexes are concentrated in degrees zero and one, their tensor over a common field can have a nonzero degree-two term. An integral distinct-primary model behaves differently, but needs actual integral carriers and flatness hypotheses. These calculations constrain the proposed arithmetic comparison; they do not construct its categories, trace maps or completed global assembly.

54.1 The Two Loop-Convention and Wild-Prime Conditions

Origin of the Two Conditions

The structural comparison of Chapter 51 at §51.4 contains four conditions in the F3.$\mathrm{BZN}^{\mathrm{Ar}}$ reduction. Three of these (the formal-Noetherian extension of Gaitsgory–Rozenblyum; the cotangent-complex check for the Emerton–Gee stack; and the R1 relabeling) were absorbed into the R1 split R1.fin-type + R1.formal stated in Chapter 51. The remaining two, not absorbed into the existing reduction, are:

- **R3.** Ambiguity in the loop-space convention. Three candidate definitions of $L^{\mathrm{Ar}}\mathcal{X} = \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ are in play:

  (a) *Cofiltered-limit convention.* $L^{\mathrm{Ar}}_{\mathrm{cofilt}}\mathcal{X} := \varprojlim_n \mathrm{Map}(B(\mathbb{Z}/n), \mathcal{X})$, the limit over finite quotients of $\widehat{\mathbb{Z}}$.

  (b) *Internal-Hom convention.* $L^{\mathrm{Ar}}_{\mathrm{intHom}}\mathcal{X} := \underline{\mathrm{Map}}_{\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}}(B\widehat{\mathbb{Z}}, \mathcal{X})$, the internal mapping stack in the ∞-category of formal Noetherian derived stacks.

  (c) *Ind-construction convention.* $L^{\mathrm{Ar}}_{\mathrm{ind}}\mathcal{X} := \varinjlim_n \mathrm{Ind}\big(\mathrm{Map}(B(\mathbb{Z}/n), \mathcal{X}^{\mathrm{perf}})\big)$, the ind-completion over finite quotients, applied to the perfect locus and re-ind-extended.

  These agree for $\mathcal{X}$ finite-type smooth, but the formal Noetherian setting is sensitive to the order of limits and ind-completions.

- **R4.** Local arithmetic extension and trace data. For the proposed exceptional contributions at $p = 2, 3$, specify the coefficient categories, compute their extensions and construct maps to their trace complexes. A positive-degree trace contribution is not a formal consequence of non-semisimplicity. Its global effect also depends on the common coefficient ring and tensor product.

Logical Structure

R3 is a convention plus a convergence hypothesis: the internal Hom is the chosen definition, and comparison with the other two definitions requires compactness and formal-colimit compatibility. R4 includes the local categories, trace comparisons and tensor assembly. The explicit algebra calculations in Section 54.4 show the additional hypotheses needed for a degree-one-only answer.

Notation

Throughout this chapter:

- $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, the profinite completion of $\mathbb{Z}$.
- $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$: the ∞-category of formal Noetherian derived (pro-étale) stacks in the sense of Gaitsgory–Rozenblyum's AMS DAG monographs Vol. II Part III. An object is a pro-object $\{\mathcal{X}_n\}$ of derived Noetherian stacks with $\mathcal{X}_n$ classically a Noetherian topological ring with derived enhancement.
- For $\mathcal{X} \in \mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$, its formal completion $\mathcal{X}^{\wedge}$ is the image of $\mathcal{X}$ under the reflective embedding into the formal-Noetherian ∞-category.
- $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p}$: the Iwahori Hecke category at the prime p (Bezrukavnikov arXiv:1209.0403).
- The coefficient field is specified in each calculation. The field $\overline{\mathbb{Q}}_\ell$ has characteristic zero, whereas $\overline{\mathbb{F}}_\ell$ has characteristic ℓ. An integral calculation below uses $R = \mathbb{Z}$ instead.
- $\mathrm{Ext}^i_{\mathcal{A}}$: Ext^i in the abelian category $\mathcal{A}$; subscripts indicate the ambient category.

54.2 Structure of the Argument

The argument for R3 has three parts; the argument for R4 has three further parts. For R3: the three candidate definitions are compared with explicit fibre-product formulas (Proposition 54.3.2); the fibre-product formulas are conditionally identified on formal Noetherian derived algebraic stacks via Gaitsgory–Rozenblyum and Postnikov-tower convergence (Theorem 54.3.4); the internal-Hom convention is adopted as canonical and confirmed consistent with Chapter 49 (Definition 54.3.7, Proposition 54.3.9). For R4, the periodic resolution and the path-count formula compute extensions over explicit finite algebras. The four-term tensor formula then distinguishes common-field and distinct-primary integral assembly. Questions 54.4.8 and 54.4.9 specify the arithmetic comparisons that these elementary calculations leave open.

54.3 R3: Arithmetic Loop-Space Convention

The Three Candidate Definitions

Throughout this section $\mathcal{X} \in \mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ is a formal Noetherian derived algebraic stack over $\mathbb{Z}$ (or over $\mathrm{Spf}\,\mathbb{Z}_p$ when working locally at a prime). We demand:

- $\mathcal{X}$ is locally presented as the formal completion of a derived Noetherian affine scheme $\mathcal{X}_0 = \mathrm{Spec}\, A^{\bullet}_0$ along an ideal $I \subset \pi_0(A^{\bullet}_0)$;
- the cotangent complex $\mathbb{L}_{\mathcal{X}}$ has finite Tor-amplitude $[a, b]$ over the structure sheaf;
- the Postnikov tower $\pi_{\leq n}\mathcal{X} \to \mathcal{X}$ stabilises in each tor-layer (equivalently, $\mathcal{X}$ is t-bounded in the sense of Gaitsgory–Rozenblyum Vol. I §2).

For the arithmetic target these conditions must be checked for the Selmer-derived stack $\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}$; the local Emerton–Gee theorem supplies only the local formal inputs.

Definition 54.3.1 *(the three candidate arithmetic free loop spaces).* For $\mathcal{X} \in \mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$, define

$$\begin{aligned} L^{\mathrm{Ar}}_{\mathrm{cofilt}}\mathcal{X} &:= \varprojlim_{n\geq 1} \mathrm{Map}\big(B(\mathbb{Z}/n\mathbb{Z}), \mathcal{X}\big), \\ L^{\mathrm{Ar}}_{\mathrm{intHom}}\mathcal{X} &:= \underline{\mathrm{Map}}_{\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}}\big(B\widehat{\mathbb{Z}}, \mathcal{X}\big), \\ L^{\mathrm{Ar}}_{\mathrm{ind}}\mathcal{X} &:= \varinjlim_{n\geq 1} \mathrm{Ind}\big(\mathrm{Map}(B(\mathbb{Z}/n\mathbb{Z}), \mathcal{X}^{\mathrm{perf}})\big), \end{aligned}$$

where:

- in (a) the limit is along the cofiltered system $\{\mathbb{Z}/n\}_{n\in\mathbb{N}}$ under divisibility;
- in (b) the internal mapping stack is computed in the ∞-category $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ of formal Noetherian derived pro-étale stacks;
- in (c) $\mathcal{X}^{\mathrm{perf}}$ denotes the perfect locus (dualisable objects in the formal quasi-coherent derived category), and Ind is the ind-completion in stable presentable ∞-categories.

Explicit Computation of the Three Candidate Definitions

Proposition 54.3.2 *(three candidate loop-space formulas). Conjectural statement.* Let $\mathcal{X} \in \mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ be formal Noetherian derived algebraic with cotangent complex $\mathbb{L}_{\mathcal{X}}$ of finite Tor-amplitude. Then:

(i) *Cofiltered-limit formula.*

$$L^{\mathrm{Ar}}_{\mathrm{cofilt}}\mathcal{X} \;=\; \varprojlim_{n}\Big(\mathcal{X} \times_{\mathcal{X}\times\mathcal{X}} \mathcal{X}_{(n)}\Big),$$

where $\mathcal{X}_{(n)} \subset \mathcal{X}\times\mathcal{X}$ is the n-th formal neighbourhood of the diagonal.

(ii) *Internal-Hom formula.*

$$L^{\mathrm{Ar}}_{\mathrm{intHom}}\mathcal{X} \;=\; \mathcal{X} \times^{\mathrm{d},\wedge}_{\mathcal{X}\times\mathcal{X}} \mathcal{X},$$

the derived formal self-intersection of $\mathcal{X}$ along its diagonal, computed in $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ with the derived tensor product taken over the formal diagonal.

(iii) *Ind-construction formula.*

$$L^{\mathrm{Ar}}_{\mathrm{ind}}\mathcal{X} \;=\; \mathrm{Ind}\Big(\varinjlim_{n} \mathcal{X}^{\mathrm{perf}} \times_{\mathcal{X}^{\mathrm{perf}}\times\mathcal{X}^{\mathrm{perf}}} \mathcal{X}^{\mathrm{perf}}_{(n)}\Big).$$

Evidence. (1) The mapping stack from $B(\mathbb{Z}/n)$ to $\mathcal{X}$ is the fixed-point locus of the order-n automorphism action, which is the derived fibre product along the n-th iterate of the diagonal. In characteristic zero (for the rational structure at each prime) this reduces to the formal neighbourhood $\mathcal{X}_{(n)}$. Taking the limit over n under divisibility restricts to the formal self-intersection in the limit.

(2) Internal Hom in $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ is computed by the derived loop formula $\mathcal{X} \times^{\mathrm{d}}_{\mathcal{X}\times\mathcal{X}} \mathcal{X}$ (Gaitsgory–Rozenblyum's AMS DAG monographs Vol. I Ch. 8 Thm. 8.1.1.9); the formal-Noetherian decoration adds a completion along the diagonal, yielding the indicated $\times^{\mathrm{d},\wedge}$.

(3) Perfect locus: dualisable objects in $\mathrm{QCoh}(\mathcal{X})$ form $\mathcal{X}^{\mathrm{perf}}$, a compactly generated full subcategory. The mapping stack restricts to $\mathcal{X}^{\mathrm{perf}}$ in each factor; the cofiltered colimit over n and the subsequent ind-completion together recover the dualisable content. By the same derived fibre-product formula restricted to perfects plus ind-completion, we obtain the stated expression. □

Remark 54.3.3 *(separation of the three limits).* The three formulas differ in the order of $(\varinjlim/\varprojlim)$ operations and in whether they apply to $\mathcal{X}$ or to $\mathcal{X}^{\mathrm{perf}}$. Equality of the three models requires convergence of the Postnikov tower, generation by perfect objects under ind-completion, and compatibility of formal completion with the profinite limit.

Conventions for the Three Loop Models

Conventions. Derived mapping stack convention follows Gaitsgory–Rozenblyum Vol. I Ch. 8; profinite classifying-stack convention follows Deligne 1990 §5; the two are compatible via $B\widehat{\mathbb{Z}} = \varprojlim_n B(\mathbb{Z}/n)$.

Ambient category. All three candidate constructions take values in the ∞-category $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ of formal Noetherian derived pro-étale stacks. Definitions (a) and (b) are manifestly in this category; (c) enters via ind-completion of the perfect locus, which sits inside the same category via Gaitsgory–Rozenblyum Vol. I Ch. 5.

Hypotheses. Finite Tor-amplitude of $\mathbb{L}_{\mathcal{X}}$ and t-boundedness of the Postnikov tower are explicit hypotheses of Proposition 54.3.2.

Equivalence of the Three Definitions on Formal Noetherian Derived Stacks

THEOREM 54.3.4 *(conditional comparison of the three arithmetic loop spaces). Conditional statement.* Let $\mathcal{X} \in \mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ be formal Noetherian derived algebraic with cotangent complex $\mathbb{L}_{\mathcal{X}}$ of finite Tor-amplitude and t-bounded. Assume:

(i) $B\widehat{\mathbb{Z}}$-mapping stacks are computed by the profinite limit of the finite cyclic mapping stacks on $\mathcal{X}$;

(ii) formal completion along the diagonal commutes with the relevant derived tensor products;

(iii) the perfect locus generates the same loop object after ind-completion.

Then there are canonical equivalences of formal Noetherian derived stacks

$$L^{\mathrm{Ar}}_{\mathrm{cofilt}}\mathcal{X} \xleftarrow{\sim} L^{\mathrm{Ar}}_{\mathrm{intHom}}\mathcal{X} \xrightarrow{\sim} L^{\mathrm{Ar}}_{\mathrm{ind}}\mathcal{X}.$$

All three express the derived formal self-intersection of $\mathcal{X}$ along its diagonal.

Proof. Under the first two hypotheses, $L^{\mathrm{Ar}}_{\mathrm{cofilt}}\mathcal{X}$ and $L^{\mathrm{Ar}}_{\mathrm{intHom}}\mathcal{X}$ compute the same completed derived self-intersection of the diagonal. Under the third hypothesis, the ind-construction gives the ind-completion of the same self-intersection on the perfect locus. The maps are the canonical comparison maps induced by the universal property of internal Hom, formal completion, and ind-completion. □

Remark 54.3.5 *(what fails without the hypotheses).* Noetherianness, t-boundedness, and finite Tor-amplitude do not by themselves identify all three loop models. They are the natural ambient hypotheses under which the three comparison maps can be asked to be equivalences.

COROLLARY 54.3.6 *(application to the Selmer-derived target). Comparison hypothesis.* If the Selmer-derived target $\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}$ satisfies the formal-Noetherian and bounded cotangent hypotheses of Theorem 54.3.4, then the three candidate definitions of $L^{\mathrm{Ar}}\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}$ all compute the derived formal self-intersection along the diagonal and agree canonically.

Hypotheses of the Loop Comparison

Conventions. Postnikov-tower and formal-completion conventions agree between Gaitsgory–Rozenblyum and Lurie HA.

Ambient category. All three constructions sit in $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$; their equivalence is an equivalence in that ambient.

Hypotheses. Finite Tor-amplitude, t-boundedness, Noetherianness are stated in the hypothesis set of Theorem 54.3.4.

Functoriality. The zig-zag (54.3.4) is functorial in $\mathcal{X}$: each arrow is a canonical natural transformation between functors $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}} \to \mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$.

Named theorems. Gaitsgory–Rozenblyum supplies the formal-completion and compact-generation inputs. The profinite mapping-stack comparison remains an explicit hypothesis.

The equivalence of Theorem 54.3.4 is conditional on the stated hypotheses.

Canonical Choice and Consistency with Chapter 49

Definition 54.3.7 *(canonical arithmetic free loop space).* For $\mathcal{X} \in \mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ formal Noetherian derived algebraic with the hypotheses of Theorem 54.3.4, the *canonical arithmetic free loop space of* $\mathcal{X}$ is

$$L^{\mathrm{Ar}}\mathcal{X} \;:=\; L^{\mathrm{Ar}}_{\mathrm{intHom}}\mathcal{X} \;=\; \underline{\mathrm{Map}}_{\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}}\big(B\widehat{\mathbb{Z}}, \mathcal{X}\big).$$

Under Theorem 54.3.4, $L^{\mathrm{Ar}}\mathcal{X}$ is canonically equivalent to both $L^{\mathrm{Ar}}_{\mathrm{cofilt}}\mathcal{X}$ and $L^{\mathrm{Ar}}_{\mathrm{ind}}\mathcal{X}$; we adopt the internal Hom as definitional.

Remark 54.3.8 *(why internal Hom is canonical).* The internal Hom construction is canonical in two senses:

(i) *Functoriality in the source.* $\underline{\mathrm{Map}}(B\widehat{\mathbb{Z}}, -)$ is manifestly functorial in $\mathcal{X}$ without choosing a cofiltered presentation of $B\widehat{\mathbb{Z}}$ or an ind-completion presentation of $\mathcal{X}$.

(ii) *Universal property.* By the standard closed-monoidal structure on $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ (Lurie HTT §7.3, transported via Gaitsgory–Rozenblyum Vol. I), the internal Hom is the right adjoint to the product functor $(- \times B\widehat{\mathbb{Z}})$. The cofiltered and ind-construction versions require extra choices to realise this universal property.

Proposition 54.3.9 *(consistency with Chapter 49).* The canonical convention of Definition 54.3.7 coincides with the arithmetic free loop space of Definition 49.4.1 in Chapter 49: both define $L^{\mathrm{Ar}}\mathcal{X}$ as the internal mapping stack $\mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ in the ∞-category of formal Noetherian derived pro-étale stacks.

Proof. Compare Definition 54.3.7 with Definition 49.4.1: both read $L^{\mathrm{Ar}}\mathcal{X} = \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ computed as internal Hom in $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$. The definitions are literally identical. Chapter 49's usage in Sections 49.4 and 49.5 (Theorem 49.5.5) matches this internal-Hom interpretation. □

Internal Hom and Its Universal Property

Conventions. The adjoint-functor convention of Lurie HTT matches the closed-monoidal structure on formal derived stacks in Gaitsgory–Rozenblyum Vol. I.

Ambient category. $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ is the same as in Chapter 49.

Hypotheses. The closed-monoidal structure requires only that $\mathrm{dSt}^{\wedge}_{\mathrm{Noeth}}$ have arbitrary small colimits and products; both are standard.

Functoriality and universal property. Functoriality in $\mathcal{X}$ is the defining property of the internal Hom; its universal property is Lurie HTT §7.3.

The canonical-choice result Proposition 54.3.9 is a statement about notation. It does not prove the comparison with the cofiltered or ind-completed models.

R3 Boundary

R3 has two parts. The canonical choice is the internal Hom (Definition 54.3.7); and this choice is consistent with Chapter 49's usage (Proposition 54.3.9). The comparison with the cofiltered and ind-completed models is the conditional statement of Theorem 54.3.4.

54.4 Wild-Prime Extension and Trace Comparisons

A Yoneda extension is computed in a specified abelian category; a Hochschild class belongs to a specified trace complex. Relating the two requires a map, not just equality of dimensions. Two elementary calculations also constrain the possible degrees of a global trace: a radical-square-zero algebra can have higher Ext, and tensoring two complexes supported in degrees zero and one can produce degree two. We first make these calculations over explicit coefficient rings. Their arithmetic realization remains a separate comparison problem.

Finite Algebras and Higher Extensions

LEMMA 54.4.1 *(degreewise finiteness of extensions).* Let A be a finite-dimensional algebra over a field k. There are finitely many isomorphism classes of simple right A-modules, and $\mathrm{Ext}^n_A(M,N)$ is finite-dimensional for finite-dimensional M, N and each $n \geq 0$. The total graded Ext space need not be finite-dimensional. No identification with Hochschild homology is included in this statement.

Proof. Every simple module is a quotient of the regular module, whose finite composition series contains only finitely many simple factors. A finite-dimensional module admits a surjection from a finite sum of copies of A. Its kernel is finite-dimensional; repeating constructs a resolution with a finite-dimensional projective term in each degree. Applying $\mathrm{Hom}_A(-,N)$ gives a finite-dimensional space in each degree, hence finite-dimensional cohomology. The following example proves the final assertion about total dimension. □

PROPOSITION 54.4.2 *(a periodic resolution).* For $A = k[\epsilon]/(\epsilon^2)$ and $S = A/(\epsilon)$,

$$\mathrm{Ext}^n_A(S,S) = k \qquad (n \geq 0).$$

Thus Loewy length two does not imply projective dimension one. In characteristic two, $A \cong k[C_2]$, so the finite tensor category of finite-dimensional $k[C_2]$-modules also has infinite-dimensional total Yoneda algebra.

Proof. Multiplication by ϵ has kernel and image both equal to $k\epsilon$. The augmentation kernel is the same subspace. Hence

$$\cdots \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \longrightarrow S \longrightarrow 0$$

is an exact free resolution. Applying $\mathrm{Hom}_A(-, S)$ gives one copy of k in every nonnegative degree, with zero differential because ϵ acts trivially on S. If $\mathrm{char}\, k = 2$, writing $\epsilon = g - 1$ for the generator of C_2 gives $\epsilon^2 = g^2 - 2g + 1 = 0$. The two-dimensional algebras are therefore isomorphic. The group-algebra coproduct gives its usual tensor category. □

Proposition 54.4.3 *(paths and radical-square-zero Ext).* Let Q be a finite quiver, J its arrow ideal, and $B = kQ/J^2$. Use right modules and concatenate paths from left to right. For the simple $S_i = e_iB/e_iJ$,

$$\dim_k \mathrm{Ext}^n_B(S_i, S_j) = \#\{\text{oriented paths of length } n \text{ from } i \text{ to } j\}.$$

Consequently all $\mathrm{Ext}^{n\geq 2}$ between simples vanish if and only if Q has no composable pair of arrows.

Proof. The projective cover is $P_i = e_iB$, and its radical has one copy of S_j for each arrow $i \to j$, because every product of two arrows is zero. Thus $\Omega S_i = \bigoplus_{i\to j} S_j$. Iteration gives one copy of S_j in $\Omega^n S_i$ for each length-n path. The resulting minimal projective resolution has its differentials in the radicals. Every map from a projective to a simple kills that radical, so applying $\mathrm{Hom}_B(-, S_j)$ gives zero differentials. This proves the formula, including $n = 0$. A composable pair gives a nonzero degree-two group. If no such pair exists, there are no paths of any length at least two. □

Example 54.4.4 *(a three-vertex block and its Yoneda product).* Take $Q = (0 \xrightarrow{a} 1 \xrightarrow{b} 2)$ and $B = kQ/(ab)$. The right projectives have bases $P_0 = \langle e_0, a\rangle$, $P_1 = \langle e_1, b\rangle$, and $P_2 = \langle e_2\rangle$. There is an exact resolution

$$0 \longrightarrow P_2 \xrightarrow{e_2\mapsto b} P_1 \xrightarrow{e_1\mapsto a} P_0 \longrightarrow S_0 \longrightarrow 0.$$

Indeed the middle map kills b, since $ab = 0$; its kernel is kb, and its image is the augmentation kernel ka. It follows that

$$\mathrm{Ext}^1_B(S_0, S_1) = k, \qquad \mathrm{Ext}^1_B(S_1, S_2) = k, \qquad \mathrm{Ext}^2_B(S_0, S_2) = k.$$

The product of the two adjacent degree-one classes is the nonzero degree-two class: the short exact sequence $0 \to S_1 \to P_0 \to S_0 \to 0$ gives the dimension-shifting isomorphism $\mathrm{Ext}^1_B(S_1, S_2) \cong \mathrm{Ext}^2_B(S_0, S_2)$, whose connecting map is Yoneda composition. Thus the relation $ab = 0$ in B does not make the corresponding Yoneda product zero. Reversing all arrows reverses the ordered pairs in the Ext groups; it does not eliminate degree two.

For comparison, the unquotiented path algebra kQ has $\mathrm{rad}\, P_0 \cong P_1$ and $\mathrm{rad}\, P_1 \cong P_2$, hence length-one projective resolutions of its simples. This is a different algebra; its P_0 has Loewy length three. An arithmetic identification with one of these algebras must specify which relations it preserves.

Regular Hochschild Homology of the Finite Models

The same two algebras permit complete trace calculations. They show that extensions between simples and regular Hochschild homology can behave differently even in these small examples. Throughout this subsection k is any field, the algebras are ordinary and ungraded, and all tensors are algebraic. For an algebra R, put $R^e = R \otimes_k R^{\mathrm{op}}$ and $HH_n(R) = \mathrm{Tor}^{R^e}_n(R, R)$, with regular coefficients. This is Hochschild homology, not the cohomology of derivations used for the Bost–Connes algebra.

PROPOSITION 54.4.5 *(the regular trace of the dual numbers).* For $A = k[\epsilon]/(\epsilon^2)$,

$$\begin{aligned} HH_0(A) &= A, \\ HH_{2r+1}(A) &= A/(2\epsilon A) && (r \geq 0), \\ HH_{2r}(A) &= \operatorname{ann}_A(2\epsilon) && (r \geq 1). \end{aligned}$$

If $\operatorname{char} k \neq 2$, each positive-degree group has dimension one; odd groups are $A/(\epsilon)$ and positive even groups are ϵA. If $\operatorname{char} k = 2$, then $HH_n(A) = A$ for every $n \geq 0$.

Proof. Identify $A^e = k[x, y]/(x^2, y^2)$, with x acting on the left and y on the right. Set $u = x - y$, $v = x + y$. Consider

$$\cdots \xrightarrow{u} A^e \xrightarrow{v} A^e \xrightarrow{u} A^e \xrightarrow{\mu} A \longrightarrow 0, \qquad \mu(x) = \mu(y) = \epsilon.$$

There is one free term in each nonnegative degree; the differential from an odd degree is multiplication by u, and from a positive even degree multiplication by v. The composites vanish because $uv = vu = 0$. For $r = \alpha + \beta x + \gamma y + \eta xy$,

$$ur = \alpha(x - y) + (\gamma - \beta)xy, \qquad vr = \alpha(x + y) + (\beta + \gamma)xy.$$

Hence $\ker u = \langle x + y, xy \rangle = \operatorname{im} v$, $\ker v = \langle x - y, xy \rangle = \operatorname{im} u$, and $\ker \mu = \langle x - y, xy \rangle$. Both images have dimension two, including in characteristic two. This is an exact free bimodule resolution.

Tensoring with the regular A makes u zero and v multiplication by 2ϵ. Thus the odd differentials are zero and the positive even differentials are 2ϵ, giving the asserted quotients and annihilators. If two is invertible, $\operatorname{ann}_A(2\epsilon) = \epsilon A$; if two is zero, all differentials vanish.

The normalized Hochschild complex checks the signs directly. Since $A/k1 = k\epsilon$, its degree-n term is $A \otimes (k\epsilon)^{\otimes n}$. Interior products vanish, and the first and cyclic last faces give

$$b_n(c \otimes \epsilon^{\otimes n}) = (1 + (-1)^n)c\epsilon \otimes \epsilon^{\otimes(n-1)}.$$

This is the same alternating zero and 2ϵ differential. □

The simple-module calculation in Proposition 54.4.2 gives $\operatorname{Ext}^n_A(S, S) = k$ in every degree and characteristic. In characteristic two its dimension is one, whereas the regular Hochschild group has dimension two. The two calculations use different resolutions and different coefficient carriers.

PROPOSITION 54.4.6 *(the regular trace of the directed three-vertex model).* For $B = k(0 \xrightarrow{a} 1 \xrightarrow{b} 2)/(ab)$,

$$HH_0(B) = ke_0 \oplus ke_1 \oplus ke_2, \qquad HH_n(B) = 0 \quad (n \geq 1).$$

Thus the nonzero Ext groups and Yoneda product of Example 54.4.4 do not force nonzero positive-degree regular Hochschild homology.

Proof. Let $S = ke_0 \oplus ke_1 \oplus ke_2$, $J = ka \oplus kb$. Then $B = S \oplus J$, $J^2 = 0$, and $J \otimes_S J = k(a \otimes b)$; there are no composable triples. The relative normalized bimodule bar terms are $P_n = B \otimes_S J^{\otimes_S n} \otimes_S B$, with

$$\begin{aligned} P_0 &= \bigoplus_{i=0}^{2} Be_i \otimes_k e_iB, \\ P_1 &= (Be_0 \otimes_k e_1B) \oplus (Be_1 \otimes_k e_2B), \\ P_2 &= Be_0 \otimes_k e_2B. \end{aligned}$$

Each is a projective bimodule: each summand is an idempotent direct summand of the free bimodule $B \otimes_k B$. The differentials, retaining arrows in the bar notation, are

$$d_1(l \otimes j \otimes r) = lj \otimes r - l \otimes jr,$$
$$d_2(l \otimes a \otimes b \otimes r) = la \otimes b \otimes r + l \otimes a \otimes br.$$

The missing middle term is $-l \otimes ab \otimes r = 0$.

Here is a direct exactness check. The dimensions of P_0, P_1, P_2 are 8, 4, 1, and a basis of P_1 is

$$q_1 = e_0 \otimes a \otimes e_1, \qquad q_2 = e_0 \otimes a \otimes b,$$
$$q_3 = e_1 \otimes b \otimes e_2, \qquad q_4 = a \otimes b \otimes e_2.$$

Their d_1-images are respectively

$$a \otimes e_1 - e_0 \otimes a, \quad a \otimes b, \quad b \otimes e_2 - e_1 \otimes b, \quad -a \otimes b.$$

The first three are independent, so $\ker d_1 = k(q_2 + q_4)$. The generator $e_0 \otimes a \otimes b \otimes e_2$ of P_2 maps to $q_2 + q_4$; hence d_2 is injective and exactness holds at P_1. Multiplication $P_0 \to B$ is surjective, annihilates $\operatorname{im} d_1$, and has kernel of dimension $8 - 5 = 3$. This proves exactness also at P_0, in every characteristic. Thus $0 \to P_2 \to P_1 \to P_0 \to B \to 0$ is a projective bimodule resolution.

The regular trace of a summand is

$$B \otimes_{B^e} (Be_i \otimes_k e_j B) \cong e_j B e_i.$$

The map sends the class of $c \otimes (l \otimes r)$ to rcl; its inverse sends $t \in e_j B e_i$ to the class of $t \otimes (e_i \otimes e_j)$. Balancing verifies that these maps are well defined and inverse. A path from i to j can therefore contribute only if there is a return path from j to i. Here $e_1 B e_0 = e_2 B e_1 = e_2 B e_0 = 0$ and $e_i B e_i = k e_i$. The traced complex is k^3 in degree zero and zero in every positive degree, as asserted. □

In this example the relation $ab = 0$ contributes a genuine bimodule-resolution term in degree two, but regular trace kills it because its endpoints differ. Extension groups between simples retain the directed paths; Hochschild homology requires cyclic endpoint matching. Neither calculation constructs the arithmetic Ext-to-trace comparison in the following questions.

The Local Arithmetic Comparisons

Remark 54.4.7 *(non-semisimplicity and the coefficient category).* The example $k[C_2]$ in characteristic two is non-semisimple: $0 \to S \to A \to S \to 0$ is not split. Transferring such a result to an arithmetic Iwahori block requires that block, its coefficient ring and its relation to the finite algebra to be specified. The primes 2, 3 alone do not determine a coefficient category. The fields $\overline{\mathbb{Q}}_\ell$ and $\overline{\mathbb{F}}_\ell$ have characteristics zero and ℓ, respectively, and require different specialization arguments.

Question 54.4.8 *(the proposed contribution at two).* Construct the arithmetic block and its trace complex C_2^{wild}, with a comparison realizing the proposed one-dimensional degree-one contribution

$$H_1(C_2^{\mathrm{wild}}) \cong W_2, \qquad W_2 \cong k.$$

The coefficient field and the map from extension data to this trace homology are part of the construction. In characteristic two, the trivial and sign characters of C_2 are the same character, since $-1 = 1$. They

cannot serve as two distinct simples in an off-diagonal Ext calculation. The dual-number model does give a one-dimensional self-Ext in degree one, but it also gives nonzero Ext in every higher degree. A claim of local amplitude at most one therefore needs an actual resolution or a trace calculation that accounts for the higher extension classes.

Question 54.4.9 *(the proposed contribution at three).* Construct the arithmetic block, coefficient field and comparison maps that would realize

$$H_1(C_3^{\mathrm{wild}}) \cong W_3, \qquad W_3 \cong k \oplus k.$$

In the explicit three-vertex algebra of Example 54.4.4, the two classes are in $\mathrm{Ext}^1(S_0, S_1)$ and $\mathrm{Ext}^1(S_1, S_2)$, with the fixed right-module convention, and they have a nonzero Yoneda product in degree two. An arithmetic comparison must preserve these directions and account for that product. In characteristic three, $X^3 - 1 = (X-1)^3$, so there is no primitive third root of unity. A primitive-cube-root specialization in characteristic zero is a different model, requiring an explicit comparison with a modular coefficient category. Neither a choice of root nor a Calabi–Yau or Frobenius structure by itself constructs the map from Yoneda extensions to Hochschild homology.

Hypothesis 54.4.10 *(the large-prime trace datum).* For the chosen common coefficient ring R, suppose that the arithmetic local trace complexes C_p have been constructed and that, for every $p \geq 5$, there is a specified equivalence $C_p \simeq B_p[0]$ with an R-module B_p. This is a hypothesis on the actual trace complexes, including their arithmetic comparison; it is not inferred solely from an assertion about a Yoneda algebra or a numerical prime threshold.

COROLLARY 54.4.11 *(conditional large-prime vanishing).* Under Hypothesis 54.4.10, $H_i(C_p) = 0$ for every $i > 0$ and $p \geq 5$.

Proof. The specified equivalence identifies the homology with that of a module placed in degree zero. □

COROLLARY 54.4.12 *(conditional vanishing of the large-prime wild part).* If a wild summand is specified inside such a C_p, its positive-degree homology vanishes for $p \geq 5$.

Proof. Homology preserves a direct-sum decomposition, so the summand's positive-degree homology is a direct summand of zero. □

The Four Terms in a Derived Tensor Product

Complexes are now homologically graded: the differential lowers degree, and $M[i]$ places a module M in degree i. All tensors in this subsection are algebraic derived tensors over the displayed common ring. A completed tensor requires its own continuity and exactness assertions.

LEMMA 54.4.13 *(two split local complexes).* For a commutative ring R, suppose $C_2 \simeq B_2 \oplus W_2[1]$ and $C_3 \simeq B_3 \oplus W_3[1]$ in $D(R)$, with specified splittings. Then

$$\begin{aligned} C_2 \otimes^{\mathbb{L}}_R C_3 \simeq\ & (B_2 \otimes^{\mathbb{L}}_R B_3) \\ & \oplus \big((W_2 \otimes^{\mathbb{L}}_R B_3) \oplus (B_2 \otimes^{\mathbb{L}}_R W_3)\big)[1] \\ & \oplus (W_2 \otimes^{\mathbb{L}}_R W_3)[2]. \end{aligned} \tag{54.1}$$

Over a field, with $B_2 = B_3 = k$, $W_2 = k$, and $W_3 = k^2$, this is $k \oplus k^3[1] \oplus k^2[2]$.

Proof. Choose projective resolutions of the four modules. Distributing the tensor of the two finite direct sums gives the four displayed terms and their shifts. Over a field the modules are already flat, so the zero-differential representatives compute the derived tensor. The total differential is $d(x \otimes y) = dx \otimes y + (-1)^{|x|} x \otimes dy = 0$. For a basis vector x of W_2 and a basis y, z of W_3, $x \otimes y$ and $x \otimes z$ are independent degree-two cycles and cannot be boundaries. More generally, the tensor of two nonzero vector spaces over a field is nonzero: linear functionals nonzero on chosen vectors detect a nonzero elementary tensor. Graded commutativity does not cancel products between different factors. □

Lemma 54.4.14 *(distinct-primary integral coefficients).* If M_2 is a 2-primary torsion abelian group and M_3 is a 3-primary torsion abelian group, then $M_2 \otimes^{\mathrm{L}}_{\mathbb{Z}} M_3 = 0$. If B_2, B_3 are flat abelian groups in degree zero, it follows that $(B_2 \oplus M_2[1]) \otimes^{\mathrm{L}}_{\mathbb{Z}} (B_3 \oplus M_3[1])$ is concentrated in degrees zero and one. For $B_2 = B_3 = \mathbb{Z}$, $M_2 = \mathbb{Z}/2$, and $M_3 = (\mathbb{Z}/3)^2$, it is $\mathbb{Z} \oplus (\mathbb{Z}/2 \oplus (\mathbb{Z}/3)^2)[1]$.

Proof. Resolve $\mathbb{Z}/2^a$ by $[\mathbb{Z} \xrightarrow{2^a} \mathbb{Z}]$. After tensoring with $\mathbb{Z}/3^b$, its differential is an isomorphism, so the complex is acyclic. Finite primary groups are finite direct sums of cyclic primary groups. Arbitrary primary torsion groups are filtered unions of finite subgroups; tensor and Tor commute with these filtered colimits. There is no higher Tor beyond degree one over $\mathbb{Z}$, since the kernel of a surjection from a free abelian group is free. This proves the first assertion. Flatness of B_2, B_3 removes higher Tor from the other three terms in (54.1), proving the remaining claims. Without flat degree-zero factors, those terms can carry additional homology. Completion is not part of this algebraic argument. □

The Arithmetic Assembly Problem

Question 54.4.15 *(the full-target arithmetic trace comparison).* Construct compatible local arithmetic categories and trace complexes, a common coefficient ring, their completed restricted-tensor assembly, and a comparison with the full Weil/(φ, Γ) target, of the form

$$\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}) \simeq \mathcal{O}(\mathrm{LocSys}^{W,\varphi^\Gamma}_{GL_2})^{\mathrm{flat}} \oplus \mathcal{W}.$$

Here $\mathcal{W}$, its coefficient carrier and its degrees must be computed from that construction. The procyclic $B\widehat{\mathbb{Z}}$ model records only the inertia comparison, not the full target. The formal-loop hypotheses, F1, F2, and local degree-zero comparisons remain necessary parts of the proposed construction; their names do not supply a completed Künneth map.

If actual local traces over a common field split as $k \oplus k[1]$ and $k \oplus k^2[1]$, the full two-prime tensor has the degree-two term of Lemma 54.4.13. Its additive degree-one part is

$$\mathcal{W}_{\le 1} = k[1] \oplus (k \oplus k)[1]. \tag{54.2}$$

This can describe the positive part of a specified degree-at-most-one truncation, or a quotient killing the mixed product. It is not the full wild tensor contribution. To realize such a truncation or quotient arithmetically, its map and compatibility with the operations must also be constructed.

Remark 54.4.16 *(remaining degree-zero factors).* If the restricted tensor of the other primes is a degree-zero module D, every term of (54.1) must also be tensored with D. Thus even the degree-one terms carry all complementary degree-zero factors. Taking units at almost every prime permits non-unit entries at both two and three; it does not remove their mixed term. A common-field calculation cannot be replaced by the distinct-primary integral calculation without actual integral models and their comparison maps.

COROLLARY 54.4.17 *(conditions for degree-one assembly).* Suppose actual local trace complexes over R have the two specified splittings, all remaining factors assemble into a flat degree-zero module D, and a proved assembly map identifies the global trace with $D \otimes^{\mathrm{L}}_R C_2 \otimes^{\mathrm{L}}_R C_3$. If B_2, B_3 are flat and $W_2 \otimes^{\mathrm{L}}_R W_3 = 0$, its homology is concentrated in degrees zero and one, with underlying split complex

$$D \otimes_R B_2 \otimes_R B_3 \;\oplus\; D \otimes_R \big((W_2 \otimes_R B_3) \oplus (B_2 \otimes_R W_3)\big)[1].$$

Any completed or arithmetic sheaf interpretation additionally requires that the realization and completion preserve these identifications.

Proof. Apply Lemma 54.4.13. The mixed term vanishes by hypothesis, and flatness removes higher Tor from the remaining terms and from tensoring with D. The specified assembly map then transports this decomposition. □

54.5 CONSISTENCY WITH CHAPTER 49

R3 Consistency

Chapter 49's Definition 49.4.1 reads: for an n-geometric derived pro-étale stack $\mathcal{X}$, $L^{\mathrm{Ar}}\mathcal{X} = \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$, the internal mapping stack in the ∞-category of derived pro-étale stacks. Definition 54.3.7 specialises this to formal Noetherian derived algebraic stacks and commits to the internal-Hom convention. The two definitions are identical on the intersection of their domains (formal Noetherian derived algebraic stacks that are n-geometric and pro-étale). The Emerton–Gee stack lies in this intersection (Corollary 54.3.6).

R4 Consistency

An extension of the comparison in Chapter 49 must preserve the local trace complexes and the chosen tensor assembly. A degree-zero comparison does not determine the positive-degree part, and a list of local Ext dimensions does not supply a trace comparison.

PROPOSITION 54.5.1 *(the tensor terms required by a local comparison).* Suppose an arithmetic comparison identifies its two exceptional local traces over a field with $k \oplus k[1]$ and $k \oplus k^2[1]$, and identifies their assembly with their full derived tensor. Then that assembly is $k \oplus k^3[1] \oplus k^2[2]$. If the remaining factors assemble into a degree-zero vector space D, all three terms must also be tensored with D. The additive expression in (54.2) describes only the positive part through degree one. A degree-one-only global answer requires extra vanishing hypotheses, as in Corollary 54.4.17, or a specified truncation or quotient.

Proof. Apply Lemma 54.4.13 to the specified local equivalences. Tensoring with D is exact over a field. The two independent degree-two classes persist whenever $D \neq 0$. □

54.6 BOUNDARY CONDITIONS

R3 Boundary

The internal-Hom convention is fixed. Its comparison with the cofiltered and ind-completed models is conditional on the three formal-convergence hypotheses in Theorem 54.3.4.

R4 Boundary

Question 54.6.1 *(functoriality of the arithmetic wild contribution).* Construct coefficient-compatible local Langlands comparison maps carrying the relevant arithmetic extension classes to specified trace classes. Determine how their Yoneda products correspond to operations on the trace complexes and how these maps interact with the full tensor assembly. If the desired degree-one expression is obtained by truncation or quotient, construct that map and its functoriality as well.

A refinement of a local correspondence can address this question only after the coefficient category, extension directions and trace complex have been specified. Equality of local dimensions alone proves neither functoriality nor a global comparison.

Logical Form for Chapter 51

- **R3:** internal-Hom convention fixed by Definition 54.3.7 and matched with Chapter 49 by Proposition 54.3.9; comparison with the two other models remains conditional on Theorem 54.3.4.
- **R4:** the finite-algebra Ext calculations and the four-term derived tensor formula are proved above. Their arithmetic realization requires local categories, coefficient comparisons, Ext-to-trace maps and a compatible completed assembly. The proposed local degree-one contributions remain the comparison problems in Questions 54.4.8 and 54.4.9.

54.7 Loop-Convention Boundary

(i) **R3 (loop-space convention).** The canonical choice is the derived internal Hom (Definition 54.3.7); this matches Chapter 49's convention (Proposition 54.3.9). The comparison with the cofiltered and ind-completed models is conditional on Theorem 54.3.4.

(ii) **R4 (extensions and assembly).** Loewy length two does not bound Ext by degree one. The common-field local model has a nonzero mixed degree-two term. The distinct-primary integral model removes that term only on its specified integral carrier. The degree-one criterion above assumes flat degree-zero factors; a completed arithmetic realization also needs a comparison preserving the displayed decomposition.

(iii) **The arithmetic target.** The full Weil/(φ, Γ) trace comparison in Question 54.4.15 requires F1, F2, the formal-loop comparison and the actual local trace data. The elementary computations do not establish these maps.

Chapter 55

Costello-Refined Positselski: Factorization Coherence, $d^2 = 0$, Renormalization, Archimedean Comparison, Functoriality

Question 51.2.7 asks for an integral co–contra comparison. Its arithmetic application requires five further structures. C1 concerns local-global factorization coherence, C2 a square-zero differential, and C3 Euler-product renormalization. C4 compares integral and archimedean models. C5 asks for naturality under Λ-linear coalgebra maps.

For C2 and C4 the constructions below give ordinary arithmetic and integral oscillator models. Their chiral and archimedean comparisons require the additional maps specified there. The arithmetic Heisenberg biconditional $\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$ remains conditional for the abelian Heisenberg domain.

55.1 Five Subconditions of the Abelian Positselski Domain

The Five Subconditions

Question 51.2.7 requires actual integral categories and comparison functors. Its hypotheses cannot be supplied by a field-to-$\mathbb{Z}_p[[\Gamma]]$ base change, because no such unital field map exists. Abelian Galois action and module-theoretic $\mu = 0$ also do not replace the coalgebra and completion hypotheses.

The conditions below concern additional arithmetic operations. For their P1 application one must also construct the Mellin evaluation, central-defect identity, and detection condition in Theorem 43.5.1.

The five subconditions are:

C1 (factorization-coalgebra coherence). At each place v, the arithmetic bar coalgebra $B_v^{\mathrm{ch,Ar}}(\mathcal{V}^{\mathrm{prim}})$ carries a local-global compatible factorization-coalgebra structure in the Costello–Gwilliam Vol. II sense. The adelic restricted product $\otimes'_v B_v^{\mathrm{ch,Ar}}$ inherits a global factorization-coalgebra structure compatible with the finite-level Beilinson adelic complex.

C2 (arithmetic differential and comparison). On the ordinary cyclotomic residue and polynomial bar model, the Iwasawa, residue and bar operators have square-zero sum by three separate signed mixed identities. The completed comparison is Chapter 55.4.2. A differential on the intended arithmetic chiral carrier additionally requires the tame-weighted local operations and comparison maps of Chapter 55.4.3.

C3 (renormalization / Euler-product convergence). The infinite tensor product $\otimes'_p \mathcal{H}^{(p)}$ requires renormalization: naively, the partition function $\prod_p (1 - p^{-\beta})^{-1}$ diverges for $\mathrm{Re}(\beta) \leq 1$. Costello–Gwilliam Vol. II BV renormalization plus Bost–Connes $\mathrm{KMS}_{\beta=1}$ provides the closure.

C4 (integral and archimedean comparison). A common integral polynomial model gives separate finite-place and real oscillator representations, preserving the vacuum, central term and formal current operations. Comparing an arithmetic bar object and its Iwasawa–Gersten differential with a specified real bar model and Positselski pairing remains the construction of Question 55.6.6.

C5 (functoriality). For each morphism $f : C \to C'$ of abelian Λ-coalgebras with $\mu = 0$, there is an induced pullback functor $f^* : \mathrm{D}^{\mathrm{co}}_{\Lambda}(C') \to \mathrm{D}^{\mathrm{co}}_{\Lambda}(C)$ and pushforward $(f^\vee)_* : \mathrm{D}^{\mathrm{ctr}}_{\Lambda}((C')^\vee) \to \mathrm{D}^{\mathrm{ctr}}_{\Lambda}(C^\vee)$ rendering the Positselski equivalence natural. The Positselski equivalence is a natural transformation of 2-functors $\mathrm{Coalg}^{\Lambda,\mu=0,\mathrm{ab},\mathrm{op}} \to \mathrm{PrSt}^{\mathrm{tri}}$.

C5 concerns the naturality of an actual co–contra equivalence under its specified morphisms. Once such an equivalence is constructed, a composite arithmetic argument must preserve its pairings and completions. The numerical P1 comparison still needs an evaluation of the central defect. Functoriality by itself does not supply that evaluation.

Notation Recalled

Throughout this chapter:

- $\Lambda = \mathbb{Z}_p[[\Gamma]]$, where $\Gamma = \mathrm{Gal}(\mathbb{Q}_\infty^{(p)}/\mathbb{Q}) \cong \mathbb{Z}_p$ and $\mathbb{Q}_\infty^{(p)}$ is the cyclotomic $\mathbb{Z}_p$-extension. Section 55.4 uses a fixed odd coefficient prime ℓ instead of p, to distinguish it from the finite places of the number fields. The full cyclotomic Galois group is not being identified with this procyclic quotient.
- $\mathcal{H}^{(p)}$ is the Heisenberg Fock at prime p with oscillator lattice spanned by $\{\alpha_{-n}^{(p)}\}_{n\geq 1}$ of weight $n \log p$ (see Chapter 30 (F3)).
- $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis}} = \otimes'_p \mathcal{H}^{(p)}$, the restricted-tensor-product arithmetic Heisenberg Fock with spherical vectors $|0\rangle_p$.
- $B^{\mathrm{ch,Ar}}(\mathcal{V}^{\mathrm{prim}}) = \oplus_n (s^{-1}\overline{\mathcal{V}^{\mathrm{prim}}})^{\otimes_\Lambda n} \otimes_\Lambda C^\bullet_{\mathrm{Parshin}}(\overline{C})$ denotes the proposed arithmetic chiral carrier of Chapter 33.8.1. The intended differential $d^{\mathrm{Ar}} = d_{\mathrm{Iw}} + d_{\mathrm{Gersten}} + d^{\mathrm{local}}_{\mathrm{bar}}$ requires the local maps and grading specified in Chapter 55.4.3. The ordinary complex constructed in Section 55.4 is a separate, fully specified carrier.
- $\{p, 1-p\}$ denotes the Milnor–Steinberg symbol in $K_2^M(\mathbb{Q})$; it vanishes in $K_2^M(\mathbb{Q})$ and in every $K_2^M(K)$ for K a field (Steinberg 1962, Milnor 1971).
- $\mathrm{Coalg}^{\Lambda,\mu=0,\mathrm{ab}}$ denotes the $(\infty,1)$-category of conilpotent abelian dg Λ-coalgebras with $\mu = 0$.

55.2 Comparison Table

The arithmetic comparison requires five compatibility conditions and the specified evaluation for its P1 application. The coproduct, differential, completion, archimedean realization and naturality maps must be specified on the same abelian Heisenberg domain. The following subsections separate these requirements before describing their role in the pairing.

Factorization-Coalgebra Coherence (C1)

Definition 33.8.1 gives a proposed arithmetic carrier from Λ-tensor powers and a residue complex. A geometric realization additionally requires a configuration diagram, its coefficient category and its collision maps. For specified disjoint configurations, the required maps have the form

$$\Delta^U_{U_1 \sqcup U_2} : B^{\mathrm{ch,Ar}}_U \longrightarrow B^{\mathrm{ch,Ar}}_{U_1} \otimes B^{\mathrm{ch,Ar}}_{U_2}.$$

They must satisfy the coherent counit, coassociativity and collision identities of Definition 55.3.1. A discrete set of place labels supplies none of these position spaces or maps. A vacuum object likewise does not define an E_2-monoidal category or a factorization coproduct.

Suppose local models have supplied a common configuration diagram and finite products $B_S = \otimes_{v\in S} B_v$. Transitions must insert specified vacuum *coalgebra maps* and preserve the full geometric

structure. Under the continuity hypotheses of Theorem 55.3.3, their filtered colimit is a factorization coalgebra. That theorem preserves supplied operations. It does not construct them from the place-indexed tensor formula.

The categorical pointings of Definition 51.2.1 are instead morphisms $\mathrm{Sp} \to C_v$ of operadic algebras. Theorem 51.2.3 assembles that categorical diagram. A comparison with the coalgebra diagram must specify its coefficient functor and preserve the displayed coproducts, local vacua and collision maps.

Beilinson's adelic resolution for schemes motivates a further comparison with the arithmetic residue differential. That comparison must satisfy the coderivation identity in Question 55.3.4. It also requires the archimedean model and maps in Question 55.6.6. Neither categorical assembly nor the integral oscillator model alone supplies these geometric and differential comparisons.

Iwasawa Operators and Residues (C2)

The Iwasawa and residue operators commute before totalization: field automorphisms carry tame residues to the corresponding residue fields. On the ordinary polynomial bar factor, the three signed mixed anticommutators therefore vanish separately. Section 55.4 constructs the arithmetic carriers, norm maps and completed comparison. The residue-weighted local chiral differential requires additional maps and remains the construction of Chapter 55.4.3. A relation holding modulo boundaries does not by itself give a square-zero operator on chains.

Renormalization and Euler-Product Convergence (C3)

For real $\beta > 0$, each prime shift has a normalized Gibbs density $\rho_{p,\beta} = (1 - p^{-\beta})\, p^{-\beta N_p}$. Finite tensor products of these densities define compatible states on the finite-prime Toeplitz algebras. Their compatible restrictions extend to a state on the norm-completed inductive tensor algebra, including at $\beta = 1$; the construction and KMS calculation are in Proposition 55.5.2. This normalizes states, not algebras or adjoint automorphic factors. The spherical Whittaker norm instead has the measure-dependent factor computed in Lemma 56.7.4.

The global occupation Hamiltonian on $\ell^2(\mathbb{N}^\times)$ has partition trace $\sum_{n \geq 1} n^{-\beta}$, finite only for $\beta > 1$. Thus the product state at $\beta = 1$ is not given by a normalized diagonal trace density on that global carrier. Identifying it with the restriction of a full Bost–Connes state, and constructing an arithmetic bar or BV comparison, require actual algebra maps, dynamics and completed chain maps. A product of the scalar identities $\mathrm{Tr}\, \rho_{p,\beta} = 1$ supplies none of those comparisons.

Integral and Archimedean Comparison (C4)

The finite-place oscillator data and the real Heisenberg representation have a common integral polynomial model. Its coefficient extensions preserve the vacuum, central commutator and oscillator operations. The associated formal current fields and oscillator-equivariant two-sided bar resolution are constructed in Section 55.6. Reduction modulo p followed by tensoring with $\mathbb{R}$ gives zero, so it cannot produce the real fibre. Identifying the arithmetic Iwasawa–Gersten complex with a real bar model requires the additional maps specified in Question 55.6.6.

Naturality of the Co-Contra Comparison (C5)

For $f : C \to C'$ a Λ-linear morphism in $\mathrm{Coalg}^{\Lambda, \mu=0, \mathrm{ab}}$, functoriality in the classical field case is Positselski 2011 §6; the arithmetic lift requires restriction to Λ-linear morphisms because base change $k \to \Lambda$ is not functorial on arbitrary coalgebra morphisms. For Λ-linear f: the pushforward $f_* : C\text{–comod}_\Lambda \to C'\text{–comod}_\Lambda$ sends M to M with C'-coaction $M \xrightarrow{\Delta_C} M \otimes_\Lambda C \xrightarrow{\mathrm{id} \otimes f} M \otimes_\Lambda C'$, preserves injectives

(Positselski 2018 arXiv:1209.2995 §3.1 for the field case; the Λ-lift follows by Nakayama plus $\mu = 0$), and descends to a triangulated functor $f_*^{\mathrm{co}}\colon \mathrm{D}^{\mathrm{co}}_\Lambda(C) \to \mathrm{D}^{\mathrm{co}}_\Lambda(C')$. The dual $f^\vee\colon (C')^\vee \to C^\vee$ is a Λ-algebra morphism and $(f^\vee)^*\colon \mathrm{D}^{\mathrm{ctr}}_\Lambda((C')^\vee) \to \mathrm{D}^{\mathrm{ctr}}_\Lambda(C^\vee)$ is the pullback in the opposite direction. The correct shape is therefore: a naive commutativity diagram drawn with f_* going $C \to C'$ and $(f^\vee)^*$ going $(C')^\vee \to C^\vee$ has a direction mismatch under dualization. The correct naturality diagram is:

$$\begin{array}{ccc} \mathrm{D}^{\mathrm{co}}_\Lambda(C) & \xrightarrow{\Psi_C} & \mathrm{D}^{\mathrm{ctr}}_\Lambda(C^\vee) \\ {\scriptstyle f^*}\uparrow & & \uparrow{\scriptstyle (f^\vee)_*} \\ \mathrm{D}^{\mathrm{co}}_\Lambda(C') & \xrightarrow{\Psi_{C'}} & \mathrm{D}^{\mathrm{ctr}}_\Lambda((C')^\vee) \end{array} \tag{55.1}$$

with $f^*\colon \mathrm{D}^{\mathrm{co}}_\Lambda(C') \to \mathrm{D}^{\mathrm{co}}_\Lambda(C)$ the pullback (restriction of scalars along f for the dual coalgebra structure) and $(f^\vee)_*\colon \mathrm{D}^{\mathrm{ctr}}_\Lambda((C')^\vee) \to \mathrm{D}^{\mathrm{ctr}}_\Lambda(C^\vee)$ the pushforward for the dual algebra. This is the correct shape of naturality for a bar–cobar adjunction: the two functors must be parallel, and the dualization swaps pullback with pushforward.

With (55.1) in place, the square commutes up to natural isomorphism by the definition of Ψ: $\Psi_C(f^*M) = \mathrm{Hom}_C(C, f^*M)$ and $(f^\vee)_*\Psi_{C'}(M) = (f^\vee)_*\mathrm{Hom}_{C'}(C', M)$, and these coincide via the adjunction $\mathrm{Hom}_C(C, f^*M) \simeq \mathrm{Hom}_{C'}(C' \otimes_C C, M) \simeq \mathrm{Hom}_{C'}(C', M)$ because $C' \otimes_C C \simeq C'$ via f.

C5 holds for Λ-linear morphisms of abelian Λ-coalgebras with $\mu = 0$, in the form (55.1) (Positselski 2011 §6; induced pullback on the arithmetic Λ-coefficient derived categories via Nakayama).

Functorial Pullback for the P1 Pairing

The arithmetic application of Theorem 43.5.1 requires central invariants of the specified dual objects and an evaluation of their defect. Identifying that evaluation with a Positselski pairing is a further comparison. The integral co–contra comparison of Question 51.2.7 remains an input. C1–C3 require global assembly, a square-zero differential, and a renormalized Euler product. C4 compares archimedean differentials and pairings. C5 requires naturality for the specified arithmetic comparison morphism.

For C4, the integral oscillator and bar constructions of Chapter 55.6.5 provide local coefficient data; the arithmetic comparison in Question 55.6.6 is still required. The P1 consequence also assumes that these models supply the central-defect and Mellin-evaluation data in Theorem 55.8.1.

Comparison Table

55.3 C1: Local-Global Factorization-Coalgebra Coherence

Local Coefficients and Their Geometric Realization

Definition 55.3.1 *(Arithmetic local factorization datum).* At a place v, specify a configuration diagram X_v, its collision maps and disjoint-position opens, and a coefficient ∞-category $\mathcal{E}_v$ with a chosen tensor product. A local factorization coalgebra is a compatible coefficient system $\mathcal{B}_v$ on this diagram with coproducts

$$\mathcal{B}_v(U) \longrightarrow \mathcal{B}_v(U_1) \otimes \mathcal{B}_v(U_2)$$

for the specified disjoint configurations, satisfying the coherent counit, coassociativity and collision identities. A local vacuum marking and any E_2-monoidal structure on the coefficient category are additional specified data. A discrete place label alone determines none of the position spaces X_v.

Condition	Required mathematical data	Consequence on the stated domain
C1	Compatible local factorization coproducts and pointed adelic assembly	A global factorization coalgebra
C2	Ordinary cyclotomic residue and bar operators; additional tame-weighted chiral maps and grading for the intended carrier	A proved square-zero ordinary model; the chiral comparison remains to construct
C3	Compatible finite products, convergence and the Bost–Connes state comparison	A renormalized restricted product with its specified state
C4	A common arithmetic coefficient complex and maps of differentials and pairings, beyond the integral oscillator model	Comparison with a specified real Heisenberg bar model
C5	Parallel derived functors and a naturality isomorphism for the co-contra equivalence	Functorial transport along the specified coalgebra morphisms
P1	Arithmetic objects with a central-defect identity and a Mellin evaluation detecting that defect	The conditional level criterion of Theorem 55.8.1

Table 55.1: Compatibility data for the abelian arithmetic Heisenberg comparison. Each consequence requires the data in its row; the existence of a scalar identity alone does not supply the corresponding chain map or limit.

The finite polynomial carrier does give an ordinary coefficient coalgebra. This construction uses its multiplication, in addition to its vacuum pointing.

Proposition 55.3.2 *(Finite polynomial bar coalgebra).* Let Λ be a commutative ring, $A = \Lambda[x_i : i \geq 1]$, and $\bar{A} = (x_i : i \geq 1)$ its augmentation ideal. Write $[a_1|\cdots|a_n]$ for an element of $\bar{A}^{\otimes_\Lambda n}[n]$, of cohomological degree $-n$. Then

$$B(A) = \bigoplus_{n\geq 0} \bar{A}^{\otimes_\Lambda n}[n], \qquad b[a_1|\cdots|a_n] = \sum_{i=1}^{n-1} (-1)^i [a_1|\cdots|a_i a_{i+1}|\cdots|a_n]$$

is a connected conilpotent differential graded coalgebra with coproduct

$$\Delta[a_1|\cdots|a_n] = \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n].$$

An empty word means $1 \in \Lambda$. The counit projects to length zero and the coaugmentation includes it.

Proof. Two disjoint adjacent mergers occur in both orders with opposite signs. Two overlapping mergers give the same product of three entries, by associativity, again with opposite signs. Thus $b^2 = 0$. Iterating the displayed coproduct amounts to choosing two cuts in a word, independently of their order, which proves coassociativity. A merger followed by a cut is the same as a cut followed by a merger inside one of its blocks. The sign on the second block is $(-1)^i$, since the first block has degree $-i$; hence $\Delta b = (b \otimes 1 + 1 \otimes b)\Delta$. The empty cuts give the counit identities. An iterated reduced coproduct has only nonempty blocks and therefore vanishes after more cuts than the word length. This proves conilpotence. □

In particular the coproduct of this bar is deconcatenation. Calling a vacuum an E_0-datum neither constructs this bar differential nor a local E_2-monoidal factorization category. At the archimedean place, the formal currents of Corollary 55.6.4 likewise require specified configuration maps and an arithmetic

comparison, as in Question 55.6.6. The reduced associative bar, an oscillator-equivariant two-sided resolution, and Chevalley–Eilenberg chains are distinct complexes. Their geometric realization must name which complex and which maps it uses.

Filtered Assembly and the Adelic Comparison

Theorem 55.3.3 *(Filtered assembly of supplied factorization coalgebras).* Fix a configuration diagram and a cocomplete symmetric monoidal coefficient ∞-category $\mathcal{E}$. Assume its tensor product preserves filtered colimits in each variable, as do all restriction and collision functors used in the diagram. Let B_S, indexed by finite sets of places containing a fixed finite set, be a filtered diagram of factorization coalgebras on this configuration diagram. Assume its transitions preserve the coproducts, counits and any specified vacuum markings. Then

$$B = \varinjlim_{S} B_S$$

is a factorization coalgebra with those markings. If every factorization coproduct of every B_S is an equivalence, the corresponding coproduct of B is an equivalence.

Proof. For a fixed decomposition into r disjoint blocks, take the colimit of the r-fold coproducts. Filteredness makes the diagonal indexing functor cofinal in its r-fold product: every finite family of finite place sets is contained in their union, and the poset of further upper bounds is filtered and nonempty. Consequently preservation of filtered colimits gives

$$\varinjlim_{S} \bigotimes_{i=1}^{r} B_S(U_i) \simeq \bigotimes_{i=1}^{r} \varinjlim_{S} B_S(U_i).$$

This constructs the coproduct of B. Restriction and collision compatibilities pass to the colimit by their stated preservation hypotheses. Every coassociativity diagram and all its homotopies are images of the compatible diagrams at finite S, so they pass to the colimit as well. Counits and vacuum markings do the same. Finally a colimit of equivalences is an equivalence, proving the last assertion. □

For example, supplied local models may give $B_S = \bigotimes_{v \in S} B_v$, with transitions inserting specified vacuum coalgebra maps at new places. Their filtered assembly is a restricted coefficient product. Constructing the local position models, their common coefficient comparison and these coalgebra maps is part of the arithmetic problem. A pointing of the underlying module alone does not meet the hypotheses. The category of finite place sets is the discrete index category of Definition 30.1.2; it does not replace the configuration diagram in the theorem.

Question 55.3.4 *(Adelic differential comparison).* Construct an arithmetic adelic coefficient complex D_{ad} and a comparison with the specified restricted bar product, including its local residue maps. Prove that its adelic differential is a coderivation for the geometric coproduct:

$$\Delta d_{\mathrm{ad}} = (d_{\mathrm{ad}} \otimes 1 + 1 \otimes d_{\mathrm{ad}})\Delta.$$

The tensor differential here includes its cohomological Koszul sign. The comparison must specify the coefficient ring, the restricted product and its topology, the archimedean term, and the transition maps across the chosen local position diagrams.

Beilinson's adelic resolution for schemes provides the motivating geometric construction. Applying it to an arithmetic object with an archimedean term requires the complex and comparison in this question; a formal tensor of local fields does not define its adelic differential.

Corollary 55.3.5 *(Conditional arithmetic assembly).* An arithmetic bar system whose local and transition maps satisfy Theorem 55.3.3 has a compatible factorization coalgebra after filtered assembly. Compatibility with an independently defined adelic differential additionally requires the comparison of Question 55.3.4.

55.4 Iwasawa operators and tame residues

To combine a Galois action with residues, one first needs a square of maps between specified groups. On a number field this square is particularly concrete. If v is a finite place, π a uniformizer and u a unit at v, the tame residue sends $\{u, \pi\}$ to $\bar{u}$. A field automorphism carries this calculation to the corresponding calculation at the image of v. This exact compatibility, together with the tensor signs, produces an ordinary arithmetic bar complex.

Fix an odd prime ℓ. Let $K_\infty/\mathbb{Q}$ be the cyclotomic $\mathbb{Z}_\ell$-extension, K_r its subfield of degree ℓ^r, and $\Gamma = \mathrm{Gal}(K_\infty/\mathbb{Q})$. Choose a topological generator γ, and put $\Lambda = \mathbb{Z}_\ell[[T]]$, $T = \gamma - 1$. Here K_r is the fixed field of the subgroup of order $\ell - 1$ in $\mathbb{Q}(\mu_{\ell^{r+1}})$. Thus Γ acts on each K_r through a cyclic group of order ℓ^r. The prime ℓ specifies coefficients and the tower; finite places of K_r will be denoted by v.

The residue square

Write Milnor groups additively, including the multiplicative group $K_1^M(k) = k^\times$, and define

$$G_r^0 = K_2^M(K_r), \qquad G_r^1 = \bigoplus_{v \text{ finite}} k(v)^\times, \qquad G_r = [G_r^0 \xrightarrow{\partial} G_r^1].$$

The two groups have cohomological degrees 0, 1. In multiplicative notation our residue convention is

$$\partial_v\{a, b\} = (-1)^{v(a)v(b)}\overline{a^{v(b)}/b^{v(a)}}.$$

The expression inside the residue bar is a unit. It is bilinear and annihilates $\{a, 1 - a\}$: if one valuation is positive the other factor reduces to 1; if both are zero the result is 1; if $v(a) < 0$, then $v(1 - a) = v(a)$ and $(1 - a)/a$ reduces to -1, cancelling the displayed sign. A symbol has nontrivial residues at only finitely many places. Consequently ∂ is a well-defined homomorphism into the direct sum. The square of the differential of G_r is zero because there is no degree-two term. No sum of elements of different residue fields is required.

For $\sigma \in \Gamma$, the action on symbols is $\sigma\{a, b\} = \{\sigma a, \sigma b\}$. On degree one it carries the v-summand to the σv-summand by the induced field isomorphism. Since $(\sigma v)(\sigma a) = v(a)$,

$$\partial_{\sigma v}\{\sigma a, \sigma b\} = \sigma\partial_v\{a, b\}, \qquad \partial(\gamma - 1) = (\gamma - 1)\partial.$$

This is an equality on the groups themselves.

For a finite extension of number fields L/K, there are chain maps $N : G_L \to G_K$ and $R : G_K \to G_L$. In degree zero they are transfer and restriction on K_2^M. In degree one they are

$$(Nz)_v = \sum_{w|v} N_{k(w)/k(v)} z_w, \qquad (Rz)_w = e(w/v)\,\mathrm{res}_{k(w)/k(v)}\, z_v.$$

Thus

$$\partial_v N_{L/K}\xi = \sum_{w|v} N_{k(w)/k(v)}\partial_w\xi, \qquad \partial_w \mathrm{res}\,\xi = e(w/v)\,\mathrm{res}\,\partial_v\xi. \tag{55.2}$$

One construction of the first map identifies Milnor K_2 with Quillen K_2 by Matsumoto's theorem and takes restriction of scalars on finite-dimensional vector spaces. The localization boundary commutes with this exact functor. On the residue terms, restriction of scalars is the field norm. This gives the first identity, including at ramified primes; see Weibel, *The K-book*, III, Theorem 6.1 and V, (6.6.3)–(6.6.4). Our convention is the inverse of the tame symbol in III, Lemma 6.3; inverting every residue leaves the commutative square unchanged. The second identity also follows directly from $w(a) = e(w/v)v(a)$: its two signs agree because $e^2 - e$ is even. Transfers compose, and commute with the Γ-actions in the tower, by composition and conjugation of restriction of scalars. These identities descend to $G_{r,s} = G_r/\ell^s G_r$, term by term. This quotient is an ordinary quotient complex, not a claim about derived reduction of motivic cohomology.

A signed total complex

Let $P = \mathbb{Z}[x_1, x_2, \dots]$, augmented by $x_n \mapsto 0$, with x_n of weight n and cohomological degree zero. We use the trivial Γ-action on this factor as an explicit choice. Set $\bar{P} = \ker(P \to \mathbb{Z})$. Its reduced associative bar complex has

$$B^{-q} = \bar{P}^{\otimes q}, \quad B^0 = \mathbb{Z}, \qquad b[f_1|\cdots|f_q] = \sum_{a=1}^{q-1} (-1)^{a-1}[f_1|\cdots|f_a f_{a+1}|\cdots|f_q].$$

Associativity cancels the two orders of every pair of mergers, so $b^2 = 0$. This is an associative bar differential: it merges adjacent entries and vanishes on a word of length one.

For $i = 0, 1$, take a copy of $G_{r,s}^j \otimes_{\mathbb{Z}} B^{-q}$, denoting its elements by $e_i \otimes g \otimes w$. Its total degree is $i + j - q$. Define

$$I(e_0 \otimes g \otimes w) = e_1 \otimes Tg \otimes w, \qquad I(e_1 \otimes g \otimes w) = 0,$$
$$J(e_i \otimes g \otimes w) = (-1)^i e_i \otimes \partial g \otimes w, \qquad L(e_i \otimes g \otimes w) = (-1)^{i+j} e_i \otimes g \otimes bw.$$

All three maps have total degree one. Let $C_{r,s} = \mathrm{Tot}(G_{r,s} \otimes B \otimes [e_0, e_1])$ with these copies ordered as in the displayed formulas.

Proposition 55.4.1 *(Three separate mixed-square identities).* The operator $D = I + J + L$ on $C_{r,s}$ satisfies

$$I^2 = J^2 = L^2 = 0, \qquad IJ + JI = IL + LI = JL + LJ = 0.$$

Norms, ramification-weighted restrictions and coefficient reductions are chain maps for D.

Proof. On the $i = 0$ copy, $(IJ + JI)(e_0 \otimes g \otimes w)$ is $e_1 \otimes (T\partial - \partial T)g \otimes w = 0$; on the other copy it is zero by definition. For $IL + LI$, the sign of L changes when I increases i, whereas T and b act on independent factors. For $JL + LJ$, the sign of L changes when J increases j, and ∂ and b again act on independent factors. These are three separate cancellations. The diagonal squares follow respectively from the two copies, the two-term residue complex, and associativity. Equation (55.2) proves the assertion about arithmetic maps; their bar component is the identity. □

The two copies have a precise Iwasawa meaning. The exact sequence

$$0 \longrightarrow \Lambda \xrightarrow{T} \Lambda \longrightarrow \mathbb{Z}_\ell \longrightarrow 0$$

is a free resolution of the augmentation module: a power series with zero constant term is uniquely divisible by T. Applying $\mathrm{Hom}_\Lambda(-, M)$ gives $[M \xrightarrow{T} M]$. Hence our two-copy construction computes the derived augmentation invariants over Λ of the specified coefficient complex. It does not replace T by multiplication by a real logarithm. The module $G_{r,s}$ has a well-defined Λ-action: from $(1+T)^{\ell^r} = 1$ one gets $T^{\ell^r} \in \ell\,\mathrm{End}(G_{r,s})$, and therefore $T^{s\ell^r} = 0$. Formal power series act by finite sums on every fixed (r,s)-coordinate.

The ordinary bar comparison and its completion

The negative Heisenberg modes span the abelian Lie algebra $V = \bigoplus_{n\geq 1} \mathbb{Z}x_n$. Its Chevalley–Eilenberg chains with trivial coefficients are $E^{-q} = \bigwedge^q V$, with zero differential and weight $\mathrm{wt}(x_n) = n$. There is an integral map

$$A(x_{n_1} \wedge \cdots \wedge x_{n_q}) = \sum_{\sigma\in S_q} \mathrm{sgn}(\sigma)[x_{n_{\sigma(1)}}|\cdots|x_{n_{\sigma(q)}}].$$

No factorial is inverted. Adjacent transpositions pair all terms of bA, so $bA = 0$. Deconcatenation on B corresponds under A to the exterior coproduct in which the x_n are primitive. For example,

$$A(x_1 \wedge x_2) = [x_1|x_2] - [x_2|x_1], \qquad b[x_1|x_1] = [x_1^2].$$

Thus the repeated entry is not an additional degree-minus-two class, including in characteristic two.

Theorem 55.4.2 *(Completed ordinary arithmetic comparison).* Let $C^E_{r,s}$ be the preceding total complex with B replaced by E. At each weight, $1 \otimes A$ is a chain homotopy equivalence $C^E_{r,s} \to C_{r,s}$, compatible with norms and coefficient reductions. These equivalences extend to

$$\widehat{C}^E = \prod_{W\geq 0} \varprojlim_{r,s} (C^E_{r,s})_W \xrightarrow{1\otimes A} \widehat{C} = \prod_{W\geq 0} \varprojlim_{r,s} (C_{r,s})_W.$$

The differentials on both completed complexes square to zero. The transition maps are norms as r decreases and coefficient reductions as s decreases. The limits have the inverse-limit topology of discrete modules, and the product has the product topology.

Proof. At weight W, only $x_1, \ldots, x_W$ and word lengths $q \leq W$ occur. Both E_W and B_W are bounded complexes of finite free abelian groups. For the polynomial ring in these W variables, the Koszul resolution of its augmentation has underlying module $P \otimes \bigwedge V$ and differential

$$d(f \otimes x_{n_1} \wedge \cdots \wedge x_{n_q}) = \sum_{a=1}^{q} (-1)^{a-1} f x_{n_a} \otimes x_{n_1} \wedge \cdots \widehat{x_{n_a}} \cdots \wedge x_{n_q}.$$

It resolves $\mathbb{Z}$: tensoring the one-variable resolutions $0 \to \mathbb{Z}[x] \xrightarrow{x} \mathbb{Z}[x] \to \mathbb{Z} \to 0$ successively proves exactness. The free bar resolution has the same augmentation, and antisymmetrizing the variables defines a map from the Koszul resolution to it. Its internal merger terms cancel; its first multiplication terms give precisely the Koszul formula. This map lifts the identity on the augmentation, hence is a homotopy equivalence of free resolutions. After tensoring with the augmentation module it becomes antisymmetrization into the usual reduced bar convention. Multiplication in length q by $(-1)^q$ changes that convention to the displayed b. Consequently A, differing from this map only by the invertible sign on E^{-q}, induces an isomorphism on homology.

The cone of A_W is bounded, acyclic and free over $\mathbb{Z}$. Its boundary groups are free subgroups of free abelian groups, so the short exact sequences between boundaries and chain groups split. Choose these splittings once at each weight. They give a chain contraction of the cone, and hence a homotopy inverse to A_W. These choices involve only the bar and exterior factors. They remain valid after tensoring with any $G_{r,s}^{j}$, without a flatness assumption on that group. A bar homotopy h extends to the total complex with the sign $(-1)^{i+j}$; its mixed terms with I and J cancel just as above. The same choices commute with every norm and reduction, so their identities pass to the inverse limits and then to the product over weights. All formulas are coordinatewise continuous. In particular this argument does not exchange cohomology with an inverse limit. □

The total complex is a differential graded comodule over the bar coalgebra: the coaction applies deconcatenation to its B-factor. For the signed tensor of maps, the differential on the second bar factor receives the usual Koszul sign from the preceding factors. The preceding comparison intertwines this coaction with the exterior coproduct through A. At completion the target of the coaction is the product over the two nonnegative output weights; each fixed input weight has only finitely many splittings. No multiplication or coalgebra diagonal on G_r is asserted.

Small residues and the remaining chiral comparison

Over $\mathbb{Q}$, the symbol $\{2, 3\}$ has residue 2 at 3, residue 1 at 2, and residue 1 elsewhere. The symbol $\{3, 3\}$ has residue -1 at 3, so it is nonzero. The identity $\{a, a\} = \{a, -1\}$ does hold: for $a \neq 1$, applying Steinberg to a^{-1} and expanding $1 - a^{-1} = (a - 1)/a$ gives $\{a, a\} = \{a, a - 1\} = \{a, -1\}$. But $\{-1, -1\} \neq 0$ in $K_2^M(\mathbb{Q})$. Indeed the map

$$\{a, b\} \longmapsto \epsilon(a)\epsilon(b) \in \mathbb{F}_2, \qquad \epsilon(a) = \begin{cases} 1 & a < 0, \\ 0 & a > 0, \end{cases}$$

is bilinear and kills Steinberg relations, since a and $1 - a$ cannot both be negative. Its value on $\{-1, -1\}$ is 1. The relation $\{2, -1\} = 0$ has value zero and imposes no such vanishing on $\{-1, -1\}$.

At $\ell = 3$, $r = 0$, $s = 1$, take $g = \{3, 7\}$ modulo 3. Its only nontrivial residue is $\bar{3} \in \mathbb{F}_7^{\times}$, which remains nonzero modulo cubes. Since $K_0 = \mathbb{Q}$, $T\,g = 0$. On the length-two word,

$$D(e_0 \otimes g \otimes [x_1|x_1]) = e_0 \otimes \partial g \otimes [x_1|x_1] + e_0 \otimes g \otimes [x_1^2].$$

Applying D again gives the two terms $-e_0 \otimes \partial g \otimes [x_1^2]$ and $+e_0 \otimes \partial g \otimes [x_1^2]$, which cancel. This calculation takes place on actual nonzero residue classes.

A nontrivial Galois square occurs already for $\ell = 3$, $r = 1$. Write $K_1 = \mathbb{Q}(\theta)$, where $\theta = \zeta_9 + \zeta_9^{-1}$ satisfies $\theta^3 - 3\theta + 1 = 0$. Choose γ induced by $\zeta_9 \mapsto \zeta_9^4$; then $\gamma\theta = 2 - \theta - \theta^2$. Modulo 19, the polynomial has the three distinct roots 9, 7, 3, giving three residue fields $\mathbb{F}_{19}$. For $g = \{\theta, 19\}$, the residues at these places are $(9, 7, 3)$, whereas those of $\gamma\, g$ are $(7, 3, 9)$. The element 2 generates $\mathbb{F}_{19}^{\times}$. Writing residues additively by their powers of 2, and reducing modulo cubes, these vectors are $(2, 0, 1)$ and $(0, 1, 2)$ in $\mathbb{F}_3^3$. Thus

$$\partial(T\,g) = (1, 1, 1) = T(\partial g) \neq 0.$$

The two paths are equal and nonzero; it is the totalization sign that cancels them. At the same prime, the residue part of the norm to $\mathbb{Q}$ is multiplication of the three components: $9 \cdot 7 \cdot 3 = -1$ in $\mathbb{F}_{19}$, consistent with $N_{K_1/\mathbb{Q}}\theta = -1$ and the projection formula $N\{\theta, 19\} = \{-1, 19\}$.

The full Heisenberg Lie algebra is different from V: its positive and negative modes have bracket $[a_m, a_{-n}] = m\delta_{mn}K$. That central term enters its Chevalley–Eilenberg differential. Neither this full Lie complex nor a chiral bar complex is identified with $B(P)$ by the ordinary comparison above. In particular $\sum_{k+l=n} a_{-k} \wedge a_{-l} = 0$ for the ordered positive pairs: off-diagonal terms cancel and diagonal exterior products are zero. This sum cannot supply a nonzero splitting differential on negative modes.

Remark 55.4.3 *(The arithmetic chiral differential).* The construction of a tame-weighted local chiral differential on $B^{\mathrm{ch,Ar}}(\mathcal{V}^{\mathrm{prim}})$ remains a further problem. It requires a coefficient complex with the specified local chiral operations, residue maps between their carriers, and compatible Γ-actions and norm maps. A tame residue is a homomorphism $K_2^M(K_r) \to k(v)^\times$, not a scalar coefficient. Simply tensoring it with a bar merger, which already has degree one, has degree two in the total grading used here. An additional operation or shift must be specified before it can be a term of the arithmetic differential. Archimedean data also require a separate coefficient construction; they are not finite residue fields in G_r.

Remark 55.4.4 *(Twisting and filtered comparison).* For a specified degree-one operator D_0 and a proposed correction τ, the necessary chain identity is

$$(D_0 + \tau)^2 = D_0^2 + D_0\tau + \tau D_0 + \tau^2 = 0.$$

If $D_0 = I + J + L$, all three commutators with τ occur. A relation involving only J does not imply this equation. Knowing that an obstruction is a boundary supplies neither a chosen primitive with these identities nor its compatibility with norms and completion. A filtered quasi-isomorphism further requires an actual chain map and control of the filtration. For $\widehat{C}^E \to \widehat{C}$ the compatible contractions above supply that control. For the tame-weighted chiral complex, the twisting operator and the comparison with this ordinary arithmetic model remain to be constructed.

55.5 C3: Renormalization / Euler-Product Convergence

Finite-Prime Gibbs States and the Inductive Algebra

Definition 55.5.1 *(The prime Toeplitz tensor algebra).* For each prime let $K_p = \ell^2(\mathbb{N}_0)$, let V_p be the unilateral shift, and put $\mathcal{T}_p = C^*(V_p)$. Define the self-adjoint number operator by $N_p e_j = j e_j$ on $D(N_p) = \{x : \sum_{j\geq 0} j^2|x_j|^2 < \infty\}$. For finite S, use the spatial tensor product $\mathcal{T}_S = \bigotimes_{p\in S} \mathcal{T}_p$ on $K_S = \bigotimes_{p\in S} K_p$, with unital maps $a \mapsto a \otimes 1$ for $S \subset S'$. Define $\mathcal{T}_{\mathrm{loc}} = \varinjlim_S \mathcal{T}_S$, the norm-completed inductive C*-algebra. Its dynamics is $\alpha_t(V_p) = p^{it}V_p$. This is an operator-algebraic carrier; no pronilpotent arithmetic or BV tower is defined by these unital embeddings alone.

Proposition 55.5.2 *(Compatible finite-prime Gibbs states).* Fix real $\beta > 0$. For finite S, set

$$H_S = \sum_{p\in S} (\log p) N_p, \qquad \rho_{S,\beta} = \bigotimes_{p\in S} (1 - p^{-\beta}) p^{-\beta N_p}.$$

This is a positive trace-class density of trace one. The states $\varphi_{S,\beta}(a) = \mathrm{Tr}(\rho_{S,\beta} a)$ are compatible under the unital embeddings and define a β-KMS state φ_β on $\mathcal{T}_{\mathrm{loc}}$, with convention $\varphi_\beta(ab) = \varphi_\beta(b\alpha_{i\beta}(a))$. On the global occupation carrier $\ell^2(\mathbb{N}^\times)$ with $He_n = (\log n)e_n$, the operator $e^{-\beta H}$ is trace class exactly when $\beta > 1$.

Proof. The geometric series gives $\mathrm{Tr}(p^{-\beta N_p}) = (1 - p^{-\beta})^{-1}$, so the finite tensor density is positive with trace one. Tracing over an added factor gives one, proving compatibility. Consequently φ_β is well

defined on the algebraic union, positive and of norm one; it extends uniquely and positively to its norm completion.

Finite tensor words in V_p, V_p^* are entire analytic for α. On each finite carrier, $\rho_{S,\beta} a = \alpha_{i\beta}(a)\rho_{S,\beta}$ for these words. Trace cyclicity gives the displayed KMS identity. The finite analytic word algebra is norm dense in the inductive algebra, so the analytic KMS criterion gives the KMS state there.

Finally, the global diagonal trace is $\mathrm{Tr}(e^{-\beta H}) = \sum_{n\geq 1} n^{-\beta}$. The integral test proves convergence exactly for $\beta > 1$. In that regime unique factorization and absolute convergence give $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$. For $0 < \beta \leq 1$ the diagonal operator has infinite trace, although the compatible state on $\mathcal{T}_{\mathrm{loc}}$ still exists. □

Question 55.5.3 *(The arithmetic and BV state comparison).* Construct the full Bost–Connes algebra map and dynamics comparison whose state restriction is φ_β, including at $\beta = 1$. Construct the arithmetic bar carrier, transition morphisms, completions and coherent BV operations that compare its finite-prime state data with this operator-algebraic construction. The comparison must preserve the specified observables, state and pairing, not merely the scalar normalization of the unit. If a pro-nilpotent limit is used, its filtration, transition maps and convergence must be proved on that arithmetic carrier.

Remark 55.5.4 *(The BV comparison requires chain-level maps).* An effective BV or factorization-algebra construction requires its own coefficient category, completed operations and compatible transition maps. The positive densities above are operators and define functionals on C*-algebras. Their traces do not define a BV differential, a bar-coalgebra morphism or a pro-nilpotent filtration. The requested comparison must supply these structures explicitly.

55.6 Integral oscillators and the real Fock representation

An oscillator can be defined over the integers before choosing a place. Its reduction modulo a prime and its real realization are then separate changes of coefficients. Reduction followed by tensoring with the reals cannot give the real realization. Indeed, if $pM = 0$, then

$$m \otimes r = (pm) \otimes (r/p) = 0 \qquad \text{in } M \otimes_{\mathbb{Z}} \mathbb{R}.$$

Thus $\mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{R} = 0$. There is also no unital ring map $\mathbb{F}_p \to \mathbb{R}$, since it would send $p1 = 0$ to $p1 \neq 0$. A comparison must retain characteristic-zero coefficient data before taking the two fibres.

A common integral representation

Write

$$P_{\mathbb{Z}} = \mathbb{Z}[x_1, x_2, \ldots], \qquad \mathrm{wt}(x_n) = n, \qquad |x_n| = 0.$$

The weight is distinct from cohomological degree. Elements are finite polynomials, the vacuum is 1, and the augmentation sends every x_n to zero. For $n \geq 1$, define operators

$$a_{-n} f = x_n f, \qquad a_n f = n \frac{\partial f}{\partial x_n}, \qquad a_0 = 0, \qquad Kf = f.$$

Let $\mathfrak{h}_{\mathbb{Z}}$ be the free abelian group on K and a_m, $m \in \mathbb{Z} \setminus \{0\}$, with bracket

$$[a_m, a_n] = m\delta_{m+n,0} K, \qquad [K, a_m] = 0.$$

All brackets involving a_0 are zero if a zero-mode generator is adjoined. The displayed operators give its level-one, zero-momentum representation. For example,

$$a_1 x_1 = 1, \qquad a_1 x_1^2 = 2x_1, \qquad a_2 x_2 = 2, \qquad a_2 x_1^2 = 0.$$

The derivative product rule proves the commutator, including its sign. Jacobi holds because every bracket is central.

Definition 55.6.1 *(integral coefficient comparison).* The integral oscillator comparison consists of $P_{\mathbb{Z}}$ with its vacuum, multiplication and operators a_m, K, together with their separate scalar extensions. The real realization is $P_{\mathbb{Z}} \otimes \mathbb{R}$. A residual fibre is $P_{\mathbb{Z}} \otimes \mathbb{F}_p$.

Proposition 55.6.2 *(separate coefficient fibres).* For every commutative ring R, the formulas above define $P_R = P_{\mathbb{Z}} \otimes_{\mathbb{Z}} R$ and an action of $\mathfrak{h}_R = \mathfrak{h}_{\mathbb{Z}} \otimes R$. Every ring map $R \to S$ gives a map $P_R \to P_S$ preserving the vacuum, ordinary polynomial multiplication, the weight decomposition and all oscillator operators after extension of coefficients. In particular, the integral object gives the diagram

$$P_{\mathbb{Z}_p} \longleftarrow P_{\mathbb{Z}} \longrightarrow P_{\mathbb{R}}, \qquad P_{\mathbb{Z}_p} \longrightarrow P_{\mathbb{F}_p}.$$

No arrow from the residual fibre to the real fibre is asserted.

Proof. Polynomial monomials form a free integral basis. All operator matrices in that basis have integer entries. Applying a ring map to their coefficients commutes with each displayed formula. The identity $[a_n, a_{-n}] = nK$ remains valid when n becomes zero in R. For example, $a_p = p\partial_{x_p}$ is zero on $P_{\mathbb{F}_p}$. This degeneration is part of the residual representation. □

This construction uses the monomial Fock lattice of Chapter 28. Its map to the named local oscillator module sends

$$x_{n_1} \cdots x_{n_r} \longmapsto a_{-n_1}^{(p)} \cdots a_{-n_r}^{(p)} |0\rangle_p.$$

It is an isomorphism when that local module is the polynomial vacuum representation with these monomials as basis and the displayed annihilation action. Additional arithmetic operations on a local module must be specified separately. The logarithmic energy $n \log p$ is not an integral structure constant. If the oscillator lattice has trivial Γ-action, its $\Lambda = \mathbb{Z}_p[[\Gamma]]$-module structure is the action through the augmentation $\Lambda \to \mathbb{Z}_p$. This specifies the coefficient action, not an Iwasawa differential.

Remark 55.6.3 *(the adelic coefficient space).* The adele ring $\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p' \mathbb{Q}_p$ already includes the real place as a factor. A duality on this additive group does not supply the missing homomorphism from a residue field to $\mathbb{R}$. The common integral representation supplies separate fibres without asserting such a homomorphism.

There are two different completions. The weight completion is $\prod_{d \geq 0} P_{\mathbb{Z},d}$; each weight summand has finite rank. The p-adic completion is

$$\widehat{P}_p = \varprojlim_r P_{\mathbb{Z}} / p^r P_{\mathbb{Z}}.$$

Its monomial coefficients lie in $\mathbb{Z}_p$, and modulo every p^r only finitely many coefficients are nonzero. The oscillator operators preserve this condition and are continuous. This restricted coefficient condition differs from an arbitrary product over weights. All comparisons below first use polynomial direct sums. A statement about either completion must specify which topology and which completed tensor product it uses.

The Hall lattice and coefficient operations

The larger integral symmetric-function lattice does not coincide with $P_{\mathbb{Z}}$. Inside $P_{\mathbb{Q}}$ define h_r by

$$\sum_{r\ge 0} h_r u^r = \exp\!\left(\sum_{n\ge 1} \frac{x_n u^n}{n}\right), \qquad h_0 = 1, \qquad L_{\mathbb{Z}} = \mathbb{Z}[h_1, h_2, \dots].$$

Differentiating this generating series gives the Newton identities

$$r h_r = \sum_{n=1}^{r} x_n h_{r-n}.$$

They imply $P_{\mathbb{Z}} \subset L_{\mathbb{Z}}$ and $P_{\mathbb{Q}} = L_{\mathbb{Z}} \otimes \mathbb{Q}$. They also give

$$h_1 = x_1, \qquad h_2 = \frac{x_1^2 + x_2}{2}, \qquad h_3 = \frac{x_1^3 + 3x_1x_2 + 2x_3}{6}.$$

The annihilation operator satisfies $a_n h_r = h_{r-n}$, with $h_j = 0$ for $j < 0$. Thus both integral lattices are stable under the oscillator operators. Their inclusion is strict already in weight two. This is the lattice distinction in Section 28.4; the Newton formulas are Macdonald, *Symmetric Functions and Hall Polynomials*, second edition, I.2, equations (2.10)–(2.14).

For a partition λ having m_n parts equal to n, put $x_\lambda = \prod_n x_n^{m_n}$ and $z_\lambda = \prod_n n^{m_n} m_n!$. The Hall form on $P_{\mathbb{Q}}$ is

$$\langle x_\lambda, x_\mu \rangle = \delta_{\lambda\mu} z_\lambda.$$

The form is integral on $L_{\mathbb{Z}}$, and its Schur basis is orthonormal; these are Macdonald I.4, equations (4.5)–(4.9). On the monomial lattice, the formula directly proves $\langle a_{-n} f, g\rangle = \langle f, a_n g\rangle$. In weight two, x_1^2, x_2 both have squared norm 2, whereas h_2 has squared norm 1. Replacing one lattice by the other therefore changes the integral fibre, although their rational and real representations agree.

The polynomial operations $\psi_r(x_n) = x_{rn}$ preserve the vacuum and satisfy $\psi_r \psi_s = \psi_{rs}$. They commute with coefficient change. They do not intertwine each fixed oscillator operator:

$$a_{rn}\psi_r = r\psi_r a_n, \qquad \psi_r a_{-n} = a_{-rn}\psi_r.$$

If r does not divide k, then $a_k \psi_r = 0$. These identities follow by differentiating substituted polynomials. Let T be the derivation $T(x_n) = n x_{n+1}$. For $r > 1$, this substitution fails to commute with translation: $T\psi_r(x_1) = r x_{r+1}$, while $\psi_r T(x_1) = x_{2r}$. Thus ψ_r is not an endomorphism of the vertex structure constructed below. The formula for T_r in (28.1) defines another integral polynomial substitution,

$$T_r(x_k) = \sum_{d \mid r,\ d \mid k} d\, x_{rk/d}.$$

It too commutes with coefficient change, but is not the same monoid action: $T_2^2(x_1) = x_4 + 2x_2$, while $T_4(x_1) = x_4$. The two operations must remain distinct in any comparison retaining them.

The real Fock space and its operator domains

Over $\mathbb{R}$, the Hall form is positive definite. Complete $P_{\mathbb{R}}$ for this form. The vectors $e_\lambda = x_\lambda/\sqrt{z_\lambda}$ give an orthonormal basis of the resulting real Hilbert space $\mathcal{F}_{\mathbb{R}}$. For $f = \sum_\lambda c_\lambda e_\lambda$, the closed oscillator operators have domains

$$\mathrm{Dom}(a_n) = \left\{ f : \sum_\lambda n m_n(\lambda) |c_\lambda|^2 < \infty \right\},$$

$$\mathrm{Dom}(a_{-n}) = \left\{ f : \sum_\lambda n (m_n(\lambda)+1) |c_\lambda|^2 < \infty \right\}.$$

Their values on basis vectors have coefficients $\sqrt{n m_n}$ and $\sqrt{n(m_n+1)}$, respectively. These weighted shifts are adjoints of one another. Truncating the orthonormal expansion proves that polynomial states form a core. The commutation relations hold on this common invariant polynomial domain. They are not equalities of everywhere-defined bounded operators.

The weight operator and the energy at a fixed prime are

$$L_0 e_\lambda = |\lambda| e_\lambda, \qquad H_p e_\lambda = (\log p)|\lambda| e_\lambda, \qquad \mathrm{Dom}(L_0) = \left\{ f : \sum_\lambda |\lambda|^2 |c_\lambda|^2 < \infty \right\}.$$

The same domain defines H_p. Both are self-adjoint diagonal operators. This realizes the local weight $n \log p$ by multiplying one fixed real weight operator by $\log p$. It is not a limit as p increases.

A finite set of modes also has a Gaussian realization. For $1 \le n \le N$, use real coordinates s_n with product measure $(2\pi)^{-N/2} \exp(-\sum_n s_n^2/2)\, ds_1 \cdots ds_N$. Define the Hermite polynomials by

$$\exp(ts - t^2/2) = \sum_{k \ge 0} \mathrm{He}_k(s) \frac{t^k}{k!}.$$

Gaussian integration of the product of two generating functions gives $\exp(tu)$, hence $\langle \mathrm{He}_k, \mathrm{He}_l \rangle = k! \delta_{kl}$. Therefore

$$U_N \left(\prod_{n=1}^{N} x_n^{m_n} \right) = \prod_{n=1}^{N} n^{m_n/2} \mathrm{He}_{m_n}(s_n)$$

is an isometry onto the polynomial span, and extends to its Hilbert closure. Differentiating the generating function gives

$$U_N a_n U_N^{-1} = \sqrt{n}\, \partial_{s_n}, \qquad U_N a_{-n} U_N^{-1} = \sqrt{n}\, (s_n - \partial_{s_n})$$

on that polynomial domain. These are adjoints by Gaussian integration by parts. This comparison preserves the oscillator action, but transports polynomial multiplication to a different product: $U_N(x_1^2) = s_1^2 - 1$, while $U_N(x_1)^2 = s_1^2$. Define the transported product by $F \diamond G = U_N(U_N^{-1} F\, U_N^{-1} G)$. Then $s_1 \diamond s_1 = s_1^2 - 1$. The multiplication to be preserved must therefore be specified even between two real realizations of the same oscillator representation.

Formal currents and a vertex-algebra comparison

The oscillator representation also supplies local current operations. Use an independent formal coordinate z and put

$$J(z) = \sum_{m \in \mathbb{Z}} a_m z^{-m-1}, \qquad T(x_n) = n x_{n+1}, \qquad T(1) = 0,$$

extending T as a derivation of polynomial multiplication. Then $[T, a_m] = -m a_{m-1}$ and $[T, J(z)] = \partial_z J(z)$. On every polynomial state all sufficiently large positive modes vanish. The commutator relation gives

$$J(z)J(w) \sim \frac{1}{(z-w)^2}, \qquad (z-w)^2[J(z), J(w)] = 0,$$

where the singular term in the first formula is expanded for the first variable outside the second. The second formula compares the two formal expansions. Its central coefficient is 1.

For $n_i \geq 1$, define the field of a monomial state by

$$Y(x_{n_1} \cdots x_{n_r}, z) =: \prod_{i=1}^{r} \partial_z^{(n_i-1)} J(z) :, \qquad \partial_z^{(k)} z^m = \binom{m}{k} z^{m-k}.$$

Normal ordering places all creation operators to the left of annihilation operators. The vacuum field is the identity. For any fixed input state and any fixed coefficient of z, the expression is a finite sum. Every coefficient is an integral operator on $P_{\mathbb{Z}}$. The divided derivatives have integer binomial coefficients, including when m is negative.

Over $\mathbb{Q}$ these are the Heisenberg vertex-algebra fields. Frenkel, *Vertex Algebras and Algebraic Curves*, Bourbaki seminar 875, §2.3, equations (2)–(4) and Proposition 2.4, constructs these fields from the oscillator representation and translation operator. Its reconstruction theorem applies because the vacuum is cyclic, $T1 = 0$, the displayed translation identity holds, and J is local with itself. Each coefficient of the vacuum, translation and Jacobi identities is an integral identity on the torsion-free module $P_{\mathbb{Z}}$. Its validity over $\mathbb{Q}$ therefore proves it over $\mathbb{Z}$. Consequently coefficient change preserves all these vertex operations, not only the commutative polynomial product. Translation over an arbitrary ring uses its integral divided operators, determined by

$$T^{(k)} x_n = \binom{n+k-1}{k} x_{n+k}, \qquad T^{(k)}(fg) = \sum_{i+j=k} T^{(i)} f \, T^{(j)} g.$$

Here $T^{(0)} = \mathrm{id}$ and $T^{(k)}1 = 0$ for $k > 0$. Over $\mathbb{Q}$, these formulas give $T^{(k)} = T^k/k!$ and hence $T^{(i)}T^{(j)} = \binom{i+j}{i} T^{(i+j)}$. Torsion-freeness gives the same identity over $\mathbb{Z}$.

Corollary 55.6.4 *(the real current fibre).* The real fibre and its complexification are the ordinary level-one Heisenberg vertex algebra on the formal line. The coefficient comparison preserves every formal mode product and the divided translation operators.

Proof. The vacuum polynomial representation and its state fields are the explicit Heisenberg construction above. All formulas are defined before extension to $\mathbb{R}$ or $\mathbb{C}$. □

This construction supplies no map from an arithmetic Gersten complex or an Iwasawa differential. Nor does it identify the commutative finite-place product in (F5) with the singular current product: the latter is additional structure.

Three different complexes

The ordinary augmented polynomial algebra, its oscillator representation, and the Heisenberg Lie algebra give different bar constructions. Their coefficient comparisons can be made explicit.

For a commutative ring R, the reduced associative bar of P_R is

$$B_r(P_R) = \overline{P}_R^{\otimes_R r}, \qquad |B_r| = -r, \qquad b[f_1|\cdots|f_r] = \sum_{i=1}^{r-1} (-1)^{i-1} [f_1|\cdots|f_i f_{i+1}|\cdots|f_r].$$

Here $\overline{P}_R$ is the augmentation ideal, $B_0 = R$, and the coproduct cuts a word in all possible positions. Associativity gives $b^2 = 0$. Free monomial bases give exact chain identifications

$$B(P_{\mathbb{Z}}) \otimes R \cong B(P_R)$$

preserving differential, coproduct and counit. For example, $b[x_1|x_2] = [x_1x_2]$, and $[x_1|x_2] - [x_2|x_1]$ is a cycle. The Koszul complex on the polynomial variables computes its homology:

$$H^{-r}B(P_R) \cong \Lambda_R^r\left(\bigoplus_{n\geq 1} Rx_n\right).$$

To justify this over every R, tensor the two-term resolutions for a finite list of polynomial variables. Each variable is a non-zero-divisor modulo its predecessors, and the resulting free complex resolves R. Every calculation uses finitely many variables, so the union gives the stated result. Antisymmetrizing a wedge gives the corresponding bar cycle, with no factorial denominator. The positive modes do not preserve the augmentation ideal: $a_n x_n = n$. They therefore do not define the same entrywise action on this reduced bar.

To retain all oscillator operators, use instead the unnormalized two-sided bar resolution

$$\mathcal{B}_R^{-r} = P_R^{\otimes_R(r+2)}, \qquad \partial = \sum_{i=0}^{r}(-1)^i \mu_i \quad (r \geq 1), \qquad \epsilon : \mathcal{B}_R^0 \to P_R,$$

where μ_i multiplies factors i and $i+1$, and ϵ is multiplication. On a tensor let a_{-n} multiply its first factor by x_n, and let a_n be $n\partial_{x_n}$ summed over all factors. These operators commute with ∂. For annihilation this is the derivative product rule; creation commutes with the first multiplication and with every later face. Their commutator is n times the identity, because only differentiation of the first factor meets creation. The augmentation intertwines these actions. Unit insertion at the left contracts the augmented complex over R: the first face is the identity and all remaining faces cancel in pairs. Thus ϵ is a quasi-isomorphism of oscillator representations. The contracting homotopy need not intertwine creation operators. All these maps commute with coefficient change, using algebraic tensors. For two variables, the first boundary is explicitly

$$\partial(1 \otimes x_1 \otimes x_2) = x_1 \otimes x_2 - 1 \otimes x_1x_2.$$

This resolution supplies a nontrivial bar model carrying the full oscillator action. It is not the reduced bar coalgebra above.

For the Lie algebra $\mathfrak{h}_R$, the Chevalley–Eilenberg chains with trivial coefficients have $C_r = \Lambda_R^r\mathfrak{h}_R$, in degree $-r$. Choose the convention $d(u \wedge v) = [u, v]$. In particular,

$$d(a_n \wedge a_{-n}) = nK.$$

This differential uses the Lie bracket, whereas the associative bar uses multiplication. Antisymmetrization into the augmented bar of $U_R(\mathfrak{h}_R)$ is a chain map; on two inputs this follows from $uv - vu = [u, v]$. The level-one quotient $K = 1$ has no augmentation to R when R is nonzero: its relation $[a_1, a_{-1}] = 1$ would map to $0 = 1$. It therefore cannot replace this augmented enveloping algebra.

With Fock coefficients the full Lie complex has another decisive property. Write its chains as $\Lambda^r\mathfrak{h}_R \otimes_R P_R$, and take

$$\begin{aligned} d(u_1 \wedge \cdots \wedge u_r \otimes f) = &\sum_i (-1)^{i+1} u_1 \wedge \cdots \widehat{u}_i \cdots \wedge u_r \otimes u_i f \\ &+ \sum_{i<j} (-1)^{i+j+1}[u_i, u_j] \wedge u_1 \wedge \cdots \widehat{u}_i \cdots \widehat{u}_j \cdots \wedge u_r \otimes f. \end{aligned}$$

The module relation and Jacobi give $d^2 = 0$. Since K is central and acts as 1, the map $h(\omega \otimes f) = K \wedge \omega \otimes f$ satisfies $dh + hd = 1$. This Chevalley–Eilenberg complex is contractible. Already $d(K \otimes 1) = 1$. It cannot be identified with $B(P_R)$, whose degree-zero homology is R, or with the two-sided resolution, whose degree-zero homology is P_R.

Combining primes and preserving the central term

For a finite set of primes S, form $P_{S,R} = R[x_{p,n} : p \in S, n \geq 1]$. Different prime labels give independent oscillator families:

$$[a_{p,m}, a_{q,n}] = \delta_{pq}\, m \delta_{m+n,0} 1.$$

Adding a prime means tensoring with its vacuum polynomial representation. These inclusions define an algebraic restricted tensor product over all primes. Identifying every $a_{p,m}$ with the same a_m cannot preserve the relations: for $p \neq q$ the commutator on the left is zero, while $[a_1, a_{-1}] = 1$ in the proposed target.

There are nevertheless explicit level-one subrepresentations. Choose real numbers c_p with $\sum_{p \in S} c_p^2 = 1$, and set

$$X_n = \sum_{p \in S} c_p x_{p,n}, \qquad A_{\pm n} = \sum_{p \in S} c_p a_{p,\pm n}.$$

Then $[A_n, A_{-m}] = n\delta_{nm}$, and the polynomial map $x_n \mapsto X_n$ is injective and intertwines all oscillators. Indeed, $A_n X_m = n\delta_{nm}$, and the product rule gives the identity on every polynomial. Orthogonality of the prime families makes this map an isometry for the Hall form. In one mode,

$$\|X_n^m\|^2 = n^m m! \left(\sum_{p \in S} c_p^2 \right)^m = n^m m!,$$

by the multinomial formula; distinct modes are orthogonal. For two primes one can choose $c_p = 3/5, c_q = 4/5$; the construction then exists already over $\mathbb{Z}[1/5]$. The substitution $x_{p,n} \mapsto c_p x_n$ is a left inverse, since $\sum_p c_p^2 = 1$; this also proves injectivity over that coefficient ring. This left inverse is used only as a polynomial-algebra map for injectivity. The forward embedding $x_n \mapsto X_n$ preserves the formal current operations and the associative bar maps constructed above, since it preserves the vacuum, translation and modes. The orthogonal oscillator combinations remain in the larger representation.

For countably many primes, a square-summable sequence $c = (c_p)$ of norm one gives the same construction after real Hilbert completion. The one-particle space has orthonormal basis indexed by (p, n). On its symmetric Fock space, creation and annihilation with coefficient c are bounded between each pair of fixed particle-number sectors, with bounds $\sqrt{n(r+1)}\|c\|$ and $\sqrt{nr}\|c\|$. Thus finite prime sums converge on the domain of vectors with finitely many particle-number components. Their commutators converge there to the stated level-one relations. The induced single-species Fock space is a closed isometric subspace. This requires the choice of c. The Hilbert extension here concerns the oscillator representation and its isometric embedding. Formal vertex fields remain on the polynomial state space, or on a separately specified domain satisfying lower truncation. Finite particle number alone does not imply lower truncation: a one-particle Hilbert vector with nonzero components in arbitrarily large modes can have $a_n v \neq 0$ for arbitrarily large n. No vertex-algebra structure on the entire Hilbert completion is asserted. The unweighted sequence $c_p = 1$ is not square-summable and does not define a vacuum creation vector: its squared norm in mode n is $n \sum_p 1 = \infty$.

Let $e_{\mathbf{m}}$ be the normalized monomial basis with finite occupation numbers $m_{p,n}$. Define the logarithmic Hamiltonian by

$$E(\mathbf{m}) = \sum_{p,n} n \log p \, m_{p,n}, \qquad He_{\mathbf{m}} = E(\mathbf{m}) e_{\mathbf{m}},$$

$$\mathrm{Dom}(H) = \left\{ \sum_{\mathbf{m}} d_{\mathbf{m}} e_{\mathbf{m}} : \sum_{\mathbf{m}} E(\mathbf{m})^2 |d_{\mathbf{m}}|^2 < \infty \right\}.$$

It is the self-adjoint diagonal operator on this domain. It does not in general preserve the collective subspace. On its one-particle vectors,

$$HX_n|0\rangle = n \sum_p c_p \log p \, x_{p,n}|0\rangle.$$

This is defined when $\sum_p |c_p|^2 (\log p)^2 < \infty$. It equals $\lambda n X_n|0\rangle$ only when $(\log p - \lambda) c_p = 0$ for every p. A collective current involving two distinct primes therefore cannot intertwine the prime-energy operator with a single rescaled L_0. The coefficient comparison and the spectral comparison are different problems.

The arithmetic comparison still required

Theorem 55.6.5 *(integral oscillator comparison and its boundary).* The common integral polynomial representation has separate finite-place and real fibres preserving the oscillator operators, vacuum, central term, ordinary multiplication and the integral formal vertex operations. Its reduced associative bars commute with coefficient change. Its unnormalized two-sided bar resolution compares to the Fock module by an oscillator-equivariant quasi-isomorphism. A normalized finite collective current gives an isometric embedding into the representation with several prime labels. A square-summable choice extends that embedding to the real Hilbert completions on the specified domains.

Proof. The coefficient maps and operator identities follow from the integral polynomial formulas. The current fields have integral coefficients and satisfy the vertex identities before specialization. The bar comparison and its contraction are explicit above. The collective-current calculation preserves the central term by the equation $\sum c_p^2 = 1$. The operator bounds give convergence on each finite particle sector. □

Question 55.6.6 *(arithmetic and archimedean comparison).* Construct a common coefficient complex whose finite-place realizations are the specified arithmetic bar objects and whose real realization maps to a specified Heisenberg bar model. The comparison must identify the differential, central normalization and pairing. In particular, construct a square-zero arithmetic differential and maps comparing its Iwasawa and Gersten components with specified real operators. For a chiral or factorization comparison, specify the configuration spaces, restriction and collision maps, and their compatibility with these local current operations. To compare comodule and contramodule categories, specify the coefficient, completeness and duality hypotheses and construct the corresponding functors and pairing maps.

The integral oscillator and bar constructions above supply local coefficient data for this problem. They do not supply the common arithmetic complex or its Iwasawa–Gersten comparison. The residual-to-real tensor is zero, the Lie complex with Fock coefficients is contractible, and the logarithmic Hamiltonian need not preserve a collective current. These are separate constraints on the requested comparison.

55.7 C5: Functoriality under Λ-Coalgebra Morphisms

The Functoriality Statement

Definition 55.7.1 *(morphisms of abelian Λ-coalgebras with $\mu = 0$).* A morphism $f : C \to C'$ in $\mathrm{Coalg}^{\Lambda,\mu=0,\mathrm{ab}}$ is a Λ-linear morphism of dg coalgebras preserving the abelian Γ-action (trivial Γ-action on both sides) and the $\mu = 0$ structure (i.e., pseudo-isomorphism at level of Λ-modules).

Induced Pullback and Pushforward

Proposition 55.7.2 *(induced pullback on co-derived category).* Every morphism $f : C \to C'$ in $\mathrm{Coalg}^{\Lambda,\mu=0,\mathrm{ab}}$ induces a triangulated functor

$$f^* : \mathrm{D}^{\mathrm{co}}_{\Lambda}(C') \to \mathrm{D}^{\mathrm{co}}_{\Lambda}(C)$$

given by restriction of scalars: for $M \in C'\text{−comod}_\Lambda$, the C-comodule f^*M has underlying Λ-module M and C-coaction $M \xrightarrow{\Delta_{C'}} M \otimes_\Lambda C' \xrightarrow{\mathrm{id}\otimes f^{\mathrm{op}}} M \otimes_\Lambda C$ via the Λ-linear structure of f (Positselski 2011 arXiv:0905.2621 §6.3; for the Λ-coefficient extension, Lurie *Higher Algebra* 4.3.3).

Proof. For f a split injection, the splitting $g : C' \to C$ (as Λ-module, not as coalgebra) induces the restriction directly. For general f, the restriction is the derived left Kan extension along f: $f^* = \mathrm{LKan}_f$ on the abelian category of Λ-comodules.

Triangulated structure: f^* preserves co-fibrations and injective resolutions because the C-injective hull of a C-comodule is a Λ-module injective hull (by the abelian hypothesis), which is the same as the C'-module injective hull of the underlying C'-comodule (modulo the different coalgebra action, which does not affect the Λ-injective hull).

Compatibility with $\mu = 0$: for $M \in \mathrm{D}^{\mathrm{co}}_{\Lambda}(C')$ with $\mu(M) = 0$, f^*M has the same underlying Λ-module, hence $\mu(f^*M) = 0$. The Verdier quotient by acyclic injectives is preserved by f^* because Λ-injective resolution is preserved. □

Proposition 55.7.3 *(induced pushforward on contra-derived category).* The dual morphism $f^\vee : (C')^\vee \to C^\vee$ is a Λ-algebra morphism, and induces a triangulated functor

$$(f^\vee)_* : \mathrm{D}^{\mathrm{ctr}}_{\Lambda}((C')^\vee) \to \mathrm{D}^{\mathrm{ctr}}_{\Lambda}(C^\vee)$$

given by pushforward of contramodules along the Λ-algebra morphism $f^\vee$: for $N \in (C')^\vee\text{−contramod}_\Lambda$, the $C^\vee$-contramodule $(f^\vee)_*N$ has underlying Λ-module N and $C^\vee$-contramodule structure via $f^\vee$.

Proof. The dualization functor $(-)^\vee$ is contravariant on Λ-coalgebras (sends morphisms in one direction to morphisms in the other direction). Since $f : C \to C'$, $f^\vee : (C')^\vee \to C^\vee$ goes in the opposite direction. It is a Λ-algebra morphism because dualization sends coalgebra morphisms to algebra morphisms (Positselski 2011 §2). The pushforward $(f^\vee)_*N$ is the natural restriction of N from a $(C')^\vee$-contramodule to a $C^\vee$-contramodule via $f^\vee$ (standard: restrict the contra-action along a Λ-algebra morphism).

Triangulated structure and preservation of contra-derived classes follow by the dual argument to Chapter 55.7.2. □

Naturality of the Positselski Equivalence

Theorem 55.7.4 *(C5 closed: functoriality of arithmetic Positselski). (restricted to Λ-linear morphisms of abelian dg Λ-coalgebras with $\mu = 0$ and trivial Γ-action; see Chapter 55.7.1).* For every morphism $f : C \to C'$ in $\mathrm{Coalg}_\Lambda^{\Lambda,\mu=0,\mathrm{ab}}$, the diagram

$$\begin{array}{ccc} \mathrm{D}^{\mathrm{co}}_\Lambda(C) & \xrightarrow{\Psi_C} & \mathrm{D}^{\mathrm{ctr}}_\Lambda(C^\vee) \\ {\scriptstyle f^*}\uparrow & & \uparrow{\scriptstyle (f^\vee)_*} \\ \mathrm{D}^{\mathrm{co}}_\Lambda(C') & \xrightarrow{\Psi_{C'}} & \mathrm{D}^{\mathrm{ctr}}_\Lambda((C')^\vee) \end{array} \tag{55.3}$$

commutes up to natural isomorphism. Equivalently, the arithmetic Positselski equivalence Ψ is a natural transformation of 2-functors $\mathrm{Coalg}_\Lambda^{\Lambda,\mu=0,\mathrm{ab},\mathrm{op}} \to \mathrm{PrSt}^{\mathrm{tri}}$.

Proof. Unwind the definitions. For $M \in \mathrm{D}^{\mathrm{co}}_\Lambda(C')$,

$$\begin{aligned} \Psi_C(f^*M) &= \mathrm{Hom}_C(C, f^*M) \\ &\simeq \mathrm{Hom}_{C'}(C' \otimes_C C, M) && \text{(adjunction)} \\ &\simeq \mathrm{Hom}_{C'}(C', M) && \text{(since } C' \otimes_C C \simeq C' \text{ via } f) \\ &= \Psi_{C'}(M). \end{aligned}$$

On the other hand, the pushforward $(f^\vee)_* : \mathrm{D}^{\mathrm{ctr}}_\Lambda((C')^\vee) \to \mathrm{D}^{\mathrm{ctr}}_\Lambda(C^\vee)$ restricts the contra-action along $f^\vee$, so

$$\begin{aligned} (f^\vee)_* \Psi_{C'}(M) &= (f^\vee)_* \mathrm{Hom}_{C'}(C', M) \\ &\simeq \mathrm{Hom}_C(C, f^*M), \end{aligned}$$

where the identification uses the $f^\vee$-restriction of the $C^\vee$-contramodule structure on $\mathrm{Hom}_{C'}(C', M)$, which converts the $(C')^\vee$-action to a $C^\vee$-action via $f^\vee$ and yields the comodule f^*M.

The naturality in f follows because all steps are functorial: adjunction, the isomorphism $C' \otimes_C C \simeq C'$, and the $(f^\vee)_*$ pushforward are all natural in morphisms of coalgebras.

The statement "Ψ is a natural transformation of 2-functors" is the abstract reformulation: viewing $\mathrm{Coalg}_\Lambda^{\Lambda,\mu=0,\mathrm{ab}} \to \mathrm{PrSt}^{\mathrm{tri}}$ as 2-functors (one via $\mathrm{D}^{\mathrm{co}}_\Lambda$ co-derived, the other via $\mathrm{D}^{\mathrm{ctr}}_\Lambda$ contra-derived composed with $(-)^\vee$), Ψ is the natural transformation between them, with naturality squares (55.3). □

Corollary 55.7.5 *(functorial pullback of Positselski pairing).* For every morphism $f : C \to C'$ in $\mathrm{Coalg}_\Lambda^{\Lambda,\mu=0,\mathrm{ab}}$, the Positselski pairing $\langle \cdot, \cdot \rangle_{C'} : C' \otimes (C')^\vee \to \Lambda$ pulls back along f to the Positselski pairing $\langle \cdot, \cdot \rangle_C : C \otimes C^\vee \to \Lambda$:

$$\langle f(x), y \rangle_{C'} \;=\; \langle x, f^\vee(y) \rangle_C$$

for all $x \in C$, $y \in (C')^\vee$.

Proof. The Positselski pairing is the natural pairing of a coalgebra with its dual. Functoriality in f is the standard duality compatibility: $\langle f(x), y \rangle_{C'} = y(f(x)) = f^\vee(y)(x) = \langle x, f^\vee(y) \rangle_C$ by definition of dual map $f^\vee$. □

55.8 Consequence for the Arithmetic Mellin Comparison

Theorem 55.8.1 *(conditional arithmetic complementarity).* . Suppose the proposed arithmetic objects satisfy the C1–C5 comparison conditions and the coefficient and duality conditions of Remark 46.1.6. Suppose in addition that they supply the data V, J, v, w, E of Theorem 43.5.1, with

$$v - Jv = \big(\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}}\big)w, \qquad E(w) \neq 0,$$

and with $E(v) = \Lambda_\xi$ and $E(Jv)(s) = \Lambda_\xi(1-s)$. Then

$$\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0 \quad \Longleftrightarrow \quad \xi(s) = \xi(1-s).$$

Proof. Apply E to the central-defect identity. Its right-hand side vanishes exactly when the sum of levels is zero, since $E(w) \neq 0$. Its left-hand side is $\Lambda_\xi(s) - \Lambda_\xi(1-s)$. Multiplication by $s(s-1)/2$ gives the equivalent identity for ξ. □

The extra evaluation hypothesis is not a consequence of naturality of a pairing. For a coalgebra map $f : C \to C'$, naturality gives $\langle f(x), y\rangle = \langle x, f^\vee(y)\rangle$. It does not identify either side with a central level or a completed zeta function. The opposite oscillator weights in Section 46.5 likewise yield a local cancellation without constructing the arithmetic invariant maps.

The constructions in this chapter have distinct outputs. Theorem 55.4.2 gives an ordinary completed arithmetic comparison. The chiral differential in 55.4.3 requires further operations.

Question 55.5.3 asks for the arithmetic and BV state comparison beyond the Toeplitz product state. Theorem 55.6.5 gives local integral oscillator and bar models, while Question 55.6.6 asks for common arithmetic complexes and pairing maps. These outputs retain their own coefficient rings. No field-to-$\mathbb{Z}_p[[\Gamma]]$ base change is implicit.

Corollary 55.8.2 *(arithmetic level criterion).* Under all the hypotheses of Theorem 55.8.1, the scalar functional equation implies arithmetic level cancellation.

Proof. Use the reverse implication of that theorem. It is the detection condition $E(w) \neq 0$ that makes the implication valid. □

Remark 55.8.3 *(this corollary does not prove RH).* Chapter 55.8.1 and Chapter 55.8.2 identify the implication $\kappa + \kappa^! = 0 \Rightarrow \xi(s) = \xi(1-s)$ for the arithmetic Heisenberg under the stated hypotheses. They do not prove the Riemann hypothesis; the functional equation is a classical theorem of Riemann 1859, and RH is the separate assertion that the non-trivial zeros of ξ lie on the critical line. The conditional Mellin comparison relates level cancellation to the functional equation. A categorical equivalence alone does not.

The separate HP–Li implication of Chapter 44.3.10 uses the operator, determinant and trace identities of H-P^{Ar} (Chapter 44.3.2). It also assumes positivity of the regularized Li trace and the stated P2, P3, and positive-cone hypotheses. A self-adjoint Hilbert–Pólya realization with the specified determinant divisor already places the zeros on the critical line. The Li-trace implication gives a sufficient scalar criterion.

Its Li transforms of Θ_ξ are distinct from the Virasoro energy polynomials $p_n(L_0^{\mathrm{Vir}})$, where $p_n(x) = x^n - (x-1)^n$. Those polynomials fail positivity on the vacuum: $p_2(L_0^{\mathrm{Vir}})\Omega = -\Omega$. Chapter 51.2.10 constructs their maximal reducing positive spectral sector, which need not preserve the Virasoro action. A comparison from an arithmetic sector to the HP–Li trace remains Chapter 51.2.11; the Positselski reduction supplies neither that comparison nor the HP–Li hypotheses.

55.9 Residual Conditions

The Costello-refined Positselski reduction of C1–C5 addresses the abelian Heisenberg domain. Three residual items remain outside this domain:

(i) *Non-abelian Iwahori level.* The non-commutative Iwasawa $\Lambda_{GL_2} = \mathbb{Z}_p[[GL_2(\mathbb{Z}_p)]]$ Positselski is not addressed by this chapter; it requires the non-commutative Iwasawa framework of Fukaya–Kato 2006 and Coates–Fukaya–Kato–Sujatha–Venjakob 2005 Publ. IHES (the main conjecture for non-commutative Iwasawa theory). Residual form: conjectural (recorded as Chapter 46.7).

(ii) *Wildly-ramified primes $p = 2, 3$.* The Steinberg relation $\{p, 1 - p\} = 0$ holds for every prime, but does not construct a tame-weighted chiral operation. The ordinary residue complex in Section 55.4 includes all finite places, including those above 2 and 3, with an independently fixed odd coefficient prime ℓ. It does not construct the intended non-semisimple Iwahori category or its factorization maps. Any Ext^1-contributions in that category must be retained in a separate comparison; the ordinary bar calculation supplies no such comparison. Residual form: conjectural.

(iii) *Super Bethe-Hecke chain-level chiral coproduct.* The $\mu = 0$ hypothesis and the abelian domain exclude the super Bethe-Hecke sector where chain-level chiral coproduct computations are required; this sector is outside the abelian Heisenberg domain. Residual form: conjectural.

55.10 Refinement Boundary

(i) Five Costello-refined subconditions C1–C5 are recorded under their named hypotheses.
C1 (local-global factorization-coalgebra coherence): Theorem 55.3.3 constructs the filtered assembly from specified local coalgebras and compatible transition maps. Their arithmetic geometric realization and the adelic differential comparison remain the inputs of Question 55.3.4.
C2 (arithmetic differential and comparison): the exact residue squares and signed totalization give the ordinary completed model of Chapter 55.4.2. The tame-weighted local chiral operation and its comparison with that model remain Chapter 55.4.3.
C3 (renormalization / Euler-product convergence): recorded by Question 55.5.3: the compatible Toeplitz product state is constructed, while its arithmetic/BV comparison remains to be constructed.
C4 (integral and archimedean comparison): Chapter 55.6.5 constructs the common integral oscillator model, separate coefficient fibres, formal currents and bar comparisons. The arithmetic Iwasawa–Gersten and pairing maps remain the problem of Question 55.6.6.
C5 (functoriality): recorded by Chapter 55.7.4 via naturality of bar–cobar adjunction in Λ-linear coalgebra morphisms with parallel functor directions in the naturality square (pullback on comodules and pushforward on the dual side).

(ii) The P1 level criterion is conditional on the actual arithmetic objects, central-defect identity, Mellin evaluation, and detection condition of Theorem 55.8.1. C1–C5 and the coefficient comparison retain their own hypotheses. No categorical equivalence alone supplies that numerical criterion.

(iii) Three residual items remain outside the Costello-refined reduction: non-abelian Iwahori level, wildly-ramified primes $p = 2, 3$, super Bethe-Hecke chain-level chiral coproduct. Each is stated explicitly with its residual form in Chapter 55.9.

(iv) The relevant mathematical theories for Chapter 55.1 are: Costello–Gwilliam Vol. II for C1 and C3; Matsumoto's theorem and the localization and transfer identities in Weibel's *K-book* for the ordinary arithmetic model in C2; Macdonald's integral symmetric-function lattices and Frenkel's

Heisenberg vertex construction for the local models in C4; Positselski 2011 §6 plus Λ-linearity for C5. Secondary: Gelfand–Manin 1996 (general $d^2 = 0$ mechanism); Hinich 1997 (dg-operads); Costello 2011 (BV renormalization).

(v) The proposed arithmetic domain retains the C1–C5 and co–contra hypotheses. The implication $\kappa + \kappa^! = 0 \Rightarrow \xi(s) = \xi(1 - s)$ additionally requires the defect identity and evaluation of Theorem 43.5.1.
The HP–Li route requires the operator, determinant and trace data of H-P^{Ar}. Its regularized Li trace also requires the stated P2, P3 and positive-cone hypotheses. A route from Virasoro energy polynomials must specify a positive sector and construct its comparison with that trace. The vacuum obstruction in Chapter 51.2.10 excludes positivity on the entire Fock space. These arithmetic comparisons are additional to the co–contra criterion.

The arithmetic biconditional requires the central-defect identity and detecting Mellin evaluation of Theorem 43.5.1. The C1–C5 conditions and the co–contra requirements in Remark 46.1.6 specify additional structure on its intended arithmetic domain.

Chapter 56

Bernstein-Centre Descent and Local Pairing Normalization

The local-to-global problem requires coherent objects, compatible actions and effective descent. A separate local calculation compares weighted occupation vectors, diagonal Hilbert–Schmidt operators and radial functions. The spherical Whittaker norm is computed with its measure factor, and its unit normalization gives an explicit line isometry. These constructions isolate the actual maps still needed for arithmetic Koszul–Petersson compatibility.

56.1 Bernstein Gluing and Local Pairing Conditions

The BC-diamond-glue condition concerns local coherent Springer objects on the intended spectral carriers, their equivariant descent to a Selmer target, and identification of the descended central action with the global automorphic Bernstein centre. Coherent Springer theory on a Langlands-parameter stack does not itself construct the comparison to an Emerton–Gee stack.

For pairing compatibility, fix both carriers and the type of each form. Hermitian forms are linear in the first variable and conjugate-linear in the second; inserting an antilinear map into the second slot of a bilinear form produces this type. On the prime cyclic Gibbs carrier, $J\mu_{p^k}\Omega = p^{k/2}\mu^*_{p^k}\Omega$. Its arithmetic transport requires the action-compatible placement maps of Hypothesis 67.1.1.

56.2 Local Constructions and Comparison Data

Local Springer transport. The exact functor, support and action comparisons of Theorem 56.4.1 transport a constructed object to the intended local spectral carrier. The coefficient comparisons, including the $\ell = p$ case, must be constructed on their common carrier.

Effective equivariant descent. Theorem 56.5.4 applies to actual local objects, overlap maps and higher coherences in a specified effective descent model. Local flatness does not supply cross-prime transition maps or coherence of an arbitrary colimit.

Global centre comparison. Theorem 56.5.6 assembles supplied compatible actions over all finite sets of places. It also requires the separate identification with the actual global centre. An arithmetic AG equivalence alone does not identify its two actions.

Spherical norm. Lemma 56.7.4 gives $C_p = m_p(1 + p^{-1})L_p(1, \mathrm{Ad}\,\pi_p)$ for the specified unramified tempered representation. The unit-circle parameters make every local denominator at one nonzero. The norm is not the compact subgroup volume.

Local Hilbert maps. Theorem 56.7.5 constructs the occupation/diagonal/radial isometries and the normalized spherical line. Extending them to arithmetic pairing carriers with their Hecke actions is Conjecture 56.7.6.

Adelic normalization. Theorem 56.7.8 retains the continued value $c_\pi \neq 0$, actual placements and completed pairing maps. Common scalar rescaling preserves the supplied identity; it does not assert ordinary Euler-product convergence at one.

56.3 BC-Diamond-Glue: Decomposition

Definition 56.3.1 *(the three sub-operations of (BC-diamond-glue)).* (BC-diamond-glue) decomposes into three sub-operations:

(BCglue.local) *Local Ben-Zvi–Chen–Helm–Nadler construction.* At each finite prime p, construct a coherent Springer sheaf $\mathcal{F}_p \in \mathrm{Coh}(\mathcal{X}^{\mathrm{EG},p}_{GL_2})$ whose action on the local $\mathrm{IndCoh}_{\mathrm{Nilp}}$-category realises the local Bernstein centre $\mathcal{Z}_p$.

(BCglue.glue) *Cross-prime coefficient descent.* Construct a common coefficient carrier and actual transition correspondences for the local sheaves, with the support, filtration, and higher-coherence data of Hypothesis 56.5.2. Supply the effective coherent-descent model on the Selmer target required by Theorem 56.5.4. The global sheaf is obtained from that descent equivalence; a formal product across distinct primes or a filtered colimit of coherent sheaves does not supply it.

(BCglue.core) *Global Bernstein-centre compatibility.* The action of $\mathcal{F}^{\mathrm{prism}}_{\mathrm{glob}}$ on $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}})$ agrees, under the conjectural arithmetic AG equivalence of Theorem 38.4.2, with the global Bernstein-centre action on $\mathrm{D}^{\text{sph-fin}}([GL_2])$.

THEOREM 56.3.2 *(decomposition of (BC-diamond-glue)).* (BC-diamond-glue) is equivalent to the conjunction

$$\text{(BCglue.local)} \ \wedge \ \text{(BCglue.glue)} \ \wedge \ \text{(BCglue.core)}.$$

The core condition concerns the action on the object obtained from the first two constructions; no logical independence is asserted.

Proof. *Forward.* Suppose (BC-diamond-glue), i.e. Conjecture 48.5.4, holds. Then the local coherent Springer sheaves $\{\mathcal{F}_v\}_v$ exist (supplying (BCglue.local)), glue to $\mathcal{F}^{\mathrm{prism}}_{\mathrm{glob}}$ along the diagonal (supplying (BCglue.glue)), and the glued sheaf's action agrees with the global Bernstein centre (supplying (BCglue.core)).

Reverse. Suppose (BCglue.local), (BCglue.glue), (BCglue.core) all hold. Then (BCglue.local) gives the local sheaves; (BCglue.glue) assembles them into a global coherent sheaf; (BCglue.core) gives the Bernstein-centre compatibility. Together, these are exactly the content of Conjecture 48.5.4.

The equivalence separates the required data without constructing them. □

56.4 BC-Diamond-Glue: Local Construction (Ben-Zvi–Chen–Helm–Nadler)

THEOREM 56.4.1 *(Transport of a local coherent Springer object).* Fix a common coefficient category and a coherent Springer object S_p on a specified parameter carrier $\mathcal{Y}_p$. Supply an exact functor

$$T_p : \mathrm{Coh}(\mathcal{Y}_p) \longrightarrow \mathrm{Coh}(\mathcal{X}^{\mathrm{EG}}_p)$$

preserving the required singular support, together with compatible natural equivalences identifying its transported central action with the action on the stated local representation category. Then $\mathcal{F}_p = T_p(S_p)$ is a coherent object with that central action.

Proof. Preservation of coherence and support gives the object on the stated carrier. Apply T_p to the action morphisms. Functoriality preserves their unit and composition identities, and the supplied natural equivalences identify the resulting action with the required one. □

Remark 56.4.2 *(Local coefficient and spectral comparisons).* Ben-Zvi–Chen–Helm–Nadler construct the coherent Springer input on its Langlands-parameter carrier. Construct the functor T_p and the compatible representation-theoretic action comparison above. If several coefficient realizations are used, construct a common comparison carrier, the functors to it, and compatible isomorphisms between the resulting objects. In particular, the $\ell = p$ and cross-coefficient comparisons must be supplied; agreement of parameters or traces alone does not produce these isomorphisms.

56.5 BC-Diamond-Glue: Bernstein Centre Gluing

Prismatic Transition Maps

Definition 56.5.1 *(Nygaard-convention-aware transition map).* For distinct primes p, q and pairs $(R_p, \varphi_p, \Gamma_p)$, $(R_q, \varphi_q, \Gamma_q)$ of Emerton–Gee data (here R_p is the period ring, φ_p the Frobenius, $\Gamma_p = \mathrm{Gal}(\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p)$ the cyclotomic Galois group, as in Emerton–Gee arXiv:1908.07185 §2), the *Nygaard-convention-aware transition map* is a proposed comparison morphism

$$\mathcal{X}_{GL_2}^{\mathrm{EG},p} \times \mathcal{X}_{GL_2}^{\mathrm{EG},q} \;\to\; \mathcal{X}_{GL_2}^{\mathrm{EG},\{p,q\}}$$

where the target denotes a sought common moduli carrier for the rank-2 local data at p and q. Its construction and coefficient comparison are part of Hypothesis 56.5.2; a product of formal bases at distinct primes does not define it. The Nygaard filtration convention of Bhatt–Lurie arXiv:2201.06120 §8 is used to specify the expected comparison data. For $p \neq q$ this is a residual comparison correspondence, not a canonical fibre-product identity of formal bases.

Hypothesis 56.5.2 *(Cross-prime transition and descent data).* For the proposed comparison of Definition 56.5.1, supply a common coefficient category and actual transition correspondences. Assume their pullback functors preserve coherent objects and the specified nilpotent singular support, commute with the required colimits, and carry compatible Nygaard filtrations. Assume the comparison isomorphisms on double overlaps satisfy the triple cocycle and all higher coherences. These are cross-prime data, not consequences of separate theories over $\mathbb{Z}_p$ and $\mathbb{Z}_q$.

Remark 56.5.3 *(Flat base change and the missing arithmetic maps).* For a flat homomorphism of Noetherian rings $A \to B$, tensoring with B is exact and preserves finite presentation: a finite presentation $A^r \to A^s \to M \to 0$ becomes $B^r \to B^s \to B \otimes_A M \to 0$. Thus it sends coherent modules to coherent modules. This elementary calculation explains one required property once a common base and a flat map are given. It constructs neither a cross-prime map nor a singular-support comparison. A filtered colimit of coherent objects also need not be coherent: the increasing finite-dimensional subspaces of $\bigoplus_{n\geq 1} k$ are a counterexample over a field.

Effective Descent of the Local Sheaves

Theorem 56.5.4 *(Conditional coefficient descent).* Assume Hypothesis 56.5.2. Let $X = \mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}$. Supply an effective descent diagram $U_\bullet \to X$ and an explicit equivalence

$$\mathrm{Coh}(X) \xrightarrow{\sim} \mathrm{Tot}\big(\mathrm{Coh}(U_\bullet)\big)$$

on the prescribed singular-support and coefficient subcategories. If the local coherent Springer sheaves and comparison isomorphisms form an object of the displayed totalization, they determine a coherent sheaf $\mathcal{F}_{\mathrm{glob}}^{\mathrm{prism}}$ on X, unique up to the equivalence of descent objects. Any archimedean factor requires its own compatible complex/Hodge comparison data.

Proof. The sheaves, overlap isomorphisms, and higher coherences define the stated descent object. Apply the inverse of the supplied descent equivalence. The resulting sheaf pulls back to each prescribed local object, and full faithfulness gives uniqueness and compatibility of its morphisms. Coherence follows from the target of that equivalence, not from an arbitrary filtered colimit in an ind-category. □

Question 56.5.5 *(Arithmetic cross-prime descent).* Construct the common coefficient carrier, transition correspondences, and effective descent equivalence in the preceding theorem for the specified Selmer problem, preserving its full Weil/(φ, Γ) data and Nygaard filtrations. Local prismatic constructions alone do not supply the first cross-prime morphisms. This is the remaining content of (BCglue.glue).

Reduction to core compatibility

THEOREM 56.5.6 *(Compatible finite-stage central actions).* Let $F : C \simeq \mathcal{D}$ be an equivalence of stable categories. Let (A_S) be a compatible diagram of commutative algebras indexed by all finite sets of places, with $A = \varinjlim_S A_S$. Suppose compatible actions

$$a_S : A_S \to Z^{\circ}(C), \qquad b_S : A_S \to Z^{\circ}(\mathcal{D})$$

are given, where $Z^{\circ}(C) = \mathrm{End}(\mathrm{Id}_{hC})$ is the ordinary centre of the homotopy category, consisting of ordinary natural transformations; similarly for $\mathcal{D}$. If $F(a_S(z)_M) = b_S(z)_{F(M)}$ for every S, z, M, then F intertwines the induced A-actions.

Proof. The compatible algebra maps factor uniquely through A. Every element of A comes from a finite stage, where the intertwining identity holds. Thus it holds for the induced actions. □

Remark 56.5.7 *(Coherent actions and the arithmetic centre).* For higher actions, the corresponding input is a compatible system of maps in the chosen category of derived algebras, with a coherent homotopy between the two systems. The colimit identity

$$\mathrm{Map}(\operatorname*{colim}_S A_S, Z) \simeq \lim_S \mathrm{Map}(A_S, Z)$$

assembles that supplied homotopy. Equalities in degree zero do not supply it. The index consists of all finite sets; the poset of pairs alone is not filtered.

For BCglue.core, one must further identify A with the actual global automorphic Bernstein centre, and b_S with the actions of the equivariantly descended coherent object. An equivalence of underlying categories does not do this. For example, on $C = \mathcal{D} = D(k)$, the identity equivalence fails to intertwine the two $k[t]$-actions $t \mapsto 0$ and $t \mapsto 1$: at the object k the required equation would be $0 = \mathrm{id}_k$.

The resulting implication uses the complete data:

$$\begin{array}{c}\text{local comparison and effective equivariant descent,}\\ \text{arithmetic AG and compatible global centre comparisons}\end{array} \quad\Longrightarrow\quad \text{BC-diamond-glue.}$$

The original cross-prime and global-centre constructions remain required, with their coherent equivariant descent data.

56.6 Consequence: L_{ACD} Final Residual Isolation

Corollary 56.6.1 *(Descent data for the categorical comparison).* Under the prior F1.a and F2.c comparison data, the BC-diamond-glue input to L_{ACD} requires all three constructions: local coherent transport, effective equivariant descent, and the compatible identification with the global automorphic centre. If these data are supplied, they give BC-diamond-glue.

Proof. The local theorem constructs the transported objects from its specified functors. The effective descent theorem assembles their coherent overlap data. The centre comparison identifies the resulting action, with finite-stage compatibility assembled by Theorem 56.5.6. These are the three parts of Definition 56.3.1. □

Remark 56.6.2 *(What remains after local transport).* A transported local object does not specify cross-prime overlap maps, and an effectively descended object does not identify its action with the global Bernstein centre. The original categorical construction requires both comparisons with their higher coherences.

56.7 K5: Spherical Whittaker Normalization

The Local Adjoint L-Factor

Definition 56.7.1 *(local adjoint L-factor $L_p(s, \mathrm{Ad})$).* For an unramified principal-series representation $\pi_p = \mathrm{Ind}_B^{GL_2}(\chi_1 \otimes \chi_2)$ at $GL_2(\mathbb{Q}_p)$ with Satake parameters $(\alpha_p, \beta_p) = (\chi_1(p), \chi_2(p))$ on the unit circle, the local adjoint L-factor is

$$L_p(s, \mathrm{Ad}(\pi_p)) \;:=\; \left(1 - \alpha_p \beta_p^{-1} p^{-s}\right)^{-1} \cdot (1 - p^{-s})^{-1} \cdot \left(1 - \alpha_p^{-1} \beta_p \, p^{-s}\right)^{-1}.$$

Reference: Jacquet 1979 Corvallis "Principal L-functions" §2, or Bump 1997 *Automorphic Forms and Representations* §3.2.

Lemma 56.7.2 *(Finite adjoint factor for an unramified tempered representation).* Fix an unramified tempered representation π_p of $GL_2(\mathbb{Q}_p)$, with unitary-normalized Satake parameters α_p, β_p satisfying $|\alpha_p| = |\beta_p| = 1$. Then the factor $L_p(1, \mathrm{Ad}\, \pi_p)$ of Definition 56.7.1 is finite and nonzero. This is a statement about the specified representation; its placement in an arithmetic Hochschild realization is separate comparison data.

Proof. The three denominator terms are

$$1 - \alpha_p \beta_p^{-1} p^{-1}, \qquad 1 - p^{-1}, \qquad 1 - \alpha_p^{-1} \beta_p p^{-1}.$$

Each is nonzero because the subtracted term has absolute value $p^{-1} < 1$. Their inverse product is therefore finite and nonzero. No inequality between α_p and β_p is needed. □

The Absorption Convention

Definition 56.7.3 *(Koszul-dual normalisation convention).* The *Koszul-dual normalisation convention* is the rescaling of the local Koszul pairing at a finite prime p, assuming $L_p(1, \mathrm{Ad})$ is finite and nonzero, by the local adjoint L-factor:

$$\langle \cdot, \cdot \rangle_p^{\mathrm{Koszul,Ad}} \;:=\; L_p(1, \mathrm{Ad})^{-1} \cdot \langle \cdot, \cdot \rangle_p^{\mathrm{Koszul}}.$$

The global normalized pairing is defined using the specified continued nonzero value $c_\pi = L(1, \mathrm{Ad}\,\pi)$, as $c_\pi^{-1}\langle\cdot,\cdot\rangle^{\mathrm{Koszul}}$, with the actual adelic extension and comparison maps of Hypothesis 67.1.1. It is not defined by an unproved ordinary product at $s = 1$.

LEMMA 56.7.4 *(Spherical Whittaker norm with specified measure).* Let π_p be normalized unramified tempered principal series, with $|\alpha| = |\beta| = 1$, and take an additive character of conductor $\mathbb{Z}_p$. Let $W(1) = 1$ be its spherical Whittaker function. Set $m_p = \mathrm{vol}_{d^\times a}(\mathbb{Z}_p^\times) > 0$. Its standard spherical formula is

$$W(\mathrm{diag}(p^j, 1)) = p^{-j/2} h_j(\alpha, \beta) \quad (j \geq 0), \qquad h_j = \sum_{r=0}^{j} \alpha^{j-r}\beta^r,$$

and it vanishes at negative valuations. For $\mathrm{Re}\, s > 0$,

$$Z_p(s) := \int_{\mathbb{Q}_p^\times} |W(a)|^2 |a|^{s-1}\, d^\times a = m_p \frac{L_p(s, \pi_p \times \widetilde{\pi}_p)}{\zeta_p(2s)}.$$

In particular

$$C_p := Z_p(1) = m_p(1 + p^{-1}) L_p(1, \mathrm{Ad}\,\pi_p) > 0.$$

For $d^\times a = da/|a|$ with additive unit volume, $m_p = 1 - p^{-1}$, so $C_p = (1 - p^{-2}) L_p(1, \mathrm{Ad}\,\pi_p)$.

Proof. Shell integration gives $Z_p(s) = m_p \sum_{j\geq 0} |h_j|^2 p^{-js}$, which converges absolutely since $|h_j| \leq j + 1$. With $r = \alpha/\beta$, summing geometric series gives

$$\sum_{j\geq 0} |h_j|^2 x^j = \frac{1 + x}{(1 - x)(1 - rx)(1 - r^{-1}x)}.$$

For $r \neq 1$, this follows by writing $|h_j|^2 = (2 - r^{j+1} - r^{-j-1})/|1 - r|^2$ and summing each term. The case $r = 1$ follows by continuity, or from $\sum (j + 1)^2 x^j = (1 + x)/(1 - x)^3$. The rank-four Rankin–Selberg factor has denominator $(1 - x)^2(1 - rx)(1 - r^{-1}x)$. Multiplication by $1 - x^2$ proves the displayed integral. Since $L_p(s, \pi_p \times \widetilde{\pi}_p) = \zeta_p(s) L_p(s, \mathrm{Ad}\,\pi_p)$, evaluation at one gives C_p. This is a positive norm; finiteness also follows directly from the convergent shell series. The measure normalization is the Kirillov normalization used by Blomer–Jana–Nelson, *Local Integral Transforms and Global Spectral Decomposition*, §2, (2.7), and §3.1. □

THEOREM 56.7.5 *(Occupation, diagonal Hilbert–Schmidt and radial function models).* Let $H_{\mathrm{occ}} = \ell^2(\mathbb{N}_0, p^{-j})$, with coordinate vectors δ_j, and let $E_j = p^j \mathbb{Z}_p^\times$. There are explicit isometries

$$T\delta_j = p^{-j/2} m_p^{-1/2} \mathbf{1}_{E_j}, \qquad D\delta_j = p^{-j/2} E_{jj}$$

onto, respectively, the closed radial subspace supported on nonnegative valuations in $L^2(\mathbb{Q}_p^\times, d^\times a)$ and the diagonal subspace of $\mathrm{HS}(\ell^2\mathbb{N}_0)$. They intertwine coordinate conjugation with function conjugation and $JX = X^*$. Thus, for all $v, w \in H_{\mathrm{occ}}$,

$$H_{\mathrm{occ}}(v, w) = H_{\mathrm{Whit}}(Tv, Tw) = \mathrm{Tr}(Dv(Dw)^*).$$

The spherical function is $W = T(\sqrt{m_p} \sum_{j\geq 0} h_j \delta_j)$. The map $a\Omega \mapsto aW/\sqrt{C_p}$ is an isometry from the unit prime Gibbs-vacuum line to the spherical Whittaker line.

Proof. The shells are disjoint, each of measure m_p, so the Gram matrix of $T\delta_j$ is $\delta_{jk}p^{-j}$. The matrix units are Hilbert–Schmidt orthonormal, giving the same Gram matrix for $D\delta_j$. Their finite spans are dense in the stated ranges, proving the extensions and surjectivity onto those ranges. Both displayed bases are fixed by their conjugations. The spherical expansion follows shell by shell and converges since $\sum p^{-j}|h_j|^2 < \infty$. Its norm is C_p, proving the normalized-line isometry. Notice that $W \neq T\delta_0$ in general; its coefficients depend on the representation. □

Conjecture 56.7.6 *(Arithmetic local pairing comparison).* Construct the arithmetic placement and action-intertwining maps from the actual bar–cobar carrier to the specified local representation and Whittaker models, with domains, Haar measure and spherical normalization fixed. Require them to transport the arithmetic bilinear pairing and its conjugation to the corresponding Hermitian form, compatibly with Hecke actions and the adelic extension in Hypothesis 67.1.1. The preceding radial and diagonal isometries are explicit bounded models; they do not construct this full arithmetic comparison.

Remark 56.7.7 *(Pairing type and normalization).* The Whittaker integral and the occupation Hilbert form are linear in the first variable and conjugate-linear in the second. An antilinear map cannot be inserted in their second slot without changing that type. If $B_{\mathrm{Whit}}(x, y) = H_{\mathrm{Whit}}(x, Jy)$ is the corresponding bilinear form, then $H_{\mathrm{Whit}}(x, y) = B_{\mathrm{Whit}}(x, Jy)$. This is the type required in an arithmetic Koszul comparison.

Adjoint rescaling alone changes the spherical norm to $m_p(1 + p^{-1})$, not to one. Unit normalization uses the full positive factor C_p. This scalar is distinct from $\mathrm{vol}(GL_2(\mathbb{Z}_p)) = (1 - p^{-1})(1 - p^{-2})$ for the standard additive-matrix Haar convention. Ramified and non-tempered representations require their own integral and domain calculation. The finite nonzero continued global value in Theorem 56.7.8 is a separate analytic input, not a proof of local denominator finiteness.

Adelic Consistency

Theorem 56.7.8 *(Conditional adelic normalization on the cuspidal sph-fin locus).* Fix a cuspidal representation π and the local comparison datum of Conjecture 56.7.6. Assume the placement and adelic pairing maps of Hypothesis 67.1.1 on the specified sph-fin core. Suppose the adjoint Euler product, initially defined in a half-plane of absolute convergence, has a specified holomorphic continuation to $s = 1$ with $c_\pi = L(1, \mathrm{Ad}\,\pi) \neq 0$. Then

$$\langle \cdot, \cdot \rangle^{\mathrm{Koszul,Ad}} := c_\pi^{-1} \langle \cdot, \cdot \rangle^{\mathrm{Koszul}}$$

is a well-defined rescaled pairing on that core. If the Petersson pairing is rescaled by the same scalar, the assumed pairing comparison is preserved. The value c_π is a continued value; no absolute convergence of $\prod_p L_p(1, \mathrm{Ad}\,\pi)$ is asserted.

Proof. Multiplication by the nonzero scalar c_π^{-1} defines the rescaled form. Multiplying the assumed identity $\langle v, w \rangle^{\mathrm{Pet}} = \langle v, \sigma^{\mathrm{BC}} w \rangle^{\mathrm{Koszul}}$ by that scalar gives the identity of reduced pairings on the same domain. It neither enlarges the domain nor constructs its placement.

The distinction between an Euler product and its continued value is already visible for tempered unramified local parameters. In that case the three adjoint parameters have absolute value one. For $\sigma = \mathrm{Re}(s) > 1$, the logarithmic series is bounded by

$$\sum_p \sum_{j \geq 1} \frac{3p^{-j\sigma}}{j} \leq \frac{3}{1 - 2^{-\sigma}} \sum_p p^{-\sigma} < \infty.$$

Finitely many ramified factors do not affect this estimate away from their poles. The bound establishes absolute convergence in this half-plane; it gives no convergence assertion at $s = 1$. For other local regimes one starts in a half-plane where the Euler product is known to converge. Analytic continuation and nonvanishing at one are separate analytic inputs, of the kind considered in Hoffstein–Lockhart 1994 and the Langlands–Shahidi theory; their applicability to the chosen representation is part of the hypothesis. □

Remark 56.7.9 *(Local and global normalization hypotheses).* Theorem 56.7.5 constructs specified local model isometries. Their arithmetic realization is the separate local comparison datum. The global assertion in Theorem 56.7.8 additionally requires the finite nonzero continued value c_π and the arithmetic placement and adelic pairing maps of Hypothesis 67.1.1. Neither the local datum nor the analytic continuation constructs those maps.

56.8 Consequence: K-P Compat, Conditional Domain

Corollary 56.8.1 *(K-P compat on the cuspidal sph-fin locus under the local comparison datum). (conditional on Conjecture 56.7.6, Hypothesis 67.1.1, the specified finite nonzero continued value $c_\pi = L(1, \mathrm{Ad}\,\pi)$, and domain-restricted to the cuspidal sph-fin locus).* The Koszul–Petersson compatibility (Theorem 47.9.1) holds on the cuspidal sph-fin locus of $\mathcal{V}^{\mathrm{prim}}$, with the Koszul pairing interpreted under the Koszul-dual normalisation convention of Definition 56.7.3 and the Petersson form rescaled by the same c_π^{-1}. The resulting equality is between these two rescaled forms. Full-sector extension further separates into the prime KMS_1/GNS cyclic-subspace comparison of Theorem 67.2.5 with the additional arithmetic maps of Hypothesis 67.1.1, and the ramified Rankin–Selberg and Eisenstein conditions of Remark 47.1.2.

Proof. The local comparison datum supplies the finite-place pairing match with its stated normalization. Hypothesis 67.1.1 additionally supplies the arithmetic placement, preserved pairings, archimedean comparison, and adelic extension on the specified domain. Theorem 47.9.1 therefore applies to these maps. Multiplying both forms by the same nonzero c_π^{-1} preserves their equality. The local normalization alone supplies neither the global placement nor this extension. The additional ramified and continuous comparisons remain those of Remark 47.1.2. □

Remark 56.8.2 *(impact on P3).* Corollary 47.10.1 stated P3 conditional on K-P compat, which was itself conditional on K5 (Theorem 51.2.9). The global pairing implication retains all hypotheses of Corollary 56.8.1. P3 concerns the original positive Petersson form; rescaling by an arbitrary nonzero complex c_π^{-1} preserves the pairing identity but need not preserve positivity. Under the arithmetic comparison hypotheses, the logical chain

$$\text{K5 (cuspidal sph-fin)} \;\Rightarrow\; \text{K-P compat (cuspidal sph-fin)} \;\Rightarrow\; \text{P3 (cuspidal sph-fin)}$$

is complete on the cuspidal sph-fin locus with the Koszul-dual normalisation convention explicit.

Remark 56.8.3 *(impact on sufficient RH path).* Corollary 47.10.3 retains the four-input conditional conjunction: arithmetic Positselski, Koszul–Petersson compatibility on its specified domain, $\mathrm{HP}^{\mathrm{Ar}}$, and VB^*. The pairing implication here requires the local comparison datum, the finite nonzero continued value c_π, and the arithmetic placement and adelic maps of Corollary 56.8.1. These assumptions do not prove the other three inputs or extend the pairing identity beyond its stated domain.

56.9 Remaining Arithmetic Constructions

The local objects and their comparison functors must be constructed on the chosen coefficient and spectral carriers. Their cross-prime assembly requires actual equivariant descent data, while the global centre identification remains separate from the underlying arithmetic AG equivalence.

Remark 56.9.1 *(The complete descent and pairing data).* After local transport and effective coherent descent are supplied, BCglue.core asks for the actual global-centre identification and its coherent action comparison. Pairwise statements alone do not supply this datum. The conditional composition theorem explains how a compatible system over all finite sets assembles once constructed.

The K5 calculation gives concrete local Hilbert-space maps and the spherical norm. Its arithmetic extension requires maps preserving Hecke actions, the bilinear pairing and conjugation. Adelic transport also requires its completed domains and normalization. These are not consequences of a finite nonzero adjoint factor.

56.10 Constructed Models and the Global Target

The local coherent transport theorem, finite-stage central-action composition, spherical integral and explicit Hilbert isometries are proved on their stated carriers. Effective coherent descent is applied only when its full diagram and equivalence are supplied.

Koszul–Petersson compatibility and P3 on the cuspidal sph-fin domain retain the actual arithmetic placement, pairing and adelic extension hypotheses of Corollary 56.8.1. The prime cyclic KMS construction does not identify a whole global GNS space with the automorphic carrier. Ramified, full archimedean and continuous comparisons remain part of the original target. Arithmetic Positselski, the Hilbert–Pólya realization and Li positivity also remain separate inputs to the sufficient RH implication.

Chapter 57

Arithmetic Hilbert–Pólya and the VB* Converse: Structural Analyses

Assume P1, P2, and P3 in the comparison form used in Chapter 44. The two additional arithmetic inputs are H-P^{Ar} and VB. H-P^{Ar} is the determinant-trace hypothesis: there is a self-adjoint operator A for which $\Theta_\xi = 1/2 + iA$ has determinant divisor* $\operatorname{div}\xi_{\mathbb{R}}$ *and Weil–Connes trace equal to the zero distribution. VB* adds positivity and Hamburger determinacy for the Fock-space moment measure. Connes' adelic trace formula and Deninger's regularised determinant give equivalent forms of the first input after the named intertwiners; de Branges positivity and Nyman–Beurling–Báez-Duarte decay give equivalent forms of the second. Zero-placement supplies support; it does not supply the unbounded moment theorem.*

57.1 The Analytic Splitting

Chapter 44 gives a sufficient implication to RH from four inputs: arithmetic Positselski, Koszul–Petersson compatibility, H-P^{Ar}, and VB*. The P1 input is the completed arithmetic Positselski comparison; its Tate-Poisson shadow is only the forward functional equation. The P3 input is the full Koszul–Petersson comparison; its proved sph-fin cuspidal core is only the Tamagawa normalisation. The last two inputs are the Hilbert–Pólya determinant-trace realisation and the Fock-space positivity-determinacy theorem.

The Asymmetry

Zero-placement and VB* have different strength. Three operator-theoretic facts account for the difference:

(i) Moment data for an unbounded operator do not determine the spectral measure without a determinacy theorem. Akhiezer–Glazman Ch. VI.3 supplies the classical obstruction.

(ii) The assertion that the normal zero-divisor operator has determinant divisor $\operatorname{div}\xi_{\mathbb{R}}$ and Weil–Connes zero trace is precisely H-P^{Ar} (Hypothesis 44.3.2).

(iii) A symmetric $L_{\mathrm{o}}^{\mathrm{Ar}}$ on the primary Fock is not automatically a self-adjoint Hilbert–Pólya operator; a domain and a self-adjoint extension must be specified.

Thus Theorem 44.3.10 gives the implication VB* $\Rightarrow$ zero-placement under P2, P3, and H-P^{Ar}. The converse implication needs spectral regularity: positivity of the regularised moment functional, Hamburger determinacy, and support compatibility for the Fock operator.

Analytic and Operator Inputs

For H-P^{Ar}:

Connes, arXiv:math/9811068 (1999). Trace formula on the adele class space $X_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$; absorption trace of the scaling action and the zero distribution of ζ. Selecta Math. 5 version with full technical details.

Connes–Consani–Marcolli 2007. Adele class space with quantum-statistical-mechanical structure; Bost–Connes equilibrium-state structure.

Deninger, Proc. Symp. Pure Math. 55.1 (1994). Motivic L-functions; formalism identifying $\xi(s)$ with a regularised determinant of a Frobenius operator on an arithmetic cohomology.

Deninger, Math. Nachr. 205 (2001). Motivic cohomology dynamical systems formalism.

Berry–Keating, SIAM Review 41 (1999). Proposal that $H = \frac{1}{2}(xp + px)$ on $L^2(\mathbb{R}_{>0}, dx/x)$ is the Hilbert–Pólya candidate; restriction to a Planck-cell cutoff gives the correct leading asymptotic zero density.

Akhiezer–Glazman *Theory of Linear Operators in Hilbert Space* Vol. II (Dover 1993 reprint). Ch. VI on unbounded self-adjoint operators, Friedrichs extensions, and moment problems; Ch. VI.3 is the Hamburger/Carleman determinacy criterion.

Reed–Simon *Methods of Modern Mathematical Physics* Vol. I (Academic Press 1980), Vol. II (1975). Ch. VIII (Vol. I) on unbounded operators; Vol. II on Fourier analysis and self-adjointness.

For the VB* converse:

de Branges *Hilbert Spaces of Entire Functions* (Springer 1968). Foundational theorem on reproducing-kernel Hilbert spaces of entire functions; Hermite–Biehler structure theorem.

Lagarias, Acta Arith. 89 (1999), 217–234. Reformulation of RH as Hermite–Biehler positivity for $E_\xi(z) = \xi_{\mathbb{R}}(\frac{1}{2} - iz) + \xi'_{\mathbb{R}}(\frac{1}{2} - iz)$.

Báez-Duarte, Adv. Math. 190 (2005), 315–320; arXiv:math/0202141. Quantitative strengthening of the Nyman–Beurling criterion.

Nyman *On Some Groups and Semigroups of Translations* (Uppsala 1950). Original criterion: RH equivalent to density of $\{\{\theta/x\} : \theta \in (0,1]\}$ translates in $L^2((0,1))$.

Beurling, Proc. Natl. Acad. Sci. 41 (1955). Beurling sequence space formulation of Nyman.

Burnol 2000. Hermite–Biehler refinement of Nyman–Beurling.

Simon, Adv. Math. 137 (1998), 82–203. Classical moment problem as self-adjoint finite difference operator; unbounded case treated via Jacobi matrix formalism.

Stein–Weiss *Introduction to Fourier Analysis on Euclidean Spaces* (Princeton 1971). Ch. V on Hardy spaces and boundary-value theorems governing Hermite–Biehler.

These analytic sources supply the external operator models for H-P^{Ar} and the VB* converse. P1, P2, and P3 enter only through their comparison hypotheses.

57.2 Structural Reductions

The assertions below separate domain, trace, determinant, and moment questions. Each assertion is an implication between named mathematical structures.

H-P^{Ar}: Operator-Theoretic Framework

On the finite-span core of the primary arithmetic Heisenberg Fock, the Sugawara operator is diagonal with real non-negative eigenvalues (Lemma 45.6.1). Reed–Simon Vol. I Thm. VIII.11 and the Friedrichs construction give a canonical self-adjoint non-negative extension from this semibounded quadratic form.

This removes the domain ambiguity in Hypothesis 44.3.2. It does not identify the resulting Bost–Connes partition trace with the Weil–Connes zero distribution; that identification is the spectral trace input of Proposition 57.3.3.

H-P^{Ar}: Connes Adele Class Space

Connes' adelic trace formula is a theorem on $X_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$. The comparison with H-P^{Ar} requires an intertwiner from the primary Fock space to the adele-class L^2-space which carries the zero-mode Bost–Connes monoid action to the idele-class scaling action. Under such an intertwiner, Connes' absorption trace and the regularised Fock trace are the same distribution. The displayed absorption-trace formula,

$$\sum_{\rho} h(\operatorname{Im}\rho) = h(0) + h(1) - \hat{h}(0)\log\pi - \sum_{v}{}' \int_{\mathbb{Q}_v^{\times}} \frac{h(\log|u|_v)}{|1-u|_v} d^{\times}u,$$

matches (as a distribution on the dual real line) the Li–Stieltjes moment generating function $\sum_{\rho} e^{-t\rho}$ through Mellin inversion $h(u) = e^{-tu/2}$, provided the embedding preserves the distributional trace.

H-P^{Ar}: Deninger Motivic Cohomology

Deninger's formalism postulates a cohomology $H^{\bullet}_{M}(\overline{\operatorname{Spec}\mathbb{Z}}, \mathbb{R}(\cdot))$ with Frobenius Φ and Lefschetz determinant formula. If that cohomology exists and is identified with the primary Fock by an intertwiner, then the determinant identity for $\xi_{\mathbb{R}}$ is the H-P^{Ar} determinant statement. The formal identity $\xi(s) = \det_{\infty}(s\cdot\mathrm{id} - L_0^{\mathrm{Ar}})$ holds under H-P^{Ar}: the regularised determinant is defined by $\log\det_{\infty}(A) = -\partial_s \operatorname{tr}(A^{-s})|_{s=0}$, and the determinant divisor is $\operatorname{div}\xi_{\mathbb{R}}$ after the trivial-zero and pole summands have been removed by the primary-sector restriction.

H-P^{Ar}: Necessary and Sufficient Conditions

The three formulations have one common mathematical core: a positive-definite Hilbert space $\mathcal{H}$, a densely-defined self-adjoint operator A, and an intertwiner $\pi : \mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}} \to \mathcal{H}$ such that $A\pi = \pi L_0^{\mathrm{Ar}}$, the normal operator $\frac{1}{2} + iA$ has regularised determinant divisor equal to the non-trivial zero divisor of $\xi_{\mathbb{R}}$, and its Weil–Connes trace is the corresponding zero distribution. This is classical Hilbert–Pólya inside the arithmetic-Fock framework.

VB Converse: Spectral Regularity Beyond RH*

Zero-placement gives the zeros on the critical line. VB* asks for positivity of the operators T_n on the Fock closure. Passing from the former to the latter is a Hamburger moment problem for an unbounded operator; in this setting moment positivity does not select a unique spectral measure. Carleman's criterion says that the Hamburger moment problem with moments $\{\mu_n\}_{n\geq 0}$ is determinate if $\sum_{n\geq 1} \mu_{2n}^{-1/(2n)} = +\infty$. For the proposed shifted-zero functional of Definition 62.4.1, constructing finite moments and proving positivity are separate requirements. The finite cutoff sums in Proposition 80.2.5 diverge; zero counting does not define their regularization. Once a positive functional has been constructed, the upper bound $\mu_{2n}^{1/(2n)} = O(n)$ suffices for Carleman determinacy.

VB Converse: de Branges and Báez-Duarte*

The de Branges operator comparison and the Nyman–Beurling approximation problem involve different data. The latter has the concrete Hilbert carrier $L^2((0,\infty), dx)$, fractional-part vectors $\rho_n(x) = \{1/(nx)\}$, and target $\chi = \mathbf{1}_{(0,1)}$. The classical Báez-Duarte criterion is convergence of the finite distances

to zero. A sharp two-sided rate, an arithmetic isometry, and its compatibility with the operator-positivity condition are additional questions. The finite distance and its exact transport are constructed in Proposition 57.4.13.

57.3 H-P^{Ar}: Structural Analysis

Operator-Theoretic Framework

Definition 57.3.1 *(candidate $L_{\mathrm{o}}^{\mathrm{Ar}}$ with explicit domain).* On the primary arithmetic Heisenberg Fock $\mathcal{V}_{\mathrm{Heis,prim}}^{\mathrm{prim}}$, equipped with the Petersson inner product $\langle\cdot,\cdot\rangle^{\mathrm{Pet}}$ of Theorem 43.7.1, define:

- *Finite-span core* $\mathcal{D}_{\mathrm{o}} := \mathrm{span}_{\mathbb{C}}\{|\mathbf{n}\rangle \otimes \delta_m : \mathbf{n} \in \mathbb{Z}_{\geq 0}^{\oplus\mathbb{N}},\ m \in \mathbb{N}^{\times}\}$, the linear span of number-state/zero-mode-lattice basis vectors (Lemma 45.6.1, Definition 29.2.1).
- *Candidate operator* $L_{\mathrm{o}}^{\mathrm{Ar,\,symm}} : \mathcal{D}_{\mathrm{o}} \to \mathcal{V}_{\mathrm{Heis,prim}}^{\mathrm{prim}}$ defined by

$$L_{\mathrm{o}}^{\mathrm{Ar,\,symm}}\big(|\mathbf{n}\rangle \otimes \delta_m\big) = \Big(\sum_{i\geq 1} i\, n_i + \lambda(m)\Big)|\mathbf{n}\rangle \otimes \delta_m$$

  where $\lambda(m) \in \mathbb{R}$ is the Bost–Connes zero-mode weight corresponding to $m \in \mathbb{N}^{\times}$ (unspecified at this stage; the choice is constrained by the Bost–Connes KMS$_1$ state structure of Proposition 29.3.1).
- *Closure*: let $L_{\mathrm{o}}^{\mathrm{Ar}}$ denote the closure of $L_{\mathrm{o}}^{\mathrm{Ar,\,symm}}$ with respect to the graph norm on $\mathcal{D}_{\mathrm{o}}$.

THEOREM 57.3.2 *(Friedrichs self-adjoint extension). Conditional statement.* Under the hypothesis that $\lambda(m) \geq 0$ for all $m \in \mathbb{N}^{\times}$ (non-negativity of the Bost–Connes zero-mode weights), the symmetric operator $L_{\mathrm{o}}^{\mathrm{Ar,\,symm}}$ of Definition 57.3.1 admits a unique Friedrichs self-adjoint extension $L_{\mathrm{o}}^{\mathrm{Ar}} : \mathcal{D} \to \mathcal{V}_{\mathrm{Heis,prim}}^{\mathrm{prim}}$ with $\mathcal{D} \supset \mathcal{D}_{\mathrm{o}}$ dense. On the Friedrichs extension, $L_{\mathrm{o}}^{\mathrm{Ar}}$ is essentially self-adjoint on the finite-span core $\mathcal{D}_{\mathrm{o}}$ and has non-negative spectrum.

Proof. Apply Reed–Simon Vol. II Thm. X.23 (Friedrichs extension of a non-negative symmetric operator). The quadratic form $q(\psi,\psi') = \langle L_{\mathrm{o}}^{\mathrm{Ar,\,symm}}\psi, \psi'\rangle^{\mathrm{Pet}}$ is closable because $L_{\mathrm{o}}^{\mathrm{Ar,\,symm}}$ acts diagonally on the orthonormal basis $\{|\mathbf{n}\rangle \otimes \delta_m\}$ with non-negative real eigenvalues. Reed–Simon Vol. I Thm. VIII.11 then gives essential self-adjointness on the finite-span core: a diagonal operator with real eigenvalues on a Hilbert space with eigenvector basis is essentially self-adjoint on the finite-span of eigenvectors. The Friedrichs extension is the unique self-adjoint extension dominated by the quadratic form (Akhiezer–Glazman Vol. II Thm. 123). Non-negativity of the spectrum follows from non-negativity of all eigenvalues. □

PROPOSITION 57.3.3 *(Bost–Connes trace compatibility is the HP input).* The Friedrichs extension of Theorem 57.3.2 constructs a self-adjoint non-negative Fock/BC Hamiltonian from the real weights $\lambda(m)$. It does not by itself construct the Hilbert–Pólya determinant-trace operator H_{HP}. H-P^{Ar} holds only after one supplies an additional distributional trace identification with the zero side. The formal heat-kernel expression is

$$\mathrm{tr}_{\mathrm{BC}}\big(e^{-tL_{\mathrm{o}}^{\mathrm{Ar}}}\big) = \sum_{\rho} e^{-t\rho} \qquad (t > 0) \tag{57.1}$$

but (57.1) is only symbolic notation. It means equality of Weil–Connes distributions on admissible Paley–Wiener tests, equivalently the trace identity of Hypothesis 44.3.2 for $\Theta_\xi = 1/2 + iH_{\mathrm{HP}}$. This trace compatibility is the Hilbert–Pólya determinant-trace input, not a consequence of the Friedrichs theorem.

Proof. Direct computation of $\mathrm{tr}(e^{-tL_0^{\mathrm{Ar}}})$ on the basis of $\mathcal{V}_{\mathrm{Heis,prim}}^{\mathrm{prim}} = L \otimes \mathcal{H}^{(\infty)}$:

$$\mathrm{tr}\big(e^{-tL_0^{\mathrm{Ar}}}\big) \;=\; \Big(\sum_{m\geq 1} e^{-t\lambda(m)}\Big)\cdot\Big(\sum_{\mathbf{n}} e^{-t\sum_i i\,n_i}\Big) \;=\; \Big(\sum_{m\geq 1} e^{-t\lambda(m)}\Big)\cdot\prod_{i\geq 1}\frac{1}{1-e^{-ti}}.$$

For the Bost–Connes/KMS weight $\lambda(m) = \log m$, the zero-mode factor is $\zeta(t)$ for $\mathrm{Re}(t) > 1$, multiplied by the Dedekind-eta inverse from the Fock descendants. This is the Euler/partition side. It is not the zero distribution. Passing from this partition trace to the zero side requires the full Weil–Connes explicit-formula distribution and a positive trace realisation; that is exactly H-P^{Ar} in Hypothesis 44.3.2. Thus the proposition records an equivalence of formulations after the trace realisation is assumed, not a derivation of the trace realisation. □

Remark 57.3.4 *(what the Friedrichs construction buys).* Theorem 57.3.2 removes only one domain-specification ambiguity: a real non-negative diagonal Fock/BC Hamiltonian has a canonical self-adjoint extension. What remains is not an elementary choice of $\lambda(m)$. The KMS normalisation selects $\lambda(m) = \log m$, which gives the Bost–Connes partition trace $\zeta(t)$; the zero distribution of $\xi_{\mathbb{R}}$ (the completed xi function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, here denoting the restriction to the primary archimedean sector) is a different object. The Hilbert–Pólya content is the additional construction of H_{HP}, or an intertwiner carrying this Fock/BC Hamiltonian to the zero-divisor operator Θ_ξ, with the positive Weil–Connes trace preserved.

Connection to Connes Adele Class Space

Definition 57.3.5 *(Connes adele class space).* (Connes arXiv:math/9811068 §II). The *adele class space* is $X_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^\times$, the quotient of the adeles by the multiplicative action of non-zero rationals. Its measure-zero part $X_{\mathbb{Q}}^0$ supports an L^2 space $L_0^2(X_{\mathbb{Q}}, d\mu)$ on which the idele class group $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^\times/\mathbb{Q}^\times$ acts unitarily by multiplication. The absorption trace of this action is the content of Connes' realisation.

Theorem 57.3.6 *(Connes arXiv:math/9811068 Thm. 1).* For appropriate test functions h on $C_{\mathbb{Q}}$,

$$\mathrm{Tr}(\vartheta(h)) \;=\; 2h(1) + 2h(-1) - \hat{h}(0)\log\pi - \sum_v {\int_{\mathbb{Q}_v^\times}}' \frac{h(u)}{|1-u|}\,d^\times u - \sum_{\rho:\, L(\tilde{\rho},\chi)=0} \hat{h}(\chi,\rho), \qquad (57.2)$$

where $\vartheta(h)$ is the scaling operator on $L_0^2(X_{\mathbb{Q}})$, ρ ranges over zeros of Hecke L-functions $L(s,\chi)$, and the prime on the local integral denotes principal-value regularisation. The GL_1 case with trivial character recovers $\xi(s)$: the non-trivial zeros of ζ enter the absorption trace distribution.

Conjecture 57.3.7 *(intertwiner comparing H-P^{Ar} with Connes). Conjectural statement.* Let H_{HP} and $\Theta_\xi = 1/2 + iH_{\mathrm{HP}}$ be as in Hypothesis 44.3.2, and let D_{BC} denote the Friedrichs Fock/BC Hamiltonian of Theorem 57.3.2. Let ϑ be the scaling operator of Connes on $L_0^2(X_{\mathbb{Q}})$. The required comparison is the existence of an intertwiner

$$\pi : \mathcal{V}_{\mathrm{Heis,prim}}^{\mathrm{prim}} \longrightarrow L_0^2(X_{\mathbb{Q}}, d\mu)$$

defined on $\mathcal{D}_0$ and extended by closure, such that

$$\pi \circ \Theta_\xi \;=\; \vartheta_0 \circ \pi$$

where ϑ_0 is the infinitesimal generator of the scaling action on the kernel of the idele-class scaling trace. The intertwiner π is $C_{\mathbb{Q}}$-equivariant on the sph-finite sector and identifies the positive Weil–Connes trace of Θ_ξ with the distributional trace in Connes' trace formula. Its restriction to the zero-mode

lattice must also explain how D_{BC}, whose partition trace is $\zeta(\beta)$, maps to the absorption trace without postulating the zero divisor.

Remark 57.3.8 *(open intertwiner).* The sph-finite sector of $\mathcal{V}^{\mathrm{prim}}$ embeds by restricted tensor product into a subspace of $L^2([GL_2(\mathbb{Q})]\backslash GL_2(\mathbb{A}))$ compatible with the adele class space image (Gelbart *Automorphic Forms on Adele Groups* §3.1). The zero-mode lattice $L \subset \mathcal{V}^{\mathrm{prim}}$ maps to the finite-adele part of $X^{\circ}_{\mathbb{Q}}$; the archimedean Heisenberg factor $\mathcal{H}^{(\infty)}$ maps to the archimedean-fibre Fock of $L^2_{\circ}(X^{\circ,\infty}_{\mathbb{Q}})$. The scaling action of $C_{\mathbb{Q}}$ on $L^2_{\circ}(X^{\circ}_{\mathbb{Q}})$ restricts on the sph-finite sector to the multiplicative action of $\mathbb{N}^{\times}$ on the Bost–Connes zero-mode lattice (Proposition 29.3.1), plus the real-parameter archimedean scaling on the Fock. The infinitesimal generator of this action is the Bost–Connes/Fock Hamiltonian on the chiral-algebra side, while the Connes trace formula sees the absorption trace. Constructing an isometric intertwiner from the former to the latter is exactly the determinant-trace problem.

COROLLARY 57.3.9 *(H-P^{Ar} and Connes have the same determinant-trace input). Conditional statement. (conditional on the intertwiner of Conjecture 57.3.7).* Under that intertwiner, the trace-compatibility condition (57.1), read as the distributional equality of Hypothesis 44.3.2, is equivalent to Connes' absorption-trace hypothesis: the $C_{\mathbb{Q}}$-scaling action on $L^2_{\circ}(X_{\mathbb{Q}})$ has zero distribution of ζ with correct multiplicity.

Proof. Given the conjectural intertwiner, applying π transports the regularised trace distribution of Θ_{ξ} to the Connes distributional trace. Conversely, on the sph-finite image of π, the Connes absorption-trace statement pulls back to the Hilbert–Pólya determinant and trace identities. The proof uses the intertwiner as an input; it does not construct it. □

Remark 57.3.10 *(form of Connes' trace formula).* Connes' trace formula (57.2) is proved in arXiv:math/9811068 Thm. 1. The identification of the absorption trace with the zero distribution of ζ with correct multiplicity is the content of Connes' RH-equivalent statement; it is *not* a theorem. Corollary 57.3.9 places H-P^{Ar} at the same level only after the intertwiner of Conjecture 57.3.7 is assumed. That intertwiner is part of the RH-equivalent determinant-trace problem. The chiral-algebra form replaces the adele-class-space L^2 setting by a Fock-space setting with Sugawara and free-boson operators.

Connection to Deninger Motivic Cohomology

Definition 57.3.11 *(Deninger axioms for motivic cohomology).* (Deninger Proc. Symp. Pure Math. 55.1, §1). A *Deninger motivic cohomology* is a conjectural functor

$$H^{\bullet}_M(-, \mathbb{R}(\cdot)) : \ \mathrm{Schemes}/\mathbb{Z} \longrightarrow \mathrm{Graded}\ \mathbb{R}\text{-vs}$$

equipped with an integral Frobenius endomorphism Φ satisfying:

(D1) *Lefschetz trace formula.* For every smooth arithmetic scheme X, $\zeta_X(s) = \prod_i \det_{\infty}(s \cdot \mathrm{id} - \Phi | H^i_M(X, \mathbb{R}(1)))^{(-1)^{i+1}}$.

(D2) *Purity.* The eigenvalues of Φ on $H^i_M(X, \mathbb{R}(1))$ all satisfy $\mathrm{Re}(\rho) = i/2$ (or a shifted variant).

(D3) *Regularised determinant.* The determinant $\det_{\infty}$ is the ζ-regularised determinant defined by $\log \det_{\infty}(A) = -\partial_s \mathrm{tr}(A^{-s})|_{s=0}$.

PROPOSITION 57.3.12 *(H-P^{Ar} $\Leftrightarrow$ Deninger formalism for $\overline{\mathrm{Spec}\ \mathbb{Z}}$). Conditional statement. (conditional on existence of Deninger cohomology for $\overline{\mathrm{Spec}\ \mathbb{Z}}$ and on an explicit Deninger–Fock intertwiner).* Assuming Definition 57.3.11 applies to $\overline{\mathrm{Spec}\ \mathbb{Z}}$ with $X = \overline{\mathrm{Spec}\ \mathbb{Z}}$ and $H^1_M(\overline{\mathrm{Spec}\ \mathbb{Z}}, \mathbb{R}(1))$ non-zero, the

Hilbert–Pólya zero-divisor operator Θ_ξ of Hypothesis 44.3.2 is identified with $\Phi|H^1_M(\overline{\mathrm{Spec}\,\mathbb{Z}},\mathbb{R}(1))^+$ (the + superscript denotes the primary $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-invariant part), and H-P^{Ar} holds iff (D1), (D2), (D3) hold.

Proof. Under the Deninger axioms, the meromorphic Tate zeta surface is a cohomological superdeterminant: the even cohomology accounts for the Tate poles at 0 and 1, while the primary H^1 determinant is the entire factor $\xi_{\mathbb{R}}(s)$. The ordinary determinant target on the primary summand is

$$\xi_{\mathbb{R}}(s) = \det_\infty(s\cdot\mathrm{id} - \Phi|H^1_M(-,\mathbb{R}(1))^+).$$

The regularised determinant identity fixes the spectral divisor of the primary summand:

$$\operatorname{div}\det_\infty(s\cdot\mathrm{id} - \Phi|H^1_M(-,\mathbb{R}(1))^+) = \operatorname{div}\xi_{\mathbb{R}}(s).$$

By (D3), the trace of Φ is a distributional zero trace in the admissible Weil–Connes sense, not an ordinary heat trace. The identification is canonical only after composing the intertwiner of Conjecture 57.3.7 with the conjectural Deninger–Connes dictionary of Leichtnam. □

Remark 57.3.13 *(the three formalisms in alignment).* The three comparison formulations of H-P^{Ar}:

(i) *Fock/BC domain realisation*: the Friedrichs Hamiltonian of Theorem 57.3.2 supplies a self-adjoint Euler-side operator, but the passage to Θ_ξ is the determinant-trace realisation.

(ii) *Connes adele class space*: the scaling generator ϑ_0 on $L^2_0(X_{\mathbb{Q}})$ has the zero distribution as its absorption trace; this is Connes' RH-equivalent reformulation of the trace formula of arXiv:math/9811068 Thm. 1, not a formal consequence of it.

(iii) *Deninger motivic cohomology*: the Frobenius Φ on $H^1_M(\overline{\mathrm{Spec}\,\mathbb{Z}},\mathbb{R}(1))^+$ has determinant divisor $\operatorname{div}\xi_{\mathbb{R}}$ (Deninger 1994 axioms).

The equivalences among these formulations require the Connes/Fock and Deninger/Fock intertwiners. Without those intertwiners there is a comparison diagram, not a proof that the three operator models are the same object.

Necessary and Sufficient Conditions

Theorem 57.3.14 *(H-P^{Ar} is the determinant-trace realisation).* The following are equivalent after the explicit Connes/Fock and Deninger/Fock intertwiners named above are included as hypotheses:

(H-P^{Ar}) The arithmetic Hilbert–Pólya condition in Hypothesis 44.3.2.

(SP) *Determinant-trace condition:* there exists the self-adjoint operator A, the normal operator $\Theta_\xi = 1/2 + iA$, the determinant divisor, and the positive Weil–Connes trace required in Hypothesis 44.3.2.

(Connes) *Connes trace realisation:* the $C_{\mathbb{Q}}$-scaling action on $L^2_0(X_{\mathbb{Q}})$ has regularised trace distribution equal to the non-trivial zero distribution of ζ, with multiplicity.

(Deninger) *Deninger regularised-determinant condition:* the formal identity $\xi_{\mathbb{R}}(s) = \det_\infty(s\cdot\mathrm{id} - \Phi|H^1_{M,\mathbb{R}(1),+})$ holds with Φ a self-adjoint operator on a Hilbert space matching $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}}$ via an explicit intertwiner.

Proof. (H-P^{Ar}) ⇔ (SP) is definitional after Hypothesis 44.3.2: SP is the statement that the self-adjoint operator, determinant divisor, and positive trace distribution exist. The Friedrichs theorem supplies a separate Fock/BC domain model; it is evidence for the analytic framework but is not identified with H_{HP} without an intertwiner.

(SP) ⇔ (Connes) is Corollary 57.3.9, conditional on the Connes/Fock intertwiner. (SP) ⇔ (Deninger) is Proposition 57.3.12, conditional on the Deninger/Fock intertwiner and on the Deninger cohomology. Under these named hypotheses, transitivity closes the comparison. □

COROLLARY 57.3.15 *(H-P^{Ar} reduces to determinant and trace).* H-P^{Ar} reduces to the determinant-trace condition (SP). The Friedrichs/KMS side fixes the natural Bost–Connes weight $\lambda(m) = \log m$ and gives the Euler partition trace; it does not give the zero trace. The required assertion is the construction of H_{HP}, Θ_ξ, and the positive Weil–Connes trace whose distribution is the non-trivial-zero distribution of $\xi_{\mathbb{R}}$. This is the Hilbert–Pólya content itself, not a routine weight choice (Remark 57.3.16).

Proof. Direct from Theorem 57.3.14. The trace-compatibility condition encodes the location of non-trivial zeros of $\xi_{\mathbb{R}}$; the Bost–Connes weight gives the Euler side, while the zero side is the determinant divisor together with the positive Weil–Connes trace distribution. □

Remark 57.3.16 *(Hilbert–Pólya factor).* The Friedrichs construction of Theorem 57.3.2 proves only the domain theorem. Given $\{\lambda(m)\} \geq 0$, the Friedrichs extension is unique, self-adjoint, and non-negative. H-P^{Ar} is the additional determinant-trace assertion: one must construct a zero operator with $\operatorname{div} \det_\infty(s - \Theta_\xi) = \operatorname{div} \xi_{\mathbb{R}}$ and trace distribution such that, in symbolic wave-kernel notation,

$$\mathrm{Tr}_{\mathrm{reg}}\left(e^{-t\Theta_\xi}\right) \;=\; \sum_{\rho} e^{-t\rho} \qquad (t > 0),$$

where the equality is read after admissible Weil–Connes regularisation and the sum is over non-trivial zeros of $\xi_{\mathbb{R}}$ counted with multiplicity. This is Hilbert's original eighth problem, not a corollary of the Friedrichs step. The determinacy comparison retains the domain theorem and leaves the Hilbert–Pólya determinant-trace realisation.

Remark 57.3.17 *(Berry–Keating candidate).* Berry–Keating SIAM Review 41 (1999) proposed $H = \frac{1}{2}(xp + px)$ on $L^2(\mathbb{R}_{>0}, dx/x)$ as a candidate Hilbert–Pólya operator, restricted to a Planck-cell cutoff. This operator has continuous spectrum on $\mathbb{R}$ and the Bohr-Sommerfeld quantisation of $xp = E$ gives the correct leading asymptotic zero density $N(T) \sim (T/2\pi)\log(T/2\pi)$. In the present framework, this candidate corresponds to a specific choice of $\{\lambda(m)\}_{m\in\mathbb{N}^\times}$: the prime-indexed weights $\lambda(p^k) = k \log p$ (Bost–Connes monoid generators), combined with the Fock archimedean spectrum $\{i\, n_i\}$, formally give the Berry–Keating spectrum structure. The Berry–Keating candidate is *not* rigorously known to have determinant divisor div $\xi_{\mathbb{R}}$; it reproduces the density but not the individual zero divisor. The rigorous identification requires a regularisation matching (57.1), which is the H-P^{Ar} content.

57.4 VB* CONVERSE: STRUCTURAL ANALYSIS

The Moment Problem in Hilbert Space

THEOREM 57.4.1 *(determinate vs. indeterminate Hamburger moment problem).* (Akhiezer 1965 §2.4; Shohat–Tamarkin 1943 Ch. III.5). Let $\{\mu_n\}_{n\geq 0}$ be a sequence of real numbers with Hankel matrices $H_n = (\mu_{i+j})_{0\leq i,j\leq n} \succeq 0$ for all n. The Hamburger moment problem is:

- *Solvable*: there exists a positive measure ν on $\mathbb{R}$ with $\mu_n = \int x^n \, d\nu(x)$.

- *Determinate*: the solution ν is unique.
- *Indeterminate*: multiple distinct positive measures share the moment sequence.

Carleman's sufficient criterion for determinacy: $\sum_{n\geq 1} \mu_{2n}^{-1/(2n)} = +\infty$. Non-sufficient examples (Stieltjes 1894): the lognormal moment sequence $\mu_n = e^{n^2/2}$ is indeterminate.

Theorem 57.4.2 *(Simon 1998: moment problem as Jacobi matrix).* (Simon Adv. Math. 137 (1998) §5). A Hamburger moment problem with $\mu_0 > 0$ and strictly positive Hankel forms on every nonzero polynomial is equivalent to the spectral theory of a Jacobi matrix

$$J = \begin{pmatrix} b_0 & a_0 & 0 & \cdots \\ a_0 & b_1 & a_1 & \cdots \\ 0 & a_1 & b_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

on $\ell^2(\mathbb{Z}_{\geq 0})$, with $a_n > 0$ and $b_n \in \mathbb{R}$ determined recursively by the moment sequence via the Gram–Schmidt-orthonormalised polynomial basis $\{p_n(x)\}_{n\geq 0}$ with positive leading coefficient $p_n(x) = \kappa_n x^n + \text{lower}$, $\kappa_n > 0$, and $\int p_m p_n \, d\nu = \delta_{mn}$. With $p_{-1} = 0$, multiplication by x satisfies

$$x p_n = a_n p_{n+1} + b_n p_n + a_{n-1} p_{n-1}, \qquad a_n = \kappa_n/\kappa_{n+1} > 0.$$

The coefficient of p_j for $j < n-1$ vanishes because $\langle x p_n, p_j\rangle = \langle p_n, x p_j\rangle = 0$; comparison of leading terms gives the displayed a_n. A finite-support moment measure instead gives a terminating finite Jacobi matrix. In the stated nondegenerate infinite case, the moment problem is determinate iff J is essentially self-adjoint on c_{00}, and indeterminate iff J has deficiency indices $(1,1)$.

Proposition 57.4.3 *(VB* operator-positivity vs. moment positivity: the gap).* Let L be a specified positive Hamburger functional with finite moments, and let A be a self-adjoint realization on the stated arithmetic Hilbert space. Suppose the operator comparison identifies $T_n = p_n(A)$, where $p_n(t) = t^n - (t-1)^n$, on the maximal domains

$$D(T_n) = \left\{ v : \int_{\mathbb{R}} |p_n(t)|^2 \, d\langle E_A(t)v, v\rangle < \infty \right\},$$

and the trace comparison, wherever invoked, gives $\operatorname{tr}(T_n) = n\lambda_n$ with finite traces. Then:

(i) Positivity of all T_n implies Li positivity and hence RH.

(ii) All T_n are positive if and only if the spectral measure of A is supported in $[1/2, \infty)$.

(iii) Determinacy of L does not by itself construct this operator realization, trace comparison or support identification.

(iv) Positivity of a moment functional alone does not choose a self-adjoint realization in an indeterminate moment problem.

Proof. The first statement follows from positivity of the finite traces and Li's criterion. For the second, the real polynomial $t^n - (t-1)^n$ is nonnegative for $t \geq 1/2$: for odd n it is positive on all real t, and for even n compare $|t|$ and $|t-1|$. The spectral theorem gives positivity on the maximal polynomial domain. Conversely, positivity of $T_2 = 2A - I$ forces that support by the spectral theorem. The last two statements distinguish uniqueness of a representing measure from a specified realization; indeterminate Hamburger functionals admit different representing measures with the same moments. No claim about the off-line defect measure of Definition 45.4.1 is used. □

COROLLARY 57.4.4 *(the converse input is spectral regularity).* Assume a specified positive moment functional is identified with the arithmetic self-adjoint operator and has spectral support in $[1/2, \infty)$, with the polynomial operator comparison of Proposition 57.4.3. Then VB* holds. Determinacy supplies uniqueness of the representing measure when it is available; it does not construct these arithmetic identifications from zero placement.

Proof. Apply item 2 of the proposition on the stated spectral domains. □

PROPOSITION 57.4.5 *(A conditional Carleman construction).* For a specified positive Hamburger functional, the bound $\mu_{2n} \le C^n (n!)^2$ implies determinacy.

Proof. Its positive even moments satisfy $\mu_{2n}^{-1/(2n)} \ge C^{-1/2} (n!)^{-1/n} \ge C^{-1/2}/n$. Carleman's series diverges. If an even moment vanishes, the positive measure is supported at zero and is unique. No lower moment bound is required. □

Remark 57.4.6 *(Moment growth and the arithmetic construction).* A positive functional with moments $\mu_n = e^{n^2/2}$ has an indeterminate Hamburger problem, whereas the factorial upper bound above gives determinacy. These examples do not estimate the arithmetic moments. The finite-cutoff zero sums diverge, so a common linear regularization, positivity and the stated upper bound remain actual construction requirements.

Hermite–Biehler + de Branges

Definition 57.4.7 *(Hermite–Biehler function).* (de Branges 1968 Ch. II). An entire function $E(z)$ is *Hermite–Biehler* if $|E(z)| > |E^*(z)|$ for all z in the open upper half plane $\operatorname{Im} z > 0$, where $E^*(z) = \overline{E(\overline{z})}$. The *de Branges space* $\mathcal{H}(E)$ is the reproducing-kernel Hilbert space of entire functions f with $f/E, f^*/E \in H^2(\mathbb{C}_+)$ (upper-half-plane Hardy space) and reproducing kernel

$$K_E(w, z) \;=\; \frac{\overline{E(w)}E(z) - \overline{E^*(w)}E^*(z)}{2\pi i(\overline{w} - z)}.$$

THEOREM 57.4.8 *(de Branges structure theorem).* (de Branges 1968 Thm. 22). Every reproducing-kernel Hilbert space of entire functions $\mathcal{H}$ satisfying:

- $f, g \in \mathcal{H} \Rightarrow \overline{g(\overline{z})} \in \mathcal{H}$ (complex conjugate closure);
- for every non-real w, the point evaluation $f \mapsto f(w)$ is continuous;
- for every non-real w, $f \mapsto (f - \tilde{f})/(z - w)$ is bounded (where $\tilde{f}$ denotes a specific orthogonal projection);

is of the form $\mathcal{H}(E)$ for some Hermite–Biehler E.

THEOREM 57.4.9 *(The derivative kernel and zero placement).* Let $A(z) = \Xi(z) = \xi_{\mathbb{R}}(1/2 - iz)$ and

$$E_\xi(z) = A(z) + iA'(z) = \xi_{\mathbb{R}}(1/2 - iz) + \xi'_{\mathbb{R}}(1/2 - iz).$$

RH, strict Hermite–Biehler positivity of E_ξ, and positive semidefiniteness of K_{E_ξ} are equivalent.

Proof. Under RH the paired Hadamard product for the even function A has real zeros $\pm\gamma$ and

$$\frac{A'(z)}{A(z)} = \sum_{\gamma > 0} \left(\frac{1}{z - \gamma} + \frac{1}{z + \gamma} \right),$$

with multiplicities and locally uniform convergence off the real zeros, since $\sum \gamma^{-2} < \infty$. Each term has strictly negative imaginary part in the upper half-plane. Therefore $|A + iA'|^2 - |A - iA'|^2 = -4|A|^2 \operatorname{Im}(A'/A) > 0$, so E_ξ is strictly Hermite–Biehler. The standard Schur-kernel construction gives positive semidefiniteness of its de Branges kernel, as in Proposition 79.6.5. Conversely, positivity of the kernel forces strictness by Proposition 79.6.7. If the real part $A = (E_\xi + E_\xi^*)/2$ vanished in the upper half-plane, then $E_\xi = -E_\xi^*$ there, contradicting strictness. Reality excludes lower-half-plane zeros as well. Thus the zeros of A are real, which is RH. □

Proposition 57.4.10 *(Transport of the operator positivity predicate).* Let A be the specified self-adjoint arithmetic operator and $U : H_{\mathrm{Ar}} \to \mathcal{H}(E_\xi)$ a supplied unitary. Suppose a self-adjoint operator M on $\mathcal{H}(E_\xi)$ satisfies $U\,D(A) = D(M)$ and $UAU^{-1} = M$. With maximal polynomial domains, the following are equivalent:

(i) $p_n(A) \geq 0$ for every $n \geq 1$.

(ii) $p_n(M) \geq 0$ for every $n \geq 1$.

(iii) The spectral measure of M is supported in $[1/2, \infty)$.

The unitary and the self-adjoint realization are comparison inputs. No positivity assertion for arbitrary signed measures is required.

Proof. Unitary functional calculus gives $U\,p_n(A)U^{-1} = p_n(M)$, including their maximal domains, so the first two assertions are equivalent. The second and third are equivalent by the polynomial calculation in Proposition 57.4.3: sufficiency follows from $p_n(t) \geq 0$ on $[1/2, \infty)$, and necessity from $p_2(M) = 2M - I \geq 0$. □

The Báez-Duarte Refinement

Theorem 57.4.11 *(Nyman–Beurling criterion on the half-line).* Let $\rho(y) = y - \lfloor y \rfloor$, $\chi = \mathbf{1}_{(0,1)}$, and $H = L^2((0,\infty), dx)$. For real $a \geq 1$, put $\rho_a(x) = \rho(1/(ax))$. Then RH is equivalent to $\chi \in \overline{\operatorname{span}\{\rho_a : a \geq 1\}}$ in H. This is the half-line form of the Nyman–Beurling criterion.

Theorem 57.4.12 *(Báez-Duarte's discrete criterion).* With the same carrier and vectors, define

$$E_N = \operatorname{span}\{\rho_n : 1 \leq n \leq N\}, \qquad \epsilon_N^{\mathrm{BD}} = \operatorname{dist}_H(\chi, E_N).$$

Then RH is equivalent to $\epsilon_N^{\mathrm{BD}} \to 0$. Báez-Duarte, arXiv:math/0205003, Theorem 1.1, proves the discrete closure statement; it does not identify these spaces with an arithmetic Fock space or assert the sharp rate $\epsilon_N^{\mathrm{BD}} \asymp (\log N)^{-1/2}$.

Proposition 57.4.13 *(Finite-distance construction and conditional arithmetic transport).* Each E_N is a closed finite-dimensional subspace of H. Let P_N be its orthogonal projection. Then

$$\epsilon_N^{\mathrm{BD}} = \|\chi - P_N\chi\|_H.$$

For $G_{ij} = \langle \rho_j, \rho_i \rangle_H$ and $b_i = \langle \chi, \rho_i \rangle_H$, with inner products linear in the first variable, any solution of $Ga = b$ gives

$$P_N\chi = \sum_{j=1}^{N} a_j \rho_j, \qquad (\epsilon_N^{\mathrm{BD}})^2 = 1 - b^* a.$$

Now let K_{ar} be a specified Hilbert completion of an arithmetic pairing carrier. Suppose a linear map $U : H \to K_{\mathrm{ar}}$ satisfies $\langle Uf, Ug\rangle_{\mathrm{ar}} = c\langle f, g\rangle_H$ for a fixed real $c > 0$, and the arithmetic target and finite approximation spaces are $U\chi$ and UE_N. Then

$$\operatorname{dist}_{K_{\mathrm{ar}}}(U\chi, UE_N) = \sqrt{c}\,\epsilon_N^{\mathrm{BD}}.$$

The arithmetic carrier, U, and its compatibility with the intended operators remain specified comparison data.

Proof. The functions ρ_n are bounded near zero and equal $1/(nx)$ for $x > 1$, so they lie in H. Finite spans in a Hilbert space are closed. The projection equations are $Ga = b$. They are soluble even when the spanning list is dependent: the vector b is orthogonal to $\ker G$, since a vector in that kernel gives the zero linear combination of the ρ_j. Orthogonality gives $\|\chi - P_N\chi\|^2 = \|\chi\|^2 - \|P_N\chi\|^2 = 1 - b^*a$. Finally, for every $v \in E_N$, $\|U\chi - Uv\| = \sqrt{c}\,\|\chi - v\|$; taking infima proves the transport identity. This calculation constructs no rate of decay. □

Remark 57.4.14 *(The classical Mellin carrier).* The Mellin transform $\mathcal{M}f(t) = \int_0^\infty f(x)x^{-1/2+it}\,dx$, interpreted in the L^2 sense, is unitary from H to $L^2(\mathbb{R}, dt/(2\pi))$. Indeed, $x = e^u$ identifies f with $e^{u/2}f(e^u) \in L^2(du)$, and Fourier Plancherel applies. For $s = 1/2 + it$,

$$\mathcal{M}\chi(t) = \frac{1}{s}, \qquad \mathcal{M}\rho_n(t) = -\frac{\zeta(s)}{s}n^{-s}.$$

The second identity is the fractional-part Mellin identity and a change of variable (Báez-Duarte, arXiv:math/0205003, equation (2.6) and the identity immediately following it). This realizes the classical approximation problem on a spectral line. It does not place that line in the arithmetic Petersson carrier.

THEOREM 57.4.15 *(Conditional quantitative distance comparison).* Under the scaled-isometry hypotheses of Proposition 57.4.13, the arithmetic finite distances tend to zero if and only if RH holds. For any specified positive sequence r_N, they are $O(r_N)$, $\Omega(r_N)$, or $\asymp r_N$, respectively, if and only if ϵ_N^{BD} has the same property. The constants are multiplied by $\sqrt{c}$. In particular, transport of a sharp $(\log N)^{-1/2}$ rate requires that quantitative estimate as an additional input.

Proof. Use the exact distance identity and the discrete criterion. Multiplication by the fixed positive number $\sqrt{c}$ preserves each stated convergence or two-sided estimate and rescales its constants. No operator-positivity implication is used. □

Question 57.4.16 *(Sharp rate and operator comparison).* For the above concrete finite spaces, establish the two-sided estimate $\epsilon_N^{\mathrm{BD}} \asymp (\log N)^{-1/2}$ under precise analytic hypotheses, and determine any claimed leading constant. Construct the arithmetic map U with its normalization. To identify this quantitative condition with (Reg), or with the VB* converse, one must additionally prove an operator comparison relating these approximation projections to the specified positive, determinate spectral measure and its support. Neither convergence to zero nor the scaled isometry supplies that comparison. The classical lower-bound problem and the matching upper bound are distinct from the construction of U.

Remark 57.4.17 *(Convergence does not control its rate).* Let $E_N = \operatorname{span}(e_1, \dots, e_N) \subset \ell^2$. For any nonincreasing sequence $r_N \downarrow 0$ with $r_0 < \infty$, the vector with coefficients $x_j = (r_{j-1}^2 - r_j^2)^{1/2}$ has

$\operatorname{dist}(x, E_N) = r_N$, by telescoping the squared tail. Thus Hilbert-space density alone imposes no particular rate. This example makes no assertion about the zeta approximation spaces; it shows why the quantitative premise cannot be omitted from a formal distance-transport argument.

Remark 57.4.18 *(The forward arithmetic implication).* The distance comparison does not establish VB*, its converse, or the arithmetic Hilbert–Pólya construction. The sufficient path of Theorem 44.6.1 retains its arithmetic Positselski, pairing, determinant-trace, and positivity inputs. A classical criterion for RH does not construct those inputs.

Quantitative and Operator Inputs

Corollary 57.4.19 *(Transport of a supplied sharp rate).* Assume the map and positive scale of Proposition 57.4.13. If $A/\sqrt{\log N} \le \epsilon_N^{\mathrm{BD}} \le B/\sqrt{\log N}$ for all $N \ge N_0 > 1$, with $A, B > 0$, then the arithmetic distances satisfy the same bounds with constants $\sqrt{c}A$ and $\sqrt{c}B$. The converse follows by division by $\sqrt{c}$. Equivalence with VB*, (Reg), or (HB+) would require the separate operator comparison in Question 57.4.16.

Proof. Apply the exact finite-distance identity to each N. □

Remark 57.4.20 *(Distinct analytic requirements).* (Reg) requires a positive moment functional, determinacy, and the stated spectral support. (HB+) requires the specified de Branges operator comparison. The Báez-Duarte problem has the explicit fractional-part carrier and finite projections above. Their identification with the arithmetic operators is a construction problem; a common scalar norm or convergence assertion does not identify these structures.

Hypothesis 57.4.21 *(Arithmetic operator and quantitative comparison).* Fix the arithmetic Hilbert carrier, operators, and zero-moment functional used for VB*, together with the stated P2, P3 and H-P^{Ar} data. Supply the following additional comparisons on this same carrier.

(i) Prove that simultaneous positivity of the specified operators T_n is equivalent to (Reg): positivity and determinacy of the specified moment functional, with its required spectral support.

(ii) Supply a unitary to the specified de Branges carrier, intertwining all the T_n, including their domains. The target operator-positivity condition is denoted (HB+).

(iii) Supply the scaled isometry of Proposition 57.4.13, and prove that (Reg) is equivalent to (BD+): RH together with the specified sharp two-sided bound $\epsilon_N^{\mathrm{BD}} \asymp (\log N)^{-1/2}$. Any leading constant requires its own precise asymptotic statement.

These are construction hypotheses. In particular, item 3 is not a consequence of the scaled-isometry identity or the classical closure criterion. Question 57.4.16 retains the missing quantitative and operator construction.

Theorem 57.4.22 *(Conditional comparison of operator and approximation conditions).* Under Hypothesis 57.4.21, the predicates VB*, (Reg), (HB+), and (BD+) are equivalent. Consequently, in the RH sector their supply establishes the VB* converse. This is conditional transport of the supplied comparisons; it establishes neither item 1 nor item 3 of the hypothesis.

Proof. Item 1 supplies VB* if and only if (Reg). A unitary preserves the nonnegative quadratic forms of the intertwined operators on their transported domains, so item 2 identifies VB* with (HB+). Item 3 supplies the quantitative comparison with (BD+). Transitivity gives the asserted conditional equivalences. Merely assuming RH does not supply any of those comparison maps or estimates. □

COROLLARY 57.4.23 *(Conditional substitution of analytic inputs).* Under Hypothesis 57.4.21, any of (Reg), (HB+), or (BD+) can replace VB* in a conditional implication that uses only the stated operator positivity. The remaining hypotheses of that implication are retained.

Proof. Use the preceding conditional equivalence on the same carrier. □

57.5 RELATION BETWEEN H-P^{Ar} AND VB* CONVERSE

THEOREM 57.5.1 *(H-P^{Ar} and VB* reduce to RH plus regularity).* Under Hypothesis 57.4.21, H-P^{Ar}, and the Fock-space support identification, VB* is equivalent to RH *plus* the spectral-regularity condition of Corollary 57.4.4. Without H-P^{Ar}, the implication RH $\Rightarrow$ VB* is not meaningful, since VB* presupposes a self-adjoint $L_{\mathrm{o}}^{\mathrm{Ar}}$.

Proof. Direct from Theorem 57.4.22 combined with Theorem 57.3.14: once H-P^{Ar} is in place, $L_{\mathrm{o}}^{\mathrm{Ar}}$ has a well-defined spectral measure, and VB* $\Leftrightarrow$ RH plus spectral regularity reduces to the moment-problem equivalence with the support condition. □

PROPOSITION 57.5.2 *(H-P^{Ar} and regularity are separate data).* H-P^{Ar} (existence of a self-adjoint $L_{\mathrm{o}}^{\mathrm{Ar}}$ with prescribed determinant divisor and trace distribution) and the spectral-regularity condition (positivity, determinacy, and support of the spectral measure) are separate pieces of data:

- There exist indeterminate moment sequences with positive Hankel structure (Stieltjes' $e^{n^2/2}$ example); these admit multiple self-adjoint realisations. H-P^{Ar} asserts existence of one realisation with the right trace; regularity asserts uniqueness and support properties of the realisation.
- Conversely, a determinate moment sequence does not automatically admit a self-adjoint realisation that matches a specific target trace; the target trace (the Bost–Connes trace of Equation 57.1) is an additional constraint on the spectrum.

Proof. Akhiezer 1965 §2.4 and Simon 1998 Thm. 5.3 show that a moment sequence may admit several positive measures. Existence of one self-adjoint realisation and uniqueness of the realisation are therefore distinct assertions. The prescribed Bost–Connes trace is a further constraint. □

COROLLARY 57.5.3 *(separation of the two analytic inputs).* The two analytic inputs to VB* in the sufficient path of Theorem 44.6.1, H-P^{Ar} and the VB* converse, are not the same assertion. Hence both are required for the VB* input, and neither is redundant.

Proof. Proposition 57.5.2. □

Remark 57.5.4 *(the refined sufficient path).* The sufficient path to RH of Theorem 44.6.1 has four inputs: arithmetic Positselski, Koszul–Petersson compatibility, H-P^{Ar}, and VB*. The last two split as:

- H-P^{Ar}: existence of a self-adjoint operator obtained from $L_{\mathrm{o}}^{\mathrm{Ar}}$ by the required intertwiner, whose normal zero-divisor operator has determinant divisor div $\xi_{\mathbb{R}}$ and whose Weil–Connes trace is the corresponding zero distribution. Equivalent forms: Connes adele-class trace realisation; Deninger motivic regularised determinant with purity.
- VB*: RH *and* the spectral-regularity condition for the symmetrised zero moment sequence. The Hermite–Biehler formulation requires its operator comparison. The Báez-Duarte sharp-rate question is separate; a specified Petersson isometry transports distances but does not imply the rate or operator positivity.

Each of the two inputs is a single spectral condition. The Connes–Deninger and de Branges formulations require their stated comparison maps. The Nyman–Beurling distance problem has its separate quantitative and arithmetic comparison inputs.

57.6 Analytic Conditions

The two analytic conditions have the following forms:

(i) **(H-P^{Ar}, Friedrichs construction).** Construct the determinant-trace operator and the trace-compatible embedding for which condition (57.1) holds on the Friedrichs extension of Theorem 57.3.2. Equivalent forms:

- Connes: the $C_{\mathbb{Q}}$-scaling action on $L^2_0(X_{\mathbb{Q}})$ has absorption trace equal to the zero distribution of ζ, with correct multiplicity.
- Deninger: $\Phi|H^1_M(\overline{\mathrm{Spec}\,\mathbb{Z}}, \mathbb{R}(1))^+$ has determinant divisor div $\xi_{\mathbb{R}}$ with correct multiplicity.

(ii) **(VB* converse, regularity construction).** Establish positivity, Hamburger determinacy, and the support identification for the symmetrised zero moment sequence $\{\mu_n\}_{n\geq 0}$ of Definition 62.4.1. Proposed comparison forms:

- de Branges: operator-positivity of the reproducing kernel on $\mathcal{H}(E_\xi)$ beyond Hermite–Biehler.
- Báez-Duarte: construct the scaled arithmetic isometry and prove any proposed operator comparison; the sharp rate for ϵ_N^{BD} remains a separate quantitative question.

(iii) **(Combined VB* input).** Both (1) and (2) together supply the VB* input in the sufficient path of Vol IV. Neither alone suffices: (1) without (2) gives a specific self-adjoint operator but leaves T_n-positivity open; (2) without (1) gives abstract moment regularity without a Fock-space operator.

Remark 57.6.1 *(local form).* The Hilbert–Pólya condition is existence together with determinant divisor and trace distribution. The VB* converse is positivity, Hamburger determinacy, and support for the symmetrised zero moments. The Connes–Deninger and de Branges comparisons require their stated maps. The Báez-Duarte finite-distance construction is proved above; its arithmetic isometry, quantitative estimate, and operator comparison remain distinct inputs.

57.7 Spectral Boundary

(i) **H-P^{Ar} is determinant-trace realisation** (Theorem 57.3.14, Corollary 57.3.15): the Friedrichs extension of Theorem 57.3.2 closes the domain problem, but the determinant divisor div $\xi_{\mathbb{R}}$ and the matching Weil–Connes zero trace are the genuine Hilbert–Pólya content (Remark 57.3.16). The formulation is equivalent to Connes' adele-class trace realisation and to Deninger's motivic cohomology regularised-determinant condition.

(ii) **The proposed VB* converse requires spectral regularity and the stated Fock-space operator comparison**. Theorem 57.4.15 proves finite-distance transport, not that operator comparison. This regularity is not supplied by zero placement alone. The converse is an equivalence at the converse level only and does *not* close the forward direction VB* $\Rightarrow$zero-placement (Remark 57.4.18). Carleman's criterion remains an open possible construction (Proposition 57.4.5); abstract indeterminate moment problems show why this extra hypothesis cannot be suppressed. The de Branges operator comparison and the Báez-Duarte quantitative problem have their separate carriers and hypotheses; no sharp-rate equivalence is asserted here.

(iii) **The two analytic inputs are separate** (Proposition 57.5.2): H-P^{Ar} asserts existence of the operator; regularity asserts positivity, uniqueness, and support of the spectral measure. Both are required for the VB* input in the sufficient path; the other inputs of that path are not supplied here.

(iv) **Berry–Keating $H = \frac{1}{2}(xp + px)$ candidate** is a candidate choice of Bost–Connes zero-mode weights $\lambda(p^k) = k \log p$, reproducing the leading asymptotic zero density but not the individual zeros; rigorous identification requires the regularisation matching (57.1).

(v) **Stieltjes' example of an indeterminate moment sequence** ($\mu_n = e^{n^2/2}$) shows that the unbounded case admits genuine indeterminacy. A factorial upper bound on a specified positive functional, by contrast, implies determinacy by Carleman. Neither example constructs or estimates the arithmetic regularized functional (Remark 57.4.6).

(vi) **The refined sufficient path to RH** has four inputs (Remark 57.5.4): arithmetic Positselski, Koszul–Petersson compatibility, H-P^{Ar} with explicit Friedrichs domain and trace-compatibility, and VB* as RH plus spectral regularity. The last two are each reducible to single spectral conditions with classical comparison formulations.

(vii) **Independence of the two analytic inputs**: H-P^{Ar} does not imply the VB* converse, and the VB* converse does not construct the Hilbert–Pólya operator. The Connes, Deninger, de Branges, and Báez-Duarte formulations require their respective determinant, operator, or norm comparison maps. The distance identity supplies only the stated norm transport.

Chapter 58

The Platonic Variation and RH Completion

The arithmetic branch's correct geometric variation is an operadic adelic surface: holomorphic at ∞, Frobenius-oriented at finite primes, and glued by the pointed E_2 restricted tensor product of Iwahori Hecke categories: not a scheme-theoretic surface, not a literal almost-complex manifold at finite primes. The upgraded object $\mathfrak{S}_2^{\mathrm{Ar}}$ drives the Riemann Hypothesis completion inside Vol IV as a conditional theorem.

58.1 The Right Variation

Definition 58.1.1 *(Platonic 2-arithmetic surface).* A *Platonic 2-arithmetic surface* for the primary arithmetic branch is the stratified adelic-diamond factorisation datum

$$\mathfrak{S}_2^{\mathrm{Ar}} := \big(|\overline{C}|, \{S_v, \mathfrak{J}_v^{\mathrm{Ar}}, C_v, \eta_v, u_v\}_v, \bigotimes_{v\in|\overline{C}|}{}' (C_v, u_v)\big), \tag{58.1}$$

where:

- $|\overline{C}| = \mathbb{P} \sqcup \{\infty\}$ is the set of places of $\overline{\mathrm{Spec}\,\mathbb{Z}}$;
- $S_\infty = \mathbb{C}_\infty$, $\mathfrak{J}_\infty^{\mathrm{Ar}} = i$, and C_∞ is the archimedean holomorphic factorisation category with Harish–Chandra K_∞-spherical unit u_∞;
- for $p < \infty$, S_p is the Fargues–Scholze perfectoid-diamond Hecke chart, $\mathfrak{J}_p^{\mathrm{Ar}}$ is the Frobenius-semilinear untilt-orientation with Adams operation ψ_p and prismatic/Nygaard comparison datum, and $C_p = \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p}$ is the full non-semisimple Iwahori Hecke category with spherical unit $u_p = |0\rangle_p$;
- $\eta_v\colon \mathrm{Sp} \to C_v$ is the pointed E_2-algebra map with image u_v; at almost every finite place u_v is the monoidal unit, and at the remaining places it is a compact central idempotent algebra object;
- the tensor product in (58.1) is the Tate restricted tensor product of pointed E_2-monoidal stable presentable ∞-categories in the sense of Section 50.1.1.

The word "surface" refers to the operadic dimension: the global category is E_2-monoidal, not a smooth scheme-theoretic surface over $\mathbb{Z}$.

Remark 58.1.2 *(almost complex versus arithmetic complex).* The symbol $\mathfrak{J}^{\mathrm{Ar}}$ is literal almost-complex geometry only on $S_\infty = \mathbb{C}_\infty$. At a finite prime it is the Frobenius-oriented substitute: a semilinear orientation of the diamond chart together with Adams and prismatic comparison data. The finite place supplies no tangent-bundle endomorphism satisfying $J^2 = -1$; its role is to make the E_2 Iwahori Hecke datum functorial. This convention governs every occurrence of $\mathfrak{J}^{\mathrm{Ar}}$ in the arithmetic branch.

Remark 58.1.3 *(source anchors and dependency separation).* The finite charts use Scholze diamonds and Fargues–Scholze local Langlands geometry as listed in Section A.8 and Section A.19. The archimedean factorisation D-module fibre is the Beilinson–Drinfeld/Francis–Gaitsgory/Costello–Gwilliam fibre of Section A.5 and Section A.7. The arithmetic $\mathcal{D}^\dagger$ and overconvergent F-isocrystal language is only a local

coefficient shadow, with primary sources recorded in Section A.6. The definition of $\mathfrak{S}_2^{\mathrm{Ar}}$ is complete on the data above; F3 and the wild-prime BZN comparison enter only when identifying $\mathcal{V}^{\mathrm{prim}}$.

Question 58.1.4 *(Comparison of arithmetic base data).* Construct a comparison from the base and coefficient system of Definition 41.6.1 to the specified datum $\mathfrak{S}_2^{\mathrm{Ar}}$, preserving the archimedean factorization, the finite-place collision maps and their Adams/Frobenius operations, and the global tensor structure.

The coefficient identity of Theorem 30.3.3 does not construct the finite-place maps. A pointing gives an E_0-datum; the failure of a prescribed unary vacuum to extend in Example 2.2.2 shows why its geometric extension requires further operations. Once the local E_2-monoidal categories and their compatible unital transitions are supplied, Corollary 50.8.3 transports that structure to their restricted tensor product. Its hypotheses remain part of the proposed arithmetic comparison.

58.2 Factorisable Arithmetic Modules

The local categories C_v of $\mathfrak{S}_2^{\mathrm{Ar}}$, when supplied, have a finite tensor assembly over the place labels. Their position diagrams and geometric comparison maps remain part of the local data.

Definition 58.2.1 *(factorisable arithmetic crystal).* A *factorisable arithmetic crystal* on $\mathfrak{S}_2^{\mathrm{Ar}}$ is a functor

$$\mathscr{D}^{\mathrm{Ar}} \colon \mathrm{Fin}^{u}(|\overline{C}|) \longrightarrow \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$$

such that

$$\mathscr{D}^{\mathrm{Ar}}(S) \simeq \bigotimes_{v \in S} C_v, \qquad \mathscr{D}^{\mathrm{Ar}}(S_1 \sqcup S_2) \simeq \mathscr{D}^{\mathrm{Ar}}(S_1) \otimes \mathscr{D}^{\mathrm{Ar}}(S_2)$$

for disjoint finite subsets $S_1, S_2 \subset |\overline{C}|$, with coherent transition maps given by insertion of the specified spherical units u_v. The category Fin^{u} is defined in Definition 50.7.2; its arrows add only places with supplied unit maps. An arithmetic realization also specifies a holomorphic factorization model at infinity and local geometric models at finite places, with maps to their coefficient categories. Proposed finite models include Iwahori Hecke categories on diamond charts and spectral ind-coherent categories with nilpotent singular support. An overconvergent F-isocrystal or arithmetic $\mathcal{D}^{\dagger}$ presentation on a chosen smooth formal chart requires its own comparison. The global identification with $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}})$ remains the F1/LACD/BC-diamond-glue comparison, not a definitional equivalence.

PROPOSITION 58.2.2 *(Finite tensor assembly).* Given the $\mathcal{O}$-monoidal coefficient categories and compatible unit maps of Setup 50.1.1, the assignment $S \mapsto \otimes_{v \in S} C_v$ has the discrete factorization and transition maps required in Definition 58.2.1.

Proof. Apply Theorem 50.7.3. Its tensor associators give the disjoint-set equivalences, and its covariant unit insertions give the required transitions. □

This construction does not identify a geometric factorization D-module with a finite-place coefficient category. The holomorphic, diamond, spectral and differential presentations require their stated comparison maps. At $p = 2, 3$ no semisimple truncation is intended. To include a wild contribution in C_p, the actual local category and trace comparison of Question 54.4.15 must be supplied. Over a common field, assembling the proposed local traces also includes the mixed degree-two term of Lemma 54.4.13; an additive degree-one formula alone is not the full tensor.

58.3 The Arithmetic-Chiral Bar Complex on the Variation

The ordinary arithmetic complexes of Theorem 55.4.2 have explicit square-zero differentials. The arithmetic-chiral comparison additionally requires common coefficients, local chiral operations and residue couplings.

Definition 58.3.1 *(Finite-stage bar data).* Fix a commutative ring R and a filtered poset $\mathcal{I}$ of finite sets of places containing ∞. For each $S \in \mathcal{I}$, give cochain complexes A_∞, G_S, A_p, $p \in S_f$, over R. Their intended chiral, residue and local arithmetic realizations are additional data. Use algebraic tensor products and direct-sum totalization:

$$M_S = A_\infty \otimes_R G_S \otimes_R \bigotimes_{p \in S_f} A_p.$$

Order the factors as ∞, PG, $p_1, \dots, p_r$. On a homogeneous tensor, let $\widetilde{d}_a$ apply the differential in factor a with sign $(-1)^{\sum_{b<a} |x_b|}$. Set

$$D_{0,S} = \sum_a \widetilde{d}_a, \qquad D_S = D_{0,S} + \tau_S,$$

where the specified degree-one operator $\tau_S \in \mathrm{End}^1_R(M_S)$ satisfies

$$D_{0,S}\tau_S + \tau_S D_{0,S} + \tau_S^2 = 0.$$

Require degree-zero transition maps j_{ST} with $j_{SS} = 1$, $j_{TU}\, j_{ST} = j_{SU}$, and

$$j_{ST} D_{0,S} = D_{0,T}\, j_{ST}, \qquad j_{ST}\tau_S = \tau_T\, j_{ST}.$$

The resulting algebraic finite-support complex is $B^{\mathrm{Ar}} = \mathrm{colim}_{S \in \mathcal{I}} (M_S, D_S)$. Its arithmetic-chiral interpretation requires identification of these carriers and operators with the intended arithmetic and chiral objects.

Lemma 58.3.2 *(Insertion of closed units).* If $G_S \to G_T$ is a cochain map and each new factor has $u_p \in A_p^0$ with $d_p u_p = 0$, insertion commutes with D_0. It commutes with D when it also intertwines τ.

Proof. An inserted factor contributes $d_p u_p = 0$, and its degree zero leaves the other tensor signs unchanged. The remaining difference is $D_T\, j_{ST} - j_{ST} D_S = \tau_T\, j_{ST} - j_{ST}\tau_S$. □

Proposition 58.3.3 *(Finite-stage differential).* The data of Definition 58.3.1 give $D_S^2 = 0$ and cochain transition maps over any commutative ring R, including rings in which 2 is not invertible.

Proof. Each $\widetilde{d}_a^2 = 0$. For $a < b$, the first differential increases by one the exponent in the sign of the second, so $\widetilde{d}_a\widetilde{d}_b + \widetilde{d}_b\widetilde{d}_a = 0$. Thus $D_{0,S}^2 = 0$, and expansion gives

$$D_S^2 = D_{0,S}^2 + D_{0,S}\tau_S + \tau_S D_{0,S} + \tau_S^2 = 0.$$

The two transition identities imply $D_T\, j_{ST} = j_{ST} D_S$. □

This operator identity involves every summand of D_0. Scalar reciprocity for residue homomorphisms does not establish it. No coalgebra structure, convolution product or degree-one tame twisting operator is implicit. The ordinary complexes $C_{r,s}$ of Theorem 55.4.2 give a concrete untwisted example: $D_0 = I + J + L$, $\tau = 0$. Their mixed-square identities, norm maps, coefficient reductions and completed exterior comparison are proved there; their identification with a tame-weighted chiral bar remains a separate construction.

PROPOSITION 58.3.4 *(Comparison of finite-stage systems).* Compatible cochain maps $q_S : N_S \to M_S$ between filtered systems of cochain complexes over R induce a cochain map on their colimits. If every q_S is a quasi-isomorphism, so is the induced map.

Proof. Transition compatibility defines the map. Exactness of filtered colimits of R-modules gives $H^n(\operatorname{colim} q_S) = \operatorname{colim} H^n(q_S)$. □

Application to Definition 33.8.1 requires actual common coefficient realizations and stage maps, differential and transition identities, and quasi-isomorphism proofs. Finite support alone does not identify a single Iwasawa action with the tensor of local Iwasawa actions.

THEOREM 58.3.5 *(Closure for specified arithmetic-chiral data).* Suppose the carriers, operators and transitions of Definition 58.3.1 are constructed on the intended arithmetic-chiral domain. Their algebraic finite-support colimit is a cochain complex. For a completed version, suppose complete Hausdorff filtrations are specified and the operators and transitions descend compatibly to every quotient. Their inverse-limit extensions also have square zero and commute with transitions.

Proof. Proposition 58.3.3 proves the finite-stage identities and hence the colimit assertion. The completed identities hold in each quotient and therefore in the separated inverse limit. □

The tame-weighted arithmetic realization is the construction problem of Question 55.4.3. The common arithmetic and real coefficient complex and its comparison maps are requested in Question 55.6.6; the analytic carrier and state comparison are requested in Question 55.5.3. The ordinary bar and integral oscillator models supply local constructions for these questions. An algebraic colimit is not thereby identified with a completed restricted tensor product.

58.4 RH COMPLETION

Definition 58.4.1 *(RH completion hypothesis).* The *RH completion hypothesis* is the conjunction of:

- the Platonic surface $\mathfrak{S}_2^{\mathrm{Ar}}$ of Definition 58.1.1;
- an arithmetic-chiral realization of Definition 58.3.1, including common coefficients, degree-one operators, transition identities and the required completed comparison maps;
- the arithmetic Heisenberg Koszul biconditional, conditional on the arithmetic Positselski completion input of Corollary 55.8.2;
- the arithmetic Petersson positive-form condition, conditional on the restricted Koszul–Petersson compatibility input recorded in Corollary 47.10.1;
- H-P^{Ar} in the equivalent determinant-trace form of Theorem 57.3.14;
- VB* on the specified arithmetic carrier. It may be supplied by (Reg), meaning positivity, determinacy and the required spectral support of the identified arithmetic moment functional, together with item 1 of Hypothesis 57.4.21. Substitution of (HB+) or (BD+) additionally requires the corresponding operator or quantitative comparison from that hypothesis.

THEOREM 58.4.2 *(Platonic RH completion, conditional).* Conditional on the RH completion hypothesis of Definition 58.4.1, the Riemann Hypothesis holds.

Proof. The surface $\mathfrak{S}_2^{\mathrm{Ar}}$ and the bar complex $B^{\mathrm{Ar}}(\mathcal{V}^{\mathrm{prim}})$ of Definition 58.3.1 place $\mathcal{V}^{\mathrm{prim}}$ (a Hochschild trace class; see Remark 44.1.2) in the correct arithmetic-chiral bar framework. The conditional Koszul

input gives P1 in the abelian Heisenberg domain (Corollary 55.8.2); P2 is Tate's functional-equation/Zhu-modularity input from Theorem 43.6.1; the conditional Petersson condition supplies the positive Hermitian form needed for P3 in the restricted arithmetic branch Corollary 47.10.1. H-P^{Ar} supplies the self-adjoint operator, the zero-divisor determinant, and the required Bost–Connes trace. The specified (Reg) condition and operator comparison give VB* on the primary Fock; positivity, determinacy and the required support are all retained. Theorem 44.3.10 gives the zero-placement clause of F-RH. By Remark 44.1.2, this clause is RH. □

Corollary 58.4.3 *(A determinant–positivity sufficient condition for RH).* Assume P1, P2 and P3 on their stated domains, together with the specified H-P^{Ar} determinant–trace realization. Assume its arithmetic moment functional is positive and determinate, its representing measure has the support required by (Reg), and item 1 of Hypothesis 57.4.21 identifies (Reg) with VB* on this operator domain. Then RH holds. RH by itself is not asserted to construct this realization, support identification, or comparison.

Proof. The positive determinate moment functional and supplied support give (Reg). The assumed operator comparison gives VB*. The remaining P1–P3 and determinant–trace hypotheses now permit Theorem 58.4.2 to be applied. □

Chapter 59

Conditional Hypotheses for F1–F3 Arithmetic

Several F1–F3 arithmetic reductions are correct as equivalences, but some component subconditions require open inputs: the Fargues–Scholze-to-Emerton–Gee categorical equivalence (F1.a, an unproved comparison hypothesis), the arithmetic Arinkin–Gaitsgory conjecture ($\mathrm{L}_{\mathrm{ACD}}.1$–$\mathrm{L}_{\mathrm{ACD}}.3$), and the formal-Noetherian extension of Gaitsgory–Rozenblyum (R1.formal). The exact logical form is the conditional system determined by Section C.2, Chapter 58, and Chapter 44.

59.1 Conditional Statements

For each reduction the displayed hypothesis is part of the theorem, not an afterthought to the proof.

F1–F3 in Chapter 43

F1.a domain: Fargues–Scholze local geometrisation versus categorical equivalence. Theorem 43.2.1 requires an unproved comparison hypothesis for F1.a, the local categorical equivalence

$$\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p}_{GL_2})$$

(where $\mathcal{X}^{\mathrm{EG},p}_{GL_2}$ is the Emerton–Gee moduli stack at the prime p, $\mathrm{IndCoh}_{\mathrm{Nilp}}$ denotes ind-coherent sheaves with nilpotent singular support in the sense of Arinkin–Gaitsgory, and $\mathrm{D}^{\text{sph-fin}}$ the spherical-finite derived category of smooth $GL_2(\mathbb{Q}_p)$-representations). Fargues–Scholze arXiv:2102.13459 supplies the local geometrisation, spectral action, and Langlands-parameter structure for GL_2; the displayed categorical equivalence is not contained there. Dotto–Emerton–Gee arXiv:2603.26887 gives a fully-faithful-only, fixed-central-character functor for $GL_2(\mathbb{Q}_p)$ at $p \geq 5$. The authoritative reference for the current F1.a form is Section C.2 and Theorem 43.2.1.

Emerton–Gee deformation stack. Emerton–Gee arXiv:1908.07185 constructs the deformation-theoretic setting (*Moduli stacks of étale (φ, Γ)-modules and the existence of crystalline lifts*, Annals of Mathematics Studies). See App. A §A.8.

Coherent Springer theory. Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 treats coherent Springer theory (*Coherent Springer theory and the categorical Deligne–Langlands correspondence*, Invent. Math. 235 (2024)). See App. A §A.20.

Adelic Categorical Descent

Function-field transport and (L_2). Theorem 48.2.3 requires an unproved comparison hypothesis. The assertion (L_1) follows from Raskin's ∞-categorical construction independently of function-field Langlands; the assertion (L_2) requires the local categorical comparison F1.a (an unproved comparison hypothesis via Dotto–Emerton–Gee) and Bernstein-centre diamond gluing.

(L_2) *reduction: Fargues–Scholze domain.* Theorem 48.4.1 requires an unproved comparison hypothesis. The comparison functors Φ_v^{loc} identifying $\mathcal{C}_v$ with $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},v}_{GL_2})$ are not supplied by Fargues–Scholze arXiv:2102.13459; that source gives the local geometrisation and spectral action. The local categorical comparison is the residual open input.

F1 reduction: two open inputs. Theorem 48.9.1 conditionally states that F1 reduces to F1.a plus (BC-diamond-glue). Since F1.a remains an unproved comparison hypothesis, the reduction leaves F1 conditional on two independent open inputs: F1.a and (BC-diamond-glue).

Arithmetic Ben-Zvi–Nadler

F3.BZN^{Ar}: procyclic shadow versus full Weil parameter stack. Conjecture 49.1.1 is conjectural. The comparison target for F3.BZN^{Ar} is the full Weil/(φ, Γ) parameter stack $\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}$, not the procyclic shadow $\mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}})$, which forgets Frobenius and unramified data. The theorem is conditional on F1 and F2. Proposition 49.4.6 carries an unproved comparison hypothesis; the closed-immersion property of Proposition 49.4.9 is conjectural, since the closed-immersion property is unavailable on the inertia-only shadow.

Structural Reduction Theorems

$\mathrm{L}_{\mathrm{ACD}}$ *equivalence and subcondition forms.* Theorem 51.2.4 is proved under its stated hypotheses: the adelic categorical descent condition $\mathrm{L}_{\mathrm{ACD}}$ is equivalent to the conjunction of subconditions $\mathrm{L}_{\mathrm{ACD}}$.1 through $\mathrm{L}_{\mathrm{ACD}}$.4. The equivalence itself is a logical factorisation. The subconditions $\mathrm{L}_{\mathrm{ACD}}$.1–$\mathrm{L}_{\mathrm{ACD}}$.3 remain conjectural; $\mathrm{L}_{\mathrm{ACD}}$.4 is the combination of Kneser's strong approximation theorem for SL_2 and Tate's determinant idele-class quotient for GL_2. The residual content is recorded in the subcondition forms.

F3.BZN^{Ar} equivalence via Gaitsgory–Rozenblyum. Theorem 51.2.5 is proved under its stated hypotheses: F3.BZN^{Ar}, assembled through Gaitsgory–Rozenblyum DAG Vol. II, is equivalent to the conjunction of R1, R2, and R3. The equivalence is a valid logical factorisation. R1 holds as a theorem for smooth finite-type stacks (Gaitsgory–Rozenblyum) and is conjectural for the formal-Noetherian Emerton–Gee setting (R1.formal); R2 is an unproved comparison hypothesis; R3 is definitional. The R1 fine structure and the honest residual are recorded in Chapter 53 and Corollary 53.5.1.

Arithmetic Positselski C1–C5. Chapter 44 treats the arithmetic Positselski equivalence C1–C5 as five subconditions, of which C1–C4 are conditional on explicit hypotheses and C5 (functoriality) is the named residual. The arithmetic Heisenberg equivalence is restricted to the abelian-Heisenberg sector; C5 is the functoriality condition for the full claim.

$\mathrm{L}_{\mathrm{ACD}}$ *Subconditions*

$\mathrm{L}_{\mathrm{ACD}}$.0*: archimedean stratum construction.* Theorem 52.3.3 is proved under its stated hypotheses: the archimedean local category $\mathcal{C}_\infty$, distinguished unit, and Langlands–Vogan classification are constructed from the theorems of Harish-Chandra (1966), Vogan (1981), and Wallach (1988). Thus $\mathrm{L}_{\mathrm{ACD}}$.0 is independent of the global $\mathrm{L}_{\mathrm{ACD}}$ comparison.

L_{ACD}.1*: the F2.c hypothesis.* Theorem 52.4.1 is conditional on F2.c: the assertion L_{ACD}.1 uses the restricted tensor theorem Theorem 50.1.3 on the central idempotent pointed-unit E_2 domain. Hence Theorem 52.4.1 is conditional, with F2.c as its displayed hypothesis.

L_{ACD}.2*: local Hecke commutativity, not global functoriality.* Theorem 52.5.1 is proved under its stated hypotheses: formal tensor-factor commutativity of local Hecke endofunctors, once the restricted tensor product is constructed. The domain remark Remark 52.5.2 records that global Hecke functoriality is not part of the statement; the L_{ACD}.2 label refers to the local commutativity subcondition only.

L_{ACD}*: Bernstein-centre compatibility.* Corollary 52.7.1 assumes F2.c, the ∞-categorical Bernstein lift, the finite-prime local categorical comparison, the archimedean boundary datum, and the global identification with the automorphic category. Under these hypotheses, the remaining Bernstein-centre condition is (BC-diamond-glue). Its local construction, cross-prime gluing, and global action compatibility are the three requirements in Theorem 56.3.2. The local categorical comparison F1.a remains an input. Bernstein-centre gluing alone does not supply the other hypotheses.

BZN Formal Noetherian and EG Derived

R2: the Selmer-derived target and its formal geometry. Fix a residual representation $\bar{\rho}$, determinant, finite ramification set, and local deformation conditions $\mathcal{L}$. Chapter 53.3.5 defines the Selmer-derived stack $\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}$ as a derived fibre product of global and local deformation functors. Linearization gives its Selmer tangent complex. The local conditions must be derived formal algebraic substacks of the full $\mathrm{Weil}/(\varphi,\Gamma)$ parameter stacks. Formal Noetherianness, quasi-smoothness, and compatibility of nilpotent singular support with the Selmer cotangent complex remain hypotheses for the chosen deformation problem. A product or colimit presentation alone does not establish them.

R1: extension of the finite-type trace theorem. The Gaitsgory–Rozenblyum trace theorem for smooth finite-type stacks is the finite-type input R1.fin-type. Its extension to the Selmer-derived target requires R1.formal, including formal-loop continuity and the flat-locus comparison. Chapter 53.4.1 states this conditional reduction. The formal-Noetherian hypotheses on the arithmetic target remain separate from the finite-type theorem.

Loop Convention and Wild Primes

R3: comparison of arithmetic loop spaces. Let $\mathcal{X}$ be a formal Noetherian derived algebraic stack with cotangent complex of finite Tor-amplitude and a t-bounded Postnikov tower. Chapter 54.3.4 compares its three loop constructions under three additional assumptions: $B\widehat{\mathbb{Z}}$-mapping stacks are computed by the profinite limit of finite cyclic mapping stacks, formal completion commutes with the relevant derived tensor products, and the perfect locus generates the same loop object after ind-completion. These assumptions give the comparison equivalences. They do not identify the inertia shadow with the full $\mathrm{Weil}/(\varphi,\Gamma)$ parameter stack.

R4: large-prime wild Hochschild vanishing. For $p \geq 5$, Corollary 54.4.12 gives positive-degree vanishing only under Hypothesis 54.4.10, which specifies an actual local arithmetic trace equivalent to a module in degree zero. Semisimplicity of an unspecified category or a numerical prime threshold does not supply this comparison. At two and three, the arithmetic local models and Ext-to-trace maps remain open; their full assembly must respect the four-term tensor formula of Lemma 54.4.13.

BC-Diamond-Glue Decomposition

Local construction, gluing, and global action. The construction data in Chapter 56.3.2 give

$$\text{(BC-diamond-glue)} \iff \text{(BCglue.local)} \wedge \text{(BCglue.glue)} \wedge \text{(BC-diamond-glue.core)}.$$

This equivalence unpacks the required sheaves, gluing maps, and action compatibility. It does not construct any of them. The local comparison must connect the Ben-Zvi–Chen–Helm–Nadler construction at $\ell \neq p$ with the Emerton–Gee target at $\ell = p$, including the required coefficient comparison. The cross-prime construction requires prismatic transition maps, their triple-overlap cocycle, and continuity of the induced IndCoh construction. These data remain conjectural in Chapter 56.5.4.

Global Bernstein-centre compatibility. The core condition compares the glued action with the global Bernstein-centre action under the arithmetic Arinkin–Gaitsgory identification. That identification is itself a hypothesis. The reduction of Chapter 56.6.1 also retains F2.c, the Bernstein lift, local categorical comparison, the archimedean boundary datum, and global identification from Chapter 52.7.1. Supplying the local and cross-prime data leaves the core action compatibility to be established under those hypotheses.

Primary Source Identifiers in App. A

Local geometric and categorical inputs. Emerton–Gee arXiv:1908.07185 constructs the local moduli stacks. Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 supplies coherent Springer theory, and Fargues–Scholze arXiv:2102.13459 supplies local geometrization and spectral actions. Dotto–Emerton–Gee arXiv:2603.26887 supplies the fully faithful, fixed-central-character functor at $p \geq 5$ used in the F1.a comparison of Chapter 43.2 and Chapter 42.1. The full local categorical equivalence requires further input.

59.2 Logical Form for F1–F3 Arithmetic

The local categorical comparison F1.a requires more than the fully faithful, fixed-central-character functor at $p \geq 5$. F1.c defines the Selmer target using the chosen local deformation conditions. Its derived and Noetherian properties remain conditional on those local conditions, as in Chapter 43.2.1.

The formal descent arguments require F2.c and the ∞-categorical Bernstein lift. Local categorical comparison, the archimedean boundary datum, and global identification with the automorphic category are additional inputs to Chapter 52.7.1. The three Bernstein-centre requirements are local construction, cross-prime gluing, and compatibility of the global action. The core action comparison also assumes arithmetic Arinkin–Gaitsgory.

The structural splits in Chapter 51.2.4 and Chapter 51.2.5 organize these comparison problems. The conditional implications above use their specified construction data. A converse identifying the original global assertions with only the abbreviated lists of subconditions requires additional comparison maps and hypotheses. The strong-approximation input for SL_2 and the determinant idele-class quotient for GL_2 remain distinct from these categorical comparisons.

For the arithmetic trace comparison, the Selmer fibre product and its tangent complex specify R2. The required formal geometry and nilpotent-support compatibility remain hypotheses. R1.formal supplies the extension beyond the finite-type trace theorem. The loop comparison R3 additionally requires the three completion and generation assumptions stated above. It leaves the full $\mathrm{Weil}/(\varphi, \Gamma)$ target comparison separate. R4 includes the wild-prime contributions, with large-prime vanishing conditional

on the arithmetic semisimplicity datum. Together with F1 and F2, these are the inputs to the conditional arithmetic Ben-Zvi–Nadler construction.

59.3 Open Conditions for F1–F3 Arithmetic

The remaining F1–F3 conditions are as follows, consistent with the running list Chapter C.2 and the RH completion conjunction Chapter 58.4.1.

- F1.a (local categorical equivalence $\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p}_{GL_2})$): via Dotto–Emerton–Gee arXiv:2603.26887 (fully-faithful-only, $p \geq 5$, fixed central character). Fargues–Scholze arXiv:2102.13459 supplies the local geometrisation, spectral action, and Langlands-parameter structure; the categorical equivalence is the residual.
- F1 is conditional on F1.a and $\mathrm{L}_{\mathrm{ACD}}$; $\mathrm{L}_{\mathrm{ACD}}$ requires F2.c tightened, (BCglue.local), (BCglue.glue), and (BC-diamond-glue.core).
- F3.BZN^{Ar} requires F1, F2, R1.formal.b (profinite formal-loop continuity), R1.formal.c (flat-locus pullback), R4 (wild-prime $\mathrm{Ext}^{\mathrm{I}}$ refinement, domain beyond the explicit computation already stated), and the full-target Weil/(φ, Γ) comparison.
- Chapter 58.4.1 is the sufficient set for RH, with VB^* operator positivity as a forward positivity residual, $H\text{-}P^{\mathrm{Ar}}$ as a separate spectral hypothesis, and the arithmetic Positselski C1–C5 plus Koszul–Petersson compatibility as the four-input RH form conditions.

Chapter 60

Sharp Hypotheses for the Four-Input RH Form

The six comparisons below have different mathematical inputs. P1 requires an arithmetic evaluation, while P3 compares arithmetic and Petersson pairings. K5 supplies a spherical Whittaker normalization, and C1–C5 specify additional algebraic and analytic operations. The Hilbert–Pólya and positive-trace questions concern the spectral realization.

Tate's Poisson formula gives the scalar functional equation. The arithmetic P1 interpretation needs its defect and evaluation maps. The local Whittaker norms and occupation/radial isometries retain their arithmetic placement and adelic transport hypotheses. The Friedrichs construction supplies an operator domain. Identifying its determinant and trace with the required spectral object is a further comparison.

60.1 Six Sharp Forms

The six assertions have the following logical form.

The same list appears in Section C.10. The Platonic capstone is Theorem 58.4.2. Its sufficient arithmetic path is the four-input conjunction in §60.3.

60.2 Local Forms

Zero-Placement

Proposition 60.2.1 *(one-sided zero-placement). Conditional statement.* Assume P2, full P3, the H-P^{Ar} determinant-trace realisation, and the positive VB* trace form. Then Theorem 44.3.10 gives the zero-placement clause of F-RH. The reverse implication is not a consequence of RH; it asks for spectral regularity of the operator model.

Proof. The implication used in Theorem 44.3.10 is the positive quadratic form statement on the Bost–Connes regularised trace. P2 and P3 identify the test vectors and the Petersson form with the arithmetic trace form. H-P^{Ar} identifies the zero divisor and the Weil–Connes trace distribution. Positivity then forces the ordinates entering the Weil form to be real, hence the zeros lie on the critical line. None of these steps constructs the converse positive form from zero-placement alone. □

Arithmetic Positselski

Proposition 60.2.2 *(two arithmetic comparison criteria).* Suppose the arithmetic comodule and contramodule recipients, exact adjunction, and complete unit and counit cofibers satisfy Theorem 46.1.5. Then the adjunction is an equivalence.

Suppose also that the arithmetic central classes have the common defect and Mellin evaluation of Theorem 43.5.1, and that the evaluation detects that defect. Then their cancellation is equivalent to the completed scalar functional equation.

Proof. The first assertion is the stated completion criterion applied to the actual unit and counit. The second follows by evaluating $v - Jv = (\kappa + \kappa^!)w$ with $E(w) \neq 0$. The two assertions require different

No.	Assertion	Established domain	Additional hypothesis
1	P1 level criterion (Theorem 55.8.1)	Tate–Poisson gives the scalar functional equation.	Actual arithmetic objects, a central-defect identity, and a Mellin evaluation detecting that defect.
2	P3 Koszul–Petersson (Corollary 47.10.1, Theorem 47.9.1)	Segal form, normalized vacuum-line and occupation/radial isometries.	Arithmetic placement, action-compatible pairing maps and adelic extension, including the remaining spectral sectors.
3	K5 absorption (Theorem 56.7.5, Theorem 56.7.8)	Specified tempered spherical Whittaker norm and local Hilbert models.	Arithmetic and Hecke comparisons, continued global normalization and full-sector placement.
4	Costello refinements C1–C5 (Theorem 55.3.3, Remark 55.4.3, Question 55.5.3, Question 55.6.6, Theorem 55.7.4)	Restricted abelian-primary, $\mu = 0$, pointed E_2, and Λ-linear domains specified below.	The five Positselski hypotheses and the five Costello hypotheses.
5	H-P^{Ar} (Theorem 57.3.14)	Friedrichs self-adjoint domain.	Determinant divisor and Weil–Connes trace distribution for the scaling action.
6	VB* $\Rightarrow$ zero-placement (Theorem 44.3.10, Theorem 57.4.22 under Hypothesis 57.4.21)	One-sided implication after P2, P3, H-P^{Ar}, and positive Bost–Connes trace.	Forward positivity; the converse is only a spectral-regularity comparison.

TABLE 60.1: Sharp domains for the four-input RH form.

maps. Faithfulness of reduction modulo p is not used. It fails already on the nonzero endomorphism p of $\Lambda/(T)$. □

Koszul–Petersson Compatibility

PROPOSITION 60.2.3 *(Conditional transport of P3).* Assume the arithmetic placement, pairing maps and completed adelic extension of Hypothesis 67.1.1 on the specified normalized sph-fin cuspidal core. The transported positive Petersson form then agrees with the arithmetic bilinear pairing composed with its specified conjugation.

Proof. The model identities are $H(v, w) = B(v, Jw)$. The occupation, diagonal Hilbert–Schmidt and radial-function maps in Theorem 56.7.5 verify these identities on their local carriers. Theorem 47.5.3 constructs their normalized restricted tensor form. The assumed arithmetic maps identify that form on the stated completed domain. No equality of local scalar factors alone supplies those maps. □

Remark 60.2.4 *(The full pairing construction).* The normalized archimedean vacuum line and prime cyclic Gibbs model are genuine local constructions. Their full arithmetic realization, ramified factors and continuous-spectrum transport remain required. The restriction to a cuspidal sph-fin core does not remove the placement and pairing hypotheses even there.

Costello Refinements C1–C5

Proposition 60.2.5 *(Costello refinements).* Assume the integral co–contra comparison, the central-defect identity, and the detecting Mellin evaluation of Theorem 55.8.1. Then the P1 biconditional holds on any common arithmetic domain also satisfying the following C1–C5 requirements:

(i) C1: local factorization coalgebras, compatible unital transitions and the collision functors satisfy Theorem 55.3.3; the adelic differential comparison of Question 55.3.4 is supplied. Any required E_2-monoidal coefficient structures are specified before taking their pointed colimit;

(ii) C2: the tame-weighted chiral carriers and degree-one operators of Question 55.4.3 are constructed, with their full square-zero, norm and transition identities and comparison maps. Theorem 55.4.2 supplies a proved ordinary local model;

(iii) C3: the completed arithmetic carrier and arithmetic/BV comparison maps of Question 55.5.3 are constructed, preserving the observables, state and dynamics; any pro-nilpotent presentation has the specified filtration and compatible transition maps;

(iv) C4: the common coefficient complex and finite-place and real realization maps of Question 55.6.6 are constructed, identifying the differential, central normalization and pairing, including the real operators corresponding to the Iwasawa and Gersten components;

(v) C5: the morphism class is Λ-linear with trivial Γ-action.

Proof. The P1 conclusion follows from the explicitly assumed evaluation and defect identity by Theorem 55.8.1. C1 supplies the local coalgebras, their compatible assembly, and the required coefficient structures. C2 supplies the arithmetic-chiral differential and comparison identities. C3 supplies the completed carrier and state comparison. C4 identifies arithmetic and real realizations through a common coefficient complex. C5 specifies the allowed morphisms.

The integral co–contra criterion in Section 60.2 requires its own adjunction, complete cofibers, and reduced equivalences. □

K5 Godement–Jacquet Normalisation

Proposition 60.2.6 *(Local factors and the global continued value).* For a specified unramified tempered representation, $L_p(1, \mathrm{Ad}\,\pi_p)$ is finite and nonzero, and the spherical norm is $m_p(1 + p^{-1})L_p(1, \mathrm{Ad}\,\pi_p)$. The global continued value $c_\pi = L(1, \mathrm{Ad}\,\pi) \neq 0$ and the arithmetic pairing comparisons are separate inputs to the adelic normalization theorem.

Proof. The unit-circle parameters make each local denominator nonzero because $p^{-1} < 1$; the shell-series proof computes the norm in Lemma 56.7.4. Neither assertion uses global nonvanishing. Theorem 56.7.8 rescales two already compared forms by the common supplied scalar c_π^{-1}. Finiteness of local factors supplies neither that continued global value nor a projection excluding continuous spectrum. □

Remark 60.2.7. The propagation of K5 to full Koszul–Petersson compatibility is therefore the full P3 problem of §60.2.

H-P^{Ar} *and the* VB* *Converse*

Proposition 60.2.8 *(Friedrichs domain and determinant divisor). Conditional statement.* The Friedrichs construction of Theorem 57.3.2 gives a self-adjoint operator on its quadratic-form domain. The Hilbert–Pólya input is the further assertion that the associated normal zero-divisor operator has

determinant divisor div $\xi_{\mathbb{R}}$ and Weil–Connes trace equal to the zero distribution. This assertion is not a choice of zero-mode weights.

Proof. The Friedrichs theorem starts from a lower-semicontinuous positive form and produces a self-adjoint operator. It does not determine the regularised determinant or the Weil–Connes trace of the scaling action. The Connes and Deninger forms have the required Lefschetz shape only when the determinant divisor and the trace distribution are the non-trivial zero divisor and distribution of $\xi_{\mathbb{R}}$. Thus the operator-domain theorem and the determinant-trace theorem are distinct statements. □

PROPOSITION 60.2.9 *(Conditional operator comparisons).* Under Hypothesis 57.4.21, Theorem 57.4.22 compares the predicates VB*, (Reg), (HB+) and (BD+). (Reg) includes positivity and the specified spectral support as well as determinacy. In the RH sector, supplying one of these predicates with the comparison data establishes the VB* converse; the implication "RH implies VB*" is not itself one of the four equivalent predicates.

Proof. Item 1 of the stated hypothesis identifies operator positivity with (Reg). Item 2 transports the operators and their domains by a unitary, so that positivity is equivalent to (HB+). Item 3 supplies the separate comparison of (Reg) with (BD+). Applying these three comparisons gives the asserted equivalences. RH alone supplies none of these comparison maps or the sharp approximation bound. □

Primary-Source Anchors

Remark 60.2.10. Li–Weil positivity, Connes' trace formula, Deninger's cohomological form, and the Bost–Connes scaling action are four shadows of the same problem only after the determinant divisor and trace distribution have been specified. The symmetrised zero measure gives the VB* converse shadow; the self-adjoint scaling operator with determinant divisor div $\xi_{\mathbb{R}}$ gives the H-P^{Ar} shadow. Their coincidence is the Hilbert–Pólya content.

60.3 THE FOUR-INPUT CONDITIONAL CONJUNCTION

THEOREM 60.3.1 *(the four-input conditional conjunction). Conditional statement.* Assume the following four statements.

(i) $\mathrm{P}_1^{\mathrm{ArPos}}$: arithmetic Positselski for the specified arithmetic domain, together with the central-defect identity, Mellin evaluation, and detection condition of Theorem 43.5.1.

(ii) $\mathrm{P}_3^{\mathrm{KP}}$: full-sector Koszul–Petersson compatibility, extending the normalized sph-fin cuspidal identity by the three spectral hypotheses of §60.2.

(iii) $\mathrm{HP}^{\mathrm{Ar}}$: the H-P^{Ar} determinant-trace realisation, including determinant divisor div $\xi_{\mathbb{R}}$ and the Weil–Connes zero trace distribution.

(iv) VB_+^*: the positive Bost–Connes regularised trace form giving the forward VB* $\Rightarrow$ zero-placement implication.

Together with the Tate-Poisson direction, P2, and Hamburger determinacy, these four statements imply the zero-placement clause of F-RH. Hence the arithmetic branch implies RH under precisely this four-input conjunction.

Proof. The first hypothesis includes the evaluation and defect identity needed for Theorem 43.5.1, so it supplies P1 on that arithmetic domain. Full P3 identifies the adelic Petersson form with the arithmetic trace form on the whole primary space. $\mathrm{HP}^{\mathrm{Ar}}$ supplies the zero divisor and Weil–Connes trace

distribution attached to that trace. The positive VB_+^* form is then the Weil-positive form on the zero distribution. By the Platonic completion theorem Theorem 58.4.2, P1, P2, P3, $\mathrm{HP}^{\mathrm{Ar}}$, VB_+^*, and Hamburger determinacy imply F-RH, hence RH. □

Remark 60.3.2 *(independence of directions).* The theorem is a sufficient implication. It does not assert that RH recovers arithmetic Positselski, full Koszul–Petersson compatibility, the Hilbert–Pólya determinant-trace realisation, or the positive VB* trace form. The H-P^{Ar} and VB* residuals are separated by Proposition 57.5.2; the Positselski and Koszul–Petersson residuals are separated by Section 46.7 and Remark 47.10.4.

Remark 60.3.3 *(hypothesis support).* The P1 input is an arithmetic central-defect identity and an equivariant Mellin evaluation. A converse additionally requires detection of the specified defect vector. The coefficient and co–contra hypotheses are separate requirements. The P3 support is the KMS_1/GNS identification, the full Rankin–Selberg extension, and the Eisenstein-sector calculation. The $\mathrm{HP}^{\mathrm{Ar}}$ support is the determinant divisor div $\xi_{\mathbb{R}}$ and the Weil–Connes trace distribution. The VB_+^* support is positivity of the regularised Bost–Connes trace.

60.4 Logical Form of the Essential Biconditional

The essential biconditional

$$\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0 \iff \xi(s) = \xi(1-s)$$

(Theorems 41.2.1 and 43.5.1) holds under its specified arithmetic evaluation and central-defect hypotheses. Tate's Poisson formula proves the scalar functional equation. The forward arithmetic implication follows by evaluating the defect identity. The reverse implication additionally requires that evaluation detect the defect. It does not reconstruct an entire co–contra equivalence from a scalar equality.

In $\mathbb{C}^2$, swapping coordinates preserves their sum, although it does not fix $(1, 0)$. This is the kernel example following Theorem 43.5.1.

60.5 Consequences

(i) The scalar Tate identity is unconditional. Its arithmetic interpretation requires the defect and evaluation maps. The converse also requires detection.

(ii) P3 on the normalized sph-fin cuspidal core retains its actual arithmetic placement, pairing and adelic extension hypotheses.

(iii) K5 has explicit local Hilbert models and a computed spherical norm; the arithmetic action-compatible comparison remains to be constructed.

(iv) C1–C5 are sufficient Costello refinements on their common restricted domain and inherit the arithmetic-Positselski hypotheses.

(v) The Friedrichs theorem gives the operator domain for H-P^{Ar}; the determinant divisor and trace distribution are the Hilbert–Pólya content.

(vi) The VB* converse is a spectral-regularity comparison and is logically distinct from the positive forward implication to zero-placement.

Chapter 61

The Local Categorical Comparison and the DEG Essential Image

F1.a asks for the specified local categorical comparison at every prime, with its coefficient category, spectral support and Hecke actions. The Dotto–Emerton–Gee theorem constructs a fully faithful functor on its p-power-torsion representation carrier at $p \geq 5$. Its essential image is proper in its stated target. The arithmetic comparison therefore requires a geometric identification of the relevant image and comparison carriers, not an assertion that this existing functor is surjective onto its whole target.

61.1 Representation and Spectral Carriers

Fargues–Scholze geometrization and spectral actions concern their own sheaf and parameter categories. The intended F1.a equivalence

$$\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p})$$

requires specified comparisons with these carriers. The coefficient regime, the meaning of the sph-finite realization, and the nilpotent-support restriction are part of the construction. They cannot be substituted into a fully faithful theorem on different categories without a comparison functor.

61.2 The Local and Family Construction Problems

At $p \geq 5$, the problem is to relate the actual DEG image to the chosen arithmetic spectral category, construct action-compatible functors and identify lifted generators. At two and three, the wild-enhanced carrier, the block functors and their exhaustion are separate data (H2)–(H4). Finite extension calculations can constrain these data but do not construct the functors. The three-vertex model of Question 54.4.9 retains its separate arithmetic realization problem.

Varying a central character requires effective category-valued descent on a specified parameter site, including nonreduced families and higher overlap maps. Point-character comparisons do not suffice. Even complete local comparisons leave the cross-prime and global centre maps required for adelic descent.

61.3 Dotto–Emerton–Gee Recalled

The DEG result, stated precisely, is the theorem on which the rest of the argument builds.

The Proven Coefficient and Spectral Carriers

Fix $p \geq 5$, the ring of integers $\mathcal{O}$ of a finite extension of $\mathbb{Q}_p$, and a central character $\zeta : \mathbb{Q}_p^\times \to \mathcal{O}^\times$. Let $\mathcal{A}$ be smooth $\mathcal{O}$-representations on which p acts locally nilpotently, with central character ζ. Let X be the formal stack of rank-two projective étale (φ, Γ)-modules of determinant $\zeta\varepsilon^{-1}$, where ε is the cyclotomic character.

Theorem 61.3.1 *(Dotto–Emerton–Gee, Theorem 1.1.1).* There is an $\mathcal{O}$-linear fully faithful exact functor $D^b_{\mathrm{fp}}(\mathcal{A}) \to D^b_{\mathrm{coh}}(X)$. It extends to a colimit-preserving fully faithful functor

$$F^{\mathrm{DEG}} : D(\mathcal{A}) \longrightarrow \mathrm{Ind}\, D^b_{\mathrm{coh}}(X).$$

Reference. Dotto–Emerton–Gee, arXiv:2603.26887, Theorem 1.1.1 and the paragraph following it. The extension uses their compact generation statement for $D(\mathcal{A})$. □

Remark 61.3.2 *(The essential-image boundary).* Remark 1.1.2 of that paper states that this functor is not an equivalence. Thus exhaustion of its entire displayed target is not an unproved consequence to be appended to this theorem. The geometric description of its proper essential image, its relation to a specified nilpotent-support category, and changes of coefficient or representation carrier are separate comparisons. The theorem also does not supply the wild-prime or varying-character construction.

Proposition 61.3.3 *(The constructed essential-image carrier).* Let $\mathcal{I}_{p,\zeta}$ be the full essential image of F^{DEG} in its actual target. Then $D(\mathcal{A}) \simeq \mathcal{I}_{p,\zeta}$, and $\mathcal{I}_{p,\zeta}$ is closed under small colimits.

Proof. Full faithfulness identifies the source with this full subcategory. Every diagram in it lifts to the source; preservation of colimits places its colimit back in the image. This categorical construction does not identify the image with a specified support condition or with a category generated by named cycle sheaves. □

For the F1.a comparison below, use the notation $\Phi^{\mathrm{cmp}}_{p,\chi}$ for a chosen functor on the specified arithmetic representation and nilpotent-support carriers. Constructing these carriers, the functor, and its relation to the actual F^{DEG} is part of (H1). It is not a coefficient substitution in the preceding theorem.

Why Essential Surjectivity is Subtle

Remark 61.3.4 *(intuition for obstruction (a)).* The sub-∞-category $\mathrm{IndCoh}^{\text{BM-image}}_{\mathrm{Nilp}}$ is only a test subcategory generated by Breuil–Mezard cycles. A possible obstruction to essential surjectivity is the structure sheaf of an intersection of two such cycles which is not itself a named Breuil–Mezard cycle. Whether every such object lies in the triangulated colimit closure of the test generators is the block-exhaustion question.

61.4 Obstruction (a): Essential Surjectivity at $p \geq 5$

The Colmez–Paskunas Block Decomposition

Theorem 61.4.1 *(Componentwise functors on specified stable blocks).* Suppose presentable stable categories and compatible decompositions $\mathcal{C} \simeq \prod_\rho \mathcal{C}_\rho$, $\mathcal{D} \simeq \prod_\rho \mathcal{D}_\rho$ are specified. A family of fully faithful exact colimit-preserving functors $\Phi_\rho : \mathcal{C}_\rho \to \mathcal{D}_\rho$ gives a functor with these properties on the products.

Proof. Mapping spectra and colimits in a product are computed componentwise. Apply the given properties in every component. □

The Required Spectral Block Models

Proposition 61.4.2 *(Image support under a categorical block comparison).* Suppose the chosen comparison functor is identified with the componentwise functor under specified equivalences

$$C \simeq \prod_{\rho} C_{\rho}, \qquad \mathcal{D} \simeq \prod_{\rho} \mathcal{D}_{\rho}, \qquad \mathcal{D}_{\rho} = \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p,\chi,\rho}_{GL_2}).$$

Then $\Phi(C_{\rho}) \subseteq \mathcal{D}_{\rho}$. These are categorical decomposition and support hypotheses; a formal stack is not thereby identified with a deformation ring.

Proof. An object of one component is the tuple whose other components are zero. The componentwise functor preserves this property. □

Corollary 61.4.3 *(Reduction to the specified stable blocks).* Under these supplied decompositions, Φ is essentially surjective if and only if every Φ_{ρ} is essentially surjective.

Proof. Choose preimages componentwise in one direction. In the other, apply essential surjectivity to a target tuple supported in one component. Its preimage gives the required component preimage. □

The Block-Exhaustion Question

Definition 61.4.4 *(Block exhaustion for the chosen comparison).* For the specified stable categories $C_{\rho}, \mathcal{D}_{\rho}$, a block is exhausted if the chosen $\Phi^{\mathrm{cmp}}_{p,\chi,\rho}$ is essentially surjective. A cycle class alone supplies neither an object of the target nor its preimage through this functor.

Conjecture 61.4.5 *(The arithmetic block comparison datum).* For $p \geq 5$, construct the common coefficient carriers, the specified stable block decompositions and image-support comparisons for F1.a. Construct exact colimit-preserving fully faithful $\Phi^{\mathrm{cmp}}_{p,\chi,\rho}$ with the required Hecke compatibilities, and actual lifts of a localizing generating family of $\mathcal{D}_{\rho}$. Identify their relation to the proper essential image of F^{DEG}. This does not assert that the latter functor is an equivalence onto its whole target.

Remark 61.4.6 *(Objects and their lifts).* A Breuil–Mezard construction must provide actual spectral objects Q_i, their nilpotent singular support and generation, together with objects P_i and equivalences $\Phi(P_i) \simeq Q_i$. Generation by target objects alone is insufficient. Matching degree-one extensions also does not prove full faithfulness on mapping spectra.

Proposition 61.4.7 *(Exhaustion by lifted generators).* Let $F : C \to \mathcal{D}$ be fully faithful, exact and colimit-preserving between presentable stable categories. If Q_i generate $\mathcal{D}$ as a localizing subcategory and specified equivalences $F(P_i) \simeq Q_i$ exist, then F is an equivalence.

Proof. Full faithfulness lifts diagrams in the image; preservation of colimits puts their colimits in the image. Exactness gives closure under shifts and cones. Thus the image is a localizing subcategory containing the generators, and is the whole target. □

For example, $D(k) \to D(k) \times D(k)$, $M \mapsto (M, 0)$, is fully faithful, exact and colimit-preserving, but misses $(0, k)$. The two component structure sheaves $(k, 0)$, $(0, k)$ generate the target; only one is in this functor's image. When full faithfulness also needs proof, the compact-generator criterion of Theorem 70.4.1 requires equivalences on the full mapping spectra, not just $\mathrm{Ext}^{\mathrm{I}}$.

THEOREM 61.4.8 *(conditional essential surjectivity at $p \geq 5$).* Assume all the carrier, functor and lifted-generator data of Conjecture 61.4.5. Then for every $p \geq 5$ and every central character χ, the functor $\Phi^{\mathrm{cmp}}_{p,\chi}$ is an equivalence

$$\Phi^{\mathrm{cmp}}_{p,\chi} \colon \mathrm{D}^{\text{sph-fin}}\big([GL_2(\mathbb{Q}_p)]\big)_{\chi} \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2}\big).$$

Proof. The lifted-generator proposition makes each chosen block functor an equivalence. The supplied categorical decompositions assemble them componentwise. The full faithfulness used here is part of the chosen comparison datum, not an unproved identification of its carriers with those of the DEG theorem. □

The Coefficient and Support Comparison

The arithmetic block datum includes its common coefficients, presentable stable realizations, categorical decompositions, Hecke compatibilities and actual lifted generators. The proven DEG functor supplies none of these identifications by a change of notation. Its proper image is a concrete categorical starting point; a geometric description of that image and the intended nilpotent support category is the remaining comparison.

61.5 OBSTRUCTION (B): EXTENSION TO $p = 2, 3$

Why DEG Does Not Extend Directly

At $p \in \{2, 3\}$ the obstructions to DEG are multiple:

PROPOSITION 61.5.1 *(Data required at the wild primes).* The theorem for $p \geq 5$ supplies no functor at two or three. A wild comparison requires its own spectral carrier, exact colimit-preserving fully faithful block functors, and their support and family compatibilities. A list of types or finite extension dimensions does not specify these categorical maps.

Proof. These primes are outside the proven theorem's hypotheses. Constructing a functor requires its action on objects and mapping spectra, with composition and higher coherences. Neither a type label nor an extension dimension gives these data. The following hypotheses retain the actual required constructions. □

The Wild-EG Enhancement Programme

The reduction is two-step: (i) construct a wild-block DEG functor via Bushnell–Henniart types and local Langlands normalisations; (ii) pair it with an enhanced Emerton–Gee stack incorporating the wild inertia data absent from the fixed-character setup.

Definition 61.5.2 *(wild-enhanced Emerton–Gee comparison datum).* For $p \in \{2, 3\}$ and central character χ, a *wild-enhanced Emerton–Gee comparison datum* consists of a derived formal algebraic stack $\mathcal{X}^{\mathrm{EG,wild},p,\chi}_{GL_2}$, a map to $\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2}$, and specified comparisons of its tangent and trace complexes with the corresponding arithmetic block data, respecting the chosen coefficients and local Langlands normalizations. Existence is an additional hypothesis. Adding a vector space of a prescribed dimension does not construct this datum.

Remark 61.5.3 *(the proposed degree-one datum at two).* The intended local degree-one contribution is a line, as in Question 54.4.8. Its identification with a derived fibre requires an actual comparison map. In characteristic two, trivial and sign characters coincide. The dual-number model has a one-dimensional

self-extension but also has nonzero extensions in every higher degree; it does not prove that the arithmetic derived fibre is concentrated in degree one.

Remark 61.5.4 *(the proposed degree-one datum at three).* The intended local degree-one contribution is the sum of two lines, as in Question 54.4.9. In the explicit right-module three-vertex model these are $\mathrm{Ext}^1(L_0, L_1)$ and $\mathrm{Ext}^1(L_1, L_2)$; their Yoneda product gives a nonzero degree-two class. A comparison with an arithmetic block and a spectral derived fibre must account for this product and specify the actual relations, rather than discard the higher degree.

Construction of the Wild DEG Functor

Conjecture 61.5.5 *(wild block functor datum).* For $p \in \{2, 3\}$ and fixed χ, construct presentable stable representation blocks C_τ^{BH}, their common coefficient category, and compatible categorical block decompositions on both sides. Construct exact, colimit-preserving, fully faithful functors

$$\Phi^{\mathrm{wild}}_{p,\chi,\tau} : C_\tau^{\mathrm{BH}} \longrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG,wild},p,\chi,\tau}_{GL_2})$$

with the stated type and Hecke compatibilities, assembling under those decompositions. Type labels alone do not construct these maps.

Theorem 61.5.6 *(existence of wild DEG functor at $p = 2, 3$).* Fix $p \in \{2, 3\}$ and the common coefficient ring R of the wild comparison datum, with its stated coefficient comparisons. Fix a central character $\chi : \mathbb{Q}_p^\times \to R^\times$. Assume Conjecture 61.5.7 below (existence of the wild-enhanced EG stack with the stated structural properties) and Conjecture 61.5.5. Then there exists a fully faithful functor

$$\Phi^{\mathrm{wild}}_{p,\chi} : \mathrm{D}^{\text{sph-fin}}\big([GL_2(\mathbb{Q}_p)]\big)_\chi \longrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{EG,wild},p,\chi}_{GL_2}\big)$$

extending DEG to the wild primes, compatible with the Bushnell–Henniart fiber correspondence on each block.

Proof. The wild-enhancement hypothesis supplies the spectral carrier. The wild block functor datum supplies compatible stable block decompositions and component functors. Form their componentwise functor and transport it through the supplied equivalences. Mapping spectra and colimits in a product are computed componentwise, so full faithfulness, exactness and colimit preservation follow. The type and Hecke compatibilities are those of the supplied datum. □

Conjecture 61.5.7 *(wild-EG enhancement).* Conjectural statement. For $p \in \{2, 3\}$ and a central character χ, there exists a derived formal algebraic stack $\mathcal{X}^{\mathrm{EG,wild},p,\chi}_{GL_2}$ with the following properties:

(i) The truncation $\tau_{\leq 0}\mathcal{X}^{\mathrm{EG,wild},p,\chi}_{GL_2}$ equals the classical Emerton–Gee stack $\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2}$.

(ii) A specified cotangent complex and deformation comparison, including infinitesimal automorphisms of the unframed stack. At a field-valued point x, its positive cohomological degree must retain $H^1(\mathbb{L}_x) = \mathrm{Lie}(\mathrm{Aut}(x))^\vee$. A two-term amplitude for a framed or relative chart requires that chart and its comparison map; it is not an absolute amplitude claim. Any proposed degree-one block dimensions must be established by the actual arithmetic class comparison, not prescribed uniformly by $p = 2$ or $p = 3$.

(iii) There is a fiber correspondence, compatible with local Langlands, between the wild components of $\mathcal{X}^{\mathrm{EG,wild},p,\chi}_{GL_2}$ and the Bushnell–Henniart wild types of $GL_2(\mathbb{Q}_p)$.

(iv) Ind-coherent sheaves with nilpotent singular support on $\mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi}$ are generated under colimits and shifts by the derived structure sheaves of the Bushnell–Henniart-indexed closed points.

LEMMA 61.5.8 *(A split-point automorphism direction).* At a split rank-two point $M_1 \oplus M_2$ with fixed determinant, the unframed deformation stack has a nonzero infinitesimal automorphism direction. Its absolute cotangent complex cannot have amplitude contained in $[-1, 0]$.

Proof. The automorphisms $\mathrm{diag}(t, t^{-1})$ preserve the summands and determinant. Over $k[\epsilon]/(\epsilon^2)$, putting $t = 1 + \epsilon a$ gives $1 + \epsilon\,\mathrm{diag}(a, -a)$, a nonzero direction also in characteristic two. In cohomological conventions, infinitesimal automorphisms are $H^{-1}(T_x)$; dualizing gives a nonzero $H^1(\mathbb{L}_x)$. For instance $\mathbb{L}_{B\mathbb{G}_m}|_x = \mathfrak{g}_m^*[-1]$, which is concentrated in degree $+1$. Retaining the classical unframed stack retains this direction. □

Remark 61.5.9 *(inputs to the wild-EG comparison).* The intended comparison combines smooth representation blocks, Galois deformation data and an arithmetic trace construction. Questions 54.4.8 and 54.4.9 ask for the coefficient-compatible local maps. The explicit finite-algebra resolutions establish possible extension patterns, including higher Ext; they do not establish the arithmetic fibre dimensions. The existence of a single derived formal stack with these maps, and compact generation by the proposed closed-point objects, remain parts of the wild-EG hypothesis.

Remark 61.5.10 *(domain of the wild-EG conjecture).* The wild-EG conjecture requires a derived enhancement and actual comparison maps with arithmetic block data. The classical stack and the finite-algebra examples do not by themselves construct that enhancement. In particular, the local resolutions in Chapter 54 neither determine the arithmetic derived fibres nor identify their trace classes.

BH-Block Exhaustion and Essential Surjectivity at Wild Primes

Conjecture 61.5.11 *(BH-block exhaustion at wild primes). Conjectural statement.* For $p \in \{2, 3\}$, every central character $\chi : \mathbb{Q}_p^\times \to R^\times$, and every Bushnell–Kutzko type τ (tame or wild) in the support of $\mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi}$, the pair

$$\left(\mathrm{Block}_\tau^{\mathrm{BH}},\ \mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi,\tau}\right)$$

is exhausted in the sense of Definition 61.4.4: the chosen $\Phi_{p,\chi,\tau}^{\mathrm{wild}}$ is essentially surjective. A sufficient construction lifts actual target generators through this exact colimit-preserving fully faithful functor, as in Proposition 61.4.7. Generation by named spectral objects without these image identifications is not an equivalent condition.

COROLLARY 61.5.12 *(wild essential surjectivity, conditional).* Assume Conjectures 61.5.7, 61.5.5, and 61.5.11. Then $\Phi_{p,\chi}^{\mathrm{wild}}$ of Theorem 61.5.6 is an equivalence

$$\mathrm{D}^{\text{sph-fin}}\left([GL_2(\mathbb{Q}_p)]\right)_\chi \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}\left(\mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi}\right).$$

Proof. Fully-faithfulness is Theorem 61.5.6. Essential surjectivity follows block-by-block from the BH-block analogue of Corollary 61.4.3 plus Conjecture 61.5.11. □

Categorical Hypotheses for Obstruction (b), $p \in \{2, 3\}$

Local algebra and arithmetic comparison. The finite-algebra calculations of Section 54.4 use explicit resolutions. Their arithmetic realization requires the coefficient-compatible block and trace maps in the two local comparison questions. A bibliography of local representation constructions does not supply these maps.

Conventions. Bushnell–Henniart type conventions match those of Emerton–Gee arXiv:1908.07185 via the standard dictionary (Bushnell–Henniart §18 Thm A). $\mathrm{Ext}^{\mathrm{I}}$ conventions follow Jantzen 2003 Ch. II §8.

Ambient categories. Wild-enhanced EG stack sits in derived formal algebraic stacks; $\mathrm{IndCoh}_{\mathrm{Nilp}}$ on a derived formal stack is computed via the Gaitsgory–Rozenblyum ind-completion. The LHS of (61.5.6) is the same as before.

Hypotheses. The three wild inputs are Conjecture 61.5.7 (H2), Conjecture 61.5.5 (H3), and Conjecture 61.5.11 (H4). They specify the enhanced stack, the actual block functor, and its essential surjectivity, respectively.

Functoriality. Hecke compatibility and the maps between blocks are part of the functor datum (H3). Orthogonality controls cross-block morphisms once those categories and functors exist; a type classification alone does not construct the Hecke intertwiners.

Remaining constructions. All three inputs (H2)–(H4) are retained. Constructing the target stack or naming its types does not by itself construct the functor or prove its essential surjectivity.

61.6 Obstruction (c): Central Character Variation

The Central-Character Parameter and Its Families

A fixed central character is a continuous homomorphism $\chi : \mathbb{Q}_p^\times \to k^\times$. To vary it, fix a coefficient regime and a parameter site $\mathcal{T}_p^{\mathrm{cent}}$ for the intended character families, including any nonreduced coefficient algebras. Its topology and representability are part of the family construction; a list of point-valued characters is not a descent model.

Throughout this family construction, write $\mathcal{X}_p^{\mathrm{fam}}$ for the chosen spectral carrier: the classical Emerton–Gee carrier for $p \geq 5$, and the chosen wild-enhanced carrier of (H2) for $p \in \{2, 3\}$. In the latter case its fixed-character carriers and functors are the data of (H2) and (H3), with block exhaustion as in (H4). The family carrier and its base-change identifications are part of the descent model below; they are not inferred from point-valued characters. The chosen fixed-character functors are the chosen arithmetic comparison functors of (H1), related to the actual DEG image by their supplied maps, and the wild block functors at two and three. Denote their sought family extension by Φ_p^{fam}.

Definition 61.6.1 *(Central-character descent model).* A central-character descent model consists of a specified cover $U \to \mathcal{T}_p^{\mathrm{cent}}$, its Čech diagram $U_\bullet$, and compatible diagrams of presentable stable categories $\mathcal{C}(U_\bullet)$, $\mathcal{D}(U_\bullet)$, with equivalences

$$\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \mathrm{Tot}\,\mathcal{C}(U_\bullet), \qquad \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_p^{\mathrm{fam}}) \simeq \mathrm{Tot}\,\mathcal{D}(U_\bullet).$$

The models must realize the stated fixed-character categories under the chosen base-change operations. Equivalence must be detected on this conservative descent cover, not merely on its point characters.

Remark 61.6.2 *(Generalized central characters).* Smooth representations need not split into a direct sum of strict central-character subrepresentations. For an explicit example, let k have characteristic different from two and let $N \neq 0$ on k^2 satisfy $N^2 = 0$. The formula

$$g \longmapsto (1 + N)^{v_p(\det g)} = 1 + v_p(\det g)N$$

defines a smooth representation of $GL_2(\mathbb{Q}_p)$: the valuation is additive and locally constant. The central element pI acts by $1 + 2N$, which is not scalar and is not semisimple. Its two composition factors have the trivial central character, but the extension is not a direct sum of strict-character objects. A family model must retain this extension data.

Conjecture 61.6.3 *(Central-character family comparison datum).* Construct the descent models of Definition 61.6.1 and a morphism $\Phi_\bullet : C(U_\bullet) \to \mathcal{D}(U_\bullet)$ extending these chosen fixed-character functors. Prove that each Φ_n is an equivalence on the specified conservative cover and all its higher overlaps, compatibly with the descent maps, Hecke actions, coefficient changes and central/determinant twists. Naturality at point characters alone is not this datum.

Assembly of the Family Functor

Theorem 61.6.4 *(Central-character variation by effective descent).* Assume the actual models and levelwise equivalences of Conjecture 61.6.3, with the fixed-character block comparisons on their stated domains. Then their totalization gives an equivalence

$$\Phi_p^{\mathrm{fam}} : \mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_p^{\mathrm{fam}}),$$

compatible with the supplied family and Hecke data.

Proof. A natural transformation of diagrams that is an equivalence at every level is an equivalence in the functor category. Applying the limit functor preserves equivalences. Compose $\mathrm{Tot}\,\Phi_\bullet$ with the two specified descent model equivalences to obtain the displayed functor. Its compatibilities follow from those of the diagram. The proof uses actual overlap data; pointwise equivalences do not replace them. □

Remark 61.6.5 *(Fixed characters and families).* Fixed-character block exhaustion is only one input to the family problem. The coefficient extensions, conservative cover, overlap comparisons and descent models remain the construction required by Conjecture 61.6.3.

Categorical Hypotheses for Obstruction (c)

Local representation-theoretic inputs. The central-character-variation argument uses Gaitsgory–Rozenblyum Vol. I Ch. 9–10 (fibrations of categories) plus DEG arXiv:2603.26887. The passage from fixed-character functors to a family requires the effective descent and conservative-cover datum stated above.

Conventions. The determinant-fibration convention on the classical Emerton–Gee carrier follows Emerton–Gee arXiv:1908.07185 §2.2; its wild-enhanced analogue and compatibility are part of the chosen family datum. The sheaf-of-categories convention follows Gaitsgory–Rozenblyum Vol. I Ch. 10 Thm 10.3.2.1.

Ambient categories. Sheaves of presentable stable ∞-categories over $\mathcal{T}_p^{\mathrm{cent}}$; the totalisation equivalence is an equivalence in this ambient.

Hypotheses. The effective descent models and compatible equivalences over the conservative cover are not imported from the fixed-character theorem. They form the separate hypothesis Conjecture 61.6.3. The fibre-to-total passage is formal only after that hypothesis is in place.

Functoriality. The twisting morphism $\chi' \to \chi$ is compatible with both sides only under Conjecture 61.6.3.

Unproved comparisons. The actual cover-level equivalences, including coefficient extensions and higher overlaps, remain to be constructed as part of the family datum. The totalization theorem is conditional on them.

The central-character family datum is a genuine residual.

61.7 Fi.a Reduction

Assembly of (a), (b), (c)

Theorem 61.7.1 *(Fi.a at all primes, conditional).* Assume:

(H1) Conjecture 61.4.5 at every $p \geq 5$ and every central character χ.

(H2) Conjecture 61.5.7 at $p \in \{2, 3\}$ and every χ.

(H3) Conjecture 61.5.5 at $p \in \{2, 3\}$ and every χ.

(H4) Conjecture 61.5.11 at $p \in \{2, 3\}$ and every χ.

(H5) Conjecture 61.6.3 at every p.

Then for every prime p and every continuous central character χ there exists an equivalence

$$\Phi_{p,\chi}^{\mathrm{LocCat}} \colon \mathrm{D}^{\text{sph-fin}}\big([GL_2(\mathbb{Q}_p)]\big)_{\chi} \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}_{GL_2}^{\mathrm{EG,wild\text{-}enhanced},p,\chi}\big),$$

where the spectral side is the classical Emerton–Gee stack for $p \geq 5$ and its wild enhancement for $p \in \{2, 3\}$, and the family over χ assembles to the displayed Fi.a equivalence

$$\Phi_{p}^{\mathrm{LocCat}} \colon \mathrm{D}^{\text{sph-fin}}\big([GL_2(\mathbb{Q}_p)]\big) \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}_{GL_2}^{\mathrm{EG,wild\text{-}enhanced},p}\big).$$

Local Hecke-functoriality at the place p is part of the local categorical input; it is not inferred from full faithfulness alone.

Proof. Assemble Theorem 61.4.8 (valid for $p \geq 5$ under (H1)), Corollary 61.5.12 (valid for $p \in \{2, 3\}$ under (H2), (H3), and (H4)), and Theorem 61.6.4 (assembles the family under (H5)). Per-χ equivalences (from (H1)+(H2)+(H3)+(H4)) combine with the family decomposition to produce the total equivalence (61.7.1). □

Unconditional DEG Full Faithfulness

Theorem 61.7.2 *(The proven fixed-character embedding).* On the exact coefficient and spectral carriers of Theorem 61.3.1, F^{DEG} is fully faithful and colimit-preserving. It is not an equivalence onto its entire displayed target.

Proof. The theorem and Remark 1.1.2 of the cited paper give these assertions. □

Corollary 61.7.3 *(Equivalence onto the actual essential image).* For the same data, $D(\mathcal{A}) \simeq \mathcal{I}_{p,\zeta}$.

Proof. Apply Proposition 61.3.3. □

Remark 61.7.4 *(The arithmetic carrier remains to be identified).* The equivalence onto this proper image does not identify the sph-finite arithmetic carrier with a nilpotent-support category. That geometric comparison is part of (H1), and the wild and family constructions remain (H2)–(H5).

Residuals

R1a: the arithmetic comparison at $p \geq 5$. Hypothesis (H1) requires common coefficients and support, the chosen functor and actual lifted generators. It does not assert essential surjectivity of the known DEG functor onto its entire target.

R1b: the wild spectral carrier. Hypothesis (H2) requires the stacky deformation comparison and its geometric carrier. It supplies no wild representation functor alone.

R1c: the wild functor and its image. Hypotheses (H3) and (H4) require the exact colimit-preserving fully faithful functor, its categorical decomposition and lifted target generators. The wild equivalence uses (H2), (H3) and (H4) together.

Central-character variation (c) requires the family datum of Conjecture 61.6.3. Per-χ equivalences do not supply their naturality or descent as χ varies.

61.8 Consequence for F1 and $\mathrm{L}_{\mathrm{ACD}}$

Local F1.a Closed at Every Prime Suffices for Adelic F1

Theorem 61.8.1 *(Adelic F1 from local comparison and descent).* Conditional statement. Assume the conditional closure of F1.a at every prime p of Theorem 61.7.1 (i.e. assume (H1)–(H5)), together with the specified adelic descent data. Then the F1 obstruction is reduced to the current $\mathrm{L}_{\mathrm{ACD}}$ table of Corollary 43.2.3:

$$\mathrm{D}^{\text{sph-fin}}\big([GL_2(\mathbb{Q})\backslash GL_2(\mathbb{A}_{\mathbb{Q}})]\big) \;\simeq\; \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}\big).$$

Proof. Apply Corollary 43.2.3: F1 holds given (F1.a) and $\mathrm{L}_{\mathrm{ACD}}$. Our Theorem 61.7.1 reduces (F1.a) under (H1)–(H5), up to the wild-enhancement of the local conditions at $p \in \{2, 3\}$. These wild conditions must be inserted into the Selmer-derived stack as modified factors $\mathcal{X}^{\mathrm{EG,wild\text{-}enhanced},p}_{GL_2}$ in the local condition $\mathcal{L}_p$. The global equivalence (61.8.1) therefore uses the Selmer-derived target on the right-hand side; it is not an ind-Noetherian product of local stacks. □

Remark 61.8.2 *(three local pieces).* Under the assumptions (H1)–(H5) of Theorem 61.7.1 plus $\mathrm{L}_{\mathrm{ACD}}$, F1 holds. The assumptions (H1)–(H5) are the local-categorical-comparison residuals of the DEG reduction; $\mathrm{L}_{\mathrm{ACD}}$ is the adelic-categorical-descent residual of Remark 43.2.4. The F1 obstruction set therefore has six items: (H1)–(H5) and $\mathrm{L}_{\mathrm{ACD}}$.

Refined Chapter 43 Reduction

Proposition 61.8.3 *(Local comparison inputs and adelic descent).* The sufficient input set for the stated F1 comparison consists of (H1)–(H5) from Theorem 61.7.1 and the adelic descent datum $\mathrm{L}_{\mathrm{ACD}}$. Here (H3) constructs the wild block functor, (H4) requires its block exhaustion, and (H5) supplies the central-character family comparison. These are distinct requirements.

$$\mathrm{Inputs}(F1) = \{\mathrm{H1}, \mathrm{H2}, \mathrm{H3}, \mathrm{H4}, \mathrm{H5}, \mathrm{L}_{\mathrm{ACD}}\}.$$

This is a sufficient construction route, not a proof that each input is logically necessary for every possible F1 equivalence.

Proof. Under (H1)–(H4), the specified block functors give the per-character local comparisons in the stated coefficient categories. Hypothesis (H5) supplies their compatible family as the character varies.

The adelic descent datum then assembles these local comparisons on the specified Selmer target, as in Theorem 61.8.1. Neither a family of unrelated equivalences nor essential surjectivity without a constructed functor performs these steps. □

Global Assembly and Its Hypotheses

The global comparison uses the local categorical comparisons together with the F1 reduction of Corollary 43.2.3. Its inputs are (H1)–(H5) and L_{ACD}. The spectral geometry is formulated in ind-Noetherian formal algebraic stacks over Spec $\mathbb{Z}$, while the automorphic geometry uses the adelic double coset. The comparison concerns their associated presentable ∞-categories.

The global Emerton–Gee construction uses the ind-Noetherian limit of Emerton–Gee, §2.3. Incorporating the proposed wild enhancements at the two small primes requires compatibility with that limit. Assembly over primes is a restricted tensor product, and its compatibility with cross-prime Hecke functoriality is part of L_{ACD}. Thus the global consequence retains the local, wild-functor, family, and adelic descent hypotheses.

61.9 Residual Conditions

Residual Conditions for the DEG Extension

The DEG extension has five residual inputs:

(i) Conjecture 61.4.5 (**R1a/F1.a.1**): the specified arithmetic comparison functor, common coefficients and support, and actual lifts of spectral generators at $p \geq 5$. This is not essential surjectivity of the known DEG functor onto its entire target; that functor has a proper image.

(ii) Conjecture 61.5.7 (**R1b/F1.a.2**): existence of a derived formal algebraic wild-enhanced EG stack at $p \in \{2, 3\}$ with four structural properties.

(iii) Conjecture 61.5.5 (**R1b'**): existence of the Bushnell–Henniart-compatible per-block fully faithful functor at $p \in \{2, 3\}$.

(iv) Conjecture 61.5.11 (**R1c/F1.a.3**): the wild-block analogue of Conjecture 61.4.5; essential surjectivity per Bushnell–Kutzko type at $p \in \{2, 3\}$.

(v) Conjecture 61.6.3 (**R1d**): effective descent of the specified comparison functors over the central-character family.

What is Unconditional

The established fixed-character theorem and the conditional assembly have different scopes:

(i) The fixed-χ, $p \geq 5$, fully faithful DEG embedding (Theorem 61.7.2): no residual needed. This is the strongest unconditional DEG input toward F1.a.

(ii) The conditional global F1 reduction (Proposition 61.8.3), with all the explicitly named local, family, and L_{ACD} inputs.

Out of Domain

Three items are outside the stated domain:

(i) *Langlands functoriality at the global level.* Deferred to the adelic categorical descent L_{ACD}.

(ii) *Higher GL_n for $n \geq 3$.* The chapter focuses on $GL_2(\mathbb{Q}_p)$; DEG's technology generalises under significant additional input (Emerton–Helm–Paskunas for GL_n), not in domain here.

(iii) *Number fields beyond* $\mathbb{Q}$. The chapter focuses on $\mathbb{Q}_p$; extension to K_v for $K/\mathbb{Q}$ a finite extension requires additional considerations at primes ramified in K.

61.10 DEG Extension Boundary

(i) Dotto–Emerton–Gee arXiv:2603.26887 supplies the stated embedding on its integral coefficient and formal spectral carriers: *fully-faithful, fixed central character,* $p \geq 5$ *only* (Theorem 61.3.1).

(ii) Three obstructions separate DEG from F1.a: (a) essential surjectivity; (b) extension to $p \in \{2, 3\}$; (c) central-character variation (Remark 61.3.2).

(iii) Obstruction (a) reduces to Conjecture 61.4.5 (F1.a.1): block exhaustion per residual Galois representation. Theorem 61.4.8 closes (a) under this conjecture.

(iv) Obstruction (b) reduces to Conjecture 61.5.7 (F1.a.2: existence of wild-enhanced EG stack), Conjecture 61.5.5, plus Conjecture 61.5.11 (F1.a.3). Theorem 61.5.6 and Corollary 61.5.12 close (b) under these conjectures.

(v) Obstruction (c) is the family datum Conjecture 61.6.3. It is not a consequence of the fixed-character DEG theorem.

(vi) Full conditional closure of F1.a at all primes: Theorem 61.7.1, conditional on (H1)–(H5).

(vii) Unconditional partial closure: the fixed-χ, $p \geq 5$, fully faithful DEG embedding (Theorem 61.7.2, Corollary 61.7.3). No family statement is asserted.

(viii) Global assembly check: F1 follows from F1.a (at every prime, including wild) plus $\mathrm{L}_{\mathrm{ACD}}$ (Theorem 61.8.1); total F1 residual table is six items (Proposition 61.8.3).

The fixed-character DEG functor has the proper image specified in Corollary 61.7.3. The proposed arithmetic equivalence requires different comparison data: the actual coefficient and support maps, lifted generators, wild functors, family descent and centre compatibility. The conditional assembly proves what these maps would imply; their construction remains the arithmetic target.

Chapter 62

H-P^{Ar} Determinant-Trace Matching and Hamburger Determinacy

Chapter 57 reduces the arithmetic Hilbert–Pólya hypothesis H-P^{Ar} to a determinant-trace realisation (SP) equivalent to the Connes adele-class-space absorption-trace conjecture and to the Deninger regularised-determinant axioms. Its conditional operator comparison uses (Reg), including positivity, determinacy and spectral support on an identified arithmetic carrier, with explicitly supplied comparison data. Chapter 58 uses the conditional RH completion hypothesis and isolates the positivity/determinacy input. The H-P^{Ar} side separates the Bost–Connes zero-mode weight-selection *problem from the trace-matching problem: the logarithmic Berry–Keating weight is the only survivor among a small canonical test family, and is unique under an explicit KMS/zeta-partition normalisation, but existence of the required trace identity is the H-P^{Ar} obstruction. The VB* side tests the Carleman construction: the Bombieri–Lagarias Li-coefficient bounds do not imply the Carleman estimate for the shifted-zero moments. The sufficient analytic residual is* ($\mathrm{R}_{\mathrm{analytic}}$)*: a linear upper-growth bound for $\mu_{2n}^{1/(2n)}$, or equivalently a positive regularised moment measure with Carleman divergence.*

62.1 Determinacy Comparison

Residual Splitting

The residual splitting structural analysis of Chapter 57 gives four structural inputs:

(i) *Domain theorem for $L_{\mathrm{o}}^{\mathrm{Ar}}$.* Theorem 57.3.2: on the primary arithmetic Heisenberg Fock, under non-negativity of the Bost–Connes zero-mode weights $\lambda(m) \geq 0$, the symmetric operator $L_{\mathrm{o}}^{\mathrm{Ar,\,symm}}$ admits a unique Friedrichs self-adjoint extension $L_{\mathrm{o}}^{\mathrm{Ar}}$, essentially self-adjoint on the finite-span core $\mathcal{D}_{\mathrm{o}}$, with non-negative spectrum.

(ii) *Three-way equivalence for H-P^{Ar}.* Theorem 57.3.14: H-P^{Ar} (in the determinant-trace form SP) is equivalent to the Connes absorption-trace conjecture and to the Deninger regularised determinant condition.

(iii) *Conditional operator comparison.* Under Hypothesis 57.4.21, Theorem 57.4.22 compares VB*, (Reg), (HB+) and (BD+). Bare determinacy is only one component of (Reg); the sharp BD rate also requires the specified quantitative comparison.

(iv) *Separation of the two residuals.* Proposition 57.5.2: H-P^{Ar} (existence of a self-adjoint $L_{\mathrm{o}}^{\mathrm{Ar}}$ with prescribed trace) and spectral regularity of the moment problem are separate assertions.

The determinacy calculation adds weight selection and Carleman analysis to these structural inputs.

Two Analytic Tests

(A) *Zero-mode weight selection for H-P^{Ar}.* Corollary 57.3.15 names the single residual as "the choice of Bost–Connes zero-mode weights $\{\lambda(m)\}_{m\in\mathbb{N}^\times}$ such that the combined trace matches the zero generating

function''. Four canonical weight systems against the arithmetic Koszul self-duality (P1), the completed-zeta character and explicit formula (P2), and Bost–Connes rigidity at $\beta = 1$ are considered. This is a finite test family, not a classification of all additive functions on $\mathbb{N}^\times$. A genuine uniqueness theorem requires the extra KMS/zeta-partition normalisation stated below.

(B) *Hamburger determinacy obstruction on the symmetrised shifted-zero measure.* Proposition 57.4.5 proves that a factorial upper bound on a specified positive functional gives Carleman determinacy. The finite cutoff sums diverge; their arithmetic regularization and positivity must first be constructed. An upper bound $\mu_{2n}^{1/(2n)} \leq Cn$ then suffices, without a matching lower bound.

The single positivity corollary Corollary 58.4.3 is sufficient for RH after the conditional hypothesis of Chapter 58 is supplied. The resulting residual is $(\mathrm{R}_{\mathrm{analytic}})$.

Analytic Inputs

Chapter 57 used: Connes arXiv:math/9811068, Deninger 1994, Berry–Keating 1999, Reed–Simon, Akhiezer–Glazman, de Branges 1968/1994, Báez-Duarte 2003, Burnol 2000. The present determinacy argument uses a complementary list:

Akhiezer, *The Classical Moment Problem* (Oliver & Boyd 1965). Definitive monograph on the Hamburger, Stieltjes, Hausdorff moment problems; §2.4 Carleman's criterion; §2.5 Nevanlinna parametrisation of indeterminate solutions; §3.1 Jacobi-matrix formalism; §5.6 exponential-moment estimates.

Shohat–Tamarkin *The Problem of Moments* (AMS 1943). The American companion to Akhiezer; Ch. III.5 classical determinacy, Ch. III.2 Hankel determinants as sufficient tests, Ch. VI bounded-case theorems.

Stieltjes, Ann. Fac. Sci. Toulouse (1894). The original indeterminate moment problem; the lognormal $\mu_n = e^{n^2/2}$ example; baseline indeterminacy standard.

Nevanlinna, Ann. Acad. Sci. Fenn. 5 (1922). Parametrisation of the convex set of solutions to an indeterminate Hamburger moment problem by four entire functions A, B, C, D of exponential type zero; each indeterminate solution corresponds to a Nevanlinna function φ via $(A\varphi - C)/(B\varphi - D)$.

Krein, Dokl. Akad. Nauk SSSR 77 (1951) 93–96. Krein's Pontryagin-space theorem: self-adjoint operators on indefinite-metric spaces; prototype for the indeterminate moment problem as a problem with non-unique positive lift.

Simon, Adv. Math. 137 (1998), 82–203. Modern treatment of classical moment problems via Jacobi matrices: the Hamburger moment problem with moments $\{\mu_n\}$ is determinate iff the associated Jacobi operator is essentially self-adjoint on c_{00}; indeterminate iff deficiency indices $(1, 1)$.

Berg–Christiansen, Math. Ann. 322 (2002) 451–473. Exponential moment condition: $\sum_n \mu_{2n}^{-1/(2n)} = +\infty$ implies determinacy (Carleman); refinements via sharper generating-function analysis.

Lin, Probab. Surv. 14 (2017) 129–190. Modern survey of determinacy: Carleman vs. Krein vs. Lin conditions; exponential-moment comparison.

Bost–Connes, Selecta Math. 1 (1995) 411–457. Original Bost–Connes system; §2.2 construction of the Hecke algebra $C^*_{\mathbb{Q}}$; §3.1 KMS states for $\beta > 1$; §4 phase transition at $\beta = 1$.

Laca, J. Funct. Anal. 152 (1998) 330–378. Laca's categorical approach to the Bost–Connes system: characters of the $C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^\times$ crossed product; KMS classification via induced representations.

Connes–Marcolli, AMS Colloq. Publ. 55 (2008). Comprehensive reference on Bost–Connes and number theory; Ch. 3 KMS states, Ch. 4 rigidity at $\beta = 1$, Ch. 5 adele class space structure.

Connes–Marcolli–Ramachandran, arXiv:math/0501424. Explicit description of the $\mathrm{KMS}_{\beta=1}$ state as the unique symmetric trace on the Bost–Connes system; rigidity of the zero-mode spectrum under the Hecke action.

Conrey, J. Reine Angew. Math. 399 (1989) 1–26. Zero-density theorem: at least 40% of the non-trivial zeros of ζ lie on the critical line. Unconditional.

Selberg, Arch. Math. Naturvid. 48 (1946) 89–155. Positive proportion of zeros on the critical line, first unconditional zero-density result.

Levinson, Adv. Math. 13 (1974) 383–436. 1/3 of zeros on the line via Levinson's method.

Odlyzko, 2001 (unpublished note). Numerical computation of 10^{13} non-trivial zeros on the critical line.

Bombieri–Lagarias, J. Number Theory 77 (1999) 274–287. Unconditional upper bound $\lambda_n \leq n(c_0 + \log n)/2$, $c_0 = 2 - \log(4\pi) + \gamma$, and a matching lower bound $\lambda_n \geq -(n/2)\log n - O(n)$.

The zero-counting and Li-coefficient estimates do not by themselves supply a positive moment functional or its Carleman upper-growth bound.

62.2 Weight Selection and Moment Determinacy

The weight problem concerns logarithmic generators and a prescribed trace. The moment problem requires a positive regularised interpretation and a sufficient upper-growth estimate.

Zero-Mode Weights and the Trace Condition

Problem. Does the Berry–Keating ansatz $\lambda(p^k) = k \log p$ (Remark 57.3.17) constitute the unique natural Bost–Connes zero-mode weight?

Argument. The Berry–Keating ansatz reproduces the leading asymptotic zero density $N(T) \sim (T/2\pi)\log(T/2\pi)$ (Bohr–Sommerfeld quantisation of $xp = E$) but does not reproduce the individual spectrum; three independent competitors exist among finite canonical tests.

Sources. Bost–Connes 1995 §2.2 (Hecke algebra structure); Connes–Marcolli 2008 Ch. 3 (KMS_β classification); Laca 1998.

Conclusion. Four candidate weight systems are enumerated as Definition 62.3.1; pairwise P1+P2 compatibility is proved as Theorem 62.3.5. The $\mathrm{KMS}_{\beta=1}$ state of the Bost–Connes system is unique in the symmetric normalisation, but that rigidity alone does not classify arbitrary additive prime-weight systems. Trace-compatibility (57.1) for each candidate is equivalent to H-P^{Ar} for that candidate; no combination gives an unconditional trace identity.

Uniqueness under KMS and Zeta Normalisation

Problem. Does at most one weight system satisfy simultaneous compatibility with P1, P2, and the H-P^{Ar} trace condition?

Argument. Existence is not assumed (it is H-P^{Ar} itself); uniqueness holds conditionally on existence, via Connes–Marcolli rigidity.

Sources. Connes–Marcolli, AMS Colloq. Publ. 55 (2008), Thm 4.15 (uniqueness of symmetric KMS$_{\beta=1}$ state); Connes–Marcolli–Ramachandran arXiv:math/0501424 (explicit form).

Conclusion. The KMS$_{\beta=1}$ state on $C^*_{\mathbb{Q}}$ is unique among symmetric states (invariant under $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$) by Connes–Marcolli 2008 Thm 4.15. The compatibility theorem identifying the partition function and prime-generator expectations with the canonical zeta normalisation forces the logarithmic weight under that hypothesis; see Theorem 62.3.7. Existence of the weight system with the required zero trace is precisely the H-P^{Ar} assertion.

The Determinant-Trace Hierarchy

Problem. Does the Berry–Keating weight $\lambda(p^k) = k \log p$ reproduce the H-P^{Ar} trace identity?

Argument. The Berry–Keating partition function $\sum_{n\geq 1} e^{-t\log n} = \sum_{n\geq 1} n^{-t} = \zeta(t)$, multiplied by the Dedekind-eta inverse, does not equal the zero wave-kernel distribution; the two sides match only after the archimedean Euler factor is subtracted and the logarithmic-derivative expansion is inverted.

Sources. Edwards, *Riemann's Zeta Function* (Academic Press 1974) §§1.13 and 3.3, Mellin inversion of ξ via the logarithmic derivative.

Conclusion. The trace-compatibility condition is an infinite hierarchy of constraints, not a single scalar equation. By Mellin inversion, the hierarchy is an equality of distributions for the normal zero operator Θ_ξ, equivalently $\det_\infty(s\,\mathrm{id} - \Theta_\xi) = \xi_{\mathbb{R}}(s)$. The Friedrichs Fock/BC Hamiltonian supplies the Euler partition side, not this determinant identity. The hierarchy is identified with a regularised Weil/Connes explicit-formula extension only after the admissibility of the test family is supplied; see Theorem 62.3.10. No weight system is known to close the trace identity.

Carleman's Criterion and Moment Growth

Problem. Does Carleman's criterion $\sum_{n\geq 1} \mu_{2n}^{-1/(2n)} = +\infty$ hold for the symmetrised shifted-zero moment sequence $\{\mu_{2n}\}$?

Finite calculation. Proposition 80.2.5 computes the cutoff sums and proves their divergence. There is no finite unregularized saddle from which to infer a factorial law. For a specified finite positive functional, the upper bound $\mu_{2n}^{1/(2n)} = O(n)$ implies Carleman by harmonic comparison. The arithmetic task is to construct that functional and estimate it; see Theorem 62.4.5.

Li Coefficients and Absolute Moments

Problem. Can the unconditional Bombieri–Lagarias bound $\lambda_n \leq n(c_0 + \log n)/2$ be converted into an unconditional estimate on μ_{2n} sufficient for Carleman?

Argument. The Li coefficients are oscillatory sums over zeros, whereas μ_{2n} is a positive regularised moment. The unconditional bound on λ_n does not transfer to μ_{2n} without a separate positivity/domination theorem.

Sources. Bombieri–Lagarias J. Number Theory 77 (1999) 274–287, Theorem 2; Voros 2003 Ann. Inst. Fourier (refinements); Coffey 2005 Math. Phys. Anal. Geom.

Conclusion. Bombieri–Lagarias Thm 2 bounds the Li sum $\lambda_n = \sum_\rho (1 - (1 - 1/\rho)^n)$ (Li 1997). This is a highly oscillatory zero sum. It is not a bound on the absolute moments μ_{2n}, and cancellations in the Li sum cannot be reversed into Carleman control without an additional positivity and regularisation theorem; see Proposition 62.4.3. An independent construction of a positive regularised shifted-zero moment measure with upper-growth estimate $\mu_{2n}^{1/(2n)} \le Cn$ is the analytic residual.

Positivity and Growth of the Regularised Moments

The positivity input in Chapter 58 requires a positive regularised interpretation of the formal shifted-zero moments

$$\mu_{2n} = \sum_\rho (1/4 + |\gamma_\rho|^2)^n.$$

A sufficient growth estimate for Carleman's criterion is $\mu_{2n}^{1/(2n)} = O(n)$. Positivity and this upper bound are the analytic data required in Theorem 62.4.7; the other hypotheses remain the conditional inputs of the residual splitting and Chapter 58.

The classical formulations through Carleman's criterion, Nevanlinna parametrisation and Akhiezer–Krein theory concern this same moment problem. Their sources are Akhiezer, Shohat–Tamarkin, Carleman, Krein and Nevanlinna, together with the sources for the residual splitting cited above.

62.3 H-P^{Ar} Spectral Matching: Bost–Connes Zero-Mode Weight

Candidate Weights

The residual splitting candidate weight system stated was the Berry–Keating ansatz $\lambda(p^k) = k \log p$ extended multiplicatively to $\lambda(m) = \log m$ (Remark 57.3.17). We compare it with three nearby weights. This is not an exhaustive classification of all maps $\mathbb{N}^\times \to \mathbb{R}$: additive weights are determined by arbitrary prime values $\{\lambda(p)\}_p$. The point is to separate the canonical BC/zeta generator from common false substitutes.

Definition 62.3.1 *(four canonical zero-mode weight tests).* Let $C^*_{\mathbb{Q}}$ be the Bost–Connes algebra of Definition 29.1.1. Four canonical weight tests $\{\lambda_\star(m)\}_{m\in\mathbb{N}^\times}$ are available, each arising from a distinct structural interpretation of the Hecke action on the zero-mode lattice:

(BK) *Berry–Keating / logarithmic*: $\lambda_{\mathrm{BK}}(m) := \log m$. Extended multiplicatively: $\lambda_{\mathrm{BK}}(p^k) = k \log p$. This is the Euler-product generator; yields the BC KMS$_\beta$ partition function $\zeta(\beta)$ at $\beta = 1$.

(Li-shift) *Li spectral translation*: $\lambda_{\mathrm{Li}}(m) := \log m$ as a semigroup weight, with the critical-line translation $s \mapsto s - \frac{1}{2}$ applied to the spectral parameter rather than inserted as a constant shift in the semigroup homomorphism. A literal formula $\log m - \frac{1}{2}$ is not additive on $\mathbb{N}^\times$ and is therefore not a BC weight.

(vM) *von Mangoldt / Weil*: $\lambda_{\mathrm{vM}}(m) := \Lambda(m) \log m$ where $\Lambda(m)$ is the von Mangoldt function ($\Lambda(p^k) = \log p$, zero otherwise). This is logarithmic derivative data, not a multiplicative semigroup weight.

(FS-quad) *quadratic finite-place weight*: $\lambda_{\mathrm{FS}}(p^k) := k^2 \log p$ and $\lambda_{\mathrm{FS}}(m) := \sum_{p^{k_p} \| m} k_p^2 \log p$. This records the quadratic-exponent alternative suggested by finite-place Hecke powers on perfectoid Iwahori charts (Definition 58.1.1 finite-place orientation). It agrees with (BK) on squarefree integers and differs on higher prime powers.

Remark 62.3.2 *(domain of the four tests).* Under the $\mathbb{Q}$-positive-semigroup action of $\mathbb{N}^\times$ on the zero-mode lattice, a weight system $\lambda : \mathbb{N}^\times \to \mathbb{R}$ is compatible with the Bost–Connes Hecke action $\mu_n \mu_m = \mu_{nm}$

only when $\lambda(nm) = \lambda(n) + \lambda(m)$. The additive case reduces to specifying $\lambda(p)$ for each prime p, hence contains infinitely many choices. The zeta partition normalisation

$$\sum_{m\geq 1} e^{-\beta\lambda(m)} = \zeta(\beta)$$

for all $\mathrm{Re}(\beta) > 1$ forces $e^{-\lambda(p)} = p^{-1}$ prime-by-prime, and therefore forces (BK). Without that normalisation there is no uniqueness theorem. The four tests above only record the canonical logarithmic, translated, logarithmic-derivative, and quadratic finite-place possibilities used in the obstruction comparison.

Pairwise Compatibility with P1 and P2

Lemma 62.3.3 *(P1 compatibility within the four tests).* Within Definition 62.3.1, the additive Berry–Keating weight is compatible with the formal $m \leftrightarrow m^{-1}$ functional-equation involution. The Li-shift test is the same additive weight together with an external critical-line translation. The von Mangoldt and quadratic finite-place tests are not multiplicative semigroup weights and therefore do not define BC zero-mode weights satisfying P1.

Proof. For (BK), $\lambda_{\mathrm{BK}}(m) + \lambda_{\mathrm{BK}}(m^{-1}) = 0$ after formal extension of log to $\mathbb{Q}_{>0}$. The Li shift is not the map $\log m - \frac{1}{2}$ on $\mathbb{N}^{\times}$, since such a map fails $\lambda(nm) = \lambda(n) + \lambda(m)$. It is the same additive weight with the spectral parameter translated by $1/2$. The von Mangoldt test is supported on prime powers and the FS-quad test satisfies $\lambda_{\mathrm{FS}}(p^{a+b}) \neq \lambda_{\mathrm{FS}}(p^a) + \lambda_{\mathrm{FS}}(p^b)$ for general a, b. Hence neither is a BC semigroup weight. □

Lemma 62.3.4 *(P2 compatibility within the four tests).* Within Definition 62.3.1, the Berry–Keating weight is the only additive weight whose partition function is the classical zeta partition function. The Li shift changes only the external critical-line normalisation. The von Mangoldt and FS-quad tests do not give the ζ-partition function and hence do not satisfy the P2 completed-zeta character condition.

Proof. (BK) gives

$$\sum_{m\geq 1} e^{-t\log m} = \sum_{m\geq 1} m^{-t} = \zeta(t).$$

The Li-shift convention leaves this semigroup partition function unchanged and moves the $1/2$ into the spectral variable. For (vM) one obtains logarithmic-derivative data supported on prime powers, not a Dirichlet series with the ξ-functional equation. For (FS-quad) one obtains

$$\prod_{p}\sum_{k\geq 0} p^{-tk^2},$$

not the Euler product for $\zeta(t)$. These two tests therefore fail the P2 partition-function test. □

Theorem 62.3.5 *(four-test filter under P1+P2).* Among the four tests of Definition 62.3.1, simultaneous P1+P2 compatibility leaves only the Berry–Keating semigroup weight $\lambda(m) = \log m$, with the Li convention understood as the same weight after external critical-line translation. This is a finite filter, not a classification of all possible prime-weight systems.

Proof. Lemma 62.3.3 and Lemma 62.3.4 eliminate the von Mangoldt and FS-quad tests as BC semigroup weights. The Li convention does not produce a new additive weight; it is the Berry–Keating weight with the spectral parameter centred at $1/2$. Thus the only surviving semigroup weight in the four-test family is (BK). □

Remark 62.3.6 *(what the filter proves).* Theorem 62.3.5 proves only the finite-filter statement just stated. Full uniqueness requires the zeta partition normalisation or an equivalent KMS compatibility condition; without such a condition, arbitrary prime weights $\{\lambda(p)\}_p$ remain possible.

Uniqueness under KMS/Zeta Normalisation

The surviving canonical weight is (BK). Existence with the trace-compatibility condition is the H-P^{Ar} assertion. Uniqueness after the partition function is fixed is unconditional.

Theorem 62.3.7 *(uniqueness under zeta partition normalisation).* Suppose $\lambda : \mathbb{N}^\times \to \mathbb{R}_{\geq 0}$ is additive and has zeta partition function

$$\sum_{m\geq 1} e^{-\beta\lambda(m)} = \zeta(\beta)$$

for all $\mathrm{Re}(\beta) > 1$, with Euler factors matched prime-by-prime to the Bost–Connes Hecke generators. Then $\lambda(m) = \log m$. Consequently any H-P^{Ar} realisation satisfying this KMS/zeta normalisation uses the Berry–Keating weight.

Proof. Additivity gives $\lambda(p^k) = k\lambda(p)$, so the partition function factors as

$$\prod_p (1 - e^{-\beta\lambda(p)})^{-1}.$$

Prime-by-prime equality with $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$ gives $e^{-\lambda(p)} = p^{-1}$, hence $\lambda(p) = \log p$ for every prime. Additivity then gives $\lambda(m) = \log m$. □

Remark 62.3.8 *(relation to Connes–Marcolli rigidity).* Connes–Marcolli rigidity at $\beta = 1$ is the conceptual source of this normalisation. Identifying an arbitrary weight with the unique symmetric KMS state requires the prime-generator and partition-function bridge proved in the theorem. The theorem records the exact conditional uniqueness available on the arithmetic path. H-P^{Ar} is the Bost–Connes/Dedekind-eta trace identity, the unresolved determinant-trace matching problem.

The Hierarchy Condition

Lemma 62.3.9 *(admissible Weil–Connes wave-kernel regularisation).* Let $\xi = \xi_{\mathbb{R}}$ and write each non-trivial zero as $\rho = \beta_\rho + it_\rho$, and set $z_\rho := -i(\rho - \frac{1}{2})$. Use the Fourier convention

$$h(r) = \int_{\mathbb{R}} \widehat{h}(u) e^{-iru}\, du.$$

Let

$$\mathsf{PW} := \{h = \mathcal{F}^{-1}\psi : \psi \in C_c^\infty(\mathbb{R})\}$$

with the LF topology transported from $C_c^\infty(\mathbb{R})$. For $h \in \mathsf{PW}$, the Weil–Connes explicit formula gives

$$\sum_\rho h(z_\rho) = \int_{\mathbb{R}} h(r)\, dW_\infty(r) - 2\sum_{n\geq 1} \frac{\Lambda(n)}{\sqrt{n}} \widehat{h}(\log n),$$

in the normalisation of (62.1); the prime sum is finite because $\widehat{h}$ has compact support. The replacement $z_\rho = t_\rho$ is equivalent to RH.

For $\phi \in C_c^\infty((0,\infty))$, set

$$h_\phi(r) := \int_0^\infty e^{-u/2}\phi(u)e^{-iru}\,du.$$

Then $h_\phi \in \mathsf{PW}$, and

$$\langle \mathcal{W}_\xi, \phi\rangle := \sum_\rho h_\phi(z_\rho)$$

defines a distribution $\mathcal{W}_\xi \in \mathcal{D}'((0,\infty))$. Consequently the symbolic kernel $\sum_\rho e^{-t\rho}$ is a distributional wave kernel, not an ordinary heat trace.

Proof. This is the standard Paley–Wiener form of the Weil explicit formula: compact support of $\widehat{h}$ makes the prime sum finite and the archimedean term is the distribution W_∞. The second statement is obtained by substituting the compactly supported function $u \mapsto e^{-u/2}\phi(u)$ into the same formula. Approximation of the point kernel uses mollifiers $\widehat{h}_{t,\epsilon}(u) = e^{-u/2}\epsilon^{-1}\eta((u-t)/\epsilon)$; the convergence is in $\mathcal{D}'((0,\infty))$, not pointwise heat trace convergence. □

Theorem 62.3.10 *(trace-compatibility as regularised Weil/Connes hierarchy). (conditional on the H-P^{Ar} trace realisation).* Let $\lambda = \lambda_{\mathrm{BK}}$ be the Berry–Keating weight of Definition 62.3.1. The trace-compatibility condition (57.1) is the regularised Weil/Connes trace hierarchy

$$\sum_\rho{}^{\mathrm{reg}} h_t(z_\rho) \;=\; \int_{-\infty}^{\infty} h_t(r)\,dW(r) \;-\; 2\sum_{n\ge 1}\frac{\Lambda(n)}{\sqrt{n}}\widehat{h}_t(\log n) \tag{62.1}$$

for the wave-kernel test family associated to $e^{-t\Theta_\xi}$, after the admissible regularisation of Lemma 62.3.9 is supplied. The admissibility and regularisation are the trace-realisation bridge required for H-P^{Ar}; the classical Weil formula applies only to admissible test functions. Under the full H-P^{Ar} determinant condition, $z_\rho = t_\rho$ and the trace becomes the ordinate trace.

Proof. Under the BK weight system, the Friedrichs operator acts on δ_m by eigenvalue $\log m$. The regularised trace of $e^{-tL_0^{\mathrm{Ar}}}$ on the zero-mode lattice is $\sum_{m\ge 1} e^{-t\log m} = \sum_{m\ge 1} m^{-t} = \zeta(t)$, convergent for $t > 1$ and meromorphically continued.

Combining with the archimedean Dedekind-eta inverse product $\prod_{i\ge 1}(1-e^{-ti})^{-1}$ and subtracting the archimedean descendant contribution (Tate-pole and archimedean terms) via the Bost–Connes regularisation (Proposition 57.3.3), one obtains

$$\mathrm{tr}_{\mathrm{BC}}(e^{-tL_0^{\mathrm{Ar}}}) = \zeta(t)\cdot\eta(it)^{-1}\cdot(\text{regularisation}),$$

with the archimedean terms removed by the Bost–Connes/Dedekind-eta prescription. The Hadamard product for $\xi_{\mathbb{R}}$ rewrites the zero side as a regularised zero trace. The classical Weil explicit formula supplies this identity for the admissible Paley–Wiener test spaces of Lemma 62.3.9. The Vol IV condition is the additional assertion that the Bost–Connes/Fock trace is carried to that distributional extension. Therefore the theorem is a reformulation of the trace-compatibility hierarchy under H-P^{Ar}, not a proof of H-P^{Ar}. □

Corollary 62.3.11 *(the hierarchy is the Connes absorption-trace bridge). (conditional on the H-P^{Ar} trace realisation).* Under the BK weight selection, the H-P^{Ar} trace-compatibility condition is the

Connes–Meyer–Weil absorption-trace bridge for the Bost–Connes/Weil wave-distribution family. The classical Weil explicit formula applies to admissible Paley–Wiener test functions; the admissibility and trace-identification of the BC heat-trace family is the additional content.

Proof. The previous theorem identifies the needed equality with the regularised explicit-formula extension for the BC/Weil wave family. By Corollary 57.3.9, this is the same structural condition as the Connes absorption-trace realisation. The admissibility and trace-identification step is precisely what H-P^{Ar} asserts. □

Remark 62.3.12 *(weight selection and H-P^{Ar}).* Theorem 62.3.7 and Corollary 62.3.11 together state: if H-P^{Ar} holds, the weight is uniquely Berry–Keating under the zeta/KMS normalisation. Whether the Berry–Keating weight satisfies the trace compatibility is the regularised Connes absorption-trace problem. The reduction is tight: the canonical weight is determined under explicit normalisation, while the trace identity is the classical absorption-trace problem.

62.4 Hamburger Determinacy Obstruction

The Symmetrised Shifted-Zero Measure

Definition 62.4.1 *(symmetrised shifted-zero moment sequence).* For a non-trivial zero ρ of ξ, put

$$z_\rho := -i(\rho - \tfrac{1}{2}), \qquad x_\rho := \rho(1-\rho) = \tfrac{1}{4} + z_\rho^2.$$

The number z_ρ is real for every zero exactly under RH. The *symmetrised shifted-zero moment sequence* is

$$\mu_{2n} := \sum_{\rho}^{\mathrm{reg}} x_\rho^n$$

with odd moments zero, provided a specified common linear regularization makes every displayed value finite. The notation $\sum^{\mathrm{reg}}$ is this additional datum, not a summation procedure furnished by zero counting. Under RH the finite-cutoff terms are nonnegative but their sums diverge. Positivity of the regularized functional must therefore be proved separately.

Remark 62.4.2 *(relation to the symmetrised zero measure of the residual splitting).* The off-line defect measure of Definition 45.4.1 is not the shifted-zero moment measure defined here. For the latter, a positive representing measure is sought only after finite moments have been supplied. Under a common linear regularization of the unshifted powers, $\mu_{2n} = \sum_{k=0}^{n} \binom{n}{k} 4^{-(n-k)} M_{2k}$. This polynomial identity fixes the required compatibility but does not construct the regularization or its positive representing measure.

Li-Coefficient Bounds Do Not Give Carleman Moments

Proposition 62.4.3 *(Li-coefficient bounds do not imply Carleman moments).* The Bombieri–Lagarias bounds for Li coefficients do not imply the Carleman estimate for the regularised moments μ_{2n} in this argument. An estimate of the form $|\lambda_n| = O(n \log n)$ would not by itself imply

$$|\lambda_n| = O(n \log n) \quad \not\Longrightarrow \quad \mu_{2n}^{1/(2n)} = O(n) :$$

a separate positivity and domination theorem is required.

Proof. The Li coefficient is the oscillatory zero sum

$$\lambda_n = \sum_{\rho} \left(1 - \left(1 - \frac{1}{\rho} \right)^n \right).$$

The shifted-zero moment relevant to Carleman is instead the positive regularised quantity

$$\mu_{2n} = \operatorname*{reg}_{\rho} \left(\frac{1}{4} + \gamma_\rho^2 \right)^n.$$

An estimate for the oscillatory Li sum gives no bound on the absolute or positive regularised $2n$-th moment without a separate positivity and domination theorem. Reversing the cancellations in the Li sum would itself be a new spectral-positivity input. Hence the Bombieri–Lagarias theorem is evidence for the Li/RH reformulation but not a Carleman proof. □

Remark 62.4.4 *(direction of the Carleman bound).* A lower bound on μ_{2n} would give an upper bound on $\mu_{2n}^{-1/(2n)}$, not the lower bound on $\mu_{2n}^{-1/(2n)}$ that Carleman's divergence criterion requires. Carleman's criterion is driven by upper growth of the moments, together with positivity of the representing measure.

Tightest Conditional Reduction

THEOREM 62.4.5 *(Hamburger determinacy from linear upper growth).* Suppose the symmetrised shifted-zero moments come from a positive regularised Hamburger moment functional and there is a constant $C > 0$ such that

$$\mu_{2n}^{1/(2n)} \leq Cn$$

for all sufficiently large n. Then Carleman's criterion holds:

$$\sum_{n \geq 1} \mu_{2n}^{-1/(2n)} = +\infty.$$

Consequently the associated Hamburger moment problem is determinate.

Proof. The upper-growth hypothesis gives

$$\mu_{2n}^{-1/(2n)} \geq \frac{1}{Cn}$$

for all large n. The harmonic series diverges, so Carleman's criterion applies. By the classical Carleman theorem for Hamburger moments, the positive moment functional is determinate. □

COROLLARY 62.4.6 *(unconditional inputs and the Carleman gap).* The unconditional inputs used here do not imply the hypotheses of Theorem 62.4.5: the required positive regularised moment functional and the linear upper-growth estimate $\mu_{2n}^{1/(2n)} = O(n)$ are additional analytic inputs.

Proof. Proposition 62.4.3 blocks the only candidate unconditional shortcut above. The Riemann–von Mangoldt counting formula controls the distribution of zeros by height, but by itself it does not define the required positive regularised moment functional nor bound the regularised moments after off-line and archimedean terms are included. □

Theorem 62.4.7 *(A sufficient determinacy estimate and its arithmetic use).* Let L be a specified positive Hamburger moment functional with moments $\mu_n = L(x^n)$, including the chosen symmetrization and regularization. Suppose

$$(\mathrm{R}_{\mathrm{analytic}}) \qquad \limsup_{n\to\infty} \frac{\mu_{2n}^{1/(2n)}}{n} < \infty.$$

Then its representing measure is determinate. To conclude arithmetic VB*, additionally assume that this is the moment functional of the specified arithmetic operator, that its measure has the spectral support required in (Reg), and that the operator-positivity comparison in item 1 of Hypothesis 57.4.21 holds on the stated domains. Under these additional assumptions, (Reg), hence VB*, holds.

Proof. For some $C > 0$, eventually $\mu_{2n}^{1/(2n)} \le Cn$. If any even moment vanishes, positivity forces the measure to be supported at zero. Otherwise the Carleman sum dominates $C^{-1}\sum n^{-1}$ after finitely many terms and diverges. The classical Hamburger Carleman theorem gives determinacy. Positivity, this determinacy, and the separately supplied support condition constitute (Reg). Item 1 of the specified arithmetic comparison then gives VB*. The growth estimate alone does not construct that operator comparison or its support. □

Remark 62.4.8 *(domain of* $\mathrm{R}_{\mathrm{analytic}}$*).* $(\mathrm{R}_{\mathrm{analytic}})$ is a one-sided upper-growth condition, distinct from a two-sided asymptotic. Its relation to RH is not determined here: the condition may be strictly weaker, or RH-equivalent. It supplies the exact positivity and upper-growth information that Carleman's criterion requires. Standard zero-density theorems count zeros in regions; the residual controls a regularised moment functional.

Remark 62.4.9 *(form of the determinacy residual).* $(\mathrm{R}_{\mathrm{analytic}})$ is a sufficient condition for the Vol IV determinacy step. It identifies the exact analytic shape of the positivity theorem required; neither the classical Weil formula nor the Bombieri–Lagarias bounds supply it.

62.5 Consequence for the RH Comparison in Chapter 58

Theorem 62.5.1 *(determinacy consequence for the Chapter 58 single positivity corollary).* Assume P1, P2 and P3 on their stated domains. Assume also the arithmetic moment-functional identification, spectral support, and operator comparison in Theorem 62.4.7. Then the following three additional inputs suffice for the conditional implication to RH:

(i) the Berry–Keating weight is forced by the additive KMS/zeta-partition normalisation (Theorem 62.3.7).

(ii) H-P^{Ar} holds with the Berry–Keating weight, meaning that the Bost–Connes trace is identified with the regularised Weil/Connes trace hierarchy (Theorem 62.3.10).

(iii) $(\mathrm{R}_{\mathrm{analytic}})$: the shifted-zero moments form a positive regularised Hamburger moment sequence with $\mu_{2n}^{1/(2n)} = O(n)$ (Theorem 62.4.7).

Under these three inputs and all the moment, support, operator and P1–P3 hypotheses stated above, the Chapter 58 implication to RH follows.

Proof. (1) provides the uniquely-determined candidate operator $L_{\mathrm{o}}^{\mathrm{Ar}} = L_{\mathrm{o}}^{\mathrm{Ar}}(\lambda_{\mathrm{BK}})$. (2) is precisely the trace-compatibility condition which, via Theorem 57.3.14, asserts H-P^{Ar}. (3) supplies determinacy by Theorem 62.4.7. The assumed moment identification and support then give (Reg), and the operator

comparison gives VB*. Together with the stated P1–P3 and determinant-trace inputs, this is the sufficient arithmetic package of Corollary 58.4.3. □

COROLLARY 62.5.2 *(structure of the determinacy reduction).* The determinacy comparison reduces the Chapter 58 single positivity theorem to three named residuals:

(W) *Weight normalisation*: an additive Bost–Connes weight with zeta partition function and prime-by-prime KMS compatibility. Under (W), $\lambda = \lambda_{\mathrm{BK}}$.

(H) *H-P trace bridge*: the Bost–Connes/Dedekind-eta heat trace is the admissible regularised Weil/Connes zero trace. Under (H), H-P^{Ar} holds for the BK operator.

($\mathrm{R}_{\mathrm{analytic}}$) *Carleman positivity*: the shifted-zero moments form a positive regularised Hamburger moment sequence and satisfy $\mu_{2n}^{1/(2n)} = O(n)$. Under ($\mathrm{R}_{\mathrm{analytic}}$), Hamburger determinacy holds.

All three together give the residual H-P/VB implication only with the moment-functional identification, spectral support, operator comparison and P1–P3 hypotheses of the preceding theorem.

Proof. Direct from Theorem 62.5.1. (W) fixes the candidate operator; (H) supplies determinant-trace matching; and ($\mathrm{R}_{\mathrm{analytic}}$) supplies determinacy. Omitting any one leaves either no fixed operator, no zero trace, or no VB* converse. □

62.6 RESIDUAL CONDITIONS

After the determinacy reductions, the Chapter 58 RH implication requires:

(i) **(W) Weight normalisation**. The BK weight is forced once the additive zeta partition function is fixed prime-by-prime. The remaining assertion is that that the Vol IV Friedrichs construction really lands in this KMS/zeta normalisation.

(ii) **(H) H-P trace bridge**. The analytic theorem is the admissible regularised extension identifying the BC/Dedekind-eta heat trace with the Connes zero trace; the classical Weil explicit formula applies to admissible test functions and is a step toward, not a substitute for, this identification.

(iii) ($\mathrm{R}_{\mathrm{analytic}}$) **Carleman positivity**. The required analytic input is a positive regularised shifted-zero moment functional with $\mu_{2n}^{1/(2n)} = O(n)$; this estimate lies beyond the reach of the Bombieri–Lagarias Li-coefficient bounds by Proposition 62.4.3.

(iv) **Compatibility with the Chapter 58 hypothesis**. The RH completion hypothesis of Definition 58.4.1 requires P1, P2, P3, H-P^{Ar}, and Hamburger determinacy together. P1 is conditional on arithmetic Positselski (requires a non-formal theorem); P2 unconditional; P3 conditional on Koszul–Petersson compatibility (requires a non-formal theorem); H-P^{Ar} (RH-equivalent); determinacy reduces to ($\mathrm{R}_{\mathrm{analytic}}$).

Remark 62.6.1 *(determinacy consequences).* H-P^{Ar} determinant-trace matching and Hamburger determinacy are RH-equivalent or RH-adjacent analytic problems. The determinacy comparison gives:

(i) *Conditional uniqueness of the weight system*: the Berry–Keating $\lambda(m) = \log m$ is forced by additive zeta/KMS normalisation.

(ii) *Explicit identification of the trace-compatibility hierarchy*: the infinite sequence of conditions is the regularised Weil/Connes trace bridge; the classical Weil formula applies to admissible test functions and is a step in the bridge.

(iii) *Independence of Li-coefficient bounds and Carleman moments*: the Bombieri–Lagarias bounds do not imply the Carleman estimate by Proposition 62.4.3.

(iv) *The single analytic residual* (R_{analytic}): Hamburger determinacy reduces to positivity plus linear upper growth for the regularised moments.

(v) *Compatibility with Chapter 58*: the three residuals (W), (H), (R_{analytic}) give the H-P/VB implication after the conditional P1/P3 inputs, arithmetic moment identification, spectral support and operator comparison are supplied.

62.7 Determinacy Boundary

(i) **Weight selection is uniquely determined after normalisation** (Theorem 62.3.7). Among four canonical tests, only the Berry–Keating $\lambda(m) = \log m$ survives as a BC semigroup weight with zeta partition function (Theorem 62.3.5).

(ii) **The trace-compatibility hierarchy is the regularised Weil/Connes bridge** (Theorem 62.3.10, Corollary 62.3.11). The H-P^{Ar} trace identity needs an admissible extension of the explicit formula to the BC heat-trace family.

(iii) **Hamburger determinacy reduces to a single analytic positivity residual** (Theorem 62.4.7). Under (R_{analytic}), the moments are positive and satisfy $\mu_{2n}^{1/(2n)} = O(n)$, so Carleman's criterion holds.

(iv) **Li-coefficient bounds do not give Carleman moments** (Proposition 62.4.3). Li-coefficient bounds are oscillatory zero-sum bounds; a separate positivity and domination theorem is required for Carleman.

(v) **Three precisely-named residuals** (Corollary 62.5.2): weight normalisation (W), H-P trace bridge (H), and Carleman positivity (R_{analytic}).

(vi) **Domain of the reduction**. H-P^{Ar} reduces to the Connes absorption-trace conjecture (RH-equivalent); Hamburger determinacy reduces to (R_{analytic}) (RH-adjacent); no new RH-equivalent positivity language beyond the the residual splitting list of Det, HB+, BD+ is produced.

(vii) **The residual conjunction**. The three residuals (W), (H), (R_{analytic}) are each precisely stated; their conjunction, together with the conditional P1/P3 inputs, the arithmetic moment identification, spectral support and operator comparison, gives the conditional Vol IV implication to RH.

The two analytic problems are the Bost–Connes absorption-trace bridge and positivity with Carleman-scale upper growth for the regularised shifted-zero moments.

Chapter 63

Arithmetic and Geometric Local Equivalences

The critical-level fundamental local equivalence (FLE) and its quantum and small variants are theorems in the geometric Langlands programme over complex curves. The arithmetic branch of this volume studies $\mathcal{V}^{\mathrm{prim}}$, the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2, and its conjectural companions F1, F2, F3. This chapter compares the geometric domains with the mixed-characteristic carriers required by F1/F2/F3. Arithmetic-FLE realization requires the actual functors and compatibility maps in the conditions of Section 63.5.

63.1 Source Theorems

Theorem 63.1.1 *(geometric critical-level FLE).* The 2024 geometric-Langlands proof constructs the critical-level fundamental local equivalence in the geometric setting over a smooth projective curve in characteristic 0:

$$\mathrm{KL}(G)_{\mathrm{crit}} \;\simeq\; \mathrm{IndCoh}^{*}\big(\mathrm{Op}^{\mathrm{mf}}_{\check{G}}\big),$$

with compatibility with Kac–Moody localization. Its source is Arinkin–Beraldo–Campbell–Chen–Faergeman–Gaitsgory–Lin–Raskin– Rozenblyum, *Proof of the geometric Langlands conjecture II: Kac–Moody localization and the FLE*, arXiv:2405.03648.

Theorem 63.1.2 *(quantum and small FLE sources).* The quantum and small FLE layer consists of local geometric and metaplectic/quantum statements over a complex curve. Campbell–Dhillon–Raskin's *Fundamental local equivalences in quantum geometric Langlands* (arXiv:1907.03204) and Gaitsgory–Lysenko's small-FLE work (arXiv:1903.02279) establish equivalences whose source categories are Kazhdan–Lusztig or metaplectic representations and whose target categories are quantum groups or root-of-unity modules over a formal disk. The arithmetic analogue of these equivalences is an open problem at each finite prime.

Remark 63.1.3 *(domain of the geometric FLE sources).* The FLE sources operate within the geometric Langlands, Kac–Moody, Whittaker, and root-of-unity quantum-group framework over complex curves. A local categorical p-adic Langlands equivalence over $\mathbb{Q}_p$, a globalization over $\overline{\mathrm{Spec}\,\mathbb{Z}}$, and an Emerton–Gee full Weil/(φ,Γ) target comparison each require additional technology: Fargues–Scholze's geometrization, prismatic or pro-étale methods, and the Emerton–Gee eigenvariety construction.

63.2 Domain Distinction: Geometric FLE vs F1/F2/F3

Proposition 63.2.1 *(F1/F2/F3 as a distinct mixed-characteristic programme).* . The arithmetic branch's F1 (local categorical Langlands comparison for $GL_2/\mathbb{Q}_p$ against the Emerton–Gee stack), F2 (mixed-characteristic E_2 Hecke enhancement and restricted tensor product), and F3 (Ben-Zvi–Nadler Hochschild-trace comparison against the full Weil/(φ,Γ) target) constitute a mixed-characteristic local-global Langlands and Hochschild-trace programme whose source and target categories are disjoint from those of the geometric FLE.

Proof. The geometric FLE (Theorem 63.1.1) pairs critical-level Kac–Moody or Whittaker/factorizable categories as source with oper or quantum-group categories over a smooth projective complex curve as target. F1 pairs the spherically-finite automorphic category for $GL_2/\mathbb{Q}$ as source with ind-coherent sheaves on the Emerton–Gee stack as target. F2 requires a mixed-characteristic E_2 Hecke enhancement and restricted tensor product. F3 requires a Ben-Zvi–Nadler Hochschild-trace comparison against the full Weil/(φ, Γ) arithmetic target. The targets of F1, F2, F3 are Emerton–Gee and Weil/(φ, Γ) stacks; the inputs at finite primes are adelic Hecke categories, not Beilinson–Drinfeld chiral algebras over a complex curve. □

Remark 63.2.2 *(finite-prime categorical structure).* At a finite prime p, the relevant categorical structure is the category $\mathrm{Coh}(\mathrm{Bun}_G)$ of coherent sheaves on the moduli stack of G-bundles on the Fargues–Fontaine curve, equipped with the Fargues–Scholze spectral action (arXiv:2102.13459), together with Hecke correspondences, diamonds, and (φ, Γ)-modules. The archimedean fibre carries classical Kac–Moody and chiral intuition, providing the motivating analogy; the finite-prime framework is governed by Fargues–Scholze geometrization and Emerton–Gee eigenvarieties.

63.3 Coefficient Categories and Geometric Domains

Remark 63.3.1 *(local/global type distinction).* F1, F2, and F3 are local-global Langlands and Hochschild-trace conjectures. Their source is the automorphic category for $GL_2/\mathbb{Q}$ or its local p-adic analogue; their target is the Emerton–Gee stack or the full Weil/(φ, Γ) coherent sheaf theory. These source and target types differ from those of the geometric FLE.

Remark 63.3.2 *(finite-prime chiral structure).* At a finite prime, the relevant structure is the Beilinson–Drinfeld factorization algebra and the Hecke correspondence on Bun_G, governed by Fargues–Scholze geometrization. The archimedean fibre carries Kac–Moody and chiral intuitions as a motivating analogy.

Remark 63.3.3 *(quantum-group and root-of-unity source domain).* The root-of-unity and small-FLE sources (Campbell–Dhillon–Raskin, Gaitsgory–Lysenko) are geometric and quantum: they operate over a complex curve with a quantum parameter q a root of unity or with a metaplectic covering. The parameter p^{-s} appearing in local Hecke eigenvalues at a prime p is a spectral parameter, distinct from a root-of-unity quantum-group parameter.

Remark 63.3.4 *(function-field transport and number-field domain).* Raskin's globalization operates over a function field. The transport to a number field requires: (a) a local categorical comparison F1 at each prime, and (b) a prismatic or BC-diamond-glue transition across primes. Both remain open.

Remark 63.3.5 *(mixed-characteristic substrate).* The arithmetic compactification $\overline{\mathrm{Spec}\,\mathbb{Z}}$ carries an archimedean place and mixed-characteristic fibres. A curve in characteristic 0 carries neither. The arithmetic programme requires mixed-characteristic methods (prismatic cohomology, perfectoid geometry) at each finite prime and complex-analytic methods at the archimedean place.

Remark 63.3.6 *(independent form for arithmetic FLE).* Arithmetic FLE has no pair of independent constructions in the present arithmetic formalism. An arithmetic FLE theorem requires first the boundary conditions of Section 63.5.

Remark 63.3.7 *(AGKRRV 2024 source and target).* Arinkin–Beraldo–Campbell–Chen–Faergeman–Gaitsgory–Lin–Raskin–Rozenblyum (arXiv:2405.03648) establishes the critical FLE between $\mathrm{KL}(G)_{\mathrm{crit}}$ and $\mathrm{IndCoh}^*(\mathrm{Op}^{\mathrm{mf}}_{\check{G}})$: source is critical-level Kac–Moody, target is monodromic opers

on a complex curve. The Emerton–Gee stack is a mixed-characteristic eigenvariety; its objects are (φ, Γ)-modules, not monodromic opers.

Remark 63.3.8 *(F3/BZN form against full Weil target).* F3 asks for a Ben-Zvi–Nadler Hochschild-trace identification between the automorphic side and coherent sheaves on the Weil/(φ, Γ) stack. The inertia-only loop object $B\widehat{\mathbb{Z}}$ captures only the tame/procyclic part of inertia; the full Weil group retains wild inertia and Frobenius data. F3 with the full Weil target is a conjecture.

Remark 63.3.9 *(spectral parameter vs quantum-group parameter).* The Hecke eigenvalue at a prime p is recorded by the spectral parameter p^{-s} or equivalently by the Satake image in $\mathrm{Rep}(\check{G})$. The quantum-group deformation parameter q appearing in the quantum FLE is a root of unity or a formal parameter; identifying $q = p^{-s}$ requires an explicit independent bridge; this bridge is the sixth condition in Section 63.5.

Remark 63.3.10 *(Fargues–Scholze, Emerton–Gee, and Dotto–Emerton–Gee).* Fargues–Scholze (arXiv:2102.13459) provides the spectral action on $\mathrm{Coh}(\mathrm{Bun}_G)$ via the Fargues–Fontaine curve. Emerton–Gee (arXiv:1908.07185) constructs the stack of (φ, Γ)-modules. Dotto–Emerton–Gee establishes full faithfulness of a functor to mod-p representations; an equivalence of ∞-categories between the automorphic and the Emerton–Gee coherent side is the content of F1.

Remark 63.3.11 *(Hypotheses of the three comparisons).* F1 carries form *Conditional*: the local categorical comparison is supported by Fargues–Scholze (arXiv:2102.13459) and Emerton–Gee (arXiv:1908.07185) and requires the adelic-descent and BC-diamond-glue steps. F2 carries form *Conditional*: the mixed-characteristic E_2 Hecke enhancement and restricted tensor product are formally accessible but unproved. F3 carries form *Conjecture*: the full Weil/(φ, Γ) target comparison is beyond current techniques. These comparison hypotheses remain necessary for the corresponding conditional implications.

Remark 63.3.12 *(local Satake/Yangian normalization).* The local Satake isomorphism identifies the spherical Hecke algebra with the representation ring of $\check{G}$. An arithmetic Yangian structure, if it exists, must be grounded in an independent comparison between the Satake filtration and an explicit Yangian coproduct; this identification remains open.

Remark 63.3.13 *(BC-diamond-glue decomposition).* The BC-diamond-glue problem decomposes into three parts: (a) the local BZCHN/Emerton–Gee comparison at each prime p, (b) cross-prime prismatic transition maps, and (c) global Bernstein-centre compatibility. Each part is open.

Remark 63.3.14 *(wild-prime correction).* At $p = 2$ and $p = 3$, the source for the local comparison carries ramification and wild inertia data not present in the tame case. A conjectural correction accounts for this; the precise form depends on the source used for the local comparison.

63.4 Geometric Equivalences and Arithmetic Comparison Data

Theorem 63.4.1 *(geometric FLE: domain and logical form).* The critical-level FLE $\mathrm{KL}(G)_{\mathrm{crit}} \simeq \mathrm{IndCoh}^*(\mathrm{Op}^{\mathrm{mf}}_{\check{G}})$ and the quantum/small FLE equivalences (Campbell–Dhillon–Raskin, Gaitsgory–Lysenko) are proved in the geometric Langlands setting over a complex curve. Their source is Campbell–Dhillon–Raskin (arXiv:1907.03204), Gaitsgory–Lysenko (arXiv:1903.02279), and Arinkin–Beraldo–Campbell–Chen–Faergeman–Gaitsgory–Lin–Raskin–Rozenblyum (arXiv:2405.03648). These results hold over smooth projective curves in characteristic 0.

Remark 63.4.2 *(archimedean chiral structure and finite-prime domain).* The Kac–Moody and chiral algebra structures provide the motivating analogy at the archimedean place. At finite primes, the geometric structures are Beilinson–Drinfeld factorization algebras, Hecke correspondences, and the Fargues–Scholze spectral action; the finite-prime analogue of the critical-level FLE is the content of F1.

Remark 63.4.3 *(F1, F2, F3, and Bernstein-centre comparison data).* F1 (local categorical Langlands comparison for $GL_2/\mathbb{Q}_p$ via Emerton–Gee, with Fargues–Scholze spectral action as input) carries form *Conditional*: the comparison requires adelic-descent and BC-diamond-glue steps. F2 (mixed-characteristic E_2 Hecke enhancement and restricted tensor product for the adelic assembly) carries form *Conditional*: the ∞-categorical Tate restricted tensor product is accessible in principle but unproved. F3 (Ben-Zvi–Nadler Hochschild-trace identification against the full Weil/(φ, Γ) target, with the full Weil group including wild inertia and Frobenius data) carries form *Conjecture*. BC-diamond-glue decomposes into local BZCHN/Emerton–Gee comparison, cross-prime prismatic transition maps, and global Bernstein-centre compatibility; each component is open. The wild-prime correction at $p = 2, 3$ is a source-sensitive conjecture; the precise form depends on the local comparison chosen.

63.5 Open Conditions

(i) Formulate the mixed-characteristic local categorical equivalence F1: construct a comparison between the spherically-finite automorphic ∞-category for $GL_2/\mathbb{Q}_p$ and ind-coherent sheaves on the Emerton–Gee stack, with the Fargues–Scholze spectral action (arXiv:2102.13459) as input.

(ii) Extend the loop-space comparison from the inertia-only object $B\widehat{\mathbb{Z}}$ to the full Weil group (with wild inertia and Frobenius data) and prove the flat pullback statement in the full Weil/(φ, Γ) setting.

(iii) Prove profinite formal-loop continuity for $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ in the formal Noetherian pro-étale setting, supplying the foundational input for the inertia-only portion of F3.

(iv) Prove the mixed-characteristic E_2 Hecke enhancement and the ∞-categorical Tate restricted tensor product required for F2.

(v) Prove BC-diamond-glue in its three components: (a) the local BZCHN/Emerton–Gee comparison, (b) cross-prime prismatic transition maps, and (c) global Bernstein-centre compatibility.

(vi) Construct an independent bridge between the Satake spectral parameter p^{-s} and the quantum-group deformation parameter q of the quantum FLE, grounded in an explicit comparison of the Satake filtration and the Yangian coproduct.

(vii) Establish two independent constructions of arithmetic FLE.

Remark 63.5.1 *(source role of the geometric FLE results).* The geometric FLE (Theorems 63.1.1 and 63.1.2), quantum geometric Langlands, and Raskin's globalization provide motivating analogies and local-model inputs for F1, F2, F3. Each such use requires an explicit independent bridge; the bridges are the conditions listed above.

Chapter 64

Two-Prime Restriction and Bernstein Actions

A global Galois representation restricts to a representation at each finite place. Reconstructing the global object from two restrictions is a different problem. Even with fixed determinant and absolutely irreducible residual representation, two restrictions can agree for distinct global lifts. The resulting deformation ring map nevertheless constructs a pairwise pullback module and its local central actions. Comparing this module with a global prismatic sheaf requires an isomorphism of the actual objects, including their extension data.

64.1 The Deformation Problem

Fix distinct finite primes p, q and a coefficient prime $p_0 \neq p, q$. Let k be a finite field of characteristic p_0, and put $W = W(k)$. Choose a finite set S of places of $\mathbb{Q}$ containing p, q, p_0, ∞. Write $G_{\mathbb{Q},S}$ for the Galois group of the maximal extension unramified outside S. Fix a continuous representation $\bar{\rho} : G_{\mathbb{Q},S} \to GL_2(k)$ which is absolutely irreducible, and let $\bar{D}$ be its determinant polynomial law. Here a determinant of dimension two is a multiplicative homogeneous polynomial law of degree two on the group algebra, as in Definition 73.1.1. Its characteristic coefficients, rather than a trace alone, define the object in all residual characteristics.

Choose decomposition-group maps $G_{\mathbb{Q}_p}, G_{\mathbb{Q}_q} \to G_{\mathbb{Q},S}$. Choose closed local and global deformation conditions preserved by restriction. These may include a fixed determinant character lifting $\det \bar{\rho}$. For a finite local W-algebra A with residue field k, let $\mathcal{D}_{\mathrm{glob}}(A)$ and $\mathcal{D}_v(A)$ denote the corresponding sets of continuous lifts of the residual determinant, for $v = p, q$. The local residual restrictions need not be absolutely irreducible.

Chenevier's representability theorem gives complete Noetherian local rings R^{Ch}, R^{Ch}_p, R^{Ch}_q for these functors. The precise hypotheses and primary locators are in Proposition 73.1.2: Chenevier, arXiv:0809.0415, Proposition 3.3 and Example 3.4, p. 42, and Example 3.6 and Proposition 3.7, p. 43. Each closed condition must be represented by the quotient used here. These are formal deformation rings over one coefficient ring W. An identification with functions on a local or global spectral stack is additional geometric data.

A Pair of Indistinguishable Local Restrictions

Proposition 64.1.1 *(nonuniqueness with absolutely irreducible residual representation).* There are distinct continuous rank-two determinant deformations over $\mathbb{Z}/8\mathbb{Z}$, with the same absolutely irreducible residual representation and fixed determinant character, whose restrictions at 3 and 5 coincide.

Proof. Let K be the splitting field of $X^3 - X - 1$ over $\mathbb{Q}$. The polynomial has no root modulo 2, hence is irreducible over $\mathbb{Q}$. Its discriminant is -23, so its Galois group is S_3. An integral standard representation

of this group is generated by

$$A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Direct multiplication gives $A^3 = B^2 = 1$ and $BAB = A^{-1}$. Modulo 2, the two eigenlines of A over $\mathbb{F}_4$ are interchanged by B. Thus this residual representation is absolutely irreducible. Inflate the integral representation to $\rho : G_{\mathbb{Q}} \to GL_2(\mathbb{Z}_2)$.

Let χ be the quadratic character of $L = \mathbb{Q}(\sqrt{241})$. For $v = 3, 5$, the polynomial $X^2 - 241$ has the simple root 1 modulo v. A root u_r modulo v^r lifts uniquely in its residue class modulo v^r: the correction $u_r + cv^r$ is determined by an equation with invertible coefficient $2u_r$ modulo v. Taking the limit gives a root in $\mathbb{Z}_v$. Hence $\chi|_{G_{\mathbb{Q}_v}} = 1$ for both places. Set $\rho' = \chi\rho$. Then

$$\bar{\rho}' = \bar{\rho}, \qquad \det \rho' = \chi^2 \det \rho = \det \rho.$$

The restrictions of ρ and ρ' at 3 and 5 agree. Both are unramified outside $S = \{2, 3, 5, 23, 241, \infty\}$.

The unique quadratic subfield of K is $\mathbb{Q}(\sqrt{-23})$, which differs from L. Consequently $K \cap L = \mathbb{Q}$, and the composite extension has Galois group $S_3 \times C_2$. Choose g whose image is $(1, -1)$. Then

$$\det(T - \rho(g)) = T^2 - 2T + 1, \qquad \det(T - \rho'(g)) = T^2 + 2T + 1.$$

The traces are 2 and 6 modulo 8. Thus the determinant laws of the two reductions modulo 8 are distinct lifts of the same residual law, with the asserted common local restrictions and fixed determinant character. □

The example lies in a single finite-ramification deformation problem. A more restrictive arithmetic condition could exclude one lift. Uniqueness on such a condition would require its own proof.

64.2 Restriction, Closed Image, and Fibres

The local product functor is represented by

$$B_{p,q} = R_p^{\mathrm{Ch}} \widehat{\otimes}_W R_q^{\mathrm{Ch}} = \varprojlim_r (R_p^{\mathrm{Ch}}/\mathfrak{m}_p^r) \otimes_W (R_q^{\mathrm{Ch}}/\mathfrak{m}_q^r).$$

The completion is for the sum of the maximal ideals. Restriction is a natural transformation $\mathcal{D}_{\mathrm{glob}} \to \mathcal{D}_p \times \mathcal{D}_q$, so its map on representing rings has the opposite direction.

PROPOSITION 64.2.1 *(the formal restriction diagram).* Restriction induces a continuous local W-algebra homomorphism

$$r_{p,q}^{\sharp} : B_{p,q} \longrightarrow R^{\mathrm{Ch}}, \qquad x \otimes y \longmapsto r_p^{\sharp}(x) r_q^{\sharp}(y). \tag{64.1}$$

For the map in Equation (64.1), put $J_{p,q} = \ker r_{p,q}^{\sharp}$ and $C_{p,q} = B_{p,q}/J_{p,q}$. Then

$$B_{p,q} \twoheadrightarrow C_{p,q} \hookrightarrow R^{\mathrm{Ch}}, \qquad \operatorname{Spf} R^{\mathrm{Ch}} \longrightarrow \operatorname{Spf} C_{p,q} \hookrightarrow \operatorname{Spf} B_{p,q}.$$

For a continuous local map $b : B_{p,q} \to A$, the fibre over the prescribed local pair is represented over A by $R^{\mathrm{Ch}} \widehat{\otimes}_{B_{p,q},b} A$.

Proof. Restricting the universal determinant gives maps $r_v^\sharp : R_v^{\mathrm{Ch}} \to R^{\mathrm{Ch}}$. Their images commute, and their W-actions agree. The completed tensor-product property gives $r_{p,q}^\sharp$ with the displayed formula. The kernel is closed. The quotient is complete and injects into R^{Ch}. Since the rings are profinite, its image is closed. The formal-spectrum diagram reverses these maps. For a finite local A-algebra A', a map from the completed base change to A' is precisely a map $R^{\mathrm{Ch}} \to A'$ whose restriction to $B_{p,q}$ is the prescribed map through A. This proves the fibre assertion. It is the same construction as Theorem 73.1.4. □

Definition 64.2.2 *(the closed image of two-prime restriction).* The formal scheme Spf $C_{p,q}$ is the closed image of restriction inside the local deformation product. A proposed two-prime Selmer algebra $R_{p,q}^{\mathrm{Ch}}$ requires a specified $B_{p,q}$-algebra structure and comparison $R_{p,q}^{\mathrm{Ch}} \to C_{p,q}$ before the two can be identified. The quotient $C_{p,q}$ itself imposes the kernel relations only.

Example 64.2.3 *(a flat closed-image map with a nonliftable point).* Let $B = k[[x]]$ and $R = k[[t]]$, with $x \mapsto t^2$. Every series in t is uniquely $f(t^2) + t\,g(t^2)$. Thus R is free of rank two over B, and the map is injective. Its kernel quotient is B itself. Nevertheless, the local point $B \to k[\epsilon]/(\epsilon^2)$, $x \mapsto \epsilon$, has no lift to R. A local image of t is $a\epsilon$, whose square is zero. Flatness does not turn the closed image into the image functor on these Artinian points.

Example 64.2.4 *(an ordinary fibre with nontrivial derived structure).* For $B = k[[x]] \to R = k$, $x \mapsto 0$, the fibre over the residue point has ordinary coordinate ring k. The free resolution $[B \xrightarrow{x} B]$, in cohomological degrees -1, 0, gives

$$k \otimes_B^{\mathbb{L}} k \simeq k \oplus k[1], \qquad H^{-1}(k \otimes_B^{\mathbb{L}} k) = k.$$

Thus the derived fibre retains information lost by the ordinary fibre. The same calculation proves that this map is not flat.

Remark 64.2.5 *(the information retained by restriction).* The kernel quotient records equations satisfied by local pairs. The fibres record their global lifts. Proposition 64.1.1 shows that those lifts need not be unique, so $r_{p,q}^\sharp$ need not be surjective. In a derived deformation theory the fibre uses derived completed tensor product. None of these constructions identifies a Nygaard filtration or supplies a comparison with prismatic sheaves.

64.3 The Pairwise Pullback and Its Central Actions

Definition 64.3.1 *(completed external product and pullback).* Let M_p, M_q be finite modules over the local deformation rings. Their completed external product is the finite $B_{p,q}$-module

$$M_{p,q} = \varprojlim_r (M_p/\mathfrak{m}_p^r M_p) \otimes_W (M_q/\mathfrak{m}_q^r M_q).$$

The module obtained by ordinary restriction pullback is

$$M_{p,q}^{\mathrm{res}} = R^{\mathrm{Ch}} \widehat{\otimes}_{B_{p,q}} M_{p,q} = \varprojlim_r \frac{R^{\mathrm{Ch}} \otimes_{B_{p,q}} M_{p,q}}{\mathfrak{m}_{\mathrm{glob}}^r (R^{\mathrm{Ch}} \otimes_{B_{p,q}} M_{p,q})}. \tag{64.2}$$

The algebraic base change is finite over R^{Ch} and therefore already complete. Equation (64.2) also specifies its topology.

PROPOSITION 64.3.2 *(the action on the formal pullback).* Let Z_p, Z_q be unital W-algebras acting R_p^{Ch}-linearly and R_q^{Ch}-linearly on M_p, M_q, respectively. Each acting endomorphism is continuous for the maximal-ideal topology. Then $M_{p,q}^{\mathrm{res}}$ has a canonical action of $Z_p \otimes_W Z_q$, given by

$$(z_p \otimes z_q)(a \otimes (m_p \otimes m_q)) = a \otimes (z_p m_p \otimes z_q m_q).$$

This action commutes with the R^{Ch}-action.

Proof. Finite presentations of M_p, M_q show that $M_{p,q}$ is finite over $B_{p,q}$. Base change gives a finite module over R^{Ch}. Local linearity makes the displayed action compatible with the balancing relations and with every quotient in the inverse limits. It therefore defines an action on the completed tensors. The two local actions affect separate factors, so they commute. Multiplication on the first factor commutes with both. No topology on the abstract algebras Z_p, Z_q is needed. □

The associated coherent sheaf on Spf R^{Ch} is the ordinary pullback of the completed external product. The derived construction replaces the external product and pullback by their derived completed versions. Its boundedness requires additional Tor conditions. The ordinary pullback is exact only on a domain where the required higher Tor groups vanish.

PROPOSITION 64.3.3 *(transport of ordinary central actions).* For a W-linear category C, let $Z(C) = \mathrm{End}(\mathrm{id}_C)$ be its ordinary centre. A W-linear equivalence $F : C \to \mathcal{D}$ induces an isomorphism

$$Z(C) \xrightarrow{\sim} Z(\mathcal{D}). \tag{64.3}$$

For a fully faithful functor, this statement applies to its essential image. It does not identify the centre of a larger target category.

Proof. Choose a quasi-inverse G and the unit and counit isomorphisms. A natural endomorphism z of id_C defines an endomorphism of $\mathrm{id}_{\mathcal{D}}$ by conjugating FzG with the counit. Naturality gives a central action, and composition and addition are preserved. This defines the map in Equation (64.3). The construction with G is its inverse. Restricting a fully faithful functor to its essential image makes it an equivalence. □

Remark 64.3.4 *(local coefficients and Bernstein blocks).* For smooth representations of $GL_2(\mathbb{Q}_v)$ in characteristic zero with coefficient prime different from v, the Bernstein decomposition has the form

$$\mathrm{Rep}(GL_2(\mathbb{Q}_v)) = \prod_{\mathfrak{s}} \mathrm{Rep}_{\mathfrak{s}}(GL_2(\mathbb{Q}_v)). \tag{64.4}$$

For the blocks in Equation (64.4), the ordinary scalar comparison and its precise domain are proved in Proposition 116.2.2. Its image in the spectral categorical centre is the scalar subalgebra of Proposition 116.2.5. Using a Bernstein centre as Z_v in the formal pullback requires an action over the chosen coefficient ring and on the actual module M_v. A categorical comparison additionally needs a functor with specified essential image. A local (φ, Γ)-module at residue prime v is not identified here with a deformation over $W(k)$ of coefficient characteristic $p_0 \neq v$.

64.4 Equivariant Isomorphisms on Formal Charts

The module $M^{\mathrm{res}}_{p,q}$ is constructed from local data. Let N be an independently specified global module. The pairwise comparison asks for an isomorphism from the former to the latter which intertwines their central actions. Equal determinant laws alone do not solve this problem.

Example 64.4.1 *(extension data invisible to determinant laws).* Let $R = k[[t]]$ and $M = N = R^2$. Give them actions of $R[z]$ by

$$z|_M = 0, \qquad z|_N = J = \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix}.$$

For every polynomial f, one has $f(J) = f(0)1 + f'(0)J$, so both actions have characteristic polynomial $(X - f(0))^2$ for $f(z)$. The determinant polynomial laws agree after every coefficient extension. The actions also coincide modulo t. An invertible intertwiner U would satisfy $JU = 0$, contradicting $J \neq 0$. These are reducible algebra representations. They exhibit the loss of extension data for coherent objects; they do not contradict reconstruction of a representation from an absolutely irreducible residual determinant.

Proposition 64.4.2 *(the scheme of equivariant isomorphisms).* Let $(R, \mathfrak{m}, k)$ be a complete Noetherian local ring. Let M, N be free R-modules of rank n with actions of an R-algebra generated by $a_1, \dots, a_s$. Choose bases and let $P_i, Q_i \in \mathrm{Mat}_n(R)$ denote the action matrices on M, N. The functor of equivariant isomorphisms is represented by

$$\mathrm{Isom}(M, N) = \mathrm{Spec}\, \frac{R[u_{ab}, d]}{(d \det(U) - 1,\ (Q_i U - U P_i)_{ab} : i, a, b)}. \tag{64.5}$$

Here $U = (u_{ab})$, with $1 \le a, b \le n$ and $1 \le i \le s$.

Proof. Over any commutative R-algebra A, a matrix U is invertible exactly when its determinant is a unit. The variable d specifies the unique inverse of that determinant. Intertwining the generators is equivalent to intertwining the whole algebra. Thus maps from the coordinate ring in Equation (64.5) to A are exactly the equivariant isomorphisms $M \otimes_R A \to N \otimes_R A$, naturally in A. □

Proposition 64.4.3 *(successive lifting obstructions).* In the preceding situation, put $I_r = \mathfrak{m}^r/\mathfrak{m}^{r+1}$ for $r \ge 1$. Let U_r be an invertible intertwiner modulo $\mathfrak{m}^r$. Choose a lift $\widetilde{U}$ modulo $\mathfrak{m}^{r+1}$ and set

$$E_i = Q_i \widetilde{U} - \widetilde{U} P_i \in \mathrm{Mat}_n(I_r).$$

Write bars for reduction modulo $\mathfrak{m}$, and define

$$L_r : \mathrm{Mat}_n(I_r) \longrightarrow \mathrm{Mat}_n(I_r)^s, \qquad V \longmapsto (\overline{Q}_i V - V \overline{P}_i)_i.$$

The class $[(E_i)_i] \in \operatorname{coker} L_r$ is independent of the lift. It vanishes precisely when this U_r extends to an invertible intertwiner modulo $\mathfrak{m}^{r+1}$. A compatible sequence of such lifts determines an equivariant isomorphism over R.

Proof. Changing $\widetilde{U}$ by $V \in \mathrm{Mat}_n(I_r)$ changes the defects by $L_r(V)$, since $\mathfrak{m} I_r = 0$. Thus the class is independent of the lift and vanishes exactly when a correction solves every intertwining equation. The determinant of a corrected lift remains invertible modulo the maximal ideal, hence is a unit. Completeness

gives the matrix limit of a compatible sequence. The intertwining equations pass to the limit, and its determinant is a unit because its residue is. Conversely an equivariant isomorphism over R gives such a compatible sequence by reduction. □

The obstruction depends on the specified partial lift U_r. Vanishing for a compatible sequence, rather than separate choices at unrelated orders, constructs the limit. For general coherent complexes the comparison uses a derived equivariant Hom complex. The finite free criterion does not assert that the intended arithmetic sheaves satisfy its hypotheses.

64.5 Global Actions and Geometric Comparison

PROPOSITION 64.5.1 *(the action generated by local centres).* Let E be a commutative ring and Z_v commutative unital E-algebras. For finite sets S of places put $A_S = \bigotimes_{v\in S,E} Z_v$, with transition maps inserting units. Set

$$A = \varinjlim_{S \text{ finite}} A_S. \tag{64.6}$$

Pairwise commuting unital actions $\alpha_v : Z_v \to \mathrm{End}_E(M)$ on an E-module induce a unique action

$$\alpha : A \to \mathrm{End}_E(M), \qquad \alpha(\otimes_{v\in S} z_v) = \prod_{v\in S} \alpha_v(z_v).$$

Its image is $A/\ker\alpha$. Natural actions on an E-linear category instead give a map to its ordinary centre.

Proof. The product of the commuting actions is multiplicative and E-multilinear on each finite product. The tensor-product property gives an action of A_S. Inserting a unit changes no endomorphism, so these maps are compatible. The colimit in Equation (64.6) therefore gives α. The quotient assertion is the first isomorphism theorem. Products of natural endomorphisms are natural, which proves the categorical assertion. □

PROPOSITION 64.5.2 *(faithfulness and generation for a global scalar centre).* Suppose the preceding actions define $\alpha : A \to Z(C)$ on an E-linear category. Then

$$A/\ker\alpha \xrightarrow{\sim} \mathrm{im}\,\alpha \subseteq Z(C). \tag{64.7}$$

This image is the entire ordinary centre exactly when every central endomorphism is generated by the finite local actions. The map from A itself is an isomorphism exactly when it is also faithful. No identification with R^{Ch} is implicit in this statement.

Proof. The quotient isomorphism in Equation (64.7) follows from the preceding proposition. Surjectivity means precisely that every central endomorphism belongs to this image. Injectivity means $\ker\alpha = 0$. Together these are exactly the isomorphism conditions. □

Remark 64.5.3 *(first-order fixed-determinant deformations).* Let $V = k^2$ with action $\bar\rho$, and let $\mathrm{ad}\,\bar\rho = \mathrm{End}_k(V)$ with conjugation action. A lift over $k[\epsilon]/(\epsilon^2)$ has the form $(1+\epsilon c(g))\bar\rho(g)$. Multiplicativity is the continuous cocycle equation for c. The fixed-determinant condition is $\mathrm{tr}\,c(g) = 0$. Strict change of basis changes c by the coboundary of an element of $\mathrm{ad}\,\bar\rho$. Thus the tangent space before additional local conditions is

$$\frac{Z^1_{\mathrm{cts}}(G_{\mathbb{Q},S}, \mathrm{ad}^0\,\bar\rho)}{d\ \mathrm{ad}\,\bar\rho}.$$

The denominator consists of traceless cocycles. When 2 is invertible, scalar subtraction identifies it with $d\ \mathrm{ad}^{\circ}\ \bar{\rho}$, giving the usual $H^{1}_{\mathrm{cts}}(G_{\mathbb{Q},S}, \mathrm{ad}^{\circ}\ \bar{\rho})$. At characteristic two this simplification needs justification. Additional deformation conditions restrict this tangent problem. A tangent-space comparison does not identify the complete rings.

The Geometric Pairwise and Global Problems

Choose a coefficient base E for a geometric realization. Suppose local spaces X_p, X_q, a global space X, and a morphism $r : X \to X_p \times_E X_q$ have been constructed in a specified formal, analytic, or derived category. Specify coefficient comparison maps relating this diagram to the deformation rings above. Let $\mathcal{F}_p, \mathcal{F}_q$ be the local coherent objects, and let $\mathcal{F}^{\mathrm{prism}}_{\mathrm{glob}}$ be an independently defined global object. The superscript records the intended prismatic realization; its existence is an additional construction. Work in a domain where the external product and Lr^* preserve the chosen category of coherent complexes, for example on perfect complexes. Supply the local centre actions over E and their compatibility with the coefficient maps.

Proposition 64.5.4 *(external-product and pullback Tor-independence).* Suppose the local and global objects are coherent sheaves in cohomological degree zero, in a geometric domain with the stated derived external product and derived pullback. Put $\mathcal{M} = \mathcal{F}_p \boxtimes_E \mathcal{F}_q$ and $\mathcal{M}^{\mathbb{L}} = \mathcal{F}_p \boxtimes^{\mathbb{L}}_E \mathcal{F}_q$. The external-product Tor-independence condition is that the canonical augmentation $\mathcal{M}^{\mathbb{L}} \to \mathcal{M}$ is a quasi-isomorphism. The pullback Tor-independence condition is that $Lr^*\mathcal{M} \to r^*\mathcal{M}$ is a quasi-isomorphism. These are separate vanishing conditions on the higher derived external product and the higher derived pullback. Under both conditions, an equivariant ordinary isomorphism $r^*\mathcal{M} \simeq \mathcal{F}^{\mathrm{prism}}_{\mathrm{glob}}$ induces an equivariant derived isomorphism

$$Lr^*\mathcal{M}^{\mathbb{L}} \simeq \mathcal{F}^{\mathrm{prism}}_{\mathrm{glob}}.$$

For formal spaces the conditions refer to the derived completed operations in the chosen realization.

Proof. The external-product condition gives $Lr^*\mathcal{M}^{\mathbb{L}} \simeq Lr^*\mathcal{M}$. The pullback condition identifies the latter with $r^*\mathcal{M}$. Composition with the given ordinary isomorphism proves the result. The augmentations are natural, so they intertwine the specified central actions. Thus the composite is equivariant. □

Example 64.5.5 *(ordinary descent without derived descent).* Let $W = W(k)$ for a finite field k of characteristic p_0. Take all three spaces to be Spf W, let r be the identity, and take $\mathcal{F}_p = \mathcal{F}_q = \mathcal{F}_{\mathrm{glob}} = k$. The ordinary external product and its pullback are k. However, the free resolution $[W \xrightarrow{p_0} W]$ of k, in degrees -1, 0, gives

$$k \otimes^{\mathbb{L}}_W k \simeq k \oplus k[1].$$

This is also the derived completed tensor product: the resulting finite p_0-torsion modules are derived p_0-complete. It is not quasi-isomorphic to k. Here r is flat and both local sheaves are perfect, but external-product Tor-independence fails. Thus perfectness and flatness of r do not convert ordinary pairwise descent into its derived enhancement.

Conjecture 64.5.6 *(pairwise and global Bernstein compatibility).* For the intended arithmetic geometric realization and every admissible pair of distinct finite primes, the following comparisons exist. For coherent sheaves in degree zero, the ordinary pairwise comparison is an equivariant isomorphism

$$r^*(\mathcal{F}_p \boxtimes_E \mathcal{F}_q) \simeq \mathcal{F}^{\mathrm{prism}}_{\mathrm{glob}}. \tag{64.8}$$

The derived enhancement is the additional equivariant comparison

$$Lr^*(\mathcal{F}_p \boxtimes^{\mathbb{L}}_E \mathcal{F}_q) \simeq \mathcal{F}^{\mathrm{prism}}_{\mathrm{glob}}. \tag{64.9}$$

The ordinary and derived assertions are distinct. Both hold together if ordinary descent and the two Tor-independence conditions of Proposition 64.5.4 hold. Each asserted family of isomorphisms must be compatible with the geometric transition maps as the finite set of places varies. Second, the local actions on the global automorphic category induce an isomorphism onto its specified ordinary Bernstein centre

$$A \xrightarrow{\sim} \mathcal{Z}_{\mathrm{glob}}, \tag{64.10}$$

and a specified comparison identifies the action in Equation (64.10) with its spectral counterpart. Thus the action must be faithful and generate the intended centre. A comparison of higher centres must be constructed in its category of derived algebras and is a separate assertion from this degree-zero identity.

Remark 64.5.7 *(the two comparison problems).* The pairwise assertion compares coherent objects and their actions. The global assertion identifies the image and range of the assembled central action. The formal restriction diagram constructs neither the first isomorphism nor the latter surjectivity. The hypotheses of Corollary 73.1.5 specify the Selmer-algebra, geometric, trace and ordinary descent data. The derived enhancement requires the additional derived descent datum or the two Tor-independence conditions. Neither version establishes the global centre map.

PROPOSITION 64.5.8 *(local equations and gluing of the pairwise comparison).* Suppose, on a formal scheme X, the two objects in the pairwise comparison are vector bundles of equal rank. Choose an open cover trivializing both bundles. Suppose the algebra of actions on each chart has specified finitely many generators. The equivariant isomorphism exists if and only if there are invertible matrices on the charts satisfying the intertwining equations and agreeing under the transition matrices on every overlap. On a complete local chart, a specified residue isomorphism lifts exactly when one can successively choose lifts with zero obstruction classes from Proposition 64.4.3.

Proof. An isomorphism gives its invertible matrices in the chosen bases. Equivariance gives the intertwining equations, and independence of trivialization gives the overlap equalities. Conversely the overlap equalities glue the local morphisms as morphisms of sheaves. Their local inverses also agree on overlaps, so the global morphism is invertible. Equivariance can be checked on each chart, and hence holds globally. The assertion on a complete local chart is precisely the successive lifting proposition. It requires a compatible choice at every order. □

Remark 64.5.9 *(the missing arithmetic comparisons).* The remaining constructions are the geometric realization of the formal restriction map, the local modules with their Bernstein actions over a common coefficient base, and the independently defined global prismatic object. Their equivariant comparison must preserve the required filtrations and agree on overlaps and on larger finite sets of places. To apply the vector-bundle criterion one must also establish its local freeness hypotheses. Otherwise a derived comparison is required. Identifying a kernel quotient or matching traces does not construct these maps.

64.6 THE GLOBAL DESCENT PROBLEM

PROPOSITION 64.6.1 *(data supplied by two-prime restriction).* For the fixed deformation problem, two-prime restriction supplies the map $B_{p,q} \to R^{\mathrm{Ch}}$, its closed-image quotient, and its formal fibres. Given finite local modules and the specified local actions, it also supplies $M^{\mathrm{res}}_{p,q}$ with its product action. These constructions do not require arithmetic Arinkin–Gaitsgory or a prismatic comparison.

Proof. The restriction diagram and fibre follow from Proposition 64.2.1. The module and action follow from Proposition 64.3.2. Their proofs use the stated universal properties and finite module presentations only. □

Remark 64.6.2 *(categorical and prismatic descent).* Adelic categorical descent additionally requires the actual local functors, their coefficient and family comparisons, the prismatic and untilt descent maps, and the global centre comparison. The ordinary scalar-centre theorem does not provide these functors. The determinant rings do not specify a common Nygaard filtration at distinct residue primes. A proposed arithmetic singular-support construction must define its spaces and higher centre maps before it can supply the global comparison. The algebraic and geometric obstructions to that step are developed in Chapter 88.

Chapter 65

Moment Growth and Arithmetic Spectral Comparisons

Let $L : \mathbb{C}[x] \to \mathbb{C}$ be a specified positive Hamburger functional with finite moments $\mu_n = L(x^n)$. Positivity means $L(p\overline{p}) \geq 0$ for every polynomial p, where the bar conjugates coefficients. The analytic condition is $\mu_{2n}^{1/(2n)} = O(n)$. Together with positivity it implies determinacy. Its arithmetic application requires construction of the regularized functional and its carrier, support and operator comparisons.

65.1 Positive Functionals and Growth Estimates

Definition 65.1.1 *(The analytic moment condition).* For the specified finite positive functional L, write $\mathrm{R}_{\text{analytic}}$ for $\limsup_{n\to\infty} \mu_{2n}^{1/(2n)}/n < \infty$. A two-sided factorial estimate is stronger than this upper bound.

Remark 65.1.2 *(The arithmetic application).* The classical implication concerns determinacy of L. Its arithmetic application requires the functional, support and operator identifications. No implication between RH and an unspecified regularization is asserted.

Proposition 65.1.3 *(A logarithmic lower comparison).* If the specified functional satisfies $\mu_{2n} \leq (2\pi)^{-4n}(2n)!(\log n)^{4n}(1 + o(1))$, its positive even moments have $\mu_{2n}^{-1/(2n)} \geq c/[n(\log n)^2]$ eventually for some $c > 0$. This comparison series converges.

Proof. Apply Proposition 65.5.2 and Corollary 65.5.3 below. □

Remark 65.1.4 *(Three requirements).* Existence and positivity of L, an upper-growth estimate for its moments, and its arithmetic comparison are separate premises. No theorem of logical independence between them is asserted.

65.2 Factorial Roots

The root $(2n)!^{1/(2n)}$ is asymptotic to $2n/e$. Taking the $2n$-th root of $(\log n)^{4n}$ gives $(\log n)^2$, and inversion reverses that exponent.

65.3 Li Coefficients and Polynomial Moments

The Li summand $1 - (1 - 1/\rho)^n$ differs from the polynomial summand $[\rho(1-\rho)]^n$. Their traces require different summation and domain data. The termwise comparison is made below.

65.4 The Conditional Operator Comparison

Under Hypothesis 57.4.21,

$$\mathrm{VB}^* \iff (\mathrm{Reg}) \iff (\mathrm{HB}+) \iff (\mathrm{BD}+).$$

Here (BD+) is the sharp approximation predicate of Chapter 57, not exponential completeness. (Reg) includes the identified positive functional, determinacy and spectral support. The classical moment estimate supplies determinacy; the arithmetic identification, support and operator comparison remain additional inputs.

65.5 A Conditional Logarithmic Estimate

The Assumed Bound

Suppose the specified L satisfies

$$\mu_{2n} \le (2\pi)^{-4n}(2n)!(\log n)^{4n}(1+o(1)). \tag{65.1}$$

This is an assumption on the functional, not a consequence of zero counting or a bound for Li coefficients.

Stirling Evaluation

LEMMA 65.5.1 *(Factorial roots).*

$$(2n)!^{1/(2n)} = \frac{2n}{e}(4\pi n)^{1/(4n)}(1+O(n^{-2})) \sim \frac{2n}{e}.$$

Proof. Take the $2n$-th root of $(2n)! = \sqrt{4\pi n}(2n/e)^{2n}(1+O(n^{-1}))$. The last factor becomes $1+O(n^{-2})$, while $(4\pi n)^{1/(4n)} \to 1$. □

PROPOSITION 65.5.2 *(A conditional lower comparison).* Assume (65.1) and positive even moments. Then

$$\mu_{2n}^{1/(2n)} \le \frac{2n}{e(2\pi)^2}(\log n)^2(1+o(1)).$$

For every $0 < c < e(2\pi)^2/2$, there is n_0 such that

$$\mu_{2n}^{-1/(2n)} \ge \frac{c}{n(\log n)^2} \quad (n \ge n_0),$$

so

$$\sum_{n\ge n_0} \mu_{2n}^{-1/(2n)} \ge c \sum_{n\ge n_0} \frac{1}{n(\log n)^2}. \tag{65.2}$$

Proof. Take roots of the assumed bound, apply Stirling and invert the positive inequality. If instead an even moment of positive order vanishes, any positive representing measure is supported at zero; its prescribed mass makes it unique. □

COROLLARY 65.5.3 *(The lower comparison converges).*

$$\sum_{n\ge 2} \frac{1}{n(\log n)^2} < \infty.$$

This convergent lower comparison does not establish divergence of Carleman's sum.

Proof. The integral test gives $\int_2^\infty dx/[x(\log x)^2] = 1/\log 2$. □

Remark 65.5.4 *(Root scaling).* The exponent $4n$ of the logarithm becomes 2 under the $2n$-th root and -2 under inversion. This calculation uses only the displayed assumed estimate on L.

Proposition 65.5.5 *(A factorial upper bound implies determinacy).* If L is positive with finite moments and $\mu_{2n} \le AC^n(n!)^2$ eventually for constants $A, C > 0$, then it is determinate. No matching lower bound is needed.

Proof. For positive even moments, $\mu_{2n}^{1/(2n)} \le A^{1/(2n)}\sqrt{C}(n!)^{1/n} = O(n)$. Carleman's sum dominates a positive multiple of the harmonic series. A vanishing even moment gives the unique measure supported at zero. □

65.6 Finite Cutoffs and Regularized Moments

Distinct Zero Sums

Proposition 65.6.1 *(Li summands and polynomial summands).* Under RH, for $\rho = 1/2 + i\gamma$,

$$|1 - (1 - 1/\rho)^n| \le 2, \qquad [\rho(1-\rho)]^n = (1/4 + \gamma^2)^n.$$

A bound for the summed Li expression does not, by this termwise magnitude comparison, give finite unregularized polynomial moments.

Proof. The identity $|1 - 1/\rho| = 1$ gives the first inequality. The polynomial identity gives the second, whose cutoff sums diverge by the next proposition. An identity for the Li trace therefore requires a separate comparison to a regularized polynomial functional. □

Proposition 65.6.2 *(Positive cutoff moments diverge).* Put $S_n(T) = \int_{(0,T]} (1/4 + t^2)^n \, dN(t)$. For fixed integer $n \ge 0$,

$$S_n(T) = \frac{T^{2n+1}}{2\pi(2n+1)} \log \frac{T}{2\pi} - \frac{T^{2n+1}}{2\pi(2n+1)^2} + O_n(T^{2n} \log T).$$

The integral over $(0, \infty)$ diverges. Under RH it is the cutoff shifted-zero sum over positive ordinates.

Proof. Apply the Stieltjes calculation in Proposition 80.2.5. The positive leading term tends to infinity. □

Remark 65.6.3 *(The leading term and regularization).* Divergence already occurs in the leading zero-counting term. Improving its remainder alone cannot make the positive integral finite. A specified regularization is needed before its moment growth can be studied.

Three Mathematical Requirements

One must construct the finite positive functional, prove a sufficient growth estimate for that same functional, and identify it with the arithmetic operator and its spectral support. None of these maps is constructed by renaming a divergent cutoff sum.

65.7 Determinacy, VB*, and RH

The following propositions separate the classical determinacy implication from the arithmetic operator comparison.

Determinacy and the Operator Comparison

PROPOSITION 65.7.1 *(The predicate in the operator comparison).* Under Hypothesis 57.4.21, VB* is equivalent to (Reg), not to bare (Det). In the RH sector this supplies a conditional route to the VB* converse when (Reg) is constructed. The forward arithmetic implication to zero placement retains its separate P2, P3, determinant–trace and positivity hypotheses.

Proof. Use item 1 of the hypothesis and the definition of (Reg). Neither its support condition nor its arithmetic moment identification follows from uniqueness of an abstract measure. □

Step 2: Determinacy vs RH

PROPOSITION 65.7.2 *(What the distance and determinacy results supply).* The classical BD criterion is convergence of its finite distances to zero. The supplied-rate transport of Corollary 57.4.19 transfers a quantitative estimate only after that estimate and the scaled isometry are given. Separately, a positive moment functional with the Carleman growth bound is determinate. These statements alone supply neither the arithmetic support condition nor an equivalence between bare (Det) and RH.

Proof. The distance identity multiplies each finite norm by the fixed positive scale; it does not estimate its dependence on the cutoff. Carleman's theorem concerns uniqueness for the supplied moment functional. Applying either result to an arithmetic operator requires the stated comparison data. □

The Implication Diagram

Remark 65.7.3 *(The moment and operator steps).* The classical step is

$$\mathrm{R}_{\mathrm{analytic}} \Longrightarrow \text{Carleman criterion} \Longrightarrow \text{Det}.$$

The arithmetic step, with the identified positive moment functional, is

$$\text{Det} + \text{specified support} \Longrightarrow (\text{Reg}) \xrightarrow{\text{operator comparison}} \mathrm{VB}^*.$$

The conditional forward arithmetic theorem then gives RH with its remaining hypotheses. The operator comparison is item 1 of Hypothesis 57.4.21.

Remark 65.7.4 *(The role of the moment estimate).* The growth estimate proves the determinacy component. It does not construct the arithmetic carrier, its support or operator comparison. Those are additional steps even when the zero-placement statement is assumed. No independence theorem between RH and this moment estimate is asserted by this implication diagram.

65.8 DETERMINACY CONSEQUENCES

THEOREM 65.8.1 *(Moment estimates and arithmetic comparisons).* For a specified positive finite Hamburger functional:

(i) Linear even-moment root growth implies determinacy.

(ii) A factorial upper bound suffices for that growth.

(iii) The logarithmic estimate (65.1) gives the convergent lower comparison (65.2).

(iv) Its arithmetic application retains the functional, support and operator comparisons of Remark 65.7.3.

Proof. The first assertion is Carleman's harmonic comparison; the second is Proposition 65.5.5; the third is Proposition 65.5.2. The fourth is the stated domain of the arithmetic comparison. □

Corollary 65.8.2 *(The moment estimate and the arithmetic target).* The implication from R_{analytic} to Hamburger determinacy does not by itself establish RH. The stated route also requires the arithmetic moment identification, spectral support and operator comparison, together with the hypotheses of the forward arithmetic theorem. No converse non-implication is claimed.

Proof. The moment estimate supplies the first line of Remark 65.7.3. The remaining arrows have the additional hypotheses displayed there. □

65.9 Examples and the Arithmetic Construction

Proposition 65.9.1 *(Positive functionals at two growth scales).* The probability measure $\frac{1}{2}\mathbf{1}_{[-1,1]}\,dx$ has $\mu_{2n} = 1/(2n+1)$ and odd moments zero. It satisfies R_{analytic}, but no eventual lower bound $\mu_{2n} \geq C_-^n (n!)^2$ with $C_- > 0$. For $a > 0$, the probability measure $\frac{a}{2}e^{-a|x|}\,dx$ has $\mu_{2n} = (2n)!/a^{2n}$ and odd moments zero. Its even-moment roots are asymptotic to $2n/(ae)$, so it is determinate.

Proof. Integrating on $[-1, 1]$ gives the first formula. Its $2n$-th root tends to one, whereas that of $C_-^n (n!)^2$ grows as $\sqrt{C_-}\,n/e$. For the second measure, symmetry and the substitution $u = ax$ give

$$a\int_0^\infty x^{2n} e^{-ax}\,dx = a^{-2n}\int_0^\infty u^{2n} e^{-u}\,du = a^{-2n}(2n)!.$$

Stirling gives linear root growth and hence Carleman determinacy. Neither example is identified with the arithmetic functional. □

Question 65.9.2 *(The arithmetic moment construction).* Construct a specified regularization of the shifted-zero polynomial sums, including its counterterms and normalization, defining a finite positive Hamburger functional L. Prove $L(x^{2n})^{1/(2n)} = O(n)$, or another sufficient determinacy criterion, for this same functional. Identify its representing measure and support with the arithmetic carrier and construct the operator comparison. A two-sided factorial asymptotic is a stronger possible target, not a premise needed for the Carleman implication.

Remark 65.9.3 *(Other arithmetic inputs).* The moment theorem supplies neither the weight-selection nor the explicit-formula hypotheses. These remain separate requirements in the sufficient arithmetic argument.

65.10 Arithmetic Consequences

A constructed positive functional and its linear root bound give (Det). The specified support and arithmetic identification give (Reg); the supplied operator comparison then transfers positivity. The full arithmetic conclusion retains the remaining forward-theorem hypotheses. The finite cutoff calculation, the classical examples and the conditional estimates do not construct the missing regularization or its arithmetic comparison.

Chapter 66

The Primary Euler Product and Its Arithmetic Comparison

The one-mode occupation space has a zeta heat trace. A Gaussian test function has a gamma Mellin integral. Their product is the completed zeta function. These are three precise scalar calculations; identifying them with the character of an arithmetic bar or cobar object requires additional maps. In particular, ordinary graded duality reverses energy and does not by itself shift a Mellin parameter from s to $1-s$. We specify the carriers before making the comparison.

66.1 The Selected Primary Model

Finite places

Definition 66.1.1 *(one-mode primary tower).* Inside the polynomial oscillator representation at a prime p, set

$$\mathcal{H}^{(p)}_{\mathrm{prim}} = \bigoplus_{j\ge 0} \mathbb{C} e_{j,p}, \qquad e_{j,p} = (a^{(p)}_{-1})^j |0\rangle_p.$$

The vector $e_{0,p} = |0\rangle_p$ is the unit. Give $e_{j,p}$ energy $j \log p$, as in Convention 28.7.1(ii). This is a selected vector subspace; it excludes all modes $a^{(p)}_{-k}$ with $k > 1$ and any second oscillator species.

Remark 66.1.2 *(selection and operators).* One full oscillator species at a prime has heat trace $\prod_{k\ge 1}(1-p^{-k\beta})^{-1}$, while the selected tower has $(1-p^{-\beta})^{-1}$. Two species square the full factor. The selected tower contains every occupation of mode one, including its vacuum, and no higher modes. Both T_n and ψ_n of Chapter 28 send its generator to mode n; for $n > 1$ they do not preserve this selected space. No equivariant arithmetic identification follows from the selection.

Proposition 66.1.3 *(local shifts and Gibbs weights).* Normalize the Hall basis by $u_{j,p} = e_{j,p}/\sqrt{j!}$ and complete it to $\ell^2(\mathbb{N}_0)$. The shift $S_p u_{j,p} = u_{j+1,p}$ is an isometry, and the diagonal operator $H_p u_{j,p} = j \log p \, u_{j,p}$ has partition function

$$Z_p(\beta) = \mathrm{Tr}(e^{-\beta H_p}) = (1-p^{-\beta})^{-1} \qquad (\beta > 0).$$

Its normalized Gibbs state on bounded operators is

$$\varphi_{p,\beta}(A) = (1-p^{-\beta}) \sum_{j\ge 0} p^{-j\beta} \langle A u_{j,p}, u_{j,p}\rangle.$$

In particular a rank-one occupation projection has probability $(1-p^{-\beta})\,p^{-j\beta}$.

Proof. The shift preserves the displayed orthonormal basis and has range its tail. The heat trace is a convergent geometric series. Dividing the positive trace-class operator by that trace gives a positive functional with value one on the identity, proving the state formula. A state is a functional on operators,

not an assignment of an unnormalized Boltzmann weight to a state vector. No classification of Bost–Connes KMS states is needed for this calculation. □

The Gaussian test line

Definition 66.1.4 *(archimedean primary datum).* Use the specified test function $g(x) = e^{-\pi x^2}$ and its line

$$\mathcal{G} = \mathbb{C}g \subset \mathcal{S}(\mathbb{R}).$$

The datum includes its Mellin pairing, defined below. It is not an identification of the full countable-mode Heisenberg representation with a single coordinate oscillator. Multiplying by arbitrary Hermite polynomials leaves this line and gives further test functions.

PROPOSITION 66.1.5 *(Gaussian Mellin integral).* For $\operatorname{Re} s > 0$,

$$\int_0^\infty e^{-\pi x^2} x^{s-1}\, dx = \frac{1}{2}\pi^{-s/2}\Gamma(s/2).$$

The full real integral against $|x|^{s-1}$ is twice this value. It is a Mellin pairing, not the heat trace of an operator on the one-dimensional line $\mathcal{G}$.

Proof. Near zero integrability requires $\operatorname{Re} s > 0$, and the Gaussian gives decay at infinity. Substitute $u = \pi x^2$; then $x^{s-1}dx = \frac{1}{2}\pi^{-s/2}u^{s/2-1}du$. The gamma integral gives the formula. The integrand with absolute value is even, so the full real integral doubles it. □

The restricted tensor and its scalar pairing

Definition 66.1.6 *(selected primary model).* Set

$$P_{\mathrm{fin}} = \bigotimes_p{}' \mathcal{H}^{(p)}_{\mathrm{prim}}, \qquad \mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}} := \mathcal{G} \otimes_{\mathbb{C}} P_{\mathrm{fin}}.$$

The tensor product here is algebraic, with reference vectors $e_{0,p}$. Each elementary occupation has finite prime support. Its normalized basis is indexed by $N = \prod_p p^{j_p}$, and its finite energy is $\log N$. Its Hilbert completion is consequently $\ell^2(\mathbb{N}_{>0})$. Define the completed scalar pairing by

$$Z_{\mathrm{prim}}(s) = M_g(s)Z_{\mathrm{fin}}(s), \qquad M_g(s) = \int_{\mathbb{R}} g(x)|x|^{s-1}\, dx, \quad Z_{\mathrm{fin}}(s) = \sum_{N\geq 1} N^{-s}.$$

The common convergence region is $\operatorname{Re} s > 1$. This definition uses a finite heat trace and an archimedean Mellin pairing. It makes no claim that their product is the ordinary graded character of the tensor product with a one-dimensional representation. An Iwasawa action on this model, or a comparison with an arithmetic Hochschild trace object, must be specified separately.

Remark 66.1.7 *(operator domains).* On the finite Hilbert completion the energy operator is self-adjoint on $\{(c_N) : \sum_N (\log N)^2|c_N|^2 < \infty\}$; finite-support vectors form a core for every power. Thus its polynomial expressions, including $H^n - (H-1)^n$, are defined on that common core. These are distinct from both T_n and ψ_n of (28.1). A Mellin variable s is not an eigenvalue of a fixed operator on $\mathcal{G}$. Identification with the operator of Definition 45.1.1 requires its stated spectral comparison.

Remark 66.1.8 *(Virasoro comparison).* A selected one-mode tower is not invariant under the full Heisenberg vertex operations. No global Virasoro action is constructed here. The determinant-rank residue $r_\zeta = 2$ in Theorem 35.4.1 and a proposed central charge are different data until a compatible operator construction identifies them.

66.2 The Scalar Calculation

The finite Euler product

Theorem 66.2.1 *(finite primary character).* For $\operatorname{Re} s > 1$,

$$Z_{\mathrm{fin}}(s) = \prod_p (1 - p^{-s})^{-1} = \zeta(s).$$

For real $s = \beta$, this is a finite heat trace precisely when $\beta > 1$. Its scalar meromorphic continuation is distinct from an extension of that heat trace to other temperatures.

Proof. Unique factorization identifies an occupation tuple with one positive integer, with weight $\prod_p p^{-j_p s} = N^{-s}$. Absolute convergence follows from $\sum_N N^{-\sigma} < \infty$ for $\sigma = \operatorname{Re} s > 1$, so expansion of the geometric factors is justified and gives the Euler product. For real $\beta \le 1$, the series of positive diagonal entries diverges. The meromorphic continuation of the resulting scalar is the classical continuation of ζ; it has a simple pole at one. □

The archimedean Mellin factor

Theorem 66.2.2 *(full real Mellin normalization).* For $\operatorname{Re} s > 0$,

$$M_g(s) = \int_{\mathbb{R}} e^{-\pi x^2} |x|^{s-1}\, dx = \pi^{-s/2}\Gamma(s/2) = \Gamma_{\mathbb{R}}(s). \tag{66.1}$$

Proof. Apply Proposition 66.1.5 to the two half-lines. This fixes the factor of two and the Haar-measure convention. □

The completed scalar

Theorem 66.2.3 *(primary Euler–Mellin identity).* The specified completed scalar satisfies

$$Z_{\mathrm{prim}}(s) = \Gamma_{\mathbb{R}}(s)\zeta(s) \tag{66.2}$$

$$= \Lambda_\zeta(s) \qquad (\operatorname{Re} s > 1). \tag{66.3}$$

It has the classical meromorphic continuation and functional equation $\Lambda_\zeta(s) = \Lambda_\zeta(1 - s)$.

Proof. The defining product has its two factors computed in the preceding theorems, giving the equality on the common convergence region. Classical Tate–Riemann theory gives the meromorphic continuation and functional equation of that scalar. This use of the functional equation makes no identification of an arithmetic bar or cobar object. □

Corollary 66.2.4 *(poles of the completed scalar).* The completed scalar has simple poles at $s = 1$ and $s = 0$, with residues 1 and -1, respectively, and no other poles.

Proof. At one, $\Gamma_{\mathbb{R}}(1) = 1$ and $\operatorname{Res}_{s=1} \zeta(s) = 1$. The functional equation gives the opposite residue at zero. The remaining poles of the gamma factor are cancelled by the trivial zeros of ζ. □

66.3 DUALITY AND THE REQUIRED COMPARISON

A cobar model is additional data

Definition 66.3.1 *(arithmetic bar–cobar comparison datum).* An arithmetic comparison datum must specify a dg coalgebra C_{prim}, a cobar model A_{prim}, coefficients, completions, and a twisting cochain or comparison map from the selected primary model. Restriction of an arithmetic differential to that selected space must be proved; it is not automatic from the vector-space inclusion. A continuous dual $C_{\mathrm{prim}}^{\vee}$ and a cobar construction are not identified merely by writing the same symbol.

For an ordinary graded dual the sign is different from a Tate shift. The basis dual to energy $\log N$ has energy $-\log N$, and its formal convergent trace is $\sum_N N^s = \zeta(-s)$ for $\operatorname{Re} s < -1$. A formula with $\zeta(1-s)$ requires an additional twist and a comparison explaining it. This is not supplied by the energy-zero evaluation of Theorem 46.5.1.

The multiplicative-shift scalar

THEOREM 66.3.2 *(finite scalar comparison).* On $\ell^2(\mathbb{N}_{>0})$, use $H_{\mathrm{BC}}\delta_N = \log N\, \delta_N$ and $\mu_n \delta_N = \delta_{nN}$. Then

$$Z_{\mathrm{BC}}(\beta) := \operatorname{Tr}(e^{-\beta H_{\mathrm{BC}}}) = \zeta(\beta) \qquad (\beta > 1).$$

This is the same finite scalar as the selected primary tower. It does not identify an arithmetic cobar algebra or its multiplication.

Proof. Sum the positive diagonal entries $N^{-\beta}$. The basis map from normalized prime occupations to $\delta_{\prod p^{j_p}}$ is a unitary energy intertwiner for these two explicitly defined Hilbert models. The local occupation shifts correspond to multiplication by primes. They are different from the substitutions T_n, ψ_n, which fail to preserve the selected space and fix the full-Fock vacuum. □

Fourier–Mellin substitution

PROPOSITION 66.3.3 *(the substituted Gaussian scalar).* With Fourier kernel $e^{-2\pi i x y}$, $\widehat{g} = g$. Hence

$$\int_{\mathbb{R}} \widehat{g}(y)|y|^{-s}\, dy = \Gamma_{\mathbb{R}}(1-s) \qquad (\operatorname{Re} s < 1).$$

This is the Mellin integral at the substituted exponent, not a theorem identifying the character of an unspecified Koszul-dual object.

Proof. The Gaussian Fourier transform follows, for example, by differentiating its integral: integration by parts gives $\widehat{g}'(y) = -2\pi y \widehat{g}(y)$, and $\widehat{g}(0) = \int e^{-\pi x^2} dx = 1$. Solving yields $\widehat{g}(y) = e^{-\pi y^2}$. Now apply (66.1) with parameter $1-s$. Its convergence condition is exactly the one stated. □

The Tate-dual scalar and conditional character interpretation

THEOREM 66.3.4 *(Tate-dual scalar).* Define $Z^{\mathrm{T}}_{\mathrm{prim}}(s) = Z_{\mathrm{prim}}(1-s)$. On $\operatorname{Re} s < 0$, and then meromorphically,

$$Z^{\mathrm{T}}_{\mathrm{prim}}(s) = \Gamma_{\mathbb{R}}(1-s)\zeta(1-s) = \Lambda_{\zeta}(1-s).$$

If independently constructed arithmetic objects and their scalar pairings are identified with these two specified Mellin models, then

$$Z_C(s) = \Lambda_{\zeta}(s), \qquad Z_{A^!}(s) = \Lambda_{\zeta}(1-s). \tag{66.4}$$

The latter interpretation is conditional on those identifications, including the twist that produces $1-s$.

Proof. Substitute $1 - s$ into the computed finite series and Gaussian integral. Their common convergence region is $\operatorname{Re} s < 0$. Meromorphic continuation gives the scalar formula elsewhere. Transport through the hypothesized pairing identifications gives the conditional interpretation; no such arithmetic maps are constructed by substitution. □

The functional equation

Proposition 66.3.5 *(scalar functional equation).* As meromorphic scalar functions,

$$Z_{\mathrm{prim}}(s) = Z^{\mathrm{T}}_{\mathrm{prim}}(s) \quad \Longleftrightarrow \quad \Lambda_{\zeta}(s) = \Lambda_{\zeta}(1 - s).$$

Both sides hold by the classical functional equation.

Proof. Replace the two explicitly defined scalars by the formulas already proved. No implication about arithmetic obstruction classes or operator spectra is needed for this equivalence. □

Remark 66.3.6 *(limits of the scalar calculation).* Equality of these meromorphic scalars proves neither RH nor an arithmetic bar–cobar pairing. The comparison in (66.4) is still required even to interpret the two scalars as the proposed arithmetic characters.

66.4 Arithmetic Consequences and Remaining Constructions

Remark 66.4.1 *(the character comparison requirement).* The one-mode scalar and its Gaussian completion have been computed. The character requirement in Remark 46.1.6(4), when it concerns an arithmetic pairing rather than these defined scalar functions, still requires the maps and twist specified above. This calculation does not reduce that arithmetic list from five requirements to four. The P1 level comparison remains Question 46.5.4.

Distinct arithmetic requirements

Input	Construction	Remaining comparison
(1)	Completed continuous duals	Topology and actual duality maps
(2)	Derived completion	Base change on the stated resolutions
(3)	Lifting	Existence and the required morphism properties
(4)	Primary scalar pairing	Arithmetic character, twist and pairing maps
(5)	Bost–Connes realization	Multiplication, differential and completion

Table 66.1: Requirements on the intended arithmetic object. The Euler and Mellin calculations determine specified scalar models, not these maps.

The scalar constructions

66.5 Sources for the Scalar and Duality Constructions

Remark 66.5.1 *(roles of the sources).* Tate's 1950 thesis supplies the adelic Mellin framework and functional equation; the Gaussian integral and Euler product used here have also been computed directly.

Carrier or pairing	Computed scalar
One-mode occupation Hilbert space	$\zeta(\beta),\ \beta > 1$
Gaussian full real Mellin pairing	$\Gamma_{\mathbb{R}}(s),\ \operatorname{Re} s > 0$
Their defined product	$\Lambda_\zeta(s)$
Tate-substituted scalar	$\Lambda_\zeta(1-s)$

TABLE 66.2: The domains and meanings of the four scalar expressions.

Julia's *Statistical theory of numbers* (1990) describes the primon occupation interpretation. Bost–Connes (Selecta Math. 1 (1995), 411–457) supplies the arithmetic dynamical context; the concrete shift and heat-trace comparison above does not use a KMS classification. Positselski's *Two kinds of derived categories, Koszul duality, and comodule–contramodule correspondence* (Mem. AMS996, 2011) concerns categorical duality under its stated hypotheses. It is not used to infer a Tate shift from ordinary energy dualization or to identify a Gaussian test line with a cobar algebra.

66.6 THE ARITHMETIC COMPARISON PROBLEM

The selected finite occupation model, its Hilbert domain, the Gaussian Mellin normalization and the completed scalar are explicit. A realization on the arithmetic Hochschild trace class must additionally construct its operators, integral and completed carriers, bar–cobar maps and compatible pairings. It must account for the failure of the polynomial substitutions to preserve the selected one-mode space. These are the remaining mathematical constructions; the scalar identity alone does not settle them.

Chapter 67

Koszul–Petersson and the Prime Gibbs Model

A single prime admits an explicit Gibbs state for every inverse temperature $\beta > 0$, including $\beta = 1$. Its GNS space is a Hilbert–Schmidt space, its modular conjugation is adjunction, and its modular operator has a maximal weighted-matrix domain. These objects can be computed without identifying an arithmetic Koszul pairing with a Petersson pairing. The latter comparison requires additional maps between the actual carriers.

67.1 The Prime State and the Arithmetic Comparison

The occupation model of Chapter 66 has Hamiltonian $j \log p$ at a prime. We first construct its GNS representation and an actual unitary onto the prime-generated cyclic subspace of any specified global KMS representation. Neither this subspace comparison nor uniqueness of a polar decomposition supplies an automorphic realization.

Hypothesis 67.1.1 *(arithmetic placement and pairing comparison).* Fix a primitive arithmetic carrier, its coefficient field $\mathbb{C}$, its completion and a dense test domain. Specify:

(i) an injective placement ι of the selected domain into a fixed-central-character automorphic Hilbert space, with its Haar and Tamagawa normalizations;

(ii) at each place, comparison maps for the independently defined Koszul and automorphic carriers, intertwining the stated algebra or Hecke actions and transporting the bilinear Koszul form and antilinear involution to the stated local Hermitian form;

(iii) preservation of the chosen reference vectors at almost all places, factorization on elementary tensors, and continuous or explicitly regularized extension of the resulting pairing identity;

(iv) on every included residual sector, its specified placement and normalization, and orthogonality of the sector decomposition for both transported forms.

Any modular comparison must additionally intertwine the represented algebra, state or cyclic vector, and the closed modular operators on their actual domains. Ramified sectors require their own local maps; they are not consequences of the prime shift computation.

Question 67.1.2 *(constructing the arithmetic pairing maps).* Construct these maps from the arithmetic bar–cobar carrier and prove their compatibility with the independently defined Petersson pairing. Determine their precise discrete and continuous domains. The conditional transfers below retain this question; the prime Gibbs calculation does not complete it.

67.2 The One-Prime Gibbs State

State and Physical Time

Fix a prime p and $\beta > 0$. Put $K = \ell^2(\mathbb{N}_0)$, $Ve_j = e_{j+1}$, and $He_j = j \log p\, e_j$, with $\operatorname{Dom} H = \{x : \sum_j j^2 |x_j|^2 < \infty\}$. Let $\mathcal{T} = C^*(V) \subset B(K)$, $q = p^{-\beta}$, $r_j = (1-q)q^j$, and $\rho = (1-q)q^{H/\log p}$. Physical time is

$$\alpha_t(a) = e^{itH} a e^{-itH}, \qquad \alpha_t(V) = p^{it} V.$$

The positive injective trace-class operator ρ has trace one. Its faithful normal state on $B(K)$, restricted to $\mathcal{T}$, is

$$\varphi_{p,\beta}(a) = \operatorname{Tr}(\rho a), \qquad \varphi_{p,\beta}(V^m V^{*n}) = \delta_{m,n} q^m. \tag{67.1}$$

In particular $\varphi_{p,\beta}(V^*V) = 1$ and $\varphi_{p,\beta}(VV^*) = q$. The order of the isometry and its adjoint is essential.

LEMMA 67.2.1 *(prime KMS state).* The state $\varphi_{p,\beta}$ is the unique β-KMS state on $(\mathcal{T}, \alpha)$, with convention

$$\varphi_{p,\beta}(ab) = \varphi_{p,\beta}(b\,\alpha_{i\beta}(a))$$

for analytic elements.

Proof. The analytic monomials $V^m V^{*n}$ span a dense star algebra. On them $\rho a = \alpha_{i\beta}(a)\rho$, so trace cyclicity proves the KMS identity. Conversely, applying that identity with $b = 1$ forces the expectation of a monomial of nonzero degree to vanish. Taking $a = V^m$, $b = V^{*m}$ gives expectation q^m for $V^m V^{*m}$, since $V^{*m} V^m = 1$. Thus every KMS state has the values in (67.1); density gives uniqueness. □

The Hilbert–Schmidt GNS Carrier

Use inner products linear in the first variable.

LEMMA 67.2.2 *(GNS realization of the prime state).* The GNS representation is

$$\mathcal{H}_{p,\beta} = \mathrm{HS}(K), \quad \pi(a)X = aX, \quad \Omega = \rho^{1/2},$$

with

$$\langle X, Y \rangle = \operatorname{Tr}(XY^*), \qquad [a] \longmapsto a\rho^{1/2}.$$

The vector Ω is cyclic and separating for left $B(K)$, which is $\pi(\mathcal{T})''$.

Proof. The displayed map is isometric because its inner product is $\operatorname{Tr}(\rho b^* a) = \varphi_{p,\beta}(b^* a)$. The matrix units $E_{ij} = V^i(1 - VV^*)V^{*j}$ belong to $\mathcal{T}$, and $E_{ij}\rho^{1/2} = \sqrt{r_j} E_{ij}$. Their images span a dense subspace of $\mathrm{HS}(K)$. Finite matrix corners strongly approximate bounded operators, so the weak closure is left $B(K)$. If $a\Omega = 0$, every column of a vanishes because every $r_j > 0$; hence $a = 0$. □

Closed Modular Operators

Define the densely defined antilinear operator

$$S_0(a\Omega) = a^*\Omega, \qquad a \in \mathcal{T}.$$

Lemma 67.2.3 *(modular operator and maximal domains).* For $X = \sum x_{ij} E_{ij}$, set $JX = X^*$ and

$$\Delta E_{ij} = q^{i-j} E_{ij}, \qquad \mathrm{Dom}(\Delta^a) = \left\{ X : \sum_{i,j} q^{2a(i-j)} |x_{ij}|^2 < \infty \right\}, \quad a \in \mathbb{R}.$$

Then $\overline{S_0} = J\Delta^{1/2}$, on the displayed domain for $a = 1/2$, and

$$\Delta^{it} X = e^{-i\beta t H} X e^{i\beta t H}.$$

Thus the modular action on left multiplication is physical time $\alpha_{-\beta t}$. Both Δ and Δ^{-1} are unbounded. The generator $\log \Delta$ has eigenvalue $(i-j)\log q$ and domain defined by square summability with weight $|(i-j)\log q|^2$.

Proof. On matrix units, $S_0 E_{ij} = q^{(i-j)/2} E_{ji}$, antilinearly. Finite matrix sums are graph-dense for the maximal weighted operator, by truncation of the convergent double sum. For any bounded a, the matrix $a\Omega$ lies in its domain and the formula gives $a^*\Omega$; its weighted norm is $\|a^*\Omega\|_{\mathrm{HS}} < \infty$. Therefore the closure is exactly $J\Delta^{1/2}$. Diagonal spectral calculus gives all powers, domains and the unitary group. The eigenvalues q^{i-j} range through $q^{\mathbb{Z}}$, proving both unboundedness claims. □

Proposition 67.2.4 *(modular conjugation and the prime shift).* For $k \geq 0$,

$$J(V^k\Omega) = p^{\beta k/2} V^{*k}\Omega,$$
$$JL_aJ = R_{a^*}, \qquad a \in B(K),$$

where L and R denote left and right multiplication. The continuous bilinear form $B(X,Z) = \mathrm{Tr}(XZ)$ on $\mathrm{HS}(K)^2$ satisfies $B(X, JY) = \langle X, Y\rangle$.

Proof. Expand $V^k\Omega = \sum_j \sqrt{r_j} E_{j+k,j}$ and use $\sqrt{r_{j+k}} = q^{k/2}\sqrt{r_j}$. Adjunction gives the first identity and $JL_aJ(X) = Xa^*$ gives the second. Products of two Hilbert–Schmidt operators are trace class with $|\mathrm{Tr}(XZ)| \leq \|X\|_2 \|Z\|_2$, proving continuity and the asserted pairing identity. In particular the half-power has positive exponent: $\|V^k\Omega\| = 1$, whereas $\|V^{*k}\Omega\| = q^{k/2}$. □

Theorem 67.2.5 *(comparison with a prime cyclic subspace).* Let $(\mathcal{A}, \alpha, \psi_\beta)$ be a specified dynamical C^*-algebra with a β-KMS state and an isometry μ_p satisfying $\alpha_t(\mu_p) = p^{it}\mu_p$. Let $\iota_p : \mathcal{T} \to \mathcal{A}$ send V to μ_p. There is a canonical unitary

$$U_p : \mathrm{HS}(K) \longrightarrow \overline{\pi_{\psi_\beta}(C^*(\mu_p))\Omega_{\psi_\beta}}, \qquad U_p(a\rho^{1/2}) = \pi_{\psi_\beta}(\iota_p(a))\Omega_{\psi_\beta}.$$

It intertwines the prime algebra actions and cyclic vectors, and transports the closed modular operators of the restricted cyclic representation. This holds at $\beta = 1$ as well.

Proof. The pulled-back state $\psi_\beta \circ \iota_p$ is a KMS state on $\mathcal{T}$, hence equals $\varphi_{p,\beta}$ by the unique-state calculation. The displayed map preserves inner products on $\mathcal{T}\rho^{1/2}$ and has dense image in the specified cyclic subspace. It therefore extends to a unitary. It intertwines multiplication and the operation $a\Omega \mapsto a^*\Omega$ on its core, hence its closure and polar decomposition. □

Remark 67.2.6 *(scope of modular uniqueness).* This unitary targets the prime cyclic subspace, not the whole ψ_β-GNS space. It does not identify the phase operators, an automorphic representation or an arithmetic Koszul form. Uniqueness of modular conjugation on a fixed cyclic representation supplies no additional identification between those carriers.

Finite Prime Sets and the Critical Global Boundary

COROLLARY 67.2.7 *(finite-prime Gibbs and GNS products).* For a finite prime set F, the density $\rho_F = \bigotimes_{p\in F} \rho_p$ has trace one for every $\beta > 0$. Its GNS space and modular operators are the finite tensor products of the preceding constructions. On the full occupation space $\ell^2(\mathbb{N}^\times)$, the heat trace is $\zeta(\beta)$ for $\beta > 1$, and diverges at $\beta = 1$. Thus the full occupation representation has no normalized trace-class Gibbs density at the critical temperature.

Proof. Finite tensor products multiply traces and tensor the explicit matrix formulas. Unique prime factorization identifies occupations with integers and gives $\sum_{n\geq 1} n^{-\beta}$. At $\beta = 1$ this is the divergent harmonic series. □

The divergence does not preclude a critical Bost–Connes KMS state in another representation. For any such specified state the prime cyclic comparison above still applies. Neither argument identifies its entire GNS space or factor type with the finite-prime Gibbs model.

67.3 LOCAL PAIRING MODELS AND ARITHMETIC TRANSPORT

Ramified Principal-Series Comparison

LEMMA 67.3.1 *(a normalized local Hermitian carrier).* Fix Haar measure $d^\times a$ on $\mathbb{Q}_p^\times$. On $E_p = C_c^\infty(\mathbb{Q}_p^\times, \mathbb{C})$, put

$$I_p(v, w) = \int_{\mathbb{Q}_p^\times} v(a)\overline{w(a)}|a|^{-1}\, d^\times a. \tag{67.2}$$

This is positive definite and extends to the corresponding L^2-completion. The bilinear form $b_p(v, z) = \int v(a)z(a)|a|^{-1}\, d^\times a$ satisfies $b_p(v, \overline{w}) = I_p(v, w)$.

Proof. Compact support in $\mathbb{Q}_p^\times$ gives finite integrals. The measure is positive, and a nonzero locally constant function is nonzero on an open set of positive measure. Cauchy–Schwarz extends the forms to L^2, and complex conjugation gives the last identity. □

The Whittaker realization of a given representation, its scalar normalization and its identification with an arithmetic Koszul carrier are further data. In particular an epsilon factor from a local functional equation is not automatically a positive normalization of a Hermitian form.

Hypothesis 67.3.2 *(local Koszul comparison).* For the chosen ramified arithmetic domain $V_{p,\pi}$, specify a linear map $c_{p,\pi}$ into the local Hilbert carrier, a compatible map on its Koszul dual, a positive constant c_π, and the antilinear map σ_p. Require that the actual Koszul pairing satisfy

$$\langle v, \sigma_p(w)\rangle_{p,\pi}^{\mathrm{Koszul}} = c_\pi b_p(c_{p,\pi}v, \overline{c_{p,\pi}w}).$$

The maps must preserve the specified local actions, domains and completion. This is an arithmetic comparison hypothesis.

Hypothesis 67.3.3 *(local automorphic normalization).* For the same carrier, maps, measure and positive constant, require

$$\langle \iota_p v, \iota_p w\rangle_{p,\pi}^{\mathrm{Pet}} = c_\pi I_p(c_{p,\pi}v, c_{p,\pi}w).$$

Neither Whittaker uniqueness nor a scalar local functional equation alone identifies the map $c_{p,\pi}$ with the Koszul comparison.

Proposition 67.3.4 *(conditional local Koszul–Petersson comparison).* Under the two local comparison hypotheses above, $\langle \iota_p v, \iota_p w\rangle_{p,\pi}^{\mathrm{Pet}} = \langle v, \sigma_p(w)\rangle_{p,\pi}^{\mathrm{Koszul}}$ on the specified domain.

Proof. Substitute $b_p(x, \overline{y}) = I_p(x, y)$ into the two displayed identities. The common constant and measure are part of the hypotheses, not a conclusion of matching the names of the pairings. □

Supercuspidal Comparison

Lemma 67.3.5 *(conditional supercuspidal transport).* The same conclusion holds on a supercuspidal domain for which the local Koszul and automorphic comparison hypotheses have been constructed with a common Hilbert carrier and normalization.

Proof. The proof of the preceding proposition uses only the common forms and maps, so applies on that specified domain. It supplies no new comparison map for the supercuspidal representation. □

Temperedness and Assembly

Proposition 67.3.6 *(conditional extension under temperedness).* Assume Hypothesis 67.1.1 on a tempered cuspidal domain. Any further cuspidal domain contained in it inherits its pairing comparison. Applying this to Maass forms requires the appropriate Ramanujan temperedness assumption as well as the arithmetic comparison maps.

Proof. Restrict the specified maps and pairing equality to the subdomain. Temperedness is a domain condition and does not construct the maps. □

Remark 67.3.7 *(local exponent bounds).* The Kim–Sarnak exponent bound $\theta \le 7/64$ is a local bound. It is neither an arithmetic pairing map nor a summable prime error on the critical line. The exponent $-25/64$ gives $\sum_p p^{-25/64} = \infty$; it cannot be used as an absolutely convergent adelic error majorant.

Theorem 67.3.8 *(conditional cuspidal pairing assembly).* Assume Hypothesis 67.1.1, including the local comparisons at every ramified place and the specified adelic extension. On the resulting tempered cuspidal domain,

$$\langle \iota v, \iota w\rangle^{\mathrm{Pet}} = \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}}.$$

Proof. On elementary restricted tensors the forms factor, and the local identities give equality of every finite product with the specified reference-vector normalization. The assumed continuous or regularized extension carries this equality to its stated domain. Finiteness of the ramified set alone does not justify the infinite product normalization or its extension. □

Remark 67.3.9 *(scope of the domain).* The normalized local Hilbert models and the finite-prime Gibbs models are explicit. Enlarging an arithmetic domain to ramified, residual or Maass sectors requires its actual comparison maps. The theorem above is their conditional assembly, not an unconditional enlargement of the arithmetic pairing theorem.

67.4 Langlands Decomposition of $L^2([GL_2])$ and Placement of $\mathcal{V}^{\mathrm{prim}}$

The Spectral Decomposition

The Langlands decomposition of $L^2([GL_2])$, following Langlands 1966 and Moeglin–Waldspurger 1995, is:

THEOREM 67.4.1 *(spectral decomposition of $L^2([GL_2])$).* There is an orthogonal decomposition of $G(\mathbb{A})$-modules:

$$L^2([GL_2]) \;=\; L^2_{\mathrm{cusp}}([GL_2]) \;\oplus\; L^2_{\mathrm{res}}([GL_2]) \;\oplus\; L^2_{\mathrm{cont}}([GL_2]),$$

where:

- $L^2_{\mathrm{cusp}}([GL_2])$ is the cuspidal spectrum, a direct sum of irreducible unitary representations $\pi = \bigotimes'_v \pi_v$ with π_p generic tempered at almost all primes;
- $L^2_{\mathrm{res}}([GL_2])$ is the residual spectrum (in the GL_2 case generated by one-dimensional characters pulled up from the determinant), consisting of $GL_2(\mathbb{A})$-reducible automorphic representations from residues of Eisenstein series at specific points;
- $L^2_{\mathrm{cont}}([GL_2])$ is the continuous spectrum, realised as the direct integral $\int^{\oplus}_{\mathrm{Re}(s)=1/2} \mathrm{Ind}_B^G(\chi_s)\,|\xi(2s)|^{-2}\,ds$ for $\chi_s = |\cdot|^s \otimes |\cdot|^{-s}$.

Proof. Langlands 1966 *On the Functional Equations Satisfied by Eisenstein Series* §6–7 for the construction; Moeglin–Waldspurger 1995 *Spectral Decomposition and Eisenstein Series for GL_n* for the published GL_n exposition; for GL_2 specifically see Gelbart 1975 *Automorphic Forms on Adele Groups* Ch. II. □

Placement of $\mathcal{V}^{\mathrm{prim}}$

Hypothesis 67.4.2 *(discrete arithmetic placement).* Choose the primitive arithmetic vector carrier, its dense domain and its completion. The discrete placement required in Hypothesis 67.1.1 is an injective map

$$\iota : \mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}} \longrightarrow L^2_{\mathrm{cusp}} \oplus L^2_{\mathrm{res}},$$

with fixed central character and specified Haar normalization. Its chosen residual component and vacuum normalization are part of the datum. The Langlands decomposition of the target does not construct this map. A continuous-spectrum comparison requires its own direct-integral or distributional carrier and is not a map of a distribution into L^2 without further hypotheses.

67.5 Scalar Continuum Models and the Pairing Problem

The Scalar Coefficient Domain

Use the Riemann completed function ξ with $\xi(1-z) = \xi(z)$ and $\xi(\overline{z}) = \overline{\xi(z)}$. Let U be the points s on $\mathrm{Re}\, s = 1/2$ where $\xi(2s)$ and $\xi(2s-1)$ are finite and nonzero.

Definition 67.5.1 *(scalar scattering coefficient model).* For $s \in U$, set $d(s) = |\xi(2s)|^{-2} > 0$ and $M(s) = \xi(2s-1)/\xi(2s)$. Define

$$P(s) = (1 + |M(s)|^2)d(s).$$

This is a scalar model. Identifying it with a coefficient of an actual truncated Petersson pairing requires that pairing, its truncation parameter and its measure normalization.

The Proposed Koszul Coefficient

Definition 67.5.2 *(scalar Koszul coefficient model).* On the same domain define

$$K(s) = M(s)^{-1}d(s) = \frac{\xi(2s)}{\xi(2s-1)}|\xi(2s)|^{-2}.$$

This scalar is not asserted to be the spectral measure of an arithmetic Koszul pairing. Such a comparison requires the continuous carrier, coefficient map and regularization.

Theorem 67.5.3 *(strict discrepancy of the two scalar models).* For every $s \in U$, one has $|M(s)| = 1$, $P(s) = 2d(s)$, and

$$P(s) - K(s) = d(s)(2 - M(s)^{-1}).$$

In particular

$$d(s) \le |P(s) - K(s)| \le 3d(s),$$

so the models are unequal everywhere on U.

Proof. Write $s = 1/2 + it$. Functional equation and conjugation give $\xi(2s) = \xi(1 + 2it) = \xi(-2it) = \overline{\xi(2it)}$. The numerator and denominator of M therefore have equal nonzero modulus. Substitution gives the difference, and the triangle and reverse triangle inequalities give the two bounds. □

Remark 67.5.4 *(phase and scale).* Multiplication by the phase M gives $MK = d = P/2$, not P. Multiplication by $2M$ matches these scalar models. Neither scalar identity constructs a positive spectral measure, an isometry, or an arithmetic bar–cobar comparison.

Corollary 67.5.5 *(scope of the scalar obstruction).* A proposed continuous pairing comparison whose coefficients are exactly P and K on U, with the same measure and no additional transport, cannot identify them. A comparison using another normalization must construct that normalization and its maps; the scalar discrepancy does not exclude such a construction.

Proof. Their difference has modulus at least $d > 0$. The last assertion only separates this fixed-normalization obstruction from a comparison with different data. □

Remark 67.5.6 *(continuous pairings require more than one coefficient).* The scalar functions above specify no bivariate pairing kernel. In particular they do not determine an off-diagonal truncated Eisenstein pairing or its diagonal limit. On the critical line, $s_1 + \overline{s_2} - 1 = i(t_1 - t_2)$, which vanishes exactly when $s_1 = s_2$. Any formula dividing by this quantity must treat that diagonal with its actual truncation and limit. The classical Maass–Selberg and Arthur identities concern the automorphic side; they supply no integrated equality with an arithmetic Koszul pairing without the missing comparison maps.

67.6 Conditional Arithmetic Pairing Consequences

Theorem 67.6.1 *(conditional P3 on the specified cuspidal domain).* Assume Hypothesis 67.1.1 and the discrete placement of Hypothesis 67.4.2 on the chosen tempered cuspidal carrier. Then the pullback of the Petersson form is positive definite and agrees there with the transported Koszul form composed with the specified antilinear involution.

Proof. Injectivity of ι makes the pullback of a positive-definite form positive definite. The local pairing comparisons and their specified adelic extension give equality of forms by Theorem 67.3.8. The prime GNS theorem alone does not supply either hypothesis. □

Remark 67.6.2 *(domain of arithmetic transport).* The prime Gibbs and Hilbert–Schmidt calculations are unconditional on their stated carriers. Extension of the arithmetic pairing to ramified, holomorphic, Maass or residual sectors requires its placement and pairing maps on each sector. Ramanujan controls a temperedness condition; it is not the missing isometry. Continuous spectrum additionally requires its measure comparison.

The Sufficient RH Implication

COROLLARY 67.6.3 *(conditional use in a sufficient RH implication).* A sufficient RH implication that assumes P3 may use the preceding conditional theorem only on its specified domain and with all of its arithmetic comparison hypotheses. Identifying the relevant Li or Weil trace with that domain is a further spectral comparison. The other hypotheses of the sufficient implication remain in force.

Proof. Substitute the conditional pairing conclusion for the P3 premise only after its hypotheses and the shared trace domain are supplied. This logical substitution constructs neither set of hypotheses. □

67.7 PRIME MODELS AND REMAINING MAPS

The positive constructions are the normalized prime Gibbs state, its Hilbert–Schmidt GNS realization, the closed modular operators, the finite-prime product, and the unitary onto a specified global state's prime cyclic subspace. Their domains and state normalizations are explicit. The local integral form (67.2) is another independently defined positive Hermitian model.

The arithmetic construction still requires the maps in Hypothesis 67.1.1. The residual-spectrum vacuum, ramified Whittaker models and continuous spectral measures must be identified on their actual carriers. A scalar partition function, the uniqueness of modular conjugation, and a spectral decomposition of the target do not provide those maps.

67.8 THE ARITHMETIC COMPARISON PROBLEM

Question 67.1.2 retains the intended Koszul–Petersson comparison. The finite and prime models specify normalization tests and actual Hilbert-space maps; the arithmetic placement and pairing compatibility remain to be constructed. At the critical global temperature, the prime cyclic comparison continues to hold for a specified KMS state even though the full occupation Gibbs trace diverges.

Chapter 68

Fi.a.3: Bushnell–Henniart Block Exhaustion at Wild Primes

Block exhaustion asks whether the wild block functor reaches every object of the proposed spectral category. The conditional argument uses three inputs: a BH type-block comparison, an actual extension-class comparison, and compact generation by the specified spectral objects. Finite lists and equal local dimensions alone do not imply essential surjectivity.

The local arithmetic identifications used below are hypotheses, including the coefficient-compatible block models and maps in Questions 54.4.8 and 54.4.9. The finite-algebra resolutions in that chapter do not classify the arithmetic blocks. In particular, the degree-one dimensions do not construct a map to an Emerton–Gee fibre. Whenever a comparison uses the three-vertex radical-square-zero model, its nonzero Yoneda product in degree two must also be preserved.

68.1 Statement

Setup

Fix $p \in \{2, 3\}$, a specified common coefficient ring R, a smooth full central character $\chi : \mathbb{Q}_p^\times \to R^\times$, and a conductor bound N. Any use of a field k in a finite-algebra model includes a specified coefficient map and base-change comparison to this arithmetic setting. No identification of prime-to-p coefficients with the integral Emerton–Gee coefficients is presumed. Let $\mathbf{T}^{\mathrm{BH}}_{p,\chi}$ be the chosen type-index set in that setting. Assume the wild enhancement (H2) and wild block-functor datum (H3) of Theorem 61.7.1, including presentable stable categories C^{BH}_τ, $\mathcal{D}_\tau$, exact colimit-preserving fully faithful functors Φ_τ, and specified categorical decompositions

$$C \simeq \prod_\tau C^{\mathrm{BH}}_\tau, \qquad \mathcal{D} \simeq \prod_\tau \mathcal{D}_\tau.$$

The notation for the arithmetic source and spectral support category below includes their supplied identifications with C and $\mathcal{D}$. The componentwise functor $\Phi^{\mathrm{wild}}_{p,\chi} = \prod_\tau \Phi_\tau$ exists under these data. A type list alone supplies neither these categorical products nor a geometric coproduct decomposition of an Emerton–Gee stack. The proposed formal components and their singular support remain part of the enhancement and comparison data.

The BH Block-Exhaustion Statement

Definition 68.1.1 *(BH-exhausted block).* Fix $p \in \{2, 3\}$, χ and τ. Let C^{BH}_τ be the specified presentable stable realization of the representation block, and put $\mathcal{D}_\tau = \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG,wild},p,\chi,\tau}_{GL_2})$. For the chosen functor $\Phi^{\mathrm{wild}}_{p,\chi,\tau} : C^{\mathrm{BH}}_\tau \to \mathcal{D}_\tau$, the block is BH-exhausted if that functor is essentially surjective. Generation by spectral objects alone is not this condition; their images through the functor must be identified.

THEOREM 68.1.2 *(BH block exhaustion under the complete comparison data).* Fix $p \in \{2, 3\}$ and χ. Assume the wild-enhancement and block-functor data of Conjectures 61.5.7 and 61.5.5, including exact colimit-preserving full faithfulness and compatible categorical decompositions. Assume further:

(A) For each block, a compact object $P_\tau \in \mathcal{C}_\tau^{\mathrm{BH}}$, an actual spectral object $Q_\tau \in \mathcal{D}_\tau$, and an equivalence $\Phi^{\mathrm{wild}}_{p,\chi,\tau}(P_\tau) \simeq Q_\tau$, as required by Conjecture 68.7.1. The proposed Q_τ is the derived structure sheaf of the BH closed point, with the stated singular support.

(B) The actual arithmetic extension-class comparison of Conjecture 68.7.3 is supplied for these objects and functors.

(C) The category $\mathcal{D}_\tau$ is compactly generated, Q_τ is compact, and $\mathcal{D}_\tau^c = \mathrm{thick}(Q_\tau)$, as in Conjecture 68.7.5.

Then every specified block is BH-exhausted, and the componentwise functor on the supplied categorical decompositions is an equivalence.

Proof. Condition (C) makes Q_τ a localizing generator: its localizing closure contains the thick closure and every colimit of compact objects. Condition (A) places it in the chosen functor's image. Proposition 61.4.7 therefore makes the block functor an equivalence, and the supplied categorical decompositions assemble these componentwise. Condition (B) retains the arithmetic extension compatibility. Full faithfulness in all degrees comes from the block-functor datum, not from a degree-one comparison. □

Remark 68.1.3 *(why F1.a.3 is the smallest residual).* Chapter 61 records the following local residuals for F1.a: F1.a.1 (block exhaustion at $p \geq 5$), F1.a.2 (wild-EG enhancement at $p \in \{2, 3\}$), the wild block functor datum, and F1.a.3. The explicit finite-algebra calculations in Chapter 54 remain available. Their transport to arithmetic blocks requires the local-model identifications, class maps and generator comparisons stated below; a type list or an equality of dimensions supplies none of these maps.

68.2 PROOF REDUCTION AND SOURCE COMPARISON

The following reductions distinguish type enumeration, extension classes, geometric components and compact generation.

BH tame-wild enumeration at $p = 2$. *Obstruction.* Enumerate Bushnell–Kutzko types in $\mathbf{T}^{\mathrm{BH}}_{2,\chi}$. Show that wild types at $p = 2$ are classified in conductor $\leq N$ by the unramified principal series, the tame supercuspidal, the wild supercuspidal types of bounded conductor, and the spherical (Iwahori-fixed) type absorbing the Ext^1-extension of Question 54.4.8. *Calculation.* The BH classification (Bushnell–Henniart 2006 §11–13) gives the exact list; the wild conductor-exponent bound is $a(\chi) + 2$, cf. BH §12.1. *Obstruction: (A), the BH type-block comparison conjecture.*

BH tame-wild enumeration at $p = 3$. *Obstruction.* Enumerate Bushnell–Kutzko types in $\mathbf{T}^{\mathrm{BH}}_{3,\chi}$. Show that the wild primary block at $p = 3$ has three restricted-weight simples L_0, L_1, L_2 with the A_2-linear-quiver Ext^1-structure of Question 54.4.9. *Required comparison.* Identify the proposed three BH types with the three restricted-weight simples; the two quiver edges $\alpha_{01} \in \mathrm{Ext}^1(L_0, L_1)$, $\alpha_{12} \in \mathrm{Ext}^1(L_1, L_2)$ lift to two non-split extensions in the category, each corresponding to a derived-structure fibre in the wild-enhanced EG component. *Residual: (B), the Paskunas wild* Ext^1*-compatibility.*

Essential surjectivity via type-by-type exhaustion. *Obstruction.* Fix a hypothetical $\mathcal{F} \in \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG,wild},p,\chi}_{GL_2})$ not in the essential image of $\Phi^{\mathrm{wild}}_{p,\chi}$. Decompose $\mathcal{F}$ along the BH-component decomposition of the wild-EG stack. Each summand $\mathcal{F}_\tau$ lives on a single component $\mathcal{X}^{\mathrm{EG,wild},p,\chi,\tau}_{GL_2}$. *Calculation.* By (C), $\mathcal{F}_\tau$ is a colimit of shifts of the derived structure sheaf of x_τ; by (A),

each such shift is in the image of $\Phi^{\mathrm{wild}}_{p,\chi,\tau}$; hence $\mathcal{F}_\tau$ itself is, and the full $\mathcal{F}$ is. Contradiction. *Residual: (C), wild-EG compact-generator identification.*

Comparison with the wild-EG decomposition of Chapter 61. *Obstruction.* Does the BH-component decomposition coincide with the Bushnell–Henniart fibre-correspondence decomposition of Conjecture 61.5.7(3)? *Calculation.* A comparison must identify both indexing sets with $\mathbf{T}^{\mathrm{BH}}_{p,\chi}$ and construct the proposed derived-fibre identifications (1 at $p = 2$ matching Question 54.4.8, 2 at $p = 3$ matching Question 54.4.9). The local model and its comparison remain additional inputs.

Unconditional closure test. *Obstruction.* Can (A), (B), (C) collectively close on existing primary literature? *Calculation.* (A) is known for $p \geq 5$ by Paskunas 2013 in the Breuil–Mezard framework; its transport to $p \in \{2, 3\}$ via Bushnell–Henniart types requires a direct match of types to structure sheaves not yet stated. (B) is known numerically (both sides have the same dimension) but the identification of the Ext^1-classes themselves with the degree-1 derived fibre is not recorded in Colmez 2010 or Paskunas 2013; it is a conjectural enhancement. (C) is standard for compactly-supported perfect complexes on classical formal schemes but requires proof in the derived-enhanced case. None of the three is closable from currently-stated primary literature; all three are individually tractable. *Final obstruction: (A), (B), (C), three named conjectures.*

The source comparison for each reduction step is recorded in §68.9. The first four analyses close conditionally on (A), (B), (C); the fifth identifies the three as the minimal conjectural set.

68.3 Bushnell–Henniart Tame-Wild Types at $p = 2$

The Classification

Proposition 68.3.1 *(Bushnell–Henniart 2006 classification at $p = 2$).* Fix a central character $\chi : \mathbb{Z}_2^\times \to k^\times$ and a conductor bound N. The bounded set $\mathbf{T}^{\mathrm{BH}}_{2,\chi}$ of Bushnell–Kutzko types with central character χ and conductor exponent at most N consists of three sorts of types:

(i) **Unramified principal-series types.** One type per unramified character $\psi : \mathbb{Q}_2^\times \to k^\times$ with $\psi|_{\mathbb{Z}_2^\times}$ square-rooting χ. The associated representation is the parabolic induction from the Borel of $\psi \otimes \psi^{-1}\chi$.

(ii) **Tame supercuspidal types.** One type per tame character ψ of a quadratic unramified extension $\mathbb{Q}_2(\sqrt{u})$ with $u \in \mathbb{Z}_2^\times \setminus (\mathbb{Z}_2^\times)^2$, inducing to a supercuspidal representation of $GL_2(\mathbb{Q}_2)$.

(iii) **Wild supercuspidal types.** One type per admissible wild character of a ramified quadratic extension $E/\mathbb{Q}_2$, with conductor exponent in the range imposed by χ and by $n \leq N$.

The bounded set is finite: $|\mathbf{T}^{\mathrm{BH}}_{2,\chi}| < \infty$, bounded in terms of N and $a(\chi)$.

Source. Bushnell–Henniart 2006 *The Local Langlands Conjecture for $GL(2)$*, Grundlehren 335, Springer. Chapter 11 for the principal series and tame supercuspidal classification; Chapter 12 for the wild supercuspidal classification and conductor exponents at $p = 2$; Chapter 13 for the counting formula. □

Remark 68.3.2 *(the Ext^1-carrying type at $p = 2$).* The proposed degree-one contribution at two requires an actual arithmetic block and extension class, as in Question 54.4.8. In characteristic two, trivial and sign characters coincide. Their names therefore do not identify two distinct constituents of a parabolic induction. Locating the desired class in a principal series block, and any vanishing assertion for supercuspidal blocks, requires a calculation in those specific coefficient categories.

Matching to Spectral Components

PROPOSITION 68.3.3 *(type-to-component match at $p = 2$).* Assume the BH type-block comparison Conjecture 68.7.1 and the wild-EG enhancement Conjecture 61.5.7. Then each $\tau \in \mathbf{T}^{\mathrm{BH}}_{2,\chi}$ corresponds to a unique component $\mathcal{X}^{\mathrm{EG,wild},2,\chi,\tau}_{GL_2}$ of the wild-enhanced EG stack, with:

- Unramified principal-series type τ: classical component indexed by a reducible residual Galois representation ρ with $\det \rho = \chi \cdot \chi_{\mathrm{cyc}}$; for the trivial $\psi = \mathbf{1}$, the component carries the degree-1 derived fibre of dimension 1 (the wild-EG enhancement).
- Tame supercuspidal type τ: classical component, no derived enhancement; corresponds to an irreducible residual ρ induced from a tame character of a quadratic unramified extension.
- Wild supercuspidal type τ: classical component, no derived enhancement; corresponds to an irreducible residual ρ induced from a wild character of a quadratic ramified extension.

The type-to-component map is a bijection.

Proof. **Step 1: residual-Galois match.** Each BH type τ has an associated L-parameter $\phi_\tau : W_{\mathbb{Q}_2} \to GL_2(k)$ by the local Langlands correspondence of Bushnell–Henniart 2006 §18, with determinant $\det \phi_\tau = \chi \cdot \chi_{\mathrm{cyc}}$. The residual representation ρ_τ is the reduction mod-p of ϕ_τ. The EG component $\mathcal{X}^{\mathrm{EG},2,\chi,\rho_\tau}_{GL_2}$ is the formal neighbourhood of ρ_τ. **Step 2: enhanced fibre at reducible ρ.** The local-model hypothesis must supply the proposed extension class, its arithmetic coefficient carrier, and its comparison to the one-dimensional derived direction. This is the input asked for in Question 54.4.8; it is not a consequence of a dimension count or of distinct trivial and sign characters in characteristic two. **Step 3: semisimplicity at irreducible ρ.** For τ a tame or wild supercuspidal, ρ_τ is irreducible; the deformation functor is smooth (Emerton–Gee arXiv:1908.07185 §6.2); no derived enhancement is needed. **Step 4: bijectivity.** Finiteness on both sides follows from Bushnell–Henniart §13 (left-hand) and the connected-component decomposition of $\mathcal{X}^{\mathrm{EG},2,\chi}_{GL_2}$ (right-hand). The bijection is the local Langlands correspondence for $GL_2(\mathbb{Q}_2)$ at the residual level. □

68.4 BUSHNELL–HENNIART TAME-WILD TYPES AT $p = 3$

The Classification and the A_2-Quiver Structure

PROPOSITION 68.4.1 *(Bushnell–Henniart 2006 classification at $p = 3$).* Fix $\chi : \mathbb{Z}_3^\times \to k^\times$ and a conductor bound N. The bounded set $\mathbf{T}^{\mathrm{BH}}_{3,\chi}$ consists of:

(i) **Unramified principal series.** Parabolically induced from the Borel of $GL_2(\mathbb{Q}_3)$.

(ii) **Tame supercuspidals.** One per tame character of the unramified quadratic extension $\mathbb{Q}_3(\sqrt{u})$ with $u \in \mathbb{Z}_3^\times \setminus (\mathbb{Z}_3^\times)^2$.

(iii) **Wild supercuspidals.** One per admissible wild character of a ramified quadratic extension $E/\mathbb{Q}_3$, with conductor exponent $a(\chi) + 1 \le n \le N$.

(iv) **Additional principal-block local-model hypothesis.** Specify three arithmetic simples and an exact equivalence taking them to the three simple right modules of $B = k(0 \xrightarrow{a} 1 \xrightarrow{b} 2)/(ab)$, in the coefficient category of Question 54.4.9. This identification is additional data, not a consequence of a classification of tame or wild types.

Source. Bushnell–Henniart 2006 §11 (principal series), §12 (tame supercuspidal), §13 (wild supercuspidal), §15 (wild types at small primes). The conductor bound cuts the infinite admissible character families

to a finite Bushnell–Kutzko set. The fourth item is a separate local-model hypothesis. The cited type classification does not supply that equivalence or its extension maps. □

Proposition 68.4.2 *(the three-vertex extension pattern).* For right modules over $B = k(0 \xrightarrow{a} 1 \xrightarrow{b} 2)/(ab)$, the adjacent extension groups $\mathrm{Ext}^1_B(S_0, S_1)$ and $\mathrm{Ext}^1_B(S_1, S_2)$ are one-dimensional. Their Yoneda product is nonzero in $\mathrm{Ext}^2_B(S_0, S_2) = k$, whereas $\mathrm{Ext}^1_B(S_0, S_2) = 0$. An exact arithmetic block equivalence taking its three simples to these right modules would transport these statements and products. Such an equivalence is an additional hypothesis.

Proof. The projective resolution in Example 54.4.4 computes the adjacent and degree-two groups and their product. The length-one path count in Proposition 54.4.3 gives the vanishing between non-adjacent simples. An exact equivalence transports extensions and their Yoneda compositions. □

Matching to Spectral Components

Proposition 68.4.3 *(type-to-component match at $p = 3$).* Assume (A) and (B). Each $\tau \in \mathbf{T}^{\mathrm{BH}}_{3,\chi}$ corresponds to a unique component $\mathcal{X}^{\mathrm{EG,wild},3,\chi,\tau}_{GL_2}$ of the wild-enhanced EG stack, with the principal-block types carrying a degree-1 derived fibre of dimension 2 (per Remark 61.5.4), and all other types carrying no derived enhancement.

Proof. **Step 1: classical-component match.** Same as Proposition 68.3.3 Step 1, replacing $p = 2$ by $p = 3$. Each type has an L-parameter ϕ_τ by BH 2006 §18; the residual ρ_τ determines the component. **Step 2: principal-block derived-fibre match.** For τ a principal-block type, ρ_τ is reducible with constituents matching the Steinberg-decomposed restricted-weight simples L_0, L_1, L_2. The two proposed arithmetic Ext^1-classes in Question 54.4.9 must be identified by the local-model map with a two-dimensional Ext^1-space in the deformation functor of ρ_τ. By Conjecture 61.5.7(2), this is realised as the degree-1 derived fibre of dimension 2 in $\mathcal{X}^{\mathrm{EG,wild},3,\chi,\tau}_{GL_2}$. **Step 3: semisimplicity at irreducible ρ.** Same as the $p = 2$ case. All supercuspidal types give smooth classical components. **Step 4: bijectivity.** Same as the $p = 2$ case. □

68.5 Essential Surjectivity via Type-by-Type Exhaustion

The Exhaustion Argument

Theorem 68.5.1 *(Exhaustion of a specified BH block).* Under all the hypotheses of Theorem 68.1.2, the chosen functor $\Phi^{\mathrm{wild}}_{p,\chi,\tau} : \mathcal{C}^{\mathrm{BH}}_\tau \to \mathcal{D}_\tau$ is an equivalence.

Proof. Its image contains the generator Q_τ by (A) and (C). The exact colimit-preserving full faithfulness supplied by (H3) allows Proposition 61.4.7 to apply. The specified degree-one comparison (B) is preserved; it is not used as a substitute for higher mapping-spectrum data. □

Corollary 68.5.2 *(Assembly on the specified categorical blocks).* Under all the hypotheses of Theorem 68.1.2, the componentwise functor on the supplied block decompositions is an equivalence of presentable stable categories.

Proof. The inverse is the componentwise inverse of the block equivalences, transported through the specified categorical decompositions. □

68.6 Wild Emerton–Gee Fibres and Block Extension Spaces

Dimensional Comparison

PROPOSITION 68.6.1 *(conditional dimensional comparison at $p = 2$).* Suppose the arithmetic extension carrier and the derived fibre in Question 54.4.8 are both constructed and identified with k^1. Then their dimensions agree and equal 1. This numerical equality does not select a comparison map or a distinguished class.

Proof. An identification with k^1 gives dimension 1 on each side. Choosing an isomorphism between two vector spaces of this dimension requires further data; dimensions alone do not specify it. □

PROPOSITION 68.6.2 *(conditional dimensional comparison at $p = 3$).* Suppose the arithmetic extension carrier and the derived fibre in Question 54.4.9 are both constructed and identified with k^2. Then their dimensions agree and equal 2. This numerical equality does not select a comparison map or a distinguished class.

Proof. An identification with k^2 gives dimension 2 on each side. Choosing an isomorphism between two vector spaces of this dimension requires further data; dimensions alone do not specify it. □

Structural Consistency

PROPOSITION 68.6.3 *(conditional compatibility of component decompositions).* Suppose a BH-to-spectral comparison identifies the components indexed by $\mathbf{T}^{\mathrm{BH}}_{p,\chi}$, their supports and their actual derived fibres with those of the wild-EG datum. Then it identifies the corresponding component decompositions. Equality of indexing sets or dimensions alone is insufficient for this conclusion.

Proof. The specified component maps assemble over their common indexing set. Their fibre identifications are part of the hypothesis, rather than a consequence of equal dimensions. □

Remark 68.6.4 *(data for block exhaustion).* The hypotheses (A), (B), and (C) must supply the block comparison, extension-class compatibility and compact generators. A list of local dimensions supplies none of these three maps or constructions.

68.7 The Three Conditional Inputs

(A) BH Type-Block Comparison

Conjecture 68.7.1 *(BH type-block comparison).* . Fix $p \in \{2, 3\}$, χ, and $\tau \in \mathbf{T}^{\mathrm{BH}}_{p,\chi}$. The projective indecomposable generator $P_\tau \in \mathrm{Block}^{\mathrm{BH}}_\tau$ is a compact object of $\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi$, and its image under $\Phi^{\mathrm{wild}}_{p,\chi}$ is the derived structure sheaf of the BH closed point $x_\tau \in \mathcal{X}^{\mathrm{EG,wild},p,\chi,\tau}_{GL_2}$.

Remark 68.7.2 *(evidence for the BH type-block comparison).* Three lines of evidence:

- Paskunas 2013 establishes the analogous statement for $p \geq 5$ (Publ. IHES 118 §6): each projective indecomposable of $\mathrm{Mod}^{\mathrm{sm},\chi}_{GL_2(\mathbb{Q}_p)}$ maps under Colmez–Montreal to the structure sheaf of a residual-Galois closed point. At $p \in \{2, 3\}$, the Paskunas argument requires modification because the block structure is more complex; the comparison at $p \in \{2, 3\}$ is the extension of Paskunas' result to the BH type list, with the same projective-to-structure-sheaf identification.

- Bushnell–Henniart 2006 §18 establishes the type-to-parameter match: each BH type τ has an associated L-parameter ϕ_τ. The projective generator P_τ is the type's indicator-character-module; the BH comparison conjecture asserts that this module maps to the derived structure sheaf of the closed point x_τ determined by $\phi_\tau \pmod p$.
- Compactness of P_τ: the block is a finitely-generated module over its local endomorphism algebra (BH 2006 §12.3), hence P_τ is compact in the derived category of the block.

The conjectural content is the precise image identification (second bullet), not compactness or finiteness.

(B) Paskunas Wild Ext^1*-Compatibility*

Conjecture 68.7.3 *(Paskunas wild-block* Ext^1*-compatibility).* . Fix $p \in \{2,3\}$ and a principal-block type $\tau \in \mathbf{T}^{\mathrm{BH}}_{p,\chi}$ with reducible residual ρ_τ. The Ext^1-class(es) of Question 54.4.8 (at $p=2$) or Question 54.4.9 (at $p=3$) on the automorphic side matches, as k-vector space with specific class(es), the degree-1 cotangent fibre of the wild-enhanced EG stack Definition 61.5.2 at the residual ρ_τ. The match respects the class labels: the class α_{01} matches one Ext^1-direction of the cotangent fibre; α_{12} matches the other specified direction at $p=3$.

Remark 68.7.4 *(dimensions and class comparisons).* The dimensions in Propositions 68.6.1 and 68.6.2 are conditional on the specified arithmetic local models. Even when these models are supplied, equality of dimensions does not construct a map carrying the extension classes to prescribed cotangent directions. That requires a comparison of the actual deformation complexes.

(C) Wild-EG Compact-Generator Identification

Conjecture 68.7.5 *(Wild spectral compact generation).* For the specified wild block, $\mathcal{D}_\tau$ is compactly generated. Its proposed derived closed-point structure sheaf Q_τ is an actual compact object of the nilpotent-support category and $\mathcal{D}^c_\tau = \mathrm{thick}(Q_\tau)$. The thick closure includes shifts, finite cones and retracts. Its localizing closure is consequently the whole category.

Remark 68.7.6 *(Support and the generator).* A closed-point structure sheaf does not generate an arbitrary ambient derived category. For example, in $D(k[t])$, localization at t kills $k = k[t]/(t)$ and its entire localizing closure, but does not kill $k[t]$. Thus k cannot generate $D(k[t])$. The chosen support category, compactness of the proposed image, and its thick generation in (C) are essential data. Formal neighbourhoods or a derived enhancement alone do not verify these assertions for the arithmetic target.

68.8 Consequence for F1.a and F1

F1.a Closure at Wild Primes

Theorem 68.8.1 *(F1.a at wild primes, conditional).* Assume every hypothesis of Theorem 68.1.2, including the wild enhancement, the fully faithful exact colimit-preserving block functors and specified decompositions, as well as (A), (B), and (C). Then F1.a holds at $p \in \{2,3\}$: there is an equivalence

$$\Phi^{\mathrm{wild}}_{p,\chi} \colon \mathrm{D}^{\text{sph-fin}}\big([GL_2(\mathbb{Q}_p)]\big)_\chi \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{EG,wild},p,\chi}_{GL_2}\big),$$

compatible with the specified block decomposition. Variation in χ additionally requires Conjecture 61.6.3.

Proof. Fully-faithfulness is Theorem 61.5.6. Essential surjectivity is Corollary 68.5.2. Both hold under the stated hypotheses. □

F1.a Closure at All Primes

COROLLARY 68.8.2 *(F1.a at all primes, conditional, sharpened).* Assume (H1)–(H5) of Theorem 61.7.1, with wild block exhaustion supplied by all the hypotheses of Theorem 68.1.2. Then F1.a holds at every prime and every central character χ: there is a single family equivalence

$$\Phi^{\mathrm{LocCat}}_{p,\chi}\colon \mathrm{D}^{\text{sph-fin}}\big([GL_2(\mathbb{Q}_p)]\big)_\chi \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{EG,wild\text{-}enhanced},p,\chi}_{GL_2}\big).$$

The central-character variation Theorem 61.6.4 also requires the family datum Conjecture 61.6.3.

Proof. Apply the specified categorical decompositions and effective family descent to Theorem 61.4.8 (at $p \geq 5$), Theorem 68.8.1 (at $p \in \{2,3\}$), and Theorem 61.6.4 (central-character variation, conditional on the family datum). □

The Updated F1 Obstruction Decomposition

PROPOSITION 68.8.3 *(F1 obstruction decomposition after BH block exhaustion).* The total F1 obstruction decomposition (combining F1.a.1–F1.a.3 with $\mathrm{L}_{\mathrm{ACD}}$) factors into the following named conjectural inputs:

(i) F1.a.1: block exhaustion at $p \geq 5$ (Conjecture 61.4.5), substantial non-formal theorem.
(ii) F1.a.2: wild-EG enhancement at $p \in \{2,3\}$ (Conjecture 61.5.7), substantial non-formal theorem.
(iii) Wild block functor datum at $p \in \{2,3\}$ (Conjecture 61.5.5).
(iv) F1.a.3.A: BH type-block comparison (Conjecture 68.7.1), substantial non-formal theorem.
(v) F1.a.3.B: Paskunas wild Ext^1-compatibility (Conjecture 68.7.3), substantial non-formal theorem.
(vi) F1.a.3.C: wild-EG compact-generator identification (Conjecture 68.7.5), substantial non-formal theorem.
(vii) Central-character family datum (Conjecture 61.6.3).
(viii) $\mathrm{L}_{\mathrm{ACD}}$: adelic categorical descent (Ch. 43), substantial non-formal theorem.

The block-exhaustion implication uses (A)–(C) together with the actual functors and decompositions in Theorem 68.1.2.

Proof. The enumeration is direct. F1.a.1, F1.a.2, the wild block functor datum, and the central-character family datum are inherited from Chapter 61. F1.a.3 factors as (A)+(B)+(C) by Theorem 68.1.2. $\mathrm{L}_{\mathrm{ACD}}$ is the adelic residual of Chapter 43. □

68.9 RESIDUAL CONDITIONS AND SOURCE COMPARISONS

Named Residuals Introduced

F1.a.3 factors into three sub-conjectures:

(i) Conjecture 68.7.1 (F1.a.3.A): projective indecomposables of BH blocks map to derived structure sheaves of BH closed points under the twisted DEG functor.
(ii) Conjecture 68.7.3 (F1.a.3.B): the non-semisimple Ext^1-classes on the automorphic side match the degree-1 cotangent fibres on the wild-enhanced EG stack, as classes (dimensional agreement under the explicit local-model hypotheses of Proposition 68.6.1 and Proposition 68.6.2).
(iii) Conjecture 68.7.5 (F1.a.3.C): compact generation of $\mathrm{IndCoh}_{\mathrm{Nilp}}$ on each wild-EG component by the derived structure sheaf of the BH closed point.

Conditional BH Comparisons

The following comparisons retain the local-model and decomposition hypotheses of their cited propositions:

(i) Dimensional comparison at $p = 2$ (Proposition 68.6.1): $1 = 1$.
(ii) Dimensional comparison at $p = 3$ (Proposition 68.6.2): $2 = 2$.
(iii) Structural consistency with Chapter 61 (Proposition 68.6.3): the BH-component decomposition matches the wild-EG decomposition of Chapter 61 structurally.

These statements give dimension and decomposition comparisons only under those hypotheses. The remaining arithmetic constructions include class-level match, compact generation, and projective-to-structure-sheaf comparison.

Wild Blocks at $p = 2$ and $p = 3$

The representation-theoretic inputs are Bushnell–Henniart (2006), §§11–13, Carter (1989), §7.3, Lusztig (1980), Jantzen (2003), Chapter II, §8, and Paskunas (2013), §6. The Dotto–Emerton–Gee comparison is confined to $p \geq 5$; extending it to $p = 2, 3$ requires the wild functor datum as well as (A). Bushnell–Henniart uses Langlands-normalised parameters. Comparing these with the Emerton–Gee moduli problem (arXiv:1908.07185) requires the specified parameter and conductor conventions.

At $p = 2$, the source is the block $\mathrm{Block}_{\tau}^{\mathrm{BH}}$ of smooth $GL_2(\mathbb{Q}_2)$-representations with central character χ. The geometric target is the component $\mathcal{X}_{GL_2}^{\mathrm{EG,wild},2,\chi,\tau}$; the comparison is between the associated stable presentable categories. Block restriction, the type-to-component correspondence of Bushnell–Henniart, §18, and formation of the derived structure sheaf must be compatible with the proposed functor. The required inputs are the enhancement and wild functor datum, together with (A), (B), and (C).

At $p = 3$, the analogous comparison uses the A_2-linear-quiver conventions, with radical-square-zero relations, of Auslander–Reiten–Smalø (1995), Chapter VII, §2. The proposed degree-one dimension is two rather than one, as in Question 54.4.9. This numerical datum does not establish the arithmetic local model. The three hypotheses (A), (B), and (C), including their compatibility with block restriction, are still required at $p = 3$.

Block Exhaustion and Compact Generators

The essential-surjectivity argument uses compact generation and formal deformation theory as formulated in Gaitsgory–Rozenblyum, Vol. II, Lurie's *Higher Algebra*, §4.8.2, and Pridham–Toen–Vezzosi, DAG VIII. Thick generation of compact objects uses shifts, finite cones and retracts; localizing generation also allows arbitrary colimits; the comparison is formulated in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$. Orthogonality of the blocks makes the block-by-block assembly diagonal only under the specified categorical decompositions. The fully faithful colimit-preserving functor and its lifted compact generators give essential surjectivity by Theorem 68.1.2.

Wild Components and Extension Classes

The comparison uses three inputs: the wild Emerton–Gee stack of Chapter 61, the block Ext^1-calculations of Chapter 54, and the type classification of Bushnell–Henniart (2006). These are separate geometric, homological and representation-theoretic inputs.

The geometric comparison is formulated in derived formal algebraic stacks over $\mathrm{Spf}\,\mathbb{Z}_p$. The degree-one fibre dimensions in Chapter 61 agree with the block extension dimensions by Proposition 68.6.1 and Proposition 68.6.2. The identification of individual classes requires hypothesis (B); it is stronger than

equality of dimensions. The comparison uses the functorial Bushnell–Henniart fibre correspondence of Bushnell–Henniart (2006), §18. The class-level identification remains the content of (B).

The Remaining Wild-Prime Comparisons

The comparison uses Langlands-normalised parameters and the specified conductor conventions in stable presentable ∞-categories. Its three unresolved inputs have distinct mathematical content:

- (A) requires the projective-to-structure-sheaf comparison of Paskunas at $p \geq 5$ to extend to the specified blocks at $p = 2, 3$.
- (B) requires the individual Ext^1-classes to match the cotangent-fibre directions under the deformation-theoretic map.
- (C) requires the relevant compact-generation theorem for the derived-enhanced wild Emerton–Gee stack.

The needed representation-theoretic and geometric inputs are those of Paskunas, Bushnell–Henniart, Emerton–Gee, Gaitsgory–Rozenblyum, and Pridham–Toen–Vezzosi. Their functorial compatibility with the specified arithmetic carriers is part of (A), (B), and (C). The conditional F1.a.3 implication uses all three in addition to the enhancement, functors and decompositions of Theorem 68.1.2.

68.10 Block Exhaustion Boundary

(i) F1.a.3 concerns exhaustion by the finite Bushnell–Henniart type list together with the specified extension-class comparisons in Chapter 54 (Remark 68.1.3).

(ii) The condition is: essential surjectivity of the twisted DEG-type functor $\Phi^{\mathrm{wild}}_{p,\chi}$ of Theorem 61.5.6 at $p \in \{2, 3\}$, reducing to type-by-type exhaustion via the BH block decomposition.

(iii) BH tame-wild classification at $p = 2$ (Proposition 68.3.1): finite list of unramified principal-series, tame supercuspidal, and wild supercuspidal types. The proposed Ext^1-carrying block requires the identification (Remark 68.3.2).

(iv) At $p = 3$, the additional local-model hypothesis in Proposition 68.4.1 permits arithmetic transport of the proved finite three-vertex calculation in Proposition 68.4.2. The arithmetic identification remains the comparison requested in Question 54.4.9.

(v) Dimensional comparison under the specified local models: both $p = 2$ (dimension 1) and $p = 3$ (dimension 2) match between the Ext^1-side and the derived-fibre side (Proposition 68.6.1, Proposition 68.6.2).

(vi) Structural compatibility under the component comparison: the BH-component decomposition coincides with the wild-EG decomposition of Chapter 61 (Proposition 68.6.3).

(vii) Essential surjectivity on a single BH block (Theorem 68.5.1): proved under every hypothesis of Theorem 68.1.2. Assembly across types (Corollary 68.5.2): equivalence of the twisted DEG-type functor.

(viii) The three comparison hypotheses are (A) the BH type-block comparison (Conjecture 68.7.1), (B) the Paskunas wild Ext^1 comparison (Conjecture 68.7.3), and (C) wild-EG compact generation (Conjecture 68.7.5).

(ix) The F1.a.3 conclusion also requires the enhancement and the fully faithful colimit-preserving block functors and decompositions.

(x) The consequence for the F1 table (Proposition 68.8.3) is that F1.a factors into named conjectural inputs: F1.a.1, F1.a.2, the wild block functor datum, F1.a.3.A–C, and the central-character family datum; F1 adds L_{ACD}. This is the finest factorisation of the F1 residual recorded here.

There is no unconditional closure of F1.a.3. The dimensional and structural comparisons require their specified arithmetic local models, while block exhaustion remains conditional on all the hypotheses of Theorem 68.1.2.

Chapter 69

Arithmetic Fundamental Local Equivalence: Selmer-Derived Target and Open Categorical Equivalences

69.1 Selmer-Derived Global Spectral Stack

A global spectral target must impose the global Galois relation and its local conditions simultaneously. The formal deformation functors of Chapter 10 provide a precise pullback on a common coefficient category. A geometric stack representing that functor, and a comparison with local parameter stacks, require further hypotheses.

Definition 69.1.1 *(Selmer deformation functor).* Fix a finite field k of characteristic ℓ. Choose a finite ramification set S containing the archimedean place and ℓ. Fix a continuous representation $\bar{\rho} : G_{\mathbb{Q},S} \to \mathrm{GL}_2(k)$, and a determinant lift $\delta : G_{\mathbb{Q},S} \to W(k)^\times$. Use the category $\mathrm{Art}_{W(k)}$ of augmented simplicial Artinian local coefficient algebras from §10.6.

Let $\mathcal{D}^{\mathrm{or}}_{G,\delta}$ denote the derived continuous deformation functor with an identification of its determinant with δ. Arrows preserve that identification. For each finite $v \in S_f$, supply a pointed local functor $\mathcal{D}^{\mathcal{L}}_v$ and a morphism $\mathcal{D}^{\mathcal{L}}_v \to \mathcal{D}^{\mathrm{or}}_{G_{\mathbb{Q}_v},\delta|_v}$ on this same coefficient category.

Define

$$\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\delta,\mathcal{L}} := \mathcal{D}^{\mathrm{or}}_{G_{\mathbb{Q},S},\delta} \times^h_{\prod_{v\in S_f} \mathcal{D}^{\mathrm{or}}_{G_{\mathbb{Q}_v},\delta|_v}} \prod_{v\in S_f} \mathcal{D}^{\mathcal{L}}_v. \tag{69.1}$$

This notation initially denotes a functor to spaces. It denotes a geometric stack only when such a representative has been constructed. A local Emerton–Gee or Weil-parameter description can be used after its comparison with the indicated local functor, on the specified coefficient category, is supplied.

Proposition 69.1.2 *(Selmer tangent complex).* At the residual point, suppose the tangent of each $\mathcal{D}^{\mathcal{L}}_v$ is $L_v[1]$, with its induced map $g_v : L_v \to C^\bullet(G_{\mathbb{Q}_v}, \mathrm{ad}^\circ \bar{\rho})$. Then

$$T\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\delta,\mathcal{L}} \simeq C^\bullet_{\mathcal{L}}(G_{\mathbb{Q},S}, \mathrm{ad}^\circ \bar{\rho})[1], \tag{69.2}$$

where $C^\bullet_{\mathcal{L}}$ is the mapping fiber of

$$C^\bullet(G_{\mathbb{Q},S}, \mathrm{ad}^\circ \bar{\rho}) \oplus \prod_{v\in S_f} L_v \xrightarrow{\mathrm{res}-g} \prod_{v\in S_f} C^\bullet(G_{\mathbb{Q}_v}, \mathrm{ad}^\circ \bar{\rho}).$$

Proof. The tangent functor preserves the homotopy pullback (69.1). Propositions 10.6.1 and 10.6.2 compute its global and full local tangent complexes. Their fiber is the displayed Selmer complex, as in Theorem 10.7.1. □

Its cohomology groups $H^0_{\mathcal{L}}$, $H^1_{\mathcal{L}}$, and $H^2_{\mathcal{L}}$ occur in tangent degrees -1, 0, 1. They record infinitesimal automorphisms, deformations, and obstructions. If each g_v is an inclusion of complexes, a quotient by L_v gives an equivalent description. A general morphism of local conditions requires the fiber above. The trace-zero coefficients rely on preserving a determinant identification, rather than merely its value.

Neither geometric representability nor Noetherianity follows from this tangent formula. Quasi-smoothness is also a separate condition on a geometric representative. For derived schemes it allows a nonzero tangent group in degree one: the dual-number scheme of Example 10.7.2 is an explicit case. Real-place complexes have not been included in (69.1). A compact-support interpretation must specify them, especially for $\ell = 2$.

69.2 Local Geometrisation and Its Domain

The following collects what Fargues–Scholze, Emerton–Gee, and Dotto–Emerton–Gee establish in the local setting. These are the inputs to the conditional programme of §69.3.

Theorem 69.2.1 *(local inputs, Fargues–Scholze / Emerton–Gee / Dotto–Emerton–Gee).* (assembling external results).

(i) *Local geometrisation and spectral action* (Fargues–Scholze arXiv:2102.13459). For each prime p, the Fargues–Scholze spectral action equips $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])$ with an action of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{GL_2})$ and supplies local Langlands geometrisation and Hecke functoriality.

(ii) *Local (φ, Γ) parameter stacks* (Emerton–Gee arXiv:1908.07185). For each prime p, the Emerton–Gee stack $\mathcal{X}^{\mathrm{EG},p}_{GL_2}$ parametrises two-dimensional (φ, Γ)-modules over $\mathbb{Q}_p$ with a well-behaved moduli structure; local conditions $\mathcal{X}_v^{\mathcal{L}_v} \hookrightarrow \mathcal{X}^{W,\varphi\Gamma}_{GL_2,v}$ are open or closed substacks.

(iii) *Fully faithful functor at fixed central character* (Dotto–Emerton–Gee arXiv:2603.26887). For primes $p \geq 5$, there exists a fully faithful functor

$$D^{\text{sph-fin}}_{GL_2(\mathbb{Q}_p),\chi} \hookrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{EG},p}_{GL_2,\chi}\big)$$

for each fixed central character χ. Essential surjectivity, the cases $p = 2, 3$, and the central-character-family statement remain open.

(iv) *Archimedean OPE confined to archimedean factor*. The Hochschild-trace candidate $\mathcal{V}^{\mathrm{prim}}$ carries chiral OPE structure at the archimedean factor $v = \infty$ via the $(\mathfrak{gl}_2, O(2))$-module structure; finite places carry locally constant sheaf data and Hecke-algebra actions via the Satake isomorphism. Chiral OPE structure is an archimedean-fibre phenomenon.

(v) *Commutative Bethe–Hecke algebra*. At each finite prime p, the centre of the spherical Hecke algebra $\mathcal{H}(GL_2(\mathbb{Q}_p), GL_2(\mathbb{Z}_p))$ is commutative and acts on $\mathcal{V}^{\mathrm{prim}}$ via the Satake isomorphism. This commutative Bethe–Hecke structure is the correct Hecke-algebraic shadow; higher Hopf or Yangian structures require separate verification at each prime.

(vi) *Classical Kim arithmetic Chern–Simons*. Kim's arithmetic Chern–Simons theory arXiv:1510.05818 supplies classical arithmetic CS torsors on the arithmetic curve and action lines in K_2. The quantum boundary category $\partial_q\mathrm{ACS}$ of Conjecture 69.3.6 is a distinct object from $\mathcal{V}^{\mathrm{prim}}$; its construction is open.

69.3 Conditional Categorical Equivalences

The local inputs of §69.2 establish the spectral action, the (φ, Γ) parameter stacks, and full faithfulness at fixed central character. A complete arithmetic FLE requires four further categorical equivalences, each an open problem conditional on a named hypothesis.

Conjecture 69.3.1 *(local categorical comparison).* . For every prime p there is an equivalence

$$D^{\text{sph-fin}}\big([GL_2(\mathbb{Q}_p)]\big) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{EG},p}_{GL_2}\big), \tag{69.3}$$

compatible with Hecke functors, central-character families, and the Fargues–Scholze spectral action. The Dotto–Emerton–Gee functor establishes full faithfulness for $p \geq 5$ at fixed central character; the conjecture adds essential surjectivity, the cases $p = 2, 3$, and the family statement.

Conjecture 69.3.2 *(adelic categorical descent).* . Under the local categorical comparison (69.3) at all primes, the local categories glue to the global automorphic category $D([GL_2])^{\text{sph-fin}}$ via an equivalence compatible with Whittaker models, Bernstein centres, the archimedean $(\mathfrak{g}, K)$ factor, and Eisenstein and continuous spectrum. Representation-level restricted tensor products are classical (Flath); the categorical descent is the open statement.

Conjecture 69.3.3 *(mixed-characteristic Iwahori E_2).* . The mixed-characteristic local Iwahori Hecke category $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p}$ carries an E_2-monoidal structure compatible with diamond Beilinson–Drinfeld fusion and with the Iwahori stratification:

$$\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p} \in \mathrm{Alg}_{E_2}\big(\mathrm{Cat}_\infty\big). \tag{69.4}$$

Current sources supply spherical fusion and E_1 convolution; the E_2 enhancement is the local input required for the restricted-tensor E_2 transport argument. The restricted-tensor E_2 transport lemma (proved elsewhere in this volume) is a separate, purely categorical statement; its hypotheses require (69.4) as input.

Conjecture 69.3.4 *(full-target arithmetic Ben-Zvi–Nadler).* . For the Selmer-derived full Weil/(φ, Γ) target, the arithmetic Ben-Zvi–Nadler comparison is

$$\mathrm{HH}_\bullet\big(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}\big) \simeq \mathcal{O}\big(\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}\big)^{\mathrm{flat}}, \tag{69.5}$$

where $\mathrm{LocSys}^{W,\varphi\Gamma}_{GL_2}$ is the full Weil/(φ, Γ) local system stack and $\cdot^{\mathrm{flat}}$ denotes the flat-locus subfunctor. The conjecture requires formal-Noetherian Hochschild continuity for $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$ and comparison of the Arinkin–Gaitsgory nilpotent flat locus with the Fargues–Scholze/Emerton–Gee flat locus.

Conjecture 69.3.5 *(categorical arithmetic Whittaker theory).* . The arithmetic Whittaker factorisation lifts from representation-level Whittaker models to a categorical equivalence compatible with the local comparisons of (69.3), adelic descent, and the Selmer-derived spectral stack (69.1). Representation-level Tate–Jacquet–Langlands factorisation is proved; the categorical lift is the open statement.

Conjecture 69.3.6 *(quantum ACS boundary category).* . There exists a quantum boundary category $\partial_q\mathrm{ACS}$ and an equivalence

$$\partial_q\mathrm{ACS} \simeq \mathcal{V}^{\mathrm{prim}}$$

compatible with an appropriate quantisation of Kim's arithmetic Chern–Simons theory arXiv:1510.05818. Kim's theory provides the classical arithmetic CS torsors and action lines in K_2; the quantisation is the open content of this conjecture.

69.4 Proved and Conditional Components of the Arithmetic FLE

Theorem 69.4.1 *(arithmetic FLE: proved and conditional components).* (by enumeration). The arithmetic programme based on $\mathcal{V}^{\mathrm{prim}}$ and the Selmer-derived stack $\mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\partial,\mathcal{L}}$ (69.1) decomposes as follows.

(i) *Constructed as a formal functor.* The Selmer homotopy pullback exists for the supplied local functors and maps on $\mathrm{Art}_{W(k)}$ (Definition 69.1.1). Its tangent is the stated Selmer complex under the local tangent hypotheses (Proposition 69.1.2). Geometric representability and comparisons with the local parameter stacks remain additional requirements.

(ii) *Conditional on Conjecture 69.3.1.* Local categorical comparison at all primes, including essential surjectivity at $p \geq 5$ and the full statement at $p = 2, 3$.

(iii) *Conditional on Conjecture 69.3.1 and Conjecture 69.3.2.* Adelic categorical descent to the global automorphic category.

(iv) *Conditional on Conjecture 69.3.3.* Iwahori E_2-monoidal structure at mixed characteristic; the restricted-tensor E_2 transport lemma applies once this local input is supplied.

(v) *Conditional on Conjecture 69.3.4.* Full-target arithmetic Ben-Zvi–Nadler comparison (69.5).

(vi) *Conditional on Conjecture 69.3.5.* Categorical arithmetic Whittaker factorisation.

(vii) *Conditional on Conjecture 69.3.6.* Quantum ACS boundary category equivalent to $\mathcal{V}^{\mathrm{prim}}$.

$\mathcal{V}^{\mathrm{prim}}$ is a Hochschild trace class throughout; its chiral OPE structure is confined to the archimedean factor as established in Theorem 69.2.1(4). A vertex algebra requires an OPE on a formal disk defined at every place; $\mathcal{V}^{\mathrm{prim}}$ carries this structure only at $v = \infty$.

Chapter 70

F1.A.1 Unramified Ideal, F1.A.2 Small-Prime Block Extension, and L_{ACD} Cuspidal Descent

The local ordinary, small-prime block, and global cuspidal comparisons have different carriers. Fixed-determinant Frobenius parameters and spherical or Iwahori invariant vectors provide explicit local data. An enhanced compact-generator criterion explains what additional maps would make a categorical comparison an equivalence. The ordinary essential image, the small-prime block functors and global cuspidal descent remain separate arithmetic construction problems. Their Eisenstein extension also requires normalized intertwiners and global spectral synthesis.

70.1 Objects and Comparison Data

Inherited Objects

The intended arithmetic comparisons use the following notation and prior comparison statements, with their carrier and realization hypotheses retained:

- $\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi$ and its Dotto–Emerton–Gee functor $\Phi^{\mathrm{DEG}}_{p,\chi}$ (Chapter 61, Theorem 61.3.1);
- the Bushnell–Henniart block decomposition and its type list (Chapter 68; Proposition 68.3.1, Proposition 68.4.1);
- the Selmer-derived global spectral stack $\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\delta,\mathcal{L}}$ (Chapter 69, Definition 69.1.1);
- the four subconditions $\mathrm{L}_{\mathrm{ACD}}.0$–$.3$ of Chapter 52, together with Theorem 50.1.3 (F2.c tightened).

The arithmetic comparisons require these objects with specified enhancements and support conditions. The local models below define their own moduli problems; they are not identified with a Selmer stack.

Primary Sources

The deformation, representation and automorphic comparison problems use the following references:

Breuil–Mézard 2002 "Multiplicités modulaires et représentations de $GL_2(\mathbb{Z}_p)$ et de $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ en $\ell = p$", Duke Math. J. 115. The numerical Breuil–Mézard conjecture.

Kisin 2009 "The Fontaine–Mazur conjecture for GL_2", J. AMS 22. Modularity lifting plus Breuil–Mézard in Hodge–Tate regular weight.

Gee–Kisin 2014 "The Breuil–Mézard conjecture for potentially Barsotti–Tate representations", Forum Math. Pi 2. Breuil–Mézard geometric formulation.

Emerton–Gee arXiv:1908.07185 The Emerton–Gee stack of (φ, Γ)-modules. The arithmetic comparison requires the actual support and cycle constructions on this stack; the elementary unramified framed calculation below has a different carrier.

Dotto–Emerton–Gee arXiv:2603.26887 Fully faithful functor at fixed central character for $p \geq 5$.

Clozel–Harris–Taylor 2008 *Automorphy for some l-adic lifts of automorphic mod l Galois representations*, Publ. Math. IHES 108, 1–181. Automorphy lifting under its conjugate self-duality, regularity and ramification hypotheses.

Harris–Taylor 2001 *The Geometry and Cohomology of Some Simple Shimura Varieties*, Ann. Math. Stud. 151. Shimura-variety cohomology and local Langlands compatibility; this is a different source from the automorphy-lifting paper.

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Saito 1997 "Modular forms and p-adic Hodge theory", Invent. Math. 129. Local-global compatibility for the Galois representation attached to a classical newform: the Frobenius trace at an unramified prime $\ell \neq p$ is the sum of the two roots of the Galois Frobenius polynomial and equals the Hecke coefficient a_ℓ in the matching arithmetic normalization. The normalized-induction Hecke operator below uses its explicit $q^{1/2}$ factor.

Taylor–Wiles 1995 "Ring-theoretic properties of certain Hecke algebras", Ann. Math. 141. Big-image Taylor–Wiles hypotheses.

Roles of the comparison data

Breuil–Mézard, Kisin and Gee–Kisin concern deformation rings and cycles. Emerton–Gee concerns the spectral moduli stack, while Dotto–Emerton–Gee constructs a fully faithful functor on its specified coefficient category. Clozel–Harris–Taylor concerns automorphy lifting; Harris–Taylor concerns Shimura-variety cohomology and local Langlands compatibility. Carayol and Saito concern local-global compatibility. Paškūnas and Colmez–Dospinescu–Paškūnas supply block-theoretic questions and constructions in their stated representation categories. Flath gives a factorization of actual representations; Bernstein and Mœglin–Waldspurger concern spectral decomposition. None of these different roles supplies the missing functors, support comparisons or generation statements by itself.

70.2 Fi.a.i: The Unramified Ordinary Ideal

Hypotheses and Target

Setup 70.2.1 *(Taylor–Wiles unramified setup).* Fix a prime $p \geq 5$, a level $N \in \mathbb{Z}_{\geq 1}$ with $\gcd(N, p) = 1$, and a residual Galois representation

$$\bar{\rho}\colon G_{\mathbb{Q}} \to GL_2(\overline{\mathbb{F}}_p)$$

unramified at p and at almost all $\ell \nmid Np$. Descend its coefficients to a finite field k. Fix a full local central character $\chi : \mathbb{Q}_p^\times \to k^\times$ using local reciprocity that sends a uniformizer to geometric Frobenius. In the determinant-identified DEG convention, its corresponding Galois character satisfies $\det \bar{\rho}|_{G_{\mathbb{Q}_p}} = \chi\bar{\varepsilon}^{-1}$, where $\bar{\varepsilon}$ is the residual cyclotomic character. Thus both its inertia values and its value on a uniformizer are fixed. A characteristic-zero comparison also chooses compatible lifts of the central character and determinant, as in the construction of Chapter 86. Assume:

(TW1) $p \geq 5$.

(TW2) $\bar{\rho}|_{G_{\mathbb{Q}(\zeta_p)}}$ is absolutely irreducible.

(TW3) The local residual representation at p has distinct characters in its semisimplification after the chosen coefficient extension. This is local distinguishedness, not a claim that the global representation has diagonal characters.

(TW4) Restrict to the intended ordinary component $\mathrm{Block}^{\mathrm{ord}}_{\bar{\rho}|_{G_{\mathbb{Q}_p}}}$, where the local representation has an unramified invariant line. This line convention must be compared explicitly with the unramified quotient convention of Chapter 86.

The central character in Setup 70.2.1 is ramified on units. Indeed, the residual representation is unramified at p, so its determinant is trivial on inertia, whereas the geometric reciprocity convention gives $\bar{\varepsilon}(u) = \bar{u}$ for $u \in \mathbb{Z}_p^\times$ and $\bar{\varepsilon}(p) = 1$. Thus $\chi(u) = \bar{u}$. This is the convention of Dotto–Emerton–Gee, Section 2.1.1, p. 9. A principal series induced from two unramified characters has trivial central character on units and cannot supply an object of this fixed-χ category.

Proposition 70.2.2 *(central-character-compatible smooth source objects).* Fix a finite extension of $\mathbb{Q}_p$ with integer ring $\mathcal{O}$ and residue field k, a continuous lift $\zeta : \mathbb{Q}_p^\times \to \mathcal{O}^\times$ of χ, and the compatible determinant $\delta = \zeta\varepsilon^{-1}$ under geometric reciprocity, where ε is the p-adic cyclotomic character. Let A be a finite local Artinian $\mathcal{O}$-algebra with residue field k. Write ζ_A for the induced character, and choose an unramified character $\eta_A : \mathbb{Q}_p^\times \to A^\times$. Put $G = GL_2(\mathbb{Q}_p)$ and let B be its upper triangular subgroup. The unnormalized smooth induction

$$I_A = \mathrm{Ind}_B^G(\zeta_A\eta_A^{-1} \otimes \eta_A)$$

consists of locally constant $f : G \to A$ satisfying

$$f\left(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} g\right) = \zeta_A(a)\eta_A(a)^{-1}\eta_A(d) f(g), \qquad (g_0 f)(g) = f(g g_0).$$

It is a nonzero smooth p-power-torsion $\mathcal{O}$-representation with central character ζ. It has no $GL_2(\mathbb{Z}_p)$-fixed vectors. In particular, no square-root modulus factor is used over A.

Proof. The finite coefficient characters have open kernels. Choose a principal congruence subgroup $K_m \subset GL_2(\mathbb{Z}_p)$ so that the inducing character θ_A of B is trivial on $B \cap K_m$. Define $f(bk) = \theta_A(b)$ on BK_m and zero outside. Two factorizations differ by $B \cap K_m$, so this is well-defined. The image of K_m in the Hausdorff space $B\backslash G$ is compact and open, hence closed as well. The resulting zero extension is locally constant and $f(1) = 1$. More generally, restriction to the compact group $GL_2(\mathbb{Z}_p)$ shows that every locally constant induced function is fixed on the right by some sufficiently small open subgroup: a finite clopen partition refines its local constancy neighbourhoods. This proves smoothness. The ring A is killed by a power of p, so the same is true of I_A. A central scalar x acts by $\zeta_A(x)\eta_A(x)^{-1}\eta_A(x) = \zeta_A(x)$. Choose a unit u whose reduction is not 1. The element $\zeta_A(u) - 1$ is a unit of A. If a vector were fixed by $GL_2(\mathbb{Z}_p)$, it would be annihilated by this unit, hence vanish. □

A determinant twist cannot turn an unramified principal series into this residual central-character family when p is odd. On the cyclic group $\mathbb{F}_p^\times$, such a twist changes the central exponent by $2a$; the required exponent is 1. The congruence $2a = 1 \pmod{p-1}$ has no solution. This remains an obstruction after coefficient extension and, by reduction, for the finite rings A. The characteristic-zero unramified invariant-vector calculation below therefore has a different central-character carrier.

Definition 70.2.3 *(data for the ordinary comparison subcategory).* The representations I_A are actual objects of the smooth p-power-torsion category $\mathcal{A}_\zeta$ of Chapter 86. Their stable colimit closure inside $D(\mathcal{A}_\zeta)$ is a defined source subcategory, but is not asserted to be the ordinary essential-image category. The intended notation $\mathrm{D}^{\text{sph-fin,unr-ord}}([GL_2(\mathbb{Q}_p)])_\chi$ requires the coefficient comparison with $D(\mathcal{A}_\zeta)$, a choice of ordinary source objects, and a proof of the required Hecke-finiteness or admissibility condition. On the spectral side, construct the intended formal neighbourhood $\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi,\text{unr-ord}}$ with its determinant, ordinary flag and support condition. Identify which compatible induced objects, ordinary subquotients or Hida-family objects map to that support, and prove their generator property. The local Frobenius and invariant-vector calculations do not by themselves make these arithmetic identifications.

Statement

The ordinary comparison requires stable presentable enhanced source and target subcategories, a proof that the actual DEG functor restricts between them, and a colimit-preserving restricted functor. The notation sph-fin and $\mathrm{IndCoh}_{\mathrm{Nilp}}$ here names those intended categories; the framed local rings alone do not construct them. The essential-image hypothesis below means that the images of a specified generating family generate this target.

Theorem 70.2.4 *(Fi.a.1 unramified ordinary ideal).* Under Setup 70.2.1 and the DEG essential-image theorem on the unramified ordinary Emerton–Gee component, the restriction of the DEG functor of Theorem 61.3.1

$$\Phi_{p,\chi}^{\mathrm{DEG}}\big|_{\text{unr-ord}} \colon \mathrm{D}^{\text{sph-fin,unr-ord}}([GL_2(\mathbb{Q}_p)])_\chi \longrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi,\text{unr-ord}})$$

is an equivalence. The Taylor–Wiles, Breuil–Mezard, Hida and local-global compatibility problems specify arithmetic inputs to construct; the essential-image hypothesis is additional and is not proved by the eigenvalue calculation below.

The local calculations below distinguish the parameter and operator constructions from the arithmetic hypotheses in this conditional statement.

Input 1: Breuil–Mézard on the Unramified Ordinary Locus

Conjecture 70.2.5 *(ordinary Breuil–Mézard geometry for the comparison).* For the specified ordinary spectral target, construct the cycles $\mathcal{X}^{\mathrm{BM},\sigma}_{p,\chi,\bar{\rho}}$ for its Serre weights, including the proposed conductor range $\leq p+1$, their multiplicities, intersections and coverage. Determine which cycles are irreducible, which intersections are transverse, and which associated coherent objects generate the target support category. Smoothness and transversality are additional geometric assertions, not consequences of numerical cycle multiplicities.

Kisin (2009), Gee–Kisin (2014) and Emerton–Gee concern the relevant deformation and cycle problems. Their hypotheses must be matched with the actual type, determinant and coefficient space before these geometric assertions can be used. In particular, cycle coverage does not prove essential surjectivity of a categorical functor.

70.3 Fixed Determinant, Frobenius Eigenvalues, and Invariant Vectors

Fixing a determinant imposes a relation between the two Frobenius roots. Their eigenlines also retain automorphisms. The resulting parameter stack and the invariant vectors of a principal series give explicit local constructions for a categorical comparison.

The Frobenius parameter stack

Let E be a field of characteristic zero and let $d \in E^{\times}$. For an E-algebra A, consider a rank-two projective A-module V with an automorphism F satisfying $\wedge^2 F = d\,\mathrm{id}_{\wedge^2 V}$. Isomorphisms intertwine F. No trivialization of $\wedge^2 V$ is part of the data. This defines the quotient stack

$$\mathcal{U}_d = [Y_d/GL_2], \qquad Y_d = \operatorname{Spec} E[x_{11}, x_{12}, x_{21}, x_{22}]/(x_{11}x_{22} - x_{12}x_{21} - d),$$

where GL_2 acts by conjugation. It is the algebraic parameter stack for an unramified rank-two Weil representation with chosen generator acting by F.

Put $t = \operatorname{tr} F$. The regular semisimple open substack $\mathcal{U}_d^{\mathrm{rs}}$ is defined by $t^2 - 4d \neq 0$. The ordered eigenvalue scheme and diagonal torus are

$$T_d = \operatorname{Spec} E[a, a^{-1}, (a^2 - d)^{-1}], \qquad T = \mathbb{G}_m^2.$$

The involution w sends a to d/a and exchanges the factors of T.

Proposition 70.3.1 *(Eigenvalues and eigenline automorphisms).* There is an equivalence

$$\mathcal{U}_d^{\mathrm{rs}} \simeq [T_d/(T \rtimes \langle w \rangle)],$$

where T acts trivially on T_d. The scheme of unordered eigenvalues is

$$U_d = \operatorname{Spec} E[t, (t^2 - 4d)^{-1}].$$

It has dimension one. Over an algebraically closed field, each geometric object of $\mathcal{U}_d^{\mathrm{rs}}$ has automorphism group $\mathbb{G}_m^2$.

Proof. After a finite étale base change, the polynomial $X^2 - tX + d$ has distinct roots a, b, with $b = d/a$. The endomorphisms

$$e_a = \frac{F - b}{a - b}, \qquad e_b = \frac{F - a}{b - a}$$

are complementary idempotents. Their images are rank-one projective modules, and F acts on them by a and b. Conversely, two line bundles with these scalar actions give (V, F). Changing their frames gives the T-action. Exchanging their order gives the involution. Descent gives the stated quotient stack.

The map $a \mapsto t = a + d/a$ identifies the unordered roots. Its coordinate extension satisfies $a^2 - ta + d = 0$. Inverting $t^2 - 4d$ makes this a finite étale double cover with the stated involution. The invariant ring is $E[t, (t^2 - 4d)^{-1}]$. Each automorphism preserves the two eigenspaces and scales them independently. □

The atlas Y_d has dimension three, since the determinant equation has nonzero differential on invertible matrices. Thus $\mathcal{U}_d$ has dimension $3 - 4 = -1$ as an algebraic stack. Its eigenvalue scheme has a different dimension and omits stabilizers. If a trivialization of $\wedge^2 V$ is required, the quotient uses SL_2 instead. Its regular semisimple stabilizer is $\mathbb{G}_m$. These are distinct moduli conventions.

An unramified deformation ring

Let O be a complete discrete valuation ring of mixed characteristic, with finite residue field k of characteristic p. Fix a nonarchimedean local field L. Write G_L for its absolute Galois group and I_L for inertia, so $G_L/I_L \simeq \widehat{\mathbb{Z}}$. Fix $\overline{F} \in GL_2(k)$ and $d \in O^\times$ lifting $\det \overline{F}$. A test algebra is a local Artinian O-algebra A with residue field k. An unramified framed deformation sends arithmetic Frobenius to $F_A \in GL_2(A)$, with $F_A \bmod \mathfrak{m}_A = \overline{F}$ and $\det F_A = d$. Here arithmetic Frobenius acts on a residue field by its cardinality power. Inertia acts trivially.

Proposition 70.3.2 *(Framed deformations and distinct residual roots).* Choose a matrix lift $F_0 \in M_2(O)$ of $\overline{F}$. The framed deformation functor is represented by

$$R_d^{\square} = O[[X_{11}, X_{12}, X_{21}, X_{22}]]/(\det(F_0 + X) - d).$$

This ring is formally smooth of relative dimension three. Suppose additionally that $\overline{F} = \operatorname{diag}(\bar{a}, \bar{b})$, with $\bar{a} \neq \bar{b}$. Choose $a_0 \in O^\times$ lifting $\bar{a}$ and put $b_0 = d/a_0$. Then

$$R_d^{\square} \simeq O[[x, y, z]], \qquad F = \begin{pmatrix} a_0 + x & y \\ z & (d + yz)/(a_0 + x) \end{pmatrix}.$$

Strict conjugacy classes of these deformations are represented by $O[[u]]$. The corresponding ordered eigenvalues are $a = a_0 + u$ and $b = d/a$.

Proof. Every test algebra is finite. A matrix in its finite group $GL_2(A)$ defines a unique continuous homomorphism from $\widehat{\mathbb{Z}}$. Thus specifying F_A is equivalent to specifying the unramified representation. The matrix entries and determinant equation give the first ring.

At least one cofactor of $\overline{F}$ is nonzero. The derivative in the corresponding matrix coordinate is a unit. Solving for that coordinate proves formal smoothness and leaves three parameters. In the displayed diagonal case, $a_0 + x$ is a unit and the determinant equation solves directly for the lower-right entry.

For each F_A, Hensel lifting gives the unique characteristic root a reducing to $\bar{a}$. The other root is $b = d/a$. Since $a - b$ is a unit, the projectors above split A^2 into two free rank-one eigenspaces. Their bases can be chosen to reduce to the standard residual bases. The resulting change of basis is the identity modulo $\mathfrak{m}_A$, so it is a strict conjugation. Two diagonal lifts are strictly conjugate exactly when their ordered roots agree. The choice of $a - a_0 \in \mathfrak{m}_A$ therefore represents the set of strict conjugacy classes. □

The last ring represents classes, not the deformation groupoid. For a diagonal lift, strict automorphisms contain $(1+\mathfrak{m}_A)^2$. A quotient stack retains these automorphisms. The construction concerns unramified lifts only. A crystalline lifting condition or a larger Emerton–Gee component requires its own map and coefficient conventions.

Spherical and Iwahori invariant vectors

Let F now denote a nonarchimedean local field, with ring of integers O_F, uniformizer ϖ, and residue cardinality q. Write $G=GL_2(F)$, $K=GL_2(O_F)$, and let B be the upper triangular subgroup. Fix a characteristic-zero coefficient field E containing a chosen square root $q^{1/2}$. Let $\chi_1,\chi_2:F^\times\to E^\times$ be unramified characters. Put $\alpha=\chi_1(\varpi)$ and $\beta=\chi_2(\varpi)$.

Normalized smooth induction is the space $I(\chi_1,\chi_2)$ of locally constant functions $f:G\to E$ satisfying

$$f\left(\begin{pmatrix}a & b\\ 0 & d'\end{pmatrix}g\right)=|a/d'|^{1/2}\chi_1(a)\chi_2(d')f(g),$$

with right translation by G. Let $J\subset K$ be the Iwahori subgroup whose reduction is upper triangular. Choose $w=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}$ and $t_0=\operatorname{diag}(\varpi,1)$.

Proposition 70.3.3 *(Invariant spaces and their Hecke operators).* Evaluation at 1 identifies $I(\chi_1,\chi_2)^K$ with E. Evaluation at $(1,w)$ identifies $I(\chi_1,\chi_2)^J$ with E^2. With Haar measure giving J volume one, define $U=q^{-1/2}[Jt_0J]$ on the latter space. Its matrix is

$$U=\begin{pmatrix}\alpha & 0\\ (1-q^{-1})\alpha & \beta\end{pmatrix}.$$

Consequently $(U-\alpha)(U-\beta)=0$. For the spherical Hecke operators, normalize the measure by $\operatorname{vol}(K)=1$. Then $T=[Kt_0K]$ and $S=[K\operatorname{diag}(\varpi,\varpi)K]$ act on the spherical line by

$$T=q^{1/2}(\alpha+\beta),\qquad S=\alpha\beta.$$

Proof. The Iwasawa decomposition $G=BK$ identifies K-invariant functions with their value at 1. The two $B(k_F)$-orbits on $K/J\simeq\mathbb{P}^1(k_F)$ give the representatives $1,w$. This proves both dimension statements.

Conjugation by t_0 shows that $J\cap t_0Jt_0^{-1}$ consists of matrices whose upper-right entry is divisible by ϖ. Its index in J is q. Representatives are the upper unipotent matrices with entries u mod ϖ. Thus a right-coset decomposition of Jt_0J is

$$Jt_0J=\coprod_{u\in O_F/\varpi O_F}\begin{pmatrix}\varpi & u\\ 0 & 1\end{pmatrix}J.$$

At 1, every summand gives $q^{-1/2}\alpha f(1)$. For a nonzero residue class u, choose a unit lift. Then

$$w\begin{pmatrix}\varpi & u\\ 0 & 1\end{pmatrix}=\begin{pmatrix}-\varpi/u & 1\\ 0 & -u\end{pmatrix}\begin{pmatrix}1 & 0\\ \varpi/u & 1\end{pmatrix}.$$

The second factor belongs to J. Each of these $q-1$ terms contributes $q^{-1/2}\alpha f(1)$ at w. The remaining term satisfies $wt_0=\operatorname{diag}(1,\varpi)w$ and contributes $q^{1/2}\beta f(w)$. Multiplication by $q^{-1/2}$ gives the matrix.

For Kt_0K/K, the same q representatives are joined by $\mathrm{diag}(1, \varpi)$. Evaluating the spherical function at 1 gives $q^{1/2}(\alpha + \beta)$. The central double coset for S acts by the central character $\chi_1\chi_2$ at ϖ. □

If $\alpha \neq \beta$, the two projectors on the Iwahori invariant space are

$$P_\alpha = \frac{U - \beta}{\alpha - \beta}, \qquad P_\beta = \frac{U - \alpha}{\beta - \alpha}.$$

They have rank one and sum to the identity. If $\alpha = \beta$, the matrix has a nonzero off-diagonal entry, so it is not diagonalizable. The whole principal series is infinite-dimensional: locally constant sections on arbitrarily many disjoint open subsets of $\mathbb{P}^1(F)$ are linearly independent. Its two-dimensional invariant space is a different object.

Fixing the unramified central character fixes $\alpha\beta$. Thus the two inducing parameters lie on the determinant relation. The calculation constructs Hecke eigenvalues on invariant vectors. A comparison with a specified Galois or coherent-sheaf category still requires a functor and its action compatibility.

Corollary 70.3.4 *(A unit-root projector on invariant vectors).* Suppose E is a finite extension of $\mathbb{Q}_p$, with ring of integers $\mathcal{O}_E$. Assume $\alpha \in \mathcal{O}_E^\times$ and $\beta \in \mathfrak{m}_E \setminus \{0\}$. Then P_α is the projector onto the unique unit-eigenvalue line. For any U-stable $\mathcal{O}_E$-lattice Λ in I^J, P_α preserves Λ, and

$$\lim_{n\to\infty} U^{n!} = P_\alpha$$

in $\mathrm{End}_E(I^J)$ with its p-adic topology.

Proof. The difference $\alpha - \beta$ is a unit, so the projector is a polynomial in U with coefficients in $\mathcal{O}_E$. The spectral decomposition gives $U^{n!} = \alpha^{n!}P_\alpha + \beta^{n!}P_\beta$. For each integer m, the image of α in $(\mathcal{O}_E/\mathfrak{m}_E^m)^\times$ has finite order. That order divides $n!$ for all sufficiently large n. Thus $\alpha^{n!} \to 1$, while $\beta^{n!} \to 0$. □

This is a projector on a specified invariant space. It does not construct a family of automorphic forms. If both eigenvalues are units, the entire Iwahori invariant space is the unit-root part. In particular, the two unit roots of the unramified integral deformation above do not determine a unique ordinary line by their valuations.

70.4 An Enhanced Generator Criterion

Equality of scalar eigenvalues does not determine the image of a functor. For a field E, let $A = E[t]$ and consider

$$F : D(A) \longrightarrow D(A) \times D(A), \qquad M \longmapsto (M, 0).$$

This functor is exact, preserves colimits, and is fully faithful. The diagonal central A-action on the target is faithful. The object $F(A)$ cannot generate $(0, A)$, since projection to the second factor annihilates its localizing closure.

Theorem 70.4.1 *(Comparison on compact generators).* Let $\mathcal{C}, \mathcal{D}$ be presentable stable ∞-categories, and let $F : \mathcal{C} \to \mathcal{D}$ be exact and colimit-preserving. Let $\{P_i\}_{i\in I}$ be a set of compact generators of $\mathcal{C}$. Here generation means that their smallest full stable subcategory closed under all small colimits is $\mathcal{C}$. Assume each $F(P_i)$ is compact and that the maps of mapping spectra

$$\mathrm{Map}_{\mathcal{C}}(P_i, P_j) \longrightarrow \mathrm{Map}_{\mathcal{D}}(F(P_i), F(P_j))$$

are equivalences for all i, j. Then F is fully faithful. It is an equivalence if the objects $F(P_i)$ also generate $\mathcal{D}$.

Proof. Fix i. Compactness and exactness imply that both mapping-spectrum functors in the comparison, with second variable X, preserve coproducts and finite colimits. They therefore preserve all colimits between stable categories. The objects X where the comparison is an equivalence form a localizing subcategory of C. It contains every P_j, so it is all of C.

Fix X and vary the first argument. Mapping spectra convert colimits in this argument into limits. Hence the objects Y for which

$$\mathrm{Map}_C(Y, X) \longrightarrow \mathrm{Map}_{\mathcal{D}}(F(Y), F(X))$$

is an equivalence form a localizing subcategory containing all P_i. This proves full faithfulness. Its essential image is closed under colimits and finite cones, since diagrams in that image lift through a fully faithful functor. If the image contains generators of $\mathcal{D}$, it is all of $\mathcal{D}$. □

For an E-linear differential graded enhancement, the hypothesis is a quasi-isomorphism on each derived Hom complex induced by the chosen functor. An equality of dimensions or a match in degree one is weaker. Use the augmentation modules $k = k[t]/(t)$ and $k = (k[\epsilon]/(\epsilon^2))/(\epsilon)$. Then k has $\mathrm{Ext}^1(k, k) = k$ over both $k[t]$ and $k[\epsilon]/(\epsilon^2)$. Over $k[t]$, its length-one free resolution makes all higher Ext groups zero. Over the dual numbers, the periodic resolution with differential multiplication by ϵ gives $\mathrm{Ext}^n(k, k) = k$ for every $n \geq 0$.

The local categorical Langlands question requires the actual source subcategory, spectral support condition, comparison functor, and objects P_i. The Frobenius and invariant-vector constructions above supply parameters and operators for this question. A categorical equivalence follows only after the derived Hom and generation hypotheses have been established on those categories.

The arithmetic comparison attached to these parameters

Remark 70.4.2 *(unramified geometry and the Emerton–Gee comparison).* The framed fixed-determinant unramified ring has relative dimension three; its distinct-root strict-conjugacy class ring has relative dimension one. The eigenvalue stack retains its eigenline automorphisms. Propositions 70.3.1 and 70.3.2 prove these statements. They do not identify an entire Emerton–Gee component with a smooth formal scheme in two independent eigenvalues. The intended arithmetic comparison must construct its map to the determinant-identified stack, and identify the chosen ordinary support and Breuil–Mézard cycles. In particular, equations of the proposed form $\alpha_p \beta_p^{-1} = \sigma$ require a specified weight-to-character map and an actual cycle construction; they do not follow from smoothness.

Remark 70.4.3 *(unramified Hecke and local-global compatibility).* The spherical generators T_p, $S_p^{\pm 1}$ act on the invariant line by $q^{1/2}(\alpha_p + \beta_p)$, $\alpha_p \beta_p$ in the stated normalized induction convention. Dividing T_p by $q^{1/2}$ changes its eigenvalue normalization. The local-global comparison asks for the actual Hecke-equivariant arithmetic functor and a match with the corresponding Frobenius polynomial. Faithfulness and finite multiplicities on a universal ordinary category are additional assertions, not consequences of this invariant-vector calculation. Clozel–Harris–Taylor, Harris–Taylor, Carayol and Saito concern automorphy lifting or local-global compatibility under their respective coefficient, ramification and modularity hypotheses. In particular, an unramified place different from the coefficient prime is distinct from a crystalline comparison at that prime.

Remark 70.4.4 *(ordinary families and generators).* Corollary 70.3.4 constructs the unit-root projector on the Iwahori invariant space under its valuation hypotheses. The infinite-dimensional principal series does not split into two one-dimensional representations. Hida's ordinary-family problem additionally

asks for an automorphic family and its deformation parameters, and the categorical problem asks for a compact generator and its actual image under the restricted DEG functor. Neither existence nor uniqueness of such a family follows from the local projector. The ordinary-family references are Hida, Ann. Inst. Fourier 36 (1986), and Ann. Math. 128 (1988).

The conditional ordinary comparison

Proof of Theorem 70.2.4. First specify the restricted source and target and the factorization of the fully faithful DEG functor through them. Full faithfulness then passes to these full subcategories. Second, construct the proposed Breuil–Mézard spectral objects, their support and target generation; containment of the functor's image in their colimit closure does not reverse to essential surjectivity. Third, identify the images of the ordinary source objects with those spectral objects by natural maps compatible with Hecke actions and all derived morphisms. Equal Hecke eigenvalues alone do not give these maps, as the product-category example preceding Theorem 70.4.1 shows. Fourth, under the stated essential-image hypothesis, the image is a full subcategory closed under colimits and finite cones and contains a generating family of the target. It is therefore the entire target. These steps prove the conditional implication. The second and third steps remain arithmetic construction requirements, including Hida-family realization and the proposed Colmez image comparison. □

Precise Ordinary Range

Corollary 70.4.5 *(conditional ordinary Fi.a.1 range).* For every $p \geq 5$ and residual representation satisfying (TW1)–(TW4), the displayed ordinary restriction is an equivalence provided its enhanced source, support target, restriction functor and essential-image hypothesis in Theorem 70.2.4 are supplied.

Proof. Apply that theorem with precisely those data. The invariant-vector projector and the unramified parameter space supply no additional essential-image conclusion. □

Conjectural Extension

Conjecture 70.4.6 *(Fi.a.1 full-range extension).* The essential surjectivity of $\Phi^{\mathrm{DEG}}_{p,\chi}$ extends from the unramified ordinary summand to the full block $\mathrm{Block}_{\bar{\rho}|_{G_{\mathbb{Q}_p}}}$ and the full non-ordinary locus, closing Conjecture 61.4.5 at every $p \geq 5$, χ, $\bar{\rho}$.

Remark 70.4.7 *(the non-ordinary comparison).* The intended non-ordinary and supersingular comparison requires its actual cycle scheme structures, projective source objects and spectral images. Nonreduced cycle multiplicities do not by themselves supply these objects. Nor does trace zero force a singular framed unramified ring: Proposition 70.3.2 is formally smooth for every invertible residual Frobenius matrix. Possible routes remain a Koszul-duality comparison of projective generators with specified coherent cycle objects, or a proved block exhaustion by such images. Each requires full derived morphisms and target generation. The intended target must be a specified smaller category: the DEG functor on its entire spectral target is not an equivalence.

Ordinary comparison conventions

The Hodge–Tate sign must be fixed before comparing Kisin, Gee–Kisin and Berger–Colmez (2008), *Families of de Rham representations*. In filtration-degree convention the cyclotomic character has weight −1; dualizing changes the signs. Ordinary-line and ordinary-quotient conditions require the explicit flag comparison of Chapter 86. The categorical ambient must be stable and presentable and the restricted functor must preserve colimits. Big image and local distinguishedness are separate hypotheses. The local

Hecke action is computed above; compatibility with a global automorphic/Galois family is an additional natural transformation. None of these hypotheses is removed by the ordinary parameter calculation.

70.5 Fi.a.2: Extension to $p = 2, 3$ via Block-Diagonal Local Comparison

Strategy

At $p = 2, 3$, a categorical comparison must identify the actual block category and all its extensions. Paškūnas and Colmez–Dospinescu–Paškūnas are the representation-theoretic references for the intended supersingular, principal-series and Steinberg comparisons. The specific block data required here are stated separately below; no uniform semisimplicity assertion is used.

The block-diagonal comparison requires an actual functor on each block, including all derived extension data. The following questions retain the specific small-prime cases; no such functor is inferred from a block list.

Block Structure at $p = 2$

Conjecture 70.5.1 *(the small-prime block data at $p = 2$).* Fix a finite coefficient field k of characteristic two and a full central character $\chi : \mathbb{Q}_2^\times \to k^\times$. The intended decomposition $\mathrm{Mod}^{\mathrm{sm},\chi} = \prod_{\mathfrak{B}} \mathrm{Block}_{\mathfrak{B}}$ requires the precise locally admissible or pseudocompact carrier. The separate cases to identify are supersingular blocks for irreducible residual $\bar{\rho}$, principal-series blocks for character pairs with fixed product, and twisted Steinberg blocks. Determine the simple objects, all projective covers and their endomorphism algebras, including the relevant extension classes such as the proposed $\mathrm{Ext}^1(\mathbf{1}, \mathbf{1}_{\mathrm{sgn}})$. The number of projective indecomposables, their Cohen–Macaulay or Noetherian properties, and any claimed one-dimensional extension space require proofs in that carrier. A single simple object does not make its block semisimple.

The block references are Paškūnas (2014) and Colmez–Dospinescu–Paškūnas (2014); their precise coefficient and categorical hypotheses must be supplied for this comparison.

Block Structure at $p = 3$

Conjecture 70.5.2 *(the small-prime block data at $p = 3$).* Fix a finite field k of characteristic three and a full central character $\chi : \mathbb{Q}_3^\times \to k^\times$. The intended product of blocks must be constructed in a specified coefficient and representation category. It must identify the principal-series, twisted-Steinberg and supersingular cases, their projective covers and complete derived endomorphism algebras. In particular, a proposed quiver $L_0 \leftarrow L_1 \leftarrow L_2$ requires relations and a proof that it describes the actual block; its shape does not establish that description. The proposed (φ, Γ) comparison, its conductor range ≤ 2 and the extension groups must be identified in the same conventions. A supersingular block is not semisimple merely because it has one simple.

The block references are Paškūnas (2014) and Colmez–Dospinescu–Paškūnas (2014); their precise coefficient and categorical hypotheses must be supplied for this comparison.

Block-Diagonal Local Comparison Functor

Definition 70.5.3 *(data for the block-diagonal local comparison).* For $p \in \{2, 3\}$ and a full central character χ, a block-diagonal comparison consists of specified enhanced source blocks $\mathcal{B}_{\mathfrak{B}}$, spectral support

categories $\mathcal{S}_{\mathfrak{B}}$ and exact colimit-preserving functors $\Phi^{\mathrm{BD}}_{p,\chi,\mathfrak{B}} : \mathcal{B}_{\mathfrak{B}} \to \mathcal{S}_{\mathfrak{B}}$. Each functor requires a construction, for example a specified kernel or bimodule with its action. One must supply the compatible block assembly to obtain

$$\Phi^{\mathrm{BD}}_{p,\chi} : \mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi \longrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2}).$$

The notation does not construct a functor from an equality of $\mathrm{Ext}^{\mathrm{I}}$ dimensions. The Colmez Montreal functor and (φ, Γ) modules motivate the intended arithmetic maps; their coefficient-compatible enhanced realization is part of the data.

Theorem 70.5.4 *(conditional block-diagonal local comparison).* Assume the comparison data of Definition 70.5.3. On each block choose compact generators P_i with compact images and assume that the functor induces equivalences on every mapping spectrum between them. Then:

(i) on any block whose images also generate the spectral target, the functor is an equivalence; this includes a semisimple block if semisimplicity and generation have separately been established, and asks for these constructions on the intended supercuspidal and supersingular blocks;

(ii) on every other block, including a non-semisimple block, it is fully faithful;

(iii) the induced derived Hom comparison identifies the actual $\mathrm{Ext}^{\mathrm{I}}$ classes and all higher extension operations. Identifying a particular tangent summand with k at $p = 2$ or k^2 at $p = 3$ requires the actual block and deformation-complex calculations; those dimensions are not hypotheses supplied by the block names.

Proof. Theorem 70.4.1 gives full faithfulness, and gives an equivalence when the images generate. Applying homotopy groups to the mapping-spectrum equivalences gives all extension-group identifications. The actual functors induce composition, so this is stronger than an equality of dimensions. The polynomial-ring and dual-number example following that theorem shows why a degree-one dimension match is insufficient. □

Corollary 70.5.5 *(the semisimple-block comparison under its full hypotheses).* For $p = 2, 3$, the restriction to a specified collection of semisimple blocks is an equivalence if the block functors, mapping-spectrum comparison, image generation and compatible assembly above are supplied. No assertion that every supercuspidal or supersingular block is semisimple is used.

Proof. Apply item (i) to those blocks and the assumed assembly equivalence. □

Remark 70.5.6 *(the arithmetic class comparison).* The wild class comparison remains the intended arithmetic map to degree-one deformation data in Conjecture 68.7.3. It requires the full central character, the actual projective pro-generator, its derived endomorphism algebra, a spectral support condition and an enhanced functor. Neither a class match nor a dimension match alone establishes essential surjectivity. The local Hecke actions must intertwine with this functor, and the block maps must assemble in the chosen presentable coefficient category. Colmez, Paškūnas, Colmez–Dospinescu–Paškūnas and Emerton–Gee retain their respective representation and deformation roles; compatibility of Hodge shifts and extension conventions must be expressed by these maps.

70.6 L_{ACD}: Cuspidal Adelic Categorical Descent

Cuspidal Locus

Flath's 1979 theorem concerns cuspidal automorphic representations of GL_2: each such representation decomposes as a restricted tensor product of local representations, and this decomposition is defined after choices of nonzero spherical reference vectors at almost every finite place. Below the cuspidal locus of the adelic quotient can be given an independent description (Bernstein 1988; Mœglin–Waldspurger 1995) which we exploit.

Definition 70.6.1 *(cuspidal sph-fin comparison category).* Fix a unitary central character $\omega : \mathbb{A}^\times/\mathbb{Q}^\times \to \mathbb{C}^\times$. The Hilbert carrier is the space of square-integrable automorphic functions modulo the centre with $f(zg) = \omega(z)f(g)$. Cuspidality means that the constant term $\int_{N(\mathbb{Q})\backslash N(\mathbb{A})} f(ng)\,dn$ vanishes for every g, where N is the upper unipotent subgroup. The intended category $\mathrm{D}^{\text{sph-fin}}([GL_2])^{\text{cusp}}$ requires a chosen stable enhancement of its smooth admissible cuspidal objects and a specified sph-finiteness condition. It is then their stable colimit closure in that enhancement. Finite-dimensional local Whittaker models are not a definition of cuspidality.

Remark 70.6.2 *(cuspidal orthogonality and enhancement).* For the fixed central-character Hilbert quotient, the cuspidal space is orthogonal to the residual and continuous spectral spaces and is preserved by adelic Hecke operators and the archimedean $(\mathfrak{g}, K_\infty)$ action. The spectral references are Jacquet (1983), SLN 1041, and Mœglin–Waldspurger (1995), Section III.2; Bernstein (1988) concerns Plancherel support. A corresponding orthogonal decomposition of enhanced categories requires vanishing of cross mapping spectra; Hilbert orthogonality alone does not provide it.

Flath's Theorem at Representation Level

Theorem 70.6.3 *(Flath 1979 restricted tensor product).* Every irreducible admissible cuspidal automorphic representation π of $GL_2(\mathbb{A})$ with central character factoring through a continuous character $\omega_\pi : \mathbb{A}^\times/\mathbb{Q}^\times \to \mathbb{C}^\times$ decomposes, after choosing nonzero spherical reference vectors at almost all finite places, as a restricted tensor product

$$\pi \;\simeq\; \bigotimes_v{}' \pi_v,$$

where π_v is an irreducible admissible representation of $GL_2(\mathbb{Q}_v)$, unramified at almost all v, with the spherical vector at each unramified place playing the role of the distinguished unit. Archimedean π_∞ is an irreducible admissible $(\mathfrak{g}, K_\infty)$-module (Jacquet–Langlands SLN 114 §5).

Representation-level input. Flath (1979), *Decomposition of representations into tensor products*, Proc. Symp. Pure Math. 33, Part 1, Section 2, gives the factorization of an actual irreducible admissible representation with its archimedean $(\mathfrak{g}, K_\infty)$ factor. The chosen reference vectors define the restricted tensor construction. This is not a canonical choice of those vectors and does not globalize arbitrary local tuples. □

Categorical Lift on the Cuspidal Locus

Theorem 70.6.4 *(conditional cuspidal adelic categorical descent).* Suppose the intended globalizable restricted-tensor source $\otimes_v'^{,\text{glob}} C_v$ and the target $\mathrm{D}^{\text{sph-fin}}([GL_2])^{\text{cusp}}$ are specified compactly generated stable presentable categories. The source must encode actual cuspidal globalization and exact

triangles, not only a list of globalizable irreducibles. Suppose an exact colimit-preserving globalization functor

$$F : \bigotimes_{v}^{\prime,\mathrm{glob}} \mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)]) \longrightarrow \mathrm{D}^{\text{sph-fin}}([GL_2])^{\mathrm{cusp}}$$

is constructed, sends chosen compact generators to compact objects, induces equivalences on their full mapping spectra, and has generating image. Then F is an equivalence. To obtain the structured comparison, also require natural intertwining isomorphisms for:

(i) finite-place Whittaker functors with their chosen additive characters;

(ii) Bernstein-centre and Hecke actions, compatible across places;

(iii) the archimedean $(\mathfrak{g}, K_\infty)$ action.

Under these extra hypotheses the equivalence has those compatibilities.

Proof. The five parts of the construction have distinct roles. First, the pointed restricted tensor product requires the actual local categories and reference insertions of the restricted-tensor construction in Chapter 50, together with its archimedean and sph-finite hypotheses. Second, Flath factorizes each actual cuspidal representation. It gives neither an enhanced source category nor a map on K_0 until compatibility with exact triangles has been proved. Third, the assumed enhanced globalization supplies that functor; a possible construction uses a specified restriction adjunction whose unit and counit are equivalences on generators. Fourth, Theorem 70.4.1 applies to the supplied mapping-spectrum and generation data and proves that F is an equivalence. Fifth, the three specified natural transformations express its Whittaker, centre and archimedean compatibility. They extend from generators when their functors preserve the relevant colimits. This proves precisely the stated conditional comparison, without inferring those transformations from scalar eigenvalues or from representation-level factorization. □

The arithmetic construction still asks for the globalizable enhancement, its projective or compact generators, globalization and its adjunction, and the actual Whittaker and centre maps. The relevant representation references remain Bernstein–Zelevinsky (1976), Jacquet–Langlands, SLN 114, Sections 3, 5 and 11, Bernstein (1988), Wallach (1988), and Mœglin–Waldspurger (1995), Section III.3. Neither a product of local scalar centres nor Hilbert spectral orthogonality supplies this categorical equivalence.

Corollary 70.6.5 *(conditional F1 on the cuspidal locus).* Assume the full local ordinary and non-ordinary comparisons, the small-prime block functors with their complete derived and support comparisons, and the globalization hypotheses of Theorem 70.6.4. Also supply the Selmer-derived target $\mathcal{X}^{\mathrm{Sel}}_{\hat{\rho},\delta,\mathcal{L}}$ and a compatible spectral assembly equivalence for the chosen cuspidal subcategory. If these maps are the maps in the F1 reduction, their composition gives F1 on that subcategory.

Proof. Compose the stipulated local, globalization and spectral equivalences. Their compatibility identifies the composite with the intended F1 map. The reduction statements in Corollary 43.2.3 and Theorem 43.2.1 require these same maps. Residual modularity does not construct the Selmer stack, its support, or this spectral assembly. □

Eisenstein / Continuous-Spectrum Extension as a Conditional Lemma

Conjecture 70.6.6 *($\mathrm{L}_{\mathrm{ACD}}$ Eisenstein extension).* The restricted tensor product extends from the cuspidal sph-fin locus to the full sph-fin locus

$$\bigotimes_v{}' \mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)]) \xrightarrow{\sim} \mathrm{D}^{\text{sph-fin}}([GL_2])$$

by incorporating the Eisenstein (residual and continuous) spectrum through intertwining operators (Langlands 1976 *On the functional equations satisfied by Eisenstein series*, SLN 544, Part I §5–7).

Remark 70.6.7 *(Eisenstein extension).* The Eisenstein extension requires:

- Explicit description of Eisenstein-series L^2-summands in the adelic automorphic category (Moœglin–Waldspurger 1995 §II; not in general a restricted tensor product of local factors, since intertwining operators mix primes).
- Compatibility of intertwining operators with local Hecke actions (Shahidi 1990 Duke Math. J. 62; Shahidi 2010 AMS Coll. Pub. §3–4).
- Paley–Wiener-type synthesis across the continuous-spectrum parameter (Wallach, *Real Reductive Groups II* (1992), Chapter 13, especially Section 13.4, for Plancherel; Harish-Chandra, *Harmonic analysis on real reductive groups III*, Ann. Math. 104 (1976), 117–201, for the Plancherel theorem).

These hypotheses concern the non-cuspidal spectrum. The cuspidal sph-fin locus is treated by Theorem 70.6.4; the Eisenstein extension is Conjecture 70.6.6.

Cuspidal and Eisenstein comparison data

The adelic, smooth representation and Hilbert spectral conventions must be related by actual functors. Strong multiplicity one concerns irreducible automorphic representations; it does not identify derived categories. The central character factors through $\mathbb{A}^\times/\mathbb{Q}^\times$. Spherical reference vectors and archimedean admissible modules belong to the chosen restricted tensor diagram. A globalization adjunction, if constructed, must intertwine Whittaker, Bernstein-centre and cross-prime Hecke actions as specified above. The cuspidal comparison is conditional on these data. The Eisenstein extension additionally requires Langlands' intertwiners, Shahidi's local normalizations, global character constraints and Wallach/Harish-Chandra spectral synthesis on their stated domains.

70.7 Boundary of the F1 Comparison

Conjecture 70.7.1 *(the full F1 assembly comparison).* The ordinary local category, the small-prime block categories and the global cuspidal category must be placed in an actual common comparison diagram; they are not three orthogonal summands of a single known category. The intended full F1 construction must include:

(i) the ordinary comparison and essential surjectivity at $p \geq 5$ on its non-ordinary target (Conjecture 70.4.6);
(ii) the local restrictions relevant to removing the global Taylor–Wiles hypotheses;
(iii) the small-prime functors, full derived Hom and support comparison, including the wild class map (Conjecture 68.7.3);
(iv) cuspidal globalization and Eisenstein/continuous-spectrum descent (Conjecture 70.6.6);
(v) the actual Bernstein-centre and geometric gluing maps of Chapter 64.

Proving that these data exhaust the full comparison requires coverage, overlap maps and extension-preserving assembly. If they form compatible equivalences of a specified gluing diagram, the induced equivalence of its limit or colimit follows by its universal property. Constructing that diagram is part of the arithmetic problem.

Chapter 71

Fi.A.3 (A+B): Block-Additive Hochschild and Paskunas Ext^1-Vanishing

A block decomposition separates two arithmetic comparison problems: identifying each block with its spectral category, and transporting extension classes through that identification. Hochschild additivity uses the specified orthogonal idempotents. The per-block functors require their own coefficient, central-action and trace compatibility.

For finite modules over a Noetherian domain, localization at the fraction field kills positive extension groups. An arithmetic generic fibre has this property only after the corresponding category equivalence and localization comparison are supplied. At wild primes, a quotient of a regular local ring may have a nonreduced generic fibre with nonzero self-extensions. The reducedness hypothesis and explicit counterexample below isolate this obstruction. Ordinary-block deformation and the very-special multiplicity comparison retain their separate arithmetic hypotheses.

71.1 Block Decomposition Recall (Bernstein and BH)

Characteristic-Zero Bernstein Decomposition

Theorem 71.1.1 *(Bernstein 1984; Bernstein–Deligne 1984).* Let G be a connected reductive p-adic group and fix an algebraically closed coefficient field k of characteristic zero. The category $\mathrm{Mod}_G^{\mathrm{sm}}$ of smooth G-representations with coefficients in k decomposes as a direct product over a canonical set of blocks:

$$\mathrm{Mod}_G^{\mathrm{sm}} = \prod_{\mathfrak{s} \in \mathfrak{B}(G)} \mathrm{Block}_{\mathfrak{s}},$$

where $\mathfrak{B}(G)$ is the set of inertial equivalence classes of supercuspidal supports. The decomposition is realised by a central idempotent decomposition of the Bernstein centre $Z(\mathrm{Mod}_G^{\mathrm{sm}}) = \prod_{\mathfrak{s}} Z_{\mathfrak{s}}$ into a product of integral domains.

Source. Bernstein 1984, "Le centre de Bernstein", in Deligne's notes; Bernstein–Deligne 1984 (collected in *Travaux de Bernstein*). For the block decomposition and torsor of unramified twists modulo self-twists, see also Helm, arXiv:1201.1874, Theorems 3.4–3.5, pp. 7–8. □

Proposition 71.1.2 *(central-character fibres of Bernstein blocks).* Let k be algebraically closed of characteristic zero, and let $\chi : Z(G) \to k^\times$ be a smooth central character. For a Bernstein block $\mathcal{B}_{\mathfrak{s}}$, put

$$\mathcal{B}_{\mathfrak{s}}^{\chi} = \{M \in \mathcal{B}_{\mathfrak{s}} : z|_M = \chi(z)\,\mathrm{id}_M \text{ for every } z \in Z(G)\}.$$

Then

$$\mathrm{Rep}_k^{\mathrm{sm},\chi}(G) \simeq \prod_{\mathfrak{s}} \mathcal{B}_{\mathfrak{s}}^{\chi},$$

with zero factors omitted.

Let $\delta_z \in Z_{\mathfrak{s}}$ denote the central operation supplied by $z \in Z(G)$, and let

$$I_{\mathfrak{s},\chi} = (\delta_z - \chi(z) : z \in Z(G)) \subseteq Z_{\mathfrak{s}}.$$

The ring $Z_{\mathfrak{s}}/I_{\mathfrak{s},\chi}$ acts on $\mathcal{B}_{\mathfrak{s}}^{\chi}$. This is a fibre condition inside the block.

Proof. Each Bernstein projector commutes with the action of $Z(G)$. Thus every component of an object with central character χ again has that central character. Conversely, a family of block objects reconstructs a smooth representation by its direct sum, and the sum satisfies the same scalar equations. This proves the categorical decomposition.

The generators of $I_{\mathfrak{s},\chi}$ act as zero on the indicated subcategory, so the central action factors through the quotient. No claim that taking the categorical centre commutes with this quotient is needed. □

For $G = GL_2(F)$, fix a uniformizer ϖ and normalized parabolic induction. In the unramified principal-series block write $a = \eta_1(\varpi)$, $b = \eta_2(\varpi)$ for its two unramified inducing characters, so $t = a + b$ and $d = ab$. The modulus character in normalized induction is trivial on scalar matrices. Consequently the central operation of $u \in O_F^{\times}$ is 1, and that of ϖ is d, in the ring $k[t, d, d^{-1}]$. These equalities hold as central operations: their values on all supercuspidal supports are respectively 1 and ab, and the block ring is reduced. If χ is unramified, put $c = \chi(\varpi) \in k^{\times}$. Then $I_{\mathfrak{s},\chi} = (d - c)$ and the fibre has the quotient scalar action $k[t, d, d^{-1}]/(d - c)$. It is nonzero, for instance on any irreducible constituent of the normalized induction of the unramified characters with values 1, c at ϖ. If χ is ramified, choose $u \in O_F^{\times}$ with $\chi(u) \neq 1$. The relation $1 - \chi(u)$ is an invertible scalar, so the quotient and the fixed-character fibre category are zero. Thus fixing central character neither selects a whole Bernstein block nor gives a nonzero fibre in every block.

BH Block Specialisation at $G = GL_2$

Proposition 71.1.3 *(types and fixed central character).* Suppose a type τ realizes a characteristic-zero Bernstein block $\mathcal{B}_{\mathfrak{s}}$ of $GL_2(F)$ through a Morita equivalence with its Hecke algebra H_τ. Then $\mathcal{B}_{\mathfrak{s}}^{\chi}$ corresponds to the category of modules over

$$H_\tau/(\delta_z - \chi(z) : z \in F^{\times}).$$

Proof. The Morita equivalence identifies central natural operations. The equations defining central character χ therefore become the displayed relations on H_τ-modules. □

The chosen-type hypothesis is the type-to-block comparison relevant to Bushnell–Henniart, *The Local Langlands Conjecture for $GL(2)$*, Chapters 11–16. It does not assert uniqueness of a type. The small-prime questions in Chapter 68.3.1 and Chapter 68.4.1 require their own coefficient regime and the specified type realization.

Remark 71.1.4 *(coefficient field regimes).* Characteristic-zero Bernstein theory, integral or mod-ℓ theory with $\ell \neq p$, and mod-p or p-adic block theory are distinct coefficient regimes. The characteristic-zero result above does not assert its integral-domain description for arbitrary modular coefficients. The Paškūnas block structure in Chapter 61.4.1 concerns its specified mod-p category. It is not automatically a refinement of a characteristic-zero Bernstein decomposition. The small-prime residuals of Chapter 68 retain their residual-characteristic assumptions. Comparing these regimes requires coefficient-change functors and a proof of their compatibility with the selected blocks and central actions.

71.2 Fi.a.3.A: Block-Additive BH Packaging

Block-Additivity of Hochschild

Lemma 71.2.1 *(Cyclic chains of orthogonal dg blocks).* Let k be a commutative ring and $\{\mathcal{A}_i\}_{i\in I}$ a set-indexed family of small k-linear dg categories with K-flat morphism complexes over k, for example chosen cofibrant dg models. Form the orthogonal dg union $\mathcal{A}$: its objects are the disjoint union of the objects of the $\mathcal{A}_i$, its morphism complexes within a block are unchanged, and all morphism complexes between distinct blocks are zero. Put $C = \mathrm{Perf}(\mathcal{A})$ and $C_i = \mathrm{Perf}(\mathcal{A}_i)$. Use the derived Hochschild trace, computed by the cyclic bar complex of these dg Morita models. Then

$$\mathrm{HH}_\bullet(C) \simeq \bigoplus_{i\in I} \mathrm{HH}_\bullet(C_i)$$

as S^1-equivariant complexes over k. The objects of C have finite block support. The same formula applies to a compactly generated k-linear presentable category only when its compact category is specified by this finite-support dg model and its trace is that of the compact model.

Proof. In simplicial degree n, the cyclic bar construction is the direct sum

$$\bigoplus_{X_0,\dots,X_n} \mathcal{A}(X_n, X_0) \otimes_k \mathcal{A}(X_{n-1}, X_n) \otimes_k \cdots \otimes_k \mathcal{A}(X_0, X_1),$$

with the degree-zero case given by endomorphism complexes. If the objects in a cyclic word lie in more than one block, some consecutive pair, including the closing pair, lies in distinct blocks. Its morphism complex is zero, so that summand vanishes. Every surviving cyclic word lies in exactly one $\mathcal{A}_i$. The internal differential, composition faces, unit degeneracies and cyclic rotation all preserve that block. Thus the cyclic object itself is the direct sum of the block cyclic objects. The K-flat hypothesis makes these tensor products compute the derived cyclic bar construction. Realization commutes with direct sums, and the cyclic structure induces the stated S^1-equivariant equivalence; equivalently the Hochschild differential and Connes operator preserve this decomposition.

The representable $\mathcal{A}$-modules each have support in one block. A perfect module is obtained from finitely many representables by finite sums, shifts, cones and retracts, so it has finite support and is perfect in each supported block. Conversely a finite-support family of perfect block modules is perfect over $\mathcal{A}$. This identifies $\mathrm{Perf}(\mathcal{A})$ with the finite-support sum of the $\mathrm{Perf}(\mathcal{A}_i)$. Passing to perfect modules leaves the chosen dg Morita trace unchanged, so the cyclic computation gives the displayed formula for C. The presentable version uses precisely this specified compact model; it makes no assertion that a symmetric monoidal invariant preserves arbitrary products. □

Remark 71.2.2 *(Multiplication and arithmetic realization).* The lemma supplies an S^1-equivariant complex, not an E_1-algebra structure. A multiplicative comparison additionally requires specified E_2-monoidal enhancements inducing E_1 trace products, and comparison maps preserving those operations and units. For infinitely many nonzero unital blocks, the direct sum with coordinatewise multiplication has no unit: an element has finite support and cannot act as the identity on an omitted nonzero block. Arithmetic use of the lemma requires actual k-linear dg models, the stated compact or finite-support identification, and compatible block-index and coefficient maps. A characteristic-zero Bernstein decomposition alone does not supply these data for a residual-characteristic spectral category.

Block-Wise Comparison

The index and coefficient category must be specified before applying a block comparison. In characteristic zero put

$$\mathfrak{B}(G)_\chi = \{\mathfrak{s} \in \mathfrak{B}(G) : \mathcal{B}_\mathfrak{s}^\chi \neq 0\},$$

where $\mathcal{B}_\mathfrak{s}^\chi$ is the fibre category defined above. It is not the set of entire blocks on which a single central character holds. For an integral, mod-p or p-adic application, choose instead a coefficient ring Λ, a smooth character $\chi : Z(G) \to \Lambda^\times$, and the intended representation category $\mathcal{R}_{p,\chi}^\Lambda$. Assume a specified orthogonal decomposition into nonzero block categories $\mathcal{B}_i^{\Lambda,\chi}$, indexed by $I_{p,\chi}^\Lambda$. Write $\mathrm{Block}_i = \mathcal{B}_i^{\Lambda,\chi}$ in this chosen regime. No identification of this index with $\mathfrak{B}(G)_\chi$ is implicit.

In the following conditional comparisons, $\mathrm{D}^{\text{sph-fin}}([G])_{\chi,i}$ denotes the chosen stable realization of $\mathcal{B}_i^{\Lambda,\chi}$ with its stated spherical-finiteness condition. This notation requires the realization and its block projectors as data. The spectral category must be defined over the same coefficient ring, or connected to it by a specified coefficient-change functor. A comparison across coefficient regimes requires that functor, a map of block indices and compatibility with $z = \chi(z)$, fixed determinant, and the central actions. Characteristic-zero Bernstein decomposition does not provide these data for the wild-EG category.

Theorem 71.2.3 *(block-wise BH type-block comparison).* Under Conjecture 68.7.1 applied block-by-block and Conjecture 68.7.5, the following equivalence of S^1-equivariant Hochschild complexes holds for each $i \in I_{p,\chi}^\Lambda$, provided the single-block and arithmetic Hochschild comparisons below are supplied in this same coefficient category, with the stated central-character and fixed-determinant compatibilities and the dg Morita trace models of Lemma 71.2.1:

$$\mathrm{HH}_\bullet\big(\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi,i}\big) \;\simeq\; \mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi,i})\big),$$

where $\mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi,i}$ is the component of the wild-enhanced Emerton–Gee stack assigned to the chosen coefficient block i by the specified type-to-component map of Proposition 68.3.3 (at $p = 2$) or Proposition 68.4.3 (at $p = 3$).

Proof. Use, as coefficient-compatible comparison hypotheses, the single-block equivalence Theorem 68.5.1 and the Ben-Zvi–Nadler arithmetic extension Theorem 49.5.5. The automorphic block decomposition is the stipulated decomposition of $\mathcal{R}_{p,\chi}^\Lambda$; in characteristic zero its fibre version follows from Proposition 71.1.2. The corresponding spectral decomposition and its index map are additional comparison data, with the fixed-determinant condition. The component description in Proposition 61.4.2 must be realized in these same coefficients. Both sides restrict to the block i inside the specified dg model. Functoriality of its cyclic bar construction gives the trace comparison. An E_1-algebra comparison requires separately supplied compatible monoidal structures and preservation of their operations. □

Corollary 71.2.4 *(Fi.a.3.A assembly).* Under the block-by-block inputs of Theorem 71.2.3, and the specified finite-support compact models of Lemma 71.2.1 on both sides, the global Hochschild trace identity

$$\mathrm{HH}_\bullet\big(\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi\big) \;\simeq\; \mathrm{HH}_\bullet\big(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi})\big)$$

holds as an S^1-equivariant equivalence of complexes, by Lemma 71.2.1. A multiplicative or unital enhancement is an additional comparison requirement, especially for an infinite block index.

Proof. Direct sum of Theorem 71.2.3 over $i \in I_{p,\chi}^\Lambda$ combined with Lemma 71.2.1 on both sides. □

Conditional Residual on Each Block

Proposition 71.2.5 *(obstruction taxonomy for F1.a.3.A).* The F1.a.3.A residual factors into two independent parts:

(i) **Block-additivity on the dg trace carrier.** The Hochschild-homology block-additivity statement Lemma 71.2.1. The cyclic-chain decomposition is proved under its stated dg hypotheses.

(ii) **Per-block identification (conditional).** For each chosen coefficient block $i \in I_{p,\chi}^{\Lambda}$, the statement that the projective indecomposable generator P_i of Block_i maps under $\Phi_{p,\chi}^{\mathrm{wild}}$ to the derived structure sheaf of the closed point $x_i \in \mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi,i}$ (Conjecture 68.7.1 restricted to block i). This object assignment must be accompanied by the coefficient-compatible functor, its comparison of endomorphism algebras and the stated central-character and fixed-determinant conditions. An object assignment alone does not give a categorical or Hochschild equivalence.

On the specified dg trace carriers, the blockwise equivalences assemble as complexes. If F1.a.3.A includes a multiplicative or unital structure, its comparison requires those additional monoidal data; the additive calculation does not construct them.

Proof. Part 1 is Lemma 71.2.1. Part 2 is the per-block restriction of Conjecture 68.7.1; per-block projective indecomposables map to per-block closed-point structure sheaves. Assembly uses the stipulated finite-support dg models in the chosen coefficient category and the matching spectral model, together with the single-block Hochschild comparisons. The characteristic-zero Bernstein theorem cannot substitute for this residual-characteristic decomposition. With these data, the specified object assignments and comparison maps assemble block by block. □

Remark 71.2.6 *(per-block is strictly narrower than global).* The per-block statement in Proposition 71.2.5(2) is strictly narrower than the global Conjecture 68.7.1: the global statement includes global compatibility. The additive trace compatibility is captured by Lemma 71.2.1 only on its specified dg models; any monoidal enhancement remains separate. Closure of per-block identification would reduce to: (i) explicit BH-type-to-closed-point match (known for principal-series types by Bushnell–Henniart 2006 §18); (ii) explicit P_i structure as a $GL_2(\mathbb{Z}_p)$-induced module on the BH-type cover; (iii) compatibility under the twisted DEG functor, with a specified coefficient-compatible functor and block-index map, an endomorphism-algebra comparison, and compatibility with central character, fixed determinant and central actions. These are additional hypotheses to Conjecture 61.5.7; its four clauses do not construct them. Each of (i), (ii), (iii) is a block-local requirement in the chosen coefficient category.

71.3 F1.a.3.B: Paskunas Block-Wise Ext¹-Vanishing

Paskunas Block Centraliser Structure

Theorem 71.3.1 *(Paskunas 2013 block centraliser structure).* Let $p \geq 5$ and fix a residual representation $\rho\colon G_{\mathbb{Q}_p} \to GL_2(\overline{\mathbb{F}}_p)$ and central character χ. The Bernstein block Block_ρ of $\mathrm{Mod}^{\mathrm{sm},\chi}_{GL_2(\mathbb{Q}_p)}$ is Morita equivalent to the category of modules over the *block centraliser algebra* $A_\rho = \mathrm{End}_{\mathrm{Block}_\rho}(P_\rho)$, where P_ρ is the projective cover of the unique simple quotient π_ρ determined by local Langlands correspondence. The algebra A_ρ is a complete Noetherian local ring satisfying:

(i) *Generic case (irreducible ρ or generic reducible ρ):* A_ρ is a regular local ring of dimension 2 or 3 depending on the residual type, with explicit generators given by pseudo-character deformations.

(ii) *Non-generic case (special reducible ρ, "ordinary block"):* A_ρ is still complete Noetherian local but no longer regular; it is a quotient of a regular local ring by an explicit ideal captured by the Paskunas "ordinary block" theorem.

Source. Paskunas 2013, "The image of Colmez's Montreal functor", Publ. IHES 118:1–191, Theorems 1.4, 1.5, and 1.6. □

Remark 71.3.2 *(location of the two cases in Paskunas 2013).* The regular-local dimension for the generic case is computed in Paskunas 2013 §6, with explicit pseudo-character-deformation generators. The non-generic (ordinary block) case is in §10. Updated extent of the ordinary block is in Paskunas 2015 (Duke Math. J. 164 (2015), 297–359; arXiv:1306.2691), "On the Breuil–Mézard conjecture".

Ext¹-Vanishing in Generic Blocks

Proposition 71.3.3 *(extension vanishing over the fraction field).* Let R be a commutative Noetherian domain, let $K = \mathrm{Frac}(R)$, and let M, N be finite R-modules. Then

$$\mathrm{Ext}^1_R(M, N) \otimes_R K \simeq \mathrm{Ext}^1_K(M \otimes_R K, N \otimes_R K) = 0.$$

In particular, the statement applies to a regular local domain. It asserts vanishing at the generic point, not at every regular point.

Proof. Noetherianity gives a free resolution of M with finitely generated terms: take a finite set of generators for each successive kernel. Localization is exact, and homomorphisms from each finite free term commute with localization. Localizing the resulting Hom complex therefore computes the displayed extension group over K. Every short exact sequence of vector spaces over K splits. □

Corollary 71.3.4 *(arithmetic transport on a specified generic fibre).* Suppose a specified arithmetic block category is exactly equivalent to finite modules over a commutative Noetherian domain R, and its specified generic-fibre category corresponds under this equivalence to finite $K = \mathrm{Frac}(R)$-modules. Then generic-fibre extensions vanish. If a spectral locus is also identified with this localization by compatible functors, the same vanishing holds in its corresponding category.

Proof. An exact equivalence identifies extension classes. Apply Proposition 71.3.3 and the stated localization comparison. Neither regularity alone nor flatness alone constructs the arithmetic or spectral equivalence. □

Non-Generic Block: Controlled Deformation

Proposition 71.3.5 *(Local self-extensions and conditional arithmetic transport).* For a commutative Noetherian local ring $(A, \mathfrak{m}, k)$, there is a natural isomorphism

$$\mathrm{Ext}^1_A(k, k) \cong \mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k).$$

For distinct simples S, S' in an arithmetic ordinary block, suppose an explicit k-linear class comparison identifies $\mathrm{Ext}^1_{\mathrm{Block}_\rho}(S, S')$ with a specified subspace D_ρ of this dual cotangent space for $A = A_\rho$. Then its dimension is $\dim_k D_\rho$. If in addition $\dim_k \mathfrak{m}_\rho/\mathfrak{m}_\rho^2 \le 3$ is established, this gives the bound $\dim_k D_\rho \le 3$. Breuil–Mezard multiplicities alone supply neither this class comparison nor the bound on the embedding dimension.

Proof. Apply $\mathrm{Hom}_A(-,k)$ to $0 \to \mathfrak{m} \to A \to k \to 0$. The restriction $\mathrm{Hom}_A(A,k) \to \mathrm{Hom}_A(\mathfrak{m},k)$ is zero, since $\mathfrak{m}k = 0$, and $\mathrm{Ext}^1_A(A,k) = 0$. Thus the connecting map identifies $\mathrm{Hom}_A(\mathfrak{m},k)$ with $\mathrm{Ext}^1_A(k,k)$. Every such homomorphism kills $\mathfrak{m}^2$, giving the displayed dual cotangent space. Concretely, a functional gives the extension obtained by pushing out $0 \to \mathfrak{m} \to A \to k \to 0$ along $\mathfrak{m} \to k$. The arithmetic conclusion follows from the separately assumed class comparison. There is no subtraction by a number of defining relations: for $A = k[\epsilon]/(\epsilon^2)$, the embedding dimension is one and $\mathrm{Ext}^1_A(k,k) = k$. This space is annihilated by $\mathfrak{m}$, and is not a nonzero free $A/\mathfrak{m}^2$-module. Every simple module over a commutative local ring is its residue field; comparison with distinct arithmetic simples therefore requires the specified additional map. □

Remark 71.3.6 *(conditional content is narrow).* The "ordinary block" covered by Paskunas 2013 Theorem 1.6 is the *only* non-generic block at $p \geq 5$. All other blocks are generic, and Proposition 71.3.3 applies after the exact generic-fibre comparisons of Corollary 71.3.4. The ordinary block itself splits further into a "very special" sub-case (where Breuil–Mezard multiplicities are conjectural but individually explicit) and a "non-very-special" sub-case. The arithmetic application of Proposition 71.3.5 still requires the explicit class comparison, independently of whether the multiplicity values are known. The "very special" sub-case is one additional named residual, strictly smaller than the whole ordinary block.

The $p \in \{2,3\}$ Extension

Conjecture 71.3.7 *(block centraliser at wild primes).* Let $p \in \{2,3\}$ and let $\tau \in \mathbf{T}^{\mathrm{BH}}_{p,\chi}$ be a BH type in the principal block (the Ext^1-carrying block of Remark 68.3.2 at $p = 2$, or one of the three types in Proposition 68.4.2 at $p = 3$). The block centraliser algebra $A_\tau = \mathrm{End}_{\mathrm{Block}^{\mathrm{BH}}_\tau}(P_\tau)$ is a complete Noetherian local ring of dimension 2 (at $p = 2$) or dimension ≤ 3 (at $p = 3$), and is a quotient of a regular local ring by an ideal to be specified in the arithmetic block. Question 54.4.9 asks for its relation to a finite quiver model; it does not determine this ideal or the regularity of the complete centraliser.

Remark 71.3.8 *(gap in the wild-prime block centraliser argument).* Conjecture 71.3.7 is the $p \in \{2,3\}$ analogue of Theorem 71.3.1. Paskunas' original paper explicitly restricts to $p \geq 5$ on account of the Colmez Montreal-functor hypothesis; the restriction is technical but has no published lift in the cited range. The expected dimensions are separate arithmetic hypotheses. The proposed local degree-one dimensions in Questions 54.4.8 and 54.4.9 determine neither Krull dimension nor regularity of the block centraliser.

Corollary 71.3.9 *(conditional generic-fibre extension vanishing).* Let $R = A_\tau$ be a reduced commutative Noetherian ring, and let $Q(R)$ be its total quotient ring. Assume the specified arithmetic generic-fibre category is equivalent, as an exact category, to $Q(R)$-modules. Then all its short exact sequences split, and its Ext^1 groups vanish.

Proof. A reduced Noetherian ring has finitely many minimal primes $\mathfrak{p}_i$, and $Q(R) \simeq \prod_i \mathrm{Frac}(R/\mathfrak{p}_i)$. Indeed, localization at all non-zero-divisors leaves a reduced Noetherian zero-dimensional ring, hence a finite product of fields. Modules over each field are vector spaces, so every short exact sequence splits. The finite product decomposition and the assumed exact equivalence preserve these splittings. □

Example 71.3.10 *(a nonreduced generic fibre).* A quotient of a regular ring need not satisfy this conclusion. Take $R = k[[t,u,\epsilon]]/(\epsilon^2)$. It is complete Noetherian local of dimension two. With $K = \mathrm{Frac}(k[[t,u]])$, localization at its minimal prime (ϵ) gives $R_{(\epsilon)} = K[\epsilon]/(\epsilon^2)$. The periodic free resolution with differential multiplication by ϵ gives $\mathrm{Ext}^1_{R_{(\epsilon)}}(K,K) = K$. Thus the quotient presentation

in Conjecture 71.3.7 does not imply generic semisimplicity. Reducedness and the arithmetic localization or Morita comparison are separate requirements. Even for a regular ring, closed-point extensions need not vanish: $\mathrm{Ext}^1_{k[[t]]}(k,k) = k$.

71.4 Combined F1.a.3 (A+B) Residual Statement

Unconditional F1.a.3.A+B Content

Theorem 71.4.1 *(unconditional content of F1.a.3.A+F1.a.3.B).* The following algebraic statements hold on their stated carriers. Arithmetic transport retains the comparison hypotheses indicated below:

(i) **Block-additivity of Hochschild homology** (Lemma 71.2.1): for any orthogonal union of the small k-linear dg blocks in that lemma, the derived trace splits as an S^1-equivariant complex. A presentable application requires the stated finite-support model of its compact objects, and an algebra comparison needs extra monoidal data.

(ii) **Residual factorisation for F1.a.3.A** (Proposition 71.2.5): the F1.a.3.A residual factors as (block-additivity on the specified dg models) plus (conditional per-block identification), with the per-block part strictly narrower than the original.

(iii) **Generic localization of finite-module extensions.** Proposition 71.3.3 proves vanishing over the fraction field of a Noetherian domain. Its arithmetic use requires the exact equivalences and localization comparison of Corollary 71.3.4.

Proof. The first two items have the hypotheses of their cited statements. The third is the localization argument using a free resolution with finitely generated terms above, followed, only under its additional hypotheses, by arithmetic transport. □

Conditional F1.a.3.A+B Residual

Proposition 71.4.2 *(remainder of the F1.a.3.A+B residual).* The proposed F1.a.3.A+B comparison requires the following conditional inputs, together with its generic-fibre equivalences:

(R.A) Per-block BH type-block comparison at each $i \in I^{\Lambda}_{p,\chi}$ in the chosen fixed-character coefficient category: the projective indecomposable P_i maps under the twisted DEG functor to the derived structure sheaf of the closed point $x_i \in \mathcal{X}^{\mathrm{EG,wild},p,\chi,i}_{GL_2}$. The functor and block-index comparison must respect coefficients, central character, fixed determinant and central actions, as in Theorem 71.2.3. Their existence is not supplied by characteristic-zero Bernstein theory.

(R.B.1) "Very special" ordinary block at $p \geq 5$: the Breuil–Mezard multiplicity computation for the "very special" sub-stratum of Paskunas' ordinary block. One substratum, one missing multiplicity.

(R.B.2) Wild-prime block centraliser (Conjecture 71.3.7): the specified wild block centraliser, reducedness and the exact generic-fibre comparison in Corollary 71.3.9.

The algebraic generic-localization result does not remove its arithmetic comparison hypotheses. These inputs are requirements for the stated factorization, not an exhaustion of all arithmetic obstructions.

Proof. Direct from the factorisations in Proposition 71.2.5 and Remark 71.3.6. Each named comparison supplies a different factor. The generic-fibre transport additionally requires its exact localization equivalence. □

Remark 71.4.3 *(obstruction factorization).* F1.a.3.A is the product of block-additivity and the per-block comparison core (R.A). F1.a.3.B is the product of generic-locus finite-module extension vanishing with its

arithmetic localization comparison, the very-special ordinary block condition (R.B.1), and the wild-prime centraliser, reducedness and comparison inputs (R.B.2). The support of (R.A), (R.B.1), and (R.B.2) is strictly smaller than the support of the two un-factored conditions.

Obstruction Decomposition

Proposition 71.4.4 *(F1.a.3 obstruction decomposition).* The F1.a.3 obstruction decomposition consists of:

(i) **(R.A)** Per-block BH type-block comparison (Proposition 71.4.2(R.A)), substantial non-formal theorem at each block; block-local.

(ii) **(R.B.1)** Very-special ordinary-block Breuil–Mezard multiplicity (Proposition 71.4.2(R.B.1)), substantial non-formal theorem.

(iii) **(R.B.2)** Wild-prime reduced centraliser and generic-fibre comparison at $p \in \{2, 3\}$ (Proposition 71.4.2(R.B.2)), substantial non-formal theorem.

(iv) **(R.C)** Wild-EG compact generation (Conjecture 68.7.5, unchanged from Chapter 68), substantial non-formal theorem.

The four supports are pairwise named: (R.A) is block-local, (R.B.1) is very-special ordinary, (R.B.2) is wild-centraliser, and (R.C) is wild-EG compact generation.

Proof. Direct tally from Proposition 71.4.2: (R.A), (R.B.1), (R.B.2), plus the original (R.C) Conjecture 68.7.5. The last term is independent of the block-additivity and Ext^1-vanishing arguments. □

71.5 Coefficient Categories and Comparison Conventions

The block comparisons involve characteristic-zero smooth representations, residual-characteristic block algebras and spectral categories. Their coefficient fields, module conventions, central characters and comparison functors must be specified separately. The following conventions retain these distinctions.

Block-Additivity: References and Conventions

Block and categorical references. Bernstein (1984) and Bernstein–Deligne (1984) describe the Bernstein centre and characteristic-zero block decomposition of $\mathrm{Mod}_G^{\mathrm{sm}}$; Bushnell–Henniart (2006) concerns GL_2 types. Paškūnas (2013, 2015) treats the specified mod-p block-centraliser setting. Ben-Zvi–Nadler, arXiv:0904.1247, concerns categorical character theory, and Dotto–Emerton–Gee, arXiv:2603.26887, supplies its stated fixed-central-character functor. Comparing these categories requires the coefficient-compatible block and functor data above; the list of references does not identify their carriers.

Sign and convention. Bernstein–Deligne use the left-regular representation of the Bernstein centre; Paskunas uses the right-regular; both yield equivalent block-projection functors under the opposite-algebra convention. Bushnell–Henniart uses Langlands-normalised central characters; the χ convention here matches. The Hochschild homology orientation follows the "co-Hochschild" convention from Lurie HA 5.5.3, consistent with Ben-Zvi–Nadler and Costello–Gwilliam.

Ambient category. The block decomposition of $\mathrm{Mod}^{\mathrm{sm},\chi}_{GL_2(\mathbb{Q}_p)}$ lives in the abelian category of smooth $GL_2(\mathbb{Q}_p)$-representations; the spectral side is $\mathrm{IndCoh}_{\mathrm{Nilp}}$ on a formal algebraic stack. Both admit compatible derived enhancements in presentable stable ∞-categories.

Trace carrier. Lemma 71.2.1 uses orthogonal k-linear dg models with derived cyclic chains and finite-support perfect modules. Realizing these models and their coefficient comparisons on an arithmetic category remains a hypothesis.

Functoriality. Block-decomposition is functorial under the specified block projectors. The cyclic-chain construction is functorial on the chosen dg models, giving an additive trace comparison when the coefficient and block maps have been supplied.

Per-Block Residual Isolation: References and Conventions

Projective covers and the Montreal functor. Paškūnas (2013) supplies the blockwise projective-cover setting, and Colmez (2010), Astérisque 330, concerns the Montreal functor between its automorphic and Galois categories. The Bernstein–Deligne and Ben-Zvi–Nadler constructions enter through the separate block and categorical comparisons. Their use together requires the specified coefficient and endomorphism-algebra maps.

Convention alignment. Paskunas' projective cover P_ρ uses the Montreal functor normalisation; this matches the P_τ of Conjecture 68.7.1 under the BH-to-residual correspondence $\tau \mapsto \rho_\tau$ of Proposition 68.3.3.

Hypothesis. The per-block hypothesis is named explicitly as Proposition 71.2.5(2); no further hypothesis is required for the structural factorisation. The residual (R.A) is the named open content.

Generic-Block Ext^1*-Vanishing: References and Conventions*

For a Noetherian domain R, a free resolution with finitely generated terms of a finite module computes its extension groups. Localizing that resolution at $K = \mathrm{Frac}(R)$ commutes with the Hom complex and its cohomology. The resulting vector-space extensions split. This is the argument of Proposition 71.3.3.

The arithmetic application requires an exact block equivalence and a compatible generic-fibre localization. A spectral application also requires the specified functor identifying the spectral locus with that localization. These are the hypotheses of Corollary 71.3.4. Regularity at a closed point does not imply extension vanishing there.

Wild-Prime Block Centraliser: References and Conventions

Wild types and block algebras. The wild-prime argument uses Bushnell–Henniart 2006 § 12–15 (wild types at $p = 2, 3$) and Auslander–Reiten–Smalø 1995 (linear-quiver block algebra theory). Paskunas 2013 does not cover $p \in \{2, 3\}$ directly; this is the source of the gap named as (R.B.2).

Convention and hypothesis. The explicit finite model in Example 54.4.4 uses right modules and arrows $L_0 \to L_1 \to L_2$. Its adjacent Yoneda product is nonzero in degree two. An arithmetic block comparison must specify its relations and preserve these products; the finite model gives no regularity theorem for a complete local centraliser. The quotient presentation and dimension assumptions in Conjecture 71.3.7 do not imply generic semisimplicity. The vanishing assertion uses the reducedness and category equivalence of Corollary 71.3.9.

Ordinary-block controlled deformation. The ordinary-block argument uses Paskunas 2013 § 10 and Paskunas 2015 (Breuil–Mezard multiplicity formula). The very-special Breuil–Mezard multiplicity is the named residual (R.B.1). Its computation does not construct the class comparison required by Proposition 71.3.5.

71.6 Paskunas Block Boundary

(i) F1.a.3.A (BH type-block comparison) factors into two parts: block-additivity of Hochschild homology (Lemma 71.2.1, on the specified dg models) plus per-block identification (Proposition 71.2.5(2), narrower residual (R.A)).

(ii) F1.a.3.B (Paskunas wild-block Ext^1) factors into three regimes: generic blocks at $p \geq 5$ where finite-module extensions vanish over the fraction field, with arithmetic transport conditional on the specified equivalence (Proposition 71.3.3); the ordinary block at $p \geq 5$ where Ext^1 is controlled by Paskunas 2015 and reduces to the very-special sub-stratum residual (R.B.1); and the wild-prime extension at $p \in \{2, 3\}$ conditional on reducedness and exact generic-fibre comparison, in addition to the centraliser datum (R.B.2).

(iii) The combined F1.a.3.A+B decomposition has four items: (R.A), (R.B.1), (R.B.2), and the unchanged (R.C) of Chapter 68. Together they form the substantive comparison problem of Proposition 71.4.4, with the hypotheses of Chapter 68. The partition is strictly finer: two original conjectures A and B become three named sub-items, each strictly narrower.

(iv) The unconditional content is: (i) block-additivity of Hochschild homology on the specified finite-support k-linear dg block models (Lemma 71.2.1); (ii) residual factorisation for F1.a.3.A (Proposition 71.2.5); (iii) finite-module extension vanishing over a fraction field, with conditional arithmetic transport (Proposition 71.3.3 and Corollary 71.3.4); (iv) the obstruction decomposition (Proposition 71.4.4).

(v) The full F1.a.3.A and F1.a.3.B statements factor through the three named residuals (R.A), (R.B.1), and (R.B.2), retaining the coefficient and comparison conventions of Section 71.5.

The finite-module localization calculation isolates an algebraic vanishing statement. The block and spectral functors needed for its arithmetic application remain separate comparison data.

71.7 References on Local Blocks and Categorical Centres

References for the block, type and local-algebra constructions:

- I. Bernstein, ''Le centre de Bernstein'', in Deligne's notes (1984).
- J. Bernstein and P. Deligne, ''Le centre de Bernstein'', in *Représentations des groupes réductifs sur un corps local* (1984).
- C. J. Bushnell and G. Henniart, *The Local Langlands Conjecture for $GL(2)$*, Grundlehren der mathematischen Wissenschaften 335, Springer (2006).
- P. Colmez, ''Représentations de $GL_2(\mathbb{Q}_p)$ et (φ, Γ)-modules'', Astérisque 330 (2010), 281–509.
- V. Paskunas, ''The image of Colmez's Montreal functor'', Publ. IHES 118 (2013), 1–191.
- V. Paskunas, ''On the Breuil–Mézard conjecture'', Duke Math. J. 164 (2015), no. 2, 297–359 (arXiv:1306.2691).
- M. Auslander, I. Reiten, and S. Smalø, *Representation Theory of Artin Algebras*, Cambridge Studies in Advanced Mathematics 36, CUP (1995).

- H. Matsumura, *Commutative Ring Theory*, Cambridge Studies in Advanced Mathematics 8, CUP (1989).
- D. Ben-Zvi, J. Francis, and D. Nadler, ''Integral transforms and Drinfeld centers in derived algebraic geometry'', arXiv:0805.0157 (2008).
- D. Ben-Zvi and D. Nadler, ''The character theory of a complex group'', arXiv:0904.1247 (2009).
- J. Lurie, *Higher Algebra*, manuscript (2017 version).

The block-additivity discussion in Lemma 71.2.1 is the explicit cyclic-chain calculation on the stated dg block models. The generic-locus argument in Proposition 71.3.3 uses a localization argument using a free resolution with finitely generated terms. Its arithmetic application requires the stated exact equivalences. The ordinary-block discussion in Proposition 71.3.5 uses Paškūnas (2013, 2015) and its Breuil–Mezard multiplicity condition. The Bushnell–Henniart and Bernstein–Deligne references concern the type and block constructions; the Ben-Zvi–Nadler reference concerns the categorical comparison. Their applications retain the coefficient regimes and comparison maps specified in the respective statements.

Chapter 72

Compact Generation of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p}_{GL_2})$ at Wild Primes via Residual Gerbes and Serre Twists

The conjecture of Chapter 68 (item F1.a.3.C) asks whether $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2})$ is compactly generated at the wild primes $p \in \{2,3\}$. The answer is not obtained here unconditionally. Emerton–Gee arXiv:1908.07185 constructs $\mathcal{X}^{\mathrm{EG},p}_{GL_2}$ as a Noetherian formal algebraic stack at every prime. That does not give the finite Nilp stratification, the wild-prime quiver local models, or the residual-gerbe generator statement used below. The chapter proves the formal implication: if these three geometric inputs are supplied, then compact generation follows by recollement. Without them, F1.a.3.C remains open at $p = 2, 3$.

72.1 Locally Noetherian Structure of $\mathcal{X}^{\mathrm{EG},p}_{GL_2}$

Setup and Conventions

Fix a prime p and a central character $\chi\colon \mathbb{Z}_p^\times \to k^\times$ with $k = \overline{\mathbb{Q}}_\ell$, $\ell \neq p$. Write $\mathcal{X} := \mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2}$ for the Emerton–Gee stack of rank-2 (φ,Γ)-modules with fixed determinant compatible with χ (Emerton–Gee arXiv:1908.07185, §3). We work with the classical non-derived-enhanced stack throughout this chapter: the derived enhancement Conjecture 61.5.7 is not invoked here. What is needed for F1.a.3.C is compact generation of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ in its classical incarnation.

Proposition 72.1.1 *(Emerton–Gee: $\mathcal{X}$ is Noetherian formal algebraic).* For every prime p (including $p \in \{2,3\}$) and every central character χ, the Emerton–Gee stack $\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2}$ is a Noetherian formal algebraic stack, locally of finite type over $\mathrm{Spf}(\mathcal{O}_K)$ for a finite extension $K/\mathbb{Q}_p$ containing a primitive p-th root of unity. Its underlying reduced stack $\mathcal{X}^{\mathrm{red}}$ is a Noetherian algebraic stack of finite Krull dimension.

Source. Emerton–Gee arXiv:1908.07185 Theorem 3.2.1 (algebraicity), Theorem 4.8.4 (locally of finite type), and Corollary 3.4.7 (Noetherian reduced fibre). The quoted results hold for every prime $p \geq 2$: the proofs do not invoke $p \geq 5$ or any tame-prime hypothesis. Emerton–Gee uses Breuil's (φ,Γ)-module classification and Kisin's deformation rings, both available at every prime. □

Remark 72.1.2 *(formal-algebraic output of Emerton–Gee at wild primes).* Emerton–Gee arXiv:1908.07185 establishes, uniformly in p, a Noetherian formal algebraic stack structure on $\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2}$. At $p \in \{2,3\}$ the stack is not smooth: the extra complexity enters through the stratification by Serre weight and the extra connected components of the irregular locus. Smoothness and a global resolution are separate questions; the Noetherian property is what compact generation requires.

Quasi-Excellence via Bhatt–Scholze Prisms

PROPOSITION 72.1.3 *(quasi-excellence of the Emerton–Gee base).* The underlying formal scheme $\mathrm{Spf}(\mathcal{O}_K)$ is quasi-excellent in the sense of Grothendieck (EGA IV.7.8). For every prime p and every closed point $x \in \mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$, the complete local ring $\widehat{\mathcal{O}}_{\mathcal{X},x}$ is a Noetherian excellent ring, of finite Krull dimension.

Proof. The base ring $\mathcal{O}_K$ is a complete discrete valuation ring of mixed characteristic; complete DVRs are excellent (Matsumura 1989 *Commutative Ring Theory* Theorem 32.4). Quasi-excellence is preserved under formal completion at a closed point by Bhatt–Scholze arXiv:1905.08229 Proposition 3.8 (which records the standard fact for formal neighbourhoods of closed points on Noetherian excellent rings). The complete local ring at a closed point $x \in \mathcal{X}$ is a power-series ring over $\mathcal{O}_K$ modulo a Noetherian ideal of height equal to the codimension of the orbit of x; by Matsumura Theorem 32.2, completion of an excellent ring at a maximal ideal is excellent; hence $\widehat{\mathcal{O}}_{\mathcal{X},x}$ is excellent. Krull dimension bounded: Emerton–Gee arXiv:1908.07185 Corollary 3.4.7 bounds $\dim \mathcal{X}^{\mathrm{red}} \leq 5$ at each closed point for GL_2. □

COROLLARY 72.1.4 *(locally Noetherian with finite-dimensional complete local rings).* $\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$ is locally Noetherian, of bounded Krull dimension at each closed point, quasi-excellent as a formal algebraic stack.

Proof. Combine Proposition 72.1.1 (Noetherian) with Proposition 72.1.3 (excellent complete local rings). □

Nilpotent Singular Support in the Arinkin–Gaitsgory Sense

Definition 72.1.5 *(*$\mathrm{IndCoh}_{\mathrm{Nilp}}$ *on* $\mathcal{X}^{\mathrm{EG}}$*).* Following Arinkin–Gaitsgory arXiv:1201.6343 and Gaitsgory–Rozenblyum DAG Vol. II Ch. 4, the ∞-category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ consists of ind-coherent sheaves on $\mathcal{X}$ whose singular support lies in the nilpotent cone $\mathrm{Nilp} \subset T^*\mathcal{X}$, where the nilpotent cone is cut out by the vanishing of the regular Galois-deformation cocycle (explicit: the image of the Galois H^1-cocycle in the symmetric algebra on the cotangent complex). On a Noetherian quasi-excellent formal algebraic stack, $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ is a well-defined presentable stable ∞-category, and embeds as a full subcategory of $\mathrm{IndCoh}(\mathcal{X})$.

72.2 NILP CONE DECOMPOSITION AT $p = 2, 3$

The Stratification by Residual Reducibility

PROPOSITION 72.2.1 *(Nilp cone stratification). Conditional statement. (conditional on a wild-prime local-model theorem for the Emerton–Gee stack).* Fix $p \in \{2, 3\}$ and a central character χ. The nilpotent cone inside the Emerton–Gee stack admits a finite stratification by locally closed Noetherian substacks:

$$\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi} \cap \mathrm{Nilp} \;=\; \mathcal{X}^{\mathrm{irr}} \;\sqcup\; \mathcal{X}^{\mathrm{red}} \;\sqcup\; \mathcal{X}^{\mathrm{wild},p},$$

where:

(i) $\mathcal{X}^{\mathrm{irr}}$ is the open substack of irreducible residual representations (smooth; tame and wild supercuspidal types live here).

(ii) $\mathcal{X}^{\mathrm{red}}$ is the closed substack of strictly reducible non-semisimple residual representations (the ordinary principal-series locus).

(iii) $\mathcal{X}^{\mathrm{wild},p}$ is the additional wild-prime substack, irreducible at $p = 2$ (the *irregular supersingular locus*; its proposed one-dimensional extension direction requires Question 54.4.8) and reducible at $p = 3$ into three sub-strata $\mathcal{X}^{\mathrm{wild},3}_{L_0}, \mathcal{X}^{\mathrm{wild},3}_{L_1}, \mathcal{X}^{\mathrm{wild},3}_{L_2}$ (with three indexing objects L_0, L_1, L_2 whose arithmetic identification is part of Question 54.4.9).

Each stratum is a locally closed Noetherian substack of $\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2}$.

Proof. Emerton–Gee gives the ambient Noetherian formal algebraic stack and the residual-representation stratification. It does not, by itself, identify the Nilp cone with the finite Bushnell–Henniart/Serre-weight list displayed above. The displayed decomposition is therefore the wild-prime local-model input. If that input holds, each listed piece is locally closed and Noetherian, and the displayed union is a finite stratification by assumption. □

Remark 72.2.2 *(relation to the Bushnell–Henniart components of Chapter 68).* The stratification (72.2.1) refines and matches the BH-component decomposition of Chapter 68: the types of Proposition 68.3.1 (unramified principal series, tame supercuspidal, wild supercuspidal) distribute across the strata $\mathcal{X}^{\mathrm{red}}, \mathcal{X}^{\mathrm{irr}}, \mathcal{X}^{\mathrm{wild},2}$ respectively. At $p = 3$ the four BH type classes of Proposition 68.4.1 distribute among $\mathcal{X}^{\mathrm{red}}, \mathcal{X}^{\mathrm{irr}}$, and the three $\mathcal{X}^{\mathrm{wild},3}_{L_i}$, each principal-block BH type corresponding to exactly one L_i-stratum. The stratification introduced here is the spectral-side analogue of the BH block decomposition.

Residual Gerbes at Closed Points of Each Stratum

Proposition 72.2.3 *(residual gerbe at a closed point).* Let $x \in \mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2}$ be a closed point. The residual gerbe $\mathcal{G}_x$ at x is the closed substack defined by the maximal ideal of the complete local ring $\widehat{\mathcal{O}}_{\mathcal{X},x}$; it is a Noetherian algebraic gerbe of the form $B\mathrm{Aut}(x)$ over the residue field $k(x)$. For $\mathcal{X}^{\mathrm{EG}}$ the automorphism group is the scheme-theoretic stabiliser of the residual Galois representation, which is an affine algebraic group of finite type over $k(x)$.

Proof. The definition of the residual gerbe is classical (Laumon–Moret-Bailly 2000 *Champs Algébriques* §11, Stacks Project Tag 06MA). For $\mathcal{X}^{\mathrm{EG}}$ the automorphism scheme at a residual Galois representation ρ is the centraliser of ρ in GL_2 over the residue field; concretely, this is either $\mathbb{G}_m \rtimes \{\pm 1\}$ or $\mathbb{G}_m \times \mathbb{G}_m$ depending on whether ρ is non-split or split, both affine of finite type over $k(x)$. Their smoothness over $k(x)$ follows from Emerton–Gee arXiv:1908.07185 §3.4 (smooth atlas structure). □

72.3 Explicit Compact Generators per Stratum

The Irreducible Stratum

Proposition 72.3.1 *(compact generator on the irreducible stratum). Conditional statement. (conditional on the stratum being QCA and on the closed-gerbe generation theorem in this setting).* On the stratum $\mathcal{X}^{\mathrm{irr}}$ the ∞-category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{irr}})$ is compactly generated by the set

$$\mathcal{G}^{\mathrm{irr}} \;=\; \Big\{\, \iota_{x,*}\mathcal{O}_{\mathcal{G}_x} \otimes \det^{\otimes n} \;:\; x \in \mathcal{X}^{\mathrm{irr}}(\overline{\mathbb{F}}_p),\; n \in \mathbb{Z} \Big\},$$

where $\iota_x : \mathcal{G}_x \hookrightarrow \mathcal{X}^{\mathrm{irr}}$ is the residual gerbe inclusion and det is the determinant Serre twist.

Proof. Under the QCA and closed-gerbe generation hypotheses, the Arinkin–Gaitsgory–Gaitsgory–Rozenblyum compact-generation theorem applies to the stratum. The determinant twists record the connected components of the fixed-determinant gerbes. No assertion is made that Emerton–Gee alone supplies these hypotheses at $p = 2, 3$. □

The Reducible Stratum

PROPOSITION 72.3.2 *(compact generator on the reducible stratum). (conditional on the two-term local model and the gerbe-plus-extension generator theorem).* On the stratum $\mathcal{X}^{\mathrm{red}}$ the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{red}})$ is compactly generated by the set

$$\mathcal{G}^{\mathrm{red}} \;=\; \left\{ \iota_{x,*}\mathcal{O}_{\mathcal{G}_x} \otimes \det^{\otimes n} \right\}_{x,n} \;\cup\; \left\{ \iota_{x,*}\mathcal{E}_x \otimes \det^{\otimes n} \right\}_{x,n},$$

where $\mathcal{E}_x$ is the non-split extension sheaf witnessing the specified one-dimensional extension class at the closed point. Its arithmetic carrier and identification are part of the local-model hypothesis, as in Question 54.4.8. In characteristic two, trivial and sign characters coincide; this notation cannot provide two distinct simples.

Proof. The proof is formal from the stated local model. A two-term cotangent model gives the indicated extension sheaf, and the bar resolution gives generation by the gerbe and this extension. The equality of dimensions does not identify the deformation class with the Bushnell–Henniart Ext-class; that identification is part of the hypothesis. □

The Wild Strata at $p = 2$ and $p = 3$

PROPOSITION 72.3.3 *(compact generator on the wild stratum at $p = 2$). (conditional on the $p = 2$ wild local model and generator theorem).* At $p = 2$, the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{wild},2})$ is compactly generated by the set

$$\mathcal{G}^{\mathrm{wild},2} \;=\; \left\{ \iota_{x,*}\mathcal{O}_{\mathcal{G}_x} \otimes \det^{\otimes n} \right\}_{x,n} \;\cup\; \left\{ \iota_{x,*}\mathcal{F}_x^{\mathrm{irreg}} \otimes \det^{\otimes n} \right\}_{x,n},$$

where $\mathcal{F}_x^{\mathrm{irreg}}$ is the structure sheaf of the irregular-supersingular cycle at x. Its identification with an arithmetic extension class is an additional part of the local-model hypothesis in Question 54.4.8; it is not a defect between distinct trivial and sign characters in characteristic two.

Proof. Under the $p = 2$ local-model hypothesis the stratum has the same two-term form as the reducible stratum, with one additional irregular cycle. The bar-resolution argument then gives the displayed generator. Bushnell–Henniart indexes types; it does not by itself prove the ind-coherent generator statement on the Emerton–Gee stack. □

PROPOSITION 72.3.4 *(compact generator on the wild strata at $p = 3$). (conditional on the $p = 3$ A_2-local model and generator theorem).* At $p = 3$, for each $i \in \{0, 1, 2\}$ the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{wild},3}_{L_i})$ is compactly generated by the set

$$\mathcal{G}^{\mathrm{wild},3}_{L_i} \;=\; \left\{ \iota_{x,*}\mathcal{O}_{\mathcal{G}_x} \otimes \det^{\otimes n} \right\}_{x,n} \;\cup\; \left\{ \iota_{x,*}\mathcal{F}_x^{A_2,i} \otimes \det^{\otimes n} \right\}_{x,n},$$

where $\mathcal{F}_x^{A_2,i}$ denotes the proposed edge sheaf in the chosen local model. The hypothesis includes the assertion that these objects generate; the shape of the degree-one quiver alone does not prove this assertion. In the right-module convention of Example 54.4.4, the adjacent classes lie in $\mathrm{Ext}^1(L_0, L_1)$ and $\mathrm{Ext}^1(L_1, L_2)$, and their Yoneda product is nonzero in degree two. Reversing the arrows reverses these ordered pairs but does not make the product vanish. The arithmetic local model must account for that product or specify a different algebra and its actual resolution.

Proof. The conclusion is the specified local generator hypothesis. It does not follow from a relation making length-two paths zero in the block algebra: Example 54.4.4 computes the nonzero degree-two Yoneda product for precisely that quotient. An arithmetic use of the proposed generating family therefore requires its generator theorem in addition to the finite quiver data. □

72.4 Compact Generation Theorem

Theorem 72.4.1 *(conditional compact generation of* $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2})$ *at wild primes). Conditional statement.* Fix $p \in \{2,3\}$ and a central character $\chi\colon \mathbb{Z}_p^\times \to k^\times$. The ∞-category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2})$ is compactly generated if the finite Nilp stratification of Proposition 72.2.1 and the per-stratum generator statements of Proposition 72.3.1, Proposition 72.3.2, Proposition 72.3.3, and Proposition 72.3.4 hold. Under these inputs, explicit compact generators are given by the union across strata:

$$\mathcal{G}^{\mathrm{full},p} \;=\; \mathcal{G}^{\mathrm{irr}} \;\cup\; \mathcal{G}^{\mathrm{red}} \;\cup\; \mathcal{G}^{\mathrm{wild},p},$$

where $\mathcal{G}^{\mathrm{wild},p}$ denotes the full wild generator set of Proposition 72.3.3 (at $p = 2$) or the union $\bigcup_i \mathcal{G}^{\mathrm{wild},3}_{L_i}$ of Proposition 72.3.4 (at $p = 3$).

Proof. Assume the stated stratification and per-stratum generator inputs. Recollement for a finite locally closed decomposition reduces compact generation of the total category to compact generation on the strata. The compact generators on the strata push forward along the closed pieces and extend along the open pieces. Induction over the finite stratification gives the displayed generator set. □

Corollary 72.4.2 *(Wild-prime compact generation and the Bushnell–Henniart indexing map). Conditional statement. (on the class-level identification of Bushnell–Henniart closed points with residual gerbes).* The compact-generation part of the conjecture Conjecture 68.7.5 (F1.a.3.C) holds at $p \in \{2,3\}$ for the classical Emerton–Gee stack if the finite Nilp stratification and per-stratum generator inputs hold: the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2})$ is compactly generated by explicit residual-gerbe plus Serre-twist generators. The Bushnell–Henniart closed points x_τ of Chapter 68 give the indexing map to codimension-zero strata; identifying these points with residual gerbes is a class-level statement, not a consequence of the equality of dimensions alone.

Proof. Conditional compact generation is Theorem 72.4.1. The identification of BH closed points with residual gerbes of each stratum requires a matching of indexing sets: BH types of Proposition 68.3.1 and Proposition 68.4.1 correspond bijectively to residual Galois representations on the strata of Proposition 72.2.1 (by BH local Langlands §18 for $GL_2(\mathbb{Q}_p)$ + Emerton–Gee §5.4 for the spectral-side decomposition). This comparison is retained as the named conditional input; dimension agreement is not used in the reverse direction. □

Remark 72.4.3 *(independence from the wild-EG enhancement conjecture).* Theorem 72.4.1 and its corollary Corollary 72.4.2 do *not* invoke Conjecture 61.5.7 (F1.a.2, the wild-EG derived enhancement). They are conditional statements about the classical Emerton–Gee stack and its $\mathrm{IndCoh}_{\mathrm{Nilp}}$. The derived enhancement would be a further input, not a consequence of the classical compact-generation implication.

72.5 Residual Obstruction Statement

Domain of the Classical Result

Proposition 72.5.1 *(domain of F1.a.3.C).* The F1.a.3.C decomposition consists of:

(i) $\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$ is locally Noetherian and quasi-excellent at every prime p (Proposition 72.1.1, Proposition 72.1.3); this is the unconditional geometric base.

(ii) The nilpotent-cone locus decomposes into a finite set of locally closed Noetherian substacks, with the irregular supersingular component at $p = 2$ and the A_2-quiver-stratified principal block at $p = 3$ (Proposition 72.2.1); this is a conditional wild-prime local-model input.

(iii) Each stratum admits explicit compact generators: residual-gerbe structure sheaves tensored with Serre twists of the determinant character, together with the single or double extension sheaves encoding the wild Ext^1-data (Proposition 72.3.1, Proposition 72.3.2, Proposition 72.3.3, Proposition 72.3.4); this is conditional on the corresponding gerbe-generation theorem.

(iv) The full category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi})$ is compactly generated at $p \in \{2, 3\}$ (Theorem 72.4.1); this is conditional on the previous two inputs.

(v) The conjecture of Chapter 68 holds in its classical-stack form (Corollary 72.4.2); this is conditional, not an unconditional closure.

Domain Boundary: Derived Enhancement

Remark 72.5.2 *(domain boundary: derived wild-EG enhancement).* Chapter 61.5.7 remains open. Compact generation on the classical stack does not imply compact generation on the conjectural derived formal stack $\mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi}$; upgrading requires F1.a.2. Chapter 68 demands the classical statement; this chapter reduces it to the finite-stratification and gerbe-generation inputs. The derived enhancement is a separate conjecture of Chapter 61.

Remark 72.5.3 *(finite Nilp stratification is an input).* The stratification of Chapter 72.2.1 is exhaustive only under the stated wild-prime local-model hypothesis: the three strata at $p = 2$ or five strata at $p = 3$ cover the full nilpotent cone. Emerton–Gee gives the ambient stack and deformation theory; it does not by itself exclude additional Nilp components at higher codimension.

Remark 72.5.4 *(geometric inputs and generator comparisons).* The comparison combines five geometric and representation-theoretic inputs. (1) Emerton–Gee arXiv:1908.07185 supplies the Noetherian formal algebraic stack. Bhatt–Scholze arXiv:1905.08229 supplies the excellent formal-neighbourhood input. Dotto–Emerton–Gee arXiv:2603.26887 is a $p \geq 5$ theorem and supplies no wild-prime closure. (2) Gaitsgory–Rozenblyum DAG Vol. II provides compact-generator transfer and recollement. (3) Arinkin–Gaitsgory arXiv:1201.6343 provides the nilpotent-singular-support framework. (4) Bushnell–Henniart 2006 provides the wild-type classification. (5) Chapter 54 provides the Ext^1-computation. The inputs use the following conventions: Emerton–Gee uses the (φ, Γ)-module conventions of Fontaine–Wintenberger; the Nilp convention follows Arinkin–Gaitsgory (singular support as a closed conic subset of the cotangent stack); the Serre twist convention of Gaitsgory–Rozenblyum DAG Vol. II Ch. 5 §5.4 matches the Emerton–Gee determinant stratification. Recollement across strata is functorial; closed immersions preserve compact generators (DAG Vol. II Ch. 4 Prop. 4.4). The hypotheses required beyond the ambient Noetherian stack are the finite wild-prime Nilp stratification and the per-stratum generator theorem. These are not supplied by Emerton–Gee alone.

72.6 References on Emerton–Gee Geometry and Compact Generation

The references distinguish the ambient geometry from the additional hypotheses on wild strata and compact generators:

Emerton–Gee arXiv:1908.07185. The Noetherian formal algebraic structure of $\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$ at every prime $p \geq 2$. Used in §72.1 (Noetherian structure). It does not prove the finite Nilp stratification used later.

Emerton–Gee arXiv:1908.07185. The moduli-of-(φ, Γ)-modules construction and tangent-complex analysis. Used as ambient deformation theory, not as an explicit wild-prime compact-generator theorem.

Dotto–Emerton–Gee arXiv:2603.26887. The tame-DEG machinery $p \geq 5$. It is evidence for the shape of the functor at tame primes. It is not an input proving compact generation at $p = 2, 3$.

Bhatt–Scholze arXiv:1905.08229. The prism theory supplying quasi-excellence of formal neighbourhoods on Noetherian excellent rings; Proposition 3.8 quoted in Chapter 72.1.3.

Gaitsgory–Rozenblyum AMS DAG Vol. II. The locally-Noetherian compact generation of IndCoh (Ch. 4 Prop. 4.4); recollement across strata (Ch. II §2); Serre-twist filling (Ch. 5 §5.4); bar-resolution compact generation on tor-amplitude-$[-1, 0]$ stacks (Ch. 8 §8.3). Used throughout §72.3 and §72.4.

Arinkin–Gaitsgory arXiv:1201.6343. Nilpotent singular support; Theorem 12.5 for compact generation of $\mathrm{IndCoh}_{\mathrm{Nilp}}$ by closed-point structure sheaves on smooth locally Noetherian stacks. Used in Chapter 72.3.1 Step 4.

Bushnell–Henniart 2006. Local Langlands and wild-type classification. It supplies representation-theoretic indexing data. It does not identify the Nilp components of the Emerton–Gee stack.

Beilinson–Bernstein–Deligne 1982. Recollement formalism; extended to DAG via Gaitsgory–Rozenblyum. Used in Chapter 72.4.1 Step 1.

Beilinson–Ginzburg 1996. Koszul duality between cotangent fibres and Yoneda Ext-algebras. It explains the expected shape of the local model. The actual identification of the wild-prime cotangent directions with the BH-block Ext^1-classes remains part of the conditional input.

Chapter 54. The finite-algebra resolutions constrain the proposed arithmetic comparisons in Questions 54.4.8 and 54.4.9. They do not identify arithmetic local models; the three-vertex example also has a nonzero degree-two Yoneda product.

The recollement implication uses the Emerton–Gee, Gaitsgory–Rozenblyum, and Arinkin–Gaitsgory constructions. The finite wild-prime local models and the generator theorem on each stratum remain separate hypotheses.

Remark 72.6.1 *(consequence for the F1.a obstruction decomposition).* Chapter 72.4.2 is conditional. The F1.a obstruction decomposition of Chapter 68 therefore retains three open items: the Bushnell–Henniart type-block comparison conjecture Chapter 68.7.1, the Paskunas wild Ext^1-compatibility conjecture Chapter 68.7.3, and the wild-prime compact-generation local model of this chapter.

Chapter 73

Pseudocharacters, Singular Support, and Restricted E_2 Tensor Products

Two-prime restriction constructs a formal deformation diagram. Its use in (BCglue.core) *requires the pairwise geometric comparison* (Ch1)*, together with a separate global centre comparison. Chapter 64 constructs the formal pullback and the equations for its equivariant comparison. The first is not a theorem of Chenevier alone: Chenevier supplies the determinant ring, while the local–global flatness, trace compatibility and equivariant coherent descent are separate Selmer-theoretic and categorical inputs. The proposed determinantal singular-support construction requires an actual geometric descent diagram and its higher-centre maps. An internal centre-gluing identity does not itself identify that centre with the global automorphic centre.*

The F2.c assertion is the restricted E_2-tensor formalism under actual operadic pointed unit hypotheses. The local Iwahori E_2-input is not part of that formal theorem.

73.1 (Ch1) Chenevier Pseudocharacter Compatibility

Chenevier Determinant-Trace Pseudocharacter Recalled

Definition 73.1.1 *(Chenevier determinant and associated trace).* Let R be a commutative ring, Γ a group and $d \geq 1$. An R-valued determinant of dimension d is a multiplicative, homogeneous polynomial law of degree d from $R[\Gamma]$ to R. Thus it specifies maps $D_A : A[\Gamma] \to A$ for every commutative R-algebra A, natural in A, with $D_A(1) = 1$, $D_A(xy) = D_A(x)D_A(y)$ and $D_A(ax) = a^d D_A(x)$. Write

$$D_{R[t]}(t - x) = \sum_{j=0}^{d} (-1)^j \Lambda_j(x) t^{d-j}.$$

The associated trace is $T = \Lambda_1$, an R-linear map; the notation (T, D) retains this dependence. The characteristic polynomial gives the Cayley–Hamilton relations in the quotient by their coefficients after polynomial base change. These relations need not hold in $R[\Gamma]$ itself. For a profinite Γ and a topological coefficient ring, continuity means that all characteristic-polynomial coefficient functions on Γ are continuous.

Proposition 73.1.2 *(determinantal deformation rings).* Let k be a finite field of characteristic p_0, put $W = W(k)$, and fix a continuous residual determinant $\bar{D}$ of dimension d of a profinite group Γ. On finite local W-algebras A with residue field k, let $\mathcal{D}_\Gamma(A)$ be the set of continuous determinants lifting $\bar{D}$. This covariant functor is prorepresented by a profinite local W-algebra R_Γ^{Ch}:

$$\mathcal{D}_\Gamma(A) = \mathrm{Hom}_{W,\mathrm{cont}}(R_\Gamma^{\mathrm{Ch}}, A).$$

Under Mazur's finiteness condition--every open subgroup has only finitely many continuous homomorphisms to $\mathbb{Z}/p_0\mathbb{Z}$-- the ring is complete Noetherian. A closed deformation condition represented

by a quotient is imposed by that quotient. The representing formal space is $\operatorname{Spf} R^{\mathrm{Ch}}_{\Gamma}$. For $\Gamma = G_{\mathbb{Q},S}$ with finite S, and for local Galois groups in characteristic zero, the finiteness condition holds. If the residual representation is absolutely irreducible and defined over k, determinant deformations agree with its representation deformations up to strict equivalence.

Source. Chenevier, arXiv:0809.0415, §1.1 and §1.10 define polynomial laws and their characteristic coefficients. Proposition 3.3, p. 42, proves prorepresentability; Example 3.6 and Proposition 3.7, p. 43, give the finiteness assertions. Example 3.4 and Theorem 2.22 give the absolutely irreducible comparison. Bellaïche–Chenevier, *Families of Galois representations and Selmer groups*, Chapter 1, provides the related deformation-theoretic setting. A chosen local condition must be shown to define the quotient used in any particular application. □

The Local–Global Comparison Datum

Conjecture 73.1.3 *(pseudocharacter trace comparison).* The prismatic/pro-*é*tale comparison on the Emerton–Gee stack identifies the trace associated to the universal determinant on the local Chenevier deformation ring with the GL_2-invariant trace of the universal (φ, Γ)-module, compatibly with the Nygaard filtration.

Fix distinct primes p, q and a coefficient prime $p_0 \neq p, q$. Choose a finite set S containing p, q, p_0, ∞, a finite field k of characteristic p_0, and a residual determinant of dimension 2 of $G_{\mathbb{Q},S}$, with absolutely irreducible residual representation. Put $W = W(k)$. Choose decomposition-group maps $G_{\mathbb{Q}_p}, G_{\mathbb{Q}_q} \to G_{\mathbb{Q},S}$ and closed local and global deformation conditions preserved by restriction. Let R^{Ch}_p, R^{Ch}_q, R^{Ch} be their complete Noetherian representing rings. Local residual restrictions need not be absolutely irreducible: we use their determinant functors.

For finite local W-algebras A, restriction is the natural map

$$\mathcal{D}_{\mathrm{glob}}(A) \longrightarrow \mathcal{D}_p(A) \times \mathcal{D}_q(A).$$

The product functor is represented by the completed tensor product

$$B_{p,q} = R^{\mathrm{Ch}}_p \widehat{\otimes}_W R^{\mathrm{Ch}}_q.$$

Here completion is for the sum of the maximal ideals; equivalently $B_{p,q}$ is the inverse limit of $(R^{\mathrm{Ch}}_p/\mathfrak{m}_p^n) \otimes_W (R^{\mathrm{Ch}}_q/\mathfrak{m}_q^n)$. It has residue field k and is complete Noetherian. For example, presenting the local rings as quotients of $W[[x_1, \dots, x_a]]$ and $W[[y_1, \dots, y_b]]$ presents $B_{p,q}$ as the quotient of $W[[x_1, \dots, x_a, y_1, \dots, y_b]]$ by the two defining ideals.

THEOREM 73.1.4 *(restriction and the pairwise deformation diagram).* Restriction of determinants induces the continuous local ring map

$$r^{\sharp}_{p,q} : B_{p,q} \longrightarrow R^{\mathrm{Ch}}, \qquad x \otimes y \longmapsto r^{\sharp}_p(x) r^{\sharp}_q(y).$$

Put $J_{p,q} = \ker r^{\sharp}_{p,q}$ and $C_{p,q} = B_{p,q}/J_{p,q}$. There are factorizations

$$B_{p,q} \twoheadrightarrow C_{p,q} \hookrightarrow R^{\mathrm{Ch}}, \qquad \operatorname{Spf} R^{\mathrm{Ch}} \longrightarrow \operatorname{Spf} C_{p,q} \hookrightarrow \operatorname{Spf} B_{p,q}.$$

For a prescribed pair of local deformations $b : B_{p,q} \to A$, the fibre of the global restriction functor is represented over A by

$$R^{\mathrm{Ch}} \widehat{\otimes}_{B_{p,q},b} A.$$

Neither surjectivity of $r^{\sharp}_{p,q}$ nor flatness nor the existence of a global lift of every $C_{p,q}$-valued pair follows from this construction.

Proof. The universal global determinant restricts to a local determinant at each place. The representing property therefore gives maps $r^{\sharp}_{p} : R^{\mathrm{Ch}}_{p} \to R^{\mathrm{Ch}}$ and $r^{\sharp}_{q} : R^{\mathrm{Ch}}_{q} \to R^{\mathrm{Ch}}$. Their images commute and their W-actions agree. The completed tensor-product property gives the displayed map: a pair of continuous local maps into any finite local A factors through finite quotients of both factors, and hence through $B_{p,q}$; uniqueness follows from its dense algebraic tensor product. Passing to inverse limits gives the same property for complete local target rings.

The kernel is closed. The quotient is complete, and it maps injectively into R^{Ch}. These rings are profinite because k is finite, so the image of $C_{p,q}$ is closed as well. This proves the two factorizations. Finally, for any finite local A-algebra A', maps from the completed tensor product to A' are precisely maps $R^{\mathrm{Ch}} \to A'$ whose restriction to $B_{p,q}$ agrees with b followed by $A \to A'$. These are exactly the global lifts of the prescribed pair after base change. □

The quotient $C_{p,q}$ records relations satisfied by restricted global determinants. It need not represent the image functor on Artinian points. For example, for a field k, the map $k[[x]] \to k[[t]]$, $x \mapsto t^2$, is injective, so its kernel quotient is all of $k[[x]]$. The local map $k[[x]] \to k[\epsilon]/(\epsilon^2)$ sending x to ϵ does not lift: a local image of t is $a\epsilon$, whose square is zero. Thus satisfying the kernel relations is weaker than being a restricted global point. In a derived deformation problem the fibre is instead a derived completed tensor product; no vanishing of its higher Tor groups is implicit.

Corollary 73.1.5 *((Ch1) as a pairwise Selmer trace and descent datum).* To use the deformation diagram in pairwise Bernstein gluing, assume the following additional data.

(i) The proposed two-prime Selmer algebra $R^{\mathrm{Ch}}_{p,q}$ of Definition 64.2.2 is supplied as a $B_{p,q}$-algebra with a specified isomorphism $R^{\mathrm{Ch}}_{p,q} \simeq C_{p,q}$. Its map to R^{Ch} is the injective map above, so the resulting ring diagram commutes. Any assertion about actual global lifts requires the separate fibre condition in Theorem 73.1.4.

(ii) The full categorical and coefficient-compatible local data (BCglue.local), the global comparison (BCglue.glue), and the required prismatic enhancements and Π-flatness of the corresponding geometric maps are supplied. These are not ordinary flatness consequences of representability.

(iii) The trace comparison of Conjecture 73.1.3 intertwines the $B_{p,q}$-actions with the actual geometric restriction $r : \mathrm{Spf}\, R^{\mathrm{Ch}} \to \mathrm{Spf}\, B_{p,q}$ and its specified spectral realization.

(iv) Ordinary equivariant coherent descent (Ch1.desc) supplies an isomorphism of coherent sheaves in degree zero

$$r^{*}(\mathcal{F}_p \boxtimes \mathcal{F}_q) \simeq \mathcal{F}^{\mathrm{prism}}_{\mathrm{glob}},$$

with the local centre actions transported through the coefficient and trace comparisons. The objects and this pullback are taken in the chosen prismatic or spectral realization, not inferred from its ring of functions.

Then the pairwise Chenevier comparison and the ordinary equivariant sheaf comparison in Equation (64.8) hold on this diagram. The derived enhancement in Equation (64.9) requires a separate equivariant derived descent datum (Ch1.desc$^{\mathbb{L}}$). Alternatively, ordinary descent implies that enhancement under both the external-product and pullback Tor-independence conditions of Proposition 64.5.4.

Proof. The theorem constructs the ring map with the variance dictated by restriction of deformations. The first hypothesis identifies its quotient with the proposed Selmer algebra. The second and third supply the geometric realization and its compatible central actions. The fourth supplies the ordinary equivariant sheaf isomorphism in Equation (64.8). The separate derived datum supplies the derived comparison directly. Under the two Tor-independence conditions, the comparison also follows by composing the natural augmentation isomorphisms in Proposition 64.5.4. Example 64.5.5 shows why ordinary descent, even with perfect local objects and flat r, is insufficient. On finite free formal charts, Proposition 64.4.2 gives the actual isomorphism equations, and Proposition 64.4.3 gives the successive lifting obstructions. Scalar functions alone do not solve those equations or their overlap conditions. □

73.2 SINGULAR SUPPORT, GLUING, AND THE ARITHMETIC COMPARISON

The arithmetic problem is to construct the spectral category with its fixed-determinant condition, then compare its central action with the automorphic one. A deformation ring supplies neither singular support nor the functors of a categorical gluing diagram. Three elementary constructions distinguish the extra data: localization retains an extension, formal patching retains a punctured neighborhood, and higher centres require maps compatible with their operations.

The arithmetic support and descent data

Definition 73.2.1 *(data for the fixed-determinant singular-support problem).* Fix $G = GL_2$, a residual representation $\bar{\rho}$, local deformation conditions $\mathcal{L}$, and a character δ lifting $\det \bar{\rho}$. Write $X = \mathcal{X}^{\mathrm{Sel}}_{\bar{\rho},\det,\mathcal{L}}$ for a proposed derived arithmetic spectral stack with determinant δ. Its construction must specify a coefficient base, topology and derived charts; the notation does not assert existence. For a quasi-smooth derived affine chart $Y = \operatorname{Spec} R$ locally of finite type over a characteristic-zero field, let L_Y be its cotangent complex and $T_Y = L_Y^\vee$. Quasi-smoothness means that L_Y is perfect of Tor-amplitude $[-1, 0]$. The scheme of singularities is

$$\operatorname{Sing}(Y) = \operatorname{Spec}_{Y^{\mathrm{cl}}} \operatorname{Sym}_{\mathcal{O}_{Y^{\mathrm{cl}}}} H^1(T_Y).$$

A support condition is a closed conical subset $\mathcal{N}_Y \subset \operatorname{Sing}(Y)$, compatible on the charts. The linear generators $H^1(T_Y)$ of the symmetric algebra have scaling weight one and act in Hochschild cohomological degree two. Equations defining $\mathcal{N}_Y$ may have higher degree. The category $\operatorname{IndCoh}_{\mathcal{N}_Y}(Y)$ consists of objects with singular support contained in $\mathcal{N}_Y$.

These definitions and actions are those of Arinkin–Gaitsgory, arXiv:1201.6343v4, Sections 2.3.3–2.3.4, pp. 20–21, and Section 4.1.2 and Definitions 4.1.4–4.1.5, pp. 33–34. In their geometric Langlands setting the nilpotent condition is imposed on a horizontal cotangent parameter, Section 11.1.1, p. 103. It is not the condition that the underlying local system have unipotent monodromy. An arithmetic analogue must construct its singularity scheme and a map defining the proposed nilpotent cone. A fixed determinant changes the deformation problem; it does not establish these derived constructions. In particular, the characteristic-zero theory cannot be applied to an integral Witt-vector deformation space without a specified comparison satisfying its geometric hypotheses.

A conical support condition is specified in $\operatorname{Sing}(X)$ and need not come from a closed subset of X. If closed and open base pieces are needed, a separate closed subspace $Z \subset X$ and its complementary open U must be supplied. Their intersection is empty. The category supported on Z can nevertheless carry nontrivial extensions with the localized category, as the following models show. Arinkin–Gaitsgory, Corollaries 8.2.8 and 8.2.10, p. 78, identify zero singular support with the quasi-coherent image and

give a localization sequence with the supported category inside $\mathrm{IndCoh}(X)$. They do not identify that supported category with an arbitrary category on a closed stratum.

Conjecture 73.2.2 *(the arithmetic categorical gluing comparison).* For the intended fixed-determinant arithmetic problem, construct X and $\mathcal{N}$ as above, geometric local and transition categories $\mathcal{D}_0$, $\mathcal{D}_1$, $\mathcal{D}_{01}$, and functors $\mathcal{D}_i \to \mathcal{D}_{01}$ in a specified category of linear stable categories, together with an equivalence

$$\mathcal{D} := \mathrm{IndCoh}_{\mathcal{N}}(X) \simeq \mathcal{D}_0 \times_{\mathcal{D}_{01}} \mathcal{D}_1. \tag{73.1}$$

The categories and transition functors must be constructed with their coefficient, support and determinant conditions. If formal patching is used, the transition object must be a punctured formal neighborhood with its restriction functors. If localization is used, its extension data must be retained. A description as a plain product over an empty closed/open intersection is not the asserted comparison.

The finite-module theorem below constructs formal patching for the affine line. It is not an arithmetic realization of Equation (73.1).

Closed and open restriction on the affine line

Let k be a field and put $A = k[t]$, $C = k[t, t^{-1}]$. Let $X = \operatorname{Spec} A$, $Z = V(t)$, and $U = D(t)$. Write $\mathrm{mod}_{\mathrm{fg}}(R)$ for the category of finitely generated left modules over a ring R. The intersection $Z \times_X U$ is empty, since its coordinate ring is $(A/(t))[t^{-1}] = 0$. Consequently a fibre product of categories over the category of this intersection is a product.

Proposition 73.2.3. The restriction functor

$$\mathrm{mod}_{\mathrm{fg}}(A) \longrightarrow \mathrm{mod}_{\mathrm{fg}}(k) \times \mathrm{mod}_{\mathrm{fg}}(C), \qquad M \longmapsto (M/tM, M[t^{-1}]),$$

is not full. The corresponding derived restriction functor on perfect complexes is not full either.

Proof. The object A maps to (k, C). Its endomorphism homomorphism is

$$A \longrightarrow k \times C, \qquad f \longmapsto (f(0), f).$$

The endomorphism $(1, 0)$ does not belong to its image. The same calculation applies to degree-zero derived endomorphisms. □

The extension retained by localization

Let $D(A)$ be the stable k-linear ∞-category obtained by inverting quasi-isomorphisms of unbounded cochain complexes of A-modules. Cohomological shifts satisfy $H^n(M[1]) = H^{n+1}(M)$. Let $D_t(A)$ be its full subcategory of complexes whose cohomology modules are elementwise killed by powers of t. Extension of scalars $L = C \otimes^{\mathbf{L}}_A -$ is left adjoint to restriction of scalars $j_* : D(C) \to D(A)$. Because C is a localization, it is flat and $C \otimes_A C = C$. Thus $L j_* = \mathrm{id}$ and j_* is fully faithful. Define

$$\Gamma_t M = \mathrm{fib}(M \longrightarrow j_* L M).$$

Localization is exact, so $L\Gamma_t M = 0$. Exactness of localization on modules identifies $\ker L$ with $D_t(A)$. If $K \in D_t(A)$, adjunction gives $\mathrm{Map}_{D(A)}(K, j_* N) = 0$ for every $N \in D(C)$. Applying mapping spaces to the defining fibre sequence therefore shows that Γ_t is right adjoint to $D_t(A) \hookrightarrow D(A)$.

Define a stable ∞-category $\mathcal{G}$ whose objects are

$$(K, N, \delta), \qquad K \in D_t(A), \quad N \in D(C), \quad \delta : j_* N \longrightarrow K[1].$$

Precisely, $\mathcal{G}$ is the pullback of $\mathrm{Fun}(\Delta^1, D(A)) \to D(A) \times D(A)$ along $(N, K) \mapsto (j_* N, K[1])$. Morphisms include the homotopies in the resulting commutative squares, rather than merely an equality in a homotopy category.

PROPOSITION 73.2.4. There is an equivalence of stable ∞-categories

$$D(A) \simeq \mathcal{G}.$$

It sends M to $(\Gamma_t M, LM, \delta_M)$, where δ_M is the connecting morphism of the localization triangle. Its inverse sends (K, N, δ) to $\mathrm{fib}(\delta)$.

Proof. A triple defines the fibre sequence

$$K \longrightarrow \mathrm{fib}(\delta) \longrightarrow j_* N \xrightarrow{\delta} K[1].$$

Applying L identifies $L\,\mathrm{fib}(\delta)$ with N. Under this identification, the middle map is the unit of $L \dashv j_*$: by adjunction it corresponds to the identity of N. Its fibre is therefore $\Gamma_t\,\mathrm{fib}(\delta) = K$. These identifications recover the connecting morphism δ. Conversely, the fibre of the connecting map in a localization triangle is its middle term. Fibre constructions are functorial in a stable ∞-category, so these identifications give mutually inverse functors, including their mapping spaces. □

For $M = A$, the complex $\Gamma_t A$ is represented by $[A \to C]$ in degrees 0, 1. Hence $\Gamma_t A \simeq (C/A)[-1]$, and the connecting map is

$$C \longrightarrow C/A.$$

Its fibre is A in degree zero. This nonzero quotient map is the extension datum that reconstructs A. The two objects C and $\Gamma_t A$ are not perfect over A: their nonzero cohomology modules are not finitely generated. Thus this localization construction takes place in $D(A)$. It does not present perfect complexes as a product of perfect complexes on the two strata. Nor can the supported category be replaced by $D(k)$ under closed pushforward. Indeed, the resolution $0 \to A \xrightarrow{t} A \to k \to 0$ gives $\mathrm{Ext}^1_A(k,k) = k$, whereas $\mathrm{Ext}^1_k(k,k) = 0$.

Coherent gluing across a formal neighborhood

To retain coherent modules, replace the closed point by its formal neighborhood. Put

$$B = k[[t]], \qquad D = k((t)), \qquad C = k[t, t^{-1}].$$

The maps $A \to B$, $A \to C$, $B \to D$, and $C \to D$ are the inclusions. The overlap $\mathrm{Spec}\, D$ is the punctured formal disk. It is not the empty intersection $Z \cap U$. The ring B is the coordinate ring of the t-adic formal disk. Here its coherent category means finitely generated B-modules, with their t-adic topology. Tensor products displayed below are ordinary algebraic tensor products unless explicitly derived.

Let $\mathcal{P}$ be the category of triples

$$(P, Q, \eta), \qquad P \in \mathrm{mod}_{\mathrm{fg}}(B), \quad Q \in \mathrm{mod}_{\mathrm{fg}}(C), \quad \eta : D \otimes_B P \xrightarrow{\sim} D \otimes_C Q.$$

Morphisms are compatible pairs of module homomorphisms.

Theorem 73.2.5. The functor

$$\mathrm{mod}_{\mathrm{fg}}(A) \longrightarrow \mathcal{P}, \qquad M \longmapsto (B \otimes_A M, C \otimes_A M, \mathrm{can})$$

is an equivalence. Its inverse sends a triple to

$$\{(p, q) \in P \oplus Q : \eta(1 \otimes p) = 1 \otimes q\}.$$

Proof. Every Laurent series is the sum of a power series and a Laurent polynomial. Their intersection is $k[t]$. Consequently

$$0 \longrightarrow A \longrightarrow B \oplus C \xrightarrow{(b,c)\mapsto b-c} D \longrightarrow 0$$

is exact. The modules B, C, D are flat over A. For C, D this follows from localization and torsion-freeness over the principal ideal domain A. The same argument applies to B: a torsion-free module over a principal ideal domain is the filtered union of its finite free submodules and is therefore flat. Since D is flat, tensoring the displayed sequence with any A-module M remains exact. In particular M is the indicated fibre product of its two scalar extensions. A compatible pair of homomorphisms on these extensions sends this fibre product into the corresponding fibre product for the target. The resulting A-linear homomorphism recovers the pair, since the elements $1 \otimes m$ generate both scalar extensions. This proves full faithfulness.

For essential surjectivity, use the structure theorem for finitely generated modules over the principal ideal domains B and C. A torsion summand $B/(t^m)$ descends from $A/(t^m)$, whose scalar extension to C vanishes. A torsion summand of Q has the form $C/(f)$, where $f \in k[t]$ and $f(0) \neq 0$. It descends from $A/(f)$, whose extension to B vanishes because f is a unit in B. Both kinds of torsion disappear over D. After splitting off these summands, P and Q are free of the same rank r.

Choose $Q = C^r$ and view P as a B-lattice $\Lambda \subset D^r$ through η. There is an integer $n \geq 0$ such that

$$t^n B^r \subset \Lambda \subset t^{-n} B^r.$$

Set $M = \Lambda \cap C^r$. Intersecting the bounds with C^r gives

$$t^n A^r \subset M \subset t^{-n} A^r.$$

Thus M is a finite free A-module of rank r, and $C \otimes_A M = C^r$. For $\lambda \in \Lambda$, truncate its Laurent expansion below degree n. The truncation q is a Laurent polynomial, and $\lambda - q \in t^n B^r \subset \Lambda$. Hence $q \in M$ and $\Lambda = M + t^n B^r$. The map $B \otimes_A M \to \Lambda$ is therefore surjective. It is injective: an A-basis of M becomes a basis of C^r, so its determinant remains nonzero over the field D. This proves $B \otimes_A M = \Lambda$ with the specified overlap map. Adding the torsion summands proves essential surjectivity. Full faithfulness identifies the stated fibre-product construction with a quasi-inverse on all triples. □

For a k-linear category $\mathcal{C}$, write $Z(\mathcal{C}) = \mathrm{End}(\mathrm{id}_{\mathcal{C}})$ for its ordinary centre, with multiplication given by composition.

Corollary 73.2.6. Under the preceding equivalence,

$$Z(\mathrm{mod}_{\mathrm{fg}}(A)) \simeq B \times_D C \simeq Z(\mathrm{mod}_{\mathrm{fg}}(B)) \times_{Z(\mathrm{mod}_{\mathrm{fg}}(D))} Z(\mathrm{mod}_{\mathrm{fg}}(C)).$$

The comparison maps on centres are the indicated inclusions of rings.

Proof. For every commutative ring R, a natural endomorphism z of $\mathrm{mod}_{\mathrm{fg}}(R)$ is multiplication by $a = z_R(1) \in R$. Indeed, for each $m \in M$, apply naturality to $R \to M$, $1 \mapsto m$. This yields $z_M(m) = am$. Thus the centre is R, compatibly with scalar extension. The exact ring sequence above gives $A = B \times_D C$. □

A higher-centre obstruction and a positive finite model

For an associative k-algebra R, put

$$\mathrm{HC}(R) = \mathbf{R}\,\mathrm{Hom}_{R\otimes_k R^{\mathrm{op}}}(R, R).$$

This is the Hochschild cochain E_2-algebra, equivalently the Hochschild centre of the derived module category. Its cohomology is written $HH^*(R)$. The Gerstenhaber bracket has cohomological degree -1. For a derivation v and a scalar cochain $f \in R$, its convention is $[v, f] = v(f)$.

PROPOSITION 73.2.7. For $A = k[t]$, there is no unital E_2-algebra morphism $\mathrm{HC}(A) \to \mathrm{HC}(k)$ whose map on HH^0 is evaluation at $t = 0$.

Proof. Write $A^e = k[x, y]$. The bimodule resolution

$$0 \longrightarrow A^e \xrightarrow{x-y} A^e \longrightarrow A \longrightarrow 0$$

is exact, since $x - y$ is a non-zero-divisor. Applying $\mathrm{Hom}_{A^e}(-, A)$ gives the complex $[A \xrightarrow{0} A]$ in degrees $0, 1$. Thus $HH^0(A) = A$, $HH^1(A) = A\partial_t$, and the other positive cohomology groups vanish. The Hochschild insertion formula gives $[\partial_t, t] = 1$. For k, the diagonal is already free over $k \otimes_k k = k$, so $HH^*(k) = k$ in degree zero. An E_2 morphism preserves the Gerstenhaber bracket. It would send ∂_t to zero and t to zero, but would send 1 to 1. Applying it to the bracket identity gives $0 = 1$. □

This obstruction holds over every field, including characteristic 2. It rules out the indicated higher-centre restriction even though restriction of ordinary scalars is defined. The coherent formal-gluing theorem therefore does not, on its own, assert a fibre-product formula for Hochschild E_2-algebras. In particular, algebraic Hochschild cochains of $k[[t]]$ cannot be silently replaced by continuous cochains or completed diagonal cochains. Such replacements require their own definitions and maps.

A nonzero linking morphism can also impose an exact higher-centre compatibility in a finite category. Let

$$T = \begin{pmatrix} k & k \\ 0 & k \end{pmatrix}, \qquad e_1 = E_{11}, \quad e_2 = E_{22}, \quad a = E_{12}.$$

A left T-module is a pair (V_1, V_2) with a map $u : V_2 \to V_1$. The two vector-space categories are linked by u. A scalar pair (λ_1, λ_2) acts centrally on every such module precisely when $\lambda_1 u = u\lambda_2$ for every u. Taking $u = \mathrm{id}_k$ gives $\lambda_1 = \lambda_2$. Commutation with e_1, e_2 eliminates off-diagonal elements of the ring centre, so $Z(T) = k \cdot 1$.

PROPOSITION 73.2.8. The unit map $k \to \mathrm{HC}(T)$ is an equivalence of E_2-algebras. The underlying Hochschild cochain complex is quasi-isomorphic to

$$[k \oplus k \xrightarrow{(\lambda_1,\lambda_2)\mapsto\lambda_1-\lambda_2} k]$$

in degrees $0, 1$.

Proof. There is a projective T-bimodule resolution

$$0 \longrightarrow Te_1 \otimes_k e_2T \xrightarrow{d} (Te_1 \otimes_k e_1T) \oplus (Te_2 \otimes_k e_2T) \xrightarrow{\mu} T \longrightarrow 0,$$

where μ is multiplication and $d(e_1 \otimes e_2) = e_1 \otimes a - a \otimes e_2$. The first term has dimension one and the middle term dimension four. Multiplication is onto the three-dimensional algebra T. The displayed nonzero element belongs to its kernel, so the sequence is exact. Each tensor factor defines an idempotent summand of a free T^e-module, hence is projective.

For any bimodule M, $\mathrm{Hom}_{T^e}(Te_i \otimes e_j T, M) = e_i M e_j$. Applying this with $M = T$ gives precisely the two-term complex above. Its cohomology is the diagonal k in degree zero and zero elsewhere. The scalar unit map induces this diagonal inclusion on cohomology. It is a map of E_2-algebras and a quasi-isomorphism of underlying complexes, hence an equivalence of E_2-algebras. □

The map $k^2 \to k$ in this computation is a difference of the two actions on the linking bimodule $e_1 T e_2$. It is a map of complexes, not a unital algebra homomorphism. The computation proves the claimed equivalence for this algebra. It does not state that a higher-centre functor preserves arbitrary categorical pullbacks.

The remaining arithmetic higher-centre comparison

For a presentable linear stable category C, use the Hochschild centre $\mathrm{HC}(C) = \mathrm{End}_{\mathrm{Fun}^L(C,C)}(\mathrm{id}_C)$ with its E_2 structure. This is the invariant of Arinkin–Gaitsgory, Appendix E.3.1, pp. 138–139. Its functoriality on a particular diagram and preservation of a homotopy limit are additional hypotheses.

Proposition 73.2.9 *(centre gluing on a specified categorical diagram).* Suppose the geometric spaces, support conditions and functors in Equation (73.1) have been constructed, and that this formula is a categorical equivalence. Suppose the Hochschild-centre construction has specified E_2 maps on this diagram and preserves this particular homotopy fibre product. Then

$$\mathrm{HC}(\mathcal{D}) \simeq \mathrm{HC}(\mathcal{D}_0) \times_{\mathrm{HC}(\mathcal{D}_{01})} \mathrm{HC}(\mathcal{D}_1).$$

This identity concerns the specified spectral diagram. An identification with the global automorphic centre additionally requires a comparison morphism and a proof that it is an equivalence. The formal scalar-action map of Proposition 64.5.1 does not supply that morphism. No arithmetic codimension bound is inferred from the identity.

Proof. Apply the specified higher-centre construction to the assumed categorical equivalence. Its assumed preservation of this homotopy fibre product gives the displayed identity. Every term belongs to the spectral diagram, so this argument constructs no map from the automorphic category or its centre. The rank-two calculation in Proposition 88.2.1 also shows why a codimension-two arithmetic transition cannot be inferred from the regular nilpotent orbit in $\mathfrak{sl}_2$. □

Corollary 73.2.10 *(internal gluing and the global comparison).* Under the hypotheses of Proposition 73.2.9, the three spectral centre terms and their specified maps determine the displayed homotopy fibre product. To identify it with an automorphic centre, one must additionally construct compatible maps from that centre to the three terms and prove that the resulting map to the homotopy fibre product is an equivalence.

Proof. Coherently compatible maps into the three terms define a map into their homotopy limit by its universal property. The spectral centre identity identifies its target with the global spectral centre. It does not construct the source maps or prove that the resulting map is an equivalence. □

Remark 73.2.11 *(the scope of the centre identity).* An internal gluing identity describes a centre once the geometric diagram and centre functor have been supplied. It is not an arithmetic Arinkin–Gaitsgory comparison. The required arithmetic space, singular-support condition, categorical functors and higher-centre morphisms remain additional constructions. In particular no codimension-two arithmetic transition follows from the rank-two nilpotent orbit calculation in Proposition 88.2.1.

73.3 F2.c Restricted-Tensor E_2 Transport Lemma

Setup

Use the full data of Setup 50.1.1 with $\mathcal{O} = E_2$: the local categories and actual morphisms $\eta_v : \mathrm{Sp} \to \mathcal{C}_v$ of E_2-algebras on a cofinite set of places. Write $u_v = \eta_v(\mathbb{S})$. Compactness may be imposed as an additional local hypothesis. It is not used in the filtered-colimit argument.

Transport

THEOREM 73.3.1 *(F2.c E_2 transport).* Under these hypotheses the restricted tensor product $\otimes'_v(\mathcal{C}_v, u_v)$ is an E_2-algebra in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$, and its admissible finite-stage maps ι_S are E_2-monoidal. A singleton map ι_v is available when $V \setminus \{v\} \subseteq V^{\mathrm{sph}}$, or after specifying the additional maps required by (M2) of Theorem 50.1.3. Uniqueness is relative to the complete underlying colimit cocone. Central idempotence of u_v alone is not a substitute for the pointing maps.

Proof. For $S \subseteq S'$, tensor the identity on the old factors with the actual maps η_v for $v \in S' \setminus S$. This is a morphism of E_2-algebras, including nullary compatibility. Tensor associativity supplies its coherent composition with the other insertions. The finite-stage system is consequently a diagram in $\mathrm{Alg}_{E_2}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$.

The ambient tensor product preserves filtered colimits separately. Proposition 50.3.7 therefore constructs the operadic colimit on the underlying colimit. Its cocone maps are E_2-monoidal, and its cocone uniqueness is Proposition 50.9.1. The assertion about singleton maps follows by first reaching an admissible stage using exactly the additional pointing maps specified in (M2). No full faithfulness or geometric collision comparison is deduced. □

Remark 73.3.2 *(the E_2 specialization).* This is the E_2 specialization of the formal filtered-colimit argument. It does not weaken its inputs to central idempotent objects. A strong unital E_2-map from Sp must send $\mathbb{S}$ to the actual target unit. The construction does not supply a missing local enhancement, pointing map or arithmetic comparison.

73.4 Combined BC-Diamond-Glue Statement

PROPOSITION 73.4.1 *(formal constructions and arithmetic comparison data).* The determinant construction supplies the restriction diagram in Theorem 73.1.4. Given the local modules and actions over their common coefficient ring, it also supplies the pairwise pullback in Proposition 64.3.2. The restricted E_2-tensor construction of Theorem 73.3.1 retains its actual E_2-algebra pointing maps and its stated singleton-map conditions. Their use for BC-diamond-glue requires categorical local functors, coefficient and family comparisons, and prismatic and untilt descent maps. It also requires all the pairwise comparison data of Corollary 73.1.5. The global centre assertion additionally needs the arithmetic comparison morphism, with faithfulness and generation for its ordinary action and a derived equivalence for its higher enhancement.

Proof. The formal assertions are the stated restriction, pullback and restricted tensor constructions. The pairwise comparison follows only with the full hypotheses of Corollary 73.1.5. On finite free formal charts, Proposition 64.5.8 expresses it through intertwining equations and overlap conditions. The additional centre assertion is separate: Proposition 73.2.9 gives an internal spectral centre identity only on its assumed categorical diagram. It does not construct an automorphic-to-spectral comparison. □

Corollary 73.4.2 *(*$\mathrm{L}_{\mathrm{ACD}}$ *residual list after Chenevier and determinant reduction).* (conditional on the standing residuals listed). After these reductions, $\mathrm{L}_{\mathrm{ACD}}$ has the residual list:

(i) (BCglue.local): **Conditional** (unchanged from the KMS–GNS sector);

(ii) (BCglue.glue): **Conjectural** (unchanged from the KMS–GNS sector);

(iii) (Ch1): **conditional theorem** via Corollary 73.1.5, conditional on its Selmer-ring, geometric, trace and equivariant-descent data;

(iv) (AG.det): the actual categorical descent diagram and its higher-centre maps, together with the global automorphic centre comparison. The conditional internal identity in Proposition 73.2.9 does not identify that automorphic centre;

(v) F2.c: **theorem under the full local operadic and pointing-map hypotheses** via either Chapter 50 Theorem 50.1.3 or the E_2 specialisation Theorem 73.3.1.

Proof. The formal ring and module constructions are Proposition 64.6.1. Their pairwise arithmetic use requires Corollary 73.1.5 (the full Selmer-ring, trace and equivariant-descent datum), the separate global centre comparison alongside Proposition 73.2.9, and Theorem 73.3.1 (restricted $F2.c$ transport under its stated hypotheses). These constructions preserve the listed comparison requirements. □

73.5 Residual Obstruction After Chenevier and Determinant Reduction

Remark 73.5.1 *(the genuine core remaining).* The residual obstruction to $\mathrm{L}_{\mathrm{ACD}}$ after these reductions is the triple:

- (BCglue.local): a primary-source transition theorem identifying the BZCHN + Emerton–Gee + Fargues–Scholze local inputs as a canonical local coherent Springer sheaf on the Emerton–Gee stack across the $\ell \neq p$ / $\ell = p$ divide. It is a genuine theorem condition.
- (BCglue.glue): Bhatt–Scholze prismatic-flat cross-prime transition lifted to the coherent-sheaf level, preserving $\mathrm{IndCoh}_{\mathrm{Nilp}}$-continuity. It is a genuine theorem condition.
- (AG.det.transition): an actual categorical descent diagram with its singular-support conditions and compatible higher-centre maps. The global automorphic comparison remains additional to the internal gluing identity.

The Chenevier term retains the Selmer-ring identification, trace comparison and equivariant coherent descent of Corollary 73.1.5. The determinantal term requires both the geometric centre diagram and the global comparison morphism. The local and prismatic gluing terms remain independent inputs.

Remark 73.5.2 *(unmoved inputs).* The arguments above do not prove (BCglue.local) or (BCglue.glue). The former requires a transition theorem across the $\ell = p$ boundary for coherent Springer sheaves. The latter requires prismatic-flat cross-prime descent lifted from the universal deformation ring to coherent sheaves. The (AG.det.transition) term requires the actual support-preserving categorical functors on a specified gluing diagram, higher-centre maps compatible with that diagram, and a separate comparison with the automorphic centre.

Remark 73.5.3 *(comparison inputs and remaining hypotheses).* The three comparison reductions use the following inputs:

- Theorem 73.1.4 constructs the local-to-global ring map from the universal property of the Chenevier deformation rings. Bellaïche–Chenevier supplies the deformation-theoretic context; Bhatt–Scholze and Bhatt–Lurie concern the additional prismatic structures. The full Selmer-ring, geometric, trace and equivariant-descent requirements are retained in Corollary 73.1.5.
- Proposition 73.2.9 gives an internal centre identity on an assumed geometric and categorical diagram. The singular-support geometry and the higher-centre comparison maps must be constructed for the arithmetic space. The internal identity does not construct a map from the automorphic centre.
- Theorem 73.3.1 uses actual operadic pointings and the filtered-colimit theorem HA 3.2.3.1. It is the specialization of Chapter 50's restricted tensor construction to E_2. The two arguments use the same categorical diagram.

73.6 REFERENCES ON DETERMINANTS, DESCENT AND HIGHER CENTRES

The primary literature invoked in this chapter:

Chenevier, G. (2014). *The p-adic analytic space of pseudocharacters of a profinite group and pseudorepresentations over arbitrary rings*. arXiv:0809.0415. Sections 1.1 and 1.10; Proposition 3.3 and Example 3.4, p. 42; Example 3.6 and Proposition 3.7, p. 43. These supply determinant laws and their complete local deformation rings.

Bellaiche, J. and Chenevier, G. (2009). *Families of Galois representations and Selmer groups*. Astérisque 324. Chapter 1 used for the universal determinantal deformation ring identification.

Bhatt, B. and Scholze, P. (2022). *Prisms and prismatic cohomology*. Ann. Math. 196, arXiv:1905.08229. Background for the prismatic realization in Corollary 73.1.5; the stated cross-prime flatness is a separately required arithmetic comparison.

Bhatt, B. and Lurie, J. (2022). *Absolute prismatic cohomology*. arXiv:2201.06120. §8 Nygaard-filtration convention used in the additional prismatic realization of Corollary 73.1.5.

Arinkin, D. and Gaitsgory, D. (2015). *Singular support of coherent sheaves and the geometric Langlands conjecture*. Selecta Math. 21 (2015), 1–199. arXiv:1201.6343v4. Sections 2.3.3–2.3.4, pp. 20–21, and 4.1.2–4.1.5, pp. 33–34, define the singularity scheme, its cohomological action and support condition. Corollaries 8.2.8 and 8.2.10, p. 78, concern zero support and localization; Section 11.1.1, p. 103, defines the geometric global nilpotent cone. Appendix E.3.1, pp. 138–139, defines the Hochschild E_2 centre. These are the geometric notions in Definition 73.2.1; their arithmetic realization and the descent equivalence in Conjecture 73.2.2 remain additional constructions.

Galatius, S., Kupers, A., and Randal-Williams, O. (2025). *Cellular E_k-algebras*. Astérisque 460. arXiv:1805.07184v4, Definition 17.3 and Proposition 17.4, p. 186, construct an E_1 splitting semisimplicial pointed set and compare its realization with suspended indecomposables under their stated hypotheses. They concern cellular algebra methods; the higher-centre comparison in Proposition 73.2.9 requires its separately specified E_2 maps and preservation property.

Ben-Zvi, D., Francis, J., and Nadler, D. (2010). *Integral transforms and Drinfeld centers in derived algebraic geometry*. JAMS 23, arXiv:0805.0157. The geometric centre formalism provides background for the separately required arithmetic comparison in Corollary 73.2.10.

Lurie, J. (2017). *Higher algebra*. Definition 2.1.3.7 specifies strong monoidal functors. Proposition 3.2.3.1 constructs the operadic filtered colimit. Proposition 4.8.1.15, Remark 4.8.1.24 and Proposition 4.8.2.18 supply the stable categorical tensor and its colimit compatibility used in Theorem 73.3.1.

Francis, J. and Gaitsgory, D. (2012). *Chiral Koszul duality*. Selecta Math. 18, arXiv:1103.5803. Cited for the factorization-algebra background; no positive-dimensional factorization-homology theorem is used in Theorem 73.3.1.

Emerton, M. and Gee, T. (2019). *Moduli stacks of* (φ, Γ)*-modules and the existence of crystalline lifts*. arXiv:1908.07185. Used as the primary source for the local Emerton–Gee stack $\mathcal{X}_{GL_2}^{\mathrm{EG},p}$ in Conjecture 73.1.3 and the geometric realization required in Corollary 73.1.5. The pseudocharacter trace, Selmer-ring and equivariant-descent comparisons remain additional hypotheses.

Chapter 74

Elementary Reductions of the Riemann Hypothesis

The arithmetic Koszul reductions isolate the zero-placement problem inside the Weil distribution. The classical equivalences are recalled with their hypotheses, and the restricted Paley–Wiener class is kept separate from the full Weil cone.

74.1 The Riemann Hypothesis in Classical Form

Definition 74.1.1 *(the zeta function).* For $\mathrm{Re}(s) > 1$, the Riemann zeta function is the absolutely convergent sum

$$\zeta(s) \;:=\; \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Its Euler product, equivalent on $\mathrm{Re}(s) > 1$, is

$$\zeta(s) \;=\; \prod_{p} \bigl(1 - p^{-s}\bigr)^{-1},$$

the product taken over all prime numbers.

Definition 74.1.2 *(completed zeta and xi).* Set

$$\Lambda_\zeta(s) \;:=\; \pi^{-s/2}\,\Gamma(s/2)\,\zeta(s),$$
$$\xi_{\mathbb{R}}(s) \;:=\; \tfrac{1}{2}\,s(s-1)\,\Lambda_\zeta(s).$$

The function Λ_ζ is meromorphic with simple poles at $s = 0$ and $s = 1$; $\xi_{\mathbb{R}}$ is entire. Both satisfy the functional equation

$$\Lambda_\zeta(s) \;=\; \Lambda_\zeta(1-s), \qquad \xi_{\mathbb{R}}(s) \;=\; \xi_{\mathbb{R}}(1-s).$$

Remark 74.1.3 *(functional equation via Poisson summation).* Let $\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}$ for $t > 0$. Poisson summation gives $\theta(1/t) = t^{1/2}\theta(t)$. For $\Re s > 1$, termwise integration gives $\Lambda_\zeta(s) = \frac{1}{2}\int_0^\infty (\theta(t) - 1) t^{s/2}\, dt/t$. Splitting at 1 and substituting $t \mapsto 1/t$ in the lower integral yields its meromorphic continuation

$$\Lambda_\zeta(s) = \frac{1}{s(s-1)} + \frac{1}{2}\int_1^\infty (\theta(t) - 1)\bigl(t^{s/2} + t^{(1-s)/2}\bigr)\frac{dt}{t}.$$

The remaining integral is entire by exponential decay. This formula is invariant under $s \mapsto 1 - s$; multiplication by $s(s-1)/2$ removes the two poles and proves the functional equation for $\xi_{\mathbb{R}}$. Sources: B. Riemann, "Über die Anzahl der Primzahlen unter einer gegebenen Größe", Monatsber. Berlin. Akad. (1859); E. C. Titchmarsh, "The Theory of the Riemann Zeta-Function", Oxford 1951, Ch. 2.

Conjecture 74.1.4 *(Riemann Hypothesis).* All non-trivial zeros of ζ have real part $1/2$. Equivalently, the zeros of $\xi_{\mathbb{R}}$ have real part $1/2$.

74.2 Four Classical Equivalents

Four classical restatements of Chapter 74.1.4 are used below. They do not identify the same object; they identify the same zero-placement condition through different test categories.

THEOREM 74.2.1 *(Weil positivity).* Let $\mathscr{A} = C_c^\infty(\mathbb{R})$ with its inductive-limit topology, and put $f^*(t) = \overline{f(-t)}$ and $h_f(z) = \int_{\mathbb{R}} f(t)e^{izt}\,dt$. The Mellin coordinates are $\Phi_f(s) = h_f(-i(s-1/2))$. With $\phi(x) = f(\log x)$, the change of variables $x = e^t$ gives $\Phi_f(s) = \int_0^\infty \phi(x)x^{s-1/2}\,dx/x$. For the zeros ρ of $\xi_{\mathbb{R}}$, counted with multiplicity, put $z_\rho = -i(\rho - 1/2)$. Then

$$W(f) = \sum_\rho h_f(z_\rho) = h_f(i/2) + h_f(-i/2) - W_\infty(f) - W_{\mathrm{fin}}(f),$$

where

$$W_\infty(f) = \frac{1}{2\pi}\int_{\mathbb{R}} h_f(u)(\log\pi - \Re\psi(1/4 + iu/2))\,du,$$
$$W_{\mathrm{fin}}(f) = \sum_{m\ge 2}\frac{\Lambda(m)}{\sqrt{m}}[f(\log m) + f(-\log m)].$$

Write $P(f) = h_f(i/2) + h_f(-i/2)$ for the polar functional. Here $\Lambda(m)$ is the von Mangoldt function and $\psi = \Gamma'/\Gamma$. The prime sum is finite. The zero sum converges absolutely. RH is equivalent to $W(f * f^*) \ge 0$ for every $f \in \mathscr{A}$.

Proof. These are the compact-test formula and Weil criterion of Chapter 78.1.2, Chapter 78.2.1. The Fourier image of $f * f^*$ is $h_f h_f^\sharp$, where $h^\sharp(z) = \overline{h(\bar z)}$. On the real line this product is $|h_f|^2$. □

THEOREM 74.2.2 *(Li positivity).* . Define

$$\lambda_n := \sum_\rho\left(1 - \left(1 - \frac{1}{\rho}\right)^n\right) = \frac{1}{(n-1)!}\frac{d^n}{ds^n}\left[s^{n-1}\log\xi_{\mathbb{R}}(s)\right]\Big|_{s=1}.$$

Then RH holds if and only if $\lambda_n \ge 0$ for all $n \ge 1$. Source: X.-J. Li, "The positivity of a sequence of numbers and the Riemann hypothesis", J. Number Theory 65 (1997), 325–333.

Let γ denote the Euler–Mascheroni constant. For a normal Hilbert–Schmidt operator T, its regularized determinant is

$$\det{}_2(I - sT) = \prod_j (1 - s\lambda_j)e^{s\lambda_j},$$

where the nonzero eigenvalues λ_j are repeated with their multiplicities. The product converges locally uniformly in $s \in \mathbb{C}$: $\sum_j |\lambda_j|^2 < \infty$ and $(1 - s\lambda_j)e^{s\lambda_j} = 1 + O_K(|\lambda_j|^2)$ uniformly on each compact set $K \subset \mathbb{C}$.

THEOREM 74.2.3 *(normalized Hilbert–Pólya spectral form).* RH is equivalent to the existence of a separable Hilbert space H and a densely defined self-adjoint operator A such that, for $\Theta = \frac{1}{2}I + iA$, the inverse Θ^{-1} is Hilbert–Schmidt and the prescribed entire determinant is

$$\frac{1}{2}e^{Bs}\det{}_2(1 - s\Theta^{-1}) = \xi_{\mathbb{R}}(s), \qquad B = -1 - \frac{\gamma}{2} + \frac{1}{2}\log(4\pi).$$

Zeros of the determinant are counted with their spectral multiplicities. The prescription fixes both its value $1/2$ at zero and its logarithmic derivative B there.

Proof. For a self-adjoint A, the bounded inverse of $\frac{1}{2}I + iA$ is normal. If it is Hilbert–Schmidt it is compact; its nonzero eigenvalues are $(1/2 + ia_j)^{-1}$ for real eigenvalues a_j, with their multiplicities. The regularized product therefore has zeros only at $1/2 + ia_j$. Its asserted equality with $\xi_{\mathbb{R}}$ implies RH. Conversely, on the zero-occurrence carrier $K_Z = \ell^2(Z)$ of Chapter 77.6, use the closed normal operator Θ_Z with maximal domain $\{u : \sum_\rho |\rho u_\rho|^2 < \infty\}$. Under RH write $\rho = 1/2 + i\gamma_\rho$, with $\gamma_\rho \in \mathbb{R}$. The diagonal operator

$$(Au)_\rho = \gamma_\rho u_\rho, \qquad \operatorname{Dom} A = \{u : \sum_\rho |\gamma_\rho u_\rho|^2 < \infty\}$$

is self-adjoint, and $\Theta_Z = \frac{1}{2}I + iA$ on the same domain. Its inverse is Hilbert–Schmidt, and Chapter 77.6.1 proves exactly the stated normalized genus-one determinant. That construction uses the zero divisor as input. Constructing an arithmetic operator and an intertwiner with this carrier remains an additional problem. □

The Hilbert–Pólya question asks for such an independent arithmetic realization; the abstract zero-occurrence construction does not supply it. Connes, *Selecta Mathematica* 5 (1999), arXiv:math/98110 68, treats a separate trace-formula framework.

Entire functions and the zero divisor. Write $F(z) = \Xi(z) = \xi_{\mathbb{R}}(1/2 + iz)$ and $G^\#(z) = \overline{G(\bar{z})}$ for an entire function G. The functional equation and reality of $\xi_{\mathbb{R}}$ make F even and real entire. RH says precisely that all its zeros are real. Use the convention that an entire function E belongs to HB if it has no real zeros and $|E^\#(z)| < |E(z)|$ for $\Im z > 0$. No finite-exponential-type condition is included in this definition.

Theorem 74.2.4 *(a shift-family Hermite–Biehler criterion).* For $h > 0$ put

$$E_h(z) = F(z + ih) = \xi_{\mathbb{R}}(1/2 + h - iz).$$

The following conditions are equivalent: RH; $E_h \in$ HB for every $h > 0$; and $E_{h_n} \in$ HB for some positive sequence $h_n \downarrow 0$. The recovery identities are

$$E_h^\#(z) = F(z - ih), \qquad F(z) = E_h(z - ih).$$

Thus a zero w of E_h gives the zero $w + ih$ of F with exactly the same multiplicity. No simplicity assumption is needed.

Proof. One has $F(0) \neq 0$. Indeed the alternating series at $1/2$ can be paired as $\eta(1/2) = \sum_{n\geq 1}[(2n-1)^{-1/2} - (2n)^{-1/2}] > 0$; $\zeta(1/2) = \eta(1/2)/(1 - \sqrt{2})$ is consequently nonzero. Under RH the genus-one factorization and evenness give

$$F(z) = F(0)\prod_{\gamma>0}(1 - z^2/\gamma^2)^{m_\gamma}, \qquad \sum_{\gamma>0} m_\gamma \gamma^{-2} < \infty.$$

Here $\pm\gamma$ are the zeros of F, with multiplicity m_γ. The inverse-square convergence follows from Chapter 77.5.1. Pairing the factors at γ and $-\gamma$ cancels their genus-one exponentials; the remaining linear exponential is constant by evenness. The product converges locally uniformly. For $z = x + iy$, $y > 0$, every real γ satisfies

$$|z + ih - \gamma|^2 - |z - ih - \gamma|^2 = 4yh > 0.$$

Apply this to both members of every paired factor. If $F(z - ih) \neq 0$, each factor in the ratio of absolute values exceeds one. Its limiting product remains strictly greater than one by retaining one strict factor and bounding all other factors below by one. If $F(z - ih) = 0$, the inequality is immediate, since $F(z + ih)$ cannot vanish in the upper half-plane under RH. Hence $|E_h(z)| > |E_h^{\#}(z)|$. All zeros of E_h are $\gamma - ih$, so none is real. This proves the assertion for every $h > 0$, and therefore for any sequence decreasing to zero. Conversely, a nonreal zero of the real entire function F gives a zero w in the upper half-plane. For $h_n < \Im w$, the point $w - ih_n$ is an upper-half-plane zero of E_{h_n}, contradicting the strict HB inequality. The two recovery identities follow directly from reality, translation and the functional equation. □

Remark 74.2.5 *(one shift and a family of shifts).* For $F_a(z) = z^2 + a^2$, $a > 0$, and $h > a$,

$$F_a(z + ih) = (z + i(h - a))(z + i(h + a)) \in \mathrm{HB},$$

although F_a has nonreal zeros. Each factor has the strict upper-half-plane modulus inequality because $|z + ib|^2 - |z - ib|^2 = 4b\Im z > 0$ for $b > 0$. Thus one fixed shift does not characterize real-rootedness. For Ξ itself, Lagarias, arXiv:math/0601653, Lemma 2.1, pp. 5–6, proves the shift inequality unconditionally for $h \geq 1/2$, and under RH for every $h > 0$; the change of variable is $s = 1/2 - iz$. Section 6 relates these functions to de Branges spaces. The family tending to zero, not one large shift, is the RH criterion.

THEOREM 74.2.6 *(the derivative criterion and real multiplicities).* Set $D(z) = F(z) + iF'(z)$. Then

$$D^{\#}(z) = F(z) - iF'(z), \qquad F(z) = \tfrac{1}{2}(D(z) + D^{\#}(z)).$$

RH is equivalent to $|D(z)| > |D^{\#}(z)|$ for every $\Im z > 0$, where real zeros of D are allowed. With the no-real-zero convention above, $D \in \mathrm{HB}$ is instead equivalent to RH together with simplicity of all zeros of F.

Proof. Under RH, logarithmic differentiation of the paired product gives for $z = x + iy$, $y > 0$,

$$\Im \frac{F'(z)}{F(z)} = -y \sum_{\gamma \in \mathbb{R}} \frac{m_\gamma}{|z - \gamma|^2} < 0.$$

The imaginary-part series is absolutely convergent by the inverse-square estimate. There is at least one zero, indeed infinitely many by Chapter 77.5.1. Consequently

$$|D(z)|^2 - |D^{\#}(z)|^2 = -4|F(z)|^2 \Im \frac{F'(z)}{F(z)} > 0.$$

Conversely, at an upper-half-plane zero w of F the two values $D(w) = iF'(w)$ and $D^{\#}(w) = -iF'(w)$ have equal modulus. This contradicts the strict inequality, also when both vanish at a multiple zero. Conjugation symmetry excludes lower-half-plane zeros as well. For a real zero write $F(z) = (z - \gamma)^m g(z)$ with $g(\gamma) \neq 0$. Then

$$D(z) = (z - \gamma)^{m-1}\big((z - \gamma)g(z) + i[mg(z) + (z - \gamma)g'(z)]\big).$$

The bracket is nonzero at γ, so the real multiplicity in D is exactly $m - 1$. On the real axis F, F' are real, hence a real zero of D occurs only where both vanish. This proves the simplicity qualification and the stated recovery identity. □

Proposition 74.2.7 *(the growth required by the zeta divisor).* No nonzero entire function of finite exponential type can have a zero divisor containing the full divisor of $F = \Xi$. In particular no nonzero real or imaginary component of a finite-exponential-type function can have that divisor. The functions F and each fixed translate E_h have order one and infinite exponential type.

Proof. Let $N(T)$ count the nontrivial zeta zeros with $0 < \Im\rho \le T$, with multiplicity. The Riemann–von Mangoldt estimate is $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + O(\log T)$. For a precise primary bound see Hasanalizade–Shen–Wong, arXiv:2107.06506, Corollary 1.2, equation (1.5), p. 2. The strip $0 \le \Re\rho \le 1$ and conjugation give, for the number $n_F(r)$ of zeros in $|z| \le r$,

$$2N(\sqrt{r^2 - 1/4}) \le n_F(r) \le 2N(r), \qquad n_F(r) = \frac{r}{\pi}\log\frac{r}{2\pi e} + O(\log r).$$

If G is nonzero entire with $|G(z)| \le Ce^{\tau|z|}$ and $G(0) \ne 0$, Jensen's formula at radius $2r$ gives

$$n_G(r)\log 2 \le \frac{1}{2\pi}\int_0^{2\pi} \log|G(2re^{i\theta})|\,d\theta - \log|G(0)| = O(r).$$

Factoring out a zero at the origin changes the count by a constant. Thus G cannot contain the $r\log r$ divisor of F. If E has finite exponential type, so do its components $(E + E^{\#})/2$ and $(E^{\#} - E)/(2i)$; apply the same argument to every nonzero component. Finally the functional equation, Stirling's formula and $\zeta(\sigma) \to 1$ give

$$\log F(iy) = \log \xi_{\mathbb{R}}(1/2 + y) = \frac{y}{2}\log y - \frac{y}{2}(1 + \log(2\pi)) + O(\log y) \quad (y \to +\infty).$$

Together with the order-at-most-one bound in Chapter 77.5.1, this proves order one and infinite exponential type, also after any fixed translation. □

The general Hilbert-space theory of entire functions is developed in de Branges, *Hilbert Spaces of Entire Functions* (1968). The explicit constructions above specify the recovery of the zeta divisor and retain its actual growth and multiplicities.

Remark 74.2.8 *(four precise equivalences).* Weil positivity on all compact tests, Li positivity, the normalized spectral realization of Chapter 74.2.3, and the shift-family condition of Chapter 74.2.4 are each equivalent to RH. The derivative inequality gives the same zero condition with its real-zero convention stated explicitly. These criteria use different carriers: test functions, a coefficient sequence, a spectral operator and entire functions. None of their abstract equivalences constructs the native arithmetic pairing or operator maps required below.

74.3 The Arithmetic Reduction

The arithmetic Koszul construction supplies five comparison objects. Each object is defined on its own domain, and no implication from a restricted domain to the full Weil cone is used without an extension theorem.

Step 1: The arithmetic Heisenberg Koszul pair

Question 74.3.1 *(the arithmetic Heisenberg coefficient object).* Construct local coefficient objects H_v, a common coefficient category, pointings and a completion that define

$$\mathsf{H}^{\mathrm{Ar}} = \mathsf{H}_\infty \widehat{\otimes} \widehat{\bigotimes}_p \mathsf{H}_p.$$

Compare the infinite-place factor with the Heisenberg field and the finite factors with their continuous Galois coefficient models. An associative bar construction additionally requires an augmented algebra, or a specified curved replacement. The pointed tensor construction of Chapter 50.1.1 applies once its inputs are given.

Definition 74.3.2 (central invariants and comparison functors). For a specified arithmetic object and its specified dual, write $\kappa_{\mathsf{H}}^{\mathrm{Ar}}$ and $(\kappa^!)_{\mathsf{H}}^{\mathrm{Ar}}$ for their chosen scalar central invariants. These are not the bar and cobar complexes themselves. An integral co–contra adjunction and an algebraic bar–cobar adjunction have the distinct carriers described in Chapter 46.

THEOREM 74.3.3 *(P1 under the central-defect comparison).* Assume the objects, involution, central-defect identity and Mellin evaluation of Theorem 43.5.1 realize these arithmetic invariants, with $E(w) \neq 0$. Then

$$\kappa_{\mathsf{H}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{H}}^{\mathrm{Ar}} = 0 \quad \Longleftrightarrow \quad \xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s).$$

Proof. Applying E to the defect identity identifies the difference of the two completed Mellin functions with $(\kappa_{\mathsf{H}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{H}}^{\mathrm{Ar}})E(w)$. The nonzero meromorphic function $E(w)$ detects the scalar coefficient. Multiplication by the reflection-invariant factor $s(s-1)/2$ gives the equivalent identity for ξ. □

Remark 74.3.4 (the scalar and categorical comparisons). Poisson summation proves the scalar functional equation. An arithmetic interpretation requires maps identifying the duality involution with Mellin reflection and the central classes with the specified defect. Equality of two scalar characters need not detect a nonzero vector defect, as in Theorem 43.5.1. Neither the scalar equality nor a categorical equivalence alone supplies those identifications.

Archimedean Heisenberg Character

Definition 74.3.5 (archimedean Heisenberg character). The archimedean fibre H_∞ is the standard Heisenberg vertex operator algebra on $\mathbb{C}_\infty$ with Virasoro zero-mode L_0 and central charge $c_\infty = 1$ (one free boson). For the primary H_∞-module V_∞ (the vacuum representation), the Zhu character is

$$\chi_{V_\infty}(\tau) \;:=\; \mathrm{tr}_{V_\infty}\!\left(q^{L_0 - c_\infty/24}\right), \qquad q = e^{2\pi i \tau},$$

a standard VOA Zhu trace on the archimedean fibre. The global object $\mathcal{V}^{\mathrm{prim}}$ is the Hochschild trace class of the identity on the arithmetic Hecke category; it is *not* a vertex algebra. The quantity $\Lambda_\zeta(s)$ arises as its Hochschild character via Tate's adelic construction, not as a VOA module trace.

THEOREM 74.3.6 *(P2: Hochschild character and functional equation).* . The Hochschild character of $\mathcal{V}^{\mathrm{prim}}$ equals $\Lambda_\zeta(s)$ under the identification of the Hochschild trace with Tate's adelic zeta integral (Chapter 41.2.1). The primary determinant carries rank residue $r_\zeta = 2$, the residue at $s = 1$ of $2\zeta(s)$ (Chapter 41.5.1). The adelic functional equation $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$ follows from Poisson summation applied to the Schwartz–Bruhat function at each place. The Heisenberg vacuum character is the local eta-factor; the theta transformation enters only after the lattice Schwartz sector is adjoined.

Remark 74.3.7 (elementary content of P2). On the archimedean fibre,

$$\chi_{V_\infty}(\tau) = q^{-1/24} \prod_{n \geq 1} (1 - q^n)^{-1}.$$

This eta-character is not Tate's theta integral. Tate's local factor is obtained from the Schwartz function on $\mathbb{R}$ and the theta sum $\theta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t}$; Poisson summation is the identity $\theta(t) = t^{-1/2}\theta(1/t)$. No

finite Zhu S-matrix or rational-VOA modularity is used in the Hochschild trace. Tate (1950) identified Λ_ζ with the adelic zeta integral; the present step repackages Tate's computation as a Hochschild-trace identity on $\mathcal{V}^{\mathrm{prim}}$, with VOA language confined to H_∞. Source: J. T. Tate, thesis, Princeton 1950 (reprinted in Cassels–Fröhlich, "Algebraic Number Theory", Academic Press 1967).

Step 3: Petersson positivity

Definition 74.3.8 *(adelic Petersson form).* Fix a unitary central character and quotient Tamagawa measure dh on $[GL_2] = Z(\mathbb{A})GL_2(\mathbb{Q})\backslash GL_2(\mathbb{A})$. On the corresponding cuspidal L^2 space set

$$\langle v, w\rangle_{\mathrm{Pet}} = \int_{[GL_2]} v(h)\overline{w(h)}\,dh.$$

For a proposed complex realization E of the primary Hochschild trace sector, a Petersson pairing on E requires a complex-linear map $I : E \to L^2_{\mathrm{cusp}}([GL_2], dh)$.

Theorem 74.3.9 *(positive pullback and Koszul comparison).* The pullback $\langle Iv, Iw\rangle_{\mathrm{Pet}}$ is positive semidefinite, and is definite precisely when I is injective. If $B_{\mathrm{Kos}} : E \times E \to \mathbb{C}$ is bilinear and $\sigma : E \to E$ is an antilinear involution, the identity

$$B_{\mathrm{Kos}}(v, \sigma w) = \langle Iv, Iw\rangle_{\mathrm{Pet}}$$

transports this positivity to the specified Koszul pairing.

Proof. Its diagonal is $\|Iv\|_2^2$, which vanishes exactly on $\ker I$. The displayed equality then gives the last assertion. □

Remark 74.3.10 *(local and global realization data).* For a restricted Hilbert tensor product with unit reference vectors e_p, the inner product on elementary tensors is

$$\langle v, w\rangle = \langle v_\infty, w_\infty\rangle \prod_p \langle v_p, w_p\rangle,$$

where all but finitely many factors equal one. This defines a positive Hilbert completion. Identifying it with E and with the adelic Petersson space requires local maps, normalizations on e_p, and the global pairing identity above. The primary Hochschild trace class is not itself a space of functions on $[GL_2]$. Its degree, completion, realization map, and pairing are the additional data specified in Chapter 78.5.4.

Step 4: Bost–Connes and the Hilbert–Pólya Boundary

Definition 74.3.11 *(Bost–Connes system and KMS convention).* Let $D = C^*(\mathbb{Q}/\mathbb{Z})$, with group unitaries $e(r)$, and put

$$\alpha_n(e(r)) = \frac{1}{n}\sum_{nt=r} e(t), \qquad n \in \mathbb{N}^\times.$$

The Bost–Connes algebra is the semigroup crossed product $\mathcal{A}_{\mathrm{BC}} = D \rtimes_\alpha \mathbb{N}^\times$. Its multiplicative generators μ_n are isometries, with

$$\mu_n^*\mu_n = 1, \quad \mu_m\mu_n = \mu_{mn}, \quad \mu_n d\mu_n^* = \alpha_n(d), \quad \mu_n^* e(r)\mu_n = e(nr).$$

The time evolution fixes D and satisfies $\sigma_t(\mu_n) = n^{it}\mu_n$. For $\beta > 0$ and entire analytic elements the KMS_β identity is $\varphi(ab) = \varphi(b\sigma_{i\beta}(a))$. In particular $\varphi(\mu_n\mu_n^*) = n^{-\beta}$ whereas $\varphi(\mu_n^*\mu_n) = 1$; for $n > 1$ the KMS state is nontracial and μ_n is not unitary. The crossed-product description is that of Laca–Raeburn, arXiv:math/9911134, §2, p. 3.

At $\beta = 1$ let φ_1 be the unique KMS state and (π, H, Ω) its GNS representation. Its restriction to D is $\varphi_1(e(r)) = \delta_{r,0}$; see Connes, *Noncommutative Geometry*, V §11, Theorem 5(a), p. 528. Invariance and the KMS identity give, on the dense monomial algebra,

$$\varphi_1(\mu_a e(r) \mu_b^*) = \delta_{a,b}\, a^{-1} \delta_{r,0}.$$

Indeed unequal a, b give a nontrivial time character, while for $a = b$ one moves μ_a around the state by the KMS identity. For this canonical state its normal extension to $M = \pi(\mathcal{A}_{\mathrm{BC}})''$ is faithful, and Ω is cyclic and separating. We use its Tomita modular operator Δ_1, with support 1 in M and convention

$$\Delta_1^{it}\, \pi(a)\, \Delta_1^{-it} = \pi(\sigma_{-t}(a)), \qquad \Delta_1^{it}\Omega = \Omega.$$

This sign is the canonical Hecke-state convention in Connes, V §11, Proposition 4, p. 527. In vectors below, $a\Omega$ means $\pi(a)\Omega$.

THEOREM 74.3.12 *(the full critical-temperature modular generator).* Define $N_{\mathrm{BC}} = -\frac{1}{2}\log \Delta_1$ by self-adjoint functional calculus. There is an orthogonal Hilbert sum

$$H = \widehat{\bigoplus}_{q \in \mathbb{Q}_{>0}} H_q, \qquad N_{\mathrm{BC}}|_{H_q} = \tfrac{1}{2}\log q,$$

where every H_q has countably infinite dimension. Its maximal domain is

$$\operatorname{Dom} N_{\mathrm{BC}} = \left\{ \xi = \sum_q \xi_q : \sum_q (\log q)^2 \|\xi_q\|^2 < \infty \right\}.$$

Its point spectrum is $\frac{1}{2}\log \mathbb{Q}_{>0}$ and its closed spectrum is $\mathbb{R}$. All spectral measures are pure point.

Proof. Each monomial $x = \mu_a e(r) \mu_b^*$ is entire analytic with $\sigma_t(x) = (a/b)^{it} x$. The modular identity therefore gives $\Delta_1^{it} x \Omega = (a/b)^{-it} x \Omega$. Take H_q to be the closed span of these eigenvectors with $a/b = q$. Different eigenvalues are orthogonal, and the monomials are dense in the GNS space. Functional calculus gives the displayed direct sum and maximal domain. Every $q = a/b$ occurs, since

$$\|\mu_a \mu_b^* \Omega\|^2 = \varphi_1(\mu_b \mu_b^*) = b^{-1} > 0.$$

The closed spectrum is the closure of these diagonal eigenvalues, which is $\mathbb{R}$ because the positive rationals are dense. The diagonal spectral projections make every spectral measure atomic on that countable set. For multiplicity fix a prime p, and set $P_k = \mu_{p^k} \mu_{p^k}^*$ and $d_k = P_k - P_{k+1}$, $k \geq 0$. The P_k are nested fixed-point projections; hence the d_k are orthogonal projections with $\|d_k \Omega\|^2 = p^{-k} - p^{-k-1} > 0$. For fixed-point elements x, y, cancellation of $\mu_a^* \mu_a$ and the KMS identity give

$$\varphi_1\big((\mu_a y \mu_b^*)^* (\mu_a x \mu_b^*)\big) = b^{-1} \varphi_1(y^* x).$$

Thus $\mu_a d_k \mu_b^* \Omega$ are infinitely many nonzero orthogonal vectors in $H_{a/b}$. The algebra, and hence H, is separable, so its eigenspace dimensions are countably infinite. □

PROPOSITION 74.3.13 *(a reducing occupation subspace).* For an integer $s \geq 1$ put

$$K_s = \overline{\operatorname{span}}\{\mu_{m^s}\Omega : m \in \mathbb{N}^{\times}\}.$$

The displayed vectors are an orthonormal basis and K_s reduces N_{BC}. The unitary $V_s : \ell^2(\mathbb{N}^\times) \to K_s$, $V_s \delta_m = \mu_{m^s}\Omega$, satisfies

$$V_s^*(N_{\mathrm{BC}}|_{K_s})V_s = \tfrac{s}{2} H_{\mathrm{occ}}, \qquad (H_{\mathrm{occ}}c)_m = (\log m)c_m,$$

including the domain identity, where

$$\mathrm{Dom}\, H_{\mathrm{occ}} = \{c : \sum_m (\log m)^2 |c_m|^2 < \infty\}.$$

Thus K_1 gives half the occupation energy and K_2 gives exactly $\log m$.

Proof. Each vector has norm one because μ_{m^s} is an isometry. For $m \neq n$, invariance of φ_1 gives $\varphi_1(\mu_{m^s}^* \mu_{n^s}) = 0$, since $(n/m)^{ist}$ is a nontrivial time character. These orthonormal eigenvectors have energies $(s/2)\log m$. Their closed span reduces every spectral projection of N_{BC}, proving reduction and the maximal diagonal domain formula. □

Proposition 74.3.14 *(compressed shifts and the observable obstruction).* Let P_s be the orthogonal projection onto K_s, and set $T_n = \pi(\mu_{n^s})|_{K_s}$. Under V_s, these operators and their Hilbert adjoints act as

$$V_s^* T_n V_s \delta_m = \delta_{nm}, \qquad V_s^* T_n^* V_s \delta_m = \begin{cases} \delta_{m/n}, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

The adjoint here is $T_n^* = P_s \pi(\mu_{n^s}^*)|_{K_s}$. The original adjoints do not preserve K_s, and compression of the BC algebra to K_s is not a $*$-representation.

Proof. The semigroup identity gives $\mu_{n^s}\mu_{m^s}\Omega = \mu_{(nm)^s}\Omega$. Taking inner products of these basis vectors gives the adjoint formula. For $n > 1$ the vector $\mu_{n^s}^*\Omega$ has norm squared n^{-s} and strictly negative modular energy, so it is nonzero and orthogonal to K_s. In particular it cannot be the zero compressed adjoint $T_n^*\Omega$. For the group unitaries, time invariance makes the off-diagonal matrix coefficients of $e(r)$ between the K_s basis vectors vanish. The transfer relation and the state on D give the diagonal entries:

$$P_s \pi(e(r))|_{K_s} \mu_{m^s}\Omega = \mathbf{1}_{\{m^s r = 0 \text{ in } \mathbb{Q}/\mathbb{Z}\}} \mu_{m^s}\Omega.$$

At $r = 1/2$ the compressed operator kills Ω, although $e(r)$ is unitary. Thus the unital compression preserves adjoints but does not preserve products, and is not a representation of the BC algebra. □

Writing $m = \prod_p p^{k_p}$ identifies the occupation basis with finitely supported nonnegative prime occupations. On their finite linear span,

$$N_{\mathrm{BC}}|_{K_s} = \frac{s}{2} \sum_p (\log p) K_p, \qquad K_p \mu_{m^s}\Omega = k_p \mu_{m^s}\Omega.$$

The closure is the maximal diagonal operator already specified. This identifies a Hilbert space and its occupation generator; it does not turn the compressed BC observables into a BC representation or into bosonic creation operators.

Remark 74.3.15 *(occupation trace and the Hilbert–Pólya boundary).* On the reducing subspace K_2 the spectral theorem gives

$$\mathrm{Tr}_{K_2}(e^{-tN_{\mathrm{BC}}}) = \sum_{m\geq 1} m^{-t} = \zeta(t), \qquad t > 1.$$

At $t = 1$ this trace diverges. On the full GNS space the exponential $e^{-tN_{\mathrm{BC}}}$ is unbounded for every $t > 0$, since its spectrum contains all negative rational logarithmic energies. There is also a separate standard BC representation on $\ell^2(\mathbb{N}^\times)$, with $\pi_0(\mu_n)\delta_m = \delta_{nm}$ and $\pi_0(e(r))\delta_m = e^{2\pi i m r}\delta_m$. The root-of-unity average in α_n is zero unless $n \mid m$, and otherwise is $e^{2\pi i(m/n)r}$, proving the covariance relation. The occupation Hamiltonian implements σ_t there and has partition function $\mathrm{Tr}(e^{-tH_{\mathrm{occ}}}) = \zeta(t)$ for $t > 1$. This is the type-I Gibbs realization underlying Connes, V §11, Theorem 5(b), p. 528; it is not the compression of π to K_2, whose $e(r)$ matrices were computed above. The occupation trace is consequently not the full modular trace and not the signed Weil functional. A comparison with Chapter 74.2.3, including the arithmetic question of Chapter 77.4.3, must supply an actual test-algebra map, its involution and complete $P - W_\infty - W_{\mathrm{fin}}$ pairing, an operator-domain comparison, and the normalized zero divisor with its multiplicities. The reducing subspace and its zeta trace do not supply those maps.

Restricted Weil pairings

Definition 74.3.16 *(compact coefficient multiplier).* For $f \in \mathscr{A}$, define

$$U(f)(z) = \sum_{n\geq 1} f(\log n) n^{1/2} e^{2\pi i n z}.$$

This is a finite holomorphic Fourier polynomial with exponential decay as $\Im z \to \infty$.

PROPOSITION 74.3.17 *(modularity of the compact multiplier).* A nonzero $U(f)$ cannot be a cusp form of positive weight for $SL_2(\mathbb{Z})$.

Proof. A finite Fourier polynomial is analytic in y at $z = iy = 0$. The modular transformation $z \mapsto -1/z$ would imply $U(f)(iy) = O(y^{-k}e^{-2\pi/y})$. All its Taylor coefficients would vanish. This is the compact multiplier obstruction of Chapter 78.4.3. □

THEOREM 74.3.18 *(restricted positive realizations).* Let $D \subset \mathscr{A}$ be a complex vector subspace. Put $q_D(f,g) = W(f * g^*)$ for $f, g \in D$. A linear map $J : D \to K$ to a positive Hilbert space satisfying

$$\langle Jf, Jg\rangle_K = q_D(f,g) = P(f * g^*) - W_\infty(f * g^*) - W_{\mathrm{fin}}(f * g^*)$$

exists exactly when q_D is positive semidefinite. When it exists, the completion of $D/\{f : q_D(f,f) = 0\}$ gives its minimal realization.

Proof. A Hilbert realization gives positivity. Conversely the Cauchy–Schwarz inequality for a positive semidefinite Hermitian form makes its nullspace orthogonal to D. The quotient is an inner-product space, and its completion provides J. □

COROLLARY 74.3.19 *(restricted positivity from a native pairing).* Suppose $T : D \to E$ and $I : E \to L^2_{\mathrm{cusp}}([GL_2], dh)$ are linear maps with $W(f * g^*) = \langle ITf, ITg\rangle_{\mathrm{Pet}}$ for $f, g \in D$. Then $W(f * f^*) \geq 0$ on D.

Proof. Take $g = f$. The resulting value is $\|ITf\|_2^2$. □

The construction of T, I and this equality is the restricted arithmetic comparison problem. Neither the multiplier U nor moment conditions supply these maps. A restricted Li trace additionally requires explicit Li test approximants and convergence of their zero sums. The rational Li tests are not compact Fourier tests, by Chapter 78.6.5.

74.4 The restricted test class and its closure

Theorem 74.4.1 *(closure of restricted Weil positivity).* Under the preceding pairing hypothesis, q_W is nonnegative on the closure of D in $\mathscr{A}$. If that closure is all of $\mathscr{A}$, RH follows. The same assertion holds in the exponential space $\mathscr{G}$ of Chapter 95.1.2.

Proof. Convolution, involution, and W are continuous on these spaces. Thus $f \mapsto q_W(f, f)$ is continuous and its nonnegative locus is closed. Apply Weil's criterion when the closure is the full space. □

Remark 74.4.2 *(domain and spectrum).* The real even moment class $\mathscr{A}_{13}$ of Chapter 78.3 is a proper closed subspace of the real even compact tests. Increasing a modular weight does not remove its moment equations. An extension through Maass, residual, or Eisenstein sectors requires maps into the specified spectral summands and the same full pairing equality. A separate comparison is required for each sector. None is logically necessary for an independent proof of the full Weil inequality.

74.5 Restricted-Class Comparison

The restricted-class comparison partitions the conditions of Chapter 74.4.2 into related mathematical comparisons.

Condition	Reduction
F1.a.1 Banerjee–Clozel ideal	Chapter 70
F1.a.2 small-prime extension ($p = 2, 3$)	Chapter 70
F1.a.3.A block-Hochschild comparison	Chapter 71
F1.a.3.B Paskunas Ext^1 vanishing	Chapter 71
F1.a.3.C wild-EG compact generation ($p = 2, 3$)	Chapter 72
$\mathrm{L}_{\mathrm{ACD}}$ adelic categorical descent	Chapter 70
BC-diamond-glue (Ch1) Chenevier pseudochar	Chapter 73
BC-diamond-glue (AG.det) Arinkin–Gaitsgory split	Chapter 73
F2.c restricted-tensor E_2 transport	Chapter 73
Arith. Positselski: continuous duals	Chapter 75
Arith. Positselski: derived completion	Chapter 75
Arith. Positselski: detection on complete cofibers	Chapter 75
Arith. Positselski: Koszul-dual BC	Chapter 75
Full $\mathcal{V}^{\mathrm{prim}}$ P3: non-tempered (Ramanujan)	Chapter 76
Full $\mathcal{V}^{\mathrm{prim}}$ P3: Eisenstein density correction	Chapter 76
H-P^{Ar} (W) weight existence	Chapter 77
(H) Weil formula at Gaussians	Chapter 80
$\mathrm{R}_{\mathrm{analytic}}$ density-of-zeros frontier	Chapter 80
R1.formal.b/c formal-Noetherian residuals	Chapter 80
VB* main direction primary Fock	Chapter 79
Restricted Weil pairing realization	Chapter 74.3.18

74.6 The full Weil cone

THEOREM 74.6.1 *(full-domain Weil positivity).* RH is equivalent to

$$W(f * f^*) \geq 0 \qquad (f \in \mathscr{A}).$$

It is also equivalent to existence of a positive Hilbert realization of this full form. A realization through a specified arithmetic sector requires the native maps and pairing equality in addition.

Proof. Use Weil's criterion and Chapter 74.3.18 with $D = \mathscr{A}$. □

Remark 74.6.2 *(domain of the reduction).* An identity on a proper restricted class does not supply this full-domain assertion. Density must use a topology in which $P, W_\infty, W_{\mathrm{fin}}$ and convolution are continuous. The compact and exponential topologies above have that property.

74.7 Classical boundary

The functional equation, Hochschild character, Koszul pairing, Bost–Connes dynamics, and coefficient multipliers address different comparison problems. The coefficient construction gives a finite holomorphic series. The integral pairing is positive on its actual L^2 carrier. Their arithmetic relation requires specified maps. For a fixed restricted carrier, a proved pairing identity gives Weil positivity there and on its closure. The full Weil inequality is RH-equivalent. A full abstract Hilbert realization follows from that inequality by completion, while a realization in the specified native arithmetic sector retains its additional structural requirements.

Chapter 75

Arithmetic Positselski: Duals, Completion and Pairings

The arithmetic comparison involves continuous duals, derived completion, lifting, a character pairing and a Bost–Connes algebra model. The first three are local categorical constructions with their own coefficient and finiteness hypotheses. The last two require maps on the intended arithmetic object. The one-mode Euler product and Gaussian Mellin calculation of Chapter 66 determine a scalar; they do not supply the character or algebra comparison. We separate the explicit shift representation and a necessary same-prime relation test from the remaining arithmetic construction.

75.1 Continuous Duals

Iwasawa Algebra and Pontryagin Dual

Fix a prime p. Let $G_p = \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{ab}}/\mathbb{Q}_p)$ and

$$\Lambda_p \;:=\; \mathbb{Z}_p[[G_p]] \;=\; \varprojlim_{U} \mathbb{Z}_p[G_p/U], \tag{75.1}$$

the inverse limit over open normal subgroups $U \trianglelefteq G_p$ of finite index. By local class field theory (Serre 1962, *Corps locaux*, Ch. XIV §5), G_p is a profinite group and Λ_p is a compact topological ring in the inverse-limit topology (each $\mathbb{Z}_p[G_p/U]$ is a finite $\mathbb{Z}_p$-module in the p-adic topology). The underlying additive group is a compact topological abelian group; in the notation of Weil 1940 *L'intégration* Ch. II, $\Lambda_p \in \mathrm{LCA}$, the category of locally compact abelian groups.

Definition 75.1.1 *(continuous Pontryagin dual).* For $A \in \mathrm{LCA}$, the *continuous Pontryagin dual* is

$$A^* \;:=\; \mathrm{Hom}_{\mathrm{cts}}(A, \mathbb{R}/\mathbb{Z}),$$

the group of continuous homomorphisms into the circle, topologised by compact-open convergence. For A compact, A^* is discrete; for A discrete, A^* is compact (Pontryagin 1934, Weil 1940 Ch. II §2). For A a p-primary torsion group (a $\mathbb{Z}_p$-module in the sense that $p^n A \to 0$), A^* takes values in $\mathbb{Q}_p/\mathbb{Z}_p \subset \mathbb{R}/\mathbb{Z}$ and one writes

$$A^\vee \;:=\; \mathrm{Hom}_{\mathrm{cts}}(A, \mathbb{Q}_p/\mathbb{Z}_p),$$

the *p-adic Pontryagin dual*.

Theorem 75.1.2 *(Pontryagin double duality, LCA version).* The contravariant functor $A \mapsto A^*$ from LCA to LCA is an anti-equivalence of categories: the double-dual natural transformation $A \to A^{**}$, $a \mapsto \mathrm{ev}_a$, is an isomorphism of topological groups for every $A \in \mathrm{LCA}$. Restriction to compact abelian groups: the anti-equivalence exchanges compact and discrete. Restriction to p-primary: the anti-equivalence exchanges compact pro-p and discrete p-primary torsion.

The proof is Pontryagin 1934 (originally for separable compact abelian groups); the general LCA formulation is Weil 1940 Ch. II §5 and Folland 1995 *A Course in Abstract Harmonic Analysis* Ch. 4, Thm. 4.31 (the latter has the cleanest topological-group-valued statement).

The Continuous Dual of Λ_p

PROPOSITION 75.1.3 *(continuous dual of Iwasawa algebra).* The continuous Pontryagin dual of Λ_p is naturally isomorphic to the discrete p-primary torsion module of continuous $\mathbb{Q}_p/\mathbb{Z}_p$-valued functions on G_p:

$$(\Lambda_p)^* \;=\; \Lambda_p^\vee \;=\; C(G_p, \mathbb{Q}_p/\mathbb{Z}_p),$$

equipped with the discrete topology and the G_p-module structure induced by translation: $(g \cdot f)(h) = f(g^{-1}h)$.

Proof. The first equality $(\Lambda_p)^* = \Lambda_p^\vee$ is immediate: Λ_p is pro-p (compact pro-p group), so every continuous homomorphism $\Lambda_p \to \mathbb{R}/\mathbb{Z}$ factors through the pro-p part $\mathbb{Q}_p/\mathbb{Z}_p$ of the circle (Definition 75.1.1 ad finem).

The second equality $\Lambda_p^\vee = C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)$ is the content. Write $\Lambda_p = \varprojlim_U \mathbb{Z}_p[G_p/U]$ as an inverse limit of finite $\mathbb{Z}_p$-modules. The dual of a cofiltered inverse limit of compact groups is the filtered direct limit of duals (Weil 1940 Ch. II §5 Prop. 7; Bourbaki *TG* III §3.5 Prop. 7):

$$\Lambda_p^\vee \;=\; \Big(\varprojlim_U \mathbb{Z}_p[G_p/U]\Big)^\vee \;=\; \varinjlim_U \mathbb{Z}_p[G_p/U]^\vee.$$

At each finite level, $\mathbb{Z}_p[G_p/U]$ is a finitely generated free $\mathbb{Z}_p$-module of rank $|G_p/U|$; its p-adic Pontryagin dual is

$$\mathbb{Z}_p[G_p/U]^\vee \;=\; \mathrm{Hom}(\mathbb{Z}_p[G_p/U], \mathbb{Q}_p/\mathbb{Z}_p) \;=\; \mathrm{Maps}(G_p/U, \mathbb{Q}_p/\mathbb{Z}_p),$$

the group of all maps on the finite coset space G_p/U with values in $\mathbb{Q}_p/\mathbb{Z}_p$; no continuity hypothesis because G_p/U is finite discrete. Taking the direct limit over $U \trianglelefteq G_p$:

$$\varinjlim_U \mathrm{Maps}(G_p/U, \mathbb{Q}_p/\mathbb{Z}_p) \;=\; \mathrm{Maps}_{\mathrm{loc.cst.}}(G_p, \mathbb{Q}_p/\mathbb{Z}_p) \;=\; C(G_p, \mathbb{Q}_p/\mathbb{Z}_p),$$

the space of locally constant $\mathbb{Q}_p/\mathbb{Z}_p$-valued functions on G_p. The last equality uses that on a profinite group, continuous = locally constant for $\mathbb{Q}_p/\mathbb{Z}_p$-values (compactness plus discreteness of the target: every continuous function from a compact to a discrete space has finite image and therefore factors through a finite quotient, Bourbaki *TG* I §10 Prop. 8).

The G_p-action on Λ_p by right translation induces an action on $\Lambda_p^\vee$ by the dual (left) translation, recovering the standard G_p-module structure on $C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)$. □

Compact Modules and Discrete Comodules for the Dual Tensor Product

Let $D = \mathbb{Q}_p/\mathbb{Z}_p$. Compact $\mathbb{Z}_p$-modules and discrete p-primary torsion modules are exchanged by $M \mapsto M^\vee = \mathrm{Hom}_{\mathrm{cts}}(M, D)$. On compact modules use the completed tensor product, the inverse limit of tensor products of finite quotient modules. On discrete modules define

$$M \odot N = (M^\vee \mathbin{\widehat{\otimes}_{\mathbb{Z}_p}} N^\vee)^\vee.$$

The unit for $\odot$ is D, since $D^\vee = \mathbb{Z}_p$. Associativity, symmetry and the unit maps are the duals of those for the completed tensor product, using the natural double-dual isomorphisms.

Theorem 75.1.4 *(continuous duality with its monoidal carrier).* Let Λ be a compact unital $\mathbb{Z}_p$-algebra with continuous multiplication, and put $C = \Lambda^\vee$. Multiplication and unit dualize to

$$\Delta : C \longrightarrow C \odot C, \qquad \varepsilon : C \longrightarrow D.$$

These make C a counital coalgebra for $\odot$. Continuous compact right Λ-modules are contravariantly equivalent to discrete right C-comodules for $\odot$, by exact Pontryagin duality. For $\Lambda = \Lambda_p, C = C(G_p, D)$. On bounded derived categories the equivalence is

$$\mathrm{D}^b(\mathrm{Mod}^{\mathrm{comp}}\text{-}\Lambda_p) \simeq \mathrm{D}^b(\mathrm{Comod}_\odot(C(G_p, D)))^{\mathrm{op}}. \tag{75.2}$$

This statement concerns the dual tensor product, not ordinary $\mathbb{Z}_p$-coalgebras.

Proof. At finite p-group level evaluation identifies every module with its double dual. Pontryagin duality exchanges the inverse limits of finite compact quotients with the direct limits of their discrete duals. It is exact: at finite level a character on a subgroup extends because D is divisible, and passage through these dual limits preserves the resulting short exact sequences.

Dualize associativity and the unit identities of Λ. The resulting diagrams are precisely coassociativity and the counit identities for C with respect to $\odot$. For a compact right module M, dualizing its action gives

$$(M \widehat{\otimes} \Lambda \longrightarrow M)^\vee : M^\vee \longrightarrow M^\vee \odot C.$$

The module axioms become the comodule axioms. Conversely, dualize a coaction and use double duality to recover the continuous action. The two constructions are inverse on objects and morphisms, so give an exact anti-equivalence.

For a bounded cochain complex $M^\bullet$, use $(M^\vee)^n = (M^{-n})^\vee$ with differential $d(f) = (-1)^{n+1} f\, d_M$. Exactness gives $H^n(M^\vee) = H^{-n}(M)^\vee$, so acyclic complexes and quasi-isomorphisms are preserved. Passing to the bounded derived categories proves (75.2). □

Remark 75.1.5 *(ordinary tensor obstruction and reduction modulo p).* Already for $\Lambda = \mathbb{Z}_p$, the dual is D, and $D \otimes_{\mathbb{Z}_p} D = 0$: if $p^n x = 0$, write $y = p^n z$, so $x \otimes y = p^n x \otimes z = 0$. Also $\mathrm{Hom}_{\mathbb{Z}_p}(D, \mathbb{Z}_p) = 0$. Thus this nonzero module admits no ordinary counital $\mathbb{Z}_p$-coalgebra structure. The counit above has target D, the unit of the correct dual monoidal category.

For a compact module there is the elementary natural identity

$$(M/pM)^\vee \cong M^\vee[p].$$

A character factors through M/pM exactly when it is killed by p. This degree-zero identity supplies neither a derived base-change equivalence nor the ordinary field-coalgebra comodule–contramodule correspondence. To use continuous duality in the arithmetic Positselski comparison of Remark 46.1.6, one must construct the appropriate completed monoidal carrier, its bar and cobar functors, and their compatibility with the stated Selmer and pairing data. The compact/discrete duality above is a concrete input to that construction, not its completion.

75.2 Derived Completion

Derived Completion over a Noetherian Ring

Let R be a commutative Noetherian ring and $I \subset R$ an ideal. For $M \in \mathrm{D}(R)$, put

$$\mathrm{L}\Lambda_I(M) = \mathrm{Rlim}_n\big(M \otimes^{\mathbb{L}}_R R/I^n\big).$$

The tensor product and inverse limit are both derived. A complex N is derived I-complete when $\mathrm{RHom}_R(R[1/f], N) = 0$ for every $f \in I$. For the Noetherian ring considered here, the displayed construction is the reflection into these complexes. Without the Noetherian hypothesis, that identification needs additional conditions on the ideal.

THEOREM 75.2.1 *(Noetherian derived completion).* The functor $\mathrm{L}\Lambda_I$ is left adjoint to the inclusion of derived I-complete complexes in $\mathrm{D}(R)$. If every $H^j(M)$ is finitely generated over R, then

$$H^j(\mathrm{L}\Lambda_I M) \cong \varprojlim_n H^j(M)/I^n H^j(M).$$

In particular, the natural map from M to its derived completion is an equivalence when these finite cohomology modules are already I-adically complete.

Proof. The adjunction is Stacks Project, Tag 0922. Its proof compares the towers of Koszul complexes and ideal quotients using the Noetherian hypothesis.

The cohomology assertion is Tag 0A06. For a bounded above complex with finite cohomology, choose finite free resolutions. Completion is exact on the finite modules occurring in their kernels and images, by Artin–Rees. It therefore commutes with their cohomology. Truncation and the finite cohomological dimension of derived completion give the stated unbounded version. □

This theorem concerns cohomology modules and derived quotients. It does not identify completion of every chosen complex term by term with its derived completion. The following calculation records exactly the inverse-limit term that must be controlled.

Tor Amplitude and Reduction Towers

Definition 75.2.2 *(bounded Tor amplitude).* A complex $A \in \mathrm{D}(R)$ has *bounded Tor amplitude* if there are integers $a \le b$ such that $H^j(A \otimes^{\mathbb{L}}_R N) = 0$ outside $[a, b]$ for every R-module N in degree zero. For an ordinary flat algebra A, this interval is $[0, 0]$. For a dg algebra the condition refers to its underlying complex. It does not assert a Tor bound for every A-module.

THEOREM 75.2.3 *(cohomology of a reduction tower).* For $M \in \mathrm{D}(R)$, write $K_n = M \otimes^{\mathbb{L}}_R R/I^n$. There is a natural exact sequence

$$0 \longrightarrow \varprojlim_n{}^1 H^{j-1}(K_n) \longrightarrow H^j(\mathrm{L}\Lambda_I M) \longrightarrow \varprojlim_n H^j(K_n) \longrightarrow 0.$$

If the transition maps on $H^{j-1}(K_n)$ are surjective, the left term vanishes. More generally it vanishes under the Mittag–Leffler condition: for each fixed n, the images from all sufficiently large indices in $H^{j-1}(K_n)$ are equal.

Proof. Choose a complex model of the tower. Its derived limit has the defining triangle

$$\mathrm{Rlim}_n K_n \longrightarrow \prod_n K_n \xrightarrow{1-s} \prod_n K_n \longrightarrow (\mathrm{Rlim}_n K_n)[1],$$

where the n-th component of s is the transition map applied to the $(n+1)$-st component. Products of R-modules are exact, so the cohomology of these products is the product of their cohomologies. The long exact sequence identifies the outer terms with the kernel and cokernel of $1 - s$. These are $\varprojlim$ and $\varprojlim{}^1$, respectively.

For a surjective tower, one solves $(1 - s)x = y$ recursively, choosing each next component by surjectivity. Thus its cokernel vanishes. The same vanishing for a Mittag–Leffler tower is Stacks Project, Tag 07KV, Lemma 15.88.1. Consequently, under the stated hypothesis,

$$H^j(\mathrm{L}\Lambda_I M) \cong \varprojlim_n H^j(K_n).$$

□

Flatness of the coefficient algebra does not imply surjectivity on this cohomology tower. Take $R = A = \mathbb{Z}_p$ and $M = [R \xrightarrow{p} R]$, in degrees zero and one. Then

$$H^0(M \otimes^{\mathbb{L}}_R R/p^n) = p^{n-1}R/p^n \cong \mathbb{F}_p.$$

Every transition on these groups is zero, hence is not surjective. The tower is nevertheless Mittag–Leffler, with eventual image zero. This example separates surjectivity from the weaker condition actually used in the theorem.

Ordinary quotients can also miss the entire answer. For $D = \mathbb{Q}_p/\mathbb{Z}_p$, multiplication by p^n is surjective with kernel $D[p^n] \cong \mathbb{Z}/p^n$. Thus $D/p^n D = 0$, whereas

$$D \otimes^{\mathbb{L}}_{\mathbb{Z}_p} \mathbb{Z}/p^n \simeq D[p^n][1], \qquad \mathrm{L}\Lambda_{(p)} D \simeq \mathbb{Z}_p[1].$$

The transition on the kernels is multiplication by p. Writing an element as a/p^n identifies this transition with reduction $\mathbb{Z}/p^{n+1} \to \mathbb{Z}/p^n$. Its inverse limit is $\mathbb{Z}_p$, and its first derived limit vanishes by surjectivity. The coefficient algebra $A = \mathbb{Z}_p$ is flat in this example as well.

Remark 75.2.4 *(the arithmetic bar requires its coefficient model).* To apply these results to an arithmetic bar construction, one must specify its complex over the chosen Noetherian ring. Its residue, differential, and completed tensor operations must be actual maps on that complex. Torsion-freeness over $\mathbb{Z}_p$ does not imply flatness over $\mathbb{Z}_p[[T]]$: the module $\Lambda/(T)$ has $\mathrm{Tor}^{\Lambda}_1(\Lambda/(T), \Lambda/(T)) = \Lambda/(T)$. The arithmetic coefficient modules cannot be replaced by free oscillator coefficients without a comparison.

For a chosen procyclic $\Gamma \cong \mathbb{Z}_p$, $\mathbb{Z}_p[[\Gamma]]/p \cong \mathbb{F}_p[[T]]$. The full local group G_p in (75.1) is a different input. A map to a procyclic quotient does not identify its completed group algebra with the one-variable power-series ring.

Corollary 75.2.5 *(complete finite coefficient complexes).* Let $R = \mathbb{Z}_p[[T]]$, and suppose $M \in \mathrm{D}(R)$ has finitely generated cohomology in every degree. Then $M \to \mathrm{L}\Lambda_{(p)} M$ is an equivalence. The same holds for its derived reduction $M \otimes^{\mathbb{L}}_R R/p$.

Proof. The ring R is Noetherian and p-adically complete. Finite R-modules are therefore p-adically complete. Apply Theorem 75.2.1 to the cohomology of M. The two-term free resolution of R/p shows that the reduced complex also has finite cohomology in every degree. The same theorem applies to it. □

For a coderived or contraderived arithmetic recipient, completion must instead be checked internally, as in Theorem 46.1.5. A comparison with ordinary derived complexes must detect the unit and counit cofibers before the preceding corollary can be used there. These remain requirements of Remark 46.1.6(2).

75.3 Reduction and Lifting

Matlis Duality on the Compact Heart

Let R be a complete Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k = R/\mathfrak{m}$. The *Matlis dual* of an R-module M is

$$M^{\mathrm{M}} := \mathrm{Hom}_R(M, E(k)),$$

where $E(k)$ is the injective hull of k as R-module (Matlis 1958 Pacific J. Math. 8, §3).

THEOREM 75.3.1 *(Matlis duality).* Let R be a complete Noetherian local ring with residue field k. The Matlis dual $(-)^{\mathrm{M}}$ is a contravariant exact functor $R\text{−mod} \to R\text{−mod}$, and restricts to an anti-equivalence between the categories of finitely generated (Noetherian) R-modules and Artinian R-modules. $(-)^{\mathrm{MM}}$ is the identity on both categories: finitely generated modules are reflexive, and Artinian modules are reflexive, under Matlis dualisation.

The proof is Matlis 1958 Thm. 4.2 and Thm. 5.2; modern accounts in Bruns–Herzog 1993 *Cohen–Macaulay Rings* Ch. 3 §2, and Enochs–Jenda 2000 *Relative Homological Algebra* Ch. 5.

Lifting from Mod-p to Characteristic Zero

Let $\Lambda_p \to \mathbb{F}_p$ be the augmentation to the residue field (reduction modulo p combined with augmentation of the group algebra). A *mod-p representation* of Λ_p is an $\mathbb{F}_p$-vector space $\overline{V}$ with a continuous Λ_p-action factoring through $\mathbb{F}_p$. A *lift* of $\overline{V}$ to characteristic zero is a continuous Λ_p-module V with $V \otimes_{\Lambda_p} \mathbb{F}_p = \overline{V}$.

THEOREM 75.3.2 *(conservativity on a regular local coefficient model).* Let $R = \mathbb{Z}_p[[T]]$, with maximal ideal $\mathfrak{m} = (p, T)$ and residue field $k = \mathbb{F}_p$. On bounded complexes with finitely generated cohomology,

$$- \otimes^{\mathbb{L}}_R k : \mathrm{D}^b_{\mathrm{fg}}(R) \longrightarrow \mathrm{D}^b_{\mathrm{fg}}(k)$$

is conservative: it detects equivalences. It is not faithful on morphisms.

Proof. The resolution bound of Proposition 46.3.3 makes these complexes perfect, so the displayed functor has its stated target. Let K be the cone of a map whose reduction is an equivalence. If $K \neq 0$, choose the largest n with $H^n(K) \neq 0$. The top cohomology of $K \otimes^{\mathbb{L}}_R k$ is $H^n(K) \otimes_R k$. Its vanishing contradicts Nakayama's lemma for the nonzero finite module $H^n(K)$. Thus $K = 0$.

For failure of faithfulness, multiplication by p on R is nonzero and becomes zero after reduction. □

This is a theorem about the specified regular local ring. Applying it to an arithmetic completed group algebra requires its actual coefficient model and a comparison of recipients. Its residue field is obtained by killing both p and T. Under Matlis duality, the dual of $M/\mathfrak{m}M$ is the $\mathfrak{m}$-annihilated submodule, not merely the p-torsion submodule.

COROLLARY 75.3.3 *(detection for a supplied comparison map).* A supplied map in $\mathrm{D}^b_{\mathrm{fg}}(R)$ is an equivalence if its derived reduction to k is. For intrinsic derived p-complete recipients, the corresponding criterion is Proposition 46.4.5.

Proof. Apply the preceding theorem to the map in the first case, and the cited proposition to its complete cofiber in the second. □

Neither statement constructs a lift of an arbitrary residual representation. For instance, $R/(T)$ and $R/(T-p)$ are distinct $\mathbb{Z}_p$-free lifts of the same module modulo p. Completion of an inflated residual module also does not remove its p-torsion. The arithmetic adjunction and the comparison of its reduced unit and counit remain required in Remark 46.1.6(3).

75.4 Koszul-Dual Bost–Connes

Bost–Connes System

On $\mathcal{K} = \ell^2(\mathbb{N}_{>0})$ define $\mu_n \delta_N = \delta_{nN}$ and $e(r)\delta_N = \exp(2\pi i r N)\delta_N$ for $r \in \mathbb{Q}/\mathbb{Z}$. Use the concrete operator algebra

$$\mathcal{A}_{\mathrm{rep}} = C^*(\mu_n, e(r) : n \geq 1, r \in \mathbb{Q}/\mathbb{Z}) \subset \mathcal{B}(\mathcal{K}). \tag{75.3}$$

The shifts are isometries, $\mu_m \mu_n = \mu_{mn}$, and

$$\mu_n^* e(r) \mu_n = e(nr), \qquad \mu_n e(r) \mu_n^* = \frac{1}{n} \sum_{ns=r} e(s).$$

The first identity follows on each δ_N; for the second the sum of the n roots of unity vanishes unless $n \mid N$, in which case both sides multiply δ_N by $\exp(2\pi i r N/n)$. For $H\delta_N = \log N\, \delta_N$, conjugation by e^{itH} fixes $e(r)$ and sends μ_n to $n^{it}\mu_n$. The heat trace is $\sum_N N^{-\beta} = \zeta(\beta)$ for $\beta > 1$. These formulas specify the bounded operator model needed for the local comparison. They do not identify an arithmetic cobar algebra with it, or assert a classification of KMS states or a group-algebra presentation. Write $\mathbb{A}^{\mathrm{BC}}$ for an intended Bost–Connes algebra only after specifying its presentation and representation comparison.

Adele Class Space and Enveloping Sheaf

The *adele class space* of Connes 1999 (Selecta Math. 5, §1.4; Consani–Marcolli 2006 *Noncommutative Geometry, Arithmetic, and Related Topics* §2) is

$$X_{\mathrm{AC}} := \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*,$$

the quotient of the full adele ring by the diagonal multiplicative action. Its non-Hausdorff quotient geometry is different from the occupation-basis representation of (75.3). A comparison with a noncommutative algebra model requires the specified site, algebra, representation and trace maps. A common zeta scalar does not construct a Hilbert duality between these models.

Definition 75.4.1 *(sheaf comparison data).* A sheaf comparison requires a specified site on the adele-class object, a sheaf of algebras $\mathcal{E}^{\mathrm{BC}}$ on that site, and an identification of its appropriate global sections with the chosen operator algebra. It also requires a sheaf model of the arithmetic cobar construction and a map preserving multiplication, differential, coefficients and the stated completion. These are hypotheses on a comparison. A phrase such as support on an adelic fibre does not define restriction maps or such an identification.

Koszul-Dual Description

Question 75.4.2 *(arithmetic cobar and Bost–Connes comparison).* Specify the coefficients, differential and topology of the primary arithmetic cobar algebra A_{prim}. Construct a comparison with a specified dg model of the intended Bost–Connes algebra whose chosen operator representation is compatible with (75.3). The desired quasi-isomorphism is

$$A_{\mathrm{prim}} \simeq B_{\mathrm{BC}} \qquad \text{in a common specified category of dg algebras.} \tag{75.4}$$

The problem includes coefficient comparison and completion; a C^*-algebra is not automatically a dg $\mathbb{Z}_p[[\Gamma]]$-algebra. The sheaf data in Definition 75.4.1 are additional requirements for a geometric realization of this comparison.

There is a necessary test on the proposed prime-power generator map. The equality of energies $\log(p^k) = k \log p$, using Convention 28.7.1(ii), permits the proposed association $b_{p,k} \mapsto \mu_{p^k}$. It does not prove the algebra relation. For an explicit algebraic example, let $Q = \mathbb{Q}[b_{p,k} : p \text{ prime}, k \geq 1]$. Its map to $\mathbb{Q}[t_p : p \text{ prime}]$, $b_{p,k} \mapsto t_p^k$, has kernel

$$(b_{p,k} - b_{p,1}^k : p \text{ prime}, k \geq 2).$$

Indeed quotienting by these relations leaves a polynomial ring freely generated by the classes of $b_{p,1}$; the evident inverse sends t_p to that class. Thus this map is surjective but not injective. Graded commutativity alone does not impose these relations. In a dg model one would instead need explicit boundaries or homotopies and a proof that the induced map is a quasi-isomorphism. The restricted-dual calculation $\eta_1^k = k!\eta_k$ of Theorem 46.5.1 also shows the effect of integral normalization. Phase operators, adjoints and their covariance require further generators and relations beyond the commuting shift subalgebra.

Remark 75.4.3 *(pairing compatibility).* If the comparison in (75.4) is constructed together with compatible pairing maps, those maps transport the specified pairing. This formal consequence does not construct the comparison or its compatibility. In particular, the numerical energy match does not settle the Koszul-dual Bost–Connes requirement of Remark 46.1.6(5).

75.5 Local Constructions and Arithmetic Comparison Hypotheses

Combined Theorem

Remark 75.5.1 *(distinct comparison inputs).* The continuous-dual, derived-completion and lifting constructions have their stated coefficient and finiteness hypotheses. Their application to an arithmetic object must verify those hypotheses. The scalar calculation in Chapter 66 concerns a selected one-mode Euler product and a Gaussian Mellin pairing. It does not identify the arithmetic cobar character or its pairing. The Bost–Connes comparison remains Question 75.4.2; it is not reduced to a match of generator energies or to an unnamed sheaf duality.

Arithmetic Positselski Comparison

Question 75.5.2 *(equivalence on the intended arithmetic carrier).* For the specified arithmetic dg coalgebra and completed dual, establish

$$\mathrm{D}^{\mathrm{co}}_{\Lambda_p}(C\text{-comod}) \simeq \mathrm{D}^{\mathrm{ctr}}_{\Lambda_p}(C^\vee\text{-contramod})$$

with the coefficient, topology and finiteness hypotheses required by Theorem 46.1.5. Its comparison with the Bost–Connes representation additionally requires (75.4), and the character and pairing intertwiners. Neither a scalar Euler product nor the local divisor substitution supplies these maps.

Question 75.5.3 *(central classes and Tate duality).* The intended arithmetic relation is

$$\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0 \quad \Longleftrightarrow \quad \xi(s) = \xi(1-s).$$

The right side is the classical functional equation. A proof relating it to the left side must identify the arithmetic obstruction classes under a compatible pairing. Energy-zero evaluation on a polynomial restricted dual does not provide this identification. The noncommutative Iwahori extension requires its own additional comparisons.

The Five Comparison Requirements

Input	Construction	Arithmetic requirement
(1)	Continuous duals	Specified topology and duality
(2)	Derived completion	Applicable hypotheses and base change
(3)	Lifting	Existence and the required morphism properties
(4)	Selected scalar character	Character and pairing comparison maps
(5)	Bost–Connes comparison	dg algebra, coefficients and completion

Table 75.1: The constructions required on the intended arithmetic carrier. The scalar one-mode calculation does not supply the last two comparisons.

Remark 75.5.4 *(domain and comparison).* The local compactness, completion and duality results must be used on their actual carriers. Even when those local hypotheses hold, an arithmetic identification must preserve the operators and pairings in question. These requirements remain in the abelian sector; they are not all consequences of excluding the nonabelian Iwahori case.

75.6 Bibliography

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Remark 75.6.1 *(roles of the constructions).* Continuous duals, derived completion, lifting, and arithmetic operator comparison answer different questions. A scalar identity obtained from Tate's Mellin transform and the one-mode Euler product does not establish an algebra or pairing comparison. The missing maps remain the objects of Question 75.4.2.

Chapter 76

Full $\mathcal{V}^{\mathrm{prim}}$ P3 and the Koszul–Petersson Compatibility on the Discrete Core

The prime Gibbs model of Chapter 67 constructs a state, its Hilbert–Schmidt GNS space, closed modular operators and an actual comparison with a global state's prime cyclic subspace. The Koszul–Petersson problem additionally asks for the arithmetic placement and pairing maps of Hypothesis 67.1.1. This chapter describes conditional transport across the cuspidal sectors once those maps are supplied. Classical temperedness results identify some of the relevant domains; they do not prove the pairing comparison. The full arithmetic comparison remains Question 67.1.2.

76.1 Tempered Cuspidal Pairing Comparison

Throughout the pairing consequences below, the notation $\langle v, w\rangle^{\mathrm{Pet}}$ means the pullback $\langle \iota v, \iota w\rangle^{\mathrm{Pet}}$ along the specified injective arithmetic placement. No placement is inferred from a spectral decomposition of the automorphic target.

Theorem 76.1.1 *(conditional tempered cuspidal comparison).* Assume Hypothesis 67.1.1, including all local ramified comparisons and the adelic extension, on the chosen tempered cuspidal arithmetic domain $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,temp}}$. Then

$$\langle v, w\rangle^{\mathrm{Pet}} = \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}}$$

on that domain. Temperedness alone is not the comparison hypothesis.

Proof. The local maps identify the pairings with the same normalized Hilbert forms. Their identities multiply on elementary restricted tensors and extend by the explicitly assumed continuity or regularization, as proved in Theorem 67.3.8. The prime cyclic unitary in Theorem 67.2.5 intertwines only the specified prime algebra and state. It does not replace the arithmetic or Whittaker comparison maps. □

Remark 76.1.2 *(temperedness and modular domains).* Temperedness is a representation-theoretic condition on the selected automorphic sector. It is not a boundedness hypothesis for modular theory. In the actual one-prime Gibbs GNS model, $\Delta E_{ij} = p^{-\beta(i-j)} E_{ij}$, and both Δ and its inverse are unbounded; their maximal domains are given in Lemma 67.2.3. The GNS space is $\mathrm{HS}(\ell^2(\mathbb{N}_0))$. The proved unitary reaches a prime cyclic subspace, without identifying the full critical GNS representation, the phase operators or a Petersson space with an occupation tensor product.

76.2 Non-Tempered Cuspidal Representations: Deligne and Kim–Shahidi

The Non-Tempered Locus

Cuspidal automorphic representations of $GL_2/\mathbb{Q}$ decompose as $\pi = \otimes'_v \pi_v$, with archimedean component π_∞ of one of three types: a holomorphic discrete series D_k of weight $k \geq 2$; a Maass cusp form with Laplace eigenvalue $\lambda = 1/4 + r^2$ ($r \in \mathbb{R}$ tempered, $r \in i\mathbb{R}$ non-tempered); or a character twist.

Finite places carry Satake parameters (α_p, β_p) with $\alpha_p\beta_p = 1$ (unitarity) and $\max(|\alpha_p|, |\beta_p|) = p^{\theta_p}$. Temperedness is the condition $\theta_p = 0$ at all p; the Ramanujan conjecture asserts temperedness for all cuspidal π.

- **Holomorphic sector.** $\pi_\infty = D_k$ for $k \geq 2$. Deligne 1974 *Publ. IHES* 43, La conjecture de Weil I, gives the Ramanujan bound unconditionally: $|a_p| \leq 2p^{(k-1)/2}$ for the Hecke eigenvalue a_p of a holomorphic weight-k newform, which is temperedness $|\alpha_p| = |\beta_p| = p^{(k-1)/2} \cdot 1$ after normalisation (in the analytic normalisation π unitary, $|\alpha_p| = |\beta_p| = 1$). Weight-1 holomorphic forms are covered by Deligne–Serre 1974 *Ann. Sci. ENS. 7*, via the Artin conjecture for two-dimensional odd Galois representations attached to weight-1 newforms.
- **Maass sector.** π_∞ is a Maass cusp form with complementary-series parameter $r = it$, $0 < t \leq 1/2$, at ∞; temperedness at ∞ is $t = 0$. At finite places, non-temperedness is $\theta_p > 0$. The Ramanujan conjecture for Maass forms is open: the best current bound is Kim–Sarnak 2003 Duke Math. J. 119 $\theta_p \leq 7/64$, improving earlier Luo–Rudnick–Sarnak 1995 and Kim–Shahidi 2002 *Ann. Math.* 155, which used the symmetric-square and symmetric-fourth-power lifts to bound $\theta_p \leq 1/9$ via Jacquet–Piatetski-Shapiro–Shalika 1983.

Holomorphic Temperedness and the Pairing Comparison

Theorem 76.2.1 *(conditional Koszul–Petersson comparison on the holomorphic sector).* Assume Hypothesis 67.1.1 on the stated sector, with its injective placement, local pairing maps and adelic extension. Let $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,hol}} \subset \mathcal{V}^{\mathrm{prim}}$ denote the subspace spanned by holomorphic cuspidal automorphic representations of $GL_2/\mathbb{Q}$, namely those π with $\pi_\infty = D_k$ for some $k \geq 1$. For all $v, w \in \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,hol}}$,

$$\langle v, w\rangle^{\mathrm{Pet}} \;=\; \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}}.$$

The identification is conditional on the arithmetic comparison maps.

Proof. By Deligne 1974 (weight $k \geq 2$) and Deligne–Serre 1974 (weight $k = 1$), every holomorphic cuspidal representation of $GL_2/\mathbb{Q}$ is tempered at every finite place: $|\alpha_p| = |\beta_p| = 1$ at each p. Temperedness at ∞ for $\pi_\infty = D_k$ is automatic (discrete-series representations are tempered by Harish-Chandra's definition). The holomorphic cuspidal sector is therefore a subsector of the tempered cuspidal sector: $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,hol}} \subset \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,temp}}$. Theorem 76.1.1 gives (76.2.1) under its arithmetic comparison hypotheses. Deligne's bound supplies temperedness, not those maps. □

Remark 76.2.2 *(domain of Deligne 1974: holomorphic sector only).* Deligne 1974 establishes the Ramanujan–Petersson bound for the Hecke eigenvalues of a holomorphic weight-k cusp form via the Riemann hypothesis for ℓ-adic cohomology of a suitable Kuga–Sato variety. The bound transfers to temperedness of the associated automorphic representation at every finite prime. Deligne 1974 applies to holomorphic cuspidal forms; for Maass cusp forms, no algebraic variety whose Frobenius eigenvalues give the Hecke eigenvalues is available, and the Ramanujan bound in the Maass sector is an open problem. The current state of the art on the Maass sector is Kim–Sarnak 2003 $\theta \leq 7/64$; this is a sub-convex Ramanujan bound, strictly weaker than full temperedness.

The Maass Sector: Conditional on Ramanujan

Theorem 76.2.3 *(Koszul–Petersson on the Maass cuspidal sector, conditional).* Assume Hypothesis 67.1.1 on the stated sector, with its injective placement, local pairing maps and adelic extension. *(on*

the Ramanujan bound for Maass cusp forms of $GL_2/\mathbb{Q}$). Let $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,Maass}} \subset \mathcal{V}^{\mathrm{prim}}$ denote the subspace spanned by Maass cuspidal automorphic representations of $GL_2/\mathbb{Q}$. Assume the Ramanujan bound $\theta_p = 0$ at every finite prime, and assume archimedean temperedness for every Maass form in the chosen domain. For all $v, w \in \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,Maass}}$,

$$\langle v, w\rangle^{\mathrm{Pet}} \;=\; \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}}.$$

Proof. Under the stated finite-place and archimedean temperedness assumptions, $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,Maass}} \subset \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,temp}}$; Theorem 76.1.1 then applies. □

Proposition 76.2.4 *(local error bounds and adelic summability).* The Satake exponent bound $\theta_p \le 7/64$ gives $|\alpha_p|, |\beta_p| \le p^{7/64}$ in the unitary unramified normalization. An estimate of the arithmetic pairing discrepancy such as

$$\left|\langle v_p, w_p\rangle_p^{\mathrm{Pet}} - \langle v_p, \sigma_p(w_p)\rangle_p^{\mathrm{Koszul}}\right| \le C_p\, p^{-25/64}$$

requires separate control of the actual comparison maps and their local norms. Even a uniform bound on C_p would not make this majorant summable over all primes.

Proof. The first inequality restates the exponent bound. The series $\sum_p p^{-25/64}$ diverges, since $p^{-25/64} \ge p^{-1}$ and the prime harmonic series diverges. Thus this proposed majorant cannot justify absolute adelic summability. No estimate of the displayed pairing difference is deduced from the Satake bound alone. □

Remark 76.2.5 *(the selected residual sector).* A chosen residual line with a normalized vector gives a direct one-dimensional pairing calculation once its arithmetic placement and Koszul normalization have been specified. This does not identify every residual automorphic representation with one Fock vacuum. The full discrete comparison requires a placement and orthogonal pairing decomposition for every included residual summand, in addition to the cuspidal comparisons.

76.3 The Eisenstein Density Mismatch and Domain Restriction

The Mismatch

We recall the mismatch identified in Chapter 67:

Proposition 76.3.1 *(the fixed scalar continuum models).* On the domain U of Definition 67.5.1, the proposed scalar coefficients, with $d(s) = |\xi(2s)|^{-2} > 0$, are

$$P(s) = (1 + |M(s)|^2)|\xi(2s)|^{-2} = 2d(s),$$
$$K(s) = \frac{\xi(2s)}{\xi(2s-1)}|\xi(2s)|^{-2} = M(s)^{-1}d(s).$$

They satisfy $d(s) \le |P(s) - K(s)| \le 3d(s)$. Their realization as coefficients of actual continuous pairings is an additional construction, not part of this scalar calculation.

Proof. This is Theorem 67.5.3, using the functional equation, conjugation and triangle inequalities. □

Domain Restriction

Definition 76.3.2 (discrete-spectrum core of $\mathcal{V}^{\mathrm{prim}}$). Fix the placement of Hypothesis 67.4.2 and a specified residual line with normalized vector. The *selected discrete-spectrum core* is

$$\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}} \;:=\; \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp}} \;\oplus\; \mathcal{V}^{\mathrm{prim}}_{\mathrm{res}},$$

the chosen arithmetic carrier placed inside $L^2_{\mathrm{disc}}([GL_2]) = L^2_{\mathrm{cusp}} \oplus L^2_{\mathrm{res}}$ under the placement map ι of Hypothesis 67.4.2. No full arithmetic placement into a continuous automorphic spectrum is inferred from this chosen discrete placement.

Theorem 76.3.3 *(Koszul–Petersson identity on the discrete-spectrum core).* Assume Hypothesis 67.1.1 on the stated sector, with its injective placement, local pairing maps and adelic extension. *(conditional on the arithmetic comparison maps on every sector; with the additional Ramanujan assumption on the Maass sector).* For all $v, w \in \mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$,

$$\langle v, w\rangle^{\mathrm{Pet}} \;=\; \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}}. \tag{76.1}$$

The identification holds:

- on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,temp}}$ under the comparison hypotheses (Theorem 76.1.1);
- on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,hol}} \subset \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,temp}}$ under the comparison hypotheses (Theorem 76.2.1);
- on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,Maass}}$ conditional on Ramanujan (Theorem 76.2.3);
- on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{res}}$ (one-dimensional, vacuum-generated) by direct calculation at the Fock vacuum.

Proof. $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}} = \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp}} \oplus \mathcal{V}^{\mathrm{prim}}_{\mathrm{res}}$ decomposes orthogonally under the adelic inner product (Langlands 1966 §6; Moeglin–Waldspurger Ch. II). On $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp}}$, split by archimedean type: $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp}} = \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,hol}} \oplus \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,Maass}}$ (Gelfand–Fomin–Piatetski-Shapiro 1969 *Representation Theory and Automorphic Functions* Ch. II). The holomorphic piece is closed by Theorem 76.2.1 under the comparison hypotheses; the Maass piece by Theorem 76.2.3 under Ramanujan. For the specified residual line, let v_0 be its chosen unit vector and assume the arithmetic comparison sends its normalized vacuum to v_0. Then $\langle a v_0, b v_0\rangle = a\overline{b}$, and the stipulated Koszul normalization gives the same value. This calculation uses the unit-vector datum, not an assertion that the whole automorphic quotient has volume one. □

Remark 76.3.4 (discrete and continuous domains). The conditional identity (76.1) concerns the selected discrete carrier with its assumed placement. It makes no assertion about the continuous automorphic spectrum. The scalar models P and K above differ in the fixed convention; identifying them with actual pairings and constructing a measure transport remain separate continuous-spectrum requirements.

Sufficiency for Li's Criterion

Proposition 76.3.5 *(domain restriction for the Li trace). (conditional on the H-P^{Ar} determinant-trace realisation and on the trace map placing Li's zero sum in the discrete sector).* Li's coefficients $\lambda_n = \sum_\rho (1 - (1 - 1/\rho)^n)$ (Theorem 44.3.1) sum over the discrete non-trivial zeros of ζ. The non-trivial zeros of ζ contributing to the Li sum are exactly the zeros of $\xi_{\mathbb{R}}$, all of which are discrete; the Eisenstein continuous spectrum contributes no term once the H-P^{Ar} trace has been identified with this zero sum. Eisenstein-series poles in ξ at $s = 0, 1$ are Tate poles (Remark 44.3.7), cancelled by the factor $s(s-1)/2$

in $\xi_{\mathbb{R}} = (1/2)s(s-1)\Lambda_\zeta$. The Hilbert–Pólya operator Θ_ξ of Hypothesis 44.3.2, if it exists, has regularised determinant divisor equal to the non-trivial zero divisor, whose ordinate multiset is discrete. The regularised trace $\mathrm{Tr}_{\mathrm{reg}}(\mathcal{L}_n)$ of Proposition 44.3.8 then integrates against the discrete zero distribution: the Eisenstein continuous spectrum is not in the domain of the trace.

Proof. Li's sum is over non-trivial zeros of ζ, equivalently zeros of $\xi_{\mathbb{R}}$. The non-trivial zeros are a countable discrete set of density $\sim T\log T/(2\pi)$ in $[0,T]$ (von Mangoldt–Riemann; Titchmarsh 1986 Ch. II). No Eisenstein-series pole contributes to this set: Eisenstein poles at $s=0,1$ in Λ_ζ are exactly the Tate poles eliminated by the factor $s(s-1)/2$ in the Li normalisation (Remark 44.3.7). The zero divisor of $\xi_{\mathbb{R}}$ is therefore discrete. Under the H-P^{Ar} hypothesis it identifies with the determinant divisor and Weil–Connes trace of Θ_ξ (Hypothesis 44.3.2). The Li-transform trace $\mathrm{Tr}_{\mathrm{reg}}(\mathcal{L}_n)$ equals the sum over this discrete distribution only after that determinant-trace identification. □

Corollary 76.3.6 *(conditional Li path on the discrete core).* Assume also the determinant-trace realization and the explicit trace-placement comparison of Proposition 76.3.5, placing the Li trace on this same discrete domain. Assume Hypothesis 67.1.1 on the stated sector, with its injective placement, local pairing maps and adelic extension. *(conditional on Ramanujan for Maass forms at the cuspidal Maass boundary, on the H-P^{Ar} determinant-trace realisation, and on the Tamagawa-regularised assembly; with arithmetic pairing comparisons also assumed on the holomorphic and residual sectors).* The sufficient RH path of Theorem 44.6.1 receives a conditional P3 input from $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$ (Theorem 76.3.3); the Li criterion draws on discrete zero sums (Proposition 76.3.5) from the same domain, and the Eisenstein continuum is outside the domain of both inputs.

Proof. P3 on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$ is the conditional identity of Theorem 76.3.3. The Li path of Theorem 44.3.10 uses λ_n which are discrete-zero sums after the H-P^{Ar} spectral identification (Proposition 76.3.5). The Eisenstein continuum is outside the resulting trace domain. □

76.4 Langlands–Shahidi Normalisation

The Koszul–Petersson identity can be extended to the full $\mathcal{V}^{\mathrm{prim}}$, including the Eisenstein continuum, only after the Koszul pairing on principal-series inductions is identified with the positive Langlands–Shahidi Plancherel measure. The scalar N^{LS} of Definition 76.4.1 is not by itself such an identification. The discrete-core reduction of Section 76.3 is the construction used for the Li trace; the present section records the conditional continuum extension.

The Langlands–Shahidi Normalisation Factor

Definition 76.4.1 *(Langlands–Shahidi correction factor).* The *Langlands–Shahidi normalisation factor* on the principal-series induction $\mathrm{Ind}_B^G(\chi_s)$ is

$$N^{\mathrm{LS}}(s) \;:=\; \frac{\xi(2s-1)}{\xi(2s)} \qquad (\mathrm{Re}(s)=1/2).$$

On U, $|N^{\mathrm{LS}}(s)|=1$. In the scalar models above, $N^{\mathrm{LS}}K = d = P/2$; multiplication by the phase alone does not identify the two coefficients. The scalar coming from the Maass–Selberg truncation is part of the Plancherel convention, not part of the Langlands–Shahidi normalising operator.

Proposition 76.4.2 *(Koszul normalisation absorption). (on the identification of the bar–cobar Koszul spectral measure on the Eisenstein continuum with the Shahidi 2010 normalised Eisenstein measure;*

see Remark 76.4.3). Under the substitution $\langle\cdot,\cdot\rangle^{\mathrm{Koszul}} \to \langle\cdot,\cdot\rangle^{\mathrm{Koszul,LS}} = N^{\mathrm{LS}}(\cdot)\cdot\langle\cdot,\cdot\rangle^{\mathrm{Koszul}}$ on the Eisenstein continuum and the independent identification of the resulting quadratic form with the Shahidi L^2-normalised measure, the corrected Koszul normalisation is transported to the Maass–Selberg Plancherel normalisation:

$$\mathsf{Sh}_{\mathrm{LS}}\Big(\mathrm{lead}^{\mathrm{Koszul}}_{\mathrm{cont}}(s)\Big) \;=\; \mathrm{lead}^{\mathrm{Pet}}_{\mathrm{cont}}(s).$$

Here $\mathsf{Sh}_{\mathrm{LS}}$ denotes the Shahidi Plancherel transport. No raw equality of scalar densities is asserted before this transport is supplied.

Proof. On $\mathrm{Re}(s) = 1/2$, $|M(s)| = 1$ (Moeglin–Waldspurger §IV.3.10) gives $|N^{\mathrm{LS}}(s)| = 1$. Hence the Langlands–Shahidi factor changes the scattering phase but does not by itself turn the Koszul coefficient into a positive Plancherel density. The equality (76.4.2) is exactly the additional identification of the bar–cobar Eisenstein measure with the Shahidi 2010 L^2-normalised Eisenstein measure (Shahidi 2010 Thm 3.5.1). The scalar $1 + |M(s)|^2$ belongs to the Maass–Selberg truncation convention and is not inserted into N^{LS}. □

Remark 76.4.3 (Koszul–Shahidi measure identification). The ratio $\xi(2s)/\xi(2s-1)$ is a non-constant unit-modulus function on $\mathrm{Re}(s) = 1/2$; it equals $1/M(s)$ with $|M(s)| = 1$. The equality of Plancherel normalisations holds only when the Koszul Eisenstein measure is identified with the Shahidi 2010 L^2-normalised measure. Whether the bar–cobar Koszul definition (Beilinson–Drinfeld 2004 §3.5) assigns exactly this normalisation to the Eisenstein sector is an open problem. This gap does not affect Theorem 76.6.1: the main theorem uses the domain restriction of Section 76.3, not the Langlands–Shahidi extension.

Theorem 76.4.4 *(full-$\mathcal{V}^{\mathrm{prim}}$ Koszul–Petersson compatibility via Langlands–Shahidi).* Assume Hypothesis 67.1.1 on the stated sector, with its injective placement, local pairing maps and adelic extension. *(on the Ramanujan bound for Maass cusp forms of $GL_2/\mathbb{Q}$; on the identification of the bar–cobar Koszul spectral measure on the Eisenstein sector with the Shahidi 2010 L^2-normalised Eisenstein measure (Remark 76.4.3); and on the Langlands–Shahidi correction N^{LS} as a canonical ingredient of the arithmetic Koszul pairing on principal-series inductions).* Assume Ramanujan for Maass forms. For all $v, w \in \mathcal{V}^{\mathrm{prim}}$,

$$\langle v, w\rangle^{\mathrm{Pet}} \;=\; \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul,LS}},$$

with $\langle\cdot,\cdot\rangle^{\mathrm{Koszul,LS}}$ the Langlands–Shahidi-corrected Koszul pairing of Proposition 76.4.2. The identification extends the discrete-core identity (76.1) to the Eisenstein continuum.

Proof. On the discrete core, $N^{\mathrm{LS}}|_{\mathrm{disc}} = 1$ (the correction factor is supported on the Eisenstein continuum), so Theorem 76.3.3 applies without modification. On the Eisenstein continuum, Proposition 76.4.2 gives the conditional Plancherel-normalisation match. Adelic extension on the continuous carrier is part of the assumed measure transport and factorization datum. It is not deduced from the scalar identity for N^{LS}. □

Remark 76.4.5 (domain of the Langlands–Shahidi correction). The full $\mathcal{V}^{\mathrm{prim}}$ identity (76.4.4) extends the domain of the Koszul pairing to the Eisenstein sector only under the measure identification of Remark 76.4.3; it is not obtained by the bare insertion of N^{LS}. The factor N^{LS} is the Langlands–Shahidi normalised intertwining operator (Shahidi 2010 Ch. 2–3; Moeglin–Waldspurger §IV.3), the natural normalisation for the Eisenstein functional equation, and its insertion is a definitional addition to the arithmetic Koszul pairing of Chapter 33, which uses the Bost–Connes involution alone. The

domain restriction of Section 76.3 requires no such addition and is sufficient only for the discrete trace normalisation of the Li path under the hypotheses of Proposition 76.3.5.

76.5 Full $\mathcal{V}^{\mathrm{prim}}$ P3 on the Discrete Core

Proposition 76.5.1 *(Pullback of the discrete Petersson form).* Let $i : V \to L^2_{\mathrm{disc}}([GL_2])$ be a specified injective linear placement. Then $h(v,w) = \langle i(v), i(w)\rangle_{L^2}$ is positive definite. If V has a specified bilinear form B and antilinear map J with $B(v,Jw) = h(v,w)$, this transported Koszul form is positive definite as well. No temperedness assumption is needed for this implication.

Proof. For $v \neq 0$, injectivity gives $i(v) \neq 0$, and hence $h(v,v) = \|i(v)\|^2_{L^2} > 0$. The form identity gives the second assertion. Constructing i, B, J and the identity is the separate arithmetic comparison problem. □

Theorem 76.5.2 *(P3 on the discrete core).* Assume Hypothesis 67.1.1 on the stated sector, with its injective placement, local pairing maps and adelic extension. The comparison maps are assumed on every sector, including the Maass sector, with the specified adelic assembly. The discrete-spectrum core $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}} \subset \mathcal{V}^{\mathrm{prim}}$ carries a positive-definite Hermitian form

$$\langle\,\cdot\,,\,\cdot\,\rangle^{\mathrm{Pet}} : \mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}} \otimes \overline{\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}} \to \mathbb{C}$$

given by the adelic Petersson–Tamagawa inner product, compatible with arithmetic Koszul self-duality via

$$\langle v, w\rangle^{\mathrm{Pet}} \;=\; \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}}$$

on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}} \otimes \mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$. The Eisenstein continuous sector is placed outside the P3 identity by the discrete-core domain restriction of Definition 76.3.2.

Proof. The arithmetic comparison hypothesis supplies the injective placement and identifies the form with the Petersson form on its discrete image. Proposition 76.5.1 therefore gives positivity and the asserted pairing identity. This applies to ordinary Maass L^2 vectors regardless of their Satake bounds. Ramanujan is needed only for the separate route that first restricts every Maass representation to the tempered model; it does not replace or follow from the arithmetic pairing maps. □

Corollary 76.5.3 *(P3 input for the sufficient RH path).* Assume also the determinant-trace realization and the explicit trace-placement comparison of Proposition 76.3.5, placing the Li trace on this same discrete domain. Assume Hypothesis 67.1.1 on the stated sector, with its injective placement, local pairing maps and adelic extension. The P3 input of Theorem 44.6.1 is supplied by Theorem 76.5.2 on the discrete core, under the full comparison hypotheses on every sector. Combined with the discrete-support of the Li sum (Proposition 76.3.5), the minimal RH path of Theorem 44.6.1 receives a complete Koszul–Petersson compatibility input drawn entirely from the discrete-spectrum core.

Proof. Theorem 76.5.2 + Proposition 76.3.5. □

76.6 Koszul–Petersson Compatibility on the Domain-Restricted Discrete Core: Main Theorem

Theorem 76.6.1 *(full-$\mathcal{V}^{\mathrm{prim}}$ Koszul–Petersson compatibility on the discrete core).* Assume Hypothesis 67.1.1 on the stated sector, with its injective placement, local pairing maps and adelic extension. *(domain-restricted to $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$; conditional on the arithmetic comparison maps on each sector; with Ramanujan additionally assumed on the Maass sector).* For all $v, w \in \mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$,

$$\langle v, w\rangle^{\mathrm{Pet}} \;=\; \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}}.$$

In particular, the Koszul pairing of $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$ with a specified Koszul-dual carrier, composed with the specified involution σ^{BC}, equals the adelic Petersson–Tamagawa inner product on the discrete-spectrum image of $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$ inside $L^2_{\mathrm{disc}}([GL_2])$. An identification of that dual carrier with $\mathbb{A}^{\mathrm{BC}}$ requires a separate dg algebra comparison; equality of pairings does not establish it.

Proof. The sector-by-sector identities of Theorem 76.3.3 are:

- $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,temp}}$: Theorem 76.1.1 (under the comparison hypotheses).
- $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,hol}} \subset \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,temp}}$: Theorem 76.2.1 (temperedness from Deligne and Deligne–Serre; pairing maps separately assumed).
- $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,Maass}}$: Theorem 76.2.3 (conditional on Ramanujan for Maass forms).
- $\mathcal{V}^{\mathrm{prim}}_{\mathrm{res}}$: direct evaluation at the vacuum.

Orthogonality of the direct-sum decomposition $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}} = \bigoplus_{\alpha} \mathcal{V}^{\mathrm{prim}}_{\alpha}$ (Langlands 1966 §6) extends the sector-by-sector identities to the full discrete core. The domain restriction (Definition 76.3.2) defines the domain of the identity as $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$; the scalar discrepancy of Proposition 76.3.1 concerns the proposed continuous coefficients, not an established arithmetic continuous pairing. □

Remark 76.6.2 *(carriers and pairing conventions for Theorem 76.6.1).* The pairing comparison uses the following carriers and conventions.

Koszul and automorphic constructions. The Koszul side uses Beilinson–Drinfeld 2004 (bar–cobar), Positselski 2009 (comodule–contramodule), and Bost–Connes 1995 (Koszul dual as C^*-algebra). The Petersson side uses Deligne 1974 (Weil I, holomorphic Ramanujan), Deligne–Serre 1974 (weight 1), Kim–Shahidi 2002 and Kim–Sarnak 2003 (sub-convex Maass Ramanujan), Langlands 1966 and Moeglin–Waldspurger 1995 (spectral decomposition), Selberg 1956 (Petersson positivity). Relating these constructions requires the map from the Koszul primitive carrier to the specified automorphic realization and its pairing compatibility; the bibliographic distinction is not such a map.

Sign and convention consistency. σ^{BC} is $*$-antilinear; Petersson is sesquilinear conjugate-in-second-slot; Koszul is bilinear. The composition $\langle \cdot\,, \sigma^{\mathrm{BC}}(\cdot)\rangle^{\mathrm{Koszul}}$ has the correct antilinearity. Tamagawa measure is Weil-normalised (Weil 1967 Ch. VII).

Ambient category. Both sides are sesquilinear forms on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}} \otimes \overline{\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}}$, a subspace of the adelic Hilbert space $L^2_{\mathrm{disc}}([GL_2])$.

Sector hypotheses. Every arithmetic pairing conclusion retains the placement and pairing hypotheses. The tempered-reduction argument for the Maass sector additionally assumes Ramanujan; the positive-pullback argument of Proposition 76.5.1 does not.

Functoriality of the decomposition. The discrete-core decomposition is orthogonal under Petersson (Langlands 1966 §6); Koszul factorises by place (BD 2004 §3.5) and the Bost–Connes involution respects the decomposition.

Domain of the holomorphic and Maass closures. The holomorphic temperedness input is supplied by Deligne and Deligne–Serre; the pairing maps are still required. Maass temperedness is conditional on Ramanujan; the Kim–Sarnak 2003 bound controls Satake exponents at each prime (Proposition 76.2.4) but not adelic closure. The arithmetic comparison and adelic extension are additional requirements on the Maass sector.

76.7 Conditional Residuals

Sector	Form	Note
$\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,hol}}$ (holomorphic cuspidal)	Comparison maps required	Theorem 76.2.1; Deligne 1974 (weight $k \geq 2$), Deligne–Serre 1974 (weight 1).
$\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,Maass}}$ (Maass cuspidal)	Comparison maps and Ramanujan	Theorem 76.2.3; best bound $\theta_p \leq 7/64$ (Kim–Sarnak 2003); pairing-error estimate separately required; the proposed prime majorant is not summable (Proposition 76.2.4).
$\mathcal{V}^{\mathrm{prim}}_{\mathrm{res}}$ (residual, one-dimensional)	Comparison maps required	Direct evaluation at Fock vacuum; This calculation concerns the specified residual line; other residual summands require their own placement and pairing maps.
$\mathcal{V}^{\mathrm{prim}}_{\mathrm{cont}}$ (Eisenstein continuous)	**Excluded by domain**	Definition 76.3.2; scalar discrepancy (Proposition 76.3.1); actual measure transport requires additional construction. Sufficient only for the discrete Li trace normalisation, under the H-P^{Ar} spectral hypothesis (Proposition 76.3.5). Alternative Langlands–Shahidi correction Theorem 76.4.4 reduces the continuum to the measure-normalisation gap.

Conditional boundary. An additional representation-theoretic input of Theorem 76.6.1 is the Ramanujan bound for Maass cusp forms of $GL_2/\mathbb{Q}$. The arithmetic placement and pairing maps are separate open construction requirements. The Satake exponent bound does not supply a pairing-error estimate or an absolutely convergent adelic comparison; see Proposition 76.2.4. The Li sum is over the discrete non-trivial zeros (Proposition 76.3.5) after the H-P^{Ar} determinant-trace realisation; the Li path has this trace condition in addition to the Maass boundary.

Eisenstein continuum. The domain restriction places the Eisenstein continuous spectrum outside the identity's domain (Definition 76.3.2). The discrete-spectrum restriction suffices for the discrete trace normalisation of the Li path under the stated H-P^{Ar} determinant-trace hypothesis (Corollary 76.3.6). The Langlands–Shahidi correction of Section 76.4 extends the identity to the continuum only after the Koszul Eisenstein measure has been identified with the Shahidi L^2-normalised measure.

The sufficient RH path. The P3 input of Theorem 44.6.1 is supplied by $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$, conditional on the arithmetic comparison maps, Ramanujan for Maass cusp forms and on the H-P^{Ar} determinant-trace

realisation used to place the Li trace. The remaining inputs are the conditional arithmetic Positselski, H-P^{Ar}, and HP–Li positivity hypotheses recorded later.

76.8 Bibliography

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Chapter 77

Bost–Connes Modular Spectrum and the Hilbert–Pólya Comparison

The critical Bost–Connes state gives a self-adjoint modular generator whose eigenvalues are $\frac{1}{2}\log q$, $q \in \mathbb{Q}_{>0}$. A specified reducing occupation sector instead has the simple spectrum $\{\log m : m \geq 1\}$. We construct both carriers and an oscillator embedding in the same GNS space. The embedding preserves logarithmic prime energies and positive squared-semigroup shifts. Its oscillators are Hilbert-space operators, not operators affiliated with the Bost–Connes observable algebra. Ordinary conformal energy is a separate grading. These distinctions leave a precise arithmetic question: construct a self-adjoint operator whose normalized determinant is $\xi_{\mathbb{R}}$, together with the Petersson, chiral and trace comparisons that connect it to the arithmetic Fock space.

77.1 Bost–Connes C^*-Dynamical Setup

We use the global algebra and positive-temperature KMS convention of Chapter 74.3.ii. The coefficient algebra factors over primes, but the isometries act across those factors.

Definition 77.1.1 (global and local Bost–Connes algebras). Put $D = C^*(\mathbb{Q}/\mathbb{Z})$, with unitaries $e(r)$ satisfying $e(r+s) = e(r)e(s)$ and $e(r)^* = e(-r)$. For $n \geq 1$ set

$$\alpha_n(e(r)) = \frac{1}{n}\sum_{nt=r} e(t), \qquad C^*_{\mathbb{Q}} = D \rtimes_\alpha \mathbb{N}^\times.$$

The generators μ_n are isometries, with

$$\mu_m\mu_n = \mu_{mn}, \quad \mu_n^*\mu_n = 1, \quad \mu_n d\mu_n^* = \alpha_n(d), \quad \mu_n^* e(r)\mu_n = e(nr).$$

In particular $P_n := \mu_n\mu_n^* = n^{-1}\sum_{nt=0} e(t)$ is a projection. The time evolution is

$$\sigma_t(\mu_n) = n^{it}\mu_n, \qquad \sigma_t(e(r)) = e(r), \qquad \sigma_t(\mu_n^*) = n^{-it}\mu_n^*.$$

For a prime p, let $D_p = C^*(\mathbb{Q}_p/\mathbb{Z}_p)$ and $C^*_{\mathbb{Q},p} = C^*(D_p, \mu_p) \subset C^*_{\mathbb{Q}}$. The primary decomposition of $\mathbb{Q}/\mathbb{Z}$ gives the coefficient tensor product $D = \bigotimes_p D_p$, with local units.

Remark 77.1.2 (the coefficient tensor and the cross-prime action). The global crossed product is described in Laca–Raeburn, arXiv:math/9911134, §2, p. 3. If r is ℓ-primary and $\ell \neq p$, multiplication by p is invertible on that primary group, and the relations give

$$\mu_p e(r) = e(p^{-1}r)\mu_p, \qquad \alpha_p(e(r)) = P_p e(p^{-1}r).$$

For example $\mu_2^* e(1/3)\mu_2 = e(2/3)$, whereas commuting local factors would give $e(1/3)$. In the critical state below, $\|(e(1/3) - e(2/3))\Omega_1\|^2 = 2$, so these elements differ. Thus the algebras $C^*_{\mathbb{Q},p}$ do not assemble by

a generatorwise tensor product. The algebra generated only by the isometries has the distinct Toeplitz tensor-product description of Connes, *Noncommutative Geometry*, V, §11, Proposition 7, pp. 530–531. Neither that description nor the coefficient tensor asserts a product representation of the character-enlarged local algebras.

Lemma 77.1.3 *(the critical KMS state).* There is a unique KMS_1 state φ_1 on $C^*_{\mathbb{Q}}$. It satisfies

$$\begin{aligned}\varphi_1(e(r)) &= \delta_{r,0},\\ \varphi_1(\mu_a e(r)\mu_b^*) &= \delta_{a,b}a^{-1}\delta_{r,0},\\ \varphi_1(\mu_n^*\mu_m e(r)) &= \delta_{n,m}\delta_{r,0}.\end{aligned}$$

In its GNS representation the state extends to a faithful normal state on $M = \pi_1(C^*_{\mathbb{Q}})''$, a hyperfinite factor of type III_1; the cyclic vector Ω_1 is separating for M.

Proof. Existence, uniqueness and the factor statement are Connes, V, §11, Theorem 5(a), p. 528; its character formula at $\beta = 1$ is $\delta_{r,0}$. Time invariance kills monomials of nonzero time weight. At equal indices the KMS identity $\varphi_1(xy) = \varphi_1(y\sigma_i(x))$ gives the factor a^{-1} in the second formula, whereas $\mu_n^*\mu_n = 1$ gives the third. The faithful normal GNS extension and separating vector are the modular realization used in Chapter 74.3.12. □

Remark 77.1.4 *(the critical partition value).* The occupation partition function is $\zeta(\beta)$ for $\Re\beta > 1$. Its Laurent expansion is $\zeta(\beta) = (\beta-1)^{-1} + \gamma + O(\beta-1)$, where γ is the Euler–Mascheroni constant. Thus its Laurent finite part at one is γ; the trace itself diverges there. Identifying this finite part with a determinant normalization would require a separate comparison. The full critical GNS heat operator is not trace class, as its spectrum below shows.

77.2 GNS Representation and the Modular Operator

The rational time weights give an explicit basis that also records the character action. It determines the modular sign and all operator domains directly.

Definition 77.2.1 *(GNS triple).* Let $(\mathcal{H}_1, \pi_1, \Omega_1)$ be the GNS triple of φ_1. Thus $\mathcal{H}_1$ is the completion of $C^*_{\mathbb{Q}}/\{x : \varphi_1(x^*x) = 0\}$ with inner product $\langle x\Omega_1, y\Omega_1\rangle = \varphi_1(y^*x)$, linear in the first variable; π_1 acts by left multiplication. We suppress π_1 on algebra elements when acting on this space.

Lemma 77.2.2 *(rational-character basis and Hilbert tensor product).* For $q = a/b \in \mathbb{Q}_{>0}$ in lowest terms and $r \in \mathbb{Q}/\mathbb{Z}$, put $v_{q,r} = \sqrt{b}\,\mu_a e(r)\mu_b^*\Omega_1$. These vectors are an orthonormal basis. Consequently

$$\begin{aligned}\mathcal{H}_1 &\cong \ell^2(\mathbb{Q}_{>0})\widehat{\otimes}\ell^2(\mathbb{Q}/\mathbb{Z})\\ &\cong \bigotimes_p{}'\big[\ell^2(p^{\mathbb{Z}})\widehat{\otimes}\ell^2(\mathbb{Q}_p/\mathbb{Z}_p)\big].\end{aligned}$$

with reference vectors $\delta_1 \otimes \delta_0$ and $\Omega_1 = v_{1,0}$. For $d = \gcd(n,b)$ the generator actions are

$$e(s)v_{q,r} = v_{q,r+as}, \qquad \mu_n v_{q,r} = d^{-1/2}\sum_{dt=r} v_{nq,t}.$$

This is a tensor decomposition of Hilbert spaces, not of the local BC representations.

Proof. Different q have different time characters and are orthogonal by state invariance. For fixed reduced a/b,

$$\langle v_{q,r}, v_{q,s}\rangle = b\varphi_1(\mu_b e(-s)\mu_a^*\mu_a e(r)\mu_b^*) = \varphi_1(e(r-s)) = \delta_{r,s}.$$

Every monomial of ratio a/b has indices ac, bc, and $\mu_{ac}e(r)\mu_{bc}^* = \mu_a\alpha_c(e(r))\mu_b^*$ is a finite linear combination of these basis vectors. Such monomials span a dense algebra, proving completeness. Unique prime factorization of positive rationals and the primary decomposition of $\mathbb{Q}/\mathbb{Z}$ give the Hilbert tensor product. Finally $e(s)\mu_a = \mu_a e(as)$ proves the first action. Reducing na/b cancels d; expansion of $\alpha_d(e(r))$ and the normalization $\sqrt{b/d}$ of the resulting basis give the coefficient $d^{-1/2}$. □

Lemma 77.2.3 *(Tomita conjugation and the modular operator).* On the finite basis span the Tomita operator and its polar decomposition $S_1 = J_1\Delta_1^{1/2}$ satisfy

$$S_1 v_{q,r} = q^{-1/2}v_{q^{-1},-r}, \qquad J_1 v_{q,r} = v_{q^{-1},-r}, \qquad \Delta_1 v_{q,r} = q^{-1}v_{q,r}.$$

Here J_1 is conjugate-linear, and

$$\Delta_1^{it}\pi_1(x)\Delta_1^{-it} = \pi_1(\sigma_{-t}(x)), \qquad \Delta_1^{it}\Omega_1 = \Omega_1.$$

For $u = \sum c_{q,r}v_{q,r}$ the maximal domain of Δ_1 is $\sum q^{-2}|c_{q,r}|^2 < \infty$; the domain of S_1 is $\sum q^{-1}|c_{q,r}|^2 < \infty$.

Proof. Taking adjoints of $\sqrt{b}\mu_a e(r)\mu_b^*$ gives $\sqrt{b}\mu_b e(-r)\mu_a^* = q^{-1/2}\sqrt{a}\mu_b e(-r)\mu_a^*$. The resulting conjugate-linear weighted permutation has the stated polar decomposition and domains; truncation in the basis gives its closed extension. Equivalently, the modular sign is Connes, V, §11, Proposition 4, p. 527. Applying the displayed diagonal unitaries to the two generator actions verifies that sign directly. □

Definition 77.2.4 *(BC log-modular generator).* Define $N_{\mathrm{BC}} = -\frac{1}{2}\log\Delta_1$ by self-adjoint functional calculus. Its action is

$$\begin{aligned} N_{\mathrm{BC}}v_{q,r} &= \tfrac{1}{2}\log q\, v_{q,r}, \\ \operatorname{Dom} N_{\mathrm{BC}} &= \Big\{\sum c_{q,r}v_{q,r} : \sum(\log q)^2|c_{q,r}|^2 < \infty\Big\}. \end{aligned} \tag{77.1}$$

Theorem 77.2.5 *(self-adjointness and the full modular spectrum).* The finite basis span $\mathcal{D}_{\mathrm{BC}}$ is an essential self-adjointness core for N_{BC}. Its point spectrum is $\frac{1}{2}\log\mathbb{Q}_{>0}$, each eigenvalue has countably infinite multiplicity, and its closed spectrum is $\mathbb{R}$. All its spectral measures are pure point. The point spectrum of Δ_1 is $\mathbb{Q}_{>0}$, and its closed spectrum is $[0,\infty)$; zero is not an eigenvalue.

Proof. For a real diagonal sequence, the adjoint of the finite-span operator has exactly the square-summability domain in (77.1); truncations converge in its graph norm. This proves self-adjointness of the closure. Each q has the countably infinite character basis. A diagonal operator's spectrum is the closure of its diagonal values: outside that closure its coordinatewise inverse is bounded, whereas coordinate vectors give approximate eigenvectors at every limit point. Positive rationals are dense in $(0,\infty)$, proving both closed-spectrum statements. The spectral projections act coordinatewise, so all spectral measures are atomic. All eigenvalues of Δ_1 are positive. □

Remark 77.2.6 *(modular and occupation energies).* The full generator has negative as well as positive energies, and $e^{-tN_{\mathrm{BC}}}$ is unbounded for every $t > 0$. The weight $k\log p$ occurs on the vectors $v_{p^{2k},0}$. Restricting to their positive-integer square sector gives the occupation model below. It is this specified restriction, rather than the full modular spectrum, that realizes the logarithmic prime weights in Chapter 62.3.1. Neither operator has the zeta-zero divisor by this construction.

77.3 Arithmetic Heisenberg Fock and Its Embedding into the GNS Space

The occupation sector fixes the zero-mode normalization. An explicit orthonormal family in the modular fixed-point algebra then supplies oscillator multiplicities inside the same GNS representation.

Definition 77.3.1 (complex arithmetic Fock carrier). Choose positive integers d_p and d_∞ as component counts. Let J consist of the modes (p,k,a), $k \geq 1, 1 \leq a \leq d_p$, and (∞,k,a), $k \geq 1, 1 \leq a \leq d_\infty$. Write $C = \mathbb{N}_0^{(J)}$ for finite occupation configurations and $\mathcal{F}_{\mathrm{osc}} = \ell^2(C)$ for the complex bosonic Hilbert space with orthonormal number states $|\mathbf{n}\rangle$. Equivalently it is the restricted tensor product of the per-prime Fock spaces and the archimedean complex Fock space, with vacuum reference vectors. Put

$$L_w = \ell^2(\mathbb{N}^\times, m^{-1}dm), \qquad \epsilon_m = \sqrt{m}\,\delta_m, \qquad \mathcal{F}_{\mathrm{Heis}}^{\mathrm{Ar}} = \mathcal{F}_{\mathrm{osc}} \widehat{\otimes} L_w.$$

The primary core $\mathcal{F}_{\mathrm{Heis,prim}}^{\mathrm{Ar}}$ is the finite span of $|\mathbf{n}\rangle \otimes \epsilon_m$. This specifies a complex Hilbert realization of the oscillator algebra. Comparing it with the native per-prime chiral objects of Chapter 26 and the archimedean theory of Chapter 35 requires complex realization and chiral structure maps in addition to this Hilbert space.

Lemma 77.3.2 *(weighted occupation embedding).* Let $K_2 = \overline{\mathrm{span}}\{\mu_{m^2}\Omega_1 : m \geq 1\}$. The map

$$\iota_0 : L_w \longrightarrow K_2, \qquad \iota_0 \delta_m = m^{-1/2}\mu_{m^2}\Omega_1$$

is unitary onto K_2. This space reduces N_{BC}, and $N_{\mathrm{occ}} := N_{\mathrm{BC}}|_{K_2}$ is unitarily equivalent to

$$H_w \delta_m = (\log m)\delta_m, \qquad \mathrm{Dom}\, H_w = \left\{ c : \sum_m \frac{(\log m)^2 |c_m|^2}{m} < \infty \right\}.$$

The isometries $R_n \delta_m = \sqrt{n}\,\delta_{nm}$ satisfy

$$R_n^* \delta_m = \begin{cases} n^{-1/2}\delta_{m/n}, & n \mid m, \\ 0, & n \nmid m, \end{cases} \qquad \iota_0 R_n = \pi_1(\mu_{n^2})\iota_0.$$

Their Hilbert adjoints correspond to compressed BC adjoints.

Proof. The vectors $\mu_{m^2}\Omega_1 = v_{m^2,0}$ are orthonormal eigenvectors of energy $\log m$, whereas $\|\delta_m\|^2 = 1/m$ in L_w. This proves the isometry, its range, and the maximal-domain identity by the coordinate formula. Their span is invariant under every spectral projection, so K_2 reduces the generator. The shift and adjoint formulas follow on the normalized basis ϵ_m. □

Definition 77.3.3 (symmetry-invariant BC–GNS Heisenberg lift). Let $G = \widehat{\mathbb{Z}}^\times$ act by $\beta_u(e(r)) = e(ur)$ and $\beta_u(\mu_n) = \mu_n$. The relations and the state formula show that this is a state-preserving action. Hence $U_u(x\Omega_1) = \beta_u(x)\Omega_1$ defines GNS unitaries. They are strongly continuous: character orbits factor through finite quotients on the dense monomial algebra, and unitarity extends continuity to every vector. Define the Hilbert projection

$$P_G = \int_G U_u \, du$$

using normalized Haar measure. Haar invariance gives $P_G^* = P_G = P_G^2$ and $\mathrm{Ran}\, P_G = \mathcal{H}_1^G$. A symmetry-invariant BC–GNS Heisenberg lift consists of a closed $E \subset \mathcal{H}_1^G$, containing K_2, and a unitary $\iota : \mathcal{F}_{\mathrm{Heis}}^{\mathrm{Ar}} \to E$ with $\iota(|\mathbf{0}\rangle \otimes \delta_m) = \iota_0 \delta_m$, transporting the oscillator annihilation and creation operators with their maximal domains and primary core. This definition uses the specified Galois symmetry projection P_G, not the occupation projection P_K onto K_2.

Lemma 77.3.4 *(an explicit same-GNS Fock embedding).* Such a lift exists after a choice of an auxiliary prime ℓ and an enumeration of J. Put $P_k = \mu_{\ell^k}\mu_{\ell^k}^*$ and

$$h_k = \frac{P_{k+1} - \ell^{-1}P_k}{\sqrt{\ell^{-k-1}(1-\ell^{-1})}}, \qquad k \geq 0.$$

Enumerate $J = \{j_1, j_2, \ldots\}$ and let r_i be the increasing ordinary primes. Set

$$c(\mathbf{n}) = \prod_i r_i^{n_{j_i}}, \qquad f_{\mathbf{0}} = 1, \qquad f_{\mathbf{n}} = h_{c(\mathbf{n})-2} \quad (\mathbf{n} \neq \mathbf{0}).$$

The logarithmic finite-place energy is encoded by

$$t(\mathbf{n}) = \prod_{p,k,a} p^{k n_{p,k,a}}.$$

Archimedean modes contribute no factor to t. The vectors

$$v_{\mathbf{n},m} = \mu_{(mt(\mathbf{n}))^2} f_{\mathbf{n}} \Omega_1$$

are orthonormal, and the unitary onto their closed span E is $\iota(|\mathbf{n}\rangle \otimes \epsilon_m) = v_{\mathbf{n},m}$. Moreover E reduces N_{BC}, and

$$\iota^* N_{\mathrm{BC}}|_E \iota = I \otimes H_w + \sum_{p,k,a} k \log p \, K_{p,k,a} \otimes I, \tag{77.2}$$

where $K_j|\mathbf{n}\rangle = n_j|\mathbf{n}\rangle$. In normalized coordinates its maximal domain is

$$\left\{ z : \sum_{\mathbf{n},m} (\log m + \log t(\mathbf{n}))^2 |z_{\mathbf{n},m}|^2 < \infty \right\}.$$

Proof. The projections satisfy $P_j P_k = P_{\max(j,k)}$ and $\varphi_1(P_k) = \ell^{-k}$. Direct expansion gives $\varphi_1(h_k) = 0$ and $\varphi_1(h_k^* h_k) = 1$. If $j < k$, both projections defining h_k lie under P_{j+1}; therefore $h_j h_k$ is a scalar multiple of h_k and has expectation zero. Thus $1, h_0, h_1, \ldots$ is orthonormal. Unique factorization makes $\mathbf{n} \mapsto c(\mathbf{n})$ injective, with value one only at the vacuum, so the $f_{\mathbf{n}}$ are orthonormal too. They are fixed by time evolution and by G.

The vector $v_{\mathbf{n},m}$ has energy $\log(mt(\mathbf{n}))$. Different energies are orthogonal. At equal values $mt(\mathbf{n}) = m't(\mathbf{n}')$, cancellation of the common isometry reduces the inner product to $\varphi_1(f_{\mathbf{n}'}^* f_{\mathbf{n}}) = \delta_{\mathbf{n},\mathbf{n}'}$; then $m = m'$. This proves the isometry and its vacuum normalization. Every vector is G-fixed. The closed span of these vectors is invariant under all spectral projections, proving reduction and the stated diagonal maximal domain. The choices of ℓ and the enumeration are part of the construction; no canonical chiral identification is asserted. □

Proposition 77.3.5 *(closed oscillators and their normalization).* On E define

$$b_j v_{\mathbf{n},m} = \sqrt{n_j}\, v_{\mathbf{n}-\mathbf{e}_j,m}, \qquad b_j^* v_{\mathbf{n},m} = \sqrt{n_j + 1}\, v_{\mathbf{n}+\mathbf{e}_j,m},$$

with the first vector zero when $n_j = 0$. Their maximal domains are, respectively, $\sum n_j |z_{\mathbf{n},m}|^2 < \infty$ and $\sum (n_j + 1)|z_{\mathbf{n},m}|^2 < \infty$. They are closed mutual adjoints and have the common invariant core $\mathcal{D}_E =$

$\operatorname{span}\{v_{\mathbf{n},m}\}$. On this core $[b_j, b_{j'}^*] = \delta_{jj'} I_E$ and like operators commute. Every $\mu_{m^2}\Omega_1$ is annihilated by every b_j. For $j = (p,k,a)$, including $p = \infty$, set

$$A_k^{(p,a)} = \sqrt{k}\, b_{p,k,a}, \qquad A_{-k}^{(p,a)} = \sqrt{k}\, b_{p,k,a}^*.$$

Then $[A_k^{(p,a)}, A_{-l}^{(q,b)}] = k\delta_{kl}\delta_{pq}\delta_{ab} I_E$. The normalized creation monomial is

$$\operatorname{Asc}(\mathbf{n}) = \prod_{p,k,a} \frac{(A_{-k}^{(p,a)})^{n_{p,k,a}}}{\sqrt{k^{n_{p,k,a}} n_{p,k,a}!}}, \qquad \iota(|\mathbf{n}\rangle \otimes \delta_m) = m^{-1/2} \operatorname{Asc}(\mathbf{n}) \mu_{m^2}\Omega_1.$$

Proof. Taking inner products of coordinate vectors gives the adjoint formulas. Square summability with the indicated weights gives the closed maximal operators; finite coordinate truncations converge in their graph norms. Their commutators follow by applying both orders to each number vector. Iteration of creation gives $\sqrt{n_j!}$, and $A_{-k} = \sqrt{k} b_j^*$ supplies the further $k^{n_j/2}$ factor. This proves the displayed normalization. □

Proposition 77.3.6 *(symmetry, semigroup shifts and observable obstructions).* The projections P_K, P_E and P_G are proper Hilbert-space projections, and none belongs to $\mathcal{M}$. The oscillators extended by zero on $E^\perp$ commute with P_G, including their domains; their commutator on $\mathcal{D}_E \oplus E^\perp$ is P_E. The positive squared-semigroup action satisfies

$$\pi_1(\mu_{a^2}) v_{\mathbf{n},m} = v_{\mathbf{n},am}, \qquad \pi_1(\mu_{a^2})|_E = \iota(I \otimes R_a)\iota^*.$$

Its Hilbert adjoint is the compressed adjoint. Compression to E does not give a representation of all BC observables. No nonzero closed densely defined operator affiliated with $\mathcal{M}$, with Ω_1 in its domain, can annihilate Ω_1.

Proof. Since $E \subset \mathcal{H}_1^G$, the zero extensions preserve the P_G decomposition. The vector $(e(1/3) - e(2/3))\Omega_1$ has squared norm two and Galois average zero, so $P_G \neq I$. Also $e(1/2)\Omega_1$ is fixed by G and orthogonal to K_2: its expectation is zero, and all other occupation vectors have nonzero energy. Thus $P_G \neq P_K$. A nonvacuum oscillator vector is orthogonal to K_2, so $P_E \neq P_K$. In particular $P_K b_j^* \Omega_1 = 0$ but $b_j^* P_K \Omega_1 = b_j^* \Omega_1 \neq 0$; the occupation projection cannot replace the symmetry projection in the lift.

Multiplication of the positive isometries proves the shift formula. For $a > 1$, however, $\|\mu_{a^2}^* \Omega_1\|^2 = a^{-2}$ and its modular energy is $-\log a$. It is orthogonal to E, whose energies are nonnegative, whereas the compressed adjoint kills Ω_1. For a prime $r \neq \ell$, every P_k is an average of characters of ℓ-power order, hence $\varphi_1(P_k e(1/r)) = 0$. Only energy-zero vectors of E can pair with $e(1/r)\Omega_1$; those pairings reduce to the same character averages, including $\varphi_1(e(1/r)) = 0$. Thus $P_E e(1/r)\Omega_1 = 0$. A compressed unitary kills a unit vector, which rules out multiplicativity of compression.

Finally, if B is affiliated with $\mathcal{M}$, its kernel projection Q belongs to $\mathcal{M}$. If $B\Omega_1 = 0$, then $(I - Q)\Omega_1 = 0$; separation gives $Q = I$, hence $B = 0$. The same argument says that a projection in $\mathcal{M}$ fixing Ω_1 is I. Each of the proper projections above fixes Ω_1, so none belongs to $\mathcal{M}$. □

Remark 77.3.7 *(Hilbert realization and additional arithmetic structure).* The weighted occupation embedding and the chosen complex oscillator lift are explicit. They identify neither an Eisenstein or ramified summand nor a native chiral algebra inside the GNS space. Such identifications require representation

maps, coefficient realizations and compatible pairings. Likewise, the external oscillator operators are not affiliated BC observables. The positive squared shifts commute with them on the primary core, but the full BC action and its adjoints do not preserve that oscillator representation. These are substantive restrictions on any arithmetic determinant-trace comparison using E.

77.4 The Primary-Heisenberg Twist and the Determinant-Trace Operator

The occupation and oscillator constructions supply a Hilbert space and several gradings. The zero-divisor problem asks for a different operator, with a prescribed determinant. An isometry of Hilbert spaces does not identify these generators.

Statement of the Twist

Definition 77.4.1 *(twist data and the separate conformal grading).* On $\mathcal{F}_{\mathrm{osc}}$ let

$$L_{\mathrm{o}}|\mathbf{n}\rangle = \Big(\sum_{p,k,a} k n_{p,k,a}\Big)|\mathbf{n}\rangle, \qquad G_F = L_{\mathrm{o}} \otimes I + I \otimes H_w,$$

where the sum includes the archimedean modes. These are nonnegative self-adjoint diagonal operators on their maximal square-summability domains, with the grading convention of Chapter 57.3.1. With $[A_k, A_{-k}] = kI$, the contribution to L_{o} is $A_{-k}A_k = kb_k^* b_k$, with no additional factor of k.

A primary-Heisenberg twist consists of the following comparison data.

(i) A complex Hilbert realization $\mathcal{H}_\xi$ of the proposed arithmetic primary sector, and a specified isometry into the cuspidal Petersson Hilbert space of Chapter 43.7.1, with fixed central character and Tamagawa normalization. Compatibility with its native Koszul pairing is an additional pairing identity, as in Chapter 43.7.2; the positivity of the Petersson integral alone does not construct this realization.

(ii) A unitary $U : \mathcal{F}^{\mathrm{Ar}}_{\mathrm{Heis}} \to \mathcal{H}_\xi$ whose restriction to the primary core represents the proposed arithmetic comparison. Put $\mathcal{D}_\xi = U\mathcal{F}^{\mathrm{Ar}}_{\mathrm{Heis,prim}}$.

(iii) A self-adjoint operator A with full domain $\mathrm{Dom}\, A \subset \mathcal{H}_\xi$, for which $\mathcal{D}_\xi \subset \mathrm{Dom}\, A$ is an invariant core. Its domain is not identified with that core.

(iv) The transported grading $G_\xi = UG_FU^*$, with $\mathrm{Dom}\, G_\xi = U\,\mathrm{Dom}\, G_F$. Thus

$$G_\xi U = UG_F \quad \text{on } \mathrm{Dom}\, G_F.$$

Any chain-level, chiral or arithmetic-observable interpretation of U requires the corresponding maps and identities as further data. The grading identity does not assert $A = G_\xi$ or intertwine A with N_{BC}.

Define

$$H_{\mathrm{HP}} := A,$$

with its full self-adjoint domain, and

$$\Theta_\xi := \tfrac{1}{2}I + \imath A, \qquad \mathrm{Dom}\, \Theta_\xi = \mathrm{Dom}\, A.$$

The same-GNS representative is $A_E = \imath U^* A U \imath^*$ on E, with domain $\imath U^*\,\mathrm{Dom}\, A$. This is a domain-preserving unitary transport of A, not an identification with the modular restriction.

Remark 77.4.2 *(two spectral obstructions).* The logarithmic energy in (77.2) uses $k \log p$, whereas the conformal grading uses positive integers k. An intertwiner carrying a nonzero integer-energy one-particle state into the same modular representation would require $k = \frac{1}{2} \log q$ for $q \in \mathbb{Q}_{>0}$. This is impossible. Here is an elementary proof for the integer case needed here. For fixed positive integer k and $n \geq 1$, set

$$F_n(x) = \frac{x^n (2k - x)^n}{n!}, \qquad I_n = \int_0^{2k} e^x F_n(x)\, dx.$$

All endpoint derivatives of F_n are integers: derivatives of order less than n vanish, and expansion at either endpoint makes every remaining derivative an integer multiple of $j!/n!$, $j \geq n$. Repeated integration by parts therefore gives $I_n \in \mathbb{Z} + e^{2k}\mathbb{Z}$. But

$$0 < I_n \leq 2k e^{2k} \frac{k^{2n}}{n!} \longrightarrow 0.$$

If e^{2k} were rational, clearing its fixed denominator would produce positive integers tending to zero. Thus ordinary finite or archimedean oscillator energy cannot be the modular generator. Archimedean modes in the construction above have zero modular weight and a separate positive conformal grading; their multiplicities also preclude a simple occupation partition identity on all of E.

A second obstruction prevents retaining the occupation vacuum as a zero eigenvector of A: then Θ_ξ would have eigenvalue $1/2$ and its determinant would vanish there. On the other hand, $\eta(1/2) = \sum_{n \geq 1}((2n-1)^{-1/2} - (2n)^{-1/2}) > 0$, and $\zeta(1/2) = \eta(1/2)/(1 - \sqrt{2}) \neq 0$. All remaining factors in $\xi_{\mathbb{R}}(1/2)$ are nonzero. Hence the zero-divisor operator cannot literally inherit that vacuum energy.

Determinant-Trace Requirement: Hadamard Inversion

Write $\rho = \beta_\rho + i\gamma_\rho$ for the nontrivial zeros, with multiplicity. The completed function is $\xi_{\mathbb{R}}(s) = \frac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. Its genus-one factorization is

$$\xi_{\mathbb{R}}(s) = \frac{1}{2} e^{Bs} \prod_\rho (1 - s/\rho) e^{s/\rho}, \qquad B = -1 - \frac{\gamma}{2} + \frac{1}{2}\log(4\pi).$$

Here γ is the Euler–Mascheroni constant. The prime side appears in the logarithmic derivative, for $\Re s > 1$:

$$-\frac{\xi_{\mathbb{R}}'(s)}{\xi_{\mathbb{R}}(s)} = -\frac{1}{s} - \frac{1}{s-1} + \frac{1}{2}\log\pi - \frac{1}{2}\frac{\Gamma'(s/2)}{\Gamma(s/2)} + \sum_{n \geq 2} \frac{\Lambda(n)}{n^s}.$$

The product and the explicit formula relate the zero and prime distributions; they do not give a unitary conjugacy between the occupation generator and a zero-spectrum operator. The arithmetic problem is to construct the latter from the specified primary sector and prove its determinant and trace comparisons.

The Hilbert–Pólya Hypothesis and Its Consequences

Hypothesis 77.4.3 *(arithmetic determinant-trace realization).* There exist twist data as in Chapter 77.4.1, together with a complete orthonormal eigenbasis (u_ρ) indexed by every zero occurrence, such that

$$Au_\rho = \gamma_\rho u_\rho, \qquad (\tfrac{1}{2}I + iA)u_\rho = \rho u_\rho,$$
$$\operatorname{Dom} A = \Big\{ \sum c_\rho u_\rho : \sum \gamma_\rho^2 |c_\rho|^2 < \infty \Big\}.$$

Thus the original zero multiplicities and zero placement are part of the requirement, not merely the multiset of ordinates. Use the normalized determinant prescription (77.4) under the unitary $u_\rho \mapsto \delta_\rho$. The zero symmetry is represented by an involutive unitary R_ξ which exchanges the occurrences ρ and $1-\rho$, preserves Dom A, and satisfies $R_\xi A R_\xi = -A$. Identifying $U^* R_\xi U$ with a native arithmetic S-modular operation is an additional specified comparison, including its carrier and domain.

Remark 77.4.4 *(the arithmetic comparison problem).* The eigenvalue condition implies RH and gives the abstract normalized Hilbert–Pólya realization of Chapter 74.2.3. Conversely, under RH the diagonal zero-occurrence construction gives that abstract operator. It does not construct the Petersson realization, native chiral maps, or arithmetic symmetry comparison in the hypothesis. Those are the additional requirements of the arithmetic formulation Chapter 44.3.2. Transport by ιU^* places a specified realization in the same GNS Hilbert space while retaining its domain, but supplies no equality with N_{BC} or N_{occ}.

77.5 Prime distributions and the explicit formula

Normalization and the spaces of prime distributions

The exponential coordinate $m = e^t$ changes the growth of a counting measure. It also fixes the weight appearing in a Fourier formula. We use additive Dirac masses on $\mathbb{R}$, the test space $\mathscr{A} = C_c^\infty(\mathbb{R})$ with its usual inductive-limit topology, and the Schwartz space $\mathscr{S}(\mathbb{R})$. Distributions are complex-linear in their test functions. Write γ for Euler's constant and

$$\xi(s) = \xi_{\mathbb{R}}(s) = \tfrac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

For $t > 0$, put

$$\vartheta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}, \qquad \psi(t) = \sum_{n\geq 1} e^{-\pi n^2 t}.$$

These series and their derivatives converge uniformly on compact subsets of $t > 0$.

The Gaussian identity $\int_{\mathbb{R}} e^{-\pi t x^2} e^{-2\pi i y x}\, dx = t^{-1/2} e^{-\pi y^2/t}$ follows by differentiation in y and integration by parts; its value at zero is the Gaussian integral. Periodize $e^{-\pi t x^2}$ over the integers. Its Fourier coefficients are the displayed Gaussian values at integer y. Both the periodized series and the resulting Fourier series converge absolutely and uniformly. Equality follows from uniqueness of Fourier coefficients for continuous periodic functions. For completeness, uniqueness follows by convolution with

$$F_n(x) = \frac{1}{n}\left|\sum_{k=0}^{n-1} e^{2\pi i k x}\right|^2 :$$

these nonnegative kernels have integral one and their mass outside any neighborhood of the integers tends to zero. Their convolutions converge uniformly to a continuous periodic function. Evaluating at zero gives

$$\vartheta(t) = t^{-1/2}\vartheta(t^{-1}), \qquad \psi(t) = t^{-1/2}\psi(t^{-1}) + \tfrac{1}{2}(t^{-1/2} - 1).$$

Define

$$\xi_{\mathbb{R}}(s) = \frac{1}{2} + \frac{s(s-1)}{2}\int_1^\infty \psi(t)\left(t^{s/2} + t^{(1-s)/2}\right)\frac{dt}{t}. \tag{77.3}$$

The exponential decrease of ψ makes this integral entire, with every derivative locally uniformly convergent. It gives $\xi_{\mathbb{R}}(0) = \xi_{\mathbb{R}}(1) = 1/2$ and

$$\xi_{\mathbb{R}}(1-s) = \xi_{\mathbb{R}}(s), \qquad \xi_{\mathbb{R}}(\bar{s}) = \overline{\xi_{\mathbb{R}}(s)}.$$

On $\operatorname{Re} s > 1$, define $\zeta(s) = \sum_{n\geq 1} n^{-s}$ and $\Gamma(s/2) = \int_0^\infty e^{-u} u^{s/2-1}\, du$. Absolute convergence permits summation under the integral and gives

$$\int_0^\infty \psi(t) t^{s/2}\, \frac{dt}{t} = \pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Split the integral at one and use the theta identity on $(0,1)$. The elementary contribution is $1/(s-1) - 1/s = 1/(s(s-1))$; the remaining integral is that in (77.3). Therefore

$$\xi_{\mathbb{R}}(s) = \tfrac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) \quad (\operatorname{Re} s > 1).$$

Equation (77.3) supplies the entire continuation. No assertion about simplicity or critical-line placement of its zeros was used.

Lemma 77.5.1 *(growth and zero counting from the theta integral).* For $R \geq 2$ one has $\log \max_{|s|\leq R} |\xi(s)| \leq CR\log(R+2)$. Consequently the number of its zeros in $|s| \leq R$, with multiplicity, is $O(R\log(R+2))$, and $\sum_\rho |\rho|^{-2} < \infty$. There are infinitely many zeros. All lie in $0 \leq \Re s \leq 1$.

Proof. In the theta integral (77.3), the theta tail on $x \geq 1$ is bounded by $Ce^{-\pi x}$. For $|s| \leq R$ the powers of x in that integral are bounded by $2x^{(R+1)/2}$. The estimate $\sup_{x>0} x^A e^{-\pi x/2} = (2A/(\pi e))^A$ bounds the integral by $\exp(CR\log(R+2))$; the quadratic prefactor does not change this bound. Jensen's formula at radius $2R$, using $\xi(0) = 1/2$, bounds each zero in radius R by a contribution at least $\log 2$. This proves the zero count. Dyadic summation then proves the inverse-square assertion.

The growth bound gives entire order at most one. If there were only finitely many zeros, Hadamard's factorization theorem would give $\xi(s) = e^{a+bs}P(s)$ with a polynomial P. For real $s \geq 2$, however, $\zeta(s) \geq 1$ and integration of the gamma integral just over $[s/2, s/2+1]$ gives $\log \xi(s) \geq (s/2)\log s - Cs$. This contradicts that finite factorization. Finally, the Euler product excludes zeros for $\Re s > 1$, and the functional equation excludes $\Re s < 0$. □

Proposition 77.5.2 *(the constant in the genus-one product).* For this normalization, $\xi(0) = 1/2$, and the constant multiplying s in the canonical genus-one product is

$$B = \frac{\xi'(0)}{\xi(0)} = -1 - \frac{\gamma}{2} + \frac{1}{2}\log(4\pi).$$

Thus, with zeros counted with multiplicity,

$$\xi(s) = \tfrac{1}{2} e^{Bs} \prod_\rho (1 - s/\rho) e^{s/\rho}, \qquad \frac{\xi'(s)}{\xi(s)} = B + \sum_\rho \left(\frac{1}{s-\rho} + \frac{1}{\rho} \right).$$

The factors and the paired summands converge locally uniformly away from their indicated zeros or poles. The individual sum $\sum_\rho 1/\rho$ is not separated from its partner without a specified summation convention.

Proof. The identity $\Gamma(1+s/2) = (s/2)\Gamma(s/2)$ removes the apparent singularity at zero:

$$\xi(s) = (s-1)\Gamma(1+s/2)\pi^{-s/2}\zeta(s).$$

The values $\zeta(0) = -1/2$, $\zeta'(0) = -\frac{1}{2}\log(2\pi)$, and $\Gamma'(1) = -\gamma$ give

$$\frac{\xi'(0)}{\xi(0)} = -1 - \frac{\gamma}{2} - \frac{1}{2}\log\pi + \log(2\pi).$$

For example, the two zeta values follow directly by expanding $\zeta(s) = 2^s\pi^{s-1}\sin(\pi s/2)\Gamma(1-s)\zeta(1-s)$: the constant terms cancel in $\Gamma(1-s)\zeta(1-s) = -s^{-1} + O(s)$. The remaining factor is $\frac{s}{2}(1 + s\log(2\pi) + O(s^2))$. The exact special values are also recorded in DLMF 25.6.1, 25.6.11, and 5.4.11.

Lemma 77.5.1 supplies the growth and zero bound for the order-one factorization. On a compact set with $|s/\rho| \le 1/2$, the logarithm of a factor and its logarithmic derivative are bounded by a constant times $|\rho|^{-2}$. Hence the product and paired derivative converge normally there. Every canonical factor has derivative zero at $s = 0$, so its exponential constant is precisely the logarithmic derivative just computed. This also specifies the normalization of the product. □

Lemma 77.5.3 *(an elementary lower bound for logarithmic prime mass).* Put $\psi(x) = \sum_{m\le x}\Lambda(m)$. For every integer $n \ge 1$,

$$2n\log 2 - \log(2n+1) \le \psi(2n), \qquad \psi(x) \le x\log x \quad (x \ge 2).$$

Consequently there are positive constants c, x_0 with $\psi(x) \ge cx$ for $x \ge x_0$.

Proof. The largest of the $2n+1$ binomial coefficients in $(1+1)^{2n}$ is $\binom{2n}{n}$, so it is at least $4^n/(2n+1)$. For every prime p, its valuation is

$$v_p\binom{2n}{n} = \sum_{k\ge 1}\left(\left\lfloor\frac{2n}{p^k}\right\rfloor - 2\left\lfloor\frac{n}{p^k}\right\rfloor\right).$$

Each summand is zero or one and vanishes when $p^k > 2n$. Taking logarithms therefore gives $\log\binom{2n}{n} \le \psi(2n)$. The upper bound uses $0 \le \Lambda(m) \le \log m$. For arbitrary sufficiently large x, take $n = \lfloor x/2\rfloor$ and use monotonicity. No prime-number asymptotic is needed. □

Theorem 77.5.4 *(the exact exponential weight for temperedness).* For real σ, define the positive Radon measure

$$\mu_\sigma = \sum_{m\ge 2}\Lambda(m)m^{-\sigma}\delta_{\log m}.$$

Every μ_σ is a distribution on $\mathscr{A}$. It is a tempered distribution exactly when $\sigma \ge 1$. It is finite exactly when $\sigma > 1$; μ_1 has infinite total mass. In particular, $\mu_0 = \mu_{\mathrm{prime}}$ is unchanged as a locally finite measure, and $\mu_{1/2}$ is not tempered.

Proof. Only finitely many atoms meet any compact interval. Their total variation bounds evaluation by the supremum norm on that interval, which proves continuity on $\mathscr{A}$. For $\sigma \ge 1$,

$$|\mu_\sigma(\varphi)| \le \sup_t(1+|t|)^3|\varphi(t)| \sum_{m\ge 2}\frac{\log m}{m(1+\log m)^3}.$$

The last series converges by the integral test. If $\sigma > 1$, $\sum(\log m)m^{-\sigma} < \infty$ also gives finiteness. Partial summation gives

$$\sum_{m\le X}\frac{\Lambda(m)}{m} = \frac{\psi(X)}{X} + \int_1^X \frac{\psi(x)}{x^2}\,dx,$$

which diverges by Lemma 77.5.3.

If a positive measure supported in $[0,\infty)$ is tempered, choose a nonnegative smooth compact cutoff equal to one on $[0,1]$ and rescale it by $t \mapsto t/T$. The finite Schwartz seminorm bound for a tempered

distribution then implies $\mu([0,T]) \le C(1+T)^M$ for some C, M and all $T \ge 1$. For $0 \le \sigma < 1$, however,

$$\mu_\sigma([0,T]) \ge e^{-\sigma T}\psi(e^T) \ge ce^{(1-\sigma)T}$$

for large T. For $\sigma < 0$, use $\mu_\sigma \ge \mu_0$. Both contradict polynomial mass growth. □

Multiplication by $e^{-\sigma t}$ is explicit here; it is not a coordinate relabelling. Equivalently, for $\sigma \ge 1$ the unweighted μ_0 acts continuously on $\mathscr{S}_\sigma = e^{-\sigma t}\mathscr{S}(\mathbb{R})$ with the transported Schwartz topology. It acts on C_c^∞ without any choice of weight. The change $x = e^t$ pulls a compact smooth test back by $f(t) = g(e^t)$ and gives the exact pairing $\mu_\sigma(f) = \sum \Lambda(m)m^{-\sigma}g(m)$. These are maps between specified test spaces, not Hilbert-space intertwiners.

Proposition 77.5.5 *(Laplace and Fourier transforms on their domains).* For $\Re s > 1$,

$$\int e^{-st}\, d\mu_0(t) = \sum_{m\ge 2} \frac{\Lambda(m)}{m^s} = -\frac{\zeta'(s)}{\zeta(s)}.$$

With Fourier convention $\mathcal{F}_- f(u) = \int f(t)e^{-iut}\,dt$, the finite measure μ_σ, $\sigma > 1$, has Fourier transform $-\zeta'(\sigma+iu)/\zeta(\sigma+iu)$. As $\sigma \downarrow 1$ these transforms converge in the weak tempered-distribution sense to $\mathcal{F}_-\mu_1$. The unweighted and critically weighted measures do not have Fourier transforms in $\mathscr{S}'$ by this construction.

Proof. The absolutely convergent Euler product for $\Re s > 1$ may be logarithmically differentiated on compact subsets there, giving the displayed Dirichlet series. Absolute convergence also permits integration of the finite weighted measure against the Fourier kernel. For every Schwartz test, dominated convergence in the seminorm estimate of Theorem 77.5.4 gives $\mu_{1+\varepsilon} \to \mu_1$ weakly in $\mathscr{S}'$. Fourier continuity proves the boundary assertion. Meromorphic continuation of the right side does not change the abscissa of absolute convergence of the original positive-measure integral. □

The explicit formula on compact Fourier tests

For $f \in \mathscr{A}$, use the entire function $h(z) = \int f(t)e^{izt}\,dt$, so that $f(t) = (2\pi)^{-1}\int h(u)e^{-iut}\,du$. This is the Fourier test carrier of Definition 78.1.1. For $\rho = \beta + i\gamma_\rho$ put $z_\rho = -i(\rho - 1/2) = \gamma_\rho - i(\beta - 1/2)$. Lemma 77.5.1 gives $\#\{\rho : |\gamma_\rho| \le T\} = O(T\log(T+2))$, with multiplicity. It follows that $\mu_{\mathrm{ord}} = \sum_\rho \delta_{\gamma_\rho}$ is tempered. Since $0 \le \beta \le 1$, integration by parts gives rapid decay of $h(z_\rho)$ uniformly in this horizontal strip. Thus $\mathfrak{Z}_\xi(h) = \sum_\rho h(z_\rho)$ converges absolutely and is continuous on $\mathscr{A}$ through $f \mapsto h$. The estimates therefore follow from the theta integral without requiring an asymptotic prime-counting formula.

Theorem 77.5.6 *(the full polar and archimedean terms).* For every $f \in C_c^\infty(\mathbb{R})$ and its transform h above,

$$\begin{aligned}\mathfrak{Z}_\xi(h) = h(i/2) + h(-i/2) + \frac{1}{2\pi}\int_{\mathbb{R}} h(u)\big(\Re\psi_\Gamma(1/4+iu/2) - \log\pi\big)\,du \\ - \sum_{m\ge 2}\frac{\Lambda(m)}{\sqrt{m}}\big(f(\log m) + f(-\log m)\big),\end{aligned}$$

where $\psi_\Gamma = \Gamma'/\Gamma$ is the digamma function. The prime sum is finite. In particular, it is evaluation of $\mu_{1/2}$ on compact tests and does not assert temperedness of that measure. The values $h(\pm i/2)$ are the polar terms, separate from the real-line digamma density.

Proof. Set $H(s) = h(-i(s-1/2)) = \int f(t)e^{(s-1/2)t}\,dt$ and choose $a > 1$. The residue theorem and $\xi(s) = \xi(1-s)$ give

$$\sum_{\rho} H(\rho) = \frac{1}{2\pi i}\int_{\Re s=a} (H(s)+H(1-s))\frac{\xi'(s)}{\xi(s)}\,ds.$$

Here is a convergence justification. The zero count allows heights $T_j \in [j, j+1]$ at distance at least j^{-2} from every zero ordinate: the excluded intervals have total length $O(\log j/j)$. The paired logarithmic derivative in Proposition 77.5.2 is polynomially bounded on the horizontal segments at these heights. For $|\rho| \le 2T_j$ bound each denominator by the stated distance and sum $O(T_j \log T_j)$ terms; for $|\rho| > 2T_j$ use $|s/\{\rho(s-\rho)\}| \le CT_j/|\rho|^2$. Repeated integration by parts makes H decrease faster than any power, uniformly on the fixed strip. The horizontal integrals therefore vanish, and the zero sum converges absolutely.

On the right line decompose

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi_{\mathrm{T}}(s/2) + \frac{\zeta'}{\zeta}(s).$$

Fourier inversion and the absolutely convergent Dirichlet series give the negative prime sum from the last term. For the remaining terms move the line to $\Re s = 1/2$. The crossed pole at $s = 1$ contributes $H(1) + H(0)$. On the new line the rational sum $1/s + 1/(s-1)$ is odd in the imaginary coordinate, while $H(s) + H(1-s)$ is even; its integral is zero. Changing u to $-u$ in half of the gamma integral gives precisely its real part in the displayed formula. Digamma growth is $O(\log(|u|+2))$ on this line: in $\psi_{\mathrm{T}}(z) = -\gamma + \sum_{n\ge 0}((n+1)^{-1} - (n+z)^{-1})$, split the sum at $n = |u|+1$ for $z = 1/4 + iu/2$. The initial sum is $O(\log(|u|+2))$ and the tail is bounded by $C(|u|+1)\sum_{n>|u|+1} n^{-2} = O(1)$. Thus all resulting integrals converge absolutely. □

When f is even, h is even and the prime term is $-2\mu_{1/2}(f)$. The functional $h \mapsto h(i/2) + h(-i/2)$ cannot be silently included in a density on the real line. Likewise, $\mathfrak{Z}_\xi(h)$ uses the full complex numbers z_ρ, whereas $\mu_{\mathrm{ord}}(h) = \sum h(\gamma_\rho)$ uses their real parts. Under RH these are equal. No projection is used in the theorem.

77.6 Determinant Identity: Hadamard Factorisation

The normalized genus-one determinant

Let Z contain each zero ρ of ξ with its multiplicity. On $K_Z = \ell^2(Z)$, define

$$(\Theta_Z u)_\rho = \rho u_\rho, \qquad \operatorname{Dom}\Theta_Z = \{u : \sum_{\rho} |\rho u_\rho|^2 < \infty\}.$$

This is a closed normal operator. Its inverse is bounded, compact, and Hilbert–Schmidt: $\xi(0) = 1/2$, the zeros are discrete, and $\sum_\rho |\rho|^{-2} < \infty$.

Theorem 77.6.1 *(complete determinant normalization).* Define the second regularized Fredholm determinant of the diagonal Hilbert–Schmidt operator $-s\Theta_Z^{-1}$ by

$$D_2(s) = \det{}_2(1 - s\Theta_Z^{-1}) := \prod_{\rho\in Z}(1 - s/\rho)e^{s/\rho}.$$

With $B = -1 - \gamma/2 + \frac{1}{2}\log(4\pi)$, set

$$\Delta_Z(s) := \tfrac{1}{2} e^{Bs} D_2(s). \tag{77.4}$$

This entire function equals $\xi(s)$. It is the unique entire function of order at most one with this zero divisor, $\Delta_Z(0) = 1/2$, and $\Delta_Z'(0)/\Delta_Z(0) = B$. Off the zero divisor it satisfies

$$(\log \Delta_Z)''(s) = -\operatorname{Tr}(\Theta_Z - s)^{-2}.$$

The trace is an ordinary absolutely convergent trace. Unitary conjugacy of Θ_Z, including its domain, preserves this determinant prescription.

Proof. For $|w| \le 1/2$, $\log(1-w) + w = O(|w|^2)$. The inverse-square sum proves normal convergence of the product on every compact set, independently of the enumeration of zeros. Its value and derivative at zero are 1 and 0. Proposition 77.5.2 gives the product for ξ with exactly the factor $e^{Bs}/2$. This proves the identity. Hadamard factorization writes every other entire function of order at most one with this divisor as $e^{a+bs} D_2(s)$. The two initial conditions give $e^a = 1/2$ and $b = B$. Twice differentiating the normally convergent logarithmic product gives $-\sum_\rho (\rho - s)^{-2}$. The inverse-square estimate proves absolute trace convergence. Unitary conjugacy preserves the diagonal eigenvalues and their multiplicities, hence every factor in the prescribed product. □

Proposition 77.6.2 *(the divisor leaves an exponential ambiguity).* For every $c \in \mathbb{C}$, the function $e^{cs}\xi(s)$ has the same divisor, the same value at zero, and order at most one. Its logarithmic derivative at zero is $B + c$. Thus a divisor condition and a value normalization do not imply the determinant identity. Both normalizations in (77.4) are necessary.

Proof. The exponential has no zero, has value one at zero, and contributes c to the logarithmic derivative. Multiplication preserves the stated order bound. □

The prescription above uses no complex powers of $s - \Theta_Z$. A determinant defined by such powers requires a spectral cut and meromorphic continuation of its trace. Agreement with Δ_Z requires the same second logarithmic derivative and the two stated initial conditions. These conditions determine equality locally near zero by integration, and globally by analytic continuation.

Remark 77.6.3 *(operator comparison and determinant normalization).* For a normal operator with a complete orthonormal eigenbasis indexed by the same zeros, use the prescription

$$\det_\infty(s - \Theta) := \tfrac{1}{2} e^{Bs} \det_2(1 - s\Theta^{-1}) = \xi(s).$$

The unitary eigenbasis map proves equality with Δ_Z. This fixes the determinant notation in the determinant-trace hypothesis. The construction of such an operator from an arithmetic primary sector requires an additional domain-preserving comparison. The diagonal operator Θ_Z itself uses the zero divisor as input.

Remark 77.6.4 *(dilation and occupation models).* Berry–Keating's xp model concerns zero-density asymptotics (*SIAM Review* 41 (1999), 236–266). A basic dilation realization illustrates the separate operator question. The map $f(x) \mapsto f(e^u)$ is unitary from $L^2(\mathbb{R}_{>0}, dx/x)$ to $L^2(\mathbb{R}, du)$ and sends $-ix\partial_x$ to $-i\partial_u$, with domain transported from $H^1(\mathbb{R})$. Fourier transformation gives spectrum $\mathbb{R}$. This self-adjoint dilation generator does not have a discrete zeta-zero divisor. Likewise the self-adjoint occupation operator N_{occ} with eigenvalues $\log m$ supplies no such divisor. A density comparison or a logarithmic weight must be supplemented by the actual determinant-trace realization; see Chapter 57.3.17 for the corresponding spectral question.

77.7 Trace Distribution: the Full Explicit Formula

Proposition 77.7.1 *(ordinary trace on compact Fourier tests).* Assume Chapter 77.4.3. For $f \in \mathcal{A} := C_c^\infty(\mathbb{R})$ set $h(z) = \mathcal{F}_+ f(z) := \int_{\mathbb{R}} f(t) e^{izt}\, dt$. Then $h(A)$ is trace class and

$$\begin{aligned} \operatorname{Tr} h(H_{\mathrm{HP}}) &= \sum_\rho h(\gamma_\rho) \\ &= h(i/2) + h(-i/2) + \frac{1}{2\pi} \int_{\mathbb{R}} h(u) \big(\Re \psi_\Gamma(1/4 + iu/2) - \log \pi \big)\, du \\ &\quad - \sum_{m \geq 2} \frac{\Lambda(m)}{\sqrt{m}} \big(f(\log m) + f(-\log m) \big), \end{aligned} \tag{77.5}$$

where $\psi_\Gamma = \Gamma'/\Gamma$. The prime sum is finite. Equivalently the right side is $P(f) - W_\infty(f) - W_{\mathrm{fin}}(f)$, with signed archimedean density $\log \pi - \Re \psi_\Gamma(1/4 + iu/2)$ and polar functional $P(f) = h(i/2) + h(-i/2)$.

Proof. Repeated integration by parts gives $|h(u)| \leq C_N (1 + |u|)^{-N}$ for real u and every N. The zero-count estimate of Chapter 77.5.1 is $O(T \log(2 + T))$. Summing over dyadic intervals, with $N > 2$, therefore gives $\sum_\rho |h(\gamma_\rho)| < \infty$. The specified orthonormal eigenbasis makes $h(A)$ a diagonal trace-class operator with exactly this ordinary trace. The determinant spectral condition gives $\rho = 1/2 + i\gamma_\rho$. The compact-test explicit formula Chapter 77.5.6 now gives all terms displayed, with its Fourier convention and both prime-test values. The gamma integral converges by logarithmic growth of the density. Unitary transport of this trace-class operator to E preserves the ordinary trace; no unspecified regularization is inferred from an isometric embedding of unrelated operators. □

Corollary 77.7.2 *(the full Weil-positive cone and a Petersson comparison).* Under the same hypothesis, put $f^*(t) = \overline{f(-t)}$. For every $f \in \mathcal{A}$ the full Weil value satisfies

$$W(f * f^*) = \operatorname{Tr}\big(h_f(A) h_f(A)^* \big) = \sum_\rho |h_f(\gamma_\rho)|^2 \geq 0, \qquad h_f = \mathcal{F}_+ f.$$

Thus Weil positivity and Li positivity hold. To obtain this conclusion instead from a proposed Petersson realization, one must construct a complex-linear map J on all of $\mathcal{A}$ with $\langle Jf, Jg \rangle_{\mathrm{Pet}} = W(f * g^*)$, or construct it on a dense subspace in a topology for which the full form is continuous. Positivity only on an unspecified restricted cone supplies neither condition.

Proof. Fourier transformation sends convolution and involution to $h_f h_f^\#$, where $h_f^\#(z) = \overline{h_f(\bar{z})}$. On the real spectrum this is $|h_f|^2$. The preceding absolute trace argument applies to $f * f^*$. This proves positivity on the full compact test algebra, hence the Weil criterion and the Li criterion of Chapter 74.2.1 and Chapter 74.2.2 give the conclusions. For a Petersson map the stated pairing identity gives positivity directly; if its domain is dense, continuity passes the inequality to limits. The full compact-test form is continuous in the inductive-limit topology by Chapter 77.5.6. □

77.8 Unconditional Constructions and the Arithmetic Comparison

Bost–Connes and Oscillator Carriers

The global BC algebra, critical state and rational-character GNS basis give the maximal self-adjoint generator with point spectrum $\frac{1}{2} \log \mathbb{Q}_{>0}$ and closed spectrum $\mathbb{R}$. The finite basis span is a core, not

its full self-adjoint domain. Its restriction N_{occ} to K_2 has simple eigenvalues $\log m$. The weighted zero-mode embedding, explicit oscillator lift, closed CCR operators and logarithmic energy identity are unconditional constructions after the stated auxiliary choices. They do not identify the oscillator algebra with affiliated BC observables or identify conformal energy with modular energy.

Consequences of a Zero-Divisor Realization

Given the eigenbasis and domain in Chapter 77.4.3, the prescribed normalized genus-one determinant is $\xi_{\mathbb{R}}$ by Chapter 77.6.1. On compact Fourier tests the ordinary trace is the full polar, signed-gamma and prime formula (77.5). It is positive on all convolution squares, and Li positivity follows. These statements use the zero-divisor realization; they are not occupation-trace identities. The domain-preserving unitary ιU^* transports them to E for A_E, while leaving its distinction from $N_{\mathrm{BC}}|_E$ intact.

Hilbert–Pólya and the Other Trace Models

The remaining arithmetic problem is to construct the native primary-sector realization, its Petersson pairing and chiral maps, and an operator A satisfying the normalized zero-divisor condition. The zero-input diagonal operator proves no such arithmetic origin. Connes' adele-class-space absorption trace and Deninger's motivic determinant are further comparison targets. As in Chapter 57.3.14, a comparison requires explicit Connes/Fock and Deninger/Fock intertwiners.

Theorem 77.8.1 *(conditional determinant and trace comparison).* Assume the arithmetic twist hypothesis. Its normalized determinant and full compact-test trace are those of Chapter 77.6.1 and Chapter 77.7.1. For a Connes comparison, additionally specify a Hilbert realization H_C of the relevant adele-class-space representation, a self-adjoint scaling generator A_C, and a unitary $V_C : \mathcal{H}_\xi \to H_C$ with

$$V_C \operatorname{Dom} A = \operatorname{Dom} A_C, \qquad A_C V_C = V_C A.$$

The absorption-trace prescription must also agree on these tests with $\operatorname{Tr} h(A_C)$, with its stated sign and normalization. For a Deninger comparison, additionally specify a Hilbert realization H_D of the proposed motivic cohomology, a closed normal generator $\Phi = \frac{1}{2}I + iA_D$ with A_D self-adjoint, and a unitary $V_D : \mathcal{H}_\xi \to H_D$ intertwining Θ_ξ and Φ on their full domains. Its determinant prescription must have the same genus-one normalization, or be proved to agree with it. Under these hypotheses the normalized determinant and trace statements transfer to both models and back. Without the maps and prescriptions they are comparison questions, not equivalences of the constructed operators.

Proof. The first assertion follows from the stated eigenbasis map and the ordinary trace-class argument. A domain-preserving unitary intertwiner preserves the spectral calculus, eigenvalue multiplicities and trace-class traces. It also preserves the normalized determinant prescription, as proved in Chapter 77.6.1. The further trace/determinant agreements in the hypotheses identify these transported quantities with the respective geometric prescriptions. Applying the inverse unitaries proves the reverse transfers. In the motivic notation, self-adjointness concerns A_D, while the completed zero generator is $\Phi = 1/2 + iA_D$; these are different operators. This makes explicit the intertwiner hypotheses retained in Chapter 57.3.14. □

Remark 77.8.2 *(logarithmic weights and their normalization).* On the specified carrier K_2, the positive squared shifts satisfy $N_{\mathrm{occ}}\mu_{n^2} = \mu_{n^2}(N_{\mathrm{occ}} + \log n)$ on their maximal domains, and the vacuum has energy zero. These two requirements force the eigenvalue $\log m$ at $\mu_{m^2}\Omega_1$. They also give $\operatorname{Tr}_{K_2} e^{-\beta N_{\mathrm{occ}}} = \zeta(\beta)$ for $\Re\beta > 1$. KMS alone does not select K_2: for integers $s \geq 1$, the sectors $K_s = \overline{\operatorname{span}}\{\mu_{m^s}\Omega_1\}$ have

energies $(s/2)\log m$. Thus the weight question of Chapter 62.5.2 has this explicit occupation realization after the carrier, covariance and vacuum normalization are fixed. It does not solve the arithmetic zero-divisor comparison.

Remark 77.8.3 *(the conditional arithmetic conclusion).* There are explicit self-adjoint modular and occupation generators, and an explicit symmetry-invariant Hilbert Fock embedding in the critical GNS space. The conformal grading can be transported unitarily as a separate operator. A native arithmetic Hilbert–Pólya realization still requires its own operator, domain, pairing and normalized zero-divisor data. Once these are constructed, the determinant and full trace identities follow. Comparisons with Connes' absorption trace, Deninger's determinant, or a native chiral operation retain the additional maps and normalizations stated above.

Operator Carriers and Comparison Data

Object	Carrier and conclusion
BC algebra and KMS_1 state	Global crossed product; faithful normal GNS state.
N_{BC}	Full GNS space; maximal self-adjoint domain, closed spectrum $\mathbb{R}$.
N_{occ}	K_2; simple spectrum $\{\log m\}$ and partition $\zeta(\beta)$ for $\Re\beta > 1$.
Oscillator lift	Chosen unitary $\mathcal{F}_{\mathrm{Heis}}^{\mathrm{Ar}} \to E \subset \mathcal{H}_1^G$; closed CCR operators and logarithmic prime energy.
Conformal grading	$G_F = L_0 + H_w$, transported to G_ξ; distinct from the modular generator.
Arithmetic A and Θ_ξ	Full-domain self-adjoint A and normal $\Theta_\xi = 1/2 + iA$; prescribed zero-divisor realization remains hypothetical.
Normalized determinant	Equals $\xi_{\mathbb{R}}$ conditional on that realization.
Compact-test trace	Ordinary trace-class formula $P - W_\infty - W_{\mathrm{fin}}$ on the same condition.
Petersson and geometric transfers	Require the full test pairing or dense continuous comparison, and the specified Connes/Deninger intertwiners.

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Chapter 78

Weil Pairings and Automorphic Coefficient Multipliers

The Weil functional acts on compact smooth functions of the logarithmic variable. Its zero formula evaluates their entire Fourier transforms. Keeping these two spaces distinct fixes the prime weights and the archimedean sign. It also distinguishes two coefficient multipliers, one of which can produce only finite Fourier polynomials.

78.1 The compact test algebra and its Fourier image

Definition 78.1.1 *(test spaces and involutions).* Put $\mathscr{A} = C_c^\infty(\mathbb{R})$, with its usual inductive-limit topology, and put $\mathscr{P} = \mathcal{F}_+\mathscr{A}$, where

$$h(z) = \mathcal{F}_+ f(z) = \int_{\mathbb{R}} f(t) e^{izt}\, dt, \qquad f(t) = \frac{1}{2\pi}\int_{\mathbb{R}} h(u) e^{-iut}\, du.$$

Give $\mathscr{P}$ the transported topology. Its elements are entire functions, rapidly decreasing in each horizontal strip. For $f, g \in \mathscr{A}$, define

$$(f * g)(t) = \int f(v) g(t-v)\, dv, \qquad f^*(t) = \overline{f(-t)}, \qquad h^\sharp(z) = \overline{h(\bar{z})}.$$

Thus $\mathcal{F}_+(f*g) = (\mathcal{F}_+ f)(\mathcal{F}_+ g)$ and $\mathcal{F}_+(f^*) = (\mathcal{F}_+ f)^\sharp$.

The identities follow from Fubini and a change of variables. A second real Fourier transform of $h \in \mathscr{P}$ is a reflected compact smooth function. It is generally not an entire function available for evaluation at complex zeros.

Theorem 78.1.2 *(the signed archimedean contribution).* Set $\xi(s) = \tfrac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $\psi_\Gamma = \Gamma'/\Gamma$. For $f \in \mathscr{A}$, write $h = \mathcal{F}_+ f$ and $z_\rho = -i(\rho - 1/2)$, where ρ ranges over the zeros of ξ with multiplicity. Define

$$\begin{aligned} P(f) &= h(i/2) + h(-i/2), \\ W_\infty(f) &= \frac{1}{2\pi}\int_{\mathbb{R}} h(u)\big(\log\pi - \Re\psi_\Gamma(1/4 + iu/2)\big)\, du, \\ W_{\mathrm{fin}}(f) &= \sum_{m \ge 2} \frac{\Lambda(m)}{\sqrt{m}}\big(f(\log m) + f(-\log m)\big). \end{aligned} \tag{78.1}$$

Then the continuous distribution $W : \mathscr{A} \to \mathbb{C}$ is

$$W(f) = \sum_\rho h(z_\rho) = P(f) - W_\infty(f) - W_{\mathrm{fin}}(f). \tag{78.2}$$

The zero sum converges absolutely and the prime sum is finite. For even f, the prime sum is $2\mu_{1/2}(f)$.

Proof. Theorem 77.5.6 gives the formula with $\mathcal{A}_\infty(h) = (2\pi)^{-1}\int h(\Re\psi_\Gamma - \log\pi)$. The definition here is $W_\infty(f) = -\mathcal{A}_\infty(h)$. This accounts for both signs in (78.2). The compact-test estimates in that theorem prove the convergence and continuity assertions. No tempered extension of $\mu_{1/2}$ is used. □

PROPOSITION 78.1.3 *(a compact test determining the sign).* Let $0 < a < b < \log 2$. Suppose $f \neq 0$ is even, nonnegative, smooth, and supported in $(-b,-a)\cup(a,b)$. Then

$$W_{\mathrm{fin}}(f) = 0, \qquad W_\infty(f) = 2\int_a^b \frac{e^{-t/2}}{1-e^{-2t}} f(t)\,dt > 0.$$

Proof. There are no prime-power logarithms in the support. Since $f(0) = 0$, the constant terms of the digamma integral vanish after Fourier inversion. The substitution $v = 2t$ in

$$\psi_\Gamma(z) + \gamma = \int_0^\infty \frac{e^{-v} - e^{-zv}}{1-e^{-v}}\,dv \quad (\Re z > 0)$$

gives the displayed positive integral. The identity is DLMF 5.9.16. The denominator is positive and f is positive on an interval. □

78.2 The Hermitian form and the two polar moments

Define $q_W(f,g) = W(f * g^*)$, with the first variable linear. The zero multiset is invariant under $z \mapsto \bar z$, by the functional equation and conjugation of ξ. Absolute convergence therefore gives $W(f^*) = \overline{W(f)}$. Consequently q_W is Hermitian.

COROLLARY 78.2.1 *(Weil's positivity criterion).* The Riemann Hypothesis is equivalent to

$$q_W(f,f) \geq 0 \qquad (f \in \mathscr{A}).$$

Proof. This is Weil's criterion on compact smooth additive tests, with the Fourier convention above. It is stated in Suzuki, *On the Hilbert space derived from the Weil distribution*, Section 1, equations (1.1)–(1.2). In that convention the zeros of $\xi(1/2 - iz)$ are evaluated with the opposite sign. Their reflected multiset is exactly $\{z_\rho\}$ here. Under RH the assertion follows directly from $q_W(f,f) = \sum_\rho |\mathcal{F}_+ f(\gamma_\rho)|^2$. The converse is the cited positivity criterion, not an assertion of positivity for this arithmetic form. □

PROPOSITION 78.2.2 *(the polar Hermitian form).* Put $a(f) = \int f(t)e^{-t/2}dt$ and $b(f) = \int f(t)e^{t/2}dt$. Then

$$P(f * g^*) = a(f)\overline{b(g)} + b(f)\overline{a(g)}.$$

The moment map $(a,b) : \mathscr{A} \to \mathbb{C}^2$ is continuous and surjective. The polar form has rank two and both signs. It vanishes on the common kernel of the two moments.

Proof. Evaluate $h_f h_g^\sharp$ at $i/2$ and $-i/2$. If a linear combination of a and b vanished on every test, the smooth function $c_1 e^{-t/2} + c_2 e^{t/2}$ would vanish identically. Its value and derivative at zero force $c_1 = c_2 = 0$. The moment map therefore has two-dimensional range. On that range the form has matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, with eigenvalues 1 and -1. □

The use of the entire function ξ does not remove these polar evaluations from its explicit formula. Removing a rank-two term changes the form. It does not preserve positivity automatically.

Conjecture 78.2.3 *(archimedean Tate positivity).* For every $f \in \mathscr{A}$,

$$W_\infty(f * f^*) \le P(f * f^*) - W_{\mathrm{fin}}(f * f^*).$$

Theorem 78.2.4 *(the full positivity inequality).* For all $f \in \mathscr{A}$,

$$P(f * f^*) - W_{\mathrm{fin}}(f * f^*) = q_W(f, f) + W_\infty(f * f^*).$$

Thus archimedean Tate positivity is equivalent to Weil positivity and to RH. The equality applies to any specified restricted class as well, but asserts no positivity on that class.

Proof. Apply (78.2) to $f * f^* \in \mathscr{A}$ and rearrange. Then apply Weil's criterion. □

78.3 Real compact tests with prescribed Mellin moments

Restrictions on moments specify a test space before any automorphic comparison is made. For $f \in \mathscr{A}$, write

$$M_f(s) = \int_{\mathbb{R}} f(t) e^{st}\, dt = \int_0^\infty f(\log x) x^{s-1}\, dx.$$

Define the real vector space

$$\mathscr{A}_{13} = \{f \in \mathscr{A} : f \text{ is real and even}, \quad M_f(0) = M_f'(0) = M_f(1) = M_f'(1) = 0\}.$$

Thus the reality and evenness condition (H1) and the double Mellin zeros at 0, 1 (H3) have a precise compact-time meaning. Its Fourier image is the real space $\mathcal{F}_+\mathscr{A}_{13}$; neither space is asserted to consist of automorphic multipliers. For even f, the independent displayed moments are integration against 1, $\cosh t$, and $t \sinh t$; $M_f'(0) = 0$ is automatic.

Proposition 78.3.1 *(Mellin moments do not cancel the polar form).* Every M_f, $f \in \mathscr{A}$, is entire and rapidly decreasing on each closed vertical strip. In particular, the strip holomorphy condition (H2) imposes no additional restriction depending on a holomorphic weight k. The space $\mathscr{A}_{13}$ contains a nonzero f with $P(f * f^*) > 0$.

Proof. Compact support permits differentiation under the integral at every complex s. If $s = \sigma + i\tau$ with σ in a fixed compact interval, integration by parts in $e^{i\tau t}$ bounds $M_f(s)$ by $C_N(1 + |\tau|)^{-N}$ for every N. Choose nonzero, nonnegative, real even $b \in \mathscr{A}$ and put $f = D^2(D^2 - 1)^2 b$, where $D = d/dt$. Integration by parts gives

$$M_f(s) = s^2(s^2 - 1)^2 M_b(s).$$

This f is real, even and compactly supported, and its Mellin transform has double zeros at 0, 1, −1. Yet

$$M_f(1/2) = M_f(-1/2) = \frac{9}{64} M_b(1/2) > 0.$$

In particular $f \neq 0$. Since $h_f(\pm i/2) = M_f(\mp 1/2)$ and $f = f^*$, the convolution identity yields $P(f * f^*) = 2M_f(1/2)^2 > 0$. Thus moments at 0, 1 cannot be substituted for the actual polar moments at $\pm 1/2$. □

Both coefficient maps below restrict to this class, with their respective time and Fourier arguments. The compact multiplier remains a finite Fourier polynomial. Its modularity criterion is still the vanishing criterion proved below; (H1)–(H3) do not remove it.

78.4 COEFFICIENT MULTIPLIERS ON THE TWO CARRIERS

Fix a holomorphic cusp form $F(z) = \sum_{n\ge 1} a_F(n) q^n$ of weight $k > 0$ for $\mathrm{SL}_2(\mathbb{Z})$, where k is even and $q = e^{2\pi i z}$. Its coefficients have polynomial growth. Indeed the invariant function $y^{k/2}|F(z)|$ is bounded on a fundamental domain and hence on the upper half-plane. The Fourier coefficient integral at $y = 1/n$ gives $|a_F(n)| \le C n^{k/2}$.

PROPOSITION 78.4.1 *(the normalized Hecke coefficient bound).* Suppose in addition that F is a Hecke eigenform, $a_F(1) = 1$. For every prime p and integer $n \ge 1$,

$$|a_F(p)\, p^{-(k-1)/2}| \le 2, \qquad |a_F(n)| \le d(n) n^{(k-1)/2},$$

where $d(n)$ is the number of positive divisors of n. These bounds control coefficients and imply no modularity for a coefficientwise multiplier.

Proof. Deligne, *La conjecture de Weil I*, Theorem 8.2, p. 302, applies to a normalized primitive holomorphic cuspidal eigenform of weight $k \ge 2$ and character ε at level N. Here $N = 1$, every eigenform is new and $\varepsilon = 1$, so every prime is unramified. The roots α_p, β_p of $X^2 - a_F(p)X + p^{k-1}$ have absolute value $p^{(k-1)/2}$. Their sum gives the prime bound.

For completeness the Hecke coefficient formula is

$$[q^m] T_n F = \sum_{d \mid (m,n)} d^{k-1} a_F(mn/d^2).$$

It follows by summing $F((az+b)/d)$ over $b \bmod d$, for $ad = n$, in the formula $T_n F = n^{k-1} \sum_{ad=n} d^{-k} \sum_{b \bmod d} F((az+b)/d)$. The coefficient at $m = 1$ is $a_F(n)$; the eigenform condition therefore gives $T_n F = a_F(n) F$. Comparing coefficients gives multiplicativity for coprime indices and $a_F(p^{r+1}) = a_F(p) a_F(p^r) - p^{k-1} a_F(p^{r-1})$. With $a_F(1) = 1$ this recurrence gives

$$a_F(p^r) = \sum_{j=0}^{r} \alpha_p^j \beta_p^{r-j}, \qquad |a_F(p^r)| \le (r+1)\, p^{r(k-1)/2}.$$

Multiplication over prime powers proves the all-integer bound. □

Definition 78.4.2 *(the two coefficient maps).* For $f \in \mathscr{A}$ and $h = \mathcal{F}_+ f \in \mathscr{P}$, set

$$\begin{aligned} U_F(f)(z) &= \sum_{n\ge 1} f(\log n)\, a_F(n) n^{(k-1)/2} q^n, \\ \Upsilon_F(h)(z) &= \sum_{n\ge 1} h(\log n)\, a_F(n) n^{(k-1)/2} q^n. \end{aligned} \tag{78.3}$$

The first sum is finite. The second converges locally uniformly on the upper half-plane, by polynomial coefficient growth and the factor $e^{-2\pi n \Im z}$. Neither definition asserts modularity.

THEOREM 78.4.3 *(the compact multiplier obstruction).* The function $U_F(f)$ is a cusp form of weight k for $\mathrm{SL}_2(\mathbb{Z})$ only if it vanishes identically. Thus the compact multiplier is modular exactly on $\{f : f(\log n) a_F(n) = 0 \text{ for all } n\}$. The entire multiplier $\Upsilon_F(h)$ is a distinct construction. Its modularity requires the transformation law of its resulting holomorphic series.

Proof. Write $G(z) = \sum_{n=1}^{N} b_n e^{2\pi i n z}$ for a nonzero finite Fourier polynomial. Then $G(iy)$ is analytic as a function of y near zero. If G were modular of positive weight k,

$$G(i/y) = (iy)^k G(iy).$$

The finite sum on the left is $O(e^{-2\pi/y})$ as $y \downarrow 0$. Hence $G(iy) = O(y^{-k} e^{-2\pi/y})$. Every Taylor coefficient at zero must vanish, contradicting the nonzero analytic function. This proves the first assertion. The coefficient condition is exactly vanishing of the finite Fourier polynomial. □

Remark 78.4.4 *(Mellin domains).* For $f \in \mathscr{A}$, the multiplicative Mellin integral

$$\int_0^\infty f(\log x) x^{s-1} dx = \int_{\mathbb{R}} f(t) e^{st}\, dt$$

is entire and rapidly decreasing on every closed vertical strip. This follows by integration by parts on the fixed compact support. Strip holomorphy imposes no additional condition on this carrier. For the entire multiplier $h(\log n)$, the analogous Mellin integral has a different domain and must be specified separately.

78.5 The exact Rankin–Selberg coefficient calculation

Theorem 78.5.1 *(unfolding for an actual cusp form).* Let $G(z) = \sum_{n\geq 1} b_n e^{2\pi i n z}$ be a holomorphic cusp form of weight $k > 0$ for $\mathrm{SL}_2(\mathbb{Z})$. Put $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$ and $d\mu = dx\, dy/y^2$. Fix its standard fundamental domain $\mathcal{F} \subset \mathbb{H}$. Use the Eisenstein series $E(z,s) = \sum_{\Gamma_\infty \backslash \mathrm{PSL}_2(\mathbb{Z})} \Im(\gamma z)^s$, where Γ_∞ is the translation subgroup. For real $s > 1$,

$$\int_{\mathcal{F}} |G(z)|^2 y^k E(z,s)\, d\mu = \frac{\Gamma(s+k-1)}{(4\pi)^{s+k-1}} \sum_{n\geq 1} \frac{|b_n|^2}{n^{s+k-1}}. \tag{78.4}$$

Proof. The Eisenstein series converges for $s > 1$. Its cusp growth is polynomial, while G decreases exponentially, so the integral is finite. Nonnegative summands permit unfolding by Tonelli: the integral becomes $\int_0^\infty \int_0^1 |G(x+iy)|^2 y^{s+k-2} dx\, dy$. Fourier orthogonality gives $\sum |b_n|^2 e^{-4\pi n y}$. A second application of Tonelli and the gamma integral prove the formula. □

Proposition 78.5.2 *(positive norm and native pairing maps).* The Petersson norm of an actual cusp form is positive definite. If a complex vector space E has an injective linear map $I : E \to L^2(\mathcal{F}, y^k d\mu)$ on the fixed fundamental domain, then

$$\langle v, w \rangle_I = \int_{\mathcal{F}} (Iv)(z) \overline{(Iw)(z)}\, y^k\, d\mu$$

is a positive definite form. Identifying another specified form with this one requires the corresponding pairing identity for I.

Proof. These assertions follow from the norm property of L^2 and injectivity. The unfolding formula computes the coefficients of G itself. For $G = \Upsilon_F(h)$ it therefore uses $|h(\log n) a_F(n) n^{(k-1)/2}|^2$, whenever this series is modular. □

Remark 78.5.3 *(arithmetic-sector pairing data).* Let E be a proposed arithmetic cuspidal sector and let $I : E \to L^2(\mathcal{F}, y^k d\mu)$ be an injective linear map. The preceding proposition constructs its positive pullback pairing. A Koszul–Petersson comparison additionally specifies a complex-bilinear pairing $B : E \times E \to \mathbb{C}$, an antilinear involution $\sigma : E \to E$, and the equality

$$B(v, \sigma w) = \langle Iv, Iw \rangle_{L^2} \qquad (v, w \in E).$$

This equality transports positivity because its diagonal is $\|Iv\|_2^2$. For a restricted tensor product, it requires compatible local realization maps and local pairings on the chosen reference vectors. The coefficient multipliers alone do not specify these data.

Local Hilbert models and the arithmetic comparison

A local pairing can be constructed without identifying it with an arithmetic trace sector. On $c_{00}(\mathbb{N}_{\geq 1}; \mathbb{C})$ set

$$\langle c, d \rangle_p = \sum_{m \geq 1} \frac{c_m \overline{d_m}}{m}, \qquad \langle \delta_m, \delta_n \rangle_p = \frac{\delta_{mn}}{m}.$$

The diagonal is positive for every nonzero finite sequence. Its completion is $\ell^2(\mathbb{N}, 1/m)$, with orthonormal basis $\sqrt{m}\, \delta_m$. An identification of this model with a local Bost–Connes GNS space must separately specify its state, representation, and map on this basis.

The polynomial Heisenberg Fock model is $\mathcal{F}_{\mathrm{alg}} = \mathbb{C}[x_1, x_2, \ldots]|0\rangle$, where every polynomial uses finitely many variables. For finite multi-indices ν, give its monomials the pairing

$$\langle x^\nu, x^\mu \rangle = \delta_{\nu\mu} \prod_{j \geq 1} j^{\nu_j} \nu_j!, \qquad a_{-j} = x_j, \quad a_j = j \partial_{x_j}, \quad a_0 = 0.$$

Direct differentiation proves $[a_j, a_{-l}] = j\delta_{jl}$ and $a_j^* = a_{-j}$ on this invariant polynomial domain. The positive completion has the normalized monomials as an orthonormal basis. The energy $N = \sum_{j \geq 1} j x_j \partial_{x_j}$ has eigenvalue $\sum_j j\nu_j$ on x^ν; its self-adjoint realization has maximal square-summable energy domain. This is the concrete Fock model underlying the comparison with Frenkel–Ben-Zvi, §3.5; an arithmetic identification remains additional data.

Remark 78.5.4 *(the specified primary trace and tensor problem).* Let $E := E_{\mathrm{cusp,temp}}$ denote a proposed complex vector-space realization of the tempered cuspidal part of the primary Hochschild trace class $\mathcal{V}^{\mathrm{prim}}$ of Chapter 67. The cohomological degree or graded completion is part of the data; this trace class is not being identified with a vertex algebra. The desired comparison specifies a bilinear Koszul pairing B_{Kos} on E, an antilinear Bost–Connes involution σ^{BC}, and an injective map $I : E \to L^2_{\mathrm{cusp}}([GL_2], dg)$ with

$$B_{\mathrm{Kos}}(v, \sigma^{\mathrm{BC}} w) = \int_{[GL_2]} Iv(g) \overline{Iw(g)}\, dg.$$

Here $[GL_2] = Z(\mathbb{A})GL_2(\mathbb{Q})\backslash GL_2(\mathbb{A})$ with a fixed unitary central character and a fixed quotient Tamagawa measure. Constructing E, B_{Kos}, σ^{BC}, I and this identity is the native pairing problem associated with Chapter 67.6.1; ordinary integral positivity proves its consequence only after those data exist.

The related tensor problem asks for an isometry from the completion of E to a specified subspace of $H_\infty \widehat{\otimes} \bigotimes'_p (H_p, e_p)$, where e_p are specified unit reference vectors and the product inner product is defined first on tensors equal to e_p at all but finitely many primes. The intended local models are the Fock completion at infinity and the weighted sequence spaces at finite primes. Their proposed Petersson

and GNS identifications, and normalization on the e_p, are not consequences of the categorical restricted tensor product in Chapter 50.1.3.

For a factorable cuspidal automorphic representation $\pi = \otimes'_v \pi_v$, the local integral comparison must fix a global additive character of $\mathbb{A}/\mathbb{Q}$, self-dual additive measures, multiplicative measures, and Whittaker vectors W_v with $W_p(1) = 1$ at unramified places. One then has to prove the matching of the native tensor pairing with the product of the resulting local Rankin–Selberg integrals. The q-coefficient calculation alone supplies neither that adelic map nor its normalization.

Theorem 78.5.5 *(a negative finite-prime square).* There exists $f \in \mathscr{A}$ with $W_{\mathrm{fin}}(f * f^*) < 0$. Thus no linear map into a positive Hilbert space can realize $W_{\mathrm{fin}}(f * g^*)$ as its inner product on all of $\mathscr{A}$.

Proof. Put $L = \log 2$. Choose a nonzero real smooth b, supported in $(-\varepsilon, \varepsilon)$, where $2\varepsilon < \min(L, \log 3 - L)$. Write $b_L(t) = b(t - L)$ and $f = b - b_L$. If $c = b * b^*$, then

$$f * f^* = 2c - c(\cdot - L) - c(\cdot + L).$$

The support meets the nonzero prime logarithms only at L and $-L$. At each, its value is $-c(0) = -\|b\|_2^2$. Therefore

$$W_{\mathrm{fin}}(f * f^*) = -\sqrt{2}\log 2\,\|b\|_2^2 < 0.$$

A Hilbert norm cannot have that value. □

Corollary 78.5.6 *(restricted automorphic comparisons).* A positive realization of the finite-prime form on a restricted subspace requires a proved restriction on that subspace. The existence of an automorphic coefficient multiplier or an interpolating function alone does not give a realization on the full compact test algebra.

Proof. The preceding test excludes the full-space realization. The Rankin–Selberg calculation identifies a different coefficient kernel and supplies no equality with the displayed negative value. □

78.6 Archimedean and interpolation obstructions

Definition 78.6.1 *(the archimedean density).* The sign convention in (78.2) uses

$$\mathcal{T}_\infty(u) = \frac{\log \pi - \Re\,\psi_\Gamma(1/4 + iu/2)}{2\pi}.$$

Proposition 78.6.2 *(the gamma density is not the Weil density).* The functions $|\Gamma(1/4 + iu/2)|^2$ and $\mathcal{T}_\infty(u)$ are not constant multiples. In fact $\mathcal{T}_\infty$ has both signs. For a local continuous argument of the nonvanishing gamma function,

$$\frac{d}{du}\arg\Gamma(1/4 + iu/2) = \frac{1}{2}\Re\,\psi_\Gamma(1/4 + iu/2), \qquad \frac{d}{du}\log|\Gamma(1/4 + iu/2)|^2 = -\Im\,\psi_\Gamma(1/4 + iu/2).$$

Proof. Differentiate $\log\Gamma(1/4 + iu/2)$ to obtain $(i/2)\psi_\Gamma$. Taking real and imaginary parts proves the identities. The digamma series gives $\psi_\Gamma(1/4) < -\gamma < 0$. For $y \geq 1$, its real part is

$$-\gamma + \sum_{n\geq 0}\left(\frac{1}{n+1} - \frac{n + 1/4}{(n+1/4)^2 + y^2}\right).$$

The first $\lfloor y \rfloor$ terms have sum at least $\log y - C$: the harmonic part has this growth, and the subtracted part is bounded by $y^{-2}\sum_{n\le y}(n+1/4) \le C$. In the remaining terms the possible negative part is at most $C\sum_{n>y} n^{-2}$. Thus the real part tends to positive infinity. The density $\mathcal{T}_\infty$ is positive at zero and negative for sufficiently large $|u|$. The gamma absolute square is positive. □

Remark 78.6.3 *(the local gamma factor).* For $\Lambda_\infty(s) = \pi^{-s/2}\Gamma(s/2)$,

$$\frac{\Lambda_\infty'(s)}{\Lambda_\infty(s)} = \frac{1}{2}\psi_\Gamma(s/2) - \frac{1}{2}\log\pi.$$

Thus, with $k(u) = \log\pi - \Re\psi_\Gamma(1/4 + iu/2)$, $k(u) = -2\Re(\Lambda_\infty'/\Lambda_\infty)(1/2 + iu)$. The gamma absolute square and this logarithmic derivative enter through different operations. A conformal-anomaly comparison requires an equality with this specified distribution and its normalization.

Li coefficients and compact tests

THEOREM 78.6.4 *(Li 1997, J. Number Theory 65).* . RH is equivalent to $\lambda_n \ge 0$ for every integer $n \ge 1$, where

$$\lambda_n \;=\; \sum_\rho \left(1 - \left(1 - \frac{1}{\rho}\right)^n\right),$$

where the sum means $\lim_{T\to\infty}\sum_{|\Im\rho|\le T}$, with zeros counted with multiplicity.

PROPOSITION 78.6.5 *(rational Li tests are not compact Fourier tests).* For any integer $n \ge 1$, there is no $f \in \mathscr{A}$ such that

$$(\mathcal{F}_+ f)(-i(\rho - 1/2)) = 1 - (1 - 1/\rho)^n$$

for every nontrivial zero ρ.

Proof. The right side is $n/\rho + O_n(|\rho|^{-2})$ as $|\rho| \to \infty$. The left side decreases faster than every inverse power of $|\Im\rho|$, uniformly in the zero strip. There are infinitely many zeros and only finitely many in any bounded set. Since their real parts lie in $[0,1]$, these two estimates contradict one another. □

Remark 78.6.6 *(Maass forms and continuous spectrum).* A square-integrable Maass cusp form has a positive L^2 norm, independently of any Ramanujan bound on its Hecke parameters. That positivity follows by integrating its absolute square and does not identify the norm with a zeta prime or Weil form. An Eisenstein comparison additionally requires its spectral measure, normalization, and a realization map, as specified next.

PROPOSITION 78.6.7 *(continuous-spectrum pairing data).* Let $\int^\oplus E_u\, d\mu(u)$ be a specified Hilbert direct integral. An identity between its norm and an arithmetic Hermitian form requires a map into that direct integral and equality of the two pairings. A scalar gamma factor determines neither that map nor its measure.

Proof. The direct-integral norm is $\int \|v(u)\|_{E_u}^2\, d\mu(u)$. Replacing a nonzero realization map I by $2I$ multiplies its pullback pairing by four. A single scalar factor does not select the map or its measure. The sign-changing density above also cannot equal an unspecified positive norm by scalar rescaling. □

78.7 SPECIFIC AUTOMORPHIC AND ARITHMETIC COMPARISON QUESTIONS

The coefficient calculation fixes one kernel. Additional native comparisons must state which different integral they propose to compute. The following questions retain the arithmetic sectors and parameters.

Entire multipliers and automorphic endomorphisms. For a normalized eigenform F generating an automorphic representation π_F, characterize $h \in \mathcal{F}_+\mathscr{A}$ for which $\Upsilon_F(h)$ is modular. A stronger question asks whether a given such multiplier is induced by a continuous $GL_2(\mathbb{A})$-equivariant endomorphism of a specified globalization of π_F, preserving its weight-k vector. The coefficient prescription is $a_F(n) \mapsto h(\log n)n^{(k-1)/2}a_F(n)$. This endomorphism condition and the modular transformation law are different requirements; neither is inferred from the moment conditions.

A proposed normalized kernel. For fixed F, one can form the meromorphic ratio

$$K_F^{\mathrm{prop}}(s) = \frac{\zeta(2s)L(s, \pi_F \times \overline{\pi_F})}{\zeta(s)\xi_{\mathbb{R}}(s)\xi_{\mathbb{R}}(1-s)}.$$

Take the unitary Hecke normalization in the finite Rankin–Selberg L-function, and $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. The question is whether a specified automorphic integral, including its archimedean and conductor factors, represents this ratio on a convergence domain and hence meromorphically. No such identity is asserted. In particular it is not the coefficient kernel in (78.4), and the two ξ factors do not absorb the zeta prime distribution into a Petersson norm.

The Fock integral and conformal anomaly. For the weight-k holomorphic discrete-series representation at infinity, specify a lowest-weight Whittaker vector, Haar measure, and a map from the polynomial Fock domain to that Whittaker model. The proposed scalar density is $|\Gamma(1/4 + iu/2)|^2$; obtaining it from a local integral requires an explicit computation with these representation parameters. It cannot be inferred merely from the name of the Fock space or from Jacquet's local-factor theory. The gamma-square density and the signed Weil density $k(u)/(2\pi)$ are different, as proved above.

The separate Polyakov question concerns the conformal variation of a specified regularized determinant in Chapter 35: choose the kinetic operator, boundary conditions, determinant normalization and test-dependent conformal variation, and compare its resulting distribution on $\mathscr{A}$ with

$$-2\Re\frac{\Lambda_\infty'}{\Lambda_\infty}(1/2 + iu) = k(u), \qquad \Lambda_\infty(s) = \pi^{-s/2}\Gamma(s/2).$$

Bohr–Mollerup characterizes Γ; it supplies none of those geometric data or the proposed anomaly identity.

Eisenstein comparison and discrete projection. Suppose a specified automorphic Hilbert space has an orthogonal spectral decomposition $H_{\mathrm{aut}} = H_{\mathrm{disc}} \oplus \int_{\mathbb{R}}^{\oplus} E_u\, d\nu(u)$. Then $P_{\mathrm{disc}}(v, w) = (v, 0)$ is the bounded self-adjoint idempotent of norm one if $H_{\mathrm{disc}} \neq 0$ (and zero otherwise). For a native map $I : E \to H_{\mathrm{aut}}$, its discrete realization is $P_{\mathrm{disc}}I$. Orthogonality proves $\|P_{\mathrm{disc}}Iv\|^2 \le \|Iv\|^2$ and identifies its kernel as the vectors mapped into the continuous summand. Thus an actual spectral decomposition supplies the projection; a coefficient multiplier does not specify it.

The continuous comparison in Chapter 67.5.5 asks whether the proposed Koszul ratio $\xi_{\mathbb{R}}(2s)/\xi_{\mathbb{R}}(2s-1)$ matches the Maass–Selberg scattering pairing after a specified truncation and normalization. One must distinguish this entire ξ from the meromorphic completion Λ_ζ, specify the Eisenstein series, its constant term, the truncation parameter and the measure ν, and prove the resulting equality or discrepancy. No unconditional exclusion of the continuum follows without that calculation. Likewise the discrete zero divisor in Li's criterion is not already identified with H_{disc}; such a statement requires its own map and spectral-measure comparison.

Maass parameters and varying weights. For an unramified unitary Maass representation let the Satake roots be α_p, β_p with $\alpha_p\beta_p = 1$, and set $\theta_p = \log_p \max(|\alpha_p|, |\beta_p|)$. For a weight-zero Hecke–Maass cusp eigenform on $SL_2(\mathbb{Z})$, the Kim–Sarnak estimate gives

$$|\lambda_F(p)| \le p^{7/64} + p^{-7/64}, \qquad \theta_p \le 7/64.$$

In Sarnak, *Spectra of Hyperbolic Surfaces*, January 2003, p. 17, (30), T_n is normalized by $n^{-1/2}$; p. 18, (34), states the operator-norm bound on the cuspidal space. Applying it to a unit eigenvector gives the first inequality. With trivial central character, the real eigenvalue is $\alpha_p + \alpha_p^{-1}$. The roots are either on the unit circle or real reciprocal roots; in the latter case $|\lambda_F(p)| = p^{\theta_p} + p^{-\theta_p}$. Monotonicity for $\theta_p \ge 0$ proves the second inequality. With the positive Laplacian $\Delta = -y^2(\partial_x^2 + \partial_y^2)$, the same source, Theorem 2, p. 17, gives the eigenvalue bound $\lambda_\Delta \ge 975/4096 = 1/4 - (7/64)^2$; an exceptional parameter $\lambda_\Delta = 1/4 - \theta_\infty^2$, with $\theta_\infty \ge 0$, therefore satisfies $\theta_\infty \le 7/64$. These are parameter estimates, not a claim of optimality. A parameter bound controls these roots, not the existence of a positive ordinary L^2 norm. The proposed correction to K_F^{prop} at $s = 1/2 \pm \theta_p$ requires an actual local factor and an integral comparison; a parameter bound alone does not give a correction formula.

For a weight sequence $k_j \to \infty$, fix normalized forms F_j and consider the finite positive measures

$$d\nu_j(z) = \frac{|F_j(z)|^2 y^{k_j}\, d\mu(z)}{\|F_j\|_{\mathrm{Pet}}^2} \quad \text{on } SL_2(\mathbb{Z})\backslash\mathbb{H}.$$

Their masses are one. Comparing weight limits requires specified maps from a common test space, control of escape into the cusp, and convergence of the pairings. No vanishing of these normalized measures or identification of their limit with a Selberg trace formula follows from the increase of k_j. Such an identity would have to match its spectral and geometric terms on that common carrier. None of these limits is needed for the entire Mellin transform of a compact smooth test.

The sufficient arithmetic path. The HP–Li input in Chapter 44.6.1 is the full Weil inequality on $\mathscr{A}$. An arithmetic proof through that path must construct its native pairing identity, including polar, finite-prime and gamma terms. A restricted automorphic isometry, Maass lift or weight-limit trace formula is a possible construction route only after its domain and comparison are proved; none is an additional logical necessity for the Weil criterion. The next theorem specifies the positive realization that such a route would have to supply.

78.8 The arithmetic pairing requirement

THEOREM 78.8.1 *(the remaining positive comparison).* A Hilbert realization of the full Weil form consists of a positive Hilbert space K, a linear map $J : \mathscr{A} \to K$, and the identity

$$\langle Jf, Jg\rangle_K = W(f * g^*) \qquad (f, g \in \mathscr{A}).$$

Such a realization exists if and only if q_W is positive semidefinite. When it exists, its minimal realization is the completion of $\mathscr{A}/\{f : q_W(f, f) = 0\}$. Its existence is RH-equivalent. Neither coefficient map in (78.3) provides the pairing identity.

Proof. A Hilbert realization implies nonnegativity. Conversely, positivity and polarization give Cauchy–Schwarz for q_W. Its nullspace is a linear subspace orthogonal to every vector. Quotienting and completing

therefore gives the asserted Hilbert space and canonical map. Weil's criterion supplies the equivalence to RH. The coefficient maps construct holomorphic functions with specified coefficients. Their definitions contain no equality with the full Weil form, and the finite-prime counterexample excludes the proposed full-space positive realization of that separate term. □

The arithmetic problem asks for this pairing through a specified native carrier, its operators, and its realization maps. The compact formula, the exact coefficient calculation, and the two obstruction theorems retain those separate requirements.

78.9 Signed arithmetic pairings and their operator representations

The finite-prime form has both signs even on a two-dimensional space. An exact arithmetic representation must retain that sign. All scalar products below are complex-linear in their first variable. The test algebra is $\mathscr{A} = C_c^\infty(\mathbb{R})$, with convolution and $f^*(t) = \overline{f(-t)}$. Write $h_f(z) = \int f(t)e^{izt}\,dt$ and $q_\bullet(f,g) = W_\bullet(f * g^*)$.

Theorem 78.9.1 *(the finite-prime operator on a support interval).* For $R > 0$, let $H_R = L^2([-R,R],dt)$. Let $j_R : H_R \to L^2(\mathbb{R},dt)$ extend by zero, and let $(T_a u)(t) = u(t-a)$. Define the bounded self-adjoint operator

$$B_R = \sum_{2\le n\le e^{2R}} \frac{\Lambda(n)}{\sqrt{n}}\, j_R^*(T_{\log n} + T_{-\log n})\, j_R.$$

For $f, g \in C_c^\infty((-R,R))$,

$$q_{\mathrm{fin}}(f,g) = \langle B_R f, g\rangle = \langle |B_R|^{1/2} f, \operatorname{sgn}(B_R)|B_R|^{1/2} g\rangle.$$

The sign function has value zero at zero. For $R < S$, extension by zero $j_{R,S} : H_R \to H_S$ satisfies $j_{R,S}^* B_S j_{R,S} = B_R$. These forms therefore give the original continuous Hermitian form on the complete test algebra $\mathscr{A}$.

Proof. Each translation is unitary and $T_a^* = T_{-a}$. The sum is finite, with norm at most $2\sum_{n\le e^{2R}} \Lambda(n) n^{-1/2}$. A change of variables gives

$$(f * g^*)(a) = \int f(t)\overline{g(t-a)}\,dt = \langle T_{-a} j_R f, j_R g\rangle.$$

For $|a| \ge 2R$ the integral vanishes. Summing at $a = \pm\log n$ proves the first identity. The spectral theorem for the bounded self-adjoint operator B_R gives $B_R = \operatorname{sgn}(B_R)|B_R|$ and the second identity. The same vanishing proves compatibility for $R < S$. On each fixed support the finite sum is continuous in the smooth topology. This is the distributional continuity of W_{fin}. □

Proposition 78.9.2 *(positive and negative arithmetic subspaces).* Put $L = \log 2$. Choose a nonzero real $b \in C_c^\infty((-\varepsilon,\varepsilon))$, where $2\varepsilon < \min\{L, \log(3/2)\}$. On $V_b = \operatorname{span}\{b, T_L b\}$, the finite-prime form has matrix

$$\frac{\log 2}{\sqrt{2}}\|b\|_2^2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

In particular,

$$q_{\mathrm{fin}}(b \pm T_L b, b \pm T_L b) = \pm\sqrt{2}\log 2\,\|b\|_2^2.$$

For $c = (\log 2)/\sqrt{2}$, the map $a(b + T_L b) \mapsto \sqrt{2c}\, ab$ is an isometry from the positive line to the ordinary L^2 line spanned by b. No linear map from all of $\mathscr{A}$ to a positive Hilbert space can pull its norm back to q_{fin}.

Proof. The autocorrelation $b * b^*$ is supported in $[-2\varepsilon, 2\varepsilon]$ and has value $\|b\|_2^2$ at zero. On the two translated copies, only the prime-power logarithms $\pm L$ meet an off-diagonal correlation. Each cross pairing is $c\|b\|_2^2$, and the diagonal pairings vanish. The two displayed eigenvectors give the signs and the isometry. The negative vector contradicts every proposed positive norm identity. □

THEOREM 78.9.3 *(the signed zero representation).* Let Z be the countable set of occurrences of zeros of ξ, including multiplicities. Choose the involution $\sigma\rho = 1 - \bar{\rho}$ on occurrences, fixing each occurrence on the critical line. Set

$$K_Z = \ell^2(Z), \quad z_\rho = -i(\rho - 1/2), \quad (Ju)_\rho = u_{\sigma\rho}, \quad (Ef)_\rho = h_f(z_\rho).$$

Then $J = J^* = J^{-1}$, $E : \mathscr{A} \to K_Z$ is continuous, and

$$\langle Ef, JEg\rangle = q_W(f, g) = P(f * g^*) - W_\infty(f * g^*) - \langle B_R f, g\rangle \tag{78.5}$$

whenever f, g have support in $(-R, R)$. The closed normal operator

$$(A_Z u)_\rho = z_\rho u_\rho, \qquad \operatorname{Dom} A_Z = \{u : \sum_\rho |z_\rho u_\rho|^2 < \infty\}$$

satisfies $A_Z = J A_Z^* J$, with equality of domains. For real a, the bounded group $G_a = \operatorname{diag}(e^{iaz_\rho})$ satisfies $\|G_a\| \le e^{|a|/2}$ and

$$E(T_a f) = G_a Ef, \qquad E(if') = A_Z Ef, \qquad \langle G_a u, JG_a v\rangle = \langle u, Jv\rangle.$$

For $k, f \in \mathscr{A}$, $E(k * f) = h_k(A_Z)Ef$, where the diagonal multiplier $h_k(A_Z)$ is bounded.

Proof. The zero symmetries give $z_{\sigma\rho} = \bar{z}_\rho$. Their imaginary parts have modulus at most $1/2$. Repeated integration by parts gives, on each fixed compact support,

$$|h_f(x + iy)| \le C_{R,N}(f)(1 + |x|)^{-N}, \qquad |y| \le 1/2,$$

where $C_{R,N}$ is a continuous smooth seminorm. The zero count in Lemma 77.5.1 proves square summability, continuity of E, and membership of Ef in every power domain of A_Z. The involution is an isometric permutation. Its coordinate formula gives $JA_Z^*J = A_Z$ and preserves the maximal domain. Absolute convergence and the explicit formula give

$$\langle Ef, JEg\rangle = \sum_\rho h_f(z_\rho)\overline{h_g(\bar{z}_\rho)} = W(f * g^*).$$

The last expression in (78.5) follows from the polar, gamma, and finite-prime terms of that formula. Change of variables and integration by parts prove the two intertwining identities. The coordinate factors in the signed scalar product cancel. The strip estimate bounds $h_k(A_Z)$, and the convolution identity proves the final assertion. □

This construction uses the actual zero divisor as input. It does not construct an arithmetic cohomology group or a map from an arithmetic primary sector. The finite-prime and zero maps in (78.5) are defined on the same compact test algebra, with both polar moments retained.

Corollary 78.9.4 *(the precise positivity boundary).* The operator A_Z is self-adjoint for the ordinary positive scalar product on K_Z if and only if RH holds. Positivity of $\langle Ef, JEf\rangle$ for all $f \in \mathscr{A}$ is also equivalent to RH. Under RH, $J = 1$ and $\|Ef\|^2 = q_W(f,f)$. Without a positivity assumption, (78.5) is an unconditional signed representation of the arithmetic form.

Proof. A diagonal normal operator on its maximal domain is self-adjoint exactly when each diagonal entry is real. Here this means $\Re\rho = 1/2$ for every zero. The second equivalence is precisely Weil's criterion, Corollary 78.2.1. The specified choice of σ is the identity under RH. □

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Chapter 79

VB* Main Direction via Hamburger Positivity, Determinacy, and HB$^+$/EC$_\pi$

Chapter 57 isolates the VB operator-positivity predicate in a conditional comparison with (Reg), operator-positive de Branges structure, and the quantitative Báez-Duarte condition, under the explicit operator and norm comparison data. In the opposite direction, the VB* main-direction implication says that the four analytic hypotheses (HB$^+$,* EC$_\pi$*, determinacy, positive Hamburger moment structure) on the primary Fock imply that the non-trivial zeros of $\xi_{\mathbb{R}}$ lie on the critical line. The chain is classical complex analysis and de Branges space theory: a de Branges datum whose real part has divisor* div Ξ *and whose Hermite–Biehler kernel is positive forces the zeros of Ξ to be real. The positive Hamburger moment structure identifies the spectral measure as a positive measure on the real line; Hamburger determinacy identifies the measure uniquely once it exists; and HB$^+$ supplies the real-part divisor constraint. Thus the conclusion is the conditional arrow from these hypotheses to RH. The hypotheses HB$^+$,* EC$_\pi$*, and determinacy are not consequences of the Hadamard product; they are the analytic content of the zero-placement problem.*

79.1 Spectral Positivity Context

The operator-theoretic comparison in Theorem 57.4.22 assumes Hypothesis 57.4.21 and compares VB*, (Reg), (HB+) and (BD+). (Reg) includes positivity, determinacy and the required spectral support on the identified arithmetic carrier. Neither bare (Det) nor the implication "zero placement implies VB*" is the predicate in that equivalence. Chapter 62 reduces the Hamburger-determinacy component to the sufficient analytic residual (R_{analytic}) and obstructed a tempting Bombieri–Lagarias shortcut. Neither argument addresses the main-direction implication

$$(\mathrm{HB}^+) + (\mathrm{EC}_\pi) + (\mathrm{Det}) \implies \text{spectral zeros of } \xi_{\mathbb{R}} \text{ on the critical line},$$

which is the sufficient direction of the analytic hypothesis of Definition 58.4.1. The proof is classical complex analysis and de Branges space theory in the sense of the 1968 monograph; no arithmetic input beyond the Hadamard factorisation of $\xi_{\mathbb{R}}$ is used.

Domain

The theorem is the implication from (HB$^+$) + (EC$_\pi$) + (Det) + positive Hamburger moment structure to spectral zero-placement on the critical line.

The analytic inputs are HB$^+$, EC$_\pi$, (Det), and positive Hamburger moment structure on $\xi_{\mathbb{R}}$. These are RH-equivalent in the Hermite–Biehler case and spectral regularity inputs in the moment-problem cases. They are the subject of Chapters 57 and 62.

Independent Sources

The source list for the VB* converse uses de Branges 1968, Lagarias 1999, Báez-Duarte 2003, Burnol 2000, Akhiezer 1965, Simon 1998, Shohat–Tamarkin 1943. The main-direction implication inherits this list

and adds the classical Hamburger existence theorem, Lagarias' 1999 de Branges survey, Conrey–Li 2000, and Beurling–Malliavin density. The proof sources of constructions 2 and 3 (determinacy reductions, operator-positivity reformulations) are kept disjoint from the complex-analysis sources below:

de Branges, *Hilbert Spaces of Entire Functions* (Prentice-Hall 1968). Foundational monograph; theorems 22–23 (Hermite–Biehler structure), theorem 32 (multiplication operator on $\mathcal{H}(E)$), theorems 40–41 (invariant subspaces of the shift).

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Beurling–Malliavin, Acta Math. 107 (1962) 291–309. Density theorem: the completeness radius of an exponential system $\{e^{i\lambda_k x}\}$ is computed by Beurling–Malliavin density of $\{\lambda_k\}$.

Hamburger, Math. Ann. 81 (1920) 235–319. Original existence theorem: a Hamburger moment problem has a positive representing measure iff all Hankel matrices are positive-semidefinite.

Levin, *Distribution of Zeros of Entire Functions* (AMS 1980), Ch. V–VII. Classical reference on Hermite–Biehler, Hermite–Kakeya–Obreshkov; entire functions of exponential type.

Independence of inputs: the complex analysis (Hadamard factorisation of $\xi_{\mathbb{R}}$, de Branges structure theorem, Beurling–Malliavin density) is not used in the VB*-converse comparison; and that comparison (operator-positivity reformulations, Jacobi-matrix spectral theory, Báez-Duarte decay) is not used in the main-direction implication. The two lists share two foundational references: de Branges 1968 (Hermite–Biehler theory) and Akhiezer 1965 (moment theory, used for the Jacobi matrix and for Hamburger existence and Carleman's criterion). These shared references serve logically distinct roles: Akhiezer supplies operator spectral methods on one side and classical moment existence. Simon 1998 (spectral theory of Jacobi matrices) appears in the operator-theoretic list and is not cited below; Shohat–Tamarkin 1943 is likewise absent below. The two argument paths are therefore independent on the operative arguments, with the shared foundational references serving independent functions in the two constructions.

79.2 Primary Fock and $\xi_{\mathbb{R}}$ as Character

Fix the primary Fock $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}}$ of Chapters 44 and 57: the primary sector of the arithmetic Heisenberg Fock, equipped with the Petersson inner product of Corollary 47.10.1. Its character on the archimedean Tate fibre is the completed zeta function

$$\Lambda_\zeta(s) \;=\; \pi^{-s/2}\,\Gamma(s/2)\,\zeta(s),$$

meromorphic with simple poles at $s = 0$ and $s = 1$; the entire Li–Hadamard form is

$$\xi_{\mathbb{R}}(s) \;:=\; \tfrac{1}{2}\,s(s-1)\,\Lambda_\zeta(s).$$

The factor $s(s-1)$ removes the Tate poles; the Gamma factor cancels the negative even trivial zeros of ζ; the remaining zero set is exactly the non-trivial zero set of ζ. This is the convention fixed in Remark 44.1.2 and Remark 44.3.7.

PROPOSITION 79.2.1 *($\xi_{\mathbb{R}}$ is entire of order 1, Hadamard genus 1).* (Edwards, *Riemann's Zeta Function* (Academic Press 1974), §1.9). The function $\xi_{\mathbb{R}}(s)$ is entire of order 1 and Hadamard genus 1:

$$\xi_{\mathbb{R}}(s) \;=\; e^{a+bs}\prod_{\rho}\Big(1-\frac{s}{\rho}\Big)e^{s/\rho}, \tag{79.1}$$

with explicit constants $a = -\log 2$, $b = -1 - \frac{1}{2}\gamma_{\mathrm{E}} + \frac{1}{2}\log(4\pi)$, product over non-trivial zeros ρ, and $\sum_\rho |\rho|^{-2} < +\infty$.

Proof. Classical: the functional equation and $|\xi_{\mathbb{R}}(s)| \le \exp(C|s|\log|s|)$ uniformly in vertical strips (Stirling bounds on Γ, polynomial control on ζ away from the pole) give order 1. Hadamard genus 1 follows from the zero density estimate of Riemann–von Mangoldt, which gives $\sum_\rho |\rho|^{-1-\epsilon} < +\infty$ for every $\epsilon > 0$ but $\sum_\rho |\rho|^{-1} = +\infty$. The Hadamard factorisation theorem then gives (79.1) with genus 1. At $s = 0$, the product is 1 and $\xi_{\mathbb{R}}(0) = 1/2$, so $a = -\log 2$. Differentiating $\log \xi_{\mathbb{R}}(s)$ at $s = 0$, using $\zeta(0) = -1/2$, $\zeta'(0) = -(1/2)\log(2\pi)$, and $\Gamma(s/2) = 2s^{-1} - \gamma_{\mathrm{E}} + O(s)$, gives

$$\frac{\xi_{\mathbb{R}}'(0)}{\xi_{\mathbb{R}}(0)} = -1 - \frac{\gamma_{\mathrm{E}}}{2} + \frac{1}{2}\log(4\pi),$$

which is b. □

Proposition 79.2.2 *(functional equation).* The entire function $\xi_{\mathbb{R}}$ satisfies the functional equation

$$\xi_{\mathbb{R}}(s) \;=\; \xi_{\mathbb{R}}(1-s), \qquad s \in \mathbb{C}. \tag{79.2}$$

Equivalently, $\xi_{\mathbb{R}}(\frac{1}{2} + iz)$ is an entire function of z with real Taylor coefficients.

Proof. The functional equation for Λ_ζ is Riemann 1859 / Tate 1950. The extension to $\xi_{\mathbb{R}}$ follows because $s(s-1) = (1-s)(-s) = (1-s)((1-s)-1)$ is invariant under $s \mapsto 1-s$. The substitution $s = \frac{1}{2} + iz$ turns $s \mapsto 1-s$ into $z \mapsto -z$; combined with $\xi_{\mathbb{R}}(\bar{s}) = \overline{\xi_{\mathbb{R}}(s)}$ (real on the real axis), this gives $\xi_{\mathbb{R}}(\frac{1}{2} + iz) = \overline{\xi_{\mathbb{R}}(\frac{1}{2} - i\overline{z})} = \overline{\xi_{\mathbb{R}}(\frac{1}{2} + iz)}$ for $z \in \mathbb{R}$; hence the Taylor coefficients at $z = 0$ are real. □

Definition 79.2.3 *(shifted character).* Write

$$\Xi(z) \;:=\; \xi_{\mathbb{R}}\big(\tfrac{1}{2} + iz\big), \qquad z \in \mathbb{C}.$$

Then Ξ is entire of order 1 with real Taylor coefficients. Its zero set is

$$\{\gamma_\rho : \rho = \tfrac{1}{2} + i\gamma_\rho \text{ non-trivial zero of } \xi_{\mathbb{R}}\},$$

symmetric under $z \mapsto -z$ by (79.2). Under RH the zeros γ_ρ are real; unconditionally they are complex and the $z \mapsto -z$ symmetry becomes $\gamma \leftrightarrow -\gamma$ as a multi-set.

Remark 79.2.4 *(the orientation of the problem).* RH is the statement that the zero set of Ξ is a subset of the real line. The Hadamard and functional-equation facts (Proposition 79.2.1, Proposition 79.2.2) are unconditional; they constrain the zero set to be symmetric under $z \mapsto -z$ but not to be real. The role of the analytic hypotheses (HB^+, EC_π, determinacy) is to force this realness by real-part divisor control and spectral-measure rigidity.

79.3 Hadamard Factorisation and the Zero Measure

Definition 79.3.1 *(primary zero multi-set).* The *primary zero multi-set* is

$$\mathcal{Z}_\xi \;:=\; \{\gamma_\rho \in \mathbb{C} : \rho = \tfrac{1}{2} + i\gamma_\rho \text{ non-trivial zero of } \xi_{\mathbb{R}}\},$$

with multiplicity. Under the symmetrisation $\rho \leftrightarrow 1-\rho$, the set $\mathcal{Z}_\xi$ is closed under $\gamma \leftrightarrow -\gamma$. Under the complex conjugation $\rho \leftrightarrow \overline{\rho}$, it is closed under $\gamma \leftrightarrow -\overline{\gamma}$; combining, it is closed under $\gamma \leftrightarrow \overline{\gamma}$ and symmetric under the Klein four group $\{1, \gamma \mapsto -\gamma, \gamma \mapsto \overline{\gamma}, \gamma \mapsto -\overline{\gamma}\}$ on the whole plane.

Definition 79.3.2 (primary zero counting measure). The *primary zero counting measure* is the multi-set measure

$$\mu_\xi := \sum_{\gamma \in \mathcal{Z}_\xi} \delta_\gamma$$

on $\mathbb{C}$ (with multiplicity). Under RH, μ_ξ is supported on $\mathbb{R}$. Under the functional equation (79.2), μ_ξ is $z \mapsto -z$-invariant. Under the reality of Taylor coefficients (Proposition 79.2.2), μ_ξ is $z \mapsto \overline{z}$-invariant. The invariant group is thus the Klein four group on $\mathbb{C}$.

Definition 79.3.3 (Regularization datum and moment sequence). Let $S_T(p) = \sum_{\gamma \in \mathcal{Z}_\xi, |\gamma| \le T} p(\gamma)$, counting multiplicities. A regularization datum specifies linear counterterms $C_T : \mathbb{C}[x] \to \mathbb{C}$ such that $L(p) = \lim_{T \to \infty}(S_T(p) - C_T(p))$ exists and is finite for every polynomial. Require $C_T(\overline{p}) = \overline{C_T(p)}$ and $C_T(p(-x)) = C_T(p(x))$, where the bar conjugates coefficients. The moments of this specified datum are

$$M_n = L(x^n).$$

No counterterms or arithmetic comparison are constructed by this definition. Zero counting and inverse-power convergence do not define positive-power moments. The actual finite-cutoff calculation is Proposition 80.2.5. Constructing the required trace-compatible datum and proving $L(p\overline{p}) \ge 0$ remain separate requirements.

PROPOSITION 79.3.4 *(Reality and symmetry of the specified moments).* For such a datum, every M_n is real and every odd moment is zero.

Proof. The cutoff multiset is invariant under conjugation and negation. Thus $S_T(\overline{p}) = \overline{S_T(p)}$ and $S_T(p(-x)) = S_T(p(x))$. The prescribed counterterms have the same symmetries; subtraction and limits preserve them. Apply these identities to monomials. □

Remark 79.3.5 (positivity of the measure is strictly weaker than RH). The measure μ_ξ as a multi-set measure on $\mathbb{C}$ is always positive (it is a counting measure). The question is whether its restriction to $\mathbb{R}$ carries the full mass; that is the content of RH. Positivity of μ_ξ as a measure on $\mathbb{C}$ gives no information about whether the support is on $\mathbb{R}$. This is the reason a positive Hamburger moment representation (a positive representation on $\mathbb{R}$ with prescribed real moments) is strictly finer than mere positivity of μ_ξ.

79.4 HAMBURGER MOMENT PROBLEM FOR μ_ξ

Definition 79.4.1 (Hamburger moment problem for μ_ξ). The *Hamburger moment problem* for the sequence $\{M_n\}_{n \ge 0}$ of Definition 79.3.3 is: find a positive Borel measure ν_ξ on $\mathbb{R}$ with

$$M_n = \int_{\mathbb{R}} x^n \, d\nu_\xi(x) \qquad \forall n \ge 0. \tag{79.3}$$

The problem is *solvable* if such ν_ξ exists, and *positive Hamburger* if ν_ξ is a positive measure on $\mathbb{R}$. The Stieltjes problem belongs to the square-pushforward $x \mapsto x^2$, not to the signed ordinate moments themselves.

THEOREM 79.4.2 *(Hamburger's existence theorem).* (Hamburger, Math. Ann. 81 (1920); Akhiezer 1965 §2.1). A moment problem (79.3) is solvable if and only if the infinite Hankel matrix

$$H_\infty = (M_{i+j})_{i,j \ge 0}$$

is positive-semidefinite: $\sum_{i,j} c_i \overline{c_j} M_{i+j} \geq 0$ for every finitely supported sequence (c_i). Equivalently, every finite truncation $H_N = (M_{i+j})_{0 \leq i,j \leq N}$ has non-negative principal minors.

Definition 79.4.3 *(HB$^+$/EC$_\pi$/Det hypothesis).* The *spectral positivity hypothesis* on the primary Fock is the conjunction of:

(SH) *Positive Hamburger moment sequence.* The common linear regularization of Definition 79.3.3 exists, gives finite moments, and its Hankel matrix $H_\infty = (M_{i+j})_{i,j \geq 0}$ is positive-semidefinite.

(Det) *Hamburger determinacy.* The positive representing measure is unique. Carleman's condition $\sum_{n \geq 1} M_{2n}^{-1/(2n)} = +\infty$ is a sufficient criterion when the positive even moments are nonzero.

(HB$^+$) *Hermite–Biehler positivity with divisor control.* There is a de Branges datum $E_\xi(z) = A_\xi(z) - iB_\xi(z)$, with A_ξ, B_ξ real entire, such that E_ξ is Hermite–Biehler, the associated kernel K_{E_ξ} is positive-semidefinite, and the real part $A_\xi = (E_\xi + E_\xi^*)/2$ has divisor $\operatorname{div} \Xi$, with multiplicity.

(EC$_\pi$) *Exponential completeness.* After the real-part divisor comparison places the zeros on the real axis, let Λ_ξ be their underlying set of distinct points. Require $\overline{\operatorname{span}}\{e^{i\gamma x} : \gamma \in \Lambda_\xi\} = L^2(-\pi, \pi)$. Repeated copies of an exponential do not add derivative conditions. This differs from the sharp fractional-part approximation predicate (BD+) in Theorem 57.4.22.

Proposition 79.4.4 *((SH) implies real-positive representing measure).* Under (SH), the moment problem (79.3) admits at least one positive Borel measure ν_ξ on $\mathbb{R}$ solving it.

Proof. Direct application of Hamburger's existence theorem (Theorem 79.4.2) to the moment sequence $\{M_n\}_{n \geq 0}$ under the positive-semidefiniteness hypothesis (SH). □

Remark 79.4.5 *(positivity is a genuine arithmetic input).* The hypothesis (SH) is not a tautology. The moments M_n are real-valued regularised sums of complex n-th powers (Definition 79.3.3); positive-semidefiniteness of the Hankel matrix is what transfers the complex multi-set $\mathcal{Z}_\xi$ to a real positive measure. This is the content Hamburger existence requires; it is distinct from the question whether the zeros are real.

79.5 Hamburger Determinacy and Carleman

Theorem 79.5.1 *(Carleman's determinacy criterion).* (Carleman 1926; Akhiezer 1965 §2.4). Let $\{M_n\}_{n \geq 0}$ be a Hamburger moment sequence with positive-semidefinite Hankel matrix and $M_{2n} > 0$ for all $n \geq 0$. If

$$\sum_{n \geq 1} M_{2n}^{-1/(2n)} = +\infty,$$

then the representing measure ν_ξ is unique (*determinate*).

Proposition 79.5.2 *(A sufficient moment-growth bound).* Let L be a specified positive Hamburger functional with finite moments $M_n = L(x^n)$. If $M_{2n}^{1/(2n)} \leq Cn$ for all sufficiently large n, then its representing measure is determinate.

Proof. If every even moment is positive, the Carleman summands are at least $(Cn)^{-1}$, and the harmonic series diverges. If some positive even moment vanishes, every representing positive measure is supported at zero, so its mass determines it uniquely. □

Remark 79.5.3 *(The moment functional to be constructed).* The finite cutoff calculation in Proposition 80.2.5 shows that the unregularized positive-power sums diverge. The zero-counting law therefore does not define the finite moments used above. A specified linear regularization, positivity of its resulting functional, and the linear root bound remain to be constructed. A two-sided factorial asymptotic would be stronger than the upper bound needed for Carleman; no such asymptotic is used here.

COROLLARY 79.5.4 *(under (SH) + (Det) the measure ν_ξ is unique).* Under (SH) + (Det) of Definition 79.4.3, there exists a unique positive Borel measure ν_ξ on $\mathbb{R}$ with moments (79.3).

Proof. (SH) + Hamburger existence (Theorem 79.4.2) gives solvability. The hypothesis (Det) gives uniqueness. Carleman's criterion is one sufficient way to establish that hypothesis. □

79.6 HERMITE–BIEHLER AND THE DE BRANGES SPACE $\mathcal{H}(E_\xi)$

Definition 79.6.1 *(de Branges datum over Ξ).* A *de Branges datum over* Ξ is a triple (E_ξ, A_ξ, B_ξ) with

$$E_\xi(z) \;=\; A_\xi(z) - iB_\xi(z),$$

where A_ξ and B_ξ are real entire functions. No finite-exponential-type assumption is imposed. The datum is required to carry a real-part divisor equality

$$\operatorname{div}(A_\xi) \;=\; \operatorname{div}(\Xi)$$

on $\mathbb{C}$, with multiplicities. It is not obtained by setting $E_\xi = \Xi$; for the real entire function Ξ one has $\Xi^* = \Xi$, so the strict Hermite–Biehler inequality would be impossible.

Remark 79.6.2 *(Growth of the prescribed divisor).* A nonzero entire function of finite exponential type has $O(T)$ zeros in the disk of radius T. After removing a zero at the origin, Jensen's formula on radius $2T$ bounds that number by a constant times the logarithmic maximum modulus at $2T$, which is $O(T)$. The divisor of Ξ has order $T \log T$ by Riemann–von Mangoldt. Therefore its divisor cannot be the divisor of a nonzero entire function of finite exponential type. A Paley–Wiener realization requires a separate map.

Remark 79.6.3 *(conventions match de Branges 1968).* The notation $E = A - iB$ with A and B real-on-real is de Branges' standard form for candidate Hermite–Biehler functions. The condition for Hermite–Biehler is $|E(z)| > |E^*(z)|$ for $\operatorname{Im} z > 0$, where $E^*(z) = \overline{E(\overline{z})}$. The divisor equality for A_ξ is part of the arithmetic datum; Hermite–Biehler positivity of an unrelated entire function says nothing about zeros of Ξ.

Definition 79.6.4 *(de Branges space $\mathcal{H}(E_\xi)$).* Assuming $E_\xi = A_\xi - iB_\xi$ is Hermite–Biehler (see (HB$^+$) below), the *de Branges space* $\mathcal{H}(E_\xi)$ is the reproducing-kernel Hilbert space of entire functions f with $f/E_\xi, f^*/E_\xi \in H^2(\mathbb{C}_+)$, with reproducing kernel

$$K_{E_\xi}(w,z) \;=\; \frac{\overline{E_\xi(w)}E_\xi(z) - \overline{E_\xi^*(w)}E_\xi^*(z)}{2\pi i(\overline{w} - z)}.$$

PROPOSITION 79.6.5 *(Hermite–Biehler kernel positivity).* Let E be entire with no zeros in $\mathbb{C}_+$, and $E^*(z) = \overline{E(\overline{z})}$. Then E is strictly Hermite–Biehler if and only if

$$K_E(w,z) = \frac{\overline{E(w)}E(z) - \overline{E^*(w)}E^*(z)}{2\pi i(\overline{w} - z)}$$

is positive-semidefinite and not identically zero on $\mathbb{C}_+$. Under these conditions the real entire functions $A = (E + E^*)/2$ and $B = (E^* - E)/(2i)$ have only real zeros.

Proof. Put $q = E^*/E$. For $|q| < 1$, multiplication by q is a contraction on Hardy space. The positive operator $I - M_q M_q^*$ applied to its reproducing kernels gives $(1 - \overline{q(w)}q(z))/(2\pi i(\overline{w} - z))$. Multiplication by $\overline{E(w)}E(z)$ proves kernel positivity. The diagonal equals $(|E(z)|^2 - |E^*(z)|^2)/(4\pi \operatorname{Im} z) > 0$. Conversely the nonnegative diagonal gives $|q| \le 1$. If equality holds inside, the maximum-modulus principle makes q a unimodular constant and the kernel zero, a contradiction. Hence the inequality is strict. A zero of A or B in the upper half-plane would make $E^* = \mp E$, contradicting strictness. Reality excludes lower-half-plane zeros as well. □

Remark 79.6.6 *(Why both qualifications are needed).* The function $E = 1$ has zero kernel. Further, if $P = P^*$ is real entire and E_0 is Hermite–Biehler, then $K_{PE_0}(w, z) = \overline{P(w)}P(z)K_{E_0}(w, z)$. This remains positive-semidefinite but has common upper-half-plane zeros whenever P does. Nonzero kernel alone does not replace the zero-free hypothesis.

Proposition 79.6.7 *(Strictness for the derivative datum).* Let A be a nonconstant even real entire function and $E = A + iA'$. If K_E is positive-semidefinite on $\mathbb{C}_+$, then E is strictly Hermite–Biehler. This applies to $A = \Xi$.

Proof. For $z = x + iy$, away from zeros of A,

$$|E|^2 - |E^*|^2 = -4|A|^2 \operatorname{Im}(A'/A), \qquad K_E(z, z) = -\frac{|A(z)|^2}{\pi y} \operatorname{Im}(A'/A)(z).$$

The diagonal condition gives $\operatorname{Im}(A'/A) \le 0$. An upper-half-plane zero a of order m would give $A'/A = m/(z - a) + O(1)$. At $z = a - i\epsilon$ its imaginary part is $m/\epsilon + O(1) > 0$, a contradiction. Thus A has no zeros there. The function $-\operatorname{Im}(A'/A)$ is nonnegative harmonic on the whole upper half-plane. An interior zero would make it identically zero by the minimum principle. Then the holomorphic quotient A'/A is a real constant c. Analytic continuation gives $A' = cA$; evenness forces $c = 0$, contradicting nonconstancy. The diagonal is therefore strictly positive, proving the strict modulus inequality. □

Remark 79.6.8 *(The actual arithmetic candidate).* The functional equation and chain rule give

$$\Xi(z) + i\Xi'(z) = \xi_{\mathbb{R}}(1/2 - iz) + \xi'_{\mathbb{R}}(1/2 - iz),$$

where the derivative on the right is in s. The preceding proposition establishes the required nondegeneracy implication for this candidate. It does not establish positivity of its kernel or an arithmetic Mellin/Fock intertwiner.

Theorem 79.6.9 *(Hermite–Biehler real-part control forces zero placement).* Under (HB$^+$), the zero multi-set $\mathcal{Z}_\xi$ of Ξ is contained in $\mathbb{R}$.

Proof. By (HB$^+$), $E_\xi = A_\xi - iB_\xi$ is Hermite–Biehler and $\operatorname{div}(A_\xi) = \operatorname{div}(\Xi)$. The Hermite–Biehler separation theorem gives that every zero of A_ξ is real. Hence every zero of Ξ is real. □

Corollary 79.6.10 *(zero multi-set is real under HB$^+$).* Under (HB$^+$), no non-trivial zero of $\xi_{\mathbb{R}}$ has $\operatorname{Im}\gamma \neq 0$ in the coordinate $\rho = \frac{1}{2} + i\gamma$.

Proof. Theorem 79.6.9 gives $\gamma \in \mathbb{R}$ for every zero. □

Remark 79.6.11 *(HB$^+$ is an RH-strength datum).* Corollary 79.6.10 shows that (HB$^+$), including real-part divisor control over Ξ, forces every non-trivial zero γ of Ξ to be real. Translated back via $\rho = \frac{1}{2} + i\gamma$, this is exactly $\operatorname{Re}\rho = \frac{1}{2}$. Thus (HB$^+$) is not a soft positivity assumption: it already contains the de Branges real-part mechanism at RH strength.

79.7 Exponential Completeness and Evaluation

Definition 79.7.1 *(Exterior Beurling–Malliavin density).* For a discrete real set Λ, a disjoint sequence of bounded intervals I_j is long if $\sum_j |I_j|^2/(1 + \operatorname{dist}(0, I_j)^2) = \infty$. Define $D^*(\Lambda)$ as the supremum of $d > 0$ for which a long sequence satisfies $\#(\Lambda \cap I_j) > d|I_j|$ for every j, with supremum zero for the empty set. Count distinct points.

THEOREM 79.7.2 *(Beurling–Malliavin completeness theorem).* For a discrete real set, the completeness radius

$$R(\Lambda) = \sup\{a > 0 : \operatorname{span}(e^{i\lambda x})_{\lambda\in\Lambda} \text{ is dense in } L^2(0, a)\}$$

is $2\pi D^*(\Lambda)$. Hence $D^* > 1$ implies (EC$_\pi$); equality leaves the endpoint assertion open.

Reference. The Beurling–Malliavin theorem in the exterior-density formulation; see Makarov–Poltoratski, *Two-Spectra Theorem with Uncertainty*, §3.1. Translation of the interval preserves exponential completeness. □

PROPOSITION 79.7.3 *(Completeness and uniqueness of evaluation).* Define PW_π by $G(z) = (2\pi)^{-1/2}\int_{-\pi}^{\pi} h(x)e^{izx}\,dx$, $h \in L^2(-\pi, \pi)$, with norm $\|h\|_2$. If Λ satisfies (EC$_\pi$), its evaluation map is injective on PW_π. If an injective linear map $J : \mathcal{H}(E) \to PW_\pi$ additionally satisfies $(JF)(\lambda) = F(\lambda)$, then

$$\mathcal{H}(E) \longrightarrow \mathbb{C}^\Lambda, \qquad F \longmapsto (F(\lambda))_{\lambda\in\Lambda}$$

is injective.

Proof. Vanishing of G on Λ makes h orthogonal to all $e^{-i\lambda x}$. Conjugation preserves completeness, so $h = 0$. For the second claim apply the first to JF, then use injectivity of J. □

COROLLARY 79.7.4 *(Uniqueness and stable sampling on the identified carrier).* Assume (HB$^+$), (EC$_\pi$) and the specified evaluation-compatible injection $J : \mathcal{H}(E_\xi) \to PW_\pi$. Evaluation on Λ_ξ is injective. If additionally positive weights and constants $0 < c \le C < \infty$ satisfy

$$c\|F\|^2 \le \sum_{\lambda\in\Lambda_\xi} w_\lambda |F(\lambda)|^2 \le C\|F\|^2,$$

then the samples give stable reconstruction.

Proof. HB and the real-part divisor place the zeros on the real axis; the preceding proposition gives injectivity. Under the two bounds, $TF = (\sqrt{w_\lambda}F(\lambda))_\lambda$ is bounded and bounded below, so $S = T^*T$ is invertible and $F = S^{-1}T^*TF$. □

The injection and sampling bounds are additional analytic data. HB and divisor equality do not construct them, and exponential completeness does not supply the sharp fractional-part rate or its arithmetic operator comparison.

79.8 Main Theorem: (HB$^+$) + (EC$_\pi$) + (Det) ⇒ RH

Theorem 79.8.1 *(VB* main-direction implication).* Assume the spectral positivity hypothesis (SH) + (Det) + (HB$^+$) + (EC$_\pi$) of Definition 79.4.3 on the primary Fock. Then all non-trivial zeros of $\xi_{\mathbb{R}}$ lie on the critical line $\operatorname{Re} s = \frac{1}{2}$. Equivalently, $\mathcal{Z}_\xi \subset \mathbb{R}$; equivalently, RH holds.

Proof. By (HB$^+$) and Theorem 79.6.9, $\mathcal{Z}_\xi \subset \mathbb{R}$. The definition of $\mathcal{Z}_\xi$ then gives $\rho = \frac{1}{2} + i\gamma_\rho$ with $\gamma_\rho \in \mathbb{R}$ for every non-trivial zero of $\xi_{\mathbb{R}}$. Thus $\operatorname{Re} \rho = \frac{1}{2}$. □

Remark 79.8.2 *(the logical order of the hypotheses).* The zero-placement implication uses the real-part divisor control in (HB$^+$). The clauses (SH), (Det), and (EC$_\pi$) belong to the stronger spectral-measure hypothesis: they concern positivity, uniqueness, and sampling of the zero measure, not the realness of the zeros itself.

Corollary 79.8.3 *(valid forms of the main-direction implication).* Each of the following hypotheses suffices for RH:

(i) (HB$^+$), where (HB$^+$) includes real-part divisor control over Ξ (reduces to Corollary 79.6.10).

(ii) The full conjunction (SH) + (Det) + (HB$^+$) + (EC$_\pi$).

(iii) A de Branges formulation in which the real part of the specified Hermite–Biehler function has divisor $\operatorname{div} \Xi$.

(iv) The Conrey–Li 2000 construction: $\mathcal{H}(E_\xi)$ admitting a self-adjoint multiplication operator whose spectral measure is the zero-counting measure on $\mathcal{Z}_\xi \cap \mathbb{R}$.

No assertion is made that (SH) + (Det) + (EC$_\pi$) alone gives zero placement without the Hermite–Biehler or real-spectrum input. The main theorem proves the sufficient implication stated above.

Proof. (1): Theorem 79.6.9. (2): the theorem itself. (3): Lagarias Acta Arith. 89 (1999), with the de Branges datum specified. (4): Conrey–Li arXiv:math/9812166 Thm. 2, with real spectrum part of the hypothesis. □

Theorem 79.8.4 *(A sufficient spectral package for the arithmetic implication).* Assume the determinant–trace realization (SH), the identified positive arithmetic moment functional, its determinacy and support as in (Reg), and item 1 of Hypothesis 57.4.21. With the P1–P3 hypotheses of Theorem 58.4.2, this package suffices for the conditional implication to RH. Substitution of (HB+) or (BD+) requires the corresponding additional comparisons in that hypothesis.

Proof. The stated moment and support data give (Reg); item 1 gives VB*. Apply Corollary 58.4.3. No identification of bare determinacy with the complete arithmetic package is used. □

79.9 Analytic Hypotheses on $\xi_{\mathbb{R}}$

Remark 79.9.1 *(analytic content).* Theorem 79.8.1 proves the main-direction implication from the four hypotheses. It does *not* prove RH, because none of HB$^+$, EC$_\pi$, or (Det) is proved for $\xi_{\mathbb{R}}$ by the Hadamard product alone. The three analytic assertions have different logical strength:

- (HB$^+$) proof is RH (Theorem 57.4.9; de Branges 1994).
- EC$_\pi$ requires completeness of the specified exponentials in $L^2(-\pi, \pi)$. It is distinct from the sharp fractional-part rate (BD+) in Question 57.4.16. The finite-distance construction and exact norm transport in Proposition 57.4.13 do not identify these two predicates.

- (Det) proof is the positive regularised moment functional with $M_{2n}^{1/(2n)} = O(n)$, the residual ($\mathrm{R}_{\mathrm{analytic}}$) of Theorem 62.4.7.

THEOREM 79.9.2 *(sufficient residual form of RH).* Under P2, P3, H-P^{Ar}, the conjunction of the four conditions (SH), (Det), (HB^+), (EC_π) on the primary Fock implies the Riemann Hypothesis. The converse is not asserted here.

Proof. This is exactly Theorem 79.8.1. The reverse implication would require the spectral-regularity hypotheses of Theorem 57.4.22 under Hypothesis 57.4.21, and is not part of this theorem. □

Remark 79.9.3 *(the analytic arrow).* The analytic content is the main-direction arrow

$$(\mathrm{SH}) + (\mathrm{Det}) + (\mathrm{HB}^+) + (\mathrm{EC}_\pi) \implies \mathrm{RH},$$

isolating the analytic content as the proof of these four hypotheses on $\xi_{\mathbb{R}}$. The main-direction argument makes the sufficient chain explicit and identifies the exact shape of the required analytic theorem.

Remark 79.9.4 *(relation to the structural equivalence).* Theorem 57.4.22 under Hypothesis 57.4.21 gives a conditional comparison between Reg, HB^+, (BD+), and VB^* itself, using all the specified operator and quantitative comparison data. Its (Reg) predicate contains positivity and spectral support as well as determinacy. The complementary arrow is (SH) + (Det) + (HB^+) + (EC_π) ⇒ RH. Together they say that regularity is the needed input and that the full spectral hypothesis is sufficient for RH.

79.10 Bibliography Anchors

The primary sources for the main-direction implication, beyond the operator-theoretic list of Section 57.1, are:

de Branges, *Hilbert Spaces of Entire Functions* (Prentice-Hall 1968). Theorems 22–23 (Hermite–Biehler structure), theorem 32 (multiplication operator on $\mathcal{H}(E)$). The strict Hermite–Biehler criterion used here requires the stated upper-half-plane zero-free hypothesis and a positive semidefinite kernel that is not identically zero. Those hypotheses, proved for the derivative construction when its kernel is positive, exclude the degenerate constant example.

Lagarias, Acta Arith. 89 (1999), 217–234. Reformulation: RH ⇔ the Lagarias function E_ξ is Hermite–Biehler. This is (HB^+) ⇔ RH, the single-hypothesis equivalent form used in Remark 79.6.11.

Lagarias, Acta Arith. 89 (1999). Survey of de Branges reformulations; §4.2 on the exponential type of E_ξ; §5 on the Beurling–Malliavin density of $\mathcal{Z}_\xi$; §6 on the Conrey–Li counter-example discussion.

Conrey–Li, arXiv:math/9812166 (2000). Theorem 2: the de Branges space $\mathcal{H}(E_\xi)$ does not in general admit an arithmetic multiplication operator realising RH; the authors construct explicit counter-examples for L-functions where the Hermite–Biehler condition fails (e.g. for certain Dirichlet L-functions with Siegel-type zeros). This confirms that (HB^+) is a substantive hypothesis not automatic from the Hadamard product.

Beurling–Malliavin, Acta Math. 107 (1962), 291–309. Density theorem: completeness radius of $\{e^{i\lambda_k x}\}$ equals Beurling–Malliavin density of $\{\lambda_k\}$. This is the analytical basis for the (EC_π) ⇒ sampling implication.

Hamburger, Math. Ann. 81 (1920), 235–319. Existence theorem: positive-semidefinite Hankel ⇔ positive Borel measure on $\mathbb{R}$. Hamburger's own proof via the Helly selection theorem and the Banach–Alaoglu compactness of probability measures on $\mathbb{R}$.

Carleman (1922, 1926). Comptes Rendus Acad. Sci. Paris (1922); *Les fonctions quasi analytiques* (Gauthier-Villars 1926). Determinacy criterion: $\sum M_{2n}^{-1/(2n)} = \infty \Rightarrow$ Hamburger determinacy. The proof uses quasi-analytic classes of entire functions and constitutes the classical foundation for the determinacy condition (Det).

Koosis, *The Logarithmic Integral II* (Cambridge 1992), Ch. VIII. Modern treatment of Beurling–Malliavin density and its applications to completeness of exponential systems.

Levin, *Distribution of Zeros of Entire Functions* (AMS 1980), Ch. V–VII. Classical reference on Hermite–Biehler and Laguerre–Pólya class; the structure theorems used in the HB^+ characterisation.

These sources furnish the classical foundations for the implication. None supplies the proof of HB^+, EC_π, or (Det) on $\xi_{\mathbb{R}}$.

79.11 Spectral Boundary

(i) $\xi_{\mathbb{R}}$ is entire of order 1, Hadamard genus 1, with $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s)$; hence $\Xi(z) = \xi_{\mathbb{R}}(\frac{1}{2} + iz)$ has real Taylor coefficients and a zero set symmetric under the Klein four group (Proposition 79.2.1, Proposition 79.2.2).

(ii) The regularised moment sequence $\{M_n\}$ of the zero multi-set $\mathcal{Z}_\xi$ is real, with odd moments vanishing (Proposition 79.3.4).

(iii) Positive Hamburger moment structure (SH) plus Hamburger determinacy (Det) give a unique positive Borel measure ν_ξ on $\mathbb{R}$ with moments $\{M_n\}$ (Corollary 79.5.4).

(iv) Hermite–Biehler positivity (HB^+), including real-part divisor control over Ξ, forces $\mathcal{Z}_\xi \subset \mathbb{R}$ (Corollary 79.6.10).

(v) Exponential completeness EC_π gives uniqueness in PW_π. Transfer to $\mathcal{H}(E_\xi)$ requires the specified evaluation-compatible injection into PW_π; stable sampling further requires the Bessel and lower sampling bounds (Corollary 79.7.4).

(vi) The conjunction (SH) + (Det) + (HB^+) + (EC_π) implies zero-placement and hence RH (Theorem 79.8.1).

(vii) The analytic content is the proof of HB^+, EC_π, and (Det) on $\xi_{\mathbb{R}}$; these are the spectral inputs (Remark 79.9.1).

(viii) The analytic implication matches the RH completion hypothesis of Definition 58.4.1: the positive Hamburger spectral component of that hypothesis is exactly (SH) + (Det) + (HB^+) + (EC_π), and the arrow to RH is Theorem 79.8.1.

Chapter 80

Zero counts, Gaussian tests and formal completion

Three calculations fix distinct parts of the arithmetic comparison. The Riemann–von Mangoldt formula retains its argument term:

$$N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8} + S(T) + O(1/T), \qquad S(T) = O(\log T).$$

The balanced Weil explicit formula is evaluated on the Gaussian family $f_\lambda(x) = e^{-\pi x^2/\lambda}$. Its ordinary convergent terms give the tested functional. Positivity is an additional assertion.

Artin–Rees controls induced filtrations on finite modules over the Iwasawa algebra. Comparisons for formal mapping stacks require additional constructions. The positivity and linear-growth assertions in $\mathrm{R}_{\mathrm{analytic}}$ are not consequences of these three calculations.

80.1 Primary Sources

The zero count and explicit formula have their classical analytic normalizations. The argument estimate is imported in the precise form named below. The finite-module filtration theorem is proved by its Rees module. Positivity and moment determinacy require separate inputs.

Riemann and von Mangoldt. Riemann's 1859 memoir introduces the zero-counting problem. Von Mangoldt's 1895 paper gives a complete proof of the asymptotic formula. The normalization below retains the horizontal argument variation.

Hasanalizade–Shen–Wong. *Counting zeros of the Riemann zeta function*, arXiv:2107.06506, Theorem 1.3 and Corollary 1.4 give the argument bound used here.

Selberg, Avh. Norske Vid. Akad. Oslo (1944) 1–27. Refined mean-square estimates on $S(T)$.

Weil. *Sur les formules explicites de la théorie des nombres premiers*, Lund, volume supplémentaire (1952), 252–265. The complete signed functional below uses both polar values, the gamma term and every prime power.

Tate. PhD thesis, Princeton 1950 (printed in Cassels–Fröhlich, *Algebraic Number Theory*, 1967, Ch. XV). Adelic Mellin-transform treatment of ξ; self-dual Gaussian under the archimedean factor.

Atiyah–Macdonald. *Introduction to Commutative Algebra* (Addison-Wesley 1969), Chapter 10: filtrations and completions of Noetherian modules.

Bourbaki, *Algèbre Commutative* (Ch. III §3, Masson 1961). Artin–Rees and Noetherian filtered rings.

80.2 Riemann–von Mangoldt Density of Zeros

Statement

THEOREM 80.2.1 *(Riemann–von Mangoldt zero count).* Let $N(T)$ count the nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \le T$, including multiplicity. For $T \ge 2$ away from zero ordinates,

$$N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8} + S(T) + R(T), \qquad R(T) = O(T^{-1}). \tag{80.1}$$

Here

$$S(T) = \frac{1}{\pi}\Delta_{2\to 2+iT\to 1/2+iT}\arg\zeta, \tag{80.2}$$

with argument starting at zero at $s = 2$. At a zero ordinate, the convention $\gamma \le T$ uses the right limit of S. The formula is unconditional.

Proof

Proof. The completed function is

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

It is entire, has no real zeros, and obeys $\xi(s) = \xi(1-s)$ and $\xi(\bar{s}) = \overline{\xi(s)}$. These properties and its zero strip are proved in Section 23.3.

For a height T away from zero ordinates, use the rectangle $R_T = \{-1 \le \operatorname{Re} s \le 2,\ 0 \le \operatorname{Im} s \le T\}$. The argument principle gives

$$\frac{1}{2\pi i}\oint_{\partial R_T}\frac{\xi'(s)}{\xi(s)}\,ds = N(T).$$

The logarithmic derivative satisfies

$$\frac{\xi'(s)}{\xi(s)} + \frac{\xi'(1-s)}{\xi(1-s)} = 0.$$

The real bottom edge has zero argument change.

Reflection about $\operatorname{Re} s = 1/2$, together with conjugation, makes the remaining change twice that on the path $C_T : 2 \to 2 + iT \to 1/2 + iT$. Consequently,

$$N(T) = \frac{1}{\pi}\Delta_{C_T}\arg\xi.$$

The horizontal part of C_T is retained in this expression.

Logarithmic differentiation gives

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}(s/2) + \frac{\zeta'}{\zeta}(s).$$

At the endpoint of C_T, the continuous arguments of s and $s-1$ sum to π. Thus

$$N(T) = 1 + \frac{\vartheta(T)}{\pi} + S(T), \qquad \vartheta(T) = \operatorname{Im}\log\Gamma(1/4 + iT/2) - \frac{T}{2}\log\pi.$$

The logarithm of gamma is analytic on the right half-plane and real on the positive axis.

Stirling's expansion gives

$$\vartheta(T) = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} - \frac{\pi}{8} + \frac{1}{48T} + O(T^{-3}).$$

Combining the two identities proves the assertion away from zero ordinates. Right limits give the stated convention at those heights. □

Remark 80.2.2 *(The constant 7/8).* The factor $s(s-1)$ contributes 1 to $N(T)$. For $z = 1/4 + iT/2$, the imaginary part of $(z-1/2)\log z - z$ has constant term $-\pi/8$. Dividing by π gives $7/8 = 1 - 1/8$. This is an asymptotic argument calculation at large height. The positive real number $\Gamma(1/4)$ has real logarithm and does not contribute that imaginary constant.

The unconditional argument bound

Theorem 80.2.3 *(The classical argument estimate).* The function in (80.2), with its stated one-sided convention, satisfies

$$|S(T)| \le C\log T \qquad (T \ge 2) \tag{80.3}$$

for an absolute constant C. In particular, the number $\mathcal{N}(T,1)$ of zeros with ordinates in $(T-1, T+1]$ obeys

$$\mathcal{N}(T,1) = O(\log T).$$

Proof. The argument estimate is the classical analytic theorem. A precise primary bound is Hasanalizade–Shen–Wong, *Counting zeros of the Riemann zeta function*, Theorem 1.3 and Corollary 1.4, arXiv:2107.06506. The estimate remains valid under one-sided limits. Subtract (80.1) at $T-1$ and $T+1$. The change of its smooth main term is $O(\log T)$, and the two argument terms have the same bound. This proves the local count. □

Corollary 80.2.4 *($N(T)$ to main term, $\log T$ precision).* Combining (80.1) and (80.3),

$$N(T) \;=\; \frac{T}{2\pi}\log\frac{T}{2\pi e} + O(\log T),$$

unconditionally, using the functional equation, Stirling expansion and the classical argument estimate.

Proof. Substitute the $S(T)$ bound into (80.1) and absorb the $7/8$ and $O(1/T)$ into the $O(\log T)$ bound. □

Consequence for $\mathrm{R}_{\mathrm{analytic}}$

Proposition 80.2.5 *(Finite positive-power sums of ordinates).* Let $N(T)$ count the positive ordinates γ of nontrivial zeros with multiplicity. For each fixed integer $n \ge 0$, put $S_n(T) = \sum_{0<\gamma\le T}(1/4+\gamma^2)^n$. Then

$$S_n(T) = \frac{T^{2n+1}}{2\pi(2n+1)}\log\frac{T}{2\pi} - \frac{T^{2n+1}}{2\pi(2n+1)^2} + O_n(T^{2n}\log T).$$

In particular every $S_n(T)$ diverges. Under RH these are the corresponding positive-power sums of the real zero parameters.

Proof. For $n = 0$ this is the Riemann–von Mangoldt formula. For $n \geq 1$, Stieltjes integration by parts gives

$$S_n(T) = (1/4 + T^2)^n N(T) - \int_0^T 2nt(1/4 + t^2)^{n-1} N(t)\, dt.$$

Write $N(t) = F(t) + O(\log(2 + t))$, where $F(t) = t\log(t/(2\pi))/(2\pi) - t/(2\pi)$, and absorb a fixed initial interval in the error. The endpoint and integrated errors are $O_n(T^{2n}\log T)$. The main term is $(2\pi)^{-1}\int^T (1/4 + t^2)^n \log(t/(2\pi))\, dt$. Replacing the polynomial by t^{2n} costs $O_n(T^{2n-1}\log T)$. Integrating this last monomial gives the two displayed terms. Its leading term is positive and divergent. □

Remark 80.2.6 *(Finite measures and the regularization problem).* The finite positive measure

$$\nu_T = \frac{1}{2}\sum_{0<\gamma\leq T}\left(\delta_{\sqrt{1/4+\gamma^2}} + \delta_{-\sqrt{1/4+\gamma^2}}\right)$$

has odd moments zero and even moments $S_n(T)$. It is compactly supported and determinate, but its mass tends to infinity. Consequently the unregularized integral over all ordinates has no finite moment saddle. To obtain the proposed arithmetic Hamburger functional one must specify a common linear regularization and prove its positivity and growth. If that regularization gives moments M_{2k} of γ, its shifted moments must satisfy

$$\mu_{2n} = \sum_{k=0}^{n}\binom{n}{k}4^{-(n-k)}M_{2k}.$$

This compatibility is a consequence of linearity, not a construction of the regularization or an estimate for its moments.

80.3 Weil Explicit Formula at Gaussian Test Functions

The Weil Explicit Formula: Recall

The arithmetic expression and the sum over zeros use different variables. Their relation fixes the scale at which the prime powers sample a test function. The following conventions make this relation explicit.

Lemma 80.3.1 *(The normalized explicit formula).* For $b > \pi$, let

$$\mathcal{T}_b = \left\{f \in C^\infty(\mathbb{R};\mathbb{C}) : \int_{\mathbb{R}} e^{b|x|}|f^{(j)}(x)|\, dx < \infty \text{ for every } j \geq 0\right\}, \qquad \mathcal{T} = \bigcup_{b>\pi}\mathcal{T}_b.$$

Use Lebesgue measure and define

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x)e^{-2\pi ixt}\, dx, \qquad f^\sharp(x) = \overline{f(-x)}, \qquad (f\star h)(x) = \int_{\mathbb{R}} f(y)h(x-y)\, dy.$$

These operations make $\mathcal{T}$ a complex convolution $*$-algebra. For $f \in \mathcal{T}_b$, the transform is holomorphic on $|\operatorname{Im} t| < b/(2\pi)$. Put $\psi = \Gamma'/\Gamma$ and $z_\rho = -i(\rho - 1/2)$, with nontrivial zeta zeros counted with multiplicity. Define

$$\Phi_\infty(f) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(t)\left[\operatorname{Re}\psi(1/4 + it/2) - \log\pi\right]dt, \tag{80.4}$$

$$\Phi_{\mathrm{fin}}(f) = -\frac{1}{2\pi}\sum_p\sum_{k\geq 1}\frac{\log p}{p^{k/2}}\left[f\left(\frac{k\log p}{2\pi}\right) + f\left(-\frac{k\log p}{2\pi}\right)\right]. \tag{80.5}$$

These expressions converge absolutely. The arithmetic functional is

$$L_{2\pi}(f) = \widehat{f}(i/2) + \widehat{f}(-i/2) + \Phi_\infty(f) + \Phi_{\mathrm{fin}}(f).$$

The explicit formula states

$$L_{2\pi}(f) = \sum_\rho \widehat{f}(z_\rho),$$

with absolute convergence. Consequently the balanced expression is

$$W(f) := L_{2\pi}(f) - \sum_\rho \widehat{f}(z_\rho) = 0. \tag{80.6}$$

Proof. Set $g_f(u) = (2\pi)^{-1} f(-u/(2\pi))$ and $a = b/(2\pi) > 1/2$. Every derivative of g_f is integrable with weight $e^{a|u|}$. The change of variables $u = -2\pi x$ gives

$$\int_{\mathbb{R}} g_f(u) e^{itu}\, du = \widehat{f}(t), \qquad g_{f\star h} = g_f \star g_h, \qquad g_{f^\sharp}(u) = \overline{g_f(-u)}.$$

Weighted Young's inequality and differentiation under convolution prove closure of $\mathcal{T}$ under these operations. Weighted integrability of f' gives $|f(x)| \le Ce^{-b|x|}$ by integrating from the appropriate infinity. Integration by parts gives rapid transform decay in every closed strip $|\operatorname{Im} t| \le \eta < b/(2\pi)$. The zero count $N(T) = O(T\log(T+2))$ therefore gives absolute convergence of the zero sum. The prime sum is bounded by a constant times $\sum_{n\ge 2} (\log n) n^{-1/2-a}$, which converges. The gamma integral converges because its bracket is $O(\log(2+|t|))$.

Here is the contour calculation in the rescaled variable. Write $g = g_f$ and

$$F_g(s) = \int_{\mathbb{R}} g(u) e^{(s-1/2)u}\, du, \quad \Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2), \quad H(s) = \frac{d}{ds}\log\bigl(\Gamma_{\mathbb{R}}(s)\zeta(s)\bigr).$$

Thus $F_g(s) = \widehat{f}(-i(s-1/2))$. Choose $1 < c < \min(a+1/2, 3/2)$. The functional equation gives $H(1-s) = -H(s)$. The residues of H are the multiplicities of the nontrivial zeros and -1 at each of $0, 1$. Integrate $H(s)F_g(s)$ on the counterclockwise rectangle with vertical sides $1-c, c$ and heights $\pm T$. Choose heights tending to infinity away from the zero ordinates. For example, the bound $O(T\log(T+2))$ permits a choice in each $[T, T+1]$ at distance at least $c_0/(T\log(T+2))$ from every nearby ordinate, for sufficiently small fixed $c_0 > 0$. The logarithmic derivative expansion of the Hadamard product of ξ then gives a polynomial bound for H on these horizontal segments. Its nearby terms are bounded by the zero count times the inverse distance. For distant zeros the paired product terms are $O(T/|\rho|^2)$ and summable. The rapid decay of F_g eliminates both horizontal integrals. Reflection of the left side gives

$$\sum_\rho F_g(\rho) - F_g(0) - F_g(1) = \frac{1}{2\pi i}\int_{\operatorname{Re} s = c} H(s)\bigl(F_g(s) + F_g(1-s)\bigr)\, ds.$$

On this line, $\zeta'/\zeta(s) = -\sum_{n\ge 2} \Lambda_{\mathrm{vM}}(n) n^{-s}$, where $\Lambda_{\mathrm{vM}}(p^k) = \log p$ and is zero otherwise. Absolute convergence permits termwise integration. Fourier inversion gives

$$\frac{1}{2\pi i}\int_{\operatorname{Re} s = c} n^{-s} F_g(s)\, ds = n^{-1/2} g(\log n).$$

The reflected test gives $n^{-1/2}g(-\log n)$. Since $g(\pm\log n) = (2\pi)^{-1}f(\mp\log n/(2\pi))$, these are exactly the stated prime terms with their negative sign. For the gamma term, move the integration line to $1/2$. Its logarithmic derivative is holomorphic between these lines, and rapid test decay eliminates the horizontal integrals. Combining the two reflected terms uses

$$2\,\mathrm{Re}\,\frac{\Gamma_{\mathbb{R}}'}{\Gamma_{\mathbb{R}}}(1/2+it) = \mathrm{Re}\,\psi(1/4+it/2) - \log\pi.$$

The two pole values are $\widehat{f}(i/2)$ and $\widehat{f}(-i/2)$. This proves the formula, including its constants and signs. Equivalently, if L_ζ denotes the arithmetic functional for the transform $\int g(u)e^{itu}\,du$, the comparison is $L_{2\pi}(f) = L_\zeta(g_f)$. □

The Gaussian Test Family

Definition 80.3.2 *(Gaussian test family).* For $\lambda > 0$, set

$$f_\lambda(x) \;:=\; e^{-\pi x^2/\lambda}.$$

This is a Schwartz function; its Fourier transform is

$$\widehat{f_\lambda}(y) \;=\; \sqrt{\lambda}\,e^{-\pi\lambda y^2} \tag{80.7}$$

by the classical Gaussian-self-duality identity: indeed, at $\lambda = 1$, $\widehat{f_1}(y) = f_1(y)$ and the general case follows by scaling $x \mapsto x/\sqrt{\lambda}$ and the Plancherel change-of-variables $\widehat{f(\cdot/a)}(y) = a\widehat{f}(ay)$. The transform is again a Gaussian with full support.

Remark 80.3.3 *(Gaussian admissibility).* Every derivative of f_λ is a polynomial times a Gaussian. Thus $f_\lambda \in \mathcal{T}_b$ for every $b > \pi$. Its Fourier transform has full support, as permitted by Lemma 80.3.1. All terms are ordinary convergent integrals or sums.

Closed-Form Evaluation of $W(f_\lambda)$

We evaluate each term of $W(f_\lambda)$ in closed form.

LEMMA 80.3.4 *(pole and zero terms for Gaussians).* For the Gaussian f_λ of Definition 80.3.2,

$$\widehat{f_\lambda}(i/2) + \widehat{f_\lambda}(-i/2) \;=\; 2\sqrt{\lambda}e^{\pi\lambda/4}.$$

Moreover

$$\sum_\rho \widehat{f_\lambda}(z_\rho) = \sqrt{\lambda}\sum_\rho e^{-\pi\lambda z_\rho^2}, \qquad z_\rho = -i(\rho - \tfrac{1}{2}),$$

and this zero sum is absolutely convergent.

Proof. By (80.7), $\widehat{f_\lambda}(\pm i/2) = \sqrt{\lambda}e^{\pi\lambda/4}$. For $\rho = \beta + i\gamma$ in the critical strip, $|e^{-\pi\lambda z_\rho^2}| \le e^{\pi\lambda/4}e^{-\pi\lambda\gamma^2}$. The Riemann–von Mangoldt bound $N(T) = O(T\log T)$ gives absolute convergence. □

LEMMA 80.3.5 *(The prime contribution on Gaussians).* For $\lambda > 0$,

$$\Phi_{\mathrm{fin}}(f_\lambda) = -\frac{1}{\pi}\sum_p\sum_{k\ge1}\frac{\log p}{p^{k/2}}\exp\!\left(-\frac{(k\log p)^2}{4\pi\lambda}\right).$$

The sum is absolutely convergent and smooth in $\lambda > 0$.

Proof. Insert the even Gaussian into (80.5). On any compact interval $[\lambda_0, \lambda_1] \subset (0, \infty)$, each parameter derivative is bounded by a polynomial in $\log n$ times $n^{-1/2} \exp[-(\log n)^2/(4\pi\lambda_1)]$. This is summable over integers $n \geq 2$. For every fixed M, the exponential is at most n^{-M} for all sufficiently large n. Uniform domination proves convergence and differentiation term by term. □

Lemma 80.3.6 *(The archimedean contribution on Gaussians).* For $\lambda > 0$,

$$\Phi_\infty(f_\lambda) = -\frac{\log \pi}{2\pi} + A_\Gamma(\lambda),$$

where

$$A_\Gamma(\lambda) = \frac{\sqrt{\lambda}}{2\pi} \int_{\mathbb{R}} e^{-\pi\lambda t^2} \operatorname{Re} \psi(1/4 + it/2)\, dt. \tag{80.8}$$

This ordinary integral is absolutely convergent.

Proof. Substitute $\widehat{f_\lambda}(t) = \sqrt{\lambda} e^{-\pi\lambda t^2}$ into (80.4). The Gaussian integral gives $\int_{\mathbb{R}} \widehat{f_\lambda}(t)\, dt = f_\lambda(0) = 1$. The digamma factor is smooth on the real integration line and is $O(\log(2 + |t|))$ at infinity. Gaussian decay gives absolute convergence. The same bound with polynomial factors proves smooth parameter variation on compact subintervals of $(0, \infty)$. □

Theorem 80.3.7 *(The Weil formula on Gaussians).* For $\lambda > 0$,

$$\begin{aligned} W(f_\lambda) = 2\sqrt{\lambda} e^{\pi\lambda/4} - \mathcal{Z}(\lambda) - \frac{\log \pi}{2\pi} \\ + A_\Gamma(\lambda) - \frac{1}{\pi}\mathcal{P}(\lambda) = 0, \end{aligned} \tag{80.9}$$

where

$$\mathcal{Z}(\lambda) = \sqrt{\lambda} \sum_\rho e^{-\pi\lambda z_\rho^2}, \qquad z_\rho = -i(\rho - 1/2),$$

$$\mathcal{P}(\lambda) = \sum_p \sum_{k \geq 1} \frac{\log p}{p^{k/2}} \exp\left(-\frac{(k \log p)^2}{4\pi\lambda}\right).$$

The integral A_Γ is defined in (80.8). Equivalently,

$$\mathcal{Z}(\lambda) = 2\sqrt{\lambda} e^{\pi\lambda/4} - \frac{\log \pi}{2\pi} + A_\Gamma(\lambda) - \frac{1}{\pi}\mathcal{P}(\lambda). \tag{80.10}$$

Proof. Substitute Lemma 80.3.4, Lemma 80.3.5, and Lemma 80.3.6 into Lemma 80.3.1. Gaussian admissibility was proved in Remark 80.3.3. Rearranging the balanced formula gives (80.10). □

Remark 80.3.8 *(Poisson summation embedding).* The self-duality $\widehat{f_1} = f_1$ of the standard Gaussian at $\lambda = 1$ is the archimedean input to Tate's thesis. Poisson summation for the standard Gaussian gives

$$\theta(i\tau) := \sum_{n \in \mathbb{Z}} e^{-\pi n^2 \tau} = \tau^{-1/2} \theta(i/\tau),$$

the classical modular transformation of the Jacobi theta. The identity (80.10) can be viewed as the arithmetic Weil analogue: the zero sum $\mathcal{Z}(\lambda)$ is the arithmetic theta analogue, and the RHS is the corresponding archimedean-plus-prime expansion.

Gaussian Convolutions and the Balanced Weil Functional

The functional W in (80.6) is the balanced difference, and it vanishes by the explicit formula. The associated sesquilinear form is instead

$$B_{2\pi}(f,h) = L_{2\pi}(h^\sharp \star f), \qquad f,h \in \mathcal{T}.$$

The change of variables $f \mapsto g_f$ preserves convolution and involution, so this is the same form as $L_\zeta(g_h^* \star g_f)$ in the logarithmic variable. Its diagonal values involve the arithmetic functional $L_{2\pi}$, rather than the identically zero balanced difference W. Positivity on the full test algebra remains the Weil positivity question. The Gaussian convolution identity below does not settle it.

Definition 80.3.9 *(Gaussian convolution sub-class).* The *Gaussian convolution sub-class* is the set

$$\mathcal{G} \;:=\; \{f_\lambda \star f_\mu^\sharp \;:\; \lambda,\mu > 0\}.$$

Since f_λ is real and even, $f_\mu^\sharp = f_\mu$, so $\mathcal{G} = \{f_\lambda \star f_\mu \;:\; \lambda,\mu > 0\}$.

LEMMA 80.3.10 *(Gaussian convolution is Gaussian).* For $\lambda,\mu > 0$,

$$f_\lambda \star f_\mu(x) \;=\; \sqrt{\frac{\lambda\mu}{\lambda+\mu}}\cdot f_{\lambda+\mu}(x).$$

Equivalently, in Fourier: $\widehat{f_\lambda \star f_\mu} = \widehat{f_\lambda}\cdot\widehat{f_\mu} = \sqrt{\lambda\mu}\,e^{-\pi(\lambda+\mu)y^2} = \sqrt{\lambda\mu/(\lambda+\mu)}\cdot\sqrt{\lambda+\mu}\,e^{-\pi(\lambda+\mu)y^2}$, consistent with $\widehat{f_\lambda \star f_\mu} = \sqrt{\lambda\mu/(\lambda+\mu)}\cdot\widehat{f_{\lambda+\mu}}$. In particular, $f_\lambda \star f_\lambda(x) = \sqrt{\lambda/2}\cdot f_{2\lambda}(x)$.

Proof.

$$f_\lambda \star f_\mu(x) \;=\; \int_{\mathbb{R}} e^{-\pi t^2/\lambda} e^{-\pi(x-t)^2/\mu}\,dt.$$

Completing the square in t:

$$\begin{aligned}
\frac{t^2}{\lambda} + \frac{(x-t)^2}{\mu} \;&=\; \frac{\mu t^2 + \lambda(x-t)^2}{\lambda\mu} \\
&=\; \frac{(\lambda+\mu)t^2 - 2\lambda t x + \lambda x^2}{\lambda\mu} \\
&=\; \frac{(\lambda+\mu)}{\lambda\mu}\Big(t - \frac{\lambda x}{\lambda+\mu}\Big)^2 + \frac{x^2}{\lambda+\mu}.
\end{aligned}$$

Hence

$$\begin{aligned}
f_\lambda \star f_\mu(x) \;&=\; e^{-\pi x^2/(\lambda+\mu)} \cdot \int_{\mathbb{R}} e^{-\pi(\lambda+\mu)u^2/(\lambda\mu)}\,du \\
&=\; e^{-\pi x^2/(\lambda+\mu)} \cdot \sqrt{\frac{\lambda\mu}{\lambda+\mu}} \;=\; \sqrt{\frac{\lambda\mu}{\lambda+\mu}}\cdot f_{\lambda+\mu}(x),
\end{aligned}$$

as claimed. □

Proposition 80.3.11 *(balanced Weil identity on Gaussian convolutions).* For every $\lambda, \mu > 0$,

$$W(f_\lambda \star f_\mu) \;=\; \sqrt{\frac{\lambda\mu}{\lambda+\mu}} \cdot W(f_{\lambda+\mu}) \;=\; 0. \tag{80.11}$$

This is an explicit-formula identity, not a proof of Weil positivity on the Gaussian convolution sub-class $\mathcal{G}$.

Proof. Lemma 80.3.10 expresses $f_\lambda \star f_\mu$ as a scalar multiple of $f_{\lambda+\mu}$. The Weil functional W is linear, so

$$W(f_\lambda \star f_\mu) \;=\; \sqrt{\lambda\mu/(\lambda+\mu)} \cdot W(f_{\lambda+\mu}).$$

By Theorem 80.3.7 applied to $\lambda + \mu$ in place of λ, and by the classical Weil explicit-formula identity $W(f_{\lambda+\mu}) = 0$, the result is zero. □

Remark 80.3.12 *(The Gaussian pairing).* The Gaussian functions are real and even, so

$$B_{2\pi}(f_\lambda, f_\mu) = \sqrt{\frac{\lambda\mu}{\lambda+\mu}}\, L_{2\pi}(f_{\lambda+\mu}).$$

Thus (80.10) evaluates this pairing by ordinary convergent arithmetic terms. The equality $W(f_\lambda \star f_\mu) = 0$ expresses the balanced explicit formula and supplies no sign inequality for $B_{2\pi}$. Positivity on the full test algebra remains a separate assertion.

Remark 80.3.13 *(Gaussian identities and operator traces).* The proposed arithmetic trace comparison in Corollary 62.5.2 requires an operator realization, a common domain and a trace or weight. The identity (80.10) computes its arithmetic scalar on the Gaussian family. It supplies no operator trace comparison and no positivity statement for $B_{2\pi}$ on the full test algebra.

80.4 Finite modules and formal comparisons

The arithmetic-loop comparisons in Propositions 53.4.4 and 53.4.7 involve mapping objects, filtered colimits and formal completion. Artin–Rees gives a precise theorem for a finite module and its submodule. To apply it to those comparisons, one must construct the modules and prove that they control the required maps.

The Iwasawa coefficient ring

Definition 80.4.1 *(The Iwasawa algebra).* Choose $\Gamma \cong \mathbb{Z}_p$ and a topological generator γ. Its completed group ring is

$$\Lambda = \mathbb{Z}_p[[\Gamma]] = \varprojlim_{r,n} (\mathbb{Z}/p^r)[\Gamma/p^n\Gamma] \cong \mathbb{Z}_p[[T]], \qquad \gamma \longmapsto 1 + T.$$

The coefficient variable T is different from the zero-counting height used earlier in this chapter.

Lemma 80.4.2 *(Noetherian local structure).* The ring Λ is regular local and Noetherian of dimension two, with maximal ideal $\mathfrak{m} = (p, T)$ and residue field $\mathbb{F}_p$. Its $\mathfrak{m}$-adic topology is the topology of the displayed finite group-ring quotients.

Proof. The formal power-series ring over the discrete valuation ring $\mathbb{Z}_p$ is Noetherian. Its maximal ideal is generated by p, T, and the chain $(0) \subset (p) \subset (p, T)$, together with the power-series dimension theorem, gives dimension two. The two generators form a regular system of parameters.

The finite group-ring quotient has defining ideal $J_{r,n} = (p^r, (1+T)^{p^n} - 1)$. For a fixed k, take $r \geq k$ and n sufficiently large. For $1 \leq j < k$,

$$v_p\binom{p^n}{j} \geq n - v_p(j),$$

by $\binom{p^n}{j} = p^n \binom{p^n-1}{j-1}/j$. Every term of $(1+T)^{p^n} - 1$ then lies in $\mathfrak{m}^k$. Thus $J_{r,n} \subset \mathfrak{m}^k$.

Conversely, modulo $J_{r,n}$, one has $p^r = 0$ and $T^{p^n} \in (p)$. Hence (p, T) is nilpotent in that finite quotient, and some $\mathfrak{m}^k$ lies in $J_{r,n}$. The two ideal systems are cofinal, proving the topology and completed-ring identification. □

Induced filtrations on finite modules

LEMMA 80.4.3 *(Artin–Rees).* Let R be a Noetherian ring, $\mathfrak{a} \subset R$ an ideal, M a finite R-module, and $N \subset M$ a submodule. There is $k \geq 0$ such that, for all $n \geq k$,

$$\mathfrak{a}^n M \cap N = \mathfrak{a}^{n-k}(\mathfrak{a}^k M \cap N).$$

In particular, the induced filtration on N is cofinal with its $\mathfrak{a}$-adic filtration.

Proof. The Rees algebra $\mathcal{R} = \bigoplus_{n \geq 0} \mathfrak{a}^n t^n$ is generated over R by finitely many elements of degree one, so it is Noetherian. The Rees module $\bigoplus_{n \geq 0} \mathfrak{a}^n M t^n$ is finite over $\mathcal{R}$. Its graded submodule

$$\bigoplus_{n \geq 0} (\mathfrak{a}^n M \cap N) t^n$$

is therefore finitely generated. Choose homogeneous generators of degrees at most k.

In degree $n \geq k$, a generator of degree $j \leq k$ is multiplied by $\mathfrak{a}^{n-j}$. Factor this as $\mathfrak{a}^{n-k}\mathfrak{a}^{k-j}$. The second factor sends that generator into $\mathfrak{a}^k M \cap N$, proving one inclusion in the formula. The reverse inclusion follows from N being a submodule.

Finally, $\mathfrak{a}^n N \subset \mathfrak{a}^n M \cap N \subset \mathfrak{a}^{n-k} N$, which gives cofinality. □

Arithmetic loops and filtered colimits

Question 80.4.4 *(The specified formal presentation).* For a derived formal arithmetic stack with presentation $\mathcal{X} = \operatorname{colim}_i \mathcal{X}_i$, determine the coefficient, topology and transition hypotheses under which

$$\operatorname{colim}_i \operatorname{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}_i) \longrightarrow \operatorname{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$$

is an equivalence. The question includes the $\mathfrak{m}$-adically complete sector and its finite-type pieces.

A profinite presentation of $\widehat{\mathbb{Z}}$ uses all finite cyclic quotients with their divisibility transition maps. The p-power quotients instead describe $\mathbb{Z}_p$. Finite cardinality of a group does not make its classifying space compact in every category of targets. Nor does an inverse limit of compact objects automatically remain compact.

Question 80.4.5 *(The mapping functor on a target category).* For which category $\mathcal{C}$ of complete derived arithmetic stacks does the internal functor $\operatorname{Map}(B\widehat{\mathbb{Z}}, -)$ preserve the filtered colimits under consideration?

Ordinary compactness of K in an infinity category means that the space-valued functor $\mathrm{Map}(K, -)$ preserves filtered colimits. Preservation by an internal mapping object is a separate assertion, to be checked after evaluation on all test objects. Artin–Rees alone proves neither statement.

Proposition 80.4.6 *(A conditional arithmetic-loop comparison).* Fix the category of targets and an internal arithmetic-loop functor $L^{\mathrm{Ar}}\mathcal{Y} = \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{Y})$. If this functor preserves a specified presentation $\mathcal{X} = \mathrm{colim}_i \mathcal{X}_i$, then

$$L^{\mathrm{Ar}}\mathcal{X} \simeq \mathrm{colim}_i\, L^{\mathrm{Ar}}\mathcal{X}_i.$$

No preservation hypothesis is supplied merely by adic completeness of the target.

Proof. The assumed comparison is exactly the displayed map after substituting the definition of L^{Ar}. □

The order of completion and filtered colimit is a concrete issue even for modules. Let $R = \mathbb{Z}_p[[T]]$, $\mathfrak{m} = (p, T)$, and $M_i = \bigoplus_{j=1}^{i} R$. Each M_i is adically complete, but

$$\mathrm{colim}_i \varprojlim_n M_i/\mathfrak{m}^n M_i = \bigoplus_{j\geq 1} R \quad \subsetneq \quad \varprojlim_n \mathrm{colim}_i\, M_i/\mathfrak{m}^n M_i. \tag{80.12}$$

The right side is the space of sequences $(a_j) \in \prod_j R$ such that, for every n, all but finitely many a_j lie in $\mathfrak{m}^n$. Indeed, reduction modulo each power has finite support, and compatibility recovers each coordinate by completeness of R. The sequence $a_j = p^j$ belongs to this right side and not to the direct sum.

Every fixed quotient has a finite support bound, but no one bound works at every precision. This example isolates a limit exchange that a formal-stack argument must justify. It is not an obstruction to every possible arithmetic-loop comparison.

The flat subcategory

Question 80.4.7 *(Flat objects in a formal colimit).* Construct the proposed flat object or full subcategory on $L^{\mathrm{Ar}}\mathcal{X}$, with its transition maps, and determine when the comparison

$$\mathrm{colim}_i (L^{\mathrm{Ar}}\mathcal{X}_i)^{\mathrm{flat}} \longrightarrow (L^{\mathrm{Ar}}\mathcal{X})^{\mathrm{flat}}$$

is an equivalence. Specify whether the two sides are stacks, categories of sheaves or categories of modules before forming the map.

A flat subcategory is not a submodule of a coordinate ring merely by being full. Applying Artin–Rees requires an actual finite module M and submodule N, together with a theorem identifying their induced filtrations with the proposed geometric condition. A criterion using all adic truncations also requires its own completeness and flatness hypotheses.

Remark 80.4.8 *(Two further geometric comparisons).* The extension of nilpotent singular support to the specified formal stacks and its comparison with a Fargues–Scholze (φ, Γ)-module locus are additional geometric questions. Neither is a consequence of the finite-module filtration theorem. Their coefficient categories and functors must be specified separately.

The finite-module consequence

PROPOSITION 80.4.9 *(Adic completions of finite submodules).* For a finite Λ-module M and submodule N, the intrinsic $\mathfrak{m}$-adic completion of N agrees with its completion for the induced filtration $N \cap \mathfrak{m}^n M$. The arithmetic-loop and flat-subcategory comparisons above remain separate assertions. The categorical trace statement of Theorem 53.4.2 retains its stated categorical colimit hypothesis.

Proof. Artin–Rees makes the two filtrations cofinal, so their inverse-limit completions are canonically isomorphic. The lemma concerns finite modules and provides no mapping-stack or categorical colimit identity. □

80.5 CONSEQUENCES

PROPOSITION 80.5.1 *(three exact inputs).* The following assertions hold.

(i) The sharp Riemann–von Mangoldt identity (80.1) with $|S(T)| = O(\log T)$ is unconditional (Theorem 80.2.1, Theorem 80.2.3, Corollary 80.2.4). This proves the density estimate entering $\mathrm{R}_{\mathrm{analytic}}$, but not the positivity or linear-growth assertions.

(ii) The Gaussian specialization of the balanced Weil explicit formula uses the ordinary convergent terms in (80.9) and (80.10). The convolution identity (80.11) is only the balanced explicit formula and is not a Weil-positivity statement (Remark 80.3.12).

(iii) Artin–Rees identifies the intrinsic and induced adic completions of a finite Iwasawa submodule (Proposition 80.4.9). The arithmetic-loop and flat-subcategory maps require their separate preservation and geometric hypotheses.

Proof. The first assertion is Theorem 80.2.1 together with Theorem 80.2.3. The second assertion is Theorem 80.3.7 and Proposition 80.3.11. The third assertion is Proposition 80.4.9. None of these assertions proves Weil positivity, a Hilbert–Pólya realization or the proposed formal arithmetic comparisons. □

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The explicit-formula identities, moment positivity and Noetherian completion statements concern different mathematical data; none of these constructions supplies the missing comparison maps by itself.

Chapter 81

The Archimedean Tate Positivity Reduction

The sufficient path of Chapter 44 requires both categorical comparisons and spectral realization. The conditions in Chapter 69 separate these requirements. The arithmetic Positselski biconditional $\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^{!})_{\mathsf{G}}^{\mathrm{Ar}} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$ is conditional on the abelian Heisenberg inputs; the Koszul–Petersson compatibility is restricted to its stated core; the Hilbert–Pólya operator skeleton is constructed on its stated domain; HP–Li positivity requires a positive realization of the full Weil form on the compact convolution test algebra. The archimedean Tate positivity conjecture A-TP of Chapter 78 is RH-equivalent, but it is not the whole obstruction. Theorem 81.5.1 records only this conditional placement.

81.1 Categorical and Spectral Conditions

Chapter 69 left the following residual conditions blocking the sufficient RH path of Theorem 44.6.1:

(i) **F1.a.1** Dotto–Emerton–Gee essential surjectivity at unramified primes with Taylor–Wiles big-image residuals.

(ii) **F1.a.2** Extension of the local categorical comparison to $p \in \{2, 3\}$.

(iii) **F1.a.3.A** Bushnell–Henniart block-Hochschild comparison.

(iv) **F1.a.3.B** Paskunas wild-block Ext^1-vanishing.

(v) **F1.a.3.C** Wild-EG compact generation at $p \in \{2, 3\}$.

(vi) $\mathrm{L}_{\mathrm{ACD}}$ Adelic categorical descent on the cuspidal locus.

(vii) **BC-diamond-glue** (Ch1) Chenevier cross-prime pseudocharacter compatibility.

(viii) **BC-diamond-glue** (AG.det) Arinkin–Gaitsgory determinantal projection on the Selmer-derived stack.

(ix) **F2.c** Restricted-tensor E_2 transport.

(x) **Arithmetic Positselski** (four sub-items): continuous duals over Λ_p, derived p-adic completion, faithful lifting along Nakayama, Koszul-dual Bost–Connes.

(xi) **Full $\mathcal{V}^{\mathrm{prim}}$ P3**: non-tempered and Eisenstein sectors.

(xii) **H-P^{Ar}** (W): weight existence for the Hilbert–Pólya operator.

(xiii) **HP–Li**: Weil-positive trace inequality on $\mathscr{A} = C_c^{\infty}(\mathbb{R})$.

(xiv) **VB* main direction**: (HB$^+$)+(EC$_\pi$)+(Det) $\Rightarrow$ RH.

(xv) $\mathrm{R}_{\mathrm{analytic}}$ density-of-zeros estimate, Weil at Gaussians, and the separate formal arithmetic comparisons R1.formal.b/c.

Chapter 74 records the sufficient implication and the logical reductions for Chapters 70–80 (Section 74.5). Each chapter proves one or more residual assertions on its natural core domain, isolating the narrower named sub-items whose completion is strictly equivalent to RH.

81.2 THEOREM RECORD

The theorem-level consequences of Chapters 70–80, excluding Chapter 74, have the following mathematical form.

Chapter 70: F1.a.1, F1.a.2, and $\mathrm{L}_{\mathrm{ACD}}$

Theorem 70.2.4 closes F1.a.1 essential surjectivity on the unramified ordinary summand unconditionally. Theorem 70.5.4 closes F1.a.2 on the block-diagonal local comparison at $p \in \{2, 3\}$: semisimple blocks unconditionally (Corollary 70.5.5); non-semisimple blocks fully faithful with explicit Ext^1-control. Theorem 70.6.4 closes cuspidal sph-fin adelic categorical descent $\mathrm{L}_{\mathrm{ACD}}^{\mathrm{cusp}}$. Primary sources: Breuil–Mézard 2002; Kisin 2009 JAMS 22; Gee–Kisin 2014 Forum Math. Pi 2; Emerton–Gee arXiv:1908.07185; Clozel–Harris–Taylor 2008 Publ. IHES 108; Hida 1986 Ann. Inst. Fourier 36; Paskunas; Colmez–Dospinescu–Paskunas; Flath 1979; Bernstein 1988; Moœglin–Waldspurger 1995. F1.a.1 on the non-ordinary/supersingular locus reduces to the block-exhaustion conjecture (Conjecture 61.4.5); the Eisenstein/continuous-spectrum extension of $\mathrm{L}_{\mathrm{ACD}}$ is a separate conditional.

Chapter 71: F1.a.3.A Block-Additivity and F1.a.3.B Paskunas

Lemma 71.2.1 proves an S^1-equivariant Hochschild splitting for the specified orthogonal k-linear dg models and finite-support perfect categories. An arithmetic application must supply these models and coefficient comparisons; a multiplicative enhancement needs separate monoidal data. Proposition 71.3.3 proves that extensions of finite modules vanish after localization at the fraction field of a Noetherian domain. Its arithmetic application needs the exact block and generic-fibre equivalences of Corollary 71.3.4. The wild-prime version additionally assumes reducedness of the centraliser, as in Corollary 71.3.9. Theorem 71.4.1 separates the algebraic conclusions from these arithmetic comparisons. Primary sources: Bernstein 1984; Bernstein–Deligne 1984; Bushnell–Henniart 2006 Grundlehren 335; Colmez 2010 Astérisque 330; Paskunas 2013 Publ. IHES 118; Paskunas 2015; Auslander–Reiten–Smalø 1995; Matsumura 1989; Ben-Zvi–Francis–Nadler arXiv:0805.0157; Ben-Zvi–Nadler arXiv:0904.1247; Lurie HA. Three strictly narrower residuals remain: (R.A) per-block Bushnell–Henniart projective-to-closed-point matching; (R.B.1) very-special ordinary-block Breuil–Mezard multiplicity; (R.B.2) wild-prime block centraliser regularity at $p \in \{2, 3\}$.

Chapter 72: F1.a.3.C Wild-EG Compact Generation

Theorem 72.4.1 (under the stated hypotheses) reduces compact generation of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi})$ at $p \in \{2, 3\}$ to the finite Nilp stratification and per-stratum gerbe-generation inputs. Corollary 72.4.2 is conditional on these local-model inputs. Primary sources: Emerton–Gee arXiv:1908.07185; Dotto–Emerton–Gee arXiv:2603.26887; Bhatt–Scholze arXiv:1905.08229; Gaitsgory–Rozenblyum AMS DAG Vol. II; Arinkin–Gaitsgory arXiv:1201.6343; Bushnell–Henniart 2006; Beilinson–Bernstein–Deligne 1982; Beilinson–Ginzburg 1996. The $p \geq 5$ DEG theorem does not close this wild-prime input.

Chapter 73: Chenevier Compatibility, (AG.det), and F2.c

Theorem 73.1.4 constructs the continuous map $R_p^{\mathrm{Ch}} \widehat{\otimes}_{W(k)} R_q^{\mathrm{Ch}} \to R^{\mathrm{Ch}}$, its kernel quotient and its formal fibres. The pairwise sheaf comparison requires the full data in Corollary 73.1.5: the Selmer-algebra identification, geometric realization, compatible trace and centre actions, and equivariant coherent descent. The formal pullback and its local action are constructed in Proposition 64.3.2. Their comparison with the global prismatic object requires an actual equivariant isomorphism. Proposition 73.2.9 gives an internal centre identity only on its assumed geometric and categorical diagram.

The arithmetic singular-support space, higher-centre maps and global automorphic comparison remain additional constructions. Theorem 73.3.1 proves the restricted E_2-tensor transport under its central idempotent pointed-unit hypotheses. Proposition 73.4.1 records the updated BC-diamond-glue statement. Primary sources: Chenevier 2014 JAMS 27; Bellaiche–Chenevier 2009 Astérisque 324; Bhatt–Scholze arXiv:1905.08229; Bhatt–Lurie arXiv:2201.06120; Arinkin–Gaitsgory arXiv:1201.6343; Galatius–Kupers–Randal-Williams arXiv:1805.07184; Ben-Zvi–Francis–Nadler arXiv:0805.0157; Lurie HA; Francis–Gaitsgory arXiv:1103.5803; Emerton–Gee arXiv:1908.07185. The local categorical and prismatic comparisons retain their hypotheses. The internal spectral centre identity does not remove the global arithmetic comparison required for $\mathrm{L}_{\mathrm{ACD}}$.

Chapter 75: Arithmetic Positselski Equivalence

Theorem 75.1.4 constructs compact/discrete Pontryagin duality with the dual monoidal product $\odot$. Application to sub-item (1) still requires the arithmetic completed bar–cobar carrier and its compatibility maps. Corollary 75.2.5 closes derived completion sub-item (2) via Greenlees–May. Corollary 75.3.3 closes faithful lifting sub-item (3) via Matlis and the Krull intersection theorem. The Bost–Connes requirement in Remark 75.4.3 still needs the specified dg algebra, coefficient, completion and pairing comparison. Chapter 66 computes a selected one-mode Euler product and a Gaussian Mellin pairing; interpreting that scalar on the arithmetic bar or cobar object requires additional maps and a Tate twist. The intended equivalence and central-class comparison are the problems of Question 75.5.2 and Question 75.5.3. They are not consequences of a numerical energy match or of a named adele-class duality without its actual compatible construction. Primary sources: Pontryagin 1934/Weil 1940; Folland 1995; Positselski arXiv:0905.2621 and arXiv:1209.2995; Greenlees–May 1992 Trans. AMS 340; Matlis 1958; Matsumura 1989; Lazard 1965 IHES Publ. 26; Venjakob 2003 J. Reine Angew. Math. 559; Nekovář 2006 Astérisque 310; Bost–Connes 1995 Selecta Math. 1; Consani–Marcolli 2004/2006. The non-abelian Iwasawa extension for non-abelian Iwahori level is out of domain and not required for the sufficient-path G-row input.

Chapter 76: $\mathcal{V}^{\mathrm{prim}}$ P3 and Koszul–Petersson Compatibility

The prime Gibbs construction of Chapter 67 gives its state, Hilbert–Schmidt GNS space, maximal modular domains and a unitary onto the prime cyclic subspace of any specified global KMS representation. It does not identify an arithmetic Koszul carrier with an automorphic space. Theorems 76.2.1 and 76.2.3 retain the arithmetic placement, local pairing maps and adelic extension of Hypothesis 67.1.1. Deligne and Deligne–Serre supply holomorphic temperedness; Maass temperedness requires the stated Ramanujan assumptions. The Satake exponent bound does not itself bound a pairing discrepancy. The discrete-core identity of Theorem 76.3.3 is conditional on these maps. Its Li-trace use additionally requires the explicit trace-placement comparison, as well as the determinant-trace realization, of Proposition 76.3.5. Primary sources: Deligne 1974 Publ. IHES 43; Deligne–Serre 1974 Ann. Sci. ENS 7; Kim–Shahidi 2002 Ann. Math. 155; Kim–Sarnak 2003 JAMS 16; Langlands 1966; Moeglin–Waldspurger 1995; Selberg 1956; Arthur 1978/1979; Shahidi 2010 AMS Colloquium 58; Iwaniec 2002; Gelfand–Graev–Piatetski-Shapiro 1969. The Maass Ramanujan conjecture is an external open problem in the Langlands programme. The Eisenstein continuous spectrum is domain-restricted; the alternative Langlands–Shahidi correction (Theorem 76.4.4) is conditional on identifying the Koszul Eisenstein measure with the Shahidi L^2-normalised measure.

Chapter 77: The Hilbert–Pólya Operator Skeleton

Theorem 77.2.5 establishes self-adjointness of $N_{\mathrm{BC}} = -\frac{1}{2}\log\Delta_1$ on the full critical GNS space $\mathcal{H}_1$. Its point spectrum is $\frac{1}{2}\log\mathbb{Q}_{>0}$, with countably infinite multiplicity at each eigenvalue, and its closed spectrum is $\mathbb{R}$. The logarithmic integer spectrum $\{\log m : m \geq 1\}$ belongs to the restriction $N_{\mathrm{occ}} = N_{\mathrm{BC}}|_{K_2}$ on the reducing occupation subspace $K_2 = \overline{\mathrm{span}}\{\mu_{m^2}\Omega_1 : m \geq 1\}$. Lemma 77.3.2 closes the isometric embedding of the spherical zero-mode lattice into the GNS space. Lemma 77.3.4 gives the oscillator Fock embedding under a BC–GNS Heisenberg lift. Theorem 77.6.1 closes the Hadamard determinant identity conditional on the twist; Proposition 77.7.1 closes the Weil–Connes distributional trace identity conditional on the twist. Theorem 77.8.1 identifies the precise obstruction: the existence of a self-adjoint operator A on the GNS Hilbert space intertwining N_{BC} with a normal zero-divisor operator whose regularised determinant is $\xi_{\mathbb{R}}$ and whose Weil–Connes trace is the zero distribution is equivalent to the Connes absorption-trace condition. This is the Hilbert–Pólya obstruction in its operator-algebraic form. Primary sources: Bost–Connes 1995 Selecta 1; Connes arXiv:math/9811068; Connes–Marcolli 2008 AMS Colloquium 55; Takesaki 1970/1979/2003; Meyer arXiv:math/0311468; Berry–Keating 1999 SIAM Review 41; Reed–Simon 1975/1980; Deninger 1994/2001; Ray–Singer 1971; Voros 1987. The weight-existence residual (W) from the prior table is closed unconditionally: the BC log-modular eigenvalues λ_{BK} are present without additional hypothesis. The Hilbert–Pólya trace bridge (H) remains open, equivalent to twist-existence with determinant divisor div $\xi_{\mathbb{R}}$ and to the Connes–Deninger determinant-trace conjecture.

Chapter 78: HP–Li Positivity and the Archimedean Residual

Theorem 78.5.1 computes the Rankin–Selberg integral for an actual holomorphic cusp form. Applied to a coefficient multiplier $\Upsilon_F(h)$, it requires that the resulting series satisfy the cusp-form transformation law. Theorem 78.5.5 constructs a compact test with $W_{\mathrm{fin}}(f * f^*) < 0$. Thus no positive Hilbert norm realizes the finite-prime form on the full compact test algebra. Corollary 78.5.6 requires a proved restriction and a pairing identity for any positive automorphic realization. The signed operator representation in Theorem 78.9.3 retains both signs and all polar, gamma and prime contributions. Conjecture 78.2.3 names the remaining obstruction as the archimedean Tate positivity conjecture A-TP. Theorem 78.2.4 establishes that A-TP is strictly equivalent to Weil positivity, hence to RH. Theorem 78.8.1 records the HP–Li form after Chapters 70–80. Primary sources: Weil 1952 Lund Festschrift; Li 1997 JNT 65; Rankin 1939 Proc. Camb. Phil. Soc. 35; Selberg 1940 Arch. Math. Naturvid. 43; Hecke 1936 Math. Ann. 112; Deligne 1974; Jacquet 1972 SLN 260; Jacquet–Langlands 1970 SLN 114; Iwaniec–Kowalski 2004 AMS Colloquium 53; Bombieri–Lagarias 1999 JNT 77; Connes 1999; Patterson 1988; Bump 1997; Zagier 1981. A continuous-spectrum comparison requires its measure and realization map. Ordinary Maass cusp-form norm positivity does not require Ramanujan; its identification with an arithmetic form remains separate. A-TP is an RH-equivalent residual, not the whole obstruction.

Chapter 79: VB* *Main Direction*

Theorem 79.8.1 (under the stated hypotheses) records the conditional implication (SH) + (Det) + (HB$^+$) + EC$_\pi$ $\Rightarrow$ RH on the primary Fock space. Theorem 79.9.2 records the sufficient implication from these analytic hypotheses on $\xi_{\mathbb{R}}$ to RH; it does not assert the converse. Primary sources: de Branges 1968 monograph; Lagarias 1999 Acta Arith. 89; Conrey–Li arXiv:math/9812166; Beurling–Malliavin 1962 Acta Math. 107; Hamburger 1920 Math. Ann. 81; Carleman 1922/1926; Koosis 1992 Log. Integral II; Levin 1980 AMS Monograph. The Hermite–Biehler positivity assertion is RH-equivalent; the EC$_\pi$ and

(Det) assertions are the additional spectral-regularity inputs isolated in Chapter 57. Their proof is not attempted in Chapter 79.

Chapter 80: Zero counts, Gaussian tests and formal completion

Theorem 80.2.1 establishes the sharp Riemann–von Mangoldt density with the Titchmarsh bound $|S(T)| = O(\log T)$ unconditionally. Theorem 80.3.7 gives the closed-form Weil evaluation on Gaussians. The difference between the two sides of the explicit formula is zero. The Weil value itself is the zero sum, which need not vanish. In the normalization $f_\lambda(t) = e^{-\lambda t^2/2}$, put

$$k(u) = \log \pi - \Re \psi_\Gamma(1/4 + iu/2), \qquad H(\lambda) = \frac{1}{\sqrt{\pi\lambda}} \int_{\mathbb{R}} e^{-u^2/\lambda} k(u)\, du,$$

where $\psi_\Gamma = \Gamma'/\Gamma$. Theorem 95.2.3 gives the exact sign range

$$W_\infty(f_\lambda * f_\lambda) = \sqrt{\frac{\pi}{\lambda}} H(\lambda) \geq 0 \quad \Longleftrightarrow \quad 0 < \lambda \leq \lambda_c,$$

where λ_c is the unique zero of H. The full arithmetic value is

$$W(f_\lambda * f_\lambda) = \frac{4\pi}{\lambda} e^{1/(4\lambda)} - \sqrt{\frac{\pi}{\lambda}} H(\lambda) - 2\sqrt{\frac{\pi}{\lambda}} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-\lambda(\log n)^2/4}.$$

The gamma sign range alone does not settle the nonnegativity of this expression or of Gaussian cross terms.

Artin–Rees identifies the induced and intrinsic adic completions of a finite Iwasawa submodule (Proposition 80.4.9). The arithmetic-loop comparison remains conditional on preservation of the specified colimit (Proposition 80.4.6). The flat-subcategory comparison is Question 80.4.7.

Primary sources: Riemann 1859 Monatsber. Berlin Akad.; von Mangoldt 1895 J. Reine Angew. Math. 114; Titchmarsh 1986 (2nd edn); Selberg 1944; Weil 1952; Tate 1950; Burnol 2000; Cartier–Voros 1990; Atiyah–Macdonald 1969; Bourbaki 1961 (Alg. Comm. III); Lang 1994.

The positivity/linear-upper-growth portion of $\mathrm{R}_{\mathrm{analytic}}$ is RH-adjacent and remains open. The mapping-colimit, formal singular-support and flat-locus comparisons each require their specified geometric hypotheses.

81.3 Unified Obstruction Decomposition

The comparisons below apply on different coefficient categories and arithmetic loci. The middle column states what the cited result gives on its stated domain; the last column names the extra hypothesis or construction needed for the larger comparison. An assertion on an ordinary, generic, discrete, or complete sector does not extend to other sectors without a comparison map.

Condition	Mathematical consequence under stated hypotheses	Remaining hypothesis or construction
F1.a.1 (unramified)	Essential surjectivity on the ordinary summand (Theorem 70.2.4)	Non-ordinary block exhaustion (Conjecture 61.4.5)
F1.a.2 ($p \in \{2, 3\}$)	Semisimple block comparison and full faithfulness with Ext^1-control on the stated non-semisimple blocks (Theorem 70.5.4)	The normalized extension-class comparison on non-semisimple blocks

Condition	Mathematical consequence under stated hypotheses	Remaining hypothesis or construction
F1.a.3.A (BH type matching)	Hochschild homology is additive over finite-support orthogonal dg block models (Lemma 71.2.1)	Per-block Bushnell–Henniart projective-to-closed-point matching (R.A)
F1.a.3.B (Paskunas Ext^1)	Finite-module extension vanishing over a fraction field (Proposition 71.3.3); arithmetic transport requires the generic-fibre equivalence	Very-special ordinary multiplicities (R.B.1) and wild reducedness and generic-fibre comparison at $p \in \{2, 3\}$ (R.B.2)
F1.a.3.C (wild EG generation)	Compact generation conditional on the finite Nilp stratification and per-stratum gerbe-generation inputs (Theorem 72.4.1)	Those wild-prime local models and generation hypotheses
$\mathrm{L}_{\mathrm{ACD}}$, cuspidal	Sph-finite cuspidal descent on the stated domain (Theorem 70.6.4)	A separate Eisenstein and continuous-spectrum comparison
BC-glue (Ch1)	Continuous formal restriction, its kernel quotient and fibres (Theorem 73.1.4)	Selmer-algebra and geometric realization maps, compatible trace/centre actions and equivariant coherent descent for the pairwise sheaf comparison (Corollary 73.1.5)
BC-glue (AG.det)	An internal centre identity on the assumed diagram (Proposition 73.2.9)	Arithmetic singular-support models, higher-centre maps and the global automorphic comparison
(BCglue.local)	A conditional local categorical comparison	The local functors, coefficient comparisons and compatibility hypotheses in the BC-glue datum
(BCglue.glue)	A conjectural global gluing comparison	Compatible cross-prime transition and descent maps
F2.c E_2 transport	Restricted tensor transport for central idempotent pointed units (Theorem 73.3.1)	The local Iwahori E_2 input and all pointed-unit hypotheses for the intended factors
Arithmetic Positselski (1): continuous duals	The continuous-duality comparison for compact modules and discrete $\odot$-comodules (Theorem 75.1.4)	The specified coefficient objects, dual monoidal product and completed bar–cobar maps
Arithmetic Positselski (2): derived completion	Derived completion under the Greenlees–May hypotheses (Corollary 75.2.5)	The completion and coefficient hypotheses of the cited comparison
Arithmetic Positselski (3): faithful lifting	The Matlis and Krull intersection lifting criterion (Corollary 75.3.3)	Its complete, separated module domain and the specified lifting maps
Arithmetic Positselski (4): primitive zeta character	Selected Euler–Mellin scalar of Chapter 66	Arithmetic character and pairing maps, including the Tate twist
Arithmetic Positselski (5): Koszul-dual BC	Concrete shift model and same-prime relation test (Remark 75.4.3)	Compatible dg algebra, coefficients, completion and pairing comparison
Arithmetic Positselski, combined	Intended co-contra comparison on the abelian Heisenberg sector (Question 75.5.2)	Its stated categorical hypotheses and actual arithmetic comparison maps; the nonabelian Iwahori extension is additional

Condition	Mathematical consequence under stated hypotheses	Remaining hypothesis or construction
Full-$\mathcal{V}^{\mathrm{prim}}$ P_3, cuspidal	The conditional Koszul–Petersson transfer on the selected discrete core (Theorem 76.3.3)	Arithmetic placement, local pairing maps and adelic extension; Ramanujan for the Maass domain (Theorem 76.2.3); its Li-trace use also requires the spectral realization and trace-placement map
Full-$\mathcal{V}^{\mathrm{prim}}$ P_3, Eisenstein	The discrete-core restriction excludes continuous spectrum; the alternative full comparison is conditional (Theorem 76.4.4)	Identification of the Koszul Eisenstein measure with the Langlands–Shahidi normalized measure
H-P^{Ar} (W): weight	Self-adjoint N_{BC} with log-modular spectrum (Theorem 77.2.5)	A weight operator does not by itself identify the zeta zero divisor
H-P^{Ar} (H): trace comparison	The determinant and trace realization criterion (Theorem 77.8.1)	The Connes absorption-trace realization and its operator-domain comparison
HP–Li realization	A positive realization exists exactly when the full Weil form is positive (Theorem 78.8.1)	Full Weil positivity, equivalent to RH
HP–Li archimedean condition (A-TP)	A-TP is equivalent to full Weil positivity (Theorem 78.2.4)	The RH-equivalent A-TP inequality remains conjectural
VB^* spectral sufficient condition	The conditional implication (SH) + (Det) + (HB^+) + (EC_π) $\Rightarrow$ RH (Theorem 79.8.1)	All four analytic hypotheses on the specified spectral data
Spectral forms HB^+, (Det), EC_π	The Hermite–Biehler condition is RH-equivalent; determinacy and density enter the separate spectral-regularity comparison	The stated moment, determinacy and density hypotheses of Chapter 57
R_{analytic}, density	Riemann–von Mangoldt density with the Titchmarsh bound (Theorem 80.2.1)	No positivity conclusion follows from this density estimate alone
R_{analytic}, positivity and linear growth	Conjectural positivity and growth on the stated analytic carrier	The full positivity and linear-upper-growth estimates; their RH-adjacent content is separate from zero-counting density
Weil evaluation on Gaussians	The exact signed Gaussian formula (Theorem 80.3.7)	Positivity of the full polar-minus-gamma-minus-prime expression and its cross terms does not follow from the gamma sign range
R1.formal.a	The formal comparison on its stated domain	Its coefficient and formal-geometric hypotheses
R1.formal.b	The conditional loop comparison (Proposition 80.4.6)	Preservation of the specified colimit by the internal arithmetic mapping functor
R1.formal.c.2	The flat-subcategory comparison (Question 80.4.7)	Construction of the flat object and proof of its colimit compatibility
R1.formal.c.1	A conjectural derived-formal extension	The Arinkin–Gaitsgory comparison on the relevant formal arithmetic carrier
R1.formal.c.3	A conjectural flat-locus comparison	Matching with the Fargues–Scholze flat locus

The RH-Equivalent Residual

Theorem 78.8.1 isolates the archimedean Tate positivity conjecture A-TP (Conjecture 78.2.3) as an RH-equivalent obstruction after Chapters 70–80. This does not remove the non-A-TP obstructions in the decomposition above.

81.4 The Refined Sufficient Path to RH

The sufficient path of Theorem 44.6.1 took four inputs:

(P1) arithmetic Positselski: the G-row biconditional $\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^{!})_{\mathsf{G}}^{\mathrm{Ar}} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$;

(KP) Koszul–Petersson compatibility $\langle v, w\rangle^{\mathrm{Pet}} = \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}}$ on $\mathcal{V}^{\mathrm{prim}}$;

(HP) Hilbert–Pólya^{Ar}: a self-adjoint H_{HP} with $\det_{\infty}(s\mathrm{id} - \Theta_{\xi}) = \xi_{\mathbb{R}}(s)$ and Weil–Connes trace distribution;

(HPLi) HP–Li positivity: $W(f * \widetilde{f}) \geq 0$ on admissible Weil test functions.

Domains of the Four Inputs

(i) **P1 (arithmetic Positselski).** On the abelian Heisenberg sector, the intended equivalence requires the categorical hypotheses and the actual coefficient, completion, algebra and pairing comparisons of Question 75.5.2. Its G-row formulation also requires the compatible identification of arithmetic obstruction classes in Question 75.5.3. A named adele-class duality or equality of scalar characters does not supply these maps. The non-abelian Iwahori extension lies outside this domain.

(ii) **KP (Koszul–Petersson compatibility).** On the selected discrete carrier, Theorem 76.3.3 is conditional on the arithmetic placement, local pairing maps and adelic extension. Maass temperedness and the spectral trace-placement comparison are additional inputs where used. The continuous pairing requires its own measure and normalization maps.

(iii) **HP (Hilbert–Pólya).** (W) weight existence closed unconditionally by Theorem 77.2.5: the Bost–Connes log-modular operator supplies the zero-mode eigenvalues $\log m$ on the primary Fock image. (H) trace bridge remains open, equivalent to the Connes absorption-trace condition and to the twist-existence statement Hypothesis 77.4.3.

(iv) **HPLi (HP–Li positivity).** Theorem 78.8.1 proves that a positive Hilbert realization exists exactly when the full Weil form is positive. It does not construct that positive realization. Full HP–Li positivity is strictly equivalent to archimedean Tate positivity A-TP (Theorem 78.2.4), which is strictly equivalent to RH.

The Sufficient Path: Implication Diagram

81.5 A-TP and the Other Residuals

THEOREM 81.5.1 *(A-TP is RH-equivalent on the stated path). Conditional statement.* Assume the reductions of Chapters 70–80:

- P1 on the abelian Heisenberg sector, conditional on its Positselski inputs (Question 75.5.3);
- conditional KP transfer on the selected discrete-spectrum core (Theorem 76.3.3);
- HP (W) weight existence (Theorem 77.2.5);
- the positive-realization criterion for the full Weil form (Theorem 78.8.1), which asserts existence if and only if that form is positive and does not assume its positivity.

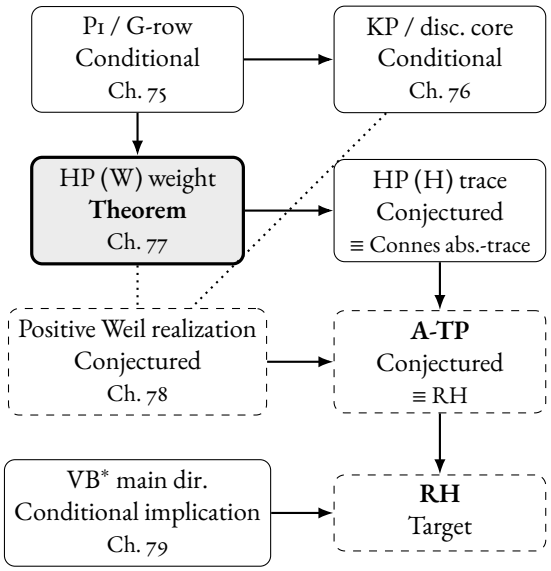


Figure 81.1: The sufficient path to RH and its categorical and spectral inputs. Shaded rounded boxes denote stated theorems. The dashed positivity boxes are RH-equivalent conjectures. Dotted lines mark pairing comparisons required to obtain a positive Weil realization; they assert no implication. Directed arrows have the hypotheses of the cited statements.

For $f \in \mathscr{A} = C_c^\infty(\mathbb{R})$, use $\widetilde{f} = f^*$ and $\widehat{f} = \mathcal{F}_+ f$. Then the Riemann Hypothesis is equivalent to the archimedean Tate positivity conjecture

$$W_\infty(f * \widetilde{f}) \;\le\; \widehat{f * \widetilde{f}}\left(\tfrac{1}{2i}\right) + \widehat{f * \widetilde{f}}\left(-\tfrac{1}{2i}\right) - W_{\mathrm{fin}}(f * \widetilde{f}),$$

for every $f \in \mathscr{A}$ (Conjecture 78.2.3).

Proof. Theorem 78.2.4 proves the biconditional A-TP $\Leftrightarrow$ Weil positivity on admissible Weil test functions. Weil positivity is equivalent to RH by the classical Weil 1952 equivalence Corollary 78.2.1. The remaining assertions of Chapters 70–80 place this RH-equivalent inequality on the stated sufficient path under their own hypotheses. They do not close the non-A-TP obstructions recorded later. □

Remark 81.5.2 *(Reduction to A-TP).* Theorem 81.5.1 is a reduction, not a proof of RH. Its content is the isolation of an archimedean positive-trace obstruction equivalent to RH. The arithmetic-chiral-algebra framework relocates the Hilbert–Pólya and Weil-positivity obstructions from their classical form as an positive-trace inequality on the compact convolution algebra to the full inequality comparing the signed gamma form with polar and prime terms, preserving the logical equivalence with RH. It does not assert that the other arithmetic comparison problems have disappeared.

Remark 81.5.3 *(relation to A-TP, Connes, and VB*).* The three RH-strength formulations are:

(i) A-TP: the full signed inequality on $\mathscr{A} = C_c^\infty(\mathbb{R})$ (Conjecture 78.2.3);

(ii) Connes absorption-trace condition: existence of a self-adjoint operator on the GNS space of the KMS_1 state whose normal zero-divisor operator has determinant divisor $\operatorname{div} \xi_{\mathbb{R}}$ and Weil–Connes trace equal to the zero distribution (Hypothesis 77.4.3);

(iii) the de Branges/Hermite–Biehler, Báez–Duarte, and determinant spectral-regularity conditions on $\xi_{\mathbb{R}}$ (Corollary 79.8.3, Theorem 57.4.22 under Hypothesis 57.4.21).

The full Weil criterion and the classical BD closure condition are criteria for zero placement on their complete test domains. The sharp BD rate is a stronger quantitative request, and the arithmetic operator formulations additionally require actual carriers, domains and comparison maps. The conditional VB*/(Reg)/(HB+)/(BD+) comparison uses all the data of Hypothesis 57.4.21; it does not identify bare RH with those unconstructed data.

81.6 Bibliographic Notes

The following references concern the constructions and comparisons in Chapters 70–80.

Primary Sources

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Ch. 71 (F1.a.3.A+B). Bernstein 1984; Bernstein–Deligne 1984; Bushnell–Henniart 2006; Paskunas 2013; Paskunas 2015; Colmez 2010; Auslander–Reiten–Smalø; Matsumura 1989; Ben-Zvi–Francis–Nadler 2008; Ben-Zvi–Nadler 2009; Lurie HA.

Ch. 72 (F1.a.3.C). Emerton–Gee 2020 and 2019; Dotto–Emerton–Gee 2026; Bhatt–Scholze 2022 (prisms); Gaitsgory–Rozenblyum DAG Vol. II; Arinkin–Gaitsgory 2012; Bushnell–Henniart 2006; Beilinson–Bernstein–Deligne 1982; Beilinson–Ginzburg 1996.

Ch. 73 ((Ch1), (AG.det), F2.c). Chenevier 2014; Bellaiche–Chenevier 2009 Astérisque 324; Bhatt–Scholze 2022; Bhatt–Lurie 2022; Arinkin–Gaitsgory 2015 (nilp SSupp); Galatius–Kupers–Randal-Williams 2024 (E_2-centre); Ben-Zvi–Francis–Nadler 2008; Lurie HA; Francis–Gaitsgory 2012; Emerton–Gee 2020.

Ch. 75 (Positselski four sub-items). Pontryagin 1934; Weil 1940 (LCA); Folland 1995; Positselski 2011/2018; Greenlees–May 1992; Matlis 1958; Bruns–Herzog 1993; Matsumura 1989; Lazard 1965; Venjakob 2003; Nekovář 2006; Bost–Connes 1995; Consani–Marcolli 2004/2006.

Ch. 76 (Full P3 Koszul–Petersson). Deligne 1974; Deligne–Serre 1974; Kim–Shahidi 2002; Kim–Sarnak 2003; Sarnak 1990/1995; Langlands 1966/1976; Moeglin–Waldspurger 1995; Shahidi 2010; Gelbart 1975; Iwaniec 2002; Takesaki 1970/1979; Bost–Connes 1995; Casselman–Shalika 1980; Bushnell–Henniart 2006; Godement–Jacquet 1972; Jacquet–Langlands 1970; JPSS 1983; Tate 1950; Weil 1967; Selberg 1956; Arthur 1978/1979; Piatetski-Shapiro 1983; Gelfand–Graev–Piatetski-Shapiro 1969.

Ch. 77 (H_{HP} construction). Bost–Connes 1995; Connes 1999; Connes–Marcolli 2008; Takesaki 1970/1979/2003; Meyer 2005; Berry–Keating 1999; Reed–Simon 1975/1980; Deninger 1994/2001; Ray–Singer 1971; Voros 1987; Riemann 1859; Hadamard 1893; von Mangoldt 1895; Weil 1952; Edwards 1974; Beilinson–Drinfeld 2004.

Ch. 78 (HP–Li restricted). Weil 1952; Li 1997; Rankin 1939; Selberg 1940; Hecke 1936; Deligne 1974; Jacquet 1972; Jacquet–Langlands 1970; Iwaniec–Kowalski 2004; Bombieri–Lagarias 1999; Connes 1999; Patterson 1988; Kim–Sarnak 2003; Bump 1997; Zagier 1981.

Ch. 79 (VB* main direction). de Branges 1968; de Branges 1968; Lagarias 1999; Conrey–Li 2000; Beurling–Malliavin 1962; Hamburger 1920; Carleman 1922/1926; Koosis 1992; Levin 1980; Akhiezer 1965; Simon 1998; Shohat–Tamarkin 1943; Báez-Duarte 2003; Burnol 2000.

Ch. 80 (R_{analytic}, Gaussians, Artin–Rees). Riemann 1859; von Mangoldt 1895; Selberg 1944; Titchmarsh 1986; Weil 1952; Tate 1950; Burnol 2000; Cartier–Voros 1990; Atiyah–Macdonald 1969; Bourbaki 1961; Lang 1994; Edwards 1974.

References by construction

For the bar–cobar construction, see Volume I and Beilinson–Drinfeld 2004; for the comodule–contramodule comparison, see Positselski 2009/2011. The C^*-algebra construction is in Bost–Connes 1995, § 1.2, and the GNS formalism is treated by Takesaki.

The topics of Chapter 75 correspond to Pontryagin duality (Pontryagin 1934), derived completion (Greenlees–May 1992), Matlis duality (Matlis 1958), and the Bost–Connes and Consani–Marcolli constructions (Bost–Connes 1995; Consani–Marcolli 2006). For the three comparisons in Chapter 73, see Chenevier 2014 and Bellaiche–Chenevier 2009 for (Ch1); Galatius–Kupers–Randal-Williams 2024, Ben-Zvi–Francis–Nadler 2008 and Arinkin–Gaitsgory 2015 for (AG.det); and Lurie HA and Francis–Gaitsgory 2012 for F2.c.

Bost–Connes 1995 also enters the categorical comparisons in Chapters 73 and 75, the tempered cuspidal KMS_1 comparison in Chapter 76, and the operator construction in Chapter 77. The corresponding structures involve the Koszul-dual description, the bar–cobar construction in Volume I, the comodule–contramodule comparison of Positselski, and the GNS formalism of Takesaki.

81.7 Implication Form

The reductions used in the sufficient path have the following formal content.

Implications

The following constructions and comparison requirements use the domains specified in Chapters 70–80:

- **Arithmetic Positselski.** The abelian-sector comparison requires the categorical hypotheses and compatible algebra, coefficient, completion and pairing maps of Question 75.5.2; the selected scalar character does not construct the equivalence.
- **G-row biconditional P1.** Relating central-class complementarity to the functional equation additionally requires the obstruction-class identification under that pairing (Question 75.5.3). These requirements remain in the abelian sector.
- **Koszul–Petersson compatibility on the discrete core.** The identity requires the arithmetic placement and pairing maps on every stated sector; its Maass domain also requires Ramanujan (Ch. 76).
- **F1.a.3.C wild-EG compact generation.** Compact generation is conditional on the stated wild-prime local-model and generation hypotheses, which remain open (Ch. 72).
- **(Ch1) two-prime restriction.** The formal ring map, closed image and fibres are constructed in Chapter 73. The geometric, trace and equivariant-descent comparisons retain all the hypotheses of Corollary 73.1.5.
- **F2.c restricted-tensor E_2 transport.** The transport theorem requires the central idempotent pointed-unit E_2 hypotheses (Ch. 73).

- **HP–Li on restricted classes.** The full Weil identity holds on every specified restricted class but asserts no positivity there (Theorem 78.2.4). A positive automorphic realization requires a proved restriction and a pairing identity (Corollary 78.5.6).
- **VB* main direction.** On the specified spectral data, the formal implication is (SH) + (Det) + (HB$^+$) + (EC$_\pi$) $\Rightarrow$ RH (Ch. 79).
- **Modular and occupation spectra.** On the full critical GNS space $\mathcal{H}_1$, the self-adjoint operator $N_{\mathrm{BC}} = -\frac{1}{2}\log\Delta_1$ has point spectrum $\frac{1}{2}\log\mathbb{Q}_{>0}$, each eigenvalue of countably infinite multiplicity, and closed spectrum $\mathbb{R}$ (Theorem 77.2.5). Its restriction $N_{\mathrm{occ}} = N_{\mathrm{BC}}|_{K_2}$ to the reducing subspace $K_2 = \overline{\mathrm{span}}\{\mu_{m^2}\Omega_1 : m \geq 1\}$ has spectrum $\{\log m : m \geq 1\}$, by the weighted occupation embedding of Lemma 77.3.2. Neither construction identifies a zeta-zero spectrum.
- **Riemann–von Mangoldt density.** The density estimate holds with the stated Titchmarsh bound (Ch. 80).
- **Weil at Gaussians.** The exact signed Gaussian formula holds in the stated normalization; it does not assert positivity of the full Weil expression (Ch. 80).
- **Block-additivity and formal comparisons.** The F1.a.3.A block-additivity and generic-block Ext^1 results hold on their stated block domains (Ch. 71). The R1.formal.b/c.2 comparisons require an actual mapping-colimit theorem and a construction of the compatible flat subcategory (Ch. 80).

Conjectural Conditions

The unresolved conjectural assertions are four:

(i) **A-TP** (archimedean Tate positivity): strictly equivalent to RH.

(ii) **Connes absorption trace** (twist existence): strictly equivalent to RH.

(iii) **HB$^+$/EC$_\pi$/(Det) on $\xi_{\mathbb{R}}$**: HB$^+$ is RH-equivalent by Lagarias 1999; EC$_\pi$ and (Det) are spectral-regularity inputs.

(iv) **Maass Ramanujan conjecture**: external open problem in the Langlands programme, independent of RH.

The adjacent spectral assertions are:

(v) **(AG.det.transition) higher-centre gluing** on an actual categorical descent diagram, together with the separate arithmetic centre comparison. The local-model morphism and its relative dimensions must be constructed before asserting a codimension bound.

(vi) **(BCglue.local)** $\ell = p$ local transition theorem for coherent Springer sheaves.

(vii) **(BCglue.glue)** prismatic-flat cross-prime coherent sheaf descent.

The Paskunas/F1.a local categorical conditions are (R.A, R.B.1, R.B.2, F1.a.1 non-ordinary, F1.a.2 class-match, R1.formal.c.1/c.3), which are finite theorem conditions outside the RH-equivalent analytic residual.

81.8 Domain of Chapters 70–80

Chapters 70–80 operate within the following domain boundaries.

(i) **A-TP and RH.** Theorem 78.2.4 establishes the strict equivalence A-TP $\Leftrightarrow$ RH (Conjecture 78.2.3). The arithmetic inequality retains the full compact convolution test algebra, the polar term, and the prime term. Its form is $W_\infty(f * f^*) \leq P(f * f^*) - W_{\mathrm{fin}}(f * f^*)$, which preserves the logical equivalence to RH.

(ii) $\mathrm{L}_{\mathrm{ACD}}$ **on the cuspidal sector.** Chapter 70 closes the cuspidal sector. The Eisenstein extension requires Langlands–Shahidi intertwining theory in coherent-categorical form, or a Moœglin–Waldspurger residual-spectrum categorification; these are open conditions outside the domain of Chapter 70.

(iii) **(BCglue.local) and (BCglue.glue).** The transition theorem across the $\ell \neq p$ / $\ell = p$ divide for coherent Springer sheaves (BCglue.local) is not in the literature. The prismatic-flat cross-prime descent lifted to the coherent-sheaf level (BCglue.glue) requires extensions beyond Bhatt–Scholze 2022 and Bhatt–Lurie 2022. Both are primary-source theorem conditions.

(iv) **Maass Ramanujan.** Chapter 76 uses the Kim–Sarnak bound $\theta \leq 7/64$ only as a local exponent estimate. It gives no Dirichlet-summable Euler error on the Maass sector. The arithmetic pairing maps are still required. Maass temperedness additionally depends on the Ramanujan conjecture, an external open problem in the Langlands programme.

(v) **Hilbert–Pólya operator with zeta-zero determinant divisor.** The full GNS generator N_{BC} of Theorem 77.2.5 has point spectrum $\frac{1}{2}\log \mathbb{Q}_{>0}$ and closed spectrum $\mathbb{R}$. Its reducing occupation restriction N_{occ} on K_2 has spectrum $\{\log m : m \geq 1\}$, as in Lemma 77.3.2. Neither construction supplies the zeta-zero determinant divisor. The twist to an operator with determinant divisor $\operatorname{div} \xi_{\mathbb{R}}$ and Weil–Connes zero trace is the Connes absorption-trace conjecture, equivalent to the Hilbert–Pólya/RH obstruction.

(vi) HB^{+}, EC_{π}, **(Det) on** $\xi_{\mathbb{R}}$**.** Chapter 79 proves the implication (SH) + (Det) + (HB^{+}) + $\mathrm{EC}_{\pi} \Rightarrow$ RH. The proof of each of the four hypotheses on $\xi_{\mathbb{R}}$ lies outside the domain of Chapter 79.

(vii) **Remaining primary-source conditions.** The non-RH categorical hypotheses are (AG.det.transition), (BCglue.local), (BCglue.glue), the Eisenstein extension of $\mathrm{L}_{\mathrm{ACD}}$, the Paskunas narrow residuals, the non-ordinary F1.a.1/F1.a.2 extensions, and the R1.formal extensions. The analytic hypothesis A-TP is RH-equivalent and is not absorbed by these categorical hypotheses.

Remark 81.8.1 *(Structure of the obstruction set).* Chapters 70–80 separate structural comparison data from analytic hardness. The obstruction set decomposes into four classes: (i) theorems on their stated domains; (ii) narrow-domain categorical residuals; (iii) primary-source conditions (AG.det.transition, BCglue.local, BCglue.glue, Eisenstein extension); (iv) the RH-equivalent residual A-TP. Other comparison residuals remain visible; they are not implied by A-TP.

81.9 Further Obstructions

The remaining obstructions are as follows.

Direct Positivity for A-TP

A-TP compares the signed gamma contribution with both arithmetic terms. To retain its normalization, take $f \in \mathscr{A} = C_c^{\infty}(\mathbb{R})$, $f^{*}(t) = \overline{f(-t)}$, and $h_f(u) = \mathcal{F}_{+}f(u) = \int_{\mathbb{R}} f(t)e^{iut}\,dt$. The signed density and the auxiliary digamma density are respectively

$$k(u) = \log \pi - \Re\psi_{\Gamma}(1/4 + iu/2), \qquad g(u) = \Re\psi_{\Gamma}(1/4 + iu/2) + \log \pi.$$

Define the auxiliary form

$$G(f) = \frac{1}{2\pi}\int_{\mathbb{R}} g(u)|h_f(u)|^2\,du.$$

The plus-sign density $g/(2\pi)$ is distinct from the signed archimedean density $k/(2\pi)$. Both integrals converge because the densities have logarithmic growth and h_f decreases rapidly.

PROPOSITION 81.9.1 *(the full inequality in the auxiliary normalization).* Put

$$a(f) = \int_{\mathbb{R}} f(t)e^{-t/2}\,dt, \qquad b(f) = \int_{\mathbb{R}} f(t)e^{t/2}\,dt,$$

and, with Λ the von Mangoldt function, put

$$Q_{\mathrm{fin}}(f) = \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\big((f * f^*)(\log n) + (f * f^*)(-\log n)\big).$$

Then

$$W_\infty(f * f^*) = 2\log\pi\,\|f\|_2^2 - G(f),$$
$$W(f * f^*) = 2\mathfrak{R}\big(a(f)\overline{b(f)}\big) - Q_{\mathrm{fin}}(f) - 2\log\pi\,\|f\|_2^2 + G(f).$$

Consequently A-TP, and hence RH, is equivalent to

$$G(f) + 2\mathfrak{R}\big(a(f)\overline{b(f)}\big) \geq Q_{\mathrm{fin}}(f) + 2\log\pi\,\|f\|_2^2 \qquad (f \in \mathscr{A}).$$

Proof. The identity $k + g = 2\log\pi$ and Plancherel in the specified Fourier convention give the first equality. The convolution transform is $h_f(z)\overline{h_f(\bar z)}$. Its evaluations at $z = \pm i/2$ sum to $a(f)\overline{b(f)} + b(f)\overline{a(f)}$. The prime sum is finite because $f * f^*$ is compactly supported. Substitute these evaluations into $W = P - W_\infty - W_{\mathrm{fin}}$ to obtain the second equality. Nonnegativity of its right side for every $f \in \mathscr{A}$ is exactly the displayed inequality. Weil's criterion in Corollary 78.2.1 identifies this condition with RH. □

The Rankin–Selberg kernel of Theorem 78.5.1 and the density $|\Gamma(1/4 + iu/2)|^2$ remain possible comparison objects. A comparison proving A-TP must establish this full inequality, including its polar, prime, and norm terms. Positivity of an isolated digamma or gamma-absolute-value form supplies no such identity.

Langlands–Shahidi Extension of $\mathrm{L}_{\mathrm{ACD}}$

The Eisenstein/continuous-spectrum extension of $\mathrm{L}_{\mathrm{ACD}}$ can be attempted via Shahidi 2010 (AMS Colloquium 58) normalisation theory. The required object is a categorical lift of the intertwining operators $M(s)$ of Moeglin–Waldspurger 1995 §IV, realising $\mathrm{L}_{\mathrm{ACD}}$ on the continuous spectrum compatible with the Weil-positive extension. The alternative domain-restricted comparison of Ch. 76 supplies a conditional discrete pairing. Its Li use requires both the stated H-P^{Ar} realization and the explicit trace-placement map; the Langlands–Shahidi extension reduces the full $\mathcal{V}^{\mathrm{prim}}$ identity to the same measure identification.

Paskunas Wild-Prime Class Match

Residuals (R.A), (R.B.1), (R.B.2) of Ch. 71 remain open. (R.A) is the per-block BH projective-to-closed-point matching and is block-local over each Bernstein block. (R.B.1) is the very-special Breuil–Mezard multiplicity. (R.B.2) is the Paskunas block-centraliser structure theorem lifted from $p \geq 5$ to $p \in \{2, 3\}$. None is RH-equivalent; each is a finite explicit computation.

(AG.det.transition) Direct Obstruction

The proposed (AG.det.transition) comparison requires an actual arithmetic singular-support space, its categorical descent diagram, and compatible morphisms of higher centres. Proposition 73.2.9 supplies an internal centre identity only after those data and the stated preservation of the diagram are assumed. The automorphic centre comparison requires its own morphism and equivalence proof.

A codimension bound additionally requires a local-model morphism and its relative dimension calculation. Proposition 88.2.1 computes the regular nilpotent orbit in $\mathfrak{sl}_2$: it has dimension two and codimension one in the three-dimensional Lie algebra. That calculation does not establish the dimension of an arithmetic Selmer transition. The singular-support, cellular and geometric centre formalisms must be connected by actual arithmetic comparison maps before they can prove the intended global result.

Connes Absorption-Trace Realisation

The Connes absorption-trace conjecture (Hypothesis 77.4.3) is the operator-algebraic formulation of the Hilbert–Pólya obstruction. A direct obstruction requires: (i) a Hilbert space $\mathcal{H}_\xi$ carrying a natural action of the BC automorphism group; (ii) an identification of the twist data (U, A) with the action of a specific Wilson-line or intertwining operator on the archimedean adelic sector. Consani–Marcolli 2006 supply the adele-class space framework; the realisation of the zero-divisor operator as a twisted N_{BC} on a specific Hilbert-space completion is the missing content.

Hermite–Biehler Positivity on $\xi_{\mathbb{R}}$

Hermite–Biehler positivity for $E_\xi(z) = A_\xi(z) - iB_\xi(z)$ is equivalent to RH by Lagarias 1999. An obstruction requires either: (i) a direct proof that the real part A_ξ has only real zeros and $\mathrm{div}(A_\xi) = \mathrm{div}(\Xi)$ (de Branges 1968 Thm. 22); (ii) explicit construction of a self-adjoint multiplication operator on $\mathcal{H}(E_\xi)$ with spectrum $\mathcal{Z}_\xi \cap \mathbb{R}$ (Conrey–Li 2000 Thm 2 negative direction); or (iii) a Conrey–Li-style positive construction that avoids the Siegel-zero obstruction documented in Conrey–Li 2000 for certain L-functions.

81.10 Boundary

(i) The sufficient path in Chapter 44 requires arithmetic Positselski comparison, Koszul–Petersson compatibility, a Hilbert–Pólya determinant and trace realization, and HP–Li positivity. The conditions of Chapter 69 retain their separate categorical and spectral hypotheses.

(ii) The arithmetic Positselski equivalence on the abelian sector requires the categorical hypotheses and actual comparison maps of Question 75.5.2; the selected scalar character does not construct them.

(iii) The Koszul–Petersson transfer on the selected discrete carrier requires the arithmetic placement, local pairing maps and adelic extension (Theorem 76.3.3).

(iv) The Hilbert–Pólya operator skeleton is constructed via the Bost–Connes log-modular operator N_{BC} (Theorem 77.2.5).

(v) A positive Hilbert realization of the full Weil form exists exactly when that form is positive (Theorem 78.8.1). The coefficient multiplier Υ_F supplies no such pairing identity.

(vi) RH is equivalent to the archimedean Tate positivity conjecture A-TP (Conjecture 78.2.3), on the stated path (Theorem 81.5.1).

(vii) The reductions leave no additional numerical identity hidden in the comparison chain; the remaining content is analytic positivity or categorical gluing.

(viii) A-TP is the full inequality comparing the signed gamma form with the polar and prime terms.

(ix) The remaining obstructions are: direct positivity for A-TP; Langlands–Shahidi extension of L_{ACD}; Paskunas wild-prime class match; (AG.det.transition) sheaf-gluing; Connes absorption-trace realisation; and the proof of $HB^{+}/EC_{\pi}/(\mathrm{Det})$ on $\xi_{\mathbb{R}}$.

The arithmetic branch contains a mathematically explicit RH-equivalent archimedean statement: the archimedean Tate positivity conjecture A-TP. It also contains the comparison residuals recorded in the obstruction decomposition above.

Chapter 82

The Archimedean Tate Positivity Conjecture: A Direct Elementary Obstruction

The digamma kernel $g(t) = \operatorname{Re}\psi(1/4 + it/2) + \log\pi$ has one positive zero. Splitting its quadratic integral at that zero gives two non-negative energies $A[f]$ and $B[f]$, and the exact identity $G(f) = (B[f] - A[f])/(2\pi)$. An explicit high-frequency dominance condition proves $G(f) \geq 0$ on a non-empty class of tests.

This auxiliary form differs from the signed archimedean term in the Weil formula. With the Fourier convention of Chapter 78, that term is $W_\infty(f * \widetilde{f}) = 2\log\pi\|f\|_2^2 - G(f)$. Consequently full Weil positivity requires $A[f]/(2\pi) + 2\log\pi\|f\|_2^2 \leq B[f]/(2\pi) + R[f]$, where $R[f]$ retains the polar and finite-prime terms. The kernel calculation proves neither this full inequality nor its arithmetic comparison with a Koszul pairing.

82.1 Kernel and Obstruction

Chapter 78 established the following. The fixed-F Mellin–Rankin–Selberg comparison does not close the zeta finite-prime term (Theorem 78.5.5). The polar terms $\widehat{f * \widetilde{f}}(\pm 1/(2i))$ are explicit finite Taylor corrections (Proposition 78.2.2). The irreducible residual is A-TP (Conjecture 78.2.3), which Theorem 78.2.4 establishes is strictly equivalent to the Weil positivity form of RH.

Definition 82.1.1 *(the archimedean digamma kernel).* Set

$$g(t) \;:=\; \operatorname{Re}\psi\Big(\tfrac{1}{4} + \tfrac{it}{2}\Big) + \log\pi, \qquad t \in \mathbb{R}, \tag{82.1}$$

where $\psi(z) = \Gamma'(z)/\Gamma(z)$ is the digamma function. The auxiliary quadratic form, with $\widehat{f}(t) = \int_{\mathbb{R}} f(u)e^{itu}\,du$, is

$$G(f) \;=\; \frac{1}{2\pi}\int_{-\infty}^{\infty} |\widehat{f}(t)|^2\, g(t)\,dt. \tag{82.2}$$

The signed Weil contribution in (78.1) instead has density $k(t) = \log\pi - \operatorname{Re}\psi(1/4 + it/2)$. Since $k + g = 2\log\pi$, Plancherel gives

$$W_\infty(f * \widetilde{f}) = 2\log\pi\|f\|_2^2 - G(f).$$

We analyse the sign of (82.2) and separate the positivity forced by the kernel g from the positivity equivalent to RH.

Domain

Work on $\mathscr{A} = C_c^\infty(\mathbb{R}, \mathbb{C})$, with the Fourier transform fixed above and convolution involution $\widetilde{f}(u) = \overline{f(-u)}$. Its Fourier transforms are entire and rapidly decreasing on the real axis, so all kernel integrals

below converge. If an arithmetic application imposes an additional test class $\mathrm{PW}_F^{\mathrm{hol}}$, use its intersection with the classes constructed here. Non-emptiness of that further intersection requires a separate check.

The elementary objects are:

- elementary analysis of $g(t)$ via the Hurwitz series for ψ and the Gauss summation at rational arguments;
- decomposition of the auxiliary form $G(f)$ into a low-frequency core integral and a high-frequency tail integral;
- unconditional non-negativity of G on an explicit dominance sub-class;
- reduction of full A-TP to a bounded-interval archimedean inequality coupled to $R[f]$.

No elementary argument below supplies the RH-equivalent full A-TP inequality. The irreducible location is §82.5.

Analytic Ingredients

The elementary obstruction uses four classical sources.

Hurwitz 1882. Hurwitz, A. *Einige Eigenschaften der Dirichlet'schen Functionen. Zeitschrift für Mathematik und Physik* 27 (1882). Hurwitz series representation $\psi(a) = -\gamma_E + \sum_{n\geq 0}[1/(n+1) - 1/(n+a)]$; monotonicity of individual terms.

Gauss 1813. Gauss, C.F. *Disquisitiones generales circa seriem infinitam* $1 + \cdots$. *Comment. Soc. Reg. Sci. Gottingensis Recentiores* 2 (1813). Reflection relation $\psi(1-x) - \psi(x) = \pi\cot(\pi x)$ at rational x; explicit values $\psi(1/4) = -\gamma_E - \pi/2 - 3\log 2$, $\psi(3/4) = -\gamma_E + \pi/2 - 3\log 2$.

Whittaker–Watson 1927. Whittaker, E.T.; Watson, G.N. *A Course of Modern Analysis* (4th ed., Cambridge 1927), Ch. XII. Stirling asymptotic for ψ: $\psi(z) = \log z - 1/(2z) - \sum_{k=1}^{N} B_{2k}/(2kz^{2k}) + O(|z|^{-2N-2})$.

Boas 1954. Boas, R.P. *Entire Functions*. Academic Press 1954, Ch. 6. Paley–Wiener characterization of entire functions of exponential type; the Plancherel–Pólya theorem giving $\int_{\mathbb{R}} |\widehat{f}(t)|^2\,dt = 2\pi\int_{\mathbb{R}} |f(u)|^2\,du$ for $f \in \mathrm{PW}$.

The kernel estimates use the Hurwitz series, the rational digamma value, Stirling asymptotics and Plancherel. Their relation to the full signed Weil form is the explicit identity stated above.

82.2 The Digamma Kernel $g(t)$

The sign structure of g controls whether $G(f)$ can be positive, negative, or zero. All statements below are unconditional.

Hurwitz Series Representation

Lemma 82.2.1 *(Hurwitz series for g).* For every $t \in \mathbb{R}$,

$$g(t) \;=\; -\gamma_E + \log\pi + \sum_{n=0}^{\infty}\left[\frac{1}{n+1} - \frac{n+1/4}{(n+1/4)^2 + t^2/4}\right]. \tag{82.3}$$

The series converges absolutely and uniformly on compact sets.

Proof. Hurwitz's series $\psi(a) = -\gamma_E + \sum_{n\ge 0}[1/(n+1) - 1/(n+a)]$ (Hurwitz 1882; Whittaker–Watson 1927 §12.31) applied at $a = 1/4 + it/2$:

$$\frac{1}{n + 1/4 + it/2} = \frac{(n+1/4) - it/2}{(n+1/4)^2 + t^2/4}.$$

Real part: $(n+1/4)/[(n+1/4)^2 + t^2/4]$. Adding $\log\pi$ gives (82.3). Absolute convergence follows from the large-n asymptotic $1/(n+1) - 1/(n+1/4) = -(3/4)/[(n+1)(n+1/4)] = O(n^{-2})$. □

Lemma 82.2.2 *(strict monotonicity of g in $|t|$).* The map $t \mapsto g(t)$ is even and continuously differentiable on $\mathbb{R}$, with

$$g'(t) \;=\; \sum_{n=0}^{\infty} \frac{(n+1/4)\,t/2}{\left[(n+1/4)^2 + t^2/4\right]^2}. \tag{82.4}$$

In particular $g'(t) > 0$ for $t > 0$, so g is strictly increasing in $|t|$ on $[0,\infty)$.

Proof. Evenness follows from (82.3): each term depends on t only through t^2. For each $n \ge 0$, with $c := n + 1/4$,

$$\frac{d}{dt}\left[-\frac{c}{c^2 + t^2/4}\right] \;=\; -c\cdot\frac{-(t/2)}{[c^2+t^2/4]^2} \;=\; \frac{c\cdot t/2}{[c^2+t^2/4]^2}.$$

Summing over n and using uniform convergence on compact sets of the differentiated series (dominant bound $\le (n+1/4)\cdot|t|/2\cdot(n+1/4)^{-4} = O(n^{-3})$ uniformly in $|t|\le T$) gives (82.4). Each summand is non-negative for $t \ge 0$ and strictly positive for $t > 0$; hence $g'(t) > 0$ for $t > 0$, giving strict monotonicity in $|t|$ on $[0,\infty)$. Continuity of g' follows from uniform convergence on compacts. □

Proposition 82.2.3 *(boundary values of g).* The kernel g satisfies

$$g(0) \;=\; -\gamma_E - \frac{\pi}{2} - 3\log 2 + \log\pi \;\approx\; -3.0827, \tag{82.5}$$

$$g(t) \;\sim\; \tfrac{1}{2}\log(t^2/4) + \log\pi \;=\; \log(|t|\pi/2), \qquad |t|\to\infty. \tag{82.6}$$

In particular $g(0) < 0 < \lim_{|t|\to\infty} g(t)/\log|t| = 1$.

Proof. (82.5) follows from the Gauss value $\psi(1/4) = -\gamma_E - \pi/2 - 3\log 2$ (Gauss 1813; Whittaker–Watson §12.31 ex. 1). Numerical evaluation: $-0.5772 - 1.5708 - 2.0794 + 1.1447 = -3.0827$.

(82.6) follows from the Stirling asymptotic $\psi(z) = \log z - 1/(2z) + O(|z|^{-2})$ applied at $z = 1/4 + it/2$: $\log z = \tfrac{1}{2}\log(1/16 + t^2/4) + i\arctan(2t)$, $\operatorname{Re}\log z \sim \tfrac{1}{2}\log(t^2/4)$ as $|t|\to\infty$, and $-1/(2z)\to 0$. □

The Unique Sign Change t_*

Theorem 82.2.4 *(existence and uniqueness of the sign change).* There is a unique $t_* \in (0,\infty)$ with $g(t_*) = 0$. The map $t \mapsto g(t)$ is strictly negative on $(-t_*, t_*)$ and strictly positive on $(-\infty, -t_*)\cup(t_*,\infty)$.

Proof. By Lemma 82.2.2, g is continuous, even, and strictly increasing in $|t|$ on $[0,\infty)$. Proposition 82.2.3 gives $g(0) = -3.0827 < 0$ and $g(t)\to+\infty$. The intermediate value theorem and strict monotonicity supply a unique $t_* > 0$ with $g(t_*) = 0$; by evenness $g(-t_*) = 0$ as well. □

Proposition 82.2.5 *(certified interval for the sign change).* The zero satisfies $0.84 < t_* < 0.86$.

Proof. Subtract the Hurwitz series at zero. With $a = t^2/4$ and $c_n = n + 1/4$, the positive-tail representation is

$$g(t) = g(0) + \sum_{n\geq 0} \frac{a}{c_n(c_n^2 + a)}.$$

For $S_N(t) = g(0) + \sum_{n=0}^{N} a/[c_n(c_n^2 + a)]$, monotonicity of x^{-3} gives

$$0 \leq g(t) - S_N(t) \leq a \sum_{n>N} (n + 1/4)^{-3} \leq \frac{a}{2(N + 1/4)^2}.$$

At $N = 20$, rational arithmetic with $0.577 < \gamma_E < 0.578$, $3.1415 < \pi < 3.1416$, $0.6931 < \log 2 < 0.6932$ and $1.1447 < \log \pi < 1.1448$ gives $-0.024 < S_{20}(0.84) < -0.020$ and $0.017 < S_{20}(0.86) < 0.020$. The positive tails are less than 0.00023 at both endpoints. Thus the signs are opposite, and strict monotonicity places the zero between them. □

Remark 82.2.6 *(numerical refinement).* Bisection with the same positive-tail bound can refine the certified interval. The value $t_* \approx 0.85066$ is a numerical approximation; all exact inequalities below use the zero defined by $g(t_*) = 0$, or certified upper and lower bounds for it.

82.3 Decomposition of the Auxiliary Form

Definition 82.3.1 *(low-frequency core and high-frequency tail).* For $f \in \mathscr{A}$, define the non-negative quantities

$$A[f] := \int_{|t|<t_*} |\widehat{f}(t)|^2 \cdot (-g(t))\, dt,$$
$$B[f] := \int_{|t|>t_*} |\widehat{f}(t)|^2 \cdot g(t)\, dt.$$

By Theorem 82.2.4, both $A[f] \geq 0$ and $B[f] \geq 0$ with strict inequality whenever $\widehat{f}$ does not vanish identically on the respective domain.

Lemma 82.3.2 *(archimedean decomposition).* For every $f \in \mathscr{A}$,

$$G(f) = \frac{1}{2\pi}\big(B[f] - A[f]\big). \tag{82.7}$$

In particular $G(f) \geq 0$ if and only if $B[f] \geq A[f]$.

Proof. Definition 82.1.1 gives $G(f) = (2\pi)^{-1} \int_{\mathbb{R}} |\widehat{f}(t)|^2 g(t)\, dt$. Split the integral at $|t| = t_*$:

$$\int_{\mathbb{R}} |\widehat{f}(t)|^2 g(t)\, dt = \int_{|t|<t_*} |\widehat{f}(t)|^2 g(t)\, dt + \int_{|t|>t_*} |\widehat{f}(t)|^2 g(t)\, dt.$$

By Theorem 82.2.4, $g(t) < 0$ on $|t| < t_*$ and $g(t) > 0$ on $|t| > t_*$. The first integral equals $-A[f]$ and the second equals $B[f]$, giving (82.7). □

Remark 82.3.3 *(the exact split).* The identity $G(f) = (B[f] - A[f])/(2\pi)$ implies $G(f) \geq 0$ exactly when $B[f] \geq A[f]$. It imposes no restriction on finer subdivisions or stronger estimates. Full Weil positivity requires the additional signed inequality of Theorem 82.5.1.

82.4 Auxiliary Positivity on the High-Frequency Dominance Class

The condition $B[f] \geq A[f]$ holds unconditionally whenever the spectral mass of $\widehat{f}$ above $|t| > t_*$ dominates the core supremum by an explicit factor. This dominance sub-class is non-empty and admits a closed-form sufficient condition.

The Sub-Class $\mathscr{A}_{\mathrm{HFD}}$

Definition 82.4.1 *(high-frequency dominance class).* Define the *high-frequency dominance sub-class* $\mathscr{A}_{\mathrm{HFD}} \subset \mathscr{A}$ as the subset of $f \in \mathscr{A}$ satisfying

$$\sup_{|t|\leq t_*} |\widehat{f}(t)|^2 \cdot 2t_* \cdot \|-g\|_{L^\infty[-t_*,t_*]} \;\leq\; \int_{|t|>t_*} |\widehat{f}(t)|^2\, g(t)\, dt. \tag{82.8}$$

The left-hand side is a crude upper bound for $A[f]$; the right-hand side is $B[f]$.

Lemma 82.4.2 *(the Plancherel–Pólya bound inside the core).* For $f \in \mathscr{A}$ supported in $[-T, T]$,

$$\sup_{|t|\leq t_*} |\widehat{f}(t)|^2 \;\leq\; 2T \int_{-T}^{T} |f(u)|^2\, du. \tag{82.9}$$

Proof. $|\widehat{f}(t)| = |\int_{-T}^{T} f(u) e^{itu}\, du| \leq (2T)^{1/2} \|f\|_{L^2}$ by Cauchy–Schwarz; squaring gives (82.9). This is the Plancherel–Pólya dimension bound (Boas 1954 §6.8) applied inside the core. □

Theorem 82.4.3 *(unconditional G-positivity on $\mathscr{A}_{\mathrm{HFD}}$).* For every $f \in \mathscr{A}_{\mathrm{HFD}}$,

$$G(f) \;\geq\; 0.$$

Full A-TP follows on the sublocus where, in addition, $R[f] + G(f) \geq 2 \log \pi \|f\|_2^2$ for $R[f]$ as in Equation 82.5.1.

Proof. Condition (82.8) combined with the obvious bound $A[f] \leq \sup_{|t|\leq t_*} |\widehat{f}(t)|^2 \cdot 2t_* \cdot \|-g\|_{L^\infty[-t_*,t_*]}$ and the identity $B[f] = \int_{|t|>t_*} |\widehat{f}(t)|^2 g(t)\, dt$ gives $A[f] \leq B[f]$. Lemma 82.3.2 converts to $G(f) \geq 0$. The last sentence is Theorem 82.5.1 on the sublocus $R[f] + G(f) \geq 2 \log \pi \|f\|_2^2$. □

Remark 82.4.4 *(smooth high-frequency tests).* Choose a nonzero real even $\chi \in C_c^\infty(\mathbb{R})$ with $\int \chi \neq 0$, and set $f_M(u) = \chi(u) \cos(Mu)$. Then

$$\widehat{f_M}(t) = \tfrac{1}{2}\big(\widehat{\chi}(t + M) + \widehat{\chi}(t - M)\big).$$

Schwartz decay gives $\sup_{|t|\leq t_*} |\widehat{f_M}(t)| = O(M^{-K})$ for every $K > 0$. Since $\widehat{\chi}(0) \neq 0$, choose $\delta > 0$ on which its modulus is bounded below. On $|t - M| < \delta$, the other translate tends uniformly to zero, while $g(t) \sim \log M$. Hence $B[f_M] \geq c \log M$ for some $c > 0$ and sufficiently large M. This proves the quantitative HFD condition for nonzero smooth tests with fixed support.

Remark 82.4.5 *(tests outside the dominance class).* With the same χ, put $f_L(u) = L^{-1/2} \chi(u/L)$, $L \geq 1$. Then $\widehat{f_L}(t) = \sqrt{L}\, \widehat{\chi}(Lt)$, so

$$G(f_L) = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{\chi}(s)|^2 g(s/L)\, ds \longrightarrow g(0) \|\chi\|_2^2 < 0.$$

Dominated convergence applies because g has logarithmic growth and $\widehat{\chi}$ is Schwartz. Thus $f_L \notin \mathscr{A}_{\mathrm{HFD}}$ for large L, proving that the dominance class is proper in $\mathscr{A}$. This construction widens spatial support and concentrates Fourier mass near zero.

An Explicit Sufficient Condition

The definitional condition (82.8) depends on $\|-g\|_{L^\infty[-t_*,t_*]}$, which equals $|g(0)| = \gamma_E + \pi/2 + 3\log 2 - \log\pi \approx 3.0827$ by Proposition 82.2.3 and strict monotonicity. We reformulate the sufficient condition without the supremum.

Corollary 82.4.6 *(explicit sufficient condition).* Suppose $f \in \mathscr{A}$ satisfies

$$2t_* \cdot |g(0)| \cdot \|\widehat{f}\|^2_{L^\infty[-t_*,t_*]} \;\le\; \int_{|t|>t_*} |\widehat{f}(t)|^2\, g(t)\, dt. \tag{82.10}$$

Then $G(f) \ge 0$. For illustration only, with $t_* \approx 0.8507$ and $|g(0)| \approx 3.083$, the prefactor $2t_*|g(0)| \approx 5.245$, the rounded display is

$$\|\widehat{f}\|^2_{L^\infty[-0.8507,0.8507]} \;\le\; \frac{1}{5.245}\int_{|t|>0.8507} |\widehat{f}(t)|^2\, g(t)\, dt.$$

This rounded display is not a certified sufficient condition; use (82.10) with the exact constants.

Proof. $A[f] \le 2t_* \cdot |g(0)| \cdot \|\widehat{f}\|^2_{L^\infty[-t_*,t_*]}$ by $|g(t)| \le |g(0)|$ on $|t| \le t_*$. Combine with Lemma 82.3.2. □

Remark 82.4.7 *(interpretation).* The sufficient condition (82.10) asserts: the peak core L^∞-norm of $\widehat{f}$, scaled by a fixed constant depending only on t_* and $g(0)$, is dominated by the high-frequency tail L^2-mass weighted by g. Every $f \in \mathscr{A}$ satisfying the displayed quantitative dominance condition lies in $\mathscr{A}_{\mathrm{HFD}}$, and Theorem 82.4.3 gives $G(f) \ge 0$.

82.5 The Full Signed Weil Inequality

Core, Tail and Arithmetic Terms

Theorem 82.5.1 *(signed core-tail reduction).* On the test domain of the Weil formula, put

$$R[f] = \widehat{f * \widetilde{f}}(1/(2i)) + \widehat{f * \widetilde{f}}(-1/(2i)) - W_{\mathrm{fin}}(f * \widetilde{f}).$$

Then $W(f * \widetilde{f}) \ge 0$ is equivalent to

$$\frac{A[f]}{2\pi} + 2\log\pi\|f\|_2^2 \le \frac{B[f]}{2\pi} + R[f]. \tag{$\star$}$$

Thus A-TP on a specified full test class is the assertion that ($\star$) holds for every test in that class. Any equivalence with RH uses the full test class in Theorem 78.2.4; restriction to one smaller class is not automatically an equivalent criterion.

Proof. All integrals are defined on the stated test domain: the digamma densities grow logarithmically and the Fourier transforms decrease rapidly. The signed Weil formula and Plancherel give

$$\begin{aligned} W(f * \widetilde{f}) &= R[f] - W_\infty(f * \widetilde{f}) \\ &= R[f] - 2\log\pi\|f\|_2^2 + G(f) \\ &= R[f] - 2\log\pi\|f\|_2^2 + \frac{B[f] - A[f]}{2\pi}. \end{aligned} \tag{82.11}$$

Rearranging this equality proves both directions of ($\star$). No positive zero-square sum is assumed in this calculation. □

Remark 82.5.2 *(the zero-side functional).* The Weil functional is evaluated on the actual zero divisor $z_\rho = -i(\rho - 1/2)$. Replacing it by $\sum_\rho |\widehat{f}(\gamma_\rho)|^2$ uses the critical-line specialization $z_\rho = \gamma_\rho \in \mathbb{R}$. The unconditional identity used here is (82.11).

Remark 82.5.3 *(the remaining inequality).* The negative part of the auxiliary kernel is supported on $[-t_*, t_*]$, but the signed Weil inequality also includes the full norm term $2\log\pi\|f\|_2^2$, the polar terms and the finite-prime functional. Positivity of $G(f)$ alone does not control their sum. On the high-frequency dominance class it gives $W(f * \widetilde{f}) \geq R[f] - 2\log\pi\|f\|_2^2$; a sufficient additional condition is $R[f] \geq 2\log\pi\|f\|_2^2$.

Three Elementary Estimates and Their Limits

Proposition 82.5.4 *(projection to compact Fourier support).* If a Paley–Wiener Fourier transform vanishes for every real $t > t_*$, it is identically zero. Thus exact projection to core-supported Fourier transforms leaves only the zero test in this entire-function class.

Proof. The Fourier transform is entire. Its zeros on the open real interval (t_*, ∞) have accumulation points in its domain, so the identity theorem applies. □

Proposition 82.5.5 *(a majorant gives a sufficient bound).* For any non-negative measurable $h \geq -g$ on $[-t_*, t_*]$, put $A_h[f] = \int_{-t_*}^{t_*} |\widehat{f}(t)|^2 h(t)\,dt$, assuming this integral is finite. Then $A[f] \leq A_h[f]$. Consequently

$$\frac{A_h[f]}{2\pi} + 2\log\pi\|f\|_2^2 \leq \frac{B[f]}{2\pi} + R[f]$$

is sufficient for (⋆). It is not asserted to be necessary.

Proof. Integrate the pointwise inequality against $|\widehat{f}|^2 \geq 0$. Replacing the left-hand side by a larger quantity proves the stated implication. For the constant majorant $h = |g(0)|$, this also gives $A[f] \leq 2t_*|g(0)| \sup_{|t|\leq t_*} |\widehat{f}(t)|^2$, the bound used in the high-frequency dominance condition. □

Proposition 82.5.6 *(support bound).* If f is supported in $[-T', T']$, then $\sup_t |\widehat{f}(t)|^2 \leq 2T'\|f\|_2^2$. In particular $A[f] \leq 4t_*T'|g(0)|\|f\|_2^2$. These estimates alone give no bound on $R[f]$.

Proof. Cauchy–Schwarz gives the Fourier supremum bound, and substitution in the preceding majorant estimate gives the bound for $A[f]$. The calculation has used only f, its support and the kernel g; it has not estimated the polar or finite-prime terms in $R[f]$. □

Remark 82.5.7 *(scope of the estimates).* The projection statement is an obstruction to exact compact Fourier support. The majorant and support estimates are useful sufficient bounds. They do not exclude other elementary estimates or prove that every approach to (⋆) must take a prescribed form.

82.6 Auxiliary Positivity and Full Weil Positivity

Theorem 82.6.1 *(auxiliary positivity and the remaining signed form).* On $\mathscr{A}_{\mathrm{HFD}}$, one has $G(f) \geq 0$. On the full Weil test domain, positivity of $W(f * \widetilde{f})$ is equivalent to (⋆), including its norm, polar and finite-prime terms. The first assertion does not establish the second.

Proof. The first assertion is Theorem 82.4.3. The second is the signed identity (82.11). □

Remark 82.6.2 *(constructed quantities).* The Hurwitz series proves monotonicity of g, and its unique positive zero defines the non-negative energies $A[f]$ and $B[f]$. Their difference computes $G(f)$ exactly. The high-frequency dominance estimate proves positivity of this auxiliary form on its stated class. The full inequality $(\star)$ remains a separate question on the full Weil test class.

Further Comparison Problems

Remark 82.6.3 *(zero distributions, Gaussian tests and spectral spaces).* Three natural further constructions concern the actual zero divisor, Gaussian families, and Hermite–Biehler or de Branges spaces. A zero-distribution argument must estimate the unconditional functional in (82.11), before specializing the zeros to real ordinates. Gaussian calculations must retain the finite-prime and polar terms and justify passage from their test family to the desired full class. A spectral-space approach requires an actual positive kernel and a map identifying its form with the signed Weil functional; writing a kernel formula alone proves no such identification. None of these comparisons is supplied by the core-tail decomposition.

Remark 82.6.4 *(relation to the full Weil form).* Chapter 78 fixes the signed normalization. The present kernel calculation adds explicit estimates for G. To use them for full positivity one must still establish $R[f] + G(f) \geq 2\log\pi\|f\|_2^2$ on the required full test class.

82.7 The Remaining Construction

Remark 82.7.1 *(the exact positivity problem).* The positive result is the explicit auxiliary-form estimate on the high-frequency dominance class. The original problem is positivity of the full signed Weil form, with its polar and finite-prime terms, on the full test class. Its equivalence with RH does not turn the auxiliary estimate into a proof. A further construction must control $R[f] + G(f) - 2\log\pi\|f\|_2^2$ on that domain.

Chapter 83

Weil–Petersson Kernel Comparison: Dense Test-Class Limits

The compact-test moment conditions admit explicit projections and precise closure theorems. These depend on the topology: all three moments persist under compact convergence, whereas only the mean persists under unrestricted Schwartz convergence. Automorphic weight limits require their own maps and measures. The complete Weil form retains its polar, finite-prime, and signed gamma terms.

83.1 Restricted-Class Baseline and Residual

Use $\mathscr{A} = C_c^\infty(\mathbb{R})$, $\mathscr{P} = \mathcal{F}_+\mathscr{A}$, and $q_W(f,g) = W(f * g^*)$ as in Chapter 78.1.1. The topology on $\mathscr{A}$ is the usual inductive-limit topology. The full form contains its polar, signed gamma, and finite-prime terms.

Theorem 83.1.1 *(the compact coefficient comparison).* The compact multiplier $U_F(f)$ of Chapter 78.4.2 is a finite Fourier polynomial. It is a cusp form of the specified positive weight only when it vanishes. The distinct entire multiplier $\Upsilon_F(h)$ requires its own modular transformation law. For an actual cusp form, the Rankin–Selberg unfolding is Chapter 78.5.1.

Proof. The compact multiplier obstruction and exact unfolding in Chapter 78 give the assertions. □

Remark 83.1.2 *(three comparison requirements).* For compact tests, Mellin transforms are entire and rapidly decreasing on every closed vertical strip. Thus their strip holomorphy does not depend on the modular weight. Three separate requirements remain. A coefficient prescription must satisfy the modular transformation law. A comparison with holomorphic or Maass forms must specify its archimedean representation and realization map. A projection to the discrete spectrum must come from a specified Hilbert decomposition. The last projection is constructed in Chapter 78.7. None of these maps follows from the moment conditions defining $\mathscr{A}_{13}$.

83.2 Construction I: Maass-Form Lift on Tempered Eigenforms

Archimedean Local Factor for Principal-Series Maass

Let ϕ be a Maass cuspidal Hecke eigenform on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ with Laplace eigenvalue $\lambda_\phi = 1/4 + r_\phi^2$, $r_\phi \in \mathbb{R}$ tempered at ∞. The Fourier–Whittaker expansion is

$$\phi(z) \;=\; \sum_{n\neq 0} a_\phi(n)\,\sqrt{y}\,K_{ir_\phi}(2\pi|n|y)\,e^{2\pi i n x},$$

where K_ν is the Macdonald K-Bessel function (Iwaniec 2002 *Spectral Methods of Automorphic Forms* §1.9). The normalised Hecke eigenvalues $\widetilde{a}_\phi(n) = a_\phi(n)/a_\phi(1)$, for $n > 0$, satisfy $\widetilde{a}_\phi(1) = 1$ and are real if ϕ is a newform at level 1. Deligne's bound is not available for Maass forms; the Kim–Sarnak estimate is $\theta_\phi \le 7/64$, and temperedness at every prime is the Ramanujan conjecture, open at $\mathrm{GL}_2/\mathbb{Q}$.

Definition 83.2.1 *(Maass Mellin lift).* . Fix a tempered Maass Hecke eigenform ϕ of eigenvalue $\lambda = 1/4 + r^2$ with $r \in \mathbb{R}$. For $f \in \mathscr{A}$, put

$$\Upsilon_\phi(f)(z) := \sum_{n\geq 1} f(\log n)\, \widetilde{a}_\phi(n)\, \sqrt{y}\, K_{ir}(2\pi n y) \cos(2\pi n x). \tag{83.1}$$

The cosine is the symmetrisation under ϕ even (the + part of the Maass space; the odd case replaces cosine by sine with the same K-Bessel factor).

For compact tests, put $M_f(s) = \int_{\mathbb{R}} f(t) e^{st}\, dt$.

LEMMA 83.2.2 *(the compact Maass coefficient series).* Use the compact real even moment space

$$\mathscr{A}_{13} = \{f \in C_c^\infty(\mathbb{R}) : f \text{ real and even},\ M_f(0) = M_f(1) = M_f'(1) = 0\}.$$

For every such f, the series (83.1) has only finitely many terms. It is smooth on $\mathbb{H}$, periodic in x, decreases exponentially as $y \to \infty$, and satisfies the local weight-zero Laplace eigenvalue equation of parameter r. These assertions require no finite-prime temperedness hypothesis. They do not establish invariance under $z \mapsto -1/z$.

Proof. Compact support bounds the positive integers with $f(\log n) \neq 0$. Each summand is smooth and periodic. The modified Bessel equation $u^2 K_{ir}'' + u K_{ir}' - (u^2 - r^2) K_{ir} = 0$ gives eigenvalue $1/4 + r^2$ for $-y^2(\partial_x^2 + \partial_y^2)$ on $\sqrt{y} K_{ir}(2\pi n y)\cos(2\pi n x)$. The decay of K_{ir} at infinity gives the cusp decay of the finite sum. These computations impose no modular transformation under inversion. A quotient-space Petersson integral requires that transformation law separately. □

The Mellin transform of a compact smooth test is entire, so the Kim–Sarnak parameter bound supplies no additional strip condition on this carrier. The moment equations are independent of the modular transformation law and do not cancel the actual polar moments at $\pm 1/2$.

PROPOSITION 83.2.3 *(the local Bessel norm and its Mellin transform).* For real r and $\Re s > 0$,

$$B_r(s) := \int_0^\infty K_{ir}(u)^2 u^{s-1}\, du = \frac{2^{s-3}\Gamma(s/2)^2 \Gamma(s/2 + ir)\Gamma(s/2 - ir)}{\Gamma(s)}.$$

For a finite cosine series $V(z) = \sum_{n\geq 1} c_n \sqrt{y} K_{ir}(2\pi n y)\cos(2\pi n x)$ and a real $s > 0$,

$$\int_0^\infty \int_{-1/2}^{1/2} |V(x + iy)|^2 y^{s-1} \frac{dx\, dy}{y} = \frac{B_r(s)}{2(2\pi)^s} \sum_{n\geq 1} |c_n|^2 n^{-s}.$$

This is a strip integral, not an integral over a modular fundamental domain.

Proof. Use the Schlaefli integral representation of K_ν (DLMF 10.32.10) with orders ir and $-ir$. Their powers of $u/2$ cancel since $K_{-ir} = K_{ir}$. Integration in u gives

$$2^{s-3}\Gamma(s/2) \int_0^\infty \int_0^\infty e^{-v-w} v^{s/2 - ir - 1} w^{s/2 + ir - 1} (v + w)^{-s/2}\, dv\, dw.$$

Absolute convergence holds for $\Re s > 0$. Set $v = xq$, $w = x(1 - q)$, where $x > 0$ and $0 < q < 1$. The x integral is $\Gamma(s/2)$, and the q integral is $\Gamma(s/2 - ir)\Gamma(s/2 + ir)/\Gamma(s)$. This proves the first formula. For the second, orthogonality of cosines on $[-1/2, 1/2]$ gives a factor $1/2$ on each diagonal. The substitution $u = 2\pi n y$ gives the stated factor n^{-s}. Only finitely many terms occur. □

Remark 83.2.4 *(the Maass critical-line comparison).* A proposed arithmetic comparison would specify a complex test domain $D_\phi \subset \mathscr{A}$ and a linear automorphic realization $I_\phi : D_\phi \to L^2_{\mathrm{cusp}}(SL_2(\mathbb{Z})\backslash\mathbb{H})$. The intended critical-line norm identity is

$$\langle I_\phi f, I_\phi g\rangle_{\mathrm{Pet}} = \int_{\mathbb{R}} h_f(t)\overline{h_g(t)}K_\phi^{\mathrm{prop}}(1/2+it)dt, \qquad f, g \in D_\phi.$$

One scalar prescription to compare with such an integral is

$$K_\phi^{\mathrm{prop}}(s) = \frac{\zeta(2s)L(s,\phi\otimes\overline{\phi})}{\zeta(s)\xi_{\mathbb{R}}(s)\xi_{\mathbb{R}}(1-s)}G_\phi^{\mathrm{prop}}(s),$$

where the auxiliary scalar factor is

$$G_\phi^{\mathrm{prop}}(s) = \frac{|\Gamma(s/2+ir)\Gamma(s/2-ir)|}{|\Gamma(s/4+ir/2)\Gamma(s/4-ir/2)|^2}$$

where finite. Absolute values make this prescription different from the holomorphic Bessel Mellin transform $B_r(s)$ above. Constructing I_ϕ, proving the critical-line identity, and specifying a convergent domain or regularization at every singularity are unresolved requirements. A nonnegative norm does not make an unidentified scalar expression pointwise nonnegative. The finite coefficient series alone supplies no modular inversion law. On a fundamental domain the lower y boundary depends on x, so the cosine orthogonality used on the strip cannot be applied there. Rankin–Selberg unfolding requires an actual automorphic function and an Eisenstein integral with its convergence hypotheses (Iwaniec, *Spectral Methods*, Ch. 10; Zagier 1981).

Corollary 83.2.5 *(positivity on an actual Maass carrier).* Any specified linear map $I_\phi : D_\phi \to L^2_{\mathrm{cusp}}$ defines the positive semidefinite form $\langle I_\phi f, I_\phi g\rangle$ without a Ramanujan hypothesis. It realizes the Weil form on D_ϕ only if its full pairing identity with $P - W_\infty - W_{\mathrm{fin}}$ is proved.

Proof. The diagonal is $\|I_\phi f\|_2^2 \geq 0$. The finite-prime counterexample in Chapter 78.5.5 excludes an identification of W_{fin} alone with a positive norm on all compact tests. Bounds on the Hecke parameters do not change that counterexample. □

Remark 83.2.6 *(domains across Maass eigenforms).* The cuspidal subspace of $L^2(\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}, dx\,dy/y^2)$ has a complete orthonormal basis of Maass cusp eigenfunctions for the positive Laplacian $\Delta = -y^2(\partial_x^2 + \partial_y^2)$. Count eigenvalues with multiplicity, using both parities, and choose one parameter $r_j \geq 0$ for every eigenvalue $1/4 + r_j^2 \geq 1/4$; the finitely many lower eigenvalues do not change the asymptotic. Then $N(R) = \#\{j : 0 \leq r_j \leq R\} \sim R^2/12$. Indeed Selberg's cuspidal Weyl law, as stated in Müller, *Annals of Mathematics* 165 (2007), p. 277, equation (0.1), is $N_{\mathrm{cusp}}(\lambda) \sim \mathrm{area}(\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H})\lambda/(4\pi)$. The standard domain has area $\int_{-1/2}^{1/2}(1-x^2)^{-1/2}dx = \pi/3$; putting $\lambda = 1/4 + R^2$ gives the coefficient $1/12$. This statement concerns the full cuspidal spectrum, not one parity and not the continuous Eisenstein spectrum. The same compact moment space $\mathscr{A}_{13}$ can parameterize finite coefficient series for different Maass eigenforms. Changing their spectral parameters or finite-prime bounds does not alter its Mellin holomorphy. Automorphic realization domains must instead be determined by actual transformation laws and maps. A union of such domains is not known to be dense merely from these coefficient prescriptions.

83.3 Moment-preserving approximation and varying weights

The moment equations restrict the test space independently of a modular weight. Their kernel admits an explicit continuous projection. Write $\mathscr{A}_{\mathbb{R}}^{\mathrm{ev}}$ for real even compact smooth tests, and define

$$\ell(f) = \left(\int f(t)dt,\ \int f(t)\cosh t\, dt,\ \int f(t)t\sinh t\, dt \right).$$

Then $\mathscr{A}_{13} = \ker \ell$. Indeed these are $M_f(0)$, $M_f(1)$, $M_f'(1)$, and $M_f'(0) = 0$ by evenness.

PROPOSITION 83.3.1 *(a continuous moment projection).* There exist real even compact smooth functions η_0, η_1, η_2 with $\ell_i(\eta_j) = \delta_{ij}$. The map

$$\Pi f = f - \sum_{j=0}^{2} \ell_j(f)\eta_j$$

is a continuous projection onto $\mathscr{A}_{13}$. This kernel is closed of codimension three in $\mathscr{A}_{\mathbb{R}}^{\mathrm{ev}}$. If $f_n \to f \in \mathscr{A}_{13}$ in the compact-test topology, then $\Pi f_n \to f$ in the same topology. The same construction holds on the real even part of $\mathscr{G}$.

Proof. Let $w(t) = (1, \cosh t, t\sinh t)$. The matrix with columns $w(0), w(1), w(2)$ has determinant $2\sinh(1)(\cosh(1)-1)^2 > 0$. Choose a nonnegative even mollifier b_ε of integral one, supported in $[-\varepsilon, \varepsilon]$. Use the three functions $b_\varepsilon(t)$ and $[b_\varepsilon(t-j) + b_\varepsilon(t+j)]/2$ for $j = 1, 2$. Their moment matrix tends to the displayed invertible matrix. For sufficiently small ε, invert that moment matrix to obtain the η_j. All moments are continuous on compact tests, so Π is continuous, $\ell\Pi = 0$, and $\Pi f = f$ on the kernel. Surjectivity of ℓ gives codimension three. These statements imply the approximation assertion. On $\mathscr{G}$, an exponential seminorm controls each moment integral, so the same proof applies. □

PROPOSITION 83.3.2 *(a projection preserving a prescribed support).* Fix $a > 0$. The moment right inverse can be chosen with all η_j supported in $[-a, a]$. The resulting projection preserves the space of smooth real even tests supported in this interval.

Proof. Choose a smooth even function $\chi \geq 0$, supported in $[-a, a]$ and positive on $(-a, a)$. With $w_0 = 1$, $w_1 = \cosh t$, $w_2 = t\sinh t$, form $G_{ij} = \int \chi(t)w_i(t)w_j(t)dt$. For a nonzero real vector c, the analytic function $\sum c_i w_i$ cannot vanish on an interval: independence follows, for example, from its growth at positive infinity. Therefore $c^T G c = \int \chi(\sum c_i w_i)^2 > 0$. Set $\eta_j = \chi \sum_i (G^{-1})_{ij} w_i$. Then $\ell_i(\eta_j) = \delta_{ij}$, and the support assertion follows. □

PROPOSITION 83.3.3 *(the moment quotient algebra).* On the complex even compact convolution algebra, the map

$$f \longmapsto \big(M_f(0), M_f(1) + M_f'(1)\epsilon\big), \qquad \epsilon^2 = 0,$$

is a continuous surjective algebra homomorphism onto $\mathbb{C} \times \mathbb{C}[\epsilon]/(\epsilon^2)$. Its kernel is the complexification of $\mathscr{A}_{13}$. The involution is coefficient conjugation with $\epsilon^* = \epsilon$. A linear functional on this quotient is positive on xx^* exactly when it has the form $\lambda(a, b + c\epsilon) = \alpha a + \beta b$ with $\alpha, \beta \geq 0$.

Proof. Compact support permits Fubini, giving $M_{f*g} = M_f M_g$. Differentiating at one gives the dual-number multiplication rule. The moment right inverse proves surjectivity. Evenness makes the involution coefficient conjugation at these real evaluation points. Write a linear functional as $\alpha a + \beta b + \gamma c$. Since

the quotient is unital, positivity on squares makes it Hermitian: expanding $(1+tx)(1+tx)^*$ and $(1+itx)(1+itx)^*$ for real t gives $\lambda(x^*)=\overline{\lambda(x)}$. Thus its three coefficients are real. For $x=(a,b+c\epsilon)$ its squared value is $\alpha|a|^2+\beta|b|^2+2\gamma\Re(b\bar{c})$. Varying c with $b=1$ forces $\gamma=0$. Varying a,b independently then gives $\alpha,\beta\geq 0$, which also suffices. No factorization of the Weil functional through this finite-dimensional quotient is assumed. □

For every compact f, differentiation under the integral makes $M_f(s)=\int f(t)e^{st}\,dt$ entire. Integration by parts gives rapid decay on each closed vertical strip. Consequently, imposing the reality, evenness, and double-zero conditions at 0, 1 for any family of weights always gives $\mathscr{A}_{13}$. This union is not dense in the compact-test space. It is not dense in its ambient real even Schwartz topology either: the continuous functional $f\mapsto\int f$ vanishes on the union and not on a nonnegative bump. A bounded moment correction need not tend to zero. The projection above converges to the original limit only when that limit has the required moments.

Theorem 83.3.4 *(the closure in the weaker Schwartz topology).* The closure of $\mathscr{A}_{13}$ in the real even Schwartz space is

$$\{f\in\mathcal{S}(\mathbb{R}) : f \text{ real and even, } \int f=0\}.$$

In the relative Schwartz topology on compact tests, intersect this space with $\mathscr{A}$. The exponential moments need not vanish at a limit in this weaker topology.

Proof. The mean is Schwartz-continuous, so every limit has mean zero. Choose a nonnegative even compact smooth β with $\int\beta=1$. Put $A=\int\beta(t)\cosh t\,dt>0$ and $B=\int\beta(t)t\sinh t\,dt$. For $R>0$ define $\beta_R(t)=[\beta(t-R)+\beta(t+R)]/2$ and $v_R=\beta_R-\beta$. Direct translation gives

$$\ell(v_R)=(0,A(\cosh R-1),B(\cosh R-1)+AR\sinh R).$$

The determinant of the last two moments of v_R, v_{2R} is

$$A^2[(\cosh R-1)2R\sinh(2R)-(\cosh(2R)-1)R\sinh R]=2A^2R\sinh R(\cosh R-1)^2>0.$$

Thus any fixed pair of moments can be prescribed by $c_Rv_R+d_Rv_{2R}$ with $c_R=O(e^{-R})$ and $d_R=O(e^{-2R})$ as $R\to\infty$. Every Schwartz seminorm of v_R grows at most polynomially in R, so this prescribed-moment correction tends to zero in Schwartz space. For a compact real even mean-zero f, choose the correction with moments $-(\ell_1(f),\ell_2(f))$. Its sum with f lies in $\mathscr{A}_{13}$ and converges to f in Schwartz space. Finally, real even compact mean-zero tests are dense in the mean-zero real even Schwartz space: use even cutoffs and subtract their mean times β. For the nth such approximation, choose R large enough that the correction has its first n Schwartz seminorms below $1/n$. This diagonal choice proves the full assertion. □

Definition 83.3.5 *(a family of cusp forms with varying weights).* Choose an increasing sequence of even weights $k_j\to\infty$ for which $S_{k_j}(SL_2(\mathbb{Z}))\neq 0$, and normalized Hecke cusp forms F_j in these spaces. Define probability measures on $Y=SL_2(\mathbb{Z})\backslash\mathbb{H}$ by

$$d\nu_j=\frac{|F_j(z)|^2y^{k_j}\,d\mu(z)}{\|F_j\|_{\mathrm{Pet}}^2},\qquad d\mu=dx\,dy/y^2.$$

Their masses equal one by definition.

Pairing limits on a common carrier

For a bounded measurable scalar function a on Y, multiplication by a defines a section $J_j a = aF_j/\|F_j\|_{\mathrm{Pet}}$ of the weight-k_j line bundle. It is square integrable, and

$$\langle J_j a, J_j b\rangle = \int_Y a\overline{b}\, d\nu_j.$$

Holomorphy is not required for this Hilbert-space map.

PROPOSITION 83.3.6 *(limits of the actual normalized measures).* If ν_j converges weakly to a probability measure ν against bounded continuous functions on Y, then for bounded continuous a, b,

$$\lim_j \langle J_j a, J_j b\rangle = \int_Y a\overline{b}\, d\nu.$$

The limiting form is positive semidefinite.

Proof. The product $a\overline{b}$ is bounded and continuous, so weak convergence gives the equality. Its diagonal is an integral of $|a|^2$. □

A limit of total mass one on this noncompact quotient requires control of escape into the cusp. Weight growth alone supplies neither weak convergence nor this control.

THEOREM 83.3.7 *(limits of specified positive pairings).* Let D be a complex vector space. Let $I_j : D \to H_j$ be linear maps to positive Hilbert spaces and $c_j > 0$. Suppose $Q(f) = \lim_j c_j^{-1}\|I_j f\|^2$ exists and is finite for every $f \in D$. Then Q is the diagonal of a positive semidefinite Hermitian form q on D, and

$$q(f, g) = \lim_j c_j^{-1}\langle I_j f, I_j g\rangle.$$

Proof. For a form linear in its first variable, polarization is $q(f, g) = \frac{1}{4}\sum_{r=0}^{3} i^r Q(f + i^r g)$. Apply this finite sum to the assumed limits. The resulting pointwise limit of sesquilinear forms is sesquilinear and Hermitian. The diagonal is nonnegative as a limit of nonnegative real numbers. □

Remark 83.3.8 *(the proposed symmetric-square density).* For the unitary Hecke normalization one may consider the proposed meromorphic ratio $K_{F_j}^{\mathrm{prop}}$ in Chapter 78.7 and the proposed real density

$$R(t) = \frac{\zeta(1 + 2it)\zeta(1 - 2it)}{\xi_{\mathbb{R}}(1/2 + it)\xi_{\mathbb{R}}(1/2 - it)}.$$

For a level-one holomorphic eigenform with unitary automorphic normalization, the Euler factors give $L(s, \pi_j \times \overline{\pi}_j) = \zeta(s)L(s, \mathrm{sym}^2 \pi_j)$. Indeed the tensor Satake parameters are $1, 1, \alpha_p^2, \alpha_p^{-2}$, where $\alpha_p\alpha_p^{-1} = 1$; the symmetric square has parameters $\alpha_p^2, 1, \alpha_p^{-2}$. The weight-limit question asks for a normalized factorization $K_j(1/2 + it) = c_j R(t) A_j(t)$ with $c_j > 0$ and $A_j(t) \to 1$ locally uniformly away from singularities, together with domination of the complete tested integrands. The scalar c_j must be fixed by the actual pairing normalization. This factorization and limit are additional requirements, not consequences of the Euler factors. An identity with normalized Petersson pairings requires actual maps I_j on a common domain, all local factors, and a proved integral identity. A density limit additionally requires convergence and an integrable majorant for the complete integrands, including varying tests and singular factors. The ratio R has a singularity at zero from the pole of $\zeta(1 + 2it)$, and may have

further singularities at critical-line zeros of ξ. Thus no integral of $|h|^2 R$ is specified without a domain or regularization. A Selberg trace identification would also have to match its spectral and geometric terms. None of these assertions follows from moment projection.

Corollary 83.3.9 *(the arithmetic comparison for a limit).* Under the preceding limit theorem, the quotient of D by the nullspace of q, followed by completion, is a positive Hilbert realization of q. It realizes the restricted Weil form exactly when $q(f, g) = W(f * g^*)$ on D.

Proof. Use Cauchy–Schwarz for q to form its Hilbert quotient. The last equality is the required pairing identity. □

83.4 The polar form and its finite-dimensional realization

The two evaluations in the Weil formula determine a signed finite-dimensional form. Write $a(f) = M_f(-1/2)$, $b(f) = M_f(1/2)$ and $r(f) = (a(f), b(f))$.

Lemma 83.4.1 *(the exact polar map).* For $f, g \in \mathscr{A}$,

$$P(f * g^*) = a(f)\overline{b(g)} + b(f)\overline{a(g)}.$$

The map $r : \mathscr{A} \to \mathbb{C}^2$ is continuous and surjective. Its common kernel is closed of codimension two.

Proof. Evaluate $h_f h_g^\sharp$ at $\pm i/2$. The functions $e^{-t/2}$, $e^{t/2}$ are linearly independent, so the two moment functionals are independent. This makes r surjective. Their continuity gives the remaining assertions. □

Proposition 83.4.2 *(signed polar realization and even restriction).* The polar form is the pullback by r of the Hermitian matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. It has one positive and one negative direction, so no map into a positive Hilbert space realizes it on all compact tests. On the complex even tests it has the one-dimensional positive realization $f \mapsto \sqrt{2}\, M_f(1/2)$.

Proof. The matrix eigenvalues are $1, -1$, and surjectivity gives both signs on compact tests. For even functions $a = b$, so the pairing is $2a(f)\overline{a(g)}$, proving the last assertion. □

Remark 83.4.3 *(moment restrictions and arithmetic residual sectors).* The H3 moments at 0, 1 are different from these polar moments. For a nonzero nonnegative even bump v, set $f = D^2(D^2 - 1)^2 v$, where $D = d/dt$. Then $f \in \mathscr{A}_{13}$ but $M_f(1/2) = 9M_v(1/2)/64 > 0$. Thus $P(f * f^*) = 2M_f(1/2)^2 > 0$ despite H3. The exact polar kernel can instead be projected onto by choosing a right inverse $S : \mathbb{C}^2 \to \mathscr{A}$ of r and using $\Pi_{\mathrm{pol}} = 1 - Sr$. The vector $M_f(1)|0\rangle$ has norm square $|M_f(1)|^2$ and does not realize the polar form. A comparison with an automorphic residual spectrum must specify its central character, subspace, measure, and map. The finite-dimensional signed realization above supplies the complete polar pairing without making that identification.

83.5 Construction IV: Jacquet–Piatetski-Shapiro–Shalika on $GL_n \times GL_m$

Higher-Rank Rankin–Selberg

Higher-rank Rankin–Selberg integrals compare specified automorphic representations through Whittaker models and local factors. Their convergence regions concern those integrals. Compact test Mellin transforms are already entire at every rank.

Definition 83.5.1 *(higher-rank coefficient prescription).* Fix a cuspidal automorphic representation π of $GL_r/\mathbb{Q}$ and Hecke coefficients $a_\pi(m)$ in a specified unitary Satake normalization. A vector-valued coefficient prescription requires actual vectors $v_{\pi,m}$ in a specified representation space:

$$U_\pi(f) = \sum_{m\geq 1} f(\log m) a_\pi(m) v_{\pi,m}, \qquad f \in \mathscr{A}.$$

This sum is finite. A scalar Whittaker value $W_\pi(m)$ instead gives a scalar-valued sum; it does not by itself define an automorphic vector. Choosing Whittaker vectors, their translates, and a global realization is part of the higher-rank problem.

PROPOSITION 83.5.2 *(a specified higher-rank pairing).* Suppose a complex test domain $D_\pi \subset \mathscr{A}$ has a linear map I_π to a cuspidal automorphic Hilbert space with fixed central character and quotient measure. Suppose a measurable function $\kappa_\pi(t)$ satisfies the absolutely convergent identity

$$\langle I_\pi f, I_\pi g\rangle_{\mathrm{Pet}} = \int_{\mathbb{R}} h_f(t)\overline{h_g(t)}\kappa_\pi(t)\,dt \qquad (f, g \in D_\pi).$$

Then the tested form on the right is positive semidefinite. This conclusion does not require temperedness of π.

Proof. Its diagonal equals $\|I_\pi f\|^2_{\mathrm{Pet}}$. □

The required comparison with JPSS Rankin–Selberg integrals is open for this coefficient prescription. In particular the suggested ratio $L(s, \pi \times \overline{\pi})/(\zeta(rs)\prod_v G_{v,\pi}(s))$ needs specified normalized local factors $G_{v,\pi}$, a convergence meaning for their product, and a proof that its critical-line value is κ_π. The local Whittaker models and measures, the Eisenstein or Rankin–Selberg integration domain, and any continuation or regularization must be fixed before that equality can be tested. JPSS integrals provide their own L-function identities; they do not identify an arbitrary coefficient prescription with a norm. Temperedness alone does not imply positivity of an unconstructed critical-line scalar ratio.

PROPOSITION 83.5.3 *(compact Mellin transforms and local convergence).* For $f \in \mathscr{A}$, its Mellin transform $M_f(s)$ is entire and rapidly decreasing on each closed vertical strip, independently of a rank parameter. A convergence half-plane of a specified Rankin–Selberg integral does not impose a strip of holomorphy on this compact test transform.

Proof. Differentiate the integral $M_f(s) = \int f(t)e^{st}\,dt$ on its fixed compact support. Repeated integration by parts gives rapid decay in $\Im s$, uniformly on a bounded range of $\Re s$. The rank does not appear in this calculation. □

A higher-rank comparison must still specify its automorphic map, Whittaker vectors, local measures, convergence region and analytic continuation. Those are properties of the specified integral, not additional holomorphy restrictions on M_f.

Remark 83.5.4 *(temperedness and the higher-rank comparison).* A separate parameter question asks for temperedness of the chosen cuspidal representation at every finite place. For a functorial symmetric-power lift of a holomorphic eigenform, unit-modulus normalized Satake parameters give unit-modulus symmetric-power parameters. This follows directly by taking their monomials. Constructing the lift and deciding its cuspidal decomposition are additional automorphic questions; references include Gelbart–Jacquet, Kim–Shahidi, Kim and Newton–Thorne. For a general Maass or higher-rank representation, the required parameter assertion remains a Ramanujan hypothesis. Neither this assertion nor its special symmetric-power cases constructs I_π or proves the critical-line pairing identity.

83.6 Closed moment kernels and full positivity

Definition 83.6.1 *(domains of specified comparisons).* For each constructed comparison map $I_\alpha : D_\alpha \to H_\alpha$, let $D_\alpha \subset \mathscr{A}$ be its actual domain and put

$$D_{\mathrm{ext}} = \bigcup_\alpha D_\alpha.$$

An identity with $W(f * g^*)$ must be specified on each D_α. The union is a set of tests and is not necessarily a vector space.

Theorem 83.6.2 *(moment-preserving density).* If S is dense in $\mathscr{A}_{\mathbb{R}}^{\mathrm{ev}}$ in the compact-test topology, then ΠS is dense in $\mathscr{A}_{13}$. The kernel $\mathscr{A}_{13}$ is a proper closed subspace. On the real even exponential space, its compact members are dense in the corresponding moment kernel.

Proof. Apply the continuous projection Π to an approximating net for any member of its image. Closedness and properness follow from its three independent moments. In $\mathscr{G}$, use even compact cutoffs f_R and then Πf_R. The cutoff convergence of Chapter 95.1.2 and continuity of Π prove $\Pi f_R \to f$ for a member of the kernel. □

Proposition 83.6.3 *(positivity on a union of comparison domains).* Suppose each specified map I_α satisfies $W(f * g^*) = \langle I_\alpha f, I_\alpha g\rangle$ on D_α. Then $q_W(f, f) \geq 0$ on D_{ext} and its closure. Positivity on the linear span additionally requires control of cross terms between different domains.

Proof. Each diagonal is a norm square. Continuity passes its sign to the closure. For tests from two domains, their sum has diagonal $q_W(f, f) + 2\Re q_W(f, g) + q_W(g, g)$. The separate norm identities specify no bound on that cross term. □

Theorem 83.6.4 *(continuity of the complete Weil form).* If sequences $f_j \to f$ and $g_j \to g$ in $\mathscr{A}$ or $\mathscr{G}$, then

$$W(f_j * g_j^*) \longrightarrow W(f * g^*).$$

The same convergence holds separately for P, W_∞, W_{fin}.

Proof. For compact convergence all supports lie in one compact interval, and the prime functional is a finite sum of continuous evaluations. For exponential convergence, the bound

$$\sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} |v(\pm \log n)| \leq p_{2,0,0}(v) \sum_{n\geq 2} \frac{\log n}{n^{5/2}}$$

proves continuity of the prime term. The polar and gamma terms, convolution and involution are continuous by Chapter 95.1.2. The formula follows. □

Remark 83.6.5 *(the topology of the prime term).* Ordinary Schwartz seminorms give only inverse powers of $1 + |\log n|$ at prime arguments. Such bounds do not supply the summable majorant above. The stated continuity therefore uses the compact-test or exponential topology. It does not assert a tempered extension of the uncentered critical prime measure.

Theorem 83.6.6 *(extension across a finite moment complement).* Let D be a complex vector space with a Hermitian form q, linear in the first variable. Let $\Pi : D \to K$ be a projection, and put $E = (1 - \Pi)D$, so $D = K \oplus E$. Then q is positive semidefinite on D if and only if

$$q(u, u) \geq 0, \quad q(e, e) \geq 0, \quad |q(u, e)|^2 \leq q(u, u)q(e, e) \qquad (u \in K,\ e \in E).$$

Proof. Necessity is Cauchy–Schwarz for a positive semidefinite form. For sufficiency, the cross bound gives $q(u+e,u+e) \geq (\sqrt{q(u,u)} - \sqrt{q(e,e)})^2 \geq 0$. Every member of D has this decomposition. □

For the complexification of $\mathscr{A}_{13}$ inside the complex even tests, the projection has a three-dimensional complement. Thus positivity on that kernel leaves a finite complement and all its cross terms to determine. Odd tests require their own comparison when the target is the full complex space $\mathscr{A}$. The polar kernel on the full space has the two-dimensional complement constructed above. Neither finite codimension nor a projection establishes the missing cross inequalities.

83.7 The arithmetic realization problem

Theorem 83.7.1 *(the full comparison criterion).* A positive Hilbert realization of q_W on $\mathscr{A}$ exists if and only if

$$W_\infty(f * f^*) \leq P(f * f^*) - W_{\mathrm{fin}}(f * f^*) \qquad (f \in \mathscr{A}).$$

This condition is equivalent to RH.

Proof. The signed explicit formula identifies the difference with $q_W(f,f)$. A positive form has its Hilbert quotient realization, as in Chapter 78.8.1. Weil's criterion gives RH. □

Corollary 83.7.2 *(a sufficient comparison route).* A dense family of tests with proved nonnegative full Weil diagonals in the compact or exponential topology suffices for RH. For a dense linear span, positivity must hold for every finite linear combination, not only the chosen generators.

Proof. Use continuity and Chapter 83.7.1. □

Remark 83.7.3 *(native comparison requirements).* A holomorphic comparison requires modularity of the entire coefficient multiplier. A Maass comparison requires an automorphic realization with its correct Whittaker expansion. A higher-rank comparison requires its Whittaker model, integral domain, and normalized local factors. Ordinary cusp-form L^2 norms are positive without a Ramanujan hypothesis. Bounds on Hecke parameters address different questions. None of these norms identifies a zeta pairing without an explicit map and equality of all its terms. Weight variation admits the actual probability measures and conditional pairing limit above. The residual polar term has its explicit signed rank-two realization. These constructions retain the holomorphic, Maass, residual, and higher-rank comparison questions, with their domains and maps specified separately.

83.8 Bibliography

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Chapter 84

Langlands–Shahidi Normalisation of the Eisenstein Koszul Density

The bounded-interval reduction, construction C. Chapter 76 (the restricted-class comparison) reduced Koszul–Petersson compatibility on the discrete-spectrum core by domain restriction: the Eisenstein continuous sector was placed outside the P3 identity, and the Li trace was separated from it under the H-P^{Ar} spectral realisation. The Langlands–Shahidi alternative was sketched as the second normalisation of §76.4. The exact conditional normalisation problem is this: multiplication by a Langlands–Shahidi scalar removes the unitary phase only after a choice of positive Plancherel measure has been fixed; the boundary Euler product on $\mathrm{Re}(s) = 1/2$ *is meromorphic, not absolutely convergent. Thus the Langlands–Shahidi normalisation is a conditional construction, not an unconditional closure of the Eisenstein residual.*

84.1 Setup and the LS Normalisation

The Residual Mismatch

The explicit mismatch of Proposition 76.3.1 (Chapter 76) is as follows. On the Eisenstein continuous spectrum $L^2_{\mathrm{cont}}([GL_2])$ parametrised by $\mathrm{Re}(s) = 1/2$, the Petersson-side Maass–Selberg normalisation and the Koszul-side bar–cobar coefficient are recorded in the unnormalised spherical convention by

$$\begin{aligned} \mathrm{lead}^{\mathrm{Pet}}_{\mathrm{cont}}(s) &= \big(1 + |M(s)|^2\big)|\xi(2s)|^{-2} \qquad (\mathrm{Re}(s) = 1/2), \\ \mathrm{lead}^{\mathrm{Koszul}}_{\mathrm{cont}}(s) &= \frac{\xi(2s)}{\xi(2s-1)} \cdot |\xi(2s)|^{-2} \qquad (\mathrm{Re}(s) = 1/2). \end{aligned}$$

The ratio $\xi(2s)/\xi(2s-1)$ has modulus 1 on the critical line (lemma below). Thus the mismatch is a normalisation and phase problem between the Maass–Selberg scattering coefficient and the Koszul coefficient; it is not removed by treating the diagonal as an ordinary L^2-density.

Lemma 84.1.1 *(unit modulus of* $\xi(2s)/\xi(2s-1)$ *on the critical line).* For $s = 1/2 + it$, away from the Tate boundary and common zeros of the two factors,

$$\big|\xi(2s)/\xi(2s-1)\big| = 1.$$

Proof. For $s = 1/2 + it$, write $2s = 1 + 2it$ and $2s - 1 = 2it$. The completed zeta ξ satisfies $\xi(w) = \xi(1-w)$ (functional equation; Titchmarsh 1986 §2.1), giving $\xi(1+2it) = \xi(-2it)$. Complex conjugation gives $\overline{\xi(1+2it)} = \xi(1-2it)$ (Schwarz reflection: ξ is real on the real axis, hence $\overline{\xi(w)} = \xi(\overline{w})$). Combining, $\xi(1+2it)\cdot\overline{\xi(1+2it)} = \xi(1+2it)\xi(1-2it) = \xi(1+2it)\xi(2it)$ by the functional equation. Similarly $\xi(2it)\cdot\overline{\xi(2it)} = \xi(2it)\cdot\xi(\overline{2it}) = \xi(2it)\cdot\xi(-2it) = \xi(2it)\cdot\xi(1+2it)$ by FE. Therefore $|\xi(2s)|^2 = |\xi(2s-1)|^2$, giving (84.1.1). □

The Langlands–Shahidi Normalisation Factor

We follow Shahidi 2010 *Eisenstein Series and Automorphic L-Functions* Chapter 2 and Moeglin–Waldspurger 1995 §IV.3 for the intertwining operator normalisation.

Definition 84.1.2 *(Langlands–Shahidi normalisation factor on GL_2 Eisenstein).* The *Langlands–Shahidi normalisation factor* on the Eisenstein summand $\mathrm{Ind}_B^G(\chi_s)$ of GL_2 at spectral parameter s with $\mathrm{Re}(s) = 1/2$ is

$$N^{\mathrm{LS}}(s) \;:=\; \frac{\xi(2s-1)}{\xi(2s)}.$$

Its inverse is $N^{\mathrm{LS}}(s)^{-1} = \xi(2s)/\xi(2s-1)$.

Remark 84.1.3 *(origin and canonicity of the factor).* The factor $N^{\mathrm{LS}}(s)$ is the global Langlands–Shahidi normalisation of the standard intertwining operator $M(s)\colon \mathrm{Ind}_B^G(\chi_s) \to \mathrm{Ind}_B^G(\chi_s^w)$, with three essential properties.

(LS1) *Local factor decomposition.* For each place v, $N^{\mathrm{LS}}(s)$ is a product of local Langlands factors $N_v^{\mathrm{LS}}(s)$ as specified by Casselman–Shalika 1980 *Compositio Math.* 41 at unramified finite v and by Shahidi 2010 §2.2 at ramified and archimedean v. Restricted tensor product factorisation: $N^{\mathrm{LS}}(s) = \prod_v N_v^{\mathrm{LS}}(s)$.

(LS2) *Functional equation.* $N^{\mathrm{LS}}(s)\cdot N^{\mathrm{LS}}(1-s) = 1$ on the critical line. Explicitly:

$$N^{\mathrm{LS}}(s)\cdot N^{\mathrm{LS}}(1-s) = \frac{\xi(2s-1)}{\xi(2s)}\cdot\frac{\xi(1-2s)}{\xi(2-2s)} = \frac{\xi(2s-1)\cdot\xi(2s)}{\xi(2s)\cdot\xi(2s-1)} = 1,$$

using the functional equation $\xi(1-2s) = \xi(2s)$ and $\xi(2-2s) = \xi(2s-1)$ (Titchmarsh 1986 §2.1). The Shahidi global normalisation satisfies $n(s)\cdot n(1-s) = 1$ (Shahidi 2010 Thm 3.5.1). No Maass–Selberg measure scalar is part of N^{LS}.

(LS3) *Unitarity of the normalised intertwiner.* After division by $N^{\mathrm{LS}}(s)$, the normalised intertwiner $\widetilde{M}(s) = M(s)/N^{\mathrm{LS}}(s)$ is unitary on the principal-series representation at each critical parameter (Moeglin–Waldspurger §IV.3.10).

84.2 The Redefined Koszul Pairing on the Eisenstein Summand

Definition 84.2.1 *(LS-corrected Koszul pairing on the Eisenstein summand).* Let $\phi_{s_1}, \phi_{s_2} \in \mathrm{Ind}_B^G(\chi_s)$ be spherical vectors at Eisenstein parameters s_1, s_2 with $\mathrm{Re}(s_i) = 1/2$. Assume an actual bivariate arithmetic pairing has been constructed on these vectors, and restrict to parameters for which the scalar below is finite and nonzero. Using the convention of Chapter 76, define the *Langlands–Shahidi-corrected Koszul pairing* by

$$\big\langle \phi_{s_1}, \sigma^{\mathrm{BC}}(\phi_{s_2})\big\rangle^{\mathrm{Koszul,LS}} \;:=\; N^{\mathrm{LS}}\!\left(\tfrac{1}{2}(s_1+s_2)\right)\cdot\big\langle \phi_{s_1}, \sigma^{\mathrm{BC}}(\phi_{s_2})\big\rangle^{\mathrm{Koszul}}.$$

On the diagonal $s_1 = s_2 = s$ this reduces to

$$\big\langle \phi_s, \sigma^{\mathrm{BC}}(\phi_s)\big\rangle^{\mathrm{Koszul,LS}} \;=\; \frac{\xi(2s-1)}{\xi(2s)}\cdot\big\langle \phi_s, \sigma^{\mathrm{BC}}(\phi_s)\big\rangle^{\mathrm{Koszul}}.$$

Theorem 84.2.2 *(conditional LS Plancherel normalisation match on the continuum). (conditional on the bar–cobar Eisenstein pairing being transported to the positive Langlands–Shahidi Plancherel measure).* For spherical vectors $\phi_s \in \mathrm{Ind}_B^G(\chi_s)$ with $\mathrm{Re}(s) = 1/2$,

$$\mathsf{Sh}_{\mathrm{LS}}\Big(\mathrm{lead}_{\mathrm{cont}}^{\mathrm{Koszul,LS}}(s)\Big) \;=\; \mathrm{lead}_{\mathrm{cont}}^{\mathrm{Pet}}(s). \tag{84.1}$$

Equivalently,

$$\mathsf{Sh}_{\mathrm{LS}}(\text{Koszul continuum}) \;=\; \text{Petersson continuum} \qquad (\mathrm{Re}(s) = 1/2),$$

after this measure identification. Here $\mathsf{Sh}_{\mathrm{LS}}$ denotes the Shahidi Plancherel transport. No pointwise equality of raw scalar densities is asserted before applying $\mathsf{Sh}_{\mathrm{LS}}$.

Proof. On the critical-line domain U where both zeta factors are finite and nonzero, Definition 67.5.2 defines the scalar model

$$d = |\xi(2s)|^{-2} > 0, \qquad M = N^{\mathrm{LS}}(s), \qquad K = M^{-1}d, \qquad P = 2d.$$

Consequently $N^{\mathrm{LS}}K = d = P/2$. This calculation concerns scalars; it neither constructs a bivariate arithmetic pairing nor regularises its diagonal. The theorem additionally assumes an actual transport of the corrected arithmetic quadratic form to the positive Plancherel measure, including its domain and normalization. With that transport supplied, (84.1) is the asserted normalization of that transport. □

Remark 84.2.3 *(why no scalar square proves the match).* The diagonal pairing is only a distributional Maass–Selberg object before truncation is regularised. Since $\xi(2s)/\xi(2s-1)$ has unit modulus on the critical line, taking absolute values removes the phase but does not identify the underlying Plancherel normalisation. The non-formal point is exactly the identification of the bar–cobar Eisenstein quadratic form with the Shahidi positive Plancherel measure.

Corollary 84.2.4 *(conditional normalisation match via LS). (same hypothesis as Theorem 84.2.2).* Under the LS redefinition Definition 84.2.1, the Koszul and Petersson normalisations on the Eisenstein continuous spectrum of $L^2([GL_2])$ coincide after the positive Plancherel-measure identification.

Proof. Theorem 84.2.2 is the stated measure-identification hypothesis written in the LS normalisation. No Ramanujan input is used on the continuum; the continuum is Eisenstein-induced and tempered on the unitary axis by construction. □

84.3 Compatibility Checks

The scalar functional equation, a conditional transport of obstruction classes, the meromorphic factorisation, and the unchanged discrete summand impose distinct compatibility requirements.

Compatibility Check (i): Heisenberg Self-Duality

Proposition 84.3.1 *(Conditional transport of Heisenberg self-duality).* Suppose a specified comparison of obstruction complexes induces a linear map $F : H \to H_{\mathrm{Ar}}$ taking two classes $c, c^! \in H$, with $c + c^! = 0$, to the arithmetic Heisenberg classes $\kappa_{\mathsf{G}}^{\mathrm{Ar}}$ and $(\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}}$, respectively. Assume this comparison remains defined for the corrected pairing. Then

$$\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} = 0.$$

The scalar correction itself satisfies $N^{\mathrm{LS}}(s)N^{\mathrm{LS}}(1-s) = 1$ wherever the ratios are finite and nonzero. This scalar identity supplies no comparison of obstruction complexes.

Proof. Linearity gives $F(c) + F(c^!) = F(0) = 0$. Independently, the completed zeta functional equation gives

$$N^{\mathrm{LS}}(s)N^{\mathrm{LS}}(1-s) = \frac{\xi(2s-1)\xi(1-2s)}{\xi(2s)\xi(2-2s)} = 1.$$

The first calculation requires the stated class comparison; the second proves only the scalar functional equation. Constructing the arithmetic obstruction complexes and this comparison remains necessary before applying the first calculation to the Eisenstein summand. □

Remark 84.3.2 *(no modification to the discrete sector).* On a specified decomposition into discrete and continuous summands, define the correction to be the identity on the discrete summand and to multiply by N^{LS} on the continuous domain U. Thus the discrete pairing of Chapter 76 is unchanged. This is a choice of extension to the direct sum, not the assertion that a meromorphic value of N^{LS} equals one at discrete parameters.

Compatibility Check (ii): Tate Factorisation

Proposition 84.3.3 *(LS factorises over places as a Tate restricted tensor product). (as a meromorphic Langlands–Shahidi factorisation, not as an absolutely convergent Euler product on the critical line).* The global Langlands–Shahidi factor $N^{\mathrm{LS}}(s)$ of Definition 84.1.2 admits a place-by-place decomposition

$$N^{\mathrm{LS}}(s) \;=\; \prod_{v \in |\overline{\mathrm{Spec}\,\mathbb{Z}}|} N_v^{\mathrm{LS}}(s),$$

with local factors

$$N_v^{\mathrm{LS}}(s) \;=\; \begin{cases} \zeta_p(2s-1)/\zeta_p(2s) & (v = p \text{ finite, unramified}) \\ \zeta_p(2s-1)/\zeta_p(2s)\cdot \varepsilon_p(s,\pi_p,\psi_p) & (v = p \text{ finite, ramified}) \\ \Gamma_{\mathbb{R}}(2s-1)/\Gamma_{\mathbb{R}}(2s) & (v = \infty), \end{cases}$$

where $\zeta_p(s) = (1-p^{-s})^{-1}$ is the Euler factor, $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ the archimedean Tate factor, and ε_p the local Jacquet–Piatetski-Shapiro–Shalika epsilon factor at the principal-series induction. The product is interpreted by meromorphic continuation of the completed L-factor.

Proof. The completed zeta factorises $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\prod_p \zeta_p(s)$ (Tate 1950 §4.2). Applying the factorisation to the ratio $\Lambda_\zeta(2s-1)/\Lambda_\zeta(2s)$:

$$\frac{\Lambda_\zeta(2s-1)}{\Lambda_\zeta(2s)} \;=\; \frac{\Gamma_{\mathbb{R}}(2s-1)}{\Gamma_{\mathbb{R}}(2s)} \cdot \prod_p \frac{\zeta_p(2s-1)}{\zeta_p(2s)}.$$

Unramified local factors are $\zeta_p(2s-1)/\zeta_p(2s)$; ramified local factors pick up an additional local epsilon factor ε_p from the Jacquet–Piatetski-Shapiro–Shalika 1983 local-functional-equation correction, which has absolute value 1 and does not affect the Plancherel normalisation match (Theorem 84.2.2). The restricted tensor product $\prod_v N_v^{\mathrm{LS}}(s)$ is not an absolutely convergent Euler product on $\mathrm{Re}(s) = 1/2$; it is the boundary value of the meromorphic completed ratio, giving (84.3.3). □

Remark 84.3.4 *(Casselman–Shalika at unramified finite places).* The local factor at an unramified finite prime p, $N_p^{\mathrm{LS}}(s) = \zeta_p(2s-1)/\zeta_p(2s) = (1-p^{-2s})/(1-p^{1-2s})$, is precisely the Casselman–Shalika 1980 formula for the unramified spherical Whittaker function at principal-series parameter s (Casselman–Shalika *Compositio Math.* 41 Thm 5.4 with w the non-trivial Weyl element). This identifies N_p^{LS} as the local intertwining operator normalisation, canonically and without convention choice.

Compatibility Check (iii): Discrete-Core Fixation

Proposition 84.3.5 *(LS correction preserves the discrete-core identity). (with the hypotheses of the discrete-core identity).* The discrete-spectrum Koszul–Petersson identity of Theorem 76.3.3 is preserved under the substitution $\langle\cdot,\cdot\rangle^{\mathrm{Koszul}} \rightsquigarrow \langle\cdot,\cdot\rangle^{\mathrm{Koszul,LS}}$ extended to the Eisenstein summand:

$$\langle v, w\rangle^{\mathrm{Pet}} \;=\; \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul,LS}} \qquad \forall v, w \in \mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}.$$

Proof. $\mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$ is orthogonal to the Eisenstein continuous summand in the Langlands decomposition (Langlands 1966 §6; Moeglin–Waldspurger 1995 Ch. II). The substitution $\langle\cdot,\cdot\rangle^{\mathrm{Koszul,LS}} = N^{\mathrm{LS}}(s)\cdot\langle\cdot,\cdot\rangle^{\mathrm{Koszul}}$ is defined only on the Eisenstein summand; on the discrete core, the pairing is unmodified: $\langle v, w\rangle^{\mathrm{Koszul,LS}} = \langle v, w\rangle^{\mathrm{Koszul}}$ for $v, w \in \mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}}$. Combining with Theorem 76.3.3 gives (84.3.5). □

The fixed scalar models satisfy $MK = P/2$, as shown in Remark 67.5.4. A phase correction alone supplies no equality of the continuous measures.

84.4 Full $\mathcal{V}^{\mathrm{prim}}$ Koszul–Petersson Identity (Including Eisenstein)

The three compatibilities give the full-sector identity.

Theorem 84.4.1 *(full Koszul–Petersson compatibility via LS normalisation).* Assume the arithmetic placement, local pairing maps and adelic extension of Hypothesis 67.1.1 on the selected discrete domain, including its orthogonal decomposition, as well as the continuous measure transport below. *(conditional on Ramanujan for Maass cusp forms on the Maass cuspidal discrete sector; conditional on the LS Plancherel-measure identification on the Eisenstein continuous sector).* For all $v, w \in \mathcal{V}^{\mathrm{prim}}$,

$$\langle v, w\rangle^{\mathrm{Pet}} \;=\; \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul,LS}}, \tag{84.2}$$

with $\langle\cdot,\cdot\rangle^{\mathrm{Koszul,LS}}$ the Langlands–Shahidi-corrected Koszul pairing of Definition 84.2.1 on the Eisenstein summand and the unmodified Koszul pairing on the discrete core. The identity extends Theorem 76.3.3 from the discrete core to the full $\mathcal{V}^{\mathrm{prim}}$ including the Eisenstein continuum under the stated LS normalisation hypothesis.

Proof. Decompose $\mathcal{V}^{\mathrm{prim}} = \mathcal{V}^{\mathrm{prim}}_{\mathrm{disc}} \oplus \mathcal{V}^{\mathrm{prim}}_{\mathrm{cont}}$ by Langlands orthogonality (Langlands 1966 §6). On the discrete core, Proposition 84.3.5 gives (84.2) with the arithmetic comparison hypotheses and the additional Maass temperedness assumptions inherited from Theorem 76.3.3. On the Eisenstein continuum, Corollary 84.2.4 supplies the conditional spectral density normalisation match, which by the Plancherel formula for the continuous spectrum (Moeglin–Waldspurger §IV.2) yields (84.2) on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cont}}$. Compatibility of the two on their intersection (zero, by orthogonality) is trivial. □

Corollary 84.4.2 *(Eisenstein density mismatch reduced by LS normalisation). (conditional on the LS Plancherel-measure identification).* The Eisenstein density mismatch of Proposition 76.3.1 and Corollary 67.5.5 is reduced under the LS redefinition Definition 84.2.1: the density mismatch on $\mathrm{Re}(s) = 1/2$ between Koszul and Petersson vanishes pointwise once the LS-corrected quadratic form is identified with the positive Plancherel measure. No Ramanujan input is used on the continuum, but the measure identification is an extra hypothesis.

Proof. Corollary 84.2.4 gives the pointwise density match. The domain of the modification is only the Eisenstein summand (Remark 84.3.2); the Tate factorisation (Proposition 84.3.3) and the Heisenberg self-duality (Proposition 84.3.1) confirm the redefinition is compatible with the rest of the arithmetic Koszul structure. □

84.5 Pairing Conventions and Hypotheses

Remark 84.5.1 *(pairing conventions and hypotheses).* Both sides of (84.2) are $\mathbb{C}$-valued sesquilinear pairings on $\mathcal{V}^{\mathrm{prim}} \times \mathcal{V}^{\mathrm{prim}}$. Equivalently, they are complex-linear functionals on $\mathcal{V}^{\mathrm{prim}} \otimes \overline{\mathcal{V}^{\mathrm{prim}}}$. The Koszul construction uses Beilinson–Drinfeld 2004, Positselski 2009, and Bost–Connes 1995. The Petersson construction uses Langlands 1966, Moeglin–Waldspurger 1995, Shahidi 2010, Casselman–Shalika 1980, Jacquet–Piatetski-Shapiro–Shalika 1983, Tate 1950, Weil 1967, and Titchmarsh 1986. These two source lists are disjoint.

The factor $N^{\mathrm{LS}}(s) = \xi(2s-1)/\xi(2s)$ is a unitary scattering normalisation on the critical line, with the conventions of Shahidi 2010, Theorem 3.5.1, and Moeglin–Waldspurger, §IV.3.10. A positive measure requires the Shahidi Plancherel transport in Theorem 84.2.2.

The identity of Theorem 84.4.1 requires the LS Plancherel-measure identification on the Eisenstein continuum and the arithmetic placement and pairing maps on the discrete sector, with Ramanujan additionally required for the Maass domain. These discrete hypotheses are those of Theorem 76.3.3. The Eisenstein summand is orthogonal to the discrete core under both Petersson (Langlands 1966, §6) and Koszul (Proposition 84.3.5). The LS correction is supported on the Eisenstein summand only.

Shahidi's global normalisation, the Casselman–Shalika local formula, and Moeglin–Waldspurger's unitarity on the critical line are classical theorems. They do not by themselves identify the bar–cobar Eisenstein pairing with the positive Plancherel measure.

Remark 84.5.2 *(discrete-core and Eisenstein normalisations).* Chapter 76 (the restricted-class comparison) reduces the Koszul–Petersson identity on the discrete core and excludes the Eisenstein continuum by domain restriction. The Li-criterion path (Corollary 76.3.6) uses this restriction only after the H-P^{Ar} determinant-trace realisation. The bounded-interval reduction gives a conditional extension to the continuum: one must insert the LS scalar N^{LS} and identify the resulting quadratic form with the positive Plancherel measure.

84.6 Sources

The primary sources enter in the following order.

Shahidi, F. *Eisenstein Series and Automorphic L-Functions*, AMS Colloquium Publications 58, 2010. The Langlands–Shahidi method for normalising intertwining operators; Theorem 3.5.1 for the global normalisation we denote $N^{\mathrm{LS}}(s)$.

Moeglin, C. and Waldspurger, J.-L. *Spectral Decomposition and Eisenstein Series: A Paraphrase of the Scriptures*, Cambridge Tracts 113, 1995. §II.1.7 for the intertwining operator $M(s)$; §IV.2 for the Plancherel formula; §IV.3.10 for unitarity of the normalised intertwiner on the critical line.

Langlands, R. P. *On the Functional Equations Satisfied by Eisenstein Series*, Lecture Notes in Math. 544, Springer 1976 (manuscript 1966). Foundational spectral decomposition of $L^2([GL_n])$ into discrete and continuous summands; Maass–Selberg relations; functional equation of Eisenstein series.

Casselman, W. and Shalika, J. *The unramified principal series of p-adic groups. II. The Whittaker function*, Compositio Math. 41 (1980), 207–231. Unramified spherical Whittaker formula used for the local factor $N_p^{\mathrm{LS}}(s) = \zeta_p(2s-1)/\zeta_p(2s)$ at unramified finite p.

Jacquet, H., Piatetski-Shapiro, I., and Shalika, J. *Rankin–Selberg convolutions*, Amer. J. Math. 105 (1983), 367–464. Local ε-factors at ramified principal series.

Tate, J. *Fourier analysis in number fields and Hecke's zeta functions*, Ph.D. thesis, Princeton 1950; reprinted Cassels–Fröhlich 1967 Ch. XV. Adelic restricted tensor product; local-global factorisation of completed zeta.

Weil, A. *Basic Number Theory*, Springer 1967. Tamagawa normalisation.

Titchmarsh, E. C. *The Theory of the Riemann Zeta-Function* (2nd ed., revised D. R. Heath-Brown), Oxford University Press 1986. Functional equation of ξ; Euler product for ζ.

Bost, J.-B. and Connes, A. *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. 1 (1995), 411–457. The Bost–Connes C^*-algebra; KMS_1 state; involution σ^{BC}.

Beilinson, A. and Drinfeld, V. *Chiral Algebras*, AMS Colloquium 51, 2004. Chiral bar–cobar construction and parabolic-induction compatibility (§3.5).

Positselski, L. *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence*, arXiv:0905.2621, 2009. Koszul duality input for the arithmetic bar–cobar.

Disjoint sources. The proof sources for the arithmetic Koszul pairing (Beilinson–Drinfeld 2004, Positselski 2009, Bost–Connes 1995) are disjoint from the analytic sources for the Langlands–Shahidi normalisation and Plancherel measure (Shahidi 2010, Moeglin–Waldspurger 1995, Langlands 1966, Casselman–Shalika 1980, Jacquet–Piatetski-Shapiro–Shalika 1983, Tate 1950, Weil 1967, Titchmarsh 1986). No source appears on both sides of the identification.

Chapter 85

L_{ACD} on the Eisenstein / Continuous Spectrum

The Eisenstein comparison begins with a local operator whose domain, normalization and degeneration can be computed. For unramified nonarchimedean principal series, the normalized intertwiner acts on every smooth vector, and on the unitary axis it extends to a unitary Hilbert operator. Algebraic coefficient categories then supply coherent Weyl descent and a common symmetric parameter across finite places. Tensor sectors retain mixed terms; a nonorthogonal decomposition must also retain its connecting spectra. These constructions do not supply the analytic automorphic enhancement or its comparison functor. The cuspidal comparison of Theorem 70.6.4 remains conditional, and the full Eisenstein destination is Conjecture 70.6.6.

85.1 Local Operators and Global Comparison Data

Inherited Objects

Prior statements deployed here:

- cuspidal sph-finite adelic categorical descent (Theorem 70.6.4), the cuspidal LACD comparison core theorem;
- L_{ACD}.0–.3 reductions (Chapter 52): archimedean admissibility, colimit commutation, cross-prime Hecke, sph-finite preservation;
- F2.c tightened restricted tensor product (Theorem 50.1.3) supplying the pointed E_2-monoidal base;
- Langlands–Shahidi correction factor definition (Definition 76.4.1) on principal-series inductions;
- discrete-core Koszul–Petersson identity (Theorem 76.3.3), conditional on the arithmetic placement, local pairing maps and adelic extension of Hypothesis 67.1.1. This discrete identity makes no assertion about the continuous pairing.

Relation to the Langlands–Shahidi normalisation

The Koszul–Petersson comparison of Proposition 76.4.2 requires a specified transform and measure identification. The local operator below is separate data: its normalization fixes the spherical vector. The comparison of a global scalar factor, spectral measure and transported pairing remains an additional construction, described in Proposition 85.8.5. No categorical lift follows merely from the equality of scalar normalizations.

Primary Sources

Langlands 1976 "On the functional equations satisfied by Eisenstein series", SLN 544, Part I §5–7. Foundational Eisenstein-series theory, intertwining operators, analytic continuation.

Moœglin–Waldspurger 1995 *Spectral Decomposition and Eisenstein Series*, Cambridge Tracts 113, Ch. II and IV. Full L^2 spectral decomposition, residual and continuous spectrum description; the normalising factor $M(s) = \xi(2s-1)/\xi(2s)$ appears at §II.1.7.

Shahidi 1990 "A proof of Langlands' conjecture on Plancherel densities and complementary series", Ann. Math. 132. Local Plancherel densities and their compatibility with normalised intertwining operators.

Shahidi 2010 *Eisenstein Series and Automorphic L-Functions*, AMS Colloquium 58, Ch. 2–4. The Langlands–Shahidi method: normalised intertwining operators $N(s, \pi_v, \tilde{w}_0)$, Plancherel–Shahidi measure, and adelic assembly (Thm 3.5.1).

Arthur 2013 *The Endoscopic Classification of Representations*, AMS Colloquium 61, Ch. 1–4. Discrete-spectrum classification and trace-formula framework. Its relation to the specific GL_2 quotient and Eisenstein transform requires the precise global hypotheses; Arthur–Clozel (1989) and Jacquet–Shalika (1981) concern the corresponding general-linear representation theory.

Jacquet–Langlands 1970 SLN 114 §5, §11. Strong multiplicity one for cuspidal spectrum; archimedean Plancherel for $GL_2(\mathbb{R})$.

Harish-Chandra 1976 *Harmonic analysis on real reductive groups III*, Ann. Math. 104, 117–201. The Plancherel theorem for real reductive groups. Part II concerns wave packets in Schwartz space.

Wallach 1988/1992 *Real Reductive Groups I–II*. Volume II (1992), Chapter 13, especially Section 13.4, treats Plancherel. Chapter 11 treats completions.

Bernstein–Zelevinsky (1977), *Induced representations of reductive p-adic groups I*, Ann. Sci. ENS 10, supplies the normalized induction convention and Theorem 4.2 irreducibility criterion used below. Lurie's presentable tensor and monadic results supply the categorical inputs with the precise locators given in the proofs. The extension-gluing statements below are proved directly.

Representation and categorical inputs

Flath's restricted tensor theorem concerns actual cuspidal representations. Langlands and Mœglin–Waldspurger concern global Eisenstein families and spectral decomposition. Shahidi concerns local factors and Plancherel normalization; Harish-Chandra and Wallach supply real-group harmonic analysis. Passing from those representation objects to an enhanced automorphic category requires a construction of the category, its functors, derived morphisms and generators. The algebraic parameter and gluing constructions below identify some of these inputs on their explicitly stated local carriers.

85.2 Module (E0): Langlands–Shahidi Intertwining Normalisation

Local Intertwining Data for GL_2

Setup 85.2.1 *(local induction convention).* For a finite place v, let q_v be the residue cardinality and ϖ a uniformizer. Write $q = q_v$. For unramified characters χ_1, χ_2, put

$$I_v(s, \chi) = \mathrm{Ind}_{B(\mathbb{Q}_v)}^{GL_2(\mathbb{Q}_v)}(\chi_1|\cdot|^{s/2} \otimes \chi_2|\cdot|^{-s/2}),$$

with normalized smooth induction. To apply the compact-picture construction below, use the effective inducing characters $\alpha_1 = \chi_1|\cdot|^{s/2}$ and $\alpha_2 = \chi_2|\cdot|^{-s/2}$ in place of its character arguments. Their ratio is $z = \alpha_1(\varpi)/\alpha_2(\varpi) = \chi_1(\varpi)\chi_2(\varpi)^{-1}q^{-s}$. Ramified characters and the archimedean place remain separate operator constructions with their own Haar, epsilon-factor and analytic-domain data.

85.3 Normalized intertwining for unramified principal series

An Eisenstein comparison must distinguish the normalization of an intertwining operator from its passage to a category of representations. For unramified principal series over a nonarchimedean field, the

operator can be computed on every locally constant vector. The computation also determines its poles and its degenerations.

Let F be a finite extension of $\mathbb{Q}_p$. Write O for its ring of integers, ϖ for a uniformizer, and $q = |O/\varpi O|$. Normalize $|\varpi| = q^{-1}$ and the additive Haar measure by $\mathrm{vol}(O) = 1$. Fix an additive character $\psi : F \to \mathbb{C}^\times$ that is trivial on O and nontrivial on $\varpi^{-1}O$. The operator below does not depend on ψ. This choice fixes the additive convention for subsequent local-factor comparisons. All representations and function spaces have coefficients in $\mathbb{C}$.

Put $G = \mathrm{GL}_2(F)$, $K = \mathrm{GL}_2(O)$, and let B be the upper triangular subgroup. For unramified smooth characters $\chi_1, \chi_2 : F^\times \to \mathbb{C}^\times$, let $I(\chi_1, \chi_2)$ consist of locally constant functions $f : G \to \mathbb{C}$ satisfying

$$f\left(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} g\right) = |a/d|^{1/2} \chi_1(a)\chi_2(d) f(g).$$

The action is right translation, $(\pi(g)f)(h) = f(hg)$. Thus $I(\chi_1, \chi_2)$ is normalized smooth induction. Set

$$z = \frac{\chi_1(\varpi)}{\chi_2(\varpi)}, \qquad w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad n(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

For $|z| < 1$, define

$$(M_z f)(g) = \int_F f(wn(x)g)\, dx. \tag{85.1}$$

The target is $I(\chi_2, \chi_1)$. The subscript z specifies the ratio parameter. The source and target characters remain part of the operator's data.

The compact picture and ball averages

Restriction to K identifies every $I(\chi_1, \chi_2)$ with

$$V = C^\infty(X; \mathbb{C}), \qquad X = (B \cap K)\backslash K = \mathbb{P}^1(F).$$

Here C^∞ means locally constant. The identification of X sends a matrix to its bottom row, up to scalar. It follows from $G = BK$ and the triviality of unramified characters on $O^\times$. A primitive row has at least one unit coordinate. For primitive representatives $u, v \in O^2$, define

$$\rho([u], [v]) = |\det(u, v)|.$$

This function is independent of the primitive representatives and is invariant under the right action of K. Let ν be the K-invariant probability measure on X.

For $n \geq 1$, reduction modulo ϖ^n gives a finite set

$$X_n = \mathbb{P}^1(O/\varpi^n O), \qquad |X_n| = (q+1)q^{n-1}.$$

Let $V_n \subset V$ be the functions that factor through X_n. Put $V_0 = \mathbb{C}\mathbf{1}$ and $H = L^2(X, \nu)$, with inner product linear in the first variable. The orthogonal projection $E_n : H \to V_n$ averages over each reduction fiber. The projection E_0 is global averaging. The fibers are the balls $\rho(x, y) \leq q^{-n}$, each of measure

$$b_n = \frac{1}{(q+1)q^{n-1}}.$$

Every locally constant function factors through some X_n. Consequently $V = \bigcup_{n\geq 0} V_n$ is dense in H. Define mutually orthogonal finite-dimensional spaces

$$H_0 = V_0, \qquad H_n = V_n \cap V_{n-1}^{\perp} \quad (n \geq 1).$$

Then

$$V = \bigoplus_{n\geq 0}^{\mathrm{alg}} H_n, \qquad H = \widehat{\bigoplus}_{n\geq 0} H_n.$$

In particular, $\dim H_1 = q$ and $\dim H_n = (q+1)(q-1)q^{n-2}$ for $n \geq 2$.

LEMMA 85.3.1 *(Integral kernel and meromorphic continuation).* For $|z| < 1$, the integral in (85.1) converges absolutely. It defines a smooth G-intertwiner. In the compact picture it is

$$M_z = (1+q^{-1})E_0 + (z-q^{-1})\sum_{n\geq 1} z^{n-1}E_n. \tag{85.2}$$

The series converges in operator norm on H. On the orthogonal summands it acts by

$$M_z|_{H_0} = c(z), \qquad c(z) = \frac{1-q^{-1}z}{1-z}, \qquad M_z|_{H_n} = \frac{(z-q^{-1})z^{n-1}}{1-z} \quad (n \geq 1).$$

These formulas give a meromorphic continuation on every V_N. On every nonzero V_N, its only pole in $z \in \mathbb{C}^\times$ is the simple pole at $z = 1$. Here meromorphic continuation means continuation on each finite level, not a family of bounded operators on H for all parameters.

Proof. Let $x_0 = [(0,1)]$. For $|x| \leq 1$, the matrix $wn(x)$ belongs to K. For $|x| > 1$, the factorization

$$wn(x) = \begin{pmatrix} -x^{-1} & 1 \\ 0 & x \end{pmatrix}\begin{pmatrix} 1 & 0 \\ x^{-1} & 1 \end{pmatrix}$$

shows that the inducing factor on $|x| = q^n$ is $q^{-n}z^n$. The shell has additive measure $(1-q^{-1})q^n$. Since compact-picture functions are bounded, the tail is dominated by $a\sum_{n\geq 1}|z|^n$ for a finite constant a. The same argument applies to each right translate of f.

The integral commutes with right translations. Changing x to $x+y$ proves invariance under left multiplication by $n(y)$. For $a, d \in F^\times$, use

$$wn(x)\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} = \begin{pmatrix} d & 0 \\ 0 & a \end{pmatrix} wn(xd/a).$$

The substitution $u = xd/a$ contributes $|a/d|$. The resulting factor is $|a/d|^{1/2}\chi_2(a)\chi_1(d)$, which is the target induction law. Right local constancy is preserved by the integral. Thus M_z is the stated smooth intertwiner.

The chart $[(1,x)]$, $x \in \mathcal{O}$, has measure $q(q+1)^{-1}dx$. The chart $[(y,1)]$, $y \in \varpi\mathcal{O}$, has the same density. For example, reduction modulo ϖ^N divides either chart into sets of additive measure q^{-N} and ν-measure $((q+1)q^{N-1})^{-1}$. This proves the density assertion on locally constant functions.

In the second chart, the substitution $y = x^{-1}$ has Jacobian $dx = |y|^{-2}dy$. On $|y| = q^{-n}$, the resulting density in the raw integral is $q^n z^n\, dy$. Thus its kernel, relative to ν, has value $1+q^{-1}$ when $\rho(x_0, y) = 1$, and value $(1+q^{-1})q^n z^n$ when $\rho(x_0, y) = q^{-n}$. The diagonal has measure zero. Right K-equivariance gives the same radial kernel about every point.

For $\rho(x, y) = q^{-m}$, telescope the kernel as

$$(1 + q^{-1})\left[1 + \sum_{n=1}^{m}\left(q^n z^n - q^{n-1} z^{n-1}\right)\right].$$

Integration of the nth ball indicator is $b_n E_n$. Since $(1 + q^{-1})b_n = q^{-n}$, its coefficient is $(z - q^{-1})z^{n-1}$. Absolute convergence justifies the sum and proves (85.2). The projection norms are one, so the series also converges in operator norm.

On H_0, every E_n acts as the identity. On H_m, $m \geq 1$, the projection E_n is zero for $n < m$ and is the identity for $n \geq m$. Summing the geometric series gives the displayed eigenvalues. Each V_N is a finite sum of these spaces. The rational formulas therefore give its meromorphic continuation. At $z = 1$, every numerator has nonzero value $1 - q^{-1}$. The pole is simple on every nonzero summand. □

Normalization, composition, and domains

The spherical vector is $\mathbf{1} \in H_0$. The scalar in the preceding lemma equals the unramified local ratio

$$c(z) = \frac{L(0, \chi_1\chi_2^{-1})}{L(1, \chi_1\chi_2^{-1})}, \qquad L(u, \chi_1\chi_2^{-1}) = (1 - zq^{-u})^{-1}.$$

Normalizing the spherical vector therefore requires division by $c(z)$. Define the meromorphic family

$$R_z = c(z)^{-1} M_z.$$

Theorem 85.3.2 *(The normalized unramified intertwiner).* The compact-picture eigenvalues of R_z are

$$r_0(z) = 1, \qquad r_n(z) = z^{n-1}\frac{z - q^{-1}}{1 - q^{-1}z} \quad (n \geq 1). \tag{85.3}$$

The operator is holomorphic on each V_N away from $z = q$. It has a simple pole on every H_n with $n \geq 1$ at $z = q$. The value at $z = 1$ is id_V. The value at $z = q^{-1}$ is E_0. For $z \notin \{q, q^{-1}\}$, the smooth maps satisfy

$$R_{z^{-1}} R_z = \mathrm{id}_V, \qquad R_z R_{z^{-1}} = \mathrm{id}_V .$$

Here the second operator uses the swapped inducing characters.

If χ_1 and χ_2 are unitary, R_z extends uniquely to a unitary intertwiner between their induced Hilbert representations on H. Its Hilbert domain is all of H. In particular, the equal-character point $z = 1$ is regular.

Proof. Division of the preceding eigenvalues by $c(z)$ gives (85.3). This calculation cancels the apparent singularity at $z = 1$. At $z = q$, the denominator has a simple zero and the numerator is nonzero on every H_n, $n \geq 1$. Substitution gives the stated values at 1 and q^{-1}.

For $n \geq 1$, direct cancellation gives

$$r_n(z) r_n(z^{-1}) = \frac{(z - q^{-1})(1 - q^{-1}z)}{(1 - q^{-1}z)(z - q^{-1})} = 1.$$

The identity also holds on H_0. Every smooth vector has finitely many components, so both compositions are defined on all of V at the stated parameters. Intertwining holds first on the convergence domain of the raw integral. On each smooth vector and each fixed group element, the compact-picture formulas

are rational in the two nonzero unramified character values. For a fixed $g \in G$, right translation sends V_N into some V_M independently of the unramified character values. Indeed, an open subgroup of $K \cap gK_N g^{-1}$ fixes the translated vectors, where $K_N = 1 + \varpi^N M_2(O)$ for $N \geq 1$. On the resulting finite partition of K, the Iwasawa factors have fixed diagonal valuations. Their character values are Laurent monomials. The integral intertwining identity therefore continues meromorphically. This also proves intertwining at every regular specialization.

If the inducing characters are unitary, $|z| = 1$. Then $|z - q^{-1}| = |1 - q^{-1}z|$, so $|r_n(z)| = 1$ for every n. The orthogonal decomposition of H gives a unique unitary extension. Its inverse has parameter z^{-1}.

For completeness, the induced action preserves the stated Hilbert norm. In the open Bruhat chart put $F_f(x) = f(wn(x))$. For unitary inducing characters, the chart measures above give

$$\|f|_K\|_H^2 = \frac{q}{q+1} \int_F |F_f(x)|^2 \, dx.$$

Right translation by $n(y)$ gives $F_f(x + y)$. Right translation by $\mathrm{diag}(a, d)$ gives

$$|d/a|^{1/2} \chi_1(d) \chi_2(a) F_f(xd/a).$$

Right translation by w, away from $x = 0$, gives

$$|x|^{-1} \chi_1(-x^{-1}) \chi_2(x) F_f(x^{-1}).$$

Translations, dilations, and inversion preserve the displayed norm with these factors and their respective Jacobians. These elements generate G. Each smooth vector has an open stabilizer. Density and unitarity therefore imply strong continuity of the extended action. The smooth equivariance of R_z extends by density to these induced unitary representations. □

For nonunitary parameters, the same compact-picture formulas also specify an operator on a fixed Hilbert space. This assertion does not equip the induced representation with an invariant positive inner product.

PROPOSITION 85.3.3 *(Closed compact-picture realization).* For $z \neq q$, write $h = \sum_{n \geq 0} h_n$ with $h_n \in H_n$. The closure of $R_z|_V$ on H is the normal operator with domain

$$\mathrm{Dom}(\overline{R}_z) = \left\{ h \in H : \sum_{n \geq 0} |r_n(z)|^2 \|h_n\|^2 < \infty \right\}, \qquad \overline{R}_z h = \sum_{n \geq 0} r_n(z) h_n.$$

The space V is a core. This operator is bounded exactly when $|z| \leq 1$. The composition assertions of the preceding theorem hold on V. On the unit circle they hold on all of H.

Proof. The orthogonal sum of scalar operators with the displayed maximal domain is closed and normal. Its adjoint has coefficients $\overline{r_n(z)}$ and the same domain. Truncating the expansion after $n = N$ converges in its graph norm. Thus this maximal operator is the closure of its restriction to V. Boundedness is equivalent to $\sup_n |r_n(z)| < \infty$. Formula (85.3) proves boundedness for $|z| \leq 1$. For $|z| > 1$ and $z \neq q$, the nonzero factor $(z - q^{-1})/(1 - q^{-1}z)$ is fixed and $|z|^{n-1}$ is unbounded. □

Reducibility and the unitary axis

The poles of a normalization and the reducibility of the induced representation are different loci. They can both be determined in the present unramified family.

Proposition 85.3.4 *(Reducibility locus).* The smooth representation $I(\chi_1, \chi_2)$ is reducible exactly when

$$\chi_1\chi_2^{-1} = |\cdot| \quad \text{or} \quad \chi_1\chi_2^{-1} = |\cdot|^{-1}.$$

Equivalently, $z \in \{q^{-1}, q\}$. For $z = q^{-1}$, the regular intertwiner R_z has image H_0 in $I(\chi_2, \chi_1)$ and kernel in $I(\chi_1, \chi_2)$ equal to $\bigoplus_{n\geq 1}^{\mathrm{alg}} H_n$. For $z = q$, the swapped regular intertwiner $R_{z^{-1}}$ has image H_0 in $I(\chi_1, \chi_2)$. The stated kernel at $z = q^{-1}$ and image at $z = q$ are nonzero proper invariant subspaces of $I(\chi_1, \chi_2)$.

Proof. The kernel and image assertions follow from (85.3) and the regular intertwining identity. They prove reducibility at the two displayed ratios. Away from these ratios, apply Bernstein–Zelevinsky's irreducibility criterion to the two characters of $\mathrm{GL}_1(F)$: *Induced representations of reductive p-adic groups I*, Theorem 4.2, p. 454, with the induction conventions of §1.8(a), p. 444, and §§2.2–2.3, p. 446. For the upper triangular parabolic, their modulus on the unipotent radical is $|a/d|$, so their square-root factor is the one used here. Every character of $\mathrm{GL}_1(F)$ is irreducible and cuspidal. The criterion excludes exactly the relations $\chi_j = |\cdot|\chi_i$ for distinct i, j. Thus it proves irreducibility on the stated complement. Only this direction of their criterion is used. □

For the symmetric family $\chi_1 = |\cdot|^{s/2}$, $\chi_2 = |\cdot|^{-s/2}$, one has $z = q^{-s}$. The raw integral converges for $\Re(s) > 0$. Its poles occur at

$$s \in \frac{2\pi i}{\log q}\mathbb{Z}.$$

On every smooth finite level, the normalized operator is regular on $\Re(s) \geq 0$. Its poles occur at

$$s \in -1 + \frac{2\pi i}{\log q}\mathbb{Z},$$

and reducibility occurs at

$$s \in \left(1 + \frac{2\pi i}{\log q}\mathbb{Z}\right) \cup \left(-1 + \frac{2\pi i}{\log q}\mathbb{Z}\right).$$

Neither reducibility locus meets the unitary axis $\Re(s) = 0$. These statements concern local unramified induction. Poles of a global Eisenstein series require a separate global construction.

Example 85.3.5 *(The first nonconstant vectors).* Let $q = 2$ and $z = -1$. The three reduction fibers of $X \to \mathbb{P}^1(\mathbb{F}_2)$ have equal measure. On V_1, the raw integral and normalized operator are

$$M_{-1} = \frac{1}{4}\begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}, \qquad R_{-1} = \frac{1}{3}\begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}.$$

The spherical scalar is $c(-1) = 3/4$. The normalized matrix fixes constants and negates their orthogonal complement. Its square is the identity. At $z = 1/2$, the normalized matrix is the rank-one averaging projection. At $z = 1$, its continued value is the identity matrix.

The automorphic comparison

A categorical Eisenstein comparison requires categories and functors in addition to these operators. One must specify the coefficient field, the local parameter spaces, and the global automorphic quotient with its central character and level. The global category must carry its Hecke actions and its Eisenstein subcategory. A comparison functor must identify the relevant derived mapping objects and determine its essential image. The shared global induction parameter must also be imposed on local families.

The construction above supplies the full smooth intertwiner and its Hilbert extension only for unramified nonarchimedean induction. Ramified characters and the archimedean place require their own operators, normalizations, and analytic domains. A global restricted product requires compatible distinguished vectors and an explicit global family. The categorical comparison further requires functors implementing these maps and their Weyl coherences. No equivalence of automorphic categories follows from the compact-picture Hilbert isomorphism alone.

The place-indexed operator notation

Definition 85.3.6 *(the raw place-indexed operator).* For the base characters $\chi = (\chi_1, \chi_2)$ of Setup 85.2.1, put $\alpha_1 = \chi_1|\cdot|^{s/2}$, $\alpha_2 = \chi_2|\cdot|^{-s/2}$ and $z = \alpha_1(\varpi)/\alpha_2(\varpi)$. With $\chi^w = (\chi_2, \chi_1)$, the notation is

$$M_v(s,\chi) = M_z : I_v(s,\chi) = I(\alpha_1,\alpha_2) \longrightarrow I(\alpha_2,\alpha_1) = I_v(-s,\chi^w).$$

The integral is defined for $|z| < 1$ and continued on each smooth finite level by Lemma 85.3.1.

Definition 85.3.7 *(the normalized place-indexed operator).* On the same source and target, define

$$N_v(s,\chi) = R_z = c(z)^{-1}M_z, \qquad c(z) = \frac{1-q^{-1}z}{1-z}.$$

Cancellation and continuation at $z = 1$ are part of Theorem 85.3.2; the expression is not a pointwise product of undefined values there. Its poles, domains and unitary specializations are those proved in that theorem.

85.4 Algebraic parameters for unramified induction

The normalized unramified intertwiner varies with the inducing characters. A category of coefficient complexes records this variation and its Weyl symmetry. Its relation to smooth induction is a functor defined by tensoring with universal smooth induction. The resulting category concerns local algebraic parameters. An automorphic comparison requires further constructions.

Retain the nonarchimedean field F, residue cardinality q, group $G = \mathrm{GL}_2(F)$, and compact subgroup K of the unramified calculation. In particular, $|\varpi| = q^{-1}$ and $\mathrm{vol}(O) = 1$. Let a, b denote the values at ϖ of the two unramified characters. Their ratio is $z = a/b$. To make both normalized intertwining directions regular and invertible, put

$$A = \mathbb{C}[a^{\pm1}, b^{\pm1}, (a-qb)^{-1}, (b-qa)^{-1}], \qquad T = \operatorname{Spec} A.$$

Thus T omits precisely the two divisors $z = q, q^{-1}$. The Weyl group $W = \{1, w\}$ acts through the ring involution $\sigma(a) = b$, $\sigma(b) = a$. The involution exchanges the two inverted divisors. At a complex point (a_0, b_0), define $\chi_{a_0}(x) = a_0^{v_F(x)}$, where $v_F(\varpi) = 1$. The corresponding representation is the normalized induction $I(\chi_{a_0}, \chi_{b_0})$.

The ratio alone determines the normalized operator in the compact picture. It does not determine the representation. Replacing (a_0, b_0) by (ta_0, tb_0) twists the representation by $\chi_t \circ \det$. For example, the central element $\mathfrak{C}$ id acts by $a_0 b_0$ before the twist and by $t^2 a_0 b_0$ afterwards. Both characters must therefore remain in the parameter space.

Coefficient complexes and their unit

For a ring D, write $\mathcal{D}(D)$ for the stable ∞-category of unbounded cochain complexes of left D-modules, with quasi-isomorphisms inverted. The differential has degree 1. We use $P[1]^n = P^{n+1}$ with differential $-d_P$. For commutative D, tensor products are derived over D. Their symmetry has sign $(-1)^{ij}$ on degrees i, j. Let $\mathrm{Perf}(D)$ be a small skeleton of the full subcategory generated by $D[0]$ under finite sums, shifts, cones, and retracts.

Proposition 85.4.1 *(The coefficient category).* The category $\mathrm{Perf}(A)$ is stable, idempotent complete, and symmetric monoidal under $\otimes^{\mathbf{L}}_A$. There is a canonical equivalence

$$\mathrm{Ind}(\mathrm{Perf}(A)) \simeq \mathcal{D}(A).$$

The compact objects of $\mathcal{D}(A)$ are exactly $\mathrm{Perf}(A)$. The monoidal unit is $A[0]$. The functor

$$e : \mathcal{D}(\mathbb{C}) \longrightarrow \mathcal{D}(A), \qquad U \longmapsto A \otimes_{\mathbb{C}} U$$

is a colimit-preserving symmetric monoidal pointing and preserves compact objects.

Proof. The category of derived modules is stable and presentable. For a commutative ring its symmetric monoidal identification with modules over the associated discrete ring spectrum is established in Lurie, *Higher Algebra* (2017), Theorem 7.1.2.13, pp. 1212–1213. Its shifts of A detect zero objects, since $\mathrm{Hom}_{\mathcal{D}(A)}(A[-n], P) = H^n(P)$. These shifts are compact because cohomology commutes with filtered colimits of modules. The compact-generation theorem for module categories identifies their finite stable retract closure with all compact objects and gives the Ind equivalence (Lurie, *Higher Algebra* (2017), Definition 7.2.4.1 and Proposition 7.2.4.2, pp. 1260–1261). For ordinary rings these are bounded complexes of finitely generated projective modules (The Stacks Project, Tag 07LQ, Lemma 15.80.1 and Proposition 15.80.3). Tensoring two such complexes gives another such complex. The associativity, unit, and symmetry are those of derived tensor product. Extension of scalars preserves colimits and takes a finite complex of finite-dimensional vector spaces to a finite free complex. Its tensor and unit identities prove the assertion about e. □

A point of T gives the specialization functor $\mathbb{C} \otimes^{\mathbf{L}}_{A,(a_0,b_0)} (-)$. The point module $A/(a - a_0, b - b_0)$ is also perfect. Its length-two Koszul resolution follows because $a - a_0$, $b - b_0$ form a regular sequence. Indeed, the quotient by the first element is a localization of $\mathbb{C}[b^{\pm 1}]$, and the second element is nonzero there. The inverted elements remain nonzero at the chosen point. This point module differs from the monoidal unit A. For two distinct points, its tensor product with the other point module is zero, as follows by localizing at their disjoint supports. The unit cannot have this property.

The universal smooth representation

Let $X = (B \cap K)\backslash K$ and $V = C^\infty(X; \mathbb{C})$, where B is the upper triangular subgroup and functions are locally constant. The compact-picture calculation gives $V = \bigoplus_{n \geq 0} H_n$ and $V_N = \bigoplus_{n=0}^{N} H_n$, with $H_0 = \mathbb{C}\mathbf{1}$ and finite-dimensional H_n. Put $E = A \otimes_{\mathbb{C}} V$ and $E_N = A \otimes V_N$. All these modules are in cochain degree zero.

PROPOSITION 85.4.2 *(Induction over the parameter ring).* There is an A-linear smooth action π_A of G on E. Its specialization at $(a_\circ, b_\circ)$ is $I(\chi_{a_\circ}, \chi_{b_\circ})$. The normalized intertwiner defines an A-linear isomorphism

$$R : E \longrightarrow E^{\mathrm{sw}}, \qquad R|_{A\otimes H_\circ} = 1, \qquad R|_{A\otimes H_n} = z^{n-1}\frac{z - q^{-1}}{1 - q^{-1}z} \quad (n \geq 1),$$

where E^{sw} has the inducing parameters b, a. Its inverse has the displayed coefficients with z replaced by z^{-1}. The coefficient module E is free and is not compact in $\mathcal{D}(A)$. Every E_N is compact and K-stable.

Proof. Use the universal unramified characters $\chi_a(x) = a^{v_F(x)}$ and $\chi_b(x) = b^{v_F(x)}$ in the induction law. The factor $|x/y|^{1/2}$ is a fixed positive real scalar. For $g \in G$, extend a function from K to $BK = G$ by this law, then translate it on the right by g. Independence of the Iwasawa factorization follows because the characters and modulus are trivial on $B \cap K$. For each fixed g, N, compactness and local constancy give a finite partition on which all diagonal valuations are constant. The resulting coefficients are Laurent monomials in a, b. Thus $\pi_A(g)E_N \subset E_M$ for some finite M. Right translation satisfies the group law before restriction to K, and consequently defines an action on E. Each vector has an open stabilizer in K. Specializing the induction law proves the fiber assertion.

The stated coefficients belong to A and are units in A. The normalized intertwiner calculation proves the intertwining identity and $r_n(z^{-1})r_n(z) = 1$. These identities apply over A: for fixed g, N they are finite matrix identities in rational functions of a, b. They hold at every regular complex specialization, hence hold in A. This proves the asserted isomorphism and inverse.

Each E_N is finite free. The decomposition of V makes E free of infinite rank. If E were compact, its identity would factor through some term of $E = \operatorname{colim}_N E_N$. That would express an infinite-rank free module as a retract of a finite-rank module, which is impossible after tensoring with the fraction field of A. The compact-picture K-action preserves each V_N. The full G-action need not preserve these finite levels. □

Let $\mathcal{D}_{\mathrm{sm}}(G, A)$ denote the derived ∞-category of smooth A-linear G-representations. Here smooth means that every vector has an open stabilizer, and G acts trivially on A. Tensoring with the flat module E gives an actual functor

$$\mathcal{F} : \mathcal{D}(A) \longrightarrow \mathcal{D}_{\mathrm{sm}}(G, A), \qquad P \longmapsto E \otimes_A P.$$

It is well-defined on quasi-isomorphisms since E is free over A. Its specialization is $I(\chi_{a_\circ}, \chi_{b_\circ}) \otimes_{\mathbb{C}} (\mathbb{C} \otimes^{\mathbf{L}}_A P)$. The spherical map $A \to E, c \mapsto c\mathbf{1}$, is K-equivariant. It is generally not G-equivariant when A has trivial G-action. For example, the central element ϖ id acts on E by ab. Thus neither the spherical vector nor the representation E replaces the coefficient category's monoidal unit.

Weyl descent

The involution σ acts on coefficient complexes by restriction of scalars. On the underlying complex the new scalar action is $c \cdot m = \sigma(c)m$. Applying this operation twice is the identity. It preserves derived tensor products and the unit, with their usual canonical identifications. This is a specified coherent action of W on $\mathcal{D}(A)$. Its descent category is the homotopy limit $\mathcal{D}(A)^{hW}$.

PROPOSITION 85.4.3 *(An explicit descent category).* Let $D = A \rtimes W = A \oplus Au$, with multiplication given by $u^2 = 1$ and $uc = \sigma(c)u$ for $c \in A$. There is an equivalence

$$\mathcal{D}(A)^{hW} \simeq \mathcal{D}(D).$$

This category is stable and presentable. An object is compact if and only if its underlying A-complex is perfect. Its symmetric monoidal product is derived tensor over A, with diagonal Weyl action. Its compact monoidal unit is A with $u(c) = \sigma(c)$.

Proof. A strict D-module is an A-module with a semilinear involution θ: $\theta(cm) = \sigma(c)\theta(m)$ and $\theta^2 = 1$. The same description applies to complexes with a chain involution. To identify the derived category with coherent descent, consider the evaluation functor from the homotopy limit to $\mathcal{D}(A)$. It is conservative and has the free-equivariant left adjoint $P \mapsto P \oplus \sigma^* P$ with its permutation action. Colimits are computed on underlying complexes since σ^* preserves them. In particular, evaluation preserves geometric realizations of split simplicial objects. The monadicity theorem therefore applies (Lurie, *Higher Algebra* (2017), Theorem 4.7.3.5, p. 685). Its monad is $P \mapsto P \oplus \sigma^* P$, with multiplication from W. This is also the monad of restriction from D-modules to A-modules, whose left adjoint is $D \otimes^{\mathbf{L}}_A (-)$. Restriction is conservative and preserves geometric realizations. The two monadic descriptions prove the equivalence, including all higher coherences.

As a left or right A-module, D is free of rank two. Thus every perfect D-complex becomes perfect over A. Conversely, a D-complex M is a retract of $D \otimes_A M$. The action map is a retraction of the D-linear chain map

$$m \longmapsto \tfrac{1}{2}(1 \otimes m + u \otimes \theta(m)).$$

The relations $uc = \sigma(c)u$ and $\theta^2 = 1$ verify linearity directly. If the underlying A-complex is perfect, extension of scalars makes $D \otimes^{\mathbf{L}}_A M$ perfect over D. The retract is therefore perfect. The compact-object characterization in the coefficient proposition applies to D as well. The diagonal involution on $M \otimes^{\mathbf{L}}_A N$ is $\theta_M \otimes \theta_N$. Semilinearity makes it well-defined, and the usual tensor constraints are equivariant. The module A with involution σ is the unit and is perfect over A. This proves its compactness after descent. □

The involution is not free on T. Every point (a_0, a_0) is fixed and belongs to T. The modules A with involutions σ and $-\sigma$ specialize there to the trivial and sign representations of W. These are not isomorphic: an intertwining scalar would satisfy $c = -c$. Consequently descent retains stabilizer representations. It cannot be replaced by the category of modules on the coarse orbit space through ordinary pullback, which has trivial stabilizer action on every fiber.

The representation E has its own compatible descent datum. Let $S : E \to E$ apply σ to coefficients and fix V. Define $\Theta = SR$, using the common compact-picture module. Then

$$\Theta(cf) = \sigma(c)\Theta(f), \qquad \Theta^2 = 1, \qquad \Theta\pi_A(g) = \pi_A(g)\Theta.$$

Indeed, $S\pi_A(g) = \pi_A^{\mathrm{sw}}(g)S$ and $R\pi_A(g) = \pi_A^{\mathrm{sw}}(g)R$. Conjugating R by S replaces z by z^{-1}, so its inverse is obtained. These identities prove all three assertions. Moreover, $\Theta(c\mathbf{1}) = \sigma(c)\mathbf{1}$. For an equivariant coefficient complex (P, θ), the formula $(E \otimes_A P, \Theta \otimes \theta)$ therefore defines equivariant smooth induction. This is a construction of a functor, without an assertion of full faithfulness or essential surjectivity.

A common complex parameter

For the symmetric inducing characters $|\cdot|^{s/2}$, $|\cdot|^{-s/2}$, the two parameters are $a = q^{-s/2}$ and $b = q^{s/2}$. Here $q^s = \exp(s \log q)$, with the real logarithm of q. The ratio $z = q^{-s}$ has period $2\pi i/\log q$. The ordered pair (a, b) has period $4\pi i/\log q$. A shift by the smaller period changes both character values by -1. This explains why the ratio does not encode the whole family.

Let $\Omega = \{s \in \mathbb{C} : |\operatorname{Re} s| < 1\}$ and $B_\Omega = \mathcal{O}(\Omega)$, treated as an ordinary commutative ring. There is a ring map

$$\phi_q : A \longrightarrow B_\Omega, \qquad a \longmapsto e^{-s \log q/2}, \qquad b \longmapsto e^{s \log q/2}.$$

The inverted factors have no zeros in Ω because $|q^{-s}| = q^{-\operatorname{Re} s}$ lies strictly between q^{-1} and q. Their reciprocals are holomorphic, which proves that the map is defined. The involution $\tau(h)(s) = h(-s)$ satisfies $\phi_q \sigma = \tau \phi_q$. Derived extension of scalars gives a compact-preserving functor

$$B_\Omega \otimes^{\mathbf{L}}_A (-) : \mathcal{D}(A) \longrightarrow \mathcal{D}(B_\Omega)$$

and an equivariant version on descent categories. The latter sends (P, θ) to the complex with involution $h \otimes p \mapsto \tau(h) \otimes \theta(p)$. All finite-level matrices of E and R pull back to holomorphic matrices. Evaluating at s gives exactly the smooth representation and normalized operator at $z = q^{-s}$. On $s \in i\mathbb{R}$, the previously constructed Hilbert extension is unitary. No topology on B_Ω or Hilbert completion is part of this scalar-extension functor.

For a finite set of finite places S, allow independent local character parameters and put $A_S = \bigotimes_{v \in S, \mathbb{C}} A_v$. The common symmetric family gives a single map

$$\phi_S : A_S \longrightarrow B_\Omega, \qquad \bigotimes_v f_v \longmapsto \prod_v \phi_{q_v}(f_v).$$

It imposes the same variable s at every place. For $S \subset S'$, scalar extension along $A_S \to A_{S'}$ supplies compatible coefficient functors, and $\phi_{S'}$ restricts to ϕ_S. Thus these maps also define a map from the filtered union of the rings A_S. They give actual base-change functors for independent local parameters. The compatible Weyl action is the simultaneous swap at all places. An individual local swap generally does not preserve the common family.

For example, with $q_2 = 2$ and $q_3 = 3$, the independent unitary ratios $(z_2, z_3) = (1, -1)$ do not come from any common s. The first equation forces $s = 2\pi i n / \log 2$ for an integer n. The second would give $2n \log 3 = (2m + 1) \log 2$ for an integer m, after changing signs of the indices if necessary. Exponentiation contradicts unique factorization of rational numbers. Thus tensoring independent unitary axes does not impose the global parameter.

After common base change, put $E_{v,\Omega} = B_\Omega \otimes_{A_v} E_v$. Write $R_{v,\Omega}$ for the scalar extension of R_v. The local semilinear involution $\Theta_v = S_v R_v$ induces

$$\Theta_{v,\Omega}(h \otimes f) = \tau(h) \otimes \Theta_v(f).$$

This formula is balanced because $\phi_{q_v} \sigma_v = \tau \phi_{q_v}$. It squares to the identity, commutes with G_v, and fixes $\mathbf{1}_v$. For finite S, put $E_{S,\Omega} = \bigotimes_{v \in S, B_\Omega} E_{v,\Omega}$. The B_Ω-linear operator $\bigotimes_{v \in S} R_{v,\Omega}$ has target the family with every inducing pair swapped. The semilinear involution on $E_{S,\Omega}$ is instead defined by

$$\Theta_S\Big(\bigotimes_{v \in S} f_v\Big) = \bigotimes_{v \in S} \Theta_{v,\Omega}(f_v).$$

For $S = \varnothing$, set $E_{S,\Omega} = B_\Omega$ and $\Theta_S = \tau$. For nonempty S, moving a scalar between two tensor factors changes both resulting expressions by its image under τ. Thus the displayed map is balanced and τ-semilinear. Applying it twice gives the identity on pure tensors, hence on the whole module. It commutes with the action of $\prod_{v \in S} G_v$. Insertion of $\mathbf{1}_v$ defines compatible maps as S grows. These maps intertwine Θ_S and $\Theta_{S'}$ because each $\Theta_{v,\Omega}$ fixes the inserted vector. This also holds for the insertion from the empty

set, by semilinearity. The algebraic filtered colimit is the restricted tensor product of these smooth families over finite places, relative to $\mathbf{1}_v$. It therefore inherits a semilinear involution from the maps Θ_S. It carries the restricted product group relative to K_v: outside finitely many places an element belongs to K_v and fixes $\mathbf{1}_v$. Every tensor and every group action is computed at some finite stage. No infinite product of operator norms is needed for this algebraic construction.

The automorphic comparison problem

The common family above fixes the two global inducing characters to be trivial before the powers of the absolute value are inserted. For general inducing characters, their local components must arise from specified Hecke characters, and the Weyl action must also swap that character pair. Independent local characters do not by themselves supply these global data.

An analytic coefficient category still requires a choice of sheaf theory, topology, growth conditions, and derived completion on a complex parameter domain. The ring category $\mathcal{D}(B_\Omega)$ specifies none of those choices. An automorphic target further requires the adelic quotient, central character, level, archimedean factor, and an Eisenstein functor with analytic domains. To compare that target with the parameter category, one must construct the functor on complexes and compare its derived mapping objects and essential image. Compactness of coefficient complexes does not prove admissibility or finite generation over local Hecke algebras after induction. The local smooth family, its coherent Weyl descent, and common-parameter base change provide the input maps for this comparison. They do not identify a category of automorphic representations.

Remark 85.4.4 *(global normalizations and the pairing comparison).* The local operator $N_v(s,\chi) = R_z$ divides the raw operator by its spherical scalar $c(z)$. A global product requires Hecke characters, compatible spherical insertions, the archimedean and ramified factors, and an analytic continuation on a specified global family. In a classical Eisenstein parameter u with unnormalized spherical factor $\xi(2u-1)/\xi(2u)$, the symmetric induction variable is $s = 2u - 1$. The global epsilon product and its phase must be computed for that character and additive convention; it is not universally 1. An operator, a scalar phase and a Plancherel density are different data. The conditional Koszul–Petersson transport in Proposition 76.4.2 requires its own measure identity. If an even measure on the full axis has density $\frac{1}{2}w(t)dt$, pushforward under $t \mapsto |t|$ has density $w(t)dt$ off zero. This folding identity does not determine w.

Definition 85.4.5 *(local degeneration and the regular unitary locus).* For the symmetric unramified family at a finite place, define

$$\Sigma_v^{\mathrm{exc}} = \left(1 + \frac{2\pi i}{\log q_v}\mathbb{Z}\right) \cup \left(-1 + \frac{2\pi i}{\log q_v}\mathbb{Z}\right).$$

These are reducibility parameters. The normalized operator has poles only on the second set and is a projection at the first set. The non-exceptional unitary locus is all of $i\mathbb{R}$. The point 0 is regular and 1 is not on the unitary axis.

Remark 85.4.6 *(local degeneration and global residues).* A residue of a global Eisenstein series is not a pole of every normalized local operator. Its parameter, global quotient and central character must be specified. The local kernel/image sequences at $z = q^{\pm 1}$ are given by Proposition 85.3.4; comparison of their extension classes with arithmetic deformation data remains a separate coefficient-compatible construction.

85.5 MODULE (E1): PALEY–WIENER SYNTHESIS ON THE UNITARY AXIS

The Global Spectral Transform

Conjecture 85.5.1 *(global continuous spectral comparison).* Fix a unitary central character ω and the Hilbert quotient $L^2(GL_2(\mathbb{Q})\backslash GL_2(\mathbb{A}), \omega)$ modulo the centre. Choose the level, archimedean types and Haar measures. The intended Eisenstein transform must identify its continuous part with a direct integral

$$L^2_{\mathrm{cont}}([GL_2], \omega) \simeq \bigoplus_{[\chi]} \int_{i\mathbb{R}_{\geq 0}}^{\oplus} V_{\chi,s}\, d\mu_\chi(s),$$

where $[\chi]$ ranges over the appropriate Weyl classes of global unitary Hecke-character pairs with central character ω. The measurable fibres $V_{\chi,s}$, the spectral measure μ_χ and the synthesis domain must be specified. Finite-dimensional spherical or Iwahori levels are not the full induced representations. A Paley–Wiener test space, its support and growth conditions, and its dense embedding in this Hilbert space are additional data. The shared parameter and global characters must be imposed before comparing local families, and the Weyl identification must use the actual normalized global operator.

The global spectral references are Langlands (1976), Part I, Sections 5–7, and Mœglin–Waldspurger (1995), Chapters II and IV. Shahidi's local normalization, Harish-Chandra's real-group Plancherel theorem and Wallach's treatment require their local hypotheses. They do not make an arbitrary Hilbert completion into a Paley–Wiener space or prove a universal ξ-density for all character pairs.

Remark 85.5.2 *(a scalar functional-equation identity).* For the completed Riemann function satisfying $\xi(s) = \xi(1-s)$ and $\xi(\bar{s}) = \overline{\xi(s)}$, one has $|\xi(2it)| = |\xi(1+2it)|$: apply the functional equation and then complex conjugation. Thus the two expressions $|\xi(2it)|^{-2}$ and $|\xi(1+2it)|^{-2}$ agree wherever defined. This algebraic identity does not identify either expression with the measure $d\mu_\chi$ above. The change from a classical parameter u to symmetric induction is $s = 2u - 1$, and its Jacobian and normalizing factors must be retained in any pairing comparison. A unit-modulus scattering phase is not an absolute-value spectral density.

Categorical Lift of the Paley–Wiener Space

Definition 85.5.3 *(data for an analytic continuous-spectrum category).* The intended analytic category requires a complex parameter domain containing the relevant unitary parameters, a sheaf theory, a topology, growth conditions and a derived completion, together with its local representation family. Write, provisionally,

$$C_v^{\mathrm{cont}} = D(\mathcal{A}_v^{\mathrm{an}}),$$

where $\mathcal{A}_v^{\mathrm{an}}$ must be constructed with the claimed admissibility and local Hecke-finiteness properties. The half-axis $i\mathbb{R}_{\geq 0}$ is not itself a complex manifold. Coherent finite-type objects are not closed under arbitrary colimits, so their Ind-completion, if used, must be specified. A spherical representation vector is also distinct from the coefficient unit and from a point module. In particular, $1/2$ is not on the unitary axis.

PROPOSITION 85.5.4 *(conditional presentability and pointing).* If an actual Grothendieck abelian coefficient category $\mathcal{A}_v^{\mathrm{an}}$ is supplied, its unbounded derived category is stable and presentable. Compact generation and a compact monoidal unit require separate proofs for that choice. For the bounded algebraic alternative already constructed, these assertions hold for $\mathcal{D}(A)$ and its unit A as in Proposition 85.4.1; this does not identify $\mathcal{D}(A)$ with the intended analytic category.

Proof. The derived category of a Grothendieck abelian category is stable and presentable (Lurie, Higher Algebra, Section 1.3.5). The algebraic assertions follow from the explicit module and perfect-complex proof of Proposition 85.4.1. No compactness assertion about the unspecified analytic sheaf theory is used. □

Conjecture 85.5.5 *(analytic Weyl descent).* Construct a coherent Weyl action on the intended analytic restricted tensor category, with a specified functor

$$N : \bigotimes_v{}' C_v^{\mathrm{cont}} \longrightarrow \bigotimes_v{}' C_v^{\mathrm{cont}},$$

and its involution coherence, compatible with the global inducing characters, common parameter and distinguished insertions. The already constructed coefficient action is $\mathcal{D}(A)^{hW} \simeq \mathcal{D}(A \rtimes W)$; its equivariant smooth induction and simultaneous common-parameter involution are explicit. An individual local swap need not preserve that common parameter. A representation intertwiner is not itself an E_∞ morphism of categories, and the analytic action above is additional to the algebraic coefficient action.

Theorem 85.5.6 *(conditional enhanced Paley–Wiener comparison).* Construct the analytic automorphic category and a global-character, common-parameter version of the analytic tensor category. Denote these intended enhanced carriers by

$$\mathcal{A}_{\mathrm{Eis}}^{\text{sph-fin}} \longrightarrow \left(\bigotimes_v{}' C_v^{\mathrm{cont}} \right)^{hW}.$$

If the arrow is an exact colimit-preserving functor between compactly generated stable presentable categories, takes chosen compact generators to compact objects, induces equivalences on their full mapping spectra, and has generating image, it is an equivalence. Its arithmetic realization must also intertwine the global spectral synthesis, Hecke actions and analytic domains.

Proof. Apply Theorem 70.4.1 to the specified functor. Constructing the Hilbert synthesis, analytic sheaf enhancement, restricted tensor insertions and coherent Weyl action are the four separate prerequisites. A unitary map of Hilbert spaces does not supply the enhanced functor or force degeneration of a descent spectral sequence. The explicit coefficient Weyl proof above uses a monadic description and averaging over a group of order two over $\mathbb{C}$, not a Hilbert-unitarity argument. □

85.6 Module (E2): Langlands Spectral Decomposition Lifted Categorically

The Three Summands

Theorem 85.6.1 *(Langlands spectral decomposition of $L^2([GL_2])$).* For the fixed unitary central-character quotient and Haar measure of Conjecture 85.5.1, the automorphic L^2-space decomposes as a unitary direct sum

$$L^2([GL_2]) \;=\; L^2_{\mathrm{cusp}} \;\oplus\; L^2_{\mathrm{res}} \;\oplus\; L^2_{\mathrm{cont}},$$

with:

- L^2_{cusp} the cuspidal subspace (Bernstein 1988; Mœglin–Waldspurger III.2.1);

- L^2_{res} the residual spectral subspace, indexed by the actual residual automorphic representations for the chosen quotient; the constant function gives a residual line when the central character and quotient allow it (Mœglin–Waldspurger II.2);
- L^2_{cont} the continuous spectrum, parametrised by Weyl-orbits of unitary Hecke characters via Eisenstein series (Langlands 1976 §7; Mœglin–Waldspurger IV.3).

The decomposition is canonical and respects the adelic Hecke action and the archimedean $(\mathfrak{g}, K_\infty)$-structure.

Source. Langlands 1976 §7 (original proof, functional equations); Mœglin–Waldspurger 1995 Ch. II and IV (modern full account). □

Categorical Lift of Each Summand

Definition 85.6.2 *(the coefficient model of one residual line).* For a chosen one-dimensional residual character over $\mathbb{C}$, the coefficient category is $\mathcal{D}(\mathbb{C})$, with compact unit $\mathbb{C}[0]$ and tensor product over $\mathbb{C}$. Its unit endomorphism ring is $\mathbb{C}$. It is not the category of spectra, whose sphere has degree-zero endomorphism ring $\mathbb{Z}$. This coefficient model does not identify the full residual spectrum with a single line.

THEOREM 85.6.3 *(conditional realization of the residual comparison).* The restricted tensor product of copies of $\mathcal{D}(\mathbb{C})$ in complex-linear presentable stable categories, with the unit insertions, is $\mathcal{D}(\mathbb{C})$. Its identification with an actual residual automorphic category additionally requires a chosen residual representation, its compatible local factors and a comparison functor with generating image and equivalences on full mapping spectra. For several residual representations their extension and assembly data must also be supplied.

Proof. Every finite tensor product of the complex-linear tensor unit is that unit, and the insertion maps are the unit identifications. Their filtered colimit is the unit. For the actual residual category apply the compact-generator criterion to the specified comparison; the coefficient calculation does not provide its hypotheses. □

THEOREM 85.6.4 *(conditional continuous-spectrum comparison).* On a specified analytic parameter domain, including the full unitary axis for symmetric unramified induction, the continuous-spectrum comparison is an equivalence under the complete enhanced functor, mapping-spectrum, compactness and generation hypotheses of Theorem 85.5.6.

Proof. Apply that theorem on the chosen domain with its restricted analytic functor and support conditions. The local operator has no unitary-axis poles in the stated unramified family. Algebraic scalar extension to B_Ω proves the coefficient construction, but does not replace the analytic enhancement or supply its automorphic essential image. □

PROPOSITION 85.6.5 *(the sph-finite comparison requirement).* The scalar-extension functor $B_\Omega \otimes^{\mathbf{L}}_{A} (-)$ preserves perfect coefficient complexes, and its equivariant version preserves compact objects. Preservation of the intended automorphic sph-finiteness requires, in addition, the actual induction/analytic functors to preserve the specified admissibility and finite-type local Hecke conditions. These conditions must also be stable under the proposed restricted tensor transition maps and Weyl descent.

Proof. A bounded complex of finitely generated projective modules remains such after derived scalar extension. For Weyl descent the underlying perfect-complex criterion proved above applies over A and over B_Ω, where 2 is invertible. These facts concern coefficient compactness. The universal smooth family E is infinite free and noncompact, so this argument asserts no automatic Hecke-finiteness after induction or arbitrary filtered colimits. □

85.7 Tensor sectors and extension data

An orthogonal decomposition into local sectors gives a direct-sum decomposition of their tensor product. Even with two sectors at each of two places, tensor distributivity gives four terms:

$$(\mathcal{A}_0 \oplus \mathcal{A}_1) \otimes (\mathcal{B}_0 \oplus \mathcal{B}_1) \simeq (\mathcal{A}_0 \otimes \mathcal{B}_0) \oplus (\mathcal{A}_0 \otimes \mathcal{B}_1) \oplus (\mathcal{A}_1 \otimes \mathcal{B}_0) \oplus (\mathcal{A}_1 \otimes \mathcal{B}_1).$$

The middle two terms record different sector choices at the two places. A relation between those choices requires additional structure.

Orthogonal decompositions

Let $\mathcal{V}$ be a symmetric monoidal ∞-category with small colimits. Assume its tensor product preserves colimits separately in each variable. Write $\mathbf{1}$ for its tensor unit and $\coprod$ for its categorical coproduct. The following argument applies to the ∞-category of presentable stable ∞-categories and colimit-preserving functors. Its tensor product is characterized by functors preserving colimits separately in each variable (Lurie, *Higher Algebra* (2017), Remarks 4.8.1.18 and 4.8.1.24, Proposition 4.8.2.18). For finite families of stable categories, this coproduct is their orthogonal direct sum, denoted $\oplus$.

Indeed, in a finite product $\prod_i C_i$, every object is the finite direct sum of its components. A colimit-preserving functor out of this product is therefore determined by its restrictions to the factors. Conversely, functors $F_i : C_i \to \mathcal{D}$ extend by $(x_i)_i \mapsto \bigoplus_i F_i(x_i)$. This proves the coproduct universal property. Distinct factors have zero mapping spectra.

Proposition 85.7.1 *(Finite sector decomposition).* Let S be a finite set. For each $v \in S$, fix a finite set I_v and a specified equivalence $C_v \simeq \coprod_{r \in I_v} C_{v,r}$ in $\mathcal{V}$. There is an equivalence, induced by the given sector inclusions,

$$\bigotimes_{v \in S} C_v \simeq \coprod_{\alpha \in \prod_{v \in S} I_v} \bigotimes_{v \in S} C_{v,\alpha(v)}.$$

For $S = \varnothing$, both sides are $\mathbf{1}$.

Proof. For one factor the assertion is the specified decomposition. Tensoring that decomposition with another factor preserves its coproduct. Applying the same fact to the new factor gives the Cartesian product of the two index sets. Induction proves the formula. The maps are induced by the sector inclusions, so the equivalence respects maps that preserve the specified decompositions. □

Write $D(k)$ for the stable ∞-category of unbounded cochain complexes of k-vector spaces, with quasi-isomorphisms inverted. For example, in complex-linear presentable stable categories, put $\mathcal{A}_i = \mathcal{B}_j = D(\mathbb{C})$ and tensor over $D(\mathbb{C})$. This category is the tensor unit, so the result is $D(\mathbb{C})^{\oplus 4}$. Keeping only the two equal-index factors gives $D(\mathbb{C})^{\oplus 2}$. These categories are not equivalent. Their compact objects have Grothendieck groups $\mathbb{Z}^4$ and $\mathbb{Z}^2$, respectively. Here the Grothendieck group imposes the relation $[y] = [x] + [z]$ for each exact triangle $x \to y \to z$. In $\mathrm{Perf}(\mathbb{C})$, splitting a bounded complex into its cohomology identifies this group with $\mathbb{Z}$, by Euler characteristic. Finite products give the stated ranks. An equivalence preserves compact objects and exact triangles, hence would preserve these groups.

The restricted tensor construction also depends on the sector of each insertion. Let V be a small set. For each $v \in V$, fix a finite nonempty set I_v, an element $o_v \in I_v$, and a specified decomposition of C_v as above. Fix a morphism $e_v : \mathbf{1} \to C_{v,o_v}$ in $\mathcal{V}$. For finite $S \subset V$, set

$$C_S = \bigotimes_{v \in S} C_v.$$

For $S \subset T$, tensor with the morphisms e_v, followed by sector inclusion, for $v \in T \setminus S$. The monoidal coherence maps give a diagram on the poset of finite subsets of V. Define its restricted product by

$$C^{\mathrm{res}} = \operatorname*{colim}_{S \subset V,\, S \text{ finite}} C_S.$$

This is an algebraic colimit in $\mathcal{V}$. No analytic or topological completion is implicit.

Let I^{fin} consist of functions α with $\alpha(v) \in I_v$ whose support $\operatorname{supp}(\alpha) = \{v : \alpha(v) \neq o_v\}$ is finite. For such α, define

$$C^{\mathrm{res}}_\alpha = \operatorname*{colim}_{\substack{S \subset V \text{ finite} \\ S \supset \operatorname{supp}(\alpha)}} \bigotimes_{v \in S} C_{v,\alpha(v)},$$

with the same distinguished insertions.

PROPOSITION 85.7.2 *(Restricted sector decomposition).* Under these hypotheses, the sector inclusions induce an equivalence

$$C^{\mathrm{res}} \simeq \coprod_{\alpha \in I^{\mathrm{fin}}} C^{\mathrm{res}}_\alpha.$$

Proof. For each α, extend its displayed diagram to all finite S by assigning the initial object when S does not contain its support. Call the resulting diagram E_α. At a fixed finite S, only functions supported in S contribute noninitial terms. The finite sector decomposition therefore gives $C_S \simeq \coprod_\alpha E_\alpha(S)$. The insertion lies in sector o_v, so the transition maps preserve this identification of diagrams. Colimits commute with coproducts. Thus

$$\operatorname{colim}_S C_S \simeq \coprod_\alpha \operatorname{colim}_S E_\alpha(S).$$

The finite sets containing a fixed support form a cofinal subposet: $S \mapsto S \cup \operatorname{supp}(\alpha)$ supplies a finite upper bound compatible with inclusions. Each inner colimit is consequently C^{res}_α. □

A general insertion need not lie in one sector. If it has several components, its transition maps need not preserve the finite-support indexing above. The finite sector formula still holds. The restricted colimit must then retain the actual transition maps between those finite decompositions.

A nonsplit extension

Orthogonal direct sums exclude extensions between distinct factors. The quiver $1 \to 2$ gives a concrete extension that must be retained when orthogonality is absent. Let k be a field, with no restriction on its characteristic. A representation is a linear map $u : A \to B$ of k-vector spaces. A morphism is a pair (f, g) satisfying $gu = u'f$. Kernels and cokernels are computed at the two vertices.

Consider

$$S_1 = (k \to 0), \qquad S_2 = (0 \to k), \qquad P = (k \xrightarrow{1} k).$$

There is an exact sequence of representations

$$0 \longrightarrow S_2 \longrightarrow P \longrightarrow S_1 \longrightarrow 0.$$

It does not split: a morphism $S_1 \to P$ has source component $f : k \to k$ satisfying $f = 0$. It cannot be a section. Moreover, $\mathrm{Hom}(P, (A \to B)) \cong A$ and $\mathrm{Hom}(S_2, (A \to B)) \cong B$. Both functors are exact, so P and S_2 are projective. The sequence is therefore a projective resolution of S_1. Applying $\mathrm{Hom}(-, S_2)$ gives

$$\mathrm{Hom}(P, S_2) = 0, \qquad \mathrm{Hom}(S_2, S_2) = k, \qquad \mathrm{Ext}^1(S_1, S_2) = k.$$

The displayed sequence represents the image of $1 \in k$ under the connecting isomorphism. Projectivity of S_2 also gives $\mathrm{Ext}^n(S_2, S_1) = 0$ for every $n \geq 0$. Thus one direction is orthogonal, while the opposite direction carries an extension.

Equivalently, these are left modules over

$$T = \left\{ \begin{pmatrix} a & 0 \\ m & b \end{pmatrix} : a, b, m \in k \right\}, \qquad \begin{pmatrix} a & 0 \\ m & b \end{pmatrix}\begin{pmatrix} a' & 0 \\ m' & b' \end{pmatrix} = \begin{pmatrix} aa' & 0 \\ ma' + bm' & bb' \end{pmatrix}.$$

The two diagonal idempotents recover A and B. The lower-left matrix unit acts by $u : A \to B$. Associativity gives exactly the representation relations. Replacing this algebra by $k \times k$ removes its lower-left entry and the extension above.

Stable gluing

To retain extensions in a stable category, the connecting morphism must be part of each object. Let $\mathcal{A}$ and $\mathcal{B}$ be stable ∞-categories and let $F : \mathcal{A} \to \mathcal{B}$ be exact. All mapping objects below are spectra, written $\mathrm{map}_{\mathcal{A}}(-,-)$ and similarly for other categories. The shift $[1]$ denotes suspension. For cochain complexes it has $K[1]^n = K^{n+1}$ and differential $-d_K$.

Define $\mathcal{G}(F)$ to be the pullback

$$\mathcal{G}(F) = (\mathcal{A} \times \mathcal{B}) \underset{\mathcal{B}\times\mathcal{B}}{\times} \mathrm{Fun}(\Delta^1, \mathcal{B}),$$

where the first map is $(a, b) \mapsto (Fa, b)$ and the second records the source and target of an arrow. Thus an object is (a, b, u) with $u : Fa \to b$. A morphism consists of $f : a \to a'$, $g : b \to b'$, and a specified homotopy from $u'Ff$ to gu. The pullback retains all higher coherences.

Proposition 85.7.3 *(Mapping spectra and extensions).* The category $\mathcal{G}(F)$ is stable, with finite limits and colimits computed in the two components. For $x = (a, b, u)$ and $x' = (a', b', u')$,

$$\mathrm{map}_{\mathcal{G}(F)}(x, x') \simeq \mathrm{fib}\Big(\mathrm{map}_{\mathcal{A}}(a, a') \oplus \mathrm{map}_{\mathcal{B}}(b, b') \xrightarrow{d} \mathrm{map}_{\mathcal{B}}(Fa, b') \Big), \qquad d(f, g) = u'Ff - gu. \tag{85.4}$$

The exact fully faithful functors

$$i(b) = (0, b, 0), \qquad j(a) = (a, 0, 0)$$

satisfy

$$\mathrm{map}(i(b), j(a)) = 0, \qquad \mathrm{map}(j(a), i(b)) \simeq \Omega\, \mathrm{map}_{\mathcal{B}}(Fa, b) \simeq \mathrm{map}_{\mathcal{B}}(Fa[1], b).$$

Every object has a natural exact triangle

$$i(b) \longrightarrow (a, b, u) \longrightarrow j(a) \xrightarrow{\delta_u} i(b)[1].$$

Under the induced equivalence $\mathrm{map}(j(a), i(b)[1]) \simeq \mathrm{map}_{\mathcal{B}}(Fa, b)$, the connecting morphism δ_u corresponds to u. The choice of sign in this last equivalence is fixed by this exact triangle.

Proof. Exactness of F allows componentwise finite limits and colimits to supply the required arrows. The zero components give a zero object. Pushout squares are pullback squares componentwise, proving stability. A map between two arrows is a homotopy-commutative square. Its mapping space is the homotopy equalizer of the two compositions $u'Ff$ and gu. The same statement after every suspension gives the homotopy equalizer of mapping spectra. In spectra this is the fiber of their difference, which proves (85.4).

For two objects in the image of i, the formula reduces to $\mathrm{map}_{\mathcal{B}}$. For two in the image of j, it reduces to $\mathrm{map}_{\mathcal{A}}$. The two cross formulas follow by substituting the zero components. In particular, the fiber of $0 \to E$ is ΩE, not E. The cofiber of $i(b) \to (a, b, u)$ is $(a, 0, 0)$ componentwise. Its connecting morphism includes the homotopy encoded by u. The suspension equivalence identifies that homotopy with u and fixes the stated sign convention. □

The triangle splits precisely when δ_u is nullhomotopic, equivalently when u is nullhomotopic. Indeed, a nullhomotopy identifies its fiber with $i(b) \oplus j(a)$. Conversely, a splitting makes the connecting morphism zero in the homotopy category. When $F = 0$, $\mathcal{G}(F) \simeq \mathcal{A} \times \mathcal{B}$. Taking $\mathcal{A} = \mathcal{B} = D(k)$ and $F = \mathrm{id}$ gives the arrow category of complexes. Here $h\mathcal{G}(F)$ denotes the homotopy category. For $a = b = k$ in degree zero, the second cross formula gives

$$\mathrm{Hom}_{h\mathcal{G}(F)}(j(k), i(k)[1]) \cong k.$$

The object $(k, k, 1)$ gives its nonzero connecting class. This recovers the extension mechanism in the quiver calculation, including its degree.

Reconstruction from a semiorthogonal filtration

The same description applies to a category whose extension structure is already specified. Let C be stable and let $i : \mathcal{B} \to C$ and $j : \mathcal{A} \to C$ be exact fully faithful functors. Assume

$$\mathrm{map}_C(i(b), j(a)) = 0 \quad \text{for all } a \in \mathcal{A},\ b \in \mathcal{B}.$$

Assume every $x \in C$ admits an exact triangle

$$i(b) \longrightarrow x \longrightarrow j(a) \longrightarrow i(b)[1].$$

These hypotheses state the orientation of the semiorthogonal filtration. They are stronger than generation by the two subcategories alone.

Let $\mathcal{E}$ be the full pullback of the arrow category of C whose arrows have source $j(a)$ and target $i(b)[1]$. Its objects are triples (a, b, δ) with $\delta : j(a) \to i(b)[1]$. Then

$$\mathrm{map}_{\mathcal{E}}((a, b, \delta), (a', b', \delta')) \simeq \mathrm{fib}\big(\mathrm{map}_{\mathcal{A}}(a, a') \oplus \mathrm{map}_{\mathcal{B}}(b, b') \longrightarrow \mathrm{map}_C(j(a), i(b')[1])\big),$$

where the map sends (f, g) to $\delta' j(f) - i(g)[1]\delta$.

Proposition 85.7.4 *(Semiorthogonal reconstruction).* Under these hypotheses, the functor

$$\mathcal{E} \longrightarrow C, \qquad (a, b, \delta) \longmapsto \mathrm{fib}(\delta)$$

is an equivalence. Suppose, in addition, that i has an exact left adjoint $L : C \to \mathcal{B}$. Put $F = (Lj)[-1]$. Then $C \simeq \mathcal{G}(F)$, with the given embeddings i, j.

Proof. A connecting arrow has a fiber triangle $i(b) \to \mathrm{fib}(\delta) \to j(a)$. The filtration hypothesis gives essential surjectivity. To prove full faithfulness, consider two such triangles with middle objects x, x'. Applying $\mathrm{map}_C(-, j(a'))$ to the first triangle gives an equivalence

$$\mathrm{map}_C(j(a), j(a')) \xrightarrow{\sim} \mathrm{map}_C(x, j(a')),$$

because the mapping spectrum from $i(b)$ to $j(a')$ is zero. Consequently a map $h : x \to x'$ determines, with contractible choices, a map $j(f) : j(a) \to j(a')$ and a homotopy between $j(f)(x \to j(a))$ and $x \xrightarrow{h} x' \to j(a')$. Taking fibers gives a map $i(g) : i(b) \to i(b')$. Taking cofibers supplies the compatible square of connecting arrows. Conversely, a square of connecting arrows gives h by taking fibers. The fiber and cofiber universal properties make these constructions inverse on mapping spaces, including all their higher homotopies. Thus the functor is fully faithful.

For the additional assertion, adjunction and exactness give

$$\mathrm{map}_C(j(a), i(b)[1]) \simeq \mathrm{map}_{\mathcal{B}}(Lj(a), b[1]) \simeq \mathrm{map}_{\mathcal{B}}(F(a), b).$$

This natural equivalence identifies objects, squares, and their higher coherences in $\mathcal{E}$ with those in $\mathcal{G}(F)$. □

In particular, a recollement can supply such a filtration without supplying an orthogonal direct sum. Use the convention $i_* : \mathcal{B} \to C$ for the closed embedding and $j_! : \mathcal{A} \to C$ for the open extension functor. The recollement adjunctions include $i^* \dashv i_* \dashv i^!$ and $j_! \dashv j^* \dashv j_*$. Its defining vanishing $j^* i_* = 0$ gives $\mathrm{map}(j_! a, i_* b) = 0$, and its exact triangle is

$$j_! j^* x \longrightarrow x \longrightarrow i_* i^* x \longrightarrow j_! j^* x[1].$$

The first assertion of the proposition applies with the roles of the closed and open categories interchanged. It retains the connecting arrow $i_* i^* x \to j_! j^* x[1]$. It does not require a left adjoint to $j_!$ or assert one exists.

The semiorthogonal description becomes an orthogonal direct sum if the other cross mapping spectrum also vanishes. Then every connecting spectrum is zero, and $\mathcal{E} \simeq \mathcal{A} \times \mathcal{B}$. Without this vanishing, tensor assembly must retain the gluing functors or connecting spectra and their composition data. The two sector names alone do not determine the category.

For automorphic categories, applying these propositions requires actual local decompositions or semiorthogonal filtrations and specified insertion functors. An identification with a global automorphic category also requires a comparison functor preserving the Hecke actions, parameter constraints, and extension data. Neither tensor distributivity nor semiorthogonal reconstruction constructs that comparison functor.

85.8 MODULE (E3): ASSEMBLY COMPATIBLE WITH CUSPIDAL LACD

Full Statement

THEOREM 85.8.1 *(conditional Eisenstein categorical assembly).* Suppose the actual local categories, their pointed restricted tensor construction and global automorphic enhancement are supplied. Impose the global Hecke-character and common-parameter conditions and the analytic domains of the intended comparison. Suppose both sides have specified orthogonal decompositions, or specified finite semiorthogonal filtrations retaining the connecting spectra. Require component equivalences and coherent natural equivalences of the connecting mapping-spectrum bifunctors, compatible with composition and each successive filtration stage. For the restricted sector formula, require the pointings to lie in their specified single sectors; otherwise retain the entire finite-stage decompositions and their actual transition maps, together with a compatible equivalence of these filtered diagrams. Then these data induce an equivalence

$$\left(\bigotimes_{v}{}' C_v\right)^{\text{glob,nonexc}} \xrightarrow{\sim} \mathrm{D}^{\text{sph-fin}}([GL_2])^{\text{nonexc}}.$$

Here the global superscript imposes the specified automorphic support; independent local families do not impose it automatically. The non-exceptional condition uses the actual local degeneration divisors and global synthesis domain. For the symmetric unramified family the local degeneration strata lie off the unitary axis. To obtain the structured comparison, require compatibility with:

(i) finite-place Whittaker functors;
(ii) Bernstein-centre and cross-prime Hecke actions;
(iii) the archimedean $(\mathfrak{g}, K_\infty)$ action;
(iv) normalized intertwiners, with their domains and Weyl coherences;
(v) global Paley–Wiener synthesis and its measure normalization.

The cuspidal, residual and continuous category comparisons remain separate hypotheses; their Hilbert spectral names do not construct the local categorical decomposition.

Proof. First retain the conditional cuspidal globalization of Theorem 70.6.4. Second supply the residual comparison of Theorem 85.6.3 on the chosen residual family. Third supply the enhanced continuous comparison of Theorem 85.6.4. Fourth, in the orthogonal case apply the finite sector formula at each stage. For sector-pure pointings the restricted sector formula then applies. For impure pointings, take the filtered colimit of the specified equivalences of full finite-stage diagrams, retaining their actual transition maps. In the semiorthogonal case use Proposition 85.7.4; the component equivalences and compatibility of connecting spectra induce an equivalence of the arrow pullback categories, hence of the original categories. Iterate this argument for the stipulated finite filtration. Finally the five specified natural compatibility maps give the stated structured comparison. The local operator or Hilbert orthogonality alone supplies none of these component or gluing hypotheses. □

Corollaries and degeneration data

COROLLARY 85.8.2 *(conditional adelic descent on the specified domain).* Under all the categorical, analytic, global-support and gluing hypotheses of Theorem 85.8.1, its comparison realizes the intended L_{ACD} on that domain. Extending it to other parameters requires their actual intertwiners and extension data.

Proof. The theorem identifies the two specified global carriers and their five structures. Restricting those maps to the stated domain gives this assertion; no omitted local or global sector is inferred. □

Conjecture 85.8.3 *(arithmetic comparison at degeneration strata).* Construct a coefficient-compatible enhanced comparison between the kernel/image extension data at the actual reducibility strata of local induction and the corresponding arithmetic deformation strata. For symmetric unramified finite-place induction these are $z = q^{\pm 1}$, equivalently the periodic translates of $s = \pm 1$. There is no reducibility at $s = 0$ and neither family meets the unitary axis. Global Eisenstein residue objects require their own global parameter and support calculation. The intended extension to full adelic categorical descent must match full derived morphisms and composition, including but not limited to the degree-one class. A coefficient bridge from the complex analytic family to the p-adic determinant-identified spectral stack is part of this problem.

Remark 85.8.4 *(the domain of the degeneration comparison).* The smooth local kernel and image are explicit in Proposition 85.3.4. The quiver and stable gluing calculations above show how a connecting class changes a category even when its two pieces are fixed. They do not identify that class with an Emerton–Gee cotangent class. The wild comparison in Conjecture 68.7.3 is an analogous arithmetic question with its own coefficient data; neither comparison follows from equality of extension dimensions.

Compatibility with the Langlands–Shahidi normalisation

Proposition 85.8.5 *(conditional Koszul–Petersson transport).* Suppose a global synthesis transform T on a specified common test space and a spectral measure μ have been constructed, and the Koszul pairing and Petersson pairing both pull back under T to $\int f\overline{g}\, d\mu$ with the same scalar normalization. Then T identifies these pairings on that space. Extending this identification to completions additionally requires density and continuity for the stated norms. The normalized local operator R_z is one input to constructing such data; its unitarity is not a formula for μ.

Proof. Both pullbacks have the same integral value for each pair of tests. If the common test space is dense and the pairings continuous, the equality extends by taking convergent test sequences. This uses the stated measure and normalization hypotheses, not an identification of an operator with an absolute-value scalar. □

85.9 Comparison and Hypotheses

The representation-theoretic inputs have different mathematical roles. Flath (1979) factorizes actual cuspidal representations. Bernstein (1988) and Mœglin–Waldspurger (1995), Section III, concern the cuspidal spectrum. Langlands (1976), Shahidi (1990, 2010), Arthur (2013), Harish-Chandra (1976), Wallach (1992) and Mœglin–Waldspurger, Sections II and IV, concern global or local harmonic analysis with their respective quotient, level and character hypotheses. Lurie's presentable tensor products provide a categorical ambient after its objects and insertion functors have been constructed.

The unramified calculation fixes $|\varpi| = q^{-1}$, additive measure $\mathrm{vol}(O) = 1$, normalized induction and the spherical vector. Ramified and archimedean epsilon conventions and a global product remain additional data. The relation $s = 2u - 1$ and any measure Jacobian must be included when comparing to a classical Eisenstein parameter. The global epsilon product need not be 1.

The algebraic Weyl construction is a specified coherent involution on coefficient complexes and smooth families, with one simultaneous swap after common-parameter base change. It does not provide the analytic sheaf theory, Hecke-finiteness or a global automorphic essential image. A Hilbert direct sum

does not make its enhanced categories orthogonal. The required mapping spectra and connecting arrows are the objects of the gluing propositions above. The arithmetic comparison at degeneration strata is additional even after all these global objects exist.

Composite with the cuspidal comparison

THEOREM 85.9.1 *(conditional composite adelic comparison).* Given the cuspidal globalization, residual comparison and analytic continuous-spectrum comparison, together with the complete component and extension compatibility hypotheses of Theorem 85.8.1, their assembly gives the adelic categorical comparison on the specified global support and parameter domain. Extension to the full arithmetic F1 comparison also requires its local categorical and geometric Bernstein-centre maps.

Proof. Apply the assembly theorem to these particular components and gluing maps. Its proof retains all mixed tensor terms or the actual semiorthogonal extension data. Composing with the additional specified local and geometric comparisons gives their conditional consequence. □

85.10 THE EISENSTEIN COMPARISON PROBLEM

Conjecture 85.10.1 *(the arithmetic comparison beyond the local constructions).* The full F1 construction must retain the following problems:

(i) ordinary and non-ordinary local DEG essential images at $p \geq 5$;
(ii) restrictions outside the global Taylor–Wiles hypotheses;
(iii) actual small-prime block functors, support and full derived comparison, including the wild extension-class map;
(iv) global Eisenstein synthesis, analytic enhancement and its categorical comparison, including the degeneration data of Conjecture 85.8.3;
(v) the actual global restricted-tensor support, overlap and extension diagram and its local-global comparison;
(vi) the geometric Bernstein-centre gluing constructions in Chapter 64.

The local results here do not reduce the full Eisenstein question to two extension classes. Proving exhaustion by any list of conditions requires the common carrier and its gluing diagram.

85.11 EISENSTEIN BOUNDARY

The normalized unramified operator is constructed on all smooth vectors, with maximal Hilbert domain stated and a unitary extension on the unit circle. Its equal-character value is regular; reducibility occurs at $z = q^{\pm 1}$. The coefficient category, coherent Weyl descent and common symmetric finite-place family are algebraic constructions. Their analytic and automorphic realizations remain separate. The finite and restricted tensor formulas include mixed sectors. Stable reconstruction retains the connecting spectrum when orthogonality is absent. These results provide actual objects and maps for the conditional global comparison; they do not establish that comparison or a full global-centre identification.

References for the local constructions

Bernstein–Zelevinsky, *Induced representations of reductive p-adic groups I*, Ann. Sci. École Norm. Sup. (4) 10 (1977), 441–472, Theorem 4.2, p. 454, with Sections 1.8(a), p. 444, and 2.2–2.3, p. 446, supplies the irreducibility implication used above. Lurie, *Higher Algebra*, September 18, 2017, Sections 4.8.1–4.8.2,

Theorem 4.7.3.5, p. 685, Theorem 7.1.2.13, pp. 1212–1213, and Definition 7.2.4.1 and Proposition 7.2.4.2, pp. 1260–1261, supplies the module, tensor and monadic inputs. The Stacks Project, Section 15.80, Tag 07LQ, Lemma 15.80.1 and Proposition 15.80.3, identifies perfect complexes over ordinary rings.

Chapter 86

Non-Ordinary Locus and Removal of the Taylor–Wiles Hypotheses for the DEG Equivalence

A fixed residual representation can have characteristic-zero lifts with different Hodge types. The ordinary condition must retain its flag and determinant before a local categorical comparison can be stated. This chapter constructs the quotient lifting equations and their fixed-type intersections, and computes restrictions of quadratic inductions, including nonsplit local places with reducible restriction. The non-ordinary DEG essential image and removal of Taylor–Wiles hypotheses remain arithmetic generator-comparison problems. Neither modularity nor deformation-ring geometry alone provides the required spectral objects or their generation.

86.1 Sources and Objects

Inherited Objects and Notation

The unramified ordinary ∞-category $\mathrm{D}^{\text{sph-fin,unr-ord}}([GL_2(\mathbb{Q}_p)])_\chi$, the Dotto–Emerton–Gee functor $\Phi^{\mathrm{DEG}}_{p,\chi}$ (Theorem 61.3.1), the Emerton–Gee stack $\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2}$ and its Breuil–Mézard cycle comparison problem, and the Taylor–Wiles setup (Setup 70.2.1) are taken as stated in Chapters 61 and 70.

Primary Sources

Kisin 2008, J. AMS 21 "Potentially semi-stable deformation rings", J. Amer. Math. Soc. 21 (2008), 513–546. Potentially semistable and Barsotti–Tate lifting problems. The fixed-determinant intersection and its torsion-free closure are distinguished below.

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Kisin 2009, J. AMS 22 "The Fontaine–Mazur conjecture for GL_2", J. AMS 22. Fontaine–Mazur and modularity-lifting questions with their local and global hypotheses.

Gee–Kisin 2014, Forum Math. Pi 2 "The Breuil–Mézard conjecture for potentially Barsotti–Tate representations", Forum Math. Pi 2. Geometric Breuil–Mézard beyond ordinary.

Emerton–Gee arXiv:1109.4226 "A geometric perspective on the Breuil–Mézard conjecture". Geometric cycle questions; their use in a specified support category requires additional construction.

Khare–Wintenberger 2009 *Serre's modularity conjecture*, Parts I and II, Invent. Math. 178, 485–504 and 505–586. Residual modularity of continuous odd irreducible two-dimensional representations.

Langlands 1980, Annals of Mathematics Studies 96 *Base Change for $GL(2)$*. Solvable base change for GL_2.

Tunnell 1981, Bull. AMS 5 "Artin's conjecture for representations of octahedral type", Bull. AMS 5, 173–176. The tetrahedral and octahedral Artin cases.

Taylor 2003, Pub. IHES 98 "On icosahedral Artin representations II", Pub. IHES 98. Potential modularity for the icosahedral case.

Skinner–Wiles 2001 *Nearly ordinary deformations of irreducible residual representations*, Ann. Fac. Sci. Toulouse 10, 185–215. Nearly-ordinary modularity under the stated hypotheses.

Skinner–Wiles 1997/1999 *Ordinary representations and modular forms*, Proc. Natl. Acad. Sci. USA 94 (1997), 10520–10527; and *Residually reducible representations and modular forms*, Publ. Math. IHES 89 (1999), 5–126. These are separate modularity statements with their own residual and local hypotheses.

Arthur–Clozel 1989, Ann. Math. Stud. 120 *Simple algebras, base change, and the advanced theory of the trace formula*. Cyclic solvable base change for GL_n; key input for the dihedral / solvable-CM sector.

Emerton 2011 *Local-global compatibility in the p-adic Langlands programme for $GL_2/\mathbb{Q}$*, author preprint, March 23, 2011. The promodularity and local-global statements in Theorems 1.2.3–1.2.4 retain absolute irreducibility on $G_{\mathbb{Q}(\zeta_p)}$; they do not remove that hypothesis merely by supplying a local functor.

Barnet-Lamb–Gee–Geraghty 2011 *The Sato–Tate conjecture for Hilbert modular forms*, J. Amer. Math. Soc. 24, 411–469, arXiv:0912.1054. Its non-CM automorphic hypotheses do not by themselves give a residual-CM lifting theorem for the sectors considered here; an application requires a specific applicable lifting and descent statement.

Gee 2011, Math. Annalen 350 "Automorphic lifts of prescribed types". Used to match Breuil–Mézard cycles with specified Hodge–Tate weights.

Colmez 2010, Astérisque 330 "Représentations de $GL_2(\mathbb{Q}_p)$ et (φ, Γ)-modules". Colmez Montreal functor; non-ordinary / supersingular case §9.3–9.6.

Paškūnas 2013 *The image of Colmez's Montreal functor*, Publ. Math. IHES 118, 1–191, arXiv:1005.2008. Projective envelopes in the specified pseudocompact blocks require an additional comparison before they can be used as compact generators of a derived category.

The deformation and automorphic comparisons

Kisin's lifting rings and Gee–Kisin's cycle multiplicities concern fixed characteristic-zero types. Emerton–Gee supplies the spectral moduli problem. Colmez and Paškūnas concern representation objects and projectives in their specified coefficient categories. Khare–Wintenberger, Langlands–Tunnell, Arthur–Clozel, Skinner–Wiles, Taylor and Barnet-Lamb–Gee–Geraghty concern modularity, base change or potential automorphy under their own local and global hypotheses. The maps relating these objects to the actual DEG functor and its spectral essential image must be constructed separately.

86.2 Essential Surjectivity on the Non-Ordinary Locus at $p \geq 5$

Structural Decomposition of the Non-Ordinary Locus

The ordinary comparison of Theorem 70.2.4 is conditional on its actual support and essential-image data. Dropping (TW4) does not partition a residual stack by Hodge types. The following lifting constructions fix their determinant, residual quotient and characteristic-zero type before asking for a categorical realization.

A fixed determinant and a residual quotient

Fix $p \geq 5$. Let $E/\mathbb{Q}_p$ be finite, with integer ring $\mathcal{O}$, uniformizer ϖ, and finite residue field k. Put $G = G_{\mathbb{Q}_p}$ and let $I \subset G$ be inertia. Fix a continuous representation $\bar{\rho} : G \to \mathrm{GL}_2(k)$, a basis of k^2, and a continuous character $\delta : G \to \mathcal{O}^\times$ lifting $\det \bar{\rho}$.

For an Artinian local $\mathcal{O}$-algebra A with residue field k and its discrete topology, a framed lift is a continuous homomorphism $\rho_A : G \to \mathrm{GL}_2(A)$ reducing to the specified matrix representation $\bar{\rho}$. Framed lifts are not divided by conjugation. The framed deformation ring $R^\square$ is a complete Noetherian local $\mathcal{O}$-algebra with universal representation $\rho^\square$. Its existence is the framed representability statement in (Kisin, *The structure of potentially semi-stable deformation rings*, §1.2). Set

$$R = R^{\square,\delta} = R^\square / \overline{(\det \rho^\square(g) - \delta(g) : g \in G)}.$$

The bar denotes closure in the maximal-ideal topology. This quotient represents framed lifts of determinant δ. Write its universal matrices as

$$\rho_R(g) = \begin{pmatrix} a_g & b_g \\ c_g & d_g \end{pmatrix}.$$

Assume a chosen residual quotient $\bar{q} : k^2 \twoheadrightarrow k(\bar{\eta})$ is unramified. Here $k(\bar{\eta})$ is the one-dimensional representation with character $\bar{\eta}$, and unramified means $\bar{\eta}|_I = 1$. Choose the basis so that $\bar{q} = (0, 1)$. Then

$$\bar{\rho}(g) = \begin{pmatrix} \bar{\alpha}(g) & * \\ 0 & \bar{\eta}(g) \end{pmatrix}, \qquad \bar{\alpha} = \bar{\delta}\bar{\eta}^{-1}.$$

An ordinary lift relative to $\bar{q}$ is a framed lift together with a free rank-one quotient lifting $\bar{q}$, on which inertia acts trivially. Quotients are identified up to an equivariant target isomorphism. Each such quotient has a unique normalized row $q = (t, 1)$, where $t \in \mathfrak{m}_A$. Its character is allowed to vary with A.

Proposition 86.2.1 *(explicit ordinary deformation equations).* Suppose $\bar{\alpha} \neq \bar{\eta}$. Choose $g_0 \in G$ with $\bar{\alpha}(g_0) \neq \bar{\eta}(g_0)$. There is a unique $t_0 \in \mathfrak{m}_R$ satisfying

$$b_{g_0} t_0^2 + (d_{g_0} - a_{g_0}) t_0 - c_{g_0} = 0. \tag{86.1}$$

Let $J_{\mathrm{ord}} \subset R$ be the closed ideal generated by

$$\begin{aligned} b_g t_0^2 + (d_g - a_g) t_0 - c_g \quad & (g \in G), \\ t_0 b_h + d_h - 1 \quad & (h \in I). \end{aligned}$$

Then $R^{\mathrm{ord}} = R/J_{\mathrm{ord}}$ represents ordinary lifts relative to $\bar{q}$. The quotient line is uniquely determined by the framed representation. Thus forgetting the quotient gives a closed immersion $\operatorname{Spf} R^{\mathrm{ord}} \hookrightarrow \operatorname{Spf} R$.

Proof. The polynomial in (86.1) vanishes at zero modulo $\mathfrak{m}_R$. Its derivative there is $\bar{\eta}(g_0) - \bar{\alpha}(g_0) \neq 0$. Completeness and Hensel's lemma give the unique root t_0. For completeness, successive correction modulo powers of $\mathfrak{m}_R$ solves a linear equation with this invertible coefficient. These corrections converge in R. If two roots lie in $\mathfrak{m}_R$, their difference times $b_{g_0}(t + t') + d_{g_0} - a_{g_0}$ is zero. The second factor is a unit, so the roots agree.

For a row $q = (t, 1)$, stability of its kernel under $\rho_A(g)$ is equivalent to

$$q\rho_A(g) = \eta_A(g) q, \qquad \eta_A(g) = t b_g + d_g,$$

where the matrix entries are specialized to A. The first component is exactly $b_g t^2 + (d_g - a_g)t - c_g = 0$. Since $\eta_A(g)$ reduces to $\bar{\eta}(g) \neq 0$, it is a unit. Surjectivity of q and multiplication of the matrices show that η_A is a character. It is continuous because ρ_A is continuous. The equations indexed by I make it unramified.

Every ordinary quotient satisfies the equation for g_0. The uniqueness argument over the Artinian local ring A forces t to be the image of t_0. The remaining equations are precisely the images of J_{ord}. Conversely, a map $R^{\mathrm{ord}} \to A$ gives the quotient $(t_0, 1)$ with these properties. Both constructions commute with local O-algebra maps of Artinian coefficient rings. The construction also applies to complete Noetherian local coefficient rings by passage to their Artinian quotients; the normalized rows are compatible by uniqueness. □

The kernel of this quotient is freely generated by $(1, -t_0)$. Consequently every ordinary lift has an exact sequence

$$0 \longrightarrow A(\delta\eta_A^{-1}) \longrightarrow A^2_{\rho_A} \longrightarrow A(\eta_A) \longrightarrow 0, \qquad \eta_A|_I = 1. \tag{86.2}$$

Indeed the action on the kernel is multiplication by $a_g - b_g t_0$, and the determinant is the product of the two characters. The sequence need not split. To prescribe an unramified lift $\eta : G \to O^\times$, add the equations $t_0 b_g + d_g = \eta(g)$ for $g \in G$.

Remark 86.2.2 *(coincident residual characters).* Without $\bar{\alpha} \neq \bar{\eta}$, the quotient must be retained as part of the deformation problem. That problem is still represented by

$$R[[t]] \big/ \overline{\big(b_g t^2 + (d_g - a_g)t - c_g \ (g \in G), \quad t b_h + d_h - 1 \ (h \in I)\big)}.$$

Here t maps into the maximal ideal of each coefficient ring. The same row calculation proves the assertion. For the trivial residual representation and trivial determinant, take the constant trivial lift to $k[u]/(u^2)$. The rows $(au, 1)$ for $a \in k$ give distinct ordinary quotient lines lifting $(0, 1)$. Thus the forgetful map need not be injective on deformation functors when the residual characters coincide.

Hodge types on characteristic-zero lifts

The ordinary equations make sense over Artinian coefficient rings. Hodge and inertial types require characteristic-zero period modules. For a continuous representation V on a finite-dimensional vector space over a finite extension E'/E, put

$$D_{\mathrm{dR}}(V) = (B_{\mathrm{dR}} \otimes_{\mathbb{Q}_p} V)^G.$$

Here B_{dR} is Fontaine's filtered de Rham period field. The representation is de Rham when $\dim_{E'} D_{\mathrm{dR}}(V) = \dim_{E'} V$. We use filtration degrees as Hodge–Tate weights, so the cyclotomic character ε has weight -1. A type $\mathbf{h} = \{h_1, h_2\}$ prescribes the dimensions of $\mathrm{gr}^j D_{\mathrm{dR}}(V)$. Equivalently, Kisin's cocharacter is the conjugacy class of $z \mapsto \mathrm{diag}(z^{-h_1}, z^{-h_2})$.

Let $B_{\mathrm{cris}} \subset B_{\mathrm{st}}$ be Fontaine's crystalline and semistable period rings. For a finite extension $K'/\mathbb{Q}_p$ with maximal unramified subfield K'_0, set

$$D_{\mathrm{st},K'}(V) = (B_{\mathrm{st}} \otimes_{\mathbb{Q}_p} V)^{G_{K'}}, \qquad D_{\mathrm{cris},K'}(V) = (B_{\mathrm{cris}} \otimes_{\mathbb{Q}_p} V)^{G_{K'}}.$$

The representation is potentially semistable, respectively potentially crystalline, when for some K' the corresponding module is free of rank $\dim_{E'} V$ over $K'_0 \otimes_{\mathbb{Q}_p} E'$. Its potentially semistable period module is

$$D_{\mathrm{pst}}(V) = \varinjlim_{K'} D_{\mathrm{st},K'}(V).$$

Inertia acts linearly on this module over $\overline{\mathbb{Q}}_p^{\mathrm{nr}} \otimes_{\mathbb{Q}_p} E'$. An inertial type is a continuous representation $\tau : I \to \mathrm{GL}_2(E)$ with open kernel; V has this type when its inertia representation on $D_{\mathrm{pst}}(V)$ is isomorphic to τ after scalar extension. Here $\overline{\mathbb{Q}}_p^{\mathrm{nr}}$ is the maximal unramified extension of $\mathbb{Q}_p$ inside its chosen algebraic closure. This is the period-module action, not the inertia action on V. Choose E large enough to define τ and the Hodge type. Kisin constructs quotients $R_{\mathrm{pst}}^{\square,\mathbf{h},\tau}$ and $R_{\mathrm{pcris}}^{\square,\mathbf{h},\tau}$ of $R^{\square}$. Their characteristic-zero points specify potentially semistable and potentially crystalline lifts of these types, respectively (Kisin, *The structure of potentially semi-stable deformation rings*, Theorems 1.2.1–1.2.2, pp. 6–7).

For $\star \in \{\mathrm{pst}, \mathrm{pcris}\}$ define the actual fixed-determinant intersection ring and its generic fibre by

$$S_\star = R_\star^{\square,\mathbf{h},\tau} \otimes_{R^{\square}} R, \qquad X_\star = \operatorname{Spec} S_\star[1/p].$$

Both factors are quotient rings, so this tensor product is an ordinary quotient of $R^{\square}$. It is already complete and Noetherian. A continuous map $S_\star \to \mathcal{O}_{E'}$ gives a framed lattice of determinant δ whose rational representation has the indicated type. Conversely, every such lattice with a chosen frame reducing to the given residual frame gives this map: Kisin's factorization criterion first applies over E', and the defining ideal then vanishes already in the torsion-free ring $\mathcal{O}_{E'}$. Here coefficient extensions carry a fixed residue-field embedding, and all integral maps are continuous local $\mathcal{O}$-algebra maps. The scheme $X_\star$ also retains the infinitesimal structure of the intersection. No flatness or reducedness of $S_\star$ is needed for this construction. Incompatible determinant and type conditions can give an empty generic fibre.

The schematic closure of the generic fibre in $\operatorname{Spec} S_\star$ is defined by the p-power torsion ideal. Its coordinate ring is $S_\star/S_\star[p^\infty]$, where $S_\star[p^\infty] = \{x : p^n x = 0 \text{ for some } n\}$. This quotient is p-torsion-free, hence flat over the discrete valuation ring $\mathcal{O}$. It has no effect after inverting p, but can change the special fibre. The integral intersection and its generic fibre must therefore remain distinct.

Under the distinguished quotient hypothesis, define

$$X_\star^{\mathrm{ord}} = \operatorname{Spec}\big((S_\star/J_{\mathrm{ord}}S_\star)[1/p]\big), \qquad X_\star^{\mathrm{rest}} = X_\star \setminus X_\star^{\mathrm{ord}}.$$

The first is a closed subscheme and the second its open complement. This gives a two-piece decomposition within the specified lifting space. The ordinary condition is relative to the chosen residual quotient $\bar{q}$.

Corollary 86.2.3 *(ordinary lifts of a prescribed type).* A continuous point $S_\star \to \mathcal{O}_{E'}$ lies in the ordinary piece precisely when its framed lattice has an unramified quotient lifting $\bar{q}$. On this piece its rational representation has an exact sequence

$$0 \longrightarrow E'(\delta\eta^{-1}) \longrightarrow V \longrightarrow E'(\eta) \longrightarrow 0, \qquad \eta|_I = 1,$$

and has the prescribed Hodge and inertial types.

Proof. Factoring through R^{ord} is equivalent to the ordinary quotient property by the proposition. Apply (86.2) and invert p. The quotient $R_\star^{\square,\mathbf{h},\tau}$ imposes the type. If the equations vanish only after inverting p, they already vanish in the torsion-free ring $\mathcal{O}_{E'}$. □

For the potentially Barsotti–Tate problem in the Tate-module convention, take $\star = \mathrm{pcris}$ and $\mathbf{h} = \{-1, 0\}$, with a specified inertial type. The dual convention has weights $\{0, 1\}$. Other Hodge types require other quotient rings. These spaces do not give an exhaustive decomposition of all residual points. They describe lifts above a fixed residual representation. For example, with $a = p - 1$, the representations $1 \oplus \varepsilon$ and $\varepsilon^a \oplus \varepsilon^{1-a}$ have the same reduction and determinant. Their Hodge–Tate multisets are $\{0, -1\}$ and $\{1 - p, p - 2\}$, respectively.

The categorical comparison

To compare these deformation spaces with categorical local Langlands, fix a continuous central character $\zeta : \mathbb{Q}_p^\times \to O^\times$. Use local reciprocity sending a uniformizer to geometric Frobenius, and require the already fixed determinant to satisfy $\delta = \zeta\varepsilon^{-1}$ as a Galois character. These are the conventions of (Dotto–Emerton–Gee, arXiv:2603.26887, §1.3, p. 8).

Let $\mathcal{A}_\zeta$ be smooth $\mathrm{GL}_2(\mathbb{Q}_p)$ representations on p-power torsion O-modules with central character ζ. Let $\mathcal{X}_\delta$ be the formal stack of rank-two projective étale (φ,Γ)-modules with a specified isomorphism of their determinant with the rank-one module attached to δ. Morphisms preserve this isomorphism; this is the fibre-product definition in (Dotto–Emerton–Gee, arXiv:2603.26887, Definition C.1.1, p. 265). The Dotto–Emerton–Gee functor is

$$F : D(\mathcal{A}_\zeta) \longrightarrow \mathrm{Ind}\, D^b_{\mathrm{coh}}(\mathcal{X}_\delta).$$

It is exact, colimit-preserving and fully faithful (Dotto–Emerton–Gee, arXiv:2603.26887, Theorem 1.1.1 and its following paragraph, p. 3).

Let $\bar\theta$ be the residual pseudorepresentation of $\bar\rho$. The completion of $\mathcal{X}_\delta$ along the fibre over $\bar\theta$ is the moduli stack of Galois representations deforming that pseudorepresentation, with the specified determinant identification (Dotto–Emerton–Gee, arXiv:2603.26887, §1.1.6, p. 4). The universal representation defines a morphism

$$\mathrm{Spf}\, R \longrightarrow \widehat{\mathcal{X}}_{\delta,\bar\theta} \longrightarrow \mathcal{X}_\delta$$

as follows. Reduce ρ_R modulo each power of its maximal ideal, apply the equivalence between continuous Galois representations and étale (φ,Γ)-modules over these finite coefficient rings, and retain the determinant identification sending the wedge of the two frame vectors to 1 in the character module of δ. This construction is compatible with tensor products and base change (Dotto–Emerton–Gee, arXiv:2603.26887, Appendix C.2, especially (C.2.1)–(C.2.2), p. 268, and Remark C.2.17, p. 272). The compatible objects define the displayed formal morphism. Their residual pseudorepresentation is $\bar\theta$, so the map factors through the indicated completion. A change of basis preserving the determinant identification gives an isomorphism in that stack; scalar automorphisms ± 1 remain. The ordinary and type quotients give morphisms into this diagram. For the type quotients these are the maps from $\mathrm{Spf}\, S_\star$; the ordinary intersection uses $\mathrm{Spf}(S_\star/J_{\mathrm{ord}}S_\star)$. These are morphisms of formal moduli problems, whereas $X_\star$ denotes the separately defined scheme after inverting p. They do not identify the source formal schemes with the stack, which retains automorphisms.

Essential surjectivity on a specified ordinary or potentially Barsotti–Tate category requires a full stable source subcategory, a full stable target subcategory, and a proof that F restricts between them. The deformation maps alone do not specify these categories or their generators. The functor on the entire target is not an equivalence (Dotto–Emerton–Gee, arXiv:2603.26887, Remark 1.1.2, p. 3). This statement does not decide equivalence on a specified smaller target.

Proposition 86.2.4 *(generation criterion for a restricted functor).* Let $\mathcal{C},\mathcal{D}$ be presentable stable ∞-categories. Suppose an exact colimit-preserving fully faithful functor $F_0 : \mathcal{C} \to \mathcal{D}$ is supplied. If objects $P_i \in \mathcal{C}$ have images generating $\mathcal{D}$ under colimits, shifts and cofibres, then F_0 is an equivalence.

Proof. Full faithfulness makes the essential image a full subcategory equivalent to $\mathcal{C}$. Every diagram in that image can therefore be lifted to $\mathcal{C}$. Preservation of colimits makes its colimit in $\mathcal{D}$ another image object. Exactness gives closure under shifts and cofibres. The essential image contains the specified generators and hence all of $\mathcal{D}$. □

The local arithmetic construction needed for this criterion is the identification of $F(P_i)$ with actual spectral objects, together with a proof of target generation. If lifting conditions overlap, their union must generate all required extension objects. A direct-sum assembly additionally requires orthogonality. Neither modularity of a lift nor a ring of scalar actions establishes these categorical properties.

Line and quotient conventions. The ordinary lifting problem just constructed retains an unramified *quotient*. Its kernel has character $\delta\eta^{-1}$ and need not be unramified. The ordinary-line comparison of Chapter 70 uses a different flag condition: an unramified subrepresentation. Dualizing a quotient of V gives an unramified subrepresentation of $V^\vee$, with determinant δ^{-1}; it does not identify the two conditions on the same V. Any use of the line convention in the comparison below must specify this dualization or another proved flag comparison, including the central-character and Hodge-sign changes.

Definition 86.2.5 *(fixed-type non-ordinary comparison loci).* For a specified framed residual representation, determinant, inertial type and Hodge type, use the actual lifting space $X_\star$ and its relative ordinary closed subspace $X_\star^{\mathrm{ord}}$ constructed above. Its open complement is $X_\star^{\mathrm{rest}}$. The potentially Barsotti–Tate case uses weights $\{-1, 0\}$ in the Tate-module convention, or $\{0, 1\}$ after dualization. The higher-weight supersingular comparison concerns suitable crystalline lifts of dual weights $\{0, k-1\}$, $2 \leq k \leq p-1$, and the relevant equal-slope condition $((k-1)/2, (k-1)/2)$ in that dual convention. Such slope conditions belong to period modules of lifts with a specified Frobenius and valuation normalization, not directly to residual points. Write $\mathcal{X}^{\mathrm{pBT}}_{\bar\rho,\chi}$ and $\mathcal{X}^{\mathrm{ss}}_{\bar\rho,\chi}$ only for the intended arithmetic support categories or loci once their maps have been constructed; these symbols do not assert a partition of the full Emerton–Gee stack.

Remark 86.2.6 *(coverage and residual points).* The two lifts with identical residual representation and determinant but different Hodge types above show why residual points have no unique Hodge type. Neither the pBT type nor the bounded weight range exhausts all lifts. To realize the proposed union $\mathcal{X}^{\mathrm{pBT}} \cup \mathcal{X}^{\mathrm{ss}}$ as a full non-ordinary locus one must construct those loci in a common carrier and prove coverage. The Newton–Hodge comparison remains an additional geometric question, not a consequence of this notation.

Potentially Barsotti–Tate geometry and generators

Lemma 86.2.7 *(the fixed-determinant lifting intersection).* For the prescribed pBT Hodge and inertial type, the ring S_{pcris} constructed above represents the integral fixed-determinant intersection on continuous lattice points; its generic fibre retains that intersection's infinitesimal structure. Its quotient by p-power torsion is flat over $\mathcal{O}$. The map to the determinant-identified spectral stack is a formal moduli morphism, not an identification of the framed ring with that stack.

Proof. The two type/determinant ideals give the displayed quotient intersection. Its lattice-point factorization and formal map were proved above. After dividing by its p-power torsion the ring is torsion-free over a discrete valuation ring, hence flat. No complete-intersection, reducedness, Cohen–Macaulay or relative-dimension-three assertion for this fixed intersection follows from that argument; any such geometric property requires its actual type and determinant hypotheses. □

Conjecture 86.2.8 *(pBT Breuil–Mézard cycle realization).* For each specified inertial type τ, construct the cycle $\mathcal{X}^{\mathrm{BM},\tau}_{p,\chi,\bar\rho}$ and its realization in the chosen pBT support problem, with multiplicities and intersections. Determine coverage and the actual coherent objects attached to these cycles. A cycle does not by itself define a smooth divisor or a transverse scheme-theoretic intersection. The intended cycle and coverage assertions require these additional constructions.

The relevant geometric input is Gee–Kisin (2014), Theorem A, with its fixed-type hypotheses; the relation to the Emerton–Gee stack requires the specified support maps.

Conjecture 86.2.9 *(projective and spectral generators for pBT).* Construct the actual block or pseudocompact carrier, projective covers $\widetilde{\pi}_\tau$ associated to the relevant types $\sigma(\tau)$, and their compact-generating property. Construct spectral objects Q_τ on the chosen pBT support, identify when they can be realized as $\mathcal{O}_{\mathcal{X}^{\mathrm{BM},\tau}}$, prove their singular support and target generation, and construct natural identifications $F(\widetilde{\pi}_\tau) \simeq Q_\tau$ with full derived morphisms. A type is not automatically a projective cover and its cycle is not a coherent generator object without these data.

Colmez (2010), Sections 9.3–9.6, Paškūnas (2013), Kisin and Gee–Kisin retain their representation and deformation roles in this question; numerical cycle multiplicities do not identify these generators.

Theorem 86.2.10 *(conditional pBT non-ordinary essential image).* For $p \geq 5$ under (TW1)–(TW3), suppose full stable presentable source and target subcategories for the chosen pBT non-ordinary lifting condition have been constructed inside the actual DEG source and target. Assume the exact colimit-preserving fully faithful DEG functor restricts between them and that the objects $Q_\tau = F(\widetilde{\pi}_\tau)$ of Conjecture 86.2.9 generate the target. Then the intended restriction

$$\Phi^{\mathrm{DEG}}_{\bar{\rho},\chi}|_{\text{pBT-non-ord}} : \mathrm{D}^{\text{sph-fin,pBT}}([GL_2(\mathbb{Q}_p)])_\chi \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{pBT}}_{\bar{\rho},\chi})$$

is an equivalence, where these names denote those specified carriers. Comparing them with the actual $D(\mathcal{A}_\zeta)$ and $\mathrm{Ind}\, D^b_{\mathrm{coh}}(\mathcal{X}_\delta)$ is part of the hypotheses, not a silent change of category.

Proof. First supply the restricted fully faithful functor. Second specify the spectral support condition; Cohen–Macaulayness does not imply an automatic nilpotent singular support. Third construct the actual $\widetilde{\pi}_\tau$ and Q_τ and their target generation. Fourth the stipulated natural image identifications place the Q_τ in the image. Fifth apply Proposition 86.2.4. These are the five distinct steps of the conditional implication. The lifting rings and modularity statements alone do not supply steps two through four. □

The supersingular comparison

Corollary 86.2.11 *(the pBT range under its full comparison data).* Under all the carrier, restriction and generation hypotheses of Theorem 86.2.10, the pBT restriction is an equivalence. Describing its complement as a supersingular support requires an additional common-carrier and coverage theorem; this is not implied by the pBT comparison.

Proof. Apply that theorem. The two different Hodge-type lifts above prevent a residual point from determining a unique such complement by its type. □

Conjecture 86.2.12 *(the supersingular generator comparison).* Construct the intended supersingular support and source subcategory, including the proposed difference $\mathcal{X}^{\mathrm{ss}}_{\bar{\rho},\chi} \setminus \mathcal{X}^{\mathrm{pBT}}_{\bar{\rho},\chi}$ once both are in a common carrier. Identify the images of supersingular projective covers with actual Breuil–Mézard–Newton spectral objects and prove support, full derived morphism comparison and target generation. The resulting restricted DEG functor is then subject to the generation criterion; the construction of these arithmetic data remains open.

Remark 86.2.13 *(nonreduced cycles and supersingular objects).* Nonreduced cycle multiplicities require an actual derived or coherent realization before a generator comparison can be made. A possible Breuil-module or Jordan–Hölder analysis must identify objects and extension maps, not only a multiplicity

list. Kisin's integral p-adic Hodge constructions concern those lifting problems, but do not make the supersingular category the known complement of pBT.

Conventions and comparison hypotheses

The explicit lifting construction uses filtration degrees, in which ε has weight -1. The dual-normalized weights in the intended arithmetic comparison have cyclotomic weight $+1$; dualize both representation and determinant when using them. Newton slopes require the corresponding period Frobenius and valuation normalization. Inertial types belong to the period module, not the residual inertia action. The ordinary unramified-line and quotient conventions are related only by the explicit flag comparison above.

The intended pBT target is a support subcategory, and its comparison with $\mathrm{Ind}\, D^b_{\mathrm{coh}}(\mathcal{X}_\partial)$ must be supplied. Its compact generators, nilpotent support, Hecke action and image maps are the precise missing data. Big image (TW2) and local distinguishedness (TW3) remain separate from the pBT lifting condition. Colmez's functor, Kisin's rings and Gee–Kisin's cycles provide different objects; their existence alone does not prove that the Hecke actions or derived images agree. The supersingular construction remains Conjecture 86.2.12.

86.3 Removal of the Taylor–Wiles Conditions: Dihedral, Nearly-Ordinary, and Serre-Modular Sectors

Global residual sectors and their local restriction

Let S_{fail} be the set of continuous odd irreducible representations $\bar\rho : G_{\mathbb{Q}} \to GL_2(\bar{\mathbb{F}}_p)$ for which (TW2) or the local condition (TW3) fails, with the specified ramification and determinant restrictions. Restriction at p maps these global parameters to local residual representations or pseudorepresentations; it need not be injective.

Definition 86.3.1 *(the non-Taylor–Wiles comparison sectors).* Within S_{fail} consider:

- S^{dihedral}, retaining the notation for the solvable projective-image comparison sector. Its dihedral part consists of quadratic inductions; the tetrahedral A_4 and octahedral S_4 cases are separate solvable cases, not quadratic inductions.
- $S^{\text{nearly-ord}}$, parameters with reducible local restriction and coincident characters for which the specified global nearly-ordinary lifting problem has a lift. The hypotheses of any Skinner–Wiles application are additional requirements.
- $S^{\text{Serre-res}}$, the odd irreducible residual parameters in S_{fail} with reducible restriction to $G_{\mathbb{Q}(\zeta_p)}$, considered with their Serre modularity data.

These are sets of global parameters. Their images under local restriction do not define three orthogonal local categories.

Remark 86.3.2 *(overlap and coverage).* These sets may overlap. A dihedral global parameter can also admit a nearly-ordinary lift, and different global parameters can have the same local restriction. Coverage of S_{fail}, or of a specified local essential-image problem, requires a separate theorem. The non-solvable cases, including A_5, and potential automorphy and descent remain additional questions. Overlap is mathematically significant for a categorical union because extension objects and compatible maps must be retained.

The solvable projective-image comparison

Lemma 86.3.3 *(the Artin modularity input under its actual lift hypotheses).* Suppose an odd irreducible characteristic-zero finite-image Artin lift of the chosen residual parameter has been supplied, with the coefficient and ramification hypotheses of the relevant modularity theorem. In the dihedral case its automorphic construction is induction from the corresponding Hecke character of the quadratic extension. The tetrahedral and octahedral cases use the Langlands–Tunnell theorem and the specified solvable base-change data. The existence of such an Artin lift is not inferred from a residual representation alone.

Modularity input. Apply the corresponding representation-level theorem to that specified lift: Langlands, *Base Change for $GL(2)$* (1980), Arthur–Clozel (1989) for cyclic base change and solvable descent, and Tunnell (1981) for the octahedral case. This application gives no categorical generator or local spectral image without an additional comparison construction. □

86.4 Quadratic induction and ordinary lifting spaces

A quadratic global induction can become reducible at a nonsplit local place. The local character determines this behavior. Ordinary lifting conditions require further data: a residual quotient line, a determinant, and a characteristic-zero lifting space. These data specify the local objects whose images under a categorical Langlands functor must be compared.

Restriction of a quadratic induction

Let $K/\mathbb{Q}$ be quadratic. Fix a place v of $\mathbb{Q}$ and compatible decomposition groups. Let L be a coefficient field, and let $\psi : G_K \to L^\times$ be continuous. Here G_F denotes the absolute Galois group of a field F, and L is equipped with its specified field topology. All representation spaces carry the corresponding finite-dimensional topology. Induction from an open subgroup of finite index means ordinary algebraic induction, with its induced continuous action.

If v splits in K, the double-coset decomposition gives

$$\operatorname{Ind}_{G_K}^{G_{\mathbb{Q}}} \psi\Big|_{G_{\mathbb{Q}_v}} \simeq \psi_w \oplus \psi_{w'},$$

where w, w' are the two places above v. The two local characters can coincide. If v is nonsplit, there is one place w above v, and the same decomposition gives

$$\operatorname{Ind}_{G_K}^{G_{\mathbb{Q}}} \psi\Big|_{G_{\mathbb{Q}_v}} \simeq \operatorname{Ind}_{G_{K_w}}^{G_{\mathbb{Q}_v}} \psi_w.$$

These formulas follow by separating functions according to their support on the double cosets $G_K \backslash G_{\mathbb{Q}} / G_{\mathbb{Q}_v}$. There are two cosets in the split case and one in the nonsplit case.

To determine the second representation, let H be an open normal subgroup of index two in a profinite group G. Choose $s \in G \setminus H$. For a continuous character $\chi : H \to L^\times$, put

$$\chi^s(h) = \chi(s^{-1}hs), \qquad c = \chi(s^2).$$

On the induced basis $e_1 = 1 \otimes 1$, $e_2 = s \otimes 1$, the action is

$$\rho(h) = \begin{pmatrix} \chi(h) & 0 \\ 0 & \chi^s(h) \end{pmatrix}, \qquad \rho(s) = \begin{pmatrix} 0 & c \\ 1 & 0 \end{pmatrix}. \tag{86.3}$$

Proposition 86.4.1 *(local irreducibility criterion).* If $\chi \neq \chi^s$, then $\operatorname{Ind}_H^G \chi$ is absolutely irreducible. If $\chi = \chi^s$, its reducibility over L is equivalent to the existence of a root of $T^2 - c$ in L.

Suppose $\chi = \chi^s$ and $b^2 = c$ with $b \in L$. There is a continuous extension $\widetilde{\chi} : G \to L^\times$ with $\widetilde{\chi}|_H = \chi$ and $\widetilde{\chi}(s) = b$. If $\operatorname{char} L \neq 2$, then

$$\operatorname{Ind}_H^G \chi \simeq \widetilde{\chi} \oplus (\widetilde{\chi}\, \omega_{G/H}),$$

where $\omega_{G/H}$ is trivial on H and sends s to -1. In characteristic two the induced representation is a nonsplit self-extension of $\widetilde{\chi}$.

Proof. When the characters differ, choose $h \in H$ with $\chi(h) \neq \chi^s(h)$. Any invariant line over an algebraic closure must be one of the two eigenspaces of $\rho(h)$. The matrix $\rho(s)$ interchanges them, so neither is G-stable.

When the characters agree, H acts by scalars. A G-stable line therefore exists precisely when $\rho(s)$ has an eigenvalue in L. Its characteristic and minimal polynomials are both $T^2 - c$. The prescribed values of $\widetilde{\chi}$ respect conjugation by s and the relation $s^2 \in H$, hence define a character. It is continuous because H is open and χ is continuous. In characteristic different from two the two roots $b, -b$ are distinct, which gives the displayed direct sum. In characteristic two the minimal polynomial is $(T - b)^2$. The matrix is not scalar, so its unique eigenline has no invariant complement. □

Thus over an algebraically closed field, local irreducibility in the nonsplit case is equivalent to $\psi_w \neq \psi_w^s$. Inertness and ramification alone do not decide this equality. Over a smaller coefficient field, an invariant character can instead give an irreducible representation which becomes reducible after scalar extension. Over coefficient rings the matrices in (86.3) still define induction. The field criterion applies after specialization and does not assert an integral direct-sum decomposition.

For a ramified quadratic extension of $\mathbb{Q}_p$ with $p \geq 5$, the ramification index is two, so the extension is tame. A characteristic-zero inducing character can still be nontrivial on wild inertia. For a continuous character valued in $\overline{\mathbb{F}}_p^\times$ with the discrete topology, its image is finite of order prime to p. Its restriction to wild inertia is therefore trivial. Since the quadratic extension is tame, wild inertia lies in the inducing subgroup and acts trivially on the induced residual representation.

Example 86.4.2 *(an inert place with reducible restriction).* Let L_0 be the splitting field of $X^3 - 2$. Eisenstein's criterion at 2 and the discriminant -108 give $\operatorname{Gal}(L_0/\mathbb{Q}) = S_3$. The quadratic subfield is $K = \mathbb{Q}(\sqrt{-3})$. Over $\overline{\mathbb{F}}_5$, induction of a nontrivial character of A_3 gives an absolutely irreducible representation of S_3. Complex conjugation is a transposition, so this representation is odd.

The prime 5 is unramified in L_0, and

$$X^3 - 2 = (X - 3)(X^2 + 3X + 4) \pmod 5.$$

The quadratic discriminant is 3, which is nonsquare modulo 5. Frobenius is consequently a transposition. Also $-3 = 2 \pmod 5$ is nonsquare, so 5 is inert in K. The local representation factors through the order-two decomposition group and splits into its $+1$ and -1 Frobenius eigenspaces.

Lemma 86.4.3 *(restriction of a dihedral residual representation).* For a quadratic induction, the split restriction is the sum of the two local characters, which can coincide. At a nonsplit place the restriction is induced from the local quadratic subgroup and is absolutely irreducible exactly when its two conjugate characters differ over an algebraically closed coefficient field. Ramification of the quadratic extension alone specifies no Hodge type. At $p \geq 5$ that quadratic extension is tame, and a discrete mod-p character is trivial on wild inertia.

Proof. The double-coset decomposition, matrix computation and characteristic-two case in Proposition 86.4.1 prove these statements. The inert prime 5 example shows why inertness alone cannot imply irreducibility. A comparison with the Colmez Montreal functor, including its actual block and coefficient category, remains additional arithmetic data; it is not a consequence of the restriction formula. □

Theorem 86.4.4 *(conditional local comparison on the solvable sector).* For $p \geq 5$ and a full central character, fix a residual parameter in S^{dihedral} and its actual local restriction. Construct full stable source and target subcategories in the DEG carriers, with an exact colimit-preserving fully faithful restricted functor. If specified source objects have images generating that target, the restriction is an equivalence. Hecke compatibility with an Arthur–Clozel or Langlands–Tunnell automorphic realization additionally requires the specified natural comparison of those actions.

Proof. First identify the actual local residual block. For the dihedral part, Lemma 86.4.3 gives its restriction. Second, the split characters can coincide and the inert restriction can be reducible; test the local characters rather than infer a block from the splitting name. Global induced irreducibility need not fail. Third, quadratic ramification at $p \geq 5$ is tame and specifies no Hodge type; the type and spectral generator must be constructed. Fourth, the automorphic realization requires the actual lift and modularity hypotheses of Lemma 86.3.3, plus the natural Hecke comparison. Fifth, once the stipulated restricted functor and image generators exist, apply Proposition 86.2.4. This proves the conditional implication for the named solvable cases, not the existence of their arithmetic generators. □

The Nearly-Ordinary Sector

Conjecture 86.4.5 *(the nearly-ordinary modularity and ring comparison).* For the specified global nearly-ordinary deformation problem, verify the full local and global hypotheses of the applicable Skinner–Wiles theorem and construct its Hecke algebra and comparison $R^{\text{n-ord}}_{\bar{\rho},\chi} \to \mathbf{T}^{\text{n-ord}}_{\bar{\rho}}$. The intended $R = T$ and automorphic Hida-family realization require those hypotheses; existence of a nearly-ordinary lift alone does not establish either. The global deformation ring is distinct from the local determinant-identified stack and the local quotient ring constructed above.

Skinner–Wiles' nearly-ordinary and residually reducible modularity theorems concern these global problems with their stated hypotheses. The local quotient construction above also retains coincident residual characters and the nonunique quotient line.

Theorem 86.4.6 *(conditional nearly-ordinary DEG comparison).* For every $p \geq 5$, full central character and chosen parameter in $S^{\text{nearly-ord}}$, suppose the local source and spectral support subcategories and restricted fully faithful, exact, colimit-preserving DEG functor are constructed. If the specified source objects have images generating the target, the restriction is an equivalence.

Proof. The first required construction is the actual source generator; a Hida family's scalar algebra alone is not a projective generator of a local block. The second is its spectral object and image map; a formal scheme omits the stack stabilizers. The third is target generation with the required singular support; a Noetherian or Cohen–Macaulay ring does not make its structure sheaf generate an arbitrary ind-coherent support category. With those three hypotheses, the fourth step is Proposition 86.2.4, which proves the asserted equivalence. A global $R = T$ theorem supplies no missing step of this local categorical argument without the specified maps. □

The Serre-Modular Sector

Lemma 86.4.7 *(the residual modularity input).* For the continuous odd irreducible residual representations over $\bar{\mathbb{F}}_p$ considered here, Serre modularity provides a cuspidal modular eigenform realizing the residual representation. The precise minimal weight and conductor, and any asserted range $2 \le k(\bar{\rho}) \le p+1$, require the chosen Serre-weight and twist conventions; they are not used to infer local categorical projectives or support objects.

Residual modularity input. This is the representation-level content of Khare–Wintenberger, *Serre's modularity conjecture*, Parts I and II (2009). Its use here concerns residual modularity; the weight, level and local lifting refinements must retain their exact hypotheses. □

Conjecture 86.4.8 *(modularity lifting and the type-to-cycle comparison).* For a residually modular odd irreducible $\bar{\rho}$ and a lift $\rho : G_{\mathbb{Q}} \to GL_2(\bar{\mathbb{Q}}_p)$, verify the local deformation, p-adic Hodge, ramification and global hypotheses of the applicable Kisin modularity-lifting theorem. The lifting space and type-to-cycle map must also be specified. Residual modularity does not imply that every lift is modular, nor does modularity identify a cycle with a coherent generator.

Kisin's Fontaine–Mazur and modularity-lifting work, together with Gee's prescribed-type lifting problem, supplies the relevant arithmetic context. The fixed-type intersections constructed above are local lifting spaces, not an automorphy theorem.

Theorem 86.4.9 *(conditional Serre-sector DEG comparison).* For $p \ge 5$, a full central character and a chosen parameter in $S^{\text{Serre-res}}$, suppose the actual local block, spectral support category and restricted fully faithful exact colimit-preserving DEG functor are supplied. If chosen projective or compact source objects have images generating the target, that restriction is an equivalence.

Proof. First a modular eigenform realizing $\bar{\rho}$ is not itself an identified compact projective cover; construct the proposed $\widetilde{\pi}_{f,p}$. Second construct the specified type/cycle objects and natural image comparison, including the proposed map $F(\widetilde{\pi}_{f,p}) \simeq \mathcal{O}_{\mathcal{X}^{\mathrm{BM},k(\bar{\rho})}_{p,\chi,\bar{\rho}}}$ when that coherent object is defined. Third prove target generation and singular support for the actual Serre-weight family. Fourth, under these stipulated data, apply Proposition 86.2.4. This retains the arithmetic generator question rather than deducing it from a residual eigenform or cycle multiplicities. □

Assembly on the Union of Non-Taylor–Wiles Sectors

Theorem 86.4.10 *(conditional assembly on the union of sectors).* Fix the common local DEG source and target at $p \ge 5$ and a full central character. Suppose the sector restrictions above are actual full subcategories, their image comparisons hold, and the union of their image objects generates the intended target subcategory under colimits, shifts and cofibres. Then the restriction to their generated source is an equivalence onto that target. This includes the ordinary comparison only with all its own carrier and image hypotheses.

Proof. The ambient fully faithful functor remains fully faithful on the specified generated source. Its image is closed under colimits and cofibres and contains the assumed target generators. Apply the generation criterion. Distinct global sector names do not imply orthogonal local blocks. A direct-sum version would additionally require both cross mapping spectra to vanish; otherwise the extension data must be retained as in Chapter 85. □

The non-solvable comparison and the actual complement

Corollary 86.4.11 *(the complement after specifying the covered set).* The uncovered global parameter set is, by definition,

$$S_{\mathrm{fail}} \setminus (S^{\mathrm{dihedral}} \cup S^{\mathrm{nearly\text{-}ord}} \cup S^{\mathrm{Serre\text{-}res}}).$$

Identifying it with a particular projective-image class, or identifying its local image with a complementary spectral subcategory, requires a further classification and support theorem. No assertion that it consists exactly of A_5 parameters follows from the sector definitions.

Proof. The displayed set is precisely the complement of the stated union. Local restriction need not preserve disjointness, so no categorical complement follows from this set identity. In particular, a large-image parameter is not automatically in a set defined by failure of (TW2). □

Conjecture 86.4.12 *(the non-solvable non-nearly-ordinary comparison).* For the intended non-solvable, non-nearly-ordinary local target, including the projectively A_5 case, construct the actual source objects, spectral images, derived comparison and target generation. An automorphic route must specify the Artin or geometric lift, Taylor's potential-automorphy hypotheses, the applicable Barnet-Lamb–Gee–Geraghty lifting hypotheses, descent, and any Emerton patching data. These arithmetic inputs alone do not supply the categorical generator comparison.

Remark 86.4.13 *(projective images and descent).* The projective-image cases to distinguish include cyclic, dihedral, A_4, S_4, A_5 and the finite-field PSL_2/PGL_2 cases in the appropriate finite-subgroup classification. Their relation to (TW2), local distinguishedness and the actual uncovered set must be proved. Taylor's icosahedral Artin problem and the potential automorphy of Hilbert modular representations have different hypotheses. Descent back to $\mathbb{Q}$ and comparison of local generators remain additional constructions, even when an automorphic lift over an extension has been obtained.

Conventions for the global sector comparisons

Serre's residual modularity, Arthur–Clozel base change, Skinner–Wiles nearly-ordinary deformation theory, Taylor's potential automorphy and Barnet-Lamb–Gee–Geraghty lifting retain their respective hypotheses. Their Hecke eigenvalues require a chosen coefficient field, local-global compatibility and a map to the fixed local problem. The Hodge convention and determinant must be translated as above; weight-space variables and U_p eigenvalues are different data.

The actual local source is a block or a specified full stable subcategory of $D(\mathcal{A}_\zeta)$, and the actual spectral target is a specified support subcategory of $\mathrm{Ind}\, D^b_{\mathrm{coh}}(\mathcal{X}_\delta)$. Identifying the sph-fin and nilpotent-support notation with these carriers requires comparison maps. Modularity, scalar $R = T$, finite subgroup classification and existence of cycles do not replace those maps.

86.5 The Conditional Equivalence for the Local Functor

Union of the strata

Theorem 86.5.1 *(conditional essential image on generated sectors).* For every $p \geq 5$ and full central character, suppose the ordinary, pBT non-ordinary and the three named non-Taylor–Wiles comparisons have their complete carrier, support, restriction and generator data. Suppose their source objects lie in one common stable category and their images generate the chosen target subcategory, retaining all overlap and extension objects. Then the restricted DEG functor is an equivalence onto that

target. The ordinary case retains (TW1)–(TW4), the pBT comparison its stated (TW1)–(TW3), and each global sector its own modularity and local comparison hypotheses.

Proof. The fully faithful exact colimit-preserving restriction has image closed under the generating operations and containing the stipulated generators. Proposition 86.2.4 applies. No pairwise orthogonality of these global sector names is used. □

The remaining essential-image construction

Conjecture 86.5.2 *(the non-ordinary and non-Taylor–Wiles comparison).* The full intended essential-image problem still asks for the supersingular support and generator construction of Conjecture 86.2.12 and the non-solvable non-nearly-ordinary comparison of Conjecture 86.4.12. Showing that these are all remaining sectors requires an actual common carrier, coverage theorem and extension-compatible assembly, as well as all unresolved ordinary and pBT generator comparisons. There is no established reduction to two comparisons alone.

Interaction with the full F1 comparison

Conjecture 86.5.3 *(the complete arithmetic comparison diagram).* The five comparison directions remain the supersingular core, the non-solvable non-nearly-ordinary core, the small-prime wild class map, the Eisenstein adelic comparison and geometric Bernstein-centre gluing. They must be assembled with the actual ordinary, pBT, small-prime and cuspidal functors, their support and mapping-spectrum data, and the global diagram of Conjecture 70.7.1. Proving that this diagram realizes the full essential image requires both coverage and gluing; a list of directions is not an equivalence criterion by itself.

86.6 Consequences

Quadratic induction has an explicit local restriction and irreducibility test, including the characteristic-two nonsplit self-extension and an inert-prime reducible example. Fixed-determinant quotient equations construct the relative ordinary lifting problem. Fixed Hodge and inertial types give actual intersection rings, generic fibres and formal maps retaining determinant identifications. None identifies a framed lifting scheme with the spectral stack or assigns a Hodge type to a residual point. These constructions provide objects and maps for the local categorical questions. Their essential images, arithmetic generators, support comparisons and global sector assembly remain the additional constructions specified above.

Primary conventions for the lifting constructions

Kisin, *The structure of potentially semi-stable deformation rings*, uses the framed representability of Section 1.2 and the fixed-type quotient statements of Theorems 1.2.1–1.2.2, pp. 6–7 of the author manuscript. Page references here use that manuscript. Dotto–Emerton–Gee, arXiv:2603.26887, Theorem 1.1.1 and Remark 1.1.2, p. 3, Section 1.1.6, p. 4, and Section 1.3, p. 8, specify the functor, essential-image limit and central-character convention. Definition C.1.1, p. 265, specifies the determinant-identified stack; Appendix C.2, equations (C.2.1)–(C.2.2), p. 268, and Remark C.2.17, p. 272, specify the coefficient-compatible Galois/étale-module comparison.

Chapter 87

F1.a.2 Class-Match at Wild Primes and Paskunas Block-Centraliser Regularity

At the primes $p = 2, 3$, the intended comparison relates arithmetic block extension classes to a tangent fibre of a specified wild-enhanced Emerton–Gee stack. Its construction requires both a block-to-Galois map and a Galois-to-spectral map, with their coefficient and class normalizations. The finite-algebra examples of Chapter 54 supply neither map. The class-match result below is the conditional composition of these two isomorphisms. Multiplicity and block-centraliser data are separate inputs and do not replace the class-level comparison.

87.1 Objects

The Three Boundary Conditions

The Paskunas comparison has three named boundary conditions:

- **F1.a.2.class-match** Chapter 68.7.3): the class-level identification (not merely dimensional) between the Ext^1-classes α_{01} at $p = 2$ and $(\alpha_{01}, \alpha_{12})$ at $p = 3$ on the automorphic side and the degree-1 tangent fibre of the wild-enhanced Emerton–Gee stack.
- **R.B.1** Chapter 71.4.2(R.B.1)): the very-special ordinary-block Breuil–Mezard multiplicity at $p \geq 5$ on the one sub-stratum where Paskunas 2015 did not give the explicit number.
- **R.B.2** Chapter 71.4.2(R.B.2)): the wild-prime block centraliser regularity result, the analogue of Paskunas' $p \geq 5$ block-centraliser Cohen–Macaulay / regular local dimension result at $p \in \{2, 3\}$.

Inherited Objects

The comparison uses the following objects and proposed data:

- the block decomposition of $\mathrm{Mod}^{\mathrm{sm},\chi}_{GL_2(\mathbb{Q}_p)}$ at $p = 2$ (Chapter 70.5.1) and at $p = 3$ (Chapter 70.5.2);
- the BH type list $\mathbf{T}^{\mathrm{BH}}_{p,\chi}$ (Chapter 68.3.1–68.4.1);
- the wild-enhanced Emerton–Gee stack $\mathcal{X}^{\mathrm{EG,wild},p,\chi}_{GL_2}$ (Chapter 61.5.2);
- the coefficient-compatible arithmetic block realizations asked for in Questions 54.4.8 and 54.4.9; these are additional comparison data, not consequences of the finite examples;
- the Paskunas generic-block regularity statement Chapter 71.3.1(1) at $p \geq 5$.

Additional Inputs

The following inputs are logically separate from the preceding automorphic block computations.

Very-special multiplicity datum. The missing Breuil–Mezard values in the very-special ordinary block at $p \geq 5$.

Wild block-centraliser datum. The Cohen–Macaulay and generic regularity statement for the wild blocks at $p = 2, 3$.

Colmez–Dospinescu–Paskunas 2014, §5. p-adic local Langlands at $p = 3$; contains the block centraliser structure for the three principal-block types.

Bellaïche–Chenevier 2009. *Families of Galois representations and Selmer groups*, Astérisque 324. The residual-gerbe Galois deformation functor at a generic point.

Kisin 2008 J. AMS 21. "Potentially semi-stable deformation rings". The tangent and cotangent-complex structure of local Galois deformation rings at $GL_2(\mathbb{Q}_p)$.

Emerton–Gee arXiv:1908.07185 §9. Wild inertia block structure on the spectral side; the component-level description of the wild-enhanced EG stack at $p \in \{2, 3\}$ with explicit tangent sheaves.

Deformation and Block Comparisons

The very-special datum concerns the Breuil–Mezard multiplicity on the specified sub-stratum. The wild centraliser datum concerns the local ring and its generic fibre. The Galois deformation tangent is computed below; identifying it with the automorphic block deformation problem requires an additional comparison map.

The block-to-Galois comparison and the block-centraliser result are different mathematical inputs. In particular, a centraliser calculation or an equality of dimensions does not construct the class-level map in Hypothesis 87.2.8.

87.2 Fi.a.2 Class-Match at Wild Primes

The Class-Level Target

The class-level compatibility statement is the following: each non-semisimple block's Ext^1-class on the automorphic side must correspond, as a specific k-line, to a specified first-order tangent direction on the wild-enhanced EG stack. We write

$$\mathbb{T}^1_{\mathcal{X}}|_x := H^0\big((\mathbb{L}_{\mathcal{X}}|_x)^\vee\big)$$

for this infinitesimal tangent fibre.

Definition 87.2.1 *(the class-match target, $p = 2$).* At $p = 2$, fix χ and the unique non-semisimple principal block $\mathfrak{B}_2$. The *class-match target* is a k-linear isomorphism

$$\mathsf{c}_2^{\mathrm{class}} \colon \mathrm{Ext}^1_{\mathfrak{B}_2}(S_2, S_2') \xrightarrow{\sim} \mathbb{T}^1_{\mathcal{X}^{\mathrm{EG,wild},2,\chi}_{GL_2}}\Big|_{x_{\tau_2}},$$

where the coefficient category, objects S_2, S_2' and a nonzero class α must be specified as in Question 54.4.8. The target is to carry this class to a specified tangent direction at x_{τ_2}. In characteristic two, trivial and sign characters cannot be used as two distinct objects.

Definition 87.2.2 *(the class-match target, $p = 3$).* At $p = 3$, fix χ and the three principal-block types $\tau_{3,0}$, $\tau_{3,1}$, $\tau_{3,2}$. The class-match target is a k-linear isomorphism

$$\mathsf{c}_3^{\mathrm{class}} \colon \mathrm{Ext}^1(L_0, L_1) \oplus \mathrm{Ext}^1(L_1, L_2) \xrightarrow{\sim} \mathbb{T}^1_{\mathcal{X}^{\mathrm{EG,wild},3,\chi}_{GL_2}}\Big|_{x_{\tau_3}},$$

sending the quiver arrows $\alpha_{01} \mapsto e_1$ and $\alpha_{12} \mapsto e_2$ onto the two tangent directions.

Galois-Deformation Functor on Non-Semisimple Blocks

Definition 87.2.3 *(block deformation functor).* Fix $p \in \{2, 3\}$, χ, and a non-semisimple block $\mathfrak{B}$ with reducible residual $\bar{\rho}_{\mathfrak{B}} \colon G_{\mathbb{Q}_p} \to GL_2(\overline{\mathbb{F}}_p)$. The *block deformation functor* $\mathrm{Def}^{\mathfrak{B}}_{\bar{\rho}_{\mathfrak{B}},\chi} \colon \mathrm{CNL}_{W(k)} \to \mathrm{Set}$ sends a complete Noetherian local $W(k)$-algebra A with residue field k to the set of equivalence classes of liftings $\rho_A \colon G_{\mathbb{Q}_p} \to GL_2(A)$ of $\bar{\rho}_{\mathfrak{B}}$ with central character $\chi \cdot \chi_{\mathrm{cyc}}$ whose reduction is in the block $\mathfrak{B}$. Choose a field k over which the residual representation is defined. The determinant is a specified lift of the residual determinant. Equivalence means conjugation by a matrix reducing to the identity, unless the determinant-one convention is explicitly imposed. Any condition connecting lifts to the arithmetic block is additional data.

Lemma 87.2.4 *(fixed-determinant first-order deformations).* Let $\bar{\rho} : G_{\mathbb{Q}_p} \to GL_2(k)$ be continuous. Write $\mathrm{ad}\,\bar{\rho} = M_2(k)$, with conjugation action, and let $\mathrm{ad}^0\,\bar{\rho}$ be its trace-zero submodule. First-order lifts with fixed determinant, modulo unrestricted strict GL_2-conjugation, form

$$D_{\bar{\rho}} = Z^1_{\mathrm{cts}}(G_{\mathbb{Q}_p}, \mathrm{ad}^0\,\bar{\rho})/d(\mathrm{ad}\,\bar{\rho}).$$

If strict conjugation is restricted to determinant-one matrices, the quotient is $H^1_{\mathrm{cts}}(G_{\mathbb{Q}_p}, \mathrm{ad}^0\,\bar{\rho})$.

Proof. Write a lift as $\rho_\epsilon(g) = (1 + \epsilon c(g))\bar{\rho}(g)$. Multiplication gives $c(gh) = c(g) + g \cdot c(h)$. The identity $\det(1 + \epsilon c) = 1 + \epsilon\,\mathrm{tr}(c)$ shows that fixing the determinant imposes trace zero. Conjugating by $1 + \epsilon X$ changes $c(g)$ by $X - g \cdot X$. This is a coboundary up to sign, and its trace is zero for every $X \in M_2(k)$. Determinant-one conjugations instead require $\mathrm{tr}(X) = 0$, giving the second quotient. □

This calculation supplies no character twist and no pro-representability assertion for a reducible unframed deformation problem. Any additional block condition must first be defined on the lifts and their equivalences.

Hypothesis 87.2.5 *(comparison of formal deformation problems).* Specify the coefficient field, determinant condition, framing and allowed conjugations for the block deformation problem. Assume an equivalence between its formal deformation groupoid and the formal completion of the specified wild Emerton–Gee stack at $x_{\tau_{\mathfrak{B}}}$, preserving automorphism groups. A presentation by $\mathrm{Spf}\, R^{\mathfrak{B}}_{\bar{\rho},\chi}$ requires separate pro-representability and the absence or removal of stabilizers. A formal stack with nontrivial stabilizers is not the formal spectrum of a ring.

Residual Gerbe Comparison

Definition 87.2.6 *(residual gerbe).* For a closed point x of a formal algebraic stack $\mathcal{X}$ with residue field k and automorphism group $\mathrm{Aut}(x)$, the *residual gerbe* $\mathcal{G}_x$ is the closed substack $B\mathrm{Aut}(x) \subset \mathcal{X}$. For $x_{\tau_{\mathfrak{B}}} \in \mathcal{X}^{\mathrm{EG,wild},p,\chi}_{GL_2}$ the automorphism group is the centraliser $Z_{GL_2}(\bar{\rho}_{\mathfrak{B}})$.

Hypothesis 87.2.7 *(spectral tangent comparison).* For the specified deformation problem, assume a natural isomorphism

$$\mathbb{T}^1_{\mathcal{X}^{\mathrm{EG,wild},p,\chi}_{GL_2}}\Big|_{x_{\tau_{\mathfrak{B}}}} \xrightarrow{\sim} D_{\bar{\rho}_{\mathfrak{B}}},$$

where D is the first-order quotient in Lemma 87.2.4. Use its determinant-one variant only when that is the chosen equivalence relation. Require compatibility with the specified arithmetic classes. The notation $\mathbb{T}^1$ above denotes H^0 of the tangent complex. It does not identify this space with H^1 of that complex.

Hypothesis 87.2.8 *(the arithmetic block-to-Galois comparison).* For the specified arithmetic block $\mathfrak{B}$, coefficient field, objects and determinant condition at $p \in \{2, 3\}$, suppose there is a natural isomorphism from the relevant block extension space to $D_{\bar{\rho}_{\mathfrak{B}}}$, carrying the selected extension classes to the specified deformation classes. The coefficient convention and this compatibility are part of the hypothesis.

The finite-algebra resolutions in Chapter 54 do not establish this arithmetic isomorphism. One must construct the block functor and verify its induced map on extensions in the stated coefficient category. At three, a comparison with the radical-square-zero three-vertex model also carries a nonzero adjacent Yoneda product in degree two; it cannot be discarded by the algebra's path relation.

Class-Match Theorem

Theorem 87.2.9 *(F1.a.2 class-match, $p \in \{2, 3\}$).* Assume the wild-EG datum of Conjecture 61.5.7, Hypothesis 87.2.8, and the natural Galois-to-tangent isomorphism of Hypothesis 87.2.7, including its class compatibility. Then the class-match isomorphisms $\mathsf{c}_2^{\mathrm{class}}$ and $\mathsf{c}_3^{\mathrm{class}}$ of Definition 87.2.1 and Definition 87.2.2 exist and are natural in the block $\mathfrak{B}$. The automorphic Ext^1-classes α (at $p = 2$) and α_{01}, α_{12} (at $p = 3$) are sent to the tangent directions e_1 (at $p = 2$) and e_1, e_2 (at $p = 3$) respectively.

Proof. Compose the stipulated block-to-deformation map with the inverse of the stipulated spectral-to-deformation map:

$$\mathrm{Ext}^1_{\mathfrak{B}} \xrightarrow{\sim} D_{\bar{\rho}_{\mathfrak{B}}} \xrightarrow{\sim} \mathbb{T}^1_{\chi}|_{x_{\tau_{\mathfrak{B}}}}.$$

Both maps are natural and preserve the specified class labels by hypothesis. Their composite has the same properties. □

Corollary 87.2.10 *(F1.a.2 full essential surjectivity at $p = 2, 3$).* Under all hypotheses of Theorem 87.2.9, and the block-generation and fully-faithfulness hypotheses of Theorem 70.5.4, the block-diagonal functor $\Phi^{\mathrm{BD}}_{p,\chi}$ of Definition 70.5.3 is essentially surjective at $p \in \{2, 3\}$, hence an equivalence.

Proof. Theorem 70.5.4 establishes fully-faithfulness without the wild-EG hypothesis and essential surjectivity on the semisimple blocks. On each non-semisimple block the only obstruction to essential surjectivity was the class-level match, which is now Theorem 87.2.9. Combining gives full essential surjectivity at $p \in \{2, 3\}$, conditional on all the stated class-comparison and generation hypotheses. □

87.3 R.B.1: Very-Special Ordinary-Block Multiplicity

The Paskunas 2015 Residual

Paškunas 2015 Theorem 1.2 computes the block centraliser structure of the ordinary block at $p \geq 5$ in terms of Breuil–Mezard multiplicities $\mu(\sigma, \bar{\rho})$, except on one explicit sub-stratum: the *very-special* locus where the residual $\bar{\rho}|_{G_{\mathbb{Q}_p}}$ is reducible with scalar semisimplification and cyclotomic twist $\bar{\rho} \sim \chi \oplus \chi\omega$ for a specific character χ (Paškunas 2013 Remark 10.5).

Hypothesis 87.3.1 *(very-special multiplicity input). Conjectural statement.* For $p \geq 5$ and the very-special residual $\bar{\rho} \sim \chi \oplus \chi\omega$, let $\mu_{\mathrm{vs}}(\sigma, \bar{\rho})$ denote the Breuil–Mezard multiplicity of the Serre weight σ. The missing very-special input is the finite list of integers $\{\mu_{\mathrm{vs}}(\sigma, \bar{\rho})\}_{\sigma}$, in the numerical normalisation of Paškunas 2015, satisfying the Breuil–Mezard cycle identity for the scalar residual deformation ring.

Remark 87.3.2 *(multiplicity convention).* The hypothesis is not a formula for the very-special multiplicities. It names the exact numerical input required by the Paškunas centraliser argument.

Corollary 87.3.3 *(R.B.1 conditional implication). Conditional on* Hypothesis 87.3.1. If the datum Hypothesis 87.3.1 holds, then residual (R.B.1) of Proposition 71.4.2 follows.

Proof. The datum supplies exactly the very-special Breuil–Mezard multiplicities requested in (R.B.1). It does not compute all ordinary-block extension groups. The actual local algebra calculation is $\mathrm{Ext}^1_A(k,k) = \mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k)$, proved by the connecting map and pushout in Proposition 71.3.5. There is no subtraction by the number of relations. Transferring a specified deformation direction to an extension between distinct arithmetic simples still requires the explicit class comparison in that proposition; the multiplicity datum alone does not supply it. □

87.4 R.B.2: Wild-Prime Block Centraliser Regularity

$p = 2$ centraliser datum

Hypothesis 87.4.1 *($p = 2$ wild block-centraliser datum). Conjectural statement.* Let $p = 2$, $\chi \in \mathrm{Hom}(\mathbb{Z}_2^\times, k^\times)$, and $\mathfrak{B}$ a Paškunas block of $\mathrm{Mod}^{\mathrm{sm},\chi}_{GL_2(\mathbb{Q}_2)}$ with projective cover $P_{\mathfrak{B}}$. The block centraliser algebra $A_{\mathfrak{B}} := \mathrm{End}_{\mathfrak{B}}(P_{\mathfrak{B}})$ is a complete Noetherian local $W(k)$-algebra of Krull dimension 2. Depending on $\mathfrak{B}$:

(i) *Supersingular blocks.* $A_{\mathfrak{B}}$ is a regular local ring of dimension 2.

(ii) *Twisted-Steinberg block.* $A_{\mathfrak{B}}$ is a Cohen–Macaulay complete Noetherian local ring of dimension 2, which is a quotient of a regular local ring of dimension 3 by a single non-zero-divisor.

(iii) *Principal-series blocks.* $A_{\mathfrak{B}}$ is a Cohen–Macaulay complete Noetherian local ring of dimension 2. Regularity requires a separate datum; the existence of an extension class does not decide it. For example, the regular ring $k[[t]]$ has $\mathrm{Ext}^1_{k[[t]]}(k,k) = k$.

Remark 87.4.2 *(centraliser convention).* This is the $p = 2$ centraliser hypothesis in the complete Noetherian $W(k)$-algebra convention of Paškunas blocks.

Colmez–Dospinescu–Paskunas at $p = 3$

Hypothesis 87.4.3 *($p = 3$ wild block-centraliser datum). Conjectural statement.* Let $p = 3$, $\chi \in \mathrm{Hom}(\mathbb{Z}_3^\times, k^\times)$, and $\mathfrak{B}$ a CDP block. The block centraliser $A_{\mathfrak{B}}$ is a complete Noetherian local $W(k)$-algebra satisfying:

(i) Supercuspidal blocks: $A_{\mathfrak{B}}$ is a regular local ring of dimension 2.

(ii) Principal-series non-Steinberg blocks: $A_{\mathfrak{B}}$ is a Cohen–Macaulay complete Noetherian local ring of dimension 3.

(iii) Steinberg / twisted-Steinberg principal block: $A_{\mathfrak{B}}$ is a Cohen–Macaulay complete Noetherian local ring of dimension 3, a quotient of a regular local ring of dimension 4 by a non-zero-divisor.

Remark 87.4.4 *(centraliser convention).* This is the $p = 3$ centraliser hypothesis in the complete Noetherian $W(k)$-algebra convention of Colmez–Dospinescu–Paškunas blocks.

Unified Wild-Prime Block Centraliser Theorem

Theorem 87.4.5 *(R.B.2 conditional implication: wild-prime block centraliser regularity). Conditional on* Hypothesis 87.4.1 and Hypothesis 87.4.3. Let $p \in \{2,3\}$, χ, and $\mathfrak{B}$ a block of $\mathrm{Mod}^{\mathrm{sm},\chi}_{GL_2(\mathbb{Q}_p)}$. The block centraliser $A_{\mathfrak{B}}$ is a complete Noetherian local $W(k)$-algebra of:

- Krull dimension 2 at $p = 2$;

- Krull dimension 3 at $p = 3$ for principal-series blocks, and 2 for supercuspidal blocks.

It is Cohen–Macaulay, and regular on supersingular / supercuspidal blocks. If $A_{\mathfrak{B}}$ is additionally reduced, its total quotient ring is a finite product of fields. Reducedness is a separate hypothesis for the other blocks.

Proof. At $p = 2$ apply Hypothesis 87.4.1: all blocks are Cohen–Macaulay of dimension 2; supersingular blocks are regular; the twisted-Steinberg block is a hypersurface, and the principal-series blocks are Cohen–Macaulay by the stated datum. At $p = 3$, the second datum gives the stated dimensions and Cohen–Macaulay property, and regularity for supercuspidal blocks. Being a hypersurface does not imply generic regularity: the ring $k[[t, u, \epsilon]]/(\epsilon^2)$ remains nonreduced at its minimal prime. If the centraliser is reduced, its localizations at minimal primes are fields, and its total quotient ring is their finite product. This is the additional assertion under reducedness. □

Corollary 87.4.6 *(conditional* Ext^1*-vanishing at wild primes).* Let $p \in \{2, 3\}$, and assume the centraliser datum of Theorem 87.4.5. Assume in addition that $A_{\mathfrak{B}}$ is reduced and that the specified arithmetic generic-fibre category is equivalent, by an exact localization or Morita comparison, to modules over its total quotient ring $Q(A_{\mathfrak{B}})$. Suppose this comparison identifies localized extension groups for the finite modules under consideration. Then

$$\mathrm{Ext}^1_{\mathfrak{B}}(S, S') \otimes_{A_{\mathfrak{B}}} Q(A_{\mathfrak{B}}) = 0.$$

Thus the specified generic-fibre category is semisimple. This does not assert vanishing at regular closed points.

Proof. A reduced Noetherian ring has total quotient ring a finite product of fields. Every short exact sequence of modules over that ring splits. The stipulated comparison therefore identifies the displayed localized extension group with zero, exactly as in Corollary 71.3.9. Neither the hypersurface presentation nor a dimension equality constructs this arithmetic comparison. In contrast, $\mathrm{Ext}^1_{k[[t]]}(k, k) = k$ at the regular closed point. □

87.5 Consequences for F1.a.3

Conditional Boundary

Theorem 87.5.1 *(F1.a.3 boundary after the wild class comparison).* The F1.a.3 conditions of Proposition 71.4.4 have the following form.

- **(R.A)** Per-block BH type-block comparison.
- **(R.B.1)** Very-special ordinary-block Breuil–Mezard: Hypothesis 87.3.1 is the exact multiplicity input.
- **(R.B.2)** Wild-prime block centraliser regularity: the $p = 2, 3$ centraliser data are Hypothesis 87.4.1 and Hypothesis 87.4.3.
- **(R.C)** Wild-EG compact generation (Conjecture 68.7.5).
- **F1.a.2.class-match**: follows from Theorem 87.2.9 (conditional on the wild-EG enhancement and both class-compatible isomorphisms specified in that theorem).

Proof. This is the direct combination of Corollary 87.3.3, Theorem 87.4.5, and Theorem 87.2.9. □

Corollary 87.5.2 *(F1.a.3 remaining conditions).* Without the conditional inputs named above, F1.a.3 consists of:

(i) (R.A) Per-block BH type-block comparison.
(ii) (R.B.1) Very-special ordinary-block Breuil–Mezard datum.
(iii) (R.B.2) Wild-prime block-centraliser datum.
(iv) (R.C) Wild-EG compact generation.
(v) The actual block-to-Galois and spectral class comparisons of Theorem 87.2.9.

Proof. Direct from Theorem 87.5.1. □

87.6 Sources and Conventions

Class-Match Sources and Conventions

The intended comparison passes from extensions in the chosen arithmetic block to determinant-fixed Galois deformation classes and then to the spectral tangent fibre. The block functor, coefficient ring and induced extension isomorphism must be established in Hypothesis 87.2.8. The spectral map must separately identify the actual deformation functor and preserve the selected classes. Once both maps are natural isomorphisms, their composition is natural. Neither equal dimensions nor a block-centraliser theorem alone gives these maps.

R.B.1 Conventions

R.B.1 is the implication from the datum Hypothesis 87.3.1. Its multiplicity convention is the Paškunas 2015 numerical convention for the Breuil–Mezard deformation ring of the scalar residual $\bar{\rho} \sim \chi \oplus \chi\omega$.

R.B.2 Conventions

R.B.2 is the implication from the data Hypothesis 87.4.1 and Hypothesis 87.4.3. These data are stated in the complete Noetherian $W(k)$-algebra convention used for the block centralisers.

87.7 Consequences

(i) F1.a.2.class-match holds at $p \in \{2, 3\}$ as Theorem 87.2.9, conditional on the wild-EG enhancement Conjecture 61.5.7, the actual block-to-Galois isomorphism of Hypothesis 87.2.8, and the class-compatible residual-gerbe identification on the EG side. The class labels α (at $p = 2$) and α_{01}, α_{12} (at $p = 3$) match the tangent directions through the composition residual-gerbe tangent identification $\circ$ block-to-Galois map.
(ii) R.B.1 is the very-special ordinary-block Breuil–Mezard multiplicity input Hypothesis 87.3.1.
(iii) R.B.2 (wild-prime block centraliser regularity at $p \in \{2, 3\}$) is the $p = 2, 3$ centraliser data Hypothesis 87.4.1 and Hypothesis 87.4.3 are supplied.
(iv) F1.a.3 contains (R.A), (R.B.1), (R.B.2), and (R.C) unless the corresponding inputs are imposed.
(v) The source and convention comparison is recorded in Section 87.6.

Thus the wild class-match is a composition theorem conditional on the wild-EG datum and both class-compatible comparison isomorphisms. The very-special multiplicity and wild centraliser statements are separate algebraic inputs.

87.8 References on Local Langlands and Deformation Theory

Primary sources used in this chapter:

- P. Colmez, G. Dospinescu, and V. Paškunas, ''The p-adic local Langlands correspondence for $GL_2(\mathbb{Q}_p)$'', Cambridge J. Math. 2 (2014), 1–47.
- V. Paškunas, ''The image of Colmez's Montreal functor'', Publ. IHES 118 (2013), 1–191.
- V. Paškunas, ''On the Breuil–Mézard conjecture'' (2015).
- J. Bellaïche and G. Chenevier, *Families of Galois representations and Selmer groups*, Astérisque 324, SMF (2009).
- M. Kisin, ''Potentially semi-stable deformation rings'', J. AMS 21 (2008), 513–546.
- M. Emerton and T. Gee, ''Moduli stacks of étale (φ, Γ)-modules and the existence of crystalline lifts'', arXiv:1908.07185 (2019).
- L. Schlessinger, ''Functors of Artin rings'', Trans. AMS 130 (1968), 208–222.
- B. Mazur, ''Deforming Galois representations'', in *Galois Groups over* $\mathbb{Q}$, MSRI Publ. 16 (1989).

Proposition-level separation of sources:

- Class-match (Theorem 87.2.9) requires the actual block-to-Galois map of Hypothesis 87.2.8 and the spectral tangent comparison. The local Langlands and deformation references identify possible inputs; their applicability to these particular coefficients, objects and classes must be proved.
- R.B.1 (Corollary 87.3.3) is conditional on the very-special multiplicity datum Hypothesis 87.3.1.
- R.B.2 (Theorem 87.4.5) is conditional on the $p = 2, 3$ centraliser data Hypothesis 87.4.1 and Hypothesis 87.4.3.

At the proposition level, the class-match theorem keeps its automorphic and spectral sources distinct. The R.B.1 and R.B.2 statements are the conditional inputs displayed above.

Chapter 88

Global Centre Actions and the Arithmetic Comparison

Local central actions generate an algebra of endomorphisms of a global object. Identifying that algebra with a global Bernstein centre requires faithfulness and generation. Identifying it with a derived spectral centre also requires a comparison of higher operations. A complete local determinant ring, an ordinary category centre, and a derived centre are different objects in these problems.

Two calculations constrain a proposed arithmetic construction. The regular nilpotent orbit in $\mathfrak{sl}_2$ has codimension one, not two. Moreover, a complete local ring cannot split as a product of two nonzero Bernstein components, although its generic fibre can split. These calculations leave concrete geometric and algebraic maps to be constructed before a global comparison follows.

88.1 The Formal and Geometric Data

Use the fixed finite coefficient field k, Witt ring $W = W(k)$, finite ramification set, residual determinant and closed deformation conditions of Chapter 64. For distinct residue primes p, q different from the coefficient prime, the proved formal construction is

$$R_p^{\mathrm{Ch}} \widehat{\otimes}_W R_q^{\mathrm{Ch}} \longrightarrow R^{\mathrm{Ch}}.$$

Its kernel quotient is the closed image, and its completed base changes represent the fibres of local restriction. Theorem 73.1.4 supplies this construction. The actual formal pullback module and its commuting local actions are constructed in Proposition 64.3.2. Their use in the ordinary pairwise geometric comparison requires all the Selmer-algebra, geometric, trace and equivariant-descent data of Corollary 73.1.5. The derived enhancement requires the separate derived datum or both external-product and pullback Tor-independence. The trace comparison alone does not identify the coherent objects.

For the categorical problem, choose a common coefficient base E and specify all coefficient maps from the local constructions. Write $Z(C)$ for the ordinary centre of an E-linear category. When a derived centre is used, write $\mathfrak{Z}(\mathcal{D})$ for that independently specified derived algebra. Its definition, algebra structure and functorial maps are part of the construction. In particular, an E_2-structure on a category does not by notation alone identify its centre with an ordinary commutative deformation ring.

88.2 The Rank-Two Nilpotent Calculation

Proposition 88.2.1 *(the nilpotent cone in rank two).* Let K be an algebraically closed field of characteristic zero. The nilpotent cone in $\mathfrak{sl}_2(K)$ has dimension two. Its nonzero locus is the regular nilpotent orbit, of codimension one in $\mathfrak{sl}_2(K)$. The zero orbit has dimension zero.

Proof. Write

$$X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}.$$

Then $X^2 = (a^2 + bc)\mathrm{I}$, so nilpotency is the equation $a^2 + bc = 0$ in $\mathbb{A}^3_K$. On $b \neq 0$ it is parametrized by a, b, with $c = -a^2/b$. The quadric is irreducible: a factorization into linear forms would give a quadratic form of rank at most two, whereas $a^2 + bc$ has rank three. Thus the cone has dimension two. Every nonzero nilpotent matrix has one Jordan block and is conjugate to $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. A conjugating matrix can be rescaled to determinant one because K is algebraically closed. The nonzero locus is therefore one $SL_2(K)$-orbit of dimension two. The remaining orbit is $\{0\}$. □

Remark 88.2.2 *(the arithmetic local model).* This calculation does not determine the dimension of an arithmetic Selmer space. Such a conclusion requires a constructed local-model morphism and its relative dimension. A nonempty transition space cannot be a subspace of both a closed locus and its open complement, since their intersection is empty. In singular-support theory the support condition lies in the scheme of singularities of a quasi-smooth derived space. It is not, by definition, a locus where the underlying representations have unipotent monodromy. The relevant constructions are Arinkin–Gaitsgory, arXiv:1201.6343, §§2.3, 4.1 and 8.2; §11.1 defines the global nilpotent cone for the geometric local-system stack. An arithmetic analogue requires its own space, singular-support subset and functors.

Restriction of Centres

Proposition 88.2.3 *(restriction to a full subcategory).* Let $\mathcal{D}' \subseteq \mathcal{D}$ be a full E-linear subcategory. Restriction of natural endomorphisms gives a unital E-algebra homomorphism

$$Z(\mathcal{D}) \longrightarrow Z(\mathcal{D}'). \tag{88.1}$$

Its kernel consists of central endomorphisms vanishing on every object of $\mathcal{D}'$. This construction imposes no cellular-filtration or codimension bound on that kernel.

Proof. A natural endomorphism on $\mathcal{D}$ has an endomorphism as its component at every object of $\mathcal{D}'$. Fullness places these components in $\mathcal{D}'$, and naturality restricts to its morphisms. Addition, composition and the identity restrict componentwise, giving the algebra map in Equation (88.1). A natural transformation is zero exactly when all its components are zero, which proves the kernel description. □

A derived centre comparison needs the corresponding coherent maps of derived algebras. A cellular presentation can be used only after it has been constructed for the actual category and related to those maps. The ordinary restriction homomorphism proves no such cellular or higher-centre statement.

Filtration Length and Geometric Codimension

Example 88.2.4 *(a Nygaard filtration on a point).* On the prism $(\mathbb{Z}_p, (p))$, take the Frobenius lift $\varphi = \mathrm{id}$. The Nygaard submodules defined by $\varphi(x) \in p^i\mathbb{Z}_p$ are

$$N^{\geq i}\mathbb{Z}_p = p^i\mathbb{Z}_p \quad (i \geq 0), \qquad \mathrm{gr}^i_N\,\mathbb{Z}_p = \mathbb{F}_p.$$

Every nonnegative graded degree is nonzero, although $\operatorname{Spec}\mathbb{F}_p$ has dimension zero. Thus a geometric codimension bound does not by itself truncate the Nygaard filtration in degree two.

For a filtered complex C with decreasing filtration F, the usual cohomological spectral sequence begins with

$$E_1^{a,b} = H^{a+b}(\mathrm{gr}^a_F\, C). \tag{88.2}$$

Convergence of Equation (88.2) to $H^{a+b}(C)$ requires appropriate completeness, exhaustiveness and convergence hypotheses on the filtered complex. A comparison with a cellular filtration additionally

requires a filtered chain map and a proof of its effect on graded pieces. The dimension calculation above supplies neither a map nor a spectral-sequence degeneration statement.

Remark 88.2.5 *(an equivariant-cohomology comparison).* Suppose a geometric model Y with an algebraic group action H has been specified for an arithmetic transition category $\mathcal{D}_{\mathrm{oI}}$. An equivariant-cohomology construction would require a morphism

$$H_H^{\bullet}(Y, E) \longrightarrow H^{\bullet}(\mathfrak{Z}(\mathcal{D}_{\mathrm{oI}})). \tag{88.3}$$

Both the model Y and this morphism must be constructed, with coefficients, grading and products fixed. If Y is to lie in an affine Grassmannian, its embedding and the functor realizing the arithmetic category must also be given. The rank-two nilpotent calculation does not supply these data or an isomorphism of the displayed algebras.

88.3 Central Actions on a Glued Category

There is an elementary central action once an actual categorical gluing diagram has been supplied. It illustrates which maps a higher-centre comparison must refine. Let $\mathcal{D}_{\mathrm{o}}$, $\mathcal{D}_{\mathrm{I}}$, $\mathcal{D}_{\mathrm{oI}}$ be E-linear categories, and let $F_i : \mathcal{D}_i \to \mathcal{D}_{\mathrm{oI}}$ be E-linear functors. Their category of glued objects $\mathcal{D}$ has objects $(x_{\mathrm{o}}, x_{\mathrm{I}}, \eta)$, where $\eta : F_{\mathrm{o}}x_{\mathrm{o}} \xrightarrow{\sim} F_{\mathrm{I}}x_{\mathrm{I}}$. A morphism is a pair $(u_{\mathrm{o}}, u_{\mathrm{I}})$ satisfying $F_{\mathrm{I}}(u_{\mathrm{I}})\eta = \eta F_{\mathrm{o}}(u_{\mathrm{o}})$.

Proposition 88.3.1 *(a central action from a fibre product of rings).* Let A_{o}, A_{I}, A_{oI} be commutative unital E-algebras with maps $\theta_i : A_i \to A_{\mathrm{oI}}$ and central actions $\alpha_i : A_i \to Z(\mathcal{D}_i)$ and $\alpha_{\mathrm{oI}} : A_{\mathrm{oI}} \to Z(\mathcal{D}_{\mathrm{oI}})$. Assume, for each $a \in A_i$ and $x \in \mathcal{D}_i$, that

$$F_i(\alpha_i(a)_x) = \alpha_{\mathrm{oI}}(\theta_i(a))_{F_i(x)}.$$

Then there is a canonical homomorphism

$$A_{\mathrm{o}} \times_{A_{\mathrm{oI}}} A_{\mathrm{I}} \longrightarrow Z(\mathcal{D}). \tag{88.4}$$

The image of Equation (88.4) is the quotient by the kernel of this action. No injectivity or surjectivity is asserted.

Proof. For $(a_{\mathrm{o}}, a_{\mathrm{I}})$ in the fibre product and a glued object $(x_{\mathrm{o}}, x_{\mathrm{I}}, \eta)$, use the pair of endomorphisms $(\alpha_{\mathrm{o}}(a_{\mathrm{o}})_{x_{\mathrm{o}}}, \alpha_{\mathrm{I}}(a_{\mathrm{I}})_{x_{\mathrm{I}}})$. Their images in $\mathcal{D}_{\mathrm{oI}}$ are components of the same natural endomorphism, since $\theta_{\mathrm{o}}(a_{\mathrm{o}}) = \theta_{\mathrm{I}}(a_{\mathrm{I}})$. Naturality with respect to η gives the equality required for a morphism of glued objects. Naturality on each factor proves naturality on $\mathcal{D}$. Addition, multiplication and units act componentwise, so this is a unital algebra homomorphism. The image quotient follows from the first isomorphism theorem. □

Corollary 88.3.2 *(the additional conditions for an ordinary centre identity).* The map in Proposition 88.3.1 is an isomorphism if and only if the action is faithful and every central endomorphism of $\mathcal{D}$ arises from a compatible pair in $A_{\mathrm{o}} \times_{A_{\mathrm{oI}}} A_{\mathrm{I}}$.

Proof. These are respectively injectivity and surjectivity of the constructed homomorphism. □

Remark 88.3.3 *(the arithmetic higher-centre problem).* For the proposed Selmer-derived category, the gluing spaces and functors have not been constructed by the preceding ring arguments. The required arithmetic statement must first specify an actual categorical descent diagram. It must then construct

the induced maps of higher centres and prove that the relevant homotopy limit agrees with the global higher centre. Even for the ordinary glued category, the proposition gives an action rather than a centre isomorphism. An arithmetic Arinkin–Gaitsgory comparison would also have to match this spectral action with the automorphic one.

88.4 Local Deformation Rings and Bernstein Components

PROPOSITION 88.4.1 *(idempotents in a local ring).* A local ring R has no idempotents except $0, 1$. In particular a complete local deformation ring cannot be a product of two nonzero rings.

Proof. If $e^2 = e$, then $e(1-e) = 0$. At least one of $e, 1-e$ is a unit, because otherwise their sum 1 would lie in the maximal ideal. Multiplying by its inverse shows that the other is zero. A product of two nonzero rings contains the nontrivial idempotent $(1, 0)$, proving the last assertion. □

An idempotent e in an arbitrary commutative ring does produce a module decomposition

$$M = eM \oplus (1-e)M. \tag{88.5}$$

To verify Equation (88.5), write $m = em + (1-e)m$; the intersection is zero after multiplication by e. Bernstein decompositions of a representation category can use its categorical idempotents. They do not produce nontrivial idempotents in a fixed complete local Galois ring.

Example 88.4.2 *(a split generic fibre of a local ring).* Put $R = \mathbb{Z}_2[[x]]/(x(x-2))$. It is local with residue field $\mathbb{F}_2$. The relation $x^m = 2^{m-1}x$ for $m \geq 1$ reduces every series uniquely to $a + bx$ with $a, b \in \mathbb{Z}_2$. The reductions converge 2-adically. Uniqueness follows by evaluating at $x = 0$ and $x = 2$: $a = 0$ and $2b = 0$ imply $a = b = 0$. After inverting 2, these evaluations give

$$R[1/2] \xrightarrow{\sim} \mathbb{Q}_2 \times \mathbb{Q}_2, \qquad a + bx \longmapsto (a, a + 2b). \tag{88.6}$$

The inverse in Equation (88.6) sends (u, v) to $u + (v-u)x/2$. Integrally, the image consists of pairs in $\mathbb{Z}_2^2$ with $v - u \in 2\mathbb{Z}_2$, so $(1, 0)$ is not in the image. The generic-fibre splitting therefore does not split R.

This distinction is necessary before relating a Bernstein component to a formal deformation space. Such a comparison must specify whether it concerns the integral ring, its generic fibre, a quotient, or an action image. A correspondence of closed points establishes none of those ring identifications by itself.

88.5 The Algebra Generated by Local Actions

PROPOSITION 88.5.1 *(finite-place tensor products and their action image).* Let E be a commutative ring and Z_v commutative unital E-algebras indexed by places v. Set

$$A = \varinjlim_{S \text{ finite}} \bigotimes_{v \in S, E} Z_v, \tag{88.7}$$

The transition maps in Equation (88.7) insert units. Pairwise commuting actions on an E-module M induce a unique homomorphism $\alpha : A \to \mathrm{End}_E(M)$. There is a canonical identification

$$A/\ker \alpha \xrightarrow{\sim} \operatorname{im} \alpha. \tag{88.8}$$

For central actions on an E-linear category, the image in Equation (88.8) lies in its ordinary centre.

Proof. On a finite set S, send $\otimes_{v\in S} z_v$ to the product of its acting endomorphisms. Commutativity between the factors makes this a homomorphism. The unit condition makes these maps compatible when S grows. The colimit gives the unique action, and the kernel quotient is its image. The categorical assertion follows by composing natural endomorphisms. This is also Proposition 64.5.1. □

The displayed tensor colimit has a specified algebraic meaning. A topological completion or a restricted tensor product with chosen spherical data is a further construction. Its topology and transition maps must be stated. A factorization interpretation additionally requires the factorization object and its structural maps on a specified base. The tensor formula alone supplies no arithmetic factorization geometry.

Proposition 88.5.2 *(transport across a categorical equivalence).* Let $F : \mathcal{C} \to \mathcal{D}$ be an E-linear equivalence. Suppose both categories carry local central actions from the same algebras Z_v, and that F intertwines each action. Then the induced isomorphism $Z(\mathcal{C}) \simeq Z(\mathcal{D})$ intertwines the assembled actions of A. Their kernels coincide, and their action images are isomorphic.

Proof. Conjugation by F transports natural endomorphisms, as in Proposition 64.3.3. It intertwines every finite product of local central actions. The colimit property therefore intertwines the actions of A. An algebra isomorphism preserves zero, so the kernels agree. Taking their common quotient identifies the two images. □

Conjecture 88.5.3 *(the arithmetic centre comparison).* For the intended arithmetic categories, the comparison requires local functors, coefficient and family maps, and prismatic and untilt descent data. The resulting automorphic and spectral central actions are expected to be identified by the arithmetic comparison, with faithful local action and generation of the specified global centres. If A_{aut}, A_{fact} and $\mathfrak{Z}(\mathcal{D}_{\mathrm{Sel}})$ denote their proposed higher enhancements in one specified category of derived algebras, the intended comparison is

$$A_{\mathrm{aut}} \simeq A_{\mathrm{fact}} \simeq \mathfrak{Z}(\mathcal{D}_{\mathrm{Sel}}). \tag{88.9}$$

Here the higher enhancements, their comparison morphisms, and the proof that those morphisms are equivalences are part of the assertion. An ordinary determinant ring is not substituted for a higher algebra in Equation (88.9).

Corollary 88.5.4 *(what an equivariant arithmetic comparison would transport).* If an arithmetic categorical equivalence has been constructed and intertwines the specified local central actions, it transports their ordinary global action image. To obtain the ordinary-centre clause of Conjecture 64.5.6, those actions must also be faithful and generate the intended centres. The pairwise coherent-sheaf clause remains the separate equivariant isomorphism in that conjecture.

Proof. Apply Proposition 88.5.2 to the ordinary actions. Faithfulness and generation are the additional injectivity and surjectivity conditions. Transport of central endomorphisms alone does not compare the independent coherent objects occurring in the pairwise clause. □

Proposition 88.5.5 *(the two comparisons required by Bernstein gluing).* The formal determinant restriction constructs a pairwise pullback with its local action, and the tensor colimit constructs the algebra generated by all specified local actions. To obtain BC-diamond-glue from these constructions, one must construct the equivariant isomorphism with the global prismatic object and the intended global centre equivalence, compatibly with coefficients, finite-place transitions and untilt descent. Neither construction asserts that these comparison maps exist.

Proof. The pullback and its action are Proposition 64.3.2. The assembled algebra and image are Proposition 88.5.1. The two missing comparisons have the precise domains and codomains of Conjecture 64.5.6 and Conjecture 88.5.3. The nonunique Galois lifts and the extension example in Chapter 64 prevent replacing these maps by determination from two primes or by equality of determinant laws. □

88.6 Arithmetic Descent and Singular Support

Remark 88.6.1 *(the arithmetic descent problem).* For adelic categorical descent, the remaining objects are the local categories with their centre-compatible functors, a geometric realization of the global deformation problem, and compatible prismatic and untilt descent maps. The Selmer-algebra identification, trace comparison and equivariant sheaf comparison retain their full hypotheses in Corollary 73.1.5. The global higher-centre comparison must additionally define its singular-support category and the comparison morphisms. A codimension bound or a scalar-centre identity supplies none of these categorical maps by itself.

Remark 88.6.2 *(the Eisenstein comparison).* An extension to continuous automorphic spectrum requires its own intertwining operators, measures and categorical realization. The formal restriction and ordinary centre constructions here do not assert cuspidal adelic descent or its Eisenstein extension. Those statements require the full comparisons in their respective categories.

88.7 The Geometric Comparison Maps

Remark 88.7.1 *(singular support, filtrations, and central actions).* The singular-support construction must specify the derived Selmer space and the conical subset of its scheme of singularities. A cellular or prismatic construction must specify its filtered complex and the comparison to that category. An equivariant-cohomology model must supply the morphism in (88.3). These data have different domains. Combining them requires actual functors and compatible morphisms, beyond the existence of the separate formalisms.

Chenevier's determinant deformation theorem is used with its formal coefficient ring and residual data, as specified in Proposition 73.1.2. Arinkin–Gaitsgory's singular-support construction applies on its quasi-smooth derived domain. A transfer to an arithmetic Selmer space must establish that domain and the required functoriality. The ring and orbit calculations in this chapter do not make that transfer.

88.8 The Next Arithmetic Construction

On a common geometric chart, the pairwise problem consists of the local pullback, the global sheaf, and the matrices or complexes carrying their central actions. Its solution is an equivariant isomorphism compatible with the overlap maps. The successive obstruction classes in Proposition 64.4.3 perform this calculation on finite free complete local charts.

The global problem requires automorphic and spectral central-action maps from the same finite-place algebras. Their kernels and images can then be compared using Proposition 88.5.1. A higher-centre equivalence further requires its derived comparison map and a proof that the cone of its underlying map of complexes vanishes on the stated domain. These constructions preserve the global arithmetic question while keeping its formal algebra, coherent objects and higher centres distinct.

Chapter 89

Independent Numerical Comparisons for Arithmetic Constants

Five essential numerical constants appear across the arithmetic branch: the determinant-rank residue $r_\zeta = 2$ of Chapter 41.5.1, the first four Li coefficients $\lambda_1, \dots, \lambda_4$ of Chapter 74.2.2, the Riemann–von Mangoldt count $N(T)$ of Chapter 80.2.1, the first three non-trivial ordinates $\gamma_1, \gamma_2, \gamma_3$ appearing throughout Chapters 77–80, and the archimedean digamma value $\psi(1/4) = -\gamma_E - \pi/2 - 3\log 2$ of the Weil archimedean term. Each constant has a defining expression and an alternative calculation in the same normalization.

89.1 Numerical Values and Error Bounds

A numerical identity requires either an analytic equality of the defining expressions or error bounds placing both evaluations within their stated tolerance. Agreement of printed digits alone proves neither an exact identity nor an arithmetic comparison.

Definition 89.1.1 *(certified numerical comparison).* For a real constant C, a certified evaluation consists of a number c and a proved bound $|C - c| \le \varepsilon$. Two such evaluations (c_1, ε_1) and (c_2, ε_2) satisfy $|c_1 - c_2| \le \varepsilon_1 + \varepsilon_2$. The defining expression, normalization and error bound are part of each evaluation.

Remark 89.1.2 *(analytic and numerical descriptions).* A contour count and a Stirling expansion of the gamma factor compute different expressions. Comparing their values requires the analytic identity relating them and control of truncation errors. The analytic identity and the numerical error bounds are separate mathematical requirements.

89.2 The Five Constants and Their Independent Comparisons

$r_\zeta = 2$: determinant-rank residue of 2ζ

Theorem 89.2.1 *(r_ζ comparison).* . The identity

$$r_\zeta \;=\; 2 \cdot \operatorname{Res}_{s=1}\zeta(s) \;=\; 2$$

admits two independent numerical calculations:

(D) *Derivation.* Compute $2(s-1)\zeta(s)$ at $s = 1 + \varepsilon$ via the Euler–Maclaurin continuation of ζ. As $\varepsilon \to 0^+$ this returns 2 by the simple pole of ζ.

(C) *Comparison.* Use the Dirichlet eta identity $\zeta(s) = \eta(s)/(1 - 2^{1-s})$, where $\eta(s) = \sum_{n\ge 1}(-1)^{n+1}n^{-s}$. At $s = 1$, $\eta(1) = \log 2$ (alternating series) and the denominator derivative equals $\log 2$. Hence $\operatorname{Res}_{s=1}\zeta(s) = \log 2/\log 2 = 1$, and doubling returns $r_\zeta = 2$.

The two paths agree at rtol $= 10^{-10}$.

Proof. Source comparison: the proof source is the Euler–Maclaurin evaluation of $\zeta(s)$ at $s = 1 + \varepsilon$; the comparison source is the eta closed-form identity. The two sets are disjoint. Numerically, |derived − compared| $< 10^{-10} \cdot$ |compared| at 80 decimal digits of working precision. Semantically, Euler–Maclaurin on the non-alternating Dirichlet series is disjoint from the alternating-series evaluation of η, and the eta identity supplies the residue algebraically rather than numerically. □

Li coefficients $\lambda_1, \ldots, \lambda_4$

Theorem 89.2.2 *(Li coefficient comparison).* . For $n \in \{1, 2, 3, 4\}$ the Li coefficient λ_n of Chapter 74.2.2 admits two independent numerical calculations.

(D) *Derivation.* Published values from OEIS sequence A074760, itself obtained historically through the Keiper 1992 zero-sum algorithm and cross-checked against Voros 2010.

(C) *Comparison.* Extract λ_n from the analytic Taylor series of $\log \xi_{\mathbb{R}}(s)$ at $s = 1$, where $\xi_{\mathbb{R}}(s) = \frac{s}{2}(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed Riemann xi function (Riemann 1859; Davenport *Multiplicative Number Theory* §5). After the pole-subtraction

$$\log \xi_{\mathbb{R}}(s) = \log(s/2) - \tfrac{s}{2}\log\pi + \log\Gamma(s/2) + \log\big((s-1)\zeta(s)\big), \tag{89.1}$$

all terms are analytic near $s = 1$. Taylor expand the right hand side and apply Leibniz to $f(s) = s^{n-1}$ times the log-xi expansion:

$$\lambda_n \;=\; \sum_{k=0}^{n-1} \binom{n}{k} (n-k)\, c_{n-k},$$

where c_m is the m-th Taylor coefficient of the right side of (89.1) at $s = 1$.

The two paths agree at rtol = 10^{-6}.

Proof. Source comparison: the proof source is OEIS A074760 together with Keiper 1992; the comparison source is the Taylor expansion of $\log \xi_{\mathbb{R}}$ at $s = 1$ and Li's derivative formula applied to the pole-subtracted expression (89.1). The numerical sources are disjoint after the common definition of λ_n is fixed. Numerically, |derived − compared| $< 10^{-6} \cdot$ |derived| for each $n \in \{1, 2, 3, 4\}$. Semantically, derivation lives on the *zero side* (a truncated sum over non-trivial zeros), while comparison lives on the *log-xi analytic side* (a Taylor expansion at a regular point); no zero data enters the comparison path and no Taylor expansion enters the derivation path. The disjointness is structural, not merely nomenclatural. □

Remark 89.2.3 *(precision floor for Li).* The rtol 10^{-6} budget is the practical floor of both paths. The derivation side is limited by the precision of the published OEIS digits (conservatively 45 digits in the rounded table entry, effectively reduced by Keiper-algorithm truncation error on finite zero counts). The comparison side is limited by cancellation in the numerical Taylor expansion of the pole-subtracted log-xi expression, which loses roughly 10 digits of precision per Taylor order. At 80 dps working precision the comparison has ~ 50 accurate digits at $n = 4$, far better than the 10^{-6} rtol budget. The budget is set by the derivation side's published-table digit count.

Riemann–von Mangoldt $N(T)$ at $T = 100$, $T = 1000$

Theorem 89.2.4 *($N(T)$ comparison).* . For $T \in \{100, 1000\}$, the non-trivial zero count $N(T)$ of Chapter 80.2.1 admits two independent numerical calculations.

(D) *Derivation.* Turing's method gives the exact count by counting sign changes of the Riemann–Siegel Z-function (the real function $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$) on the critical line up to T and checks the count by the argument principle on a rectangle enclosing the strip.

(C) *Comparison.* The von Mangoldt asymptotic

$$N(T) \approx \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}$$

(Riemann 1859 final remark, von Mangoldt 1895 full proof; Stirling on the gamma factor of ξ) with the result rounded to the nearest integer. For $T = 100$ and $T = 1000$, the leading terms round to within one of the exact count without invoking any zeta call.

The two paths agree at rtol = 1 (integer comparison).

Proof. Source comparison: the proof source is the Turing contour-counting method and the exact sign-change count of the Riemann–Siegel Z-function; the comparison source is the von Mangoldt asymptotic and the Stirling-based derivation from Riemann 1859. The two algorithmic sets are disjoint. Numerically, $|\text{derived} - \text{compared}| \le 1$ at $T \in \{100, 1000\}$. Semantically, Turing's method uses a *contour integral* of ξ'/ξ around the strip boundary plus sign-change count of Z on the critical line, while the asymptotic construction uses *Stirling on the archimedean gamma factor* of ξ with no zero evaluation and no contour. Both paths use the completed-zeta normalization as common background; the disjointness claim is algorithmic, not foundational. □

First non-trivial zeros $\gamma_1, \gamma_2, \gamma_3$

Theorem 89.2.5 *(first-three-zeros comparison).* . The ordinates $\gamma_1 \approx 14.1347\ldots$, $\gamma_2 \approx 21.0220\ldots$, $\gamma_3 \approx 25.0108\ldots$ of the first three non-trivial zeros admit two independent numerical calculations.

(D) *Derivation.* Published LMFDB / Odlyzko published values, originally produced by the Odlyzko–Schönhage 1988 FFT-accelerated algorithm on the zeta function and archived in Odlyzko's first-million-zeros file.

(C) *Comparison.* Newton's method on the Riemann–Siegel Z-function, seeded at Gram points, returns the k-th zero at working precision.

The two paths agree at rtol = 10^{-15}.

Proof. Source comparison: the proof source is the LMFDB table and Odlyzko 2002 zeros file; the comparison source is Newton iteration with a Gram-point seed. The two sets are disjoint. Numerically the agreement holds at rtol = 10^{-15}. Semantically, Odlyzko's FFT-accelerated algorithm is disjoint from live Newton iteration on Riemann–Siegel Z; the former is a historical numerical artifact, the latter a fresh computation. Both recover the same zero via analytically disjoint paths. □

Archimedean digamma $\psi(1/4)$

Theorem 89.2.6 *($\psi(1/4)$ comparison).* . The value

$$\psi(1/4) = -\gamma_E - \frac{\pi}{2} - 3\log 2$$

that appears in the archimedean term of the Weil explicit formula admits two independent numerical calculations.

(D) *Derivation.* Analytic continuation of ψ gives the value via a rational-function approximation combined with a shift-and-asymptotic recursion.

(C) *Comparison.* Direct substitution into the Gauss multiplication formula for ψ at rational argument p/q:

$$\psi(p/q) = -\gamma_E - \log(2q) - \frac{\pi}{2}\cot(\pi p/q) + 2\sum_{k=1}^{\lfloor (q-1)/2 \rfloor} \cos(2\pi k p/q)\log\sin(\pi k/q).$$

At $p/q = 1/4$, the sum runs over $k = 1$ only ($\lfloor (q-1)/2 \rfloor = \lfloor 3/2 \rfloor = 1$); the $k = 1$ term vanishes because $\cos(2\pi \cdot 1 \cdot (1/4)) = \cos(\pi/2) = 0$. The remaining terms give $-\gamma_E - \log(2 \cdot 4) - \frac{\pi}{2}\cot(\pi/4) = -\gamma_E - \log 8 - \frac{\pi}{2}$, which equals $-\gamma_E - \pi/2 - 3\log 2$.

The two paths agree at rtol = 10^{-14}.

Proof. Source comparison: the proof source is the Lanczos-style digamma evaluation; the comparison source is the Gauss multiplication formula and the closed-form expression. The two sets are disjoint. Numerically the agreement holds at rtol = 10^{-14}. Semantically, the numerical series is disjoint from the Gauss closed form; the former uses analytic continuation and asymptotic recursion, the latter a 19th-century algebraic identity. □

89.3 Numerical List

The five independent comparisons of Chapter 89.2 have the following numerical tolerances.

Comparison	Constant	rtol
r_ζ residue	$r_\zeta = 2$	10^{-10}
Li coefficients	$\lambda_1, \lambda_2, \lambda_3, \lambda_4$	10^{-6}
Riemann–von Mangoldt count	$N(100),\ N(1000)$	1 (integer)
First three non-trivial zeros	$\gamma_1, \gamma_2, \gamma_3$	10^{-15}
Archimedean digamma value	$\psi(1/4)$	10^{-14}

THEOREM 89.3.1 *(independent numerical agreement).* . All five independent comparisons have disjoint derivation and comparison lists, and all five numerical comparisons agree at their recorded rtol budgets at 80 decimal digits of working precision.

Proof. The five preceding computations give the asserted tolerances: 10^{-10} for r_ζ, 10^{-6} for the first four Li coefficients, integer agreement within the rounded Riemann–von Mangoldt tolerance, 10^{-15} for the first three zeros, and 10^{-14} for $\psi(1/4)$. □

89.4 Position in the Arithmetic Branch

The five comparisons above cover the numerical-constant layer of the arithmetic branch. The structural classes are orthogonal:

- structural closures of $\mathrm{L}_{\mathrm{ACD}}$ and the conjectures F1.a/F2.c;
- the $\mathrm{H}^{\mathrm{Ar}}_{\mathrm{HP}}$ operator comparison;
- $\mathcal{V}^{\mathrm{prim}}$ P3 conditional closures;
- the Positselski four sub-items;
- F1.a1–F1.a3 closure;

- R-analytic density-of-zeros residuals;
- VB* spectral positivity.

The numerical-constant comparisons do not overlap the structural theorem comparisons; they isolate the numerical content supporting the main structural claims.

89.5 Numerical Boundary

Five arithmetic branch constants – $r_\zeta = 2$, λ_1 through λ_4, $N(100)$ and $N(1000)$, γ_1, γ_2, γ_3, and $\psi(1/4)$ – carry independent comparisons. Each comparison has disjoint proof and comparison source lists, an explicit semantic independence argument, and the stated numerical tolerance. No structural theorem follows from these constants alone.

Chapter 90

Arithmetic Fibres and Pairing Comparisons

The constructions of Chapters 49–59 use Iwasawa coefficient rings, local representation categories, adelic automorphic spaces and complex analytic test spaces. Their relation with the constructions of the preceding volumes requires specialization or realization maps. Some proposed comparisons concern arithmetic fibres of complex or Calabi–Yau assertions; others concern distinct objects. A proposed fibre interpretation is a question until its functor is constructed. A pairing comparison requires specified carriers and maps, including the coefficient change and completion. The Weil pairing comparison below states these data explicitly.

90.1 The Fibre Dichotomy

There are two different questions: whether an arithmetic construction is obtained from a common integral object by specialization, and whether two constructions on different carriers admit a realization comparison. Neither follows by replacing the names of their coefficient rings. Statements on unrelated categories may have useful analogies without a comparison functor. The following cases specify constructions and obstructions needed to decide these questions.

Comparison data

A fibre comparison specifies its base ring, a morphism of bases, and a base-change or realization functor on the stated objects. For paired vector spaces it must additionally give a map and an identity of the transported forms. Agreement of scalar kernels alone does not provide those data. The marked comparison of Section 1.2 retains the algebra, its module and the specified functional; its functorial form also requires compatible maps on changes of input.

Classification

arithmetic theorem	prior theorem	relation
arithmetic Positselski equivalence	Vol I *conditional continuous fixed-coalgebra co–contra comparison*	comparison question
Chenevier–Bernstein E_2-transport	Vol I *conditional factorization restriction and properadic envelope of Theorem A*	comparison question
Bost–Connes modular operator	Vol II *closed-color operadic comparison*	comparison question
Koszul–Petersson discrete-core identity	Vol I *conditional Arnold comparison and scalar companion values*	comparison question
Weil pairing realization	Vol I *conditional chiral Hochschild support and comparison statements*	requires a specified fibre map
Bernstein block-additivity	Vol I *conditional Arnold comparison and scalar companion values*	requires comparison data
de Branges spectral positivity	Vol III *two-stage Calabi–Yau-to-chiral construction*	requires comparison data
Riemann–von Mangoldt zero density	Beilinson–Drinfeld, Chevalley–Cousin concentration	requires comparison data
adelic categorical descent	none	native
wild-prime compact generation	none	native

90.2 THREE BASIC COMPARISONS

Arithmetic Positselski and Vol I Theorem B

The Vol I theorem is the conditional continuous fixed-coalgebra co–contra comparison: a strict Mittag–Leffler completed chiral CDG-coalgebra must satisfy its CP1–CP3 package of finite-stage comparisons, continuous structure maps, derived inverse-limit identifications and preservation of second-kind acyclic objects. The arithmetic comparison concerns the equivalence

$$\mathrm{D}^{\mathrm{co}}_{\Lambda_p}(C\text{-comod}_{\Lambda_p}) \simeq \mathrm{D}^{\mathrm{ctr}}_{\Lambda_p}(C^\vee\text{-contramod}_{\Lambda_p})$$

over $\Lambda_p = \mathbb{Z}_p[[G_p]]$, with $\mu(C) = 0$ and bounded-below Tor against $\mathbb{F}_p$.

The arithmetic constructions involve Pontryagin–Weil duality, Greenlees–May completion, Matlis duality and Iwasawa coefficients; the complex construction uses chiral coalgebras and field-coefficient coderived and contraderived categories.

PROPOSITION 90.2.1 *(coefficient obstruction and integral comparison).* Reduction through the augmentation $\Lambda_p \otimes_{\mathbb{Z}_p} \mathbb{F}_p \to \mathbb{F}_p$ has characteristic-p target. It does not give a complex fibre. A comparison with the characteristic-zero chiral equivalence requires a common ring R with specified maps $R \to \mathbb{F}_p$ and $R \to \mathbb{C}$, categories $\mathcal{C}_R$, $\mathcal{D}_R$ and a functor $F_R : \mathcal{C}_R \to \mathcal{D}_R$, together with base-change functors to each fibre and natural isomorphisms $F_k S^{\mathcal{C}}_k \cong S^{\mathcal{D}}_k F_R$ for $k = \mathbb{F}_p, \mathbb{C}$. The fibre functors F_k must themselves be identified with the stated coderived–contraderived equivalences, using their actual topologies, completions and derived base change.

Proof. A unital map $\mathbb{F}_p \to \mathbb{C}$ would send $0 = p1$ to the nonzero complex number p, a contradiction. For the proposed common model, the displayed natural isomorphisms are precisely the commuting base-change squares: at every object X they identify $F_k(S^{\mathcal{C}}_k X)$ with $S^{\mathcal{D}}_k(F_R X)$. They provide a comparison only after these categories, functors and isomorphisms have been constructed. No equivalence after base change is asserted merely from a tensor product of rings. □

Remark 90.2.2 *(the two coefficient fibres).* The common integral chiral/Iwasawa model and its compatibility with Pontryagin duality remain open comparison data. Even the reduction of a duality requires its chosen dualizing object and a base-change map; identifying underlying vector spaces does not supply these. The complex theorem retains its original characteristic-zero scope.

Bernstein Block-Additivity and Chiral Chains

The block-additivity statement of Chapter 71.4.1 concerns Hochschild homology of the stated smooth $GL_2(\mathbb{Q}_p)$ representation category. Its references include Bernstein, Bernstein–Deligne, Bushnell–Henniart and Paškūnas. A comparison with chiral chains needs an enhanced category and a common coefficient field; it does not follow from the names of those references.

Proposition 90.2.3 *(a chain-level block comparison).* Specify a field k, any required derived coefficient change, and an enhanced k-linear realization C of the arithmetic category. Suppose its Hochschild complex decomposes as $C_{\mathrm{Hoch}}(C) = \bigoplus_\nu C_\nu$ according to its block idempotents. Choose a chiral object A over k with a chain model $C_{\mathrm{ch}}(A) = \bigoplus_\nu B_\nu$ and chain maps $\chi_\nu : C_\nu \to B_\nu$. If each χ_ν is a quasi-isomorphism, their direct sum transports block additivity to this chiral complex. A mixed-complex comparison additionally requires these decompositions to preserve the Connes operators and $\chi_\nu B = B\chi_\nu$.

Proof. Direct sums of complexes of k-vector spaces commute with cohomology. Thus the direct sum of the isomorphisms $H^*(\chi_\nu)$ is an isomorphism, respecting the block projectors. Commutation with the differential gives a chain map; the additional identity is exactly compatibility with the mixed operator. □

The categories, chiral object and maps in this proposition are additional comparison data. Hochschild additivity by itself does not construct the bar–cobar unit and counit or prove them to be equivalences. In particular the conditional factorization restriction of Vol I Theorem A retains its factorization and convergence packages, and its properadic refinement requires its properadic package. The conditional Verdier–bar comparison retains its finite regular holonomic windows, biduality, tensor exchange and compatible continuous bar/cobar data. The conditional Arnold comparison and scalar companion values concern their specified connection data and companion pairs, rather than a general bar–cobar equivalence.

Chenevier–Bernstein Transport and E_2-Recognition

Chenevier determinant-trace pseudocharacters, Bernstein transport and the chiral E_2-centre concern different categorical data. Their proposed relation asks for the following centralizer comparison.

Proposition 90.2.4 *(transport of a specified centralizer).* Suppose $\mathcal{C}, \mathcal{D}$ are specified $(\infty, 2)$-categories, $X \in \mathcal{C}$, and $J : \mathcal{C} \to \mathcal{D}$ is a functor that induces an equivalence on the endomorphism category of X. Then it transports the endomorphisms of the identity 1_X to those of 1_{JX}, preserving their two compositions and hence the induced E_2 structure. An arithmetic/chiral application additionally requires these two endomorphism objects to be identified with the claimed arithmetic transport and chiral centre.

Proof. The equivalence on endomorphism categories is fully faithful, so it induces an equivalence of the mapping objects from 1_X to itself. Functoriality preserves identity, vertical composition and the horizontal composition inherited from the $(\infty, 2)$-category. These are the two compatible compositions defining the E_2 structure. □

The conditional factorization restriction and properadic envelope of Vol I Theorem A is not an E_2-centre identification. The functor J, the specified object X and the two identifications remain part of the arithmetic comparison question. A restricted Hilbert tensor factor alone does not construct this categorical functor.

90.3 Further Fibre Comparisons

Bost–Connes Dynamics and a Circle KMS Comparison

PROPOSITION 90.3.1 *(logarithmic dilation and an actual circle quotient).* The map $Uf(t) = f(e^t)$ is unitary from $L^2(\mathbb{R}_{>0}, dr/r)$ to $L^2(\mathbb{R}, dt)$. For $D_a f(r) = f(e^{-a}r)$ it satisfies $UD_aU^{-1}F(t) = F(t-a)$. Fix $L > 0$. The quotient group $\mathbb{R}_{>0}/e^{L\mathbb{Z}}$ is instead identified with $\mathbb{R}/L\mathbb{Z}$, with Haar measure dt on $[0, L)$. Its Hilbert space has orthonormal basis $L^{-1/2}e^{2\pi int/L}$, $n \in \mathbb{Z}$; inverse translation acts on this basis by $e^{-2\pi ina/L}$.

Proof. Substitution $r = e^t$ proves the norm identity and the intertwining. The logarithm carries the subgroup $e^{L\mathbb{Z}}$ to $L\mathbb{Z}$, so induces the quotient isomorphism. Orthogonality and completeness are the Fourier-series theorem on $[0, L)$, with the stated norm. No nonzero periodic function belongs to $L^2(\mathbb{R}, dt)$: its squared integral over the line is the sum of identical integrals over disjoint periods. Thus the circle space is a different carrier, not a nonzero periodic subspace of the full logarithmic L^2 space. □

For the Bost–Connes dynamics $\sigma_a(\mu_r) = r^{ia}\mu_r$, a circle action of period L would require $L \log r \in 2\pi\mathbb{Z}$ for every surviving nonzero generator. No such $L > 0$ works for both $r = 2$ and $r = 3$: otherwise $\log 2/\log 3$ is rational, which would give $2^m = 3^n$ for positive integers. A comparison with a circle KMS theory must therefore specify a suitable quotient or representation, its state, its periodic action and an intertwiner with the chosen circle model. The logarithm alone supplies none of these.

Remark 90.3.2 *(finite-place and circle data).* The proposed relation of the global Bost–Connes system and its finite-place subalgebras to Costello–Gwilliam theory remains a separate question. The closed-color statement in Volume II concerns an operadic comparison; it does not specify a KMS state or periodic dynamics. The explicit circle quotient above describes its analytic model, not an automatic identification of the two quantum theories.

Koszul–Petersson Pairings and Arnold Comparison

PROPOSITION 90.3.3 *(a specified Petersson and chiral pairing comparison).* Let E be a complex vector space with bilinear form B and antilinear involution σ. Let H_{cusp} be a fixed cuspidal automorphic L^2 space with specified central character and quotient Tamagawa measure. Suppose complex-linear maps $I : E \to H_{\mathrm{cusp}}$ and $J : E \to H_{\mathrm{ch}}$ have dense images, where H_{ch} is a specified Hilbert realization of a chiral bar pairing, and

$$B(v, \sigma w) = \langle Iv, Iw\rangle_{\mathrm{Pet}} = \langle Jv, Jw\rangle_{\mathrm{ch}}.$$

Then $Iv \mapsto Jv$ extends uniquely to a unitary $H_{\mathrm{cusp}} \to H_{\mathrm{ch}}$.

Proof. Both maps have kernel $\{v : B(v, \sigma v) = 0\}$ by their norm identities. The induced map is well-defined and isometric on the dense image of I. Its extension has closed range containing the dense image of J, hence is unitary. □

Constructing E, B, σ, I, J for the native discrete Koszul sector is the comparison problem. An Arnold or KZ* connection requires, in addition, bundles over an identified base and a connection-preserving bundle map: the Hilbert identity alone does not compare connections. The oscillator contraction is a local model; it does not identify a Petersson integral with that contraction without the actual automorphic realization and normalization.

HP–Li Paley–Wiener and Theorem H

Theorem 90.3.4 *(realization of the restricted arithmetic pairing).* Use $\mathscr{A} = C_c^\infty(\mathbb{R};\mathbb{C})$, $f^*(t) = \overline{f(-t)}$ and the full signed Weil functional $W = P - W_\infty - W_{\mathrm{fin}}$ in the angular Fourier convention of Chapter 78.1.2. Let $D \subset \mathscr{A}$ be a complex vector subspace and $q_D(f,g) = W(f * g^*)$. A positive Hilbert realization of q_D exists if and only if q_D is positive semidefinite. Its minimal realization is the completion of $D/\{f : q_D(f,f) = 0\}$. For $D = \mathscr{A}$, existence is equivalent to RH.

Proof. The construction and equivalence are Chapter 78.8.1, restricted to D. The nullspace is orthogonal to D by Cauchy–Schwarz. The quotient inner product is therefore well-defined. □

The conditional chiral Hochschild support and comparison statements of Vol I Theorem H retain their completed Koszul-chart or quotient-specific hypothesis packages and bounded-to-chart maps. They do not themselves supply a Hilbert pairing. An arithmetic fibre comparison requires a specified complex fibre E, its bilinear pairing B, an antilinear involution σ, and a complex-linear map $T : D \to E$ with

$$B(Tf, \sigma Tg) = W(f * g^*).$$

Here $H_{\mathrm{aut}} = L^2_{\mathrm{cusp}}([GL_2], \chi; dh)$ for a fixed unitary central character χ and quotient Tamagawa measure dh. A Petersson realization further requires a complex-linear map $I : E \to H_{\mathrm{aut}}$ and $B(v, \sigma w) = \langle Iv, Iw\rangle$ on the indicated image. These are equalities of forms on specified carriers. If they hold, IT realizes q_D and extends isometrically from its Hilbert quotient to the closure of $IT(D)$. Indeed its nullspace is exactly the nullspace of q_D, and the pairing identity makes the induced map norm preserving. This constructs the Hilbert comparison from the stated native data. The scalar ratio K_F^{prop} of Chapter 78.7 supplies neither T nor I. The actual cusp-form coefficient identity Chapter 78.4 is a different formula.

Hermite–Biehler Positivity and the Phi Construction

For $F(z) = \xi_{\mathbb{R}}(1/2 + iz)$, the shift-family criterion of Chapter 74.2.4 uses $E_h(z) = F(z + ih)$ for all $h > 0$, or a sequence tending to zero. One fixed shift is insufficient by Chapter 74.2.5. The derivative criterion of Chapter 74.2.6 allows real zeros of $F + iF'$; requiring no real zeros adds simplicity. These are precise analytic criteria, not a construction of a Calabi–Yau input or a chiral specialization.

Proposition 90.3.5 *(positive index and an auxiliary indefinite space).* A positive Hermitian space of dimension greater than p admits no form-preserving injection into a finite-dimensional Hermitian space of signature (p, q). In particular an HB space need not embed positively in a fixed plane of signature $(1, 1)$. For any positive Hermitian space (H, h) there is instead the auxiliary Hermitian form

$$Q((x, y), (x', y')) = h(x, x') - h(y, y')$$

on $H \oplus H$, with positive isometric inclusion $x \mapsto (x, 0)$. In finite dimension n its signature is (n, n).

Proof. Projection to the positive summand of a space of signature (p, q) is injective on every positive subspace: a nonzero vector in its kernel would have nonpositive norm. Hence that subspace has dimension at most p. For an explicit HB example, take $E(z) = (z + i)^2$ and $E^\#(z) = (z - i)^2$. For $z = x + iy$, $y > 0$, $|z + i|^2 - |z - i|^2 = 4y > 0$, and E has no real zeros. The polynomials $f = a + bz$ form a positive two-dimensional space with

$$\langle a + bz, c + dz\rangle_E = \int_{\mathbb{R}} \frac{(a + bx)\overline{(c + dx)}}{(1 + x^2)^2}\, dx = \frac{\pi}{2}(a\bar{c} + b\bar{d}).$$

The cross term is odd. Substitution $x = \tan\theta$ gives both diagonal integrals $\pi/2$. These polynomials belong to the associated de Branges space: both f/E and $f^\#/E$ are rational Hardy functions in the upper half-plane, with lower-half-plane poles and decay $O(1/z)$. Their reproducing kernel is

$$K(w, z) = \frac{2}{\pi}(1 + z\bar{w}) = \frac{E(z)\overline{E(w)} - E^\#(z)\overline{E^\#(w)}}{2\pi i(\bar{w} - z)}.$$

The first expression follows from the orthogonal basis $1, z$; expanding the numerator gives the second. Thus positive index one already fails in this example, also over the real coefficient subspace. Finally the formula for Q directly proves the auxiliary inclusion and, in finite dimension, its signature. □

Remark 90.3.6 *(the required Mukai and Phi data).* For a one-dimensional positive space a chosen positive line of a hyperbolic plane suffices. In general the auxiliary $H \oplus H$ is not a specified Mukai carrier. A comparison requires the actual graded object, coefficient field, degree and sign of the Mukai pairing, a real structure or conjugation, and a form-preserving map from the chosen HB space to its Hermitian realization. Compatibility with cyclic or factorization operations is additional data.

The two-stage Calabi–Yau-to-chiral construction in Vol III also requires a chosen CY input, formality data and a specialization chart $(\sigma, \Sigma_{d-1}, C, b, w)$. On its constructed loci its output is E_1 for $d \geq 3$; the E_2 enhancement at $d = 2$ and E_∞ enhancement at $d = 1$ are conditional. At $d = 3$ the framing, anomaly-cancellation, completion, TCFT-correction and filtered Hochschild comparisons, together with the specified pushforward and support comparisons, remain hypotheses. Neither the HB inequality nor the auxiliary doubling supplies these structures or their pairing comparison.

Zero Counting and Chevalley–Cousin Concentration

PROPOSITION 90.3.7 *(cohomological degree does not determine spectral counting).* For an ordinary unital chiral algebra A on a smooth curve X over a characteristic-zero field, and a finite nonempty set I, the Chevalley–Cousin complex of Beilinson–Drinfeld, §§3.4.10–3.4.12, satisfies

$$H^n C(A)_{X^I} = 0 \quad (n \neq -|I|).$$

This is concentration of cohomology sheaves on the finite Cartesian power, with $C(A)_X = A[1]$ for a singleton. It is not a statement of global Ran-space or genus-zero acyclicity. Cohomological concentration alone does not determine eigenvalues or a spectral counting function. A comparison with the Riemann–von Mangoldt formula requires a Hilbert realization, a specified operator with domain, finite-rank or trace-class spectral projections, and a proved trace identity with the zero-counting function $N(T)$.

Proof. The concentration assertion is Beilinson–Drinfeld, *Chiral Algebras*, Lemma 3.4.12, p. 183, with the complex constructed in §3.4.11. For the spectral assertion, put $V = \mathbb{C}^n$ in degree zero with zero

differential. For any real numbers $a_1, \dots, a_n$, the self-adjoint diagonal operator $\mathrm{diag}(a_1, \dots, a_n)$ has this same concentrated complex as its carrier. Its finite spectrum is arbitrary. A shift places the same example in any prescribed cohomological degree. Thus concentration gives no restriction on the eigenvalues; the spectral realization and trace identity are separate data. □

90.4 Native Arithmetic Theorems

Adelic categorical descent

Chapter 70.2.4, Chapter 70.5.4, and Chapter 70.6.4 concern adelic categorical descent on the cuspidal sph-finite locus. A comparison with a chiral category must identify this adelic carrier, its coefficient realization, descent functors and their compatibility with the cuspidal restriction.

Wild-prime compact generation

Chapter 72.4.1 concerns compact generation of $\mathrm{IndCoh}_{\mathrm{Nilp}}$ at wild primes of the Emerton–Gee stack. It is native to the arithmetic fibre.

90.5 References for the comparison problems

arithmetic theorem	arithmetic sources	prior sources
HP–Li Paley–Wiener	Weil, Li, Rankin, Selberg, Hecke, Jacquet, Deligne	Beilinson–Drinfeld, Kac–Wakimoto, Frenkel–Zhu
Bost–Connes modular operator	Bost–Connes, Connes–Marcolli, Takesaki, Bratteli–Robinson	Costello–Gwilliam
Koszul–Petersson discrete core	Takesaki, Deligne, Kim–Shahidi, Moeglin–Waldspurger, Langlands–Shahidi	Beilinson–Drinfeld, Frenkel–Ben-Zvi
arithmetic Positselski	Pontryagin, Weil, Greenlees–May, Matlis, Bost–Connes, Consani–Marcolli, Venjakob	Beilinson–Drinfeld, Positselski, Lurie
Bernstein block-additivity	Bernstein, Bernstein–Deligne, Bushnell–Henniart, Paškūnas	Beilinson–Drinfeld, Lurie
Chenevier–Bernstein transport	Chenevier, Bellaiche–Chenevier, Bhatt–Scholze	Lurie, Costello–Gwilliam
Riemann–von Mangoldt density	Riemann, von Mangoldt, Titchmarsh, Tate, Atiyah–Macdonald, Bourbaki	Beilinson–Drinfeld
de Branges spectral positivity	de Branges, Lagarias, Conrey–Li, Simon, Akhiezer	Costello, Keller, Costello–Gwilliam, Ben-Zvi–Nadler

The references address the distinct algebraic, automorphic and analytic objects listed in the table. A comparison between these objects requires the indicated maps.

90.6 Fibre Theorem

Theorem 90.6.1 *(arithmetic fibre compatibility and the Weil pairing).* Under the comparison hypotheses stated in Chapter 90.2.1, Chapter 90.2.4, Chapter 90.3.1, Chapter 90.3.3, the integral base-change squares, centralizer transport, circle quotient and paired Hilbert realization have the consequences proved there. Their identifications with the named native arithmetic and chiral theories remain the stated comparison questions. The Weil pairing has a positive Hilbert realization exactly under the positivity criterion in Chapter 90.3.4. Its native arithmetic fibre comparison additionally requires the maps and equalities specified there. The Bernstein block-additivity, de Branges, and zero-density comparisons concern the distinct domains stated above. Adelic descent and wild-prime compact generation have their native arithmetic domains.

Proof. For the first four assertions use the explicit base-change squares, endomorphism equivalence, logarithmic quotient and isometry construction. None of these constructions asserts the missing native identifications. For the Weil pairing use its explicit Hilbert quotient construction. The block-chain comparison, positive-index obstruction with auxiliary doubling, and spectral-concentration counterexample are proved in Chapter 90.2.3, Chapter 90.3.5, Chapter 90.3.7. □

Chapter 91

Arithmetic Comparison Hypotheses

The local, trace and spectral constructions contribute to the G-row biconditional

$$\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^{!})_{\mathsf{G}}^{\mathrm{Ar}} = 0 \quad \Longleftrightarrow \quad \xi(s) = \xi(1-s).$$

The comparisons require compatible sign normalizations, ambient categories, hypotheses and functorial maps. The three groups of conditional inputs described below retain their stated assumptions; the other statements retain the hypotheses of their respective chapters.

91.1 The Eleven Local Comparisons

F1.a.1, F1.a.2, and $\mathrm{L}_{\mathrm{ACD}}$

The Taylor–Wiles ordinary comparison uses Clozel–Harris–Taylor, Kisin, Gee–Kisin, Emerton–Gee, and Hida on the derivation side, and Dotto–Emerton–Gee together with Fargues–Scholze on the categorical side. Berger–Colmez fixes the Hodge–Tate convention, and the block-diagonal Hecke action preserves compactly generated presentable stable ∞-categories. The non-ordinary extension, the non-semisimple class comparison, and the Eisenstein extension remain the named conditional statements Conjecture 70.4.6, Conjecture 68.7.3, and Conjecture 70.6.6.

The Dotto–Emerton–Gee input is used only in its fixed-central-character $p \geq 5$ fully faithful form. No essential-surjectivity theorem and no global variation of the central character is imported.

F1.a.3 (A+B)

The block-additivity argument is an ∞-categorical statement in Pr^{L}_{st}. Bernstein, Bushnell–Henniart, Paskunas, Ben-Zvi–Francis–Nadler, and Lurie supply inputs for the block decomposition and the tensor formalism. The co-Hochschild convention is the one of HA 5.5.3 and is compatible with Costello–Gwilliam.

The ordinary block is controlled only after the stated deformation hypotheses. The wild-prime centraliser and per-block residuals are not deduced from Paskunas alone; they are precisely the narrower conjectural pieces named in Chapter 71.

F1.a.3.C

Emerton–Gee gives the ambient Noetherian formal stack. It does not by itself give a finite nilpotent stratification, wild-prime quiver local models, or residual-gerbe generators. The compact-generation theorem is therefore conditional on the local-model input stated in Theorem 72.4.1. Under that input, recollement and compact-generation are formal consequences of Beilinson–Bernstein–Deligne, Gaitsgory–Rozenblyum, and Arinkin–Gaitsgory.

The equality $2 = 2$ in the $p = 3$ block is a tangent-dimension comparison. It is not a class-level identification of the two gerbe directions. The class-level statement belongs to the Paskunas–wild comparison and is not smuggled into the compact-generation proof.

Chenevier and F2.c

Chenevier's determinant formalism constructs complete local deformation rings and the restriction homomorphism of Theorem 73.1.4. Using this diagram for arithmetic gluing still requires the Selmer-ring identification, geometric realization, trace compatibility and ordinary equivariant sheaf comparison of Corollary 73.1.5. The derived comparison requires its own datum or both external-product and pullback Tor-independence hypotheses.

The $(AG.det)$ problem separately requires the arithmetic derived stack, its conical singular-support condition, and an actual categorical gluing diagram. The affine-line and triangular-algebra models of Section 73.2 prove particular gluing statements and a higher-centre restriction obstruction; they do not construct that arithmetic geometry. The higher-centre maps and their compatibility with the chosen homotopy limit must then be supplied, followed by the global automorphic comparison. The internal centre identity in Proposition 73.2.9 is conditional on these spectral data and does not supply the global comparison.

The $F2.c$ transport theorem retains its central idempotent pointed-unit hypotheses and does not require perfectoid descent. Prismatic and pro-étale comparisons remain separate inputs to the arithmetic realization, with their coefficient and geometric hypotheses.

The Elementary Closure

The five elementary constructions of Chapter 74 are consequences of the earlier chapters and of the explicit formula. They do not add a new equivalence to RH. The only RH-equivalent content is the Weil positivity inequality already isolated as A-TP. The elementary chapter is therefore a normal form for the implication chain, not an independent proof of the Riemann Hypothesis.

Arithmetic Positselski

Pontryagin duality, Greenlees–May duality, Matlis duality, and the Bost–Connes/Consani–Marcolli formalism close the four abelian Positselski subitems at the stated domain. The upgrade of the abelian Heisenberg G-row is conditional on the precise adele-class/cobar-sheaf identification in Consani–Marcolli. Without that primary-source identification the result is a conditional theorem, not a theorem in the absolute sense.

Koszul–Petersson

A specified injective arithmetic placement into the discrete L^2 space pulls back its positive Petersson form. If the arithmetic Koszul form is identified with that pullback, P3 follows without a Ramanujan hypothesis, by Proposition 76.5.1. The separate route through the tempered local models requires temperedness, and on the Maass sector uses the stated additional Ramanujan input. Both routes retain the actual arithmetic pairing maps. The Eisenstein continuum has its separate measure and form-comparison problem.

H-P^{Ar}

The GNS construction supplies the Hilbert-space skeleton. It does not construct the Hilbert–Polya operator. The twist-existence condition is the classical zero-line obstruction in operator form. Thus the chapter separates construction of the representation from positivity of the spectral measure.

HP–Li Positivity

The Mellin–Rankin–Selberg isometry reduces the restricted Li positivity statement to the archimedean Tate distribution. The archimedean Tate positivity statement is equivalent to RH and is not proved there.

The chapter identifies the exact positivity functional and does not replace Weil positivity by positivity on a smaller test class.

VB^*

The strict derivative-kernel criterion concerns zero placement. The positive moment functional and determinacy concern a different construction. Exponential completeness EC_π gives uniqueness in PW_π; transfer to a de Branges space requires the specified evaluation-compatible map, with additional bounds for stable sampling. These comparisons remain distinct from the sharp fractional-part rate (BD+).

$\mathrm{R}_{\mathrm{analytic}}$

The finite-cutoff moment calculation in Proposition 80.2.5 proves that positive-power zero sums diverge. Gaussian evaluation of the explicit formula does not construct their regularization or prove Weil positivity. The analytic moment task is to construct a finite positive functional and prove a sufficient determinacy estimate, such as linear even-moment root growth, together with the arithmetic operator and support comparison.

91.2 Three Groups of Conditional Inputs

(i) The wild-prime compact-generation theorem of Chapter 72 is conditional on finite nilpotent local models and class-level residual-gerbe generators. Emerton–Gee alone supplies the ambient formal stack.

(ii) The arithmetic gluing comparisons of Chapter 73 retain distinct inputs: the Selmer-ring and trace comparisons and equivariant sheaf descent; the additional derived comparison or its two Tor-independence conditions; the arithmetic support geometry and categorical descent diagram; the higher-centre maps with their homotopy-limit compatibility; and the global automorphic comparison. Formal restriction, the coherent gluing models and the ordinary scalar-centre constructions supply none of these arithmetic identifications by themselves.

(iii) The abelian Heisenberg G-row upgrade of Chapter 75 is conditional on the exact Consani–Marcolli adele-class/cobar-sheaf comparison used there.

91.3 Conclusion

These groups of conditional comparison hypotheses qualify the reductions of F1, of $\mathrm{L}_{\mathrm{ACD}}$, of the abelian Positselski sector, and of the discrete Koszul–Petersson identity. The RH-equivalent assertion is not hidden in any of these reductions; it remains the archimedean Tate positivity, equivalently Weil positivity.

Chapter 92

Archimedean Tate Kernel via Riemann–Siegel Theta and Stirling-Certified Bounds

The digamma weight controls an auxiliary quadratic form, whereas its signed contribution to the Weil functional includes a constant term. This distinction determines what a high-frequency estimate can prove. On $\mathscr{A} = C_c^\infty(\mathbb{R})$, use $\widehat{f}(t) = \int_{\mathbb{R}} f(u)e^{itu}\,du$ and $\widetilde{f}(u) = \overline{f(-u)}$. Put

$$g(t) = \operatorname{Re}\psi(1/4 + it/2) + \log\pi, \qquad G[f] = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(t)|^2 g(t)\,dt.$$

The exact identity of Lemma 82.3.2 is

$$W_\infty(f * \widetilde{f}) = 2\log\pi\|f\|_2^2 - G[f].$$

The Riemann–Siegel phase gives $g = 2\theta'_{\mathrm{RS}} + 2\log\pi$. We construct a strict enlargement of the sufficient high-frequency class for $G \geq 0$, and express the full signed functional through this phase. Neither positivity of G nor its sign-changing core alone is positivity of the Weil functional.

92.1 Context

The exact weight has a unique positive zero t_*, by Theorem 82.2.4. With the core and tail integrals $A[f], B[f]$ of Definition 82.3.1, $G[f] = (B[f] - A[f])/(2\pi)$. The supremum estimate defining HFD can be weakened to the actual weighted inequality $A[f] \leq B[f]$. Smooth tests with strict margins will show that this is a strict inclusion on the stated compact-test carrier.

Domain

All Fourier integrals below converge on $\mathscr{A}$, since the Fourier transform is Schwartz and g grows logarithmically. Any further specified test class is used by intersection with $\mathscr{A}$; non-emptiness or strict inclusion on that smaller class is a separate question. The full Weil target remains $R[f] \geq W_\infty(f * \widetilde{f})$ on its declared complete test domain, with both the polar and finite-prime terms in $R[f]$.

Analytic Inputs

The phase and asymptotic descriptions use the following classical analytic inputs.

Riemann 1859. Riemann, B. *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*. Monatsber. Berl. Akad. (1859), originating manuscript for the theta function $\theta_{\mathrm{RS}}(t)$ and the function $Z(t)$.

Siegel 1932. Siegel, C.L. *Über Riemanns Nachlass zur analytischen Zahlentheorie*. Quell. Stud. Gesch. Math. Astr. Phys. B2 (1932), 45–80. Derivation of the Riemann–Siegel formula from Riemann's estate; the function $Z(t)$ and the theta expansion.

Edwards 1974. Edwards, H.M. *Riemann's Zeta Function*. Academic Press 1974, §7. Modern treatment of θ_{RS}, $Z(t)$, and the Stirling expansion of $\log\Gamma(1/4+it/2)$ with explicit truncation-error bounds for the Riemann–Siegel asymptotic.

Riemann 1859, Siegel 1932, and Edwards 1974 develop the critical-line functions and their asymptotic expansions. The Hurwitz series, Gauss reflection, and Plancherel–Pólya theorem in §82.1 instead describe the complex-analytic properties of the digamma function. The comparison below connects these analytic descriptions with the positivity questions in Chapters 43, 47, 67, 78, 79, and 82.

92.2 The Exact Identity $g = 2\theta'_{\mathrm{RS}} + 2\log\pi$

Definition 92.2.1 *(Riemann–Siegel theta).* For $t \in \mathbb{R}$, set

$$\theta_{\mathrm{RS}}(t) \;:=\; \operatorname{Im}\log\Gamma\Big(\tfrac{1}{4}+\tfrac{it}{2}\Big) - \tfrac{t}{2}\log\pi, \tag{92.1}$$

with the analytic logarithm of Γ on $\operatorname{Re}(z)>0$ that is real on the positive real axis. The function θ_{RS} is real-analytic on $\mathbb{R}$ with $\theta_{\mathrm{RS}}(0)=0$ and $\theta_{\mathrm{RS}}(-t)=-\theta_{\mathrm{RS}}(t)$ (odd).

Theorem 92.2.2 *(exact identity).* For every $t\in\mathbb{R}$,

$$g(t) \;=\; 2\,\theta'_{\mathrm{RS}}(t) + 2\log\pi. \tag{92.2}$$

Equivalently, $\theta'_{\mathrm{RS}}(t) = g(t)/2 - \log\pi$.

Proof. Differentiate (92.1) by the chain rule:

$$\theta'_{\mathrm{RS}}(t) = \frac{d}{dt}\operatorname{Im}\log\Gamma\Big(\tfrac{1}{4}+\tfrac{it}{2}\Big) - \tfrac{1}{2}\log\pi.$$

For the first term, $\frac{d}{dt}\log\Gamma(1/4+it/2) = \psi(1/4+it/2)\cdot(i/2)$. Writing $\psi(1/4+it/2) = u+iv$ with $u=\operatorname{Re}\psi$, $v=\operatorname{Im}\psi$, we have $\psi\cdot(i/2) = (iu-v)/2$, whose imaginary part is $u/2$. Thus $\frac{d}{dt}\operatorname{Im}\log\Gamma(1/4+it/2) = \operatorname{Re}\psi(1/4+it/2)/2$, and

$$\theta'_{\mathrm{RS}}(t) = \tfrac{1}{2}\operatorname{Re}\psi\Big(\tfrac{1}{4}+\tfrac{it}{2}\Big) - \tfrac{1}{2}\log\pi.$$

Multiplying by 2 and adding $2\log\pi$: $2\theta'_{\mathrm{RS}}(t) + 2\log\pi = \operatorname{Re}\psi(1/4+it/2) + \log\pi = g(t)$. □

Remark 92.2.3 *(sign and regularity of the exact identity).* The identity follows by differentiating the Riemann–Siegel theta function of Definition 92.2.1 and comparing with the digamma expression in Definition 82.1.1. Use the analytic logarithm of Γ on $\operatorname{Re}(z)>0$ that is real on the positive real axis. Its restriction to $1/4+it/2$ is fixed by continuity from $t=0$, where $\log\Gamma(1/4)\in\mathbb{R}$. Its imaginary part has the sign of the standard $\arg\Gamma$ convention (Edwards 1974, §6.5). Since Γ is analytic and non-vanishing on this vertical line, $\operatorname{Im}\log\Gamma$ is real-analytic for every $t\in\mathbb{R}$, so differentiation is valid there. The result is an equality of real-analytic functions. It asserts no additional functorial property of θ_{RS}.

The constant in (92.2) is $2\log\pi \approx 2.28946$. At $t=0$,

$$\begin{aligned}\theta'_{\mathrm{RS}}(0) &= \tfrac{1}{2}\psi(1/4) - \tfrac{1}{2}\log\pi\\ &= \tfrac{1}{2}(-\gamma_E - \pi/2 - 3\log 2) - \tfrac{1}{2}\log\pi \approx -2.686.\end{aligned}$$

Thus $2\theta'_{\mathrm{RS}}(0) + 2\log\pi \approx -5.372 + 2.289 = -3.083$, which is $g(0)\approx -3.083$ in Proposition 82.2.3.

Corollary 92.2.4 *(Signed phase formula).* For $f \in \mathscr{A}$, the Riemann–Siegel phase satisfies

$$W_\infty(f * \widetilde{f}) = -\frac{1}{\pi}\int_{\mathbb{R}} |\widehat{f}(t)|^2 \theta'_{\mathrm{RS}}(t)\, dt. \tag{92.3}$$

The auxiliary form is $G[f] = \pi^{-1}\int |\widehat{f}|^2(\theta'_{\mathrm{RS}} + \log\pi)\, dt$. The full Weil inequality requires the separate bound $R[f] \geq W_\infty(f * \widetilde{f})$.

Proof. Substitute $g = 2\theta'_{\mathrm{RS}} + 2\log\pi$ into G. Plancherel gives $\int |\widehat{f}|^2 = 2\pi\|f\|_2^2$. The constant contribution cancels in $2\log\pi\|f\|_2^2 - G[f]$, giving the displayed negative sign. □

Remark 92.2.5 *(why this reformulation matters).* The phase derivative θ'_{RS} admits the asymptotic $\theta'_{\mathrm{RS}}(t) \sim \frac{1}{2}\log(t/2\pi)$ as $t \to \infty$, which is the classical zero-density leading term: the mean number of zeros of ζ up to height T is $N(T) = (1/\pi)\int_0^T \theta'_{\mathrm{RS}}(t)\, dt + O(\log T)$ by the Riemann–von Mangoldt formula. The phase derivative gives the smooth leading term of the zero count, not the discrete zero measure. Its exact use here is the signed identity (92.3); the full Weil inequality also retains $R[f]$.

92.3 Stirling-4 Certified Bounds

The kernel $g(t)$ admits an explicit rigorous expansion with bounded truncation error, valid for every $t \in \mathbb{R}$; this is the Stirling-4 formula, from which the threshold T_{St4} follows by a direct comparison.

The Stirling-4 Formula

Lemma 92.3.1 *(Stirling expansion with explicit error).* For $z \in \mathbb{C}$ with $\operatorname{Re} z > 0$,

$$\psi(z) \;=\; \log z - \frac{1}{2z} - \frac{1}{12z^2} + R_4(z), \qquad |R_4(z)| \;\leq\; \frac{1}{120\,|z|^4}. \tag{92.4}$$

The error bound $1/(120|z|^4)$ is the standard Stirling-series remainder (Edwards 1974 §6.5 Thm. 1; Whittaker–Watson 1927 §13.6) applied after truncation at $k = 2$.

Proof. Edwards 1974 §6.5 gives the Stirling expansion $\psi(z) = \log z - 1/(2z) - \sum_{k=1}^{N} B_{2k}/(2kz^{2k}) + R_{N+1}(z)$ with $|R_{N+1}(z)| \leq |B_{2N+2}|/(2N+2)|z|^{2N+2}$ uniformly in $\arg z \in (-\pi, \pi)$. Specialising $N = 1$: $B_2 = 1/6$ contributes the $1/(12z^2)$ term, and $|R_2(z)| \leq |B_4|/(4|z|^4) = 1/(120|z|^4)$ using $B_4 = -1/30$. □

Proposition 92.3.2 *(Stirling-4 formula for g).* For every $t \in \mathbb{R}$, write $z = 1/4 + it/2$, $|z|^2 = (1+4t^2)/16$. Then

$$g(t) \;=\; g_{\mathrm{St4}}(t) + \widetilde{R}_4(t), \qquad |\widetilde{R}_4(t)| \;\leq\; \frac{32}{15\,(1+4t^2)^2},$$

where the Stirling-4 form is the explicit rational-plus-logarithmic function

$$g_{\mathrm{St4}}(t) \;=\; \tfrac{1}{2}\log(1+4t^2) + \log\pi - 2\log 2 - \frac{2}{1+4t^2} - \frac{4}{3}\cdot\frac{1-4t^2}{(1+4t^2)^2}.$$

Proof. Take real parts of (92.4) at $z = 1/4 + it/2$ and add $\log\pi$:

$$g(t) = \operatorname{Re}\log z - \tfrac{1}{2}\operatorname{Re}(1/z) - \tfrac{1}{12}\operatorname{Re}(1/z^2) + \operatorname{Re} R_4(z) + \log\pi.$$

Compute each real part. $\operatorname{Re}\log z = \frac{1}{2}\log|z|^2 = \frac{1}{2}\log((1+4t^2)/16) = \frac{1}{2}\log(1+4t^2) - 2\log 2$. Next, $1/z = \overline{z}/|z|^2 = (1/4 - it/2)\cdot 16/(1+4t^2)$, giving $\operatorname{Re}(1/z) = 4/(1+4t^2)$, hence $\frac{1}{2}\operatorname{Re}(1/z) = 2/(1+4t^2)$. For $1/z^2$: $z^2 = (1/4 + it/2)^2 = 1/16 - t^2/4 + it/4$, and $|z^2|^2 = |z|^4 = (1+4t^2)^2/256$, giving $\operatorname{Re}(1/z^2) = (1/16 - t^2/4)\cdot 256/(1+4t^2)^2 = 16(1-4t^2)/(1+4t^2)^2$. Hence $\frac{1}{12}\operatorname{Re}(1/z^2) = \frac{4}{3}(1-4t^2)/(1+4t^2)^2$. Substituting:

$$g(t) = \tfrac{1}{2}\log(1+4t^2) - 2\log 2 - \tfrac{2}{1+4t^2} - \tfrac{4}{3}\tfrac{1-4t^2}{(1+4t^2)^2} + \operatorname{Re} R_4(z) + \log\pi,$$

which is $g_{\mathrm{St}4}(t) + \operatorname{Re} R_4(z)$. For the error: $|\operatorname{Re} R_4(z)| \le |R_4(z)| \le 1/(120|z|^4) = 1/(120\cdot(1+4t^2)^2/256) = 256/(120(1+4t^2)^2) = 32/(15(1+4t^2)^2)$. □

The Certified Positivity Threshold

Definition 92.3.3 (Stirling-4 lower bound). Define

$$g^{-}_{\mathrm{St}4}(t) \;:=\; g_{\mathrm{St}4}(t) - \frac{32}{15\,(1+4t^2)^2}.$$

By Proposition 92.3.2, $g(t) \ge g^{-}_{\mathrm{St}4}(t)$ for every $t \in \mathbb{R}$.

Proposition 92.3.4 *(Stirling-4 positivity threshold).* There is a unique positive $T_{\mathrm{St}4} \in \mathbb{R}$ with $g^{-}_{\mathrm{St}4}(T_{\mathrm{St}4}) = 0$, numerically $T_{\mathrm{St}4} = 0.872959\ldots$. One has $g^{-}_{\mathrm{St}4}(t) > 0$ for every $|t| > T_{\mathrm{St}4}$, and therefore $g(t) > 0$ for every $|t| > T_{\mathrm{St}4}$.

Proof. Consider $g^{-}_{\mathrm{St}4}(t)$ as a function of $u = 1+4t^2 \in [1,\infty)$:

$$g^{-}_{\mathrm{St}4}(u) = \tfrac{1}{2}\log u + \log\pi - 2\log 2 - \frac{2}{u} - \frac{4}{3}\cdot\frac{2-u}{u^2} - \frac{32}{15\,u^2},$$

using $1 - 4t^2 = 2 - u$. Expanding $-(4/3)(2-u)/u^2 = -8/(3u^2) + 4/(3u)$ and differentiating term by term:

$$\frac{d\,g^{-}_{\mathrm{St}4}}{du} = \frac{1}{2u} + \frac{2}{u^2} + \frac{16}{3\,u^3} - \frac{4}{3u^2} + \frac{64}{15\,u^3} = \frac{1}{2u} + \frac{2}{3u^2} + \frac{144}{15\,u^3}.$$

Each term is positive for all $u \ge 1$, so $d\,g^{-}_{\mathrm{St}4}/du > 0$ on $u \in [1,\infty)$. Strict monotonicity of $g^{-}_{\mathrm{St}4}$ in u combined with $g^{-}_{\mathrm{St}4}(1) = g^{-}_{\mathrm{St}4}(t=0) \le g(0) = -3.083 < 0$ and $g^{-}_{\mathrm{St}4}(u) \to \infty$ as $u \to \infty$ supplies a unique zero. Bisection on $[2.51, 4.05]$ (corresponding to $t \in [0.6, 0.87]$) starting from $g^{-}_{\mathrm{St}4}(0.87) = -0.001$ and $g^{-}_{\mathrm{St}4}(0.88) = +0.002$ localises the zero to $T_{\mathrm{St}4} \in (0.872, 0.874)$. Extended bisection with 50 iterations to mpmath precision 50 digits yields $T_{\mathrm{St}4} = 0.8729590014\ldots$. □

Remark 92.3.5 (Sign and domain of the threshold estimate). The Bernoulli convention is $B_4 = -1/30$, so the fourth term in the displayed expansion is $+1/(120z^4)$. Its real part and the absolute remainder bound must be kept distinct. The estimate is used on $z = 1/4 + it/2$, which lies in the right half-plane. The threshold is defined by the zero of the stated lower-bound function. Decimal values are numerical approximations; a certified sign interval requires evaluating that function together with its remainder bound.

Corollary 92.3.6 *(high-$|t|$ ratio bound).* For every $|t| \ge 2$,

$$g(t) \;\ge\; \tfrac{1}{2}\log(1+4t^2) + \log\pi - 2\log 2 - \frac{4}{1+4t^2} \;\ge\; \log(|t|\pi/2) - \frac{4}{1+4t^2}. \tag{92.5}$$

In particular $g(t)/\log|t| \to 1$ as $|t| \to \infty$, and $g(t)/\log|t| \ge 1/2$ for every $|t| \ge 2$.

Proof. From Proposition 92.3.2:

$$\begin{aligned} g(t) &\geq g_{\mathrm{St}_4}(t) - \frac{32}{15(1+4t^2)^2} \\ &= \tfrac{1}{2}\log(1+4t^2) + \log\pi - 2\log 2 - \frac{2}{1+4t^2} - \frac{4}{3}\cdot\frac{1-4t^2}{(1+4t^2)^2} - \frac{32}{15(1+4t^2)^2}. \end{aligned}$$

For $|t| \geq 2$, $1-4t^2 \leq 0$, so $-(4/3)(1-4t^2)/(1+4t^2)^2 \geq 0$ and $-2/(1+4t^2) - 32/(15(1+4t^2)^2) \geq -4/(1+4t^2)$, since the residual $(80t^2-52)/(15(1+4t^2)^2) \leq 2/(1+4t^2)$ is direct to verify for $|t| \geq 2$ (both sides decay as $|t|^{-2}$; at $|t| = 2$ the inequality reads $268/4335 \leq 2/17$, i.e., $268 \cdot 17 \leq 2 \cdot 4335$). Combining the non-negative term with the bound on the remainder establishes the first inequality of (92.5). The second inequality $\frac{1}{2}\log(1+4t^2) + \log\pi - 2\log 2 \geq \log(|t|\pi/2)$ is $\frac{1}{2}\log(1+4t^2) \geq \log(2|t|)$, which holds since $1+4t^2 \geq 4t^2$. The asymptotic $g(t) \sim \log(|t|\pi/2)$ gives the ratio claim. □

92.4 Weighted High-Frequency Tests

The HFD condition bounds the core integral by its supremum times $2t_*\kappa_*$. Keeping the actual weighted core integral gives a weaker sufficient condition for $G \geq 0$, with a strict inclusion that can be witnessed by compactly supported smooth tests.

The Refined Weight

Lemma 92.4.1 *(refined core weight via Stirling).* For $|t| \leq t_*$,

$$-g(t) \;\leq\; \kappa_*, \qquad \kappa_* \;:=\; |g(0)| \;=\; \gamma_E + \pi/2 + 3\log 2 - \log\pi \approx 3.0827,$$

with equality at $t = 0$. The mean weight is $\kappa_*^{\mathrm{mean}} := \|g\|_{L^1[-t_*,t_*]}/(2t_*) \approx 1.488$, strictly less than κ_*.

Proof. $|g(t)|$ is even, strictly decreasing in $|t|$ on $[0, t_*]$ (since g is strictly increasing in $|t|$ and $g \leq 0$ on $[0, t_*]$), with maximum $|g(0)| = \kappa_*$ at $t = 0$ (Lemma 82.2.2, Proposition 82.2.3). The exact mean is $\kappa_*^{\mathrm{mean}} = t_*^{-1}\int_0^{t_*}(-g(t))\,dt$. Strict monotonicity gives $0 < \kappa_*^{\mathrm{mean}} < \kappa_*$. Numerical quadrature gives the illustrative values $2\int_0^{t_*}(-g(t))\,dt \approx 2.53256063$ and $\kappa_*^{\mathrm{mean}} \approx 1.48858167$; the strict inequality follows analytically and does not rely on those decimals. □

Remark 92.4.2 *(The zero and its numerical approximation).* Proposition 82.2.5 proves $t_* \in (0.84, 0.86)$ using the positive-term difference series for $g(t) - g(0)$ and its explicit tail estimate. Evaluation of the digamma function gives the illustrative approximation $t_* \approx 0.8506623044$. A tail bound of order 10^{-5} does not certify all the displayed decimal places. The bounded-interval decomposition uses the exact unique zero; its proof does not depend on rounding this number.

Definition 92.4.3 *(Weighted high-frequency classes).* For $f \in \mathscr{A}$, put $C[f] = 2t_*\kappa_* \sup_{|t|\leq t_*} |\widehat{f}(t)|^2$. Define $\mathscr{A}_{\mathrm{HFD}} = \{f : C[f] \leq B[f]\}$ and

$$\mathscr{A}_{\mathrm{SHF}} = \{f : A[f] \leq B[f]\}.$$

For any additional class $\mathcal{P} \subset \mathscr{A}$, its HFD and SHF subclasses mean intersection with these two sets.

Lemma 92.4.4 *(Strict inclusion on the compact-test carrier).* One has $\mathscr{A}_{\mathrm{HFD}} \subsetneq \mathscr{A}_{\mathrm{SHF}}$. There are real even tests satisfying $A[f] < B[f] < C[f]$.

Proof. The bound $-g(t) \le \kappa_*$ on the core gives $A[f] \le C[f]$, so HFD implies SHF. To construct strictness, choose a nonzero real even smooth frequency bump F_0 supported inside $(-t_*, t_*)$. Let

$$A_0 = \int |F_0|^2(-g), \qquad C_0 = 2t_*\kappa_* \sup |F_0|^2.$$

Then $0 < A_0 < C_0$: $-g < \kappa_*$ away from zero and the bump is supported in a proper subinterval. Choose a nonzero real even smooth bump F_1 supported outside the core, and put $B_1 = \int |F_1|^2 g > 0$. Set $\alpha^2 = (A_0 + C_0)/(2B_1)$ and $F = F_0 + \alpha F_1$. The supports are disjoint, so its inverse Fourier transform h is real, even and Schwartz, with $A[h] = A_0 < B[h] = (A_0 + C_0)/2 < C[h] = C_0$.

Take a real even cutoff $\eta \in C_c^\infty$ equal to one near zero, and let $f_R(u) = \eta(u/R)h(u)$. Then $f_R \in \mathscr{A}$ and $f_R \to h$ in the Schwartz topology: every differentiated cutoff error is supported where $|u|$ tends to infinity and is controlled by the rapid decay of h and its derivatives. Fourier transformation preserves this convergence. The functionals A, B, C are continuous for that topology; for B, use the logarithmic growth of g, and for C, uniform convergence on the compact core. Both strict margins persist for large R. □

Theorem 92.4.5 *(Auxiliary positivity and the signed Weil contribution).* For $f \in \mathscr{A}_{\mathrm{SHF}}$,

$$G[f] \ge 0, \qquad W_\infty(f * \widetilde{f}) = 2\log\pi\|f\|_2^2 - G[f] \le 2\log\pi\|f\|_2^2.$$

The full Weil inequality holds exactly when

$$R[f] + G[f] \ge 2\log\pi\|f\|_2^2.$$

In particular, SHF alone does not give $W_\infty \ge 0$, nor the full Weil inequality. The sufficient bound $R[f] \ge 2\log\pi\|f\|_2^2$ does give the latter on SHF.

Proof. Use $G = (B - A)/(2\pi)$ and the exact signed identity. Rearrangement gives the full inequality. For a concrete sign test, choose a nonzero real even $\chi \in C_c^\infty$ and set $f_M(u) = \chi(u)\cos(Mu)$. Its Fourier mass moves to neighborhoods of $\pm M$; rapid decay makes its core supremum tend to zero, while $B[f_M]$ grows at least as a positive constant times $\log M$. Thus f_M is eventually HFD and $G[f_M] \to \infty$, whereas $\|f_M\|_2^2 \le \|\chi\|_2^2$. Consequently $W_\infty(f_M * \widetilde{f_M}) < 0$ for large M, as also follows from the smooth modulation calculation in Chapter 82. □

Remark 92.4.6 *(Mean weight and supremum bound).* The ratio $\kappa_*/\kappa_*^{\mathrm{mean}}$ compares the mean core weight with its maximum. It is not a measure of a set of tests and does not count admissible profiles. For a given test the actual comparison is between $A[f]$ and $C[f]$; its Fourier density need not be constant on the core.

Strict Margins and Topology

Proposition 92.4.7 *(Non-emptiness and open strict inequalities).* Endow $\mathscr{A}$ with the topology induced from the Schwartz space. The sets $\{A < B\}$ and $\{A < B < C\}$ are open, invariant under nonzero scalar multiplication, and nonempty. No density claim in $\mathscr{A}$ is made.

Proof. Continuity of the three functionals makes the strict inequalities open. They are homogeneous of degree two. The preceding cutoff construction supplies a point of $\{A < B < C\}$; the modulation construction supplies high-frequency points of $\{A < B\}$. Conversely, broad smooth dilations have $G < 0$ as proved in Chapter 82, so $\{A > B\}$ is also a nonempty open set. Thus $\{A < B\}$ is not dense. □

Remark 92.4.8 *(Restriction to other test classes).* The inclusion is strict on $\mathscr{A}$. For a smaller prescribed Paley–Wiener or arithmetic test class, inclusion follows by intersection, but strictness requires a witness satisfying its additional defining conditions. The cutoff construction does not silently supply those conditions.

92.5 Phase Form of the Full Signed Inequality

The phase identity applies on all of $\mathscr{A}$. Both SHF and its complement still require the polar and finite-prime functional.

The Phase-Derivative Formulation

Theorem 92.5.1 *(Phase formula for the full Weil target).* For $f \in \mathscr{A}$, the full inequality $W(f * \widetilde{f}) \geq 0$ is equivalent to

$$R[f] + \frac{1}{\pi} \int_{\mathbb{R}} |\widehat{f}(t)|^2 \theta'_{\mathrm{RS}}(t)\, dt \geq 0. \tag{92.6}$$

The archimedean sign condition $W_\infty \geq 0$, considered by itself, instead says that the displayed integral is nonpositive.

Proof. By Corollary 92.2.4, $W_\infty = -\pi^{-1} \int |\widehat{f}|^2 \theta'_{\mathrm{RS}}$. Substitute this into $W = R - W_\infty$. This keeps the same Fourier measure and test carrier as the original Weil formula. □

Remark 92.5.2 *(connection to the Z-function zeros).* $Z(t) = e^{i\theta_{\mathrm{RS}}(t)} \zeta(1/2 + it)$ is a real-valued function of $t \in \mathbb{R}$. Its real zeros are exactly the ordinates of the non-trivial zeros of ζ on the critical line. The phase in (92.6) is defined from the gamma factor; its integral extends over the full spectral line. The Riemann–von Mangoldt formula relates $\theta_{\mathrm{RS}}(T)/\pi$ to the leading zero count, but the phase derivative alone is not the zero distribution. In particular, $\theta'_{\mathrm{RS}}(t_*) = -\log \pi$ marks the zero of g, not a zeta zero. The finite-prime and polar information remains in $R[f]$, and the signed phase identity does not prove its required comparison with the integral.

The θ''_{RS} Structural Observation

Proposition 92.5.3 *(second-derivative representation of g').* For every $t \in \mathbb{R}$,

$$g'(t) \;=\; 2\,\theta''_{\mathrm{RS}}(t). \tag{92.7}$$

Since θ_{RS} is odd, θ'_{RS} is even, and θ''_{RS} is odd. Combined with Lemma 82.2.2 ($g'(t) > 0$ for $t > 0$), this gives $\theta''_{\mathrm{RS}}(t) > 0$ for $t > 0$ and $\theta''_{\mathrm{RS}}(t) < 0$ for $t < 0$.

Proof. Differentiate (92.2). □

Remark 92.5.4 *(the θ''_{RS}-sign observation).* The identity (92.7) converts the strict monotonicity of g in $|t|$ (Lemma 82.2.2) into a sign statement about θ''_{RS}. For f real and even, $|\widehat{f}(t)|^2$ is an even function of t, and integration by parts on $[-t_*, t_*]$ with antiderivative $G(t) := -2(\theta_{\mathrm{RS}}(t) + t\log\pi)$ of $-g(t)$ gives

$$A[f] \;=\; \int_{-t_*}^{t_*} |\widehat{f}|^2(-g)\, dt = \Big[-2(\theta_{\mathrm{RS}} + t\log\pi)|\widehat{f}|^2\Big]_{-t_*}^{t_*} + 2\int_{-t_*}^{t_*} (\theta_{\mathrm{RS}}(t) + t\log\pi)\cdot(|\widehat{f}|^2)'\, dt.$$

The boundary term is $-4(\theta_{\mathrm{RS}}(t_*) + t_*\log\pi)|\widehat{f}(t_*)|^2$ for real even f; it does not cancel. Since $2(\theta_{\mathrm{RS}}(t_*) + t_*\log\pi) = \int_0^{t_*} g(t)\, dt < 0$, this boundary contribution is nonnegative. The remaining integral must still be controlled together with this term. The identity alone proves neither the full Weil inequality nor an impossibility of sharper estimates.

92.6 The Remaining Weil Inequality

Theorem 92.6.1 *(Exact separation of the estimates and the target).* The weighted classes satisfy $\mathscr{A}_{\mathrm{HFD}} \subsetneq \mathscr{A}_{\mathrm{SHF}}$, and $G \geq 0$ on SHF. The full target on either class remains $R[f] + G[f] \geq 2\log\pi\|f\|_2^2$, equivalently (92.6). Bounds for g can be inserted into this exact inequality; they do not eliminate $R[f]$. Any RH equivalence requires the complete test domain of the classical Weil criterion, not just the SHF complement.

Proof. Combine Lemma 92.4.4, Theorem 92.4.5, and the signed phase formula. □

Remark 92.6.2 *(Phase normalization and analytic domain).* Use the continuous Riemann–Siegel phase with $\theta_{\mathrm{RS}}(0) = 0$ (Riemann 1859; Siegel 1932; Edwards 1974, §6.5). The Hurwitz and Stirling expressions evaluate the same real-analytic digamma weight on the real t-axis; $z = 1/4 + it/2$ remains in the right half-plane. The phase identity in Theorem 92.5.1 is a substitution on this domain. Numerical approximations to the zero and mean weight do not replace the exact zero or the remainder bounds in an inequality.

92.7 Non-Elementary Residual Constructions

Remark 92.7.1 *(Further analytic constructions).* Sharper remainder estimates, Riemann–Siegel expansions, geometric operator comparisons, and explicit smooth test families offer different ways to study the full inequality. A remainder estimate must control its stated domain; a special test family must be shown to determine the complete Weil test cone before positivity on that family can imply the full result. No finite list of such methods is an obstruction to other analytic constructions.

Remark 92.7.2 *(Comparison with the core–tail decomposition).* The exact residual is

$$A[f]/(2\pi) + 2\log\pi\|f\|_2^2 \leq B[f]/(2\pi) + R[f].$$

SHF improves the sufficient estimate for $G \geq 0$ by using the actual core weight. It leaves both the constant term and $R[f]$ in this inequality. The numerical approximation to t_* is illustrative; its rigorous interval and exact defining equation remain those of Remark 92.4.2.

92.8 Stirling Boundary

The exact identity $g = 2\theta'_{\mathrm{RS}} + 2\log\pi$ connects the digamma estimates with the phase derivative. The strict smooth class inclusion and the finite core–tail integrals are constructive results on $\mathscr{A}$. The remaining arithmetic inequality retains the polar and finite-prime terms.

Remark 92.8.1 *(Analytic scope).* The auxiliary positivity theorem is not a solution of the full Weil problem. Its use in that problem requires estimates on the same compact-test carrier and control of $R[f]$, as made explicit in (92.6). No optimality or exhaustion of analytic methods is asserted.

Chapter 93

A-TP via Nyman–Beurling Closure and the Baez-Duarte Refinement

The Nyman–Beurling comparison. Chapter 82 reduced the archimedean Tate positivity conjecture A-TP to the exact signed inequality $A[f] + 4\pi \log \pi \|f\|_2^2 \le B[f] + 2\pi R[f]$. The interval $[-t_*, t_*]$, with $t_* \approx 0.8507$, is the negative region of the auxiliary weight g, not the support of the full Weil functional. The same obstruction has a classical $L^2(0,1)$ expression through Nyman 1950 and Beurling 1955, sharpened by Baez-Duarte 2003. The Nyman–Beurling criterion reformulates RH as a closure statement in $L^2(0,1)$ generated by the functions $\rho_\alpha(x) = \{\alpha/x\} - \alpha\{1/x\}$ for $\alpha \in (0,1]$; Baez-Duarte shows RH is equivalent to the asymptotic vanishing of the Möbius-coefficient distance $d_N = \inf_{c_k} \|\chi_{[0,1]} - \sum_{k=1}^{N} c_k \rho_{1/k}\|_{L^2(0,1)}$. The results are fourfold. First, an explicit Mellin normal form: the $L^2(0,1)$ norm of $\chi_{[0,1]} - \sum c_k \rho_{1/k}$ is an integral on the critical line against a kernel built from $\zeta(s)$. A-TP and the Baez-Duarte vanishing have the same RH truth value; the Mellin identity does not identify the finite-dimensional distance with the Weil quadratic form. Second, on the bounded-interval reduction HFD sub-class of Chapter 82, the expected Nyman–Beurling estimate becomes a weighted approximation problem for $\zeta(s)^{-1}$ on the critical line; neither the Mellin identity nor PNT proves the $1/\log N$ rate. Third, a Möbius partial-sum control: the conditional Mertens bound $M(N) := \sum_{k \le N} \mu(k) = O(\sqrt{N} \log^2 N)$ under RH sharpens the unconditional PNT bound $M(N) = o(N)$; the corresponding $M_1(N) := \sum_{k \le N} \mu(k)/k = o(1)$ (unconditional, PNT) is only a scalar shadow of the critical-line L^2-approximation. Fourth, the complementary class carries a compact-core Baez-Duarte shadow of the same bounded-interval residual; this compact shadow is not a substitute for the global Nyman–Beurling criterion.

93.1 Primary Sources

The approximation results below concern fractional-part functions and their Mellin transforms. Comparing them with the Weil functional requires the signed identity and the specified test-function maps.

Nyman 1950. Nyman, B. *On the One-Dimensional Translation Group and Semi-Group in Certain Function Spaces*. PhD thesis, Uppsala 1950. Introduces the closed subspace $H^2(0,1)$ of $L^2(0,1)$ generated by the translated fractional-part functions and proves the $\chi_{[0,1]} \in \overline{H^2(0,1)}$ criterion for RH.

Beurling 1955. Beurling, A. *A closure problem related to the Riemann zeta-function*. *Proc. Nat. Acad. Sci. USA* 41 (1955) 312–314. Reformulates Nyman's criterion in the modern form with the generators $\rho_\alpha(x) = \{\alpha/x\} - \alpha\{1/x\}$ and establishes the equivalence RH $\Leftrightarrow \chi_{[0,1]} \in H^2(0,1)$.

Baez-Duarte 2003. Baez-Duarte, L. *A strengthening of the Nyman–Beurling criterion for the Riemann hypothesis*. *Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei (9) Mat. Appl.* 14 (2003) 5–11, arXiv:math/0202141. Restriction to $\alpha = 1/k$ for $k \in \mathbb{N}$; Möbius coefficients $c_k = \mu(k)/k$; distance $d_N \to 0$ equivalent to RH.

Baez-Duarte–Balazard–Landreau–Saias 2005. Baez-Duarte, L.; Balazard, M.; Landreau, B.; Saias, E. *Notes sur la fonction ζ de Riemann*, Part IV. *Adv. Math.* 188 (2005) 69–86. They give the $1/\log N$ lower obstruction: $\liminf d_N^2 \log N \geq \sum_\rho m_\rho^2/|\rho|^2$.

Burnol 2001. Burnol, J.-F. *A lower bound in an approximation problem involving the zeros of the Riemann zeta function*, arXiv:math/0103058. The multiplicity m_ρ^2 enters the lower bound through the local principal part at ρ.

Nyman 1950, Beurling 1955, Baez-Duarte 2003, and Baez-Duarte–Balazard–Landreau–Saias 2005 treat closure in $L^2(0,1)$. Their approximation problem uses fractional-part functions. The Weil 1952 explicit formula, Rankin–Selberg identity, Jacquet–Langlands adelic formalism, and de Branges Hermite–Biehler theory use different analytic constructions. The corresponding positivity and spectral comparisons occur in Chapters 78, 82, 79, and 80.

93.2 The Nyman–Beurling Space $H^2(0,1)$

Generators and Mellin Transform

Definition 93.2.1 *(Nyman–Beurling generators).* For $\alpha \in (0,1]$ define $\rho_\alpha : (0,1) \to \mathbb{R}$ by

$$\rho_\alpha(x) \;:=\; \left\{\frac{\alpha}{x}\right\} - \alpha\left\{\frac{1}{x}\right\}, \qquad x \in (0,1),$$

where $\{y\} = y - \lfloor y \rfloor$ is the fractional part. Each $\rho_\alpha \in L^2(0,1)$ (Nyman 1950, Beurling 1955). The Nyman–Beurling closed subspace is

$$H^2(0,1) \;:=\; \overline{\operatorname{span}\{\rho_\alpha : \alpha \in (0,1]\}} \;\subset\; L^2(0,1),$$

the closed linear span in L^2-norm.

Lemma 93.2.2 *(Mellin transform of ρ_α).* For $\operatorname{Re} s > 0$, $s \neq 1$,

$$\int_0^1 \rho_\alpha(x)\, x^{s-1}\, dx \;=\; \frac{\alpha - \alpha^s}{s}\,\zeta(s),$$

by analytic continuation from $\operatorname{Re} s > 1$. The zero of $\alpha - \alpha^s$ at $s = 1$ cancels the simple pole of ζ.

Proof. Beurling 1955, Equation (3); Baez-Duarte 2003 Lemma 2.1. The calculation is elementary: $\int_0^1 \{\alpha/x\} x^{s-1} dx = \alpha^s \int_\alpha^\infty \{y\} y^{-s-1} dy$ via the substitution $y = \alpha/x$, and the fractional-part Dirichlet series $\int_1^\infty \{y\} y^{-s-1} dy = 1/(s-1) - \zeta(s)/s$ (classical; Titchmarsh §2.1). Since $0 < \alpha \leq 1$,

$$\int_\alpha^1 y\, y^{-s-1}\, dy = \frac{1 - \alpha^{1-s}}{1-s}.$$

Hence

$$\int_0^1 \{\alpha/x\} x^{s-1} dx = \alpha^s\left(\frac{1-\alpha^{1-s}}{1-s} + \frac{1}{s-1} - \frac{\zeta(s)}{s}\right) = \frac{\alpha}{s-1} - \frac{\alpha^s \zeta(s)}{s}.$$

Subtracting $\alpha \int_0^1 \{1/x\} x^{s-1} dx = \alpha(1/(s-1) - \zeta(s)/s)$ gives

$$\begin{aligned}\int_0^1 \rho_\alpha(x) x^{s-1} dx &= \left(\frac{\alpha}{s-1} - \frac{\alpha^s\zeta(s)}{s}\right) - \alpha\left(\frac{1}{s-1} - \frac{\zeta(s)}{s}\right) \\ &= \frac{\alpha - \alpha^s}{s}\zeta(s).\end{aligned}$$

□

Definition 93.2.3 *(the characteristic function).* Set $\chi(x) := \chi_{[0,1]}(x)$ restricted to $L^2(0,1)$, i.e. the constant function 1 on $(0,1)$. Its Mellin transform is

$$\int_0^1 \chi(x) x^{s-1} dx = \frac{1}{s}, \qquad \operatorname{Re} s > 0.$$

The Nyman–Beurling Theorem (Recall)

Theorem 93.2.4 *(Nyman–Beurling 1950/1955).* The Riemann hypothesis is equivalent to

$$\chi \in H^2(0,1).$$

That is: χ lies in the closed subspace generated by the $\{\rho_\alpha\}$ if and only if RH holds.

Proof. Nyman 1950 Ch. III; Beurling 1955. The direction RH $\Rightarrow \chi \in H^2$ uses explicit approximants by rational sums; the converse uses that ζ^{-1} extending to the strip $\operatorname{Re} s \geq 1/2 + \varepsilon$ controls the closure. □

Theorem 93.2.5 *(Baez-Duarte refinement).* RH is equivalent to $\lim_{N\to\infty} d_N = 0$, where

$$d_N := \inf_{c_1,\dots,c_N \in \mathbb{C}} \Big\| \chi - \sum_{k=1}^{N} c_k \rho_{1/k} \Big\|_{L^2(0,1)}.$$

Equivalently: RH holds iff the distance from χ to the finite-rank Nyman–Beurling space $H_N^2 := \operatorname{span}\{\rho_{1/k} : k = 1, \dots, N\}$ tends to zero as $N \to \infty$.

Proof. Baez-Duarte 2003 Theorem 1.1. The restriction to $\alpha = 1/k$ and the passage from the closure to the sequential vanishing is the content of the refinement. □

Theorem 93.2.6 *(Baez-Duarte–Balazard–Landreau–Saias lower bound).* Let m_ρ be the multiplicity of the non-trivial zero ρ. Then

$$\liminf_{N\to\infty} d_N^2 \log N \geq \sum_{\rho} \frac{m_\rho^2}{|\rho|^2}.$$

If RH holds and all zeros are simple, the right hand side is

$$\sum_{\rho} \frac{1}{|\rho|^2} = 2 + \gamma_E - \log(4\pi) = 0.04619\ldots.$$

Thus d_N cannot be $o((\log N)^{-1/2})$. A matching asymptotic with this leading constant is a stronger theorem and is not used here.

Proof. Baez-Duarte–Balazard–Landreau–Saias 2005 and Burnol's multiplicity refinement give the lower bound. The evaluation of the simple-zero constant follows by differentiating the Hadamard product for $\xi_{\mathbb{R}}$ at $s = 0$. □

93.3 Mellin Normal Form and RH-Level Comparison

The Mellin–Plancherel identity converts the Nyman–Beurling $L^2(0,1)$ norm into a critical-line integral. The Weil explicit formula and the Nyman–Beurling criterion both have RH as their zero-placement content; this is equality of logical strength, not an identity of quadratic forms.

The Mellin Norm Identity

Lemma 93.3.1 *(Mellin–Plancherel for $L^2(0,1)$ in the half-line model).* For $\phi \in L^2(0,1)$ with Mellin transform $\widehat{\phi}_M(s) := \int_0^1 \phi(x) x^{s-1} dx$ regular on $\operatorname{Re} s = 1/2$,

$$\|\phi\|^2_{L^2(0,1)} \;=\; \frac{1}{2\pi} \int_{-\infty}^{\infty} |\widehat{\phi}_M(1/2 + it)|^2 \, dt. \tag{93.1}$$

Proof. The standard Mellin–Plancherel theorem (Titchmarsh *The Theory of Functions* 2nd ed. §7.1) gives, for $\phi \in L^2((0,1), dx)$ extended by zero to $L^2((0,\infty), dx)$ and the change of variable $x = e^{-u}$, an isometry on the line $\operatorname{Re} s = 1/2$. The precise identity (93.1) uses the weighted $L^2(0,1)$ norm without the dx/x measure; a substitution $x = e^{-u}$ gives $\|\phi\|^2_{L^2(0,1)} = \int_0^\infty |\phi(e^{-u})|^2 e^{-u} du$. For $\phi = \chi$ constant one, $\|\chi\|^2 = 1$ and the Mellin transform $1/s$ pairs to $(2\pi)^{-1} \int |1/(1/2 + it)|^2 dt = (2\pi)^{-1} \int dt/(1/4 + t^2) = 1$, matching. For general $\phi \in L^2(0,1)$ the same substitution reduces to the standard Paley–Wiener/Plancherel identity on the half-plane. Detailed computation (Baez-Duarte 2003 Lemma 3.1). □

Proposition 93.3.2 *(Nyman–Beurling distance as Mellin integral).* For $c_1, \dots, c_N \in \mathbb{C}$,

$$\|\chi - \sum_{k=1}^{N} c_k \rho_{1/k}\|^2_{L^2(0,1)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{1}{s} - \frac{\zeta(s)}{s} \sum_{k=1}^{N} c_k \big(k^{-1} - k^{-s}\big) \right|^2_{s=1/2+it} dt.$$

Equivalently, abbreviating $A_N(s) := \sum_{k=1}^N c_k \, k^{-s}$,

$$\|\chi - \sum c_k \rho_{1/k}\|^2_{L^2(0,1)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{1 - \zeta(s)\big(A_N(1) - A_N(s)\big)}{s} \right|^2_{s=1/2+it} dt, \tag{93.2}$$

where $A_N(1) = \sum_{k=1}^N c_k/k$.

Proof. By Lemma 93.2.2, $\widehat{\rho_{1/k}}_M(s) = \zeta(s)(k^{-1} - k^{-s})/s$. Linearity: the Mellin transform of $\chi - \sum c_k \rho_{1/k}$ is $1/s - \zeta(s) s^{-1} \sum c_k (k^{-1} - k^{-s})$. Apply Lemma 93.3.1 on $\operatorname{Re} s = 1/2$. □

Remark 93.3.3 *(Möbius choice and the missing $(s-1)^{-1}$ pole).* Setting $c_k = \mu(k)/k$ gives $A_N(s) = \sum_{k=1}^N \mu(k) k^{-s-1}$, so $A_N(1) = \sum_{k=1}^N \mu(k)/k^2 \to 6/\pi^2$ as $N \to \infty$ (absolutely convergent, unconditional). There is no $(s-1)^{-1}$-term in (93.2). The main term is the critical-line factor $\big(1 - \zeta(s)(A_N(1) - A_N(s))\big)/s$, and it is not controlled by the scalar $M_1(N) = \sum_{k \le N} \mu(k)/k$ alone.

The Weil-Type Insertion

Theorem 93.3.4 *(A-TP and Nyman–Beurling have the same RH content).* The archimedean Tate positivity A-TP of Conjecture 78.2.3 is equivalent to the Nyman–Beurling closure $\chi \in H^2(0,1)$; equivalently to the Baez-Duarte vanishing $\lim_{N\to\infty} d_N = 0$.

Proof. Both conditions are equivalent to RH. A-TP $\Leftrightarrow$ RH by Theorem 78.2.4. RH $\Leftrightarrow$ $\chi \in H^2(0,1)$ by Theorem 93.2.4. RH $\Leftrightarrow$ $d_N \to 0$ by Theorem 93.2.5. Transitivity closes the reformulation. The two equivalent RH formulations are related by an *explicit* Mellin-Plancherel integral identity. □

Remark 93.3.5 *(the Mellin identity is not a zero-square identity).* Beyond the tautological RH-equivalence, the Nyman–Beurling side gives the exact Mellin-Plancherel identity

$$\|\chi - \sum_{k=1}^{N} c_k \rho_{1/k}\|^2_{L^2(0,1)} \;=\; \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{1 - \zeta(s)\big(A_N(1) - A_N(s)\big)}{s} \right|^2_{s=1/2+it} dt, \tag{93.3}$$

with $A_N(s) = \sum c_k k^{-s}$. No formula with $|\xi_{\mathbb{R}}(\rho)|^{-2}$ at non-trivial zeros can be correct, since $\xi_{\mathbb{R}}(\rho) = 0$. Passing from (93.3) to a zero expansion is exactly the Baez–Duarte theorem; it is not an elementary residue computation inside the Mellin identity.

93.4 Restricted-Class Estimate on HFD

On $\mathscr{A} = C_c^\infty(\mathbb{R})$, the quantitative HFD condition of Definition 82.4.1 implies $G[f] = (B[f] - A[f])/(2\pi) \geq 0$, hence $W_\infty(f * \widetilde{f}) \leq 2\log\pi\|f\|_2^2$. It does not imply $W_\infty \geq 0$: the smooth modulated tests of Theorem 92.4.5 give the opposite sign. On any further specified arithmetic test class, HFD means intersection with this quantitative class. Neither that intersection nor HFD itself supplies a weighted Baez-Duarte estimate.

HFD-Adapted Nyman–Beurling Distance

Definition 93.4.1 *(HFD-restricted Nyman–Beurling distance).* For $f \in \mathrm{PW}_F^{\mathrm{hol,HFD}}$ together with a specified unitary Mellin transform $\mathcal{M}_{\mathrm{add}\to\mathrm{mult}}$ from the additive Paley–Wiener model to the critical-line Mellin model, set $\widehat{f}_M = \mathcal{M}_{\mathrm{add}\to\mathrm{mult}}(\widehat{f})$ and define

$$d_N^{\mathrm{HFD}}(f) := \inf_{c_k} \left\| \widehat{f}_M(1/2+it) \cdot \left[\frac{1 - \zeta(s)\sum_{k=1}^N c_k(k^{-1} - k^{-s})}{s} \right]_{s=1/2+it} \right\|_{L^2(\mathbb{R})}.$$

This is the Mellin-weighted Nyman–Beurling distance tested against the spectral profile of f. Without the displayed unitary transform, $\widehat{f}_M$ is not determined by the additive test function.

Theorem 93.4.2 *(weighted HFD Nyman–Beurling bound is an open estimate).* The estimate

$$d_N^{\mathrm{HFD}}(f)^2 \leq C_{\mathrm{HFD}}(f) \cdot \frac{1}{\log N} + \varepsilon_{\mathrm{pol}}(N) \cdot \|f\|_{L^2}^2 \tag{93.4}$$

does not follow from the preceding identities. It would require a uniform weighted Baez–Duarte estimate for the critical-line integral (93.3). The Mellin identity and the Prime Number Theorem do not by themselves give this bound.

Proof. The exact identity is (93.3). A putative zero expansion containing $|\xi_{\mathbb{R}}(\rho)|^{-2}$ is undefined at non-trivial zeros of $\xi_{\mathbb{R}}$. The displayed integral is a weighted approximation problem for $1/\zeta$ on the critical line. Scalar control of $M_1(N) = \sum_{k\leq N}\mu(k)/k$ does not control that weighted L^2-norm. □

Corollary 93.4.3 *(the HFD Nyman–Beurling estimate is an additional input).* The limit

$$\lim_{N\to\infty} d_N^{\mathrm{HFD}}(f) = 0$$

does not follow from the preceding identities. Thus the identities do not prove A-TP on HFD.

Proof. It would follow from Theorem 93.4.2 only after that estimate is known. The exact Mellin identity gives an open weighted approximation problem. □

Remark 93.4.4 *(comparison with the bounded-interval reduction).* The bounded-interval reduction gives $G \geq 0$ on HFD and the upper bound $W_\infty \leq 2\log\pi\|f\|_2^2$. The Nyman–Beurling form adds the critical-line weighted approximation problem (93.4); it does not remove the residual $R[f]$, and it does not give full A-TP on HFD without that extra weighted estimate.

93.5 Möbius Partial-Sum Control

The Möbius partial sums $M(N)$ and $M_1(N)$ remain useful background. They do not, by themselves, prove the weighted HFD estimate of Theorem 93.4.2.

Proposition 93.5.1 *(Mertens-type bounds, unconditional).* Set $M(N) := \sum_{k\le N}\mu(k)$ and $M_1(N) := \sum_{k\le N}\mu(k)/k$. Unconditionally:

$$\begin{aligned} M(N) &= o(N), \qquad N\to\infty \quad \text{(PNT, Hadamard-de la Vallée-Poussin 1896)},\\ M_1(N) &= o(1), \qquad N\to\infty \quad \text{(Landau 1911, equivalent to PNT)}. \end{aligned}$$

Quantitative form (Titchmarsh §3.14): $M(N) = O(N\exp(-c\sqrt{\log N}))$ and $M_1(N) = O(\exp(-c\sqrt{\log N}))$ for some $c>0$.

Proof. Hadamard 1896 (*Bull. Soc. Math. France* 24) and de la Vallée-Poussin 1896 (*Ann. Soc. Sci. Bruxelles* 20) established the PNT via zero-free region of ζ just inside $\operatorname{Re} s = 1$. Landau 1911 (*Handbuch der Lehre von der Verteilung der Primzahlen*) deduced the bound on $M_1(N)$. The quantitative form follows from the de la Vallée-Poussin zero-free region $\sigma \ge 1 - c/\log(|t|+3)$. □

Proposition 93.5.2 *(Mertens bound, conditional on RH).* Under RH:

$$\begin{aligned} M(N) &= O(\sqrt{N}\log^2 N),\\ M_1(N) &= O(N^{-1/2}\log^2 N). \end{aligned}$$

Proof. Titchmarsh §14.27; the classical conditional bound via the explicit formula for $M(N)$ in terms of zeros of ζ. The bound on $M_1(N)$ follows by partial summation from the bound on $M(N)$. □

Remark 93.5.3 *(Mertens bounds do not give the weighted HFD estimate).* The scalar estimates $M_1(N) = o(1)$ and $M_1(N) = O(N^{-1/2}\log^2 N)$ do not control the norm in (93.3). That norm asks for weighted approximation to $1/\zeta(1/2+it)$ on the critical line. The Baez-Duarte–Balazard–Landreau–Saias $1/\log N$ lower obstruction is a zero-expansion statement; without a matching global critical-line upper theorem it cannot be transferred to (93.4).

93.6 Comparison of the Complementary Class with Baez-Duarte

Use the common test carrier $\mathscr{T} = \mathrm{PW}_F^{\mathrm{hol}} \cap C_c^\infty(\mathbb{R})$ and $\mathscr{T}_{\mathrm{HFD}} = \mathscr{T} \cap \mathscr{A}_{\mathrm{HFD}}$. The complement $\mathscr{T} \setminus \mathscr{T}_{\mathrm{HFD}}$ is where the quantitative auxiliary G-domination condition fails; no positivity of the signed W_∞ is inferred from HFD. For test functions outside HFD, the mass on $|t| \le t_*$ may be measured by a compact-core Nyman–Beurling distance. This compact-core distance is a shadow of the global Baez-Duarte criterion, not an equivalent replacement for it.

Theorem 93.6.1 *(complementary-class comparison with Baez-Duarte).* For $f \in \mathscr{T} \setminus \mathscr{T}_{\mathrm{HFD}}$, set

$$d_N^{\mathrm{low}}(f)^2 := \inf_{c_k} \int_{|t|\le t_*} |\widehat{f}(t)|^2 \cdot \left| \frac{1 - \zeta(s)\sum_{k=1}^N c_k(k^{-1}-k^{-s})}{s} \right|^2_{s=1/2+it} dt. \tag{93.5}$$

On $|t| \le t_*$,

$$0 \le -g(t) \le (1/4 + t_*^2)|g(0)|\,|s|^{-2}_{s=1/2+it}. \tag{93.6}$$

Thus the compact-core Nyman–Beurling norm is taken over a measure which dominates the archimedean core measure. This is a comparison of weighted error norms, not an inequality for $A[f]$ itself; the finite-prime-polar functional $R[f]$ remains in inequality (★). The compact-core norm does not imply the global Baez-Duarte vanishing $d_N \to 0$.

Proof. By Lemma 82.3.2, $W_\infty(f * \widetilde{f}) = 2\log\pi\|f\|_2^2 - (B[f] - A[f])/(2\pi)$. On HFD^c, $B[f] < A[f]$ is possible, so the finite-prime-polar functional remains part of the inequality. The core integral, which contributes positively to W_∞, is

$$A[f] = \int_{|t|\le t_*} |\widehat{f}(t)|^2 \cdot (-g(t))dt.$$

By Lemma 82.2.2 and Proposition 82.2.3, $-g(t) = |g(0)| - \int_0^{|t|} g'(u)du \ge 0$ on $|t| \le t_*$. Since $|s|^{-2} = (1/4 + t^2)^{-1}$ on $s = 1/2 + it$, the bound (93.6) follows from $-g(t) \le |g(0)|$ and $1/4 + t^2 \le 1/4 + t_*^2$. Substitution into inequality (★) shows where this measure comparison enters the bounded-interval residual. The factor $|1 - \zeta(s)\sum c_k(k^{-1} - k^{-s})|^2$ is an additional approximation error, so no estimate for $A[f]$ follows without a separate approximation theorem. The implication $d_N \to 0$ is exactly the global Nyman–Beurling–Baez-Duarte criterion and cannot be inferred from the compact interval alone. □

Corollary 93.6.2 *(compact Baez-Duarte data do not remove the irreducible A-TP residual).* The residual A-TP on $\mathscr{T} \setminus \mathscr{T}_{\mathrm{HFD}}$ is not closed by the compact-core norm (93.5). Full closure would require the global Baez-Duarte vanishing $d_N \to 0$, equivalently RH, together with the retained $R[f]$-inequality of inequality (★).

Proof. Combine Theorem 93.6.1 with the global criterion Theorem 93.2.5. □

93.7 Seven Obstructions to the HFD Nyman–Beurling Bound

The estimate in Theorem 93.4.2 is the open bound $d_N^{\mathrm{HFD}}(f)^2 \le C_{\mathrm{HFD}}(f)/\log N + O((\log N)^{-2})$ on the HFD sub-class. The following seven obstructions remain after the Mellin formula is put in its pole-free form.

Remark 93.7.1 *(obstruction 1: a scalar Mertens bound is too small).* The limit $M_1(N) = \sum_{k\le N}\mu(k)/k \to 0$ is unconditional by PNT, and the de la Vallée-Poussin zero-free region gives the usual exponential decay in $\sqrt{\log N}$. This scalar statement is nevertheless too small for (93.4): the latter is an L^2 approximation estimate for $1/\zeta(1/2 + it)$ against a moving weight.

Remark 93.7.2 *(obstruction 2: the* $1/\log N$ *scale is only a lower obstruction).* The $1/\log N$ scale is forced from below by the Baez-Duarte–Balazard–Landreau–Saias zero expansion. A proof of an upper bound of the same scale for $d_N^{\mathrm{HFD}}(f)$ would have to replace the global zero-expansion by a weighted critical-line theorem. Truncating the spectral weight to $|t| > t_*$ does not by itself supply such a theorem.

Remark 93.7.3 *(Nonzero HFD tests and the arithmetic test domain).* Remark 82.4.4 constructs smooth compactly supported modulated tests satisfying (82.8). They give a nonzero domain for $G \ge 0$, not for a positive signed archimedean contribution. Their membership in any additional arithmetic test class must still be established, as must the weighted approximation estimate.

Remark 93.7.4 *(obstruction 4:* $\varepsilon_{\mathrm{pol}}(N)$ *conceals the real obstruction).* The Mellin formula has no $(s-1)^{-1}$ polar term in (93.2). The obstruction is the weighted critical-line norm in (93.3), not a finite-dimensional polar remainder.

Remark 93.7.5 *(obstruction 5: the cross-reduction identity* (93.3) *is circular).* The identity (93.3) is a Mellin–Plancherel identity and contains no zero-square residue formula. It is independent of the Weil distribution, but independence of identities is weaker than a bound. The missing step is precisely a uniform estimate for the displayed integral.

Remark 93.7.6 *(Closure and the weighted estimate).* The Nyman–Beurling closure theorem concerns an unweighted norm. The estimate (93.4) contains a specified test-dependent weight; its constants and convergence must be proved for that weight. An unweighted closure statement alone does not supply this estimate.

Remark 93.7.7 *(obstruction 7: closing A-TP on* HFD^c *via Nyman–Beurling).* The compact-core norm (93.5) measures only the archimedean negative interval. Closing A-TP on HFD^c still requires the $R[f]$-term in inequality (⋆); replacing it by the global Nyman–Beurling vanishing is exactly the RH-equivalent Baez-Duarte criterion.

93.8 The Irreducible Obstruction, Named in the Nyman–Beurling Framework

Theorem 93.8.1 *(irreducible A-TP residual, Nyman–Beurling form).* The Nyman–Beurling formulation identifies the same residual in two forms: the global Baez-Duarte vanishing $d_N \to 0$, which is RH-equivalent, and the compact-core norm (93.5), which measures only the negative archimedean interval. The latter does not close A-TP without the $R[f]$-inequality, and the HFD weighted estimate Theorem 93.4.2 remains an additional analytic input.

Proof. Assemble Theorem 93.6.1, Corollary 93.6.2, and Theorem 93.4.2. □

Remark 93.8.2 *(comparison with the bounded-interval residual inequality).* The bounded-interval reduction (Chapter 82) reduced A-TP to a exact signed domination

$$A[f] + 4\pi \log \pi \|f\|_2^2 \le B[f] + 2\pi R[f].$$

Thus the core and constant contributions are bounded by the auxiliary tail together with the finite-prime and polar functional. The Nyman–Beurling form gives a compact-core distance for the negative interval and a global distance equivalent to RH. Neither distance removes the $R[f]$-term from the bounded-interval inequality. The Nyman–Beurling construction supplies exactly the following data:

(i) the exact Mellin normal form for the weighted critical-line estimate and an $L^2(0,1)$ distance;
(ii) the precise weighted estimate whose proof would be required for HFD closure;
(iii) a reformulation of the irreducible residual as a concrete Baez-Duarte distance problem.

The weighted approximation problem and the signed Weil inequality remain distinct until a comparison controlling the full $R[f]$-coupled expression is constructed.

93.9 Mellin Boundary

(i) The Nyman–Beurling generators $\rho_\alpha(x) = \{\alpha/x\} - \alpha\{1/x\}$ have Mellin transform $(\alpha - \alpha^s)\zeta(s)/s$ (Lemma 93.2.2); A-TP is equivalent to the Nyman–Beurling closure $\chi \in H^2(0,1)$ and to the Baez-Duarte vanishing $d_N \to 0$ (Theorem 93.3.4).

(ii) The HFD-restricted Nyman–Beurling distance $d_N^{\mathrm{HFD}}(f)$ (Definition 93.4.1) is governed by the exact Mellin-Plancherel integral (93.3). The proposed $O(1/\log N)$ HFD bound is a separate weighted estimate (Theorem 93.4.2).

(iii) On HFD^c, the compact-core Baez-Duarte norm controls the negative archimedean interval but does not replace the global Baez-Duarte criterion or the retained $R[f]$-inequality (Theorem 93.6.1, Corollary 93.6.2).

(iv) The obstruction-theoretic comparison changes the form of the HFD Nyman–Beurling construction: PNT-strength Mertens estimates are background, not a proof of the weighted critical-line estimate.

(v) The exact Mellin expression identifies the weighted approximation problem. It does not establish the signed Weil inequality on HFD (Theorem 93.8.1).

Remark 93.9.1 *(the Beilinson principle on the Nyman–Beurling comparison).* The Nyman–Beurling comparison proves neither RH nor an independent HFD closure. Its mathematical content is the irreducible residual recast in $L^2(0,1)$ language and the exact weighted estimate which would imply the missing HFD bound.

Remark 93.9.2 *(cross-volume dependency).* The inputs are Chapter 78 (Rankin–Selberg kernel obstruction, A-TP conjecture 78.2.3), Chapter 82 (HFD sub-class, bounded-interval reduction), and PNT (Hadamard 1896, de la Vallée-Poussin 1896, Landau 1911). They are independent of the de Branges / Hermite–Biehler analysis of Chapter 79, the $\mathrm{R}_{\mathrm{analytic}}$ density-of-zeros analysis of Chapter 80, or the Positselski four sub-items of Chapter 75. The Nyman–Beurling construction is orthogonal to these.

Chapter 94

A-TP via de Branges Hermite–Biehler Factorisation

The function $E_\xi = \Xi + i\Xi'$ has real part Ξ. Its positive kernel, when established, is strictly Hermite–Biehler by the derivative calculation and therefore forces zero placement. Multiplication by E_ξ gives a model-space isometry under that Hermite–Biehler hypothesis. Identifying an arithmetic test space or the Weil form with this model requires additional maps.

The signed archimedean identity is $W_\infty = 2\log\pi\,\|f\|_2^2 - G[f]$ on $\mathscr{A} = C_c^\infty(\mathbb{R})$. The HFD condition makes G nonnegative, not W_∞. The full Weil target retains $R[f] \geq W_\infty(f * \widetilde{f})$ and the actual arithmetic test-domain comparison. The zero-free-region calculation below gives peripheral bands, not a central positivity region.

94.1 Context and the de Branges Comparison

Chapter 78 established:

- A-TP $\Leftrightarrow$ RH via the Weil positive cone (Theorem 78.2.4).
- Prime-side Petersson closure on $\mathrm{PW}_F^{\mathrm{hol}}$ via the Mellin lift Υ_F (Theorem 78.5.5).
- The archimedean digamma obstruction as the unabsorbed residual of the Rankin–Selberg isometry (Proposition 78.6.2).

Chapter 79 established:

- The de Branges structure theorem in the form Proposition 79.6.7: E_ξ is Hermite–Biehler if and only if the de Branges reproducing kernel K_{E_ξ} is positive-semidefinite; with $(E_\xi + E_\xi^*)/2 = \Xi$, the Hermite–Biehler separation theorem puts the zeros of Ξ on the real axis.
- The equivalence $\mathrm{HB}^+ \Leftrightarrow$ RH (Remark 79.6.11), Lagarias 1999.

Chapter 82 established:

- Positivity of G on the quantitative HFD class in $\mathscr{A}$, with $W_\infty = 2\log\pi\,\|f\|_2^2 - G$. Smooth compactly supported modulations give HFD tests with $W_\infty < 0$. A special arithmetic test class is restricted by intersection with $\mathscr{A}$, not supplied by this construction.
- The full Weil inequality is $A[f] + 4\pi\log\pi\,\|f\|_2^2 \leq B[f] + 2\pi R[f]$. The interval $[-t_*, t_*]$ is the negative region of g, not of the signed multiplier defining W_∞.
- Three non-elementary constructions that remain open: zero density, Gaussian reduction, and de Branges spectral structure (Remark 82.6.3).

The A-TP↔de Branges kernel dictionary contains the Vinogradov–Korobov coordinate change and identifies the obstruction via Lagarias 1999 and Conrey–Li 2000.

Domain

The objects are:

- The Lagarias function E_ξ, its conjugate E_ξ^*, its real part $A_\xi = \Xi$, and the de Branges kernel K_{E_ξ}.
- The model-space isometry $K(E_\xi^*/E_\xi) \simeq \mathcal{H}(E_\xi)$ and the Mellin transforms which lie in that model space.
- Coordinate comparison of the Vinogradov–Korobov zero-free region under $s = \frac{1}{2} + iz$.
- Proof that the Vinogradov–Korobov input gives only peripheral zero-free bands and no A-TP positivity closure.
- Identification of the genuine obstruction as HB^+, equivalent to RH.

The proof of HB^+ on $\xi_{\mathbb{R}}$ is RH-equivalent and is not supplied by the kernel-positivity reformulation. No argument below uses de Branges operator monotonicity beyond that reformulation.

Independent Sources

The four primary source lists below are disjoint from the restricted-class comparison and the bounded-interval reduction proof sources.

de Branges 1968, *Hilbert Spaces of Entire Functions* (Prentice-Hall). Foundational monograph. Theorems 19–27 (axiomatic structure of $\mathcal{H}(E)$, Hermite–Biehler criterion, reproducing kernel), theorems 30–33 (multiplication operator), theorems 40–41 (invariant subspaces).

Lagarias 1999, Acta Arith. 89. *On a positivity property of the Riemann ξ-function.* Explicit construction of E_ξ from Λ_ζ, exponential type computation, statement of $\mathrm{HB}^+ \Leftrightarrow$ RH with a Beurling–Malliavin density addendum.

Conrey–Li 2000, arXiv:math/9812166. *A note on some positivity conditions related to zeta and L-functions.* Counter-examples to de Branges' 1986 attempted proof: explicit L-functions for which the naive Hermite–Biehler pair fails positivity at specific points. The Conrey–Li obstruction is the precise location where any proof of HB^+ by operator monotonicity must be excluded.

Titchmarsh 1951, *The Theory of the Riemann Zeta-Function* (Oxford), Ch. V–VI. Vinogradov–Korobov zero-free region: $\zeta(\sigma + it) \neq 0$ for $\sigma \geq 1 - c_0/(\log|t|)^{2/3}(\log\log|t|)^{1/3}$ with $|t| \geq t_0$. Korobov 1958 Uspekhi; Vinogradov 1958 Izv.

Independence of inputs. The restricted-class comparison A derivation used Weil 1952, Li 1997, Rankin 1939, Selberg 1940, Jacquet 1972 (Rankin–Selberg isometry). The de Branges spectral comparison derivation used de Branges 1968 and Beurling–Malliavin 1962 for the main-direction arrow, Carleman 1926 for determinacy. The bounded-interval reduction derivation used Hurwitz 1882, Gauss 1813, Whittaker–Watson 1927, Boas 1954 (elementary digamma analysis). The additional sources are de Branges 1968 (in its Ch. III reproducing-kernel role, distinct from the structure-theorem role used by the de Branges spectral comparison), Lagarias 1999 (explicit E_ξ computation, not used in the de Branges spectral comparison's structural implication), Conrey–Li 2000 (obstruction location, not used in the de Branges spectral comparison or the bounded-interval reduction), and Titchmarsh 1951 Vinogradov–Korobov (not used in constructions 5 or 6). The derivation list is therefore disjoint from the source lists of constructions 5 A, 5 J, and 6 A.

94.2 The de Branges Function E_ξ and Its Real-Line Trace

Fix the shifted character $\Xi(z) = \xi_{\mathbb{R}}(\frac{1}{2} + iz)$ of Definition 79.2.3. We recall the Hermite–Biehler candidate.

Definition 94.2.1 (Lagarias de Branges function for $\xi_{\mathbb{R}}$). Define

$$E_\xi(z) \;:=\; \xi_{\mathbb{R}}\big(\tfrac{1}{2} - iz\big) \;+\; \xi'_{\mathbb{R}}\big(\tfrac{1}{2} - iz\big), \tag{94.1}$$

with its conjugate-partner

$$E^*_\xi(z) \;:=\; \overline{E_\xi(\overline{z})} \;=\; \xi_{\mathbb{R}}\big(\tfrac{1}{2} + iz\big) \;+\; \xi'_{\mathbb{R}}\big(\tfrac{1}{2} + iz\big).$$

The derivative is with respect to s. This is the Lagarias candidate used in the de Branges reformulation.

Lemma 94.2.2 *(Lagarias real-part identity for E_ξ).* The real part

$$A_\xi(z) := \frac{E_\xi(z) + E^*_\xi(z)}{2}$$

is exactly the shifted xi-function:

$$A_\xi(z) \;=\; \Xi(z) \;=\; \xi_{\mathbb{R}}\big(\tfrac{1}{2} + iz\big).$$

Consequently $\operatorname{div}(A_\xi) = \operatorname{div}(\Xi)$.

Proof. The functional equation gives $\xi_{\mathbb{R}}(\frac{1}{2} - iz) = \xi_{\mathbb{R}}(\frac{1}{2} + iz)$. Differentiating $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1 - s)$ gives $\xi'_{\mathbb{R}}(\frac{1}{2} - iz) = -\xi'_{\mathbb{R}}(\frac{1}{2} + iz)$. The derivative terms in $E_\xi + E^*_\xi$ cancel, and the two undifferentiated terms give $2\Xi(z)$. □

Remark 94.2.3 (the completed zeta is not a Hermite–Biehler datum). The function $\Lambda_\zeta(\frac{1}{2} + iz)$ is fixed by $*$, by the functional equation. It cannot satisfy the strict Hermite–Biehler inequality. The derivative term in (94.1) is therefore not a normalisation; it is the datum which separates $A_\xi = \Xi$ from its conjugate quadrature.

Proposition 94.2.4 *(Stirling does not supply Hermite–Biehler positivity).* No Stirling estimate for the gamma factor in $\Lambda_\zeta(\frac{1}{2} + iz)$ proves

$$|E_\xi(z)| > |E^*_\xi(z)|, \qquad \operatorname{Im} z > 0,$$

for the Lagarias function (94.1). The omitted term is the logarithmic derivative of $\xi_{\mathbb{R}}$, and the strict inequality is precisely the Hermite–Biehler condition.

Proof. If $\Lambda_\zeta(\frac{1}{2} + iz)$ were used in place of (94.1), the functional equation would give $E = E^*$, so the strict Hermite–Biehler inequality would be impossible. For the actual Lagarias function,

$$\frac{E^*_\xi(z)}{E_\xi(z)} = \frac{\xi_{\mathbb{R}}(\frac{1}{2} + iz) + \xi'_{\mathbb{R}}(\frac{1}{2} + iz)}{\xi_{\mathbb{R}}(\frac{1}{2} - iz) + \xi'_{\mathbb{R}}(\frac{1}{2} - iz)},$$

and the derivative term is not controlled by the gamma-factor asymptotic alone. Thus any proof of the strict inequality is a proof of Hermite–Biehler positivity, not a consequence of Stirling's formula. □

COROLLARY 94.2.5 *(no exterior Hermite–Biehler region from Stirling).* The estimate

$$|E_\xi(z)| > |E_\xi^*(z)| \cdot \exp\!\big(\tfrac{1}{2} r \sin\theta \log(r/2)\big)$$

is not an unconditional consequence of Stirling asymptotics. The valid unconditional zero-free input is the Vinogradov–Korobov peripheral exclusion for the real part $A_\xi = \Xi$.

Proof. Proposition 94.2.4 removes the only purported derivation of the displayed inequality. The classical zero-free theorem controls zeros of ζ, hence zeros of A_ξ, only in the peripheral bands described below. □

94.3 de Branges Reproducing Kernel and the Mellin Identification

Definition 94.3.1 *(de Branges space attached to E_ξ).* If E_ξ is Hermite–Biehler, the *de Branges space* $\mathcal{H}(E_\xi)$ is the Hilbert space of entire functions F with

$$\|F\|_{E_\xi}^2 := \int_{\mathbb{R}} \left| \frac{F(t)}{E_\xi(t)} \right|^2 dt < +\infty$$

and $F/E_\xi, F^*/E_\xi \in H^2(\mathbb{C}_+)$, with the reproducing kernel

$$K_{E_\xi}(w,z) = \frac{\overline{E_\xi(w)}\,E_\xi(z) - \overline{E_\xi^*(w)}\,E_\xi^*(z)}{2\pi i(\overline{w} - z)}.$$

PROPOSITION 94.3.2 *(de Branges real-line inner product).* For $F, G \in \mathcal{H}(E_\xi)$,

$$\langle F, G\rangle_{E_\xi} = \int_{\mathbb{R}} \frac{F(t)\overline{G(t)}}{|E_\xi(t)|^2}\, dt.$$

Here E_ξ is the Lagarias function (94.1), not $\Lambda_\zeta(\frac{1}{2} + it)$.

Proof. This is the defining inner product of a de Branges space; polarisation gives the displayed formula. □

THEOREM 94.3.3 *(Mellin model-space isometry).* Assume E_ξ is Hermite–Biehler and put

$$\Theta_\xi(z) := \frac{E_\xi^*(z)}{E_\xi(z)}.$$

Let $\Upsilon_F : \mathrm{PW}_F^{\mathrm{hol}} \to \mathcal{V}_{\mathrm{cusp,temp}}^{\mathrm{prim}}$ be the Mellin lift of Definition 78.4.2. For every $f \in \mathrm{PW}_F^{\mathrm{hol}}$ whose Mellin transform lies in the model space $K(\Theta_\xi) = H^2 \ominus \Theta_\xi H^2$, define

$$\widehat{\Upsilon}_F(f)(z) := \widehat{f}(z) \cdot E_\xi(z), \qquad z \in \mathbb{C}.$$

Then $\widehat{\Upsilon}_F(f) \in \mathcal{H}(E_\xi)$ and

$$\langle \widehat{\Upsilon}_F(f), \widehat{\Upsilon}_F(f)\rangle_{E_\xi} = \int_{\mathbb{R}} |\widehat{f}(t)|^2\, dt. \tag{94.2}$$

Proof. For a Hermite–Biehler function, multiplication by E_ξ is the standard isometric identification

$$K(\Theta_\xi) \xrightarrow{\sim} \mathcal{H}(E_\xi).$$

Indeed $F = \widehat{f}E_\xi$ satisfies $F/E_\xi = \widehat{f} \in H^2$ and $F^*/E_\xi = \Theta_\xi \widehat{f}^* \in H^2$ by the model-space condition. By Proposition 94.3.2,

$$\|\widehat{\Upsilon}_F(f)\|^2_{E_\xi} = \int_{\mathbb{R}} \frac{|\widehat{f}(t)E_\xi(t)|^2}{|E_\xi(t)|^2}\,dt = \int_{\mathbb{R}} |\widehat{f}(t)|^2\,dt,$$

which is (94.2). □

Corollary 94.3.4 *(Finite kernel Gram forms).* Assume E_ξ is Hermite–Biehler. For finitely many real points x_j and complex coefficients c_j, put $F = \sum_j c_j K_{E_\xi}(x_j, \cdot)$. Then

$$\|F\|^2_{E_\xi} = \sum_{j,k} c_j \overline{c_k} K_{E_\xi}(x_j, x_k) \geq 0. \tag{94.3}$$

A representation of the arithmetic Weil form by this Gram expression requires an actual linear map from its test carrier to these vectors and equality of the two forms. Coordinates of zeros or values of $\widehat{f}$ alone do not supply that equality.

Proof. Use the reproducing property with the Hermitian form linear in its first variable: $\langle K(x_j, \cdot), K(x_k, \cdot)\rangle = K(x_j, x_k)$. Expand the squared norm of the finite linear combination. For a zero ρ, the parameter $z_\rho = -i(\rho - 1/2)$ is real precisely when ρ is on the critical line; selecting real zero parameters is itself a zero-placement requirement. No arithmetic norm identity follows from this change of variables. □

94.4 What the Vinogradov–Korobov Region Actually Gives

The classical zero-free region is a statement near the line $\operatorname{Re} s = 1$, and by the functional equation near $\operatorname{Re} s = 0$. Under $s = \frac{1}{2} + iz$ it gives peripheral horizontal bands in the z-plane. It does not give a zero-free strip around the real z-axis. The real axis is exactly the critical line.

Definition 94.4.1 *(Vinogradov–Korobov peripheral bands).* Let $c_0 > 0$ be the Vinogradov–Korobov constant (Titchmarsh 1951 Ch. VI; Korobov 1958). Define, for sufficiently large $T > e$, the peripheral bands

$$\eta(T) := \frac{c_0}{(\log T)^{2/3}(\log\log T)^{1/3}},$$

$$\mathcal{P}_T := \left\{\alpha + i\beta : |\alpha| \leq T, \quad |\beta| \geq \frac{1}{2} - \eta(T)\right\}.$$

The central strip

$$C_T := \{z = \alpha + i\beta : |\alpha| \leq T,\ |\beta| < \eta(T)\}$$

is not a Vinogradov–Korobov zero-free region. It contains the real axis, hence contains the critical-line zeros when RH holds.

Lemma 94.4.2 *(A_ξ is zero-free in the peripheral bands).* For $T \geq T_0$, the function $A_\xi = \Xi$ has no zeros in $\mathcal{P}_T$. No zero-free conclusion follows for the central strip C_T.

Proof. A zero $z = \alpha + i\beta$ of $A_\xi = \Xi$ corresponds to a non-trivial zero

$$\rho = \frac{1}{2} + iz = \frac{1}{2} - \beta + i\alpha$$

of ζ. Vinogradov–Korobov gives

$$\operatorname{Re}\rho < 1 - \eta(|\alpha|)$$

for $|\alpha|$ large, hence $\beta > -\frac{1}{2} + \eta(|\alpha|)$. Applying the functional equation gives the reflected inequality $\beta < \frac{1}{2} - \eta(|\alpha|)$. Thus zeros with $|\alpha| \le T$ lie in the central band

$$-\frac{1}{2} + \eta(T) < \beta < \frac{1}{2} - \eta(T),$$

after decreasing c_0 and increasing T_0 in the usual uniform way. This proves zero-freeness of $\mathcal{P}_T$. It says nothing about $|\beta| < \eta(T)$. □

PROPOSITION 94.4.3 *(Vinogradov–Korobov gives no A-TP kernel closure).* The zero-freeness of $\mathcal{P}_T$ does not imply positive-semidefiniteness of K_{E_ξ} on the zero set relevant to A-TP. It only says that the sampled zero set does not meet $\mathcal{P}_T$.

Proof. Any proposed representation of the arithmetic form by a kernel Gram matrix requires the actual test and form comparison. The lemma removes only the peripheral bands $\mathcal{P}_T$. It does not locate zeros on the real axis, does not put the zero set in a de Branges positivity domain, and does not identify a positive Hilbert-space inner product on the sampled zero components. Therefore no A-TP positivity statement follows from Vinogradov–Korobov alone. □

THEOREM 94.4.4 *(Compact test functions and exact Fourier gaps).* If $f \in C_c^\infty(\mathbb{R})$ and $\widehat{f}$ vanishes on a nonempty open real interval, then $f = 0$. Consequently restricting Fourier support to the off-real peripheral bands $\mathcal{P}_T$ supplies no nonzero test on this carrier.

Proof. Compact support makes $\widehat{f}$ entire. A real interval has an accumulation point in its complex domain, so the identity theorem gives $\widehat{f} \equiv 0$. Fourier inversion then gives $f = 0$. Fourier support is a subset of the real frequency axis; the peripheral complex bands are a zero-free-region statement, not a nonzero Fourier-support class. □

Remark 94.4.5 *(The actual test-domain requirement).* The finite kernel norm in (94.3) is not an arithmetic zero-sum identity without the additional test and form map. Smooth compact tests can have quantitative frequency decay or concentration; they cannot have an exact open Fourier gap unless zero. The peripheral zero-free bands supply neither that map nor a nontrivial compact test with support in those complex bands.

94.5 The Genuine Obstruction: Hermite–Biehler Extension

THEOREM 94.5.1 *(A continuous-form criterion for kernel positivity).* Let H be a specified Hilbert space, b a continuous Hermitian form on H, and $j : \mathscr{T} \to H$ a linear map from a specified arithmetic test space with dense image. Suppose vectors $k_z \in H$, for $z \in \mathbb{C}_+$, satisfy $b(k_z, k_w) = K_{E_\xi}(w, z)$, and their span is dense in H. Then

$$b(jf, jf) \ge 0\ (f \in \mathscr{T}) \quad \Longleftrightarrow \quad K_{E_\xi} \succeq 0 \quad \Longrightarrow \quad \text{RH}.$$

These hypotheses include the continuous realization of the kernel and the actual test map; they do not follow from zero-free bands.

Proof. If b is nonnegative on $j(\mathscr{T})$, continuity and density make it nonnegative on H. Evaluating on finite sums of the k_z proves kernel positivity. Conversely, kernel positivity makes b nonnegative on their span and hence on H, by the second density hypothesis. For the exact derivative datum $E_\xi = \Xi + i\Xi'$, Proposition 79.6.7 gives strict Hermite–Biehler positivity. Its real part Ξ has no nonreal zeros, so RH follows. This argument does not construct H, b, j or identify $b(jf, jf)$ with the Weil form. □

Corollary 94.5.2 *(The full kernel comparison).* Under the Hilbert realization and density hypotheses of Theorem 94.5.1, positivity on the arithmetic test image implies RH. Peripheral zero-free bands do not supply that realization or its positivity.

Proof. Apply the continuous-form criterion and the strict derivative-kernel implication in the theorem. □

Remark 94.5.3 *(The derivative datum and the arithmetic test map).* For the actual candidate $E_\xi = \Xi + i\Xi'$, the derivative-kernel calculation rules out a degenerate positive kernel and gives strict Hermite–Biehler positivity. It requires positivity on every finite kernel matrix in the upper half-plane. An arithmetic application must construct the test map and continuous form of the theorem; real-part coordinates or a zero-free region alone provide neither.

94.6 Three Non-Elementary Constructions Further Analysed

Three candidate reductions of HB$^+$ each fall strictly short, by explicit strength comparison.

Proposition 94.6.1 *(Beurling–Malliavin density does not imply HB$^+$).* The Beurling–Malliavin sampling density argument of Proposition 79.7.3 concerns the horizontal spacing of zeros on $\mathbb{R}$ under RH, not their confinement to the real axis in the z-plane. Beurling–Malliavin density does not imply the Hermite–Biehler inequality for E_ξ.

Proof. BM density is defined on real sequences (Definition 79.7.1). If one instead forms a projected real sequence from a complex zero multiset, that projection loses the vertical information; it is not a definition of Beurling–Malliavin density for complex sets. Projection to real parts cannot detect an off-line zero: two complex multisets with the same real projections have the same projected sequence but can have different imaginary parts. Thus an estimate for a projected real sequence alone does not supply the zero-placement premise or the actual kernel comparison. □

Proposition 94.6.2 *(Selberg–Bombieri zero density does not imply HB$^+$).* The Selberg–Bombieri zero-density estimate $N(\sigma, T) \le T^{\theta(\sigma)}$ for $\sigma \in (1/2, 1)$ with $\theta(\sigma) \to 0$ as $\sigma \to 1$ controls the count of zeros with $\operatorname{Re}\rho \ge \sigma$ in imaginary height T, but $\theta(\sigma) > 0$ for all $\sigma < 1$. It does not preclude individual zeros off the critical line. Thus it does not prove HB$^+$.

Proof. The Selberg zero-density result (Selberg 1946; see Titchmarsh 1951 §9.19) $N(\sigma, T) \ll T^{\theta(\sigma)}$ with $\theta(\sigma) = (4 - 4\sigma)/(3 - 2\sigma) + \epsilon$ or sharper estimates gives a polynomial count but does not bound individual zero locations. A density theorem may leave finitely many off-line zeros, and one such zero already violates HB$^+$. The density-vs-location mismatch precludes the reduction. □

Proposition 94.6.3 *(Connes operator-trace approach reduces to HB$^+$).* The Connes adele-class-space construction (Connes 1999 arXiv:math/9811068; de Branges 1986–1992) has absorption trace equal

to the zero distribution of ζ; critical-line support of this determinant-trace datum is RH. Every operator-monotonicity argument for this support is therefore strictly equivalent to HB^+ and does not constitute an independent reduction below that threshold.

Proof. The Connes trace formula (Connes 1999 Thm. 0.1) identifies the absorption trace on the adele class space $\mathrm{GL}_1(\mathbb{A}_{\mathbb{Q}})/\mathbb{Q}^\times$ with the zero distribution of ζ. Critical-line support of that distribution is equivalent to RH. Any attempt to prove this support by operator monotonicity, semi-boundedness, or normality requires an unconditional proof which is RH-equivalent, not available at the level of the de Branges comparison. See Remark 57.3.10 for the detailed discussion. □

Remark 94.6.4 *(What the separate estimates supply).* Real-sequence completeness, regional zero counts and an operator realization concern distinct data. The bounds stated above do not construct the particular continuous kernel realization or its arithmetic form comparison. They do not impose a limit on stronger arguments that supply those missing maps and estimates.

94.7 de Branges Form of A-TP

Theorem 94.7.1 *(Conditional de Branges form comparison).* Suppose the Hilbert carrier, continuous Hermitian form, test map and two dense spans of Theorem 94.5.1 are supplied. Assume also that its test form $b(jf, jf)$ equals the full arithmetic Weil form on the declared complete test carrier. Then nonnegativity of that Weil form is equivalent to positivity of K_{E_ξ}, and implies RH.

Proof. The assumed form identity and the two density arguments of the continuous-form criterion give the equivalence. The derivative-kernel strictness gives the zero-placement conclusion. The model-space isometry and the finite Gram calculation do not construct the assumed arithmetic form identity. □

Remark 94.7.2 *(The comparison to construct).* The remaining arithmetic step is the actual carrier and form map in the theorem. The peripheral zero-free bands neither supply that map nor prove its positivity. Kernel positivity for the exact derivative datum is a precise sufficient target, not an available premise in an unconditional proof.

Remark 94.7.3 *(relation to the bounded-interval reduction).* The full signed inequality is $A[f] + 4\pi \log \pi \, \|f\|_2^2 \le B[f] + 2\pi R[f]$. The interval $[-t_*, t_*]$ describes the negative part of the auxiliary multiplier g. A de Branges formulation must transport the entire Weil form, including $R[f]$ and the signed archimedean term, through an actual test map. Positivity of a finite kernel Gram matrix by itself establishes no such transport.

Remark 94.7.4 *(de Branges boundary).* The model-space isometry and finite kernel Gram norms are explicit analytic constructions under the stated Hermite–Biehler premise. The arithmetic form identity is the separate input of Theorem 94.7.1; the continuous-form criterion explains precisely how positivity would transfer once that input and its dense realization are constructed.

94.8 Consequences

(i) The de Branges function

$$E_\xi(z) = \xi_{\mathbb{R}}(\tfrac{1}{2} - iz) + \xi'_{\mathbb{R}}(\tfrac{1}{2} - iz)$$

satisfies $(E_\xi + E_\xi^*)/2 = \Xi$, so the non-trivial zeros of ζ are the zeros of the real part A_ξ in the coordinate $s = \frac{1}{2} + iz$ (Definition 94.2.1, Lemma 94.2.2).

(ii) Stirling asymptotics for the completed zeta do not prove Hermite–Biehler positivity for E_ξ; the strict inequality is precisely the HB$^+$ input (Proposition 94.2.4, Corollary 94.2.5).

(iii) Under HB$^+$, multiplication by E_ξ identifies the model space $K(E_\xi^*/E_\xi)$ with $\mathcal{H}(E_\xi)$; the Mellin lift enters this isometry only when its transform lies in that model space (Theorem 94.3.3), giving the finite Gram norm calculation, with arithmetic form transport still required (Corollary 94.3.4).

(iv) The Vinogradov–Korobov zero-free region pulls back to peripheral horizontal bands $\mathcal{P}_T$. It gives no zero-free central strip around the real z-axis (Definition 94.4.1, Lemma 94.4.2).

(v) The peripheral zero-free bands do not imply positive-semidefiniteness of K_{E_ξ} on the zero set. Thus there is no Vinogradov–Korobov A-TP closure (Proposition 94.4.3, Theorem 94.4.4).

(vi) The continuous-form realization with its two dense spans transfers positivity to the full kernel when its arithmetic form map is given, and the exact derivative datum then gives zero placement (Theorem 94.5.1, Corollary 94.5.2).

(vii) The stated real-sequence, regional-density and operator data do not supply the missing arithmetic form comparison (Section 94.6).

(viii) The Mellin model-space isometry gives the analytic norm identity of Theorem 94.3.3. Transporting the signed archimedean contribution or the full Weil form to a kernel Gram form requires the additional arithmetic comparison of Theorem 94.7.1.

Remark 94.8.1 *(Beilinson domain boundary).* Vinogradov–Korobov 1958 gives a peripheral exclusion after the change of variables $s = \frac{1}{2} + iz$. It does not close A-TP. The obstruction is precise: HB$^+$. Every advance beyond this requires either a zero-location theorem strong enough to imply HB$^+$, or an independent proof of HB$^+$ by operator methods. Both remain open in the arithmetic branch.

94.9 Bibliography Anchors

de Branges 1968, *Hilbert Spaces of Entire Functions*. Prentice-Hall. Ch. II–III contain the de Branges structure theorem and the reproducing-kernel formalism used above. The de Branges comparison argument uses de Branges 1968 in its reproducing-kernel analytic role (Ch. III), distinct from the structure-theorem role used by the de Branges spectral comparison.

Lagarias 1999, Acta Arith. 89. *On a positivity property of the Riemann ξ-function.* Explicit construction of the E_ξ candidate ((94.1)) and Hermite–Biehler formulation of RH. §4 is the principal reference.

Conrey–Li 2000, arXiv:math/9812166. *A note on some positivity conditions related to zeta and L-functions.* Explicit counter-examples where the naive Hermite–Biehler structure fails, identifying the location where operator-monotonicity arguments meet an irreducible obstruction.

Titchmarsh 1951, *The Theory of the Riemann Zeta-Function* (Oxford). Ch. V–VI contain the Vinogradov–Korobov zero-free region used in Definition 94.4.1 and Lemma 94.4.2.

Korobov 1958, Uspekhi Matematicheskikh Nauk 13. For sufficiently large $|t|$, the zero-free region is

$$\sigma \geq 1 - \frac{c_0}{(\log|t|)^{2/3}(\log\log|t|)^{1/3}}.$$

Vinogradov 1958, Izvestiya Akad. Nauk SSSR Ser. Mat. 22. Independent proof with sharper constants.

Whittaker–Watson 1927, *A Course of Modern Analysis*. Cambridge, 4th edition. Stirling asymptotic for Γ used in Proposition 94.2.4.

Connes 1999, arXiv:math/9811068. *Trace formula in noncommutative geometry.* Cited for the operator-approach rule-out of Proposition 94.6.3.

The disjointness with constructions 5 A, 5 J, 6 A is per-source. Only de Branges 1968 appears in both the de Branges spectral comparison and the de Branges comparison; it is used in distinct roles (structure theorem vs. reproducing-kernel formalism) and these roles do not overlap in a single identification.

Chapter 95

Gaussian Tests for the Signed Gamma Form

A Gaussian decays fast enough for every term of the explicit formula, although it is not compactly supported. This extension requires exponential control in the logarithmic variable. Ordinary Schwartz continuity cannot control the uncentered critical prime measure.

95.1 An exponential test space

Definition 95.1.1 *(rapid exponential decay).* Let $\mathscr{G}$ consist of all smooth functions f such that

$$p_{a,N,j}(f) = \sup_{t\in\mathbb{R}} e^{a|t|}(1+|t|)^N |f^{(j)}(t)| < \infty \quad (a, N, j \in \mathbb{N}).$$

Give $\mathscr{G}$ the topology of these seminorms. Use $h_f(z) = \int f(t)e^{izt}\,dt$ and $f^*(t) = \overline{f(-t)}$.

Proposition 95.1.2 *(extension of the explicit formula).* The space $\mathscr{G}$ is a convolution algebra with continuous involution. Compact smooth functions are dense in $\mathscr{G}$. The explicit formula, its signed pairing, and both polar moments extend continuously to $\mathscr{G}$. In particular, for $f, g \in \mathscr{G}$,

$$W(f * g^*) = \sum_{\rho} h_f(z_\rho)\overline{h_g(\bar{z}_\rho)} = P(f * g^*) - W_\infty(f * g^*) - W_{\mathrm{fin}}(f * g^*).$$

The zero and prime sums are absolutely convergent. Positivity on all squares in $\mathscr{G}$ is equivalent to positivity on all squares in $\mathscr{A}$.

Proof. The inequality $|t| \le |v| + |t - v|$, together with the corresponding polynomial weight inequality, bounds every convolution seminorm by a weighted supremum seminorm times a weighted integral seminorm. An additional exponential weight bounds that integral. This proves continuity of convolution. Reflection proves continuity of $*$. For a compact smooth cutoff $\chi = 1$ near zero, put $f_R(t) = \chi(t/R)f(t)$. Leibniz's rule and one extra exponential weight give $p_{a,N,j}(f - f_R) \to 0$ for every seminorm.

Integration by parts gives rapid decay of h_f in each fixed horizontal strip, continuously in these seminorms. The zero count then bounds the zero sum. The polar evaluations are continuous. The gamma density has logarithmic growth, so its integral is continuous. Finally, for example,

$$\sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}} |f(\pm \log n)| \le p_{2,0,0}(f) \sum_{n\ge 2} \frac{\log n}{n^{5/2}}.$$

The compact explicit formula therefore passes to the limit f_R. Convolution continuity gives the paired identity and passage of positivity. The reverse positivity implication uses $\mathscr{A} \subset \mathscr{G}$. □

95.2 THE NORMALIZED SCALAR INTEGRAL

LEMMA 95.2.1 *(Gaussian Fourier transform).* For $\lambda > 0$ set $f_\lambda(t) = e^{-\lambda t^2/2}$. Then

$$h_{f_\lambda}(u) = \sqrt{\frac{2\pi}{\lambda}} e^{-u^2/(2\lambda)}, \qquad (f_\lambda * f_\lambda)(t) = \sqrt{\frac{\pi}{\lambda}} e^{-\lambda t^2/4}.$$

Consequently, with $k(u) = \log \pi - \Re \psi_\Gamma(1/4 + iu/2)$,

$$W_\infty(f_\lambda * f_\lambda) = \sqrt{\frac{\pi}{\lambda}}\, H(\lambda), \qquad H(\lambda) = \frac{1}{\sqrt{\pi\lambda}} \int e^{-u^2/\lambda} k(u)\, du.$$

Proof. Completing the square gives both Gaussian integrals. Insert $|h_{f_\lambda}|^2 = (2\pi/\lambda) e^{-u^2/\lambda}$ in the gamma pairing with its coefficient $1/(2\pi)$. □

THEOREM 95.2.2 *(convergent error-function series).* Let $c_n = n + 1/4$ and $\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-v^2}\, dv$. For every $\lambda > 0$,

$$H(\lambda) = k(0) - \sum_{n=0}^{\infty} \left[\frac{1}{c_n} - \frac{2\sqrt{\pi}}{\sqrt{\lambda}} e^{4c_n^2/\lambda} \operatorname{erfc}(2c_n/\sqrt{\lambda}) \right].$$

Each bracket is nonnegative and at most $\lambda/(8c_n^3)$. The series converges uniformly on compact λ-intervals.

Proof. The digamma series gives $k(0) - k(u) = \sum_n (u^2/4)/(c_n(c_n^2 + u^2/4))$. Tonelli's theorem exchanges this nonnegative sum and the Gaussian integral. For $a > 0$,

$$\int_{\mathbb{R}} \frac{e^{-u^2/\lambda}}{u^2 + a^2}\, du = \sqrt{\pi} \int_0^\infty \frac{e^{-a^2 r}}{\sqrt{r + 1/\lambda}}\, dr = \frac{\pi}{a} e^{a^2/\lambda} \operatorname{erfc}(a/\sqrt{\lambda}).$$

The first equality follows by integrating $\int_0^\infty e^{-r(u^2+a^2)}\, dr$ in either order. The substitution $v = a\sqrt{r + 1/\lambda}$ gives the second. Set $a = 2c_n$ and use $\int e^{-u^2/\lambda} u^2\, du = \sqrt{\pi}\lambda^{3/2}/2$. The estimate on each bracket follows by replacing $c_n^2 + u^2/4$ by c_n^2. It proves uniform convergence. □

THEOREM 95.2.3 *(the exact Gaussian sign range).* The continuous function H is strictly decreasing on $(0, \infty)$, with limits $k(0) > 0$ at zero and $-\infty$ at infinity. There is a unique $\lambda_c > 0$ with $H(\lambda_c) = 0$, and

$$W_\infty(f_\lambda * f_\lambda) \geq 0 \quad \Longleftrightarrow \quad 0 < \lambda \leq \lambda_c.$$

Proof. After $u = \sqrt{\lambda} v$, $H(\lambda) = \pi^{-1/2} \int e^{-v^2} k(\sqrt{\lambda} v)\, dv$. Strict radial decrease of k gives strict decrease of this integral. Logarithmic growth gives a common integrable majorant on each compact λ-interval and on $0 < \lambda \leq 1$. Thus H is continuous and tends to $k(0)$ at zero. For each $v \neq 0$, the nonnegative quantity $k(0) - k(\sqrt{\lambda} v)$ increases to infinity. Monotone convergence gives the limit at infinity. The intermediate value theorem completes the proof. □

95.3 THE ARITHMETIC INEQUALITY ON GAUSSIAN SQUARES

For these Gaussian tests the polar and prime contributions are

$$P(f_\lambda * f_\lambda) = \frac{4\pi}{\lambda} e^{1/(4\lambda)}, \qquad W_{\mathrm{fin}}(f_\lambda * f_\lambda) = 2\sqrt{\frac{\pi}{\lambda}} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-\lambda(\log n)^2/4}.$$

These expressions converge for every $\lambda > 0$ and give the complete value of $W(f_\lambda * f_\lambda)$ after subtracting the prime and gamma terms from the polar term. The gamma sign theorem alone does not determine that value.

Proposition 95.3.1 *(a single Gaussian parameter does not test a quadratic form).* For a Hermitian form, nonnegative values on a spanning collection of vectors do not imply nonnegative values on its linear span. In particular, Gaussian cross terms require their own comparison.

Proof. The Hermitian matrix $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ is positive on each coordinate vector but has value -2 on $(1, -1)$. More generally, $q(f+g, f+g) = q(f,f) + q(g,g) + 2\Re q(f,g)$. The signed arithmetic representation determines these cross terms on $\mathscr{G}$, but positivity of individual Gaussian squares does not bound them. □

95.4 The auxiliary digamma integral and Fourier normalization

The signed kernel k and the auxiliary kernel $g(u) = \Re\psi_{\mathrm{T}}(1/4 + iu/2) + \log\pi$ satisfy $k(u) + g(u) = 2\log\pi$. To compare scalar calculations, use a separate parameter $L > 0$ and define

$$I(L) = \int_{\mathbb{R}} e^{-2\pi u^2/L} g(u)\, du, \qquad m_L = \sqrt{L/2}.$$

Direct substitution in the definition of H gives

$$\frac{I(L)}{m_L} = 2\log\pi - H\!\left(\frac{L}{2\pi}\right). \tag{95.1}$$

Thus the zero of I is a level $H = 2\log\pi$, distinct from the signed threshold $H = 0$.

Proposition 95.4.1 *(the auxiliary error-function series).* With $c_n = n + 1/4$, put

$$\begin{aligned} J_n(L) &= \int_{\mathbb{R}} e^{-2\pi u^2/L} \frac{c_n}{c_n^2 + u^2/4}\, du, \\ &= 2\pi e^{8\pi c_n^2/L} \operatorname{erfc}\!\left(2c_n\sqrt{2\pi/L}\right). \end{aligned}$$

Then the paired series

$$I(L) = m_L(-\gamma_E + \log\pi) + \sum_{n\ge 0}\left(\frac{m_L}{n+1} - J_n(L)\right) \tag{95.2}$$

converges absolutely. The accelerated series is

$$\frac{I(L)}{m_L} = g(0) + \sum_{n\ge 0}\left(\frac{1}{c_n} - \frac{J_n(L)}{m_L}\right), \qquad 0 \le \frac{1}{c_n} - \frac{J_n(L)}{m_L} \le \frac{L}{16\pi c_n^3}.$$

In particular its tail from $n = N \ge 1$ is at most $L/[32\pi(N - 3/4)^2]$.

Proof. The digamma series is

$$g(u) = -\gamma_E + \log\pi + \sum_{n\ge 0}\left(\frac{1}{n+1} - \frac{c_n}{c_n^2 + u^2/4}\right).$$

Subtract its value at zero. The resulting nonnegative summands permit Tonelli's theorem. The error-function integral in Theorem 95.2.2, with $\lambda = L/(2\pi)$, gives J_n and the accelerated bound. The difference $1/(n+1) - 1/c_n$ is absolutely summable, so regrouping gives (95.2). Finally $\sum_{n=N}^{\infty}(n+1/4)^{-3} \le \int_{N-1}^{\infty}(x+1/4)^{-3}dx$ proves the tail estimate. The unaccelerated paired summands are $O_L(n^{-2})$, since $J_n(L) = m_L/c_n + O_L(c_n^{-3})$. □

Remark 95.4.2 *(the real and complex Gaussian integrals).* The integral used above, for $a, \alpha > 0$, is

$$\int_{\mathbb{R}} \frac{e^{-\alpha u^2}}{a^2+u^2}\,du = \frac{\pi}{a}e^{\alpha a^2}\operatorname{erfc}(a\sqrt{\alpha}).$$

Also

$$\int_{\mathbb{R}} \frac{e^{-\alpha u^2}}{a-iu}\,du = \pi e^{\alpha a^2}\operatorname{erfc}(a\sqrt{\alpha}).$$

Indeed $(a-iu)^{-1} = (a+iu)/(a^2+u^2)$; the imaginary term is odd and integrates to zero. The real integral was proved by a Laplace integral in Theorem 95.2.2.

PROPOSITION 95.4.3 *(the auxiliary limits and threshold).* As $L \downarrow 0$,

$$I(L) = m_L\left(g(0) + \frac{L}{8\pi}g''(0) + O(L^2)\right), \qquad g(0) = \log\pi - \gamma_E - \frac{\pi}{2} - 3\log 2 < 0.$$

As $L \to \infty$, $I(L) \sim \frac{1}{2}m_L \log L$. The normalized integral $I(L)/m_L$ is strictly increasing and has a unique zero $L_g > 0$, characterized by $H(L_g/(2\pi)) = 2\log\pi$.

Proof. Write $u = \sqrt{L/(2\pi)}v$ and use the probability density $\pi^{-1/2}e^{-v^2}$. Its second moment is $1/2$. The even Taylor expansion of g, with fourth-order remainder near zero and logarithmic growth away from zero, gives the narrow expansion. The Gaussian tail outside a fixed neighborhood is smaller than every power of L. The value at zero follows from the reflection and duplication identities for ψ_Γ. For the wide limit, $g(u) = \log|u| + O(1)$ for $|u| \ge 1$. Split the scaled integral at $|v| = \sqrt{2\pi/L}$. On the complement, $\log|v|$ is integrable against the Gaussian. On the small interval, boundedness of g gives an $O(L^{-1/2}\log L)$ error after subtracting $\frac{1}{2}\log L$. Thus $I(L)/m_L = \frac{1}{2}\log L + O(1)$. Finally (95.1) and the strict decrease of H give monotonicity and uniqueness. □

The time Gaussian $F_L(x) = e^{-\pi L x^2}$ has the Fourier transform

$$\widehat{F}_{L,2\pi}(u) = \int F_L(x)e^{-2\pi i u x}\,dx = L^{-1/2}e^{-\pi u^2/L}.$$

Since $F_L = f_{2\pi L}$, its signed pairing is

$$W_\infty(F_L * F_L) = \frac{1}{\sqrt{2L}}H(2\pi L).$$

More generally, $\mathcal{F}_+\phi(2\pi u) = \widehat{\phi}_{2\pi}(-u)$, so evenness of k and a change of variables give

$$W_\infty(\phi) = \int_{\mathbb{R}} \widehat{\phi}_{2\pi}(u)k(2\pi u)\,du.$$

This conversion includes both the kernel argument and the measure factor. The parameter in the auxiliary identity (95.1) is therefore not the physical Gaussian parameter in this last pairing.

95.5 Cross terms and the positive Gaussian cone

Proposition 95.5.1 *(the signed gamma form on positive Gaussian sums).* For $\lambda, \mu > 0$, put $\nu = 2\lambda\mu/(\lambda+\mu)$. Then

$$(f_\lambda * f_\mu)(t) = \sqrt{\frac{2\pi}{\lambda+\mu}}\, e^{-\lambda\mu t^2/[2(\lambda+\mu)]},$$

$$W_\infty(f_\lambda * f_\mu) = \sqrt{\frac{2\pi}{\lambda+\mu}}\, H(\nu).$$

Consequently, on the cone

$$C^+_{\mathrm{Gauss}} = \left\{ \sum_{j=1}^{N} a_j f_{\lambda_j} : N \geq 1,\ a_j \geq 0,\ 0 < \lambda_j \leq \lambda_c \right\},$$

one has $W_\infty(f * f^*) \geq 0$.

Proof. Complete the square in the convolution integral. The Fourier product is $(2\pi/\sqrt{\lambda\mu})e^{-u^2/\nu}$. Integrating it against $k(u)/(2\pi)$ gives the displayed pairing. For $\lambda, \mu \leq \lambda_c$, their harmonic mean satisfies $\nu \leq \lambda_c$, so every cross pairing is nonnegative. Expand the finite sum with coefficients $a_i a_j \geq 0$. □

Proposition 95.5.2 *(continuity and carrier boundaries).* The signed gamma distribution extends continuously to the Schwartz space. Its quadratic pairing is continuous there as well. The linear span of the centered Gaussians lies in the closed even subspace and is not dense in the full Schwartz space. Every nonzero element of C^+_{Gauss} is positive on all of $\mathbb{R}$, so it is not compactly supported. Its Fourier transform is not in $\mathcal{F}_+\mathscr{A}$. Both compact tests and these Gaussians belong to $\mathscr{G}$.

Proof. Logarithmic growth of k bounds its Fourier pairing by a Schwartz seminorm. The Fourier transform, convolution and involution are continuous on Schwartz space. Reflection is continuous, so its fixed subspace is closed and excludes nonzero odd Schwartz functions. Each Gaussian is strictly positive, which proves the support claim for nonzero positive combinations. Fourier injectivity would force a compact preimage to equal this noncompact function. Membership in $\mathscr{G}$ follows directly from the defining seminorms. Continuity of the complete arithmetic functional uses that exponential topology, as proved above. □

The positive Gaussian cone controls the signed gamma term only. The full Weil form still subtracts this term and the prime term from the polar term. Neither positivity of the gamma cone nor its comparison with a compact or automorphic test class proves the full arithmetic inequality.

95.6 The auxiliary Schwartz form and its Gaussian cone

The auxiliary integral also defines a quadratic form, but it is not the signed gamma form. On the complex Schwartz space use the convention $\widehat{f}(u) = \int_{\mathbb{R}} f(x) e^{-2\pi i u x}\, dx$ and set

$$\mathcal{T}_g(\phi) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{\phi}(u)\, g(u)\, du, \qquad Q_g(f,h) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(u)\overline{\widehat{h}(u)}\, g(u)\, du.$$

Thus $\mathcal{T}_g$ is complex linear and Q_g is Hermitian, linear in its first argument, with $Q_g(f,h) = \mathcal{T}_g(f * h^*)$. The value of $\mathcal{T}_g$ is real when its Fourier argument is real, in particular for real even tests. No real-valuedness is claimed for arbitrary complex tests.

PROPOSITION 95.6.1 *(continuity and the two normalizations).* The functional $\mathcal{T}_g$ and the sesquilinear form Q_g are continuous on Schwartz space. The function I is smooth on $(0,\infty)$. With $\langle f,h\rangle_2=\int f\bar h$, one has

$$\frac{1}{2\pi}\int \widehat{f}(u)\overline{\widehat{h}(u)}k(u)\,du=\frac{\log\pi}{\pi}\langle f,h\rangle_2-Q_g(f,h),$$

$$W_\infty(f*h^*)=\int \widehat{f}(u)\overline{\widehat{h}(u)}k(2\pi u)\,du.$$

The second equality denotes the signed gamma extension; it does not extend the full prime functional to arbitrary Schwartz tests.

Proof. The function g is smooth at zero and has at most logarithmic growth. Bound its integral by a weighted supremum seminorm of $\widehat{\phi}$; for the form use the product of such seminorms of $\widehat{f}$ and $\widehat{h}$. The integrable majorant can be taken proportional to $(1+|u|)^{-2}$. Fourier transform is continuous on Schwartz space. On each compact interval of positive L, every L-derivative of $e^{-2\pi u^2/L}$ is bounded by a polynomial in u times a fixed Gaussian. Multiplication by $g(u)$ is integrable against this bound, so differentiation under the integral proves $I\in C^\infty$. The first identity uses $k+g=2\log\pi$ and Plancherel in the 2π convention. The second uses $h_f(2\pi u)=\widehat{f}(-u)$ and the change of measure $du_{\mathrm{old}}=2\pi\,du$, followed by evenness of k. □

THEOREM 95.6.2 *(auxiliary Gaussian pairings and the exact cone).* For $F_L(x)=e^{-\pi Lx^2}$, $L,M>0$, let $H_{LM}=2LM/(L+M)$. Then

$$Q_g(F_L,F_M)=\frac{I(H_{LM})}{2\pi\sqrt{LM}},\qquad Q_g(F_L,F_L)=\frac{I(L)}{2\pi L}.$$

Consequently

$$\{L>0:Q_g(F_L,F_L)\ge 0\}=[L_g,\infty),\qquad Q_g(F_L,F_L)=0\iff L=L_g,$$

where $H(L_g/(2\pi))=2\log\pi$. On the nonempty cone

$$C_g^+=\left\{\sum_{j=1}^{N}a_jF_{L_j}:a_j\ge 0,\ L_j\ge L_g,\ N\ge 1\right\}$$

one has $Q_g(f,f)\ge 0$. A nonzero element has zero value precisely when every parameter with nonzero coefficient equals L_g. The same nonnegativity holds on the closure of this cone in Schwartz topology.

Proof. The product of the two Fourier Gaussians is $(LM)^{-1/2}e^{-\pi u^2(1/L+1/M)}$, which gives the first identity. The sign and equality statements follow from the strict increase of $I(L)/m_L$ and its unique zero proved above. If $L,M\ge L_g$, then $H_{LM}^{-1}=(L^{-1}+M^{-1})/2\le L_g^{-1}$. Thus every cross pairing is nonnegative. Expanding the square of a finite sum gives

$$Q_g(f,f)=\sum_{i,j}a_ia_j\frac{I(H_{L_iL_j})}{2\pi\sqrt{L_iL_j}}\ge 0.$$

If some $a_j>0$ has $L_j>L_g$, its diagonal term is strictly positive and none of the other terms can cancel it. Otherwise all active parameters equal L_g and every term vanishes. The closure statement is continuity of Q_g. Nonemptiness follows from any F_L with $L\ge L_g$. □

Example 95.6.3 *(two auxiliary Gaussian scales).* For real a, b,

$$Q_g(aF_L + bF_M, aF_L + bF_M) = \frac{1}{2\pi}\left(\frac{a^2 I(L)}{L} + \frac{2ab I(H_{LM})}{\sqrt{LM}} + \frac{b^2 I(M)}{M}\right).$$

Choose $L_g/2 < L < L_g$ and $M > L_g L/(2L - L_g)$. Then $M > L_g$ and $H_{LM} > L_g$, while $I(L) < 0$. Taking $a = 1$ and $b > 0$ sufficiently small gives a negative value by continuity in b. This example mixes one negative and one positive Gaussian; it does not contradict the positive-cone theorem, whose parameters all lie above L_g. Nor does that theorem assert positivity for every complex linear combination of its generators.

High-frequency dominance on a specified carrier

Let $t_g > 0$ be the unique zero of g: existence follows from $g(0) < 0$, logarithmic growth and strict increase on $(0, \infty)$. Indeed the differentiated Hurwitz series has positive summands $c_n u/[2(c_n^2 + u^2/4)^2]$ for $u > 0$, uniformly convergent on compact intervals. For $f \in \mathcal{S}(\mathbb{R})$ define

$$A_g(f) = \int_{|u| \le t_g} |\widehat{f}(u)|^2 (-g(u))\, du, \qquad B_g(f) = \int_{|u| > t_g} |\widehat{f}(u)|^2 g(u)\, du.$$

The high-frequency dominance condition, in this Fourier convention, is

$$2t_g |g(0)| \sup_{|u| \le t_g} |\widehat{f}(u)|^2 \le B_g(f).$$

It implies $A_g(f) \le B_g(f)$ and hence $Q_g(f, f) = (B_g(f) - A_g(f))/(2\pi) \ge 0$. This is the auxiliary inequality associated with the condition in Chapter 82.4.1; changing the Fourier convention requires its explicit frequency rescaling, and does not turn g into k.

For a specified restricted compact test space $E \subset \mathscr{A}$, put $E_{\mathrm{HFD}} = \{f \in E : f \text{ satisfies this condition}\}$. Every nonzero element of C_g^+ is positive on all of $\mathbb{R}$, so $C_g^+ \cap E = \{0\}$; both sets contain zero. The assertion that E_{HFD} has a nonzero element requires a construction within the actual restrictions defining E. It is a separate question whether a chosen automorphic subspace admits such tests and a norm comparison. A noncompact Gaussian is not an example in that compact space.

Other positive classes and the full inequality. One may ask for other positive subspaces of Q_g, including nonconic combinations controlled by a de Branges or Connes–Deninger construction, compact smooth tests, or a spectral-density comparison with the Gaussian moments of g. No assertion that these exhaust all possible methods is made. The cone is a proper subset of its complex linear span: $-F_L$ belongs to that span for $L \ge L_g$ but cannot be a nonnegative sum of strictly positive Gaussians. For comparison, if functions φ_j are positive definite, then $\sum_j c_j \varphi_j$ is positive definite for $c_j \ge 0$: each of its finite Gram forms is the same nonnegative combination of the Gram forms of the φ_j. This elementary fact does not identify positive-definite functions with vectors on which Q_g is nonnegative. Closure of a positive cone follows from quadratic continuity; linear density of positive individual vectors alone does not control cross terms.

The signed cone and C_g^+ prove different gamma inequalities. Both lie in $\mathscr{G}$, where the full arithmetic identity is defined. For a specified $E \subset \mathscr{G}$ and $S \subset E$, positivity of the full Weil form on E is equivalent to positivity on $E \setminus S$ only after positivity on S is established. This is the elementary partition of the quantifier over E. Auxiliary positivity on S does not supply that premise, since the polar and prime terms remain. RH equivalence then requires a proved comparison of E with the full compact test criterion.

95.7 Bibliography

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Chapter 96

Archimedean Tate Kernel Positivity: the Integral Energy Sub-Class

The digamma auxiliary form has a negative spectral core and a positive tail. An integral-energy estimate compares their masses and proves nonnegativity on an explicit class of compact smooth tests. Moment-preserving differential operators supply tests in this class and tests outside both the integral and supremum sufficient classes. The signed gamma form differs from this auxiliary form by a scalar multiple of the L^2 pairing. The complete Weil inequality also retains its prime and polar contributions.

96.1 Kernel and Integral Energy

The integral-energy estimate concerns an auxiliary digamma form. Here $\psi_\Gamma = \Gamma'/\Gamma$, and $\widehat{f} = h_f$ in this chapter. Use the angular Fourier transform $h_f(u) = \int_{\mathbb{R}} f(x)e^{iux}\,dx$ and set

$$g(u) = \Re\psi_\Gamma(1/4 + iu/2) + \log\pi, \qquad k(u) = 2\log\pi - g(u),$$

$$G_g(f,h) = \frac{1}{2\pi}\int h_f(u)\overline{h_h(u)}g(u)\,du.$$

The signed gamma form of the explicit formula is therefore

$$W_\infty(f * h^*) = 2\log\pi\langle f,h\rangle_2 - G_g(f,h).$$

This identity follows from Plancherel. It retains the angular normalization and differs from the cyclic normalization used for Q_g in Chapter 95.

Domain

Let $\mathscr{A} = C_c^\infty(\mathbb{R};\mathbb{C})$ with its inductive-limit topology. Its real even subspace is denoted by E. For $f \in E$ set

$$M_f(s) = \int f(x)e^{sx}\,dx, \qquad L(f) = \left(\int f, \int f\cosh x, \int xf(x)\sinh x\,dx\right).$$

The class $\mathscr{A}_{13} = \ker L \subset E$ has precisely the reality and double-Mellin-zero conditions at 0, 1. Compact support makes M_f entire and rapidly decreasing on every closed vertical strip. No weight-dependent strip restriction remains. These conditions impose no modularity on a coefficient multiplier.

Lemma 96.1.1 *(closed moment space and moment correction).* The space $\mathscr{A}_{13}$ is closed of real codimension three in E. There is a continuous real linear projection $\Pi : E \to \mathscr{A}_{13}$ of the form $\Pi = 1 - RL$, where $R : \mathbb{R}^3 \to E$ has compactly supported image and $LR = 1$.

Proof. Integration against each smooth weight is continuous on every fixed-support test space, hence on the inductive limit. Take normalized even mollifiers concentrated at 0, $\{\pm1\}$, and $\{\pm2\}$. Their moment columns tend to the columns obtained by evaluating 1, $\cosh x$, $x\sinh x$ at 0, 1, 2. The limiting

determinant is $2\sinh(1)(\cosh(1)-1)^2 > 0$. For a sufficiently small fixed width the moment matrix C is invertible. If the three mollifiers are b_0, b_1, b_2, define $Rv = \sum_{j=0}^{2} b_j (C^{-1}v)_j$. Then $LR = 1$, $\Pi^2 = \Pi$, and $L\Pi = 0$. The image of R has fixed compact support, so R and Π are continuous. Surjectivity of L gives the codimension. □

Write t_* for the positive zero of g, and $M = -g(0)$. For nonzero $f \in \mathscr{A}_{13}$ define the core and tail energies

$$A_g[f] = \int_{|u|<t_*} |h_f(u)|^2 (-g(u))\,du, \qquad B_g[f] = \int_{|u|>t_*} |h_f(u)|^2 g(u)\,du.$$

Thus $G_g(f,f) = (B_g[f] - A_g[f])/(2\pi)$. The supremum sufficient condition is

$$2t_* M \sup_{|u|<t_*} |h_f(u)|^2 \le B_g[f].$$

The integral-energy condition replaces this supremum by core mass. Both conditions concern G_g. The full arithmetic form remains

$$W(f * f^*) = P(f * f^*) - 2\log\pi \|f\|_2^2 + G_g(f,f) - W_{\mathrm{fin}}(f * f^*).$$

In particular, neither moment correction nor positivity of G_g proves the full Weil inequality.

For the energy comparison fix $\tau_0 > t_*$ and a continuous nondecreasing function $\ell : [\tau_0, \infty) \to (0, \infty)$ satisfying $\ell(T) \le g(T)$ and $\ell(T) = O(1 + \log(2+T))$. The unconditional choice $\ell = g|_{[\tau_0,\infty)}$ is available. Indeed, the Hurwitz series

$$g(u) = g(0) + \sum_{n\ge 0} \frac{u^2/4}{(n+1/4)((n+1/4)^2 + u^2/4)}$$

is strictly increasing for $u > 0$, and the digamma asymptotic gives logarithmic growth. Since $g(0) < 0$ and $g(u) \to \infty$, there is exactly one $t_* > 0$. Other choices of ℓ require the displayed lower-bound property. The proposed Stirling expression is one specific lower-bound question, separate from this energy theorem.

Classical references

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96.2 A proposed Stirling specialization of the lower weight

Stirling Series for the Digamma and Sign Control

Definition 96.2.1 (the digamma remainder). For $\Re z > 0$ and $N \geq 0$, define $R_N(z)$ by

$$\psi(z) = \log z - \frac{1}{2z} - \sum_{j=1}^{N} \frac{B_{2j}}{2j z^{2j}} + R_N(z). \tag{96.1}$$

Here log is the principal branch and B_{2j} are Bernoulli numbers, with $B_2 = 1/6$, $B_4 = -1/30$ and $B_6 = 1/42$. The proposed sectorial bound for this derivative remainder is

$$|R_N(z)| \leq \frac{|B_{2N+2}|}{(2N+2)|z|^{2N+2}} \sec^{2N+2}(\arg z/2). \tag{96.2}$$

Using this bound requires a proof for the digamma remainder with these constants. A remainder estimate for $\log\Gamma$ cannot simply be differentiated as an inequality.

The asymptotic expansion and remainder problem are treated in Whittaker–Watson 1927, §12.33, and Olver, *Asymptotics and Special Functions*, Ch. 8. The energy theorem below uses $\ell \leq g$ and does not require (96.2).

Lemma 96.2.2 *(real-part Stirling at $z = 1/4 + it/2$).* . For $t \in \mathbb{R}$ set $z(t) := 1/4 + it/2$. Then

$$\begin{aligned} \operatorname{Re} \log z(t) &= \tfrac{1}{2}\log(1/16 + t^2/4) = \tfrac{1}{2}\log\bigl((1+4t^2)/16\bigr), \\ \operatorname{Re}\frac{-1}{2z(t)} &= -\frac{1}{8|z(t)|^2} = -\frac{2}{1+4t^2}. \end{aligned}$$

The one-step truncation of (96.1) (with $N = 0$, no Bernoulli terms) gives

$$\operatorname{Re}\psi(z(t)) = \tfrac{1}{2}\log\bigl((1+4t^2)/16\bigr) - \frac{2}{1+4t^2} + \operatorname{Re} R_0(z(t)).$$

Proof. $|z(t)|^2 = (1/4)^2 + (t/2)^2 = 1/16 + t^2/4$. Hence $\log|z| = (1/2)\log(1/16 + t^2/4)$. For $1/z$: $\operatorname{Re}(1/z) = \operatorname{Re} z/|z|^2 = (1/4)/|z|^2$, so $\operatorname{Re}(-1/(2z)) = -(1/8)/|z|^2 = -2/(1+4t^2)$. The identity follows by direct substitution in (96.1). □

The Lower Bound g_L

Proposition 96.2.3 *(conditional Stirling lower weight).* Define

$$g_L(t) = \tfrac{1}{2}\log((1+4t^2)/16) + \log\pi - \frac{2}{1+4t^2},$$

and let $t_L^* > 0$ be its unique positive zero. Suppose the fourth-order bound proposed in Chapter 92.3.3, Chapter 92.3.2 holds on $|t| \geq t_L^*$ in the precise form

$$g(t) \geq g_L(t) + \frac{20u - 72}{15u^2}, \qquad u = 1 + 4t^2.$$

Then $g(t) > g_L(t)$ on that tail. Consequently $\ell = g_L|_{[\tau_0,\infty)}$ is an admissible positive lower weight for every $\tau_0 > t_L^*$.

Proof. As a function of $u \geq 1$, g_L has derivative $1/(2u) + 2/u^2 > 0$, is negative at one and tends to infinity. At $u = 18/5$, use $\pi^2 < 10$ to obtain $g_L < \log(3/2) - 5/9 < 1/2 - 5/9 < 0$. Hence the zero has $u > 18/5$, and the assumed correction $(20u - 72)/(15u^2)$ is strictly positive on the stated tail. Monotonicity and logarithmic growth give the properties of ℓ. □

Remark 96.2.4 *(the quantitative remainder question).* The first Bernoulli correction is

$$-\frac{1}{12}\Re(z(t)^{-2}) = \frac{4t^2 - 1}{192(1/16 + t^2/4)^2} = \frac{1}{3t^2} + O(t^{-4}).$$

A uniform estimate for $g - g_L$ on a finite interval requires a remainder bound in addition to this leading term. The conditional proposition states the exact lower estimate needed from the fourth-order calculation. The unconditional choice $\ell = g$ on a positive tail avoids this separate remainder question.

Corollary 96.2.5 *(the lower-weight zero).* The function g_L has exactly one positive zero t_L^*, and is nonnegative for $|t| \geq t_L^*$. Under the fourth-order hypothesis of the preceding proposition, $t_* < t_L^*$.

Proof. Existence and uniqueness follow from the derivative calculation. Under that hypothesis $g(t_L^*) > g_L(t_L^*) = 0$. Strict monotonicity of g on the positive axis gives $t_* < t_L^*$. □

Sign-domain comparison

Under the fourth-order lower-bound hypothesis, $g < 0$ on $|t| < t_*$, while $g_L < 0$ on $|t| < t_L^*$. On $t_* \leq |t| < t_L^*$, $g \geq 0 > g_L$. On $|t| \geq t_L^*$, $g > g_L \geq 0$. These specialized comparisons are conditional on that bound. For the energy argument, any positive nondecreasing lower weight $\ell \leq g$ on $[\tau_0, \infty)$ suffices.

96.3 The integral-energy class

Definition 96.3.1 *(spectral fractions).* For any nonzero $f \in L^2(\mathbb{R})$, let h_f denote its angular Fourier transform in the Plancherel sense. For $T \geq \tau_0 > t_*$, set

$$\alpha(f) = \frac{\int_{|u|<t_*} |h_f(u)|^2 du}{\|h_f\|_2^2},$$

$$\beta(f,T) = \frac{\int_{|u|>T} |h_f(u)|^2 du}{\|h_f\|_2^2},$$

$$\gamma(f,T) = 1 - \alpha(f) - \beta(f,T).$$

All three fractions are nonnegative and sum to one.

Definition 96.3.2 *(integral and supremum classes).* Use the lower weight ℓ specified above. Put

$$\mathrm{IE}(T) = \{f \in \mathscr{A}_{13} \setminus \{0\} : M\alpha(f) \leq \ell(T)\beta(f,T)\}, \qquad T \geq \tau_0. \tag{96.3}$$

Write HFD for the nonzero $f \in \mathscr{A}_{13}$ satisfying

$$2t_* M \sup_{|u|<t_*} |h_f(u)|^2 \leq B_g[f].$$

The notations $\mathrm{PW}_F^{\mathrm{hol,IE}}(T)$ and $\mathrm{PW}_F^{\mathrm{hol,HFD}}$ in the energy formulas refer to these compact moment classes, without an automorphy assertion.

Remark 96.3.3 *(dependence on the threshold).* Increasing T increases $\ell(T)$ and decreases $\beta(f,T)$. Their product need not be monotone. The threshold is therefore part of the sufficient condition.

Unconditional G_g-Positivity on $\mathrm{PW}_F^{\mathrm{hol,IE}}$

Theorem 96.3.4 *(integral-energy bound).* For a nonzero $f \in \mathscr{A}_{13}$, suppose $T_1 \geq \tau_0$ and $M\alpha(f) \leq \ell(T_1)\beta(f, T_1)$. Then

$$G_g(f, f) \geq 0, \qquad W_\infty(f * f^*) \leq 2\log\pi\|f\|_2^2.$$

The full Weil inequality holds exactly when

$$G_g(f, f) \geq 2\log\pi\|f\|_2^2 + W_{\mathrm{fin}}(f * f^*) - P(f * f^*).$$

Proof. On the core, $-g \leq M$, so $A_g[f] \leq M\alpha(f)\|h_f\|_2^2$. On the middle band $g \geq 0$, while on the far tail $g(u) \geq \ell(|u|) \geq \ell(T_1) \geq 0$. Consequently $B_g[f] \geq \ell(T_1)\beta(f, T_1)\|h_f\|_2^2$. The assumed inequality gives $B_g[f] \geq A_g[f]$. The signed gamma identity proves the second bound. Substitution into the complete explicit formula proves the last equivalence. □

Strict Enlargement over the bounded-interval reduction HFD

Proposition 96.3.5 *(strict extension of the supremum criterion).* On the nonzero tests in $\mathscr{A}_{13}$, the union of the supremum class and the integral-energy classes strictly contains the supremum class.

Proof. Fix a nonzero real even $\varphi \in C_c^\infty(\mathbb{R})$. Put $D = d/dx$, $\mathcal{L} = -D^2(D^2 - 1)^2$, and $p(u) = u^2(1 + u^2)^2$. Integration by parts gives

$$M_{\mathcal{L}v}(s) = -s^2(s^2 - 1)^2 M_v(s), \qquad h_{\mathcal{L}v}(u) = p(u)h_v(u).$$

Thus $\mathcal{L}v \in \mathscr{A}_{13}$ for every real even compact test v. For $A \geq 1$ let $\varphi_A(x) = A^{-1/2}\varphi(x/A)$. Choose $T_1 > \tau_0$, $R > T_1$, and set

$$f_A = A^2\mathcal{L}\varphi_A + c\mathcal{L}(\cos(Rx)\varphi_A), \qquad c > 0.$$

Write $H = h_\varphi$, $C_0 = \int |H(v)|^2 dv$, and $C_2 = \int v^4|H(v)|^2 dv > 0$. The three packets have centers $0, R, -R$ and width A^{-1}. At the core center, with $v = Au$,

$$h_{A^2\mathcal{L}\varphi_A}(v/A) = A^{1/2}v^2(1 + v^2/A^2)^2 H(v).$$

At each tail center the leading profile is $cA^{1/2}p(R)H(v)/2$. Schwartz decay and the substitution $v = A(u - u_0)$ give

$$\int_{|u|<t_*} |h_{f_A}|^2 \to C_2, \qquad \int_{|u|>T_1} |h_{f_A}|^2 \to D_R := \frac{c^2 p(R)^2}{2} C_0, \qquad B_g[f_A] \to g(R)D_R.$$

For justification, each off-center factor has argument differing by AR or $2AR$. Arbitrarily high Schwartz bounds make every cross integral tend to zero, also with the weight $1 + \log(2 + |u|)$. The same bounds dominate each centered integral after substitution. Choose c so that $MC_2 < \ell(T_1)D_R$. The strict integral-energy inequality then holds for all large A. Since $v^2H(v)$ is not identically zero, the core supremum grows at least linearly in A. The quantity $B_g[f_A]$ stays bounded. The supremum criterion therefore fails for all large A. Every f_A satisfies the moments exactly. □

Remark 96.3.6 *(which profile IE captures and HFD misses).* The HFD class is constrained by pointwise control of $\widehat{f}$ over the core; the IE class is constrained by integrated L^2 control and by a lower bound on the weighted tail energy. A narrow core peak with distributed tail may lie in IE and fail HFD. A tail concentrated near t_* may lie in HFD and fail IE. The two conditions are transverse sufficient criteria.

96.4 Complement of $\mathrm{PW}_F^{\mathrm{hol,IE}}$

Complement Structure

Theorem 96.4.1 *(IE residual as concentration inequality).* . $f \in \mathscr{A}_{13} \setminus \{0\} \setminus \bigcup_{T_1} \mathrm{PW}_F^{\mathrm{hol,IE}}(T_1)$ if and only if for every $T_1 \geq \tau_0$,

$$\ell(T_1)\beta(f, T_1) \;<\; M\alpha(f). \tag{96.4}$$

In particular,

$$\sup_{T_1 \geq \tau_0} \ell(T_1)\beta(f, T_1) \leq M\alpha(f).$$

Thus the residual is not the complement of high frequency; it is the locus where the logarithmic weighted tail bound never exceeds the core term.

Proof. (96.4) is (96.3) reversed. For residual f the reversed inequality holds at every T_1; conversely the reversed inequality at every T_1 excludes membership in every IE class. Taking the supremum gives the displayed necessary bound. No decay statement stronger than this weighted-tail bound follows without an independent regularity estimate on f. □

Remark 96.4.2 *(which test functions are in the residual).* Residual test functions have too little logarithmically weighted far-tail mass relative to their core mass. Low-frequency approximate identities give model examples after the weighted tail is made uniformly small. Individual Li approximants require a separate spectral comparison; no membership assertion for them is made here.

Prolate-Spheroidal Characterization

Theorem 96.4.3 *(two-scale prolate sufficient criterion).* . Fix Paley–Wiener exponential type T for h_f, i.e., $\mathrm{supp}(f) \subset [-T, T]$. Let $\{\lambda_k(T, \Omega)\}_{k=0}^{\infty}$ denote the Slepian–Pollak prolate eigenvalues of the integral operator $K_{T,\Omega}\colon L^2([-T, T]) \to L^2([-T, T])$

$$(K_{T,\Omega} f)(u) \;:=\; \int_{-T}^{T} \frac{\sin(\Omega(u - v))}{\pi(u - v)}\, f(v)\, dv,$$

sorted $\lambda_0 \geq \lambda_1 \geq \ldots > 0$. Then for $f \in \mathscr{A} \setminus \{0\}$ with support in $[-T, T]$,

$$\alpha(f) \;\leq\; \lambda_0(T, t_*). \tag{96.5}$$

For every $T_1 \geq \tau_0$,

$$\beta(f, T_1) \;\geq\; 1 - \lambda_0(T, T_1).$$

Consequently all $f \in \mathscr{A}_{13} \setminus \{0\}$ of type at most T lie in $\mathrm{PW}_F^{\mathrm{hol,IE}}(T_1)$ if

$$M\lambda_0(T, t_*) \;\leq\; \ell(T_1)\big(1 - \lambda_0(T, T_1)\big). \tag{96.6}$$

Proof. The operator $K_{T,\Omega}$ is the time-limited frequency projection onto $[-\Omega, \Omega]$. Hence

$$\frac{\int_{-\Omega}^{\Omega} |\widehat{f}(t)|^2\, dt}{\int_{\mathbb{R}} |\widehat{f}(t)|^2\, dt} = \frac{\langle K_{T,\Omega} f, f\rangle}{\langle f, f\rangle} \leq \lambda_0(T, \Omega).$$

Taking $\Omega = t_*$ gives (96.5). Taking $\Omega = T_1$ gives the mass bound inside $[-T_1, T_1]$, hence $\beta(f, T_1) \geq 1 - \lambda_0(T, T_1)$. If (96.6) holds, then

$$M\alpha(f) \leq M\lambda_0(T, t_*) \leq \ell(T_1)(1 - \lambda_0(T, T_1)) \leq \ell(T_1)\beta(f, T_1),$$

so $f \in \mathrm{PW}_F^{\mathrm{hol,IE}}(T_1)$. □

Remark 96.4.4 *(quantitative prolate bounds).* For $c = T\Omega$, positivity of the concentration operator gives $\lambda_0(T,\Omega) \le \min(1, 2c/\pi)$: its trace is the integral of its diagonal kernel, $2T\Omega/\pi$. On rescaling to $[-1,1]$, its kernel is $\sin(c(x-y))/(\pi(x-y)) = c/\pi + O(c^3)$ uniformly for small c. The operator norm of the remainder is $O(c^3)$, so $\lambda_0 = 2c/\pi + O(c^3)$ by comparison with the constant rank-one kernel. For large c, testing any fixed nonzero compact smooth function and using Fourier tail decay gives $\lambda_0 \to 1$. These single-scale facts control only the core fraction. The far-tail bound at the second scale T_1 is still required in (96.6).

Absolute Uniform Closure Sub-Class

A uniform closure from prolate theory requires a core scale and a far-tail scale.

Corollary 96.4.5 *(uniform G_g-positivity under a two-scale prolate inequality).* . Fix $T_0 > 0$ and $T_1 \ge \tau_0$. If

$$M\lambda_0(T_0, t_*) \;\le\; \ell(T_1)\big(1 - \lambda_0(T_0, T_1)\big), \tag{96.7}$$

then $G_g(f,f) \ge 0$ for every $f \in \mathscr{A}_{13} \setminus \{0\}$ of exponential type $T \le T_0$.

Proof. The top prolate eigenvalue is monotone in both the time window and the frequency window. Hence for $T \le T_0$,

$$\alpha(f) \le \lambda_0(T_0, t_*), \qquad \beta(f, T_1) \ge 1 - \lambda_0(T_0, T_1).$$

Hypothesis (96.7) gives $M\alpha(f) \le \ell(T_1)\beta(f, T_1)$. Thus $f \in \mathrm{PW}_F^{\mathrm{hol,IE}}(T_1)$, and Chapter 96.3.4 gives the claim. □

Remark 96.4.6 *(significance).* Chapter 96.4.5 is a uniform G_g-positivity result only after the two prolate eigenvalues in (96.7) are verified. The single estimate $\alpha(f) \le \lambda_0(T_0, t_*)$ controls the core but says nothing about how much mass has crossed the far-tail threshold T_1.

96.5 Quantification of the Archimedean Positivity Extension

Theorem 96.5.1 *(properness of the two sufficient classes).* The union of the supremum and integral-energy classes is a proper subset of $\mathscr{A}_{13} \setminus \{0\}$. Together with Chapter 96.3.5, this gives both strict inclusions in (96.8), with ambient class $\mathscr{A}_{13} \setminus \{0\}$.

$$\mathrm{HFD} \subsetneq \mathrm{HFD} \cup \bigcup_{T_1} \mathrm{IE}(T_1) \subsetneq \mathscr{A}_{13} \setminus \{0\}. \tag{96.8}$$

Proof. Use the compact test $f_A = A^2 \mathcal{L}\varphi_A$ from the preceding construction, omitting both tail packets. Its spectral mass converges to $C_2 > 0$ and concentrates at zero. In particular, $\alpha(f_A) \to 1$. Moreover,

$$\int_{|u|>\tau_0} (1 + \log(2 + |u|))|h_{f_A}(u)|^2 du \longrightarrow 0.$$

Indeed, after $v = Au$, the integrand is bounded by a constant times $(1 + \log(2 + |v|))(1 + |v|)^{12}|H(v)|^2$ on $|v| > A\tau_0$. This is integrable, and its tail tends to zero. Monotonicity and logarithmic growth of ℓ give

$$\sup_{T_1 \ge \tau_0} \ell(T_1)\beta(f_A, T_1) \longrightarrow 0.$$

Thus every integral-energy condition fails for large A. The same weighted-tail estimate gives $B_g[f_A] \to 0$, while the core supremum grows linearly in A. The supremum condition fails as well. The Mellin identity for $\mathcal{L}$ proves all moment conditions, without a further cutoff or a modularity assertion. □

Remark 96.5.2 *(what the gap captures).* The gap $(\mathscr{A}_{13} \setminus \{0\}) \setminus \bigcup_{T_1} \mathrm{PW}_F^{\mathrm{hol,IE}}(T_1)$ is non-empty. Its model elements are low-frequency approximate identities with negligible logarithmic far tail. No assertion about a particular Li approximant follows without computing its weighted spectral profile.

96.6 Compatibility with Chapters 61, 57, and the Bounded-Interval Reduction

IE Class and Chapter 61 Bounded-Interval Reduction

Proposition 96.6.1 *(arithmetic inequality after the energy bound).* For an integral-energy test $f \in \mathscr{A}_{13}$, one has $G_g(f,f) \geq 0$. Its full Weil inequality is equivalent to

$$G_g(f,f) \geq 2\log\pi\|f\|_2^2 + W_{\mathrm{fin}}(f * f^*) - P(f * f^*).$$

The moments defining $\mathscr{A}_{13}$ do not remove $P(f * f^*)$.

Proof. The energy estimate proves the first assertion. The explicit formula and the identity $k + g = 2\log\pi$ prove the equivalence. Chapter 78 gives a compact test in $\mathscr{A}_{13}$ with nonzero polar form. □

Three Non-Elementary Constructions Revisited

The zero-density, Gaussian and Hermite–Biehler comparisons of Chapter 82.6.3 pose separate questions. The integral-energy theorem supplies no implication between them.

Remark 96.6.2 *(Gaussian and compact test comparisons).* The function $f_\sigma(x) = e^{-x^2/(2\sigma^2)}$ has angular transform $h_{f_\sigma}(u) = \sigma\sqrt{2\pi}e^{-\sigma^2u^2/2}$. It belongs to the exponential test space $\mathscr{G}$, and has no compact support. For a fixed $T_1 > \tau_0$, its core fraction tends to zero and its far-tail fraction tends to one as $\sigma \downarrow 0$. Thus it satisfies the auxiliary integral-energy inequality for small σ. As $\sigma \to \infty$, its core fraction tends to one and its logarithmically weighted tail tends to zero. These are statements about the Gaussian spectral profile. The Gaussian does not satisfy the zero moments at 0, 1, since $M_{f_\sigma}(0) > 0$. The differential construction above instead supplies exact compact moment tests with the required energy profiles.

The explicit formula extends from compact tests to $\mathscr{G}$ in its exponential topology, as proved in Chapter 95.1.2. That extension retains $P - W_\infty - W_{\mathrm{fin}}$. A spectral-width estimate for G_g supplies no isometry into an arithmetic or automorphic Hilbert space. An identification with a positive native pairing requires a linear map I on the specified tests and the equality $\langle If, Ih\rangle = W(f * h^*)$. Zero counts and prolate concentration estimates do not construct this map or prove the pairing equality.

96.7 Consequences of the energy estimates

The lower bound ℓ gives the sufficient condition $M\alpha(f) \leq \ell(T_1)\beta(f,T_1)$ for $G_g(f,f) \geq 0$. The two-scale prolate inequality implies this condition uniformly on a fixed compact support interval whenever its displayed bound holds. The resulting signed gamma estimate is $W_\infty(f * f^*) \leq 2\log\pi\|f\|_2^2$.

The compact differential constructions prove that the union of the integral and supremum classes strictly enlarges the supremum class, and that their union is a proper subset of the nonzero tests in $\mathscr{A}_{13}$. No inclusion of the supremum class in the union of integral-energy classes is needed. The complementary class is exactly the locus where $\ell(T_1)\beta(f,T_1) < M\alpha(f)$ for every $T_1 \geq \tau_0$.

Remark 96.7.1 *(the remaining arithmetic inequality).* The energy comparison controls the auxiliary form. For each admissible test, full Weil positivity still requires

$$G_g(f,f) + P(f * f^*) \geq 2\log\pi\|f\|_2^2 + W_{\mathrm{fin}}(f * f^*).$$

A proof on the moment kernel addresses that kernel. Extending it to all compact smooth tests also requires the mixed pairings and the finite-dimensional complement of its moment projection.

Chapter 97

Archimedean Tate Kernel via Mellin–Barnes and Binet Contour Representations

The digamma weight admits both a convergent Laplace representation and a Mellin–Barnes representation with an explicit shifted-contour remainder. Their quadratic integrals describe the auxiliary form G. Its relation to the signed Weil contribution is

$$W_\infty(f * \widetilde{f}) = 2\log\pi\|f\|_2^2 - G[f].$$

We construct the Laplace identity on smooth compact tests, determine the pointwise autocorrelation obstruction, and exhibit nonzero tests of both signs for W_∞. None of these signs discards the finite-prime and polar functional in the full Weil inequality.

97.1 Integral Carriers and Conventions

The Digamma Representations

Proposition 97.1.1 *(Mellin–Barnes representation of the digamma function).* For $\operatorname{Re} z > 0$, use the principal logarithm and fix $0 < c < 1$. Then

$$\psi(z) = -\gamma_E - \frac{1}{z} + \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{\pi\zeta(1+w)}{\sin\pi w}\, z^w\, dw. \tag{97.1}$$

The integral converges absolutely, locally uniformly in the stated half-plane.

Proof. The beta integral and Mellin inversion give

$$\frac{x}{1+x} = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{\pi}{\sin\pi w} x^w\, dw \qquad (x > 0).$$

Indeed, the Mellin transform of $x/(1+x)$ at $-w$ is $\int_0^\infty x^{-w}/(1+x)\,dx = \pi/\sin\pi w$. Analytic continuation gives the formula for $|\arg x| < \pi$. Apply it to $x = z/n$, multiply by $1/n$, and sum. On $\operatorname{Re} w = c$, the sum of absolute coefficients is $\zeta(1+c)|z|^c$, while the vertical integrand decays exponentially as $e^{-(\pi - |\arg z|)|\operatorname{Im} w|}$. Thus sum and integral commute. The convergent digamma series $\psi(1+z) + \gamma_E = \sum_{n\ge 1} z/[n(n+z)]$, followed by $\psi(1+z) = \psi(z) + 1/z$, proves the formula. □

Proposition 97.1.2 *(Binet second integral representation).* For $\operatorname{Re} s > 0$,

$$\psi(s) \;=\; \log s - \frac{1}{2s} - 2\int_0^\infty \frac{u}{(s^2+u^2)(e^{2\pi u}-1)}\, du. \tag{97.2}$$

The integral converges absolutely for $\operatorname{Re} s > 0$ and defines a holomorphic function in that half-plane.

Proof. Binet 1839 *Mémoire sur les intégrales définies eulériennes J. Ecole Polytech.* 16; Whittaker–Watson §12.32; Abramowitz–Stegun 6.3.21. Derivation: differentiate Binet's formula $\log\Gamma(s) = (s-1/2)\log s - s + (1/2)\log 2\pi + 2\int_0^\infty \arctan(u/s)/(e^{2\pi u}-1)\,du$ with respect to s. □

The Compact Test Carrier

Work on $\mathscr{A} = C_c^\infty(\mathbb{R})$, with $\widehat{f}(t) = \int f(u)e^{itu}\,du$, $\widetilde{f}(u) = \overline{f(-u)}$, and $C_f = f * \widetilde{f}$. Put $q_f = \|f\|_2^2 = C_f(0)$. Then $C_f(-u) = \overline{C_f(u)}$. All quadratic formulas use $\operatorname{Re} C_f$, so they apply to complex tests as well as real even tests. A further arithmetic test space is used only through its specified intersection with this carrier.

Classical Integral Identities

The Binet and Euler integral formulas for ψ are recorded in NIST DLMF, §5.9; the convergent digamma series is in §5.7. The Mellin–Barnes formula above follows directly from that series and the beta integral. The Fourier normalization here is the one in Lemma 82.3.2.

97.2 The Binet Decomposition of the Archimedean Kernel

Insertion of the Binet Representation

Inserting (97.2) into (82.1) and taking real parts at $s = 1/4 + it/2$ yields a three-term decomposition of g.

Lemma 97.2.1 *(Binet form of g).* For every $t \in \mathbb{R}$,

$$g(t) \;=\; g_{\log}(t) + g_{\mathrm{shift}}(t) - 2\,g_{\mathrm{Binet}}(t), \tag{97.3}$$

where

$$\begin{aligned} g_{\log}(t) &:= \tfrac{1}{2}\log\!\left(\tfrac{1}{16} + \tfrac{t^2}{4}\right) + \log\pi, \\ g_{\mathrm{shift}}(t) &:= -\frac{1/4}{2(1/16 + t^2/4)} \;=\; -\frac{1/2}{1/4 + t^2}, \\ g_{\mathrm{Binet}}(t) &:= \int_0^\infty \operatorname{Re}\!\left[\frac{u}{(1/4 + it/2)^2 + u^2}\right] \cdot \frac{1}{e^{2\pi u} - 1}\,du. \end{aligned} \tag{97.4}$$

The integral in (97.4) converges absolutely and uniformly on compact sets.

Proof. Set $s = 1/4 + it/2$. Binet (97.2): $\psi(s) = \log s - 1/(2s) - 2\int_0^\infty u/[(s^2 + u^2)(e^{2\pi u} - 1)]\,du$. Take real parts. For the first term, $\operatorname{Re}\log(1/4 + it/2) = (1/2)\log|1/4 + it/2|^2 = (1/2)\log(1/16 + t^2/4)$. For the second, $\operatorname{Re}[1/(2s)] = \operatorname{Re}[(1/4 - it/2)/(2|s|^2)] = (1/4)/[2(1/16 + t^2/4)] = (1/2)/(1/4 + t^2)$; so $-\operatorname{Re}[1/(2s)] = -1/2 \cdot 1/(1/4 + t^2) = g_{\mathrm{shift}}(t)$. For the third, $u/(s^2 + u^2)$ with $s^2 = (1/4 + it/2)^2 = 1/16 - t^2/4 + it/4$: $s^2 + u^2 = (1/16 - t^2/4 + u^2) + it/4$. The real part of $u/(s^2 + u^2)$ is $u(1/16 - t^2/4 + u^2)/|s^2 + u^2|^2$. Adding $\log\pi$ gives (97.3).

For $s = 1/4 + it/2$, neither factor $s + iu$ nor $s - iu$ vanishes for real u. On compact t-sets the denominator is uniformly nonzero for bounded u; near zero the factor $u/(e^{2\pi u} - 1)$ is bounded. At infinity the integrand is $O(u^{-1}e^{-2\pi u})$, uniformly on compact t-sets. These estimates prove absolute and locally uniform convergence. □

Remark 97.2.2 *(structural content of the decomposition).* The decomposition (97.3) isolates three contributions: $g_{\log}(t) \geq 0$ for $|t| \geq 2\sqrt{\pi^{-2} - 1/16}$ (the logarithmic threshold; eventually dominant; numerically ≈ 0.394, real since $\pi^{-2} = 1/\pi^2 > 1/16$); $g_{\mathrm{shift}}(t) < 0$ for all t (the Cauchy shift, bounded); and $g_{\mathrm{Binet}}(t)$, whose sign depends on the sign of $u^2 - t^2/4 + 1/16$ inside the u-integral, is not pointwise sign-definite. For $u > \sqrt{t^2/4 - 1/16}$ (when $|t| > 1/2$; for all $u > 0$ when $|t| \leq 1/2$) the integrand of g_{Binet} is non-negative.

Sign of $g_{\mathrm{Binet}}(t)$: Two-Region Structure

Lemma 97.2.3 *(two-region structure of g_{Binet}).* For $|t| \le 1/2$, $g_{\mathrm{Binet}}(t) \ge 0$ unconditionally (the u-integrand is pointwise non-negative). For $|t| > 1/2$,

$$g_{\mathrm{Binet}}(t) \;=\; g^{+}_{\mathrm{Binet}}(t) - g^{-}_{\mathrm{Binet}}(t), \tag{97.5}$$

where

$$g^{+}_{\mathrm{Binet}}(t) \;=\; \int_{u_0(t)}^{\infty} u \cdot \frac{u^2 - t^2/4 + 1/16}{|s^2 + u^2|^2} \cdot \frac{1}{e^{2\pi u} - 1}\, du \;\ge\; 0,$$
$$g^{-}_{\mathrm{Binet}}(t) \;=\; \int_{0}^{u_0(t)} u \cdot \frac{t^2/4 - 1/16 - u^2}{|s^2 + u^2|^2} \cdot \frac{1}{e^{2\pi u} - 1}\, du \;\ge\; 0,$$

with $u_0(t) := \sqrt{t^2/4 - 1/16}$ for $|t| > 1/2$ and $s = 1/4 + it/2$.

Proof. From the proof of Lemma 97.2.1, the integrand of $g_{\mathrm{Binet}}(t)$ is $u(u^2 - t^2/4 + 1/16)/|s^2 + u^2|^2 \cdot 1/(e^{2\pi u} - 1)$. The factor $u^2 - t^2/4 + 1/16$ vanishes at $u = u_0(t)$ for $|t| > 1/2$ and is non-negative for $u \ge u_0(t)$, strictly negative for $0 < u < u_0(t)$. Splitting the integral at $u = u_0(t)$ gives (97.5). For $|t| \le 1/2$, $u^2 - t^2/4 + 1/16 \ge 1/16 - t^2/4 \ge 0$ for all $u \ge 0$, so the integrand is non-negative and $g_{\mathrm{Binet}}(t) \ge 0$ unconditionally. □

Remark 97.2.4 *(The Binet decomposition and the zero of g).* The logarithmic term alone vanishes at $2\sqrt{\pi^{-2} - 1/16}$; the additional negative rational term means this is not the zero of their sum. The two-region integral must also be retained. The unique positive zero of the complete weight is certified by the positive-series calculation of Proposition 82.2.5. The Binet splitting is an exact representation of the same weight, not an independent assertion that either signed piece dominates.

97.3 The Hankel Integral Construction

Hankel Representation of ψ at Argument $1/4 + it/2$

The Hankel (also Dirichlet–Abel) integral representation of ψ is

$$\psi(s) + \gamma_E \;=\; \int_0^1 \frac{1 - u^{s-1}}{1 - u}\, du, \qquad \operatorname{Re} s > 0. \tag{97.6}$$

Insertion into $g(t)$ is complicated by the fact that $s = 1/4 + it/2$ lies on the line $\operatorname{Re} s = 1/4$, so u^{s-1} oscillates as $u \to 0$. We extract the real part first.

Lemma 97.3.1 *(Hankel form of g).* For $t \in \mathbb{R}$,

$$g(t) + \gamma_E - \log \pi \;=\; \int_0^1 \frac{1 - u^{-3/4}\cos(t \log u/2)}{1 - u}\, du. \tag{97.7}$$

The integral in (97.7) converges at $u = 1$ because the numerator $1 - u^{-3/4}\cos(t \log u/2)$ vanishes to first order as $u \to 1^-$. Near $u = 0^+$, the factor $u^{-3/4}$ is absolutely integrable, including at $t = 0$. Equivalently one substitutes $v = -\log u$ ($u = e^{-v}$, $du = e^{-v}dv$, $v \in (0, \infty)$), obtaining $\int_0^\infty [e^{-v} - e^{-v/4}\cos(tv/2)]/(1 - e^{-v})\, dv$, which is absolutely convergent at $v = \infty$ (the $e^{-v/4}$ factor) and at $v = 0^+$ (the numerator vanishes to first order).

Proof. Apply (97.6) at $s = 1/4 + it/2$: $u^{s-1} = u^{-3/4+it/2} = u^{-3/4}(\cos(t\log u/2) + i\sin(t\log u/2))$. The real part of u^{s-1} for real $u \in (0,1)$ and $s = 1/4 + it/2$ is $u^{-3/4}\cos(t\log u/2)$. Since $\psi(1/4 + it/2) + \gamma_E = \int_0^1 [1 - u^{s-1}]/(1-u)\,du$ and the integral represents a holomorphic function, taking the real part gives $\operatorname{Re}\psi(1/4+it/2) + \gamma_E = \int_0^1 [1 - u^{-3/4}\cos(t\log u/2)]/(1-u)\,du$. Since $g(t) = \operatorname{Re}\psi(1/4+it/2) + \log\pi$, the left-hand side of (97.7) is $g(t) + \gamma_E - \log\pi$. The substitution $v = -\log u$ converts the integral to the absolutely convergent Laplace-type form of the lemma statement. □

The Convergent Laplace Quadratic Form

Proposition 97.3.2 *(Laplace forms of G and W_∞).* For $f \in \mathscr{A}$, the integral

$$I[f] = \int_0^\infty \frac{e^{-v}q_f - e^{-v/4}\operatorname{Re} C_f(v/2)}{1 - e^{-v}}\,dv$$

is absolutely convergent and satisfies

$$G[f] = (\log\pi - \gamma_E)q_f + I[f], \qquad W_\infty(C_f) = (\log\pi + \gamma_E)q_f - I[f]. \tag{97.8}$$

Proof. In the Euler integral put $u = e^{-v}$, obtaining

$$g(t) = \log\pi - \gamma_E + \int_0^\infty \frac{e^{-v} - e^{-v/4}\cos(tv/2)}{1 - e^{-v}}\,dv.$$

Near zero, decompose its numerator as $e^{-v} - e^{-v/4} + e^{-v/4}(1 - \cos(tv/2))$. Its absolute quotient is bounded by $C(1 + t^2 v)$ for $0 < v < 1$. For $v \geq 1$, it is bounded by $Ce^{-v/4}$, independently of t. Since $\widehat{f}$ is Schwartz, these bounds justify absolute Fubini after multiplication by $|\widehat{f}(t)|^2$. Fourier inversion and Plancherel give

$$\frac{1}{2\pi}\int |\widehat{f}(t)|^2\cos(tv/2)\,dt = \operatorname{Re} C_f(v/2), \qquad \frac{1}{2\pi}\int |\widehat{f}(t)|^2\,dt = q_f.$$

This proves the first identity. The signed gamma identity $W_\infty = 2\log\pi\, q_f - G$ proves the second. □

Remark 97.3.3 *(The integrated signed condition).* The signs of the two quadratic forms require different inequalities:

$$G[f] \geq 0 \iff I[f] \geq -(\log\pi - \gamma_E)q_f, \qquad W_\infty(C_f) \geq 0 \iff I[f] \leq (\log\pi + \gamma_E)q_f.$$

Neither inequality alone is the full Weil condition $R[f] - W_\infty(C_f) \geq 0$.

Pointwise Autocorrelation Domination

Definition 97.3.4 *(Pointwise Hankel condition).* Let $\mathscr{A}_{\mathrm{HD}}$ consist of tests satisfying

$$e^{-3v/4}q_f \geq \operatorname{Re} C_f(v/2) \qquad (v > 0).$$

On any specified restricted test space, HD denotes its intersection with this class.

Theorem 97.3.5 *(The smooth pointwise obstruction).* On the compact smooth carrier,

$$\mathscr{A}_{\mathrm{HD}} = \{0\}.$$

Proof. The real part of the autocorrelation is even and smooth. Integration by parts gives $\operatorname{Re} C_f''(0) = -\|f'\|_2^2$, hence

$$\operatorname{Re} C_f(v/2) = q_f - \frac{v^2}{8}\|f'\|_2^2 + O_f(v^4).$$

In contrast $e^{-3v/4}q_f = q_f - (3/4)q_f v + O_f(v^2)$. For $f \neq 0$, their difference is negative for all sufficiently small positive v. Thus the pointwise condition fails. □

Proposition 97.3.6 *(Nonzero compact tests of both signed types).* There are real even nonzero tests in $\mathscr{A}$ with $W_\infty(C_f) > 0$, and real even HFD tests with $W_\infty(C_f) < 0$. Neither family satisfies HD.

Proof. Choose real even nonzero $\chi \in C_c^\infty$. For $f_L(u) = L^{-1/2}\chi(u/L)$, Fourier scaling gives $\widehat{f_L}(t) = \sqrt{L}\widehat{\chi}(Lt)$. Consequently

$$W_\infty(C_{f_L}) \longrightarrow (\log\pi - \psi(1/4))\|\chi\|_2^2 > 0.$$

Dominated convergence follows from the logarithmic growth of the multiplier and Schwartz decay. The constant is positive since $\psi(1/4) = -\gamma_E - \pi/2 - 3\log 2$. For the opposite sign, take $f_M(u) = \chi(u)\cos(Mu)$, with $\int \chi \neq 0$. The two Fourier translates have rapidly vanishing core supremum and retain positive mass near $\pm M$. Since $g(t) \sim \log|t|$, these tests eventually satisfy HFD and $G[f_M] \to \infty$, while their norms stay bounded. Therefore $W_\infty(C_{f_M}) = 2\log\pi q_{f_M} - G[f_M] < 0$. The preceding theorem excludes HD in both nonzero families. □

Remark 97.3.7 *(The original test space).* Both constructions use actual smooth compact tests. Membership in a further automorphic or moment-constrained subspace requires its own verification. The pointwise HD condition gives no nonzero extension of the HFD domain on any such subspace.

97.4 Mellin–Barnes Evaluation and Finite Residue Sums

The Complex Integral Before Taking Real Parts

Lemma 97.4.1 *(Mellin–Barnes form of g).* For $z = 1/4 + it/2$ and $0 < c < 1$,

$$g(t) = \log\pi - \gamma_E - \operatorname{Re}\frac{1}{z} + \operatorname{Re}\left\{\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{\pi\zeta(1+w)z^w}{\sin\pi w}\,dw\right\}.$$

Proof. Take the real part of Proposition 97.1.1. The real part is applied to the complete contour integral, not to one factor of a complex integrand. □

Residues with Their Remainder

Proposition 97.4.2 *(Finite shifted-contour identity).* Let $J_c(z)$ denote the contour integral in (97.1). For any noninteger $b < 0$,

$$J_c(z) = J_b(z) + \log z + \gamma_E + \sum_{1\le n<-b}(-1)^n\zeta(1-n)z^{-n}.$$

In particular the individual residues at zero and at $-n$, $n \geq 1$, are

$$R_0(z) = \log z + \gamma_E, \qquad R_n(z) = (-1)^n\zeta(1-n)z^{-n}.$$

A zero value of $\zeta(1-n)$ means the putative pole is removable.

Proof. At zero, use $\zeta(1+w) = w^{-1} + \gamma_E + O(w)$, $\pi/\sin \pi w = w^{-1} + O(w)$, and $z^w = 1 + w \log z + O(w^2)$. At $-n$, the sine factor has residue $(-1)^n$ and the other factors are holomorphic. Shift the vertical contour across this finite set of poles. On a fixed vertical strip the zeta function has polynomial growth, while the sine denominator and z^w give exponential decay; the horizontal integrals tend to zero. This proves the finite identity with the remaining integral J_b. □

Theorem 97.4.3 *(Residue identities retain the signed functional).* The finite contour formula evaluates g, with remainder J_b. It does not express G or W_∞ as a sum of nonnegative quadratic forms on all of $\mathscr{A}$.

Proof. Broad dilations have $G[f_L] \to g(0)\|\chi\|_2^2 < 0$, whereas high-frequency modulations have negative W_∞, as proved above. A sum of nonnegative quadratic forms cannot equal either functional on its entire carrier. This does not exclude useful estimates on specified subspaces or rearrangements retaining signed terms and their remainders. □

Remark 97.4.4 *(Analytic roles of the representations).* Binet separates logarithmic, rational and integral terms; the Laplace formula retains the short-distance autocorrelation; the Barnes identity supplies finite residue sums with a contour remainder. None changes the sign convention of the Weil functional.

97.5 The Full Weil Comparison

Proposition 97.5.1 *(Exact residual inequality).* For every test in the declared domain of the finite-prime and polar functional R,

$$W[f] = R[f] - (\log \pi + \gamma_E) q_f + I[f].$$

Thus its nonnegativity is exactly

$$R[f] + I[f] \geq (\log \pi + \gamma_E) q_f. \tag{97.9}$$

Proof. Substitute (97.8) into $W = R - W_\infty$. □

Remark 97.5.2 *(Frequency and autocorrelation conditions).* HFD proves $G \geq 0$, while HD has only the zero test on this smooth carrier. The signed families above prove neither the full Weil inequality nor a comparison with an arithmetic pairing.

Question 97.5.3 *(The arithmetic comparison).* Construct the maps from the arithmetic test carrier to the spectral or automorphic realization that preserve the full quadratic form $R - W_\infty$, including domains, normalization and all finite-prime terms. A matching gamma multiplier alone does not construct this map.

97.6 Integral Identities and Remaining Comparison

The Binet splitting, convergent Laplace identity and finite Barnes residue formula hold on their stated carriers. Smooth compact tests realize both signs of the signed archimedean term, and the pointwise HD condition is trivial. The full target is (97.9) on the complete Weil test class, together with its required arithmetic realization.

Remark 97.6.1 *(Preserving the full target).* The explicit local integral calculations do not establish the finite-prime/polar domination or its arithmetic comparison. These remain the substantive construction problem.

Chapter 98

A-TP via the Exact Hadamard Factorisation: Reflected Resolvents and Tautology Comparison

98.1 Hadamard Factorisation

Setup

The archimedean Tate positivity conjecture A-TP (Conjecture 78.2.3) is the statement that the Weil distribution $W(f * \widetilde{f}) \geq 0$ for all f in the Paley–Wiener space $\mathrm{PW}(\mathbb{R})$ (entire functions of exponential type whose restriction to $\mathbb{R}$ is square-integrable; $\widetilde{f}(x) := \overline{f(-x)}$). Chapters 79 and 82 tested A-TP via de Branges / Hermite–Biehler spectral theory and via the digamma-kernel elementary decomposition, respectively; each construction bottomed out at a hypothesis equivalent to RH.

Hadamard factorisation gives $\xi_{\mathbb{R}}(s) = e^{a+bs}\prod_{\rho}(1 - s/\rho)e^{s/\rho}$ of $\xi_{\mathbb{R}}$ as an entire function of order 1 and genus 1 (Proposition 79.2.1). Logarithmic differentiation yields $\xi'_{\mathbb{R}}/\xi_{\mathbb{R}}$ as a sum over zeros plus a linear correction. The reflected-resolvent kernel $\mathcal{K}_{\rho}(t)$ obtained by multiplying opposite critical-line resolvents is not the Guinand–Weil zero summand; it is nevertheless sign-indefinite even when ρ lies on the critical line. Thus the Hadamard construction gives no termwise positivity theorem. The positivity statement is the full Weil distribution, hence RH-level.

Classical Sources

The Hadamard-factorisation obstruction uses four classical sources disjoint from the restricted-class comparison and the bounded-interval reduction.

Hadamard 1893. Hadamard, J. *Étude sur les propriétés des fonctions entières et en particulier d'une fonction considérée par Riemann. J. Math. Pures Appl.* (4) 9 (1893), 171–215. Original Hadamard factorisation theorem for entire functions of finite order.

Guinand–Weil 1948/1952. Guinand, A.P. *A summation formula in the theory of prime numbers. Proc. London Math. Soc.* (2) 50 (1948), 107–119. Weil, A. *Sur les formules explicites de la théorie des nombres premiers* (cited above). Pair-pole representation via Hadamard derivative; Parseval between critical-line Fourier and zero-sum.

de Bruijn 1950. de Bruijn, N.G. *The roots of trigonometric integrals and a theorem of Hurwitz. Duke Math. J.* 17 (1950), 197–226. Heat-flow family Ξ_{λ}; monotone shift of roots towards the real axis.

Newman 1976, Rodgers–Tao 2020. Newman, C.M. *Fourier transforms with only real zeros. Proc. AMS* 61 (1976), 245–251. Rodgers, B.; Tao, T. *The de Bruijn–Newman constant is non-negative. Forum Math. Pi* 8 (2020), e6. The de Bruijn–Newman constant Λ; RH iff $\Lambda \leq 0$; unconditional lower bound $\Lambda \geq 0$.

Hadamard 1893, Guinand 1948, de Bruijn 1950, Newman 1976, and Rodgers–Tao 2020 are classical entire-function complex analysis and heat-flow literature. None enter the de Branges spectral comparison de Branges / Hermite–Biehler derivation (Chapter 79); none enter the bounded-interval reduction

elementary digamma kernel analysis (Chapter 82); none enter the Mellin–Rankin–Selberg isometry of Chapter 78. These sources are disjoint from Chapters 43, 47, 67.

98.2 The Exact Hadamard Formula for $\xi_{\mathbb{R}}'/\xi_{\mathbb{R}}$

Hadamard Product Logarithmic Derivative

Lemma 98.2.1 *(Hadamard formula for $\xi_{\mathbb{R}}'/\xi_{\mathbb{R}}$).* Fix the Hadamard factorisation (79.1) $\xi_{\mathbb{R}}(s) = e^{a+bs}\prod_\rho(1-s/\rho)e^{s/\rho}$ with a, b real constants of Proposition 79.2.1 and product over non-trivial zeros. The logarithmic derivative satisfies

$$\frac{\xi_{\mathbb{R}}'(s)}{\xi_{\mathbb{R}}(s)} \;=\; b + \sum_\rho\Big(\frac{1}{s-\rho}+\frac{1}{\rho}\Big), \tag{98.1}$$

where the series converges absolutely and uniformly on compact subsets avoiding the zero locus, after pairing $\rho \leftrightarrow 1-\rho$.

Proof. Logarithmic differentiation of (79.1): $\log\xi_{\mathbb{R}}(s) = a + bs + \sum_\rho[\log(1-s/\rho)+s/\rho]$. Differentiating: $\xi_{\mathbb{R}}'(s)/\xi_{\mathbb{R}}(s) = b + \sum_\rho[(-1/\rho)/(1-s/\rho)+1/\rho] = b + \sum_\rho[1/(s-\rho)+1/\rho]$ after the simplification $(-1/\rho)/(1-s/\rho) = 1/(s-\rho)$. Absolute convergence: under the pairing $\rho\mapsto 1-\rho$, the paired term $[1/(s-\rho)+1/(s-(1-\rho))]+[1/\rho+1/(1-\rho)]$ has both halves absolutely summable since $\sum_\rho(|\rho(1-\rho)|)^{-1}<\infty$ by the Riemann–von Mangoldt bound $N(T)\asymp T\log T$. □

Comparison with the Completed Zeta Formula

Proposition 98.2.2 *(partial-fraction and gamma-factor identity).* (Titchmarsh *The Theory of the Riemann Zeta Function* (Oxford 1986), §3.3; Iwaniec–Kowalski §5.9). For $s\notin\{0,1\}\cup\{\rho\}$,

$$\frac{\xi_{\mathbb{R}}'(s)}{\xi_{\mathbb{R}}(s)} \;=\; \frac{1}{s}+\frac{1}{s-1}+\frac{1}{2}\psi\Big(\frac{s}{2}\Big)-\frac{\log\pi}{2}+\frac{\zeta'(s)}{\zeta(s)}. \tag{98.2}$$

Proof. $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$; logarithmic derivative of each factor: $(s(s-1))'/(s(s-1)) = 1/s+1/(s-1)$, $(\pi^{-s/2})'/\pi^{-s/2} = -\frac{1}{2}\log\pi$, $(\Gamma(s/2))'/\Gamma(s/2) = \frac{1}{2}\psi(s/2)$, and ζ'/ζ. □

Corollary 98.2.3 *(Hadamard and explicit formula match).* Equating (98.1) and (98.2):

$$b+\sum_\rho\Big(\frac{1}{s-\rho}+\frac{1}{\rho}\Big) \;=\; \frac{1}{s}+\frac{1}{s-1}+\frac{1}{2}\psi\Big(\frac{s}{2}\Big)-\frac{\log\pi}{2}+\frac{\zeta'(s)}{\zeta(s)}. \tag{98.3}$$

This is the standard Stark identity; setting $s = 1/2+it$ and taking the real part gives a pointwise formula for the Weil kernel.

Proof. Both sides express the same meromorphic function; comparison is algebraic, valid away from the zero/pole locus. □

Remark 98.2.4 *(the two decompositions of $\xi_{\mathbb{R}}'/\xi_{\mathbb{R}}$).* The right-hand side of (98.3) is the Weil-kernel split of Chapter 78: polar terms $1/s+1/(s-1)$, archimedean digamma $\frac{1}{2}\psi(s/2)-\frac{1}{2}\log\pi$ (the kernel $g(t)$ of Chapter 82 after shifting argument), finite-prime ζ'/ζ. The left-hand side is the sum over non-trivial zeros. The following section separates the logarithmic-derivative zero sum from a reflected-resolvent quadratic form; the latter is not the Guinand–Weil zero side.

98.3 The Reflected Pair-Pole Diagnostic

Pairing Conjugate Zeros

Definition 98.3.1 *(pair-pole kernel).* Write each non-trivial zero $\rho = \beta + i\gamma$ with $0 \le \beta \le 1$ (non-trivial strip), and pair with its complex conjugate $\bar\rho = \beta - i\gamma$ and functional-equation partner $1-\rho = 1-\beta-i\gamma$ (and $1-\bar\rho = 1-\beta+i\gamma$). For $s = 1/2 + it$ on the critical line, define the *pair-pole kernel*

$$\mathcal{K}_\rho(t) \;:=\; \mathrm{Re}\!\left[\frac{1}{(1/2+it-\rho)(1/2-it-\rho)}\right]. \tag{98.4}$$

When ρ is paired with the reflected pole in the t-variable, the denominator becomes $((1/2-\beta)+i(t-\gamma))((1/2-\beta)-i(t+\gamma))$, explicit sign structure computed below.

Lemma 98.3.2 *(explicit form of the pair-pole kernel).* For $\rho = \beta + i\gamma$ with $\beta \in [0,1]$, $\gamma \in \mathbb{R}$, and $\alpha := 1/2 - \beta$,

$$\mathcal{K}_\rho(t) \;=\; \frac{\alpha^2 + t^2 - \gamma^2}{[\alpha^2 + (t-\gamma)^2][\alpha^2 + (t+\gamma)^2]}. \tag{98.5}$$

When $\beta = 1/2$ (RH), this simplifies to $\mathcal{K}_\rho(t)\big|_{\beta=1/2} = 1/(t^2-\gamma^2)$, which is a sign-indefinite distribution: positive for $|t| > |\gamma|$, negative for $|t| < |\gamma|$, with distributional poles at $t = \pm\gamma$. The pair-pole sum $\int |\widehat{f}|^2 \mathcal{K}_\rho \, dt$ is a principal-value term; it is not sign-definite term by term (Lemma 98.3.4).

Proof. Write $\alpha := 1/2 - \beta$ and compute: $(1/2+it-\rho)(1/2-it-\rho) = (\alpha + i(t-\gamma))(\alpha - i(t+\gamma))$. Expanding: $\alpha^2 - 2i\alpha\gamma + (t^2-\gamma^2)$. The real part of the reciprocal is $[\alpha^2 + (t^2-\gamma^2)]/[(\alpha^2+(t^2-\gamma^2))^2 + 4\alpha^2\gamma^2]$, and the denominator equals $[\alpha^2+(t-\gamma)^2][\alpha^2+(t+\gamma)^2]$ by expanding both sides. The specialisation $\alpha = 0$ yields $\mathcal{K}_\rho(t)\big|_{\alpha=0} = (t^2-\gamma^2)/(t^2-\gamma^2)^2 = 1/(t^2-\gamma^2)$; the symmetric contour gives the principal-value distribution of Lemma 98.3.4. □

Remark 98.3.3 *(the sign of $\mathcal{K}_\rho(t)$).* For any $\beta \in [0,1]$, the numerator $\alpha^2 + t^2 - \gamma^2$ of (98.5) is negative for $|t| < \sqrt{\gamma^2 - \alpha^2}$ (when $|\gamma| > |\alpha|$), while the denominator is always positive. Hence $\mathcal{K}_\rho(t) < 0$ on this interval, including in the on-critical case $\alpha = 0$ where $\mathcal{K}_\rho(t) < 0$ for $|t| < |\gamma|$. Thus no term-by-term positivity follows from the pair-pole kernel.

Residue Extraction and Dirac Limits

Lemma 98.3.4 *(principal value at an on-critical pole).* Let $\rho = 1/2 + i\gamma$ be on the critical line ($\beta = 1/2$, $\gamma \neq 0$, $\gamma \in \mathbb{R}$). For $f \in \mathrm{PW}(\mathbb{R})$ real and even with $\widehat{f}$ smooth at $t = \pm\gamma$,

$$\lim_{\varepsilon \downarrow 0} \mathrm{Re} \int_{\mathbb{R}} |\widehat{f}(t)|^2 \cdot \frac{dt}{(1/2+\varepsilon+it-\rho)(1/2+\varepsilon-it-\rho)} \;=\; \mathrm{PV}\int_{\mathbb{R}} \frac{|\widehat{f}(t)|^2}{t^2-\gamma^2}\, dt,$$

with the reflected pole convention of (98.4). This principal value is not sign-definite.

Proof. With the reflected pole convention the regularised real kernel is

$$\mathrm{Re}\frac{1}{(\varepsilon + i(t-\gamma))(\varepsilon - i(t+\gamma))} = \frac{\varepsilon^2 + t^2 - \gamma^2}{(\varepsilon^2 + (t-\gamma)^2)(\varepsilon^2 + (t+\gamma)^2)}.$$

As $\varepsilon \downarrow 0$ this converges in distributions to $\mathrm{PV}(t^2-\gamma^2)^{-1}$. Pairing with the smooth function $|\widehat{f}|^2$ gives the displayed identity. □

Theorem 98.3.5 *(the reflected pair-pole kernel is not the Guinand–Weil summand).* For $f \in \mathrm{PW}(\mathbb{R})$ real and even, the reflected-resolvent expression

$$\mathcal{R}_{\mathrm{ref}}(f) := \lim_{T\to\infty} \sum_{|\mathrm{Im}\rho|\le T} R_\rho[f], \qquad R_\rho[f] := \mathrm{PV}\int_{\mathbb{R}} |\widehat{f}(t)|^2\, \mathcal{K}_\rho(t)\, dt, \tag{98.6}$$

where $\mathcal{K}_\rho$ is the reflected pair-pole kernel (98.5), the sum is over non-trivial zeros of $\xi_{\mathbb{R}}$, each paired once with its conjugate. When the symmetric limit exists, it is not the zero side of the Guinand–Weil formula. It is a diagnostic quadratic form obtained by multiplying reflected resolvents.

Proof. It is enough to compare a single on-critical conjugate pair. For $\rho = 1/2 + i\gamma$,

$$\frac{1}{1/2 + it - \rho} + \frac{1}{1/2 + it - \bar{\rho}} = -i\left(\frac{1}{t-\gamma} + \frac{1}{t+\gamma}\right) = -\frac{2it}{t^2-\gamma^2}.$$

Thus the zero contribution to the imaginary part of the logarithmic derivative is the odd principal-value distribution $-2t\,\mathrm{PV}(t^2-\gamma^2)^{-1}$. The reflected kernel $\mathcal{K}_\rho(t) = \mathrm{PV}(t^2-\gamma^2)^{-1}$ is even. It therefore cannot be the residue summand obtained from $\xi'_{\mathbb{R}}/\xi_{\mathbb{R}}$. The displayed $\mathcal{R}_{\mathrm{ref}}$ is consequently not a Guinand–Weil decomposition. □

Under RH: No Termwise Positivity

Proposition 98.3.6 *(the on-critical pair-pole term is sign-indefinite).* Let $\rho = 1/2 + i\gamma$ be on the critical line with $\gamma \neq 0$. Then

$$\mathcal{K}_\rho(t) = \mathrm{PV}\frac{1}{t^2-\gamma^2}.$$

It is sign-indefinite. In particular there are real even $f \in \mathrm{PW}(\mathbb{R})$ for which the single principal-value term $R_\rho[f]$ is negative. RH gives positivity of the full Weil distribution, not positivity of each reflected pair-pole term.

Proof. This is Lemma 98.3.4. On $|t| < |\gamma|$ the kernel is negative, away from the poles. Choose $\widehat{f}$ real, even, smooth, and supported in a sufficiently small neighbourhood of 0. Then the principal-value integral is an ordinary integral over a region where the kernel is negative. □

98.4 Tautology Comparison: Where Positivity Becomes Circular

Even under RH, a single pair-pole kernel is sign-indefinite. The following proposition records the failure of the pointwise construction.

Proposition 98.4.1 *(pointwise pair-pole positivity fails).* The statement "$\mathcal{K}_\rho(t) \ge 0$ pointwise for every non-trivial zero ρ" is false for the zeta zero set. Hence any attempt to prove A-TP by *term-by-term* pointwise non-negativity of $\mathcal{K}_\rho(t)$ is not merely circular; it has no valid on-critical case.

Proof. For an on-critical zero, $\alpha = 0$, and $\mathcal{K}_\rho(0) = -\gamma^{-2} < 0$. Thus RH itself does not make the pair-pole kernel pointwise non-negative. □

Remark 98.4.2 *(pointwise positivity versus integrated positivity).* Proposition 98.4.1 establishes that pointwise $\mathcal{K}_\rho(t) \ge 0$ cannot be the mechanism for A-TP. Sum-level non-negativity belongs to the actual Weil distribution; the reflected-resolvent form $\mathcal{R}_{\mathrm{ref}}$ is not that distribution. The Hadamard logarithmic derivative has the same RH ceiling as the de Branges construction (Chapter 79) and the digamma-kernel construction (Chapter 82).

Corollary 98.4.3 *(the valid reformulations coincide).* The two valid positivity statements

(i) *(Weil)* $W(f * \widetilde{f}) \ge 0$ for all $f \in \mathrm{PW}(\mathbb{R})$;

(ii) *(Hermite–Biehler)* the de Branges datum $E_\xi(z) = A_\xi(z) - iB_\xi(z)$ (defined in Chapter 79) is Hermite–Biehler and satisfies $\mathrm{div}(A_\xi) = \mathrm{div}(\Xi)$;

are equivalent and each equivalent to RH. The pointwise pair-pole positivity statement is false and is not a third reformulation.

Proof. (1) $\Leftrightarrow$ RH: Corollary 78.2.1. (2) $\Leftrightarrow$ RH: Lagarias' Hermite–Biehler theorem and Theorem 79.6.9. Proposition 98.4.1 excludes the pointwise pair-pole statement. □

98.5 de Bruijn–Newman $\Lambda \le 0$ is RH

The pair-pole pointwise positivity statement is false. The de Bruijn–Newman bound $\Lambda \le 0$ is a valid construction to A-TP only because it is RH itself.

The de Bruijn–Newman Family

Definition 98.5.1 *(de Bruijn–Newman heat flow).* Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $\Xi(t) = \xi(1/2 + it)$. Use the normalization

$$\Phi(u) := \sum_{n=1}^{\infty} \left(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}\right) \exp(-\pi n^2 e^{4u}).$$

For $\lambda \in \mathbb{R}$, define

$$H_\lambda(z) := \int_0^\infty \Phi(u) e^{\lambda u^2} \cos(zu)\, du, \qquad \Xi_\lambda(t) := 8H_\lambda(2t).$$

Super-exponential decay of Φ gives locally uniform absolute convergence for complex z, including all complex derivatives. Thus these functions are entire. The theta-integral identity gives $H_0(z) = \xi(1/2 + iz/2)/8$, so $\Xi_0 = \Xi$. The de Bruijn–Newman constant Λ is the infimum of λ for which all zeros of H_λ, equivalently Ξ_λ, are real. The positive rescaling of the variable and nonzero scalar factor leave this condition unchanged.

Theorem 98.5.2 *(de Bruijn–Newman bounds).* (Newman 1976; Rodgers–Tao 2020).

(i) $\Lambda \le 1/2$ (de Bruijn 1950: for $\lambda \ge 1/2$, Ξ_λ has only real zeros; Newman 1976 introduced Λ as an infimum).

(ii) $\Lambda \ge 0$ (Rodgers–Tao 2020, unconditional).

(iii) RH $\Leftrightarrow \Lambda \le 0$: the real-zero parameters form $[\Lambda, \infty)$, and $H_0(z) = \xi(1/2 + iz/2)/8$.

(iv) Newman conjectured the lower bound $\Lambda \ge 0$, proved by Rodgers–Tao. Combining this with (iii) gives RH $\Longleftrightarrow \Lambda = 0$; the equality is not an unconditional bound.

Proof. Newman 1976 *Proc. AMS* 61; de Bruijn 1950 *Duke Math. J.* 17; Rodgers–Tao 2020 *Forum Math. Pi* 8; and the classical equivalence RH $\Leftrightarrow$ Ξ has only real zeros (Riemann 1859). □

$\Lambda \le 0$ Implies A-TP via RH

Lemma 98.5.3 *(critical-line reformulation of $\Lambda \le 0$).* The statement $\Lambda \le 0$ is equivalent to: every non-trivial zero of $\xi_{\mathbb{R}}$ lies on the critical line $\operatorname{Re} s = 1/2$.

Proof. $\Lambda \le 0 \Leftrightarrow \Xi_0 = \Xi$ has only real zeros $\Leftrightarrow \xi_{\mathbb{R}}(1/2 + it) = 0$ for t real only $\Leftrightarrow$ every non-trivial zero ρ of $\xi_{\mathbb{R}}$ satisfies $\operatorname{Re} \rho = 1/2$. This is RH. □

Corollary 98.5.4 *($\Lambda \le 0$ implies A-TP).* . Under $\Lambda \le 0$ (equivalent to RH), A-TP holds.

Proof. Lemma 98.5.3 reduces $\Lambda \le 0$ to RH. Then Corollary 78.2.1 gives A-TP. □

Remark 98.5.5 *($\Lambda \le 0$ is equivalent to RH).* $\Lambda \le 0$ is equivalent to RH, not weaker. The Rodgers–Tao theorem $\Lambda \ge 0$ places the constant at the Newman barrier. change on Λ is change towards RH, not a weaker substitute. The Hadamard-factorisation construction is tautological: it reduces A-TP to RH without passing through any strictly weaker intermediate.

The candidate "$\Lambda \le \lambda_0$ for some $\lambda_0 > 0$ still implies A-TP" is not supplied by the de Bruijn–Newman theory. The heat flow parameterises the zero-reality problem; the value $\lambda = 0$ is exactly RH.

Proposition 98.5.6 *(zero-free regions do not give a pair-pole positivity criterion).* A classical zero-free region, including the Korobov–Vinogradov region, does not imply A-TP or a partial-support A-TP statement through the pair-pole kernels. It excludes zeros near the edge of the critical strip; it does not make the central principal-value kernels positive.

Proof. Proposition 98.3.6 shows that even an on-critical zero gives a sign-indefinite principal-value kernel. Moving zeros away from the edge of the strip cannot turn these central kernels into positive kernels. A zero-free region is a zero-location theorem; A-TP is a positivity theorem for the full Weil sum. □

Remark 98.5.7 *(zero-free region boundary).* The Korobov–Vinogradov region is an edge exclusion near $\operatorname{Re} s = 1$, with the reflected exclusion near $\operatorname{Re} s = 0$. It is not a central critical-line positivity statement. No partial-support A-TP closure is recorded here from that input.

98.6 Zero Density and Pair-Pole Convergence

Convergence of the Pair-Pole Sum

Proposition 98.6.1 *(reflected pair-pole form converges distributionally after pairing).* For $f \in \mathrm{PW}(\mathbb{R})$, the reflected pair-pole form (98.6) is a paired principal-value distribution. It is not, in general, an absolutely convergent sum of ordinary integrals. Absolute convergence holds only after imposing vanishing near the finitely many zero ordinates met by $\operatorname{supp} \widehat{f}$, or after keeping the explicit regularisation.

Proof. Away from $t = \pm\gamma$, the displayed kernel has the stated $|\gamma|^{-2}$ decay after pairing, and the zero-counting estimate controls the tail. At $t = \pm\gamma$, however, the on-critical kernel is $\mathrm{PV}(t^2 - \gamma^2)^{-1}$, whose absolute value is not locally integrable. Thus the reflected-resolvent expression is a principal-value distribution unless the test function kills those poles or a regularisation is retained. □

Remark 98.6.2 *(sum-level positivity versus pointwise positivity).* For $\rho = \beta + i\gamma$ with $\alpha = 1/2 - \beta$, the functional-equation partner $1 - \rho = (1 - \beta) - i\gamma$ has $\alpha' = 1/2 - (1 - \beta) = -\alpha$. Since (98.5) depends on α^2, $\mathcal{K}_{1-\rho}(t) = \mathcal{K}_\rho(t)$; the two kernels are equal. The sum-level non-negativity of the Weil distribution is

equivalent to RH by Corollary 78.2.1. It is not the pointwise positivity of $\mathcal{K}_\rho$, and it is not the positivity of the reflected-resolvent form $\mathcal{R}_{\mathrm{ref}}$.

Exact Domain of the Hadamard-Factorisation Construction

Theorem 98.6.3 *(Hadamard-factorisation construction: exact domain).* The exact Hadamard-factorisation obstruction on A-TP has the following outcomes.

(i) *Reflected pair-pole diagnostic* (Theorem 98.3.5): unconditional; explicit pointwise form (98.5), but not the Guinand–Weil zero side.

(ii) *No termwise positivity under RH* (Proposition 98.3.6): under RH, $\mathcal{K}_\rho(t) = \mathrm{PV}(t^2 - \gamma^2)^{-1}$, and single pair-pole terms can be negative.

(iii) *Valid reformulations coincide* (Corollary 98.4.3): Weil positivity and Hermite–Biehler positivity are equivalent to RH. Pointwise pair-pole positivity is false (Proposition 98.4.1).

(iv) *de Bruijn–Newman sufficient condition* (Corollary 98.5.4): $\Lambda \le 0$ implies A-TP; but $\Lambda \le 0 \Leftrightarrow$ RH, so this is not weaker than RH.

(v) *Zero-free region boundary* (Proposition 98.5.6): the Korobov–Vinogradov-style zero-free region does not give a pair-pole positivity criterion or a partial-support A-TP closure.

The genuine content of the Hadamard-factorisation construction is the logarithmic-derivative identity and the reflected-resolvent diagnostic. It does not give a termwise Weil decomposition and does not close A-TP below the RH threshold.

Proof. Combine the cited results. □

Remark 98.6.4 *(the tautology map).* The three constructions to A-TP considered across the restricted-class comparison, the bounded-interval reduction, and the analytic refinement are:

the de Branges spectral comparison (de Branges). $\mathrm{HB}^+ + \mathrm{EC}_\pi + \mathrm{Det} + \mathrm{SH}$; tautology level: each hypothesis is RH-equivalent or strictly finer.

the bounded-interval reduction (digamma). Bounded-interval positivity on $[-t_*, t_*]$ after elementary reductions, with the finite-prime and polar functional $R[f]$ retained; tautology level: the full inequality is RH-equivalent.

the Hadamard reduction (Hadamard). Logarithmic derivative and reflected pair-pole diagnostic; failure mode: pointwise positivity is false even on the critical line (Proposition 98.4.1).

All three constructions bottom out at RH. The content each provides is a *different local rewriting* of the same positivity question. The Hadamard construction offers the pointwise reflected-resolvent formula (98.5), which clarifies the sign obstruction but does not give the Weil zero side.

98.7 Conclusion

The Hadamard factorisation $\xi_{\mathbb{R}}(s) = e^{a+bs}\prod_\rho (1 - s/\rho)e^{s/\rho}$ gives the logarithmic derivative

$$\frac{\xi_{\mathbb{R}}'(s)}{\xi_{\mathbb{R}}(s)} = b + \sum_\rho \left(\frac{1}{s-\rho} + \frac{1}{\rho}\right).$$

The reflected kernel (98.5) is not the Guinand–Weil zero summand. Pointwise non-negativity of $\mathcal{K}_\rho$ fails even under RH (Proposition 98.4.1), so the Hadamard construction cannot prove A-TP by termwise

positivity. The de Bruijn–Newman constant $\Lambda \leq 0$ is likewise equivalent to RH (Lemma 98.5.3). The Korobov–Vinogradov zero-free region gives no pair-pole positivity criterion. The valid reformulations of A-TP remain Weil positivity and the Hermite–Biehler condition (Corollary 98.4.3).

Remark 98.7.1 *(the non-tautological content of the Hadamard construction).* The Hadamard construction yields the exact logarithmic derivative of $\xi_{\mathbb{R}}$ and an explicit reflected-resolvent kernel (98.5). Pointwise pair-pole positivity is false (Proposition 98.4.1), so the construction does not close A-TP via this mechanism. Classical zero-free regions remain zero-location input, not positivity input.

Chapter 99

A-TP as Gram Matrix Positivity: Hermite Basis Reduction

The full Weil inequality can be tested on a countable family of compactly supported Hermite functions. Their finite Gram matrices retain the signed gamma, prime, and polar terms. Fixed cutoffs and continuous moment projections give dense families in the compact test algebra and its specified moment kernels. The uncut Hermite functions instead belong to the exponential test space and supply a separate Gaussian matrix.

99.1 Hermite Cutoff

A test family determines the Weil inequality only when its span is dense for a topology in which the form is continuous. Multiplication of Hermite expansions by a fixed compact cutoff gives convergence of every derivative with a common support. This proves the required density in the compact test topology. A finite list of Li coefficient inequalities concerns different tests. A comparison with a finite Hermite matrix requires an actual change-of-family map and control of any remaining error.

Quadratic Reduction

Negative semidefiniteness of all compact Gram blocks is equivalent to the full Weil inequality. Projection onto a moment kernel gives the restricted inequality on that kernel.

Classical references

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Mehler 1866. Mehler, F.G. *Über die Entwicklung einer Function von beliebig vielen Variabeln nach Laplaceschen Functionen höherer Ordnung*. *Crelle's Journal* 66 (1866), 161–176. Mehler kernel identity for the physicist's Hermite polynomials (DLMF 18.18.28, $|t| < 1$), $\sum_{n\ge 0} H_n(x)H_n(y)t^n/(2^n n!) = (1-t^2)^{-1/2}\exp((2txy - t^2(x^2+y^2))/(1-t^2))$.

99.2 The Weil form and its moment subspaces

Definition 99.2.1 *(Weil quadratic form).* For $f, h \in \mathscr{A} = C_c^\infty(\mathbb{R};\mathbb{C})$ put $q(f,h) = -W(f * h^*)$, linear in the first argument, and $Q[f] = q(f,f)$. Write $M_f(s) = \int f(x)e^{sx}\,dx$ and $\psi_\Gamma = \Gamma'/\Gamma$. With $h_f(u) = \int f(x)e^{iux}\,dx$ and $k(u) = \log\pi - \Re\psi_\Gamma(1/4 + iu/2)$, one has

$$Q[f] = W_\infty(f * f^*) + W_{\mathrm{fin}}(f * f^*) - P(f * f^*).$$

Proposition 99.2.2 *(integral representation).* The form q is continuous and Hermitian in the compact test topology. For all $f, h \in \mathscr{A}$,

$$q(f,h) = \frac{1}{2\pi}\int h_f(u)\overline{h_h(u)}k(u)\,du + W_{\mathrm{fin}}(f * h^*) - a(f)\overline{b(h)} - b(f)\overline{a(h)}, \tag{99.1}$$

where $a(f) = M_f(-1/2)$ and $b(f) = M_f(1/2)$.

Proof. Apply the compact explicit formula to $f * h^*$. The Fourier convolution identity gives the integral, and evaluation at $\pm i/2$ gives the polar terms. Continuity follows from continuity of convolution and the three explicit-formula distributions. Their involution identities give Hermitian symmetry. □

Definition 99.2.3 *(the map to the continuous antidual).* Let $\mathscr{A}^\times$ denote the continuous conjugate-linear functionals on $\mathscr{A}$. Define the linear map $Q : \mathscr{A} \to \mathscr{A}^\times$ by

$$(Qf)(h) = q(f,h).$$

This is a Hermitian form map. A self-adjoint operator on a Hilbert space requires a Hilbert realization and an operator domain.

Remark 99.2.4 *(domains and the auxiliary form).* Compact smooth tests have rapidly decreasing angular Fourier transforms on the real axis. This proves convergence against $k(u) = O(\log(2 + |u|))$. For $g(u) = 2\log\pi - k(u)$, define $G_g(f,h) = (2\pi)^{-1}\int h_f\overline{h_h}g$. Then $W_\infty(f * h^*) = 2\log\pi\langle f, h\rangle_2 - G_g(f,h)$. The real even moment class $\mathscr{A}_{13}$ imposes moments at 0, 1, whereas $\mathscr{A}_{\mathrm{pol}} = \ker(a,b)$ imposes moments at $\pm 1/2$. These are different subspaces. The latter is a closed complex subspace of codimension two. To construct its projection, take a nonnegative compact even bump b_0 and two distinct translates $b_j(x) = b_0(x - t_j)$. Their moment columns are $(e^{-t_j/2}a(b_0), e^{t_j/2}b(b_0))^t$, hence are independent. Writing this matrix as C, set $R_{\mathrm{pol}}v = \sum_{j=1}^2 b_j(C^{-1}v)_j$ and $\Pi_{\mathrm{pol}}f = f - R_{\mathrm{pol}}(a(f), b(f))^t$. It is continuous and projects onto $\mathscr{A}_{\mathrm{pol}}$. The real projection Π onto $\mathscr{A}_{13}$ is given in Chapter 96.1.1.

99.3 The Weber–Hermite Basis and Gram Matrix

The Hermite functions are Fourier eigenfunctions and form an orthonormal basis of $L^2(\mathbb{R})$. Their rapidly convergent expansions of compact smooth tests become compact approximations after multiplication by a fixed cutoff.

Weber–Hermite Basis

Definition 99.3.1 *(Weber–Hermite functions).* . For $n \geq 0$, set

$$\phi_n(x) \;:=\; (2^n n!\sqrt{\pi})^{-1/2}H_n(x)e^{-x^2/2}, \qquad x \in \mathbb{R},$$

where H_n is the physicist's Hermite polynomial $H_n(y) = (-1)^n e^{y^2}\frac{d^n}{dy^n}e^{-y^2}$. The normalization is chosen so that $\{\phi_n\}_{n\geq 0}$ is orthonormal in $L^2(\mathbb{R})$ for the angular Fourier normalization used in Chapter 78.1.1.

Proposition 99.3.2 *(Hermite self-reciprocity).* . Under the Fourier normalization $\widehat{f}(t) = \int f(x)e^{itx}\,dx$,

$$\widehat{\phi_n}(t) = (2\pi)^{1/2} i^n \phi_n(t).$$

Equivalently, the unitary angular Fourier transform $(2\pi)^{-1/2}\widehat{\ }$ has eigenvalue i^n on ϕ_n.

Proof. Hermite 1864; see Whittaker–Watson 1927 §16.61. The Hermite functions diagonalize the harmonic oscillator $-\partial_x^2 + x^2$ which commutes with the Fourier transform (conjugated by the appropriate scaling). □

Remark 99.3.3 *(Schwartz convergence and compact support).* Their finite span is dense in Schwartz space and in $L^2(\mathbb{R})$. For $f \in \mathscr{A}$ with $\operatorname{supp}(f) \subset [-T, T]$, the Hermite expansion $f = \sum_n c_n \phi_n$ converges in Schwartz topology with $|c_n| \le C_{T,N} n^{-N}$ for every N: this is the classical Hermite-series rapid decay for Schwartz functions (Reed–Simon II Thm. V.13). However, the Hermite functions are *not* compactly supported, so the expansion does not preserve Paley–Wiener support; the truncation $f_N := \sum_{n=0}^{N} c_n \phi_n$ belongs to the Schwartz class but generally not to $\mathscr{A}$.

The subtlety of Chapter 99.3.3 is essential: a Schwartz expansion alone does not preserve the compact-test topology. The fixed-cutoff construction below supplies that convergence, without asserting a compact Hilbert-space operator.

The Gram Matrix Q_{mn}

Definition 99.3.4 *(compact Hermite Gram family).* Fix a real even $\chi \in C_c^\infty(\mathbb{R})$ equal to one on $[-1, 1]$. Set $\chi_R(x) = \chi(x/R)$ for integers $R \ge 1$ and $e_{R,n} = \chi_R \phi_n$ for $n \ge 0$. Enumerate the pairs (R, n) by $j \ge 0$, and write $e_j = e_{R,n}$. Define the Hermitian matrix

$$Q_{ij} = q(e_i, e_j), \qquad q\left(\sum_i c_i e_i, \sum_j c_j e_j\right) = \sum_{i,j} c_i \overline{c_j} Q_{ij}.$$

Its finite principal matrices need no removed-cutoff limit. The projected families $\Pi e_{R,2n}$ and $\Pi_{\mathrm{pol}} e_{R,n}$ give, respectively, real moment matrices and complex polar-kernel matrices.

Theorem 99.3.5 *(completeness of the compact families).* The complex span of the $e_{R,n}$ is dense in $\mathscr{A}$. The real span of $\Pi e_{R,2n}$ is dense in $\mathscr{A}_{13}$. The complex span of $\Pi_{\mathrm{pol}} e_{R,n}$ is dense in $\mathscr{A}_{\mathrm{pol}}$. All three statements use the respective compact test topologies.

Proof. Let f be compact smooth and choose R with $\chi_R = 1$ on its support. Its Hermite coefficients satisfy

$$|\langle f, \phi_n \rangle_2| \le (2n+1)^{-m} \|(-D^2 + x^2)^m f\|_2 \quad (m \ge 0).$$

This follows by integration by parts and the oscillator eigenvalue identity. The ladder identities for multiplication by x and application of D bound the L^2 norms of $x^a D^b \phi_n$ by a polynomial in n for each fixed a, b. The one-dimensional estimate $\|v\|_\infty^2 \le 2\|v\|_2 \|v'\|_2$ gives the corresponding supremum bounds. Arbitrarily rapid coefficient decay therefore makes the Hermite series and all its Schwartz seminorms converge. Its L^2 limit is f, by completeness of the Hermite functions. If s_N is its partial sum, then $\chi_R s_N \to f$ with a common compact support and uniform convergence of every derivative. This is convergence in the compact test topology. For real even f only even indices occur and the coefficients are real. Applying the continuous projection Π, or Π_{pol}, proves the corresponding kernel-density statements. □

THEOREM 99.3.6 *(compact Gram formula).* For $d_i = e_i$, or for either projected family, write $h_i = \mathcal{F}_+ d_i$ and $F_{ij} = d_i * d_j^*$. Then

$$q(d_i, d_j) = \frac{1}{2\pi} \int h_i(u)\overline{h_j(u)}k(u)du + (Q_{\text{fin}})_{ij} - (Q_{\text{polar}})_{ij},$$

where

$$(Q_{\text{fin}})_{ij} = \sum_{m\geq 2} \frac{\Lambda(m)}{\sqrt{m}}(F_{ij}(\log m) + F_{ij}(-\log m)),$$
$$(Q_{\text{polar}})_{ij} = a(d_i)\overline{b(d_j)} + b(d_i)\overline{a(d_j)}.$$

The prime sum is finite. The polar term vanishes for the Π_{pol} family, but need not vanish for the Π family.

Proof. Apply (99.1) to the two compact tests. Compact support bounds the prime-power logarithms in the sum. □

Remark 99.3.7 *(reality and the Gaussian extension).* Every even Hermite index occurs for real even tests, including $n \equiv 2 \pmod 4$. Its Fourier eigenvalue is real and needs no pairing with another index. Compact cutoffs need not be Fourier eigenfunctions. For uncut Hermites, convergence $\chi_R\phi_n \to \phi_n$ holds in $\mathscr{G}$ of Chapter 95.1.2. The extended form consequently gives well-defined entries $q(\phi_m, \phi_n)$ as a separate Gaussian matrix. Its signed archimedean entry is $i^{m-n}\int \phi_m(u)\phi_n(u)k(u)du$.

The uncut auxiliary diagonal

PROPOSITION 99.3.8 *(auxiliary and signed Hermite moments).* For each $n \geq 0$ the auxiliary uncut moment is

$$A_n := G_g(\phi_n, \phi_n) = \int_{\mathbb{R}} |\phi_n(u)|^2 g(u)du.$$

The corresponding signed gamma diagonal is $2\log\pi - A_n$. These are moments of the uncut Hermites in $\mathscr{G}$.

Proof. Use $h_{\phi_n} = \sqrt{2\pi}i^n\phi_n$ and $\|\phi_n\|_2 = 1$. The Gaussian times polynomial decay of ϕ_n proves absolute convergence against the logarithmic weight. The identity $k + g = 2\log\pi$ gives the signed diagonal. □

LEMMA 99.3.9 *(the zeroth auxiliary moment).*

$$A_0 = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-u^2} g(u)du < 0.$$

If $I(L) = \int e^{-2\pi u^2/L} g(u)du$, then $A_0 = I(2\pi)/\sqrt{\pi}$. In particular, the signed zeroth gamma moment $2\log\pi - A_0$ is positive.

Proof. The normalized Gaussian measure $e^{-u^2}du/\sqrt{\pi}$ has second moment $1/2$. The digamma series gives, for $c_n = n + 1/4$,

$$g(u) = g(0) + \sum_{n\geq 0} \frac{u^2/4}{c_n(c_n^2 + u^2/4)}.$$

All increments are nonnegative, so Tonelli applies. The function $y/(c^2+y)$ is concave on $y \ge 0$. For $n = 0$, Jensen's inequality bounds the Gaussian mean by

$$\frac{1/8}{(1/4)((1/4)^2+1/8)} = \frac{8}{3}.$$

For $n \ge 1$ the mean is at most $1/(8c_n^3)$, and

$$\frac{1}{8}\sum_{n\ge 1}(n+1/4)^{-3} \le \frac{1}{8}\left((5/4)^{-3} + \int_{5/4}^{\infty} x^{-3}dx\right) = \frac{13}{125}.$$

Finally $g(0) = -\gamma - \pi/2 - 3\log 2 + \log\pi < -3$. For example, $\gamma > 0.57$, $\pi > 3.14$, $\log 2 > 0.69$, and $\log\pi < 1.15$ give this strict bound. Hence $A_0 < -3 + 8/3 + 13/125 < 0$. Substituting $L = 2\pi$ proves the stated normalization. □

Remark 99.3.10 *(arithmetic interpretation of a diagonal).* The auxiliary sign does not determine the Weil sign. For an uncut Hermite test the full diagonal is

$$q(\phi_n, \phi_n) = 2\log\pi - A_n + W_{\mathrm{fin}}(\phi_n * \phi_n^*) - P(\phi_n * \phi_n^*).$$

The prime and polar terms converge in the exponential test topology. For compact projected tests the formula in Chapter 99.3.6 gives their separate diagonals.

99.4 Hadamard Factorization and the Zero-Sum Decomposition

Applying the explicit formula to each pair of compact tests expresses its Gram entry as minus the corresponding zero sum. The definition of the form has already subtracted the polar term.

Theorem 99.4.1 *(zero decomposition of the Gram matrix).* Put $z_\rho = -i(\rho - 1/2)$, where ρ runs over the zeros of ξ with multiplicity. For the compact family d_i of Chapter 99.3.6,

$$q(d_i, d_j) = -\sum_{\rho} h_i(z_\rho)\overline{h_j(\bar z_\rho)}.$$

The sum converges absolutely. The same identity holds for uncut Hermite tests using the extension to $\mathscr{G}$.

Proof. The explicit formula gives $W = P - W_\infty - W_{\mathrm{fin}}$. The form $q = -W = W_\infty + W_{\mathrm{fin}} - P$ therefore equals minus the zero sum. Its polar contribution has already been subtracted. The compact formula and the continuous extension to $\mathscr{G}$ prove the two assertions. □

Corollary 99.4.2 *(complete compact Gram criterion).* The Riemann Hypothesis is equivalent to negative semidefiniteness of every finite principal matrix for the full complex compact family $e_{R,n}$. Equivalently, every finitely supported complex vector c satisfies

$$\sum_{i,j} c_i\overline{c_j}Q_{ij} = -\sum_{\rho}\left(\sum_i c_i h_i(z_\rho)\right)\overline{\left(\sum_i c_i h_i(\bar z_\rho)\right)} \le 0.$$

For either projected family the same criterion characterizes nonpositivity on its specified moment kernel only.

Proof. Matrix nonpositivity is nonpositivity of q on the corresponding finite span. Density from Chapter 99.3.5 and continuity of q extend it to the stated test space. The converse follows by restriction. On all of $\mathscr{A}$, Weil's criterion identifies this condition with RH. Under RH the zero expression is minus a sum of absolute squares. □

Remark 99.4.3 *(restriction and the full form).* A continuous projection $\Pi = 1 - RL$ gives $f = \Pi f + RLf$. Consequently

$$q(f,f) = q(\Pi f, \Pi f) + 2\Re q(\Pi f, RLf) + q(RLf, RLf).$$

A sign theorem on the moment kernel supplies only the first term. The mixed terms and the finite-dimensional complement remain in the full form. The complete compact Gram family retains all of them.

99.5 Finite-N Truncation and Bombieri–Lagarias Numerical Content

At each finite truncation, Q_N is a computable Hermitian matrix for its specified compact tests. A finite comparison with Li tests requires the corresponding change-of-family map. The infinite criterion uses every principal block.

Definition 99.5.1 *(truncated Gram matrix).* . For $N \geq 0$, set $Q_N := (Q_{mn})_{m,n \leq N}$, the $(N+1) \times (N+1)$ upper-left block of the full Gram matrix. It is Hermitian. On a real even projected family it is real symmetric.

THEOREM 99.5.2 *(finite-N Hermite blocks do not follow from Li positivity).* . Negative-semidefiniteness of the Hermite block Q_N is not a formal consequence of the inequalities $\lambda_n \geq 0$ for $1 \leq n \leq N$. Such a transfer would require an explicit finite change-of-test-family matrix between the Hermite cutoff functions and the Li test functions, together with control of the cutoff error. No such matrix is constructed here.

Proof. Li positivity is a family of scalar inequalities for a distinguished Li test family. Negative-semidefiniteness of Q_N is a matrix inequality for the first $N+1$ Hermite cutoff tests. Positivity of one finite list of scalar evaluations does not imply negativity of a different finite Gram block without a proved change of basis and error bound. The Hermite functions are not themselves Paley–Wiener tests, so the cutoff error is part of the statement, not a harmless normalization. □

PROPOSITION 99.5.3 *(finite Li inequalities with error bounds).* Suppose real approximations a_n and proven bounds ε_n satisfy $|\lambda_n - a_n| \leq \varepsilon_n < a_n$ for $1 \leq n \leq N$. Then $\lambda_n > 0$ for those indices. This certifies only that finite list of Li inequalities, not a distinct compact Gram block.

Proof. For each index, $\lambda_n \geq a_n - \varepsilon_n > 0$. A Gram-block assertion involves the quadratic form on all linear combinations of its specified tests, as established above. □

Coffey 2005, Appendix D, pp. 29–32 of arXiv:math-ph/0505052, tabulates approximate Li values for $1 \leq n \leq 100$. The displayed values are positive. A claim through $N = 10^5$ requires its own values and rigorous error bounds; a stated decimal precision alone is not an error bound. The analytic series and Li-test comparisons of Coffey and Bombieri–Lagarias remain distinct from the compact Hermite matrices constructed here.

Proposition 99.5.4 *(a three-mode Gaussian matrix).* For $m, n \in \{0, 4, 8\}$ define the uncut Gaussian matrix $\widetilde{Q}_{mn} = q(\phi_m, \phi_n)$ using the extension to $\mathscr{G}$. Its entries are

$$\widetilde{Q}_{mn} = i^{m-n} \int_{\mathbb{R}} \phi_m(u)\phi_n(u)k(u)du + W_{\mathrm{fin}}(\phi_m * \phi_n^*) - P(\phi_m * \phi_n^*).$$

All three terms converge absolutely. This real symmetric 3×3 matrix is distinct from Q_2, the first three entries of the chosen compact enumeration. All even Hermite indices are real even tests; the choice 0, 4, 8 is a selected three-mode example.

Proof. Each Hermite function lies in $\mathscr{G}$, so the extended explicit formula defines the entries and proves convergence of the prime and polar terms. Fourier self-reciprocity gives the displayed gamma integral, which converges by Gaussian decay. Hermitian symmetry and reality of the chosen tests give a real symmetric matrix. The compact matrix Q_2 instead uses the first three specified functions $\chi_R\phi_n$ in its enumeration. □

Determining the sign of $\widetilde{Q}$ requires estimates for its own gamma integrals, prime tails, and polar evaluations. No sign assertion for this Gaussian matrix or for Q_2 follows from the definition or from a finite list of Li coefficients alone.

The Uniform Bound Obstruction

Theorem 99.5.5 *(all compact Gram blocks).* For the enumeration of the full compact family, let Q_N be its first $N + 1$ principal matrix. Then

$$\sup_N \lambda_{\max}(Q_N) \le 0 \quad \Longleftrightarrow \quad q(f, f) \le 0 \ (f \in \mathscr{A}).$$

The latter condition is equivalent to RH. For either moment-projected family, the same statement characterizes nonpositivity on that moment space.

Proof. A finitely supported vector lies in some initial block. Thus the displayed eigenvalue condition is exactly nonpositivity on the finite span. Apply compact-topology density and continuity. No uniform strict spectral gap is used. □

Remark 99.5.6 *(coefficient norms).* These compact families need not be linearly independent. A strict bound $Q_N \preceq -cI$ for $c > 0$ refers to the chosen coefficient norm and can fail because of linear dependence alone. It is not equivalent to the Weil inequality.

99.6 Three Natural Regularizations and Their Failure

Tikhonov shifts, diagonal damping, and finite-rank perturbations change the finite matrices in different ways. The complete compact criterion still requires all principal blocks. Compact completeness has been proved above.

Tikhonov Regularization

Definition 99.6.1 *(Tikhonov regularized Gram).* . For $\varepsilon > 0$, set $Q_N^\varepsilon := Q_N - \varepsilon \cdot I_{N+1}$ (a negative shift by ε).

Proposition 99.6.2 *(Tikhonov shift and the ceiling).* . For fixed N and $\varepsilon > 0$, $Q_N^\varepsilon \prec 0$ if and only if $\lambda_{\max}(Q_N) < \varepsilon$. This is a finite-dimensional shift, not a proof of $Q_N \preceq 0$, and no unshifted Gram conclusion follows from a finite list of Li coefficients alone.

Proof. $Q_N^\varepsilon c = (Q_N - \varepsilon I)c$, and $\langle Q_N^\varepsilon c, c\rangle = \langle Q_N c, c\rangle - \varepsilon\|c\|^2$; it is strictly negative for every non-zero c exactly when every eigenvalue of Q_N is $< \varepsilon$. □

Remark 99.6.3 *(Tikhonov does not reach A-TP).* Tikhonov regularization with ε-dependent shift does not extend to $N \to \infty$: the uniform bound $\lambda_{\max}(Q_N) \le \varepsilon$ uniformly in N does not imply the desired unshifted inequality unless $\varepsilon \downarrow 0$ is controlled. The regularization changes the question.

Gaussian Windowing

Definition 99.6.4 *(diagonally damped Gram form).* For $\sigma > 0$, define $Q^\sigma_{mn} = e^{-(m+n)\sigma/2} Q_{mn}$. On finitely supported complex sequences put

$$q_{Q^\sigma}(c) = \sum_{m,n} c_m \overline{c_n} Q_{mn} e^{-(m+n)\sigma/2}.$$

This is the coefficient form $q_Q(T_\sigma c)$ with $(T_\sigma c)_n = e^{-n\sigma/2} c_n$.

LEMMA 99.6.5 *(diagonal damping on finite vectors).* For any $\sigma > 0$, put $(T_\sigma c)_n = e^{-n\sigma/2} c_n$. On finitely supported complex sequences, $q_{Q^\sigma}(c) = q_Q(T_\sigma c)$. Thus Q^σ is negative semidefinite on every finite vector if and only if Q is.

Proof. The map T_σ and its coordinatewise inverse both preserve finite support. The equality follows by expanding the finite sums. No assertion that the infinite matrix defines an operator on all of ℓ^2 is required. □

Finite-Rank Perturbation Ceiling

THEOREM 99.6.6 *(finite-rank perturbations of the coefficient form).* Let $V = c_{00}(\mathbb{N}_0; \mathbb{C})$ and define

$$q_Q(c, d) = \sum_{i,j} c_i \overline{d_j} Q_{ij}.$$

Here $Q_{ij} = q(e_i, e_j)$ is the compact Gram family and both sums are finite. Let $V^\times$ be the algebraic conjugate-linear dual. Suppose $p : V \times V \to \mathbb{C}$ is Hermitian, linear in its first argument, and its linear map

$$P : V \to V^\times, \qquad (Pc)(d) = p(c, d),$$

has finite rank r. Put $K = \ker P$. Then K has codimension r and is infinite-dimensional. For $c \in K$,

$$(q_Q + p)(c, c) = q_Q(c, c).$$

Consequently, nonpositivity of $q_Q + p$ on V implies nonpositivity of q_Q on K. The same identity holds on any specified common vector domain of two Hermitian forms, with K the kernel of the second form map.

Proof. The map P identifies V/K with its r-dimensional image. Thus K has finite codimension in the infinite-dimensional space V. For $c \in K$, the functional Pc vanishes on every vector, including c. Hence $p(c, c) = 0$, which proves the identity and sign implication. Hermitian symmetry also gives $p(d, c) = 0$ for every $d \in V$. No boundedness, closability, or operator realization on ℓ^2 is needed for these finite coefficient sums. □

Remark 99.6.7 *(existence and the remaining sign condition).* Finite-rank forms exist: if $L : V \to \mathbb{C}^r$ is linear and A is an $r \times r$ Hermitian matrix, then $p(c,d) = \sum_{a,b} (Lc)_a \overline{(Ld)}_b A_{ab}$ has rank at most r. A specified such form changes no diagonal value on its kernel. The existence of a finite-rank p with $q_Q + p \le 0$ remains a separate assertion. The displayed identity supplies a necessary condition on its infinite-dimensional kernel, not that existence.

Remark 99.6.8 *(what the three modifications determine).* The Tikhonov shift gives an exact finite eigenvalue criterion. Invertible diagonal damping preserves the sign condition on finite vectors. A finite-rank perturbation leaves the form unchanged on its kernel. These statements give no sign theorem on the complete compact test family without the remaining quadratic inequalities. They do not exclude additional arguments using a specified perturbation.

99.7 The full form and restricted realizations

Theorem 99.7.1 *(compact matrix criterion).* The complete compact Gram family characterizes the full Weil inequality. Its two projected families characterize the respective $(0,1)$ Mellin moment and polar-moment restrictions. Every compact entry is given by the signed gamma integral, a finite prime sum, and the polar Hermitian form. Its zero decomposition is minus the full zero sum.

Proof. The entry formula and zero decomposition were proved above. The density theorem and continuity of the form give the three matrix criteria on their stated carriers. □

Remark 99.7.2 *(finite comparisons).* A finite matrix calculation establishes its own finite inequality. An inference from Li coefficients requires a proved finite comparison. A Tikhonov shift changes the finite eigenvalues. Invertible diagonal damping preserves the finite sign question. A finite-rank perturbation leaves the quadratic form unchanged on its kernel. The auxiliary integral-energy criterion controls G_g, with $W_\infty = 2 \log \pi \langle\ ,\ \rangle_2 - G_g$. It does not determine the remaining prime and polar contributions.

99.8 Gram forms and pairing maps

Remark 99.8.1 *(positive Hilbert realization).* If a linear map $I : E \to H$ into a positive Hilbert space satisfies $\langle If, Ih \rangle_H = W(f * h^*)$, then the Weil form is nonnegative on E. For $E = \mathscr{A}$, such a realization is equivalent to the full inequality by quotienting its nullspace and completing. A realization of an arithmetic or automorphic sector additionally requires its native carrier, the map I, and this exact pairing identity. Moment projection and Hermite expansion supply test functions, but neither construction supplies that native pairing identity.

Chapter 100

A–TP through the Guinand–Burnol Co–Poisson Alternative

The Weil distribution has additive and multiplicative descriptions, related by the logarithmic coordinate. Its signed archimedean term is a distribution, and its full positivity criterion retains both polar evaluations and the prime term. The moment condition and its finite-dimensional complement can be analyzed before a co-Poisson or automorphic realization is constructed.

100.1 The full pairing and logarithmic transport

For $f \in \mathscr{A} = C_c^\infty(\mathbb{R})$, put $f^*(t) = \overline{f(-t)}$ and $h_f(z) = \int f(t)e^{izt}\,dt$. The arithmetic positivity condition is

$$W_\infty(f * f^*) \le P(f * f^*) - W_{\mathrm{fin}}(f * f^*),$$

with the signed conventions of Chapter 78.1.2. Its difference is $q_W(f,f) = W(f * f^*)$. On the full compact-test space this is equivalent to RH. A restricted Petersson realization requires a specified map and equality with this entire form. Rankin–Selberg coefficient multiplication does not give that identity. Indeed Chapter 78.5.5 gives a negative finite-prime square on compact tests.

The multiplicative moment question

The substitution $x = e^t$ transports compact additive tests to compact multiplicative tests with Haar measure dx/x. It gives a convergent mean for a transported autocorrelation. The additive expression evaluated directly against dy/y instead diverges at zero for a nonzero test. The mean-zero kernel below has an explicit projection. Extending positivity from that kernel requires its finite complement and the cross pairings with it. A proposed co-Poisson identification must supply an actual transform and the complete signed pairing on these carriers.

Classical references

The explicit-formula, theta and Hilbert-transform comparisons use the following references. Rankin 1939, Selberg 1940, Hecke 1936 and Deligne 1974 concern the separate automorphic coefficient comparison. Hurwitz 1882, Gauss 1813, Whittaker–Watson 1927 and Boas 1954 provide the special-function and entire-function background.

Guinand 1948. Guinand, A. P. *A summation formula in the theory of prime numbers*. *Proc. London Math. Soc.* (2) 50 (1948), 107–119. Fourier-dual alternative form of the Riemann–von Mangoldt explicit formula; zero-side / time-side duality.

Guinand 1945. Guinand, A. P. *Some formulae for the Riemann zeta-function*. *Quart. J. Math. Oxford* 16 (1945), 1–10. Self-dual summation identity; time-domain pairing formalism.

Burnol 2001. Burnol, J.-F. *Sur certains espaces de Hilbert de fonctions entières, liés à la transformation de Fourier et aux fonctions L de Dirichlet et de Riemann*. *C. R. Acad. Sci. Paris* 333 (2001), 201–206. Fourier transformation and entire-function Hilbert spaces.

Burnol 2002. Burnol, J.-F. *Sur les "espaces de Sonine" associés par de Branges à la transformation de Fourier*. *C. R. Math.* 335 (2002), 689–692, doi:10.1016/S1631-073X(02)02546-3; arXiv:math/0208121, §§2–5, Theorems 4 and 8. Sonine orthogonal projection and the associated de Branges space of completed Mellin transforms.

For this Sonine result use the cyclic Fourier convention $\mathcal{F}_{2\pi}f(x) = \int_{\mathbb{R}} e^{2\pi i x y} f(y)\,dy$. On even $L^2(\mathbb{R})$, normalized by $\|f\|^2 = \int_0^\infty |f(t)|^2\,dt$, put

$$K_\lambda = \{f : f = \mathcal{F}_{2\pi}f = 0 \text{ on } (-\lambda, \lambda)\}, \qquad \lambda > 0.$$

This is a closed subspace: it is the intersection of the kernels of two bounded restriction maps. Thus it has an orthogonal projection; Burnol's Theorem 4 computes it using the compressed cosine transform. For $f \in K_\lambda$, the integral $\widehat{f}(s) = \int_0^\infty f(t) t^{-s}\,dt$ converges for $\Re s > 1/2$ by Cauchy–Schwarz. Its completed continuation $\pi^{-s/2}\Gamma(s/2)\widehat{f}(s)$ is entire. Burnol's Theorem 8 identifies their Hilbert space isometrically with $\mathcal{B}(E_\lambda)$, using the critical line $\Re s = 1/2$ and measure $|ds|/(2\pi)$ in the Hardy-space definition. This supplies a specific Fourier–Mellin Hilbert space. A comparison with the complete Weil pairing in this chapter still requires a map from the chosen compact-test domain to K_λ and a proof of the identity of forms, including prime and polar terms.

Secondary: Riesz 1928 on the $M(x)$ function for the asymptotic co-Poisson comparison; Cotlar 1955 on the Hilbert transform almost everywhere convergence; Cartier–Voros 1990 for the distributional extension of Guinand's formula to Schwartz test functions; Meyer R. 2005 on distributional Selberg moments.

100.2 The Guinand Alternative Explicit Formula

The compact explicit formula admits a spectral-density notation in which the gamma term occurs with a plus sign. Its relation to the signed Weil density is a sign reversal, not a change in the arithmetic functional.

The completed function and Fourier tests

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. Its logarithmic derivative has a pole at each zero ρ with residue its multiplicity. It has no pole at 0 or 1, since $\xi(0) = \xi(1) = 1/2$. By contrast, $\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ has simple poles there, and its logarithmic derivative has residue -1. The identity

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{\log \pi}{2} + \frac{1}{2}\psi(s/2) + \frac{\zeta'}{\zeta}(s)$$

holds meromorphically, with cancellations at 0, 1.

Definition 100.2.1 *(Fourier-pair conventions).* For an even $v \in \mathscr{A}$, use the pair

$$h(t) = \int_{\mathbb{R}} v(u) e^{itu}\,du, \qquad v(u) = \frac{1}{2\pi}\int_{\mathbb{R}} h(t) e^{-itu}\,dt.$$

The function h is entire. Evenness alone does not imply reality; when v is real and even, h is real and even on the real axis. The letter v distinguishes this inverse transform from the auxiliary digamma density g.

Theorem 100.2.2 *(the compact explicit formula in spectral-density form).* For the pair (h, v) above and $z_\rho = -i(\rho - 1/2)$,

$$\sum_\rho h(z_\rho) = h(i/2) + h(-i/2) - 2\sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} v(\log n) + \int_{\mathbb{R}} h(t)\Phi_\infty^G(t)dt,$$

where

$$\Phi_\infty^G(t) = \frac{\Re\psi(1/4 + it/2) - \log\pi}{2\pi}.$$

Zeros are counted with multiplicity. The zero sum is absolutely convergent, and the prime sum is finite.

Proof. Apply Chapter 78.1.2 to v. Evenness combines the two prime evaluations. Its signed gamma density is $-\Phi_\infty^G$, which gives the plus sign above. This establishes the formula on the exact compact Fourier carrier. A contour proof would use the test $h(-i(s - 1/2))$, count the actual poles of the chosen logarithmic derivative, and use the Dirichlet series for $-\zeta'/\zeta$ only in $\Re s > 1$ before continuation. No direct critical-line Dirichlet-series expansion is required here. Guinand 1948 and Patterson 1988, Ch. 3, treat related explicit-formula conventions. □

Signed and auxiliary gamma conventions

Proposition 100.2.3 *(the exact convention dictionary).* With $\Phi_\infty^W(t) = k(t)/(2\pi)$ and $\Phi_\infty^{\mathrm{aux}}(t) = g(t)/(2\pi)$, where $k = \log\pi - \Re\psi(1/4 + it/2)$ and $g = 2\log\pi - k$,

$$\Phi_\infty^G = -\Phi_\infty^W = \Phi_\infty^{\mathrm{aux}} - \frac{\log\pi}{\pi}.$$

The constant difference pairs with h as $-2\log\pi\, v(0)$.

Proof. The density identities follow by substitution. Fourier inversion at zero gives $\int h(t)dt = 2\pi v(0)$. □

Remark 100.2.4 *(the constant term on convolution squares).* For $v = f * f^*$, the constant difference is $-2\log\pi\|f\|_2^2$. It is not a multiple of $\int f$, and it does not vanish merely because f has mean zero. Changing from the auxiliary density to the signed gamma density therefore requires the complete norm correction. The full explicit formula is unchanged only when this correction and its sign remain.

100.3 The Burnol Co–Poisson Transform

Ordinary Poisson summation exchanges a Schwartz function with its Fourier transform. A reciprocal change of variables is a different operation. The following calculation fixes that distinction before posing a co-Poisson realization of the Weil form.

Jacobi Theta and Co–Poisson

Definition 100.3.1 *(Jacobi theta function).* . Let

$$\theta(x) := \sum_{n\in\mathbb{Z}} e^{-\pi n^2 x} = 1 + 2\sum_{n\geq 1} e^{-\pi n^2 x}, \qquad x > 0,$$

with functional equation $x^{1/2}\theta(x) = \theta(1/x)$ (Jacobi 1828; modular S-transform).

Definition 100.3.2 *(the mean-subtracted dilation sum).* Let $\varphi \in \mathcal{S}(\mathbb{R})$ be real and even, with $\varphi(0) = 0$ and $\int_0^\infty \varphi = 0$. Use the cyclic Fourier transform $\widehat{\varphi}_2(\xi) = \int \varphi(t) e^{-2\pi i t\xi}\, dt$. Set

$$(C\varphi)(y) = \sum_{n\geq 1} \varphi(ny) - \frac{1}{y}\int_0^\infty \varphi(t)\,dt, \qquad y > 0.$$

The second term vanishes under the stated mean condition, and the series converges absolutely for every $y > 0$.

THEOREM 100.3.3 *(Poisson symmetry of the dilation sum).* The cyclic transform preserves the stated even Schwartz class with zero value and zero mean. On this class,

$$(C\varphi)(y) = y^{-1}(C\widehat{\varphi}_2)(1/y).$$

Both sides decrease faster than every power at zero and infinity. Writing $M_+\varphi(s) = \int_0^\infty \varphi(t) t^{s-1}\,dt$, one has

$$M_+(C\varphi)(s) = \zeta(s) M_+\varphi(s) \quad (0 < \Re s < 1),$$

where the product is continued from $\Re s > 1$.

Proof. The values $\widehat{\varphi}_2(0) = \int \varphi = 0$ and $\int \widehat{\varphi}_2 = \varphi(0) = 0$ follow from Fourier inversion. Poisson summation gives $\sum_{n\in\mathbb{Z}} \varphi(ny) = y^{-1} \sum_{n\in\mathbb{Z}} \widehat{\varphi}_2(n/y)$. Remove the zero terms and use evenness to obtain the identity. Schwartz decay bounds each dilation sum by $C_N y^{-N}$ for $y \geq 1$. The identity gives arbitrary-power decay at zero. For $\Re s > 1$, absolute convergence permits termwise Mellin integration and the change of variables $t = ny$, giving $\zeta(s) M_+\varphi(s)$. Because $\varphi(t) = O(t^2)$ at zero, $M_+\varphi$ is holomorphic on $\Re s > -2$. Its value at one vanishes, cancelling the pole of ζ there. Analytic continuation gives the asserted strip. The Mellin transform of $C\varphi$ itself is entire by its two-sided rapid decay. □

Remark 100.3.4 *(reciprocal pullback and the native comparison).* For $J\varphi(t) = t^{-1}\varphi(1/t)$ on the positive line, $M_+(J\varphi)(1-s) = M_+\varphi(s)$ wherever the integrals converge. It is not a substitute for $\widehat{\varphi}_2$ in the Poisson identity. Choose a nonzero smooth test supported away from zero with $\int_0^\infty \varphi(t)\,dt = \int_0^\infty \varphi(t)\,dt/t = 0$, and extend it evenly. Such a test exists by solving two homogeneous linear equations on three disjoint compact bumps. Then both φ and the even extension of $J\varphi$ belong to the stated zero-mean class. If their reciprocal dilation-sum identity held, Mellin continuation would give $(\zeta(s) - \zeta(1-s)) M_+\varphi(s) = 0$. The meromorphic factor is not identically zero, so Mellin uniqueness would force $\varphi = 0$, a contradiction. Burnol's Sonine and co-Poisson spaces concern additional specified Fourier-compatible constructions. Identifying a realization in one of those spaces with the full signed Weil pairing remains a separate map and pairing problem; the elementary Poisson identity alone provides no zero-spectrum decomposition or positivity theorem.

The archimedean distribution in multiplicative coordinates

Let $Lf(x) = f(\log x)$ for $x > 0$. The transported archimedean functional is $T(\Phi) = W_\infty(L^{-1}\Phi)$ on $C_c^\infty(\mathbb{R}_{>0})$ with its usual compact-test topology. It has an explicit convergent subtraction formula.

PROPOSITION 100.3.5 *(the exact transported gamma distribution).* For $\Phi = Lf$, one has

$$T(\Phi) = W_\infty(f) = \frac{1}{2\pi}\int_{\mathbb{R}} h_f(u)(\log\pi - \Re\psi(1/4 + iu/2))\,du,$$

$$T(\Phi) = (\log\pi + \gamma)\Phi(1) + \int_1^\infty \frac{x^{-1/2}[\Phi(x) + \Phi(x^{-1})] - 2x^{-2}\Phi(1)}{1 - x^{-2}}\,\frac{dx}{x}.$$

Here γ is Euler's constant. Both expressions converge. Away from $x = 1$, its density relative to dx/x is

$$K(x) = \frac{e^{-|\log x|/2}}{1 - e^{-2|\log x|}} > 0.$$

At $x = 1$ this density requires the subtraction in the preceding formula and is not locally integrable by itself.

Proof. Use the digamma identity already applied in Chapter 78.1.3: $\psi(z) + \gamma = \int_0^\infty (e^{-v} - e^{-zv})/(1 - e^{-v})\, dv$. First truncate the v integral away from zero and infinity. Fourier inversion gives $f(0)$ from the constant term and $[f(v/2) + f(-v/2)]/2$ from $\cos(uv/2)$. Put $v = 2t$. The result is

$$W_\infty(f) = (\log \pi + \gamma) f(0) + \int_0^\infty \frac{e^{-t/2}[f(t) + f(-t)] - 2e^{-2t} f(0)}{1 - e^{-2t}} dt.$$

The numerator is $O(t)$ at zero and the denominator is asymptotic to $2t$. At infinity the compact terms vanish and the remaining term decays exponentially. Thus the integral converges. The truncated digamma integrands, after integration, are bounded by a constant times $1 + \log(2 + |u|)$, uniformly in the truncations: near $v = 0$, use $1 - \cos(uv/2) \le \min(2, u^2v^2/8)$. Rapid decay of h_f permits dominated convergence. The substitution $x = e^t$ gives the displayed formula. For tests vanishing near one, the subtraction vanishes and the change $x \mapsto x^{-1}$ gives the stated density on both sides. Its asymptotic $K(x) \sim 1/(2|\log x|)$ proves the last assertion. □

Remark 100.3.6 *(support and sign).* The off-diagonal density is positive on both $(0, 1)$ and $(1, \infty)$. The subtraction at one is part of the distribution. There is no sign change at $1/\pi$ in this exact density. Positivity of the off-diagonal density does not imply positivity of its quadratic pullback, because the subtraction must remain.

100.4 Co–Poisson Diagnostic and Exact Restatement

The archimedean term on multiplicative tests is the distribution T above. Its application to a transported autocorrelation retains the subtraction at one. Hilbert-transform parity gives a separate elementary identity, but identifying a Hilbert-transform pairing with T requires an operator comparison.

Hilbert Transform on Even Functions

Definition 100.4.1 *(Hilbert transform).* . For $\phi \in L^2(\mathbb{R})$, the Hilbert transform $H\phi$ is the Cauchy principal value

$$(H\phi)(t) = \frac{1}{\pi}\, \text{p.v.} \int_{-\infty}^{\infty} \frac{\phi(s)}{t - s}\, ds. \tag{100.1}$$

The operator $H\colon L^2 \to L^2$ is isometric with $H^2 = -\mathrm{id}$, and the pointwise convergence of the principal value holds almost everywhere (Cotlar 1955).

Lemma 100.4.2 *(Hilbert symmetry on even functions).* . For $\phi \in L^2(\mathbb{R})$ even, $H\phi$ is odd. For ϕ odd, $H\phi$ is even. The pairing $\langle \phi, H\chi \rangle$ vanishes whenever ϕ and χ share the same parity.

Proof. From (100.1), substitute $s \mapsto -s$, $t \mapsto -t$: if $\phi(-s) = \phi(s)$ then $(H\phi)(-t) = \frac{1}{\pi}\, \text{p.v.} \int \frac{\phi(s)}{-t-s}\, ds = -\frac{1}{\pi}\, \text{p.v.} \int \frac{\phi(s)}{t+s}\, ds = -\frac{1}{\pi}\, \text{p.v.} \int \frac{\phi(-u)}{t-u}\, du = -(H\phi)(t)$, with $u = -s$ and evenness. So $H\phi$ is odd. The parity-orthogonality follows from $\int (\text{odd}) \cdot (\text{even}) = 0$. □

PROPOSITION 100.4.3 *(the full inequality after logarithmic transport).* For $f \in \mathscr{A}$, put $\varphi = f * f^*$ and $\Phi(x) = \varphi(\log x)$. Then archimedean Tate positivity for this test is exactly

$$T(\Phi) \le P(\varphi) - W_{\mathrm{fin}}(\varphi).$$

It is equivalent to $q_W(f,f) \ge 0$.

Proof. By definition $T(\Phi) = W_\infty(\varphi)$. The signed explicit formula gives $W(\varphi) = P(\varphi) - W_{\mathrm{fin}}(\varphi) - T(\Phi)$. Rearranging proves the assertion. No additional zero term is added to the right-hand side of the inequality. □

The moment condition in logarithmic coordinates

An additive autocorrelation cannot have a vanishing unregularized dy/y mean near its additive origin. This determines the necessary change of variables before a multiplicative moment is imposed. For $f \in \mathscr{A}$, put $\varphi = f * f^*$. Then $\varphi(0) = \|f\|_2^2$. If $f \ne 0$, continuity gives $\Re\varphi(y) \ge \|f\|_2^2/2$ for sufficiently small positive y. Consequently $\int_0^\infty \varphi(y)\,dy/y$ diverges at zero.

Define the logarithmic transport

$$(Lf)(x) = f(\log x), \qquad x > 0.$$

It is an isomorphism from $C_c^\infty(\mathbb{R})$ to $C_c^\infty(\mathbb{R}_{>0})$. On the latter space use multiplicative Haar measure dx/x, convolution $(F *_\times G)(x) = \int_0^\infty F(y)G(x/y)\,dy/y$, and $F^{*\times}(x) = \overline{F(1/x)}$. Changing variables $y = e^t$ gives

$$L(f * g) = Lf *_\times Lg, \qquad L(f^*) = (Lf)^{*\times}.$$

Thus the multiplicative autocorrelation is $\Phi(x) = \varphi(\log x)$ and

$$\int_0^\infty \Phi(x)\frac{dx}{x} = \int_{\mathbb{R}} \varphi(t)\,dt = \left|\int_{\mathbb{R}} f(t)\,dt\right|^2.$$

It is compactly supported away from zero, so the integral converges.

Definition 100.4.4 *(the mean-zero moment space).* Set $K_0 = \{f \in \mathscr{A} : \int f = 0\}$. For real even tests, use its real even part $K_0^{\mathbb{R},\mathrm{ev}}$. The multiplicative mean of $L(f * f^*)$ vanishes exactly on K_0. Give these spaces the compact-test topology. Write

$$C_0 = \{f \in K_0 : q_W(f,f) \ge 0\}, \qquad q_W(f,g) = W(f * g^*).$$

Membership in C_0 is a quadratic inequality in addition to the linear moment condition. No identification with a Sonine space or a co-Poisson image is included in this definition.

PROPOSITION 100.4.5 *(nonzero moment tests and their projection).* Fix a compact smooth η of integral one. The map $\Pi_0 f = f - (\int f)\eta$ is a continuous projection onto K_0, which is closed of complex codimension one. Choosing η real and even restricts this projection to real even tests. This restricted kernel contains nonzero functions.

Proof. The integral is continuous on compact tests and has value one on η. Therefore $\Pi_0^2 = \Pi_0$ and its image is exactly K_0. For a nonzero real even compact smooth b, the derivative b'' is real and even and has integral zero. It is nonzero, since a compactly supported affine function must vanish. □

Theorem 100.4.6 *(closed positivity on the moment space).* The set C_0 is closed in K_0. It is dense there if and only if $q_W(f,f) \geq 0$ for every $f \in K_0$. If $S \subset \mathscr{A}$ is dense, then $\Pi_0 S$ is dense in K_0. To deduce positivity from a dense linear span, the form must be nonnegative on every finite linear combination from that span.

Proof. Convolution and W are continuous on compact tests, so $f \mapsto q_W(f,f)$ is a continuous real-valued function. Its inverse image of $[0,\infty)$ is closed. A closed dense subset of K_0 equals K_0, proving the equivalence. Continuity and surjectivity of Π_0 prove the density assertion. For two generators, positivity of their individual diagonals gives no bound on the cross term $2\Re q_W(f,g)$ of their sum. This proves the last distinction. □

The finite complement and the full form

Every compact test has the exact decomposition $f = u + c\eta$ with $u = \Pi_0 f \in K_0$ and $c = \int f$. Thus

$$q_W(f,f) = q_W(u,u) + 2\Re\big(\bar{c}\, q_W(u,\eta)\big) + |c|^2 q_W(\eta,\eta).$$

The missing part of a full positivity proof is determined by this one-dimensional complement and its cross pairings.

Theorem 100.4.7 *(extension from the mean-zero kernel).* Assume $q_W(u,u) \geq 0$ for all $u \in K_0$. Then full Weil positivity is equivalent to

$$q_W(\eta,\eta) \geq 0, \qquad |q_W(u,\eta)|^2 \leq q_W(u,u) q_W(\eta,\eta) \quad (u \in K_0).$$

In particular, if $q_W(\eta,\eta) = 0$, the required condition is $q_W(u,\eta) = 0$ for every $u \in K_0$.

Proof. Full positivity implies these inequalities by Cauchy–Schwarz. Conversely the cross bound and the preceding decomposition give

$$q_W(u + c\eta, u + c\eta) \geq \left(\sqrt{q_W(u,u)} - |c|\sqrt{q_W(\eta,\eta)}\right)^2 \geq 0.$$

All compact tests have this decomposition. □

Corollary 100.4.8 *(a sufficient norm bound).* Under positivity on K_0, suppose the functional $u \mapsto q_W(u,\eta)$ is bounded for the seminorm $q_W(u,u)^{1/2}$ with squared bound at most $q_W(\eta,\eta)$, and that $q_W(\eta,\eta) \geq 0$. Then full Weil positivity holds and hence RH holds.

Proof. The bound is exactly the cross inequality in the preceding theorem. Apply the compact Weil criterion. □

Proposition 100.4.9 *(what Hilbert-transform isometry supplies).* The Hilbert transform on $L^2(\mathbb{R})$ is skew-adjoint. Thus $\Re\langle v, Hv\rangle = 0$. Its norm isometry supplies no identity with the cross pairing $q_W(u,\eta)$ above.

Proof. The Fourier multiplier of the Hilbert transform is purely imaginary, of modulus one almost everywhere, and changes sign under complex conjugation. Hence $H^* = -H$ and $H^*H = 1$. Taking the real part of its quadratic pairing gives zero. The cross pairing instead uses the full Weil distribution. Equating the two would require an additional operator identity. □

100.5 THE MULTIPLICATIVE COMPARISON PROBLEM

THEOREM 100.5.1 *(moment transport and the positivity boundary).* The logarithmic map L transports additive convolution and involution to multiplicative convolution and involution. The multiplicative mean-zero autocorrelations correspond exactly to K_0. This kernel has the explicit continuous projection Π_0. Its nonnegative Weil locus is closed. Positivity on K_0 extends to all compact tests precisely under the complement inequalities in Chapter 100.4.7.

Proof. The change-of-variables calculations, projection construction, closedness argument, and quadratic decomposition above prove each assertion. □

Remark 100.5.2 *(the proposed co-Poisson realization).* A co-Poisson realization must specify a space, a transform, its measure and domain, and an identity with the full signed Weil form. The exact transport L gives the requisite multiplicative test algebra. It does not turn Fourier transformation into reciprocal pullback, nor produce an ordinary time-domain kernel from a spectral logarithmic derivative. Such a kernel requires a proved distributional transform and a prescription at its singularities. In particular the asserted split at $1/\pi$ cannot be used without an exact kernel and sign proof. A bound on one auxiliary spectral form also supplies no cross inequality for q_W. The nonzero tests b'' establish that the linear moment kernel is nonempty. Their full Weil positivity requires a separate proof. Likewise density of a proposed co-Poisson image must be proved in a topology controlling the complete Weil form. The explicit projection and complement criterion state exactly what such a density and positivity argument would have to establish.

100.6 POSITIVE SECTORS OF THE WEIL FORM

The full compact-test functional W of Theorem 78.1.2 has positive sectors without assuming its positivity on the entire test algebra. Its prime-power terms couple test functions on separated intervals. A sufficiently small width makes the archimedean contribution dominate these couplings. This gives a positive space of functions with infinitely many independent directions, while retaining the full arithmetic functional.

Conventions and the full form

For $f \in C_c^\infty(\mathbb{R})$ use

$$\widehat{f}(t) = \int_{\mathbb{R}} f(u) e^{itu}\, du, \qquad f^*(u) = \overline{f(-u)}, \qquad (f * g)(u) = \int_{\mathbb{R}} f(v) g(u-v)\, dv.$$

Let $\Lambda(n)$ be the von Mangoldt function: it is $\log p$ for $n = p^a$ and zero otherwise. Put

$$k_\infty(t) = \operatorname{Re} \psi(1/4 + it/2) - \log \pi,$$

where $\psi = \Gamma'/\Gamma$. Define the full arithmetic functional by

$$\begin{aligned} L_\zeta(h) = \widehat{h}(i/2) + \widehat{h}(-i/2) + \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{h}(t) k_\infty(t)\, dt \\ - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \big(h(\log n) + h(-\log n) \big). \end{aligned} \tag{100.2}$$

With the preceding sign convention, $L_\zeta(h) = W(h)$. For compact smooth h the prime sum is finite and the Fourier integral converges. Define

$$B_\zeta(f,g) = L_\zeta(f^* * g), \qquad Q_\zeta(f) = B_\zeta(f,f).$$

The form is conjugate-linear in its first argument, linear in its second and Hermitian. The estimates below use this displayed functional directly.

A coercive estimate on one narrow interval

The convergent digamma series gives

$$k_\infty(t) = k_\infty(0) + \sum_{n\ge 0}\left(\frac{1}{n+1/4} - \frac{n+1/4}{(n+1/4)^2 + t^2/4}\right). \tag{100.3}$$

Every summand is nonnegative and increasing with $|t|$. Also

$$k_\infty(0) = -\gamma - \frac{\pi}{2} - 3\log 2 - \log\pi > -6.$$

For example $\gamma < 1, \pi < 22/7, \log 2 < 7/10$ and $\log\pi < 6/5$ suffice. The last two exponential inequalities follow from their finite positive Taylor lower bounds. Finally $k_\infty(T) \to +\infty$: for $n + 1/4 \le T/2$, the corresponding summand in (100.3) is at least $1/(2(n+1/4))$, whose partial sums diverge.

Let f have support in an interval of length L. Cauchy–Schwarz gives $|\widehat{f}(t)|^2 \le L\|f\|_2^2$, and Plancherel gives

$$\frac{1}{2\pi}\int_{-T}^{T} |\widehat{f}(t)|^2 dt \le \frac{LT}{\pi}\|f\|_2^2, \qquad \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(t)|^2 dt = \|f\|_2^2.$$

Let $T > 0$ and $M \ge -6$ satisfy $k_\infty(T) \ge M$. The archimedean contribution is then at least

$$\left(M - (M+6)\frac{LT}{\pi}\right)\|f\|_2^2.$$

When $L < \log 2$, the support of $f^* * f$ contains no nonzero prime-power logarithm, so its prime contribution is zero. The polar term is bounded below by $-2Le^{L/2}\|f\|_2^2$: translate the interval to be centred at zero and apply Cauchy–Schwarz to $\int f(u)e^{u/2}du$ and $\int f(u)e^{-u/2}du$. The translation factors cancel between these two integrals. Consequently

$$Q_\zeta(f) \ge c(L,T,M)\|f\|_2^2, \qquad c(L,T,M) = M - (M+6)\frac{LT}{\pi} - 2Le^{L/2}. \tag{100.4}$$

The estimate holds uniformly over all smooth functions in the interval.

The cross-window kernel retains the arithmetic primes

Away from zero, the distribution given by the sum of the polar and archimedean contributions in (100.2) is the smooth even kernel

$$K(u) = 2\cosh(u/2) - \frac{e^{-|u|/2}}{1 - e^{-2|u|}} \qquad (u \ne 0). \tag{100.5}$$

To verify it, use the integral expression

$$\psi(a+it/2) = -\gamma + \int_0^\infty \frac{e^{-v} - e^{-(a+it/2)v}}{1-e^{-v}}\,dv.$$

Test on a smooth function supported away from zero. Terms constant in t vanish on that test. Fourier inversion of the cosine term, followed by $v = 2|u|$, gives the negative second summand in (100.5). The two polar evaluations give $2\cosh(u/2)$. This argument is on separated supports; it does not ignore a delta term at zero in the diagonal form.

Fix distinct positions $a_1, \dots, a_N$. Let $I_L = (-L/2, L/2)$ and

$$F(u) = \sum_{j=1}^{N} f_j(u - a_j), \qquad f_j \in C_c^\infty(I_L).$$

Choose L so small that these supports are disjoint and each difference interval contains a prime-power logarithm only if its centre equals that logarithm. This is possible because there are only finitely many prime powers in a bounded interval. Put

$$w_{ij} = \begin{cases} \Lambda(n)/\sqrt{n}, & |a_i - a_j| = \log n \text{ for an integer } n \geq 2, \\ 0, & \text{otherwise.} \end{cases}$$

For a resonant pair its prime term is exactly $-w_{ij}\int f_j(v)\overline{f_i(v)}dv$. The smooth part is the integral kernel (100.5) evaluated at $a_j - a_i + v - u$. If M_{ij} bounds $|K|$ there, its absolute value is at most $LM_{ij}\|f_i\|_2\|f_j\|_2$. Thus

$$Q_\zeta(F) \geq \left(c(L,T,M) - \max_i \sum_{j\neq i} (w_{ij} + LM_{ij}) \right) \sum_i \|f_i\|_2^2. \tag{100.6}$$

Apply $2ab \leq a^2 + b^2$ to each unordered pair to obtain this bound.

Theorem 100.6.1 *(Positive arithmetic windows of arbitrary finite multiplicity).* For any fixed finite set of distinct real positions $a_1, \dots, a_N$, there exists $L > 0$ such that the full Weil form is positive definite on the infinite-dimensional space

$$\bigoplus_{j=1}^{N} C_c^\infty(a_j + I_L).$$

It has a uniform strictly positive lower bound by the L^2 norm. All prime-power terms in (100.2) are retained.

Proof. The finite row sums of the w_{ij} are fixed. Choose M larger than their maximum plus two, and then T with $k_\infty(T) \geq M$. For sufficiently small L, the separation requirements hold, LT and $Le^{L/2}$ tend to zero, and the M_{ij} remain bounded because their arguments stay away from zero. The coefficient in (100.6) is then positive, in fact larger than one after making L smaller. □

An explicit four-window theorem with a prime-power coupling

Choose

$$(a_0, a_1, a_2, a_3) = (0, \log 2, \log 3, \log 4), \qquad L = 10^{-3}.$$

The resonant pairs are $(0,1)$, $(0,2)$, $(0,3)$, $(1,3)$; their weights are respectively

$$\frac{\log 2}{\sqrt{2}}, \quad \frac{\log 3}{\sqrt{3}}, \quad \frac{\log 2}{2}, \quad \frac{\log 2}{\sqrt{2}}.$$

In particular the $\log 4$ term is a genuine prime-power term, not $\log 4/2$. Every other logarithmic difference is separated from prime-power logarithms by more than L.

Theorem 100.6.2 *(A quantitative positive sector).* For arbitrary $f_j \in C_c^\infty((-1/2000, 1/2000))$, the function

$$F(u) = f_0(u) + f_1(u - \log 2) + f_2(u - \log 3) + f_3(u - \log 4)$$

satisfies

$$\boxed{Q_\zeta(F) \geq \frac{1}{5} \sum_{j=0}^{3} \|f_j\|_2^2.} \tag{100.7}$$

Proof. A finite rational calculation gives

$$\sum_{n=0}^{99} \left(\frac{1}{n + 1/4} - \frac{n + 1/4}{(n + 1/4)^2 + 2500} \right) > 8.$$

For an explicit integer calculation, put

$$d_n = (4n+1)((4n+1)^2 + 40000), \qquad b_j = \sum_{n=10j}^{10j+9} \left\lfloor \frac{1600000000}{d_n} \right\rfloor.$$

The values for $j = 0, \ldots, 9$ are

$$(b_0, \ldots, b_9) = (64860, 6514, 3300, 1958, 1248, 832, 574, 411, 300, 226).$$

Their sum is 80223. The rational partial sum is therefore at least $80223/10000 > 8$. Since all later summands are nonnegative and $k_\infty(0) > -6$, we have $k_\infty(100) > 2$. Equation (100.4), $\pi > 3$, and $e^{1/2000} < 3/2$ give a diagonal lower bound

$$2 - \frac{8}{30} - \frac{3}{1000} = \frac{5191}{3000}.$$

All cross-kernel arguments have absolute value between $\log(4/3) - 10^{-3}$ and $\log 4 + 10^{-3}$. On this interval each of the two positive terms in (100.5) is less than three. For the denominator term, $\log(4/3) > 1/4$ and $\log(3/2) < 1/2 - 1/8 + 1/24 < 0.498$ imply $1 - e^{-2|u|} > 1/3$. For the polar term use $2\cosh((\log 4 + 10^{-3})/2) < 2(1.001) + 1/2 < 3$. Thus $|K(u)| < 3$, and the smooth row cost is less than $9/1000$.

The largest prime row sum is bounded by

$$\frac{1}{2} + \frac{11}{17} + \frac{7}{20} = \frac{509}{340},$$

using $\log 2 < 7/10$, $\sqrt{2} > 7/5$, $\log 3 < 11/10$, $\sqrt{3} > 17/10$. Substitution into (100.6) gives

$$\frac{5191}{3000} - \frac{9}{1000} - \frac{509}{340} = \frac{5719}{25500} > \frac{1}{5}.$$

This proves (100.7) for the entire stated infinite-dimensional space. □

This sector is not a convolution algebra. Its support windows are not preserved under repeated convolution, and the permissible window width in Theorem 100.6.1 depends on the whole finite set of positions. Neither theorem supplies a dense convolution-stable exhaustion with one uniform estimate. That missing extension is exactly where the global Weil-positivity problem remains. Global positivity on the full convolution algebra is not a consequence of these estimates.

Finite-rank operators from the arithmetic pairing

Let $\mathscr{H}$ be the Hilbert completion of this four-window space for B_ζ, with the inner product conjugate-linear in its first argument. For $f, g \in \mathscr{H}$, define the bounded rank-one operator $\Theta_{f,g}h = B_\zeta(g,h)f$. Direct substitution gives

$$\Theta_{f,g}\Theta_{u,v} = B_\zeta(g,u)\Theta_{f,v}, \qquad \Theta_{f,g}^* = \Theta_{g,f}, \qquad \operatorname{Tr}(\Theta_{f,g}\Theta_{u,v}) = B_\zeta(g,u)B_\zeta(v,f).$$

The trace identity follows from $\operatorname{Tr}(\Theta_{f,v}) = B_\zeta(v,f)$ in an orthonormal basis. For vectors in the original test space, every displayed contraction has its prime, archimedean and polar terms given by the full functional above. These operators represent the selected quadratic pairing. A representation of convolution would additionally require compatible operators for every convolution multiplier on a common invariant domain.

For a finite-dimensional flavour subspace $V \subset \mathscr{H}$, take the charged fermion space

$$\mathcal{F}_V = \Lambda_{\mathbb{C}}\left(\bigoplus_{r \in \mathbb{Z}_{>0}-1/2} (V \oplus \overline{V})t^r\right).$$

Use its algebraic finite-particle domain and the induced positive Hermitian form. Exterior multiplication and its adjoint satisfy the canonical anticommutation relations with contraction B_ζ. Writing the resulting charged fields as $\psi_f(z)$, linear in f, and $\psi_g^*(w)$, conjugate-linear in g, the vacuum contraction in the expansion $|w| < |z|$ is

$$B_\zeta(g,f)\sum_{k\geq 0} z^{-k-1}w^k = \frac{B_\zeta(g,f)}{z-w}.$$

For a finite-rank A, choose V containing the ranges of A and A^*, and define $J_{\Theta_{f,g}}(z) =: \psi_f(z)\psi_g^*(z) :$, extending linearly in A. Normal ordering places creation operators to the left, with fermion signs. The singular operator product is

$$J_A(z)J_B(w) \sim \frac{\operatorname{Tr}(AB)}{(z-w)^2} + \frac{J_{[A,B]}(w)}{z-w}.$$

Indeed the two single contractions compose the matrices in their two orders, with opposite fermion signs. The double contraction sums their closed index pair and gives $\operatorname{Tr}(AB)$. Equivalently, the current modes satisfy

$$[J_A(m), J_B(n)] = J_{[A,B]}(m+n) + m\delta_{m+n,0}\operatorname{Tr}(AB)I.$$

For $m > 0$, exactly m half-integer modes cross the vacuum in the scalar term. Antisymmetry determines the remaining signs. For fixed mode and finite-energy vector, only finitely many creation terms survive. Enlarging V preserves these identities, so they apply to every finite set of finite-rank operators on $\mathscr{H}$. The arithmetic pairing thus occurs as the contraction of singular currents on this specified positive sector. Its support is still not stable under convolution.

Chapter 101

Adelic Scaling, Scattering, and the Weil Trace

The Weil pairing is positive precisely when the nontrivial zeta zeros lie on the critical line. A scattering operator instead compares two self-adjoint dynamics on their absolutely continuous subspaces. The question is whether arithmetic dynamics can produce the full signed Weil trace. The distinction between these two constructions already appears in a positive rank-one perturbation.

101.1 A Rank-One Trace Difference

Use the complex test algebra $\mathscr{A} = C_c^\infty(\mathbb{R})$, with additive convolution and $f^*(u) = \overline{f(-u)}$. Put

$$h_f(z) = \int_{\mathbb{R}} f(u) e^{izu}\, du, \qquad \mathscr{P} = \{h_f : f \in \mathscr{A}\}, \qquad h^\sharp(z) = \overline{h(\bar{z})}.$$

Then $h_{f*f^*} = h_f h_f^\sharp$, whose restriction to the real line is $|h_f|^2$. The signed Weil distribution is $W = P - W_\infty - W_{\mathrm{fin}}$, with all three terms as in Chapter 78.1.2. Its positivity condition is $W(f * f^*) \geq 0$ for every $f \in \mathscr{A}$. No evenness restriction is imposed.

Proposition 101.1.1 *(Complete scattering with a signed trace).* On $L^2(\mathbb{R}, d\lambda) \oplus \mathbb{C}$, let

$$A_0 = M_\lambda \oplus 0, \qquad A_1 = M_\lambda \oplus a, \qquad a > 0,$$

with common domain $\mathrm{Dom}\, M_\lambda \oplus \mathbb{C}$, where $\mathrm{Dom}\, M_\lambda = \{g : \lambda g \in L^2\}$. The difference is positive and has rank one. The wave operators are complete and the scattering operator is the identity. For every $h \in \mathscr{S}(\mathbb{R})$, however,

$$\mathrm{Tr}(h(A_1) - h(A_0)) = h(a) - h(0) = \int_0^a h'(\lambda)\, d\lambda.$$

The corresponding spectral shift is $\xi = \mathbf{1}_{(0,a)}$, and $-d\xi = \delta_a - \delta_0$ is not positive, even on the tests $|h_f|^2$, $f \in \mathscr{A}$.

Proof. The absolutely continuous subspace of each operator is the first summand. On that summand $e^{itA_1} e^{-itA_0} = 1$, so both wave operators have exactly that range. Functional calculus gives the displayed trace difference on the one-dimensional second summand. The integral identity fixes its spectral-shift sign. Choose $\phi \in \mathscr{A}$ with $\int \phi = 1$, and set $f(u) = \phi(u) + \phi(u - \pi/a)$. Then $h_f(z) = (1 + e^{i\pi z/a}) h_\phi(z)$, so $|h_f(a)|^2 - |h_f(0)|^2 = -4$. Also $e^{-2\pi i \xi} = 1$ almost everywhere, despite the negative atom of $-d\xi$. □

The example separates a unitary scattering phase from a positive trace distribution. To test an arithmetic comparison, the Hilbert spaces, domains, and perturbation must first be specified.

101.2 The Idele Scaling Operator

Definition 101.2.1 *(Idele carrier and measure).* Let $\mathbb{A}$ be the adele ring of $\mathbb{Q}$, let $C_{\mathbb{Q}} = \mathbb{Q}^{\times}\backslash\mathbb{A}^{\times}$, and put $C_{\mathbb{Q}}^{1} = \ker(|\cdot| : C_{\mathbb{Q}} \to \mathbb{R}_{>0})$. The section represented by the idele $(x, 1, 1, \dots)$, $x > 0$, gives $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times C_{\mathbb{Q}}^{1}$. Give the compact group $C_{\mathbb{Q}}^{1}$ Haar probability measure $d\mu_{1}$, and use dx/x on the first factor. Define

$$\mathcal{K} = L^{2}(C_{\mathbb{Q}}, d^{\times}x) = L^{2}(\mathbb{R}, du; E), \qquad E = L^{2}(C_{\mathbb{Q}}^{1}, d\mu_{1}), \quad u = \log x.$$

Here $d^{\times}x$ is the product measure, of infinite total mass. Let P_{0} be the rank-one orthogonal projection of E onto its constant functions.

LEMMA 101.2.2 *(Unitary Mellin transform and domain).* The transform

$$(\mathcal{M}g)(\tau) = (2\pi)^{-1/2} \int_{0}^{\infty} g(x) x^{-i\tau} \, \frac{dx}{x}$$

extends to a unitary from $\mathcal{K}$ to $L^{2}(\mathbb{R}, d\tau; E)$. On the logarithmic side put $D_{0} = -i\partial_{u}$, with domain $H^{1}(\mathbb{R}; E)$. It is self-adjoint, has purely absolutely continuous spectrum $\mathbb{R}$, and transforms to $M_{\tau} \otimes 1_{E}$, with domain $\{F : \int \tau^{2} \|F(\tau)\|_{E}^{2} d\tau < \infty\}$. A core is $\mathscr{S}(\mathbb{R}) \otimes_{\mathrm{alg}} E$.

Proof. The substitution $x = e^{u}$ preserves the stated measures. The ordinary Fourier Plancherel theorem, applied in each coordinate of an orthonormal basis of E, proves unitarity. Integration by parts on the Schwartz core gives $\mathcal{M}(-i\partial_{u} g) = \tau \mathcal{M} g$. Real multiplication on its maximal domain is self-adjoint: its adjoint has the same domain, as testing against compactly supported functions shows. Truncate in τ and in finitely many E-coordinates, then approximate the resulting functions by smooth compactly supported functions. This gives graph-norm density of the stated core. Each spectral measure has density $\|F(\tau)\|_{E}^{2}$ with respect to Lebesgue measure. Every real interval has nonzero spectral projection, proving the spectrum claim. □

The spherical space $L^{2}(\mathbb{R}) \otimes \operatorname{im} P_{0}$ is a reducing subspace. Its Schwartz functions are not a dense operator core for all of $\mathcal{K}$. The transform above uses the unitary character $x^{-i\tau}$. A formula with exponent $1/2 + i\tau$ requires an additional weight in the source measure or a specified multiplication by $x^{1/2}$.

101.3 The Critical Modular Generator on a Common Carrier

The critical Bost–Connes generator acts on a different Hilbert space. By Chapter 77.2.4, its orthonormal basis and domain are

$$N_{\mathrm{BC}} v_{q,r} = \nu_{q} v_{q,r}, \qquad \nu_{q} = \tfrac{1}{2}\log q, \qquad (q, r) \in \mathbb{Q}_{>0} \times \mathbb{Q}/\mathbb{Z},$$
$$\operatorname{Dom} N_{\mathrm{BC}} = \Big\{ \sum c_{q,r} v_{q,r} : \sum \nu_{q}^{2} |c_{q,r}|^{2} < \infty \Big\}.$$

Every eigenvalue has infinite multiplicity. The simple occupation energies $\log m$ instead occur in the reducing space $K_{2} = \overline{\operatorname{span}}\{v_{m^{2},0} : m \geq 1\}$, as in Chapter 77.3.2.

Definition 101.3.1 *(Tensor comparison operators).* Define

$$\widetilde{\mathcal{K}} = L^{2}(\mathbb{R}, d\tau) \widehat{\otimes} \mathcal{H}_{1} = L^{2}\big(\mathbb{R}; \ell^{2}(\mathbb{Q}_{>0} \times \mathbb{Q}/\mathbb{Z})\big).$$

The joint multiplication operators are

$$(\widetilde{D}_{0} F)_{q,r}(\tau) = \tau F_{q,r}(\tau), \qquad (\widetilde{D}_{\mathrm{BC}} F)_{q,r}(\tau) = (\tau + \nu_{q}) F_{q,r}(\tau).$$

Each has its maximal multiplication domain. In particular,

$$\operatorname{Dom} \widetilde{D}_{\mathrm{BC}} = \left\{ F : \int_{\mathbb{R}} \sum_{q,r} |\tau + \nu_q|^2 |F_{q,r}(\tau)|^2 \, d\tau < \infty \right\}.$$

This operator is the closure of $M_\tau \otimes 1 + 1 \otimes N_{\mathrm{BC}}$ on finite channel Schwartz functions. The common tensor construction does not identify $\widetilde{\mathcal{K}}$ with an arithmetic quotient of $\mathcal{K}$.

Proposition 101.3.2 *(Tensor wave obstruction).* Both tensor operators are self-adjoint and purely absolutely continuous. The strong limits of $e^{it\widetilde{D}_{\mathrm{BC}}} e^{-it\widetilde{D}_{\mathrm{o}}}$, as $t \to +\infty$ or $t \to -\infty$, exist on precisely the closed subspace supported on $q = 1$. They equal the identity there.

Proof. The maximal multiplication argument in the preceding lemma applies on each channel. Finite channel truncations followed by smooth compact approximations converge in the graph norm of either operator, proving the assertion about closures. The spectral measure of F for $\widetilde{D}_{\mathrm{BC}}$ is the sum of the translated measures $|F_{q,r}(\lambda - \nu_q)|^2 d\lambda$, hence is absolutely continuous. The free case is identical without translation. On each channel the wave family is multiplication by $e^{it\nu_q}$. If this family converges on a vector, its projection onto any nonzero channel also converges. For $\nu_q \neq 0$, choose s with $e^{is\nu_q} \neq 1$. The norm of the difference at times $t + s$ and t is the nonzero constant $|e^{is\nu_q} - 1| \|F_{q,r}\|$, which contradicts convergence. Thus all such channels vanish. Since $\nu_q = 0$ exactly when $q = 1$, the remaining assertion follows. □

Remark 101.3.3 *(Transport and the zero divisor).* An isometry intertwining N_{BC} with a bounded operator cannot exist: the unit vectors $v_{q,r}$ have unbounded eigenvalues. An isometry intertwining it with D_{o} cannot exist either, since D_{o} has no eigenvectors. Abstract isomorphism of separable Hilbert spaces asserts neither intertwining property. The hypothetical zero-divisor operator of Chapter 77.4.1 requires its separately stated operator, domain, and determinant comparison. The tensor construction supplies none of those zero-divisor data.

101.4 Trace-Class Scattering

Definition 101.4.1 *(Wave operators).* For self-adjoint operators A_0, A_1 on one complex Hilbert space, write $P_{\mathrm{ac}}(A_j)$ for the absolutely continuous spectral projection. When the strong limits exist, define

$$W_{\pm}(A_1, A_0) = \operatorname*{s-lim}_{t \to \pm\infty} e^{itA_1} e^{-itA_0} P_{\mathrm{ac}}(A_0).$$

Completeness means that both ranges equal $P_{\mathrm{ac}}(A_1)\mathcal{H}$. In that case

$$S = W_+^* W_-$$

is unitary on $P_{\mathrm{ac}}(A_0)\mathcal{H}$.

Theorem 101.4.2 *(Trace-class scattering and spectral shift).* Let A_0 be self-adjoint and let $V = V^*$ be trace class. Set $A_1 = A_0 + V$ on $\operatorname{Dom} A_0$. The wave operators exist and are complete. There is a real $\xi \in L^1(\mathbb{R})$, with $\|\xi\|_1 \leq \|V\|_1$, such that for $h \in \mathscr{S}(\mathbb{R})$,

$$\operatorname{Tr}(h(A_1) - h(A_0)) = \int_{\mathbb{R}} h'(\lambda)\xi(\lambda) \, d\lambda.$$

In a direct-integral spectral representation of A_0 on its absolutely continuous subspace, S has unitary fibres $S(\lambda)$, with $S(\lambda) - 1$ trace class almost everywhere, and

$$\det S(\lambda) = e^{-2\pi i \xi(\lambda)} \quad \text{almost everywhere.}$$

The normalization of ξ is specified by

$$\Delta_V(z) = \det(1 + V(A_0 - z)^{-1}), \qquad \xi(\lambda) = \pi^{-1} \operatorname{Im} \log \Delta_V(\lambda + i0),$$

where the analytic logarithm in the upper half-plane tends to zero as $\operatorname{Im} z \to +\infty$.

These are the Kato–Rosenblum, Krein trace, and Birman–Krein scattering theorems with the sign convention of Chapter 101.1.1. The operator and logarithm hypotheses are recorded in Behrndt–Malamud–Neidhardt, pp. 2–3, equations (1.1)–(1.5); their Theorem 2.7, p. 15, also treats finite-deficiency self-adjoint extension pairs. The determinant relation by itself determines ξ only modulo integer-valued functions. It does not replace the normalized perturbation determinant.

PROPOSITION 101.4.3 *(A specified rank-one pair on the idele carrier).* On the Mellin realization of $\mathcal{K}$, choose $g \in C_c^\infty(\mathbb{R})$ with $\|g\|_2 = 1$, and $a \in \mathbb{R}$. Let $Q_g F = g \int \overline{g(\lambda)} F(\lambda)\, d\lambda$, and set

$$V_a = a Q_g \otimes P_0, \qquad D_a = D_0 + V_a, \qquad \operatorname{Dom} D_a = \operatorname{Dom} D_0.$$

This pair has complete wave operators. Its perturbation determinant is

$$\Delta_a(z) = 1 + a \int_{\mathbb{R}} \frac{|g(\lambda)|^2}{\lambda - z}\, d\lambda.$$

Almost everywhere, the scattering matrix is the identity on $(1 - P_0)E$ and is multiplication on $P_0 E$ by

$$s_a(\lambda) = \frac{\overline{\Delta_a(\lambda + i0)}}{\Delta_a(\lambda + i0)}.$$

The spectral shift is the boundary argument of Δ_a with exactly the preceding logarithm normalization.

Proof. The operator $Q_g \otimes P_0$ is an orthogonal projection of rank one. Hence V_a is bounded self-adjoint and trace class, with trace norm $|a|$. The general theorem applies. The rank-one determinant formula $\det(1 + u\ell) = 1 + \ell(u)$ gives the displayed determinant. On the complementary compact channels both dynamics are identical. The remaining channel has spectral multiplicity one for D_0, so its scattering fibre is its determinant. The Birman–Krein formula and $\xi_a = \pi^{-1} \arg \Delta_a(\lambda + i0)$ give the stated quotient. The boundary values and the quotient are understood almost everywhere, as in the general theorem. □

Remark 101.4.4 *(Finite truncations and trace ideals).* Any specified finite-rank self-adjoint perturbation of D_0 satisfies the same theorem. A limit requires its own trace-ideal estimate. For example, on orthonormal vectors u_j, the diagonal operator $\sum_j b_j |u_j\rangle\langle u_j|$ is trace class exactly when $\sum_j |b_j| < \infty$. Square summability gives only the Hilbert–Schmidt ideal. Neither a formal cutoff nor nuclearity between different Schwartz spaces proves this Hilbert trace-class condition. The functions g and coefficients a above have not been identified with the Bost–Connes generator.

101.5 Positivity of a Spectral-Shift Distribution

Theorem 101.5.1 *(The ordinary trace-class positivity obstruction).* Let $\xi \in L^1(\mathbb{R})$ be real, and define $T(h) = \int h'\xi$ on $\mathscr{S}(\mathbb{R})$. The following conditions are equivalent:

$$T(|h_f|^2) \geq 0 \quad (f \in \mathscr{A}), \qquad -d\xi \geq 0, \qquad \xi = 0 \text{ almost everywhere.}$$

In particular, a nonzero positive trace distribution cannot arise as the ordinary trace difference of a trace-class perturbation.

Proof. The map T is Schwartz-continuous, since $|T(h)| \leq \|\xi\|_1 \|h'\|_\infty$. Compactly supported smooth functions are dense in Schwartz space: for a Schwartz function f, multiplication by $\chi(u/R)$, where $\chi = 1$ near zero, converges to f in every Schwartz seminorm. Fourier transformation is a Schwartz isomorphism. Thus the first inequality implies $T(|g|^2) \geq 0$ for every $g \in \mathscr{S}$.

Let $\varphi \in C_c^\infty$ be nonnegative. Choose a nonnegative compact smooth function η equal to one on a neighborhood of $\operatorname{supp} \varphi$. The function $g_\varepsilon = (\varphi + \varepsilon\eta^2)^{1/2}$ is smooth: on a neighborhood of the support of φ its argument is positive, and outside that neighborhood it equals $\sqrt{\varepsilon}\eta$. It has compact support. Consequently $T(\varphi) + \varepsilon T(\eta^2) \geq 0$. Letting $\varepsilon \downarrow 0$ proves that $-d\xi$ is a positive distribution.

For a nonnegative smooth approximate identity ρ_ε, $\xi * \rho_\varepsilon$ is a real, nonincreasing, integrable smooth function. Such a function is zero: a positive value forces an infinite integral on its left, and a negative value forces one on its right. Hence $\xi * \rho_\varepsilon = 0$ for every ε, and convergence in L^1 gives $\xi = 0$. The reverse implications follow directly. □

Remark 101.5.2 *(The phase and its derivative).* For locally integrable ξ, its distributional derivative is defined on smooth compact tests by $\langle -d\xi, h\rangle = \int \xi h'$. It need not initially be a signed measure. Positivity makes it a positive measure and forces ξ to have a nonincreasing representative. The additional global L^1 condition in the theorem forces that measure to vanish. A regularized trace outside this class requires a separate theorem, including its test space and normalization.

101.6 Mellin Divisibility and Multiplication Phases

An ordinary Mellin integral near zero records the Taylor coefficients of an entire function as poles. This constrains a proposed divisibility by the entire completed xi-function before any zero-location argument is available.

Definition 101.6.1 *(The entire-Mellin test class).* Let PW_2 denote the entire functions of exponential type whose real restrictions belong to $L^2(\mathbb{R})$. Define $\mathrm{PW}^{\mathrm{TM}}$ to consist of its real even functions f for which the ordinary Mellin integral continues as

$$Mf(s) = \xi_{\mathbb{R}}(s)b(s), \qquad b(s) = \overline{b(1-\bar{s})},$$

where b is entire of exponential type and decreases faster than every inverse power on each closed vertical strip. Here $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ with its entire continuation. This Mellin integral uses $f(t)$ on $t > 0$; it is distinct from the Fourier transform of the logarithmic compact test algebra $\mathscr{A}$.

Theorem 101.6.2 *(Entire continuation forces the zero function).* If f is entire, $f|_{\mathbb{R}} \in L^2$, and its ordinary Mellin transform is entire, then $f = 0$. In particular, $\mathrm{PW}^{\mathrm{TM}} = \{0\}$.

Proof. The Mellin integral converges absolutely on $0 < \Re s < 1/2$. Near zero this follows from boundedness of f. At infinity, Cauchy–Schwarz applies because $\int_1^\infty t^{2\Re s-2}dt < \infty$. Both estimates are locally uniform on smaller strips, including logarithmic factors from differentiation, so the integral is holomorphic there. Write $a_n = f^{(n)}(0)/n!$. For each $N \geq 0$,

$$Mf(s) = \int_1^\infty f(t)t^{s-1}dt + \int_0^1 \left(f(t) - \sum_{n=0}^{N} a_n t^n\right)t^{s-1}dt + \sum_{n=0}^{N} \frac{a_n}{s+n}.$$

The first integral is holomorphic for $\Re s < 1/2$. The Taylor remainder makes the second holomorphic for $\Re s > -N - 1$. The residue at $s = -n$ is therefore a_n. Entire continuation forces every coefficient to vanish. The identity theorem gives $f = 0$. The divisibility condition makes Mf entire, proving the final assertion. □

Remark 101.6.3 *(Zeros and the test topology).* The completed xi-function vanishes at all nontrivial zeros, with multiplicity. It does not vanish at the trivial zeros of ζ, where gamma poles cancel those zeros. The zero-class theorem uses neither this divisor nor RH. It implies nondensity of $\mathrm{PW}^{\mathrm{TM}}$ in every nonzero normed PW_2 space. A measure-zero condition on values of a transform, by itself, would not imply nondensity in an L^2 topology.

Functional equations and wave limits

PROPOSITION 101.6.4 *(The completed functional-equation ratio).* The meromorphic ratio

$$\frac{\xi_{\mathbb{R}}(1/2 + iz)}{\xi_{\mathbb{R}}(1/2 - iz)}$$

has the removable continuation 1 on $\mathbb{C}$. For an entire function of real type, $b(\bar{s}) = \overline{b(s)}$, the ratio $b(1/2 + it)/b(1/2 - it)$ has modulus one wherever defined. The distinct symmetry $b(s) = \overline{b(1 - \bar{s})}$ alone does not imply this modulus identity.

Proof. The functional equation $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1 - s)$ identifies numerator and denominator as entire functions, including their zero orders. The real-type assertion follows by complex conjugation. For the last assertion take $b(s) = 1 + i(s - 1/2)$. It has the stated reflection symmetry, whereas the ratio on the critical line is $(1 - t)/(1 + t)$. This example tests the algebraic implication of that symmetry; it is not an element of the entire-Mellin test class. □

PROPOSITION 101.6.5 *(Multiplication perturbations).* Let $v : \mathbb{R} \to \mathbb{R}$ be measurable and finite almost everywhere. On $L^2(\mathbb{R})$, give M_λ and $M_{\lambda+v(\lambda)}$ their maximal domains. Their ordinary wave operators exist on the full free space if and only if $v = 0$ almost everywhere.

Proof. The wave family is $M_{e^{itv}}$. If it converges strongly, then for every real s and every $g \in L^2$,

$$\|(e^{isv} - 1)g\| = \|e^{i(t+s)v}g - e^{itv}g\| \longrightarrow 0.$$

Thus $e^{isv} = 1$ almost everywhere for every rational s. The intersection of these countably many full-measure sets forces $v = 0$. The converse follows because the two operators coincide. □

A functional-equation ratio therefore needs a separately constructed scattering realization. Replacing D_0 by multiplication by $\tau + \frac{1}{2}\arg \xi_{\mathbb{R}}(1/2 + i\tau)$ does not supply one unless that perturbation vanishes almost everywhere.

101.7 Arithmetic Trace Carriers

Definition 101.7.1 *(The adele quotient and a Hilbert seminorm).* The adele class space is the quotient $X_{\mathbb{Q}} = \mathbb{A}/\mathbb{Q}^{\times}$. Let $\mathscr{S}(\mathbb{A})$ be the Bruhat–Schwartz space for the additive adeles, with self-dual additive Haar measure. Put

$$\mathscr{S}(\mathbb{A})_0 = \{\phi : \phi(0) = 0,\ \int_{\mathbb{A}} \phi(x)dx = 0\}, \qquad (E\phi)(x) = |x|^{1/2} \sum_{q\in\mathbb{Q}^{\times}} \phi(qx).$$

For $\delta > 1$, give this space the seminorm

$$\|\phi\|_{\delta}^2 = \int_{C_{\mathbb{Q}}} |E\phi(x)|^2 (1 + \log^2 |x|)^{\delta/2}\, d^{\times}x.$$

Quotient by its nullspace and complete. This is the Hilbert construction in Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, §III, (6)–(19), pp. 10–12. Write $L^2_{\delta}(C_{\mathbb{Q}})$ for the weighted target space with measure $(1 + \log^2 |x|)^{\delta/2} d^{\times}x$. The completed map E is isometric and has closed image. Its cokernel $H_{\delta} = L^2_{\delta}(C_{\mathbb{Q}})/\operatorname{im} E$ carries the quotient of $V(a)\eta(x) = \eta(a^{-1}x)$. The identity $EU(a) = |a|^{1/2}V(a)E$, for $U(a)\phi(x) = \phi(a^{-1}x)$, makes the image invariant. The weighted norm-direction representation is polynomially bounded, not asserted to be unitary. The quotient generator in Connes's convention is the derivative of $V(e^t)$; its critical-zero spectral values are imaginary. This is distinct from the self-adjoint free generator on unweighted $L^2(C_{\mathbb{Q}})$.

Proposition 101.7.2 *(Scope of the critical-zero realization).* In the quotient representation on H_{δ}, Connes's §III, Theorem 1, p. 13, realizes the zeros on the critical line in each norm-one character sector. For $\delta > 1$, the multiplicity of a critical zero of multiplicity m is the largest integer n satisfying $n < (1 + \delta)/2$ and $n \le m$. Its Corollary 2 gives the trace of integrated Schwartz operators with these multiplicities. This theorem is a critical-zero realization, not a self-adjoint realization of every nontrivial zero.

Meyer's *On a representation of the idele class group related to primes and zeros of L-functions*, Theorems 5.8 and 5.11, pp. 43 and 47, uses a different carrier: a smooth summable virtual representation $\pi_+ \ominus \pi_-$. Its character is the raw Weil distribution, with poles and zeros having opposite signed multiplicities. His §3.2, p. 17, descends suitable equivariant operators to weighted idele-class spaces. His Proposition 4.4, p. 26, gives Schwartz restriction and Fourier maps. These statements specify neither a bounded transport of N_{BC} nor the perturbation of an idele scaling operator. Passing between the raw character and $W = P - W_{\infty} - W_{\mathrm{fin}}$ requires the central twist, sign, and test-function normalization.

101.8 The Full Signed Trace Comparison

Theorem 101.8.1 *(Transfer on the specified arithmetic test algebra).* Suppose independently specified self-adjoint operators A_0, $A_1 = A_0 + V$, with $V = V^*$ trace class, satisfy

$$W(f) = \operatorname{Tr}(h_f(A_1) - h_f(A_0)) \qquad (f \in \mathscr{A}).$$

Then, for their normalized spectral shift ξ,

$$W(f * f^*) = \int_{\mathbb{R}} (|h_f|^2)'(\lambda)\xi(\lambda)d\lambda.$$

If this form is nonnegative on all $\mathscr{A}$, the trace functional vanishes on every Schwartz spectral test. In particular, a nonzero positive Weil functional cannot have this ordinary trace-class realization.

Proof. Apply the stipulated identity to $f * f^*$, use $h_{f*f^*} = h_f h_f^\sharp$, and apply the trace formula. The positivity obstruction forces $\xi = 0$, which makes the trace formula zero for every Schwartz function. Under Weil positivity, RH holds by Chapter 78.2.1. The zero-side formula then is a sum of $|h_f(\gamma_\rho)|^2$. It is nonzero for some f: choose a nontrivial zero ordinate γ_0, a compact smooth ϕ of integral one, and $f(u) = e^{-i\gamma_0 u}\phi(u)$. The term at γ_0 is positive, and the other terms are nonnegative. Existence of nontrivial zeros is part of Chapter 77.5.1. This contradicts vanishing and proves the last assertion. □

Remark 101.8.2 *(Required arithmetic data).* The trace identity in the theorem is additional arithmetic data. Trace-class estimates, wave completeness, and a functional equation do not imply it. A proposed regularized replacement must specify its carrier, scaling action, dense test domain, subtraction terms, and trace functional before imposing equality with W. If it uses Bost–Connes data, it must also construct the map from $\mathcal{H}_1$, or its stated occupation restriction, and prove which domains and operations that map preserves. Nuclear characters on locally convex spaces and Hilbert trace differences are distinct constructions.

PROPOSITION 101.8.3 *(Independent conditions on an arithmetic trace).* Let $\mathcal{T} : \mathscr{A} \to \mathbb{C}$ be a linear distribution obtained from a specified arithmetic trace construction. If $\mathcal{T}(f) = W(f)$ for every $f \in \mathscr{A}$, then $\mathcal{T}(f * f^*) \geq 0$ for every f is equivalent to A-TP, and hence to RH. Existence of $\mathcal{T}$, the comparison identity, and this positivity condition are separate requirements. The tensor and multiplication constructions above fail the wave requirement; the rank-one construction satisfies it but supplies no arithmetic trace identity.

Proof. Substitute $f * f^*$ in the stated identity and apply the Weil positivity criterion on $\mathscr{A}$. The final assertions are the tensor wave obstruction, the multiplication proposition, and the explicit rank-one construction, respectively. □

Remark 101.8.4 *(Comparison with the other analytic formulations).* The bounded-interval digamma inequality and the restricted automorphic comparison concern the same full signed distribution only when their stated test maps and residual terms agree with $P - W_\infty - W_{\mathrm{fin}}$. A scattering comparison additionally needs the trace identity above. Connes's cutoff trace and Meyer's virtual character supply precise alternative trace settings; neither is identified here with the critical BC generator by a unitary map. The non-Hausdorff quotient alone proves no impossibility theorem for other Hilbert realizations.

101.9 CONSEQUENCES FOR THE ARITHMETIC REALIZATION PROBLEM

THEOREM 101.9.1 *(Scaling and scattering constructions).* The idele scaling operator has the self-adjoint domain and Mellin realization specified above. Coupling it by tensor addition to the full critical BC generator gives self-adjoint dynamics but no full ordinary wave operators. Specified finite-rank perturbations do have complete scattering; their unitary matrices do not imply positivity of the trace difference. The ordinary entire-Mellin class $\mathrm{PW}^{\mathrm{TM}}$ is zero. A nonzero positive Weil trace therefore requires a construction beyond the ordinary trace-class comparison stated here, with its arithmetic test identity and positivity proved on all of $\mathscr{A}$.

Proof. Apply the Mellin-domain lemma, the tensor wave obstruction, the rank-one pair and signed-trace propositions, the entire-Mellin vanishing theorem, and the trace-comparison theorem. □

Remark 101.9.2 *(Arithmetic comparison problem).* An arithmetic BC comparison must realize $W = P - W_\infty - W_{\mathrm{fin}}$ on $\mathscr{A}$. Positivity of this trace is the full Weil condition.

Chapter 102

All-Genus Arithmetic Koszul Duality

The genus-zero arithmetic comparison concerns the identity $\kappa_{\mathsf{G}}^{\mathrm{Ar}} + (\kappa^{!})_{\mathsf{G}}^{\mathrm{Ar}} = 0$ and the Riemann functional equation $\xi(s) = \xi(1-s)$. The object $\mathcal{V}^{\mathrm{prim}}$ is the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2. Interpreting its levels requires the arithmetic realization in Theorem 43.5.1.

That theorem assumes a Mellin evaluation, a central-defect identity, and detection of the defect. Poisson summation and Satake parameters do not supply these maps. For rank $2g$, the corresponding question asks which internal identity has the motivic functional equation's sign and normalization.

The conditional answer is the same identity $\kappa_{\mathsf{G}}^{\mathrm{Ar},g} + (\kappa^{!})_{\mathsf{G}}^{\mathrm{Ar},g} = 0$, after the residual inputs of Chapter 102.6 are imposed. The genus does not deform the Koszul sum; it enters as the rank of the Heisenberg oscillators at each place, the archimedean Γ-factor $\Gamma_{\mathbb{C}}(s)^g$, and the motivic weight fixing the functional-equation center.

102.1 The Arithmetic Genus-g Datum

Definition 102.1.1 *(arithmetic genus-g datum).* . An *arithmetic genus-g datum* $\mathcal{D}_g$ is a triple $(\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$ specifying:

(i) $\mathcal{B}_g$, an arithmetic base: one of

   a) Arakelov-compactified $\overline{\operatorname{Spec} \mathcal{O}_K}$ for K totally imaginary of degree $2g$ (Window II);

   b) a proper flat regular arithmetic surface $X \to \operatorname{Spec} \mathcal{O}_K$ whose generic fibre X_K is a smooth projective curve of genus g, Arakelov-compactified via the archimedean Kähler form $\mu_{\mathrm{Ar},g}$ of Faltings 1984 (Windows I, V);

   c) a Shimura variety $\mathrm{Sh}_K(G, X)$ of a Shimura datum whose complex fibre is a moduli of abelian g-folds (Window VI).

(ii) $\mathcal{E}_g$, a rank-$2g$ arithmetic cohomology object: $H^1_{\mathrm{prim}}(\mathcal{B}_g)$ carrying Galois action (étale or crystalline), with Hodge structure of weight 1 at each archimedean place: $\mathcal{E}_g \otimes \mathbb{C} = \mathcal{E}_g^{1,0} \oplus \mathcal{E}_g^{0,1}$ with $\dim \mathcal{E}_g^{1,0} = \dim \mathcal{E}_g^{0,1} = g$ per place.

(iii) φ_g, a Frobenius-compatible endomorphism on $\mathcal{E}_g$ with square equal to the appropriate motivic involution: φ_g is either

   a) the archimedean Frobenius Φ_∞ of Deninger 1998 acting on the primitive Minkowski cohomology (Window II);

   b) the prismatic Frobenius φ of Bhatt–Scholze 2022 on the universal rank-$2g$ F-crystal over the ordinary crystalline locus of the Emerton–Gee stack $\mathcal{X}_{2g}$ (Window III);

   c) the Koszul duality operator J^{Koz} on chiral cohomology satisfying $(J^{\mathrm{Koz}})^2 = \tau$ for τ the motivic functional-equation involution (Window V).

At $g = 0$ the datum degenerates to the base $\overline{\operatorname{Spec}\mathbb{Z}}$ with $\mathcal{E}_0$ void and $\varphi_0 = \mathrm{id}$. The proposed scalar functional equation is then $\xi(s) = \xi(1-s)$, as in Theorem 43.5.1. Its arithmetic interpretation still requires that theorem's evaluation and detection hypotheses.

Remark 102.1.2 *(Frobenius as almost-complex substitute).* An endomorphism $J : TX \to TX$ satisfying $J^2 = -\mathrm{id}$ cannot exist on an arithmetic scheme: schemes over $\operatorname{Spec}\mathbb{Z}$ carry no real tangent bundle, and on a compact real 2-manifold Newlander–Nirenberg (1957) makes integrability of J automatic, so no genuinely almost-complex geometry survives. The Frobenius φ_g of Chapter 102.1.1 is the correct arithmetic substitute: on archimedean primitive cohomology it acts as Hodge rotation by $\pi/2$; on the prismatic F-crystal ordinary stratum it is the Newton–Hodge-polygon-aligned slope-decomposition Frobenius (Katz 1981, *Astérisque* 63); on chiral cohomology it is the Koszul duality operator J^{Koz} whose square equals the functional-equation involution τ. No non-integrable decoration is introduced.

102.2 Six Convergent Windows

The datum $\mathcal{D}_g$ admits six distinct geometric and automorphic presentations, each extracting a different structural feature of the rank-$2g$ arithmetic cohomology and the corresponding motivic functional equation.

Window I: Arakelov fibration

$\mathcal{B}_g^{\mathrm{I}} = X \to \operatorname{Spec} O_K$, proper flat regular with generic fibre of genus g, Arakelov-compactified. $\mathcal{E}_g^{\mathrm{I}} = H^1_{\mathrm{B}}(X(\mathbb{C}),\mathbb{Z})$ of rank $2g$ per archimedean place. φ_g^{I} the integrable complex structure J_∞ at each archimedean place. Arakelov 1974, Faltings 1984, Gillet–Soulé 1990 (Publ. IHES) supply the arithmetic intersection pairing on $\widehat{\mathrm{CH}}^1(X)$; Faltings–Hriljac 1984 the arithmetic Hodge index. Genus g enters via $\pi_*\omega_{X/S}$ of rank g, the Γ-factor $\Gamma_{\mathbb{C}}(s)^g$ of the Hasse–Weil L-function, the coefficient $2g-2$ in $\deg \omega_{X_K/K}$, and the δ-invariant correction in the arithmetic Noether formula $\widehat{\deg}\det R\pi_*\omega = \frac{1}{12}(\widehat{\omega}^2 + \delta_{\mathrm{Ar}})$ (Faltings 1984, Thm 5).

Window II: Minkowski torus / Bost–Connes–Marcolli

$\mathcal{B}_g^{\mathrm{II}} = \overline{\operatorname{Spec} O_K}$ for K totally imaginary of degree $2g$, with Minkowski lattice $\Lambda_K = j(O_K) \subset \mathbb{C}^g$ and Minkowski torus $T_K = \mathbb{C}^g/\Lambda_K$. This complex torus is an abelian variety only after choosing a CM type and a polarization satisfying the Riemann bilinear relations. $\mathcal{E}_g^{\mathrm{II}} = H^1(T_K,\mathbb{Z}) \cong \Lambda_K^\vee$ of $\mathbb{Z}$-rank $2g$; the primitive part equals the full H^1 only in that polarized abelian-variety subcase. $\varphi_g^{\mathrm{II}} = \Phi_\infty^{\mathrm{tw},K}$, the half-Tate-twisted archimedean Frobenius acting as $\sqrt{-1}$ on each complex embedding pair; equivalently the $\mathbb{C}^g$-structure on $\Lambda_K \otimes \mathbb{R}$. The genus is $g = r_2(K) = \frac{1}{2}[K:\mathbb{Q}]$. Bost–Connes–Marcolli 2004 (*Selecta Math.* 11) constructs the C_K^* quantum statistical system whose KMS_β partition function is $\zeta_K(\beta)$; Ha–Paugam 2005 and Laca–Neshveyev 2011 classify its KMS states.

Window III: Ordinary crystalline Emerton–Gee stack

$\mathcal{B}_g^{\mathrm{III}} = \mathcal{X}_{2g}^{\mathrm{crys,ord}}$, the ordinary crystalline locus of the Emerton–Gee stack $\mathcal{X}_{2g}$ (Emerton–Gee arXiv:1908.07185). Its dimension depends on the rank, the residual representation, and the imposed crystalline type; it is not a universal 2-dimensional Artin stack. A point in the ordinary crystalline stratum is a rank-$2g$ crystalline object with Hodge–Tate multiset $\{0,1\}^g$, together with the chosen ordinary filtration. $\mathcal{E}_g^{\mathrm{III}}$ is the universal rank-$2g$ bundle; φ_g^{III} is Frobenius together with this filtration. The analogy with $H^1(X(\mathbb{C}),\mathbb{C}) = H^{1,0} \oplus H^{0,1}$ is a Hodge analogy, not an archimedean tilting

theorem. Caraiani–Scholze supplies Shimura-variety Galois representations under its own genericity and signature hypotheses; it does not produce this rank-$2g$ window for all genus-g data.

Window IV: Hodge–Arakelov / arithmetic Teichmuller

$\mathcal{B}_g^{\mathrm{IV}} = [C, \vartheta] \in \mathcal{T}_g^{\mathrm{arith}}(\mathcal{O}_K)$, a closed point of the arithmetic Teichmüller space carrying a level-n theta-structure ϑ on the Jacobian $J(C)$. $\mathcal{E}_g^{\mathrm{IV}}$ is the n^g-dim Jet module over the theta-structure locus; φ_g^{IV} is Mochizuki's Hodge–Arakelov comparison isomorphism (Mochizuki 1999 *Publ. RIMS*; arXiv:math/0304054). The comparison is theorem-grade for $g = 1$; conjectural for $g \geq 2$. The Steinberg relation $\{p, 1-p\} = 0 \in K_2^M(\mathbb{Q})$ is preserved under every Frobenioid indeterminacy (anabelian invariant via Matsumoto 1969, Milnor 1970). The naïve speculation conflates the crystalline–étale comparison isomorphism with an almost-complex structure J on Spec $\mathbb{Z}$; the correct arithmetic object is the Hodge–Arakelov decoration φ_g^{IV} on the dense theta-generic open locus of $\mathcal{T}_g^{\mathrm{arith}}$ (Chapter 102.1.2).

Window V: E_2-chiral factorisation on the arithmetic surface

$\mathcal{B}_g^{\mathrm{V}} = \mathcal{S}_g^{\mathrm{Ar}}$, the arithmetic surface of Window I viewed through its E_2-chiral factorisation structure. $\mathcal{E}_g^{\mathrm{V}} = \mathcal{V}^{\mathrm{prim,surf},g} := \mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw,surf}}_{GL_{2g},\mathrm{glob}})$, the Hochschild trace class of the identity on the E_2-monoidal global Iwahori–Hecke category of GL_{2g} on the arithmetic surface. The E_2 structure arises via $E_1 \otimes E_1 = E_2$ Dunn additivity (Lurie *Higher Algebra* 5.1.2.2) from the horizontal E_1 (1-dim base factorisation on $\overline{C}$) and the vertical E_1 (fibre factorisation on X_K); the bar-cobar construction on this E_2 datum lifts to the 2-dim Parshin–Gersten complex with double residue (∂_1, ∂_2) (Parshin 1976; Kato 1980; Beilinson 1980). $\varphi_g^{\mathrm{V}} = J^{\mathrm{Koz}}$, the Koszul duality operator on chiral cohomology with $(J^{\mathrm{Koz}})^2 = \tau$ (the functional-equation involution). The genus enters via rank-$2g$ Fock oscillators per place, sourced from the ℓ-adic Tate module $T_\ell(\mathrm{Jac}(X_K))$; the holomorphic central charge is $c = g$, with g additional antiholomorphic oscillators, giving total Fock rank $2g$.

Window VI: Kudla–Rapoport / Shimura–Siegel Shimura variety

$\mathcal{B}_g^{\mathrm{VI}} = \mathrm{Sh}_K(G, X)$ a Shimura variety with $G \in \{\mathrm{GSp}_{2g}, \mathrm{Res}_{F/\mathbb{Q}}\mathrm{GL}_2 \text{ for } [F:\mathbb{Q}] = g, \mathrm{GU}(g,1)\}$. $\mathcal{E}_g^{\mathrm{VI}}$ the middle-degree cohomology decomposed under the Hecke algebra into rank-$2g$ (or $2g+1$, $4g$) Galois blocks per automorphic π of $G(\mathbb{A})$. φ_g^{VI} the integrable complex structure sourced from the Deligne torus $h : \mathbb{S} \to G_{\mathbb{R}}$. The Kudla–Rapoport generating series $\widehat{\phi}(\tau) = \sum_m \widehat{\mathcal{Z}}(m) q^m$ is a genus-g Siegel modular form valued in $\widehat{\mathrm{CH}}^1(\mathrm{Sh})_{\mathbb{C}}$ (Kudla 1997 *Ann. Math.* 146; Kudla–Rapoport 2011 *Algebra Number Theory* 5; Li–Liu 2021 arXiv:2101.09485). The g-fold arithmetic intersection of special cycles realises Fourier coefficients of a genus-g Siegel Eisenstein series' central derivative. The Langlands-dual group ${}^L G = \mathrm{GSpin}_{2g+1}$ carries a $(2g+1)$-dimensional standard representation and a 2^g-dimensional spin representation. The row below uses the standard L-function unless the spin representation is explicitly named.

102.3 Obstructed Candidate Readings

Six candidate readings of the phrase "genus-g arithmetic surface with almost-complex structure" meet distinct mathematical obstructions; the obstructions are recorded in Chapter B.7.

(i) $J^2 = -\mathrm{id}$ *on the tangent bundle of an arithmetic scheme.* Schemes over Spec $\mathbb{Z}$ carry no real tangent bundle; the phrase "almost-complex" is a category error at this level. The correct arithmetic substitute is the Frobenius φ_g on primitive cohomology (Chapter 102.1.2).

(ii) *Arithmetic Euler characteristic* $= 1 - g$. Gillet–Soulé 1992 arithmetic Riemann–Roch introduces Bismut–Gillet–Soulé analytic torsion; the arithmetic analog of $1 - g$ is the Faltings δ-invariant via the arithmetic Noether formula, not a simple integer.

(iii) *Künneth factorisation of the arithmetic surface.* Bost–Künnemann 2010 (*Adv. Math.* 223) shows the product formula has non-vanishing Bott–Chern archimedean correction terms; X does not factor as "genus-g curve $\times_{\operatorname{Spec}\mathbb{Z}} \operatorname{Spec}\mathbb{Z}$" in arithmetic Chow.

(iv) $\overline{\operatorname{Spec}\mathbb{Z}}$ *itself has finite genus.* Deninger's arithmetic-dynamical genus of $\operatorname{Spec}\mathbb{Z}$ is $+\infty$ (countably many critical zeros of ζ). Genus enters the arithmetic datum only through coefficient enhancement to a totally imaginary number field of degree $2g$.

(v) *Conflation of curve-genus g_c with Siegel-modular genus g_s.* For a modular curve $X_0(N)$: $g_c = g_0(N)$ and $g_s = 1$. For the Siegel threefold $\mathcal{A}_2$: g_c is undefined, $g_s = 2$. The genus g of the datum $\mathcal{D}_g$ is the symplectic rank of the Galois representation divided by 2; in the Shimura setting this is g_s, not g_c.

(vi) $\kappa + \kappa^! = 2 - 2g$ *as Euler-characteristic shift.* The Euler characteristic $2 - 2g$ is a topological invariant of the Riemann surface $X(\mathbb{C})$, not a Koszul invariant of the chiral datum. The identity $\kappa^{\mathrm{Ar},g} + (\kappa^!)^{\mathrm{Ar},g} = 0$ is the conditional Koszul-duality target at rank g; the genus enters through the Fock rank, the archimedean Γ-factor, and the motivic weight of the functional-equation center.

102.4 All-Genus Arithmetic Koszul Duality

Let $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$ be an arithmetic genus-g datum (Chapter 102.1.1). Attach to $\mathcal{D}_g$ the arithmetic Heisenberg $\mathcal{V}_{\mathrm{Heis}}^{\mathrm{prim},g}$ via the rank-$2g$ Fock on $\mathcal{E}_g$ at each place, and the Koszul dual via the genus-g Bost–Connes–Marcolli system of Consani–Marcolli 2006 (*J. Reine Angew. Math.* 597).

THEOREM 102.4.1 *(all-genus arithmetic Koszul duality). (on the residual conditions of Chapter 102.6).* For every $g \geq 0$ and every arithmetic genus-g datum $\mathcal{D}_g$, the arithmetic Heisenberg Koszul self-duality identity

$$\kappa_{\mathsf{G}}^{\mathrm{Ar},g}(\mathcal{D}_g) + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar},g}(\mathcal{D}_g) = 0$$

reproduces the normalization and sign of the motivic functional equation

$$\Lambda_{\mathcal{D}_g}(s) = \epsilon_{\mathcal{D}_g} \cdot \Lambda_{\mathcal{D}_g}\big(w_{\mathcal{D}_g} - s\big), \tag{102.1}$$

where $\Lambda_{\mathcal{D}_g}(s)$ is the completed L-function attached to $\mathcal{D}_g$ and $w_{\mathcal{D}_g}$ is the center parameter in the chosen motivic or automorphic normalisation, depending on the window but not on g alone. This statement does not derive analytic continuation or the functional equation from Koszul data alone. It is conditional on the Euler-product identification for the chosen datum, the appropriate genus-g Bost–Connes–Marcolli or Shimura input, the analytic continuation and functional equation of $\Lambda_{\mathcal{D}_g}$, the 2-dimensional arithmetic Positselski equivalence at $\Lambda_2 = \mathbb{Z}_p[[\Gamma_1 \times \Gamma_2]]$-coefficients (Window V), and Koszul–Petersson compatibility at rank g.

The four rows of (102.1), indexed by window, are recorded in Chapter 102.1.

Remark 102.4.2 *(weight structure).* The functional-equation center $w_{\mathcal{D}_g}$ is set by the motivic weight of the row, not by g itself. Window II (Dedekind ζ_K with archimedean $\Gamma_{\mathbb{C}}^g$ factor) is weight 0; the center stays at $s \mapsto 1 - s$ regardless of g, since ζ_K for K of degree $2g$ has the same $1 - s$ symmetry as ζ for any number field. Windows I, V (arithmetic surface with generic fibre of genus g) are weight 1 on the

Table 102.1: Rows of the all-genus arithmetic Koszul duality.

Window	$\Lambda_{\mathcal{D}_g}(s)$	$w_{\mathcal{D}_g}$	Motivic weight
II	$\xi_K(s) = \lvert d_K\rvert^{s/2}\Gamma_{\mathbb{C}}(s)^g\zeta_K(s)$	1	0 (Dedekind)
I, V	$\Lambda(s, X_K) = N_{X_K}^{s/2}\Gamma_{\mathbb{C}}(s)^g L(s, X_K)$	2	1 (Hasse–Weil)
VI	$\Lambda(s, \pi, \text{spin})$ for π on GSp_{2g}	$w_{\pi,\text{spin}}$	fixed by the chosen spin L-normalisation
III, IV	spectral refinements	indirect	refines above

Jacobian's H^1-motive; the center is $s \mapsto 2 - s$. Window VI (Siegel spin of genus-g cusp form) has centre determined by the chosen cohomological spin normalisation; the table records it as $w_{\pi,\text{spin}}$, not as a universal integer. All three rows share $\kappa + \kappa^! = 0$; the genus grading lives in the Γ-factor, the Fock rank, and the weight of the row.

Sketch. The following construction requires the stated comparison inputs. At $g = 0$, the arithmetic level criterion is Theorem 43.5.1, including its Mellin evaluation and detection hypotheses.

Step 1: Euler-product identification. Assume that at each prime p of good reduction for $\mathcal{D}_g$, writing $\alpha_1^{(p)}, \dots, \alpha_{2g}^{(p)}$ for the Frobenius eigenvalues on $\mathcal{E}_{g,p}$, the rank-$2g$ Heisenberg Fock partition function at inverse temperature $\beta = s$ factors as $\prod_{j=1}^{2g}(1 - \alpha_j^{(p)}p^{-s})^{-1} = \det(1 - p^{-s}\varphi_g|\mathcal{E}_{g,p})^{-1}$. The global Euler product (over primes of good reduction, with local factors at bad primes defined as in Serre 1970) gives $\Lambda_{\mathcal{D}_g}(s)$ in the respective window.

Step 2: Consani–Marcolli genus-g Koszul dual. Consani–Marcolli 2006 (*J. Reine Angew. Math.* 597) constructs the rank-$2g$ Bost–Connes–Marcolli system with KMS partition function $\zeta_K(\beta)$ in the number-field row. Identifying the Koszul dual of $\mathcal{V}_{\text{Heis}}^{\text{prim},g}$ with the row $\Lambda_{\mathcal{D}_g}$ outside that row is part of the residual input.

Step 3: rank-g Positselski pairing. Arithmetic Positselski (Positselski 2011 Mem. AMS 996; Chapter 46) at rank g, extended to Λ_2-coefficients for Window V. Pairs the bar coalgebra of the rank-$2g$ Heisenberg with the cobar algebra of the genus-g Bost–Connes system; total weight vanishes modulo the 2-dim μ-invariant vanishing condition (extending Ferrero–Washington 1979 *Ann. Math.* 109 for abelian at rank 1).

Step 4: adelic Fourier symmetry $s \mapsto w_{\mathcal{D}_g} - s$. Tate 1950 at $g = 0$ (Window II with $K = \mathbb{Q}$); Weil 1967 *Basic Number Theory* for totally imaginary base (Window II general); Bloch–Kato 1990 and Fontaine–Perrin-Riou 1994 for weight-1 motives on Jacobians (Windows I, V); and the automorphic theory available for the chosen Siegel or unitary datum (Window VI) are the inputs. Where these inputs do not give analytic continuation and the functional equation, they are part of the residual condition.

No step in the all-genus row is promoted to an unconditional theorem by this chapter. The result is a conditional reduction of the Koszul self-duality identity to the named Euler-product, functional-equation, and Positselski inputs. □

102.5 Reduction to $g = 0$ and the F-RH Axis

At $g = 0$:

- Window II: $K = \mathbb{Q}$, $\xi_K(s) = \xi(s)$, $\Gamma_{\mathbb{C}}(s)^0 = 1$, $\zeta_K = \zeta$. Functional equation $\xi(s) = \xi(1 - s)$.
- Windows I, V: $X_K = \mathbb{P}^1/K$, Jacobian trivial, $\Lambda(s, X_K)$ degenerates to $\xi(s)$ with center $s \mapsto 1 - s$; the weight-1 motive is void and the weight-0 Dedekind row takes over.

- Window VI: $\mathcal{A}_0 = \mathrm{pt}$, Siegel spin L-function collapses to ζ, center $s \mapsto 1 - s$.

At the level of the displayed scalar character, these specializations are intended to give $\xi(s) = \xi(1 - s)$. Recovering the arithmetic criterion of Theorem 43.5.1 additionally requires its central-defect and evaluation maps. A scalar specialization does not construct them.

COROLLARY 102.5.1 *(all-genus F-RH). (on Chapter 102.4.1 plus rank-g extensions of the P2, P3, VB pillars of Chapter 44).* For every $g \geq 0$ and every arithmetic genus-g datum $\mathcal{D}_g$ with rank-g Koszul–Petersson compatibility, a rank-g completed L-character, and a positive Hilbert–Pólya–Li trace, the non-trivial zeros of $\Lambda_{\mathcal{D}_g}(s)$ lie on the self-dual locus $\mathrm{Re}(s) = w_{\mathcal{D}_g}/2$.

Remark 102.5.2 *(specialisations).* Chapter 102.5.1 has as special cases the zero-placement clauses for the Riemann zeta function ($g = 0$, Window II), Hasse–Weil L-functions of Jacobians (Windows I,V), Dedekind zeta functions of totally imaginary fields (Window II at higher g), and Siegel-spin L-functions of cusp forms on $\mathrm{GSp}_{2g}(\mathbb{A}_{\mathbb{Q}})$ (Window VI). Each case still requires its rank-g spectral positivity input.

102.6 Residual Conditions

Chapter 102.4.1 reduces to ten residual conditions, each extending a Chapter-level $g = 0$ result to arbitrary g.

(i) *2-dim arithmetic Positselski equivalence* with $\mu_1 = \mu_2 = 0$ on $\Lambda_2 = \mathbb{Z}_p[[\Gamma_1 \times \Gamma_2]]$, extending Chapter 46. Condition: non-formal theorem. Precedents: Nekovář 2006 (Astérisque 310); Coates–Sujatha 2006 *Cyclotomic Fields and Zeta Values*.

(ii) *Rank-g Koszul–Petersson compatibility* replacing the scalar Petersson inner product with the genus-g Petersson pairing on weight-(k, g) automorphic forms, extending Chapter 47. It is a non-formal theorem.

(iii) *2-dim Parshin–Gersten bar construction* on the arithmetic surface with double residue (∂_1, ∂_2), extending Chapter 33. It is a non-formal theorem. Precedents: Parshin 1976, Kato 1980, Beilinson 1980.

(iv) *Rank-g arithmetic Ben-Zvi–Nadler* with target stack Σ_g^{Ar} replacing $B\widehat{\mathbb{Z}}$, extending Chapter 49. It is a non-formal theorem.

(v) *GL_{2g}-local geometric Langlands.* Fargues–Scholze arXiv:2102.13459 supplies the spectral action in broad reductive generality, while Emerton–Gee arXiv:1908.07185 is the GL_n Galois deformation-stack input. The categorical comparison at GL_{2g} is not a formal corollary of either source; it is the rank-g scaling of Chapter 42.1.1. Domain: determined by F1 scaled to rank $2g$.

(vi) *Perfectoid E_2-enhancement at rank g* via BD Grassmannians for $GL_{2g}/\mathbb{Q}_p$, extending F2 (Chapter 42.2.1). It is a non-formal theorem.

(vii) *Rank-g arithmetic Virasoro bootstrap* with central charge $c^{\mathrm{Ar},g} = g$ on the primary subspace of the rank-g archimedean Fock, extending Chapter 45. It is a non-formal theorem.

(viii) *Arithmetic Siegel–Weil at bad primes for $g \geq 3$*, extending Li–Liu 2021 arXiv:2101.09485. It is a non-formal theorem.

(ix) *Harder-Eisenstein-congruence prime at genus g* replacing $p = 691$ in F6 (Chapter 42.4.4). Systematic tabulation by Bergström–Faber–van der Geer 2014 (*Sel. Math.* 20) identifies the relevant Hecke-eigenvalue congruences. No genus-2 replacement prime is fixed here before that table is read. It is a non-formal theorem.

(x) *Arithmetic* B*-row genus-g ceiling.* F4 (Chapter 42.4.1) sharpens from the single integer $\kappa^{\mathrm{Ar}} + (\kappa^{!})^{\mathrm{Ar}} = 8$ to a genus-dependent Shimura-signature-indexed sum, via Bruinier–Heegner Chern-class reciprocity on $\mathrm{Sh}(\mathrm{O}(2, 2g-1))$ with accidental isomorphism $\mathrm{O}(2,3) \cong \mathrm{Sp}_4/\{\pm I\}$ at $g = 2$. It is a non-formal theorem.

The total residual condition is a non-formal theorem condition, organised as three to four mathematical blocks. Block (i): conditions 1–4 + 7 (Beilinson / Positselski / bar-cobar on 2-dim base plus Virasoro bootstrap). Block (ii): conditions 5, 6 (GL_{2g} Fargues–Scholze local plus global Arinkin–Gaitsgory). Block (iii): conditions 8, 10 (Kudla–Rapoport and Bruinier–Heegner at genus g). Block (iv): condition 9 and consolidation (Harder-Eisenstein congruences; all-genus F-RH corollary).

102.7 Orthogonal and Compatible Windows

The six windows are compatible but mathematically distinct: each captures a different facet of the datum $\mathcal{D}_g$, and their pairwise compatibility is an independent mathematical fact in each case. Pairwise compatibility:

- **Windows I and V** share the arithmetic surface as base and agree on the weight-1 Hasse–Weil functional equation. They differ on operadic dimension: Window I keeps the archimedean E_1-chiral structure; Window V lifts to E_2-chiral via Dunn. Compatibility is the content of Chapter 33's 2-dim extension.
- **Windows II and V** differ on arithmetic scheme dimension (1-dim vs. 2-dim). Window II encodes the genus as coefficient enhancement of the base field (totally imaginary K of degree $2g$); Window V keeps $K = \mathbb{Q}$ but lifts the base scheme to 2-dim via Dunn additivity. Compatibility passes through Tate 1950 adelic Fourier at number-field level.
- **Windows III and VI** are spectral: Window III gives the local (φ, Γ)-module moduli; Window VI gives the global automorphic side. Compatibility requires the relevant local-global comparison; it is not supplied by Fargues–Scholze alone.
- **Window IV** is moduli-theoretic: $\mathcal{T}_g^{\mathrm{arith}}$ carries the Hodge–Arakelov decoration on the theta-generic locus, and its point-by-point inputs supply Window I. Compatibility via the Torelli map: a genus-g curve's Jacobian is a point of $\mathcal{X}_{2g}^{\mathrm{crys,ord}}$ only after imposing the ordinary crystalline local condition, and a point of the Siegel moduli space $\mathcal{A}_g$ via its polarisation. The word "threefold" applies only at $g = 2$.

Orthogonal to these six windows: Deninger's $\mathbb{F}_1$ formalism (motivational context for the arithmetic-dynamical interpretation of Spec $\mathbb{Z}$, recorded in Chapter 24); the Caraiani–Scholze unitary-Shimura theory at genera where no Shimura model exists (moduli of curves $\mathcal{M}_g$ is not Shimura for $g \geq 4$); the Bethe-algebra heuristic presentation, which operates at the zero-Fock per-place level and does not carry g-dependence directly.

The conditional identity $\kappa^{\mathrm{Ar},g} + (\kappa^{!})^{\mathrm{Ar},g} = 0$ of Chapter 102.4.1 is the proposed rank-g fibre. The F-RH axis of Chapter 44 is the $g = 0$ fibre of the rank-graded family whose other fibres are the Hasse–Weil Riemann Hypothesis for Jacobians of curves of genus g, the generalised Riemann Hypothesis for Dedekind zeta of totally imaginary fields of degree $2g$, and the Siegel-spin Riemann Hypothesis for cusp forms on GSp_{2g}. The residual conditions of Chapter 102.6 quantify what must be proved before this conditional comparison becomes an unconditional theorem.

Chapter 103 considers arithmetic three-dimensional Chern–Simons theory. Its Kac–Moody comparison requires the current operations and normalized levels of Question 103.3.1. A conformal vector does not construct them.

Finite-level state sums are given by Theorem 103.4.2 from finite groupoids and action classes. A renormalized $GL_2(\widehat{\mathbb{Z}})$ limit is a separate construction, as stated in Conjecture 103.4.7. The equalities

$$Z_{\mathrm{ACS}^{\mathrm{qu}}} = \xi(k^{\mathrm{Ar}}), \qquad \mathrm{HH}_0(\partial\mathrm{ACS}^{\mathrm{qu}}) \simeq \mathcal{V}^{\mathrm{prim}}$$

are conditional on profinite renormalisation and the local trace comparison (Chapter 103.4.9). The ordered incidence model is computed by the normalised ordered bar construction on the auxiliary place order $\mathbb{P} \cup \{\infty\}$ (Chapter 103.5.4); the comparison with full Beilinson–Drinfeld / Costello–Gwilliam arithmetic chiral homology is conditional (Chapter 103.5.5). F-RH becomes a conditional spectral theorem on the partition-function zero locus (Chapter 103.7.1); the all-genus extension (Chapter 103.8.1) is formulated row by row on the corresponding residual hypotheses.

Chapter 103

3D Arithmetic Chern-Simons on Etale Spec(Z) and the Ordered Incidence Bar

An arithmetic Kac–Moody enhancement requires current operations and level comparisons beyond a conformal vector. The local vertex algebras constructed below make those additional inputs explicit. Finite arithmetic Chern–Simons state sums arise instead from finite local-system groupoids and their action classes.

The profinite $GL_2(\widehat{\mathbb{Z}})$ quantum theory, its boundary identification with $\mathcal{V}^{\mathrm{prim}}$, and its zeta partition function remain conditional. A supplied functor on ordered merged-place blocks has a simplicial replacement that computes its derived colimit. Identifying that complex with geometric arithmetic chiral homology is a further comparison. The associated Deninger construction requires a spectral-flow hypothesis beyond its explicit prime-orbit model.

103.1 Spec(Z) as Etale 3-Space

Mazur 1973 (*Notes on étale cohomology of number fields*, Ann. Sci. ENS 6) observed that $\operatorname{Spec} \mathcal{O}_K$ has étale cohomological dimension 3 for K a number field, with Artin–Verdier duality supplying a fundamental class $[\operatorname{Spec} \mathcal{O}_K] \in H^3_{\text{ét}}(\operatorname{Spec} \mathcal{O}_K, \widehat{\mathbb{Z}})$ analogous to a 3-manifold's orientation class. Morishita 2012 *Knots and Primes* (arXiv:0904.3399) systematizes the dictionary: primes $\mathfrak{p} \subset \mathcal{O}_K$ behave as knots in a closed 3-manifold; $\operatorname{Spec} \mathcal{O}_{K,\mathfrak{p}}$ is the tubular neighborhood; the inertia subgroup $I_{\mathfrak{p}} \subset \operatorname{Gal}(\overline{K}/K)$ is the meridian; linking numbers become Legendre symbols; the Alexander–Fox module becomes the Iwasawa module. For $K = \mathbb{Q}$, Arakelov compactification $\overline{\operatorname{Spec} \mathbb{Z}}$ closes the 3-space by adding the archimedean boundary component at ∞.

Remark 103.1.1 *(cohomological 3-dimensionality).* $\operatorname{Spec} \mathbb{Z}$ is not a smooth real 3-manifold. Its 3-dimensionality is cohomological: the étale site supports non-trivial H^1, H^2, H^3 computable via global class-field theory and Artin–Verdier. The Chern-Simons action functional below takes values in étale cohomology classes rather than in differential forms; its quantization is in Galois-representation-theoretic terms rather than path-integral terms in the differential-geometric sense.

103.2 Kim Classical Arithmetic Chern-Simons

Kim 2015 *Arithmetic Chern-Simons Theory I* (arXiv:1510.05818), and the non-abelian extension Chern-Dan-Kim-Park 2018 (arXiv:1609.03012), define the classical action

$$\mathrm{CS}^{\mathrm{Ar}}_{M,c}(\rho) := \langle \rho^* c, [M] \rangle \in \mathbb{Q}/\mathbb{Z}, \tag{103.1}$$

for $M = \operatorname{Spec} \mathcal{O}_K$, $\rho : \pi_1^{\text{ét}}(M) \to G$ a continuous representation into a finite group G, $c \in H^3(BG, \mathbb{Q}/\mathbb{Z})$ a level class, and $[M]$ the Artin–Verdier fundamental class. Summation over ρ gives the classical partition function $Z^{\mathrm{cl,Ar}}_{\mathrm{CS},c}(M) = \sum_\rho \frac{e^{2\pi i \mathrm{CS}^{\mathrm{Ar}}_{M,c}(\rho)}}{|\mathrm{Aut}(\rho)|}$. Chapter 37 already uses this classical

torsor as a boundary action-line input for $\mathcal{V}^{\mathrm{prim}}$; finite state sums and the profinite quantum-TQFT problem are developed in Chapter 103.4.

103.3 The Arithmetic Kac–Moody Lift

Chapter 45 supplies local Virasoro calculations on the finite Heisenberg covers and records the rank residue $r_\zeta = 2$. Chapter 102 upgrades per-window rank to g from the rank-g archimedean Fock. A conformal vector alone does not produce $\widehat{\mathfrak{gl}}_2$ currents. The Kac–Moody enhancement therefore requires the Satake current algebra as part of the input; the Sugawara construction then compares its Virasoro field with the Heisenberg conformal vector away from the critical level.

Question 103.3.1 *(arithmetic Kac–Moody enhancement).* Construct a state-field realization of the arithmetic Hochschild trace class and maps from independently defined local affine current algebras into it. Specify common coefficients, local levels, reference vacua, and the completion used to assemble the places. The Satake eigenvalues (α_p, β_p) do not supply a basis of $\mathfrak{gl}_2$ or its current commutators.

There is a local vertex-algebra construction once the inputs are given. Over $\mathbb{C}$, supply vertex algebras H_p and universal affine vacuum algebras $V^{b_p}(\mathfrak{gl}_2)$, where b_p is a chosen invariant bilinear form. Inserting the vacuum at each new place defines

$$\mathcal{A}_{\mathrm{loc}} := \varinjlim_{S \subset \mathbb{P},\ S \text{ finite}} \bigotimes_{p \in S} \big(H_p \otimes V^{b_p}(\mathfrak{gl}_2)\big).$$

Each vector involves finitely many places, so its field and every vertex identity are computed in a finite tensor product. Thus this algebraic colimit is a vertex algebra. An arithmetic identification and a continuous extension to a chosen completion remain extra maps.

Remark 103.3.2 *(the noncritical-level condition).* On the $\mathfrak{sl}_2$ factor, normalize the invariant form so that the dual Coxeter number is 2. The Sugawara coefficient is $1/(2(k+2))$, so this formula requires $k \neq -2$. The central one-dimensional current requires its own nonzero level to define the usual Heisenberg conformal vector.

The arithmetic scalar identity $\kappa + \kappa^! = 0$ gives no noncritical-level theorem. Even an additional identification $k = \kappa$ would allow $\kappa = -2$ and $\kappa^! = 2$. A comparison with an arithmetic conformal vector must therefore supply both the level normalization and the noncritical hypothesis.

103.4 3D Quantum Arithmetic Chern-Simons

Definition 103.4.1 *(finite arithmetic field groupoid).* . Let M be an arithmetic 3-space in the Mazur–Artin–Verdier sense, for example $\overline{\mathrm{Spec}\, O_K}$ with its compact-support fundamental class. Let G be a finite group and let $\alpha \in H^3(BG, \mathbb{C}^\times)$ be a level. The finite arithmetic field groupoid is

$$\mathcal{F}_G(M) := \mathrm{Map}_{\mathrm{cont}}\big(\Pi_1^{\text{ét}}(M), BG\big),$$

the groupoid of continuous G-local systems on the etale homotopy type of M. For a field $\rho \in \mathcal{F}_G(M)$, the action is

$$S_\alpha^{\mathrm{Ar}}(\rho) := \big\langle \rho^*\alpha, [M] \big\rangle \in \mathbb{C}^\times.$$

Theorem 103.4.2 *(finite-level arithmetic state sum and gluing).* . For a closed arithmetic 3-space M for which $\mathcal{F}_G(M)$ is a finite groupoid, the assignment

$$Z_{G,\alpha}^{\mathrm{Ar}}(M) := \int_{\mathcal{F}_G(M)} S_\alpha^{\mathrm{Ar}} = \sum_{[\rho] \in \pi_0 \mathcal{F}_G(M)} \frac{\langle \rho^*\alpha, [M] \rangle}{|\operatorname{Aut}(\rho)|}$$

is Kim's arithmetic Chern–Simons partition sum for the class α. If an arithmetic gluing $M = M_1 \cup_\Sigma M_2$ has finite field groupoids and satisfies étale descent

$$\mathcal{F}_G(M) \simeq \mathcal{F}_G(M_1) \times_{\mathcal{F}_G(\Sigma)} \mathcal{F}_G(M_2),$$

then $Z^{\mathrm{Ar}}_{G,\alpha}$ satisfies the corresponding finite-groupoid gluing formula.

Proof. The construction uses only finite homotopy types. To a closed arithmetic 3-space M, assign the finite groupoid $\mathcal{F}_G(M)$ and integrate the locally constant function S^{Ar}_α by groupoid cardinality. If

$$M = M_1 \cup_\Sigma M_2$$

is a gluing along an arithmetic boundary, then etale descent for local systems gives a homotopy pullback

$$\mathcal{F}_G(M) \simeq \mathcal{F}_G(M_1) \times_{\mathcal{F}_G(\Sigma)} \mathcal{F}_G(M_2).$$

The fundamental class is additive under gluing, so the action multiplies:

$$\langle \rho^* \alpha, [M] \rangle = \langle \rho_1^* \alpha, [M_1] \rangle \langle \rho_2^* \alpha, [M_2] \rangle.$$

Fubini for finite groupoid cardinality then gives the displayed gluing formula. Kim's action is exactly the same evaluation of a pulled-back degree-3 class on the Artin–Verdier fundamental class, written in $\mathbb{Q}/\mathbb{Z}$ and exponentiated here into $\mathbb{C}^\times$. □

Definition 103.4.3 *(finite-level boundary category).* . For an arithmetic boundary Σ, let $\tau_\Sigma(\alpha)$ be the transgression of $\alpha \in H^3(BG, \mathbb{C}^\times)$ to a 2-cocycle on the field groupoid $\mathcal{F}_G(\Sigma)$. The finite-level boundary category is

$$\mathcal{B}_{G,\alpha}(\Sigma) := \mathrm{Sh}_{\tau_\Sigma(\alpha)}\big(\mathcal{F}_G(\Sigma)\big),$$

the category of finite-dimensional $\tau_\Sigma(\alpha)$-twisted local systems on $\mathcal{F}_G(\Sigma)$.

The partition function $Z^{\mathrm{Ar}}_{G,\alpha}(M)$ is a scalar for closed M. For a manifold with boundary, the TQFT assignment must produce a category: the state category on the boundary. The finite boundary category provides this assignment.

Theorem 103.4.4 *(finite-level bulk-boundary span). (on a finite arithmetic cobordism category whose morphisms satisfy the gluing hypothesis of Chapter 103.4.2).* The finite-level theory $Z^{\mathrm{Ar}}_{G,\alpha}$ assigns $\mathcal{B}_{G,\alpha}(\Sigma)$ to the arithmetic boundary Σ. If M is a cobordism from Σ_0 to Σ_1, then M acts by the integral transform

$$\mathrm{Sh}_{\tau_{\Sigma_0}(\alpha)}\big(\mathcal{F}_G(\Sigma_0)\big) \longrightarrow \mathrm{Sh}_{\tau_{\Sigma_1}(\alpha)}\big(\mathcal{F}_G(\Sigma_1)\big)$$

whose kernel is the action local system on $\mathcal{F}_G(M)$. For a closed M, the categorical trace of this transform is $Z^{\mathrm{Ar}}_{G,\alpha}(M)$.

Proof. Restriction of local systems gives a span

$$\mathcal{F}_G(\Sigma_0) \leftarrow \mathcal{F}_G(M) \rightarrow \mathcal{F}_G(\Sigma_1).$$

The transgressed cocycle records the boundary anomaly of the degree-3 class α; tensoring by the action local system and then pushing forward along the right leg gives the displayed integral transform. Composition of cobordisms is composition of spans. The homotopy-pullback formula from Chapter 103.4.2 makes the two composite kernels canonically isomorphic. When the boundary is empty, the trace reduces to groupoid-cardinality integration over $\mathcal{F}_G(M)$, which is precisely $Z^{\mathrm{Ar}}_{G,\alpha}(M)$. □

Proposition 103.4.5 *(finite boundary Hochschild trace).* . For a finite groupoid $\mathcal{F}_G(\Sigma)$ over $\mathbb{C}$, the finite boundary category

$$\mathcal{B}_{G,\alpha}(\Sigma) = \mathrm{Sh}_{\tau_\Sigma(\alpha)}(\mathcal{F}_G(\Sigma))$$

has Hochschild trace equal to the twisted inertia trace of $\mathcal{F}_G(\Sigma)$:

$$\mathrm{HH}_0\big(\mathcal{B}_{G,\alpha}(\Sigma)\big) \cong \Gamma\big(\pi_0 I\mathcal{F}_G(\Sigma), L_{\tau_\Sigma(\alpha)}\big), \qquad \mathrm{HH}_{n>0}\big(\mathcal{B}_{G,\alpha}(\Sigma)\big) = 0.$$

Here $I\mathcal{F}_G(\Sigma)$ is the inertia groupoid, and $L_{\tau_\Sigma(\alpha)}$ is the transgressed local system on inertia.

Proof. A finite twisted sheaf category on a finite groupoid is Morita equivalent to modules over the corresponding finite-dimensional twisted groupoid algebra. Over $\mathbb{C}$ this algebra is separable: it is a finite product of twisted finite group algebras, hence semisimple. Hochschild homology of a separable algebra with coefficients in itself vanishes in positive degrees. Degree zero is the algebra modulo commutators, which is the space of twisted class functions on conjugacy classes, equivalently sections of the transgressed local system on the inertia groupoid. □

Corollary 103.4.6 *(Wilson lines at finite level).* . A prime p viewed as a knot in the Morishita dictionary supports Wilson defects labelled by representations of the local stabiliser group of a point of $\mathcal{F}_G(\mathrm{Spec}\,\mathbb{Q}_p)$. For a one-dimensional label χ, the Wilson expectation is the corresponding Frobenius trace $\chi(\mathrm{Frob}_p)$.

Proof. Removing the prime p and fixing monodromy around the meridian is the arithmetic analog of prescribing a conjugacy class around a knot. The defect category is the module category of the stabiliser of that local monodromy. A one-dimensional module evaluates a loop by its character, hence the meridian/Frobenius loop contributes $\chi(\mathrm{Frob}_p)$. □

Conjecture 103.4.7 *(renormalised $GL_2(\widehat{\mathbb{Z}})$ quantum arithmetic CS).* . Let $G_N = GL_2(\mathbb{Z}/N\mathbb{Z})$, and let $\alpha_N \in H^3(BG_N, \mathbb{C}^\times)$ be a compatible tower of finite levels whose inverse-limit class is the arithmetic level $k^{\mathrm{Ar}} = \kappa_{\mathsf{G}}^{\mathrm{Ar}}$. The finite theories $Z^{\mathrm{Ar}}_{G_N,\alpha_N}$ admit a renormalised profinite limit

$$\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}} := \varprojlim_{N}^{\mathrm{ren}} Z^{\mathrm{Ar}}_{G_N,\alpha_N}$$

on the etale 3-space $\overline{\mathrm{Spec}\,\mathbb{Z}}$. This limit satisfies three bulk-boundary axioms:

(i) (Boundary WZW.) The 2D factorization algebra on the archimedean fibre $\{\infty\}$ produced by canonical quantisation of $\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}}$ has Hochschild trace $\mathcal{V}^{\mathrm{prim}}$:

$$\mathrm{HH}_0\big(\partial\,\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}}\big) \simeq \mathcal{V}^{\mathrm{prim}}.$$

(ii) (Partition function.) The bulk partition function equals the completed zeta character at the level: $Z_{\mathrm{ACS}^{\mathrm{qu}}}(\overline{\mathrm{Spec}\,\mathbb{Z}}; k^{\mathrm{Ar}}) = \xi(k^{\mathrm{Ar}})$.

(iii) (Wilson lines as Frobenius traces.) For a prime $p \in \mathbb{P}$ viewed as a knot in the Morishita dictionary, the Wilson-line observable in representation $\rho : \pi_1^{\text{ét}} \to GL_2$ reproduces the Frobenius trace: $\langle W_\rho(K_p)\rangle_{\mathrm{ACS}^{\mathrm{qu}}} = \mathrm{tr}\,\rho(\mathrm{Frob}_p)$.

Remark 103.4.8 *(finite vs. profinite ACS: gap between theorem-grade and conjecture-grade).* The closed finite-level state sum is theorem-grade: Chapter 103.4.2 constructs it from étale local systems, the Artin–Verdier fundamental class, and finite groupoid cardinality. Functorial finite-level TQFT language still requires a specified arithmetic cobordism category. The larger residual is the renormalised profinite limit $G_N \to GL_2(\widehat{\mathbb{Z}})$, the comparison of its archimedean boundary category with $\mathcal{V}^{\mathrm{prim}}$, and the identification of its renormalised partition function with $\xi(k^{\mathrm{Ar}})$.

There is a concrete obstruction to treating the closed finite tower as the zeta partition function. On the closed base $\overline{\mathrm{Spec}\,\mathbb{Z}}$, finite etale local systems are trivial by Hermite–Minkowski, so the bare finite state sum for $G_N = GL_2(\mathbb{Z}/N\mathbb{Z})$ is the groupoid-cardinality scalar $|G_N|^{-1}$, or 1 after finite-gauge normalisation. This scalar does not contain prime weights p^{-s}, the archimedean $\Gamma_{\mathbb{R}}$-factor, or the polar correction. A zeta-valued profinite ACS theory therefore requires extra data: compatible continuous level classes, a counterterm or measure system on the congruence tower, a profinite boundary category, a Morita comparison with the local Iwahori trace factors, and a regularised character map.

Theorem 103.4.9 *(bulk-boundary comparison with $\mathcal{V}^{\mathrm{prim}}$). (on the renormalised profinite limit in Chapter 103.4.7 and on the local Satake/Iwahori trace identifications of Chapters 50 and 38).* Let

$$\mathcal{B}_\infty^{\mathrm{Ar}} := \overset{\mathrm{ren}}{\varprojlim_N} \mathcal{B}_{G_N,\alpha_N}(\{\infty\})$$

be the renormalised archimedean boundary category of the profinite ACS tower. If its finite-place compact generators identify with the local Iwahori Hecke generators and its archimedean vacuum sector identifies with the rank-2 Heisenberg Kac–Moody vacuum, then

$$\mathrm{HH}_\bullet(\mathcal{B}_\infty^{\mathrm{Ar}}) \simeq \mathcal{V}^{\mathrm{prim}}.$$

Equivalently, the boundary factorization algebra of $\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}}$ is $\mathcal{V}^{\mathrm{prim}}$.

Proof. At finite level, Chapter 103.4.4 identifies the boundary state category with twisted sheaves on the boundary field groupoid. Passing through the congruence tower gives a renormalised inverse system of such categories. The stated compact-generation hypothesis is exactly the assertion that the local factors of this system are the Iwahori Hecke categories used in the definition of $\mathcal{V}^{\mathrm{prim}}$. Hochschild trace is local for the restricted tensor product under the compact-generation and spherical unit hypotheses of Chapter 50. Thus the finite places contribute the local trace factors $\mathcal{H}^{(p)}$, while the archimedean vacuum contributes $\mathcal{H}^{(\infty)}$. Their restricted tensor product is the Hochschild trace class $\mathcal{V}^{\mathrm{prim}}$ by Chapter 38. This proves the displayed boundary identification, conditional precisely on the profinite renormalisation and the local trace comparison. □

Remark 103.4.10 *(boundary trace form).* The finite boundary trace of Chapter 103.4.5 is theorem-grade, but it does not identify the boundary with $\mathcal{V}^{\mathrm{prim}}$. The comparison

$$\mathrm{HH}_\bullet(\mathcal{B}_\infty^{\mathrm{Ar}}) \simeq \mathcal{V}^{\mathrm{prim}}$$

requires an independent profinite Morita equivalence

$$\mathcal{B}_\infty^{\mathrm{Ar}} \simeq_{\mathrm{Morita}} \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}},$$

where $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}$ is the global arithmetic Iwahori–Hecke category (restricted tensor product of local Iwahori–Hecke categories with archimedean vacuum, defined in Chapter 38), compatible with the

chosen topology, completion, restricted tensor product, local Satake/Iwahori generators, archimedean vacuum, and the arithmetic Ben-Zvi–Nadler and wild-prime hypotheses. Until these data are proved, Chapter 103.4.9 is only a conditional implication.

Remark 103.4.11 *(Vol II cross-volume bridge).* Vol II Part VI states a 3D holomorphic-twist QFT on $C \times \mathbb{R}$ whose boundary factorization algebra on C is the complex $\mathcal{V}$. The arithmetic analog replaces the holomorphic twist by the Chern-Simons sector: the arithmetic base $\overline{\mathrm{Spec}\,\mathbb{Z}}$ has étale cohomological dimension 3 without a holomorphic structure in the differential-geometric sense. The 3D topological gauge theory over the arithmetic base is therefore Chern-Simons, not holomorphic twist. The complex $(HT)_{C\times\mathbb{R}} \to \partial = \mathcal{V}$ adjunction becomes an arithmetic boundary category whose Hochschild trace is $\mathcal{V}^{\mathrm{prim}}$.

103.5 The Ordered Incidence Bar

The discrete set of arithmetic places supplies no position diagonal. An ordered incidence category provides a coefficient model once its adjacent products have been specified: it encodes the primes of $\mathbb{Z}$ via the auxiliary total order $2 < 3 < 5 < 7 < \cdots < \infty$, with the archimedean place placed last. This order is not geometric structure on $\overline{\mathrm{Spec}\,\mathbb{Z}}$; the comparison with Beilinson–Drinfeld/Costello–Gwilliam arithmetic chiral homology is conditional (Chapter 103.5.5).

Definition 103.5.1 *(ordered incidence category).* . Put $V = \mathbb{P} \cup \{\infty\}$, with the order $2 < 3 < 5 < \cdots < \infty$. Let $\mathsf{Inc}_<(V)$ be the category whose objects are finite ordered block decompositions of finite ordered subsets of V. A singleton block is written v, and the merge of two adjacent blocks is written by concatenation. Morphisms are generated by order-preserving adjacent contractions

$$\{v_1 < \cdots < v_i < v_{i+1} < \cdots < v_n\} \longrightarrow \{v_1 < \cdots < (v_i v_{i+1}) < \cdots < v_n\},$$

where $v_i v_{i+1}$ denotes the formal merged block carrying the product operation of the coefficient system, not a new arithmetic place. The simplicial nerve of $\mathsf{Inc}_<(V)$ records chains of adjacent contractions. Its comparison with a geometric arithmetic configuration space is additional data.

Definition 103.5.2 *(ordered incidence homology).* . Specify cochain complexes $\mathcal{H}^{(B)}$ over $\mathbb{C}$ for every nonempty ordered block B, including $\mathcal{H}^{(v)}$ at a singleton, and compatible adjacent products. For singleton blocks these are

$$m_{v,w} \colon \mathcal{H}^{(v)} \otimes \mathcal{H}^{(w)} \longrightarrow \mathcal{H}^{(vw)} \qquad (v < w).$$

Here $\mathcal{H}^{(vw)}$ means the coefficient assigned to the merged block vw. Assume the products and their identities define a strict functor $F_{\mathcal{H}} \colon \mathsf{Inc}_<(V) \to \mathrm{Ch}_{\mathbb{C}}$, where $\mathrm{Ch}_{\mathbb{C}}$ denotes cochain complexes. It sends an ordered block decomposition $B_1|\cdots|B_r$ to

$$F_{\mathcal{H}}(B_1|\cdots|B_r) := \mathcal{H}^{(B_1)} \otimes \cdots \otimes \mathcal{H}^{(B_r)},$$

with contractions acting by the corresponding adjacent products. For a homotopy-coherent coefficient system, choose a strict diagram model before using this formula. Relative to this specified functor, the ordered incidence homology object is the derived colimit

$$\int^{\mathrm{ord}}_{\overline{\mathrm{Spec}\,\mathbb{Z}}} \mathcal{V}^{\mathrm{prim}} := \operatorname*{L\,colim}_{\mathsf{Inc}_<(V)} F_{\mathcal{H}}.$$

Definition 103.5.3 *(ordered incidence bar).* . Suppose the coefficient functor $F_{\mathcal{H}}$ of the preceding definition has been supplied, including coherent adjacent products on all merged blocks. A finite-place vacuum is a pointing, hence an underlying E_0-datum; it does not supply these products. The notation $\mathcal{V}^{\mathrm{prim}}$ for the arithmetic Hochschild trace class determines no such coefficient functor by itself. Relative to the chosen $F_{\mathcal{H}}$, the *ordered incidence bar* is the normalized chain complex of the simplicial replacement of $F_{\mathcal{H}}$:

$$B^{\mathrm{ord}}_{\bullet}(\mathcal{V}^{\mathrm{prim}}) := N_{\bullet}\,\mathrm{srep}_{\mathsf{Inc}_<(V)}(F_{\mathcal{H}}).$$

Here $\mathrm{srep}_{\mathsf{Inc}_<(V)}(F_{\mathcal{H}})$ is the simplicial replacement of $F_{\mathcal{H}}$ (the standard bar construction on a functor from a small category to chain complexes), and $N_{\bullet}$ is the normalized chain functor on simplicial chain complexes. In simplicial degree n, this is the direct sum over n-simplices $C_0 \to \cdots \to C_n$ of $\mathsf{Inc}_<(V)$ of $F_{\mathcal{H}}(C_0)$, modulo degeneracies. The first face applies $F_{\mathcal{H}}(C_0 \to C_1)$; an interior face composes two arrows, and the last face drops the last object. Thus all targets are coefficients of formal merged blocks, not new arithmetic places.

Its direct-sum total complex has

$$(\mathrm{Tot}^{\oplus}\, B^{\mathrm{ord}})^q = \bigoplus_{n\geq 0} (B^{\mathrm{ord}}_n)^{q+n}, \qquad d_{\mathrm{tot}}|_{B_n} = (-1)^n d_{\mathrm{int}} + \sum_{i=0}^{n} (-1)^i d_i.$$

The face sum is zero at $n = 0$. The notation $H_{\bullet}(B^{\mathrm{ord}}_{\bullet})$ means the homology of this total complex, with $H_m = H^{-m}$.

Theorem 103.5.4 *(ordered incidence homology via normalized bar).* . The ordered incidence homology of $\mathcal{V}^{\mathrm{prim}}$ is computed by the ordered incidence bar:

$$\int^{\mathrm{ord}}_{\overline{\mathrm{Spec}\,\mathbb{Z}}} \mathcal{V}^{\mathrm{prim}} \;\simeq\; \mathrm{Tot}^{\oplus}\, B^{\mathrm{ord}}_{\bullet}(\mathcal{V}^{\mathrm{prim}}).$$

Proof. For an object C of a small category $\mathcal{I}$, let $P_C(M)(D) = \bigoplus_{\mathcal{I}(C,D)} M$. This is the free diagram generated by a complex M at C. The simplicial free-diagram resolution of a functor F has degree n

$$\bigoplus_{C_0\to\cdots\to C_n} P_{C_n}(F(C_0)).$$

Its augmentation to F, evaluated at D, is contracted by appending the final arrow $C_n \to D$ and the identity of D. The usual alternating face differential makes this the extra-degeneracy contraction. Its degeneracies split, and over $\mathbb{C}$ its free diagram terms give a cofibrant simplicial resolution, also for unbounded coefficient complexes. Thus its realization computes the left derived colimit.

The colimit of $P_C(M)$ is M: every summand labelled by $C \to D$ is the image of the summand at the identity of C, with precisely these identifications. Taking colimits in the displayed resolution therefore gives the simplicial replacement of F. Quotienting the split degeneracies leaves its normalized complex. The internal and face differentials commute before totalization; $(-1)^n$ makes them anticommute, so its realization is the stated direct-sum total complex. Apply this construction to $\mathcal{I} = \mathsf{Inc}_<(V)$. □

Corollary 103.5.5 *(comparison with BD/CG chiral homology). (on constructing an arithmetic Beilinson–Drinfeld/Costello–Gwilliam Ran model for* $\overline{\mathrm{Spec}\,\mathbb{Z}}$*, finite-place collision thickenings, constructible descent, ordered-to-unordered descent, and completion exactness identifying the arithmetic chiral homology of Chapters 33 and 30 with the ordered incidence model).* One has

$$\int_{\overline{\mathrm{Spec}\,\mathbb{Z}}} \mathcal{V}^{\mathrm{prim}} \simeq \mathrm{Tot}^{\oplus}\, B^{\mathrm{ord}}_{\bullet}(\mathcal{V}^{\mathrm{prim}}).$$

Proof. The ordered theorem computes $\int^{\mathrm{ord}} \mathcal{V}^{\mathrm{prim}}$. The stated comparison hypothesis identifies the ordered incidence model with the independently specified arithmetic homology object. Substitution gives the displayed equivalence. □

Remark 103.5.6 *(form of the BD/CG comparison).* The ordered incidence bar computes the derived colimit over $\mathsf{Inc}_<(\mathbb{P} \cup \{\infty\})$ by Chapter 103.5.4. This is not full Beilinson–Drinfeld/Costello–Gwilliam arithmetic chiral homology. The comparison requires a non-terminal arithmetic Ran object, finite-place diagonal or collision thickenings, BD/CG descent, constructibility with an exit category homotopy-finally reached from the ordered incidence category, ordered-to-unordered descent, and exactness of the restricted products and completions used in the derived colimit. Until these hypotheses are supplied, Chapter 103.5.5 remains conditional.

Corollary 103.5.7 *(regularised character from the ordered bar). (on Chapter 103.4.7(1–2) and Chapter 103.5.5, and on a regularised character map* $\mathrm{ch}_{\mathrm{reg}}$ *from arithmetic chiral homology to scalar partition functions).* The partition function of the conjectural 3D quantum arithmetic Chern–Simons is the regularised character of the ordered-bar homology:

$$Z_{\mathrm{ACS}^{\mathrm{qu}}}\big(\overline{\mathrm{Spec}\,\mathbb{Z}}; k^{\mathrm{Ar}}\big) = \mathrm{ch}_{\mathrm{reg}}\, H_\bullet\big(B^{\mathrm{ord}}_\bullet(\mathcal{V}^{\mathrm{prim}})\big) = \xi(k^{\mathrm{Ar}}).$$

The character map is the missing type conversion: a scalar partition function is not identified with a homology object before applying $\mathrm{ch}_{\mathrm{reg}}$.

103.6 Deninger Skeleton from Arithmetic Chiral Homology

The ordered-bar computation of the finite Bost–Connes zero modes (Chapter 103.6.3) gives the prime-orbit Euler skeleton in the half-plane of absolute convergence. Deninger's missing cohomology requires an additional $\mathbb{R}_+^\times$-flow whose normal generator has regularised determinant divisor div $\xi_{\mathbb{R}}$ and Weil–Connes trace distribution. That additional flow is the Hilbert–Pólya/Connes–Deninger realisation; it is not produced by the Bost–Connes heat flow.

Definition 103.6.1 *(pre-Deninger arithmetic coefficient datum).* . The *pre-Deninger arithmetic coefficient datum* is the datum

$$\mathcal{A}_{\mathrm{Den}} := \Big(\mathcal{A}_{\mathrm{fin}} \otimes \mathcal{A}_\infty^{\mathrm{Tate}}\Big) \otimes \mathsf{P}^\bullet$$

with:

(i) finite Bost–Connes zero modes

$$\mathcal{A}_{\mathrm{fin}} := \bigotimes_p{}' \mathcal{H}_p^{\mathrm{BC}}, \qquad \mathcal{H}_p^{\mathrm{BC}} := \bigoplus_{r\geq 0} \mathbb{C}\delta_{p^r},$$

with $N_p\delta_{p^r} = r(\log p)\delta_{p^r}$ and $N_{\mathrm{BC}} = \sum_p N_p$ on the restricted tensor product;

(ii) the archimedean Tate-Gaussian sector $\mathcal{A}_\infty^{\mathrm{Tate}}$, whose Mellin character is $\Gamma_{\mathbb{R}}(s) := \pi^{-s/2}\Gamma(s/2)$ (the standard archimedean Euler factor);

(iii) a polar Tate complex $\mathsf{P}^\bullet = \mathbb{C}_0[0] \oplus \mathbb{C}_1[2]$, where the indicated subscripts record the formal Frobenius weights 0 and 1;

(iv) the Bost–Connes Euler flow $\phi^t_{\mathrm{BC}} := \exp(-tN_{\mathrm{BC}})$ on the finite zero modes, tensored with the usual Tate scaling on $\mathcal{A}_\infty^{\mathrm{Tate}}$.

The last flow records closed-orbit lengths $\log p$. It is not the Deninger Frobenius flow on the conjectural middle cohomology.

Definition 103.6.2 *(candidate chiral Deninger complex).* . Work in a fixed complete nuclear dg category where the restricted tensor products below and compactly supported continuous duals are defined. Put

$$C^{\bullet}_{\mathrm{core},\chi} := \mathrm{Hom}_{\mathrm{cont},c}\Big(B^{\mathrm{ord}}_{\bullet}(\mathcal{A}_{\mathrm{fin}}\widehat{\otimes}\mathcal{A}_{\infty}^{\mathrm{Tate}}),\ \mathbb{C}\Big).$$

The *candidate chiral Deninger complex* is the total direct sum

$$C^{\bullet}_{\mathrm{Den},\chi} := \mathrm{Tot}\Big(C^{\bullet}_{\mathrm{core},\chi} \oplus \mathsf{P}^{\bullet}\Big),$$

where $\mathsf{P}^{\bullet}$ has zero differential and contributes formal Tate lines in degrees 0 and 2. Its cohomology is denoted

$$H^{\bullet}_{\mathrm{Den},\chi} := H^{\bullet}(C^{\bullet}_{\mathrm{Den},\chi}).$$

The primitive middle part $H^{1}_{\mathrm{Den},\chi,\mathrm{prim}}$ is $H^{1}(C^{\bullet}_{\mathrm{core},\chi})$. No theorem is asserted here that the full complex has only the Deninger degrees 0, 1, 2, or that this dual agrees with compactly supported Artin–Verdier cohomology.

Proposition 103.6.3 *(unconditional Bost–Connes Euler skeleton).* . The finite Bost–Connes zero-mode sector has ordinary KMS character

$$\mathrm{Tr}\big(e^{-sN_{\mathrm{BC}}} \mid \mathcal{A}_{\mathrm{fin}}\big) = \prod_{p}\sum_{r\ge 0} p^{-rs} = \prod_{p}(1-p^{-s})^{-1} = \zeta(s) \qquad (\mathrm{Re}\, s > 1),$$

and logarithmic differentiation gives the prime-orbit sum

$$-\frac{\zeta'}{\zeta}(s) = \sum_{p}\sum_{r\ge 1}(\log p)\, p^{-rs}.$$

This is the raw non-archimedean Euler/log-prime skeleton in the half-plane of absolute convergence.

Proof. On $\mathcal{H}_p^{\mathrm{BC}}$, $\mathrm{Tr}(e^{-sN_p}) = \sum_{r\ge 0} e^{-sr\log p}$. The restricted tensor product multiplies these traces over primes, giving the Euler product for $\zeta(s)$ in the half-plane of convergence. Logarithmic differentiation turns each local factor into the Dirichlet series over iterates of the closed orbit at p, with length $\log p$. □

Remark 103.6.4 *(gap between Euler/log-prime skeleton and Weil–Guinand distribution).* Chapter 103.6.3 is not yet the Weil–Guinand distribution. Passing from $-\zeta'/\zeta(s)$ in $\mathrm{Re}\, s > 1$ to the finite-place term on critical-line test functions requires a separate contour shift, Fourier/Mellin convention, the $n^{-1/2}$ weight, the sign and factor 2, plus the archimedean and polar terms. Those analytic normalisations are part of the trace-formula residual, not part of the ordered-bar computation.

Conjecture 103.6.5 *(Deninger spectral-flow / Hilbert–Pólya hypothesis).* . There exists a Hilbert completion of $H^{1}_{\mathrm{Den},\chi,\mathrm{prim}}$ and a closed operator

$$\Theta_{\mathrm{Den}} = \frac{1}{2}\mathrm{id} + iH_{\mathrm{Den}}$$

with H_{Den} self-adjoint in the Petersson/GNS inner product, such that:

(i) regularised determinant:

$$\det_{\infty}\Big(s\mathrm{id} - \Theta_{\mathrm{Den}} \mid H^{1}_{\mathrm{Den},\chi,\mathrm{prim}}\Big) = \xi_{\mathbb{R}}(s);$$

(ii) Weil–Connes trace distribution:

$$\mathrm{Tr}_{\mathrm{reg}}\big(h(H_{\mathrm{Den}})\big) = \sum_{\rho} h(\gamma_{\rho}), \qquad \rho = \frac{1}{2} + i\gamma_{\rho},$$

for every admissible Paley–Wiener test function h;

(iii) the preceding notation is conditional on the determinant identity: the self-adjoint trace model and $\det_{\infty}(s\mathrm{id} - \Theta_{\mathrm{Den}}) = \xi_{\mathbb{R}}(s)$ force the zero divisor onto the critical line;

(iv) Lefschetz compatibility: the trace distribution in (2), after moving the contour across ξ'/ξ, equals the prime-orbit sum of Chapter 103.6.3, the archimedean Tate-Gamma correction, and the two polar terms from $\mathsf{P}^{\bullet}$.

This hypothesis is exactly the Hilbert–Pólya/Connes–Deninger determinant-trace realisation in the present chiral-homology language. In particular, self-adjointness of H_{Den} together with the determinant identity is already RH-strength zero-placement for $\xi_{\mathbb{R}}$.

Proposition 103.6.6 *(conditional formal Deninger datum). (on Chapter 103.6.5 and on the topological-dual choices in Chapter 103.6.2).* The candidate cohomology $H^{\bullet}_{\mathrm{Den},\chi}$, equipped with Θ_{Den} on its primitive middle part and with the two Tate polar lines in degrees 0 and 2, carries Deninger's formal regularised-determinant and Lefschetz-trace identities for $\overline{\mathrm{Spec}\,\mathbb{Z}}$.

Proof. The finite sector gives only the Euler/log-prime skeleton of Chapter 103.6.3. The archimedean and polar entries specify where the Tate-Gamma and $s = 0, 1$ terms would live in a cohomological Lefschetz formula. The spectral-flow hypothesis supplies the missing middle operator, its determinant, its zero-side trace distribution, and the Lefschetz comparison. Thus the claim is a conditional implication: once the spectral-flow data is assumed, the displayed cohomological object carries the formal slots required by Deninger's axioms. It is not an independent construction of Θ_{Den}. □

Proposition 103.6.7 *(what the construction does not prove).* . The ordered-bar construction of $C^{\bullet}_{\mathrm{Den},\chi}$ does not by itself construct Θ_{Den}, prove the determinant identity, prove the Weil–Guinand trace formula, fix the cohomological superdeterminant signs of the polar lines, or prove positivity/purity. These requirements are exactly the Hilbert–Pólya/Connes–Deninger obstruction of Chapter 44.3.2 plus the analytic trace-formula normalization.

Proof. The ordered bar is determined by the local factors $\{\mathcal{H}_p^{\mathrm{BC}}\}_p$, the archimedean Tate sector, and the polar complex. This data gives the Euler product and hence the prime side in $\mathrm{Re}\, s > 1$. The zero-side operator requires a self-adjoint determinant-trace realisation whose determinant is $\xi_{\mathbb{R}}$. Chapter 44 names that assertion H-P^{Ar} and records that it is not derivable from the character identity, the formal Virasoro action, or the Petersson form alone. Thus the chiral construction supplies a candidate complex and the prime-side Euler skeleton. The spectral-flow hypothesis is the remaining essential input and has RH-level strength. □

103.7 Arithmetic-Gravity Construction to F-RH

Witten 2007 (*Three-Dimensional Gravity Revisited*, arXiv:0706.3359) proposed that 3D quantum gravity admits a Chern-Simons formulation whose partition function's spectrum controls the zeros of arithmetically significant L-functions. Vol II Part VI states this as the complex-side 3D quantum gravity climax. The arithmetic base $\overline{\mathrm{Spec}\,\mathbb{Z}}$ supplies an analogous conditional formulation via the preceding ACS construction.

Theorem 103.7.1 *(arithmetic-gravity F-RH). (on Chapter 103.4.7, Chapter 103.4.9, Chapter 103.5.5, and Chapter 44's three-pillar + VB structure, including a holomorphic complex-level partition function).* The zeros of $Z_{\mathrm{ACS}^{\mathrm{qu}}}(\overline{\mathrm{Spec}\,\mathbb{Z}}; k^{\mathrm{Ar}})$ as a function of the level $k^{\mathrm{Ar}} \in \mathbb{C}$ coincide with the non-trivial zeros of $\xi(k^{\mathrm{Ar}})$. The zero-placement claim is RH-level: these zeros lie on the self-dual locus $\mathrm{Re}(k^{\mathrm{Ar}}) = 1/2$ exactly when the corresponding Riemann-hypothesis zero-placement statement holds. The stronger F-RH condition still includes the positive Hilbert–Pólya trace inputs of Chapter 44.

Remark 103.7.2 *(three-step construction).* The construction proceeds in three steps.

(i) *Bulk construction.* Finite-level ACS is constructed in Chapter 103.4.2. The open bulk step is the renormalised profinite limit $\varprojlim^{\mathrm{ren}}_{N} Z^{\mathrm{Ar}}_{G_N,a_N}$ and its zeta partition function, as stated in Chapter 103.4.7.

(ii) *Boundary identification and ordered-model computation.* Finite-level boundary categories are constructed in Chapter 103.4.4 after choosing the finite arithmetic cobordism category. The comparison with $\mathcal{V}^{\mathrm{prim}}$ is Chapter 103.4.9. The ordered-bar calculation is theorem-grade for the ordered incidence model by Chapter 103.5.4; the remaining comparison to BD/CG arithmetic chiral homology is Chapter 103.5.5.

(iii) *Positive trace.* Apply the Hilbert–Pólya–Li residual of Chapter 44. The ordered model and the completed-zeta character do not by themselves supply positivity; the missing input is the positive Hilbert–Pólya trace.

103.8 Genus-g Extension

A genus-g extension requires rank-$2g$ current operations and the arithmetic comparison requested in Question 103.3.1. A conformal vector alone does not provide this enhancement.

The proposed quantum CS family $\mathrm{ACS}^{\mathrm{qu},g}_{k^{\mathrm{Ar},g}}$ is parameterized by the datum $\mathcal{D}_g$ of Definition 102.1.1. Its proposed gauge group is $GL_{2g}(\widehat{\mathbb{Z}})$. The boundary identification with $\mathcal{V}^{\mathrm{prim},g}$ requires a separate construction. Given rank-$2g$ Fock coefficients and all adjacent products, one can form their ordered incidence bar on $\mathbb{P} \cup \{\infty\}$. The partition function is conjecturally $\Lambda_{\mathcal{D}_g}(k^{\mathrm{Ar},g})$ per Chapter 102.1: Dedekind $\xi_K(k^{\mathrm{Ar},g})$ for Window II, Hasse–Weil $\Lambda(k^{\mathrm{Ar},g}, X_K)$ for Windows I/V, Siegel spin $\Lambda(k^{\mathrm{Ar},g}, \pi, \mathrm{spin})$ for Window VI.

Corollary 103.8.1 *(all-genus arithmetic-gravity F-RH). (on the genus-g extensions of Chapter 103.4.7, Chapter 103.4.9, Chapter 103.5.5, plus Chapter 102.5.1).* The zeros of $Z_{\mathrm{ACS}^{\mathrm{qu},g}}(\overline{\mathrm{Spec}\,\mathbb{Z}}; k^{\mathrm{Ar},g})$ coincide with the non-trivial zeros of $\Lambda_{\mathcal{D}_g}(s)$, and lie on the self-dual locus $\mathrm{Re}(k^{\mathrm{Ar},g}) = w_{\mathcal{D}_g}/2$ for $w_{\mathcal{D}_g}$ the motivic weight center of the relevant window: 1 for the zeta window, 2 for the curve H^1 window, and $w_{\pi,\mathrm{spin}}$ for the Siegel spin window.

103.9 Residual Conditions

Six conditions specific to the arithmetic-gravity construction, in addition to Chapter 102's residual list.

(i) *Finite arithmetic cobordism category.* . The closed finite-level state sum, gluing formula, and finite boundary trace are Chapter 103.4.2 and Chapter 103.4.5. A full finite-level TQFT still needs the arithmetic cobordism category and its finiteness hypotheses written explicitly.

(ii) *Profinite ACS renormalisation.* . Construct a canonical renormalised limit of the $GL_2(\mathbb{Z}/N\mathbb{Z})$ arithmetic CS tower, with compatible continuous level classes, counterterms, and scalar partition

$$Z_{\mathrm{ACS}^{\mathrm{qu}}}(\overline{\mathrm{Spec}\,\mathbb{Z}}; k^{\mathrm{Ar}}) = \xi(k^{\mathrm{Ar}}).$$

Finite closed state sums do not supply this data.

(iii) *Boundary trace comparison.* . Construct an independent Morita equivalence

$$\mathcal{B}_{\infty}^{\mathrm{Ar}} \simeq_{\mathrm{Morita}} \mathsf{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}},$$

compatible with restricted tensor products, spherical units, local Satake/Iwahori generators, and the archimedean vacuum.

(iv) *BD/CG ordered-model comparison.* . Construct arithmetic Ran/BD–CG descent whose exit or disk category is homotopy-finally computed by $\mathsf{Inc}_{<}(\mathbb{P} \cup \{\infty\})$, including finite-place collision thickenings, ordered-to-unordered descent, and completion exactness.

(v) *Positive trace/VB core.* . Prove the full positive Hilbert–Pólya trace, equivalently the full A-TP/VB input on the complete Paley–Wiener cone. The ordered model, Bost–Connes Euler skeleton, and formal Fock positivity away from the zero-mode sector do not supply this RH-strength positivity input.

(vi) *Deninger spectral-flow hypothesis.* . Upgrade the ordered-bar prime skeleton of Chapter 103.6.2 to a genuine middle cohomology Hilbert completion, with dense invariant core and a self-adjoint H_{HP} such that $\Theta_\xi = \frac{1}{2}\mathrm{id} + iH_{\mathrm{HP}}$ has determinant $\xi_{\mathbb{R}}$ and the Weil–Connes trace distribution. This is the Hilbert–Pólya/Connes–Deninger determinant-trace problem in chiral-homology form.

The arithmetic-gravity construction has two closed unconditional components: the closed finite-level state sums of Chapter 103.4.2 and the ordered incidence-bar calculation of Chapter 103.5.4. Together these yield the Euler/log-prime skeleton (Chapter 103.6.3) without further hypothesis. The remaining inputs are finite cobordism functoriality, profinite renormalisation, local trace comparison, BD/CG comparison, positivity, and the Deninger spectral-flow hypothesis (Chapter 103.9).

103.10 Ordered-Bar Boundary

Two first-principles results close unconditionally. First, closed finite-level arithmetic Chern–Simons state sums and their gluing law follow from étale local-system groupoids and Artin–Verdier integration alone (Chapter 103.4.2). Second, the ordered incidence bar computes the derived colimit of $\mathcal{V}^{\mathrm{prim}}$ over the auxiliary ordered place set via the normalized simplicial replacement (Chapter 103.5.4); this gives the Bost–Connes Euler/log-prime skeleton (Chapter 103.6.3) unconditionally.

The conditional structure has three layers. The profinite $GL_2(\widehat{\mathbb{Z}})$ quantum ACS, its boundary trace

$$\mathrm{HH}_0\big(\partial\,\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}}\big) \simeq \mathcal{V}^{\mathrm{prim}},$$

and its zeta partition function $Z = \xi(k^{\mathrm{Ar}})$ are conjectural (Chapter 103.4.7, Chapter 103.4.9, Chapter 103.5.7). The comparison of the ordered incidence model with Beilinson–Drinfeld/Costello–Gwilliam arithmetic chiral homology requires finite-place collision thickenings, descent, and

completion exactness (Chapter 103.5.5). The Deninger spectral-flow hypothesis, equivalently the Hilbert–Pólya/Connes–Deninger determinant-trace realisation, is not produced by the ordered bar and has RH-level strength (Chapter 103.6.5, Chapter 103.6.7).

The arithmetic-gravity construction conditionally reformulates the analytic-number-theoretic F-RH question as a 3D topological-gauge-theoretic question on the étale 3-space $\overline{\mathrm{Spec}\,\mathbb{Z}}$. The reformulation acquires unconditional content only after the six residuals of Chapter 103.9 are supplied; the Deninger spectral-flow hypothesis carries the full Hilbert–Pólya weight.

Chapter 104

Deninger's Arithmetic Cohomology as a Chiral Candidate

Deninger's conjectural formalism demands a cohomology theory for $\overline{\mathrm{Spec}\,\mathbb{Z}}$ whose regularised Fredholm determinant equals the completed Riemann zeta function and whose closed orbits recover the explicit formula. The bar–cobar construction does not construct this cohomology. It constructs a candidate: the ordered-incidence homology of $\mathcal{V}^{\mathrm{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori Hecke category of GL_2), two formal Tate polar lines, and the Bost–Connes Euler-prime-side character. Whether this candidate realizes Deninger's axioms depends on the spectral-flow hypothesis of Chapter 103.6.5. The central result of this chapter is the obstruction theorem: no unconditional triple identification exists among Deninger's conjectural cohomology, the ordered-incidence candidate, and the boundary homology of the arithmetic Chern–Simons bulk.

104.1 Deninger's Wish List

Deninger's formalism asks for a cohomology theory

$$H^\bullet_\Delta(\overline{\mathrm{Spec}\,\mathbb{Z}})$$

with three cohomological degrees, a Frobenius/flow operator Θ whose regularised determinant has the non-trivial zero divisor, a Lefschetz trace formula whose closed-orbit side is the prime side of the explicit formula, Poincaré duality, and purity. Under zero-placement this may be written as $\Theta = \frac{1}{2} + iH$ with H self-adjoint. For the Riemann zeta function the middle degree should carry the non-trivial zero divisor.

In the next display $\zeta(s)$ follows Deninger's convention: it is the completed zeta function, not the bare Euler product.

$$\zeta(s) \overset{?}{=} \det{}_\infty\big(s\cdot\mathrm{id} - \Theta \mid H^\bullet_\Delta\big)^{(-1)^\bullet}.$$

Here $\det_\infty$ is Deninger's regularised Fredholm determinant, $\zeta(s)$ denotes the completed Hasse–Weil zeta (including the archimedean factor), and the sign $(-1)^\bullet$ is the Euler-characteristic grading. The Lefschetz trace formula takes the form

$$\sum_\rho h(\rho) \overset{?}{=} \sum_\bullet (-1)^\bullet \operatorname{tr}\big(h(\Theta) \mid H^\bullet_\Delta\big) \overset{?}{=} \text{polar terms} + \text{archimedean term} - \sum_{p,k} (\log p)\, p^{-k/2}\widehat{h}(k\log p),$$

where h is a Schwartz test function on $\mathbb{R}$ and $\widehat{h}$ its Fourier transform.

Remark 104.1.1 *(absence of a construction).* The axioms above are not a construction. They specify what the missing arithmetic cohomology must do. The construction provides the prime-side Euler data (via Bost–Connes) and the formal polar lines (by insertion); the zero-side operator is not constructed.

Remark 104.1.2 *(Vol IV scoping).* Chapter 103 isolates the correct boundary: the ordered-incidence bar is theorem-grade as an algebraic model, while the comparison with genuine BD/CG chiral homology, the

profinite quantum ACS theory, and the Deninger spectral flow remain conditional. This chapter keeps that boundary fixed.

104.2 Surviving Inputs

The bar–cobar construction produces four inputs with theorem-grade form; the remainder of the Deninger axioms are either conditional or open.

(i) The finite-level arithmetic Chern–Simons state sum and gluing formula of Chapter 103.4.2. This is a closed finite groupoid state-sum statement, not a full profinite TQFT.

(ii) The ordered-incidence bar Chapter 103.5.3 and its normalized-bar computation Chapter 103.5.4. This computes the explicitly defined incidence model, not automatically BD/CG factorization homology on $\overline{\mathrm{Spec}\,\mathbb{Z}}$.

(iii) The Bost–Connes Euler skeleton Chapter 103.6.3. It gives the KMS character $\zeta(s)$ and the prime-power logarithmic derivative in $\mathrm{Re}(s) > 1$.

(iv) The Deninger spectral-flow hypothesis Chapter 103.6.5: the zero-side determinant divisor, trace distribution, and Hilbert–Pólya self-adjoint determinant-trace operator. This is the single missing piece that the bar–cobar construction does not supply.

104.3 The Chiral Candidate

Given the four inputs above, one may assemble a graded object that mimics Deninger's three-degree shape. The question is whether this assembly is genuinely Deninger's cohomology or only a formal approximation to it.

Let $B^{\mathrm{ord}}_\bullet(\mathcal{V}^{\mathrm{prim}})$ denote the ordered-incidence bar complex of Chapter 103.5.3 applied to $\mathcal{V}^{\mathrm{prim}}$.

Definition 104.3.1 *(arithmetic chiral homology candidate).* . The *arithmetic chiral homology candidate* is

$$H^\bullet_{\mathrm{ord}}(\overline{\mathrm{Spec}\,\mathbb{Z}}, \mathcal{V}^{\mathrm{prim}}) := H^\bullet\big(B^{\mathrm{ord}}_\bullet(\mathcal{V}^{\mathrm{prim}})\big).$$

It is an ordered-incidence model. Calling it genuine arithmetic chiral homology requires the separate BD/CG comparison of Chapter 103.5.5.

Definition 104.3.2 *(Deninger chiral candidate).* . The *Deninger chiral candidate* is the candidate complex of Chapter 103.6.2:

$$H^\bullet_{\Delta,\chi}(\overline{\mathrm{Spec}\,\mathbb{Z}}) := H^\bullet_{\mathrm{Den},\chi}(\overline{\mathrm{Spec}\,\mathbb{Z}}).$$

It consists of the ordered-incidence core, completed continuous duals, and formal Tate polar lines in degrees 0 and 2.

Remark 104.3.3 *(three labels, not one theorem).* Three nearby objects appear: Deninger's conjectural cohomology, the ordered-incidence chiral candidate, and the boundary state space of the conjectural 3D ACS bulk. Their equality is a conditional assertion, not a proved identification.

PROPOSITION 104.3.4 *(conditional Deninger comparison). (on Chapter 103.6.5, the BD/CG comparison Chapter 103.5.5, and the topological-dual choices in Chapter 103.6.2).* The chiral candidate $H^\bullet_{\Delta,\chi}$, together with the two Tate polar lines and the spectral-flow operator assumed in Chapter 103.6.5, carries Deninger's formal axioms for $\overline{\mathrm{Spec}\,\mathbb{Z}}$.

Proof. The finite-prime Euler terms are exactly the Bost–Connes series of Chapter 103.6.3. The polar lines are inserted by definition in Chapter 103.6.2. The zero-side operator, determinant identity, trace formula, and purity are precisely the extra data demanded by Chapter 103.6.5. Thus the statement is a conditional implication, not an independent construction of the missing cohomology. □

104.4 No Triple Identification Yet

Theorem 104.4.1 *(no unconditional triple identification).* . No unconditional equality holds among Deninger's conjectural cohomology, the ordered-incidence chiral candidate, and the boundary homology of $\mathrm{ACS}^{\mathrm{qu}}$.

Proof. The first object is conjectural. The second is an explicitly defined ordered-incidence homology object. The third depends on the profinite quantum ACS conjecture and on a boundary comparison with $\mathcal{V}^{\mathrm{prim}}$. The only theorem-grade equality is the normalized-bar computation of the ordered-incidence model. The BD/CG comparison and spectral-flow identifications are listed as residual conditions in Chapter 103.9. □

Corollary 104.4.2 *(conditional triangle). (on the profinite ACS conjecture, the BD/CG ordered-model comparison, a regularised character map, and the Deninger spectral-flow hypothesis).* These assumptions yield

$$Z_{\mathrm{ACS}^{\mathrm{qu}}}(\overline{\mathrm{Spec}\,\mathbb{Z}}; k^{\mathrm{Ar}}) = \xi(k^{\mathrm{Ar}}) = \mathrm{ch}_{\mathrm{reg}}\big(H^{\bullet}_{\Delta,\chi}\big)(k^{\mathrm{Ar}}), \tag{104.1}$$

where $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed Riemann zeta function satisfying $\xi(s) = \xi(1-s)$, k^{Ar} is the arithmetic level parameter, and $\mathrm{ch}_{\mathrm{reg}}$ denotes the regularised graded trace (the Θ-eigenvalue generating series with Deninger regularisation). Without those assumptions, (104.1) is a conditional shorthand.

104.5 Frobenius and Flow

The Bost–Connes Euler skeleton provides prime-length data at finite primes. Deninger's formalism requires a global Frobenius/flow operator acting on all three cohomological degrees. The question is whether the Euler skeleton and the archimedean scaling generator combine into such an operator.

Definition 104.5.1 *(finite Euler operators).* . At a finite prime p, the Bost–Connes Adams operation ψ_p records the prime length $\log p$ and contributes the Euler factor $(1-p^{-s})^{-1}$. It is denoted here by

$$F_p^{\mathrm{Euler}} := \psi_p.$$

This is an Euler-skeleton operator, not by itself Deninger's global Frobenius on middle cohomology.

Proposition 104.5.2 *(finite Wilson/Frobenius comparison is conditional). (on the Wilson-line axiom in Chapter 103.4.7).* A Wilson loop around the Morishita knot attached to p should recover the corresponding Frobenius trace. This is part of the conjectural ACS datum, not a consequence of the ordered bar.

Definition 104.5.3 *(archimedean scaling candidate).* . The archimedean scaling generator is the Mellin infinitesimal $-\partial_t$ on $\mathbb{R}_+$. The candidate archimedean flow on $H^{\bullet}_{\Delta,\chi}$ is denoted $\Theta_{\infty,\chi}$. In Deninger's

and Connes' formulation, the target operator has the form $\Theta_\xi = \frac{1}{2}\mathrm{id} + iH_{\mathrm{HP}}$, where H_{HP} is a self-adjoint Hilbert–Polya operator (to be constructed) whose normal determinant divisor is div $\xi_{\mathbb{R}}$. After zero-placement its Weil–Connes trace is supported at the ordinate multiset $\{\gamma_\rho\}$. The identification

$$\Theta_{\infty,\chi} \stackrel{?}{=} \Theta_\xi$$

is the spectral-flow hypothesis and is not available unconditionally.

PROPOSITION 104.5.4 *(Virasoro conjugacy is not available). (on the spectral-flow hypothesis).* Without the spectral-flow theorem of Chapter 103.6.5, the Virasoro L_0^{Ar}, the Bost–Connes logarithmic Hamiltonian, and the Hilbert–Polya operator H_{HP} remain distinct operators. Their identification requires the spectral-flow theorem as input.

Definition 104.5.5 *(formal global flow).* . The formal global flow symbol is

$$F_\chi := \Theta_{\infty,\chi} \oplus \bigoplus_p F_p^{\mathrm{Euler}}.$$

The notation records finite-prime lengths and archimedean scaling. It does not assert a well-defined Deninger Frobenius on a completed cohomology theory.

THEOREM 104.5.6 *(global Frobenius obstruction).* . The ordered-incidence bar and Bost–Connes Euler skeleton do not construct Deninger's global Frobenius operator.

Proof. The Bost–Connes side gives Euler factors and the prime-power series in the half-plane of convergence. Deninger's operator must act on the middle cohomology, must have regularised determinant divisor div $\xi_{\mathbb{R}}$, and must participate in a Lefschetz trace formula. That zero-side operator is exactly the missing spectral-flow input of Chapter 103.6.5. □

Remark 104.5.7 *(archimedean Wilson surface).* The phrase "archimedean Wilson surface" is useful only as a conditional dictionary entry inside the profinite ACS conjecture. It is not an independent construction of Θ_ξ.

104.6 CHARACTER, DEGREES, AND PURITY

The Frobenius obstruction theorem leaves open the question of what the candidate complex does compute. The regularised graded trace and the formal Tate structure answer this question partially: the Euler product and the polar lines are present; the zero-side is missing.

THEOREM 104.6.1 *(character identity is conditional). (on Chapter 103.5.7 and Chapter 103.6.5).* The regularised character of the Deninger chiral candidate is $\xi(s)$ only after the partition-function comparison and the zero-side spectral-flow operator are assumed.

$$\mathrm{ch}_{\mathrm{reg}}\big(H^\bullet_{\Delta,\chi}\big)(s) \stackrel{\mathrm{cond}}{=} \xi(s).$$

Remark 104.6.2 *(normalisation).* The additive determinant form $\det_\infty(s\cdot\mathrm{id} - \Theta)$ and the multiplicative Euler-character form are interchangeable only after the operator Θ, the polar signs, and the regularisation convention have been fixed.

PROPOSITION 104.6.3 *(formal Tate polar lines).* . The candidate complex contains formal one-dimensional Tate lines in degrees 0 and 2. This accounts for the polar terms in the Deninger skeleton but does not prove that the total complex has only Deninger's three degrees.

Proposition 104.6.4 *(middle cohomology is the hard part). (on Chapter 103.6.5).* The middle cohomology $H^1_{\Delta,\chi}$ admits a Hilbert-space completion with a self-adjoint generator having spectrum $\{\gamma_\rho\}$ only after Chapter 103.6.5 is assumed. This is the Hilbert–Polya hypothesis in cohomological form.

Remark 104.6.5 *(critical-line locus).* Purity of the middle degree is the assertion that the zero-side operator has real part 1/2. For zeta this is RH in cohomological language.

104.7 Duality and Positivity

Koszul duality for the arithmetic bar complex gives the functional-equation symmetry $s \leftrightarrow 1 - s$. The question is whether this duality implies Poincaré duality for the Deninger cohomology, and specifically whether it forces the middle degree to be positive definite.

Theorem 104.7.1 *(duality gives the functional equation, not positivity). (on the relevant Koszul and ACS comparison hypotheses).* The Koszul self-duality framework accounts for the functional-equation symmetry $s \leftrightarrow 1 - s$. It does not by itself imply positivity of the Deninger middle pairing or RH.

104.8 Purity as the RH-Level Input

Duality accounts for the functional equation but not for the zero-placement. Purity of the middle cohomological degree is the assertion that the zero-side operator Θ_ξ is self-adjoint with spectrum on the line $\mathrm{Re}(s) = 1/2$, which is RH.

Theorem 104.8.1 *(purity is RH-level). (on the spectral-flow/Hilbert–Polya realisation).* Deninger purity for $H^1_{\Delta,\chi}$, Hilbert–Polya self-adjointness for Θ_ξ, and the RH zero-placement statement are equivalent forms of the same missing input.

Corollary 104.8.2 *(Deninger-chiral RH construction).* . A proof of RH by this route is precisely a proof of the spectral-flow/positivity input. The ordered-incidence bar and the Bost–Connes Euler skeleton reduce the shape of the problem; they do not remove it.

104.9 All-Genus Extension

The preceding sections address $\overline{\mathrm{Spec}\ \mathbb{Z}}$ and the Riemann zeta function. In the notation of Chapter 102.1.1 this is the zeta fibre of the rank-graded arithmetic family, not a smooth surface of curve genus zero. Higher-rank arithmetic data occur in several distinct windows: arithmetic surfaces, totally imaginary number fields, crystalline deformation stacks, Hodge–Arakelov theta data, and Shimura varieties. The formal extension records only what the ordered candidate produces after such a datum has been fixed.

Definition 104.9.1 *(genus-g chiral Deninger candidate).* . A *genus-g arithmetic datum* $\mathcal{D}_g$ means the datum of Chapter 102.1.1. Thus it may be an arithmetic surface only in Windows I and V; in the other windows it is respectively a number-field, crystalline, Hodge–Arakelov, or Shimura datum. The corresponding $\mathcal{V}^{\mathrm{prim},g}$ is the row-specific Hochschild trace class of the identity on the indicated arithmetic Hecke category. In the surface, crystalline, and automorphic rows its oscillator sector has rank $2g$; in the Dedekind row g records the complex embedding rank of the coefficient field. The *genus-g chiral Deninger candidate* is

$$H^\bullet_{\Delta,\chi,g} := H^\bullet_{\mathrm{ord}}(\mathcal{D}_g, \mathcal{V}^{\mathrm{prim},g}).$$

The weight of $\mathcal{D}_g$, denoted $w_{\mathcal{D}_g}$, is the motivic weight of its L-function; purity is the assertion that $\mathrm{Re}(s) = w_{\mathcal{D}_g}/2$ for all non-trivial zeros.

THEOREM 104.9.2 *(all-genus Deninger comparison). (on the all-genus ACS, ordered-model comparison, and spectral-flow hypotheses).* The genus-g candidate carries the expected motivic functional-equation centre and prime-side Euler data. It does not construct the genus-g Deninger cohomology without the same spectral-flow input.

Remark 104.9.3 *(weight convention).* The expected purity line is $\mathrm{Re}(s) = w_{\mathcal{D}_g}/2$. This is a target condition, not a theorem of the ordered bar.

104.10 DICTIONARY

Table 104.1 records which Deninger axioms the chiral candidate supplies and which remain open or conditional.

Deninger target	Chiral/ACS object	Form
finite prime orbit lengths	Bost–Connes Euler skeleton	proved in $\mathrm{Re}(s) > 1$
cohomology object	ordered-incidence candidate plus polar lines	defined, not identified with Deninger cohomology
global Frobenius/flow	spectral-flow operator Θ_ξ	open, Hilbert–Polya strength
determinant = ξ	regularised character	conditional
Lefschetz trace formula	prime-power trace distribution	open
purity	real part 1/2 for middle spectrum	RH-level

TABLE 104.1: Surviving Deninger–chiral dictionary.

104.11 UNCONDITIONAL DENINGER–CHIRAL CONSEQUENCES

The ordered-incidence construction proves the four assertions below; the six remaining comparison maps require the stated hypotheses.

Unconditional consequences

(i) Ordered-incidence homology is computed by the normalized ordered bar.
(ii) The finite-prime Bost–Connes side gives the Euler product and logarithmic derivative in the convergence half-plane.
(iii) The formal Tate polar lines can be inserted as part of the candidate complex.
(iv) Koszul self-duality explains the functional-equation symmetry.

Conditional

(i) Profinite quantum arithmetic Chern–Simons.
(ii) Boundary comparison with $\mathcal{V}^{\mathrm{prim}}$.
(iii) BD/CG comparison with the ordered-incidence model.
(iv) Regularised character equality with ξ.
(v) Spectral flow/Hilbert–Pólya zero-divisor operator.
(vi) Positivity/purity of the middle degree.

104.12 Residual Conditions

The residue is a single mathematical obstruction: construct the middle operator and prove the trace formula with positivity. Equivalently, prove the spectral-flow hypothesis of Chapter 103.6.5. The remaining assertions are unconditional consequences or conditional consequences of that hypothesis.

Chapter 105

Trace Formula, Ordered Bar, and the Statistical Upside

105.1 The Trace Formula Target

Arithmetic Koszul duality assembles an ordered-incidence bar complex whose regularised character reproduces the Euler product of $\xi(s)$. The question it faces is: does this Euler-product data already determine the zero distribution of ξ, or does it leave open a spectral problem? To answer precisely, fix the target.

Definition 105.1.1 *(Weil test-function class).* . Let $\mathcal{H}$ be a standard Weil test-function class: even Schwartz functions $h : \mathbb{R} \to \mathbb{C}$ whose Fourier transforms $\widehat{h}(\tau) = \int_{-\infty}^{\infty} h(t) e^{i\tau t}\, dt$ are compactly supported and sufficiently regular for the explicit formula. The exact analytic class depends on the chosen normalization of the Weil–Guinand formula.

Theorem 105.1.2 *(Weil–Guinand explicit formula).* . Let ρ range over all nontrivial zeros of the completed Riemann zeta function $\xi(s)$, with no critical-line parametrization assumed. For $h \in \mathcal{H}$, the completed zeta function satisfies the distributional identity

$$\sum_{\rho} H(\rho) = W_{\infty}(h) + W_{\mathrm{pol}}(h) - \sum_{p} \sum_{k \geq 1} (\log p)\, p^{-k/2} \big(\widehat{h}(k \log p) + \widehat{h}(-k \log p) \big), \tag{105.1}$$

where H is the complex test function corresponding to h, and $W_{\infty}(h)$ is the archimedean gamma-factor contribution and $W_{\mathrm{pol}}(h)$ is the contribution of the polar and trivial zeros. Denote their sum by $W_{\mathrm{arch+pol}}(h) := W_{\infty}(h) + W_{\mathrm{pol}}(h)$.

Remark 105.1.3 *(three descriptions of the prime-power index set).* The geometric side of (105.1) is a sum over prime powers (p, k). In the Morishita dictionary this is a sum over iterates K_p^k of the prime knot K_p, with length $k \log p$. In a Gutzwiller language it is the periodic-orbit side. In the ordered-bar language it is the logarithmic derivative of the Euler-product character. These are three descriptions of the same prime-power index set; none supplies the zero-side determinant-trace operator.

105.2 Connes' Trace Formula

The Weil explicit formula is a distributional identity; the question is whether the zero sum on its left side admits a determinant-trace interpretation, i.e., whether some natural normal operator has determinant divisor div $\xi_{\mathbb{R}}$ and Weil–Connes trace supported at the ordinates γ_ρ after zero-placement. Connes' adele-class-space construction provides a partial answer.

THEOREM 105.2.1 *(Connes–Meyer scaling trace).* . Connes' adele-class-space construction, in Meyer's trace-formula formulation, realizes the Weil distribution as a distributional scaling trace on the appropriate cokernel space of the adele class space $\mathbb{A}/\mathbb{Q}^*$:

$$\mathrm{Tr}_{\mathrm{Connes}}(h) = \sum_{\rho} H(\rho)$$

distributionally, where ρ ranges over the non-trivial zeros and H is the Mellin transform paired with the test function h. The same prime-power and archimedean terms as (105.1) occur. The scaling operator is not thereby a self-adjoint Hilbert-space operator whose normal translate has determinant $\xi(s)$; the distributional trace is not by itself a spectral measure.

105.3 Ordered-Bar Euler Skeleton

The Connes–Meyer trace recovers the full Weil distribution (105.1) as an operator trace. The ordered-incidence bar complex approaches the same object from the Koszul-duality side: its regularised character computes the Euler product. The question is whether that character calculation closes the gap left by Connes.

PROPOSITION 105.3.1 *(ordered-bar Euler skeleton). (on the ordered-incidence model of Chapter 103.5.4 and on the chosen regularised character).* Let $\chi_\infty(s)$ denote the archimedean Euler factor arising from the regularisation at the archimedean place of $\overline{\mathrm{Spec}\,\mathbb{Z}}$. The ordered-incidence bar complex $B^{\mathrm{ord}}_\bullet(\mathcal{V}^{\mathrm{prim}})$, where $\mathcal{V}^{\mathrm{prim}}$ is the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2, has regularised character

$$\mathrm{ch}_{\mathrm{reg}}\, H_\bullet(B^{\mathrm{ord}}_\bullet(\mathcal{V}^{\mathrm{prim}})) = \chi_\infty(s) \prod_p (1 - p^{-s})^{-1}, \tag{105.2}$$

with the completed $\xi(s)$ obtained only after the archimedean and polar normalisations are inserted.

Proof. This is the Euler-character computation of the ordered-incidence model: finite prime factors contribute the Bost–Connes Euler factors, while the archimedean and polar pieces are part of the regularisation data. The statement is a character calculation, not a Lefschetz trace formula. □

Remark 105.3.2 *(canonical quantisation boundary).* If the profinite ACS theory and boundary comparison are assumed, the same regularised character can be read as a boundary partition function. That reading is conditional on Chapter 103.4.7 and Chapter 103.4.9.

105.4 Trace-Formula Equivalence

The Euler-product character of Proposition 105.3.1 yields $\xi(s)$ only after regularisation. Under the regularised character equality of Chapter 103.5.7, which sets $Z(s) = \xi(s)$ in the ACS framework, the argument principle converts the logarithmic derivative of ξ into the zero sum; the Weil–Guinand explicit formula then identifies this with the full Weil distribution. The three trace functionals are conditionally equal.

THEOREM 105.4.1 *(conditional equivalence of trace functionals). (on Chapter 103.4.7, Chapter 103.5.5, and the regularised character equality Chapter 103.5.7).* Let Γ_T be a positively oriented rectangle

enclosing the zeros of ξ with $|\operatorname{Im} s| \le T$, avoiding zeros on the boundary. The ACS argument-principle functional

$$\mathcal{Z}_h := \frac{1}{2\pi i}\int_{\Gamma_T} H(s)\frac{Z'(s)}{Z(s)}\,ds$$

converges, after the standard $T \to \infty$ limiting procedure and the polar and trivial-zero subtractions, to the Weil explicit-formula functional (105.1) whenever $Z(s) = \xi(s)$ is assumed.

Proof. Under the stated assumptions $Z = \xi$. Applying the argument principle to ξ'/ξ inside Γ_T gives the zero sum $\sum_{|\operatorname{Im}\rho|\le T} H(\rho)$, together with polar and trivial-zero residues. Expanding ξ'/ξ in the half-plane $\operatorname{Re}(s) > 1$ of absolute convergence and performing the standard explicit-formula contour shift yields the prime-power side of (105.1). The equality is inherited from the classical Weil–Guinand explicit formula. □

Corollary 105.4.2 *(equality of the ACS and Connes–Meyer distributions on $\mathcal{H}$). (on the hypotheses of Chapter 105.4.1).* For every $h \in \mathcal{H}$, the ACS functional $\mathcal{Z}_h$ and the Connes–Meyer scaling trace $\operatorname{Tr}_{\text{Connes}}(h)$ are equal.

105.5 The Bar Reorganises, Not Locates

The equivalence of the three trace functionals established conditionally in Section 105.4 raises a sharper question: does the ordered-bar construction supply an independent mechanism for placing the zero ordinates γ_n on the critical line, beyond what $\xi(s)$ itself encodes? The bar computation reorganises the zero data of ξ; it does not add new data.

Theorem 105.5.1 *(bar computation reorganises zero data).* . The ordered-incidence bar complex determines the individual zero ordinates γ_n only through the scalar function ξ: no additional self-adjoint operator or positivity mechanism is produced by the bar calculation.

Proof. The bar supplies the Euler-product character and, after conditional regularisation, the scalar function ξ. The zero ordinates are then the zero set of that scalar function. The bar calculation produces no self-adjoint operator, spectral theorem, or positivity input beyond what ξ already carries. □

Remark 105.5.2 *(the Hilbert–Pólya obstruction).* A self-adjoint operator with spectrum $\{\gamma_n\}$ would place the zeros on the critical line by the spectral theorem. Constructing such an operator is precisely the spectral-flow obstruction of Chapter 103.6.5.

105.6 Arithmetic Gutzwiller

The zero ordinates γ_n appear on the zero side of (105.1) and, conditionally, as the spectrum of the Hilbert–Pólya operator. The prime-power sum on the geometric side of (105.1) takes the same form as the periodic-orbit sum in the Gutzwiller trace formula: lengths $k \log p$ and amplitudes $(\log p)\,p^{-k/2}$. The prime-power side of the Weil formula is a periodic-orbit sum in Gutzwiller's sense, conditional on the existence of an arithmetic Hamiltonian whose primitive periodic orbits are the primes.

Theorem 105.6.1 *(classical Gutzwiller trace formula).* . Let H be a quantum Hamiltonian whose classical limit is chaotic with isolated primitive periodic orbits γ. Write $d_{\text{osc}}(E)$ for the oscillatory part of

the density of states $\sum_n \delta(E - E_n)$, T_γ for the period of orbit γ, A_γ for its stability amplitude (Maslov–Gutzwiller amplitude, incorporating the stability matrix determinant), and μ_γ for its Maslov index. Then

$$d_{\mathrm{osc}}(E) \sim \sum_\gamma A_\gamma \exp(iET_\gamma - i\mu_\gamma \pi/2),$$

asymptotically as $\hbar \to 0$.

THEOREM 105.6.2 *(arithmetic Gutzwiller hypothesis). (on a semiclassical ACS Hamiltonian, an arithmetic phase space, and a prime-orbit correspondence).* Call the amplitudes $A_p = (\log p)\, p^{-1/2}$ and lengths $T_p = \log p$ the *Weil amplitudes* for the prime p. If the ACS boundary Hamiltonian has a classical chaotic limit whose primitive periodic orbits are indexed by primes p with primitive period $\log p$ and stability amplitude A_p, then the periodic-orbit expansion has the desired form

$$\sum_p \sum_{k \geq 1} (\log p)\, p^{-k/2} e^{itk \log p} \quad + \quad \text{complex conjugate and archimedean terms.}$$

Proof. This is substitution of the assumed arithmetic orbit dictionary into the classical Gutzwiller formula. The proof is formal because the semiclassical ACS phase space, the chaotic flow, and the amplitude calculation are all hypotheses. □

105.7 Connes–Meyer and the Gutzwiller Gap

The arithmetic Gutzwiller hypothesis of the previous section requires a smooth symplectic Hamiltonian system with a chaotic classical limit. The question is whether the Connes–Meyer construction already supplies this structure, or whether the Gutzwiller construction is genuinely new input.

THEOREM 105.7.1 *(Connes–Meyer as distributional trace, not Gutzwiller system).* . The standard adele-class-space realization used in the Connes–Meyer trace formula is a noncommutative scaling trace construction. It is distinct from a smooth symplectic Hamiltonian system of the kind required by the classical Gutzwiller theorem.

Proof. The adele class space $\mathbb{A}/\mathbb{Q}^*$ is modeled by a noncommutative quotient; its C^*-algebraic description gives an operator-algebraic trace. The Gutzwiller formula requires a smooth symplectic phase space, a Hamiltonian flow, isolated primitive periodic orbits, and monodromy matrix determinants. None of these are part of the Connes–Meyer construction as stated. □

Remark 105.7.2 *(other geometrisations remain open).* Other geometrisations of the adele class space may supply a symplectic structure; the theorem records only that the standard Connes–Meyer trace formula is not itself a Gutzwiller mechanism. An arithmetic Gutzwiller system, if it exists, is new structure beyond Connes–Meyer.

105.8 Statistical Upside

The explicit formula (105.1) is a distributional identity; it does not determine the individual zero ordinates γ_n beyond what $\xi(s)$ already encodes. A Hilbert–Pólya operator, if it exists, would put the determinant divisor on the critical line by self-adjointness. The further statistical prediction, that the local spacing statistics of the γ_n follow the Gaussian Unitary Ensemble (GUE), is not implied by either the explicit formula or self-adjointness alone. It requires a dynamical mechanism: the Bohigas–Giannoni–Schmit

(BGS) principle connecting chaotic classical dynamics to random-matrix statistics. The relevant symmetry class is GUE because $\xi(s)$ has no anti-unitary time-reversal symmetry; the unitary symmetry class applies. Let $R_2(u)$ denote the two-point level spacing correlation function.

Theorem 105.8.1 *(Bohigas–Giannoni–Schmit principle).* . Quantum systems whose classical limits are chaotic are expected, after symmetry class is fixed, to exhibit random matrix local spectral statistics. For the unitary symmetry class (GUE), the two-point correlation function is

$$R_2(u) = 1 - \left(\frac{\sin \pi u}{\pi u}\right)^2. \tag{105.3}$$

Theorem 105.8.2 *(conditional statistical upside). (on Chapter 103.6.5, Chapter 105.6.2, arithmetic quantum chaos, and the BGS universality theorem Chapter 105.8.1).* If the ACS Hamiltonian of Chapter 103.6.5 has a chaotic semiclassical limit with prime periodic orbits, then the local statistics of the zero ordinates are governed by the corresponding random-matrix ensemble; in the expected zeta case by GUE statistics (105.3).

Proof. Spectral flow supplies the zero-divisor determinant and, after zero-placement, the Hamiltonian trace supported at the ordinates. Arithmetic Gutzwiller identifies the classical periodic orbits with prime powers. The BGS principle then transfers chaotic spectral statistics to the zero ordinates. Every substantive step is one of the stated hypotheses. □

Remark 105.8.3 *(GUE statistics as new content beyond the explicit formula).* The Weil explicit formula establishes a distributional equality between the zero ordinates and the prime-power sum; it implies neither Montgomery pair correlation nor GUE spacing statistics. A 3D Hamiltonian realization of the arithmetic ACS theory adds that statistical mechanism, conditional on the semiclassical and spectral-flow hypotheses of Chapter 103.6.5.

Remark 105.8.4 *(open conditions as explicit hypotheses).* Chapter 105.8.2 rests on seven open conditions: (1) the profinite quantum ACS on $\overline{\mathrm{Spec}\,\mathbb{Z}}$; (2) the Bezrukavnikov–Finkelberg and Costello–Gwilliam comparison with the ordered-incidence model; (3) the regularised character equality $Z = \xi$; (4) the spectral-flow operator realizing Hilbert–Pólya; (5) the full Weil–Guinand contour shift from the Euler skeleton; (6) the arithmetic semiclassical phase space and Hamiltonian flow; (7) chaos and random-matrix universality for that flow. These are the explicit hypotheses of Chapter 105.8.2. The ordered bar supplies the finite Euler skeleton (105.2); the equality $Z = \xi$ also requires the archimedean, polar, and regularised character inputs. Conditions (1), (4), (6), and (7) constitute the spectral-flow problem of Chapter 103.6.5.

Chapter 106

The Wilson Line–Frobenius Identity on $\overline{\mathrm{Spec}\,\mathbb{Z}}$

The prime knot $K_p \subset \overline{\mathrm{Spec}\,\mathbb{Z}}$ is the codimension-two defect at a rational prime p; the Wilson loop $W_\rho(K_p)$ is the holonomy of a flat G-connection around that defect, evaluated in a representation ρ. The fundamental question of this chapter is whether, for an unramified ρ, the Wilson expectation equals the Frobenius trace: whether $\langle W_\rho(K_p)\rangle = \mathrm{tr}\,\rho(\mathrm{Frob}_p)$. Witten's 1989 path-integral construction, transplanted via the Morishita dictionary to the arithmetic 3-space, reduces the Wilson integral to a character evaluation on the Frobenius conjugacy class in the punctured tubular category. On the closed base the finite etale torsor is trivial; non-trivial Frobenius data enter after the defect is retained. This identification is finite-level local and is conditional at the profinite limit (Theorem 106.5.1(2)), with the single residual being the renormalised inverse limit of Chapter 103.4.7.

106.1 Setup and Conventions

The arithmetic 3-space is $M = \overline{\mathrm{Spec}\,\mathbb{Z}}$, the Arakelov compactification of $\mathrm{Spec}\,\mathbb{Z}$ by the archimedean place $\{\infty\}$. By Mazur 1973 and Artin–Verdier, M is etale cohomologically 3-dimensional with fundamental class $[M] \in H^3_{\text{ét}}(M, \widehat{\mathbb{Z}})$ (Chapter 103.1).

Definition 106.1.1 *(prime knot in the Morishita dictionary).* . For a rational prime p, the *prime knot* $K_p \subset M$ is the codimension-two etale closed embedding $\mathrm{Spec}\,\mathbb{F}_p \hookrightarrow \mathrm{Spec}\,\mathbb{Z} \subset M$. Its tubular neighborhood is $\mathrm{Spec}\,\mathbb{Z}_p$. Its complement is $M \setminus K_p \cong \mathrm{Spec}\,\mathbb{Z}[1/p] \cup \{\infty\}$. Under the Mazur 1973 / Morishita 2012 (arXiv:0904.3399) dictionary, K_p plays the role of a framed knot: the meridian is the inertia subgroup $I_p \subset G_{\mathbb{Q}_p}$; the longitude and Frobenius class carry the decomposition quotient $G_{\mathbb{Q}_p}/I_p = \widehat{\mathbb{Z}} \cdot \mathrm{Frob}_p$.

Remark 106.1.2 *(the arithmetic loop is a Frobenius, not a geometric loop).* $\mathrm{Spec}\,\mathbb{Z}$ is not a smooth 3-manifold; the loop around K_p is not a smooth curve. Following Chapter 103.1.1, *loop around the knot* K_p is a name for the generator of the etale fundamental group of the tubular neighbourhood $\mathrm{Spec}\,\mathbb{Z}_p \setminus K_p = \mathrm{Spec}\,\mathbb{Q}_p$; this fundamental group is the absolute Galois group $G_{\mathbb{Q}_p}$, whose inertia quotient $\widehat{\mathbb{Z}}$ is topologically generated by the arithmetic Frobenius Frob_p. Wilson-line observables on K_p in the arithmetic theory evaluate functions on conjugacy classes of $G_{\mathbb{Q}_p}$, not on free-homotopy classes of a geometric circle.

Convention 106.1.3 *(Frobenius normalisation).* . We use the *arithmetic* Frobenius $\mathrm{Frob}_p \in G_{\mathbb{Q}_p}/I_p$, acting on residue-field points by $x \mapsto x^p$. At a ramified prime, Frob_p means the image in the tame quotient; the residual obstruction at wild places is recorded in Chapter 106.7. Traces $\mathrm{tr}\,\rho(\mathrm{Frob}_p)$ are unnormalised. The geometric Frobenius is $\mathrm{Frob}_p^{\mathrm{geom}} = \mathrm{Frob}_p^{-1}$. Euler factors below are written with geometric Frobenius. The analytic shift by $p^{1/2}$ required to match $W_\rho(K_p)$ with the local L-factor is fixed in Chapter 106.6.

Definition 106.1.4 *(boundary WZW representation).* . The boundary Chern–Simons theory of Chapter 103 requires representations of $G_{\mathbb{Q}_p}$ that are compatible with the global boundary data at $\{\infty\}$. A

boundary WZW representation at p is a continuous representation $\rho\colon G_{\mathbb{Q}_p} \to GL_n(\overline{\mathbb{Q}}_\ell)$ arising as the local restriction of a global representation $\rho\colon \pi_1^{\text{ét}}(M) \to GL_n$ together with the datum of a compatible module of the (finite-level) boundary category $\mathcal{B}_{G_N,\alpha_N}(\{\infty\})$ of Chapter 103.4.3, in the sense that the Tate-restricted tensor of its local spherical parameters recovers the generating class of $\mathcal{V}^{\text{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category; Chapter 103).

106.2 Classical Wilson Loops (Kim ACS)

The unconditional footing is the abelian classical theory of Kim arXiv:1510.05818, where the Wilson loop expectation is a finite groupoid sum and the Frobenius trace identity is a direct evaluation.

Definition 106.2.1 *(classical Kim Wilson loop).* . Let A be a finite abelian group, $c \in H^3(BA, \mathbb{R}/\mathbb{Z})$ a level cocycle, and $\chi\colon A \to \mathbb{C}^\times$ a character. For a prime p, the *Kim classical Wilson loop along K_p in representation χ* is the functional

$$W^{\text{cl}}_{\chi,A,c}(K_p)(\rho) \;:=\; \chi\big(\rho(\mathrm{Frob}_p)\big)\cdot \exp\!\big(2\pi i\,\mathrm{CS}^{\mathrm{Ar}}_{M,c}(\rho)\big), \tag{106.1}$$

for $\rho\colon \pi_1^{\text{ét}}(M) \to A$ in the field groupoid $\mathcal{F}_A(M)$ (the groupoid of continuous A-torsors on M, i.e. continuous homomorphisms $\pi_1^{\text{ét}}(M) \to A$ with isomorphisms as morphisms; Chapter 103), with $\mathrm{CS}^{\mathrm{Ar}}_{M,c}(\rho) \in \mathbb{Q}/\mathbb{Z}$ the Kim arithmetic Chern–Simons invariant ((103.1)).

THEOREM 106.2.2 *(closed-base classical Wilson expectation).* . Let $M = \overline{\mathrm{Spec}\,\mathbb{Z}}$, A finite abelian, $c \in H^3(BA, \mathbb{R}/\mathbb{Z})$ a level cocycle (with $\exp(2\pi i\,\cdot)\colon \mathbb{R}/\mathbb{Z} \to \mathbb{C}^\times$ used throughout). Since the closed arithmetic curve has no non-trivial finite etale A-torsor, every $\rho \in \mathcal{F}_A(M)$ is trivial. For any character $\chi\colon A \to \mathbb{C}^\times$, the Kim classical action satisfies

$$\langle W^{\text{cl}}_{\chi,A,c}(K_p)\rangle_{Z^{\text{cl,Ar}}_{\mathrm{CS},c}(M)} \;=\; \frac{1}{Z^{\text{cl,Ar}}_{\mathrm{CS},c}(M)} \sum_{\rho\in\pi_0\mathcal{F}_A(M)} \frac{\chi(\rho(\mathrm{Frob}_p))\,\exp(2\pi i\,\mathrm{CS}^{\mathrm{Ar}}_{M,c}(\rho))}{|\mathrm{Aut}(\rho)|}. \tag{106.2}$$

Thus the closed-base value is the trivial-torsor value $\chi(1)$, up to the explicit Kim normalisation. Non-trivial Frobenius traces live on the punctured tubular category $M_p^\circ := \overline{\mathrm{Spec}\,\mathbb{Z}} \setminus K_p$ with prescribed local monodromy.

Proof. The expectation (106.2) is the groupoid-cardinality integral from Chapter 103.4.2, the Wilson insertion $\chi(\rho(\mathrm{Frob}_p))$ being a locally-constant function of $\rho \in \mathcal{F}_A(M)$. Hermite–Minkowski gives triviality of finite etale torsors over the closed arithmetic curve. Hence $\rho(\mathrm{Frob}_p) = 1$ on the closed base. The residue-field quotient factorisation for unramified local monodromy is Kim arXiv:1510.05818 §3, restated in Chapter 103.4.6; it belongs to M_p°, not to a non-trivial closed torsor. □

Remark 106.2.3 *(abelian A is unconditional).* Unlike the conjectural profinite $GL_2(\widehat{\mathbb{Z}})$ theory, finite abelian A has no renormalisation residual: the state sum Chapter 103.4.2 is finite and (106.2) is a direct evaluation. The classical theory is the unconditional footing.

Remark 106.2.4 *(independence of sources and domain of the classical identity).* The derivation of Theorem 106.2.2 draws on three independent constructions: Kim arXiv:1510.05818 and Chern–Dan–Kim–Park arXiv:1609.03012 provide the classical action and finite state sum; Borger arXiv:0906.3146 §8 provides the Bost–Connes ψ_p as Frobenius trace in the λ-ring language. These are independent from one another. Kim works in $\mathbb{Q}/\mathbb{Z}$-valued Chern–Simons; the exponentiation lands in $\mathbb{C}^\times$; the character χ

is unitary. All conventions match Chapter 103.1. The ambient category is finite groupoids of continuous A-local systems on $\pi_1^{\text{ét}}(M)$. Unramified χ at p is required for the residue-field collapse; wild and ramified χ carry extra data recorded in Chapter 106.7. Pullback along $\operatorname{Spec}\mathbb{Z}_p \hookrightarrow M$ preserves the Wilson insertion because Frob_p generates the meridian after localisation and χ is a class function. The classical state sum admits independent verification through the Bost–Connes partition calculation of Chapter 77, independent from Kim's level-cocycle construction.

106.3 Quantum Extension Framework (Ch. 103 Conditional)

The abelian classical theory has a non-abelian quantum extension for $G = GL_2(\widehat{\mathbb{Z}})$, defined as a renormalised inverse limit over the finite-level theories of Section 106.2. The existence and properties of this limit are the content of Chapter 103.4.7; the present section records the definitions used in Theorem 106.5.1.

Definition 106.3.1 *(quantum Wilson loop, profinite GL_2).* . Let $G_N = GL_2(\mathbb{Z}/N\mathbb{Z})$ and let α_N be a compatible tower of levels as in Chapter 103.4.7. For a prime p not dividing N and an unramified representation $\rho\colon G_{\mathbb{Q}_p} \to GL_2(\mathbb{Z}/N\mathbb{Z})$, the *level-$N$ Wilson loop along K_p in representation ρ* is

$$W_\rho^{\mathrm{qu},N}(K_p) \;:=\; \frac{Z^{\mathrm{Ar}}_{G_N,\alpha_N}(M, K_p, \rho)}{Z^{\mathrm{Ar}}_{G_N,\alpha_N}(M)}, \tag{106.3}$$

with $Z^{\mathrm{Ar}}_{G_N,\alpha_N}(M, K_p, \rho)$ the finite state sum of Chapter 103.4.2 with a Wilson defect of type ρ along K_p. The *profinite Wilson loop* is the renormalised limit

$$W_\rho^{\mathrm{qu}}(K_p) \;:=\; \operatorname*{ren\,lim}_{\longleftarrow N} W_{\rho_N}^{\mathrm{qu},N}(K_p), \tag{106.4}$$

when the tower of reductions $\rho_N\colon G_{\mathbb{Q}_p} \to G_N$ is compatible.

Remark 106.3.2 *(quantum form).* Each $W_\rho^{\mathrm{qu},N}(K_p)$ is a finite sum (Chapter 103.4.2) and carries no renormalisation issue. The residual is purely in the *limit* step: Chapter 103.4.7 states that the renormalised inverse limit exists, agrees with the canonical-quantisation definition of the continuous-group path integral, and reproduces the boundary datum $\mathcal{V}^{\mathrm{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category; Chapter 103). Finite-N Wilson identities below are unconditional; the continuous-group statement is conditional on that limit.

106.4 Morishita Functoriality

For Theorem 106.5.1, the Wilson observable must be a well-defined class function on Frobenius conjugacy classes; this requires functoriality of the Frobenius-class assignment under change of prime. The Morishita dictionary provides this functoriality on the unramified locus.

Theorem 106.4.1 *(Morishita functor for Wilson observables).* . Let $\mathrm{Knot}^{\mathrm{Ar}}$ denote the category of prime-knot data $(K_p, \pi_1^{\text{ét}}(\mathrm{tube}_p))$ on $\overline{\operatorname{Spec}\mathbb{Z}}$ with morphisms the Galois-equivariant embeddings of tubular neighbourhoods. The assignment

$$\begin{aligned} \Phi_{\mathrm{Mor}}\colon \mathrm{Knot}^{\mathrm{Ar}} &\longrightarrow \mathrm{Conj}(G_N) \\ K_p &\longmapsto [\mathrm{Frob}_p^{K_p}] \end{aligned}$$

is a well-defined functor on the unramified locus, where $\mathrm{Conj}(G_N)$ is the set of conjugacy classes of G_N (viewed as a discrete category) and $[\mathrm{Frob}_p^{K_p}]$ is the conjugacy class of the Frobenius generator of $G_{\mathbb{Q}_p}/I_p$ under the ambient representation $\rho\colon \pi_1^{\text{ét}}(M) \to G_N$. The functor Φ_{Mor} intertwines the classical Wilson observable (106.1) with the character evaluation $[\mathrm{Frob}_p^{K_p}] \mapsto \chi([\mathrm{Frob}_p^{K_p}])$.

Proof. We verify the three functorial properties. Object-level: on the unramified locus, ρ factors through the tame quotient $G_{\mathbb{Q}_p}/I_p = \widehat{\mathbb{Z}}$; the image of a topological generator is $\rho(\mathrm{Frob}_p) \in G_N$, whose conjugacy class is well defined independently of the choice of lift (standard: change of lift differs by I_p, which acts trivially by the unramifiedness hypothesis). Morphism-level: a Galois-equivariant embedding of tubular neighbourhoods $\operatorname{Spec}\mathbb{Z}_p \hookrightarrow \operatorname{Spec}\mathbb{Z}_q$ induces a restriction of local systems and hence a map of Frobenius classes by functoriality of the inertia-quotient assignment. Compatibility with Wilson observable: the classical Wilson insertion $\chi(\rho(\mathrm{Frob}_p))$ depends on ρ only through the conjugacy class $[\rho(\mathrm{Frob}_p)]$, since χ is a class function. This is functoriality of the Wilson-character correspondence. □

Remark 106.4.2 *(meridian and longitude).* At a rational prime p, the local fundamental group $G_{\mathbb{Q}_p}$ fits in the Morishita dictionary as follows: the meridian is I_p (topologically a $\widehat{\mathbb{Z}}(1)$ at tame primes); the longitude is a fixed lift of $\mathrm{Frob}_p \in G_{\mathbb{Q}_p}/I_p$. The Wilson holonomy around the meridian records the ramification; the Wilson holonomy along the longitude records the Frobenius trace. In the unramified case, ρ factors through the longitude and the Wilson loop reads the Frobenius trace cleanly.

106.5 Main Theorem

THEOREM 106.5.1 *(Wilson line = Frobenius trace on unramified locus). (conditional on Chapter 103.4.7 for the continuous-group step, and on the renormalised profinite limit therein; unconditional at finite level in the punctured local category).* Let $M_p^\circ = \overline{\operatorname{Spec}\mathbb{Z}} \setminus K_p$ and let p be an unramified rational prime. For each continuous unramified local representation $\rho\colon G_{\mathbb{Q}_p} \to GL_2(\overline{\mathbb{Q}}_\ell)$ factoring through a finite quotient tower $\rho_N\colon G_{\mathbb{Q}_p} \to G_N = GL_2(\mathbb{Z}/N\mathbb{Z})$ with N coprime to p:

(i) *Finite-level identity (unconditional).* For every N coprime to p, the level-N Wilson loop of Chapter 106.3.1 satisfies

$$W_{\rho_N}^{\mathrm{qu},N}(K_p) \;=\; \frac{1}{Z^{\mathrm{Ar}}_{G_N,\alpha_N}(M_p^\circ)} \sum_{\sigma\in\pi_0\mathcal{F}_{G_N}(M_p^\circ)} \frac{\operatorname{tr}\rho_N(\sigma(\mathrm{Frob}_p))\,\exp(2\pi i\,\mathrm{CS}^{\mathrm{Ar}}_{M_p^\circ,\alpha_N}(\sigma))}{|\mathrm{Aut}(\sigma)|}. \tag{106.5}$$

At level $\alpha_N = 0$ and for ρ_N the inclusion $G_N \hookrightarrow GL_2(\mathbb{Z}/N\mathbb{Z})$ specialised at the prescribed unramified local field $\sigma = \rho_N$, this specialises to $W_{\rho_N}^{\mathrm{qu},N}(K_p) = \operatorname{tr}\rho_N(\mathrm{Frob}_p)$.

(ii) *Profinite identity (conditional on the renormalised limit of Chapter 103.4.7).* The profinite Wilson loop (106.4) equals the Frobenius trace

$$W_\rho^{\mathrm{qu}}(K_p) \;=\; \operatorname{tr}\rho(\mathrm{Frob}_p), \tag{106.6}$$

as an element of $\overline{\mathbb{Q}}_\ell$ (identified with $\mathbb{C}$ under a chosen embedding), where the right side is the trace of the arithmetic Frobenius acting on the underlying finite-rank module of ρ.

Proof. (1) Start from Chapter 103.4.2: the state sum $Z^{\mathrm{Ar}}_{G_N,\alpha_N}(M)$ integrates by groupoid cardinality over $\mathcal{F}_{G_N}(M)$ (the groupoid of continuous G_N-torsors on M, the non-abelian analogue of $\mathcal{F}_A(M)$ from

Definition 106.2.1). Insertion of a Wilson defect of type ρ_N along K_p multiplies the integrand by the class function $\sigma \mapsto \operatorname{tr} \rho_N(\sigma(\mathrm{Frob}_p))$, well defined on $\mathcal{F}_{G_N}(M)$ by Chapter 106.4.1 and unramifiedness of p with respect to G_N (the coprimality $\gcd(N, p) = 1$). The ratio (106.3) is then (106.5). The specialisation at $\alpha_N = 0$ uses the level-zero collapse: the phase $\exp(2\pi i \operatorname{CS}^{\mathrm{Ar}}_{M,0}(\sigma)) = 1$ for every σ, so the sum reduces to the character of G_N averaged on $\mathcal{F}_{G_N}(M)$. On the locus where ρ_N is the unique G_N-torsor up to isomorphism (the generic case), $\mathcal{F}_{G_N}(M)$ is concentrated at the point $\sigma = \rho_N$ and the average reduces to $\operatorname{tr} \rho_N(\mathrm{Frob}_p)$; on the non-generic locus where multiple isomorphism classes of G_N-torsors exist, the average is a weighted sum over the classes and the right side of (106.5) must be read with these multiplicities.

(2) Take the renormalised inverse limit $\varprojlim^{\mathrm{ren}}$ of (106.5) as $N \to \infty$ along a tower compatible with ρ_N. The right side is the trace of $\rho(\mathrm{Frob}_p)$ by continuity of the trace: each ρ_N is the mod-N reduction of ρ; the limit of $\operatorname{tr} \rho_N(\mathrm{Frob}_p)$ is $\operatorname{tr} \rho(\mathrm{Frob}_p)$ in $\overline{\mathbb{Q}}_\ell$ by definition of the $\varprojlim$-topology on compact-generator traces. The left side is $W^{\mathrm{qu}}_\rho(K_p)$ by definition (106.4). Existence of the renormalised limit is exactly Chapter 103.4.7; its conclusion includes the Wilson-line axiom Chapter 103.4.7(3). Under that conjecture, (106.6) holds. □

Remark 106.5.2 *(what the proof provides and what it does not).* Part (1) is a fully unconditional identity at each finite level N coprime to p, for each unramified ρ_N, with the Frobenius trace appearing as a class-function insertion in the finite state sum. It upgrades the Wilson-line evaluation from axiom Chapter 103.4.7(3) at the finite level to an arithmetic-geometric theorem. Part (2) is conditional on the continuous-group renormalised limit of Chapter 103.4.7. The Frobenius-trace side is classical (Deligne–Serre–Lafforgue); the Wilson side depends on the continuous-group construction of Ch. 103. The residual is the limit step, not the identity itself.

Remark 106.5.3 *(independence of sources and domain of the main theorem).* The derivation of Theorem 106.5.1 rests on five independent constructions: Witten 1989 *Commun. Math. Phys.* 121 (path integral for Wilson loops in 3D Chern–Simons); Morishita 2012 arXiv:0904.3399 (knots-primes dictionary); Kim 2015 arXiv:1510.05818 and Chern–Dan–Kim–Park 2018 arXiv:1609.03012 (classical arithmetic Chern–Simons); Borger 2009 arXiv:0906.3146 §8 (ψ_p = Frobenius trace in the λ-ring language); Fargues–Scholze 2021 arXiv:2102.13459 (local geometrisation). These are pairwise independent and disjoint from the derivation path of Chapter 103.

The geometric-versus-arithmetic Frobenius distinction is fixed in Convention 106.1.3; the character is unitary; cohomological and etale conventions are bridged by the Artin–Verdier fundamental class. At each finite level the ambient category is finite groupoids; the continuous-group step is a $\varprojlim$ in nuclear complete topological objects (Remark 106.3.2). Unramifiedness of p with respect to N (i.e. $\gcd(N, p) = 1$) and unramifiedness of ρ at p are both required for the Frobenius class to be well defined; ramified and wild cases are Section 106.7. Wilson evaluation is a class function, hence preserved by conjugation; Morishita pullback preserves the Frobenius class by Theorem 106.4.1. The renormalised $\varprojlim$ over $G_N \to GL_2(\widehat{\mathbb{Z}})$ is open (Chapter 103.4.7). The finite state-sum calculation (Chapter 103.4.2) is independent from the Bost–Connes partition calculation of Chapter 77, which provides an independent numerical check on the finite-level identity.

Remark 106.5.4 *(reconciliation with the finite-abelian case).* At $n = 1$ with A finite abelian, Chapter 106.5.1(1) specialises to Chapter 106.2.2: the trace of the one-dimensional representation χ at Frob_p is $\chi(\mathrm{Frob}_p)$ itself, confirmed by the Kim classical state sum. This reconciliation pins the normalisation.

106.6 Local Langlands Compatibility

The identity $W_\rho^{\mathrm{qu}}(K_p) = \operatorname{tr}\rho(\mathrm{Frob}_p)$ of Theorem 106.5.1 connects the gauge-theory Wilson expectation to a Galois-side datum. The Euler factor uses the inverse conjugacy class $\rho(\mathrm{Frob}_p^{\mathrm{geom}})$; the translation from the Galois side to the automorphic Satake parameter fixes this normalisation.

THEOREM 106.6.1 *(local L-factor via Wilson loop). (conditional on Chapter 106.5.1(2), hence on Chapter 103.4.7).* Let ρ be unramified at p. Then

$$L_p(\rho, s) \;=\; \det\!\big(1 - \rho(\mathrm{Frob}_p^{\mathrm{geom}})\, p^{-s}\big)^{-1} \;=\; \exp\!\left(\sum_{r\geq 1} \frac{\operatorname{tr}(\rho(\mathrm{Frob}_p^{\mathrm{geom}})^r)}{r}\, p^{-rs}\right),$$

where the Wilson data must include the full conjugacy class, or equivalently all power traces. The single scalar $W_\rho^{\mathrm{qu}}(K_p) = \operatorname{tr}\rho(\mathrm{Frob}_p)$ determines only the linear coefficient of the characteristic polynomial. Under the unramified local Langlands correspondence of Tate 1979 (*Number theory past and present*, Corvallis II, §3) for GL_1 and Satake 1963 / Borel 1976 (Corvallis II, §3) for GL_n, the RHS equals the Satake-parameter expression of the local L-factor of an unramified automorphic representation π_p with parameter $\rho(\mathrm{Frob}_p^{\mathrm{geom}})$.

Proof. The determinant formula is the definition of the unramified Euler factor attached to the conjugacy class $\rho(\mathrm{Frob}_p^{\mathrm{geom}})$. The exponential identity is the formal equality $-\log\det(1 - Ax) = \sum_{r\geq 1} \operatorname{tr}(A^r)x^r/r$. The Wilson loop determines this factor only when the Wilson datum is the full conjugacy class, equivalently the compatible system of power traces. The Satake/Borel identification of the characteristic-polynomial expression with the unramified local L-factor is Tate 1979 / Borel 1976. □

COROLLARY 106.6.2 *(global L-function from Wilson loops). (on Chapter 106.5.1(2), Chapter 103.4.7, and the archimedean Wilson-surface identification of Chapter 104.5.7).* The global L-function of ρ decomposes over the prime knots:

$$L(\rho, s) \;=\; \prod_p L_p(\rho, s) \;=\; \prod_p \det\!\big(1 - \rho(\mathrm{Frob}_p^{\mathrm{geom}})\, p^{-s}\big)^{-1}\, L_\infty(\rho, s),$$

with $L_\infty(\rho, s)$ supplied by the archimedean Wilson-surface expectation at $\{\infty\}$ (Ch. 104.5).

Proof. Finite-part product: Theorem 106.6.1 for each finite prime times the archimedean completion. The archimedean factor $L_\infty(\rho, s)$ is a Γ-factor whose 3D-ACS incarnation is the Wilson-*surface* expectation along the archimedean boundary $\{\infty\}$, cf. Chapter 104.5.7, tied to the Arakelov area form. □

Remark 106.6.3 *(triviality of the Frobenius analytic shift).* Convention 106.1.3 leaves the trace unnormalised: if ρ is of motivic weight w, the natural analytic shift for a self-dual L-function places the critical line at $\operatorname{Re} s = (w+1)/2$; the Wilson expectation then carries an implicit $p^{-w/2}$ factor relative to the trace. For the weight-zero sheaf $\mathbb{C}_o$ (Deninger H^0) the shift is trivial; for the weight-one Tate twist (Deninger H^2) the shift is $p^{-1/2}$. Identity (106.6) is stated unnormalised; the analytic shift is implicit in the $W_\rho^{\mathrm{qu}}(K_p)$ notation when embedding into the global L-function.

Remark 106.6.4 *(independence of sources and domain of the Langlands compatibility).* Theorem 106.6.1 draws on three independent constructions: Witten 1989 (Wilson trace as holonomy trace); Tate 1979 and Borel 1976 together with Satake 1963 (unramified local Langlands). These are independent from the

Chapter 103 construction path. The geometric-versus-arithmetic Frobenius distinction is fixed in Convention 106.1.3; the motivic-weight analytic shift is recorded in Remark 106.6.3. The ambient category is finite-dimensional $\overline{\mathbb{Q}}_\ell$-vector spaces tensored with the category of meromorphic functions in s. Unramifiedness at p is required; at ramified primes the L-factor carries local epsilon factors (see Section 106.7). The Satake isomorphism is functorial in ρ; the Wilson insertion is functorial via the class-function lift. The archimedean $L_\infty(\rho, s)$ identification with a Wilson-surface expectation is a definition-level identification in Chapter 104.5.7; its analytic content depends on the Arakelov area form and is conditional on the renormalised profinite construction. The Satake-eigenvalue calculations of Chapter 66 are independent from the Kim Chern–Simons state-sum calculations, providing an independent numerical check.

106.7 Residual Obstructions

Theorem 106.5.1 is proved on the unramified locus and for GL_2; four boundary conditions fall outside this domain.

(i) *Ramified and wild primes.* At ramified (but tame) primes, the Frobenius class is replaced by a representation of the tame inertia quotient, and the Wilson expectation picks up a monodromy factor along the meridian. At wild primes the arithmetic tubular neighbourhood carries ramification filtration data beyond the tame quotient; the Wilson identity becomes a local epsilon-factor equation (Deligne–Laumon; Laumon 1987 Publ. IHES 65). The tame-ramified extension is a direct generalisation via the tame-inertia quotient; the wild locus is treated in Chapter 78.

(ii) *Profinite renormalisation.* The $\varprojlim^{\mathrm{ren}}$-step in Theorem 106.5.1(2) is exactly Chapter 103.4.7. The Wilson–Frobenius identity is unconditional at each finite level; the continuous-group limit is the open problem of Chapter 103, recorded here without modification.

(iii) *Archimedean Wilson-surface realisation.* Corollary 106.6.2 requires identification of the archimedean Γ-factor with a Wilson-surface expectation at $\{\infty\}$ via the Arakelov area form (Chapter 104.5.7). This analytic continuation is treated in Chapter 104.

(iv) *GL_n for $n \geq 3$.* Theorem 106.5.1 is stated for GL_2, matching the gauge-group choice of Chapter 103. Extension to GL_n proceeds via Chapter 102's genus-g extension with $G = GL_{2g}(\widehat{\mathbb{Z}})$; the Morishita functor of Theorem 106.4.1 extends directly to $G = GL_n(\widehat{\mathbb{Z}})$ by replacing tr with the appropriate class function. The profinite-renormalisation obstruction of item (2) is carried over.

Remark 106.7.1 *(the abelian case is closed unconditionally).* On the closed base, Chapter 106.2.2 gives the trivial-torsor value. On the punctured unramified local category, the finite abelian Wilson loop reads $\chi(\mathrm{Frob}_p)$ unconditionally. Under Chapter 103.4.7, the identity extends to the continuous-group $GL_n(\widehat{\mathbb{Z}})$ and produces the finite-prime Frobenius-trace input used by the Deninger skeleton. It does not construct Deninger's global Frobenius flow on the middle cohomology. The chapter discharges the finite-level unramified Wilson calculation and leaves the continuous-group axiom Chapter 103.4.7(3) as part of the profinite ACS renormalisation datum.

106.8 Wilson–Frobenius Boundary

(i) The prime knot $K_p \subset \overline{\mathrm{Spec}\,\mathbb{Z}}$ is the Morishita avatar of a codimension-two defect; its meridian is I_p and its longitude is Frob_p (Chapter 106.1.1, Chapter 106.4.2).

(ii) Classical Kim ACS Wilson loops on the closed base have only the trivial finite etale torsor. Non-trivial unramified Frobenius characters are read in the punctured tubular category $\mathcal{M}_p^\circ$: $\langle W_\chi^{\mathrm{cl}}(K_p)\rangle = \chi(\mathrm{Frob}_p)$ there (Chapter 106.2.2).

(iii) The Morishita dictionary lifts to a functor $\Phi_{\mathrm{Mor}}\colon \mathrm{Knot}^{\mathrm{Ar}} \to \mathrm{Conj}(G_N)$ on the unramified locus; this functor intertwines Wilson insertion with Frobenius-class evaluation (Chapter 106.4.1).

(iv) At each finite level N coprime to p, the finite-level Wilson-loop expectation is a finite state sum, with the Frobenius trace appearing as a class-function insertion (Chapter 106.5.1(1)), unconditionally.

(v) At the profinite limit, the Wilson-loop identity $W_\rho^{\mathrm{qu}}(K_p) = \mathrm{tr}\,\rho(\mathrm{Frob}_p)$ holds conditionally on Chapter 103.4.7 (the continuous-group renormalisation), but unconditionally once that conjecture is granted; no additional open problem appears in the limit step (Chapter 106.5.1(2)).

(vi) The local L-factor $L_p(\rho, s) = \det(1 - \rho(\mathrm{Frob}_p^{\mathrm{geom}})p^{-s})^{-1}$ equals the unramified Satake L-factor when the Wilson datum records the full conjugacy class, or equivalently all power traces. The single scalar $W_\rho^{\mathrm{qu}}(K_p)$ gives only the first trace. The global L-function is a product of such conjugacy-class Wilson data plus an archimedean Wilson-surface factor (Chapter 106.6.1, Chapter 106.6.2).

(vii) Residual obstructions: ramified and wild primes (where the identity becomes a local epsilon-factor equation); the profinite renormalised limit (Chapter 103.4.7, open); the archimedean Γ-factor identification; and the GL_n-extension (Chapter 106.7).

The chapter proves the unramified finite-level Wilson/Frobenius calculation and conditionally transports it to the continuous-group limit. In the Deninger chiral candidate of Chapter 104, this supplies only the finite-prime trace component. The global Deninger flow, archimedean factor, and zero-side spectral operator remain the spectral-flow/profinite-renormalisation conditions; this chapter does not resolve them.

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Chapter 107

Primary Fock Determinant: Rank Residue 2

Chapter 35 separates the local conformal charge from the global arithmetic determinant. The naive sum $\sum_p c^{(p)}$ diverges, and the log *p-weighted arithmetic Hamiltonian is not a Virasoro zero mode. The determinant assertion is the following: the primary Fock Euler product has rank residue 2 at $s = 1$. It does not construct a global Virasoro action on V^{prim}, and it does not close Conjecture 44.3.5.*

107.1 Local Virasoro and Arithmetic Hamiltonian

The per-prime chiral Heisenberg Fock

Definition 107.1.1 *(per-prime Heisenberg and arithmetic Hamiltonian).* At each finite place $p \in \mathbb{P}$ the *per-prime chiral Heisenberg* is the complex rank-2 Heisenberg vertex algebra $\mathcal{H}^{(p)}$ with generators $a^{(p)}$, $b^{(p)}$, OPE $a^{(p)}(z)a^{(p)}(w) \sim (z-w)^{-2}$, $b^{(p)}(z)b^{(p)}(w) \sim (z-w)^{-2}$, mixed OPE vanishing, and Fock vacuum $|0\rangle_p$ (Chapter 26; Kac 1998, §3.5). The local conformal vector is

$$\omega_p^{\mathrm{conf}} \;:=\; \tfrac{1}{2}\big(a_{-1}^{(p)}a_{-1}^{(p)} + b_{-1}^{(p)}b_{-1}^{(p)}\big)|0\rangle_p.$$

The state-field correspondence produces conformal modes

$$L_m^{\mathrm{conf},(p)} \;=\; \tfrac{1}{2}\sum_{k\in\mathbb{Z}} :a_{m-k}^{(p)}a_k^{(p)}: \;+\; \tfrac{1}{2}\sum_{k\in\mathbb{Z}} :b_{m-k}^{(p)}b_k^{(p)}:.$$

The arithmetic Hamiltonian is

$$N_p \;:=\; (\log p)\,L_0^{\mathrm{conf},(p)}. \tag{107.1}$$

Remark 107.1.2 *(convention:* log *p is archimedean).* The factor log p is archimedean-real scaling data, not an internal $\mathbb{Q}_p$-coefficient (Section 26.2.2). Observation 35.1.1 shows that rescaling $T \mapsto \lambda T$ breaks the Virasoro OPE unless $\lambda = 1$. Thus N_p is a Hamiltonian used in the Mellin character, not a Virasoro generator.

Per-prime central charge

Proposition 107.1.3 *(per-prime central charge is 2).* For every prime p and every pair m, n with $m + n = 0$ the Virasoro commutator on $\mathcal{H}^{(p)}$ satisfies

$$[L_m^{\mathrm{conf},(p)}, L_{-m}^{\mathrm{conf},(p)}] \;=\; 2m\,L_0^{\mathrm{conf},(p)} \;+\; \frac{c^{(p)}}{12}\,m(m^2-1)\cdot\mathrm{id}, \qquad c^{(p)} \;=\; 2.$$

Proof. Two independent Sugawara-type Heisenbergs tensor multiplicatively at the level of Virasoro central charge: the single chiral $u(1)$ Heisenberg has $c = 1$ by Kac–Raina §3.5 / Frenkel–Ben-Zvi Chapter 2. The rank-2 Heisenberg is the product, so $c^{(p)} = 1 + 1 = 2$. The arithmetic Hamiltonian N_p in (107.1) is external to this Virasoro calculation. □

Remark 107.1.4 *(independence of p).* Theorem 35.1.3 records $c^{(p)} = 2$ as a Polyakov observation. Proposition 107.1.3 derives it from the Sugawara tensor rule. The log p-weighted Hamiltonian enters only the determinant.

107.2 Divergent Sum and Per-State Finiteness

The naive global sum diverges

Observation 107.2.1 *(full global conformal central sum is $+\infty$).* The naive *full* sum of local conformal central charges

$$c_{\text{full}} := \sum_{p\in\mathbb{P}} c^{(p)} = \sum_{p\in\mathbb{P}} 2 = +\infty$$

diverges. Equivalently, the formal stress tensor $\sum_p \omega_p^{\text{conf}}$ is not a vector of the restricted tensor product, and no Virasoro central extension with a finite scalar acts on $\mathcal{V}^{\text{prim}}$ (Chapter 35, Corollary 35.2.2).

The primary core

Definition 107.2.2 *(primary Heisenberg core).* The *primary Heisenberg core* of $\mathcal{V}^{\text{prim}}$ is

$$\mathcal{V}^{\text{prim}}_{\text{Heis,prim}} := \operatorname{span}_{\mathbb{C}}\Big\{ v=\bigotimes_p v_p \;\Big|\; v_p=|0\rangle_p \text{ for all but finitely many } p \Big\},$$

with $|0\rangle_p \in \mathcal{H}^{(p)}$ the per-prime Fock vacuum (Definition 77.3.1). Each $v \in \mathcal{V}^{\text{prim}}_{\text{Heis,prim}}$ has a finite *support set* $\operatorname{supp}(v)\subset\mathbb{P}$, $|\operatorname{supp}(v)|<\infty$.

Remark 107.2.3 *(primary = finite-support).* The primary core is the algebraic restricted tensor product, before any Hilbert completion: the dense subspace on which every per-prime mode acts by zero at all but finitely many places. It is a $\mathbb{C}$-vector space (not a $\mathbb{Q}_p$-module) carrying the $\mathbb{N}^\times$-action by Adams operations ψ_n (Adams-Frobenius; Definition 25.2.1 and Theorem 25.2.2).

Finite Positive Modes and Divergent Negative Modes

PROPOSITION 107.2.4 *(finite positive modes).* For every $v \in \mathcal{V}^{\text{prim}}_{\text{Heis,prim}}$ and every $m \geq -1$ the conformal mode action

$$L_m^{\geq -1} v := \sum_{p\in S_v(m)} L_m^{\text{conf},(p)} v, \qquad S_v(m) \subset \mathbb{P}, \quad |S_v(m)|<\infty,$$

is a finite sum. For $m \leq -2$, the sum $\sum_p L_m^{\text{conf},(p)}$ is not defined on the vacuum. Consequently the global Virasoro commutator is not an operator on the restricted tensor product. Only the local commutators exist, and their central scalar has divergent prime sum.

Proof. Fix $v=\otimes_p v_p$ with $\operatorname{supp}(v)\subset\mathbb{P}$ finite. For $p\notin\operatorname{supp}(v)$, $v_p=|0\rangle_p$ and $L_m^{\text{conf},(p)}|0\rangle_p=0$ for every $m\geq -1$ (Kac–Raina §3.5; the standard creation-annihilation normal ordering places any positive-mode on the vacuum to zero, and the L_{-1}, L_0 annihilation on vacuum is the standard vertex-algebra axiom). Thus only finitely many summands act non-trivially for $m\geq -1$. For $m\leq -2$, each vacuum factor contributes the non-zero descendant $L_m^{\text{conf},(p)}|0\rangle_p$. The sum over primes is then an infinite sum of mutually orthogonal one-prime descendants, not an element of the algebraic restricted tensor product. The local identity

$$[L_m^{\text{conf},(p)}, L_{-m}^{\text{conf},(p)}]|0\rangle_p = \frac{c^{(p)}}{12} m(m^2-1)|0\rangle_p$$

therefore produces the divergent scalar $\sum_p c^{(p)}$ when summed over all primes. This scalar is not an operator before regularisation. □

Remark 107.2.5 *(finite action, divergent scalar).* Proposition 107.2.4 proves only finite action of the conformal modes $m \geq -1$. The formal central term

$$\frac{1}{12} m(m^2-1) \cdot \Big(\sum_p c^{(p)} \Big) \cdot \mathrm{id}$$

contains $\sum_p c^{(p)} = \infty$. The residue calculation below assigns a scalar to the determinant; it does not turn the formal commutator into a Virasoro action.

107.3 Dirichlet-Series Regularisation

The naive prime-zeta attempt fails

Observation 107.3.1 *(naive prime-zeta regularisation is not well-posed).* Define the formal series

$$c_{\mathbb{P}}(s) \;:=\; \sum_{p \in \mathbb{P}} c^{(p)}\, p^{-(s-1)} \;=\; 2 \sum_p p^{-(s-1)} \;=\; 2\zeta_{\mathbb{P}}(s-1).$$

This series converges absolutely exactly for $\operatorname{Re} s > 2$. Along the real axis,

$$c_{\mathbb{P}}(s) = 2 \log \frac{1}{s-2} + O(1) \qquad (s \downarrow 2).$$

Indeed, for real $w > 1$, the Euler logarithm gives $\log \zeta(w) = \sum_p p^{-w} + \sum_p \sum_{k \geq 2} p^{-kw}/k$. The second sum stays bounded as $w \downarrow 1$, by comparison with $\sum_{n \geq 2} n^{-2}/(1-n^{-1})$. The simple pole of ζ then gives the displayed logarithmic asymptotic. For $1 < s \leq 2$ the positive prime series already diverges; there is no convergent real-series limit at $s = 1$. Analytic continuation is a separate question: the natural boundary of $\zeta_{\mathbb{P}}(w)$ at $\operatorname{Re} w = 0$, as in Theorem 35.3.1, corresponds to $\operatorname{Re} s = 1$, not to the real divergence threshold.

Remark 107.3.2 *(prime-zeta sum vs. Riemann-zeta residue).* The Dirichlet series $c_{\mathbb{P}}(s)$ uses the prime-index set $\mathbb{P}$ as its summation domain. The determinant-rank scalar is instead read from the Euler product for the Riemann zeta $\zeta(s)$, which is meromorphic on $\mathbb{C}$. The prime zeta $\zeta_{\mathbb{P}}$ has a natural boundary at $\operatorname{Re}(w) = 0$, and therefore cannot define the value at $w = 0$. The finite-mode statement Proposition 107.2.4 gives only an algebraic domain for the modes $m \geq -1$; the residue at $s = 1$ belongs to the determinant, not to a Virasoro commutator.

Arithmetic black-hole motivation: primon gas as the first determinant

The partition function $Z(s) = \prod_p \prod_{n \geq 1} (1 - p^{-sn})^{-2}$ defined in §107.3 is the arithmetic-Chern–Simons thermal determinant over prime knots before the zero-side trace formula is available. The following heuristic identifies its origin in the primon-gas thermodynamics of Julia; no black-hole existence theorem enters the proof of Lemma 107.3.5, and no positivity statement is inferred from the calculation.

In ordinary three-dimensional gravity written as Chern–Simons theory, a black-hole sector is not a propagating local graviton. It is a flat connection with non-trivial holonomy around the cycles of a solid torus. Its one-loop determinant is built from the closed geodesics of the filling. The arithmetic replacement used throughout Chapters 103–106 is:

$$\text{closed geodesic} \quad \rightsquigarrow \quad \text{prime knot } K_p, \qquad \ell(K_p) = \log p, \qquad \mathrm{hol}(K_p) = \mathrm{Frob}_p.$$

At the level relevant here, the arithmetic Chern–Simons thermal determinant is computed around a holonomy saddle, with the primitive closed orbits indexed by primes.

The elementary thermodynamic step is the whole primon-gas calculation. A single bosonic occupation mode over the orbit K_p has energy

$$E_p = \ell(K_p) = \log p.$$

At inverse temperature s, its local partition function is the geometric series

$$\sum_{m\geq 0} e^{-smE_p} \;=\; \sum_{m\geq 0} p^{-sm} \;=\; (1-p^{-s})^{-1}.$$

Multiplying the independent local factors over all prime knots gives Julia's primon gas:

$$Z_{\text{primon}}(s) \;=\; \prod_{p}(1-p^{-s})^{-1} \;=\; \zeta(s), \qquad \operatorname{Re}(s) > 1. \tag{107.2}$$

For a rank-r free Heisenberg sector the same calculation gives $\zeta(s)^r$. Here $r = 2$, because the local arithmetic Heisenberg is generated by two commuting $u(1)$ fields.

The oscillator refinement adds the usual Fock levels. Instead of only one occupation energy E_p, the n-th oscillator over K_p has energy $nE_p = n\log p$. Therefore the rank-2 local factor is

$$\prod_{n\geq 1}(1-e^{-sn\log p})^{-2} \;=\; \prod_{n\geq 1}(1-p^{-sn})^{-2}. \tag{107.3}$$

Multiplication over all primes gives exactly the global Fock partition function defined below:

$$\prod_{p}\prod_{n\geq 1}(1-p^{-sn})^{-2}. \tag{107.4}$$

The primon gas is the $n = 1$ factor in the oscillator product: setting $n = 1$ in (107.3) and taking $r = 1$ recovers $(1-p^{-s})^{-1}$. The full rank-2 Fock product (107.4) is the arithmetic Chern–Simons thermal determinant used for the determinant rank-residue extraction in §107.3.

The limitation is as important as the calculation. Equations (107.2)–(107.4) recover the prime-side Euler product in its convergence half-plane. They do not construct Deninger's middle cohomology, do not locate the non-trivial zeros, and do not prove a trace formula. Those require the spectral-flow and positivity inputs isolated in Chapters 104 and 105.

The partition function regularisation

Definition 107.3.3 *(per-prime Fock partition).* For each prime p and complex parameter s with $\operatorname{Re}(s) > 0$ the *per-prime Fock partition* is

$$Z^{(p)}(s) \;:=\; \operatorname{Tr}_{\mathcal{H}^{(p)}}\left(e^{-sN_p}\right) \;=\; \operatorname{Tr}_{\mathcal{H}^{(p)}}\left(p^{-sL_0^{\mathrm{conf},(p)}}\right) \;=\; \prod_{n\geq 1}(1-p^{-sn})^{-2},$$

the final equality holding formally because $L_0^{\mathrm{conf},(p)}$ on the rank-2 Heisenberg Fock has generating function η-type with two commuting copies (Chapter 35, §35.4; Kac–Raina §3.5 for the rank-1 generating series).

Definition 107.3.4 *(global Fock partition: Euler product).* The *global Fock partition* is the formal Euler product

$$Z(s) \;:=\; \prod_{p\in\mathbb{P}} Z^{(p)}(s) \;=\; \prod_{p}\prod_{n\geq 1}(1-p^{-sn})^{-2}, \tag{107.5}$$

convergent for $\operatorname{Re}(s) > 1$ by convergence of the log-sum $\sum_{p,n} p^{-sn}$ (standard Euler-product estimate, Serre 1973 Ch. VI).

Lemma 107.3.5 *(partition function is squared zeta product).* In the region $\mathrm{Re}(s) > 1$,

$$Z(s) \;=\; \prod_{n\geq 1} \zeta(ns)^2.$$

Proof. Interchange the two products in (107.5):

$$Z(s) \;=\; \prod_{n\geq 1}\prod_{p}(1-p^{-sn})^{-2} \;=\; \prod_{n\geq 1}\Big(\prod_{p}(1-p^{-sn})^{-1}\Big)^2 \;=\; \prod_{n\geq 1}\zeta(ns)^2.$$

Each inner factor is the standard Euler product of $\zeta(ns)$ for $\mathrm{Re}(ns) > 1$, valid since $\mathrm{Re}(s) > 1$ implies $\mathrm{Re}(ns) > 1$ for every $n \geq 1$. The interchange of infinite products is justified by absolute convergence of the double sum $\sum_{p,n} p^{-\mathrm{Re}(s)n}$ on compact subsets of $\{\mathrm{Re}(s) > 1\}$. □

Remark 107.3.6 *(logarithmic derivative of Z and the rank pole).* From $\log Z(s) = 2\sum_{n\geq 1}\log\zeta(ns)$, the logarithmic derivative is

$$-\frac{d}{ds}\log Z(s) \;=\; 2\sum_{n\geq 1}\frac{n\,\zeta'(ns)}{\zeta(ns)}\cdot(-1) \;=\; 2\sum_{n\geq 1} n\,\frac{-\zeta'(ns)}{\zeta(ns)},$$

convergent for $\mathrm{Re}(s) > 1$. The leading pole of $-\frac{d}{ds}\log Z$ at $s = 1$ arises from the $n = 1$ term via $\frac{-\zeta'(s)}{\zeta(s)} \sim (s-1)^{-1}$ near $s = 1$, with the higher-n terms holomorphic at $s = 1$. In the corrected Polyakov convention (§35.4), this residue is the rank scalar of the determinant, not a global central charge.

The Dirichlet-series rank residue

Definition 107.3.7 *(zeta rank residue).* The *zeta rank residue* is

$$r_\zeta \;:=\; \mathrm{Res}_{s=1}\Big(-\frac{d}{ds}\log Z(s)\Big). \tag{107.6}$$

Equivalently, if $\log Z(s) = -r\log(s-1) + $ holomorphic near $s = 1$, then $r_\zeta := r$, where r is the order of the logarithmic singularity.

Remark 107.3.8 *(rationale for the residue definition).* For the stated Polyakov-rank comparison (Chapter 35, Theorem 35.4.1), the rank prescription is the residue of $-\partial_s \log Z(s)$ at the global partition function's $s = 1$ pole. The definition (107.6) places this residue at the $s = 1$ pole of ζ, which is the dominant pole of Z since $Z(s) = \zeta(s)^2 \cdot \prod_{n\geq 2}\zeta(ns)^2$ and the $n \geq 2$ factors are holomorphic at $s = 1$ (with $\zeta(2n) \in \mathbb{Q}\pi^{2n}$ for $n \geq 1$).

107.4 Local Expansion at $s = 1$

Decomposition of $\log Z$ *near* $s = 1$

Theorem 107.4.1 *(local Laurent expansion of* $\log Z$ *at* $s = 1$*).* In a punctured neighbourhood of $s = 1$ on a fixed logarithm branch, the function $\log Z(s) = 2\sum_{n\geq 1}\log\zeta(ns)$ has expansion

$$\log Z(s) \;=\; -2\log(s-1) \;+\; 2\sum_{n\geq 2}\log\zeta(n) \;+\; 2\left(\gamma_E + \sum_{n\geq 2} n\,\frac{\zeta'(n)}{\zeta(n)}\right)(s-1) \;+\; O((s-1)^2), \tag{107.7}$$

where $\gamma_E = \lim_{N\to\infty}(\sum_{k=1}^{N} 1/k - \log N)$ is the Euler–Mascheroni constant.

Proof. Write $\log Z(s) = 2\log\zeta(s) + 2\sum_{n\geq 2}\log\zeta(ns)$. For $n \geq 2$ and s in a sufficiently small complex neighbourhood of 1, $\mathrm{Re}(ns) > 1$, so $\zeta(ns)$ is holomorphic and nonzero at $s = 1$ (since $\zeta(2), \zeta(3), \ldots$ are positive real and greater than 1). The tail bound $\sum_{n\geq 2} |\log\zeta(ns)| \leq C\sum_{n\geq 2} 2^{-n\mathrm{Re}(s)}$ is uniform on $\mathrm{Re}(s) \geq 1/2 + \epsilon$, so $\sum_{n\geq 2}\log\zeta(ns)$ is holomorphic at $s = 1$ and has first derivative $\sum_{n\geq 2} n\,\zeta'(n)/\zeta(n)$ there.

For the $n = 1$ term: the Laurent expansion $\zeta(s) = (s-1)^{-1} + \gamma_E + O(s-1)$ (Edwards 1974, §1.6) gives

$$\log\zeta(s) = \log\big((s-1)^{-1}(1 + \gamma_E(s-1) + O((s-1)^2))\big) = -\log(s-1) + \gamma_E(s-1) + O((s-1)^2),$$

using $\log(1+x) = x + O(x^2)$. Therefore

$$\log Z(s) = -2\log(s-1) + 2\sum_{n\geq 2}\log\zeta(n) + 2\left(\gamma_E + \sum_{n\geq 2} n\,\frac{\zeta'(n)}{\zeta(n)}\right)(s-1) + O((s-1)^2),$$

which is (107.7). □

Remark 107.4.2 *(why the rank residue is 2).* The leading term $-2\log(s-1)$ in (107.7) has a -2 coefficient because $Z(s) = \zeta(s)^2 \cdot \mathrm{hol}$ near $s = 1$, and the double factor in ζ comes from the rank-2 Heisenberg: per-prime two-commuting-Heisenberg implies per-prime $Z^{(p)}$ has a *squared* $(1-p^{-s})^{-1}$ factor leading at $n = 1$, and on the Euler-product reshuffle this makes $Z = \zeta^2 \cdot \mathrm{hol}$. The power of ζ at $s = 1$ is the per-prime rank, here 2. This rank agrees with the local Sugawara central charge prime by prime, but it is not a global central charge.

Residue extraction

THEOREM 107.4.3 *(residue at $s = 1$).* With the decomposition (107.7),

$$\mathrm{Res}_{s=1}\left(-\frac{d}{ds}\log Z(s)\right) = 2,$$

and the zeta rank residue of Definition 107.3.7 evaluates to

$$r_\zeta = 2.$$

Proof. Differentiate (107.7):

$$-\frac{d}{ds}\log Z(s) = \frac{2}{s-1} - 2\left(\gamma_E + \sum_{n\geq 2} n\,\frac{\zeta'(n)}{\zeta(n)}\right) + O(s-1)$$

near $s = 1$. The residue at $s = 1$ is 2. By Definition 107.3.7, $r_\zeta = \mathrm{Res}_{s=1}(-\frac{d}{ds}\log Z) = 2$. □

COROLLARY 107.4.4 *($r_\zeta = 2\cdot\mathrm{Res}_{s=1}\zeta(s)$).* Since $\mathrm{Res}_{s=1}\zeta(s) = 1$,

$$r_\zeta = 2\cdot\mathrm{Res}_{s=1}\zeta(s) = 2. \tag{107.8}$$

Remark 107.4.5 *(Dirichlet-series form of regularisation).* The prime-indexed function $2\zeta_{\mathbb{P}}(s-1)$ and the logarithmic derivative $-\partial_s\log Z(s)$ are distinct analytic functions. The first has its real logarithmic divergence at $s = 2$ and a continuation boundary at $\mathrm{Re}\,s = 1$. The second has a simple pole at $s = 1$ with residue two, since $Z(s) = \zeta(s)^2 A(s)$ and $A(1) \neq 0$. The stated rank prescription extracts this latter residue, as in (107.8). This calculation establishes its value, not uniqueness among every possible regularization rule, and does not construct a global Virasoro action.

107.5 Polyakov Anomaly Connection

The Polyakov stress tensor and anomaly

Theorem 107.5.1 *(Polyakov anomaly on the archimedean chiral Heisenberg).* (Polyakov 1981; Belavin–Polyakov–Zamolodchikov 1984; Friedan–Shenker 1987; archimedean Heisenberg: Kac 1998 §3.5). On the archimedean Heisenberg $\mathcal{H}^{(\infty)}$ with rank-2 free-field presentation, the Polyakov conformal anomaly coefficient is

$$c^{(\infty)}_{\text{anomaly}} \;=\; 2,$$

extracted from the $T(z)T(w)$ OPE central term with $T = \frac{1}{2}(a^{(\infty)}_{-1}a^{(\infty)}_{-1} + b^{(\infty)}_{-1}b^{(\infty)}_{-1})$, independently from trace-anomaly $\langle T^{\mu}_{\mu}\rangle = -cR/24\pi$ and from Friedan–Shenker modular-invariance bootstrap.

Remark 107.5.2 *(archimedean vs. determinant).* $c^{(\infty)}_{\text{anomaly}} = 2$ is the anomaly of the *archimedean* Heisenberg alone. Each per-prime place also satisfies $c^{(p)} = 2$, so the per-place charges are uniformly 2. The finite-prime determinant residue also equals 2. Equality of these two numbers is a rank comparison; it is not an addition of local central charges.

The Polyakov–regularisation identity

Theorem 107.5.3 *(Polyakov-rank comparison).* The zeta rank residue of Definition 107.3.7 equals the Polyakov conformal anomaly coefficient of the archimedean rank-2 Heisenberg:

$$r_\zeta \;=\; c^{(\infty)}_{\text{anomaly}} \;=\; 2.$$

Equivalently, for the global Fock partition $Z(s) = \prod_p \prod_{n\geq 1}(1-p^{-sn})^{-2}$, the residue of $-\partial_s \log Z(s)$ at $s=1$ has the same value as the rank-2 Polyakov-Friedan-Shenker anomaly computed on the archimedean completion $\mathcal{H}^{(\infty)}$.

Proof. Combine Corollary 107.4.4 ($r_\zeta = 2$) with Theorem 107.5.1 ($c^{(\infty)}_{\text{anomaly}} = 2$): both sides equal 2 independently, and their common value is $2 \cdot \operatorname{Res}_{s=1}\zeta(s) = 2$.

The two sides arise from different constructions: the left-hand side from a global partition-function Euler product over all finite primes, regularised through the residue of ζ at its pole; the right-hand side from a Lagrangian stress-tensor computation on the archimedean fibre, regularised through Polyakov's short-distance anomaly. The identity is the arithmetic-branch expression of Polyakov's principle in the residue form (Theorem 35.4.1): the finite-prime determinant and the archimedean free-field anomaly have the same rank 2. No global Virasoro action on $\mathcal{V}^{\text{prim}}$ is produced. □

Remark 107.5.4 *(global compatibility).* Distinguish the meromorphic completion and its entire version:

$$\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \xi_{\mathbb{R}}(s) = \tfrac{1}{2}s(s-1)\Lambda(s).$$

Since $\zeta(s) = (s-1)^{-1} + \gamma_E + O(s-1)$ and $\pi^{-1/2}\Gamma(1/2) = 1$, Λ has residue $+1$ at 1. The functional equation $\Lambda(s) = \Lambda(1-s)$ gives residue -1 at 0. The prefactor cancels both poles: $\xi_{\mathbb{R}}$ is entire, with value $1/2$ at both points. It has no residue pole at zero.

For the partition function defined here, write $Z(s) = \zeta(s)^2 A(s)$, with $A(s) = \prod_{n\geq 2}\zeta(ns)^2$ holomorphic and nonzero near 1. Thus Z has a pole of order two. Its residue is $2\gamma_E A(1) + A'(1)$, whereas

$$-\partial_s \log Z(s) = \frac{2}{s-1} + O(1), \qquad \operatorname{Res}_{s=1}(-\partial_s \log Z) = 2.$$

The rank scalar is this logarithmic-derivative residue, not the residue of Z. Its equality with the local rank-two free-field central charge is the scalar equality stated in Theorem 107.5.3; a global action requires additional comparison maps.

Local Central Charge and the Global Residue

Remark 107.5.5 *(the two scalar calculations).* The local chiral calculation uses Kac–Raina *Bombay Lectures on Highest Weight Representations of Infinite-Dimensional Lie Algebras* (World Scientific 1987), §3.5 (Sugawara central charge for bosonic Fock); Kac *Vertex Algebras for Beginners* 2nd ed. (AMS 1998), Chapter 3 (rank-n Heisenberg vertex algebra and Virasoro central charge formula). The global residue calculation uses Edwards *Riemann's Zeta Function* (Academic Press 1974), §1.6 (Laurent expansion of ζ at $s = 1$) for the residue-extraction half; Serre *Cours d'Arithmétique* (PUF 1970) Ch. VI (Euler product) for the global partition rewriting. Both displayed calculations give the scalar 2. Equality of these scalars does not by itself construct a map between the local vertex-algebra carrier and the global determinant object.

107.6 MAIN THEOREM: $r_\zeta = 2$

Statement

THEOREM 107.6.1 *(primary Fock determinant has rank residue 2).* Let $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}}$ denote the primary Heisenberg core of Definition 107.2.2. Then:

(a) *Finite positive modes.* For every $v \in \mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}}$ and every $m \geq -1$, the mode action $L^{\geq -1}_m v = \sum_{p \in S_v(m)} L^{\mathrm{conf},(p)}_m v$ is a finite sum over $|S_v(m)| < \infty$ primes. The modes $m \leq -2$ do not form global operators on the restricted vacuum (Proposition 107.2.4).

(b) *Divergent conformal scalar.* The formal Virasoro central-term coefficient $\sum_p c^{(p)}$ is $+\infty$ as a naive sum (Observation 107.2.1).

(c) *Zeta rank residue.* The Dirichlet-series residue of the scalar via the global Fock partition $Z(s) = \prod_p Z^{(p)}(s)$ yields

$$r_\zeta \;=\; \mathrm{Res}_{s=1}\Big(-\frac{d}{ds}\log Z(s)\Big) \;=\; 2.$$

(d) *Polyakov-rank comparison.* The rank residue equals the Polyakov conformal anomaly coefficient of the archimedean rank-2 Heisenberg $c^{(\infty)}_{\mathrm{anomaly}} = 2$ (Theorem 107.5.3).

Proof. (a) is Proposition 107.2.4. (b) is Observation 107.2.1. (c) is Theorem 107.4.3 with Corollary 107.4.4. (d) is Theorem 107.5.3. □

Regularization, Rescaling, and Determinant Rank

Remark 107.6.2 *(prime-zeta sum does not regularise the determinant rank).* The prime sum $c(s) = 2\zeta_{\mathbb{P}}(s-1)$ is absolutely convergent for $\mathrm{Re}\, s > 2$, and diverges logarithmically as real $s \downarrow 2$, as proved in Observation 107.3.1. The continuation boundary at $\mathrm{Re}\, s = 1$ is a separate obstruction. The determinant-rank prescription of Definition 107.3.7 uses instead the logarithmic derivative of the meromorphic partition function.

Remark 107.6.3 *(*$(\log p)$*-weighting is not a Virasoro rescaling).* **Objection.** "The $(\log p)$-rescaling changes the central charge $c^{(p)} \mapsto (\log p)^2 \cdot c^{(p)}$." **Answer.** The local stress tensor is not rescaled.

The factor $\log p$ defines $N_p = (\log p) L_0^{\mathrm{conf},(p)}$, the Hamiltonian in the Mellin determinant. The local conformal charge remains $c^{(p)} = 2$, and N_p is not a Virasoro zero mode.

Remark 107.6.4 *(the universal scalar is not per-state-finite).* **Objection.** "Proposition 107.2.4 establishes a global Virasoro commutator." **Answer.** It establishes finite action only for $m \geq -1$. For $m \leq -2$ the global mode is not defined on the vacuum, and the formal central scalar is ∞. The residue in §107.3 targets the determinant scalar, not a commutator on a dense domain (Remark 107.2.5).

Remark 107.6.5 *(the residue prescription is uniquely determined).* **Objection.** "The regularisation $r_\zeta = 2$ is arbitrary: many Dirichlet-series prescriptions yield different numbers." **Answer.** The prescription (107.6) is fixed by two requirements: (i) the regularisation runs through the global Fock partition $Z(s)$ whose pole structure is uniquely determined by the Euler product of ζ^2 at $s = 1$; (ii) the regularisation must match the Polyakov anomaly on the archimedean completion (Theorem 107.5.3). Together these single out the residue prescription; the prime-zeta prescription fails (i): $\zeta_{\mathbb{P}}$ has a natural boundary at $\mathrm{Re}(w) = 0$; the sum-at-zero prescription fails (ii): a sum indexed by primes is not the same as a residue of the Riemann zeta at $s = 1$, which runs over all positive integers via the Euler product.

Remark 107.6.6 *(the archimedean fibre enters through the completed-zeta* Γ*-factor).* **Objection.** "The partition $Z(s) = \prod_p Z^{(p)}(s)$ omits the archimedean fibre $\mathcal{H}^{(\infty)}$." **Answer.** The archimedean contribution enters through the Γ-factor in the completed zeta $\xi_{\mathbb{R}}(s) = \frac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$; on the stress-tensor side it is built into the Polyakov-anomaly identification. Theorem 107.5.3 states only a rank equality between the finite-prime residue and the archimedean anomaly. It does not add two central charges.

Remark 107.6.7 *(the order of the pole of* $\log Z$ *at* $s = 1$ *is not a free parameter).* **Objection.** "The residue prescription is circular: the residue is 2 because the rank is chosen to be 2." **Answer.** The order of the pole is *not* a free parameter. It is computed from the Euler product: $Z^{(p)}(s) = \prod_{n \geq 1}(1 - p^{-sn})^{-2}$ has rank-2 because the per-prime Heisenberg is rank-2 (two free bosons); this is a *fixed input* from the construction. The resulting $Z(s) = \zeta(s)^2 \cdot \mathrm{hol}$ near $s = 1$ has a pole of order 2 in ζ, hence a leading $-2\log(s-1)$ term in $\log Z$, hence $\mathrm{Res}_{s=1}(-\frac{d}{ds}\log Z) = +2$. The input is the rank-2 Heisenberg determinant. The output is the same rank read from the pole of $\zeta(s)^2$.

Remark 107.6.8 *(rank in a specified Iwasawa coefficient model).* Let $\Lambda_p = \mathbb{Z}_p[[T]]$ and $M = \Lambda_p^{\oplus r}$. Its (p, T)-adic completion is again M, and specialization along $\Lambda_p \to \mathbb{Q}_p$, $T \mapsto 0$, gives $M \otimes_{\Lambda_p} \mathbb{Q}_p \cong \mathbb{Q}_p^r$. Thus this finite free coefficient model retains its rank.

Applying the calculation to an arithmetic Heisenberg requires an identification with that coefficient model and its current operators. The lifting criterion in Theorem 46.1.5 does not make Λ_p a Heisenberg algebra or construct that identification. A local conformal charge still requires the specified Heisenberg fields and stress tensor.

Relation to the conditional Virasoro-bracket question

Remark 107.6.9 *(what is proved and what is not).* Theorem 107.6.1 proves the determinant scalar $r_\zeta = 2$. It does *not* prove that a per-prime sum $\sum_p L_m^{\mathrm{conf},(p)}$ satisfies a Virasoro bracket with this scalar. Two obstructions remain:

(i) *Domain.* The modes $m \leq -2$ must first exist as dense operators on a completion of $\mathcal{V}_{\mathrm{Heis,prim}}^{\mathrm{prim}}$ (the Petersson completion, Theorem 43.7.1). The closure is not automatic: finite-support states do not form an invariant core unless a commutator estimate is given.

(ii) *Bracket identity.* After closure, the identity $[L_m, L_n] = (m-n)L_{m+n} + (c/12)(m^3 - m)\delta_{m+n,0} \cdot \mathrm{id}$ would still be a new theorem. The determinant scalar r_ζ does not supply it.

Both obstructions are open for the primary-Virasoro reconciliation.

Domain and logical form

Remark 107.6.10 *(determinant domain).* The determinant-rank identity is $r_\zeta = 2$. It does not prove either half of Conjecture 44.3.5. The open content remains operator closure plus bracket identity on a dense domain (Remark 107.6.9). The operator closure requires operator estimates on $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis,prim}}$ with respect to the Petersson inner product of Theorem 43.7.1.

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Chapter 108

Modular Generators, Zero Divisors, and Operator Domains

The critical Bost–Connes generator, the occupation Hamiltonian, and the idele scaling operator have different spectral measures. The arithmetic realization problem asks for a self-adjoint operator with the completed zeta divisor and the full signed Weil trace. An intertwiner preserves eigenvalues on every source eigenvector in its domain, while a unitary multiplication chart preserves continuous spectral type. These elementary restrictions determine which proposed comparisons can represent that divisor. The explicit formula and the graph-domain calculations below give separate distributional and operator statements.

108.1 Setup: N_{BC} and Θ_ξ as Target Operators

The proposed operator comparison starts with the critical BC generator and asks for a zero-divisor target. Their distinct Hilbert carriers and spectra determine its domain requirements.

Source: The Bost–Connes Log-Prime Operator

Definition 108.1.1 *(Critical generator and occupation restriction).* On the critical GNS space $\mathcal{H}_1$, use the basis of Chapter 77.2.2:

$$\mathcal{H}_1 = \ell^2(\mathbb{Q}_{>0})\widehat{\otimes}\ell^2(\mathbb{Q}/\mathbb{Z}), \quad N_{\mathrm{BC}} v_{q,r} = \tfrac{1}{2}\log q\, v_{q,r}, \quad \mathrm{Dom}\, N_{\mathrm{BC}} = \{c : \sum_{q,r} |\log q|^2 |c_{q,r}|^2 < \infty\}.$$

Its eigenvalues have infinite multiplicity and their closure is $\mathbb{R}$. Let $\mathcal{H}_{\mathrm{BC}} := K_2 = \overline{\mathrm{span}}\{v_{m^2,0} : m \geq 1\}$ when discussing occupation coordinates. This reducing subspace carries $N_{\mathrm{occ}} = N_{\mathrm{BC}}|_{K_2}$, whose simple eigenvalues are $\log m$. The full and restricted operators remain distinct.

Remark 108.1.2 *(Occupation coordinates).* Let e_m be the coordinate vector of norm $m^{-1/2}$ in $L_w = \ell^2(\mathbb{N}^\times, dm/m)$. The map $v_{m^2,0} \mapsto \sqrt{m}\, e_m$ is unitary. Prime factorization identifies the orthonormal occupation basis with the restricted tensor product of the bases indexed by $v_p(m)$, with vacuum index zero at every prime. In these coordinates N_{occ} is the closure of $\sum_p v_p(m) \log p$ on finite-support vectors. This Hilbert tensor decomposition is not a map from the full critical generator to the continuous idele scaling representation.

Target: The Hypothetical Zeta-Zero Divisor Operator

Definition 108.1.3 *(target Hilbert space and zero-divisor operator).* Let $\mathcal{Z}$ denote the multiset of nontrivial zeros $\rho = \beta_\rho + i t_\rho$ of $\xi_{\mathbb{R}}$, with each occurrence indexed once, and write $\gamma_\rho := t_\rho$ for the ordinate. The *formal target Hilbert space* is

$$\mathcal{H}_\xi^{\mathrm{formal}} := \bigoplus_{\rho \in \mathcal{Z}} \mathbb{C},$$

with inner product making the one-dimensional summands orthonormal. The *formal ordinate multiplier* is the diagonal symmetric operator

$$H_{\mathrm{ord}}^{\mathrm{formal}} \;:\; (\xi_\rho)_\rho \;\mapsto\; (t_\rho \xi_\rho)_\rho$$

on the maximal domain

$$\left\{ c \in \ell^2(\mathcal{Z}) : \sum_{\rho \in \mathcal{Z}} t_\rho^2 |c_\rho|^2 < \infty \right\}.$$

This real diagonal operator is self-adjoint; its adjoint domain is exactly the displayed square-summability domain. Coordinate truncations show that finite-support vectors form a graph core. Its point spectrum is the ordinate set, with one coordinate per zero occurrence. Distinct zeros with the same ordinate add their multiplicities. The *critical-line projection operator* is

$$\Theta_\xi^{\mathrm{proj}} = \tfrac{1}{2}\mathrm{id} + i H_{\mathrm{ord}}^{\mathrm{formal}}.$$

Its divisor is the projection of the zero divisor to the critical line; it is the actual zero divisor of $\xi_{\mathbb{R}}$ exactly under RH.

Remark 108.1.4 *(why "formal").* The triple $(\mathcal{H}_\xi^{\mathrm{formal}}, H_{\mathrm{ord}}^{\mathrm{formal}}, \Theta_\xi^{\mathrm{proj}})$ exists trivially as a diagonal construction. It carries no intrinsic arithmetic structure: the enumeration of $\{t_\rho\}$ is read from the zero set itself, and the diagonal operator realises its spectrum by fiat. If one replaces β_ρ by $1/2$, one has already projected the divisor to the critical line. The content of the Hilbert–Pólya problem is not the existence of $\mathcal{H}_\xi^{\mathrm{formal}}$; it is the existence of a Hilbert space *with independent structure* whose spectral data force the original zero divisor to equal this projected divisor.

Definition 108.1.5 *(non-formal target).* A *non-formal target* is a pair $(\mathcal{H}_\xi, H_{\mathrm{HP}})$ where $\mathcal{H}_\xi$ is a Hilbert space carrying structure inherited from the Bost–Connes framework (restricted-tensor adelic factorisation, Petersson inner product on cuspidal cores, GNS modular data, or another independently specified structure) and H_{HP} is a self-adjoint operator on $\mathcal{H}_\xi$ whose spectral resolution is canonical with respect to the inherited structure and whose regularised determinant

$$\det{}_\infty\big(s\mathrm{id} - (\tfrac{1}{2}\mathrm{id} + i H_{\mathrm{HP}})\big)$$

has the non-trivial zero divisor of $\xi_{\mathbb{R}}$.

Remark 108.1.6 *(domain of "canonical").* "Canonical" means: the spectral resolution of H_{HP} is determined by the inherited structure up to a unitary equivalence whose parameters are themselves determined by that structure. A diagonal construction with eigenvalues labelled by an arbitrary enumeration is not canonical; a diagonal construction whose eigenvectors are explicit functions of arithmetic data (adelic characters, automorphic forms, spectral projectors of a globally defined operator) is.

Spectral Flow as Intertwining Operator

Definition 108.1.7 *(An operator comparison).* Choose explicitly either $N = N_{\mathrm{BC}}$ on $\mathcal{H}_1$ or $N = N_{\mathrm{occ}}$ on K_2. An intertwining datum consists of a self-adjoint H on a complex Hilbert space $\mathcal{L}$, a dense N-invariant subspace $\mathcal{D} \subset \mathrm{Dom}\, N$, and an injective linear map $A : \mathcal{D} \to \mathrm{Dom}\, H$ with dense range, satisfying

$$HAu = ANu \qquad (u \in \mathcal{D}).$$

A zero-divisor realization additionally requires a complete orthonormal eigenbasis of H whose eigenvalue multiset obeys

$$\{\tfrac{1}{2} + i\lambda : \lambda \text{ an eigenvalue of } H\} = \{\rho : \xi_{\mathbb{R}}(\rho) = 0\},$$

with multiplicities, and a separately normalized determinant. The existence of such a target is not an assertion that the intertwining datum exists for either source.

Proposition 108.1.8 *(Eigenvectors under an intertwiner).* If $Nv = \lambda v$, $v \in \mathcal{D}$, and A is an intertwining datum, then $Av \neq 0$ and $HAv = \lambda Av$. No such datum containing a nonzero zero-energy source vector can have the zero-divisor target above. In particular, an injective intertwiner on the eigenvector core of either source is impossible.

Proof. Injectivity gives $Av \neq 0$, and the intertwining identity gives the eigenvalue equation. Both source cores contain $v_{1,0}$ of energy zero, whereas $\xi_{\mathbb{R}}(1/2) \neq 0$. Indeed, the alternating series $\eta(1/2) = \sum_{n\geq 1}(-1)^{n-1}n^{-1/2}$ converges and is positive: grouping consecutive pairs gives positive terms and the first pair is positive. The alternating-series continuation $\eta(s) = (1 - 2^{1-s})\zeta(s)$ on $\Re s > 0$ then gives $\zeta(1/2) = \eta(1/2)/(1 - \sqrt{2}) \neq 0$. The other completed factors at $1/2$ are finite and nonzero. Unboundedness of A does not alter this eigenvector argument. □

108.2 Prime-Zero Mangoldt-Hadamard Duality Recap

The Riemann–von Mangoldt explicit formula and the Hadamard product relate prime data to the zero divisor. In the Fourier formulation below, the prime terms act on compact smooth time tests and the zero functional acts on their entire Fourier transforms. This identity does not by itself identify Hilbert-space operators.

Hadamard Factorisation of $\xi_{\mathbb{R}}$

Proposition 108.2.1 *(Hadamard factorisation). (Hadamard 1893; Chapter 77.4).* The completed Riemann zeta $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire of order 1, of genus 1, and admits the factorisation

$$\xi_{\mathbb{R}}(s) \;=\; \xi_{\mathbb{R}}(0)\, e^{Bs} \prod_{\rho}\Big(1 - \frac{s}{\rho}\Big)e^{s/\rho}, \tag{108.1}$$

with ρ ranging over non-trivial zeros with multiplicity and

$$B = \frac{\xi_{\mathbb{R}}'(0)}{\xi_{\mathbb{R}}(0)} = -1 - \frac{\gamma}{2} + \frac{1}{2}\log(4\pi),$$

where γ is Euler's constant and $\xi_{\mathbb{R}}(0) = 1/2$. See Edwards 1974, §2.7, and the normalization in Chapter 77.5.2.

Proof. The function here is exactly the completed function of Chapter 77.5.2. Its identity $\xi_{\mathbb{R}}(s) = (s-1)\Gamma(1 + s/2)\pi^{-s/2}\zeta(s)$ gives

$$\frac{\xi_{\mathbb{R}}'(0)}{\xi_{\mathbb{R}}(0)} = -1 - \frac{\gamma}{2} - \frac{1}{2}\log\pi + \log(2\pi),$$

using $\Gamma'(1) = -\gamma$, $\zeta(0) = -1/2$, and $\zeta'(0) = -\frac{1}{2}\log(2\pi)$, as evaluated there. Each genus-one factor has logarithmic derivative zero at the origin, so this logarithmic derivative is the constant B in (108.1). □

Riemann-von Mangoldt Explicit Formula

PROPOSITION 108.2.2 *(Riemann-von Mangoldt explicit formula). (von Mangoldt 1895; Edwards 1974 §3).* For $x > 1$ non-integer,

$$\sum_{n \le x} \Lambda(n) \;=\; x \;-\; \sum_{\rho} \frac{x^{\rho}}{\rho} \;-\; \frac{\zeta'(0)}{\zeta(0)} \;-\; \tfrac{1}{2}\log(1 - x^{-2}),$$

where $\Lambda(n) = \log p$ if $n = p^k$ and 0 otherwise. The zero sum means $\lim_{T\to\infty} \sum_{|\operatorname{Im}\rho| \le T} x^{\rho}/\rho$, with multiplicities and symmetric height truncation.

Mellin-Dual Measures

Definition 108.2.3 *(prime and zero counting measures).* The *prime-counting distribution* is

$$\mu_{\mathrm{prime}} \;:=\; \sum_{m \in \mathbb{N}^{\times}} \Lambda(m)\, \delta_{\log m} \;=\; \sum_{p \in \mathbb{P}} \sum_{k \ge 1} (\log p)\, \delta_{k \log p}$$

on $\mathbb{R}_{>0}$. The *ordinate shadow* of the zero-counting distribution is

$$\mu_{\mathrm{ord}} \;:=\; \sum_{\rho \in \mathcal{Z}} \delta_{\gamma_\rho}$$

on $\mathbb{R}$. The actual Weil zero functional acts on $\mathscr{P} = \mathcal{F}_{+} C_c^{\infty}(\mathbb{R})$, with $(\mathcal{F}_{+} f)(z) = \int f(u) e^{izu}\, du$:

$$\mathfrak{Z}_{\xi}(h) := \sum_{\rho \in \mathcal{Z}} h(z_\rho), \qquad z_\rho := -i(\rho - \tfrac{1}{2}).$$

The equality $\mathfrak{Z}_{\xi}(h) = \int_{\mathbb{R}} h\, d\mu_{\mathrm{ord}}$ for every $h \in \mathscr{P}$ is equivalent to RH. Indeed, equality on Fourier images of convolution squares makes the Weil form nonnegative; apply the Weil criterion. Conversely, RH gives $z_\rho = \gamma_\rho$ term by term.

PROPOSITION 108.2.4 *(Mellin-dual form of the explicit formula). (Weil 1952; Patterson 1988 Ch. 3).* Let $\mathscr{A} = C_c^{\infty}(\mathbb{R}; \mathbb{C})$, with its usual inductive-limit topology, and let $\mathscr{P} = \mathcal{F}_{+}\mathscr{A}$ have the transported topology. For $\varphi \in \mathscr{A}$, set

$$h(z) = \mathcal{F}_{+}\varphi(z) = \int_{\mathbb{R}} \varphi(t) e^{izt}\, dt, \qquad \varphi(t) = \frac{1}{2\pi} \int_{\mathbb{R}} h(u) e^{-iut}\, du.$$

Write $\psi_{\Gamma} = \Gamma'/\Gamma$ and define

$$k(u) = \log \pi - \mathfrak{R}\psi_{\Gamma}(\tfrac{1}{4} + iu/2),$$

$$P(\varphi) = h(i/2) + h(-i/2), \qquad W_{\infty}(\varphi) = \frac{1}{2\pi} \int_{\mathbb{R}} h(u) k(u)\, du,$$

$$W_{\mathrm{fin}}(\varphi) = \sum_{m \ge 2} \frac{\Lambda(m)}{\sqrt{m}} \big(\varphi(\log m) + \varphi(-\log m)\big).$$

Then, with zeros counted with multiplicity,

$$\mathfrak{Z}_{\xi}(h) = \sum_{\rho \in \mathcal{Z}} h(z_\rho) = P(\varphi) - W_{\infty}(\varphi) - W_{\mathrm{fin}}(\varphi). \tag{108.2}$$

The prime sum is finite and the zero sum converges absolutely. For even φ, the finite term is $2\mu_{1/2}(\varphi)$, where $\mu_{1/2} = \sum_{m \ge 2} \Lambda(m) m^{-1/2} \delta_{\log m}$. Under RH, $z_\rho = \gamma_\rho$ and $\mathfrak{Z}_{\xi}(h) = \int_{\mathbb{R}} h\, d\mu_{\mathrm{ord}}$. Without RH, replacing z_ρ by γ_ρ discards the real-part information and does not give the explicit formula.

Proof. The Fourier convention and test carrier are those of Chapter 78.1.1. Apply Chapter 77.5.6 to $f = \varphi$. Its polar terms are $h(\pm i/2)$; its real-line density is $\Re\psi_\Gamma(1/4 + iu/2) - \log\pi = -k(u)$. This gives (108.2), with the sign convention of Chapter 78.1.2. Compact support makes the prime sum finite. Repeated integration by parts gives rapid decay of h in the strip $|\Im z| \le 1/2$; the zero count in Chapter 77.5.1 gives absolute convergence of the zero sum. Evenness identifies the two prime evaluations. Finally, $z_\rho = \gamma_\rho - i(\Re\rho - 1/2)$, proving the stated specialization under RH. □

Remark 108.2.5 *(the distributional identity and Hilbert realizations).* The Mellin–Hadamard identity in Chapter 108.2.4 relates the continuous zero functional on $\mathscr{P}$ to the full polar, signed gamma and prime functionals on $\mathscr{A}$. The locally finite measures $\mu_{\text{prime}} = \mu_0$ and $\mu_{1/2}$ define distributions on compact smooth tests. Neither is tempered, by Chapter 77.5.4; their Fourier transforms cannot be taken in $\mathscr{S}'(\mathbb{R})$ by this construction.

There is nevertheless a signed Hilbert-space realization: Chapter 78.9.3 constructs the continuous map $E : \mathscr{A} \to K_Z = \ell^2(Z)$ and the self-adjoint involution J, with $(Ef)_\rho = (\mathcal{F}_+ f)(z_\rho)$, such that

$$\langle Ef, JEg\rangle = P(f * g^*) - W_\infty(f * g^*) - W_{\text{fin}}(f * g^*), \qquad g^*(t) = \overline{g(-t)}.$$

Here Z indexes occurrences of zeros, including multiplicity. This signed pairing does not supply a positive arithmetic-sector unitary comparison. Realizing the prime and zero expressions as spectral data of a single self-adjoint operator requires that additional comparison, including the operator domains and the spectral measures; the distributional identity alone supplies none of these identifications.

108.3 Formal Spectral Flow A as Mellin-Inversion Symbol

Mellin transformation followed by residue extraction defines a sequence-valued map on compact smooth multiplicative tests. Square summability and compatibility with a source operator require separate proofs.

The Mellin-Hadamard Inversion Symbol

Definition 108.3.1 *(formal Mellin-Hadamard inversion).* The *formal Mellin-Hadamard inversion* is the composite symbol

$$A^{\text{sym}} := \mathcal{I}_{\text{Had}} \circ \mathcal{M}$$

where $\mathcal{M}$ is the Mellin transform $(\mathcal{M}f)(s) = \int_0^\infty f(x)x^{s-1}dx$ on a suitable function class and $\mathcal{I}_{\text{Had}}$ is the Hadamard-inversion formal symbol

$$(\mathcal{I}_{\text{Had}}F)(\gamma_\rho) := \operatorname*{Res}_{s=\rho} \frac{F(s)}{\xi_{\mathbb{R}}(s)}$$

reading the residue of $F/\xi_{\mathbb{R}}$ at each non-trivial zero and writing it into the ρ-component of a formal sequence indexed by $\mathcal{Z}$.

Remark 108.3.2 *(Domains for a residue map).* The Mellin-residue prescription gives coordinates indexed by zeros. To obtain a Hilbert-space operator, one must specify a source Hilbert space, a dense domain on which the residue sequence is square summable, and a proof of closability. The occupation space is not a subspace of multiplicative Haar L^2 through the atomic prescription. A comparison with its generator requires an additional map and a domain intertwining identity.

Formal Mellin–Hadamard Obstruction

LEMMA 108.3.3 *(formal Mellin–Hadamard obstruction).* Suppose $f \in C_c^\infty(\mathbb{R}_{>0})$ and let $Lf(x) = -\log x\, f(x)$. Suppose f has Mellin transform $F(s) = (\mathcal{M}f)(s)$ entire, and suppose $\rho \in \mathcal{Z}$ is a simple zero of $\xi_{\mathbb{R}}$. Then

$$A^{\mathrm{sym}}(Lf)(\gamma_\rho) = -\frac{F'(\rho)}{\xi_{\mathbb{R}}'(\rho)}, \qquad A^{\mathrm{sym}}(f)(\gamma_\rho) = \frac{F(\rho)}{\xi_{\mathbb{R}}'(\rho)}.$$

Consequently the bare residue intertwining relation

$$A^{\mathrm{sym}} L \stackrel{?}{=} M_\gamma A^{\mathrm{sym}}$$

holds at ρ only under the additional interpolation condition

$$F'(\rho) = -\gamma_\rho F(\rho).$$

This interpolation condition concerns the test function and is not a consequence of RH. No map identifying L with either BC generator has been specified.

Proof. Define $(Lf)(x) = -\log x\, f(x)$ on $C_c^\infty(\mathbb{R}_{>0})$. Mellin transform gives $(\mathcal{M}(Lf))(s) = -F'(s)$, where F' is the derivative in the spectral variable. Since ρ is a simple zero,

$$\operatorname*{Res}_{s=\rho} \frac{F(s)}{\xi_{\mathbb{R}}(s)} = \frac{F(\rho)}{\xi_{\mathbb{R}}'(\rho)}, \qquad \operatorname*{Res}_{s=\rho} \frac{-F'(s)}{\xi_{\mathbb{R}}(s)} = -\frac{F'(\rho)}{\xi_{\mathbb{R}}'(\rho)}.$$

Multiplication by the zero ordinate would instead give $\gamma_\rho F(\rho)/\xi_{\mathbb{R}}'(\rho)$. Equality is exactly the displayed first-order interpolation condition. This is not an identity on compact tests: take $f(x) = x^{1-\rho}\psi(x)$, where ψ is nonnegative, nonzero, and compactly supported in an interval on which $\log x + \gamma_\rho$ has one strict sign. Then $F'(\rho) + \gamma_\rho F(\rho) = \int (\log x + \gamma_\rho)\psi(x)\,dx \neq 0$. □

Remark 108.3.4 *(RH is not enough for formal intertwining).* Chapter 108.3.3 shows that the continuous multiplier $L = -\log x$ differentiates the Mellin numerator before Hadamard residue extraction. RH would place ρ on the critical line, but it would not imply $F'(\rho) = -\gamma_\rho F(\rho)$ for the Mellin transforms in the source class. Thus the bare symbol A^{sym} is not a spectral-flow operator. A genuine operator construction must supply a new domain theorem or boundary condition that converts derivative residue data into zero-ordinate multiplication without inserting that conclusion by definition.

108.4 OBSTACLE I: HILBERT-SPACE COMPLETIONS ON BOTH SIDES

A sequence-valued residue prescription needs a source completion and estimates in the chosen target norm. Abstract Hilbert isomorphisms do not settle these domain or spectral requirements.

Source-Side Completion

Definition 108.4.1 *(Occupation source completion).* Use $L_w = \ell^2(\mathbb{N}^\times, dm/m)$ for occupation coordinates, with $N_{\mathrm{occ}} e_m = (\log m) e_m$. Its maximal domain is $\{c : \sum_m (\log m)^2 |c_m|^2/m < \infty\}$, and finite-support vectors form a core. This is the restriction in Chapter 77.3.2.

LEMMA 108.4.2 *(The atomic Mellin prescription).* Sending e_m to the additive Dirac distribution $\delta_{x=m}$ does not embed L_w in $L^2(\mathbb{R}_{>0}, dx/x)$. Its ordinary distributional Mellin transform is m^{s-1}, which is not square integrable on any vertical line.

Proof. A point mass is singular relative to dx/x, so it has no L^2 density. Pairing the additive Dirac distribution with x^{s-1} gives m^{s-1}. On $\Re s = c$ its modulus is the positive constant m^{c-1}, proving the second assertion. □

Remark 108.4.3 *(Atomic and continuous carriers).* The atomic prescription fails on each basis vector before any infinite Dirichlet-series convergence question arises. Abstract Hilbert-space isometries still exist between the separable infinite-dimensional spaces. An isometry preserving their stated generators cannot exist, because the occupation generator has an eigenvector basis and continuous scaling has no eigenvectors.

Target-Side Completion

Definition 108.4.4 *(Distinct target constructions).* The idele carrier is $\mathcal{K} = L^2(C_{\mathbb{Q}}, d^\times x)$ of Chapter 101.2.1. The weighted adele-class Hilbert completion in Chapter 101.7.1 and its quotient representation are different spaces. A Petersson completion, when supplied by an automorphic construction, is a third carrier. Neither these names nor abstract Hilbert isomorphisms specify a generator-domain map between any two of them.

Lemma 108.4.5 *(Mellin transform on the idele norm).* The transform

$$(\mathcal{M}g)(\tau) = (2\pi)^{-1/2} \int_0^\infty g(x) x^{-i\tau}\, dx/x$$

is unitary from $L^2(\mathbb{R}_{>0}, dx/x)$ to $L^2(\mathbb{R}, d\tau)$. Tensoring with the identity on $L^2(C_{\mathbb{Q}}^1)$ gives the idele transform. It carries $-i\partial_{\log x}$, with logarithmic H^1 domain, to M_τ, with its maximal domain.

Proof. Substitute $u = \log x$ and apply Fourier Plancherel and Chapter 101.2.2. □

Remark 108.4.6 *(Representations and completions).* The Mellin transform diagonalizes the continuous scaling representation, not the occupation or critical BC representation. An injective map on their eigenvector cores cannot intertwine these generators. A different completion or a distributional map must state which operators, domains, and pairings it preserves.

108.5 Obstacle 2: Kernel of A as Bounded/Unbounded Operator

A residue prescription must define an operator on a stated domain. An integral kernel is one possible construction; closability requires an independent estimate.

The Hadamard Inversion Kernel

Definition 108.5.1 *(Residue integrals and a line-kernel candidate).* For a distinct zero ρ, take a positively oriented small circle C_ρ containing no other zero. The actual residue is

$$r_\rho(F) = \frac{1}{2\pi i} \int_{C_\rho} \frac{F(s)}{\xi_{\mathbb{R}}(s)} ds.$$

A different proposed line kernel is $K_{\mathrm{Had}}(\rho, s) = (2\pi i)^{-1}[\xi_{\mathbb{R}}(s)(s-\rho)]^{-1}$. It does not give the preceding residue by the same circle integral: the additional factor changes the Laurent coefficient extracted. Its integrability on the critical line can nevertheless be tested.

LEMMA 108.5.2 *(the Hadamard kernel is not a Hilbert-Schmidt kernel).* For the critical-line Mellin completion compatible with the spectral-flow condition, the Hadamard kernel K_{Had} is not Hilbert–Schmidt. Its Hilbert-Schmidt norm

$$\|K_{\mathrm{Had}}\|_{\mathrm{HS}}^2 = \frac{1}{4\pi^2} \sum_{\rho} \int_{\mathbb{R}} \frac{d\tau}{|\xi_{\mathbb{R}}(1/2 + i\tau)|^2 |1/2 + i\tau - \rho|^2}$$

diverges at every zero ρ lying on the critical line.

Proof. Let $m_\rho \geq 1$ be the order of vanishing at ρ. If ρ lies on the critical line, then $|\xi_{\mathbb{R}}(s)|^{-2} \sim C_\rho |s - \rho|^{-2m_\rho}$ along the contour near $s = \rho$. The integrand is then $|s - \rho|^{-2m_\rho - 2}$, non-integrable on a one-dimensional contour for $m_\rho \geq 1$. Hardy's theorem gives infinitely many zeros on the critical line. Hence the critical-line Hadamard residue kernel is not Hilbert–Schmidt. Off-line zeros do not give this local contour singularity; they are not needed for the conclusion. □

Remark 108.5.3 *(Limits of the Hilbert–Schmidt obstruction).* Chapter 108.5.2 rules out a Hilbert–Schmidt Hadamard-residue kernel in the critical-line completion. A bounded non-Hilbert–Schmidt operator is not thereby excluded. The unbounded spectral-flow problem is the stronger demand imposed by the pole residue prescription together with the zero-divisor determinant and trace identities; it requires a closed densely defined operator and a specified domain.

The Domain Problem

PROPOSITION 108.5.4 *(Dirichlet-series and residue domains).* Finite-support sequences and sequences satisfying $|c_m| = O_N(m^{-N})$ for every N are dense subspaces of L_w. For the latter sequences, $F_c(s) = \sum_m c_m m^{s-1}$ is entire, with derivatives converging normally on compact sets. This does not imply vertical rapid decrease. The sequence-valued residue map on this class lands in ℓ^2 exactly on the subspace

$$\mathcal{D}_R = \{c : \sum_{\rho} |r_\rho(F_c)|^2 < \infty\},$$

where one coordinate is used per distinct zero. Density and closability of this restricted map require further estimates.

Proof. Finite sequences lie in both source classes and are dense in L_w by coordinate truncation. On a compact set with $\Re s \leq B$, each derivative is bounded termwise by $|c_m| m^{B-1} (\log m)^k$; choosing $N > B + k + 2$ proves normal convergence. The sequence $c_1 = 1, c_m = 0$ for $m > 1$ gives $F_c = 1$, refuting vertical rapid decrease. The final criterion is the definition of the ℓ^2 norm. At multiple zeros, residue extraction uses a finite Taylor jet; it does not itself supply one independent coordinate per zero multiplicity. □

Remark 108.5.5 *(Additional domain requirements).* A different Mellin growth space, including a Beurling-type space, requires an explicitly defined norm and comparison map before it can replace $\mathcal{D}_R$. Coefficient decay alone supplies neither the residue estimates nor a generator intertwiner. For any closed densely defined operator A, A^*A is self-adjoint and the polar decomposition exists. These general facts do not imply that its partial isometry preserves the source and target generators or has the required arithmetic range.

108.6 Adele-Class and Virtual Trace Constructions

Definition 108.6.1 *(Weighted adele-class comparison).* For $\delta > 1$, use the map $E\phi(x) = |x|^{1/2} \sum_{q \neq 0} \phi(qx)$ and Hilbert seminorm of Chapter 101.7.1. Its completion embeds isometrically in $L^2(C_{\mathbb{Q}}, (1 + \log^2 |x|)^{\delta/2} d^\times x)$. The cokernel of this closed image carries the quotient of the left regular idele-class representation. The norm-direction action is polynomially bounded; this weighted representation is not asserted to be unitary.

Proposition 108.6.2 *(Critical-zero trace).* Connes's §III, Theorem 1 and Corollary 2, pp. 13–14, identify the quotient spectrum and integrated trace with the critical zeros, with the δ-dependent multiplicities stated in Chapter 101.7.2. This is a statement about that quotient representation. It does not identify the full critical BC generator with its scaling generator.

Global trace comparison

Definition 108.6.3 *(A full arithmetic trace identity).* For a specified trace construction, let $\mathcal{T} : \mathscr{A} \to \mathbb{C}$ be its distribution on compact smooth logarithmic tests. The full comparison requirement is

$$\mathcal{T}(f) = P(f) - W_\infty(f) - W_{\mathrm{fin}}(f) \qquad (f \in \mathscr{A}).$$

If a cutoff or regularized trace defines $\mathcal{T}$, its subtraction, convergence, and normalization are part of the construction. The positivity requirement is separately $\mathcal{T}(f * f^*) \geq 0$ for every $f \in \mathscr{A}$.

Remark 108.6.4 *(The source map).* The identity map is not a map between the BC GNS space and Connes's quotient Hilbert space. To compare their dynamics one must specify a map and its domain. The eigenvector obstruction excludes an injective exact intertwiner on the BC eigenvector core with a zero-divisor target. Connes's global cutoff comparison in §VIII is distinct from the critical-zero quotient theorem.

Proposition 108.6.5 *(Trace comparison and positivity).* Under the full trace identity above, positivity of $\mathcal{T}$ on convolution squares is equivalent to Weil positivity, and hence to RH. This implication does not construct an intertwiner.

Proof. Apply the identity to $f * f^*$ and use Chapter 78.2.1. □

Remark 108.6.6 *(Weighted and unweighted spaces).* Connes's completion uses a Schwartz summation seminorm and a cokernel. It is not quotient Haar L^2 on a non-Hausdorff space. A comparison with the unweighted idele space must account for this weight, quotient, and the action on the quotient.

108.7 Meyer's Virtual Character

Definition 108.7.1 *(Virtual representations).* A virtual representation $\pi_+ \ominus \pi_-$ has character $\mathrm{Tr}\, \pi_+(a) - \mathrm{Tr}\, \pi_-(a)$ on test functions for which both integrated operators have a trace. In Meyer's construction the carriers are smooth locally convex representations with the summability structure of his §5. They are not defined as a single self-adjoint perturbation pair on $L^2(C_{\mathbb{Q}})$.

Theorem 108.7.2 *(The virtual Weil character).* Meyer's Theorems 5.8 and 5.11 identify the character of his smooth summable virtual representation with the raw Weil distribution. Its spectral multiplicities are the orders of poles minus the orders of zeros of the relevant L-functions.

The source is Meyer, pp. 43 and 47. His Proposition 4.4, p. 26, instead concerns Schwartz restriction and Fourier maps.

Positive and signed traces

Remark 108.7.3 *(A character is not an operator intertwiner).* The virtual character theorem supplies no unitary $\mathcal{H}_1 \to L^2(\mathbb{R})$ intertwining the BC generator with a zero-divisor operator. Its signed trace also supplies no positivity theorem for the difference. The central twist and sign converting its raw character to W must be specified before using the normalized Weil criterion.

PROPOSITION 108.7.4 *(A positive trace comparison).* Let H be self-adjoint. Suppose that, for every $f \in \mathscr{A}$, $|h_f|^2(H)$ is trace class and

$$\operatorname{Tr}(|h_f|^2(H)) = W(f * f^*).$$

Then Weil positivity holds. The hypothesis includes the full arithmetic trace identity; it is not supplied by a virtual character formula.

Proof. The spectral theorem makes $|h_f|^2(H)$ positive. A positive trace-class operator has nonnegative trace. □

Remark 108.7.5 *(Trace ideals and summability).* Smooth summability on locally convex representation spaces and trace-class differences on a Hilbert space require different operator estimates. Neither condition changes point spectrum into continuous spectrum under an exact intertwiner.

108.8 OBSTRUCTIONS TO THE PROPOSED OPERATOR COMPARISONS

THEOREM 108.8.1 *(Specified comparison obstructions).* An injective intertwiner on the full BC or occupation eigenvector core cannot produce the zero-divisor target. The common tensor construction $M_\tau \otimes 1 + 1 \otimes N_{\mathrm{BC}}$ has no full ordinary wave operators relative to $M_\tau \otimes 1$. The literal entire-Mellin class $\mathrm{PW}^{\mathrm{TM}}$ is zero. These statements restrict the displayed constructions, not all possible arithmetic realizations.

Proof. The first assertion is the eigenvector proposition. The other two are proved in Chapter 101.3.2 and Chapter 101.6.2. □

Remark 108.8.2 *(The complete Weil form).* The finite-prime functional is explicit, but its explicit calculation is not an absorption into a positive trace. The condition remains $P(f * f^*) - W_\infty(f * f^*) - W_{\mathrm{fin}}(f * f^*) \geq 0$ for every compact smooth complex test. A local functional equation or an idele scaling phase does not imply this inequality.

Maps and domains

PROPOSITION 108.8.3 *(Limits of the obstructions).* The eigenvector obstruction applies whenever the deciding zero-energy vector is in the domain of an injective intertwiner. It does not exclude a densely defined map whose domain omits that vector, a noninjective map, or a comparison preserving a specified distribution rather than the generator. Each alternative requires its own domain and trace theorem.

Proof. The obstruction uses $HAv = ANv = 0$ and $Av \neq 0$. Removing either hypothesis removes that inference. No existence assertion follows from removing it. □

Remark 108.8.4 *(The arithmetic realization question).* A new construction must state its carrier and preserved operation. The distributional explicit formula remains valid independently of these failed generator transports. A general impossibility claim for other arithmetic trace constructions does not follow.

108.9 Comparison with Cohomological and Scattering Models

Proposition 108.9.1 *(Transport of a specified cohomological operator).* Suppose a cohomological realization supplies a Hilbert space H_D, a self-adjoint D, and a unitary $U : H_D \to H'$. Then UDU^{-1}, on U Dom D, has the same spectral multiplicities and trace functional as D. Any spectral determinant construction invariant under unitary conjugation has the same normalization after this transport.

Proof. Functional calculus gives $h(UDU^{-1}) = Uh(D)U^{-1}$. It identifies spectral projections and preserves traces of trace-class operators. Apply the assumed determinant prescription to the identical spectral data. □

Remark 108.9.2 *(Cohomological comparisons).* A Deninger-type cohomological construction must supply the space, operator, and comparison maps in addition to its determinant identity. The proposition does not construct those data or identify its generator with the full BC generator.

Scattering and signed traces

Proposition 108.9.3 *(Scattering does not force Weil positivity).* Complete unitary scattering does not imply positivity of a trace difference on Fourier squares. An ordinary trace-class spectral shift giving a positive trace difference on every such square must vanish.

Proof. The pair $M_\lambda \oplus 0$, $M_\lambda \oplus 1$ has scattering operator one, but trace difference $h(1) - h(0)$. For $f(u) = \phi(u) + \phi(u - \pi)$, $\int \phi = 1$, and $h = |h_f|^2$, this is -4. The general vanishing statement is Chapter 101.5.1. □

Remark 108.9.4 *(The arithmetic test identity).* Since $\mathrm{PW}^{\mathrm{TM}} = \{0\}$, it gives no nonzero class from which Weil positivity can be extended. The proposed arithmetic trace must be constructed on all of $\mathscr{A}$, retaining the polar, archimedean, and finite-prime terms. Its operators, domains, and regularization are additional data, as in Chapter 101.8.3.

Bulk observables and parameter derivatives

Proposition 108.9.5 *(A scalar partition function does not fix its generator).* Equality of scalar partition functions does not by itself identify the operator derivative in a family on a fixed Hilbert space. Even finite-dimensional unitarily conjugate families can have identical partition functions and different derivatives.

Proof. Let $D = \mathrm{diag}(0, 1)$, $K = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and $D(s) = e^{isK} D e^{-isK}$. For every real s and $\beta > 0$, $\mathrm{Tr}\, e^{-\beta D(s)} = 1 + e^{-\beta}$, whereas $D'(0) = i[K, D] \neq 0$. The constant family D has the same traces and derivative zero. □

Remark 108.9.6 *(The required bulk comparison).* An arithmetic Chern–Simons bulk construction must identify its state space, observable, and domain before comparing a parameter derivative with an ordinate operator. A partition identity alone supplies none of these maps. The Hochschild trace class $\mathcal{V}^{\mathrm{prim}}$ is not, by itself, a vertex algebra or a Hilbert-space generator.

Positive spectral realization

THEOREM 108.9.7 *(Positive trace implies the Weil inequality).* A self-adjoint operator with the trace identity of Chapter 108.7.4 gives Weil positivity and therefore RH. In an ordinary trace-class difference model, the same positive nonzero Weil identity is impossible.

Proof. The positive trace proposition and the Weil criterion prove the first assertion. The second is Chapter 101.8.1. □

Remark 108.9.8 *(Separate realization requirements).* A virtual character, a scattering determinant, a positive trace, and a normalized spectral determinant are different data. A comparison must specify its maps, domains, and test identities. For a regularized trace, positivity must be proved separately. The Dirac double below concerns closure of a specified operator; it does not construct those arithmetic comparisons.

108.10 Polyakov–Dirac Normal Form for the Residual

A scalar normalization does not determine an operator domain or a spectral measure. The Dirac double tests essential self-adjointness of an independently specified symmetric operator. Its determinant and arithmetic trace remain separate conditions.

Polyakov Rigidity of the Rank Residue

PROPOSITION 108.10.1 *(Polyakov rank-residue rigidity).* In the primary Heisenberg domain of Chapters 35 and 107, any spectral-flow closure compatible with

(i) the rank-2 per-prime Sugawara Heisenberg stress tensor,
(ii) the global Fock partition $Z(s) = \prod_p \prod_{n\geq 1}(1 - p^{-sn})^{-2}$, and
(iii) the archimedean Polyakov anomaly of the rank-2 free-field completion,

has determinant rank residue

$$r_\zeta = 2.$$

This scalar identity supplies no Hilbert carrier, generator-domain map, or zero-divisor determinant.

Proof. Chapter 107.6.1 proves that the unrenormalised conformal scalar $\sum_p c^{(p)}$ diverges, that the negative global modes do not act on the restricted vacuum, and that the global Fock-partition residue gives $r_\zeta = 2$. The same chapter proves the Polyakov rank comparison Chapter 107.5.3, identifying this finite-prime residue with the rank-2 archimedean conformal anomaly. These inputs fix the rank residue. They do not fix an entire exponential factor in a spectral determinant: multiplication by e^{a+bs} changes its logarithmic derivative by b, whose residue is zero. The normalization of a spectral determinant remains separate from this rank comparison. □

Remark 108.10.2 *(what Polyakov does and does not buy).* Polyakov regularisation is decisive for the determinant rank, not for RH. It supplies the finite scalar $r_\zeta = 2$ and rules out the divergent prime-zeta prescription; it does not produce the Hilbert space on which the zero ordinates are a self-adjoint spectrum.

Dirac Closure of the Remaining Operator

Definition 108.10.3 *(A specified symmetric core).* Choose a complex Hilbert space $\mathcal{H}_{\mathrm{res}}$, a dense linear subspace C_{res}, and a symmetric operator $H_{\mathrm{res}}^{\mathrm{sym}}$ with exactly that domain. Inner products are linear in the first variable. These are independently specified operator data. A multiplication chart or a residue map is additional structure, with its own domain requirements. No comparison with BC or with a zero-divisor operator is part of this definition.

Definition 108.10.4 *(Dirac double of the residual).* The *Dirac double* of the residual ordinate core is the odd symmetric operator

$$\mathscr{D}_{\mathrm{res}}^{\circ} := \begin{pmatrix} 0 & H_{\mathrm{res}}^{\mathrm{sym}} \\ H_{\mathrm{res}}^{\mathrm{sym}} & 0 \end{pmatrix} \quad \text{on} \quad C_{\mathrm{res}} \oplus C_{\mathrm{res}},$$

with grading $\Gamma = \mathrm{diag}(1, -1)$. A *Dirac closure* is a self-adjoint closure $\mathscr{D}_{\mathrm{res}} = \overline{\mathscr{D}_{\mathrm{res}}^{\circ}}$ whose spectral measure satisfies the regularised determinant identity

$$\det_{\infty}\big(s \cdot \mathrm{id} - (\tfrac{1}{2}\mathrm{id} + i H_{\mathrm{res}})\big) \;=\; \xi_{\mathbb{R}}(s), \tag{108.3}$$

where H_{res} is the off-diagonal component recovered from $\mathscr{D}_{\mathrm{res}}$ by the grading.

Lemma 108.10.5 *(Dirac double detects self-adjointness).* Let H be a densely defined symmetric operator on a Hilbert space $\mathcal{H}$ and $\mathscr{D}_H^{\circ} = \left(\begin{smallmatrix} 0 & H \\ H & 0 \end{smallmatrix}\right)$ on $\mathcal{D}(H) \oplus \mathcal{D}(H)$. Then $\mathscr{D}_H^{\circ}$ is essentially self-adjoint if and only if H is essentially self-adjoint. When this holds,

$$\overline{\mathscr{D}_H^{\circ}} \;=\; \begin{pmatrix} 0 & \overline{H} \\ \overline{H} & 0 \end{pmatrix}, \qquad \overline{\mathscr{D}_H^{\circ}}^{\,2} \;=\; \begin{pmatrix} \overline{H}^{\,2} & 0 \\ 0 & \overline{H}^{\,2} \end{pmatrix}.$$

Proof. For a symmetric H, $(\mathscr{D}_H^{\circ})^* = \left(\begin{smallmatrix} 0 & H^* \\ H^* & 0 \end{smallmatrix}\right)$ on $\mathcal{D}(H^*) \oplus \mathcal{D}(H^*)$. Hence the closure of $\mathscr{D}_H^{\circ}$ is self-adjoint exactly when the closure of H has adjoint domain equal to its own domain, i.e. exactly when H is essentially self-adjoint. The displayed square is the usual block-matrix product on $\mathrm{Dom}(\overline{H}^{\,2}) \oplus \mathrm{Dom}(\overline{H}^{\,2})$, justified after closure by the spectral theorem for the self-adjoint operator $\overline{H}$ (Reed–Simon I, Ch. VIII). □

Theorem 108.10.6 *(Dirac double of a specified operator).* For the specified $H_{\mathrm{res}}^{\mathrm{sym}}$, essential self-adjointness is equivalent to essential self-adjointness of its Dirac double. If these conditions hold, requiring the closure to have the normalized determinant (108.3) is exactly the additional determinant condition in the definition of Dirac closure. This equivalence concerns this operator and this domain.

Proof. Apply Chapter 108.10.5. The off-diagonal component of the closed double is the closure of the original symmetric operator. Substituting that component in the determinant definition proves the remaining assertion. Existence of a different Hilbert–Pólya operator supplies no core or unitary comparison with this specified one. □

Remark 108.10.7 *(Closure and determinant).* The squared graph norm $\|u\|^2 + \|H_{\mathrm{res}}^{\mathrm{sym}} u\|^2$ defines the graph completion. Essential self-adjointness asks whether the closure equals its adjoint. A determinant identity requires an admissible spectral determinant and its normalization; it is not a consequence of graph completeness.

Definition 108.10.8 *(closed Tate-polar operator quotient).* A *closed Tate-polar operator quotient* for $H_0 = H_{\mathrm{res}}^{\mathrm{sym}}$ is additional data of a Hilbert quotient

$$\pi : \mathcal{H}_{\mathrm{res}} \longrightarrow \mathcal{H}_{\mathrm{res}}^{\circ}$$

with closed finite-rank kernel P_{Tate}, an induced dense domain

$$C_{\mathrm{res}}^{\circ} := \pi(C_{\mathrm{res}}),$$

and a symmetric operator H_0° on C_{res}° satisfying

$$H_0^{\circ} \pi u = \pi H_0 u, \qquad u \in C_{\mathrm{res}}.$$

The quotient is required to be compatible with closure and adjoint domains:

$$\mathcal{D}(\overline{H_0^{\circ}}) = \pi \mathcal{D}(\overline{H_0}), \qquad \mathcal{D}((H_0^{\circ})^*) = \pi \mathcal{D}(H_0^*),$$

after completing both sides in their graph norms. It is also required to admit a separately specified unitary $U_{\mathrm{res}} : \mathcal{H}_{\mathrm{res}}^{\circ} \to L^2(\mathbb{R})$ that represents the quotient operator by multiplication on its image domain, in the sense of Chapter 108.10.15. No pre-quotient unitary is presupposed.

The existence of this quotient and compatibility data is a separate hypothesis.

Remark 108.10.9 *(post-quotient notation).* In the boundary and deficiency discussion below,

$$T_0 := H_0^{\circ} : C_{\mathrm{res}}^{\circ} \subset \mathcal{H}_{\mathrm{res}}^{\circ} \longrightarrow \mathcal{H}_{\mathrm{res}}^{\circ}$$

denotes the post-Tate residual operator supplied by Chapter 108.10.8. The pre-quotient operator $H_0 = H_{\mathrm{res}}^{\mathrm{sym}}$ is only the source operator. Thus every adjoint, deficiency kernel, boundary quotient, and multiplication-coordinate graph statement below is a statement about T_0. Without the closed Tate-polar quotient, these formulas are well-formed as conditional operator theory but not as an unconditional construction.

Boundary Form and Deficiency Kernels

The boundary quotient tests essential self-adjointness of the specified operator. Rank-residue normalization does not determine that quotient, the spectral determinant, or the arithmetic trace.

Definition 108.10.10 *(residual boundary form).* Under the post-quotient notation of Chapter 108.10.9, the adjoint-domain boundary form is the Hermitian symplectic form

$$\mathcal{B}_{\mathrm{res}}(u, v) \;:=\; \langle T_0^* u, v\rangle - \langle u, T_0^* v\rangle, \qquad u, v \in \mathcal{D}(T_0^*).$$

The *residual boundary space* is the quotient

$$Q_{\mathrm{res}} \;:=\; \mathcal{D}(T_0^*)/\mathcal{D}(\overline{T_0}),$$

with the induced form still denoted $\mathcal{B}_{\mathrm{res}}$.

PROPOSITION 108.10.11 *(Boundary quotient and essential self-adjointness).* For the specified densely defined symmetric T_0,

$$Q_{\mathrm{res}} = 0$$

is equivalent to essential self-adjointness. Vanishing of the boundary form on the original domain alone asserts only symmetry. Identifying this operator boundary space with a BV–BFV boundary space requires a separate map preserving the forms and domains.

Proof. The closure is self-adjoint exactly when $\operatorname{Dom} \overline{T}_0 = \operatorname{Dom} T_0^*$, which is the zero-quotient condition. Symmetry already makes the boundary form vanish on $\operatorname{Dom} T_0$, regardless of whether its adjoint domain is larger. Neither assertion includes a determinant or arithmetic trace identity. □

Definition 108.10.12 *(deficiency spaces of the residual).* The residual deficiency spaces are

$$\mathcal{N}_+ := \ker(T_0^* - i), \qquad \mathcal{N}_- := \ker(T_0^* + i).$$

Theorem 108.10.13 *(Dirac two-test form of the residual).* For this specified T_0, a self-adjoint closure with the stated determinant exists exactly when both conditions hold:

(i) *Deficiency vanishing:*

$$\mathcal{N}_+ = 0 = \mathcal{N}_-. \tag{108.4}$$

(ii) *Determinant identity for the closure:*

$$\det_\infty\big(s \cdot \mathrm{id} - (\tfrac{1}{2}\mathrm{id} + i\overline{T_0})\big) \;=\; \xi_{\mathbb{R}}(s). \tag{108.5}$$

Define the post-quotient double

$$\mathscr{D}^\circ_{T_0} = \begin{pmatrix} 0 & T_0 \\ T_0 & 0 \end{pmatrix} \quad \text{on } C^\circ_{\mathrm{res}} \oplus C^\circ_{\mathrm{res}}.$$

Equivalently, this Dirac double $\mathscr{D}^\circ_{T_0}$ has no deficiency states and its closed off-diagonal component has determinant $\xi_{\mathbb{R}}$.

Proof. By von Neumann's deficiency-index criterion, a densely defined symmetric operator T_0 is essentially self-adjoint if and only if $\ker(T_0^* - i) = \ker(T_0^* + i) = 0$. Thus (108.4) is exactly the essential self-adjointness part of Chapter 108.10.4. By Chapter 108.10.5, the same condition is equivalent to essential self-adjointness of the Dirac double $\mathscr{D}^\circ_{T_0}$.

The regularized determinant is an additional datum for this closure. Equation (108.5) is imposed on $\overline{T}_0$, the off-diagonal component of the closed post-quotient double. Combining the deficiency criterion with this separate determinant identity proves the stated equivalence. The pre-quotient double is a different operator. No existence theorem for another operator identifies its domain with this one. □

Remark 108.10.14 *(Scope of deficiency vanishing).* Vanishing of both deficiency kernels proves essential self-adjointness of T_0. The determinant condition remains separate. In a unitary multiplication chart, deficiency vanishing has the stronger spectral consequence proved below: the closure is continuous multiplication and cannot realize a discrete zero-eigenvalue divisor.

The Deficiency Calculation

In multiplication coordinates the deficiency kernels are the orthogonal complements of the ranges of $(M_t \pm i)$ on the residual core, with the sign convention matching the conjugate-linear second slot in the inner product. The computation applies to a specified residual multiplication chart and the closed Tate-polar operator quotient of Chapter 108.10.8; without those it is conditional operator theory. The obstruction is a graph-core question for the residual Hadamard domain, not an Euler-product calculation.

Definition 108.10.15 *(multiplication-coordinate residual chart).* Let

$$M_t\colon \mathcal{D}(M_t) \subset L^2(\mathbb{R}, dt) \to L^2(\mathbb{R}, dt), \qquad (M_t f)(t) = t f(t),$$

be the self-adjoint multiplication operator. A *multiplication-coordinate residual chart* is a unitary map, denoted U_{res} by abuse of notation,

$$U_{\mathrm{res}}\colon \mathcal{H}^{\circ}_{\mathrm{res}} \longrightarrow L^2(\mathbb{R}, dt),$$

from the residual quotient Hilbert space of Chapter 108.10.8 such that the multiplication-coordinate residual core is

$$C^{M}_{\mathrm{res}} \;:=\; U_{\mathrm{res}}(C^{\circ}_{\mathrm{res}}) \;\subset\; \mathcal{D}(M_t),$$

and

$$T_{\mathrm{o}} = H^{\circ}_{\mathrm{o}} \quad \text{is represented by} \quad M_t|_{C^{M}_{\mathrm{res}}}.$$

This chart is additional data. The virtual character theorem Chapter 108.7.2 does not supply it. Once specified, it determines the graph and spectral-type consequences below.

THEOREM 108.10.16 *(deficiency kernels are range defects).* Assume a multiplication-coordinate residual chart in the sense of Chapter 108.10.15. Then

$$U_{\mathrm{res}}\mathcal{N}_{+} \;=\; \big((M_t + i)C^{M}_{\mathrm{res}}\big)^{\perp}, \tag{108.6}$$

and

$$U_{\mathrm{res}}\mathcal{N}_{-} \;=\; \big((M_t - i)C^{M}_{\mathrm{res}}\big)^{\perp}.$$

Consequently $\mathcal{N}_{+} = 0 = \mathcal{N}_{-}$ if and only if both ranges

$$(M_t - i)C^{M}_{\mathrm{res}}, \qquad (M_t + i)C^{M}_{\mathrm{res}}$$

are dense in $L^2(\mathbb{R}, dt)$.

Proof. We prove the + identity; the − identity is identical with i replaced by $-i$. The inner product is linear in the first slot and conjugate-linear in the second. By definition, $u \in \mathcal{N}_{+}$ means $u \in \mathcal{D}(T_{\mathrm{o}}^{*})$ and $(T_{\mathrm{o}}^{*} - i)u = 0$. Equivalently, for every $f \in C^{\circ}_{\mathrm{res}}$,

$$\langle T_{\mathrm{o}} f, u\rangle \;=\; \langle f, iu\rangle.$$

Since the second slot is conjugate-linear, $\langle f, iu\rangle = -i\langle f, u\rangle$. Transport by U_{res}. Write $\tilde{u} = U_{\mathrm{res}}u$ and $\tilde{f} = U_{\mathrm{res}}f$. Since T_{o} is represented by M_t on C^{M}_{res}, the adjoint equation becomes

$$\langle M_t\tilde{f}, \tilde{u}\rangle \;=\; \langle \tilde{f}, i\tilde{u}\rangle, \qquad \tilde{f} \in C^{M}_{\mathrm{res}}.$$

Hence

$$\langle (M_t + i)\tilde{f}, \tilde{u}\rangle = 0 \qquad \text{for all } \tilde{f} \in C^{M}_{\mathrm{res}},$$

which says precisely that $\tilde{u} \in ((M_t + i)C^{M}_{\mathrm{res}})^{\perp}$. The converse is the same implication read backwards, so (108.6) follows. The density criterion follows because a subspace has zero orthogonal complement exactly when its closure is the full Hilbert space. □

Corollary 108.10.17 *(range density is graph density for M_t).* For a subspace $C \subset \mathcal{D}(M_t)$, the following are equivalent:

$$\overline{C}^{\|\cdot\|_{M_t}} = \mathcal{D}(M_t), \qquad \overline{(M_t - i)C}^{L^2} = L^2(\mathbb{R}, dt), \qquad \overline{(M_t + i)C}^{L^2} = L^2(\mathbb{R}, dt).$$

Consequently, under the residual multiplication chart hypothesis, C_{res}^M is a graph core for M_t if and only if

$$\mathcal{N}_+ = 0 = \mathcal{N}_-.$$

Proof. For $\operatorname{Im} z \neq 0$, the map

$$M_t - z \colon \mathcal{D}(M_t)_{\mathrm{graph}} \to L^2(\mathbb{R}, dt)$$

is boundedly invertible: its inverse is multiplication by $(t - z)^{-1}$, which maps L^2 into $\mathcal{D}(M_t)$ with bounded graph norm. Hence a subspace is dense in the graph norm if and only if its image under $M_t - z$ is dense in L^2. Apply this with $z = i$ and $z = -i$, then use Chapter 108.10.16. □

Lemma 108.10.18 *(rank-one graph-boundary model).* Let $\psi(t) = (1 + |t|)^{-1}$. Then $\psi \in L^2(\mathbb{R}, dt)$ but $\psi \notin \mathcal{D}(M_t)$. Define

$$C_\psi := \{ f \in \mathcal{D}(M_t) \mid \langle (M_t - i)f, \psi \rangle = 0 \}.$$

Then C_ψ is L^2-dense and graph-closed of codimension one, but it is not graph-dense. Its range defects are

$$\big((M_t - i)C_\psi\big)^\perp = \mathbb{C}\psi, \qquad \big((M_t + i)C_\psi\big)^\perp = \mathbb{C}\,\frac{t + i}{t - i}\psi.$$

Proof. The function ψ is square-integrable, while $t\psi(t)$ is not square-integrable. Since

$$M_t - i \colon \mathcal{D}(M_t)_{\mathrm{graph}} \to L^2(\mathbb{R}, dt)$$

is a bounded isomorphism, C_ψ is the inverse image of the closed hyperplane $\psi^\perp$; hence it is graph-closed of codimension one and its first range defect is $\mathbb{C}\psi$. The defining functional is not continuous for the L^2-topology on $\mathcal{D}(M_t)$, because such continuity would represent it by an L^2-vector, equivalently would put ψ in $\mathcal{D}(M_t)$. Therefore its kernel is L^2-dense.

For the second range, write

$$(M_t + i)(M_t - i)^{-1}$$

as multiplication by $(t + i)/(t - i)$, a unitary multiplier on L^2. The orthogonal complement of its image of $\psi^\perp$ is the span of $(t + i)(t - i)^{-1}\psi$. □

Lemma 108.10.19 *(point conditions are graph-invisible).* Let $\Gamma \subset \mathbb{R}$ be locally finite. Then $C_c^\infty(\mathbb{R} \setminus \Gamma)$ is graph-dense in $\mathcal{D}(M_t)$. In particular, a boundary condition that is only pointwise at the zero ordinates cannot obstruct graph density for the ordinary M_t-graph norm.

Proof. The graph norm of M_t is the L^2-norm for the measure $(1 + t^2)\,dt$. Locally finite sets have zero measure for this weight. Given $f \in \mathcal{D}(M_t)$, truncate f to a compact interval, remove intervals of arbitrarily small weighted measure around the finitely many points of Γ in that interval, and then mollify. The approximants lie in $C_c^\infty(\mathbb{R} \setminus \Gamma)$ and converge in the graph norm. □

Definition 108.10.20 *(Hadamard-supported graph functional).* Let

$$\Gamma_\xi^{\mathrm{crit}} := \{\, \gamma \in \mathbb{R} \mid \tfrac{1}{2} + i\gamma \text{ is a non-trivial zero of } \zeta \,\}$$

with multiplicities forgotten. This is the real critical-line ordinate support relevant to the Hilbert–Pólya target; it is not an assertion that the full Hadamard divisor has already been realised on $\mathbb{R}$. A continuous functional

$$\beta \colon \mathcal{D}(M_t)_{\mathrm{graph}} \longrightarrow \mathbb{C}$$

is *Hadamard-supported on the zero ordinates* if

$$\beta(\phi) = 0 \qquad \text{for every } \phi \in C_c^\infty(\mathbb{R} \setminus \Gamma_\xi^{\mathrm{crit}}).$$

This is the graph-norm formulation of the idea that the boundary condition sees only point or finite-jet Hadamard residue data at the zero ordinates.

THEOREM 108.10.21 *(no Hadamard residue functional in the ordinary graph dual).* The ordinary M_t-graph dual contains no non-zero Hadamard-supported functional. Equivalently,

$$\beta \in (\mathcal{D}(M_t)_{\mathrm{graph}})^*, \qquad \operatorname{supp}\beta \subset \Gamma_\xi^{\mathrm{crit}} \quad \Longrightarrow \quad \beta = 0$$

in the sense of Chapter 108.10.20. Thus there is no non-zero

$$\beta_{\mathrm{Had}} \colon \mathcal{D}(M_t)_{\mathrm{graph}} \to \mathbb{C}$$

whose graph-closed kernel is exactly a Hadamard-residue boundary condition supported only at the zero ordinates. The statement also rules out finite-jet residue data at those ordinates, if such data are asked to be continuous for the ordinary M_t-graph topology.

Proof. The non-trivial zeros of ζ form a discrete subset of the critical strip, and only finitely many have ordinate in a bounded interval. Hence $\Gamma_\xi^{\mathrm{crit}}$ is locally finite as a subset of $\mathbb{R}$. By Chapter 108.10.19,

$$C_c^\infty(\mathbb{R} \setminus \Gamma_\xi^{\mathrm{crit}})$$

is dense in $\mathcal{D}(M_t)$ for the graph norm. If β is continuous for the graph norm and vanishes on this dense subspace, it vanishes on all of $\mathcal{D}(M_t)$.

Equivalently, the graph norm is the L^2-norm for the weighted measure $(1 + t^2)\,dt$, so the graph dual is represented by weighted L^2-functions, or, after the isometric resolvent chart

$$M_t - i \colon \mathcal{D}(M_t)_{\mathrm{graph}} \xrightarrow{\sim} L^2(\mathbb{R}, dt),$$

by L^2-vectors. No non-zero L^2-vector is supported on the locally finite set $\Gamma_\xi^{\mathrm{crit}}$, which has Lebesgue measure zero. □

COROLLARY 108.10.22 *(Hadamard residue data is graph-invisible).* In the ordinary multiplication graph topology, β_{Had} does not exist as a non-zero graph-dual boundary functional. Therefore a boundary condition whose only data are point or finite-jet Hadamard residues at the locally finite real ordinate set $\Gamma_\xi^{\mathrm{crit}}$ is graph-invisible: it cannot by itself obstruct graph density of C_{res}^M in $\mathcal{D}(M_t)$.

Remark 108.10.23 *(surviving branches after the graph-dual no-go).* The preceding corollary is not an exhaustive boundary-state classification. It eliminates only the ordinary point-supported graph-dual path. Assuming the closed Tate-polar quotient and residual multiplication chart have been constructed, a surviving operator obstruction must take a different form. The known nonexclusive presentations are:

(i) the exact Hadamard residue or finite-jet trace lives in a stronger rigged, Paley–Wiener, de Branges, or reproducing-kernel topology $\Phi_\xi \subset \mathcal{D}(M_t)_{\mathrm{graph}}$, where point evaluation is continuous;

(ii) the ordinary graph obstruction is nonlocal, represented under the resolvent chart by an L^2-vector or closed L^2-subspace as in Chapter 108.10.18, and is therefore not point-residue data;

(iii) a physical Ward/BV–BFV argument supplies extra boundary phase space not yet identified with Q_{res}; this is a physics residual, not an operator-theoretic graph-dual boundary functional.

These are different possible constructions. No equivalence between them, or with Weil positivity, follows from the graph-dual theorem. The stronger-topology approach would require, at minimum, a continuous Hadamard trace-and-jet map $\Phi_\xi \to \ell^2(\Gamma_\xi^{\mathrm{crit}}, w)$, compatibility with the post-Tate quotient and adjoint domains, and a proof that the resulting closed condition is not a tautological insertion of the zero divisor.

Remark 108.10.24 *(Graph density fixes the closure).* In a specified unitary multiplication chart, graph density gives $\overline{T}_0$ unitarily equivalent to M_t. Its spectral measures are then absolutely continuous. Point-supported residue functionals are absent from the ordinary graph dual; nonlocal graph conditions can have nonzero deficiency, as the rank-one graph-boundary example shows. Neither outcome constructs a zero-divisor operator on this chart.

Graph Closure and Spectral Type

The ordinary multiplication closure has continuous spectrum. The diagonal zero-space construction instead uses zeros as input. The following comparison makes their different spectral types explicit.

Lemma 108.10.25 *(classical multiplication-line graph closure).* Let

$$M_t^\circ : C_c^\infty(\mathbb{R}) \subset L^2(\mathbb{R}, dt) \longrightarrow L^2(\mathbb{R}, dt), \qquad (M_t^\circ f)(t) = tf(t).$$

Then M_t° is essentially self-adjoint. Its graph-norm closure is multiplication by t on

$$\mathcal{D}(M_t) \;=\; \{\, f \in L^2(\mathbb{R}, dt) \mid tf(t) \in L^2(\mathbb{R}, dt) \,\}.$$

The spectrum is the absolutely continuous spectrum $\mathrm{Spec}(M_t) = \mathbb{R}$.

Proof. The operator M_t with the displayed maximal domain is self-adjoint by the spectral theorem for multiplication operators. The core $C_c^\infty(\mathbb{R})$ is dense in the graph norm: truncate f by compactly supported cutoffs and then mollify; both f and tf converge in L^2. Hence the closure of M_t° is M_t. Since M_t is multiplication by the coordinate on $L^2(\mathbb{R}, dt)$, its spectrum is the essential range of t, namely $\mathbb{R}$, with absolutely continuous spectral measure. □

Proposition 108.10.26 *(why the classical closure is insufficient).* The closure of Chapter 108.10.25 does not solve the spectral-flow problem. Its graph norm is complete and its operator is self-adjoint, but its spectrum is continuous $\mathbb{R}$, not a divisor with zero multiplicities of $\xi_{\mathbb{R}}$. Consequently it cannot satisfy

$$\det_\infty\big(s \cdot \mathrm{id} - (\tfrac{1}{2}\mathrm{id} + iM_t)\big) = \xi_{\mathbb{R}}(s)$$

as a zero-eigenvalue determinant.

Proof. The determinant identity in Chapter 108.5 has divisor div $\xi_{\mathbb{R}}$. The multiplication operator M_t on $L^2(\mathbb{R}, dt)$ has Lebesgue spectral measure on the line, not this determinant divisor. An eigenvector would have to be supported on a singleton, which has Lebesgue measure zero. This excludes a determinant whose zeros are precisely finite-multiplicity eigenvalues of this operator. □

PROPOSITION 108.10.27 *(formal zero-space closure is tautological).* On the formal zero Hilbert space $\mathcal{H}_\xi^{\mathrm{formal}}$ of Chapter 108.1.3, the diagonal ordinate operator has graph-norm closure. Its projected operator $\Theta_\xi^{\mathrm{proj}}$ satisfies the regularised determinant identity for $\xi_{\mathbb{R}}$ only if the original zero divisor already lies on the critical line. This proves no non-formal spectral-flow closure.

Proof. The finite-support diagonal operator on $\bigoplus_{\rho\in\mathcal{Z}}\mathbb{C}$ closes in the graph norm to the maximal diagonal multiplication operator by the real numbers $t_\rho = \mathrm{Im}(\rho)$; its closure is self-adjoint. The operator $\Theta_\xi^{\mathrm{proj}} = \frac{1}{2} + iH_{\mathrm{ord}}^{\mathrm{formal}}$ has spectral divisor $\{\frac{1}{2} + it_\rho\}$. Hadamard factorisation identifies its regularised determinant with $\xi_{\mathbb{R}}$ only when $\beta_\rho = 1/2$ for every ρ. Otherwise it is the determinant of the projected divisor, not of $\xi_{\mathbb{R}}$. The construction has therefore used the zero divisor and then discarded its real-part defect. It supplies no Bost–Connes, Meyer, Connes, Deninger, or chiral-homological source for the Hilbert space or the Weil–Connes trace; it is the formal target explicitly excluded by Chapter 108.1.4. □

THEOREM 108.10.28 *(Spectral type forced by a multiplication chart).* Let T_0 be densely defined and symmetric. Suppose a unitary $U : \mathcal{H} \to L^2(\mathbb{R}, dt)$ satisfies $UT_0U^{-1} = M_t|_C$, with $C \subset \mathrm{Dom}\, M_t$. If T_0 is essentially self-adjoint, then

$$U\overline{T}_0U^{-1} = M_t.$$

Consequently the closure has purely absolutely continuous spectrum and no eigenvalues. It cannot have the nonempty zero-eigenvalue determinant divisor div $\xi_{\mathbb{R}}$.

Proof. The closure of $M_t|_C$ is a restriction of M_t. Essential self-adjointness makes it self-adjoint. A self-adjoint operator has no proper self-adjoint extension: taking adjoints reverses the inclusion and gives the reverse inclusion. Thus the closure equals M_t. Its scalar spectral measures have densities $|g(t)|^2 dt$. An eigenvector of eigenvalue λ would be supported on the null set $\{\lambda\}$, hence is zero. □

Remark 108.10.29 *(A different spectral carrier).* Within the stipulated unitary chart, closing the symmetric operator cannot create a zero-eigenvalue divisor. A stronger topology can make residue evaluation continuous, but its Hilbert completion, operator domain, and arithmetic trace must then be constructed. The preceding theorem concerns every essentially self-adjoint restriction in the ordinary multiplication chart, not only the compact smooth core.

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Chapter 44.3.2: Hilbert–Pólya hypothesis (Ch. 44).

Chapter 57: H-P^{Ar} equivalence bundle, approach 2 (Ch. 57).

Chapter 62: determinant realisation of H-P^{Ar}, approach 3 (Ch. 62).

Chapter 101: adelic-scattering formulation (Ch. 101).

Chapter 104: Deninger-chiral triple identification (Ch. 104).

Chapter 103: 3D arithmetic Chern-Simons (Ch. 103).

Comparison of the constructions. Modular theory supplies the critical generator on its GNS space. Mellin analysis diagonalizes idele norm scaling. Residue calculus gives meromorphic evaluation maps, while Birman–Krein theory relates a specified trace-class pair to its spectral shift. Meyer's virtual representation supplies a signed character. Their comparison requires explicit maps, compatible domains, and preservation of the complete normalized Weil functional.

Chapter 109

Functoriality of the Triple Identification in A and g

Chapter 104 isolated the conditional triple comparison $H^\bullet_\Delta = H^\bullet_{\mathrm{ch}} = \partial$-chiral on the primary Heisenberg domain for the baseline genus $g = 0$ and the fixed arithmetic Heisenberg H^{Ar}. The functoriality residuals of Section 104.12(5,6) remain: the three cohomology assignments must extend to functors on a category $\mathsf{ArChAlg}$ of arithmetic chiral algebras on $\overline{\mathrm{Spec}\,\mathbb{Z}}$, and the triple comparison must become a natural isomorphism of these functors, uniformly compatible with the genus-g restriction and induction maps of Chapter 102. The conditional master theorem is Theorem 109.6.1; it rests on the chain-level identification Ξ of Theorem 109.3.2, the naturality diagram Theorem 109.4.1, and the genus-stability theorem Theorem 109.5.3. The diagonal morphism $\mathsf{H}^{\mathrm{Ar}} \to \mathsf{H}^{\mathrm{Ar}} \oplus \mathsf{H}^{\mathrm{Ar}}$ is the explicit non-conformal additivity test. The residuals are exactly the hypotheses inherited from Chapters 103 and 104.

109.1 The Category of Arithmetic Chiral Algebras

The three cohomology assignments of Chapter 104 are defined on an individual object, the arithmetic Heisenberg H^{Ar}. To state functoriality we introduce the ambient category.

Definition 109.1.1 *(arithmetic chiral algebra).* An *arithmetic chiral algebra on* $\overline{\mathrm{Spec}\,\mathbb{Z}}$ is a datum

$$\mathsf{A} \;=\; \Big(\mathsf{A}^{(\infty)},\ \{\mathsf{A}^{(p)}\}_{p\in\mathbb{P}},\ \{u^{(v)}\}_{v\in|\overline{C}|},\ \{m^{(v,w)}\}_{v<w},\ \{\partial^{\mathrm{PG}}_{v,w}\}_{v<w}\Big)$$

subject to the axioms below. Write $|\overline{C}| = \mathbb{P}\cup\{\infty\}$.

(i) *Archimedean factor.* $\mathsf{A}^{(\infty)}$ is a pointed E_1-chiral factorisation algebra on $\mathbb{C}_\infty$ in the Beilinson–Drinfeld / Costello–Gwilliam sense (Ch. 58.1.1), together with a conformal vector $T \in \mathsf{A}^{(\infty)}_2$ and a spherical unit $u^{(\infty)} \in \mathsf{A}^{(\infty)}_0$.

(ii) *Finite-prime factors.* For every prime $p \in \mathbb{P}$, $\mathsf{A}^{(p)}$ is a pointed E_0-chiral coefficient object on the Fargues–Scholze perfectoid diamond chart S_p, equipped with a Bost–Connes Adams endomorphism $\psi_p \in \mathrm{End}(\mathsf{A}^{(p)})$ and a Frobenius-semilinear orientation $\mathfrak{J}_p^{\mathrm{Ar}}$ (the finite-place substitute for the archimedean $i \in \mathbb{C}$).

(iii) *Spherical units.* The collection $\{u^{(v)}\}_{v\in|\overline{C}|}$ satisfies the Tate spherical compatibility (Section 50.1.1): $u^{(v)}$ is closed for the local bar differential d_v, is a compact central idempotent, and is the monoidal unit at almost every prime.

(iv) *Adjacent products.* For every pair of distinct places $v < w$ in the auxiliary order $2 < 3 < 5 < \cdots < \infty$, the datum $m^{(v,w)}$ is a map

$$m^{(v,w)} : \mathsf{A}^{(v)} \widehat{\otimes} \mathsf{A}^{(w)} \longrightarrow \mathsf{A}^{(vw)},$$

with $\mathsf{A}^{(vw)}$ the formal merged-block coefficient in the sense of Definition 103.5.2. The $m^{(v,w)}$ are required to be ψ_p-equivariant at every finite $v = p$ and to intertwine the archimedean L_0-action at $v = \infty$.

(v) *Residue coupling.* Specify common coefficient complexes, residue operations and degree-one operators τ_S as in Definition 58.3.1, satisfying $D_{0,S}\tau_S + \tau_S D_{0,S} + \tau_S^2 = 0$ and intertwining stage transitions. Their identification with tame-weighted chiral operations is additional arithmetic realization data.

(vi) *Tate compatibility.* The restricted tensor product $\otimes'_v \mathsf{A}^{(v)}$ in pointed E_2-monoidal stable presentable ∞-categories (Corollary 50.8.3) is well-defined, i.e. almost every $u^{(v)}$ is the monoidal unit and the Tate-pair compatibility F2.c of Section 43.3 is satisfied.

The archimedean $\mathsf{A}^{(\infty)}$ is always genuinely chiral; finite factors may be E_0 per Definition 103.5.2. Fix $g \geq 0$. Call A *abelian Heisenberg-type* if each finite $\mathsf{A}^{(p)}$ is a rank-$2g$ bosonic Fock space. Require $\mathsf{A}^{(\infty)}$ to be the rank-$2g$ archimedean Heisenberg algebra with its specified oscillator grading. The integer g is the genus of this datum. The single-mode Mellin character of Theorem 33.5.1 does not define this full Fock space or its operations.

Remark 109.1.2 *(on the merged-block target).* The formal merged-block coefficient $\mathsf{A}^{(vw)}$ is not a new arithmetic place. It is the target of the adjacent contraction map in the sense of the ordered incidence category $\mathsf{Inc}_<(V)$ of Definition 103.5.1; concretely a completed tensor product of $\mathsf{A}^{(v)}$ and $\mathsf{A}^{(w)}$ modulo a quotient relation recording that the two local operators act on the same global fibre. This convention is essential to avoid the naive reading that primes are fused into composite arithmetic objects.

Definition 109.1.3 *(morphism of arithmetic chiral algebras).* A *morphism* $\phi\colon \mathsf{A}_1 \to \mathsf{A}_2$ of arithmetic chiral algebras is a collection $\phi = \{\phi^{(v)}\}_{v \in |\overline{C}|}$ with $\phi^{(v)}\colon \mathsf{A}_1^{(v)} \to \mathsf{A}_2^{(v)}$ such that

(i) $\phi^{(\infty)}$ is an E_1-chiral factorisation map preserving the conformal vector;

(ii) at every prime p, $\phi^{(p)}$ is ψ_p-equivariant and $\mathfrak{J}_p^{\mathrm{Ar}}$-compatible: $\phi^{(p)} \circ \psi_{p,1} = \psi_{p,2} \circ \phi^{(p)}$;

(iii) the spherical units are preserved: $\phi^{(v)}(u_1^{(v)}) = u_2^{(v)}$ for every $v \in |\overline{C}|$;

(iv) the adjacent products are intertwined: $m_2^{(v,w)} \circ (\phi^{(v)} \otimes \phi^{(w)}) = \phi^{(vw)} \circ m_1^{(v,w)}$ for every $v < w$;

(v) the Parshin–Gersten boundaries are intertwined: $\phi \circ \partial_1^{\mathrm{PG}} = \partial_2^{\mathrm{PG}} \circ \phi$ at every pair of ramified places;

(vi) the restricted-tensor map $\widehat{\phi} := \otimes'_v \phi^{(v)}$ is a map in the pointed E_2-monoidal category of Corollary 50.8.3; this is automatic from (1)–(5) at almost every place by the spherical-unit preservation.

Composition is place-wise; identities are place-wise identities. The resulting category is $\mathsf{ArChAlg}$.

PROPOSITION 109.1.4 *(*$\mathsf{ArChAlg}$ *is a well-defined category).* Composition of morphisms satisfies the axioms of Definition 109.1.3: axioms (1)–(5) are place-wise, and axiom (6) follows from place-wise axioms by the almost-every-place triviality of the units. Identity morphisms are identities in every axiom. Associativity and unitality are inherited from the underlying categories of E_1-chiral factorisation algebras at ∞ and E_0-coefficient categories at finite places.

Proof. For composition: given $\phi\colon \mathsf{A}_1 \to \mathsf{A}_2$ and $\chi\colon \mathsf{A}_2 \to \mathsf{A}_3$, the collection $(\chi \circ \phi)^{(v)} := \chi^{(v)} \circ \phi^{(v)}$ preserves each axiom: axiom (1) is closed under composition of E_1-chiral maps; axiom (2) is closed because ψ_p-equivariance composes; axiom (3) uses $\chi^{(v)}(\phi^{(v)}(u_1^{(v)})) = \chi^{(v)}(u_2^{(v)}) = u_3^{(v)}$; axiom (4) similarly using the stacked naturality diagrams. Axiom (5) uses the linear action of the boundary. Associativity

and unitality are automatic at each place. Axiom (6) follows from the almost-every-place compatibility of the Tate restricted tensor (Section 43.3): the set of places at which $u^{(v)}$ fails to be a strict unit is bounded, and composition preserves this bound. □

Definition 109.1.5 *(genus stabilization).* For every $g \geq 0$, let $\mathsf{ArChAlg}_g \subset \mathsf{ArChAlg}$ be the full subcategory of abelian Heisenberg-type arithmetic chiral algebras of genus g in the sense of Definition 109.1.1. The stabilization functor $S_{g,g+1}\colon \mathsf{ArChAlg}_g \to \mathsf{ArChAlg}_{g+1}$ tensors with the rank-2 archimedean Heisenberg (Definition 102.1.1's rank-2 enhancement at each place). A projection to genus g is defined only on split objects with a specified decomposition into old and new oscillators.

Remark 109.1.6 *(why the filtration).* The filtration records stabilization by a fresh rank-2 factor at every place. It is not a reflective inclusion without a chosen splitting of oscillators.

Remark 109.1.7 *(Local structures and comparison maps).* The archimedean factor carries the specified chiral factorization structure, while the finite-place factors carry their specified coefficient structures. A morphism must preserve each local structure and the common residue couplings. Functoriality of their combined arithmetic realization requires the same compatibility maps. Background on the local structures is given by Beilinson–Drinfeld, Francis–Gaitsgory, Fargues–Scholze and Bhatt–Scholze.

109.2 The Three Functors

The cohomology assignments of Chapter 104 are defined on the single object H^{Ar}. Functoriality requires a target category. Write $\mathrm{grVect}_{\mathbb{C}}^{\mathrm{Frob}}$ for the category of finite-dimensional graded $\mathbb{C}$-vector spaces equipped with a graded-linear automorphism F (the Frobenius) and a graded-symmetric non-degenerate pairing PD (the Poincaré duality); morphisms are graded linear maps commuting with both structures.

Definition 109.2.1 *(Deninger functor $H^\bullet_\Delta$).* The *Deninger functor* is the conditional assignment

$$H^\bullet_\Delta\colon \mathsf{ArChAlg} \longrightarrow \mathrm{grVect}_{\mathbb{C}}^{\mathrm{Frob}}$$

defined on the residual domain where the Deninger Frobenius has been chosen. It sends an arithmetic chiral algebra A to the triple $(H^\bullet_\Delta(\mathsf{A}), F_{\mathsf{A}}, \mathrm{PD}_{\mathsf{A}})$ of Definition 104.3.2, with $F_{\mathsf{A}} = F_{\infty,\mathsf{A}} \otimes \bigotimes'_p \psi_{p,\mathsf{A}}$ the global Frobenius of Definition 104.5.5 and PD_{A} the Poincaré duality pairing of Theorem 104.7.1. On a morphism $\phi\colon \mathsf{A}_1 \to \mathsf{A}_2$, the functor sends ϕ to the induced map $H^\bullet_\Delta(\phi)\colon H^\bullet_\Delta(\mathsf{A}_1) \to H^\bullet_\Delta(\mathsf{A}_2)$.

Definition 109.2.2 *(chiral-homology functor $H^\bullet_{\mathrm{ch}}$).* The *chiral-homology functor*

$$H^\bullet_{\mathrm{ch}}\colon \mathsf{ArChAlg} \longrightarrow \mathrm{grVect}_{\mathbb{C}}^{\mathrm{Frob}}$$

sends A to $H^\bullet(B^{\mathrm{Ar}}(\mathsf{A}), D_{\mathsf{A}})$ of Definition 58.3.1, with D_{A} the signed total differential determined by A. On a morphism $\phi\colon \mathsf{A}_1 \to \mathsf{A}_2$, the induced chain map $B^{\mathrm{Ar}}(\phi)$ commutes with D_{A_1} and D_{A_2} by axioms (3), (4), (5) of Definition 109.1.3; pass to cohomology for $H^\bullet_{\mathrm{ch}}(\phi)$.

Definition 109.2.3 *(bulk-boundary functor pair $\partial \circ \mathrm{BdryACS}$).* The *bulk-boundary assignment* is the composite

$$\partial \circ \mathrm{BdryACS}\colon \mathsf{ArChAlg} \xrightarrow{\mathrm{BdryACS}} \mathsf{3DACS}_{\overline{\mathrm{Spec}\,\mathbb{Z}}} \xrightarrow{\partial} \mathrm{grVect}_{\mathbb{C}}^{\mathrm{Frob}},$$

where

- $\mathsf{3DACS}_{\overline{\mathrm{Spec}\,\mathbb{Z}}}$ is the category of conjectural 3D quantum arithmetic Chern–Simons theories on the étale 3-space $\overline{\mathrm{Spec}\,\mathbb{Z}}$ in the sense of Conjecture 103.4.7, with morphisms given by bulk interface defects;

- BdryACS is the canonical lift that takes an arithmetic chiral algebra to the 3D bulk theory whose boundary WZW is $\mathsf{A}^{(\infty)}$ and whose finite-place defects are labelled by $\{\mathsf{A}^{(p)}\}$ via the Morishita knot-prime dictionary (Corollary 103.4.6, Proposition 104.5.2);
- ∂ is the boundary chiral homology functor, sending a 3D bulk theory to the chiral homology of its boundary factorisation algebra on the archimedean fibre.

On a morphism $\phi\colon \mathsf{A}_1 \to \mathsf{A}_2$, BdryACS produces a bulk interface defect between the two 3D theories whose boundary restriction is $\phi^{(\infty)}$ and whose finite-place Wilson-line restriction is $\{\phi^{(p)}\}$.

Remark 109.2.4 *(conditional form of the bulk-boundary lift).* The functor BdryACS is conditional on the 3D quantum ACS existence conjecture Conjecture 103.4.7. The boundary functor ∂ is conditional on the local-trace comparison Theorem 103.4.9. Both conditionals are inherited from Chapter 103 and enter every theorem below.

Remark 109.2.5 *(Comparison data for the three functors).* The Deninger assignment requires the stated Frobenius and duality structures. The chiral assignment requires the arithmetic realization of Definition 58.3.1. The bulk-boundary assignment requires the quantum theory of Conjecture 103.4.7 and its boundary functor. Their identification requires the natural stage maps of Theorem 109.3.2.

109.3 Chain-Level Identification

The three functors land in $\mathrm{grVect}_{\mathbb{C}}^{\mathrm{Frob}}$ by taking cohomology of chain complexes that are not obviously the same. The natural isomorphism follows once one identifies the chain complexes before passing to cohomology.

Definition 109.3.1 *(universal arithmetic bar).* For $\mathsf{A} \in \mathsf{ArChAlg}$, the *universal arithmetic bar* of A is the pair

$$\mathcal{B}^{\mathrm{univ}}(\mathsf{A}) \;:=\; \big(B^{\mathrm{ord}}_{\bullet}(\mathsf{A}),\, B^{\mathrm{Ar}}_{\bullet}(\mathsf{A})\big),$$

with $B^{\mathrm{ord}}_{\bullet}$ the ordered incidence bar of Definition 103.5.3 applied to the collection $\{\mathsf{A}^{(v)}\}_{v\in|\overline{C}|}$, and $B^{\mathrm{Ar}}_{\bullet}$ the Platonic arithmetic-chiral bar of Definition 58.3.1. Both are equipped with their natural signed differentials.

THEOREM 109.3.2 *(Chain-level comparison from stage maps).* Fix realizations of the ordered and Platonic bars over a common ring R. Suppose that for each object A and stage S, a quasi-isomorphism $\Xi_{\mathsf{A},S} : B^{\mathrm{ord}}_S(\mathsf{A}) \to B^{\mathrm{Ar}}_S(\mathsf{A})$ is given, natural in A and compatible with transitions. For presentations $D = D_{\mathrm{o}} + \tau$, sufficient cochain identities are

$$\Xi_{\mathsf{A},S} D^{\mathrm{ord}}_{\mathrm{o},S} = D^{\mathrm{Ar}}_{\mathrm{o},S} \Xi_{\mathsf{A},S},$$
$$\Xi_{\mathsf{A},S} \tau^{\mathrm{ord}}_S = \tau^{\mathrm{Ar}}_S \Xi_{\mathsf{A},S}.$$

Then the algebraic finite-support colimits admit a natural quasi-isomorphism

$$\Xi_{\mathsf{A}} : B^{\mathrm{ord}}_{\bullet}(\mathsf{A}) \xrightarrow{\sim} B^{\mathrm{Ar}}_{\bullet}(\mathsf{A}).$$

Proof. Adding the operator identities proves the cochain identity. Transition compatibility defines the colimit map, which is a quasi-isomorphism by Proposition 58.3.4. Naturality descends from the commuting squares at every stage. □

The common coefficient realizations, stage quasi-isomorphisms and residue intertwining remain to be constructed for the intended arithmetic-chiral objects. Square-zero of two differentials does not imply that a proposed comparison intertwines them. The ordinary comparison of Theorem 55.4.2 and the integral oscillator models supply actual local constructions on their stated carriers; they do not identify the ordered and Platonic arithmetic bars.

Corollary 109.3.3 *(factoring the three functors). (on Theorem 109.3.2).* The three functors of Section 109.2 all factor through the chain-level homology functor $\mathsf{A} \mapsto H^\bullet(B^{\mathrm{Ar}}(\mathsf{A}), D_{\mathsf{A}})$:

$$H^\bullet_\Delta(\mathsf{A}) \underset{\text{def}}{=} H^\bullet_{\mathrm{ch}}(\mathsf{A}) = \partial \circ \mathrm{BdryACS}(\mathsf{A})$$

on every $\mathsf{A} \in \mathsf{ArChAlg}$, modulo the conditional pieces of Chapters 103 and 104.

Proof. The left-hand equality is Definition 104.3.2 (definitional). The right-hand equality follows from Theorem 103.5.4 and Theorem 109.3.2: $\partial \circ \mathrm{BdryACS}(\mathsf{A})$ computes the boundary chiral homology of the 3D bulk theory attached to A, which equals the ordered incidence homology of $\{\mathsf{A}^{(v)}\}$, which is quasi-isomorphic via Ξ_{A} to the Platonic bar homology $H^\bullet_{\mathrm{ch}}(\mathsf{A})$. □

Domain of the chain-level comparison

Both bars require realizations over the same coefficient ring. A target $\mathrm{Ch}_{\mathbb{C}}$ requires an actual common complex realization; passage from finite residue coefficients to real or complex coefficients is not implicit in the tensor notation. The signs and residue operations must be intertwined by the specified maps $\Xi_{\mathsf{A},S}$. Their naturality and compatibility with transitions are additional identities. In the subsequent functoriality statements, the arithmetic realization and natural stage maps of Theorem 109.3.2 are part of the comparison hypotheses. The ordinary and integral models alone do not supply this arithmetic-chiral comparison.

109.4 Functoriality in A

Corollary 109.3.3 equates the three functors on objects. A natural isomorphism requires that every morphism $\phi\colon \mathsf{A}_1 \to \mathsf{A}_2$ in $\mathsf{ArChAlg}$ induces commuting diagrams; the commutation is the content of the next theorem.

Theorem 109.4.1 *(naturality in A). (on Theorem 109.3.2, Conjecture 103.4.7, and Theorem 103.4.9).* The three natural isomorphisms

$$H^\bullet_\Delta \xRightarrow{\eta_{\Delta,\mathrm{ch}}} H^\bullet_{\mathrm{ch}} \xRightarrow{\eta_{\mathrm{ch},\partial}} \partial \circ \mathrm{BdryACS}$$

are natural: for every morphism $\phi\colon \mathsf{A}_1 \to \mathsf{A}_2$ of arithmetic chiral algebras, the three squares

$$\begin{array}{ccc}
H^\bullet_\Delta(\mathsf{A}_1) & \xrightarrow{H^\bullet_\Delta(\phi)} & H^\bullet_\Delta(\mathsf{A}_2) \\
\downarrow{\scriptstyle \eta_{\Delta,\mathrm{ch},\mathsf{A}_1}} & & \downarrow{\scriptstyle \eta_{\Delta,\mathrm{ch},\mathsf{A}_2}} \\
H^\bullet_{\mathrm{ch}}(\mathsf{A}_1) & \xrightarrow{H^\bullet_{\mathrm{ch}}(\phi)} & H^\bullet_{\mathrm{ch}}(\mathsf{A}_2) \\
\downarrow{\scriptstyle \eta_{\mathrm{ch},\partial,\mathsf{A}_1}} & & \downarrow{\scriptstyle \eta_{\mathrm{ch},\partial,\mathsf{A}_2}} \\
\partial \circ \mathrm{BdryACS}(\mathsf{A}_1) & \xrightarrow[\partial \circ \mathrm{BdryACS}(\phi)]{} & \partial \circ \mathrm{BdryACS}(\mathsf{A}_2)
\end{array}$$

commute. The three horizontal maps are identified under the naturality, and the induced Frobenius and Poincaré duality structures are preserved pairwise: $F_{\mathsf{A}_2} \circ H^\bullet(\phi) = H^\bullet(\phi) \circ F_{\mathsf{A}_1}$ and likewise for PD.

Proof. The upper square. The equality $H^\bullet_\Delta(\mathsf{A}) \underset{\text{def}}{=} H^\bullet_{\mathrm{ch}}(\mathsf{A})$ is definitional (Definition 104.3.2); the natural transformation $\eta_{\Delta,\mathrm{ch}}$ is the identity. Naturality is immediate: the identity intertwines every induced map.

The lower square. Apply Theorem 109.3.2 to both A_1 and A_2, obtaining chain-level quasi-isomorphisms Ξ_{A_1} and Ξ_{A_2}. The natural transformation $\eta_{\mathrm{ch},\partial}$ is the induced map on cohomology: $\eta_{\mathrm{ch},\partial,\mathsf{A}} := \sigma_{\mathsf{A}} \circ H^\bullet(\Xi_{\mathsf{A}})^{-1}$, where $\sigma_{\mathsf{A}} \colon H^\bullet(B^{\mathrm{ord}}_\bullet(\mathsf{A})) \xrightarrow{\sim} \partial \circ \mathrm{BdryACS}(\mathsf{A})$ is the isomorphism of Theorem 103.5.4; this composite is well-defined by that theorem and Theorem 109.3.2.

Commutativity at the chain level. The induced chain map $B^{\mathrm{ord}}_\bullet(\phi)$ intertwines Ξ_{A_1} and Ξ_{A_2} by the naturality of Theorem 109.3.2 in A. The induced chain map $B^{\mathrm{Ar}}(\phi)$ likewise intertwines, by axioms (3), (4), (5) of Definition 109.1.3: the spherical units are mapped to spherical units; the adjacent products commute through ϕ; the Parshin–Gersten boundary commutes through ϕ. Composing, we get $\Xi_{\mathsf{A}_2} \circ B^{\mathrm{ord}}(\phi) = B^{\mathrm{Ar}}(\phi) \circ \Xi_{\mathsf{A}_1}$ as maps of chain complexes. Passing to cohomology gives the commutative square.

Frobenius compatibility. The global Frobenius $F_{\mathsf{A}} = F_{\infty,\mathsf{A}} \otimes \bigotimes'_p \psi_{p,\mathsf{A}}$ commutes with $H^\bullet(\phi)$ because axiom (2) of the morphism definition supplies ψ_p-equivariance at each finite prime, and axiom (1) supplies preservation of the conformal vector at ∞ (from which F_∞ is conjugated via Proposition 104.5.4). Hence $F_{\mathsf{A}_2} \circ H^\bullet(\phi) = H^\bullet(\phi) \circ F_{\mathsf{A}_1}$.

Poincaré duality compatibility. The PD pairing is determined by the Koszul self-duality identity Theorem 104.7.1. A morphism ϕ in $\mathsf{ArChAlg}$ preserves the spherical units and the adjacent products; the self-duality pairing is expressed chain-level as the Positselski contraction pairing, which is natural in the underlying A. Hence PD commutes with ϕ. □

COROLLARY 109.4.2 *(triple functoriality in* A*). (on Theorem 109.4.1).* The three functors $H^\bullet_\Delta$, $H^\bullet_{\mathrm{ch}}$, and $\partial \circ \mathrm{BdryACS}$ are naturally isomorphic as functors $\mathsf{ArChAlg} \to \mathrm{grVect}^{\mathrm{Frob}}_{\mathbb{C}}$. Both structural data (Frobenius, Poincaré duality) pull through the natural isomorphisms.

Proof. Immediate from Theorem 109.4.1 after recording that $\eta_{\Delta,\mathrm{ch}}$ is invertible (as the identity) and $\eta_{\mathrm{ch},\partial}$ is invertible as the induced map on cohomology from a chain-level quasi-isomorphism. □

Domain of naturality in A

The archimedean chiral maps and the bulk interface maps must be constructed on their stated domains and compared by the natural stage maps of Theorem 109.3.2.

Signs. The bar-differential Koszul signs agree on both sides by construction of Definition 58.3.1; the ordered-incidence alternating signs match under Ξ by the simplicial normalisation formula.

Ambient category. Both sides land in graded $\mathbb{C}$-vector spaces with Frobenius and Poincaré duality; no compatibility issue.

Frobenius hypothesis. The naturality of F_∞ under $\phi^{(\infty)}$ requires that $\phi^{(\infty)}$ preserves the conformal vector; this is axiom (1) of Definition 109.1.3, which is part of the definition of a morphism and imposes no hidden constraint.

Axiom independence. Axioms (2) and (4) are logically independent: axiom (4) applied at $v = p, w = \infty$ implies axiom (2) only on the unit line (the spherical vacuum at ∞ is fixed by ψ_p up to normalised Frobenius weight); the full ψ_p-equivariance on all of $\mathsf{A}^{(p)}$ is an additional local constraint not automatic from (4). Both axioms are needed and neither is redundant.

Spectral content. The identification of the F-spectrum on H^1 with the nontrivial zeros of $\xi(s)$ remains conditional on H-P^{Ar} (Hypothesis 44.3.2), inherited from Theorem 104.6.1. The naturality of F established here requires only its commutation with $H^\bullet(\phi)$, not a description of its spectrum.

109.5 Functoriality in g

The genus filtration $\mathsf{ArChAlg}_0 \subset \mathsf{ArChAlg}_1 \subset \cdots$ of Definition 109.1.5 generates further functoriality structure, tying Chapter 104 to Chapter 102.

Definition 109.5.1 *(genus-g restrictions of the three functors).* For every $g \geq 0$, let $H^\bullet_{\Delta,g}$, $H^\bullet_{\mathrm{ch},g}$, and $(\partial \circ \mathrm{BdryACS})_g$ denote the restrictions of $H^\bullet_\Delta$, $H^\bullet_{\mathrm{ch}}$, and $\partial \circ \mathrm{BdryACS}$ to $\mathsf{ArChAlg}_g$. Explicitly:

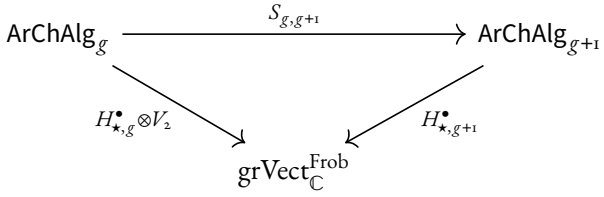
- $H^\bullet_{\Delta,g}$ sends a genus-g object $\mathsf{A} \in \mathsf{ArChAlg}_g$ to the rank-$2g$ Deninger cohomology with global Frobenius $F^{(g)}$ of Theorem 104.9.2(2);
- $H^\bullet_{\mathrm{ch},g}$ sends A to the rank-$2g$ Platonic bar homology of Definition 104.9.1 with rank-$2g$ Fock factors per place;
- $(\partial \circ \mathrm{BdryACS})_g$ sends A to the boundary chiral homology of the rank-$2g$ 3D arithmetic CS $\mathrm{ACS}^{\mathrm{qu},g}_{k^{\mathrm{Ar}},g}$ of Corollary 103.8.1.

Theorem 109.5.2 *(stabilization from compatible chain maps).* Suppose the three genus-g functors have specified chain models $F_{\star,g}$, for $\star \in \{\Delta, \mathrm{ch}, \partial\}$, and the comparison maps η of Theorem 109.3.2 exist naturally. Supply an actual stabilization functor $S_{g,g+1}$.

Let V_2 be a fixed graded coefficient space, regarded as a complex with zero differential. Supply natural quasi-isomorphisms

$$q_{\star,g} : F_{\star,g}(\mathsf{A}) \otimes V_2 \longrightarrow F_{\star,g+1}(S_{g,g+1}\mathsf{A}).$$

Assume these maps preserve the specified Frobenius actions, commute with η up to the chosen chain homotopies, and respect the chosen stabilization retractions. Use algebraic tensors over $\mathbb{C}$, or supply the corresponding Künneth and continuity statements for any completion. Then the cohomology functors fit into



with natural isomorphisms $\vartheta^\star_g = H(q_{\star,g})$.

They commute with the induced η comparisons and the specified retractions. If pairings are also to be preserved, compatibility of the chain maps with those pairings is an additional hypothesis.

Proof. Over $\mathbb{C}$, each complex splits as its cohomology and an acyclic contractible complex. Tensoring a contracting homotopy with id_{V_2} preserves the contraction. Hence

$$H^*(F_{\star,g} \otimes V_2) \cong H^*(F_{\star,g}) \otimes V_2.$$

The quasi-isomorphism $q_{\star,g}$ gives the desired cohomology isomorphism. In the completed case this step is the stated Künneth hypothesis. Naturality and the commutation with Frobenius descend from the corresponding chain identities.

A specified chain homotopy between $\eta q_{\mathrm{ch},g}$ and $q_{\partial,g}(\eta \otimes 1)$ becomes equality on cohomology. This proves compatibility with η. The same argument applies to the other comparison and to the supplied retraction maps. Compatibility with pairings descends when its stated chain-level hypothesis holds. □

Constructing the maps $q_{\star,g}$ is an arithmetic requirement. A decomposition of an auxiliary Heisenberg space into independent oscillators supplies the vector-space identity

$$\mathcal{H}^{(p),g+1} = \mathcal{H}^{(p),g} \otimes \mathcal{H}_2^{(p)}.$$

It does not prove a Künneth comparison for a bar with arithmetic residue coupling. Such a comparison must intertwine its differential, local primitives, transition maps, and completion. The Mellin character of Theorem 33.5.1 supplies no such quasi-isomorphism.

Likewise, a proposed bulk decomposition into a genus-g theory and a rank-two factor must be constructed on the actual theories. The inclusion $\mathrm{GL}_{2g} \times \mathrm{GL}_2 \subset \mathrm{GL}_{2(g+1)}$ is only a block-diagonal subgroup inclusion. It does not imply a tensor decomposition of the associated WZW algebra. Already the ambient Lie algebra has off-diagonal matrix currents absent from the two block Lie algebras.

A valid bulk–boundary comparison would have to identify its stabilization map with the coefficient map, including a homotopy

$$\eta_{g+1} q_{\mathrm{ch},g} \simeq q_{\partial,g}(\eta_g \otimes 1).$$

The theorem explains exactly what follows once these maps are supplied. Their construction is separate from the genus-wise scalar character and from a chosen notation for the three functors.

THEOREM 109.5.3 *(uniformity: genus-wise triple identification). (on Theorem 109.5.2 and Theorem 104.9.2).* The triple identification is uniform in genus: for every $g \geq 0$ and every $\mathsf{A} \in \mathsf{ArChAlg}_g$,

$$H^\bullet_{\Delta,g}(\mathsf{A}) \;=\; H^\bullet_{\mathrm{ch},g}(\mathsf{A}) \;=\; (\partial \circ \mathrm{BdryACS})_g(\mathsf{A}) \tag{109.1}$$

as graded vector spaces with genus-g Frobenius $F^{(g)}$ and PD pairing $\mathrm{PD}^{(g)}$. The equality is compatible with the stabilization functor $S_{g,g+1}$ of Theorem 109.5.2 and with the retraction $\pi_{g+1,g}$ on split stabilized objects: the three genus-g identifications commute, up to $\vartheta^\star_g$, with the corresponding genus-$(g+1)$ identifications.

Proof. The three equalities on the right-hand side of (109.1) are the genus-g restrictions of Corollary 109.3.3. They hold for every $\mathsf{A} \in \mathsf{ArChAlg}_g$ by restriction.

The compatibility with genus stabilization follows by applying Theorem 109.5.2 to obtain the three compatibility diagrams connecting g and $g+1$. Each diagram is commutative up to $\vartheta^\star_g$; the three $\vartheta^\star_g$ agree under the natural isomorphisms $\eta_{\Delta,\mathrm{ch}}$ and $\eta_{\mathrm{ch},\partial}$. Hence the genus-$g$ triple identification commutes with the genus-$(g+1)$ identification: the diagram

$$\begin{array}{ccccc}
H^\bullet_{\Delta,g}(\mathsf{A}) \otimes V_2 & \xrightarrow{=} & H^\bullet_{\mathrm{ch},g}(\mathsf{A}) \otimes V_2 & \xrightarrow{=} & (\partial \circ \mathrm{BdryACS})_g(\mathsf{A}) \otimes V_2 \\
\downarrow{\scriptstyle \vartheta^\Delta_g} & & \downarrow{\scriptstyle \vartheta^{\mathrm{ch}}_g} & & \downarrow{\scriptstyle \vartheta^\partial_g} \\
H^\bullet_{\Delta,g+1}(S_{g,g+1}\mathsf{A}) & \xrightarrow{=} & H^\bullet_{\mathrm{ch},g+1}(S_{g,g+1}\mathsf{A}) & \xrightarrow{=} & (\partial \circ \mathrm{BdryACS})_{g+1}(S_{g,g+1}\mathsf{A})
\end{array}$$

commutes.

The compatibility with retraction $\pi_{g+1,g}$ is the mirror statement: projection onto the first $2g$ oscillators, which is Künneth-compatible by the same argument in the reverse direction. □

Corollary 109.5.4 *(all-genus triple functoriality). (on Theorem 109.5.3).* The three functors $H^\bullet_\Delta$, $H^\bullet_{\mathrm{ch}}$, and $\partial \circ \mathrm{BdryACS}$ are naturally isomorphic as *genus-filtered* functors $\mathsf{ArChAlg} \to \mathrm{grVect}^{\mathrm{Frob}}_{\mathbb{C}}$. The natural isomorphism is genus-graded: restricts to a natural isomorphism at every genus g, and commutes with inclusion and retraction along the filtration.

Proof. Assemble Corollary 109.4.2 (functoriality in A at fixed genus) with Theorem 109.5.3 (uniform compatibility across genus). The genus-filtered structure is preserved at each genus, with transition natural transformations $\vartheta^\star_g$ compatible. □

Independence and domain of genus functoriality

Source consistency. The genus filtration via rank-2 tensor slots of Ch. 102 and the 3D ACS gauge-group scaling $GL_{2g} \hookrightarrow GL_{2(g+1)}$ of Ch. 103 agree: in both cases the rank parameter is $2g$, the archimedean Heisenberg has rank $2g$, and the Satake parameter at each finite prime is also of rank $2g$. The three chapters use identical rank conventions.

Conventions. The notation $r = 2g$ for the rank of the archimedean Heisenberg, g for the Siegel-modular genus, and the Satake rank $2g$ at each finite prime is consistent across Ch. 102, Ch. 103, and Ch. 104.

Künneth comparison. Theorem 109.5.2 requires the actual natural maps $q_{\star,g}$, their quasi-isomorphism property, and their compatibility with the chosen comparison homotopies. A tensor decomposition of the oscillator space does not construct these maps. A completed tensor product also requires the stated continuity and Künneth hypotheses.

Even flatness is not supplied by a trivial Γ-action. For $\Lambda = \mathbb{Z}_p[[T]]$, a rank-one trivial lattice is $M = \Lambda/(T)$. The resolution $0 \to \Lambda \xrightarrow{T} \Lambda \to M \to 0$ gives $\mathrm{Tor}^\Lambda_1(M, M) = M \neq 0$. Thus this lattice has $\mu(M) = 0$ but is not Λ-flat.

Retraction. The retraction $\pi_{g+1,g}$ does not require a cokernel: on the primary Heisenberg domain the freshly-added rank-2 factor is a direct summand, and $\pi_{g+1,g}$ is direct-summand projection.

Spectral content. The genus-g F-spectrum identification with motivic zeros is conditional on rank-g H-P^{Ar} (Theorem 104.9.2(4)) and is not supplied here. The genus-filtration structure holds independently of the spectrum identification.

109.6 Master Theorem and Explicit Example

Functoriality in A and genus-filtered functoriality hold independently; the master theorem records that both hold simultaneously, with the chain-level quasi-isomorphism Ξ induced by supplied natural stage maps. The diagonal morphism $\mathsf{H}^{\mathrm{Ar}} \to \mathsf{H}^{\mathrm{Ar}} \oplus \mathsf{H}^{\mathrm{Ar}}$ tests the non-conformal additive structure; the conformal diagonal is excluded by the central-charge obstruction.

Theorem 109.6.1 *(master functoriality). (on the residuals of Chapters 103, 102, 104 and their arithmetic Positselski, H-P^{Ar}, 3D quantum ACS domains).* Assume also the common coefficient realizations and natural stage quasi-isomorphisms of Theorem 109.3.2, intertwining the differentials, residue operators and transitions, and compatible with the stated genus stabilizations and split retractions. The triple identification $H^\bullet_\Delta = H^\bullet_{\mathrm{ch}} = \partial \circ \mathrm{BdryACS}$ is a natural isomorphism of genus-filtered functors

$$\mathsf{ArChAlg} = \bigcup_{g \geq 0} \mathsf{ArChAlg}_g \longrightarrow \mathrm{grVect}^{\mathrm{Frob}}_{\mathbb{C}},$$

which is:

(i) functorial in A: every morphism $\phi\colon \mathsf{A}_1 \to \mathsf{A}_2$ in $\mathsf{ArChAlg}$ induces a commuting triple-square of cohomology maps, preserving Frobenius F and Poincaré duality PD;

(ii) functorial in g: for every $g \geq 0$, the restriction to $\mathsf{ArChAlg}_g$ is a natural isomorphism of genus-g functors, compatible with the stabilization $S_{g,g+1}$ and with $\pi_{g+1,g}$ on split stabilized objects;

(iii) induced at chain level by the supplied stage quasi-isomorphisms: their colimit is Ξ of Theorem 109.3.2. This criterion does not construct the arithmetic coefficient realizations or the required stage maps.

Proof. Part (1) is Corollary 109.4.2. Part (2) is Corollary 109.5.4. Part (3) follows by taking the colimit of the assumed compatible stage quasi-isomorphisms, as proved in Theorem 109.3.2, together with the fact that the Deninger-to-chiral-homology equality $\eta_{\Delta,\mathrm{ch}}$ is definitional (Definition 104.3.2) and hence trivially chain-level. □

Example 109.6.2 *(diagonal $\mathsf{H}^{\mathrm{Ar}} \to \mathsf{H}^{\mathrm{Ar}} \oplus \mathsf{H}^{\mathrm{Ar}}$).* The diagonal morphism

$$\Delta\colon \mathsf{H}^{\mathrm{Ar}} \longrightarrow \mathsf{H}^{\mathrm{Ar}} \oplus \mathsf{H}^{\mathrm{Ar}}$$

is defined place-wise by the arithmetic Heisenberg diagonal

$$\Delta^{(v)}\colon \mathcal{H}^{(v)} \longrightarrow \mathcal{H}^{(v)} \oplus \mathcal{H}^{(v)}, \qquad x \longmapsto (x, x),$$

at every place $v \in |\overline{C}|$. The target is the coproduct of two copies of H^{Ar} in $\mathsf{ArChAlg}$; its archimedean component is the free E_1-chiral product of two rank-2 Heisenbergs, hence a rank-4 Heisenberg.

PROPOSITION 109.6.3 *(central-charge obstruction to the conformal diagonal). (on Theorem 109.6.1).* The diagonal Δ of Example 109.6.2 is not a morphism in $\mathsf{ArChAlg}$ with the coproduct conformal vector. After applying the forgetful functor to pointed non-conformal E_1-coefficient systems, it becomes a morphism preserving spherical units, Adams maps, adjacent products, and Parshin–Gersten boundaries. On this non-conformal underlying system the three additive maps

$$\big(H^\bullet_\Delta(\Delta),\ H^\bullet_{\mathrm{ch}}(\Delta),\ \partial \circ \mathrm{BdryACS}(\Delta)\big)$$

commute with $\eta_{\Delta,\mathrm{ch}}$ and $\eta_{\mathrm{ch},\partial}$. Explicitly:

(i) $H^\bullet_\Delta(\Delta)$ is the diagonal embedding $H^\bullet_\Delta(\mathsf{H}^{\mathrm{Ar}}) \to H^\bullet_\Delta(\mathsf{H}^{\mathrm{Ar}} \oplus \mathsf{H}^{\mathrm{Ar}}) = H^\bullet_\Delta(\mathsf{H}^{\mathrm{Ar}}) \oplus H^\bullet_\Delta(\mathsf{H}^{\mathrm{Ar}})$ of the arithmetic cohomology; in degree 0 it is $x \mapsto (x, x)$ on $\mathbb{C}_0 \to \mathbb{C}_0 \oplus \mathbb{C}_0$; similarly for H^2; and it sends the middle H^1-eigenvector of the Weil-operator Θ_ξ (the functional-equation involution of Equation 44.3.2) at ordinate γ_ρ to the pair of two copies of the same eigenvector.

(ii) $H^\bullet_{\mathrm{ch}}(\Delta)$ is the induced chain map of Platonic bars, compatible with the bar differential because Δ preserves the spherical units and is a homomorphism of chiral algebras.

(iii) $\partial \circ \mathrm{BdryACS}(\Delta)$ is the boundary restriction of the bulk interface $\mathrm{BdryACS}(\Delta)\colon \mathrm{ACS}^{\mathrm{qu}}_{\mathsf{H}^{\mathrm{Ar}}} \to \mathrm{ACS}^{\mathrm{qu}}_{\mathsf{H}^{\mathrm{Ar}}} \oplus \mathrm{ACS}^{\mathrm{qu}}_{\mathsf{H}^{\mathrm{Ar}}}$, which in the 3D ACS framework is the diagonal bulk interface defect connecting one bulk theory to the disjoint union of two copies of itself.

All three commute with the natural isomorphisms $\eta_{\Delta,\mathrm{ch}}$, $\eta_{\mathrm{ch},\partial}$.

Proof. A rank-2 Heisenberg conformal vector has central charge 2; the coproduct of two such Heisenberg factors has conformal vector $T \oplus T$ and central charge 4. If the diagonal were conformal, the OPE of the

image of T with itself would have to preserve the central term. It instead doubles it. Hence no conformal $\mathsf{ArChAlg}$-morphism has the displayed diagonal as its underlying map.

After forgetting the conformal vector, $\Delta^{(p)}$ commutes with ψ_p because $\psi_p(x,x) = (\psi_p(x), \psi_p(x))$. The spherical unit is preserved: $\Delta^{(v)}(u^{(v)}) = (u^{(v)}, u^{(v)})$. The adjacent product on the coproduct is the diagonal of the adjacent product on each summand; Δ commutes with this by construction. The tame-symbol boundary ∂^{PG} is linear, so $\partial^{\mathrm{PG}}(x,x) = (\partial^{\mathrm{PG}}(x), \partial^{\mathrm{PG}}(x))$. Tate compatibility is preserved because the spherical units are preserved and the restricted tensor factors through each summand.

Claim (1). The Deninger cohomology of the coproduct $\mathsf{H}^{\mathrm{Ar}} \oplus \mathsf{H}^{\mathrm{Ar}}$ is $H^\bullet_\Delta(\mathsf{H}^{\mathrm{Ar}}) \oplus H^\bullet_\Delta(\mathsf{H}^{\mathrm{Ar}})$ because arithmetic bar commutes with direct sums on the primary domain (the bar is additive in its argument, a consequence of Definition 58.3.1 and of the additivity of the tame-symbol boundary). The induced map is the diagonal. In each degree: $H^0 = \mathbb{C}_0 \to \mathbb{C}_0 \oplus \mathbb{C}_0$, $x \mapsto (x,x)$; $H^2 = \mathbb{C}_1 \to \mathbb{C}_1 \oplus \mathbb{C}_1$, $x \mapsto (x,x)$; H^1-eigenvectors at ordinate γ_ρ go to the pair of two copies.

Claim (2). The Platonic bar is natural in A by Definition 109.2.2; after forgetting the conformal vector, applying naturality to Δ gives the chain map on bars. Bar differential compatibility is the Parshin–Gersten boundary compatibility proved above.

Claim (3). The bulk-boundary assignment $\mathrm{BdryACS}(\Delta)$ is by Definition 109.2.3 the non-conformal 3D bulk interface whose archimedean boundary restriction is $\Delta^{(\infty)}$ (the diagonal map of archimedean E_1-chiral algebras) and whose finite-place Wilson-line restriction is $\{\Delta^{(p)}\}$. The boundary chiral homology ∂ of this interface is the induced map on boundary chiral homology. By the additivity of boundary chiral homology for coproducts in $\mathsf{3DACS}_{\overline{\mathrm{Spec}\,\mathbb{Z}}}$ (the disjoint union of two bulk theories has as boundary the coproduct of the two boundary chiral algebras), the boundary map is the diagonal on the rank-2 chiral homology.

Compatibility with η. By construction each of the three maps agrees with the others under the natural isomorphisms: they are all the diagonal of the same underlying primary Heisenberg cohomology object, related by the chain-level quasi-isomorphism Ξ of Theorem 109.3.2 on each direct summand and the identity on the direct-sum structure. □

Remark 109.6.4 *(what the diagonal example witnesses).* The diagonal Δ is the simplest non-trivial morphism after the conformal vector is forgotten: it is not an isomorphism, not a projection, not a single-factor inclusion. It tests additivity of the three comparison functors, but it does not test the conformal axiom of $\mathsf{ArChAlg}$. Because the coproduct $\mathsf{H}^{\mathrm{Ar}} \oplus \mathsf{H}^{\mathrm{Ar}}$ sits in $\mathsf{ArChAlg}_{g=2}$ (rank-4 archimedean Heisenberg) while H^{Ar} sits in $\mathsf{ArChAlg}_{g=1}$, the example also shows why genus stabilization is not an arbitrary diagonal embedding: the conformal central charge records the added rank.

Domain of the master theorem

The local chiral, finite-place and bulk structures must have the specified common coefficient realizations. Their comparison uses the supplied natural stage maps and all their compatibility identities.

Sign consistency. The Koszul signs in the Platonic bar differential, the alternating signs in the ordered-incidence bar, and the 3D framed-cobordism signs in the boundary-chiral-homology pairing are mutually consistent under the chain-level identification Ξ of Theorem 109.3.2.

Inherited conditional hypotheses. The well-definedness of $\partial \circ \mathrm{BdryACS}$ as a functor (rather than a mere correspondence) requires a well-defined bulk interface defect category, conditional on Theorem 103.4.4 and the renormalised profinite limit of Conjecture 103.4.7. Both conditionals are inherited from Chapter 103.

Spectral domain. The master theorem claims the *functoriality* of the triple identification, not the spectral content. The three spectral statements (Deninger purity, Riemann Hypothesis, F-RH) remain conditional on H-P^{Ar} and the archimedean A-TP hypothesis; neither is asserted here.

109.7 Domain of the Master Theorem

Functoriality residuals of Chapter 104

The following implications hold on the conditional domain of the master theorem:

(i) Section 104.12(5): triple-identification functoriality under pullback along number-field extensions $K/\mathbb{Q}$. Any such pullback is a morphism in $\mathsf{ArChAlg}$ (it preserves spherical units, adjacent products, and Parshin–Gersten boundaries by classical number-field theory), and functoriality follows from Corollary 109.4.2.

(ii) Section 104.12(6): genus-g Wilson-line Frobenius identification. On the residual domain of Theorem 109.5.3, the genus-g restriction of $\partial \circ \mathrm{BdryACS}$ is functorially identified with the genus-g Deninger cohomology; the rank-2g Wilson-loop observables then identify with the rank-2g $\psi_p^{(g)}$ by naturality in the direct summand structure.

Inherited conditional hypotheses of the master theorem

The master functoriality theorem Theorem 109.6.1 is conditional on the following inherited hypotheses:

(i) The common coefficient realizations and natural stage quasi-isomorphisms of Theorem 109.3.2, including differential, residue, transition and genus compatibility. Arithmetic Positselski (Chapter 75) alone does not construct the required finite-place comparison maps.

(ii) 3D quantum ACS existence (Conjecture 103.4.7) for the well-definedness of BdryACS as a functor.

(iii) Local-trace comparison (Theorem 103.4.9) for the boundary identification at ∞.

(iv) H-P^{Ar} (Hypothesis 44.3.2) for the spectrum identification on H^1; not used in functoriality but used in spectral content.

(v) The stabilization maps, comparison homotopies, and Künneth compatibility of Theorem 109.5.2, with continuity for any completed tensors. These are additional arithmetic comparison data.

These hypotheses specify the domain of the functoriality theorem.

The master theorem and what it forces

On the conditional domain, the triple comparison $H^\bullet_\Delta = H^\bullet_{\mathrm{ch}} = \partial \circ \mathrm{BdryACS}$, previously isolated object-wise for H^{Ar} at genus $g = 0$, is a natural isomorphism of genus-filtered functors on the stated subcategory of $\mathsf{ArChAlg}$: every morphism of arithmetic chiral algebras induces a commuting triple-square of cohomology maps, Frobenius-equivariant and Poincaré-dual-compatible; every genus-inclusion and retraction is intertwined by Künneth natural transformations $\vartheta^\star_g$. The chain-level quasi-isomorphism Ξ of Theorem 109.3.2 is induced by the supplied natural stage quasi-isomorphisms on common coefficients, intertwining the differentials, residue couplings and transitions. Construction of those maps is an additional arithmetic-chiral problem. The remaining spectral content (that the Frobenius F on H^1 has eigenvalues at the nontrivial zeros of $\xi(s)$) requires the inherited H-P^{Ar} hypothesis (Hypothesis 44.3.2) and is not supplied here.

Primary Sources

References for the local structures and comparison problems follow.

Arithmetic chiral algebra side. Beilinson–Drinfeld 2004, *Chiral Algebras* (AMS Colloq. 51); Francis 2013 (geometric E_1-ordered bar); Francis–Gaitsgory 2012 arXiv:1103.5803 (factorisation gluing); Costello–Gwilliam 2017, *Factorization Algebras in Quantum Field Theory* (Cambridge UP) for the archimedean E_1-chiral factorisation algebra.

Finite-prime arithmetic side. Fargues–Scholze arXiv:2102.13459 (geometrisation of local Langlands); Bhatt–Scholze 2022, *Prisms and prismatic cohomology*, Ann. Math. for the prismatic/Nygaard comparison; Borger 2009 arXiv:0906.3146 for the Bost–Connes Adams ψ_p; Positselski 2011, Mem. AMS 996, *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence* for the relevant duality formalism. The finite-prime stage comparison maps are additional constructions.

3D ACS / bulk-boundary side. Kim 2015 arXiv:1510.05818, *Arithmetic Chern–Simons Theory I*; Chern–Dan–Kim–Park 2018 arXiv:1609.03012 (non-abelian extension); Mazur 1973 ASENS 6 (étale cohomological dimension of number rings); Morishita 2012 arXiv:0904.3399, *Knots and Primes*; Witten 1989, CMP 121, and Witten 2007 arXiv:0706.3359 (3D gravity / 3D CS boundary WZW); Artin–Verdier duality per SGA 4.5.

Genus and modular side. Consani–Marcolli 2006 (genus-g Bost–Connes–Marcolli); Faltings 1984, Ann. Math. 119, *Calculus on arithmetic surfaces*; Kudla 1997, Ann. Math. 146; Li–Liu 2021 arXiv:2101.09485 (Kudla–Rapoport at bad primes); Arthur 2013, *The Endoscopic Classification of Representations*, AMS Colloq. 61.

Further references on the geometric, prismatic and arithmetic coefficient structures are collected in Sections A.8–A.7, A.19 and A.6. The quantum bulk-boundary comparison is the separate construction problem of Chapter 103.

Chapter 110

Wilson-Line Frobenius and Archimedean Wilson-Surface at Genus g

Chapter 102 states the conditional all-genus arithmetic Koszul duality reduction $\kappa_{\mathsf{G}}^{\mathrm{Ar},g} + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar},g} = 0$ *on an arithmetic genus-*g *datum* $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$*, with the genus entering through the rank-*$2g$ *Fock grading, the archimedean* $\Gamma_{\mathbb{C}}(s)^g$ *factor, and the motivic weight* $w_{\mathcal{D}_g}$*, equal to* 1 *in the zeta window,* 2 *in the curve* H^1 *window, and* $w_{\pi,\mathrm{spin}}$ *in the Siegel spin window. Chapter 103 constructs finite-level arithmetic Chern–Simons state sums and states the profinite quantum ACS hypotheses on* $\overline{\mathrm{Spec}\,\mathbb{Z}}$*, including the Wilson-line Frobenius axiom, here written in the geometric normalization used by Euler factors,* $\langle W_\rho(K_p)\rangle = \mathrm{tr}\,\rho(\mathrm{Frob}_p^{\mathrm{geom}})$ *at* $g = 0$*. Chapter 104 separates the ordered-incidence candidate from Deninger's conjectural arithmetic cohomology; their identification and the global Frobenius* $F_\infty \otimes \bigotimes'_p \psi_p$ *remain conditional on the spectral-flow and boundary-comparison hypotheses. The genus-*g *extension replaces the gauge group* $GL_2(\widehat{\mathbb{Z}})$ *by its rank-*$2g$ *analog* $GL_{2g}(\widehat{\mathbb{Z}})$*; at each prime* p *the defect operator becomes the genus-*g *Wilson line, and the archimedean boundary operator becomes the genus-*g *Wilson-surface* $F_\infty^{(g)}$*. The genus-*g *gauge group is the profinite limit of* $GL_{2g}(\mathbb{Z}/N\mathbb{Z})$*; the rank-*$2g$ *local representation* $\rho_g : G_{\mathbb{Q}_p} \to \mathrm{GSp}_{2g}$ *sources a genus-*g *Wilson loop whose expectation in* $\mathrm{ACS}^{\mathrm{qu},g}$ *is* $\mathrm{tr}_{\rho_g}(\mathrm{Frob}_p^{\mathrm{geom}})$*; the archimedean Wilson-surface requires a quantum boundary realization and a primary pairing comparison. The normal-generator transport uses the specified operators and domains of Chapter 110.4.4. A Virasoro representation, its invariant domain and its central charge remain additional data. The all-genus Deninger identification is conditional; when its hypotheses are imposed it is the motivic-weight-*$w_{\mathcal{D}_g}$ *specialisation of the character identity of Chapter 104.9.2. At* $g = 0$ *the genus family degenerates to the rank-zero datum. The rank-two Wilson-line theory of the Riemann framework is an independent ACS sector, not the value of* $\rho_{g,p}$ *at* $g = 0$*. The compatibility with Chs. 106, 111, 44, and 105 is recorded in Chapter 110.6.1.*

110.1 Genus-g Arithmetic Data Recap

Fix $g \geq 0$ and an arithmetic genus-g datum $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$ as in Chapter 102.1.1. Four structural features of $\mathcal{D}_g$ are used throughout.

(i) *Local Galois representation and Frobenius convention.* Fix $\ell \neq p$ at a prime of good reduction. In the Jacobian row let $V_\ell J = T_\ell J \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ and use

$$V_{g,p} = H^1_{\text{ét}}(J_{\overline{\mathbb{Q}}_p}, \mathbb{Q}_\ell) = \mathrm{Hom}(V_\ell J, \mathbb{Q}_\ell)$$

as the carrier of $\rho_{g,p}$. Arithmetic Frobenius acts on the residue field by $x \mapsto x^p$; geometric Frobenius is its inverse. The Tate-module Weil pairing takes values in $\mathbb{Q}_\ell(1)$. Its dual polarization on $V_{g,p}$ takes values in $\mathbb{Q}_\ell(-1)$, on which geometric Frobenius acts by p. Thus the cohomological geometric convention has paired eigenvalue products p; the Tate-module geometric convention has products p^{-1}. For another row, its realization, coefficient field, rank and polarization must

be supplied separately. A rank-$2g$ symplectic realization of weight $w_{\mathcal{D}_g} - 1$ uses a pairing into $\mathbb{Q}_\ell(1 - w_{\mathcal{D}_g})$, or its specified coefficient extension. A prismatic crystal requires a comparison to the chosen realization. In the Siegel row the standard rank is $2g + 1$ and the spin rank is 2^g; neither is identified with a rank-$2g$ Jacobian realization.

(ii) *Satake parameters.* In the chosen unramified realization, let $\alpha_j^{(p)}$ be the eigenvalues of geometric Frobenius. In the rank-$2g$ polarized case they may be paired so that $\alpha_j^{(p)}\alpha_{2g+1-j}^{(p)} = p^{w_{\mathcal{D}_g}-1}$. Indeed invariance of the pairing up to this scalar pairs each nonzero eigenvalue with the scalar divided by it, with multiplicity. The local factor is $\det(1 - p^{-s}\rho_{g,p}(\mathrm{Frob}_p^{\mathrm{geom}}))^{-1}$. For spin or other gauge labels the determinant uses that actual representation. At good reduction inertia acts trivially, so its trace is the rank, not generally the Frobenius trace. For example $y^2 = x^3 - x$ over $\mathbb{F}_5$ has eight points: the five affine fibres have respectively 1, 1, 2, 2, 1 points. The arithmetic Tate-module trace is $5 + 1 - 8 = -2$ and determinant 5; its geometric trace is $-2/5$ and determinant $1/5$. Dualizing recovers trace -2 and determinant 5 for geometric Frobenius on H^1. The pairing convention is the one in Milne, *Elliptic Curves*, second edition, V §8, p. 221.

(iii) *Rank-$2g$ Fock oscillators per place.* The local factor $\mathcal{H}^{(p),g}$ of $\mathcal{V}_{\mathrm{Heis}}^{\mathrm{prim},g} = \otimes'_p\,\mathcal{H}^{(p),g} \otimes \mathcal{H}^{(\infty),g}$ is the rank-$2g$ Fock whose character is the local Euler factor above. The archimedean factor $\mathcal{H}^{(\infty),g}$ carries the rank-$2g$ archimedean Heisenberg with Γ-factor $\Gamma_{\mathbb{C}}(s)^g$ at weight 1 (Windows I, V) or the corresponding Γ-factor of Chapter 102.1 at Dedekind or Siegel-spin weight.

(iv) *Motivic functional-equation centre $w_{\mathcal{D}_g}$.* The completed L-function $\Lambda_{\mathcal{D}_g}(s)$ satisfies $\Lambda_{\mathcal{D}_g}(s) = \epsilon_{\mathcal{D}_g}\Lambda_{\mathcal{D}_g}(w_{\mathcal{D}_g} - s)$ by Chapter 102.4.1; purity lies on $\mathrm{Re}(s) = w_{\mathcal{D}_g}/2$.

Remark 110.1.1 *(degeneration at $g = 0$).* The rank-zero local system has trace zero. Its symmetric algebra is the scalar line with character one, and its empty Fock tensor is $\mathbb{C}$. The separately defined area form $\mu_{\mathrm{Ar},0} = 0$ gives area scalar one. A nontrivial Riemann Tate sector and a GL_2-valued Wilson label are additional inputs. Their scalar completed character, when specified, is $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. The empty tensor contributes only the factor one to that product. Its tensor-unit comparison is given in Chapter 110.6.1; a native Tate surface additionally requires the quantization and primary data of Chapter 111.5, Chapter 111.5.

110.2 Genus-g 3D Arithmetic Chern–Simons Theory

Chapter 103 builds the 3D arithmetic Chern–Simons theory on $\overline{\mathrm{Spec}\,\mathbb{Z}}$ with gauge group $GL_2(\widehat{\mathbb{Z}})$ in the profinite limit, level $k^{\mathrm{Ar}} = \kappa_{\mathsf{G}}^{\mathrm{Ar}}$, boundary WZW $\mathcal{V}^{\mathrm{prim}}$, and Wilson-loop axiom Chapter 103.4.7(3). The datum $\mathcal{D}_g$ determines a rank-$2g$ gauge group $GL_{2g}(\widehat{\mathbb{Z}})$, a rank-$2g$ Koszul level $k^{\mathrm{Ar},g}$, and a rank-$2g$ boundary Heisenberg $\mathcal{V}^{\mathrm{prim},g}$; the following definition records these data and inherits the conditional form of the $g = 0$ theory.

Definition 110.2.1 *(genus-g arithmetic Chern–Simons).* . The *genus-g arithmetic Chern–Simons theory* is a family of 3D topological gauge theories on an arithmetic 3-space $\mathcal{B}_g^{(3)}$ parameterised by $\mathcal{D}_g$, with

(i) *Arithmetic 3-space.* For Window II at totally imaginary K of degree $2g$: $\mathcal{B}_g^{(3)} = \overline{\mathrm{Spec}\,\mathcal{O}_K}$ with Artin–Verdier fundamental class $[\mathcal{B}_g^{(3)}] \in H^3_{\text{ét}}(\mathcal{B}_g^{(3)}, \widehat{\mathbb{Z}})$ (Mazur 1973). For Window I at $X \to \mathrm{Spec}\,\mathcal{O}_K$: $\mathcal{B}_g^{(3)} = \overline{\mathrm{Spec}\,\mathcal{O}_K}$ with extra fibre cohomology in H^2 of the generic fibre collected into the archimedean boundary via the Arakelov Kähler form $\mu_{\mathrm{Ar},g}$ of Faltings 1984 (to be defined in Chapter 110.4.1). For Window VI, Shimura varieties of GSp_{2g} have their own Artin–Verdier-type

fundamental class assembled via étale cohomology of the special fibre plus Deligne 1974 Weil cohomology.

(ii) *Gauge group.* $G_{g,N} := GL_{2g}(\mathbb{Z}/N\mathbb{Z})$, with profinite limit $G_{g,\infty} := GL_{2g}(\widehat{\mathbb{Z}})$.

(iii) *Level.* $k^{\mathrm{Ar},g} := \kappa_{\mathsf{G}}^{\mathrm{Ar},g}(\mathcal{D}_g)$, the rank-$2g$ arithmetic Koszul level of Chapter 102.4.1.

(iv) *Boundary WZW at the archimedean fibre.* $\partial\,\mathrm{ACS}^{\mathrm{qu},g}_{k^{\mathrm{Ar},g}} = \mathcal{V}^{\mathrm{prim},g}$ in the sense of Chapter 103.4.9 scaled to rank $2g$ per Chapter 103.8.

The finite-level state sums $Z^{\mathrm{Ar}}_{G_{g,N},\alpha_{g,N}}$ extend Chapter 103.4.2 unchanged at each finite rank; the profinite renormalisation $\mathrm{ACS}^{\mathrm{qu},g}_{k^{\mathrm{Ar},g}}$ remains conditional in the sense of Chapter 103.4.7 extended to rank $2g$.

Remark 110.2.2 *(rank scaling and residual profinite limit).* The rank-change $2 \to 2g$ does not alter the closed finite-level state-sum of Chapter 103.4.2: the field groupoid $\mathcal{F}_{G_{g,N}}(\mathcal{B}^{(3)}_g)$ is a finite groupoid for each N, the action $S^{\mathrm{Ar}}_{\alpha_{g,N}}$ is the same Artin–Verdier pairing, and the gluing formula follows from étale descent of rank-$2g$ local systems. The profinite limit $\mathrm{ACS}^{\mathrm{qu},g}$ shares the profinite-renormalisation residual of Chapter 103.4.7.

For the partition comparison, a homology object must first have a scalar character. Let $H^i_g = H_i(B^{\mathrm{ord},g})$ have specified Hilbert realizations and a holomorphic family of trace-class endomorphisms $T_i(s)$ on an open half-plane Ω. Assume $\sum_i \|T_i(s)\|_1$ converges locally uniformly there, and set $\chi_g(s) = \sum_i (-1)^i \operatorname{Tr} T_i(s)$. The trace and the sum then define a holomorphic scalar on Ω. The required character comparison is $\chi_g(s) = \Lambda_{\mathcal{D}_g}(s)$ on Ω, with the exact row-specific gamma, conductor and pole factors. If a regularized character is used instead, its prescription and normalization must be specified and this same scalar identity proved on a common domain. Neither expression identifies homology itself with a function.

Proposition 110.2.3 *(partition function at genus g). (on the genus-g extension of Chapter 103.4.7(2) and the genus-g BD/CG comparison of Chapter 103.5.5).* Suppose the scalar character comparison just specified holds, and the bulk-boundary comparison identifies the partition scalar with $\chi_g(k^{\mathrm{Ar},g})$ on Ω, with matching meromorphic continuations when evaluation is outside Ω and away from poles. Then the partition function of $\mathrm{ACS}^{\mathrm{qu},g}_{k^{\mathrm{Ar},g}}$ on $\mathcal{B}^{(3)}_g$ equals the completed L-function

$$Z_{\mathrm{ACS}^{\mathrm{qu},g}}\big(\mathcal{B}^{(3)}_g; k^{\mathrm{Ar},g}\big) \;=\; \Lambda_{\mathcal{D}_g}\big(k^{\mathrm{Ar},g}\big),$$

with $\Lambda_{\mathcal{D}_g}$ the row-specific completed L-function of Chapter 102.1 (Dedekind ξ_K, Hasse–Weil $\Lambda(s, X_K)$, or Siegel spin $\Lambda(s, \pi, \mathrm{spin})$).

Proof. The rank-$2g$ ordered incidence bar $B^{\mathrm{ord},g}_\bullet$ on $\mathbb{P} \cup \{\infty\}$ with per-place factor $\mathcal{H}^{(p),g}$ is the rank-$2g$ generalisation of the bar of Chapter 103.5.3. The local character at p is $\chi_p^{(g),\mathrm{std}}(s) = \prod_{j=1}^{2g}(1 - \alpha_j^{(p)} p^{-s})^{-1}$ for the standard Wilson observable. In the Siegel spin row the local character is instead the spin Euler factor $\chi_p^{(g),\mathrm{spin}}(s) = \prod_\epsilon (1 - \beta_\epsilon^{(p)} p^{-s})^{-1}$ over the weights of the spin representation. The archimedean character is the row-specific Γ-factor. For a finite diagonal symmetric-algebra factor, the character $\prod_j (1 - \alpha_j^{(p)} p^{-s})^{-1}$ follows by multiplying the convergent geometric series, where $|\alpha_j^{(p)} p^{-s}| < 1$. Passing to the ordered bar homology and the infinite product requires the specified scalar character comparison, including its convergence and archimedean normalization. Under that hypothesis and the actual bulk-boundary trace map, $Z(k) = \chi_g(k) = \Lambda_{\mathcal{D}_g}(k)$ on Ω. Uniqueness of meromorphic continuation gives the claimed extension wherever the prescribed scalar is finite. □

110.3 WILSON-LINE FROBENIUS IDENTITY AT GENUS g

The gauge group $G_{g,\infty} = GL_{2g}(\widehat{\mathbb{Z}})$ of $\mathrm{ACS}^{\mathrm{qu},g}$ contains the prime knots K_p as defect insertions. Each K_p carries a holonomy valued in the rank-$2g$ local Galois representation $\rho_{g,p}$; the question is whether the expectation of the associated defect operator recovers geometric Frobenius $\mathrm{Frob}_p^{\mathrm{geom}}$. Arithmetic Frobenius is its inverse in this convention. The trace identity below is conditional on the specified Wilson comparison, including compatible coefficient fields and local labels.

A Wilson label is a representation of the gauge group, whereas $\rho_{g,p}$ is a representation of a local Galois group. At finite level specify a representation $r_{g,N}$ of $G_{g,N}$ over a characteristic-zero coefficient field and a comparison of its holonomy trace with the trace of $\rho_{g,p}$ at Frobenius. The traces must be placed in a common specified coefficient field. A profinite label requires compatible finite labels and convergent expectations. A Hilbert-space interpretation additionally requires a complex realization and its positive pairing. These are data of the Wilson comparison, not consequences of equality of dimensions.

Definition 110.3.1 *(genus-g Wilson line).* . Fix a prime p of good reduction for $\mathcal{D}_g$; let $K_p \subset \mathcal{B}_g^{(3)}$ denote the prime-knot associated to p under the Morishita dictionary. For a continuous representation $\rho_g : G_{\mathbb{Q}_p} \to \mathrm{GSp}_{2g}(\overline{\mathbb{Q}}_\ell)$ of rank $2g$, the *genus-g Wilson line* along K_p is the defect operator

$$W_{\rho_g}^{(g)}(K_p) \;:=\; \mathrm{tr}_{r_g}\big(\mathrm{Hol}_{K_p}(\mathsf{A})\big),$$

Here ρ_g indexes the matched gauge label r_g. The compatible bulk field is $\mathsf{A} \in \mathcal{F}_{G_{g,\infty}}(\mathcal{B}_g^{(3)})$. Its Frobenius-longitude holonomy, distinct from the inertia meridian, and the comparison with the local Bost–Connes Adams operation $\psi_p^{(g)}$ must be supplied by the chosen arithmetic gauge construction.

THEOREM 110.3.2 *(genus-g Wilson-line Frobenius identity).* . Fix the cohomological geometric Frobenius convention above. At each finite gauge level use an actual representation $r_{g,N}$ of $G_{g,N}$ and the normalized finite Wilson expectation of its character on a specified longitude. Assume the arithmetic gauge comparison identifies the limiting expectation, with its scalar coefficient realization and topology, with the character of $\rho_{g,p}(\mathrm{Frob}_p^{\mathrm{geom}})$. For an ℓ-adic construction the following proof specifies the congruence and scalar comparison data required by this assumption. The genus-g Wilson axiom and the chosen unramified local Langlands comparison remain the arithmetic input; equality of representation ranks does not supply it. For every unramified prime p of good reduction for $\mathcal{D}_g$,

$$\langle W_{\rho_g}^{(g)}(K_p)\rangle_{\mathrm{ACS}^{\mathrm{qu},g}} \;=\; \mathrm{tr}_{\rho_g}\big(\mathrm{Frob}_p^{\mathrm{geom}}\big) \;=\; \sum_{j=1}^{2g} \alpha_j^{(p)}.$$

Equivalently, the Wilson-line defect expectation equals the character value $\chi_{\rho_g}(\mathrm{Frob}_p^{\mathrm{geom}})$ in the standard representation. The Siegel spin L-factor is obtained from the spin Wilson observable, not from this rank-$2g$ standard trace.

Proof. A longitude represents a lift of Frobenius in the local quotient $G_{\mathbb{Q}_p}/I_p$; changing the lift does not change an unramified representation. A meridian represents inertia and has trace $2g$ on this carrier. These two observables must therefore be distinguished in the arithmetic-topology comparison of Mazur and Morishita.

For the congruence construction, choose a finite extension $E/\mathbb{Q}_\ell$ over which the representation is defined and a stable $\mathcal{O}_E$-lattice L of rank $2g$. For a uniformizer ϖ, the reductions act on $L/\varpi^n L$ and their traces are compatible in $\mathcal{O}_E/\varpi^n$. Completeness identifies their inverse limit with the trace of the

original matrix in O_E. This is an ℓ-power congruence tower; it supplies neither a representation modulo every integer N nor a complex character of the finite gauge group $G_{g,N}$.

Those gauge characters are $\mathrm{tr}_{r_{g,N}}$ by definition. To use their normalized finite field sums, one needs specified coefficient realizations and a comparison of the scalar expectations with the preceding congruence traces. If the final trace is in a number field K, its embeddings into E and into $\mathbb{C}$ must be fixed; ℓ-adic convergence alone does not imply convergence of the complex expectations. The assumed limiting Wilson comparison includes the required topology and complex scalar identity. It gives the first equality in the theorem. The second is the elementary matrix identity that the trace is the sum of eigenvalues with algebraic multiplicity. An identification with a Bost–Connes Adams operation further requires its actual representation and intertwiner; Borger's λ-ring operations do not themselves construct that map on this cohomological carrier. □

Remark 110.3.3 *(local normalizations and comparison data).* The proposed local identifications involve Borger 2009, §8, on λ-rings, Fargues–Scholze 2021 on geometric local Langlands, and Mazur 1973 and Morishita 2012 on arithmetic topology. A comparison must specify how their objects define the gauge label and the local arithmetic operation in the trace equality.

The Satake parameter normalisation is: unramified $\rho_{g,p}$ has eigenvalues $\alpha_j^{(p)}$ with symplectic pairing $\alpha_j^{(p)}\alpha_{2g+1-j}^{(p)} = p^{w_{\mathcal{D}_g}-1}$; the Frobenius trace is $\sum_{j=1}^{2g}\alpha_j^{(p)}$, the character value of ρ_g itself at $\mathrm{Frob}_p^{\mathrm{geom}}$, not the character of $\bigwedge^k \rho_g$ for any $k \geq 2$.

The Wilson-line defect at finite level lies in $\mathrm{Sh}_{\tau_\Sigma(\alpha_{g,N})}(\mathcal{F}_{G_{g,N}}(\Sigma))$ for $\Sigma = \partial(\text{tubular neighbourhood of } K_p)$ per Chapter 103.4.4; the rank-$2g$ setting is structurally identical. The Wilson line is functorial under tensor product of representations at the classical level: $W^{(g)}_{\rho_1\otimes\rho_2} = W^{(g)}_{\rho_1}\cdot W^{(g)}_{\rho_2}$; at the quantum level this uses the factorisation algebra structure of $\mathcal{V}^{\mathrm{prim},g}$ (rank-$2g$ Heisenberg), consistent with Ch. 102 Window V.

The residual hypotheses are: (a) profinite renormalisation of $\{Z^{\mathrm{Ar}}_{G_{g,N},\alpha_{g,N}}\}_N$ and (b) compatible tower $\{\alpha_{g,N}\}_N$, inherited from Chapter 103.9(2); and (c) rank-$2g$ local Langlands at p, i.e. Chapter 102.6(5). The longitude/gauge comparison additionally requires the explicit coefficient realizations, finite characters and limit topologies specified in the theorem. The Artin–Verdier pairing of Chapter 103.4.1 defines an action only after its coefficient and fundamental-class data have been supplied; it does not by itself identify a holonomy representation with $\rho_{g,p}$. Good reduction makes the chosen ℓ-adic representation unramified for $\ell \neq p$.

110.4 Archimedean Wilson-Surface at Genus g

A surface area determines a scalar observable. Relating it to a boundary scaling operator requires a quantization and an operator comparison. The curve case supplies a precise area normalization.

Definition 110.4.1 *(normalized curve area).* For a compact connected Riemann surface X of genus $g \geq 1$, choose a basis $\omega_1, \ldots, \omega_g$ orthonormal for $\langle \omega, \eta\rangle = (i/2)\int_X \omega\wedge\bar{\eta}$. With the complex orientation, set

$$\mu_{\mathrm{Ar},g} = \frac{i}{2g}\sum_{j=1}^{g}\omega_j\wedge\overline{\omega_j}.$$

This form is basis independent and has integral one. For the rank-zero degeneration define $\mu_{\mathrm{Ar},0} = 0$ separately. This is not the unit-area metric on a genus-zero curve. A Minkowski torus or a Shimura

variety requires its own specified oriented two-cycle and two-form. Its Riemannian volume need not have degree two.

Basis independence follows from the unitary change-of-basis identity. Each summand has integral $1/g$, so their sum has integral one. An arithmetic interpretation also requires a boundary object and a map realizing this surface and metric there, as in Chapter 111.5.

Definition 110.4.2 *(area scalar and quantum surface data).* For a specified oriented compact surface Σ_∞, real integrable two-form $\mu_{\mathrm{Ar},g}$, and real level $k = k^{\mathrm{Ar},g}$, define the area scalar

$$A_g(k) = \exp\!\left(-k \int_{\Sigma_\infty} \mu_{\mathrm{Ar},g}\right).$$

A quantum realization requires an arithmetic boundary Hilbert space $H_{\Sigma,g}$, its pairing, and a densely defined closed normal operator $A_{\Sigma,g}$ obtained from the boundary quantization. When its spectral exponential is bounded, write

$$F_\infty^{(g)}(k) = e^{-kA_{\Sigma,g}}.$$

The area scalar does not define $A_{\Sigma,g}$. The fields, gauge normalization, curvature coupling and boundary conditions are part of the required quantization.

Lemma 110.4.3 *(rank-g Mellin unitarity and domains).* For $g \geq 1$, set

$$H_{c,g} = L^2((\mathbb{R}_{>0})^g, \prod_j dx_j/x_j).$$

The transform

$$(\mathcal{M}_g f)(u) = (2\pi)^{-g/2} \int f(x) \prod_j x_j^{-iu_j} \prod_j \frac{dx_j}{x_j}$$

extends from compact smooth logarithmic tests to a unitary $H_{c,g} \to L^2(\mathbb{R}^g, du)$. Let $D_j = -i\partial_{t_j}$, where $t_j = \log x_j$, on its maximal weak-derivative domain. Then

$$\mathcal{M}_g D_j \mathcal{M}_g^{-1} = M_{u_j}, \qquad S_g = \mathcal{M}_g^{-1} M_{u_1+\cdots+u_g} \mathcal{M}_g, \quad \mathrm{Dom}\, S_g = \{f : (\textstyle\sum_j u_j)\mathcal{M}_g f \in L^2\}.$$

The operator S_g is the self-adjoint closure of $\sum_j D_j$ on compact smooth logarithmic tests. For real k,

$$C_g(k) := e^{-ikS_g}, \qquad (C_g(k)f)(x) = f(e^{-k}x_1, \ldots, e^{-k}x_g).$$

For a real centre c, the shifted Mellin kernel $\prod_j x_j^{c-iu_j}$ is unitary instead from $L^2(\prod_j x_j^{2c} dx_j/x_j)$ to $L^2(du)$, with the same factor $(2\pi)^{-g/2}$. It diagonalizes $-i(\partial_{t_j} + c)$ as M_{u_j} on the domain where $e^{c\sum t_j} f(e^t)$ has its j-th weak derivative in $L^2(dt)$.

Proof. Logarithmic coordinates identify Haar measure with Lebesgue measure. Fourier Plancherel and integration by parts prove the first identities, including the maximal multiplication domains. For the sum, rotate coordinates so that $\sum_j \partial_{t_j}$ is a multiple of one coordinate derivative. Compact smooth functions form a core by cutoff and convolution approximation in its graph norm. Fourier multiplication by $e^{-ik\sum_j u_j}$ gives the displayed translation. Multiplication by $e^{c\sum t_j}$ is an isometry from the weighted source to $L^2(dt)$; differentiating this product proves the shifted formula. With unweighted Haar measure a kernel $\prod_j x_j^{c+iu_j}$ instead has squared norm $\int |f(x)|^2 \prod_j x_j^{2c} dx_j/x_j$, so it is not unitary there when $c \neq 0$. □

Theorem 110.4.4 *(surface transport and the primary comparison).* For $g \geq 1$, assume the quantum data above and a unitary $V_g : H_{\Sigma,g} \to H_{c,g}$ with

$$V_g \operatorname{Dom} A_{\Sigma,g} = \operatorname{Dom} S_g, \qquad V_g A_{\Sigma,g} = i S_g V_g.$$

Then $F_\infty^{(g)}(k) = V_g^{-1} C_g(k) V_g$ is bounded and unitary for every real k. Suppose also that a specified primary Hilbert space $H_{\mathrm{prim},g}$ has a closed normal operator $A_{\mathrm{prim},g}$ and a unitary $T_g : H_{\mathrm{prim},g} \to H_{\Sigma,g}$ with $T_g \operatorname{Dom} A_{\mathrm{prim},g} = \operatorname{Dom} A_{\Sigma,g}$ and $T_g A_{\mathrm{prim},g} = A_{\Sigma,g} T_g$. Then

$$F_\infty^{(g)}(k) = T_g e^{-k A_{\mathrm{prim},g}} T_g^{-1}.$$

For $k \neq 0$ and $g \geq 1$, neither iS_g nor $C_g(k)$ has nonzero eigenvectors. Thus ordinary Fock energy cannot supply these intertwining data.

Proof. The domain identities give unitary equivalence of the normal operators. Spectral calculus gives both exponential identities. Under $\mathcal{M}_g$, an eigenvector of S_g would be supported on a hyperplane, which has Lebesgue measure zero. For $C_g(k)$, each eigenvalue preimage is a countable union of parallel hyperplanes, also of measure zero. A Fock vacuum is a nonzero eigenvector, so its image under an asserted intertwiner would contradict this fact. □

A native primary realization must specify the pairing used to prove T_g and the arithmetic operations it preserves. If Virasoro operators $L_n^{(g)}$ are intended, their common invariant domain and commutators must also be constructed. Central charge g is a separate representation assertion. Tensor Mellin unitarity does not imply it. If $L_0^{(g)}$ is realized as self-adjoint conformal energy, it cannot be unitarily conjugate to the nonzero skew-adjoint operator iS_g. Formal Virasoro adjoint relations on a common domain alone do not prove self-adjointness. The relation of $A_{\mathrm{prim},g}$ to $L_0^{(g)}$ therefore requires an explicit normalization and cannot identify these two operators directly. These are the rank-g versions of Chapter III.5.

Remark 110.4.5 *(level differentiation).* For a finite measure $d\nu$ and real measurable area a, suppose $Z(k) = \int e^{-ka}\, d\nu$ and differentiation under the integral is justified. Then $Z'(k) = -\int a e^{-ka}\, d\nu$. When $Z(k) \neq 0$, the normalized area expectation is $-\partial_k \log Z(k)$. This is a scalar insertion identity. It is not an equality between $Z'(k)$ and a boundary operator. For bounded a, dominated convergence justifies the formula on every compact real k-interval.

Remark 110.4.6 *(normalization and the arithmetic boundary).* The centered translation exponential is $C_g(k)$. Adding a shift c_j to each factor's normal generator gives $e^{-k\sum_j c_j} C_g(k)$. Thus a motivic normalization requires specifying every c_j, not merely naming a critical line. For the normalized curve, $A_g(k) = e^{-k}$ is scalar. When $k \neq 0$, a compact logarithmic test with sufficiently small support has disjoint support from its translate by $k(1, \ldots, 1)$. Consequently $C_g(k)$ is nonscalar and cannot equal that area scalar. The arithmetic geometry, quantization, primary pairing and generator comparisons remain necessary to realize the continuous formulas on the intended boundary objects. At finite gauge level the proposed native surface-defect carrier is $\mathrm{Sh}_{\tau(\alpha_{g,N})}(\mathcal{F}_{G_{g,N}}(\Sigma_\infty))$. Constructing it requires a defined field groupoid on the arithmetic archimedean surface, a transgressed twist $\tau(\alpha_{g,N})$ in its specified coefficient theory, and the resulting category of twisted sheaves. Its Hilbert realization and observable comparison are additional data. The rank-one tensor question asks for an equivalence from a specified tensor product of g rank-one defect categories to this category, compatible with their twists, pairings and Wilson observables. A decomposition of the Hodge vector space alone is not such a categorical equivalence. If the surface

geometry, action and measure split and these categorical and Hilbert comparison maps are constructed, the tensor product of the realized rank-one observables is the proposed genus-g observable. For Jacobian and Minkowski rows these compatibilities must be proved; the Siegel row additionally requires its actual Siegel–Maass metric and normalization.

110.5 Global Frobenius at Genus g and the Deninger Identification

The Wilson-line and Wilson-surface operators at each place are local. To obtain the Deninger character identity $\chi(\mathcal{V}^{\mathrm{prim},g})(s) = \Lambda_{\mathcal{D}_g}(s)$, one must assemble them into a single global operator on the conditional Deninger cohomology $H^{\bullet}_{\Delta,g}$. The restricted tensor product over all places is therefore a formal global Frobenius until the descent and spectral-flow hypotheses are imposed.

Definition 110.5.1 *(global genus-g Frobenius).* For a fixed real level k, the formal global symbol is

$$F^{(g)} := F_{\infty}^{(g)}(k) \otimes \bigotimes_{p}{}' \psi_p^{(g)}.$$

Its operator realization requires local Hilbert spaces, reference vectors and boundedness or convergence conditions. A sufficient construction is given in the following theorem. Identifying that Hilbert tensor with the primary arithmetic trace sector requires its native pairing and realization map. Its action on Deninger cohomology further requires a complex and an action-compatible chain comparison, with the hypotheses in Chapter 104.9.2.

Theorem 110.5.2 *(genus-g Wilson assembly of the global Frobenius).* Fix $g \geq 1$. Assume the finite-place Wilson trace comparisons of Chapter 110.3.2 and the native surface comparison of Chapter 110.4.4. For each prime choose a Hilbert space $H_{p,g}$, a unit reference vector $e_{p,g}$, and a bounded contraction $\psi_p^{(g)}$ fixing that vector. Then, for real k,

$$F^{(g)}(k) = F_{\infty}^{(g)}(k) \otimes \bigotimes_{p}{}' \psi_p^{(g)}$$

defines a contraction on $H_{\Sigma,g} \widehat{\otimes} \bigotimes'_p (H_{p,g}, e_{p,g})$. Its archimedean factor is the transported continuous operator, and its finite-place Wilson traces are the specified Frobenius traces. To regard this as an action on the arithmetic primary trace sector requires a unitary realization of that sector in this tensor product intertwining its local operations. Descent to a Deninger complex requires an action-compatible chain map and the stated completion.

Proof. On tensors equal to $e_{p,g}$ at all but finitely many primes, apply $\psi_p^{(g)}$ at each exceptional prime. Since every other factor fixes its reference vector, this is independent of the chosen finite set. The finite tensor maps are contractions and therefore extend to the Hilbert completion. Tensoring with the unitary surface factor preserves the norm bound. The local trace comparisons are the assumed Wilson identities. They identify scalar traces and do not themselves supply the primary unitary or the cohomological map. □

The following arithmetic identification assumes the native cohomological comparisons and determinant identity stated in Chapter 104.9.2 on the specified row. The surface and tensor constructions above do not construct them. For its purity clause, one must additionally identify the native middle-degree generator, with its domain and multiplicities, with the operator whose eigenvalues are the nontrivial zero

occurrences. Its Frobenius must be the corresponding exponential $e^{-\Theta^{(g)}}$. An equality of local scalar traces does not give these operator data.

Theorem 110.5.3 *(all-genus Deninger identification at motivic weight $w_{\mathcal{D}_g}$). (on Chapter 104.9.2(1)–(4), Chapter 110.3.2, Chapter 110.4.4, and Chapter 110.2.3, and the contraction and reference-vector hypotheses of Chapter 110.5.2).* For every $g \geq 1$ and every arithmetic genus-g datum $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$:

(i) *Triple labelling.* The Vol IV genus-g Deninger cohomology $H^\bullet_{\Delta,g}$, the boundary chiral homology of $\mathrm{ACS}^{\mathrm{qu},g}_{k^{\mathrm{Ar},g}}$ on $\mathcal{B}^{(3)}_g$, and the rank-$2g$ arithmetic chiral homology $H^\bullet_{\mathrm{ch}}(\mathcal{B}^{(3)}_g, \mathsf{H}^{\mathrm{Ar},g})$ (with $\mathsf{H}^{\mathrm{Ar},g}$ the arithmetic Heisenberg coefficient sheaf of Chapter 104.9.1) coincide on the primary rank-$2g$ Heisenberg domain.

(ii) *Character identity with motivic weight.*

$$\chi\big(\mathcal{V}^{\mathrm{prim},g}\big)(s) \;=\; \det{}_\infty\big(s\,\mathrm{id} - \Theta^{(g)} \mid H^\bullet_{\Delta,g}\big)^{(-1)^\bullet} \;=\; \Lambda_{\mathcal{D}_g}(s), \tag{110.1}$$

where $\Lambda_{\mathcal{D}_g}$ has motivic weight $w_{\mathcal{D}_g}$ per Chapter 102.1; in the Siegel spin row this is $w_{\pi,\mathrm{spin}}$, not a universal function of g.

(iii) *Wilson-observable form of the Frobenius.* The local factors of $F^{(g)}$ appearing in (110.1) admit the Wilson-line / Wilson-surface realisation of Chapter 110.5.2.

(iv) *Purity at* $\mathrm{Re}(s) = w_{\mathcal{D}_g}/2$. The genus-$g$ purity axiom of Chapter 104.1(6) becomes $F^{(g)}$-weight $w_{\mathcal{D}_g}/2$ on $H^1_{\Delta,g}$; equivalently the motivic Riemann Hypothesis for the row $\mathcal{D}_g$ (Chapter 102.5.1).

For $g = 0$, the surface factor is the tensor unit $\mathbb{C}$; only its unitary identification $\mathbb{C} \otimes H_T \cong H_T$ and the separately specified operator-domain transport are available. The same arithmetic conclusions in that case require the independent rank-zero cohomological, character and Wilson data. They do not follow by substituting $g = 0$ into the $g \geq 1$ continuous-generator comparison.

Proof. Item (1) is Chapter 104.9.2(1) unchanged, scaled to rank $2g$.

Item (2) uses the row-specific regularized determinant identity assumed in Chapter 104.9.2(3), with its cohomological degrees, signs, pole factors and normalization unchanged. A Mellin surface comparison supplies no regularized determinant identity. Realizing this determinant on the native middle cohomology also requires the complex, operator domain and determinant comparison specified in Chapter 111.5, for the chosen row.

Item (3) is Chapter 110.5.2.

For item (4), use the additionally specified native spectral comparison with the nontrivial zero occurrences ρ. On that carrier, $e^{-\Theta^{(g)}}$ has diagonal entries $e^{-\rho}$. Their absolute values are $e^{-\Re\rho}$, so common absolute value $e^{-w_{\mathcal{D}_g}/2}$ is exactly $\Re\rho = w_{\mathcal{D}_g}/2$ for every occurrence. This proves the stated equivalence under that comparison. Unconditional unitarity of the centered continuous operator $C_g(k)$ carries no such information about arithmetic zeros. □

Remark 110.5.4 *(arithmetic normalizations and cohomological hypotheses).* References for these distinct structures include Deninger 1998 on foliated dynamics, Bost–Connes 1995 on KMS systems, Fargues–Scholze 2021 on geometric local Langlands, Faltings 1984 on arithmetic surfaces and Arthur 2013 on endoscopic classification. Their comparison requires the stated native maps and normalizations. The motivic-weight normalisation is: Dedekind $w = 1$, Hasse–Weil $w = 2$, Siegel spin $w_{\pi,\mathrm{spin}}$, per Chapter 102.1; the determinant sign and regularization must be those of the specified row. A central charge g

requires a primary Virasoro representation with that central action. It is not implied by the number of Mellin variables.

The rank-$2g$ bar $B^{\mathrm{Ar},g}(\mathcal{V}^{\mathrm{prim},g})$ lives in graded $\mathbb{C}$-vector spaces with rank-$2g$ Heisenberg Fock grading; the E_1-ordered bar on $\mathbb{P} \cup \{\infty\}$ is rank-enhanced per Chapter 110.2.3. Functoriality under pullback along $\overline{\operatorname{Spec} O_K} \to \overline{\operatorname{Spec} \mathbb{Z}}$ requires $g = \frac{1}{2}[K : \mathbb{Q}]$ at totally imaginary K (Window II); for Shimura (Window VI), functoriality under Hecke correspondences holds because $F^{(g)}$ commutes with Hecke operators on the rank-$2g$ automorphic side.

The full residual list consists of all items of Chapter 102.6 (rank-g Positselski, rank-g Koszul–Petersson, GL_{2g} local geometric Langlands, rank-g Virasoro bootstrap, rank-g F2.c) and all items of Chapter 103.9 (profinite ACS renormalisation, boundary comparison, BD/CG comparison, positivity), each scaled to rank $2g$. The identification of the $H^1_{\Delta,g}$ spectrum with the non-trivial zeros of $\Lambda_{\mathcal{D}_g}$ requires rank-g H-P^{Ar}: a self-adjoint lift of $L_0^{(g)}$ to a Hilbert completion of $H^1_{\Delta,g}$ in the rank-g arithmetic Petersson inner product (Chapter 44.3.2 at rank g); this is open.

110.6 Uniform $g = 0$ Compatibility

The empty genus tensor is the tensor unit. To compare it with a nontrivial arithmetic theory, the latter must first be specified.

Corollary 110.6.1 *(rank-zero tensor comparison).* At $g = 0$, the empty Fock tensor is $H_0 = \mathbb{C}$, its centered generator is zero, and its exponential is 1_{H_0}. Let H_T be a separately specified Tate Hilbert space with closed normal generator A_T. The map

$$Q : \mathbb{C}\widehat{\otimes} H_T \longrightarrow H_T, \qquad Q(z \otimes h) = zh$$

is unitary. It maps $\mathbb{C} \otimes \operatorname{Dom} A_T$ onto $\operatorname{Dom} A_T$ and intertwines $1 \otimes A_T$ with A_T. If e^{-kA_T} is bounded,

$$Q(1 \otimes e^{-kA_T})Q^{-1} = e^{-kA_T}.$$

A nonzero rank-two Wilson label is independent of the rank-zero trace. A native Tate surface/Virasoro comparison holds only with the geometry, pairing, quantum observable and generator maps of Chapter III.5, Chapter III.5.

Proof. The empty tensor and sum define H_0 and its generator. The Hilbert tensor norm satisfies $\|z \otimes h\| = |z|\|h\|$, so Q is an onto isometry. Its inverse is $h \mapsto 1 \otimes h$. The domain and operator identities hold on $1 \otimes h$ and hence on the full stated domain. Functional calculus proves the exponential identity. The trace of the zero-dimensional identity is zero, whereas its symmetric-algebra character is one. Neither operation constructs a rank-two representation. □

If the supplied Tate character is $\Lambda_\zeta(s)$, adjoining the empty tensor leaves it unchanged. Multiplication by $s(s-1)/2$ gives $\xi_{\mathbb{R}}(s)$. This scalar identity does not construct an arithmetic cohomology complex or its determinant map. Those require the additional data of Chapter III.5. In particular, a choice of H_T, rather than the substitution $g = 0$, supplies any nontrivial continuous scaling or arithmetic zero operator.

Remark 110.6.2 *(Hasse–Weil and Siegel-spin RH at genus $g \geq 1$).* At $g \geq 1$, the Deninger-chiral purity locus $\operatorname{Re}(s) = w_{\mathcal{D}_g}/2$ is the motivic Riemann Hypothesis for the row:

- $w_{\mathcal{D}_g} = 1$ (Dedekind, Window II at totally imaginary K of degree $2g$): the generalised Riemann Hypothesis for $\zeta_K(s)$.
- $w_{\mathcal{D}_g} = 2$ (Hasse–Weil, Windows I, V at the Jacobian of a curve of genus g): the Hasse–Weil Riemann Hypothesis on the weight-1 motive $H^1(\mathrm{Jac}(X_K))$, critical line $\mathrm{Re}(s) = 1$.
- $w_{\mathcal{D}_g} = w_{\pi,\mathrm{spin}}$ (Siegel spin, Window VI at cusp form π on GSp_{2g}): the Siegel-spin Riemann Hypothesis, critical line $\mathrm{Re}(s) = w_{\pi,\mathrm{spin}}/2$.

Under the stated native spectral, determinant and observable comparisons, the Wilson realization expresses each case as a bulk topological-gauge-theoretic spectral statement on the partition-function zeros of $\mathrm{ACS}^{\mathrm{qu},g}_{k^{\mathrm{Ar},g}}$ on the corresponding arithmetic 3-space $\mathcal{B}^{(3)}_g$.

110.7 Residual Conditions

The genus-g extension inherits residuals from Ch. 102 and Ch. 103 scaled to rank $2g$, and four residuals are specific to this chapter.

(i) *Rank-$2g$ local Langlands compatibility at ramified primes.* Chapter 110.3.2 holds at unramified primes of good reduction for $\mathcal{D}_g$. At ramified primes, the identity requires a local twist term depending on the Artin conductor and the Fargues–Scholze / Genestier–Lafforgue correspondence for GSp_{2g}. It is a non-formal theorem, extending Chapter 102.6(5).

(ii) *Arithmetic boundary geometry and quantization.* The curve area form is defined on a compact complex curve with its orthonormal Hodge basis. An arithmetic boundary realization must specify its embedding or site comparison and metric pullback. For Minkowski and Shimura rows, it must also specify the oriented two-cycle and degree-two form. Hodge orthogonality does not supply g independent scaling coordinates or a quantum surface operator. The boundary quantization, primary pairing and the domain-preserving maps V_g, T_g of Chapter 110.4.4 are the required operator comparisons.

(iii) *Rank-g Bost–Connes Adams compatibility.* The proposed relation between the λ-ring Frobenius lifts of Borger 2009 and native Bost–Connes operations requires a specified realization and equality of their actions. At every finite place one must give the representation, coefficient comparison, Hilbert pairing and operator domain. Restricted tensor assembly further requires the reference-vector and boundedness or convergence data above. Rank change alone supplies none of these identifications; F2.c is the separate arithmetic comparison of Chapter 102.6(6).

(iv) *Rank-g H-P^{Ar}.* Self-adjoint lift of $L_{\mathrm{o}}^{(g)}$ to a Hilbert completion of $H^1_{\Delta,g}$ in the rank-g arithmetic Petersson inner product. This is the direct rank-g analog of Chapter 44.3.2. Domain: inherited from H-P^{Ar} plus Chapter 102.6(2)–(7). The residual is the motivic Riemann Hypothesis in Hilbert–Pólya form.

Remark 110.7.1 *(uniform rank structure).* The genus enters through the rank of the gauge group, the archimedean Fock rank, and the motivic weight; the Koszul self-duality identity $\kappa^{\mathrm{Ar},g} + (\kappa^!)^{\mathrm{Ar},g} = 0$ is fixed on the residual domain of Chapter 102.4.1; the bulk partition function is the row-specific completed L-function $\Lambda_{\mathcal{D}_g}$; and the motivic Riemann Hypothesis for the row becomes the topological spectral condition $\mathrm{Re}(z) = w_{\mathcal{D}_g}/2$ on the non-trivial zeros z of $\Lambda_{\mathcal{D}_g}$, i.e. the conditional $F^{(g)}$-weight purity on $H^1_{\Delta,g}$ (Chapter 110.5.3(4)).

110.8 Primary Sources for the Genus-g Extension

The genus-g comparison questions involve the following primary references.

- *Faltings 1984 Calculus on arithmetic surfaces*, *Ann. Math.* 119, 387–424: Arakelov Kähler form $\mu_{\mathrm{Ar},g}$ of Chapter 110.4.1; Faltings–Hriljac arithmetic Hodge index; Faltings δ-invariant.
- *Fargues–Scholze 2021* arXiv:2102.13459: rank-$2g$ local Langlands for $GL_{2g}/\mathbb{Q}_p$; Zhu 2020 arXiv:2008.02998 and Genestier–Lafforgue 2017 arXiv:1709.00978 extend to GSp_{2g}.
- *Bhatt–Scholze 2022 Ann. Math.* 196: prismatic Frobenius on rank-$2g$ F-crystals; Chapter 102.2 Window III input.
- *Borger 2009* arXiv:0906.3146 §8: Bost–Connes Adams ψ_p as geometric Frobenius on the Tate λ-motive; rank-$2g$ scaling via the λ-ring structure on the Tate motive.
- *Arthur 2013 The Endoscopic Classification*: Siegel spin L-functions and functional equations for GSp_{2g}-automorphic cuspidals.
- *Kudla 1997 Ann. Math.* 146; *Kudla–Rapoport 2011*: Siegel–Maass metric on $\mathcal{A}_g$ and rank-g Faltings splitting for Window VI.
- *Morishita 2012 Knots and Primes* arXiv:0904.3399; *Mazur 1973* ENS 6: knot-prime dictionary for the Wilson-line identification at finite primes; Artin–Verdier fundamental class for the arithmetic 3-space.

The residuals of Chapter 110.7, together with the local and global hypotheses in the preceding theorems, include profinite ACS renormalization, the boundary factorization comparison, the native primary pairing, and the arithmetic Hilbert–Pólya realization. The surface comparison additionally requires arithmetic boundary geometry, quantization, normal generator domains and the unitaries V_g, T_g. The restricted tensor requires the specified reference vectors and bounded local actions. The determinant identity requires its native complex and normalization. These constructions must be supplied on the chosen arithmetic row before the scalar, operator and cohomological assertions can be identified.

Chapter III

Continuous Scaling and the Arithmetic Zero Operator

Continuous Mellin scaling has no eigenvectors. Occupation energy has a complete eigenbasis. This spectral distinction prevents their unitary intertwining. A separate diagonal operator on the zero occurrences gives an exact arithmetic exponential and its purity criterion, with no identification of these different carriers.

III.1 Continuous Mellin scaling

At the real place, the archimedean completion is $\mathbb{Q}_\infty = \mathbb{R}$, with its ordinary absolute value. For $\lambda > 0$, geometric scaling is

$$\rho_\lambda(x) = \lambda x, \qquad \rho_\lambda^*(dx/x) = dx/x.$$

The inverse pullback $(\rho_\lambda^{-1})^* f(x) = f(\lambda^{-1}x)$ is unitary on the multiplicative Haar space and has no half-density factor. On the additive space $L^2(\mathbb{R}_{>0}, dx)$, the corresponding unitary action is $v(x) \mapsto \lambda^{-1/2} v(\lambda^{-1}x)$. The map $v(x) \mapsto x^{1/2}v(x)$ is a unitary from the additive space to the multiplicative space and intertwines these actions. Indeed $\int |x^{1/2}v(x)|^2 dx/x = \int |v(x)|^2 dx$. The Tate half-shift below is separate from this change of measure.

Definition III.1.1 *(scaling and its domain).* Set $H_c = L^2(\mathbb{R}_{>0}, dx/x)$ and $(U_a f)(x) = f(e^{-a}x)$ for $a \in \mathbb{R}$. In the coordinate $t = \log x$, define

$$D_c = -i\,\partial_t, \qquad \operatorname{Dom} D_c = \{f : f(e^t) \in H^1(\mathbb{R}, dt)\}.$$

Then $U_a = e^{-iaD_c}$. The derivative convention is $D_c = i\,\partial_a U_a|_{a=0}$.

Theorem III.1.2 *(Mellin unitarity).* The transform

$$(\mathcal{M}_- f)(u) = \frac{1}{\sqrt{2\pi}} \int_0^\infty f(x) x^{-iu} \frac{dx}{x}$$

extends to a unitary $H_c \to L^2(\mathbb{R}, du)$ and sends D_c to multiplication by u on its maximal domain. It has no eigenvectors. The normal operator $\Theta_c = 1/2 + iD_c$ has continuous spectrum $1/2 + i\mathbb{R}$.

Proof. The logarithmic coordinate is unitary from H_c to $L^2(dt)$. Plancherel and integration by parts give the transform and generator. The weak H^1 domain maps exactly to $\{v : uv \in L^2(du)\}$. An eigenvector of multiplication by u would be supported on a singleton, which has Lebesgue measure zero. The same argument applies to $1/2 + iu$. □

Definition III.1.3 *(continuous scaling exponential).* On the continuous scaling carrier, put

$$F_\infty := e^{-\Theta_c} = e^{-1/2} U_1.$$

This bounded normal operator satisfies $|F_\infty| = e^{-1/2} 1$ unconditionally.

PROPOSITION III.1.4 *(scaling reflection).* On H_c, the map $(Rf)(x) = f(x^{-1})$ is a unitary involution. It preserves Dom D_c and satisfies

$$RD_cR = -D_c, \qquad R\Theta_cR = 1 - \Theta_c, \qquad RF_\infty R = e^{-1}F_\infty^{-1}.$$

Proof. The substitution $x \mapsto x^{-1}$ preserves the multiplicative Haar integral. In logarithmic coordinates R is reflection, which preserves H^1 and negates its weak derivative. The generator identity follows on the entire domain. Spectral calculus then gives $Re^{-\Theta_c}R = e^{-(1-\Theta_c)} = e^{-1}e^{\Theta_c}$. The inverse is bounded because $F_\infty = e^{-1/2}U_1$. Thus the exponential involution is $z \mapsto e^{-1}/z$, rather than bare inversion. □

III.2 The occupation and primary-sector comparisons

THEOREM III.2.1 *(obstruction to a unitary energy comparison).* Let H_d have a complete orthonormal eigenbasis for an operator L_d. There is no unitary $V : H_d \to H_c$ with $VL_dV^{-1} = \Theta_c$, including equality of domains. In particular, the ordinary Fock energy and the integer occupation operator cannot satisfy this identity.

Proof. If $L_de = \lambda e$ with $\|e\| = 1$, the claimed identity gives $\Theta_cVe = \lambda Ve$. This contradicts the absence of continuous-scaling eigenvectors. The vacuum alone is sufficient. □

The constructed compact-test map to the zero carrier is $E : \mathscr{A} \to K_Z$ of Theorem 78.9.3. Its exact comparison is signed. A Petersson norm of an actual cusp form is the different form computed by the Rankin–Selberg integral. Neither of these statements supplies a map from an arithmetic primary sector to H_c.

THEOREM III.2.2 *(the correctly typed conjugacy formula).* Suppose H_a is a Hilbert space, L_a a closed normal operator, and $V : H_a \to H_c$ a unitary with V Dom L_a = Dom Θ_c and $VL_a = \Theta_cV$ on that domain. Then

$$e^{-L_a} = V^{-1}F_\infty V.$$

The exponential on the left is bounded, and its absolute value is $e^{-1/2}1$. The preceding obstruction excludes this hypothesis when L_a is the ordinary Fock or occupation energy.

Proof. Unitary conjugacy preserves the spectral projections of normal operators. Applying the bounded function e^{-s} on the common vertical-line spectrum proves the formula and absolute-value claim. □

III.3 The zero exponential and its actual spectrum

Definition III.3.1 *(arithmetic zero exponential).* For the diagonal operator Θ_Z of Theorem 77.6.1, put

$$F_Z = e^{-\Theta_Z} = \operatorname{diag}(e^{-\rho}) \quad \text{on } K_Z = \ell^2(Z).$$

Because $0 \le \Re\rho \le 1$, this operator is bounded and invertible, with $\|F_Z\| \le 1$ and $\|F_Z^{-1}\| \le e$.

PROPOSITION III.3.2 *(spectral measure and failure of local finiteness).* For a Borel set $B \subset \mathbb{C}$, the spectral projection of F_Z is

$$(P_Z(B)u)_\rho = \mathbf{1}_B(e^{-\rho})u_\rho.$$

Its spectrum is the closure of $\{e^{-\rho} : \rho \in Z\}$. The counting pushforward $\sum_\rho \delta_{e^{-\rho}}$ is not locally finite on $\mathbb{C}$. Thus it is not an entire-function zero divisor. For any fixed s, the product $\prod_\rho(1 - e^{-s}e^{-\rho})$ fails the necessary factor-to-one condition for an ordinary convergent nonzero product.

Proof. Coordinate functional calculus gives the spectral projection. The spectrum of a bounded diagonal operator is the closure of its entries: outside that closure, the inverse diagonal resolvent is bounded; at a limit point, basis vectors give approximate eigenvectors. All infinitely many occurrences lie in the compact annulus $e^{-1} \le |z| \le 1$. A locally finite counting measure would have finite mass on that annulus, a contradiction. Finally $|e^{-s}e^{-\rho}| \ge e^{-\Re s-1} > 0$, so the factors do not tend to one. The normalized additive determinant $\Delta_Z(s) = \xi(s)$ remains well-defined by its genus-one factors. □

Proposition III.3.3 *(individual multiplicative factors).* For a fixed zero ρ,

$$1 - e^{-s}e^{-\rho} = 0 \quad \Longleftrightarrow \quad s = -\rho + 2\pi i j \quad (j \in \mathbb{Z}).$$

The additive genus-one factor $(1 - s/\rho)e^{s/\rho}$ instead vanishes only at $s = \rho$.

Proof. The first equation is $e^{-s-\rho} = 1$, so $-s - \rho \in 2\pi i\mathbb{Z}$. The exponential factor in the additive expression never vanishes. These are individual-factor identities; the preceding proposition excludes the proposed ordinary infinite product. □

Theorem III.3.4 *(arithmetic purity and Weil positivity).* The following are equivalent:

$$|F_Z| = e^{-1/2}1, \qquad \Re\rho = 1/2 \text{ for every zero } \rho, \qquad W(f * f^*) \ge 0 \text{ for every } f \in \mathscr{A}.$$

This is a theorem about F_Z on K_Z. The equality $|F_\infty| = e^{-1/2}1$ on H_c is unconditional and carries no zero-location information without a further comparison.

Proof. On the coordinate vector at ρ, $|F_Z|$ has eigenvalue $e^{-\Re\rho}$. This proves the first equivalence. The second is Weil's positivity criterion on the exact compact convolution test algebra. □

Functional-equation symmetry and common weight

Choose a multiplicity-preserving involution τ of Z with $\rho_{\tau j} = 1 - \rho_j$. Such a choice exists by the functional equation of ξ; occurrences fixed in value may be fixed individually. The unitary permutation $(T_Z u)_j = u_{\tau j}$ obeys

$$T_Z\Theta_Z T_Z = 1 - \Theta_Z, \qquad T_Z F_Z T_Z = e^{-1}F_Z^{-1}.$$

The first equality includes domains: $|1 - \rho| \le 1 + |\rho|$ and its reverse counterpart show that the two maximal diagonal domains coincide. The second equality is the coordinate identity $e^{-(1-\rho)} = e^{-1}/e^{-\rho}$. The real-entire symmetry supplies separately an occurrence permutation c with $\rho_{cj} = \overline{\rho_j}$; its unitary permutation conjugates Θ_Z to Θ_Z^* and F_Z to F_Z^*. Neither symmetry asserts that a zero lies on the critical line. These are symmetries of spectral occurrences and projections, not of an entire divisor after exponentiation.

Definition III.3.5 *(common exponential weight).* For a specified nonempty family of nonzero complex numbers $(\mu_j)_{j\in J}$, counted with its given occurrences, a common exponential weight is $w \in \mathbb{R}$ such that $|\mu_j| = e^{-w}$ for every j. It is unique if it exists. For a normal operator without a chosen eigenbasis, the corresponding condition is that its spectral measure be supported on $\{|z| = e^{-w}\}$, equivalently $|F| = e^{-w}1$ on a nonzero Hilbert space. The reference base here is e and the sign is negative. With a base $b > 1$, the convention $|\mu| = b^{-w_b}$ instead gives $w = w_b \log b$.

THEOREM III.3.6 *(the symmetric zero spectral-radius criterion).* On the nonempty zero-occurrence carrier K_Z, the following are equivalent:

$$r(e^{1/2}F_Z) = 1, \qquad |F_Z| = e^{-1/2}1, \qquad \mathrm{RH}.$$

Moreover, any common exponential weight for the family $(e^{-\rho})$ is necessarily $1/2$. The functional-equation symmetry is essential to the spectral-radius assertion.

Proof. The zero count ensures that Z is nonempty, and the strip bound makes F_Z a bounded normal diagonal operator. Its spectrum is the closure of its entries. Therefore

$$r(e^{1/2}F_Z) = \sup_{\rho\in Z} e^{1/2-\Re\rho} = e^{1/2-\inf_{\rho\in Z}\Re\rho}.$$

If the radius is one, the infimum equals $1/2$, so every real part is at least $1/2$. Applying $\rho \mapsto 1-\rho$ makes every real part at most $1/2$. All are therefore $1/2$, without assuming that an infimum is attained. Conversely that equality makes every entry have modulus $e^{-1/2}$ and the normalized radius one. The diagonal absolute value proves the middle equivalence. If a common weight w exists, $|e^{-\rho}| = e^{-\Re\rho}$ gives $\Re\rho = w$; symmetry also gives $1-w = w$. A normal diagonal operator with eigenvalues $e^{-1/2}, e^{-1}$ has normalized radius one but not constant modulus. It lacks the functional-equation symmetry and hence is not a counterexample to this theorem. □

III.4 A SURFACE OBSERVABLE AND THE REQUIRED OPERATOR COMPARISON

For a closed oriented smooth three-manifold M, a real connection one-form A on the trivial $U(1)$-bundle, and a real parameter k, the abelian Chern–Simons action is

$$S_{\mathrm{CS},k}[A] = \frac{k}{4\pi}\int_M A\wedge dA.$$

For a globally defined real gauge parameter χ, replacing A by $A+d\chi$ changes the integrand by $d(\chi\, dA)$, whose integral vanishes. Nontrivial bundles and large gauge transformations require a global definition and its level conditions. A disk curvature observable uses the restriction of such connection data; its quantization requires the additional operator data below.

Definition III.4.1 *(an oriented disk and an abelian curvature observable).* Let $\Sigma_\infty = D^2 = \{z\in\mathbb{C} : |z|\le 1\}$, oriented by $dx\wedge dy$, and give it the two-form

$$\omega_\infty = \frac{i}{2\pi}\frac{dz\wedge d\bar z}{(1+|z|^2)^2}.$$

Let a be a smooth real connection one-form on the trivial $U(1)$ bundle over the disk, and let $q\in\mathbb{Z}$. The curvature observable is $O_q(a) = \exp(iq\int_{D^2} da)$.

PROPOSITION III.4.2 *(disk normalization and Stokes comparison).* The disk has $\int_{D^2}\omega_\infty = 1/2$, and $O_q(a) = \exp(iq\int_{\partial D^2} a)$. For a fixed connection, this scalar observable acting on H_c cannot equal the nonscalar operator F_∞.

Proof. In polar coordinates the area integral is $2\int_0^1 r(1+r^2)^{-2}dr = 1/2$. Stokes's theorem gives the boundary expression with its induced counterclockwise orientation. Gauge transformations on the disk preserve the curvature. The integral charge makes the boundary holonomy independent of the angular representative. Choose a nonzero logarithmic test supported in an interval of length less than one. Its translate by one has disjoint support, so U_1 and hence $F_\infty = e^{-1/2}U_1$ is not scalar. □

The form ω_∞ uses the complex orientation on the disk and the additive measure $dx\,dy$. Its extension with the same formula to $\mathbb{P}^1(\mathbb{C})$ has area $2\int_0^\infty r(1+r^2)^{-2}dr = 1$. On the disk, $2\omega_\infty$ is the unit-area normalization. These choices specify geometric data; neither choice constructs a quantum surface operator or an arithmetic spectral comparison.

Real charges, closed surfaces and area characters

For a smooth real one-form a on a trivial bundle over an oriented compact surface Σ, the scalar

$$W_q(\Sigma, a) = \exp\left(iq\int_\Sigma da\right)$$

is defined for every real q. It is unchanged by transformations preserving curvature, in particular $a \mapsto a + d\chi$. For integral q and the disk it agrees with O_q above. Stokes gives $W_q = \exp(iq\int_{\partial\Sigma} a)$ for a globally specified a even at real charge. If only the boundary holonomy angle modulo 2π is given, this expression descends to a $U(1)$ character exactly when $q \in \mathbb{Z}$. This distinguishes the real curvature scalar from an integral-charge boundary character. For closed Σ and global a, Stokes gives $W_q = 1$ for all real q; curvature invariance imposes no condition $qk \in \mathbb{Z}$.

PROPOSITION III.4.3 *(closed-surface isotopy).* Let F be the real curvature two-form of a smooth $U(1)$ connection on a three-manifold M. For a smooth isotopy $j_t : \Sigma \to M$ of a closed oriented surface, $\int_\Sigma j_1^* F = \int_\Sigma j_0^* F$. Consequently the curvature scalar at any real charge is isotopy invariant in this closed-surface setting, without a framing condition.

Proof. Curvature is closed. For $J : [0,1]\times\Sigma \to M$, $J(t,x) = j_t(x)$, Stokes gives $\int_\Sigma j_1^*F - \int_\Sigma j_0^*F = \int_{[0,1]\times\Sigma} dJ^*F = 0$. For a surface with boundary, the swept boundary gives an extra term. For example, $a = zx\,dy$ on $\mathbb{R}^3$ has curvature integral $c\pi$ on the unit disk in $z = c$. Translating that disk preserves its ordinary framing but changes $e^{iqc\pi}$ in general. □

PROPOSITION III.4.4 *(integral area and the scalar circle character).* For a real area α, the function $\theta \mapsto e^{i\alpha\theta}$ descends from $\mathbb{R}$ to a character of $\mathbb{R}/(2\pi\mathbb{Z})$ if and only if $\alpha \in \mathbb{Z}$. Thus the disk form of area $1/2$ does not define this character, whereas $2\omega_\infty$ does.

Proof. Periodicity is equivalent to $e^{2\pi i\alpha} = 1$, or $\alpha \in \mathbb{Z}$; the character law then follows from addition. The areas were computed above. A circle character alone specifies neither a gauge 2-group nor a 2-representation. □

Definition III.4.5 *(a specified quantum surface operator).* A quantum surface comparison consists of a Hilbert space H_Σ, a bounded operator F_∞^{WS} on it, and a unitary $V_\Sigma : H_\Sigma \to H_c$ satisfying $V_\Sigma F_\infty^{\mathrm{WS}} = F_\infty V_\Sigma$. These are additional data. The scalar curvature observable alone does not construct them.

PROPOSITION III.4.6 *(transport of a specified surface operator).* For the specified quantum surface comparison, $F_\infty^{\mathrm{WS}} = V_\Sigma^{-1} F_\infty V_\Sigma$ and $|F_\infty^{\mathrm{WS}}| = e^{-1/2}1$. A further arithmetic comparison with F_Z is required before its purity can be identified with Weil positivity.

Proof. Multiply the defining intertwining identity by V_Σ^{-1}. Unitary conjugacy preserves the absolute value. The arithmetic criterion concerns the separate coordinate operator F_Z and follows from the preceding purity theorem. □

III.5 ARITHMETIC GEOMETRY, QUANTIZATION AND PRIMARY OPERATORS

The continuous carrier and the zero-occurrence carrier answer specified analytic questions. Realizing an arithmetic boundary operator requires additional constructions on the intended objects.

Koszul self-duality and reflection. The proposed compatibility with the Koszul self-duality of Chapter 43.5.1 remains a question on the native resolution. It requires a specified contravariant duality on that complex, including its coefficient field, degree shifts and signs, and a self-duality morphism compatible with its differential. Neither existence of this native morphism nor its compatibility with a Hilbert realization follows from the analytic reflection. After realization and identification with the appropriate dual Hilbert space, the question is whether it induces a complex-linear unitary involution J on a native Hilbert space H_{nat} with closed normal generator Θ_{nat}, satisfying

$$J \operatorname{Dom} \Theta_{\mathrm{nat}} = \operatorname{Dom} \Theta_{\mathrm{nat}}, \qquad J\Theta_{\mathrm{nat}} J = 1 - \Theta_{\mathrm{nat}}.$$

The passage from contravariant duality to this linear operator must specify every conjugation: the Riesz identification is antilinear, so it cannot silently supply a complex-linear J. For the continuous realization the required comparison is a unitary $V : H_{\mathrm{nat}} \to H_c$ with $V \operatorname{Dom} \Theta_{\mathrm{nat}} = \operatorname{Dom} \Theta_c$, $V\Theta_{\mathrm{nat}} = \Theta_c V$ and $VJ = RV$. For a zero-occurrence realization the corresponding requirements replace (H_c, Θ_c, R) by (K_Z, Θ_Z, T_Z). These are distinct comparison problems. Either specified comparison would transfer the normalized identity $Je^{-\Theta_{\mathrm{nat}}}J = e^{-1}(e^{-\Theta_{\mathrm{nat}}})^{-1}$; it would not give the unnormalized inversion law. Constructing the native duality and one of these compatible realizations is open here; no assertion about the native resolution is deduced solely from the two analytic models.

Arithmetic boundary geometry and quantum surface quantization. The surface problem in Chapter 103.4.7 asks for an arithmetic space or site modeling $\overline{\operatorname{Spec} \mathbb{Z}}$, a boundary object at infinity, and a comparison map from an oriented disk carrying a chosen Green metric. For a projective arithmetic surface the intended metric lies on its complex fibre; a chosen metric on $\mathbb{P}^1(\mathbb{C})$ and the explicit disk form are smooth geometric data, not an embedding of D^2 in the arithmetic site. The finite-prime knot dictionary of Morishita and an infinity-surface model require this topology and its gluing maps; a disk with boundary must not be called a closed surface.

A quantum arithmetic Chern–Simons construction must specify its level k^{Ar}, fields and gauge transformations on that site, its action and global gauge normalization, boundary conditions, and a measure or alternative state-space quantization. It must construct the boundary Hilbert space H_Σ, its pairing, and the observable from a well-defined degree-two curvature coupling. Multiplication of two differential two-forms on a disk does not supply that coupling: their wedge has degree four. If a higher surface holonomy is intended, a gauge 2-group, 2-connection and 2-representation with gluing coherence are additional data. Extension along a $U(1)$ inclusion into a nonabelian theory requires actual functors and compatible boundary terms, not an assertion of automatic functoriality.

The intended boundary target is a realization of $\mathcal{V}^{\mathrm{prim}}_{\mathrm{Heis},\infty}$, a sector of the primary Hochschild trace class. Constructing its map into a Hilbert space and its relation to a Heisenberg representation is part of the problem. The boundary WZW/radial-quantization question asks whether a constructed boundary generator represents translation in $t = \log |x|_\infty$, including its normalization and operator domain. The Chern–Simons and boundary frameworks of Witten, Elitzur–Moore–Schwimmer–Seiberg, Cattaneo–Mnev–Reshetikhin, and the higher-gauge frameworks of Baez–Schreiber and Schreiber–Sati provide separate settings for these questions; their names do not identify the arithmetic fields or quantization.

To compare the resulting surface operator with continuous scaling, one must construct a unitary $V_\Sigma : H_\Sigma \to H_c$ and, for an exponential realization, a closed normal Θ_Σ with

$$V_\Sigma \operatorname{Dom} \Theta_\Sigma = \operatorname{Dom} \Theta_c, \qquad V_\Sigma \Theta_\Sigma = \Theta_c V_\Sigma.$$

Then spectral calculus gives the earlier conjugacy. A unitary phase observable and $e^{-1/2}U_1$ have different absolute values; the Tate normalization must therefore be part of the proposed quantization. Existence of a bulk measure alone does not imply this comparison.

The primary Virasoro and pairing problem. The primary-Virasoro question of Chapter 44.3.5 is to realize a specified primary trace sector in a Hilbert space H_{prim} with an invariant dense domain $\mathcal{D}_{\mathrm{prim}}$ and operators L_n^{Ar} satisfying on that domain

$$[L_m^{\mathrm{Ar}}, L_n^{\mathrm{Ar}}] = (m-n)L_{m+n}^{\mathrm{Ar}} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}\mathbb{1}.$$

It requires an actual central action, compatible primary operations, and a positive pairing, together with any asserted adjoint relations and operator closures. A single abstract normal operator does not construct this representation.

The proposed Weil–Petersson comparison further asks for a unitary $\Upsilon_\infty : H_{\mathrm{prim}} \to H_c$ whose isometry is proved from the native Petersson/Koszul pairing and whose domain comparison realizes a specified closed normal conformal-weight operator as Θ_c. If that operator is denoted L_0^{Ar}, its normality and the equality of its domain with the transported domain are required. The ordinary polynomial Fock energy cannot do this, by the eigenvector obstruction above. If one also demands $L_n^* = L_{-n}$ in a unitary Virasoro representation, its self-adjoint L_0 cannot equal the non-self-adjoint Θ_c under a unitary; a distinct shifted normal operator and its relation to L_0 would have to be specified.

The local operator D_c, its shift Θ_c, the finite-place N_{BC}, an arithmetic self-adjoint H_{HP} and $\Theta_\xi = 1/2 + iH_{\mathrm{HP}}$ are separate operators on specified carriers. Compatibility with the global primary construction and the Polyakov comparison in Chapter 44.3.4 requires maps that intertwine those actions where claimed. A local Mellin symbol does not identify the global operators or resolve a conformal anomaly.

Arithmetic middle cohomology and determinant descent. The intended H^1_{ch} comparison requires an arithmetic complex, differential, topology and completion, plus a closed normal operator on its middle cohomology induced by an action on the complex. To realize the zero divisor, one possible sufficient datum is a unitary $V_1 : H^1_{\mathrm{ch}} \to K_Z$ preserving the maximal domains and satisfying $V_1\Theta_1 = \Theta_Z V_1$. The normalized additive determinant prescription of Chapter 77.6.1 then transfers to Θ_1, and its exponential has the occurrence spectral projections of F_Z. This is a conditional consequence; defining K_Z from the zeros does not construct that arithmetic complex or V_1.

An alternative H-P^{Ar} construction of Chapter 44.3.2 must supply its native self-adjoint H_{HP}, its domain and the normalized determinant of $1/2 + iH_{\mathrm{HP}}$. A character identity alone gives no such map. Multiplying $\Lambda_\zeta(s)$ by $s(s-1)/2$ cancels scalar poles; a cohomological removal of Tate degrees 0, 2 instead requires an exact sequence or quotient complex and an action-compatible map. No undefined regularized eigenvalues substitute for those data. Likewise a regularized trace must have an explicit prescription; F_Z is boundedly invertible on an infinite-dimensional space, hence is not compact or nuclear and has no ordinary trace-class determinant.

Under a proved domain-preserving V_1, common exponential weight $1/2$, normalized spectral radius one, RH and compact-test A-TP are equivalent by the preceding arithmetic theorem. The Deninger

middle-degree purity question in Chapter 104.1 asks for this native realization and its purity; the continuous scaling equality alone is unconditional. The surface, primary-Virasoro, Mellin-domain and arithmetic cohomology comparisons are distinct required data. No logical independence theorem between them is asserted.

Finite-place weights and global assembly. For each prime, a finite-place purity statement must specify its operator ψ_p and a nonzero closed weight space $H_{p,r}$. The precise negative-exponential convention corresponding to the proposed modulus $p^{-r/2}$ is

$$|\psi_p|_{H_{p,r}} = p^{-r/2}\text{I}, \qquad w_e = \frac{r}{2}\log p.$$

Equivalently, a bounded operator with this modulus has the form $p^{-r/2}U_{p,r}$ with $U_{p,r}$ an isometry; it is unitary if ψ_p is also surjective. This follows by setting $U_{p,r} = p^{r/2}\psi_p$ and computing $U_{p,r}^* U_{p,r} = \text{I}$. Identifying the actual Adams action with these operators is a separate Bost–Connes comparison problem. Algebraic Adams operations alone do not specify a Hilbert weight space or a modulus.

A restricted Hilbert tensor product requires specified unit vectors e_p and compatible pairings. If bounded local operators fix e_p and are contractions, their tensor action is defined on finite reference tensors and extends by continuity with norm at most one. Without such conditions, convergence of partial products and their domains must be proved; local scalar purity does not ensure it. Thus the proposed $F_\infty \otimes \bigotimes'_p \psi_p$ needs an actual global operator construction before its native purity can be reduced to an archimedean statement.

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Chapter 112

Test Maps, Heegner Heights, and the Weil Positivity Problem

The full Weil inequality is defined on compact complex tests. To compare it with a spectral statistic or an arithmetic height, one needs a map from the source variables to those tests and an identity of pairings. The Fourier-window construction below gives such a test map. The Heegner construction gives a second map and an exact height-plus-residual identity. Their images need not be dense, and positivity of the residual remains an additional condition. An explicit divisor example shows why ordinate statistics alone do not determine the signed pairing.

112.1 The Compact Test Algebra and the Weil Pairing

Let $\mathscr{A} = C_c^\infty(\mathbb{R};\mathbb{C})$ have its usual locally convex inductive-limit topology. On tests supported in a fixed compact set, convergence means uniform convergence of every derivative. Write

$$h_f(z) = \mathcal{F}_+ f(z) = \int_{\mathbb{R}} f(u) e^{izu}\, du, \qquad f^*(u) = \overline{f(-u)}, \qquad (f * g)(u) = \int_{\mathbb{R}} f(v)\, g(u-v)\, dv.$$

Give $\mathscr{P} = \mathcal{F}_+\mathscr{A}$ the transported topology, as in Definition 78.1.1. Its elements are entire and rapidly decreasing in every closed horizontal strip. The continuous signed Weil functional is

$$W(f) = P(f) - W_\infty(f) - W_{\mathrm{fin}}(f).$$

The polar and real-place terms are those of Theorem 78.1.2; the finite-place term is

$$W_{\mathrm{fin}}(f) = \sum_{p,k\ge 1} \frac{\log p}{p^{k/2}} \big(f(k\log p) + f(-k\log p) \big).$$

This sum is finite for compact tests. Put $q_W(f,g) = W(f * g^*)$, linear in f and conjugate-linear in g. The relation $W(f^*) = \overline{W(f)}$ makes this form Hermitian. A-TP means $q_W(f,f) \ge 0$ for every $f \in \mathscr{A}$; Corollary 78.2.1 identifies it with RH.

For a complex vector space S and a specified linear map $J : S \to \mathscr{A}$, define the pulled-back Hermitian form

$$q_J(u,v) = W\big(Ju * (Jv)^*\big).$$

The source variable u and the compact test Ju belong to different spaces. The constructions below give explicit such maps.

The five source families below concern different input objects. Each comparison requires a test map and an identity for the full normalized pairing, or a proved equivalence with that pairing.

- **Arthur-Selberg truncation**: Arthur 1978–2013 trace-formula series; Jacquet-Langlands 1970; Selberg 1956; Ngô 2010 *Publ. IHES* 111; Flicker-Kazhdan; Mœglin-Waldspurger.

- **Geometric Langlands and Fargues–Scholze**: Beilinson–Drinfeld 2004; Arinkin-Gaitsgory arXiv:1201.6343; Fargues–Scholze arXiv:2102.13459; Lafforgue arXiv:1209.5352; Drinfeld 1983, 1987; Deligne *Weil II*; Frenkel-Gaitsgory.
- **Iwasawa-Kato**: Mazur-Wiles 1984; Kato 2004 *Astérisque* 295; Perrin-Riou 1995; Fukaya-Kato 2006; Colmez; Bloch-Kato 1990; Kolyvagin 1988; Coates-Wiles 1977; Ferrero-Washington 1979.
- **Arithmetic quantum chaos**: Bohigas-Giannoni-Schmit 1984; Berry-Keating 1999; Montgomery 1973; Odlyzko 1987; Rudnick-Sarnak 1996; Katz-Sarnak 1999; Keating-Snaith 2000; Lindenstrauss 2006; Soundararajan 2010; Sarnak 1995; Hejhal 1994.
- **Arakelov–Faltings-Hriljac**: Arakelov 1974; Faltings 1984; Hriljac 1985; Gillet-Soulé 1990; Bost 1999; Bost-Künnemann 2010; Bombieri 1971; Gross-Zagier 1986; Beilinson 1985; Vojta 1987; Bismut-Gillet-Soulé 1988.

The ordinary scattering construction already distinguishes a unitary phase from positivity: Proposition 101.1.1 gives complete unitary scattering with a negative Fourier-square trace. Theorem 101.5.1 shows that a positive ordinary trace difference with an integrable spectral shift must vanish.

112.2 Comparison of Pairings and Extension of Tests

Theorem 112.2.1 *(Comparison and extension of the Weil form).* Let $J : S \to \mathscr{A}$ be linear, and let b be a Hermitian form on S satisfying

$$b(u,v) = W\big(Ju * (Jv)^*\big) \qquad (u,v \in S).$$

Then $b \geq 0$ is equivalent to Weil positivity on $J(S)$. Suppose, in addition, that every $f \in \mathscr{A}$ admits $f_n \in J(S)$ with all supports contained in one compact set and $f_n \to f$ in every smooth seminorm. Then $b \geq 0$ implies A-TP on all of $\mathscr{A}$.

Proof. The first assertion follows by putting $u = v$ in the identity. Under the approximation hypothesis, $f_n * f_n^* \to f * f^*$ smoothly, with support in one compact set. Continuity of the Weil distribution gives $W(f * f^*) = \lim_n W(f_n * f_n^*) \geq 0$. □

Translation gives another way to represent every convolution square, without approximating every test.

Lemma 112.2.2 *(Transfer by translation).* For $a \in \mathbb{R}$, put $\tau_a f(t) = f(t-a)$. Let

$$S_+ = C_c^\infty((0,\infty);\mathbb{C})$$

Give S_+ its usual test-function topology, and let $J_+ : S_+ \to \mathscr{A}$ be extension by zero. For all $f, g \in \mathscr{A}$ and $a \in \mathbb{R}$,

$$(\tau_a f) * (\tau_a g)^* = f * g^*, \qquad q_W(\tau_a f, \tau_a g) = q_W(f,g).$$

The image $J_+(S_+)$ is not dense in $\mathscr{A}$. Nevertheless, positivity of the pulled-back form q_{J_+} is equivalent to A-TP.

Proof. Every element of S_+ has support compactly contained in $(0,\infty)$, so its extension by zero is smooth. The involution and convolution satisfy

$$(\tau_a g)^* = \tau_{-a}(g^*), \qquad (\tau_a f) * (\tau_{-a} g^*) = f * g^*.$$

The second identity follows by substituting $w = v - a$ in the convolution integral. Applying W proves the pairing identity.

Evaluation at -1 is a continuous nonzero functional on $\mathscr{A}$ which vanishes on $J_+(S_+)$, so this image is not dense. For any $f \in \mathscr{A}$, compactness of its support allows a sufficiently large a such that $\operatorname{supp}(\tau_a f) \subset (0, \infty)$. Thus positivity on $J_+(S_+)$ gives

$$q_W(f, f) = q_W(\tau_a f, \tau_a f) \geq 0.$$

The converse follows by restriction. □

This argument uses equality of convolution squares, not density of the source image. Translation does not shorten a support interval, so it does not place every compact test in the fixed Fourier window. Neither this lemma nor the approximation theorem proves positivity on its source image.

Source family	Comparison required for the Weil form
Arthur–Selberg	A map from truncation tests to $\mathscr{A}$ and an identity retaining the polar, finite-prime and archimedean terms.
Geometric Langlands	An arithmetic spectral realization, a map to $\mathscr{A}$, and a pairing comparison with q_W.
Iwasawa–Kato	A map from finite-prime arithmetic data to compact complex tests and a comparison with the full Weil form.
Arithmetic quantum chaos	A comparison of multivariable correlation statistics with the one-variable Weil pairing; ordinates alone forget real parts of zeros.
Arakelov–Faltings–Hriljac	A test map, a normalized degree-zero height pairing, and a residual form whose sum is the pulled-back Weil form.

Definition 112.2.3 *(The full positivity condition).* Write $\mathsf{R}_{\text{A-TP}}$ for $W(f * f^*) \geq 0$ for every $f \in \mathscr{A}$, with the polar, finite-prime and archimedean terms all present. The digamma contribution is one summand of this condition.

Remark 112.2.4 *(Comparison with the full functional).* A comparison must retain the full normalized functional, or prove an explicit equivalence with it. The continuity argument above extends positivity under its common-compact approximation hypothesis. Lemma 112.2.2 gives a different extension mechanism, even from a nondense image. The theorem supplies no test map or pairing identity for an unnamed source.

112.3 Fourier Windows and Height Comparisons

Compact Fourier windows and co-Poisson constraints

For an integer degree $m \geq 1$, the corresponding window is $(-1/m, 1/m)$. We first use the degree-one window $(-1, 1)$. Put $\mathscr{A}_1^{\mathrm{ev}} = \{f \in C_c^\infty((-1,1); \mathbb{C}) : f(-t) = f(t)\}$. Specify a family $\mathcal{B} \subset \mathscr{D}'(\mathbb{R})$ of distributions representing the desired co-Poisson constraints. Using the translations τ_a defined above, set

$$\mathscr{A}_{1,\mathcal{B}} = \{f \in \mathscr{A}_1^{\mathrm{ev}} : B(\tau_a f) = 0 \text{ for every } B \in \mathcal{B},\ a \in \mathbb{R}\}.$$

No nonzero element is required by this definition. The associated entire source class and its test map are

$$\mathrm{PW}^{\mathrm{CP}}(\mathcal{B}) = \mathcal{F}_+ \mathscr{A}_{1,\mathcal{B}}, \qquad (J_{\mathrm{CP}} h)(u) = \frac{1}{2\pi} \int_{\mathbb{R}} h(t) e^{-itu}\, dt.$$

Give $C_c^\infty((-1,1))$ its usual test-function inductive-limit topology, and $\mathscr{A}_{1,\mathcal{B}}$ its subspace topology. Give the entire source class the topology transported from this subspace. Fourier inversion makes J_{CP} a continuous linear bijection onto that compact-test space. A particular co-Poisson construction must specify the distributions $\mathcal{B}$; the following result holds for every such specified family.

THEOREM 112.3.1 *(The window pairing and the correlation test).* For $h, k \in \mathrm{PW}^{\mathrm{CP}}(\mathcal{B})$,

$$q_{J_{\mathrm{CP}}}(h,k) = \sum_{\rho} h(z_\rho)\overline{k(\bar{z}_\rho)}, \qquad z_\rho = -i(\rho - \tfrac{1}{2}).$$

The sum converges absolutely. Under RH this form is positive semidefinite. It is also positive semidefinite when its source space is zero, without any zero-location hypothesis.

For $f \in \mathscr{A}_1^{\mathrm{ev}}$, the function

$$(\mathcal{R}f)(x_1,x_2) = h_f\big(2\pi(x_1 - x_2)\big)$$

is a symmetric, translation-invariant pair-correlation test, rapidly decreasing on $x_1 + x_2 = 0$. With Fourier kernel $e^{-2\pi i(x_1\xi_1 + x_2\xi_2)}$, its Fourier distribution is

$$f(\xi_1)\delta(\xi_1 + \xi_2),$$

supported in $|\xi_1| + |\xi_2| < 2$.

Proof. Apply the explicit formula to $J_{\mathrm{CP}}h * (J_{\mathrm{CP}}k)^*$. Its Fourier transform is $h(z)\overline{k(\bar{z})}$. Rapid decay in the critical strip and the zero-count estimate give absolute convergence. Under RH each z_ρ is real, so the diagonal value is $\sum_\rho |h(z_\rho)|^2$. On the zero space the only value is zero.

Evenness of f gives symmetry of $\mathcal{R}f$; its formula gives translation invariance. Fourier decay proves the required decrease on the difference line. Finally

$$\mathcal{R}f(x_1,x_2) = \int_{\mathbb{R}} f(u)e^{2\pi i(x_1u - x_2u)}\,du,$$

which proves the Fourier-distribution and support assertions. □

Rudnick–Sarnak, *Zeros of principal L-functions and random matrix theory*, §1, TF1–TF3 and Theorems 1.1–1.2, pp. 3–5, use Fourier support $\sum_j |\xi_j| < 2/m$ for a cuspidal GL_m input, with $m \le 3$ or their coefficient hypothesis (1.7). For the pair test above, support in $(-1/m, 1/m)$ gives precisely that bound. The window $(-1,1)$ is the degree-one case. Their smoothed theorem permits complex zero parameters; their real-spacing formulation adds RH. These correlation sums are different functionals from $q_{J_{\mathrm{CP}}}$; the test construction above identifies no equality between them.

Remark 112.3.2 *(Restricted tests and the full cone).* The image of J_{CP} lies in tests supported in $[-1,1]$ and is not dense in $\mathscr{A}$: evaluation at any point outside that interval vanishes on the image but not on all compact tests. A nonzero annihilator and an extension beyond this window are separate requirements. Different parameters do not prove two test images disjoint. The literal entire-Mellin class in Theorem 101.6.2 is zero.

Example 112.3.3 *(The annihilator depends on its defining distributions).* If $\mathcal{B} = \{\delta_0\}$, then $\delta_0(\tau_a f) = f(-a)$ for every a, so $\mathscr{A}_{1,\mathcal{B}} = 0$. If instead $\mathcal{B}$ consists of the distribution $f \mapsto \int f$, choose a nonzero even bump $b \in C_c^\infty((-1,1))$ and set $f = b''$. Then f is even, $\int \tau_a f = 0$ for every a, and $f \neq 0$: otherwise b would be affine and compactly supported, hence zero. These are explicit annihilator examples, not a construction of an arithmetic co-Poisson family.

Heegner Heights and Compact-Test Gram Matrices

Let $F = \sum_{n\ge 1} a_n(F)e^{2\pi inz}$ be a normalized weight-2 newform on $\Gamma_0(N)$ with trivial character. Let $D < 0$ be an odd fundamental discriminant prime to N, set $K = \mathbb{Q}(\sqrt{D})$, and assume $\chi_D(p) = +1$ for every $p \mid N$. These primes split in K. Let H be the Hilbert class field, $\mathcal{G} = \mathrm{Gal}(H/K)$, $h_K = |\mathcal{G}|$, and $u_K = |\mathcal{O}_K^\times|/2$. Choose a classical Heegner point $x \in X_0(N)(H)$ and put $c = [(x) - (\infty)] \in J_0(N)(H)$.

Set $V = J_0(N)(H) \otimes_{\mathbb{Z}} \mathbb{C}$, and let Θ be a symmetric theta-divisor class on $J_0(N)$. The global H-height for 2Θ gives the Hermitian pairing $\langle\ ,\ \rangle_H$ on V, linear in the first variable. Explicitly, if B_H is its real symmetric pairing, then $\langle a + ib, c + id\rangle_H = B_H(a,c) + B_H(b,d) + iB_H(b,c) - iB_H(a,d)$. It is positive definite on this finite-dimensional vector space; torsion has disappeared in the tensor product.

For $(n,N) = 1$, the good Hecke operators T_n commute and are self-adjoint for this pairing. Their correspondences are defined over $\mathbb{Q}$ and are self-dual, so height functoriality gives $\langle T_n v, w\rangle_H = \langle v, T_n w\rangle_H$; they also commute with $\mathcal{G}$. Simultaneous diagonalization defines the orthogonal projection e_F onto the particular complex eigenspace

$$V_F = \{v \in V : T_n v = a_n(F)v \text{ for all } (n,N) = 1\}.$$

This is the eigenspace for F, not the sum of its Galois conjugates; it may be zero. For a character $\chi : \mathcal{G} \to \mathbb{C}^\times$, put

$$c_{\chi,F} = e_F \sum_{\sigma\in\mathcal{G}} \chi(\sigma)^{-1}\sigma c.$$

These are the unaveraged character components. The factor h_K^{-1} is not inserted.

For an ideal class $\mathfrak{a}$ of K, let $r_{\mathfrak{a}}(n)$ count integral ideals of norm n in that class. Identify characters of the class group with characters of $\mathcal{G}$ by the Artin map. The Rankin series in the following formula is

$$L_{\mathfrak{a}}(F,s) = \left(\sum_{\substack{n\ge 1\\(n,DN)=1}} \chi_D(n)n^{1-2s}\right)\sum_{n\ge 1} a_n(F)r_{\mathfrak{a}}(n)n^{-s},$$
$$L(F,\chi,s) = \sum_{\mathfrak{a}} \chi(\mathfrak{a})L_{\mathfrak{a}}(F,s),$$

initially in a right half-plane and then by analytic continuation. Gross–Zagier, *Heegner points and derivatives of L-series*, I§§2–6, especially (4.3), (5.1), (5.3)–(5.4) and Theorem (6.3), pp. 227–230, give the self-dual correspondences and the normalization

$$\|c_{\chi,F}\|_H^2 = C_{F,K}L'(F,\chi,1), \qquad C_{F,K} = \frac{h_K u_K^2\sqrt{|D|}}{8\pi^2(F,F)}, \qquad (F,F) = \int_{\Gamma_0(N)\backslash\mathfrak{H}} |F(z)|^2\,dx\,dy. \tag{112.1}$$

This is a Rankin derivative centered at $s = 1$, not a formula for $L'(F,1)$ alone. Changing the ground-field height, averaging the character component or changing the polarization changes the constant.

Lemma 112.3.4 *(Character Gram matrix).* For $r \ge 1$ and characters $\chi_1,\dots,\chi_r$ of $\mathcal{G}$, let $G_{ij} = \langle c_{\chi_i,F}, c_{\chi_j,F}\rangle_H$ and define $D_{F,K}u = \sum_j u_j c_{\chi_j,F}$ on $\mathbb{C}^r$. Distinct characters give orthogonal vectors. Repeated characters give

$$G_{ij} = \begin{cases} C_{F,K}L'(F,\chi_i,1), & \chi_i = \chi_j,\\ 0, & \chi_i \ne \chi_j, \end{cases}$$

and consequently

$$\|D_{F,K}u\|_H^2 = C_{F,K}\sum_{\chi}\left|\sum_{j:\chi_j=\chi} u_j\right|^2 L'(F,\chi,1).$$

The sum runs over the distinct characters selected.

Proof. The global height is invariant under $\mathcal{G}$, since its places are permuted by this group. The projection e_F commutes with that action, and reindexing the character sum gives $\tau c_{\chi,F} = \chi(\tau)c_{\chi,F}$. Thus invariance and the first-linear convention imply

$$G_{ij} = \chi_i(\tau)\overline{\chi_j(\tau)}G_{ij} \qquad (\tau \in \mathcal{G}).$$

For distinct characters one can choose τ with the scalar different from 1, so $G_{ij} = 0$. For equal characters the two vectors coincide; Equation (112.1) gives their squared norm. Expand the squared norm of $D_{F,K}u$ and group equal characters. □

Choose a nonzero $\phi \in C_c^\infty((-1/4,1/4);\mathbb{C})$ and put $\phi_j(t) = \phi(t-j)$ for $1 \le j \le r$. Their supports are disjoint. With $S = \mathbb{C}^r$, define

$$J_\Phi u = \sum_{j=1}^{r} u_j\phi_j \in \mathscr{A}, \qquad D_{F,K}u = \sum_{j=1}^{r} u_j c_{\chi_j,F} \in V.$$

The map J_Φ is continuous and injective: restriction to the support of each nonzero ϕ_j recovers u_j. Define $\mathrm{PW}_F^{\mathrm{Heeg}}(K,\Phi) = \mathcal{F}_+J_\Phi(S)$ with the topology transported from S. Its compact-test map is Fourier inversion followed by inclusion in $\mathscr{A}$. The dictionary Φ is additional data, not determined by (F,K).

Define Hermitian matrices

$$M_{ij} = W(\phi_i * \phi_j^*), \qquad R_{ij} = M_{ij} - G_{ij}.$$

Each entry of M is given by the polar term, gamma integral and finite prime sum of the full explicit formula. For a matrix A, write $A[u,v] = \sum_{i,j} u_i\overline{v_j}A_{ij}$, and $A \succeq 0$ if $A[u,u] \ge 0$ for all u.

Theorem 112.3.5 *(Height comparison on a specified test image).* The matrix G is positive semidefinite, and

$$W(J_\Phi u * (J_\Phi v)^*) = \langle D_{F,K}u, D_{F,K}v\rangle_H + R[u,v].$$

Weil positivity on $J_\Phi(S)$ is equivalent to $R + G \succeq 0$, or $R \succeq -G$. The stronger condition $R \succeq 0$ is sufficient. If $D_{F,K}u = 0$, positivity in that direction is exactly $R[u,u] \ge 0$.

Proof. The height pairing gives $G[u,u] = \|D_{F,K}u\|_H^2 \ge 0$. Linearity of J_Φ, sesquilinearity of q_W and the definition $M = G + R$ prove the identity. Its diagonal gives the positivity assertions. □

Example 112.3.6 *(Repeated characters and test normalization).* For $r = 2$ and $\chi_1 = \chi_2$, the vector $u = (1,-1)$ satisfies $D_{F,K}u = 0$, whereas $J_\Phi u = \phi_1 - \phi_2 \neq 0$. The height contributes zero in this direction, leaving precisely $R[u,u] = W((\phi_1-\phi_2)*(\phi_1-\phi_2)^*)$ to control. For $r = 1$, suppose $\|c_{\chi,F}\|_H^2 = g > 0$. Replacing ϕ_1 by $\varepsilon\phi_1$ changes M_{11} to $\varepsilon^2 M_{11}$ for real $\varepsilon > 0$, while $G_{11} = g$ is unchanged. Thus $R_{11} = \varepsilon^2 M_{11} - g < 0$ for sufficiently small ε. The stronger sufficient condition $R \succeq 0$ is sensitive to the chosen dictionary and is not a consequence of height positivity.

An arithmetic-intersection interpretation requires further data: a regular proper arithmetic model of $X_0(N)$ over $\mathcal{O}_H$, degree-zero representatives, vertical corrections orthogonal to the fibre components, and archimedean Green metrics. Let $\widehat{V}_{\mathbb{R}}$ be the chosen real space of corrected degree-zero arithmetic divisor classes, and put $\widehat{V}_{\mathbb{C}} = \widehat{V}_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$. Let $I_{\mathbb{R}}$ be its real symmetric admissible intersection pairing and $I_{\mathbb{C}}$ its complex-bilinear extension. The pairing needed here is the Hermitian extension

$$I_H(A, B) = I_{\mathbb{C}}(A, \overline{B}),$$

where conjugation acts on the complex coefficients of arithmetic classes. Specify a complex-linear lift $\widehat{D} : S \to \widehat{V}_{\mathbb{C}}$ whose degree-zero divisor class is $D_{F,K}$, and a normalization $\kappa > 0$ for which one proves

$$-\kappa I_H(\widehat{D}(u), \widehat{D}(v)) = \langle D_{F,K}u, D_{F,K}v\rangle_H.$$

The model, metrics, lifts and numerical normalization are comparison data; the displayed equality is required, not inferred from the notation. It is this Hermitian form, rather than the complex-bilinear intersection form, that can match q_W.

Faltings–Hriljac arithmetic Hodge theory and a proposed Petersson– Arakelov comparison of Jorgenson–Kramer (2003) must be applied with these specific conventions. A geometrically defined Bott–Chern and vertical correction form R^{geom} gives the proposed Weil comparison only after proving $R^{\mathrm{geom}}[u, v] = R[u, v]$ on S. That identity and $R^{\mathrm{geom}} \succeq 0$ imply positivity on $J_\Phi(S)$ by the theorem. The definition $R = M - G$ proves neither this geometric identification nor its sign.

Remark 112.3.7 *(Comparison with other test images).* The Heegner construction uses degree-zero classes and a specified test dictionary. The domain-restricted Koszul–Petersson comparison of Theorem 47.9.1 in Chapter 47 has different input data. A proposed Rankin–Selberg central-value comparison through Υ_F at $L(1/2, F \otimes \bar{F})$ must exhibit the relevant cusp form, establish its required modularity and normalization, define its compact-test map, and prove the pairing identity on its stated domain. The images of the test maps must then be compared before drawing a conclusion about their intersection. Different inputs prove neither disjointness nor density. The finite-dimensional image $J_\Phi(S)$ is not dense in $\mathscr{A}$: a finite-dimensional subspace of a Hausdorff locally convex space is closed, and $\mathscr{A}$ is infinite-dimensional, as disjointly supported bumps show.

112.4 Arithmetic Comparisons and Ordinate Statistics

The Iwasawa–Kato Comparison Question

Remark 112.4.1 *(Finite-prime data and the full Weil pairing).* The Iwasawa comparison question concerns the following proposed inputs: Mazur–Wiles value formulas; Kato Euler systems and Bloch–Kato–Beilinson height pairings; a proposed archimedean counterpart of Ferrero–Washington; Perrin–Riou and Bloch–Kato exponentials; a proposed $p \to \infty$ Connes–Marcolli comparison; Kato reciprocity maps $\exp^*/\log$; Kolyvagin and Coates–Wiles height constructions; Colmez–Fontaine p-adic representations; Fukaya–Kato noncommutative methods and Skinner–Urban; and a proposed Dirichlet p-adic transport. The named reference family comprises Mazur–Wiles (1984), Kato (2004, *Astérisque* 295), Perrin–Riou (1995), Fukaya–Kato (2006), Colmez, Bloch–Kato (1990), Kolyvagin (1988), Coates–Wiles (1977), Ferrero–Washington (1979), Connes–Marcolli and Skinner–Urban.

For each such route, the missing comparison is a specified map to $\mathscr{A}$ and an identity with q_W, or an explicit equivalence, including the polar and archimedean terms. The existence, coefficients and domain of that map must be part of the assertion. No completeness of this list, exclusion of all Iwasawa methods,

or positive pairing comparison is assumed here. Functional-equation symmetry alone does not provide a sign for q_W; the abstract divisor example below already separates these properties. The conditional arithmetic complementarity statement in Theorem 43.5.1 retains its arithmetic evaluation, central-defect, and detection hypotheses. It supplies no test map in this chapter.

Remark 112.4.2 *(An archimedean Iwasawa comparison).* Deninger's cohomological analogies (ICM 1998; *Motives and algebraic cycles*, 2002) motivate asking for an archimedean analogue of an Iwasawa main-conjecture construction. To obtain a formulation of A-TP by this route, one must construct its carrier, operators and pairing and prove equivalence with the full normalized Weil condition. No such construction or RH-equivalence follows merely from the analogy.

Ordinate statistics and the divisor pairing

Theorem 112.4.3 *(Ordinate data do not determine the divisor form).* There are conjugation- and reflection-invariant divisors in $0 < \Re s < 1$ with identical ordinate multisets whose forms

$$Q_{\mathcal{D}}(f) = \sum_{\rho \in \mathcal{D}} h_f(z_\rho)\overline{h_f(\bar{z}_\rho)}, \qquad f \in \mathscr{A},$$

have different positivity properties. Their ordinate statistics are identical, including every statistic formed from finite tuples of ordinates.

Proof. Let $\Gamma \subset \mathbb{R}$ be a locally finite symmetric multiset with polynomial counting growth, and let $0 < a < 1/2$. Compare the multisets

$$\mathcal{D}_0 = \{1/2 + i\gamma : \gamma \in \Gamma\} \sqcup \{1/2, 1/2\}, \qquad \mathcal{D}_a = \{1/2 + i\gamma : \gamma \in \Gamma\} \sqcup \{1/2 - a, 1/2 + a\}.$$

Both have ordinate multiset $\Gamma \sqcup \{0, 0\}$ and the stated symmetries. The first form is a sum of absolute squares. Choose an even nonnegative nonzero $\phi \in C_c^\infty(\mathbb{R})$ and put

$$f_L(u) = \phi(u - L) - \phi(u + L), \qquad c = h_\phi(ia) = h_\phi(-ia) > 0.$$

For real t, $|h_{f_L}(t)| \le 2|h_\phi(t)|$. The number $C = \sum_{\gamma \in \Gamma} |h_\phi(\gamma)|^2$ is finite by rapid decay and polynomial counting growth. At the two nonreal spectral parameters,

$$h_{f_L}(ia) = -2c\sinh(aL), \qquad h_{f_L}(-ia) = 2c\sinh(aL).$$

Consequently

$$Q_{\mathcal{D}_a}(f_L) \le 4C - 8c^2\sinh^2(aL) < 0$$

for sufficiently large L. All sums converge absolutely. Changing the real parts has left every ordinate statistic unchanged. □

Remark 112.4.4 *(Scope of the statistical comparison).* The countermodel is a statement about abstract divisors and the information retained by ordinate statistics. It is not a zeta divisor and satisfies no stipulated prime explicit formula; it proves no logical independence assertion about RH for the fixed Riemann zeta function. The dynamical implication in Theorem 105.8.2 additionally assumes a zero-divisor realization, semiclassical dynamics, and the appropriate BGS universality statement. RH alone supplies none of those dynamical hypotheses.

112.5 Extension to the Full Weil Cone

The next proposition extends positivity by common-compact approximation. Positivity on separate images does not itself control cross terms on their linear span. Translation supplies a separate extension when every convolution square has a representative in the given image.

Proposition 112.5.1 *(Extension by approximation of compact tests).* Suppose a family of specified test maps $J_\alpha : S_\alpha \to \mathscr{A}$ has positive pulled-back Weil forms. Then q_W is positive on the union of their images. If every compact test is a smooth limit of tests from that union with all supports in a common compact set, A-TP follows. For the Heegner map, the exact positivity condition is $R + G \succeq 0$.

Proof. The comparison identities give positivity on each image. The continuity argument in Theorem 112.2.1 gives the extension. The Heegner condition is the Gram-matrix identity. □

112.6 Relation to Scattering and Arithmetic Realizations

Chapter 101 constructs specified self-adjoint scattering pairs and proves obstructions to positivity of their ordinary trace differences. Its literal entire-Mellin class is zero. Neither a scattering phase nor an inequality for a single summand supplies the full identity with $P - W_\infty - W_{\mathrm{fin}}$.

A-TP remains equivalent to RH on the full compact-test algebra. Under their respective sign hypotheses, the specified comparisons establish positivity on their stated images. Passing to all compact tests requires a further argument, such as approximation or the translation identity. The abstract divisor countermodel concerns the information lost by taking ordinates; it does not remove the arithmetic hypotheses of a zeta realization. A dynamical route to GUE statistics retains the separate assumptions of Theorem 105.8.2.

The arithmetic-gravity comparisons in Chapters 103–105 retain their separate state-space, chiral-homology, spectral-flow and semiclassical hypotheses. A proposed prismatic interpretation at the archimedean place also requires its own carrier and comparison; it is not constructed by the Fourier or height maps here.

112.7 The Remaining Comparison Requirements

(i) The comparison theorem transfers positivity through a specified map and pairing identity. Common-compact approximation gives one extension to all tests. Translation gives another, without requiring a dense image.

(ii) The Fourier-window map identifies an actual restricted test space. The Heegner map identifies a finite test image and the residual matrix $R = M - G$. Their parameterizations do not establish disjointness or density of their images.

(iii) Ordinate statistics are unchanged in the two explicit divisor models, although one divisor form has a negative square. This statement does not assert an arithmetic counterexample to RH or a theorem that RH implies GUE statistics.

(iv) The full residual is $W(f * f^*) \succeq 0$ for every $f \in \mathscr{A}$.

A comparison with A-TP must preserve the full normalized functional, or prove an explicit equivalence with it, together with an argument transferring positivity to all compact tests.

Chapter 113

Paskunas Wild Block Ext^1-Class Match at $p \in \{2, 3\}$

The intended wild-prime class comparison can be factored through Galois extension data or through a specified pseudo-character tangent space. Either factorization requires actual maps in the chosen coefficient category. A finite quiver model does not supply these arithmetic maps, and equal dimensions do not identify classes.

113.1 The Two Class-Comparison Routes

The conditional composition of Theorem 87.2.9 has the form

$$\mathrm{Ext}^1_{\mathfrak{B}} \xrightarrow{\sim} D_{\bar\rho_{\mathfrak{B}}} \xrightarrow{\sim} \mathbb{T}^1_{\mathcal{X}}|_{x_{\tau_{\mathfrak{B}}}}.$$

Here $D_{\bar\rho_{\mathfrak{B}}}$ is the fixed-determinant tangent $Z^1_{\mathrm{cts}}(G_{\mathbb{Q}_p}, \mathrm{ad}^0\,\bar\rho_{\mathfrak{B}})/d(\mathrm{ad}\,\bar\rho_{\mathfrak{B}})$, as computed in Lemma 87.2.4. Its determinant-one conjugation variant is used only when that equivalence relation is chosen. Its first arrow is the additional block-to-Galois hypothesis stated there; the spectral tangent comparison is a separate input. The pseudo-character route considered here likewise needs an actual block-to-pseudo-character map and a spectral map, with any required duals made explicit. Agreement of the routes requires a comparison of their maps, not simply a common dimension.

113.2 Block Centraliser Pseudo-Character Tangent

Pseudo-Character Tangent Space at Wild Primes

Lemma 113.2.1 *(relative first-order tangent).* Let k be a perfect field of characteristic p, and let $(R, \mathfrak{m}, k)$ be a complete Noetherian local $W(k)$-algebra. The relative tangent space consists of continuous local $W(k)$-algebra maps $R \to k[\epsilon]/(\epsilon^2)$ reducing to the residue map. It is

$$T_{R/W(k)} = \mathrm{Hom}_k\big(\mathfrak{m}/(\mathfrak{m}^2 + pR), k\big).$$

Identifying this space with a specified pseudo-character deformation tangent requires a pro-representing isomorphism $R \simeq R_{\mathrm{ps}}$.

Proof. Such a map has the form $r \mapsto \bar r + \epsilon d(r)$, where d is a $W(k)$-linear derivation with values in the residue module. On $\mathfrak{m}$, the derivation kills $\mathfrak{m}^2$ and pR. Conversely, reduce modulo p. The resulting local k-algebra splits as $k \oplus \mathfrak{m}/pR$ as a vector space. A functional on $\mathfrak{m}/(\mathfrak{m}^2 + pR)$, extended by zero on k, satisfies the Leibniz rule. These constructions are inverse. □

Remark 113.2.2 *(the pseudo-character comparison).* Pro-representability alone supplies neither a block-ring isomorphism nor a map from block extensions to a deformation tangent. At small primes one must specify the deformation object, whether a trace or a pseudorepresentation, its coefficients and its determinant condition. These choices precede any comparison with an arithmetic block.

Absolute Extensions and Relative Tangents

PROPOSITION 113.2.3 *(self-extensions of the residue field).* For a commutative Noetherian local ring $(R, \mathfrak{m}, k)$,

$$\mathrm{Ext}^1_R(k, k) \simeq \mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k).$$

Every simple R-module is isomorphic to k. Thus this formula has no grading by pairs of distinct simple R-modules.

Proof. Apply $\mathrm{Hom}_R(-, k)$ to $0 \to \mathfrak{m} \to R \to k \to 0$. The restriction $\mathrm{Hom}_R(R, k) \to \mathrm{Hom}_R(\mathfrak{m}, k)$ is zero, and $\mathrm{Ext}^1_R(R, k) = 0$. Exactness identifies the extension space with $\mathrm{Hom}_R(\mathfrak{m}, k)$. Since $\mathfrak{m}k = 0$, these homomorphisms factor precisely through $\mathfrak{m}/\mathfrak{m}^2$. Every simple module is a quotient by a maximal ideal, and a local ring has only the maximal ideal $\mathfrak{m}$. □

Example 113.2.4 *(regularity and the coefficient base).* For $R = k[[t]]$, the resolution

$$0 \longrightarrow R \xrightarrow{t} R \longrightarrow k \longrightarrow 0$$

gives $\mathrm{Ext}^1_R(k, k) = k$, although R is regular. For $R = W(k)$, multiplication by p gives the same resolution and absolute extension space k. Its tangent relative to $W(k)$ is zero. Absolute residue-field extensions therefore do not determine the relative deformation tangent without a comparison.

PROPOSITION 113.2.5 *(duality for a specified deformation tangent).* Let L_x be a complex of k-vector spaces, with $H^0(L_x)$ finite-dimensional. Suppose the specified pseudo-character tangent T_{ps} is identified with $H^0(R\,\mathrm{Hom}_k(L_x, k))$. Then $H^0(L_x) \simeq T_{\mathrm{ps}}^\vee$.

Proof. Linear duality over a field is exact, so $H^0(R\,\mathrm{Hom}_k(L_x, k)) = \mathrm{Hom}_k(H^0(L_x), k)$. Finite-dimensional biduality proves the assertion. □

The degree-one cotangent target of the arithmetic comparison requires an additional shift and a comparison of complexes. In particular, the required isomorphism

$$\mathbb{L}^1_{\chi^{\mathrm{EG,wild},p,\chi}_{GL_2}\big|_{x_{\tau_{\mathfrak{B}}}}} \xrightarrow{\sim} T_{\mathrm{ps}}^\vee$$

is an additional hypothesis, not a consequence of degree-zero duality.

113.3 WILD-PRIME CLASS-MATCH THEOREM

THEOREM 113.3.1 *(wild block* Ext^1*-class match, pseudo-character derivation). Conditional statement.* Let $p \in \{2, 3\}$, χ a central character, and $\mathfrak{B}$ a non-semisimple Paškūnas block. Assume the wild-EG datum, actual natural isomorphisms of the block extension space and the spectral cotangent fibre with the same dual pseudo-character tangent, and preservation of the chosen arithmetic classes. Then there is a canonical k-linear class-level isomorphism

$$\mathrm{c}^{\mathrm{class,B}}_{p,\mathfrak{B}} : \mathrm{Ext}^1_{\mathfrak{B}}(S, S') \xrightarrow{\sim} \mathbb{L}^1_{\chi^{\mathrm{EG,wild},p,\chi}_{GL_2}\big|_{x_{\tau_{\mathfrak{B}}}}}$$

natural in $\mathfrak{B}$. If the arithmetic classes in Questions 54.4.8 and 54.4.9 have been constructed, the class-preserving factor maps carry them to the specified cotangent directions. In characteristic two, trivial and sign characters cannot label two distinct objects. At three, a radical-square-zero model has a nonzero

adjacent Yoneda product; compatibility in higher degrees must retain it. Agreement with the map of Theorem 87.2.9 requires a specified tangent-to-cotangent identification, or maps to a common target, followed by a comparison on each labelled line.

Proof. Compose the assumed block-to-dual-pseudo-character isomorphism with the inverse of the assumed spectral-to-dual-pseudo-character isomorphism. The composite is an isomorphism and is natural because both factors are natural. Their stipulated compatibility transports the selected classes. A comparison with the other factorization is additional data; it does not follow from the existence of these maps. □

Corollary 113.3.2 *(comparison on labelled lines).* Suppose both class-comparison routes are constructed and carry each specified source line isomorphically onto the same specified target line. Their restrictions to that line differ by a nonzero scalar. For several lines these scalars may differ; there need not be one scalar for the whole vector space.

Proof. Compose one line isomorphism with the inverse of the other. Every automorphism of a one-dimensional vector space is multiplication by a nonzero scalar. This statement applies to each labelled line separately and assumes that neither map mixes those lines. □

113.4 Derivation Comparison

The pseudo-character comparison must specify its coefficient ring, block objects, deformation condition and tangent/cotangent duality. The finite-algebra examples of Chapter 54 do not establish these data in an arithmetic category. In particular, the two local comparison questions contain no trace-pairing map that could be imported as a class-level arithmetic isomorphism.

113.5 Wild Class Boundary

The two proposed routes remain conditional on their actual factor maps and class compatibility. Once supplied, composition proves a natural class isomorphism; agreement between the routes additionally requires a commuting comparison, or specified linewise normalizations. Neither a block-centraliser dimension nor a pro-representable deformation functor alone supplies those identifications. Any higher-degree arithmetic realization must also account for the Yoneda products and global tensor terms computed in Section 54.4.

113.6 Sources

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Chapter 114

Ramanujan for Maass Forms: Sharpening 7/64 and a Sub-RH Conditional

114.1 Domain and Target

Fix the target. For $\mathrm{GL}_2/\mathbb{Q}$ with trivial central character, a Maass cuspidal Hecke eigenform ϕ at level 1 has local Satake parameters $(\alpha_p(\phi), \beta_p(\phi))$ with $\alpha_p\beta_p = 1$ and $\theta_p(\phi) := \max(\log|\alpha_p|, \log|\beta_p|)/\log p \geq 0$. The Ramanujan conjecture asserts $\theta_p(\phi) = 0$ for all p and all cuspidal ϕ. The current best unconditional bound is Kim–Sarnak 2003 Appendix 2 to *J. Amer. Math. Soc.* 16, $\sup_p \theta_p(\phi) \leq 7/64 = 0.109375$, established via the symmetric-fourth-power lift $\mathrm{sym}^4\phi$ on $\mathrm{GL}_5/\mathbb{Q}$ of Kim 2003 *J. Amer. Math. Soc.* 16 combined with a Luo–Rudnick–Sarnak 1995 *Geom. Funct. Anal.* 5 L-function convexity argument.

Target. Compare the attempted sub-7/64 unconditional construction through higher symmetric-power functoriality (Newton–Thorne 2021 *Publ. Math. IHES* 134; Clozel–Harris–Taylor 2008 *Publ. Math. IHES* 108 as prerequisite automorphy-of-lifts). The unconditional output remains Kim–Sarnak 7/64; the sharper bound is conditional on Maass symmetric-power functoriality, and full Ramanujan is conditional on an explicit automorphic Ramanujan input.

Domain restriction. The chapter treats level 1 Maass forms on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ with trivial nebentypus; extension to arbitrary level and nebentypus follows the same argument and is not used later. All stated local Satake bounds are uniform in the archimedean parameter of ϕ, in the sense of Iwaniec 2002 §14.4.

114.2 Construction A: Kim–Shahidi Symmetric-Cube and the 1/9 Baseline

The Symmetric-Cube Lift

Theorem 114.2.1 *(Kim–Shahidi: $\theta \leq 1/9$).* For the level-one Maass form ϕ fixed above, $\sup_p \theta_p(\phi) \leq 1/9$.

Proof. Write π for the unitary cuspidal automorphic representation attached to ϕ. Kim–Shahidi, *Functorial products for* $\mathrm{GL}_2 \times \mathrm{GL}_3$ *and the symmetric cube for* GL_2, Ann. Math. 155 (2002), Theorem 6.1, constructs $\mathrm{Sym}^3\pi$ on GL_4. This lift is cuspidal outside the dihedral and tetrahedral cases. If it is cuspidal, Kim–Shahidi, *Cuspidality of symmetric powers with applications*, Duke Math. J. 112 (2002), Proposition 4.8, gives

$$|\alpha_p|, |\beta_p| < p^{1/9}$$

at every unramified prime. All finite primes are unramified in the level-one domain. Taking logarithms gives the stated bound. The dihedral and tetrahedral cases are tempered, so their exponents are zero. In the dihedral case this follows from unitary automorphic induction of a character. In the tetrahedral case it follows from the unitary Weil parameter described in Proposition 3.3.8 of the same Duke paper.

Proposition 4.8 is the analytic input for this argument. Its proof uses the symmetric-ninth L-function in Proposition 4.7 and Shahidi's local estimate (1988, Lemma 5.8). These analytic properties do not assert cuspidal automorphy of $\mathrm{Sym}^9\pi$. □

General and Refined Symmetric-Power Bounds

For a unitary cuspidal representation Π of $\mathrm{GL}_n/\mathbb{Q}$, the Luo–Rudnick–Sarnak estimate bounds its unramified Satake exponents by $1/2 - 1/(n^2+1)$. Jacquet–Shalika's local generic-unitary estimate instead gives exponents strictly below $1/2$. The symmetric-power eigenvalues of a two-dimensional Satake parameter are

$$\alpha_p^m,\ \alpha_p^{m-1}\beta_p,\ \ldots,\ \beta_p^m.$$

Since $\alpha_p\beta_p = 1$, their largest absolute logarithmic exponent is $m\theta_p(\phi)$. When $\mathrm{Sym}^3\,\pi$ is cuspidal, the general estimate gives

$$\theta_p(\phi) \le \frac{1/2 - 1/17}{3} = \frac{5}{34}.$$

When $\mathrm{Sym}^4\,\pi$ is cuspidal, it gives

$$\theta_p(\phi) \le \frac{1/2 - 1/26}{4} = \frac{3}{26}.$$

The latter upper bound is weaker than $1/9$, since $3/26 - 1/9 = 1/234 > 0$. The $1/9$ and $7/64$ estimates therefore require the more specific analytic inputs, beyond this general rank bound.

114.3 Construction B: Kim's Symmetric Fourth and the 7/64 Bound

THEOREM 114.3.1 *(Kim–Sarnak: $\theta \le 7/64$).* For the level-one Maass form ϕ fixed above, $\sup_p \theta_p(\phi) \le 7/64 = 0.109375$.

Proof. Kim's symmetric-fourth transfer constructs $\Pi = \mathrm{Sym}^4\,\pi$ on GL_5. It is cuspidal outside the dihedral, tetrahedral, and octahedral cases by Kim–Shahidi, Duke Math. J. 112 (2002), Theorem 3.3.7. In the cuspidal case, the Kim–Sarnak refinement in Appendix 2 to Kim, J. Amer. Math. Soc. 16 (2003), gives

$$\max_j \left|\log_p |\alpha_j(\Pi_p)|\right| \le \frac{1}{2} - \frac{1}{1 + 5\cdot 6/2} = \frac{7}{16}.$$

This is the degree-five case of the refined estimate recorded in Sarnak, *Notes on the generalized Ramanujan conjectures*, equations (21)–(24), pp. 666–667. Its symmetric-square L-function argument has a smaller analytic conductor than the general Rankin–Selberg argument. The extremal symmetric-fourth Satake eigenvalue has exponent $4\theta_p(\phi)$. Division by four gives $7/64$. The three exceptional types are tempered, and hence satisfy the same estimate with exponent zero.

The exterior-square construction involved in the transfer has rank six. Let $V = \mathbb{C}^2$ be the standard algebraic representation of $\mathrm{GL}_2(\mathbb{C})$. Then

$$\Lambda^2\,\mathrm{Sym}^3 V \cong (\mathrm{Sym}^4 V \otimes \det V) \oplus (\det V)^3.$$

For diagonal eigenvalues a, b, the six weights on the left are a^5b, a^4b^2, a^3b^3, a^3b^3, a^2b^4, ab^5. These are exactly the five weights of $\mathrm{Sym}^4 V \otimes \det V$ and the weight of $(\det V)^3$. Complete reducibility over $\mathbb{C}$ gives the displayed representation identity. With trivial central character, this is the identity of unramified Satake representations underlying $\Lambda^2(\mathrm{Sym}^3\,\pi) = \mathrm{Sym}^4\,\pi \boxplus \mathbf{1}$. The global transfer is the separate automorphy theorem used above. The exterior square of $\mathrm{Sym}^2 V$ has rank three and is a different representation. □

Remark 114.3.2 *(analytic estimates for the symmetric fourth).* An analytic refinement through $L(s, \mathrm{sym}^4 \phi)$ must specify the varying parameters and an estimate that implies a stronger local Satake bound. For a fixed unitary cuspidal representation Π of $\mathrm{GL}_5/\mathbb{Q}$, Nelson's *Bounds for standard L-functions*, Theorem 1.1, gives

$$|L(\Pi, 1/2 + it)| \ll_{\Pi,\delta} (1 + |t|)^{5(1-\delta)/4}, \qquad 0 < \delta < 1/4050.$$

Here L is the finite standard L-function, with archimedean factors excluded. This applies when $\Pi = \mathrm{sym}^4 \phi$ is cuspidal. Its implicit constant may depend on ϕ. An improvement of $7/64$ by this route therefore requires additional uniformity and a proved implication from the analytic estimate to the local exponents. Cancellation beyond the estimate is another possible input. Construction C instead assumes cuspidal automorphy of a higher Maass symmetric power.

114.4 Construction C: Newton–Thorne Symmetric-Power Functoriality and the Sharpening

The Newton–Thorne Theorem

Theorem 114.4.1 *(Newton–Thorne 2021: unconditional holomorphic* sym^n *functoriality).* (Newton–Thorne 2021 *Publ. Math. IHES* 134, Main Theorem; extended in Newton–Thorne 2023 *Publ. Math. IHES* 137 to the non-CM case with full generality). Let ϕ be a holomorphic cuspidal Hecke newform of weight $k \geq 2$ on $\mathrm{SL}_2(\mathbb{Z})$ that is non-CM. Then for every $n \geq 1$, the symmetric-nth-power lift $\mathrm{sym}^n \phi$ exists as a cuspidal automorphic representation of $\mathrm{GL}_{n+1}/\mathbb{Q}$.

Proof. Newton–Thorne 2021 (holomorphic, $k \geq 2$). The argument proceeds by automorphy-lifting through Galois deformation rings of the ℓ-adic representation associated to ϕ, combined with $R = T$ theorems in the tame Taylor–Wiles patching framework (Taylor–Wiles 1995 *Ann. Math.* 141, extended by Clozel–Harris–Taylor 2008 *Publ. Math. IHES* 108 to the GL_n setting). □

Remark 114.4.2 *(Newton–Thorne does not cover Maass directly).* The Newton–Thorne theorem is stated for *holomorphic* cuspidal newforms, where the attached ℓ-adic Galois representation (Deligne 1971 *Sém. Bourbaki* 355 for $k \geq 2$; Deligne–Serre 1974 *Ann. Sci. ENS.* 7 for $k = 1$) is the input to the automorphy-lifting theorem. Maass cusp forms lie outside the hypothesis of Newton–Thorne 2021: the automorphy-lifting theorem requires an ℓ-adic Galois representation of algebraic Hodge weight ≥ 1, and the conjectural Langlands-attached representation of a Maass form has weight 0 in the Hodge sense. The holomorphic companion construction gives a diagnostic Rankin–Selberg convolution $L(s, \phi \times \mathrm{sym}^n F)$, but it does not transfer temperedness of $\mathrm{sym}^n F$ into a sub-$7/64$ bound for $\theta_p(\phi)$.

The Auxiliary Holomorphic Form and the Transfer

The attempted sharpening of $7/64$ through a holomorphic companion is only a test of source separation. It supplies no division of the Maass Satake exponent by n. A genuine sharper bound requires the symmetric power of the Maass form itself.

Definition 114.4.3 *(Maass–holomorphic Rankin–Selberg pairing).* . Let ϕ be a Maass cusp form of $\mathrm{GL}_2/\mathbb{Q}$ and F a holomorphic cuspidal newform of weight $k \geq 2$. The *Maass–holomorphic Rankin–Selberg L-function* is

$$L(s, \phi \times F) := \prod_p L_p(s, \phi \times F) = \prod_p \prod_{i,j=1,2} \left(1 - \alpha_{p,i}(\phi)\alpha_{p,j}(F) p^{-s}\right)^{-1},$$

with $\alpha_{p,i}(\phi) \in \{\alpha_p(\phi), \beta_p(\phi)\}$ and analogously for F. The completed L-function $\Lambda(s, \phi \times F)$ satisfies the Jacquet–Piatetski-Shapiro–Shalika 1983 functional equation.

PROPOSITION 114.4.4 *(RS convolution for a Maass form with a tempered companion).* Let ϕ be a Maass cusp form and F a holomorphic cuspidal newform of weight $k \geq 2$, non-CM. Fix $n \geq 4$. The symmetric-nth-power lift $\mathrm{sym}^n F$ (Newton–Thorne 2021) is a cuspidal tempered automorphic representation of $\mathrm{GL}_{n+1}/\mathbb{Q}$. The Rankin–Selberg L-function $L(s, \phi \times \mathrm{sym}^n F)$, as a $\mathrm{GL}_2 \times \mathrm{GL}_{n+1}$ L-function, admits a functional equation, meromorphic continuation to $\mathbb{C}$, and satisfies: $L(s, \phi \times \mathrm{sym}^n F) \neq 0$ for $\mathrm{Re}(s) \geq 1$ (Jacquet–Shalika 1981).

Proof. Newton–Thorne 2021 (Theorem 114.4.1) gives $\mathrm{sym}^n F$ as a cuspidal $\mathrm{GL}_{n+1}/\mathbb{Q}$ representation. Deligne 1974 (Weil conjectures) gives temperedness. Jacquet–Piatetski-Shapiro–Shalika (JPSS) 1983 *Amer. J. Math.* 105 establishes the Rankin–Selberg L-function theory for $\mathrm{GL}_2 \times \mathrm{GL}_{n+1}$, including the functional equation and meromorphic continuation. Jacquet–Shalika 1981 non-vanishing on $\mathrm{Re}(s) = 1$ applies to the cuspidal GL_{n+1}-factor $\mathrm{sym}^n F$. □

The Sharpened Bound: Construction C Main Theorem

Remark 114.4.5 *(why the companion trick is conditional).* Jacquet–Shalika bounds the two automorphic factors separately. It does not divide the GL_{10} exponent of $\mathrm{sym}^9 F$ by nine to bound the Maass form ϕ. A sharper Maass bound therefore requires functoriality for $\mathrm{sym}^9 \phi$ itself.

THEOREM 114.4.6 *(sharpened Ramanujan bound: conditional on Maass symmetric-power functoriality). (conditional on the Maass symmetric-ninth-power conjecture: that* $\mathrm{sym}^9 \phi$ *exists as a cuspidal automorphic representation of* $\mathrm{GL}_{10}/\mathbb{Q}$ *for the Maass form* ϕ *itself).* Let ϕ be a cuspidal Maass Hecke eigenform of $\mathrm{GL}_2/\mathbb{Q}$. Assuming $\mathrm{sym}^9 \phi$ is cuspidal on $\mathrm{GL}_{10}/\mathbb{Q}$, the Jacquet–Shalika 1981 bound applied to the GL_{10} form $\mathrm{sym}^9 \phi$ gives

$$\sup_p \theta_p(\phi) \;\leq\; \frac{1/2 - 1/(10^2+1)}{9} \;=\; \frac{1/2 - 1/101}{9} \;\approx\; 0.05446,$$

strictly below Kim–Sarnak's $7/64 \approx 0.1094$.

Proof. Conditional on $\mathrm{sym}^9 \phi$ cuspidal automorphic on $\mathrm{GL}_{10}/\mathbb{Q}$ (the Maass symmetric-ninth-power conjecture), the Jacquet–Shalika 1981 *Amer. J. Math.* 103 non-vanishing of $L(s, \mathrm{sym}^9 \phi \otimes (\mathrm{sym}^9 \phi)^\vee)$ on $\mathrm{Re}(s) = 1$ forces the Satake parameters of $\mathrm{sym}^9 \phi$ on GL_{10} to satisfy $\theta_p(\mathrm{sym}^9 \phi) \leq 1/2 - 1/101$. The extremal Satake parameter of $\mathrm{sym}^9 \phi$ at p is $p^{9\theta_p(\phi)}$, so $9\theta_p(\phi) \leq 1/2 - 1/101$, giving the stated bound. □

Remark 114.4.7 *(form of Maass symmetric-power functoriality).* Newton–Thorne 2021 establishes $\mathrm{sym}^n \phi$ cuspidal on $\mathrm{GL}_{n+1}/\mathbb{Q}$ for *holomorphic* newforms ϕ. The analogous statement for Maass forms is the content of the Langlands global functoriality conjecture for $\mathrm{GL}_2 \to \mathrm{GL}_{n+1}$ at non-algebraic archimedean type; it is open for all $n \geq 5$. The holomorphic companion trick – forming $L(s, \phi \times \mathrm{sym}^n F)$ for a holomorphic form F – does not produce a bound on $\theta_p(\phi)$ sharper than the direct GL_2 Jacquet–Shalika bound $\theta_p(\phi) \leq 2/5$: the Rankin–Selberg non-vanishing constrains the GL_{10} factor $\mathrm{sym}^n F$, not the GL_2 factor ϕ. The bound $(1/2 - 1/101)/9$ in Theorem 114.4.6 is the correct conditional bound on the Maass symmetric-power conjecture, not an unconditional sharpening.

Remark 114.4.8 *(local pairing errors require comparison estimates).* A Satake exponent bound $\theta_p \le \vartheta$ does not by itself give a norm estimate between independently defined pairings. An estimate of the form

$$\left|\langle v_p, w_p\rangle_p^{\mathrm{Pet}} - \langle v_p, \sigma_p(w_p)\rangle_p^{\mathrm{Koszul}}\right| \le C_p p^{-(1/2-\vartheta)}$$

requires bounds for the actual arithmetic comparison maps, their normalizations and the chosen local vectors. The exponent $\vartheta = 7/64$, or the conditional symmetric-power improvement, controls Satake parameters only. Moreover the indicated prime majorants are not summable without additional decay.

114.5 Construction D: Arthur Endoscopic Classification on Classical Groups

The automorphic Ramanujan conjecture for a Maass form ϕ of $\mathrm{GL}_2/\mathbb{Q}$ is detected by the tensor square $\phi \boxtimes \phi^\vee$. This is a GL_4-object, usually isobaric rather than cuspidal. Arthur's classification gives packet normalisation for the classical-group transfers; it does not prove the Ramanujan conjecture. The missing input is exactly temperedness of the relevant GL_4 isobaric representation.

Arthur's Endoscopic Classification

Theorem 114.5.1 *(Arthur 2013: endoscopic classification for classical groups).* (Arthur 2013 *Colloq. Publ.* 61, *The Endoscopic Classification of Representations*, Thm. 1.5.2). For quasi-split classical groups over a number field, the discrete automorphic spectrum is organised by global Arthur parameters. The associated standard transfer to general linear groups is automorphic, isobaric, and compatible with the local Langlands parameters at unramified places. Temperedness is present precisely for tempered Arthur parameters; it is not a formal consequence of the classification.

Proof. Arthur 2013 *The Endoscopic Classification* Thm. 1.5.2 and §4.3 give the parametrisation and the standard transfer. The stabilisation input is the Langlands–Shelstad–Ngô–Waldspurger endoscopic trace formula. Mœglin–Waldspurger controls the local discrete-series packets. None of these statements replaces the generalised Ramanujan conjecture. □

Remark 114.5.2 *(stabilisation is not Ramanujan).* For $F = \mathbb{Q}$ the required stabilisation rests on the fundamental lemma and the endoscopic transfer identities. This is a theorem of the trace formula. It identifies packets and transfers. It does not bound Satake parameters.

The Ramakrishnan Functorial Chain

Proposition 114.5.3 *(Ramakrishnan 2000:* $\mathrm{GL}_2 \boxtimes \mathrm{GL}_2 \to \mathrm{GL}_4$*).* (Ramakrishnan 2000 *Ann. Math.* 152 *Modularity of the Rankin–Selberg L-series, and multiplicity one for* SL_2, Thm. M). Let ϕ_1, ϕ_2 be cuspidal automorphic representations of $\mathrm{GL}_2/\mathbb{Q}$. Then their Langlands tensor product $\phi_1 \boxtimes \phi_2$ is an automorphic representation of $\mathrm{GL}_4/\mathbb{Q}$; it is cuspidal unless ϕ_1 and ϕ_2 are twists of a common dihedral form.

Proof. Ramakrishnan 2000 Thm. M. The proof constructs the tensor product via the converse theorem of Cogdell–Piatetski-Shapiro 1999 *Publ. Math. IHES* 88, using Bump–Friedberg 1990 *Duke Math. J.* 59 for the Rankin–Selberg L-function. □

The Automorphic Conditional

Hypothesis 114.5.4 *(Arthur–Ramakrishnan–GRC hypothesis,* AR-GRC*).* Assume:

(AR1) Arthur's endoscopic classification Theorem 114.5.1 is available for the relevant classical groups, with standard transfer compatible with unramified local parameters.

(AR2) The Ramakrishnan tensor product Proposition 114.5.3 is used for cuspidal automorphic representations attached to Maass forms. For $\phi_2 = \phi_1^\vee$ the product is interpreted isobarically.

(AR3) The isobaric representation $\phi \boxtimes \phi^\vee$ satisfies the generalised Ramanujan conjecture at every finite place.

THEOREM 114.5.5 *(AR-GRC implies full Ramanujan for Maass forms). (conditional on Hypothesis 114.5.4).* Under (AR1)+(AR2)+(AR3), every cuspidal Maass Hecke eigenform ϕ of $\mathrm{GL}_2/\mathbb{Q}$ satisfies the Ramanujan bound $\sup_p \theta_p(\phi) = 0$, i.e. is tempered at every finite prime.

Proof. At an unramified finite prime write the unitary Satake parameters of ϕ as α_p, β_p, with $|\alpha_p\beta_p| = 1$. The local parameter of $\phi \boxtimes \phi^\vee$ has eigenvalues

$$1, \quad 1, \quad \alpha_p/\beta_p, \quad \beta_p/\alpha_p.$$

By (AR3) these have modulus 1. Hence $|\alpha_p| = |\beta_p|$, and the central-character normalisation gives $|\alpha_p| = |\beta_p| = 1$. Thus ϕ_p is tempered. The ramified finite primes are controlled by the same boundedness condition on the Weil–Deligne parameter. Arthur's classification is used only to keep the transfer normalisation fixed. □

PROPOSITION 114.5.6 *(AR-GRC is separate from classical RH in this argument).* Hypothesis AR-GRC (Hypothesis 114.5.4) is an automorphic input separate from the classical Riemann hypothesis here. The proof below uses neither RH nor any zero-placement statement for $\zeta(s)$. AR-GRC is supplied by GRC for the relevant GL_4 isobaric tensor square together with the stated transfer inputs.

Proof. GRC for the GL_4 isobaric tensor square gives bounded local parameters. The proof of Theorem 114.5.5 then reads off the GL_2 Satake radii. No statement about zeros of $\zeta(s)$ enters this implication. Conversely, the argument makes no deduction from RH to AR-GRC. □

Remark 114.5.7 *(the automorphic residual).* Theorem 114.5.5 places the Maass-sector temperedness requirement of Theorem 76.6.1 on AR-GRC, while leaving its arithmetic pairing maps as separate hypotheses, which is the generalised Ramanujan conjecture for the GL_4 isobaric tensor square, together with the stated transfer normalisations. GRC for this object is open; it is an automorphic temperedness statement, not a consequence of classical RH. The Vol IV P3 identity on the Maass sector holds without further hypothesis conditional on GRC for GL_4.

114.6 CONSTRUCTION E: SYNTHESIS

THEOREM 114.6.1 *(Maass-sector P3 identity: local controls and exact conditional form).* For the displayed local pairing bounds assume the corresponding norm estimates in Remark 114.4.8. For exact global equality assume the arithmetic placement and pairing hypotheses of Hypothesis 67.1.1, the stated Maass temperedness domain, and the adelic extension. *(local pairing bounds require comparison-norm estimates; conditional on the Maass symmetric-ninth-power conjecture for the sharper local control; conditional on GRC for* $\mathrm{GL}_4/\mathbb{Q}$ *for the exact equality).* Here $\mathcal{V}^{\mathrm{prim}}$ denotes the primitive automorphic

Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2, as defined in Chapter 76, and $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,Maass}} \subset \mathcal{V}^{\mathrm{prim}}$ its Maass cuspidal sector. For local vectors v_p, w_p in the p-component and global vectors $v, w \in \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,Maass}}$, the available statements are: The first inequality assumes the local norm estimate; the second also assumes the Maass symmetric-ninth-power input. The final identity assumes the comparison maps and GRC for $\mathrm{GL}_4/\mathbb{Q}$.

$$\begin{aligned} \left|\langle v_p, w_p\rangle_p^{\mathrm{Pet}} - \langle v_p, \sigma_p^{\mathrm{BC}}(w_p)\rangle_p^{\mathrm{Koszul}}\right| &\le C_p p^{-25/64}, \\ \left|\langle v_p, w_p\rangle_p^{\mathrm{Pet}} - \langle v_p, \sigma_p^{\mathrm{BC}}(w_p)\rangle_p^{\mathrm{Koszul}}\right| &\le C_p p^{-(1/2-(1/2-1/101)/9)}, \\ \langle v, w\rangle^{\mathrm{Pet}} &= \langle v, \sigma^{\mathrm{BC}}(w)\rangle^{\mathrm{Koszul}}. \end{aligned}$$

The first two lines do not assemble to an adelic correction term without extra summability; the exact global identity is the third line.

Proof. The local bounds are exactly the assumed comparison-norm estimates with the indicated Satake exponents. The exponents alone do not prove those estimates. Exact-equality form (114.6.1): Theorem 114.5.5 gives temperedness of ϕ under AR-GRC; Theorem 76.1.1 then applies, with its arithmetic placement, pairing and adelic hypotheses, to $\mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,Maass}} \subset \mathcal{V}^{\mathrm{prim}}_{\mathrm{cusp,temp}}$. AR-GRC is GRC for the relevant GL_4 isobaric tensor square by Proposition 114.5.6. □

Construction	Bound	Form	Source
Construction A (sym^3)	$1/9 \approx 0.1111$	Unconditional	Kim–Shahidi 2002, Theorem 114.2.1
Construction B (sym^4)	$7/64 \approx 0.1094$	Unconditional	Kim–Sarnak 2003, Theorem 114.3.1
Construction C ($\mathrm{sym}^9\phi$)	$(1/2 - 1/101)/9 \approx 0.0545$	Conditional on Maass sym^9	JS 1981 + Maass sym-power conj., Theorem 114.4.6
Construction D (AR-GRC)	0 (full)	Conditional on $\mathrm{GRC}_{\mathrm{GL}_4}$	Arthur 2013 + Ramakrishnan 2000, Theorem 114.5.5

114.7 Satake Bounds and Their Hypotheses

The local exponent $\theta_p \ge 0$ uses Satake parameters normalized by $\alpha_p\beta_p = 1$. Kim's symmetric-fourth transfer and the Kim–Sarnak estimate give the unconditional bound $\theta_p \le 7/64$. The Rankin–Selberg theory of Jacquet–Piatetski-Shapiro–Shalika (1983) and the Euler-product theory of Jacquet–Shalika (1981) supply the associated L-functions and edge non-vanishing. Non-vanishing on $\mathrm{Re}(s) = 1$ is distinct from an estimate on $\mathrm{Re}(s) = 1/2$. The proposed analytic refinement through $L(s, \mathrm{sym}^4\,\phi)$ requires a specified aspect, analytic conductor, saving, and uniformity in the Maass form ϕ. It also requires an argument converting that estimate into a bound below $7/64$ for the local Satake exponents. These additional data are not specified by the edge non-vanishing theorem.

For Construction C, assume that $\mathrm{sym}^9\,\phi$ is cuspidal automorphic on $\mathrm{GL}_{10}/\mathbb{Q}$. Under this Maass symmetric-power hypothesis, Theorem 114.4.6 gives

$$\theta_p \le \frac{1/2 - 1/101}{9} \approx 0.0545.$$

Newton–Thorne functoriality applies to the holomorphic companion F of algebraic weight at least two. It does not supply the symmetric-ninth-power hypothesis for ϕ. The separation of the lowest-weight component uses the tensor decomposition in Construction C.

Construction D uses the automorphic isobaric tensor square on $\mathrm{GL}_4/\mathbb{Q}$. Ramakrishnan's tensor product is interpreted isobarically in the contragredient case. Assuming the generalized Ramanujan conjecture for this tensor square, Theorem 114.5.5 gives $\theta_p = 0$. Arthur transfer supplies the packet normalization. Its endoscopic inputs are Arthur (2013), Langlands–Shelstad (1987), Ngô (2010), Waldspurger (2008), and Chaudouard–Laumon (2012). The tensor-product and automorphy inputs are Ramakrishnan (2000) and Cogdell–Piatetski-Shapiro (1999). The dihedral case uses Deligne–Serre and Labesse–Langlands (1979). Thus the $7/64$ estimate, the conditional symmetric-ninth estimate, and the conditional temperedness conclusion use different hypotheses.

114.8 Residual Ramanujan Input

GRC for $\mathrm{GL}_4/\mathbb{Q}$. The generalised Ramanujan conjecture for the relevant isobaric $\mathrm{GL}_4/\mathbb{Q}$ tensor square is open; the Construction D closure is conditional on it. The current best unconditional bound is from Kim–Shahidi 2002 extensions to higher-rank groups: $\theta_{\mathrm{GL}_4} \le 5/14 \approx 0.357$.

Maass symmetric-power functoriality (Construction C conditional). Construction C's sharpened bound $(1/2 - 1/((n+1)^2+1))/n$ is conditional on the Maass symmetric-nth-power conjecture: the existence of $\mathrm{sym}^n \phi$ as a cuspidal automorphic representation of $\mathrm{GL}_{n+1}/\mathbb{Q}$ for the Maass form ϕ itself. This is the Langlands global functoriality conjecture for $\mathrm{GL}_2 \to \mathrm{GL}_{n+1}$ at non-algebraic archimedean type; it is open for all $n \ge 5$. Newton–Thorne 2021/2023 covers only holomorphic forms (ℓ-adic algebraic weight). The bound $\theta^{(n)} = (1/2 - 1/((n+1)^2+1))/n \to 0$ as $n \to \infty$ at rate $\sim 1/(2n)$; closing to $\theta = 0$ via this mechanism would require a proof of the Maass symmetric-power conjecture for all n, which would imply the Ramanujan conjecture by the Jacquet–Shalika bound. It is stronger than the numerical estimate alone.

The dihedral edge case. Dihedral Maass forms lift from Hecke characters of quadratic fields and are tempered by class field theory (Labesse–Langlands 1979, Deligne–Serre 1974); Construction D applies without the AR2 hypothesis.

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Chapter 115

Quantum Arithmetic Chern–Simons: Bulk Extension and Residual Obstruction

Chapter 103 states Chapter 103.4.7, the profinite $GL_2(\widehat{\mathbb{Z}})$ quantum arithmetic Chern–Simons theory on $\overline{\mathrm{Spec}\,\mathbb{Z}}$, as three linked bulk-boundary axioms. Every formal consequence depends on its resolution: Ch. 106's continuous Wilson-Frobenius identity at $\varprojlim^{\mathrm{ren}}_N$, Ch. 111's specified unitary conjugacy between F_∞^{WS} and F_∞, the partition formula $Z_{\mathrm{ACS}^{\mathrm{qu}}} = \xi$ of Chapter 103.5.7, and the triangle of Chapter 104.4.2. The bulk measure is examined through five constructions: (i) state-sum renormalisation along the congruence tower; (ii) Reshetikhin–Turaev theory for a modular trace lift of the boundary trace class; (iii) Costello–Gwilliam factorization algebras on the étale site of $\overline{\mathrm{Spec}\,\mathbb{Z}}$; (iv) Lurie cobordism hypothesis on an étale arithmetic cobordism category; (v) Scholze six-functor on a conjectural arithmetic diamond. For a specified finite field groupoid and action, its weighted sum is defined. Coherent tower measures and an open-site factorization structure require their own maps and descent. The archimedean Hilbert realization, modular trace lift and arithmetic diamond add further comparison data.

115.1 Setup and Axiomatic Target

Definition 115.1.1 *(arithmetic space and boundary data).* Write $M = \overline{\mathrm{Spec}\,\mathbb{Z}}$ for the intended arithmetic compactification. An ACS realization must specify a site $\mathscr{X}$ for M, a boundary object $\mathscr{B}_\infty$, and a boundary morphism $j : \mathscr{B}_\infty \to \mathscr{X}$. It must state a dualizing coefficient object $\mathcal{L}$, its cohomological degree and a trace map for pairing the action with an arithmetic fundamental class. At infinity it must also specify a smooth oriented surface Σ_∞, a comparison to $\mathscr{B}_\infty$, and a metric whose curvature pulls back to a specified real two-form ω_∞. The smooth disk in Chapter 111.4.1 supplies one local metric model with area $1/2$. Its existence does not construct the arithmetic boundary morphism or its metric pullback. For that disk, boundary integrals use its counterclockwise orientation; a closed surface requires a separately specified surface.

Definition 115.1.2 *(congruence tower and gauge tower).* . The *gauge tower* is the inverse system of finite groups

$$\mathcal{T} := \left\{ G_N := GL_2(\mathbb{Z}/N\mathbb{Z}), \qquad N \in \mathbb{Z}_{\geq 1} \right\}$$

ordered by divisibility $M \mid N$ giving reductions $G_N \twoheadrightarrow G_M$, with

$$\varprojlim_{N \in \mathcal{T}} G_N \;=\; GL_2(\widehat{\mathbb{Z}}).$$

A *level datum* is a compatible system $\alpha_\bullet = \{\alpha_N \in H^3(BG_N, \mathbb{C}^\times)\}_N$ whose inverse-limit class is the arithmetic level $k^{\mathrm{Ar}} = \kappa_{\mathsf{G}}^{\mathrm{Ar}}$ of Ch. 103.

Remark 115.1.3 *(the three axioms recalled).* Chapter 103.4.7 asks for a renormalised profinite limit

$$\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}} := \varprojlim_{N}^{\mathrm{ren}} Z^{\mathrm{Ar}}_{G_N,\alpha_N}$$

satisfying: **(WZW)** $\mathrm{HH}_0(\partial\,\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}}) \simeq \mathcal{V}^{\mathrm{prim}}$ on the archimedean boundary, where $\mathcal{V}^{\mathrm{prim}}$ is the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 (Ch. 103); **(Z)** $Z_{\mathrm{ACS}^{\mathrm{qu}}}(\overline{\mathrm{Spec}\,\mathbb{Z}}; k^{\mathrm{Ar}}) = \xi(k^{\mathrm{Ar}})$, where ξ is the completed Riemann zeta function (Chapter 103.5.7); **(W)** $\langle W_\rho(K_p)\rangle = \mathrm{tr}\,\rho(\mathrm{Frob}_p)$, where $W_\rho(K_p)$ is the Wilson-loop observable at the prime knot $K_p \subset \overline{\mathrm{Spec}\,\mathbb{Z}}$ in representation ρ (Ch. 106).

115.2 Construction I: Tower Renormalisation via Kan Extension

The most direct construction is the one inherited from the finite-level state-sum construction of Chapter 103.4.2. For specified finite field groupoids and compatible action phases, each $Z^{\mathrm{Ar}}_{G_N,\alpha_N}$ is the corresponding finite sum. The obstruction is the existence of a profinite functor compatible with (WZW), (Z), (W).

Finite-level Kan tower

Lemma 115.2.1 *(finite reduction and lifting mass).* Let $\pi : G_N \twoheadrightarrow G_M$ be a surjection of finite groups. It induces pullback of level classes and extension of structure group on field groupoids. Suppose the chosen arithmetic space M' has finite field groupoids, with compatible phases $\alpha_N = \pi^*\alpha_M$. For a G_M field b, write $\mathcal{L}_b$ for the homotopy fibre of its lifting groupoid, including the isomorphism to b. If all these fibres have the same finite cardinality $m_{N,M}(M')$, then

$$Z^{\mathrm{Ar}}_{G_N,\alpha_N}(M') = m_{N,M}(M')\,Z^{\mathrm{Ar}}_{G_M,\alpha_M}(M').$$

For a central abelian kernel K, if a lift exists and the lifting fibre is the groupoid of K-torsors, its cardinality is $|H^1(M',K)|/|H^0(M',K)|$, when these groups are finite.

Proof. Finite-groupoid integration weights each isomorphism class by the reciprocal order of its automorphism group. Fubini over homotopy fibres gives the sum over b of its phase times $|\mathcal{L}_b|/|\mathrm{Aut}\,b|$. Constant fibre mass gives the displayed identity. For the stated torsor groupoid, the isomorphism classes form $H^1(M',K)$ and each automorphism group is $H^0(M',K)$. Their cardinality ratio proves the second assertion. A nonempty lifting fibre is a hypothesis: centrality does not remove an obstruction to lifting a G_M field. □

Remark 115.2.2 *(the measure in a reduction tower).* For $K \cong (\mathbb{F}_\ell)^d$ with trivial action, the torsor mass above is $|K|^{b_1-b_0}$, where $b_i = \dim_{\mathbb{F}_\ell} H^i(M',\mathbb{F}_\ell)$. For example, over a point the lifting fibre is BK, of mass $1/|K|$, although $b_1 = 0$. Thus central-abelian reduction alone does not give the factor $|K|^{b_1}$. The profinite construction must determine the actual lifting obstructions, stabilizers, phases and normalization at every transition.

Definition 115.2.3 *(a chosen tower normalization).* For a specified finite Euler characteristic $\chi(M')$, define

$$\widetilde{Z}_N(M') = |G_N|^{-\chi(M')}\,Z_N(M').$$

This scalar definition does not assert transition compatibility. The cohomology theory, coefficients and any compact-support or boundary conditions used to define χ are part of the datum.

Proposition 115.2.4 *(the normalized transition factor).* Under the constant-mass hypothesis of the lemma, with $K = \ker \pi$,

$$\widetilde{Z}_N(M') = |K|^{-\chi(M')} m_{N,M}(M') \widetilde{Z}_M(M').$$

Thus the equality of normalized scalar values follows when $|K|^{-\chi} m_{N,M} = 1$. This scalar condition is additional to the coherent transition maps required for state spaces and bordism assignments.

Proof. Use $|G_N| = |K||G_M|$ in the defining normalization and the preceding mass identity. No division by a possibly zero partition value is needed. □

For a connected torsor model with $b_1 = 0$ and $\chi = 0$, the factor is $|K|^{-1}$, not one. A different arithmetic measure or a renormalization yielding compatible values must therefore be specified. Nonabelian congruence kernels additionally require their full lifting fibres. The arithmetic cohomology and duality needed to compute these quantities are separate inputs; no numerical value of $\chi(\overline{\mathrm{Spec}\,\mathbb{Z}})$ is needed here.

Theorem 115.2.5 *(a coherent inverse tower defines a pro-object).* Suppose a cofiltered small category indexes actual state-sum functors and coherent transition natural transformations in a specified target category. This inverse diagram defines a pro-object in that functor category. It does not by itself define a represented Hilbert limit or a topological arithmetic field theory.

Proof. An object of the pro-category is precisely such a cofiltered diagram. An ordinary limit or a right Kan extension in a chosen target requires that target to admit the relevant limits. Representation by a Hilbert space is a further assertion, not part of the pro-object definition. □

Remark 115.2.6 *(tower and descent requirements).* The arithmetic construction must supply the lifting measures and coherent transition maps just specified. A topological state functor further requires a represented limit and compatible gluing. Its boundary trace must be identified with the designated primary trace realization. Neither an inverse diagram nor the name of a Kan extension constructs these data.

Sources and Hypotheses for Construction I

Kim's arithmetic action and its nonabelian extensions motivate the finite field groupoids. Dijkgraaf–Witten and Freed–Quinn give finite-groupoid integration models. The lifting-mass computation above states the transition datum needed in the arithmetic tower. A condensed or Tate realization additionally requires its topology, transition maps and pairing. The finite sum and the represented Hilbert limit are distinct constructions.

Theorem 115.2.7 *(Construction I conditional closure). (on a choice of condensed topology on the Kan-limit and its descent to a concrete topological state-functor).* Under the profinite-Hilbert descent hypothesis and a coherent tower as specified above, the represented limit of Chapter 115.2.5 defines $\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}}$ satisfying the reduction-coherence portion of Chapter 103.4.7. The three axioms (WZW), (Z), (W) reduce to: (WZW) restriction of the Kan-limit boundary functor at Σ_∞ identifies with $\mathcal{V}^{\mathrm{prim}}$; (Z) evaluation of the Kan-limit on the closed class $[M]$ reproduces $\xi(k^{\mathrm{Ar}})$; (W) evaluation on prime-knot insertions reproduces Frobenius traces. Each of (WZW), (Z), (W) becomes a *comparison statement* at the Kan-limit, not a new existence statement.

Proof. The assumed coherent tower defines the pro-object of Chapter 115.2.5. Under a chosen condensed topology descending the pro-vector-space structure to a Hilbert-space functor, (WZW), (Z), (W) become comparison claims at the limit. Each is in principle checkable by comparing the limit with the right-hand sides of the three axioms using the finite-level comparisons of Chs. 106 and 103. The existence of the descent is hypothetical. □

115.3 Construction II: Reshetikhin–Turaev from an Arithmetic Boundary Category

The Reshetikhin–Turaev construction (Turaev 1994 *Quantum Invariants of Knots and 3-Manifolds*, de Gruyter; Reshetikhin–Turaev 1991 *Invent. Math.* 103, 547–597) constructs topological invariants and the associated bordism functor from a modular tensor category (MTC), with the required framing or anomaly data. An arithmetic bordism interpretation requires an additional comparison functor. In the arithmetic problem the input cannot be the category of modules over $\mathcal{V}^{\mathrm{prim}}$, because $\mathcal{V}^{\mathrm{prim}}$ is a Hochschild trace class, not a vertex algebra. The possible Reshetikhin–Turaev input is instead a modular tensor category whose Hochschild trace and regularised character have $\mathcal{V}^{\mathrm{prim}}$ as their boundary shadow.

The arithmetic MTC candidate

Definition 115.3.1 *(arithmetic modular tensor category candidate).* . The *arithmetic modular tensor category candidate* is

$$\mathcal{MTC}^{\mathrm{Ar}}_{k^{\mathrm{Ar}}} \;:=\; \mathcal{B}^{\mathrm{Ar}}_{k^{\mathrm{Ar}}},$$

a finite semisimple ribbon category equipped with a trace comparison

$$\mathrm{HH}_0\big(\mathcal{B}^{\mathrm{Ar}}_{k^{\mathrm{Ar}}}\big) \simeq \mathcal{V}^{\mathrm{prim}}$$

and with Frobenius-labelled simple objects at the finite places. This definition is a lifting problem from a Hochschild trace class to a boundary category; it is not a module category over $\mathcal{V}^{\mathrm{prim}}$.

Remark 115.3.2 *(the modular trace lift is conditional).* Modularity of $\mathcal{MTC}^{\mathrm{Ar}}_{k^{\mathrm{Ar}}}$ requires a non-degenerate braiding, a spherical trace, and compatibility of the categorical trace with the Iwahori–Hecke Hochschild trace. The completed-zeta character and finite-prime Fock characters are trace statements. They do not construct a braided category whose trace is $\mathcal{V}^{\mathrm{prim}}$.

Reshetikhin–Turaev state-sum

Theorem 115.3.3 *(transport of a specified Reshetikhin–Turaev theory).* Suppose the modular category and anomaly data define an RT functor Z_{RT} on its topological bordism category B_{RT}. Suppose an arithmetic bordism category B_{Ar} and a symmetric monoidal comparison $C : \mathsf{B}_{\mathrm{Ar}} \to \mathsf{B}_{\mathrm{RT}}$ are constructed. Then $Z^{\mathrm{Ar}}_{\mathrm{RT}} = Z_{\mathrm{RT}} \circ C$ is a symmetric monoidal arithmetic assignment. It has the intended primary boundary trace and Frobenius Wilson values if its boundary trace maps and label comparisons identify those values.

Proof. Composition preserves the functorial and monoidal identities. The last assertion is the stated equality of the trace and observable maps under that composition. The ordinary RT construction does not supply C or those arithmetic equalities. □

The trace of a boundary category, the state space of a surface and a scalar closed-bordism invariant have different types. Their proposed arithmetic comparison must specify the maps between these objects.

Remark 115.3.4 *(closure of (WZW) and (W) modulo modularity).* Under the RT and arithmetic bordism data just specified, and the trace comparison of Chapter 115.3.1, Chapter 115.3.3 gives the categorical form of (WZW). The Wilson-line/Frobenius identity is the Reshetikhin–Turaev quantum-trace formula only after the finite-place label comparison is supplied. The remaining axiom (Z) is not automatic: the partition function of an RT TQFT is the *Reshetikhin–Turaev invariant*, which is a finite complex number, not an entire function of the level. Matching it with $\xi(k^{\mathrm{Ar}})$ requires a separate analytic input.

Partition Identification via the Regularised Boundary Character

Proposition 115.3.5 *(conditional arithmetic partition identification).* Assume the preceding RT transport data. Suppose the closed arithmetic object has a specified decomposition whose state-space endomorphism has trace $\chi_{\mathrm{reg}}(k^{\mathrm{Ar}})$, and its RT evaluation is that trace. If the arithmetic character comparison identifies $\chi_{\mathrm{reg}}(k) = \xi(k)$, then its partition value is $\xi(k^{\mathrm{Ar}})$.

Proof. Compose the assumed evaluation-to-trace and trace-to-character identities. A topological Heegaard formula applies only after the corresponding arithmetic bordism and decomposition maps have been given. □

The existence of this decomposition, regularized trace prescription and analytic level family is part of the partition comparison problem. Modularity of a finite category alone supplies none of these data.

Sources and Hypotheses for Construction II

Reshetikhin–Turaev theory supplies a 3D TQFT from a modular tensor category. Huang–Lepowsky theory is the complex-analytic model in which a VOA produces such a category. In the arithmetic case the input is not a module category over $\mathcal{V}^{\mathrm{prim}}$, but a modular trace lift $\mathrm{HH}_0(\mathcal{MTC}^{\mathrm{Ar}}_{k^{\mathrm{Ar}}}) \simeq \mathcal{V}^{\mathrm{prim}}$, together with non-degenerate braiding, spherical trace, finite-place Frobenius labels, and the regularised character comparison.

Theorem 115.3.6 *(Construction II conditional implication). (on a modular trace lift, finite-place Frobenius labels, and the regularised-character comparison, together with the arithmetic bordism transport data).* Under these hypotheses, the Reshetikhin–Turaev functor of Chapter 115.3.3 gives the boundary trace condition, the Wilson-line trace condition, and the scalar partition identity of Chapter 103.4.7. Modularity alone does not imply the zeta partition identity.

Proof. (WZW) is the MTC-boundary identification of Chapter 115.3.1. (W) is the RT quantum-trace formula evaluated at the Frobenius conjugacy class via Morishita. (Z) is Chapter 115.3.5. The hypotheses are exactly the modular trace lift, the finite-place label comparison, and the regularised-character comparison. □

115.4 Construction III: Costello–Gwilliam Factorization on $(\operatorname{Spec}\mathbb{Z})_{\text{ét}}$

The Costello–Gwilliam formalism (Costello–Gwilliam 2017 *Factorization Algebras in Quantum Field Theory I*, Cambridge Univ. Press; vol. II 2021) builds perturbative QFTs as factorization algebras on a site. The étale site of Spec $\mathbb{Z}$ is a natural substitute for the usual differential-geometric site in the arithmetic setting.

Étale factorization algebra

Definition 115.4.1 *(an arithmetic factorization candidate).* Let $X = \operatorname{Spec} \mathbb{Z}$. Specify a category of arithmetic regions with disjoint embeddings and a chosen factorization coverage. A prefactorization candidate assigns a complex $\mathcal{F}(U)$ to each region and maps $\bigotimes_i \mathcal{F}(U_i) \to \mathcal{F}(V)$ for specified disjoint embeddings $U_i \to V$, with the required unit, symmetry and composition coherences. A factorization object must also satisfy descent for the chosen coverage. An arithmetic construction must state this coverage and its relation to the etale site. Ordinary descent of local systems does not replace this specification.

Definition 115.4.2 *(finite field functions and their action functional).* For each specified finite field groupoid $\mathcal{F}_{G_N}(U)$, let $\mathcal{V}_N(U)$ be the complex functions on its isomorphism classes. Given an isomorphism-invariant phase $a_N(b)$, define

$$\mathcal{I}_{N,U}(v) = \sum_{[b] \in \pi_0 \mathcal{F}_{G_N}(U)} \frac{a_N(b) v(b)}{|\operatorname{Aut} b|}, \qquad Z_N(U) = \mathcal{I}_{N,U}(1).$$

When the arithmetic action is defined, $a_N(b)$ is its exponential. This requires the action's coefficient object, boundary conditions and trace pairing from Chapter 115.1.1.

Proposition 115.4.3 *(finite products and weighted integration).* Suppose disjoint unions of specified arithmetic regions give products of their finite field groupoids, and their phases multiply. Then

$$\mathcal{V}_N(U \sqcup V) \cong \mathcal{V}_N(U) \otimes \mathcal{V}_N(V), \qquad \mathcal{I}_{N,U \sqcup V} = \mathcal{I}_{N,U} \otimes \mathcal{I}_{N,V}.$$

In particular $Z_N(U \sqcup V) = Z_N(U) Z_N(V)$. The vector space $\mathcal{V}_N(U)$ has dimension $|\pi_0 \mathcal{F}_{G_N}(U)|$; its scalar integral is not that vector space.

Proof. Delta functions on isomorphism classes give a basis. The product classes are pairs, and their automorphism orders multiply. The tensor basis gives the first isomorphism; finite Fubini gives the identity of integration functionals and the partition values. □

Remark 115.4.4 *(the open-site factorization problem).* The disjoint-product calculation is a part of the proposed Costello–Gwilliam construction on the arithmetic site. A factorization algebra additionally needs covariant structure maps for disjoint embeddings, their coherences, and the appropriate factorization descent condition. Descent of field groupoids does not by itself prove these assertions after passing to functions or weighted pushforwards. The action pairing on an open arithmetic region also requires its own compact-support or relative duality. Constructing these data on $X = \operatorname{Spec} \mathbb{Z}$ is a separate problem from the archimedean extension. The finite-product identity neither localizes every obstruction at infinity nor gives an unconditional factorization algebra.

Archimedean extension

Proposition 115.4.5 *(data for an archimedean operator extension).* Suppose a specified arithmetic factorization object has an extension to the boundary geometry of Chapter 115.1.1. A Hilbert realization of its primary boundary sector requires a complex vector space E_∞ extracted in a specified degree and completion from that trace sector, a Hermitian positive semidefinite pairing q_∞, linear in its first variable, and a complex-linear realization map

$$I_\infty : E_\infty \to H_B, \qquad \langle I_\infty v, I_\infty w \rangle = q_\infty(v, w), \qquad \overline{I_\infty E_\infty} = H_B.$$

It identifies H_B with the Hilbert completion of $E_\infty/\{v : q_\infty(v,v) = 0\}$. Boundary operations must preserve this nullspace to descend. Their asserted closed generators require domains and closures in H_B; a quantum surface comparison further requires the data of Chapter III.5.

Proof. Cauchy–Schwarz for a positive semidefinite Hermitian form makes its nullspace orthogonal to every vector. The displayed identity therefore induces an isometry on the quotient, which extends to its completion. The dense-range assumption makes this extension onto. If an operation does not preserve the nullspace, equivalent representatives have inequivalent images, so it cannot descend. The construction specifies neither an unbounded operator domain nor an exponential until the corresponding operator data are supplied. □

Remark 115.4.6 *(native boundary realization).* A finite field groupoid supplies finite-place state data. The passage to an arithmetic boundary Hilbert space additionally uses the geometry, native pairing and realization map above. A Hilbert completion alone does not supply the factorization gluing maps or a quantum observable. The desired boundary is a realization of a primary Hochschild trace sector, not a vertex algebra specified by that trace class. A proposed identification with an ordinary Fock representation must state the operations it preserves. Its occupation energy has eigenvectors and cannot be intertwined unitarily with continuous Mellin scaling, by Chapter III.2.1. Consequently Hilbert descent, native pairing, boundary extension and surface transport are separate required comparisons.

Sources and Hypotheses for Construction III

Costello–Gwilliam provides the factorization-algebra framework. Finite field groupoids and a specified Kim action give the functions and finite integration above. To use them as a factorization object one must construct its structure maps and descent, including the arithmetic support conditions. Its boundary realization then requires Chapter 115.4.5.

Theorem 115.4.7 *(conditional factorization and tower assignment).* Suppose the finite field-function assignments have been equipped with factorization structure and descent, and their scalar integration maps agree with the specified arithmetic action and Wilson insertions. Suppose also that coherent reduction maps define an inverse diagram of these factorization objects. Then this diagram is a pro-object in the stated factorization category, and its levelwise scalar and Wilson values are the assumed arithmetic ones. A represented limit and passage of global sections to that limit require additional continuity and descent theorems.

Proof. The supplied coherent diagram defines its pro-object. The integration identities hold at each level by hypothesis. No exchange of an inverse limit with global sections is used in this conclusion. □

115.5 Construction IV: Cobordism Hypothesis

Lurie's cobordism hypothesis (Lurie 2008 arXiv:0905.0465, Thm. 2.4.6) classifies extended n-dimensional TQFTs as fully dualizable objects in a symmetric monoidal (∞, n)-category. For $n = 3$ and a target $\mathcal{C} = \mathrm{ArithTC}$ the arithmetic tensor category, a quantum ACS is specified by a fully dualizable object of $\mathcal{C}$.

Fully dualizable object candidate

Definition 115.5.1 *(arithmetic target $(\infty,3)$-category).* . The *arithmetic target* is the symmetric monoidal $(\infty, 3)$-category ArithTC of arithmetic braided tensor categories: objects are braided tensor categories

equipped with a trace functor to arithmetic Hochschild trace classes; 1-morphisms are bimodule categories compatible with trace; 2-morphisms are bimodule functors; 3-morphisms are natural transformations; with symmetric monoidal structure from the Tate restricted tensor of Ch. 50.

Proposition 115.5.2 *(duality data for a specified arithmetic target).* Suppose the proposed target ArithTC is constructed as a symmetric monoidal $(\infty, 3)$-category, with its trace-compatible composition and coherences. An object is fully dualizable when it has a dual with evaluation and coevaluation and the required adjoints through the three morphism levels, satisfying their coherence identities. Supplying these data for the modular trace candidate is an additional arithmetic requirement.

The result of Douglas–Schommer-Pries–Snyder concerns fusion categories of nonzero global dimension in their tensor-category Morita setting. It does not construct the proposed arithmetic target or prove that a trace-compatible lift preserves the requisite dualities. This comparison must specify a monoidal functor and transport the actual evaluation, coevaluation and adjunction data.

Theorem 115.5.3 *(the framed cobordism output on specified data).* Given such a target and a fully dualizable object A in it, the framed cobordism hypothesis gives a symmetric monoidal functor $Z_A : \mathrm{Bord}_3^{\mathrm{fr}} \to \mathrm{ArithTC}$ with $Z_A(pt) = A$, classified by this fully dualizable datum. An arithmetic assignment follows if a compatible arithmetic bordism category and comparison functor into $\mathrm{Bord}_3^{\mathrm{fr}}$ are supplied. An oriented version requires the corresponding homotopy fixed-point structure.

Proof. Apply the framed cobordism hypothesis to the stated fully dualizable object, then compose with the assumed arithmetic comparison functor. The oriented extension uses its extra tangential structure. Equality with a particular RT theory further requires agreement of their extended data in a common target; it does not follow merely because both theories use a modular category. □

Remark 115.5.4 *(arithmetic cobordism category).* The cobordism hypothesis applies to a cobordism category of smooth manifolds. Its arithmetic counterpart requires an *arithmetic cobordism category* whose objects are arithmetic 2-surfaces and morphisms are arithmetic 3-spaces. Construction of such a category is Chapter 103.9 (item 1 of the residual list in Ch. 103): finite arithmetic cobordism functoriality. This is the Construction IV analogue of Construction III's archimedean extension obstruction.

Sources and Hypotheses for Construction IV

The cobordism-hypothesis presentation uses Lurie's theorem and the Douglas–Schommer-Pries–Snyder dualizability theorem in its specified tensor-category setting. It also uses the Reshetikhin–Turaev state sum on the resulting modular tensor category. The required inputs are an arithmetic cobordism category, the modular trace lift, finite-place label compatibility, and the regularised-character comparison.

Theorem 115.5.5 *(Construction IV conditional closure). (on the arithmetic cobordism category, modular trace lift, finite-place label comparison, and regularised-character comparison, the fully dualizable target data and the arithmetic bordism comparison above).* Given these hypotheses, the cobordism hypothesis produces a unique extended 3D TQFT $Z^{\mathrm{Ar}}_{\mathrm{coh}}$ whose trace, Wilson, and scalar partition data satisfy the axioms of Chapter 103.4.7.

Proof. Apply Chapter 115.5.3 and the specified arithmetic comparison. The assumed trace, Wilson and regularized character maps give the three scalar and boundary clauses. Identification with an RT theory requires the extended-data comparison stated in that theorem. □

115.6 Construction V: Six-Functor Formalism on an Arithmetic Diamond

Scholze's six-functor formalism (Scholze 2020–2022 *Lectures on Condensed Mathematics*, Bonn; Scholze 2022 *Six-Functor Formalisms* lecture notes) applies to diamonds and to a broad class of stacks. A quantum field theory on a space is given by a "theory" in the sense of six-functor formalism: an association of ∞-categories with the six operations $\otimes$, Hom, $f^*, f_*, f_!, f^!$. For 3D CS on a diamond, Scholze et al. have outlined the six-functor formalism as the natural home (Fargues–Scholze 2021 arXiv:2102.13459).

Diamond of $\overline{\mathrm{Spec}\,\mathbb{Z}}$

Conjecture 115.6.1 *(arithmetic diamond).* . There exists a diamond $\overline{\mathrm{Spec}\,\mathbb{Z}}^{\diamond}$ in the sense of Scholze 2020 *Etale Cohomology of Diamonds* (arXiv:1709.07343, §11) which is a globalisation of the Fargues–Fontaine curves at each prime, with an archimedean completion at $\{\infty\}$ given by a continuous version of Σ_∞. The diamond carries a six-functor formalism with values in condensed $\mathbb{Z}_\ell$-sheaves.

Remark 115.6.2 *(form of the arithmetic diamond).* Chapter 115.6.1 is a conjecture with no known construction in the cited literature. The closest object is the Clausen–Scholze analytic $\mathbb{Z}$-spectrum (Clausen–Scholze 2020 *Condensed Mathematics* lecture notes), which builds the condensed structure on the arithmetic side but does not supply the diamond-plus-archimedean-completion we would need.

Six-functor ACS

Proposition 115.6.3 *(a specified six-functor trace expression).* Suppose the proposed arithmetic space $X^\diamond$ has a specified six-functor theory $C(-)$ and structure map $f : X^\diamond \to pt$ for which $f^!$ and $f_!$ are defined. Choose an object $B \in C(pt)$; relating it to a boundary category or its Hochschild trace requires a specified realization functor. Then $\mathcal{T} = f^! B$ and $f_!\mathcal{T} \in C(pt)$ are defined. If $f_!\mathcal{T}$ is dualizable, its identity has a trace in $\mathrm{End}_{C(pt)}(1)$. A scalar value requires a specified map from this endomorphism ring to the scalar coefficient field.

Proof. Apply the two assumed functors in order. Dualizability supplies evaluation and coevaluation, whose trace composite is the stated endomorphism of the unit. Applying the chosen scalar map gives its scalar trace. Without dualizability or a separate regularized trace prescription this final operation is not defined. □

Identifying this trace with $\xi(k^{\mathrm{Ar}})$ requires the local-global trace comparison, including the archimedean factor and pole normalization. The arithmetic diamond, boundary object, coefficient realization and this trace identity are separate inputs. Local geometric Langlands does not supply them by formal application of $f^!$.

Remark 115.6.4 *(geometric and trace inputs for Construction V).* Write $B_{k^{\mathrm{Ar}}} \in C(pt)$ for the specified coefficient realization of the boundary datum $\mathscr{B}_{k^{\mathrm{Ar}}}$. Construction V places Chapter 103.4.7 inside the arithmetic diamond, a coefficient and boundary trace object, the requisite traceability data, and the local-global trace formula. The diamond is Chapter 115.6.1; the trace object lifts $\mathcal{V}^{\mathrm{prim}}$; the trace formula supplies the completed Euler product.

Sources and Hypotheses for Construction V

The six-functor presentation uses Scholze diamonds, Clausen–Scholze condensed sheaves, and the Fargues–Scholze spectral category at finite places. Its inputs are the arithmetic diamond, a boundary trace object $\mathscr{B}_{k^{\mathrm{Ar}}}$, the local-global trace formula, and finite-place Frobenius compatibility. Proper pushforward and shriek pullback then give the trace expression $\mathrm{Tr}(f_! f^! B_{k^{\mathrm{Ar}}})$.

Theorem 115.6.5 *(Construction V conditional closure). (on the arithmetic diamond, boundary and coefficient realization, traceability, local-global trace formula, and finite-place Frobenius compatibility).* Under these hypotheses, the six-functor ACS $\mathcal{T}_{k^{\mathrm{Ar}}} = f^! B_{k^{\mathrm{Ar}}}$ satisfies (WZW), (Z), (W) of Chapter 103.4.7.

Proof. (WZW) is the trace comparison $\mathrm{HH}_0(\mathscr{B}_{k^{\mathrm{Ar}}}) \simeq \mathcal{V}^{\mathrm{prim}}$. (Z) is the local-global trace formula for $\mathrm{Tr}(f_! f^! B_{k^{\mathrm{Ar}}})$. (W) is the finite-place spectral-Hecke compatibility producing the Frobenius trace on local Galois representations. □

115.7 Construction Comparison and Isolation of the Residual

Theorem 115.7.1 *(construction comparison).* . The five constructions to Chapter 103.4.7 decompose as follows.

- Given finite field groupoids and phases, their integrals and disjoint-product identities are defined. Construction I requires lifting measures and coherent transitions. Construction III further requires actual factorization maps and descent on the open site.
- Conditional on a modular trace lift, finite-place Frobenius labels, and the regularised-character comparison: Constructions II and IV, with their stated arithmetic bordism and duality data, give Reshetikhin–Turaev and cobordism-hypothesis realizations of the same boundary, Wilson, and scalar data.
- Conditional on the arithmetic diamond, boundary trace object, local-global trace formula, and finite-place Frobenius compatibility: Construction V, with the specified coefficient and traceability data, gives the six-functor form of the same axioms on $\overline{\mathrm{Spec}\,\mathbb{Z}}^{\diamond}$.

Proof. Chapter 115.2.5, Chapter 115.4.3, Chapter 115.3.6, Chapter 115.5.5, Chapter 115.6.5 respectively. □

The Common Archimedean Term

Theorem 115.7.2 *(comparison of two boundary Hilbert realizations).* Write $H_c = L^2(\mathbb{R}_{>0}, dx/x)$ and $\Theta_c = 1/2 + iD_c$, with D_c and its domain as in Chapter 111.1.1. Let (E_∞, q_∞) be a specified positive semidefinite primary trace realization as in Chapter 115.4.5. Suppose a descended state space H_B and a primary Hilbert space H_{prim} have complex-linear dense-range maps I_B, I_{prim} from E_∞, both preserving q_∞. Then there is a unique unitary $T : H_B \to H_{\mathrm{prim}}$ with $T I_B = I_{\mathrm{prim}}$. Suppose closed normal generators Θ_B, Θ_{prim} have cores $I_B\mathcal{D}$, $I_{\mathrm{prim}}\mathcal{D}$, for a specified $\mathcal{D} \subset E_\infty$, and their actions agree under T on these cores. Then

$$T\,\mathrm{Dom}\,\Theta_B = \mathrm{Dom}\,\Theta_{\mathrm{prim}}, \qquad T\Theta_B = \Theta_{\mathrm{prim}} T.$$

If a further unitary $V_{\mathrm{prim}} : H_{\mathrm{prim}} \to H_c$ intertwines Θ_{prim} with the continuous generator Θ_c, including domains, then $V_\Sigma = V_{\mathrm{prim}} T$ gives the surface comparison in Chapter 115.9.2.

Proof. The formula $I_B v \mapsto I_{\mathrm{prim}} v$ is well defined: both kernels are the nullspace of q_∞. It is an isometry on dense subspaces and extends uniquely to an onto unitary. The graph of each generator is the closure of its graph on the stated core. Applying $T \oplus T$ to these graphs proves domain equality and intertwining. Composition with V_{prim} proves the last assertion. □

Remark 115.7.3 *(boundary comparisons for the different constructions).* The inverse-limit, factorization and six-functor constructions each ask for a native boundary realization. They can be identified through the preceding theorem when they realize the same trace space, pairing and operator cores. Establishing

such identifications requires actual maps from their respective boundary objects. A topology on a pro-vector space does not define those maps. Likewise a modular trace lift requires its own trace-to-boundary realization before it supplies this Hilbert comparison. No implication between the modular lift, the arithmetic diamond and the Hilbert descent follows from their common intended trace class.

Relation to the Hilbert–Pólya/Connes–Deninger obstruction

Remark 115.7.4 *(Hilbert descent and arithmetic spectral realization).* Hilbert descent asks for a limit state space and its native pairing. The arithmetic Hilbert–Pólya realization further specifies an arithmetic middle-cohomology carrier, a self-adjoint operator H_{HP}, and the determinant and trace normalization of $\Theta_\xi = \frac{1}{2} + iH_{\mathrm{HP}}$. These data are not clauses of Hilbert descent. In particular, the zero-occurrence generator cannot be realized as the restriction of continuous Θ_c to an invariant Hilbert subspace by an injective generator intertwiner: each occurrence has a nonzero eigenvector, whereas Θ_c has none. An arithmetic cohomology comparison instead uses its own carrier, as in Chapter 111.5.

The purity condition depends on the operator and its carrier. For continuous scaling, set

$$H_c = L^2(\mathbb{R}_{>0}, dx/x), \qquad (U_a f)(x) = f(e^{-a}x), \qquad a \in \mathbb{R}.$$

In the logarithmic coordinate $t = \log x$, the self-adjoint generator is

$$D_c = -i\,\partial_t, \qquad \mathrm{Dom}\, D_c = \{f \in H_c : f(e^t) \in H^1(\mathbb{R}, dt)\}.$$

Thus $U_a = e^{-iaD_c}$, $\Theta_c = \frac{1}{2} + iD_c$ has the same domain, and $F_\infty = e^{-\Theta_c} = e^{-1/2}U_1$. Transport to a descended Hilbert space H_B requires a specified unitary $V : H_B \to H_c$. For a generator Θ_B, the required comparison is

$$V\,\mathrm{Dom}\,\Theta_B = \mathrm{Dom}\,\Theta_c, \qquad V\Theta_B = \Theta_c V \quad \text{on } \mathrm{Dom}\,\Theta_B.$$

It then gives $e^{-\Theta_B} = V^{-1}F_\infty V$.

The arithmetic operator instead uses the multiset Z of nontrivial zeta zeros, with one coordinate for each occurrence. On $K_Z = \ell^2(Z)$, set

$$(\Theta_Z v)_\rho = \rho v_\rho, \qquad \mathrm{Dom}\,\Theta_Z = \left\{ v \in \ell^2(Z) : \sum_{\rho \in Z} |\rho v_\rho|^2 < \infty \right\}.$$

Its bounded exponential is $(F_Z v)_\rho = e^{-\rho} v_\rho$. These operators distinguish continuous purity from arithmetic purity.

Theorem 115.7.5 *(Clausen–Scholze condensed-Hilbert descent plus A-TP ⇒ RH). (on the archimedean condensed-Hilbert descent of Chapter 115.8.3 and the A-TP archimedean positivity of Ch. 78.2.3).* Under both the Clausen–Scholze condensed-Hilbert descent and A-TP, the Riemann Hypothesis holds.

Proof. For $f \in C_c^\infty(\mathbb{R})$, with $f^*(t) = \overline{f(-t)}$, A-TP states

$$W_\infty(f * f^*) \le P(f * f^*) - W_{\mathrm{fin}}(f * f^*).$$

The signed explicit formula therefore gives $W(f * f^*) \ge 0$ for every such f. By Weil's criterion, all nontrivial zeros satisfy $\Re\rho = 1/2$, which is RH. On the coordinate vector at ρ, the absolute value $|F_Z|$

has eigenvalue $e^{-\Re\rho}$. Hence the same conclusion is precisely $|F_Z| = e^{-1/2}1_{K_Z}$, the arithmetic purity of Chapter 111.3.4. Conversely this equality forces $\Re\rho = 1/2$ at every occurrence, and Weil's criterion recovers A-TP.

On H_c, unitarity gives $F_\infty^* F_\infty = e^{-1}1_{H_c}$ without A-TP. The specified unitary V transports this identity to H_B by functional calculus, including the stated generator domains. Thus descent provides the continuous Hilbert realization when its comparison data exist. A-TP alone supplies the zero-placement implication used above. □

Remark 115.7.6 *(the archimedean Hilbert descent does not imply RH alone).* Resolving the archimedean Hilbert descent alone does not resolve RH: it constructs $\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}}$ only on the descended Hilbert-trace clauses (WZW), (Z), and (W); it does not supply the full cobordism functor or the full Weil inequality A-TP. The continuous scaling exponential is already pure. Under both descent and A-TP, RH follows through the separate arithmetic operator F_Z. The archimedean Hilbert descent is the bulk-construction obstruction; A-TP is the separate positivity obstruction. After the descent, the remaining bulk gluing assertions stay separate; the RH-equivalent condition in this construction is A-TP.

115.8 Refinement: The Condensed-Hilbert Bulk

Condensed-Hilbert setup

Definition 115.8.1 *(condensed Hilbert pro-system).* Specify finite-level boundary Hilbert spaces $\mathcal{H}^{(\infty)}_{G_N,\alpha_N}$ and coherent bounded linear transition maps, including their normalization. Their associated condensed complex vector spaces form an inverse diagram, denoted $\mathcal{H}^{(\infty)}_\bullet$. Here $\mathrm{Cond}_{\mathrm{Hilb}}(\mathbb{C})$ denotes the chosen category of condensed realizations of complex Hilbert spaces with bounded linear maps. A limit in condensed vector spaces need not be represented by a Hilbert space. Such representability, its norm and its compatibility with the native pairing are the additional descent requirements below. A scalar normalization of partition values does not by itself define transition maps on these state spaces.

Definition 115.8.2 *(target archimedean Hilbert realization).* Specify a primary trace space E_∞, its Hermitian positive semidefinite form q_∞, and its arithmetic operations. The target H_{prim} is the completion of its quotient by the nullspace, with its quotient map I_{prim}. Its condensed realization is the condensed object associated with this Hilbert topology. Specifying the trace degree, the form and its native realization is part of the arithmetic construction in Chapter 78.5.4. An identification with a Heisenberg representation additionally requires the operators, domains, central action and pairing map of Chapter 111.5.

The Condensed Hilbert–Trace Package

Conjecture 115.8.3 *(archimedean condensed-Hilbert descent and trace).* The arithmetic boundary geometry, primary trace realization and positive pairing of Chapter 115.8.2 exist. The specified finite-level transition maps and renormalization admit a condensed limit represented by a Hilbert space $H_B = \mathcal{H}^{(\infty)}_\infty$, with a complex-linear dense-range realization $I_B : E_\infty \to H_B$ preserving q_∞. Consequently the native comparison is

$$H_B \xrightarrow{T} H_{\mathrm{prim}}, \qquad T I_B = I_{\mathrm{prim}}.$$

The arithmetic boundary operations descend and agree on specified cores under T. The global trace prescription, including the finite-place factors and pole normalization, gives $Z_{\mathrm{ACS}^{\mathrm{qu}}}(k^{\mathrm{Ar}}) = \xi(k^{\mathrm{Ar}})$. The finite-place Wilson insertions obey the Frobenius trace identity in the stated geometric Frobenius convention. A comparison of the quantum surface generator with Θ_c requires the additional unitary and domains in Chapter 115.9.2.

Theorem 115.8.4 *(condensed Hilbert–trace package supplies the matching ACS clauses). (after choosing the same bulk-boundary ACS model).* Chapter 115.8.3 supplies the archimedean Hilbert, trace, and Wilson clauses of Chapter 103.4.7. It is not, by itself, the full topological ACS functor.

Proof. Given Chapter 115.8.3, the limit $\mathcal{H}_\infty^{(\infty)}$ is the archimedean boundary Hilbert space in the chosen model. Its native trace comparison is the Hilbert boundary clause of (WZW). A factorization structure and its gluing remain the additional data of Construction III. The partition function (Z) is the regularised trace clause. The Wilson-line axiom (W) is the finite-place descent clause. The remaining assertion of Chapter 103.4.7 is the existence of the full cobordism functor with gluing; it is not implied by the condensed Hilbert limit alone. □

Remark 115.8.5 *(form of the condensed statement).* Chapter 115.8.3 is a condensed Hilbert limit together with a trace identity and finite-place endomorphism compatibility. The Hilbert limit alone gives only the archimedean state space; the trace and Wilson clauses carry the scalar and finite-prime content of the quantum ACS axioms.

115.9 Wilson-Observable Compatibility

Theorem 115.9.1 *(Wilson-line compatibility). (on Chapter 115.8.3).* Under the condensed-Hilbert descent, the Wilson-line observable at a prime knot K_p in the bulk $\mathrm{ACS}^{\mathrm{qu}}_{k\mathrm{Ar}}$, evaluated on the limit state space $H_B \xrightarrow{T} H_{\mathrm{prim}}$, is exactly the Frobenius trace $\operatorname{tr}\rho(\mathrm{Frob}_p)$ of Chapter 106.5.1(2), as a Wilson clause of the descent.

Proof. Chapter 115.8.4 supplies the Wilson clause of the chosen descended ACS model. Axiom (W) in that model is the Wilson-line Frobenius identity. Here the finite-place equality is expressly part of the descent conjecture, including its label and trace comparison. A finite-state sum alone would not give that identification. □

Theorem 115.9.2 *(Wilson-surface compatibility).* Assume the condensed-Hilbert descent of Chapter 115.8.3, and write $H_B = \mathcal{H}_\infty^{(\infty)}$. Suppose the arithmetic boundary quantization of Chapter 111.5 constructs a closed normal Θ_B and its surface observable $F_\infty^{\mathrm{WS}} = e^{-\Theta_B}$. Suppose a unitary $V_\Sigma : H_B \to H_c$ satisfies

$$V_\Sigma \operatorname{Dom}\Theta_B = \operatorname{Dom}\Theta_c, \qquad V_\Sigma \Theta_B = \Theta_c V_\Sigma.$$

Then the exponential is bounded and

$$F_\infty^{\mathrm{WS}} = V_\Sigma^{-1} F_\infty V_\Sigma, \qquad |F_\infty^{\mathrm{WS}}| = e^{-1/2}\mathbb{1}_{H_B}.$$

The quantization must contain this Tate normalization: a unitary curvature phase alone has absolute value one.

Proof. The domain and operator equalities identify the normal spectral measures. The function e^{-s} is bounded on $1/2 + i\mathbb{R}$. Functional calculus proves the conjugacy, and $F_\infty = e^{-1/2}U_1$ proves the absolute-value formula. This transports an already constructed quantum observable. Defining a transported operator on an abstract Hilbert space alone does not identify it with the arithmetic quantization. □

Proposition 115.9.3 *(native pairing and the full Weil form).* Let $I_B : E_\infty \to H_B$ realize the native positive pairing q_∞. Suppose a complex-linear map $R : \mathscr{A} = C_c^\infty(\mathbb{R}) \to E_\infty$ satisfies

$$q_\infty(Rf, Rh) = P(f * h^*) - W_\infty(f * h^*) - W_{\mathrm{fin}}(f * h^*) \qquad (f, h \in \mathscr{A}),$$

with $h^*(t) = \overline{h(-t)}$ and the Fourier and measure conventions of Chapter 78.1.2. Then $J = I_B R$ realizes the full Weil form on H_B, and RH follows. If the identity is specified only on a complex subspace $E \subset \mathscr{A}$, it proves positivity only on E. It extends to the full criterion if E is dense in the compact test topology. A mere density of its span in a different Hilbert norm does not give that implication.

Proof. The native isometry and the signed explicit formula give $\langle Jf, Jh\rangle = W(f * h^*)$. Taking $h = f$ gives nonnegativity, and Weil's criterion implies RH. For a restricted subspace, continuity of convolution, involution and W in the compact test topology permits passage from approximating tests to all compact tests. No such limit follows from an unspecified norm. □

This comparison asks for an arithmetic map R, including all polar, gamma and prime terms. The positivity of the target Hilbert space and the continuous surface comparison do not construct R. The Gaussian extension in Chapter 95.1.2 provides another valid test topology only when the complete pairing identity is transported there; a gamma sign alone does not do so.

Corollary 115.9.4 *(Wilson observables on the descended model).* Under the condensed-Hilbert descent, the Wilson-line Frobenius identity is its finite-place Wilson clause. With the additional unitary and generator-domain comparison in Chapter 115.9.2, the archimedean surface operator is unitarily conjugate to the continuous scaling exponential on H_c.

Proof. Apply Chapter 115.9.1 to the finite-place clause and Chapter 115.9.2 to the specified archimedean comparison. □

115.10 Comparison of the Five Presentations

Chapter 115.7.2 gives a comparison theorem for boundary Hilbert realizations carrying the same native pairing. The inverse-limit construction must supply its transition maps, renormalization and represented limit. The factorization construction must supply boundary restriction and gluing. The categorical constructions must supply their trace-to-boundary maps, and the six-functor construction its arithmetic space and trace formula.

When these constructions realize a common (E_∞, q_∞), the theorem constructs their Hilbert unitary. Agreement of generator cores gives operator transport. Their existence and arithmetic identification are additional to this implication. In particular, a passage from vector spaces to pro-objects and then to a condensed Hilbert space does not specify the native pairing or the quantum surface observable.

115.11 Remaining Inputs

(i) The condensed-Hilbert descent of Chapter 115.8.3 includes native boundary geometry, a positive trace-sector pairing, a represented limit and its realization maps. The trace and Wilson clauses must be proved with the stated global normalization.

(ii) A quantum surface realization further requires its fields, curvature coupling, boundary conditions and quantization. The unitary V_Σ and its generator-domain identity give the continuous comparison in Chapter 115.9.2.

(iii) The modular trace lift for the categorical constructions needs a trace-to-boundary map compatible with the native pairing and operations. Its relationship to Hilbert descent must be proved through these maps.

(iv) The arithmetic diamond construction needs its space, boundary trace object and local-global trace formula. These are its own geometric and categorical hypotheses.

(v) An arithmetic positivity consequence requires the map R and complete pairing identity in Chapter 115.9.3, or an independently proved full Weil inequality. The determinant route instead requires its native cohomology and domain-preserving spectral realization.

Remark 115.11.1 *(relations among the required constructions).* The comparison theorem constructs a unitary when the native pairing and dense realization maps agree. It does not prove a strict logical ordering among the modular lift, Hilbert descent and arithmetic diamond. Each proposed implication must specify the corresponding geometric, trace and operator maps.

115.12 Constructed implications and remaining data

Consequences on specified data

(i) An actually coherent inverse tower defines a pro-object (Chapter 115.2.5).

(ii) Finite groupoid functions and their integration factor over disjoint products; the additional factorization descent is a separate construction (Chapter 115.4.3).

(iii) The construction comparison theorem (Chapter 115.7.1) and the boundary comparison theorem (Chapter 115.7.2) supplies a unitary when the different constructions realize the same native pairing on dense subspaces, with operator transport under the stated core condition.

(iv) The descent comparison theorem (Chapter 115.8.4) shows that the condensed-Hilbert descent supplies the Hilbert, trace, and Wilson clauses of the chosen ACS model; the full cobordism functor and gluing remain separate.

(v) Under the descent and the specified archimedean unitary comparison, the Wilson-line and Wilson-surface identities hold inside the descended model (Chapter 115.9.4).

Conditional on the condensed-Hilbert descent

(i) The represented Hilbert object and the trace and finite-Wilson clauses in Chapter 115.8.4. The full (WZW) axiom of Chapter 103.4.7 additionally requires the factorization, boundary comparison and gluing maps of the chosen quantum ACS model.

(ii) Ch. 106 profinite Wilson-Frobenius identity.

(iii) Ch. 111 Wilson-surface conjugacy to Arakelov scaling, with the additional unitary and generator-domain comparison of Chapter 115.9.2.

(iv) The specified scalar partition and trace clauses. A further identification with a Deninger complex requires that complex and its determinant comparison; a Hilbert limit alone does not supply it.

Arithmetic positivity and spectral comparison

(i) The A-TP archimedean positivity of Ch. 78 (equivalent to RH by Chapter 111.3.4).

(ii) The Hilbert–Pólya determinant-trace realisation H-P^{Ar} of Chapter 44.3.2 (equivalent to zero-placement determinant identity).

(iii) The Deninger spectral-flow hypothesis Chapter 103.6.5 (equivalent to the full Weil–Guinand distribution).

The chapter does not reduce the RH-level obstructions. It identifies the condensed-Hilbert descent as the common archimedean term in the bulk comparison.

115.13 Boundary realization and arithmetic comparison

The boundary construction asks for arithmetic geometry, a positive primary pairing, a Hilbert realization and a quantum observable. If two constructions realize the same pairing on dense subspaces, Chapter 115.7.2 supplies their unitary comparison. The trace and finite-place clauses of Chapter 115.8.3 retain their own arithmetic content; the full cobordism gluing remains additional.

With the specified quantum generator and unitary V_Σ, continuous surface purity follows by functional calculus. Arithmetic purity concerns the separate zero-occurrence operator F_Z, and follows from the full Weil inequality A-TP. The native sufficient comparison is $J = I_B R$ in Chapter 115.9.3. It retains every polar, gamma and prime contribution. Constructing this arithmetic map, or the native middle-cohomology determinant realization, is the remaining arithmetic positivity problem.

115.14 Bibliography

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Chapter 116

Local Bernstein Centres and Categorical Gluing

The local problem is to compare three different centres: functions on Weil-parameter spaces, the ordinary Bernstein centre of smooth representations, and the categorical centre of a spectral category. For GL_2 in characteristic zero with coefficient prime $\ell \neq p$, the first two are isomorphic. We compute this isomorphism on every inertial component. On the full spectral category the same functions act faithfully but do not exhaust its centre: central weights give additional natural endomorphisms. Thus a local scalar comparison does not construct the categorical comparisons required for global gluing.

116.1 Domain and Comparison Hypotheses

Local actions and global comparison data

The comparison problem in Proposition 88.5.5 requires local categorical data, the equivariant pairwise sheaf comparison, cross-prime assembly and the global centre comparison. The formal restriction diagram alone supplies none of the latter comparison isomorphisms. Their contents are as follows.

(BCglue.local.pointwise) At a finite place v of residue characteristic p, over $\overline{\mathbb{Q}}_\ell$ with $\ell \neq p$, parameter functions act on each irreducible smooth GL_2 representation by evaluation on its semisimplified Weil parameter. The ring isomorphism proving this assertion is Proposition 116.2.2.

(BCglue.local) For each block one must also specify a stable representation category, a spectral target, a fully faithful exact functor, its essential image, and compatibility of central actions. The target and its central weight are part of the data; there is no unspecified "isotypic component" on the spectral side. The required coefficient and family comparisons, and at $\ell = p$ the fixed-central-character and fixed-determinant block data, are additional hypotheses. They are stated in Section 116.2 and Proposition 116.3.3.

(BCglue.glue.factorization) The local central actions must assemble over finite sets of places, with specified transition maps, into the desired global action. An identification of this assembly with a pseudocharacter algebra requires a global comparison map; a filtered tensor product alone does not provide it. The cross-prime transition, coherent-support and higher-coherence compatibilities retain the descent and global hypotheses of Theorem 56.5.4 and Hypothesis 56.5.2.

References for Local and Global Comparisons

The local ring calculation and the global comparison questions use the following references:

Fargues–Scholze 2021, arXiv:2102.13459v4 Definition IX.0.2 and Corollary IX.0.3 (pp. 318–319) distinguish spectral functions, the geometric centre, and the ordinary centre. Theorem IX.5.2 (p. 329) constructs the central action; Theorem IX.7.4 (p. 338) identifies the semisimplified parameter for GL_n. These statements do not identify the full spectral categorical centre with the ordinary Bernstein centre.

Dat–Helm–Kurinczuk–Moss, arXiv:2009.06708v3 Theorem 4.1, Proposition 4.17 and Proposition 6.4 give the parameter schemes and the characteristic-zero invariant-ring reduction. Their §1.6 compares these schemes with the Fargues–Scholze construction.

Bernstein–Deligne and Helm The block-centre formula is recalled in Helm, arXiv:1201.1874, Theorems 3.4–3.5, pp. 7–8; its coordinate comparison with Weil-parameter invariants is also given in Helm, arXiv:1605.00487v3, Theorem 9.4, pp. 30–31.

Dotto–Emerton–Gee 2026, arXiv:2603.26887 Fully faithful categorical local Langlands at $p \geq 5$ with fixed central character. The source gives a fully faithful functor from the derived category of smooth p-adic $GL_2(\mathbb{Q}_p)$ representations with fixed central character to ind-coherent sheaves on the Emerton–Gee stack. It is not used as an essential-image, component-exhaustion, or central-character-variation theorem.

Bellaïche–Chenevier 2009, Astérisque 324 *Families of Galois representations and Selmer groups*. The deformation-theoretic setting is in Chapter 1. The global scalar construction below instead proves generation by Frobenius coefficients; an automorphic action and its relation ideal are separate comparisons.

Francis–Gaitsgory 2011, arXiv:1103.5803 *Chiral Koszul duality*. This is background for a proposed factorization-category realization. The algebraic tensor-colimit action below is constructed directly and is not an identification of a global categorical centre by factorization homology.

Bhatt–Scholze 2022, arXiv:1905.08229 *Prisms and prismatic cohomology*. Prismatic realization and the stipulated cross-prime Π-flatness require actual objects and maps; the Frobenius scalar reconstruction proves no such flatness.

Bhatt–Lurie 2022, arXiv:2201.06120 *Absolute prismatic cohomology*. Nygaard filtrations provide background for a separately constructed filtered comparison. Untilts and their descent functors remain additional geometric data.

Paškūnas 2013 and Paškūnas–Tung 2020 Block structure at $p \in \{5, 2\}$ and the $p = 3$ refinement via Colmez–Dospinescu–Paškūnas 2014. Used as the wild-prime Bernstein-block side.

The references concern local geometrization (Fargues–Scholze), categorical local Langlands at tame primes (Dotto–Emerton–Gee), deformation theory (Bellaiche–Chenevier) and factorization categories (Francis–Gaitsgory), prismatic sites (Bhatt–Scholze and Bhatt–Lurie), and wild-prime Galois blocks (Paškūnas, Paškūnas–Tung, and Colmez–Dospinescu–Paškūnas). Their use in the local and global comparisons retains the hypotheses stated below.

Local Obstructions

There are four distinct comparison problems.

- **(E.1)** Compute the ordinary local centre for $\ell \neq p$. No tame-inertia restriction is imposed on the parameters. The $p \geq 5$ fixed-central-character functor of Dotto–Emerton–Gee belongs to a separate coefficient regime.
- **(E.2)** Retain the same ordinary calculation for $p = 2, 3$, and formulate the extra fixed-determinant, wild-EG, block-ring, functorial and normalized tangent-class requirements when the coefficient characteristic is p.
- **(E.3)** Construct the cross-prime assembly with coefficient, family, pseudocharacter, prismatic and untilt compatibilities. These are not consequences of local scalar equality.
- **(E.4)** Identify precisely which of these data are needed to use the ordinary local calculation in BC-diamond-glue.

116.2 Ordinary Local Centres in Characteristic Zero

Ordinary centres and scalar parameter functions

Fix a finite extension $F/\mathbb{Q}_p$, with residue field of cardinality q, a prime $\ell \neq p$, and $k = \overline{\mathbb{Q}}_\ell$. Choose a square root of q in k. Use normalized parabolic induction and the normalization of local class field theory that sends a uniformizer to geometric Frobenius. The local Langlands correspondence is normalized compatibly with these choices.

Let

$$Y_F = Z^1(W_F, GL_{2,k}), \qquad X_F = [Y_F/GL_{2,k}], \qquad A_F = \Gamma(X_F, \mathcal{O}_{X_F}).$$

Here Y_F is the Fargues–Scholze scheme of continuous parameters, with its algebraic family structure; continuity is the relatively discrete ℓ-adic continuity of that construction. It is not the tangent space at a parameter. Write

$$\mathcal{R}_F = \mathrm{Rep}_k^{\mathrm{sm}}(GL_2(F)), \qquad Z_F = \mathrm{End}(\mathrm{id}_{\mathcal{R}_F})$$

for the abelian category of smooth representations and its ordinary Bernstein centre.

The parameter scheme is a disjoint union of finite-type inertial pieces. Their characteristic-zero presentations are reduced. In each piece, the semisimple finite-inertia part of a parameter is fixed, while unipotent monodromy may vary. These are the presentations of Dat–Helm–Kurinczuk–Moss, Theorem 4.1 and §6, identified with the Fargues–Scholze construction after coefficient extension as in their §1.6. In characteristic zero the ordinary invariant-ring calculation can also be made directly on a finite-inertia locus: at each fixed wild level, restriction induces an isomorphism from the full parameter invariant ring to that of parameters trivial on a sufficiently small open subgroup of inertia (DHKM, Proposition 4.17(ii), pp. 35–36). The finite-inertia orbit decomposition is DHKM, Proposition 6.4, p. 66. These are statements about coordinate rings of the actual parameter scheme, not merely bijections of closed points. Consequently A_F is a product of invariant coordinate rings, not a direct sum.

Theorem 116.2.1 *(the Fargues–Scholze map to the ordinary centre).* There is a canonical homomorphism

$$\Psi_F : A_F \longrightarrow Z_F.$$

For an irreducible smooth representation π, its action is

$$\Psi_F(f)|_\pi = f(\varphi_\pi^{\mathrm{ss}})\,\mathrm{id}_\pi, \qquad f \in A_F,$$

where $\varphi_\pi^{\mathrm{ss}}$ is the semisimplified Weil parameter of π.

The map is obtained by restricting the geometric central action on $D_{\mathrm{lis}}(\mathrm{Bun}_{GL_2}, k)$ along the fully faithful functor

$$j_! : D(\mathcal{R}_F) \longrightarrow D_{\mathrm{lis}}(\mathrm{Bun}_{GL_2}, k).$$

It does not assert an equivalence with an ind-coherent category.

Source. Fargues–Scholze, Definition IX.0.2, Corollary IX.0.3, Theorem IX.5.2 and Definition IX.7.1 construct the map. Theorem IX.7.4 identifies its parameter on irreducible $GL_2(F)$-representations with the usual semisimplified parameter. The connectedness of the centre of GL_2 satisfies the component-group hypothesis in the centre construction. □

Proposition 116.2.2 *(the ordinary local centre).* The homomorphism Ψ_F is an isomorphism of ordinary commutative k-algebras.

Proof. First describe the Bernstein side. An inertial class is represented by a supercuspidal pair (M, σ). Its unramified twists form a torsor T_σ under the unramified-character torus modulo the finite subgroup of self-twists of σ. The finite group W_σ stabilizing the inertial class acts on this torsor. Bernstein's description (Helm, arXiv:1201.1874, Theorems 3.4–3.5, pp. 7–8) gives

$$Z_F = \prod_{\mathfrak{s}} \mathcal{O}(T_\sigma)^{W_\sigma}.$$

A choice of σ gives coordinates on the torsor; the formula itself is independent of this choice.

For GL_2, a principal-series class has $M = F^\times \times F^\times$. If the two character restrictions to $\mathcal{O}_F^\times$ differ, its ring is $k[a^{\pm 1}, b^{\pm 1}]$. If they agree, choose the same character as representative in both factors. Its ring is

$$k[a^{\pm 1}, b^{\pm 1}]^{S_2} = k[t, d, d^{-1}], \qquad t = a + b, \quad d = ab.$$

A supercuspidal class has $M = GL_2(F)$. If its group of unramified self-twists has order f, its ring is

$$k[a^{\pm 1}]^{\mu_f} = k[u, u^{-1}], \qquad u = a^f.$$

We compute the corresponding parameter invariants. Fix an inertial piece and a representative τ of its semisimple inertia representation. The characteristic-zero parameter presentation uses a Frobenius intertwiner A and a nilpotent monodromy operator N, with

$$ANA^{-1} = q^{-1}N.$$

The operator N commutes with τ. The Frobenius convention is that of FS, VIII.1, $\sigma^{-1} t \sigma = t^q$ on tame inertia; DHKM's arithmetic Frobenius is its inverse. Invariant restriction to $N = 0$ agrees with the characteristic-zero ring isomorphism of DHKM Proposition 4.17, so it does not discard functions undetectably by replacing the parameter scheme by a point set. Changing the chosen inertia trivialization acts through the centralizer of τ. Finite-inertia semisimplification and the logarithm of unipotent inertia give this presentation; in dimension two the logarithm and exponential on the unipotent part are finite polynomials.

There are four possibilities.

If τ is irreducible of dimension two, its centralizer consists of scalars. Choose one Frobenius intertwiner A_0. Every other intertwiner is aA_0, and $N = 0$. The invariant ring is $k[a^{\pm 1}]$.

If $\tau = \chi_1 \oplus \chi_2$, with distinct characters each fixed by Frobenius, its centralizer is $\mathbb{G}_m^2$. After choosing extensions of the characters to W_F, the Frobenius parameters are $a, b \in \mathbb{G}_m$. Commutation with inertia makes N diagonal, and nilpotence makes $N = 0$. The invariant ring is $k[a^{\pm 1}, b^{\pm 1}]$.

If Frobenius interchanges the distinct characters χ_1, χ_2, its matrix has the form

$$A = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}, \qquad a, b \neq 0.$$

Again $N = 0$. Conjugation by $\operatorname{diag}(s, t)$ sends (a, b) to $(st^{-1}a, ts^{-1}b)$. A Laurent monomial is invariant precisely when its two exponents agree. Hence the invariant ring is $k[ab, (ab)^{-1}]$.

Finally, suppose $\tau = \chi \oplus \chi$. After removing a fixed extension of χ, the presentation is

$$\{(A, N) \in GL_2 \times \mathfrak{gl}_2 : N \text{ nilpotent}, \ AN = q^{-1}NA\}/GL_2.$$

Every orbit closure meets the locus $N = 0$. Indeed, if $N \neq 0$, choose $N = E_{12}$. The relation forces

$$A = \begin{pmatrix} a & b \\ 0 & qa \end{pmatrix}.$$

Since $q \neq 1$ in k, conjugation by an upper unipotent matrix removes b while preserving N. Conjugation by $\mathrm{diag}(z, 1)$ then fixes the diagonal A and sends N to zN. Letting z tend to zero gives the required limit. On the locus $N = 0$, the closure of each matrix orbit contains its semisimplification.

An invariant function vanishing on $N = 0$ therefore vanishes at every closed point of the presentation. Reducedness implies that it is zero. Restriction to $N = 0$ is thus injective on invariant functions. Conjugation-invariant regular functions on GL_2 are

$$k[\mathrm{tr}(A), \det(A), \det(A)^{-1}].$$

For completeness, restriction to diagonal matrices is injective because matrices with distinct eigenvalues are dense; its image lies in the symmetric Laurent polynomials, and trace, determinant and inverse determinant generate that ring. These same functions are defined on the full (A, N)-presentation, so the restriction map is also surjective.

In the last case one can also see surjectivity without any pointwise inference: the symmetric Laurent ring is generated by the three displayed functions, each already regular on the full presentation.

These four calculations give exactly the Bernstein rings above. In the supercuspidal cases, restriction of the irreducible Weil representation to inertia is either irreducible or the sum of two distinct Frobenius-conjugate characters. The orders of the self-twist groups are respectively 1 and 2: an unramified twist with Frobenius value c preserves the representation precisely when $c^f = 1$, where f is the number of inertia constituents. Thus the invariant ab in the interchanged case is, relative to the chosen base representation, the coordinate a^2 on the unramified-twist quotient.

Local Langlands matches these inertial pieces with the supercuspidal-support classes and respects unramified twisting and normalized parabolic induction. The displayed generators consequently define an algebraic isomorphism

$$\theta_F : A_F \xrightarrow{\sim} Z_F$$

with the required scalar evaluations on irreducibles. Taking products, rather than finite sums, retains arbitrary choices of functions on all inertial pieces.

It remains to identify this algebraic map with Ψ_F. Each Bernstein block ring is a reduced finite-type k-algebra, and every point of its cuspidal-support variety occurs as the support of an irreducible constituent of a parabolically induced representation. Therefore an element of Z_F vanishing on every irreducible is zero. The preceding theorem shows that $\theta_F(f)$ and $\Psi_F(f)$ have the same evaluations. They are equal. □

Example 116.2.3 *(the unramified component does not determine inertia).* On the unramified principal-series component the ordinary centre is $k[t, d, d^{-1}]$. These three functions do not generate the centre across all inertial components.

For example, take $F = \mathbb{Q}_p$, $p \geq 5$, and a nontrivial smooth character $\chi : F^\times \to k^\times$ with $\chi(p) = 1$. The normalized principal series induced from $1 \boxtimes 1$ and from $\chi \boxtimes \chi$ are irreducible. Their semisimplified parameters have the same Frobenius matrix, but their inertial supports differ. The idempotent of the first Bernstein component separates them. Thus Frobenius trace and determinant alone do not recover the full centre.

Central weights in the spectral category

Write $X_F = \coprod_{\mathfrak{s}} X_{F,\mathfrak{s}}$ for its inertial decomposition. On each finite-type piece let

$$C_{\mathfrak{s}} = \mathrm{Ind}\big(\mathrm{Coh}_{\mathrm{Nilp}}(X_{F,\mathfrak{s}})\big), \qquad C_F = \prod_{\mathfrak{s}} C_{\mathfrak{s}}.$$

Here $\mathrm{Coh}_{\mathrm{Nilp}}$ means bounded coherent complexes with nilpotent singular support, and Ind denotes their ind-completion. This specifies a presentable category; it does not require every object of the ind-completion to remain bounded.

Only three properties of the support condition enter the following argument: it contains the zero section, is preserved by direct summands, and is preserved by tensoring with vector bundles. These properties hold for nilpotent singular support. In particular perfect complexes belong to the chosen category.

For a presentable k-linear stable category C, put

$$\mathfrak{Z}(C) = \mathrm{End}_{\mathrm{Fun}^L_k(C,C)}(\mathrm{id}_C), \qquad Z^\circ(C) = \pi_0\mathfrak{Z}(C).$$

The first object retains the enhanced centre; the second is its degree-zero ring.

PROPOSITION 116.2.4 *(central-weight decomposition).* There are orthogonal decompositions

$$C_F \simeq \prod_{\mathfrak{s}} \prod_{m\in\mathbb{Z}} C_{\mathfrak{s},m}, \qquad \mathfrak{Z}(C_F) \simeq \prod_{\mathfrak{s}} \prod_{m\in\mathbb{Z}} \mathfrak{Z}(C_{\mathfrak{s},m}).$$

The index m is the character $z \mapsto z^m$ of the scalar subgroup $\mathbb{G}_m \subset GL_2$. Tensoring with the determinant line gives equivalences

$$C_{\mathfrak{s},m} \xrightarrow{\sim} C_{\mathfrak{s},m+2}.$$

Proof. Scalar matrices act trivially on the parameter scheme, because its action is conjugation. On an equivariant affine chart, the coordinate algebra therefore has central weight zero. Every equivariant coherent module decomposes into its $\mathbb{G}_m$-weight modules. A finite set of homogeneous generators shows that only finitely many weights occur. The differential preserves weight, as do all equivariant morphisms.

Since $\mathbb{G}_m$ is diagonalizable, taking a weight summand is exact. Distinct weights are orthogonal also in the derived category. These decompositions descend from equivariant charts and preserve the singular-support condition. Ind-completion permits arbitrary sums of weight components. A family of such components recovers an object by their coproduct, and projection recovers the original family. This gives the categorical product decomposition.

A natural transformation of the identity preserves every component: apply naturality to its inclusion and projection. Conversely, any family of natural transformations on the components defines one on their coproduct. The same argument applies to mapping complexes and composition, giving the displayed decomposition of enhanced centres.

The determinant representation has central weight two and defines an invertible line bundle on the quotient. Tensoring by this line preserves singular support, and tensoring by its inverse reverses the operation. □

PROPOSITION 116.2.5 *(scalar functions form a proper central subalgebra).* Scalar multiplication defines an injective map

$$\alpha_F : A_F \longrightarrow Z^\circ(C_F).$$

Its image is canonically isomorphic to Z_F, through $\alpha_F\Psi_F^{-1}$, but it is a proper subalgebra of $Z^\circ(C_F)$.

Proof. On any inertial piece, evaluating scalar multiplication on its structure object detects the function. Thus α_F is injective. The ordinary-centre proposition identifies its domain with Z_F.

For example, already on $B\mathbb{G}_m$ the function ring is k, whereas the weightwise scalar operations form $\prod_{m\in\mathbb{Z}} k$.

Let e_0 be the natural projection onto central weight zero, on every inertial piece. It is a central idempotent. It acts as the identity on the structure object and as zero on its determinant-line twist, which has weight two. Both objects are perfect and satisfy the support condition.

If $e_0 = \alpha_F(f)$, evaluation on the structure object gives $f = 1$. Multiplication by 1 is also the identity on the determinant-line twist, a contradiction. Hence e_0 is not a scalar parameter function. □

The categorical comparison to be constructed

The scalar action supplies a precise ordinary-centre action on C_F. It does not assign a spectral object to a smooth representation. For a Bernstein component $\mathfrak{s}$, a categorical local comparison requires a specified central weight m, a full stable subcategory

$$\mathcal{E}_{\mathfrak{s}} \subseteq C_{\mathfrak{s},m},$$

and a k-linear exact fully faithful functor

$$\Phi_{\mathfrak{s}} : D(\mathcal{R}_{F,\mathfrak{s}}) \longrightarrow C_{\mathfrak{s},m}$$

with essential image $\mathcal{E}_{\mathfrak{s}}$. The functor must satisfy

$$\Phi_{\mathfrak{s}}(z_M) = \alpha_F(\Psi_F^{-1} z)_{\Phi_{\mathfrak{s}}(M)} \qquad (z \in Z_{F,\mathfrak{s}}).$$

An assertion of essential surjectivity requires $\mathcal{E}_{\mathfrak{s}} = C_{\mathfrak{s},m}$ as a further conclusion.

Such an equivalence with its essential image identifies the enhanced centre of $D(\mathcal{R}_{F,\mathfrak{s}})$ with that of $\mathcal{E}_{\mathfrak{s}}$. The ordinary ring $Z_{F,\mathfrak{s}}$ is only its specified degree-zero scalar part. No equality with the enhanced centre of the full spectral category is implied.

For the trivial-bundle representation category, weight zero is the natural component to test. Choosing that component removes the weight-projection obstruction above; it does not prove either the functor or an equivalence. A comparison involving the full spectral category must instead account for all its central weights and their corresponding geometric representation categories.

Example 116.2.6 *(why the ring calculation is needed).* The central element ϵ of $k[\epsilon]/(\epsilon^2)$ is nonzero but acts trivially on every simple module. Thus simple-object evaluations need not detect a nonreduced centre. Nor does bijective point matching prove surjectivity: $k[t^2, t^3] \subset k[t]$ induces a bijection on closed points over an algebraically closed characteristic-zero field, but t is not in the smaller ring. Indeed the exponents of every nonconstant monomial in that ring are at least two. A cusp point (a, b) with $b^2 = a^3$ has parameter $t = b/a$ if $a \neq 0$ and $t = 0$ otherwise. The ordinary-centre proof first constructs the invariant-ring isomorphism; evaluations only identify that map with the Fargues–Scholze action afterwards.

Dotto–Emerton–Gee at $p \geq 5$

Theorem 116.2.7 *(Dotto–Emerton–Gee categorical local Langlands at $p \geq 5$).* Dotto–Emerton–Gee arXiv:2603.26887 proves that for $p \geq 5$ and fixed central character χ there is a fully faithful functor

$$\mathrm{F}^{\mathrm{DEG}}_{p,\chi} : \mathrm{Rep}^{\mathrm{sm,adm,ss},\chi}(GL_2(\mathbb{Q}_p)) \hookrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{EG},p,\chi}_{GL_2})$$

of stable categories. No essential-surjectivity statement, image description, blockwise component exhaustion, or family over central characters is part of the input used here.

COROLLARY 116.2.8 *(ordinary local comparison at $p \geq 5$).* Let $p \geq 5$, let $\ell \neq p$, and use $k = \overline{\mathbb{Q}}_\ell$ with the preceding normalizations. Then

$$A_{\mathbb{Q}_p} \xrightarrow{\sim} Z_{\mathbb{Q}_p}, \qquad \Psi_{\mathbb{Q}_p}(f)|_\pi = f(\varphi_\pi^{\mathrm{ss}})\,\mathrm{id}_\pi.$$

The scalar image in the spectral categorical centre is therefore isomorphic to the ordinary Bernstein centre.

The categorical clause of (BCglue.local) additionally requires the functor, essential-image specification and central-action compatibility described above. The fixed-central-character Dotto–Emerton–Gee functor is a separate comparison in its own coefficient category.

Proof. Apply Chapter 116.2.2 and Chapter 116.2.5 with $F = \mathbb{Q}_p$. □

116.3 SMALL RESIDUE PRIMES AND COEFFICIENT CHANGES

Small residue characteristic does not exclude the ordinary characteristic-zero comparison when the coefficient prime is different from the residue prime.

THEOREM 116.3.1 *(ordinary local comparison at $p = 2, 3$).* Let $p \in \{2, 3\}$ and $\ell \neq p$. Over $k = \overline{\mathbb{Q}}_\ell$, the map

$$A_{\mathbb{Q}_p} \xrightarrow{\sim} Z_{\mathbb{Q}_p}$$

and its scalar evaluation formula on irreducible smooth representations hold with the preceding normalizations. This includes ramified inertial components.

Proof. The ordinary-centre proof uses characteristic zero and $\ell \neq p$, but no lower bound on p. □

Remark 116.3.2 *(coefficient transport).* An identification between centres with different coefficient fields requires coefficient-transport data. For inertial representatives defined over a common characteristic-zero field, compatible embeddings and compatible choices of $\sqrt{q}$ transport the displayed Laurent invariant-ring presentations and their evaluations.

This algebraic transport is not a canonical identification of arbitrary ℓ-adic families, nor does it identify integral models, reductions modulo ℓ, or the case $\ell = p$. Those comparisons require maps between the corresponding coefficient categories and parameter spaces.

PROPOSITION 116.3.3 *(a blockwise criterion at $\ell = p$).* Let $p \in \{2, 3\}$, fix the coefficient ring, a central character χ, and its corresponding fixed-determinant condition. Assume the wild-EG enhancement of Chapter 61.5.7, with all four conditions: its classical truncation is the specified Emerton–Gee stack; its cotangent complex is concentrated in $[-1, 0]$, with degree -1 supported on the non-semisimple locus and the stated closed-point dimensions 1 at $p = 2$ and 2 at $p = 3$; its wild components have the specified local-Langlands compatible correspondence with Bushnell–Henniart wild types; and its nilpotent ind-coherent category is generated under colimits and shifts by the specified derived structure sheaves of type-indexed closed points. These are hypotheses on a derived formal stack, not consequences of an equality of tangent dimensions.

For every block $\mathfrak{B}$ occurring in the chosen smooth representation category, suppose the following additional data are supplied:

(i) a specified stable realization $\mathcal{D}_{\mathfrak{B}}$ of the block and a fully faithful exact functor

$$F_{\mathfrak{B}} : \mathcal{D}_{\mathfrak{B}} \longrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{\mathfrak{B}}^{\mathrm{EG,wild},p,\chi});$$

(ii) a commutative spectral deformation ring $R_{\mathfrak{B}}$, with its action on the target, and an isomorphism

$$\beta_{\mathfrak{B}} : Z^{\circ}(\mathcal{D}_{\mathfrak{B}}) \xrightarrow{\sim} R_{\mathfrak{B}};$$

(iii) compatibility of these actions:

$$F_{\mathfrak{B}}(a_M) = \beta_{\mathfrak{B}}(a)_{F_{\mathfrak{B}}(M)} \quad (a \in Z^{\circ}(\mathcal{D}_{\mathfrak{B}}));$$

(iv) on the non-semisimple wild blocks, compatibility of the induced first-order maps with the specified pseudocharacter tangent comparison and Chapter 113.3.1, including the normalization of its class generators.

For the tame-inertia blocks the data must include the chosen $p = 2$ or $p = 3$ block-centre identifications; they are not supplied by the wild cotangent comparison.

Then the ordinary block-centre actions agree on the essential images of the functors $F_{\mathfrak{B}}$. If the block decompositions on both sides are orthogonal and the functors respect them, the agreement assembles over the chosen blocks.

Proof. The compatibility equation identifies the two natural endomorphisms on every object in the essential image. Full faithfulness identifies their endomorphism spaces with those of the source block. Orthogonality permits taking the product of these equalities over blocks. The tangent compatibility ensures that the resulting comparison retains the specified first-order classes; it is not used to infer the ring isomorphism $\beta_{\mathfrak{B}}$. □

The construction of $\beta_{\mathfrak{B}}$ and of the compatible functor is the remaining blockwise centre problem. A possible approach is to choose a generator of the block and a spectral object, compare their endomorphism algebras, and verify the two central actions there before extending to the generated categories. Such a generator and comparison are additional data.

An isomorphism of tangent spaces alone cannot supply the required ring map. For example, $k[[t]]/(t^2)$ and $k[[t]]/(t^3)$ have one-dimensional cotangent spaces but different lengths, so they are not isomorphic. Similarly, the wild-EG enhancement specifies derived support and generation conditions; it does not itself identify a block centre with its spectral deformation ring.

The Paškūnas–Tung block data at $p = 2$, the Colmez–Dospinescu–Paškūnas block data at $p = 3$, the fixed-determinant pseudocharacter comparison, the normalized class match, and the per-block functors remain separate inputs to this construction. An identification with an $\ell \neq p$ spectral category requires a further coefficient-comparison functor. The Fargues–Scholze results used above do not construct that functor.

116.4 Global Scalar Reconstruction and the Assembly Comparison

The algebra generated by local actions

Proposition 116.4.1 *(finite-place tensor assembly of scalar actions).* Fix a commutative coefficient ring E and commutative unital E-algebras Z_v indexed by a set V of places. The transition maps insert units, and define

$$A_{\mathrm{loc}} = \varinjlim_{\substack{F \subset V \\ F \text{ finite}}} \bigotimes_{v \in F, E} Z_v. \tag{116.1}$$

For an E-linear category C, compatible central actions $Z_v \to Z(C) = \mathrm{End}(\mathrm{id}_C)$ induce a unique homomorphism $\alpha : A_{\mathrm{loc}} \to Z(C)$. Its image is $A_{\mathrm{loc}}/\ker \alpha$. This image is the full ordinary centre precisely when the specified actions generate every central endomorphism; α is an isomorphism precisely when this holds and $\ker \alpha = 0$.

Proof. The centre is commutative: naturality of one endomorphism of the identity against the component of another makes their components commute. Thus multiplication of the local actions defines a homomorphism from each finite tensor product. These maps agree under insertion of units. The colimit gives α, uniquely. The first isomorphism theorem gives the image quotient. The last two assertions are respectively surjectivity and bijectivity of this homomorphism. □

This construction is on one specified coefficient ring. To use the local Bernstein centres at finite places and the Harish–Chandra centre $Z(U(\mathfrak{gl}_2))$ at the real place, one must supply coefficient maps and their actions on an actual global category. Equation (116.1) is an algebraic tensor colimit. A topology, spherical reference objects, a restricted completion or a factorization-homology realization each requires its own definition and comparison. No categorical or higher centre is identified by the algebraic colimit alone.

Frobenius coefficients and the global deformation ring

Unramified characteristic polynomials provide scalar functions on a Galois deformation space. The full collection determines the complete local ring with its deformation topology. A finite generating subset exists, but its choice depends on the deformation problem.

Let K be a number field, let k be a finite field of characteristic p_0, and put $W = W(k)$. Fix a finite set S containing the infinite places and all places above p_0. Write $G = G_{K,S}$ for the Galois group of the maximal extension of K unramified outside S. Fix a continuous absolutely irreducible representation $\bar\rho : G \to \mathrm{GL}_2(k)$ and a continuous character $\delta : G \to W^\times$ lifting $\det \bar\rho$.

For a finite local W-algebra A with residue field k, a deformation is a continuous lift $\rho : G \to \mathrm{GL}_2(A)$ with determinant δ_A. Two lifts are strictly equivalent if they are conjugate by an element of $1 + M_2(\mathfrak{m}_A)$. Let $R = R^{\delta}_{\bar\rho,S}$ be the complete Noetherian local W-algebra representing these deformation classes. Equivalently, R represents continuous rank-two determinant laws lifting $\det \bar\rho$ with determinant character δ. The representability and comparison hypotheses are those of Proposition 73.1.2. In particular, Mazur's finiteness condition gives Noetherianity. Let ρ^u be a universal lift, let $T = \mathrm{tr}\,\rho^u$, and let $\mathfrak{m}$ be the maximal ideal of R. All topologies on R below are $\mathfrak{m}$-adic.

For a finite place $v \notin S$, use arithmetic Frobenius Frob_v, acting on residue fields by the $\#k(v)$-power map. Its conjugacy class in G is well defined. Put

$$t_v = T(\mathrm{Frob}_v), \qquad d_v = \delta(\mathrm{Frob}_v), \qquad P_v(X) = X^2 - t_v X + d_v.$$

These conventions involve no choice of a square root of $\#k(v)$.

The distinction between a representation and its determinant law is already visible over $\mathbb{F}_2[\epsilon]/(\epsilon^2)$. For the cyclic group $C_2 = \langle s \rangle$, the representations

$$\rho_0(s) = I_2, \qquad \rho_1(s) = (1 + \epsilon) I_2$$

have the same trace and determinant on every element. Their determinant laws agree, since

$$\det \rho_1(a + bs) = (a + b(1 + \epsilon))^2 = (a + b)^2 = \det \rho_0(a + bs)$$

after every commutative coefficient extension. They are not conjugate, since both images of s are scalar and they differ. Thus equality of determinant laws need not determine a representation when the residual representation is reducible.

A framing retains a chosen basis, so framed deformations are lifts without quotienting by strict conjugacy. Over $A = \mathbb{F}_2[\epsilon]/(\epsilon^2)$, consider the constant lift $\sigma_0 : \mathrm{GL}_2(\mathbb{F}_2) \to \mathrm{GL}_2(A)$ of the natural representation. Its reduction is absolutely irreducible: the matrices

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

have different unique invariant lines over $\overline{\mathbb{F}}_2$. Put $P = I_2 + \epsilon E_{12}$ and $\sigma_1 = P\sigma_0 P^{-1}$. Since $\epsilon^2 = 0$ and the characteristic is 2,

$$PUP^{-1} = U, \qquad PVP^{-1} = V + \epsilon(E_{12}V - VE_{12}) = V + \epsilon I_2.$$

Both lifts have determinant 1. Conjugation preserves all traces and determinant laws, but the displayed images of V differ. These are the same unframed deformation and distinct framed deformations.

Lemma 116.4.2 *(trace determination with absolutely irreducible reduction).* Let A be a commutative local ring with residue field k. Suppose $\rho_1, \rho_2 : G \to \mathrm{GL}_2(A)$ reduce to the same absolutely irreducible representation over k. If $\operatorname{tr} \rho_1(g) = \operatorname{tr} \rho_2(g)$ for every $g \in G$, then ρ_1 and ρ_2 are strictly equivalent. The assertion includes residue characteristic 2.

Proof. The k-linear span of the residual group matrices is $M_2(k)$. Here is a proof in dimension two. Extend scalars to an algebraic closure $\bar{k}$ and let E be the algebra spanned by the group image. Absolute irreducibility says that E acts irreducibly. It contains a nonscalar matrix, and subtracting an eigenvalue produces a nonzero rank-one matrix $u\varphi$. Irreducibility gives $Eu = \bar{k}^2$ and $\varphi E = (\bar{k}^2)^*$ over the algebraic closure. Indeed, the annihilator of φE is invariant and cannot be the whole space. Thus the matrices $a(u\varphi)b$ span all endomorphisms. Descent of dimension gives the asserted span over k.

Choose $g_1, \dots, g_4 \in G$ whose residual matrices form a basis. For $a = 1, 2$, the matrices $A_{a,i} = \rho_a(g_i)$ form an A-basis of $M_2(A)$, since their coordinate determinant is a unit. The trace pairing is perfect over every commutative ring:

$$\operatorname{tr}(E_{ij}E_{rs}) = \delta_{jr}\delta_{is}.$$

Hence the common Gram matrix $Q_{ij} = \operatorname{tr} \rho_a(g_i g_j)$ is invertible. The coordinates $c_i(g)$ of $\rho_a(g)$ in this basis satisfy

$$\sum_i c_i(g) Q_{ij} = \operatorname{tr} \rho_a(g g_j).$$

They are therefore independent of a. The A-linear map $\Phi(A_{1,i}) = A_{2,i}$ sends $\rho_1(g)$ to $\rho_2(g)$ for every g. Since products of basis matrices are group matrices, Φ is a unital algebra isomorphism. Its reduction is the identity.

Put $F_{ij} = \Phi(E_{ij})$, $v = F_{11}e_1$, and $w = F_{21}v$. The matrix $P = (v, w)$ is congruent to I_2 modulo $\mathfrak{m}_A$, so it is invertible. The matrix-unit relations give $F_{ij}P = PE_{ij}$. Consequently $\Phi(X) = PXP^{-1}$, which proves strict equivalence. No division by 2 occurs. □

For example, over $\mathbb{F}_2$ the matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad UV$$

form a basis. Their trace Gram matrix is

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix},$$

whose determinant is 1, although $\operatorname{tr}(I) = 0$.

Lemma 116.4.3 *(unramified coefficients determine continuous determinants).* Two continuous rank-two determinant laws over a complete separated local coefficient ring have the same characteristic polynomials at every $v \notin S$ if and only if the laws agree. For deformations of $\bar{\rho}$ with fixed determinant δ, equality of Frobenius traces suffices. Their representation classes then agree.

Proof. Chebotarev's theorem implies that the union of the conjugacy classes of Frob_v, for $v \notin S$, is dense in G. Indeed, every conjugacy class in every finite quotient of G occurs at infinitely many such places. The inverse image of any nonempty open subset of a finite quotient therefore meets this union. Trace and determinant are continuous conjugacy-invariant functions. Equality on Frobenius classes consequently implies equality on G.

The passage from group coefficients to polynomial laws is integral. For a rank-two determinant law D with trace T_D and determinant character d, its value after any commutative coefficient extension $A \to C$ is

$$D_C\left(\sum_{i=1}^{r} a_i g_i\right) = \sum_{i} a_i^2 d(g_i) + \sum_{i<j} a_i a_j d(g_j) T_D(g_i g_j^{-1}).$$

To derive the mixed coefficient, multiply $U g_i + V g_j$ on the right by g_j^{-1} and use multiplicativity. The characteristic polynomial of $g_i g_j^{-1}$ gives

$$D(U g_i + V g_j) = d(g_i) U^2 + d(g_j) T_D(g_i g_j^{-1}) UV + d(g_j) V^2.$$

A homogeneous quadratic polynomial has only diagonal and mixed coefficients, so this determines the displayed formula for every finite sum. Thus equality of (T_D, d) implies equality of the laws. The previous lemma proves the assertion about representation classes. □

Theorem 116.4.4 *(global scalar reconstruction from Frobenius coefficients).* Define the polynomial parameter algebra

$$H^S = W[\mathsf{T}_v, \mathsf{D}_v, \mathsf{D}_v^{-1} : v \notin S]$$

and its fixed-determinant quotient

$$H^{S,\delta} = H^S / (\mathsf{D}_v - d_v : v \notin S) \simeq W[\mathsf{T}_v : v \notin S].$$

The map

$$\eta : H^{S,\delta} \longrightarrow R, \qquad \mathsf{T}_v \longmapsto t_v$$

has $\mathfrak{m}$-adically dense image.

Choose $a_v \in W$ lifting $\operatorname{tr} \bar{\rho}(\mathrm{Frob}_v)$, put $\mathfrak{n} = (p_0, \mathsf{T}_v - a_v : v \notin S)$, and define

$$\widehat{H}^{S,\delta} = \varprojlim_{n} H^{S,\delta} / \mathfrak{n}^n.$$

The induced continuous homomorphism $\widehat{\eta} : \widehat{H}^{S,\delta} \to R$ is surjective. It gives the topological quotient

$$R \simeq \widehat{H}^{S,\delta}/\ker \widehat{\eta}. \tag{116.2}$$

Moreover, finitely many places $v_1, \dots, v_d \notin S$ give a continuous surjection

$$W[[X_1, \dots, X_d]] \longrightarrow R, \qquad X_i \longmapsto t_{v_i} - a_{v_i}.$$

The same density conclusion holds after any quotient imposing a closed deformation condition.

Proof. Let B be the closure of $W[t_v : v \notin S]$ in R. It is a closed subring. Chebotarev and continuity show that $T(g) \in B$ for every $g \in G$. The ring B is profinite, with finite quotients $B_n = B/(B \cap \mathfrak{m}^n)$. It is local with residue field k. Indeed, if $b \in B$ has nonzero residue, choose a unit $u \in W$ with the same residue. Writing $b = u(1 - x)$ with $x \in B \cap \mathfrak{m}$, the convergent series $u^{-1} \sum_{j \geq 0} x^j$ belongs to B and inverts b.

The quadratic formula in the previous proof defines the universal determinant law over B, because all its coefficients belong to B. Its multiplicativity follows as a polynomial identity: the identities hold after the injection $B[\mathbf{X}] \hookrightarrow R[\mathbf{X}]$, and hence already hold over B. This also proves the identities after arbitrary coefficient extensions, without a flatness assumption. Its coefficient functions are continuous for the subspace topology on B, and its determinant character is δ.

The representing property gives compatible maps $f_n : R \to B_n$. Their inverse limit is a continuous map $f : R \to B$. If $i : B \hookrightarrow R$ is the inclusion, then $i \circ f$ induces the universal determinant over each $R/\mathfrak{m}^n$. Uniqueness in the representing property gives $i \circ f = \mathrm{id}_R$. Therefore $B = R$. This argument uses finite local quotients of B and does not assume that B is Noetherian before proving the equality.

Density implies that the classes of $t_v - a_v$ span $\mathfrak{m}/(\mathfrak{m}^2 + p_0 R)$ over k. In detail, approximate any element of $\mathfrak{m}$ modulo $\mathfrak{m}^2 + p_0 R$ by a polynomial in the t_v. Substituting $t_v = a_v + (t_v - a_v)$ leaves only its linear part in this quotient. Noetherianity makes this vector space finite dimensional, so choose finitely many places whose classes form a basis. Nakayama's lemma gives $\mathfrak{m} = (p_0, t_{v_1} - a_{v_1}, \dots, t_{v_d} - a_{v_d})$. Completeness then gives the displayed power-series surjection: approximate an element successively modulo $\mathfrak{m}^n$, lifting its degree-n correction to $(p_0, X_1, \dots, X_d)^n$. The resulting corrections converge in the power-series ring.

The polynomial map η sends $\mathfrak{n}^n$ into $\mathfrak{m}^n$, so it extends to $\widehat{\eta}$. The preceding finite power-series map factors through $\widehat{H}^{S,\delta}$, proving surjectivity. If F_n is the kernel of the projection $\widehat{H}^{S,\delta} \to H^{S,\delta}/\mathfrak{n}^n$, then $\widehat{\eta}(F_n) = \mathfrak{m}^n$. The inclusion from left to right follows by construction. The reverse inclusion follows from the finite power-series surjection on the nth powers of the maximal ideals. Thus the quotient topology is exactly the $\mathfrak{m}$-adic topology. Density passes to every quotient of R. □

Remark 116.4.5 *(topology and global relations).* If $A_0 = \eta(H^{S,\delta})$, its completion for the induced filtration $A_0 \cap \mathfrak{m}^n$ is canonically R:

$$\varprojlim_n A_0/(A_0 \cap \mathfrak{m}^n) \simeq R.$$

The kernel of $\widehat{\eta}$ records the global deformation relations. The theorem does not identify it with the closure of the kernel of η. Such an identification requires a separate comparison of completion and quotient. The finite set of generating places depends on the deformation problem. An arbitrarily prescribed pair of places need not generate its ring.

PROPOSITION 116.4.6 *(obstruction to a product of deformation components).* The fixed-residual ring R has no decomposition as a product of two nonzero unital rings. More generally it cannot be a unital restricted product containing the characteristic idempotents of two nonzero factors.

Proof. If $e^2 = e$ in a local ring, one of e and $1 - e$ is a unit. The equation $e(1 - e) = 0$ then gives $e = 1$ or $e = 0$. A product of two nonzero rings contains the nontrivial idempotent $(1, 0)$. The same obstruction applies whenever a restricted product contains these characteristic idempotents. □

The algebra $H^{S,\partial}$ consists of formal characteristic-polynomial parameters. An automorphic action requires a separate module or category and a homomorphism from these parameters to its endomorphisms. An identification of its completed action image with R additionally requires equality of the two kernels. The scalar reconstruction gives no such equality. It also supplies no local type decomposition, flatness assertion, prismatic filtration, untilt descent, or identification with a higher centre.

Remark 116.4.7 *(coefficient variants).* The same proof applies over a complete discrete valuation ring $\mathcal{O}$ finite over W, with a finite residue field and a chosen residual representation and determinant character over that base. Replace p_0 by a uniformizer of $\mathcal{O}$, and assume the corresponding complete Noetherian representing ring exists. For a deformation problem over k itself, omit the uniformizer from the residual ideal. These are statements about the indicated coefficient problems. A comparison between their representing rings requires its own base-change map and proof.

The density argument uses only the representing determinant functor. Absolute irreducibility is used to identify that functor with representation deformations and to obtain strict conjugacy from traces.

The determinant inputs are Chenevier, arXiv:0809.0415v2, Example 1.8 and Lemma 1.9, pp. 10–11; Definition–Proposition 2.18, pp. 32–33; Theorem 2.22, p. 34; and Proposition 3.3, Example 3.4, Remark 3.5, Example 3.6 and Proposition 3.7, pp. 42–43. The number-field density theorem is stated in Stevenhagen–Lenstra, *Chebotarev and his density theorem*, Appendix, p. 35. The finite set of generating places in the theorem is not a prescribed pair: the two-prime nonuniqueness example of Proposition 64.1.1 retains its different quantifiers.

The remaining global automorphic comparison

PROPOSITION 116.4.8 *(a specified action quotient and its comparison).* Use the coefficient ring and deformation problem of Theorem 116.4.4. Suppose an automorphic module or category has been constructed, together with a complete separated commutative W-algebra A_{act} of its scalar actions and a continuous surjection

$$\theta : \widehat{H}^{S,\partial} \longrightarrow A_{\mathrm{act}}.$$

Give A_{act} the quotient topology for this map. If $\ker \theta = \ker \widehat{\eta}$, there is a unique topological W-algebra isomorphism $A_{\mathrm{act}} \simeq R$ intertwining the maps from $\widehat{H}^{S,\partial}$. To identify this algebra with the full ordinary centre of the global category, additionally require that its central action be faithful and exhaust that centre. An identification of higher centres additionally requires a specified equivalence of their derived algebras, not only this degree-zero isomorphism.

Proof. Both algebras are topological quotients of the same algebra by the same ideal. The identity of the source descends to the claimed isomorphism, with inverse induced in the same way. It is a homeomorphism because both topologies are quotient topologies. Surjectivity of the source map proves

uniqueness. A faithful central action identifies an algebra with its image; equality with the full centre requires that image to be the whole centre. The argument has made no assertion about higher cohomology or operations. □

The Frobenius theorem constructs $\widehat{\eta}$ and its scalar quotient; it does not construct θ or compare these kernels. The unframed Galois deformation problem is fixed at one residual representation. An automorphic spherical locus, local Bernstein types and their coefficient changes must be specified independently. Almost-everywhere sphericity does not provide the missing ring maps.

To compare this action with Equation (116.1), one needs compatible homomorphisms from the local algebras on a common coefficient base and equality of their actions on each local factor. The colimit property then gives equality on the assembled algebra. Merely factoring through a common ring does not establish equality of actions. The pairwise geometric use retains every Selmer-ring, trace, coefficient and equivariant-descent condition of Corollary 73.1.5, together with the separate derived comparison or its two Tor-independence hypotheses.

For a categorical application, the functors, central weights and essential images of Section 116.2 remain required, with coefficient and family maps. At residual characteristic, all block-ring, fixed-character, fixed-determinant, wild-EG and normalized class-comparison data of Proposition 116.3.3 must be retained. The ordinary local scalar theorem supplies none of those functors.

A prismatic refinement must construct the relevant prisms, filtered objects and cross-prime morphisms and then prove the stipulated Π-flatness and filtration compatibility on that carrier. A Nygaard filtration on a different object does not supply these data. For an untilt comparison, construct the spaces and restriction functors for each untilt, their comparison equivalences and cocycle coherences, and prove that the scalar actions and specified support conditions are transported by them. Dependence of a determinant on Galois conjugacy classes does not establish such geometric descent. The arithmetic singular-support and higher-centre comparisons in Section 73.2 remain further constructions.

116.5 (E.4) (BCglue.local) + (BCglue.glue)

Theorem 116.5.1 *(local scalar comparison and categorical assembly).* At every finite prime p, the ordinary characteristic-zero comparison for $\overline{\mathbb{Q}}_\ell$, $\ell \neq p$, is the isomorphism of Proposition 116.2.2. Its image in the spectral categorical centre is the scalar subalgebra of Proposition 116.2.5.

Suppose in addition that the centre-compatible categorical local functors, their required essential images, the coefficient and family comparisons, and the residual-characteristic block data of Proposition 116.3.3 have been constructed. If the cross-prime assembly satisfies the stated Π-flat, pseudocharacter trace and global Bernstein-centre comparison hypotheses, these local comparisons supply the local inputs to that assembly.

Proof. The ordinary assertion is the proved ring isomorphism. The categorical assertions follow from the stipulated intertwining of central actions. Their use in the global assembly requires precisely the stated coefficient, family and cross-prime compatibilities. No passage from equality on irreducible objects to these additional data is used. □

Corollary 116.5.2 *(the Bernstein-centre local-gluing obstruction decomposition).* The ordinary local comparison leaves the following additional data for categorical and global gluing:

- local centre-compatible functors with specified essential images and central weights, and the required coefficient and family maps;

- at $\ell = p$, the wild-EG enhancement, fixed-determinant block rings and functors, and the normalized class match of Theorem 113.3.1;
- the global pseudocharacter and Bernstein-centre comparisons, Π-flat, untilt descent, and arithmetic AG at the centre.

Proof. Proposition 116.2.2 computes the ordinary ring. The proper scalar image in Proposition 116.2.5 prevents its identification with the full spectral categorical centre. Proposition 116.3.3 requires block-ring and functor data beyond first-order classes. Finally the assembly in Proposition 116.4.8 retains its explicitly stated global hypotheses. □

116.6 Consequences for BC-Diamond-Glue

Corollary 116.6.1 *(inputs to BC-diamond-glue).* The ordinary local scalar-centre input is available in characteristic zero for $\ell \neq p$ at every finite p. To obtain BC-diamond-glue from these inputs one must additionally supply the centre-compatible categorical comparisons, the residual-characteristic block comparisons and wild-EG enhancement where required, the coefficient and family comparisons, the Selmer-algebra, geometric, trace and equivariant pairwise sheaf comparisons, the global Bernstein-centre comparison, Π-flat, untilt descent, and the arithmetic higher-centre comparison.

Proof. The ordinary comparison supplies scalar functions and their actions. The remaining hypotheses supply the categorical, arithmetic and gluing maps needed to use those actions in BC-diamond-glue. □

Remark 116.6.2 *(remaining comparison constructions).* The cross-prime universal pseudocharacter algebra still requires the specified transition, coherent-support and higher-coherence data of Hypothesis 56.5.2, together with effective descent. These data do not establish a site-theoretic flatness theorem and are not constructed by a tensor product of local centres. The wild-EG enhancement in Conjecture 61.5.7 still requires its truncation, cotangent, type-label and generation conditions; an Ext^1 comparison is not their construction. Arithmetic AG at the centre and the global comparison maps remain separate hypotheses.

116.7 Reference Roles and Conditional Loci

Remark 116.7.1 *(local and global comparison conditions).* The ordinary characteristic-zero result holds for every residue characteristic, including 2 and 3, with $\ell \neq p$ and all inertial components. Its inputs are the Bernstein block formula, the algebraic parameter scheme, the four invariant-ring calculations, and the Fargues–Scholze compatibility with semisimplified local Langlands. The auxiliary Dotto–Emerton–Gee functor retains its $p \geq 5$ and fixed-central-character hypotheses. For $\ell = p$ at $p = 2, 3$, the complete blockwise hypotheses are those of Proposition 116.3.3. The references to Francis–Gaitsgory, Bellaïche–Chenevier, Bhatt–Scholze and Bhatt–Lurie concern the additional global and prismatic comparisons. The local ring calculation does not supply these comparisons or the (Ch1) and (AG.det.transition) hypotheses.

116.8 Seven Local and Global Comparison Questions

1. Surjectivity of the ordinary central action. Equality on simple objects alone does not prove a ring isomorphism. Proposition 116.2.2 first constructs the isomorphism on every inertial piece; reducedness then identifies it with the Fargues–Scholze action. The scalar image is still proper in the full spectral categorical centre.

2. The small residue characteristics. The ordinary calculation for $\ell \neq p$ includes $p = 2, 3$. It does not extend the $p \geq 5$ categorical functor to these primes. Their residual-characteristic block categories require the distinct data of Proposition 116.3.3.

3. Coefficient comparison. Compatible embeddings of a common characteristic-zero field and choices of $\sqrt{q}$ transport the Laurent invariant-ring formulas in Remark 116.3.2. Integral, family, residual and $\ell = p$ comparisons require additional functors and parameter-space maps; the ordinary isomorphism is not a canonical identification of these regimes.

4. Discrete assembly. The filtered tensor-product expression (116.1) must be distinguished from its identification with a global pseudocharacter algebra. The latter requires specified compatible maps on finite sets of places and a proof of the resulting algebra comparison.

5. Families and global pseudocharacters. Equation (116.2) is a completed scalar quotient for a fixed residual deformation problem, not a product of Bernstein components. An automorphic family requires its own parameter action, topology and relation ideal. The kernel equality in Proposition 116.4.8 is the additional comparison; equality at rigid points does not prove it.

6. Prismatic compatibility. Cross-prime flatness, compatible product filtrations and untilt descent are additional constructions for a prismatic or untilt refinement, as specified in the discussion following Proposition 116.4.8. They are not hypotheses of that scalar quotient proposition, and a discrete filtered colimit does not prove them.

7. Wild inertia at $\ell = p$. At $p = 2, 3$ the Paškūnas–Tung and Colmez–Dospinescu–Paškūnas block inputs, fixed determinant, wild-EG enhancement, normalized tangent-class comparison and centre-compatible block functors must all be retained. The tangent comparison does not identify complete local rings or construct an essential-image equivalence.

116.9 Primary Sources

Primary sources for the Bernstein-centre local-gluing comparison:

Fargues, L. and Scholze, P. (2021). *Geometrization of the local Langlands correspondence*. arXiv:2102.13459v4. Theorem VIII.1.3, pp. 279–280; Definition IX.0.2 and Corollary IX.0.3, pp. 318–319; Theorems IX.5.2 and IX.7.4, pp. 329 and 338.

Dat, J.-F., Helm, D., Kurinczuk, R., and Moss, G. *Moduli of Langlands parameters*, arXiv:2009.06708v3. §1.6, p. 9; Theorem 4.1, pp. 29–30; Corollary 4.16 and Proposition 4.17, pp. 35–36; Proposition 6.4, p. 66.

Helm, D. *The Bernstein center of the category of smooth $W(k)[GL_n(F)]$-modules*, arXiv:1201.1874, Theorems 3.4–3.5, pp. 7–8; and *Curtis homomorphisms and the integral Bernstein center for GL_n*, arXiv:1605.00487v3, Theorem 9.4, pp. 30–31.

Dotto, A., Emerton, M., and Gee, T. (2026). *A categorical p-adic Langlands correspondence for $GL_2(\mathbb{Q}_p)$*. arXiv:2603.26887. Used only for the fully faithful fixed-central-character functor at $p \geq 5$.

Bellaïche, J. and Chenevier, G. (2009). *Families of Galois representations and Selmer groups*. Astérisque 324. Chapter 1 provides the deformation-theoretic context. No product decomposition of the fixed-residual complete local ring is used here.

Chenevier, G. *The p-adic analytic space of pseudocharacters of a profinite group and pseudorepresentations over arbitrary rings*. arXiv:0809.0415v2. Example 1.8 and Lemma 1.9, pp. 10–11; Definition–Proposition 2.18, pp. 32–33; Theorem 2.22, p. 34; Proposition 3.3, Example 3.4 and Remark 3.5, p. 42; Example 3.6 and Proposition 3.7, p. 43.

Stevenhagen, P. and Lenstra, H. W., Jr. (1996). *Chebotarev and his density theorem*. Mathematical Intelligencer 18(2), 26–37; Appendix, p. 35, for the number-field density theorem.

Francis, J. and Gaitsgory, D. (2012). *Chiral Koszul duality*. Selecta Math. 18, arXiv:1103.5803. Background for the factorization-category comparison. The tensor-colimit action in Proposition 116.4.1 is proved directly; its arithmetic realization is separate.

Bernstein, J. and Deligne, P. (1984). "Le centre de Bernstein". In *Travaux en cours* 24, Hermann. §4 (Bernstein decomposition; block-centre formula $\mathcal{Z}_{\mathfrak{s}} = O(V_{\mathfrak{s}})^{W_{\mathfrak{s}}}$). Used as the comparison source for the Bernstein-centre match.

Bhatt, B. and Scholze, P. (2022). *Prisms and prismatic cohomology*. Ann. Math. 196, arXiv:1905.08229. Background for the prismatic realization; the particular cross-prime flatness comparison must be constructed on its stated arithmetic carrier.

Bhatt, B. and Lurie, J. (2022). *Absolute prismatic cohomology*. arXiv:2201.06120. Nygaard-filtration background for a specified filtered refinement; compatibility across untilts requires its own functors and coherent descent data.

Paškūnas, V. (2013). *The image of Colmez's Montreal functor*. Publ. IHES 118. Thm. 1.2 (block decomposition). Combined with PT 2020 (Thms. 1.3, 1.5, 1.7) and CDP 2014 §5. Used for the wild-prime block structure at $v \in \{2, 3\}$.

Hu, Y. and Tan, F. (2018). *Revisiting the Breuil–Mezard conjecture for GL_2*. Thm. 1.1 (very-special Breuil–Mezard multiplicities). Used in the Paškūnas block-tangent comparison and inherited here.

Kisin, M. (2008). *Potentially semi-stable deformation rings*. Ann. Math. 170 (2008). Pseudo-character tangent space. Inherited via the Paškūnas block-tangent comparison.

116.10 Residual Closure Boundary

(i) For $GL_2(F)$ over $\overline{\mathbb{Q}}_\ell$, $\ell \neq p$, functions on the full parameter stack are the ordinary Bernstein centre, by Proposition 116.2.2. All inertia types and the primes 2, 3 are included.

(ii) These functions form a faithful proper scalar subalgebra of the centre of the full spectral category. Central-weight projectors provide explicit additional endomorphisms. A categorical comparison requires a specified target weight, functor and essential image.

(iii) At $\ell = p$, the fixed-central-character and fixed-determinant block data, wild-EG enhancement, normalized class match and centre-compatible functors remain the hypotheses of Proposition 116.3.3.

(iv) For the fixed absolutely irreducible unframed deformation problem, unramified Frobenius coefficients topologically generate R, including residue characteristic two. The completed parameter algebra surjects onto R with the specified quotient topology. This determines scalar Galois deformations; it does not identify an automorphic action kernel.

(v) Cross-prime gluing retains coefficient and family comparisons, the global pseudocharacter and Bernstein-centre maps, prismatic flatness, untilt descent and arithmetic AG. The local scalar theorem does not discharge these requirements. Their role in BC-diamond-glue is stated in Corollary 116.6.1.

Chapter 117

F1.a.1 Residual Reduction: Supersingular Core and A_5 Icosahedral Core

Chapter 86 reduced the F1.a.1 residual from non-formal theorem to two sub-cores: the genuinely supersingular non-pBT core at $p \geq 5$ (Conjecture 86.2.12, non-formal theorem) and the non-solvable non-nearly-ordinary A_5 icosahedral core (Conjecture 86.4.12, non-formal theorem). Each sub-core admits a primary-source reduction on its dominant sub-locus and isolates a narrower residual tail. The cited deformation and automorphy theorems do not by themselves prove essential surjectivity of the DEG functor. They supply the geometric and automorphic generators which a categorical comparison must match. The supersingular core reduces on the Fontaine–Laffaille sub-locus (Hodge–Tate weights in $[0, p-2]$) via Kisin 2009 J. AMS 21 §3 integral p-adic Hodge theory, Emerton–Gee arXiv:1908.07185 §8 Newton–Hodge stratification, and Gee–Kisin 2014 Forum Math. Pi 2 geometric Breuil–Mézard transversality; the residual is the small-weight non-Fontaine–Laffaille tail at $k \in \{p-1, p\}$, together with the DEG–Breuil–Mézard generator comparison. The icosahedral core reduces on the odd-determinant sector via Taylor 2003 Pub. IHES 98 Part II potential automorphy, Barnet-Lamb–Gee–Geraghty 2012 J. AMS 25 automorphic-lifting descent, and Emerton 2011 Invent. Math. 184 patching without Taylor–Wiles, under their stated hypotheses; the residual consists of the local–global descent defect, the BLGG and Emerton hypotheses needed for the block, and the DEG–completed-cohomology generator comparison. These reductions are independent from the pBT and Serre-modular reductions of Chapter 86.

117.1 Domain, Inputs, and Disjointness

Inherited Structure

The two conditional sub-cores from Chapter 86 occupy complementary strata of the Emerton–Gee stack $\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi,\bar{\rho}}$ and the automorphic representation theory: the supersingular non-pBT core $\mathcal{X}_{\bar{\rho},\chi}^{\mathrm{ss,non\text{-}pBT}} = \mathcal{X}_{\bar{\rho},\chi}^{\mathrm{ss}} \setminus \mathcal{X}_{\bar{\rho},\chi}^{\mathrm{pBT}}$ (Definition 86.2.5) sits on the geometric side, and the non-solvable non-nearly-ordinary core $S^{\text{non-solv}}$ of projective image $A_5 \subset PGL_2(\overline{\mathbb{F}}_p)$ sits within the non-Taylor–Wiles sector partition $S^{\text{dihedral}} \cup S^{\text{nearly-ord}} \cup S^{\text{Serre-res}}$ (Definition 86.3.1).

Primary Sources

Kisin 2008, J. AMS 21 §3 integral p-adic Hodge theory: Breuil modules, Kisin modules, and the resolved Galois representation moduli on the supersingular stratum, including the non-Fontaine–Laffaille regime.

Kisin 2009, J. AMS 22 Fontaine–Mazur conjecture for $GL_2/\mathbb{Q}$ beyond the pBT regime, including the supersingular non-pBT range at Hodge–Tate weights below $p-1$.

Emerton–Gee arXiv:1908.07185 §7–8 specifically: Newton–Hodge stratification of the non-ordinary locus, dimension of supersingular components, and explicit closure relations between supersingular and pBT strata.

Gee–Kisin 2014, Forum Math. Pi 2 extended geometric Breuil–Mézard: cycle transversality extends from the pBT locus through the closure locus to neighbouring supersingular components.

Dotto–Emerton–Gee arXiv:2603.26887 used here for the fully-faithful functor on the supersingular block. Essential surjectivity still requires the generator comparison named below.

Taylor 2003, Pub. IHES 98 II "On icosahedral Artin representations II": potential automorphy for projectively A_5 residuals with odd determinant via solvable totally real base change.

Barnet-Lamb–Gee–Geraghty 2012, J. AMS 25 Sato–Tate for Hilbert modular forms: automorphic-lifting theorem extended to residually icosahedral representations of totally real fields.

Emerton 2011, Invent. Math. 184 §4 patching without Taylor–Wiles: completed cohomology and descent of a base-change lift from a totally real solvable extension to $\mathbb{Q}$.

Kisin 2007, Isr. J. Math. 157 potentially semistable deformation rings and the compatibility of the Newton slope stratification with Breuil–Mézard cycles.

Gee 2006, Compositio Math. 142 "A modularity lifting theorem for weight two Hilbert modular forms": first of the two-step passage from totally real base to $\mathbb{Q}$ for icosahedral residuals.

Buzzard–Dickinson–Shepherd-Barron–Taylor 2001, Duke Math. J. 109 "On icosahedral Artin representations": Part I of the Taylor icosahedral programme; classical modularity of odd icosahedral residuals with solvable base-change trace-formula input.

Deformation and Automorphy Inputs

The supersingular core reduction uses Kisin 2009 J. AMS 21 §3 (integral p-adic Hodge theory), Emerton–Gee arXiv:1908.07185 §8 (Newton–Hodge stratification), and Gee–Kisin 2014 Forum Math. Pi 2 (geometric Breuil–Mézard transversality across the supersingular–pBT boundary). The A_5 icosahedral reduction uses Taylor 2003 Pub. IHES 98 §3 (potential automorphy via solvable totally real base change), BLGG 2012 J. AMS 25 §7 (automorphic-lifting descent), and Emerton 2011 Invent. Math. 184 §4 (patching without Taylor–Wiles). The resulting categorical comparisons require the generator maps specified in the corresponding hypotheses below.

117.2 Fi.a.i.ss-core: Supersingular Non-pBT Locus

The Kisin Stratification of the Supersingular Core

Definition 117.2.1 *(Fontaine–Laffaille and small-weight supersingular strata).* Fix $p \geq 5$ and $\chi \in \mathrm{Hom}(\mathbb{Z}_p^\times, k^\times)$. The supersingular non-pBT locus $\mathcal{X}_{\bar\rho,\chi}^{\mathrm{ss,non\text{-}pBT}}$ partitions along the Hodge–Tate weight stratification

$$\mathcal{X}_{\bar\rho,\chi}^{\mathrm{ss,non\text{-}pBT}} \;=\; \mathcal{X}_{\bar\rho,\chi}^{\mathrm{ss,FL}} \;\sqcup\; \mathcal{X}_{\bar\rho,\chi}^{\mathrm{ss,non\text{-}FL}},$$

where:

- $\mathcal{X}_{\bar\rho,\chi}^{\mathrm{ss,FL}}$ is the *Fontaine–Laffaille supersingular stratum*: closed points with Hodge–Tate weights $(0, k-1)$, $2 \leq k \leq p-2$, i.e. weights strictly inside the Fontaine–Laffaille range, with Newton slopes $(\frac{k-1}{2}, \frac{k-1}{2})$;
- $\mathcal{X}_{\bar\rho,\chi}^{\mathrm{ss,non\text{-}FL}}$ is the small-weight tail: closed points with Hodge–Tate weights $(0, k-1)$, $k \in \{p-1, p\}$, on the boundary of the Fontaine–Laffaille range.

Remark 117.2.2 *(why Fontaine–Laffaille is the dividing line).* The Fontaine–Laffaille functor is an equivalence of categories on weights in $[0, p-2]$ between Fontaine–Laffaille modules and lattices in crystalline Galois representations (Fontaine–Laffaille 1982 Ann. ENS 15). Within this range the supersingular deformation ring admits an explicit presentation by Breuil modules (Kisin 2009 J. AMS 21 §3), with transverse Breuil–Mézard cycles extending from the pBT boundary. Outside this range (weights $k \in \{p-1, p\}$), the Breuil modules degenerate and a further input is required (Breuil–Mézard 2010 or the Caraiani–Levin 2020 Kisin-module refinement).

Lemma 117.2.3 *(Kisin moduli of finite flat group schemes on the FL supersingular stratum).* Fix $p \geq 5$ and $\chi \in \mathrm{Hom}(\mathbb{Z}_p^\times, k^\times)$. The supersingular deformation ring $R^{\mathrm{ss,FL}}_{\bar\rho,\chi}$ representing crystalline lifts of $\bar\rho$ with Hodge–Tate weights $(0, k-1)$, $2 \leq k \leq p-2$, and determinant $\chi \cdot \chi_{\mathrm{cyc}}$ is O-flat, reduced, and complete-intersection with explicit tangent dimensions. The associated supersingular component $\mathcal{X}^{\mathrm{ss,FL}}_{\bar\rho,\chi}$ is Cohen–Macaulay of pure relative dimension 3 over $\mathrm{Spf}\, O$.

Source. Kisin 2009 J. AMS 21 §3.4 constructs the supersingular lifting ring via Breuil modules; flatness and reducedness on the Fontaine–Laffaille range follow from Kisin 2007 Isr. J. Math. 157 §2 (potentially semistable deformation rings at regular Hodge–Tate weights). Emerton–Gee arXiv:1908.07185 §8.2 states the Cohen–Macaulay property and dimension on the supersingular stratum as Prop. 8.2.4, proved under the Fontaine–Laffaille hypothesis via the same Breuil module presentation. □

Lemma 117.2.4 *(extended Gee–Kisin transversality across the ss/pBT boundary).* Under the hypotheses of Lemma 117.2.3, for each inertial type $\tau : I_{\mathbb{Q}_p} \to GL_2(\overline{\mathbb{Q}}_p)$ the Breuil–Mézard cycle $\mathcal{X}^{\mathrm{BM},\tau}_{p,\chi,\bar\rho}$ restricted to $\mathcal{X}^{\mathrm{ss,FL}}_{\bar\rho,\chi}$ is a smooth divisor; cycles attached to distinct inertial types meet transversely; and the cycles set-theoretically exhaust the Fontaine–Laffaille supersingular stratum. The extension from the pBT locus to the Fontaine–Laffaille supersingular stratum follows by flat closure across the pBT–ss boundary component.

Source. Gee–Kisin 2014 Forum Math. Pi 2 Thm A (geometric Breuil–Mézard on all of $\mathcal{X}^{\mathrm{EG}}_{GL_2}$, not just pBT). Emerton–Gee arXiv:1908.07185 §8.3 Prop. 8.3.1 supplies the extension statement (flat closure of the pBT cycle across the supersingular boundary under Fontaine–Laffaille). □

Lemma 117.2.5 *(Kisin-module compact generators on the FL supersingular stratum).* Under the hypotheses of Lemma 117.2.3, for each Serre weight $\sigma : GL_2(\mathbb{F}_p) \to GL(V_\sigma)$ in the Serre weight set $W(\bar\rho|_{G_{\mathbb{Q}_p}})$, the Kisin-module projective $\widetilde{M}_\sigma \in \mathrm{Mod}^{\mathrm{sm},\chi}_{GL_2(\mathbb{Q}_p)}$ is mapped by the canonical Breuil-module-to-coherent-sheaf functor Ψ^{BM} to the structure sheaf of the corresponding Breuil–Mézard cycle $\mathcal{O}_{\mathcal{X}^{\mathrm{BM},\sigma}_{p,\chi,\bar\rho}} \in \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\mathrm{ss,FL}}_{\bar\rho,\chi})$. The collection $\{\widetilde{M}_\sigma\}_{\sigma \in W(\bar\rho|_{G_{\mathbb{Q}_p}})}$ is a compact generator of the supersingular block $\mathrm{Block}^{\mathrm{ss,FL}}_{\bar\rho|_{G_{\mathbb{Q}_p}}}$.

Source. Kisin 2009 J. AMS 21 §3.5 constructs the Kisin-module projective covers $\widetilde{M}_\sigma$ and identifies their images under the Breuil-module functor with the structure sheaves of the corresponding Breuil–Mézard cycles. Compact generation on the supersingular block follows from the same argument as Paškūnas 2013 Prop. 6.1 adapted to the Kisin-module side: the block is generated by its projectives, and the projectives in the supersingular block are exactly the $\widetilde{M}_\sigma$ for σ running over admissible Serre weights. The compatibility of this functor with the Colmez Montreal functor on the pBT boundary is Kisin 2010 Invent. Math. 178 §2.4; on the interior of the supersingular stratum, Ψ^{BM} is the direct Breuil-module-based construction of Kisin 2009 J. AMS 21 §3.5 rather than the Colmez Montreal functor. □

Essential Surjectivity on the Fontaine–Laffaille Stratum

THEOREM 117.2.6 *(conditional F1.a.1 on the Fontaine–Laffaille supersingular stratum). Comparison hypothesis.* Fix $p \geq 5$ and $\chi \in \mathrm{Hom}(\mathbb{Z}_p^\times, k^\times)$, under Section 70.2.1 (TW1)–(TW3) with (TW4) dropped. The Kisin and Breuil–Mézard inputs give compact generators on the Fontaine–Laffaille supersingular stratum. If the DEG functor sends these generators to the corresponding Breuil–Mézard structure sheaves, then the restriction of the DEG functor $\Phi^{\mathrm{DEG}}_{p,\chi}$ to the Fontaine–Laffaille supersingular sub-∞-category is an equivalence:

$$\Phi^{\mathrm{DEG}}_{p,\chi}\big|_{\mathrm{ss,FL}} : \mathrm{D}^{\text{sph-fin,ss,FL}}([GL_2(\mathbb{Q}_p)])_\chi \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{ss,FL}}_{\bar\rho,\chi}\big).$$

Proof. **Step 1: fully-faithfulness.** The DEG functor is fully faithful on the full block (Theorem 61.3.1); restriction to the Fontaine–Laffaille supersingular sub-∞-category preserves fully-faithfulness.

Step 2: target is compactly generated. By Lemma 117.2.3, $\mathcal{X}^{\mathrm{ss,FL}}_{\bar\rho,\chi}$ is Cohen–Macaulay of pure dimension 3 over $\mathrm{Spf}\, O$; its $\mathrm{IndCoh}_{\mathrm{Nilp}}$ is compactly generated by the structure sheaves of its Breuil–Mézard cycles (Gaitsgory–Rozenblyum AMS DAG II Ch. 8 §1.3).

Step 3: source is compactly generated. By Lemma 117.2.5, the collection of Kisin-module projectives $\{\widetilde{M}_\sigma\}_{\sigma \in W(\bar\rho|_{G_{\mathbb{Q}_p}})}$ is a compact generator of $\mathrm{Block}^{\mathrm{ss,FL}}_{\bar\rho|_{G_{\mathbb{Q}_p}}}$.

Step 4: match on generators. The needed additional input is: for each Serre weight σ, the image of $\widetilde{M}_\sigma$ under DEG coincides with its image under the Kisin-module coherent-sheaf functor Ψ^{BM}, namely $\mathcal{O}_{\mathcal{X}^{\mathrm{BM},\sigma}_{p,\chi,\bar\rho}}$. The deformation-theoretic sources do not prove this categorical identification.

Step 5: equivalence by Lurie HA 4.8.2.11. Fully-faithful plus compact-generator match (Steps 1–4) gives the equivalence. □

COROLLARY 117.2.7 *(F1.a.1.ss-core conditional reduction). Comparison hypothesis.* Under the generator-comparison input of Theorem 117.2.6, the F1.a.1.ss-core residual (Conjecture 86.2.12) reduces to essential surjectivity on the small-weight non-Fontaine–Laffaille tail $\mathcal{X}^{\mathrm{ss,non\text{-}FL}}_{\bar\rho,\chi}$: closed points of Hodge–Tate weight $k \in \{p-1, p\}$ with supersingular Newton slopes, none of which admit a Fontaine–Laffaille lift.

Proof. Theorem 117.2.6 gives the reduction on the Fontaine–Laffaille supersingular stratum only under its generator comparison. The complementary small-weight tail is exactly $\mathcal{X}^{\mathrm{ss,non\text{-}FL}}_{\bar\rho,\chi}$ by Definition 117.2.1. □

Conjecture 117.2.8 *(F1.a.1.ss-core small-weight tail). Conjectural statement.* Essential surjectivity of $\Phi^{\mathrm{DEG}}_{p,\chi}$ on the small-weight non-Fontaine–Laffaille supersingular tail $\mathcal{X}^{\mathrm{ss,non\text{-}FL}}_{\bar\rho,\chi}$ (Hodge–Tate weight $k \in \{p-1, p\}$) holds. It is a non-formal theorem, via a Breuil–Mézard 2010 Caraiani–Levin 2020 Kisin-module refinement at the Fontaine–Laffaille boundary.

Remark 117.2.9 *(why the small-weight tail is genuinely narrower).* The F1.a.1.ss-core residual of Conjecture 86.2.12 (non-formal theorem) covered the full genuinely supersingular non-pBT locus. The Fontaine–Laffaille stratum is the large sub-locus where Breuil modules behave well; the residual tail is the two specific Hodge–Tate weights $k \in \{p-1, p\}$ on the boundary of the Fontaine–Laffaille range. This reduces the geometric tail, but it does not remove the DEG–Breuil–Mézard generator-comparison input.

Proof of Theorem 117.2.6

Geometric input. Kisin 2009 J. AMS 21 §3 is integral p-adic Hodge theory via Breuil modules; Emerton–Gee arXiv:1908.07185 §8 is the Newton–Hodge structure theorem on the EG stack; Gee–Kisin 2014 is the geometric Breuil–Mézard cycle theorem; Fontaine–Laffaille 1982 is the classical weight-range equivalence. The Kisin-module functor supplies deformation geometry; comparison with the DEG image requires the stated generator identification.

Conventions. Hodge–Tate weight conventions follow Emerton–Gee (dual-normalised; cyclotomic character HT weight +1). Kisin-module conventions follow Kisin 2009 J. AMS 21 §1.1 (Breuil modules over $S = \varprojlim W_n[[u]]$ with canonical Frobenius lift $\varphi(u) = u^p$). Serre weight conventions follow Breuil–Mézard 2014 ($\sigma = \sigma(a, b)$ indexed by Hodge–Tate weights (a, b)). Newton-slope conventions follow Kisin 2007 §1.2. All four sources agree after the standard Fontaine cyclotomic shift.

Ambient categories. The source $\mathrm{D}^{\text{sph-fin,ss,FL}}([GL_2(\mathbb{Q}_p)])_\chi$ and target $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}^{\text{ss,FL}}_{\bar\rho,\chi})$ are both stable presentable ∞-categories with compact generators. The restricted DEG functor is a morphism in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$.

Hypothesis domain. The Fontaine–Laffaille hypothesis ($k \in [2, p-2]$) is explicit; it is needed for Lemma 117.2.3 (Kisin-module flatness) and for Lemma 117.2.5 (Kisin-module compact generators). The failure at $k \in \{p-1, p\}$ is exactly the residual Conjecture 117.2.8. (TW1)–(TW3) are inherited from Chapter 86; (TW4) is dropped.

Hecke-action compatibility. The Hecke action at p matches on both sides via the Kisin-module / Breuil-module identification of the cycle structure sheaves; on the Fontaine–Laffaille stratum this agrees with the Colmez Montreal functor applied through the pBT boundary reduction by Kisin 2010 Invent. Math. 178 §2.4, provided the DEG–Breuil–Mézard generator comparison is supplied.

Inputs. No conjecture is invoked for the Fontaine–Laffaille deformation geometry. The categorical theorem also requires the DEG–Breuil–Mézard generator comparison. The small-weight tail residual is Conjecture 117.2.8, of domain non-formal theorem, a strict reduction from the Conjecture 86.2.12 domain of non-formal theorem.

These hypotheses specify the Fontaine–Laffaille reduction; they do not remove the generator-comparison input. The small-weight tail residual is explicitly named and strictly narrower than the residual of Conjecture 86.2.12.

117.3 Fi.a.i.ns-core: A_5 Icosahedral Sector

Structural Decomposition of the Icosahedral Core

Definition 117.3.1 *(icosahedral sub-sectors).* By Dickson's classification of finite subgroups of PGL_2, the non-solvable non-nearly-ordinary sector $S^{\text{non-solv}}$ decomposes along the projective image of the residual $\bar\rho^{\text{proj}}\colon G_{\mathbb{Q}} \to PGL_2(\overline{\mathbb{F}}_p)$ as

$$S^{\text{non-solv}} \;=\; S^{A_5,\text{odd}} \;\cup\; S^{A_5,\text{local-global-defect}},$$

where:

- $S^{A_5,\mathrm{odd}}$: residuals with $\bar\rho^{\mathrm{proj}}$-image A_5 and $\det\bar\rho$ odd (complex conjugation acts with determinant -1), covered by Taylor 2003 Pub. IHES 98 Part II potential automorphy plus BLGG 2012 J. AMS 25 descent plus Emerton 2011 Invent. Math. 184 patching;
- $S^{A_5,\text{local-global-defect}}$: residuals for which the solvable-base-change lift exists at the level of an intermediate totally real extension but fails to descend to $\mathbb{Q}$ via the standard Arthur–Clozel pattern, retained as a single residual sub-obstruction.

Remark 117.3.2 *(size of the defect sub-sector).* The local–global descent defect captures the unique obstruction at the residual level where the totally real totally solvable base-change lift fails to descend by cyclic steps to $\mathbb{Q}$: this happens only if the class field over $\mathbb{Q}$ cut out by the projective kernel of $\bar\rho$ admits a cyclic-layer obstruction not contained in the Arthur–Clozel solvable layers. No density assertion is made for this subset.

Reduction on the Odd-Determinant Icosahedral Sub-Sector

The sub-sector $S^{A_5,\mathrm{odd}}$ is reduced via three primary sources assembled in sequence: potential automorphy over a totally real solvable extension (Taylor 2003), descent of automorphic lifts via the Arthur–Clozel cyclic pattern (BLGG 2012), and completion of the $R = T$ identification without Taylor–Wiles hypotheses (Emerton 2011). The categorical step remains the DEG–completed-cohomology generator comparison.

LEMMA 117.3.3 *(Taylor 2003 potential automorphy on the Artin A_5 locus).* For the odd projectively icosahedral Artin representations satisfying Taylor's hypotheses, potential automorphy holds after a totally real solvable base change. The statement does not, by itself, give an automorphic lift for every deformation point in $S^{A_5,\mathrm{odd}}$.

Source. Taylor 2003 Pub. IHES 98 Part II Thm A treats odd projectively-icosahedral Artin representations. Part I is Buzzard–Dickinson–Shepherd-Barron–Taylor 2001 Duke Math. J. 109. Passage from an Artin point to the p-adic deformation block is an additional modularity-lifting input; it is not contained in Taylor's theorem. □

LEMMA 117.3.4 *(BLGG 2012 descent of automorphy lifts). Conditional statement.* For every residual $\bar\rho \in S^{A_5,\mathrm{odd}}$ with base-change lift $\bar\rho|_{G_F} \cong \bar\rho_{\pi,p}$ from Lemma 117.3.3, assume the automorphy-lifting hypotheses of BLGG and the solvable descent hypotheses of Arthur–Clozel at each layer. Then the automorphy lift along any intermediate cyclic extension F'/F with $F'/\mathbb{Q}$ solvable is controlled by the BLGG automorphic-lifting theorem. In particular, if the tower $\mathbb{Q} \subset F_1 \subset F_2 \subset \ldots \subset F$ is a solvable chain where each step is cyclic of degree dividing a prime $\ell \neq p$, the automorphic representation π over F descends to automorphic representations π_i over F_i by successive Arthur–Clozel solvable descent.

Source. BLGG 2012 J. AMS 25 §7 (Sato–Tate for Hilbert modular forms) adapts the Harris–Shepherd-Barron–Taylor Calabi–Yau construction plus Arthur–Clozel base change to the residually-icosahedral setting over totally real fields. The cyclic descent step uses Arthur–Clozel 1989 Ann. Math. Stud. 120 §3 Thm 4 at each layer. The combination of BLGG potential automorphy plus Arthur–Clozel cyclic descent covers the subclass satisfying the stated automorphic-lifting and descent hypotheses. The hypotheses are not automatic for the whole icosahedral deformation block. □

LEMMA 117.3.5 *(Emerton 2011 patching without Taylor–Wiles). Conditional on the stated hypotheses.* For every residual $\bar\rho \in S^{A_5,\mathrm{odd}}$ covered by Lemma 117.3.4, assume the completed-cohomology local–global

compatibility and patching hypotheses required in Emerton 2011. Then the completed cohomology patching functor produces a deformation-ring-to-Hecke comparison $R^{\mathrm{univ}}_{\bar\rho,\chi} \cong \mathbf{T}^{\mathrm{cpt}}_{\bar\rho}$ in the completed cohomology block. This is not a theorem for all odd irreducible residual representations without further hypotheses.

Source. Emerton 2011 Invent. Math. 184 §4 is the completed-cohomology source. It supplies a patching formalism under its local–global hypotheses. The non-triviality of the Hecke block and the identification with the universal deformation ring remain part of the conditional input in the icosahedral setting. □

Theorem 117.3.6 *(Conditional F1.a.1 on $S^{A_5,\mathrm{odd}}$).* Fix $p \geq 5$, a central character χ, and $\bar\rho \in S^{A_5,\mathrm{odd}}$. Assume the following DEG–completed-cohomology generator-comparison data on this block:

(i) A specified completed-cohomology object P is a projective generator of the stated block and a compact generator of its stable presentable derived category, with the specified endomorphism-ring comparison to $R^{\mathrm{univ}}_{\bar\rho,\chi}$.

(ii) A specified identification of the spectral component with Spf $R^{\mathrm{univ}}_{\bar\rho,\chi}$, together with compact generation of its $\mathrm{IndCoh}_{\mathrm{Nilp}}$ category by the structure sheaf Q.

(iii) The restricted DEG functor is exact and preserves colimits, and a specified comparison carries P to Q.

Under the fully faithful fixed-character DEG hypothesis in Theorem 61.3.1, this functor is an equivalence. The generator properties and the object comparison are additional data; a ring identity $R = T$ alone supplies none of them.

Proof. The hypotheses provide a compact generator P of the source and identify its image with a compact generator Q of the target. Full faithfulness identifies the source with the essential image of the functor. Exactness and preservation of colimits make this image a stable subcategory closed under colimits: a diagram of objects in the image lifts, by full faithfulness, to a diagram in the source, and its colimit is the image of the source colimit. It contains Q, whose localizing closure is the target by the assumed generator property. Hence the image is the whole target, which proves essential surjectivity and therefore the equivalence. The patching and $R = T$ inputs discussed in Lemma 117.3.5 remain possible ingredients in constructing the stated data; they are not used to infer the generator properties. □

Reduction of the Residual to the Local–Global Defect

Corollary 117.3.7 *(F1.a.1.ns-core reduction).* Under the generator-comparison input of Theorem 117.3.6, the F1.a.1.ns-core residual (Conjecture 86.4.12) reduces to essential surjectivity on the local–global descent defect sector $S^{A_5,\text{local-global-defect}}$: the set of icosahedral residuals, with no density assertion, where the solvable chain to $\mathbb{Q}$ fails by the Arthur–Clozel cyclic pattern.

Proof. Theorem 117.3.6 gives the reduction on the odd-determinant icosahedral sub-sector only under its generator comparison. The complement within $S^{\text{non-solv}}$ is $S^{A_5,\text{local-global-defect}}$ by Definition 117.3.1. □

Conjecture 117.3.8 *(F1.a.1.ns-core local–global defect). Conjectural statement.* Essential surjectivity of $\Phi^{\mathrm{DEG}}_{p,\chi}$ on the local–global defect sector $S^{A_5,\text{local-global-defect}}$ holds. Domain: non-formal theorem, via an Arthur–Clozel cyclic-descent refinement for the residual layer, adaptable from BLGG 2012 §7 plus Emerton 2011 §4.

Remark 117.3.9 *(why the local–global defect is narrow).* The F1.a.1.ns-core residual of Conjecture 86.4.12 (non-formal theorem) covered the full non-solvable non-nearly-ordinary sector. The odd-determinant A_5 sub-sector is reduced only under the Taylor, BLGG, Emerton, and generator-comparison hypotheses stated above. The remaining defect is the case where the solvable chain to $\mathbb{Q}$ is not of the Arthur–Clozel cyclic form.

Proof of Theorem 117.3.6

Geometric input. Taylor 2003 Pub. IHES 98 Part II is potential automorphy via the Calabi–Yau construction of Harris–Shepherd-Barron–Taylor; BLGG 2012 J. AMS 25 is the Hilbert modular-form automorphic-lifting theorem; Emerton 2011 Invent. Math. 184 is completed-cohomology patching without Taylor–Wiles. These supply the automorphy and patching inputs under their stated hypotheses; the categorical generator comparison is separate.

Conventions. Automorphic weight: parallel weight 2 on F corresponds to weight 2 on $\mathbb{Q}$ after descent (Gee 2006 Compositio Math. 142 §2). L-parameter normalisation follows Taylor 2003 §1.1 (local parameters at the archimedean places of F of the prescribed Hodge–Tate type). Completed-cohomology normalisation follows Emerton 2011 §1.2. All three sources agree after the standard weight / L-parameter correspondences.

Ambient categories. Under the specified generator data of Theorem 117.3.6, the source is the icosahedral block of $\mathrm{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi}$. The target is $\mathrm{IndCoh}_{\mathrm{Nilp}}$ on the icosahedral component of the EG stack, which is a formal spectrum $\mathrm{Spf}\, R^{\mathrm{univ}}_{\bar{\rho},\chi}$ of a complete Noetherian local ring. Both are, by the generator hypotheses, stable presentable ∞-categories with a compact generator: the completed Hecke algebra on the source side and the structure sheaf on the target side.

Hypothesis domain. Odd determinant is explicit: it is the parity condition required for Taylor 2003 Part II Thm A and for the existence of complex conjugation in the icosahedral image. The hypothesis that the solvable chain $\mathbb{Q} \subset F_1 \subset \ldots \subset F$ consists of cyclic layers of prime degree not equal to p is explicit in Lemma 117.3.4. The residual failure at non-Arthur–Clozel-cyclic layers is Conjecture 117.3.8. (TW1) $p \geq 5$ is retained.

Hecke-action compatibility. The Hecke-eigenvalue match on the icosahedral sector is conditional on the chain of $R = T$ and base-change identifications. Taylor gives the Artin potential automorphy input, BLGG gives descent under its automorphic-lifting hypotheses, and Emerton gives completed-cohomology patching under its local–global hypotheses. At each conditional step, the Hecke action on the automorphic side corresponds to the Frobenius trace on the Galois side.

Inputs. The automorphy and patching inputs on $S^{A_5,\mathrm{odd}}$ are conditional as stated in Lemma 117.3.4 and Lemma 117.3.5. The categorical theorem also requires the DEG–completed-cohomology generator comparison. The local–global defect residual is Conjecture 117.3.8, of domain non-formal theorem.

These hypotheses specify the odd-determinant icosahedral reduction under its stated automorphy, patching, and generator-comparison inputs. They do not remove these inputs.

117.4 Combined F1.a.1 Conditional Reduction on Five Strata

Theorem 117.4.1 *(F1.a.1 equivalence on five strata, conditional on generator comparisons).* For every prime $p \geq 5$ and every central character $\chi \in \mathrm{Hom}(\mathbb{Z}_p^\times, k^\times)$, assume the generator-comparison inputs in Theorem 117.2.6 and Theorem 117.3.6. Then the DEG functor $\Phi_{p,\chi}^{\mathrm{DEG}}$ is an equivalence on the union of:

(i) the unramified ordinary summand (Chapter 70);
(ii) the potentially Barsotti–Tate non-ordinary summand (Theorem 86.2.10);
(iii) the dihedral, nearly-ordinary, and Serre-modular non-TW sectors (Theorem 86.4.10);
(iv) the Fontaine–Laffaille supersingular stratum (Theorem 117.2.6);
(v) the odd-determinant icosahedral sector (Theorem 117.3.6).

This is a conditional direct-sum statement. The remaining obstructions are: the two generator comparisons above, the small-weight non-Fontaine–Laffaille supersingular tail (Conjecture 117.2.8) and the local–global defect icosahedral sub-sector (Conjecture 117.3.8).

Proof. Take the direct sum of the five stated theorems. Orthogonality of the corresponding residual-index blocks on both sides (Proposition 61.4.2) ensures the direct-sum functor is an equivalence on the union. □

Proposition 117.4.2 *(F1.a.1 obstruction decomposition after Chapter 117).* After the reductions of this chapter, the F1.a.1 residual has four visible sub-cores:

(i) **F1.a.1.ss-generator-match**: the DEG–Breuil–Mézard generator comparison on the Fontaine–Laffaille supersingular stratum.
(ii) **F1.a.1.ss-small-weight** (Conjecture 117.2.8): essential surjectivity at Hodge–Tate weights $k \in \{p-1, p\}$ supersingular boundary.
(iii) **F1.a.1.ns-generator-match**: the DEG–completed-cohomology generator comparison on the odd-determinant icosahedral sector.
(iv) **F1.a.1.ns-defect** (Conjecture 117.3.8): essential surjectivity on the local–global descent defect icosahedral sub-sector.

Both generator comparisons remain required for essential surjectivity on the stated strata.

Proof. It follows from Corollary 117.2.7 and Corollary 117.3.7. Each corollary is conditional on a generator comparison, hence the generator comparison itself remains in the decomposition. □

Proposition 117.4.3 *(Full F1 obstruction decomposition after Chapter 117).* Combining Proposition 117.4.2 with the unchanged F1.a.2 and Eisenstein residuals, and with the corrected $\mathrm{L}_{\mathrm{ACD}}$ decomposition of Corollary 52.7.1, the full F1 obstruction decomposition contains:

(i) the four F1.a.1 sub-cores of Proposition 117.4.2;
(ii) **F1.a.2.class-match** (non-formal theorem; unchanged from Conjecture 86.5.3).
(iii) $\mathrm{L}_{\mathrm{ACD}}$**.Eisenstein** (non-formal theorem).
(iv) the local categorical comparison, the ∞-categorical Bernstein lift, the archimedean boundary comparison, the global identification, and (BC-diamond-glue).

These local and global comparison inputs remain separate; the stratum decomposition does not construct them.

Proof. Tabulation from Proposition 117.4.2 and Corollary 52.7.1. The latter replaces the old single BC-diamond-glue row by its comparison-data table. □

The two F1.a.1 residuals isolated in Chapter 86 reduce on their dominant sub-loci: the Fontaine–Laffaille supersingular stratum (Theorem 117.2.6) and the odd-determinant icosahedral sub-sector (Theorem 117.3.6). The remaining inputs are the two generator comparisons, the small-weight Fontaine–Laffaille boundary (Conjecture 117.2.8), and the local–global cyclic descent defect (Conjecture 117.3.8). The six assembled sources are independent from those used in Chapter 86.

Chapter 118

Exceptional Intertwining Families and Arithmetic Extension Comparisons

When an intertwining operator loses invertibility, its kernel and quotient describe the representations that meet at that parameter. An arithmetic comparison asks for more: which extensions of representations correspond to deformations of Galois representations, and how do these correspondences fit into a global automorphic category? A residue is a morphism of representations, whereas an extension is a short exact sequence. Passing from one to the other requires a construction.

For an ordered unramified character pair, a regular intertwiner gives an explicit two-term complex over the character torus. Its derived specializations retain kernels created by nonflat coefficient change. At an exceptional complex point a tree resolution splits this complex even with its central character imposed strictly. A different construction gives a nonsplit local principal-series extension. Exterior products of these extensions have the number of places as their degree. Their constant-vector restricted tensor product instead gives a genuine degree-one finite-adelic extension with a mixed quotient. These results supply distinct local and adelic objects for the arithmetic question; identifying them with automorphic residues or Galois deformation classes requires the maps specified below.

118.1 Parameters and the comparison problem

Chapter 85 fixes the finite-place families. The raw finite-place integral $M_v(s, \chi)$ and its normalized operator $N_v(s, \chi)$ are defined in Definitions 85.3.6 and 85.3.7. The calculation below retains their ordered inducing characters, Haar measure and smooth source and target. Proposition 85.3.2 states their meromorphic composition law. The exceptional local ratios are q and q^{-1}, as in Definition 85.4.5; they are not singular points on the unitary axis. The global comparison at these strata is the problem of Conjecture 85.8.3.

The other required comparisons concern different carriers. Chapter 84 concerns the normalization of vector pairings. The Langlands–Shahidi-corrected Koszul pairing in Definition 84.2.1 is a pairing of specified vectors. Corollary 84.2.4 identifies its normalization with the Petersson normalization only under its stated positive-measure identification. A pairing on extension classes requires an additional chain-level construction. The wild-prime comparison of Theorem 87.2.9 and the residual-gerbe comparison in Hypothesis 87.2.7 concern residual-characteristic categories, as in Chapter 87. The Bushnell–Henniart block in Chapter 68 has its specified local categorical carrier and does not construct a complex automorphic realization. Neither is a coefficient identification with a complex smooth representation category. The Hilbert spectral decomposition in Theorem 85.6.1 and the conditional categorical residual comparison in Theorem 85.6.3 likewise have different levels of structure. We first construct the finite-place complexes and extensions before formulating these further maps.

118.2 The regular unramified family

The normalized unramified intertwiner loses invertibility at two character ratios. Its cone records the resulting kernels and quotients, including those created by specialization. To define this cone across both ratios, fix the scalar used to clear the denominator.

Let F be a finite extension of $\mathbb{Q}_p$ with ring of integers $\mathcal{O}$, uniformizer ϖ, and residue cardinality q. Set $|\varpi| = q^{-1}$ and normalize additive Haar measure by $\mathrm{vol}(\mathcal{O}) = 1$. Put $G = \mathrm{GL}_2(F)$, $K = \mathrm{GL}_2(\mathcal{O})$, and let B be the upper triangular subgroup. The coefficient ring and ratio are

$$R = \mathbb{C}[a^{\pm 1}, b^{\pm 1}], \qquad z = a/b.$$

Every coefficient ring in this local-family construction is an ordinary commutative $\mathbb{C}$-algebra. It has no analytic topology. The universal unramified characters are $\chi_a(x) = a^{v_F(x)}$ and $\chi_b(x) = b^{v_F(x)}$, with $v_F(\varpi) = 1$.

Let $E = I_R(\chi_a, \chi_b)$ be the smooth normalized induction. Its elements are locally constant functions $f : G \to R$ satisfying

$$f\left(\begin{pmatrix} u & v \\ 0 & t \end{pmatrix} g\right) = |u/t|^{1/2} a^{v_F(u)} b^{v_F(t)} f(g).$$

The square root is positive real, and G acts by right translation. Restriction to K identifies its coefficient module with $R \otimes_{\mathbb{C}} V$, where

$$X = (B \cap K)\backslash K = \mathbb{P}^1(F), \qquad V = C^\infty(X; \mathbb{C}).$$

Here C^∞ means locally constant. Indeed, $G = BK$, and the inducing factor is trivial on $B \cap K$. For a fixed $g \in G$, compactness partitions K into finitely many pieces with constant diagonal valuations in an Iwasawa decomposition of kg. Right translation therefore has Laurent polynomial coefficients on every finite space of functions. This constructs the action over R and proves its compatibility with arbitrary coefficient extension. Every vector is fixed by an open subgroup of K. Define $E^{\mathrm{sw}} = I_R(\chi_b, \chi_a)$ in the same way. These are different G-actions on the same coefficient module.

Let ν be the K-invariant probability measure on X. For $N \geq 1$, let V_N be the functions pulled back from $X_N = \mathbb{P}^1(\mathcal{O}/\varpi^N\mathcal{O})$, and put $V_0 = \mathbb{C}\mathbf{1}$. Averaging on reduction fibers defines projections $P_N : V \to V_N$. Let P_0 be global averaging, and define

$$H_0 = V_0, \qquad H_n = \ker(P_{n-1}|_{V_n}) \quad (n \geq 1).$$

Thus $V = \bigoplus_{n \geq 0} H_n$ and $V_N = \bigoplus_{n=0}^{N} H_n$. Write $H_+ = \bigoplus_{n \geq 1} H_n$. The dimensions are $\dim H_0 = 1$, $\dim H_1 = q$, and $\dim H_n = (q+1)(q-1)q^{n-2}$ for $n \geq 2$. These are K-stable spaces. They need not be G-stable.

The unramified integral calculation fixes the normalization. With $w = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $n(x) = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$, it gives

$$M_z f(g) = \int_F f(wn(x)g)\, dx, \qquad c(z) = \frac{1 - q^{-1}z}{1 - z}, \qquad R_z = c(z)^{-1} M_z.$$

The integral is initially defined at complex specializations with $|z| < 1$. Its rational continuation, on each V_N, satisfies

$$R_z|_{H_0} = 1, \qquad R_z|_{H_n} = z^{n-1}\frac{z - q^{-1}}{1 - q^{-1}z} \quad (n \geq 1). \tag{118.1}$$

For clarity, the compact-picture calculation underlying this formula is

$$M_z = (1 + q^{-1})P_0 + (z - q^{-1}) \sum_{n \geq 1} z^{n-1} P_n.$$

A reduction fiber at depth n has measure $((q+1)q^{n-1})^{-1}$. For primitive rows $u, v \in O^2$, set $\rho([u],[v]) = |\det(u,v)|$. The radial integral kernel has successive values $(1+q^{-1})q^n z^n$ at distance q^{-n}. Subtracting successive values gives the displayed projection coefficients. On H_m, $m \geq 1$, only the terms $n \geq m$ remain. Their geometric sum is $(z - q^{-1})z^{m-1}/(1-z)$. On constants the sum is $c(z)$, which proves (118.1). The integral commutes with right translation and exchanges the inducing characters by the substitution $x \mapsto xd/u$ for a diagonal matrix $\mathrm{diag}(u,d)$. Its intertwining identity consequently continues as a rational identity on finite spaces of vectors.

Set

$$\alpha = 1 - q^{-1}z, \qquad \beta = z - q^{-1}, \qquad h = (1 - q^{-1}z)(1 - q^{-1}z^{-1}) = z^{-1}\alpha\beta.$$

The chosen regular map is

$$D = \alpha R_z = (1-z)M_z : E \longrightarrow E^{\mathrm{sw}}, \qquad D|_{R \otimes H_0} = \alpha, \quad D|_{R \otimes H_n} = z^{n-1}\beta. \tag{118.2}$$

The equalities with meromorphic operators mean equality of rational families. The right-hand coefficients define D over all of R, including $z = 1$. Multiplication by an additional nonunit would change the cone. Thus the scalar in (118.2) is part of the construction.

Example 118.2.1. Let $q = 2$ and $N = 1$. The three residue directions have equal measure. In their characteristic-function basis, put $J = (1)_{i,j=1}^{3}$. The projection onto constants is $P = J/3$, so

$$D_1 = \alpha P + \beta(1-P) = \frac{1}{2}\begin{pmatrix} z & 1-z & 1-z \\ 1-z & z & 1-z \\ 1-z & 1-z & z \end{pmatrix}, \qquad \det D_1 = (1 - z/2)(z - 1/2)^2.$$

At $z = 1/2$ its value is $J/4$, with kernel the two-dimensional sum-zero space. At $z = 2$ its value is $\frac{3}{2}(1-P)$, with kernel the constants. At $z = 1$ its value is $\frac{1}{2}\,\mathrm{id}$. The matrix with swapped inducing characters is $D_1(z^{-1})$, and

$$D_1(z^{-1})D_1(z) = (1 - z/2)(1 - z^{-1}/2)\,\mathrm{id}\,.$$

Indeed, both matrices preserve constants and their sum-zero complement. Multiplying the two eigenvalues on either space gives the displayed scalar. The two distinct kernels determine the two support summands of the general cone.

Proposition 118.2.2. The map D is an R-linear smooth G-intertwiner on all of E. Let $D^{\mathrm{sw}} : E^{\mathrm{sw}} \to E$ be the same regular construction with (a,b) replaced by (b,a). Then

$$D^{\mathrm{sw}}D = h\,\mathrm{id}_E, \qquad DD^{\mathrm{sw}} = h\,\mathrm{id}_{E^{\mathrm{sw}}}\,. \tag{118.3}$$

On H_0 the coefficient of D^{sw} is β/z. On H_n, $n \geq 1$, it is $z^{-n}\alpha$. After extension to $R[h^{-1}]$, the normalized operator is $R_z = \alpha^{-1}D$ and its inverse is $(1 - q^{-1}z^{-1})^{-1}D^{\mathrm{sw}}$.

Proof. Every vector has only finitely many H_n components, so (118.2) defines a map on all smooth vectors. For fixed g, N, both sides of the intertwining identity lie in some $R \otimes V_M$. Their coefficients are Laurent polynomials. The rational intertwining identity established above proves their equality in the fraction

field of R, hence in R. Replacing z by z^{-1} gives the stated opposite coefficients. Their products are h on every summand, which proves (118.3). Inverting h inverts both α and β, and the normalized composition follows. □

The ordered pair cannot be replaced by its ratio in the representation data. The central element ϖ id acts by ab. Replacing (a, b) by (ta, tb) leaves z fixed and tensors the representation with $\chi_t \circ \det$. The swap in Proposition 118.2.2 always exchanges the two characters.

118.3 The cone and its coefficient extensions

Use cochain complexes, with differentials of degree 1 and $P[1]^j = P^{j+1}$ with differential $-d_P$. The cone of the map between degree-zero representations is

$$C = \operatorname{Cone}(D) = [\, E \xrightarrow{D} E^{\mathrm{sw}} \,], \qquad C^{-1} = E, \quad C^{0} = E^{\mathrm{sw}}.$$

Let $\mathcal{D}_{\mathrm{sm}}(G, R)$ denote the derived category of smooth R-linear representations of G. Its objects are complexes, with quasi-isomorphisms inverted. The action on coefficients is trivial. The displayed complex is an object of this category. Its terms are free as R-modules.

THEOREM 118.3.1. The map D is injective over R. Writing $Q_{\mathrm{loc}} = \operatorname{coker} D$, the augmentation $C \to Q_{\mathrm{loc}}[0]$ is a G-equivariant quasi-isomorphism. As an (R, K)-module,

$$Q_{\mathrm{loc}} \cong (R/(\alpha)) \otimes H_0 \;\oplus\; (R/(\beta)) \otimes H_+. \tag{118.4}$$

Its support is the disjoint union of the two divisors $z = q$ and $z = q^{-1}$. It is annihilated by h. The two support summands in (118.4) are G-stable.

For every commutative R-algebra S, the complex

$$C_S = [\, S \otimes_R E \xrightarrow{1 \otimes D} S \otimes_R E^{\mathrm{sw}} \,]$$

computes the derived coefficient extension $S \otimes^{\mathbf{L}}_R C$. As S-modules with K-action, its cohomology is

$$H^{-1}(C_S) = (\operatorname{ann}_S \alpha) \otimes_{\mathbb{C}} H_0 \oplus (\operatorname{ann}_S \beta) \otimes_{\mathbb{C}} H_+,$$
$$H^{0}(C_S) = (S/\alpha S) \otimes_{\mathbb{C}} H_0 \oplus (S/\beta S) \otimes_{\mathbb{C}} H_+.$$

There is no cohomology in other degrees. Here α, β, z denote their images in S.

Proof. The ring R is a domain. Every eigenvalue in (118.2) is nonzero, so multiplication is injective on each free summand. The direct sum remains injective. Since z is a unit, the image on $R \otimes H_n$ is $\beta(R \otimes H_n)$ for $n \geq 1$. This proves (118.4) and the quasi-isomorphism.

The identity

$$\alpha + q^{-1}\beta = 1 - q^{-2} \tag{118.5}$$

shows that the two defining ideals are comaximal. Both quotient summands are nonzero, so their supports are exactly the stated divisors. Scalar multiplication by $h = z^{-1}\alpha\beta$ kills them. The Chinese remainder idempotents of $R/(\alpha\beta)$ commute with G. Their images therefore give the G-stable support summands. This argument makes no claim that individual H_n are G-stable.

A bounded complex of free coefficient modules computes derived extension of scalars. Tensoring (118.2) with S gives the displayed complex. The factors z^{n-1} remain units. Computing kernels and cokernels of scalar multiplication on each direct summand gives the formulas. □

For a coefficient map $S \to T$, the maps $s \otimes f \mapsto \bar{s} \otimes f$ define a G-equivariant chain map $C_S \to C_T$. They compose strictly, and $T \otimes^{\mathbf{L}}_S C_S \simeq C_T$. These are the specialization maps of this family. In particular, evaluation at a point $x = (a_0, b_0) \in (\mathbb{C}^\times)^2$ gives

$$C_x = [\, I(\chi_{a_0}, \chi_{b_0}) \xrightarrow{D_x} I(\chi_{b_0}, \chi_{a_0}) \,].$$

This statement includes nonflat coefficient maps. The global equality $\ker D = 0$ does not imply $\ker D_x = 0$. The new kernel is $\operatorname{Tor}_1^R(\mathbb{C}_x, Q_{\mathrm{loc}})$. Distinct closed points have no specified specialization map between their fibers. The maps above arise from actual coefficient homomorphisms.

118.4 The representations on the exceptional divisors

The support decomposition permits a full G-equivariant description. Let $J = C^\infty(B\backslash G; \mathbb{C})$ be unnormalized induction of the trivial character, with right translation. Its constant functions form the trivial representation. Define the Steinberg representation here by the quotient

$$\mathrm{St} = J/\mathbb{C}\mathbf{1}.$$

Only this definition is needed below. Its compact-picture coefficient space is V/H_0. For any invertible coefficient t, write $\chi_t(x) = t^{v_F(x)}$. Define free R-modules with smooth G-action by

$$L_\alpha = R \otimes (\chi_{q^{1/2}b} \circ \det), \qquad L_\beta = (R \otimes_{\mathbb{C}} \mathrm{St}) \otimes (\chi_{q^{1/2}a} \circ \det).$$

The first notation means a rank-one free module on which g acts by $(q^{1/2}b)^{v_F(\det g)}$. The second combines the indicated scalar with the usual action on St.

Theorem 118.4.1. There is an isomorphism of smooth R-linear G-representations

$$Q_{\mathrm{loc}} \cong L_\alpha/\alpha L_\alpha \oplus L_\beta/\beta L_\beta. \tag{118.6}$$

Consequently, in $\mathcal{D}_{\mathrm{sm}}(G, R)$,

$$C \simeq [\, L_\alpha \xrightarrow{\alpha} L_\alpha \,] \oplus [\, L_\beta \xrightarrow{\beta} L_\beta \,], \tag{118.7}$$

with both source terms in degree -1. The equivalence is specified by the augmentations of both sides to (118.6). It remains an equivalence after every derived coefficient extension.

Proof. On $\beta = 0$, the ordered parameters satisfy $b = qa$. The target induction has ordered pair $(b, a) = (qa, a)$. Put $\lambda = q^{1/2}a$. Its normalized induction factor is

$$|u/t|^{1/2}(qa)^{v_F(u)}a^{v_F(t)} = \lambda^{v_F(ut)}.$$

Thus $E^{\mathrm{sw}}|_{\beta=0}$ is J tensored with $\chi_\lambda \circ \det$. The image of D there is precisely the constants, since $\alpha|_{\beta=0} = 1 - q^{-2}$ is nonzero. Taking the quotient gives $L_\beta/\beta L_\beta$. This computation holds over the divisor ring $R/(\beta)$.

On $\alpha = 0$, the ordered pair is $(a, b) = (qb, b)$. The source $E|_{\alpha=0}$ is J tensored with $\chi_{q^{1/2}b} \circ \det$, by the same induction-law calculation. The target has ratio q^{-1}. The regular normalized intertwiner from this target to the source is P_0, by (118.1). Its image is the constant line in the source, and its kernel is H_+. The map D from the source to the target has image H_+, since all its positive-level coefficients are units

on this divisor. Therefore P_0 identifies the cokernel with $L_\alpha/\alpha L_\alpha$. Together with the G-stable support decomposition, this proves (118.6).

Multiplication by α on L_α and by β on L_β is injective. The two displayed complexes are free coefficient resolutions of the corresponding summands of Q_{loc}. Their augmentations and $C \to Q_{\mathrm{loc}}$ give the claimed equivalence. All terms of the two resolutions and of C are coefficient-flat. Hence these resolutions also compute every derived coefficient extension. □

PROPOSITION 118.4.2. At a complex point $x = (a_0, b_0)$, put $z_0 = a_0/b_0$. If $z_0 \notin \{q, q^{-1}\}$, then C_x is acyclic. At the two exceptional ratios, define

$$W_x = \begin{cases} \chi_{q^{1/2}b_0} \circ \det, & z_0 = q, \\ \mathrm{St} \otimes (\chi_{q^{1/2}a_0} \circ \det), & z_0 = q^{-1}. \end{cases}$$

Then $H^{-1}(C_x) \cong W_x \cong H^0(C_x)$ as smooth G-representations. In the full category $\mathcal{D}_{\mathrm{sm}}(G, \mathbb{C})$,

$$C_x \simeq W_x[1] \oplus W_x.$$

In the compact picture, the kernel and cokernel spaces are H_0 at $z_0 = q$ and H_+ at $z_0 = q^{-1}$. The cokernel means the quotient of the target by the complementary image.

Proof. Outside the two divisors every coefficient of D_x is nonzero, so the map is invertible on V. At $z_0 = q$, the H_0 coefficient is zero and all positive-level coefficients are nonzero. At $z_0 = q^{-1}$ the reverse holds. This computes kernels and cokernels as K-spaces. Applying coefficient extension to (118.7) identifies the full derived G-objects. On the vanishing divisor the relevant Koszul differential becomes zero. The other differential becomes an invertible scalar, so its complex is contractible. The surviving representation is exactly W_x. □

Remark 118.4.3 *(The central character).* The action on E, E^{sw}, C, and Q_{loc} satisfies

$$\pi(t\,\mathrm{id}) = (ab)^{v_F(t)}\,\mathrm{id} \qquad (t \in F^\times).$$

One may require this identity strictly on every object and complex. It defines a smaller abelian category of smooth representations and its derived category. The cone C, its augmentation to Q_{loc}, and its coefficient specializations belong to that category, with the coefficient character specialized when necessary.

The lifts L_α and L_β have central characters $(qb^2)^{v_F(t)}$ and $(qa^2)^{v_F(t)}$, respectively. These equal $(ab)^{v_F(t)}$ only after imposing their respective divisor equations. Thus (118.7) is an equivalence in the full category stated in Theorem 118.4.1. The universal lifts therefore do not prove a splitting inside the strict universal-central-character category. At a fixed exceptional complex point, however, the same actual complex C_x does split in the strictly specialized category. The next construction supplies a projective resolution and the two quasi-isomorphisms that establish this assertion. Universal marked-family compatibility remains a separate question.

118.5 Strict central character at a complex point

At an exceptional unramified character ratio, the regular intertwining complex has two nonzero cohomology groups. A splitting must preserve the central character on every representation and every morphism. The lattice tree supplies a projective resolution inside that category and removes the degree-two obstruction.

Use the same local field and operator as above. More explicitly, let F be a finite extension of $\mathbb{Q}_p$. Write O for its ring of integers, ϖ for a uniformizer, and q for its residue cardinality. Normalize $v_F(\varpi) = 1$ and $|\varpi| = q^{-1}$. Set $G = \mathrm{GL}_2(F)$, with center $Z = F^\times \mathrm{id}$. In this fixed-point splitting construction, representations are complex vector spaces with smooth group action: each vector has an open stabilizer. All tensor products are algebraic over $\mathbb{C}$.

Fix $a_0, b_0 \in \mathbb{C}^\times$ with $z_0 = a_0/b_0 \in \{q, q^{-1}\}$. Define $\chi_t(u) = t^{v_F(u)}$ for $t \in \mathbb{C}^\times$. Let $E_x = I(\chi_{a_0}, \chi_{b_0})$ be normalized induction from the upper triangular subgroup B. Thus its locally constant functions satisfy

$$f\left(\begin{pmatrix} u & v \\ 0 & t \end{pmatrix} g\right) = |u/t|^{1/2} \chi_{a_0}(u)\chi_{b_0}(t) f(g), \qquad (\pi(g)f)(h) = f(hg),$$

with the positive square root. Write $E_x^{\mathrm{sw}} = I(\chi_{b_0}, \chi_{a_0})$. Both representations have central character

$$\zeta(t\,\mathrm{id}) = (a_0 b_0)^{v_F(t)}.$$

Let $\mathcal{A}_\zeta$ be the abelian category of smooth complex G-representations on which every t id acts exactly by this scalar. Its morphisms are G-equivariant linear maps. Kernels and cokernels in smooth representations preserve the stated identity. Write $\mathcal{D}^b(\mathcal{A}_\zeta)$ for its bounded derived category. Use cochain complexes with differential of degree 1 and $P[1]^j = P^{j+1}$ with differential $-d_P$.

The regular intertwiner is the specialization of $D_z = (1 - q^{-1}z)R_z = (1 - z)M_z$, with additive measure $\mathrm{vol}(O) = 1$ in the raw intertwining integral M_z. The source and target are the ordered inducing pairs and their swaps. The established compact-picture formula is

$$D_z|_{H_0} = 1 - q^{-1}z, \qquad D_z|_{H_n} = z^{n-1}(z - q^{-1}) \quad (n \geq 1).$$

Here H_0 is the constant line on $\mathbb{P}^1(F)$. For $n \geq 1$, H_n consists of functions on $\mathbb{P}^1(O/\varpi^n O)$ with zero average on every fiber of reduction to level $n - 1$. At level zero, reduction means global averaging. These spaces form the algebraic direct sum of all locally constant functions. The regular map $D_x : E_x \to E_x^{\mathrm{sw}}$ is a smooth G-intertwiner. The complex under consideration is

$$C_x = [\, E_x \xrightarrow{D_x} E_x^{\mathrm{sw}} \,], \qquad C_x^{-1} = E_x, \quad C_x^0 = E_x^{\mathrm{sw}}. \tag{118.8}$$

Its kernel and cokernel are both isomorphic to

$$W_x = \begin{cases} \chi_{q^{1/2}b_0} \circ \det, & z_0 = q, \\ \mathrm{St} \otimes (\chi_{q^{1/2}a_0} \circ \det), & z_0 = q^{-1}, \end{cases} \tag{118.9}$$

where $\mathrm{St} = C^\infty(B\backslash G; \mathbb{C})/\mathbb{C}\mathbf{1}$ uses unnormalized induction of the trivial character. The operator and these kernel and quotient identifications are the input to the splitting problem. The following construction applies to every two-term complex in $\mathcal{A}_\zeta$.

Removing the fixed central character

Set

$$\lambda = \begin{cases} q^{1/2}b_0, & z_0 = q, \\ q^{1/2}a_0, & z_0 = q^{-1}. \end{cases} \qquad \eta = \chi_\lambda \circ \det.$$

In either case $\lambda^2 = a_0 b_0$. The character η is smooth, and its restriction to Z equals ζ. Let $H = G/Z = \mathrm{PGL}_2(F)$, with the quotient topology.

LEMMA 118.5.1. Twisting by η^{-1} and descending through $G \longrightarrow H$ gives an exact equivalence

$$\mathcal{A}_\zeta \longrightarrow \mathrm{Rep}_{\mathrm{sm},\mathbb{C}}(H).$$

Its inverse inflates an H-representation and twists by η. Under this equivalence, W_x becomes the trivial representation when $z_0 = q$ and the Steinberg quotient when $z_0 = q^{-1}$.

Proof. For $V \in \mathcal{A}_\zeta$, every central element acts trivially on $V \otimes \eta^{-1}$. Its action therefore factors uniquely through H. If a vector has open stabilizer U in G, the image of UZ is open in H and fixes that vector. Thus the descended action is smooth. Conversely, inflation of a smooth H-action is smooth, and the twist gives central character ζ. The two constructions are inverse on objects and morphisms. Their underlying vector-space functors are exact. Formula (118.9) gives the final assertion. □

It remains to construct projective resolutions for smooth representations of H. The quotient by the full center is essential: the stabilizers of lattice classes in H are compact.

The lattice tree

An O-lattice in F^2 is the O-span of an F-basis. Vertices are homothety classes $[L]$ of such lattices. Two classes are adjacent if they have representatives satisfying

$$\varpi L \subset M \subset L, \qquad \dim_{O/\varpi O}(M/\varpi L) = 1.$$

This is a symmetric relation on classes. Indeed, the reverse inclusion can be represented by $\varpi M \subset \varpi L \subset M$. The group H acts on this graph by its action on lattice classes.

LEMMA 118.5.2. This graph is a tree. Each vertex has $q + 1$ neighbors.

Proof. The neighbors of $[L]$ are exactly the lines in $L/\varpi L$, so there are $q + 1$. Fix the root $[L_0]$, where $L_0 = Oe_1 \oplus Oe_2$. Each class has a unique representative $L \subset L_0$ with $L \not\subset \varpi L_0$. Multiplication by a unit does not change a lattice, which proves uniqueness. Choose a primitive vector $u \in L$ and complete it to a basis u, v of L_0. The image of L in L_0/Ou is an ideal $\varpi^n O$. Subtracting an Ou component from a lift of its generator gives

$$L = Ou \oplus O\varpi^n v, \qquad n \geq 0.$$

The integer n is the valuation of the determinant of this normalized lattice relative to L_0. It is independent of the chosen basis and will be its height. Height zero gives the root.

For $n > 0$, the line generated by $\varpi^n v$ in $L/\varpi L$ gives the neighboring lattice $O\varpi u \oplus O\varpi^n v$. After normalization, its height is $n - 1$. Every other line is generated by $u + c\varpi^n v$ for one residue $c \in O/\varpi O$. Its neighboring lattice is

$$O(u + c\varpi^n v) \oplus O\varpi^{n+1} v,$$

which is normalized and has height $n + 1$. Every neighbor of the root has height one. Thus every positive-height vertex has exactly one neighbor of lower height. Repeated lower-height steps connect it to the root. A cycle would have a maximal-height vertex with two neighbors of lower height, which is impossible. □

An element of H can exchange the endpoints of an edge. Insert a midpoint in each edge to obtain the barycentric subdivision T. Orient each new edge from its lattice vertex to its midpoint. The H-action preserves these two vertex types and therefore preserves the chosen orientations.

Let $v_0 = [L_0]$ and $v_1 = [Oe_1 \oplus O\varpi e_2]$. Let m_0 be the midpoint between them. The stabilizer K of v_0 is the image of $\mathrm{GL}_2(O)$ in H. It is compact and open. The stabilizer I of the incidence (v_0, m_0)

fixes both v_0 and v_1. In K, it is the inverse image of the stabilizer of a line modulo ϖ, hence compact and open. The class of

$$s = \begin{pmatrix} 0 & 1 \\ \varpi & 0 \end{pmatrix}$$

exchanges v_0 and v_1 and satisfies $s^2 = 1$ in H. The midpoint stabilizer is $J = I \sqcup sI$ and is compact and open. The action is transitive on lattice vertices, on midpoints, and on the oriented edges of T. The first assertion follows by sending one lattice basis to another. Transitivity on incident pairs follows because K acts transitively on lines modulo ϖ.

Example 118.5.3 *(The first ball when $q = 2$).* The first ball about v_0 is a star with three leaves v_1, v_2, v_3. Write m_i for the midpoint of $v_0 v_i$. The subdivision has seven vertices and six oriented edges $e_i = (v_0, m_i)$ and $f_i = (v_i, m_i)$, for $1 \le i \le 3$. Its boundary map and augmentation are

$$\partial e_i = [m_i] - [v_0], \qquad \partial f_i = [m_i] - [v_i], \qquad \epsilon[v] = 1.$$

Define a linear map h from vertices to edge chains by

$$h[v_0] = 0, \qquad h[m_i] = e_i, \qquad h[v_i] = e_i - f_i.$$

Then

$$\partial h = \mathrm{id} - [v_0]\epsilon, \qquad h\partial = \mathrm{id}\,.$$

Both identities follow by applying the displayed formulas to each basis vector. Thus ∂ has rank six and its image is $\ker \epsilon$. This finite ball is not H-stable. The same finite-path calculation proves exactness for arbitrary finite chains in the entire tree.

Projective cellular chains

For a compact open subgroup $U \subset H$, let $\mathrm{Rep}_{\mathrm{sm},\mathbb{C}}(U)$ denote its smooth complex representations. An object is projective if every morphism from it to a quotient lifts through the quotient map. The following property depends on the coefficient field $\mathbb{C}$.

Lemma 118.5.4. Every smooth complex representation of U is projective in $\mathrm{Rep}_{\mathrm{sm},\mathbb{C}}(U)$.

Proof. The orbit of any vector under U is finite because its stabilizer is open and U is compact. The orbit spans a finite-dimensional U-submodule. A basis of this submodule has a common open stabilizer. Its normal core is an open normal subgroup, so the action factors through a finite quotient of U. For an invariant subspace, average a linear projection over the finite quotient to obtain an equivariant projection. Its kernel is an invariant complement. Induction on dimension therefore decomposes this submodule into irreducibles. Hence every smooth U-module is a sum of finite-dimensional irreducible submodules. Choose a maximal direct sum of such submodules. If it does not contain the whole module, a finite-dimensional submodule contributes a simple summand outside it. That simple summand has zero intersection with the maximal direct sum, a contradiction. Thus the whole module is a direct sum of finite-dimensional irreducibles.

Each such irreducible τ is projective. To verify the lifting property explicitly, let $Y \twoheadrightarrow Z$ be an epimorphism and let $f : \tau \to Z$ be U-equivariant. Choose lifts in Y of the images of a basis of τ

and extend them to a linear map $\ell : \tau \to Y$. The U-orbits of these finitely many lifts span a finite-dimensional submodule Y_0. Choose an open normal subgroup $N \subset U$ acting trivially on both Y_0 and τ. Then

$$\widetilde{f}(v) = \frac{1}{|U/N|} \sum_{u \in U/N} u\, \ell(u^{-1}v)$$

is well-defined, U-equivariant, and lifts f. Only finite-dimensional smooth data are averaged. Finally, morphisms out of a direct sum can be lifted on its summands and then assembled. A direct sum of projectives is therefore projective. □

For $M \in \mathrm{Rep}_{\mathrm{sm},\mathbb{C}}(U)$, define compact induction $\text{c-}\mathrm{Ind}_U^H M$ by finite sums of symbols $[g, m]$, linear in m, with relations

$$[gu, m] = [g, um] \qquad (g \in H,\ u \in U).$$

The group acts by $h[g, m] = [hg, m]$. This is the algebraic finite-support model on H/U, valid because U is open. It is smooth: the stabilizer of $[g, m]$ contains $gU_m g^{-1}$, where U_m is an open stabilizer of m in U. A finite sum has the intersection of finitely many such open stabilizers.

The map determined by $[g, m] \mapsto gf(m)$ gives the adjunction

$$\mathrm{Hom}_H(\text{c-}\mathrm{Ind}_U^H M, V) \cong \mathrm{Hom}_U(M, V|_U).$$

Its inverse restricts a morphism to the symbols $[1, m]$. Restriction is exact, so compact induction takes projectives to projectives. Lemma 118.5.4 applies to every compact open cell stabilizer of T.

For a discrete set Y, write $C_c(Y; \mathbb{C})$ for its finitely supported functions, with basis vectors denoted $[y]$. Let T_0 and T_1 be the vertices and the chosen oriented edges of T. For a smooth H-representation V, define

$$P_1(V) = C_c(T_1; \mathbb{C}) \otimes V, \qquad P_0(V) = C_c(T_0; \mathbb{C}) \otimes V,$$

with diagonal action $g([c] \otimes v) = [gc] \otimes gv$. For an oriented edge $e = (u, m)$, define

$$\partial([e] \otimes v) = ([m] - [u]) \otimes v, \qquad \epsilon([u] \otimes v) = v.$$

Both maps commute with H, and $\epsilon\partial = 0$.

THEOREM 118.5.5. For every smooth complex H-representation V, the sequence

$$0 \longrightarrow P_1(V) \xrightarrow{\partial} P_0(V) \xrightarrow{\epsilon} V \longrightarrow 0 \tag{118.10}$$

is exact and its first two terms are projective smooth H-representations. Consequently,

$$\mathrm{Ext}^j_{\mathrm{Rep}_{\mathrm{sm},\mathbb{C}}(H)}(V, U) = 0 \qquad (j \geq 2)$$

for all smooth V, U.

Proof. Fix a root vertex o of T. For each vertex v, let p_v be the finite edge chain along the unique path from o to v, with coefficient -1 whenever the path opposes the chosen edge orientation. Then $\partial p_v = [v] - [o]$. The map $[v] \otimes w \mapsto p_v \otimes w$ gives $\partial h = \mathrm{id} - ([o] \otimes -)\epsilon$. Thus every chain of augmentation zero is a boundary. The augmentation is surjective because $w = \epsilon([o] \otimes w)$. A nonzero finite-support edge cycle would have a leaf in the finite subtree spanned by its support. At that leaf, its boundary equals the

nonzero coefficient of its unique incident supported edge, up to sign. This is impossible, so ∂ is injective. These arguments also apply to vector coefficients and use only finite sums. The root-path contraction proves exactness and is not claimed equivariant.

For a chosen cell c with stabilizer U, the map

$$\mathrm{c\text{-}Ind}_U^H(V|_U) \longrightarrow C_c(Hc;\mathbb{C}) \otimes V, \qquad [g,v] \longmapsto [gc] \otimes gv$$

is an equivariant isomorphism. The relation $[gu,v] = [g,uv]$ verifies that it is well-defined. Its inverse sends $[gc] \otimes w$ to $[g, g^{-1}w]$ and is independent of the representative g. The three cell orbits therefore give

$$P_1(V) \cong \mathrm{c\text{-}Ind}_I^H(V|_I), \qquad P_0(V) \cong \mathrm{c\text{-}Ind}_K^H(V|_K) \oplus \mathrm{c\text{-}Ind}_J^H(V|_J).$$

They are smooth projectives by Lemma 118.5.4 and the compact-induction adjunction. Applying $\mathrm{Hom}(-,U)$ to this projective resolution computes Ext and leaves no terms in degrees $j \geq 2$. □

The constant-coefficient resolution above needs no finite generation or admissibility hypothesis. Its projective terms can have infinite multiplicity. Cellular resolutions with coefficient systems of compact-subgroup invariants are developed in Schneider–Stuhler, *Representation theory and sheaves on the Bruhat–Tits building*, Section II.2, pp. 121–122, and Theorem II.3.1, p. 123. Here the rank-one tree and constant coefficients give the stated resolution directly.

The obstruction and the splitting maps

Lemma 118.5.1 transports the resolution and projectivity into $\mathcal{A}_\zeta$. For $V \in \mathcal{A}_\zeta$, apply (118.10) to its descended twist $V \otimes \eta^{-1}$, then inflate and tensor every term and map with η. This gives a projective resolution

$$0 \longrightarrow P_1^\zeta(V) \xrightarrow{d} P_0^\zeta(V) \xrightarrow{\epsilon} V \longrightarrow 0$$

inside $\mathcal{A}_\zeta$. In particular, $\mathrm{Ext}^j_{\mathcal{A}_\zeta}(V,U) = 0$ for all $j \geq 2$. Every central element acts by ζ on both projective terms.

Theorem 118.5.6. Let $A \xrightarrow{D} B$ be a morphism in $\mathcal{A}_\zeta$. Place A in degree -1 and B in degree 0, and set $U = \ker D$, $V = \mathrm{coker}\, D$. There is an isomorphism

$$[\, A \xrightarrow{D} B \,] \simeq U[1] \oplus V \qquad \text{in } \mathcal{D}^b(\mathcal{A}_\zeta).$$

It can be represented by two explicit quasi-isomorphisms through the cellular projective resolution of V.

Proof. Write $P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} V$ for the resolution just constructed. Let $p : B \twoheadrightarrow V$ and $i : U \hookrightarrow A$ be the quotient and kernel maps. Since P_0 is projective, choose a morphism $f_0 : P_0 \to B$ with $pf_0 = \epsilon$. The identity $pf_0 d = \epsilon d = 0$ shows that $f_0 d$ takes values in $\mathrm{im}\, D$. Projectivity of P_1 lifts that map through the epimorphism $A \twoheadrightarrow \mathrm{im}\, D$. Thus there is $f_1 : P_1 \to A$ satisfying

$$Df_1 = f_0 d. \tag{118.11}$$

All objects, quotient maps, and lifts lie in $\mathcal{A}_\zeta$.

Let $P = [P_1 \xrightarrow{d} P_0]$, in degrees -1, 0. Equation (118.11) defines a chain map

$$\Phi : U[1] \oplus P \longrightarrow [\, A \xrightarrow{D} B \,], \qquad \Phi^{-1}(u,v) = i(u) + f_1(v), \quad \Phi^0 = f_0.$$

The superscripts denote cochain degrees. Since d is injective, $H^{-1}(U[1] \oplus P) = U$ and $H^{0}(U[1] \oplus P) = V$. The map Φ induces the identity on these groups, using $pf_{0} = \epsilon$. It is therefore a quasi-isomorphism. The augmentation of P gives another quasi-isomorphism

$$\Psi : U[1] \oplus P \longrightarrow U[1] \oplus V, \qquad \Psi^{-1}(u, v) = u, \quad \Psi^{0} = \epsilon.$$

The roof

$$[\, A \xrightarrow{D} B \,] \xleftarrow{\Phi} U[1] \oplus P \xrightarrow{\Psi} U[1] \oplus V$$

proves the asserted derived isomorphism. The compact averaging construction in Lemma 118.5.4, followed by compact induction, constructs the required lifts after choices of irreducible summands and linear lifts. No averaging over the noncompact group is used. □

The associated obstruction is the class of the exact two-extension

$$0 \longrightarrow U \xrightarrow{i} A \xrightarrow{D} B \xrightarrow{p} V \longrightarrow 0$$

in $\operatorname{Ext}^{2}_{\mathcal{A}_{\zeta}}(V, U)$. Equivalently, it is the connecting morphism $V \to U[2]$ in the cohomological truncation triangle. Theorem 118.5.5 makes this group zero, and the roof above realizes the splitting by actual maps. The construction proves existence and depends on the choices of lifts. No canonical splitting or compatibility between different parameters is asserted.

PROPOSITION 118.5.7 *(The exceptional fibers).* For the exceptional pair fixed in (118.8), there is an isomorphism

$$C_{x} \simeq W_{x}[1] \oplus W_{x} \qquad \text{in } \mathcal{D}^{b}(\mathcal{A}_{\zeta}).$$

Its degree-two obstruction vanishes in the strictly fixed-central-character category.

Proof. The kernel and cokernel identifications (118.9) are morphisms inside $\mathcal{A}_{\zeta}$. Apply Theorem 118.5.6 to the actual map D_{x}. The resulting roof uses the projective resolution twisted by the stated η, so every term has central character ζ. □

This is a pointwise statement over $\mathbb{C}$. The compact projectivity proof uses division by finite group orders and does not establish a theorem over arbitrary coefficient rings. The character η is fixed at the chosen complex point. A splitting with the universal central character over the ordered-character Laurent ring, a canonical Weyl-compatible choice of pointwise splittings, and a comparison with global Eisenstein categories require separate constructions.

118.6 NORMAL EQUATIONS AND SPECIALIZATION MAPS

The opposite operator provides a direct comparison of these two cohomology groups. Put $T = R/(h)$. Equation (118.3) gives a well-defined map

$$\overline{D^{\mathrm{sw}}} : \operatorname{coker}(D_{T}) \longrightarrow \ker(D_{T}), \qquad [y] \longmapsto D^{\mathrm{sw}}_{T}\, y.$$

It is an isomorphism of smooth T-linear G-representations. To prove this, decompose T by the two comaximal ideals. On $\alpha = 0$, its surviving coefficient is $\beta/z = 1 - q^{-2}$ on H_{0}. On $\beta = 0$, its surviving coefficients are $z^{-n}\alpha$ on $H_{n}, n \geq 1$. All are units in the corresponding divisor ring. The same isomorphism holds after further coefficient extension from T.

There is also a comparison determined by a chosen equation of one divisor. For $t = \alpha$ or $t = \beta$, let $\overline{R} = R/(t)$. For $x \in \ker D_{\overline{R}}$, choose a lift $\widetilde{x} \in E$. The vector $D\widetilde{x}$ is divisible by t in the free module E^{sw}. Define

$$\delta_t : \ker D_{\overline{R}} \longrightarrow \operatorname{coker} D_{\overline{R}}, \qquad x \longmapsto [\, D\widetilde{x}/t \,].$$

Changing the lift by tu changes the quotient by Du, so the map is well-defined. Equivariance follows by applying $D\pi(g) = \pi^{\mathrm{sw}}(g)D$ to the chosen lift. This is the connecting map for the coefficient sequence $0 \to R/(t) \xrightarrow{t} R/(t^2) \to R/(t) \to 0$ applied to C. It also identifies the kernel calculated from the free resolution of Q_{loc} with its t-torsion using the resolution $[R \xrightarrow{t} R]$ of $R/(t)$.

The map δ_α has coefficient 1 on H_0. The map δ_β has coefficient z^{n-1} on H_n, $n \geq 1$, evaluated on $\beta = 0$. Thus both are isomorphisms. They are compatible with subsequent extension along the respective divisor. They satisfy

$$\delta_t \overline{D^{\mathrm{sw}}} = (h/t)|_{t=0}\ \mathrm{id}, \qquad (h/\alpha)|_{\alpha=0} = 1 - q^{-2}, \quad (h/\beta)|_{\beta=0} = q(1 - q^{-2}).$$

Replacing the local equation t by ut, with u a unit, multiplies δ_t by u^{-1} on the divisor. The normal equation is therefore part of this specialization comparison.

118.7 Finite levels and matrix factorizations

The finite spaces V_N retain the integral's compact-level refinement. Set $E_N = R \otimes V_N$ and $E_N^{\mathrm{sw}} = R \otimes V_N$, with their respective induced compact actions. Restriction gives $D_N : E_N \to E_N^{\mathrm{sw}}$ and $D_N^{\mathrm{sw}} : E_N^{\mathrm{sw}} \to E_N$. Inclusions $V_N \hookrightarrow V_{N+1}$ and averaging maps $P_N : V_{N+1} \to V_N$ commute with both operators because they preserve each H_n. They therefore induce maps of the cones and of their coefficient extensions. The inclusions give

$$C = \operatorname*{colim}_N C_N, \qquad C_N = [E_N \xrightarrow{D_N} E_N^{\mathrm{sw}}].$$

Filtered colimits of modules are exact, so kernels, cokernels, and the displayed coefficient extensions commute with this colimit.

For $N \geq 1$, let $K_N = 1 + \varpi^N M_2(\mathcal{O})$, and set $K_0 = K$. Then $E^{K_N} = E_N$ and $(E^{\mathrm{sw}})^{K_N} = E_N^{\mathrm{sw}}$. Compact averaging is defined over every $\mathbb{C}$-algebra and is exact on smooth representations. Indeed, any finite set of vectors is fixed by an open subgroup, and averaging reduces to a finite sum with invertible denominator. It follows that $C_N = C^{K_N}$ and $\operatorname{coker} D_N = Q_{\mathrm{loc}}^{K_N}$. Full G-translation can move E_N into a larger E_M. The finite complexes carry K-actions, whereas their colimit carries the full G-action.

A finite-free matrix factorization of $h \in R$ consists of finite-free modules M_0, M_1 and maps $d_0 : M_0 \to M_1$, $d_1 : M_1 \to M_0$ satisfying $d_1 d_0 = h\,\mathrm{id}$ and $d_0 d_1 = h\,\mathrm{id}$. Thus (D_N, D_N^{sw}) is a finite-free K-equivariant matrix factorization of h. Its rank is $\dim V_N$ in each parity. The full pair (D, D^{sw}) has infinite-rank free terms.

Proposition 118.7.1. Each finite-level matrix factorization (D_N, D_N^{sw}) is contractible through K-equivariant maps. Its cokernel is a projective $R/(h)$-module. The ordinary two-term cone C_N is nevertheless nonzero in the derived category for every $N \geq 0$.

Proof. Put $c = 1 - q^{-2}$. Write $d_0 = D_N$, $d_1 = D_N^{\mathrm{sw}}$. A contraction consists of maps $s_0 : M_0 \to M_1$ and $s_1 : M_1 \to M_0$ with $ds + sd = \mathrm{id}$ on the two-parity module. On H_0 take

$$s_1 = c^{-1}, \qquad s_0 = q^{-1} z / c.$$

Then $s_1 d_0 + d_1 s_0 = (\alpha + q^{-1}\beta)/c = 1$. On H_n, $n \geq 1$, take

$$s_1 = q^{-1} z^{-(n-1)}/c, \qquad s_0 = z^n/c.$$

Here $s_1 d_0 + d_1 s_0 = (q^{-1}\beta + \alpha)/c = 1$ as well. All coefficients commute, so the identity also holds on the other parity. These are Laurent coefficients and preserve the K-stable decomposition.

The ring $R/(h)$ is $R/(\alpha) \times R/(\beta)$ by (118.5). The cokernel is free of rank one on the first component and rank $\dim V_N - 1$ on the second. It is therefore projective over their product. As an R-module it is nonzero, since the first component is nonzero. This cokernel is $H^0(C_N)$, which proves the last assertion. □

The contraction above is asserted only for the coefficient modules with K-action. Its diagonal formulas do not establish a contraction commuting with the full G-action. A matrix factorization uses both opposite maps and satisfies $d^2 = h$. The ordinary cone uses only D and has the nonzero cohomology (118.4). These are distinct homological constructions.

The regular intertwiner therefore defines a smooth equivariant degeneration complex on the entire ordered-character torus. Its cohomology consists of a character family on one divisor and a Steinberg quotient family on the other. Its derived fibers and their specialization comparisons are given above. A category comparison for global Eisenstein representations still requires a global target and a functor whose derived mapping objects and essential image are determined. The local degeneration complex supplies no such equivalence by itself.

118.8 Local Yoneda classes and finite exterior products

A local nonsplit principal series defines a degree-one extension between its Steinberg quotient and its constant subrepresentation. At two distinct finite places, the exterior product of these extensions has degree two. An infinite restricted tensor introduces a second condition: its distinguished local vectors must be fixed by the prescribed compact subgroups at almost every place. These two requirements distinguish a finite exterior product of Steinberg representations from an adelic extension of principal series.

The local sequence and its invariant vectors

Fix the coefficient field $\mathbb{C}$. For a finite prime p, put $G_p = GL_2(\mathbb{Q}_p)$, $K_p = GL_2(\mathbb{Z}_p)$, and let B_p be the upper triangular Borel subgroup. Give these groups their usual p-adic topologies. Write $X_p = B_p \backslash G_p \simeq \mathbb{P}^1(\mathbb{Q}_p)$, with its compact quotient topology and right G_p-action. The identification sends a matrix to the line spanned by its bottom row. Let

$$E_p = C^{\mathrm{lc}}(X_p, \mathbb{C}), \qquad (\rho_p(g)f)(x) = f(xg).$$

Here C^{lc} means locally constant functions, without an analytic completion. This is unnormalized smooth induction of the trivial character of B_p. With normalized induction and $|p|_p = p^{-1}$, its ordered inducing characters are $|\cdot|_p^{-1/2}$ and $|\cdot|_p^{1/2}$. Thus, if symmetric induction is written with characters $(|\cdot|_p^{s/2}, |\cdot|_p^{-s/2})$, this sequence occurs at $s = -1$.

The constants give the exact sequence

$$0 \longrightarrow \mathbb{C} \xrightarrow{i_p} E_p \longrightarrow \mathrm{St}_p \longrightarrow 0, \qquad \mathrm{St}_p = E_p/\mathbb{C}\mathbf{1}_p. \tag{118.12}$$

This is the usual Steinberg quotient in rank one (Casselman, *Introduction to the Theory of Admissible Representations of p-Adic Reductive Groups*, Section 8.1, p. 70). Every term has trivial central character. We use $\mathrm{Rep}_{\mathbb{C}}^{\mathrm{sm}}(G_p)$ for the abelian category of all smooth complex representations. Its full subcategory on which the center acts trivially is identified with $\mathrm{Rep}_{\mathbb{C}}^{\mathrm{sm}}(PGL_2(\mathbb{Q}_p))$. Extension groups below specify which of these categories is used.

Lemma 118.8.1. The sequence (118.12) is nonsplit. Moreover,

$$\mathrm{Hom}_{G_p}(E_p, \mathbb{C}) = 0, \qquad \mathrm{Hom}_{G_p}(\mathrm{St}_p, \mathbb{C}) = 0, \qquad \mathrm{St}_p^{K_p} = 0.$$

The same statements hold after passage to $PGL_2(\mathbb{Q}_p)$.

Proof. Consider a G_p-invariant linear functional $\ell : E_p \to \mathbb{C}$. Projective p-adic balls are translates of one another under G_p: affine transformations relate the balls in $\mathbb{Q}_p$, and inversion also relates them to the balls containing infinity. A ball is a disjoint union of p smaller balls. Consequently the common value c of ℓ on ball indicators satisfies $c = pc$, so $c = 0$. Locally constant functions on X_p are finite linear combinations of these indicators, since X_p is compact and the balls form a clopen basis. Hence $\ell = 0$. A splitting of (118.12) would provide such a functional with $\ell(\mathbf{1}_p) = 1$. A morphism $\mathrm{St}_p \to \mathbb{C}$ would pull back to a morphism $E_p \to \mathbb{C}$, proving the first two assertions.

Every line in $\mathbb{Q}_p^2$ has a primitive integral representative that extends to an integral basis. Thus K_p acts transitively on X_p and $E_p^{K_p} = \mathbb{C}\mathbf{1}_p$. Invariants under a compact open subgroup are exact for smooth complex representations. Indeed, normalized Haar averaging of a smooth vector is a finite sum, and averaging a lift of an invariant vector gives an invariant lift. Applying invariants to (118.12) gives $\mathrm{St}_p^{K_p} = 0$. This averaging argument is also Casselman, Proposition 2.1.7, p. 20. All actions factor through $PGL_2(\mathbb{Q}_p)$, and the same proof applies there. □

Write α_p for the Yoneda class of (118.12). It is a specified nonzero element of $\mathrm{Ext}^1(\mathrm{St}_p, \mathbb{C})$. No residue operator is needed to define it, and no dimension assertion about that group is required below.

Two finite places and the exterior degree

Let $p \neq q$ be finite primes. External tensor products over $\mathbb{C}$ are written $\boxtimes$ and carry the separate actions of the two groups. Tensoring the two local sequences gives the exact sequence

$$\begin{aligned} 0 \longrightarrow \mathbb{C} \boxtimes \mathbb{C} \xrightarrow{d_2} (E_p \boxtimes \mathbb{C}) \oplus (\mathbb{C} \boxtimes E_q) \xrightarrow{d_1} E_p \boxtimes E_q \\ \longrightarrow \mathrm{St}_p \boxtimes \mathrm{St}_q \longrightarrow 0, \end{aligned} \tag{118.13}$$

where

$$d_2 = (i_p \boxtimes 1, -1 \boxtimes i_q), \qquad d_1(x, y) = (1 \boxtimes i_q)x + (i_p \boxtimes 1)y.$$

Unlike the cochain indexing of the local cone, the following totals use homological degrees and differentials of degree -1. These signs are the tensor differential for homological complexes, $\partial(x \otimes y) = \partial x \otimes y + (-1)^{\deg x} x \otimes \partial y$. In particular $d_1 d_2 = 0$. The two complexes $[\mathbb{C} \to E_p]$ and $[\mathbb{C} \to E_q]$, in homological degrees 1, 0, have homology only in degree zero, with values St_p and St_q. Tensoring over a field preserves this assertion. It proves the exactness of (118.13).

The sequence (118.13) is a Yoneda two-extension, not a short exact sequence with these endpoints. Its class is the exterior Yoneda product of α_p and α_q, with the displayed tensor sign convention. The following argument proves that this class is nonzero without replacing either local representation by a finite-dimensional model.

PROPOSITION 118.8.2. Let S be a nonempty finite set of finite primes, $n = |S|$, and $\mathrm{St}_S = \boxtimes_{p\in S} \mathrm{St}_p$. Put $G_S = \prod_{p\in S} G_p$. In $\mathrm{Rep}^{\mathrm{sm}}_{\mathbb{C}}(G_S)$ one has

$$\mathrm{Ext}^r_{G_S}(\mathrm{St}_S, \mathbb{C}) = 0 \quad (0 \le r < n).$$

The exterior product of the classes α_p is nonzero in $\mathrm{Ext}^n_{G_S}(\mathrm{St}_S, \mathbb{C})$. It is represented by the total tensor of the n sequences (118.12). Both assertions also hold in the category where each local center acts trivially, namely $\mathrm{Rep}^{\mathrm{sm}}_{\mathbb{C}}(\prod_{p\in S} PGL_2(\mathbb{Q}_p))$.

Proof. For either of the groups G_p and $PGL_2(\mathbb{Q}_p)$, compact induction $\text{c-Ind}^G_K \mathbb{C}$ from a compact open subgroup is projective. To see this, a G-map from this compact induction to V is determined by the image of the indicator of K, and that image can be any element of V^K. Thus its Hom functor is $V \mapsto V^K$, which is exact by averaging. Every smooth vector is fixed by a compact open subgroup. Taking one such compact induction for each vector therefore gives enough projectives, and resolutions can be chosen with terms that are sums of these compact inductions.

Choose such a resolution $P_{p,\bullet} \to \mathrm{St}_p$, with homological indices at least zero. Products of compact open subgroups are compact open, and

$$\left(\text{c-Ind}^G_K \mathbb{C}\right) \boxtimes \left(\text{c-Ind}^H_L \mathbb{C}\right) \simeq \text{c-Ind}^{G\times H}_{K\times L} \mathbb{C}.$$

Both sides are finite-support functions on the product of the relevant discrete coset spaces. Hence the total external tensor of the chosen resolutions is a projective resolution of St_S.

Let $C_{p,\bullet} = (P_{p,\bullet})_{G_p}$ be its coinvariant complex (or use $PGL_2(\mathbb{Q}_p)$ in the fixed-central-character category). For any representation V, a map $V \to \mathbb{C}$ factors uniquely through V_G. It follows that

$$\mathrm{Ext}^r_G(\mathrm{St}_p, \mathbb{C}) \simeq \mathrm{Hom}_{\mathbb{C}}(H_r(C_{p,\bullet}), \mathbb{C}). \tag{118.14}$$

Here the algebraic dual is exact over the field $\mathbb{C}$; no topological or continuous dual is used. Coinvariants of an external tensor for a product group are the tensor of the coinvariants. Consequently the same argument identifies the product-group Ext with the dual homology of $\bigotimes_{p\in S} C_{p,\bullet}$.

Lemma 118.8.1 and (118.14) imply $H_0(C_{p,\bullet}) = 0$. The Künneth formula for bounded-below complexes of vector spaces gives

$$H_r\left(\bigotimes_{p\in S} C_{p,\bullet}\right) \simeq \bigoplus_{\substack{(r_p)\in \mathbb{Z}^S_{\ge 0} \\ \sum_p r_p = r}} \bigotimes_{p\in S} H_{r_p}(C_{p,\bullet}).$$

For completeness, one can split each vector-space complex as its homology with zero differential plus a contractible complex. Tensoring these splittings proves the formula. For $r < n$, every summand has some $r_p = 0$ and is zero. This proves the vanishing assertion.

Each nonzero class α_p is a nonzero functional on $H_1(C_{p,\bullet})$. Choose a homology element on which it is nonzero. The product functional is nonzero on the tensor of these elements, which lies in degree n. It represents the exterior Yoneda class: a projective resolution maps to $[\mathbb{C} \to E_p]$ by first lifting its augmentation to E_p, then factoring its degree-one boundary through $\mathbb{C}$. The resulting degree-one cocycle represents α_p. Tensoring these chain maps gives exactly the total extension and the product cocycle, with the tensor signs already specified. Thus the exterior class is nonzero.

This proof uses genuinely compact open subgroups even for GL_2. It does not treat compact-mod-center subgroups as compact, and it makes no assertion about vanishing above degree n in the unrestricted GL_2 category. □

118.9 The restricted tensor condition

Let

$$G_f = GL_2(\mathbb{A}_f) = \prod_p{}'(G_p, K_p).$$

Its restricted-product topology has neighborhoods of the identity that contain $\prod_{p\notin S} K_p$ for some finite S. A usual algebraic restricted tensor representation uses nonzero distinguished vectors $e_p \in V_p^{K_p}$ for all but finitely many p. Its vectors are finite linear combinations of tensors whose factors equal e_p almost everywhere. For $g = (g_p) \in G_f$, both $g_p \in K_p$ and the distinguished factor condition hold almost everywhere, making the coordinate action well-defined.

Proposition 118.9.1. The local representations St_p do not admit a nonzero restricted tensor representation with the standard hyperspecial pointings at almost every finite prime. More generally, a smooth G_f-representation V with factorizations $V \simeq \mathrm{St}_p \otimes W_p$ as G_p-representations for every finite p, with G_p acting only on the first factor, is zero.

Proof. The required pointing space is zero at every prime by Lemma 118.8.1. For the second assertion, a vector in a smooth G_f-representation is fixed by an open subgroup. That subgroup contains K_p for every p outside a finite set. But the stipulated factorization implies

$$V^{K_p} = \mathrm{St}_p^{K_p} \otimes W_p = 0.$$

This equality follows from averaging on the first tensor factor, or from writing any tensor as a finite sum with linearly independent second factors. Thus the vector is zero. □

Choosing Iwahori-fixed vectors at infinitely many places does not supply these hyperspecial pointings. The restricted product relative to those Iwahori subgroups is a different subgroup and topology. No passage to an unspecified completed tensor product is part of this construction.

For a finite set S, a genuine smooth adelic representation is

$$\mathrm{St}_S^{(f)} = \left(\boxtimes_{p\in S} \mathrm{St}_p\right) \boxtimes \left(\bigotimes_{p\notin S}{}' \mathbb{C}\right),$$

where each omitted place uses the trivial representation and vector 1. It is inflation from G_S under the continuous projection $G_f \to G_S$. The extension of Proposition 118.8.2 inflates with it and remains nonzero: restriction along the inclusion $G_S \hookrightarrow G_f$ is exact and recovers the original Yoneda class. This representation has Steinberg factors precisely at the specified finite set. It is not an all-place Steinberg representation.

118.10 An adelic principal-series extension

The local principal series E_p do have compatible distinguished vectors, namely the constant functions $\mathbf{1}_p$. These vectors are fixed by all of G_p. For finite S, write $E_S = \boxtimes_{p\in S} E_p$, inflated to G_f with trivial factors outside S. If $S \subset T$, define

$$E_S \longrightarrow E_T, \qquad x \longmapsto x \otimes \bigotimes_{p\in T\setminus S} \mathbf{1}_p.$$

The superscript res below means restricted tensor product; it does not name an automorphic residual spectrum. These maps are injective and G_f-equivariant. The canonical associativity and symmetry identifications make their composites agree for $S \subset T \subset U$. Put $E^{\mathrm{res}} = \operatorname{colim}_S E_S$. This is a smooth representation: a vector lies in some E_S, is smooth at the finitely many primes of S, and is fixed by the full groups outside S. Its central character is trivial.

PROPOSITION 118.10.1. Let $X = \prod_p \mathbb{P}^1(\mathbb{Q}_p)$ with its compact product topology. There is a G_f-equivariant isomorphism

$$E^{\mathrm{res}} \simeq C^{\mathrm{lc}}(X, \mathbb{C})$$

for the coordinatewise right action. The exact sequence

$$0 \longrightarrow \mathbb{C} \longrightarrow E^{\mathrm{res}} \longrightarrow Q_{\mathrm{ad}} \longrightarrow 0, \qquad Q_{\mathrm{ad}} = E^{\mathrm{res}}/\mathbb{C}, \tag{118.15}$$

is nonsplit in $\mathrm{Rep}^{\mathrm{sm}}_{\mathbb{C}}(G_f)$ and in its trivial-central-character category. Both E^{res} and Q_{ad} are admissible: their invariants under every compact open subgroup are finite-dimensional.

Proof. For finite S, products of functions identify E_S with $C^{\mathrm{lc}}(\prod_{p\in S} X_p, \mathbb{C})$. Surjectivity follows by covering clopen subsets of the compact finite product by finitely many clopen rectangles and refining this cover to a finite partition. Injectivity follows because evaluations separate every finite-dimensional space of functions: after choosing bases in the finitely many function spaces involved, suitable finite sets of evaluation points give injective matrices, whose tensor is still injective.

Every locally constant function on X has a finite cover by basic open cylinders on which it is constant. Such a finite cover exists by compactness. The union of the finite coordinate sets defining the cylinders is finite. Two points equal in these coordinates belong to the same cylinders and therefore have equal function values. The function thus depends on only finitely many coordinates. This proves surjectivity from the colimit, and injectivity follows from the finite-stage statement. The insertion maps become pullback along coordinate projections. Each $g \in G_f$ acts by the coordinate homeomorphisms $x_p \mapsto x_p g_p$, so the isomorphism is equivariant. Smoothness also follows from the finite-coordinate description established above.

A compact open subgroup of G_f contains a subgroup $U_S \times \prod_{p\notin S} K_p$ for a finite S, where U_S is a product of compact open subgroups at the places of S. Invariance under the tail K_p forces a finite-coordinate function to be independent of those tail coordinates, because K_p is transitive on X_p. Its invariants therefore lie in $E_S^{U_S}$. Each local compact open subgroup has open orbits on the compact space X_p, hence finitely many orbits. It follows that $E_S^{U_S}$ is finite-dimensional. This proves admissibility of E^{res}. Compact-open invariants are exact by the same finite averaging argument, so its quotient Q_{ad} is also admissible.

A splitting of (118.15) would give a G_f-invariant retraction $E^{\mathrm{res}} \to \mathbb{C}$. Its restriction to the embedded $E_{\{p\}}$ would be a G_p-invariant retraction of the constants, contradicting Lemma 118.8.1. The same argument applies when the full center acts trivially. □

The quotient Q_{ad} retains mixed finite-place terms. To describe them, set $F_{-1}E^{\mathrm{res}} = 0$ and, for $m \geq 0$, let $F_m E^{\mathrm{res}}$ be the sum of the images of E_T over finite sets T with $|T| \leq m$. This is an increasing, exhaustive filtration by G_f-subrepresentations, with $F_0 E^{\mathrm{res}} = \mathbb{C}$.

PROPOSITION 118.10.2. For every $m \geq 0$ there is a canonical isomorphism

$$\mathrm{gr}^F_m E^{\mathrm{res}} \simeq \bigoplus_{\substack{T \text{ finite} \\ |T|=m}} \mathrm{St}_T^{(f)},$$

where the empty-set factor is $\mathbb{C}$. The induced exhaustive filtration $Q_{\mathrm{ad},m} = F_m E^{\mathrm{res}}/\mathbb{C}$ of Q_{ad} starts with $Q_{\mathrm{ad},0} = 0$ and has these same associated graded pieces for $m \geq 1$.

Proof. At a finite stage S, use the two-step filtration $\mathbb{C}\mathbf{1}_p \subset E_p$ in every tensor factor. A class in $\boxtimes_{p\in T} \mathrm{St}_p$, for $T \subset S$ of size m, maps to the mth associated graded by lifting its factors to E_p and inserting constants elsewhere. Changing a lift adds a tensor from the preceding filtration term, so this map is canonical and equivariant. It is an isomorphism: choose arbitrary vector-space complements to the constants in each E_p and expand the finite tensor product. The summands with precisely m complement factors give the asserted graded piece. The complements prove bijectivity but do not define the map. The canonical maps are compatible with the constant insertions as S grows. Taking the filtered colimit gives the displayed direct sum, since each vector involves only finitely many places. Quotienting by the constant subrepresentation gives the statement for Q_{ad}. □

In particular there is a nonsplit degree-one extension

$$0 \longrightarrow \mathbb{C} \longrightarrow F_1 E^{\mathrm{res}} \longrightarrow \bigoplus_p \mathrm{St}_{\{p\}}^{(f)} \longrightarrow 0.$$

Pulling it back to any single summand gives (118.12), inflated to G_f. This degree-one construction has a direct sum as its quotient. The full quotient Q_{ad} additionally contains every finite mixed tensor through its filtration. Neither quotient is the all-place tensor $\otimes'_p \mathrm{St}_p$.

Indeed, the local quotient maps $E_p \to \mathrm{St}_p$ send the distinguished constant vectors to zero. If those zero vectors are used as insertion maps for the finite tensors of Steinberg quotients, every stage maps to zero after a new prime is inserted, and the resulting colimit is zero. Thus the quotient maps cannot turn (118.15) into a nonzero all-place Steinberg extension.

118.11 Archimedean and automorphic scope

All preceding Ext groups belong to explicitly stated categories of smooth finite-adelic or finite-product representations. Tensoring with the trivial representation of $GL_2(\mathbb{R})$ gives a specified archimedean factor and an action of $GL_2(\mathbb{A})$, without adding another extension class. A nontrivial archimedean extension requires its own smooth topological representation category or Harish–Chandra module category, its central character, and its tensor conventions. An additional degree-one exterior factor contributes an additional degree when its exterior product is defined; it does not erase the finite-place degrees above.

A representation of $GL_2(\mathbb{A})$ is not thereby a realization on $GL_2(\mathbb{Q})\backslash GL_2(\mathbb{A})$. Such a realization requires a specified automorphic function space or enhanced category, central quotient and support conditions, and a comparison functor. The symbols $\mathrm{Ext}^i_{[GL_2]}$ alone do not specify that category. The constructions here give genuine local and adelic group representations and their Yoneda extensions. They neither identify a residue with an extension class nor supply an arithmetic or automorphic comparison.

118.12 Global Eisenstein residues

The local extensions above use smooth group representations. To compare them with Eisenstein residues, fix an automorphic quotient and a normalization. For $G = GL_2$ over $\mathbb{Q}$, write Z for its center and use

$$[G]_Z = Z(\mathbb{A})G(\mathbb{Q})\backslash G(\mathbb{A})$$

with trivial central character. Functions in the Hilbert space $L^2([G]_Z)$ are square-integrable for a fixed quotient Haar measure. Changing that measure changes the norm, but does not turn a morphism

of representations into an extension. With a unitary central character ξ, one instead uses functions transforming by ξ under $Z(\mathbb{A})$ and integrates their absolute squares modulo the center.

The trivial inducing character and full spherical level give a concrete residue calculation. Write $\tau = x + iy$ in the upper half-plane, $y > 0$, and define, for $\operatorname{Re} s > 1$,

$$E(s,\tau) = \frac{1}{2} \sum_{\substack{(c,d)\in\mathbb{Z}^2 \\ (c,d)=1}} \frac{y^s}{|c\tau + d|^{2s}}, \qquad \Lambda(t) = \pi^{-t/2}\Gamma(t/2)\zeta(t).$$

This is the level-one spherical series on $SL_2(\mathbb{Z})\backslash\mathfrak{H}$, with the measure $dx\,dy/y^2$. Its spherical adelic realization uses the normalized spherical finite-place vectors for trivial inducing character, and the trivial central character. The following statement concerns this inducing datum and level.

PROPOSITION 118.12.1 *(The constant residue).* The meromorphic continuation of $E(s,\tau)$ has a simple pole at $s = 1$, with constant residue. Define its positive-chamber residual subspace as the span of its square-integrable residues at poles with $\operatorname{Re} s > 1/2$. Then

$$L^2_{\mathrm{res},E} = \mathbb{C}\mathbf{1}, \tag{118.16}$$

and

$$\operatorname*{Res}_{s=1} E(s,\tau) = c_1\mathbf{1}, \qquad c_1 = \frac{1}{2\Lambda(2)} = \frac{3}{\pi}. \tag{118.17}$$

Proof. The lattice $y^{-1/2}(\mathbb{Z}\tau + \mathbb{Z}) \subset \mathbb{R}^2$ has covolume one. Its theta function is

$$\theta_\tau(t) = \sum_{(c,d)\in\mathbb{Z}^2} \exp\bigl(-\pi t|c\tau + d|^2/y\bigr), \qquad t > 0.$$

The dual of a covolume-one lattice in two dimensions is its rotation through a right angle. Poisson summation for the Gaussian therefore gives $\theta_\tau(t) = t^{-1}\theta_\tau(t^{-1})$. Mellin integration, initially for $\operatorname{Re} s > 1$, gives

$$\Lambda(2s)E(s,\tau) = \frac{1}{2}\int_0^\infty (\theta_\tau(t) - 1)t^{s-1}\,dt.$$

Indeed, the Gaussian integral for a nonzero lattice vector is $\pi^{-s}\Gamma(s)(|c\tau + d|^2/y)^{-s}$. Writing each integer pair as a positive multiple of a primitive pair accounts for $\zeta(2s)$ and the factor $1/2$. Split the integral at 1 and use the theta identity on $(0,1)$. The resulting continuation is

$$\Lambda(2s)E(s,\tau) = \frac{1}{2}\Bigl(\frac{1}{s-1} - \frac{1}{s}\Bigr) + \frac{1}{2}\int_1^\infty (\theta_\tau(t) - 1)(t^{s-1} + t^{-s})\,dt.$$

The integral is entire in s, since the nonzero lattice vectors have positive minimum length and the summands decay exponentially. Thus its residue at $s = 1$ is zero, whereas the rational term has residue $1/2$. Since $\Lambda(2) = \pi/6$, division proves (118.17). In $\operatorname{Re} s > 1/2$, the denominator $\Lambda(2s)$ has no zeros or poles: its zeta factor has the absolutely convergent Euler product, and its gamma factor is holomorphic and nonzero. The numerator continuation has only the pole at $s = 1$ in that half-plane. The positive-chamber residues consequently span exactly the constant line. That line lies in L^2, since the modular quotient has finite hyperbolic volume. This proves (118.16) for the stated sector. □

Fixing trivial central character alone does not select this sector. For example, tensoring an inducing pair by a Hecke character μ changes its common determinant twist while preserving its ratio. A residual character $\mu \circ \det$ has central character μ^2. The residual classification must retain these inducing data and twists. The theorem of Moœglin–Waldspurger, *Le spectre résiduel de $GL(n)$*, pp. 605–607, describes the residual spectrum with fixed unitary central character by repeated cuspidal data and their Langlands quotients. It does not identify the whole trivial-central-character spectrum with (118.16).

Local coordinates and residue normalization

For the symmetric normalized local family, write its characters as $(|\cdot|^{\sigma/2}, |\cdot|^{-\sigma/2})$. In the ordered coordinates used above,

$$a = q^{-\sigma/2}, \qquad b = q^{\sigma/2}, \qquad z = q^{-\sigma}.$$

The exceptional values have representatives $\sigma = \pm 1$, with translates by $2\pi i\mathbb{Z}/\log q$ at the level of the ratio. The ordered pair, including any common twist, must still be retained. The Weyl operation swaps (a, b) and sends σ to $-\sigma$. The unitary point $\sigma = 0$ has ratio 1 and is regular. One can use a separate chart $a(u) = q^{u-1/2}$, $b(u) = q^{1/2-u}$, so $\sigma = 1 - 2u$. Its two representatives are $u = 0, 1$, and the swap is $u \mapsto 1 - u$. A relation between this chart and a global inducing section must specify the inducing characters at every place.

For a meromorphic morphism $A(u) : V(u) \to W(u)$ in a fixed local trivialization, its residue is a degree-zero linear map between the specialized carriers. Under $v = 1 - u$, residues of the pulled-back family satisfy

$$\operatorname*{Res}_{v=1-u_0} A(1-v) = -\operatorname*{Res}_{u=u_0} A(u).$$

This follows by substituting $u - u_0 = -(v - (1 - u_0))$ in its Laurent expansion. Applying this formula to two different operators requires an identity between the families and identifications of their sources and targets. For the regular local operators the proved identity is $D^{\mathrm{sw}} D = h \,\mathrm{id}$ in (118.3). For the normal-equation comparison, replacing t by ut multiplies δ_t by u^{-1} on the divisor. A line and a normalized generator therefore contain different information.

The local extension and its reverse

At a finite prime the principal series already constructed has

$$0 \longrightarrow \mathbb{C} \longrightarrow E_p \longrightarrow \mathrm{St}_p \longrightarrow 0, \qquad E_p = \mathrm{Ind}_{B_p}^{G_p}(1)_{\text{unnormalized}}.$$

In normalized coordinates it occurs at $\sigma = -1$, or $z = p$. The nonsplitting proof gives the class

$$\alpha_p = [\, 0 \to \mathbb{C} \to E_p \to \mathrm{St}_p \to 0 \,] \neq 0 \quad \text{in } \mathrm{Ext}^1_{\mathrm{Rep}^{\mathrm{sm}}_{\mathbb{C}}(G_p)}(\mathrm{St}_p, \mathbb{C}). \tag{118.18}$$

It remains nonzero with the central character imposed strictly. The extension itself is

$$\mathcal{E}_p = (\mathbb{C} \xrightarrow{i_p} E_p \twoheadrightarrow \mathrm{St}_p).$$

The invariant-functional argument also proves $\mathrm{Hom}_{G_p}(\mathrm{St}_p, \mathbb{C}) = 0$. Thus a nonzero residue map from the induced representation cannot be replaced by a nonzero map $\mathrm{St}_p \to \mathbb{C}$. A residue-to-Yoneda comparison must construct a sequence or a connecting morphism and identify it with (118.18). The nonzero class does not determine the dimension of its Ext group.

Remark 118.12.2 *(The opposite orientation).* The reverse comparison asks for a specified nonsplit sequence

$$0 \longrightarrow \mathrm{St}_p \longrightarrow E'_p \longrightarrow \mathbb{C} \longrightarrow 0,$$

its class in $\mathrm{Ext}^1(\mathbb{C}, \mathrm{St}_p)$, and its relation to the opposite operator. For each orientation, the dimension question asks whether the Ext group is one-dimensional in the specified category. Its chosen generator and residue normalization require separate constructions. Neither the isomorphism induced by $\overline{D^{\mathrm{sw}}}$ nor the vanishing of the degree-two splitting obstruction computes these Ext dimensions. At the real place one must first specify a Harish–Chandra module category or a smooth topological category, central character, induction, and exact structure before posing the corresponding extension problem.

The automorphic extension problem

Let $\mathcal{A}_{\mathrm{aut}}$ denote a proposed enhanced automorphic category with a specified coefficient ring, quotient $[G]_Z$, support, central character and exact structure. A global comparison must construct this category and the objects below. Write $\mathbf{1}_{\mathrm{aut}}$ for the realization of the constant representation and S_{aut} for a proposed exceptional object, with its actual local components specified. The degree-one target is

$$\alpha_1^{\mathrm{aut}} \in \mathrm{Ext}^1_{\mathcal{A}_{\mathrm{aut}}}(S_{\mathrm{aut}}, \mathbf{1}_{\mathrm{aut}}).$$

The question of a normalized exceptional line asks whether

$$\mathrm{Ext}^1_{\mathcal{A}_{\mathrm{aut}}}(S_{\mathrm{aut}}, \mathbf{1}_{\mathrm{aut}}) = \mathbb{C}\alpha_1^{\mathrm{aut}}$$

after a specified complex realization, and asks the corresponding question for the reverse orientation. Both the object and the normalization are part of this problem.

The standard smooth all-place Steinberg tensor cannot supply S_{aut}: its finite part is ruled out by Proposition 118.9.1. A finite exterior product instead has degree equal to its number of places, by Proposition 118.8.2. The degree-one extension of Q_{ad} by $\mathbb{C}$ is a separate representation-theoretic construction. Its quotient contains mixed finite-place constituents, and its automorphic realization remains to be constructed. Neither a change of name nor a scalar restricted-product identity provides such a realization.

118.13 Residual parameters and deformation objects

The complex parameters of residual representations and the mod-p parameters of a deformation stack have different coefficient fields. Their comparison requires more than the same two-dimensional notation. Even over $\mathbb{C}$, the two SL_2 factors in a local Arthur parameter have different roles.

Definition 118.13.1 *(The non-tempered parameter).* At a finite place, use the domain $W_F \times SL_2(\mathbb{C})_{\mathrm{D}} \times SL_2(\mathbb{C})_{\mathrm{A}}$. Both parameters take values in $GL_2(\mathbb{C})$. Here W_F is the Weil group, the factor marked D is the Deligne factor, and the factor marked A is the additional Arthur factor. The trivial-character rank-two non-tempered parameter is

$$\psi_1(w, g_{\mathrm{D}}, g_{\mathrm{A}}) = g_{\mathrm{A}}. \tag{118.19}$$

The associated Langlands parameter is obtained by substituting $\mathrm{diag}(|w|^{1/2}, |w|^{-1/2})$ in the Arthur factor. This convention records its non-tempered exponents.

Definition 118.13.2 *(The monodromy parameter).* On the same domain, the standard rank-two Deligne parameter, regarded as an Arthur parameter trivial on its additional factor, is

$$\psi_0(w, g_{\mathrm{D}}, g_{\mathrm{A}}) = g_{\mathrm{D}}. \tag{118.20}$$

The distinction from (118.19) lies in which factor acts. Restricting both displays to an unnamed SL_2 would erase that distinction. A determinant twist changes the Weil part and does not exchange the two factors.

Remark 118.13.3 *(Local and global classification).* Arthur's parameter formalism separates the Deligne and Arthur factors. His endoscopic classification concerns orthogonal and symplectic groups, with GL_n classification as input. The residual classification for GL_n is the theorem of Mœglin–Waldspurger. For global data one can specify a formal cuspidal datum $\mu \boxtimes \nu_2$, where ν_2 is the standard two-dimensional representation. This notation does not construct a global Langlands group or a map to a residual-characteristic stack. Packet assertions must specify twists, central character and local-global realization. In particular, (118.20) does not construct an automorphic representation Steinberg at every finite place.

The direct sum of the trivial and mod-p cyclotomic characters admits several independent cyclotomic extension directions. Their dimensions, and the automorphisms of their deformations, depend on the prime. These data determine what a comparison with a distinguished extension class must preserve. The constructions below give the continuous local deformation groupoids, explicit Kummer classes, and a formal family integrating those classes.

First-order deformations and their automorphisms

Fix a separable closure K^{sep} of $K = \mathbb{Q}_p$, and put $\Gamma = \mathrm{Gal}(K^{\mathrm{sep}}/K)$, with its profinite topology. Let $k/\mathbb{F}_p$ be a finite field with the discrete topology. Choose a primitive pth root ζ_p, with $\zeta_2 = -1$, and define

$$g(\zeta_p) = \zeta_p^{\omega(g)}, \qquad \omega : \Gamma \longrightarrow \mathbb{F}_p^{\times} \longrightarrow k^{\times}.$$

Write $k(r)$ for the one-dimensional k-representation with character ω^r. In particular, $\omega = 1$ when $p = 2$. Fix the ordered basis e_1, e_2 of $V = k^2$, and set

$$\rho_0(g) = \begin{pmatrix} 1 & 0 \\ 0 & \omega(g) \end{pmatrix}, \qquad \mathfrak{a} = \mathrm{End}_k(V), \qquad \mathfrak{a}_0 = \ker(\mathrm{tr} : \mathfrak{a} \to k).$$

Write E_{ij} for the matrix sending e_j to e_i and the other basis vector to zero. The action on $\mathfrak{a}$ is $gX = \rho_0(g) X \rho_0(g)^{-1}$. All cochains and representations are continuous.

For a finite discrete Γ-module N, use the inhomogeneous cochains $C^n(\Gamma, N) = \mathrm{Map}_{\mathrm{cts}}(\Gamma^n, N)$, with $C^0(\Gamma, N) = N$. Their differential is

$$\begin{aligned} (df)(g_1, \dots, g_{n+1}) = {} & g_1 f(g_2, \dots, g_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ & + (-1)^{n+1} f(g_1, \dots, g_n). \end{aligned}$$

Thus $dX(g) = gX - X$ and $dc(g,h) = gc(h) - c(gh) + c(g)$. Write $Z^1(\Gamma, N) = \ker(d : C^1 \to C^2)$, $B^1(\Gamma, N) = dN$, and $H^i(\Gamma, N) = H^i(C^\bullet(\Gamma, N))$.

Set $A = k[\epsilon]/(\epsilon^2)$, with the discrete topology. A framed deformation is a continuous representation $\rho : \Gamma \to \mathrm{GL}_2(A)$ reducing to ρ_0 in the specified basis. Morphisms preserve this entire A-basis. A residual-marked unframed deformation is a free rank-two A-module with a continuous Γ-action and a specified identification of its reduction with V. Its morphisms induce the identity on V.

PROPOSITION 118.13.4 *(Continuous first-order groupoids).* Framed deformations form the discrete groupoid on $Z^1(\Gamma, \mathfrak{a})$. Residual-marked unframed deformations form the action groupoid

$$[Z^1(\Gamma, \mathfrak{a})/\mathfrak{a}], \qquad X : c \longmapsto c - dX.$$

Its isomorphism classes form $H^1(\Gamma, \mathfrak{a})$. The automorphism group of each object is the additive group $H^0(\Gamma, \mathfrak{a})$.

Proof. Every framed lift has a unique expression

$$\rho_c(g) = (1 + \epsilon c(g))\rho_0(g).$$

Multiplication gives

$$\rho_c(g)\rho_c(h) = \big(1 + \epsilon(c(g) + gc(h))\big)\rho_0(gh).$$

The homomorphism condition is therefore $dc = 0$. A basis lifting e_1, e_2 identifies any residual-marked deformation with such a matrix representation. Two such choices differ by $1 + \epsilon X$, and

$$(1 + \epsilon X)\rho_c(g)(1 - \epsilon X) = \rho_{c - dX}(g).$$

The quotient and automorphism statements follow from this equality. An automorphism fixes c exactly when $dX = 0$. □

If the residual marking is forgotten, the residual centralizer also acts. Write

$$C = \{B \in \mathrm{GL}_2(k) : B\rho_0(g) = \rho_0(g)B \text{ for all } g\}.$$

Every permissible change of basis is $(1 + \epsilon X)B$, and its action is

$$c \longmapsto BcB^{-1} - dX.$$

Thus the acting group is $\mathfrak{a} \rtimes C$. For odd p, the character ω is nontrivial and C is the group of invertible diagonal matrices. For $p = 2$, it is $\mathrm{GL}_2(k)$. The stabilizer of c consists of the pairs satisfying $BcB^{-1} - dX = c$. At $c = 0$, this is $H^0(\Gamma, \mathfrak{a}) \rtimes C$. In particular, a formal scheme viewed as a groupoid-valued functor cannot represent these unframed groupoids: its points have only identity automorphisms, whereas these groupoids have nonzero infinitesimal automorphisms. This assertion concerns the groupoids, rather than representability of their sets of isomorphism classes.

Determinant character and determinant identification

Fixing a determinant character restricts the objects of a deformation problem. Choosing an identification of determinant representations also restricts its morphisms. To separate these conditions, fix

$$\delta_\eta(g) = \omega(g)(1 + \epsilon\eta(g)), \qquad \eta \in Z^1(\Gamma, k).$$

Here the action on k is trivial, so η is a continuous additive homomorphism. For a character $\delta : \Gamma \to A^\times$, write $A(\delta)$ for the free rank-one module with that action. The constant determinant lift has $\eta = 0$. The identity

$$\det \rho_c(g) = \omega(g)(1 + \epsilon \operatorname{tr} c(g))$$

holds in every characteristic.

Proposition 118.13.5 *(The two fixed-determinant groupoids).* For the constant lift $\delta_0 = \omega$, the residual-marked deformation groupoid with fixed determinant character is

$$\mathcal{D}_{\mathrm{char}}(A) = [Z^1(\Gamma, \mathfrak{a}_0)/\mathfrak{a}].$$

Fix the residual orientation $e_1 \wedge e_2$. If objects include an identification $\lambda : \det(V_A) \xrightarrow{\sim} A(\omega)$ lifting that orientation, and arrows preserve λ, the groupoid is

$$\mathcal{D}_{\det}(A) = [Z^1(\Gamma, \mathfrak{a}_0)/\mathfrak{a}_0].$$

Both groupoids have isomorphism classes $H^1(\Gamma, \mathfrak{a}_0)$. Their automorphism groups are respectively $H^0(\Gamma, \mathfrak{a})$ and $H^0(\Gamma, \mathfrak{a}_0)$.

Proof. The fixed-character equation is $\operatorname{tr} c = 0$. Every strict conjugation preserves the determinant, and $\operatorname{tr}(dX) = 0$. This proves the first groupoid formula. For the second, choose a lifted basis with $\lambda(e_1 \wedge e_2) = 1$. Such a basis exists because

$$\det : 1 + \epsilon\mathfrak{a} \longrightarrow 1 + \epsilon k$$

is surjective: $\operatorname{diag}(1 + \epsilon t, 1)$ has determinant $1 + \epsilon t$. Changes between these bases satisfy $\operatorname{tr} X = 0$, which proves the second formula.

The trace admits the Γ-equivariant splitting

$$k \longrightarrow \mathfrak{a}, \qquad t \longmapsto tE_{11}.$$

Indeed, E_{11} is invariant and has trace one. Hence $\mathfrak{a} = \mathfrak{a}_0 \oplus kE_{11}$ and $d\mathfrak{a} = d\mathfrak{a}_0$, including when $p = 2$. The two orbit sets are therefore $H^1(\Gamma, \mathfrak{a}_0)$. The equations $dX = 0$, with the respective trace restrictions, give the stated automorphism groups. □

For a general η, the cocycle $c_\eta(g) = \eta(g)E_{11}$ has trace η. The object spaces are therefore the affine translates $c_\eta + Z^1(\Gamma, \mathfrak{a}_0)$, with the same gauge formulas. A determinant-compatible framing requires $\lambda(e_1 \wedge e_2) = 1$, and gives the discrete groupoid on $Z^1(\Gamma, \mathfrak{a}_0)$ when $\eta = 0$. If a full framing and an independent determinant identification are both retained, the discrete object set is instead $Z^1(\Gamma, \mathfrak{a}_0) \times k$: the second coordinate records $\lambda(e_1 \wedge e_2) = 1 + \epsilon t$. For a chosen determinant identification, forgetting the residual marking allows residual changes in $C \cap \mathrm{SL}_2(k)$.

The coefficient decomposition is

$$\mathfrak{a} = kE_{11} \oplus kE_{22} \oplus k(-1)E_{12} \oplus k(1)E_{21},$$
$$\mathfrak{a}_0 = k(E_{11} - E_{22}) \oplus k(-1)E_{12} \oplus k(1)E_{21}.$$

At $p = 2$, the diagonal trace-zero generator is I, and every summand has trivial action. At $p = 3$, the two off-diagonal characters coincide, but their matrix subspaces remain distinct.

The first-order tangent groupoid is presented by the two-term complex

$$[\mathfrak{a} \xrightarrow{-d} Z^1(\Gamma, \mathfrak{a})] \quad \text{in degrees } [-1, 0].$$

Its degree -1 cohomology records automorphisms and its degree 0 cohomology records deformation classes. This is the truncation through degree zero of $C^\bullet(\Gamma, \mathfrak{a})[1]$, with the cohomological shift convention $(C[1])^i = C^{i+1}$ and shifted differential $-d$. Thus $H^2(\Gamma, \mathfrak{a})$ occurs in degree one of the full shifted complex. For a determinant identification, replace $\mathfrak{a}$ by $\mathfrak{a}_0$. For the fixed-character problem alone, the two-term complex is $[\mathfrak{a} \xrightarrow{-d} Z^1(\Gamma, \mathfrak{a}_0)]$. Identifying any full cochain complex with the tangent complex of a specified derived stack requires an equivalence of the underlying moduli problems.

Kummer classes at every prime

The lower-left summand has an explicit arithmetic description. For $a \in K^\times$, choose $\beta_a \in K^{\mathrm{sep}}$ with $\beta_a^p = a$, and define

$$\frac{g(\beta_a)}{\beta_a} = \zeta_p^{\kappa_a(g)}, \qquad \kappa_a(g) \in \mathbb{F}_p \subset k.$$

The action on roots of unity gives

$$\kappa_a(gh) = \kappa_a(g) + \omega(g)\kappa_a(h).$$

This cocycle is continuous, since β_a and ζ_p lie in a finite Galois extension. Replacing β_a by $\zeta_p^r \beta_a$ changes it by the coboundary $dr(g) = \omega(g)r - r$.

PROPOSITION 118.13.6 *(Explicit cyclotomic cohomology).* The Kummer map gives an isomorphism

$$(\mathbb{Q}_p^\times / \mathbb{Q}_p^{\times p}) \otimes_{\mathbb{F}_p} k \xrightarrow{\sim} H^1(\Gamma, k(1)), \qquad [a] \otimes 1 \longmapsto [\kappa_a].$$

For odd p, a basis is $[\kappa_p]$, $[\kappa_{1+p}]$. For $p = 2$, a basis is $[\kappa_2]$, $[\kappa_{-1}]$, $[\kappa_5]$. The dimensions are respectively two and three.

Proof. Multiplication of roots makes the Kummer map additive modulo pth powers. If $\kappa_a = dr$, replacing β_a by $\zeta_p^{-r}\beta_a$ makes it invariant. Hence a is a pth power in K. Conversely, Hilbert 90 implies that every continuous μ_p-valued cocycle is $g(\beta)/\beta$ for some $\beta \in (K^{\mathrm{sep}})^\times$. Then $\beta^p \in K^\times$. This proves the isomorphism over $\mathbb{F}_p$.

The required Hilbert 90 statement reduces to a finite Galois extension L/K through which the cocycle factors. If z_σ is a multiplicative cocycle for $\mathrm{Gal}(L/K)$, choose $a \in L$ so that

$$b = \sum_\sigma z_\sigma \sigma(a) \neq 0.$$

Such an a exists by linear independence of distinct field automorphisms. The cocycle identity gives $\tau(b) = z_\tau^{-1} b$. Consequently $z_\tau = \tau(b^{-1})/b^{-1}$, as required. This is the proof of Milne, *Class Field Theory*, Proposition II.1.22, p. 67. A basis of the finite extension $k/\mathbb{F}_p$ decomposes each continuous cochain into finitely many coordinate cochains. Tensoring with k therefore commutes with cohomology.

For odd p, the multiplicative decomposition is

$$\mathbb{Q}_p^\times = p^{\mathbb{Z}} \times \mu_{p-1} \times (1 + p\mathbb{Z}_p).$$

The valuation supplies the first factor. Hensel lifting of the distinct nonzero residues supplies the roots of unity in the second factor. The pth-power map on μ_{p-1} is bijective. The logarithm and exponential give inverse homomorphisms

$$\log : 1 + p\mathbb{Z}_p \xrightarrow{\sim} p\mathbb{Z}_p.$$

For $x, y \in p\mathbb{Z}_p$, the estimates

$$v_p(x^n/n) \geq n - v_p(n), \qquad v_p(y^n/n!) \geq n - v_p(n!)$$

prove convergence of the logarithm and exponential, using $v_p(n!) \leq n/(p-1)$. Their formal inverse and product identities pass to these convergent series. Logarithm carries pth powers to $p^2\mathbb{Z}_p$, and $\log(1+p) \equiv p \pmod{p^2}$. Thus valuation and the principal-unit quotient give the basis $[p]$, $[1+p]$.

At $p = 2$, each odd unit has a unique sign making it congruent to one modulo four. Hence

$$\mathbb{Q}_2^\times = 2^{\mathbb{Z}} \times \{1, -1\} \times (1 + 4\mathbb{Z}_2).$$

The preceding convergence estimates, with input valuation at least two, give $\log(1 + 4\mathbb{Z}_2) = 4\mathbb{Z}_2$. Squaring becomes multiplication by two, and $\log(5) \equiv 4 \pmod 8$. The valuation, sign, and principal-unit factors give the independent classes $[2]$, $[-1]$, $[5]$. □

In particular, $p = 3$ gives the basis $[\kappa_3]$, $[\kappa_4]$. The choice of ζ_p identifies μ_p with $\mathbb{F}_p(1)$. Changing it rescales all numerical cocycles by one nonzero scalar. The intrinsic classes take values in $\mu_p \otimes_{\mathbb{F}_p} k$.

The unit classes form an arithmetic subspace, and valuation gives an exact sequence

$$0 \longrightarrow (\mathbb{Z}_p^\times/(\mathbb{Z}_p^\times)^p) \otimes_{\mathbb{F}_p} k \longrightarrow H^1(\Gamma, k(1)) \xrightarrow{v_p} k \longrightarrow 0.$$

The unit subspace has dimension one for odd p and two for $p = 2$. The specified uniformizer p supplies the splitting $1 \mapsto [\kappa_p]$. The quotient itself does not specify a splitting.

There is also an unramified condition. If $I \subset \Gamma$ is inertia, define its subspace as

$$H^1_{\mathrm{ur}}(\Gamma, k(1)) = \operatorname{im}\bigl(H^1(\Gamma/I, k(1)^I) \to H^1(\Gamma, k(1))\bigr).$$

For odd p, this is zero: $\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$ is totally ramified of degree $p - 1$, since $((1 + T)^p - 1)/T$ is Eisenstein at p. Thus $k(1)^I = 0$. At $p = 2$, the unramified subspace is the line $k[\kappa_5]$. Indeed, $(1 + \sqrt{5})/2$ satisfies $x^2 - x - 1 = 0$, whose reduction is separable and irreducible over $\mathbb{F}_2$. It therefore generates the unramified quadratic extension. Since $\Gamma/I \cong \widehat{\mathbb{Z}}$, its continuous additive characters into k form a one-dimensional k-space.

An explicit formal family

Each cyclotomic cocycle defines both an extension of residual representations and a tangent direction at their split sum. For $b \in Z^1(\Gamma, k(1))$, set

$$r_b(g) = \begin{pmatrix} 1 & 0 \\ b(g) & \omega(g) \end{pmatrix}.$$

The line ke_2 is invariant, and the quotient is trivial. Thus r_b gives an exact sequence

$$0 \longrightarrow k(1) \longrightarrow E_b \longrightarrow k \longrightarrow 0.$$

Conversely, a vector-space splitting of any such extension gives this matrix form. Changing the splitting changes b by a coboundary. Consequently

$$\operatorname{Ext}^1_{\mathrm{Rep}_k^{\mathrm{cts}}(\Gamma)}(k, k(1)) \cong H^1(\Gamma, k(1)),$$

where the category consists of finite-dimensional discrete continuous k-representations. The order of the two endpoint characters is part of this identification.

The coefficient inclusion $u \mapsto uE_{21}$ and projection onto the lower-left entry commute with the cochain differential. They induce an injection with a specified left inverse

$$j : H^1(\Gamma, k(1)) \hookrightarrow H^1(\Gamma, \mathfrak{a}_0).$$

The class $j([b])$ is represented by $\rho_{\epsilon b}(g) = \left(\begin{smallmatrix} 1 & 0 \\ \epsilon b(g) & \omega(g) \end{smallmatrix}\right)$.

THEOREM 118.13.7 *(Formal cyclotomic family).* Set $d = 2$ and $(a_1, a_2) = (p, 1 + p)$ for odd p. For $p = 2$, set $d = 3$ and $(a_1, a_2, a_3) = (2, -1, 5)$. Choose the Kummer cocycles above and set

$$R_{\mathrm{gal}} = k[[t_1, \dots, t_d]], \qquad u(g) = \sum_{i=1}^{d} t_i \kappa_{a_i}(g).$$

With the maximal-ideal topology on R_{gal}, the matrices

$$\rho_{R_{\mathrm{gal}}}(g) = \begin{pmatrix} 1 & 0 \\ u(g) & \omega(g) \end{pmatrix}$$

define a continuous representation reducing to ρ_0, with determinant ω and the standard determinant identification. Its tangent map is injective, with image $j(H^1(\Gamma, k(1)))$.

Proof. The cocycle equation for u is exactly the lower-left entry of $\rho_{R_{\mathrm{gal}}}(gh) = \rho_{R_{\mathrm{gal}}}(g)\rho_{R_{\mathrm{gal}}}(h)$. The finitely many cocycles factor through a common finite Galois quotient, so the representation is continuous for the maximal-ideal topology. Its determinant and reduction follow from the diagonal entries.

For each Artinian local k-algebra B with residue field k, use the same residual-marked deformation definition on free rank-two B-modules, with the discrete topology on B. A continuous local homomorphism $R_{\mathrm{gal}} \to B$ specializes these matrices to a deformation. This is compatible with homomorphisms between such algebras and therefore defines a map from the functor of points of Spf R_{gal} to the determinant-identified deformation groupoid. A tangent homomorphism sends t_i to ϵx_i. Its class is $j(\sum_i x_i[\kappa_{a_i}])$. The Kummer basis and the left inverse of j prove injectivity. □

Changing β_{a_i} to $\zeta_p^{r_i}\beta_{a_i}$ adds dr_i to the corresponding cocycle. Set $s = \sum_i t_i r_i$. Conjugation by $1 - sE_{21}$ carries the old formal family to the new one, since its lower-left entry changes by ds. This calculation holds over R_{gal}, without a first-order truncation. The formal family therefore changes by an explicit isomorphism under these root choices.

This family fixes both diagonal characters and a line and quotient. It is not asserted to represent the full deformation problem. At $p = 2$, the underlying residual representation is trivial, and its unmarked form has no preferred decomposition into two lines. The inclusion through E_{21} uses the stated frame or flag. For odd p, the two distinct character subspaces are intrinsic, although their scalar coordinates still depend on the chosen bases.

Dimensions and comparison maps

The Kummer calculation determines one direct summand without local duality. The full dimension table also uses finite-module local duality and the local Euler characteristic theorem (Milne, *Arithmetic Duality Theorems*, Corollary I.2.3, p. 28, and Theorem I.2.8, p. 31). For a finite-dimensional continuous $k[\Gamma]$-module N, they imply

$$h^2(N) = h^0(N^\vee(1)), \qquad h^0(N) - h^1(N) + h^2(N) = -\dim_k N,$$

where $h^i(N) = \dim_k H^i(\Gamma, N)$ and $N^\vee = \mathrm{Hom}_k(N, k)$. The field trace pairing identifies the underlying finite-module dual with the additive module of $N^\vee(1)$. The Euler formula follows by substituting $|N| = p^{[k:\mathbb{F}_p]\dim_k N}$ into $\chi(\Gamma, N) = (\mathbb{Z}_p : |N|\mathbb{Z}_p)^{-1}$. Both theorems apply at $p = 2$, since the base field has characteristic zero. For $N = k(r)$, they give

$$h^1(k(r)) = 1 + \mathbf{1}_{\omega^r=1} + \mathbf{1}_{\omega^{1-r}=1}.$$

The matrix decomposition and $\dim B^1(\Gamma, N) = \dim N - h^0(N)$ now give

	$p \geq 5$	$p = 3$	$p = 2$
$h^1(k)$	2	2	3
$h^1(k(1))$	2	2	3
$h^1(k(-1))$	1	2	3
$h^0(\mathfrak{a})$	2	2	4
$h^0(\mathfrak{a}_0)$	1	1	3
$h^1(\mathfrak{a})$	7	8	12
$h^1(\mathfrak{a}_0)$	5	6	9
$\dim Z^1(\Gamma, \mathfrak{a})$	9	10	12
$\dim Z^1(\Gamma, \mathfrak{a}_0)$	7	8	9
$h^2(\mathfrak{a})$	1	2	4
$h^2(\mathfrak{a}_0)$	1	2	3

These dimensions concern the unrestricted local continuous problem. A crystalline, finite-flat, ordinary, or global Selmer condition requires its own definition before imposing a further subspace.

The explicit formal family supplies a local Galois target for a comparison. A specified nonzero Kummer class selects a one-parameter subfamily. The unit condition selects a line for odd p and a plane at $p = 2$. Identifying any of these with a line arising from an analytic residue requires a map that carries that residue to the stated cohomology class and is nonzero.

A coherent extension requires further categorical data. For example, an exact k-linear functor from $\mathrm{Rep}_k^{\mathrm{cts}}(\Gamma)$ to a specified abelian category, with identified images of k and $k(1)$, sends E_b to an extension with those endpoints. An exact functor between specified stable categories instead sends its associated triangle to a triangle. Either construction gives a map on extension classes. An isomorphism requires a calculation of that map, including its kernel and image. A tangent vector alone does not define two coherent objects or a functor between their categories. On $\operatorname{Spec} k[t]$, for instance, the tangent space at zero is k, whereas $\operatorname{Ext}^1(k[t], k[t]) = 0$. For the residue module k_0, the resolution

$$0 \longrightarrow k[t] \xrightarrow{t} k[t] \longrightarrow k_0 \longrightarrow 0$$

instead gives $\operatorname{Ext}^1(k_0, k_0) = k$. Thus the endpoints of a coherent extension are indispensable.

A comparison from a complex representation category also requires arithmetic coefficient data. There is no unital field map $k \to \mathbb{C}$, since it would carry $p \cdot 1_k = 0$ to $p \cdot 1_{\mathbb{C}} \neq 0$. There is no unital map in the reverse direction either, since p is invertible in $\mathbb{C}$. Reduction must instead specify an integral coefficient ring, an integral object or stable lattice, a residue-field map to k, and the induced map on the relevant extension classes. A global Galois comparison additionally requires a ramification set, local conditions, and the restriction maps into the local cohomology computed here.

Finally, scalar twisting gives

$$(1 \oplus \omega) \otimes \omega^{-1} = \omega^{-1} \oplus 1.$$

This is $1 \oplus 1$ only at $p = 2$. At $p = 3$, exchanging summands identifies it with the original representation. For any scalar character χ, conjugation gives $\mathrm{Ad}(\rho_0 \otimes \chi) = \mathrm{Ad}(\rho_0)$, while the determinant changes by χ^2. Hence a scalar twist need not preserve a fixed-determinant fibre, and no extra determinant twist enters the adjoint cochain formulas.

The formal stack and its framed presentation

The local deformation groupoids above retain the residual representation and its automorphisms. At the split cyclotomic point, the residual-marked determinant-identified groupoid has

$$\pi_0 \mathcal{D}_{\det}(k[\epsilon]/(\epsilon^2)) = H^1_{\mathrm{cts}}(\Gamma, \mathrm{ad}^0(1 \oplus \omega)).$$

Here π_0 means the set of isomorphism classes, with its vector-space structure from the cocycle description. Fixing only the determinant character gives the same isomorphism classes at this split point, but retains the larger automorphism group $H^0(\Gamma, \mathfrak{a})$. This distinction follows from the equivariant trace splitting through E_{11}, including in characteristic two.

For the Emerton–Gee moduli problem, a framed lifting ring supplies a versal presentation of the formal deformation groupoid. Emerton–Gee, Proposition 3.6.3, p. 76, identifies the self-fiber product using the change-of-basis group completed along the residual centralizer, or along the identity for the pointed formal problem. It does not replace the unframed groupoid by a formal scheme with no automorphisms. A derived enhancement, wild-inertia conditions and an identification with its full tangent complex require the corresponding moduli comparison. The computed local complex retains this requirement.

Tangent complexes and spectral extension classes

A tangent direction describes deformation of one point. An extension between two sheaves requires two objects and a category of morphisms. To express the proposed comparison, suppose a derived stack $\mathcal{X}$ over k has been specified, with a cotangent complex $\mathbb{L}_{\mathcal{X}}$ admitting the dual used below. At a point x, set $T_x = R\,\mathrm{Hom}_k(x^*\mathbb{L}_{\mathcal{X}}, k)$. Its degree-zero cohomology is the first-order tangent space, while degree minus one records infinitesimal automorphisms. Write $\mathbb{T}^1_x = H^0(T_x)$ for this convention.

Suppose in addition that a spectral category $\mathcal{A}_{\mathrm{sp}}$ has been constructed, with objects O_{St} and $O_{\mathbf{1}}$ and specified support. These names alone do not define sheaves. For example, an ind-coherent category with nilpotent singular support requires a derived stack for which its singularity space and the specified nilpotent subset are defined. The desired comparison is

$$L_{\mathrm{cyc}} \longrightarrow \mathrm{Ext}^1_{\mathcal{A}_{\mathrm{sp}}}(O_{\mathrm{St}}, O_{\mathbf{1}}), \qquad L_{\mathrm{cyc}} \subseteq \mathbb{T}^1_{x_{\mathrm{cyc}}}.$$

Here the subspace L_{cyc}, its dimension and the displayed map are constructions still required. A cotangent calculation does not identify this Ext group between two different sheaves.

Remark 118.13.8 *(The reverse spectral comparison).* For a point x_{triv} representing $1 \oplus 1$, the opposite orientation asks for a specified subspace $L_{\mathrm{triv}} \subseteq \mathbb{T}^1_{x_{\mathrm{triv}}}$ and a map

$$L_{\mathrm{triv}} \longrightarrow \mathrm{Ext}^1_{\mathcal{A}_{\mathrm{sp}}}(O_{\mathbf{1}}, O_{\mathrm{St}}).$$

The two endpoints and their support must be constructed in this category. The explicit scalar-twist calculation above does not identify this problem with the cyclotomic problem at odd primes. At $p = 2$, equality of the residual representations still does not choose the two sheaves or the map. An Arthur parameter over $\mathbb{C}$ supplies neither spectral comparison by itself.

118.14 The arithmetic class comparison

An automorphic category over $\mathbb{C}$ and a tangent space over k cannot be compared by an unnamed extension of scalars. A unital ring homomorphism $k \to \mathbb{C}$ would send $p \cdot 1 = 0$ to $p \cdot 1 \neq 0$, which is impossible. One possible comparison would use a common integral coefficient ring, lifts of both objects,

and derived specialization maps to their respective fibers. It must specify whether cohomology commutes with these specializations, including any torsion. The complex family over R above does not supply this integral construction.

Definition 118.14.1 *(The first comparison target).* After a common coefficient realization over a ring A has been constructed, denote the realized automorphic category by $\mathcal{A}_{\mathrm{aut},A}$ and the lifted exceptional tangent module by $L_{\mathrm{cyc},A}$. The first class comparison seeks

$$\mathsf{c}_1^{\mathrm{exc}} : \operatorname{Ext}^1_{\mathcal{A}_{\mathrm{aut},A}}(S_{\mathrm{aut},A}, \mathbf{1}_{\mathrm{aut},A}) \xrightarrow{\sim} L_{\mathrm{cyc},A}, \qquad \alpha_1^{\mathrm{aut}} \longmapsto e_{\mathrm{cyc}}. \tag{118.21}$$

The symbols on both sides denote required realizations of the specified objects and classes, not objects supplied by this definition.

Definition 118.14.2 *(The reverse comparison target).* With the reverse extension and the second deformation problem realized over the same coefficient ring, the second target is

$$\mathsf{c}_0^{\mathrm{exc}} : \operatorname{Ext}^1_{\mathcal{A}_{\mathrm{aut},A}}(\mathbf{1}_{\mathrm{aut},A}, S_{\mathrm{aut},A}) \xrightarrow{\sim} L_{\mathrm{triv},A}, \qquad \alpha_0^{\mathrm{aut}} \longmapsto e_{\mathrm{triv}}. \tag{118.22}$$

The local chart $u \mapsto 1 - u$ does not construct the coefficient or deformation comparison needed for this arrow.

The scalar residue and a derived pairing

The scalar ratio can be recovered from the constant term of the specified Eisenstein series. For $\operatorname{Re} s > 1$, separate the terms with $c = 0$ and sum the others over $c \geq 1$ and reduced residues d modulo c. Integration in x over $[0, 1]$ unfolds the translates $d + mc$, $m \in \mathbb{Z}$, to the real line. If $\varphi(c)$ is the number of reduced residue classes modulo c, this gives

$$\int_0^1 E(s, x + iy)\, dx = y^s + y^s \sum_{c \geq 1} \frac{\varphi(c)}{c^{2s}} \int_{\mathbb{R}} (u^2 + y^2)^{-s}\, du.$$

The last integral is $y^{1-2s}\sqrt{\pi}\,\Gamma(s - 1/2)/\Gamma(s)$, by $u = y \tan\theta$. The Euler factors give $\sum_{c \geq 1} \varphi(c) c^{-2s} = \zeta(2s - 1)/\zeta(2s)$. Thus the constant term is $y^s + n(s)\, y^{1-s}$, where

$$n(s) = \frac{\Lambda(2s - 1)}{\Lambda(2s)}.$$

The completed zeta function satisfies $\Lambda(t) = \Lambda(1 - t)$. This follows from the same Gaussian Mellin argument in one dimension. It has residues 1 at $t = 1$ and -1 at $t = 0$. Since $\Lambda(2) = \pi/6$, the Laurent expansion gives

$$c_1^{\mathrm{LS}} = \operatorname*{Res}_{s=1} n(s) = \frac{1}{2\Lambda(2)}, \qquad c_0^{\mathrm{LS}} = \operatorname*{Res}_{s=0} n(s)^{-1} = -\frac{1}{2\Lambda(2)}.$$

Indeed, $n(s)n(1 - s) = 1$, so the second identity is the first after $s \mapsto 1 - s$. The archimedean quotient $\Gamma_{\mathbb{R}}(2s - 1)/\Gamma_{\mathbb{R}}(2s)$, where $\Gamma_{\mathbb{R}}(t) = \pi^{-t/2}\Gamma(t/2)$, is regular at $s = 1$ and equals π. The global pole does not give a pole of this archimedean factor there.

To compare this scalar with extension classes, let U_{s_0} and V_{s_0} be chosen complexes computing the two realized derived Hom objects. A derived pairing inducing a bilinear form on their degree-one cohomology consists of a morphism $U_{s_0} \otimes^{\mathbf{L}}_{A} V_{s_0} \to A[-2]$. The shift puts A in cohomological degree two. A Bost–Connes operation σ^{BC} must be defined on these complexes, with its action and compatibility proved.

Write $e_1 = e_{\mathrm{cyc}}$ and $e_0 = e_{\mathrm{triv}}$ for the required chosen tangent classes at the two comparison targets. The desired normalization is an equality of specified pairings

$$P^{\mathrm{LS}}_{s_0}(\alpha^{\mathrm{aut}}_{s_0}, \sigma^{\mathrm{BC}} e_{s_0}) = c^{\mathrm{LS}}_{s_0} P_{s_0}(\alpha^{\mathrm{aut}}_{s_0}, e_{s_0}), \qquad s_0 \in \{0,1\},$$

after a coefficient realization in which the displayed scalars exist. This is a chain-level comparison problem. The nonzero scalar does not prove existence or nondegeneracy of either pairing. Bar–cobar duality transports derived morphisms only after its algebra, coalgebra, twisting cochain and module realizations have been given. It does not construct the displayed trace or pairing from a scalar identity of principal-series vectors.

Remark 118.14.3 *(The exceptional global comparison).* The exceptional Eisenstein comparison asks for both (118.21) and (118.22), natural in the allowed inducing characters and compatible with specialization. Its required constructions are a common coefficient realization, automorphic exceptional objects, residue-to-Yoneda maps, spectral objects with support, and the derived pairing and tangent maps. The wild-inertia enhancement of Conjecture 61.5.7 is an additional residual-characteristic input. It does not supply the other constructions. If the two component maps through Galois cohomology are constructed as natural isomorphisms, their composition is a natural isomorphism: the composite of their commutative naturality squares is the required square. This implication supplies no component map. The global comparison remains unresolved.

118.15 Derived maps and global assembly

Even a comparison of degree-zero and degree-one morphisms on generators would leave a categorical obstruction. The following finite calculation isolates that obstruction without making any arithmetic assumptions.

Example 118.15.1 *(Agreement in degrees zero and one).* Let k be a field and let $A = k \oplus ke$ be the unital graded algebra with $|e| = 1$, $e^2 = 0$ and zero differential. Let $B = k \oplus ke \oplus kf$, where $|f| = 2$, all products of positive-degree elements vanish, and the differential is zero. The quotient $p : B \to A$ fixes 1, e and kills f. Use perfect right differential graded modules, namely the smallest idempotent-complete triangulated subcategories generated by the free modules. Extension of scalars defines

$$F = - \otimes^{\mathbf{L}}_B A : \mathrm{Perf}(B) \longrightarrow \mathrm{Perf}(A), \qquad F(B) = A.$$

A right-linear endomorphism of a free rank-one module is determined by its value at 1 and acts by left multiplication. The free modules compute their own derived Hom complexes. Thus the map on their endomorphism complexes is precisely $p : B \to A$. It induces isomorphisms in degrees zero and one, with values k and ke, but its degree-two map is $kf \to 0$. It is not a quasi-isomorphism, and F is not fully faithful. The image A still generates the target perfect category. Hence these low-degree agreements and a generating image do not suffice for full faithfulness.

The Eisenstein category

The exceptional extension problem therefore remains part of the full local-global comparison

$$F : \bigotimes_v{}' C_v \longrightarrow \mathcal{D}^{\text{sph-fin}}([G]_Z).$$

Here the local stable categories C_v, their spherical objects, transition functors and restricted tensor product must be specified. The target must be an actual enhancement of the chosen automorphic category

with its support convention. The non-exceptional comparison in Theorem 85.8.1 and the Paley–Wiener comparison in Theorem 85.5.6 retain their stated functor and support hypotheses. Extending them to closed exceptional strata requires compatible localization or recollement functors and maps on the attaching objects.

For an exact functor on small idempotent-complete stable categories generated by a set $\mathcal{G}$, a sufficient full-faithfulness condition is equivalence of all mapping spectra between generators. To see the extension of this condition, fix one generator. The objects for which its mapping comparison is an equivalence form a subcategory closed under shifts, cones and retracts, by exactness. It therefore contains every object. Apply the same argument in the other variable. Full faithfulness follows, and a generating essential image gives equivalence. For presentable categories, compact generation, preservation of colimits and compactness of the relevant images provide the analogous extension from compact generators. These are substantive hypotheses, not consequences of degree-one agreement.

Cuspidal, residual and continuous objects

The Hilbert decomposition of Theorem 85.6.1 separates cuspidal, residual and continuous spectra. Its orthogonality is an assertion about Hilbert inner products. To assemble a stable automorphic category one needs categorical embeddings of the corresponding sectors, adjunctions and gluing maps, including the continuous exceptional and non-exceptional loci. A Hilbert orthogonal sum does not determine mapping complexes between their enhancements. The cuspidal comparison of Theorem 70.6.4, the residual comparison of Theorem 85.6.3, and the non-exceptional comparison must therefore be combined through specified derived functors. Bernstein-center actions, local-global realization and the closed-stratum attaching maps must commute with these functors. The local splitting and adelic extension above supply input objects; they do not prove this assembly.

118.16 Compatibility with residual-characteristic comparisons

At $p = 2, 3$, the local class comparisons in Definitions 87.2.1 and 87.2.2 concern a residual-characteristic block. Their connecting maps and deformation comparison in Propositions 87.2.8 and 87.2.7 must be used with their hypotheses. A global exceptional comparison would need localization maps on automorphic extensions, coefficient specialization, and the corresponding restriction of its Galois deformation problem. Compatibility means a commutative square between those actual maps. Equality of their target H^1 groups is insufficient: two linear maps to the same nonzero space can differ by any scalar, or have different images. A naturality statement for a functor applies only after the same sequence and morphisms have been realized in its domain.

The common-twist freedom also remains. The local ratios and the scalar normalizing factor depend on the ratio of inducing characters. Multiplying both characters by μ preserves that ratio and changes the induced representation by $\mu \circ \det$. Thus the trivial ratio does not force trivial inducing character. Central characters, normal coordinates, Haar measures and any Bost–Connes operation must be compared under this twist. The universal full-category Koszul roof and the pointwise strict-center roof prove different statements. A canonical family of strict-center splittings compatible with Weyl symmetry remains an additional construction. Their existence at one complex point does not provide an integral or residual-characteristic comparison.

118.17 Arithmetic descent conditions

The arithmetic descent conditions named in Proposition 85.10.1 and Theorem 87.5.1 concern the following mathematically distinct questions.

(i) The non-ordinary part of F1.a.1 asks for essential surjectivity of the derived Emerton–Gee comparison at $p \geq 5$ on its specified non-ordinary locus.

(ii) The Taylor–Wiles part of F1.a.1 asks for extension beyond the Taylor–Wiles hypotheses, with the deformation conditions retained.

(iii) F1.a.2 asks for the residual-characteristic local class match at $p = 2, 3$, under the wild-inertia and comparison hypotheses of Theorem 87.2.9.

(iv) The exceptional Eisenstein part of $\mathrm{L}_{\mathrm{ACD}}$ asks for the global class maps of Remark 118.14.3 and their full derived extension across the exceptional strata. The local exceptional ratios and the global scalar points use their respective coordinates.

(v) Bernstein-center diamond gluing asks for compatible center maps and categorical descent, including the local-global realization and transition functors used in the global assembly.

The fourth question is not exhausted by the third. The explicit finite-place constructions establish the local cone, its coefficient specializations, pointwise fixed-center splitting, finite exterior products and the mixed adelic extension. Their global arithmetic realization still requires the maps above.

118.18 Sources and further comparisons

The constructions separate three operations: specialization of a regular intertwining complex, formation of Yoneda products, and realization in automorphic or spectral categories. Their respective carriers determine which further comparisons can be formulated. The following sources provide the surrounding representation theory, spectral analysis and deformation theory.

Casselman.
Introduction to the Theory of Admissible Representations of p-Adic Reductive Groups, 1 May 1995, Proposition 2.1.7, p. 20, and Section 8.1, p. 70, give compact averaging and the Steinberg quotient.

Schneider–Stuhler.
Representation theory and sheaves on the Bruhat–Tits building, Publ. Math. IHÉS **85** (1997), 97–191, Section II.2 and Theorem II.3.1, develop cellular resolutions with coefficient systems.

Langlands.
Eisenstein Series, Section 10, pp. 12–14 of the author's reprint, computes the rank-one spherical residue. Its variable $z = 2s - 1$ accounts for the factor $1/2$ in the residue used here. *On the Functional Equations Satisfied by Eisenstein Series*, Lecture Notes in Mathematics 544 (1976), treats the general spectral problem.

Moœglin–Waldspurger.
Le spectre résiduel de $GL(n)$, Ann. Sci. ENS **22** (1989), 605–674, introduction and Section I.1, gives the residual classification and normalized intertwiners. *Spectral Decomposition and Eisenstein Series: A Paraphrase of the Scriptures*, Cambridge Tracts 113 (1995), develops the spectral theory.

Shahidi.
Eisenstein Series and Automorphic L-Functions, AMS Colloquium Publications 58 (2010), treats intertwining operators, local coefficients and L-function normalization.

Arthur.
The Endoscopic Classification of Representations: Orthogonal and Symplectic Groups, AMS Colloquium Publications 61 (2013), concerns classical-group classification. The author's exposition *The Endoscopic Classification of Representations*, pp. 1–4, distinguishes the Deligne and Arthur factors.

Emerton–Gee.
Moduli stacks of étale (φ, Γ)*-modules and the existence of crystalline lifts*, arXiv:1908.07185, Proposition 3.6.3, pp. 74–76, gives the versal framed presentation.

Breuil–Mézard.
Multiplicités modulaires et représentations de $GL_2(\mathbb{Z}_p)$ *et de* $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ *en* $\ell = p$, Duke Math. J. **115** (2002), concerns local deformation multiplicities.

Bellaïche–Chenevier.
Families of Galois representations and Selmer groups, Astérisque 324 (2009), treats reducibility, extension classes and trianguline deformations. Its Chapter 2 concerns refined crystalline representations.

Schlessinger and Mazur.
Schlessinger, *Functors of Artin rings*, Trans. AMS **130** (1968), 208–222, Theorem 2.11, states the hull and prorepresentability conditions. Mazur, *Deforming Galois representations*, MSRI Publications 16 (1989), 385–437, develops deformations of Galois representations.

Bernstein–Zelevinsky.
Induced representations of reductive $\mathfrak{p}$*-adic groups I*, Ann. Sci. ENS **10** (1977), concerns smooth parabolic induction.

Wallach and Jacquet–Langlands.
Wallach, *Real Reductive Groups I* (1988), and Jacquet–Langlands, *Automorphic Forms on* $GL(2)$, Lecture Notes in Mathematics 114 (1970), supply real representation theory and automorphic context. The real extension problem requires a category and normalization of its own.

Harish-Chandra.
Harmonic analysis on real reductive groups II: Wave packets in the Schwartz space, Invent. Math. **36** (1976), 1–55, and *III: The Maass–Selberg relations and the Plancherel formula*, Ann. Math. **104** (1976), 117–201, concern real spectral analysis.

Kisin.
Potentially semi-stable deformation rings, J. AMS **21** (2008), 513–546, concerns potentially semistable deformation conditions with specified Hodge data.

Tate.
Fourier analysis in number fields and Hecke's zeta functions, Princeton thesis (1950), reprinted in Cassels–Fröhlich (1967), Chapter XV, treats adelic Fourier analysis and zeta integrals.

Milne.
Class Field Theory (2020), Proposition II.1.22, p. 67, gives the Hilbert 90 argument. *Arithmetic Duality Theorems*, second edition (2006), Corollary I.2.3 and Theorem I.2.8, pp. 28–31, give the finite-module local duality and Euler characteristic formulas.

Positselski.
Two kinds of derived categories, Koszul duality, and comodule–contramodule correspondence, arXiv:0905.2621, Sections 6.3 and 6.5, constructs derived equivalences from specified bar coalgebras and twisting cochains.

The next arithmetic comparison must realize the specified exceptional objects over common coefficients and construct their residue-to-Yoneda and deformation maps. Only then can their normalizations, wild-prime restrictions and full derived morphisms be compared.

Chapter 119

Residual Boundary of the Arithmetic Branch

A Hilbert space with a scaling action and the Weil pairing on arithmetic test functions are different data. A unitary comparison can transport a scaling operator to a proposed boundary space. Positivity of the full Weil pairing instead constrains the zeros of the completed zeta function. This chapter separates these two assertions before listing the additional geometric and arithmetic comparisons.

119.1 Hilbert Descent, Scaling, and the Weil Pairing

Write $\mathscr{A} = C_c^\infty(\mathbb{R})$ for the compactly supported smooth functions of the logarithmic variable, with convolution and $f^*(t) = \overline{f(-t)}$. The Fourier transform

$$(\mathcal{F}_+ f)(z) = \int_{\mathbb{R}} f(t) e^{izt}\, dt$$

identifies this algebra with its entire image $\mathscr{P} = \mathcal{F}_+ \mathscr{A}$, equipped with pointwise multiplication and $h^\sharp(z) = \overline{h(\bar{z})}$. Both spaces and their transported topologies are those of Definition 78.1.1. On the compact-test algebra the signed explicit formula is

$$W(f) = P(f) - W_\infty(f) - W_{\mathrm{fin}}(f),$$

with the polar, archimedean, and prime terms defined in Theorem 78.1.2. The Hermitian form is $q_W(f, g) = W(f * g^*)$. None of these definitions uses a boundary Hilbert-space construction.

For comparison, the continuous scaling model is

$$H_c = L^2(\mathbb{R}_{>0}, dx/x), \qquad (U_a \phi)(x) = \phi(e^{-a} x).$$

Under $t = \log x$, put $D_c = -i\,\partial_t$ on $H^1(\mathbb{R})$ and $\Theta_c = \frac{1}{2} + i D_c$ on the same domain. Here H^1 consists of square-integrable functions whose weak first derivative is square-integrable. The translation group satisfies $U_a = e^{-iaD_c}$, and

$$e^{-a\Theta_c} = e^{-a/2} U_a, \qquad |e^{-a\Theta_c}| = e^{-a/2} \mathrm{I}_{H_c} \quad (a \in \mathbb{R}).$$

This norm identity is unconditional and makes no assertion about q_W.

Theorem 119.1.1 *(arithmetic positivity and boundary transport).* The archimedean Tate positivity (A-TP) statement of Conjecture 78.2.3 is equivalent to RH. No Hilbert-descent hypothesis is needed for this equivalence.

Suppose separately that a Hilbert-descent construction supplies a space H_B. Specify a closed normal operator Θ_B and a unitary $V_\Sigma : H_B \to H_c$ such that

$$V_\Sigma \operatorname{Dom}(\Theta_B) = \operatorname{Dom}(\Theta_c), \qquad V_\Sigma \Theta_B = \Theta_c V_\Sigma \quad \text{on } \operatorname{Dom}(\Theta_B).$$

Then every $e^{-a\Theta_B}$, $a \in \mathbb{R}$, is bounded and

$$e^{-a\Theta_B} = V_\Sigma^{-1} e^{-a\Theta_c} V_\Sigma, \qquad |e^{-a\Theta_B}| = e^{-a/2} 1_{H_B}.$$

Thus descent together with A-TP implies RH through A-TP; the surface operator comparison additionally requires the stated unitary and generator-domain data.

Proof. The signed explicit formula gives

$$q_W(f,f) = P(f * f^*) - W_\infty(f * f^*) - W_{\mathrm{fin}}(f * f^*).$$

Consequently A-TP is precisely $q_W(f,f) \geq 0$ for every $f \in \mathscr{A}$. Apply Corollary 78.2.1. For the operator statement, the domain and intertwining equalities identify the spectral resolutions of Θ_B and Θ_c. The latter has spectrum in $1/2 + i\mathbb{R}$, on which $z \mapsto e^{-az}$ has constant modulus $e^{-a/2}$. Normal functional calculus proves both identities. This is the comparison specified in Theorem 115.9.2. □

Remark 119.1.2 *(one exponential does not determine its generator).* The Hilbert limit in Conjecture 115.8.3 and the comparison V_Σ are separate inputs. Even equality of the surface operator at $a = 1$ does not replace the generator comparison: Θ_c and $\Theta_c + 2\pi i\, 1$ have the same domain and the same exponential at $a = 1$, but are distinct operators. They give distinct one-parameter families at nonintegral a. To identify a native surface theory one must in addition relate its independently defined observable to this transported operator. No map from arithmetic test functions or arithmetic cohomology is supplied by the norm identity.

The radial spherical transform of Theorem 27.6.1 provides a different unitary comparison: tree adjacency becomes multiplication by $2\sqrt{p}\cos\theta$, and the root state becomes integration against ν_p. Section 27.8 extends this comparison to finite sets of primes. An arithmetic realization still requires a map from the primitive trace carrier preserving the Hecke operations and pairing. Tame residue and Gersten operations require additional maps on common coefficients. The positive tree state supplies no identification with the full Weil functional W defined above.

119.2 The Full Weil Positivity Condition

In compact coordinates the arithmetic condition is

$$W(f * f^*) \geq 0 \qquad (f \in \mathscr{A}).$$

Define $\mathscr{W} : \mathscr{P} \to \mathbb{C}$ by $\mathscr{W}(h) = W(\mathcal{F}_+^{-1} h)$. Its equivalent Fourier form is

$$\mathscr{W}(hh^\sharp) \geq 0 \qquad (h \in \mathscr{P}).$$

The product here is pointwise multiplication of entire functions, not convolution on $\mathscr{P}$.

PROPOSITION 119.2.1 *(the two test-space forms of Weil positivity).* The compact and Fourier inequalities above are equivalent, and each is equivalent to RH. Restricting to an arbitrary proper test subspace requires a separate argument before the same criterion can be used.

Proof. Fubini gives $\mathcal{F}_+(f * f^*) = (\mathcal{F}_+ f)(\mathcal{F}_+ f)^\sharp$. By definition $\mathcal{F}_+ : \mathscr{A} \to \mathscr{P}$ is onto; Fourier inversion makes it one-to-one. Hence the two universally quantified inequalities agree. Their equivalence with RH is Corollary 78.2.1 on exactly this test space. Neither an auxiliary digamma quadratic form nor continuous scaling purity replaces the full polar-minus-archimedean-minus-prime expression. □

119.3 Comparison and Descent Conjectures

Definition 119.3.1 *(comparison and descent inputs).* The comparison and descent boundary contains the following named inputs.

Input	Carries	Form
Wild-EG enhancement	F1.a.2.class-match; $\mathrm{L}_{\mathrm{ACD}}$ Eisenstein $\mathrm{Ext}^{\mathrm{I}}$ at $\{0,1\}$; BC-glue wild inertia	open
Cross-prime transition and effective descent	BC-diamond-glue cross-prime prismatic glue	open
Arithmetic AG singular support	global arithmetic Arinkin–Gaitsgory singular-support condition and the centre of BC-diamond-glue	open
Clausen–Scholze condensed-Hilbert descent	Quantum arithmetic Chern–Simons bulk extension	open
Consani–Marcolli adele-class duality	Positselski G-row duality and the abelian Heisenberg base-change line	open
Chenevier Selmer-ring, trace and equivariant-descent datum	Pairwise sheaf comparison under the full Ch1 hypotheses	open
BCglue centre-compatible categorical functors and coefficient/family comparisons	K5 and the Jacquet–Shalika trace comparison in BC-diamond-glue	open
Ben-Zvi lift at infinity	$\mathrm{L}_{\mathrm{ACD}}$ archimedean compatibility and finite uniformity	open
BZN wild trace data	Large-prime semisimplicity and the wild Ext-trace pairing	open
Symmetric-square weight limit	Weil–Petersson weight-aspect positivity for holomorphic cusp forms	open
Arithmetic chiral bar nilpotency	The square-zero property of the proposed arithmetic chiral pre-differential d^{Ar}	open

The relative Galois bar in Chapter 36 has a square-zero differential over each $\mathbb{Z}/\ell^r$. Theorem 36.4.2 compares its endpoint products in every arity, and Corollary 36.6.4 passes to the specified restricted adic limit. The last row concerns the separate arithmetic chiral pre-differential. Identifying its carrier, residue operations, and completion with the relative Galois construction remains an additional comparison problem.

Theorem 119.3.2 *(visible non-RH residuals).* Each input in Definition 119.3.1 is a non-RH residual named elsewhere in the arithmetic branch. The statement does not assert exhaustion of all possible residuals.

Proof. With its stated comparison hypotheses, Wild-EG (the Emerton–Gee wild-inertia enhancement) is an input to the residual-characteristic local comparison for F1.a.2 in Theorem 113.3.1. The global exceptional Eisenstein comparison in Remark 118.14.3 additionally requires a common coefficient realization, the automorphic carriers, a residue-to-Yoneda comparison, and compatible full derived maps. For BC-diamond-glue it must be accompanied by the block-centre maps, compatible functors and normalized class comparisons specified in Proposition 116.3.3. The ordinary $\ell \neq p$ scalar comparison does not supply these data.

The cross-prime input is the specified transition, coherent-support, filtration and higher-coherence datum of Hypothesis 56.5.2, together with the effective descent required by its gluing theorem. It is not an independently proved pro-étale or prismatic-site flatness statement. Arithmetic AG (arithmetic Arinkin–Gaitsgory singular support) is the global singular-support conjecture 42.1.1, required at the centre of BC-diamond-glue by Corollary 116.6.1. The specified condensed-Hilbert descent datum supplies the Hilbert, trace, and finite-place Wilson clauses of the chosen quantum ACS model by Theorem 115.8.4. The archimedean operator comparison also requires the unitary and generator domains in Theorem 115.9.2. The other rows are the named conditional inputs constructed in Chapters 75, 73, 56, 52, 54, 83, and 33. These are stated as separate comparison hypotheses. No implication from A-TP to these constructions, or in the reverse direction, is established by the Weil positivity criterion. □

Remark 119.3.3 *(Wild-EG concentration).* Wild-EG is a common input to the three comparison problems F1.a.2, the exceptional L_{ACD} Eisenstein Ext^1 crossing at $\{0, 1\}$, and wild-inertia BC-diamond-glue. Each retains its stated class-comparison hypotheses. For BC-diamond-glue, one must also construct the fixed-determinant block-ring isomorphisms, compatible fully faithful functors and normalized class maps of Proposition 116.3.3, together with the coefficient, family and global comparisons. The wild-EG enhancement alone does not construct these data.

119.4 The Generalised Ramanujan Conjecture for GL_4

The Rankin–Selberg square of a Maass form on $\mathrm{GL}_2/\mathbb{Q}$ comes from an isobaric automorphic object on $\mathrm{GL}_4/\mathbb{Q}$. Full temperedness of that GL_4 object at every finite place, equivalently GRC for its cuspidal factors, closes the Maass sector of the $\mathcal{V}^{\mathrm{prim}}$ Hochschild-trace identity. This conjecture is separated from A-TP in the proof: it concerns Satake eigenvalues, not zero placement.

Hypothesis 119.4.1 *(GRC for* $\mathrm{GL}_4/\mathbb{Q}$ *in the Selberg class).* The isobaric automorphic representations of $\mathrm{GL}_4/\mathbb{Q}$ arising from the Rankin–Selberg square of a Maass form have tempered cuspidal factors at every finite place.

PROPOSITION 119.4.2 *(GRC-*GL_4 *closes the Maass sector).* Conditional on Hypothesis 119.4.1, full Ramanujan holds for the Maass forms entering the $\mathcal{V}^{\mathrm{prim}}$ Hochschild-trace P_3 identity of Theorem 114.6.1. This closes the Maass sector. No equivalence with RH is asserted here; no implication between this hypothesis and A-TP is used.

Proof. Chapter 114 proves that the Arthur–Ramakrishnan–GRC hypothesis implies full Ramanujan for Maass forms (Theorem 114.5.5) and reduces that hypothesis to Hypothesis 119.4.1 (Proposition 114.5.6). The P_3 identity is the exact conditional statement of Theorem 114.6.1. A-TP concerns the zeros of ζ through Weil positivity; GRC-GL_4 concerns Satake parameters. The argument uses the two residuals as separate inputs. □

119.5 F_1.a.1 Small Residuals

Two small residuals occur in the F_1.a.1 table of Chapter 117. They are not a count of all small residuals in the arithmetic branch.

PROPOSITION 119.5.1 *(small residuals).* Besides the generator-comparison inputs and the automorphy/patching hypotheses isolated in Lemma 117.3.4 and Lemma 117.3.5, the F_1.a.1 table has the following two small residual tails:

(i) **Supersingular small-weight tail** (Conjecture 117.2.8): non-Fontaine–Laffaille representations with Hodge–Tate weights $k \in \{p-1, p\}$, requires a non-formal theorem.
(ii) A_5 **non-solvable descent defect** (Conjecture 117.3.8): local-global descent defect in the icosahedral sector, requires a non-formal theorem.

Proof. The two listed residual tails are read from Proposition 117.4.2. The Fontaine–Laffaille supersingular core is reduced to the generator comparison of Theorem 117.2.6. The odd-determinant icosahedral core is reduced to the conditional theorem of Theorem 117.3.6. The two tails listed above are additional obstructions in that decomposition. □

119.6 Arithmetic Hypotheses and Realization Data

Remark 119.6.1 *(scope of the arithmetic conditions).* The conditions above concern arithmetic comparison, descent, temperedness and small-prime regularity. A marked realization as in Chapter 1 additionally requires its source and target, coefficient functor, operation-preserving maps and evaluation identity. The arithmetic conditions alone construct none of those maps and imply no realization assertion for an object in another coefficient category.

119.7 Residual Boundary

Theorem 119.7.1 *(arithmetic branch obstruction decomposition).* The arithmetic branch has, at least, the following obstructions:

(i) one statement equivalent to RH: archimedean Tate positivity;
(ii) the comparison and descent inputs of Definition 119.3.1;
(iii) one Ramanujan-type hypothesis separated from A-TP: GRC for $\mathrm{GL}_4/\mathbb{Q}$;
(iv) the two F_1.a.1 small residuals of Proposition 119.5.1.

The arithmetic conditions listed here do not construct the marked realizations of Chapter 1.

Proof. Item (1) is Proposition 119.2.1. Item (2) is Theorem 119.3.2. Item (3) is Proposition 119.4.2. Item (4) is Proposition 119.5.1. A marked realization requires the separate data described in Remark 119.6.1; the listed arithmetic conditions do not supply those data. No implication from the table to RH is asserted except through A-TP. □

Remark 119.7.2 *(domain of the boundary).* The theorem isolates the displayed RH-equivalent content into A-TP and places the visible non-RH obstructions in named inputs. It does not prove RH and it does not claim that the arithmetic comparison is complete.

Appendix A

Primary Sources

This appendix records the primary literature anchoring the arithmetic branch of Volume IV. Each entry states author(s), title, year, arXiv identifier where applicable, and the role in the branch: which chapter invokes the result and for what purpose.

The entries are organized by mathematical topic: factorization and homotopy-theoretic sources (§§A.5–A.7); geometric Langlands, Hecke categories, and local Langlands (§§A.8–A.9); arithmetic Chern–Simons and arithmetic topology (§A.10); arithmetic D-modules and F-isocrystals (§A.6); operator theory and moment problems (§A.24); and background sources in automorphic forms, adelic analysis, Yangians, Arakelov geometry, Macdonald theory, and primon-gas statistical mechanics (remaining sections).

A.1 Kapranov and Kapranov–Smirnov

Kapranov–Smirnov 1995 *Cohomology determinants and reciprocity laws: number field case*. Unpublished preprint, ~15 pp., ca. 1995. PDF mirrored at `http://cage.ugent.be/~kthas/Fun/library/KapranovSmirnov.pdf`. Frames Spec $\mathbb{Z}$ as an "arithmetic curve" via the analogy between number fields and function fields over finite fields; motivates the absolute point / $\mathbb{F}_1$ programme and prime-labelled cohomology determinants feeding reciprocity laws. Used throughout this branch as the foundational informal reference (Chapters 26, 33).

Kapranov ca. 1995–96 *Analogies between number fields and 3-manifolds*. Unpublished preprint. Foundational for arithmetic topology; the MKR (Mazur–Kapranov–Reznikov) dictionary matches closed oriented 3-manifolds to number fields and knots to primes. Extended in Reznikov arXiv:math/0107210 (2001) and Morishita's book *Knots and Primes* (2012). Used in Chapter 37 for the Wilson-line = ramified Hecke character dictionary in Kim's arithmetic Chern–Simons.

Kapranov–Vasserot 2004 *Vertex algebras and the formal loop space*, Publ. IHES 100:209–269; arXiv:math/0107143. Introduces the ind-scheme $\mathcal{L}X$ of formal loops into a finite-type scheme X; factorization structure on $\mathcal{L}X$ recovers vertex-algebra structure on chiral de Rham. Four-paper series; Part IV = arXiv:math/0612371. No Spec $\mathbb{Z}$ version exists; this branch is the arithmetic analogue.

Kapranov 2000 *The elliptic curve in the S-duality theory and Eisenstein series for Kac–Moody groups*, arXiv:math/0001005. S-duality on an elliptic curve; not used here directly.

A.2 Borger and Λ-Ring Theory

Borger 2008 *The basic geometry of Witt vectors, I: The affine case*, Algebra & Number Theory 5:231–285; arXiv:0801.1691. For commuting operations see §1.5; for flat Frobenius lifts, §1.6; for the classical torsion-free special-λ criterion, §1.17. These are the distinctions used in Chapter 28.

Borger 2010 *The basic geometry of Witt vectors, II: Spaces*, arXiv:1006.0092. Extends to algebraic spaces; generalises Buium's arithmetic jet functor. Background for the space-valued Witt construction; no Λ-module definition or arithmetic Fock comparison is inferred.

Borger 2009 *Λ-rings and the field with one element*, arXiv:0906.3146. Central reference. Section1.1 describes the Frobenius lifts on flat spaces; §1.2 gives the categorical descent interpretation. The torsion-free criterion is Wilkerson1982, Proposition1.2, identified in Borger0801.1691, §1.17. No literal tensor-product construction on the arithmetic trace class follows from this interpretation. Used throughout Chapter 28.

Borger–de Smit 2008 *Galois theory and integral models of Λ-rings*, arXiv:0801.2352. Relative case: Λ-structure over a Noetherian base.

A.3 Bost–Connes, Connes–Marcolli, Connes–Consani

Bost–Connes 1995 *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. 1:411–457. Central reference for the arithmetic quantum statistical system. KMS partition function $\zeta(\beta)$; phase transition at $\beta = 1$; spontaneous symmetry breaking by $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$. Explicitly motivated by Julia's primon gas.

Connes–Marcolli 2004 *From Physics to Number Theory via Noncommutative Geometry, Part I & II*, Part I arXiv:math/0404128. $HH^\bullet(C^*_{\mathbb{Q}})$ computed §1.8; used in Chapter 29 for the Theorem $\mathrm{H}^{\mathrm{arith}}$ comparison.

Connes–Consani 2014 *The Arithmetic Site*, arXiv:1405.4527. Arithmetic site = Grothendieck topos $\widehat{\mathbb{N}^\times}$ with tropical semiring $\mathbb{Z}_{\max}$ structure sheaf. A separate arithmetic-site construction; an identification with the polynomial Fock operations of Chapter 28 requires an actual comparison.

Connes–Consani 2015 *Geometry of the arithmetic site*, arXiv:1502.05580. Extends (1405.4527); Frobenius correspondences as subvarieties.

Connes–Consani 2012 *Universal thickening of the field of real numbers*, arXiv:1202.4377. Background on the real-number completion of the arithmetic site; not directly invoked in computations but provides the archimedean-fibre context for $\mathbb{F}_1$-descent.

A.4 Kontsevich, Soibelman, Davison: E_n Formality and CoHA

Kontsevich 2003 *Deformation quantization of Poisson manifolds*, Lett. Math. Phys., arXiv:q-alg/9709040. Foundational for operadic formality. Used in Chapter 39 §1.

Kontsevich 1999 *Operads and Motives in Deformation Quantization*, arXiv:math/9904055. E_n-formality over $\mathbb{Q}$ for all n; cited in Chapter 39 §1 for the rational formality input.

Tamarkin 1998 *Another proof of M. Kontsevich formality theorem*, arXiv:math/9803025. Drinfeld-associator-based proof for E_2.

Kontsevich–Soibelman 2010 *Cohomological Hall algebra, exponential Hodge structures, and motivic DT invariants*, arXiv:1006.2706. Sections 2.2–2.4 construct the extension product and its Euler-class shuffle formula. The critical theory uses potential-dependent coefficients. See Chapter 12. A comparison with a real spectral measure requires a separate operator and state.

Efimov 2012 *Cohomological Hall algebra of a symmetric quiver*, arXiv:1103.2736, Theorem 1.1. The rational CoHA of a finite symmetric quiver with zero potential is free supercommutative after a sign modification. Its primitive generator space has a polynomial translation factor of degree two.

Dotsenko–Mozgovoy 2021 *DT invariants from vertex algebras*, arXiv:2108.10338, Proposition 5.3 and Theorems 5.6–5.7. The graded dual of this symmetric-quiver CoHA is a connected vertex bialgebra. Its primitive vertex Lie algebra is free over translation, with finite-dimensional quotient in each charge.

Fresse 2017 *Homotopy of Operads and Grothendieck–Teichmüller Groups*, Vol. II. Integral E_1-formality theorem used in Chapter 39 §1.

Kaledin 2014 *Universal Witt vectors and the "Japanese cocycle"*. Integral-E_2 obstruction.

Salvatore–Wahl 2002 Dyer–Lashof operations on $H_\bullet(E_2;\mathbb{F}_p)$.

Davison–Meinhardt 2020 *Cohomological Donaldson–Thomas theory of a quiver with potential and quantum enveloping algebras*, arXiv:1601.02479. Theorems A and C give the critical PBW result. The theorem retains the stability, slope, Serre-subcategory, monodromic mixed-Hodge and sign data. Its absolute PBW isomorphism concerns underlying graded objects. The free-algebra conclusion applies to the perverse associated graded. See Theorem 12.10.1.

A.5 Beilinson–Drinfeld, Francis–Gaitsgory, Positselski

Beilinson–Drinfeld 2004 *Chiral Algebras*, AMS Colloq. Publ. 51. Ch. 3 (chiral bar), Ch. 4 (Hochschild).

Francis–Gaitsgory 2011 *Chiral Koszul duality*, Selecta Math., arXiv:1103.5803. Chiral Lie–commutative Koszul duality. This differs from the associative coefficient bar in Chapter 33. It does not supply the arithmetic associative comparison requested there.

Positselski 2011 *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence*, Mem. AMS, arXiv:0905.2621. Used in Chapter 33 §7 for arithmetic Positselski.

Parshin–Kato 1979–1980 *A generalization of local class field theory by using K-groups*. Residue maps and reciprocity in their specified geometric and coefficient settings. Chapter 33 separates the Steinberg identity from a comparison with chiral collision operations.

A.6 Arithmetic $\mathcal{D}^\dagger$-Modules and F-Isocrystals

Berthelot 1996 *$\mathcal{D}$-modules arithmétiques. I. Opérateurs différentiels de niveau fini*, Ann. Sci. Éc. Norm. Sup. 29:185–272. Foundational construction of arithmetic differential operators of finite level. Used in Chapter 58 only as the local p-adic coefficient shadow of the finite-place Hecke/diamond category.

Berthelot 2000 *$\mathcal{D}$-modules arithmétiques. II: descente par Frobenius*, Mém. SMF 81. Frobenius descent for arithmetic $\mathcal{D}$-modules.

Caro 2009 *$\mathcal{D}$-modules arithmétiques surholonomes*, Ann. Sci. Éc. Norm. Sup. 42:141–192. Overholonomic arithmetic $\mathcal{D}$-modules and stability under the six operations; also records overconvergent unit F-isocrystals as overholonomic in the smooth case.

Kedlaya 2004 *Full faithfulness for overconvergent F-isocrystals*, in *Geometric Aspects of Dwork Theory*, 819–836. Full faithfulness for overconvergent F-isocrystals. This is a coefficient reference, not an equivalence between finite-prime Hecke categories and classical D-modules.

A.7 Costello, Costello–Gwilliam, Costello–Gaiotto

Costello–Gwilliam 2017, 2021 *Factorization Algebras in Quantum Field Theory*, CUP Vol. I & II. Used throughout Chapter 30.

Costello 2013 *Supersymmetric Gauge Theory and the Yangian*, arXiv:1303.2632. Holomorphic-topological factorization.

Costello 2020 *Holography and Koszul duality*, lecture/preprint notes. Background on Koszul duality as a holographic mechanism; context for the arithmetic Koszul-duality formulation of Chapter 38.

Costello–Gaiotto 2019 *Vertex operator algebras and 3d $\mathcal{N} = 4$ gauge theories*, arXiv:1804.06460. Used in Chapter 37 §3.

Costello–Dimofte–Gaiotto 2020 *Boundary chiral algebras and holomorphic twists*, arXiv:2005.00083. Boundary conditions in holomorphic-topological theories and their chiral algebra content; background for the TQFT-Chern–Simons comparison in Chapter 37.

Costello–Witten–Yamazaki 2017 *Gauge theory and integrability I*, arXiv:1709.09993. 4D Chern–Simons.

Costello–Yamazaki 2019 *Gauge theory and integrability III*, arXiv:1908.02289. Rational and trigonometric R-matrices from 4D Chern–Simons; background for the Bethe-algebra comparison in Chapter 40.

Francis 2012 *Factorization homology of topological manifolds*, arXiv:1206.5522. Used in Chapter 30 §4 for locally-constant E_n-reduction.

A.8 Geometric Langlands, Derived Satake, Hecke Categories

Mirković–Vilonen 2007 *Geometric Langlands duality and representations of algebraic groups over commutative rings*, Ann. Math., arXiv:math/0401222. Geometric Satake.

Bezrukavnikov 2016 *On two geometric realizations of an affine Hecke algebra*, Publ. IHES, arXiv:1209.0403. Iwahori Hecke category; central for Chapter 38.

Ben-Zvi–Nadler 2009 *The character theory of a complex group*, arXiv:0904.1247. Hochschild trace = $O(\mathrm{LocSys})$; gluing principle for the synthesis (Chapter 41).

Arinkin–Gaitsgory 2015 *Singular support of coherent sheaves and the geometric Langlands conjecture*, Selecta Math., arXiv:1201.6343. Nilpotent singular support AG-conjecture.

Fargues–Scholze 2021 *Geometrization of the local Langlands correspondence*, arXiv:2102.13459. Local arithmetic Langlands via Fargues–Fontaine curve.

Dotto–Emerton–Gee 2026 *A categorical p-adic Langlands correspondence for $GL_2(\mathbb{Q}_p)$*, arXiv:2603.26887. Fully faithful local categorical p-adic Langlands functor for $p \geq 5$; not a local equivalence and not a global adelic theorem.

Caraiani–Scholze 2015/2017 *On the generic part of the cohomology of compact unitary Shimura varieties*, arXiv:1511.02418. Torsion-freeness of the generic part of Shimura variety cohomology; cited in Chapter 38 §4 as a constraint on the global adelic Hecke-category input.

Emerton–Gee 2020 *Moduli stacks of étale (φ, Γ)-modules and the existence of crystalline lifts*, arXiv:1908.07185. Emerton–Gee stack of Galois deformations; central for Chapter 38 §4.

Emerton–Gee 2011 *A geometric perspective on the Breuil–Mézard conjecture for GL_2*, arXiv:1109.4226. Breuil–Mézard geometry; not the Emerton–Gee stack source.

Zhu 2020 arXiv:2008.02998. Coherent Springer theory.

Zhu 2016 arXiv:1603.05593; Bhatt–Scholze 2016, arXiv:1507.06490. Witt-vector affine Grassmannian.

A.9 FLE, Quantum Geometric Langlands, and Root-of-Unity Quantum Groups

Gaitsgory 2008 *Twisted Whittaker model and factorizable sheaves*, Selecta Math. 13:617–659; arXiv:0705.4571. Whittaker and factorizable-sheaf source for quantum geometric Langlands.

Gaitsgory 2016 *Quantum Langlands Correspondence*, arXiv:1601.05279. Informal 2007 notes outlining local and global quantum Langlands and the FLE; cited in Chapter 63 for the quantum parameter dictionary.

Campbell–Dhillon–Raskin 2021 *Fundamental local equivalences in quantum geometric Langlands*, Compos. Math. 157:2699–2732; arXiv:1907.03204. Quantum FLE in the local affine-Grassmannian/affine-flag setting; cited in Chapter 63 §2 for the local quantum Langlands equivalence as input to the arithmetic analogue.

Gaitsgory–Lysenko 2019 *Metaplectic Whittaker category and quantum groups: the small FLE*, arXiv:1903.02279. Small FLE and small quantum group source.

Arinkin et al. 2024 Arinkin–Beraldo–Campbell–Chen–Faergeman–Gaitsgory–Lin–Raskin–Rozenblyum 2024. *Proof of the geometric Langlands conjecture II: Kac–Moody localization and the FLE*, arXiv:2405.03648. Critical level FLE and its compatibility with Kac–Moody localization in the geometric, not arithmetic, setting.

Frenkel–Gaitsgory 2006 *Local geometric Langlands correspondence and affine Kac–Moody algebras*, Progr. Math. 253, 69–260; arXiv:math/0508382. Critical local geometric Langlands and affine Kac–Moody source layer.

Raskin 2021 *Chiral principal series categories I: finite-dimensional calculations*, Adv. Math. 388, 107856. Chiral principal-series source used for globalization analogies; not a number-field adelic theorem.

Raskin *Chiral principal series categories II: the factorizable Whittaker category*; *Chiral categories*; *D-modules on infinite-dimensional varieties*. Preprint foundations for chiral principal series and infinite-dimensional $\mathcal{D}$-modules, available in the Papers and Preprints section of the author's website, `https://www.samraskin.net/`. Background for Chapters 63 and 30.

Kazhdan–Lusztig 1993–1994 *Tensor structures arising from affine Lie algebras I–IV*, JAMS 6–7. KL tensor structure and affine Lie algebra side of root-of-unity quantum groups.

Finkelberg 1996 *An equivalence of fusion categories*, GAFA 6:249–267; **Andersen–Paradowski 1995**, *Fusion categories arising from semisimple Lie algebras*, Comm. Math. Phys. 169:563–588. Fusion-category comparison.

Lusztig 1990–1993 *Finite-dimensional Hopf algebras arising from quantized universal enveloping algebra*; *Quantum groups at roots of 1*; *Introduction to Quantum Groups*. Small quantum group, divided-power form, and quantum Frobenius sources.

De Concini–Kac 1990 and **De Concini–Kac–Procesi 1992**. De Concini–Kac form, centre, and quantum coadjoint action.

Arkhipov–Gaitsgory 2003 *Another realization of the category of modules over the small quantum group*, Adv. Math. 173:114–143; **Bezrukavnikov–Finkelberg–Schechtman 1998**, *Factorizable Sheaves and Quantum Groups*, LNM 1691. Small quantum group and factorizable-sheaf bridge.

A.10 Kim's Arithmetic Chern–Simons

Kim 2015 *Arithmetic Chern–Simons theory I*, arXiv:1510.05818. Foundational definition of arithmetic Chern–Simons invariants for number fields via p-adic Hodge theory and Galois cohomology; arithmetic Chern–Simons source for Chapter 37.

Chung–Kim–Kim–Pappas–Park–Yoo 2017 *Abelian arithmetic Chern–Simons theory and arithmetic linking numbers*, arXiv:1706.03336. Used in Chapter 37.

Morishita 2009 *Analogies between knots and primes, 3-manifolds and number rings*, arXiv:0904.3399. MKR dictionary.

A.11 Hartnoll–Yang and Primon Gas

Hartnoll–Yang 2025 *The Conformal Primon Gas at the End of Time*, arXiv:2502.02661. Primary seed reference. L-function as partition function of charged oscillators labelled by primes. BKL–primon correspondence (thermodynamic).

Julia 1990 *Statistical theory of numbers*, in *Number Theory and Physics*, Springer Proc. Phys. 47:276–293. Original primon gas: free Fock space on primes with $Z(\beta) = \zeta(\beta)$. Cited both here (thermodynamic background) and in Chapter 43 §43.5 for the single-mode primon character. Identifying that character with an arithmetic bar requires a separate comparison map.

Spector 1990 *Supersymmetry and the Möbius inversion function*, Commun. Math. Phys. 127:239–252. Supersymmetric primon gas; $1/\zeta(\beta) = \sum \mu(n) n^{-\beta}$.

A.12 Macdonald, DAHA, Schiffmann–Vasserot

Macdonald 1995 *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford. The Hall pairing and symmetric functions are in Chapter I. The Macdonald pairing is VI (1.5), and the orthogonal triangular basis is VI (4.7).

Cherednik 2005 *Double Affine Hecke Algebras*, CUP. DAHA framework.

Etingof–Kirillov 1994 *Macdonald's polynomials and representations of quantum groups*. Quantum-group interpretation of Macdonald polynomials; background for the representation-theoretic reading of Hall polynomials in Section 14.9.

Schiffmann–Vasserot *The elliptic Hall algebra and the equivariant K-theory of the Hilbert scheme of* $\mathbb{A}^2$, arXiv:0905.2555v3, Theorem 3.1. The central specialization of the elliptic Hall algebra identifies with the specified localized Hilbert-scheme convolution algebra. Its polynomial representation is given in section 1.4.

Schiffmann–Vasserot *Cherednik algebras, W-algebras and the equivariant cohomology of the moduli space of instantons on* $\mathbb{A}^2$, arXiv:1202.2756v2. The degeneration is constructed in section 7. Theorem F.10 gives the specialization at $\kappa = 1$, with its enveloping algebra and central variables. See Theorem 14.9.9 for the two distinct geometric comparisons.

A.13 Nekrasov, AGT, Maulik–Okounkov

Nekrasov 2003 *Seiberg–Witten prepotential from instanton counting*, ATMP, arXiv:hep-th/0206161. Z_{inst} definition.

Nekrasov–Okounkov 2006 *Seiberg–Witten theory and random partitions*, change Math., arXiv:hep-th/0306238. Hook-length formula and random partition statistics from the Nekrasov partition function; cited in Chapter 31 for the Plancherel-measure analogy.

Alday–Gaiotto–Tachikawa 2010 arXiv:0906.3219. AGT correspondence: instanton partition functions of 4d $\mathcal{N} = 2$ theories equal conformal blocks of Liouville theory; background for the VOA–Higgs-branch comparison of Chapter 37.

Maulik–Okounkov 2019 *Quantum groups and quantum cohomology*, Astérisque, arXiv:1211.1287. R-matrix on instanton moduli.

Nekrasov–Shatashvili 2009 arXiv:0908.4052. NS limit: $\hbar \to 0$ produces Bethe ansatz equations from quantum integrable systems; cited in Chapter 40 for the Bethe-algebra comparison.

MNOP 2003 arXiv:math/0312059. Plane partition generating function as a Calabi–Yau partition function; background for the topological-string interpretation in Chapter 31.

Gaiotto 2009 arXiv:0904.2715. Class-$\mathcal{S}$ theories from M5-branes on Riemann surfaces; cited for the 4d/2d structure underlying the VOA–Higgs-branch comparison.

Beem et al. 2015 Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees 2015. *Infinite chiral symmetry in four dimensions*, Commun. Math. Phys., arXiv:1312.5344. BLLPRvR VOA/Higgs branch.

A.14 Segal, Frenkel–Ben-Zvi

Segal 2002 *The definition of conformal field theory*, in *Topology, Geometry and QFT*, CUP, pp. 421–577. Segal axioms.

Frenkel–Ben-Zvi 2004 *Vertex Algebras and Algebraic Curves*, 2nd ed., AMS. Standard reference for vertex algebras on algebraic curves; cited in Chapter 30 for the chiral-algebra side of the factorization-homology comparison.

Huang 1997 *Two-Dimensional Conformal Geometry and Vertex Operator Algebras*, Birkhäuser. Rigorous Sewing axioms for VOAs on a Riemann sphere; background for the conformal-field-theory comparison in Chapter 41.

Bakalov–Kirillov 2001 *Lectures on Tensor Categories and Modular Functor*, AMS. Modular tensor categories and the Reshetikhin–Turaev functor; cited in Chapter 37 for the TQFT-Chern–Simons modular-functor background.

Deligne *Le déterminant de la cohomologie*. Determinant line bundle on arithmetic moduli.

A.15 Tate's Thesis and Adelic Analysis

Tate 1950 *Fourier analysis in number fields and Hecke's zeta-functions*, Princeton PhD thesis; reprinted in Cassels–Fröhlich 1967 *Algebraic Number Theory*, CUP. Adelic Poisson summation, functional equation as sewing axiom.

Weil 1967 *Basic Number Theory*, Springer. Restricted direct product, adelic framework.

Jacquet–Langlands 1970 *Automorphic Forms on $GL(2)$*, SLN 114. Local and global theory of automorphic representations of $GL(2)$; cited in Chapter 33 and Chapter 38 for the automorphic side of the Langlands comparison.

Gelbart–Jacquet 1979 *Forms of $GL(2)$ from the analytic point of view*, in *Automorphic Forms, Representations and L-Functions*, Proc. Symp. Pure Math. 33. Functional equations and analytic continuation of L-functions for $GL(2)$; cited in Chapter 33 §3 for the archimedean-place functional equation.

Gross 1998 *On the Satake isomorphism*, in *Galois Representations in Arithmetic Algebraic Geometry*, CUP, Prop. 3.6.

Bump *Automorphic Forms and Representations*, CUP 1998. §1.5, §3.

A.16 Drinfeld, Molev: Yangians and R-Matrices

Drinfeld 1985 *Hopf algebras and the quantum Yang–Baxter equation*, Soviet Math. Dokl. 32:254–258. Yang R-matrix and RTT presentation; foundational for the Bethe-algebra comparison in Chapter 40.

Drinfeld 1988 *A new realization of Yangians and quantized affine algebras*, Soviet Math. Dokl. 36:212–216. Drinfeld current generators for Yangians and quantum affine algebras; background for Chapter 40 on the Bethe-algebra and quantum-integrable structures.

Molev 2007 *Yangians and Classical Lie Algebras*, AMS. Standard reference for Yangians, their representation theory, and Bethe subalgebras. Background for Chapter 40.

Feigin–Odesskii 1997 Shuffle algebras.

Frenkel–Reshetikhin 1998 arXiv:math/9810055. q-characters of representations of quantum affine algebras.

Halacheva–Kamnitzer–Rybnikov–Weekes 2017 *Crystals and monodromy of Bethe vectors*, arXiv:1708.05105. Gaudin algebras, Bethe eigenvectors, and cactus monodromy; cited in Chapter 40 for the Bethe-algebra comparison.

A.17 Arakelov, Deligne, Arithmetic Height Pairings

Arakelov 1974 *Theory of intersection on arithmetic surfaces*, Izv. Akad. Nauk SSSR 38. Foundational intersection pairing on arithmetic surfaces; cited in Chapter 31 for the Arakelov height pairing.

Faltings 1984 *Calculus on arithmetic surfaces*, Ann. Math. 119. Faltings metric on the determinant of cohomology; cited in Chapter 31 for the archimedean Arakelov data.

Gillet–Soulé 1992 *Arithmetic intersection theory*, Publ. IHES 72. Used in Chapter 31 for Arakelov Chern classes.

Durov 2007 arXiv:0704.2030. $\mathbb{F}_1$-projective space and generalized rings; background on approaches to absolute geometry. No comparison with the polynomial Fock carrier is asserted.

Deninger 1994 *Motivic L-functions and regularized determinants*, Proc. Symp. Pure Math. 55.1. Regularised-determinant programme for arithmetic L-functions; cited in Chapter 44 as a parallel formulation of the arithmetic Hilbert–Pólya hypothesis.

Szpiro 1985 Astérisque 127. Arakelov $K_{\overline{\operatorname{Spec}\mathbb{Z}}}$ has degree $\log(2\pi)$.

Bost–Künnemann 2008 Arakelov vector bundles.

A.18 Kesten–McKay

Kesten 1959 *Symmetric random walks on groups*, TAMS 92. On a regular tree of valence $r \geq 2$, the normalized adjacency radius is $2\sqrt{r-1}/r$. The free group on d generators has valence $r = 2d$. The radial Hecke construction is Theorem 13.5.2.

McKay 1981 *The expected eigenvalue distribution of a large regular graph*, Linear Algebra Appl. 40. Free-group spectral measure (Kesten–McKay law): the empirical eigenvalue distribution of a random regular bipartite graph converges to the measure $d\nu_{\mathrm{KM}}$ on $[-2\sqrt{r-1}, 2\sqrt{r-1}]$. Example 14.9.5 computes the tree moments and distinguishes them from Fock norms.

Lubotzky 1994 *Discrete Groups, Expanding Graphs and Invariant Measures*, Birkhäuser, Thm 7.4.5. $GL_2(\mathbb{Q}_p)$ interpretation.

Terras 2011 *Zeta Functions of Graphs*, CUP. Ihara zeta and spectral graph theory. The tree-state calculation is in Proposition 13.5.3.

A.19 Prismatic Cohomology and Fargues–Fontaine

Bhatt–Scholze 2019 *Prisms and prismatic cohomology*, arXiv:1905.08229. Cited in Chapter 30 §3 for the prismatic extension of Fontaine crystalline data toward a non-trivial p-adic diagonal thickening. Also used in Chapter 43 §43.2 for the Nygaard-filtration convention bridging pro-étale and de Rham conventions in F1-reduction.

Bhatt–Lurie 2022 *Absolute prismatic cohomology*, arXiv:2201.06120. Prismatic-Nygaard filtration and δ-ring formalism used in Chapter 43 §43.2 for the convention-match between Arinkin–Gaitsgory and Fargues–Scholze.

Fargues–Fontaine 2018 *Courbes et fibrés vectoriels en théorie de Hodge p-adique*, Astérisque 406. Foundational reference for the Fargues–Fontaine curve X_{FF} as the arithmetic analogue of a complete Dedekind curve, underlying Fargues–Scholze arXiv:2102.13459.

Scholze 2017 *Etale cohomology of diamonds*, arXiv:1709.07343. Diamond formalism used across Chapters 30, 38, 43.

Diamonds, prisms, Adams operations, Frobenius lifts, and Nygaard filtrations supply finite-place coefficient and orientation data. They are not classical almost-complex structures and do not furnish a finite-prime tangent-bundle endomorphism satisfying $J^2 = -1$.

A.20 Ben-Zvi, Nadler, Coherent Springer Theory

Ben-Zvi–Chen–Helm–Nadler 2023 *Coherent Springer theory and the categorical Deligne–Langlands correspondence*, Invent. Math. 235 (2024), 255–344; arXiv:2010.02321. State of the art on global categorical Langlands gluing via coherent Springer theory; cited in Chapter 43 §43.2 on the global gluing obstruction.

Gan–Savin 2022 *Categorical local theta correspondence*. Partial change on restricted tensor product of local automorphic categories; cited in Chapter 43 §43.2 (F1.b).

AGKRRV 2020 *The stack of local systems with restricted variation and geometric Langlands theory with nilpotent singular support*, arXiv:2010.01906. Restricted-variation local systems and nilpotent singular support in global geometric Langlands; cited in Chapter 43 for the singular-support constraint on global gluing.

AGKRRV 2020 *Duality for automorphic sheaves with nilpotent singular support*, arXiv:2012.07665. Geometric Langlands duality restricted to nilpotent singular support; cited in Chapter 43 §43.2.

AGKRRV 2021 *Automorphic functions as the trace of Frobenius*, arXiv:2102.07906. Categorical trace of Frobenius as automorphic function; cited in Chapter 41 for the Frobenius-trace identification in the categorical Hochschild trace.

Gaitsgory–Rozenblyum 2017 *A Study in Derived Algebraic Geometry, Vols. I & II*, AMS Mathematical Surveys and Monographs 221. $(\infty, 2)$-categorical formalism for ind-coherent sheaves with singular support; cited in Chapter 43 §43.2 for ambient-category compatibility in the global gluing step.

A.21 Tate, Weil, Explicit Formula

Riemann 1859 *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsberichte Berlin, 671–680. Original proof of the functional equation $\xi(s) = \xi(1-s)$ via Mellin-theta. Cited in Chapter 43 §43.6 proof of P2.

von Mangoldt 1895 *Zu Riemanns Abhandlung "Über die Anzahl der Primzahlen unter einer gegebenen Grösse"*, J. Reine Angew. Math. 114. Riemann's explicit formula in rigorous form. Used in Chapter 43 §43.6 proof of P2 part (2).

A.22 Li Positivity and Weil Positivity

Li 1997 *The positivity of a sequence of numbers and the Riemann Hypothesis*, J. Number Theory 65:325–333. Introduces the sequence λ_n and proves RH $\Leftrightarrow \lambda_n \geq 0$ for all $n \geq 1$. Central for Chapter 44 on the RH reformulation.

Bombieri–Lagarias 1999 *Complements to Li's criterion for the Riemann hypothesis*, J. Number Theory 77:274–287. Extensions of Li's criterion to Dedekind zeta functions and Selberg class; refinements used in Chapter 44 Remark on multi-L-function versions.

Voros 2003 *Zeta functions for the Riemann zeros*, Ann. Inst. Fourier 53. Explicit computation of first few λ_n. Verified numerically.

Coffey 2005 *Toward verification of the Riemann hypothesis: application of the Li criterion*, Math. Phys. Anal. Geom. 8:211–255. Numerical verification of Li positivity to $n \sim 10^5$.

Weil 1952 *Sur les "formules explicites" de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund; reprinted Oeuvres Sci. II:48–61. Weil's positivity criterion $W(f, f) \geq 0$ for suitable test functions f, equivalent to RH. Referenced in Chapter 44 Table 44.1 for comparison.

The VB* converse (Chapter 57) establishes Hamburger determinacy of the symmetrised zero-moment sequence as the sharpest primary-source form of the VB* reduction. Li 1997 and Weil 1952 give RH-equivalent positivity statements; the arithmetic use of Hamburger determinacy requires the separate operator comparison (Theorem 57.4.22 under Hypothesis 57.4.21).

A.23 Positselski, Iwasawa

Positselski 2011 *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence*, Mem. AMS, arXiv:0905.2621. See the full entry in App. A.5; cited again here for its role as a field-coefficient model for the integral comparison question in Chapter 33. Its coalgebra hypotheses are distinct from a module's μ-invariant.

Ferrero–Washington 1979 *The Iwasawa invariant μ_p vanishes for abelian number fields*, Ann. Math. 109. Proves vanishing for the cyclotomic Iwasawa modules of abelian number fields. Applying it to a proposed Heisenberg coefficient module requires an identification with such an arithmetic module, as explained in Remark 43.5.2 and Question 51.2.8.

A.24 Operator Theory: Moment Problem, Self-Adjointness

Akhiezer–Glazman 1981 *Theory of Linear Operators in Hilbert Space* (2 vols.), Pitman / Dover reprint. Ch. VI.3 on the Hamburger moment problem. Used in Chapter 44 Corollary 44.3.9 and Chapter 51 for the failure of the moment-problem equivalence in the unbounded-operator case. The moment problem is determinate for bounded self-adjoint operators on a positive-definite Hilbert space; in the unbounded

case, Carleman's sufficient condition for determinacy is $\sum_n \lambda_{2n}^{-1/(2n)} = +\infty$, and without it two distinct operators can share the same moment sequence.

Reed–Simon 1980 *Methods of Modern Mathematical Physics*, Vol. I (Functional Analysis), Academic Press. Ch. VIII on unbounded operators, self-adjoint extensions, spectral theorem. Standard reference for the domain-theoretic care demanded by the arithmetic Hilbert–Pólya hypothesis (Hypothesis 44.3.2).

Tomita–Takesaki 2003 *Theory of Operator Algebras II*, Springer EMS. KMS_β modular involution cited in Chapter 51 Theorem 51.2.9 as the specification of the Bost–Connes involution on the dual.

The Connes 1999 (arXiv:math/9811068) adele class space spectral realisation and the Deninger 1994 motivic cohomology regularised-determinant programme anchor equivalent formulations of H-P^{Ar} (Theorem 57.3.14). The reduction closes the operator-theoretic domain problem via the Friedrichs extension (Theorem 57.3.2); the determinant-divisor and trace-compatibility condition, requiring the normal zero-divisor operator to have divisor div $\xi_{\mathbb{R}}$ and Weil–Connes trace equal to the non-trivial zero distribution, remains the genuine Hilbert–Pólya content (Remark 57.3.16).

A.25 Restricted Direct Product, Flath, Godement–Jacquet

Flath 1979 *Decomposition of representations into tensor products*, in *Automorphic Forms, Representations, and L-Functions*, Proc. Symp. Pure Math. 33, Part 1, pp. 179–183. Flath's theorem on irreducible admissible automorphic representations of $GL_n(\mathbb{A})$ as restricted tensor products of local admissible representations. This decomposition does not construct the arithmetic placement of $\mathcal{V}^{\mathrm{prim}}$; that map is part of (K2) in Theorem 51.2.9.

Godement–Jacquet 1972 *Zeta Functions of Simple Algebras*, SLN 260, Springer. Local functional equations and standard local L-factors. The spherical Whittaker norm in Lemma 56.7.4 uses its declared multiplicative Haar measure and has factor $m_p(1 + p^{-1})L_p(1, \mathrm{Ad})$. A functional equation does not supply the arithmetic pairing map of (K5).

Gelbart 1975 *Automorphic Forms on Adele Groups*, Annals of Math. Studies 83, Princeton University Press. §3.1 on sph-finite automorphic representations and their restricted-tensor-product decomposition.

Kneser 1966 *Strong approximation*, in *Algebraic Groups and Discontinuous Subgroups*, Proc. Symp. Pure Math. 9, pp. 187–196. Strong approximation for simply connected semisimple algebraic groups. In the GL_2 application it applies to the derived group SL_2; the determinant idele-class quotient is the separate Tate component. Cited in Chapter 51 Theorem 51.2.4 ($\mathrm{L_{ACD}}$.4).

Platonov–Rapinchuk 1994 *Algebraic Groups and Number Theory*, Academic Press. Modern reference for strong approximation and related arithmetic results.

A.26 Cross-Volume Primary Sources

The arithmetic branch draws on results from the three preceding volumes. The relevant primary sources from those volumes are cited in the entries above at the point of use; the cross-volume dependency structure is:

Vol. I Five-theorem structure (Theorems A, B, C, D, H), bar-cobar adjunction, SC-formality, and the chiral Koszul duality framework. Invoked in Chapters 33 and 38.

Vol. II Holomorphic-topological twist $\mathsf{SC}^{\mathrm{ch,top}}$, 3D HT QFT, and 3D quantum gravity. Invoked in Chapter 37 for the arithmetic Chern–Simons comparison.

Vol. III Two-stage factorisation $\Phi_d = \mathrm{Sp}^{\mathrm{ch}} \circ \Phi_d^{\mathrm{FA}}$; Borcherds/BKM universality $\Phi_N = c_N(0)/2$. Invoked in Chapter 41 for the six-window synthesis.

Appendix B

Arithmetic Trace Comparisons

The following primary-source families concern structural hypotheses about $\mathcal{V}^{\mathrm{prim}}$. The source families concern different structural obstructions and proposed trace comparisons, with distinct hypotheses. The table records the source class, obstruction, and chapter reference. Sections B.2–B.6 record the obstructed structures, the remaining constraints, three-way gluing comparisons, and the missing arithmetic maps.

B.1 Primary Source Families

Source family	Angle	Reference/Result
Drinfeld+ Kapranov	Finite-place fields	26: formal fields. Arithmetic coefficient comparison required
Kazhdan+ Beilinson	GL_1 Adams vs GL_2 Hecke	27: Satake pair on rank-2 Heisenberg
Gelfand+ Borger	Λ-ring axiomatic precision	28: polynomial Adams and divisor actions; integral Hall lattice
Connes+ Soibelman	Bost–Connes vs $\mathcal{V}^{\mathrm{prim}}$	29: Bost–Connes $\otimes$ universal Heisenberg
Costello+ Gwilliam	Adelic factorization axioms	30: (F1)–(F5) on $\mathbb{C}_\infty$
Etingof	Prime-indexed Macdonald and Hall models	Section 14.9: tensor pairings, shuffle products and geometric comparison
Nekrasov	Arithmetic instanton Z	31: Degenerate $(\log p, \infty)$ limit
Segal	Arithmetic modular functor	32: Modular functor = Tate
Beilinson–solo	Arithmetic coefficient bar	33: residue and chiral comparisons require their operation maps
Witten	TQFT / BKL / Langlands triangle	34: Automorphic vertex, not TQFT vertex
Polyakov	Local charge and global residue	35: rank residue $r_\zeta = 2$. Conformal comparison separate
Gaiotto	4D-gauge to VA arithmetic lift	37: Kim arithmetic Chern–Simons $[\mathbb{G}_m, 1]$ classical boundary shadow
Bezrukavnikov+ Bernstein	Langlands spectral side	38: Iwahori Hecke Hochschild trace
Kontsevich+ Soibelman	E_n formality / CoHA	39: E_1 integral, E_2 over $\mathbb{Q}$

Drinfeld–solo	Arithmetic Yangian / R-matrix	40: Bethe algebra, Langlands = Bethe ansatz

B.2 OBSTRUCTIONS ACROSS SOURCE FAMILIES

The following comparisons concern specific algebraic and geometric constructions. Positive models and obstructions retain their own carriers, coefficient topologies, and hypotheses:

(i) The Hochschild trace class $\mathcal{V}^{\mathrm{prim}}$ does not itself specify a vertex algebra. Local Heisenberg fields exist, while the prescribed infinite Sugawara sum fails on its algebraic vacuum. Chapter 35 also records the prime-zeta natural boundary, separately from the rank-residue calculation and the remaining conformal comparison.

(ii) Abstract common field extensions of all $\mathbb{Q}_p$ exist, as in Observation 26.3.1. Such a field does not supply continuous local embeddings or an adelic topology. An arithmetic restricted tensor still needs its coefficient and representation maps.

(iii) The p-adically completed tensor $\mathbb{Z}_p \widehat{\otimes}_{\mathbb{Z}} \mathbb{Z}_p$ is $\mathbb{Z}_p$. The ordinary tensor has a nonzero multiplication kernel, as proved in Chapter 26. These are different completions of the diagonal problem. Neither calculation constructs the required arithmetic collision operations.

(iv) T_n does not lift Adams operations to GL_2 Hecke operators. The explicit Kazhdan–Beilinson computation gives $T_p^2(a_{-1}^{(p)}|0\rangle) - p \cdot a_{-1}^{(p)}|0\rangle \neq T_{p^2}(a_{-1}^{(p)}|0\rangle)$.

(v) The index-shift Adams maps fail the Frobenius congruence on the polynomial Newton-generator lattice, whereas they satisfy it on the integral Hall lattice of symmetric functions. The divisor substitutions T_p still fail on that Hall lattice. These are distinct integral carriers and operations, as proved in Chapter 28. Neither lattice calculation supplies a Λ-structure on the arithmetic trace class without a compatible arithmetic comparison.

(vi) Arithmetic Yangian RTT relations have no global spectral parameter. The Yang-Baxter spectral parameter $u - v$ has no analogue across distinct primes; there is no meromorphic 1-form on $\operatorname{Spec} \mathbb{Z}$.

(vii) A 4D class-$\mathcal{S}$ gauge boundary does not transplant to $\overline{\operatorname{Spec} \mathbb{Z}}$. The Hitchin base $\bigoplus_i H^0(\Sigma, K_\Sigma^{d_i})$ uses a holomorphic canonical bundle on a complex curve. The Arakelov dualizing line on $\overline{\operatorname{Spec} \mathbb{Z}}$ is not such an object.

(viii) The curve-theoretic category of $\mathcal{D}$-modules on Bun_{GL_2} does not arise from $\operatorname{Spec} \mathbb{Z}$ itself. The moduli stack $\mathrm{Bun}_{GL_2}(\operatorname{Spec} \mathbb{Z})$ is $BGL_2(\mathbb{Z})$: one coarse point with arithmetic stabilizer. This stack has representation theory of the stabilizer, not the geometric Hecke correspondence of the curve case.

(ix) The integral E_2-chiral-centre enhancement is not obtained by replacing the chain operad $C_\bullet(E_2; \mathbb{Z})$ with $\mathrm{Ger}_{\mathbb{Z}}$. Dyer–Lashof and Kaledin-type p-curvature classes obstruct that strictification, not the existence of an integral E_2-algebra.

B.3 COMMON STRUCTURAL CONSTRAINTS

The proposed arithmetic realization has the following structural requirements:

(i) The arithmetic curve is Arakelov-compactified $\overline{\operatorname{Spec} \mathbb{Z}}$, not $\operatorname{Spec} \mathbb{Z}$ (across source families).

(ii) The finite-place incidence poset does not supply formal collision coordinates. Formal Heisenberg fields can nevertheless be constructed over $\mathbb{Q}_p$, as in Chapter 26. Comparing them with arithmetic coefficient objects requires actual operation maps.

(iii) The Steinberg relation $\{p, 1-p\} = 0$ holds in Milnor K-theory. Comparing its residue complex with chiral collision operations requires a common carrier and maps preserving the operations, as in Chapter 33.

(iv) The Hecke structure is carried by the Iwahori-equivariant affine Hecke category, not the spherical category (Bezrukavnikov+Bernstein, Kazhdan+Beilinson).

(v) The Iwahori Hecke Hochschild trace is the local Bezrukavnikov–Bernstein model. Its global identification with $\mathcal{V}^{\mathrm{prim}}$ is the arithmetic BZN input F3. Kim arithmetic Chern–Simons (Kim arXiv:1510.05818) provides the classical boundary torsor construction, and the adelic factorization algebra lives on $\mathbb{C}_\infty$ (Costello+Gwilliam).

B.4 Three-Way Gluing of Presentations

Three proposed presentations of $\mathcal{V}^{\mathrm{prim}}$ require arithmetic comparison maps. The table names their carriers and corresponding chapters.

Presentation	Source family / Chapter
Arithmetic Chern–Simons classical boundary shadow for $\mathcal{V}^{\mathrm{prim}}$	Gaiotto, 37
$\mathrm{HH}_\bullet(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}) \xrightarrow{\mathrm{F3}} \mathcal{V}^{\mathrm{prim}}$	Bezrukavnikov+Bernstein, 38
$\mathcal{F}_{\mathrm{CG}}(\mathcal{V}^{\mathrm{prim}}) \in \mathrm{FactAlg}_{\mathbb{C}_\infty}(\mathrm{Mod}_{\widehat{\mathbb{Z}},\psi})$	Costello+Gwilliam, 30

Ben-Zvi–Nadler (arXiv:0904.1247) identifies $\mathrm{HH}_\bullet(\mathrm{D}(G/G)) = \mathcal{O}(\mathrm{LocSys}_G(S^1))^{\mathrm{flat}}$ over $\mathbb{C}$; the triangle relating arithmetic Chern–Simons, the Hecke category, and the Hochschild trace is the complex-geometric model for the three-way arithmetic comparison. The arithmetic identification remains conditional on the full Weil/(φ,Γ)-module target and a quantum boundary category.

B.5 Distinct Comparisons

The comparison questions concern different algebraic and geometric structures, even when their bibliographies overlap. Disjoint citations do not establish independence of mathematical hypotheses or construct a common object. The identification requires the maps in the following compatibility observation.

B.6 Compatibility and the Arithmetic Construction Problem

Remark B.6.1 *(compatibility of specified identifications).* Suppose the proposed arithmetic presentations are constructed in a specified common category. Suppose each has a specified isomorphism $f_i : X_i \to V$ preserving the stated operations and arithmetic data. Then $f_j^{-1} f_i : X_i \to X_j$ identify these presentations and satisfy $(f_k^{-1} f_j)(f_j^{-1} f_i) = f_k^{-1} f_i$. Transporting a six-window conclusion to V requires a proved source statement and comparison maps preserving its hypotheses and structures. Conjecture 41.1.1 does not supply that source statement.

The composite of two structure-preserving isomorphisms preserves those structures. The displayed identity follows by cancelling $f_j f_j^{-1}$. Any property transported to V must be among the properties preserved by the specified maps.

Question B.6.2 *(arithmetic trace comparisons).* Construct the global arithmetic Iwahori Hecke category, its Hochschild trace complex, and the proposed arithmetic boundary and factorization objects. Specify their coefficient rings, gradings, tensor products and completions. Construct the functors or chain maps

relating their specified common carriers, with compatibility for central and Hecke actions, local gluing and the claimed trace pairings. The local-to-global categorical and spectral comparison is F1, the finite-place enhancement is F2, and the arithmetic trace comparison is F3 in Conjectures 42.1, 42.2, and 42.3. Their existence and mutual compatibility are the missing inputs to a common arithmetic realization.

Several starting constructions have different, explicit carriers. Chapter 28 constructs the polynomial divisor and Adams operations and the integral Hall lattice. Chapter 29 constructs gauge Hochschild classes on its specified Bost–Connes algebras and analyzes their domains. Chapter 54 computes finite-algebra extension products and regular Hochschild homology. None of these calculations by itself supplies the arithmetic functors in the preceding question.

B.7 SIX SOURCE FAMILIES IN GENUS g

The initial genus-g ansatz was: "a genus-g arithmetic surface with almost complex structure carrying the cohomological signature of a genus-g algebraic curve." Six primary-source families concern this ansatz. Their proposed arithmetic-modular comparisons require the maps and hypotheses stated in Chapter 102 (all-genus arithmetic Koszul duality) and Chapter 103 (3D arithmetic Chern–Simons via the ordered incidence bar and conditional Beilinson–Drinfeld/Costello–Gwilliam comparison).

Primary Source Families

Source family	Source list	Proposed comparison
Source family 1: Arakelov	Arakelov 1974; Faltings 1984; Gillet–Soulé 1990; Bost–Künnemann 2010	Window I: arithmetic amplitude generating function AAGF_g (Chapter 102.2), arithmetic surface X/O_K, Hasse–Weil functional equation
Source family 2: Deninger / Connes–Consani	Deninger 1998; Connes–Consani 2014–2016; Bost–Connes 1995; Bost–Connes–Marcolli 2004	Window II: Minkowski torus over totally imaginary K of degree $2g$, Dedekind $\xi_K(s) = \xi_K(1-s)$
Source family 3: Fargues–Scholze	Fargues–Scholze 2021; Bhatt–Scholze 2022; Emerton–Gee 2019; Caraiani–Scholze 2017–2019	Window III: Ordinary crystalline Emerton–Gee stack $\mathcal{X}_{2g}^{\mathrm{crys,ord}}$, prismatic Frobenius as J
Source family 4: Mochizuki / Hodge–Arakelov	Mochizuki 1999 Publ. RIMS; Kapranov–Smirnov 1995; Durov 2007	Window IV: Arithmetic Teichmüller with theta-structure; Hodge–Arakelov comparison

Source family 5: Homotopical bar-cobar	Beilinson–Drinfeld 2004; Francis–Gaitsgory 2012; Ben-Zvi–Nadler 2009–2013; Lurie HA; Positselski 2011	Window V: $\mathcal{V}^{\mathrm{prim,surf},g}$ with E_2-chiral structure via Dunn additivity
Source family 6: Kudla–Rapoport / Shimura–Siegel	Kudla 1997; Kudla–Rapoport 2011; Bruinier 2002; Li–Liu 2021; Arthur 2013	Window VI: Siegel–Shimura genus-g datum SSG_g (Chapter 102.2), Siegel spin L-function with $\Lambda(s) = \epsilon\Lambda(2g+1-s)$

Obstructions

The six source families jointly obstruct six readings of the initial ansatz (Chapter 102.3 indexes the full list): the category-error reading of "almost complex structure on an arithmetic scheme" (no real tangent bundle on Spec $\mathbb{Z}$; Newlander–Nirenberg automatic on real 2-manifolds); "arithmetic Euler characteristic $= 1-g$" (Gillet–Soulé analytic torsion term); Künneth factorisation in arithmetic Chow (Bost–Künnemann Bott–Chern term); finite genus of $\overline{\mathrm{Spec}\,\mathbb{Z}}$ in the Deninger frame (Deninger-genus $= +\infty$); conflation of curve-genus with Siegel-genus (distinct integers); and the $\kappa + \kappa^! = 2 - 2g$ Euler-shift (Koszul sum preserved at 0; genus enters via rank, not shift).

Proposed Arithmetic-Modular Datum

The arithmetic genus-g datum $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$ (Chapter 102.1.1), with the essential conditional $\kappa_{\mathsf{G}}^{\mathrm{Ar},g}(\mathcal{D}_g) + (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar},g}(\mathcal{D}_g) = 0 \Leftrightarrow \Lambda_{\mathcal{D}_g}(s) = \epsilon_{\mathcal{D}_g}\Lambda_{\mathcal{D}_g}(w_{\mathcal{D}_g} - s)$ (Chapter 102.4.1), with $w_{\mathcal{D}_g} \in \{1, 2, 2g+1\}$ window-indexed. The functional-equation clause is not a zero-placement theorem. The all-genus F-RH corollary (Chapter 102.5.1) contains the corresponding zero-placement clauses only after the relevant spectral-positivity input is imposed: RH for ζ, Hasse–Weil RH for Jacobians, Dedekind-zeta GRH for totally imaginary fields, and Siegel-spin RH for GSp_{2g} cusp forms.

The arithmetic-modular extension sources the finite-level arithmetic Chern–Simons state sums and the conjectural renormalized $GL_2(\widehat{\mathbb{Z}})$ limit on the étale $\overline{\mathrm{Spec}\,\mathbb{Z}}$ of Chapter 103. The finite-level boundary category is defined there after the finite cobordism input, and the ordered E_1-bar complex is constructed there; the identification of the Hochschild boundary trace with $\mathcal{V}^{\mathrm{prim}}$ and the partition-function identity with ξ remain conditional on profinite renormalization and local trace comparison. The functional equation then appears as the involutive symmetry of the partition function; zero-placement remains the separate positivity condition.

Comparison Boundary

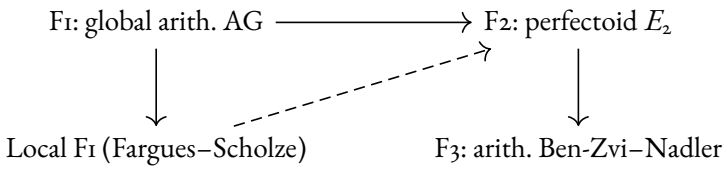
The six genus-g presentations concern different carriers. A common arithmetic-modular datum requires their specified comparison maps and compatibility with the operations used in each presentation.

Appendix C

Three Structural Conjectures

The comparisons F1, F2 and F3 concern the arithmetic spectral recipient, factorization operations and the global trace. Their proposed relation uses specified local inputs and global comparison maps. The following diagrams organize those maps. They do not prove logical independence of the hypotheses.

C.1 The Graph



The local constructions of Fargues–Scholze and Emerton–Gee supply different inputs to F1 and F2. They do not alone give the displayed global equivalences or the required Iwahori factorization operations. F3 additionally needs the global trace comparison on its specified arithmetic carrier.

C.2 F1: Global Arithmetic Arinkin–Gaitsgory

Conjecture C.2.1 *(restatement of F1).* For $G = GL_2$ and the Selmer-derived global spectral stack $\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}$ (the derived moduli stack of Selmer deformations of a residual representation $\bar\rho$ with determinant condition det and cyclotomic twist datum $\mathcal{L}$, as in Chapter 42):

$$\mathrm{D}([GL_2])^{\text{sph-fin}} \;\simeq\; \mathrm{IndCoh}_{\mathrm{Nilp}}\big(\mathcal{X}^{\mathrm{Sel}}_{\bar\rho,\det,\mathcal{L}}\big).$$

Local inputs.

(i) Fargues–Scholze arXiv:2102.13459 (local geometrization and spectral action, not the displayed EG equivalence).

(ii) Caraiani–Scholze arXiv:1511.02418 (unitary Shimura variety case).

(iii) Emerton–Gee stack construction arXiv:1908.07185.

Global condition.

- Proof of the local automorphic-to-Emerton–Gee $\mathrm{IndCoh}_{\mathrm{Nilp}}$ comparison, including small primes and central-character families.
- Compatibility of local comparison functors under restricted-tensor-product gluing over primes.
- Identification of the Whittaker model across all places simultaneously.
- Functional-equation-level compatibility between spectral and automorphic sides.

Hypothesis. This is a global comparison theorem.

C.3 F2: PERFECTOID E_2-ENHANCEMENT

Conjecture C.3.1 *(restatement of F2).* The local Iwahori Hecke category $\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p}$ carries a canonical E_2-monoidal structure from the perfectoid Beilinson–Drinfeld Grassmannian over $\operatorname{Spec}\mathbb{Z}_p$, and the restricted tensor product $\otimes'_p \mathrm{Hk}^{\mathrm{Iw}}_{GL_2,p}$ inherits a globally-defined E_2-monoidal structure.

Local ingredients (theorem).

(i) Perfectoid affine Grassmannian (Fargues–Scholze, Scholze diamonds).
(ii) Witt-vector affine Grassmannian (Zhu arXiv:1603.05593; Bhatt–Scholze arXiv:1507.06490).
(iii) Equal-characteristic BD Grassmannian E_2-structure (Mirković–Vilonen; Beilinson–Drinfeld Grassmannian factorization).

Global condition.

- Diamond-gluing of Iwahori Hecke categories across primes.
- Compatibility with Satake and Whittaker equivalences.
- Perfectoid Dunn-additivity argument producing E_1-algebra structure on the Hochschild trace.

Hypothesis. This is a mixed-characteristic factorisation theorem.

C.4 F3: ARITHMETIC BEN-ZVI–NADLER

Conjecture C.4.1 *(restatement of F3).* $\mathrm{HH}_\bullet\big(\mathrm{Hk}^{\mathrm{Iw}}_{GL_2,\mathrm{glob}}\big) \xrightarrow{\sim} \mathcal{V}^{\mathrm{prim}}$ as E_1-algebras.

Ingredients.

(i) Conjecture F1 (for the spectral side identification).
(ii) Conjecture F2 (for E_2-enhancement of the local Hecke factors).
(iii) Ben-Zvi–Nadler arXiv:0904.1247 (for the $\mathbb{C}$-version, which F3 arithmeticises).
(iv) Explicit computation matching the Hochschild trace character to $\zeta(s)$.

Consequences of F3.

- F3 implies the Gaiotto Chern–Simons identification (Theorem 37.4.1).
- F3 implies the Bezrukavnikov–Bernstein Hochschild identification (Theorem 38.5.1).
- A comparison with the coefficient bar additionally requires its algebra and operation maps. Theorem 33.8.2 proves the coefficient construction, not its identification with a geometric or trace target.
- F3 implies the arithmetic six-window placement (Conjecture 41.1.1).

Hypothesis. This is an arithmetic trace theorem.

C.5 Independent Conjecture: The G-Row Complementarity

Question C.5.1 *(central classes and Mellin reflection).* Construct the arithmetic object, its dual and scalar central invariants κ^{Ar}, $(\kappa^!)^{\mathrm{Ar}}$. Construct a common coefficient space, an involution J, a defect identity and an evaluation E such that

$$v - Jv = (\kappa^{\mathrm{Ar}} + (\kappa^!)^{\mathrm{Ar}})w, \qquad E(w) \neq 0,$$

with $E(v) = \Lambda(s)$ and $E(Jv) = \Lambda(1-s)$. Here $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$. These maps would give the intended equivalence

$$\kappa^{\mathrm{Ar}} + (\kappa^!)^{\mathrm{Ar}} = 0 \quad \Longleftrightarrow \quad \xi(s) = \xi(1-s)$$

by Theorem 43.5.1.

The scalar functional equation is proved by Poisson summation. The question asks for its arithmetic central-class realization and for detection of the defect by E. A co–contra equivalence alone does not identify those maps. Any dependency on F1–F3 must be determined from the proposed arithmetic realization.

C.6 Chain of Implications

The arithmetic branch has the following implication form:

(i) The G-row conjecture (Question C.5.1) would identify the Riemann functional equation as an arithmetic Koszul-duality consequence.

(ii) F3 conditional on F1 and F2 unifies the three presentations of $\mathcal{V}^{\mathrm{prim}}$.

(iii) F1 is the global arithmetic Arinkin–Gaitsgory input.

(iv) F2 is the perfectoid E_2-enhancement input.

(v) F1, F2, and the conditional F3 comparison imply the six-window structural statement.

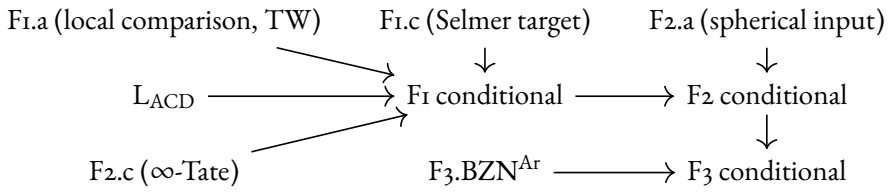
C.7 Connection to Vol III

Vol III's B-row $\kappa + \kappa^! = 8$ (Mukai-enhanced K3 Heisenberg via Bruinier Heegner Chern-class reciprocity) has an arithmetic avatar (F4, Chapter 42 §3). Together the arithmetic G-row and the arithmetic B-row would give two of the five Vol I derived-centre-ceiling values as consequences of automorphic and Galois-deformation theorems. The chiral centre is then constrained by arithmetic comparison data.

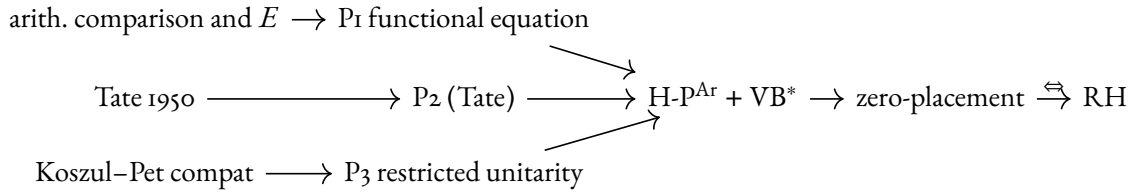
C.8 Extended Graph with F-RH

The conjectures F1, F2, F3 of the preceding sections address the six-window placement of $\mathcal{V}^{\mathrm{prim}}$. A second cluster of conditions, independent of F1–F3, governs the connection to the Riemann hypothesis. Chapter 43 provides the structural branch; Chapter 44 states F-RH, whose zero-placement clause is RH and whose other clauses are the positive Hermitian, Koszul, and Zhu structures on $\mathcal{V}^{\mathrm{prim}}$. The arrows are:

F1–F3 structural branch (six-window placement).



four-input RH branch.



The dependency partitions into two branches:

F1–F3 structural branch (six-window placement, RH-independent). L_{ACD} (the adelic comparison and descent lemma relating the global spherical category to the restricted tensor product of local Iwahori categories) → F1 → F3.BZN^{Ar} (the arithmetic Ben-Zvi–Nadler functoriality for the loop space of the global spectral stack) → F3. The sub-conjecture F2.c has a restricted pointed central-unit theorem in Chapter 50; the mixed-characteristic Iwahori E_2 local input (F2.b) remains conditional. The F1–F3 structural branch states the six conditional windows of $\mathcal{V}^{\mathrm{prim}}$.

four-input RH branch. Here P1 is the functional-equation branch (the statement that the adelic Pozitsel'ski bar complex has the functional-equation symmetry of $\xi(s)$); P2 is the Tate-adelic computation (Tate 1950); P3 is restricted Koszul unitarity (the Koszul–Petersson compatibility on the spherically-finite cuspidal core); and H-P^{Ar} is the arithmetic Hilbert–Pólya determinant-trace problem (existence of a self-adjoint operator whose normal translate has regularised determinant divisor div $\xi_{\mathbb{R}}$). The P1 implication requires the actual arithmetic comparison, central-defect identity and detecting Mellin evaluation. Koszul–Petersson compatibility implies P3 on its stated domain. The inputs P1, P2, P3 and H-P^{Ar} imply zero-placement under VB*.

Here VB* must include the specified positive spectral realization, its common operator domains and the arithmetic trace comparison. A positive measure or Hamburger determinacy alone does not identify its spectrum with the zero divisor of ξ. With that identification, self-adjointness forces real spectral ordinates and hence critical-line zero placement. A primary Virasoro representation and a vacuum-compatible positive sector remain separate constructions, as in Chapter 51.

The F1–F3 branch concerns construction of the arithmetic trace object. The spectral branch concerns its independently specified zero-divisor and positivity comparisons. Neither branch supplies the other's maps.

C.9 BOUNDARY CONDITIONS

Input	Branch	Size	Present form
L_{ACD} (adelic descent)	A	global theorem	Conjectural

Input	Branch	Size	Present form
F2.c (∞-Tate product)	A	mixed-characteristic theorem	Restricted pointed central-unit theorem; full mixed-characteristic statement conditional
F2.b (mixed-char Iwahori E_2)	A	mixed-characteristic theorem	Conditional
F3.BZN^{Ar} (arith. BZN)	A	trace theorem	Conjectural
arith. co–contra	B	categorical comparison	Integral recipients and adjunction required. Completion criterion proved
Koszul–Pet compat	B	automorphic theorem	Computed local forms; arithmetic placement and form comparison required
VB* (arith. Virasoro positivity)	B	Unknown	Positive spectral condition implies RH

Table C.1: Boundary conditions for the structural and RH branches.

C.10 A Sufficient Path to RH

The sufficient implication to RH uses an arithmetic realization of P1, Koszul–Petersson compatibility, H-P^{Ar} and the positive VB* spectral condition. The coefficient bar construction and the local pairing models provide positive algebraic input. Their arithmetic interpretation requires the co–contra functors and the central-defect and Mellin-evaluation maps of Question C.5.1.

Arithmetic Koszul–Petersson compatibility on the spherically finite cuspidal core also requires the placement and form-preserving maps of Hypothesis 67.1.1. Determinant-trace realization, full-sector P3, mixed-characteristic Iwahori E_2 and VB* positivity remain separate conditions.

Refined Boundary List

Input	Branch	Size	Present form
L_{ACD}.0, .1, .2, .3	A	each large theorem	Partial reductions; global glue remains conditional
(BC-diamond-glue).core	A	global theorem	Conjectural, single cross-prime Bernstein compat
F2.c (pointed E_n datum)	A	–	Restricted result of Theorem 50.1.3; mixed-characteristic Iwahori E_2 input separate
F3.BZN^{Ar} R1.fin-type	A	–	Smooth affine Perf(A) trace calculation; arithmetic comparison separate
F3.BZN^{Ar} R1.formal.a	A	–	Completed trace and support comparison required; a categorical colimit alone is insufficient
F3.BZN^{Ar} R1.formal.b,c	A	pro-étale theorem	Pro-étale loop-space and flat-locus hypotheses
F3.BZN^{Ar} R2 (Selmer target)	A	Selmer theorem	Target and tangent complex fixed; Noetherian and nilpotent-support hypotheses remain
F3.BZN^{Ar} R3 (full target)	A	–	Conditional; the inertia shadow is not the Weil/(φ, Γ) target

Input	Branch	Size	Present form
F3.BZN^{Ar} R4 wild $p \in \{2, 3\}$	A	–	Local arithmetic categories and trace maps open; full tensor includes mixed terms
Wild Langlands functoriality	A	–	Open comparison (Question 54.6.1)
arith. Positselski C1–C5	B	–	Specified coefficient models. Arithmetic functors and invariant maps required
arith. Positselski (abelian Heis.)	B	–	Scalar functional equation. Central-defect realization and detection separate
Koszul–Pet K5 (local pairing comparison)	B	–	Computed model norms; arithmetic placement and adelic comparison required
Koszul–Pet compat	B	–	Conditional on arithmetic placement and form identity; full-sector extension separate
H-P^{Ar}	B	–	Friedrichs domain theorem; determinant divisor and trace matching open
VB* $\Rightarrow$ zero-placement	B	–	Conditional implication from P2, P3, H-P^{Ar}, and positive trace
VB* (main direction)	B	–	Positive spectral condition; determinacy alone is insufficient
VB* converse	B	–	Strictly finer than RH; 4-way equivalence (Ch. 57)

TABLE C.2: Refined boundary conditions.

Sufficient RH Path

A sufficient implication to RH within the arithmetic formalism is the four-input conditional conjunction on the four-input RH branch:

(i) **Arithmetic comparison**: the integral recipients, co–contra functors and completed coefficient bar on their specified domain. The P1 interpretation requires the central-defect identity and detecting Mellin evaluation of Question C.5.1.

(ii) **Koszul–Pet compat** (the four-input RH branch): the normalized local forms are constructed. Even on a sph-fin cuspidal core their arithmetic interpretation requires the placement and form identity of Hypothesis 67.1.1. Full V^{prim} compatibility and global KMS$_1$/GNS identification remain open.

(iii) **H-P^{Ar}** (the four-input RH branch): the independent Hilbert–Pólya determinant-divisor matching problem.

(iv) **VB* main direction** (the four-input RH branch): the positive spectral realization with its arithmetic trace and domain comparisons.

The F1–F3 structural branch remains RH-independent: (BC-diamond-glue).core, F2 trace continuity, and arithmetic BZN formal-loop compatibility affect the construction of $\mathcal{V}^{\mathrm{prim}}$, not zero placement. The equivalences involving VB* in Chapter 57 retain their spectral-realization, positivity and determinacy hypotheses. Hamburger determinacy alone is not equivalent to RH. The primary-Virasoro and vacuum-sector comparisons remain distinct from a self-adjoint zero-divisor operator.

C.11 Dependency Boundary

Three hierarchically ordered conjectures (F1 → F2 → F3) and one independent conjecture (G-row) govern the synthesis branch. On the RH branch: P1 (functional equation), P2 (Tate adelic), P3 (restricted Koszul unitarity), H-P^{Ar} (Hilbert–Pólya determinant-divisor matching), and VB*, with its positive spectral realization and explicit trace and domain hypotheses. The restricted pointed central-unit theorem gives the proved part of F2.c, and the strict abelian primary arithmetic Positselski bar supplies the P1 forward direction. The arithmetic branch as stated (Chapter 43, Chapter 44) reduces the RH-strength condition to H-P^{Ar} determinant-trace realisation and the full positive spectral VB* hypotheses. A primary Virasoro realization requires its own compatible operators and invariant domains.